Administration Guide | PUBLIC
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SAP Cloud Identity Services

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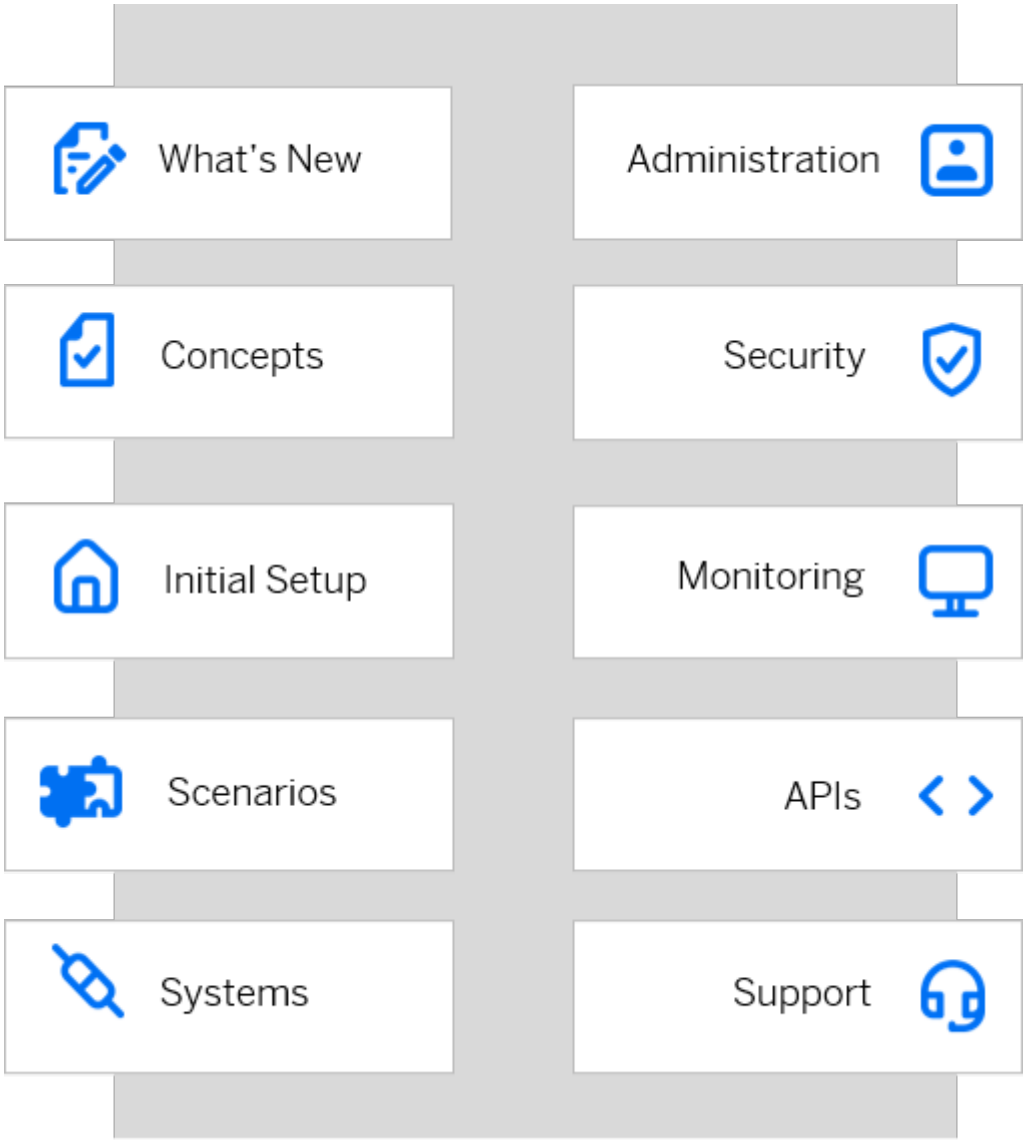
1 SAP Cloud Identity Services



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- #unique_1/unique_1_Connect_42_subsection-im10 [page 7]

Hover over the image and click the highlighted areas for more information on each documentation section of SAP Cloud Identity Services.

Before you start, see What Are Cloud Identity Services? [page 7]

What's New

Stay informed about the newest features and updates of SAP Cloud Identity Services.

For more information, see What's New for Cloud Identity Services [page 26].

Concepts

Understand the concepts relevant for SAP Cloud Identity Services.

For more information, see Concepts [page 26].

Initial Setup

Explore the required steps to get started with SAP Cloud Identity Services.

For more information, see Initial Setup [page 523].

Scenarios

Understand the scenarios supported by SAP Cloud Identity Services.

For more information, see Scenarios [page 544].

Systems

Familiarize yourself with the supported provisioning systems (connectors) by Identity Provisioning.

For more information, see Supported Systems [page 585].

Administration

Learn how to configure and administer SAP Cloud Identity Services.

For more information, see Administration [page 1918].

Security

Use the security features of SAP Cloud Identity Services to support the security policies of your organization.

For more information, see Security [page 2896].

Monitoring

Explore monitoring and reporting to get information about audit logs, job logs, usage statistics, and more.

For more information, see Monitoring and Troubleshooting [page 2964].

APIs

Get to know the APIs of SAP Cloud Identity Services delivered on the SAP Business Accelerator Hub.

For more information, see SAP Cloud Identity Services APIs.

Support

Get support when you have questions or encounter issues while working with SAP Cloud Identity Services.

For more information, see Getting Support [page 3141].

1.1 What Are Cloud Identity Services?

Manage identities, authorizations and SSO across cloud and on-premise solutions.

SAP Cloud Identity Services are a group of services of SAP Business Technology Platform (SAP BTP), which enable you to integrate identity and access management between systems. The goal is to provide a seamless single sign-on experience across systems while ensuring that system and data access are secure. SAP Cloud Identity Services include Identity Authentication, Identity Provisioning, Identity Directory, and Authorization Management.

Environment

SAP Cloud Identity Services run on several underlying Infrastructure-as-a-Service technologies and regions. Regions are provided either by SAP or by SAP's Infrastructure-as-a-Service (IaaS) partners Amazon Web Services (AWS), Microsoft Azure, Google Cloud, and Alibaba Cloud. The third-party region providers operate the infrastructure layer of the regions, whereas SAP operates the platform layer. For more information, see Regional Availability [page 11].

SAP Cloud Identity Services are available for consumption with the following SAP BTP environments:

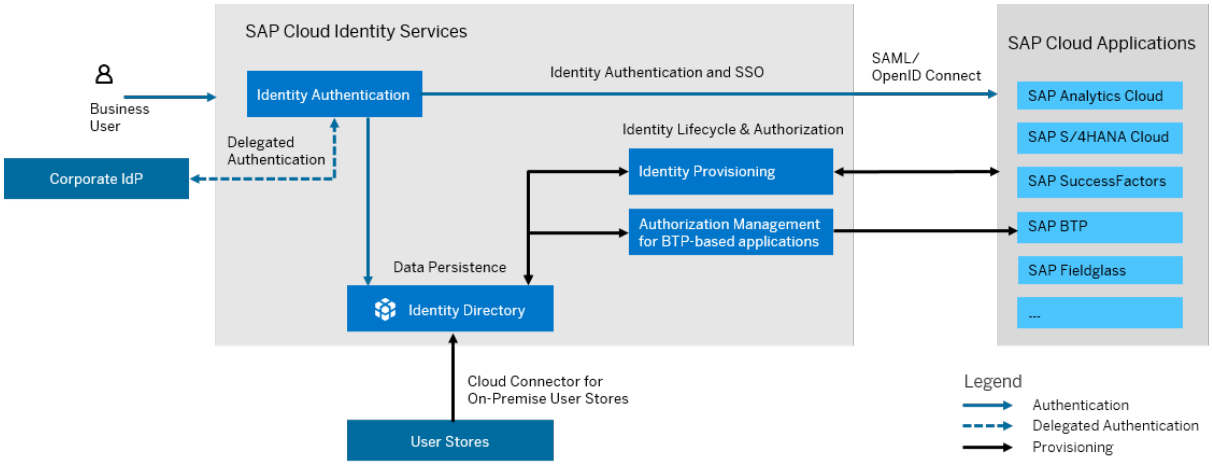
- Cloud Foundry
- Kyma
- Neo

Features

Authentication and SSO	Choose one of the supported authentication methods to control access to your application, like Form, SPNEGO, Social, or 2FA. Use OpenID Connect or SAML 2.0 to provide single sign-on. Integrate your application programmatically using authentication via API.

Data Persistence	Store and manage users and groups in Identity Directory - the user store of SAP Cloud Identity Services.
Delegate Authentication	Delegate authentication to a 3rd party or on-premise IdP, as default or based on a condition like IdP, email domain, user type or user group, and thus enable SSO across on-premise and the cloud.
Job Logging and Notifications	View and manage job logs and real-time provisioning logs. Subscribe to source systems and receive notifications for the status of provisioning jobs.
Policy-Based Authorizations	Develop SAP BTP applications with instance-based authorizations. Configure those instance-based authorizations centrally as policies within SAP Cloud Identity Services.
Risk-Based Authentication	Help enforce two-factor authentication based on IP ranges, user groups, user type, or authentication method to manage access to a business application.
User and Group Management via API	Use SCIM REST API to manage users and groups, invite users, customize end-user UI texts in any language.
User and Group Provisioning	Synchronize users and groups between multiple supported cloud and on-premise systems, both SAP and non-SAP. Filter data. Run jobs in full and delta read mode.

Overview Graphic



Identity Authentication

Identity Authentication provides you with controlled cloud-based access to business processes, applications, and data. It simplifies your user experience through authentication mechanisms, single sign-on, on-premise integration, and convenient self-service options. Identity Authentication provides the following benefits:

- Authentication: All SAP cloud applications can offer their users the same authentication mechanisms , as well as strong authentication with configurable multi-factor (MFA) enforcement; easy separation mechanism for multiple user stores and flexible configuration where to validate user's credentials.

- Single Sign-On: Identity Authentication offers central SSO endpoint for all SAP cloud applications and pre-configured or semi-automated trust configuration.
- Integrating SAP applications: Identity Authentication offers common identity for users, as well as a unified way for user management and security token service for protection of system-to-system communication. Data across applications can be correlated (precondition for central foundation services).

Identity Provisioning

Identity Provisioning offers easier and more secure identity lifecycle management as a service with identity and authorization provisioning and deprovisioning. It enables customers to set up faster and more efficient administration of user onboarding and offboarding. Identity Provisioning supports a centralized lifecycle of corporate identities in the cloud. In addition, it can handle automated provisioning of existing on-premise identities to cloud applications.

Identity Directory

Identity Directory is the persistency layer of SAP Cloud Identity Services. It offers a central place for storing and managing users and groups. Its SCIM 2.0 REST API allows customers to define their own custom schemas with own attributes. The directory generates the Global User ID attribute - the unique user identifier across landscape. This attribute is distributed by Identity Provisioning to SAP cloud applications, like SAP Task Center, which need the common user identifier in their integration scenarios.

Authorization Management

Authorization Management enables you to refine authorization policies that give access to resources in enabled SAP BTP-based business applications. Restrict policies based on the values of user or business object attributes. Assign policies to users with the group management capabilities of the identity directory. Use the corresponding user interfaces of SAP Cloud Identity Services or the SCIM API of the identity directory.

Use Cases

SAP Cloud Identity Services offers the following scenarios:

- Identity Authentication Scenarios [page 544] for consumers (business-to-consumer scenarios), partners (business-to-business scenarios), and employees (business-to-employee scenarios).
- Authorization Management Scenario for configuring and assigning authorization policies to users.
- Identity Directory Scenario for storing and provisioning users and groups.
- Identity Provisioning Scenario for using bundled tenants in provisioning and deprovisioning.

Prerequisites

To use SAP Cloud Identity Services, you must obtain a tenant. It's delivered to you as part of a bundle with an SAP cloud solution or as part of a self-service request in SAP BTP cockpit.

For more information, see Tenants [page 27] and Bundles [page 35].

Tools

The central point of entry to SAP Cloud Identity Services is the administration console, where you can manage tenant configurations, user authentication and provisioning. It is a Fiori-based user interface adaptive to most browsers.

For more information, see Operation Guide.

> → Tip
>
> The English version of this guide is open for contributions and feedback using GitHub. This allows you to get in contact with responsible authors of SAP Help Portal pages and the development team to discuss documentation-related issues. To contribute to this guide, or to provide feedback, choose the corresponding option on SAP Help Portal:
>
> - *Feedback* > *Create issue*: Provide feedback about a documentation page. This option opens an issue on GitHub.
> - *Feedback* > *Edit page*: Contribute to a documentation page. This option opens a pull request on GitHub.
>
> You need a GitHub account to use these options.
>
> More information:
>
> - Contribution Guidelines
> - Introduction Video
> - Introduction Blog Post

Regional Availability [page 11]
: Tenants are deployed on the productive domains `accounts.ondemand.com` and `accounts.cloud.sap`.

Disaster Recovery/High Availability [page 21]
: Disaster recovery (DR) and high availability (HA) are based on the capabilities of the underlying infrastructure.

Web-Based Logon Interface [page 22]
: Service providers that delegate authentication to Identity Authentication can use two types of visualization of the web-based user interfaces for the logon pages of their applications.

Browser Support [page 23]
: Information on the supported browser version for the administration console, and the end user screens of SAP Cloud Identity Services.

Supported Languages [page 24]
: Information on the supported languages for the administration console, and the end user screens of Identity Authentication.

Accessibility Features in SAP Cloud Identity Services [page 25]
: To optimize your experience of SAP Cloud Identity Services, SAP Cloud Identity Services tools provide features and settings that help you use the software efficiently.

1.1.1 Regional Availability

Tenants are deployed on the productive domains `accounts.ondemand.com` and `accounts.cloud.sap`.

The productive domain is available on a country/regional basis, where each country/region represents the locations of data centers. The productive domain represents the productive environment. It can be used by customer and partner accounts only.

A customer or partner account is associated with a particular data center and this is independent of your own location. For example, you could be located in North America, but operate your account in Europe. Thus, you use a data center that is situated in Europe.

Country/regions with more than one data centers operate in high availability (HA) mode among the respective data centers. Tenants located in these country/regions are distributed among the data centers there.

The country/region, domain, data center, and IP address are listed below:

Productive Environment

Country/ Region	Domain	Data Center	Infrastructure	LB IPs	NAT IPs	First IP - Last IP
Australia	`accounts.ondemand.com` `accounts.cloud.sap`	Australia (Sydney)	SAP / Azure	**157.133.168.73, 4.200.99.225**	**157.133.168.32/27, 130.214.240.32/27, 13.70.93.153/32**	**157.133.168.32-157.133.168.63, 130.214.240.32-130.214.240.63, 13.70.93.153-13.70.93.153**
Brazil	`accounts.ondemand.com` `accounts.cloud.sap`	Brazil (São Paulo)	AWS	**54.232.33.83, 54.207.203.50, 54.207.116.12, 52.67.178.118, 54.207.2.140, 54.94.82.121**	**18.228.21.223/32, 54.94.238.246/32, 54.94.95.71/32, 18.231.202.16/32, 18.230.68.1/32, 54.232.108.119/32**	**18.228.21.223-18.228.21.223, 54.94.238.246-54.94.238.246, 54.94.95.71-54.94.95.71, 18.231.202.16, 18.230.68.1-18.230.68.1, 54.232.108.119-54.232.108.119**

Country/ Region	Domain	Data Center	Infrastructure	LB IPs	NAT IPs	First IP - Last IP
China	accounts.sapcloud.cn	China (Shanghai)	SAP / Azure	**40.162.206.196, 121.91.104.205**	**40.162.31.41/32, 121.91.104.32/27**	**40.162.31.41-40.162.31.41, 121.91.104.32-121.91.104.62**
Europe	accounts.ondemand.com accounts.cloud.sap	Netherlands (Amsterdam)	Azure	**20.61.154.58, 20.54.146.218**	**20.8.221.111/32, 20.76.156.208/32, 20.61.154.58/32, 20.4.165.94/32, 20.126.231.246/32, 135.236.116.226/32, 4.231.72.242/32, 132.220.115.155/32, 20.54.146.218/32, 48.222.214.58/32**	20.8.221.111-20.8.221.111, 20.76.156.208-20.76.156.208, 20.61.154.58-20.61.154.58, 20.4.165.94-20.4.165.94, 20.126.231.246-20.126.231.246, 135.236.116.226-135.236.116.226, 4.231.72.242-4.231.72.242, 132.220.115.155-132.220.115.155, 20.54.146.218-20.54.146.218, 48.222.214.58-48.222.214.58

Country/ Region	Domain	Data Center	Infrastructure	LB IPs	NAT IPs	First IP - Last IP
Europe	`accounts.ondemand.com` `accounts.cloud.sap`	Germany (Frankfurt)	AWS	**3.125.77.225, 3.126.218.72, 3.64.78.167, 3.78.87.109, 18.192.230.105, 3.126.8.63**	**35.159.253.27/32, 18.192.171.14/32, 63.176.37.171/32, 3.65.162.41/32, 18.193.241.90/32, 18.199.198.32/32**	**35.159.253.27-35.159.253.27, 18.192.171.14-18.192.171.14, 63.176.37.171-63.176.37.171, 3.65.162.41-3.65.162.41, 18.193.241.90-18.193.241.90, 18.199.198.32-18.199.198.32**
Europe	`accounts.ondemand.com` `accounts.cloud.sap`	Europe (Frankfurt / Amsterdam)	SAP	**155.56.128.137, 157.133.170.72, 130.214.144.214**	**157.133.160.32/27, 130.214.226.32/27, 157.133.170.32/27, 130.214.230.32/27, 130.214.228.32/27**	**157.133.160.32-157.133.160.63, 130.214.226.32-130.214.226.63, 157.133.170.32-157.133.170.63, 130.214.230.32-130.214.230.63, 130.214.228.32-130.214.228.63**

Country/ Region	Domain	Data Center	Infrastructure	LB IPs	NAT IPs	First IP - Last IP
India	accounts.ondemand.com accounts.cloud.sap	India (Mumbai)	AWS	3.109.7.59, 52.66.148.12, 43.205.45.235, 3.6.178.171, 43.205.3.38, 3.6.141.120	3.108.210.169/32, 13.127.105.96/32, 13.232.175.70/32, 3.7.5.154/32, 43.205.169.10/32, 65.2.62.24/32	3.108.210.169-3.108.210.169, 13.127.105.96-13.127.105.96, 13.232.175.70-13.232.175.70, 3.7.5.154-3.7.5.154, 43.205.169.10-43.205.169.10, 65.2.62.24-65.2.62.24
Israel	accounts.ondemand.com accounts.cloud.sap	Israel (Tel Aviv)	Google Cloud	34.165.59.107, 34.165.82.240	34.165.48.136/32, 34.165.90.82/32, 34.165.0.183/32	34.165.48.136 - 34.165.48.136, 34.165.90.82 - 34.165.90.82, 34.165.0.183 - 34.165.0.183
Japan	accounts.ondemand.com accounts.cloud.sap	Japan (Tokyo / Osaka)	SAP	157.133.182.83, 130.214.246.74	157.133.182.32/27, 130.214.244.32/27, 157.133.184.32/27, 130.214.246.32/27	157.133.182.32-157.133.182.63, 130.214.244.32-130.214.244.63, 157.133.184.32-157.133.184.63, 130.214.246.32-130.214.246.63

Country/ Region	Domain	Data Center	Infrastructure	LB IPs	NAT IPs	First IP - Last IP
North America (Canada Central)	accounts.ondemand.com accounts.cloud.sap	Canada (Toronto)	Azure	20.151.9.145, 20.43.19.31, 52.139.41.10	4.206.67.217/32, 4.205.230.84/32	4.206.67.217-4.206.67.217, 4.205.230.84-4.205.230.84
Saudi Arabia (SA1)	accounts.ondemand.com accounts.cloud.sap	Saudi Arabia (Dammam)	Google Cloud	34.166.140.252, 34.166.98.190	34.166.143.220/32, 34.166.36.253/32	34.166.143.220-34.166.143.220, 34.166.36.253-34.166.36.253
Saudi Arabia (SA2)	accounts.ondemand.com accounts.cloud.sap	Saudi Arabia (Dammam) - Regulated	Google Cloud	34.166.65.197, 34.166.56.10	34.166.19.58/32, 34.166.58.219/32, 34.166.53.185/32, 34.166.10.70/32	34.166.19.58-34.166.19.58, 34.166.58.219-34.166.58.219, 34.166.53.185-34.166.53.185, 34.166.10.70-34.166.10.70
Singapore	accounts.ondemand.com accounts.cloud.sap	Singapore	AWS	18.138.93.141, 13.251.80.194, 52.221.66.111, 13.215.237.53, 13.251.19.109, 3.0.227.4	54.254.59.23/32, 52.76.191.74/32, 47.129.67.102/32, 18.140.41.153/32, 47.129.16.236/32, 13.213.203.26/32	54.254.59.23-54.254.59.23, 52.76.191.74-52.76.191.74, 47.129.67.102-47.129.67.102, 18.140.41.153-18.140.41.153, 47.129.16.236-47.129.16.236, 13.213.203.26-13.213.203.26

Country/ Region	Domain	Data Center	Infrastructure	LB IPs	NAT IPs	First IP - Last IP
South Korea	accounts.ondemand.com accounts.cloud.sap	South Korea (Seoul)	AWS	3.34.214.12, 52.78.91.176, 15.164.154.86, 3.35.109.127, 3.37.25.236, 15.164.195.144	3.39.112.56/32, 54.180.170.146/32, 3.38.127.188/32, 43.202.135.25/32, 3.36.27.20/32, 52.79.160.50/32	3.39.112.56-3.39.112.56, 54.180.170.146-54.180.170.146, 3.38.127.188-3.38.127.188, 43.202.135.25-43.202.135.25, 3.36.27.20-3.36.27.20, 52.79.160.50-52.79.160.50
Switzerland	accounts.ondemand.com accounts.cloud.sap	Switzerland (Zürich)	Azure	20.250.104.188, 20.250.104.193	172.161.63.46/32, 172.161.59.62/32	172.161.63.46-172.161.63.46,
United Arab Emirates	accounts.ondemand.com accounts.cloud.sap	United Arab Emirates (Dubai)	SAP	130.214.250.103, 130.214.250.118	130.214.80.0/26, 130.214.250.32/27	130.214.80.0-130.214.80.63, 130.214.250.32-130.214.250.63
United Kingdom	accounts.ondemand.com accounts.cloud.sap	United Kingdom (London)	Azure	4.250.195.23, 145.133.42.3	51.11.159.33/32, 20.90.185.26/32	51.11.159.33 - 51.11.159.33, 20.90.185.26 - 20.90.185.26

Country/ Region	Domain	Data Center	Infrastructure	LB IPs	NAT IPs	First IP - Last IP
US (North America East)	`accounts.ondemand.com` `accounts.cloud.sap`	USA (Sterling, VA / N. Virginia)	SAP / AWS	`157.133.166.69, 3.92.131.87, 52.200.183.196, 35.168.205.166`	`157.133.166.32/27, 130.214.234.32/27, 157.133.176.32/27, 52.44.60.92/32, 54.211.101.173/32, 54.225.47.27/32`	`157.133.166.32-157.133.166.63, 130.214.234.32-130.214.234.63, 157.133.176.32-157.133.176.63, 52.44.60.92-52.44.60.92,54.211.101.173-54.211.101.173, 54.225.47.27-54.225.47.27`
US West	`accounts.ondemand.com` `accounts.cloud.sap`	USA (Quincy, WA)	Azure	`52.143.72.52, 40.91.81.47`	`4.246.74.117/32, 13.66.186.117/32`	`4.246.74.117-4.246.74.117, 13.66.186.117-13.66.186.117/32`

Trial Environment

Country/ Region	Domain	Data Center	Infrastructure	LB IPs	NAT IPs	First IP - Last IP
US East	`trial-accounts.ondemand.com`	Trial (USA)	Azure	`20.62.179.159, 20.62.181.214, 20.62.180.136`	`172.171.209.145/32, 172.190.167.36/32`	172.171.209.145-172.171.209.145, 172.190.167.36-172.190.167.36

⚠ Restriction

The `accounts.cloud.sap` domain is not relevant for China (Shanghai) and Government Cloud (US) regions.

ⓘ Note

- LB (load balancer) IPs - ingress, for incoming requests
- NAT (network address translation) IPs - egress, IPs for requests from Identity Authentication app.

Parent topic: What Are Cloud Identity Services? [page 7]

Related Information

Disaster Recovery/High Availability [page 21]
Web-Based Logon Interface [page 22]
Browser Support [page 23]
Supported Languages [page 24]
Accessibility Features in SAP Cloud Identity Services [page 25]

1.1.1.1 Updating Host URLs for Shanghai Tenants

If your Identity Authentication tenant host is in the Asia-Pacific region, precisely Shanghai (China), you must update it to the new host: `accounts.sapcloud.cn`.

Context

The former domain on which you've accessed your Shanghai (China) tenants is: `accounts.ondemand.com`. This domain will soon stop working, thus we recommend that you start using the new one. Effective September 2020, tenants from Shanghai (China) should be accessed via the following domain: `accounts.sapcloud.cn`.

> **Note**
>
> The other productive domains, available on a country/regional basis, are working as usual.

The migration procedure for Identity Authentication tenants affects the following scenarios:

- SAML 2.0 Application Configuration
- OpenID Connect Application Configuration
- Real-Time User Provisioning
- Delegated Authentication
- API Usage
- IdP-Initated SSO

For more information about the migration procedure for SAP Cloud Identity Services - Identity Provisioning, see Updating Host URLs for Shanghai Tenants.

SAML 2.0 Application Configuration

Context

When you set the trust between Identity Authentication, acting as identity provider (IdP), and the service provider, make sure that you use the Identity Authentication metadata downloaded via the new host `https://<tenant Id>.accounts.sapcloud.cn/admin`.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services via the new URL `https://<tenant Id>.accounts.sapcloud.cn/admin`.
2. Access *Applications and Resources* > *Tenant Settings* > *SAML 2.0 Configuration*.
3. Download the new SAML metadata of Identity Authentication.

   > **Note**
   >
   > If you do not have access to the trust configuration on the application side report an incident on SAP Support Portal Home with a component `BC-IAM-IDS`.

4. Configure the new metadata of the Identity Provider (IdP) on the application (service provider) side. See Tenant SAML 2.0 Configurations [page 2103].

   If you have set trusts with more than one service provider, configure the metadata in every service provider. For more information about how to configure the metadata, see the documentation of the respective service providers.

OpenID Connect Application Configuration

Context

If your application an OpenID Connect one, update the configurations in the administration console for SAP Cloud Identity Services to match the new host.

Procedure

1. In the administration console for SAP Cloud Identity Services, access *Applications and Resources* > *Tenant Settings* > *OpenID Connect Configuration*.
2. Update the name of the identity provider to match the new host name (https:// <tenant ID>.accounts.sapcloud.cn).
3. Update the endpoints on the relying party (RP) side to match the new host.

Real Time User Provisioning

Context

When you want to perform real-time provisioning through the administration console for SAP Cloud Identity Services, make sure you use the correct domain when you construct your OAuth URL and SCIM URL.

Procedure

1. Access theIdentity Authentication tenant via the new domain `<tenant Id>.accounts.sapcloud.cn`.
2. In the administration console, access *User Provisioning*.
3. Update the *host* part in the *SCIM URL* and *OAuth URL* of the Identity Provisioning target system with the new domain `dispatcher.cn1.platform.sapcloud.cn`. See Real-Time Provisioning: Identity Authentication

   SCIM URL Pattern: `https://ipsproxy<provider_account>-<consumer_subaccount>.<host>/ipsproxy/api/v1/systems/<Identity_Authentication_ID>/entities/user`

   OAuth URL Pattern: `https://oauthasservices-<consumer_subaccount>.<host>/oauth2/api/v1/token`

Delegated Authentication

Context

In this scenario Identity Authentication acts as a proxy to the corporate identity provider.

When you set the trust between Identity Authentication, acting as a proxy), and the corporate IdP, make sure that you use the Identity Authentication metadata downloaded via the new host `https://<tenant Id>.accounts.sapcloud.cn/admin`.

For more information about the scenario, Corporate Identity Providers [page 2556].

Procedure

1. Access theIdentity Authentication tenant via the new domain `<tenant Id>.accounts.sapcloud.cn/admin`.
2. In the administration console, access *Applications and Resources* > *Tenant Settings* > *SAML 2.0 Configuration* .
3. Download the new SAML metadata of the identity provider (IdP).

   > **Note**
   >
   > If you don't have access to the trust configuration on the application side report an incident on SAP Support Portal Home with a component `BC-IAM-IDS`.

4. Update the trust configuration on the corporate identity provider side.

API Usage

Procedure

Use the new domain in the URLs. See API Documentation [page 2790].

URL: https://<tenant ID>.accounts.sapcloud.cn/<...>

Identity Authentication IdP-Initated SSO

Procedure

Inform the end users about the new IdP-initiated links, updated with the new domain.

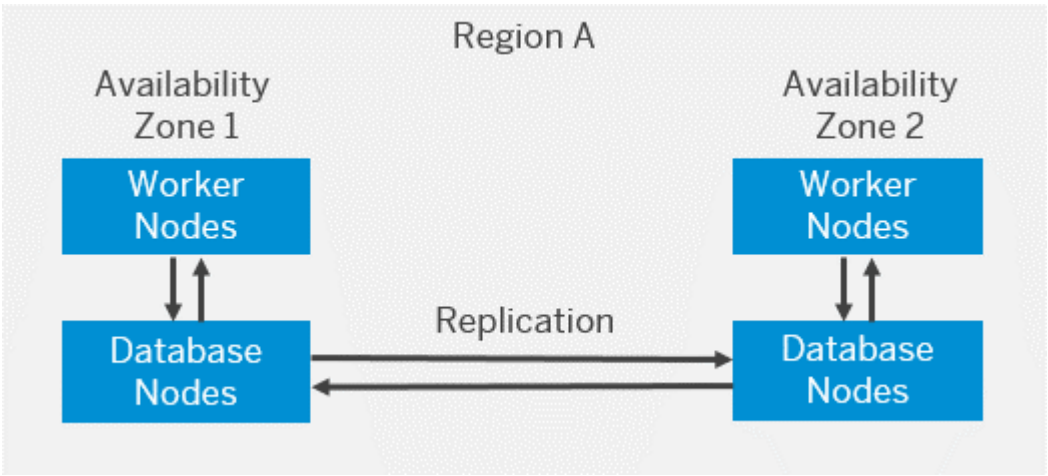
1.1.2 Disaster Recovery/High Availability

Disaster recovery (DR) and high availability (HA) are based on the capabilities of the underlying infrastructure.

Identity Authentication is a multi-tenant system where tenants share the hardware and software and use dedicated (and isolated) database instances for persistence.

High Availability - Single Region Setup

All deployments that have one data center support replication of the data between two zones within the same region.



Backup and Recovery

A full backup is performed everyday at 01:00 UTC. The backup is kept for 90 days.

If a restore is required due to an issue on infrastructure level, then SAP DevOps team takes care of the recovery procedure.

If a restore is required due to an issue on application level, report an incident on SAP Support Portal Home with a component `BC-IAM-IDS`.

Parent topic: What Are Cloud Identity Services? [page 7]

Related Information

Regional Availability [page 11]
Web-Based Logon Interface [page 22]
Browser Support [page 23]
Supported Languages [page 24]
Accessibility Features in SAP Cloud Identity Services [page 25]
Cloud Service Status

1.1.3 Web-Based Logon Interface

Service providers that delegate authentication to Identity Authentication can use two types of visualization of the web-based user interfaces for the logon pages of their applications.

The two types of visualization of the web-based user interfaces are on a fullscreen of the window and with overlay on top of the application page.

Overlay

The use of overlays maintains the application context, by keeping the application page as dimmed background, to provide for minimum disturbance to the work flow. The logon page is open in an iframe over the public page of the application.

Fullscreen

The logon page is displayed on fullscreen in the browser when at least one of the following conditions is fulfilled:

- the overlay is not integrated in the application

- the application does not provide a public page
- the user opens directly a protected page of the application

Parent topic: What Are Cloud Identity Services? [page 7]

Related Information

Regional Availability [page 11]
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Supported Languages [page 24]
Accessibility Features in SAP Cloud Identity Services [page 25]
Add Logon Overlays in Customer Applications [page 2877]

1.1.4 Browser Support

Information on the supported browser version for the administration console, and the end user screens of SAP Cloud Identity Services.

Parent topic: What Are Cloud Identity Services? [page 7]

Related Information

Regional Availability [page 11]
Disaster Recovery/High Availability [page 21]
Web-Based Logon Interface [page 22]
Supported Languages [page 24]
Accessibility Features in SAP Cloud Identity Services [page 25]

Supported Browser Versions for Administration Console

The Administration Console uses the SAPUI5 libraries. For more information on the supported browsers, see Browser and Platform Support.

Supported Browser Versions for End User Screens

Supported Desktop Browsers

The end user screens, such as logon, forgot password, and so on, of the applications that use Cloud Identity Services for authentication, support the following browsers:

Web Browser	Supported Version
Microsoft Edge	Latest Version
Mozilla Firefox	Latest Version
Google Chrome	Latest Version
Safari	Latest Version

> **Note**
>
> You cannot use versions 7 and 8 of Microsoft Internet Explorer for some user interfaces (responsive user interfaces).

Supported Mobile Browsers

Web Browser	Supported Version
Mobile or iOS Safari	Latest Version
Android Browser	Latest Version
Google Chrome for Android	Latest Version
Internet Explorer Mobile	11 or higher
Blackberry Browser	10.0 or higher

1.1.5 Supported Languages

Information on the supported languages for the administration console, and the end user screens of Identity Authentication.

Parent topic: What Are Cloud Identity Services? [page 7]

Related Information

Regional Availability [page 11]

Disaster Recovery/High Availability [page 21]
Web-Based Logon Interface [page 22]
Browser Support [page 23]
Accessibility Features in SAP Cloud Identity Services [page 25]

Supported Languages for Administration Console

The administration console for SAP Cloud Identity Services supports only English (en) language.

Supported Languages for Identity Authentication End User Screens

The end user screens of the applications that use Cloud Identity Services for authentication, support the following languages: Arabic (ar), Bulgarian (bg), Catalan (ca), Chinese PRC (zh_CN), Chinese Taiwan (zh_TW), Croatian (hr), Czech (cs), Danish (da), Dutch (nl), English United Kingdom (en_GB), English United States (en_US), Estonian (et), Finnish (fi), French Standard (fr), French Canada (fr_CA), German Standard (de), Greek (el), Hebrew (iw), Hungarian (hu), Italian (it), Japanese (ja), Korean (ko), Latvian (lv), Lithuanian (lt), Malay (s), Norwegian (no), Polish (pl), Portuguese Portugal (pt), Romanian (ro), Russian (ru), Serbian (sr), Slovak (sk), Slovenian (sl), Spanish Standard (es), Spanish Mexico (es_MX), Swedish (sv), Thai (th), Turkish (tr), Ukrainian (uk), Vietnamese (vi), Welsh (cy).

1.1.6 Accessibility Features in SAP Cloud Identity Services

To optimize your experience of SAP Cloud Identity Services, SAP Cloud Identity Services tools provide features and settings that help you use the software efficiently.

> **Note**
>
> SAP Cloud Identity Services is based on SAPUI5. For this reason, accessibility features for SAPUI5 also apply. See the accessibility documentation for SAPUI5 on SAP Help Portal at Accessibility for End Users.

For more information on screen reader support and keyboard shortcuts, see Keyboard Handling for SAPUI5 UI Elements and Screen-Reader Support for SAPUI5 Controls.

Parent topic: What Are Cloud Identity Services? [page 7]

Related Information

Regional Availability [page 11]

Disaster Recovery/High Availability [page 21]
Web-Based Logon Interface [page 22]
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1.2 What's New for Cloud Identity Services

See the latest release notes for SAP Cloud Identity Services. It contains updates about new features and enhancements.

What's New Viewer

1.3 Concepts

This section describes basic concepts that will help you familiarize yourself with the SAP Cloud Identity Services.

Tenants [page 27]
: A tenant refers to your (customer-specific) instance of SAP Cloud Identity Services. It's delivered to you as part of a bundle with an SAP cloud solution or as part of a self-service request in SAP BTP cockpit.

Bundles [page 35]
: A bundle is a group of preconfigured products and services which are sold together.

Applications [page 61]
: An application is associated with a consumer of Identity Authentication as an identity provider. This consumer could be for example an SAP cloud solution, a third-party application, SAP BTP subaccount, or the SAP Cloud Identity Services administration console.

Provisioning Systems [page 69]
: Identity Provisioning provides connectors to various business applications for provisioning and deprovisioning of users and groups. These business applications are set up as provisioning systems in the administration console of SAP Cloud Identity Services.

Properties [page 73]
: Properties hold the configuration of a provisioning system.

Transformations [page 388]
: Transformations help you transform user and group attributes from the data model of the source system to the data model of the target system.

Users [page 509]
: Users in SAP Cloud Identity Services fall into three categories: administrators, end users, and technical users.

Agents [page 511]

Agent identity represents AI and automation agents in SAP Cloud Identity Services. Agents can authenticate, receive authorization, and be managed like human users while retaining agent-specific capabilities and security controls.

Groups [page 514]
: SAP Cloud Identity Services offers groups to organize users based on common characteristics, authorization, or application. Use them to efficiently manage user access and permissions within your organization's SAP Cloud Identity Services environment.

Authorization Policies [page 518]
: Authorization Management lets administrators control who can access what in business applications.

SAP-Managed SAP Cloud Identity [page 520]
: SAP Cloud Identity Services support both SAP-managed and customer-managed integration models, depending on who is responsible for maintaining and operating the tenant, application and provisioning configurations.

1.3.1 Tenants

A tenant refers to your (customer-specific) instance of SAP Cloud Identity Services. It's delivered to you as part of a bundle with an SAP cloud solution or as part of a self-service request in SAP BTP cockpit.

When SAP Cloud Identity Services are bundled with an SAP cloud solution, you are entitled to one productive and one test tenant preconfigured with the SAP cloud solution. If you get another solution that also bundles SAP Cloud Identity Services, you don't get additional tenant. Your existing one is reused. When your tenant is created as a result of a self-service request, it isn't preconfigured out of the box. You must set it up manually for your specific authentication and provisioning scenarios.

For more information, see Get Your Tenant [page 528] and Bundles [page 35].

Standalone tenants refer to SAP Cloud Identity Services - Identity Provisioning tenants delivered and used as a separate (standalone) product. As of October 20, 2020, Identity Provisioning is no longer sold as a standalone product. Existing customers of standalone tenants can use them as-is until the end of their contracts.

For more information, see Standalone Tenant.

Licensing and Usage

From a usage perspective, Identity Authentication, Identity Provisioning, Identity Directory, and SAP Authorization and Trust Management service are commercially included for all SAP-branded cloud solutions sold under the general terms and conditions for SAP Cloud Services, and all applications running on SAP BTP. However, you are charged for every successful authentication to SAP Cloud Identity Services from a third-party solution regardless of the protocol used. Only logon requests coming from an SAP cloud or SAP on-premise solutions are free of charge.

Logon requests are independent of the authentication mechanism and user type.

> ✤ Example
>
> *SAP Applications*

- A user wants to access SAP Build Workzone, an application that runs on SAP BTP. The application delegates authentication to Identity Authentication which is configured to act as a proxy and redirects to the corporate identity provider (IdP) for authentication. After successful authentication at the corporate IdP, Identity Authentication sends the single-sign-on (SSO) token to SAP Build Workzone. This request counts as a free of charge logon to Identity Authentication.
- A user wants to access SAP Integrated Business Planning (IBP). The user goes to Identity Authentication via a direct link. After providing the required credentials, Identity Authentication authenticates the user and sends assertion about the user to the service provider, which, in turn, validates the assertion and gives access rights to the user. Identity Authentication redirects the user to SAP IBP. This request counts as a free of charge logon to Identity Authentication.

Third-Party Applications

- A user wants to access a third-party application, which delegates authentication to Identity Authentication. The user provides a username and password and is authenticated by Identity Authentication. This request counts as a charged logon to Identity Authentication.

To get a tenant, in addition to the test and productive tenants you're already entitled to, use self-service request. You must have a global account that uses one of the consumption-based commercial models: Cloud Platform Enterprise Agreement (CPEA) or Pay-As-You-Go for SAP BTP.

For more information, see SAP Business Technology Platform Service Description Guide .

Tenant Types

Before going live, your organization most probably wants to test various tenant configurations. SAP Cloud Identity Services provide three types of tenants - productive, test, and trial.

Test tenants are used for testing purposes. Productive and test tenants have no difference from a technical and functional perspective. The tenant type flag is just a means for structuring your landscape. This means, an SAP Cloud Identity Services bundle tenant and an SAP cloud solution bundle tenant are created of one particular type: test or productive.

A trial tenant is also intended for testing purposes. However, it differs from the test one as it can be used for a limited period with a limited number of users and some other restrictions.

For more information, see Tenant Types.

Tenant Infrastructure

SAP Cloud Identity Services run on several underlying Infrastructure-as-a-Service technologies and regions. Regions are provided either by SAP or by our Infrastructure-as-a-Service (IaaS) partners Amazon Web Services (AWS), Microsoft Azure, Google Cloud. The third-party region providers operate the infrastructure layer of the regions, whereas SAP operates the platform layer.

For more information, see Regional Availability.

SAP Cloud Identity Services - Identity Authentication and Identity Provisioning tenants run on the infrastructure of SAP Cloud Identity Services.

> **Note**
>
> Identity Provisioning bundle tenants delivered before March 15, 2022, and Identity Provisioning standalone tenants delivered before September 1, 2020, run on SAP BTP, Neo environment. Customers with tenants on Neo environment are encouraged to migrate them to the SAP Cloud Identity Services infrastructure.
>
> For more information, see Migrate Identity Provisioning Bundle Tenant.

Data Center Mapping

Below is the mapping between the multi-environment subaccount in the Cloud Foundry region and the region of the Identity Authentication tenant:

Region Mapping: Cloud Foundry - Identity Authentication

IaaS Provider	Cloud Foundry - Region Name	Cloud Foundry - Region	Identity Authentication - Data Center	Infrastructure
SAP Cloud Infrastructure	UAE (Dubai)	ae01	United Arab Emirates (Dubai)	SAP
SAP Cloud Infrastructure	Australia (Sydney)	ap01	Australia (Sydney)	SAP / Azure
Amazon Web Services	Australia (Sydney)	ap10	Australia (Sydney)	SAP / Azure
Amazon Web Services	Asia Pacific (Singapore)	ap11	Singapore	AWS
Amazon Web Services	Asia Pacific (Seoul)	ap12	South Korea (Seoul)	AWS
Microsoft Azure	Australia (Sydney)	ap20	Australia (Sydney)	SAP / Azure
Microsoft Azure	Singapore	ap21	Singapore	AWS
Google Cloud	Australia (Sydney)	ap30	Australia (Sydney)	SAP / Azure
Amazon Web Services	Brazil (São Paulo)	br10	Brazil (São Paulo)	AWS
Microsoft Azure	Brazil (São Paulo)	br20	Brazil (São Paulo)	AWS
Google Cloud	Brazil (São Paulo)	br30	Brazil (São Paulo)	AWS
Amazon Web Services	Canada (Montreal)	ca10	Canada (Toronto)	Azure
Microsoft Azure	Canada Central (Toronto)	ca20	Canada (Toronto)	Azure
Microsoft Azure	Switzerland (Zurich)	ch20	Switzerland (Zürich)	Azure
Microsoft Azure	China (North 3)	cn20	China (Shanghai)	SAP / Azure

IaaS Provider	Cloud Foundry - Region Name	Cloud Foundry - Region	Identity Authentication - Data Center	Infrastructure
Alibaba Cloud	China (Shanghai)	cn40	China (Shanghai)	SAP / Azure
SAP Cloud Infrastructure	Europe (Frankfurt) - EU Access Only	eu01	Europe (Frankfurt / Amsterdam)	SAP
SAP Cloud Infrastructure	Europe (Rot) - SAP EU Access	eu02	Europe (Frankfurt / Amsterdam)	SAP
Amazon Web Services	Europe (Frankfurt)	eu10	Germany (Frankfurt)	AWS
Amazon Web Services	Europe (Frankfurt)	eu11	Germany (Frankfurt)	AWS
Amazon Web Services	Europe (Milan)	eu13	Germany (Frankfurt)	AWS
Microsoft Azure	Europe (Netherlands)	eu20	Germany (Frankfurt)	AWS
Google Cloud	Europe (Frankfurt)	eu30	Germany (Frankfurt)	AWS
Google Cloud	Israel (Tel Aviv)	il30	Israel (Tel Aviv)	Google Cloud
Google Cloud	India (Mumbai) GCP	in30	India (Mumbai)	AWS
SAP Cloud Infrastructure	Japan (Tokyo)	jp01	Japan (Tokyo / Osaka)	SAP
Amazon Web Services	Japan (Tokyo)	jp10	Japan (Tokyo / Osaka)	SAP
Microsoft Azure	Japan (Tokyo)	jp20	Japan (Tokyo / Osaka)	SAP
Google Cloud	Japan (Osaka)	jp30	Japan (Tokyo / Osaka)	SAP
Google Cloud	Japan (Tokyo)	jp31	Japan (Tokyo / Osaka)	SAP
Google Cloud	KSA (Dammam)	sa30	Saudi Arabia (Dammam) - Regulated	Google Cloud
Google Cloud	KSA (Dammam)	sa31	Saudi Arabia (Dammam)	SAP
Microsoft Azure	UK South (London)	uk20	United Kingdom (London)	Azure
SAP Cloud Infrastructure	US (Sterling)	us01	USA (Sterling, VA / N. Virginia)	SAP
SAP Cloud Infrastructure	US West (Colorado)	us02	USA (Quincy, WA)	Azure
Amazon Web Services	US East (VA)	us10	USA (Sterling, VA / N. Virginia)	SAP

IaaS Provider	Cloud Foundry - Region Name	Cloud Foundry - Region	Identity Authentication - Data Center	Infrastructure
Amazon Web Services	US West (Oregon)	us11	USA (Quincy, WA)	Azure
Microsoft Azure	US West (WA)	us20	USA (Quincy, WA)	Azure
Microsoft Azure	US East (VA)	us21	USA (Sterling, VA / N. Virginia)	SAP
Google Cloud	US Central (IA)	us30	USA (Quincy, WA)	Azure

Tenant Names

SAP Cloud Identity Services tenant, Identity Authentication tenant and Identity Provisioning tenant are sometimes used interchangeably in the documentation. However, there are some differences.

- *SAP Cloud Identity Services* tenant is an Identity Authentication tenant where Identity Provisioning service is enabled. The Identity Provisioning functionality is embedded in the SAP Cloud Identity Services admin console. The tenant runs on the SAP Cloud Identity Services infrastructure and is accessed at: `https://<ias-tenant-host>/admin`.

  > **Note**
  >
  > As of March 15, 2022, *SAP Cloud Identity Services* is the default name of the tenant.

- *Identity Authentication* tenant is a tenant where the Identity Provisioning service isn't enabled. The Identity Provisioning functionality isn't embedded in the SAP Cloud Identity Services admin console. The tenant runs on the SAP Cloud Identity Services infrastructure and is accessed at: `https://<ias-tenant-host>/admin`.
- *Identity Provisioning* tenant is a standalone or bundle tenant. It runs on the SAP BTP, Neo environment and its administration console is accessed at: `https://ips-<consumer_account>.dispatcher.<region_host>/webapp/index.html`.

Parent topic: Concepts [page 26]

Related Information

Bundles [page 35]
Applications [page 61]
Provisioning Systems [page 69]
Properties [page 73]
Transformations [page 388]
Users [page 509]
Agents [page 511]

Groups [page 514]
Authorization Policies [page 518]
SAP-Managed SAP Cloud Identity [page 520]

1.3.1.1 Tenant Types

Identity Authentication provides three types of tenants - productive, test, and trial

> **Note**
>
> For more information about obtaining a tenant of Identity Authentication, see Tenants [page 27].

Productive and Test Tenants

Identity Authentication provides one productive and one test tenant per customer, regardless of the number of contracts signed in which Identity Authentication is included or bundled. The productive and test tenants are identical from technical and functional perspective, they only differ in the naming. Test tenants are provided to customers in addition to their productive tenants where they can test new features before implementing them productively. The test tenant is also the default identity provider for the test landscape.

> **Note**
>
> In some cases, SAP order fulfillment considers the tenant type. For example, for some test instances the trust can only be configured to Identity Authentication test tenants.

> **Restriction**
>
> A customer can't change the type of the tenant.

Trial Tenants

The trial tenant is intended for testing purposes only. It allows you to try out and explore SAP Cloud Identity Services – Identity Authentication and Identity Provisioning. It's free of charge and limited to a trial period in accordance with the SAP BTP trial terms and conditions.

> **Restriction**
>
> - You can have only one SAP Cloud Identity Services trial tenant per SAP BTP global account regardless of how many subaccounts you create. If you have multiple subaccounts, and you have trial tenants for them, they use one and the same trial tenant.
> - You can have up to 50 users in a trial tenant.
> - You can't connect to on-premise systems using the cloud connector. It's not possible to subscribe to the Cloud Identity Services connectivity plan in your trial subaccount.

- The `Invitation Rest API` is not available. For more information about the API, see Invitation REST API [page 2802].
- Email template customization is not possible.

1.3.1.2 Tenant Lifecycle

Tenants of SAP Cloud Identity Services - Identity Authentication and Identity Provisioning go through onboarding and offboarding in their lifecycle.

Onboarding

Onboarding involves all the processes needed to instantiate a tenant, configure and provision it to the customer for test or productive usage.

Tenants	Onboarding Process

• Tenants delivered as part of a bundle with an SAP cloud solution
• Tenants delivered for applications running on SAP-managed subaccounts
	1. Search for existing tenant of the customer.
	  • If one or more existing tenants are found, the existing default or the oldest one is reused.
	  The oldest (non-additional) productive and test tenants are considered default.
	  • If no existing tenant is found, a new one is created by SAP.
	2. Tenant is preconfigured for the customer.
	3. Tenant is provisioned to the customer.
	For more information, see Bundles [page 35].

Tenants delivered for applications running on customer-managed subaccounts	Tenant delivery is triggered by the customer in SAP BTP cockpit. The tenant is not preconfigured and must be configured by the customer. For more information, see Get Your Tenant [page 528] .
Tenants delivered by triggering an upgrade process in the admin tool of SAP cloud solutions **Note** This is valid for tenants bundled with SAP SuccessFactors, SAP SuccessFactors Learning and others. For more information, see Bundle Tenants and Connectors.	Tenant delivery is triggered by the customer in the administration tool of the cloud solution. In the case of SAP SuccessFactors, this is the Upgrade Center. The tenant is preconfigured for the customer. For more information, see Initiating the Upgrade to SAP Cloud Identity Services - Identity Authentication Service.

Tenants	Onboarding Process
Tenants delivered by opening a ticket to the technical component of SAP cloud solutions **Note** This is valid for tenants bundled with SAP Concur, SAP Ariba and older versions of SAP Marketing Cloud, SAP S/4HANA Cloud Public Edition. For more information, see Bundle Tenants and Connectors.	1. Customer opens a ticket to the respective technical component. 2. The support team reuses an existing tenant or creates a new one. 3. Tenant is not preconfigured and must be configured by the customer. For more information, see SAP Concur Integration Scenario.

Tenant Usage

Following the onboarding phase, tenants are provisioned to customers. They can use them as-is or configure them for their customer specific needs.

Offboarding

Offboarding leads to tenant deletion when all customer contracts for SAP cloud solutions are terminated. Depending on how the tenant is delivered, you can expect the following:

- The tenant is delivered as part of a bundling with SAP cloud solutions
  Offboarding is triggered if the customer contracts for all SAP cloud solutions are terminated or expired.

  Note

  If all of your SAP licenses are terminated, and you have a valid **SAP Concur** license, if you do not use your SAP Cloud Identity Services for 30 days, your tenant may be deleted.
- The tenant (additional tenant) is delivered as part of self-service request in SAP BTP cockpit
  Offboarding is triggered if the subscription to the Cloud Identity Services is deleted in SAP BTP cockpit. The following takes place:
  - If the tenant is obtained only for SAP BTP and no other SAP cloud solutions are reusing it, this tenant will be deleted immediately.
  - If the tenant is reused for other SAP cloud solutions, this tenant will not be deleted until all SAP cloud solutions offboard it.

  In the case of trial tenants, offboarding is triggered if the subscription to the Cloud Identity Services is deleted in SAP BTP cockpit or the SAP BTP developer trial account expired.

1.3.2 Bundles

A bundle is a group of preconfigured products and services which are sold together.

SAP Cloud Identity Services refer to a bundle as a group of preconfigured SAP cloud solutions and Cloud Identity Services. You purchase the cloud solution and get the services free of charge. The reason for bundling is to offer you out-of-the-box integration between SAP solutions.

> **Note**
>
> As a customer, you need at least one licensed SAP cloud solution which triggers SAP Cloud Identity Services tenant delivery.

The bundling process results in provisioning an SAP Cloud Identity Services tenant to your organization, where Identity Authentication and Identity Provisioning are preconfigured for the relevant SAP cloud solution. In the majority of cases, this means that in the SAP Cloud Identity Services admin console, a bundled application is created and connected to the cloud solution, where your users will log on. The trust between Identity Authentication and the given solution is established.

For Identity Provisioning, this means that communication and authentication with the provisioning systems relevant for the SAP cloud solution are set up. The transformations of those systems might be different from the default ones and adapted for specific scenarios. You are ready to schedule and run jobs to synchronize your users and groups.

Note that all provisioning systems (connectors) are enabled for the SAP Cloud Identity Access Governance bundle tenant, and all source and proxy connectors are enabled for the SAP Jam Collaboration bundle tenant.

> **Note**
>
> Note the availability of the following connectors within the bundle tenants running in the infrastructure of SAP Cloud Identity Services:
>
> - SCIM, LDAP Server, Microsoft AD, Microsoft Entra ID and Google G suite are not supported as target connectors in bundle tenants.
> - Cloud Foundry UAA Server is not supported as source, target and proxy connector in bundle tenants.

For more information, see Get Your Tenant [page 528].

SAP Cloud Solutions Bundled with Cloud Identity Services

The following SAP cloud solutions bundle with SAP Cloud Identity Services which come preconfigured for the given cloud solutions:

SAP (A-F)	SAP (G-Z)
• SAP Build Work Zone, advanced edition • SAP Business Technology Platform • SAP Central Business Configuration • Sales Cloud – Analytics & AI • SAP Cloud ALM • SAP Cloud Identity Access Governance • SAP Commerce Cloud • SAP Concur • SAP Fieldglass • SAP Field Service Management	• SAP Integrated Business Planning • SAP Jam Collaboration • SAP Marketing Cloud • SAP Market Communication for Utilities • SAP S/4HANA Cloud Public Edition • SAP SuccessFactors • SAP SuccessFactors Learning • SAP SuccessFactors Incentive Management

> **Note**
>
> Some SAP cloud solutions bundle with Identity Authentication only. There could be various reasons for that. For example, the solution may not have a local user store and therefore doesn't need to bundle with Identity Provisioning for managing identity lifecycle. Or the provisioning service doesn't offer a connector to the given SAP cloud solution.

Parent topic: Concepts [page 26]

Related Information

Tenants [page 27]
Applications [page 61]
Provisioning Systems [page 69]
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SAP-Managed SAP Cloud Identity [page 520]

1.3.2.1 SAP Build Work Zone, advanced edition Bundle

SAP Build Work Zone, advanced edition bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning

How to Obtain

After purchasing SAP Build Work Zone, advanced edition, you can obtain the Identity Provisioning tenant by connecting your subaccount to the service in Work Zone Manager. As the subaccount administrator, proceed as follows:

1. Connect your subaccount to Identity Provisioning as described in: Post Booster Configuration
2. Open the URL https://iamtenants.accounts.cloud.sap/ and login with your S-user.
3. View the Identity Provisioning and Identity Authentication tenants that are assigned to your customer ID.

For the complete onboarding process, see Getting Started

Related Information

SAP Build Work Zone, advanced edition

1.3.2.2 SAP Cloud Identity Access Governance Bundle

SAP Cloud Identity Access Governance bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

How to Obtain

After purchasing SAP Cloud Identity Access Governance, if your license includes only SAP Cloud Identity Access Governance, you can obtain the Identity Provisioning as well.

> **Note**
>
> You won't be charged any extra fee as Identity Provisioning has been officially integrated in the SAP Cloud Identity Access Governance license.

You already have Identity Authentication (a mandatory service), which enables you to sign in and authenticate in the Identity Provisioning UI and within SAP Business Technology Platform.

To obtain Identity Provisioning tenant, you need to create an incident. Follow the steps:

1. Create an incident to component *GRC-IAG-OPS* (*SAP Cloud Identity Access Governance*).
2. Explain that you've purchased an SAP Cloud Identity Access Governance product and you require access to the Identity Provisioning.
3. Specify the S-user to be assigned as the first administrator of the Identity Provisioning tenants. Later, this S-user can add other users as administrators.
   For complete information on what you need to specify in the incident, see Connecting Identity Provisioning Tenant in *SAP Cloud Identity Access Governance Admin Guide*

Related Information

SAP Cloud Identity Access Governance

1.3.2.3 SAP Central Business Configuration Bundle

SAP Central Business Configuration bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

Prerequisites

- You have credentials to log in to SAP for Me. For more information, see Access and Authorizations.

How to Obtain

After purchasing SAP Central Business Configuration, you must initiate its provisioning via SAP for Me by following the procedure described in SAP for Me Online Help. If you don't have an existing SAP Cloud Identity Services tenant, this will automatically trigger its creation. If you already have a tenant, it will be reused. For more information, see Activate Your Account.

How to Use

This bundle tenant is provisioned to your organization with the following preconfigured source and target systems:

- For groups provisioning: SAP Central Business Configuration as a source system and Identity Authentication as a target system.

> **Note**
>
> Note that the default user groups for SAP Central Business Configuration are already created in Identity Authentication.

- For users provisioning: Identity Authentication as a source system and SAP Central Business Configuration as a target system.

The communication between SAP Central Business Configuration and Identity Provisioning is set up.

You receive also a preconfigured technical user in Identity Authentication for system-to-system communication.

You can review the provisioning system configurations, adjust them if needed and schedule provisioning jobs. For more information about SAP Central Business Configuration provisioning systems, see:

- SAP Central Business Configuration (Source) [page 706]
- SAP Central Business Configuration (Target) [page 1076]

The other provisioning systems in the scope of this bundle are enabled. This means that you can start adding and configuring them in the Identity Provisioning UI. See: Add New Systems [page 2415]

Related Information

SAP Central Business Configuration

1.3.2.4 SAP Commerce Cloud Bundle

SAP Commerce Cloud bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

How to Obtain

After purchasing SAP Commerce Cloud, the technical contact person of your organization receives two onboarding e-mails from SAP. Each of them provides a tenant URL for accessing the SAP Cloud Identity Services administration console. One of the tenant URLs is for testing purposes, the other one is for productive usage. The technical contact person is granted the administrator permissions of the tenants and performs the initial logon to the SAP Cloud Identity Services administration console.

How to Use

This bundle tenant is provisioned to your organization with preconfigured source and target systems. Identity Authentication system is preconfigured as a source and SAP Commerce Cloud solution is preconfigured as a target. Identity Authentication default user groups are created for SAP Commerce Cloud.

You can review the provisioning system configurations, adjust them if needed and schedule read jobs.

1.3.2.5 SAP Concur Bundle

SAP Concur bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

Customers can integrate SAP Cloud Identity Services no matter whether the Concur instance is integrated with their own corporate identity provider or not.

Prerequisites

- New or existing Concur Expense, Concur Request, Concur Invoice, and/or Concur Travel & Expense customers that have at least one other SAP cloud solution.
- Identity Authentication and Identity Provisioning tenants must be created or configured by SAP:
  1. You need to check with your SAP Concur account team to determine if you are eligible for provisioning of a tenant with Identity Authentication and Identity Provisioning services.
  2. If eligible, create a support case to initiate the provisioning of an SAP Cloud Identity Services tenant with Identity Authentication and Identity Provisioning services. For customers with an existing Identity Authentication and Identity Provisioning tenants, the tenants will be reused.

     > **Note**
     >
     > You must create a case to set up Identity Authentication and Identity Provisioning tenant provisioning for your SAP Concur entity even if you already have existing Identity Authentication and Identity Provisioning tenants from other SAP solutions.

     To create a case, you can choose between creating an SAP for Me case (recommended) or an SAP Concur support case.
     Ensure your support case is directed correctly by using the following subject line in the short description: **`Set up SAP Cloud Identity Services for Concur`**.
     To help avoid delays due to clarification, please answer the following questions:
     - Do you have any other SAP products? (Please answer Yes or No.)
     - Are you currently using SAP Cloud Identity Services - Identity Authentication with existing SAP solutions, or is this a new setup for you?
     - What is your customer number? You can find it on the SAP for Me portal. For help accessing the portal and locating your customer number, refer to Access and Authorizations and User Action Menu (Profile).

For more information about creating an SAP for Me case, see this SAP Note: 1296527 - How to create a support case (contact SAP Product Support) - SAP for Me. Use the Component = CON-INT-SSO-SD.
For more information about creating an SAP Concur support case, see the *How to Navigate the SAP Concur Support Portal* section on the SAP Concur Support page for instructions on creating a support case. Use the Case Type = Single Sign On.

How to Use

Your approach to using SAP Cloud Identity Services for authenticating and managing users in SAP Concur solutions relies on knowing which components are automated and which ones require manual configuration.

As of June 23, 2025, SAP Concur has automated its integration with Identity Authentication, enabling you to access the SAP Cloud Identity Services administration console within SAP Concur. This automation sets up the trust between Identity Authentication and SAP Concur, including password migration.

For more information about the automated migration of user authentication and password policies from SAP Concur to SAP Cloud Identity Services, refer to Process.

For details on manual password migration, refer to Configure Authentication Provider To Migrate User Passwords from SAP Concur to Identity Authentication.

Related Information

Overview

1.3.2.6 SAP Fieldglass Bundle

SAP Fieldglass bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

How to Obtain

After purchasing SAP Fieldglass, the technical contact person of your organization receives two onboarding e-mails from SAP. Each of them provides a tenant URL for accessing the SAP Cloud Identity Services administration console. One of the tenant URLs is for testing purposes, the other one is for productive usage. The technical contact person is granted the administrator permissions of the tenants and performs the initial logon to the SAP Cloud Identity Services administration console.

How to Use

This bundle tenant is provisioned to your organization with preconfigured source and target systems.

- If you are a new SAP Fieldglass customer (greenfield scenario), you get Identity Authentication preconfigured as a source system and SAP Fieldglass preconfigured as a target.
- If you are an existing SAP Fieldglass customer (brownfield scenario), in addition to preconfigured Identity Authentication source system and SAP Fieldglass target system, you get preconfigured SAP Fieldglass source system and Identity Authentication target system. To use brownfield scenario, your SAP Fieldglass administrator must trigger an upgrade process from the SAP Fieldglass UI.

You can review the provisioning system configurations, adjust them if needed and schedule read jobs.

Related Information

SAP Fieldglass and SAP Cloud Identity Services Business Synopsis

1.3.2.7 SAP Field Service Management Bundle

SAP Field Service Management bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

Prerequisites

- You have credentials to log in to SAP for Me. For more information, see Access and Authorizations.

How to Obtain

After purchasing SAP Field Service Management, you must initiate its provisioning via SAP for Me by following the procedure described in Provisioning Environments and SAP for Me Online Help. If you don't have an existing SAP Cloud Identity Services tenant, this will automatically trigger its creation. If you already have a tenant, it will be reused. For more information, see Get Your Tenant.

> **Note**
>
> If the SAP Cloud Identity Services - Identity Authentication tenant is newly created the technical contact person of your organization is added as administrator of this tenant. If the tenant is reused, the technical contact person of your organization is added as a regular user. In case a user with matching email address already exists in the SAP Cloud Identity Services - Identity Authentication tenant, no additional user is created. In that case no activation email is sent from SAP Cloud Identity Services.

When the provisioning is complete, the technical contact person of your organization receives an email which contains the URL to the console of Field Service Management and the Account Name. For more information, see Initial System Access.

How to Use

This bundle tenant is provisioned to your organization with preconfigured provisioning systems. Also, the communication between SAP Field Service Management and Identity Provisioning is set up. You have preconfigured technical users in Identity Authentication and SAP Field Service Management for system-to-system communication.

The authentication flow from SAP Field Service Management is set up through OpenID Connect with the Identity Authentication.

You have the following provisioning systems preconfigured:

- For external users provisioning: SAP Field Service Management as a source system and Identity Authentication as a target system.
- For employee users provisioning: Identity Authentication as a source system and SAP Field Service Management as a target system. The source system has set filter to provision only employee users assigned to groups with `groups.display` starting with `"FSM_"`.

Additionally, you have automatically enabled real-time provisioning for external users from SAP Field Service Management to Identity Authentication and a technical user for accessing the real-time provisioning API.

You can review the provisioning system configurations, adjust them if needed and schedule provisioning jobs. For more information about SAP Field Service Management provisioning systems, see:

- SAP Field Service Management (Source) [page 757]
- SAP Field Service Management (Target) [page 1140]

The other provisioning systems in the scope of this bundle are enabled. This means that you can start adding and configuring them in the Identity Provisioning UI. See: Add New Systems [page 2415]

Related Information

SAP Field Service Management
Leverage IPS Scenarios

1.3.2.8 SAP Integrated Business Planning Bundle

SAP Integrated Business Planning (in short, SAP IBP) bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

How to Obtain

Since November 2022, the Identity Provisioning service is provided together with SAP IBP by default. For information about accessing your Identity Provisioning tenant as an administrator, see Initial Administrator User. If your system has been provisioned before November 2022, you can obtain an Identity Provisioning bundle tenant by creating an incident as follows:

1. Create an incident to component *SCM-IBP-OPS-SRV* (*Cloud Operations Service Requests*).
2. Specify the S-user to be assigned as the first administrator of the Identity Provisioning tenants. Later, this S-user can add other users as administrators.
3. Specify the URLs to your SAP IBP **Quality** and **Productive** systems.

In the reply of your incident, you'll receive two URLs related to two Identity Provisioning tenants.

- The first URL will be bound to your **Quality** instance, and you can use it for testing purposes.
- The second URL will be bound to your **Productive** instance, and you can use it for productive provisioning configurations and jobs. This bounding principle is applied to your Identity Authentication tenants as well.

If you encounter issues with accessing your Identity Provisioning UI, create an incident to component *BC-IAM-IPS*.

How to Use

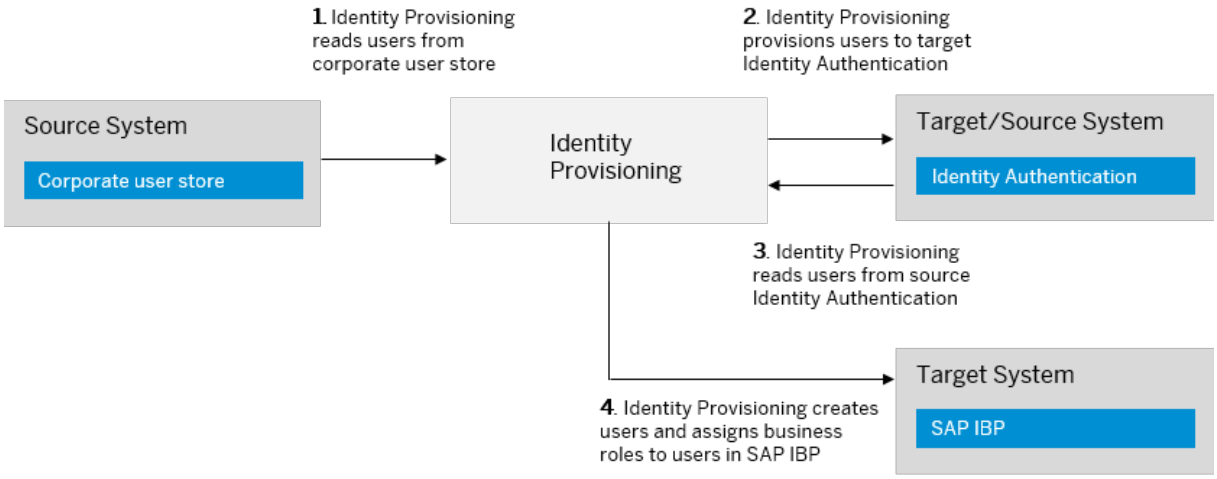
The scope of the SAP IBP bundle tenant includes the provisioning systems (connectors) displayed in the diagram above.

By default, the communication between SAP IBP and Identity Provisioning is automatically set up. Also, Identity Authentication is preconfigured as a source system and SAP IBP is preconfigured as a target system. This means that you can start running provisioning jobs manually or set a time interval for scheduled jobs. See: Start and Stop Provisioning Jobs [page 2472]

The other provisioning systems in the scope of this bundle are enabled. This means that you can start adding and configuring them in the Identity Provisioning UI. See: Add New Systems [page 2415]

> **Note**
>
> Users and groups in Identity Authentication correspond to business users and business roles in SAP IBP. The user groups defined in SAP IBP are not considered by the Identity Provisioning.

> **Example**
>
> In this example scenario, users are first provisioned to Identity Authentication (target system) from a corporate user store, such as Microsoft Azure AD (source system). Afterwards, those users are provisioned from Identity Authentication (this time set up as a source system) to SAP IBP (target system).



For more information about SAP IBP integration scenarios, see: Integrating Identity Provisioning Service with SAP IBP

For more information about the provisioning systems (source, target and proxy) relevant to SAP IBP bundle, see: Supported Systems [page 585]

Related Information

SAP Integrated Business Planning

1.3.2.9 SAP Jam Collaboration Bundle

SAP Jam Collaboration bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

How to Obtain

- **Your license contains Identity Provisioning**
  After purchasing SAP Jam Collaboration, you'll receive two e-mails from SAP. According to your contract with SAP, a technical contact person has been chosen as the first user of the Identity Provisioning service, who is granted with *Administrator* permissions. In these e-mails from SAP, you'll find the ID of this administrator (their P- or S-user) and their e-mail address. They can access the Identity Provisioning UI with their user credentials.
  Each e-mail from SAP contains also a URL link that you, as an administrator, can use to directly access the Identity Provisioning UI. These two URLs are related to two different Identity Provisioning tenants – the first one you can use for *testing* purposes, and the second one – for *productive* provisioning configurations and jobs.

If you encounter issues with accessing your Identity Provisioning UI, create an incident to component *BC-IAM-IPS*.

- **Your license does not contain Identity Provisioning**
  After the successful purchase, if your license includes only SAP Jam Collaboration, you can request a tenant for the Identity Provisioning service.

  > **Note**
  >
  > You won't be charged any extra fee as Identity Provisioning service has been officially integrated in the SAP Jam license.
  >
  > See the following blog post: SAP Jam now comes with the Identity Provisioning service

  You already have Identity Authentication (a mandatory service), which enables you to sign in and authenticate in the Identity Provisioning UI and within SAP Business Technology Platform.
  To obtain Identity Provisioning, you need to create an incident. Follow the steps:
  1. Create an incident to component *BC-IAM-IPS*.
  2. Explain that you've purchased an SAP Jam Collaboration product and you require access to the Identity Provisioning service.
  3. Specify the S-user to be assigned as the first administrator of the Identity Provisioning tenants. Later, this S-user can add other users as administrators.

How To Use

If your SAP Jam Collaboration license includes Identity Provisioning, this bundle tenant is provisioned to your organization with preconfigured source and target systems. Identity Authentication system is preconfigured as a source system and SAP Jam Collaboration is preconfigured as a target. For more information, see Identity Authentication and Identity Provisioning in the *SAP Jam Collaboration Administrator Guide*.

You can review the provisioning system configurations, adjust them if needed and schedule read jobs.

Related Information

SAP Jam Collaboration

1.3.2.10 SAP Marketing Cloud Bundle

SAP Marketing Cloud bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

You have purchased the product before release 1911

If you've bought SAP Marketing Cloud before the **1911** release, you have to request access to the Identity Provisioning service. To do this, follow the steps:

1. Create an incident to component *XX-S4C-OPR-SRV* (*S/4HANA Cloud service requests*).
2. Explain that you've purchased an SAP Marketing Cloud product before **1911** release.
3. Specify the S-user to be assigned as the first administrator of the Identity Provisioning tenants. Later, this S-user can add other users as administrators.
4. Specify the URLs to your SAP Marketing Cloud **Quality** and **Productive** systems.

In the reply of your incident, you'll receive two URLs related to two Identity Provisioning tenants. The first URL will be bound to your **Quality** instance, and you can use it for *testing* purposes. The second URL will be bound to your **Productive** instance, and you can use it for *productive* provisioning configurations and jobs. This bounding principle is applied to your Identity Authentication tenants as well.

You have purchased the product after release 1911

If you've bought SAP Marketing Cloud with release **1911** or higher, you'll receive two onboarding e-mails from SAP. According to your contract with SAP, a technical contact person has been chosen as the first user of the Identity Provisioning service, who is granted with *Administrator* permissions. Each onboarding e-mail contains a URL link that you, as an administrator, can use to directly access the Identity Provisioning UI. The relevant URLs are related to two different Identity Provisioning tenants – the first one you can use for *testing* purposes, and the second one – for *productive* provisioning configurations and jobs.

If you encounter issues with accessing your Identity Provisioning UI, create an incident to component *BC-IAM-IPS*.

Related Information

SAP Marketing Cloud

1.3.2.11 SAP Market Communication for Utilities Bundle

SAP Market Communication for Utilities bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

How to Obtain

After purchasing SAP Market Communication for Utilities, the technical contact person of your organization receives two onboarding e-mails from SAP. Each of them provides a tenant URL for accessing the SAP Cloud

Identity Services administration console. One of the tenant URLs is for testing purposes, the other one is for productive usage. The technical contact person is granted the administrator permissions of the tenants and performs the initial logon to the SAP Cloud Identity Services administration console.

How to Use

This bundle tenant is provisioned to your organization with preconfigured source and target systems.

The Identity Authentication system (named **IAS for MaCo – source**) is configured as the source, while the SAP Market Communication for Utilities system (named **MaCo-<subdomain>**) is configured as the target that reads from this source.

Default user groups for SAP Market Communication for Utilities are created in Identity Authentication. Communication between SAP Market Communication for Utilities and Identity Provisioning is already established, using **ClientCertificateAuthentication**. For more information, see What has the system automatically prepared for the synchronization?.

> **Note**
>
> A single SAP Cloud Identity Services tenant with configured Identity Authentication source system is used for provisioning to all SAP Market Communication for Utilities tenants that you obtain.

The transformations in the preconfigured systems (provided below) differ from the default Identity Provisioning transformations in Identity Authentication (source system) [page 586] and SAP Market Communication for Utilities (target system) [page 1187], as they are adapted to the scenarios relevant to this bundle.

Identity Authentication (source system)

Code Syntax

```
{
       "user": {
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extens
ion:custom:2.0:User']['userId']",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']['userUuid']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.userName",
                "optional": true,
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional": true,
                "targetPath":
"$.name.middleName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional": true,
                "targetPath":
"$.name.familyName"
            },
            {
```

SAP Market Communication for Utilities (target system)

Code Syntax

```
{
        "user": {
        "condition":
"$.groups[?(@.display==
'MaCo_BU_<subdomain>' ||
@.display== 'MaCo_PW_<subdomain>'
|| @.display==
'MaCo_DPP_<subdomain>')] EMPTY
false ",
        "mappings": [
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.personExternalID"
            },
            {
                "sourcePath":
"$.externalId",
                "optional": true,
                "targetPath":
"$.personExternalID"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['employeeNumber']",
                "optional": true,
                "targetPath":
"$.personExternalID"
            },
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.personID"
            },
            {
                "constant":
"false",
                "targetPath":
"$.markedForArchivingIndicator"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']['userUuid']",
                "optional": true,
                "targetPath":
"$.user.globalUserID"
            },
            {
                "targetPath":
"$.businessPartnerRoleCode",
                "type":
"valueMapping",
                "sourcePaths": [
                    "$.userType"
                ],
```

Identity Authentication (source system)

```
                "sourcePath":
"$.name.honorificPrefix",
                "optional": true,
                "targetPath":
"$.name.honorificPrefix"
            },
            {
                "sourcePath":
"$.emails[*].value",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.emails[?(@.value)]"
            },
            {
                "sourcePath":
"$.emails[?(@.primary==
true)].value",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.active",
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.userType",
                "optional": true,
                "targetPath":
"$.userType"
            },
            {
                "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.addresses"
            },
            {
                "sourcePath":
"$.locale",
                "optional": true,
                "targetPath":
"$.locale"
            },
            {
                "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.phoneNumbers"
            },
```

SAP Market Communication for Utilities (target system)

```
                "defaultValue":
"BUP003",
                "valueMappings": [
                    {
                        "key": [

"Employee"
                        ],

"mappedValue": "BUP003"
                    },
                    {
                        "key": [

"Freelancer"
                        ],

"mappedValue": "BBP010"
                    },
                    {
                        "key": [

"Service Performer"
                        ],

"mappedValue": "BBP005"
                    }
                ]
            },
            {
                "sourceVariable":
"currentDate",
                "targetPath":
"$.validityPeriod.startDate",
                "scope":
"createEntity"
            },
            {
                "constant":
"9999-12-31",
                "targetPath":
"$.validityPeriod.endDate",
                "scope":
"createEntity"
            },
            {
                "sourceVariable":
"currentDate",
                "targetPath":
"$.user.validityPeriod.startDate",
                "scope":
"createEntity"
            },
            {
                "constant":
"9999-12-31",
                "targetPath":
"$.user.validityPeriod.endDate",
                "scope":
"createEntity"
            },
            {
```

Identity Authentication (source system)	SAP Market Communication for Utilities (target system)

```
            {
                "sourcePath":
"$.timeZone",
                "optional": true,
                "targetPath":
"$.timezone"
            },
            {
                "sourcePath":
"$.displayName",
                "optional": true,
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.groups",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups"
            },
            {
                "condition":
"$.displayName EMPTY true",
                "targetPath":
"$.displayName",
                "type": "remove"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']['validFrom']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']['validFrom']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']['validTo']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']['validTo']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']
['sourceSystem']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']
['sourceSystem']"
            },
            {
```

```
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.personalInformation.firstName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "targetPath":
"$.personalInformation.lastName"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional": true,
                "targetPath":
"$.personalInformation.middleName"
            },
            {
                "sourcePath":
"$.name.formatted",
                "optional": true,
                "targetPath":
"$.personalInformation.personFullNa
me"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.user.userName"
            },
            {
                "sourcePath":
"$.locale",
                "optional": true,
                "targetPath":
"$.user.logonLanguageCode"
            },
            {
                "sourcePath":
"$.emails[0].value",
                "optional": true,
                "targetPath":
"$.workplaceInformation.emailAddres
s"
            },
            {
                "constant":
"false",
                "targetPath":
"$.user.lockedIndicator"
            },
            {
                "condition":
"$.active == false",
                "constant": "true",
                "targetPath":
"$.user.lockedIndicator"
            }
        ]
```

Identity Authentication (source system)	SAP Market Communication for Utilities (target system)

```
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']
['sourceSystemId']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']
['sourceSystemId']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']
['sourceSystem']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User']
['sourceSystem']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['employeeNumber']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['employeeNumber']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['costCenter']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['costCenter']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['organization']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['organization']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['division']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
```

```
    },
    "group": {
        "condition":
"$.displayName=='MaCo_BU_<subdomain
>' ||
$.displayName=='MaCo_PW_<subdomain>
' ||
$.displayName=='MaCo_DPP_<subdomain
>'",
        "mappings": [
            {
                "targetPath":
"$.ROLE_NAME",
                "targetVariable":
"entityIdTargetSystem",
                "type":
"valueMapping",
                "defaultValue": "",
                "sourcePaths": [
                    "$.displayName"
                ],
                "valueMappings": [
                    {
                        "key": [

"MaCo_BU_<subdomain>"
                        ],

"mappedValue":
"BR_CUSTOMER_USER_AGENT"
                    },
                    {
                        "key": [

"MaCo_PW_<subdomain>"
                        ],

"mappedValue":
"BR_CUSTOMER_POWER_USER"
                    },
                    {
                        "key": [

"MaCo_DPP_<subdomain>"
                        ],

"mappedValue":
"BR_CUSTOMER_DPP_USER"
                    }
                ]
            },
            {
                "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members[?(@.value)]",
                "functions": [
                    {
```

Identity Authentication (source system)	SAP Market Communication for Utilities (target system)

```
nsion:enterprise:2.0:User']
['division']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['department']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['department']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['manager']['value']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['manager']['value']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['manager']['displayName']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['manager']['displayName']"
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extens
ion:custom:2.0:User']",
                "optional": true,
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extens
ion:custom:2.0:User']"
            },
            {
                "sourcePath":
"$.company",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User']
['organization']"
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourcePath":
"$.id",
```

```
"function": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

Identity Authentication (source system)	SAP Market Communication for Utilities (target system)

```
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extens
ion:custom:2.0:Group']['name']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extens
ion:custom:2.0:Group']['name']"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.members",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members"
            }
        ]
    }
}
```

In addition, bulk operations are enabled for the SAP Market Communication for Utilities target. The following properties are configured:

- maco.bulk.operations.max.count = 50
- maco.support.bulk.operation = true
- maco.user.roles.overwrite = false
- maco.hr.switch.active = false
- ips.date.variable.format = yyyy-MM-dd

For more information about these properties, see List of Properties [page 80].

You can review the provisioning system configurations, adjust them if needed and schedule read jobs.

For more information on the manual tasks you can perform, see What do you need to do manually?.

Related Information

SAP Market Communication for Utilities

1.3.2.12 SAP S/4HANA Cloud Public Edition Bundle

SAP S/4HANA Cloud Public Edition bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

How to Obtain

The way you obtain Identity Provisioning bundle tenant depends on the release of your SAP S/4HANA Cloud Public Edition.

SAP S/4HANA Cloud Public Edition Purchased Before 1911 Release

If you've purchased SAP S/4HANA Cloud Public Edition before the **1911** release, you have to request access to the Identity Provisioning service. To do this, follow the steps:

1. Create an incident to component *XX-S4C-OPR-SRV* (*S/4HANA Cloud service requests*).
2. Explain that you've purchased an SAP S/4HANA Cloud Public Edition product before **1911** release.
3. Specify the S-user to be assigned as the first administrator of the Identity Provisioning tenants. Later, this S-user can add other users as administrators.

In the reply of your incident, you'll receive two URLs related to two Identity Provisioning tenants. The first URL will be bound to your **Quality** instance, and you can use it for *testing* purposes. The second URL will be bound to your **Productive** instance, and you can use it for *productive* provisioning configurations and jobs. This bounding principle is applied to your Identity Authentication tenants as well.

SAP S/4HANA Cloud Public Edition Purchased With 1911 Release or Higher

If you've purchased SAP S/4HANA Cloud Public Edition with release **1911** or higher, you'll receive two onboarding e-mails from SAP. According to your contract with SAP, a technical contact person has been chosen as the first user of the Identity Provisioning service, who is granted with *Administrator* permissions. Each onboarding e-mail contains a URL link that you, as an administrator, can use to directly access the Identity Provisioning UI. The relevant URLs are related to two different Identity Provisioning tenants – the first one you can use for *testing* purposes, and the second one – for *productive* provisioning configurations and jobs.

How to Use

The Identity Provisioning bundle tenant for SAP S/4HANA Cloud Public Edition release 1911 or higher comes with preconfigured provisioning systems (connectors) for specific integration scenarios:

- For integrating SAP S/4HANA Cloud Public Edition with the embedded SAP Analytics Cloud, SAP provides preconfigured SAP S/4HANA Cloud Public Edition source system and SAP Analytics Cloud target system. To learn more about what's been initially preconfigured and how to apply subsequent changes to the Identity Provisioning configurations, see:
  - Ensure SAP S/4HANA Cloud Public Edition users are replicated during their validity period
  - Ensure deleted SAP S/4HANA Cloud Public Edition users are also deleted in SAP Analytics Cloud
- For integrating SAP S/4HANA Cloud Public Edition with Identity Authentication, SAP provides preconfigured Identity Authentication source system and SAP S/4HANA Cloud Public Edition target

system. Also preconfigured Identity Authentication source system and SAP Central Business Configuration target system are delivered with system provisioning.

> **Note**
>
> Your initial provisioning jobs are paused. This means, you need to manually run them. To do this, open your relevant source systems, go to the *Jobs* > *Read Job* > *Schedule* and turn the job scheduler *ON*. For more information, see Read Provisioning Job [page 2474] . If you need to, correct the credentials (users and passwords) for your relevant systems first, and then start the initial job(s).

If you encounter issues with accessing your Identity Provisioning UI, create an incident to component *BC-IAM-IPS*.

Related Information

SAP S/4HANA Cloud Public Edition

SAP Identity Provisioning (IPS) is now bundled with SAP S/4HANA Cloud Public Edition!

1.3.2.13 SAP SuccessFactors Bundle

SAP SuccessFactors bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

If your SAP SuccessFactors tenant was created after December 9, 2022, Identity Authentication and Identity Provisioning have already been enabled. You do not need to complete the steps to upgrade to Identity Authentication, as described in Initiating the Upgrade to SAP Cloud Identity Services - Identity Authentication Service.

How to Obtain

After purchasing SAP SuccessFactors, if your license includes only SAP SuccessFactors, you can obtain Identity Authentication and Identity Provisioning as well. To do this, you have to enable these services via the *Upgrade Center* of SAP SuccessFactors. You need to provide an S-user and password for validation. This S-user will be assigned as the first administrator of the Identity Provisioning tenants.

> **Note**
>
> In case you already have an existing Identity Provisioning tenant, its initial administrator may be different than the S-user you've provided in the *Upgrade Center* procedure. Due to security reasons, this S-user will not be assigned as an administrator. If you want it to be granted IPS_ADMIN permissions, you need to ask the initial Identity Provisioning administrator to do that for you. To learn how, see: Manage Authorizations
>
> See also: (Guided Answers) Can't Get Admin Permissions for a Bundle

You also need to create a password for the automatically created SAP SuccessFactors API user (*IPSADMIN*), and grant it the following permissions:

- Go to *Admin Center* > *Manage Permission Roles* > *Manage Integration Tools* and choose "*Allow Admin to Access OData API through Basic Authentication*".
- Go to *Admin Center* > *Manage Permission Roles* > *Administrator Permission* > *Manage User* and choose "*User tenant OData entity*".

Proceed as follows:

1. Getting Started with Identity Authentication and SAP SuccessFactors
2. Initiate Upgrade to Identity Authentication
3. Identity Provisioning Administration Console Tasks
4. Setting an API User for Sync Jobs

Your Identity Provisioning system is created once you've initiated your upgrade to Identity Authentication, and the Identity Provisioning administrator receives an email with two links to the Identity Provisioning admin console (related to two different tenants) – one for *testing* purposes, and one – for *productive* provisioning configurations and jobs.

If the upgrade procedure fails, create an incident to component *LOD-SF-PLT-IAS*.

If you have troubles with the Identity Provisioning admin console or configurations, create an incident to component *BC-IAM-IPS*.

How to Use

This bundle tenant is provisioned to your organization with preconfigured source and target systems. SAP SuccessFactors is preconfigured as a source system and Identity Authentication is preconfigured as a target.

Integrating SAP SuccessFactors with the embedded SAP Analytics Cloud is triggered from SAP SuccessFactors *Upgrade Center*. As a result, SAP Analytics Cloud is created as a target system. It is disabled and no source system is linked to it.

> **Note**
>
> SAP SuccessFactors target system is enabled only for Identity Provisioning bundle tenants running on SAP Cloud Identity Services infrastructure.

You can review the provisioning system configurations, adjust them if needed and schedule read jobs.

Related Information

SAP SuccessFactors Platform

1.3.2.14 SAP SuccessFactors Incentive Management Bundle

SAP SuccessFactors Incentive Management, formerly known as SAP Commissions, bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

How to Obtain

After purchasing SAP SuccessFactors Incentive Management, you'll receive two e-mails from SAP. According to your contract with SAP, a technical contact person has been chosen as the first user of the Identity Provisioning service, who is granted with *Administrator* permissions. In these e-mails from SAP, you'll find the ID of this administrator (their P- or S-user) and their e-mail address. They can access the Identity Provisioning UI with their user credentials.

Each e-mail from SAP contains also a URL link that you, as an administrator, can use to directly access the Identity Provisioning UI. These two URLs are related to two different Identity Provisioning tenants – the first one you can use for *testing* purposes, and the second one – for *productive* provisioning configurations and jobs.

Related Information

SAP SuccessFactors Incentive Management

1.3.2.15 SAP SuccessFactors Learning Bundle

SAP SuccessFactors Learning bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

How to Obtain

After purchasing SAP SuccessFactors with license for SAP SuccessFactors Learning, you can obtain Identity Authentication and Identity Provisioning tenants by initiating the integration in SAP SuccessFactors *Upgrade Center* as outlined here: How to Integrate SAP SuccessFactors Learning and SAP Cloud Identity Services

The process to integrate SAP SuccessFactors Learning and SAP Cloud Identity Services - Identity Authentication and Identity Provisioning consists of the following four steps:

1. Initiating Learning Identity Authentication Migration
2. Learning Configuration Procedure
3. Migrating Learning Users to Identity Authentication
4. Identity Authentication Activation

How to Use

This bundle tenant is provisioned to your organization with preconfigured source and target systems. SAP SuccessFactors Learning is preconfigured as a source system and Identity Authentication is preconfigured as a target.

You can review the provisioning system configurations, adjust them if needed and schedule read jobs.

1.3.2.16 Sales Cloud – Analytics & AI Bundle

Sales Cloud – Analytics & AI bundles with SAP Cloud Identity Services – Identity Authentication and Identity Provisioning.

How to Obtain

After purchasing Sales Cloud – Analytics & AI, you'll receive two e-mails from SAP. According to your contract with SAP, a technical contact person has been chosen as the first user of the Identity Provisioning service, who is granted with *Administrator* permissions. In these e-mails from SAP, you'll find the ID of this administrator (their P- or S-user) and their e-mail address. They can access the Identity Provisioning UI with their user credentials.

Each e-mail from SAP contains also a URL link that you, as an administrator, can use to directly access the Identity Provisioning UI. These two URLs are related to two different Identity Provisioning tenants – the first one you can use for *testing* purposes, and the second one – for *productive* provisioning configurations and jobs.

1.3.2.17 SAP Cloud ALM Bundle

SAP Cloud ALM bundles with SAP Cloud Identity Services – Identity Authentication.

Prerequisites

You have an S-user with the role *Edit Cloud Data* on SAP for Me in order to request SAP Cloud ALM.

How to Obtain

SAP Cloud ALM helps you to implement and operate intelligent cloud and hybrid business solutions. SAP Cloud ALM is included in your **cloud** subscription containing Enterprise Support, cloud edition and in SAP Enterprise

Support. You can request SAP Cloud ALM on SAP for Me for yourself or for all entitled customers for whom you have sufficient permissions. If you don't have an existing productive SAP Cloud Identity Services tenant, this will automatically trigger its creation. If you already have a productive tenant, it will be reused, except for the cases in which it cannot be used due to some restrictions and legal requirements. For more information, see Requesting SAP Cloud ALM and Get Your Tenant [page 528].

> **Note**
>
> The user who requested SAP Cloud ALM receives permissions that are required for administrative tasks in the SAP BTP cockpit and in SAP Cloud ALM. Additionally, if the SAP Cloud Identity Services tenant is newly created this user is added as administrator of this tenant. If the tenant is reused, the user is added as a regular user. In case a user with matching email address already exists in the SAP Cloud Identity Services tenant, no additional user is created. In that case no activation email is sent from SAP Cloud Identity Services.

When the provisioning of SAP Cloud ALM is complete, the requesting user receives emails containing sign in information, configuration guidance, and links to support resources.

How to Use

This bundle tenant is provisioned to your organization in a newly created SAP BTP customer-managed subaccount. An OIDC trust is preconfigured between your Identity Authentication tenant and your SAP BTP multi-environment subaccount with enabled Cloud Foundry environment. Identity Authentication is set as trusted identity provider. This means that the way business users are signing in to SAP Cloud ALM is based on the configuration in the Identity Authentication.

You receive also the following attributes preconfigured:

- The home URL of SAP Cloud ALM is configured. For more information, see Configure an Application's Home URL [page 1937].
- The `e-mail` is configured as `Name ID` attribute in the settings of your identity provider. For more information, see Configure the Subject Name Identifier Sent to the Application [page 1988].
- The default assertion attributes for the established OIDC trust are configured. For more information, see Customizing Attribute Mappings.

After the provisioning of SAP Cloud ALM is complete, a few setup steps are required so that it can be used productively. For more information, see Required Setup for SAP Cloud ALM. To give further users access to SAP Cloud ALM, see Onboard Users in the Identity Authentication Service.

Related Information

SAP Cloud ALM
Monitoring with SAP Cloud ALM [page 2991]

1.3.3 Applications

An application is associated with a consumer of Identity Authentication as an identity provider. This consumer could be for example an SAP cloud solution, a third-party application, SAP BTP subaccount, or the SAP Cloud Identity Services administration console.

The application holds the configuration information about trust, authentication, access, and branding of the given consumer. SAP and customers create applications in SAP Cloud Identity Services administration console. The following table describes the different types of applications, who is in charge of them and whether they are configurable or not:

Created By	Application Type	Application Display Name	Specifics
SAP	*Bundled Applications*	For example: *S4HANA Cloud* - `<s4-tenant-type>`	SAP creates bundled applications and is in charge of their configuration.
	System Applications	*Administration Console*	SAP creates and preconfigures the administration console as a system application.
		User Profile	SAP creates and preconfigures the profile page with some personal information.
	Subscribed Applications	For example: *SAP BTP* `<subaccount display name>`	Subscribed applications are associated with an SAP BTP subaccount or an application registered there. These applications are automatically created for you. You can't create or delete them.
Customer	*Charged Applications*	For example: *Microsoft Entra ID*	Charged applications are created by customers for third-party (non-SAP) solutions. Customers are in charge of the configuration and are charged for integrating applications with Identity Authentication.

Created By	Application Type	Application Display Name	Specifics
	Bundled Applications	For example: *SAP Concur*	Apart from SAP, bundled applications can also be created by customers. This applies for SAP cloud applications that do not bundle the delivery and preconfiguration of the SAP Cloud Identity Services.

Applications are sometimes referred to as SAML 2.0 applications or OpenID Connect applications depending on the configured protocol. Some software-as-a-service applications and the `Administration Console` application support both protocols and display the hybrid `SAML 2.0 & OpenID Connect` protocol type. The SAML 2.0 standard defines applications as service providers, while OpenID Connect (OIDC) defines them as relying parties.

For more information, see Application Types [page 63] and Configuring Applications [page 1920].

Parent topic: Concepts [page 26]

Related Information

Tenants [page 27]
Bundles [page 35]
Provisioning Systems [page 69]
Properties [page 73]
Transformations [page 388]
Users [page 509]
Agents [page 511]
Groups [page 514]
Authorization Policies [page 518]
SAP-Managed SAP Cloud Identity [page 520]

1.3.3.1 Application Types

Application types in the administration console for SAP Cloud Identity Services.

SAP Applications

The SAP applications are created and managed by SAP, and they can't be deleted. Part of their configurations are read-only. SAP applications are: bundled, subscriptions and system applications.

Identity Authentication identifies the type of the application by the URI or SAML 2.0 endpoints.

> ⓘ Note
>
> Applications using custom domains can't be marked correctly, so if you have such applications, report an incident on SAP Support Portal Home with a component `BC-IAM-IDS`.

Bundled Applications

Bundled applications are managed and configured by SAP and can't be deleted. Part of their SAML 2.0 configurations are read-only. For more information, see Configure an Application's Type [page 1935].

Subscribed Applications

A special kind of applications are the so-called subscribed applications.

Most subscribed applications use the trust configuration of the SAP BTP subaccount where they're subscribed. For some applications, when you subscribe to them, the system automatically sets up an application in Identity Authentication for you. These applications appear as child applications of the application in SAP Cloud Identity Services that represents the SAP BTP subaccount.

You can't create or delete a subscribed application. For these applications, you can only see and configure a certain subset of their settings. The change in the following configurations is restricted, since they are managed by the application provider:

- Protocol
- OpenID Connect Configuration

  > ⓘ Note
  >
  > It's possible to add custom URIs, but those defined by the application provider remain.

- Attributes - Self-defined Attributes
  The attributes supported by subscribed applications are predefined and listed under the *Primary Attribute* section, with default mappings for source *Identity Directory*. You can see which attributes are supported, and add or remove mappings for the other sources - *Corporate Identity Provider* and *Expression*.

  > → Tip
  >
  > For some subscribed applications, you might need to add custom attribute mappings (under the *Self-Defined Attributes* section).

- Client Authentication

  > ⓘ Note
  >
  > Fully managed by the application provider.

- APIs

  > ⓘ Note
  >
  > You have no access to the APIs provided by Identity Authentication.

System Applications

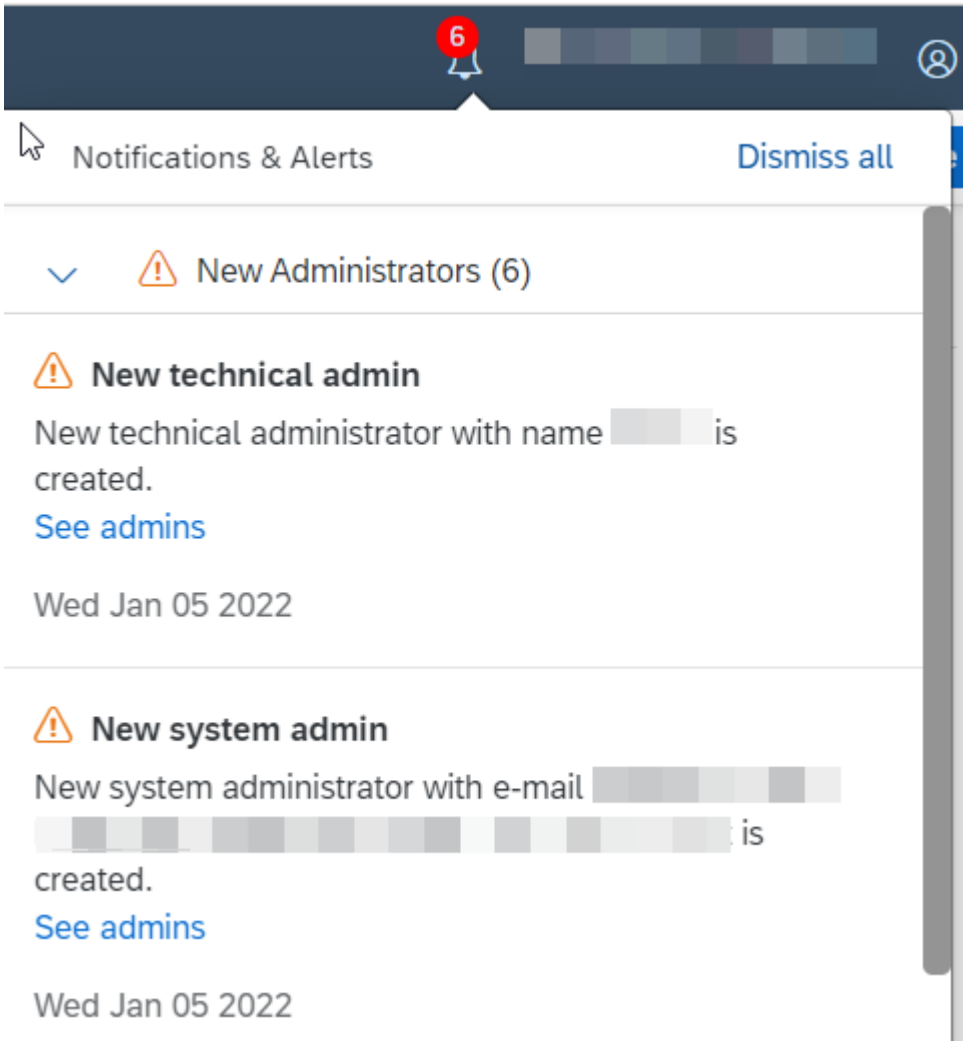
The tenant of Identity Authentication has two additional system applications. They are predefined with the creation of the tenant. These applications are: `Administration Console`, and `User Profile`, previously called `SAP Cloud Identity`.

> ⓘ Note
>
> In some tenants, the `User Profile` application still bears its previous name, `SAP Cloud Identity`.

> → Tip
>
> If `Administration Console` or `User Profile` are not in the list of the system applications you may request them. To do this, report an incident on SAP Support Portal Home under the component `BC-IAM-IDS`.

> ⚠ Caution
>
> Please be careful when you make configuration changes to the system applications. Certain configuration options for the system applications are read-only.
>
> System applications can't be parents to other applications, and can't have child applications.

The `Administration Console` application contains the configurations of the administration console for SAP Cloud Identity Services, and information about expiring certificates, system notifications and new administrators. The information is visible at the top-right corner of the administration console:



You can access the tenant's administration console for SAP Cloud Identity Services by using the console's URL.

> **Note**
>
> The URL has the following pattern:
>
> `https://<tenant ID>.accounts.ondemand.com/admin` or `https://<tenant ID>.accounts.cloud.sap/admin`
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].
>
> If you have a configured custom domain, the URL has the `<your custom domain>/admin` pattern.
>
> If you have updated the `Name` in the *SAML 2.0 Configuration* or the `URL` in the *OpenID Connect Configuration* in *Tenant Settings* make sure to access the tenant with the updated domain in the `Name`

or **URL**. For more information, see Tenant SAML 2.0 Configurations [page 2103] and Tenant OpenID Connect (OIDC) Configurations [page 2117].

As a tenant administrator you can change the default configurations:

- to configure stronger protection for the administration console for SAP Cloud Identity Services via the *Risk-Based Authentication* option.
- to configure stronger password requirements for the tenant administrators of the administration console for SAP Cloud Identity Services.
- to choose authenticating identity provider and define rules for authenticating identity provider.
- to customize the look and feel of the sign-in page of the administration console for SAP Cloud Identity Services.

→ Tip

To change the appearance of the administration console, at the top-right corner choose the *circle with your initials* > *Settings* > *Appearance* and select the theme you want. You can chose from:

- *SAP Morning Horizon (default)*
- *SAP Evening Horizon*
- *SAP Quartz Light*
- *SAP Horizon High Contrast White*
- *SAP Horizon High Contrast Black*

The `User Profile` application contains the configurations of the Profile Page.

As a tenant administrator you can change the default configurations:

- to configure stronger protection via the *Risk-Based Authentication* option.
- to choose authenticating identity provider and define rules for authenticating identity provider.
- to define custom email template sets for users created via the *Add User* option in the administration console for SAP Cloud Identity Services, or via the SCIM REST API.
- to customize the look and feel of the sign-in page of the Profile Page.

ⓘ Note

The URL of the profile page has the following pattern:

`https://<tenant ID>.accounts.ondemand.com` or `https://<tenant ID>.accounts.cloud.sap`

Tenant ID is an automatically generated ID by the system.

If you have a configured custom domain, the URL is `<your custom domain`.

→ Tip

For a quick access to your own profile page, at the top-right corner of the administration console, choose the *circle with your initials* > *Settings* > *User Account* > *Manage Profile*. Your profile page opens in a new tab.

Customer Applications

Applications that are created by the SAP customers are recognized as charged, or third-party applications. If no URI or SAML 2.0 endpoints are specified in the configuration of the application, Identity Authentication recognizes the application as a charged one.

Charged applications are created by customers for third-party (non-SAP) solutions. Customers are in charge of the configuration and are charged for integrating applications with Identity Authentication.

All the configurations of the applications are done by you as a customer.

Parent, Child Applications

SAP and non-SAP created applications can also be parent or child. An application can be defined as parent or child at its creation, or later when editing the application.

When you create or edit an application, you can select a parent for that application. Thus, the application that is created or edited becomes child, and the selected application becomes parent. When creating a new application, the parent application option in the administration console is optional.

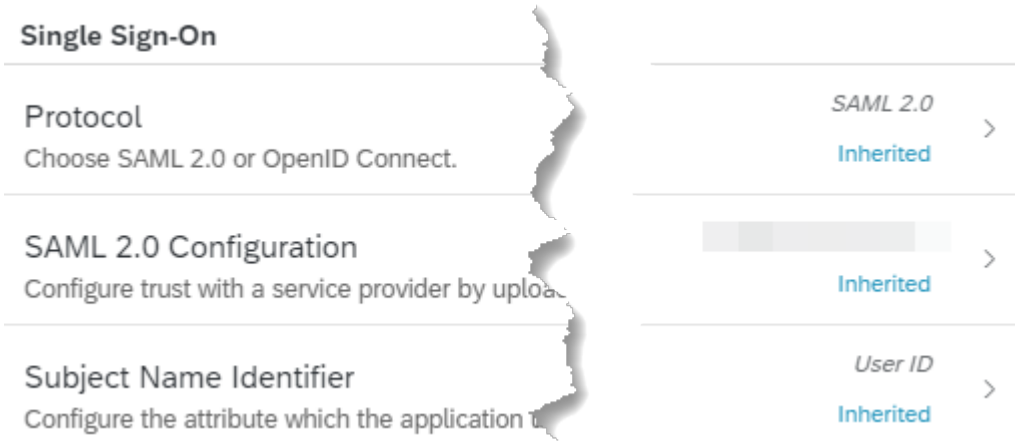
Changing child application's configuration doesn't affect the configuration of the parent.

Applications that have a parent application configured can't be selected as parent application.

An application that is chosen as a parent application to another can't have a parent application assigned for it.

Newly created applications with an assigned parent application inherits all the configurations from the parent with the except for the `Client ID` and `Secrets`.

The inherited configurations are marked as such.



Existing applications that have a parent application assigned to them will inherit only the configurations that have not been changed, including the configurations that are made at the creation of the application. The configurations that are changed, and the configurations made at the creation of the application are not inherited. See the list below for the configurations made at the creation of the application:

- Protocol

- Subject Name Identifier

  > **Note**
  >
  > When a subscribed application becomes a child application, the configuration of the `Subject Name Identifier` of that child application can be different from the parent application or from the application it is subscribed to. You can make the configuration of the `Subject Name Identifier` the same as the parent application by choosing the *Inherit from Parent* button on the top right-hand corner of the screen. You can also make the configuration of the `Subject Name Identifier` the same as the application it is subscribed to by choosing the *Reset* button on the top right-hand corner of the screen. Depending on your configuration, either the *Inherit from Parent* or the *Reset* button is active, or both buttons are active.

- Default Name ID Format
- Attributes

  > **Note**
  >
  > Attributes from different sources are inherited separately. For more information about the different user attribute sources, see User Attributes [page 1972].

- Dependencies
- Risk-Based Authentication
- Conditional Authentication
- Email Verification
- Password Policy
- Terms Of Use
- Privacy Policy
- User Application Access
- User Attributes for Access
- Token Url Separator
- Reload Parent Page

> **→ Tip**
>
> A child application can override all the configurations inherited from the parent application. If you change a configuration in the child that is inherited from the parent, and after that you decide to return to the inherited one, go to the respective configuration and choose the *Inherit from Parent* button on the top right-hand corner of the screen.

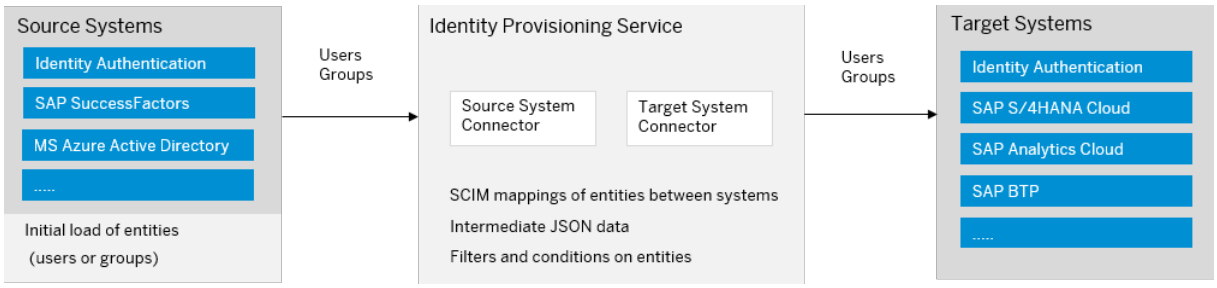
> **⚠ Caution**
>
> If you change a configuration in the child that is inherited from the parent, also change the name of the child, so that it differs from the name of the parent. Saving the configuration without changing the child application's name may result in failing authentication requests.

1.3.4 Provisioning Systems

Identity Provisioning provides connectors to various business applications for provisioning and deprovisioning of users and groups. These business applications are set up as provisioning systems in the administration console of SAP Cloud Identity Services.

Similar to applications in Identity Authentication, provisioning systems can be viewed as abstractions of the business applications, where these abstractions are used for the purposes of Identity Provisioning.

A provisioning system configured for reading users and groups is called source system. The source system is usually the user store of your organization. A provisioning system used for replicating users and groups is called target system. The target system is usually a cloud system with local user store, where entities are provisioned to from one or more source systems.



The main use case of Identity Provisioning is to read users and groups from a source system and provision them to a target system. This is the standard provisioning mode illustrated in the diagram above.

Identity Provisioning can also be used in proxy mode. In this mode, the service is used for synchronizing user data to and from a central identity management solution (for example, the on-premise SAP Identity Management) and a provisioning system - called proxy system (for example, SAP Analytics Cloud, embedded edition). Here, Identity Provisioning acts as a proxy between the identity management solution and the system with proxy configuration.

For more information, see System Types and Supported Systems.

Parent topic: Concepts [page 26]

Related Information

Tenants [page 27]
Bundles [page 35]
Applications [page 61]
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1.3.4.1 System Types

Identity Provisioning categorizes systems based on the method of creation, usage and underlying protocols.

Creation

There are two types of provisioning systems based on the method of creation: customer managed and SAP initiated.

Customer Managed

These are the provisioning systems that you as a customer create and configure in the Identity Provisioning UI. There is a limit to the number of systems you can create. It is 20 for source systems and 50 or target systems. However, if your scenarios require more provisioning systems, open an incident for component **BC-IAM-IPS** to request them.

SAP Initiated

These are the provisioning systems that are automatically created and preconfigured in the Identity Provisioning UI. There is no limit to the number of systems created for you by SAP. Furthermore, you can modify their initial configuration to make them suitable for your scenario.

Usage

There are three types of provisioning systems based on their usage: *Source*, *Target*, and *Proxy*. All three of them can be created as customer managed and SAP-initiated systems.

Source Systems

A source system is the connector used for reading entities (users, groups, roles). Source systems can be on-premise or cloud-based, SAP or non-SAP, and usually represent the corporate user store where identities are currently maintained. Identity Provisioning reads the entities from the source system and creates or updates them in the relevant target ones. The provisioning is triggered from the Jobs tab of a source system.

You can connect one source system to one or multiple target systems.

> ⓘ Note
>
> In the case of multiple (enabled) target systems, when you start a *Read* or a *Resync* job, this operation will trigger provisioning of entities from this source system to all relevant target ones.

To check the list of all supported source systems, see: Source Systems [page 585]

Target Systems

A target system is the connector used for writing (provisioning) entities. Target systems are usually clouds, where Identity Provisioning creates or updates the entities taken from the source system.

A target system can be connected to a single or multiple source systems.

> **Note**
>
> In the case of multiple source systems, we recommend that you run the provisioning jobs successively for each system, not simultaneously. This way, you'll avoid incorrect overwriting or merging of entity data, hence failed provisioning jobs.

To check the list of all supported target systems, see: Target Systems [page 936]

Proxy Systems

A proxy system is a special connector used for "hybrid" scenarios. It exposes any Identity Provisioning supported backend system as a SCIM 2.0 service provider, which can be consumed by any SCIM 2.0 compatible client application, without making a direct connection between them.

To achieve this, the Identity Provisioning service uses this special proxy system to execute provisioning operations (create, update, delete, etc.) requested by the client application.

The examples in this section cover using of SAP Identity Management as a consuming client application but you can use any other SCIM-based identity management solution. For more information, see: Hybrid Scenario: SAP Identity Management [page 2961]. To provide communication between SAP Identity Management and the back-end system, the proxy application uses a SCIM 2.0 protocol. A system can act as a proxy if it supports both read and write operations.

How a proxy system works:

1. The Identity Provisioning service exposes the back end of a supported system as a "proxy".
2. An external application (for example, SAP Identity Management) regards the proxy system as its back-end system.
3. The entities (users) exposed by the back-end system are mapped to SCIM 2.0 entities, if possible. If not possible, the SCIM standard provides a mechanism to define a new resource type with the appropriate schema. You can use the custom resource type to map the back-end entities. See: SCIM Resources
4. Finally, the external application can start sending REST web service requests to the proxy system in order to read identities from the back end of the SCIM 2.0 system.

To check the list of all supported proxy systems, see: Proxy Systems [page 1343]

Protocol

There are four types of provisioning systems based on the underlying protocol:

SCIM-based	SOAP/OData-based	LDAP-based	RFC-based
SCIM-based systems use the SCIM (System for	SOAP or OData-based systems are typically S/	LDAP-based systems use the LDAP protocol to manage	RFC-based systems use Remote Function Call (RFC)

SCIM-based	SOAP/OData-based	LDAP-based	RFC-based
Cross-domain Identity Management) standard API to manage and provision user identities.	4HANA systems which use these protocols for identity and role management within the S/4HANA environment.	and retrieve directory information.	for communication between SAP systems and between SAP and non-SAP systems.
Type: *HTTP* • Identity Authentication • Intelligent Opportunity Analyzer • Local Identity Directory • SAP Advanced Financial Closing • SAP Advanced Workflow • SAP Analytics Cloud • SAP Ariba Applications • SAP Ariba Category Management (deprecated) • SAP Ariba Central Invoice Management • SAP BTP Account Members (Neo) • SAP BTP Java/HTML5 apps (Neo) • SAP BTP Platform Members (Cloud Foundry) • SAP BTP XS Advanced UAA (Cloud Foundry) • SAP Build Work Zone, advanced edition • SAP Build Work Zone, standard edition • SAP Business Network • SAP Central Business Configuration • SAP Commerce Cloud • SAP Concur • SAP CPQ • SAP Data Custodian • SAP Datasphere • SAP Enable Now	Type: *HTTP* • SAP BTP ABAP environment • SAP Integrated Business Planning • SAP Market Communication for Utilities • SAP Marketing Cloud • SAP S/4HANA Cloud Public Edition • SAP S/4HANA On-Premise • SAP SuccessFactors version 1	Type: *LDAP* • LDAP Server • Microsoft Active Directory	Type: *RFC* • SAP AS ABAP

SCIM-based	SOAP/OData-based	LDAP-based	RFC-based
• SAP Enterprise Portal • SAP Fieldglass • SAP Field Service Management • SAP HANA Cloud, SAP HANA Database • SAP Intelligent Agriculture • SAP Jam Collaboration • SAP Master Data Integration • SAP S/4HANA for procurement planning • SAP Sales Cloud and SAP Service Cloud • SAP SuccessFactors version 2 • SAP SuccessFactors Employee Central Payroll • SAP SuccessFactors Incentive Management • SAP SuccessFactors Learning • Sales Cloud – Analytics & AI • Cloud Foundry UAA Server • Procurement Data Warehouse • SCIM System • SSH Server (Beta)			

Both Microsoft Entra ID and Google G Suite use `Type`: *HTTP*.

1.3.5 Properties

Properties hold the configuration of a provisioning system.

Properties define the connection and authentication information needed for accessing the provisioning system. They control which entities (users and groups) will be provisioned - by defining filters and unique attributes, as well as how they will be provisioned - in bulk operations or one entity at a time, with put or patch requests.

There are various types of properties depending on their usability, whether they hold mandatory or optional information, standard information in plain text or credentials in encrypted strings. They can be categorized as follows:

- Standard
- Credential
- Default
- Parameterized
- Internal

For your system provisioning goals, you can set properties in the administration console of SAP Cloud Identity Services on the *Properties* tab of the provisioning system.

Prefixes

Property names have specific prefixes that indicate the provisioning system they are applicable to. This helps you differentiate between properties designated for different systems.

- Properties starting with `<ips>` are applicable to all provisioning systems, for example `ips.failed.request.retry.attempts`.
- Properties starting with the abbreviation of a provisioning system name, such as `<s4hana.cloud>` or `<sf>`, are applicable to the respective systems—in this case, SAP S/4HANA Cloud Public Edition and SAP SuccessFactors, respectively. For example, `s4hana.cloud.support.bulk.operation` and `sf.user.filter`.
- Properties starting with `<scim>` are applicable to SCIM-based provisioning systems. To view the list of these systems, refer to System Types [page 70]. The scim-prefixed properties were used before the implementation of system-specific prefixes. Following the introduction of these prefixes, both scim and system-specific prefixes are valid.

> **Note**
>
> The scim-prefixed properties and the system-specific properties can be used interchangeably. If both are configured, Identity Provisioning will prioritize and consider the system-specific properties.

Parent topic: Concepts [page 26]

Related Information

Tenants [page 27]
Bundles [page 35]
Applications [page 61]
Provisioning Systems [page 69]
Transformations [page 388]
Users [page 509]

Agents [page 511]
Groups [page 514]
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SAP-Managed SAP Cloud Identity [page 520]
List of Properties [page 80]
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1.3.5.1 Property Types

Properties can help you filter which entities and entity attributes are read from the source system or written to the target system. According to their usability, properties can be categorized as follows:

Standard System Properties

Each source, target, and proxy system support specific types of properties. For example:

Examples:

AS ABAP System	SAP SuccessFactors
jco.client.r3name=PSE	sf.page.size=100
jco.destination.peak_limit=10	sf.user.filter=firstName John
jco.destination.pool_capacity=5	sf.user.attributes=email

Credential Properties

The values of these properties contain sensitive information that must **not** be displayed as plain text. The default credential property name is *Password*, which can represent standard passwords, private keys, or OAuth client secrets. When you add a credential property, its value is displayed as an encrypted string. For better security, the encrypted string is always displayed as 40 characters, no matter how long your password actually is.

> **Note**
>
> Properties whose values contain sensitive information can be added only as *Credential* in the Identity Provisioning UI. The values of these properties are stored as encrypted data. They are excluded from the file during system configuration export.

Examples:

SAP HANA Database	SSH Server
hana.jdbc.db.password=**********	ssh.password=**********
	ssh.private.key=**********
	ssh.totp.secret.key=**********

> **Note**
>
> When you update the URL or the host name of an existing provisioning system, you must re-enter the values of the configured credential properties. The only exception to this are the credential properties of systems that are created with a connectivity destination.

Default System Properties

These properties depend on the particular connector type. They exist in the transformations by default. It is possible to delete some of them but this may cause a loss of provisioned data. Example:

Examples:

LDAP Server
ldap.group.object.class=groupOfNames
ldap.user.object.class=inetOrgPerson
ldap.attribute.user.mobile=mobile
ldap.group.filter=<empty>
ldap.user.filter=<empty>

Parameterized System Transformations

They use parameters taken from the system property sets. The parameters consist of a *unique key* and a *value*. Like the standard properties, they can be configured in the system's *Properties* tab, and/or in the system's destination properties (in the platform cockpit). When one parameter exists in both property sets, the system properties have priority over the system destination properties. In the system transformations, the parameter is referenced by the unique key surrounded by the percent character (*%<key>%*).

Parameters can be used within almost all JSON value parts in a transformation when the following conditions are fulfilled:

- The JSON structure is not broken. For more information see, Introducing JSON .
- The structure of the mapping transformation is not broken. For more information, see Transformations [page 388].

During the transformation evaluation, each occurrence of %<...>% is interpreted as a parameter reference and is replaced by the corresponding parameter's value as text literal. If the parameter is missing from the system configuration, that literal is *null*.

The example below shows how the mapping transformations work in case of existing parameter configuration:

System Properties	Mapping Transformation	Evaluated Transformation

`ldap.attribute.user.mail=mail`
`ldap.attribute.user.givenName=givenName`
`ldap.attribute.user.groups=memberOf`
	`{`
	`"sourcePath": "$.%ldap.attribute.user.mail%[0]",`
	`"targetPath": "$.emails[0].value",`
	`"optional": true`
	`},`
	
	`{`
	`"sourcePath": "$.%ldap.attribute.user.givenName%[0]",`
	`"targetPath": "$.name.givenName",`
	`"optional": true`
	`},`
		`{`
		`"sourcePath": "$.mail[0]",`
		`"targetPath": "$.emails[0].value",`
		`"optional": true`
		`},`
		`{`
		`"sourcePath": "$.givenName[0]",`
		`"targetPath": "$.name.givenName",`
		`"optional": true`
		`}`

You can add also optional parameters to the system properties configuration. These parameters do not appear by default at system creation and can be added on demand.

The tables below show how a system transformation including an optional parameter is evaluated:

Evaluation of Mapping Transformation with an Optional Parameter

System Properties	Mapping Transformation	Evaluated Transformation

`sac.group.prefix=SAC_GROUP`	`{`
	`"sourcePath": "$.displayName",`
	`"targetPath": "$.displayName",`
	`"functions": [`
	`{`
	`"condition": "'%sac.group.prefix%' != 'null'",`
	`"function": "concatString",`
	`"prefix": "%sac.group.prefix%"`
	`}`
	`]`
	`}`
		`{`
		`"sourcePath": "$.displayName",`
		`"targetPath": "$.displayName",`
		`"functions": [`
		`{`
		`"condition": "'SAC_GROUP' !== 'null'",`
		`// the condition is true`
		`"function": "concatString",`
		`"prefix": "%sac.group.prefix%"`
		`}`
		`]`
		`}`

System Properties	Mapping Transformation	Evaluated Transformation

No optional parameter configured	<pre>{
	  "sourcePath":
	"$.displayName",
	  "targetPath":
	"$.displayName",
	  "functions": [
	  {
	  "condition":
	"'%sac.group.prefix%' !
	== 'null'",
	  "function":
	"concatString",
	  "prefix":
	"%sac.group.prefix%"
	  }
	  ]
	}</pre>
		<pre>{
		  "sourcePath":
		"$.displayName",
		  "targetPath":
		"$.displayName",
		  "functions": [
		  {
		  "condition":
		"'null' !== 'null'",
		  // the condition
		is false
		  "function":
		"concatString",
		  "prefix":
		"%sac.group.prefix%"
		  }
		  ]
		}</pre>

> **Note**
>
> A possibility for incorrect interpretation of a %<...>% character sequences occurs if a text literal already contains more than one % characters itself.

The table below includes an example of incorrect interpretation in case of a missing parameter configuration:

System Properties	Mapping Transformation	Evaluated Transformation

No parameters configured	<pre>"condition": "$.email =~ /
	^([a-zA-Z0-9._%-]+@[a-zA-Z0-9.-]+
	\\.[a-zA-Z]{2,})$/ || $.ExternalEmail[0]
	=~ /^([a-zA-Z0-9._%-]+
	@[a-zA-Z0-9.-]+\\.[a-zA-Z]
	{2,})$/"</pre>
		<pre>"condition": "$.email =~ /
		^([a-zA-Z0-9null-]+
		@[a-zA-Z0-9.-]+\\.[a-zA-Z]
		{2,})$/"</pre>

Such cases can be avoided by extracting the parts containing % into separate parameters in the system configuration, as displayed in the table below:

System Properties	Mapping Transformation

email-regex = **/^([a-zA-Z0-9._%-]+@[a-zA-Z0-9.-]+\\.[a-zA-Z]{2,})$/**	<pre>"condition": "$.email =~ %email-regex% ||
	$.ExternalEmail[0] =~ %email-regex%"</pre>

Parameter values may contain character sequences matching a parameter reference, but such sequences are not interpreted as parameters themselves. For more information, see the table below:

System Properties	Mapping Transformation	Evaluated Transformation
parameter-A=**value-A** *parameter-B*=**value-B** **%parameter-A%**	`"constant": "%parameter-A%", .. "constant": "%parameter-B%", ..`	`"constant": "value-A", .. "constant": " value-B %parameter-A%", ..`

> ### Note
>
> Avoid using special regex characters such as `()`, `[]`, `{}`, `.`, `+`, `*`, `?`, `\`, `^`, and `|` within property values. They are evaluated in transformations - specifically in conditions involving regex strings and functions with regex parameters - and may lead to errors. For example, the following configuration will result in an error due to unfulfilled conditions:
>
> - *Property*: xsuaa.group.prefix=[XSUAA]
> - *Condition*: ('%xsuaa.group.prefix%' === 'null') || ($.displayName =~ /%xsuaa.group.prefix%.*/)

Internal Properties

> ### Note
>
> Identity Provisioning uses internal properties in various cases. Internal properties must not be used by customers.

For example:

Property Name	Value
`destinationName`	The name of the destination created in SAP BTP cockpit.

Related Information

List of Properties [page 80]
Manage Properties [page 2438]

1.3.5.2 List of Properties

On this page you can find all the available properties to use in the Identity Provisioning service. You can filter them by system type name, "All Systems", by a word or only part of it.

Name	Description	System Type
`Type`	Protocol type for making a connection **Possible values:** • *HTTP* • *LDAP* • *RFC* **System Role:** Source, Target, Proxy	All systems
URL	URL needed to make an HTTP(S) connection to an on-premise system or a cloud service **Possible value:** • When the proxy type of the HTTP connection is `Internet`: **`http(s)://<host>:<port>`** • When the proxy type of the HTTP connection is `OnPremise`: **`http://<virtualhost>:<virtualport>`** **System Role:** Source, Target, Proxy	All HTTP systems
`ProxyType`	Proxy type required for HTTP connection **Possible values:** • *Internet* • *OnPremise* **System Role:** Source, Target, Proxy	All HTTP systems

Name	Description	System Type
`Authentication`	Authentication type required for HTTP connection **Possible values:** • *NoAuthentication* • *BasicAuthentication* • *ClientCertificateAuthentication* **System Role:** Source, Target, Proxy	All HTTP systems **Note** Identity Provisioning supports certificate-based authentication for secure communication with the provisioning systems (connectors) provided by the service. Refer to the documentation of the respective systems to find out how to upload Identity Provisioning certificates on their end. For example, see How to Create Communication Users in SAP BTP ABAP Environment.
`User`	It represents: • User name – used in standard destinations • Client ID – used for access token retrieval in OAuth HTTP destinations **Possible values:** Text/numeric string **System Role:** Source, Target, Proxy	All HTTP systems
`client_id`	The OAuth Client id used for access token retrieval in case of OAuth certificate authentication. Possible values: Text/numeric string **System Role:** Source, Target, Proxy	All HTTP systems
`Password`	(Credential) It represents: • Password – used in standard destinations • Client secret key – used for access token retrieval in OAuth HTTP destinations **Possible values:** Encrypted string **System Role:** Source, Target, Proxy	All HTTP systems

Name	Description	System Type
`ips.proxy.scim.total.results.strategy`	This property determines how `totalResults` is reported in the SCIM response of the concrete IPS Proxy system. **Possible values:** • *Provided* - The `totalResults` value returned by the backend system is used as-is in the proxy response. • *Calculated* - The `totalResults` value is calculated by the proxy system based on the entities that passed the transformation condition. Default value: *Calculated* **System Role:** Proxy	All systems
`fg.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users in the target system. This means, for example, that if a user attribute is modified, only these changes will be provisioned and applied in the target system. • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Fieldglass

Name	Description	System Type
`abap.user.filter`	Filters users by a regular expression on their user-name. The regex can define any kind of search pattern. **⚠ Caution** This property is rather obsolete. For newly created systems, please use `abap.user.name.filter`. **Possible values:** For example: `abap.user.filter` = *^A.** This filter returns all user names that start with capital *A*. **System Role:** Source, Proxy	SAP Application Server ABAP
`abap.role.filter`	**⚠ Caution** This property is rather obsolete. For newly created systems, please use `abap.role.name.filter`. **Possible values:** For example: `abap.role.filter` = *^order.** This filter provisions all roles that start with *order*. **System Role:** Source, Proxy	SAP Application Server ABAP

Name	Description	System Type
`abap.user.name.filter`	Filters users by a regular expression on their username. The regex can define any kind of search pattern. This property has a higher priority over `abap.user.filter`. That means, if you set both properties in a system, the value of `abap.user.name.filter` will be used. However, if the value of `abap.user.name.filter` is empty, then `abap.user.filter`'s value will be used instead. **Note** As `abap.user.filter` is obsolete, we recommend that you use `abap.user.name.filter`. **Possible values:** For example: `abap.user.name.filter` = *^MAR.** This regex reads all user names that start with *MAR*, such as *MARK*, *MARTINA*, and so on. **System Role:** Source, Proxy	SAP Application Server ABAP

Name	Description	System Type
`abap.role.name.filter`	Filters roles by a regular expression. The regex can define any kind of search pattern. This property has a higher priority over `abap.role.filter`. That means, if you set both properties in a system, the value of `abap.role.name.filter` will be used. However, if the value of `abap.role.name.filter` is empty, then `abap.role.filter`'s value will be used instead. **Note** As `abap.role.filter` is obsolete, we recommend that you use `abap.role.name.filter`. **Possible values:** For example: `abap.role.name.filter` = *^inter.** This regex reads all roles that start with *inter*, such as *internal*, *internship*, and so on. For more information about specific search patterns, refer to SAP Note 3546855. **System Role:** Source, Proxy	SAP Application Server ABAP

Name	Description	System Type
`abap.user.page.size`	Use this property to configure the number of users to be read from SAP Application Server ABAP at once. If the property is not set, the function module `SUSR_USER_GET_DETAIL_MASS` **cannot** be used, even if it is available on the ABAP server. **Note** The property can be used only when the function module `SUSR_USER_GET_DETAIL_MASS` is exposed as an accessible resource and is present on the ABAP server. Even when these conditions are fulfilled, if the property is not set, the function module **cannot** be used. For more information, see SAP Note 3631587. **Possible values**: Default value: *100* The maximum allowed number is 100. System Role: Source, Proxy	SAP Application Server ABAP

Name	Description	System Type
`abap.role.prefix`	This property distinguishes SAP Application Server ABAP (AS ABAP) roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **ABAP_** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the roles that are read from the AS ABAP source system and will be provisioned to the target system with the following name pattern: **`ABAP_<role_name>`** . This way AS ABAP roles in the target system will be easily distinguished from roles provisioned from other applications. If the property is not set, the AS ABAP roles will be read and provisioned to the target system with their actual role names. • When **set in the target system**, only roles containing the **ABAP_** prefix in their role name will be provisioned to AS ABAP. Roles without this prefix in their names won't be provisioned. If the property is not set, all roles will be provisioned to AS ABAP. **System Role:** Source and Target	SAP Application Server ABAP
`abap.user.membership.filter`	Filters users by a regular expression, based on their *Role* memberships in AS ABAP. The regex can define any kind of search pattern. **Possible values:** For example: `abap.user.membership.filter` = *(?i)^new.** This reads all users who have an assigned role which starts with *new*. This regex is case insensitive, which means the result can be roles starting with *new*, or *New*, or *NEW*, and so on. **System Role:** Source, Proxy	SAP Application Server ABAP

Name	Description	System Type
`ariba.applications.api.key`	(Credential) This property corresponds to the *Application key* for your SAP Ariba application. You obtain it during the creation of your application in the SAP Ariba developer portal. See: How to find your application's application key and OAuth client ID **Possible values:** Text/numeric string For example: This property corresponds to the SAP Ariba realm that your application has access to. To learn how to get it, see: *123abc123XYZ000abc123ABC012345* **System Role:** Source, Target, Proxy	SAP Ariba Applications
`ariba.applications.realm.id`	This property corresponds to the SAP Ariba realm that yourHow to find your SAP Ariba realm name? **Possible values:** Text/numeric string For example: *123abc123XYZ000abc123ABC012345* **System Role:** Source, Target, Proxy	SAP Ariba Applications

Name	Description	System Type
`ariba.applications.group.flatten`	This property allows or forbids reading "nested groups" (group structures) from SAP Ariba Applications. If enabled (*true*), group members of type *group* will be ignored during read in order not to be provisioned to target systems that do not support nested groups. Default value: • *Source systems:true* • *Proxy systems: false* Predefined value (during system creation): • *Source systems: true* Set it to *false* only if you are sure that the target system supports nested groups. • *Proxy systems: false* Leave the default/predefined value only if you are sure that the consuming external application (identity management system) supports nested groups. **Note** If you set this property to *true* in your proxy system, the execution of requests with parameter `membersType=group` will not be supported. **System Role:** Source, Proxy	SAP Ariba Applications

Name	Description	System Type
`ariba.applications.group.children.conversion`	This property allows provisioning of "nested groups" (group structures) of type authorization from SAP Ariba Applications source systems to the Identity Directory of SAP Cloud Identity Services or Identity Authentication (SCIM API version 2) target systems. When enabled (set to true), the members of a given group from SAP Ariba Applications source system will be provisioned to the target system with reversed parent-child relationship. That is, the child groups in the source system become parent groups in the target system, while the original parent group becomes a child group in the target. For more information, see the description of the `children` attribute in Groups. The property is not automatically added during system creation. **Possible values:** • *true* • *false* Default value: *false* **System Role**: Source	SAP Ariba Applications

Name	Description	System Type
`ariba.applications.group.prefix`	This property distinguishes SAP Ariba Applications groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`ARIBA_APPLICATIONS_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Ariba Applications source system and will be provisioned to the target system with the following name pattern: **`ARIBA_APPLICATIONS_<GroupDisplayName>`**. This way SAP Ariba Applications groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Ariba Applications groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`ARIBA_APPLICATIONS_`** prefix in their display name will be provisioned to SAP Ariba Applications. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Ariba Applications. **System Role:** Source and Target	SAP Ariba Applications

Name	Description	System Type
`ariba.applications.support.patch.operation`	The default value of this property is *false*. But for SAP Ariba Applications proxy systems, this property appears during creation and its predefined value is *true*. That means, when the Identity Provisioning identifies a changed entity in the back-end system, it will execute the updates as PATCH requests instead of PUT. That is, only changes will be written in SAP Ariba Applications, instead of provisioning the whole entity data. Note that only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". **Note** Removing empty groups returns 400 Bad Request instead of 200 OK. Default value: *false* Predefined value (during system creation): *true* **System Role:** Target, Proxy	SAP Ariba Applications
`ariba.applications.user.filter`	When specified, only those SAP Ariba Applications users matching the filter expression will be read. **Possible values:** For example: *userName eq "SmithJ"* **System Role:** Source, Proxy	SAP Ariba Applications
`ariba.applications.group.filter`	When specified, only those SAP Ariba Applications groups matching the filter expression will be read. **Possible values:** For example: *displayName eq "ProjectTeam1"* **System Role:** Source, Proxy	SAP Ariba Applications

Name	Description	System Type
`ariba.applications.content.type`	This property makes a SAP Ariba Applications connector to send a specified value for the *Content-Type* HTTP header. This is needed because SAP Ariba Applications could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	SAP Ariba Applications

Name	Description	System Type
`ariba.applications.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in SAP Ariba Applications. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value* • *userName,emails[0].value* Default value: *userName* **System Role:** Target	SAP Ariba Applications

Name	Description	System Type
`ariba.applications.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists on the SAP Ariba Applications target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for this property. **Possible values:** Default value (when not specified): *displayName* If the property is not specified, the search is done by the default attribute: *displayName* **System Role:** Target	SAP Ariba Applications
`ariba.applications.include.if.match.wildcard.header`	Makes the SAP Ariba Applications connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Ariba Applications system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Ariba Applications

Name	Description	System Type
`ariba.applications.group.members.page.size`	The number of group members that SAP Ariba Applications returns per request when reading a group. When the Identity Provisioning reads a group, it compares the number of group members with the value configured for `ariba.applications.group.members.page.size`. Depending on the number of group members, you can expect the following results: • If the number of group members is equal or exceeds the configured value, and the `ariba.applications.group.members.paging.enabled` property is set to *true*, Identity Provisioning will read all group members in separate requests. For example, if a group has 6 group members and the value of the page size is set to *2*, Identity Provisioning will send 3 requests to read the group members. • If the number of group members is less than the configured value for `ariba.applications.group.members.page.size` or the property `ariba.applications.group.members.paging.enabled` is disabled, the Identity Provisioning will read all group members via one request. For example, if a group has 2 group members and the value of the page size is set to *6*, Identity Provisioning will read all group members via one request. Setting the property `ariba.applications.group.members.paging.enabled` allows you to read groups with a large number of group members. For more information, see `ariba.applications.group.members.paging.enabled`. Default value: *50* **System Role:** Source, Proxy	SAP Ariba Applications

Name	Description	System Type
`ariba.applications.group.members.paging.enabled`	This property enables paging of group members. When it is set to *true*, all groups with members count exceeding the value defined in `ariba.applications.group.members.page.size` will be read by the Identity Provisioning via subsequent requests. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *true* **System Role:** Source, Proxy	SAP Ariba Applications
`s4hana.cloud.group.members.paging.enabled`	This property enables paging of group members. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *false* **System Role:** Source, Proxy	SAP S/4HANA Cloud Public Edition (using version 3 - System for Cross-domain Identity Management (SCIM) API)
`ibp.group.members.paging.enabled`	This property enables paging of group members. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *false* **System Role:** Source, Proxy	SAP Integrated Business Planning (using version 2 - SCIM Interface for IAM)
`ibp.user.groups.paging.enabled`	This property enables paging of user groups. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *false* **System Role:** Source, Proxy	SAP Integrated Business Planning (using version 2 - SCIM Interface for IAM)

Name	Description	System Type
`s4hana.cloud.user.groups.paging.enabled`	This property enables paging of user groups. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *false* **System Role:** Source, Proxy	SAP S/4HANA Cloud Public Edition (using version 3 - System for Cross-domain Identity Management (SCIM) API)

Name	Description	System Type
`ariba.applications.user.groups.page.size`	The number of user groups that SAP Ariba Application returns per request when reading a user. When the Identity Provisioning reads a user, it compares the number groups assigned with the value configured for `ariba.applications.user.groups.page.size`. Depending on the number of groups assigned to the user, you can expect the following results: • If the number of assigned groups is equal or exceeds the configured value, and the `ariba.applications.user.groups.paging.enabled` property is set to *true*, Identity Provisioning will read all user groups in separate requests. For example, if a user has 6 groups assigned and the value of the page size is set to *2*, Identity Provisioning will send 3 requests to read the user groups. • If the number of assigned groups is less than the configured value for `ariba.applications.user.groups.page.size` or the property `ariba.applications.user.groups.paging.enabled` is disabled, the Identity Provisioning will read all user groups via one request. For example, if a user has 2 groups assigned and the value of the page size is set to *6*, Identity Provisioning will read all user groups via one request. Setting the property `ariba.applications.user.groups.paging.enabled` allows you to read users with a large number of groups assigned. For more information, see `ariba.applications.user.groups.paging.enabled`. Default value: *50* **System Role:** Source, Proxy	SAP Ariba Applications

Name	Description	System Type
`ariba.applications.user.groups.paging.enabled`	This property enables paging of user's groups. This maximum value is specified by the property `ariba.applications.group.members.page.size`. To read a user with *'groups'* attribute over the configured value, set `ariba.applications.user.groups.paging.enabled` to *true*. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *true* **System Role:** Source, Proxy	SAP Ariba Applications
`c4c.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Sales Cloud and SAP Service Cloud

Name	Description	System Type
`c4c.api.version`	This property defines the API version that the API of your SAP Sales Cloud and SAP Service Cloud system uses. **Possible values:** • *1* • *2* • *3* • *4* By default, Identity Provisioning uses version *4*, which means - SCIM 2.0 based API. **System Role:** Source, Target, Proxy	SAP Sales Cloud and SAP Service Cloud
`btp.cf.pm.origin`	The value of this property is the origin key of your SAP Cloud Identity Services tenant. You can find it in the SAP BTP cockpit. Go to your SAP BTP subaccount, choose *Trust Configuration* and see the value under *Origin Key* for the relevant *Custom Identity Provider for Platform Users*. The value of this property is a string, which always ends with the suffix *-platform*. It will be used as the *origin* attribute in the system transformation. **Possible values:** Text/numeric string **System Role:** Source, Target, Proxy	SAP BTP Platform Members (Cloud Foundry)

Name	Description	System Type
`btp.cf.pm.landscape`	Enter the *technical key* or the *host name* of the landscape in which your multi-environment subaccount with enabled Cloud Foundry environment is located. The techical key is available only for the SAP BTP Cloud Foundry regions. For more information, see Regions and API Endpoints Available for the Cloud Foundry Environment. Please note that when the technical key contains an extension index, the correct value is displayed in the *API Endpoint* column instead of the *Technical Key* column of the table. This is an example of a *technical key*: **`cf-eu10-002`**, corresponding to the API endpoint `api.cf.eu10-002.hana.ondemand.com` and host name `cf.eu10-002.hana.ondemand.com` The host name of the landscape can be found by looking up the *API Endpoint* field in the *Overview* tab of your *Cloud Foundry Environment* on subaccount level of the SAP BTP cockpit. The API endpoint has the following format: `https://api.<landscape-hostname>` **Possible values:** Text/numeric string **System Role:** Source, Target, Proxy	SAP BTP Platform Members (Cloud Foundry)

Name	Description	System Type
`btp.cf.pm.group.prefix`	This property distinguishes SAP BTP Platform Members (Cloud Foundry) groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`BTP_CF_PM_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP BTP Platform Members (Cloud Foundry) source system and will be provisioned to the target system with the following name pattern: **`BTP_CF_PM_<GroupDisplayName>`**. This way SAP BTP Platform Members (Cloud Foundry) groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP BTP Platform Members (Cloud Foundry) groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`BTP_CF_PM_`** prefix in their display name will be provisioned to SAP BTP Platform Members (Cloud Foundry). Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP BTP Platform Members (Cloud Foundry). **System Role:** Source and Target	SAP BTP Platform Members (Cloud Foundry)
`btp.cf.pm.user.filter`	When specified, only those SAP BTP Platform Members (Cloud Foundry) users matching the filter expression will be read. **Possible values:** For example: *userName eq "SmithJ"* **System Role:**Source, Proxy	SAP BTP Platform Members (Cloud Foundry)

Name	Description	System Type
`c4c.custom.namespace.<prefix>`	**Note** Only relevant to API **v.2**. The Identity Provisioning service uses a single predefined namespace for all attributes. However, you can provision entities by defining your own (custom) namespaces for some attributes. For this purpose, you have to: 1. Specify a namespace using this property. 2. Set the custom namespace in the JSON transformation. For more information, see: SAP Sales Cloud and SAP Service Cloud [page 1211] The value of this property is the namespace URI. For **<prefix>**, enter the prefix of the custom XML namespace (for example, *a123*). Example for setting up the whole property: *c4c.custom.namespace.a123=http://sap.com/xi/AP/CustomerExtension/ABC/A123XX* **System Role:** Target	SAP Sales Cloud and SAP Service Cloud
`cbc.user.filter`	When specified, only those SAP Central Business Configuration users matching the filter expression will be read. **Note** For source systems only: Using this property makes sense only if you have set the *"ignore": true* statement to *false*. **Possible values:** For example: *name.familyName eq "Smith" and addresses.country eq "US"* **System Role:** Source, Proxy	SAP Central Business Configuration

Name	Description	System Type
`cbc.group.filter`	When specified, only those SAP Central Business Configuration groups matching the filter expression will be read. **Possible values:** For example: *displayName eq "ProjectTeam1" or "Employees2020"* **System Role:** Source, Proxy	SAP Central Business Configuration
`cbc.group.prefix`	This property distinguishes SAP Central Business Configuration groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`CBC_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Central Business Configuration source system and will be provisioned to the target system with the following name pattern: **`CBC_<GroupDisplayName>`**. This way SAP Central Business Configuration groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Central Business Configuration groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`CBC_`** prefix in their display name will be provisioned to SAP Central Business Configuration. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Central Business Configuration. **System Role:** Source and Target	SAP Central Business Configuration

Name	Description	System Type
`dc.group.filter`	This property filters groups by display name or externalId. When specified, only those groups matching the filter expression will be read. **Possible values:** • *SAP_Data_Custodian_Auditor* • *SAP_Data_Custodian_Service_Admin* • *SAP_Data_Custodian_Key_Admin* • *SAP_Data_Custodian_Key_User* For example: *displayName eq "SAP_Data_Custodian_Auditor"* **System Role:** Source, Proxy	SAP Data Custodian
`dc.group.unique.attribute`	If the service tries to create a group that already exists in the target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for the `dc.group.unique.attribute` property. If the property is not specified, the search is done by the default attribute: *displayName* **System Role:** Target	SAP Data Custodian
`dc.user.filter`	When specified, only those SAP Data Custodian users matching the filter expression will be read. You can filter users by *userName*, *displayName* or *externalId*. **Possible values:** For example: *userName eq "Smith.J"* **System Role:** Source, Proxy	SAP Data Custodian

Name	Description	System Type
`dc.user.unique.attribute`	When Identity Provisioning attempts to provision a user for the first time, it may detect that this user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user will be searched and resolved. If the service finds a user on the target system via this filter, then the conflicting user will overwrite the existing one. If the service does not find a user on the target system via this filter, the creation will fail. **Default behavior**: This property is missing during system creation. Its default value is *userName*. This means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such *userName* is not found, the creation of the conflicting user fails. Default value: *userName* **System Role:** Target	SAP Data Custodian
`dc.content.type`	Makes the connector send a specified value for the *Content-Type* HTTP header. This is needed because a SCIM system could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* If the property is not specified, the default value is taken: *application/scim+json* **System Role:** Target, Proxy	SAP Data Custodian

Name	Description	System Type
`dc.include.if.match.wildcard.header`	Makes the SAP Data Custodian connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Data Custodian system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Data Custodian

Name	Description	System Type
`dc.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Data Custodian

Name	Description	System Type
`dc.group.prefix`	This property distinguishes SAP Data Custodian groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`DC_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Data Custodian source system and will be provisioned to the target system with the following name pattern: **`DC_<GroupDisplayName>`**. This way SAP Data Custodian groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Data Custodian groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`DC_`** prefix in their display name will be provisioned to SAP Data Custodian. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Data Custodian. **System Role:** Source, Target	SAP Data Custodian
`ds.user.filter`	When specified, only those SAP Datasphere users matching the filter expression will be read. **Possible values:** For example: *userName eq "SmithJ"* **System Role:**Source, Proxy	SAP Datasphere

Name	Description	System Type
`ds.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in SAP Datasphere. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value* • *userName,emails[0].value* Default value: *userName* **System Role:** Target	SAP Datasphere

Name	Description	System Type
`ds.include.if.match.wildcard.header`	This property makes the SAP Datasphere connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. The header could be used by an SAP Datasphere system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Datasphere

Name	Description	System Type
`ds.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a **PATCH** request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. • If set to *false*, **PUT** operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Datasphere

Name	Description	System Type
`ds.bulk.operations.max.count`	This property sets the number of operations to be performed in one bulk request. Default value: *30* **Note** The value must not exceed the number of entities defined by the SAP Datasphere system as a SCIM service provider. Otherwise, the provisioning job will fail with HTTP response code 413 (*Payload Too Large*). **System Role:** Target	SAP Datasphere
`ds.api.csrf.protection`	Specifies whether to fetch a CSRF token when sending requests to the system. The property is automatically added in the system, with default value: *enabled*. **Possible values:** • *enabled* • *disabled* Default value: *enabled* **Note** If the property is missing, its default value is **disabled**. **System Role:** Target, Proxy	SAP Datasphere

Name	Description	System Type
`hcp.application.names`	Enter a comma-separated list of application names. That could be applications deployed on your account, or applications for which your account has subscribed. The property returns the roles assigned to these applications. **Possible values:** Use the following format (no spaces): *<app_name1>,<app_name2>,<provider_subaccount>:<provider_app>* For example: *myapp1,myapp2,provider1:app123,provider2:cloud789,mynewapp* **⚠ Caution** You must not leave this property with an empty value. **System Role:** Source	SAP BTP Java/HTML5 apps (Neo)
`hcp.read.group.roles`	If you set this property to *true*, the Identity Provisioning will read the following additional attributes for a SAP Business Technology Platform group: - Application roles - Group mappings, defined by your identity provider **⚠ Restriction** This property is applicable only if SAP Business Technology Platform and the external SCIM-based system belong to one and the same region. **Possible values:** - *true* - *false* Default value: *false* **System Role:** Proxy	SAP BTP Java/HTML5 apps (Neo)

Name	Description	System Type
`hcp.group.prefix`	This property distinguishes groups from Java/HTML5 applications running on SAP BTP, Neo environment by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`HCP_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Business Technology Platform source system and will be provisioned to the target system with the following name pattern: **`HCP_<GroupDisplayName>`**. This way groups from Java/HTML5 applications running on SAP BTP, Neo environment in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the groups from Java/HTML5 applications running on SAP BTP, Neo environment will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`HCP_`** prefix in their display name will be provisioned to SAP Business Technology Platform. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Business Technology Platform. **System Role:** Source and Target	SAP BTP Java/HTML5 apps (Neo)

Name	Description	System Type
`hcp.patch.response.with.resource`	Use this property when you execute hybrid scenarios with SAP Business Technology Platform (Neo) as a SCIM proxy system, and you update an entity (mostly relevant to groups, like when you change the members of a group) via a PATCH request. If you set this property to *true*, the successful **PATCH** request will return a response code 200 (*OK*) back to the consumer client application with a payload body containing the updated attributes of the relevant group. If you don't specify the property (or it's set to *false*), the successful **PATCH** request will return a response code 204 (*No Content*) indicating successful group update but with no payload body. For more information, see: SCIM 2.0: Modifying with PATCH . **Possible values:** • *true* • *false* Default value: *false* **System Role:** Proxy	SAP BTP Java/HTML5 apps (Neo)
`ia.group.filter`	When specified, only those SAP Intelligent Agriculture groups matching the filter expression will be read. **Possible values:** • *intagri_Data_Scientist* • *intagri_Farm_Manager* • *intagri_Operator* • *intagri_Operations_Planner* • *intagri_Master_Data_Manager* • *intagri_Administrator* For example: *displayName eq "intagri_Master_Data_Manager"* **System Role:** Source, Proxy	SAP Intelligent Agriculture

Name	Description	System Type
`ia.user.filter`	When specified, only those SAP Intelligent Agriculture users matching the filter expression will be read. Supported operators: `eq` (equal), `ne` (not equal), `sw` (starts with), `ew` (ends with) and `co` (contains) **Possible values:** For example: *userName eq "Smith.J"* *emails eq "julie.armstrong@example.com"* **System Role:** Source, Proxy	SAP Intelligent Agriculture
`ia.group.unique.attribute`	If the service tries to create a group that already exists in the target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). This property defines by which unique attribute(s) the existing group will be searched and resolved. The default value is *displayName*. Currently, it is the only unique attribute that is supported. When set, you can expect the following behavior: • If a group with the given *displayName* is found in the target system, the group that you try to provision will overwrite the existing one. • If a group with the given *displayName* is not found in the target system, the group that you try to provision will not be created in the target system. If the property is not specified, the search is done by the default attribute: `displayName` **System Role:** Target	SAP Intelligent Agriculture

Name	Description	System Type
`ia.user.unique.attribute`	When Identity Provisioning attempts to provision a user for the first time, it may detect that this user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user will be searched and resolved. If the service finds a user on the target system via this filter, then the conflicting user will overwrite the existing one. If the service does not find a user on the target system via this filter, the creation will fail. **Default behavior**: This property is missing during system creation. Its default value is *userName* . This means, if the service finds an existing user by a *userName* , it updates this user with the data of the conflicting one. If a user with such *userName* is not found, the creation of the conflicting user fails. **Possible values:** • *userUuid* • *emails* • *userName* • *externalId* Default value: *userName* **System Role:** Target	SAP Intelligent Agriculture

Name	Description	System Type
`ia.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. This means, for example, that if a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Intelligent Agriculture

Name	Description	System Type
`ia.group.prefix`	This property distinguishes SAP Intelligent Agriculture groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`IA_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Intelligent Agriculture source system and will be provisioned to the target system with the following name pattern: **`IA_<GroupDisplayName>`**. This way SAP Intelligent Agriculture groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Intelligent Agriculture groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`IA_`** prefix in their display name will be provisioned to SAP Intelligent Agriculture. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Intelligent Agriculture. **System Role:** Source, Target	SAP Intelligent Agriculture
`ia.include.if.match.wildcard.header`	Makes the connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Intelligent Agriculture system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Intelligent Agriculture

Name	Description	System Type
`idds.group.filter`	This property filters groups by display name. You can set a single display name or multiple ones as filter criteria. If you enter multiple display names (using **OR** operator), the filter will search for any of them. Value pattern (single): *displayName eq "<group_name>"* Value pattern (multiple): *displayName eq "<group_name1>" or displayName eq "<group_name2>"* **Possible values:** For example: • Single: *displayName eq "FellowshipTeam1"* • Multiple: *displayName eq "FellowshipTeam1" or displayName eq "JuniorTest3"* **System Role:** Source, Proxy	Local Identity Directory

Name	Description	System Type
`idds.user.filter`	This property filters users by particular attributes. You can set a single attribute or multiple ones as search criteria. Value pattern (single): *<user_attribute> eq "<value>"* Value pattern (multiple): *<user_attribute1> eq "<value1>" and/or <user_attribute2> eq "<value2>"* **Possible values:** For example: • Single: *userName eq "Sebastian"* • Multiple (with **OR**): *userName eq "Sebastian" or addresses.country eq "France"* • Multiple (with **AND**): *userName eq "Sebastian" and addresses.country eq "France"* • Multiple (with brackets): *userName eq "Sebastian" or (addresses.country eq "France" and emails.value eq "sebastian123@mail.com")* • Multiple (enterprise attributes): *urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:department eq "Dev" and urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:organization eq "Technology"* **System Role:** Source, Proxy	Local Identity Directory
`idds.group.members.paging.enabled`	This property enables paging of group members. The maximum number of group members returned per request is 20 000. To read more than 20 000 group members, paging must be enabled. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *false* **System Role:** Source, Proxy	Local Identity Directory

Name	Description	System Type
`idds.user.groups.paging.enabled`	This property enables paging of user's groups. The maximum number of user's groups returned per request is 1 000. To read more than 1 000 user's groups, paging must be enabled. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *false* **System Role:** Source, Proxy	Local Identity Directory
`idds.content.type`	Makes the connector send a specified value for the *Content-Type* HTTP header. This is needed because a SCIM system could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* If the property is not specified, the default value is taken: *application/scim+json* **System Role:** Target, Proxy	Local Identity Directory

Name	Description	System Type
`idds.group.unique.attribute`	If you try to provision a group that already exists in a target system, the group creation will fail. In such cases, the existing group only needs to be updated. This property defines the unique attribute(s) by which the existing group will be searched and resolved. The default value is *['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']*. When set, you can expect the following behavior: • If a group with the given *['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']* is found in the target system, the group that you try to provision will overwrite the existing one. • If a group with the given *['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']* is not found in the target system, the group that you try to provision will not be created in the target system. In scenarios where groups are provisioned from SAP Integrated Business Planning version 2 and SAP S/4HANA Cloud Public Edition version 3, the uniqueness of the group is resolved by a combination of three attributes: *['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type'],* *['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName'],* *['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']* For the conflict to be resolved, an existing group must match all three attributes. If the property is not specified, the search is done by the default attribute: *['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']* **System Role:** Target	Local Identity Directory

Name	Description	System Type
`idds.include.if.match.wildcard.header`	Makes the connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by a SCIM system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	Local Identity Directory

Name	Description	System Type
`idds.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with "scope": "createEntity", such as: `{ "constant":true, "targetPath":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mailVerified']", "scope":"createEntity" }` • If set to *false*, Identity Provisioning sends a PUT request to the user or group resource in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. Users and groups can be updated in the target system in various cases, such as: • In the source system, some user or group attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users or groups not to be read anymore. • A user or a group is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**.	Local Identity Directory

Name	Description	System Type
	Possible values: • *true* • *false* Default value: *false* **System Role:** Target, Proxy	

Name	Description	System Type

`idds.user.automatic.conflict.resolution`	Controls whether automatic conflict resolution is switched on or off in Local Identity Directory (target system) when provisioning is triggered from source systems containing different users with the same user identifiers (IDs).
	
	For example, when SAP SuccessFactors and SAP SuccessFactors Learning are configured as source systems for provisioning users to Local Identity Directory, it could happen that different SAP SuccessFactors and SAP SuccessFactors Learning users have the same user IDs. In this case, when the first user is created in Local Identity Directory, after triggering a provisioning job, the second (conflicting) user will either overwrite the already existing one (automatic conflict resolution is switched on) or will fail and won't be created (automatic conflict resolution is switched off).
	
	To control this behavior, you can use the `idds.user.automatic.conflict.resolution` property in the target Local Identity Directory system. This property is not added by default.
	
	Possible values:
	
	• *true* - If the property is set to true, or is not set at all, the automatic conflict resolution is switched on. This means that Identity Provisioning takes into account the unique attribute(s) defined on the `idds.user.unique.attribute` property and tries to find an already existing user in Local Identity Directory matching these attributes.
	  ◦ If a user is found, the provisioning of a new (conflicting) user is resolved as follows: the conflicting user overwrites the existing one.
	  ◦ If a user is not found, the provisioning of a conflicting user fails, and it is not created in Local Identity Directory.
	• *false* - If the property is set to false, the automatic conflict resolution is switched off. This means that Identity
		Local Identity Directory

Name	Description	System Type
	Provisioning does not take into account the unique attribute(s) defined on the `idds.user.unique.attribute` property and fails the provisioning of a conflicting user. This user is not created in Local Identity Directory and does not overwrite the existing one. In the Job Log, an error code `409, uniqueness` will be displayed. Default value: *true* **System Role:** Target	

Name	Description	System Type
`idds.user.unique.attribute`	When Identity Provisioning attempts to provision a user for the first time, it may detect that this user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user will be searched and resolved. **If the service finds a user on the target system via this filter, then the conflicting user will overwrite the existing one.** If the service does not find a user on the target system via this filter, the creation will fail. According to your use case and system type, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *phoneNumbers[0].value*. If the service finds an existing user with such *phoneNumber*, it updates this user with the data of the conflicting one. If a user with such *phoneNumber* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName, phoneNumbers[0].value*. If the service finds an existing user with both these *userName* and *phoneNumber*, it updates this user with the data of the conflicting one. If such a user is not found, that	Local Identity Directory

Name	Description	System Type
	means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName, emails[0].value, phoneNumbers[0].value*. If the service finds an existing user with these *userName*, *email* and *phoneNumber*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value* • *userName,emails[0].value* • *phoneNumbers[0].value* • *userName, phoneNumbers[0].value* • *userName, emails[0].value, phoneNumbers[0].value* • *externalId*, or another SCIM attribute, or a conjunction of SCIM attributes Default value: *userName* **System Role:** Target	

Name	Description	System Type
`idds.user.update.instead.delete`	`PATCH` request in Local Identity Directory target system and to preserve the user record instead of deleting it. This behavior is supported only when the scope of the attribute is set to **deleteEntity**. In addition to configuring this property, you also need to adapt the write transformation. For example, if you want to disable a user account in Local Identity Directory, you need to do the following: 1. Set `idds.user.update.instead.delete`=*true* 2. Adapt the write transformation as follows: `{ "user":{ "mappings":[{ "constant":"urn:ietf:params:scim:api:messages:2.0:PatchOp", "targetPath":"$.schemas[0]", "scope":"deleteEntity" }, { "constant":"replace", "targetPath":"$.Operations[0].op", "scope":"deleteEntity" }, { "constant":"active", "targetPath":"$.Operations[0].path", "scope":"deleteEntity" }, { "constant":false, "targetPath":"$.Operations[0].value", "scope":"deleteEntity" }, ...`	Local Identity Directory

Name	Description	System Type
	In this case, the PATCH operation will replace *true* with *false* as the value of the `active` user attribute. As a result, when the PATCH operation is executed, the user record in the target system will no longer be managed by Identity Provisioning as it is considered deleted. For more information, see: Transformation Expressions [page 391] → *Scope* → *deleteEntity* **Possible values:** • *true* • *false* Default value: *false* When the property is set to *true*, adapt the write transformation with the attribute name and the attribute value you want to update: `{ "user":{ "mappings":[{ "constant":"urn:ietf:params:scim:api:messages:2.0:PatchOp", "targetPath":"$.schemas[0]", "scope":"deleteEntity" }, { "constant":"replace", "targetPath":"$.Operations[0].op", "scope":"deleteEntity" }, { "constant":"<attribute_name>", "targetPath":"$.Operations[0].path", "scope":"deleteEntity" }, { "constant":<attribute_value>, "targetPath":"$.Operations[0].value",`	

Name	Description	System Type

	`"scope":"deleteEntity"`
	`},`
	`...`
	
	System Role: Target	

`ips.date.variable.format`	This is a default property that the Identity Provisioning UI automatically adds to the configuration of every newly created system. The property allows you to change the default date format to another, more suitable for your scenario. See also: Transformation Variables [page 490]. Default value: *yyyy-MM-dd HH:mm:ss.SSS* **System Role:** Source, Target, Proxy	All systems
`ips.delete.existedbefore.entities`	**Use case:** An entity exists on the target system, and then a provisioning job reads the same entity from a source system and updates it on the target. If later on you delete this entity from the source system, the next provisioning job will recognize it as a "previously existed one" and will **not delete** it from the target. If you want such *recognized* entities to be deleted from the target as well, open the relevant target system and set this property to **true**. For more information, see Manage Deleted Entities [page 2463]. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	All systems

Name	Description	System Type
`ips.overwrite.existedbefore.assignments`	This property defines whether or not the Identity Provisioning service to overwrite user/group assignments that have existed in the target system before you start provisioning entities to that system. **Example:** Let's say there is a group in the target system that contains some assignments (users and subgroups). In the source system there is a matching group, which contains different assignments. • If you start a provisioning job without setting this property (by default, it's *true*), all assignments from the source group will overwrite the ones from the target group. • If you set the property to *false*, all existing assignments will be kept in the target system group, and the new ones will just be added. **Possible values:** • *true* • *false* Default value: *true* **System Role:** Target	SAP Application Server ABAP
`ips.failed.request.retry.attempts`	If an entity operation fails due to an occurred exception (rate limit, bad gateway, missing authorization, or timeout), you can specify a number of retries for this operation. Use this property to set the number of retries. > **Note** > Rate limit is the controlled rate of requests sent to a system. Some systems implement rate limit to avoid overloading and performance issues. **Possible values:** Default value: *2* Maximum value: *3* **System Role:** Source, Target, Proxy	All systems > **Note** > For more information about the cases in which the retry is supported, see Handle Failed Operations [page 2482].

Name	Description	System Type
`ips.failed.request.retry.attempts.interval`	Specify a time interval (in seconds) between the retries, in case an operation fails due to an occurred exception. This property is related to `ips.failed.request.retry.attempts`. **Possible values:** Default value: *30* Maximum value: *60* **Note** Following an HTTP 502 Bad Gateway server error, the time interval for SAP BTP XS Advanced UAA (Cloud Foundry) and SAP Sales Cloud and SAP Service Cloud must not exceed 50 (seconds). **System Role:** Source, Target, Proxy	All systems **Note** For more information about the cases in which the retry is supported, see Handle Failed Operations [page 2482].

Name	Description	System Type
`ips.job.notification.repeat.on.failure`	If you have activated notifications for a source system and a provisioning job fails, you'll receive notification e-mails with subject *Provisioning Finished with Error*. You can also receive an e-mail if you manually stop a running job. With property `ips.job.notification.repeat.on.failure`, you can control the frequency of the received notifications. • If you set the property to *true*, you will receive notification e-mails every time a job fails. • If you want to stop or control the notification frequency, set the property to *false* (default value). This property has a higher priority than `ips.job.notification.ignored.consecutive.failures`. See also: Manage Job Notifications [page 3018]. **Possible values:** • *true* • *false* Default value: *false*. That means, when a job fails, only one notification e-mail will be sent. **System Role:** Source	All systems

Name	Description	System Type
`ips.job.notification.ignored.consecutive.failures`	If you have activated notifications for a source system and a provisioning job fails, you'll receive notification e-mails with subject *Provisioning Finished with Error*. You can also receive an e-mail if you manually stop a running job. With property `ips.job.notification.ignored.consecutive.failures`, you can control the number of the received consecutive notifications. > **Note** > Property `ips.job.notification.repeat.on.failure` must be set to *false* or not specified at all. **Example:** If you set `ips.job.notification.ignored.consecutive.failures` = *3* and the job is constantly failing, the first three times you'll not receive a notification. On the fourth job fail, you will receive one notification e-mail. No subsequent e-mails will be sent by the service until the first successful run of the job. See also: Manage Job Notifications [page 3018]. Default value: *0*. That means, a notification e-mail will be sent after the first job fail. **System Role:** Source	All systems

Name	Description	System Type
`ips.job.notification.skip.intermediate.notifications`	If you have activated notifications for a source system, and an entity fails during the provisioning job, you'll receive one notification e-mail with subject *Provisioning Running with Error*. Property `ips.job.notification.skip.intermediate.notifications` controls whether you will receive a notification or not. • If the property is set to *true*, no notifications will be sent. • If the property is not specified or is set to *false* (default), you'll receive one notification e-mail. No subsequent e-mails will be sent by the service until the first successful run of the job. See also: Manage Job Notifications [page 3018]. **Possible values:** • *true* • *false* Default value: *false*. That means, after the first failed entity, a notification e-mail will be sent. **System Role:** Source	All systems

Name	Description	System Type
`ips.trace.failed.entity.content`	If a provisioning job repeatedly fails and you need problem investigation, you can enable logging and tracing for the personal and sensitive data of your provisioned entities. To do this, set the property to *true* to enable logging and tracing for the personal and sensitive data of your provisioned entities by the next executed job. After the job has finished, the property is automatically set to *false*. If the property is not set, in the logs you see: `content = <hidden content>` To learn more about personal and sensitive data, see: • Glossary for Data Protection and Privacy • Customer Data → **Data Storage Security** **Possible values:** • *true* • *false* Default value: *false* **System Role:** Source	All systems

Name	Description	System Type
`ips.trace.skipped.entity.content`	If a provisioning job results in skipping entities from source or target systems, you can view the details for each skipped user and group. To do this, set the property to *true* to enable logging and tracing for the personal and sensitive data of your provisioned entities by the next executed job. After the job has finished, the property is automatically set to *false*. If the property is not set, in the logs you see: `content = <hidden content>` To learn more about personal and sensitive data, see: • Glossary for Data Protection and Privacy • Customer Data → **Data Storage Security** **Possible values:** • *true* • *false* Default value: *false* **System Role:** Source	All systems

Name	Description	System Type
`ips.trace.created.entity.content`	If a provisioning job results in created entities in target or proxy systems, you can view the details for each created user and group. To do this, set the property to *true* to enable logging and tracing for the personal and sensitive data of your provisioned entities by the next executed job. After the job has finished, the property is automatically set to *false*. If the property is not set, in the logs you see: `content = <hidden content>` To learn more about personal and sensitive data, see: • Glossary for Data Protection and Privacy • Customer Data → **Data Storage Security** **Possible values:** • *true* • *false* Default value: *false* **System Role:** Source	All systems

Name	Description	System Type
`ips.trace.updated.entity.content`	If a provisioning job results in updated entities in target or proxy systems, you can view the details for each updated user and group. To do this, set the property to *true* to enable logging and tracing for the personal and sensitive data of your provisioned entities by the next executed job. After the job has finished, the property is automatically set to *false*. If the property is not set, in the logs you see: `content = <hidden content>` To learn more about personal and sensitive data, see: • Glossary for Data Protection and Privacy • Customer Data → **Data Storage Security** **Possible values:** • *true* • *false* Default value: *false* **System Role:** Source	All systems
`ips.trace.skipped.entity`	This property allows you to download and view the details of all skipped entities for a given job in a `zip` archive. For more information, see Manage Provisioning Job Logs [page 3010] **Possible values:** • *true* - The downloaded `zip` file contains all skipped entities for the given job, the systems they are skipped from, the reason behind this, as well as the content of the entities (if `ips.trace.skipped.entity.content` is set to **`true`**). • *false* - The downloaded `zip` file is empty. Default value: *false* **System Role:** Source	All systems

Name	Description	System Type
`ips.trace.deleted.entity`	This property allows you to download and view the details of all deleted entities for a given job in a `zip` archive. For more information, see Manage Provisioning Job Logs [page 3010] **Possible values:** • *true* - The downloaded `zip` file contains all deleted entities for the given job and the systems from which they are deleted. • *false* - The downloaded `zip` file is empty. Default value: *false* **System Role:** Source	All systems
`ips.trace.created.entity`	This property allows you to download and view the details of all created entities for a given job in a `zip` archive. For more information, see Manage Provisioning Job Logs [page 3010] **Possible values:** • *true* - The downloaded `zip` file contains all created entities for the given job, the systems in which they are created, as well as the content of the entities (if `ips.trace.created.entity.content` is set to **`true`**). • *false* - The downloaded `zip` file is empty. Default value: *false* **System Role:** Source	All systems

Name	Description	System Type
`ips.trace.updated.entity`	This property allows you to download and view the details of all updated entities for a given job in a `zip` archive. For more information, see Manage Provisioning Job Logs [page 3010] **Possible values:** • *true* - The downloaded `zip` file contains all updated entities for the given job, the systems in which they are updated, as well as the content of the entities (if `ips.trace.updated.entity.content` is set to **`true`**). • *false* - The downloaded `zip` file is empty. Default value: *false* **System Role:** Source	All systems

Name	Description	System Type
`ips.http.header.<header_name>`	Use this property to pass additional information with the HTTP requests. The provisioning system may override your custom HTTP headers, if specific header settings are implemented in the system. **Possible values:** • Example for an authorization header: `ips.http.header.authorization` = *Basic VDAwdfhjgHGSzmfnNA==* > **Note** > If you provide credentials for the provisioning system, this property will not take effect. Its value (token) will be overridden by the token generated by the system implementation. • Example for a header that allows you to control whether teams (meaning groups) should keep their role assignments in SAP Analytics Cloud (SCIM API version 1) upon update: `ips.http.header.x-ignore-roles-if-missing` Possible values: • *true* - Teams will keep their role assignments when no roles are provisioned. • *false* - Teams will lose their role assignments when no roles are provisioned. This is the default value. For more information, see SAP note 3092730. In SAP Analytics Cloud (SCIM API version 2), this behavior is managed by `sac.support.patch.operation` property. **System Role:** Source, Target, Proxy	All HTTP systems

Name	Description	System Type
`ips.delta.read`	If this property is enabled,the provisioning job does not retrieve the entire amount of source system data but only the last changed entities. This property can be used in combination with `ips.full.read.force.count`. For more information refer to the description of the property or Manage Full and Delta Read [page 2459]. **Possible values:** • *enabled* • *disabled* **System Role:** Source	• Identity Authentication • Intelligent Opportunity Analyzer • Local Identity Directory • Microsoft Active Directory • SAP Advanced Financial Closing • SAP Advanced Workflow • SAP Central Business Configuration • SAP Concur • SAP CPQ • SAP Data Custodian • SAP Enterprise Portal • SAP Field Service Management • SAP HANA Cloud Database Services • SAP Intelligent Agriculture • SAP Signavio Process Transformation Suite • SAP SuccessFactors • SAP SuccessFactors Employee Central Payroll • SAP SuccessFactors Learning • SCIM System • SAP Sales Cloud and SAP Service Cloud

Name	Description	System Type
`ips.full.read.force.count`	If your system (connector) works in *delta read* mode, it's recommended to enforce full reads from time to time. To achieve this, set this property to an integer number. This property is used in combination with `ips.delta.read`. When the provisioning system works in *delta read* mode, the Identity Provisioning triggers a full read job after executing the defined number of delta read jobs. If the full read job finishes successfully, the system switches to *delta read* mode. If it fails, the system continues triggering full read jobs until one finishes successfully. **Possible values:** For example: *10* This value results in alternating full reads after every 10 delta reads are performed. In case the property is not set, only delta read jobs will be executed. For more information, see Manage Full and Delta Read [page 2459]. **System Role:** Source	• Identity Authentication • Local Identity Directory • SAP Advanced Financial Closing • SAP Advanced Workflow • SAP Analytics Cloud • SAP Ariba Applications • SAP BTP ABAP environment • SAP BTP Account Members (Neo) • SAP BTP Java/HTML5 apps (Neo) • SAP BTP XS Advanced UAA (Cloud Foundry) • SAP Build Work Zone, advanced edition • SAP Business Network • SAP Central Business Configuration • SAP Commerce Cloud • SAP SuccessFactors Incentive Management • SAP Concur • SAP CPQ • SAP Data Custodian • SAP Enetrprise Portal • SAP Fieldglass • SAP Field Service Management • SAP Integrated Business Planning • SAP Jam Collaboration • SAP Market Communication for Utilities • SAP Marketing Cloud • SAP Master Data Integration • SAP S/4HANA Cloud Public Edition • SAP S/4HANA for procurement planning • SAP S/4HANA On-Premise • SAP Sales Cloud and SAP Service Cloud • SAP SuccessFactors • SAP SuccessFactors Learning • Sales Cloud – Analytics & AI • Cloud Foundry UAA server • Google G Suite • LDAP Server • Microsoft Active Directory • Microsoft Entra ID • SCIM System

Name	Description	System Type
`ips.application.id`	This property holds the value of the *Application ID* of the application, created in the SAP Cloud Identity Services tenant. Its main purpose is to show that a group is associated with a particular application. This is an optional property set by the customer. The *application ID* is the identifier of an application configured in SAP Cloud Identity Services - Identity Authentication that corresponds to a particular source system configured in the administration console for SAP Cloud Identity Services. It is used to distinguish groups provisioned from various source systems to the Identity Directory of SAP Cloud Identity Services. When the property is set, the groups are provisioned with their *applicationId* attribute which is internally mapped to the name of the corresponding application. As a result, the name of the associated application with the group is displayed in the *Application Name* field under *Users&Authorizations* > *Groups*. **System Role:** Source, Target	• Identity Authentication (SCIM API version 2) • Local Identity Directory • Microsoft Entra ID • SAP Advanced Financial Closing • SAP Analytics Cloud • SAP Application Server ABAP • SAP Ariba Applications • SAP Ariba Central Invoice Management • SAP BTP ABAP environment • SAP BTP Account Members (Neo) • SAP Build Work Zone, advanced edition • SAP BTP Platform Members (Cloud Foundry) • SAP BTP XS Advanced UAA (Cloud Foundry) • SAP Central Business Configuration • SAP Commerce Cloud • SAP Concur • SAP CPQ • SAP Data Custodian • SAP Enable Now • SAP Fieldglass • SAP Field Service Management • SAP HANA Cloud Database Services • SAP Integrated Business Planning • SAP Intelligent Agriculture • SAP Jam Collaboration • SAP Marketing Cloud • SAP Market Communication for Utilities • SAP Sales Cloud and SAP Service Cloud • SAP Signavio Process Transformation Suite • SAP S/4 HANA Cloud • SAP SuccessFactors Employee Central Payroll • SAP Revenue Growth Management • Cloud Foundry UAA Server • Procurement Data Warehouse

Name	Description	System Type
`OAuth2TokenServiceURL`	If you need to make OAuth authentication to the system, enter the URL to the access token provider service. **Possible values:** Access token URL **System Role:** Source, Target, Proxy	• Cloud Foundry UAA Server • Google G Suite • Intelligent Opportunity Analyzer • Microsoft Entra ID • Sales Cloud – Analytics & AI • SAP Analytics Cloud • SAP Ariba Applications • SAP Ariba Category Management • SAP Ariba Central Invoice Management • SAP BTP Account Members (Neo) • SAP BTP Java/HTML5 apps (Neo) • SAP BTP XS Advanced UAA (Cloud Foundry) • SAP Build Work Zone, advanced edition • SAP Build Work Zone, standard edition • SAP Business Network • SAP Central Business Configuration • SAP Commerce Cloud • SAP CPQ • SAP Data Custodian • SAP Datasphere • SAP Fieldglass • SAP Field Service Management • SAP HANA Cloud Database Services • SAP Intelligent Agriculture • SAP Jam Collaboration • SAP Master Data Integration • SAP Advanced Financial Closing • Procurement Data Warehouse • SAP Sales Cloud and SAP Service Cloud • SAP S/4HANA for procurement planning • SCIM System

Name	Description	System Type
`OAuth2TokenScope`	If your backend system is OAuth protected and requires an access token with scope, use this property to specify the scope. It defines the token's level of access to protected resources. This is an optional property. Providing a value only makes sense if the backend system and its trusted OAuth server requires it. For more information, see OAuth Scopes . OAuth does not define particular values for scopes as they are system-specific. Both single and multiple scope values are accepted. When using multiple scopes, space is the standard delimiter, as defined in the OAuth 2.0 Authorization Framework . **Possible values**: • Example value of Google G Suite OAuth single scope: **`https://www.googleapis.com/auth/admin.directory.user`**. Although Google G Suite OAuth scopes look like URLs, they are not web pages. They are access token permissions. • Example value of multiple space-delimited scopes: **`read write delete`** **System Role:** Source, Target, Proxy	• Cloud Foundry UAA Server • Google G Suite • Microsoft Entra ID • Sales Cloud – Analytics & AI • SAP Analytics Cloud • SAP Ariba Applications • SAP BTP Account Members (Neo) • SAP BTP Java/HTML5 apps (Neo) • SAP BTP XS Advanced UAA (Cloud Foundry) • SAP Build Work Zone, advanced edition • SAP Build Work Zone, standard edition • SAP Business Network • SAP Central Business Configuration • SAP Commerce Cloud • SAP Data Custodian • SAP Fieldglass • SAP Jam Collaboration • SAP Master Data Integration • SAP Advanced Financial Closing • SAP S/4HANA for procurement planning • SCIM System
`jco.client.user`	Enter the user for AS ABAP. **System Role:** Source, Target, Proxy	SAP Application Server ABAP
`jco.client.passwd`	(Credential) Enter the password for the AS ABAP user. **System Role:** Source, Target, Proxy	SAP Application Server ABAP
`jco.client.ashost`	Enter the virtual host entry that you have configured in the Cloud connector → *Access Control* configuration. **Possible values:** For example: *abapserver.hana.cloud* **System Role:** Source, Target, Proxy	SAP Application Server ABAP

Name	Description	System Type
`jco.client.client`	Enter the client to be used in the ABAP system. Valid format is a three-digit number. **Possible values:** For example: *001* **System Role:** Source, Target, Proxy	SAP Application Server ABAP
`jco.client.r3name`	Enter the three-character system ID of the ABAP system to be addressed. **Possible values:** For example: *WPE* **System Role:** Source, Target, Proxy	SAP Application Server ABAP
`jco.client.sysnr`	Enter the "system number" of the ABAP system. **Possible values:** For example: *42* **System Role:** Source, Target, Proxy	SAP Application Server ABAP
`jco.destination.proxy_type`	Defines the proxy type of the connection you need to provide for your ABAP system. The proxy type *OnPremise* requires the Cloud Connector to access resources within your on-premise network. **Possible values:***OnPremise* **System Role:** Source, Target, Proxy	SAP Application Server ABAP
`jco.destination.peak_limit`	Represents the maximum number of active connections that can simultaneously be created for a destination. **Possible values:** For example: *10* **System Role:** Source, Target, Proxy	SAP Application Server ABAP
`jco.destination.pool_capacity`	Represents the maximum number of idle connections kept open by the destination. **Possible values:** For example: *5* **System Role:** Source, Target, Proxy	SAP Application Server ABAP

Name	Description	System Type
`jco.client.mshost`	Represents the message server host to be used. **System Role:** Source, Target, Proxy	SAP Application Server ABAP
`X-ConsumerKey`	(Credential) Enter the Concur access token needed for the connection. **System Role:** Target, Proxy	SAP Concur
`jwt.subject`	Enter the Google G Suite user on behalf of which the Google Directory API is called. **System Role:** Target, Proxy	Google G Suite
`jwt.scope`	Enter space-separated Google Directory API authorization scopes. **System Role:** Target, Proxy	Google G Suite
`ldap.url`	URL needed to make an LDAP connection to an on-premise system or a cloud service **Possible values:** `ldap://<host><port>` **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory
`ldap.proxyType`	Proxy type for the LDAP connection **Possible values:** *OnPremise* **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory
`ldap.authentication`	Authentication type for the LDAP connection **Possible values:** *BasicAuthentication* **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory
`ldap.user`	User name for the LDAP Server **Possible values:** Text/numeric string **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory
`ldap.password`	(Credential) Password for the LDAP Server user **Possible values:** Encrypted string **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory

Name	Description	System Type
`ldap.group.path`	Enter the complete path to the node containing the groups in the LDAP tree. → Remember We strongly recommend that you enter different paths for LDAP users and groups. That means, the value of `ldap.group.path` should be different than the value of `ldap.user.path`. **Possible values:** For example: `ldap.group.path`=*OU=Groups,OU=IAS,DC=global,DC=corp,DC=mycompany* **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory
`ldap.user.path`	Enter the complete path to the users in the LDAP Server or Microsoft AD. → Remember We strongly recommend that you enter different paths for LDAP users and groups. That means, the value of `ldap.users.path` should be different than the value of `ldap.group.path`. **Possible values:** For example: `ldap.user.path`=*OU=Users,OU=IAS,DC=global,DC=corp,DC=mycompany* **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory

Name	Description	System Type
`ldap.user.attributes`	Shows which user attributes to be read from the LDAP server (and respectively, from the intermediate JSON data). Separate the attributes by comma (,). **Possible values:** • LDAP server version 1 - Even though you specify which user attributes you want to read from the LDAP server , all user attributes are read. • LDAP server version 2 - Only the specified attributes will be read from the LDAP server. If nothing is set, all attributes are read. **System Role:** Source, Proxy	• LDAP Server • Microsoft Active Directory
`ldap.group.attributes`	Shows which group attributes to be read from the LDAP server (and respectively, from the intermediate JSON data). Separate the attributes by comma (,). **Possible values:** • LDAP server version 1 - Even though you specify which group attributes you want to read from the LDAP server, all user and group attributes are read. • LDAP server version 2 - Only the specified attributes will be read from the LDAP server. If nothing is set, all attributes are read. **System Role:** Source, Proxy	• LDAP Server • Microsoft Active Directory
`ldap.user.object.class`	Criteria for user. In the intermediate JSON data, the following LDAP filter is used: `(objectClass=user)` For target LDAP systems: this property defines the set of supported and required attributes for an LDAP user entity. Default value: *inetOrgPerson* **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory

Name	Description	System Type
`ldap.group.object.class`	Criteria for group. In the intermediate JSON data the following LDAP filter is used: `(objectClass=group)` For target LDAP systems: this property defines the set of supported and required attributes for an LDAP group entity. Default value: *groupOfNames* **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory
`ldap.group.uniquename.attribute`	By default, the *memberOf* array in the source JSON data contains the CN part of the complete distinguished name of the groups to which the entity belongs. An administrator can use this property to change the default behavior and specify an attribute name to be used instead of CN. **Note** • Any group that doesn't have the attribute specified, will not be part of the resulting *memberOf* JSON array. • Any group that doesn't match the `ldap.group.path` property, will not be part of the resulting *memberOf* JSON array. **Possible values:** • *cn* (default for LDAP) • *displayName* – this will produce a *memberOf* array which contains the *displayName* attribute value of the groups to which the entity belongs. **Note** Whatever value you choose for this property, it should correspond to the one in the JSON transformation of your LDAP system (the *group* mapping). **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory

Name	Description	System Type
`ldap.attribute.group.id`	This property denotes the ID of a group. • When a user is a member of a group, this group is returned in the *memberOf* array for this user. This property evaluates the attribute used as ID of this group. • When a group is a member of another group, this property evaluates the attribute used as ID of the "parent group". In this case, the `ldap.attribute.group.id` property has a higher priority than `ldap.group.uniquename.attribute`. **Possible values:** • *cn* (default) • *distinguishedName* – this will produce a *memberOf* array which contains the `distinguishedName` attribute value of the groups to which the entity belongs. **Note** Whatever value you choose for this property, it should correspond to the one in the JSON transformation of your LDAP system (the *group* mapping). **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory

Name	Description	System Type
`ldap.attribute.dn`	This property denotes the distinguished name of a user or a group. The distinguished name is auttomatically assigned and cannot be configured. The behavior described below is valid only when Microsoft Active Directory is used as target system: When the Identity Provisioning attempts to provision a user or a group to Microsoft Active Directory for the first time, it may detect that such a user or group already exists on the target system. Thus, the service needs to retrieve the entityId of the existing user or group by using this property for conflict resolution. If the service finds such a user on the target system via this filter, the creation will fail. In this case, the conflicting user will overwrite the existing one. If the service finds such a group on the target system via this filter, the creation will fail. In this case, the existing group only needs to be updated. If the service does not find a user or a group via this filter, the creation will fail. Default and only possible value: *distinguishedName* **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory

Name	Description	System Type
`ldap.member.uniquename.attribute`	Determines the value of the `member` attribute of groups in the intermediate JSON data. This property can return either the common name (CN) of the user or the entire distinguished name (DN). **Possible values:** • *cn* (default for Microsoft AD) • *uid* (default for LDAP Server) • *distinguishedName* **Note** Whatever value you choose for this property, it should correspond to the one in the JSON transformation of your system (the *user* mapping). **System Role:** Source	• LDAP Server • Microsoft Active Directory
`ldap.attribute.user.id`	This property denotes the ID of a user. When a user is a member of a group, this property evaluates the attribute used as ID of this member. In this case, the `ldap.attribute.user.id` or `ldap.attribute.group.id` property has a higher priority than `ldap.member.uniquename.attribute`. Possible values for LDAP Server: • *cn* (default for Microsoft AD) • *uid* (default for LDAP Server) • *distinguishedName* **Note** Whatever value you choose for this property, it should correspond to the one in the JSON transformation of your system (the *user* mapping). **System Role:** Source, Target, Proxy	• LDAP Server • Microsoft Active Directory
`ldap.attribute.group.member`	Default value: *member* **System Role:** Source, Target, Proxy	LDAP Server
`ldap.attribute.user.mail`	Default value: *mail* **System Role:** Source, Target, Proxy	LDAP Server

Name	Description	System Type
`ldap.attribute.user.mobile`	Default value: *mobile* **System Role:** Source, Target, Proxy	LDAP Server
`ldap.attribute.user.givenName`	Default value: *givenName* **System Role:** Source, Target, Proxy	LDAP Server
`ldap.attribute.user.surname`	Default value: *sn* **System Role:** Source, Target, Proxy	LDAP Server
`ldap.attribute.user.groups`	Default value: *memberOf* **System Role:** Source, Target, Proxy	LDAP Server
`ldap.attribute.user.telephoneNumber`	Default value: *telephoneNumber* **System Role:** Source, Target, Proxy	LDAP Server
`ldap.attribute.group.object.class.required`	The LDAP object classes have attributes required for creation of entities. • For Open LDAP Server, the required attribute is the common name (CN) of the group. • For other implementations, it could be another attribute. Default value: *cn* **System Role:** Target, Proxy	LDAP Server
`ldap.attribute.user.object.class.required`	The LDAP object classes have attributes required for creation of entities. • For Open LDAP Server, the required attribute is the common name (CN) of the user. • For other implementations, it could be another attribute. Default value: *cn* **System Role:** Target, Proxy	LDAP Server
`ldap.respond.with.resource.after.create`	When set to *true*, the SCIM `create` operation will read the created entity from the LDAP server. Value *true* is required because the SCIM `create` operation must return the created entity. Default value: *true* **System Role:** Proxy	LDAP Server

Name	Description	System Type
`ldap.respond.with.resource.after.update`	When set to *true*, the SCIM `update` operation will read the modified entity from the LDAP server. When set to *false* or the property is missing, the `update` operation will respond with error *204* (no content). Default value: *true* **System Role:** Proxy	LDAP Server
`ldap.user.filter`	You can optimize the search to return only particular users. To enter correct user filters, stick to the standard LDAP specification. See: LDAP Representation of Filters – Examples . **Possible values:** For example: Value *(cn=123*)* will return only users whose UID starts with "123" (such as *1234567689* or *1230011*). By default, this filter is empty. That is, if the property is not specified, the filter will search for every user. **System Role:** Source	• LDAP Server • Microsoft Active Directory
`ldap.group.filter`	You can optimize the search to return only particular groups. To enter correct group filters, stick to the standard LDAP specification. See: LDAP Representation of Filters – Examples . **Possible values:** For example: Value *(cn=mar*)* will return only groups whose CN starts with "mar" (such as *marked*, *March*, or *Marketing*). By default, this filter is empty. That is, if the property is not specified, the filter will search for every group. **System Role:** Source, Proxy	• LDAP Server • Microsoft Active Directory

Name	Description	System Type
`ldap.page.size`	Use this property to configure the paging (pagination). That means, the number of entities to be read from the LDAP server at once. Default value: *100* **Note** It is not recommended to exceed *1000*. **System Role:** Source	• LDAP Server • Microsoft Active Directory
`ldap.api.version`	Defines the version of LDAP Server API. **Possible values:** • **1** - Indicates that LDAP Server API version 1 is used. • **2** - Indicates that LDAP Server API version 2 is used. If the property is not defined - LDAP Server API version 1 is used. **System Role:** Source, Target, Proxy	LDAP Server
`concur.page.size`	Use this property to configure the paging. That means, the number of entities to be read from Concur at once. Default value: *100* **Note** The maximum allowed number is 100. **System Role:** Source	SAP Concur (Using User v1 API)

Name	Description	System Type
`concur.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with PATCH requests, and below which they are provisioned with PUT request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. **Note** Use this property when your SAP Concur connector is configured for version 3. The expected value is a positive integer without any separators, such as a space, comma, or period. **Possible values**: integer Default value: *1000* Minimum value: *1* Maximum value: *1000* For example: • PATCH requests: If you have a group with 700 members and you update the group by adding another 1 200 members, setting this property to 900 results in the following: As 1900 (the target count of the members) is above the threshold number of 900, 2 PATCH requests will be sent to the SAP Concur target system. The first request will add 900 group members and the second request will add 300 group members. The threshold number you set defines the maximum number of group members processed per batch. • PUT request: If you have a group with 700 members and you update the group by adding another 100 members, setting this property to 900 results in the following: As 800 (the target count of the members) is below the threshold number of 900, 1 PUT request with 800 group members will be sent to the SAP Concur target system to update the group.	SAP Concur (using SCIM v4 API)

Name	Description	System Type
	Note If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members. **Note** Regardless of the threshold number you define, when removing group members in SAP Concur, the maximum number of members which can be removed per one PATCH request is 1000. **System Role:** Target	

Name	Description	System Type
`concur.group.members.page.size`	Defines the number of group members that SAP Concur returns per request when reading a group. **Note** Use this property when your SAP Concur connector is configured for version 3. When Identity Provisioning reads a group, it compares the number of group members with the value configured for `concur.group.members.page.size`. • If the number of group members is equal to or greater than the configured page size and the `concur.group.members.paging.enabled` property is set to *true*, Identity Provisioning will read all group members in separate requests. For example, if a group has 6 members and the page size is set to *2*, Identity Provisioning will send 3 requests to read the group members. • If the number of group members is less than the page size or `concur.group.members.paging.enabled` is set to false, Identity Provisioning will read all group members in a single request. For example, if a group has 2 members and the page size is set to *6*, Identity Provisioning will read all group members in a single request. Setting the property `concur.group.members.paging.enabled` allows you to read groups with a large number of group members. For more information, see `concur.group.members.paging.enabled`. **Default value:** 1000 **System Role:** Source, Proxy	SAP Concur (using SCIM v4 API)

Name	Description	System Type
`concur.group.members.paging.enabled`	This property enables paging of group members. **Note** Use this property when your SAP Concur connector is configured for version 3. The maximum number of group members returned per request is 1000. To read more than 1000 group members, paging must be enabled. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *true* **System Role:** Source, Proxy	SAP Concur (using SCIM v4 API)

Name	Description	System Type
`ibp.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. **Note** Use this property when your SAP Integrated Business Planning connector is configured for version 2. - If set to *true*, PATCH operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. - If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** - *true* - *false* Default value: *false* **System Role:** Target, Proxy	SAP Integrated Business Planning (using version 2 - SCIM Interface for IAM)

Name	Description	System Type
`s4hana.cloud.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. **Note** Use this property when your SAP S/4HANA Cloud Public Edition connector is configured for version 3. - If set to *true*, **PATCH** operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. - If set to *false*, **PUT** operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** - *true* - *false* Default value: *false* **System Role:** Target, Proxy	SAP S/4HANA Cloud Public Edition (using version 3 - System for Cross-domain Identity Management (SCIM) API)

Name	Description	System Type
`concur.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. **Note** Use this property when your SAP Concur connector is configured for version 3. - If set to *true*, PATCH operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. - If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** - *true* - *false* Default value: *false* **System Role:** Target, Proxy	SAP Concur (using SCIM v4 API)

Name	Description	System Type
`concur.group.prefix`	This property distinguishes SAP Concur groups by specific prefix. It is an optional property which does not appear by default at system creation. **Note** Use this property when your SAP Concur connector is configured for version 3. Example value: **`Concur_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Concur source system and will be provisioned to the target system with the following name pattern: **`Concur_<GroupDisplayName>`**. This way SAP Concur groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Concur groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`Concur_`** prefix in their display name will be provisioned to SAP Concur. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Concur. **System Role:** Source and Target	SAP Concur (using SCIM v4 API)

Name	Description	System Type
`concur.group.unique.attribute`	This property defines by which unique attribute(s) the existing group will be searched and resolved. **Note** Use this property when your SAP Concur connector is configured for version 3. The default value is *displayName*. If the service finds an existing group by this unique attribute, the group that you try to provision will overwrite the existing one. If such a group is not found, the group that you try to provision will not be created in the target system. **System Role:** Target	SAP Concur (using SCIM v4 API)
`concur.group.filter`	When specified, only those groups matching the filter expression will be read. **Note** Use this property when your SAP Concur connector is configured for version 3. For example: *displayName* eq "ProjectTeam1" **System Role:** Source and Proxy	SAP Concur (using SCIM v4 API)
`msgraph-filter(Deprecated)`	Use this property to filter users and groups by specific criteria, according to the API syntax of Microsoft Entra ID. **Note** This property is deprecated. Use `aad.user.filter` and `aad.group.filter` instead. Default value: *null* To set a particular value, see Microsoft Graph: filter parameter . **System Role:** Source	Microsoft Entra ID

Name	Description	System Type
`gsuite.page.size`	Use this property to configure the paging. That means, the number of entities to be read from Google G Suite at once. Default value: *100* **Note** The maximum allowed number is 500. **System Role:** Source	Google G Suite
`gsuite.get.deleted`	This property determines whether recently deleted entities should be read. **Note** You can apply this property only for **users**. For groups it will be ignored. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Source	Google G Suite
`gsuite.domain`	This property determines whether entities from a particular domain should be read. **Possible values:** For example: *myaccount.ondemand.com* **System Role:** Source	Google G Suite
`gsuite.customer.id`	This property determines whether entities for a particular customer ID to be read. This property takes precedence over `gsuite.domain`. **Possible values:** Customer ID number For more information, see Google G Suite API: User Accounts . **System Role:** Source	Google G Suite

Name	Description	System Type
`com.sun.jndi.ldap.read.timeout`	Use this property if you want to specify the read timeout (in milliseconds) for an LDAP connection. The expected value is a positive integer without any separators, such as a space, comma, or period. **Possible values:** For example: *5000* This value causes the LDAP service provider to abort the read attempt if the server does not respond within 5 seconds. **System Role:** Source	• LDAP Server • Microsoft Active Directory
`com.sun.jndi.ldap.connect.timeout`	Use this property if you want to set the timeout (in milliseconds) for connecting to the LDAP server. The expected value is a positive integer without any separators, such as a space, comma, or period. **Possible values:** For example: *500* This value causes the LDAP service provider to abort the connection attempt if a connection cannot be established in half a second. **System Role:** Source	• LDAP Server • Microsoft Active Directory
`aad.oauth2.resource.name`	Specifies the OAuth2 resource name for Microsoft Graph API. **Default value:** `https://graph.microsoft.com` You can change the Microsoft Graph endpoint to match your organization's cloud environment. For more information, see Microsoft Graph national cloud deployments . **System role:** Source, Target, Proxy	Microsoft Entra ID

Name	Description	System Type
`ad.group.flatten`	There are target systems that do not support nested groups (group structures). Therefore, if your Microsoft AD system contains such groups, they will not be resolved properly during the provisioning job. Such target systems are: • *SAP Jam Collaboration* • *Identity Authentication* To enable reading of group structures, you can use the `ad.group.flatten` property and set it to *true*. It will read the group structure recursively and will "flatten" it so that all users from all groups and subgroups will be resolved and written in the target system as members of the main parent group. For best results, we recommend you also set the system property `ldap.group.filter` whose value is one or multiple Microsoft AD parent groups. **Possible values:** • *true* • *false* Default value: *false* Examples for filtering: • If your Microsoft AD system contains a parent group "*Canteen*", which contains nested subgroups, you have to set the filter like this: `ldap.group.filter` = *(cn=Canteen)* The Identity Provisioning will resolve all the direct users and groups of "*Canteen*", along with all the users of its subgroups (and their subgroups). In the target system, all users will be written in one parent group named also "*Canteen*". • If you have multiple parent groups (for example, *Canteen*, *Finances*, and *Support_Team*) that contain nested subgroups, you have to set the filter like this: `ldap.group.filter` = *(|(cn=Canteen)(cn=Finance)(cn=Support_Team))* **System Role:** Source	Microsoft Active Directory

Name	Description	System Type
`aad.domain.name`	Enter one of the verified domain names from the corresponding Microsoft Entra ID tenant. **System Role:** Source, Target, Proxy	Microsoft Entra ID
`aad.group.member.attributes`	This property defines the attributes of a group member to be read by the Identity Provisioning. By default, it always reads the *type* and the *id* of a member. If you prefer the Identity Provisioning to read additional attributes, you can add them as a single or a comma-separated value. For example: **Example** • If you want to read the e-mails too, enter: `aad.group.member.attributes` =*mail* This will read a member's type, ID and e-mail. • If you want to read multiple additional attributes, enter: `aad.group.member.attributes` =*mail,mobilePhone,displayName* This will read a member's type, ID, e-mail, phone and display name. See: Microsoft Entra ID [page 914] **Possible values:** • *type* (default) • *id* (default) • Any valid Microsoft Entra ID attribute of a group member • A comma-separated list of valid Microsoft Entra ID attributes of a group member **→ Remember** The Identity Provisioning service always retrieves the *id* and *type* attributes of a group member, regardless of the additional attributes you specify. **System Role:** Source, Proxy	Microsoft Entra ID

Name	Description	System Type
`aad.user.attributes.membership.active`	Use this property if you want to get information about all the groups to which the users are assigned (if any). • If the property is missing, or is set to *false* – group membership details for the users will not be extracted. • If the property is set to *true* – group membership details for the users will be extracted. If you set the property to *true*, you will get information about the group ID and its entity type (group) – default result. However, if you also set a value for property `aad.group.attributes`, you will get additional information relevant to this value. For example: If you set `aad.user.attributes.membership.active` = *true* and `aad.group.attributes` = *displayName*, you will receive the following exemplary data for a group as part of the user object: `"groups": [ { "displayName": "Microsoft Entra ID Group 1", "id": "aaa111999-0000-444-123-777fff000", "type": "group" } ]` **Possible values:** • *true* • *false* (default) **System Role:** Source, Proxy	Microsoft Entra ID

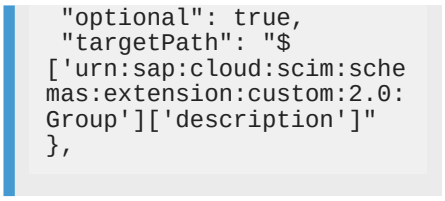
Name	Description	System Type
`aad.user.filter`	Via this property, you can filter users by specific criteria, according to the syntax of Microsoft Graph REST API . You can also filter out users with advanced query parameters, as described in Advanced query capabilities on Microsoft Entra ID objects **Note** This property replaces the deprecated `msgraph-filter` property. **Possible values:** Text/numeric string For example: • Value = *Department eq 'Finance'* • Value = *displayName eq 'John Smith' and city eq 'Sofia'* • Value = *userPrincipalName ne 'Julie Armstrong'* **System Role:** Source, Proxy	Microsoft Entra ID
`aad.group.filter`	Via this property, you can filter groups by specific criteria, according to the syntax of Microsoft Graph REST API . You can also filter out groups with advanced query parameters, as described in Advanced query capabilities on Microsoft Entra ID objects **Possible values:** Text/numeric string For example: • Value = *displayName eq 'Employees 2020'* • Value = *displayName eq 'Service Administrators' and mail eq 'serviceadmins@abcd.onmicrosoft.com'* • Value = *startsWith(displayName, 'ABC_')* • Value = *displayName ne 'Service Administrators'* **System Role:** Source, Proxy	Microsoft Entra ID

Name	Description	System Type
`aad.user.attributes`	Defines which user attributes are read from Microsoft Entra ID system. The property is set during system creation with the following default value: *id,mail,userPrincipalName,displayName,mailNickname,givenName,surname,mobilePhone,businessPhones* This means that by default, Identity Provisioning will read from Microsoft Entra ID the user attributes defined in the property value. Those attributes are also used in the default read transformation. To check the complete set of user attributes (properties) supported by Microsoft Entra ID, see: Microsoft Graph: User Properties If you want the Identity Provisioning to read additional user attributes, add them to the default list of attributes separated by comma and adapt the transformations. For example, to read the *employeeId* of the Microsoft Entra ID users in addition to the default list of attributes, and provision them to Identity Authentication, proceed as follows: 1. Add the attribute in the property value: id,mail,userPrincipalName,displayName,mailNickname,givenName,surname,mobilePhone,businessPhones,*employeeId* 2. Extend the Microsoft Entra ID read transformation by adding the following mapping for the user resource: Sample Code `{ "sourcePath": "$.employeeId", "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['employeeNumber']", "optional": true },`	Microsoft Entra ID

Name	Description	System Type
	3. Make sure the following mapping is present in the Identity Authentication write transformation: **Sample Code** `{ "sourcePath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['employeeNumber']", "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['employeeNumber']", "optional": true },` In case you remove the default list of attributes from the value of this property and only add the additional attributes, Identity Provisioning will return the additional user attributes plus the mandatory ones: *id,mail, userPrincipalName*. **System Role:** Source, Proxy	

Name	Description	System Type

`aad.group.attributes`	Defines which group attributes are read from Microsoft Entra ID system.
	
	The property is set during system creation with the following default value: *id,displayName,mailNickname*
	
	This means that by default, Identity Provisioning will read from Microsoft Entra ID the group attributes defined in the property value and will also return the *members* attribute. Those attributes are used in the default read transformation.
	
	To check the complete set of group attributes (properties) supported by Microsoft Entra ID, see: Microsoft Graph: Group Properties
	
	If you want the Identity Provisioning to read additional group attributes, add them to the default list of attributes separated by comma and adapt the transformations.
	
	For example, to read the *description* of the Microsoft Entra ID groups in addition to the default list of attributes, and provision them to Identity Authentication, proceed as follows:
	
	1. Add the attribute in the property value: id,displayName,mailNickname,*description*
	2. Extend the Microsoft Entra ID read transformation by adding the following mapping for the group resource:
	
	Code Syntax
	`{ "sourcePath": "$.description", "optional": true, "targetPath": "$.description" },`
	
	3. Extend the Identity Authentication write transformation by adding the following mapping for the group resource:
	
	Code Syntax
	`{ "sourcePath": "$.description",`
		Microsoft Entra ID

Name	Description	System Type
	 In case you remove the default list of attributes from the value of this property and only add the additional attributes, Identity Provisioning will read from Microsoft Entra ID the additional group attributes, the group *id*, *displayName*, *mailNickname* and will also return the *members* attribute. **System Role:** Source, Proxy	
`aad.entities.top`	This property defines the number of entities to be read per page. Default value: *100* **System Role:** Source, Proxy	Microsoft Entra ID
`csrf.token.path`	Path which is appended to the URL to retrieve the CSRF token. The property can be configured for the following provisioning systems: • SCIM System • SAP Analytics Cloud **System Role:** Target, Proxy	• SAP Analytics Cloud • SCIM System

Name	Description	System Type
`sac.csrf.token.path`	Path which is appended to the URL to retrieve the CSRF token. The property is automatically added during system creation, when the SAP Analytics Cloud system consumes SCIM API version 2. Based on the version of the SCIM API consumed, the default value for the property is: • SAP Analytics Cloud SCIM API version 1: `/api/v1/scim/Users?count=1` **Note** The property is present only for systems created before December 3, 2024. Systems created after this date use the default value for the connector version without adding the property. • SAP Analytics Cloud SCIM API version 2: `/api/v1/scim2/Users?count=1` **Note** If you delete the property, Identity Provisioning will use the default value for the property for SAP Analytics Cloud using SCIM API **version 1**. **System Role:** Target, Proxy	SAP Analytics Cloud
`scim.csrf.token.path`	Specifies the path which is appended to the URL to retrieve the CSRF token when `scim.api.csrf.protection` is enabled. The property is not automatically added during system creation. **Note** A CSRF token will be fetched upon request only when `scim.api.csrf.protection` is set to *enabled*. **System Role:** Target, Proxy	SCIM System

Name	Description	System Type
`s4hana.cloud.hr.switch.active`	Defines whether the human resources (HR) integration is active or not. The property is set to `true` by default. This means that the synchronization of business partners is managed by the HR system and SAP S/4HANA Cloud Public Edition. Identity Provisioning can only be used for managing business users (login users and their login information, such as date/time preferences) and role assignments. When set to `false`, synchronization is handled by the identity management system instead. **Possible values:** • *true* (default) • *false* **System Role:** Target, Proxy	SAP S/4HANA Cloud Public Edition (using version 1 - SAP S/4HANA Cloud Public Edition API: Business User)
`s4hana.cloud.roles.filter`	Enter OData filtering for reading roles in the SAP S/4HANA Cloud Public Edition system. To learn what criteria you can use, see: OData URI Conventions → 4.5 Filter System Query Option **System Role:** Source, Proxy	SAP S/4HANA Cloud Public Edition (using version 1 - SAP S/4HANA Cloud Public Edition API: Business User)
`s4hana.cloud.api.version`	This property defines the API version that your SAP S/4HANA Cloud Public Edition system uses. By default, Identity Provisioning uses version *1*. Version *1* means your SAP S/4HANA Cloud Public Edition system consumes SAP S/4HANA Cloud Public Edition API: Business User. Version *3* means your SAP S/4HANA Cloud Public Edition system consumes System for Cross-domain Identity Management (SCIM) API. **System Role:** Source, Target, Proxy	SAP S/4HANA Cloud Public Edition

Name	Description	System Type
`s4hana.cloud.group.filter`	When specified, only those SAP S/4HANA Cloud Public Edition groups matching the filter expression will be read. Supported operators: eq (equal), ne (not equal), sw (starts with), ew (ends with) and co (contains) **Possible values**: • `displayName` • `externalId` • `urn:ietf:params:scim:schemas:extension:sap:2.0:Group:type` For example: • *displayName co "ProjectTeam1" or "Students2023"* • *externalId eq "12345678" or externalId sw "123"* • *urn:ietf:params:scim:schemas:extension:sap:2.0:Group:type eq "authorization" and externalId eq "12345678"* **Note** These combinations are valid for both 'or' and 'and' operators. **System Role:** Source, Proxy	SAP S/4HANA Cloud Public Edition (using version 3 - System for Cross-domain Identity Management (SCIM) API)

Name	Description	System Type
`s4hana.cloud.group.prefix`	This property distinguishes SAP S/4HANA Cloud Public Edition groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`S4HANACLOUD_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP S/4HANA Cloud Public Edition source system and will be provisioned to the target system with the following name pattern: **`S4HANACLOUD_<GroupDisplayName>`**. This way SAP S/4HANA Cloud Public Edition groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP S/4HANA Cloud Public Edition groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`S4HANACLOUD_`** prefix in their display name will be provisioned to SAP S/4HANA Cloud Public Edition. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP S/4HANA Cloud Public Edition. **System Role:** Source and Target	SAP S/4HANA Cloud Public Edition (using version 3 - System for Cross-domain Identity Management (SCIM) API)

Name	Description	System Type
`s4hana.cloud.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists in the SAP S/4HANA Cloud Public Edition target system, the creation will fail. In this case, the existing group only needs to be updated. In SAP S/4HANA Cloud Public Edition the uniqueness of a group is defined by the combination of the following two attributes, which are automatically filled in when the target system is created: *externalId, ['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']*. To resolve the conflict, an existing group must match both of these attributes. If the group matches only one of the attributes, the conflict is not resolved, and group creation will fail. **Note** Although you can define your own unique combination of attributes (for example, *externalId,displayName*), we recommend that you do not modify the automatically filled value of the `s4hana.cloud.group.unique.attribute` property. If the property is not specified, the search is done by the default attribute *displayName*. **System Role:** Target	SAP S/4HANA Cloud Public Edition (using version 3 - System for Cross-domain Identity Management (SCIM) API)

Name	Description	System Type
`s4hana.cloud.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with PATCH requests, and below which they are provisioned with PUT request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. The expected value is a positive integer without any separators, such as a space, comma, or period. **Possible values**: integer Default value: *20000* Minimum value: *1* Maximum value: *20000* For example: • PATCH requests: If you have a group with 700 members and you update the group by adding another 1 200 members, setting this property to 900 results in the following: As 1900 (the target count of the members) is above the threshold number of 900, 2 PATCH requests will be sent to the SAP S/4HANA Cloud Public Edition target system. The first request will add 900 group members and the second request will add 300 group members. The threshold number you set defines the maximum number of group members processed per batch. • PUT request: If you have a group with 700 members and you update the group by adding another 100 members, setting this property to 900 results in the following: As 800 (the target count of the members) is below the threshold number of 900, 1 PUT request with 800 group members will be sent to the SAP S/4HANA Cloud Public Edition target system to update the group. **Note** If this property is not configured, the default value is applied automatically. The value	SAP S/4HANA Cloud Public Edition (using version 3 - System for Cross-domain Identity Management (SCIM) API)

Name	Description	System Type
	must be greater than *0* to ensure proper provisioning of group members. **Note** Regardless of the threshold number you define, when removing group members in SAP S/4HANA Cloud Public Edition, the maximum number of members which can be removed per one **PATCH** request is 50. **System Role:** Target	
`s4hana.cloud.user.filter`	When specified, only those SAP S/4HANA Cloud Public Edition users matching the filter expression will be read. You can filter users by **list of attributes** according to the API syntax of SAP S/4HANA Cloud Public Edition. You can set a single attribute or multiple ones as filter criteria. If you enter multiple attributes (using OR operator), the filter will search for any of them. Supported operators: eq (equal), ne (not equal), sw (starts with), ew (ends with) and co (contains) **Possible values:** • `userName` • `externalId` • `emails[0].value` For example: • *userName eq "johnbrown" and externalId eq "P000252"* • *userName sw "J" or externalId eq "P000252"* • *userName eq "johnbrown" and emails[0].value eq "johnbrown@email.com"* • *userName eq "johnbrown" and emails[0].value ne "johnbrown@email.com" or externalId eq "P000252"* **Note** These combinations are valid for both 'or' and 'and' operators. **System Role:** Source, Proxy	SAP S/4HANA Cloud Public Edition (using version 3 - System for Cross-domain Identity Management (SCIM) API)

Name	Description	System Type
`s4hana.cloud.hr.switch.dependent.role.codes`	A default property. Add the codes of the roles maintained by the HR integration. Make sure these role codes are part of your read and write transformations. **Possible values:** For example: *BUP003, BBP005* That means, your HR integration will support *employees* and *contingent worker*. **System Role:** Target, Proxy	SAP S/4HANA Cloud Public Edition (using version 1 - SAP S/4HANA Cloud Public Edition API: Business User)
`s4hana.cloud.skip.read.archived`	In the event of archived (disabled) entities in a source SAP S/4HANA Cloud Public Edition system, you can choose whether the provisioning jobs to continue reading such entities or to skip them. In the source and proxy systems, this property is activated by default. If you want to always read disabled entities, set the property to *false*, or delete it. **Possible values:** • *true* • *false* Default value: *true* **System Role:** Source, Proxy	SAP S/4HANA Cloud Public Edition (using version 1 - SAP S/4HANA Cloud Public Edition API: Business User)
`s4hana.onprem.hr.switch.dependent.role.codes`	Add the codes of the roles maintained by the HR integration. Make sure these role codes are part of your read and write transformations. This property is applicable only if `s4hana.onprem.hr.switch.active` = *true* **Possible values:** For example: *BUP003, BBP005, BUP012, WFM001* That means, your HR integration will support *employees*, *contingent workers*, *collaboration users*, and *resources*. **System Role:** Target, Proxy	SAP S/4HANA On-Premise

Name	Description	System Type
`s4hana.onprem.hr.switch.active`	Defines whether the system should include HR integration or not. This property is related to `s4hana.onprem.hr.switch.dependent.role.codes`. **Possible values:** • *true* – HR integration is enabled for your system • *false* – HR integration is disabled for your system Default value: *false* **System Role:** Target, Proxy	SAP S/4HANA On-Premise
`s4hana.onprem.skip.read.archived`	In the event of archived (disabled) entities in a source SAP S/4HANA On-Premise system, you can choose whether the provisioning jobs to continue reading such entities or to skip them. In the source and proxy systems, this property is activated by default. If you want to always read disabled entities, set the property to *false*, or delete it. **Possible values:** • *true* • *false* Default value: *true* **System Role:** Source, Proxy	SAP S/4HANA On-Premise
`s4hana.onprem.sap-client`	Use this property if you want to specify a particular AS ABAP client to use as the `sap-client` URL parameter. AS ABAP client. To learn more, see: Specifying the Client For more information about `sap-client`, see: SAP URL Parameters **Possible values:** A three-digit integer number For example: *102* **System Role:** Source, Target, Proxy	SAP S/4HANA On-Premise

Name	Description	System Type
`s4hana.onprem.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: APIs for Business User Management **Possible values:** • *true* • *false* Default value: *false* **System Role:** Source, Target, Proxy	SAP S/4HANA On-Premise
`s4hana.onprem.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of users to be provisioned per request. **Possible values:** Default value: *20* Maximum value: *100* If you enter a number larger than 100, the service will replace it with the default value (20). **System Role:** Target	SAP S/4HANA On-Premise

Name	Description	System Type
`fg.group.prefix`	This property distinguishes SAP Fieldglass groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **FG_** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Fieldglass source system and will be provisioned to the target system with the following name pattern: **FG_<GroupDisplayName>**. This way SAP Fieldglass groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Fieldglass groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **FG_** prefix in their display name will be provisioned to SAP Fieldglass. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Fieldglass. **System Role:** Source and Target	SAP Fieldglass
`fg.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning users and groups. This means, the Identity Provisioning service can write, update, and delete multiple users in a single request. For more information, see: SAP Fieldglass Identity Management API **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP Fieldglass

Name	Description	System Type
`fg.bulk.operations.max.count`	This property sets the number of operations to be performed in one bulk request. **Possible values:** Default value: *20* Minimum value: *10* Maximum value: *100* If you provide a value outside of the minimum and maximum range, the service will replace it with the default value (20). **System Role:** Target	SAP Fieldglass
`a4c.skip.read.archived`	In the event of archived (disabled) entities in a source SAP BTP ABAP environment system, you can choose whether the provisioning jobs to continue reading such entities or to skip them. In the source and proxy systems, this property is activated by default. If you want to always read disabled entities, set the property to *false*, or delete it. **Possible values:** • *true* • *false* Default value: *true* **System Role:** Source, Target, Proxy	SAP BTP ABAP environment
`a4c.roles.filter`	Enter OData filtering for reading roles in the SAP BTP ABAP environment system. To learn what criteria you can use, see: OData URI Conventions → 4.5 Filter System Query Option **System Role:** Source, Proxy	SAP BTP ABAP environment

Name	Description	System Type
`a4c.roles.prefix`	This property distinguishes SAP BTP ABAP environment roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`A4C_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the roles that are read from the SAP BTP ABAP environment source system and will be provisioned to the target system with the following name pattern: **`A4C_<role_name>`** . This way SAP BTP ABAP environment roles in the target system will be easily distinguished from roles provisioned from other applications. If the property is not set, the SAP BTP ABAP environment roles will be read and provisioned to the target system with their actual role names. • When **set in the target system**, only roles containing the **`A4C_`** prefix in their role name will be provisioned to SAP BTP ABAP environment. Roles without this prefix in their names won't be provisioned. If the property is not set, all roles will be provisioned to SAP BTP ABAP environment. **System Role:** Source and Target	SAP BTP ABAP environment

Name	Description	System Type
`a4c.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP BTP ABAP environment target or proxy system. See also: Extended Explanation of the *user.roles.overwrite Properties **Possible values:** • *true* – the current user roles will be deleted in the proxy system, and the user will be updated only with the roles provisioned by the service. • *false* – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the proxy system. Default value (if the property is missing during system creation): *true* Default value (if the property appears during system creation): *false* **System Role:** Target, Proxy	SAP BTP ABAP environment

Name	Description	System Type
`ibp.roles.prefix`	This property distinguishes SAP Integrated Business Planning roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`IBP_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the roles that are read from the SAP Integrated Business Planning source system and will be provisioned to the target system with the following name pattern: **`IBP_<role_name>`** . This way SAP Integrated Business Planning roles in the target system will be distinguished from roles provisioned from other applications. If the property is not set, the SAP Integrated Business Planning roles will be read and provisioned to the target system with their actual role names. • When **set in the target system**, only roles containing the **`IBP_`** prefix in their role name will be provisioned to SAP Integrated Business Planning. Roles without this prefix in the role name won't be provisioned. If the property is not set, all roles will be provisioned to SAP Integrated Business Planning. **System Role:** Source and Target	SAP Integrated Business Planning (using version 1 - SAP Integrated Business Planning API: Business User)

Name	Description	System Type
`ibp.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP IBP target or proxy system. See also: Extended Explanation of the *user.roles.overwrite Properties **Possible values:** • *true* – the current user roles will be deleted in the target or proxy system, and the user will be updated only with the roles provisioned by the service. • *false* – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the target or proxy system. Default value (if the property is missing during system creation): *true* Default value (if the property appears during system creation): *false* **System Role:** Target, Proxy	SAP Integrated Business Planning (using version 1 - SAP Integrated Business Planning API: Business User)
`marketing.cloud.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP Marketing Cloud target or proxy system. See also: Extended Explanation of the *user.roles.overwrite Properties **Possible values:** • *true* – the current user roles will be deleted in the proxy system, and the user will be updated only with the roles provisioned by the service. • *false* – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the proxy system. Default value (if the property is missing during system creation): *true* Default value (if the property appears during system creation): *false* **System Role:** Target, Proxy	SAP Marketing Cloud

Name	Description	System Type
`s4hana.cloud.roles.page.size`	This property indicates how many business roles (considered as *groups*) per page to be read from your SAP S/4HANA Cloud Public Edition source system. **Possible values:** Integer number For example, if you set the property's value = *30*, the Identity Provisioning will read 30 roles (groups) at once, then – another 30, and so on. **System Role:** Source, Proxy	SAP S/4HANA Cloud Public Edition (using version 1 - SAP S/4HANA Cloud Public Edition API: Business User)
`s4hana.cloud.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: APIs for Business User Management **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP S/4HANA Cloud Public Edition (using version 1 - SAP S/4HANA Cloud Public Edition API: Business User)
`s4hana.cloud.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of users to be provisioned per request. **Possible values:** Default value: *20* Maximum value: *100* If you enter a number larger than 100, the service will replace it with the default value (20). **System Role:** Target	SAP S/4HANA Cloud Public Edition (using version 1 - SAP S/4HANA Cloud Public Edition API: Business User)

Name	Description	System Type
`marketing.cloud.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: APIs for Business User Management **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP Marketing Cloud
`marketing.cloud.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of users to be provisioned per request. **Possible values:** Default value: *20* Maximum value: *100* If you enter a number larger than 100, the service will replace it with the default value (20). **System Role:** Target	SAP Marketing Cloud
`a4c.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: APIs for Business User Management **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP BTP ABAP environment

Name	Description	System Type
`a4c.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of users to be provisioned per request. **Possible values:** Default value: *20* Maximum value: *100* If you enter a number larger than 100, the service will replace it with the default value (20). **System Role:** Target	SAP BTP ABAP environment
`a4c.roles.page.size`	This property indicates how many business roles (considered as *groups*) per page to be read from your SAP BTP ABAP environment source system. **Possible values:** Integer number For example, if you set the property's value = *30*, the Identity Provisioning will read 30 roles (groups) at once, then – another 30, and so on. **System Role:** Source, Proxy	SAP BTP ABAP environment
`ibp.roles.page.size`	This property indicates how many business roles (considered as *groups*) per page to be read from your SAP IBP source system. **Possible values:** Integer number For example, if you set the property's value = *30*, the Identity Provisioning will read 30 roles (groups) at once, then – another 30, and so on. **System Role:** Source, Proxy	SAP Integrated Business Planning (using version 1 - SAP Integrated Business Planning API: Business User)
`ibp.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: APIs for Business User Management **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP Integrated Business Planning (using version 1 - SAP Integrated Business Planning API: Business User)

Name	Description	System Type
`ibp.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of users to be provisioned per request. **Possible values:** Default value: *20* Maximum value: *100* If you enter a number larger than 100, the service will replace it with the default value (20). **System Role:** Target	SAP Integrated Business Planning (using version 1 - SAP Integrated Business Planning API: Business User)
`ibp.api.version`	This property defines the API version which is consumed by the SAP Integrated Business Planning system. **Possible values:** • *1* - SAP Integrated Business Planning API: Business User is used. • *2* - SCIM Interface for IAM is used. By default, Identity Provisioning uses version *1*. **System Role:** Source, Target, Proxy	SAP Integrated Business Planning

Name	Description	System Type
`ibp.group.filter`	When specified, only those SAP Integrated Business Planning groups matching the filter expression will be read. Supported operators: eq (equal), ne (not equal), sw (starts with), ew (ends with) and co (contains) **Possible values**: • `displayName` • `externalId` • `urn:ietf:params:scim:schemas:extension:sap:2.0:Group:type` For example: • *displayName co "ProjectTeam1" or "Students2023"* • *externalId eq "12345678" or externalId sw "123"* • *urn:ietf:params:scim:schemas:extension:sap:2.0:Group:type eq "authorization" and externalId eq "12345678"* **Note** These combinations are valid for both 'or' and 'and' operators. **System Role:** Source, Proxy	SAP Integrated Business Planning (using version 2 - SCIM Interface for IAM)

Name	Description	System Type
`ibp.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists in the SAP Integrated Business Planning target system, the creation will fail. In this case, the existing group only needs to be updated. In SAP Integrated Business Planning the uniqueness of a group is defined by the combination of the following two attributes, which are automatically filled in when the target system is created: *externalId, ['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']*. To resolve the conflict, an existing group must match both of these attributes. If the group matches only one of the attributes, the conflict is not resolved, and group creation will fail. **Note** Although you can define your own unique combination of attributes (for example, *externalId,displayName*), we recommend that you do not modify the automatically filled value of the `ibp.group.unique.attribute` property. If the property is not specified, the search is done by the default attribute *displayName*. **System Role:** Target	SAP Integrated Business Planning (using version 2 - SCIM Interface for IAM)

Name	Description	System Type
`ibp.group.prefix`	This property distinguishes SAP Integrated Business Planning groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`IBP_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Integrated Business Planning source system and will be provisioned to the target system with the following name pattern: **`IBP_<GroupDisplayName>`**. This way SAP Integrated Business Planning groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Integrated Business Planning groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`IBP_`** prefix in their display name will be provisioned to SAP Integrated Business Planning. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Integrated Business Planning. **System Role:** Source and Target	SAP Integrated Business Planning (using version 2 - SCIM Interface for IAM)

Name	Description	System Type
`ibp.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with PATCH requests, and below which they are provisioned with PUT request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. The expected value is a positive integer without any separators, such as a space, comma, or period. **Possible values**: integer Default value: *20000* Minimum value: *1* Maximum value: *20000* For example: • PATCH requests: If you have a group with 700 members and you update the group by adding another 1 200 members, setting this property to 900 results in the following: As 1900 (the target count of the members) is above the threshold number of 900, 2 PATCH requests will be sent to the SAP Integrated Business Planning target system. The first request will add 900 group members and the second request will add 300 group members. The threshold number you set defines the maximum number of group members processed per batch. • PUT request: If you have a group with 700 members and you update the group by adding another 100 members, setting this property to 900 results in the following: As 800 (the target count of the members) is below the threshold number of 900, 1 PUT request with 800 group members will be sent to the SAP Integrated Business Planning target system to update the group. **Note** If this property is not configured, the default value is applied automatically. The value	SAP Integrated Business Planning (using version 2 - SCIM Interface for IAM)

Name	Description	System Type
	must be greater than *0* to ensure proper provisioning of group members. **Note** Regardless of the threshold number you define, when removing group members in SAP Integrated Business Planning, the maximum number of members which can be removed per one **PATCH** request is 50. **System Role:** Target	
`ibp.user.filter`	When specified, only those SAP Integrated Business Planning users matching the filter expression will be read. You can filter users by **list of attributes** according to the API syntax of SAP Integrated Business Planning. You can set a single attribute or multiple ones as filter criteria. If you enter multiple attributes (using OR operator), the filter will search for any of them. Supported operators: eq (equal), ne (not equal), sw (starts with), ew (ends with) and co (contains) **Possible values:** • `userName` • `externalId` • `emails[0].value` For example: • *userName eq "johnbrown" and externalId eq "P000252"* • *userName sw "J" or externalId eq "P000252"* • *userName eq "johnbrown" and emails[0].value eq "johnbrown@email.com"* • *userName eq "johnbrown" and emails[0].value ne "johnbrown@email.com" or externalId eq "P000252"* **Note** These combinations are valid for both 'or' and 'and' operators. **System Role:** Source, Proxy	SAP Integrated Business Planning (using version 2 - SCIM Interface for IAM)

Name	Description	System Type
`marketing.cloud.roles.page.size`	This property indicates how many business roles (considered as *groups*) per page to be read from your SAP Marketing Cloud source system. **Possible values:** Integer number For example, if you set the property's value = *30*, the Identity Provisioning will read 30 roles (groups) at once, then – another 30, and so on. **System Role:** Source, Proxy	SAP Marketing Cloud
`marketing.cloud.roles.prefix`	This property distinguishes SAP Marketing Cloud roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`SMKC_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the roles that are read from the SAP Marketing Cloud source system and will be provisioned to the target system with the following name pattern: **`SMKC_<role_name>`** . This way SAP Marketing Cloud roles in the target system will be distinguished from roles provisioned from other applications. If the property is not set, the SAP Marketing Cloud roles will be read and provisioned to the target system with their actual role names. • When **set in the target system**, only roles containing the **`SMKC_`** prefix in their role name will be provisioned to SAP Marketing Cloud. Roles without this prefix in the role name won't be provisioned. If the property is not set, all roles will be provisioned to SAP Marketing Cloud. **System Role:** Source and Target	SAP Marketing Cloud

Name	Description	System Type
`s4hana.cloud.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP S/4HANA Cloud Public Edition target or proxy system. See also: Extended Explanation of the *user.roles.overwrite Properties **Possible values:** • *true* – the current user roles will be deleted in the proxy system, and the user will be updated only with the roles provisioned by the service. • *false* – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the proxy system. Default value (if the property is missing during system creation): *true* Default value (if the property appears during system creation): *false* **System Role:** Target, Proxy	SAP S/4HANA Cloud Public Edition (using version 1 - SAP S/4HANA Cloud Public Edition API: Business User)
`ibp.skip.read.archived`	In the event of archived (disabled) entities in a source SAP IBP system, you can choose whether the provisioning jobs to continue reading such entities or to skip them. In the source systems, this property is activated by default. If you want to always read disabled entities, set the property to *false*, or delete it. **Possible values:** • *true* • *false* Default value: *true* **System Role:** Source, Target, Proxy	SAP Integrated Business Planning
`ibp.roles.filter`	Enter OData filtering for reading roles in the SAP IBP system. To learn what criteria you can use, see: OData URI Conventions → 4.5 Filter System Query Option **System Role:** Source, Proxy	SAP Integrated Business Planning (using version 1 - SAP Integrated Business Planning API: Business User)

Name	Description	System Type
`marketing.cloud.skip.read.archived`	In the event of archived (disabled) entities in a source SAP Marketing Cloud system, you can choose whether the provisioning jobs to continue reading such entities or to skip them. In the source and proxy systems, this property is activated by default. If you want to always read disabled entities, set the property to *false*, or delete it. **Possible values:** • *true* • *false* Default value: *true* **System Role:** Source, Target, Proxy	SAP Marketing Cloud
`marketing.cloud.roles.filter`	Enter OData filtering for reading roles in the SAP Marketing Cloud system. To learn what criteria you can use, see: OData URI Conventions → 4.5 Filter System Query Option **System Role:** Source, Proxy	SAP Marketing Cloud
`scim.api.csrf.protection`	Specifies whether to fetch a CSRF token when sending requests to the system. If enabled, this property requires the configuration of the `scim.csrf.token.path` property to specify the path where the CSRF token can be retrieved. **Possible values:** • *enabled* • *disabled* > **Note** > If the property is missing, its default value is **disabled**. **System Role:** Target, Proxy	SCIM System

Name	Description	System Type
`sac.api.csrf.protection`	Specifies whether to fetch a CSRF token when sending requests to the system. The property is automatically added in the system, with default value: *enabled*. **Possible values:** • *enabled* • *disabled* Default value: *enabled* **Note** If the property is missing, its default value is *disabled*. **System Role:** Target, Proxy	SAP Analytics Cloud
`sac.support.bulk.operation`	This property enables bulk operations for users. When bulk operations are enabled (set to *true*), Identity Provisioning service creates, updates, and deletes multiple users in one request. When bulk operations are not enabled (set to *false*), Identity Provisioning service creates, updates, and deletes one user at a time. For more information, see: SCIM Protocol: Bulk Operations **Note** SCIM bulk operations are not supported for provisioning groups to SAP Analytics Cloud. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP Analytics Cloud

Name	Description	System Type
`sac.bulk.operations.max.count`	If you have enabled the SCIM bulk operations, you can use this property to set the number of users to be provisioned per request. **Possible values:** Default value: • *100* - when using SAP Analytics Cloud SCIM API version 1. • *30* - when using SAP Analytics Cloud SCIM API version 2. **Note** The value must not exceed the number of entities defined by the SAP Analytics Cloud system as a SCIM service provider. Otherwise, the provisioning job will fail with HTTP response code 413 (*Payload Too Large*). **System Role:** Target	SAP Analytics Cloud
`scim.user.filter`	When specified, only those users matching the filter expression will be read. **Possible values:** For example: *name.familyName eq "SmithJ" and addresses.country eq "US"* **System Role:** Source	All SCIM-based systems

Name	Description	System Type
`scim.group.filter`	When specified, only those groups matching the filter expression will be read. **Possible values:** For example: *displayName eq "ProjectTeam1" or "Students2018"* **System Role:** Source	• Identity Authentication • Local Identity Directory • SAP Advanced Financial Closing • SAP Analytics Cloud • SAP Ariba Applications • SAP Ariba Central Invoice Management • SAP BTP Account Members (Neo) • SAP BTP Java/HTML5 apps (Neo) • SAP BTP XS Advanced UAA (Cloud Foundry) • SAP Build Work Zone, advanced edition • SAP Build Work Zone, standard edition • SAP Business Network • SAP Central Business Configuration • SAP Commerce Cloud • SAP CPQ • SAP Data Custodian • SAP Enable Now • SAP Enterprise Portal • SAP Fieldglass • SAP Field Service Management • SAP HANA Cloud Database Services • SAP Intelligent Agriculture • SAP Jam Collaboration • SAP Sales Cloud and SAP Service Cloud • SAP SuccessFactors version 2 • SAP SuccessFactors Employee Central Payroll • SAP SuccessFactors Incentive Management • Sales Cloud – Analytics & AI • Cloud Foundry UAA Server • Procurement Data Warehouse • SCIM System • SSH Server (Beta)

Name	Description	System Type
`scim.content.type`	Makes the connector send a specified value for the *Content-Type* HTTP header. This is needed because a SCIM system could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* If the property is not specified, the default value is taken: *application/scim+json* **System Role:** Target, Proxy	All SCIM-based systems

Name	Description	System Type
`scim.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). **If the service finds such a user on the target system via this filter, then the conflicting user will overwrite the existing one.** If the service does not find such a user, the creation will fail. According to your use case and system type, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. **Note** Relevant only for *Identity Authentication* and *SAP Analytics Cloud*: • For systems created **before** `April 7, 2020`, this property is missing during system creation, and it has the default value, *userName*. If the service does not find an existing user with such a *userName*, it will try again to resolve the conflicting user – by *email*. If the second attempt for resolution is unsuccessful too, the creation of the conflicting user fails. • For systems created **after** `April 7, 2020`, this property appears by default during system creation, and its value is set to *userName*. If the service does not find an existing user with such a *userName*, the creation of the conflicting user fails. However, if you delete the property, the service will try again to resolve the conflicting user – by *email*. If the second attempt for resolution is un-	All SCIM-based systems

Name	Description	System Type
	successful too, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value* • *userName,emails[0].value* • *externalId*, or another SCIM attribute, or a conjunction of SCIM attributes **Note** • *phoneNumbers[0].value* - supported unique attribute for Local Identity Directory (when Identity Provisioning is running on SAP Cloud Identity Infrastructure) Default value: *userName* **System Role:** Target	

Name	Description	System Type
`scim.group.unique.attribute`	If the service tries to create a group that already exists in the target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for the `scim.group.unique.attribute` property. If the property is not specified, the search is done by the default attribute: *displayName* **System Role:** Target	• Identity Authentication • Local Identity Directory • SAP Advanced Financial Closing • SAP Analytics Cloud • SAP Ariba Applications • SAP Ariba Central Invoice Management • SAP BTP Account Members (Neo) • SAP BTP Java/HTML5 apps (Neo) • SAP BTP XS Advanced UAA (Cloud Foundry) • SAP Build Work Zone, advanced edition • SAP Build Work Zone, standard edition • SAP Business Network • SAP Central Business Configuration • SAP Commerce Cloud • SAP CPQ • SAP Data Custodian • SAP Enable Now • SAP Enterprise Portal • SAP Fieldglass • SAP Field Service Management • SAP HANA Cloud Database Services • SAP Intelligent Agriculture • SAP Jam Collaboration • SAP Sales Cloud and SAP Service Cloud • SAP SuccessFactors version 2 • SAP SuccessFactors Employee Central Payroll • SAP SuccessFactors Incentive Management • Sales Cloud – Analytics & AI • Cloud Foundry UAA Server • Procurement Data Warehouse • SCIM System • SSH Server (Beta)

Name	Description	System Type
`scim.group.members.additional.attributes`	Defines additional attributes you can request from an **Identity Authentication** source system when reading groups. If you read groups through REST API, use the **GET** request. Add the additional attributes (coma-separated) as a value of the URL parameter `membersAdditionalAttributes`. **Possible values:** a coma-separated list of attribute names You can add the following attributes: • *emails* • *userName* • *displayName* • *urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:employeeNumber* **System Role:** Source	Identity Authentication (SCIM API version 1)
`scim.include.if.match.wildcard.header`	Makes the connector send the `If-Match` HTTP header with a value of "*" for every request to the target system. This header could be used by a SCIM system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	All SCIM-based systems

Name	Description	System Type
`scim.support.patch.operation`	If your target or proxy system is among the SCIM-based ones listed under *System Type* and supports PATCH operations, set this property to *true*. This way, when the Identity Provisioning identifies a changed entity in the source system, it will execute the updates as PATCH requests instead of PUT. That means, only the changes will be written in the target system, instead of provisioning the whole entity data. Note that only attributes without `"scope"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with scope, such as: `{ "constant": "xsuaa-dummy-value", "targetPath": "$.id", "scope": "createEntity" }` **Additional Information:** There are different cases when an entity should be updated in the target system: • In the source system, some of the entity attributes have been changed, or new attributes have been added. • In the source system, a condition or a filter is set for this entity not to be read anymore. • The whole entity has been deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false*	• Identity Authentication (SCIM API version 2) • Local Identity Directory • Cloud Foundry UAA Server • SAP BTP XS Advanced UAA (Cloud Foundry) • SCIM System • SAP Ariba Applications • SAP Ariba Category Management • SAP Advanced Workflow • SAP Business Network • SAP Commerce Cloud • SAP Build Work Zone, standard edition • SAP Ariba Central Invoice Management • SAP Data Custodian • SAP Datasphere • SAP Jam Collaboration • SAP SuccessFactors Employee Central Payroll • SAP Field Service Management • SAP HANA Cloud Database Services • SAP Intelligent Agriculture • SAP SuccessFactors (SCIM API version 2) • SAP SuccessFactors Learning • SAP Advanced Financial Closing • SAP Analytics Cloud • SAP Build Work Zone, advanced edition

Name	Description	System Type
	System Role: Target, Proxy	
`jam.group.prefix`	This property distinguishes SAP Jam Collaboration groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`SJC_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Jam Collaboration source system and will be provisioned to the target system with the following name pattern: **`SJC_<GroupDisplayName>`**. This way SAP Jam Collaboration groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Jam Collaboration groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`SJC_`** prefix in their display name will be provisioned to SAP Jam Collaboration. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Jam Collaboration. **System Role:** Source and Target	SAP Jam Collaboration
`scp.user.userbase`	This property specifies the host to the identity provider to be used with this target system. All provisioned users can be authenticated only by this identity provider. If you use another IdP, enter its value as configured in the SAP BTP cockpit. For example: `<account_ID>.accounts.ondemand.com` Default value: *accounts.sap.com* **System Role:** Target, Proxy	SAP BTP Account Members (Neo)

Name	Description	System Type
`AuthType`	Enter the type of authentication used for access token retrieval for OAuth HTTP destinations. **Possible values:** • *Basic* • *Form* Default value: *Basic* **System Role:** Source, Target, Proxy	• SCIM System • SAP Analytics Cloud • SAP SuccessFactors Incentive Management • SAP Jam Collaboration • Identity Authentication • Local Identity Directory • Cloud Foundry UAA Server • SAP BTP XS Advanced UAA (Cloud Foundry) • Sales Cloud – Analytics & AI • SAP BTP Account Members (Neo) • SAP Fieldglass • SAP Ariba Applications
`CloudConnectorLocationId`	Relevant when the `ProxyType` property is set to *OnPremise*. Use it in case your SAP Business Technology Platform account uses more than one Cloud Connector or when the Cloud Connector uses `LocationID`. **Possible values:** String **System Role:** Source, Target, Proxy	• SSH Server (Beta) • SAP HANA Database • LDAP Server • Microsoft Active Directory • SAP S/4HANA On-Premise • All HTTP systems
`hana.jdbc.db.user`	Name of the SAP HANA Database user **System Role:** Target	SAP HANA Database
`hana.jdbc.db.password`	(Credential) **System Role:** Target	SAP HANA Database
`hana.jdbc.db.host`	SAP HANA Database host **System Role:** Target	SAP HANA Database
`hana.jdbc.db.port`	SAP HANA Database port **Default value:** *30015* **System Role:** Target	SAP HANA Database

Name	Description	System Type
`hana.jdbc.db.encrypt`	This property enables or disables TLS encryption. Possible values: • **true** (Default value) • **false** **Note** For existing systems, the property value defaults to **false**. **System Role:** Target	SAP HANA Database
`hana.jdbc.db.validateCertificate`	When set to true, this property specifies that the server certificate is validated. When set to false, the certificate is not validated. Possible values: • **true** (Default value) • **false** **System Role:** Target	SAP HANA Database
`sales.cloud.analytics_ai.user.filter`	Use this property to filter users by specific criteria, according to the API syntax of SCAAI. **Possible values:** Text/numeric string For example: *externalId eq "John123"* **System Role:** Source, Proxy	Sales Cloud – Analytics & AI
`sales.cloud.analytics_ai.group.filter`	Use this property to filter groups by specific criteria, according to the API syntax of SCAAI. **Possible values:** Text/numeric string For example: *displayName eq "first_group"* **System Role:** Source, Proxy	Sales Cloud – Analytics & AI
`ssh.read.users.command`	Path to the bash command you need to execute to read users. **System Role:** Source	SSH Server (Beta)
`ssh.create.user.command`	Path to the bash command you need to execute to create a user. **System Role:** Source, Target	SSH Server (Beta)

Name	Description	System Type
`ssh.update.user.command`	Path to the bash command you need to execute to update a user. **System Role:** Source, Target	SSH Server (Beta)
`ssh.delete.user.command`	Path to the bash command you need to execute to delete a user. **System Role:** Source, Target	SSH Server (Beta)
`ssh.read.groups.command`	Path to the bash command you need to execute to read groups. **System Role:** Source	SSH Server (Beta)
`ssh.create.group.command`	Path to the bash command you need to execute to create a group. **System Role:** Source, Target	SSH Server (Beta)
`ssh.update.group.command`	Path to the bash command you need to execute to update a group. **System Role:** Source, Target	SSH Server (Beta)
`ssh.delete.group.command`	Path to the bash command you need to execute to delete a group. **System Role:** Source, Target	SSH Server (Beta)
`ssh.create.user.command.exit.code.already.exists`	An exit code number **System Role:** Source, Target	SSH Server (Beta)
`ssh.update.user.command.exit.code.not.found`	An exit code number **System Role:** Source, Target	SSH Server (Beta)
`ssh.delete.user.command.exit.code.not.found`	An exit code number **System Role:** Source, Target	SSH Server (Beta)
`ssh.create.group.command.exit.code.already.exists`	An exit code number **System Role:** Source, Target	SSH Server (Beta)
`ssh.update.group.command.exit.code.not.found`	An exit code number **System Role:** Source, Target	SSH Server (Beta)

Name	Description	System Type
`ssh.delete.group.command.exit.code.not.found`	An exit code number **System Role:** Source, Target	SSH Server (Beta)
`ssh.auth.type`	Supported SSH authentication types: • *key* • *pwd* • *otp* • *key+otp* • *key+pwd* • *pwd+otp* • *key+pwd+otp* **System Role:** Source, Target	SSH Server (Beta)
`ssh.host`	**System Role:** Source, Target	SSH Server (Beta)
`ssh.port`	**Possible values:** *22* **System Role:** Source, Target	SSH Server (Beta)
`ssh.username`	**System Role:** Source, Target	SSH Server (Beta)
`ssh.password`	(Credential) Taken into account only if the authentication type includes **pwd**. That means any of the following: • `ssh.auth.type` = *pwd* • `ssh.auth.type` = *pwd+otp* • `ssh.auth.type` = *key+pwd* • `ssh.auth.type` = *key+pwd+otp* **System Role:** Source, Target	SSH Server (Beta)
`ssh.totp.secret.key`	(Credential) Taken into account only if the authentication type includes **otp**. That means any of the following: • `ssh.auth.type` = *otp* • `ssh.auth.type` = *key+otp* • `ssh.auth.type` = *pwd+otp* • `ssh.auth.type` = *key+pwd+otp* **System Role:** Source, Target	SSH Server (Beta)

Name	Description	System Type
`ssh.private.key`	(Credential) Taken into account only if the authentication type includes **key**. That means any of the following: • `ssh.auth.type` = *key* • `ssh.auth.type` = *key+pwd* • `ssh.auth.type` = *key+otp* • `ssh.auth.type` = *key+pwd+otp* **System Role:** Source, Target	SSH Server (Beta)
`ssh.private.key.type`	The format of SSH private key. **Possible values:** • *ssh-rsa* • *ssh-dsa* Default value: *ssh-rsa* **System Role:** Source, Target	SSH Server (Beta)
`sf.page.size`	Use this property to configure the paging. That means, the number of entities to be read from SAP SuccessFactors at once. Default value: *100* **System Role:** Source, Proxy	SAP SuccessFactors (using version 1 - SAP SuccessFactors HCM Suite OData API)

Name	Description	System Type
`sf.group.filter`	The possible values of this property depend on the API version which your SAP SuccessFactors system consumes. Use this property to filter dynamic groups from SAP SuccessFactors. The filter obtains values as described in the OData 2.0 syntax, except any statements with attribute `lastModifiedDateTime`. To learn more, see: • OData version 2 → **4.5. Filter System Query Option ($filter)** • SAP SuccessFactors HCM Suite OData API → **DynamicGroup** • syntax, except anySAP SuccessFactors Workforce SCIM API **Possible values:** If your system consumes SAP SuccessFactors Workforce SCIM API, you can filter groups only by `displayName`. For example: *groupType eq 'permission'* **System Role:** Source, Proxy	• SAP SuccessFactors (using version 1 - SAP SuccessFactors HCM Suite OData API) • SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)

Name	Description	System Type
`sf.group.prefix`	This property distinguishes SAP SuccessFactors groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **SF_** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP SuccessFactors source system and will be provisioned to the target system with the following name pattern: **`SF_<GroupDisplayName>`**. This way SAP SuccessFactors groups in the target system will be distinguished from groups provisioned from other applications. • When **set in the target system**, only groups containing the **SF_** prefix in their display name will be provisioned to SAP SuccessFactors. Groups without this prefix in the display name won't be provisioned. If the property is not set, the SAP SuccessFactors groups will be read and provisioned to the target system with their actual display names. **System Role:** Source, Target	SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)

Name	Description	System Type
`sf.user.filter`	The possible values of this property depend on the API version which your SAP SuccessFactors system consumes. This property takes values as described in the OData version 2 `lastModifiedDateTime`. **⚠ Caution** Attribute `lastModifiedDateTime` is used internally by the Identity Provisioning service, for calculating the delta load from the SAP SuccessFactors system. You must not use it in your filter statements. If you do, your provisioning jobs will fail. **→ Tip** By default, only **active** users are read from SAP SuccessFactors. If you want to filter by another user status, you can set it in the value of this property, using either the *status value* or the *status text* parameters. See: SAP SuccessFactors HCM Suite OData API → **Querying Different Types of Users** **Possible values:** • If your system consumes SAP SuccessFactors Workforce SCIM API, you can use the supported filter attributes described in SAP SuccessFactors Workforce SCIM API For example: *userName eq "Sebastian"* • If your system consumes SAP SuccessFactors HCM Suite OData API, you can only use attributes supported as filterable by this API. See SAP SuccessFactors HCM Suite OData API. Some of these filterable attributes are: `firstName`, `lastName`, `department`, `division`, `jobCode`, `location`, `status`, `userId`, `username`. For example: *division eq 'Manufacturing (MANU)'* **System Role:** Source, Proxy	• SAP SuccessFactors (using version 1 - SAP SuccessFactors HCM Suite OData API) • SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)

Name	Description	System Type
`sf.user.attributes`	The value of this property is a comma-separated list of user attributes that have to be loaded from/to the SAP SuccessFactors system. Default value: *userId,username,status,email,lastName,firstName,lastModifiedDateTime,personKeyNav* SAP SuccessFactors supports a huge amount of user information, which requires a lot of memory processing time and may even lead to time-out errors. That's why we recommend that you keep the default list of attributes, or specify only a few (the most significant attributes) for your provisioning scenario. **Note** If you want to add more attributes, make sure you have added: • the relevant extra attributes to the value of this property, separated by commas • extra mappings for these attributes in the *user* transformation • extra mappings for these attributes in the write transformation of the relevant target system **Note** Always make sure that attribute `lastModifiedDateTime` is in the list. If you don't specify it, the provisioning from/to SAP SuccessFactors will fail. **System Role:** Source, Target, Proxy	SAP SuccessFactors (using version 1 - SAP SuccessFactors HCM Suite OData API)

Name	Description	System Type

`sf.user.attributes.expand`	This property reads/writes additional user data related to *complex (navigation)* attributes, which are specified in the `sf.user.attributes` property.
	
	Default value: *personKeyNav,personKeyNav/userAccountNav*
	
	For example: If you also need to read the *username* of the manager of a company employee, enter the following configuration in the *Properties* tab:
	
	`sf.user.attributes`
	= *username,firstName,lastName,manager/username*
	
	`sf.user.attributes.expand`
	= *personKeyNav,personKeyNav/userAccountNav,manager*
	
	System Role: Source, Target, Proxy
		SAP SuccessFactors (using version 1 - SAP SuccessFactors HCM Suite OData API)

Name	Description	System Type
`aad.user.attributes.expand`	This property allows you to expand the list of attributes specified in the `aad.user.attributes` property with additional user attributes. Once you provide the value of the additional attributes in the `aad.user.attributes.expand` property, you need to extend the read transformation of Microsoft Entra ID with attribute mappings based on the given value. Currently, the read transformation of Microsoft Entra ID is extended with the attribute mappings for manager **id** and **displayName** as follows: Code Syntax `{ "sourcePath": "$.manager.id", "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']['value']", "optional": true }, { "sourcePath": "$.manager.displayName", "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']['displayName']", "optional": true }`  To read the manager of the user, you need to provide the manager as a value of the `aad.user.attributes.expand` property in the following format: **`manager($select=id,displayName)`** For more information on the attributes (relationships) that support the **$expand** query parameter, refer to Microsoft Graph REST API v1.0 → *Relationships*. **System Role:** Source, Proxy	Microsoft Entra ID

Name	Description	System Type
`aad.group.attributes.expand`	This property allows you to expand the list of attributes specified in the `aad.group.attributes` property with additional group attributes. Once you provide the value of the additional attributes in the `aad.group.attributes.expand` property, you need to extend the read transformation of Microsoft Entra ID with attribute mappings based on the given value. For more information on the attributes (relationships) that support the **$expand** query parameter, refer to Microsoft Graph REST API v1.0 → *Relationships*. **System Role:** Source, Proxy	Microsoft Entra ID
`RemoteSystemID`	**Note** Only relevant to API **v.1**. Enter the system instance ID, configured for the communication system setting in the SAP Sales Cloud and SAP Service Cloud system. **Possible values:** For example: *IPS* **System Role:** Target	SAP Sales Cloud and SAP Service Cloud
`RecipientPartyID`	**Note** Only relevant to API **v.2**. Enter the recipient system name. **Possible values:** For example: *0011SAP* **System Role:** Target	SAP Sales Cloud and SAP Service Cloud

Name	Description	System Type
`SenderPartyID`	**Note** Only relevant to API **v.2**. Enter the name of the sender system name. It's equal to the value of property `RemoteSystemID` from API v.1. **Possible values:** For example: *IPS* **System Role:** Target	SAP Sales Cloud and SAP Service Cloud

Name	Description	System Type
`TrustAll`	Use this property when you create a connectivity destination in SAP BTP cockpit with authentication type `BasicAuthentication` to configure your provisioning system. Use cases: • If this property is not specified or set to *false*, you need to add a truststore certificate to check for SSL connections. You can either use the default JDK truststore, or provide a custom truststore certificate – if you use a custom domain instead of the default Identity Authentication one. For more information, see: Use Destination Certificates (Cockpit) Use Custom Domain in Identity Authentication • If the property is enabled (set to *true*), the server certificate will be ignored, thus – not checked for SSL connections. → Remember For productive scenarios, we recommend that you do not use this property (or set it to *false*) because the SSL server certificate cannot be verified, and thus the server is not authenticated. Enable the property only for testing purposes. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Source, Target, Proxy	All systems

Name	Description	System Type
`uaa.origin`	It denotes the `origin` attribute in the system transformation. The value of this property is the location of your Cloud Foundry identity provider. If not sure about the value, ask your Cloud Foundry system administrator. **Possible values:** Text/numeric string Source, Target, Proxy	Cloud Foundry UAA Server
`uaa.origin.filter.enabled`	This flag property depends on `uaa.origin`. If the flag is set to *true*, the Identity Provisioning service will read only users whose identity provider is set as a value of `uaa.origin`. **Possible values:** *true* or *false* - If set to *true*, the Identity Provisioning service will read only users whose identity provider is set as a value of `uaa.origin`. - If set to *false*, the Identity Provisioning service will read all users, regardless of their origin. - If set to *true* but the `uaa.origin` property is missing, the provisioning job will fail. **System Role:** Source, Proxy	Cloud Foundry UAA Server
`uaa.patch.response.with.resource`	Use this property if you want to retrieve a group whose membership was modified. **Note** This property is usable only when you have configured membership modifications via `Add/Remove Member` UAA endpoints. That is, when the `scim.support.patch.operation` property is set to *false*. **Possible values:** - *true* – the Identity Provisioning service will return the modified group via the `GET /Groups` endpoint of UAA. To learn how, see Retrieve . - *false* – no modified groups will be returned by the service. **System Role:** Proxy	Cloud Foundry UAA Server

Name	Description	System Type
`uaa.group.prefix`	This property distinguishes Cloud Foundry UAA Server groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **UAA_** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the Cloud Foundry UAA Server source system and will be provisioned to the target system with the following name pattern: **`UAA_<GroupDisplayName>`**. This way Cloud Foundry UAA Server groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the Cloud Foundry UAA Server groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **UAA_** prefix in their display name will be provisioned to Cloud Foundry UAA Server. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to Cloud Foundry UAA Server. **System Role:** Source and Target	Cloud Foundry UAA Server
`xsuaa.origin`	Holds the identity provider of a user in SAP BTP XS Advanced UAA (Cloud Foundry). You can find it in the SAP BTP cockpit. Go to your Cloud Foundry subaccount, choose *Trust Configuration* and see the value under *Origin Key*. For more information, see Configure Single and Multiple Origins [page 1051] **Possible values:** Text/numeric string For example: *myaccount-xsuaa.accounts.ondemand.com* **System Role:** Source, Target, Proxy	SAP BTP XS Advanced UAA (Cloud Foundry)

Name	Description	System Type
`xsuaa.origin.filter.enabled`	This flag property depends on `xsuaa.origin`. If the flag is set to *true*, the Identity Provisioning service will read only users whose identity provider is set as a value of `xsuaa.origin`. Possible values: *true* or *false* • If set to *true*, the Identity Provisioning service will read only users whose identity provider is set as a value of `xsuaa.origin`. • If set to *false*, the Identity Provisioning service will read all users, regardless of their origin. • If set to *true* but the `xsuaa.origin` property is missing, the provisioning job will fail. **System Role:** Source, Proxy	SAP BTP XS Advanced UAA (Cloud Foundry)
`xsuaa.patch.response.with.resource`	Use this property if you want to retrieve a group whose membership was modified. **Note** This property is usable only when you have configured membership modifications via *Add/Remove Member* UAA endpoints. That is, when the `scim.support.patch.operation` property is set to *false*. **Possible values:** • *true* – the Identity Provisioning service will return the modified group via the `GET /Groups` endpoint of UAA. To learn how, see Retrieve . • *false* – no modified groups will be returned by the service. **System Role:** Proxy	SAP BTP XS Advanced UAA (Cloud Foundry)
`workzone.user.filter`	When specified, only those SAP Build Work Zone, advanced edition users matching the filter expression will be read. **Possible values:** For example: *userName eq "SmithJ"* **System Role:** Source	SAP Build Work Zone, advanced edition

Name	Description	System Type
`workzone.group.filter`	When specified, only those SAP Build Work Zone, advanced edition groups matching the filter expression will be read. **Possible values:** For example: *displayName eq "ProjectTeam1"* **System Role:** Source, Proxy	SAP Build Work Zone, advanced edition
`workzone.group.prefix`	This property distinguishes SAP Build Work Zone, advanced edition groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: WORKZONE_ You can use the example value or provide your own. • When set in the source system, the prefix will be prepended to the name of the groups that are read from the SAP Build Work Zone, advanced edition source system and will be provisioned to the target system with the following name pattern: WORKZONE_GroupDisplayName. This way SAP Build Work Zone, advanced edition groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Build Work Zone, advanced edition groups will be read and provisioned to the target system with their actual display names. • When set in the target system, only groups containing the WORKZONE_ prefix in their display name will be provisioned to SAP Build Work Zone, advanced edition. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Build Work Zone, advanced edition. System Role: Source and Target	SAP Build Work Zone, advanced edition

Name	Description	System Type
`workzone.content.type`	This property makes a SAP Build Work Zone, advanced edition connector to send a specified value for the *Content-Type* HTTP header. This is needed because SAP Build Work Zone, advanced edition could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	SAP Build Work Zone, advanced edition

Name	Description	System Type
`workzone.support.patch.operation`	The default value of this property is *false*. But for SAP Build Work Zone, advanced edition proxy systems, this property appears during creation and its predefined value is *true*. That means, when the Identity Provisioning identifies a changed entity in the back-end system, it will execute the updates as PATCH requests instead of PUT. That is, only changes will be written in SAP Build Work Zone, advanced edition, instead of provisioning the whole entity data. **Additional Information:** There are different cases when an entity should be updated in the target system: • In the source system, some of the entity attributes have been changed, or new attributes have been added. • In the source system, a condition or a filter is set for this entity not to be read anymore. • The whole entity has been deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value for proxy systems: *true* Default value for target systems: *false* **System Role:** Target, Proxy	SAP Build Work Zone, advanced edition

Name	Description	System Type
`workzone.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in SAP Build Work Zone, advanced edition. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value* • *userName,emails[0].value* • *externalId*, or another SCIM attribute, or a conjunction of SCIM attributes Default value: *userName* **System Role:** Target	SAP Build Work Zone, advanced edition

Name	Description	System Type
`workzone.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists on the SAP Build Work Zone, advanced edition target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for this property. **Possible values:** Default value (when not specified): *displayName* If the property is not specified, the search is done by the default attribute: `displayName` **System Role:** Target	SAP Build Work Zone, advanced edition
`fsm.group.filter`	When specified, only those SAP Field Service Management groups matching the filter expression will be read. **Possible values:** For example: *displayName eq "ProjectTeam1"* **System Role:** Source, Proxy	SAP Field Service Management
`fsm.user.filter`	When specified, only those SAP Field Service Management users matching the filter expression will be read. **Possible values:** For example: *userName eq "SmithJ"* **System Role:** Source, Proxy	SAP Field Service Management
`fsm.content.type`	This property makes the SAP Field Service Management connector to send a specified value for the *Content-Type* HTTP header. This is needed because SAP Field Service Management could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	SAP Field Service Management

Name	Description	System Type
`fsm.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists in the SAP Field Service Management target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for this property. Default value (when not specified): *displayName* If the property is not specified, the search is done by the default attribute: `displayName`. **System Role:** Target	SAP Field Service Management
`fsm.include.if.match.wildcard.header`	Makes the SAP Field Service Management connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Field Service Management system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Field Service Management

Name	Description	System Type
`fsm.support.patch.operation`	The default value of this property is *false*. But for SAP Field Service Management proxy systems, this property appears during creation and its predefined value is *true*. That means, when the Identity Provisioning identifies a changed entity in the back-end system, it will execute the updates as PATCH requests instead of PUT. That is, only changes will be written in SAP Field Service Management, instead of provisioning the whole entity data. Note that only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". **Possible values:** Default value: *false* Predefined value (during system creation): *true* **System Role:** Target, Proxy	SAP Field Service Management

Name	Description	System Type
`fsm.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in SAP Field Service Management. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value* • *userName,emails[0].value* Default value: *userName* **System Role:** Target	SAP Field Service Management

Name	Description	System Type
`ias.api.version`	Defines the version of Identity Authentication SCIM API. **Possible values:** • *1* - Identity Authentication SCIM API is used. • *2* - Identity Directory SCIM API is used. Default value: *2* **System Role:** Source, Target, Proxy	Identity Authentication

Name	Description	System Type
`ias.user.filter`	This property filters users by attributes from the SCIM core schema, the Enterprise user resource schema and the Custom defined schema. For example: `userName`, `emails.value`, `addresses.country`, `employeeNumber`, `costCenter`, `department` and others. For more information on the attributes defined in the SCIM core schema and the Enterprise user resource schema, see Identity Directory Service Schema View You can set a single attribute or multiple ones as search criteria in the following value pattern: Single attribute: *<user_attribute> eq "<value>"* Multiple attributes: *<user_attribute1> eq "<value1>" and/or <user_attribute2> eq "<value2>"* **Possible values:** For example: • Single attribute: *userName eq "Sebastian"* • Multiple attributes (with **OR**): *userName eq "Sebastian" or addresses.country eq "France"* • Multiple attributes (with **AND**): *userName eq "Sebastian" and addresses.country eq "France"* • Multiple attributes (with brackets): *userName eq "Sebastian" or (addresses.country eq "France" and emails.value eq "sebastian123@mail.com")* • Multiple attributes (enterprise attributes): *urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:department eq "Dev" and urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:organization eq "Technology"* **System Role:** Source, Proxy	Identity Authentication (SCIM API version 2)

Name	Description	System Type
`ias.group.filter`	This property filters groups by display name. You can set a single display name or multiple ones as filter criteria. If you enter multiple display names (using **OR** operator), the filter will search for any of them. Single attribute: *displayName eq "<group_name>"* Multiple attributes: *displayName eq "<group_name1>" or displayName eq "<group_name2>"* **Possible values:** For example: • Single attribute: *displayName eq "FellowshipTeam1"* • Multiple attributes: *displayName eq "FellowshipTeam1" or displayName eq "JuniorTest3"* **System Role:** Source, Proxy	Identity Authentication (SCIM API version 2)

Name	Description	System Type
`ias.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with "scope": "createEntity", such as: `{ "constant":true, "targetPath":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mailVerified']", "scope":"createEntity" }` • If set to *false*, Identity Provisioning sends a PUT request to the user or group resource in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. Users and groups can be updated in the target system in various cases, such as: • In the source system, some user or group attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users or groups not to be read anymore. • A user or a group is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**.	Identity Authentication (SCIM API version 2)

Name	Description	System Type
	Possible values: • *true* • *false* Default value: *false* **System Role:** Target, Proxy	
`ias.user.groups.paging.enabled`	This property enables paging of user's groups. The maximum number of user's groups returned per request is 1 000. To read more than 1 000 user's groups, paging must be enabled. **Possible values:** • *true* - Paging is enabled. You can read more than 1 000 user's groups in one request. • *false* - Paging is disabled. You can read up to 1 000 user's groups in one request. Default value: *false* **System Role:** Source, Proxy	Identity Authentication (SCIM API version 2)
`ias.group.members.paging.enabled`	This property enables paging of group members. The maximum number of group members returned per request is 20 000. To read more than 20 000 group members, paging must be enabled. **Possible values:** • *true* - Paging is enabled. You can read more than 20 000 group members in one request. • *false* - Paging is disabled. You can read up to 20 000 group members in one request. Default value: *false* **System Role:** Source, Proxy	Identity Authentication (SCIM API version 2)

Name	Description	System Type
`ias.include.if.match.wildcard.header`	Makes the connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by a SCIM system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	Identity Authentication (SCIM API version 2)

Name	Description	System Type
`ias.user.unique.attribute`	When Identity Provisioning attempts to provision a user for the first time, it may detect that this user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user will be searched and resolved. **If the service finds a user on the target system via this filter, then the conflicting user will overwrite the existing one.** If the service does not find a user on the target system via this filter, the creation will fail. According to your use case and system type, choose how to set up this property: • Default behavior: This property is set during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *phoneNumbers[0].value*. If the service finds an existing user with such *phoneNumber*, it updates this user with the data of the conflicting one. If a user with such *phoneNumber* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName, phoneNumbers[0].value*. If the service finds an existing user with both these *userName* and *phoneNumber*, it updates this user with the data of the conflicting one. If such a user is not found, that	Identity Authentication (SCIM API version 2)

Name	Description	System Type
	means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName, emails[0].value, phoneNumbers[0].value*. If the service finds an existing user with these *userName*, *email* and *phoneNumber*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value* • *userName,emails[0].value* • *phoneNumbers[0].value* • *userName, phoneNumbers[0].value* • *userName, emails[0].value, phoneNumbers[0].value* • *externalId*, or another SCIM attribute, or a conjunction of SCIM attributes Default value: *userName* **System Role:** Target	

Name	Description	System Type
`ias.group.unique.attribute`	If you try to provision a group that already exists in a target system, the group creation will fail. In such cases, the existing group only needs to be updated. This property defines the unique attribute(s) by which the existing group will be searched and resolved. The default value is *['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']*. When set, you can expect the following behavior: • If a group with the given *['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']* is found in the target system, the group that you try to provision will overwrite the existing one. • If a group with the given *['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']* is not found in the target system, the group that you try to provision will not be created in the target system. In scenarios where groups are provisioned from SAP Integrated Business Planning version 2 and SAP S/4HANA Cloud Public Edition version 3, the uniqueness of the group is resolved by a combination of three attributes: *['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type'],* *['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName'],* *['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']* For the conflict to be resolved, an existing group must match all three attributes. If the property is not specified, the search is done by the default attribute: *['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']* **System Role:** Target	Identity Authentication (SCIM API version 2)

Name	Description	System Type
`ias.content.type`	Makes the connector send a specified value for the *Content-Type* HTTP header. This is needed because a SCIM system could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* If the property is not specified, the default value is taken: *application/scim+json* **System Role:** Target, Proxy	Identity Authentication (SCIM API version 2)

Name	Description	System Type
`aad.user.filter.group.filter.combine`	Filters Microsoft Entra ID users based on their group assignments. When set to *true*, this property combines user and group filters defined on the `aad.user.filter` and `aad.group.filter` properties to further narrow the search results. This way, only users that meet the following filtering criteria are returned: • Users that match the user filter and at the same time are members of groups that match the group filter. • Members of the filtered groups that match the user filter. **Note** To make the `aad.user.filter.group.filter.combine` property work, ensure that both user and group entities are read, that is, neither of them is ignored in the transformation code. **Example** 1. You have the following users, located in two cities with one or more assigned groups: **User**: David Thompson from London with assigned **Groups**: Marketing and Sales **User**: Julie Armstrong from New York with assigned **Groups**: Employee **User**: John Smith from New York with assigned **Groups**: Marketing and Sales 2. You have defined the following filtering criteria: `aad.user.filter` = *city eq "New York"* `aad.group.filter` = *displayName eq “Marketing”* `aad.user.filter.group.filter.combine` = *true* 3. You get the following result: Only user John Smith is returned as it matches	Microsoft Entra ID

Name	Description	System Type
	the user filter and at the same time is a member of the group that matches the group filter. Although David Thompson matches the group filter, he doesn't match the user filter. Although Julie Armstrong matches the user filter, shed doesn't match the group filter. When set to *false*, user and group filters are not combined. For more information, see: Identity Provisioning: How to Get Users Based on Group Assignments from MS Entra ID **Possible values:** • *true* When this property is set to **true**, it is expected that filtering criteria are defined for `aad.user.filter` and `aad.group.filter` properties. If one or both are not defined in the UI, be aware of the following behavior: • Only `aad.user.filter` is defined: Users that match the user filter and are members of any group will be returned. If a user matches the user filter but is not a member of any group, this user will not be returned. • Only `aad.group.filter` is defined: Users that are members of the groups matching the group filter will be returned. • None of the properties are defined: Users that are members of any group will be returned. When this property is set to *true* and filtering criteria are defined for `aad.user.filter` and `aad.group.filter` properties, if you are searching for a specific user or group using Identity Provisioning service API, be aware of the following behavior: • When searching for specific user with `GET .../Users/UserId` request,	

Name	Description	System Type
	filtering criteria defined on both properties are not considered. The user is returned with all the groups he or she is a member of. • When searching for specific group with `GET ...Groups/GroupId` request, filtering criteria defined on both properties are not considered. The group is returned with all its group members. • *false* - default value **System Role**: Source, Proxy	
`s4hana.pp.user.filter`	When specified, only those SAP S/4HANA for procurement planning users matching the filter expression will be read. **Possible values:** Example: *name.familyName eq "Smith" and addresses.country eq "US"* **System Role:** Source, Proxy	SAP S/4HANA for procurement planning
`s4hana.pp.content.type`	Makes the connector send a specified value for the *Content-Type* HTTP header. This is needed because a SCIM system could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* If the property is not specified, the default value is taken: *application/scim+json* **System Role:** Target, Proxy	SAP S/4HANA for procurement planning
`s4hana.pp.include.if.match.wildcard.header`	Makes the connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by a SCIM system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP S/4HANA for procurement planning

Name	Description	System Type
`s4hana.pp.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). **If the service finds such a user on the target system via this filter, then the conflicting user will overwrite the existing one.** If the service does not find such a user, the creation will fail. According to your use case and system type, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value* • *userName,emails[0].value* • *externalId*, or another SCIM attribute, or a conjunction of SCIM attributes Default value: *userName* **System Role:** Target	SAP S/4HANA for procurement planning

Name	Description	System Type

`ias.user.automatic.conflict.resolution`	Controls whether automatic conflict resolution is switched on or off in Identity Authentication (target system) when provisioning is triggered from source systems containing different users with the same user identifiers (IDs).
	
	For example, when SAP SuccessFactors and SAP SuccessFactors Learning are configured as source systems for provisioning users to Identity Authentication, it could happen that different SAP SuccessFactors and SAP SuccessFactors Learning users have the same user IDs. In this case, when the first user is created in Identity Authentication, after triggering a provisioning job, the second (conflicting) user will either overwrite the already existing one (automatic conflict resolution is switched on) or will fail and won't be created (automatic conflict resolution is switched off).
	
	To control this behavior, you can use the `ias.user.automatic.conflict.resolution` property in the target Identity Authentication system. This property is not added by default.
	
	Possible values:
	
	• *true* - If the property is set to true, or is not set at all, the automatic conflict resolution is switched on. This means that Identity Provisioning takes into account the unique attribute(s) defined on the `scim.user.unique.attribute` property (when using SCIM API version 1) or `ias.user.unique.attribute` property (when using SCIM API version 2) and tries to find an already existing user in Identity Authentication matching these attributes.
	  • If a user is found, the provisioning of a new (conflicting) user is resolved as follows: the conflicting user overwrites the existing one.
	  • If a user is not found, the provisioning of a conflicting user fails, and it is not created in Identity Authentication.
		Identity Authentication

Name	Description	System Type
	• *false* - If the property is set to false, the automatic conflict resolution is switched off. This means that Identity Provisioning does not take into account the unique attribute(s) defined on the `scim.user.unique.attribute` property (when using SCIM API version 1) or `ias.user.unique.attribute` property (when using SCIM API version 2) and fails the provisioning of a conflicting user. This user is not created in Identity Authentication and does not overwrite the existing one. In the Job Log, an error code `409, uniqueness` will be displayed. Default value: *true* **System Role:** Target	

Name	Description	System Type
`ias.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with PATCH requests, and below which they are provisioned with PUT request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. The expected value is a positive integer without any separators, such as a space, comma, or period. **Note** You can use this property when Identity Authentication is based on Identity Directory SCIM API (in short, SCIM API version 2). **Possible values**: integer Default value: *20000* Minimum value: *1* Maximum value: *200000* For example: • PATCH requests: If you have a group with 700 members and you update the group by adding another 1 200 members, setting this property to 900 results in the following: As 1900 (the target count of the members) is above the threshold number of 900, 2 PATCH requests will be sent to the Identity Authentication target system. The first request will add 900 group members and the second request will add 300 group members. The threshold number you set defines the maximum number of group members processed per batch. • PUT request: If you have a group with 700 members and you update the group by adding another 100 members, setting this property to 900 results in the following: As 800 (the target count of the members) is below the threshold number of 900, 1 PUT request with 800 group members will	Identity Authentication

Name	Description	System Type
	be sent to the Identity Authentication target system to update the group. **Note** If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members. **Note** Regardless of the threshold number you define, when removing group members in Identity Authentication, the maximum number of members which can be removed per one **PATCH** request is 50. **System Role:** Target	

Name	Description	System Type
`idds.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with PATCH requests, and below which they are provisioned with PUT request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. The expected value is a positive integer without any separators, such as a space, comma, or period. The expected value is a positive integer without any separators, such as a space, comma, or period. **Note** You can use this property when Identity Authentication and Identity Provisioning (where Local Identity Directory is configured), are running on the same infrastructure, that is, the infrastructure of Identity Authentication. **Possible values**: integer Default value: *20000* Minimum value: *1* Maximum value: *200000* For example: • PATCH requests: If you have a group with 700 members and you update the group by adding another 1 200 members, setting this property to 900 results in the following: As 1900 (the target count of the members) is above the threshold number of 900, 2 PATCH requests will be sent to the Local Identity Directory target system. The first request will add 900 group members and the second request will add 300 group members. The threshold number you set defines the maximum number of group members processed per batch. • PUT request: If you have a group with 700 members and you update the group by add-	Local Identity Directory

Name	Description	System Type
	ing another 100 members, setting this property to 900 results in the following: As 800 (the target count of the members) is below the threshold number of 900, 1 **PUT** request with 800 group members will be sent to the Local Identity Directory target system to update the group. **Note** If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members. **Note** Regardless of the threshold number you define, when removing group members in Local Identity Directory, the maximum number of members which can be removed per one **PATCH** request is 50. **System Role:** Target	

Name	Description	System Type
`xsuaa.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with PATCH requests, and below which they are provisioned with PUT request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. The expected value is a positive integer without any separators, such as a space, comma, or period. **Possible values**: integer Default value: *20000* Minimum value: *1* Maximum value: *200000* For example: • PATCH requests: If you have a group with 700 members and you update the group by adding another 1 200 members, setting this property to 900 results in the following: As 1 900 (the target count of the members) is above the threshold number of 900, 2 PATCH requests will be sent to the XSUAA target system. The first request will add 900 group members and the second request will add 300 group members. The threshold number you set defines the maximum number of group members processed per batch. • PUT request: If you have a group with 700 members and you update the group by adding another 100 members, setting this property to 900 results in the following: As 800 (the target count of the members) is below the threshold number of 900, 1 PUT request with 800 group members will be sent to the XSUAA target system to update the group. **Note** If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members.	SAP BTP XS Advanced UAA (Cloud Foundry)

Name	Description	System Type
	System Role: Target	

Name	Description	System Type
`ias.user.update.instead.delete`	When using SCIM API version 2, this property allows you to update user attributes with PATCH request in Identity Authentication target system and to preserve the user record instead of deleting it. This behavior is supported only when the scope of the attribute is set to **deleteEntity**. In addition to configuring this property, you also need to adapt the write transformation. For example, if you want to disable a user account in Identity Authentication, you need to do the following: 1. Set `ias.user.update.instead.delete` =*true* 2. Adapt the write transformation as follows: `{ "user":{ "mappings":[{ "constant":"urn:ietf:params:scim:api:messages:2.0:PatchOp", "targetPath":"$.schemas[0]", "scope":"deleteEntity" }, { "constant":"replace", "targetPath":"$.Operations[0].op", "scope":"deleteEntity" }, { "constant":"active", "targetPath":"$.Operations[0].path", "scope":"deleteEntity" }, { "constant":false, "targetPath":"$.Operations[0].value", "scope":"deleteEntity"`	Identity Authentication

Name	Description	System Type

```
            },
...
```

In this case, the PATCH operation will replace *true* with *false* as the value of the `active` user attribute. As a result, when the PATCH operation is executed, the user record in the target system will no longer be managed by Identity Provisioning as it is considered deleted.

For more information, see: Transformation Expressions [page 391] → *Scope* → *deleteEntity* → *Identity Authentication (SCIM API version 2)*

Possible values:

- *true*
- *false*

Default value: *false*

When the property is set to *true*, adapt the write transformation with the attribute name and the attribute value you want to update:

```
{
   "user":{
      "mappings":[
         {

"constant":"urn:ietf:params:sci
m:api:messages:2.0:PatchOp",

"targetPath":"$.schemas[0]",

"scope":"deleteEntity"
         },
         {

"constant":"replace",

"targetPath":"$.Operations[0].o
p",

"scope":"deleteEntity"
         },
         {

"constant":"<attribute_name>",

"targetPath":"$.Operations[0].p
ath",

"scope":"deleteEntity"
         },
         {

"constant":<attribute_value>,
```

Name	Description	System Type

	<pre>"targetPath":"$.Operations[0].value",
	"scope":"deleteEntity"
	},
	...</pre>
	System Role: Target	

`lms.user.filter`	When specified, only those users matching the filter expression will be read. **Possible values:** • *userName eq "testName"* • *externalID eq "testID"* • *active eq "true"* • *sourceSystem eq "Learning"* - indicates that the user is created directly in SAP SuccessFactors Learning with no involvement of Identity Provisioning. • *sourceSystem eq "Identity Provisioning"* - indicates that the user is created in SAP SuccessFactors Learning by Identity Provisioning. **System Role:** Source	SAP SuccessFactors Learning
`lms.content.type`	This property makes the SAP SuccessFactors Learning connector to send a specified value for the *Content-Type* HTTP header. This is needed because SAP SuccessFactors Learning could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	SAP SuccessFactors Learning

Name	Description	System Type
`lms.user.unique.attribute`	When Identity Provisioning attempts to provision a user for the first time, it may detect that such user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). If the service finds such user on the target system via this filter, then the conflicting user will overwrite the existing one. If the service does not find such a user, the creation will fail. **Default behavior**: The property is missing during system creation. Its default value is *userName*. This means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such *userName* is not found, the creation of the conflicting user fails. **Possible values:** Default value: *userName* **System Role:** Target	SAP SuccessFactors Learning
`lms.include.if.match.wildcard.header`	Makes the connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by a SCIM system for entity versioning. **Possible values:** • *true* • *false* **System Role:** Target, Proxy	SAP SuccessFactors Learning

Name	Description	System Type
`lms.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP SuccessFactors Learning

Name	Description	System Type
`lms.instance.host`	Enter the host of your SAP SuccessFactors Learning instance. This property must be configured if you what to use client certificate authentication for the communication between Identity Provisioning and SAP SuccessFactors Learning. **System Role:** Source, Target, Proxy	SAP SuccessFactors Learning
`ias.support.bulk.operation`	This property enables bulk operations for users and groups. When bulk operations are enabled, Identity Provisioning creates, updates, and deletes multiple users and groups in one request. When bulk operations are not enabled, Identity Provisioning creates, updates, and deletes one user at a time. For more information, see: Identity Directory SCIM API. **Possible values:** • *true* - bulk operations are enabled • *false* - bulk operations are not enabled Default value: *false* **System Role:** Target	Identity Authentication (using SCIM API version 2)
`ias.bulk.operations.max.count`	This property sets the number of operations to be performed in one bulk request. **Possible values:** Default value: *20* Maximum value: *100* If you enter a number larger than 100, Identity Provisioning will replace it with the default value (20). **System Role:** Target	Identity Authentication (using SCIM API version 2)

Name	Description	System Type
`idds.support.bulk.operation`	This property enables bulk operations for users and groups. When bulk operations are enabled, Identity Provisioning creates, updates, and deletes multiple users and groups in one request. When bulk operations are not enabled, Identity Provisioning creates, updates, and deletes one user at a time. For more information, see: Identity Directory SCIM API. **Possible values:** • *true* - bulk operations are enabled • *false* - bulk operations are not enabled Default value: *false* **System Role:** Target	Local Identity Directory
`idds.bulk.operations.max.count`	This property sets the number of operations to be performed in one bulk request. **Possible values:** Default value: *20* Maximum value: *100* If you enter a number larger than 100, Identity Provisioning will replace it with the default value (20). **System Role:** Target	Local Identity Directory

Name	Description	System Type
`concur.user.filter`	When specified, only those users matching the filter expression will be read. **Possible values:** Examples when SAP Concur connector is configured for version 2: • *userName eq "johnsmith@example.com"* As the userName must be unique across SAP Concur, this filter returns only the user matching this userName. • *companyId eq "aa067ada-71a9-4f57-8e98-9300b1c3171d"* This filter returns all users in the company with this companyId. • *externalId eq "0fe44868-31a7-4930-9ah30-757tg2513b64"* This filter returns a user with the specified value, that is, the userUUID generated for the user in Identity Authentication. • *employeeNumber eq "Concur Administrator"* This filter returns a user with the specified employee number. The employeeNumber could also be a number having six or more digits. Examples when SAP Concur connector is configured for version 3: • *userName eq "JohnSmith"* • *userName sw "John"* • *externalId eq "0fe44868-31a7-4930-9ah30-757tg2513b64"* • *urn:ietf:params:scim:schemas:extension:enterprise:2.0:employeeNumber eq "00102345"* **System Role:** Source, Proxy	SAP Concur (using SAP Concur Identity v4 API)

Name	Description	System Type
`concur.content.type`	Makes the connector send a specified value for the *Content-Type* HTTP header. This is needed because a SCIM system could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* If the property is not specified, the default value is taken: *application/scim+json* **System Role:** Target, Proxy	SAP Concur (using SAP Concur Identity v4 API)
`concur.include.if.match.wildcard.header`	Makes the connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by a SCIM system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Concur (using SAP Concur Identity v4 API)

Name	Description	System Type
`concur.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). **If the service finds such a user on the target system via this filter, then the conflicting user will overwrite the existing one.** If the service does not find such a user, the creation will fail. You can set any combination of the supported unique attributes, separated by commas: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value=*userName,urn:ietf:params:scim:schemas:extension:enterprise:2.0:employeeNumber*. If the service finds an existing user with both these userName and employeeNumber, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values** for version 2: • *userName* • *externalId* or another SCIM attribute, or a conjunction of SCIM attributes **Possible values** for version 3: • *userName* • *externalId* • *urn:ietf:params:scim:schemas:extension:enterprise:2.0:employeeNumber* Default value: *userName* **System Role:** Target	SAP Concur (using SAP Concur Identity v4 API and SAP Concur SCIM API)

Name	Description	System Type
`concur.datacenter`	Specify the SAP Concur data center your Identity Provisioning tenant belongs to. The following SAP Concur data centers are available: • APJ1 • EU2 • US2 **System Role:** Source, Target, Proxy	SAP Concur
`concur.api.version`	Defines the version of SAP Concur API. **Possible values:** • *1* - User v1 API is used. • *2* - Identity v4 API is used. • *3* - SCIM v4 API is used. Default value: *3* **System Role:** Source, Target, Proxy	SAP Concur

Name	Description	System Type
`concur.authorization.code`	(Credential) Enter the *Company Request Token* and run a provisioning job within 24 hours from generating the token in the SAP Concur Company Request Token self-service tool. Otherwise, the token will expire, and you'll need a new one. After the first run of the job, Identity Provisioning fills in automatically a refresh token as the value of the concur.refresh.token property. If a provisioning job has not been run for six months, you'll again need to generate a new token. → Remember The company request token has a 24 hour validity. If this token expires, you must request a new token. The refresh token has a six month validity. Every time you run a provisioning job, the validity of the refresh token is extended with six months starting from the date of the last run. If you haven't run a provisioning job for six months, your refresh token will expire and you must request a new company request token. The Company Request Token is generated in the SAP Concur Company Request Token self-service tool. **System Role:** Source, Target, Proxy	SAP Concur
`concur.company.id`	Your company UUID The Company ID is generated in the SAP Concur Company Request Token self-service tool. **System Role:** Target, Proxy	SAP Concur

Name	Description	System Type
`concur.company.domain`	Your company domain The username and the company domain are concatenated in the SAP Concur default transformations in the following format: user@domain Your company domain is the part of your username behind the @ symbol. For example: *johnsmith@example.com* **System Role:** Target, Proxy	SAP Concur
`ariba.applications.patch.group.members.of.nested.groups`	If you set this property to *true*, Identity Provisioning will update only user members of a group in SAP Ariba Applications target system. The update will be executed on batches via PATCH requests. This will preserve the group hierarchy with nested groups in the SAP Ariba Applications backend. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP Ariba Applications

Name	Description	System Type
`ariba.applications.patch.group.members.above.threshold`	This property is relevant only when `ariba.applications.patch.group.members.of.nested.groups` is set to *true*. It defines the maximum number of user members of a group that are included in one PATCH request. If the maximum value of 200 000 is exceeded, the system sets automatically the default value. The expected value is a positive integer without any separators, such as a space, comma, or period. **Possible values**: integer Default value: *50* Minimum value: *1* Maximum value: *50* **Note** If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members. **System Role:** Target	SAP Ariba Applications
`cflp.providerId`	Your SAP Build Work Zone, standard edition provider ID The provider ID is specified in the Channel Manager of theSAP Build Work Zone, standard edition when defining a new content provider. For more information about configuring the content provider to use the Identity Provisioning service, see Configure Integration with the Identity Provisioning Service **Possible values:** The value of your SAP Build Work Zone, standard edition provider ID For example: *ABC123* **System Role:** Target, Proxy	SAP Build Work Zone, standard edition

Name	Description	System Type
`cflp.user.filter`	When specified, only those SAP Build Work Zone, standard edition users matching the filter expression will be read. By default, users are always filtered by the *providerId*. If another filtering attribute is defined, for example the email of the user, both filters are combined. **Possible values:** • *emails.value eq "john.smith@example.com"* > **Note** > Although, the email is supported as a filtering attribute, it is not returned when searching for the user. • *urn:ietf:params:scim:schemas:extension:2.0:mapping.providerId eq "ABC123"* Combining filter attributes is allowed using the and operator, for example: *emails.value eq "john.smith@example.com" and urn:ietf:params:scim:schemas:extension:2.0:mapping.providerId eq "ABC123"* **System Role:** Proxy	SAP Build Work Zone, standard edition
`cflp.group.filter`	When specified, only those SAP Build Work Zone, standard edition groups matching the filter expression will be read. By default, groups are always filtered by the *providerId*. **Possible values:** • *externalId eq 12345678* • *urn:ietf:params:scim:schemas:extension:2.0:mapping.providerId eq 'ABC123'* • *meta.isIAG eq true* This filtering attribute indicates whether the group will be used in a hybrid scenario with SAP Cloud Identity Access Governance. **System Role:** Proxy	SAP Build Work Zone, standard edition

Name	Description	System Type
`cflp.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. **Possible values:** SAP Build Work Zone, standard edition supports the following unique attributes which are automatically filled in when the target system is added in the service UI: **`emails[0].value, ['urn:ietf:params:scim:schemas:extension:2.0:mapping'] ['providerId'],externalId`** **Note** SAP Build Work Zone, standard edition, only supports **`externalId`** of type UUID. - If the user has an **`externalId`**, the conflict is resolved by **`externalId`** and **`providerId`**. - If the user doesn't have an **`externalId`**, the conflict is resolved by **`email`** and **`providerId`**. For the conflict to be resolved, an existing user matching both unique attributes should be found. Whether the existing user will be updated or a new one will be created by Identity Provisioning depends on the following conditions - was the user created by the Identity Provisioning service, what combination of unique attributes exist for the user, and has any of the provisioning systems (source or target) been reset. For more information, see Step 4 of the relevant SAP Build Work Zone, standard edition documentation. **→ Recommendation** We recommend that you do not modify the value of the `cflp.user.unique.attribute` property. Otherwise, user creation fails.	SAP Build Work Zone, standard edition

Name	Description	System Type
	System Role: Target	
`cflp.group.unique.attribute`	If Identity Provisioning tries to provision a group that already exists in the target system (a conflicting group), this property defines the unique attributes by which the existing group will be searched and resolved. **Possible values:** SAP Build Work Zone, standard edition supports a pair of unique attributes which is automatically filled in when the target system is added in the service UI: *externalId, ['urn:ietf:params:scim:schemas:extension:2.0:mapping']['providerId']* For the conflict to be resolved, an existing group matching both unique attributes should be found. In this case, Identity Provisioning updates the group. This means, the conflicting group overwrites the existing one. If the group matches only one of the unique attributes, the conflict is not resolved, and the group creation fails. → Recommendation We recommend that you do not modify the value of the `cflp.group.unique.attribute` property. Otherwise, the group creation fails. **System Role:** Target	SAP Build Work Zone, standard edition

Name	Description	System Type
`cflp.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with PATCH requests, and below which they are provisioned with PUT request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. The expected value is a positive integer without any separators, such as a space, comma, or period. **Possible values:** integer Default and maximum value: *5000* Minimum value: *1* For example: • PATCH requests: If you have a group with 700 members and you update the group by adding another 1 200 members, setting this property to 900 results in the following: As 1 900 (the target count of the members) is above the threshold number of 900, 2 PATCH requests will be sent to the SAP Build Work Zone, standard edition target system. The first request will add 900 group members and the second request will add 300 group members. The threshold number you set defines the maximum number of group members processed per batch. • PUT request: If you have a group with 700 members and you update the group by adding another 100 members, setting this property to 900 results in the following: As 800 (the target count of the members) is below the threshold number of 900, 1 PUT request with 800 group members will be sent to the SAP Build Work Zone, standard edition target system to update the group. **Note** If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members.	SAP Build Work Zone, standard edition

Name	Description	System Type
	Note If the maximum value of *5000* is exceeded, the system will automatically use the default value. **System Role:** Target	
`cflp.support.bulk.operation`	This property enables bulk operations for users and groups. When the property is enabled (set to *true*), the following operations can be executed by Identity Provisioning service in a single request: • Create or delete multiple users • Create, update, or delete multiple groups When the property is disabled (set to *false*), the following operations can be executed by Identity Provisioning service for a single entity at a time: • Create or delete a user • Create, update, or delete a group **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP Build Work Zone, standard edition
`cflp.bulk.operations.max.count`	This property sets the number of operations to be performed in one bulk request. **Possible values:** Default value: *20* Maximum value: *100* If you enter a number larger than 100, the service will replace it with the default value (20). **System Role:** Target	SAP Build Work Zone, standard edition

Name	Description	System Type
`xsuaa.user.unique.attribute`	When Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). If the service finds such a user on the target system via this filter, then the conflicting user will overwrite the existing one. If the service does not find such a user, the creation will fail. According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both attributes: *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,origin*. If the service finds an existing user with both attributes: *userName* and *origin*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value*	SAP BTP XS Advanced UAA (Cloud Foundry)

Name	Description	System Type
	• *userName,emails[0].value* • *userName,origin* • *externalId*, or another SCIM attribute, or a conjunction of SCIM attributes Default value: *userName* **System Role:** Target	
`fsm.group.prefix`	This property distinguishes SAP Field Service Management groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`FSM_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Field Service Management source system and will be provisioned to the target system with the following name pattern: **`FSM_<GroupDisplayName>`**. This way SAP Field Service Management groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Field Service Management groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`FSM_`** prefix in their display name will be provisioned to SAP Field Service Management. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Field Service Management. **System Role:** Source and Target	SAP Field Service Management

Name	Description	System Type
`sf.user.read.deactivatedafter`	This property filters SAP SuccessFactors inactive users from a particular date on. It is an optional property which does not appear by default at system creation. It accepts a value in the **`yyyy-MM-dd`** format. For example: **`2023-07-17`** The `sf.user.read.deactivatedafter` property is supported for SAP SuccessFactors version 2 using SAP SuccessFactors Workforce SCIM API. It works together with the `sf.user.filter` property which is added at system creation with the default value: *active eq true*. Using it can further narrow down the filtering results. To filter active users along with inactive ones from a particular date on, the following configuration must be in place: • Set the `sf.user.read.deactivatedafter` value to a date in the expected format. For example: *2023-07-17* • `sf.user.filter` = *active eq true* As a result, Identity Provisioning reads SAP SuccessFactors active users and the users set to inactive from that date on using the 2023-07-17T00:00:00Z date-time format. Depending on the value you define for `sf.user.filter`, expect the following results: • `sf.user.filter` = *active eq false* All inactive users will be returned. • `sf.user.filter` = *active eq false and userName sw "Test_"* All inactive users with username starting with Test_ will be returned. **Note** When you filter by `sf.user.filter` = *active eq false* along with the property `sf.user.read.deactivatedafter`, the users that match the two critera will be read twice.	SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)

Name	Description	System Type
	• `sf.user.filter` = *active eq true and userName sw "Test_"* Inactive users from the provided date on and all active users with username starting with Test_ will be returned. **System Role:** Source, Proxy	
`cc.user.filter`	When specified, only those users matching the filter expression will be read. You can filter users by **`userName`**, **`emails.value`**, and **`externalId`**, according to the API syntax of SAP Commerce Cloud. **Possible values:** text/ numeric string For example: • *userName eq "johnbrown" and externalId eq "P000252"* • *userName eq "johnbrown" and emails.value eq "johnbrown@email.com"* • *userName eq "johnbrown" and emails.value eq "johnbrown@email.com" and externalId eq "P000252"* > **Note** > These combinations are valid for both 'or' and 'and' operators. **System Role:** Source, Proxy	SAP Commerce Cloud
`cc.group.filter`	When specified, only those groups matching the filter expression will be read. **Possible values:** For example: *displayName eq "ProjectTeam1" or "Students2018"* **System Role:** Source, Proxy	SAP Commerce Cloud

Name	Description	System Type
`cc.group.prefix`	This property distinguishes SAP Commerce Cloud groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`CC_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Commerce Cloud source system and will be provisioned to the target system with the following name pattern: **`CC_<GroupDisplayName>`**. This way SAP Commerce Cloud groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Commerce Cloud groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`CC_`** prefix in their display name will be provisioned to SAP Commerce Cloud. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Commerce Cloud. **System Role:** Source, Target	SAP Commerce Cloud

Name	Description	System Type
`cc.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Note** When executing a remove operation, all members of a group are removed without considering the number of deleted group members in the source system. **Additional Information:** There are different cases when an entity should be updated in the target system: • In the source system, some of the entity attributes have been changed, or new attributes have been added. • In the source system, a condition or a filter is set for this entity not to be read anymore. • The whole entity has been deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of	SAP Commerce Cloud

Name	Description	System Type
	your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value for proxy systems: *true* Default value for target systems: *false* **System Role:** Proxy, Target	
`cc.include.if.match.wildcard.header`	Makes the SAP Commerce Cloud connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Commerce Cloud system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Commerce Cloud

Name	Description	System Type
`cc.group.unique.attribute`	If you try to provision a group that already exists in a target system, the group creation will fail. In this case, the existing group only needs to be updated. This property defines by which unique attribute(s) the existing group will be searched and resolved. The default value is *displayName*. Currently, it is the only unique attribute that is supported. When set, you can expect the following behavior: • If a group with the given *displayName* is found in the target system, the group that you try to provision will overwrite the existing one. • If a group with the given *displayName* is not found in the target system, the group that you try to provision will not be created in the target system. If the property is not specified, the search is done by the default attribute: `displayName` **System Role:** Target	SAP Commerce Cloud

Name	Description	System Type
`cc.user.unique.attribute`	When Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). If the service finds such a user on the target system via this filter, then the conflicting user will overwrite the existing one. If the service does not find such a user, the creation will fail. The property is automatically added during system creation. If the service finds an existing user by at least one of the uniqueness criteria, which are *email*, *userName*, or *externalId*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the update of the conflicting user fails. If more than one users with these unique attributes are found, the update fails. **Possible values:***emails[0].value, userName, externalId* Default value: *emails[0].value, userName, externalId* **System Role:** Target	SAP Commerce Cloud
`cc.content.type`	Makes the connector send a specified value for the *Content-Type* HTTP header. This is needed because a SCIM system could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* If the property is not specified, the default value is taken: *application/scim+json* **System Role:** Target, Proxy	SAP Commerce Cloud

Name	Description	System Type
`cc.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with PATCH requests, and below which they are provisioned with PUT request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. The expected value is a positive integer without any separators, such as a space, comma, or period. **Note** You can use this property when SAP Commerce Cloud is based on SAP Commerce Cloud SCIM API (in short, SCIM API version 2). **Possible values**: integer Default value: *20000* Minimum value: *1* Maximum value: *200000* For example: • PATCH requests: If you have a group with 700 members and you update the group by adding another 1 200 members, setting this property to 900 results in the following: As 1900 (the target count of the members) is above the threshold number of 900, 2 PATCH requests will be sent to the SAP Commerce Cloud target system. The first request will add 900 group members and the second request will add 300 group members. The threshold number you set defines the maximum number of group members processed per batch. • PUT request: If you have a group with 700 members and you update the group by adding another 100 members, setting this property to 900 results in the following: As 800 (the target count of the members) is below the threshold number of 900, 1 PUT request with 800 group members will be	SAP Commerce Cloud

Name	Description	System Type
	sent to the SAP Commerce Cloud target system to update the group. **Note** If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members. **Note** Regardless of the threshold number you define, when removing group members in SAP Commerce Cloud, the maximum number of members which can be removed per one **PATCH** request is 98. **System Role:** Target	
`cc.patch.group.members.of.nested.groups`	If you set this property to *true*, Identity Provisioning will update only user members of a group in SAP Commerce Cloud target system. The update will be executed on batches via PATCH requests. This will preserve the group hierarchy with nested groups in the SAP Commerce Cloud backend. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP Commerce Cloud

Name	Description	System Type
`maco.roles.prefix`	This property distinguishes SAP Market Communication for Utilities roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`SMC_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the roles that are read from the SAP Market Communication for Utilities source system and will be provisioned to the target system with the following name pattern: **`SMC_<role_name>`** . This way SAP Market Communication for Utilities roles in the target system will be distinguished from roles provisioned from other applications. If the property is not set, the SAP Market Communication for Utilities roles will be read and provisioned to the target system with their actual role names. • When **set in the target system**, only roles containing the **`SMC_`** prefix in their role name will be provisioned to SAP Market Communication for Utilities. Roles without this prefix in the role name won't be provisioned. If the property is not set, all roles will be provisioned to SAP Market Communication for Utilities. **System Role:** Source and Target	SAP Market Communication for Utilities

Name	Description	System Type
`maco.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP Market Communication for Utilities target or proxy system. See also: Extended Explanation of the *user.roles.overwrite Properties **Possible values:** • *true* – the current user roles will be deleted in the proxy system, and the user will be updated only with the roles provisioned by the service. • *false* – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the proxy system. Default value (if the property is missing during system creation): *true* Default value (if the property appears during system creation): *false* **System Role:** Target, Proxy	SAP Market Communication for Utilities
`maco.roles.filter`	Enter OData filtering for reading roles in the SAP Market Communication for Utilities system. To learn what criteria you can use, see: OData URI Conventions → 4.5 Filter System Query Option **System Role:** Source, Proxy	SAP Market Communication for Utilities
`maco.roles.page.size`	This property indicates how many business roles (considered as *groups*) per page to be read from your SAP Market Communication for Utilities source system. **Possible values:** Integer number For example, if you set the property's value = *30*, the Identity Provisioning will read 30 roles (groups) at once, then – another 30, and so on. **System Role:** Source, Proxy	SAP Market Communication for Utilities

Name	Description	System Type
`maco.skip.read.archived`	In the event of archived (disabled) entities in a source SAP Market Communication for Utilities system, you can choose whether the provisioning jobs to continue reading such entities or to skip them. In the source systems, this property is activated by default. If you want to always read disabled entities, set the property to *false*, or delete it. **Possible values:** • *true* • *false* Default value: *true* **System Role:** Source, Target, Proxy	SAP Market Communication for Utilities
`maco.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: APIs for Business User Management **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP Market Communication for Utilities
`maco.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of users to be provisioned per request. **Possible values:** Default value: *20* Maximum value: *100* If you enter a number larger than 100, the service will replace it with the default value (20). **System Role:** Target	SAP Market Communication for Utilities

Name	Description	System Type
`sf.api.version`	Handles the version of the API which is consumed by the SAP SuccessFactors system. **Possible values:** • *1* - SAP SuccessFactors HCM Suite OData API (in short, OData API) is used. • *2* - SAP SuccessFactors Workforce SCIM API (in short, SCIM API) is used. Default value: *1* **System Role:** Source, Target, Proxy	• SAP SuccessFactors (using version 1 - SAP SuccessFactors HCM Suite OData API) • SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)
`sf.company.id`	Enter the Company ID of your SAP SuccessFactors system. The Company ID is a short string of characters that identifies each SAP SuccessFactors system. It is like a username for your organization. All users of the same system share the same Company ID. This property must be configured if you what to use client certificate authentication for the communication between Identity Provisioning and SAP SuccessFactors. **System Role:** Source, Target, Proxy	• SAP SuccessFactors (using version 1 - SAP SuccessFactors HCM Suite OData API) • SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)

Name	Description	System Type
`sf.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a `PATCH` request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, `PUT` operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. **Note** Updating the user attributes: `name` and `manager`, is not supported with PATCH operation. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false*	SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)

Name	Description	System Type
	System Role: Target	
`sf.include.if.match.wildcard.header`	Makes the connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by a SCIM system for entity versioning. **Possible values:** • *true* • *false* **System Role:** Target, Proxy	SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)
`sf.group.unique.attribute`	If the service tries to create a group that already exists in the target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for the `sf.group.unique.attribute` property. If the property is not specified, the search is done by the default attribute: *displayName* **System Role:** Target	SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)

Name	Description	System Type
`sf.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). **If the service finds such a user on the target system via this filter, then the conflicting user will overwrite the existing one.** If the service does not find such a user, the creation will fail. According to your use case and system type, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value* • *userName,emails[0].value* • *externalId*, or another SCIM attribute, or a conjunction of SCIM attributes Default value: *userName* **System Role:** Target	SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)

Name	Description	System Type
`sf.content.type`	This property makes the SAP SuccessFactors connector to send a specified value for the *Content-Type* HTTP header. This is needed because SAP SuccessFactors could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	SAP SuccessFactors (using version 1 - SAP SuccessFactors HCM Suite OData API)

Name	Description	System Type
`sf.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with PATCH requests, and below which they are provisioned with PUT request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. The expected value is a positive integer without any separators, such as a space, comma, or period. **Note** You can use this property when SAP SuccessFactors is based on SAP SuccessFactors Workforce SCIM API (in short, SCIM API). **Possible values**: integer Default value: *20000* Minimum value: *1* Maximum value: *200000* For example: • PATCH requests: If you have a group with 700 members and you update the group by adding another 1 200 members, setting this property to 900 results in the following: As 1 900 (the target count of the members) is above the threshold number of 900, 2 PATCH requests will be sent to the SAP SuccessFactors target system. The first request will add 900 group members and the second request will add 300 group members. The threshold number you set defines the maximum number of group members processed per batch. • PUT request: If you have a group with 700 members and you update the group by adding another 100 members, setting this property to 900 results in the following: As 800 (the target count of the members) is below the threshold number of 900, 1 PUT request with 800 group members will be sent to the SAP SuccessFactors target system to update the group.	SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)

Name	Description	System Type
	Note If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members. **System Role:** Target	
`sf.group.members.paging.enabled`	This property enables paging of group members. The maximum number of group members returned per request is 100. To read more than 100 group members, paging must be enabled. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *true* **System Role:** Source, Proxy	SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)
`sf.patch.group.members.of.nested.groups`	If you set this property to *true*, Identity Provisioning will update only user members of a group in SAP SuccessFactors target system. The update will be executed on batches via PATCH requests. This will preserve the group hierarchy with nested groups in the SAP SuccessFactors backend. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP SuccessFactors (using version 2 - SAP SuccessFactors Workforce SCIM API)

Name	Description	System Type
`sac.group.prefix`	This property distinguishes SAP Analytics Cloud groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`SAC_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Analytics Cloud source system and will be provisioned to the target system with the following name pattern: **`SAC_<GroupDisplayName>`**. This way SAP Analytics Cloud groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Analytics Cloud groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`SAC_`** prefix in their display name will be provisioned to SAP Analytics Cloud. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Analytics Cloud. **System Role:** Source and Target	SAP Analytics Cloud

Name	Description	System Type
`cpq.group.prefix`	This property distinguishes SAP CPQ groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`CPQ_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP CPQ source system and will be provisioned to the target system with the following name pattern: **`CPQ_<GroupDisplayName>`**. This way SAP CPQ groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP CPQ groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`CPQ_`** prefix in their display name will be provisioned to SAP CPQ. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP CPQ. **System Role:** Source and Target	SAP CPQ
`cpq.group.members.paging.enabled`	This property enables the complete retrieval of group members through additional sequential requests. The maximum number of group members returned per request is 100 as defined in `cpq.group.members.page.size`. For more information, see `cpq.group.members.page.size`. Possible Values: • *true* - Paging is enabled. • *false* - Paging is disabled. Default Value: *true* **System Role:** Source, Proxy	SAP CPQ

Name	Description	System Type
`cpq.group.members.page.size`	When the `cpq.group.members.paging.enabled` property is set to true, this setting defines the size of the paged members request. **Note** If you configure `cpq.group.members.page.size` to a number greater than the default value, the complete member retrieval is disabled. Default Value: *100* The number of group members that SAP CPQ returns per request when reading a group. **System Role:** Source, Proxy	SAP CPQ
`cpq.user.filter`	When specified, only those SAP CPQ users matching the filter expression will be read. Example: **name.familyName eq "Smith" and addresses.country eq "US"** **System Role:** Source, Proxy	SAP CPQ

Name	Description	System Type
`scp.group.prefix`	This property distinguishes SAP BTP Account Members (Neo) groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`SCP_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP BTP Account Members (Neo) source system and will be provisioned to the target system with the following name pattern: **`SCP_<GroupDisplayName>`**. This way SAP BTP Account Members (Neo) groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP BTP Account Members (Neo) groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`SCP_`** prefix in their display name will be provisioned to SAP BTP Account Members (Neo). Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP BTP Account Members (Neo). **System Role:** Source and Target	SAP BTP Account Members (Neo)

Name	Description	System Type
`c4c.user.filter`	When specified, only those SAP Sales Cloud and SAP Service Cloud, formerly known as SAP Cloud for Customer (in short, C4C) users matching the filter expression will be read. Supported user filters in connector version 4: • userName eq "Smith" • userName sw "Sm" • userName eq "Smith" and emails.value eq "smith@example.com" Supported user filter in connector version 3: • addresses.country eq "USA" **System Role:** Source, Proxy	SAP Sales Cloud and SAP Service Cloud
`c4c.group.filter`	When specified, only those SAP Sales Cloud and SAP Service Cloud groups matching the filter expression will be read. SAP Sales Cloud and SAP Service Cloud is formerly known as SAP Cloud for Customer (in short, C4C). Example: *displayName eq "ProjectTeam1"* **System Role:** Source, Proxy	SAP Sales Cloud and SAP Service Cloud

Name	Description	System Type
`c4c.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with PATCH requests, and below which they are provisioned with PUT request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. The expected value is a positive integer without any separators, such as a space, comma, or period. **Possible values**: integer Default value: *1000* Minimum value: *1* Maximum value: *1000* For example: • PATCH requests: If you have a group with 700 members and you update the group by adding another 1 200 members, setting this property to 900 results in the following: As 1 900 (the target count of the members) is above the threshold number of 900, 2 PATCH requests will be sent to the SAP Sales Cloud and SAP Service Cloud target system. The first request will add 900 group members and the second request will add 300 group members. The threshold number you set defines the maximum number of group member changes processed per batch. • PUT request: If you have a group with 1 000 members and the threshold is not set or its value is above *1000*, one PUT request will be sent to SAP Sales Cloud and SAP Service Cloud. But if you set the threshold to 500, 2 PATCH requests will be sent instead, each adding 500 members. **Note** If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members.	SAP Sales Cloud and SAP Service Cloud

Name	Description	System Type
	Note If the maximum value of *1000* is exceeded, the system will automatically use the default value. **System Role:** Target	
`xsuaa.group.prefix`	This property distinguishes SAP BTP XS Advanced UAA (Cloud Foundry) groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **XSUAA_** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP BTP XS Advanced UAA (Cloud Foundry) source system and will be provisioned to the target system with the following name pattern: **`XSUAA_<GroupDisplayName>`**. This way SAP BTP XS Advanced UAA (Cloud Foundry) groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP BTP XS Advanced UAA (Cloud Foundry) groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **XSUAA_** prefix in their display name will be provisioned to SAP BTP XS Advanced UAA (Cloud Foundry). Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP BTP XS Advanced UAA (Cloud Foundry). **System Role:** Source and Target	SAP BTP XS Advanced UAA (Cloud Foundry)

Name	Description	System Type
`cflp.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Note** Updating users and groups returns 404 Not Found. Users and groups can be updated in the target system in various cases, such as: • In the source system, some user or group attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users or groups not to be read anymore. • A user or a group is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**.	SAP Build Work Zone, standard edition

Name	Description	System Type
	Possible values: • *true* • *false* Default value for proxy systems: *true* Default value for target systems: *false* **System Role:** Target, Proxy	
`sac.api.version`	Defines the version of SAP Analytics Cloud SCIM API. **Possible values:** • *1* - SAP Analytics Cloud SCIM API version 1 is used. • *2* - SAP Analytics Cloud SCIM API version 2 is used. Default value: *1* **System Role:** Source, Target, Proxy	SAP Analytics Cloud

Name	Description	System Type
`sac.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. **Note** You can use this property when SAP Analytics Cloud is based on SCIM API version 2. - If set to *true*, PATCH operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. - If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** - *true* - *false* Default value: *false* **System Role:** Target, Proxy	SAP Analytics Cloud

Name	Description	System Type
`sac.include.if.match.wildcard.header`	This property makes the SAP Analytics Cloud connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. The header could be used by an SAP Analytics Cloud system for entity versioning. **Note** You can use this property when SAP Analytics Cloud is based on SCIM API version 2. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Analytics Cloud
`sac.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists in the SAP Analytics Cloud target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for this property. **Note** You can use this property when SAP Analytics Cloud is based on SCIM API version 2. Default value: *displayName* If the property is not specified, the search is done by the default attribute: `displayName`. **System Role:** Target	SAP Analytics Cloud

Name	Description	System Type
`sac.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in SAP Analytics Cloud. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value* • *userName,emails[0].value* Default value: *userName* **System Role:** Target	SAP Analytics Cloud

Name	Description	System Type
`sac.content.type`	Makes the connector send a specified value for the *Content-Type* HTTP header. This is needed because a SCIM system could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Note** You can use this property when SAP Analytics Cloud is based on SCIM API version 2. **Possible values:** For example: *application/json* If the property is not specified, the default value is taken: *application/scim+json* **System Role:** Target, Proxy	SAP Analytics Cloud
`sac.user.filter`	When specified, only those SAP Analytics Cloud users matching the filter expression will be read. **Note** You can use this property when SAP Analytics Cloud is based on SCIM API version 2. **Possible values:** For example: *userName eq "SmithJ"* **System Role:** Source	SAP Analytics Cloud
`sac.group.filter`	When specified, only those SAP Analytics Cloud groups matching the filter expression will be read. **Note** You can use this property when SAP Analytics Cloud is based on SCIM API version 2. **Possible values:** For example: *displayName eq "ProjectTeam1"* **System Role:** Source, Proxy	SAP Analytics Cloud

Name	Description	System Type
`sac.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with PATCH requests, and below which they are provisioned with PUT request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. The expected value is a positive integer without any separators, such as a space, comma, or period. **Note** You can use this property when SAP Analytics Cloud is based on SCIM API version 2. **Possible values**: integer Default value: *20000* Minimum value: *1* Maximum value: *200000* For example: • PATCH requests: If you have a group with 700 members and you update the group by adding another 1 200 members, setting this property to 900 results in the following: As 1900 (the target count of the members) is above the threshold number of 900, 2 PATCH requests will be sent to the SAP Analytics Cloud target system. The first request will add 900 group members and the second request will add 300 group members. The threshold number you set defines the maximum number of group members processed per batch. • PUT request: If you have a group with 700 members and you update the group by adding another 100 members, setting this property to 900 results in the following: As 800 (the target count of the members) is below the threshold number of 900, 1 PUT request with 800 group members will be sent to the SAP Analytics Cloud target system to update the group.	SAP Analytics Cloud

Name	Description	System Type
	Note If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members. **Note** Regardless of the threshold number you define, when removing group members in SAP Analytics Cloud, the maximum number of members which can be removed per one **PATCH** request is 98. **System Role:** Target	
`sac.patch.group.members.of.nested.groups`	If you set this property to *true*, Identity Provisioning will update only user members of a group in SAP Analytics Cloud target system. The update will be executed on batches via PATCH requests. This will preserve the group hierarchy with nested groups in the SAP Analytics Cloud backend. **Note** You can use this property when SAP Analytics Cloud is based on SCIM API version 2. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP Analytics Cloud
`ep.user.filter`	When specified, only those SAP Enterprise Portal users matching the filter expression will be read. For more information, see Filtering.	SAP Enterprise Portal
`ep.group.filter`	When specified, only those SAP Enterprise Portal groups matching the filter expression will be read. For more information, see Filtering.	SAP Enterprise Portal

Name	Description	System Type
`bn.user.filter`	When specified, only those SAP Business Network users matching the filter expression will be read. **Possible values:** • *userName eq "Julie Armstrong"* • *userName sw "J"* • *name.familyName eq "Armstrong"* • *emails eq "julie.armstrong@example.com"* **System Role:** Source	SAP Business Network
`bn.group.filter`	When specified, only those SAP Business Network groups matching the filter expression will be read. **Possible values:** *displayName eq "Employees"* **System Role:** Source, Proxy	SAP Business Network
`bn.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. The property is not added automatically at system creation. Default value: *userName* If the service finds an existing user by userName, it updates this user with the data of the conflicting one. If the service does not find an existing user by userName, the creation of the conflicting user fails. **System Role**: Target	SAP Business Network
`bn.include.if.match.wildcard.header`	Makes the connector send the `If-Match` HTTP header with a value of "*" for every request to the target system. This header could be used by a SCIM system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Business Network

Name	Description	System Type
`bn.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set `"scope": "createEntity"`. • If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Note** Updating a group by removing all its members doesn't work. When executing a remove operation, groups must have at least one member. Users and groups can be updated in the target system in various cases, such as: • In the source system, some user or group attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users or groups not to be read anymore. • A user or a group is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity scope` in the transformation of your target or	SAP Business Network

Name	Description	System Type
	proxy system. See:Transformation Expressions [page 391] → *deleteEntity*. **Possible values:** • *true* • *false* Default value for proxy systems: *true* Default value for target systems: *false* **System Role**: Target, Proxy	
`bn.content.type`	This property makes SAP Business Network connector to send a specified value for the *Content-Type* HTTP header. This is needed because SAP Business Network could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	SAP Business Network
`bn.api.key`	An API Key represents the unique key that identifies a particular application as a legitimate consumer of an API. **System Role:**Source, Target, Proxy	SAP Business Network
`bn.realm.id`	The realm name is part of the URL you use to access SAP Business Network. **System Role:**Source, Target, Proxy	SAP Business Network

Name	Description	System Type
`abap.host.timezone`	Specifies the time zone of SAP Application Server ABAP on-premise systems. The value is used for calculating the correct assignments validity in case your SAP AS ABAP and Identity Provisioning tenant are running in different time zones. **Possible values:** The value should be provided in the following format: **`UTC+/- offset`**. For example: **`UTC+02:00`**, **`UTC-04:00`**, **`UTC+03:30`**. Internet Assigned Numbers Authority (IANA) Time Zone database format is also supported. For more information, see RFC 6557: Procedures for Maintaining the Time Zone Database . **System Role:** Source	SAP Application Server ABAP

Name	Description	System Type
`pdw.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in Procurement Data Warehouse. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. Default value: **userName** Possible values: • **emails[0].value** • **userName,emails[0].value** **Note** If this property is missing, or you've deleted it, and the service does not find such a *userName*, it will try again to resolve the conflicting user – by *email*. If the second attempt for resolution is unsuccessful too, the creation of the conflicting user fails. **System Role:** Target	Procurement Data Warehouse

Name	Description	System Type
`pdw.group.prefix`	This property distinguishes procurement data warehouse groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`PDW_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the procurement data warehouse source system and will be provisioned to the target system with the following name pattern: **`PDW_<GroupDisplayName>`**. This way procurement data warehouse groups in the target system will be distinguished from groups provisioned from other applications. • When **set in the target system**, only groups containing the **`PDW_`** prefix in their display name will be provisioned to procurement data warehouse. Groups without this prefix in the display name won't be provisioned. If the property is not set, the procurement data warehouse groups will be read and provisioned to the target system with their actual display names. **System Role:** Source, Target	Procurement Data Warehouse
`pdw.support.bulk.operation`	Set this property to *true* if you want to enable SCIM bulk operations for provisioning users. That means, the Identity Provisioning service can write, update, and delete a potentially large collection of users in a single request. For more information, see: SCIM Protocol: Bulk Operations If not specified, the default value is *false*. **Note** SCIM bulk operations are not supported for provisioning groups to procurement data warehouse. **System Role:** Target	Procurement Data Warehouse

Name	Description	System Type
`pdw.bulk.operations.max.count`	If you have enabled the SCIM bulk operations, you can use this property to set the number of users to be provisioned per request. Default value: **`20`** **Note** The value must not exceed the number of entities defined by the procurement data warehouse system as a SCIM service provider. Otherwise, the provisioning job will fail with HTTP response code 413 (*Request Entity Too Large*). **System Role:** Target	Procurement Data Warehouse
`pdw.user.filter`	When specified, only those procurement data warehouse users matching the filter expression will be read. For example: - *userName eq "Julie Armstrong"* - *userName sw "J"* - *emails eq "julie.armstrong@example.com"* **System Role:** Source	Procurement Data Warehouse
`pdw.content.type`	Makes the connector send a specified value for the *Content-Type* HTTP header. This is needed because a SCIM system could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* If the property is not specified, the default value is taken: *application/scim+json* **System Role:** Target, Proxy	Procurement Data Warehouse

Name	Description	System Type
`s4hana.afc.group.filter`	When specified, only those SAP Advanced Financial Closing groups matching the filter expression will be read. Supported operators: **eq** (equal), **sw** (starts with) and **co** (contains) **Possible values:** For example: *displayName eq "Administrators"* **System Role:** Source, Proxy	SAP Advanced Financial Closing
`s4hana.afc.user.filter`	When specified, only those SAP Advanced Financial Closing users matching the filter expression will be read. Supported operators: **eq** (equal), **sw** (starts with) and **co** (contains) **Possible values:** For example: • *userName eq "Julie Armstrong"* • *emails eq "julie.armstrong@example.com"* • *name.familyName sw "A"* • *name.givenName co "Ju"* **System Role:** Source, Proxy	SAP Advanced Financial Closing
`s4hana.afc.content.type`	This property makes a SAP Advanced Financial Closing connector to send a specified value for the *Content-Type* HTTP header. This is needed because SAP Advanced Financial Closing could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	SAP Advanced Financial Closing

Name	Description	System Type
`s4hana.afc.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists on the SAP Advanced Financial Closing target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for this property. Default value: *displayName* If the property is not specified, the search is done by the default attribute: *displayName* **System Role:** Target	SAP Advanced Financial Closing
`s4hana.afc.include.if.match.wildcard.header`	Makes the SAP Advanced Financial Closing connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Advanced Financial Closing system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Advanced Financial Closing

Name	Description	System Type
`s4hana.afc.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Advanced Financial Closing

Name	Description	System Type
`s4hana.afc.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP Advanced Financial Closing target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. According to your use case, choose how to set up this property: • Default behavior: This property does not appear in the UI during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. Possible values: • *userName* • *emails[0].value* • *userName,emails[0].value* Default value: *userName* **System Role:** Target	SAP Advanced Financial Closing

Name	Description	System Type
`s4hana.afc.group.prefix`	This property distinguishes SAP Advanced Financial Closing groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`AFC_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Advanced Financial Closing source system and will be provisioned to the target system with the following name pattern: **`AFC_<GroupDisplayName>`**. This way SAP Advanced Financial Closing groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Advanced Financial Closing groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`AFC_`** prefix in their display name will be provisioned to SAP Advanced Financial Closing. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Advanced Financial Closing. **System Role:** Source and Target	SAP Advanced Financial Closing

Name	Description	System Type
`s4hana.pp.group.prefix`	**Note** SAP S/4HANA for procurement planning does not support groups. Therefore, configuring the `s4hana.pp.group.prefix` property is irrelevant. This property distinguishes the groups of a given provisioning system by specific prefix that you provide. It is optional and does not appear by default at system creation. • When **set in the source system**, the prefix is prepended to the name of the groups and they are provisioned to the target system with the following name pattern: **`<YOUR_PREFIX>_<GroupDisplayName>`**. This way those groups are distinguished from groups provisioned from other applications. If the property is not set, the groups are read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing **`YOUR_PREFIX_`** prefix in their display name are provisioned there. Groups without this prefix in the display name are not provisioned. If the property is not set, all groups are provisioned to the target system. **System Role:** Source and Target	SAP S/4HANA for procurement planning

Name	Description	System Type
`s4hana.cloud.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP S/4HANA Cloud Public Edition target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. **Note** You can configure this property only when the integration between a human resource (HR) system and SAP S/4HANA Cloud Public Edition target system is **OFF**. According to your use case, choose how to set up this property: • Default behavior: This property does not appear in the UI during system creation. Its default value is *personExternalID*. That means, if the service finds an existing user by a *personExternalID* , it updates this user with the data of the conflicting one. If a user with such a *personExternalID* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user matching both unique attributes *email* and *personExternalID*, it updates this user with the data of the conflicting one. If the service finds an existing user matching only the *email*, the update of the existing user fails. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. Possible values: • *personExternalID* • *emails[0].value* Default value: *personExternalID* **System Role:** Target	SAP S/4HANA Cloud Public Edition

Name	Description	System Type
`s4hana.onprem.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP S/4HANA On-Premise target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. According to your use case, choose how to set up this property: • Default behavior: This property does not appear in the UI during system creation. Its default value is *personExternalID*. That means, if the service finds an existing user by a *personExternalID*, it updates this user with the data of the conflicting one. If a user with such a *personExternalID* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user matching both unique attributes *email* and *personExternalID*, it updates this user with the data of the conflicting one. If the service finds an existing user matching only the *email*, the update of the existing user fails. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. Possible values: • *personExternalID* • *emails[0].value* Default value: *personExternalID* **System Role:** Target	SAP S/4HANA On-Premise

Name	Description	System Type
`ibp.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP Integrated Business Planning target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. According to your use case, choose how to set up this property: • Default behavior: This property does not appear in the UI during system creation. Its default value is *personExternalID*. That means, if the service finds an existing user by a *personExternalID* , it updates this user with the data of the conflicting one. If a user with such a *personExternalID* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user matching both unique attributes *email* and *personExternalID*, it updates this user with the data of the conflicting one. If the service finds an existing user matching only the *email*, the update of the existing user fails. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. Possible values: • *personExternalID* • *emails[0].value* Default value: *personExternalID* **System Role:** Target	SAP Integrated Business Planning

Name	Description	System Type
`maco.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP Market Communication for Utilities target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. According to your use case, choose how to set up this property: • Default behavior: This property does not appear in the UI during system creation. Its default value is *personExternalID*. That means, if the service finds an existing user by a *personExternalID* , it updates this user with the data of the conflicting one. If a user with such a *personExternalID* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user matching both unique attributes *email* and *personExternalID*, it updates this user with the data of the conflicting one. If the service finds an existing user matching only the *email*, the update of the existing user fails. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. Possible values: • *personExternalID* • *emails[0].value* Default value: *personExternalID* **System Role:** Target	SAP Market Communication for Utilities

Name	Description	System Type
`marketing.cloud.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP Marketing Cloud target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. According to your use case, choose how to set up this property: • Default behavior: This property does not appear in the UI during system creation. Its default value is *personExternalID*. That means, if the service finds an existing user by a *personExternalID* , it updates this user with the data of the conflicting one. If a user with such a *personExternalID* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user matching both unique attributes *email* and *personExternalID*, it updates this user with the data of the conflicting one. If the service finds an existing user matching only the *email*, the update of the existing user fails. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. Possible values: • *personExternalID* • *emails[0].value* Default value: *personExternalID* **System Role:** Target	SAP Marketing Cloud

Name	Description	System Type
`a4c.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP BTP ABAP environment target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. According to your use case, choose how to set up this property: • Default behavior: This property does not appear in the UI during system creation. Its default value is *personExternalID*. That means, if the service finds an existing user by a *personExternalID* , it updates this user with the data of the conflicting one. If a user with such a *personExternalID* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user matching both unique attributes *email* and *personExternalID*, it updates this user with the data of the conflicting one. If the service finds an existing user matching only the *email*, the update of the existing user fails. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. Possible values: • *personExternalID* • *emails[0].value* Default value: *personExternalID* **System Role:** Target	SAP BTP ABAP environment

Name	Description	System Type
`c4c.group.prefix`	This property distinguishes SAP Sales Cloud and SAP Service Cloud groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`C4C_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Sales Cloud and SAP Service Cloud source system and will be provisioned to the target system with the following name pattern: **`C4C_<GroupDisplayName>`**. This way SAP Sales Cloud and SAP Service Cloud groups in the target system will be distinguished from groups provisioned from other applications. • When **set in the target system**, only groups containing the **`C4C_`** prefix in their display name will be provisioned to SAP Sales Cloud and SAP Service Cloud. Groups without this prefix in the display name won't be provisioned. If the property is not set, the SAP Sales Cloud and SAP Service Cloud groups will be read and provisioned to the target system with their actual display names. **System Role:** Source, Target	SAP Sales Cloud and SAP Service Cloud

Name	Description	System Type
`s4hana.cloud.roles.prefix`	This property distinguishes SAP S/4HANA Cloud Public Edition roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`S4HANA_CLOUD_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the roles that are read from the SAP S/4HANA Cloud Public Edition source system and will be provisioned to the target system with the following name pattern: **`S4HANA_CLOUD_<role_name>`** . This way SAP S/4HANA Cloud Public Edition roles in the target system will be distinguished from roles provisioned from other applications. If the property is not set, the SAP S/4HANA Cloud Public Edition roles will be read and provisioned to the target system with their actual role names. • When **set in the target system**, only roles containing the **`S4HANA_CLOUD_`** prefix in their role name will be provisioned to SAP S/4HANA Cloud Public Edition. Roles without this prefix in the role name won't be provisioned. If the property is not set, all roles will be provisioned to SAP S/4HANA Cloud Public Edition. **System Role:** Source and Target	SAP S/4HANA Cloud Public Edition

Name	Description	System Type
`ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see Thresholds Prevent Unintended Deletion of Users when Provisioning with SAP Cloud Identity Services. The property is optional and is defined on the target systems only. It is not added at system creation. The expected value is a positive integer without any separators, such as a space, comma, or period. Controlling the deletion works as follows: 1. You add the `ips.delete.threshold.users` property on the target system before running a job and define a threshold value, for example: *500*. 2. You run a read or resync job that is expected to delete a huge number of users. As a result, expect the following behavior: • If the number of the users to be deleted is lower or equal (for example: *400* or *500*) to the defined threshold value of *500*, Identity Provisioning continues with the deletion. • If the number of the users to be deleted is greater (for example: *1000*) than the defined threshold value of *500*, Identity Provisioning does not delete any users. Instead, the service marks the expected to be deleted users as failed in the job statistics. An error message in the job logs explains that users cannot be deleted as the defined threshold is exceeded. **System Role:** Target	All systems

Name	Description	System Type
`ips.delete.threshold.groups`	Use this property to control the number of groups or group assignments to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups or group assignments, for example by adding a filter or condition. For more information, see Thresholds Prevent Unintended Deletion of Users when Provisioning with SAP Cloud Identity Services. The property is optional and is defined on the target systems only. It is not added at system creation. The expected value is a positive integer without any separators, such as a space, comma, or period. Controlling the deletion works as follows: 1. You add the `ips.delete.threshold.groups` property on the target system before running a job and define a threshold value, for example: *500*. 2. You run a read or resync job that is expected to delete a huge number of groups. As a result, expect the following behavior: • If the number of the groups to be deleted is lower or equal (for example: *400* or *500*) to the defined threshold value of *500*, Identity Provisioning continues with the deletion. • If the number of the groups to be deleted is greater (for example: *1000*) than the defined threshold value of *500*, Identity Provisioning does not delete any groups. Instead, the service marks the expected to be deleted groups as failed in the job statistics. An error message in the job logs explains that groups cannot be deleted as the defined threshold is exceeded. **System Role:** Target	• Identity Authentication • Local Identity Directory • SAP Advanced Financial Closing • SAP Advanced Workflow • SAP Analytics Cloud • SAP Ariba Applications • SAP BTP Account Members (Neo) • SAP BTP Java/HTML5 apps (Neo) • SAP BTP XS Advanced UAA (Cloud Foundry) • SAP Build Work Zone, advanced edition • SAP Build Work Zone, standard edition • SAP Business Network • SAP Central Business Configuration • SAP Commerce Cloud • SAP Concur • SAP CPQ • SAP Data Custodian • SAP Document Center • SAP Fieldglass • SAP Field Service Management • SAP HANA Database • SAP Jam Collaboration • SAP S/4HANA for procurement planning • SAP Sales Cloud and SAP Service Cloud • SAP SuccessFactors • SAP SuccessFactors Learning • Sales Cloud – Analytics & AI • Cloud Foundry UAA Server • Google G Suite • LDAP Server • Microsoft Active Directory • Microsoft Entra ID • SCIM System • SSH Server (Beta)

Name	Description	System Type
`ips.delete.threshold.roles`	Use this property to control the number of roles or user assignments of a role to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of roles or user assignments of a role, for example by adding a filter or condition. For more information, see Thresholds Prevent Unintended Deletion of Users when Provisioning with SAP Cloud Identity Services. The property is optional and is defined on the target systems only. It is not added at system creation. The expected value is a positive integer without any separators, such as a space, comma, or period. Controlling the deletion works as follows: 1. You add the `ips.delete.threshold.roles` property on the target system before running a job and define a threshold value, for example: *500*. 2. You run a read or resync job that is expected to delete a huge number of roles. As a result, expect the following behavior: • If the number of the roles to be deleted is lower or equal (for example: *400* or *500*) to the defined threshold value of *500*, Identity Provisioning continues with the deletion. • If the number of the roles to be deleted is greater (for example: *1000*) than the defined threshold value of *500*, Identity Provisioning does not delete any roles. Instead, the service marks the expected to be deleted roles as failed in the job statistics. An error message in the job logs explains that roles cannot be deleted as the defined threshold is exceeded. **System Role:** Target	SAP BTP ABAP environment SAP Application Server ABAP SAP Integrated Business Planning SAP Market Communication for Utilities SAP Marketing Cloud SAP S/4HANA Cloud Public Edition SAP S/4HANA On-Premise

Name	Description	System Type
`ioa.content.type`	This property makes an intelligent opportunity analyzer connector to send a specified value for the *Content-Type* HTTP header. This is needed because intelligent opportunity analyzer could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	Intelligent Opportunity Analyzer
`ioa.include.if.match.wildcard.header`	Makes the intelligent opportunity analyzer connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an intelligent opportunity analyzer system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	Intelligent Opportunity Analyzer
`ioa.user.filter`	When specified, only those intelligent opportunity analyzer users matching the filter expression will be read. Supported operators: eq (equal) and sw (starts with). For example: • *userName eq "SmithJ"* • *emails.value eq "john.doe@example.com"* • **System Role:** Source, Proxy	Intelligent Opportunity Analyzer

Name	Description	System Type
`ioa.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in intelligent opportynity analyzer. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attributes. This property defines by which unique attributes the existing user to be searched and resolved. Intelligent opportynity analyzer supports two user unique attributes for managing conflict resolution: *username* and *email*. Both of the attributes are mandatory. The email is a multivalue attribute, indicating that a user may have multiple unique emails. According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[*].value*. If the service finds an existing user with such emails (at least one identical email is needed), it updates this user with the data of the conflicting one. If a user with such emails is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName, emails[*].value*. If the service finds an existing user with both userName and email, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Note** The uniqueness of a user in the target system is determined not only by the individual attributes or the combination of both, but also by considering whether one or multiple existing users in the target system share the same values for these unique attributes. For more information, see **User Update and Unique-**	Intelligent Opportunity Analyzer

Name	Description	System Type
	ness in SAP Ariba Category Management [page 1012]. **Possible values:** • *userName* • *emails[*].value* • *userName, emails[*].value* Default value: *userName* **System Role:** Target	
`awf.domain`	The domain name is the name of your SAP Advanced Workflow tenant. If you don't know your tenant name, contact your supervisor or administrator, or refer to the email notification you received when your account was created. **System Role:** Source, Target, Proxy	SAP Advanced Workflow
`awf.user.filter`	When specified, only those SAP Advanced Workflow users matching the filter expression will be read. **Possible values:** • *userName eq "Julie Armstrong"* • *userName sw "J"* • *name.familyName eq "Armstrong"* • *emails eq "julie.armstrong@example.com"* • *userName eq "Julie Armstrong" and emails.value eq "julie.armstrong@example.com"* **System Role:** Source	SAP Advanced Workflow

Name	Description	System Type
`awf.include.if.match.wildcard.header`	Makes the connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Advanced Workflow system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Advanced Workflow

Name	Description	System Type
`awf.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Advanced Workflow

Name	Description	System Type
`awf.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in SAP Advanced Workflow. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). **Default behavior**: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. Default value: *userName* **System Role:** Target	SAP Advanced Workflow
`cim.user.filter`	When specified, only those users matching the filter expression will be read. **Possible values:** For example: *name.familyName eq "SmithJ"* **System Role:** Source, Proxy	SAP Ariba Central Invoice Management
`cim.group.filter`	When specified, only those groups matching the filter expression will be read. **Possible values:** For example: *displayName eq "ProjectTeam1"* **System Role:** Source, Proxy	SAP Ariba Central Invoice Management

Name	Description	System Type
`cim.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in SAP Ariba Central Invoice Management. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[*].value*. If the service finds an existing user with at least one of the emails of the user in the source system, it updates this existing user with the data of the conflicting one. If a user with such an email is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[*].value*. If the service finds an existing user by *userName* and at least one of the emails of the user in the source system, it updates this user with the data of the conflicting one. If a user with such *userName* and *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • **Possible values:** • *userName* • *emails[*].value* • *userName,emails[*].value* Default value: *userName* **System Role:** Target	SAP Ariba Central Invoice Management

Name	Description	System Type
`cim.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists on the SAP Central Invoice Management target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for this property. Default value (when not specified): *displayName* If the property is not specified, the search is done by the default attribute: *displayName* **System Role:** Target	SAP Ariba Central Invoice Management

Name	Description	System Type
`cim.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Ariba Central Invoice Management

Name	Description	System Type
`cim.include.if.match.wildcard.header`	Makes the SAP Central Invoice Management connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Central Invoice Management system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Ariba Central Invoice Management
`cim.content.type`	This property makes a SAP Central Invoice Management connector to send a specified value for the *Content-Type* HTTP header. This is needed because SAP Central Invoice Management could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	SAP Ariba Central Invoice Management

Name	Description	System Type
`cim.group.prefix`	This property distinguishes SAP Central Invoice Management groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`CIM_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Central Invoice Management source system and will be provisioned to the target system with the following name pattern: **`CIM_<GroupDisplayName>`**. This way SAP Central Invoice Management groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Central Invoice Management groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`CIM_`** prefix in their display name will be provisioned to SAP Central Invoice Management. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Central Invoice Management. **System Role:** Source and Target	SAP Ariba Central Invoice Management
`cim.bulk.operations.max.count`	This property sets the number of operations to be performed in one bulk request. **Possible values:** Default value: *20* Maximum value: *100* If you enter a number larger than 100, Identity Provisioning will replace it with the default value (20). **System Role:** Target	SAP Ariba Central Invoice Management

Name	Description	System Type
`cim.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: User Management (SCIM) for SAP Ariba Central Invoice Management **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP Ariba Central Invoice Management
`ecp.user.filter`	When specified, only those SAP SuccessFactors Employee Central Payroll users matching the filter expression will be read. Supported operators: eq (equal) and sw (starts with). For example: • Value = *userName eq "JohnSmith"* • Value = *userName sw "J"* • Value = *emails.value eq "john.smith@example.com"* **System Role:** Source, Proxy	SAP SuccessFactors Employee Central Payroll
`ecp.group.filter`	When specified, only those groups matching the filter expression will be read. Supported operators: eq (equal) and sw (starts with). For example: • Value = *displayName eq "ProjectTeam1"* • Value = *displayName sw "P"* **System Role:** Source, Proxy	SAP SuccessFactors Employee Central Payroll

Name	Description	System Type
`ecp.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in SAP SuccessFactors Employee Central Payroll. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. Possible values: *userName* **System Role:** Target	SAP SuccessFactors Employee Central Payroll
`ecp.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists on the SAP SuccessFactors Employee Central Payroll target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for this property. Default value (when not specified): *displayName* If the property is not specified, the search is done by the default attribute: *displayName* **System Role:** Target	SAP SuccessFactors Employee Central Payroll

Name	Description	System Type
`ecp.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP SuccessFactors Employee Central Payroll

Name	Description	System Type
`ecp.include.if.match.wildcard.header`	Makes the SAP SuccessFactors Employee Central Payroll connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP SuccessFactors Employee Central Payroll system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP SuccessFactors Employee Central Payroll
`ecp.content.type`	This property makes a SAP SuccessFactors Employee Central Payroll connector to send a specified value for the *Content-Type* HTTP header. This is needed because SAP SuccessFactors Employee Central Payroll could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	SAP SuccessFactors Employee Central Payroll

Name	Description	System Type
`ecp.group.prefix`	This property distinguishes SAP SuccessFactors Employee Central Payroll groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **ECP_** You can use the example value or provide your own. • When set in the source system, the prefix will be prepended to the name of the groups that are read from the SAP SuccessFactors Employee Central Payroll source system and will be provisioned to the target system with the following name pattern: ECP_GroupDisplayName. This way SAP SuccessFactors Employee Central Payroll groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP SuccessFactors Employee Central Payroll groups will be read and provisioned to the target system with their actual display names. • When set in the target system, only groups containing the ECP_ prefix in their display name will be provisioned to SAP SuccessFactors Employee Central Payroll. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP SuccessFactors Employee Central Payroll. **System Role:** Source and Target	SAP SuccessFactors Employee Central Payroll
`ecp.sap-client`	Use this property to specify the SAP client number (a three-digit number) of your SAP SuccessFactors Employee Central Payroll system. For example: **102**	SAP SuccessFactors Employee Central Payroll

Name	Description	System Type
`cm.user.filter`	When specified, only those SAP Ariba Category Management users matching the filter expression will be read. Supported operators: eq (equal) and sw (starts with). For example: • *userName eq "SmithJ"* • *emails.value eq "john.doe@example.com"* • *emails.value eq "john.doe@example.com" or emails.value eq "john.d@example.com" or emails.value eq "j.doe@example.com"* **System Role:** Source, Proxy	SAP Ariba Category Management

Name	Description	System Type
`cm.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in SAP Ariba Category Management. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attributes. This property defines by which unique attributes the existing user to be searched and resolved. SAP Ariba Category Management supports two user unique attributes for managing conflict resolution: *username* and *email*. The username is a mandatory attribute, while the email is an optional, multivalue attribute, indicating that a user may have multiple unique emails. According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[*].value*. If the service finds an existing user with such emails (at least one identical email is needed), it updates this user with the data of the conflicting one. If a user with such emails is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName, emails[*].value*. If the service finds an existing user with both userName and email, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Note** The uniqueness of a user in the target system is determined not only by the individual attributes or the combination of both, but also by considering whether one or multiple existing users in the target system share the same values for these unique attributes. For more information, see **User Update and Unique-**	SAP Ariba Category Management

Name	Description	System Type
	ness in SAP Ariba Category Management [page 1012]. **Possible values:** • *userName* • *emails[*].value* • *userName, emails[*].value* Default value: *userName* **System Role:** Target	

Name	Description	System Type
`cm.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Ariba Category Management

Name	Description	System Type
`cm.include.if.match.wildcard.header`	Makes the SAP Ariba Category Management connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Ariba Category Management system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Ariba Category Management
`cm.content.type`	This property makes a SAP Ariba Category Management connector to send a specified value for the *Content-Type* HTTP header. This is needed because SAP Ariba Category Management could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	SAP Ariba Category Management
`hana.cloud.db.group.filter`	When specified, only those SAP HANA Cloud Database Services groups matching the filter expression will be read. For example: *displayName eq "ProjectTeam1"* **System Role:** Source, Proxy	SAP HANA Cloud Database Services
`hana.cloud.db.user.filter`	When specified, only those SAP HANA Cloud Database Services users matching the filter expression will be read. For example: *userName eq "SmithJ"* **System Role:** Source, Proxy	SAP HANA Cloud Database Services

Name	Description	System Type
`hana.cloud.db.content.type`	This property makes a SAP HANA Cloud Database Services connector to send a specified value for the *Content-Type* HTTP header. This is needed because SAP HANA Cloud Database Services could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	SAP HANA Cloud Database Services
`hana.cloud.db.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists on the SAP HANA Cloud Database Services target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for this property. Default value (when not specified): *displayName* If the property is not specified, the search is done by the default attribute: *displayName* **System Role:** Target	SAP HANA Cloud Database Services
`hana.cloud.db.include.if.match.wildcard.header`	Makes the SAP HANA Cloud Database Services connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP HANA Cloud Database Services system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP HANA Cloud Database Services

Name	Description	System Type
`hana.cloud.db.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP HANA Cloud Database Services

Name	Description	System Type
`hana.cloud.db.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in SAP HANA Cloud Database Services. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *urn:ietf:params:scim:schemas:extension:sap:2.0:U ser:userId*. If the service finds an existing user with such *urn:ietf:params:scim:schemas:extension:sap:2.0:U ser:userId*, it updates this user with the data of the conflicting one. If a user with such attribute value is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,urn:ietf:params:scim:schemas:extension:sap:2.0:User:userId*. If the service finds an existing user with such *userName* or *urn:ietf:params:scim:schemas:extension:sap:2.0:User:userId*, it updates this user with the data of the conflicting one. If a user with such *userName* or *urn:ietf:params:scim:schemas:extension:sap:2.0:User:userId* is not found, the creation of the conflicting user fails. **Possible values:** • *userName* • *urn:ietf:params:scim:schemas:extension:sap:2.0:User:userId* • *userName,urn:ietf:params:scim:schemas:extension:sap:2.0:User:userId* Default value: *userName* **System Role:** Target	SAP HANA Cloud Database Services

Name	Description	System Type
`hana.cloud.db.authenticationType`	Specifies the method by which users authenticate when connecting to the database. **Possible values:** JWT (Users authenticate via JWT tokens issued by an identity provider.) **System Role:** Target, Proxy	SAP HANA Cloud Database Services
`hana.cloud.db.instance.id`	Refers to a unique identifier associated with a specific database instance within the SAP HANA Cloud. **System Role:** Source, Target	SAP HANA Cloud Database Services
`hana.cloud.db.instance.type`	Refers to the type of instance being configured or used within the SAP HANA Cloud. The value can be set to: • **hdb** - SAP HANA Database • **hdlre** - SAP HANA Data Lake Relational Engine Default value: **hdb** **System Role:** Source, Target	SAP HANA Cloud Database Services
`hana.cloud.db.providerName`	Specifies the identity provider (IdP) that is responsible for handling authentication requests. **System Role:** Target	SAP HANA Cloud Database Services

Name	Description	System Type
`hana.cloud.db.group.prefix`	(Optional) This property distinguishes SAP HANA Cloud Database Services groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`HANA_Cloud_DB_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP HANA Cloud Database Services source system and will be provisioned to the target system with the following name pattern: **`HANA_Cloud_DB_<GroupDisplayName>`**. This way SAP HANA Cloud Database Services groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP HANA Cloud Database Services groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`HANA_Cloud_DB_`** prefix in their display name will be provisioned to SAP HANA Cloud Database Services. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP HANA Cloud Database Services. **System Role:** Source, Target	SAP HANA Cloud Database Services
`en.user.filter`	When specified, only those SAP Enable Now users matching the filter expression will be read. **Possible values:** For example: *userName eq "SmithJ"* **System Role:**Source, Proxy	SAP Enable Now

Name	Description	System Type
`en.group.filter`	When specified, only those SAP Enable Now groups matching the filter expression will be read. **Possible values:** For example: *displayName eq "ProjectTeam1"* **System Role:** Source, Proxy	SAP Enable Now
`en.content.type`	This property makes a SAP Enable Now connector to send a specified value for the *Content-Type* HTTP header. This is needed because SAP Enable Now could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* Default value: *application/scim+json* **System Role:** Target, Proxy	SAP Enable Now
`en.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. The property is not added automatically at system creation. Default value: *userName* If the service finds an existing user by userName, it updates this user with the data of the conflicting one. If the service does not find an existing user by userName, the creation of the conflicting user fails. **System Role**: Target	SAP Enable Now

Name	Description	System Type
`en.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists on the SAP Enable Now target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for this property. Default value (when not specified): *displayName* If the property is not specified, the search is done by the default attribute: *displayName* **System Role:** Target	SAP Enable Now
`en.include.if.match.wildcard.header`	Makes the SAP Enable Now connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Enable Now system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Enable Now

Name	Description	System Type
`en.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target	SAP Enable Now

Name	Description	System Type
`en.group.prefix`	(Optional) This property distinguishes SAP Enable Now groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`EN_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Enable Now source system and will be provisioned to the target system with the following name pattern: **`EN_<GroupDisplayName>`**. This way SAP Enable Now groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Enable Now groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`EN_`** prefix in their display name will be provisioned to SAP Enable Now. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Enable Now. **System Role:** Source, Target	SAP Enable Now
`sig.user.filter`	When specified, only those SAP Signavio Process Transformation Suite users matching the filter expression will be read. **Possible values:** For example: *userName eq "SmithJ"* **System Role:**Source, Proxy	SAP Signavio Process Transformation Suite
`sig.group.filter`	When specified, only those SAP Signavio Process Transformation Suite groups matching the filter expression will be read. **Possible values:** For example: *displayName eq "ProjectTeam1"* **System Role:**Source, Proxy	SAP Signavio Process Transformation Suite

Name	Description	System Type
`sig.user.unique.attribute`	When Identity Provisioning attempts to provision a user for the first time, it may detect that this user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user will be searched and resolved. If the service finds a user on the target system via this filter, then the conflicting user will overwrite the existing one. If the service does not find a user on the target system via this filter, the creation will fail. **Default behavior**: This property is missing during system creation. Its default value is *userName*. This means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such *userName* is not found, the creation of the conflicting user fails. Default value: *userName* **System Role:** Target	SAP Signavio Process Transformation Suite
`sig.group.unique.attribute`	If the service tries to create a group that already exists in the target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for the `sig.group.unique.attribute` property. If the property is not specified, the search is done by the default attribute: *displayName* **System Role:** Target	SAP Signavio Process Transformation Suite

Name	Description	System Type
`sig.include.if.match.wildcard.header`	Makes the SAP Signavio Process Transformation Suite connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Signavio Process Transformation Suite system for entity versioning. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Signavio Process Transformation Suite
`sig.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Signavio Process Transformation Suite

Name	Description	System Type
`sig.content.type`	Makes the connector send a specified value for the *Content-Type* HTTP header. This is needed because an SAP Signavio Process Transformation Suite system could potentially not implement the protocol in the specification, which states that a system must accept *application/scim+json* as a value of the *Content-Type* header. **Possible values:** For example: *application/json* If the property is not specified, the default value is taken: *application/scim+json* **System Role:** Target, Proxy	SAP Signavio Process Transformation Suite
`sig.tenant.id`	The identifier of your SAP Signavio Process Transformation Suite tenant. **System Role:** Source, Target, Proxy	SAP Signavio Process Transformation Suite
`sig.group.members.paging.enabled`	Enables paging of group members. **Possible values:** • *true* • *false* **Default value:** *true* **System Role:** Source, Proxy	SAP Signavio Process Transformation Suite
`sig.user.groups.paging.enabled`	Enables paging of user's groups. **Possible values:** • *true* • *false* **Default value:** *true* **System Role:** Source, Proxy	SAP Signavio Process Transformation Suite
`sig.group.members.page.size`	The number of group members that SAP Signavio Process Transformation Suite returns per request when reading a group. **Default value:** 1000 **System Role:** Source, Proxy	SAP Signavio Process Transformation Suite

Name	Description	System Type
`s4hana.cloud.group.members.page.size`	The number of group members that SAP S/4HANA Cloud Public Edition returns per request when reading a group. **Default value:** 20000 **System Role:** Source, Proxy	SAP S/4HANA Cloud Public Edition (using version 3 - System for Cross-domain Identity Management (SCIM) API)
`ibp.group.members.page.size`	The number of group members that SAP Integrated Business Planning returns per request when reading a group. **Default value:** 20000 **System Role:** Source, Proxy	SAP Integrated Business Planning (using version 2 - SCIM Interface for IAM)
`ibp.user.groups.page.size`	The number of user groups that SAP Integrated Business Planning returns per request when reading a group. **Default value:** 20000 **System Role:** Source, Proxy	SAP Integrated Business Planning (using version 2 - SCIM Interface for IAM)
`s4hana.cloud.user.groups.page.size`	The number of user groups that SAP S/4HANA Cloud Public Edition returns per request when reading a group. **Default value:** 20000 **System Role:** Source, Proxy	SAP S/4HANA Cloud Public Edition (using version 3 - System for Cross-domain Identity Management (SCIM) API)
`sig.user.groups.page.size`	The number of user groups that SAP Signavio Process Transformation Suite returns per request when reading a user. **Default value:** 1000 **System Role:** Source, Proxy	SAP Signavio Process Transformation Suite

Name	Description	System Type
`sig.group.prefix`	This property distinguishes SAP Signavio Process Transformation Suite groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`SIG_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Signavio Process Transformation Suite source system and will be provisioned to the target system with the following name pattern: **`SIG_<GroupDisplayName>`**. This way SAP Signavio Process Transformation Suite groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Signavio Process Transformation Suite groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`SIG_`** prefix in their display name will be provisioned to SAP Signavio Process Transformation Suite. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Signavio Process Transformation Suite. **System Role:** Source, Target	SAP Signavio Process Transformation Suite

Name	Description	System Type
`rgm.user.unique.attribute`	When Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). SAP Revenue Growth Management supports the following unique attributes which are automatically filled in during system creation: **`userName,emails[0].value,externalId`**. If the service finds an existing user by at least one of the unique attributes, it updates this user with the data of the conflicting one. If such a user is not found, the update of the conflicting user fails. If more than one users with these unique attributes are found, the update fails. Default value: *userName,emails[0].value,externalId* System Role: Target	SAP Revenue Growth Management
`rgm.user.filter`	When specified, only those users matching the filter expression will be read. You can filter users by **`userName`**, **`emails[0].value`**, and **`externalId`**, according to the API syntax of SAP Revenue Growth Management. **Possible values:** text/ numeric string For example: • *userName eq "johnbrown" and externalId eq "c167254d-25fd-5fac-af32-0b8c35e0de27"* • *userName sw "J" and externalId eq "c167254d-25fd-5fac-af32-0b8c35e0de27"* • *userName eq "johnbrown" and emails.value eq "johnbrown@email.com"* • *userName eq "johnbrown" and emails.value eq "johnbrown@email.com" and externalId eq "c167254d-25fd-5fac-af32-0b8c35e0de27"* > **Note** > These combinations are valid for both 'or' and 'and' operators. **System Role:** Source, Proxy	SAP Revenue Growth Management

Name	Description	System Type
`rgm.group.unique.attribute`	If you try to provision a group that already exists in a target system, the group creation will fail. In this case, the existing group only needs to be updated. This property defines by which unique attribute(s) the existing group will be searched and resolved. The default value is **`['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']`**. Currently, it is the only unique attribute that is supported. When set, you can expect the following behavior: • If a group with the given unique attribute is found in the target system, the group that you try to provision will overwrite the existing one. • If a group with the given unique attribute is not found in the target system, the group that you try to provision will not be created in the target system. **Possible values:** If the property is not specified, the search is done by the default attribute: `['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']` **System Role:** Target	SAP Revenue Growth Management
`rgm.group.filter`	When specified, only those SAP Revenue Growth Management groups matching the filter expression will be read. **Possible values:** For example: *urn:sap:cloud:scim:schemas:extension:custom:2.0:Group:name eq "ProjectTeam1" or "Students2018"* **System Role:** Source, Proxy	SAP Revenue Growth Management

Name	Description	System Type
`rgm.group.prefix`	This property distinguishes SAP Revenue Growth Management groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`RGM_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Revenue Growth Management source system and will be provisioned to the target system with the following name pattern: **`RGM_<GroupDisplayName>`**. This way SAP Revenue Growth Management groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Revenue Growth Management groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`RGM_`** prefix in their display name will be provisioned to SAP Revenue Growth Management. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Revenue Growth Management. **System Role:** Source and Target	SAP Revenue Growth Management

Name	Description	System Type
`rgm.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** • *true* • *false* Default value: *false* **System Role:** Target, Proxy	SAP Revenue Growth Management
`rgm.userImport.api.dependency.name`	This property holds the value of the dependency name used in the configuration of the SAP Revenue Growth Management consumer application, created in the SAP Cloud Identity Services tenant. It is used for access token retrieval from Identity Authentication. For more information, see Integrating Applications. Possible values: Text/numeric string **System Role:** Source, Target, Proxy	SAP Revenue Growth Management

Name	Description	System Type
`c4c.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in SAP Sales Cloud and SAP Service Cloud. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • userName • emails[0].value • userName,emails[0].value Default value: *userName* **System Role:** Target	SAP Sales Cloud and SAP Service Cloud

Name	Description	System Type
`c4c.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists on the SAP Sales Cloud and SAP Service Cloud target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for this property. **Possible values:** Default value (when not specified): *displayName* If the property is not specified, the search is done by the default attribute: *displayName* **System Role:** Target	SAP Sales Cloud and SAP Service Cloud
`ariba.applications.api.dependency.name`	This property holds the value of the dependency name used in the configuration of the SAP Ariba Applications consumer application, created in the SAP Cloud Identity Services tenant. It is used for access token retrieval from Identity Authentication. For more information, see Integrating Applications. This property is mandatory when using the next-generation SCIM API provided by SAP Ariba Applications. Possible values: Text/numeric string **System Role:** Source, Target, Proxy	SAP Ariba Applications

Name	Description	System Type
`ad.group.flatten.page.size`	This property controls the page size used during Microsoft Active Directory group member flattening. When flattening is enabled (`ad.group.flatten=true` - meaning all nested group members, including users and subgroups, are retrieved), this property specifies how many users and groups are requested from AD in a single page. **Possible values**: • Minimum value: 100 • Maximum value: 1000 If the value is missing, null, or invalid, it defaults to 1000. Default value: 1000 **System Role:** Source	Microsoft Active Directory

1.3.6 Transformations

Transformations help you transform user and group attributes from the data model of the source system to the data model of the target system.

Identity Provisioning provides a default transformation logic for every supported provisioning system. Source systems apply read transformations, target systems apply write transformations, while proxy systems apply both – read and write proxy transformations.

During a provisioning job, the read transformation of the source system converts the system specific JSON representation of users and groups to one common JSON format (also called internal JSON). This common JSON format is then used as a source for passing the data to the target system, which applies the write transformation.

The following examples illustrates how the last name user attribute is read from the SAP S/4HANA Cloud Public Edition source system, transformed and prepared as the family name user attribute for the Identity Authentication target system using both - graphical and JSON text editor.

Graphical Editor

Source System-Specific JSON	Common JSON

Common JSON	Target System-Specific JSON

JSON Text Editor

Source System-Specific JSON	Common JSON	Target System-Specific JSON
Code Syntax { "sourcePath":"$.personalInformation.lastName", "targetPath":"$.name.familyName" },	Code Syntax { "sourcePath":"$.name.familyName", "targetPath":"$.name.familyName" },	Code Syntax { "sourcePath":"$.name.familyName", "targetPath":"$.name.familyName" },

You can adapt the transformation logic, for example, to read only the entities that should be provisioned to the target system. This filter can speed up the processing of the entities and their provisioning to the target system.

Every time you modify the transformation of a provisioning system and save your changes, a new version is created and displayed in the *Transformation History* screen. It lists all versions - from the initial to the current one, and gives you additional information about the last modification date, the person who modified it and a description.

Identity Provisioning provides graphical and JSON text editor for managing provisioning system transformations.

Transformation Types [page 390]
Learn about the types of JSON transformations needed for the provisioning jobs.

Transformation Expressions [page 391]

Transformation Functions [page 428]

Transformation Variables [page 490]

Transformation Conditions [page 496]
A condition defines a JSON filter expression that can be applied to entity types (such as users, groups, and roles) or within attribute mappings. It can be combined with strings, constants, variables, and functions, or used within functions.

Transformation Examples [page 501]
The following examples explain how transformations work.

Transformation Editors [page 505]
Identity Provisioning provides graphical and JSON text editor for managing provisioning system transformations.

Parent topic: Concepts [page 26]

Related Information

Tenants [page 27]
Bundles [page 35]
Applications [page 61]
Provisioning Systems [page 69]
Properties [page 73]
Users [page 509]
Agents [page 511]
Groups [page 514]
Authorization Policies [page 518]
SAP-Managed SAP Cloud Identity [page 520]

1.3.6.1 Transformation Types

Learn about the types of JSON transformations needed for the provisioning jobs.

Context

Two types of transformations occur before the provisioning of entities:

- **Read Transformation** – from the *source system* to the provisioning framework. It reads the data in the source system and transfers it to an intermediate JSON data in the provisioning framework. The reading of entities from the source system can be complete (full read) or partial (delta read). For more information, see Manage Full and Delta Read [page 2459].
- **Write Transformation** – from the provisioning framework to the *target system*. It prepares the data to be written to the target system.

Both transformations result in JSON data.

Every supported system holds and requires specific JSON data. To convert the source JSON data to an intermediate JSON version (which can be used for transformation to a supported target system), the Identity

Provisioning administrator can use the suggested JSON transformation logic on the *Transformations* tab, and adapt it to the required transformation.

All transformations from the source systems transform their specific JSON data to intermediate data according to the *System for Cross-domain Identity Management* (SCIM 2.0) specification.

> ⓘ Note
>
> *Proxy systems* contain both *Read* and *Write* transformations.

Example

If the source JSON data contains the attribute **name**, the read transformation converts this attribute to *name23* in the intermediate JSON data. Then, the write transformation should use the attribute *name23* (instead of **name**) as *sourcePath* attribute.

Parent topic: Transformations [page 388]

Related Information

Transformation Expressions [page 391]
Transformation Functions [page 428]
Transformation Variables [page 490]
Transformation Conditions [page 496]
Transformation Examples [page 501]
Transformation Editors [page 505]
Manage Transformations [page 2439]
Transformation Expressions [page 391]
Transformation Functions [page 428]

1.3.6.2 Transformation Expressions

The transformation logic is based on JSON. The order of the expressions in the file is decisive for how the transformation is executed. The transformation actions are performed in the sequence defined in the transformation logic.

There is a different transformation logic for every entity (users, groups, roles). Below are listed some of the JSON expressions you can use.

Transformation Expression	Usage

applyOnElements	Use this JSON expression when the value referenced by the *sourcePath* is a multivalue structure (a list or an array), and you want to apply a function to its elements instead of to the whole structure.
	
	You can use the *applyOnElements* expression in any function, for any provisioning system.
	
	For example, you can see it in the default write transformation of the *SAP Application Server ABAP* proxy system. In this case, the **decode** function reads all members of a user list and converts this list into a byte array. Then, the **toString** function reads this byte-array and converts it back into a string (a list of usernames).
	
	Example
	
	`// Function toString will be applied to all members read from a user list.`
	`...`
	`{`
	`"sourcePath": "$.members[*].value",`
	`"preserveArrayWithSingleElement": true,`
	`"optional": true,`
	`"targetPath": "$.USERLIST[?(@.USERNAME)]",`
	`"functions": [`
	`{`
	`"function": "decode",`
	`"algorithm": "base32",`
	`"skipPadding": true`
	`},`
	`{`
	`"function": "toString",`
	`"applyOnElements": true`
	`}`
	`]`
	`}`
	`...`
	
	Possible values:
	
	• *true*
	• *false*
	
	Data type: Boolean

Transformation Expression	Usage
applyOnJsonAttribute	This expression is deprecated. You can use the new one – *applyOnAttribute*. **Data type**: String

Transformation Expression	Usage

applyOnAttribute	Use this JSON expression when the structure referenced by the *sourcePath* is a complex one (a map), and you want to apply a function to a single member of this map.
	
	In the example below, all occurrences of "SAP" in the values of attribute COMPOSITE_PRIVILEGE are replaced with "test" in the transformation result.
	
	Example
	Privileges in the source system:
	
	`{ "PRIVILEGES": [ { "COMPOSITE_PRIVILEGE": "SAP_01_COMP" }, { "COMPOSITE_PRIVILEGE": "SAP_02_COMP" } ] }`
	
	Example
	Source system transformation:
	
	`{ "mappings": [ { "sourcePath": "$.PRIVILEGES", "targetPath": "$.groups", "functions": [ { "function": "replaceString", "applyOnElements": true, "target": "SAP", "replacement": "test", "applyOnAttribute": "COMPOSITE_PRIVILEGE" } ] } ] }`

Transformation Expression	Usage
	Example Transformation result: `{ "groups": [ { "COMPOSITE_PRIVILEGE": "test_01_COMP" }, { "COMPOSITE_PRIVILEGE": "test_02_COMP" } ] }` **Data type**: String

Transformation Expression	Usage

assignToAttribute	Use this JSON expression when the structure referenced by *targetPath* is a complex one (a map), and you want to assign the result of this mapping to a specific map entry.
	
	In the example below, all occurrences of "SAP" in the values of attribute COMPOSITE_PRIVILEGE are replaced with "test", and the result is assigned to the value of attribute "ASSIGNMENT".
	
	Example
	Privileges in the source system:
	
	`{ "PRIVILEGES": [ { "COMPOSITE_PRIVILEGE": "SAP_01_COMP" }, { "COMPOSITE_PRIVILEGE": "SAP_02_COMP" } ] }`
	
	Example
	Source system transformation:
	
	`{ "mappings": [ { "sourcePath": "$.PRIVILEGES", "targetPath": "$.groups", "functions": [ { "function": "replaceString", "applyOnElements": true, "target": "SAP", "replacement": "test", "applyOnAttribute": "COMPOSITE_PRIVILEGE", "assignToAttribute": "ASSIGNMENT" } ] } ] }`

Transformation Expression	Usage
	Example Transformation result: `{ "groups": [ { "ASSIGNMENT": "test_01_COMP", "COMPOSITE_PRIVILEGE": "SAP_01_COMP" }, { "ASSIGNMENT": "test_02_COMP", "COMPOSITE_PRIVILEGE": "SAP_02_COMP" } ] }` **Data type**: String

Transformation Expression	Usage

condition	A condition specifies a JSON filter expression. It can be applied on a single mapping entity or on the whole entity type. You can use conditions on strings, constants, variables and functions.
	• Condition with **strings**
	Example
	`{ "mappings": [ … { "condition": "$.memberOf contains 'group1'", "constant": "NewDisplayName", "targetPath": "$.displayName" } ] }`
	• Condition with **constants** – AS ABAP (proxy system)
	Example
	`{ "condition": "($.emails.length() > 0) && ($.name.familyName EMPTY false)", "mappings": [ { "sourcePath": "$", "targetPath": "$" }, … }`
	• Condition with format **variables** – SAP S/4HANA Cloud Public Edition (source system)
	Example
	`{ "user": { "condition": "($.validityPeriod.startDate <= '${currentDate}') && ($.validityPeriod.endDate > '${currentDate}')", "mappings": [ {`

Transformation Expression	Usage

```
                "sourcePath":
"$.personID",

"targetVariable":
"entityIdSourceSystem"
            },
```

- Conditions in **functions** – Microsoft Active Directory (target system)

Example

```
"group": {
...
      {
        "sourcePath":
"$.members[*]",
        "targetVariable":
"membersVariable",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "functions": [
          {
            "condition":
"@.type != 'Group'",
            "entityType":
"user",
            "applyOnElements":
true,
            "type":
"resolveEntityIds"
          },
          {
            "condition":
"@.type == 'Group'",
            "entityType":
"group",
            "applyOnElements":
true,
            "type":
"resolveEntityIds"
          },
          {
            "condition":
"(@.type != 'Group') &&
('%ldap.attribute.user.id%' !=
'%ldap.attribute.dn%')",
            "function":
"concatString",
            "applyOnElements":
true,
            "applyOnAttribute":
"value",
            "prefix":
"%ldap.attribute.user.id%=",
            "suffix":
",%ldap.user.path%"
```

Transformation Expression	Usage

```
          },
          {
            "condition":
"(@.type == 'Group') &&
('%ldap.attribute.group.id%' !=
'%ldap.attribute.dn%')",
            "function":
"concatString",
            "applyOnElements":
true,
            "applyOnAttribute":
"value",
            "prefix":
"%ldap.attribute.group.id%=",
            "suffix":
",%ldap.group.path%"
          }
        ]
...
```

- Conditions with **functions**

  ### Example

  ```
  {
    "user": {
      "condition": "($.emails
  EMPTY false) &&
  isValidEmail($.emails[0].value)"
  ,
      "mappings": [
        {
          "sourceVariable":
  "entityIdTargetSystem",
          "targetPath": "$.id"
        },
   .....
  }
  ```

- Conditions and **skipOperations**

  ### Example

  #### Code Syntax

  ```
  "condition":"$.groups empty
  false",
  "skipOperations":[
     "update"
  ],
  ```

  First, the condition is evaluated. Users and groups who do not match the condition are skipped. After the condition is evaluated, the skip operation determines

Transformation Expression	Usage
	whether the remaining users will be skipped or provisioned. **Data type**: String

constant	Set a constant if the target system requires attributes that are not defined in the source system.
	
	You can also use schemas to organize and combine multiple constants.
	
	Example
	
	``
	{
	    "targetPath":
	"$.emails[0].type",
	    "constant": "work"
	},
	``
	
	Data type:
	
	- String
	- Integer
	- Boolean
	- JSONObject
	- StringArray
	- IntegerArray
	- BooleanArray
	- JSONObjectArray

Transformation Expression	Usage

correlationAttribute	Correlation attributes are used in the *"user"* mappings of the *Read Transformations* (in source and proxy systems). They mark attributes that are unique for a user in multiple source systems where this user is read from.
	
	The purpose of `correlationAttribute` is to properly count managed identities (users) during the reading process. This is essential because the calculated value is compared to the customer's licensing quota.
	
	• *Use Case 1* – Same attribute in different systems
	
	Example
	Mapping from the *Microsoft Active Directory* read transformation:
	
	`{ "sourcePath": "$.sAMAccountName[0]", "targetPath": "$.userName", "correlationAttribute": true },`
	
	Example
	Mapping from the *SAP S/4HANA On-Premise* read transformation:
	
	`{ "sourcePath": "$.userAssignment.userID", "optional": true, "targetPath": "$.userName", "correlationAttribute": true },`
	
	Example
	Mapping from the *LDAP Server* read transformation:
	
	`{ "sourcePath": "$.%ldap.attribute.user.id%[0]", "targetPath": "$.userName", "correlationAttribute": true`

Transformation Expression	Usage

```
    },
```

Result: If the Identity Provisioning reads a user whose username is one and the same in all the three systems (as listed above), the service will consider and count these "3 users" as one managed identity.

- *Use Case 2* – Different attributes from several systems

Example

Mapping from the *Microsoft Active Directory* read transformation:

```
      {
        "sourcePath":
"$.sAMAccountName[0]",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
```

Example

Mapping from the *SAP Analytics Cloud* read transformation:

```
    {
       "sourcePath": "$.emails",
       "targetPath": "$.emails",

"preserveArrayWithSingleElement"
: true
    },
    {
       "sourcePath": "$.emails[?
(@.primary== true)].value",
       "correlationAttribute":
true
    },
```

Example

Mapping from the *Identity Authentication* read transformation:

```
    {
       "sourcePath":
"$.userName",
       "targetPath":
"$.userName",
       "optional": true,
       "correlationAttribute":
true
```

Transformation Expression	Usage

	<pre> },
	     ...
	    {
	       "sourcePath":
	"$.emails[*].value",
	
	"preserveArrayWithSingleElement"
	: true,
	       "targetPath": "$.emails[?
	(@.value)]"
	    },
	    {
	       "sourcePath": "$.emails[?
	(@.primary== true)].value",
	       "correlationAttribute":
	true
	    },</pre>
	Result: If the Identity Provisioning reads a user with a certain username from one system, then it reads a user with a particular e-mail from another system, and then finds a user with the same username and e-mail in a third system (as listed above), the service will consider and count these "3 users" as one managed identity.
	
	Possible values:
	
	• *true*
	• *false*
	
	Data type: Boolean

Transformation Expression	Usage

defaultValue	This JSON expression returns the default value of an attribute when it is set as optional and there is no specific value provided in the `sourcePath`. The **defaultValue** expression comes in handy in the following cases:
	• When an attribute of a previously provisioned entity is later deleted from the source system. After a new provisioning job, the target system will try to get some value for this missing attribute, and thus it will write its default one.
	• When the system transformation contains a *valueMapping* operation. If the value of the mapped attribute is not found as a key in the `"valueMappings"` definitions, or some of the source paths does not return a value, the JSON transformation will use the default one.
	The default value of an attribute can be a *string*, *integer*, *Boolean*, *array*, or empty (null).
	You can add `"defaultValue"` for any attribute, in the transformation of any provisioning system. For example, you can find it in the default transformation of the **Identity Authentication** target system:
	Example
	`...`
	`{`
	`"sourcePath": "$.userType",`
	`"targetPath": "$.userType",`
	`"defaultValue": "employee",`
	`"optional": true`
	`},`
	`...`
	Data type:
	• String
	• Integer
	• Boolean
	• JSONObject
	• StringArray
	• IntegerArray
	• BooleanArray
	• JSONObjectArray

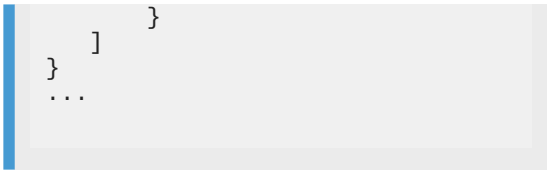
Transformation Expression	Usage

ignore	Use the *ignore* expression if you would like to exclude certain parts of the transformation from being considered during the provisioning. Similar to *condition*, you can set *ignore* on various levels, including on entire entity types such as users or roles, or specific mapping entities. This is applicable for both source and target systems.
	
	Note
	Note that if an entity type is set to be ignored in the read transformations, you won't get any data in the job log statistics for this entity type.
	
	Example
	`"group": { "ignore": true, "mappings": [ { "sourcePath": "$.sAMAccountName[0]", "targetVariable": "entityIdSourceSystem" }, { "sourcePath": "$.sAMAccountName[0]", "targetPath": "$.displayName" }, ...`
	
	Example
	`"user": { "mappings": [ { "ignore": true, "sourcePath": "$.sAMAccountName[0]", "targetVariable": "entityIdSourceSystem" }, { "sourcePath": "$.sAMAccountName[0]", "targetPath": "$.userName" }, ...`
	
	Possible values:
	• *true*

Transformation Expression	Usage
	• *false* **Data type**: Boolean

Transformation Expression	Usage

preserveArrayWithSingleElement	This JSON expression converts an array into a single element. That means, when the Identity Provisioning service reads an array that contains only one element, then in the target system:
	• If **"preserveArrayWithSingleElement"** is set to *true*, this expression will keep the array with the single element as is.
	Example
	The following mapping keeps the array of group members, resolving their IDs, even if the array contains only one member.
	`{ "sourcePath": "$.members[*].value", "preserveArrayWithSingleElement": true, "optional": true, "targetPath": "$.members[?(@.value)]", "functions": [ { "type": "resolveEntityIds" } ] } ...`
	• If **"preserveArrayWithSingleElement"** is set to *false*, this expression will convert the array into the relevant single element.
	Example
	The following mapping converts the array into a single element (a group member), and resolves its ID.
	`{ "sourcePath": "$.members[*].value", "preserveArrayWithSingleElement": false, "optional": true, "targetPath": "$.members[?(@.value)]", "functions": [ { "type": "resolveEntityIds"`

Transformation Expression	Usage
	 **Possible values:** • *true* • *false* **Data type**: Boolean

Transformation Expression	Usage
scope	You can set a scope for an entity attribute, based on its lifecycle. A scope can have the following values: • *createEntity* – An entity attribute is processed only during creation. Transformation mappings without scope are always processed during the creation. **Note** The `createEntity` scope currently applies only to **write** and **proxy write** transformations. **Example** The following mapping provides an initial password during the creation of a user. `{ "user": { "mappings": [ { "scope": "createEntity", "targetPath": "$.Password", "constant": "Initial1" } ] }, ...` • *patchEntity* – When using the *ipsproxy* application and a system that supports SCIM PATCH operation, you may need to perform certain transformations over the PATCH request. To do this, use the `patchEntity` scope in the system transformation. Only mappings with this scope will be processed. **Note** • The Identity Provisioning service supports PATCH requests only for groups, group members (assignments), and users. • The `patchEntity` scope is only applicable for write transformations in **proxy systems**.

Transformation Expression	Usage

	Example
	The following mapping decodes the group ID and preserves the patch request body (payload) into the target system.
	`{ "scope": "patchEntity", "sourceVariable": "entityIdTargetSystem", "targetVariable": "entityIdTargetSystem", "functions": [ { "type": "decode", "algorithm": "base32", "skipPadding": true }, { "function": "toString" } ] }, { "scope": "patchEntity", "sourcePath": "$", "targetPath": "$" },`
	• *deleteEntity* – If an entity has been deleted from the source system or has been set a condition for it not to be read anymore, this entity can "stay" in the target system for the following reasons:
	  • The target system does not support deletion of entities.
	  • You do not want to delete it but only temporary disable/deactivate it.
	  • You want to neither delete it, nor deactivate it but only remove its permissions, or exclude it from some corporate groups.
	If you have to fulfill some of these scenarios for an entity, use the `deleteEntity` scope. It prevents from deleting the entity from the target or proxy system as only updating its status instead. Also, bear in mind the following:
	  • For the affected entity, all transformation mappings that do not contain this scope will be ignored.
	  • If a condition exists on entity type level, it will be ignored as well.
	  • If the `deleteEntity` scope is included in a proxy write transformation, it is taken into account only

Transformation Expression	Usage
	when used in the transformation mapping for the **entityIdTargetSystem** variable. Use this scope for the following systems: • Cloud Foundry UAA server • Identity Authentication • Local Identity Directory • Microsoft Entra ID • Sales Cloud – Analytics & AI • SAP Advanced Financial Closing • SAP Analytics Cloud • SAP Ariba Applications • SAP BTP Account Members (Neo) • SAP BTP XS Advanced UAA (Cloud Foundry) • SAP Business Network • SAP Build Work Zone, advanced edition • SAP Build Work Zone, standard edition • SAP Central Business Configuration • SAP Commerce Cloud • SAP SuccessFactors Incentive Management • SAP Concur version 2 • SAP CPQ • SAP Data Custodian • SAP Enterprise Portal • SAP Fieldglass • SAP Field Service Management • SAP Intelligent Agriculture • SAP Jam Collaboration • SAP Sales Cloud and SAP Service Cloud • SAP SuccessFactors Learning • SAP SuccessFactors version 2 • SAP S/4HANA for procurement planning • SCIM System **Example** **Concur**: The following mapping disables the user account: `{ "user": { "mappings": [ { "scope": "deleteEntity", "constant": "US",`

Transformation Expression	Usage

```
          "targetPath":
"$.Custom21"
        },
        {
          "scope":
"deleteEntity",
          "constant": "",
          "targetPath":
"$.Password"
        },
        {
          "scope":
"deleteEntity",
          "constant": "DEFAULT",
          "targetPath":
"$.LedgerCode"
       },
       {
          "constant": "N",
          "targetPath":
"$.Active",
          "scope": "deleteEntity"
       },
...
```

Example

Microsoft Entra ID: The following mapping disables the user account:

```
{
   "user": {
    "mappings": [

      {
          "constant": false
          "targetPath":
"$.accountEnabled",
          "scope":
"deleteEntity",
        },
...
```

Example

Identity Authentication (SCIM API version 1): The following mapping disables the user account and unassigns the user from all the groups it was a member of:

```
{
   "user": {
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
```

Transformation Expression	Usage

```
        "targetPath": "$.id",
        "scope": "deleteEntity"
      },
      {
         "constant": false,
         "targetPath":
"$.active",
         "scope": "deleteEntity"
       },
       {
         "constant": [],
         "targetPath":
"$.corporateGroups",
         "scope": "deleteEntity"
       },
       {
         "constant": [],
         "targetPath":
"$.groups",
         "scope": "deleteEntity"
       },
...
```

Example

Identity Authentication (SCIM API version 2): The following mapping disables the user account.

Disabling a user account is done with a PATCH operation which replaces the value of the `active` user attribute **`true`** with **`false`**. For this, you also need to set `ias.user.update.instead.delete=true` in the Identity Authentication target system.

For more information, see: List of Properties [page 80]

```
{
   "user":{
      "mappings":[
         {

"constant":"urn:ietf:params:scim
:api:messages:2.0:PatchOp",

"targetPath":"$.schemas[0]",

"scope":"deleteEntity"
         },
         {

"constant":"replace",

"targetPath":"$.Operations[0].op
",
```

Transformation Expression	Usage

```
"scope":"deleteEntity"
        },
        {
           "constant":"active",

"targetPath":"$.Operations[0].pa
th",

"scope":"deleteEntity"
        },
        {
           "constant":false,

"targetPath":"$.Operations[0].va
lue",

"scope":"deleteEntity"
        },
...
```

Example

SAP Jam: The following mapping disables the user account in the target system:

```
 "user": {
       "mappings": [

           {

"sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id",
                "scope":
"deleteEntity"
           },
           {
                "constant":
false,
                "targetPath":
"$.active",
                "scope":
"deleteEntity"
           },
...
```

Possible values:

- *createEntity*
- *patchEntity*
- *deleteEntity*

Data type: String

Transformation Expression	Usage

skipOperations	If you want the provisioning job to not execute operations of a certain type on groups and users , use the `skipOperations` expression. You can apply it when you need to avoid creating, deleting, or updating entities.
	
	You can use `skipOperations` only in target system transformations.
	
	• In the following example, the transformation does not allow updating groups in the target system. This approach is valuable when you have done group changes in the target system, and after a new *Read* or *Resync* job, you want these changes to be kept instead of being overwritten by the groups in the source system.
	
	Example
	
	`"group": { "skipOperations": [ "update" ], "mappings": [ {`
	
	• In the following example, the transformation does not allow creating and deleting users in the target system:
	
	Example
	
	`{ "user": { "skipOperations": [ "create", "delete" ], "mappings": [ {`
	
	You can set the transformation to skip the "`create`" operation if you're sure that all entities from the source system already exist in the target – independently from the Identity Provisioning service. That means, they have been created in the target system manually or via a script. Thus, by skipping the "`create`" operation, the identity provisioning job will only provision the changes instead of creating the users all over again in the target. The Identity Provisioning service will skip creating the existing entities and will try to update them. There are two options for resolving the existing users by the service.
	Option 1: If the update operation is performed using a PATCH request, or a PUT request when the system

Transformation Expression	Usage

	is SCIM-based, the Identity Provisioning service will search and resolve existing users based on the system's default or predefined unique attributes.
	For more information about the SCIM-based systems, see System Types [page 70].
	For more information about the unique attributes of your system, see List of Properties [page 80].
	Option 2: In all other cases that differ from the above mentioned (the target system is LDAP-based, RFC-based, SOAP or OData-based), the existing users are resolved by retrieving their IDs. In order for the *skip create* operation to be executed without an error, you need to make a few adjustments in your source and target transformations. This way, the service will search and retrieve the IDs of the entities correctly. An example how to modify your source and target transformations is included below.
	In the following detailed example, *SAP SuccessFactors* is a source system and *SAP S/4HANA Cloud Public Edition*, which is SOAP-based system, is a target. You want all users that exist on both systems to be skipped during creation and to only be updated on the target. If there are users existing only in the source system, they will not be created in the target. To do that:
	1. Open your *SAP SuccessFactors* admin console. For every user that you want to be provisioned and updated in *SAP S/4HANA Cloud Public Edition*, you need to set up one of its predefined custom attributes (*custom01* – *custom15*), entering the *SAP S/4HANA Cloud Public Edition* unique identifier for that user (which is the value of the `personID` attribute). For example, you can do that in the **custom10** attribute.
	2. Then go to the administration console for SAP Cloud Identity Services, and in the *Properties* tab update the `sf.user.attributes` property, adding **custom10** to the list of attributes.
	3. In the *SAP SuccessFactors* source system, add the following extra user mapping. It maps **custom10** to a custom attribute in *SAP S/4HANA Cloud Public Edition*, for example **custom_user_ID**:
	✤ Example
	`{`
	`"user": {`

Transformation Expression	Usage

```
        "mappings": [
           {
            "sourcePath":
"$.custom10",
            "targetPath":
"$.custom_user_ID",
            "optional": true
           },
...
```

4. In the *SAP S/4 HANA Cloud* target system, replace the following user mapping:

Example

```
{
    "user": {
        "mappings": [
          {
             "targetPath":
"$.personID",

"sourceVariable":
"entityIdTargetSystem"
           },
...
```

with:

Example

```
    {
        "sourcePath":
"$.custom_user_ID",
        "targetVariable":
"entityIdTargetSystem"
      },
      {
        "targetPath":
"$.personID",
        "sourcePath":
"$.custom_user_ID"
      },
```

5. (Optional) If there are users that exist only in the source system and you want them to be reported as skipped directly, without being processed further by the Identity Provisioning, adjust the *SAP S/4 HANA Cloud* target transformation by adding the following condition:

Example

Transformation Expression	Usage

	<pre>{
	"user": {
	"condition":
	"$.custom_user_ID EMPTY
	false",
	"mappings": [
	. . .</pre>
	6. Run the first *Read* provisioning job.
	7. Check in the administration console for SAP Cloud Identity Services if the relevant "marked" users are updated, and that the rest are not. Also, check that users that exist only in *SAP SuccessFactors* (if any) are not created in *SAP S/4 HANA Cloud*.
	Such transformations adjustmants are not needed if your target system is SAP Application Server ABAP. In that case the Identity Provisioning service will search and resolve the existing users in the target system by `userName`.
	Note
	If an entity does not exist in the target system (which means, a retrieved ID does not match any target entity), it will neither be created, nor updated.
	Possible values:
	• *create*
	• *update*
	• *delete*
	Data type: String

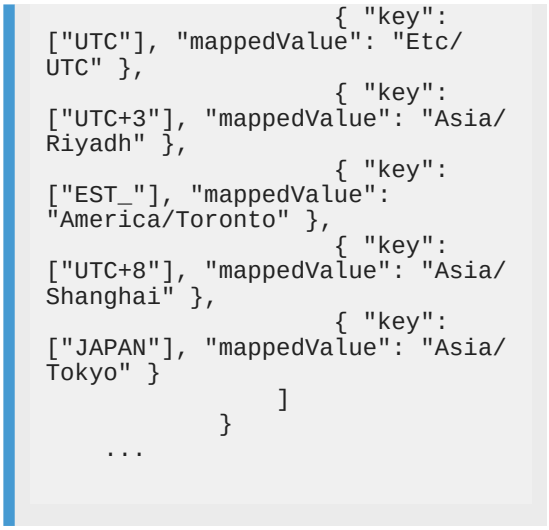
sourcePath	The *sourcePath* expression denotes the path to an attribute in the input JSON message (could be the source system JSON data or the intermediate one).
	Example
	<pre>{
	"sourcePath":
	"$.name.familyName",
	"targetPath": "$.name.familyName"
	},</pre>
	Data type: String

Transformation Expression	Usage
targetPath	The *targetPath* expression denotes the path where the attribute should be stored in the output JSON message (could be the intermediate or the target system JSON data). **Example** `{ "targetPath": "$.name.familyName", "sourcePath": "$.sn[0]" },` **Data type**: String

Transformation Expression	Usage

type	The type of action to be performed in the mapping. It could be *set*, *remove*, *rename* and *valueMapping*.
	• The *set* type maps an attribute from the source system to an attribute in the target JSON data. If no type is defined, `"type": "set"` is used by default.
	Example
	`{ "type": "set", "targetPath": "$.groups" }`
	• The *remove* type deletes an attribute during transformation. This attribute is not present in the target JSON data.
	Example
	`{ "type": "remove", "targetPath": "$.groups" }`
	• The *rename* type renames the last element of a complex targetPath.
	In the example below, the last element `.lastName` of the targethPath `$.name.lastName` will be renamed with the provided constant value `familyName`. This will result in the following targetPath: `"$.name.familyName",`.
	Code Syntax
	`{ "targetPath": "$.name.lastName", "constant": "familyName", "type": "rename", "optional": true }`
	• The *valueMapping* type allows multiple entity attributes (read from the source system) to be mapped to a single target attribute.
	For example, you can set a mapping condition for a user attribute **user.timeZoneCode**. After the provisioning job, its value will be mapped to a new attribute –

Transformation Expression	Usage
	timezone. The example below provides a number of world locations and their relevant timezone. **Note** If the value of **user.timeZoneCode** is not found as a key in the `valueMappings` definitions, or some of the source paths does not return a value, the JSON transformation will use the default one. (In the example below, the default value is *Europe/Berlin*.) **Example** JSON code for mapping user timezone: `"user": { ... { "type": "valueMapping", "sourcePaths": ["$.user.timeZoneCode"], "targetPath": "$.timezone", "defaultValue": "Europe/Berlin", "valueMappings": [{ "key": ["WDFT"], "mappedValue": "Europe/Berlin" }, { "key": ["ISRAEL"], "mappedValue": "Asia/Jerusalem" }, { "key": ["RUS03"], "mappedValue": "Europe/Moscow" }, { "key": ["AUSNSW"], "mappedValue": "Australia/Sydney" }, { "key": ["UTC+4"], "mappedValue": "Asia/Dubai" }, { "key": ["BRAZIL"], "mappedValue": "America/Sao_Paulo" }, { "key": ["BRZLEA"], "mappedValue": "America/Sao_Paulo" }, { "key": ["MSTNO"], "mappedValue": "America/Phoenix" }, { "key": ["EST"], "mappedValue": "America/New_York" },`

Transformation Expression	Usage
	 **Possible values:** • *set* • *remove* • *rename* • *valueMapping* **Data type**: String

Transformation Expression	Usage
valueMappings	Provides a list with mapped values. It can only be used when *type* is set to **`valueMapping`**. The list contains the keys, which are read from the source system and mapped to values in a single target attribute. The example below provides the list of world locations and their relevant timezone. **Example** JSON code for mapping user timezone: `"user": { ... { "type": "valueMapping", "sourcePaths": ["$.user.timeZoneCode"], "targetPath": "$.timezone", "defaultValue": "Europe/Berlin", "valueMappings": [{ "key": ["WDFT"], "mappedValue": "Europe/Berlin" }, { "key": ["ISRAEL"], "mappedValue": "Asia/Jerusalem" }, { "key": ["RUS03"], "mappedValue": "Europe/Moscow" }, { "key": ["AUSNSW"], "mappedValue": "Australia/Sydney" }, { "key": ["UTC+4"], "mappedValue": "Asia/Dubai" }, { "key": ["BRAZIL"], "mappedValue": "America/Sao_Paulo" }, { "key": ["BRZLEA"], "mappedValue": "America/Sao_Paulo" }, { "key": ["MSTNO"], "mappedValue": "America/Phoenix" }, { "key": ["EST"], "mappedValue": "America/New_York" }, { "key": ["UTC"], "mappedValue": "Etc/UTC" }, { "key": ["UTC+3"], "mappedValue": "Asia/Riyadh" },`

Transformation Expression	Usage
	 Data type: JSONObjectArray

Transformation Expression	Usage
sourcePaths	Specifies paths to attributes in the input JSON message from the source system. The *sourcePaths* can only be used with the *valueMappings* expression, which requires the type to be set to **`valueMapping`**. The number of values in the key arrays must correspond to the number of sourcePaths. In the following example, the value pairs US and New York, US and Dallas, Germany and Berlin in the key arrays correspond to the two sourcePaths: country and city. The *mappedValue* is the value to be written to the *targetPath* attribute, such as timezone, provided that each key value matches the corresponding attribute value (e.g., **`US`** for **`country`** and **`New York`** for **`city`**). **Code Syntax** `{ "sourcePaths": ["$.country", "$.city"], "defaultValue": "UTC", "targetPath": "$.timezone", "type": "valueMapping", "valueMappings": [{ "key": ["US", "New York"], "mappedValue": "EST" }, { "key": ["US", "Dallas"], "mappedValue": "CST" }, { "key": ["Germany", "Berlin"], "mappedValue": "CET" }] },` **Note** Paths to the country and city attributes are provided as exemplary values. These attribute paths might be system-specific. For example, in SAP AS ABAP, the "$.country" path is "$.ADDRESS.COUNTRY". **Data type**: StringArray

Transformation Expression	Usage
optional	Specifies whether attribute's availability is mandatory or optional for a given system. • If set to *true*, this means that the attribute is optional and even if it is missing, provisioning won't fail. • If set to *false* or not set at all, this means that the attribute is mandatory and if it is missing, provisioning will fail. For example, the default read transformation of SAP SuccessFactors shows that if the first name of a user is missing in the source system, this user will be read and provisioned to a target system. However, if the last name of a user is missing, provisioning will fail as this is a mandatory attribute. **Code Syntax** `{ "sourcePath": "$.firstName", "optional": true, "targetPath": "$.name.givenName" }, { "sourcePath": "$.lastName", "targetPath": "$.name.familyName" },` **Possible values:** • *true* • *false* **Data type**: Boolean
sourceVariable	The *sourceVariable* expression denotes a variable that is valid only within a given provisioning system. It shows where the variable will be read from. For more information, see Transformation Variables [page 490] **Data type**: String
targetVariable	The *targetVariable* expression denotes a variable that is valid only within a given provisioning system. It shows where the variable will be written to. For more information, see Transformation Variables [page 490] **Data type**: String

Parent topic: Transformations [page 388]

Related Information

Transformation Types [page 390]
Transformation Functions [page 428]
Transformation Variables [page 490]
Transformation Conditions [page 496]
Transformation Examples [page 501]
Transformation Editors [page 505]

1.3.6.3 Transformation Functions

A function is a hardcoded piece of transformation logic that receives a value denoted by the input specified by a source path or a source variable. As a result, the value is replicated into the target path or target variable, accordingly.

The Identity Provisioning service uses two types of functions: *conditional* [page 429] and *mapping* [page 434] functions. The conditional functions are functions which can be used only in conditional statements. The mapping functions are used in entity transformations and are included as mappings.

Functions can also be chained – that means, the output of one function is the input for the next one. See an example with such functions in the **encode / decode** section below.

For more information, see the tables below.

Conditional Functions

Function	Parameters	Description

isValidEmail	• JSON path attribute which holds the email value
	Required: Yes
	Type: String
	• Possible values:
	  • `$.emails[0].value`
	  • `$.emails[*].value`
		This function verifies whether an e-mail address is valid by checking if the given String value matches the following regex pattern:
		`"[a-zA-Z0-9!#$%&‘'*+/=?^_`{\|}~-]+(?>\\.[a-zA-Z0-9!#$%&‘'*+/=?^_`{|}~-]+)*@(?>[a-zA-Z0-9][a-zA-Z0-9-]*[a-zA-Z0-9]?\\.)+[a-zA-Z0-9](?>[a-zA-Z0-9-]*[a-zA-Z0-9])?"`
		After the check, the function returns a Boolean result.
		This function can be used only as part of a conditional statement.
		EXAMPLE 1:
		In this example, the source JSON code contains the following e-mail address of a user:
		`{ "userName": "test", "emails": [ { "value": "example@company.com", "primary": true } ] }`
		The condition with function *isValidEmail* is the following:
		`{ "condition": "isValidEmail($.emails[0].value)", "mappings": [ {`

Function	Parameters	Description

		<pre>"sourcePath":
		"$.userName",
		
		"targetPath":
		"$.userName"
		},
		{
		
		"sourcePath":
		"$.emails[*].value",
		
		"preserveArrayWithSi
		ngleElement": true,
		
		"targetPath":
		"$.emails[?
		(@.value)]"
		}
		]
		}</pre>
		EXAMPLE 2:
		
		In cases where users have multiple emails, the following condition will validate each email in the emails array:
		
		<pre>{
		"condition":
		"isValidEmail($.emai
		ls[*].value)",
		"mappings": [
		{
		......</pre>

Function	Parameters	Description
isApplicationSpecificGroup	Not applicable	This function verifies whether a group is associated with a particular application configured in SAP Cloud Identity Services - Identity Authentication. It checks if the property `ips.application.id` is set for the provisioning system and whether its value matches the group attribute `$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']`. After the check, the function returns a Boolean result. This function can be used only as part of a conditional statement. The function is part of the default write transformation of some target systems (like SAP Advanced Financial Closing).
isRegularGroup	Not applicable	This function verifies whether a group is regular user group. It checks consecutively the values of the group attributes `supportedOperations` and `applicationId` and returns a Boolean result. If one or both of the attributes are missing, the returned result is `true`. Otherwise, the returned value is `false`. This function can be used only as part of a conditional statement. The function is part of the default write transformation of some target systems (like SAP Advanced Financial Closing).

Function	Parameters	Description

isAttributeWithOptionalPrefix	• `attribute-jsonpath` - JSON path attribute
	Required: Yes
	Type: String
	• `property-key` - specifies a property in the system configuration
	Required: Yes
	Type: String
		This function checks whether an attribute value starts with an optional prefix.
		
		The syntax used by the function is as follows:
		
		`isAttributeWithOptionalPrefix(attribute-jsonpath, property-key)`
		
		For example:
		
		`"condition": "isAttributeWithOptionalPrefix($.displayName, s4hana.afc.group.prefix)"`
		
		After the check, the function returns a Boolean result.
		
		The returned result is `true` when one of the following conditions is fulfilled:
		
		• the attribute `attribute-jsonpath` is missing
		• the property `property-key` is missing
		• the value of the attribute `attribute-jsonpath` starts with the value of the property `property-key`
		
		Otherwise, the returned value is `false`.
		
		The function is part of the default write transformation of some target systems (like SAP Advanced Financial Closing).

Function	Parameters	Description

isAttributeWithMandatoryPrefix	• `attribute-jsonpath` - JSON path attribute
	Required: Yes
	Type: String
	• `property-key` - specifies a property in the system configuration
	Required: Yes
	Type: String
		This function checks whether an attribute value starts with a predefined mandatory prefix.
		
		The syntax used by the function is the following:
		
		`isAttributeWithMandatoryPrefix(attribute-jsonpath, property-key)`
		
		For example:
		
		`"condition": "isAttributeWithMandatoryPrefix($.displayName, s4hana.afc.group.prefix)"`
		
		After the check, the function returns a Boolean result.
		
		The returned result is `true` when all of the following conditions are fulfilled:
		
		• the attribute `attribute-jsonpath` is present
		• the property `property-key` is present
		• the value of the attribute `attribute-jsonpath` starts with the value of the property `property-key`
		
		If one or more of the conditions are not fulfilled the returned value is `false`.
		
		The function is part of the default write transformation of some target systems (like SAP Advanced Financial Closing).

Mapping Functions

Function	Parameters	Description

concatString	• **prefix**
	Required: No
	Type: String
	• **suffix**
	Required: No
	Type: Integer; String; Boolean
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function concatenates a string with a prefix or a suffix.
		
		EXAMPLE 1:
		
		In the following transformation example, function *concatString* will be applied to all source logon names (SAM-account user names) read from *Microsoft AD*, adhering a prefix **ips_** and a suffix **123** in the target system. For example, a *sAMAccountName* name **johnsmith** from the source system will be provisioned as **ips_johnsmith123** in the target system.
		
		Example
		
		`{ "user": { "mappings": [ { "sourcePath": "$.sAMAccountName[0]", "targetPath": "$.userName", "functions": [ { "function": "concatString", "prefix": "ips_", "suffix": 123 } ] } ... }`
		
		EXAMPLE 2:

Function	Parameters	Description
		In the following transformation example, function *concatString* will be applied to all source user IDs in a source system, converting them into uniform e-mails in the *Microsoft Entra ID* target system. In this case, the *concatString* function takes the value of the *aad.domain.name* property (which is the name of a verified Microsoft Entra ID domain) and adheres it to the *userId*. For example, a source user ID **johnsmith123** can produce a target principal name **johnsmith123@mail.acme.com**. Example `{ "sourcePath": "$.userId", "targetPath": "$.emails[0].value", "correlationAttribute": true, "functions": [ { "function": "concatString", "suffix": "@%aad.domain.name%" } ] }, ...`

Function	Parameters	Description

convertCountryCode	• **outputFormat**
	Required: Yes
	Possible values:
	  • `fullName`
	  • `alpha2`
	  • `alpha3`
	  • `numeric`
	• **inputAttributes**
	Required: Yes
	Possible values:
	  • Single value or list of address attributes which will be used. The function requires attribute `country`.
	  • `issuingCountry`
	• **outputAttribute**
	Required: Yes
	Possible values: `country`
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function converts the countries into format compliant with *ISO 3166-1*.
		
		`{ "sourcePath": "$.addresses", "preserveArrayWithSingleElement": true, "optional": true, "targetPath": "$.addresses", "functions": [ { "type": "convertCountryCode", "outputFormat": "alpha2", "inputAttributes": ["country"], "outputAttribute": "country" } ] }`
		
		EXAMPLE:
		
		In the following transformation example, function *convertCountryCode* will be applied to all source countries read from the source system, converting them in the wanted ISO format. For example, the `country` name *United States* from the source system will be provisioned in the target system as:
		• *US* if the `outputFormat` is *alpha2*;
		• *USA* if the `outputFormat` is *alpha3*;
		• *840* if the `outputFormat` is *numeric*;
		• *United States* if the `outputFormat` is *fullName*.

Function	Parameters	Description
convertCountryRegion	• **outputFormat** Required: Yes Possible values: • `fullName` • `alpha2` • **inputAttributes** Required: Yes Possible values: • `country` • `region` • **outputAttribute** Required: Yes Possible values: `region` • **condition** Required: No Type: String • **applyOnElements** Required: No Type: Boolean Possible values: • true • false • **applyOnAttribute** Required: No Type: String • **assignToAttribute** Required: No Type: String	This function converts the countries into format compliant with *ISO 3166-2*. `{ "condition": "($.addresses[*].region EMPTY false) && ($.addresses[*].country EMPTY false)", "sourcePath": "$.addresses", "preserveArrayWithSingleElement": true, "optional": true, "targetPath": "$.addresses", "functions": [{ "type": "convertCountryRegion", "outputFormat": "alpha2", "inputAttributes": ["region","country"], "outputAttribute": "region" }] }`  EXAMPLE:  In the following transformation example, function *convertCountryRegion* will be applied to all source regions read from the source system, converting them in the wanted ISO format. For example, the `region` name *US-AL* or only *AL* from the source system will be provisioned to the target system as:  • *US-AL* if the `outputFormat` is *alpha2*;

Function	Parameters	Description
		• *Alabama* if the `outputFormat` is *fullName*.

Function	Parameters	Description

convertLocaleCode	• **outputFormat**
	Required: Yes
	Type: String
	Possible values:
	• `bcp47`
	• `iso639`
		This function converts the locale code from the source format into the format required by the target system.
		
		For example, SAP Concur (connector version 3) expects locale codes in *BCP 47* format (like `en-US`, `de-DE`), while Identity Authentication (connector version 2) uses *ISO 639* language codes (such as `EN`, `DE`).
		
		Code Syntax
		`{`
		`"sourcePath":"$.locale",`
		`"targetPath":"$.locale",`
		`"optional":true,`
		`"functions":[`
		`{`
		`"type":"convertLocaleCode",`
		`"outputFormat":"bcp47"`
		`}`
		`]`
		`},`
		
		When converting from or to *ISO 639*, the function works only with the default language codes used in Identity Authentication. You can view them in the SAP Cloud Identity Services administration console under: *Applications & Resources* > *Master Data* > *Languages*. For more information, see Languages Default Master Data
		
		Languages that do not have a defined mapping, such as EO (Esperanto), IA (Interlingua), IO (In-

Function	Parameters	Description
		terlingue), are mapped to English by default.

Function	Parameters	Description

convertTimezoneCode	• **outputFormat**
	Required: Yes
	Type: String
	Possible values:
	• `iana`
	• `sap`
		This function converts the value of a `timezone` attribute into another format, specified by the JSON modifier `"outputFormat"`.
		
		The `"outputFormat"` modifier has two possible values:
		
		• `iana`, which follows the convention **`{AREA}/{LOCATION}`**
		• `sap`, which uses abbreviations
		
		The function retrieves the value of the `timezone` from the specified source path and searches for it within a predefined list of timezones and their corresponding world locations:
		
		Code Syntax
		
		`WET,Europe/Lisbon`
		`WET,Atlantic/Canary`
		`WET,Atlantic/Faroe`
		`WET,Atlantic/Madeira`
		`WAT,Africa/Lagos`
		`WAT,Africa/Algiers`
		`WAT,Africa/Bangui`
		`WAT,Africa/Brazzaville`
		`WAT,Africa/Douala`
		`WAT,Africa/Kinshasa`
		`WAT,Africa/Libreville`
		`WAT,Africa/Luanda`
		`WAT,Africa/Malabo`
		`WAT,Africa/Ndjamena`
		`WAT,Africa/Niamey`
		`WAT,Africa/Porto-Novo`
		`ART,America/Argentina/Buenos_Aires`
		`ART,America/Argentina/Catamarca`

Function	Parameters	Description
		ART,America/Argentina/Cordoba ART,America/Argentina/Jujuy ART,America/Argentina/La_Rioja ART,America/Argentina/Mendoza ART,America/Argentina/Rio_Gallegos ART,America/Argentina/Salta ART,America/Argentina/San_Juan ART,America/Argentina/San_Luis ART,America/Argentina/Tucuman ART,America/Argentina/Ushuaia PST,America/Los_Angeles PST,America/Tijuana PST,America/Vancouver PKT,Asia/Karachi URUGUA,America/Montevideo TURKEY,Europe/Istanbul SYRIA,Asia/Damascus RUS12,Asia/Kamchatka RUS12,Asia/Anadyr RUS11,Asia/Sakhalin RUS11,Asia/Magadan RUS11,Asia/Srednekolymsk RUS10,Asia/Vladivostok RUS10,Asia/Ust-Nera RUS09,Asia/Yakutsk RUS09,Asia/Chita RUS09,Asia/Khandyga RUS08,Asia/Irkutsk RUS07,Asia/Novosibirsk RUS07,Asia/Barnaul

Function	Parameters	Description
		RUS07,Asia/Krasnoyarsk RUS07,Asia/Novokuznetsk RUS07,Asia/Tomsk RUS06,Asia/Omsk RUS05,Asia/Yekaterinburg RUS04,Europe/Astrakhan RUS04,Europe/Samara RUS04,Europe/Saratov RUS04,Europe/Ulyanovsk RUS03,Europe/Moscow RUS03,Europe/Kirov RUS03,Europe/Volgograd RUS02,Europe/Kaliningrad PIERRE,America/Miquelon PARAGY,America/Asuncion NZST,Pacific/Auckland NZST,Antarctica/McMurdo NZCHA,Pacific/Chatham NST,America/St_Johns NORFLK,Pacific/Norfolk NEPAL,Asia/Kathmandu MSTNO,America/Phoenix MSTNO,America/Creston MSTNO,America/Dawson_Creek MSTNO,America/Fort_Nelson MSTNO,America/Hermosillo MSTNO,America/Mazatlan MSTNO,America/Whitehorse MSTNO,America/Dawson MST,America/Denver MST,America/Boise MST,America/Cambridge_Bay

Function	Parameters	Description
		MST,America/Ciudad_Juarez MST,America/Edmonton MST,America/Inuvik MST,America/Yellowknife MOROCC,Africa/Casablanca MOLDVA,Europe/Chisinau MAU,Indian/Mauritius LBANON,Asia/Beirut JORDAN,Asia/Amman JAPAN,Asia/Tokyo ISRAEL,Asia/Jerusalem IRAQ,Asia/Baghdad IRAN,Asia/Tehran INDIA,Asia/Kolkata HAW,Pacific/Honolulu GSTW,America/Thule GSTE,America/Scoresbysund GST,America/Nuuk GMTUK,Europe/London GMTUK,Europe/Dublin GMTUK,Europe/Guernsey GMTUK,Europe/Isle_of_Man GMTUK,Europe/Jersey FLKLND,Atlantic/Stanley FIJI,Pacific/Fiji ESTNO,America/Atikokan ESTNO,America/Cancun EST_NA,America/Havana EST,America/New_York EST,America/Detroit EST,America/Grand_Turk EST,America/Indiana/Indianapolis EST,America/Indiana/Marengo

Function	Parameters	Description
		EST,America/Indiana/Petersburg EST,America/Indiana/Vevay EST,America/Indiana/Vincennes EST,America/Indiana/Winamac EST,America/Iqaluit EST,America/Kentucky/Louisville EST,America/Kentucky/Monticello EST,America/Nassau EST,America/Nipigon EST,America/Pangnirtung EST,America/Port-au-Prince EST,America/Thunder_Bay EST,America/Toronto EGYPT,Africa/Cairo EET,Europe/Athens EET,Asia/Gaza EET,Asia/Hebron EET,Europe/Bucharest EET,Europe/Helsinki EET,Europe/Kiev EET,Europe/Mariehamn EET,Europe/Riga EET,Europe/Sofia EET,Europe/Tallinn EET,Europe/Uzhgorod EET,Europe/Vilnius EET,Europe/Zaporozhye CYPRUS,Asia/Nicosia CYPRUS,Asia/Famagusta CSTNO,America/Mexico_City CSTNO,America/Bahia_Banderas CSTNO,America/Chihuahua

Function	Parameters	Description
		CSTNO,America/Merida CSTNO,America/Monterrey CSTNO,America/Regina CSTNO,America/Swift_Current CST,America/Chicago CST,America/Indiana/Knox CST,America/Indiana/Tell_City CST,America/Matamoros CST,America/Menominee CST,America/North_Dakota/Beulah CST,America/North_Dakota/Center CST,America/North_Dakota/New_Salem CST,America/Ojinaga CST,America/Rainy_River CST,America/Rankin_Inlet CST,America/Resolute CST,America/Winnipeg CHILEM,America/Punta_Arenas CHILEM,America/Coyhaique CHILEE,Pacific/Easter CHILE,America/Santiago CET,Europe/Berlin CET,Africa/Ceuta CET,Arctic/Longyearbyen CET,Europe/Amsterdam CET,Europe/Andorra CET,Europe/Belgrade CET,Europe/Bratislava CET,Europe/Brussels CET,Europe/Budapest

Function	Parameters	Description
		CET,Europe/Busingen CET,Europe/Copenhagen CET,Europe/Gibraltar CET,Europe/Ljubljana CET,Europe/Luxembourg CET,Europe/Madrid CET,Europe/Malta CET,Europe/Monaco CET,Europe/Oslo CET,Europe/Paris CET,Europe/Podgorica CET,Europe/Prague CET,Europe/Rome CET,Europe/San_Marino CET,Europe/Sarajevo CET,Europe/Skopje CET,Europe/Stockholm CET,Europe/Tirane CET,Europe/Vaduz CET,Europe/Vatican CET,Europe/Vienna CET,Europe/Warsaw CET,Europe/Zagreb CET,Europe/Zurich CAT,Africa/Johannesburg CAT,Africa/Blantyre CAT,Africa/Bujumbura CAT,Africa/Gaborone CAT,Africa/Harare CAT,Africa/Juba CAT,Africa/Khartoum CAT,Africa/Kigali CAT,Africa/Lubumbashi CAT,Africa/Lusaka CAT,Africa/Maputo CAT,Africa/Maseru CAT,Africa/Mbabane CAT,Africa/Tripoli CAT,Africa/Windhoek BRZLWE,America/Boa_Vista BRZLWE,America/Campo_Grande

Function	Parameters	Description
		BRZLWE,America/Cuiaba BRZLWE,America/Manaus BRZLWE,America/Porto_Velho BRZLAN,America/Eirunepe BRZLAN,America/Rio_Branco BRAZIL,America/Sao_Paulo BRAZIL,America/Araguaina BRAZIL,America/Bahia BRAZIL,America/Belem BRAZIL,America/Fortaleza BRAZIL,America/Maceio BRAZIL,America/Recife BRAZIL,America/Santarem BDT,Asia/Dhaka AZT,Asia/Baku AZOREN,Atlantic/Azores AUSWA,Australia/Perth AUSVIC,Australia/Melbourne AUSTAS,Australia/Hobart AUSSA,Australia/Adelaide AUSSA,Australia/Broken_Hill AUSQLD,Australia/Brisbane AUSQLD,Australia/Lindeman AUSNT,Australia/Darwin AUSNSW,Australia/Sydney AUSNSW,Antarctica/Macquarie AUSLHI,Australia/Lord_Howe AUSEUC,Australia/Eucla AST,America/Halifax AST,America/Glace_Bay AST,America/Goose_Bay AST,America/Moncton

Function	Parameters	Description
		AST,Atlantic/Bermuda ALAW,America/Adak ALA,America/Anchorage ALA,America/Juneau ALA,America/Metlakatla ALA,America/Nome ALA,America/Sitka ALA,America/Yakutat AFGHAN,Asia/Kabul UTC+12,Etc/GMT-12 UTC+12,Pacific/Funafuti UTC+12,Pacific/Kwajalein UTC+12,Pacific/Majuro UTC+12,Pacific/Nauru UTC+12,Pacific/Tarawa UTC+12,Pacific/Wake UTC+12,Pacific/Wallis UTC+11,Etc/GMT-11 UTC+11,Antarctica/Casey UTC+11,Pacific/Bougainville UTC+11,Pacific/Efate UTC+11,Pacific/Guadalcanal UTC+11,Pacific/Kosrae UTC+11,Pacific/Noumea UTC+11,Pacific/Pohnpei UTC+10,Etc/GMT-10 UTC+10,Pacific/Bougainville UTC+10,Pacific/Efate UTC+10,Pacific/Guadalcanal UTC+10,Pacific/Kosrae UTC+10,Pacific/Noumea UTC+10,Pacific/Pohnpei UTC+10,Antarctica/DumontDUrville UTC+9,Etc/GMT-9 UTC+9,Asia/Dili

Function	Parameters	Description
		UTC+9,Asia/Jayapura UTC+9,Asia/Pyongyang UTC+9,Asia/Seoul UTC+9,Pacific/Palau UTC+8,Etc/GMT-8 UTC+8,Asia/Brunei UTC+8,Asia/Choibalsan UTC+8,Asia/Hong_Kong UTC+8,Asia/Kuala_Lumpur UTC+8,Asia/Kuching UTC+8,Asia/Macau UTC+8,Asia/Makassar UTC+8,Asia/Manila UTC+8,Asia/Shanghai UTC+8,Asia/Singapore UTC+8,Asia/Taipei UTC+8,Asia/Ulaanbaatar UTC+7,Etc/GMT-7 UTC+7,Antarctica/Davis UTC+7,Asia/Bangkok UTC+7,Asia/Ho_Chi_Minh UTC+7,Asia/Hovd UTC+7,Asia/Jakarta UTC+7,Asia/Phnom_Penh UTC+7,Asia/Pontianak UTC+7,Asia/Vientiane UTC+7,Indian/Christmas UTC+6,Etc/GMT-6 UTC+6,Indian/Chagos UTC+6,Antarctica/Vostok UTC+6,Asia/Bishkek UTC+6,Asia/Thimphu UTC+6,Asia/Urumqi UTC+63,Asia/Yangon UTC+63,Indian/Cocos UTC+5,Etc/GMT-5

Function	Parameters	Description
		UTC+5,Antarctica/Mawson UTC+5,Asia/Almaty UTC+5,Asia/Anadyr UTC+5,Asia/Aqtau UTC+5,Asia/Aqtobe UTC+5,Asia/Ashgabat UTC+5,Asia/Atyrau UTC+5,Asia/Dushanbe UTC+5,Asia/Oral UTC+5,Asia/Qostanay UTC+5,Asia/Qyzylorda UTC+5,Asia/Samarkand UTC+5,Asia/Tashkent UTC+5,Indian/Kerguelen UTC+5,Indian/Maldives UTC+53,Asia/Colombo UTC+4,Etc/GMT-4 UTC+4,Asia/Dubai UTC+4,Asia/Muscat UTC+4,Asia/Tbilisi UTC+4,Asia/Yerevan UTC+4,Indian/Mahe UTC+4,Indian/Reunion UTC+3,Etc/GMT-3 UTC+3,Africa/Addis_Ababa UTC+3,Africa/Asmara UTC+3,Africa/Dar_es_Salaam UTC+3,Africa/Djibouti UTC+3,Africa/Kampala UTC+3,Africa/Mogadishu UTC+3,Africa/Nairobi UTC+3,Antarctica/Syowa UTC+3,Asia/Aden UTC+3,Asia/Bahrain UTC+3,Asia/Kuwait UTC+3,Asia/Qatar UTC+3,Asia/Riyadh UTC+3,Europe/Minsk

Function	Parameters	Description
		UTC+3,Europe/Simferopol UTC+3,Indian/Antananarivo UTC+3,Indian/Comoro UTC+3,Indian/Mayotte UTC+2,Etc/GMT-2 UTC+1,Etc/GMT-1 UTC+1,Africa/El_Aaiun UTC+1,Africa/Tunis UTC+1,Africa/Dar_es_Salaam UTC+1,Africa/Djibouti UTC+1,Africa/Kampala UTC+1,Africa/Mogadishu UTC,Etc/UTC UTC,Africa/Abidjan UTC,Africa/Accra UTC,Africa/Bamako UTC,Africa/Banjul UTC,Africa/Bissau UTC,Africa/Conakry UTC,Africa/Dakar UTC,Africa/Freetown UTC,Africa/Lome UTC,Africa/Monrovia UTC,Africa/Nouakchott UTC,Africa/Ouagadougou UTC,Africa/Sao_Tome UTC,America/Danmarkshavn UTC,Antarctica/Troll UTC,Atlantic/Reykjavik UTC,Atlantic/St_Helena UTC,Etc/GMT UTC,Etc/GMT+0 UTC,Etc/GMT-0 UTC,Etc/GMT0 UTC,Etc/Greenwich UTC,Etc/Universal UTC,Etc/Zulu UTC-12,Etc/GMT+12 UTC-11,Etc/GMT+11 UTC-11,Pacific/Midway

Function	Parameters	Description
		UTC-11,Pacific/Niue UTC-11,Pacific/Pago_Pago UTC-10,Etc/GMT+10 UTC-10,Pacific/Rarotonga UTC-10,Pacific/Tahiti UTC-9,Etc/GMT+9 UTC-9,Pacific/Gambier UTC-8,Etc/GMT+8 UTC-8,Pacific/Pitcairn UTC-7,Etc/GMT+7 UTC-6,Etc/GMT+6 UTC-6,America/Belize UTC-6,America/Costa_Rica UTC-6,America/El_Salvador UTC-6,America/Guatemala UTC-6,America/Managua UTC-6,America/Tegucigalpa UTC-6,Pacific/Galapagos UTC-5,Etc/GMT+5 UTC-5,America/Bogota UTC-5,America/Cayman UTC-5,America/Guayaquil UTC-5,America/Jamaica UTC-5,America/Lima UTC-5,America/Panama UTC-4,Etc/GMT+4 UTC-4,America/Anguilla UTC-4,America/Antigua UTC-4,America/Aruba UTC-4,America/Barbados UTC-4,America/Blanc-Sablon UTC-4,America/Caracas UTC-4,America/Curacao UTC-4,America/Dominica

Function	Parameters	Description
		UTC-4,America/Grenada UTC-4,America/Guadeloupe UTC-4,America/Guyana UTC-4,America/Kralendijk UTC-4,America/La_Paz UTC-4,America/Lower_Princes UTC-4,America/Marigot UTC-4,America/Martinique UTC-4,America/Montserrat UTC-4,America/Port_of_Spain UTC-4,America/Puerto_Rico UTC-4,America/Santo_Domingo UTC-4,America/St_Barthelemy UTC-4,America/St_Kitts UTC-4,America/St_Lucia UTC-4,America/St_Thomas UTC-4,America/St_Vincent UTC-4,America/Tortola UTC-3,Etc/GMT+3 UTC-3,America/Cayenne UTC-3,America/Paramaribo UTC-3,Antarctica/Palmer UTC-3,Antarctica/Rothera UTC-2,Etc/GMT+2 UTC-2,America/Noronha UTC-2,Atlantic/South_Georgia UTC-1,Etc/GMT+1 UTC-1,Atlantic/Cape_Verde Depending on the search result, you can expect the following behavior: • If the value of the `timezone` attribute is

Function	Parameters	Description

		found in the predefined list of mapped values, it is provisioned in the specified format.
		
		Note
		When the conversion is set to `iana` output format, the value of the `timezone` attribute is converted to its default world region mapping value set for the system, which is the first matching timezone entry from the list. This could result in a changed timezone value during the initial provisioning, in which the `"outputFormat"` modifier is set to `iana` output format.
		
		• If there is not `timezone` attribute value in the specified source path or it is not found in the predefined list, the function returns the default value for the specified output format, as follows:
		  • for `iana` output format - the value `Europe/Berlin`
		  • for `sap` output format - the value `CET`
		
		EXAMPLE 1:
		
		In the following transformation example, function *convertTimezoneCode* will be applied to the timezone attribute values from the specified source path, converting them in `sap` output format.

Function	Parameters	Description

		For example, the time zone **America/Phoenix** will be provisioned as MSTNO.
		
		`{ "mappings": [ { "sourcePath": "$.timezone", "targetPath": "$.timezone", "functions": [ { "type": "convertTimezoneCode", "outputFormat": "sap" } ] } ] }`
		
		EXAMPLE 2:
		
		In the following transformation example, function *convertTimezoneCode* will be applied to the timezone attribute values from the specified source path, converting them in iana output format.
		
		For example, the time zone **AST** will be provisioned as America/Halifax, which is the default value set for the relevant timezone.
		
		`{ "mappings": [ { "sourcePath": "$.timezone", "targetPath": "$.timezone", "functions": [ { "type": "convertTimezoneCode", "outputFormat": "iana"`

Function	Parameters	Description

		<pre>}
		]
		}
		]
		}</pre>

Function	Parameters	Description

compositeId	• **separator**
	Required: Yes
	Possible values: : (colon)
	• **subId**
	Required: Yes
	Type: String; JSONPath expression
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function appends the value referred by the source path to the value of the `subId` parameter, separated by ":".
		
		Example
		Input JSON code before transformation:
		`{ "qualifierId": "johnsmith" }`
		
		Example
		Transformation using the function:
		`{ "sourcePath": "$.qualifierId", "targetPath": "$['composite_id.custom_separator.colon']", "functions": [ { "function": "compositeId", "separator": ":", "subId": "e0123456" } ] }, ...`
		
		Example
		After the function execution, the output JSON results to:
		`{ "composite_id.default_separator": "johnsmith:e0123456",`

Function	Parameters	Description
		"composite_id.custom_separator.colon": "johnsmith:e0123456", "composite_id.custom_separator.dash": "johnsmith-e0123456" }

Function	Parameters	Description

copyMapEntry	• **sourceKey**
	Required: Yes
	Type: String
	• **targetKey**
	Required: Yes
	Type: String
		This function copies a key-value pair from the source JSON and creates a similar one in the target JSON. The nelwy created pair consists of the specified `targetKey` String and the value copied from the initial pair. The two key-value pairs are written into the target system.
		
		EXAMPLE:
		
		For example, if the source entity contains the following roles:
		
		`{ "role": [{"roleName": "ROLE1"}, {"roleName": "ROLE2"}] }`
		
		After applying the function:
		
		`{ "sourcePath": "$.user.role", "optional": true, "targetPath": "$.groups", "preserveArrayWithSingleElement": true, "functions": [ { "function": "copyMapEntry", "sourceKey": "roleName", "targetKey": "value" } ] }`
		
		Two pairs with `targetKey` **`"value"`** are added to the target JSON and the result is:
		
		`{ "groups": [{"roleName": "ROLE1", "value":"ROLE1"}, {"roleName": "ROLE2", "value":"ROLE2"}] }`

Function	Parameters	Description

decode	• **algorithm**
	Required: Yes
	Possible values:
	• base32
	• base64
	• **skipPadding**
	Required: No
	Type: Boolean
	Possible values:
	• true
	• false
	Default value: false
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	• true
	• false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		The function **decode** is part of the Write transformation of some proxy systems (like AS ABAP).It decodes the string value specified by the sourcePath or sourceVariable mapping attribute, by using specified decoding algorithm.
		
		Function result: Byte array
		
		EXAMPLE:
		
		For example, if the input JSON code contains the following user name:
		
		`{ "USERNAME": "JOHN*SMITH/Junior" }`
		
		The base32 encoded result will be the following id:
		
		Example
		
		`{ "id": "JJHUQTRKKNGUSVCIF5FHK3TJN5ZA====" }`
		
		The *decode* function reads the **id** and converts it into a byte array. Afterward, the *toString* function reads this byte array and converts it into the original **USERNAME** string. The base32 padding symbols (==) are skipped.
		
		Example
		
		`// Write Transformation { "user": { "mappings": [ { "sourcePath": "$.id",`

Function	Parameters	Description

		<pre>"targetPath": "$.USERNAME",
		"functions": [
		{
		"function": "decode",
		"algorithm": "base32",
		"skipPadding": true
		},
		{
		"function": "toString"
		}
		]
		},
		...</pre>

Function	Parameters	Description

elementAt	• **index**
	Required: Yes
	Type: Integer
		This function is used to retrieve a specific element from an array based on its index. The index indicates the element you want to access.
		
		Using the **elementAt** function with index 0 in this example results in requesting the first email within an array of primary emails. After evaluating the sourcePath, it returns all primary emails. Applying the **elementAt** function with index 0 then retrieves the first primary email from this array.
		
		Code Syntax
		
		`{`
		
		`"condition":"$.emails[?(@.primary == true)].value != []",`
		
		`"sourcePath":"$.emails[?(@.primary == true)].value",`
		
		`"preserveArrayWithSingleElement":true,`
		
		`"optional":true,`
		
		`"targetPath":"$.email",`
		`"functions":[`
		`{`
		
		`"function":"elementAt",`
		
		`"index":0`
		`}`
		`]`
		`},`

Function	Parameters	Description

encode	• **algorithm**
	Required: Yes
	Possible values:
	  • base32
	  • base64
	• **skipPadding**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	Default value: false
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		The function **encode** is part of the Read transformation of some proxy systems (like AS ABAP). It encodes the value specified by the sourcePath or sourceVariable mapping attribute, by using specified encoding algorithm.
		
		The function can accept a Byte array or everything that has a string representation. The string is assumed in UTF-8 character encoding.
		
		Function result: String
		
		EXAMPLE:
		
		For example, if the input JSON code contains the following user name:
		
		`{ "USERNAME": "JOHN*SMITH/Junior" }`
		
		The base32 encoded result will be the following id:
		
		Example
		
		`{ "id": "JJHUQTRKKNGUSVCIF5FHK3TJN5ZA====" }`
		
		The *encode* function reads the **USERNAME** string and writes the encoded value into the **id** path. The base32 padding symbols (==) are skipped.
		
		Example
		
		`// Read Transformation { "user": {`

Function	Parameters	Description
		"mappings": [{ "sourcePath": "$.USERNAME", "targetPath": "$.id", "targetVariable": "entityIdSourceSystem", "functions": [{ "function": "encode", "algorithm": "base32", "skipPadding": true }] }, ...

Function	Parameters	Description

getMatchedRegexGroup	• **regex**
	Required: Yes
	Type: String
	• **groupIndex**
	Required: Yes
	Type: Integer
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function checks whether a given string matches the provided regular expression (regex), and returns the required group.
		
		For example, when the value of groupIndex is:
		• *0* – the function returns the entire string
		• Positive number – the function returns a specific group from the string. For instance, `groupIndex` = *3* will return the third group.
		
		Example
		Initial user data:
		`{ "distinguishedName": [ "CN=test10,OU=subUsers,OU=testUsers" ] }`
		
		Example
		Transformation in the source system:
		`{ "group": { "mappings": [ { ... "sourcePath": "$.distinguishedName[0]", "functions": [ { "function": "getMatchedRegexGroup", "regex": "(?`

Function	Parameters	Description

		<pre>i)cn=.*?,
		(.*?),OU=testUser
		s",
		
		"groupIndex": 1
		}
		],
		
		"targetPath": "$
		['urn:sap:cloud:s
		cim:schemas:exten
		sion:ad:2.0:User'
		]['nestedPath']"
		}
		]
		}</pre>
		Example
		Nested path extracted from the user DN, after the transformation:
		<pre>{
		
		"urn:sap:cloud:sc
		im:schemas:extens
		ion:ad:2.0:User":
		{
		
		"nestedPath":
		"OU=subUsers"
		}
		}</pre>

Function	Parameters	Description

manipulateDate	• **sourceDateFormat**
	Required: No
	Type: Date; Time
	• **targetDateFormat**
	Required: No
	Type: Date; Time
	• **years**
	Required: No
	Type: Integer
	• **months**
	Required: No
	Type: Integer
	• **days**
	Required: No
	Type: Integer
	• **hours**
	Required: No
	Type: Integer
	• **minutes**
	Required: No
	Type: Integer
	• **seconds**
	Required: No
	Type: Integer
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function converts one date format into another after JSON-Path transformations.
		
		Use cases:
		• A Java date format can be converted into another Java date format.
		Example: "**2018-02-28 11:00:00.000**" to "**02/28/2018**"
		• A date format based on *Unix Time Stamp* can be converted into a Java one. That means, if the source system stores a date as a number of milliseconds, after the transformations this number will be converted and written in the target system as a human readable date.
		Example: "**Date(1519809649123-0240)**" to "**2018-02-28 UTC+1**"
		
		Note
		Bear in mind the following restrictions about *Unix Time Stamp* format:
		• It is mainly applicable for SAP SuccessFactors connectors.
		• If the source date format contains a timezone (GMT, EST, ACT, etc.), after converting from *Unix Time Stamp*, the date will be displayed as a UTC offset.
		• During calculation, the timezone is ignored – the milliseconds are converted to a "pure"

Function	Parameters	Description
		date. The timezone is displayed (as UTC offset) but not taken into account. The *manipulateDate* function supports the following operations: • (**Java**) Incrementing the date by the "**+**" sign or when there is no sign • (**Java**) Decrementing the date by the "**–**" sign • (**Unix Time Stamp**) Converting a number of milliseconds into a human readable date **Note** The parameters `sourceDateFormat` and `targetDateFormat` accept values in short or full date & time format. For example: *yyyy-MM-dd'T'HH:mm:ss'Z'*, **`d MMM uuuu`**, or **`HH:mm dd/MM/yyyy`**. **Example** **Reads and writes the current date in standard Java date format** `{ "targetPath": "$.EmployeeType.ValidityPeriod.StartDate", "sourceVariable": "currentDate", "functions": [ { "function": "manipulateDate",`

Function	Parameters	Description

		"targetDateFormat" : "MM/dd/yyyy",
		
		"sourceDateFormat" : "yyyy-MM-dd HH:mm:ss.SSS",
		
		"years":
		
		"months":
		// You can also, for example, increment the date with 3 days and 2 hours
		
		"days": "3"
		
		"hours": "+2"
		
		"minutes":
		
		"seconds":
		}
		...
		}

		Example **Reads a given date in Unix Time Stamp format (in milliseconds) and writes the converted value in the target system as a standard Java date format** { "targetPath": "$.EmployeeType.ValidityPeriod.StartDate", "sourcePath": "$date", "functions": [{ "function": "manipulateDate", "sourceDateFormat": "Date(milliseconds)",

Function	Parameters	Description

		<pre>"targetDateFormat": "yyyy-MM-dd HH:mm:ss.SSS"
		}
		...
		}</pre>

Function	Parameters	Description

matchRegex	• **regex**
	Required: Yes
	Type: String
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function checks whether a given string matches the provided regular expression and returns a Boolean result.
		
		Example
		Checks whether the value of attribute "COMPOSITE_PRIVILEGE" matches the regular expression "*_COMP", and assigns the result (true or false) to attribute "composite".
		
		`{ "mappings": [ { "sourcePath": "$.PRIVILEGES", "targetPath": "$.groups", "functions": [ { "function": "matchRegex", "regex": ".*_COMP", "applyOnAttribute": "COMPOSITE_PRIVILEGE", "assignToAttribute": "composite" } ] } ] }`

Function	Parameters	Description

putIfAbsent	• **key**
	Required: Yes
	Type: String
	• **defaultValue**
	Required: Yes
	Type: String; Array; Integer; another value type
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function works on user attributes of type array (list of elements), as each element consists of key-value pairs. If an element key misses a value, the putIfAbsent function will set a default value for this key.
		
		Example
		This function is part of the default **Identity Authentication** target and proxy write transformation. In this case, the user attribute "**addresses**" is a list of addresses. The **putIfAbsent** function is applied on every single address (element) of this list.
		
		`...`
		`{`
		`"sourcePath": "$.addresses",`
		`"targetPath": "$.addresses",`
		`"preserveArrayWithSingleElement": true,`
		`"defaultValue": [],`
		`"optional": true,`
		`"functions": [`
		`{`
		`"function": "putIfAbsent",`
		`"key": "type",`
		`"defaultValue": "work"`
		`}`
		`]`
		`},`
		`...`

Function	Parameters	Description

putIfPresent	• **key**
	Required: Yes
	Type: String
	• **defaultValue**
	Required: Yes
	Type: String; Array; Integer; another value type
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function works on user attributes of type **array** (list of elements), as each element consists of *key-value* pairs.
		
		• If an element key has a value but it's different than the default one, the **putIfPresent** function will set a default value for this key.
		• If an element has a value and it's equal to the default one, the **putIfPresent** function keeps this value as is.
		
		Example
		This function is part of the default **Identity Authentication** target and proxy write transformation. In this case, the user attribute "**addresses**" is a list of addresses. The **putIfPresent** function is applied on every single address (element) of this list.
		
		`...`
		`{`
		`"sourcePath": "$.addresses",`
		`"targetPath": "$.addresses",`
		`"preserveArrayWithSingleElement": true,`
		`"defaultValue": [],`
		`"optional": true,`
		`"functions": [`
		`{`
		`"condition": "(@.type NIN ['work', 'home'])",`
		`"function": "putIfPresent",`

Function	Parameters	Description
		"key": "type", "defaultValue": "work" }] }, ...

Function	Parameters	Description

randomPassword	• **passwordLength**
	Required: Yes
	Type: Integer
	• **minimumNumberOfLowercaseLetters**
	Required: Yes
	Type: Integer
	• **minimumNumberOfUppercaseLetters**
	Required: Yes
	Type: Integer
	• **minimumNumberOfDigits**
	Required: Yes
	Type: Integer
	• **minimumNumberOfSpecialSymbols**
	Required: Yes
	Type: Integer
	• **specialSymbols**
	Required: No
	Possible values: List of allowed special symbols (no comma separation)
	Default value: ~!@#$%^&*()_+
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function generates a random password. It picks characters from four character sets - digits, lowercase letters, uppercase letters, and special symbols.
		
		Bear in mind the following tips:
		• The password length must be supplied along with the number of characters from each set. If a value "**0**" is supplied for a given parameter, no characters will be picked from the corresponding character set.
		• If the summed up number of characters (from all sets) exceeds the total password length, the function execution will result in error.
		• If the summed up number of characters (from all sets) is less than the total password length, the remaining characters will be randomly picked from all character sets.
		• A custom character set is supplied by the `specialSymbols` parameter.
		• If a custom set of special symbols is supplied, the parameter `minimumNumberOfSpecialSymbols` cannot have a value of "**0**".
		
		Note
		The **randomPassword** function does not require `sourcePath`, `sourceVariable`, or `constant` to be specified in the mapping.
		
		Example

Function	Parameters	Description

		{
		"user": {
		"mappings": [
		{
		"targetPath": "$.password",
		"functions": [
		{
		"function": "randomPassword",
		"passwordLength": 16,
		"minimumNumberOfLowercaseLetters": 4,
		"minimumNumberOfUppercaseLetters": 4,
		"minimumNumberOfDigits": 4,
		"minimumNumberOfSpecialSymbols": 4,
		"specialSymbols": ",.<>/?~`!@#"
		}
		...
		}

Function	Parameters	Description

renameMapEntry	• **sourceKey**
	Required: Yes
	Type: String
	• **targetKey**
	Required: Yes
	Type: String
		This function reads the `sourceKey` String from the source JSON and renames it with the given `targetKey` String in the target JSON.
		
		EXAMPLE:
		
		For example, if the source entity contains the following roles:
		
		`{ "role": [{"roleName": "ROLE1"}, {"roleName": "ROLE2"}] }`
		
		After applying the function:
		
		`{ "sourcePath": "$.user.role", "optional": true, "targetPath": "$.groups", "preserveArrayWithSingleElement": true, "functions": [ { "function": "renameMapEntry", "sourceKey": "roleName", "targetKey": "value" } ] }`
		
		The `sourceKey` String in the target JSON is replaced with the one given in the `targetKey` and the result is:
		
		`{ "groups": [{"value":"ROLE1"}, {"value":"ROLE2"}] }`

Function	Parameters	Description

replaceString	• **target**
	Required: Yes
	Type: String; number
	• **replacement**
	Required: Yes
	Type: String; number
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function reads strings from the source system and then replaces each substring that matches the value of the target parameter with the replacement one.
		
		For example, let's say the source system contains a string "*National College of Linguistics and History*". The `target` substring is "**National**" and the `replacement` substring is "**European**". Then, in the target system the main string will be written as "*European College of Linguistics and History*".
		
		Example
		
		`{ "user": { "mappings": [ { "sourcePath": "$.sAMAccountName[0]", "targetPath": "$.userName", "functions": [ { "function": "replaceString", "target": "iag", "replacement": "ips" } ] } ... }`

Function	Parameters	Description

replaceFirstString	• **regex**
	Required: Yes
	Type: String; number
	• **replacement**
	Required: Yes
	Type: String; number
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function replaces the first substring of a given string that matches the provided regex with the string in replacement.
		
		Example
		
		`{ "user": { "mappings": [ { "sourcePath": "$.sAMAccountName[0]", "targetPath": "$.userName", "functions": [ { "function": "replaceFirstString", "regex": "14\\d{1}", "replacement": 123 } ] } ... }`

Function	Parameters	Description

replaceLastString	• **regex**
	Required: Yes
	Type: String; number
	• **replacement**
	Required: Yes
	Type: String; number
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function replaces the last substring of a given string that matches the provided regex with the string in replacement.
		
		Example
		
		`{ "user": { "mappings": [ { "sourcePath": "$.sAMAccountNam e[0]", "targetPath": "$.userName", "functions": [ { "function": "replaceLastStrin g", "regex": "14\ \d{1}", "replacement": 123 } ] } ... }`

Function	Parameters	Description

replaceAllString	• **regex**
	Required: Yes
	Type: String; number
	• **replacement**
	Required: Yes
	Type: String; number
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function replaces each sub-string of the given string that matches the provided regex with the string in replacement.
		
		Example
		
		`{ "user": { "mappings": [ { "sourcePath": "$.sAMAccountName[0]", "targetPath": "$.userName", "functions": [ { "function": "replaceAllString", "regex": "14\\d{1}", "replacement": 123 } ] } ... }`

Function	Parameters	Description

resolveEntityIds	**entityType**
	Required: No
	Possible values:
	• `user`
	• `group`
	Default value: `user`
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	• true
	• false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
	• **sourceSystemName**
	Required: No
	Type: String
	• **targetSystemName**
	Required: No
	Type: String
		This function resolves the value of a source system attribute to an existing back-end key in the target system. It should be used only in the transformation mappings of the target system.
		
		For example, it can resolve the value of a source system member attribute to the ID of an existing SCIM resource that represents this member in a SCIM target system.
		
		Example
		`{`
		`"sourcePath": "$.member",`
		`"preserveArrayWithSingleElement": true,`
		`"optional": true,`
		`"targetPath": "$.members[?(@.value)]",`
		`"functions": [`
		`{`
		`"entityType": "group",`
		`"function": "resolveEntityIds"`
		`}`
		`]`
		`...`
		`}`
		
		The `sourceSystemName` and the `targetSystemName` parameters are typically used within this function to resolve group members.
		
		In the following example, we assume that you have created two

Function	Parameters	Description
		source systems reading entities from one and the same user store (for example, Microsoft Entra ID). You provision users from *SourceSystem1* and groups from *SourceSystem2* to one and the same target system. By specifying the `sourceSystemName` as *SourceSystem1*, the user IDs of the group members in *SourceSystem2* will be resolved against the user IDs in the *SourceSystem1*. **Code Syntax** `"functions": [ { "sourceSystemName": "SourceSystem1", "function": "resolveEntityIds" } ] ...`

Function	Parameters	Description

splitStringToArray	• **separator**
	Required: Yes
	Type: String
		This function splits the initial String using the defined separator and produces as a result an array of Strings or objects.
		
		In the following examples, function *splitStringToArray* is applied to the `sourcePath`, read from the source system. Based on the `targetPath` defined, the initial String will be split into array of Strings or objects.
		
		Example 1
		
		In the first example, the initial String we use is:
		
		`"countries": "Serbia,Germany,Spain"`
		
		The transformation using the function is the following:
		
		`{ "sourcePath": "$.countries", "targetPath": "$.countryList", "functions": [ { "function": "splitStringToArray", "separator": "," } ] }`
		
		After the function execution, the output results to the following array of Strings:
		
		`"countryList": [ "Serbia", "Germany",`

Function	Parameters	Description

```
        "Spain"
    ],
```

Example 2

In the second example, we use the same initial String:

```
"countries":
"Serbia,Germany,Spai
n"
```

In this case, the `targetPath` in the transformation using the function supports multiple values:

```
 {

"sourcePath":
"$.countries",

"targetPath":
"$.countryList[?
(@.name)]",

"functions": [

{

    "function":
"splitStringToArray"
,

    "separator":
","


}
                ]
            },
```

After the function execution, the output results to the following array of objects:

```
"countryList": [
        {
            "name":
"Serbia"
        },
        {
            "name":
"Germany"
        },
        {
```

Function	Parameters	Description
		<pre>"name": "Spain" }], </pre>

substring	• **beginIndex**
	Required: Yes
	Type: Integer
	• **endIndex**
	Required: No
	Type: Integer
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function returns a string if `endIndex` is not provided. It begins at the specified `beginIndex` and extends either to the character at index `endIndex - 1` or to the end of this string.
		
		Example
		<pre>{ "user": { "mappings": [{ "sourcePath": "$.sAMAccountName[0]", "targetPath": "$.userName", "functions": [{ "function": "substring", "beginIndex": 3, "endIndex": 5 }] } ... }</pre>

Function	Parameters	Description

toString	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function accepts a Byte array or a string representation. The string is assumed in UTF-8 character encoding. Function result: **String**.
		
		To see a relevant example, go to function encode and decode.

toUpperCaseString	• **locale**
	Required: No
	Possible values: Abbreviation of locale, such as `de_DE`, `ja_JP`, `ru_RU` and so on.
	Default value: `en_EN`
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function converts all the characters in the given string to upper case, using the provided locale, or if nothing defined – English.
		
		Example
		`{ "user": { "mappings": [ { "sourcePath": "$.sAMAccountName[0]", "targetPath": "$.userName", "functions": [ { "function": "toUpperCaseString", "locale": "en_EN" } ] } ... }`

Function	Parameters	Description

toLowerCaseString	• **locale**
	Required: No
	Possible values: Abbreviation of locale, such as `de_DE`, `ja_JP`, `ru_RU` and so on.
	Default value: `en_EN`
	• **condition**
	Required: No
	Type: String
	• **applyOnElements**
	Required: No
	Type: Boolean
	Possible values:
	  • true
	  • false
	• **applyOnAttribute**
	Required: No
	Type: String
	• **assignToAttribute**
	Required: No
	Type: String
		This function converts all the characters in the given string to lower case, using the provided locale, or if nothing defined – English.
		
		Example
		`{ "user": { "mappings": [ { "sourcePath": "$.sAMAccountName[0]", "targetPath": "$.userName", "functions": [ { "function": "toLowerCaseString", "locale": "en_EN" } ] } ... }`

Parent topic: Transformations [page 388]

Related Information

Transformation Types [page 390]
Transformation Expressions [page 391]
Transformation Variables [page 490]
Transformation Conditions [page 496]
Transformation Examples [page 501]
Transformation Editors [page 505]

1.3.6.4 Transformation Variables

System variables specify particular attributes of the read and written entities. The variables are used for temporary saving of data in the transformation mappings of the provisioning systems. They do not have representation in the output JSON file. You can define a variable in the context of one provisioning system and specific entity. The value of the defined variable cannot be used in another provisioning system or entity.

The table below includes information about variables that have specific behavior:

Transformation Variables

Variable	Definition & Example	System Types
assignGroup	An optional variable, which enables you to assign groups to a user in the target system. It can be used for standard provisioning with or without configured bulk operation on the target, and real-time provisioning. To enable the group assignment, you need to modify the target system transformation by adding a mapping under the user resource containing the following configuration parts: • **Condition** - Defines for which users the condition will apply. For example, all users with emails. • **Constant** - Holds the IDs of the groups in the target system. Currently, you can only identify groups by ID. • **targetVariable** - Specifies whether you want to assign users to groups or unassign users from groups. Example `{ "user":{ "mappings":[ <Existing transformation> { "condition":"($.emails EMPTY false)", "constant":[ { "id":"00f8ab94-a732-48fa-9169-e51f87b8dcd5" }, { "id":"01231139-4711-4a28-8f9d-6745843ef716" } ], "targetVariable":"assignGroup" } ] }, "group":{ "mappings":[` This mapping will result in assigning all users with emails to two groups in a target system. For more information, see Enabling Group Assignment [page 2445].	Target systems

Variable	Definition & Example	System Types
entityIdSourceSystem	Mandatory for every read transformation (in source and proxy systems). It specifies which attribute of a read entity to be considered as a unique ID in the source system. Use this mapping with caution, as it can overwrite the value of the ID returned by the source system. It is not recommended to change the attribute specified in the source path of the mapping. Such change may cause incorrect deletion of entities from the target system. In case you should change the mapping for the variable, you must reset the source system. For more information, see Reset Identity Provisioning System [page 2485]. **Example** `{ "targetVariable": "entityIdSourceSystem", "sourcePath": "$.name" }`	• Source systems • Proxy systems
entityIdTargetSystem	Mandatory for every write transformation (in target and proxy systems). It specifies which attribute of a written entity to be considered as a unique ID in the target system. This variable is defined by the target system according to the system response during entity creation, or is read from the Identity Provisioning database during entity modification or deletion. It is not recommended to change the attribute specified in the target path of the mapping. Such change may cause incorrect deletion of entities from the target system. In case you should change the mapping for the variable, you must reset the target system. For more information, see Reset Identity Provisioning System [page 2485]. **Example** `{ "sourceVariable": "entityIdTargetSystem", "targetPath": "$.id" }`	• Target systems • Proxy systems

Variable	Definition & Example	System Types
entityBaseLocation	Mandatory only for read transformations in proxy systems. It contains the proxy application URL featuring the entity type endpoint: **https://ipsproxy<proxy_provider_account>-<consumer_account>.<neo_landscape>:443/ipsproxy/api/v1/scim/<system_ID>/Users** Example `{ "sourceVariable": "entityBaseLocation", "targetVariable": "entityLocationSourceSystem", "targetPath": "$.meta.location", "functions": [ { "type": "concatString", "suffix": "${entityIdSourceSystem}" } ] }`	Proxy systems
entityLocationSource-System	Mandatory only for read transformations in proxy systems. It contains the proxy application URL featuring the SCIM 2.0 resource endpoint for an entity: **https://ipsproxy<proxy_provider_account>-<consumer_account>.<neo_landscape>:443/ipsproxy/api/v1/scim/<system_ID>/Users/<user_ID>** Example `"sourceVariable": "entityBaseLocation", "targetVariable": "entityLocationSourceSystem", "targetPath": "$.meta.location", "functions": [ { "type": "concatString", "suffix": "${entityIdSourceSystem}" } ] }`	Proxy systems

Variable	Definition & Example	System Types
currentDate	An optional variable, which contains the current date, in format: *yyyy-MM-dd HH:mm:ss.SSS* You can configure this variable by using property `ips.date.variable.format`, according to the Java Class DateTimeFormatter . **Example** `{ "targetPath": "$.PersonalDetails.ValidityPeriod.StartDate", "sourceVariable": "currentDate", "functions": [ { "type": "manipulateDate", "targetDateFormat": "yyyy-MM-dd" } ] }`	• Source systems • Target systems • Proxy systems

Variable	Definition & Example	System Types
unassignGroup	An optional variable, which enables you to remove the group assignments of a deleted user from the source system. In that case, you need to modify the target system transformation by adding a mapping under the user resource containing the following configuration parts:. • **Constant** - Holds the IDs of the groups in the target system which will be updated. Currently, you can only identify groups by ID. • **targetVariable** - Use **`"unassignGroup"`** to specify that you want to unassign users from the defined groups. • **scope** - Use **`"deleteEntity"`** to define that the deleted user from the source system should be persisted in the target system. **Example** `{ "constant": [ { "id": "groupID" } ], "targetVariable": "unassignGroup", "scope": "deleteEntity" }` This mapping will result in updating the assignments to the defined group in the target system. When using the group unassignment for users that are deleted from the source system, these are two possible scenarios based on the system where the user was created and the value of the property `ips.delete.existedbefore.entities`. For more information, see Enabling Group Assignment [page 2445].	Target systems

Parent topic: Transformations [page 388]

Related Information

Transformation Types [page 390]
Transformation Expressions [page 391]
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1.3.6.5 Transformation Conditions

A condition defines a JSON filter expression that can be applied to entity types (such as users, groups, and roles) or within attribute mappings. It can be combined with strings, constants, variables, and functions, or used within functions.

You can add conditions in both source and target system transformations. When applied in the source transformation, entities that do not meet the condition are skipped after being read. When applied in the target transformation, entities that do not meet the condition are skipped before being written.

You can define conditions using the JSONPath specification in the Identity Provisioning editors: JSON editor and Graphical editor.

Operators and Checks	Description
==	Left-hand value is equal to right-hand value. For example: `"condition": "$.emails[?(@.primary == true)].value == []",` This condition checks if the user has no primary email defined. It evaluates to `true` if the `emails` array contains no entry where `primary` is set to `true`. This condition is used in the default write transformations of SAP Commerce Cloud, where if the user has no primary email, the first email in the list is used instead. The following explains each part of the filter expression: • `$` - Indicates the starting point for the JSONPath expressions. • `emails` - Accesses the emails array within the user entity. • `[?()]` - Starts a filter expression to evaluate the array. • `@` - Refers to the current element within the array. • `@.primary == true` - Filters the array to retrieve only the elements marked as the primary email. • `.value` - After applying the filter condition, this extracts the actual email address. • `== []` - Checks if the value on the left-hand side is equal to an empty array `[]`. → Tip Use this pattern `[?(@. ...)]` for filtering elements within an array based on defined conditions.

Operators and Checks	Description
`!=` or `!==`	Left-hand value is not equal to right-hand value. For example: `"condition": "$.name.familyName != 'Smith'",` This condition checks if the user's `familyName` is not equal to `Smith`. Only users whose `familyName` is different from `Smith` fulfill this condition.
`=~`	Left-hand value matches the regular expression on the right-hand side. For example: `"condition": "$.emails[0].value =~ /.*@example.com/",` This condition checks if the user's first email in the `emails` array ends with `@example.com`. Only users whose first email (`emails[0].value`) contains this domain fulfill this condition. • `/` - Marks the beginning and the end of a regex. • `.*` - Matches any sequence of characters, including an empty sequence.
`&&`	AND: Returns `true` if both expressions are `true`. For example: `"condition": "($.user.validityPeriod.startDate <= '${currentDate}') && ($.user.validityPeriod.endDate > '${currentDate}')",` This condition checks whether the user is within their validity period. Only users whose validity period includes the current date will meet this condition, indicating that they are considered active. This condition is used in the default read transformations of SAP S/4HANA Cloud Public Edition source system to process only active users.

Operators and Checks	Description
`||`	OR: Returns `true` if at least one expression is `true`. For example: `"condition": "((('%c4c.group.prefix%' === 'null') || ($.groups[?(@.display =~ /%c4c.group.prefix%.*/)] empty false)) && $.userName empty false && ($.userType === 'Employee' || $.userType === 'employee'))",` Simplified pattern: `(<group prefix check> || <group presence check>) && <username presence check> && (<user type 'Employee' check> || <user type 'employee' check>)` This condition checks if the group prefix is null, or if there is at least one group in the user's groups array whose display name starts with the specified prefix. Additionally, it ensures that the `userName` is not empty and that the `userType` is either `"Employee"` or `"employee"`, allowing case insensitivity in the first letter only. This condition is used in the default write transformations of SAP Sales Cloud and SAP Service Cloud connector version 4 target system.
`empty true`	Left-hand value is considered empty (null, empty string, empty array). For example: `"condition": "($.emails EMPTY true) \|\| isValidEmail($.emails[0].value)",` This condition checks if an email is either missing (that is, the `emails` array is empty) or the first email in the array is valid. This condition is used in the default write transformations of SAP Advanced Workflow target system to ensure that an email is either provided and valid or missing.

Operators and Checks	Description
`empty false`	Left-hand value is not considered empty. For example: `"condition": "($.emails EMPTY false) && ($.userName EMPTY false) && isValidEmail($.emails[0].value)",` This condition checks if all parts are `true`. It expects at least one email to be present, the `userName` field to be filled, and the first email address in the `emails` array to be valid. This condition is used for the user entity in the default write transformations of Identity Authentication version 2 and Local Identity Directory target systems.

The following examples show how you can use conditions for filtering users based on group assignments.

Check if Users are Assigned to a Group with Specific Prefix

Example

Code Syntax

```
{
   "user":{
      "condition":"$.groups[?(@.display =~ /ABC.*/)] empty false",
```

This condition evaluates to `true` if the user is a member of at least one group whose display name begins with ABC.

Check if Users are Not Assigned to a Particular Group

Example

Code Syntax

```
{
   "user":{
      "condition":"$.groups EMPTY true || $.groups[?(@.display === 'XYZ' )]
empty true",
```

This condition evaluates to `true` if the user has no group memberships at all, or if they are not a member of any group named XYZ. You can use it to exclude (skip) users who are specifically assigned to the XYZ group.

Check if Users are Assigned to One or Two Particular Groups

Example

Code Syntax

```
{
   "user":{
      "condition":"($.groups EMPTY false) && ( ($.groups[?(@.display
===ABC' )] != []) || ($.groups[?(@.display ===XYZ')] != []))",
```

This condition evaluates to `true` if the user has at least one group assigned to them, and if they are a member of either the ABC group or the XYZ group or both.

Check for Particular Custom Attribute of Users

Example

Code Syntax

```
"sourcePath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']
['attributes'][?(@.name=='customAttribute1')]['value']",
```

While not a condition itself, this source path expression can be used to retrieve the value of a given custom attribute (`customAttribute1`) from the user's data. It works by filtering the `attributes` array to find the attribute with the name `customAttribute1`, and then returns the value of that attribute.

Assign User to a Group Based on UserName

Example

Code Syntax

```
{
    "condition": "$.userName =~ /.*explorer.*/",
    "constant": "Explorers",
    "targetPath": "$.groups[0].value"
},
{
    "condition": "$.userName =~ /.*scifi.*/",
    "constant": "Scientists",
    "targetPath": "$.groups[1].value"
}
```

This configuration assigns users to groups based on their `userName`: those with `explorer` in their `userName` are assigned to the `Explorers` group, and those with `scifi` are assigned to the `Scientists` group.

> **Note**
>
> This mapping works for systems that support group assignment via the user entity, for example, Identity Authentication SCIM API version 1. Make sure that the `group ID` in the target system matches the value defined in the constant field (in this case, `Explorers` and `Scientists`).

Parent topic: Transformations [page 388]

Related Information

Transformation Types [page 390]
Transformation Expressions [page 391]
Transformation Functions [page 428]
Transformation Variables [page 490]
Transformation Examples [page 501]
Transformation Editors [page 505]

1.3.6.6 Transformation Examples

The following examples explain how transformations work.

Basic Transformation

The basic transformation copies all attributes from the input JSON messages to the output JSON ones.

> **Code Syntax**

```
{
    "sourcePath": "$",
    "targetPath": "$"
},
```

Source, Intermediate, and Target Data

In this example, the source system JSON data contains the **sn[0]** attribute. The read transformation converts this attribute to **name.familyName** in the intermediate JSON data. Then, the write transformation reads the **name.familyName** attribute and maps it to **name.familyName** in the target system.

Source entity data (from Microsoft Active Directory)

```
{
    "sAMAccountName": ["jsmith"],
    "mail": ["john.smith@company.com"],
    "givenName": ["John"],
    "sn": ["Smith"],
    "memberOf": ["group1"],
    "memberOf_2": ["group21", "group22"],
    "memberOf_3": ["group31", "group32", "group33"]
}
```

Read transformation (intermediate JSON Data)

```
{
    "user": {
        "mappings": [
...
            {
                "sourcePath": "$.%ldap.attribute.user.id%[0]",
                "targetVariable": "entityIdSourceSystem",
                "correlationAttribute": true
            },
            {
                "condition": "('%ldap.attribute.user.id%' ==
'%ldap.attribute.dn%')",
                "constant": "",
                "targetVariable": "nestedPathRegex",
                "functions": [
                    {
                        "function": "concatString",
                        "prefix": "(?i)cn=.*?,(.*?)",
                        "suffix": ",%ldap.user.path%"
                    }
                ]
            },
            {
                "condition": "('%ldap.attribute.user.id%' ==
'%ldap.attribute.dn%')",
                "sourcePath": "$.%ldap.attribute.user.id%[0]",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:ad:2.0:User']['nestedPath']",
                "functions": [
                    {
                        "function": "getMatchedRegexGroup",
                        "regex": "${nestedPathRegex}",
                        "groupIndex": 1
                    }
                ],
                "defaultValue": ""
            },
            {
                "sourcePath": "$.sAMAccountName[0]",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.displayName[0]",
                "optional": true,
                "targetPath": "$.displayName"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.mail[0]",
```

```
                "optional": true,
                "targetPath": "$.emails[0].value",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.givenName[0]",
                "optional": true,
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.sn[0]",
                "optional": true,
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.memberOf",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.groups[?(@.value)]"
            },
...
```

Write transformation (intermediate JSON Data)

```
{
    "user": {
        "condition": "($.emails.length() > 0) && ($.name.familyName EMPTY
false)",
        "mappings": [
            {
                "sourcePath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.corporateGroups"
            },
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
                "targetPath": "$.schemas[0]"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
                "targetPath": "$.schemas[1]"
            },
            {
                "sourcePath": "$.userName",
                "optional": true,
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$.emails[*].value",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails[?(@.value)]"
            },
            {
                "sourcePath": "$.userType",
                "optional": true,
                "targetPath": "$.userType"
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.name.givenName"
```

```
            },
            {
                "sourcePath": "$.name.middleName",
                "optional": true,
                "targetPath": "$.name.middleName"
            },
            {
                "sourcePath": "$.name.familyName",
                "optional": true,
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.displayName",
                "optional": true,
                "targetPath": "$.displayName"
            },
...
```

Target entity data (result in Identity Authentication)

```
{
    "schemas": [
        "urn:ietf:params:scim:schemas:core:2.0:User",
        "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User"
    ],
    "id": "P000100",
    "userName": "jsmith",
    "name": {
        "familyName": "Smith",
        "givenName": "John"
    },
    "emails": [
        {
            "value": "john.smith@company.com",
            "primary": "true",
            "type": "work"
        }
    ],
    "groups": [
        {
            "value": "group1"
        }
    ],
    "groups_2": [
        {
            "value": "group21"
        },
        {
            "value": "group22"
        }
    ],
    "groups_3": [
    {
            "value": "group31"
        },
        {
            "value": "group32"
        },
        {
            "value": "group33"
        }
    ]
}
```

For more information about condition examples, see Transformation Conditions.

Parent topic: Transformations [page 388]

Related Information

Transformation Types [page 390]
Transformation Expressions [page 391]
Transformation Functions [page 428]
Transformation Variables [page 490]
Transformation Conditions [page 496]
Transformation Editors [page 505]
Manage Transformations [page 2439]

1.3.6.7 Transformation Editors

Identity Provisioning provides graphical and JSON text editor for managing provisioning system transformations.

Graphical Editor

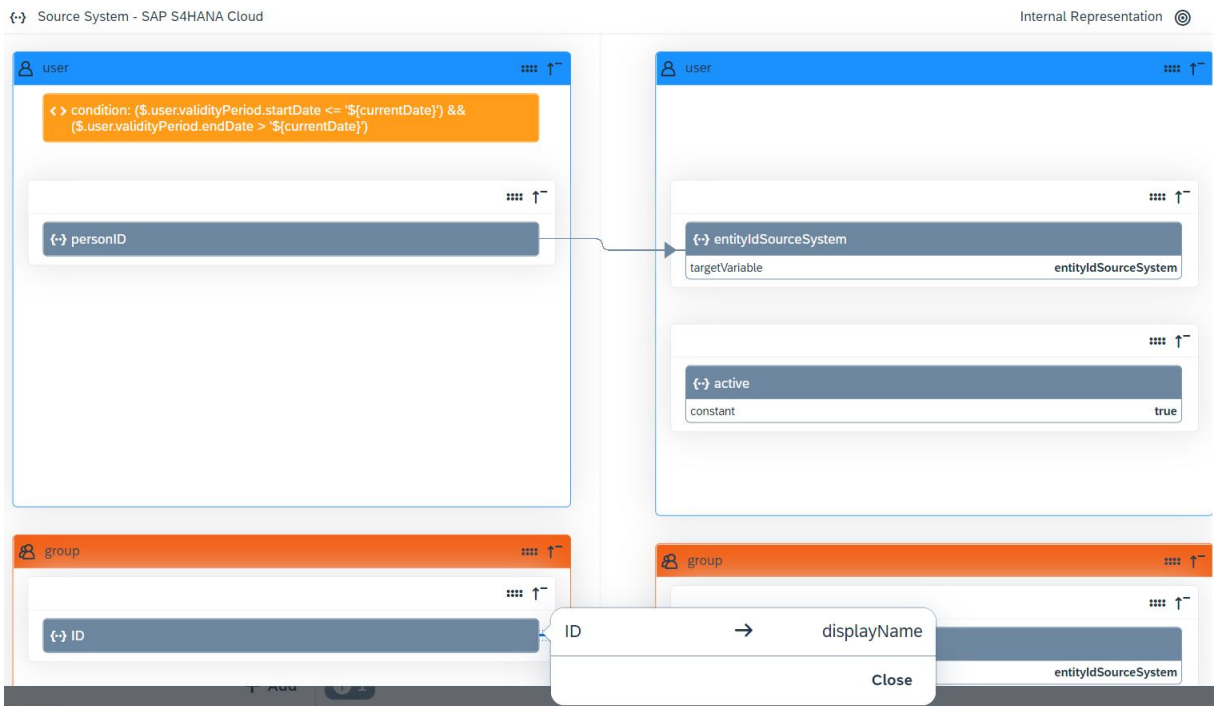
This editor allows you to graphically model the entities (users, groups, roles) and their content (attribute mappings and transformation expressions, like conditions, functions, constants, and others). It offers the following advantages:

- Visualization
  Entities are organized in colored boxes. Every entity has a dedicated color. Users are always blue, groups are orange and roles are green. Source and target path attributes are displayed horizontally (side-by-side). They are connected with an arrow which indicates the direction of reading and writing the data. Color coding makes it easy to identify specific configurations. Ignored entities are greyed out, skipped operations and conditions are displayed in orange.
- Simplification
  Typing code and following JSON syntax rules are no longer needed, except in cases when conditions are defined. Expressions and functions along with their possible values are prefilled and available for selection in dropdown lists. The editor takes care of adding the dot notation, that is, separating child elements with a period (.), enclosing string values within double quotes, prepending the dollar sign ($) to the JSONPath expression.
- Validation
  The JSON validator and the preview ensure that the input is in the expected format.

> ⓘ Note
>
> The graphical editor is available only for Identity Provisioning tenants running on SAP Cloud Identity infrastructure. It is the default editor.

> **Note**
>
> By default, the transformation opens in the graphical editor. However, the availability of the graphical editor depends on the total number of mappings across all entities. The JSON editor is always available.
>
> - 500 mappings or fewer: Both the graphical and JSON editors are available, and you can switch between them.
> - More than 500 mappings: The transformation can be viewed and edited only in the JSON editor. Switching to the graphical editor is not supported.

The following interactive screenshot shows the main elements of the graphical editor. Hover over the image and click highlighted areas for more information.



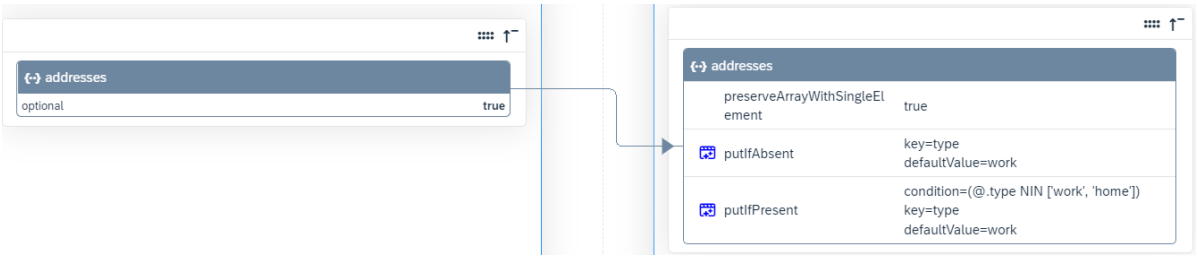
- #unique_111/unique_111_Connect_42_subsection-im0 [page 507]
- #unique_111/unique_111_Connect_42_subsection-im8 [page 507]
- #unique_111/unique_111_Connect_42_subsection-im1 [page 507]
- #unique_111/unique_111_Connect_42_subsection-im9 [page 507]
- #unique_111/unique_111_Connect_42_subsection-im2 [page 507]
- #unique_111/unique_111_Connect_42_subsection-im4 [page 507]
- #unique_111/unique_111_Connect_42_subsection-im3 [page 507]
- #unique_111/unique_111_Connect_42_subsection-im10 [page 507]
- #unique_111/unique_111_Connect_42_subsection-im11 [page 507]
- #unique_111/unique_111_Connect_42_subsection-im5 [page 507]
- #unique_111/unique_111_Connect_42_subsection-im12 [page 507]
- #unique_111/unique_111_Connect_42_subsection-im13 [page 507]

- #unique_111/unique_111_Connect_42_subsection-im6 [page 507]
- #unique_111/unique_111_Connect_42_subsection-im7 [page 507]

Hover over each element for a description. Click the element for more information.

The following example illustrates how the address attribute mapping from the Identity Authentication write transformation is displayed in the graphical editor. Compare the same attribute with its representation in the JSON editor below.

Attribute Mapping in Graphical Editor



The source path attribute **addresses** in the internal representation (on the left-hand side) and the target path attribute **addresses** in the target system (on the right-hand side) are displayed horizontally. The arrow shows the direction of reading and writing the data. The transformation expressions and functions are listed under the respective `sourcePath` and `targetPath` attribute they are applicable for.

JSON Editor

This editor allows you to work with entities and their transformation mappings in text mode. To add or modify transformations, you need to type. You can also perform operations like select, cut, copy, paste and others. The editor provides line numbering, syntax coloring and syntax validation.

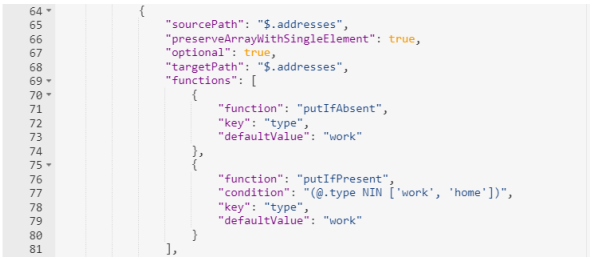
The following example illustrates how the address attribute mapping from the Identity Authentication write transformation is displayed in the JSON editor:

Attribute Mapping in JSON Editor

The source path attribute **addresses** in the internal representation and the target path attribute **addresses** in the target system are displayed vertically - one below the other.

The transformation expressions and functions are listed in individual lines below the respective `sourcePath` and `targetPath` attribute they are applicable for.

Syntax coloring and syntax validation are enabled by default.



```
64      {
65          "sourcePath": "$.addresses",
66          "preserveArrayWithSingleElement": true,
67          "optional": true,
68          "targetPath": "$.addresses",
69          "functions": [
70              {
71                  "function": "putIfAbsent",
72                  "key": "type",
73                  "defaultValue": "work"
74              },
75              {
76                  "function": "putIfPresent",
77                  "condition": "(@.type NIN ['work', 'home'])",
78                  "key": "type",
79                  "defaultValue": "work"
80              }
81          ],
```

Although the graphical and the JSON editors differ in the user experience they provide, other tasks, such as: export and reset transformations remain the same.

Parent topic: Transformations [page 388]

Related Information

Transformation Types [page 390]
Transformation Expressions [page 391]

Transformation Functions [page 428]
Transformation Variables [page 490]
Transformation Conditions [page 496]
Transformation Examples [page 501]
Manage Transformations [page 2439]
Working with Graphical Editor [page 2440]

1.3.7 Users

Users in SAP Cloud Identity Services fall into three categories: administrators, end users, and technical users.

Administrators

SAP Cloud Identity Services distinguish between two types of administrators: **user administrator** and **system administrator**.

- User administrators are real persons. They manage tenant configurations, applications, corporate identity providers, identity provisioning, users, and groups. The initial administrator of SAP Cloud Identity Services tenants is created as a user administrator.
- System administrators are technical users. They are used for establishing system-to-system communication in provisioning scenarios, accessing real-time provisioning and proxy system APIs, as well as accessing tenant API for running Identity Provisioning jobs. System administrators can also act as user administrators because they have the same authorizations.

End Users

End users use business applications that consume Identity Authentication as identity provider. The service authenticates the users with credentials maintained in Identity Authentication or delegates the authentication request to a 3rd-party identity provider. End users are initially created, imported, or provisioned to the user store of SAP Cloud Identity Services.

There are six user types of end users:

User Type	Description
Customer	External users that are customers for the company. They may need access to some applications.
Employee	Internal company users, mainly users with long-term contracts.

User Type	Description
Partner	External users from a partner company that will need some kind of access.
Public	External users that are not controlled by the company. For example, self-registered.
External	External users that can be a temporary hired employees that need access to some applications, or external users that need access to learning materials and training managed by the company.
Onboardee	An employee user that needs a different authentication, because, for example the user may not exist in the corporate identity provider.
Alumni	External users that were once employed by the company but now they are no more part of it. They may need some kind of access.

Technical Users

Technical users are non-human identities that enable automated system-to-system communication and integration. Unlike administrators and end users, technical users authenticate using credentials such as secrets or certificates rather than passwords. Technical users are managed separately in the administration console via the *Technical Users* tile that appears under *Users & Authorizations*. The following prerequisites must be met, so that an administrator or technical user can manage the technical users:

- You have enabled the authorizations based on policies option in the admin console for SAP Cloud Identity Services. See Configure Authorizations Based on Policies [page 2311].
- The administrator or end user is assigned with the respective authorization policy, depending on the tasks they need to perform. For more information, see Configure Technical User Authorizations [page 2318].

> → Remember
>
> The minimal authorization policy required for the creation of a technical user is `CREATE_TECHNICAL_USERS`. The `MANAGE_TECHNICAL_USERS` allows all CRUD operations with a technical user.

For more information about how to create and manage the technical users, see Managing Technical Users [page 2329].

Parent topic: Concepts [page 26]

Related Information

Tenants [page 27]
Bundles [page 35]
Applications [page 61]
Provisioning Systems [page 69]
Properties [page 73]
Transformations [page 388]
Agents [page 511]
Groups [page 514]
Authorization Policies [page 518]
SAP-Managed SAP Cloud Identity [page 520]
List and Edit User Details [page 2361]
Add Administrators [page 2303]

1.3.8 Agents

Agent identity represents AI and automation agents in SAP Cloud Identity Services. Agents can authenticate, receive authorization, and be managed like human users while retaining agent-specific capabilities and security controls.

> ⚠ Caution
>
> This page is still work in progress. This page contains information about product features that are part of an SAP Early Adopter Care (EAC) program. Products or features that are part of an EAC program are not available to users outside of this program.

Agent identity combines multiple identity and access management (IAM) artifacts to provide comprehensive identity management for agents within SAP-centric environments.

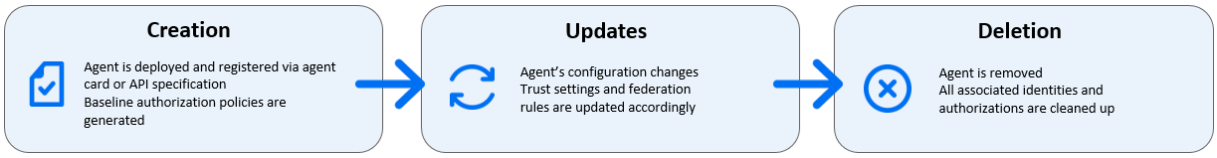
When an agent is built in Joule Studio, the agent automatically appears as an agent identity in the SAP Cloud Identity Services administration console. This allows user administrators to review the agent's authorizations and define who can access the agent.

In development landscapes, Joule Studio includes a feature that automatically assign access, reducing the need to contact user administrators for routine development tasks.

From an identity and access management perspective, an agent is an application registered in SAP Cloud Identity Services. Each deployed agent appears in the *Agents* tile of the SAP Cloud Identity Services administration console, where you can review its authorization policies, inspect which APIs it can call, and control which users or groups can invoke it.

You can't create or delete agents through the administration console. The SAP platform automatically manages their lifecycle in response to deployment events. When an agent is deployed, SAP Cloud Identity Services registers it using the agent's own metadata (its agent card or API specification) and generates baseline authorization policies at that point. When the agent's configuration changes, the system updates trust settings and federation rules in a controlled, traceable manner. When an agent is removed, the system fully cleans up

all associated identities and authorizations. To control which users or user groups can invoke an agent, use the *Agents* tile in the administration console.



Lifecycle of an Agent

SAP supports three categories of agents, each with a different level of administrator control:

- LOB agents
  The system provisions LOB agents as part of SAP line-of-business solutions such as SAP S/4HANA or SAP SuccessFactors. You can adjust authorization rules for these agents but cannot modify the agent registration itself.
- Custom agents
  Customers or partners build agents, for example using Joule Studio. The full range of configuration options is available, including trust settings and federation rules.
- Third-party agents
  Third-party agents are external AI tools that connect to the SAP landscape through the agent gateway. They authenticate and authorize against SAP Cloud Identity Services in the same way as SAP-native agents.

The system manages authorizations in two phases for increased security:

- Central Authorization in SAP Cloud Identity Services: Defines who can access an agent. This allows you to restrict agent access and provides visibility into the authorizations the agent has.
- Principal Propagation: The agent works on behalf of the current user. Each application and back end checks the user's authorizations, ensuring that the agent can only access what the user has permission to access.

This two-phase approach ensures security even when workflows span multiple runtime types.

Agents and Assistants

In Joule, an assistant is a higher-level AI entity that translates user intent into actions by coordinating agents. An assistant and each agent it invokes are separate registered identities in SAP Cloud Identity Services. Each requires its own user access assignment.

Access assignments are not inherited. If an assistant invokes one or more agents, assigning a user to the assistant does not automatically grant them access to those agents. You must assign the user to the assistant and to each agent it invokes separately. If any assignment is missing, the user receives an authorization error when the assistant attempts to invoke the unassigned agent, even if the user has access to all the other entities.

As an IAM administrator, plan your access assignments across the full chain of entities a user needs to reach, not just the entry point. For more information about how to perform these assignments, see Enable Users to Access Agents [page 2338].

What is an Agent Identity?

An agent identity in SAP Cloud Identity Services consists of the following components:

Application Artifact

- Represents a trusted client or system, such as S/4HANA or a dedicated agent application.
- Includes metadata such as redirect URIs, keys, and trust configuration.
- Establishes the technical connection for the SAP landscape.

Agent User (Technical User)

- Every agent has an associated technical user in the directory. This user is not a human user and can't authenticate through a browser (front channel). It authenticates only through back-channel methods such as client credentials, JWT bearer, and token exchange.
- Agents always act on behalf of a human user. The agent user is used to assign authorization policies and user groups to the agent. These assignments define what the agent can do.
- In a future release, agents will be able to act on their own behalf without a human user context.

Dual Authorization Model

- Agents act on behalf of a human user. For an action to succeed, both the agent and the user must have the necessary authorizations. The agent can access only the APIs defined by its authorization policies. The target application also applies the user's own authorization checks. An action succeeds only when both the agent's scope and the user's permissions allow it.

The system creates or deletes agents automatically when you subscribe or unsubscribe to an application or use an AI tool to create an agent.

Core Use Cases

Agent Acting for Present Human Users

When a human user is present and delegates permissions:

- The human-in-the-loop flow requests user approval.
- The agent requests a user token through the token exchange flow.
- The token includes user context for authorized actions.

> ⓘ Planned
>
> Additional use cases are planned for future releases. These use cases include agents acting on your behalf when you're offline (using stored delegation relationships) and unattended agents acting on their own behalf without you in the loop. These use cases aren't currently supported.

Parent topic: Concepts [page 26]

Related Information

Tenants [page 27]
Bundles [page 35]
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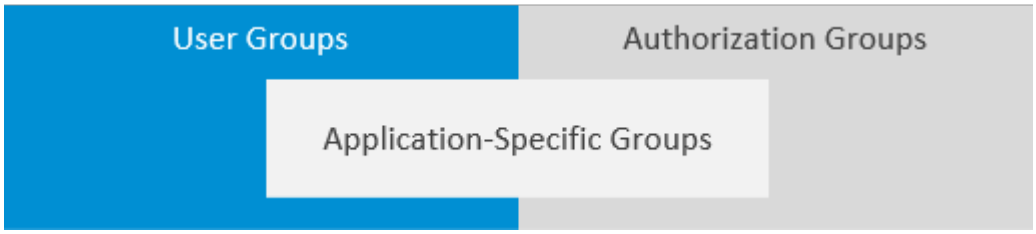
Groups [page 514]
Authorization Policies [page 518]
SAP-Managed SAP Cloud Identity [page 520]
Managing Agents [page 2333]
Understanding AI Agents Across SAP

1.3.9 Groups

SAP Cloud Identity Services offers groups to organize users based on common characteristics, authorization, or application. Use them to efficiently manage user access and permissions within your organization's SAP Cloud Identity Services environment.

There are two kinds of groups in SAP Cloud Identity Services - *user groups* and *authorization groups*, and one special group - *application-specific groups*.



The following sections explain the different groups.

User Groups

The user groups serve to create sets of users who have something in common. An example for a user group is a group of users who work in the same department, or users who have similar tasks in a company.

The type of these groups in the administration console for SAP Cloud Identity Services is *User Group*.

For more information, see Managing Groups [page 2393].

Authorization Groups

The authorization groups are related to authorization of users, groups or applications.

The groups that result from the creation of authorization represent the authorization policies that are available in the administration console for SAP Cloud Identity Services. When you create a new authorization, a group with the same name is also created and appears in the list of the groups. Initially, only the authorization groups related to the base authorization policies are visible.

> ⚠ Restriction
>
> You can't modify or delete an authorization group that is related to an authorization policy via the administration console.

The type of these groups in the administration console is *Authorization*

For more information, see Managing Groups [page 2393], Configure Authorizations Based on Policies [page 2311] and Configuring Authorization Policies [page 2244].

Application-Specific Groups

The application-specific groups are a special kind of groups which are bound to an application and can be created in the Identity Directory of SAP Cloud Identity Services or Identity Authentication (SCIM API version 2) target system by running provisioning jobs, or via the administration console. For more information, see Start and Stop Provisioning Jobs [page 2472] and Create a Group [page 2398].

The extension schema of the Identity Directory API *urn:ietf:params:scim:schemas:extension:sap:2.0:Group* defines the following attributes:

- `applicationId` - supported for the Local Identity Directory and Identity Authentication (SCIM API version 2) default write and proxy write transformations as mandatory attribute. The value of this attribute is provided from the source system by setting the property `ips.application.id`. For more information, see List of Properties [page 80] →`ips.application.id`.
- `type` - supported by the Local Identity Directory and Identity Authentication (SCIM API version 2) source, target, and proxy provisioning systems as optional attribute. If no value is specified in the write or proxy write default transformations, the default value `userGroup` is set.
- `supportedOperations` - supported by Local Identity Directory and Identity Authentication (SCIM API version 2) source, target, and proxy provisioning systems as optional attribute. If no value is specified in the write or proxy write default transformations, the default value `readWrite` is set. The attribute defines the supported update options for the application-specific group.
- `externalName` - unique within the application. If this attribute is present, it is sent with the other configured attributes in the OpenID Connect (OIDC) token or SAML 2.0 assertion. The value is immutable.

  > ⓘ Note
  >
  > Two groups with same `externalName` for two different applications can be created, but the same scenario for one application is not allowed.

- `children` - a list of child groups used to represent hierarchical group structures. By default, this attribute is not included in the response when retrieving group data. The `children` attribute is supported only for single group retrieval.

  > **→ Remember**
  >
  > To include the `children` attribute in the response for a single group, you must explicitly request it using the `attributes` query parameter. For example:
  >
  > ```
  > /scim/Groups/{id}?
  > attributes=urn:ietf:params:scim:schemas:extension:sap:2.0:group:children
  > ```

  > **⚠ Restriction**
  >
  > When retrieving multiple groups, the `children` attribute can't be requested. If you attempt to include this attribute in the `attributes` query parameter for multiple groups, the Identity Directory API will return a status code 400, along with a message indicating that requesting the `children` attribute is only supported for single group retrieval.

- `externalLocation` - Specifies the location of the source group. The value is immutable.

  > **ⓘ Note**
  >
  > In the Administration Console you can navigate from the group with `externalLocation` attribute to the same group in the application from which the group comes : *sign in to the administration console* *select the relevant group* *Go to source application*.

The application-specific groups appear with their name in the *Application Name* column in the administration console for SAP Cloud Identity Services under the *Groups* tile. The *Application Name* column for the groups that aren't application-specific is empty.

> **⚠ Restriction**
>
> You must be assigned to a policy that allows reading applications, or to the *Manage Applications* role to view *Application Name* column. For more information, see Configure User Authorizations [page 2312].

The application-specific groups can be of the following types:

Application-Specific Groups Types

Group Type Technical Name	Group Type in Administration Console	Description
`userGroup`	*User Group*	A group of users who have something in common.
`authorization`	*Authorization*	A group related to authorization of users, groups or applications.
`deepLinkActivationPermission`	*Deep Link Activation*	A group of applications that support deep linking.

The application-specific groups support the following operations:

Application-Specific Groups Supported Operations

Supported Operation Technical Name	Supported Operation in Administration Console	You can
`readOnly`	*Read*	• View the group details and group members. • Edit the *Display Name* and *Description* of the group. • Delete the group.
`readWrite`	*Read & Write*	• View the group details and group members. • Edit the *Display Name* and *Description* of the group. • Delete the group. • Add new entities - users, and groups, as group members. • Remove entities - users, and groups, from the group members.
`userOnlyMembership`	*User Membership Only*	• View the group details and group members. • Edit the *Display Name* and *Description* of the group. • Add new entities - users, as group members. • Remove the users from group members.
`membership`	*Membership Only*	• View the group details and group members. • Edit the *Display Name* and *Description* of the group. • Add new entities - users, and groups, as group members. • Remove entities - users, and groups, from the group members.

Parent topic: Concepts [page 26]

Related Information

Tenants [page 27]
Bundles [page 35]
Applications [page 61]
Provisioning Systems [page 69]
Properties [page 73]
Transformations [page 388]

Users [page 509]
Agents [page 511]
Authorization Policies [page 518]
SAP-Managed SAP Cloud Identity [page 520]
List and Edit Groups [page 2400]
List and Search Users in Groups [page 2405]
Add Users to a Group [page 2406]
Remove Users from a Group [page 2408]
Assign Groups to a User [page 2409]
Unassign Groups from a User [page 2410]
Blog Post: Taking Groups to the Next Level with Application-Specific Groups

1.3.10 Authorization Policies

Authorization Management lets administrators control who can access what in business applications.

Authorizations can be functional and instance-based. Administrators refine authorization policies that give access to resources in enabled business applications. They can restrict predefined and custom authorization policies based on user and business object attributes and assign them to users with the group management capabilities of the identity directory. See Configuring Authorization Policies [page 2244].

Prerequisites

In line with SAP's identity and access management (IAM) strategy, an identity directory with user profiles is a prerequisite for using Authorization Management. For more information, see CIO Guide: Identity Lifecycle in SAP Landscapes.

Overview

Authorizations are a critical component of any business domain. For cloud business applications to be enterprise-grade and ready, they must incorporate a robust and highly customizable authorization concept. This involves not only checking permissions for business actions but also ensuring these actions are performed on the correct business objects, known as resources.

Functional Authorizations

Functional authorizations are mandatory because they define what actions a user can perform in an application, such as reading data, creating records, or administrative tasks.

Following the principle of least privilege, each user receives only the permissions required for their work, with separate authorizations for personas such as administrators, support staff, auditors, key users, and specific roles like accountants or production planners. Authorization must be enforced consistently across all access paths, including user interfaces, APIs, batch processes, and search results.

Instance-Based Authorizations

Instance-based authorizations control who can perform specific actions on specific resources. They restrict access to specific data objects within functional authorizations. For example, in a financial application, a user might be authorized to make journal entries, but only within certain company codes. Conversely, the same user might be allowed to execute the closing process at the end of periods for different company codes. This nuanced approach ensures that permissions are tailored to specific business needs and contexts.

Using Authorization Policies in Custom Applications

For developers of SAP BTP based business applications, handling authorizations should be straightforward. The authorization framework enables developers to easily declare and enforce their business-specific authorization models.

There are libraries for developers of SAP BTP based business applications to declare and enforce functional and instance-based authorization. Authorizations are declared as code-based "authorization policies" of the business application and enforced with client libraries for Node.js, Java, and Go, and they provide value help for authorization administration. Authorization Management is fully integrated in the Cloud Application Programming model (Authorization and Access Control) and supports typical SAP BTP developments (Developing Authorizations [page 2876]), such a Cloud Foundry and Kyma.

Parent topic: Concepts [page 26]

Related Information

Tenants [page 27]
Bundles [page 35]
Applications [page 61]
Provisioning Systems [page 69]
Properties [page 73]
Transformations [page 388]
Users [page 509]
Agents [page 511]
Groups [page 514]
SAP-Managed SAP Cloud Identity [page 520]

1.3.11 SAP-Managed SAP Cloud Identity

SAP Cloud Identity Services support both SAP-managed and customer-managed integration models, depending on who is responsible for maintaining and operating the tenant, application and provisioning configurations.

Overview

SAP-managed refers to a delivery approach in which SAP is responsible for key technical and operational aspects of the solution. This approach reduces complexity for you and provides a consistent, best-practice experience. SAP manages the software lifecycle, including provisioning, technical integration, monitoring, automatic updates, backups, and offboarding. For more information, see SAP Note 3709792 and SAP's Responsibilities.

> **Note**
>
> To use the SAP-managed delivery approach, your tenant must run on the SAP Cloud Identity Services infrastructure. If your tenant is still running on the Neo environment, you must first migrate it to the SAP Cloud Identity Services infrastructure.
>
> For more information, see Migrate Identity Provisioning Bundle Tenant.

Customer-managed refers to a delivery approach in which you are responsible for configuring and maintaining your tenants according to your business requirements. This approach allows you to adapt features and functionality to your specific needs. For more information, see Your Responsibilities.

SAP-managed SAP Cloud Identity Services support the following delivery approaches: SAP can manage both the tenant and the integrated applications. You can also manage your own tenants, with SAP managing the integrated applications connected to those tenants.

What does SAP-managed mean for the individual services within SAP Cloud Identity Services?

Identity Authentication

- SAP manages the trust between SAP Cloud Identity Services and the connected SAP applications.
- In the SAP Cloud Identity Services administration console, the *Conditional Authentication* and *Authentication and Access* settings for these applications are read-only and can't be modified.
- The *Provisioning* tab, which is used to enable central-store based provisioning, is not available.
- The *Authorization Policies* tab remains available and can be configured as needed.
- Most settings under *Branding and Layout* can be modified, with the exception of *Token URL Separator* and login page behavior, which remain SAP-managed.

Identity Provisioning

- SAP-managed source and target systems are automatically preconfigured and maintained by SAP. As a result, these systems are not displayed in the administration console. You cannot manually schedule or run provisioning jobs for them. Also, you cannot access their job logs.
- Data replication is automated. Provisioning jobs run automatically every 30 minutes.
- You can still create and configure provisioning systems, as well as schedule and run provisioning jobs according to your business requirements.

> ⚠ Caution
>
> Do not create configurations that duplicate or overlap with SAP-managed provisioning flows. This includes configuring jobs and modifying transformations. Running both SAP-managed and customer-managed provisioning for the same back-end systems within the same SAP-managed flow, results in data inconsistencies.

Monitoring and Support

For SAP Cloud Identity Services, you can monitor and troubleshoot the parts you manage. For more information, see Monitoring and Troubleshooting [page 2964].

If you encounter issues, you can report an incident to SAP Support. For information, see Getting Support [page 3141].

SAP-Managed IAM for Joule

> ⚠ Caution
>
> This section is still work in progress. This page contains information about product features that are part of an SAP Early Adopter Care (EAC) program. Products or features that are part of an EAC program are not available outside this program.

SAP-managed IAM for Joule represents an initial step toward the same unified identity model used for the SAP Business Suite. In the Business Suite model, SAP Cloud Identity Services is the authoritative source for user and role data. Assignments are distributed outward to connected applications. The current Joule model adopts a pragmatic starting point: user records and group assignments originate in each line-of-business application. Identity provisioning jobs read this data from the line-of-business system into SAP Cloud Identity Services. The jobs then write the global user ID back to the line-of-business system to link the records. The line-of-business system remains the authoritative source for user and role data in the current release.

This approach enables Joule to reuse existing user and role data without requiring you to restructure your landscapes. Fully centralized assignment management in SAP Cloud Identity Services (the same harmonized model as the Business Suite) is planned for a future release.

For the EAC program, the following restrictions and requirements apply to identity provisioning for SAP-managed Joule, in addition to those described in the *Identity Provisioning* section above:

- You must manually import SAP-managed provisioning job templates from SAP Note 3784171. The system doesn't yet create jobs automatically when you activate Joule for a line-of-business application. (If you are part of the EAC program and need access to this SAP Note, open an incident under BC-IAM-IPS.)

  > ⚠ Caution
  >
  > Do not modify the imported SAP-managed provisioning job templates. If issues occur, do not create duplicate jobs or change the template configurations. Contact SAP Support for assistance.

- You must set up the SAP-managed provisioning jobs after the application representing the line-of-business system already exists in SAP Cloud Identity Services. This ensures that application-specific groups created by the provisioning job are associated with the correct application.
- Before you run the provisioning jobs, every user in scope must have logged in at least once to the line-of-business application using SAP Cloud Identity Services as the identity provider. The system can't automatically consolidate users who haven't signed in yet.
- For SAP S/4HANA Cloud, the communication arrangement for SAP-managed identity provisioning uses communication scenario `SAP_COM_1265`.
- You don't need identity provisioning jobs for SAP LeanIX. The access token issued by LeanIX already contains the required group information.

Parent topic: Concepts [page 26]

Related Information

Tenants [page 27]
Bundles [page 35]
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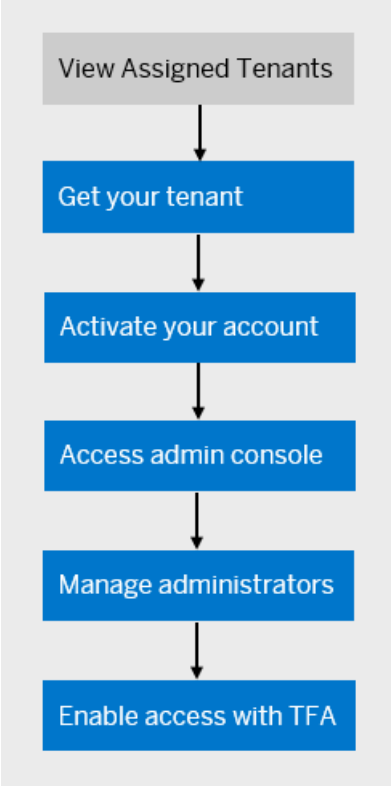
Agents [page 511]
Groups [page 514]
Authorization Policies [page 518]

1.4 Initial Setup

The initial setup of SAP Cloud Identity Services involves the following steps:

Quickstart



- [page 523]
- [page 524]
- [page 524]
- [page 524]
- [page 524]
- [page 524]

The tenant administrator performs all the steps in the *Initial Setup* section. Hover over the image and click the highlighted areas for more information on each individual step.

View Assigned Tenants and Admins

Before getting a tenant, check if there is an existing one assigned to your customer ID.

- If there is an existing tenant and you are among its administrators, skip this step and access the administration console [page 542]. Your existing tenant will be reused.
- If there is no existing tenant, follow the steps below.

See View Assigned Tenants and Administrators [page 525]

Get Your Tenant

As a customer, you are entitled to one productive and one test SAP Cloud Identity Services tenant. You are also allowed to request an additional tenant. Trial tenants are open to everyone.

See Get Your Tenant [page 528]

Activate Your Account

Once you get your tenant, you receive an email with instructions to activate your account for the administration console of SAP Cloud Identity Services. This tool is the central point for configuring identity authentication and provisioning scenarios.

See Activate Your Account [page 540]

Access Admin Console

Once you activate your account and sign in successfully, you open the SAP Cloud Identity Services administration console. You are the initial administrator user listed under *Users & Authorizations* > *Administrators*.

See Access Admin Console [page 542]

Manage Administrators

As the initial administrator user of SAP Cloud Identity Services, you can view and manage your predefined authorizations, add new administrators, as well as define authorization policies to further control their data access.

See Manage Administrators [page 543]

Enable Two-Factor Authentication

We recommend that you protect the access to the SAP Cloud Identity Services administration console with two-factor authentication (TFA). Compared to single-factor authentication, two-factor authentication provides enhanced security.

See Enable Two-Factor Authentication [page 544]

1. View Assigned Tenants and Administrators [page 525]
   View all SAP Cloud Identity Services tenants that are assigned to your customer ID with the *iamtenants* application or the Cloud Identity Services Administrators Card on the SAP for Me portal.
2. Get Your Tenant [page 528]
   Get your SAP Cloud Identity Services tenant and become productive.
3. Activate Your Account [page 540]
   Open the email from your SAP team and follow the instructions to activate your account.
4. Access Admin Console [page 542]
   Sign in to the administration console of SAP Cloud Identity Services.

5. Manage Administrators [page 543]
   View and configure the administrator authorizations. Add additional user administrators.
6. Enable Two-Factor Authentication [page 544]
   Enable two-factor authentication for all administrators accessing SAP Cloud Identity Services administration console.

1.4.1 View Assigned Tenants and Administrators

View all SAP Cloud Identity Services tenants that are assigned to your customer ID with the *iamtenants* application or the Cloud Identity Services Administrators Card on the SAP for Me portal.

Prerequisites

You have an *S-user*. For more information, see About the User, Authorization, and Administrator Concept.

Context

The SAP Cloud Identity Services - Tenants application shows which are the Identity Authentication and Identity Provisioning tenants that are assigned to a customer ID and who are the tenant administrators of these tenants. The information also includes the status of the administrator. Only administrators that have the *Manage Tenant Configuration* role are displayed.

The default tenants, one test and productive tenant per customer, are provided regardless of the number of contracts signed in which SAP Cloud Identity Services is included or bundled. Additional productive or test tenants beyond the initial ones must be purchased separately. For more information, see Tenants [page 27].

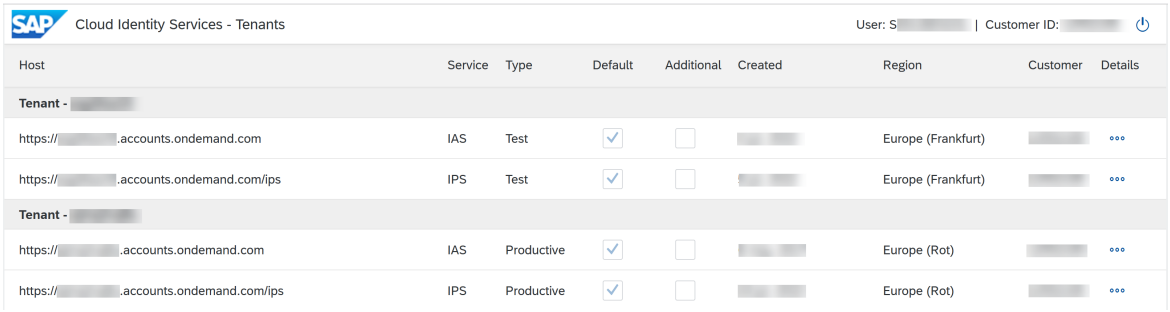
> → Tip
>
> As an alternative, you can view the tenant administrators of the tenants that are assigned to you and add new administrators in SAP Cloud Identity Services via the SAP for Me portal. For more information see Cloud Identity Services Administrators Card.

Procedure

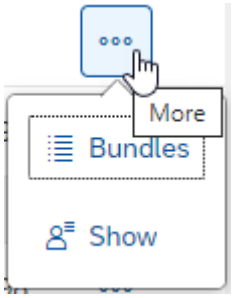
1. Access the SAP Cloud Identity Services - Tenants application via the following URL:https://iamtenants.accounts.cloud.sap/
2. Sign in with your *S-user*.

   > ⚠ Restriction
   >
   > Not signing in with an *S-user* returns *HTTP Status 500 - Internal Server Error*.

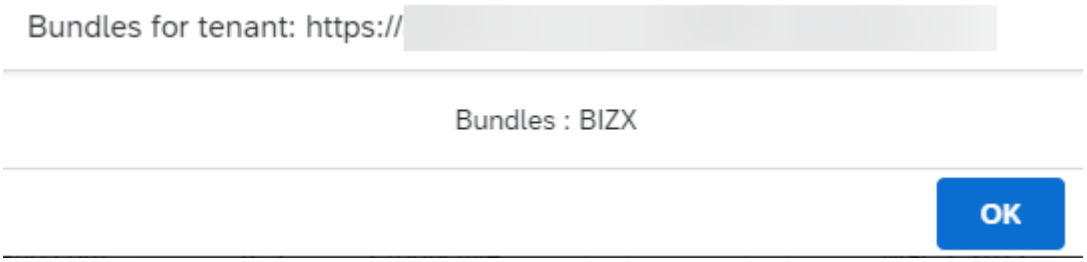
You can view the Identity Authentication and Identity Provisioning tenants that are assigned to your customer ID.



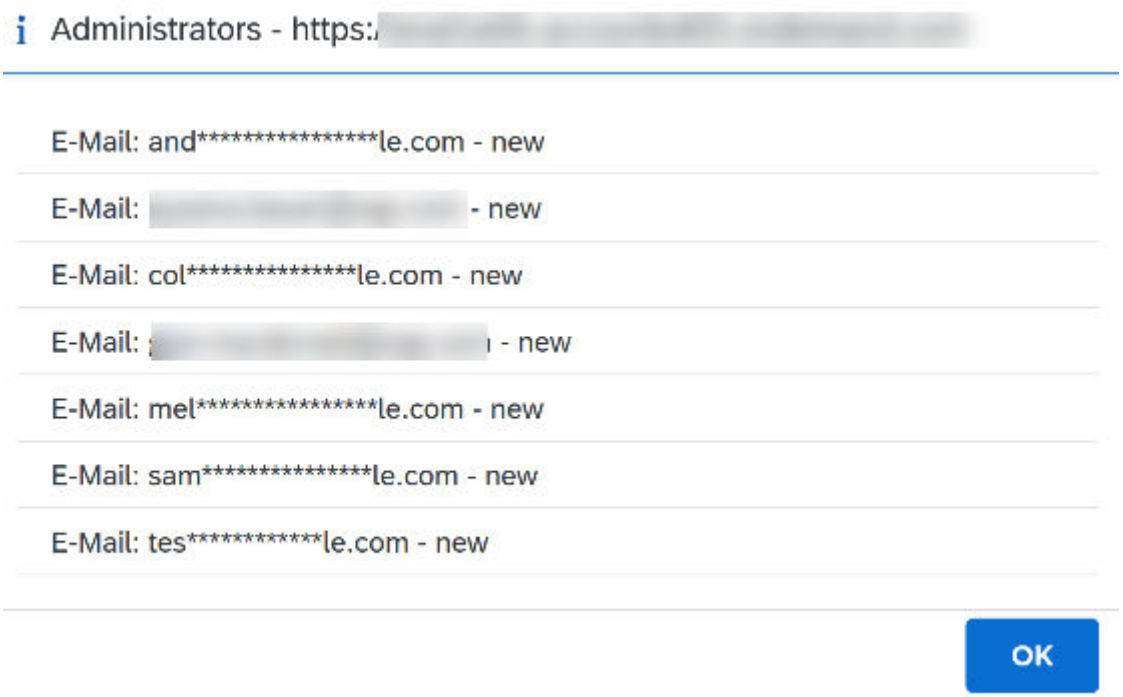
3. Choose details next to the tenant to view the bundles or tenant administrators for that tenant.



- Bundles - shows the bundles for the tenant.



- Show - shows the administrators for the tenant.



> **Note**
>
> The users and emails are masked if the domain of the authenticated user is different from the domain of the tenant administrators.

Task overview: Initial Setup [page 523]

Next: Get Your Tenant [page 528]

Related Information

Tenants [page 27]

1.4.2 Get Your Tenant

Get your SAP Cloud Identity Services tenant and become productive.

Check Your Tenant Availability

Check whether you already have an assigned SAP Cloud Identity Services tenant or need to request one. Use one of the following tools to check your tenant availability:

- SAP Cloud Identity Services - Tenants (*iamtenants*) application:
  Go to https://iamtenants.accounts.cloud.sap/. For more information, see Viewing Assigned Tenants and Administrators.
- SAP for Me:
  Check the Public Cloud Systems card. For more information, see Public Cloud Systems Card.

Based on the result:

- If a tenant exists and you are one of its administrators, go to Access Admin Console [page 542].
- If no tenant exists, follow the steps below to request one according to your needs.

You receive an SAP Cloud Identity Services tenant either as part of a bundle with an SAP cloud solution or through a self-service request in SAP for Me or SAP BTP cockpit.

Bundle with an SAP Cloud Solution

When you purchase an SAP cloud solution that bundles with SAP Cloud Identity Services, you are entitled to one productive and one test tenant. Depending on the specific cloud solution, SAP may provision only one productive tenant, only one test tenant, or both.

If you already have a tenant and purchase additional SAP cloud solutions, you will not get additional tenants. Your existing one will be reused. However, there are some special cases and exceptions to this rule. For example:

- Your business operates in multiple regions, and your SAP cloud solutions are delivered across these regions. You can request additional SAP Cloud Identity Services tenants in the particular region where other SAP cloud solutions have been licensed.
- You have multiple instances of SAP SuccessFactors. It is not recommended to bundle them with the same SAP Cloud Identity Services tenant. Instead, you can request a dedicated tenant for each SAP SuccessFactors instance through the SAP SuccessFactors Upgrade Center.

> → Recommendation
>
> Using a single productive SAP Cloud Identity Services tenant for your SAP cloud solutions is important for the integration and out-of-the-box Single Sign-On (SSO) between these solutions. If your company is already using a productive SAP Cloud Identity Services tenant, we strongly recommend reusing it for any additional SAP cloud solutions you plan to purchase and adopt.

Getting a bundled tenant is generally an automatic process that requires no customer interaction. SAP takes care of creating and provisioning the tenant based on your cloud solution and regional choices. If you don't have a tenant, a new one will be created. If you have multiple tenants, the default tenant—typically the first one created for you—is selected and provisioned with the cloud solution. This automatic process is also used to create additional tenants when it is necessary to comply with specific regulations and in scenarios involving SAP SuccessFactors.

There are cases where customer interaction is required. If multiple tenants are available for selection, for example when using SAP for Me, Boosters, SAP SuccessFactors Upgrade Center, or other tools, you will need to choose the most appropriate one.

> → Tip
>
> If you have multiple SAP Cloud Identity Services tenants and are unsure which one to select, consider these recommendations:
>
> - Default Tenant: A tenant marked as `Default` is typically the first one created for the customer. Many SAP applications automatically integrate with this tenant when provisioned. Choose it if these specifics matter to you. To check which tenant is your default one, refer to View Assigned Tenants and Administrators [page 525].
> - Regional Provisioning: For customers with an `EU Access` contract, choose an SAP Cloud Identity Services tenant located in Europe if your SAP cloud solution is provisioned in an `EU Access` region. This ensures better performance and compliance with the European regulations. For more information, see EU Access.
> - Multi-Regional Requirements: If your business operates in multiple regions and needs additional tenants, you can request them by following the process described in Get Additional Tenant [page 533]. Keep in mind that having a dedicated tenant per region might mean SSO isn't guaranteed out-of-the-box.
> - SAP SuccessFactors Use Case: If you have multiple instances of SAP SuccessFactors, you can request a separate SAP Cloud Identity Services tenant for each instance from the SAP SuccessFactors Upgrade Center. This ensures that each instance, representing the system of origin for employees, will have a dedicated identity provider and proper handling of users with the same login name. For more information, see Initiating the Upgrade to SAP Cloud Identity Services - Identity Authentication Service.

When tenants delivered as part of the bundling process are created, you get their tenant URLs in a welcome email from SAP. For more information, see Bundles [page 35].

Self-Service Request in SAP for Me

To request a tenant of SAP Cloud Identity Services, you can use SAP for Me, the SAP's central customer portal for managing products, systems, and entitlements.

In SAP for Me, there are two starting points for tenant provisioning: Portfolio & Products and Systems & Provisioning. Portfolio & Products serves as the commercial catalog and shows what you purchased, such as entitlements, licenses, and product roadmaps. Systems & Provisioning is the area where you view, request, activate, and manage your cloud tenants.

Get Tenant in SAP for Me (Portfolio & Products)

Use this option to request a test or production tenant of SAP Cloud Identity Services for integration with SAP-managed cloud ERP applications, Joule, and other supported solutions. The tenant is provisioned in your selected region and comes with SAP-managed integration configurations.

For more information, see Provisioning and Identity and Access Management in SAP Managed Cloud ERP Applications.

> ### Note
>
> New tenants are created only if there are no existing tenants bound to the customer ID.

1. Log on to SAP for Me with your S-user ID and password and open the Portfolio & Products dashboard.
2. Go to the *My Product Packages* tab.
3. Select the *SAP Cloud Identity Services* tile.
4. Go to the *Applications* tab.
5. In the *SAP Cloud Identity Services* tile, choose *Start Provisioning*.
   The *Create Solution* wizard is launched.
6. In the *General Information* section, provide the following data for the solution:
   - *Entitlement System*: Displays the system ID of SAP Cloud Identity.
   - *Name*: **<Name>**

     > ### Caution
     >
     > You can't rename solutions after you create them.

     > ### Note
     >
     > Names must meet these guidelines:
     >
     > - Limit names to 63 characters.
     > - Use only lowercase letters (a-z), numbers (0-9), hyphens (-), and periods (.).
     > - Start and end with a letter or number.
     > - Use a name that's unique within the selected resource group.

   - *Path*: Use the *Value Help* ( ) to select a path to a resource group.
   - *Business Type*: Select *Test* or *Production* from the dropdown menu.

   Choose *Next Step*.
7. In the *Configure Parameters* section, select a region. Choose *Review*.
8. Review your entries and choose *Finish*.
   Your provisioning request for the solution has started and its resources are being set up in the selected resource group. The provisioning process can take a few moments to complete.
   The new solution resource will achieve the ready status when all tenants are successfully provisioned and integrations are fully established. To view the status of the generated resources, go to the *Resources* tab.

After the SAP Cloud Identity tenant is provisioned, you can proceed with provisioning your SAP cloud applications. During this process, the newly provisioned SAP Cloud Identity tenant, associated with the application tenant, will be available for selection.

Get Tenant in SAP for Me (Systems & Provisioning)

Use this option to request a test or production tenant of SAP Cloud Identity Services bundled with an SAP cloud solution. The tenant is provisioned in the same region as the bundled solution and preconfigured accordingly.

There is no explicit option to request a tenant as part of this provisioning flow. When you provision your application, and no SAP Cloud Identity Services tenant is available for selection, a tenant is automatically created when you submit the provisioning request.

In some situations, automatic tenant creation is not available. In these cases, you must first request a tenant manually in the SAP BTP cockpit, as described in the *Self-Service Request in SAP BTP Cockpit* section below. After the tenant is created, return to SAP for Me and restart the provisioning process so the new tenant becomes available for selection. You can recognize this situation when the *Start Provisioning* button (step 3 below) is disabled.

1. Log on to SAP for Me with your S-user ID and password and open the Systems & Provisioning dashboard.
2. Go to the *Provisioning* tab.
3. Under *Systems to be Provisioned*, choose *Start Provisioning* for the SAP cloud solution you want to bundle with SAP Cloud Identity Services.
4. In the *Start New Provisioning Request* dialog, provide the following information:
    1. *Region*: The geographic location of the data center where the SAP cloud solution tenant will be deployed.
    This option is not available for all applications. In some cases, the region may be system-determined based on the contract.
    2. *Integration with SAP Cloud Identity Services*: Select either a test or production SAP Cloud Identity tenant.
    By default, you can select either a productive or a test SAP Cloud Identity tenant, depending on the type of application tenant being provisioned. For some applications, you may also have the option to integrate a test application tenant with a productive SAP Cloud Identity tenant.
    For more information, see Onboard to SAP Cloud Identity Services.
    3. *Owner of SAP Cloud Identity Services*: Choose one of the following:
    Own systems – SAP Cloud Identity tenants that belong to the same Customer ID as the SAP cloud solution being provisioned.
    Shared systems – SAP Cloud Identity tenants that belong to a different Customer ID but are shared with your company.
    For more information, see KBA 2632518 and Reuse SAP Cloud Identity Services Tenants for Different Customer IDs.
5. Choose *Submit*.

For more information on getting started and the SAP for Me dashboards, see SAP for Me Online Help.

Self-Service Request in SAP BTP Cockpit

Products delivered on SAP Business Technology Platform (SAP BTP) do not trigger automatic creation of SAP Cloud Identity Services productive and test tenant. You need to request the tenant in the SAP BTP cockpit.

This is true in most cases, but not for all SAP BTP applications, and depends on the subaccount where the application runs on. In SAP BTP, there are customer-managed and SAP-managed subaccounts.

- If the application runs on SAP-managed subaccount, then the SAP Cloud Identity Services tenant is automatically provisioned with the application. In this case, the administrator of the customer doesn't have access to the SAP BTP subaccount.
- If the application runs on customer-managed subaccount, then in most cases the administrator of the customer has to manually create SAP Cloud Identity Services tenant via SAP BTP cockpit. However, there are also exceptions, like SAP Cloud ALM (application lifecycle management), where SAP creates and provisions the SAP Cloud Identity Services tenant.

Additional tenants and trial tenants are also created through self-service request.

Get Productive and Test Tenant

Get your productive and test tenant as part of a self-service request in SAP BTP cockpit. Productive and test tenants have no difference from technical and functional perspective.

Prerequisites

- You have a global account in SAP BTP cockpit with a multi-environment subaccount.

Procedure

1. Access the SAP BTP cockpit, choose your subaccount and navigate to it by choosing its tile.
2. In the navigation area, choose *Entitlements*. You have two options:
   - if you see *default (Application)* for Cloud Identity Services in the entitlements page of your subaccount, skip steps 4 to 6, and go to step 7.
   - if you don't see *default (Application)* for Cloud Identity Services in the entitlements page of your subaccount, continue with step 4.

   > → Tip
   >
   > - If you use directories entitlement management in your global account, you must assign the *default (Application)* plan to the directory first, and after that to the subaccount. Once you assign the *default* plan to your directory, it will appear in your subaccount entitlements too. For more information, see Configure Entitlements and Quotas for Directories.

3. Choose *Edit* > *Add Service Plans*, and select *Cloud Identity Services* from the list of entitlements available for this subaccount.
4. Mark the *default (Application)* plan. The *0* number in the *Add 0 Service Plan* button increments to *1*.
5. Choose the *Add 1 Service Plan* button.
6. Save your changes on the *Entitlements* page.
7. Navigate to *Services* > *Service Marketplace* and select the *Cloud Identity Services* tile. Under *Application Plans* you should see the *default* plan.
8. In the top-right corner, choose *Create* and follow the steps in the wizard to configure your new Cloud Identity Services tenant:
   1. From the dropdown list, choose *Cloud Identity Services*.
   2. Select the *default* plan and then choose *Next*.
   3. From the *Cloud Service Type* dropdown, choose the tenant type: *TEST* or *PRODUCTIVE* (default value).

4. Choose *Next*, review your configuration and choose *Create*.

> **Note**
>
> For each service plan you must have a separate subaccount.
>
> The default plan allows you to consume SAP Cloud Identity Services. New tenants are created only if there aren't already existing tenants of the same type (productive or test) bound to your customer ID, regardless of the region.
>
> The default test and productive tenants will be created in the same region.

9. If you initially create a test tenant and then you want to proceed with a productive one (or vice versa), repeat the preceding steps in a separate subaccount of the same region.

Next Steps

Go to your email inbox and activate your account. For more information, see Activate Your Account [page 540].

> **Note**
>
> **SAP BTP, Neo subaccount**
>
> If you have a global account with Neo subaccount, you can get a productive tenant and use it for free under the General Terms and Conditions for SAP Cloud Services. For this, you need to open your Neo subaccount, navigate to *Services* > *Identity Authentication* tile and enable the service. The new tenants are created only if there aren't existing productive tenants bound to your customer ID already.

Get Additional Tenant

Get an additional tenant as part of a self-service request in SAP BTP cockpit. SAP customers and partners can use SAP Cloud Identity Services additional tenants via CPEA (Cloud Platform Enterprise Agreement) and Pay-As-You-Go consumption-based commercial models. They can request one additional tenant per SAP BTP subaccount.

> **Note**
>
> The additional tenant doesn't add additional logon requests. Licensed logon requests are distributed across all tenants.

Prerequisites

- You have a global account in SAP BTP cockpit with a multi-environment subaccount.
- You have an existing default tenant. A tenant marked as default is typically the first one created for the customer. Many SAP applications automatically integrate with this tenant when provisioned. To check which tenant is your default one, refer to View Assigned Tenants and Administrators [page 525].

Procedure

To get additional tenant using SAP BTP multi-environment subaccount, proceed as follows:

1. Access the SAP BTP cockpit, choose your subaccount and navigate to it by choosing its tile.
2. In the navigation area, choose *Entitlements*. You have two options:
   - if you see *additional-tenant (Application)* for Cloud Identity Services in the entitlements page of your subaccount, skip steps 3 to 6, and go to step 7.
   - if you don't see *additional-tenant (Application)* for Cloud Identity Services in the entitlements page of your subaccount, continue with step 3.

   → Tip

   - If you use directories entitlement management in your global account, you must assign the *additional-tenant (Application)* plan to the directory first, and after that to the subaccount. Once you assign the *additional-tenant* plan to your directory, it will appear in your subaccount entitlements too. For more information, see Configure Entitlements and Quotas for Directories.
3. Choose *Edit* > *Add Service Plans*, and select *Cloud Identity Services* from the list of entitlements available for this subaccount.
4. Mark the *additional-tenant (Application)* plan. The *0* number in the *Add 0 Service Plan* button increments to *1*.
5. Choose the *Add 1 Service Plan* button.
6. Save your changes on the *Entitlements* page.
7. Navigate to *Services* > *Service Marketplace* and select the *Cloud Identity Services* tile. Under *Application Plans* you should see the *additional-tenant (Application)* plan.
8. In the top-right corner, choose *Create* and follow the steps in the wizard to configure your new Cloud Identity Services tenant:
   1. From the dropdown list, choose *Cloud Identity Services*.
   2. Select the *additional-tenant* plan and then choose *Next*.
   3. From the *Cloud Service Type* dropdown, choose the tenant type: *TEST* or *PRODUCTIVE* (default value).
   4. Choose *Next*, review your configuration and choose *Create*.

   ⓘ Note

   For each service plan you must have a separate subaccount.

   The *additional-tenant* plan allows the customer to create a new SAP Cloud Identity Services additional tenant. The type of the created tenant may be test or productive depending on the customer selection. The new instance is created only if there is an existing default instance bound to your customer ID.

   You can delete the additional tenant by choosing the *Delete* button, which initiates the deletion of the subscription to the additional-tenant plan of Cloud Identity Services in your subaccount. The tenant will be deactivated for 30 days before being permanently deleted. After deletion, it cannot be restored.

Next Steps

Go to your email inbox and activate your account. For more information, see .Activate Your Account [page 540]

ⓘ Note

SAP BTP, Neo subaccount

If you have a global account with an SAP BTP, Neo subaccount, you can get an additional tenant, too.

Open your existing CPEA Neo account or create new CPEA Neo account, navigate to *Services* > *Identity Authentication - Additional Tenant* tile and enable the service.

Identity Authentication additional productive tenant will be created and the account member who triggered the creation will get an activation email. Note that test tenants cannot be created from a Neo subaccount.

You can delete the additional tenant by choosing the *Delete* button, which initiates the deletion of the subscription to the additional-tenant plan of Cloud Identity Services in your subaccount. The tenant will be deactivated for 30 days before being permanently deleted. After deletion, it cannot be restored.

Migrate Additional Tenant from Neo to Multi-Environment

Migrate an additional tenant from a Neo Subaccount to a multi-environment subaccount as part of a self-service request in SAP BTP cockpit.

→ Remember

SAP Business Technology Platform, Neo environment will sunset on **December 31, 2028**, subject to terms of customer or partner contracts.

For more information, see SAP Note 3351844.

⚠ Caution

The SAP NetWeaver AS JAVA user store scenario is only supported in the Neo environment. For more information, see Corporate User Store (Neo Environment) [page 2531].

Prerequisites

- You have a global account in SAP BTP cockpit with a multi-environment subaccount.
- You have an existing additional tenant on SAP BTP, Neo subaccount.

ⓘ Note

The additional tenant doesn't add additional logon requests. Licensed logon requests are distributed across all tenants.

Procedure

To migrate an additional tenant from a Neo subaccount to a multi-environment subaccount, proceed as follows:

1. Access the SAP BTP cockpit, choose your multi-environment subaccount and navigate to it by choosing its tile.
2. In the navigation area, choose *Entitlements*. You have two options:
   - If you see *additional-tenant (Application)* for Cloud Identity Services in the entitlements page of your subaccount, skip steps 3 to 6, and go to step 7.
   - If you don't see *additional-tenant (Application)* for Cloud Identity Services in the entitlements page of your subaccount, continue with step 3.

→ Tip

- If you use directories entitlement management in your global account, you must assign the *additional-tenant (Application)* plan to the directory first, and after that to the subaccount. Once you assign the *additional-tenant* plan to your directory, it will appear in your subaccount entitlements too. For more information, see Configure Entitlements and Quotas for Directories.

3. Choose *Edit* > *Add Service Plans*, and select *Cloud Identity Services* from the list of entitlements available for this subaccount.
4. Mark the *additional-tenant (Application)* plan. The *0* number in the *Add 0 Service Plan* button increments to *1*.
5. Choose the *Add 1 Service Plan* button.
6. Save your changes on the *Entitlements* page.
7. Navigate to *Services* > *Service Marketplace* and select the *Cloud Identity Services* tile. Under *Application Plans* you should see the *additional-tenant (Application)* plan.
8. In the top-right corner, choose *Create* and follow the steps in the wizard to configure your new Cloud Identity Services tenant:
    1. From the dropdown list, choose *Cloud Identity Services*.
    2. Select the *additional-tenant* plan and then choose *Next*.
    3. Under the *Form* tab, from the *Cloud Service Type* dropdown, choose the tenant type: *TEST* or *PRODUCTIVE* (default value).
    4. Under the *JSON* tab, select an already prepared JSON file or specify the parameters in the following JSON format:

        {·} Sample Code

        ```
        {
            "cloud_service": "<the tenant type, TEST or PRODUCTIVE as selected
        in the previous step>",
            "tenant_id": "<the ID of the tenant that will be migrated>",
            "subaccount": "<the ID of the Neo subaccount>"
        }
        ```

    5. Choose *Next*, review your configuration, and choose *Create*.

Next Steps

If the migration is successful, delete the additional-tenant plan from the Neo subaccount. Go to your Neo subaccount, navigate to *Services* > *Instances and Subscriptions* > *Acctions (...)* > *Delete*.

Migrate Additional Tenant from Multi-Environment to Multi-Environment

Migrate an additional tenant from a multi-environment subaccount to another multi-environment subaccount as part of a self-service request in SAP BTP cockpit.

⚠ Caution

The SAP NetWeaver AS JAVA user store store scenario is only supported in the Neo environment. For more information, see Corporate User Store (Neo Environment) [page 2531].

Prerequisites

- You have a global account in SAP BTP cockpit with a multi-environment subaccount.
- You have an existing additional tenant on a multi-environment subaccount.
- The source and target subaccounts must be in the same SAP Cloud Identity Services region.

> **Note**
>
> The additional tenant doesn't add additional logon requests. Licensed logon requests are distributed across all tenants.

> **Note**
>
> The target global account can be different from the source global account, as long as it belongs to the same customer ID.

Procedure

To migrate an additional tenant from a multi-environment subaccount to another multi-environment subaccount, proceed as follows:

1. Access the SAP BTP cockpit, choose your target multi-environment subaccount and navigate to it by choosing its tile.
2. In the navigation area, choose *Entitlements*. You have two options:
   - If you see *additional-tenant (Application)* for Cloud Identity Services in the entitlements page of your subaccount, skip steps 3 to 6, and go to step 7.
   - If you don't see *additional-tenant (Application)* for Cloud Identity Services in the entitlements page of your subaccount, continue with step 3.

   > **→ Tip**
   >
   > - If you use directories entitlement management in your global account, you must assign the *additional-tenant (Application)* plan to the directory first, and after that to the subaccount. Once you assign the *additional-tenant* plan to your directory, it will appear in your subaccount entitlements too. For more information, see Configure Entitlements and Quotas for Directories.

3. Choose *Edit* > *Add Service Plans*, and select *Cloud Identity Services* from the list of entitlements available for this subaccount.
4. Mark the *additional-tenant (Application)* plan. The *0* number in the *Add 0 Service Plan* button increments to *1*.
5. Choose the *Add 1 Service Plan* button.
6. Save your changes on the *Entitlements* page.
7. Navigate to *Services* > *Service Marketplace* and select the *Cloud Identity Services* tile. Under *Application Plans* you should see the *additional-tenant (Application)* plan.
8. In the top-right corner, choose *Create* and follow the steps in the wizard to configure your new Cloud Identity Services tenant:
   1. From the dropdown list, choose *Cloud Identity Services*.
   2. Select the *additional-tenant* plan and then choose *Next*.
   3. Under the *Form* tab, from the *Cloud Service Type* dropdown, choose the tenant type: *TEST* or *PRODUCTIVE* (default value).

4. Under the *JSON* tab, select an already prepared JSON file or specify the parameters in the following JSON format:

   Sample Code

   ```
   {
       "cloud_service": "<the tenant type, TEST or PRODUCTIVE as selected
   in the previous step>",
       "tenant_id": "<the ID of the tenant that will be migrated>",
       "subaccount": "<the ID of the source multi-environment subaccount>"
   }
   ```

5. Choose *Next*, review your configuration, and choose *Create*.

Next Steps

If the migration is successful, delete the additional-tenant plan from the source multi-environment subaccount. Go to your source subaccount, navigate to *Services* > *Instances and Subscriptions* > *Acctions (...)* > *Delete*.

Get Trial Tenant

Create a trial tenant from an SAP BTP trial account.

A trial tenant is intended for testing purposes. It allows you to try out and explore SAP Cloud Identity Services – Identity Authentication and Identity Provisioning. It is free of charge and limited to a trial period in accordance with the SAP BTP trial terms and conditions. A trial tenant differs from a test tenant. Trial tenants can be used for a limited period, while test tenants are provided to customers in addition to their productive tenants where they can test new features before implementing them productively.

Restriction

- You can have only one SAP Cloud Identity Services trial tenant per SAP BTP global account regardless of how many subaccounts you create. If you have multiple subaccounts, and you have trial tenants for them, they use one and the same trial tenant.
- You can have up to 50 users in a trial tenant.
- You can't connect to on-premise systems using the cloud connector. It's not possible to subscribe to the cloud Identity Services connectivity plan in your trial subaccount.
- The `Invitation Rest API` is not available. For more information about the API, see Invitation REST API [page 2802].
- Email template customization is not possible.

Prerequisites

- You have a free account on SAP BTP Trial, as described in Get a Free Account on SAP BTP Trial

Procedure

1. Go to the SAP BTP Trial page and click *Log On*.
2. Choose *Go to Your Trial Account* and open your subaccount.

3. In the navigation area, choose *Entitlements*. You have two options:
   - if you see *default (Application)* for Cloud Identity Services in the entitlements page of your subaccount, skip steps 4 to 7, and go to step 8.
   - if you don't see *default (Application)* for Cloud Identity Services in the entitlements page of your subaccount, continue with step 4.
4. Choose *Edit* > *Add Service Plans*, and select *Cloud Identity Services* from the list of entitlements available for this subaccount.
5. Mark the *default (Application)* plan. The *0* number in the *Add 0 Service Plan* button increments to *1*.
6. Choose the *Add 1 Service Plan* button.
7. Choose *Save*.
8. In the navigation area, choose *Services* > *Instances and Subscriptions*.
9. In the top-right corner, choose *Create*, and follow the steps in the wizard to configure your new Cloud Identity Services tenant:
   1. Choose *Cloud Identity Services*.
   2. Select the *default* plan.
   3. Choose *Create*.

Next Steps

Activate your account for Cloud Identity Services.

1. Go to your inbox and open the account activation email.
2. Click the tenant URL provided in the email.
3. Enter a password and choose *Continue*.
   You are informed that your account is successfully activated.
4. Choose *Continue*.
   This opens the administration console of SAP Cloud Identity Services.

> ### Note
>
> You can delete your trial tenant by deleting the subaccount or by deleting the Cloud Identity Services application under the Subscriptions section. For more information, see Delete a Subaccount.

> ### Note
>
> If you encounter any issues and you need support, send an email to `SAPCPTrialSupport@sap.com` or start a discussion in SAP Community.

Related Information

- **Blog Post**: SAP Cloud Identity Services offered as Trial Version
- Trial Accounts and Free Tier

Assign Existing Tenant to SAP BTP Subaccount

Assign your existing tenant to SAP BTP subaccount.

You can assign existing tenant to an SAP BTP subaccount via the default plan. Activating the default plan does not trigger the creation of new tenants if tenant with the same type already exists.

Prerequisites

- You have an existing tenant delivered as part of a bundle with another SAP product.
- You have a global account in SAP BTP cockpit with a Cloud Foundry subaccount.

Procedure

1. In the SAP BTP cockpit, choose your subaccount and navigate to *Services* > *Service Marketplace*.
2. Select the *Cloud Identity Services* tile. Under *Application Plans* you should see the default plan.
3. In the top-right corner, choose *Create* and follow the steps in the wizard to configure your new instance:
    1. From the dropdown list, choose *Cloud Identity Services*.
    2. Select the *default* plan and then choose *Next*.
    3. From the *Cloud Service Type* dropdown, choose the tenant type: *Test* or *Productive* (default value).
    4. Choose *Next*, review your configuration and choose *Create*.

> ### Note
>
> Although bundled and self-requested tenants differ in the way you get them, from technical and functional perspective they have no differences.

> ### Note
>
> It will take some time until the tenant becomes accessible. If you try to access the tenant right after creating it, you will receive `Error 404 Not Found`.

For more information, see Tenants [page 27].

Parent topic: Initial Setup [page 523]

Previous task: View Assigned Tenants and Administrators [page 525]

Next: Activate Your Account [page 540]

1.4.3 Activate Your Account

Open the email from your SAP team and follow the instructions to activate your account.

The first person who gets an account activation email for the administration console of SAP Cloud Identity Services is the initial admin user.

- If your tenant is obtained as part of a bundle with an SAP cloud solution, **the technical contact person** (also known as Contact Person IT) of the customer receives an email with the tenant URL. He or she is granted the administrator permissions of the tenant and performs the initial logon.

In addition, the technical contact person receives an email from the team of the bundled SAP cloud solution. It contains the tenant URL of the cloud solution and may contain the tenant URL of SAP Cloud Identity Services.

> **Note**
>
> When a customer with existing SAP Cloud Identity Services tenant purchases more bundled SAP cloud solutions, no new tenants are created and the existing one is reused.
>
> - If the contact person of the customer has a user account in this existing tenant, no account activation email is sent. This person can access the SAP Cloud Identity Services - Tenants application at the following URL: https://iamtenants.accounts.cloud.sap/ and check the customer tenants and administrators.
> - If the contact person of the customer has no user account in this existing tenant, and such an account is needed by the solution (application), he or she is created as a regular user and receives an account activation email for the user profile. This user doesn't get access to the admin console unless an administrator grants him or her the necessary permissions.

- If your tenant is obtained as part of the self-service request, **the logged-in user in SAP BTP cockpit** is granted the administrator permissions of the tenant. He or she receives the account activation email and performs the initial logon.

  > **Note**
  >
  > When a customer with existing SAP Cloud Identity Services tenant subscribes to the Cloud Identity Services default plan in SAP BTP cockpit, no new tenant is created and the existing one is reused. The logged-in user doesn't receive an email notification.

1. Click the tenant URL provided in the email.
2. Enter a password and choose *Continue*.
   You are informed that your account is successfully activated.
3. Choose *Continue*.
   This opens the administration console of SAP Cloud Identity Services.
   For regular users, this opens the User Profile page of SAP Cloud Identity Services

Parent topic: Initial Setup [page 523]

Previous: Get Your Tenant [page 528]

Next task: Access Admin Console [page 542]

1.4.4 Access Admin Console

Sign in to the administration console of SAP Cloud Identity Services.

Prerequisites

You have activated your account for the administration console of SAP Cloud Identity Services.

Procedure

1. Open the SAP Cloud Identity Services tenant URL.

   > **Note**
   >
   > The URL has the following pattern:
   >
   > `https://<tenant ID>.accounts.ondemand.com/admin` or `https://<tenant ID>.accounts.cloud.sap/admin`
   >
   > *Tenant ID* is an automatically generated ID by the system.
   >
   > If you have a configured custom domain, the URL has the `<your custom domain>/admin` pattern.

2. Sign in to the administration console of SAP Cloud Identity Services.

   The login credentials are set when the administrator logs in for the first time.

   > **Note**
   >
   > If you experience troubles in accessing your tenant, check the Accessing the Administration Console [page 3024] troubleshooting page or report an incident on SAP Support Portal Home with a component `BC-IAM-IDS`.

Task overview: Initial Setup [page 523]

Previous: Activate Your Account [page 540]

Next task: Manage Administrators [page 543]

1.4.5 Manage Administrators

View and configure the administrator authorizations. Add additional user administrators.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services.
2. Navigate to *Users & Authorizations* > *Administrators* and select your user.
3. Under *Configure Authorizations* view your predefined authorizations and manage them according to your needs.

   For more information, see List Administrators [page 2302] and Edit Administrator Authorizations [page 2309].
4. **Required:** Create additional user administrators.

   > **Note**
   >
   > When the SAP Cloud Identity Services tenant is initially provisioned to your organization, only one user is added as a tenant administrator. After that, due to possible legal and security issues, SAP adds additional tenant administrators only in exceptional cases (for example, the existing administrator left the company, or for some reason there is no active administrator for this tenant). To avoid access related issues in such cases, it is always a good practice for you to assign more than one administrators. Adding additional ones is exclusively in the responsibility of the current tenant administrators.

   For more information, see Add User as Administrator [page 2304]
5. **Optional:** Manage the predefined authorizations of the additional administrators.

   For more information, see Edit Administrator Authorizations [page 2309]
6. **Optional:** Define authorization policies to further control what data administrators can access.

   For more information, see Configure Authorizations Based on Policies [page 2311]
7. **Optional:** Create system administrators.

   For more information, see Add System as Administrator [page 2305]

Task overview: Initial Setup [page 523]

Previous task: Access Admin Console [page 542]

Next task: Enable Two-Factor Authentication [page 544]

1.4.6 Enable Two-Factor Authentication

Enable two-factor authentication for all administrators accessing SAP Cloud Identity Services administration console.

Context

We recommend that you protect the access to the administration console with two-factor authentication (TFA). Requesting additional means of identity verification provides enhanced security.

The initial administrator configures the access to this application under *Applications & Resources* > *Applications* > *System Applications* > *Administration Console* > *Authentication and Access*, where the TFA method must be defined as a default authentication rule under *Risk-Based Authentication*.

For more information, see Configure Risk-Based Authentication for an Application [page 2007].

Task overview: Initial Setup [page 523]

Previous task: Manage Administrators [page 543]

1.5 Scenarios

Define your user engagement scenario - whether business-to-consumer, business-to-business, or business-to-employee - and explore onboarding and provisioning scenarios for secure authentication and identity management with SAP Cloud Identity Services.

User Engagement Scenarios [page 545]
Identify your relevant user engagement scenario supported by Identity Authentication: consumer (business-to-consumer scenario), partner (business-to-business scenario), and employee (business-to-employee scenario).

Onboarding and Provisioning [page 552]
Onboard your users and groups into the Identity Directory to enable SAP Cloud Identity Services for secure authentication, authorization, and identity lifecycle management.

Identity and Access Management for Joule [page 580]
Identity and access management architecture required for Joule, including the requirement for a single SAP Cloud Identity Services tenant, SAP-managed provisioning, and global user ID assignment. Understand this foundation before you implement Joule to ensure consistent authentication and authorization across all connected SAP applications.

1.5.1 User Engagement Scenarios

Identify your relevant user engagement scenario supported by Identity Authentication: consumer (business-to-consumer scenario), partner (business-to-business scenario), and employee (business-to-employee scenario).

Business-to-Consumer [page 545]
: The business-to-consumer scenario is related to any actions performed by the consumer, such as registration to applications and consumer retailing. In this scenario, administrators facilitate the consumer processes, but they do not act on the consumer's behalf.

Business-to-Business [page 548]
: The business-to-business scenario is related to services for business partners. Unlike the business-to-consumer scenario, consumer self-registration is not allowed, and the administrator of the company is usually the one that triggers the user registration process.

Business-to-Employee [page 550]
: The business-to-employee scenario is related to services for employees of an organization. Employees can access various applications with one logon. Furthermore, administrators can upload employees data by using the user import functionality.

Parent topic: Scenarios [page 544]

Related Information

Onboarding and Provisioning [page 552]
Identity and Access Management for Joule [page 580]

1.5.1.1 Business-to-Consumer

The business-to-consumer scenario is related to any actions performed by the consumer, such as registration to applications and consumer retailing. In this scenario, administrators facilitate the consumer processes, but they do not act on the consumer's behalf.

This scenario includes the following features:

- Authentication with user name and password
- A secure single sign-on to cloud applications
- Social sign-on to cloud applications

  > ⓘ Note
  >
  > The content in this section is not relevant for China (Shanghai) region.

- Self-registration of consumers
- Invitation of consumers
- Branding elements on all the forms for logon, registration, password update, and account activation

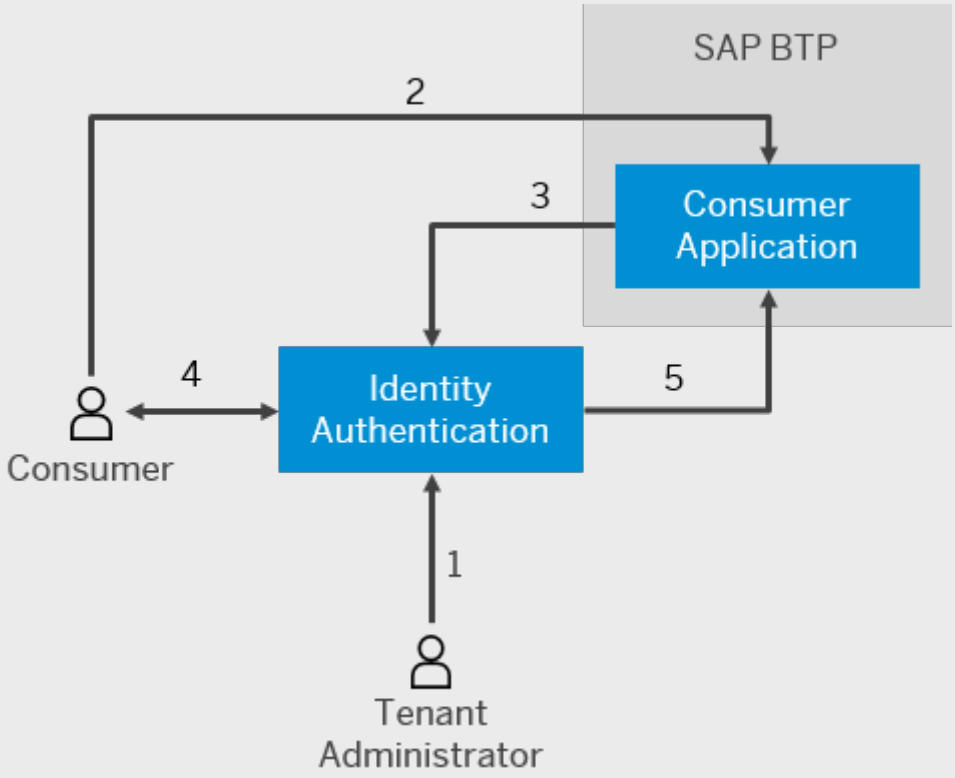
- Customized privacy policy and terms of use
- Consumer security policy
- User import and export

Example

Michael Adams is an administrator at retail company A. He would like to configure his system such that users can register on their own and then purchase from the company's site. Michael also wants to allow users to access his *Company A Purchasing* application by self-registration. To do this, Michael logs on to the administration console for SAP Cloud Identity Services, navigates to *Applications* > *Company A Purchasing*, and chooses *Authentication and Access* > *User Application Access*. He selects the *Public* radio button to allow user self-registration for his *Company A Purchasing* application. Michael also allows users to authenticate through a social provider by providing the keys and secrets for the social providers after he chooses the *Social Identity Providers* tile. He then enables social sign-on by choosing the *Applications* tile and navigating to *Company A Purchasing* > *Authentication and Access*. Furthermore, Michael configures custom terms of use and privacy policy for the *Company A Purchasing* application. To do this, he chooses the *Terms of Use Documents* and *Privacy Policy Documents* tiles and adds the new plain text files in English language versions. He then returns to *Applications* > *Company A Purchasing* view and sets the new terms of use and privacy policy documents for the application under the *Authentication and Access* tab.

Michael also decides to use a custom branding style for the buttons on his logon and registration forms. To do this, he opens the *Branding Style* page under the *Branding and Layout* tab in the *Application* view and selects the branding styles.

Donna Moore is a customer who wants to purchase goods from company A for the first time. When she accesses company A's application, she is redirected to the company's logon page. Because she is not registered yet, she has to choose the *Register Now* link to start the registration process. A registration form then appears, prompting Donna to enter her names, email, and address and to accept the organization's terms of use and privacy policy. When she submits the form, she receives an email with instructions on how to activate her registration. Once she activates her registration, she is able to log on to the retailing application with her user credentials.



1. Configures user access to the application.
2. Registers or provides credentials.
3. Delegates self-registration or authentication.
4. Activates account.
5. Confirms authentication.

Parent topic: User Engagement Scenarios [page 545]

Related Information

Business-to-Business [page 548]
Business-to-Employee [page 550]
Administration [page 1918]
Development [page 2779]
Configure User Access to the Application [page 2025]
Configuring Privacy Policies [page 2235]
Configuring Terms of Use [page 2273]
Choose and Configure a Branding Style for an Application [page 2035]

1.5.1.2 Business-to-Business

The business-to-business scenario is related to services for business partners. Unlike the business-to-consumer scenario, consumer self-registration is not allowed, and the administrator of the company is usually the one that triggers the user registration process.
The administrator invites partners or registers them on their behalf.

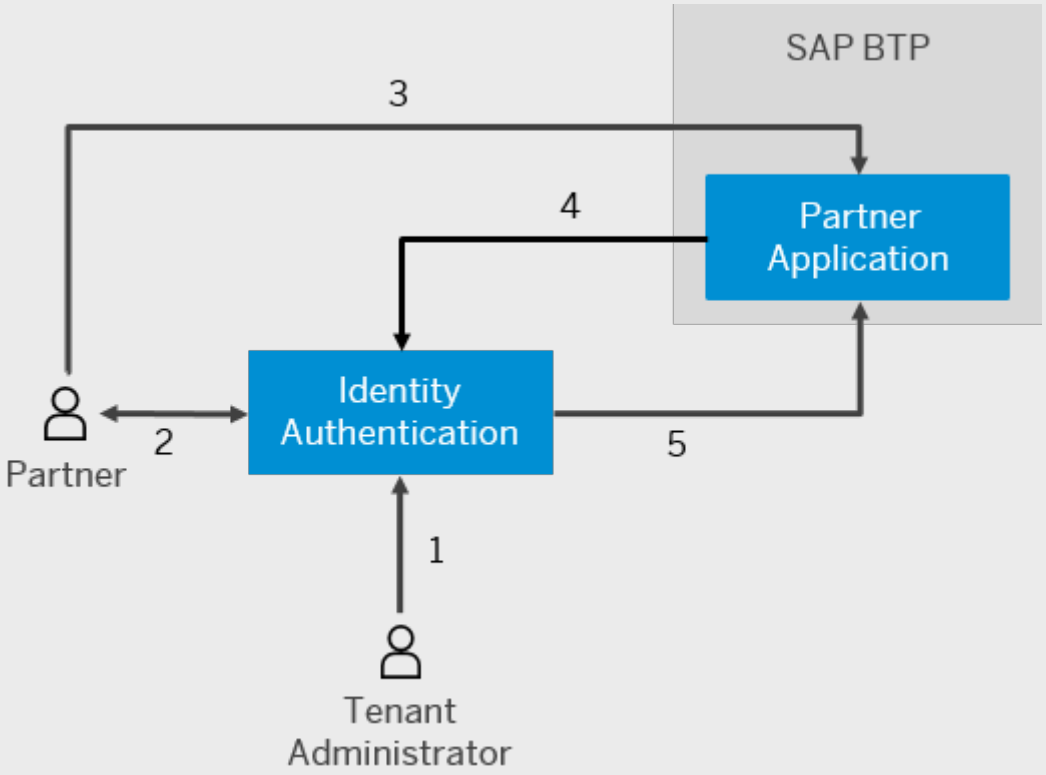
This scenario includes the following features:

- Authentication with user name and password
- A secure SSO to cloud applications
- Social sign-on to cloud applications

  > **Note**
  >
  > The content in this section is not relevant for China (Shanghai) region.

- Invitation of partners by administrators
- On-behalf registration of partners by administrators
- Branding elements on all the forms for logon, password update, and account activation
- Customized privacy policy and terms of use documents
- Partner security policy
- User import and export

> **Example**
>
> Donna Moore is a tenant administrator at company A. This company is a goods and services retailer. She would like to invite five transportation companies to join her organization in helping the distribution of goods and services to distant locations. The distributors will purchase from the *Company A Distribution* application. For this purpose, Donna registers these distributors on their behalf, logs on to the administration console for SAP Cloud Identity Services, navigates to *Applications* > *Company A Distribution* page, and chooses *Authentication and Access* > *User Application Access*. She selects the *Private* radio button in order to restrict access to just these users. The partners then activate their registration via the on-behalf registration email and can log on to the *Company A Distribution* application.



1. Invites or registers partner.
2. Activates account.
3. Provides credentials.
4. Delegates authentication.
5. Confirms authentication.

Parent topic: User Engagement Scenarios [page 545]

Related Information

Business-to-Consumer [page 545]
Business-to-Employee [page 550]
Administration [page 1918]
Development [page 2779]
Configure User Access to the Application [page 2025]

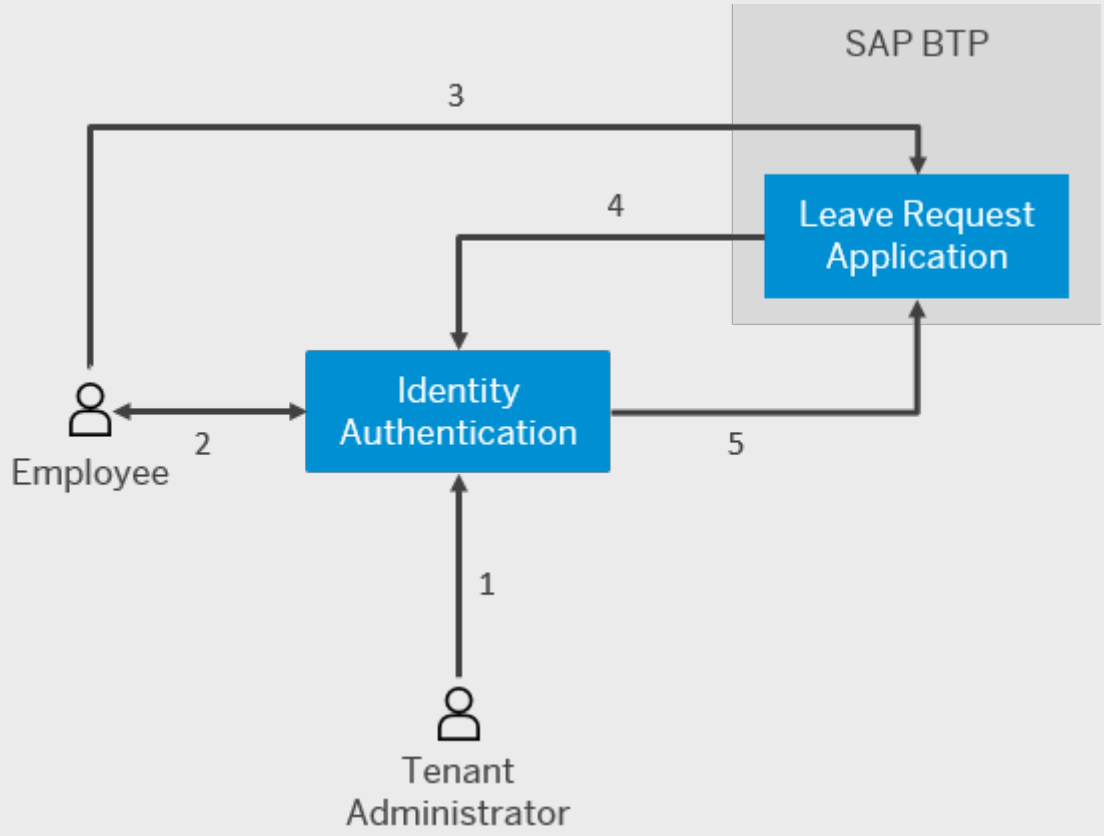
1.5.1.3 Business-to-Employee

The business-to-employee scenario is related to services for employees of an organization. Employees can access various applications with one logon. Furthermore, administrators can upload employees data by using the user import functionality.

The scenario includes the following features:

- Authentication with user name and password
- A secure SSO to cloud applications
- Branding elements on all the forms for logon and password update
- Customized privacy policy and terms of use documents
- Employee security policy
- User import and export
- Database restricted for employees only

> **Example**
>
> Julie Armstrong is an administrator at company B. She wants to configure a leave request application to be used by the employees of the company. For this purpose, she imports the employees by opening the *Import Users* page in the administration console for SAP Cloud Identity Services and selecting a CSV file containing the employees. Once she has imported all the new users into the system, she sends them an email with instructions how to activate their accounts. She also configures the trust on SAP BTP.
>
> Julie wants only the employees to access the application. She selects the *Internal* radio button after she chooses *Authentication and Access* > *User Application Access* for the leave request application in the administration console.
>
> As an employee of company B, Michael accesses an SAP BTP application to make a leave request. When he opens the application, he has to choose the *Forgot Password* link to activate his account. After activation, Michael provides a user name and password to log on to the leave request application with. He is redirected to Identity Authentication for authentication. Identity Authentication verifies his credentials and sends a response back to the SAP BTP application. As a result, Michael logs on and enters his leave request.



1. Imports users.
2. Activates account.
3. Provides credentials.
4. Delegates authentication.
5. Confirms authentication.

Parent topic: User Engagement Scenarios [page 545]

Related Information

Business-to-Consumer [page 545]
Business-to-Business [page 548]
Import or Update Users for a Specific Application [page 2348]
Configure User Access to the Application [page 2025]
Tenant SAML 2.0 Configurations [page 2103]

1.5.2 Onboarding and Provisioning

Onboard your users and groups into the Identity Directory to enable SAP Cloud Identity Services for secure authentication, authorization, and identity lifecycle management.

This section describes the available onboarding and provisioning scenarios and the corresponding configuration steps. It explains how to create and manage users and groups in the Identity Directory, configure centralized provisioning scenarios that use the Identity Directory as the central identity store, and automatically provision application-specific groups from the Identity Directory to target systems whenever changes occur.

User and Group Creation in Identity Directory [page 552]
Learn how to create users and groups in the Identity Directory — the persistency layer of SAP Cloud Identity Services.

Centralized Provisioning with Identity Directory [page 554]
Identity Directory is the persistency layer of SAP Cloud Identity Services, providing a central place for storing and managing users and groups. You can use it in centralized provisioning scenarios for managing user access to SAP cloud applications from a single, central location.

Central Store-Based Provisioning [page 579]
Central store-based provisioning enables the automatic provisioning of application-specific groups from the Identity Directory to the target systems whenever changes occur. These changes include user assignments or modifications of group attributes.

Parent topic: Scenarios [page 544]

Related Information

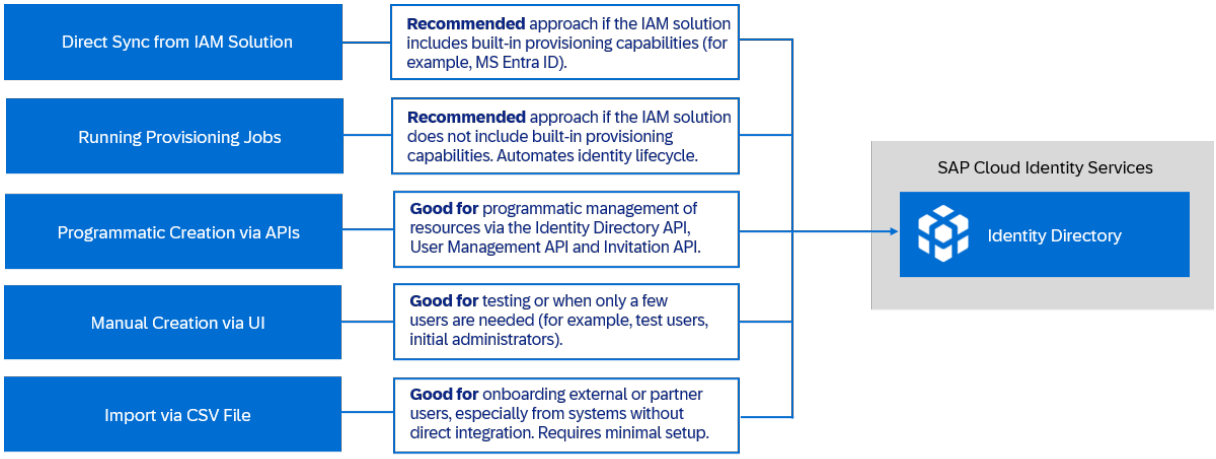
User Engagement Scenarios [page 545]
Identity and Access Management for Joule [page 580]

1.5.2.1 User and Group Creation in Identity Directory

Learn how to create users and groups in the Identity Directory — the persistency layer of SAP Cloud Identity Services.

We strongly recommend creating users in SAP Cloud Identity Services. In certain scenarios, such as when using Joule, SAP Task Center, or SAP SuccessFactors People Analytics, having users in SAP Cloud Identity Services is mandatory. Integrations with Joule and SAP Task Center require user persistence in the directory to enable cross-application correlation.

The diagram below outlines the available methods for onboarding users and groups, depending on your use case and goals.



Direct Sync from IAM Solution

Users and groups are provisioned through synchronization with the identity and access management (IAM) solution (direct sync), provided that both systems support the integration.

The IAM solution (for example, Microsoft Entra ID) includes built-in capabilities to provision users directly to the Identity Directory of SAP Cloud Identity Services. This allows users to authenticate via SAP Cloud Identity Services and access other SAP applications. For more information, see Configure SAP Cloud Identity Services for automatic user provisioning with Microsoft Entra ID .

Running Provisioning Jobs

Users and groups are created by executing scheduled or manual provisioning jobs from source systems in Identity Provisioning.

The Identity Provisioning service provides connectors to configure the synchronization of users and groups to the Identity Directory, and from there, transfer them to other SAP applications. For more information, see Centralized Provisioning with Identity Directory→ Source-Target System Provisioning.

Programmatic Creation via APIs

Users and groups can be provisioned programmatically using the Identity Directory SCIM API. Additionally, SAP Cloud Identity Services provide the Invitation REST API and the User Management REST API, which allow you to manage specific tasks programmatically, such as sending user invitations, registering users for specific applications.

For more information, see:

- Using the Identity Directory SCIM: Identity Directory API
- Using the Invitation REST API: Invitation REST API
- Using User Management REST API: User Registration

Manual Creation via UI

Users and groups are created manually through the directory's user interface in the SAP Cloud Identity Services administration console.

For more information, see Create a New User and Create a Group [page 2398].

Import via CSV File

Users and groups are imported using CSV file uploads in the SAP Cloud Identity Services administration console.

User import is supported at both the tenant level and the application level.

- **Tenant-level import** is typically used for users who will be managed centrally across multiple applications. For more information, see Import CSV File with Full User Profile
- **Application-level import** is used when only a subset of users needs access to a specific application. For more information, see Import or Update Users for a Specific Application

Group import is only supported at the tenant level.

For more information, see Import Groups via CSV File [page 2395]

Parent topic: Onboarding and Provisioning [page 552]

Related Information

Centralized Provisioning with Identity Directory [page 554]
Central Store-Based Provisioning [page 579]

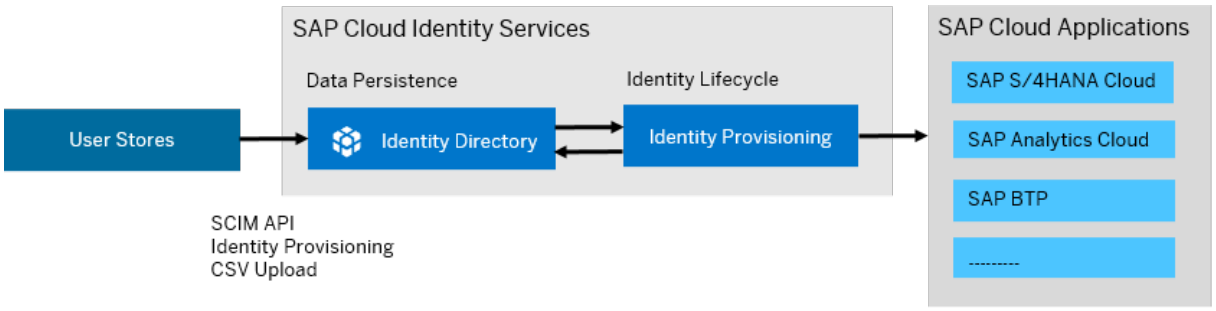
1.5.2.2 Centralized Provisioning with Identity Directory

Identity Directory is the persistency layer of SAP Cloud Identity Services, providing a central place for storing and managing users and groups. You can use it in centralized provisioning scenarios for managing user access to SAP cloud applications from a single, central location.

In a typical scenario, illustrated in the diagram below, the Identity Provisioning service reads identities from the customer's user store, like Microsoft Entra ID, and replicates them to the directory. From there, they are

provisioned further to various SAP cloud systems like SAP S/4HANA Cloud Public Edition, SAP Analytics Cloud and more.

For every newly created user, the directory generates a Global User ID, which serves as unique identifier across the entire landscape. This unique correlation attribute is then distributed by Identity Provisioning to the SAP cloud applications that need it for their integration with the SAP Task Center, for example. The unique identifier can also be generated externally and persisted in the identity directory during the initial replication from the customer's user store. For more information, see Global User ID in Integration Scenarios.



You can access the Identity Directory in the following ways:

- Use the Identity Provisioning service to create, update and delete users and groups by running provisioning jobs. The service provides a connector, known as Local Identity Directory, which can be set up as the source, target or proxy system for provisioning scenarios. There is no need to configure connectivity and authentication properties when connecting the Identity Directory and Identity Provisioning. Both services share the same SAP Cloud Identity Services tenant that you are currently logged in.
  For more information, see Local Identity Directory [page 957].

  > **Note**
  >
  > The Local Identity Directory connector is available for bundle and standalone tenants running on SAP Cloud Identity Services infrastructure.

- Use the SCIM 2.0 API to programmatically access and manage identities in the directory. The API allows you to view predefined schemas and to define custom schemas with your own attributes. For more information, see Identity Directory API.
- Use the SAP Cloud Identity Services administration console to access and manage identities in the directory via the user interface. For more information, see Managing Users [page 2342] and Managing Groups [page 2393].

Source-Target System Provisioning

Configure Local Identity Directory, both as a target and as a source system.

In this scenario, the Local Identity Directory is initially set up as a target system for user and group provisioning. Typically, identities are initially provisioned from the customer's user store to the directory, thus establishing the user store as the leading source system for identity management. Following this, users and groups from other source systems are provisioned to the directory. Finally, the identities in the directory are replicated to the target systems.

The following configuration example uses a number of systems, including: Microsoft Entra ID, Local Identity Directory, and SAP S/4HANA Cloud Public Edition.

1. Sign in to SAP Cloud Identity Services administration console and navigate to *Identity Provisioning*.
2. Add Microsoft Entra ID [page 914] as a source system.
3. Add Local Identity Directory [page 957] as a target system.
4. **Optional**: Configure properties that are relevant for the Local Identity Directory target system, for example `idds.support.bulk.operation` and `idds.bulk.operations.max.count`. For more information, see List of Properties [page 80].
5. **Optional**: Review and adapt the default write transformations as necessary.

Code Syntax

```
{
  "user": {
    "condition": "($.emails EMPTY false) && ($.userName EMPTY false) &&
isValidEmail($.emails[0].value)",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:core:2.0:User","urn:ietf:params:scim:schemas
:extension:enterprise:2.0:User","urn:ietf:params:scim:schemas:extension:sap
:2.0:User","urn:sap:cloud:scim:schemas:extension:custom:2.0:User"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['emails']",
        "scope": "createEntity",
        "functions": [
           {
                 "function": "putIfAbsent",
                 "key": "verified",
                 "defaultValue": true
           }
         ]
      },
      {

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['emails'][*]['type']",
        "type": "remove"
      },
      {
        "sourcePath": "$.emails[*].value",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.emails[?(@.value)]"
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
```

```
        "sourcePath": "$.name.middleName",
        "targetPath": "$.name.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.name.honorificPrefix",
        "targetPath": "$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath": "$.addresses",
        "targetPath": "$.addresses",
        "preserveArrayWithSingleElement": true,
        "defaultValue": [],
        "optional": true,
        "functions": [
          {
            "function": "putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          },
          {
            "condition": "(@.type NIN ['work', 'home'])",
            "function": "putIfPresent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "sourcePath": "$.phoneNumbers",
        "targetPath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.userType",
        "targetPath": "$.userType",
        "defaultValue": "employee",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.timezone",
        "targetPath": "$.timezone",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "defaultValue": true,
        "optional": true
      },
      {
```

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional" : true,
        "functions": [
            {
```

```
              "function": "resolveEntityIds"
            }
          ]
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "optional" : true
      },
      {
        "constant": false,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sendMail']",
        "scope": "createEntity"
      },
      {
        "constant": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mailVerified']",
        "scope": "createEntity"
      },
      {
        "constant": "disabled",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['passwordDetails']['status']",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['attributes']",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['attributes']"
      },
      {
        "constant": "<your-initial-password>",
        "targetPath": "$.password",
        "scope": "createEntity",
        "ignore": true
      },
      {
        "constant": "<your-source-system-type-code>",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
        "scope": "createEntity",
        "ignore": true
      },
      {
        "constant": "<your-source-system-id>",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystemId']",
        "scope": "createEntity",
        "ignore": true
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
```

```
      {
        "constant":
["urn:ietf:params:scim:schemas:core:2.0:Group","urn:sap:cloud:scim:schemas:
extension:custom:2.0:Group","urn:ietf:params:scim:schemas:extension:sap:2.0
:Group"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
            {
              "entityType": "user",
              "type": "resolveEntityIds"
            }
          ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
            {
              "entityType": "group",
              "type": "resolveEntityIds"
            }
          ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
```

```
        "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false",

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      }
    ]
  }
}
```

6. Schedule or run manually a provisioning job [page 2472] from the *Microsoft Entra ID* source system.
7. View the provisioning job logs [page 2982] and subscribe to receive job notifications [page 3018]. We recommend that you do this for each source system where a provisioning job is initiated.
8. Add Local Identity Directory [page 957] as a source system.
   It already contains all the users and groups provisioned from *Microsoft Entra ID*.
9. **Optional**: Configure properties that are relevant for the Local Identity Directory source system, for example `idds.user.filter` and `idds.group.filter`. For more information, see List of Properties [page 80].
10. **Optional**: Review and adapt the default read transformations as necessary.

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['userId']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userId']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userId']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.userName",
```

```
                "targetPath": "$.userName",
                "optional": true,
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName",
                "optional": true
            },
            {
                "sourcePath": "$.name.middleName",
                "targetPath": "$.name.middleName",
                "optional": true
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName",
                "optional": true
            },
            {
                "sourcePath": "$.name.honorificPrefix",
                "targetPath": "$.name.honorificPrefix",
                "optional": true
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails"
            },
            {
                "sourcePath": "$.emails[?(@.primary== true)].value",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.userType",
                "targetPath": "$.userType",
                "optional": true
            },
            {
                "sourcePath": "$.addresses",
                "targetPath": "$.addresses",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.locale",
                "targetPath": "$.locale",
                "optional": true
            },
            {
                "sourcePath": "$.phoneNumbers",
                "targetPath": "$.phoneNumbers",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.timezone",
                "targetPath": "$.timezone",
                "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
```

```
            },
            {
                "sourcePath": "$.groups",
                "targetPath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "condition": "$.displayName EMPTY true",
                "type": "remove",
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystemId']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystemId']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {
```

```
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']",
                "optional": true
            },
            {
                "sourcePath": "$.company",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "optional": true
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName"
            },
            {
```

```
                "sourcePath": "$.members",
                "targetPath": "$.members",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            }
        ]
    }
}
```

11. Add SAP S/4HANA Cloud Public Edition [page 833] as a target system.
12. Start another provisioning job, this time from the *Local Identity Directory* source system.

Results

The Microsoft Entra ID users and groups are provisioned to the identity directory. It now becomes the central point of truth for identity information. Once in the directory, the users and groups are replicated to the SAP S/4HANA Cloud Public Edition target.

Merging Data from Multiple Sources

The personal and technical information of the user varies across different systems. To manage a unified and up-to-date user data, you can provision the identities from multiple sources and merge them into one target identity directory.

You can choose your own approach when merging data from various sources. This could include defining the unique attribute or a combination of them when resolving conflicts, as well as deciding on how to add and update user and group attributes. All of these require changes in the identity directory configuration.

> ⚠ Caution
>
> When reading user data from multiple sources, we strongly recommend that you perform **consecutive** provisioning jobs. Simultaneous jobs may lead to inconsistent or overwritten user data in the target system.

> ### Note
>
> To execute PATCH instead of PUT requests when updating an entry, set the `idds.support.patch.operation` = *true* on the *Properties* tab of the Local Identity Directory target system.

Merge Users by Email as Correlation Attribute

By default, Identity Provisioning determines the user uniqueness based on the `userName` attribute. This means that if a user with a particular `userName` already exists in the identity directory, the service will match any new user, provisioned from other source systems with the same `userName`, to this existing user and update it accordingly.

To merge the user data by email instead of userName, you only need to set the `idds.user.unique.attribute` to **`emails[0].value`**. For more information about the identity directory unique attributes supported for conflict resolution, see List of Properties [page 80] → `idds.user.unique.attribute`.

Merge User Attributes

Following the provisioning of your identities from the leading source system, you may want to replicate additional user attributes from the second source system which were not available in the first one.

To achieve this, we suggest that you configure two pairs of source-target systems. Each source system should point to the same identity directory instance, although both will have different configurations. For example, if Microsoft Entra ID (your leading source system) and Local Identity Directory 1 (target) make up your first pair, then the second pair could consist of SAP SuccessFactors (source) and Local Identity Directory 2 (target). This setup allows you to control which user attribute comes from which source system. The overall process is the following:

1. Create the first pair of source-target systems: *Microsoft Entra ID* > *Local Identity Directory 1*.
2. On the *Properties* tab of the target system set the `idds.support.patch.operation` = *true* and the `idds.user.unique.attribute` to a unique attribute of your choice.

   > ### Note
   >
   > The unique attribute you set for the `idds.user.unique.attribute` property can either be the same or different in each of your target systems. However, it must be unique in each case. For example, if userName and email are unique attributes, you can set **`userName`** in the *Local Identity Directory 1* and **`emails[0].value`** in the *Local Identity Directory 2*.

3. Modify the write transformation of the target system to provision particular user attributes. For example, keep the required `id`, `userName` and `emails`, as well as the `givenName`, `familyName`, `displayName` and `addresses`.

   > ### Code Syntax
   >
   ```
   {
    "user":{
       "condition":"($.emails EMPTY false) && ($.userName EMPTY false) &&
   isValidEmail($.emails[0].value)",
       "mappings":[
          {
             "sourceVariable":"entityIdTargetSystem",
             "targetPath":"$.id"
          },
   ```

```
         {
            "constant":[
               "urn:ietf:params:scim:schemas:core:2.0:User",
               "urn:ietf:params:scim:schemas:extension:sap:2.0:User",
               "urn:sap:cloud:scim:schemas:extension:custom:2.0:User"
            ],
            "targetPath":"$.schemas"
         },
         {
            "sourcePath":"$.userName",
            "targetPath":"$.userName"
         },
         {
            "sourcePath":"$.emails",
            "preserveArrayWithSingleElement":true,
            "targetPath":"$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['emails']",
            "scope":"createEntity",
            "functions":[
               {
                  "function":"putIfAbsent",
                  "key":"verified",
                  "defaultValue":true
               }
            ]
         },
         {

"targetPath":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['emails'][*]['type']",
            "type":"remove"
         },
         {
            "sourcePath":"$.emails[*].value",
            "preserveArrayWithSingleElement":true,
            "targetPath":"$.emails[?(@.value)]"
         },
         {
            "sourcePath":"$.name.givenName",
            "targetPath":"$.name.givenName",
            "optional":true
         },
         {
            "sourcePath":"$.name.familyName",
            "targetPath":"$.name.familyName",
            "optional":true
         },
         {
            "sourcePath":"$.addresses",
            "targetPath":"$.addresses",
            "preserveArrayWithSingleElement":true,
            "defaultValue":[

            ],
            "optional":true,
            "functions":[
               {
                  "function":"putIfAbsent",
                  "key":"type",
                  "defaultValue":"work"
               },
               {
                  "condition":"(@.type NIN ['work', 'home'])",
                  "function":"putIfPresent",
                  "key":"type",
                  "defaultValue":"work"
               }
            ]
```

```
         },
         {
            "sourcePath":"$.displayName",
            "targetPath":"$.displayName",
            "optional":true
         },
         {
            "constant":true,
            "targetPath":"$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mailVerified']",
            "scope":"createEntity"
         }
      ]
   },
   "group":{
      "mappings":[

 .....
```

4. Run the provisioning job from Microsoft Entra ID and replicate the identities to the identity directory instance.
5. Create the second pair of source-target systems: *SAP SuccessFactors* > *Local Identity Directory 2* >.
6. Set the `idds.support.patch.operation` = *true* and the `idds.user.unique.attribute` to a unique attribute of your choice.
7. Modify the write transformation of the target system to provision another set of user attributes. For example, keep the required `id`, `userName` and `emails`, as well as the `userType`, `department` and `division` and map attributes from the SAP SuccessFactors extension schema `perPersonUuid` and `personIdExternal` to custom attributes.

Code Syntax

```
{
   "user":{
      "condition":"($.emails EMPTY false) && ($.userName EMPTY false) &&
isValidEmail($.emails[0].value)",
      "mappings":[
         {
            "sourceVariable":"entityIdTargetSystem",
            "targetPath":"$.id"
         },
         {
            "constant":[
               "urn:ietf:params:scim:schemas:core:2.0:User",

"urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
               "urn:ietf:params:scim:schemas:extension:sap:2.0:User",
               "urn:sap:cloud:scim:schemas:extension:custom:2.0:User"
            ],
            "targetPath":"$.schemas"
         },
         {
            "sourcePath":"$.userName",
            "targetPath":"$.userName"
         },
         {
            "sourcePath":"$.emails",
            "preserveArrayWithSingleElement":true,
            "targetPath":"$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['emails']",
            "scope":"createEntity",
            "functions":[
               {
                  "function":"putIfAbsent",
```

```
                  "key":"verified",
                  "defaultValue":true
               }
            ]
         },
         {

"targetPath":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['emails'][*]['type']",
            "type":"remove"
         },
         {
            "sourcePath":"$.emails[*].value",
            "preserveArrayWithSingleElement":true,
            "targetPath":"$.emails[?(@.value)]"
         },
         {
            "sourcePath":"$.userType",
            "targetPath":"$.userType",
            "defaultValue": "employee",
            "optional":true
         },
         {
            "sourcePath":"$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
            "targetPath":"$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
            "optional":true
         },
         {
            "sourcePath":"$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
            "targetPath":"$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
            "optional":true
         },
         {
            "sourcePath":"$
['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['perPersonUuid']",
            "targetPath":"$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['attributes'][0]
['value']",
            "optional":true
         },
         {
            "targetPath":"$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['attributes'][0]
['name']",
            "condition":"$
['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['perPersonUuid'] EMPTY false",
            "constant":"customAttribute1"
         },
            "sourcePath":"$
['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['personIdExternal']",
            "targetPath":"$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['attributes'][1]
['value']",
            "optional":true
         },
         {
```

```
            "targetPath":"$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['attributes'][1]
['name']",
            "condition":"$
['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['personIdExternal'] EMPTY false",
            "constant":"customAttribute2"
         }
   ]
},
"group":{
      "mappings":[

 .....
```

8. Run a second provisioning job, this time from SAP SuccessFactors, to replicate the identities to the same identity directory instance.
   This provisioning job will create new users in the Identity Directory from SAP SuccessFactors and update existing users with additional attributes from SAP SuccessFactors, specifically userType, department, division, perPersonUuid, and personIdExternal.

As a result, the identities from the two source systems are provisioned to a single target identity directory. Continuous provisioning jobs from both source systems run without overwriting attributes. Any updates are executed as patch requests.

Proxy System Provisioning

Configure *Local Identity Directory* as a proxy system in the Identity Provisioning.

You can use the identity directory as a SCIM 2.0 based proxy connector. Before exposing it to an external system, you should first populate it with users and groups. To learn how to read and then provision them to the identity directory, see the *Source-Target System Provisioning* section above.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API → *Filters*.

1. Sign in to SAP Cloud Identity Services administration console and create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning. If you already have a technical user, skip this step.
   - For **Certificate-based authentication**, follow the procedure in Manage Certificates for Inbound Connection [page 2456] → *SAP Cloud Identity Infrastructure*.
     For **Basic authentication**, navigate to *Users & Authorizations* > *Administrators* and add an admin user of type **System**. Configure a client ID and secret for this user and enable the *Access Proxy System API* permission.
2. Navigate to *Identity Provisioning* > *Proxy Systems*.
3. Add Local Identity Directory [page 1382] as a proxy system.
4. **Optional**: Configure properties applicable for the Local Identity Directory proxy system, such as `idds.user.filter` and `idds.group.filter`. For more information, see List of Properties [page 80].

5. **Optional**: Adapt the default proxy read and proxy write transformations:

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.emails[*].value",

"preserveArrayWithSingleElement":
true,
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:cor
e:2.0:User","urn:ietf:params:scim:
schemas:extension:enterprise:2.0:U
ser","urn:ietf:params:scim:schemas
:extension:sap:2.0:User"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "optional": true
      },
      {
        "sourcePath":
"$.emails[*].value",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails[?
(@.value)]"
      },
      {
        "sourcePath":
"$.userType",
        "targetPath":
"$.userType",
        "defaultValue":
"employee",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
```

Read Transformation

```
        "targetPath": "$.emails[?
(@.value)]"
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value"
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.userType",
        "targetPath":
"$.userType",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath":
"$.addresses",
        "targetPath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
```

Write Transformation

```
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath":
"$.addresses",
        "targetPath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "defaultValue": [],
        "optional": true,
        "functions": [
          {
            "function":
"putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          },
          {
            "condition": "(@.type
NIN ['work', 'home'])",
            "function":
"putIfPresent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
```

Read Transformation

```
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.timezone",
        "targetPath":
"$.timezone",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath":
"$.sourceSystem",
        "targetPath":
"$.sourceSystem",
        "ignore": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "condition":
"$.displayName EMPTY true",
        "type": "remove",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
```

Write Transformation

```
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true
```

Read Transformation

```
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
```

Write Transformation

```
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "defaultValue": true,
        "optional": true
      },
      {
        "constant": false,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['sendMail']",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails']",
        "scope": "createEntity",
        "functions": [
          {
            "function":
"putIfAbsent",
            "key": "verified",
            "defaultValue": true
```

Read Transformation

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']",
        "optional": true
      },
      {
        "sourcePath": "$.company",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']"
      },
      {
        "sourcePath":
"$.displayName",
```

Write Transformation

```
          }
        ]
      },
      {
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails'][*]
['type']",
        "type": "remove"
      },
      {
        "constant": "disabled",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['passwordDetails']['status']",
        "scope": "createEntity"
      },
      {
        "constant": 39,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['sourceSystem']",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.timezone",
        "targetPath":
"$.timezone",
        "optional": true
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
```

Read Transformation

```
          "targetPath":
"$.displayName"
        },
        {
          "sourcePath":
"$.members",
          "targetPath":
"$.members",

"preserveArrayWithSingleElement":
true,
          "optional": true
        },
        {
          "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
          "targetPath": "$.schemas"
        },
        {
          "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId']",
          "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId']",
          "optional": true
        },
        {
          "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
          "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
          "optional": true
        },
        {
          "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
          "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
          "optional": true
        }
      ]
    }
}
```

Write Transformation

```
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:cor
e:2.0:Group","urn:sap:cloud:scim:s
chemas:extension:custom:2.0:Group"
,"urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "scope": "createEntity",
        "functions": [
          {
            "type":
"replaceAllString",
            "regex": "[\\s\
\p{Punct}]",
            "replacement": "_"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "optional": true,
```

Read Transformation	Write Transformation

```
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']
['description']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']
['description']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId'] EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId'] EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId'] EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']"
      }
    ]
  }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in *Step 1*. If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in *Step 1* and the SCIM assignment method as described below:
   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ⓘ Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Results

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

Parent topic: Onboarding and Provisioning [page 552]

Related Information

User and Group Creation in Identity Directory [page 552]
Central Store-Based Provisioning [page 579]

1.5.2.3 Central Store-Based Provisioning

Central store-based provisioning enables the automatic provisioning of application-specific groups from the Identity Directory to the target systems whenever changes occur. These changes include user assignments or modifications of group attributes.

Prerequisites

To ensure the groups immediate provisioning to the target systems, the following application configurations are required:

- An application must be created for the relevant target system under *Applications & Resources* > *Applications*. See Create a New Application [page 1922].
- The application ID must be set as the value for the `ips.application.id` property in the target system. See List of Properties [page 80].
- The *Central Store-Based Provisioning* option must be enabled under *Applications & Resources* > *Applications* > *Provisioning*. See Enable or Disable Central Store-Based Provisioning [page 2401].

Context

You can initially create application-specific groups in the Identity Directory in the following ways: by running a provisioning job from a given source system, manually through the directory's user interface in the SAP Cloud Identity Services administration console or programmatically using the Identity Directory SCIM API.

For more information, see Groups [page 514] and User and Group Creation in Identity Directory [page 552].

When provisioning application-specific groups from the Identity Directory to target systems, groups that do not already exist in the target will be created. However, if you assign users to an application-specific group in the directory, this assignment will only be provisioned if the users already exist in the target system. Note that central store-based provisioning does not create users in the target system.

Procedure

Here is the necessary configuration to enable central store-based provisioning. The following example uses the Local Identity Directory as the source system and SAP Advanced Financial Closing as the target system.

1. Sign in to SAP Cloud Identity Services administration console and navigate to *Identity Provisioning*.
2. Add Local Identity Directory [page 604] as a source system.
3. **Optional**: Configure properties that are relevant for the Local Identity Directory source system, for example `idds.user.filter` and `idds.group.filter`. For more information, see List of Properties [page 80].
4. **Optional**: Review and adapt the default read transformations as necessary.
5. Add a target system of your choice, for example *SAP Advanced Financial Closing*.

6. Select the *Local Identity Directory* as relevant source system.
7. Set the property `ips.application.id` for the target system. For more information, see List of Properties [page 80].
8. Start a read job from the *Local Identity Directory* source system.
9. Navigate to your application in the administration console for SAP Cloud Identity Services under *Applications & Resources* > *Applications* and enable the *Central Store-Based Provisioning*. For more information, see Enable or Disable Central Store-Based Provisioning [page 2401].

Result

Once the procedure is executed, any change to an attribute value or a member of an application-specific group in the Local Identity Directory will trigger instant provisioning of the modified entity to the configured target system.

Parent topic: Onboarding and Provisioning [page 552]

Related Information

User and Group Creation in Identity Directory [page 552]
Centralized Provisioning with Identity Directory [page 554]
Blog Post: Taking Groups to the Next Level with Application-Specific Groups

1.5.3 Identity and Access Management for Joule

Identity and access management architecture required for Joule, including the requirement for a single SAP Cloud Identity Services tenant, SAP-managed provisioning, and global user ID assignment. Understand this foundation before you implement Joule to ensure consistent authentication and authorization across all connected SAP applications.

> ⚠ Caution
>
> This page is still work in progress. This page contains information about product features that are part of an SAP Early Adopter Care (EAC) program. Products or features that are part of an EAC program are not available to users outside of this program.

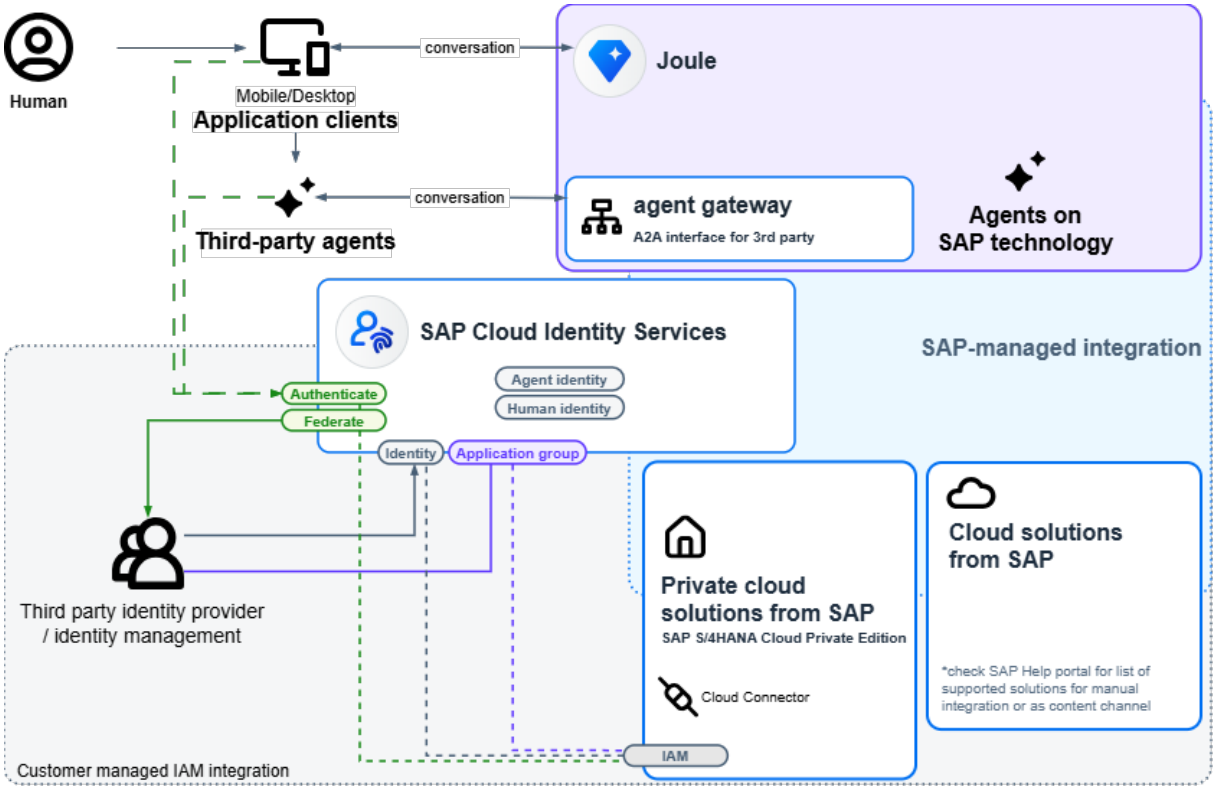
Joule, SAP's AI assistant, works across multiple connected SAP applications. For Joule to deliver a consistent experience, all of those applications must authenticate users through a single SAP Cloud Identity Services tenant. This makes SAP Cloud Identity Services the central identity and access management (IAM) layer across your SAP landscape. It forms the foundation on which you build Joule and any AI agents you deploy alongside it.

This topic explains the IAM architecture required for Joule. It covers why we designed it this way, what you need to put in place before you activate Joule, and how agent authentication and authorization fit into your

existing IAM model. The same principles apply whether your landscape is new or already has existing IAM configuration: a single SAP Cloud Identity Services tenant, users with global user IDs in the central directory, and trust established between each SAP application and that tenant.

> **Note**
>
> SAP Cloud Identity Services is included with SAP cloud applications at no additional charge. You don't need a separate subscription to use it for Joule.

SAP Cloud Identity Services is available to all customers for these scenarios automatically. No additional purchase is required.



SAP-Managed IAM Landscape for Joule

SAP-Managed IAM

SAP Cloud Identity Services supports SAP-managed identity integration across your SAP landscape. Without this approach, you must connect each SAP application to your identity layer individually. This means configuring trust, running provisioning jobs, and managing role assignments system by system. This costs you significant effort. With SAP-managed, SAP handles that configuration and keeps it running across software updates.

In the SAP-managed model, SAP takes responsibility for the technical integration. This includes setting up trust between SAP Cloud Identity Services and each connected SAP application, maintaining connector compatibility across product releases, running identity provisioning jobs, and replicating the global user ID back to each line-of-business (LoB) system. You retain control of business decisions. This includes choosing the identity provider, defining and assigning business roles, managing HR data, and configuring authentication policies.

SAP-managed IAM for Joule represents an initial step toward the same unified identity model used for the SAP Business Suite. In the Business Suite model, SAP Cloud Identity Services acts as the authoritative source for user and role data. The system distributes assignments outward to applications. The current Joule model adopts a pragmatic starting point. User records and group assignments originate in each LoB application. SAP reads them into SAP Cloud Identity Services and uses them to issue authorization tokens for Joule. The system then writes the global user ID back to the LoB system to link the records. The LoB system remains the authoritative source for user and role data in the current release. If you already follow the Business Suite model, this approach is consistent with the direction you are already heading. Fully centralized assignment management in SAP Cloud Identity Services is planned for a future release.

For Joule and the Business AI Platform, SAP-managed is the only supported integration path for public cloud applications. SAP handles the provisioning job configuration. You don't need any Identity Provisioning credentials for these systems. Identity Provisioning remains available for other applications in your landscape.

For more information, see SAP-Managed SAP Cloud Identity [page 520].

Shared Responsibilities

SAP Manages	You Manage
Connector compatibility	Identity provider selection
Infrastructure	Business role definition
Availability	User assignment
Trust setup	HR data management
Identity Provisioning job execution	Authentication policies
Global user ID replication	Agent access authorization

User Prerequisites and Business User Consolidation

For Joule to identify and authorize a user consistently across all connected SAP applications, that user must have a global user ID assigned in SAP Cloud Identity Services. The global user ID acts as the common identifier that links the user's records across different SAP systems. Joule can't recognize users without a global user ID in the central directory.

To get users into SAP Cloud Identity Services with their global user IDs and group assignments, SAP-managed identity provisioning runs two coordinated jobs for each connected LoB system:

> → Remember
>
> Before running the SAP-managed provisioning jobs, every user in scope must have logged in at least once to the LoB application using SAP Cloud Identity Services as the identity provider. This creates the application-user record that the provisioning job requires to perform consolidation correctly. Users who haven't yet logged in will have consolidatable user records created, but the system can't automatically merge those records with existing user profiles until the user signs in.

> ⚠ Caution
>
> EAC customers must currently import SAP-managed provisioning job templates manually. For more information, see . Job templates are available from SAP Note 3784171. (If you are part of the EAC program and need access to this SAP Note, open an incident under BC-IAM-IPS.)

1. A **read job** runs from the LoB system to SAP Cloud Identity Services. It reads user records and application-specific groups from the LoB system and creates corresponding entries in the SAP Cloud Identity Services user directory. Users created by this job are **consolidatable users**: shadow records that represent a user known to the LoB system but who hasn't yet logged in through SAP Cloud Identity Services. They are internal records maintained per application. A user with accounts in multiple LoB systems will have a separate consolidatable record for each. When the user logs in for the first time, the system merges the consolidatable user record with their full user profile and confirms the global user ID.
2. A **write-back job** runs from SAP Cloud Identity Services to the LoB system. It writes the global user ID assigned in SAP Cloud Identity Services back to the user record in the LoB system. This establishes the permanent link between the two records.

> ⓘ Note
>
> During user replication, only a minimal user representation is provisioned to and from the Identity Directory, indicating the source application from which the user originates. The primary purpose of this replication is to provide users with a consistent identity across connected SAP applications by assigning a Global User ID.

After the read job runs, you may need to resolve group conflicts before the write-back job can run. Group conflicts arise when the same group exists in both the Identity Directory and the LoB application and the system cannot automatically reconcile them.

After initial provisioning, your landscape is in **Stage 1**: connected applications and their roles appear in SAP Cloud Identity Services, consolidatable user records are present in the background, and Joule is available as a conversational AI assistant within the embedded user interface of each connected Business Application. In Stage 1, agents aren't available. They appear only when users have real identities in SAP Cloud Identity Services.

In **Stage 2**, real user identities exist in SAP Cloud Identity Services and are linked to application groups. Users with identities can be assigned to assistants and agents from the SAP Cloud Identity Services administration console or with the SCIM API. Stage 2 also unlocks the full Joule experience. LoB systems remain the authoritative source for user and role data until a future release.

> ⚠ Caution
>
> Agents are not available to users in Stage 1. If your SAP Cloud Identity Services tenant is new and contains no pre-existing user identities, your users will be limited to conversational Joule functions until Stage 2 is reached. We recommend creating user identities in SAP Cloud Identity Services before deploying Joule so that users have immediate access to agents after provisioning.

Whether you can advance to Stage 2 immediately after provisioning depends on your existing landscape:

- If SAP Cloud Identity Services already contains user identities with matching attributes (such as email address), the SAP-managed jobs match and link them automatically when they run. In this case, you can advance to Stage 2 as soon as provisioning completes.
- If SAP Cloud Identity Services is new with no existing user records, create user identities before deploying Joule. You can upload a .csv file, use the SCIM API, or create users manually in the SAP Cloud Identity

Services administration console. Once identities exist, the SAP-managed jobs match and link them automatically, and you reach Stage 2 after provisioning.

The user consolidation tool is a dedicated capability for transitioning users from Stage 1 to Stage 2 after initial provisioning. It presents administrators with suggested matches between consolidatable user records and real identities across applications, user by user, and requires explicit approval before creating the linked identity. This tool is not available at general availability (GA) but is planned for a future release. Until this tool becomes available, customers who deployed Joule with a new SAP Cloud Identity Services tenant and no pre-existing identities remain in Stage 1 and do not have access to agents.

If your landscape already includes user synchronization, for example, from SAP SuccessFactors to SAP Cloud Identity Services, those existing jobs continue independently. SAP-managed jobs for Joule create application-specific groups alongside existing configuration without conflicting with it.

> **Note**
>
> SAP SuccessFactors provisioning jobs create full user profiles rather than consolidatable users. Users provisioned from SuccessFactors are therefore treated differently during consolidation than users coming from other LoB systems.

Authentication Considerations

When you connect SAP applications to Joule, the subject name identifier (SNI) used for authentication in each application must align with the SAP-managed identity provisioning configuration. For most SAP applications, the SNI is the user's email address. If an application uses a different value as the SNI, for example, a login name or worker ID, you must adapt the identity provisioning job transformation to ensure users are matched correctly across systems. Application-specific SNI requirements are documented in the setup guidance for each LoB application.

If users in your landscape access SAP applications across different domain patterns, for example, some applications use the `ondemand.com` domain while others use `cloud.sap`, single sign-on sessions may not carry over automatically between domains. To resolve this, enable the *Sync SSO Cookie Between Domains* option in your SAP Cloud Identity Services tenant. You must also register any custom domains used in your landscape in the Joule Admin Center.

If you use a third-party identity provider such as Microsoft Entra ID or Okta, SAP Cloud Identity Services acts as a proxy between your identity provider and the connected SAP applications. All SAP applications connected to Joule must use the same SAP Cloud Identity Services tenant as their identity provider. Connecting a LoB application to a different SAP Cloud Identity Services tenant from the one used for Joule isn't supported.

Parent topic: Scenarios [page 544]

Related Information

User Engagement Scenarios [page 545]
Onboarding and Provisioning [page 552]
Agents [page 511]

SAP-Managed SAP Cloud Identity [page 520]
Managing Agents [page 2333]

1.6 Supported Systems

Identity Provisioning synchronizes users and groups between multiple supported cloud and on-premise systems, both SAP and non-SAP.

For more inforomation about the provisioning systems (connectors) available for use as *source*, *target*, and *proxy* system types, refer to the related topics within this section.

> **Note**
>
> This documentation refers to Identity Provisioning in the SAP Cloud Identity Services infrastructure. To access the service documentation in the Neo environment, go to Identity Provisioning Service in the Neo Environment.

Related Information

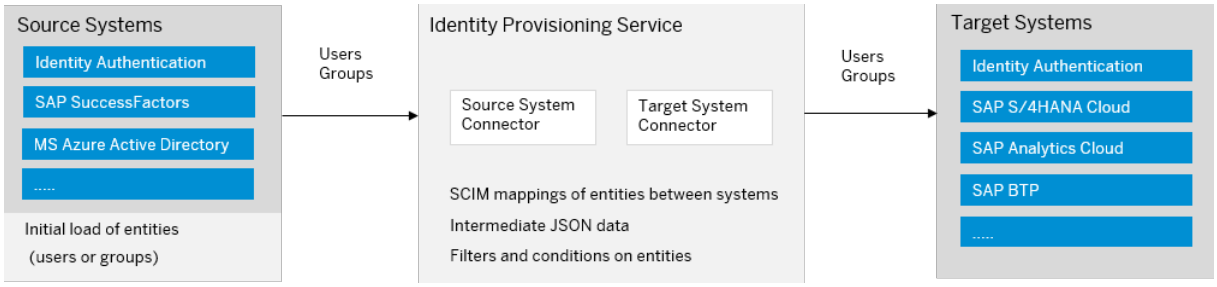
Source Systems [page 585]
Target Systems [page 936]
Proxy Systems [page 1343]

1.6.1 Source Systems

A source system is the connector used for reading entities (users, groups, roles).

Source systems can be on-premise or cloud-based, SAP or non-SAP, and usually represent the corporate user store where identities are currently maintained. Identity Provisioning reads the entities from the source system and creates or updates them in the relevant target ones. The provisioning is triggered from the *Jobs* tab of a source system.

You can connect one source system to one or multiple target systems. In the case of multiple (enabled) target systems, when you start a *Read* or a *Resync* job, this operation will trigger provisioning of entities from this source system to all relevant target ones.



> ⚠ Restriction
>
> By default, the maximum number of productive source systems you are allowed to add for your tenant is **20**. This restriction is valid for *Customer Managed* systems only.
>
> If your business requires using more systems, create an incident for component *BC-IAM-IPS* to request them. Describe your scenarios and provide a reason why you need the additional systems.

Related Information

System Types [page 70]

1.6.1.1 Identity Authentication

Follow this procedure to set up SAP Cloud Identity Service – Identity Authentication as a source system.

Prerequisites

To establish the connection between Identity Provisioning and Identity Authentication, you need to set up the technical user (of type *System*) in Identity Authentication and assign this user the necessary authorizations. You have the option to do it either now, as a prerequisite, or while configuring Identity Authentication as a source system, as described in step 3.

Context

Identity Authentication provides authentication and single sign-on for users in the cloud. The user store of Identity Authentication can manage different type of users (employees, partners, customers and public) as well as groups. Using Identity Provisioning, you can read those users (self-registered, imported, or manually created) and groups and provision them to various target systems.

For this, you need to configure the Identity Authentication as a source system in the Identity Provisioning UI. This source system consumes SCIM 2.0 API provided by Identity Authentication.

There are two versions of the Identity Authentication SCIM API. They are handled by the `ias.api.version` property as follows:

- When the value is set to **1** or the property is not defined (typical for systems created before versioning was introduced on July 9, 2021) - Identity Authentication SCIM API (in short, SCIM API version 1) is used.

  > ⓘ Note
  >
  > The SAP Cloud Identity Services – Identity Authentication SCIM API (version 1) will be decommissioned on November 30, 2028. You must use its successor, the Identity Directory SCIM API.

This requires updating your Identity Authentication connector from version 1 to version 2, as described in Update Connector Version [page 2424].

- When the value is set to **2** - Identity Directory SCIM API (in short, SCIM API version 2) is used. This value is set automatically for all manually created systems in the as a source system. For more information, see Identity Provisioning UI after versioning was introduced on July 9, 2021.
  SCIM API version 2 is enhanced to support paging for group members and user's groups, custom attributes, delta read mode for users. Also, the group resource mapping in the transformation is not ignored by default, as it is in SCIM API version 1.

  > **Note**
  >
  > Identity Authentication (using SCIM API version 2) and Identity Directory are sometimes used interchangeably. Identity Directory is the persistency layer of SAP Cloud Identity Services – Identity Authentication.

To create Identity Authentication as a source system, proceed as follows:

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *Identity Authentication* as a source system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning as a source system and Identity Authentication and configure your authentication method (basic or certificate-based).

   > **Note**
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added Identity Authentication source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Manage Certificates [page 2450].

      Skip step **a.** if you want to use basic authentication.

      In SAP Cloud Identity Services administration console, perform the next steps. They are relevant for both basic and certificate-based authentication.

   b. Add System as administrator and provide the respective credentials.

      For basic authentication, provide a password. The user ID will be generated automatically when you set the password for the first time.

      For certificate-based authentication, upload the certificate you have generated in SAP Cloud Identity Services administration console on the previous step.

   c. Save your changes.

   d. Make sure *Manage Users* (or at least *Read Users*) and *Manage Groups* authorization roles are enabled for the technical user. This way, you can read users and groups from the Identity Authentication user store.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL of the Identity Authentication tenant of your company. For example: **https://mytenant.accounts.ondemand.com** **Note** If your Identity Authentication Shanghai (China) tenants reside on SAP BTP, Neo, you should use the following URL pattern: *https://<tenant_ID>.accounts.sapcloud.cn/*
`ProxyType`	Enter: *Internet* The Identity Authentication service is a cloud solution and is outside of your company on-premise infrastructure.
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the Client ID (previously User ID) of the Identity Authentication technical user. It is generated automatically for the administrator of type system, when choosing *Secrets* > *Add* > *Save*. For example: *1ab7c243-5de5-4530-8g14-1234h26373ab* If your technical user was created before *January 2020*, enter the T-user. For example: *T000003*
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the Client Secret (previously Password) of the Identity Authentication technical user. It is generated automatically for the administrator of type system, when choosing *Secrets* > *Add* > *Save*.

Optional Properties

Property Name	Description & Value
• `ias.<property_name>` • `scim.<property_name>`	When using SCIM API version 2, property names start with `ias` prefix, for example: `ias.user.unique.attribute`. When using SCIM API version 1, property names start with `scim` prefix, for example: `scim.user.unique.attribute`. For more information, see List of Properties [page 80]. Use the main search or filter properties by *Name* or *System Type* columns.
`ias.user.filter`	When specified, only those users matching the filter expression will be read. For example: **name.familyName eq "Smith" and addresses.country eq "US"** This filter will read only users whose name is "*Smith*" and are living in the *United States*. For more information, see Identity Directory SCIM API: User Search.
`ias.group.filter`	When specified, only those groups matching the filter expression will be read. For example: **displayName eq "ProjectTeam1"** This filter will read only groups, whose display name is "*ProjectTeam1*". For more information, see Identity Directory SCIM API: Group Search.
`ips.failed.request.retry.attempts`	Predefined value: *2*
`ips.failed.request.retry.attempts.interval`	Predefined value: *60*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Identity Authentication* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   When Identity Authentication is configured as a source system, the default transformation logic reads all the user attributes from the Identity Authentication user store. The logic is provided by the Identity Authentication SCIM REST API, which then maps the attributes to the internal SCIM representation.

Handling Application-Specific Groups

In case you want to provision application-specific groups between the following pairs of provisioning systems:

Source System	Target System
Identity Authentication (SCIM API version 2)	Identity Authentication (SCIM API version 2)
Identity Authentication (SCIM API version 2)	Local Identity Directory
Local Identity Directory	Identity Authentication (SCIM API version 2)

The Identity Provisioning will search and try to resolve the groups by the attribute values of `displayName`, `applicationId`, and `type`. To ensure successful provisioning, you must define value mappings for the attribute `applicationId` in the transformation mappings of the source or the target system. You should define to which `applicationId` in the target system will be mapped the `applicationId` read from the source system.

For example:

```
{
"sourcePaths": [
    "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId']"
],
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId']",
"valueMappings": [
    {
        "key": [
            "<your_source_system_applicationId>"
        ],
        "mappedValue": "<your_target_system_applicationId>"
    }
    ],
    "condition": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId'] EMPTY false",
    "type": "valueMapping",
    "defaultValue": ""
}
```

For more information, see Transformation Expressions [page 391] → section **valueMapping**.

You can change the default transformation mapping rules to reflect your current setup of entities in your Identity Authentication system. For more information, see: Manage Transformations [page 2439]

SCIM API version 1: Identity Authentication: SCIM REST API

SCIM API version 2: Identity Directory SCIM API

> **Note**
>
> When a user is deleted from the Identity Authentication, the deletion status is considered by it during the read processes. Depending on the off-boarding user handling in the target system, a user can be deleted, or can be set to *inactive*.

Default transformation for SCIM API version 1:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['userId']"
            },
            {
                "sourcePath": "$.userUuid",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "optional": true,
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName",
                "optional": true
            },
            {
                "sourcePath": "$.name.middleName",
                "targetPath": "$.name.middleName",
                "optional": true
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName",
                "optional": true
            },
            {
                "sourcePath": "$.name.honorificPrefix",
                "targetPath": "$.name.honorificPrefix",
                "optional": true
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails"
            },
            {
                "sourcePath": "$.emails[?(@.primary== true)].value",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true,
                "defaultValue": true
            },
            {
                "sourcePath": "$.userType",
```

```
                "targetPath": "$.userType",
                "optional": true
            },
            {
                "sourcePath": "$.addresses",
                "targetPath": "$.addresses",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.locale",
                "targetPath": "$.locale",
                "optional": true
            },
            {
                "sourcePath": "$.phoneNumbers",
                "targetPath": "$.phoneNumbers",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.timeZone",
                "targetPath": "$.timezone",
                "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
            },
            {
                "sourcePath": "$.sourceSystemId",
                "targetPath": "$.sourceSystemId",
                "ignore": true
            },
            {
                "sourcePath": "$.groups",
                "targetPath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "condition": "$.displayName EMPTY true",
                "type": "remove",
                "targetPath": "$.displayName"
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "optional": true
            },
            {
```

```
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']",
                "optional": true
            },
            {
                "sourcePath": "$.company",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "optional": true
            }
        ]
    },

    // By default, the group mapping is inactive (ignored) but groups are
supported.
    // To start reading groups, either delete the statement "ignore":
true, or set its value to false.
```

```
    "group": {
        "ignore": true,
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
                "targetPath": "$.displayName"
            },
            {
                "optional": true,
                "preserveArrayWithSingleElement": true,
                "sourcePath": "$.members",
                "targetPath": "$.members"
            },
            {
                "ignore": true,
                "constant":
"urn:sap:cloud:scim:schemas:extension:custom:2.0:Group",
                "targetPath": "$.schemas[1]"
            },
            {
                "ignore": true,
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']"
            },
            {
                "ignore": true,
                "optional": true,
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']"
            }
        ]
    }
}
```

Default transformation for SCIM API version 2:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['userId']"
            },
            {
```

```
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userId']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userId']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "optional": true,
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName",
                "optional": true
            },
            {
                "sourcePath": "$.name.middleName",
                "targetPath": "$.name.middleName",
                "optional": true
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName",
                "optional": true
            },
            {
                "sourcePath": "$.name.honorificPrefix",
                "targetPath": "$.name.honorificPrefix",
                "optional": true
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails"
            },
            {
                "sourcePath": "$.emails[?(@.primary== true)].value",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.userType",
                "targetPath": "$.userType",
                "optional": true
            },
            {
                "sourcePath": "$.addresses",
                "targetPath": "$.addresses",
                "preserveArrayWithSingleElement": true,
                "optional": true
```

```
            },
            {
                "sourcePath": "$.locale",
                "targetPath": "$.locale",
                "optional": true
            },
            {
                "sourcePath": "$.phoneNumbers",
                "targetPath": "$.phoneNumbers",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.timezone",
                "targetPath": "$.timezone",
                "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
            },
            {
                "sourcePath": "$.groups",
                "targetPath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "condition": "$.displayName EMPTY true",
                "type": "remove",
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystemId']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystemId']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
```

```
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']",
                "optional": true
```

```
            },
            {
                "sourcePath": "$.company",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['workAssignment']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['workAssignment']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['jobDetails']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['jobDetails']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['workforcePersonId']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['workforcePersonId']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['jobClassification']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['jobClassification']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['organizationalUnit']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['organizationalUnit']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['costCenter']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['costCenter']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
```

```
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['plant']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['plant']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['companyCode']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['companyCode']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['generalLedgerAccount']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['generalLedgerAccount']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingOrganization']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingOrganization']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingGroup']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingGroup']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName"
            },
```

```
            {
                "sourcePath": "$.members",
                "targetPath": "$.members",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']",
                "optional": true
            }
        ]
    }
}
```

6. Add a target system where to provision users and groups. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Identity Authentication: Documentation
Identity Authentication: SCIM REST API

1.6.1.2 Intelligent Opportunity Analyzer

Follow this procedure to set up intelligent opportunity analyzer as a source system.

Prerequisites

You have created an instance and generated a service key for the *scim* service plan of *Intelligent Opportunity Analyzer*. For more information, see Assigning the Entitlements for Intelligent Opportunity Analyzer to Your Subaccounts.

Context

Intelligent opportunity analyzer is a service that comes integrated with SAP Ariba Category Management for identifying and managing opportunities that lead to improving the performance of a purchasing category.

You can use Identity Provisioning to configure intelligent opportunity analyzer as a source system where you can read users from and provision them to a target system.

> **Note**
>
> Intelligent opportunity analyzer does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *Intelligent Opportunity Analyzer* as a source system. See: Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL provided by the service key under the *scim-v1* field without adding the path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*

Property Name	Value
`User`	Enter the value from the *clientid* field of the service key.
`Password`	(Credential) Enter the value from the *clientsecret* field of the service key.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. This is the value from the *url* field of the service key. Ensure to append the value `/oauth/token?grant_type=client_credentials`.
(Optional) `ioa.user.filter`	When specified, only those intelligent opportunnity analyzer users matching the filter expression will be read. For example: • *userName eq "SmithJ"* • *emails.value eq "john.doe@example.com"*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Intelligent Opportunity Analyzer* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your system. For more information, see:

   Manage Transformations [page 2439]

   Mapping logic – the behavior of the default transformation logic is to read all user attributes from the source intelligent opportunity analyzer system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

   > ### Note
   >
   > The documentation for the Intelligent Opportunity Analyzer API and its integration with Identity Provisioning is still under development. Refer to the Intelligent Opportunity Analyzer Setup for updates.

   Default transformation:

   ### Code Syntax

   ```
   {
     "user":{
       "mappings":[
         {
           "sourcePath":"$.id",
           "targetVariable":"entityIdSourceSystem"
         },
         {
   ```

```
        "sourcePath":"$.schemas",
        "targetPath":"$.schemas",
        "preserveArrayWithSingleElement":true
      },
      {
        "sourcePath":"$.userName",
        "targetPath":"$.userName",
        "correlationAttribute":true
      },
      {
        "sourcePath":"$.loginName",
        "targetPath":"$.loginName",
        "optional":true
      },
      {
        "sourcePath":"$.name.givenName",
        "targetPath":"$.name.givenName",
        "optional":true
      },
      {
        "sourcePath":"$.name.familyName",
        "targetPath":"$.name.familyName",
        "optional":true
      },
      {
        "sourcePath":"$.name.formatted",
        "targetPath":"$.name.formatted",
        "optional":true
      },
      {
        "sourcePath":"$.name.middleName",
        "targetPath":"$.name.middleName",
        "optional":true
      },
      {
        "sourcePath":"$.name.honorificPrefix",
        "targetPath":"$.name.honorificPrefix",
        "optional":true
      },
      {
        "sourcePath":"$.name.honorificSuffix",
        "targetPath":"$.name.honorificSuffix",
        "optional":true
      },
      {
        "sourcePath":"$.emails",
        "targetPath":"$.emails",
        "preserveArrayWithSingleElement":true
      },
      {
        "sourcePath":"$.emails[?(@.primary== true)].value",
        "correlationAttribute":true
      },
      {
        "sourcePath":"$.active",
        "targetPath":"$.active"
      },
      {
        "sourcePath":"$.displayName",
        "targetPath":"$.displayName",
        "optional":true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
```

```
        }
      ]
    }
}
```

5. Add a target system to provision users to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.3 Local Identity Directory

Follow this procedure to set up Local Identity Directory as a source system.

Prerequisites

> **Note**
>
> The *Local Identity Directory* connector is available for both bundle and standalone tenants running on SAP Cloud Identity Services infrastructure.

Context

Identity directory is the user store of SAP Cloud Identity Services. It provides a central place for storing and managing users, groups and custom schemas through the System for Cross-domain Identity Management 2.0 REST API, in short Identity Directory SCIM API.

You can read users and groups from the identity directory and provision them to various target systems by configuring the Local Identity Directory connector as a source system in the SAP Cloud Identity Services administration console. The entities are read from the identity directory of your current SAP Cloud Identity Services tenant. Therefore, you don't have to set up connectivity and authentication properties for the source system. In case you want to read entities from the identity directory in another tenant, add Identity Authentication (version 2) as a source system and configure the respective connectivity and authentication properties.

To create Local Identity Directory as a source system, proceed as follows:

Procedure

1. Sign in to SAP Cloud Identity Services administration console and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *Local Identity Directory* as a source system.

   For more information, see Add New Systems [page 2415].
3. **Optional:** If needed, choose the *Properties* tab to configure filtering and paging.

   Optional Properties

Property Name	Value
`idds.user.filter`	This property filters users by particular attributes. You can set a single attribute or multiple ones as search criteria. • Value pattern for single attribute: *<user_attribute> eq "<value>"* • Value pattern for multiple attributes: *<user_attribute1> eq "<value1>" and/or <user_attribute2> eq "<value2>"*
`idds.group.filter`	This property filters groups by display name. You can set a single display name or multiple ones as filter criteria. If you enter multiple display names (using **OR** operator), the filter will search for any of them. • Value pattern for single attribute: *displayName eq "<group_name>"* • Value pattern for multiple attributes: *displayName eq "<group_name1>" or displayName eq "<group_name2>"*
`idds.user.groups.paging.enabled`	This property enables paging of user's groups. The maximum number of user's groups returned per request is 1000. To read more than 1000 user's groups, paging must be enabled.
`idds.group.members.paging.enabled`	This property enables paging of group members. The maximum number of group members returned per request is 20 000. To read more than 20 000 group members, paging must be enabled.

   Local Identity Directory properties are prefixed with `idds.<property_name>`. For more information, see List of Properties [page 80].
4. **Optional:** Configure transformations.

Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the Local Identity Directory source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

When Local Identity Directory is configured as a source system, the default transformation logic reads the user attributes from the user store of SAP Cloud Identity Services. The logic is provided by the Identity Directory SCIM API, which then maps the attributes to the internal SCIM representation.

Handling Application-Specific Groups

In case you want to provision application-specific groups between the following pairs of provisioning systems:

Source System	Target System
Identity Authentication (SCIM API version 2)	Identity Authentication (SCIM API version 2)
Identity Authentication (SCIM API version 2)	Local Identity Directory
Local Identity Directory	Identity Authentication (SCIM API version 2)

The Identity Provisioning will search and try to resolve the groups by the attribute values of `displayName`, `applicationId`, and `type`. To ensure successful provisioning, you must define value mappings for the attribute `applicationId` in the transformation mappings of the source or the target system. You should define to which `applicationId` in the target system will be mapped the `applicationId` read from the source system.

For example:

```
{
"sourcePaths": [
    "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId']"
],
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId']",
"valueMappings": [
    {
        "key": [
            "<your_source_system_applicationId>"
        ],
        "mappedValue": "<your_target_system_applicationId>"
    }
    ],
    "condition": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId'] EMPTY false",
    "type": "valueMapping",
    "defaultValue": ""
}
```

For more information, see Transformation Expressions [page 391] → section **valueMapping**.

You can change the default transformation mapping rules to reflect your current setup of entities in your Local Identity Directory. For more information, see:

- Manage Transformations [page 2439]
- SCIM API version 2: Identity Directory SCIM API

Default transformation for Local Identity Directory:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['userId']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userId']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userId']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "optional": true,
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName",
                "optional": true
            },
            {
                "sourcePath": "$.name.middleName",
                "targetPath": "$.name.middleName",
                "optional": true
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName",
                "optional": true
            },
            {
                "sourcePath": "$.name.honorificPrefix",
                "targetPath": "$.name.honorificPrefix",
                "optional": true
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails"
            },
            {
                "sourcePath": "$.emails[?(@.primary== true)].value",
                "correlationAttribute": true
```

```
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.userType",
                "targetPath": "$.userType",
                "optional": true
            },
            {
                "sourcePath": "$.addresses",
                "targetPath": "$.addresses",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.locale",
                "targetPath": "$.locale",
                "optional": true
            },
            {
                "sourcePath": "$.phoneNumbers",
                "targetPath": "$.phoneNumbers",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.timezone",
                "targetPath": "$.timezone",
                "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
            },
            {
                "sourcePath": "$.groups",
                "targetPath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "condition": "$.displayName EMPTY true",
                "type": "remove",
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
```

```
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystemId']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystemId']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "optional": true
            },
```

```
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']",
                "optional": true
            },
            {
                "sourcePath": "$.company",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['workAssignment']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['workAssignment']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['jobDetails']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['jobDetails']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['workforcePersonId']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['workforcePersonId']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['jobClassification']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['jobClassification']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['organizationalUnit']",
```

```
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['organizationalUnit']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['costCenter']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['costCenter']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['plant']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['plant']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['companyCode']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['companyCode']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['generalLedgerAccount']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['generalLedgerAccount']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingOrganization']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingOrganization']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingGroup']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingGroup']",
                "preserveArrayWithSingleElement": true,
                "optional": true
            }
        ]
    },
    "group": {
```

```
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.members",
                "targetPath": "$.members",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']",
                "optional": true
            }
        ]
    }
}
```

5. Add a target system where to provision users and groups. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].

- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.4 SAP Advanced Financial Closing

Follow this procedure to set up a SAP Advanced Financial Closing as a source system.

Prerequisites

You have created an instance and generated a service key for the standard service plan of SAP Advanced Financial Closing. For more information, see: How to Manage User Access Using the SCIM API Provided.

The service key contains the API URL and the OAuth credentials (`clientid` and `clientsecret`) under the `uaa` property.

Context

SAP Advanced Financial Closing allows you to define, automate, process, and monitor the financial closing tasks for the entities of your organization. It is an SAP BTP application that runs in an SAP BTP subaccount.

You can use Identity Provisioning to configure SAP Advanced Financial Closing as a source system where you can read users, user groups and user roles from, and provision them to a target system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Advanced Financial Closing* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
Type	Enter: *HTTP*

Property Name	Value
`URL`	Enter the URL provided by the service key under *endpoints* > *scim2* without adding the path information. For example: `https://afc-production-afc-api.cfapps.eu10.hana.ondemand.com`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the user ID provided by the service key under *uaa* > *clientid*.
`Password`	(Credential) Enter the password provided by the service key under *uaa* > *clientsecret*.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL provided by the service key of your SAP Advanced Financial Closing instance. It follows the pattern: **`<uaa.url>/oauth/token`**, where: • `<uaa.url>` is the URL provided by the service key under *uaa* > *url*. • **`/oauth/token`** is the suffix you need to add.
(Optional) `s4hana.afc.user.filter`	When specified, only those SAP Advanced Financial Closing users matching the filter expression will be read. Supported operators: **`eq`** (equal), **`sw`** (starts with) and **`co`** (contains) For example: • *userName eq "Julie Armstrong"* • *emails eq "julie.armstrong@example.com"* • *name.familyName sw "A"* • *name.givenName co "Ju"*
(Optional) `s4hana.afc.group.filter`	When specified, only those SAP Advanced Financial Closing groups matching the filter expression will be read. Supported operators: **`eq`** (equal), **`sw`** (starts with) and **`co`** (contains) For example: *displayName eq "Administrators"*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the SAP Advanced Financial Closing source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Advanced Financial Closing. For more information, see Manage Transformations [page 2439].

How to Manage User Access Using the SCIM API Provided

SAP Business Accelerator Hub: SAP Advanced Financial Closing

> ### Note
>
> When configuring SAP Advanced Financial Closing as a source system, there may be a mismatch between optional user attributes in the source system and required user attributes in the target system. For example, the email is an optional attribute in SAP Advanced Financial Closing but a required one in Identity Authentication. In such cases, you need to review and adapt your source and/or target system transformations and configurations to reflect your needs.

Default transformation:

> ### Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.name.formatted",
        "targetPath": "$.name.formatted",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
```

```
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "functions": [
          {
            "condition": "'%s4hana.afc.group.prefix%' !== 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "applyOnAttribute": "display",
            "prefix": "%s4hana.afc.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.roles",
        "targetPath": "$.roles",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.phoneNumbers",
        "targetPath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%s4hana.afc.group.prefix%' !== 'null'",
```

```
            "function": "concatString",
            "prefix": "%s4hana.afc.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "functions": [
          {
            "condition": "'%s4hana.afc.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%s4hana.afc.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      }
    ]
  }
}
```

5. Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].

- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.5 SAP Advanced Workflow

Follow this procedure to set up a source connector for SAP Advanced Workflow.

Prerequisites

- You have technical credentials for SAP Advanced Workflow. Note that SAP Advanced Workflow is available to SAP SuccessFactors Incentive Management customers as an optional add-on. You create an Admin user in SAP SuccessFactors Incentive Management, which is synchronized with SAP Advanced Workflow. For more information, see: Adding an Admin User and Commissions User Synchronization.
- You have set up SSO between Identity Authentication and SAP Advanced Workflow. For more information, see Integration with SAP IdP.

Context

SAP Advanced Workflow enables you to analyse, organize, and execute business processes to connect people, data, and daily activities. Workflow provides you the tools you need to configure and customize your business processes based on your specific business needs.

After fulfilling the prerequisites, follow the procedure below to add a source system for SAP Advanced Workflow where you can read **users** from and provision them to a target system. This source system consumes Workflow SCIM API.

> **Note**
>
> SAP Advanced Workflow does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Advanced Workflow* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the API of your SAP Advanced Workflow system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the user for your SAP Advanced Workflow system.
`Password`	(Credential) Enter the password for your SAP Advanced Workflow user.
`awf.domain`	The domain name is the name of your SAP Advanced Workflow tenant. If you don't know your tenant name, contact your supervisor or administrator, or refer to the email notification you received when your account was created.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Advanced Workflow* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SCIM system. For more information, see:

   Manage Transformations [page 2439]

   The behavior of the default transformation logic is to read all user attributes from the source SAP Advanced Workflow system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

   Default transformation:

   Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.schemas",
                "targetPath": "$.schemas",
                "preserveArrayWithSingleElement": true
            },
            {
```

```
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.name.middleName",
                "optional": true,
                "targetPath": "$.name.middleName"
            },
            {
                "sourcePath": "$.name.familyName",
                "optional": true,
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.phoneNumbers[0].value",
                "optional": true,
                "targetPath": "$.phoneNumbers[0].value"
            },
            {
                "sourcePath": "$.profileUrl",
                "targetPath": "$.profileUrl",
                "optional": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true
            },
            {
                "sourcePath": "$.emails[0].value",
                "optional": true,
                "targetPath": "$.emails[0].value"
            },
            {
                "sourcePath": "$.timezone",
                "targetPath": "$.timezone",
                "optional": true
            }
        ]
    }
}
```

5. Now, add a target system to provision users and their group assignments to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.6 SAP Analytics Cloud

Follow this procedure to set up SAP Analytics Cloud as a source system.

Prerequisites

- In SAP Analytics Cloud, you have enabled a custom SAML Identity Provider, for which *User Attribute* is set to **Custom SAML User Mapping**. To learn how, see: Enabling a Custom SAML Identity Provider
- Add an OAuth client with authorization grant **Client Credentials**. To learn how, see: Managing OAuth Clients and Trusted Identity Providers

Context

SAP Analytics Cloud is an all-in-one cloud product offered as software as a service for business intelligence, planning, and predictive analytics.

You can use Identity Provisioning to configure SAP Analytics Cloud as a source system where you can read users and groups and provision them to target systems of your choice. The source system consumes SCIM 2.0 API provided by SAP Analytics Cloud.

There are two versions of the SAP Analytics Cloud SCIM API. They are handled by the `sac.api.version` property as follows:

- When the value is set to **1** or the property is not defined (typical for systems created before versioning was introduced on April 10, 2023), SAP Analytics Cloud SCIM API version 1 is used. This is the default value.
- When the value is set to **2** - SAP Analytics Cloud SCIM API version 2 is used. This version is released with enhancements, such as: reading groups is not ignored in the default read transformation.

For more information on the differences between SAP Analytics Cloud SCIM API version 1 and 2, see Managing Users and Teams.

For more information on how to update to version 2, see Update Connector Version [page 2424].

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Analytics Cloud* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP Analytics Cloud system without adding the path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the client ID to retrieve the OAuth access token for SAP Analytics Cloud.
`Password`	(Credential) Enter the client secret to retrieve the OAuth access token for SAP Analytics Cloud.
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Analytics Cloud instance. This token URL is listed in the *OAuth Clients* section of the *App Integration* page. For more information, refer to *Authorize API Access for OAuth Clients* in Manage OAuth Clients
(Optional) `sac.api.version`	Handles the version of SAP Analytics Cloud SCIM API. **Possible values:** • *1* - Indicates that SAP Analytics Cloud SCIM API version 1 is used. • *2* - Indicates that SAP Analytics Cloud SCIM API version 2 is used. Default value: *1*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Analytic Cloud* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Analytic Cloud system. For more information, see: Manage Transformations [page 2439]

   SAP Analytics Cloud REST API: Managing Users and Teams

   Managing Users and Teams → api/v1/scim

   Managing Users and Teams → api/v1/scim2

   Default transformation for SAP Analytic Cloud SCIM API version 1:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "targetPath": "$.schemas",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
              "sourcePath": "$.name.givenName",
              "targetPath": "$.name.givenName",
              "optional": true
            },
            {
              "sourcePath": "$.name.middleName",
              "targetPath": "$.name.middleName",
              "optional": true
            },
            {
              "sourcePath": "$.name.familyName",
              "targetPath": "$.name.familyName",
              "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.emails[?(@.primary== true)].value",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.roles",
                "targetPath": "$.roles",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.groups",
                "targetPath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "functions": [
                    {
                        "condition": "'%sac.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "applyOnElements": true,
                        "applyOnAttribute": "display",
                        "prefix": "%sac.group.prefix%"
```

```
                    }
                ]
            },
            {
                "sourcePath": "$
['urn:scim:schemas:extension:enterprise:1.0']['manager']['managerId']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']"
            }
        ]
    },
    "group": {
        "ignore": true,
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "targetPath": "$.schemas",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "'%sac.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "prefix": "%sac.group.prefix%"
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "sourcePath": "$.members",
                "targetPath": "$.members",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.roles",
                "targetPath": "$.roles",
                "preserveArrayWithSingleElement": true
            },
            {   "condition": "'%ips.application.id%' !== 'null'",
                "constant": "%ips.application.id%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
```

```
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            }
        ]
    }
}
```

Default transformation for SAP Analytic Cloud SCIM API version 2:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "targetPath": "$.schemas",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
              "sourcePath": "$.name.givenName",
              "targetPath": "$.name.givenName",
              "optional": true
            },
            {
              "sourcePath": "$.name.middleName",
              "targetPath": "$.name.middleName",
              "optional": true
            },
            {
              "sourcePath": "$.name.familyName",
              "targetPath": "$.name.familyName",
              "optional": true
            },
            {
                "sourcePath": "$.externalId",
                "targetPath": "$.externalId",
                "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "preserveArrayWithSingleElement": true
            },
            {
```

```
                "sourcePath": "$.emails[?(@.primary== true)].value",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:user-custom-parameters']",
                "targetPath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:user-custom-parameters']"
            },
            {
                "sourcePath": "$.roles",
                "targetPath": "$.roles",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath": "$.groups",
                "targetPath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "functions": [
                    {
                        "condition": "'%sac.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "applyOnElements": true,
                        "applyOnAttribute": "display",
                        "prefix": "%sac.group.prefix%"
                    }
                ]
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "targetPath": "$.schemas",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "'%sac.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "prefix": "%sac.group.prefix%"
                    }
```

```
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "sourcePath": "$.members",
                "targetPath": "$.members",
                "optional": true,
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:group-roles']",
                "targetPath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:group-roles']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:group-custom-parameters']",
                "targetPath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:group-custom-parameters']",
                "optional": true
            },
            {   "condition": "'%ips.application.id%' !== 'null'",
                "constant": "%ips.application.id%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            }
        ]
    }
}
```

5. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.7 SAP Application Server ABAP

Follow this procedure to set up SAP Application Server ABAP (AS ABAP) as a source system.

Prerequisites

- You have installed the Cloud Connector in your corporate environment and have done the initial configuration. For more information, see: Cloud Connector.
  For more information on how to make a connection to this on-premise system, follow the procedure in Connect to On-Premise Systems [page 2468].
- You have credentials of a technical user with read permissions for AS ABAP, which plays the role of a user data source. Via this user, the Identity Provisioning service will call the ABAP public API in order to execute a number of function modules. These function modules are listed in **step 1** from the procedure below.
- The technical user must have the required RFC and security authorizations, as described in SAP Note 3666721.

> **Note**
>
> If the technical user has insufficient security authorization, no users or roles can be read.

Context

SAP Application Server ABAP (AS ABAP) offers a user store and user administration capabilities for maintaining users and their authorizations for AS ABAP applications. You can configure AS ABAP as a source system for your identity provisioning process, in the following cases:

- Use AS ABAP as a central store for the identity data of your business users.
- Reuse the permission model, implemented in your AS ABAP client, as a permission model for cloud applications. For example, you can provision roles and permission assignments to SAP BTP.

> **Note**
>
> During a provisioning job (*Read* or *Resync*), only active ABAP users are read. That means, users that have been created before the job is started, and whose expiration date is after the end of the job.

Procedure

1. Open the Cloud Connector to add an access control system mapping for **AS ABAP**. This is needed to allow the Identity Provisioning service to access AS ABAP as a back-end system on the intranet. To learn how, see: Configure Access Control (RFC)

Go to *Cloud To On-Premise* > *Access Control* > tab and select protocol *RFC SNC*. Then, expose the following *exact names* as accessible resources:

- `PRGN_ROLE_GETLIST`
- `BAPI_USER_GETLIST`
- `BAPI_USER_GET_DETAIL`
- `BAPI_USER_CREATE1`
- `BAPI_USER_ACTGROUPS_ASSIGN`
- `IDENTITY_MODIFY`
- `BAPI_USER_DELETE`
- `PRGN_ACTIVITY_GROUPS_LOAD_RFC`
- `SUSR_USER_GET_DETAIL_MASS` - available as of `SAP_BASIS 7.58`. For more information, see SAP Note 3631587 and 3677459 .

These are Business Application Programming Interface (BAPI) functional modules designed to perform certain tasks in the SAP AS ABAP system, such as retrieving, creating, updating or deleting user data.

For more information about each function module and its parameters, refer to the documentation provided in the SAP system. Logon to your SAP system and run transaction code **SE37**. In the *Function Builder: Initial Screen*, enter the function name, choose *Display* and then *Functional Module Documentation*.

2. Open SAP BTP cockpit, and in your Identity Provisioning subaccount create a destination for the AS ABAP system. To learn how, see: Create RFC Destinations

   The destination configuration is required by the Identity Provisioning service to find the back-end system to be used for reading data. It also provides the credentials of the technical user, needed for the connection to the ABAP public API.

   Below are the fields you have to fill in the cockpit destination before using an AS ABAP client as a source system.

   > ### Note
   >
   > In the RFC destination, you can set only JCo properties. Properties starting with `abap` are connector specific and must be set in the Identity Provisioning UI.

Field/Property Name	Value
Name	Enter a destination name.
Type	Select *RFC*.
User	Enter the user for AS ABAP. The *User* field corresponds to property `jco.client.user` in the exported RFC destination.
Password	(Credential) Enter the password for the AS ABAP user. The *Password* field corresponds to property `jco.client.passwd` in the exported RFC destination.

Field/Property Name	Value
jco.client.client	The property is configured during destination creation in SAP BTP cockpit. Provide the client to be used in the ABAP system. Valid format is a three-digit number.
Proxy Type	Defines the proxy type of the connection you need to provide for your ABAP system. The value is set during destination creation in SAP BTP cockpit. The proxy type *OnPremise* requires the Cloud Connector to access resources within your on-premise network. Enter: *OnPremise*
Direct Connection	
jco.client.ashost	Provide the virtual host entry that you have configured in the Cloud Connector → *Access Control* configuration. At least one property specifying the target host must be set during destination creation.
jco.client.sysnr	Provide the "system number" of the ABAP system.
Load Balancing Connection	
jco.client.mshost	Represents the message server host to be used. At least one property specifying the target host must be set during destination creation.
jco.client.r3name	Provide the three-character system ID of the ABAP system to be addressed.
jco.client.msserv	Provide the port on which the message server is listening for incoming requests. You can use this property as an alternative to `jco.client.r3name`.
Optional Properties	
`jco.destination.peak_limit`	The value represents the maximum number of active connections that can simultaneously be created for a destination. For example: *10*
`jco.destination.pool_capacity`	The value represents the maximum number of idle connections kept open by the destination. For example: *5*

Field/Property Name	Value
`abap.user.name.filter`	Filters user names by a regular expression. The regex can define any kind of search pattern. For example, `abap.user.name.filter` = **^MAR.*** reads all user names that start with *MAR*, such as **MARK**, **MARTINA**, and so on. **Note** This property has a higher priority over `abap.user.filter`. That means, if you set both properties in a system, the value of **abap.user.name.filter** will be used. However, if the value of `abap.user.name.filter` is empty, then **abap.user.filter**'s value will be used instead.
`abap.role.name.filter`	Filters user roles by a regular expression. The regex can define any kind of search pattern. For example, `abap.role.name.filter` = **^inter.*** reads all roles which start with *inter*, such as **internal**, **internship**, and so on. **Note** This property has a higher priority over `abap.role.filter`. That means, if you set both properties in a system, the value of **abap.role.name.filter** will be used. However, if the value of `abap.role.name.filter` is empty, then **abap.role.filter**'s value will be used instead.
`abap.user.membership.filter`	Filters users by a regular expression, based on their *Role* memberships in AS ABAP. The regex can define any kind of search pattern. For example, `abap.user.membership.filter` = **(?i)^new.*** reads all users who have an assigned role which starts with *new*. This regex is case insensitive, which means the result can be roles starting with **new**, or **New**, or **NEW**, and so on.

Field/Property Name	Value
`abap.user.page.size`	Use this property to configure the number of users to be read from SAP Application Server ABAP at once. If the property is not set, the function module `SUSR_USER_GET_DETAIL_MASS` **cannot** be used, even if it is available on the ABAP server. **Note** The property can be used only when the function module `SUSR_USER_GET_DETAIL_MASS` is exposed as an accessible resource and is present on the ABAP server. Even when these conditions are fulfilled, if the property is not set, the function module **cannot** be used. For more information, see SAP Note 3631587. **Possible values**: Default value: *100* The maximum allowed number is 100.

Note

If connection properties, like `User` and `Password`, are configured both in the destination (SAP BTP cockpit) and on the *Properties* tab (Identity Provisioning User Interface), the values set in the destination are considered with higher priority.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

3. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
4. Add *SAP Application Server ABAP* as a source system. To learn how, see: Add New Systems [page 2415]
5. From the *Destination Name* dropdown, choose the RFC destination you have created in *step 2*.
6. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Application Server ABAP* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in AS ABAP. For more information, see Manage Transformations [page 2439].

> ### Note
>
> The value of the **entityIdSourceSystem** attribute stores the unique ID of the entity. Do not delete or change this statement! You can exchange the default attribute **USERNAME** (which is used as a source) with another one, but make sure it is unique.

When AS ABAP is configured as a source system for the Identity Provisioning service, the ABAP public API is used to retrieve the identity data from the AS ABAP system. During the reading process, the JSON data generated by the Identity Provisioning service, is following the structure of the API export parameters list and tables. Every BAPI table is represented as a JSON array and every BAPI structure is represented as a child JSON object.

Default transformation:

> ### Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.USERNAME",
                "targetVariable": "entityIdSourceSystem"
            },

// The USERNAME attribute is also used as userName value for the internal
JSON representation.
            {
                "sourcePath": "$.USERNAME",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.ALIAS.USERALIAS",
                "optional": true,
                "targetPath": "$.externalId",
                "correlationAttribute": true
            },

// The constant urn:ietf:params:scim:schemas:core:2.0:User is required as
a value for the
// schemas definition in the Identity Authentication SCIM REST API.
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
                "targetPath": "$.schemas[0]"
            },

// The ADDRESS.E_MAIL attribute is used also as a first array value in the
emails JSON array.
            {
                "sourcePath": "$.ADDRESS.E_MAIL",
                "optional": true,
                "targetPath": "$.emails[0].value",
                "correlationAttribute": true
            },
            {
                "condition": "$.ADDRESS.E_MAIL EMPTY false",
                "constant": true,
                "targetPath": "$.emails[0].primary"
            },
            {
                "condition": "$.ADDRESS.E_MAIL EMPTY false",
                "constant": "work",
                "targetPath": "$.emails[0].type"
```

```
            },

            {
                "sourcePath": "$.ADDRESS.FIRSTNAME",
                "optional": true,
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.ADDRESS.LASTNAME",
                "optional": true,
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.ADDRESS.MIDDLENAME",
                "optional": true,
                "targetPath": "$.name.middleName"
            },
            {
                "sourcePath": "$.ADDRESS.NICKNAME",
                "optional": true,
                "targetPath": "$.nickName"
            },
            {
                "sourcePath": "$.ADDRESS.TITLE_P",
                "optional": true,
                "targetPath": "$.name.honorificPrefix"
            },
            {
                "sourcePath": "$.ADDRESS.COUNTRY",
                "optional": true,
                "targetPath": "$.addresses[0].country"
            },
            {
                "condition": "$.ADDRESS.COUNTRY EMPTY false",
                "constant": true,
                "targetPath": "$.addresses[0].primary"
            },
            {
                "condition": "$.ADDRESS.COUNTRY EMPTY false",
                "constant": "work",
                "targetPath": "$.addresses[0].type"
            },
            {
                "sourcePath": "$.ADDRESS.TEL1_NUMBR",
                "optional": true,
                "targetPath": "$.phoneNumbers[0].value"
            },
            {
                "condition": "$.ADDRESS.TEL1_NUMBR EMPTY false",
                "constant": true,
                "targetPath": "$.phoneNumbers[0].primary"
            },
            {
                "condition": "$.ADDRESS.TEL1_NUMBR EMPTY false",
                "constant": "work",
                "targetPath": "$.phoneNumbers[0].type"
            },
// The Identity Provisioning reads the specific ABAP language codes and
mapped them as locales in the target system.
// The transformation provides an example with key = "W", which in the
target system is mapped as "bg". The default language is en.
// To see all languages and codes supported by AS ABAP, see the Related
Information section below.
            {
                "optional": true,
                "targetPath": "$.locale",
                "type": "valueMapping",
                "sourcePaths": [
```

```
                    "$.DEFAULTS.LANGU"
                ],
                "defaultValue": "en",
                "valueMappings": [
                    {
                        "key": [
                            "W"
                        ],
                        "mappedValue": "bg"
                    }
                ]
            },
            {
        "sourcePath": "$.ADDRESS.LANGUP_ISO",
        "optional": true,
        "targetPath": "$.preferredLanguage",
        "functions": [
          {
            "function": "toLowerCaseString"
          }
        ]
      },
// The Identity Provisioning reads standard timezone codes, which are
supported by the AS ABAP BAPI.
// However, the standard SCIM API does not support these codes, thus the
target system can only accept values in format "<region>/<city>".
// The transformation provides an example with key = "EET", which in the
target system is mapped as "Europe/Sofia". The default timezone is Berlin.
            {
                "optional": true,
                "targetPath": "$.timezone",
                "type": "valueMapping",
                "sourcePaths": [
                    "$.LOGONDATA.TZONE"
                ],
                "defaultValue": "Europe/Berlin",
                "valueMappings": [
                    {
                        "key": [
                            "EET"
                        ],
                        "mappedValue": "Europe/Sofia"
                    }
                ]
            },
            {
                "constant": false,
                "targetPath": "$.active"
            },
            {
                "condition": "($.ISLOCKED.LOCAL_LOCK != 'L') &&
($.ISLOCKED.NO_USER_PW != 'L') && ($.ISLOCKED.GLOB_LOCK != 'L') &&
($.ISLOCKED.WRNG_LOGON != 'L')",
                "constant": true,
                "targetPath": "$.active"
            },

// The attribute ACTIVITYGROUPS (SAP ABAP roles) is transformed by default
into groups attribute of the SCIM internal representation.
            {
                "sourcePath": "$.ACTIVITYGROUPS[*].AGR_NAME",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.groups[?(@.value)]"
            }
        ]
     },
    "group": {
```

```
        "mappings": [
            {
                "sourcePath": "$.ROLE_NAME",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.ROLE_NAME",
                "targetPath": "$.displayName"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.USERLIST[*].USERNAME",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]"
            }
        ]
    }
}
}
```

Default transformation supporting User UUID attribute:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.USERNAME",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.USERNAME",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.ALIAS.USERALIAS",
        "optional": true,
        "targetPath": "$.externalId",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.SAPUSER_UUID.SAP_UID",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.ADDRESS.E_MAIL",
        "optional": true,
        "targetPath": "$.emails[0].value",
        "correlationAttribute": true
      },
      {
        "condition": "$.ADDRESS.E_MAIL EMPTY false",
        "constant": true,
```

```
      "targetPath": "$.emails[0].primary"
    },
    {
      "condition": "$.ADDRESS.E_MAIL EMPTY false",
      "constant": "work",
      "targetPath": "$.emails[0].type"
    },
    {
      "sourcePath": "$.ADDRESS.FIRSTNAME",
      "optional": true,
      "targetPath": "$.name.givenName"
    },
    {
      "sourcePath": "$.ADDRESS.LASTNAME",
      "targetPath": "$.name.familyName"
    },
    {
      "sourcePath": "$.ADDRESS.MIDDLENAME",
      "optional": true,
      "targetPath": "$.name.middleName"
    },
    {
      "sourcePath": "$.ADDRESS.NICKNAME",
      "optional": true,
      "targetPath": "$.nickName"
    },
    {
      "sourcePath": "$.ADDRESS.TITLE_P",
      "optional": true,
      "targetPath": "$.name.honorificPrefix"
    },
    {
      "sourcePath": "$.ADDRESS.COUNTRY",
      "optional": true,
      "targetPath": "$.addresses[0].country"
    },
    {
      "condition": "$.ADDRESS.COUNTRY EMPTY false",
      "constant": true,
      "targetPath": "$.addresses[0].primary"
    },
    {
      "condition": "$.ADDRESS.COUNTRY EMPTY false",
      "constant": "work",
      "targetPath": "$.addresses[0].type"
    },
    {
      "sourcePath": "$.ADDRESS.TEL1_NUMBR",
      "optional": true,
      "targetPath": "$.phoneNumbers[0].value"
    },
    {
      "condition": "$.ADDRESS.TEL1_NUMBR EMPTY false",
      "constant": true,
      "targetPath": "$.phoneNumbers[0].primary"
    },
    {
      "condition": "$.ADDRESS.TEL1_NUMBR EMPTY false",
      "constant": "work",
      "targetPath": "$.phoneNumbers[0].type"
    },
    {
      "type": "valueMapping",
      "sourcePaths": [
        "$.DEFAULTS.LANGU"
      ],
      "optional": true,
      "targetPath": "$.locale",
```

```
        "defaultValue": "en",
        "valueMappings": [
          {
            "key": [
              "W"
            ],
            "mappedValue": "bg"
          }
        ]
      },
      {
        "sourcePath": "$.ADDRESS.LANGUP_ISO",
        "optional": true,
        "targetPath": "$.preferredLanguage",
        "functions": [
          {
            "function": "toLowerCaseString"
          }
        ]
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.LOGONDATA.TZONE"
        ],
        "optional": true,
        "targetPath": "$.timezone",
        "defaultValue": "Europe/Berlin",
        "valueMappings": [
          {
            "key": [
              "EET"
            ],
            "mappedValue": "Europe/Sofia"
          }
        ]
      },
      {
        "constant": false,
        "targetPath": "$.active"
      },
      {
        "condition": "($.ISLOCKED.LOCAL_LOCK != 'L') &&
($.ISLOCKED.GLOB_LOCK != 'L') && ($.ISLOCKED.WRNG_LOGON != 'L')",
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.ACTIVITYGROUPS[*].AGR_NAME",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups[?(@.value)]"
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.ROLE_NAME",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.ROLE_NAME",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%abap.role.prefix%' !== 'null'",
            "function": "concatString",
```

```
            "prefix": "%abap.role.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.ROLE_NAME",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.USERLIST[*].USERNAME",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.members[?(@.value)]",
        "optional": true
      },
      {  "condition": "'%ips.application.id%' !== 'null'",
         "constant": "%ips.application.id%",
         "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "constant": "authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {
        "constant": "readWrite",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      }
    ]
  }
}
```

7. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Example

How to transform roles, assigned to AS ABAP users, into corporate groups in the Identity Authentication?

The AS ABAP roles are represented as groups in your Identity Authentication tenant. That is, when you configure AS ABAP as a source and Identity Authentication as a target system, the default transformations helps you to use the ABAP roles assignment of the users as source data and to automatically create corporate group assignments for the users in the Identity Authentication. When a user is assigned to one or more AS ABAP roles, the technical names of these roles (their ABAP attribute name is **AGR_NAME**) will become corporate groups value in the Identity Authentication.

1. Transforming source data into the intermediate JSON representation.

The following example demonstrates how the sample roles, read from the AS ABAP system, will become groups in the intermediate JSON data, as a result from the transformation statement:

Data read from AS ABAP user store	Intermediate JSON data

Sample Code
…
"ACTIVITYGROUPS": [
{
"AGR_TEXT": "FICO 03",
"AGR_NAME": "ZFICO_03",
"FROM_DAT": "27.04.2016",
"TO_DAT": "31.12.9999"
},
{
"AGR_TEXT": "CASH 01",
"AGR_NAME": "ZCASH_01",
"FROM_DAT": "16.05.2016",
"TO_DAT": "31.12.9999"
}
]
…
	Sample Code
	…
	"groups":[
	{
	"value": "ZFICO_03"
	},
	{
	"value": “ZCASH_01”
	},
	] …

2. The mapping statement in the default transformation, available when the Identity Authentication service is configured as a target system:

Sample Code

```
{
"sourcePath": "$.groups",
"preserveArrayWithSingleElement": true,
"optional": true,
"targetPath": "$.corporateGroups"
}
```

3. The following example demonstrates how the groups from the intermediate JSON are transformed into corporate groups, using the transformation statement:

Intermediate JSON Data	Transformation output result
Sample Code … "groups":[{ "value": "ZFICO_03" }, { "value": “ZCASH_01” },] …	Sample Code … "corporateGroups":[{ "value": "ZFICO_03" }, { "value": “ZCASH_01” },] …

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

ABAP: Supported Languages and Code Pages

1.6.1.8 SAP Ariba Applications

Follow this procedure to set up SAP Ariba Applications as a source system.

Prerequisites

> **Note**
>
> This topic describes customer-managed configurations for the SAP Ariba Applications source system. For SAP-managed configurations, refer to the SAP-Managed SAP Cloud Identity [page 520] documentation.

Next-Generation SCIM API

If you are using the next-generation SCIM API provided by SAP Ariba Applications:

- The Identity Provisioning service must be configured to use the OAuth 2.0 client-credentials flow to obtain access tokens from the Identity Authentication service for calling the SAP Ariba Applications next generation SCIM API. The required dependency must be defined in the configuration of the consumer application. For more information, see Configure Form-Based OAuth Authentication [page 2457].

Current-Generation SCIM API

If you are using the current-generation SCIM API provided by SAP Ariba Applications:

- You have created a client application on SAP Ariba APIs Portal for your integration with Identity Provisioning. This is needed to enable SCIM API for user and group for Ariba Applications. For more information, see Register an Application in SAP Ariba Developer Portal .

  > **Note**
  >
  > If you don't have an SAP Ariba APIs account, go to the SAP Ariba Developer Portal and follow the instructions for sign up. For more information, see Get an SAP Ariba APIs Account.

To find your *realm* name, log in to your SAP Ariba system. It is embedded in your login URL, as illustrated by the following examples:

- SAP Ariba Buyer example: `https://s1.ariba.com/Buyer/Main/ad/loginPage/...&realm=`*mycompany-t*
- SAP Ariba Sourcing example: `http://`*mycompany*`.sourcing.ariba.com/`

- You have enabled the client application for Identity Provisioning on the SAP Ariba Developer Portal.
- You have generated an OAuth secret for your application. For more information, see Generating the OAuth Secret and Base64 Encoded Client ID and Secret.

To set up your SAP Ariba Applications provisioning system, you'll need to map your SAP Ariba application parameters to the corresponding Identity Provisioning properties. The mapping is as follows:

SAP Ariba	Identity Provisioning	Values
SCIM API URL	`URL`	Examples: • US: **https://openapi.ariba.com** • Europe: **https://eu.openapi.ariba.com** • UAE: **https://mn1.openapi.ariba.com** For more information, see SAP Ariba APIs Portal Discover Strategic Sourcing SCIM API for User and Group Master Data Environment Details.
SAP Ariba OAuth 2.0 Token URL	`OAuth2TokenServiceURL`	Examples: • US: **https://api.ariba.com/v2/oauth/token** • Europe: **https://api-eu.ariba.com/v2/oauth/token** • UAE: **https://api.mn1.ariba.com/v2/oauth/token** For more information, see SAP Ariba APIs Portal Discover Strategic Sourcing SCIM API for User and Group Master Data Environment Details.
OAuth Client ID	`User`	Alphanumeric string Example: **aaaa12345-1111-3333-cccc-1234567890**
OAuth Secret	`Password`	Alphanumeric string Example: **aaaGGG1eee12abcdefGHIJK123lmnopTTT**

SAP Ariba	Identity Provisioning	Values
Application key	`ariba.applications.api.key`	Alphanumeric string Example:**123abc123XYZ000abc123ABC012345**
AN-ID	`ariba.applications.realm.id`	AN<numeric_string> Example: **AN000111222333**

Context

You can read users and groups from SAP Ariba Applications source system and provision them to a target system of your choice.

These source systems can consume either the current-generation SCIM API or the next-generation SCIM API provided by SAP Ariba Applications. Identity Provisioning distinguishes between the two by the API URL pattern (the current SCIM API contains **openapi** in the URL, while the next-generation API does not) and by the `AuthType=Form` property, which must be configured only for the next-generation API.

For more information about the SAP Ariba SCIM API scope of support, see SAP Ariba SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Ariba Applications* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the SCIM API URL for your SAP Ariba application (see the **Prerequisites** section). The current-generation SCIM API uses the **openapi** pattern in the URL.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*

Property Name	Description & Value
`AuthType`	Enter: *Form* **Relevant for next-generation SCIM API**
`User`	Enter the *Client ID* for the current-generation or the next-generation SCIM API, as described in the corresponding Prerequisites section.
`Password`	(Credential) Enter the *Client Secret* for the current-generation or the next-generation SCIM API, as described in the corresponding Prerequisites section.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL for the current-generation or the next-generation SCIM API, as described in the corresponding Prerequisites section.
`ariba.applications.api.dependency.name`	This property holds the value of the dependency name used in the configuration of the SAP Ariba Applications consumer application, created in the SAP Cloud Identity Services tenant. It is used for access token retrieval from Identity Authentication. **Relevant for next-generation SCIM API**
`ariba.applications.api.key`	(Credential) Enter your application key (see the **Prerequisites** section). **Relevant for current-generation SCIM API**
`ariba.applications.realm.id`	Enter your AN-ID (see the **Prerequisites** section). **Relevant for current-generation SCIM API**
(Optional) `ariba.applications.group.flatten`	This property allows or forbids reading "nested groups" (group structures) from SAP Ariba Applications. If enabled (**true**), group members of type *group* will be ignored during read in order not to be provisioned to target systems that do not support nested groups. Thus, set it to **false** only if you are sure that the target system supports nested groups. Default value: *false* Predefined value (during system creation): *true* Set it to *false* only if you are sure that the target system supports nested groups.

Property Name	Description & Value
(Optional) `ariba.applications.group.children.conversion`	This property allows provisioning of "nested groups" (group structures) of type authorization from SAP Ariba Applications source systems to the Identity Directory of SAP Cloud Identity Services or Identity Authentication (SCIM API version 2) target systems. When enabled (set to true), the members of a given group from SAP Ariba Applications source system will be provisioned to the target system with reversed parent-child relationship. That is, the child groups in the source system become parent groups in the target system, while the original parent group becomes a child group in the target. For more information, see the description of the `children` attribute in Groups. The property is not automatically added during system creation. **Possible values:** • *true* • *false* Default value: *false*

Property Name	Description & Value
`ariba.applications.group.members.page.size`	SAP Ariba Applications has a hardcoded limit of the number of group members returned per request when reading the group SCIM resources. When the Identity Provisioning reads a group, it compares the number of group members with the value configured for `ariba.applications.group.members.page.size`. Depending on the value you defined for `ariba.applications.group.members.paging.enabled`, you can expect the following results: • In case the number of group members equals or exceeds the configured limit and the property `ariba.applications.group.members.paging.enabled` is enabled, the Identity Provisioning will perform full read of all group members via subsequent requests. • In case the number of group members is less than the configured limit or the property `ariba.applications.group.members.paging.enabled` is disabled, the Identity Provisioning will get the returned group members from the target system. Setting the property `ariba.applications.group.members.paging.enabled` allows you to read groups with a large number of group members. For more information, see ariba.applications.group.members.paging.enabled. Default value: *50*
`ariba.applications.group.members.paging.enabled`	This property enables the full read of group members. When it is set to *true*, all groups with members count exceeding the value defined in `ariba.applications.group.members.page.size` will be fully read by the Identity Provisioning via subsequent requests. **Possible values:** • *true* - Full read of group members is enabled. • *false* - Full read of group members is disabled. Default value: *true*

Property Name	Description & Value
`ariba.applications.user.groups.paging.enabled`	SAP Ariba Applications has a hardcoded limitation over the number of user's groups returned per request. This limit is specified by the property `ariba.applications.group.members.page.size`. To trigger a full read of a user with *'groups'* attribute over the configured value, set `ariba.applications.user.groups.paging.enabled` to *true*. **Possible values:** • *true* - the full read of user's groups is enabled. • *false* - the full read of user's groups is disabled. Default value: *true*

Property Name	Description & Value
`ariba.applications.user.groups.page.size`	The number of user groups that SAP Ariba Application returns per request when reading a user. When the Identity Provisioning reads a user, it compares the number groups assigned with the value configured for `ariba.applications.user.groups.page.size`. Depending on the number of groups assigned to the user, you can expect the following results: • If the number of assigned groups is equal or exceeds the configured value, and the `ariba.applications.user.groups.paging.enabled` property is set to *true*, Identity Provisioning will read all user groups in separate requests. For example, if a user has 6 groups assigned and the value of the page size is set to *2*, Identity Provisioning will send 3 requests to read the user groups. • If the number of assigned groups is less than the configured value for `ariba.applications.user.groups.page.size` or the property `ariba.applications.user.groups.paging.enabled` is disabled, the Identity Provisioning will read all user groups via one request. For example, if a user has 2 groups assigned and the value of the page size is set to *6*, Identity Provisioning will read all user groups via one request. Setting the property `ariba.applications.user.groups.paging.enabled` allows you to read users with a large number of groups assigned. For more information, see `ariba.applications.user.groups.paging.enabled`. Default value: *50*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination (property configuration):

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*https://openapi.ariba.com*

`User`=*aaaa12345-1111-3333-cccc-1234567890*

`Password`=**********

`OAuth2TokenServiceURL`=*https://api.ariba.com/v2/oauth/token*

`ariba.applications.group.flatten`=*true*

`ariba.applications.api.key`=*123abc123XYZ000abc123ABC012345*

`ariba.applications.realm.id`=*AN000111222333*

---

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Ariba Applications* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   > ### Note
   >
   > When provisioning from SAP Ariba Applications source system to the Identity Directory of SAP Cloud Identity Services or Identity Authentication (SCIM API version 2) target system, you can preserve the group hierarchy with nested **authorization** groups by setting the `ariba.applications.group.children.conversion` property to *true*.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Ariba Applications source system. For more information, see:

   Manage Transformations [page 2439]

   SCIM API for User and Group Master Data

   Default transformation:

   > ### Code Syntax
   >
   ```
   {
     "user": {
       "mappings": [
         {
           "sourcePath": "$.id",
           "targetVariable": "entityIdSourceSystem"
         },
         {
           "sourcePath": "$.schemas",
           "preserveArrayWithSingleElement": true,
           "targetPath": "$.schemas"
         },
         {
           "sourcePath": "$.userName",
   ```

```
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.emails[?(@.primary == true)].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['companyCode']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['companyCode']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['costCenter']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingGroup']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingGroup']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['generalLedgerAccount']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['generalLedgerAccount']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingOrganization']",
```

```
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingOrganization']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['plant']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['plant']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']['currency']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']['currency']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['deliverTo']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['deliverTo']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['purchasingUnit']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['purchasingUnit']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']['network']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']['network']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['addresses']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['addresses']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['passwordAdapter']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['passwordAdapter']",
        "optional": true
      },
      {
```

```
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['approvalLimit']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['approvalLimit']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['expenseApprovalLimit']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['expenseApprovalLimit']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:sap:cloud:scim:schemas:extension:custom:2.0:profile:User']
['alternativeDisplayNames']",
        "targetPath":
"$['urn:sap:cloud:scim:schemas:extension:custom:2.0:profile:User']
['alternativeDisplayNames']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.title",
        "targetPath": "$.title",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.phoneNumbers"
      },
      {
        "sourcePath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups",
        "functions": [
          {
            "condition": "'%ariba.applications.group.prefix%' !== 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "applyOnAttribute": "display",
            "prefix": "%ariba.applications.group.prefix%"
          }
        ]
      },
```

```
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%ariba.applications.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%ariba.applications.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalLocation']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalLocation']",
        "optional": true
```

```
      },
      {
        "sourcePath":
"$['urn:sap:cloud:scim:schemas:extension:custom:2.0:profile:Group']
['alternativeDisplayNames']",
        "targetPath":
"$['urn:sap:cloud:scim:schemas:extension:custom:2.0:profile:Group']
['alternativeDisplayNames']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members"
      }
    ]
  }
}
```

5. Now, add a target system to provision users into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

The SAP Ariba developer portal
Video: Create application and API approval process

1.6.1.9 SAP Ariba Category Management

Follow this procedure to set up SAP Ariba Category Management as a source system.

Prerequisites

You have created an instance and generated a service key for the scim service plan of SAP Ariba Category Management. For more information, see Creating a SAP Ariba Category Management SCIM Service Instance in Your Subaccount.

> **Note**
>
> This connector has been deprecated as of August 25, 2025, and is no longer needed. Customers with existing configurations should stop running jobs to or from SAP Ariba Category Management provisioning systems.

Context

SAP Ariba Category Management is a cloud-based solution that enables organizations to define and manage purchasing categories and provides actionable market intelligence, category insights, and various tools and templates for developing and operationalizing category strategies that drive business value and performance. This business solution is built on SAP Business Technology Platform (BTP). You can use Identity Provisioning to configure SAP Ariba Category Management as a source system where you can read users from and provision them to a target system.

> **Note**
>
> SAP Ariba Category Management does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Ariba Category Management* as a source system. See: Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL provided by the service key under the *scim-v1* field without adding the path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the value from the *clientid* field of the service key.
`Password`	(Credential) Enter the value from the *clientsecret* field of the service key.

Property Name	Value
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. This is the value from the *url* field of the service key plus `/oauth/token`
(Optional) `cm.user.filter`	When specified, only those SAP Ariba Category Management users matching the filter expression will be read. For example: • *userName eq "SmithJ"* • *emails.value eq "john.doe@example.com"* • *emails.value eq "john.doe@example.com" or emails.value eq "john.d@example.com" or emails.value eq "j.doe@example.com"*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Ariba Category Management* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your system. For more information, see:

   Manage Transformations [page 2439]

   SCIM APIs for SAP Ariba Category Management

   Mapping logic – the behavior of the default transformation logic is to read all user attributes from the source SAP Ariba Category Management system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

   Default transformation:

   Code Syntax

```
{
   "user":{
      "mappings":[
         {
            "sourcePath":"$.id",
            "targetVariable":"entityIdSourceSystem"
         },
         {
            "sourcePath":"$.schemas",
            "targetPath":"$.schemas",
            "preserveArrayWithSingleElement":true
         },
         {
            "sourcePath":"$.userName",
            "targetPath":"$.userName",
            "correlationAttribute":true
         },
         {
```

```
            "sourcePath":"$.externalId",
            "targetPath":"$.externalId",
            "optional":true
         },
         {
            "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
            "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
            "optional": true
         },
         {
            "sourcePath":"$.name.givenName",
            "targetPath":"$.name.givenName",
            "optional":true
         },
         {
            "sourcePath":"$.name.familyName",
            "targetPath":"$.name.familyName",
            "optional":true
         },
         {
            "sourcePath":"$.emails[*].value",
            "targetPath":"$.emails[?(@.value)]",
            "optional":true,
            "preserveArrayWithSingleElement":true
         },
         {
            "sourcePath":"$.emails[?(@.primary== true)].value",
            "optional":true,
            "correlationAttribute":true
         },
         {
            "sourcePath":"$.active",
            "targetPath":"$.active"
         },
         {
            "sourcePath":"$.displayName",
            "targetPath":"$.displayName",
            "optional":true
         }
      ]
   }
}
```

5. Add a target system to provision users to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

User Replication Setup

1.6.1.10 SAP Ariba Central Invoice Management

Follow this procedure to set up SAP Ariba Central Invoice Management as a source system.

Prerequisites

You have created an instance and generated a service key for the scim service plan of SAP Ariba Central Invoice Management. This step is automated by the booster *Set Up SAP Ariba Central Invoice Management*. For more information, see Setting Up Invoice Processing with SAP Ariba Central Invoice Management (4N6) –> *Set up Configuration Using Booster in SAP BTP Cockpit*.

Context

SAP Ariba Central Invoice Management is an SAP BTP SaaS application running on SAP BTP, Cloud Foundry environment. This application enables central management of supplier invoices from multiple connected systems, such as SAP S/4HANA Cloud Public Edition systems. You can use Identity Provisioning to configure SAP Ariba Central Invoice Management as a source system where you can read users and groups from and provision them to a target system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Ariba Central Invoice Management* as a source system. See: Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
URL	Enter the URL provided by the service key under the *ips-connector* field without adding the path information. For example: `https://eu10.cim.cloud.sap`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*

Property Name	Value
`User`	Enter the value from the *clientid* field of the service key.
`Password`	(Credential) Enter the value from the *clientsecret* field of the service key.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. This is the value from the *url* field of the service key plus `/oauth/token` For example: `https://<my tenant>.authentication.eu10.hana.ondemand.com/oauth/token`
(Optional) `cim.user.filter`	When specified, only those SAP Ariba Central Invoice Management users matching the filter expression will be read. Example: *name.familyName eq "Smith"* and *addresses.country eq "US"*
(Optional) `cim.group.filter`	When specified, only those SAP Ariba Central Invoice Management groups matching the filter expression will be read. Example: *displayName eq "ProjectTeam1"*
(Optional) `cim.group.prefix`	This property distinguishes SAP Ariba Central Invoice Management groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`CIM_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Ariba Central Invoice Management source system and will be provisioned to the target system with the following name pattern: **`CIM_<GroupDisplayName>`**. This way SAP Ariba Central Invoice Management groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Ariba Central Invoice Management groups will be read and provisioned to the target system with their actual display names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Ariba Central Invoice Management* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your system. For more information, see:

Manage Transformations [page 2439]

User Management (SCIM) for SAP Ariba Central Invoice Management

Mapping logic – the behavior of the default transformation logic is to read all user attributes from the source SAP Ariba Central Invoice Management system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

Default transformation:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.name.middleName",
        "targetPath": "$.name.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.name.formatted",
        "targetPath": "$.name.formatted",
        "optional": true
      },
      {
        "sourcePath": "$.name.honorificPrefix",
```

```
        "targetPath": "$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath": "$.name.honorificSuffix",
        "targetPath": "$.name.honorificSuffix",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "optional": true,
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.emails[?(@.primary== true)].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.userType",
        "targetPath": "$.userType",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.nickName",
        "targetPath": "$.nickName",
        "optional": true
      },
      {
        "sourcePath": "$.title",
        "targetPath": "$.title",
        "optional": true
      },
      {
        "sourcePath": "$.preferredLanguage",
        "targetPath": "$.preferredLanguage",
        "optional": true
      },
      {
        "sourcePath": "$.timezone",
        "targetPath": "$.timezone",
        "optional": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "functions": [
          {
            "condition": "'%cim.group.prefix%' !== 'null'",
            "function": "concatString",
```

```
            "applyOnElements": true,
            "applyOnAttribute": "display",
            "prefix": "%cim.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {  "condition": "'%ips.application.id%' !== 'null'",
         "constant": "%ips.application.id%",
         "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%cim.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%cim.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
```

```
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "optional": true,
        "preserveArrayWithSingleElement": true
      }
    ]
  }
}
```

5. Add a target system to provision users to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Ariba Central Invoice Management

1.6.1.11 SAP BTP ABAP environment

Follow this procedure to set up SAP BTP ABAP environment as a source system.

Prerequisites

To establish the connection between Identity Provisioning and SAP BTP ABAP environment, you need to set up the communication (user, system and arrangement) on SAP BTP ABAP environment. You have the option to do it either now, as a prerequisite, or while configuring SAP BTP ABAP environment as a source system, as described in step 3.

Context

You can use SAP BTP ABAP environment to read entities from it and provision them to a target system. This scenario supports reading **business users** (Employee), **user assignments**, and **business roles** (which are considered as *groups*).

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP BTP ABAP environment* as a source system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP BTP ABAP environment and configure your authentication method (basic or certificate-based).

   > ### Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP BTP ABAP environment source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP BTP ABAP environment backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Create a communication system and assign the created user to the communication system.

      For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

   d. Create a communication arrangement with the created system.

      For your Identity Provisioning scenario, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

      For more information, see Maintain a Communication Arrangement for Inbound Communication

      > ### Note
      >
      > The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
      >
      > The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the API URL to your SAP BTP ABAP environment system. You can take the URL from the communication scenario SAP_COM_0193.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`a4c.skip.read.archived`	In the event of archived (disabled) entities in a source SAP BTP ABAP environment system, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to **false**, or delete it.
`ips.date.variable.format`	*yyyy-MM-dd*

Property Name	Description & Value
(Optional) `a4c.roles.filter`	Enter OData filtering for reading roles in the SAP BTP ABAP environment system. To learn what criteria you can use, see: OData URI Conventions → **4.5 Filter System Query Option**
(Optional) `a4c.roles.page.size`	Indicate how many business roles (considered as *groups*) per page to be read from your SAP BTP ABAP environment system. The value must be an integer number.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://12345-aaaaa-3333.abap.hana.ondemand.com
User=MyABAPEnvUser
Password=************
ips.date.variable.format=yyyy-MM-dd
a4c.skip.read.archived=true
a4c.roles.filter=startswith(ID, 'EMPLOYEE_LEVEL_3') eq true
a4c.roles.page.size=30
```

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP ABAP environment* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules depending on your setup of entities in your SAP BTP ABAP environment. For more information, see:

   Manage Transformations [page 2439]

   Inbound Service: Business User

   Default transformation:

Code Syntax

```
{
  "user": {
    "condition": "($.validityPeriod.startDate <= '${currentDate}') &&
($.validityPeriod.endDate > '${currentDate}')",
    "mappings": [
      {
        "sourcePath": "$.personID",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.personalInformation.firstName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.lastName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.middleName",
        "targetPath": "$.name.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.personFullName",
        "targetPath": "$.name.formatted",
        "optional": true
      },
      {
        "sourcePath": "$.user.userName",
        "targetPath": "$.userName",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "condition": "$.user.lockedIndicator == 'true'",
        "constant": false,
        "targetPath": "$.active",
        "optional": true
      },
// The following condition states that if a business user is outside its
validity period, it will be set as inactive.
// That means, this user will not be able to log into the SAP BTP ABAP
environment system.
      {
        "condition": "($.user.validityPeriod.startDate > '${currentDate}')
|| ('${currentDate}' > $.user.validityPeriod.endDate)",
        "constant": false,
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.workplaceInformation.emailAddress",
        "targetPath": "$.emails[0].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.user.logonLanguageCode",
        "optional": true,
```

```
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.PersonExternalID",
        "optional": true,
        "correlationAttribute": true
      },
// The Identity Provisioning reads both users and user assignments from
SAP BTP ABAP environment.
      {
        "sourcePath": "$.user.role",
        "optional": true,
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "functions": [
          {
            "condition": "'%a4c.roles.prefix%' !== 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "%a4c.roles.prefix%",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "display"
          },
          {
            "condition": "'%a4c.roles.prefix%' === 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "display"
          },
          {
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "value"
          }
        ]
      },
      {
        "type": "remove",
        "targetPath": "$.groups[*].roleName"
      },
      {
        "sourcePath": "$.user.globalUserID",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$.user.timeZoneCode",
        "optional": true,
        "targetPath": "$.timezone",
        "functions": [
          {
            "type": "convertTimezoneCode",
            "outputFormat": "iana"
          }
        ],
        "defaultValue": "Europe/Berlin"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.businessPartnerRoleCode"
        ],
        "targetPath": "$.userType",
```

```
        "defaultValue": "Employee",
        "valueMappings": [
          {
            "key": [
              "BUP003"
            ],
            "mappedValue": "Employee"
          }
        ]
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.ID",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.ID",
        "functions": [
          {
            "condition": "'%a4c.roles.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%a4c.roles.prefix%"
          }
        ],
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.ID",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.to_BusinessUserAssignment.results",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.members"
      },
      {
        "type": "remove",
        "targetPath": "$.members[*].__metadata"
      },
      {
        "type": "remove",
        "targetPath": "$.members[*].UserName"
      },
      {
        "type": "rename",
        "constant": "value",
        "targetPath": "$.members[*].PersonID"
      },
      {
        "constant": "User",
        "targetPath": "$.members[*].type"
      },
      {  "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "constant": "authorization",
```

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {
        "constant": "readWrite",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      }
    ]
  }
}
```

By default, Identity Provisioning reads group IDs and members. If you want the service to also read group descriptions, you can add an extra mapping to the *"group"* resource. To learn how, see Guided Answers: Business Role Description.

6. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

User Provisioning in SAP BTP ABAP environment

1.6.1.12 SAP BTP Account Members (Neo)

Follow this procedure to set up your SAP Business Technology Platform as a source system, from which you can read and manage members in your Neo account.

Prerequisites

You have created a new Platform API OAuth client, with API *Account Member Management* and scopes *Manage Account Members* and *Read Account Members*.

Save the *Client ID* and *Client Secret* as you'll need them when you configure your source system. Make sure you save the client secret as you cannot retrieve it later.

For more information, see Create a Platform API Client.

Context

The Identity Provisioning service helps companies to automatically manage the user-to-platform roles assignments for SAP Business Technology Platform subaccounts. For this scenario, as a source system are used SAP BTP subaccounts. The target system can be a solution supported by the Identity Provisioning service with write access for user and group artifacts.

This provisioning scenario is based on the Platform Authorization Management API of SAP BTP. For more information, see Platform Authorization Management API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP BTP Account Members (Neo)* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter: **`https://api.<SAP_BTP_host>/authorization/v1/platform/accounts/<SAP_BTP_account>`** Examples: • (Europe – Rot) *https://api.hana.ondemand.com/authorization/v1/platform/accounts/abc123xyz* • (Japan – Tokyo) *https://api.jp1.hana.ondemand.com/authorization/v1/platform/accounts/abc123xyz*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the Client ID of the new Platform API OAuth client created for the **Account Member Management** API (see the prerequisites).
`Password`	(Credential) Enter the Client Secret of the new Platform API OAuth client created for the **Account Member Management** API (see the prerequisites).

Property Name	Description & Value
`OAuth2TokenServiceURL`	Enter: **`https://api.<SAP_BTP_host>/oauth2/apitoken/v1`** Examples: • (Europe – Rot) *https://api.hana.ondemand.com/oauth2/apitoken/v1* • (US East – Sterling) *https://api.us3.hana.ondemand.com/oauth2/apitoken/v1*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP Account Members (Neo)* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   Using the default transformation, all users and platform roles available in the SAP Business Technology Platform source system will be created as users and groups in the chosen target system.

   You can change the default transformation mapping rules to reflect the data to be written in your source system. For more information, see:

   Manage Transformations [page 2439]

   SAP BTP: Authorization Management API

   Default transformation:

   Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
```

```
      },
      {
        "sourcePath": "$.emails[0].value",
        "optional": true,
        "targetPath": "$.emails[0].value"
      },
      {
        "sourcePath": "$.description",
        "optional": true,
        "targetPath": "$.description"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.roles[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups[?(@.value)]"
      },
      {
        "type": "remove",
        "targetPath": "$.meta"
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%scp.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%scp.group.prefix%"
          }
        ]
      },
      {
```

```
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]"
      },
      {
        "constant": "User",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[*].type"
      }
    ]
  }
}
```

5. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

What is SAP BTP

1.6.1.13 SAP BTP Platform Members (Cloud Foundry)

Follow this procedure to set up the SAP BTP Platform Members (Cloud Foundry) as a source system.

Prerequisites

- You have a global account in SAP BTP with at least one multi-environment subaccount with enabled Cloud Foundry environment.

- You have established trust between your SAP Cloud Identity Services tenant as custom identity provider for platform users and your global account(s) containing these subaccounts. For more information, see Establish Trust and Federation of Custom Identity Providers for Platform Users.
- You have created a CF provisioning user (which is a regular user of type *Employee* in the local identity directory of your SAP Cloud Identity Services tenant) that will be used for provisioning. Give this user an email address with the following pattern `cf-user-provisioning-<origin_key>@sap.invalid`, where `<origin_key>` is the origin key of the trust configuration for BTP platform users which points to your SAP Cloud Identity Services tenant. You have set an initial password for the user and marked its email address as *verified*. For more information, see Create a New User and List and Edit User Details.

  > **Note**
  >
  > The password lifetime for the CF provisioning user depends on the password policy set for the *SAP Business Technology Platform* application in the administration console of SAP Cloud Identity Services. For more information, see Set a Password Policy for an Application.

- You have activated the account of the CF provisioning user by opening the *Profile Page* of SAP Cloud Identity Services, in a separate browser session, and changing its initial password. The URL has the following pattern: `https://<tenant ID>.accounts.ondemand.com` or `https://<tenant ID>.accounts.cloud.sap`
- You have created the group *cf-user-provisioning* in your SAP Cloud Identity Services tenant and added the CF provisioning user to it. For more information, see Create a New Group and Add Users to a Group.
- You have added the CF provisioning user as org member with role **Org Manager** to each Cloud Foundry organization where you want to provision users, in the SAP BTP cockpit. For more information, see Add Org Members and About Roles in the Cloud Foundry Environment.

Context

The Cloud Foundry environment enables you to create polyglot cloud applications in Cloud Foundry. It contains the SAP BTP, Cloud Foundry runtime service, which is based on the open-source application platform managed by the Cloud Foundry Foundation. For more information, see Cloud Foundry Environment.

When you enable the Cloud Foundry environment in your subaccount, the system automatically creates a Cloud Foundry organization for you. You can add platform users as org members and space members and assign roles to grant these users platform access. For more information, see Valid role types .

In this scenario, your SAP Cloud Identity Services tenant acts as your identity provider and SAP BTP acts as your service provider. Identity Provisioning in between provides the SAP BTP Platform Members (Cloud Foundry) connector to replicate the org and space members with their role assignments from the identity provider to the local user storage of SAP BTP. For more information, see SAP BTP Integration Scenario.

SAP BTP Platform Members (Cloud Foundry) connector manages org and space members, as well as their role assignments, in the Cloud Foundry environment of a multi-environment subaccount, where a **single** SAP Cloud Identity Services tenant acts as custom identity provider. We recommend that you use the Identity Provisioning service enabled in this SAP Cloud Identity Services tenant.

Follow the steps below to create SAP BTP Platform Members (Cloud Foundry) as a source system to read users (org members and space members), roles (organization and space), and role assignments from your Cloud Foundry environment, which can then be provisioned to target systems of your choice. In SAP BTP Platform Members (Cloud Foundry), groups correspond to roles in particular Cloud Foundry orgs or spaces, thus group

members are user assignments of a role in a specific Cloud Foundry org or space. Group names must follow a defined pattern, explained below, to ensure the correct mapping and provisioning of users and their role assignments to the relevant Cloud Foundry organization or space.

> **Note**
>
> The attribute *name*, defined within the *urn:sap:cloud:scim:schemas:extension:custom:2.0:Group* schema, is used to map the users and user assignments of a role to the relevant Cloud Foundry organization or space. The group names follow the respective patterns:
>
> - organization group name: `<org_ID> <org_role>`
> - space group name: `<org_ID> <space_ID> <space_role>`, where `space_ID` can be retrieved using the Cloud Foundry Command Line Interface by executing the command **`cf space --guid`**.
>
> Whereas the group display names adhere to the following patterns:
>
> - organization group display name: `<btp.cf.pm.group.prefix><btp.cf.pm.landscape>:<org_name>:<org_role>`
> - space group display name: `<btp.cf.pm.group.prefix><btp.cf.pm.landscape>:<org_name>:<space_name>:<space_role>`

> **Note**
>
> In case you have Cloud Foundry environment enabled in multiple Cloud Foundry landscapes or more than one SAP Cloud Identity Services tenants, you must configure a separate SAP BTP Platform Members (Cloud Foundry) connector for each of them.

The source system consumes `User Account and Authentication API` and `Cloud Foundry V3 API` provided by Cloud Foundry.

Procedure

1. To enable provisioning of platform users and user role assignments to and from Cloud Foundry environment, create a Support Ticket on component `BC-CP-CF-SEC-IAM`.

   Specify your SAP Cloud Identity Services tenant ID and the origin key of the trust configuration for BTP platform users which points to your SAP Cloud Identity Services tenant. You can find the origin key value in the SAP BTP cockpit. Go to your SAP BTP subaccount, choose *Trust Configuration* and see the value under *Origin Key* for the relevant *Custom Identity Provider for Platform Users*.

2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
3. Add *SAP BTP Platform Members (Cloud Foundry)* as a source system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the `email` of the CF provisioning user (see **Prerequisites**).
`Password`	(Credential) Enter the password for the CF provisioning user (see **Prerequisites**).
`btp.cf.pm.origin`	Enter the origin key of your SAP Cloud Identity Services tenant (see **Step 1**). The value of this property is a string, which always ends with the suffix *-platform*. It will be used as the *origin* attribute in the system transformation.

Property Name	Description & Value
`btp.cf.pm.landscape`	Enter the *technical key* or the *host name* of the landscape in which your multi-environment subaccount with enabled Cloud Foundry environment is located. The technical key is available only for the SAP BTP Cloud Foundry regions. For more information, see Regions and API Endpoints Available for the Cloud Foundry Environment. Please note that when the technical key contains an extension index, the correct value is displayed in the *API Endpoint* column instead of the *Technical Key* column of the table. This is an example of a *technical key*: **`cf-eu10-002`**, corresponding to the API endpoint `api.cf.eu10-002.hana.ondemand.com` and host name `cf.eu10-002.hana.ondemand.com` The host name of the landscape can be found by looking up the *API Endpoint* field in the *Overview* tab of your *Cloud Foundry Environment* on subaccount level of the SAP BTP cockpit. The API endpoint has the following format: `https://api.<landscape-hostname>` **Note** If your provisioning job finishes with error *'Name or service not known'*, the value used for technical key is not correct. In that case you can use the host name of the landscape instead.
(Optional) `btp.cf.pm.user.filter`	When specified, only those SAP BTP Platform Members (Cloud Foundry) users matching the filter expression will be read. For example: *userName eq "SmithJ"*

Property Name	Description & Value
(Optional) `btp.cf.pm.group.prefix`	This property distinguishes SAP BTP Platform Members (Cloud Foundry) groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`BTP_CF_PM_`** When set in the source system, the prefix will be prepended to the name of the groups that are read from the SAP BTP Platform Members (Cloud Foundry) source system and will be provisioned to the target system with the following name pattern: **`BTP_CF_PM_<GroupDisplayName>`**. This way SAP BTP Platform Members (Cloud Foundry) groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP BTP Platform Members (Cloud Foundry) groups will be read and provisioned to the target system with their actual display names.
(Optional) `ips.trace.failed.entity.content`	If a provisioning job repeatedly fails and you need problem investigation, you can enable logging and tracing for the personal and sensitive data of your provisioned entities. To do this, set this property to *true*. For more information, see List of Properties [page 80].

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP Platform Members (Cloud Foundry)* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP BTP Platform Members (Cloud Foundry) system. For more information, see:

   Manage Transformations [page 2439]

   Cloud Foundry V3 API

   - **Mapping logic** – The behavior of the default transformation logic is to map all attributes from a SAP BTP Platform Members (Cloud Foundry) entity to the intermediate Identity Provisioning representation.
   - **User offboarding** – If a user or a user assignment of a role has been deleted from the SAP BTP Platform Members (Cloud Foundry), this change is recognized, and the user/user assignment of a role is deleted from the target system too.

⚠ Caution

Keep in mind that when you add or delete users and user role assignments in SAP BTP Platform Members (Cloud Foundry), the operations order must adhere to the Cloud Foundry Environment hierarchy. In this case, you can expect the following behavior:

- When adding a user as space member, it must already be present as member of the relevant organization with assigned role **Org User**. When deleting an org member, it should be first removed from the spaces of the relevant organization. In case you end up with erroneous entity structure, you can manually add or delete users via the administration console for SAP Cloud Identity Services.
- When adding user role assignments, you should first add the org roles and afterwards proceed with the space roles. Also, when deleting user role assignments, you should first delete the space roles and then proceed with the org roles. If this condition is not fulfilled the job will finish with error. In that case, you can rerun the job until it finishes successfully.

Default transformation:

{·} Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.name.formatted",
        "targetPath": "$.name.formatted",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.emails[?(@.primary== true)].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
```

```
        "optional": true
      },
      {
        "sourcePath": "$.verified",
        "targetPath": "$.verified",
        "optional": true
      },
      {
        "sourcePath": "$.origin",
        "targetPath": "$.origin",
        "optional": true
      },
      {
        "sourcePath": "$.zoneId",
        "targetPath": "$.zoneId",
        "optional": true
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.id",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']"
      },
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%btp.cf.pm.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%btp.cf.pm.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "constant": "authorization",
```

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {
        "constant": "userOnlyMembership",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      }
    ]
  }
}
```

6. Now, add a target system to which to provision users and user assignment of a role. For example: Local Identity Directory [page 957]
7. (Optional) Run a *Simulate* and/or a *Validate* job before you run the actual provisioning job to verify that Identity Provisioning configurations produce the desired result in the target systems. For more information, see Simulate Provisioning Jobs [page 2477] and Validate Provisioning Jobs [page 2478].

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP BTP Integration Scenario

1.6.1.14 SAP BTP Java/HTML5 apps (Neo)

Follow this procedure to set up SAP Business Technology Platform as a source system, from which you can read user-to-groups assignments for Java/HTML5 applications that run on SAP BTP, Neo environment.

Prerequisites

You have created a new Platform API OAuth client, with *Authorization Management* scopes. Save the *Client ID* and *Client Secret* as you'll need them when you configure your source system. Make sure you save the client secret as you cannot retrieve it later.

For more information, see Create a Platform API Client.

Context

The Identity Provisioning service helps companies to automatically manage the user-to-groups assignments for Java/HTML5 applications running on SAP BTP, Neo environment. For this scenario, SAP BTP is the source system. The target system can be a solution supported by the Identity Provisioning service with write access for user and group artifacts.

This provisioning scenario is based on the Authorization Management REST API of the SAP Business Technology Platform. For more information, see Using the Authorization Management REST API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP BTP Java/HTML5 apps (Neo)* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter: **`https://api.<SAP_BTP_host>/authorization/v1/accounts/<SAP_BTP_account>`** Examples: • (Europe – Rot) *https://api.hana.ondemand.com/authorization/v1/accounts/abc123xyz* • (Japan – Tokyo) *https://api.jp1.hana.ondemand.com/authorization/v1/accounts/abc123xyz*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the Client ID of the new Platform API OAuth client created for the Authorization Management API (see the prerequisites).
`Password`	(Credential) Enter the Client Secret of the new Platform API OAuth client created for the Authorization Management API (see the prerequisites).

Property Name	Description & Value
`OAuth2TokenServiceURL`	Enter: **`https://api.<SAP_BTP_host>/oauth2/apitoken/v1`** Examples: • (Europe – Rot) *https://api.hana.ondemand.com/oauth2/apitoken/v1* • (US East – Sterling) *https://api.us3.hana.ondemand.com/oauth2/apitoken/v1*
`hcp.application.names`	Enter a comma-separated list of application names. That could be applications deployed on your account, or applications for which your account has subscribed. The property returns the roles assigned to these applications. Use the following format (no spaces): *<app_name1>,<app_name2>,<provider_subaccount>:<provider_app>* **⚠ Caution** You must not leave this property with an empty value.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP Java/HTML5 apps (Neo)* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   Using the default transformation, all groups and their members (with roles) available in the SAP Business Technology Platform source system will be created as groups and respective members (with roles) in the chosen target system.

   You can change the default transformation mapping rules to reflect the data to be written in the target system. For more information, see:

   Manage Transformations [page 2439]

   SAP BTP: Authorization Management API

   Default transformation:

   {·} Code Syntax

   ```
   // To read the roles assigned to your applications, add property
   hcp.application.names to your system configuration.
   {
     "role": {
       "mappings": [
         {
   ```

```
        "targetPath": "$",
        "sourcePath": "$"
      },
      {
        "targetVariable": "entityIdSourceSystem",
        "constant": "-",
        "functions": [
          {
            "type": "concatString",
            "prefix": "$.applicationName",
            "suffix": "$.name"
          }
        ]
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "targetPath": "$.id",
        "sourcePath": "$.name"
      },
      {
        "targetVariable": "entityIdSourceSystem",
        "sourcePath": "$.name"
      },
      {
        "targetPath": "$.displayName",
           "sourcePath": "$.name",
        "functions": [
          {
            "condition": "'%hcp.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%hcp.group.prefix%"
          }
        ]
      },
      {
        "targetPath": "$.schemas[0]",
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group"
      },
      {
        "sourcePath": "$.members",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.members[?(@.value)]",
        "optional": true
      },
      {
        "constant": "User",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.members[*].type",
        "optional": true
      }
    ]
  },
  "user": {
    "mappings": [
      {
        "sourcePath": "$.name",
        "targetPath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "targetPath": "$.displayName",
        "sourcePath": "$.name"
      },
      {
        "targetPath": "$.userName",
```

```
        "sourcePath": "$.name",
        "correlationAttribute": true
      },
      {
        "targetPath": "$.schemas[0]",
        "constant": "urn:ietf:params:scim:schemas:core:2.0:User"
      },
      {
        "sourcePath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.groups[?(@.value)]",
        "optional": true
      },
      {
        "constant": "direct",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.groups[*].type",
        "optional": true
      }
    ]
  }
}
```

5. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

What is SAP BTP

1.6.1.15 SAP BTP XS Advanced UAA (Cloud Foundry)

Follow this procedure to set up the SAP BTP XS Advanced UAA (running on SAP BTP, Cloud Foundry environment) as a source system.

Prerequisites

You have created credentials to call the REST APIs of the SAP Authorization and Trust Management service on global account, subaccount, or directory level. For more information, see: Get Access to the APIs and btp create security/api-credential.

Context

The SAP BTP XS Advanced UAA connector enables you to read platform users, business users, groups and group assignments from the local user stores of global accounts, directories, and multi-environment subaccounts on SAP BTP. These users and groups (considered role collections in SAP BTP), can then be provisioned to target systems of your choice.

Create a separate source system for each global account, subaccount and directory, and configure the respective credentials and connectivity properties. Depending from where you want to read your entities, you need to:

- Create a source system for the global account to provision **platform users** and groups.
- Create a source system for the directory to provision **platform users** and groups.
- Create a source system for the subaccount to provision **platform users**, **business users** and groups.

> **Note**
>
> Reading of platform users as **org members** and **space members** requires the usage of a different connector: SAP BTP Platform Members (Cloud Foundry). For more information, see SAP BTP Platform Members (Cloud Foundry) [page 1034].

Follow the steps below to create SAP BTP XS Advanced UAA as a source system to read SAP BTP users and groups from global accounts, directories, and multi-environment subaccounts on SAP BTP.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP BTP XS Advanced UAA (Cloud Foundry)* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the API URL in the following format: `https://api.authentication.<region>.hana.ondemand.com` or For more information, see Call an API.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*

Property Name	Description & Value
`OAuth2TokenServiceURL`	Enter the URL to the OAuth2 token service in the following format: • `https://<global_account_subdomain>.authentication.<region>.hana.ondemand.com/oauth/token` or • `https://<subaccount_subdomain>.authentication.<region>.hana.ondemand.com/oauth/token` or • `https://<directory_subdomain>.authentication.<region>.hana.ondemand.com/oauth/token` or For more information, see Call an API.
`User`	Enter the respective OAuth client ID that was generated when you created credentials for each global account, subaccount and directory. For more information, see Get Access to the APIs.
`Password`	(Credential) Enter the respective OAuth client secret that was generated when you created credentials for each global account, subaccount and directory. For more information, see Get Access to the APIs.
`xsuaa.origin.filter.enabled`	This flag property depends on `xsuaa.origin`. Possible values: **true** or **false** • If set to *true*, the Identity Provisioning service will read only users whose identity provider is set as a value of `xsuaa.origin`. • If set to *false*, the Identity Provisioning service will read all users, regardless of their origin. • If set to *true* but the `xsuaa.origin` property is missing, the provisioning job will fail.

Property Name	Description & Value
`xsuaa.origin`	Enter the Origin Key of your identity provider. This is needed only when the `xsuaa.origin.filter.enabled` property is set to *true*. • Identity provider for platform users: In the SAP BTP cockpit, go to your global account (see Navigate in the Cockpit), choose *Security* > *Trust Configuration* and copy the *Origin Key* value used for platform users. This origin key always ends with `-platform`. • Identity provider for business users: In the SAP BTP cockpit, go to your subaccount (see Navigate in the Cockpit), choose *Security* > *Trust Configuration* and copy the *Origin Key* value used for business (application) users. For example: sap.custom **Note** If multiple origins are configured, a user will be read multiple times - once per origin - but only one user will be created in the target. For more information, see Configure Single and Multiple Origins [page 693]
`xsuaa.group.prefix`	This property distinguishes SAP BTP XS Advanced UAA (Cloud Foundry) groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **XSUAA_** You can use the example value or provide your own. When **set in the target system**, only groups containing the **XSUAA_** prefix in their display name will be provisioned to SAP BTP XS Advanced UAA (Cloud Foundry). Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be be provisioned to SAP BTP XS Advanced UAA (Cloud Foundry).

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://api.authentication.eu10.hana.ondemand.com
OAuth2TokenServiceURL=https://myaccount.authentication.eu10.hana.ondemand.com/oauth/token
User=sb-full-access!a76125
Password=************
xsuaa.origin=sap.custom
xsuaa.origin.filter.enabled=true
```

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP Advanced UAA* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   - **Mapping logic** – The behavior of the default transformation logic is to map all attributes from an SAP BTP XS Advanced UAA entity to the intermediate Identity Provisioning representation.
   - **User offboarding** – If a user or group has been deleted from the SAP BTP XS Advanced UAA, this change is recognized and the user/group is deleted in the target system too.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP BTP XS Advanced UAA system. For more information, see Manage Transformations [page 2439] and the API documentation.

   - User Management (System for Cross-domain Identity Management (SCIM))- SCIM groups - role collections
   - User Management (System for Cross-domain Identity Management (SCIM)) - SCIM users - shadow users

   Default transformation:

   Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath":"$.name.givenName",
```

```
        "targetPath":"$.name.givenName",
        "optional":true
      },
      {
        "sourcePath":"$.name.familyName",
        "targetPath":"$.name.familyName",
        "optional":true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.emails[?(@.primary== true)].value",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "functions": [
          {
            "condition": "'%xsuaa.group.prefix%' !== 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "applyOnAttribute": "display",
            "prefix": "%xsuaa.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.phoneNumbers",
        "targetPath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.verified",
        "targetPath": "$.verified",
        "optional": true
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.origin",
        "targetPath": "$.origin",
        "optional": true
      },
      {
        "sourcePath": "$.zoneId",
        "targetPath": "$.zoneId",
        "optional": true
      }
    ]
  },
  "group": {
    "mappings": [
      {
```

```
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%xsuaa.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%xsuaa.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "sourcePath": "$.description",
        "targetPath": "$.description",
        "optional": true
      },
      {
        "sourcePath": "$.zoneId",
        "targetPath": "$.zoneId",
        "optional": true
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "constant": "authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      }
    ]
  }
}
```

5. Add a target system to which to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Authorization and Trust Management Service
btp CLI Command Reference
SAP BTP Integration Scenario

1.6.1.15.1 Configure Single and Multiple Origins

Configure and provision users with single or multiple origins in SAP BTP XS Advanced UAA (Cloud Foundry) source system.

Overview

An origin tells you which is the identity provider of a user in SAP BTP XS Advanced UAA (Cloud Foundry). It is defined in the trust configuration in the SAP BTP cockpit under *Security* > *Trust Configuration* > *Origin Key*.

The origin itself is not a concept of Identity Provisioning. The role of the service is to ensure that you can read and provision users with their identity providers. Once you find the identity provider in the Origin Key, you need to set it in the `xsuaa.origin` property. You can configure it in source, target and proxy SAP BTP XS Advanced UAA (Cloud Foundry) systems. Both single and multiple values are supported. The value is a string that usually specifies the name of the identity provider or its location.

For example: `xsuaa.origin=`*ldap* and `xsuaa.origin=` *ldap;myaccount-xsuaa.accounts.ondemand.com*, where the ";" (semicolon) is the only supported delimiter.

Provisioning Users with Single Origin

You want to provision a user with a single origin from SAP BTP XS Advanced UAA (Cloud Foundry) source system to a given target system.

1. On the *Properties* tab of the source system, set the `xsuaa.origin.filter.enabled` property to *true*. This is a prerequisite for enabling the `xsuaa.origin` property in source systems.
2. Enter the value for the `xsuaa.origin` property, for example: *idp1*.
   The value of this property acts like a filter as only users with the values specified there are read.

3. Run a provisioning job.

As a result, one user entry with single origin is read and created in the target system.

Provisioning Users with Multiple Origins

You want to provision a user with multiple origins from SAP BTP XS Advanced UAA (Cloud Foundry) source system to a given target system.

1. On the *Properties* tab of the source system, set the `xsuaa.origin.filter.enabled` property to *true*.
   This is a prerequisite for enabling the `xsuaa.origin` property in source systems.
2. Enter multiple values for the `xsuaa.origin` property, for example: *idp1;idp2*.
   The value of this property acts like a filter as only users with the values specified there are read.
3. Set the `ips.delete.existedbefore.entities` property to *true* on the target system.
   This is needed before you run the provisioning job. It will ensure that previously existed users in the target system will be deleted if they were deleted in the source. For more information, see Manage Deleted Entities [page 2463]
4. Run a provisioning job.

As a result, two user entries for one and the same user are read. Only one user is created in the target system, because it does not support the origin concept and recognizes both user entries as one user. The job logs show that two users are read, one is created, and one is updated on the target system. A deletion of one of the origins in the source system would result in the following statistics: one user is read and one user is updated.

> **Note**
>
> Multiple origins are not supported in provisioning scenarios between SAP BTP XS Advanced UAA (Cloud Foundry) source system and SAP BTP XS Advanced UAA (Cloud Foundry) target system.

1.6.1.16 SAP Build Work Zone, advanced edition

Follow this procedure to set up SAP Build Work Zone, advanced edition as a source system.

Prerequisites

You have OAuth credentials for SAP Build Work Zone, advanced edition. To learn how, see SAP Build Work Zone, advanced edition: Add an OAuth Client

Context

After fulfilling the prerequisites, follow the procedure below to create a source SAP Build Work Zone, advanced edition system to read users and groups.

These source systems consume SCIM 2.0 API provided by SAP Build Work Zone, advanced edition.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Build Work Zone, advanced edition* as a source system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP Build Work Zone, advanced edition and configure the authentication method (basic or OAuth certificate-based).

   > ### Note
   >
   > We recommend that you use OAuth certificate-based authentication in the regions where it is supported by SAP Build Work Zone, advanced edition. This authentication method is supported for subaccounts created on the recommended data centers for new customers according to the SAP Build Work Zone, advanced edition documentation.

   a. In your newly added SAP Build Work Zone, advanced edition source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection [page 2452].

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in the SAP Build Work Zone, advanced edition Admin Console and are relevant for certificate-based authentication only.

   b. Login to the SAP Build Work Zone, advanced edition Admin Console and execute substep *2. f. iii. Use X.509 Certificate-Based Authentication* from *Step 3: Configure the Identity Authentication service and the Identity Provisioning service* in Option 1: Create a new tenant .
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
URL	Enter the URL related to your SAP Build Work Zone, advanced edition system, in format: **`https://<account><sap_wz_domain>.workzone.ondemand.com`** For example: *https://mytenant.mydomain123.workzone.ondemand.com*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*

Property Name	Description & Value
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the OAuth client key, created for your SAP Build Work Zone, advanced edition tenant (see **Prerequisites**).
`Password`	Valid if *BasicAuthentication* is configured as authentication method. (Credential) Enter the OAuth client secret, created for your SAP Build Work Zone, advanced edition tenant (see **Prerequisites**).
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Build Work Zone, advanced edition instance, based on the configured authentication method: • *BasicAuthentication* - **`https://<account><sap_wz_domain>.workzone.ondemand.com/api/v1/auth/token`** For example: *https://myaccountmydomain123.workzone.ondemand.com/api/v1/auth/token* • *ClientCertificateAuthentication* - **`https://<company_uuid>.api-<region_host>.dws.workzone.ondemand.com/api/v2/auth/token`** **Note** The `api-` prefix before `<region_host>` is necessary for using *ClientCertificateAuthentication* authentication. For example: *https://mycompanyuuid123.api-eu-20.dws.workzone.ondemand.com/api/v2/auth/token*
`client_id`	Valid if *ClientCertificateAuthentication* is configured as authentication method. Enter the OAuth Client id used for access token retrieval.
Optional Properties	
(Optional) `workzone.group.filter`	Enter a SCIM-based search criteria for filtering groups. Example: **displayName eq "Project123"**
(Optional) `workzone.user.filter`	Enter a SCIM-based search criteria for filtering users. Example: **userName eq "SmithJ"**

Property Name	Description & Value
(Optional) `ips.failed.request.retry.attempts`	Predefined value: *2*
(Optional) `ips.failed.request.retry.attempts.interval`	Predefined value: *30*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Build Work Zone, advanced edition* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Build Work Zone, advanced edition system. For more information, see:

   Manage Transformations [page 2439]

   SAP Build Work Zone SCIM API

   Default transformation:

   Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.emails[?(@.primary== true)].value",
                "optional": true,
                "correlationAttribute": true
            },
            {
                "type": "remove",
                "targetPath": "$.id"
            },
            {
                "type": "remove",
                "targetPath": "$.meta"
            },
            {
```

```
                "sourcePath":"$.name.givenName",
                "targetPath":"$.name.givenName",
                "optional":true
            },
            {
                "sourcePath":"$.name.familyName",
                "targetPath":"$.name.familyName",
                "optional":true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true
            },
            {
                "sourcePath": "$.title",
                "targetPath": "$.title",
                "optional": true
            },
            {
                "sourcePath": "$.locale",
                "targetPath": "$.locale",
                "optional": true
            },
            {
                "sourcePath": "$.timezone",
                "optional": true,
                "targetPath": "$.timezone"
            },
            {
                "sourcePath": "$.addresses",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.addresses"
            },
            {
                "sourcePath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.groups"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "optional": true
```

```
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "optional": true
            }
        ]
    },
    "group": {
        "ignore": true,
        "mappings": [
            {
```

```
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "function": "concatString",
                        "condition": "'%workzone.group.prefix%' !==
'null'",
                        "prefix": "%workzone.group.prefix%"
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "sourcePath": "$.members",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
                "optional": true,
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']"
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']",
                "optional": true,
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
            },
            {
                "condition": "'%ips.application.id%' !== 'null'",
                "constant": "%ips.application.id%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
            }
        ]
```

```
    }
}
```

6. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

1. Before starting a provisioning job, you can first subscribe to the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during your jobs. For more information, see Manage Job Notifications [page 3018].
2. Now, start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.17 SAP Business Network

Follow this procedure to set up SAP Business Network as a source system.

Context

> **Note**
>
> Currently, the SAP Business Network connector cannot be used by customers due to API access constraints. SAP is working on resolving these constraints to make the connector fully operational and accessible.

SAP Business Network, formerly known as Ariba Network, is a cloud-based offering that makes it possible for buyers and suppliers to collaborate on transactions, strengthen their relationships, and discover new business opportunities.

You can use Identity Provisioning to configure SAP Business Network as a source system where you can read users and groups and provision them to target systems of your choice.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Business Network* as a source system. See: Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the SAP Business Network API URL.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, created for your SAP Business Network system.
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP Business Network system.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL.
`bn.api.key`	An API Key represents the unique key that identifies a particular application as a legitimate consumer of an API.
`bn.realm.id`	The realm name is part of the URL you use to access SAP Business Network.
(Optional) `bn.user.filter`	When specified, only those SAP Business Network users matching the filter expression will be read. For example: • *userName eq "Julie Armstrong"* • *userName sw "J"* • *emails eq "julie.armstrong@example.com"* For more information, see List of Properties [page 80]
(Optional) `bn.group.filter`	When specified, only those SAP Business Network groups matching the filter expression will be read. For example: *displayName eq "Employees"* For more information, see List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Business Network* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your system. For more information, see:

   Manage Transformations [page 2439]

Mapping logic – the behavior of the default transformation logic is to read all user attributes from the source SAP Business Network system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

Default transformation:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sendMail']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sendMail']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap.business.network:2.0:User']
['userStatus']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap.business.network:2.0:User']
['userStatus']",
        "optional": true
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "correlationAttribute": true,
        "optional": true
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.middleName",
        "targetPath": "$.name.middleName",
        "optional": true
      },
      {
```

```
      "sourcePath": "$.name.familyName",
      "targetPath": "$.name.familyName"
    },
    {
      "sourcePath": "$.name.formatted",
      "targetPath": "$.name.formattedName",
      "optional": true
    },
    {
      "sourcePath": "$.name.honorificPrefix",
      "targetPath": "$.name.honorificPrefix",
      "optional": true
    },
    {
      "sourcePath": "$.name.honorificSuffix",
      "targetPath": "$.name.honorificSuffix",
      "optional": true
    },
    {
      "sourcePath": "$.displayName",
      "targetPath": "$.displayName",
      "optional": true
    },
    {
      "sourcePath": "$.emails",
      "targetPath": "$.emails",
      "preserveArrayWithSingleElement": true
    },
    {
      "sourcePath": "$.emails[0].value",
      "correlationAttribute": true
    },
    {
      "sourcePath": "$.active",
      "targetPath": "$.active"
    },
    {
      "sourcePath": "$.preferredLanguage",
      "targetPath": "$.preferredLanguage",
      "optional": true
    },
    {
      "sourcePath": "$.addresses",
      "targetPath": "$.addresses",
      "preserveArrayWithSingleElement": true,
      "optional": true
    },
    {
      "sourcePath": "$.locale",
      "targetPath": "$.locale",
      "optional": true
    },
    {
      "sourcePath": "$.phoneNumbers",
      "targetPath": "$.phoneNumbers",
      "preserveArrayWithSingleElement": true,
      "optional": true
    },
    {
      "sourcePath": "$.timezone",
      "targetPath": "$.timezone",
      "optional": true
    },
    {
      "sourcePath": "$.groups",
      "targetPath": "$.groups",
      "preserveArrayWithSingleElement": true,
      "optional": true
```

```
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true
      }
    ]
  }
}
```

5. Add a target system to provision users and groups to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Business Network

1.6.1.18 SAP Central Business Configuration

Follow this procedure to set up SAP Central Business Configuration (in short, CBC) as a source system.

Prerequisites

You have created a technical user with administrator permissions that will be used to call the API of SAP Central Business Configuration for reading user and group information.

Context

Create a CBC source system to read users, groups, and group members from it.

> **Note**
>
> Reading users is skipped by default. To enable it, go to the transformation and set the *"ignore": true* statement to **false**.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Central Business Configuration* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the API of your CBC system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Specify the technical user for your CBC system.
`Password`	(Credential) Specify the password for your technical user.

Property Name	Value
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. For example: **https://mycbcaccount.authentication.us2.hana.ondemand.com/oauth/token**
(Optional) `cbc.user.filter`	When specified, only those CBC users matching the filter expression will be read. Example: **name.familyName eq "Smith" and addresses.country eq "US"** **Note** Using this property makes sense only if you have set the *"ignore": true* statement to **false**.
(Optional) `cbc.group.filter`	When specified, only those CBC groups matching the filter expression will be read. Example: **displayName eq "ProjectTeam1" or "Employees2020"**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Central Business Configuration* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your CBC system. For more information, see Manage Transformations [page 2439].

   Mapping logic – the behavior of the default transformation logic is to read all user attributes from the source CBC system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

   Note

   Provisioning of user's group assignments from an SAP Central Business Configuration source system to a given target system require that users already exist in the target system. These users should be initially created by a previously executed job from that source system.

   Afterwards, the provisioning of user assignments can be triggered by running a read or resync job.

   Default transformation:

   Code Syntax

   ```
   {
       "group": {
   ```

```
        "mappings": [
            {
                "sourcePath": "$.schema",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schema"
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "'%cbc.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "prefix": "%cbc.group.prefix%"
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            },
            {
                "condition": "'%ips.application.id%' !== 'null'",
                "constant": "%ips.application.id%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
            },
            {
                "sourcePath": "$.members",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.members"
            }
        ]
    },
    "user": {
        "ignore": true,
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
```

```
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.userType",
                "optional": true,
                "targetPath": "$.userType"
            },
            {
                "sourcePath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.groups",
                "functions": [
                    {
                        "condition": "'%cbc.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "applyOnElements": true,
                        "applyOnAttribute": "display",
                        "prefix": "%cbc.group.prefix%"
                    }
                ]
            }
        ]
    }
}
```

5. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Central Business Configuration – Collection

1.6.1.19 SAP Commerce Cloud

Follow this procedure to set up SAP Commerce Cloud as a source system.

Prerequisites

- In SAP Commerce Cloud, you have added an OAuth client with authorization grant **Client Credentials**. To learn how, see: Configuring OAuth Client.

Context

SAP Commerce Cloud is a highly flexible and modular platform for delivering modern customer experiences. It provides a standardized, automated, end-to-end solution that allows your projects to release code from repository to cloud.

You can use Identity Provisioning to configure SAP Commerce Cloud as a source system to read users and groups. These source systems consume SCIM 2.0 API provided by SAP Commerce Cloud. For more information, see Commerce Cloud SCIM Web Services API Documentation.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Commerce Cloud* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP Commerce Cloud system. The URL follows the pattern: `https://backoffice.<tenant>. model-t.cc.commerce.ondemand.com` You can find the correct URL in the *Environments* section of SAP Cloud Portal.
`ProxyType`	Enter: *Internet*

Property Name	Description & Value
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the client ID to retrieve the OAuth access token for SAP Commerce Cloud.
`Password`	(Credential) Enter the client secret to retrieve the OAuth access token for SAP Commerce Cloud.
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Commerce Cloud instance.
(Optional) `cc.user.filter`	When specified, only those users matching the filter expression will be read. You can filter users by **`userName`**, **`emails.value`**, and **`externalId`**, according to the API syntax of SAP Commerce Cloud. For example: • *userName eq "johnbrown" and externalId eq "P000252"* • *userName eq "johnbrown" and emails.value eq "johnbrown@email.com"* • *userName eq "johnbrown" and emails.value eq "johnbrown@email.com" and externalId eq "P000252"* **Note** These combinations are valid for both 'or' and 'and' operators.
(Optional) `cc.group.filter`	When specified, only those groups matching the filter expression will be read. For example: *displayName eq "ProjectTeam1" or "Students2018"*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Commerce Cloud* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Commerce Cloud system. For more information, see:

   Manage Transformations [page 2439]

   Commerce Cloud SCIM Web Services API Documentation

> ### Note
>
> When configuring SAP Commerce Cloud as a source system, note that there may be mismatches between optional and required user attributes in the given target systems. For example, the email is an optional attribute in SAP Commerce Cloud but a required one in Identity Authentication. In such cases, you need to review and adapt your source and/or target system transformations and configurations to reflect your needs.

> ### Note
>
> Users that are created in SAP Commerce Cloud through the UI, don't have an ID and a username. As the ID is a mandatory attribute by SCIM specification, provisioning of those users to target systems will fail.
>
> To solve this, create the users in SAP Commerce Cloud through API or implement ImpEx (import and export) for user ID provisioning, as described in Configuring SAP Commerce Cloud.

Default transformation:

> ### Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.userName",
                "optional": true,
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath": "$.externalId",
                "optional": true,
                "targetPath": "$.externalId"
            },
            {
                "sourcePath": "$.displayName",
                "optional": true,
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.nickName",
                "optional": true,
                "targetPath": "$.nickName"
            },
```

```
            {
                "sourcePath": "$.name.familyName",
                "optional": true,
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.name.middleName",
                "optional": true,
                "targetPath": "$.name.middleName"
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.name.honorificSuffix",
                "optional": true,
                "targetPath": "$.name.honorificSuffix"
            },
            {
                "sourcePath": "$.addresses",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.addresses",
                "functions": [
                    {
                        "type": "convertCountryCode",
                        "outputFormat": "alpha2",
                        "inputAttributes": [
                            "country"
                        ],
                        "outputAttribute": "country"
                    }
                ]
            },
            {
                "sourcePath": "$.userType",
                "optional": true,
                "targetPath": "$.userType"
            },
            {
                "sourcePath": "$.active",
                "optional": true,
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.emails"
            },
            {
                "sourcePath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.groups"
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.schemas",
```

```
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "condition": "'%ips.application.id%' !== 'null'",
                "constant": "%ips.application.id%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
            },
            {
                "constant": "authorization",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                   {
                        "function": "concatString",
                        "condition": "'%cc.group.prefix%' !== 'null'",
                        "prefix": "%cc.group.prefix%"
                   }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "sourcePath": "$.members",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members"
            }
        ]
    }
}
```

5. Now, add a target system to provision users and groups into it. Choose from:Target Systems [page 936].

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Cloud Identity Services Integration Architecture → cloudscimwebservices extension

1.6.1.20 SAP Concur

Follow this procedure to set up SAP Concur as a source system.

Prerequisites

- You are using User v1 API, Identity v4 API or SCIM v4 API.

  > **Note**
  >
  > The **User v1 API** has been decommissioned. For more information, see Concur User API Decommission.

- Your SAP Concur admin user has obtained a *Company Request Token* and a *Company UUID* from the SAP Concur Company Request Token self-service tool.
  For more information, see SAP Concur Company Request Token.

Context

Companies that use SAP Concur for managing and controlling travel expenses, invoices and other can use Identity Provisioning service to automate the identity and access management for the SAP Concur solution. This service allows you to read identity data from SAP Concur and provision it to a target system of your choice (corporate user stores or SAP cloud user stores).

SAP Concur provides different APIs for integration with Identity Provisioning, resulting in different connector versions. Each connector version supports specific attribute mappings within the transformations and requires particular property values. The value of the `concur.api.version` property controls which API you use.

- When the value is set to **3**, the SCIM v4 API is used. It supports reading of users and groups.
  By default, the Identity Provisioning service uses version 3 for all newly created systems starting December 2, 2025, when version 3 was introduced. The default version is the latest released connector version at the time the system is created.
- When the value is set to **2**, the Identity v4 API is used. This applies to systems created between December 8, 2021, and December 1, 2025. Identity v4 API supports provisioning of users with the `userUUID` attribute, which is generated by Identity Authentication at user creation.
- When the value is set to **1**, or the property is not defined, the User v1 API is used. This applies to systems created before versioning was introduced on December 8, 2021. Version 1 is deprecated.

For more information on how to update to the connector version you need, see: Update Connector Version [page 2424].

To create SAP Concur as a source system, proceed as follows:

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Concur* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Connector Version 1 Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	For EU2 data center, enter: **`eu2.concursolutions.com`** For US2 data center, enter: **`www.concursolutions.com`**
`ProxyType`	Enter: **`Internet`**
`concur.api.version`	Defines the version of SAP Concur API. Set the value to *1* to use the User v1 API.
`Authentication`	Enter: *BasicAuthentication*
`User`	Deprecated property. Do not use.
`Password`	Deprecated property. Do not use.
`X-ConsumerKey`	Deprecated property. Do not use.
`concur.datacenter`	Specify the SAP Concur data center your Identity Provisioning tenant belongs to. The following SAP Concur data centers are available: • EU2 • US2 Based on the provided data center, Identity Provisioning configures the URL of the User v1 API or Identity v4 API as follows: • If you provide **`us2`**, the URL is: `us.api.concursolutions.com`

Property Name	Description & Value
`concur.authorization.code`	(Credential) Enter the *Company Request Token* and run a provisioning job within 24 hours from generating the token in the SAP Concur Company Request Token self-service tool. Otherwise, the token will expire, and you'll need a new one. After the first run of the job, Identity Provisioning fills in automatically a refresh token as the value of the concur.refresh.token property. If a provisioning job has not been run for six months, you'll again need to generate a new token. **→ Remember** The company request token has a 24 hour validity. If this token expires, you must request a new token. The refresh token has a six month validity. Every time you run a provisioning job, the validity of the refresh token is extended with six months starting from the date of the last run. If you haven't run a provisioning job for six months, your refresh token will expire and you must request a new company request token.
`concur.company.id`	Enter the *Company UUID* as described in the *Prerequisites* section.
`concur.company.domain`	Enter your company domain. The username and the company domain are concatenated in the default transformation in the following format: user@domain Your company domain is the part of your username behind the @ symbol. For example: johnsmith@example.com

Connector Version 2 Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`ProxyType`	Enter: **`Internet`**
`Authentication`	Enter: *BasicAuthentication*

Property Name	Description & Value
`concur.datacenter`	Specify the SAP Concur data center your Identity Provisioning tenant belongs to. The following SAP Concur data centers are available: • APJ1 • EU2 • US2 Based on the provided data center, Identity Provisioning configures the URL of the User v1 API or Identity v4 API as follows: • If you provide **`us2`**, the URL is: `us.api.concursolutions.com`
`concur.api.version`	Defines the version of SAP Concur API. Set the value to *2* to use the Identity v4 API. This was the default value for systems created between December 8, 2021, and December 1, 2025.
`concur.authorization.code`	(Credential) Enter the *Company Request Token* and run a provisioning job within 24 hours from generating the token in the SAP Concur Company Request Token self-service tool. Otherwise, the token will expire, and you'll need a new one. After the first run of the job, Identity Provisioning fills in automatically a refresh token as the value of the concur.refresh.token property. If a provisioning job has not been run for six months, you'll again need to generate a new token. → Remember The company request token has a 24 hour validity. If this token expires, you must request a new token. The refresh token has a six month validity. Every time you run a provisioning job, the validity of the refresh token is extended with six months starting from the date of the last run. If you haven't run a provisioning job for six months, your refresh token will expire and you must request a new company request token.
`concur.company.id`	Enter the *Company UUID* as described in the *Prerequisites* section.

Property Name	Description & Value
`concur.company.domain`	Enter your company domain. The username and the company domain are concatenated in the default transformation in the following format: user@domain Your company domain is the part of your username behind the @ symbol. For example: johnsmith@example.com
(Optional) `concur.user.filter`	When specified, only those users matching the filter expression will be read. For example: • *userName* eq "johnsmith@example.com" As the userName must be unique in SAP Concur, this filter returns only the user matching this userName. • *companyId* eq "aa067ada-71a9-4f57-8e98-9300b1c3171d" This filter returns all users in the company with this companyId.

Connector Version 3 Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`ProxyType`	Enter: **`Internet`**
`Authentication`	Enter: *BasicAuthentication*
`concur.datacenter`	Specify the SAP Concur data center your Identity Provisioning tenant belongs to. The following SAP Concur data centers are available: • APJ1 • EU2 • US2 Based on the provided data center, Identity Provisioning configures the URL of the User v1 API or Identity v4 API as follows: • If you provide **`us2`**, the URL is: `us.api.concursolutions.com`

Property Name	Description & Value
`concur.api.version`	Defines the version of SAP Concur API. Set the value to *3* to use the SCIM v4 API. By default, the Identity Provisioning service uses version 3 for all newly created systems starting December 2, 2025, when version 3 was introduced.
`concur.authorization.code`	(Credential) Enter the *Company Request Token* and run a provisioning job within 24 hours from generating the token in the SAP Concur Company Request Token self-service tool. Otherwise, the token will expire, and you'll need a new one. After the first run of the job, Identity Provisioning fills in automatically a refresh token as the value of the concur.refresh.token property. If a provisioning job has not been run for six months, you'll again need to generate a new token. → Remember The company request token has a 24 hour validity. If this token expires, you must request a new token. The refresh token has a six month validity. Every time you run a provisioning job, the validity of the refresh token is extended with six months starting from the date of the last run. If you haven't run a provisioning job for six months, your refresh token will expire and you must request a new company request token.
`concur.company.id`	Enter the *Company UUID* as described in the *Prerequisites* section.
`concur.company.domain`	Enter your company domain. The username and the company domain are concatenated in the default transformation in the following format: user@domain Your company domain is the part of your username behind the @ symbol. For example: johnsmith@example.com

Property Name	Description & Value
(Optional) `concur.user.filter`	When specified, only those users matching the filter expression will be read. For example: • *userName* eq "JohnSmith" • *userName* sw "John" • *externalId eq "0fe44868-31a7-4930-9ah30-757tg2513b64"* • *urn:ietf:params:scim:schemas:extension:enterprise:2.0:employeeNumber* eq "00102345"
(Optional) `concur.group.filter`	When specified, only those groups matching the filter expression will be read. For example: *displayName* eq "ProjectTeam1"
(Optional) `concur.group.members.page.size`	The number of group members that SAP Concur returns per request when reading a group. Default value: 1000
(Optional) `concur.group.members.paging.enabled`	This property enables paging of group members. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *true*
`concur.group.prefix`	This property distinguishes SAP Concur groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`Concur_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Concur source system and will be provisioned to the target system with the following name pattern: **`Concur_<GroupDisplayName>`**. This way SAP Concur groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Concur groups will be read and provisioned to the target system with their actual display names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Concur* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Concur source system. For more information, see:

   Manage Transformations [page 2439]

   SAP Business Accelerator Hub: Concur (Users)

   User v1 API: User v1

   Identity v4 API: Identity v4

   SCIM v4 API: SCIM v4 API

   Default transformation when using User v1 API (connector version 1):

   Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.EmployeeID",
                "targetVariable": "entityIdSourceSystem",
                "functions": [
                    {
                        "type": "compositeId",
                        "subId": "$.LoginID"
                    }
                ]
            },
            {
                "sourcePath": "$.EmployeeID",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.PrimaryEmail",
                "targetPath": "$.emails[0].value",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.FirstName",
                "optional": true,
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.LastName",
                "optional": true,
                "targetPath": "$.name.familyName"
            },
```

```
            {
                "sourcePath": "$.CellPhoneNumber",
                "optional": true,
                "targetPath": "$.phoneNumbers[0].value"
            },
            {
                "sourcePath": "$.LoginID",
                "correlationAttribute": true
            }
        ]
    }
}
```

Default transformation when using Identity v4 API (connector version 2):

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.emails"
      },
      {
        "type": "remove",
        "targetPath": "$.emails[*].notifications"
      },
      {
        "sourcePath": "$.name",
        "targetPath": "$.name"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.displayName",
        "optional": true,
```

```
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.addresses",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.addresses"
      },
      {
        "sourcePath": "$.title",
        "optional": true,
        "targetPath": "$.title"
      },
      {
        "sourcePath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.phoneNumbers"
      },
      {
        "sourcePath": "$.emergencyContacts",
        "targetPath": "$.emergencyContacts",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['companyId']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['companyId']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']"
      },
      {
```

```
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['division']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']"
      },
      {
        "sourcePath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups"
      }
    ]
  }
}
```

Default transformation when using SCIM v4 API (connector version 3):

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
```

```
      "sourcePath": "$.name",
      "targetPath": "$.name"
    },
    {
      "sourcePath": "$.active",
      "targetPath": "$.active"
    },
    {
      "sourcePath": "$.emails",
      "preserveArrayWithSingleElement": true,
      "optional": true,
      "targetPath": "$.emails"
    },
    {
      "type": "remove",
      "targetPath": "$.emails[*].notifications"
    },
    {
      "sourcePath": "$.displayName",
      "optional": true,
      "targetPath": "$.displayName"
    },
    {
      "sourcePath": "$.addresses",
      "preserveArrayWithSingleElement": true,
      "optional": true,
      "targetPath": "$.addresses"
    },
    {
      "sourcePath": "$.timezone",
      "optional": true,
      "targetPath": "$.timezone"
    },
    {
      "sourcePath": "$.phoneNumbers",
      "preserveArrayWithSingleElement": true,
      "optional": true,
      "targetPath": "$.phoneNumbers"
    },
    {
      "targetPath": "$.phoneNumbers[*].issuingCountry",
      "type": "remove"
    },
    {
      "targetPath": "$.phoneNumbers[*].notifications",
      "type": "remove"
    },
    {
      "targetPath": "$.phoneNumbers[*].workExtension",
      "type": "remove"
    },
    {
      "sourcePath": "$.emergencyContacts",
      "targetPath": "$.emergencyContacts",
      "preserveArrayWithSingleElement": true,
      "optional": true
    },
    {
      "sourcePath": "$.preferredLanguage",
      "optional": true,
      "targetPath": "$.preferredLanguage"
    },
    {
      "sourcePath": "$.title",
      "optional": true,
      "targetPath": "$.title"
    },
    {
```

```
        "sourcePath": "$.nickName",
        "optional": true,
        "targetPath": "$.nickName"
      },
      {
        "sourcePath": "$.externalId",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true,
        "functions": [
          {
            "function": "convertLocaleCode",
            "outputFormat": "iso639"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['contactPreferences']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['contactPreferences']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['companyId']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['companyId']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
```

```
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['division']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']"
      },
      {
        "sourcePath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups"
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
```

```
        "targetPath": "$.schemas",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%concur.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%concur.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "optional": true,
        "preserveArrayWithSingleElement": true
      }
    ]
  }
}
```

5. Now, add a target system to provision users into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.21 SAP CPQ

Follow this procedure to set up SAP CPQ (Configure, Price, and Quote) as a source system.

Prerequisites

You have created a technical user with administrator permissions that will be used to call the API of SAP CPQ for reading user and group information.

Context

SAP CPQ is a highly configurable system designated to help sales representatives to configure products, apply pricing and generate quotes. SAP CPQ is part of the SAP Sales Cloud portfolio.

Create an SAP CPQ source system to read users and groups from it.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP CPQ* as a source system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP CPQ and configure the authentication method (basic or OAuth certificate-based).

   > ### Note
   >
   > We recommend that you use OAuth certificate-based authentication.

   a. In your newly added SAP CPQ source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection [page 2452].

   Skip this step if you use basic authentication. The next steps are relevant for OAuth certificate-based authentication only. For more information, see Configure OAuth Certificate Authentication [page 2456].

   b. Convert your certificate file from `.crt` to `.cer`.

   c. Login to SAP CPQ and navigate to *Security* > *Certificate Management* > *User Certificates* > *New Certificate* and upload the certificate to the technical user of SAP CPQ.

   Make sure the *Enable User Certificates* option is turned on. Additionally, the technical user must be set to *Active*.
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`cpq.user.filter`	(Optional) When specified, only those SAP CPQ users matching the filter expression will be read. Example: **name.familyName eq "Smith" and addresses.country eq "US"**
`OAuth2TokenServiceURL`	Valid if *ClientCertificateAuthentication* is configured as authentication method. Enter the OAuth 2.0 Token Service URL. For example: `https://<cpq_url>/oauth2/token`
`Password`	Valid if *BasicAuthentication* is configured as authentication method. (Credential) Specify the password for your technical user.
`ProxyType`	Enter: *Internet*
`Type`	Enter: *HTTP*
`URL`	Enter the URL to the API of your SAP CPQ system. It is the same as your SAP CPQ tenant URL. It must contain the domain name. For example: **https://sample1234.mycpqdomain.com**
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the Username of your SAP CPQ technical user.
`cpq.group.members.page.size`	When the `cpq.group.members.paging.enabled` property is set to true, this setting defines the size of the paged members request. **Note** If you configure `cpq.group.members.page.size` to a number greater than the default value, the complete member retrieval is disabled. Default Value: **100** The number of group members that SAP CPQ returns per request when reading a group.

Property Name	Value
`cpq.group.members.paging.enabled`	This property enables the complete retrieval of group members through additional sequential requests. The maximum number of group members returned per request is 100 as defined in `cpq.group.members.page.size`. For more information, see `cpq.group.members.page.size`. Possible Values: • **true** - Paging is enabled. • **false** - Paging is disabled. Default Value: **true**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP CPQ* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   The behavior of the default transformation logic is to read all user attributes from the source SAP CPQ system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity. You can change the default transformation mapping rules to reflect your current setup of entities in your SAP CPQ system. For more information, see:

   Manage Transformations [page 2439]

   SAP CPQ: SCIM API

   Default transformation:

   Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
```

```
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']"
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails"
            },
            {
                "sourcePath": "$.emails[?(@.primary == true)].value",
                "optional": true,
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.addresses",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.addresses"
            },
            {
                "sourcePath": "$.phoneNumbers",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.phoneNumbers"
            },
            {
                "sourcePath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.groups",
                "functions": [
                    {
                        "condition": "'%cpq.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "applyOnElements": true,
                        "applyOnAttribute": "display",
                        "prefix": "%cpq.group.prefix%"
                    }
                ]
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
```

```
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "'%cpq.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "prefix": "%cpq.group.prefix%"
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "sourcePath": "$.members",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.members"
            },
            {  "condition": "'%ips.application.id%' !== 'null'",
                "constant": "%ips.application.id%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
            },
            {
                "constant": "authorization",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            }
        ]
    }
}
```

6. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.22 SAP Data Custodian

Follow this procedure to set up SAP Data Custodian as a source system.

Prerequisites

- You have created an SAP Data Custodian tenant.
- You have created a user within your tenant with the required roles for your scenario.
- You have added your user to SAP Identity Service Management (SAP ISM).

Context

SAP Data Custodian is a robust Software as a Service (SaaS) solution that protects sensitive data stored in public, private, hybrid, and multicloud environments. This solution integrates with partnered public hyperscalers, SAP applications, and SAP managed clouds.

After fulfilling the prerequisites, follow the procedure below to add a source system for SAP Data Custodian to read users and user assignments to groups. This source system consumes SCIM 2.0 API provided by SAP Data Custodian.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Data Custodian* as a source system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP Data Custodian and configure the authentication method (basic or certificate-based).

   > **Note**
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP Data Custodian source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip this step if you use basic authentication. The next steps are performed in SAP Data Custodian and are relevant for certificate-based authentication only.

   b. Login to SAP Data Custodian tenant and create a tenant operations technical user (TOTU), as described in Create a Tenant Operations Technical User in SAP Data Custodian.

   c. Upload the outbound certificate that you downloaded in **substep a.** to the TOTU, as described in Register the Outbound Certificate to Your Tenant Operations Technical User.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the SAP Data Custodian SCIM API excluding the path information. `https://<SCIM_API_URL>` For more information, see Identity Provisioning.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* Enter the OAuth client key, created for your SAP Data Custodian tenant. For more information, see Basic Authentication Prerequisites.
`Password`	Valid if *BasicAuthentication* (Credential) Enter the OAuth client secret, created for your SAP Data Custodian tenant. For more information, see Certificate Authentication Prerequisites.
`OAuth2TokenServiceURL`	Specify the URL of the access token provider service for your SAP Data Custodian instance: `https://<SCIM_API_URL>/kms/scim/v1/Oauth2/Token`
`dc.group.filter`	This property filters groups by display name or externalId. When specified, only those groups matching the filter expression will be read. **Possible values:** • *SAP_Data_Custodian_Auditor* • *SAP_Data_Custodian_Service_Admin* • *SAP_Data_Custodian_Key_Admin* • *SAP_Data_Custodian_Key_User* For example: *displayName eq "SAP_Data_Custodian_Auditor"*

Property Name	Value
`dc.user.filter`	When specified, only those SAP Data Custodian users matching the filter expression will be read. You can filter users by *userName*, *displayName* or *externalId*. **Possible values:** For example: *userName eq "Smith.J"*
`dc.group.prefix`	This property distinguishes SAP Data Custodian groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`DC_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Data Custodian source system and will be provisioned to the target system with the following name pattern: **`DC_<GroupDisplayName>`**. This way SAP Data Custodian groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Data Custodian groups will be read and provisioned to the target system with their actual display names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Data Custodian* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   > ### Note
   >
   > During user creation, the expected format for `userName` is email. Hence, the primary email of the user is set for `userName`.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SCIM system. For more information, see:

   Manage Transformations [page 2439]

   SAP Data Custodian SCIM 2.0 API

   Default transformation:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "targetPath": "$.schemas",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
            },
            {
              "sourcePath": "$.displayName",
              "optional":true,
              "targetPath": "$.displayName"
            },
            {
              "sourcePath": "$.name.givenName",
              "optional":true,
              "targetPath": "$.name.givenName"
            },
            {
              "sourcePath": "$.name.familyName",
              "optional":true,
              "targetPath": "$.name.familyName"
            },
            {
              "sourcePath": "$.name.formatted",
              "optional":true,
              "targetPath": "$.name.formatted"
            },
            {
              "sourcePath": "$.name.honorificPrefix",
              "optional":true,
              "targetPath": "$.name.honorificPrefix"
            },
            {
              "sourcePath": "$.name.honorificSuffix",
              "optional":true,
              "targetPath": "$.name.honorificSuffix"
            },
            {
              "sourcePath": "$.name.middleName",
              "optional":true,
              "targetPath": "$.name.middleName"
            },
            {
                "sourcePath": "$.externalId",
                "targetPath": "$.externalId",
                "optional": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
```

```
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.groups",
                "targetPath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "functions": [
                    {
                        "condition": "'%dc.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "applyOnElements": true,
                        "applyOnAttribute": "display",
                        "prefix": "%dc.group.prefix%"
                    }
                ]
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "targetPath": "$.schemas",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "'%dc.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "prefix": "%dc.group.prefix%"
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "sourcePath": "$.members",
                "targetPath": "$.members",
                "optional": true,
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.externalId",
                "targetPath": "$.externalId",
                "optional": true
            },
            {
                "condition": "'%ips.application.id%' !== 'null'",
                "constant": "%ips.application.id%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
            },
```

```
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            }
        ]
    }
}
```

6. Now, add a target system to provision users and their group assignments to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Data Custodian

1.6.1.23 SAP Datasphere

Follow this procedure to set up SAP Datasphere as a source system.

Prerequisites

- In SAP Datasphere, you have enabled a custom SAML Identity Provider, for which *User Attribute* is set to **Custom SAML User Mapping**. To learn how, see: Enabling a Custom SAML Identity Provider
- In SAP Datasphere, you have added an OAuth client with authorization grant **Client Credentials**. To learn how, see: Create OAuth2.0 Clients to Authenticate Against SAP Datasphere

Context

SAP Datasphere is a solution that harmonizes data management, analytics, and intelligent planning, including capabilities of SAP Data Intelligence Cloud and SAP Analytics Cloud. The solution enables SAP customers to consolidate their SAP data and analytics needs into a single SAP solution.

You can use Identity Provisioning to configure SAP Datasphere as a source system where you can read users and provision them to target systems of your choice. The source system consumes SCIM 2.0 API provided by SAP Datasphere.

> **Note**
>
> SAP Datasphere does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Datasphere* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP Datasphere system without adding the path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the client ID to retrieve the OAuth access token for SAP Datasphere.
`Password`	(Credential) Enter the client secret to retrieve the OAuth access token for SAP Datasphere.
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Datasphere instance. This token URL is listed in the *Trusted Identity Providers* section of the *App Integration* page. For more information, see Create OAuth2.0 Clients to Authenticate Against SAP Datasphere

Property Name	Description & Value
(Optional) `ds.user.filter`	When specified, only those SAP Datasphere users matching the filter expression will be read For example: *userName eq "SmithJ"*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Datasphere* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Datasphere system. For more information, see: Manage Transformations [page 2439]

   SAP Datasphere SCIM 2.0 API

   Default transformation:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "targetPath": "$.schemas",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
              "sourcePath": "$.name.givenName",
              "optional": true,
              "targetPath": "$.name.givenName"
            },
            {
              "sourcePath": "$.name.familyName",
              "optional": true,
              "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.externalId",
                "targetPath": "$.externalId",
                "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName"
```

```
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.emails[?(@.primary== true)].value",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:user-custom-parameters']",
                "targetPath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:user-custom-parameters']"
            },
            {
                "sourcePath": "$.photos",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.photos",
                "optional": true
            },
            {
                "sourcePath": "$.preferredLanguage",
                "targetPath": "$.preferredLanguage",
                "optional": true
            },
            {
                "sourcePath": "$.roles",
                "targetPath": "$.roles",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "optional": true
            }
        ]
    }
}
```

5. Now, add a target system to provision users into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.24 SAP Enable Now

Follow this procedure to set up SAP Enable Now as a source system.

Prerequisites

- You have created a technical user in SAP Enable Now with the necessary permissions for establishing a connection with Identity Provisioning.

Context

The SAP Enable Now solution is SAP's digital adoption platform. It can help you improve the adoption, productivity, and efficiency of your software by providing in-application help and e-learning content to users.

You can use Identity Provisioning to configure SAP Enable Now as a source system where you can read users and provision them to target systems of your choice. The source system consumes SCIM 2.0 API provided by SAP Enable Now.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Enable Now* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP Enable Now system. It follows the pattern: • `https://<company-name>.enable-now.cloud.sap` • `https://<company-name>.sap-enable-now.cn` (relevant for China)
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*

Property Name	Description & Value
`User`	Enter the username of the SAP Enable Now technical user.
`Password`	(Credential) Enter the password of the SAP Enable Now technical user.
(Optional) `en.user.filter`	When specified, only those SAP Enable Now users matching the filter expression will be read For example: *userName eq "SmithJ"*
(Optional)`en.group.prefix`	This property distinguishes SAP Enable Now groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`EN_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Enable Now source system and will be provisioned to the target system with the following name pattern: **`EN_<GroupDisplayName>`**. This way SAP Enable Now groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Enable Now groups will be read and provisioned to the target system with their actual display names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Enable Now* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Enable Now system. For more information, see: Manage Transformations [page 2439]

   Default transformation:

   Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
```

```
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.name.middleName",
                "optional": true,
                "targetPath": "$.name.middleName"
            },
            {
                "sourcePath": "$.name.familyName",
                "optional": true,
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.emails"
            },
            {
                "sourcePath": "$.active",
                "optional": true,
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.groups",
                "targetPath": "$.groups",
                "optional": true,
                "preserveArrayWithSingleElement": true,
                "functions": [
                    {
                        "condition": "'%en.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "applyOnElements": true,
                        "applyOnAttribute": "display",
                        "prefix": "%en.group.prefix%"
                    }
                ]
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
```

```
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "'%en.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "prefix": "%en.group.prefix%"
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "condition": "'%ips.application.id%' !== 'null'",
                "constant": "%ips.application.id%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            },
            {
                "sourcePath": "$.members",
                "targetPath": "$.members",
                "preserveArrayWithSingleElement": true,
                "optional": true
            }
        ]
    }
}
```

5. Now, add a target system to provision users into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Enable Now

1.6.1.25 SAP Enterprise Portal

Follow this procedure to set up SAP Enterprise Portal as a source system.

Prerequisites

- You are using SAP NetWeaver 7.5 Patch Level 22 and higher.
- You have prepared AS Java to use the SCIM application for REST API calls, as described in Initial Setup. This means, you have activated the SCIM application and created and set user authorization.

Context

SAP Enterprise Portal uses and builds on the user management of SAP NetWeaver AS for Java. Using SAP Enterprise Portal, organizations can give their employees, customers, partners, and suppliers a single point of access to the company applications, services, and information needed for conducting daily work. The portal offers business users the capability to easily create and manage portal pages and to generate their own portal content.

You can use Identity Provisioning to configure SAP Enterprise Portal as a source system where you can read users, groups and group assignments and provision them to target systems of your choice. In SAP Enterprise Portal, those entities correspond to users, Portal roles and user assignments to Portal roles, respectively.

> ⓘ Note
>
> When reading groups from SAP Enterprise Portal source system, only the exposed Portal roles are returned. When reading users, only user members of the exposed roles are returned.

In a typical scenario, SAP Enterprise Portal is configured as a source system and SAP Build Work Zone, standard edition [page 1062] is configured as a target system. The SCIM Portal application for AS Java is developed to satisfy the SAP Enterprise Portal and SAP Build Work Zone, standard edition requirements for synchronization of exposed Portal roles and the user assignments to Portal roles.

For more information, see SCIM Portal Application for AS Java

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Enterprise Portal* as a source system. See: Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the SCIM API of your SAP Enterprise Portal system. It follows the pattern: `http://<host>:<port>` of SAP NetWeaver where the SAP Enterprise Portal runs on.
`ProxyType`	Enter: *OnPremise*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the user name of the technical user you have created following the *Create and Set User Authorization* step in the *Prerequisites* section.
`Password`	(Credential) Enter the password of the technical user you have created following the *Create and Set User Authorization* step in the *Prerequisites* section.
`ep.user.filter`	When specified, only those SAP Enterprise Portal users matching the filter expression will be read. For more information, see Filtering.
`ep.group.filter`	When specified, only those SAP Enterprise Portal groups matching the filter expression will be read. For more information, see Filtering.

   To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map user and group attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Enterprise Portal* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your system. For more information, see:

   Manage Transformations [page 2439]

SCIM Implementation for AS Java

Mapping logic – the behavior of the default transformation logic is to read all user attributes from the source SAP Enterprise Portal system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

Default transformation:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails",
                "optional": true
            },
            {
                "sourcePath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.groups"
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.externalId",
                "targetPath": "$.externalId",
                "optional": true
            },
            {
                "sourcePath": "$.members",
                "targetPath": "$.members",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
```

```
                "targetPath": "$.schemas"
            }
        ]
    }
}
```

5. Add a target system to provision users and groups to it. For example: SAP Build Work Zone, standard edition [page 1062].

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Enterprise Portal

1.6.1.26 SAP Fieldglass

Follow this procedure to set up a source connector for SAP Fieldglass.

Prerequisites

You have created an API application key and a web service. To do that, follow the steps on page: Create API Application Key or Web Service and Web Services Setup

You will need the values of *Virtual Person Name (Username)* and *License Key* for the configuration of your source system (**step 3** below).

Context

After fulfilling the prerequisites, follow the procedure below to add a source system for SAP Fieldglass to read users and groups from it. This source system consumes SCIM 2.0 API provided by SAP Fieldglass.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Fieldglass* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify your SAP Fieldglass environment URL. For example: *https://abc123.fgvms.com*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter your *Virtual Person Name (Username)* – see the **Prerequisites** section above.
`Password`	(Credential) Enter your *License Key* – see the **Prerequisites** section above.
`OAuth2TokenServiceURL`	Enter your OAuth token URL in the following format: `https://<Environment_URL>/api/oauth2/v2.0/token` For example: *https://abc123.fgvms.com/api/oauth2/v2.0/token*

Property Name	Value
(Optional) `fg.group.prefix`	This property distinguishes SAP Fieldglass groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **FG_** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Fieldglass source system and will be provisioned to the target system with the following name pattern: **FG_<GroupDisplayName>**. This way SAP Fieldglass groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Fieldglass groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **FG_** prefix in their display name will be provisioned to SAP Fieldglass. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be be provisioned to SAP Fieldglass.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Fieldglass* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   The behavior of the default transformation logic is to read all user attributes from the source SAP Fieldglass system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SCIM system. For more information, see:

   Manage Transformations [page 2439]

   SAP Business Accelerator Hub: SAP Fieldglass

   Default transformation:

   Code Syntax

   ```
   {
   ```

```
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.formatted",
        "optional": true,
        "targetPath": "$.name.formatted"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.title",
        "targetPath": "$.title",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.emails[?(@.primary== true)].value",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.addresses",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.addresses"
      },
      {
        "sourcePath": "$.groups",
```

```
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups",
        "functions": [
          {
            "condition": "'%fg.group.prefix%' !== 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "applyOnAttribute": "display",
            "prefix": "%fg.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$['resourceExtensions']
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
      },
      {
        "sourcePath": "$['resourceExtensions']
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "optional": true
      },
      {
        "sourcePath": "$['resourceExtensions']
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath": "$['resourceExtensions']
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "optional": true
      },
      {
        "sourcePath": "$['resourceExtensions']
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "optional": true
      },
      {
        "sourcePath": "$['resourceExtensions']
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
```

```
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true
      },
      {
        "sourcePath": "$['resourceExtensions']
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "optional": true
      },
      {
        "sourcePath": "$['resourceExtensions']
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "optional": true
      }
    ]
  },
// By default, the group mapping is inactive (ignored) but groups are
supported.
// To start provisioning groups, either delete the statement "ignore":
true, or set its value to false.
  "group": {
    "ignore": true,
    "mappings": [
      {
       "sourcePath": "$.id",
       "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
            {
                "condition": "'%fg.group.prefix%' !== 'null'",
                "function": "concatString",
                "prefix": "%fg.group.prefix%"
            }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "sourcePath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members"
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
```

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      }
    ]
  }
}
```

5. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.27 SAP Field Service Management

Follow this procedure to set up SAP Field Service Management as a source system.

Prerequisites

You have OAuth credentials for SAP Field Service Management. For more information, see: Generating Client ID & Secret

Context

SAP Field Service Management is a cloud-based solution that is used to resolve customers issues with end-to-end field service management. For example, it helps customers overcome resource limitations, such as having enough skilled technicians in all locations.

You can use the Identity Provisioning user interface (UI) to configure SAP Field Service Management as a source system where you can read users, groups, and group members.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Field Service Management* as a source system. See: Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the API of your SAP Field Service Management system. It follows the pattern: `https://<cluster>.fsm.cloud.sap`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, created for your SAP Field Service Management system.
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP Field Service Management system.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. For example: `https://<cluster>.fsm.cloud.sap/api/oauth2/v2/token`
(Optional) `fsm.user.filter`	When specified, only those SAP Field Service Management users matching the filter expression will be read. Example: **name.familyName eq "Smith" and addresses.country eq "US"** **Note** Using this property makes sense only if you have set the *"ignore": true* statement to **false**.

Property Name	Value
(Optional) `fsm.group.prefix`	This property distinguishes SAP Field Service Management groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`FSM_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Field Service Management source system and will be provisioned to the target system with the following name pattern: **`FSM_<GroupDisplayName>`**. This way SAP Field Service Management groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Field Service Management groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`FSM_`** prefix in their display name will be provisioned to SAP Field Service Management. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be be provisioned to SAP Field Service Management.
(Optional) `fsm.group.filter`	When specified, only those SAP Field Service Management groups matching the filter expression will be read. Example: **displayName eq "ProjectTeam1" or "Employees2020"**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Field Service Management* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your system. For more information, see:

   Manage Transformations [page 2439]

   SAP Field Service Management - SCIM API

Mapping logic – the behavior of the default transformation logic is to read all user attributes from the source SAP Field Service Management system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

Default transformation:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.name.familyName",
                "optional": true,
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.emails"
            },
            {
                "sourcePath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.groups",
                "functions": [
                    {
                        "condition": "'%fsm.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "applyOnElements": true,
                        "applyOnAttribute": "display",
                        "prefix": "%fsm.group.prefix%"
                    }
                ]
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "optional": true,
```

```
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "condition": "'%ips.application.id%' !== 'null'",
                "constant": "%ips.application.id%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "'%fsm.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "prefix": "%fsm.group.prefix%"
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "sourcePath": "$.members",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members"
            }
        ]
    }
}
```

5. Add a target system to provision users and groups to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Field Service Management - Federated Authentication

1.6.1.28 SAP HANA Cloud Database Services

Follow this procedure to set up SAP HANA Cloud Database Services as a source system.

Prerequisites

You have created a technical user for Identity Provisioning in the SAP HANA database with the OPERATOR and ROLE ADMIN system privileges. For more information, see SAP Identity Provisioning Service → *Prerequisites*.

Context

SAP HANA Cloud allows you to consume the SAP HANA database from cloud applications running on SAP Business Technology Platform, as well as from applications running elsewhere using the standard SAP HANA clients. Every instance of SAP HANA Cloud has its own single SAP HANA Database.

You can use Identity Provisioning to configure SAP HANA Cloud Database Services as a source system where you can read users, roles and user role assignments from and provision them to target systems of your choice.

The SAP HANA Cloud Database Services connector supports provisioning of entities to the following two database types:

- SAP HANA Database (HDB)
- SAP HANA Data Lake Relational Engine (HDLRE / IQ Database)

You can define the target database type by setting the `hana.cloud.db.instance.type` property to either `hdb` or `hdlre`.

For more information on the supported database types, see SAP HANA Cloud, SAP HANA Database and SAP HANA Cloud, Data Lake.

> **Note**
>
> In SAP HANA Cloud Database Services, groups correspond to roles.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP HANA Cloud Database Services* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Authentication`	Enter: *BasicAuthentication*
`hana.cloud.db.instance.type`	Refers to the database type configured or used within SAP HANA Cloud. The value can be set to: • **hdb** - SAP HANA Database • **hdlre** - SAP HANA Data Lake Relational Engine Default value: **hdb**
`hana.cloud.db.instance.id`	Refers to a unique identifier associated with a specific database instance within the SAP HANA Cloud.
`hana.cloud.db.user.filter`	(Optional) When specified, only those SAP HANA Cloud Database Services users matching the filter expression will be read. Example: **name.familyName eq "Smith"**
`hana.cloud.db.group.filter`	(Optional) When specified, only those SAP HANA Cloud Database Services roles matching the filter expression will be read. Example: **displayName eq "ProjectTeam1" or "Employees2020"**

Property Name	Description & Value
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. For example: `https://demo-hc-3-test.authentication.sap.hana.ondemand.com/oauth/token` For more information on how to obtain the value, refer to the procedure in the User Provisioning with Identity Provisioning Service.
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP HANA Database system. For more information on how to obtain the value for the technical user, refer to the procedure in the User Provisioning with Identity Provisioning Service.
`ProxyType`	Enter: *Internet*
`Type`	Enter: *HTTP*
`URL`	Enter the SAP HANA Cloud Database Services API URL. For example: `https://api.gateway.orchestration.<cluster>-<datacenter>.hanacloud.ondemand.com`
`User`	Enter the OAuth Client Id, created for your SAP HANA Database system. For more information on how to obtain the value for the technical user, refer to the procedure in the User Provisioning with Identity Provisioning Service.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP HANA Cloud Database Services* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   Manage Transformations [page 2439]

   > ### Note
   >
   > Given that the SAP HANA Database username attribute is immutable and the email is not supported, refer to Handling Specific Attributes for your provisioning scenarios, particularly with Identity Authentication.

Default transformation:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userId']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap.hana.db:2.0:User']
['authenticationDetails']['providerName']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap.hana.db:2.0:User']
['authenticationDetails']['providerName']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap.hana.db:2.0:User']
['userCategory']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap.hana.db:2.0:User']
['userCategory']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap.hana.db:2.0:User']
['authenticationDetails']['authenticationType']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap.hana.db:2.0:User']
['authenticationDetails']['authenticationType']",
        "optional": true
```

```
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "functions": [
          {
            "condition": "'%hana.cloud.db.group.prefix%' !== 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "applyOnAttribute": "display",
            "prefix": "%hana.cloud.db.group.prefix%"
          }
        ]
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%hana.cloud.db.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%hana.cloud.db.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
```

```
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "optional": true,
        "preserveArrayWithSingleElement": true
      }
    ]
  }
}
```

By default, Identity Provisioning reads group IDs and members. If you want the service to also read group descriptions, you can add an extra mapping to the *"group"* resource. To learn how, see Guided Answers: Business Role Description.

5. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

User Provisioning

1.6.1.29 SAP Integrated Business Planning

Follow this procedure to set up SAP Integrated Business Planning (in short, SAP IBP) as a source system.

Prerequisites

To establish the connection between Identity Provisioning and SAP Integrated Business Planning, you need to set up the communication (user, system and arrangement) on SAP Integrated Business Planning. You can do it now (as a prerequisite) or in the process of configuring SAP Integrated Business Planning as a source system, as described in step 3.

Context

SAP Integrated Business Planning is a cloud-based solution that combines sales and operations planning (S&OP), forecasting and demand, response and supply, demand-driven replenishment, and inventory planning.

You can use Identity Provisioning to configure SAP IBP as a source system where you can read entities from and provision them to a target system.

SAP Integrated Business Planning provides different APIs for integration with Identity Provisioning, resulting in different connector versions for each API. Each connector version supports specific attribute mappings within the transformations and requires particular property values. By default, the Identity Provisioning service uses API version *1*. The value of the `ibp.api.version` property controls which API you use.

- When the value is set to *1* or the property is not present (typical for systems created before versioning was introduced on November 6, 2025), the SOAP API provided by the communication scenario SAP_COM_0193 is used. It supports reading **business users** (employees), **user assignments to business roles**, and **business roles** (considered as *groups*).
  For more information on how to update to version *2*, see Update SAP Integrated Business Planning [page 2432].
- When the value is set to *2*, the SCIM interface provided by the communication scenario SAP_COM_0465 is used. It supports provisioning of **business users**, including their **assignments** to groups of type **userGroup** and **authorization** (roles). For more information, see Groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Integrated Business Planning* as a source system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP Integrated Business Planning and configure your authentication method (basic or certificate-based).

   > ⓘ Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP Integrated Business Planning source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP Integrated Business Planning backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

c. Create a communication system and assign the created user to the communication system.

   For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

d. Create a communication arrangement with the created system.

   - When `ibp.api.version=1` is set in Identity Provisioning for the SAP IBP source system, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

     > **Note**
     >
     > The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
     >
     > The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

   - When `ibp.api.version=2` is set in Identity Provisioning for the SAP IBP source system, choose *Scenario ID* SAP_COM_0465 (System for Cross-domain Identity Management Integration).

     > **Note**
     >
     > The communication scenario *SAP_COM_0465* allows you to maintain data via SCIM (System for Cross-domain Identity Management). For more information, see Integrate Cross-Domain Identity Management.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value SAP Integrated Business Planning
`Type`	Enter: *HTTP*
`URL`	Specify the URL to your SAP IBP system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **Restriction** Do not use special symbol ',' (comma) as it is not supported.

Property Name	Description & Value SAP Integrated Business Planning
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`ibp.api.version`	The API version which is consumed by the SAP Integrated Business Planning system. By default, Identity Provisioning uses version *1*.
`ibp.skip.read.archived`	In the event of archived (disabled) entities in a source SAP IBP system, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to **false**, or delete it. **Relevant for connector version 1**
`ips.date.variable.format`	*yyyy-MM-dd*
(Optional) `ibp.roles.filter`	Enter OData filtering for reading roles in the SAP IBP system. To learn what criteria you can use, see: OData URI Conventions → **4.5 Filter System Query Option** **Relevant for connector version 1**
(Optional) `ibp.group.filter`	When specified, only those SAP Integrated Business Planning groups matching the filter expression will be read. Supported operators: eq (equal), co (contains), and sw (starts with). **Possible values:**: • `displayName` • `externalId` • `urn:ietf:params:scim:schemas:extension:sap:2.0:Group:type` For more information, see List of Properties [page 80]. **Relevant for connector version 2**

Property Name	Description & Value SAP Integrated Business Planning
(Optional) `ibp.roles.page.size`	Indicate how many business roles (considered as *groups*) per page to be read from your SAP IBP system. The value must be an integer number. **Relevant for connector version 1**
(Optional) `ibp.roles.prefix`	This property distinguishes SAP Integrated Business Planning roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`IBP_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be pre-pended to the name of the roles that are read from the SAP Integrated Business Planning source system and will be provisioned to the target system with the following name pattern: **`IBP_<role_name>`** . This way SAP Integrated Business Planning roles in the target system will be distinguished from roles provisioned from other applications. If the property is not set, the SAP Integrated Business Planning roles will be read and provisioned to the target system with their actual role names. **Relevant for connector version 1**

Property Name	Description & Value SAP Integrated Business Planning
(Optional) `ibp.group.prefix`	This property distinguishes SAP Integrated Business Planning groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`IBP_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Integrated Business Planning source system and will be provisioned to the target system with the following name pattern: **`IBP_<GroupDisplayName>`**. This way SAP Integrated Business Planning groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Integrated Business Planning groups will be read and provisioned to the target system with their actual display names. For more information, see List of Properties [page 80]. **Relevant for connector version 2**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*https://my1234567-api.scmibp.ondemand.com*

`User`=*MyIBPuser*

`Password`=**********

`ips.date.variable.format`=*yyyy-MM-dd*

`ibp.skip.read.archived`=*true*

`ibp.roles.filter`=*startswith(ID, 'EMPLOYEE_LEVEL_3') eq true*

`ibp.roles.page.size`=*30*

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default

transformation for the *SAP IBP* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules depending on your setup of entities in your SAP IBP. For more information, see:

Manage Transformations [page 2439]

SAP Integrated Business Planning API: Business User

SAP Business Accelerator Hub: SAP IBP

Integrate Cross-Domain Identity Management

SCIM Interface for IAM

Default transformation for connector version 1:

Code Syntax

```
{
  "user": {
    "condition": "($.validityPeriod.startDate <= '${currentDate}') &&
($.validityPeriod.endDate > '${currentDate}')",
    "mappings": [
      {
        "sourcePath": "$.personID",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.personalInformation.firstName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.lastName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.middleName",
        "targetPath": "$.name.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.personFullName",
        "targetPath": "$.name.formatted",
        "optional": true
      },
      {
        "sourcePath": "$.user.userName",
        "targetPath": "$.userName",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "condition": "$.user.lockedIndicator == 'true'",
        "constant": false,
        "targetPath": "$.active",
        "optional": true
```

```
      },
      {
        "condition": "($.user.validityPeriod.startDate > '${currentDate}')
|| ('${currentDate}' > $.user.validityPeriod.endDate)",
        "constant": false,
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.workplaceInformation.emailAddress",
        "targetPath": "$.emails[0].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.user.logonLanguageCode",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.personExternalID",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.user.globalUserID",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$.user.role",
        "optional": true,
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "functions": [
          {
            "condition": "'%ibp.roles.prefix%' !== 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "%ibp.roles.prefix%",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "display"
          },
          {
            "condition": "'%ibp.roles.prefix%' === 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "display"
          },
          {
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "value"
          }
        ]
      },
      {
        "type": "remove",
        "targetPath": "$.groups[*].roleName"
      },
      {
        "sourcePath": "$.user.timeZoneCode",
        "optional": true,
```

```
        "targetPath": "$.timezone",
        "functions": [
          {
            "type": "convertTimezoneCode",
            "outputFormat": "iana"
          }
        ],
        "defaultValue": "Europe/Berlin"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.businessPartnerRoleCode"
        ],
        "targetPath": "$.userType",
        "defaultValue": "Employee",
        "valueMappings": [
          {
            "key": [
              "BUP003"
            ],
            "mappedValue": "Employee"
          }
        ]
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.ID",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.ID",
        "functions": [
          {
            "condition": "'%ibp.roles.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%ibp.roles.prefix%"
          }
        ],
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.ID",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schemas[0]"
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "constant": "authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
```

```
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "sourcePath": "$.to_BusinessUserAssignment.results",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.members"
      },
      {
        "type": "remove",
        "targetPath": "$.members[*].__metadata"
      },
      {
        "type": "remove",
        "targetPath": "$.members[*].UserName"
      },
      {
        "type": "rename",
        "constant": "value",
        "targetPath": "$.members[*].PersonID"
      },
      {
        "constant": "User",
        "targetPath": "$.members[*].type"
      }
    ]
  }
}
```

By default, Identity Provisioning reads group IDs and members. If you want the service to also read group descriptions, you can add an extra mapping to the *"group"* resource. To learn how, see Guided Answers: Business Role Description.

Default transformation for connector version 2:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.externalId",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
```

```
        "sourcePath": "$.name.formatted",
        "optional": true,
        "targetPath": "$.name.formatted"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.displayName",
        "optional": true,
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.phoneNumbers"
      },
      {
        "sourcePath": "$.userType",
        "optional": true,
        "targetPath": "$.userType"
      },
      {
        "condition": "$.preferredLanguage !== 'not_unique'",
        "sourcePath": "$.preferredLanguage",
        "optional": true,
        "targetPath": "$.preferredLanguage"
      },
      {
        "condition": "$.locale !== 'not_unique'",
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
```

```
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['language']['correspondence']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['language']['correspondence']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['numericalRepresentation']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['numericalRepresentation']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['date']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['date']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['time']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['time']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['userUuidHistory']",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuidHistory']"
      },
      {
        "sourcePath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups",
        "functions": [
          {
            "function": "concatString",
            "condition": "'%ibp.group.prefix%' !== 'null'",
            "applyOnElements": true,
            "applyOnAttribute": "display",
            "prefix": "%ibp.group.prefix%"
```

```
          }
        ]
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%ibp.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%ibp.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type'] ==
'userGroup'",
        "sourcePath": "$.externalId",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "functions": [
          {
            "function": "compositeId",
            "separator": ":",
            "subId": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
          },
          {
```

```
            "condition": "'%ibp.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%ibp.group.prefix%"
          }
        ]
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type'] ==
'authorization'",
        "sourcePath": "$.externalId",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "functions": [
          {
            "condition": "'%ibp.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%ibp.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      }
    ]
  }
}
```

> ### Note
>
> If you are using SAP Integrated Business Planning version 2511, note that the `timeZone` value is not provided in the `IANA` format required by SCIM. To fix this, you need to manually add the `convertTimezoneCode` function within the `timeZone` mapping, as shown below:
>
> ```
> {
>    "sourcePath":"$.timeZone",
>    "optional":true,
>    "targetPath":"$.timezone",
>    "functions":[
>       {
>          "type":"convertTimezoneCode",
>          "outputFormat":"iana"
>       }
>    ],
>    "defaultValue":"Europe/Berlin"
> },
> ```

Group Uniqueness and Provisioning

In the most common scenario, SAP Integrated Business Planning connector version 2 is used as the source system for groups and Identity Authentication connector version 2 or Local Identity Directory is the target system.

In SAP Integrated Business Planning version 2, the uniqueness of a group is defined by the combination of `externalId` and `type` attributes. The system supports two group types: `userGroup` and `authorization` (role). As a result, two different groups can share the same `externalId` while belonging to different types, potentially causing conflicts when provisioned to the target system.

To avoid conflicts, ensure the following:

- Your SAP Integrated Business Planning system is updated to version 2, as described in Update SAP Integrated Business Planning [page 2432].
- Your target system (Identity Authentication version 2 or Local Identity Directory) has the respective group uniqueness property: `ias.group.unique.attribute` or `idds.group.unique.attribute`, set with the following combination of values: `['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']`, `['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']`, `['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']`. For more information, see List of Properties [page 80].
  Because the `externalId` attribute in SAP Integrated Business Planning version 2 connector is mapped to the `name` attribute in the target, when a group of type `userGroup` is provisioned from the source to the target, the suffix `:userGroup` is automatically appended to its `name`. This ensures that even if two groups share the same `externalId`, their names remain unique in the target system.

6. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Integrated Business Planning

1.6.1.30 SAP Intelligent Agriculture

Follow this procedure to set up SAP Intelligent Agriculture as a source system.

Prerequisites

- You have onboarded your SAP Intelligent Agriculture tenant as described in the Onboarding with the SAP BTP Cockpit section of the *Administration Guide for SAP Intelligent Agriculture*.

- You have OAuth credentials for SAP Intelligent Agriculture. For more information, see: Getting the Credentials from the SAP Intelligent Agriculture Instance.

Context

SAP Intelligent Agriculture enables agribusinesses to sustainably increase farming efficiency by digitizing their farming processes and services from plan-to-harvest, taking advantage of data science and machine learning capabilities.

After fulfilling the prerequisites, follow the procedure below to create a source SAP Intelligent Agriculture system to read users and groups.

These source systems consume SCIM 2.0 API provided by SAP Intelligent Agriculture.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Intelligent Agriculture* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL related to your SAP Intelligent Agriculture system, in format: **`https://*.prod-<region>.ia.agribusiness.cloud.sap`** , where *: is the name of your subaccount. For example: **`https://custom-domain.prod-eu10.ia.agribusiness.cloud.sap`**
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, created for your SAP Intelligent Agriculture instance (see **Prerequisites**).
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP Intelligent Agriculture instance (see **Prerequisites**).

Property Name	Description & Value
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Intelligent Agriculture instance, in format: **`https://<TENANT>.authentication.<REGION>.hana.ondemand.com/oauth/token`**
(Optional) `ia.group.filter`	When specified, only those SAP Intelligent Agriculture groups matching the filter expression will be read. **Possible values:** • *intagri_Data_Scientist* • *intagri_Farm_Manager* • *intagri_Operator* • *intagri_Operations_Planner* • *intagri_Master_Data_Manager* • *intagri_Administrator* For example: *displayName eq "intagri_Master_Data_Manager"* For more information, see List of Properties [page 80]
(Optional) `ia.user.filter`	When specified, only those SAP Intelligent Agriculture users matching the filter expression will be read. For example: *userName eq "Smith.J"* For more information, see List of Properties [page 80]
(Optional)`ia.group.prefix`	This property distinguishes SAP Intelligent Agriculture groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`IA_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Intelligent Agriculture source system and will be provisioned to the target system with the following name pattern: **`IA_<GroupDisplayName>`**. This way SAP Intelligent Agriculture groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Intelligent Agriculture groups will be read and provisioned to the target system with their actual display names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Intelligent Agriculture* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Intelligent Agriculture system. For more information, see:

   Manage Transformations [page 2439]

   SAP Intelligent Agriculture SCIM API

   Default transformation:

   Code Syntax

```
{
    "user": {
        "condition": "($.userName != 'ANONYMOUS' && $.userUuid !=
'ANONYMOUS' && $.externalId != 'ANONYMOUS' )",
//When a user is deleted from the SAP Intelligent Agriculture system, its
unique attributes are set to value 'ANONYMOUS'. Such users are skipped by
the read provisioning jobs.
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "targetPath": "$.schemas",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.userUuid",
                "targetPath": "$.userUuid"
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.name.middleName",
                "optional": true,
                "targetPath": "$.name.middleName"
            },
            {
                "sourcePath": "$.name.familyName",
                "optional": true,
```

```
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.externalId",
                "targetPath": "$.externalId"
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.emails[0].value",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.groups",
                "targetPath": "$.groups",
                "optional": true,
                "preserveArrayWithSingleElement": true,
                "functions": [
                    {
                        "function": "concatString",
                        "condition": "'%ia.group.prefix%' !== 'null'",
                        "applyOnElements": true,
                        "applyOnAttribute": "display",
                        "prefix": "%ia.group.prefix%"
                    }
                ]
            }
        ]
    },
    "group": {
    "mappings": [
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "function": "concatString",
            "condition": "'%ia.group.prefix%' !== 'null'",
            "prefix": "%ia.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
```

```
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "optional": true,
        "preserveArrayWithSingleElement": true
      }
    ]
  }
}
```

5. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

1. Before starting a provisioning job, you can first subscribe to the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during your jobs. For more information, see Manage Job Notifications [page 3018].
2. Now, start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.31 SAP Jam Collaboration

Follow this procedure to set up SAP Jam Collaboration as a source system.

Prerequisites

You get OAuth credentials for SAP Jam Collaboration. If your SAP Jam tenant is of "SCIM provisioning" type, an OAuth client is automatically created for it, with the name **SCIM API Client**. To find this client:

1. Go to the SAP Jam Collaboration admin panel.
2. Choose *Integrations* > *OAuth Clients*.

3. For *SCIM API Client*, choose *View*.
4. Save the *Key* and *Secret* values – you'll need them later while configuring your SAP Jam Collaboration provisioning system.

To learn more, see: SAP Jam: Add an OAuth Client

Context

After fulfilling the prerequisites, follow the procedure below to create a source SAP Jam Collaboration system to read users and groups.

These source systems consume SCIM 2.0 API provided by SAP Jam Collaboration.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Jam Collaboration* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL related to your SAP Jam system, in format: **`https://<SAP_Jam_datacenter>.sapjam.com`** For example: *https://jam4.sapjam.com*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth client key, created for your SAP Jam tenant (see **Prerequisites**).
`Password`	(Credential) Enter the OAuth client secret, created for your SAP Jam tenant (see **Prerequisites**).
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Jam instance, in format: **`https://<SAP_Jam_datacenter>/api/v1/auth/token`** For example: *https://jam4.sapjam.com/api/v1/auth/token*

Property Name	Description & Value
(Optional) `jam.group.prefix`	This property distinguishes SAP Jam Collaboration groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`SJC_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Jam Collaboration source system and will be provisioned to the target system with the following name pattern: **`SJC_<GroupDisplayName>`**. This way SAP Jam Collaboration groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Jam Collaboration groups will be read and provisioned to the target system with their actual display names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Jam Collaboration* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Jam Collaboration system. For more information, see:

   Manage Transformations [page 2439]

   SAP Business Accelerator Hub: SAP Jam Collaboration

   Default transformation:

   Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "targetPath": "$",
                "sourcePath": "$"
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.userName",
                "correlationAttribute": true
            },
```

```
            {
                "sourcePath": "$.emails[?(@.primary== true)].value",
                "optional": true,
                "correlationAttribute": true
            },
            {
                "type": "remove",
                "targetPath": "$.id"
            },
            {
                "type": "remove",
                "targetPath": "$.meta"
            }
        ]
    },
    "group": {
        "ignore": true,
        "mappings": [
            {
                "targetPath": "$",
                "sourcePath": "$"
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "'%jam.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "prefix": "%jam.group.prefix%"
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "type": "remove",
                "targetPath": "$.id"
            },
            {
                "type": "remove",
                "targetPath": "$.meta"
            },
            {
                "condition" : "'%ips.application.id%'!== 'null'",
                "constant" : "%ips.application.id%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true,
```

```
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
            }
        ]
    }
}
```

5. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

1. Before starting a provisioning job, you can first subscribe to the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during your jobs. For more information, see Manage Job Notifications [page 3018].
2. Now, start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.32 SAP Market Communication for Utilities

Follow this procedure to set up SAP Market Communication for Utilities as a source system.

Prerequisites

To establish the connection between Identity Provisioning and SAP Market Communication for Utilities, you need to set up the communication user in SAP BTP ABAP environment. You can do it now (as a prerequisite) or in the process of configuring SAP Market Communication for Utilities as a source system, as described in step 3.

Context

The SAP Market Communication for Utilities application is based on SAP BTP ABAP environment. You can use Identity Provisioning to configure SAP Market Communication for Utilities as a source system to read entities from and provision them to a target system.

This scenario supports reading **business users** (Employee), **user assignments**, and **business roles** (which are considered as *groups*).

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Market Communication for Utilities* as a source system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP Market Communication for Utilities and configure your authentication method (basic or certificate-based).

   > ### Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP Market Communication for Utilities source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection [page 2452].

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP BTP ABAP environment backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Create a communication system and assign the created user to the communication system.

      For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

   d. Create a communication arrangement with the created system.

      For your Identity Provisioning scenario, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

      For more information, see Maintain a Communication Arrangement for Inbound Communication

      > ### Note
      >
      > The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
      >
      > The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the API URL to your SAP Market Communication for Utilities system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`maco.skip.read.archived`	In the event of archived (disabled) entities in a source SAP Market Communication for Utilities system, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to **false**, or delete it.
`ips.date.variable.format`	*yyyy-MM-dd*
(Optional) `maco.roles.filter`	Enter OData filtering for reading roles in the SAP Market Communication for Utilities system. To learn what criteria you can use, see: OData URI Conventions → **4.5 Filter System Query Option**

Property Name	Description & Value
(Optional) `maco.roles.page.size`	Indicate how many business roles (considered as *groups*) per page to be read from your SAP Market Communication for Utilities system. The value must be an integer number.
(Optional) `maco.roles.prefix`	This property distinguishes SAP Market Communication for Utilities roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`SMC_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the roles that are read from the SAP Market Communication for Utilities source system and will be provisioned to the target system with the following name pattern: **`SMC_<role_name>`** . This way SAP Market Communication for Utilities roles in the target system will be distinguished from roles provisioned from other applications. If the property is not set, the SAP Market Communication for Utilities roles will be read and provisioned to the target system with their actual role names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*https://12345-aaaaa-3333.abap.hana.ondemand.com*

`User`=*MyMaCoUser*

`Password`=**********

`ips.date.variable.format`=*yyyy-MM-dd*

`maco.skip.read.archived`=*true*

`maco.roles.filter`=*startswith(ID, 'EMPLOYEE_LEVEL_3') eq true*

`maco.roles.page.size`=*30*

5. (Optional) Configure the transformations.

Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Market Communication for Utilities* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules depending on your setup of entities in your SAP Market Communication for Utilities. For more information, see:

Manage Transformations [page 2439]

SAP S/4HANA Cloud Public Edition API: Business User

Default transformation:

Code Syntax

```
{
  "user": {
    "condition": "($.validityPeriod.startDate <= '${currentDate}') &&
($.validityPeriod.endDate > '${currentDate}')",
    "mappings": [
      {
        "sourcePath": "$.personID",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.personalInformation.firstName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.lastName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.middleName",
        "targetPath": "$.name.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.personFullName",
        "targetPath": "$.name.formatted",
        "optional": true
      },
      {
        "sourcePath": "$.user.userName",
        "targetPath": "$.userName",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "condition": "$.user.lockedIndicator == 'true'",
        "constant": false,
        "targetPath": "$.active",
        "optional": true
      },
      {
        "condition": "($.user.validityPeriod.startDate > '${currentDate}')
|| ('${currentDate}' > $.user.validityPeriod.endDate)",
        "constant": false,
```

```
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.workplaceInformation.emailAddress",
        "targetPath": "$.emails[0].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.user.logonLanguageCode",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.PersonExternalID",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.user.role",
        "optional": true,
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "functions": [
          {
            "condition": "'%maco.roles.prefix%' !== 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "%maco.roles.prefix%",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "display"
          },
          {
            "condition": "'%maco.roles.prefix%' === 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "display"
          },
          {
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "value"
          }
        ]
      },
      {
        "type": "remove",
        "targetPath": "$.groups[*].roleName"
      },
      {
        "sourcePath": "$.user.globalUserID",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$.user.timeZoneCode",
        "optional": true,
        "targetPath": "$.timezone",
        "functions": [
          {
            "type": "convertTimezoneCode",
            "outputFormat": "iana"
```

```
          }
        ],
        "defaultValue": "Europe/Berlin"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.businessPartnerRoleCode"
        ],
        "targetPath": "$.userType",
        "defaultValue": "Employee",
        "valueMappings": [
          {
            "key": [
              "BUP003"
            ],
            "mappedValue": "Employee"
          }
        ]
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.ID",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.ID",
        "functions": [
          {
            "condition": "'%maco.roles.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%maco.roles.prefix%"
          }
        ],
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.ID",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.to_BusinessUserAssignment.results",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.members"
      },
      {
        "type": "remove",
        "targetPath": "$.members[*].__metadata"
      },
      {
        "type": "remove",
        "targetPath": "$.members[*].UserName"
      },
      {
        "type": "rename",
        "constant": "value",
        "targetPath": "$.members[*].PersonID"
      },
      {
```

```
        "constant": "User",
        "targetPath": "$.members[*].type"
      },
      {
        "condition": "'%ips.application.id%'!== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "constant": "authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

By default, Identity Provisioning reads group IDs and members. If you want the service to also read group descriptions, you can add an extra mapping to the *"group"* resource. To learn how, see Guided Answers: Business Role Description.

6. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP S/4HANA Cloud Public Edition Documentation

1.6.1.33 SAP Marketing Cloud

Follow this procedure to set up SAP Marketing Cloud as a source system.

Prerequisites

To establish the connection between Identity Provisioning and SAP Marketing Cloud, you need to set up the communication (user, system and arrangement) on SAP Marketing Cloud. You have the option to do it either now, as a prerequisite, or while configuring SAP Marketing Cloud as a source system, as described in step 3.

Context

You can use SAP Marketing Cloud to read entities from it and provision them to a target system. This scenario supports reading **users**, **user assignments**, and **Business roles** (which are considered as *groups*).

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Marketing Cloud* as a source system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP Marketing Cloud and configure your authentication method (basic or certificate-based).

   > ### Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP Marketing Cloud source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP Marketing Cloud backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Create a communication system and assign the created user to the communication system.

      For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

d. Create a communication arrangement with the created system.

   For your Identity Provisioning scenario, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

   > **Note**
   >
   > The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
   >
   > The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to your SAP Marketing Cloud system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **Restriction** Do not use special symbol ',' (comma) as it is not supported.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **Restriction** Do not use special symbol ',' (comma) as it is not supported.

Property Name	Description & Value
`marketing.cloud.skip.read.archived`	In the event of archived (disabled) entities in a source SAP Marketing Cloud system, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to **false**, or delete it.
`ips.date.variable.format`	*yyyy-MM-dd*
(Optional) `marketing.cloud.roles.filter`	Enter OData filtering for reading roles in the SAP Marketing Cloud system. To learn what criteria you can use, see: OData URI Conventions → **4.5 Filter System Query Option**
(Optional) `marketing.cloud.roles.page.size`	Indicate how many business roles (considered as *groups*) per page to be read from your SAP Marketing Cloud system. The value must be an integer number.
(Optional) `marketing.cloud.roles.prefix`	This property distinguishes SAP Marketing Cloud roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`SMKC_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the roles that are read from the SAP Marketing Cloud source system and will be provisioned to the target system with the following name pattern: **`SMKC_<role_name>`** . This way SAP Marketing Cloud roles in the target system will be distinguished from roles provisioned from other applications. If the property is not set, the SAP Marketing Cloud roles will be read and provisioned to the target system with their actual role names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://my1234567-api.s4hana.ondemand.com
User=MyMarketingCloudUser
Password=************
ips.date.variable.format=yyyy-MM-dd
marketing.skip.read.archived=true
marketing.cloud.roles.filter=startswith(ID, 'EMPLOYEE_LEVEL_3') eq true
marketing.cloud.roles.page.size=30
```

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Marketing Cloud* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules depending on your setup of entities in your SAP Marketing Cloud. For more information, see:

   Manage Transformations [page 2439]

   SAP Marketing Cloud API: Business User

   Default transformation:

   Code Syntax

```
{
  "user": {
    "condition": "($.validityPeriod.startDate <= '${currentDate}') &&
($.validityPeriod.endDate > '${currentDate}')",
    "mappings": [
      {
        "sourcePath": "$.personID",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.personalInformation.firstName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.lastName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.middleName",
        "targetPath": "$.name.middleName",
```

```
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.personFullName",
        "targetPath": "$.name.formatted",
        "optional": true
      },
      {
        "sourcePath": "$.user.userName",
        "targetPath": "$.userName",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "condition": "$.user.lockedIndicator == 'true'",
        "constant": false,
        "targetPath": "$.active",
        "optional": true
      },
// The following condition states that if a business user is outside its
validity period, it will be set as inactive.
// That means, this user will not be able to log into the SAP Marketing
Cloud system.
      {
        "condition": "($.user.validityPeriod.startDate > '${currentDate}')
|| ('${currentDate}' > $.user.validityPeriod.endDate)",
        "constant": false,
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.workplaceInformation.emailAddress",
        "targetPath": "$.emails[0].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.user.logonLanguageCode",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.PersonExternalID",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.user.globalUserID",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
// The Identity Provisioning reads both users and user assignments from
SAP Marketing Cloud.
      {
        "sourcePath": "$.user.role",
        "optional": true,
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "functions": [
          {
            "condition": "'%marketing.cloud.roles.prefix%' !== 'null'",
            "function": "concatString",
            "applyOnElements": true,
```

```
            "prefix": "%marketing.cloud.roles.prefix%",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "display"
          },
          {
            "condition": "'%marketing.cloud.roles.prefix%' === 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "display"
          },
          {
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "value"
          }
        ]
      },
      {
        "type": "remove",
        "targetPath": "$.groups[*].roleName"
      },
      {
        "sourcePath": "$.user.timeZoneCode",
        "optional": true,
        "targetPath": "$.timezone",
        "functions": [
          {
            "type": "convertTimezoneCode",
            "outputFormat": "iana"
          }
        ],
        "defaultValue": "Europe/Berlin"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.businessPartnerRoleCode"
        ],
        "targetPath": "$.userType",
        "defaultValue": "Employee",
        "valueMappings": [
          {
            "key": [
              "BUP003"
            ],
            "mappedValue": "Employee"
          },
          {
            "key": [
              "BBP005"
            ],
            "mappedValue": "Contingent Worker"
          }
        ]
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.ID",
        "targetVariable": "entityIdSourceSystem"
      },
      {
```

```
        "sourcePath": "$.ID",
        "functions": [
          {
            "condition": "'%marketing.cloud.roles.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%marketing.cloud.roles.prefix%"
          }
        ],
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.ID",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.to_BusinessUserAssignment.results",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.members"
      },
      {
        "type": "remove",
        "targetPath": "$.members[*].__metadata"
      },
      {
        "type": "remove",
        "targetPath": "$.members[*].UserName"
      },
      {
        "type": "rename",
        "constant": "value",
        "targetPath": "$.members[*].PersonID"
      },
      {
        "constant": "User",
        "targetPath": "$.members[*].type"
      },
      {  "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "constant": "authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

By default, Identity Provisioning reads group IDs and members. If you want the service to also read group descriptions, you can add an extra mapping to the *"group"* resource. To learn how, see Guided Answers: Business Role Description.

6. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP S/4HANA Cloud Documentation

1.6.1.34 SAP Master Data Integration

Follow this procedure to set up SAP Master Data Integration (in short, MDI) as a source system.

Prerequisites

- You have created a tenant in a subaccount on SAP BTP, Cloud Foundry environment. You can create your own tenant (free of charge), or integrate with an existing one.
- You have created a service instance for MDI in the subaccount in order to connect a new system to your tenant and read user account information from it. To learn how, see: Creating Service Instances
- You have created a service key in this instance, which contains the necessary credentials to connect to the MDI service. Creating multiple service keys in the same service instance is not supported. To learn how, see: Creating service Instances

> → Tip
>
> The *serviceKey* payload provides you with the following properties that you will later need for your system configuration:
>
> - **uri** =`URL`
> - **uaa.url** = `OAuth2TokenServiceURL`
> - **uaa.clientid** = `User`
> - **clientsecret** = `Password`

Context

As part of SAP's data model and integration unification strategy, SAP BTP Integration Suite has *Master Data Integration* to enable a harmonized integration and distribution of different master data objects and data between SAP solutions. This includes master data for business partners, cost centers, and workforce data. Workforce data is provided for integration scenarios that need data from SAP SuccessFactors Employee Central or other HR systems.

To learn more, see: Integrating SAP SuccessFactors Employee Central with SAP Master Data Integration

You can use Identity Provisioning to configure SAP Master Data Integration as a source system for provisioning of users and HR events to a target system of your choice. The HR events reflect the various stages of an employee's lifecycle from hire to retire. For example: hire, termination, job change, position change. For more information, see Events in Employee Central.

HR events are read with delta load marker so that the target system is aware of the previous and current instance of the event. The delta load marker is deleted during system reset.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Master Data Integration* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to the relevant integration application running in the relevant region of SAP Business Technology Platform. See the **Prerequisites** section → *serviceKey* tip.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the technical user that has access to the API of your MDI service. See the **Prerequisites** section → *serviceKey* tip.
`Password`	(Credential) Enter the password for this technical user. See the **Prerequisites** section → *serviceKey* tip.

Property Name	Description & Value
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. See the **Prerequisites** section → *serviceKey* tip.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Master Data Integration* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your MDI source system. For more information, see:

   Manage Transformations [page 2439]

   Field Mapping Between Employee Central and SAP Master Data Integration

   Default transformation:

   Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:sfsf:2.0:User']['personGUID']",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.externalId",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
      },
      {
        "sourcePath":
"$.profileDetail[0].content.scriptedProfileDetails[0].firstName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.profileDetail[0].content.scriptedProfileDetails[0].lastName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.profileDetail[0].content.scriptedProfileDetails[0].middleName",
        "targetPath": "$.name.middleName",
        "optional": true
```

```
      },
      {
        "sourcePath": "$.userAccount.userName",
        "targetPath": "$.userName",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.emails[?(@.isDefault == true)].address",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "targetPath": "$.emails[*].usage",
        "type": "remove"
      },
      {
        "targetPath": "$.emails[*].address",
        "type": "rename",
        "constant": "value"
      },
      {
        "targetPath": "$.emails[*].isDefault",
        "type": "rename",
        "constant": "primary"
      },
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:User",
          "urn:sap:cloud:scim:schemas:extension:sfsf:2.0:User",
          "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User"
        ],
        "targetPath": "$.schemas"
      }
    ]
  },
  "event": {
    "ignore": true,
    "condition": "($.instance.id EMPTY false) || ($.instanceId EMPTY
false)",
    "mappings": [
      {
        "sourcePath": "$.instance.id",
        "optional": true,
        "targetPath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.instanceId",
        "optional": true,
        "targetPath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$",
        "targetPath": "$"
      }
    ]
```

```
    }
}
```

5. Now, add a target system to provision users into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Community: The New Master Data Integration Service for SAP SuccessFactors

1.6.1.35 SAP Revenue Growth Management

Follow this procedure to set up SAP Revenue Growth Management as a source system.

Prerequisites

- You have created an SAP Revenue Growth Management consumer application in your SAP Cloud Identity Services tenant and defined a dependency to its API, as described in Configure Integration Between Applications. *UserImport* must be selected for the dependency *API*.
- You have generated API credentials for the consumer application, as described in Generate Credentials to Access the APIs of an Application.

Context

SAP Revenue Growth Management is an SAP BTP SaaS application that provides customers in the Consumer Products industry an easy way to efficiently create, monitor, and maintain account plans.

You can use Identity Provisioning to configure SAP Revenue Growth Management as a source system to read users and groups. These source systems consume SCIM 2.0 API provided by SAP Revenue Growth Management. For more information, see SAP Revenue Growth Management SCIM API.

> **Note**
>
> Support for groups is intended for future use.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Revenue Growth Management* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the base URL of your SAP Revenue Growth Management system's SCIM API, excluding any path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the client ID generated when creating API credentials for the consumer application. For more information, see Generate Credentials to Access the APIs of an Application.
`Password`	(Credential) Enter the client secret generated when creating API credentials for the consumer application. For more information, see Generate Credentials to Access the APIs of an Application.
`rgm.userImport.api.dependency.name`	Enter the value of the dependency name used in the configuration of the SAP Revenue Growth Management consumer application, created in the SAP Cloud Identity Services tenant. It is used for access token retrieval from Identity Authentication. For more information, see Integrating Applications.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL.

Property Name	Description & Value
(Optional) `rgm.user.filter`	When specified, only those users matching the filter expression will be read. You can filter users by **`userName`**, **`emails[0].value`**, and **`externalId`**, according to the API syntax of SAP Revenue Growth Management. For example: • *userName eq "johnbrown" and externalId eq "c167254d-25fd-5fac-af32-0b8c35e0de27"* • *userName sw "J" and externalId eq "c167254d-25fd-5fac-af32-0b8c35e0de27"* • *userName eq "johnbrown" and emails.value eq "johnbrown@email.com"* • *userName eq "johnbrown" and emails.value eq "johnbrown@email.com" and externalId eq "c167254d-25fd-5fac-af32-0b8c35e0de27"* **Note** These combinations are valid for both 'or' and 'and' operators.
(Optional) `rgm.group.filter`	When specified, only those groups matching the filter expression will be read. For example: *urn:sap:cloud:scim:schemas:extension:custom:2.0:Group: name eq "ProjectTeam1" or "Students2018"*

Property Name	Description & Value
(Optional) `rgm.group.prefix`	This property distinguishes SAP Revenue Growth Management groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`RGM_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Revenue Growth Management source system and will be provisioned to the target system with the following name pattern: **`RGM_<GroupDisplayName>`**. This way SAP Revenue Growth Management groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Revenue Growth Management groups will be read and provisioned to the target system with their actual display names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Revenue Growth Management* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Revenue Growth Management system. For more information, see:

   Manage Transformations [page 2439]

   SAP Revenue Growth Management SCIM API

   Default transformation:

   Code Syntax

```
{
  "user":{
    "mappings":[
      {
        "sourcePath":"$.id",
        "targetVariable":"entityIdSourceSystem"
      },
      {
        "sourcePath":"$.schemas",
        "targetPath":"$.schemas",
        "preserveArrayWithSingleElement":true
      },
```

```
        {
          "sourcePath":"$.userName",
          "targetPath":"$.userName",
          "correlationAttribute":true
        },
        {
          "sourcePath":"$.emails",
          "targetPath":"$.emails",
          "optional":true,
          "preserveArrayWithSingleElement":true
        },
        {
          "sourcePath":"$.emails[?(@.primary == true)].value",
          "correlationAttribute":true
        },
        {
          "sourcePath":"$.externalId",
          "targetPath":"$.externalId",
          "optional":true
        },
        {
          "sourcePath":"$.name.givenName",
          "targetPath":"$.name.givenName",
          "optional":true
        },
        {
          "sourcePath":"$.name.familyName",
          "targetPath":"$.name.familyName",
          "optional":true
        },
        {
          "sourcePath":"$.displayName",
          "targetPath":"$.displayName",
          "optional":true
        },
        {
          "sourcePath":"$.userType",
          "targetPath":"$.userType",
          "optional":true
        },
        {
          "sourcePath":"$.preferredLanguage",
          "targetPath":"$.preferredLanguage",
          "optional":true
        },
        {
          "sourcePath":"$.locale",
          "targetPath":"$.locale",
          "optional":true
        },
        {
          "sourcePath":"$.timezone",
          "targetPath":"$.timezone",
          "optional":true
        },
        {
          "sourcePath":"$.active",
          "targetPath":"$.active",
          "optional":true
        },
        {
          "sourcePath":"$.addresses",
          "preserveArrayWithSingleElement": true,
          "optional": true,
          "targetPath":"$.addresses"
        },
        {
```

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "functions": [
            {
                "function": "concatString",
                "condition": "'%rgm.group.prefix%' !== 'null'",
                "applyOnElements": true,
                "applyOnAttribute": "display",
                "prefix": "%rgm.group.prefix%"
            }
        ]
      }
    ]
  },
  "group": {
    "mappings":[
      {
        "sourcePath":"$.id",
        "targetVariable":"entityIdSourceSystem"
      },
      {
        "sourcePath":"$.schemas",
        "targetPath":"$.schemas",
        "preserveArrayWithSingleElement":true
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "functions": [
            {
                "condition": "'%rgm.group.prefix%' !== 'null'",
                "function": "concatString",
                "prefix": "%rgm.group.prefix%"
            }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
            {
                "condition": "'%rgm.group.prefix%' !== 'null'",
                "function": "concatString",
                "prefix": "%rgm.group.prefix%"
            }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "optional": true,
        "preserveArrayWithSingleElement": true
```

```
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      }
    ]
  }
}
```

5. Now, add a target system to provision users and groups into it. Choose from:Target Systems [page 936].

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Revenue Growth Management

1.6.1.36 SAP Sales Cloud and SAP Service Cloud

Follow this procedure to set up SAP Sales Cloud and SAP Service Cloud, formerly known as *SAP Cloud for Customer* (in short, C4C), as a source system.

Prerequisites

To integrate SAP Sales Cloud and SAP Service Cloud with Identity Provisioning, you need to use SAP Cloud Integration (SAP CI). This service provides a package with integration flows (iFlows) for enabling the creation of users and assignment of users to groups via SCIM API in SAP Sales Cloud and SAP Service Cloud.

- To configure SAP Cloud Integration and SAP Sales Cloud and SAP Service Cloud, see: Identity Provisioning in SAP Cloud for Customer using System for Cross-Domain Identity Management (SCIM)
- To set up and use the *SAP Cloud for Customer Integration with Identity Provisioning via System for Cross-domain Identity Management* package, see: SAP Business Accelerator Hub: SAP Cloud for Customer Integration with Identity Provisioning via System for Cross-domain Identity Management

Context

SAP Sales Cloud and SAP Service Cloud is a cloud-based solution that helps customers manage day-to-day sales and service interactions by sending and receiving signals between front- and back-office solutions and providing a single view of the customer.

You can use Identity Provisioning to configure SAP Sales Cloud and SAP Service Cloud as a source system where you can read *business users*, *employee users*, and *groups* from and provision them to a target system.

The cloud-based solution provides different APIs for integration with Identity Provisioning, leading to different connector versions for each API. Currently, the following connector versions are available: Version 1 (SOAP-based API), which is deprecated, Version 2 (SOAP-based API), Version 3 (SCIM 2.0 based API) and Version 4 (SCIM 2.0 based API), which is applicable for SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 only.

Each connector version supports specific attribute mappings in transformations and requires particular properties. The latest version is set as the default.

Version 4 (Applicable for SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 only)

When using connector version 4 (SCIM 2.0 based API), only users of type `Employee` are provisioned. This version introduces an enhanced SCIM API that no longer requires SAP Cloud Integration. It supports the delta read method for retrieving changed data, the **`sw`** filtering operator for user searches, and reading of application-specific groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.

2. Add *SAP Sales Cloud and SAP Service Cloud* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter SAP Cloud Integration runtime URL. See: How to Get SAP Cloud Integration Runtime URL
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`OAuth2TokenServiceURL`	Enter the URL to the access token provider service. For example: `https://<c4c_instance>/auth/token` SAP Sales Cloud and SAP Service Cloud support OAuth token-based authentication, enabling you to obtain an access token using either basic authentication (by setting `User` and `Password`) or certificate authentication. **Relevant for connector version 4**
`User`	Enter SAP Cloud Integration user ID to connect to SAP Cloud Integration. See: • Setting Up Inbound HTTP Connections (with Basic Authentication), Neo Environment • Basic Authentication of IdP User for Integration Flow Processing (Cloud Foundry environment) **Relevant for connector version 3**
`Password`	(Credential) Enter SAP Cloud Integration password to connect to SAP Cloud Integration. See: • Setting Up Inbound HTTP Connections (with Basic Authentication), Neo Environment • Basic Authentication of IdP User for Integration Flow Processing (Cloud Foundry environment) **Relevant for connector version 3**

Property Name	Description & Value
`c4c.api.version`	The version of the SAP Sales Cloud and SAP Service Cloud API you use. By default, the Identity Provisioning service uses version **4** - the SCIM 2.0 based API.
(Optional) `c4c.user.filter`	When specified, only those users matching the filter expression will be read. Supported user filters in connector version 4: • userName eq "Smith" • userName sw "Sm" • userName eq "Smith" and emails.value eq "smith@example.com" Supported user filter in connector version 3: • addresses.country eq "USA"
(Optional) `c4c.group.filter`	When specified, only those groups matching the filter expression will be read. Example: **displayName eq "ProjectTeam1"**
(Optional)`c4c.group.prefix`	This property distinguishes SAP Sales Cloud and SAP Service Cloud groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`C4C_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Sales Cloud and SAP Service Cloud source system and will be provisioned to the target system with the following name pattern: **`C4C_<GroupDisplayName>`**. This way SAP Sales Cloud and SAP Service Cloud groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Sales Cloud and SAP Service Cloud groups will be read and provisioned to the target system with their actual display names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

The Identity Provisioning offers a default transformation for the *SAP Sales Cloud and SAP Service Clouds* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in C4C. For more information, see Manage Transformations [page 2439].

Default transformation for connector version 3

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.emails[?(@.primary == true)].value",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.userType",
        "optional": true,
        "targetPath": "$.userType"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.middleName",
        "optional": true,
        "targetPath": "$.name.middleName"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
```

```
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "functions": [
          {
            "condition": "'%c4c.group.prefix%' !== 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "applyOnAttribute": "display",
            "prefix": "%c4c.group.prefix%"
          }
        ]
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%c4c.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%c4c.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {  "condition": "'%ips.application.id%' !== 'null'",
         "constant": "%ips.application.id%",
         "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
```

```
        "sourcePath": "$.members",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.members",
        "optional": true
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      }
    ]
  }
}
```

Default transformation for connector version 4

(Applicable for SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 only)

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.emails[?(@.primary == true)].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "optional": true,
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.userType",
        "targetPath": "$.userType"
      },
      {
        "sourcePath": "$.displayName",
        "optional": true,
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.externalId",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
```

```
      },
      {
        "sourcePath": "$.name.middleName",
        "optional": true,
        "targetPath": "$.name.middleName"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.formatted",
        "optional": true,
        "targetPath": "$.name.formatted"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['status']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['status']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
```

```
      {
        "sourcePath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups",
        "functions": [
          {
            "function": "concatString",
            "condition": "'%c4c.group.prefix%' !== 'null'",
            "applyOnElements": true,
            "applyOnAttribute": "display",
            "prefix": "%c4c.group.prefix%"
          }
        ]
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "function": "concatString",
            "condition": "'%c4c.group.prefix%' !== 'null'",
            "prefix": "%c4c.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      },
      {
        "sourcePath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members"
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
```

```
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      }
    ]
  }
}
```

5. Add a target system to provision users to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.37 SAP Signavio Process Transformation Suite

Follow this procedure to set up SAP Signavio Process Transformation Suite as a source system.

Prerequisites

> **Note**
>
> The initial setup of SAP Signavio Process Transformation Suite is completed during the onboarding process. This setup includes establishing a connection between Identity Provisioning and SAP Signavio Process Transformation Suite using certificate-based authentication. Any changes needed after the initial setup should be processed through a support incident.

Context

SAP Signavio Process Transformation Suite helps you understand, improve, and transform your business processes – fast and at scale – with a cloud-based process management platform.

You can use Identity Provisioning to configure SAP Signavio Process Transformation Suite as a source system where you can read users and groups from and provision them to a target system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP Signavio Process Transformation Suite* as a source system. See: Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	The URL to the SCIM API of your SAP Signavio Process Transformation Suite system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *ClientCertificateAuthentication*
`sig.tenant.id`	Enter the identifier of your SAP Signavio Process Transformation Suite tenant.
`sig.group.members.paging.enabled`	This property enables paging of group members. Default value: *true*
`sig.user.groups.paging.enabled`	This property enables paging of user's groups. Default value: *true*
`sig.group.members.page.size`	The number of group members that SAP Signavio Process Transformation Suite returns per request when reading a group. Default value = 1000
`sig.user.groups.page.size`	The number of user groups that SAP Signavio Process Transformation Suite returns per request when reading a user. Default value = 1000
(Optional) `sig.user.filter`	When specified, only those SAP Signavio Process Transformation Suite users matching the filter expression will be read. For example: • Value = *userName eq "JohnSmith"* • Value = *userName sw "J"*

Property Name	Value
(Optional) `sig.group.filter`	When specified, only those SAP Signavio Process Transformation Suite groups matching the filter expression will be read. For example: • Value = *displayName eq "ProjectTeam1"* • Value = *displayName sw "P"*
`sig.group.prefix`	This property distinguishes SAP Signavio Process Transformation Suite groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: SIG_ You can use the example value or provide your own. When set in the source system, the prefix will be prepended to the name of the groups that are read from the SAP Signavio Process Transformation Suite source system and will be provisioned to the target system with the following name pattern: SIG_. This way SAP Signavio Process Transformation Suite groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Signavio Process Transformation Suite groups will be read and provisioned to the target system with their actual display names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Signavio Process Transformation Suite* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your system. For more information, see:

   Manage Transformations [page 2439]

   Mapping logic – the behavior of the default transformation logic is to read all user attributes from the source SAP Signavio Process Transformation Suite system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

   Default transformation:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.externalId",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.middleName",
        "optional": true,
        "targetPath": "$.name.middleName"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.honorificPrefix",
        "optional": true,
        "targetPath": "$.name.honorificPrefix"
      },
      {
        "sourcePath": "$.name.honorificSuffix",
        "optional": true,
        "targetPath": "$.name.honorificSuffix"
      },
      {
        "sourcePath": "$.displayName",
        "optional": true,
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.nickName",
        "optional": true,
        "targetPath": "$.nickName"
      },
      {
        "sourcePath": "$.profileUrl",
        "optional": true,
        "targetPath": "$.profileUrl"
      },
      {
        "sourcePath": "$.title",
        "optional": true,
```

```
        "targetPath": "$.title"
      },
      {
        "sourcePath": "$.userType",
        "optional": true,
        "targetPath": "$.userType"
      },
      {
        "sourcePath": "$.preferredLanguage",
        "optional": true,
        "targetPath": "$.preferredLanguage"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['emails']",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['emails']"
      },
      {
        "sourcePath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.phoneNumbers"
      },
      {
        "sourcePath": "$.ims",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.ims"
      },
      {
        "sourcePath": "$.photos",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.photos"
      },
      {
        "sourcePath": "$.addresses",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.addresses"
      },
      {
        "sourcePath": "$.entitlements",
        "preserveArrayWithSingleElement": true,
```

```
        "optional": true,
        "targetPath": "$.entitlements"
      },
      {
        "sourcePath": "$.roles",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.roles"
      },
      {
        "sourcePath": "$.x509Certificates",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.x509Certificates"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['division']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']"
      },
```

```
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['userUuidHistory']",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuidHistory']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['status']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['status']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mfaEnabled']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mfaEnabled']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['totpEnabled']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['totpEnabled']"
      },
      {
```

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['loginTime']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['loginTime']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['contactPreferences']",
        "preserveArrayWithSingleElement": true,
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['contactPreferences']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['passwordDetails']",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['passwordDetails']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sncName']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sncName']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['groupDomains']",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['groupDomains']"
      },
      {
        "sourcePath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups"
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%sig.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%sig.group.prefix%"
          }
```

```
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "sourcePath": "$.externalId",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "sourcePath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members"
      }
    ]
  }
}
```

5. Add a target system to provision users to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Signavio Process Transformation Suite

1.6.1.38 SAP S/4HANA Cloud Public Edition

Follow this procedure to set up SAP S/4HANA Cloud Public Edition as a source system.

Prerequisites

To establish the connection between Identity Provisioning and SAP S/4HANA Cloud Public Edition, you need to set up the communication (user, system and arrangement) on SAP S/4HANA Cloud Public Edition. You have the option to do it either now, as a prerequisite, or while configuring SAP S/4HANA Cloud Public Edition as a source system, as described in step 3.

Context

SAP S/4HANA Cloud Public Edition is a complete enterprise resource planning (ERP) system with built-in intelligent technologies and advanced analytics.

You can use Identity Provisioning to configure SAP S/4HANA Cloud Public Edition as a source system where you can read entities from and provision them to target systems of your choice.

SAP S/4HANA Cloud Public Edition provides different APIs for integration with Identity Provisioning, resulting in different connector versions for each API. Each connector version supports specific attribute mappings within the transformations and requires particular property values. The value of the `s4hana.cloud.api.version` property controls which API you use. By default, the Identity Provisioning service uses version *1*.

- When the value is set to *1* or the property is not present (typical for systems created before versioning was introduced on November 6, 2025), the SOAP API provided by the communication scenario SAP_COM_0193 is used. It supports reading **business users** (Employee, Contingent Worker), **user assignments**, and **business roles** (which are considered as *groups*).
- When the value is set to *3*, the SCIM interface provided by the communication scenario SAP_COM_0465 is used. It supports provisioning of **business users** with their **assignments** to groups of type **userGroup** and **authorization**. For more information, see Groups.

For more information on how to update to version *3*, see Update SAP S/4HANA Cloud Public Edition [page 2435].

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP S/4HANA Cloud Public Edition* as a source system. For more information, see Add New Systems [page 2415].

3. Set up the communication between Identity Provisioning and SAP S/4HANA Cloud Public Edition and configure your authentication method (basic or certificate-based).

> ### Note
>
> We recommend that you use certificate-based authentication.

   a. In your newly added SAP S/4HANA source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP S/4HANA backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Create a communication system and assign the created user to the communication system.

      For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

   d. Create a communication arrangement with the created system.

      - When `s4hana.cloud.api.version=1` is set in Identity Provisioning for the SAP S/4HANA Cloud Public Edition source system, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

        > ### Note
        >
        > The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
        >
        > The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

      - When `s4hana.cloud.api.version=3` is set in Identity Provisioning for the SAP S/4HANA Cloud Public Edition source system, choose *Scenario ID* SAP_COM_0465 (System for Cross-domain Identity Management Integration).

        > ### Note
        >
        > The communication scenario *SAP_COM_0465* allows you to maintain data via SCIM (System for Cross-domain Identity Management). For more information, see Integrate Cross-Domain Identity Management.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the SAP S/4HANA Cloud Public Edition API URL. You can find the correct URL in the *API-URL* field of the communication arrangement set up for the communication scenario used. For example: `https://my123456-api.s4hana.ondemand.com`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`s4hana.cloud.skip.read.archived`	In the event of archived (disabled) entities in a source SAP S/4HANA Cloud Public Edition system, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to **false**, or delete it. **Relevant for connector version 1**
`ips.date.variable.format`	*yyyy-MM-dd*

Property Name	Description & Value
`s4hana.cloud.api.version`	The version of the system API you use. By default, Identity Provisioning uses version *1*. Version *1* means your SAP S/4HANA Cloud Public Edition system consumes SAP S/4HANA Cloud Public Edition API: Business User. Version *3* means your SAP S/4HANA Cloud Public Edition system consumes System for Cross-domain Identity Management (SCIM) API.
(Optional) `s4hana.cloud.roles.filter`	Enter OData filtering for reading roles in the S/4HANA system. To learn what criteria you can use, see: OData URI Conventions → **4.5 Filter System Query Option** **Relevant for connector version 1**
(Optional) `s4hana.cloud.group.filter`	When specified, only those SAP S/4HANA Cloud Public Edition groups matching the filter expression will be read. Supported operators: eq (equal), co (contains), and sw (starts with). **Possible values:**: • `displayName` • `externalId` • `urn:ietf:params:scim:schemas:extension:sap:2.0:Group:type` For more information, see List of Properties [page 80]. **Relevant for connector version 3**

Property Name	Description & Value
(Optional) `s4hana.cloud.user.filter`	When specified, only those SAP S/4HANA Cloud Public Edition users matching the filter expression will be read. You can filter users by **list of attributes** according to the API syntax of SAP S/4HANA Cloud Public Edition. You can set a single attribute or multiple ones as filter criteria. If you enter multiple attributes (using OR operator), the filter will search for any of them. Supported operators: eq (equal), ne (not equal), sw (starts with), ew (ends with) and co (contains) **Possible values:** • `userName` • `externalId` • `emails[0].value` For example: • *userName eq "johnbrown" and externalId eq "P000252"* • *userName sw "J" or externalId eq "P000252"* • *userName eq "johnbrown" and emails[0].value eq "johnbrown@email.com"* • *userName eq "johnbrown" and emails[0].value ne "johnbrown@email.com" or externalId eq "P000252"* **Note** These combinations are valid for both 'or' and 'and' operators. **Relevant for connector version 3**
(Optional) `s4hana.cloud.roles.page.size`	Indicate how many business roles (considered as *groups*) per page to be read from your SAP S/4HANA Cloud Public Edition system. The value must be an integer number. **Relevant for connector version 1**

Property Name	Description & Value
(Optional) `s4hana.cloud.group.prefix`	This property distinguishes SAP S/4HANA Cloud Public Edition groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`S4HANACLOUD_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP S/4HANA Cloud Public Edition source system and will be provisioned to the target system with the following name pattern: **`S4HANACLOUD_<GroupDisplayName>`**. This way SAP S/4HANA Cloud Public Edition groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP S/4HANA Cloud Public Edition groups will be read and provisioned to the target system with their actual display names. For more information, see List of Properties [page 80]. **Relevant for connector version 3**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://my1234567-api.s4hana.ondemand.com
User=MyS4HANAuser
Password=************
s4hana.cloud.api.version=1
ips.date.variable.format=yyyy-MM-dd
s4hana.cloud.skip.read.archived=true
s4hana.cloud.roles.filter=startswith(ID, 'EMPLOYEE_LEVEL_3') eq true
s4hana.cloud.roles.page.size=30
```

5. (Optional) Configure the transformations.

Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP S/4HANA Cloud Public Edition* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules depending on your setup of entities in your SAP S/4HANA Cloud Public Edition. For more information, see:

Manage Transformations [page 2439]

SAP S/4HANA Cloud API: Business User

System for Cross-domain Identity Management (SCIM)

Integrate Cross-Domain Identity Management

Default transformation for connector version 1:

Code Syntax

```
{
  "user": {
    "condition": "($.user.validityPeriod.startDate <= '${currentDate}') &&
($.user.validityPeriod.endDate > '${currentDate}')",
    "mappings": [
      {
        "sourcePath": "$.personID",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.personalInformation.firstName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.lastName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.middleName",
        "targetPath": "$.name.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.personalInformation.personFullName",
        "targetPath": "$.name.formatted",
        "optional": true
      },
      {
        "sourcePath": "$.user.userName",
        "targetPath": "$.userName",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "condition": "$.user.lockedIndicator == 'true'",
        "constant": false,
        "targetPath": "$.active",
        "optional": true
```

```
      },
      {
        "sourcePath": "$.workplaceInformation.emailAddress",
        "targetPath": "$.emails[0].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.user.logonLanguageCode",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.PersonExternalID",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.user.globalUserID",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$.user.role",
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "functions": [
          {
            "condition": "'%s4hana.cloud.roles.prefix%' !== 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "%s4hana.cloud.roles.prefix%",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "display"
          },
          {
            "condition": "'%s4hana.cloud.roles.prefix%' === 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "display"
          },
          {
            "function": "concatString",
            "applyOnElements": true,
            "prefix": "",
            "applyOnAttribute": "roleName",
            "assignToAttribute": "value"
          }
        ]
      },
      {
        "type": "remove",
        "targetPath": "$.groups[*].roleName"
      },
      {
        "sourcePath": "$.user.timeZoneCode",
        "optional": true,
        "targetPath": "$.timezone",
        "functions": [
          {
            "type": "convertTimezoneCode",
            "outputFormat": "iana"
          }
        ],
        "defaultValue": "Europe/Berlin"
```

```
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.businessPartnerRoleCode"
        ],
        "targetPath": "$.userType",
        "defaultValue": "Employee",
        "valueMappings": [
          {
            "key": [
              "BUP003"
            ],
            "mappedValue": "Employee"
          },
          {
            "key": [
              "BBP005"
            ],
            "mappedValue": "Contingent Worker"
          }
        ]
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.ID",
        "targetPath": "$.displayName",
        "targetVariable": "entityIdSourceSystem",
        "functions": [
          {
            "condition": "'%s4hana.cloud.roles.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%s4hana.cloud.roles.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.ID",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schemas[0]"
      },
      {
        "constant": "authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {
        "constant": "readWrite",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      },
      {  "condition": "'%ips.application.id%' !== 'null'",
         "constant": "%ips.application.id%",
         "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "sourcePath": "$.to_BusinessUserAssignment.results",
        "optional": true,
        "preserveArrayWithSingleElement": true,
```

```
        "targetPath": "$.members"
      },
      {
        "type": "remove",
        "targetPath": "$.members[*].__metadata"
      },
      {
        "type": "remove",
        "targetPath": "$.members[*].UserName"
      },
      {
        "type": "rename",
        "constant": "value",
        "targetPath": "$.members[*].PersonID"
      },
      {
        "constant": "User",
        "targetPath": "$.members[*].type"
      }
    ]
  }
}
```

By default, Identity Provisioning reads group IDs and members. If you want the service to also read group descriptions, you can add an extra mapping to the *"group"* resource. To learn how, see Guided Answers: Business Role Description.

Default transformation for connector version 3:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.name.formatted",
        "optional": true,
        "targetPath": "$.name.formatted"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
```

```
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.displayName",
        "optional": true,
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.phoneNumbers",
        "optional": true
      },
      {
        "sourcePath": "$.userType",
        "optional": true,
        "targetPath": "$.userType"
      },
      {
        "condition": "$.preferredLanguage !== 'not_unique'",
        "sourcePath": "$.preferredLanguage",
        "optional": true,
        "targetPath": "$.preferredLanguage"
      },
      {
        "condition": "$.locale !== 'not_unique'",
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['language']['correspondence']",
```

```
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['language']['correspondence']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['numericalRepresentation']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['numericalRepresentation']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['date']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['date']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['time']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['time']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['userUuidHistory']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['userUuidHistory']",
        "optional": true,
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "functions": [
            {
                "function": "concatString",
                "condition": "'%s4hana.cloud.group.prefix%' !== 'null'",
                "applyOnElements": true,
                "applyOnAttribute": "display",
                "prefix": "%s4hana.cloud.group.prefix%"
            }
        ]
      }
    ]
  },
  "group": {
    "mappings": [
```

```
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
            {
                "condition": "'%s4hana.cloud.group.prefix%' !== 'null'",
                "function": "concatString",
                "prefix": "%s4hana.cloud.group.prefix%"
            }
        ]
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type'] ==
'userGroup'",
        "sourcePath": "$.externalId",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "functions": [
          {
            "function": "compositeId",
            "separator": ":",
            "subId": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
          },
          {
            "condition": "'%s4hana.cloud.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%s4hana.cloud.group.prefix%"
          }
        ]
      },
      {
```

```
        "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type'] ==
'authorization'",
        "sourcePath": "$.externalId",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "functions": [
          {
            "condition": "'%s4hana.cloud.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%s4hana.cloud.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      }
    ]
  }
}
```

Group Uniqueness and Provisioning

In the most common scenario, SAP S/4HANA Cloud Public Edition connector version 3 is used as the source system for groups and Identity Authentication version 2 or Local Identity Directory is the target system.

In SAP S/4HANA Cloud Public Edition version 3, the uniqueness of a group is defined by the combination of `externalId` and `type` attributes. The system supports two group types: `userGroup` and `authorization`. As a result, two different groups can share the same `externalId` while belonging to different types, potentially causing conflicts when provisioned to the target system.

To avoid conflicts, ensure the following:

- Your SAP S/4HANA Cloud Public Edition system is updated to version 3, as described in Update SAP S/4HANA Cloud Public Edition [page 2435].
- Your target system (Identity Authentication version 2 or Local Identity Directory) has the respective group uniqueness property: `ias.group.unique.attribute` or `idds.group.unique.attribute`, set with the following combination of values: `['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']`, `['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']`, `['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']`. For more information, see List of Properties [page 80].
  Because the `externalId` attribute in SAP S/4HANA Cloud Public Edition connector version 3 is mapped to the `name` attribute in the target, when a group of type `userGroup` is provisioned from the source to the target, the suffix `:userGroup` is automatically appended to its `name`. This ensures that even if two groups share the same `externalId`, their names remain unique in the target system.

6. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP S/4HANA Cloud Documentation

1.6.1.39 SAP S/4HANA for procurement planning

Follow this procedure to set up SAP S/4HANA for procurement planning as a source system.

Prerequisites

You have technical credentials for SAP S/4HANA for procurement planning. See: Onboadring

Context

SAP S/4HANA for procurement planning is a cloud-based solution designed to help you plan procurement activities with regard to the time schedule, as well as the investment planning of items based on a central bill of material.

You can use Identity Provisioning to configure SAP S/4HANA for procurement planning as a source system where you can read **users** from and provision them to a target system.

> **Note**
>
> SAP S/4HANA for procurement planning does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.

2. Add *SAP S/4HANA Procurement Planning* as a source system. See: Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the SCIM API of your SAP S/4HANA for procurement planning system without path information. For example: `https://procplanning-api.cfapps.eu10.hana.ondemand.com`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, created for your SAP S/4HANA for procurement planning system.
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP S/4HANA for procurement planning system.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. For example: **https://procplansecurity.authentication.eu10.hana.ondemand.com/oauth/token**
(Optional) `s4hana.pp.user.filter`	When specified, only those SAP S/4HANA for procurement planning users matching the filter expression will be read. Example: **name.familyName eq "Smith" and addresses.country eq "US"**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP S/4HANA for procurement planning* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your system. For more information, see:

   Manage Transformations [page 2439]

   SAP Business Accelerator Hub: SAP S/4HANA for Procurement Planning

Mapping logic – the behavior of the default transformation logic is to read all user attributes from the source SAP S/4HANA for procurement planning system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

Default transformation:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                 "sourcePath": "$.userName",
                 "targetPath": "$.userName",
                 "correlationAttribute": true
            },
            {
                 "sourcePath": "$.displayName",
                 "targetPath": "$.displayName"
            },
            {
                 "sourcePath": "$.active",
                 "targetPath": "$.active"
            },
            {
                 "sourcePath": "$.emails",
                 "targetPath": "$.emails",
                 "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.emails[?(@.primary== true)].value",
                "optional": true,
                "correlationAttribute": true
            },
            {
                "type": "remove",
                "targetPath": "$.meta"
            }
        ]
    }
}
```

5. Add a target system to provision users to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP S/4HANA for procurement planning – Product Page

1.6.1.40 SAP S/4HANA On-Premise

Follow this procedure to set up SAP S/4HANA on-premise (also valid for SAP S/4HANA Cloud, private edition) as a source system.

Prerequisites

> **Note**
>
> If you have purchased the Identity Provisioning service between **September 1, 2020** and **October 20, 2020**, and you want to make a connection to this on-premise system, follow the procedure in: Connect to On-Premise Systems in SAP Cloud Identity Infrastructure.

- You have installed the Cloud Connector in your corporate environment and have done the initial configuration. For more information, see: Cloud Connector (Neo)
- You have technical credentials (user and password) for SAP S/4HANA on-premise.
- The SAP S/4HANA on-premise system is version **1809** or higher.
- You have configured your SOA Manager to directly call the following Web services:
  - **ManageBusinessUserIn**
  - **QueryBusinessUserIn**

  For more information, see: Setting Up SOA Manager.

Context

You can use Identity Provisioning to configure SAP S/4HANA on-premise as a source system where you can read **business users** (employee, collaboration user, contingent worker, and resource) from and provision them to a target system.

In SAP S/4HANA, a business user is defined as a natural person who is represented by a business partner and a link to a user in the system. Actually, the business user is an AS ABAP (SU01) user who also has a one-to-one relation to a corresponding business partner. For more information on the identity model for business users, see SAP Note 2570961.

SAP S/4HANA on-premise supports reading of users with *User UUID* attribute which is generated by Identity Authentication at user creation. The attribute mapping is handled by the default transformation of SAP AS ABAP connector. Therefore, in order to read a user with *User UUID* from S/4HANA on-premise, the user should first be created (provisioned) in AS ABAP and then linked to its corresponding business user in S/4HANA on-premise. For more information, see: SAP Application Server ABAP [page 628].

Reading of **roles** (considered as *groups*) is also handled by the default transformation SAP AS ABAP connector.

Procedure

1. Open the Cloud Connector to add an access control system mapping for **SAP S/4HANA On-Premise**. This is needed to allow the Identity Provisioning service to access SAP S/4HANA On-Premise as a back-end system on the intranet. To learn how, see: Configure Access Control (HTTP)
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
3. Add *SAP S/4HANA On-Premise* as a source system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to your SAP S/4HANA On-Premise system.
`ProxyType`	Enter: *OnPremise*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the technical user for SAP S/4HANA On-Premise.
`Password`	(Credential) Enter the password for the SAP S/4HANA On-Premise technical user.
`s4hana.onprem.skip.read.archived`	In the event of archived (disabled) entities in a source SAP S/4HANA On-Premise system, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to **false**, or delete it.
`ips.date.variable.format`	*yyyy-MM-dd*
(Optional) `s4hana.onprem.sap-client`	Use this property if you want to specify a particular AS ABAP client to use as the **sap-client** URL parameter. If this property is not specified, the URL will open your default AS ABAP client. To learn more, see: Specifying the Client For more information about **sap-client**, see: SAP URL Parameters

Property Name	Description & Value
(Optional)`CloudConnectorLocationId`	Relevant when the `ProxyType` property is set to *OnPremise*. Use it in case your SAP Business Technology Platform account uses more than one Cloud Connector or when the Cloud Connector uses `LocationID`.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*OnPremise*

`URL`=*http://aaa777.myhost:1234*

`User`=*MYS4HANAUSER*

`Password`=**********

`ips.date.variable.format`=*yyyy-MM-dd*

`s4hana.onprem.skip.read.archived`=*true*

`s4hana.onprem.sap-client`=*101*

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP S/4HANA On-Premise* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules depending on your setup of entities in SAP S/4HANA On-Premise. For more information, see:

   Manage Transformations [page 2439]

   APIs for Business User Management

   Default transformation:

   Code Syntax

```
{
  "user": {
    "condition": "($.validityPeriod.startDate <= '${currentDate}') &&
($.validityPeriod.endDate > '${currentDate}')",
    "mappings": [
      {
        "sourcePath": "$.personID",
        "targetVariable": "entityIdSourceSystem"
      },
```

```
        {
          "sourcePath": "$.personalInformation.firstName",
          "targetPath": "$.name.givenName",
          "optional": true
        },
        {
          "sourcePath": "$.personalInformation.lastName",
          "targetPath": "$.name.familyName",
          "optional": true
        },
        {
          "sourcePath": "$.personalInformation.middleName",
          "targetPath": "$.name.middleName",
          "optional": true
        },
        {
          "sourcePath": "$.personalInformation.personFullName",
          "targetPath": "$.name.formatted",
          "optional": true
        },
        {
          "sourcePath": "$.personalInformation.nickName",
          "targetPath": "$.nickName",
          "optional": true
        },
        {
          "sourcePath": "$.userAssignment.userID",
          "targetPath": "$.userName",
          "optional": true,
          "correlationAttribute": true
        },
        {
          "constant": true,
          "targetPath": "$.active"
        },
        {
          "sourcePath": "$.workplaceInformation.emailAddress",
          "targetPath": "$.emails[0].value",
          "optional": true,
          "correlationAttribute": true
        },
        {
          "sourcePath": "$.PersonExternalID",
          "optional": true,
          "correlationAttribute": true
        },
        {
          "type": "valueMapping",
          "sourcePaths": [
            "$.businessPartnerRoleCode"
          ],
          "targetPath": "$.userType",
          "defaultValue": "Employee",
          "valueMappings": [
            {
              "key": [
                "BUP003"
              ],
              "mappedValue": "Employee"
            },
            {
              "key": [
                "BBP005"
              ],
              "mappedValue": "Contingent Worker"
            },
            {
              "key": [
```

```
              "BUP012"
            ],
            "mappedValue": "Collaboration User"
          },
          {
            "key": [
              "WFM001"
            ],
            "mappedValue": "Resource"
          }
        ]
      }
    ]
  }
}
```

6. Now, add a target system to provision users into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP S/4HANA On-Premise
APIs for Business User Management
Maintain Collaboration Users

1.6.1.41 SAP SuccessFactors

Follow this procedure to set up SAP SuccessFactors as a source system.

Prerequisites

- You are using **SAP SuccessFactors HCM Suite OData API**
  You have created a technical user with permissions to **call** the SAP SuccessFactors HCM Suite OData API and to **export** employee data from the SAP SuccessFactors system. For more information, see Permissions and Setting Up an API User for Sync Jobs in SAP SuccessFactors. The following permissions are required:
  - *Manage User* > *Employee Export*

- *Manage User* > *User Account OData API Entity*
- *Manage Integration Tools* > *Allow Admin to Access OData API*
- *Manage Role-Based Permission Access* > *Role-Based Permission Admin*

- You are using **SAP SuccessFactors Workforce SCIM API**
  Consumers of the SAP SuccessFactors Workforce SCIM API can either create a technical user and assign the necessary permissions, or use the predefined technical user with built-in permissions for communication between SAP SuccessFactors and Identity Provisioning using certificate authentication. For more information, see Upgrade to X.509 Certificate-Based Authentication for Incoming Calls and Permissions.

Context

Companies that manage their employees using SAP SuccessFactors HCM Suite can use the Identity Provisioning service to automatically create accounts for these employees and manage their permissions for the cloud applications. When the hiring process of a new employee is completed in the SAP SuccessFactors HCM solution, a user record with the employee identity data is created in the SAP SuccessFactors system. The Identity Provisioning service can use this data for the identity and authorization provisioning processes.

SAP SuccessFactors provides two APIs for its integration with Identity Provisioning: SAP SuccessFactors HCM Suite OData API and SAP SuccessFactors Workforce SCIM API. The value of `sf.api.version` property controls which API you use.

- When the value is set to **`1`**, or the property is not defined - SAP SuccessFactors HCM Suite OData API (in short, OData API) is used. This is the default value. SAP SuccessFactors source systems created before the introduction of `sf.api.version` property, use OData API.
- When the value is set to **`2`** - SAP SuccessFactors Workforce SCIM API (in short, SCIM API) is used. For more information on how to update to version 2, see Update Connector Version [page 2424].

Identity Provisioning supports reading of SAP SuccessFactors users, as well as static and dynamic groups of type Permission.

When users from SAP SuccessFactors source system are provisioned to Identity Authentication target system, newly created and updated users get the *userUUID* attribute generated by Identity Authentication. The Identity Provisioning in its turn sends the *userUUID* back to SAP SuccessFactors to update the users.

To create SAP SuccessFactors as a source system, proceed as follows:

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP SuccessFactors* as a source system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP SuccessFactors and configure the authentication method (basic or certificate-based).

> ### Note
>
> We recommend that you use certificate-based authentication.

a. In your newly added SAP SuccessFactors source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

   Skip this step if you use basic authentication. The next steps are performed in SAP SuccessFactors Admin Center and are relevant for certificate-based authentication only.

b. Login to SAP SuccessFactors and go to *Admin Center*. Follow the procedure described in Upgrade to X.509 Certificate-Based Authentication for Incoming Calls.

   Make sure you select *Identity Provisioning Service* in the *Integration Name* field.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
Type	Enter: *HTTP*

Property Name	Description & Value

URL	Specify the URL to your SAP SuccessFactors API.
	
	For example:
	• For version 1:
	  • When *BasicAuthentication* is configured as authentication method: **`https://apitest.successfactors.com/odata/v2`**
	  • When *ClientCertificateAuthentication* is configured as authentication method: **`https://apitest.cert.successfactors.com/odata/v2`**
	• For version 2:
	  • When *BasicAuthentication* is configured as authentication method: **`https://apitest.successfactors.com`**
	  • When *ClientCertificateAuthentication* is configured as authentication method: **`https://apitest.cert.successfactors.com`**
	
	To see the list of all SAP SuccessFactors data centers, see: SAP SuccessFactors API Reference Guide (OData V2): List of SAP SuccessFactors API Servers and System for Cross-domain Identity Management for Workforce in SuccessFactors

`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
(Optional) `sf.api.version`	Handles the version of the API which is consumed by the SAP SuccessFactors system. sf.user.attributes.expand **Possible values:** • *1* - Indicates that SAP SuccessFactors HCM Suite OData API (in short, OData API) is used. • *2* - Indicates that SAP SuccessFactors Workforce SCIM API (in short, SCIM API) is used. Default value: *1*

Property Name	Description & Value
`User`	Valid if *BasicAuthentication* Enter the *userID* of your SAP SuccessFactors technical user in the following format: is configured as authentication method.**<user_ID>@<company_ID>**
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the password for your SAP SuccessFactors technical user.
`sf.company.id`	Valid if *ClientCertificateAuthentication* is is configured as authentication method. Enter the Company ID of your SAP SuccessFactors system. The Company ID is a short string of characters that identifies each SAP SuccessFactors system. It is like a username for your organization. All users of the same system share the same Company ID.

Property Name	Description & Value
`sf.user.attributes`	Default property. It's a string representing a comma-separated list of user attributes that have to be loaded (read) from SAP SuccessFactors. You can leave the default property value (all listed attributes), leave only some of them, or add some more. **Note** If you want to add more attributes, make sure you have added: • the relevant extra attributes to the value of this property, separated by commas • extra mappings for these attributes in the *user* transformation • extra mappings for these attributes in the write transformation of the relevant target system **→ Remember** • Always make sure that attribute `lastModifiedDateTime` is in the list of values. If you delete it (don't specify it), the provisioning from SAP SuccessFactors will fail. • If a user in SAP SuccessFactors is missing the `lastModifiedDateTime` attribute, it will break the provisioning. As a solution, you can exclude such users from the provisioning – either by using the `sf.user.filter` property, or by setting a condition in the transformation logic. **Connector version**: SAP SuccessFactors version 1

Property Name	Description & Value
(Optional) `sf.user.attributes.expand`	This property reads additional user data related to *complex (navigation) attributes*, which are specified in `sf.user.attributes`. Default value: *personKeyNav,personKeyNav/userAccountNav* **Note** If you want to add more complex attributes, enter the correct "paths" to them, specifying all parent attribute levels. For example, if you want to read information about user's skills and competency, add *positionSkillMappings/skill*. In this case, make sure you have added: • **positionSkillMappings** to the value of property `sf.user.attributes` • a relevant mapping for this attribute in the *user* transformation • a relevant mapping for this attribute in the write transformation of the relevant target system **Connector version**: SAP SuccessFactors version 1
(Optional) `sf.user.filter`	The possible values of this property depend on the API version which your SAP SuccessFactors system consumes. Use this property to filter users from SAP SuccessFactors. The filter obtains values as described in the OData 2.0 syntax, except any statements with attribute `lastModifiedDateTime`. Find below example syntax for filtering users depending of the API version: • OData API - **`username eq 'cbraun'`** • SCIM API - **`userName eq "cbraun"`** To learn more, see: • OData version 2 → **4.5. Filter System Query Option ($filter)**. • SAP SuccessFactors API Reference Guide (OData V2) • SAP SuccessFactors Workforce SCIM API and System for Cross-domain Identity Management for Workforce in SuccessFactors

Property Name	Description & Value
(Optional) `sf.user.read.deactivatedafter`	This property filters SAP SuccessFactors inactive users from a particular date on. It is an optional property which does not appear by default at system creation. It accepts a value in the **`yyyy-MM-dd`** format. For example: **`2023-07-17`** The `sf.user.read.deactivatedafter` property works together with the `sf.user.filter` property which is added at system creation with the default value: *active eq true*. Using it can further narrow down the filtering results. To filter active users along with inactive ones from a particular date on, the following configuration must be in place: • Set the `sf.user.read.deactivatedafter` value to a date in the expected format. For example: *2023-07-17* • `sf.user.filter` = *active eq true* As a result, Identity Provisioning reads SAP SuccessFactors active users and the users set to inactive from that date on using the 2023-07-17T00:00:00Z date-time format. Depending on the value you define for `sf.user.filter`, expect the following results: • `sf.user.filter` = *active eq false* All inactive users will be returned. • `sf.user.filter` = *active eq false and userName sw "Test_"* All inactive users with username starting with Test_ will be returned. **Note** When you filter by `sf.user.filter` = *active eq false* along with the property `sf.user.read.deactivatedafter`, the users that match the two critera will be read twice. • `sf.user.filter` = *active eq true and userName sw "Test_"* Inactive users from the provided date on and all active users with username starting with Test_ will be returned.

Property Name	Description & Value
	Connector version: SAP SuccessFactors version 2
(Optional) `sf.group.filter`	The possible values of this property depend on the API version which your SAP SuccessFactors system consumes. Use this property to filter dynamic groups from SAP SuccessFactors. The filter obtains values as described in the OData 2.0 syntax, except any statements with attribute `lastModifiedDateTime`. To learn more, see: • OData version 2 → **4.5. Filter System Query Option ($filter)** • SAP SuccessFactors API Reference Guide (OData V2) → **DynamicGroup** • SAP SuccessFactors Workforce SCIM API and System for Cross-domain Identity Management for Workforce in SuccessFactors
(Optional) `sf.group.prefix`	This property distinguishes SAP SuccessFactors groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **SF_** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP SuccessFactors source system and will be provisioned to the target system with the following name pattern: **SF_<GroupDisplayName>**. This way SAP SuccessFactors groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP SuccessFactors groups will be read and provisioned to the target system with their actual display names. **Connector version**: SAP SuccessFactors version 2
(Optional) `sf.page.size`	Defines the paging size. • Default value: **100** • Maximum value: **1000** **Connector version**: SAP SuccessFactors version 1

Property Name	Description & Value
(Optional)`sf.group.members.paging.enabled`	This property enables paging of group members. The maximum number of group members returned per request is 100. To read more than 100 group members, paging must be enabled. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *true* **Connector version**: SAP SuccessFactors version 2

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination (configuration):

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*https://apitest.successfactors.com/odata/v2*

`User`=*sfsf_admin@mycompany.com*

`Password`=**********

`sf.user.attributes`=*userId,username,status,email,lastName,firstName,lastModifiedDateTime,personKeyNav,positionSkillMappings*

=*personKeyNav,personKeyNav/userAccountNav,positionSkillMappings/skill*

`sf.user.filter`=*department ne 'Manufacturing'*

`sf.group.filter`=*groupType eq 'permission'*

`sf.page.size`=*70*

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP SuccessFactors* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   When the SAP SuccessFactors system is configured as a source, the Identity Provisioning service will read all the attributes of the user records supported by the SAP SuccessFactors API.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP SuccessFactors. For more information, see:

Manage Transformations [page 2439]

SAP SuccessFactors HCM Suite OData API

SAP SuccessFactors Workforce SCIM API

Default transformation for SAP SuccessFactors HCM Suite OData API version 1:

Code Syntax

```
// The value of entityIdSourceSystem is used for storing the unique ID of
each user.
// You can change the default source attribute perPersonUuiD, but make
sure the new source attribute is also unique.
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.personKeyNav.perPersonUuid",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "constant": false,
        "targetPath": "$.active"
      },
      {
        "condition": "$.personKeyNav.userAccountNav.accountStatus
=='ACTIVE'",
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.personKeyNav.userAccountNav.username",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.username",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "constant": "employee",
        "targetPath": "$.userType"
      },
      {
        "sourcePath": "$.firstName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.lastName",
        "targetPath": "$.name.familyName"
      },
// The email attribute is used as a first value for the emails array of
the intermediate JSON data.
      {
        "sourcePath": "$.email",
        "targetPath": "$.emails[0].value",
        "correlationAttribute": true
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
```

```
        "targetPath": "$.schemas[0]"
      },
      {
        "constant": "urn:sap:cloud:scim:schemas:extension:sfsf:2.0:User",
        "targetPath": "$.schemas[1]"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
        "targetPath": "$.schemas[2]"
      },
      {
        "sourcePath": "$.personKeyNav.perPersonUuid",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:sfsf:2.0:User']['personGUID']"
      },
      {
        "sourcePath": "$.userId",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:sfsf:2.0:User']['userSysID']"
      },
      {
        "sourcePath": "$.personKeyNav.personIdExternal",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "correlationAttribute": true
      }
    ]
  },
// By default, the group mapping is inactive (ignored) but groups are
supported.
// To start provisioning groups, either delete the statement "ignore":
true, or set its value to false.
  "group": {
    "ignore": true,
    "mappings": [
      {
        "sourcePath": "$.groupID",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.groupName",
        "targetPath": "$.displayName"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.users[*].personGUID",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]"
      }
    ]
  }
}
```

Default transformation for SCIM API version 2:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.userType",
        "targetPath": "$.userType"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.middleName",
        "optional": true,
        "targetPath": "$.name.middleName"
      },
      {
        "sourcePath": "$.name.honorificPrefix",
        "optional": true,
        "targetPath": "$.name.honorificPrefix"
      },
      {
        "sourcePath": "$.name.honorificSuffix",
        "optional": true,
        "targetPath": "$.name.honorificSuffix"
      },
      {
        "sourcePath": "$.name.formatted",
        "optional": true,
        "targetPath": "$.name.formatted"
      },
      {
        "sourcePath": "$.nickName",
        "optional": true,
        "targetPath": "$.nickName"
      },
      {
        "sourcePath": "$.preferredLanguage",
        "optional": true,
        "targetPath": "$.preferredLanguage"
      },
      {
        "sourcePath": "$.displayName",
        "optional": true,
        "targetPath": "$.displayName"
```

```
      },
      {
        "sourcePath": "$.title",
        "optional": true,
        "targetPath": "$.title"
      },
      {
        "sourcePath": "$.externalId",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['perPersonUuid']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['perPersonUuid']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['loginMethod']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['loginMethod']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['personIdExternal']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['personIdExternal']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['customFields']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['customFields']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
```

```
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['division']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['$ref']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['$ref']"
      },
      {
        "sourcePath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.phoneNumbers"
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath": "$.emails[?(@.primary == true)].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups",
        "functions": [
          {
```

```
            "condition": "'%sf.group.prefix%' !== 'null'",
            "function": "concatString",
            "applyOnElements": true,
            "applyOnAttribute": "display",
            "prefix": "%sf.group.prefix%"
          }
        ]
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%sf.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%sf.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members"
      }
    ]
  }
}
```

> ### Note
>
> When SAP SuccessFactors users are provisioned to Identity Authentication, the service generates a user UUID for every newly created user. Within the same provisioning job, Identity Provisioning returns the user UUID back to SAP SuccessFactors and updates the *sapGlobalUserId* attribute.
>
> Identity Provisioning updates *sapGlobalUserId* when its value is **`null`** or when its value already exists. The latter is valid in cases where, for example, a user is deleted in the Identity Authentication target system and after running a Resync Job, the deleted user is created again. A new user UUID is generated which updates the old value in SAP SuccessFactors.
>
> Returning the user UUID back to SAP SuccessFactors and updating the *sapGlobalUserId* attribute is supported for standard and real-time provisioning scenarios. It is also supported in bulk scenarios, when bulk operations are enabled on the Identity Authentication target system.
>
> In case you encounter error **statusCode 432** during userUUID synchronization, see: Guided Answers: Error statusCode: 432

6. Now, add a target system to provision users into it. Choose from: Target Systems [page 936]

> ⚠ Caution
>
> The **email** attribute is not unique for *SAP SuccessFactors* but it's unique for systems like *Identity Authentication* or *SAP Analytics Cloud*. That's why, if you provision users to any of these systems, make sure that there are no *SAP SuccessFactors* users with duplicate e-mails. If there are such, then in the target system all affected users will be created as a single user, with merged user data.
>
> To learn more, see *Guided Answers*: Multiple Users from a Source System Are Created as One in the Target

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

URI Conventions (OData Version 2.0)
SAP SuccessFactors HCM Suite OData API
SAP SuccessFactors Workforce SCIM API
(Guided Answers) Multiple Users from SAP SuccessFactors Are Created as One in Identity Authentication

1.6.1.42 SAP SuccessFactors Employee Central Payroll

Follow this procedure to set up SAP SuccessFactors Employee Central Payroll as a source system.

Context

SAP SuccessFactors Employee Central Payroll is a cloud product for payroll processing. It's a service offering where SAP SuccessFactors Employee Central is used for maintenance of core HR master data and SAP HCM Payroll system hosted in an SAP or Hyperscaler data center is used for payroll.

You can use Identity Provisioning to configure SAP SuccessFactors Employee Central Payroll as a source system where you read users and groups from and then provision them to a target system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP SuccessFactors Employee Central Payroll* as a source system. See: Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP SuccessFactors Employee Central Payroll and configure your authentication method (basic or certificate-based).

   > ### Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP SuccessFactors Employee Central Payroll source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next step is performed in SAP SuccessFactors Employee Central Payroll backend system and is relevant for both basic and certificate-based authentication.

   b. For certificate-based authentication, import the certificate generated in the Identity Provisioning UI in your SAP SuccessFactors Employee Central Payroll system, as described in Prerequisite: Finding the Access URL → step 5.

      For basic authentication, provide *User* and *Password*, as described in Prerequisite: Finding the Access URL → step 5.

   c. Save your changes.
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
URL	Enter the URL to your SAP SuccessFactors Employee Central Payroll system. For more information, see Prerequisite: Finding the Access URL → step 8.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication* For more information, see Prerequisite: Finding the Access URL → step 5.

Property Name	Value
`User`	Valid if *BasicAuthentication* is configured as authentication method. For more information, see Prerequisite: Finding the Access URL → step 5.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. For more information, see Prerequisite: Finding the Access URL → step 5.
`ecp.sap-client`	Enter the SAP client number (a three-digit number) of your SAP SuccessFactors Employee Central Payroll system. For example: **`102`**
(Optional) `ecp.user.filter`	When specified, only those SAP SuccessFactors Employee Central Payroll users matching the filter expression will be read. Supported operators: eq (equal) and sw (starts with). For example: • Value = *userName eq "JohnSmith"* • Value = *userName sw "J"* • Value = *emails.value eq "john.smith@example.com"*
(Optional) `ecp.group.filter`	When specified, only those SAP SuccessFactors Employee Central Payroll groups matching the filter expression will be read. Supported operators: eq (equal) and sw (starts with). For example: • Value = *displayName eq "ProjectTeam1"* • Value = *displayName sw "P"*

Property Name	Value
(Optional) `ecp.group.prefix`	This property distinguishes SAP SuccessFactors Employee Central Payroll groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`ECP_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP SuccessFactors Employee Central Payroll source system and will be provisioned to the target system with the following name pattern: **`ECP_<GroupDisplayName>`**. This way SAP SuccessFactors Employee Central Payroll groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP SuccessFactors Employee Central Payroll groups will be read and provisioned to the target system with their actual display names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP SuccessFactors Employee Central Payroll* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your system. For more information, see:

   Manage Transformations [page 2439]

   Mapping logic – the behavior of the default transformation logic is to read all user attributes from the source SAP SuccessFactors Employee Central Payroll system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

   > ### Note
   >
   > When configuring SAP SuccessFactors Employee Central Payroll as a source system, there may be a mismatch between optional user attributes in the source system and required user attributes in the target system. For example, the email is an optional attribute in SAP SuccessFactors Employee Central Payroll but a required one in Identity Authentication. In such cases, you need to review and adapt your source and/or target system transformations and configurations to reflect your needs.

   Default transformation:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "targetPath": "$.schemas",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
              "sourcePath": "$.name.givenName",
              "targetPath": "$.name.givenName",
              "optional": true
            },
            {
              "sourcePath": "$.name.familyName",
              "targetPath": "$.name.familyName",
              "optional": true
            },
            {
              "sourcePath": "$.name.middleName",
              "targetPath": "$.name.middleName",
              "optional": true
            },
            {
                "sourcePath": "$.nickName",
                "targetPath": "$.nickName",
                "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
            },
            {
                "sourcePath": "$.phoneNumbers",
                "targetPath": "$.phoneNumbers",
                "optional": true,
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "optional": true,
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true
            },
            {
                "sourcePath": "$.groups",
                "targetPath": "$.groups",
                "optional": true,
                "preserveArrayWithSingleElement": true,
                "functions": [
```

```
                    {
                        "function": "concatString",
                        "condition": "'%ecp.group.prefix%' !== 'null'",
                        "applyOnElements": true,
                        "applyOnAttribute": "display",
                        "prefix": "%ecp.group.prefix%"
                    }
                ]
            }
        ]
    },
    "group": {
        "condition": "$.displayName EMPTY false",
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "condition" : "'%ips.application.id%'!== 'null'",
                "constant" : "%ips.application.id%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "'%ecp.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "prefix": "%ecp.group.prefix%"
                    }
                ]
            },
            {
              "sourcePath": "$.displayName",
              "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "sourcePath": "$.members",
                "targetPath": "$.members",
                "optional": true,
                "preserveArrayWithSingleElement": true
            }
        ]
```

```
    }
}
```

6. Add a target system to provision users to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Identity Provisioning for Employee Central Payroll

1.6.1.43 SAP SuccessFactors Incentive Management

Follow this procedure to set up a source connector for SAP SuccessFactors Incentive Management, formerly known as SAP Commissions.

Prerequisites

You have technical user credentials for an SAP SuccessFactors Incentive Management system with read and write access permissions.

Context

After fulfilling the prerequisites, follow the procedure below to add a source system for SAP SuccessFactors Incentive Management to read users and user assignments to groups. This source system consumes SCIM 2.0 API provided by SAP SuccessFactors Incentive Management.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP SuccessFactors Incentive Management* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the SAP SuccessFactors Incentive Management SCIM API portal.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the user for your SAP SuccessFactors Incentive Management system.
`Password`	(Credential) Enter the password for your SAP SuccessFactors Incentive Management user.

   To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

   Exemplary destination:

   `Type`=*HTTP*

   `Authentication`=*BasicAuthentication*

   `ProxyType`=*Internet*

   `URL`=*https://mycommissions.callidus.run/CallidusPortal*

   `User`=*MyCommissionsUser*

   `Password`=**********

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP SuccessFactors Incentive Management* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SCIM system. For more information, see:

   Manage Transformations [page 2439]

   SAP SuccessFactors Incentive Management REST API

The behavior of the default transformation logic is to read all user attributes from the source SAP SuccessFactors Incentive Management system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

Default transformation:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "targetVariable": "entityIdSourceSystem",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "optional": true
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.name.middleName",
        "targetPath": "$.name.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.name.honorificPrefix",
        "targetPath": "$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "optional": true,
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.emails[?(@.primary== true)].value",
```

```
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.addresses",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.addresses"
      },
      {
        "sourcePath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups"
      },
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale",
        "functions": [
          {
            "type": "substring",
            "beginIndex": 0,
            "endIndex": 2
          }
        ]
      }
    ]
  }
}
```

5. Now, add a target system to provision users and their group assignments to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP SuccessFactors Incentive Management: Integration with SAP IdP

1.6.1.44 SAP SuccessFactors Learning

Follow this procedure to set up SAP SuccessFactors Learning as a source system.

Prerequisites

You have created a technical user with administrator permissions that will be used to call the API of SAP SuccessFactors Learning for reading user information.

Context

SAP SuccessFactors Learning is a learning solution which helps organizations to improve employee skills and talent management, align learning outcomes with performance goals, boost compliance, and train external audiences.

You can use Identity Provisioning to configure SAP SuccessFactors Learning as a source system where you can read **users** from and provision them to a target system.

When users from SAP SuccessFactors Learning source system, are provisioned to Identity Authentication target system, all newly created users get the *userUUID* attribute generated by Identity Authentication. The Identity Provisioning in its turn sends the *userUUID* back to SAP SuccessFactors Learning to update the users.

> ⚠ Restriction
>
> Reading *groups* is not supported.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SAP SuccessFactors Learning* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
Type	Enter: *HTTP*

Property Name	Value
`URL`	Specify the URL to the API of your SAP SuccessFactors Learning system. It follows the pattern: `https://<root URL>/learning/public-api/rest/admin/Integration.svc/ias`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the technical user ID for SAP SuccessFactors Learning.
`Password`	(Credential) Enter the password for the SAP SuccessFactors Learning technical user. For more information, see Learning Technical User.
(Optional) `lms.user.filter`	When specified, only those users matching the filter expression will be read. **Possible values:** • *userName eq "testName"* • *externalID eq "testID"* • *active eq "true"* • *sourceSystem eq "Learning"* - indicates that the user is created directly in SAP SuccessFactors Learning with no involvement of Identity Provisioning. • *sourceSystem eq "Identity Provisioning"* - indicates that the user is created in SAP SuccessFactors Learning by Identity Provisioning.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   The Identity Provisioning offers a default transformation for the *SAP SuccessFactors Learning* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   The behavior of the default transformation logic is to read all user attributes from the source SAP SuccessFactors Learning system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

   You can change the default transformation mapping rules depending on your setup of entities in your SAP SuccessFactors Learning. For more information, see:

   Manage Transformations [page 2439]

   SAP Business Accelerator Hub: SAP SuccessFactors Learning

   Default transformation:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.schemas",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.externalId",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['siteID']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['siteID']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['sourceSystem']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['sourceSystem']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['applicationID']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['applicationID']"
      },
```

```
      {
"sourcePath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']
['customColumns']['110']",
        "optional": true,
        "targetPath": "$.custom-column-path-1"
      },
      {
"sourcePath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']
['customColumns']['120']",
        "optional": true,
        "targetPath":  "$.custom-column-path-2"
      },
      {
        "sourcePath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.phoneNumbers"
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath": "$.emails[?(@.primary== true)].value",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"

      }
    ]
  }
}
```

→ Tip

Your main scenario with this source connector might be provisioning SAP SuccessFactors Learning users to SAP Cloud Identity Service – Identity Authentication. In this case, you can add the following **condition** at the beginning of the "user" resource since *emails* and *name.familyName* are mandatory attributes for Identity Authentication users.

```
{
  "user": {
"condition": "($.emails[0].value EMPTY false) && ($.name.familyName EMPTY
false)",
    "mappings": [
      ...
```

ⓘ Note

When SAP SuccessFactors Learning users are provisioned to Identity Authentication, the service generates a user UUID for every newly created user. Within the same provisioning job, Identity Provisioning returns the user UUID back to SAP SuccessFactors Learning and updates the *userUUID* attribute.

> Identity Provisioning updates *userUUID* when its value is *null* or when its value already exists. The latter is valid in cases where, for example, a user is deleted in the Identity Authentication target system and after running a Resync Job, the deleted user is created again. A new user UUID is generated which updates the old value in SAP SuccessFactors Learning.
>
> Returning the user UUID back to SAP SuccessFactors Learning and updating the *userUUID* attribute is supported for standard and real-time provisioning scenarios. It is also supported in bulk scenarios, when bulk operations are enabled on the Identity Authentication target system.

5. Now, add a target system to provision users to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP SuccessFactors Learning

1.6.1.45 Sales Cloud – Analytics & AI

Follow this procedure to set up Sales Cloud – Analytics & AI as a source system.

Prerequisites

You have technical user credentials for an Sales Cloud – Analytics & AI (in short, *SCAAI*) system with read and write access permissions.

Context

After fulfilling the prerequisites, follow the procedure to add a source system for Sales Cloud – Analytics & AI to read users and user assignments to groups. This source system consumes SCIM 2.0 API provided by SCAAI.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *Sales Cloud – Analytics & AI* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the SCIM API portal of your SCAAI system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the user for your SCAAI system.
`Password`	(Credential) Enter the password for your SCAAI user.
`OAuth2TokenServiceURL`	Enter the URL to the OAuth2 token service. If not sure about the exact URL, ask your SCAAI administrator.
(Optional)`sales.cloud.analytics_ai.group.filter`	Enter a group filter criteria, according to the API syntax of SCAAI. For example: **displayName eq "first_group"**
(Optional)`sales.cloud.analytics_ai.user.filter`	Enter a user filter criteria, according to the API syntax of SCAAI. For example: **externalId eq "John123"**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=http://myscaai:8080/scim_services
User=MySCAAIUser
Password=************
OAuth2TokenServiceURL=http://myscaai:8080/gateway_services/api/auth/ips/token
```

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Sales Cloud – Analytics & AI* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SCAAI system. For more information, see Manage Transformations [page 2439].

   The behavior of the default transformation logic is to read all user attributes from the source SCAAI system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity.

   Default transformation:

   Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.groups",
                "targetPath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional" : true
            },
            {
                "type": "remove",
                "targetPath": "$.groups[*].display"
            },
```

```
            {
                "type": "remove",
                "targetPath": "$.groups[*].ref"
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName"
            },
            {
                "optional": true,
                "preserveArrayWithSingleElement": true,
                "sourcePath": "$.members",
                "targetPath": "$.members"
            },
            {
                "type": "remove",
                "targetPath": "$.members[*].$ref"
            },
            {
                "type": "remove",
                "targetPath": "$.members[*].display"
            }
        ]
    }
}
```

5. Now, add a target system to provision users and their group assignments to it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.46 Cloud Foundry UAA Server

Follow this procedure to set up the Cloud Foundry UAA server as a source system.

Prerequisites

> ⚠ Restriction
>
> This system is available for standalone tenants running on SAP Cloud Identity infrastructure and SAP BTP, Neo environment. Bundle tenants running on SAP Cloud Identity Services infrastructure and Neo environment can use it only through **SAP Jam Collaboration** and **SAP Identity Access Governance** bundle options.

- You have technical user credentials for a Cloud Foundry system with read access permissions.
  In case OAuth is used for authentication, client ID and secret are required when creating a destination for access token retrieval. You need Cloud Foundry UAA version **4.2** or higher.
- (Optional) You have installed the Cloud Connector in your corporate environment and have done the initial configuration. You need to do this only if the Cloud Foundry UAA server is exposed in a private corporate network. For more information, see Cloud Connector.

Context

User Account and Authentication Service (UAA) is an OAuth2 server that you can use for centralized identity management. It owns the user accounts and authentication sources, and supports standard protocols (such as *SAML*, *LDAP*, and *OpenID Connect*) to provide SSO and delegated authorization to Web applications. For more information, see Cloud Foundry: Overview .

Cloud Foundry UAA is responsible for the SAP ID service to create and manage platform users (platform administrators and platform developers) in Cloud Foundry.

> → Tip
>
> This connector is meant for reading users and groups from **general** Cloud Foundry systems (they could be non-SAP ones). If you want to trigger provisioning of entities from SAP Business Technology Platform Cloud Foundry applications, you'd better use SAP BTP XS Advanced UAA (Cloud Foundry) [page 686] source system.

These source systems consume SCIM 1.1 API provided by Cloud Foundry UAA.

> → Remember
>
> You can read Cloud Foundry users and groups on **application** level only. You cannot read or manage them on a *subaccount* level.

Follow the steps below to create Cloud Foundry UAA as a source system to read users and groups, which can then be provisioned to a certain target system.

Procedure

1. (Optional) If the Cloud Foundry UAA server is exposed in a private corporate network, add an access control system mapping in Cloud Connector. For more information, see Configure Access Control (HTTP).
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
3. Add *Cloud Foundry UAA Server* as a source system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the Cloud Foundry UAA SCIM API. If not sure about the exact URL, ask your Cloud Foundry UAA administrator.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`OAuth2TokenServiceURL`	As you need to make OAuth authentication to the UAA system, enter the URL to the OAuth2 token service. If not sure about the exact URL, ask your Cloud Foundry UAA administrator.
`User`	Enter the OAuth client ID of the Cloud Foundry UAA technical user.
`Password`	(Credential) Enter the OAuth client secret of the technical user.
`uaa.origin`	Enter the location of your Cloud Foundry identity provider. If not sure about the value, ask your Cloud Foundry UAA administrator. The value of this property is a string, which will be used as the *origin* attribute in the system transformation.
`uaa.origin.filter.enabled`	This flag property depends on `uaa.origin`. Possible values: **true** or **false** • If set to *true*, the Identity Provisioning service will read only users whose identity provider is set as a value of `uaa.origin`. • If set to *false*, the Identity Provisioning service will read all users, regardless of their origin. • If set to *true* but the `uaa.origin` property is missing, the provisioning job will fail.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://api.authentication.hana.ondemand.com
OAuth2TokenServiceURL=https://MyCFaccount.authentication.hana.ondemand.com/oauth/token
User=MyCFuser
Password=************
uaa.origin=my_UAA_location
uaa.origin.filter.enabled=true
```

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Cloud Foundry UAA Server* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   - **Mapping logic** – The behavior of the default transformation logic is to map all attributes from a Cloud Foundry UAA entity to the intermediate Identity Provisioning representation.
   - **User offboarding** – If a user or group has been deleted from the Cloud Foundry UAA server, this change is recognized, and the user/group is deleted from the target system too.

   You can change the default transformation mapping rules to reflect your current setup of entities in your Cloud Foundry UAA server. For more information, see:

   Manage Transformations [page 2439]

   Cloud Foundry UAA API: Users

   Cloud Foundry UAA API: Groups

   Default transformation:

   Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "correlationAttribute": true
```

```
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.emails[?(@.primary== true)].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.emails[?(@.primary== true)].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.phoneNumbers",
        "targetPath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.verified",
        "targetPath": "$.verified",
        "optional": true
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.origin",
        "targetPath": "$.origin",
        "optional": true
      },
      {
        "sourcePath": "$.zoneId",
        "targetPath": "$.zoneId",
        "optional": true
      }
    ]
  },
  "group": {
    "mappings": [
      {
```

```
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "'%uaa.group.prefix%' !== 'null'",
            "function": "concatString",
            "prefix": "%uaa.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "sourcePath": "$.description",
        "targetPath": "$.description",
        "optional": true
      },
      {
        "sourcePath": "$.zoneId",
        "targetPath": "$.zoneId",
        "optional": true
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true
      }
    ]
  }
}
```

6. Now, add a target system to which to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Cloud Foundry UAA: Users
Cloud Foundry UAA: Groups

1.6.1.47 Google G Suite

Follow this procedure to set up Google G Suite as a source system.

Prerequisites

1. Sign in to the Google API console (https://console.developers.google.com) and create a project.
2. Enable the Admin SDK. To do this, go to *Dashboard* > *ENABLE API* > *Admin SDK* > *ENABLE*.
3. Create a service account for your project. We recommend that you select *Enable G Suite Domain-wide Delegation* during the creation. If you skip this option, you can set it later. For more information, see Creating a service account .
4. Then, in the Google admin console (https://admin.google.com), a user with **Super Admin** role can delegate domain-wide authority to your service account. This way, it will have access to the Google Admin SDK on behalf of your user. For more information, see Delegating domain-wide authority .

> **Note**
>
> When specifying the scopes, the administrator has to enter the following:
>
> `https://www.googleapis.com/auth/admin.directory.user, https://www.googleapis.com/auth/admin.directory.group`

Context

A Google service account with delegated domain-wide authority is required for authentication and authorization of the Identity Provisioning service to G Suite domain. The authentication is based on OAuth 2.0 protocol with JSON Web Token (JWT). The private key for the signature is distributed by Google via one-time

downloadable JSON data, which is accessible by the domain administrator. The private key is encoded in PKCS8 format and is in the *private_key* field of the JSON data. For more information, see JSON Web Token (JWT) .

- When using it as a source system, you can read both users and groups from Google G Suite and provision them to any target system you have added in the Identity Provisioning user interface.
- When using it as a target system, you can write both users and groups, read from any source system you have added in the Identity Provisioning user interface. Google G Suite can automatically create accounts for your users in the Google Cloud Datastore.

The Identity Provisioning service supports user and group operations based on the following Google Directory API. See the table below.

User Operations	Group Operations
Create a user	Create a group
Retrieve a user	Retrieve a group's properties
Update a user	Update a group's properties
Delete a user	Delete a group

> ⚠ Caution
>
> You can only provision users whose e-mails are from verified domains.

If you have successfully finished with the initial setup (described in the **Prerequisites** section), continue with the procedure below.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems* .
2. Add *Google G Suite* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
Type	Enter: *HTTP*
URL	Specify the service URL: `https://www.googleapis.com/admin/directory`

Property Name	Description & Value
`ProxyType`	Enter: **`Internet`**
`Authentication`	Enter: *BasicAuthentication* The authentication type in use is actually **OAuth** with JWT. But for any provisioning system based on OAuth, **BasicAuthentication** is used along with the `OAuth2TokenServiceURL` additional property.
`User`	Enter the service account's ID. You can take it from the *"client_email"* field in the JSON data, downloaded during the setup of Google service account.
`Password`	(Credential) Enter the service account's private key, which represents a long string in PKCS8 format. You can take it from the *"private key"* field in the JSON data, downloaded during the setup of Google service account.
`OAuth2TokenServiceURL`	To make OAuth authentication to the Google G Suite system, enter the URL to the access token provider service. For more information, see Using OAuth 2.0 to Access Google APIs .
`jwt.subject`	Enter the Google G Suite user on behalf of which the Google Directory API is called. This user has been assigned the role **User Management Admin**. This property corresponds to "sub" claim in JWT being generated during access token request: JWT: "sub" (Subject) Claim
(Optional) `jwt.scope`	Enter space-separated Google Directory API authorization scopes. For example: **`https://www.googleapis.com/auth/admin.directory.user`**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary Configuration:

`ProxyType`=*Internet*

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`URL`=*https://www.googleapis.com/admin/directory*

`User`=*1234567890-compute@developer.gserviceaccount.com*

`Password`=*-----BEGIN PRIVATE KEY-----\n123ABCDEFG123456789...*

... /123456789ABCDEFG123=\n-----END PRIVATE KEY-----\n

`OAuth2TokenServiceURL`=*https://www.googleapis.com/oauth2/v4/token*

`jwt.subject`=*john.smith@me123.accounts.ondemand.com*

`jwt.scope`=*https://www.googleapis.com/auth/admin.directory.user*

---

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Google G Suite* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   Transformation principles for the source system integration:

   - **Mapping logic** – The provisioning framework reads all attributes from the Google G Suite source system and transfers them to the intermediate JSON data, which then tries to create consistent records in the target system, using all the available attributes accepted by the target system API. When a required attribute is missing, the default transformation is designed with a condition that will exclude the inconsistent records.
   - **User off-boarding** – Identity Provisioning service is handling the deletion status of the users. When a user is deleted from Google G Suite, this deletion will be enforced into the target system as well.

   You can change the default transformation mapping rules to reflect your current setup of entities in your Google G Suite. For more information, see:

   Manage Transformations [page 2439]

   Google Directory API: Users

   Google Directory API: Groups

   Default transformation:

   Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
```

```
                "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.primaryEmail",
                "targetPath": "$.emails[0].value",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.primaryEmail",
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$.name",
                "targetPath": "$.name"
            },
            {
                "constant": true,
                "targetPath": "$.active"
            },
            {
                "condition": "$.suspended == true",
                "constant": false,
                "targetPath": "$.active"
            }
        ]
    },
      "group": {
        "ignore": true,
        "mappings": [
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.name",
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.members[?((@.type == 'USER') && (@.status
== 'ACTIVE'))]",
                "targetPath": "$.members",
                "optional": true,
                "preserveArrayWithSingleElement": true
            },
            {
                "targetPath": "$.members[*].status",
                "type": "remove"
            },
            {
                "targetPath": "$.members[*].id",
                "type": "rename",
                "constant": "value"
            },
            {
                "constant": "display",
                "targetPath": "$.members[*].email",
                "type": "rename"
            },
            {
                "targetPath": "$.members[*].kind",
                "type": "remove"
            },
            {
```

```
                "targetPath": "$.members[*].etag",
                "type": "remove"
            },
            {
                "targetPath": "$.members[*].role",
                "type": "remove"
            }
        ]
    }
}
```

If the **displayName** attribute in the source system transformation does not provide group e-mails, you can modify the transformation the following ways:

- Map **email** to another attribute that contains a unique group e-mail.
- Concatenate the **displayName** attribute with your domain. For example:

Sample Code

```
        {
            "sourcePath": "$.displayName",
            "targetPath": "$.email",
            "scope": "createEntity",
            "functions": [
                {
                    "type": "concatString",
                    "suffix": "@test.myaccount.ondemand.com"
                }
            ]
        }
```

5. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.48 LDAP Server

Follow this procedure to set up LDAP Server as a source system.

Prerequisites

> **Note**
>
> If you have purchased the Identity Provisioning service between **September 1, 2020** and **October 20, 2020**, and you want to make a connection to this on-premise system, follow the procedure in: Connect to On-Premise Systems in SAP Cloud Identity Infrastructure.

- You have installed the Cloud Connector in your corporate environment and have done the initial configuration. For more information, see: Cloud Connector (Neo) or Cloud Connector (Cloud Foundry)
- **For tenants running on the infrastructure of SAP Cloud Identity Services**: You have a multi-environment subaccount in the Cloud Foundry region that maps the region of your Identity Authentication tenant and it is subscribed to the *Cloud Identity Services* application. For more information, see Connect to On-Premise Systems in SAP Cloud Identity Infrastructure.
- You have the credentials of a technical user in the LDAP Server, which is used to call the LDAP Server API to read the users and their attributes.

Context

You can use LDAP Server to read entities from it and provision them to a target system. This scenario supports reading **users** and **groups**.

There are two versions of the LDAP Server connector. Both consume the LDAP Server API to read and write users and groups. The versions are handled by the `ldap.api.version` property as follows:

- When the value is set to *1* or the property is not defined (typical for systems created before versioning was introduced on May 25, 2023) LDAP Server API version 1 is used. This is the default value of `ldap.api.version`.
  When using this version of the connector, the entities (users and groups) are read with all attributes.
- When the value is set to *2* – LDAP Server API version 2 is used.
  This version of the connector comes with improved performance of the read operation for user and group attributes. You are now able to define which user and group attributes to be read. This is possible by adding values to the properties `ldap.user.attributes` or `ldap.group.attributes`.
  Via these properties, you are able to add also user and group operational attributes (attributes which the directory organizes for internal use). For more information, refer to the official LDAP server documentation.
  After the additional values of the properties are set, the default read or proxy read transformations should also be adjusted accordingly.
  For more information on how to update to LDAP Server connector version 2, see Update Connector Version [page 2424].

Procedure

1. Add an access control system mapping for the **LDAP Server** in the Cloud Connector. This is needed to allow the Identity Provisioning service to access the LDAP server as a back-end system on the intranet. For more information, see Configure Access Control (LDAP).
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
3. Add *LDAP Server* as a source system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *LDAP*
`ldap.url`	Specify the destination URL. It must be in the following format: `ldap://<external_host>:<external_port>`
`ldap.proxyType`	Enter: *OnPremise*
`ldap.authentication`	Enter: *BasicAuthentication*
`ldap.user`	Enter the *distinguishedName* of the technical LDAP user. This is the user you need to establish the connection and to perform all queries.
`ldap.password`	(Credential) Enter the password for the LDAP technical user.
`ldap.group.path`	Enter the complete path to the node containing the groups in the LDAP tree.
`ldap.user.path`	Enter the complete path to the users in the LDAP tree.
(Optional)`ldap.api.version`	Defines the version of LDAP Server API. **Possible values:** • **1** - Indicates that LDAP Server API version 1 is used. • **2** - Indicates that LDAP Server API version 2 is used. If the property is not defined - LDAP Server API version 1 is used.

Property Name	Description & Value
`ldap.user.attributes`	Shows which user attributes from the source system to be included in the LDAP read result (and respectively, in the intermediate JSON data). Separate the attributes by comma (,). **Possible values:** • LDAP Server version 1 version 1 - Even though you specified the attributes to be included in the LDAP read result, Identity Provisioning will return all attributes from the LDAP server. • LDAP Server version 2 - Only the specified attributes will be included in the LDAP server read result.
`ldap.group.attributes`	Shows which group attributes from the source system to be included in the LDAP read result (and respectively, in the intermediate JSON data). Separate the attributes by comma (,). **Possible values:** • LDAP Server version 1 - Even though you specified the attributes to be included in the LDAP read result, Identity Provisioning will return all attributes from the LDAP server. • LDAP Server version 2 - Only the specified attributes will be included in the LDAP server read result. If nothing is set, all attributes are included.

> → Remember
>
> We strongly recommend that you enter different paths for LDAP users and groups. That means, the value of `ldap.user.path` should be different than the value of `ldap.group.path`.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

The LDAP Server source system is created by default with the properties listed below:

Default LDAP Properties

```
ldap.user.attributes=
ldap.group.attributes=
ldap.user.object.class= inetOrgPerson
ldap.group.object.class= groupOfNames
ldap.group.uniquename.attribute= cn
ldap.member.uniquename.attribute=uid
ldap.attribute.group.id=cn
ldap.attribute.group.member= member
ldap.attribute.user.id= uid
ldap.attribute.dn=distinguishedName
ldap.user.filter=
ldap.group.filter=
ldap.page.size= 100
ldap.attribute.user.mail= mail
ldap.attribute.user.mobile=mobile
ldap.attribute.user.givenName= givenName
ldap.attribute.user.surname= sn
ldap.attribute.user.groups= memberOf
ldap.attribute.user.telephoneNumber= telephoneNumber
```

> ### Note
>
> The **ldap.attribute.*** properties are used as parameterized properties in the default transformation. That is, if a property used in the transformation doesn't have a value, the provisioning job will fail when the transformation is loaded on runtime and the property value is substituted.
>
> Also, you can change a property and use a new one (with a new name). In this case, you must replace the old property with the new one at all corresponding places in the transformation.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *LDAP Server* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your LDAP server. For more information, see Manage Transformations [page 2439].

Currently, the default read transformations of the two connector versions have no differences, so there is no need to update them.

Before the read transformation, the LDAP Server attributes are represented as arrays (single-element arrays, or multi-value arrays separated by comma (,)). After read transformation (in the intermediate JSON data), the attributes are in SCIM format. For more information, see the official documentation for LDAP Server schema attributes in the **Related Information** section.

> ### Note
>
> When a user is deleted from LDAP Server, the deletion status is considered by the Identity Provisioning service during the read processes. Depending on the offboarding handling of the users in the target system, the user can be deleted, or can be set to *inactive*.

Default transformation:

Code Syntax

```
{
    "user": {
        "mappings": [

/* The value of entityIdSourceSystem is used to store the unique ID of the
identity. You should not delete this statement.
You could exchange the default attribute, resolved from
ldap.attribute.user.id system property (which is used as a source) with
another one but make sure the new source attribute is unique as well. */
        {
            "sourcePath": "$.%ldap.attribute.user.id%[0]",
            "targetVariable": "entityIdSourceSystem",
            "correlationAttribute": true
        },

/* The value of the attribute resolved from ldap.attribute.user.id
system property is used also as userName value for the internal JSON
representation. */
        {
            "sourcePath": "$.%ldap.attribute.user.id%[0]",
            "targetPath": "$.userName",
            "correlationAttribute": true
        },

/* The constant urn:ietf:params:scim:schemas:core:2.0:User is required as
a value for the schemas definition in the Identity Authentication service
SCIM REST API. */
        {
            "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
            "targetPath": "$.schemas[0]"
        },

/* The value of the attribute resolved from ldap.attribute.user.mail
system property is used also as a first array value in the emails JSON
array. */
        {
            "sourcePath": "$.%ldap.attribute.user.mail%[0]",
            "targetPath": "$.emails[0].value",
            "optional": true,
```

```
            "correlationAttribute": true
        },

/* The value of the attribute resolved from ldap.attribute.user.givenName
system property is used for the name.givenName value in internal JSON
representation. */
        {
            "sourcePath": "$.%ldap.attribute.user.givenName%[0]",
            "targetPath": "$.name.givenName",
            "optional": true
        },

/* The value of the attribute resolved from ldap.attribute.user.surname
system property is used for the name.familyName value in internal JSON
representation. */
        {
            "sourcePath": "$.%ldap.attribute.user.surname%[0]",
            "targetPath": "$.name.familyName",
            "optional": true
        },

/* The attribute resolved from ldap.attribute.user.groups system property
is transformed by default into groups attribute of the SCIM internal
representation: */
        {
            "sourcePath": "$.%ldap.attribute.user.groups%",
            "preserveArrayWithSingleElement": true,
            "optional": true,
            "targetPath": "$.groups[?(@.value)]"
        },

        {
            "sourcePath": "$.%ldap.attribute.user.mobile%[0]",
            "optional": true,
            "targetPath": "$.phoneNumbers[0].value"
        },
        {
            "condition": "$.%ldap.attribute.user.mobile%.length() > 0",
            "constant": "mobile",
            "targetPath": "$.phoneNumbers[0].type"
        },
        {
            "sourcePath": "$.%ldap.attribute.user.telephoneNumber%[0]",
            "optional": true,
            "targetPath": "$.phoneNumbers[1].value"
        },
        {
            "condition": "$.%ldap.attribute.user.telephoneNumber%.length()
> 0",
            "constant": "work",
            "targetPath": "$.phoneNumbers[1].type"
        }
      ]
    },
      "group": {
        "ignore": true,
        "mappings": [
            {
                "sourcePath": "$.%ldap.attribute.group.id%[0]",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.%ldap.attribute.group.id%[0]",
                "targetPath": "$.displayName"
            },
```

```
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.%ldap.attribute.group.member%",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]"
            }
        ]
    }
}
```

As result of this mapping, this is how the data from LDAP Server looks like before and after the read transformation:

Transformation Snippet (from the user mapping)	Source JSON Data (as read from LDAP Server)	Intermediate JSON Data (as result from the transformation)
Sample Code `{ "user": { "mappings": [ ... { "sourcePath": "$.%ldap.attribute.user.groups%[0]", "preserveArrayWithSingleElement": true, "optional": true, "targetPath": "$.groups[?(@.value)]" }, ...`	Sample Code `... "memberOf": [ "SALES_US", "SALES_EU", "SALES_JA" ] …`	Sample Code `... "groups":[ { "value": "SALES_US" }, { "value": "SALES_EU" }, { "value": "SALES_JA" } ] …`

> **Note**
>
> By default, the **cn** attribute is returned for every read group. An administrator can change this behavior by setting the following properties:
>
> - `ldap.group.uniquename.attribute` – the value can be either the CN or the whole DN (**distinguishedName**) of the group.
> - `ldap.attribute.group.id` – the value can be CN or another attribute to be used as a group ID instead (for example, **displayName** or **description**).
>
> For more information about these properties, see: List of Properties [page 80]

6. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Technical Documents
Setting Timeout for Ldap Operations
Connection Pooling Configuration

1.6.1.49 Microsoft Active Directory

Follow this procedure to set up Microsoft Active Directory as a source system.

Prerequisites

> **Note**
>
> If you have purchased the Identity Provisioning service between **September 1, 2020** and **October 20, 2020**, and you want to make a connection to this on-premise system, follow the procedure in: Connect to On-Premise Systems in SAP Cloud Identity Infrastructure.

- You have installed the Cloud Connector in your corporate environment and have done the initial configuration. For more information, see: Cloud Connector (Neo)
- You have the credentials of a technical user in the Microsoft Active Directory, which is used to call the Microsoft Active Directory API to read the users and their attributes.

Context

You can configure Microsoft Active Directory (Microsoft AD) as a source system to read users, groups and permission assignments.

There are two versions of the Microsoft AD connector. Both consume the LDAP Server API to read and write users and groups. The versions are handled by the `ldap.api.version` property as follows:

- When the value is set to *1* or the property is not defined (typical for systems created before versioning was introduced on June 19, 2023) LDAP Server API version 1 is used. This is the default value of `ldap.api.version`.
  When using this version of the connector, the entities (users and groups) are read with all attributes.
  In this version, the `group members` attribute mapping in the proxy read transformation does not include `type` sub-attribute. In this case, all members are considered of type `User`, which is the sub-attribute fallback value. As a consequence, if the external system includes nested groups, they will not be handled properly.
- When the value is set to *2* – LDAP Server API version 2 is used.
  This version of the connector is with improved performance of the read operation for user and group attributes. You are now able to define which user and group attributes to be read. This is possible by adding values to the properties `ldap.user.attributes` or `ldap.group.attributes`.
  Via these properties, you are able to add also user and group operational attributes (attributes which the directory organizes for internal use). For more information, refer to the official LDAP server documentation.
  After the additional values of the properties are set, the default read or proxy read transformations should also be adjusted accordingly.
  In this version, the `group members` attribute mapping in the proxy read transformation is enhanced with `type` sub-attribute. The sub-attribute has two possible values – `User` and `Group`. This allows you to read and preserve nested groups.
  For more information on how to update to Microsoft Active Directory version 2, see Update Connector Version [page 2424].

Procedure

1. Add an access control system mapping for the **Microsoft Active Directory** in the Cloud Connector. This is needed to allow the Identity Provisioning service to access Microsoft AD as a back-end system on the intranet. For more information, see Configure Access Control (LDAP).
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
3. Add *Microsoft Active Directory* as a source system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *LDAP*
`ldap.url`	Specify a destination URL. It must be in the following format: `ldap://<ext_host>:<ext_port>`
`ldap.proxyType`	Enter: *OnPremise*

Property Name	Description & Value
`ldap.authentication`	Enter: *BasicAuthentication*
`ldap.attribute.user.id`	Default property, which denotes the ID of a user. By default, it's set to: *cn*
`ldap.attribute.group.id`	Default property, which denotes the ID of a group. By default, it's set to: *cn*
`ldap.attribute.dn`	Default property, which denotes the distinguished name of a user or a group. Only possible value: *distinguishedName*
`ldap.user`	Enter the *distinguishedName* or the *userPrincipalName* of the Microsoft AD technical user. This is the user you need to establish the connection and to perform all queries.
`ldap.password`	(Credential) Enter the password for the Microsoft AD technical user.
`ldap.group.path`	Enter the complete path to the node containing the groups in Microsoft AD.
`ldap.user.path`	Enter the complete path to the users in Microsoft AD.
`ldap.api.version`	Defines the version of LDAP Server API. **Possible values:** • **1** - Indicates that LDAP Server API version 1 is used. • **2** - Indicates that LDAP Server API version 2 is used. If the property is not defined - LDAP Server API version 1 is used.
`ldap.user.attributes`	Shows which user attributes from the source system to be included in the Microsoft AD read result (and respectively, in the intermediate JSON data). Separate the attributes by comma (,). **Possible values:** • Microsoft AD version 1 - Even though you specified the attributes to be included in the Microsoft AD read result, Identity Provisioning will return all attributes from the Microsoft AD. • Microsoft AD version 2 - Only the specified attributes will be included in the Microsoft AD read result. *sAMAccountName* is always included as mandatory user attribute for Microsoft AD version 2.

Property Name	Description & Value
`ldap.group.attributes`	Shows which group attributes from the source system to be included in the Microsoft AD read result (and respectively, in the intermediate JSON data). Separate the attributes by comma (,). **Possible values:** • Microsoft AD version 1 - Even though you specified the attributes to be included in the Microsoft AD read result, Identity Provisioning will return all attributes from the Microsoft AD. • Microsoft AD version 2 - Only the specified attributes will be included in the or Microsoft AD. If nothing is set, all attributes are included.
(Optional) `ad.group.flatten`	There are target systems that do not support nested groups (group structures). Therefore, if your Microsoft AD system contains such groups, they will not be resolved properly during the provisioning job. Such target systems are: • *SAP Jam Collaboration* • *Identity Authentication* To enable reading of group structures, you can use the `ad.group.flatten` property and set it to *true*. It will read the group structure recursively and will "flatten" it so that all users from all groups and subgroups will be resolved and written in the target system as members of the main parent group. **Note** Performance might be affected when `ad.group.flatten` is set to true and group structure is very complex (contains a lot of nested groups). For best results, we recommend you also set the system property `ldap.group.filter` whose value is one or multiple Microsoft AD parent groups. For example, if a parent group is named **Canteen**, the property should be set as: `ldap.group.filter`= *(cn=Canteen)* See the example below with multiple parent groups.

Example for a destination or a set of properties:

```
Type=LDAP
Name=MyADDestination
ldap.user=john.smith@some.dummy.domain.com
ldap.password=********
ldap.attribute.user.id=cn
ldap.attribute.group.id=cn
ldap.attribute.dn=distinguishedName
ldap.url=ldap://abcd:123
ldap.proxyType=OnPremise
ldap.authentication=BasicAuthentication
ldap.group.path=OU=Groups,OU=IAS,DC=global,DC=corp,DC=mycompany
ldap.user.path=OU=Users,OU=IAS,DC=global,DC=corp,DC=mycompany
ldap.group.filter=(|(cn=Canteen)(cn=Finance)(cn=Support_Team))
ad.group.flatten=true
```

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Microsoft Active Directory* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   Before the read transformation, the Microsoft Active Directory attributes are represented as arrays – single-element arrays, or multi-value arrays separated by comma (,). After the read transformation (in the intermediate JSON data), the attributes are in SCIM format. For more information about Microsoft AD schema attributes, see the **Related Information** section.

   Currently, the default read transformations of the two connector versions have no differences, so there is no need to update them.

   You can change the default transformation mapping rules to reflect your current setup of entities in Microsoft Active Directory. For more information, see:

   Manage Transformations [page 2439]

   Default transformation:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.%ldap.attribute.user.id%[0]",
        "targetVariable": "entityIdSourceSystem",
        "correlationAttribute": true
      },
      {
        "condition": "('%ldap.attribute.user.id%' ==
'%ldap.attribute.dn%')",
        "constant": "",
        "functions": [
          {
            "function": "concatString",
            "prefix": "(?i)cn=.*?,(.*?)",
            "suffix": ",%ldap.user.path%"
          }
        ],
        "targetVariable": "nestedPathRegex"
      },
// If a user is not a direct member of the configured user base path,
then function getMatchedRegexGroup extracts the nested path from the
distinguishedName of this user,
// and maps the nested path to attribute
['urn:sap:cloud:scim:schemas:extension:ad:2.0:User']['nestedPath']
      {
        "condition": "('%ldap.attribute.user.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath": "$.%ldap.attribute.user.id%[0]",
        "functions": [
          {
            "function": "getMatchedRegexGroup",
            "regex": "${nestedPathRegex}",
            "groupIndex": 1
          }
        ],
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:ad:2.0:User']['nestedPath']",
        "defaultValue": ""
      },
      {
        "sourcePath": "$.sAMAccountName[0]",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.displayName[0]",
        "targetPath": "$.displayName",
        "optional": true
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.mail[0]",
        "targetPath": "$.emails[0].value",
        "optional": true,
        "correlationAttribute": true
      },
      {
```

```
        "sourcePath": "$.givenName[0]",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.sn[0]",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.memberOf",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups[?(@.value)]"
      },
      {
        "sourcePath": "$.mobile[0]",
        "optional": true,
        "targetPath": "$.phoneNumbers[0].value"
      },
      {
        "condition": "$.mobile.length() > 0",
        "constant": "mobile",
        "targetPath": "$.phoneNumbers[0].type"
      },
      {
        "sourcePath": "$.telephoneNumber[0]",
        "optional": true,
        "targetPath": "$.phoneNumbers[1].value"
      },
      {
        "condition": "$.telephoneNumber.length() > 0",
        "constant": "work",
        "targetPath": "$.phoneNumbers[1].type"
      }
    ]
  },
  "group": {
    "ignore": true,
    "mappings": [
      {
        "sourcePath": "$.%ldap.attribute.group.id%[0]",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "condition": "('%ldap.attribute.group.id%' ==
'%ldap.attribute.dn%')",
        "constant": "",
        "functions": [
          {
            "function": "concatString",
            "prefix": "(?i)cn=.*?,(.*?)",
            "suffix": ",%ldap.group.path%"
          }
        ],
        "targetVariable": "nestedPathRegex"
      },
// If a group is not a direct member of the configured group base
path, then function getMatchedRegexGroup extracts the nested path from the
distinguishedName of this group,
// and maps the nested path to attribute
['urn:sap:cloud:scim:schemas:extension:ad:2.0:Group']['nestedPath']
      {
        "condition": "('%ldap.attribute.group.id%' ==
'%ldap.attribute.dn%')",
```

```
        "sourcePath": "$.%ldap.attribute.group.id%[0]",
        "functions": [
          {
            "function": "getMatchedRegexGroup",
            "regex": "${nestedPathRegex}",
            "groupIndex": 1
          }
        ],
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:ad:2.0:Group']['nestedPath']",
        "defaultValue": ""
      },
      {
        "sourcePath": "$.sAMAccountName[0]",
        "targetPath": "$.displayName"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.member",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.members[?(@.value)]",
        "optional": true
      }
    ]
  }
}
```

As result of this mapping, that is how the data from Microsoft Active Directory looks like before and after the read transformation:

Source JSON Data (as read from Microsoft Active Directory)	Intermediate JSON Data (as a result from the transformation)

Sample Code
`...`
`"memberOf": [`
`"SALES_US",`
`"SALES_EU"`
`]`
`…`
	Sample Code
	`...`
	`"groups":[`
	`{`
	`"value": "SALES_US"`
	`},`
	`{`
	`"value": "SALES_EU"`
	`},`
	`]`
	`…`

6. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].

- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Technical Documents
Setting Timeout for Ldap Operations
Connection Pooling Configuration
Active Directory Domain Services Overview

1.6.1.50 Microsoft Entra ID

Follow this procedure to set up Microsoft Entra ID (formerly known as Microsoft Azure Active Directory) as a source system.

Prerequisites

- You've logged on to Microsoft Azure Portal, with credentials for a user with directory role **Global administrator**. For more information, see Microsoft Entra built-in roles .
- In *Azure Active Directory* > *App registrations*, you've registered an application with a secret key and permissions for Microsoft Graph API. These permissions must be consented by an administrator. For more information, see Microsoft Graph permissions reference .
- (Relevant to target systems) Your registered application is assigned the **User Account Administrator** role. This role allows you to deprovision users. For more information, see Add-MsolRoleMember .

> **Note**
>
> If this role isn't assigned, you can only disable users. To do that, set the `accountEnabled` property to **false**. For more information, see MS Graph: user resource type

Permissions

Assign the following permissions to your application, according to your scenario. Also, the permissions have to be of type *Application*.

- Users – *User.Read.All*
- Groups – *Group.Read.All*

For more information, see MS Graph: Users and MS Graph: Groups

Context

When using it as a source system, you can read both users and groups from Azure AD and provision them to any target system you've added in the Identity Provisioning user interface (if it supports groups).

If you've successfully finished with the initial setup (described in the **Prerequisites** section), continue with the procedure.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *Microsoft Entra ID* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter: **https://graph.microsoft.com**
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the application ID registered in your Azure AD subscription (see the **Prerequisites** section).
`Password`	(Credential) Enter the secret key associated to your app registration.
`aad.domain.name`	Enter one of the verified domain names from the corresponding Azure AD tenant. On this domain, you perform the provisioning operations. For more information, see Managing custom domain names in your Microsoft Entra ID.
`OAuth2TokenServiceURL`	Enter: **https://login.microsoftonline.com/<your_domain>/oauth2/token**, where `<your_domain>` is the domain name you have set in the **`aad.domain.name`** property.
(Optional) `aad.oauth2.resource.name`	Specifies the OAuth2 resource name for Microsoft Graph API. Enter: **https://graph.microsoft.com** You can change the Microsoft Graph endpoint to match your organization's cloud environment. For more information, see Microsoft Graph national cloud deployments.

Property Name	Description & Value
(Optional) `aad.group.member.attributes`	This property defines the attributes of a group member to be read by the Identity Provisioning. By default, it always reads the **type** and the **id** of a member. If you want the Identity Provisioning to read additional attributes, enter them as a single or a comma-separated value. For example: **Example** • If you want to read the e-mails too, enter: `aad.group.member.attributes=`*mail* This reads a member's type, ID, and e-mail. • If you want to read multiple additional attributes, enter: `aad.group.member.attributes=`*mail,mobilePhone,displayName* This reads a member's type, ID, e-mail, phone, and display name.
(Optional) `aad.user.attributes.membership.active`	Use this property if you want to retrieve information about all the groups to which the users are assigned (if any). • If the property is missing, or is set to *false* – group membership details for the users will not be extracted. • If the property is set to *true* – group membership details for the users will be extracted. To learn more, see: List of Properties [page 80]
(Optional) `aad.user.filter`	Use this property to filter users by specific criteria, according to the Microsoft Graph REST API . **Note** This property replaces the deprecated `msgraph-filter` property. To learn more, see: List of Properties [page 80]
(Optional) `aad.group.filter`	Use this property to filter groups by specific criteria, according to the Microsoft Graph REST API .

Property Name	Description & Value
(Optional) `aad.user.filter.group.filter.combine`	Use this property to filter users based on their group assignments. When set to **true**, this property combines user and group filters defined on the `aad.user.filter` and `aad.group.filter` properties to further narrow the search results. This way, only users that meet the following filtering criteria are returned: • Users that match the user filter and at the same time are members of groups that match the group filter. • Members of the filtered groups that match the user filter. When set to **false**, user and group filters are not combined. To learn more, see: List of Properties [page 80] Identity Provisioning: How to Get Users Based on Group Assignments from MS Azure AD
(Optional) `aad.user.attributes`	Defines which user attributes are read from Microsoft Entra ID system. The property is set during system creation with the following default value: *id,mail,userPrincipalName,displayName,mailNickname,givenName,surname,mobilePhone,businessPhones* This means that by default, Identity Provisioning will read from Microsoft Entra ID the user attributes defined in the property value. Those attributes are used in the default read transformation. To check the complete set of user attributes (properties) supported by Microsoft Entra ID, see: Microsoft Graph: User Properties To learn more, see: List of Properties [page 80]

Property Name	Description & Value
(Optional) `aad.group.attributes`	Defines which group attributes are read from Microsoft Entra ID system. The property is set during system creation with the following default value: *id,displayName,mailNickname* This means that by default, Identity Provisioning will read from Microsoft Entra ID the group attributes defined in the property value and will also return the *members* attribute. Those attributes are used in the default read transformation. To check the complete set of group attributes (properties) supported by Microsoft Entra ID, see: Microsoft Graph: Group Properties To learn more, see: List of Properties [page 80]
(Optional) `aad.entities.top`	This property defines the number of entities to be read per page. Default value: *100*
(Deprecated) `oauth.resource.name`	Enter: **https://graph.microsoft.com** **Note** The **oauth.resource.name** property has been deprecated, as it is no longer needed by the MS Entra ID connector. Use the **aad.oauth2.resource.name** property instead.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Microsoft Entra ID* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules depending on your setup of entities in your Microsoft Entra ID. For more information, see:

   Manage Transformations [page 2439]

   MS Graph: Users

   MS Graph: Groups

   Default transformation:

Code Syntax

```
{
  "user": {
    "condition": "$.userPrincipalName EMPTY false",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.mailNickname",
        "optional": true,
        "targetPath": "$.externalId",
        "correlationAttribute": true
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.mail",
        "targetPath": "$.emails[0].value",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.userPrincipalName",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.displayName",
        "optional": true,
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.surname",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.mobilePhone",
        "optional": true,
        "targetPath": "$.phoneNumbers[0].value"
      },
      {
        "condition": "$.mobilePhone EMPTY false",
        "constant": "mobile",
        "targetPath": "$.phoneNumbers[0].type"
      },
      {
        "sourcePath": "$.businessPhones[0]",
        "optional": true,
        "targetPath": "$.phoneNumbers[1].value"
      },
      {
        "condition": "$.businessPhones.length() > 0",
        "constant": "work",
        "targetPath": "$.phoneNumbers[1].type"
      },
```

```
      {
        "sourcePath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups"
      },
      {
        "sourcePath": "$.manager.id",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true
      },
      {
        "sourcePath": "$.manager.displayName",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "optional": true
      }
    ]
  },
  "group": {
    "ignore": true,
    "mappings": [
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.id",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.mailNickname",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "sourcePath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members"
      },
      {
        "targetPath": "$.members[*].id",
        "constant": "value",
        "type": "rename",
        "optional": true
      },
      {
        "condition": "'%ips.application.id%' !== 'null'",
        "constant": "%ips.application.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
```

```
        "constant": "userGroup",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {
        "constant": "readWrite",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      }
    ]
  }
}
```

Custom Configurations

Goal	Action	Result
You want Identity Provisioning to read the additional user attributes specified in property `aad.user.attributes` and write them successfully in the target system.	Extend the "*user*" mapping as follows: `{ "user": { "condition": "$.userPrincipalName EMPTY false", "mappings": [ { "sourcePath": "$", "targetPath": "$" }, { "sourcePath": "$.id", "targetVariable": "entityIdSourceSystem" }, ...`	For example, you specify the `aad.user.attributes` property and set its value to: *id,mail,userPrincipalName,city,department,companyName* As a result, every user in the target system will have the following attributes populated – ID, e-mail, user principle name, city, department, and company name. Returned information of an exemplary user: `... { "Resources":[ { "id":"555-aaaa-333-abcd-111222333", "mail":"john.smith@domain.com", "userPrincipalName":"abc@something.onmicrosoft.com", "city": "Sofia", "department":"029", "companyName":"SAP" } ... ] }`

Goal	Action	Result
You want Identity Provisioning to read the additional group attributes specified in property `aad.group.attributes` and write them successfully in the target system.	Extend the "*group*" mapping as follows: `{ "group": { "ignore": false, "mappings": [ { "sourcePath": "$", "targetPath": "$" }, { "constant": "urn:ietf:params:scim:schemas:core:2.0:Group", "targetPath": "$.schemas[0]" }, ...`	Example: Specify the `aad.group.attributes` property and set its value to: *id,displayName,recommendation,isSubscribedByMail* As a result, every group in the target system will have the following attributes populated – ID, display name, date and time of the last renewal, and information if it's subscribed by e-mail or not. Returned information of an exemplary group: `... { "Resources":[ { "id":"12345-ccc-000-xyz-777888999", "displayName":"ImportantGroup3", "renewedDateTime":"2018-01-01T00:00:00Z", "isSubscribedByMail":"true" } ... ] }`

Goal	Action	Result

You want the returned value of a group member to be not the ID but a different attribute.	In the `"group"` mapping, replace **id** with the new attribute. For example:
	
	If you replace *id* with *mail*, the transformation will look like this:
	
	`...`
	`{`
	`"sourcePath": "$.members",`
	`"preserveArrayWithSingleElement": true,`
	`"optional": true,`
	`"targetPath": "$.members"`
	`},`
	`{`
	`"targetPath": "$.members[*].mail",`
	`"constant": "value",`
	`"type": "rename",`
	`"optional": true`
	`}`
	`...`
	
	⚠ Caution
	Make sure that you've added this attribute as a value of property *aad.group.member.attributes*.
		Returned information of an exemplary group member:
		
		`...`
		`{`
		`"members":[`
		`{`
		`"id":"5555555-aaaa-333-abcd-1111122223333",`
		`"type":"user",`
		`"value":"johnsmith@mail.acme.com"`
		`}`
		`]`
		`}`
		`...`

5. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.51 Procurement Data Warehouse

Follow this procedure to set up Procurement Data Warehouse as a source system.

Prerequisites

- You obtain credentials to access the Procurement Data Warehouse API.

  > **Note**
  >
  > To obtain credentials to access the Procurement Data Warehouse API, create a case to the component *BNS-ARI-PRI*.
  >
  > In response to the case, SAP provides the certificate needed to access the API as well as the URL for the procurement data warehouse enrollment service, which is landscape-specific and starts with `*.cloud.sap`.

- You established trust between your SAP BTP subaccount and the SAP Cloud Identity Services - Identity Authentication service. For more information, see Establishing Trust Automatically.
- To configure your *Procurement Data Warehouse* provisioning system (see the procedure below), you will need to map your SAP Ariba application parameters to the relevant Identity Provisioning properties. The property mapping between the two systems is as follows:

Procurement Data Warehouse Application	Identity Provisioning	Values
API URL	URL	Examples: • US: **https://openapi.ariba.com** • Europe: **https://eu.openapi.ariba.com**
procurement data warehouseOAuth Token URL	`OAuth2TokenServiceURL`	Examples: • US: **https://api.ariba.com/v2/oauth/token** • Europe: **https://api-eu.ariba.com/v2/oauth/token**
OAuth Client ID	`client_id`	Enter the OAuth Client id used for access token retrieval.

- To load transaction data into procurement data warehouse, you must have one or more of the following SAP Ariba applications:
  - SAP Ariba Buying
  - SAP Ariba Buying and Invoicing
  - Guided sourcing capability for SAP Ariba Sourcing
  - SAP Ariba Contracts
  - SAP Ariba Spend Analysis

Context

The Procurement Data Warehouse is a capability that feeds analytics information from SAP Ariba applications, including guided sourcing capability for SAP Ariba Sourcing, SAP Ariba Buying and Invoicing, and SAP Spend Control Tower, to the embedded SAP Analytics Cloud. It provides data transfer, storage, loading, and reporting, and it drives visualization through advanced analytics dashboards.

You can use Identity Provisioning to configure procurement data warehouse as a source system where you can read users and groups and provision them to target systems of your choice for custom reporting. The source system consumes SCIM 2.0 API provided by procurement data warehouse.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *Procurement Data Warehouse* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the SAP Analytics Cloud URL for your procurement data warehouse.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *ClientCertificateAuthentication*
`client_id`	Enter the OAuth client id, generated for your procurement data warehouse. (see the **Prerequisites** section)
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your procurement data warehouse instance. This token URL is listed in the *OAuth Clients* section of the *App Integration* page.
(Optional) `pdw.user.filter`	When specified, only those procurement data warehouse users matching the filter expression will be read. For example: • *userName eq "Julie Armstrong"* • *userName sw "J"* • *emails eq "julie.armstrong@example.com"*

Property Name	Description & Value
(Optional)`pdw.group.prefix`	This property distinguishes procurement data warehouse groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`PDW_`** You can use the example value or provide your own. When **set in the source system**, the prefix will be pre-pended to the name of the groups that are read from the procurement data warehouse source system and will be provisioned to the target system with the following name pattern: **`PDW_<GroupDisplayName>`**. This way procurement data warehouse groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the procurement data warehouse groups will be read and provisioned to the target system with their actual display names.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Procurement Data Warehouse* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your Procurement Data Warehouse system. For more information, see: Manage Transformations [page 2439]

   > ### Note
   >
   > To access the Procurement Data Warehouse API documentation, contact the Procurement Data Warehouse technical support.

   Default transformation:

   > ### Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "targetPath": "$.schemas",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
```

```
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.preferredLanguage",
                "optional": true,
                "targetPath": "$.preferredLanguage"
            },
            {
              "sourcePath": "$.name.givenName",
              "targetPath": "$.name.givenName",
              "optional": true
            },
            {
              "sourcePath": "$.name.familyName",
              "targetPath": "$.name.familyName",
              "optional": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.emails[?(@.primary == true)].value",
                "optional": true,
                "correlationAttribute": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:user-custom-parameters:1.0']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:user-custom-parameters:1.0']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:scim:schemas:extension:enterprise:1.0']",
                "optional":true,
                "targetPath": "$
['urn:scim:schemas:extension:enterprise:1.0']"
            },
            {
                "sourcePath": "$.roles",
                "targetPath": "$.roles",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.photos",
                "preserveArrayWithSingleElement": true,
```

```
                "targetPath": "$.photos",
                "optional": true
            },
            {
                "sourcePath": "$.groups",
                "targetPath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "functions": [
                    {
                        "condition": "'%pdw.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "applyOnElements": true,
                        "applyOnAttribute": "display",
                        "prefix": "%pdw.group.prefix%"
                    }
                ]
            }
        ]
    },
    "group": {
        "ignore": true,
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "targetPath": "$.schemas",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "'%pdw.group.prefix%' !== 'null'",
                        "function": "concatString",
                        "prefix": "%pdw.group.prefix%"
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "sourcePath": "$.members",
                "targetPath": "$.members",
                "optional": true,
                "preserveArrayWithSingleElement": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:group-custom-
parameters:1.0']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:group-custom-
parameters:1.0']",
                "optional": true
             },
            {
                "condition": "'%ips.application.id%' !== 'null'",
                "constant": "%ips.application.id%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
```

```
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            }
        ]
    }
}
```

5. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Configure OAuth Certificate Authentication [page 2456]

1.6.1.52 SCIM System

Follow this procedure to set up a SCIM system as a source system.

Prerequisites

- (Optional) You have installed the Cloud Connector in your corporate environment and have done the initial configuration. You need this only if the SCIM system is exposed in a private corporate network. For more information, see Cloud Connector.

- You have technical user credentials for a SCIM system, with read/write access permissions, depending on the scenario you want to implement. In case OAuth is used for authentication, client ID and secret are required when creating a destination for access token retrieval.

Context

Create a general SCIM 2.0 based source system to read users and groups from it.

Procedure

1. (Optional) If the SCIM system is exposed in a private corporate network, add an access control system mapping in Cloud Connector. For more information, see Configure Access Control (HTTP).
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
3. Add *SCIM* as a source system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the service URL. For example: **`http://<cloudfoundry_server>.com/api/uaa/`**
`ProxyType`	Depending on your network exposure, enter one of the following: • *Internet* • *OnPremise*
`Authentication`	Enter: *BasicAuthentication*
`User`	You can specify one of the following: • Technical user ID • Client ID for OAuth HTTP destinations. It's used for retrieving of the access token.
`Password`	(Credential) You can enter one of the following: • Technical user password • Client secret for OAuth HTTP destinations. It's used for retrieving of the access token.

Property Name	Value
`OAuth2TokenServiceURL`	If you need to make OAuth authentication to the system, enter the URL to the access token provider service for OAuth HTTP destinations. For example: **`https://<token_provider>.com/api/oauth2/v2.0/token`**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SCIM* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SCIM system. For more information, see Manage Transformations [page 2439].

   - **Mapping logic** – the behavior of the default transformation logic is to read all user attributes from the source SCIM system, and then map them to the internal SCIM representation. It uses `entityIdSourceSystem` to store the unique ID of the identity. The ID is removed by default, because it is specific for the source system.
   - **User off-boarding** – it depends on the target system API. When a user is deleted from the SCIM system, the deletion status is considered and depends on the user status handling of the target system. The user will be either deleted or set as **inactive**.

   Default transformation:

   Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "targetPath": "$",
                "sourcePath": "$"
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "sourcePath": "$.userName",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.emails[?(@.primary== true)].value",
                "optional": true,
                "correlationAttribute": true
            },
            {
                "type": "remove",
                "targetPath": "$.id"
```

```
            },
            {
                "type": "remove",
                "targetPath": "$.meta"
            }
        ]
    },
    "group": {
        "ignore": true,
        "mappings": [
            {
                "targetPath": "$",
                "sourcePath": "$"
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            },
            {
                "type": "remove",
                "targetPath": "$.id"
            },
            {
                "type": "remove",
                "targetPath": "$.meta"
            }
        ]
    }
}
```

6. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.1.53 SSH Server (Beta)

Follow this procedure to set up an SSH server (Beta) as a source system.

Prerequisites

- You have credentials for a tenant in SAP Business Technology Platform. For more information, see: Accounts
- (Optional) You have installed the Cloud Connector in your corporate environment and have done the initial configuration. You need this only when your SSH server resides in a remote system, outside your Neo environment. For more information, see Cloud Connector.

> ### Note
>
> This is a beta feature available on SAP Business Technology Platform. For more information, see: *Enable beta features* in Change Subaccount Details

Context

SSH Server is a system (connector) in beta state. It helps you execute bash scripts through SSH connection. The configuration allows you to attach separate scripts per entity lifecycle callback (such as user create, group create/update, and so on). This system helps you connect to remote machines via SSH tunnel, with or without use of the Cloud Connector, depending on whether the SSH port is visible or not.

The bash scripts can take as parameters fields that are coming from the entity JSON data. For example: *sudo su - vcap /home/myscript.sh $.userName $.email*

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Add *SSH Server (Beta)* as a source system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Below are listed all available SSH Server properties. Some of them can be mandatory and others – optional, depending on your scenario.

   Mandatory Properties

Property Name	Description & Value
`ProxyType`	Possible values: • **Internet** – if the SSH port is visible in your Neo environment • **OnPremise** – if the SSH port is not directly accessible, and you have to use theCloud Connector. You have to configure TCP protocol connection to the SSH host and port (specify the configuration properties `ssh.host` and `ssh.port`).
`CloudConnectorLocationId`	Relevant when the proxy type is *OnPremise*. Use it only if your SAP Business Technology Platform account uses more than one Cloud Connector.

Property Name	Description & Value
`ssh.auth.type`	Supported SSH authentication types: • **key** • **pwd** • **otp** • **key+otp** • **key+pwd** • **pwd+otp** • **key+pwd+otp**
`ssh.host`	
`ssh.port`	22
`ssh.username`	
`ssh.password`	(Credential) Taken into account only if the authentication type includes **pwd**. That means any of the following: • `ssh.auth.type` = *pwd* • `ssh.auth.type` = *pwd+otp* • `ssh.auth.type` = *key+pwd* • `ssh.auth.type` = *key+pwd+otp*
`ssh.totp.secret.key`	(Credential) Taken into account only if the authentication type includes **otp**. That means any of the following: • `ssh.auth.type` = *otp* • `ssh.auth.type` = *key+otp* • `ssh.auth.type` = *pwd+otp* • `ssh.auth.type` = *key+pwd+otp*
`ssh.private.key.type`	The type of the SSH private key. Possible values: • **ssh-rsa** • **ssh-dsa** Default value: *ssh-rsa* **Note** If you choose *ssh-rsa*, the key should be in format **PKCS #8**, non-encrypted.
`ssh.private.key`	(Credential) Taken into account only if the authentication type includes **key**. That means any of the following: • `ssh.auth.type` = *key* • `ssh.auth.type` = *key+pwd* • `ssh.auth.type` = *key+otp* • `ssh.auth.type` = *key+pwd+otp*

Property Name	Description & Value
`ssh.read.groups.command`	Path to the bash command you need to execute to read groups.
`ssh.read.users.command`	Path to the bash command you need to execute to read users.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SSH Server (Beta)* source system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SSH server. For more information, see Manage Transformations [page 2439].

   Default transformation:

   Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "targetPath": "$",
                "sourcePath": "$"
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem",
                "correlationAttribute": true
            },
            {
                "sourcePath": "$.userName",
                "optional": true,
                "correlationAttribute": true
            }
        ]
    },
    "group": {
        "ignore": true,
        "mappings": [
            {
                "targetPath": "$",
                "sourcePath": "$"
            },
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdSourceSystem"
            }
        ]
    }
}
```

5. Now, add a target system to provision users and groups into it. Choose from: Target Systems [page 936]

Next Steps

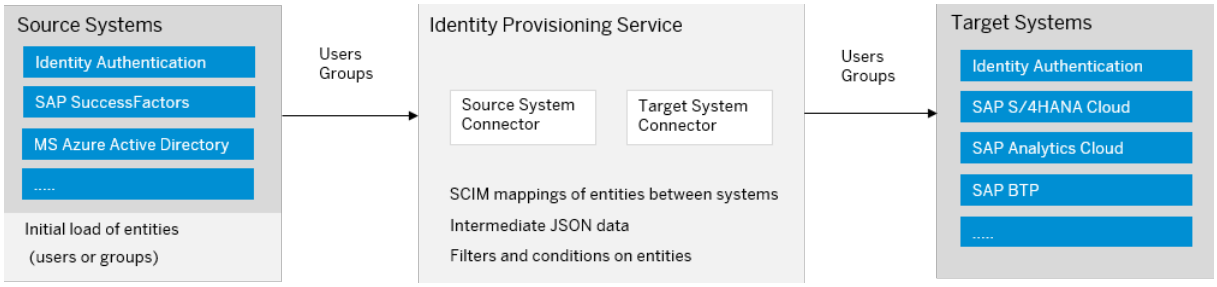
- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2 Target Systems

A target system is the connector used for writing entities.

Target systems are usually cloud systems, where Identity Provisioning creates or updates the entities read from the source system.

A target system can be connected to a single or multiple source systems. In the case of multiple source systems, we recommend that you run the provisioning jobs **successively** for each system, not simultaneously. This way, you'll avoid incorrect overwriting or merging of entity data, hence failed provisioning jobs.



> ⚠ Restriction
>
> By default, the maximum number of productive target systems you are allowed to add for your tenant is **50**. This restriction is valid for *Customer Managed* systems only.
>
> If your business requires using more systems, create an incident for component *BC-IAM-IPS* to request them. Describe your scenarios and provide a reason why you need the additional systems.

Related Information

System Types [page 70]

1.6.2.1 Identity Authentication

Follow this procedure to set up SAP Cloud Identity Service – Identity Authentication as a target system.

Prerequisites

To establish the connection between Identity Provisioning and Identity Authentication, you need to set up the technical user (of type *System*) in Identity Authentication and assign this user the necessary authorizations. You have the option to do it either now, as a prerequisite, or while configuring Identity Authentication as a target system, as described in step 3.

Context

Identity Authentication provides authentication and single sign-on for users in the cloud.

Using Identity Provisioning, you can read corporate users from on-premise or cloud systems, and provision them to the Identity Authentication user store. This way, you can implement secure authentication, single sign-on (SSO), strong authentication and mobile SSO so that the provisioned users have access to the business applications of your company. For example, you can implement two-factor authentication and mobile SSO for SAP SuccessFactors users.

You can use the Identity Provisioning UI to configure Identity Authentication as a target system where you can provision users, groups and group members. The target system consumes SCIM 2.0 API provided by Identity Authentication.

There are two versions of the Identity Authentication SCIM API. They are controlled by the `ias.api.version` property as follows:

- When the value is set to **1** or the property is not defined (typical for systems created before versioning was introduced on July 9, 2021) - Identity Authentication SCIM API (in short, SCIM API version 1) is used.

  > ### Note
  >
  > The SAP Cloud Identity Services – Identity Authentication SCIM API (version 1) will be decommissioned on November 30, 2028. You must use its successor, the Identity Directory SCIM API. This requires updating your Identity Authentication connector from version 1 to version 2, as described in Update Connector Version [page 2424].

- When the value is set to **2** - Identity Directory SCIM API (in short, SCIM API version 2) is used. This value is set automatically for all manually created systems in the Identity Provisioning UI after versioning was introduced on July 9, 2021.
  SCIM API version 2 does not support managing of group assignments via the SCIM user resource. The "groups" attribute of the user is read-only. This means that the user group assignments should be managed via the SCIM group resources using the "members" attribute (as it is defined by the SCIM standard). Also, the group resource mapping in the transformation is not ignored by default, as it is in SCIM API version 1.

> **Note**
>
> Identity Authentication (using SCIM API version 2) and Identity Directory are sometimes used interchangeably. Identity Directory is the persistency layer of SAP Cloud Identity Services – Identity Authentication.

To create Identity Authentication as a target system, proceed as follows:

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *Identity Authentication* as a target system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and Identity Authentication and configure your authentication method (basic or certificate-based).

   > **Note**
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added Identity Authentication target system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Manage Certificates [page 2450].

      Skip step **a.** if you want to use basic authentication.

      In SAP Cloud Identity Services administration console, perform the next steps. They are relevant for both basic and certificate-based authentication.

   b. Add System as administrator and provide the respective credentials.

      For basic authentication, provide a password. The user ID will be generated automatically when you set the password for the first time.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Save your changes.

   d. Make sure *Manage Users* and *Manage Groups* authorization roles are enabled for the technical user. This way, you can create, edit and delete users and groups in the Identity Authentication user store.

4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
Type	Enter: *HTTP*

Property Name	Description & Value
URL	Specify the URL of the Identity Authentication tenant of your company. For example: **https://mytenant.accounts.ondemand.com** **Note** If your Identity Authentication Shanghai (China) tenants reside on SAP BTP, Neo, you should use the following URL pattern: *https://<tenant_ID>.accounts.sapcloud.cn/*
ProxyType	Enter: *Internet* The Identity Authentication is a cloud solution and is outside of your company on-premise infrastructure.
Authentication	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
User	Valid if *BasicAuthentication* is configured as authentication method. Enter the Client ID (previously User ID) of the Identity Authentication technical user. It is generated automatically for the administrator of type system, when choosing *Secrets* > *Add* > *Save*. For example: *1ab7c243-5de5-4530-8g14-1234h26373ab* If your technical user was created before *January 2020*, enter the T-user. For example: *T000003*
Password	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the Client Secret (previously Password) of the Identity Authentication technical user. It is generated automatically for the administrator of type system, when choosing *Secrets* > *Add* > *Save*.

Optional Properties

Property Name	Description & Value
• `ias.<property_name>` • `scim.<property_name>`	When using SCIM API version 2, property names start with `ias` prefix, for example: `ias.user.unique.attribute`. When using SCIM API version 1, property names start with `scim` prefix, for example: `scim.user.unique.attribute`. For more information, see List of Properties [page 80]. Use the main search or filter properties by *Name* or *System Type* columns.
`ias.user.unique.attribute`	This property defines by which unique attribute(s) an existing user will be resolved in the event of conflicting users. It appears by default when the system is created, and its value is set to *userName*. Other possible values: • *emails[0].value* • *userName,emails[0].value* • *phoneNumbers[0].value* • *externalId*, or another SCIM attribute, or a conjunction of SCIM attributes If the service finds a user on the target system through this filter, then the conflicting user will overwrite the existing one. If the service does not find a user on the target system through this filter, the creation will fail. For more information, see: List of Properties [page 80]
`ips.failed.request.retry.attempts`	Predefined value: *2*
`ips.failed.request.retry.attempts.interval`	Predefined value: *60*
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

5. (Optional) Configure the transformations.

Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Identity Authentication* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

When Identity Authentication is configured as a target system, the default transformation logic writes all the user attributes in the Identity Authentication user store. The logic is provided by the Identity Authentication SCIM REST API, which then maps the attributes to the internal SCIM representation.

Handling Application-Specific Groups

In case you want to provision application-specific groups between the following pairs of provisioning systems:

Source System	Target System
Identity Authentication (SCIM API version 2)	Identity Authentication (SCIM API version 2)
Identity Authentication (SCIM API version 2)	Local Identity Directory
Local Identity Directory	Identity Authentication (SCIM API version 2)

The Identity Provisioning will search and try to resolve the groups by the attribute values of `displayName`, `applicationId`, and `type`. To ensure successful provisioning, you must define value mappings for the attribute `applicationId` in the transformation mappings of the source or the target system. You should define to which `applicationId` in the target system will be mapped the `applicationId` read from the source system.

For example:

```
{
"sourcePaths": [
    "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId']"
],
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId']",
"valueMappings": [
    {
        "key": [
            "<your_source_system_applicationId>"
        ],
        "mappedValue": "<your_target_system_applicationId>"
    }
    ],
    "condition": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId'] EMPTY false",
    "type": "valueMapping",
    "defaultValue": ""
}
```

For more information, see Transformation Expressions [page 391] → section **valueMapping**.

SCIM API version 1: Identity Authentication: SCIM REST API

SCIM API version 2: Identity Directory SCIM API

When Identity Authentication is configured as a target system, the default transformation logic:

- Reads all user attributes from the intermediate SCIM representation.
- Skips some of the attributes from the identity records.

This way, the transformation logic ensures that the identity data, sent to the Identity Provisioning SCIM REST API, is consistent.

Default transformation for SCIM API version 1:

Code Syntax

```
{
  "user": {
    "condition": "($.emails.length() > 0) && ($.name.familyName EMPTY
false) && isValidEmail($.emails[0].value)",
    "mappings": [
      {
        "sourcePath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.corporateGroups"
      },
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
        "targetPath": "$.schemas[0]"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
        "targetPath": "$.schemas[1]"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "optional": true
      },
      {
        "sourcePath": "$.emails[*].value",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.emails[?(@.value)]"
      },
      {
        "sourcePath": "$.userType",
        "targetPath": "$.userType",
        "defaultValue": "employee",
        "optional": true
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.name.middleName",
        "targetPath": "$.name.middleName",
        "optional": true
      },
      {
```

```
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.name.honorificPrefix",
        "targetPath": "$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath": "$.addresses",
        "targetPath": "$.addresses",
        "preserveArrayWithSingleElement": true,
        "defaultValue": [],
        "optional": true,
        "functions": [
          {
            "function": "putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          },
          {
            "condition": "(@.type NIN ['work', 'home'])",
            "function": "putIfPresent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.phoneNumbers",
        "targetPath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "optional" : true
      },
      {
```

```
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional" : true,
        "functions": [
            {
                "function": "resolveEntityIds"
            }
        ]
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "optional" : true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "defaultValue": true,
        "optional": true
      },
      {
        "constant": "false",
        "targetPath": "$.sendMail",
        "scope": "createEntity"
      },
      {
        "constant": "true",
        "targetPath": "$.mailVerified",
        "scope": "createEntity"
      },
```

```
      {
        "constant": "disabled",
        "targetPath": "$.passwordStatus",
        "scope": "createEntity"
      },
      {
        "constant": "39",
        "targetPath": "$.sourceSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.timezone",
        "targetPath": "$.timeZone",
        "optional": true
      }
    ]
  },
  "group": {
    "ignore": true,
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName"
      },
      {
        "scope": "createEntity",
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "functions": [
          {
            "type": "replaceAllString",
            "regex": "[\\s\\p{Punct}]",
            "replacement": "_"
          }
        ]
      },
      {
        "scope": "createEntity",
        "optional": true,
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']"
      },
      {
        "optional": true,
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']"
      },
      {
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "sourcePath": "$.members[*].value",
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "type": "resolveEntityIds"
          }
```

```
        ]
      }
    ]
  }
}
```

Default transformation for SCIM API version 2:

Code Syntax

```
{
  "user": {
    "condition": "($.emails EMPTY false) && ($.userName EMPTY false) &&
isValidEmail($.emails[0].value)",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:core:2.0:User","urn:ietf:params:scim:schemas
:extension:enterprise:2.0:User","urn:ietf:params:scim:schemas:extension:sap
:2.0:User","urn:sap:cloud:scim:schemas:extension:custom:2.0:User"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['emails']",
        "scope": "createEntity",
        "functions": [
           {
                 "function": "putIfAbsent",
                 "key": "verified",
                 "defaultValue": true
           }
         ]
      },
      {

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['emails'][*]['type']",
        "type": "remove"
      },
      {
        "sourcePath": "$.emails[*].['value','display','primary']",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.name.middleName",
```

```
      "targetPath": "$.name.middleName",
      "optional": true
    },
    {
      "sourcePath": "$.name.familyName",
      "targetPath": "$.name.familyName",
      "optional": true
    },
    {
      "sourcePath": "$.name.honorificPrefix",
      "targetPath": "$.name.honorificPrefix",
      "optional": true
    },
    {
      "sourcePath": "$.addresses",
      "targetPath": "$.addresses",
      "preserveArrayWithSingleElement": true,
      "defaultValue": [],
      "optional": true,
      "functions": [
        {
          "function": "putIfAbsent",
          "key": "type",
          "defaultValue": "work"
        },
        {
          "condition": "(@.type NIN ['work', 'home'])",
          "function": "putIfPresent",
          "key": "type",
          "defaultValue": "work"
        }
      ]
    },
    {
      "sourcePath": "$.phoneNumbers",
      "targetPath": "$.phoneNumbers",
      "preserveArrayWithSingleElement": true,
      "optional": true
    },
    {
      "sourcePath": "$.displayName",
      "targetPath": "$.displayName",
      "optional": true
    },
    {
      "sourcePath": "$.userType",
      "targetPath": "$.userType",
      "defaultValue": "employee",
      "optional": true
    },
    {
      "sourcePath": "$.locale",
      "targetPath": "$.locale",
      "optional": true
    },
    {
      "sourcePath": "$.timezone",
      "targetPath": "$.timezone",
      "optional": true
    },
    {
      "sourcePath": "$.active",
      "targetPath": "$.active",
      "defaultValue": true,
      "optional": true
```

```
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "optional" : true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
```

```
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional" : true,
        "functions": [
            {
              "function": "resolveEntityIds"
            }
          ]
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "optional" : true
      },
      {
        "constant": false,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sendMail']",
        "scope": "createEntity"
      },
      {
        "constant": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mailVerified']",
        "scope": "createEntity"
      },
      {
        "constant": "disabled",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['passwordDetails']['status']",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['attributes']",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['attributes']"
      },
      {
        "constant": "<your-initial-password>",
        "targetPath": "$.password",
        "scope": "createEntity",
        "ignore": "true"
      },
      {
        "constant": "<your-source-system-type-code>",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
        "scope": "createEntity",
        "ignore": true
      },
      {
        "constant": "<your-source-system-id>",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystemId']",
        "scope": "createEntity",
        "ignore": true
```

```
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:core:2.0:Group","urn:sap:cloud:scim:schemas:
extension:custom:2.0:Group","urn:ietf:params:scim:schemas:extension:sap:2.0
:Group"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
            {
              "entityType": "user",
              "type": "resolveEntityIds"
            }
          ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
            {
              "entityType": "group",
              "type": "resolveEntityIds"
            }
          ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']"
      },
```

```
      {
        "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false",

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false",

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalLocation']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalLocation']",
        "scope": "createEntity"
      }
    ]
  }
}
```

→ Remember

If you set `$.userType` to "**public**", all passwords will be written by default in the Identity Authentication. Thus, all provisioned users will successfully sign in to Identity Authentication target system.

When `$.userType` is set to "**employee**", the sign-in behavior of the provisioned users depends on whether users have been created with or without a password, and where these passwords are stored. Thus, you need to modify the target transformations accordingly in order for the users to successfully sign in to the Identity Authentication console.

To learn more, see *Guided Answers*: Identity Authentication: Provisioned Users Can't Sign In

6. Add a source system from which to read users and groups. Choose from: Source Systems [page 585]

> ⚠ Caution
>
> By default, the **email** attribute is unique for *Identity Authentication* but it might not be unique for other systems, such as *SAP SuccessFactors*. That's why, when you choose a source system, make sure that there are no users with duplicate e-mails. If there are such, then after the provisioning job all users with the same e-mail address will be created as a single user in *Identity Authentication*.
>
> To learn more about how to fix this issue or prevent it from happening, see *Guided Answers*: Multiple Users from a Source System Are Created as One in the Target
>
> To learn more about excluding users from Identity Authentication target, see: Exclude Users from Provisioning to Target Systems

If you have configured the **email** attribute to NOT be unique in *Identity Authentication*, then ignore this **Caution** note and the guided answer. For more information about configurable attributes, see: Identity Authentication: Configure User Identifier Attributes

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Identity Authentication: Documentation
Identity Authentication: SCIM REST API
(Guided Answers) Multiple Users from a Source System Are Created as One in the Target
(Guided Answers) Identity Authentication: Provisioned Users Can't Sign In

1.6.2.2 Intelligent Opportunity Analyzer

Follow this procedure to set up intelligent opportunity analyzer as a target system.

Prerequisites

You have created an instance and generated a service key for the *scim* service plan of *Intelligent Opportunity Analyzer*. For more information, see Assigning the Entitlements for Intelligent Opportunity Analyzer to Your Subaccounts.

Context

Intelligent opportunity analyzer is a service that comes integrated with SAP Ariba Category Management for identifying and managing opportunities that lead to improving the performance of a purchasing category.

You can use Identity Provisioning to configure intelligent opportunity analyzer as a target system where you can provision users from source systems.

> **Note**
>
> Intelligent opportunity analyzer does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *Intelligent Opportunity Analyzer* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL provided by the service key under the *scim-v1* field without adding the path information.
`ProxyType`	Enter: *Internet*

Property Name	Value
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the value from the *clientid* field of the service key.
`Password`	(Credential) Enter the value from the *clientsecret* field of the service key.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. This is the value from the *url* field of the service key. Ensure to append the value `/oauth/token?grant_type=client_credentials`.
(Optional) `ioa.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. The property is not added automatically at system creation. Possible values: *userName* and *emails[*].value* For more information, see: **User Update and Uniqueness** section below and List of Properties [page 80].
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Intelligent Opportunity Analyzer* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your intelligent opportunity analyzer system. For more information, see:

   Manage Transformations [page 2439]

   Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target intelligent opportunity analyzer entity.

   > ### Note
   >
   > The documentation for the Intelligent Opportunity Analyzer API and its integration with Identity Provisioning is still under development. Refer to the Intelligent Opportunity Analyzer Setup for updates.

User Update and Uniqueness

The write transformation of intelligent opportunity analyzer supports two user unique attributes for managing conflict resolution: *username* and *email*. Both of the attributes are mandatory. The email is a multivalue attribute, indicating that a user may have multiple unique emails.

The uniqueness of a user in the target system is determined not only by the individual attributes or the combination of both, but also by considering whether one or multiple existing users in the target system share the same values for these unique attributes. Have in mind the following specifics when updating an existing user in the target system, in accordance with the configuration of the `ioa.user.unique.attribute` property:

- When `ioa.user.unique.attribute`= *userName*
  If a user in the source system, with a specific username and an email, matches an existing user in the target system by the same email, but a different userName, the user provisioning will fail. In this case, Identity Provisioning will try to create a new user but will fail, since a user with the same email already exists.
- When `ioa.user.unique.attribute`= *emails[*].value*
  If a user in the source system, with a specific username and an email, matches an existing user in the target system by the same username, but a different email, the user provisioning will fail. In this case, Identity Provisioning will try to create a new user but will fail, since a user with the same username already exists.
- When `ioa.user.unique.attribute`= *userName, emails[*].value*
  - If a user in the source system, with a specific username and multiple emails, matches an existing user in the target system by the same username and at least one identical email, the user in the target system will be updated.
  - If a user in the source system, with a specific username and multiple emails, matches multiple existing users in the target system (each sharing one of the multiple emails but having different usernames), the user provisioning will fail. In this case, the user will neither be created nor updated.

Default transformation:

Code Syntax

```
{
  "user":{
    "mappings":[
     {
       "constant": [
         "urn:ietf:params:scim:schemas:core:2.0:User"
       ],
       "targetPath": "$.schemas"
     },
     {
       "sourceVariable": "entityIdTargetSystem",
       "targetPath": "$.id"
     },
     {
       "sourcePath": "$.userName",
       "targetPath": "$.userName"
     },
     {
       "sourcePath": "$.emails",
       "targetPath": "$.emails",
       "preserveArrayWithSingleElement": true
     },
```

```
        {
          "sourcePath":"$.loginName",
          "targetPath":"$.loginName",
          "optional":true
        },
        {
          "sourcePath":"$.name.givenName",
          "targetPath":"$.name.givenName",
          "optional":true
        },
        {
          "sourcePath":"$.name.familyName",
          "targetPath":"$.name.familyName",
          "optional":true
        },
        {
          "sourcePath":"$.name.formatted",
          "targetPath":"$.name.formatted",
          "optional":true
        },
        {
          "sourcePath":"$.name.middleName",
          "targetPath":"$.name.middleName",
          "optional":true
        },
        {
          "sourcePath":"$.name.honorificPrefix",
          "targetPath":"$.name.honorificPrefix",
          "optional":true
        },
        {
          "sourcePath":"$.name.honorificSuffix",
          "targetPath":"$.name.honorificSuffix",
          "optional":true
        },
        {
          "sourcePath":"$.active",
          "targetPath":"$.active"
        },
        {
          "sourcePath":"$.displayName",
          "targetPath":"$.displayName",
          "optional":true
        },
        {
          "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
          "optional": true,
          "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
        }
      ]
    }
}
```

5. Add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].

- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.3 Local Identity Directory

Follow this procedure to set up Local Identity Directory as a target system.

Prerequisites

> **Note**
>
> The *Local Identity Directory* connector is available for both bundle and standalone tenants running on SAP Cloud Identity Services infrastructure.

Context

Identity directory is the user store of SAP Cloud Identity Services. It provides a central place for storing and managing users, groups and custom schemas through the System for Cross-domain Identity Management 2.0 REST API, in short Identity Directory SCIM API.

You can provision users and groups to the identity directory from various source systems by configuring the Local Identity Directory connector as a target system in the SAP Cloud Identity Services administration console. The entities are written to the identity directory of your current SAP Cloud Identity Services tenant. Therefore, you don't have to set up connectivity and authentication properties for the target system. In case you want to write entities to the identity directory in another tenant, add Identity Authentication (version 2) as a target system and configure the respective connectivity and authentication properties.

To create Local Identity Directory as a target system, proceed as follows:

Procedure

1. Sign in to SAP Cloud Identity Services administration console and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *Local Identity Directory* as a target system.

   For more information, see Add New Systems [page 2415].
3. **Optional:** If needed, choose the *Properties* tab to configure patch and bulk operations.

Optional Properties

Property Name	Value
`idds.bulk.operations.max.count`	This property sets the number of operations to be performed in one bulk request. • Default value: *20* • Maximum value: *100*
`idds.support.bulk.operation`	This property enables bulk operations for users and groups. • When bulk operations are enabled, Identity Provisioning creates, updates, and deletes multiple users and groups in one request. • When bulk operations are not enabled, Identity Provisioning creates, updates, and deletes one user at a time.
`idds.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with **PATCH** requests, and below which they are provisioned with **PUT** request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. • Default value: *20 000* • Minimum value: *1* • Maximum value: *200 000* **Note** If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members.

Local Identity Directory properties are prefixed with `idds.<property_name>`. For more information, see List of Properties [page 80].

4. **Optional:** Configure transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the Local Identity Directory target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   When Local Identity Directory is configured as a target system, the default transformation logic writes the user attributes in the user store of SAP Cloud Identity Services. The logic is provided by the Identity Directory SCIM API, which then maps the attributes to the internal SCIM representation.

   ### Handling Application-Specific Groups

In case you want to provision application-specific groups between the following pairs of provisioning systems:

Source System	Target System
Identity Authentication (SCIM API version 2)	Identity Authentication (SCIM API version 2)
Identity Authentication (SCIM API version 2)	Local Identity Directory
Local Identity Directory	Identity Authentication (SCIM API version 2)

The Identity Provisioning will search and try to resolve the groups by the attribute values of `displayName`, `applicationId`, and `type`. To ensure successful provisioning, you must define value mappings for the attribute `applicationId` in the transformation mappings of the source or the target system. You should define to which `applicationId` in the target system will be mapped the `applicationId` read from the source system.

For example:

```
{
"sourcePaths": [
    "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId']"
],
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId']",
"valueMappings": [
    {
        "key": [
            "<your_source_system_applicationId>"
        ],
        "mappedValue": "<your_target_system_applicationId>"
    }
    ],
    "condition": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId'] EMPTY false",
    "type": "valueMapping",
    "defaultValue": ""
}
```

For more information, see Transformation Expressions [page 391] → section **valueMapping**.

You can change the default transformation mapping rules to reflect your current setup of entities in your Local Identity Directory. For more information, see:

- Manage Transformations [page 2439]
- SCIM API version 2: Identity Directory SCIM API

Default transformation for Local Identity Directory:

Code Syntax

```
{
    "user": {
        "condition": "($.emails EMPTY false) && ($.userName EMPTY false)
&& isValidEmail($.emails[0].value)",
        "mappings": [
            {
                "targetPath": "$.id",
```

```
                "sourceVariable": "entityIdTargetSystem"
            },
            {
                "targetPath": "$.schemas",
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:User",

"urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
                    "urn:ietf:params:scim:schemas:extension:sap:2.0:User",
                    "urn:sap:cloud:scim:schemas:extension:custom:2.0:User"
                ]
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['emails']",
                "preserveArrayWithSingleElement": true,
                "scope": "createEntity",
                "functions": [
                    {
                        "function": "putIfAbsent",
                        "key": "verified",
                        "defaultValue": true
                    }
                ]
            },
            {

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['emails'][*]['type']",
                "type": "remove"
            },
            {
                "sourcePath": "$.emails[*].value",
                "targetPath": "$.emails[?(@.value)]",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName",
                "optional": true
            },
            {
                "sourcePath": "$.name.middleName",
                "targetPath": "$.name.middleName",
                "optional": true
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName",
                "optional": true
            },
            {
                "sourcePath": "$.name.honorificPrefix",
                "targetPath": "$.name.honorificPrefix",
                "optional": true
            },
            {
                "sourcePath": "$.addresses",
                "targetPath": "$.addresses",
                "optional": true,
```

```
                "preserveArrayWithSingleElement": true,
                "functions": [
                    {
                        "function": "putIfAbsent",
                        "key": "type",
                        "defaultValue": "work"
                    },
                    {
                        "function": "putIfPresent",
                        "condition": "(@.type NIN ['work', 'home'])",
                        "key": "type",
                        "defaultValue": "work"
                    }
                ],
                "defaultValue": []
            },
            {
                "sourcePath": "$.phoneNumbers",
                "targetPath": "$.phoneNumbers",
                "optional": true,
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
            },
            {
                "sourcePath": "$.userType",
                "targetPath": "$.userType",
                "optional": true,
                "defaultValue": "employee"
            },
            {
                "sourcePath": "$.locale",
                "targetPath": "$.locale",
                "optional": true
            },
            {
                "sourcePath": "$.timezone",
                "targetPath": "$.timezone",
                "optional": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true,
                "defaultValue": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
                "optional": true
            },
            {
```

```
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "optional": true,
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
```

```
                "optional": true
            },
            {
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sendMail']",
                "constant": false,
                "scope": "createEntity"
            },
            {
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mailVerified']",
                "constant": true,
                "scope": "createEntity"
            },
            {

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['passwordDetails']['status']",
                "constant": "disabled",
                "scope": "createEntity"
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['attributes']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['attributes']",
                "optional": true,
                "preserveArrayWithSingleElement": true
            },
            {
                "targetPath": "$.password",
                "ignore": true,
                "constant": "<your-initial-password>",
                "scope": "createEntity"
            },
            {
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
                "ignore": true,
                "constant": "<your-source-system-type-code>",
                "scope": "createEntity"
            },
            {
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystemId']",
                "ignore": true,
                "constant": "<your-source-system-id>",
                "scope": "createEntity"
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "targetPath": "$.id",
                "sourceVariable": "entityIdTargetSystem"
            },
            {
                "targetPath": "$.schemas",
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:Group",

"urn:sap:cloud:scim:schemas:extension:custom:2.0:Group",
                    "urn:ietf:params:scim:schemas:extension:sap:2.0:Group"
                ]
```

```
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.members[*].value",
                "targetPath": "$.members[?(@.value)]",
                "optional": true,
                "preserveArrayWithSingleElement": true,
                "functions": [
                    {
                        "function": "resolveEntityIds",
                        "entityType": "user"
                    }
                ]
            },
            {
                "sourcePath": "$.members[*].value",
                "targetPath": "$.members[?(@.value)]",
                "optional": true,
                "preserveArrayWithSingleElement": true,
                "functions": [
                    {
                        "function": "resolveEntityIds",
                        "entityType": "group"
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
                "scope": "createEntity"
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
                "optional": true,
                "scope": "createEntity"
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']",
                "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']",
                "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
```

```
                "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false",
                "optional": true
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
                "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false",
                "optional": true
            },
            {
                "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false",
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']"
            },
            {
                "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']
EMPTY false",

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalLocation']",
                "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalLocation']",
                "scope": "createEntity"
            }
        ]
    }
}
```

5. Add a source system from which to read users and groups. Choose from: Source Systems [page 585]

> ⚠ Caution
>
> By default, the **email** attribute is unique for Local Identity Directory but it might not be unique for other systems, such as *SAP SuccessFactors*. That's why, when you choose a source system, make sure that there are no users with duplicate e-mails. If there are such, then after the provisioning job all users with the same e-mail address will be created as a single user in Local Identity Directory.
>
> To learn more about how to fix this issue or prevent it from happening, see *Guided Answers*: Multiple Users from a Source System Are Created as One in the Target
>
> To learn more about excluding users from Local Identity Directory target, see: Exclude Users from Provisioning to Target Systems

If you have configured the **email** attribute to NOT be unique in Local Identity Directory, then ignore this **Caution** note and the guided answer. For more information about configurable attributes, see: Configure User Identifier Attributes

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

(Guided Answers) Multiple Users from a Source System Are Created as One in the Target
(Guided Answers) Identity Authentication: Provisioned Users Can't Sign In

1.6.2.4 SAP Advanced Financial Closing

Follow this procedure to set up SAP Advanced Financial Closing as a target system.

Prerequisites

You have created an instance and generated a service key for the standard service plan of SAP Advanced Financial Closing. For more information, see: How to Manage User Access Using the SCIM API Provided.

The service key contains the API URL and the OAuth credentials (`clientid` and `clientsecret`) under the `uaa` property.

Context

SAP Advanced Financial Closing allows you to define, automate, process, and monitor the financial closing tasks for the entities of your organization. It is an SAP BTP application that runs in an SAP BTP subaccount.

You can use Identity Provisioning to configure SAP Advanced Financial Closing as a target system for provisioning users and user groups from various source systems.

> **Note**
>
> The last active user in a user group cannot be deleted, as specified in the restriction described in How to Delete Users.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Advanced Financial Closing* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL provided by the service key under *endpoints* > *scim2* without adding the path information. For example: `https://afc-production-afc-api.cfapps.eu10.hana.ondemand.com`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the user ID provided by the service key under *uaa* > *clientid*.
`Password`	(Credential) Enter the password provided by the service key under *uaa* > *clientsecret*.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL provided by the service key of your SAP Advanced Financial Closing instance. It follows the pattern: **`<uaa.url>/oauth/token`**, where: • `<uaa.url>` is the URL provided by the service key under *uaa* > *url*. • **`/oauth/token`** is the suffix you need to add.

Property Name	Value
`s4hana.afc.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. Possible values: • *userName* • *emails[0].value* • *userName,emails[0].value*
`s4hana.afc.group.unique.attribute`	If Identity Provisioning tries to provision a group that already exists in the target system (a conflicting group), this property defines the unique attributes by which the existing group will be searched and resolved. The default value is *displayName*.
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map user and group attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Advanced Financial Closing* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Advanced Financial Closing system. For more information, see: Manage Transformations [page 2439].

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   To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

   Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP Advanced Financial Closing entity.

Default transformation:

Code Syntax

```
{
  "user": {
    "condition": "('%s4hana.afc.group.prefix%' === 'null') || ($.groups[?
(@.display =~ /%s4hana.afc.group.prefix%.+/)] empty false)",
    "mappings": [
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:User",
          "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
          "urn:ietf:params:scim:schemas:extension:sap:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.name.formatted",
        "targetPath": "$.name.formatted",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "condition": "$.emails[0].length() > 0",
        "targetPath": "$.emails[0].primary",
        "constant": true
```

```
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails",
        "functions": [
          {
            "function": "putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "sourcePath": "$.phoneNumbers",
        "targetPath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
s4hana.afc.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
s4hana.afc.group.prefix) && (isRegularGroup() ||
isApplicationSpecificGroup())",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:Group",
          "urn:sap:cloud:scim:schemas:extension:custom:2.0:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
s4hana.afc.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%s4hana.afc.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
s4hana.afc.group.prefix)",
            "function": "replaceFirstString",
```

```
            "regex": "%s4hana.afc.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
s4hana.afc.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%s4hana.afc.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

5. Add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.5 SAP Advanced Workflow

Follow this procedure to set up a target connector for SAP Advanced Workflow.

Prerequisites

- You have technical credentials for SAP Advanced Workflow. Note that SAP Advanced Workflow is available to SAP SuccessFactors Incentive Management customers as an optional add-on. You create an Admin user in SAP SuccessFactors Incentive Management, which is synchronized with SAP Advanced Workflow. For more information, see: Adding an Admin User and Commissions User Synchronization.
- You have set up SSO between Identity Authentication and SAP Advanced Workflow. For more information, see Integration with SAP IdP.

Context

SAP Advanced Workflow enables you to analyse, organize, and execute business processes to connect people, data, and daily activities. Workflow provides you the tools you need to configure and customize your business processes based on your specific business needs.

After fulfilling the prerequisites, follow the procedure below to add a target system for SAP Advanced Workflow where you can provision **users** from source systems. This target system consumes Workflow SCIM API.

> ⓘ Note
>
> SAP Advanced Workflow does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Advanced Workflow* as a target system. For more information, see Add New Systems [page 2415].

3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the API of your SAP Advanced Workflow system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the user for your SAP Advanced Workflow system.
`Password`	(Credential) Enter the password for your SAP Advanced Workflow user.
`awf.domain`	The domain name is the name of your SAP Advanced Workflow tenant. If you don't know your tenant name, contact your supervisor or administrator, or refer to the email notification you received when your account was created.
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Advanced Workflow* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Advanced Workflow. For more information, see:

Manage Transformations [page 2439]

The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target entity. If the entity has more than one e-mail addresses, only one of them is used. It is either the e-mail set as primary in the source system, or if no primary e-mail is set, the one that comes first is used.

Default transformation:

Code Syntax

```
{
    "user": {
        "condition": "($.emails EMPTY true) ||
isValidEmail($.emails[0].value)",
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": ["urn:ietf:params:scim:schemas:core:2.0:User",
"urn:ietf:params:scim:schemas:extension:sap.spm.workflow:2.0:User"],
                "targetPath": "$.schemas"
            },
                    {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName",
                "optional": true
            },
            {
                "sourcePath": "$.name.middleName",
                "targetPath": "$.name.middleName",
                "optional": true
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName",
                "optional": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.phoneNumbers",
                "targetPath": "$.phoneNumbers",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.timezone",
                "targetPath": "$.timezone",
                "optional": true
            }
        ]
    }
}
```

5. Now, add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.6 SAP Analytics Cloud

Follow this procedure to set up SAP Analytics Cloud as a target system.

Prerequisites

- In SAP Analytics Cloud, you have enabled a custom SAML Identity Provider, for which *User Attribute* is set to **Custom SAML User Mapping**. To learn how, see: Enabling a Custom SAML Identity Provider
- Add an OAuth client with authorization grant **Client Credentials**. To learn how, see: Managing OAuth Clients and Trusted Identity Providers

Context

SAP Analytics Cloud is an all-in-one cloud product offered as software as a service for business intelligence, planning, and predictive analytics.

You can use Identity Provisioning to configure SAP Analytics Cloud as a target system where you can provision users and groups. The target system consumes SCIM 2.0 API provided by SAP Analytics Cloud. For more information, see SAP Analytics Cloud: User and Team Provisioning API.

There are two versions of the SAP Analytics Cloud SCIM API. They are handled by the `sac.api.version` property as follows:

- When the value is set to **1** or the property is not defined (typical for systems created before versioning was introduced on April 10, 2023), SAP Analytics Cloud SCIM API version 1 is used. This is the default value.
- When the value is set to **2** - SAP Analytics Cloud SCIM API version 2 is used. This version is released with enhancements, such as: support for patch operations.

For more information on the differences between SAP Analytics Cloud SCIM API version 1 and 2, see Managing Users and Teams.

For more information on how to update to version 2, see Update Connector Version [page 2424].

> **Note**
>
> In SAP Analytics Cloud, groups are considered teams.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Analytics Cloud* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP Analytics Cloud system without adding the path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the client ID to retrieve the OAuth access token for SAP Analytics Cloud.
`Password`	(Credential) Enter the client secret to retrieve the OAuth access token for SAP Analytics Cloud.
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Analytics Cloud instance. This token URL is listed in the *OAuth Clients* section of the *App Integration* page. For more information, refer to *Authorize API Access for OAuth Clients* in Manage OAuth Clients
`sac.api.csrf.protection`	Specifies whether to fetch a CSRF token when sending requests to the system. This property is automatically added to the system, with default value: **enabled** **Note** If the property is missing, its default value is **disabled**.

Property Name	Description & Value
(Optional) `sac.csrf.token.path`	Path which is appended to the URL to retrieve the CSRF token. The property is automatically added during system creation, when the system in use is SAP Analytics Cloud consuming SCIM API version 2. Based on the version of the SCIM API consumed, the default value for the property is: • SAP Analytics Cloud SCIM API version 1: `/api/v1/scim/Users?count=1` **Note** The property is present only for systems created before December 3, 2024. Systems created after this date use the default value for the connector version without adding the property. • SAP Analytics Cloud SCIM API version 2: `/api/v1/scim2/Users?count=1` **Note** If you delete the property, the Identity Provisioning will use the default value for the property for SAP Analytics Cloud using SCIM API **version 1**.

Property Name	Description & Value
`sac.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists in SAP Analytics Cloud. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). According to your use case, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Note** If this property is missing, or you've deleted it, and the service does not find such a *userName*, it will try again to resolve the conflicting user – by *email*. If the second attempt for resolution is unsuccessful too, the creation of the conflicting user fails. For more information, see: List of Properties [page 80]
(Optional) `sac.api.version`	Handles the version of SAP Analytics Cloud SCIM API. **Possible values:** • *1* - Indicates that SAP Analytics Cloud SCIM API version 1 is used. • *2* - Indicates that SAP Analytics Cloud SCIM API version 2 is used. Default value: *1*

Property Name	Description & Value
(Optional) `sac.support.bulk.operation`	Set this property to *true* if you want to enable SCIM bulk operations for provisioning users. That means, the Identity Provisioning service can write, update, and delete a potentially large collection of users in a single request. For more information, see: SCIM Protocol: Bulk Operations If not specified, the default value is *false*. **Note** SCIM bulk operations are not supported for provisioning groups to SAP Analytics Cloud.
(Optional) `sac.bulk.operations.max.count`	If you have enabled the SCIM bulk operations, you can use this property to set the number of users to be provisioned per request. Default value: • **`100`** - when using SAP Analytics Cloud SCIM API version 1. • **`30`** - when using SAP Analytics Cloud SCIM API version 2. **Note** The value must not exceed the number of entities defined by the SAP Analytics Cloud system as a SCIM service provider. Otherwise, the provisioning job will fail with HTTP response code 413 (*Payload Too Large*).
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Property Name	Description & Value
(Optional) `ips.http.header.x-ignore-roles-if-missing`	Use this property to control whether users or teams (referred to as groups) should keep their role assignments in SAP Analytics Cloud (SCIM API version 1) when no roles are provisioned. Possible values: • true - Users or teams will keep their role assignments when no roles are provisioned. • false - Users or teams will lose their role assignments when no roles are provisioned. This is the default value. For more information, see SAP note 3092730. In SAP Analytics Cloud (SCIM API version 2), this behavior is managed by `sac.support.patch.operation` property.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Analytics Cloud* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Analytic Cloud system. For more information, see: Manage Transformations [page 2439]

   SAP Analytics Cloud REST API: Managing Users and Teams

   Managing Users and Teams → api/v1/scim

   Managing Users and Teams → api/v1/scim2

   Default transformation for SAP Analytics Cloud SCIM API version 1:

   Code Syntax

```
{
    "user": {
        "condition": "isValidEmail($.emails[0].value) &&
(('%sac.group.prefix%' === 'null') || ($.groups[?(@.display =~ /
%sac.group.prefix%.*/)] empty false))",
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.schemas"
            },
            {
                "sourceVariable": "entityIdTargetSystem",
```

```
                "targetPath": "$.id"
            },
            {
                "sourcePath": "$.emails[0].value",
                "targetPath": "$.userName"
            },
            {
                "condition": "$.emails[?(@.primary == true)].value != []",
                "sourcePath": "$.emails[?(@.primary == true)].value",
                "preserveArrayWithSingleElement": false,
                "optional": true,
                "targetPath": "$.userName"
            },
            {
              "sourcePath": "$.name.givenName",
              "targetPath": "$.name.givenName",
              "optional": true
            },
            {
              "sourcePath": "$.name.middleName",
              "targetPath": "$.name.middleName",
              "optional": true
            },
            {
              "sourcePath": "$.name.familyName",
              "targetPath": "$.name.familyName",
              "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "optional": true,
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails"
            },
            {
                "condition": "$.emails[0].length() > 0",
                "targetPath": "$.emails[0].primary",
                "constant": true
            },
            {
                "sourcePath": "$.roles",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.roles"
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "optional": true,
                "targetPath": "$
['urn:scim:schemas:extension:enterprise:1.0']['manager']['managerId']",
                "functions": [
                    {
                        "type": "resolveEntityIds"
                    }
                ]
            }
        ]
```

```
    },
    "group": {
        "condition": "isAttributeWithOptionalPrefix($.displayName,
sac.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
sac.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
        "mappings": [
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.schemas"
            },
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.id",
                "scope": "createEntity",
                "functions": [
                    {
                        "condition": "isAttributeWithMandatoryPrefix(@,
sac.group.prefix)",
                        "function": "replaceFirstString",
                        "regex": "%sac.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "isAttributeWithMandatoryPrefix(@,
sac.group.prefix)",
                        "function": "replaceFirstString",
                        "regex": "%sac.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$.roles",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.roles"
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

Default transformation for SAP Analytics Cloud SCIM API version 2:

Code Syntax

```
{
  "user": {
    "condition": "isValidEmail($.emails[0].value) &&
(('%sac.group.prefix%' === 'null') || ($.groups[?(@.display =~ /
%sac.group.prefix%.*/)] empty false))",
    "mappings": [
      {
        "constant": [
          "urn:sap:params:scim:schemas:extension:sac:2.0:user-custom-
parameters",
          "urn:ietf:params:scim:schemas:core:2.0:User",
          "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
          "urn:ietf:params:scim:schemas:extension:sap:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.emails[0].value",
        "targetPath": "$.userName"
      },
      {
        "condition": "$.emails[?(@.primary == true)].value != []",
        "sourcePath": "$.emails[?(@.primary == true)].value",
        "preserveArrayWithSingleElement": false,
        "optional": true,
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.userName",
        "optional": true,
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.middleName",
        "optional": true,
        "targetPath": "$.name.middleName"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.displayName",
        "optional": true,
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.externalId",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
```

```
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.emails"
      },
      {
        "condition": "$.emails[0].length() > 0",
        "constant": true,
        "targetPath": "$.emails[0].primary"
      },
      {
        "sourcePath": "$.groups[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups[?(@.value)]",
        "functions": [
          {
            "entityType": "group",
            "type": "resolveEntityIds"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:user-custom-parameters']",
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:user-custom-parameters']"
      },
      {
        "sourcePath": "$.emails[0].value",
        "optional": true,

"targetPath": "$['urn:sap:params:scim:schemas:extension:sac:2.0:user-
custom-parameters']['idpUserId']"
      },
      {
        "condition": "$.emails[?(@.primary == true)].value != []",
        "sourcePath": "$.emails[?(@.primary == true)].value",
        "preserveArrayWithSingleElement": false,
        "optional": true,

"targetPath": "$['urn:sap:params:scim:schemas:extension:sac:2.0:user-
custom-parameters']['idpUserId']"
      },
      {

"sourcePath": "$['urn:sap:params:scim:schemas:extension:sac:2.0:user-
custom-parameters']['idpUserId']",
        "optional": true,

"targetPath": "$['urn:sap:params:scim:schemas:extension:sac:2.0:user-
custom-parameters']['idpUserId']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "functions": [
          {
            "type": "resolveEntityIds"
          }
        ]
```

```
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
sac.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
sac.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "mappings": [
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:Group",
          "urn:sap:params:scim:schemas:extension:sac:2.0:group-roles",
          "urn:sap:params:scim:schemas:extension:sac:2.0:group-custom-
parameters"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.id",
        "scope": "createEntity",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
sac.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%sac.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
sac.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%sac.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.externalId",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.roles",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.roles"
      },
      {
```

```
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "entityType": "user",
            "type": "resolveEntityIds"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:group-roles']",
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:group-roles']"
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:group-custom-parameters']",
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:group-custom-parameters']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['description']",
        "optional": true,

"targetPath": "$['urn:sap:params:scim:schemas:extension:sac:2.0:group-
custom-parameters']['description']"
      }
    ]
  }
}
```

⚠ Caution

When running a provisioning job to update users or teams (referred to as groups) with role assignments in SAP Analytics Cloud, these role assignments will be removed as a result of the update. This behavior, which can lead to permission issues, is expected because role assignments in SAP Analytics Cloud are not available as group parameters in some source systems, such as Identity Authentication. To preserve existing role assignments when updating users or teams, proceed as follows based on the version of the SAP Analytics Cloud SCIM API:

- For SCIM API version 1, set the `ips.http.header.x-ignore-roles-if-missing` property to *true* in the SAP Analytics Cloud target system. For more information, see the properties table above and the SAP note 3092730.
- For SCIM API version 2, set the `sac.support.patch.operation` property to *true* in the SAP Analytics Cloud target system.

ⓘ Note

Updating a user in SAP Analytics Cloud using SCIM API version 2 depends on whether user attributes in SAP Analytics Cloud are mapped to SAML attributes in your identity provider. If this is the case, the values of those attributes are populated by the identity provider and cannot be changed by the SAP Analytics SCIM API or the UI. For more information, see Mapping SAML Attributes to Users

5. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

> ⚠ Caution
>
> The **email** attribute is unique for *SAP Analytics Cloud* but it might not be unique for other systems, such as *SAP SuccessFactors*. That's why, when you choose a source system, make sure that there are no users with duplicate e-mails. If there are such, then after the provisioning job all users with the same e-mail address will be created as a single user in *SAP Analytics Cloud*.
>
> To learn more, see *Guided Answers*: Multiple Users from a Source System Are Created as One in the Target
>
> To learn more about excluding users from SAP Analytics Cloud target, see: Exclude Users from Provisioning to Target Systems

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

(Guided Answers) Multiple Users from a Source System Are Created as One in the Target

1.6.2.7 SAP Application Server ABAP

Follow this procedure to set up SAP Application Server ABAP (AS ABAP) as a target system.

Prerequisites

- You have installed the Cloud Connector in your corporate environment and have done the initial configuration. For more information, see: Cloud Connector.
  For more information on how to make a connection to this on-premise system, follow the procedure in Connect to On-Premise Systems [page 2468].
- You have credentials of a technical user with write permissions for AS ABAP. Via this user, the Identity Provisioning service will call the ABAP public API in order to execute a number of function modules. These function modules are listed in **step 1** from the procedure below.
- The technical user must have the required RFC and security authorizations, as described in SAP Note 3666721.

> **Note**
>
> If the technical user has insufficient security authorization, no users or roles can be read.

Context

SAP Application Server ABAP (AS ABAP) offers a user store and user administration capabilities for maintaining users and their authorizations for AS ABAP applications. You can configure AS ABAP as a target system so as to provision new entities from another on-premise or cloud system.

> **Caution**
>
> You can't create or delete groups in AS ABAP. That means:
>
> - On the attempt to create a group in AS ABAP, Identity Provisioning will only add new members or update existing ones. Also, when the service reads a group from a source system, there must be a group with the exact same display name (case sensitive) in the AS ABAP target system. Otherwise, an error will be thrown and the group members will not be updated.
>   If group names in the source system differ from the ones in AS ABAP, you can define value mappings for these group names so the provisioning will still work. To learn how, see Transformation Expressions [page 391] → section **valueMapping**.
> - On the attempt to delete a group in AS ABAP, Identity Provisioning will only remove its members (group assignments). And this can happen only if the relevant group assignments have been provisioned/are present in the target system.

Groups in source systems are mapped to roles in AS ABAP target systems.

> **Note**
>
> Role unassignments are handled differently in AS ABAP version 7.31 and lower, and versions higher than 7.31. For example: If a child role R2 is assigned to a user – first, indirectly through a parent role R1, and then directly to the user, when the parent role is unassigned, expect the following results:
>
> AS ABAP **7.31 and lower**: Directly assigned parent role R1 and indirectly assigned child role R2 are unassigned. Directly assigned child role R2 is still assigned to the user.
>
> AS ABAP **higher than 7.31**: All roles are unassigned from the user.

Procedure

1. Open the Cloud Connector to add an access control system mapping for **AS ABAP**. This is needed to allow the Identity Provisioning service to access AS ABAP as a back-end system on the intranet. To learn how, see: Configure Access Control (RFC)

Go to *Cloud To On-Premise* > *Access Control* tab and select protocol *RFC SNC*. Then, expose the following *exact names* as accessible resources:

- `PRGN_ROLE_GETLIST`
- `BAPI_USER_GETLIST`
- `BAPI_USER_GET_DETAIL`
- `BAPI_USER_CREATE1`
- `BAPI_USER_ACTGROUPS_ASSIGN`
- `IDENTITY_MODIFY`
- `BAPI_USER_DELETE`
- `PRGN_ACTIVITY_GROUPS_LOAD_RFC`

These are Business Application Programming Interface (BAPI) functional modules designed to perform certain tasks in the SAP AS ABAP system, such as retrieving, creating, updating or deleting user data.

For more information about each function module and its parameters, refer to the documentation provided in the SAP system. Logon to your SAP system and run transaction code **SE37**. In the *Function Builder: Initial Screen*, enter the function name, choose *Display* and then *Functional Module Documentation*.

2. Open SAP BTP cockpit, and in your Identity Provisioning subaccount create a destination for the AS ABAP system. To learn how, see: Create RFC Destinations

   The destination configuration is required by the Identity Provisioning service to find the back-end system to be used for writing data. It also provides the credentials of the technical user, needed for the connection to the ABAP public API.

   Below are the fields you have to fill in the cockpit destination before using an AS ABAP client as a target system.

   > ### Note
   >
   > In the RFC destination, you can set only JCo properties. Properties starting with `abap` are connector specific and must be set in the Identity Provisioning UI.

Field/Property Name	Value
Name	Enter a destination name.
Type	Select *RFC*.
User	Enter the user for AS ABAP. The *User* field corresponds to property `jco.client.user` in the exported RFC destination.
Password	(Credential) Enter the password for the AS ABAP user. The *Password* field corresponds to property `jco.client.passwd` in the exported RFC destination.

Field/Property Name	Value
jco.client.client	The property is configured during destination creation in SAP BTP cockpit. Provide the client to be used in the ABAP system. Valid format is a three-digit number.
Proxy Type	Defines the proxy type of the connection you need to provide for your ABAP system. The value is set during destination creation in SAP BTP cockpit. The proxy type *OnPremise* requires the Cloud Connector to access resources within your on-premise network. Enter: *OnPremise*
Direct Connection	
jco.client.ashost	Provide the virtual host entry that you have configured in the Cloud Connector → *Access Control* configuration. At least one property specifying the target host must be set during destination creation.
jco.client.sysnr	Provide the "system number" of the ABAP system.
Load Balancing Connection	
jco.client.mshost	Represents the message server host to be used. At least one property specifying the target host must be set during destination creation.
jco.client.r3name	Provide the three-character system ID of the ABAP system to be addressed.
jco.client.msservt	Provide the port on which the message server is listening for incoming requests. You can use this property as an alternative to `jco.client.r3name`.
Optional Properties	
`jco.destination.peak_limit`	The value represents the maximum number of active connections that can simultaneously be created for a destination. For example: *10*
`jco.destination.pool_capacity`	The value represents the maximum number of idle connections kept open by the destination. For example: *5*

Field/Property Name	Value
`ips.overwrite.existedbefore.assignments`	This property defines whether or not the Identity Provisioning service to overwrite user/group assignments that have existed in the target system before you start provisioning entities to that system. • If you start a provisioning job without setting this property (by default, it's *true*), all assignments from the source group will overwrite the ones from the target group. • If you set the property to *false*, all existing assignments will be kept in the target system group, and the new ones will just be added.
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of group assignments to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of group assignments, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
`ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

> **Note**
>
> If connection properties, like `User` and `Password`, are configured both in the destination (SAP BTP cockpit) and on the *Properties* tab (Identity Provisioning User Interface), the values set in the destination are considered with higher priority.

3. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
4. Add *SAP Application Server ABAP* as a target system. To learn how, see: Add New Systems [page 2415]
5. From the *Destination Name* dropdown, choose the RFC destination you have created in *step 2*.
6. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Application Server ABAP* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in AS ABAP. For more information, see Manage Transformations [page 2439].

> **Note**
>
> The value of the **entityIdTargetSystem** attribute stores the entity's unique ID, which is written in the target system as the value of the **USERNAME** attribute. Do not delete or change this statement!

When AS ABAP is configured as a target system for the Identity Provisioning service, the ABAP public API is used to write the identity data retrieved from the relevant source system. During the writing process, the JSON data is following the structure of the export parameters and will turn them into list and tables. Therefore, every JSON array from the intermediate transformation will be represented as a BAPI table, and every child JSON object will be represented as a BAPI structure.

Default transformation:

- Use the default transformation if the version of your AS ABAP system is *7.40* or higher. If your system version is *7.31*, you can still use this transformation – just apply SAP Note 1695883.
  Note: *Active* and *inactive* users are handled as follows:
  - If a user is *inactive* in the source system, Identity Provisioning creates it as *inactive* (locked) in AS ABAP target system too.
  - If a user is *active* in the source system, Identity Provisioning sets the `LOCK` value to `"U"` when updating the user in the AS ABAP target system.
  - If a user is *inactive* in the source system, Identity Provisioning sets the `LOCK` value to `"L"` when updating the user in the AS ABAP target system.

> **Code Syntax**

```
{
    "user": {
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.USERNAME"
            },

// The userName attribute is used also as USERNAME value for the
internal JSON representation.
            {
                "sourcePath": "$.userName",
                "targetPath": "$.USERNAME"
            },
            {
                "sourcePath": "$.externalId",
                "optional": true,
                "targetPath": "$.ALIAS.USERALIAS",
                "defaultValue": ""
            },
            {
                "condition": "$.emails[?(@.primary == true)].value !=
[]",
                "sourcePath": "$.emails[?(@.primary == true)].value",
                "preserveArrayWithSingleElement": false,
                "optional": true,
                "targetPath": "$.ADDRESS.E_MAIL"
            },
            {
                "condition": "$.emails[?(@.type == 'work')].value !=
[]",
                "sourcePath": "$.emails[?(@.type == 'work')].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.ADDRESS.E_MAIL",
                "functions": [
```

```
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "sourcePath": "$.emails[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.ADDSMTP[?(@.E_MAIL)]"
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.ADDRESS.LASTNAME"
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.ADDRESS.FIRSTNAME"
            },
            {
                "sourcePath": "$.name.middleName",
                "optional": true,
                "targetPath": "$.ADDRESS.MIDDLENAME"
            },
            {
                "sourcePath": "$.nickName",
                "optional": true,
                "targetPath": "$.ADDRESS.NICKNAME"
            },
            {
                "sourcePath": "$.name.honorificPrefix",
                "optional": true,
                "targetPath": "$.ADDRESS.TITLE_P"
            },
            {
                "condition": "$.phoneNumbers[?(@.primary ==
true)].value != []",
                "sourcePath": "$.phoneNumbers[?(@.primary ==
true)].value",
                "preserveArrayWithSingleElement": false,
                "optional": true,
                "targetPath": "$.ADDRESS.TEL1_NUMBR"
            },
            {
                "condition": "$.phoneNumbers[?(@.type ==
'work')].value != []",
                "sourcePath": "$.phoneNumbers[?(@.type ==
'work')].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.ADDRESS.TEL1_NUMBR",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "sourcePath": "$.phoneNumbers[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.ADDTEL[?(@.TELEPHONE)]"
            },
// The Identity Provisioning reads the locales from the source system
and map them as specific ABAP language codes in the target ABAP system.
```

```
// The transformation provides an example with key = "bg", which in the
ABAP system is mapped as "W". The default language is en.
// To see all languages and codes supported by AS ABAP, see the Related
Information section below.
            {
                "optional": true,
                "targetPath": "$.DEFAULTS.LANGU",
                "type": "valueMapping",
                "sourcePaths": [
                    "$.locale"
                ],
                "defaultValue": "E",
                "valueMappings": [
                    {
                        "key": [
                            "bg"
                        ],
                        "mappedValue": "W"
                    }
                ]
            },
            {
                "sourcePath": "$.preferredLanguage",
                "optional": true,
                "targetPath": "$.ADDRESS.LANGUP_ISO",
                "functions": [
                    {
                        "function": "toUpperCaseString"
                    }
                ]
            },
// The Identity Provisioning writes standard timezone codes, which are
supported by the AS ABAP BAPI.
// However, the standard SCIM API does not support these codes, thus
the source system can only provide values in format "<region>/<city>".
// The transformation provides an example with key = "Europe/Sofia", which
in the target system is mapped as "EET". The default timezone is CET.
            {
                "optional": true,
                "targetPath": "$.LOGONDATA.TZONE",
                "type": "valueMapping",
                "sourcePaths": [
                    "$.timezone"
                ],
                "defaultValue": "CET",
                "valueMappings": [
                    {
                        "key": [
                            "Europe/Sofia"
                        ],
                        "mappedValue": "EET"
                    }
                ]
            },
             {
                "targetPath": "$.PASSWORD.BAPIPWD",
                "scope": "createEntity",
                "functions": [
                    {
                        "function": "randomPassword",
                        "passwordLength": 24,
                        "minimumNumberOfLowercaseLetters": 1,
                        "minimumNumberOfUppercaseLetters": 1,
                        "minimumNumberOfDigits": 1,
                        "minimumNumberOfSpecialSymbols": 0
                    }
                ]
            },
```

```
            {
              "constant": "updateEntity",
              "targetVariable": "operationTypeVariable"
            },
            {
              "constant": "createEntity",
              "targetVariable": "operationTypeVariable",
              "scope": "createEntity"
            },
            {
              "condition": "$.active == false && '$
{operationTypeVariable}' == 'createEntity'",
              "constant": "X",
              "targetPath": "$.LOCK_LOCALLY"
            },
            {
              "condition": "'${operationTypeVariable}' ==
'updateEntity'",
              "constant": "U",
              "targetPath": "$.LOCK"
            },
            {
              "condition": "$.active == false && '$
{operationTypeVariable}' == 'updateEntity'",
              "constant": "L",
              "targetPath": "$.LOCK"
            }
        ]
  },
    "group": {
        "mappings": [
            {
                "sourcePath": "$.displayName",
                "targetVariable": "entityIdTargetSystem",
                "scope": "createEntity"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.ROLE_NAME"
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.USERLIST[?(@.USERNAME)]"
            }
        ]
    }
}
```

- If your AS ABAP version is *7.30* or lower, use the transformation below.
  Note: This transformation creates all users as *active* (unlocked) in AS ABAP, regardless if they are *active* or *inactive* in the source system.

  Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.USERNAME"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.USERNAME"
```

```
            },
            {
                "sourcePath": "$.externalId",
                "optional": true,
                "targetPath": "$.ALIAS.USERALIAS",
                "defaultValue": ""
            },
            {
                "condition": "$.emails[?(@.primary == true)].value !=
[]",
                "sourcePath": "$.emails[?(@.primary == true)].value",
                "preserveArrayWithSingleElement": false,
                "optional": true,
                "targetPath": "$.ADDRESS.E_MAIL"
            },
            {
                "condition": "$.emails[?(@.type == 'work')].value !=
[]",
                "sourcePath": "$.emails[?(@.type == 'work')].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.ADDRESS.E_MAIL",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "sourcePath": "$.emails[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.ADDSMTP[?(@.E_MAIL)]"
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.ADDRESS.LASTNAME"
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.ADDRESS.FIRSTNAME"
            },
            {
                "sourcePath": "$.name.middleName",
                "optional": true,
                "targetPath": "$.ADDRESS.MIDDLENAME"
            },
            {
                "sourcePath": "$.nickName",
                "optional": true,
                "targetPath": "$.ADDRESS.NICKNAME"
            },
            {
                "sourcePath": "$.name.honorificPrefix",
                "optional": true,
                "targetPath": "$.ADDRESS.TITLE_P"
            },
            {
                "condition": "$.phoneNumbers[?(@.primary ==
true)].value != []",
                "sourcePath": "$.phoneNumbers[?(@.primary ==
true)].value",
                "preserveArrayWithSingleElement": false,
                "optional": true,
                "targetPath": "$.ADDRESS.TEL1_NUMBR"
            },
```

```
            {
                "condition": "$.phoneNumbers[?(@.type ==
'work')].value != []",
                "sourcePath": "$.phoneNumbers[?(@.type ==
'work')].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.ADDRESS.TEL1_NUMBR",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "sourcePath": "$.phoneNumbers[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.ADDTEL[?(@.TELEPHONE)]"
            },
            {
                "optional": true,
                "targetPath": "$.DEFAULTS.LANGU",
                "type": "valueMapping",
                "sourcePaths": [
                    "$.locale"
                ],
                "defaultValue": "E",
                "valueMappings": [
                    {
                        "key": [
                            "bg"
                        ],
                        "mappedValue": "W"
                    }
                ]
            },
            {
                "sourcePath": "$.preferredLanguage",
                "optional": true,
                "targetPath": "$.ADDRESS.LANGUP_ISO",
                "functions": [
                    {
                        "function": "toUpperCaseString"
                    }
                ]
            },
            {
                "optional": true,
                "targetPath": "$.LOGONDATA.TZONE",
                "type": "valueMapping",
                "sourcePaths": [
                    "$.timezone"
                ],
                "defaultValue": "CET",
                "valueMappings": [
                    {
                        "key": [
                            "Europe/Sofia"
                        ],
                        "mappedValue": "EET"
                    }
                ]
            },
             {
                "targetPath": "$.PASSWORD.BAPIPWD",
                "scope": "createEntity",
```

```
                "functions": [
                    {
                        "function": "randomPassword",
                        "passwordLength": 24,
                        "minimumNumberOfLowercaseLetters": 1,
                        "minimumNumberOfUppercaseLetters": 1,
                        "minimumNumberOfDigits": 1,
                        "minimumNumberOfSpecialSymbols": 0
                    }
                ]
            },
            {
              "constant": "updateEntity",
              "targetVariable": "operationTypeVariable"
            },
            {
              "constant": "createEntity",
              "targetVariable": "operationTypeVariable",
              "scope": "createEntity"
            },
            {
              "condition": "'${operationTypeVariable}' ==
'updateEntity'",
              "constant": "U",
              "targetPath": "$.LOCK"
            },
            {
              "condition": "$.active == false && '$
{operationTypeVariable}' == 'updateEntity'",
              "constant": "L",
              "targetPath": "$.LOCK"
            }
        ]
  },
    "group": {
        "mappings": [
            {
                "sourcePath": "$.displayName",
                "targetVariable": "entityIdTargetSystem",
                "scope": "createEntity"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.ROLE_NAME"
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.USERLIST[?(@.USERNAME)]"
            }
        ]
    }
}
```

- Default transformation supporting User UUID attribute:
  The following SAP Note must be implemented in the SAP AS ABAP 3003462 *Interface enhancement for global user ID*.

Code Syntax

```
{
  "user": {
    "condition": "(($.emails EMPTY true) ||
isValidEmail($.emails[0].value)) && (('%abap.role.prefix%' === 'null')
|| ($.groups[?(@.display =~ /%abap.role.prefix%.*/)] empty false))",
```

```
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.USERNAME"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.USERNAME"
      },
      {
        "sourcePath": "$.externalId",
        "optional": true,
        "targetPath": "$.ALIAS.USERALIAS",
        "defaultValue": ""
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$.SAPUSER_UUID.SAP_UID"
      },
      {
        "condition": "$.emails[?(@.primary == true)].value != []",
        "sourcePath": "$.emails[?(@.primary == true)].value",
        "preserveArrayWithSingleElement": false,
        "optional": true,
        "targetPath": "$.ADDRESS.E_MAIL"
      },
      {
        "condition": "$.emails[?(@.type == 'work')].value != []",
        "sourcePath": "$.emails[?(@.type == 'work')].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.ADDRESS.E_MAIL",
        "functions": [
          {
            "function": "elementAt",
            "index": 0
          }
        ]
      },
      {
        "sourcePath": "$.emails[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.ADDSMTP[?(@.E_MAIL)]"
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.ADDRESS.LASTNAME"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.ADDRESS.FIRSTNAME"
      },
      {
        "sourcePath": "$.name.middleName",
        "optional": true,
        "targetPath": "$.ADDRESS.MIDDLENAME"
      },
      {
        "sourcePath": "$.nickName",
        "optional": true,
        "targetPath": "$.ADDRESS.NICKNAME"
      },
      {
        "sourcePath": "$.name.honorificPrefix",
```

```
          "optional": true,
          "targetPath": "$.ADDRESS.TITLE_P"
        },
        {
          "condition": "$.phoneNumbers[?(@.primary == true)].value != []",
          "sourcePath": "$.phoneNumbers[?(@.primary == true)].value",
          "preserveArrayWithSingleElement": false,
          "optional": true,
          "targetPath": "$.ADDRESS.TEL1_NUMBR"
        },
        {
          "condition": "$.phoneNumbers[?(@.type == 'work')].value != []",
          "sourcePath": "$.phoneNumbers[?(@.type == 'work')].value",
          "preserveArrayWithSingleElement": true,
          "optional": true,
          "targetPath": "$.ADDRESS.TEL1_NUMBR",
          "functions": [
            {
              "function": "elementAt",
              "index": 0
            }
          ]
        },
        {
          "sourcePath": "$.phoneNumbers[*].value",
          "preserveArrayWithSingleElement": true,
          "optional": true,
          "targetPath": "$.ADDTEL[?(@.TELEPHONE)]"
        },
        {
          "type": "valueMapping",
          "sourcePaths": [
            "$.locale"
          ],
          "optional": true,
          "targetPath": "$.DEFAULTS.LANGU",
          "defaultValue": "E",
          "valueMappings": [
            {
              "key": [
                "bg"
              ],
              "mappedValue": "W"
            }
          ]
        },
        {
          "sourcePath": "$.preferredLanguage",
          "optional": true,
          "targetPath": "$.ADDRESS.LANGUP_ISO",
          "functions": [
            {
              "function": "toUpperCaseString"
            }
          ]
        },
        {
          "type": "valueMapping",
          "sourcePaths": [
            "$.timezone"
          ],
          "optional": true,
          "targetPath": "$.LOGONDATA.TZONE",
          "defaultValue": "CET",
          "valueMappings": [
            {
              "key": [
                "Europe/Sofia"
```

```
            ],
            "mappedValue": "EET"
          }
        ]
      },
      {
        "targetPath": "$.PASSWORD.BAPIPWD",
        "scope": "createEntity",
        "functions": [
          {
            "function": "randomPassword",
            "passwordLength": 24,
            "minimumNumberOfLowercaseLetters": 1,
            "minimumNumberOfUppercaseLetters": 1,
            "minimumNumberOfDigits": 1,
            "minimumNumberOfSpecialSymbols": 0
          }
        ]
      },
      {
        "constant": "updateEntity",
        "targetVariable": "operationTypeVariable"
      },
      {
        "constant": "createEntity",
        "targetVariable": "operationTypeVariable",
        "scope": "createEntity"
      },
      {
        "condition": "$.active == false && '${operationTypeVariable}'
== 'createEntity'",
        "constant": "X",
        "targetPath": "$.LOCK_LOCALLY"
      },
      {
        "condition": "'${operationTypeVariable}' == 'updateEntity'",
        "constant": "U",
        "targetPath": "$.LOCK"
      },
      {
        "condition": "$.active == false && '${operationTypeVariable}'
== 'updateEntity'",
        "constant": "L",
        "targetPath": "$.LOCK"
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
abap.role.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
abap.role.prefix) && (isRegularGroup() ||
isApplicationSpecificGroup())",
    "mappings": [
      {
        "scope": "createEntity",
        "sourcePath": "$.displayName",
        "targetVariable": "entityIdTargetSystem",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
abap.role.prefix)",
            "function": "replaceFirstString",
            "regex": "%abap.role.prefix%",
            "replacement": ""
          }
        ]
      },
```

```
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.ROLE_NAME",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
abap.role.prefix)",
            "function": "replaceFirstString",
            "regex": "%abap.role.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.USERLIST[?(@.USERNAME)]",
        "optional": true,
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

7. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

ABAP: Supported Languages and Code Pages

1.6.2.8 SAP Ariba Applications

Follow this procedure to set up SAP Ariba Applications as a target system.

Prerequisites

> **Note**
>
> This topic describes customer-managed configurations for the SAP Ariba Applications target system. For SAP-managed configurations, refer to the SAP-Managed SAP Cloud Identity [page 520] documentation.

Next-Generation SCIM API

If you are using the next-generation SCIM API provided by SAP Ariba Applications:

- The Identity Provisioning service must be configured to use the OAuth 2.0 client-credentials flow to obtain access tokens from the Identity Authentication service for calling the SAP Ariba Applications next generation SCIM API. The required dependency must be defined in the configuration of the consumer application. For more information, see Configure Form-Based OAuth Authentication [page 2457].

Current-Generation SCIM API

If you are using the current-generation SCIM API provided by SAP Ariba Applications:

- You have created a client application on SAP Ariba APIs Portal for your integration with Identity Provisioning. This is needed to enable SCIM API for user and group for Ariba Applications. For more information, see Register an Application in SAP Ariba Developer Portal .

  > **Note**
  >
  > If you don't have an SAP Ariba APIs account, go to the SAP Ariba Developer Portal and follow the instructions for sign up. For more information, see Get an SAP Ariba APIs Account.

  To find your *realm* name, log in to your SAP Ariba system. It is embedded in your login URL, as illustrated by the following examples:
  - SAP Ariba Buyer example: `https://s1.ariba.com/Buyer/Main/ad/loginPage/...&realm=`*`mycompany-t`*
  - SAP Ariba Sourcing example: `http://`*`mycompany`*`.sourcing.ariba.com/`
- You have enabled the client application for Identity Provisioning on the SAP Ariba Developer Portal.
- You have generated an OAuth secret for your application. For more information, see Generating the OAuth Secret and Base64 Encoded Client ID and Secret.
  To set up your SAP Ariba Applications provisioning system, you'll need to map your SAP Ariba application parameters to the corresponding Identity Provisioning properties. The mapping is as follows:

SAP Ariba	Identity Provisioning	Values
SCIM API URL	URL	Examples:

SAP Ariba	Identity Provisioning	Values
		• US: **https://openapi.ariba.com** • Europe: **https://eu.openapi.ariba.com** • UAE: **https://mn1.openapi.ariba.com** For more information, see SAP Ariba APIs Portal Discover Strategic Sourcing SCIM API for User and Group Master Data Environment Details.
SAP Ariba OAuth 2.0 Token URL	`OAuth2TokenServiceURL`	Examples: • US: **https://api.ariba.com/v2/oauth/token** • Europe: **https://api-eu.ariba.com/v2/oauth/token** • UAE: **https://api.mn1.ariba.com/v2/oauth/token** For more information, see SAP Ariba APIs Portal Discover Strategic Sourcing SCIM API for User and Group Master Data Environment Details.
OAuth Client ID	`User`	Alphanumeric string Example: **aaaa12345-1111-3333-cccc-1234567890**
OAuth Secret	`Password`	Alphanumeric string Example: **aaaGGG1eee12abcdefGHIJK123lmnopTTT**
Application key	`ariba.applications.api.key`	Alphanumeric string Example:**123abc123XYZ000abc123ABC012345**
AN-ID	`ariba.applications.realm.id`	AN<numeric_string> Example: **AN000111222333**

Context

After fulfilling the prerequisites, you can create an SAP Ariba Applications target system to provision users and groups.

These target systems can consume either the current-generation SCIM API or the next-generation SCIM API provided by SAP Ariba Applications. Identity Provisioning distinguishes between the two by the API URL pattern (the current SCIM API contains **openapi** in the URL, while the next-generation API does not) and by the `AuthType=Form` property, which must be configured only for the next-generation API.

For more information about the SAP Ariba SCIM API scope of support, see SAP Ariba SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Ariba Applications* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the SCIM API URL for your SAP Ariba application (see the **Prerequisites** section). The current-generation SCIM API uses the **openapi** pattern in the URL.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`AuthType`	Enter: *Form* **Relevant for next-generation SCIM API**
`User`	Enter the *Client ID* for the current-generation or the next-generation SCIM API, as described in the corresponding Prerequisites section.
`Password`	(Credential) Enter the *Client Secret* for the current-generation or the next-generation SCIM API, as described in the corresponding Prerequisites section.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL for the current-generation or the next-generation SCIM API, as described in the corresponding Prerequisites section.

Property Name	Description & Value
`ariba.applications.api.dependency.name`	This property holds the value of the dependency name used in the configuration of the SAP Ariba Applications consumer application, created in the SAP Cloud Identity Services tenant. It is used for access token retrieval from Identity Authentication. **Relevant for next-generation SCIM API**
`ariba.applications.api.key`	(Credential) Enter your application key (see the **Prerequisites** section). **Relevant for current-generation SCIM API**
`ariba.applications.realm.id`	Enter your AN-ID (see the **Prerequisites** section). **Relevant for current-generation SCIM API**
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Exemplary destination (property configuration):

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://openapi.ariba.com
User=aaaa12345-1111-3333-cccc-1234567890
Password=**********
OAuth2TokenServiceURL=https://api.ariba.com/v2/oauth/token
ariba.applications.api.key=123abc123XYZ000abc123ABC012345
ariba.applications.realm.id=AN000111222333
```

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Ariba Applications* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Ariba Applications. For more information, see:

   Manage Transformations [page 2439]

   SCIM API for User and Group Master Data

   To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

   Default transformation:

   Code Syntax

```
{
  "user": {
    "condition": "('%ariba.applications.group.prefix%' === 'null') ||
($.groups[?(@.display =~ /%ariba.applications.group.prefix%.*/)] empty
false)",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:User",
          "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
          "urn:ietf:params:scim:schemas:extension:sap:2.0:User",
          "urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User",
          "urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User",
          "urn:sap:cloud:scim:schemas:extension:custom:2.0:profile:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "condition": "$.emails[0].length() > 0",
        "constant": true,
        "targetPath": "$.emails[0].primary"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
```

```
      {
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['companyCode']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['companyCode']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']['costCenter']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingGroup']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingGroup']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['generalLedgerAccount']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['generalLedgerAccount']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingOrganization']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap.odm:2.0:User']
['purchasingOrganization']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']['currency']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']['currency']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['deliverTo']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['deliverTo']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['purchasingUnit']",
```

```
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['purchasingUnit']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']['network']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']['network']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['addresses']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['addresses']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['passwordAdapter']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['passwordAdapter']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['approvalLimit']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['approvalLimit']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['expenseApprovalLimit']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap:ariba:2.0:User']
['expenseApprovalLimit']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:sap:cloud:scim:schemas:extension:custom:2.0:profile:User']
['alternativeDisplayNames']",
        "targetPath":
"$['urn:sap:cloud:scim:schemas:extension:custom:2.0:profile:User']
['alternativeDisplayNames']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
```

```
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.phoneNumbers",
        "functions": [
          {
            "function": "putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true,
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
ariba.applications.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
ariba.applications.group.prefix) && (isRegularGroup() ||
isApplicationSpecificGroup())",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:Group",
          "urn:ietf:params:scim:schemas:extension:sap:2.0:Group",
          "urn:sap:cloud:scim:schemas:extension:custom:2.0:profile:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
ariba.applications.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%ariba.applications.group.prefix%",
```

```
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:sap:cloud:scim:schemas:extension:custom:2.0:profile:Group']
['alternativeDisplayNames']",
        "targetPath":
"$['urn:sap:cloud:scim:schemas:extension:custom:2.0:profile:Group']
['alternativeDisplayNames']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "type": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

5. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

1. Before starting a provisioning job, you can first subscribe to the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during your jobs. For more information, see Manage Job Notifications [page 3018].
2. Now, start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

The SAP Ariba developer portal
Video: Create application and API approval process

1.6.2.9 SAP Ariba Category Management

Follow this procedure to set up SAP Ariba Category Management as a target system.

Prerequisites

You have created an instance and generated a service key for the scim service plan of SAP Ariba Category Management. For more information, see Creating a SAP Ariba Category Management SCIM Service Instance in Your Subaccount.

> **Note**
>
> This connector has been deprecated as of August 25, 2025, and is no longer needed. Customers with existing configurations should stop running jobs to or from SAP Ariba Category Management provisioning systems.

Context

SAP Ariba Category Management is a cloud-based solution that enables organizations to define and manage purchasing categories and provides actionable market intelligence, category insights, and various tools and templates for developing and operationalizing category strategies that drive business value and performance. This business solution is built on SAP Business Technology Platform (BTP). You can use Identity Provisioning to configure SAP Ariba Category Management as a target system where you can provision users from source systems.

> **Note**
>
> SAP Ariba Category Management does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.

2. Add *SAP Ariba Category Management* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL provided by the service key under the *scim-v1* field without adding the path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the value from the *clientid* field of the service key.
`Password`	(Credential) Enter the value from the *clientsecret* field of the service key.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. This is the value from the *url* field of the service key plus `/oauth/token`
(Optional) `cm.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. The property is not added automatically at system creation. Possible values: *userName* and *emails[*].value* For more information, see: **User Update and Uniqueness** section below and List of Properties [page 80].
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

   To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Ariba Category Management* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Ariba Category Management system. For more information, see:

Manage Transformations [page 2439]

SCIM APIs for SAP Ariba Category Management

Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP Ariba Category Management entity.

User Update and Uniqueness

The write transformation of SAP Ariba Category Management supports two user unique attributes for managing conflict resolution: *username* and *email*. The username is a mandatory attribute, while the email is an optional, multivalue attribute, indicating that a user may have multiple unique emails.

The uniqueness of a user in the target system is determined not only by the individual attributes or the combination of both, but also by considering whether one or multiple existing users in the target system share the same values for these unique attributes. Have in mind the following specifics when updating an existing user in the target system, in accordance with the configuration of the `cm.user.unique.attribute` property:

- When `cm.user.unique.attribute`= *userName*
  If a user in the source system, with a specific username and an email, matches an existing user in the target system by the same email, but a different userName, the user provisioning will fail. In this case, Identity Provisioning will try to create a new user but will fail, since a user with the same email already exists.
- When `cm.user.unique.attribute`= *emails[*].value*
  If a user in the source system, with a specific username and an email, matches an existing user in the target system by the same username, but a different email, the user provisioning will fail. In this case, Identity Provisioning will try to create a new user but will fail, since a user with the same username already exists.
- When `cm.user.unique.attribute`= *userName, emails[*].value*
  - If a user in the source system, with a specific username and multiple emails, matches an existing user in the target system by the same username and at least one identical email, the user in the target system will be updated.
  - If a user in the source system, with a specific username and multiple emails, matches multiple existing users in the target system (each sharing one of the multiple emails but having different usernames), the user provisioning will fail. In this case, the user will neither be created nor updated.

Default transformation:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "targetPath": "$.id",
                "sourceVariable": "entityIdTargetSystem"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
                "targetPath": "$.schemas[0]",
```

```
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            },
            {
                "sourcePath":"$.externalId",
                "targetPath":"$.externalId",
                "optional":true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "optional": true
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName",
                "optional": true
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName",
                "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true,
                "defaultValue": true
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "optional":true,
                "preserveArrayWithSingleElement": true
            }
        ]
    }
}
```

5. Add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

User Replication Setup

1.6.2.10 SAP Ariba Central Invoice Management

Follow this procedure to set up SAP Ariba Central Invoice Management as a target system.

Prerequisites

You have created an instance and generated a service key for the scim service plan of SAP Ariba Central Invoice Management. This step is automated by the booster *Set Up SAP Ariba Central Invoice Management*. For more information, see Setting Up Invoice Processing with SAP Ariba Central Invoice Management (4N6) –> *Set up Configuration Using Booster in SAP BTP Cockpit*.

Context

SAP Ariba Central Invoice Management is an SAP BTP SaaS application running on SAP BTP, Cloud Foundry environment. This application enables central management of supplier invoices from multiple connected systems, such as SAP S/4HANA Cloud Public Edition systems. You can use Identity Provisioning to configure SAP Ariba Central Invoice Management as a target system where you can provision users and groups from source systems.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Ariba Central Invoice Management* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
Type	Enter: *HTTP*

Property Name	Value
`URL`	Enter the URL provided by the service key under the *ips-connector* field without adding the path information. For example: `https://eu10.cim.cloud.sap`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the value from the *clientid* field of the service key.
`Password`	(Credential) Enter the value from the *clientsecret* field of the service key.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. This is the value from the *url* field of the service key plus `/oauth/token` For example: `https://<my tenant>.authentication.eu10.hana.ondemand.com/oauth/token`
(Optional) `cim.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. The property is not added automatically at system creation. **Possible values:** • *userName* - default value • *emails[*].value* • *userName,emails[*].value* For more information, see: List of Properties [page 80]
(Optional) `cim.group.prefix`	This property distinguishes SAP Ariba Central Invoice Management groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`CIM_`** You can use the example value or provide your own. When **set in the target system**, only groups containing the **`CIM_`** prefix in their display name will be provisioned to SAP Ariba Central Invoice Management. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be be provisioned to SAP Ariba Central Invoice Management.

Property Name	Value
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Ariba Central Invoice Management* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Ariba Central Invoice Management system. For more information, see:

   Manage Transformations [page 2439]

   User Management (SCIM) for SAP Ariba Central Invoice Management

   Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP Ariba Central Invoice Management entity.

   Default transformation:

   Code Syntax

```
{
  "user": {
    "condition": "(($.emails EMPTY true) ||
(isValidEmail($.emails[*].value))) && (('%cim.group.prefix%' === 'null')
|| ($.groups[?(@.display =~ /%cim.group.prefix%.*/)] empty false))",
    "mappings": [
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:User"
        ],
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      },
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
```

```
      "sourcePath": "$.userName",
      "targetPath": "$.userName"
    },
    {
      "sourcePath": "$.externalId",
      "targetPath": "$.externalId",
      "optional": true
    },
    {
      "sourcePath": "$.name.givenName",
      "targetPath": "$.name.givenName",
      "optional": true
    },
    {
      "sourcePath": "$.name.middleName",
      "targetPath": "$.name.middleName",
      "optional": true
    },
    {
      "sourcePath": "$.name.familyName",
      "targetPath": "$.name.familyName",
      "optional": true
    },
    {
      "sourcePath": "$.name.formatted",
      "targetPath": "$.name.formatted",
      "optional": true
    },
    {
      "sourcePath": "$.name.honorificPrefix",
      "targetPath": "$.name.honorificPrefix",
      "optional": true
    },
    {
      "sourcePath": "$.name.honorificSuffix",
      "targetPath": "$.name.honorificSuffix",
      "optional": true
    },
    {
      "sourcePath": "$.emails",
      "preserveArrayWithSingleElement": true,
      "optional": true,
      "targetPath": "$.emails"
    },
    {
      "condition": "$.emails[?(@.primary == true)] empty true",
      "constant": true,
      "targetPath": "$.emails[0].primary"
    },
    {
      "sourcePath": "$.active",
      "targetPath": "$.active",
      "optional": true
    },
    {
      "sourcePath": "$.displayName",
      "targetPath": "$.displayName",
      "optional": true
    },
    {
      "sourcePath": "$.nickName",
      "targetPath": "$.nickName",
      "optional": true
    },
    {
      "sourcePath": "$.profileUrl",
      "targetPath": "$.profileUrl",
      "optional": true
```

```
      },
      {
        "sourcePath": "$.title",
        "targetPath": "$.title",
        "optional": true
      },
      {
        "sourcePath": "$.userType",
        "targetPath": "$.userType",
        "optional": true
      },
      {
        "sourcePath": "$.preferredLanguage",
        "targetPath": "$.preferredLanguage",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.timezone",
        "targetPath": "$.timezone",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
cim.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
cim.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
cim.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%cim.group.prefix%",
            "replacement": ""
          }
        ]
```

```
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

5. Add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Ariba Central Invoice Management

1.6.2.11 SAP BTP ABAP environment

Follow this procedure to set up SAP BTP ABAP environment as a target system.

Prerequisites

To establish the connection between Identity Provisioning and SAP BTP ABAP environment, you need to set up the communication (user, system and arrangement) on SAP BTP ABAP environment. You can do it now (as a prerequisite) or in the process of configuring SAP BTP ABAP environment as a target system, as described in step 3.

Context

You can use SAP BTP ABAP environment as a target system to provision entities from a certain source system. This scenario supports writing **users** and **assignments**. In SAP BTP ABAP environment, groups correspond to roles, thus group members are user assignments of a role.

> ### Note
>
> Identity Provisioning cannot create and delete roles in SAP BTP ABAP environment target system. It can only create, update and delete user assignments of a role. Therefore, roles must have been created in SAP BTP ABAP environment target system before you run a provisioning job.
>
> For example, if you try to create or delete a role in SAP BTP ABAP environment, Identity Provisioning will only add or remove the user assignments of that role, respectively.

The Identity Provisioning service manages the complete set of **business partners** and their relevant **business users** (Employee).

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP BTP ABAP environment* as a target system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP BTP ABAP environment and configure your authentication method (basic or certificate-based).

   > ### Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP BTP ABAP environment target system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP BTP ABAP environment backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Create a communication system and assign the created user to the communication system.

      For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

   d. Create a communication arrangement with the created system.

For your Identity Provisioning scenario, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

For more information, see Maintain a Communication Arrangement for Inbound Communication

> **Note**
>
> The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
>
> The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the API URL to your SAP BTP ABAP environment system. You can take the URL from the communication scenario SAP_COM_0193.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **Restriction** Do not use special symbol ',' (comma) as it is not supported.

Property Name	Description & Value
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`ips.date.variable.format`	*yyyy-MM-dd*
`a4c.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP BTP ABAP environment target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. According to your use case, choose how to set up this property: • Default behavior: This property does not appear in the UI during system creation. Its default value is *personExternalID*. That means, if the service finds an existing user by a *personExternalID* , it updates this user with the data of the conflicting one. If a user with such a *personExternalID* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user matching both unique attributes *email* and *personExternalID*, it updates this user with the data of the conflicting one. If the service finds an existing user matching only the *email*, the update of the existing user fails. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. Possible values: • *personExternalID* • *emails[0].value* Default value: *personExternalID*

Property Name	Description & Value
`a4c.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP BTP ABAP environment target system. • **true** – the current user roles will be deleted in the target system, and the user will be updated only with the roles provisioned by the service. • **false** – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the target system. Default value (if the property is missing during system creation): *true* Default value (if the property appears during system creation): *false* See also: Extended Explanation of the *.user.roles.overwrite Properties
(Optional) `a4c.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: APIs for Business User Management If not specified, the default value is *false*.
(Optional) `a4c.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of users to be provisioned per request. The default value, if not specified, is *20*. The maximum value is **100**. If you enter a number larger than 100, the service will replace it with the default value (20).
(Optional) `ips.delete.threshold.roles`	Use this property to control the number of user assignments of a role to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of user assignments of a role, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Property Name	Description & Value
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*https://12345-aaaaa-3333.abap.hana.ondemand.com*

`User`=*MyABAPEnvUser*

`Password`=**********

`ips.date.variable.format`=*yyyy-MM-dd*

`a4c.user.roles.overwrite`=*false*

`a4c.support.bulk.operation` = *true*

`a4c.bulk.operations.max.count` = *50*

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP ABAP environment* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in SAP BTP ABAP environment. For more information, see:

   Manage Transformations [page 2439]

   Inbound Service: Business User

   Default transformation:

   Code Syntax

```
{
  "user": {
    "condition": "isValidEmail($.emails[0].value) &&
(('%a4c.roles.prefix%' === 'null') || ($.groups[?(@.display =~ /
%a4c.roles.prefix%.*/)] empty false))",
```

```
    "mappings": [
      {
        "sourcePath": "$.userName",
        "targetPath": "$.personExternalID"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.personExternalID",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$.personExternalID",
        "optional": true
      },
      {
        "targetPath": "$.personID",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "targetPath": "$.markedForArchivingIndicator",
        "constant": "false"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$.user.globalUserID"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.user.timeZoneCode",
        "functions": [
          {
            "type": "convertTimezoneCode",
            "outputFormat": "sap"
          }
        ],
        "defaultValue": "CET"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.userType"
        ],
        "targetPath": "$.businessPartnerRoleCode",
        "defaultValue": "BUP003",
        "valueMappings": [
          {
            "key": [
              "Employee"
            ],
            "mappedValue": "BUP003"
          }
        ]
      },
      {
        "scope": "createEntity",
        "targetPath": "$.validityPeriod.startDate",
        "sourceVariable": "currentDate"
      },
      {
        "scope": "createEntity",
        "targetPath": "$.validityPeriod.endDate",
        "constant": "9999-12-31"
```

```
      },
      {
        "scope": "createEntity",
        "sourceVariable": "currentDate",
        "targetPath": "$.user.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
        "targetPath": "$.user.validityPeriod.endDate"
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.personalInformation.firstName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.personalInformation.lastName"
      },
      {
        "sourcePath": "$.name.middleName",
        "targetPath": "$.personalInformation.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.name.formatted",
        "targetPath": "$.personalInformation.personFullName",
        "optional": true
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.user.userName"
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.user.logonLanguageCode",
        "optional": true
      },
      {
        "sourcePath": "$.emails[0].value",
        "targetPath": "$.workplaceInformation.emailAddress"
      },
      {
        "targetPath": "$.user.lockedIndicator",
        "constant": "false"
      },
      {
        "condition": "$.active == false",
        "targetPath": "$.user.lockedIndicator",
        "constant": "true"
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
a4c.roles.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
a4c.roles.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "mappings": [
      {
        "sourcePath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
a4c.roles.prefix)",
            "function": "replaceFirstString",
            "regex": "%a4c.roles.prefix%",
```

```
            "replacement": ""
          }
        ],
        "targetVariable": "entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
a4c.roles.prefix)",
            "function": "replaceFirstString",
            "regex": "%a4c.roles.prefix%",
            "replacement": ""
          }
        ],
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

See also: Extended Explanation of the *user.roles.overwrite Properties

6. Now, add a source system from which to read users and roles. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

User Provisioning in SAP BTP ABAP environment

1.6.2.12 SAP BTP Account Members (Neo)

Follow this procedure to set up your SAP Business Technology Platform as a target system, to which you can provision and write members in your Neo account.

Prerequisites

You have created a new Platform API OAuth client, with API *Account Member Management* and scopes *Manage Account Members* and *Read Account Members*.

Save the *Client ID* and *Client Secret* as you'll need them when you configure your target system. Make sure you save the client secret as you cannot retrieve it later.

For more information, see Create a Platform API Client.

Context

The Identity Provisioning service helps companies to automatically manage the user-to-platform roles assignments for SAP Business Technology Platform subaccounts. For this aim, the service reuses data from an active company user store. For this scenario, as a target system are used SAP Business Technology Platform subaccounts. A source system can be a solution supported by the Identity Provisioning service with read access for user artifacts.

This provisioning scenario is based on the Platform Authorization Management API of SAP BTP. For more information, see Platform Authorization Management API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP BTP Account Members (Neo)* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
Type	Enter: *HTTP*

Property Name	Description & Value
URL	Enter: **`https://api.<SAP_BTP_host>/authorization/v1/platform/accounts/<SAP_BTP_account>`** Examples: • (Europe – Rot) *https://api.hana.ondemand.com/authorization/v1/platform/accounts/abc123xyz* • (Japan – Tokyo) *https://api.jp1.hana.ondemand.com/authorization/v1/platform/accounts/abc123xyz*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the Client ID of the new Platform API OAuth client created for the **Account Member Management** API (see the prerequisites).
`Password`	(Credential) Enter the Client Secret of the new Platform API OAuth client created for the **Account Member Management** API (see the prerequisites).
`OAuth2TokenServiceURL`	Enter: **`https://api.<SAP_CP_host>/oauth2/apitoken/v1`** Examples: • (Europe – Rot) *https://api.hana.ondemand.com/oauth2/apitoken/v1* • (US East – Sterling) *https://api.us3.hana.ondemand.com/oauth2/apitoken/v1*
`scp.user.userbase`	This property specifies the host to the identity provider to be used with this proxy system. All provisioned users can be authenticated only by this identity provider. Default value: *accounts.sap.com* If you use another IdP, enter its value as configured in the SAP BTP cockpit. For example: **`<account_ID>.accounts.ondemand.com`**
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Property Name	Description & Value
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP Account Members* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   Using the default transformation, all groups that are available in the source system and their respective members (as identifiers) are assigned as platform roles to the users. These users are added to the SAP Business Technology Platform subaccount via the target system.

   You can change the default transformation mapping rules to reflect your current setup of entities in your target system. For more information, see:

   Manage Transformations [page 2439]

   SAP BTP: Authorization Management API

   Default transformation:

   Code Syntax

```
{
  "user": {
    "condition": "('%scp.group.prefix%' === 'null') || ($.groups[?
(@.display =~ /%scp.group.prefix%.*/)] empty false)",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
// The roles in the conditions below are placeholder names mapped to the
SAP BTP roles. You can replace these names with the actual role names used
in your source system.
// You can delete some of these roles (if redundant), or add new mappings
for additional roles.
      {
        "condition": " (('%scp.group.prefix%' === 'null')
&& ($.groups[*].value contains 'AccountAdministrator')) ||
(('%scp.group.prefix%' !== 'null') && ($.groups[*].value contains
'%scp.group.prefix%AccountAdministrator'))",
        "constant": "AccountAdministrator",
        "targetPath": "$.roles[0].value"
      },
```

```
      {
        "condition": " (('%scp.group.prefix%' === 'null') &&
($.groups[*].value contains 'Developer')) || (('%scp.group.prefix%' !==
'null') && ($.groups[*].value contains '%scp.group.prefix%Developer'))",
        "constant": "Developer",
        "targetPath": "$.roles[1].value"
      },
      {
        "condition": " (('%scp.group.prefix%' === 'null')
&& ($.groups[*].value contains 'CloudConnectorAdministrator')) ||
(('%scp.group.prefix%' !== 'null') && ($.groups[*].value contains
'%scp.group.prefix%CloudConnectorAdministrator'))",
        "constant": "CloudConnectorAdministrator",
        "targetPath": "$.roles[2].value"
      },
      {
        "condition": " (('%scp.group.prefix%' === 'null')
&& ($.groups[*].value contains 'ApplicationUserAdministrator')) ||
(('%scp.group.prefix%' !== 'null') && ($.groups[*].value contains
'%scp.group.prefix%ApplicationUserAdministrator'))",
        "constant": "ApplicationUserAdministrator",
        "targetPath": "$.roles[3].value"
      },
      {
        "condition": " (('%scp.group.prefix%' === 'null') &&
($.groups[*].value contains 'ReadOnly')) || (('%scp.group.prefix%' !==
'null') && ($.groups[*].value contains '%scp.group.prefix%ReadOnly'))",
        "constant": "ReadOnly",
        "targetPath": "$.roles[4].value"
      },
      {
        "constant": "%scp.user.userbase%",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:UserExt']['userbase']"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
        "targetPath": "$.schemas[0]"
      },
      {
        "constant":
"urn:sap:cloud:scim:schemas:extension:custom:2.0:UserExt",
        "targetPath": "$.schemas[1]"
      }
    ]
  },
 "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
scp.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
scp.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "skipOperations": [
      "delete"
    ],
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
scp.group.prefix)",
            "function": "replaceFirstString",
```

```
            "regex": "%scp.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "sourcePath": "$.members[*].value",
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "type": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

5. Now, add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

What is SAP BTP

1.6.2.13 SAP BTP Platform Members (Cloud Foundry)

Follow this procedure to set up a SAP BTP Platform Members (Cloud Foundry) as a target system.

Prerequisites

- You have a global account in SAP BTP with at least one multi-environment subaccount with enabled Cloud Foundry environment.
- You have added SAP BTP Platform Members (Cloud Foundry) as a source system for the same Cloud Foundry landscape and executed successfully a read job, typically to the Identity Directory of the SAP Cloud Identity Services tenant that is established as a trusted custom identity provider.

This is required to preserve the specific group naming convention (see the defined name pattern below) used by SAP BTP Platform Members (Cloud Foundry). Reading groups from this source into the Identity Directory, assigning users to the groups, and synchronizing the assignments back to SAP BTP Platform Members (Cloud Foundry) ensures the correct provisioning of user group assignments. For more information, see SAP BTP Platform Members (Cloud Foundry) source system [page 674] and Local Identity Directory target system [page 957].

Context

The Cloud Foundry environment enables you to create polyglot cloud applications in Cloud Foundry. It contains the SAP BTP, Cloud Foundry runtime service, which is based on the open-source application platform managed by the Cloud Foundry Foundation. For more information, see Cloud Foundry Environment.

When you enable the Cloud Foundry environment in your subaccount, the system automatically creates a Cloud Foundry organization for you. You are able to add platform users as org members and space members and assign roles to grant these users platform access. For more information, see Valid role types .

SAP BTP Platform Members (Cloud Foundry) connector manages org and space members, as well as their role assignments, in the Cloud Foundry environment of a multi-environment subaccount, where a **single** SAP Cloud Identity Services tenant acts as custom identity provider. We recommend that you use the Identity Provisioning service enabled in this SAP Cloud Identity Services tenant.

In SAP BTP Platform Members (Cloud Foundry), groups correspond to roles in particular Cloud Foundry orgs or spaces, thus group members are user assignments of a role in a specific Cloud Foundry org or space. Group names must follow a defined pattern, explained below, to ensure the correct mapping and provisioning of users and their role assignments to the relevant Cloud Foundry organization or space.

> **Note**
>
> The attribute *name*, defined within the *urn:sap:cloud:scim:schemas:extension:custom:2.0:Group* schema, is used to map the users and user assignments of a role to the relevant Cloud Foundry organization or space. The group names follow the respective patterns:
>
> - organization group name: `<org_ID> <org_role>`
> - space group name: `<org_ID> <space_ID> <space_role>`, where `space_ID` can be retrieved using the Cloud Foundry Command Line Interface by executing the command **`cf space --guid`**.
>
> Whereas the group display names adhere to the following patterns:
>
> - organization group display name: `<btp.cf.pm.group.prefix><btp.cf.pm.landscape>:<org_name>:<org_role>`
> - space group display name: `<btp.cf.pm.group.prefix><btp.cf.pm.landscape>:<org_name>:<space_name>:<space_role>`

The target system consumes `User Account and Authentication API` and `Cloud Foundry V3 API` provided by Cloud Foundry.

→ Remember

This connector enables you to write users and user role assignments to Cloud Foundry on *subaccount* level. For provisioning of users and groups to Cloud Foundry on *application* level, refer to SAP BTP XS Advanced UAA (Cloud Foundry) [page 1045]. For more information, see SAP BTP Integration Scenario.

ⓘ Note

The number of platform users you can provision is limited, as specified in the restriction described in User Provisioning.

Follow the steps below to create SAP BTP Platform Members (Cloud Foundry) as a target system to provision users and user role assignments.

ⓘ Note

In case you have Cloud Foundry environment enabled in multiple Cloud Foundry landscapes or more than one SAP Cloud Identity Services tenants, you must configure a separate SAP BTP Platform Members (Cloud Foundry) connector for each of them.

Procedure

1. To enable provisioning of platform users and user role assignments to and from Cloud Foundry environment, create a Support Ticket on component `BC-CP-CF-SEC-IAM`.

   Specify your SAP Cloud Identity Services tenant ID and the origin key of the trust configuration for BTP platform users which points to your SAP Cloud Identity Services tenant. You can find the origin key value in the SAP BTP cockpit. Go to your SAP BTP subaccount, choose *Trust Configuration* and see the value under *Origin Key* for the relevant *Custom Identity Provider for Platform Users*.
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
3. Add *SAP BTP Platform Members (Cloud Foundry)* as a target system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the `email` of the CF provisioning user (see **Prerequisites**).

Property Name	Description & Value
`Password`	(Credential) Enter the password for the CF provisioning user (see **Prerequisites**).
`btp.cf.pm.origin`	Enter the origin key of your SAP Cloud Identity Services tenant (see **Step 1**). The value of this property is a string, which always ends with the suffix *-platform*. It will be used as the *origin* attribute in the system transformation.
`btp.cf.pm.landscape`	Enter the *technical key* or the *host name* of the landscape in which your multi-environment subaccount with enabled Cloud Foundry environment is located. The technical key is available only for the SAP BTP Cloud Foundry regions. For more information, see Regions and API Endpoints Available for the Cloud Foundry Environment. Please note that when the technical key contains an extension index, the correct value is displayed in the *API Endpoint* column instead of the *Technical Key* column of the table. This is an example of a *technical key*: **`cf-eu10-002`**, corresponding to the API endpoint `api.cf.eu10-002.hana.ondemand.com` and host name `cf.eu10-002.hana.ondemand.com` The host name of the landscape can be found by looking up the *API Endpoint* field in the *Overview* tab of your *Cloud Foundry Environment* on subaccount level of the SAP BTP cockpit. The API endpoint has the following format: `https://api.<landscape-hostname>` **Note** If your provisioning job finishes with error *'Name or service not known'*, the value used for technical key is not correct. In that case you can use the host name of the landscape instead.

Property Name	Description & Value
(Optional) `btp.cf.pm.group.prefix`	This property distinguishes SAP BTP Platform Members (Cloud Foundry) groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`BTP_CF_PM_`** When set in the target system, only groups containing the **`BTP_CF_PM_`** prefix in their display name will be provisioned to SAP BTP Platform Members (Cloud Foundry). Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP BTP Platform Members (Cloud Foundry).
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.trace.failed.entity.content`	If a provisioning job repeatedly fails and you need problem investigation, you can enable logging and tracing for the personal and sensitive data of your provisioned entities. To do this, set this property to *true*. For more information, see List of Properties [page 80].

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP Platform Members (Cloud Foundry)* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP BTP Platform Members (Cloud Foundry) system. For more information, see:

   Manage Transformations [page 2439]

   Cloud Foundry V3 API

- **Mapping logic** – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the SAP BTP Platform Members (Cloud Foundry) target attributes.
- **User offboarding** – If a user has been deleted from the source system, this change is recognized, and the user is deleted from the SAP BTP Platform Members (Cloud Foundry) target system too.

> ### Note
>
> Update operation is skipped for users in the default write transformation.

> ### Caution
>
> Keep in mind that when you add or delete users and user role assignments in SAP BTP Platform Members (Cloud Foundry), the operations order must adhere to the Cloud Foundry Environment hierarchy. In this case, you can expect the following behavior:
>
> - When adding a user as space member, it must already be present as member of the relevant organization with assigned role **Org User**. When deleting an org member, it should be first removed from the spaces of the relevant organization. In case you end up with erroneous entity structure, you can manually add or delete users via the administration console for SAP Cloud Identity Services.
> - When adding user role assignments, you should first add the org roles and afterwards proceed with the space roles. Also, when deleting user role assignments, you should first delete the space roles and then proceed with the org roles. If this condition is not fullfiled the job will finish with error. In that case, you can rerun the job until it finishes successfully.

Default transformation:

> ### Code Syntax

```
{
  "user": {
    "skipOperations": [
      "update"
    ],
    "condition": "(('%btp.cf.pm.group.prefix%' === 'null') || ($.groups[?
(@.display =~ /%btp.cf.pm.group.prefix%%btp.cf.pm.landscape%.+/)] empty
false)) && $.groups[?(@.display === 'cf-user-provisioning')] empty true",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.name.formatted",
```

```
        "targetPath": "$.name.formatted",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "defaultValue": true,
        "optional": true
      },
      {
        "scope": "createEntity",
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.verified",
        "targetPath": "$.verified",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mailVerified']",
        "targetPath": "$.verified",
        "optional": true
      },
      {
        "scope": "createEntity",
        "constant": true,
        "targetPath": "$.verified"
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true
      },
      {
        "condition": "$.emails[?(@.primary == true)].value == []",
        "targetPath": "$.emails[0].primary",
        "constant": true
      },
      {
        "constant": [
          "urn:scim:schemas:core:1.0"
        ],
        "targetPath": "$.schemas"
      }
    ]
  },
  "group": {
    "condition": "('%btp.cf.pm.group.prefix%' === 'null') ||
($.displayName =~ /%btp.cf.pm.group.prefix%%btp.cf.pm.landscape%.+/)",
    "mappings": [
      {
        "scope": "createEntity",
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "targetPath": "$.id",
        "targetVariable": "entityIdTargetSystem"
      },
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.members[*].value",
        "targetPath": "$.members[?(@.value)]",
        "optional": true,
        "preserveArrayWithSingleElement": true,
```

```
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

6. Now, add a source system from which to read users and user role assignments. For example: Local Identity Directory [page 604]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP BTP Integration Scenario

1.6.2.14 SAP BTP Java/HTML5 apps (Neo)

Follow this procedure to set up SAP Business Technology Platform as a target system, to which you can provision user-to-groups assignments for Java/HTML5 applications that run on SAP BTP, Neo environment.

Prerequisites

You have created a new Platform API OAuth client, with *Authorization Management* scopes. Save the *Client ID* and *Client Secret* as you'll need them when you configure your target system. Make sure you save the client secret as you cannot retrieve it later.

For more information, see Create a Platform API Client.

Context

The Identity Provisioning service helps companies to automatically manage the user-to-groups assignments for Java/HTML5 applications running on SAP BTP, Neo environment. For this aim, the service reuses data from

an active company user store. For this scenario, SAP BTP is the target system. The source system can be a solution supported by the Identity Provisioning service with read access for user and group artifacts.

This provisioning scenario is based on the Authorization Management REST API of SAP BTP. For more information, see Using the Authorization Management REST API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP BTP Java/HTML5 apps (Neo)* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter: **`https://api.<SAP_BTP_host>/authorization/v1/accounts/<SAP_BTP_account>`** Examples: • (Europe – Rot) *https://api.hana.ondemand.com/authorization/v1/accounts/abc123xyz* • (Japan – Tokyo) *https://api.jp1.hana.ondemand.com/authorization/v1/accounts/abc123xyz*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the Client ID of the new Platform API OAuth client created for the Authorization Management API (see the prerequisites).
`Password`	(Credential) Enter the Client Secret of the new Platform API OAuth client created for the Authorization Management API (see the prerequisites).
`OAuth2TokenServiceURL`	Enter: **`https://api.<SAP_BTP_host>/oauth2/apitoken/v1`** Examples: • (Europe – Rot) *https://api.hana.ondemand.com/oauth2/apitoken/v1* • (US East – Sterling) *https://api.us3.hana.ondemand.com/oauth2/apitoken/v1*

Property Name	Description & Value
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP Java/HTML5 apps (Neo)* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   Using the default transformation, all groups that are available in the source system and their respective members (as identifiers) will be created as groups and members in the SAP BTP account. In the target system, they will be configured and assigned to the same list of users (as identities) that are available as members for these roles in the source system.

   > ### Note
   >
   > Make sure the *group* mapping in the source system is enabled. If it contains **`"ignore": true`**, change it to **`"ignore": false`**.

   You can change the default transformation mapping rules to reflect your current setup of entities in your target system. For more information, see:

   Manage Transformations [page 2439]

   SAP BTP: Authorization Management API

   Default transformation:

   > ### Code Syntax

```
{
    "user": {
        "ignore": true,
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetVariable": "entityIdTargetSystem"
            }
        ]
    },
```

```
    "group": {
    "condition": "('%hcp.group.prefix%' === 'null') || ($.displayName =~ /
%hcp.group.prefix%.*/)",`
        "mappings": [
// Attribute entityIdTargetSystem stores the displayName attribute as a
unique value of the group.
                "sourcePath": "$.displayName",
                "targetVariable": "entityIdTargetSystem",
                "functions": [
                  {
                    "condition": "('%hcp.group.prefix%' !== 'null') && (@
=~ /%hcp.group.prefix%.*/)",
                    "function": "replaceFirstString",
                    "regex": "%hcp.group.prefix%",
                    "replacement": ""
                  }
              ]
            },
// All members of a source group will be transformed, by default, into
users for a new group.
// This group will be created in SAP BTP when the JSON data is prepared to
be sent to the target system.
            {
                "sourcePath": "$.members[*].value",
                "optional": true,
                "targetPath": "$.users"
            }
        ]
    }
}
```

5. Now, add a source system from which to read groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

What is SAP BTP

1.6.2.15 SAP BTP XS Advanced UAA (Cloud Foundry)

Follow this procedure to set up the SAP BTP XS Advanced UAA (running on SAP BTP, Cloud Foundry environment) as a target system.

Prerequisites

You have created credentials to call the REST APIs of the SAP Authorization and Trust Management service on global account, subaccount, or directory level. For more information, see: Get Access to the APIs and btp create security/api-credential.

Context

The SAP BTP XS Advanced UAA connector enables you to provision platform users, business users, groups and group assignments to the local user stores of global accounts, directories, and multi-environment subaccounts on SAP BTP. These users and groups (considered role collections in SAP BTP), are typically read from source systems, such as identity providers like SAP Cloud Identity Services.

Create a separate target system for each global account, subaccount and directory, and configure the respective credentials and connectivity properties. Depending on where you want to provision your entities, you need to:

- Create a target system for the global account to provision **platform users** and groups.
- Create a target system for the directory to provision **platform users** and groups.
- Create a target system for the subaccount to provision **platform users**, **business users** and groups.

> **Note**
>
> Provisioning of platform users as **org members** and **space members** requires the usage of a different connector: SAP BTP Platform Members (Cloud Foundry). For more information, see SAP BTP Platform Members (Cloud Foundry) [page 1034].

Creating, updating, and deleting groups follow the standard processes. If groups exist in the source but do not exist in the target, Identity Provisioning will create them. If groups have been changed or deleted in the source, Identity Provisioning will update or delete them accordingly. Deleting involves deassigning the group members and deleting the group itself.

Follow the steps below to create SAP BTP XS Advanced UAA as a target system to provision SAP BTP users and groups to global accounts, directories, and multi-environment subaccounts on SAP BTP.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.

2. Add *SAP BTP Advanced UAA (Cloud Foundry)* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the API URL in the following format: `https://api.authentication.<region>.hana.ondemand.com` For more information, see Call an API.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`OAuth2TokenServiceURL`	Enter the URL to the OAuth2 token service in the following format: • `https://<global_account_subdomain>.authentication.<region>.hana.ondemand.com/oauth/token` • `https://<subaccount_subdomain>.authentication.<region>.hana.ondemand.com/oauth/token` • `https://<directory_subdomain>.authentication.<region>.hana.ondemand.com/oauth/token` For more information, see Call an API.
`User`	Enter the respective OAuth client ID that was generated when you created credentials for each global account, subaccount and directory. For more information, see Get Access to the APIs.
`Password`	(Credential) Enter the respective OAuth client secret that was generated when you created credentials for each global account, subaccount and directory. For more information, see Get Access to the APIs.

Property Name	Description & Value
`xsuaa.origin`	Enter the Origin Key of your identity provider. • Identity provider for platform users: In the SAP BTP cockpit, go to your global account (see Navigate in the Cockpit), choose *Security* > *Trust Configuration* and copy the *Origin Key* value used for platform users. This origin key always ends with `-platform`. • Identity provider for business users: In the SAP BTP cockpit, go to your subaccount (see Navigate in the Cockpit), choose *Security* > *Trust Configuration* and copy the *Origin Key* value used for business (application) users. For example: sap.custom **Note** If multiple origins are configured, the user will be created multiple times, once per origin. For more information, see Configure Single and Multiple Origins [page 1051]
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://api.authentication.eu10.hana.ondemand.com
OAuth2TokenServiceURL=https://myaccount.authentication.eu10.hana.ondemand.com/oauth/token
User=sb-full-access!a76125
Password=************
xsuaa.origin=sap.custom
```

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP Advanced UAA* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   - **Mapping logic** – The behavior of the default transformation logic is to map all attributes from the intermediate Identity Provisioning representation to the SAP BTP XS Advanced UAA target attributes.
   - **User offboarding** – If a user has been deleted from the source system, this change is recognized and the user is deleted in the SAP BTP XS Advanced UAA target system too.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP BTP XS Advanced UAA system. For more information, see Manage Transformations [page 2439] and the API documentation.

   - User Management (System for Cross-domain Identity Management (SCIM))- SCIM groups - role collections
   - User Management (System for Cross-domain Identity Management (SCIM)) - SCIM users - shadow users

   To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

   Default transformation:

   Code Syntax

```
{
  "user": {
    "condition": "isValidEmail($.emails[0].value) &&
(('%xsuaa.group.prefix%' === 'null') || ($.groups[?(@.display =~ /
%xsuaa.group.prefix%.*/)] empty false))",
    "mappings": [
      {
        "constant": "xsuaa-dummy-value",
        "targetPath": "$.id",
        "scope": "createEntity"
      },
      {
        "sourceVariable": "entityIdTargetSystem",
```

```
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath":"$.name.givenName",
        "targetPath":"$.name.givenName",
        "optional":true
      },
      {
        "sourcePath":"$.name.familyName",
        "targetPath":"$.name.familyName",
        "optional":true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "defaultValue": true,
        "optional": true
      },
      {
        "scope": "createEntity",
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.verified",
        "targetPath": "$.verified",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mailVerified']",
        "targetPath": "$.verified",
        "optional": true
      },
      {
        "scope": "createEntity",
        "constant": true,
        "targetPath": "$.verified"
      },
      {
        "constant": "%xsuaa.origin%",
        "targetPath": "$.origin"
      },
      {
        "sourcePath": "$.phoneNumbers",
        "targetPath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true
      },
      {
        "condition": "$.emails[?(@.primary == true)].value == []",
        "targetPath": "$.emails[0].primary",
        "constant": true
      },
      {
        "constant": "urn:scim:schemas:core:1.0",
        "targetPath": "$.schemas[0]"
      }
    ]
```

```
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
xsuaa.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
xsuaa.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "mappings": [
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "targetVariable": "entityIdTargetSystem",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
xsuaa.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%xsuaa.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.description",
        "targetPath": "$.description",
        "optional": true
      },
      {
        "constant": "urn:scim:schemas:core:1.0",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      },
      {
        "constant": "USER",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.members[*].type",
        "optional": true
      }
    ]
  }
}
```

5. Add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].

- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Authorization and Trust Management Service
btp CLI Command Reference
SAP BTP Integration Scenario

1.6.2.15.1 Configure Single and Multiple Origins

Configure and provision users with single or multiple origins in SAP BTP XS Advanced UAA (Cloud Foundry) target system.

Context

An origin tells you which is the identity provider of a user in SAP BTP XS Advanced UAA (Cloud Foundry). It is defined in the trust configuration in the SAP BTP cockpit under *Security* > *Trust Configuration* > *Origin Key*.

The origin itself is not a concept of Identity Provisioning. The role of the service is to ensure that you can read and provision users with their identity providers. Once you find the identity provider in the Origin Key, you need to set it in the `xsuaa.origin` property. You can configure it in source, target and proxy SAP BTP XS Advanced UAA (Cloud Foundry) systems. Both single and multiple values are supported. The value is a string that usually specifies the name of the identity provider or its location.

For example: `xsuaa.origin=`*ldap* and `xsuaa.origin=` *ldap;myaccount-xsuaa.accounts.ondemand.com*, where the ";" (semicolon) is the only supported delimiter.

Provisioning Users with Single Origin

You want to provision a user from a given source system to SAP BTP XS Advanced UAA (Cloud Foundry) target system where a single origin is configured.

1. On the *Properties* tab of the target system, enter the value for the `xsuaa.origin` property, for example: *idp1*.
2. (Optional) Ensure the property resolving the uniqueness of the attributes is configured. By default, it is set to: `xsuaa.user.unique.attribute=`*userName*.
3. Run a provisioning job from the source system.

As a result, users are created with a single origin in the target system.

Provisioning Users with Multiple Origins

You want to provision a user from a given source system to SAP BTP XS Advanced UAA (Cloud Foundry) target system where multiple origins are configured.

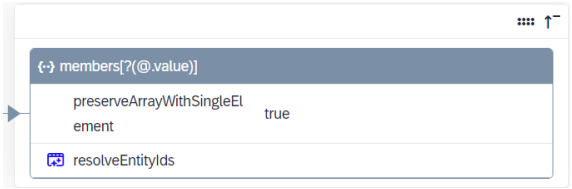
1. On the *Properties* tab of the target system, enter multiple values for the `xsuaa.origin` property, for example: *idp1;idp2*.
2. Ensure the property resolving the uniqueness of the attributes is configured and contains the *origin* value. For example, set it to: `xsuaa.user.unique.attribute`=*userName,origin*.
3. On the *Transformations* tab, update the SAP BTP XS Advanced UAA (Cloud Foundry) target system transformation by adding "[]" (square brackets) to ensure that the following constant supports multiple values:

JSON Text Editor	Graphical Editor
Code Syntax `{ "constant":"%xsuaa.origin%[]", "targetPath":"$.origin" },`	

> ### Note
>
> In case you use the `resolveEntityIds` function in your SAP BTP XS Advanced UAA (Cloud Foundry) target system transformation to resolve the value of one group member in the `sourcePath` to exactly one group member in the `targetPath`, this won't work with multiple origin values. You need to change the `targetPath` in the transformation so that the value of one group member is resolved to multiple group members. See the example below.

JSON Text Editor	Graphical Editor
Current mapping: Code Syntax `{ "sourcePath":"$.members[0].value", "preserveArrayWithSingleElement":true, "optional":true, "targetPath":"$.members[0].value", "functions":[ { "function":"resolveEntityIds" } ] },`	Current mapping:

JSON Text Editor	Graphical Editor

Expected mapping:
Code Syntax
`{`
`"sourcePath":"$.members[0].value",`
`"preserveArrayWithSingleElement":true,`
`"optional":true,`
`"targetPath":"$.members[?(@.value)]",`
`"functions":[`
`{`
`"function":"resolveEntityIds"`
`}`
`]`
`},`
	Expected mapping:
	 

4. Run a provisioning job.

As a result, two entries for one and the same user are created. Both entries contain the same user attributes (for example, username and email address) and differ only in their origins. The job logs show that one user is read and two users are created.

If the provisioned user is assigned to a group, this group will have two user entries assigned as group members which differ only in their origins.

> **Note**
>
> Multiple origins are not supported in provisioning scenarios between SAP BTP XS Advanced UAA (Cloud Foundry) source system and SAP BTP XS Advanced UAA (Cloud Foundry) target system.

Change From Single to Multiple Origins

You have provisioned users to SAP BTP XS Advanced UAA (Cloud Foundry) with a single origin and now you want to provision them with multiple ones.

Follow the steps in the *Provisioning Users with Multiple Origins* above. Before you run the provisioning job, you must reset the target system. For more information, see Reset Identity Provisioning System [page 2485]

1.6.2.16 SAP Build Work Zone, advanced edition

Follow this procedure to set up SAP Build Work Zone, advanced edition as a target system.

Prerequisites

You have OAuth credentials for SAP Build Work Zone, advanced edition. To learn how, see SAP Build Work Zone, advanced edition: Add an OAuth Client

Context

After fulfilling the prerequisites, follow the procedure below to create a target SAP Build Work Zone, advanced edition system to provision users and groups.

These target systems consume SCIM 2.0 API provided by SAP Build Work Zone, advanced edition.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Build Work Zone, advanced edition* as a target system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP Build Work Zone, advanced edition and configure the authentication method (basic or OAuth certificate-based).

   > ### Note
   >
   > We recommend that you use OAuth certificate-based authentication in the regions where it is supported by SAP Build Work Zone, advanced edition. This authentication method is supported for subaccounts created on the recommended data centers for new customers according to the SAP Build Work Zone, advanced edition documentation.

   a. In your newly added SAP Build Work Zone, advanced edition target system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection [page 2452].

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in the SAP Build Work Zone, advanced edition Admin Console and are relevant for certificate-based authentication only.

   b. Login to the SAP Build Work Zone, advanced edition Admin Console and execute substep *2. f. iii. Use X.509 Certificate-Based Authentication* from *Step 3: Configure the Identity Authentication service and the Identity Provisioning service* in Option 1: Create a new tenant .

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL related to your SAP Build Work Zone, advanced edition system, in format: **`https://<account><sap_wz_domain>.workzone.ondemand.com`** For example: *https://mytenant.mydomain123.workzone.ondemand.com*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the OAuth client key, created for your SAP Build Work Zone, advanced edition tenant (see **Prerequisites**).
`Password`	Valid if *BasicAuthentication* is configured as authentication method. (Credential) Enter the OAuth client secret, created for your SAP Build Work Zone, advanced edition tenant (see **Prerequisites**).

Property Name	Description & Value
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Build Work Zone, advanced edition instance, based on the configured authentication method: • *BasicAuthentication* - **`https://<account><sap_wz_domain>.workzone.ondemand.com/api/v1/auth/token`** For example: *https://myaccountmydomain123.workzone.ondemand.com/api/v1/auth/token* • *ClientCertificateAuthentication* - **`https://<company_uuid>.api-<region_host>.dws.workzone.ondemand.com/api/v2/auth/token`** **Note** The `api-` prefix before `<region_host>` is necessary for using *ClientCertificateAuthentication* authentication. For example: *https://mycompanyuuid123.api-eu-20.dws.workzone.ondemand.com/api/v2/auth/token*
`client_id`	Valid if *ClientCertificateAuthentication* is configured as authentication method. Enter the OAuth Client id used for access token retrieval.
Optional Properties	
(Optional) `workzone.content.type`	This property makes the SAP Build Work Zone, advanced edition target system to send the specified value for the *Content-Type* HTTP header. Example: **application/json** Default value (when not specified): *application/scim+json*
(Optional) `workzone.support.patch.operation`	This is a default property – it appears during system creation. Its default value is *true*. That means, when the Identity Provisioning identifies a changed entity in the source system, it will execute the updates as PATCH requests instead of PUT. That means, only the changes will be written in SAP Build Work Zone, advanced edition, instead of provisioning the whole entity data.

Property Name	Description & Value
(Optional) `workzone.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists on the SAP Build Work Zone, advanced edition target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). Default value (when not specified): *userName* To learn more, see: List of Properties
(Optional) `workzone.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists on the SAP Build Work Zone, advanced edition target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for this property. Default value (when not specified): *displayName* To learn more, see: List of Properties
(Optional) `workzone.include.if.match.wildcard.header`	This property makes the SAP Build Work Zone, advanced edition target system to send the *If-Match* HTTP header with a value of "*" for every request to SAP Build Work Zone, advanced edition. This header could be used for entity versioning. Default value (when not specified): *false*
(Optional) `ips.failed.request.retry.attempts`	Predefined value: *2*
(Optional) `ips.failed.request.retry.attempts.interval`	Predefined value: *30*
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Build Work Zone, advanced edition* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Build Work Zone, advanced edition. For more information, see:

   Manage Transformations [page 2439]

   SAP Build Work Zone SCIM API

   Default transformation:

   Code Syntax

```
{
    "user": {
        "condition": "('%workzone.group.prefix%' === 'null') || ($.groups[?
(@.display =~ /%workzone.group.prefix%.*/)] empty false)",
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id",
                "scope": "deleteEntity"
            },
// When a user is supposed to be deleted from SAP Work Zone, it actually
has its status set to inactive instead of being deleted.
            {
                "constant": false,
                "targetPath": "$.active",
                "scope": "deleteEntity"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
                "targetPath": "$.schemas[0]"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
                "targetPath": "$.schemas[1]"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:extension:sap:2.0:User",
                "targetPath": "$.schemas[2]"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas",
                "optional": true
            },
            {
                 "sourcePath": "$.userType",
```

```
                "optional":true,
                "targetPath": "$.userType"
            },
            {
                "condition": "$.groups[?(@.value
=~ /.*Workzone_User_Type_public/)] EMPTY false || $.groups[?(@.display
=~ /.*Workzone_User_Type_public/)] EMPTY false",
                "constant": "public",
                "targetPath": "$.userType"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "targetPath": "$.userName",
                "optional": true
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "condition": "$.emails[0].length() > 0",
                "constant": true,
                "targetPath": "$.emails[0].primary"
            },
            {
                "condition": "($.locale EMPTY false) && ($.addresses[?
(@.type == 'work')].country EMPTY false)",
                "sourcePath": "$.locale",
                "targetPath": "$.locale",
                "functions": [
                    {
                        "function": "toLowerCaseString"
                    },
                    {
                        "function": "concatString",
                        "suffix": "_"
                    },
                    {
                        "function": "concatString",
                        "suffix": "$.addresses[?(@.type ==
'work')].country"
                    }
                ]
            },
            {
              "sourcePath":"$.name.givenName",
              "targetPath":"$.name.givenName",
              "optional":true
            },
            {
              "sourcePath":"$.name.familyName",
              "targetPath":"$.name.familyName",
              "optional":true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
            },
            {
                "sourcePath": "$.active",
```

```
                "targetPath": "$.active",
                "optional": true
            },
            {
                "sourcePath": "$.title",
                "targetPath": "$.title",
                "optional": true
            },
            {
                "sourcePath": "$.timezone",
                "optional": true,
                "targetPath": "$.timezone"
            },
            {
                "sourcePath": "$.addresses",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.addresses"
            },
            {
                "condition": "$.addresses[?(@.type == 'work')].country
EMPTY false",
                "constant": true,
                "targetPath": "$.addresses[1].primary"
            },
            {
                "condition": "$.groups[?(@.value =~ /.*Workzone_Admin/)]
EMPTY false || $.groups[?(@.display =~ /.*Workzone_Admin/)] EMPTY false",
                "constant": "Administrator",
                "targetPath": "$.roles[0].value"
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "optional": true
            },
            {
```

```
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "optional": true
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "optional": true,
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['attributes']",
                "targetPath":
"$['urn:sap:cloud:scim:schemas:extension:custom:2.0:JamCustomUser']
['attributes']",
                "optional": true
            }
        ]
    },
    "group": {
        "condition": "isAttributeWithOptionalPrefix($.displayName,
workzone.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
workzone.group.prefix) && (isRegularGroup() ||
isApplicationSpecificGroup())",
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas",
                "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
```

```
                        "condition": "isAttributeWithMandatoryPrefix(@,
workzone.group.prefix)",
                        "function": "replaceFirstString",
                        "regex": "%workzone.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "type": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

6. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

1. Before starting a provisioning job, you can first subscribe to the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during your jobs. For more information, see Manage Job Notifications [page 3018].
2. Now, start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.17 SAP Build Work Zone, standard edition

Follow this procedure to set up SAP Build Work Zone, standard edition as a target system.

Prerequisites

You have created an instance and generated a service key for the standard service plan of SAP Build Work Zone, standard edition. For more information, see: Initial Setup.

The service key contains the API URL and the OAuth credentials (`clientid` and `clientsecret`) under the `uaa` property.

Context

The SAP Build Work Zone, standard edition simplifies access to SAP, custom-built, and third party applications and extensions (both on the cloud and on premise) by establishing a central launchpad.

You can use the Identity Provisioning UI to configure SAP Build Work Zone, standard edition as a target system for provisioning users, groups and users' group assignments from various source systems.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Build Work Zone, standard edition* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the SAP Build Work Zone, standard edition API URL from the service key of your SAP Build Work Zone, standard edition instance under `endpoints [portal-service]`. It follows the pattern: `https://portal-service.cfapps.sap.hana.ondemand.com`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, from the service key of your SAP Build Work Zone, standard edition instance under `uaa.clientid`.
`Password`	(Credential) Enter the OAuth Client Secret, from the service key of your SAP Build Work Zone, standard edition instance under `uaa.clientsecret`.

Property Name	Value
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL from the service key of your SAP Build Work Zone, standard edition instance. It follows the pattern: `<uaa.url>/oauth/token`. For example: `https://ips-cflp-woaealle.authentication.sap.hana.ondemand.com/oauth/token`
`cflp.providerId`	Enter a valid providerID value. The provider ID is specified in the Channel Manager of the SAP Build Work Zone, standard edition when defining a new content provider. For more information about configuring the content provider to use the Identity Provisioning service, see: Manage Content Providers (Cloud), Manage Content Providers (On Premise). **Note** All users and groups are provisioned to the target SAP Build Work Zone, standard edition system with the providerID defined for this target system. If you want to use different providerIDs, you need to create separate target systems for every providerID.

Property Name	Value
`cflp.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. SAP Build Work Zone, standard edition supports the following unique attributes which are automatically filled in when the target system is added in the service UI: **`emails[0].value, ['urn:ietf:params:scim:schemas:extension:2.0:mapping'] ['providerId'],externalId`** **Note** SAP Build Work Zone, standard edition, only supports **`externalId`** of type UUID. • If the user has an **`externalId`**, the conflict is resolved by **`externalId`** and **`providerId`**. • If the user doesn't have an **`externalId`**, the conflict is resolved by **`email`** and **`providerId`**. The general rule is that to resolve the conflict, there should be an existing user who matches both unique attributes. However, when resolving conflicts, Identity Provisioning decides whether to update an existing user or create a new one based on several criteria: whether the user was created by the Identity Provisioning service, if any of the provisioning systems (source or target) have been reset, and last but not least, the unique combination of user attributes. For more information, see **Step 4**. → Recommendation We recommend that you do not modify the value of the `cflp.user.unique.attribute` property. Otherwise, user creation fails.

Property Name	Value
`cflp.group.unique.attribute`	If Identity Provisioning tries to provision a group that already exists in the target system (a conflicting group), this property defines the unique attributes by which the existing group will be searched and resolved. SAP Build Work Zone, standard edition supports a pair of unique attributes which is automatically filled in when the target system is added in the service UI: **`externalId, ['urn:ietf:params:scim:schemas:extension:2.0:mapping']['providerId']`** For the conflict to be resolved, an existing group matching both unique attributes should be found. In this case, Identity Provisioning updates the group. This means, the conflicting group overwrites the existing one. If the group matches only one of the unique attributes, the conflict is not resolved, and the group creation fails. → Recommendation We recommend that you do not modify the value of the `cflp.group.unique.attribute` property. Otherwise, the group creation fails.
`cflp.support.bulk.operation`	This property enables bulk operations for users and groups. When the property is enabled (set to *true*), the following operations can be executed by Identity Provisioning service in a single request: • Create or delete multiple users • Create, update, or delete multiple groups When the property is disabled (set to *false*), the following operations can be executed by Identity Provisioning service for a single entity at a time: • Create or delete a user • Create, update, or delete a group **Possible values:** • *true* • *false* Default value: *false*

Property Name	Value
`cflp.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of operations to be provisioned per request. **Possible values:** Default value: *20* Maximum value: *100* If you enter a number larger than 100, the service will replace it with the default value (20).
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map user and group attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Build Work Zone, standard edition* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP Build Work Zone, standard edition entity.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Build Work Zone, standard edition system. For more information, see: Manage Transformations [page 2439].

   To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

   ### User Update and Uniqueness

   The write transformation of SAP Build Work Zone, standard edition supports three user unique attributes: *externalId*, *email* and *providerId*. Only providerId is mandatory in combination with at least one of the other attributes. Based on the combinations of these attributes, there can be distinguished three types of users:

   - User 1 - with *email* and *providerId*, but no *externalId*

- User 2 - with *externalId* and *providerId*, but no *email*
- User 3 - with all three attributes: *externalId*, *email* and *providerId*

The uniqueness of the user is ensured by the combination of two unique attributes: the *providerId* and *externalId* or the *providerId* and the *email*.

Updating an existing user in the target system depends on how the user is identified there (its combination of unique attributes) and whether Identity Provisioning is aware of this user:

- The existing user was created by a provisioning job (that is, the service is aware of this user).
  If the following attributes are changed in the source system: the *email* of user 1, the *externalId* of user 2, or the *email* or *externalId* of user 3, after running a new job, Identity Provisioning will update the existing user.
- The existing user was not created by a provisioning job or was created by one but afterwards the source and the target systems have been reset (that is, the service is not aware of this user).
  If the following attributes are changed in the source system, expect the subsequent results after running a new job:
  - Changing the *email* of user 1 will result in creating a new user in the target system, because the user won't match the combination of two unique attributes: *email* and *providerId*.
  - Changing the *externalId* of user 2 will result in creating a new user in the target, because the user won't match the combination of two unique attributes: *externalId* and *providerId*.
  - Changing the *email* or the *externalId* of user 3 will result in updating the user in the target system, because the user will still match a combination of two unique attributes. However, changing both, the *email* and the *externalId*, of user 3 will result in creating a new user in the target system, because the user will match only one unique attribute - the *providerId*.

Default transformation:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:User",
                    "urn:ietf:params:scim:schemas:extension:sap:2.0:User",
                    "urn:ietf:params:scim:schemas:extension:2.0:mapping"
                ],
                "targetPath": "$.schemas"
            },
            {
                "constant": "%cflp.providerId%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:2.0:mapping']['providerId']"
            },
            {
                "condition": "$.emails[?(@.primary == true)].value == []",
                "sourcePath": "$.emails[0].value",
                "optional": true,
                "targetPath": "$.emails[0].value"
            },
            {
                "condition": "$.emails[?(@.primary == true)].value != []",
```

```
                "sourcePath": "$.emails[?(@.primary == true)].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.emails[0].value",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "condition": "$.emails[0].length() > 0",
                "constant": true,
                "targetPath": "$.emails[0].primary"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "optional": true,
                "targetPath": "$.externalId"
            },
            {
                "sourcePath": "$.groups[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.groups[?(@.value)]",
                "functions": [
                    {
                        "entityType": "group",
                        "function": "resolveEntityIds"
                    }
                ]
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:Group",
                    "urn:ietf:params:scim:schemas:extension:2.0:mapping"
                ],
                "targetPath": "$.schemas"
            },
            {
                "constant": "%cflp.providerId%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:2.0:mapping']['providerId']"
            },
            {
                "sourcePath": "$.displayName",
                "optional": true,
                "targetPath": "$.externalId"
            },
            {
                "sourcePath": "$.externalId",
                "optional": true,
                "targetPath": "$.externalId",
                "functions": [
                    {
                        "function":"replaceAllString",
                        "regex":"(?i)(^pcd:)",
                        "replacement":""
```

```
                    },
                    {
                        "function": "replaceString",
                        "target": "/",
                        "replacement": ":"
                    },
                    {
                        "function": "replaceString",
                        "target": "(",
                        "replacement": "@"
                    },
                    {
                        "function": "replaceString",
                        "target": ")",
                        "replacement": "+"
                    }
                ]
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

5. Add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Configure Integration with the Identity Provisioning Service

1.6.2.18 SAP Business Network

Follow this procedure to set up SAP Business Network as a target system.

Context

> **Note**
>
> Currently, the SAP Business Network connector cannot be used by customers due to API access constraints. SAP is working on resolving these constraints to make the connector fully operational and accessible.

SAP Business Network, formerly known as Ariba Network, is a cloud-based offering that makes it possible for buyers and suppliers to collaborate on transactions, strengthen their relationships, and discover new business opportunities.

You can use Identity Provisioning to configure SAP Business Network as a target system where you can provision users and update group members.

> **Note**
>
> Identity Provisioning cannot create and delete groups in SAP Business Network target system. It can only update existing groups by adding or removing group members. Therefore, groups must have been created in SAP Business Network before you run a provisioning job.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Business Network* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the SAP Business Network API URL.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*

Property Name	Value
`User`	Enter the OAuth Client Id, created for your SAP Business Network system.
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP Business Network system.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL.
`bn.api.key`	An API Key represents the unique key that identifies a particular application as a legitimate consumer of an API.
`bn.realm.id`	The realm name is part of the URL you use to access SAP Business Network.
(Optional) `bn.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. The property is not added automatically at system creation. Default value: *userName* If the service finds an existing user by userName, it updates this user with the data of the conflicting one. If the service does not find an existing user by userName, the creation of the conflicting user fails.
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of group assignments to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of group assignments, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Business Network* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Business Network system. For more information, see:

Manage Transformations [page 2439].

To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP Business Network entity.

Default transformation:

Code Syntax

```
{
  "user": {
    "condition": "($.emails.length() > 0) && ($.name.givenName EMPTY
false) && ($.name.familyName EMPTY false)",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:User",
          "urn:ietf:params:scim:schemas:extension:sap:2.0:User",

"urn:ietf:params:scim:schemas:extension:sap.business.network:2.0:User",
          "urn:com.sap.ariba.framework.scim2.common.types.UserResourceV2"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sendMail']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sendMail']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:sap.business.network:2.0:User']
['userStatus']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap.business.network:2.0:User']
['userStatus']",
        "optional": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
```

```
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true,
        "defaultValue": true
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.middleName",
        "targetPath": "$.name.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.formatted",
        "targetPath": "$.name.formattedName",
        "optional": true
      },
      {
        "sourcePath": "$.name.honorificPrefix",
        "targetPath": "$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath": "$.name.honorificSuffix",
        "targetPath": "$.name.honorificSuffix",
        "optional": true
      },
      {
        "sourcePath": "$.emails[0].value",
        "targetPath": "$.emails[0].value"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "condition": "$.addresses[?(@.type == 'work')].country EMPTY
false",
        "sourcePath": "$.addresses[?(@.type == 'work')]",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.addresses[0]"
      },
      {
        "sourcePath": "$.phoneNumbers",
        "targetPath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.preferredLanguage",
        "targetPath": "$.preferredLanguage",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
```

```
      {
        "sourcePath": "$.timezone",
        "targetPath": "$.timezone",
        "optional": true
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": ["urn:ietf:params:scim:schemas:core:2.0:Group"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "entityType": "group",
            "type": "resolveEntityIds"
          }
        ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "entityType": "user",
            "type": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

5. Add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Business Network

1.6.2.19 SAP Central Business Configuration

Follow this procedure to set up SAP Central Business Configuration (in short, CBC) as a target system.

Prerequisites

You have created a technical user with administrator permissions that will be used to call the API of SAP Central Business Configuration for creating or updating user and group member information.

Context

Create a CBC target system to provision users and group members to it.

> ⚠ Caution
>
> You can't create or delete groups on CBC. That means:
>
> - On the attempt to create a group on CBC, Identity Provisioning will only add new members or update existing ones. Also, when the service reads a group from a source system, there must be a group with the exact same display name (case sensitive) in the CBC target system. Otherwise, an error will be thrown and the group members will not be updated.
> - On the attempt to delete a group on CBC, Identity Provisioning will only remove its members (group assignments). And this can happen only if the relevant group assignments have been provisioned/are present in the target system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Central Business Configuration* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the API of your CBC system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Specify the technical user for your CBC system.
`Password`	(Credential) Specify the password for your technical user.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. For example: **https://mycbcaccount.authentication.us2.hana.ondemand.com/oauth/token**
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of group assignments to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of group assignments, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Central Business Configuration* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your CBC system. For more information, see Manage Transformations [page 2439].

   Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target CBC entity.

   Default transformation:

   Code Syntax

   ```
   {
   ```

```
    "group": {
        "condition": "isAttributeWithOptionalPrefix($.displayName,
cbc.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
cbc.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
        "mappings": [
            {
                "targetPath": "$.id",
                "sourceVariable": "entityIdTargetSystem"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schema[0]"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "isAttributeWithMandatoryPrefix(@,
cbc.group.prefix)",
                        "function": "replaceFirstString",
                        "regex": "%cbc.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            }
        ]
    },
    "user": {
    "condition": "('%cbc.group.prefix%' === 'null') || ($.groups[?
(@.display =~ /%cbc.group.prefix%.*/)] empty false)",
        "mappings": [
            {
                "targetPath": "$.id",
                "sourceVariable": "entityIdTargetSystem"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
                "targetPath": "$.schemas[0]"
            },
            {
                "constant":
"urn:sap:cloud:scim:schemas:extension:custom:2.0:User",
                "targetPath": "$.schemas[1]"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName"
```

```
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails"
            },
            {
                "targetPath": "$.emails[0].primary",
                "condition": "$.emails[?(@.primary == true)].value == []",
                "constant": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.userType",
                "optional": true,
                "targetPath": "$.userType"
            }
        ]
    }
}
```

5. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Central Business Configuration – Collection

1.6.2.20 SAP Commerce Cloud

Follow this procedure to set up SAP Commerce Cloud as a target system.

Prerequisites

- In SAP Commerce Cloud, you have added an OAuth client with authorization grant **Client Credentials**. To learn how, see: Configuring OAuth Client.

Context

SAP Commerce Cloud is a highly flexible and modular platform for delivering modern customer experiences. It provides a standardized, automated, end-to-end solution that allows your projects to release code from repository to cloud.

You can use Identity Provisioning to configure SAP Commerce Cloud as a target system to provision users and groups. These target systems consume SCIM 2.0 API provided by SAP Commerce Cloud. For more information, see Commerce Cloud SCIM Web Services API Documentation.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Commerce Cloud* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP Commerce Cloud system. The URL follows the pattern: `https://backoffice.<tenant>. model-t.cc.commerce.ondemand.com` You can find the correct URL in the *Environments* section of SAP Cloud Portal.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the client ID to retrieve the OAuth access token for SAP Commerce Cloud.
`Password`	(Credential) Enter the client secret to retrieve the OAuth access token for SAP Commerce Cloud.
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Commerce Cloud instance.

Property Name	Description & Value
(Optional) `cc.user.unique.attribute`	This property defines by which unique attribute(s) an existing user in the target system will be searched and resolved in case Identity Provisioning tries to provision a conflicting user. SAP Commerce Cloud supports the following unique attributes which are automatically filled in during system creation: **`emails[0].value, userName, externalId`**. If the service finds an existing user by at least one of the unique attributes, it updates this user with the data of the conflicting one. If such a user is not found, the update of the conflicting user fails. If more than one users with these unique attributes are found, the update fails.
(Optional) `cc.group.unique.attribute`	If you try to provision a group that already exists in a target system, the group creation will fail. In this case, the existing group only needs to be updated. This property defines by which unique attribute(s) the existing group will be searched and resolved. The default value is **`displayName`**. If the service finds an existing group by this unique attribute, the group that you try to provision will overwrite the existing one. If such a group is not found, the group that you try to provision will not be created in the target system.
(Optional) `cc.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. Default value for target systems: *false*
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Property Name	Description & Value
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Commerce Cloud* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   > ### Note
   >
   > Users that are created in SAP Commerce Cloud through the UI, don't have an ID and a username. When synchronizing such users from a source system to SAP Commerce Cloud target system, Identity Provisioning detects that they already exist. To resolve the conflict, the service tries to retrieve them from the target, but as they have no IDs, the users are skipped.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Commerce Cloud system. For more information, see:

   Manage Transformations [page 2439].

   Commerce Cloud SCIM Web Services API Documentation

   To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

   Default transformation:

   ### Code Syntax

```
{
    "user": {
        "condition": "$.userName EMPTY false && (('%cc.group.prefix%' ===
'null') || ($.groups[?(@.display =~ /%cc.group.prefix%.*/)] empty false))",
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$.schemas",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
```

```
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userId']",
                "targetPath": "$.externalId"
            },
            {
                "sourcePath": "$.displayName",
                "optional": true,
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.nickName",
                "optional": true,
                "targetPath": "$.nickName"
            },
            {
                "sourcePath": "$.name.familyName",
                "optional": true,
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.name.middleName",
                "optional": true,
                "targetPath": "$.name.middleName"
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.name.honorificSuffix",
                "optional": true,
                "targetPath": "$.name.honorificSuffix"
            },
            {
                "condition": "($.addresses[*].region EMPTY false) &&
($.addresses[*].country EMPTY false)",
                "sourcePath": "$.addresses",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.addresses",
                "functions": [
                    {
                        "type": "convertCountryRegion",
                        "outputFormat": "alpha2",
                        "inputAttributes": [
                            "region",
                            "country"
                        ],
                        "outputAttribute": "region"
                    }
                ]
            },
            {
                "sourcePath": "$.userType",
                "optional": true,
                "targetPath": "$.userType"
            },
            {
                "sourcePath": "$.active",
```

```
                "optional": true,
                "targetPath": "$.active"
            },
            {
                "condition": "$.emails[?(@.primary == true)].value == []",
                "sourcePath": "$.emails[0].value",
                "optional": true,
                "targetPath": "$.emails[0].value"
            },
            {
                "condition": "$.emails[?(@.primary == true)].value != []",
                "sourcePath": "$.emails[?(@.primary == true)].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.emails[0].value",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "condition": "$.emails[0].length() > 0",
                "constant": true,
                "targetPath": "$.emails[0].primary"
            }
        ]
    },
    "group": {
         "condition": "isAttributeWithOptionalPrefix($.displayName,
cc.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
cc.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
         "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "isAttributeWithMandatoryPrefix(@,
cc.group.prefix)",
                        "function": "replaceFirstString",
                        "regex": "%cc.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "condition": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'] EMPTY
false",
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "isAttributeWithMandatoryPrefix(@,
cc.group.prefix)",
                        "function": "replaceFirstString",
```

```
                        "regex": "%cc.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

5. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585].

Related Information

SAP Cloud Identity Services Integration Architecture → cloudscimwebservices extension

1.6.2.21 SAP Concur

Follow this procedure to set up SAP Concur as a target system.

Prerequisites

- You are using User v1 API, Identity v4 API or SCIM v4 API.

  > **Note**
  >
  > The **User v1 API** has been decommissioned. For more information, see Concur User API Decommission.

- Your SAP Concur admin user has obtained a *Company Request Token* and a *Company UUID* from the SAP Concur Company Request Token self-service tool.
  For more information, see SAP Concur Company Request Token.

Context

Companies that use SAP Concur for managing and controlling travel expenses, invoices and other can use Identity Provisioning service to automate the identity and access management for the SAP Concur solution.

This service allows you to provision identity data from your existing corporate identity stores, such as SAP AS ABAP or Microsoft Active Directory, or provision employee user data from different SAP cloud user stores, such as SAP SuccessFactors or SAP Cloud Identity Service – Identity Authentication.

The SAP Concur integration with Identity Provisioning is supported with Standard and Professional editions.

SAP Concur provides different APIs for integration with Identity Provisioning, resulting in different connector versions. Each connector version supports specific attribute mappings within the transformations and requires particular property values. The value of the `concur.api.version` property controls which API you use.

- When the value is set to **3**, the SCIM v4 API is used. It supports the provisioning of users and group assignments. However, group assignments are ignored by the default write transformations and are provisioned only if the `ignore` parameter is set to `false`.

  > **Note**
  >
  > Identity Provisioning cannot create and delete groups in SAP Concur target system. It can only create, update and delete user assignments of a group. Therefore, groups must have been created in SAP Concur system before you run a provisioning job.

  By default, the Identity Provisioning service uses version 3 for all newly created systems starting December 2, 2025, when version 3 was introduced. The default version is the latest released connector version at the time the system is created.
- When the value is set to **2**, the Identity v4 API is used. This applies to systems created between December 8, 2021, and December 1, 2025. Identity v4 API supports provisioning of users with the `userUUID` attribute, which is generated by Identity Authentication at user creation.
- When the value is set to **1**, or the property is not defined, the User v1 API is used. This applies to systems created before versioning was introduced on December 8, 2021. Version 1 is deprecated.

For more information on how to update to the connector version you need, see: Update Connector Version [page 2424].

To create SAP Concur as a target system, proceed as follows:

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Concur* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Connector Version 1 Mandatory Properties

Property Name	Description & Value
Type	Enter: *HTTP*

Property Name	Description & Value

`URL`	For EU2 data center, enter:
	`eu2.concursolutions.com`
	For US2 data center, enter:
	`www.concursolutions.com`

`ProxyType`	Enter: **`Internet`**
`concur.api.version`	Defines the version of SAP Concur API. Set the value to *1* to use User v1 API.
`Authentication`	Enter: *BasicAuthentication*
`User`	Deprecated property. Do not use.
`Password`	Deprecated property. Do not use.
`X-ConsumerKey`	Deprecated property. Do not use.
`concur.datacenter`	Specify the SAP Concur data center your Identity Provisioning tenant belongs to. The following SAP Concur data centers are available: • EU2 • US2 Based on the provided data center, Identity Provisioning configures the URL of the User v1 API or Identity v4 API as follows: • If you provide **`us2`**, the URL is: `us.api.concursolutions.com`

Property Name	Description & Value
`concur.authorization.code`	(Credential) Enter the *Company Request Token* and run a provisioning job within 24 hours from generating the token in the SAP Concur Company Request Token self-service tool. Otherwise, the token will expire, and you'll need a new one. After the first run of the job, Identity Provisioning fills in automatically a refresh token as the value of the concur.refresh.token property. If a provisioning job has not been run for six months, you'll again need to generate a new token. → Remember The company request token has a 24 hour validity. If this token expires, you must request a new token. The refresh token has a six month validity. Every time you run a provisioning job, the validity of the refresh token is extended with six months starting from the date of the last run. If you haven't run a provisioning job for six months, your refresh token will expire and you must request a new company request token.
`concur.company.id`	Enter the *Company UUID* as described in the *Prerequisites* section.
`concur.company.domain`	Enter your company domain. The username and the company domain are concatenated in the default transformation in the following format: user@domain Your company domain is the part of your username behind the @ symbol. For example: johnsmith@example.com
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Connector Version 2 Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`ProxyType`	Enter: **`Internet`**
`Authentication`	Enter: *BasicAuthentication*
`concur.datacenter`	Specify the SAP Concur data center your Identity Provisioning tenant belongs to. The following SAP Concur data centers are available: • APJ1 • EU2 • US2 Based on the provided data center, Identity Provisioning configures the URL of the User v1 API or Identity v4 API as follows: • If you provide **`us2`**, the URL is: `us.api.concursolutions.com`
`concur.api.version`	Defines the version of SAP Concur API. Set the value to *2* to use the Identity v4 API. This was the default value for systems created between December 8, 2021, and December 1, 2025.
`concur.authorization.code`	(Credential) Enter the *Company Request Token* and run a provisioning job within 24 hours from generating the token in the SAP Concur Company Request Token self-service tool. Otherwise, the token will expire, and you'll need a new one. After the first run of the job, Identity Provisioning fills in automatically a refresh token as the value of the concur.refresh.token property. If a provisioning job has not been run for six months, you'll again need to generate a new token. → Remember The company request token has a 24 hour validity. If this token expires, you must request a new token. The refresh token has a six month validity. Every time you run a provisioning job, the validity of the refresh token is extended with six months starting from the date of the last run. If you haven't run a provisioning job for six months, your refresh token will expire and you must request a new company request token.

Property Name	Description & Value
`concur.company.id`	Enter the *Company UUID* as described in the *Prerequisites* section.
`concur.company.domain`	Enter your company domain. The username and the company domain are concatenated in the default transformation in the following format: user@domain Your company domain is the part of your username behind the @ symbol. For example: johnsmith@example.com
`concur.user.unique.attribute`	This property defines by which unique attribute(s) an existing user in the target system will be searched and resolved in case Identity Provisioning tries to provision a conflicting user. This property appears by default when the system is created. Its value is set to *userName*. Other possible values: • userName • Another unique attribute that supports filtering.
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Connector Version 3 Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`ProxyType`	Enter: **`Internet`**
`Authentication`	Enter: *BasicAuthentication*

Property Name	Description & Value
`concur.datacenter`	Specify the SAP Concur data center your Identity Provisioning tenant belongs to. The following SAP Concur data centers are available: • APJ1 • EU2 • US2 Based on the provided data center, Identity Provisioning configures the URL of the User v1 API or Identity v4 API as follows: • If you provide **`us2`**, the URL is: `us.api.concursolutions.com`
`concur.api.version`	Defines the version of SAP Concur API. Set the value to *3* to use the SCIM v4 API. By default, the Identity Provisioning service uses version 3 for all newly created systems starting December 2, 2025, when version 3 was introduced.
`concur.authorization.code`	(Credential) Enter the *Company Request Token* and run a provisioning job within 24 hours from generating the token in the SAP Concur Company Request Token self-service tool. Otherwise, the token will expire, and you'll need a new one. After the first run of the job, Identity Provisioning fills in automatically a refresh token as the value of the concur.refresh.token property. If a provisioning job has not been run for six months, you'll again need to generate a new token. → Remember The company request token has a 24 hour validity. If this token expires, you must request a new token. The refresh token has a six month validity. Every time you run a provisioning job, the validity of the refresh token is extended with six months starting from the date of the last run. If you haven't run a provisioning job for six months, your refresh token will expire and you must request a new company request token.
`concur.company.id`	Enter the *Company UUID* as described in the *Prerequisites* section.

Property Name	Description & Value
`concur.company.domain`	Enter your company domain. The username and the company domain are concatenated in the default transformation in the following format: user@domain Your company domain is the part of your username behind the @ symbol. For example: johnsmith@example.com
`concur.user.unique.attribute`	This property defines by which unique attribute(s) an existing user in the target system will be searched and resolved in case Identity Provisioning tries to provision a conflicting user. Possible values: • userName (Default value) • externalId • urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:employeeNumber **Note** In SAP Concur, *userName*, *externalId*, and *employeeNumber* are unique user attributes, but only *userName* is required. To ensure proper conflict resolution, *externalId* and *employeeNumber* must also be present in the source.
`concur.group.unique.attribute`	This property defines by which unique attribute(s) the existing group will be searched and resolved. The default value is *displayName*. If the service finds an existing group by this unique attribute, the group that you try to provision will overwrite the existing one. If such a group is not found, the group that you try to provision will not be created in the target system.

Property Name	Description & Value
`concur.group.prefix`	This property distinguishes SAP Concur groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`Concur_`** You can use the example value or provide your own. When **set in the target system**, only groups containing the **`Concur_`** prefix in their display name will be provisioned to SAP Concur. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Concur. For more information, see List of Properties.
(Optional)`concur.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. Default value for target systems: *false*
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Concur* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Concur. For more information, see:

Manage Transformations [page 2439]

SAP Business Accelerator Hub: Concur (Users)

User v1 API: User v1

Identity v4 API: Identity v4

SCIM v4 API: SCIM v4 API

Default transformation when using User v1 API (connector version 1):

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.userName",
                "targetPath": "$.EmpId"
            },
/* The first array value of the SCIM attribute emails will be used as an
e-mail address (EmailAddress) for the user record in SAP Concur. */
            {
                "sourcePath": "$.emails[0].value",
                "targetPath": "$.EmailAddress"
            },
            {
                "sourcePath": "$.emails[0].value",
                "targetPath": "$.LoginId"
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.FirstName"
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.LastName"
            },
            {
                "constant": "N",
                "targetPath": "$.Active"
            },
            {
                "condition": "$.active == true",
                "constant": "Y",
                "targetPath": "$.Active"
            },
            {
                "constant": "N",
                "targetPath": "$.ExpenseApprover"
            },
            {
                "constant": "N",
                "targetPath": "$.ExpenseUser"
            },
            {
                "constant": "N",
                "targetPath": "$.InvoiceApprover"
            },
```

```
            {
                "constant": "N",
                "targetPath": "$.InvoiceUser"
            },
            {
                "constant": "N",
                "targetPath": "$.IsTestEmp"
            },
            {
                "constant": "N",
                "targetPath": "$.TripUser"
            },
/* An initial password setup is mandatory for all newly provisioned user
accounts. To enable the provisioning to SAP Concur,
enable the statement for the Password attribute and make sure its value is
not empty. For more information, see the Caution box above. */
            {
                "ignore": true,
                "constant": "",
                "targetPath": "$.Password"
            },
            {
                "constant": "USD",
                "targetPath": "$.CrnKey"
            },
            {
                "sourcePath": "$.addresses[?(@.type == 'home')].country",
                "targetPath": "$.CtryCode"
            },
            {
                "constant": "US",
                "targetPath": "$.Custom21"
            },
            {
                "constant": "DEFAULT",
                "targetPath": "$.LedgerName"
            },
            {
                "constant": "DEFAULT",
                "targetPath": "$.LedgerCode"
            },
            {
                "constant": "en_US",
                "targetPath": "$.LocaleName"
            }
        ]
    }
}
```

- **Mapping logic** – When the SAP Concur system is configured as a target, the default transformation logic offered by the Identity Provisioning service contains the minimum of required properties for the successful provisioning of the users. You can change the default transformation mapping rules to reflect your current setup of entities in the source system. Before you start extending the default transformation, you have to get familiar with the requirements of the SAP Concur API to avoid inconsistencies. For more information, see User v1.
- **User off-boarding** – Identity Provisioning service handles the end-to-end lifecycle of the users, including their off-boarding. For some source systems, the deletion of a user or inactive user status is the final step of this lifecycle process. The SAP Concur solution, however, does not allow user accounts to be deleted. The offboarding of SAP Concur user accounts is always performed by setting them as disabled. When a user is deleted or set with status **inactive** in a system configured as a source for

user data provisioning to SAP Concur, the user account in SAP Concur will be disabled (the attribute `"targetPath": "$.Active"` gets a value "**N**").

⚠ Caution

The User v1 API requires an initial password setup for all newly provisioned user accounts. The default transformation offers a statement with an empty string as a value for the password configuration. However, it is ignored in order to prevent from a default setup of a wrong initial password for your systems. While the password statement is ignored, the provisioning will not be working. To enable the provisioning to SAP Concur, you need to perform the following operations:

1. Enable the password statement. To do this, either delete **"ignore": true,** or set it as **"ignore": false**.
2. Set a proper statement for the password attribute value (**"targetPath": "$.Password"**).

(Optional) You can leave the default empty string, or you can use the **randomPassword** function to calculate a random value for the initial password of the newly created SAP Concur accounts. If you choose one of these two options and if you are not using single sign-on solution for SAP Concur, you have to also arrange a password reset support process in your company. This will securely offer an initial password to your corporate users for their newly created SAP Concur accounts. For more information, see Transformation Expressions [page 391] → **Transformation Functions**.

The Identity Provisioning does not provide group resource mapping in the SAP Concur write transformation. Although you cannot provision groups to SAP Concur, you can group the users into organizational units by enhancing your target transformation as follows:

{·} Sample Code

```
...

    {
            "constant": "<provided by SAP Concur>",
            "targetPath": "$.LedgerCode"
        },
        {
            "constant": "<obtain from SAP Concur API>",
            "targetPath": "$.Custom21"
        },
        {
            "constant": "<obtain from SAP Concur API>",
            "targetPath": "$.OrgUnit1"
        },
        {
            "constant": "<obtain from SAP Concur API>",
            "targetPath": "$.OrgUnit2"
        }
    {
            "constant": "DEFAULT" or "<obtain from SAP Concur API>",
            "targetPath": "$.LedgerKey"
        },
```

Default transformation when using Identity v4 API (connector version 2):

{·} Code Syntax

```
{
```

```
  "user": {
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:core:2.0:User","urn:sap:cloud:scim:schemas:e
xtension:custom:2.0:User","urn:ietf:params:scim:schemas:extension:sap:2.0:U
ser", "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "functions": [
          {
            "type": "concatString",
            "suffix": "@%concur.company.domain%"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.emails",
        "functions": [
          {
            "function": "putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "targetPath": "$.emails[*].primary",
        "type": "remove"
      },
      {
        "sourcePath": "$.name",
        "targetPath": "$.name"
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.givenName",
```

```
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.displayName",
        "optional": true,
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.addresses",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.addresses"
      },
      {
        "targetPath": "$.addresses[*].primary",
        "type": "remove"
      },
      {
        "sourcePath": "$.title",
        "optional": true,
        "targetPath": "$.title"
      },
      {
        "sourcePath": "$.phoneNumbers[?(@.type != 'mobile' || (@.type ==
'mobile' && @.primary == false))]",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.phoneNumbers[0]"
      },
      {
        "targetPath": "$.phoneNumbers[*].primary",
        "type": "remove"
      },
      {
        "sourcePath": "$.phoneNumbers[?(@.type == 'mobile' && @.primary ==
true)]",
        "optional": true,
        "targetPath": "$.phoneNumbers[0]"
      },
      {
        "sourcePath": "$.emergencyContacts",
        "targetPath": "$.emergencyContacts",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "constant": "%concur.company.id%",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['companyId']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
```

```
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['division']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']"
      }
    ]
  }
}
```

Note

The default transformation for Concur connector version 2 (based on Identity v4 API) provides a mapping that generates the Concur User Login_ID based on the username from the source (Login Name in SAP Cloud Identity Services) and the domain configured as the `concur.company.domain` property.

In case you want to do the following:

- Provision users with usernames that are already in email format
- Use the user email address as SAP Concur Login_ID
- Use other attribute with value in email format as SAP Concur Login_ID

you have to update this mapping and remove the function that appends the domain part. For example:

Default

Code Syntax

```
{
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "functions": [
          {
            "type":
"concatString",
            "suffix":
"@%concur.company.domain%"
          }
        ]
      },
```

Changed

Code Syntax

```
{
   "condition":"$.emails[?
(@.primary== true)] empty
false",
   "sourcePath":"$.emails[?
(@.primary== true)].value",
   "targetPath":"$.userName",
   "optional":true
},
{
   "condition":"$.emails[?
(@.primary== true)] empty true",

"sourcePath":"$.emails[0].value"
,
   "targetPath":"$.userName",
   "optional":true
},
```

In addition, make sure that this mapping is aligned with the configuration of Name ID attribute in Identity Authentication, as described in Appendix – SAP Cloud Identity Services - Identity Authentication Service (SAP IAS) Setup.

Default transformation when using SCIM v4 API (connector version 3):

Code Syntax

```
{
  "user": {
    "condition": "('%concur.group.prefix%' === 'null') || ($.groups[?
(@.display =~ /%concur.group.prefix%.*/)] empty false)",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
```

```
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:User",
          "urn:ietf:params:scim:schemas:extension:sap:2.0:User",
          "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "functions": [
          {
            "type": "concatString",
            "suffix": "@%concur.company.domain%"
          }
        ]
      },
      {
        "sourcePath": "$.name",
        "targetPath": "$.name"
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName"
      },
      {
        "condition": "$.name.academicTitle NIN ['Dr.', 'Doctor',
'Prof.','Professor', 'Prof. Dr.', 'Professor Doctor', 'B.A.', 'Bachelor
of Arts', 'MBA', 'Master of Business Administration', 'Ph.D', 'Doctor of
Philosophy']",
        "type": "remove",
        "targetPath": "$.name.academicTitle"
      },
      {
        "sourcePath": "$.addresses[?(@.type NIN ['work', 'home', 'other',
'billing', 'bank', 'shipping'])]",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.addresses",
        "functions": [
          {
            "condition": "(@.country empty false)",
            "type": "convertCountryCode",
            "outputFormat": "alpha2",
            "inputAttributes": [
              "country"
            ],
            "outputAttribute": "country"
          }
        ]
      },
      {
        "targetPath": "$.addresses[*].type",
        "type": "remove"
      },
      {
        "sourcePath": "$.addresses[?(@.type IN ['work', 'home', 'other',
'billing', 'bank', 'shipping'])]",
```

```
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.addresses[0]",
        "functions": [
          {
            "condition": "(@.country empty false)",
            "type": "convertCountryCode",
            "outputFormat": "alpha2",
            "inputAttributes": [
              "country"
            ],
            "outputAttribute": "country"
          }
        ]
      },
      {
        "targetPath": "$.addresses[*].primary",
        "type": "remove"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone",
        "functions": [
          {
            "type": "convertTimezoneCode",
            "outputFormat": "iana"
          }
        ]
      },
      {
        "sourcePath": "$.emergencyContacts[?(@.relationship IN ['Spouse',
'Brother', 'Parent', 'Sister', 'Life Partner', 'Other'])]",
        "targetPath": "$.emergencyContacts",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "functions": [
          {
            "condition": "(@.country empty false)",
            "type": "convertCountryCode",
            "outputFormat": "alpha2",
            "inputAttributes": [
              "country"
            ],
            "outputAttribute": "country"
          }
        ]
      },
      {
        "sourcePath": "$.preferredLanguage",
        "optional": true,
        "targetPath": "$.preferredLanguage"
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true,
        "functions": [
          {
            "type": "convertLocaleCode",
            "outputFormat": "bcp47"
          }
        ]
      },
      {
        "sourcePath": "$.title",
```

```
        "optional": true,
        "targetPath": "$.title"
      },
      {
        "sourcePath": "$.nickName",
        "optional": true,
        "targetPath": "$.nickName"
      },
      {
        "sourcePath": "$.emails[?(@.type IN ['work', 'home', 'work2',
'other', 'other2'])]",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath": "$.emails[?(@.type empty true)]",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails[0]",
        "functions": [
          {
            "function": "putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "targetPath": "$.emails[*].primary",
        "type": "remove"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['emails']",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['emails']"
      },
      {

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['emails'][*]['primary']",
        "type": "remove"
      },
      {

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['emails'][*]['type']",
        "type": "remove"
      },
      {
        "sourcePath": "$.phoneNumbers[?(@.type IN ['home', 'fax', 'pager',
'other', 'work2'])]",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.phoneNumbers"
      },
      {
        "targetPath": "$.phoneNumbers[*].primary",
        "type": "remove"
      },
      {
        "targetPath": "$.phoneNumbers[*].issuingCountry",
        "type": "remove"
```

```
      },
      {
        "targetPath": "$.phoneNumbers[*].notifications",
        "type": "remove"
      },
      {
        "sourcePath": "$.phoneNumbers[?((@.type == 'mobile') || (@.type
empty true))]",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.phoneNumbers[0]",
        "functions": [
          {
            "function": "putIfAbsent",
            "key": "type",
            "defaultValue": "mobile"
          },
          {
            "condition": "(@.issuingCountry empty false)",
            "type": "convertCountryCode",
            "outputFormat": "alpha2",
            "inputAttributes": [
              "issuingCountry"
            ],
            "outputAttribute": "issuingCountry"
          }
        ]
      },
      {
        "targetPath": "$.phoneNumbers[*].workExtension",
        "type": "remove"
      },
      {
        "sourcePath": "$.phoneNumbers[?(@.type == 'work')]",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.phoneNumbers[0]"
      },
      {
        "condition": "$.phoneNumbers[?(@.type == 'work')] empty false",
        "targetPath": "$.phoneNumbers[0].primary",
        "type": "remove"
      },
      {
        "condition": "$.phoneNumbers[?(@.type == 'work')] empty false",
        "targetPath": "$.phoneNumbers[0].issuingCountry",
        "type": "remove"
      },
      {
        "condition": "$.phoneNumbers[?(@.type == 'work')] empty false",
        "targetPath": "$.phoneNumbers[0].notifications",
        "type": "remove"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$.externalId"
      },
```

```
      {
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['contactPreferences']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['contactPreferences']",
        "optional": true
      },
      {
"condition": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['contactPreferences']['emailFormat'] NIN ['HTTP', 'plain']",
        "type": "remove",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['contactPreferences']['emailFormat']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {
        "constant": "%concur.company.id%",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['companyId']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['division']"
      },
      {
```

```
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  },
  "group": {
    "ignore": true,
    "condition": "isAttributeWithOptionalPrefix($.displayName,
concur.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
concur.group.prefix) && (isRegularGroup() ||
isApplicationSpecificGroup())",
    "mappings": [
      {
        "targetPath": "$.id",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:Group",
          "urn:ietf:params:scim:schemas:extension:sap:2.0:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
concur.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%concur.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
```

```
      }
    ]
  }
}
```

In SAP Concur connector version 3, *phoneNumbers* and *emails* require a type (such as *work*, *home*, mobile). One phone number and one email are allowed per type, except for *mobile* phone numbers, where multiple entries are permitted. If multiple mobile phone numbers are present, at least one must be primary. When a phone number has no type, Identity Provisioning sets it to *mobile*. When an email has no type, the service sets it to *work*.

A user may have multiple phone numbers, emails and addresses in a given source system. By using the *elementAt* function, only the first element in the array will be retrieved. This function can be applied to phone numbers, emails, or addresses:

Code Syntax

```
{
   "condition":"$.phoneNumbers[?(@.type == 'work')] empty false",
   "sourcePath":"$.phoneNumbers[?(@.type == 'work')]",
   "optional":true,
   "preserveArrayWithSingleElement":true,
   "targetPath":"$.phoneNumbers[0]",
   "functions":[
      {
         "function":"elementAt",
         "index":0
      }
   ]
}
```

The *address* attribute does not require a type. If a type is provided, only one address per type is allowed.

5. Now, add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.22 SAP CPQ

Follow this procedure to set up SAP CPQ (Configure, Price, and Quote) as a target system.

Prerequisites

- You have created a technical user with administrator permissions that will be used to call the API of SAP CPQ for creating and updating user and group information.
- Make sure all users that need to be read from the source system have an organization unit name. This unit name must correspond to an existing company system ID in SAP CPQ. The *organization unit* and the *company system ID* must be exactly the same. Users with empty (missing) organization units will not be provisioned, as well as users whose organization units don't match any of the SAP CPQ company system IDs for the relevant tenant.
- In order for a created user to be active in SAP CPQ, it should be assigned to an SAP CPQ target group, whose ID ends with suffix *–USERTYPE*. To learn more, see SAP CPQ: SCIM API → section **Mappings between SCIM API and SAP CPQ** → `groups`.

Context

SAP CPQ is a highly configurable system designated to help sales representatives to configure products, apply pricing and generate quotes. SAP CPQ is part of the SAP Sales Cloud portfolio.

Create an SAP CPQ target system to provision users and groups to it.

> ⚠ Caution
>
> You can't create or delete groups on SAP CPQ. That means:
>
> - On the attempt to create a group on SAP CPQ, Identity Provisioning will only add new members or update existing ones. Also, if you read a group from a source system, there must be a group with the exact same display name (case sensitive) in the SAP CPQ target system. Otherwise, an error will be thrown and the group members will not be updated.
> - On the attempt to delete a group on SAP CPQ, Identity Provisioning will only remove its members (group assignments). And this can happen only if the relevant group assignments have been provisioned/are present in the target system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP CPQ* as a target system. For more information, see Add New Systems [page 2415].

3. Set up the communication between Identity Provisioning and SAP CPQ and configure the authentication method (basic or OAuth certificate-based).

> **Note**
>
> We recommend that you use OAuth certificate-based authentication.

   a. In your newly added SAP CPQ target system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection [page 2452].

      Skip this step if you use basic authentication. The next steps are relevant for OAuth certificate-based authentication only. For more information, see Configure OAuth Certificate Authentication [page 2456].

   b. Convert your certificate file from `.crt` to `.cer`.

   c. Login to SAP CPQ and navigate to *Security* > *Certificate Management* > *User Certificates* > *New Certificate* and upload the certificate to the technical user of SAP CPQ.

      Make sure the *Enable User Certificates* option is turned on. Additionally, the technical user must be set to *Active*.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`ips.delete.existedbefore.entities`	Enter: *true* SAP CPQ API does not allow creation and deletion of groups. Thus, if a previously read group is later deleted from the source system, this property will delete the relevant members from SAP CPQ (not the group itself).
`ips.delete.threshold.groups`	(Optional) Use this property to control the number of group assignments to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of group assignments, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
`ips.delete.threshold.users`	(Optional) Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Property Name	Value
`OAuth2TokenServiceURL`	Valid if *ClientCertificateAuthentication* is configured as authentication method. Enter the OAuth 2.0 Token Service URL. For example: `https://<cpq_url>/oauth2/token`
`Password`	Valid if *BasicAuthentication* is configured as authentication method. (Credential) Specify the password for your technical user.
`ProxyType`	Enter: *Internet*
`Type`	Enter: *HTTP*
`URL`	Enter the URL to the API of your SAP CPQ system. It is the same as your SAP CPQ tenant URL. It must contain the domain name. For example: **https://sample1234.mycpqdomain.com**
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the Username of your SAP CPQ technical user.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Identity Provisioning offers a default transformation for the *SAP CPQ* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP CPQ entity. You can change the default transformation mapping rules to reflect your current setup of entities in your SAP CPQ. For more information, see:

   Manage Transformations [page 2439]

   SAP CPQ: SCIM API

   Default transformation:

   Code Syntax

```
{
    "user": {
        "condition": "isValidEmail($.emails[0].value) &&
(('%cpq.group.prefix%' === 'null') || ($.groups[?(@.display =~ /
%cpq.group.prefix%.*/)] empty false))",
        "mappings": [
            {
                "targetPath": "$.id",
                "sourceVariable": "entityIdTargetSystem"
            },
```

```
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
                "targetPath": "$.schemas[0]"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
                "targetPath": "$.schemas[1]"
            },
            {
                "sourcePath": "$.externalId",
                "optional": true,
                "targetPath": "$.externalId"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails"
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.addresses",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.addresses"
            },
            {
                "sourcePath": "$.phoneNumbers",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.phoneNumbers"
            },
            {
                "scope": "createEntity",
                "constant": [],
                "targetPath": "$.groups"
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']"
            },
            {
                "constant": true,
                "targetPath": "$
['urn:sap:cpq:scim:schemas:extension:custom:2.0:User']['IsSsoUser']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
```

```
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            }
        ]
    },
    "group": {
        "condition": "isAttributeWithOptionalPrefix($.displayName,
cpq.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
cpq.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
        "mappings": [
            {
                "targetPath": "$.id",
                "sourceVariable": "entityIdTargetSystem"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "isAttributeWithMandatoryPrefix(@,
cpq.group.prefix)",
                        "function": "replaceFirstString",
                        "regex": "%cpq.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
Powered by Gitiles
```

6. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.23 SAP Data Custodian

Follow this procedure to set up SAP Data Custodian as a target system.

Prerequisites

> ⚠ Restriction
>
> This system is available for all standalone tenants and bundle tenants running on SAP Cloud Identity Services infrastructure. Bundle tenants running on Neo environment can use it only through **SAP Identity Access Governance** bundle option.

- You have created an SAP Data Custodian tenant.
- You have created a user within your tenant with the required roles for your scenario.
- You have added your user to SAP Identity Service Management (SAP ISM).

Context

SAP Data Custodian is a robust Software as a Service (SaaS) solution that protects sensitive data stored in public, private, hybrid, and multicloud environments. This solution integrates with partnered public hyperscalers, SAP applications, and SAP managed clouds.

After fulfilling the prerequisites, follow the procedure below to create a target SAP Data Custodian system to provision users and groups.

These target systems consume SCIM 2.0 API provided by SAP Data Custodian.

SCIM API 2.0 does not support managing of group assignments via the SCIM user resource. The "groups" attribute of the user is read-only. This means that the user group assignments should be managed via the SCIM group resources using the "members" attribute (as it is defined by the SCIM standard).

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Data Custodian* as a target system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP Data Custodian and configure the authentication method (basic or certificate-based).

   > ⓘ Note
   >
   > We recommend that you use certificate-based authentication.

a. In your newly added SAP Data Custodian source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

   Skip this step if you use basic authentication. The next steps are performed in SAP Data Custodian and are relevant for certificate-based authentication only.

b. Login to SAP Data Custodian tenant and create a tenant operations technical user (TOTU), as described in Create a Tenant Operations Technical User in SAP Data Custodian.

c. Upload the outbound certificate that you downloaded in **substep a.** to the TOTU, as described in Register the Outbound Certificate to Your Tenant Operations Technical User.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the SAP Data Custodian SCIM API excluding the path information. `https://<SCIM_API_URL>` For more information, see Identity Provisioning.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* Enter the OAuth client key, created for your SAP Data Custodian tenant. For more information, see Basic Authentication Prerequisites.
`Password`	Valid if *BasicAuthentication* (Credential) Enter the OAuth client secret, created for your SAP Data Custodian tenant. For more information, see Certificate Authentication Prerequisites.
`OAuth2TokenServiceURL`	Specify the URL of the access token provider service for your SAP Data Custodian instance: `https://<SCIM_API_URL>/kms/scim/v1/Oauth2/Token`

Property Name	Description & Value
`dc.group.prefix`	This property distinguishes SAP Data Custodian groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **DC_** You can use the example value or provide your own. When **set in the target system**, only groups containing the **DC_** prefix in their display name will be provisioned to SAP Data Custodian. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be be provisioned to SAP Data Custodian.
`dc.group.unique.attribute`	If the service tries to create a group that already exists in the target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for the `dc.group.unique.attribute` property. If the property is not specified, the search is done by the default attribute: *displayName*
`dc.user.unique.attribute`	When Identity Provisioning attempts to provision a user for the first time, it may detect that this user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user will be searched and resolved. If the service finds a user on the target system via this filter, then the conflicting user will overwrite the existing one. If the service does not find a user on the target system via this filter, the creation will fail. **Default behavior**: The property is missing during system creation. Its default value is *userName*. This means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such *userName* is not found, the creation of the conflicting user fails. **Possible values:** Default value: *userName*

Property Name	Description & Value
`dc.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users in the target system. This means, for example, that if a user attribute is modified, only this change will be provisioned and applied in the target system. • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false*
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Data Custodian* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Jam Collaboration. For more information, see:

   Manage Transformations [page 2439]

   SAP Data Custodian SCIM 2.0 API

   To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

   Default transformation:

   Code Syntax

```
{
    "user": {
        "condition": "isValidEmail($.emails[0].value) &&
(('%dc.group.prefix%' === 'null') || ($.groups[?(@.display =~ /
%dc.group.prefix%.*/)] empty false))",
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:User",

"urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
                    "urn:ietf:params:scim:schemas:extension:sap:2.0:User",
                    "urn:sap:cloud:scim:schemas:extension:custom:2.0:User"
                ],
                "targetPath": "$.schemas"
            },
            {
                "condition": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid'] EMPTY
false",
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "targetPath": "$.externalId"
            },
            {
                "condition": "$.externalId EMPTY false",
                "sourcePath": "$.externalId",
                "targetPath": "$.externalId"
            },
            {
                "condition": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['userId'] EMPTY
false",
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['userId']",
                "targetPath": "$.externalId"
            },
            {
                "condition": "$.emails[?(@.primary == true)].value == []",
```

```
                "sourcePath": "$.emails[0].value",
                "targetPath": "$.userName"
            },
            {
                "condition": "$.emails[?(@.primary == true)].value != []",
                "sourcePath": "$.emails[?(@.primary == true)].value",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.userName",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "optional": true,
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.name.familyName",
                "optional": true,
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.name.formatted",
                "optional": true,
                "targetPath": "$.name.formatted"
            },
            {
                "sourcePath": "$.name.honorificPrefix",
                "optional": true,
                "targetPath": "$.name.honorificPrefix"
            },
            {
                "sourcePath": "$.name.honorificSuffix",
                "optional": true,
                "targetPath": "$.name.honorificSuffix"
            },
            {
                "sourcePath": "$.name.middleName",
                "optional": true,
                "targetPath": "$.name.middleName"
            },
            {
                "sourcePath": "$.active",
                "optional": true,
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails"
            }
        ]
    },
    "group": {
        "condition": "isAttributeWithOptionalPrefix($.displayName,
dc.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
dc.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
        "mappings": [
```

```
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "isAttributeWithMandatoryPrefix(@,
dc.group.prefix)",
                        "function": "replaceFirstString",
                        "regex": "%dc.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

6. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

> ### Note
>
> When Identity Authentication is configured as a source system, the default transformation logic:
>
> - Provisions only groups and user group assignments if they are part of the following predefined group list:
>   - *SAP_Data_Custodian_Auditor*
>   - *SAP_Data_Custodian_Service_Admin*
>   - *SAP_Data_Custodian_Key_Admin*
>   - *SAP_Data_Custodian_Key_User*
> - Skips some of the attributes from the identity records.
> - Sets primary email for `userName` of the user.
>
> This way, the transformation logic ensures that the identity data, sent to the Identity Provisioning SCIM REST API, is consistent.

Next Steps

1. Before starting a provisioning job, you can first subscribe to the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during your jobs. For more information, see Manage Job Notifications [page 3018].
2. Now, start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Data Custodian

1.6.2.24 SAP Datasphere

Follow this procedure to set up SAP Datasphere as a target system.

Prerequisites

- In SAP Datasphere, you have enabled a custom SAML Identity Provider, for which *User Attribute* is set to **Custom SAML User Mapping**. To learn how, see: Enabling a Custom SAML Identity Provider
- In SAP Datasphere, you have added an OAuth client with authorization grant **Client Credentials**. To learn how, see: Create OAuth2.0 Clients to Authenticate Against SAP Datasphere

Context

SAP Datasphere is a solution that harmonizes data management, analytics, and intelligent planning, including capabilities of SAP Data Intelligence Cloud and SAP Analytics Cloud. The solution enables SAP customers to consolidate their SAP data and analytics needs into a single SAP solution.

You can use Identity Provisioning to configure SAP Datasphere as a target system where you can provision users. The target system consumes SCIM 2.0 API provided by SAP Datasphere.

> **Note**
>
> SAP Datasphere does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Datasphere* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP Datasphere system without adding the path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the client ID to retrieve the OAuth access token for SAP Datasphere.
`Password`	(Credential) Enter the client secret to retrieve the OAuth access token for SAP Datasphere.
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Datasphere instance. This token URL is listed in the *Trusted Identity Providers* section of the *App Integration* page. For more information, see Create OAuth2.0 Clients to Authenticate Against SAP Datasphere
`ds.api.csrf.protection`	Specifies whether to fetch a CSRF token when sending requests to the system. This property is automatically added to the system, with default value: **enabled**

Property Name	Description & Value
`ds.user.unique.attribute`	This property appears by default when the system is created, and its value is set to *userName*. It defines by which unique attribute(s) an existing user to be resolved in the event of conflicting users. Other possible values: • **emails[0].value** • **userName,emails[0].value** • **externalId**, or another SCIM attribute, or a conjunction of SCIM attributes **Note** If this property is missing, or you've deleted it, and the service does not find such a *userName*, it will try again to resolve the conflicting user – by *email*. If the second attempt for resolution is unsuccessful too, the creation of the conflicting user fails. For more information, see: List of Properties [page 80]
`ds.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a PATCH request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** • *true* • *false* Default value: *false*

Property Name	Description & Value
(Optional) `ds.support.bulk.operation`	Set this property to *true* if you want to enable SCIM bulk operations for provisioning users. That means, the Identity Provisioning service can write, update, and delete a potentially large collection of users in a single request. For more information, see: SCIM Protocol: Bulk Operations and Bulk Operations If not specified, the default value is *false*.
(Optional) `ds.bulk.operations.max.count`	If you have enabled the SCIM bulk operations, you can use this property to set the number of users to be provisioned per request. Default value: **30** **Note** The value must not exceed the number of entities defined by the SAP Datasphere system as a SCIM service provider. Otherwise, the provisioning job will fail with HTTP response code 413 (*Payload Too Large*).
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Datasphere* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Datasphere system. For more information, see: Manage Transformations [page 2439]

   SAP Datasphere SCIM 2.0 API

   Default transformation:

   Code Syntax

```
{
  "user": {
    "condition": "($.emails[0].value EMPTY false) &&
isValidEmail($.emails[0].value) && ($.userName EMPTY false)",
    "mappings": [
```

```
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:User",
          "urn:sap:params:scim:schemas:extension:sac:2.0:user-custom-
parameters",
          "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.displayName",
        "optional": true,
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.externalId",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.preferredLanguage",
        "optional": true,
        "targetPath": "$.preferredLanguage"
      },
      {
        "sourcePath": "$.photos",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.photos",
        "optional": true
      },
      {
        "sourcePath": "$.roles",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.roles"
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.emails"
      },
      {
        "condition": "$.emails[?(@.primary == true)].value == []",
        "sourcePath": "$.emails[0].value",
        "targetPath": "$.emails[0].value"
```

```
      },
      {
        "condition": "$.emails[?(@.primary == true)].value != []",
        "sourcePath": "$.emails[?(@.primary == true)].value",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.emails[0].value",
        "functions": [
          {
            "function": "elementAt",
            "index": 0
          }
        ]
      },
      {
        "condition": "$.emails[0].length() > 0",
        "constant": true,
        "targetPath": "$.emails[0].primary"
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:user-custom-parameters']",
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:extension:sac:2.0:user-custom-parameters']"
      },
      {
        "sourcePath": "$.emails[0].value",
        "optional": true,

"targetPath": "$['urn:sap:params:scim:schemas:extension:sac:2.0:user-
custom-parameters']['idpUserId']"
      },
      {
        "condition": "$.emails[?(@.primary == true)].value != []",
        "sourcePath": "$.emails[?(@.primary == true)].value",
        "preserveArrayWithSingleElement": false,
        "optional": true,

"targetPath": "$['urn:sap:params:scim:schemas:extension:sac:2.0:user-
custom-parameters']['idpUserId']"
      },
      {

"sourcePath": "$['urn:sap:params:scim:schemas:extension:sac:2.0:user-
custom-parameters']['idpUserId']",
        "optional": true,

"targetPath": "$['urn:sap:params:scim:schemas:extension:sac:2.0:user-
custom-parameters']['idpUserId']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "functions": [
          {
            "type": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

> **Note**
>
> Updating a user in SAP Datasphere depends on whether user attributes in SAP Datasphere are mapped to SAML attributes in your identity provider. If this is the case, the values of those attributes are populated by the identity provider and cannot be changed by the SAP Datasphere SCIM API or the UI. For more information, see Map SAML Attributes to Users and Create a User.

5. Now, add a source system from which to read users. Choose from: Source Systems [page 585]

> **Caution**
>
> The **email** attribute is unique for *SAP Datasphere* but it might not be unique for other systems, such as *SAP SuccessFactors*. That's why, when you choose a source system, make sure that there are no users with duplicate e-mails. If there are such, then after the provisioning job all users with the same e-mail address will be created as a single user in *SAP Datasphere*.
>
> To learn more, see *Guided Answers*: Multiple Users from a Source System Are Created as One in the Target
>
> To learn more about excluding users from SAP Datasphere target, see: Exclude Users from Provisioning to Target Systems

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.25 SAP Document Center

Follow this procedure to set up SAP Document Center as a target system.

Prerequisites

- You have an SAP Business Technology Platform user with administration rights for the tenant.
- You have enabled the SAP Document Center service in the cockpit.

Context

SAP Document Center offers programs (apps) that can be downloaded and run on multiple independent devices. For more information, see SAP Document Center.

It plays the role of a content service for your SAP Business Technology Platform subaccount. To use it as a target system for writing users, follow the procedure below.

Procedure

1. Assign your SAP Business Technology Platform user **admin** rights for SAP Document Center. To do this, open the *SAP Document Center* service tile (in the cockpit), open link *Assign Roles & Set Destinations*, choose *Administrator*, and then – *Assign*.
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
3. Add *SAP Document Center* as a target system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: **HTTP**
`URL`	Enter the URL, generated in the cockpit for your subaccount in the *SAP Document Center* tile. You can take this URL from the *Configure SAP Document Center* link. Remove the last slash after "…**/admin"**.
`ProxyType`	Enter: **Internet**
`Authentication`	Enter: **BasicAuthentication**
`User`	Enter your SAP Business Technology Platform user (with administrator rights).
`Password`	(Credential) Enter the password for your SAP Business Technology Platform user.
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Property Name	Description & Value
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Document Center* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Document Center. For more information, see:

   Manage Transformations [page 2439]

   SAP Document Center: REST API

   Default transformation:

   Code Syntax

```
{
    "user": {
        "condition": "$.userName EMPTY false",
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.firstName"
            },
            {
                "sourcePath": "$.name.familyName",
                "optional": true,
                "targetPath": "$.lastName"
            },
            {
                "sourcePath": "$.emails[0].value",
                "optional": true,
                "targetPath": "$.email"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.logonId"
            }
        ]
    }
}
```

6. Now, add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.26 SAP Enable Now

Follow this procedure to set up SAP Enable Now as a target system.

Prerequisites

- You have created a technical user in SAP Enable Now with the necessary permissions for establishing a connection with Identity Provisioning.

Context

The SAP Enable Now solution is SAP's digital adoption platform. It can help you improve the adoption, productivity, and efficiency of your software by providing in-application help and e-learning content to users.

You can use Identity Provisioning to configure SAP Enable Now as a target system where you can provision users and groups. The target system consumes SCIM 2.0 API provided by SAP Enable Now.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Enable Now* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP Enable Now system. It follows the pattern: • `https://<company-name>.enable-now.cloud.sap` • `https://<company-name>.sap-enable-now.cn` (relevant for China)
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the username of the SAP Enable Now technical user.
`Password`	(Credential) Enter the password of the SAP Enable Now technical user.
`en.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. Default value: *userName*
`en.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. Default value: *false*

Property Name	Description & Value
(Optional) `en.group.prefix`	This property distinguishes SAP Enable Now groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **EN_** You can use the example value or provide your own. When **set in the target system**, only groups containing the **EN_** prefix in their display name will be provisioned to SAP Enable Now. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be be provisioned to SAP Enable Now.
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Enable Now* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Enable Now system. For more information, see: Manage Transformations [page 2439]

   Default transformation:

   Code Syntax

```
{
    "user": {
        "condition": "($.userName EMPTY false) &&
isValidEmail($.emails[0].value) && (('%en.group.prefix%' === 'null') ||
($.groups[?(@.display =~ /%en.group.prefix%.*/)] empty false))",
        "mappings": [
            {
                "targetPath": "$.id",
                "sourceVariable": "entityIdTargetSystem"
            },
            {
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:User",
                    "urn:ietf:params:scim:schemas:extension:sap:2.0:User"
                ],
                "targetPath": "$.schemas"
            },
            {
```

```
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            },
            {
              "sourcePath": "$.name.givenName",
              "optional": true,
              "targetPath": "$.name.givenName"
            },
            {
              "sourcePath": "$.name.middleName",
              "optional": true,
              "targetPath": "$.name.middleName"
            },
            {
              "sourcePath": "$.name.familyName",
              "optional": true,
              "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "optional": true
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "preserveArrayWithSingleElement": true
            }
        ]
    },
    "group": {
        "condition": "isAttributeWithOptionalPrefix($.displayName,
en.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
en.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
        "mappings": [
            {
                "targetPath": "$.id",
                "sourceVariable": "entityIdTargetSystem"
            },
            {
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:Group",
                    "urn:ietf:params:scim:schemas:extension:sap:2.0:Group"
                ],
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "isAttributeWithMandatoryPrefix(@,
en.group.prefix)",
                        "function": "replaceFirstString",
                        "regex": "%en.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
```

```
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
                "optional": true
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

5. Now, add a source system from which to read users. Choose from: Source Systems [page 585]

> ⚠ Caution
>
> The **email** attribute is unique for *SAP Enable Now* but it might not be unique for other systems, such as *SAP SuccessFactors*. That's why, when you choose a source system, make sure that there are no users with duplicate e-mails. If there are such, then after the provisioning job all users with the same e-mail address will be created as a single user in *SAP Enable Now*.
>
> To learn more, see *Guided Answers*: Multiple Users from a Source System Are Created as One in the Target
>
> To learn more about excluding users from SAP Enable Now target, see: Exclude Users from Provisioning to Target Systems

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Enable Now

1.6.2.27 SAP Fieldglass

Follow this procedure to set up a target connector for SAP Fieldglass.

Prerequisites

You have created an API application key and a web service. To do that, follow the steps on page: Create API Application Key or Web Service and Web Services Setup

You will need the values of *Virtual Person Name (Username)* and *License Key* for the configuration of your target system (**step 3** below).

Context

After fulfilling the prerequisites, follow the procedure below to add a target system for SAP Fieldglass to write users to it. This target system consumes SCIM 2.0 API provided by SAP Fieldglass.

> ⓘ Note
>
> The system transformation supports groups but cannot directly create new groups. However, you can map groups from the source system to groups from SAP Fieldglass by display name.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Fieldglass* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify your SAP Fieldglass environment URL. For example: *https://abc123.fgvms.com*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*

Property Name	Value
`User`	Enter your *Virtual Person Name (Username)* – see the **Prerequisites** section above.
`Password`	(Credential) Enter your *License Key* – see the **Prerequisites** section above.
`OAuth2TokenServiceURL`	Enter your OAuth token URL in the following format: `https://<Environment_URL>/api/oauth2/v2.0/token` For example: *https://abc123.fgvms.com/api/oauth2/v2.0/token*
(Optional) `fg.group.prefix`	This property distinguishes SAP Fieldglass groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **FG_** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Fieldglass source system and will be provisioned to the target system with the following name pattern: **FG_<GroupDisplayName>**. This way SAP Fieldglass groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Fieldglass groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **FG_** prefix in their display name will be provisioned to SAP Fieldglass. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be be provisioned to SAP Fieldglass.
`fg.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users in the target system. • If set to *false*, PUT operations are used to update users in the target system. Default value: *false* For more information, see: List of Properties [page 80]

Property Name	Value
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Fieldglass* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Fieldglass. For more information, see:

   Manage Transformations [page 2439]

   SAP Business Accelerator Hub: SAP Fieldglass

   - **Mapping logic** – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target entity. If the entity has e-mail addresses, the first entry will be marked as primary.
   - **User off-boarding** – Users can be deleted from the target system. Depending on the implementation, this could be done through a user interface (if such exists) or the SCIM REST API. Users could be deactivated, depending on the SAP Fieldglass system implementation. The SCIM core schema defines an attribute "**active**", whose definition depends on the service provider. For more information, see SCIM: Singular Attributes

   Default transformation:

   Code Syntax

```
{
  "user": {
    "condition": "('%fg.group.prefix%' === 'null') || ($.groups[?
(@.display =~ /%fg.group.prefix%.*/)] empty false)",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "scope": "createEntity",
```

```
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.formatted",
        "optional": true,
        "targetPath": "$.name.formatted"
      },
      {
        "sourcePath": "$.name.honorificPrefix",
        "targetPath": "$.name.honorificPrefix",
        "condition": "$.name.honorificPrefix IN ['Mr.', 'Mrs.', 'Ms.',
'Dr.']",
        "optional": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.title",
        "targetPath": "$.title",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.emails[0].value",
        "targetPath": "$.emails[0].value"
      },
      {
        "optional": true,
        "defaultValue": "work",
        "sourcePath": "$.emails[0].type",
        "targetPath": "$.emails[0].type"
      },
      {
        "defaultValue": true,
        "optional": true,
        "sourcePath": "$.emails[0].primary",
        "targetPath": "$.emails[0].primary"
      },
      {
        "sourcePath": "$.timezone",
```

```
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.addresses",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.addresses"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
        "targetPath": "$.schemas[0]"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
        "targetPath": "$.schemas[1]"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "scope": "createEntity",
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "optional": true
      },
      {
        "scope": "createEntity",
```

```
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      }
    ]
  },
// You cannot create groups but instead, you can map groups from the
source system to groups from SAP Fieldglass by display name.
// The group deletion operation is skipped.
  "group": {
    "skipOperations": [
      "delete"
    ],
     "condition": "(isAttributeWithOptionalPrefix($.displayName,
fg.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
fg.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup()))",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
            {
            "condition": "isAttributeWithMandatoryPrefix(@,
fg.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%fg.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "sourcePath": "$.members[*].value",
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",
```

```
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.schemas"
      }
    ]
  }
}
```

5. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.28 SAP Field Service Management

Follow this procedure to set up SAP Field Service Management as a target system.

Prerequisites

You have OAuth credentials for SAP Field Service Management. For more information, see: Generating Client ID & Secret

Context

SAP Field Service Management is a cloud-based solution that is used to resolve customers issues with end-to-end field service management. For example, it helps customers overcome resource limitations, such as having enough skilled technicians in all locations.

You can use the Identity Provisioning user interface (UI) to configure SAP Field Service Management as a target system where you can provision users and group members. When integrated with Identity Provisioning, the cloud solution supports the following group concept:

- A user is created with a default group assigned. A group is mapped to a company (one of the key organizational units in SAP Field Service Management). When using the Identity Provisioning service API, the group is returned as follows:

  ```
  <GroupName>_<CompanyName>
  ```

- A user can have one group assignment for each company. As in SAP Field Service Management a user can be assigned to multiple companies, this requires a group assignment for each of the companies the user can access. A user cannot be assigned to different groups mapped to one and the same company.

> ⚠ Caution
>
> You cannot create or delete groups in SAP Field Service Management. In this case, you can expect the following behavior:
>
> - If a new group is created in the source system (for example, Identity Authentication) and you run the provisioning job to SAP Field Service Management, the job will fail and no group will be created in the target system.
> - If a group exists both in the source system (for example, Identity Authentication) and the target system - SAP Field Service Management, running a provisioning job will only add new members or update existing ones. In this case, groups in the source and target systems must have the same display name (case sensitive). Otherwise, the job will fail, and no group members will be updated.
> - If a group exists in the target SAP Field Service Management system and you try to delete it, Identity Provisioning will only remove its group members. In this case, the relevant group members must exist in the target system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Field Service Management* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the API of your SAP Field Service Management system. It follows the pattern: `https://<cluster>.fsm.cloud.sap`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, created for your SAP Field Service Management system.

Property Name	Value
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP Field Service Management system.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. For example: `https://<cluster>.fsm.cloud.sap/api/oauth2/v2/token`
(Optional) `fsm.group.prefix`	This property distinguishes SAP Field Service Management groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`FSM_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the groups that are read from the SAP Field Service Management source system and will be provisioned to the target system with the following name pattern: **`FSM_<GroupDisplayName>`**. This way SAP Field Service Management groups in the target system will be distinguished from groups provisioned from other applications. If the property is not set, the SAP Field Service Management groups will be read and provisioned to the target system with their actual display names. • When **set in the target system**, only groups containing the **`FSM_`** prefix in their display name will be provisioned to SAP Field Service Management. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be be provisioned to SAP Field Service Management.
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of group assignments to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of group assignments, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Property Name	Value
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Field Service Management* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Field Service Management system. For more information, see:

   Manage Transformations [page 2439].

   Field Service Management - SCIM API

   To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

   Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP Field Service Management entity.

   Default transformation:

   Code Syntax

```
{
    "user": {
        "condition": "($.emails EMPTY false) && ($.userName EMPTY false)
&& isValidEmail($.emails[0].value) && (('%fsm.group.prefix%' === 'null')
|| ($.groups[?(@.display =~ /%fsm.group.prefix%.*/)] empty false))",
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:User",
                    "urn:ietf:params:scim:schemas:extension:sap:2.0:User"
                ],
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$.name.givenName",
```

```
                "targetPath": "$.name.givenName"
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active"
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            }
        ]
    },
    "group": {
        "condition": "(isAttributeWithOptionalPrefix($.displayName,
fsm.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
fsm.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup()))
|| ('%fsm.group.prefix%' === 'null') || ($.displayName =~ /
%fsm.group.prefix%.*/)",
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "('%fsm.group.prefix%' !== 'null') &&
(@ =~ /%fsm.group.prefix%.*/)",
                        "function": "replaceFirstString",
                        "regex": "%fsm.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

5. Add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.29 SAP HANA Cloud Database Services

Follow this procedure to set up SAP HANA Cloud Database Services as a target system.

Prerequisites

You have created a technical user for Identity Provisioning in the SAP HANA database with the OPERATOR and ROLE ADMIN system privileges. For more information, see SAP Identity Provisioning Service → *Prerequisites*.

Context

SAP HANA Cloud allows you to consume the SAP HANA database from cloud applications running on SAP Business Technology Platform, as well as from applications running elsewhere using the standard SAP HANA clients. Every instance of SAP HANA Cloud has its own single SAP HANA database.

You can use the Identity Provisioning user interface (UI) to configure SAP HANA Cloud Database Services as a target system where you can provision users and group members, that is, user assignments for a role.

The SAP HANA Cloud Database Services connector supports provisioning of entities to the following two database types:

- SAP HANA Database (HDB)
- SAP HANA Data Lake Relational Engine (HDLRE / IQ Database)

You can define the target database type by setting the `hana.cloud.db.instance.type` property to either `hdb` or `hdlre`.

For more information on the supported database types, see SAP HANA Cloud, SAP HANA Database and SAP HANA Cloud, Data Lake.

> **Note**
>
> In SAP HANA Cloud Database Services, groups correspond to roles.

> ⚠ Caution
>
> Identity Provisioning cannot create or delete roles in SAP HANA Cloud Database Services. It can only create, update, and delete user assignments for a role. Therefore, roles must be created in the SAP HANA Cloud Database Services system before running a provisioning job. Additionally, groups in the source system and roles in the target system must have identical names.

Review the expected behavior in the following scenarios:

- If a new group is created in the source system (for example, Identity Authentication) and you run the provisioning job to SAP HANA Cloud Database Services, the job will fail and no role will be created in the target system.
- If a group in the source system (Identity Authentication) and a role in the target system (SAP HANA Cloud Database Services) exist, when running a provisioning job, the user group assignments from the source will be mapped to the user role assignments in the target, provided that the group and role names are identical. If the names do not match, the job will fail, and no role assignments will be updated. You have the following options to handle these discrepancies: use the `valueMappings` expression to change the group names, as described in Transformation Expressions [page 391], or dynamically assign users to groups in SAP HANA Cloud Database Services, as described in Enabling Group Assignment [page 2445].
- If a role exists in the target SAP HANA Cloud Database Services system and you try to delete it, Identity Provisioning will only remove the existing user assignments for that role.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP HANA Cloud Database Services* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Authentication`	Enter: *BasicAuthentication*
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. For example: `https://demo-hc-3-test.authentication.sap.hana.ondemand.com/oauth/token` For more information on how to obtain the value, refer to the create a service binding step in the User Provisioning with Identity Provisioning Service.

Property Name	Value
`hana.cloud.db.instance.type`	Refers to the database type configured or used within SAP HANA Cloud. The value can be set to: • **hdb** - SAP HANA Database • **hdlre** - SAP HANA Data Lake Relational Engine Default value: **hdb**
`hana.cloud.db.instance.id`	Refers to a unique identifier associated with a specific database instance within the SAP HANA Cloud.
`hana.cloud.db.providerName`	Specifies the identity provider (IdP) that users managed through Identity Provisioning will use for authentication. The provider must be configured to support the authentication method specified in `hana.cloud.db.authenticationType`. For more information on how to obtain the value, refer to the create a JWT provider step in the User Provisioning with Identity Provisioning Service.
`hana.cloud.db.authenticationType`	Specifies the method by which users managed through Identity Provisioning authenticate when connecting to the database. The authentication method is set to *JWT* (users authenticate via JWT tokens issued by an identity provider) during system creation.

Property Name	Value
`hana.cloud.db.group.prefix`	(Optional) This property distinguishes SAP HANA Cloud Database Services roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`HANA_Cloud_DB_`** You can use the example value or provide your own. • When **set in the source system**, the prefix will be prepended to the name of the roles that are read from the SAP HANA Cloud Database Services source system and will be provisioned to the target system with the following name pattern: **`HANA_Cloud_DB_<role_name>`**. This way SAP HANA Cloud Database Services roles in the target system will be distinguished from roles provisioned from other applications. If the property is not set, the SAP HANA Cloud Database Services roles will be read and provisioned to the target system with their actual names. • When **set in the target system**, only roles containing the **`HANA_Cloud_DB_`** prefix in their role name will be provisioned to SAP HANA Cloud Database Services. Roles without this prefix in the role name won't be provisioned. If the property is not set, all roles will be be provisioned to SAP HANA Cloud Database Services.
`hana.cloud.db.user.unique.attribute`	This property defines by which unique attribute(s) an existing user in the target system will be searched and resolved in case Identity Provisioning tries to provision a conflicting user. Default value: *userName*
`ips.delete.threshold.roles`	(Optional) Use this property to control the number of role assignments to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of role assignments, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Property Name	Value
`ips.delete.threshold.users`	(Optional) Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP HANA Database system. For more information on how to obtain the value for the technical user, refer to the procedure in the User Provisioning with Identity Provisioning Service.
`ProxyType`	Enter: *Internet*
`Type`	Enter: *HTTP*
`URL`	Enter the SAP HANA Cloud Database Services API URL. For example: `https://api.gateway.orchestration.<cluster>-<datacenter>.hanacloud.ondemand.com`
`User`	Enter the OAuth Client Id, created for your SAP HANA Database system. For more information on how to obtain the value for the technical user, refer to the procedure in the User Provisioning with Identity Provisioning Service.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP HANA Cloud Database Services* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP HANA Cloud Database Services system. For more information, see:

   Manage Transformations [page 2439].

   > ### Note
   >
   > Given that the SAP HANA Database username attribute is immutable and the email is not supported, refer to Handling Specific Attributes for your provisioning scenarios, particularly with Identity Authentication.

Default transformation:

Code Syntax

```
{
  "user": {
    "condition": "('%hana.cloud.db.group.prefix%' === 'null') ||
($.groups[?(@.display =~ /%hana.cloud.db.group.prefix%.*/)] empty false)",
    "mappings": [
      {
        "targetPath": "$.id",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:User",
          "urn:ietf:params:scim:schemas:extension:sap:2.0:User",
          "urn:ietf:params:scim:schemas:extension:sap.hana.db:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userId']",
        "optional": true
      },
      {
        "condition": "'%hana.cloud.db.providerName%' !== 'null'",
        "constant": "%hana.cloud.db.providerName%",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap.hana.db:2.0:User']
['authenticationDetails']['providerName']"
      },
      {
        "condition": "'%hana.cloud.db.authenticationType%' !== 'null'",
        "constant": "%hana.cloud.db.authenticationType%",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:sap.hana.db:2.0:User']
['authenticationDetails']['authenticationType']"
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
hana.cloud.db.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
hana.cloud.db.group.prefix) && (isRegularGroup() ||
isApplicationSpecificGroup())",
```

```
    "mappings": [
      {
        "targetPath": "$.id",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
hana.cloud.db.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%hana.cloud.db.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

5. Add a source system from which to read users and user assignments for a role. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

User Provisioning

1.6.2.30 SAP HANA Database

Follow this procedure to set up SAP HANA Database as a target system.

Prerequisites

- You have credentials for a tenant in SAP Business Technology Platform. For more information, see: Accounts
- You have the necessary connection settings to reach an SAP HANA database.
- (Optional) You have installed the Cloud Connector in your corporate environment and have done the initial configuration. You need this only when your SAP HANA DB resides in a remote on-premise system, outside your Neo environment. For more information, see Cloud Connector.

Context

> **Note**
>
> SAP HANA Database is only available as a target system.

SAP HANA Database is a target system (connector), which allows you to log into remote systems that have SAP HANA installed. Only provisioning of entity type **user** is currently supported by this connector. That includes user assignments to roles and all types of catalog and repository privileges (*schema*, *analytic*, *application*). For more information about SAP HANA privileges, see:

SAP HANA: GRANT Statement (Access Control)

SAP HANA: Stored Procedures Used to Grant/Revoke Privileges on Activated Repository Objects

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP HANA Database* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Below are listed all available SAP HANA properties. Some of them can be mandatory and others – optional, depending on your scenario.

Mandatory Properties

Property Name	Description & Value
`ProxyType`	Possible values: • **Internet** • **OnPremise** The proxy type **OnPremise** requires you to provide mappings to your host and port to the Cloud Connector.
(Optional) `Type`	Enter: **HTTP** **Note** This property is mandatory only when your **ProxyType** is **OnPremise**.
`hana.jdbc.db.user`	Name of the SAP HANA Database user
`hana.jdbc.db.password`	(Credential)
`hana.jdbc.db.host`	SAP HANA Database host
`hana.jdbc.db.port`	Default value: **30015**
`hana.jdbc.db.encrypt`	This property enables or disables TLS encryption. Possible values: • true (Default value) • false **Note** For existing systems, the property value defaults to false.
`hana.jdbc.db.validateCertificate`	When set to true, this property specifies that the server certificate is validated. When set to false, the certificate is not validated. Possible values: • true (Default value) • false
(Optional) `CloudConnectorLocationId`	Relevant when the proxy type is *OnPremise*. Use it only if your SAP Business Technology Platform account uses more than one Cloud Connector.

Property Name	Description & Value
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP HANA Database* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP HANA Database. For more information, see Manage Transformations [page 2439].

   Default transformation:

   Code Syntax

```
{
    "user": {
        "condition": "$.userName EMPTY false",
        "mappings": [
            {
                "sourcePath": "$.userName",
                "targetPath": "$.username"
            },
            {
                "targetPath": "$.password_option.password",
                "scope": "createEntity",
                "functions": [
                    {
                        "type": "randomPassword",
                        "passwordLength": 24,
                        "minimumNumberOfLowercaseLetters": 1,
                        "minimumNumberOfUppercaseLetters": 1,
                        "minimumNumberOfDigits": 1,
                        "minimumNumberOfSpecialSymbols": 0
                    }
                ]
            },
            {
                "ignore": true,
                "constant": true,
```

```
                "targetPath":
"$.password_option.no_force_first_password_change",
                "scope": "createEntity"
            },
            {
                "constant": true,
                "targetPath": "$.deactivate",
                "scope": "deleteEntity"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.username",
                "scope": "deleteEntity"
            },
            {
                "constant": false,
                "targetPath": "$.deactivate"
            },
            {
                "constant": true,
                "targetPath": "$.reset_connect_attempts"
            },
            {
                "ignore": true,
                "constant": true,
                "targetPath": "$.force_password_change"
            },
            {
                "ignore": true,
                "constant": true,
                "targetPath": "$.enable_password_lifetime"
            },
            {
                "ignore": true,
                "constant": true,
                "targetPath": "$.disable_client_connect"
            },
            {
                "constant": "NOW",
                "targetPath": "$.valid_from"
            },
            {
                "constant": "FOREVER",
                "targetPath": "$.valid_to"
            },
            {
                "ignore": true,
                "constant": "1970-01-01 00:00:00.0",
                "targetPath": "$.valid_from"
            },
            {
                "ignore": true,
                "constant": "1970-01-01 00:00:00.0",
                "targetPath": "$.valid_to"
            },
            {
                "ignore": true,
                "constant": "role",
                "targetPath": "$.catalog_permissions[0].type"
            },
            {
                "ignore": true,
                "constant": "MONITORING",
                "targetPath": "$.catalog_permissions[0].name"
            },
            {
                "ignore": true,
                "constant": "ADMIN",
```

```
                "targetPath": "$.catalog_permissions[0].option"
            },
            {
                "ignore": true,
                "constant": "object_privilege",
                "targetPath": "$.catalog_permissions[1].type"
            },
            {
                "ignore": true,
                "constant": "SELECT CDS METADATA",
                "targetPath": "$.catalog_permissions[1].name"
            },
            {
                "ignore": true,
                "constant": "SYS.USERS",
                "targetPath": "$.catalog_permissions[1].on"
            },
            {
                "ignore": true,
                "constant": "role",
                "targetPath": "$.repository_permissions[0].type"
            },
            {
                "ignore": true,
                "constant":
"sap.appcore.auth.p::select_ACCESS_VIEWS_BY_USER",
                "targetPath": "$.repository_permissions[0].name"
            },
            {
                "ignore": true,
                "constant": "application_privilege",
                "targetPath": "$.repository_permissions[1].type"
            },
            {
                "ignore": true,
                "constant": "sap.hana.ide::Catalog",
                "targetPath": "$.repository_permissions[1].name"
            },
            {
                "ignore": true,
                "constant": true,
                "targetPath": "$.repository_permissions[2].revoke"
            },
            {
                "ignore": true,
                "constant": "analytic_privilege",
                "targetPath": "$.repository_permissions[2].type"
            },
            {
                "ignore": true,
                "constant": "_SYS_BI_CP_ALL",
                "targetPath": "$.repository_permissions[2].name"
            }
        ]
    }
}
```

5. Now, add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.31 SAP Integrated Business Planning

Follow this procedure to set up SAP Integrated Business Planning (in short, SAP IBP) as a target system.

Prerequisites

To establish the connection between Identity Provisioning and SAP Integrated Business Planning, you need to set up the communication (user, system and arrangement) on SAP Integrated Business Planning. You can do it now (as a prerequisite) or in the process of configuring SAP Integrated Business Planning as a target system, as described in step 3.

Context

SAP Integrated Business Planning is a cloud-based solution that combines sales and operations planning (S&OP), forecasting and demand, response and supply, demand-driven replenishment, and inventory planning.

You can use Identity Provisioning to configure SAP IBP as a target system where you can provision entities from a particular source system. The Identity Provisioning service manages the complete set of **business partners** and their relevant **business users** (employee).

SAP Integrated Business Planning provides different APIs for integration with Identity Provisioning, resulting in different connector versions for each API. Each connector version supports specific attribute mappings within the transformations and requires particular property values. By default, the Identity Provisioning service uses API version *1*. The value of the `ibp.api.version` property controls which API you use.

- When the value is set to *1* or the property is not present (typical for systems created before versioning was introduced on November 6, 2025), the SOAP API provided by the communication scenario SAP_COM_0193 is used. It supports provisioning **business users** and **role assignments**. In SAP Integrated Business Planning version 1, groups correspond to roles, thus group members are user assignments to a role.

  > **Note**
  >
  > Identity Provisioning cannot create and delete roles in SAP Integrated Business Planning target system. It can only create, update and delete user assignments of a role. Therefore, roles must have been created in SAP Integrated Business Planning system before you run a provisioning job.

For more information on how to update to version *2*, see Update SAP Integrated Business Planning [page 2432].

- When the value is set to *2*, the SCIM interface provided by the communication scenario SAP_COM_0465 is used. It supports provisioning of **business users** with their **assignments** to user groups and roles.
  This version allows you to create, update, and delete user groups, referred to as business user groups in SAP Integrated Business Planning. However, similar to connector version 1, you cannot create or delete groups of type **authorization** (business roles in SAP Integrated Business Planning). You can only update the role assignments.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Integrated Business Planning* as a target system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP Integrated Business Planning and configure your authentication method (basic or certificate-based).

   > ### Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP Integrated Business Planning target system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP Integrated Business Planning backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Create a communication system and assign the created user to the communication system.

      For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

   d. Create a communication arrangement with the created system.

      - When `ibp.api.version=1` is set in Identity Provisioning for the SAP IBP target system, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

        > ### Note
        >
        > The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.

> The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

- When `ibp.api.version=2` is set in Identity Provisioning for the SAP IBP target system, choose *Scenario ID* SAP_COM_0465 (System for Cross-domain Identity Management Integration).

> **Note**
>
> The communication scenario *SAP_COM_0465* allows you to maintain data via SCIM (System for Cross-domain Identity Management). For more information, see Integrate Cross-Domain Identity Management.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to your SAP IBP system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **Restriction** Do not use special symbol ',' (comma) as it is not supported.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **Restriction** Do not use special symbol ',' (comma) as it is not supported.

Property Name	Description & Value
`ibp.api.version`	The API version which is consumed by the SAP Integrated Business Planning system. By default, Identity Provisioning uses version *1*.
`ips.date.variable.format`	*yyyy-MM-dd*

Property Name	Description & Value

`ibp.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP Integrated Business Planning target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved.
	
	The possible values depend on the API version which your SAP Integrated Business Planning system consumes.
	
	• If your system consumes SAP Integrated Business Planning API: Business User, the service searches for an existing user by *personExternalID* and *emails[0].value*.
	According to your use case, choose how to set up the property:
	  • Default behavior: This property does not appear in the UI during system creation. Its default value is *personExternalID*. That means, if the service finds an existing user by a *personExternalID* , it updates this user with the data of the conflicting one.
	If a user with such a *personExternalID* is not found, the creation of the conflicting user fails.
	  • Value = *emails[0].value*. If the service finds an existing user matching both unique attributes *email* and *personExternalID*, it updates this user with the data of the conflicting one. If the service finds an existing user matching only the *email*, the update of the existing user fails.
	If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails.
	• If your system consumes SCIM Interface for IAM, the service searches for an existing user by *userName* , *emails[0].value*, *externalId*, or another SCIM attribute, or a conjunction of SCIM attributes.
	According to your use case, choose how to set up the property:
	  • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails.

Property Name	Description & Value
	• Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *externalId*. If the service finds an existing user with *externalId* , it updates this user with the data of the conflicting one. If a user with such *externalId* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. For more information, see List of Properties [page 80].
`ibp.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP IBP target system. • **true** – the current user roles will be deleted in the target system, and the user will be updated only with the roles provisioned by the service. • **false** – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the target system. See also: Extended Explanation of the *.user.roles.overwrite Properties **Relevant for connector version 1**
(Optional) `ibp.roles.prefix`	This property distinguishes SAP Integrated Business Planning roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **IBP_** You can use the example value or provide your own. When **set in the target system**, only roles containing the **IBP_** prefix in their role name will be provisioned to SAP Integrated Business Planning. Roles without this prefix in the role name won't be provisioned. If the property is not set, all roles will be provisioned to SAP Integrated Business Planning. **Relevant for connector version 1**

Property Name	Description & Value
(Optional) `ibp.group.prefix`	This property distinguishes SAP Integrated Business Planning groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`IBP_`** You can use the example value or provide your own. When **set in the target system**, only groups containing the **`IBP_`** prefix in their display name will be provisioned to SAP Integrated Business Planning. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Integrated Business Planning. For more information, see List of Properties [page 80]. **Relevant for connector version 2**
`ibp.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists in the SAP Integrated Business Planning target system, the creation will fail. In this case, the existing group only needs to be updated. In SAP Integrated Business Planning the uniqueness of a group is defined by the combination of the following two attributes, which are automatically filled in when the target system is created: *externalId, ['urn:ietf:params:scim:schemas:extension:sap:2.0:Group'] ['type']*. To resolve the conflict, an existing group must match both of these attributes. If the group matches only one of the attributes, the conflict is not resolved, and group creation will fail. **Note** Although you can define your own unique combination of attributes (for example, *externalId,displayName*), we recommend that you do not modify the automatically filled value of the `ibp.group.unique.attribute` property. If the property is not specified, the search is done by the default attribute *displayName*. **Relevant for connector version 2**

Property Name	Description & Value
(Optional) `ibp.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: APIs for Business User Management If not specified, the default value is *false*. **Relevant for connector version 1**
(Optional)`ibp.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to **true**, PATCH operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to **false**, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. Default value: *false* **Relevant for connector version 2**
(Optional) `ibp.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of users to be provisioned per request. The default value, if not specified, is *20*. The maximum value is **100**. If you enter a number larger than 100, the service will replace it with the default value (20). **Relevant for connector version 1**

Property Name	Description & Value
`ibp.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with **PATCH** requests, and below which they are provisioned with **PUT** request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. The expected value is a positive integer without any separators, such as a space, comma, or period. **Note** You can use this property when SAP Integrated Business Planning is based on SCIM Interface for IAM (in short, SCIM API). **Possible values**: integer Default value: *20000* Minimum value: *1* Maximum value: *20000* **Note** If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members. For more information, see List of Properties [page 80]. **Relevant for connector version 2**
(Optional) `ips.delete.threshold.roles`	Use this property to control the number of user assignments of a role to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of user assignments of a role, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://my1234567-api.scmibp.ondemand.com
User=MyIBPCloudUser
Password=************
ips.date.variable.format=yyyy-MM-dd
ibp.user.roles.overwrite = false
ibp.support.bulk.operation = true
ibp.bulk.operations.max.count = 50
```

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP IBP* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules depending on your setup of entities in your SAP IBP. For more information, see:

   Manage Transformations [page 2439]

   SAP Integrated Business Planning API: Business User

   SAP Business Accelerator Hub: SAP IBP

   Integrate Cross-Domain Identity Management

   SCIM Interface for IAM

   Default transformation for connector version 1:

   Code Syntax

```
{
  "user": {
    "condition": "isValidEmail($.emails[0].value) &&
(('%ibp.roles.prefix%' === 'null') || ($.groups[?(@.display =~ /
%ibp.roles.prefix%.*/)] empty false))",
    "mappings": [
      {
        "sourcePath": "$.userName",
        "targetPath": "$.personExternalID"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.personExternalID",
        "optional": true
      },
```

```
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$.personExternalID",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$.user.globalUserID"
      },
      {
        "targetPath": "$.personID",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "targetPath": "$.markedForArchivingIndicator",
        "constant": "false"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.user.timeZoneCode",
        "functions": [
          {
            "type": "convertTimezoneCode",
            "outputFormat": "sap"
          }
        ],
        "defaultValue": "CET"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.userType"
        ],
        "targetPath": "$.businessPartnerRoleCode",
        "defaultValue": "BUP003",
        "valueMappings": [
          {
            "key": [
              "Employee"
            ],
            "mappedValue": "BUP003"
          }
        ]
      },
      {
        "scope": "createEntity",
        "targetPath": "$.validityPeriod.startDate",
        "sourceVariable": "currentDate"
      },
      {
        "scope": "createEntity",
        "targetPath": "$.validityPeriod.endDate",
        "constant": "9999-12-31"
      },
      {
        "scope": "createEntity",
        "sourceVariable": "currentDate",
        "targetPath": "$.user.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
        "targetPath": "$.user.validityPeriod.endDate"
```

```
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.personalInformation.firstName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.personalInformation.lastName"
      },
      {
        "sourcePath": "$.name.middleName",
        "targetPath": "$.personalInformation.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.name.formatted",
        "targetPath": "$.personalInformation.personFullName",
        "optional": true
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.user.userName"
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.user.logonLanguageCode",
        "optional": true
      },
      {
        "sourcePath": "$.emails[0].value",
        "targetPath": "$.workplaceInformation.emailAddress"
      },
      {
        "targetPath": "$.user.lockedIndicator",
        "constant": "false"
      },
      {
        "condition": "$.active == false",
        "targetPath": "$.user.lockedIndicator",
        "constant": "true"
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
ibp.roles.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
ibp.roles.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "mappings": [
      {
        "sourcePath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
ibp.roles.prefix)",
            "function": "replaceFirstString",
            "regex": "%ibp.roles.prefix%",
            "replacement": ""
          }
        ],
        "targetVariable": "entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.displayName",
        "functions": [
          {
```

```
            "condition": "isAttributeWithMandatoryPrefix(@,
ibp.roles.prefix)",
            "function": "replaceFirstString",
            "regex": "%ibp.roles.prefix%",
            "replacement": ""
          }
        ],
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

See also: Extended Explanation of the *user.roles.overwrite Properties

Default transformation for connector version 2:

Code Syntax

```
{
  "user": {
    "condition": "(('%ibp.group.prefix%' === 'null') || ($.groups[?
(@.display =~ /%ibp.group.prefix%.*/)] empty false))",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:core:2.0:User","urn:ietf:params:scim:schemas
:extension:sap:2.0:User"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "functions": [
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.externalId",
        "functions": [
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
```

```
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$.externalId",
        "optional": true,
        "functions": [
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true,
        "functions": [
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath": "$.name.formatted",
        "optional": true,
        "targetPath": "$.name.formatted"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.displayName",
        "optional": true,
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails",
        "functions": [
          {
            "function": "putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "sourcePath": "$.userType",
        "optional": true,
        "targetPath": "$.userType"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.preferredLanguage"
      },
```

```
      {
        "sourcePath": "$.preferredLanguage",
        "optional": true,
        "targetPath": "$.preferredLanguage"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['language']['correspondence']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['language']['correspondence']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['numericalRepresentation']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['numericalRepresentation']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['date']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['date']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['time']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['time']",
```

```
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['userUuidHistory']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['userUuidHistory']",
        "optional": true,
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.groups[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups[?(@.value)]",
        "functions": [
          {
            "entityType": "group",
            "type": "resolveEntityIds"
          }
        ]
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
ibp.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
ibp.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "mappings": [
      {
        "targetPath": "$.id",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:core:2.0:Group","urn:ietf:params:scim:schema
s:extension:sap:2.0:Group"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
ibp.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%ibp.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.externalId",
        "functions": [
          {
```

```
            "condition": "isAttributeWithMandatoryPrefix(@,
ibp.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%ibp.group.prefix%",
            "replacement": ""
          },
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true,
        "functions": [
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "entityType": "user",
            "type": "resolveEntityIds"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']",
        "optional": true,
        "targetPath": "$.externalId",
        "functions": [
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true,
        "defaultValue": "userGroup"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

> **Note**
>
> If you are using SAP Integrated Business Planning version 2511, note that the `timeZone` value is not provided in the `IANA` format required by SCIM. To fix this, you need to manually add the `convertTimezoneCode` function within the `timeZone` mapping, as shown below:

```
{
   "sourcePath":"$.timeZone",
   "optional":true,
   "targetPath":"$.timezone",
   "functions":[
      {
         "type":"convertTimezoneCode",
         "outputFormat":"sap"
      }
   ],
   "defaultValue":"CET"
},
```

Group Uniqueness and Provisioning

In the most common scenario, Identity Authentication connector version 2 or Local Identity Directory is used as the source system for groups and SAP Integrated Business Planning connector version 2 is the target system.

In SAP Integrated Business Planning version 2, the uniqueness of a group is defined by the combination of `externalId` and `type` attributes. The system supports two group types: `userGroup` and `authorization` (role). As a result, two different groups can share the same `externalId` while belonging to different types, potentially causing conflicts when provisioned to the target system.

To avoid conflicts, ensure the following:

- Your SAP Integrated Business Planning target system must be updated to version 2, as described in Update SAP Integrated Business Planning [page 2432] and it must have the group uniqueness property: `ibp.group.unique.attribute`, set with the following combination of values: `externalId,['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']`. For more information, see List of Properties [page 80].
- The groups in your Identity Authentication connector version 2 or Local Identity Directory source systems must have a value for the `externalName` attribute, which is mapped to the `externalId` attribute.

6. Now, add a source system from which to read users and roles. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Integrated Business Planning

1.6.2.32 SAP Intelligent Agriculture

Follow this procedure to set up SAP Intelligent Agriculture as a target system.

Prerequisites

- You have onboarded your SAP Intelligent Agriculture tenant as described in the Onboarding with the SAP BTP Cockpit section of the *Administration Guide for SAP Intelligent Agriculture*.
- You have OAuth credentials for SAP Intelligent Agriculture. For more information, see: Getting the Credentials from the SAP Intelligent Agriculture Instance.

Context

SAP Intelligent Agriculture enables agribusinesses to sustainably increase farming efficiency by digitizing their farming processes and services from plan-to-harvest, taking advantage of data science and machine learning capabilities.

You can use Identity Provisioning to configure SAP Intelligent Agriculture as a target system where you can provision users and update group members.

> ⚠ Caution
>
> You can't create or delete groups on SAP Intelligent Agriculture. That means:
>
> - On the attempt to create a group on SAP Intelligent Agriculture, Identity Provisioning will only add new members or update existing ones. Also, if you read a group from the external back-end system, there must be a group with the exact same display name (case sensitive) in the SAP Intelligent Agriculture system. Otherwise, an error will be thrown and the group members will not be updated.
> - On the attempt to delete a group on SAP Intelligent Agriculture, Identity Provisioning will only remove its members (group assignments). And this can happen only if the relevant group assignments have been provisioned/are present in the target system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.

2. Add *SAP Intelligent Agriculture* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter:
`URL`	Enter the URL related to your SAP Intelligent Agriculture system, in format: **`https://*.prod-<region>.ia.agribusiness.cloud.sap`** , where * is the name of your subaccount. For example: **`https://custom-domain.prod-eu10.ia.agribusiness.cloud.sap`**
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, created for your SAP Intelligent Agriculture instance (see **Prerequisites**).
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP Intelligent Agriculture instance (see **Prerequisites**).
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Intelligent Agriculture instance, in format: **`https://<TENANT>.authentication.<REGION>.hana.ondemand.com/oauth/token`**

Property Name	Value
(Optional) `ia.user.unique.attribute`	When Identity Provisioning attempts to provision a user for the first time, it may detect that this user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user will be searched and resolved. If the service finds a user on the target system via this filter, then the conflicting user will overwrite the existing one. If the service does not find a user on the target system via this filter, the creation will fail. **Default behavior**: The property is missing during system creation. Its default value is *userName* . This means, if the service finds an existing user by a *userName* , it updates this user with the data of the conflicting one. If a user with such *userName* is not found, the creation of the conflicting user fails. **Possible values:** • *userUuid* • *emails* • *userName* • *externalId* Default value: *userName*

Property Name	Value
`ia.group.unique.attribute`	If the service tries to create a group that already exists in the target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). *displayName*. Currently, it is the only unique attribute that is supported. When set, you can expect the following behavior: • If a group with the given This property defines by which unique attribute(s) the existing group will be searched and resolved. The default value is *displayName* is found in the target system, the group that you try to provision will overwrite the existing one. • This property defines by which unique attribute(s) theIf a group with the given *displayName* is not found in the target system, the group that you try to provision will not be created in the target system. **Possible values:** If the property is not specified, the search is done by the default attribute: `displayName`
`ia.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. This means, for example, that if a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** • *true* • *false* Default value: *false*

Property Name	Value
(Optional) `ia.group.prefix`	This property distinguishes SAP Intelligent Agriculture groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`IA_`** You can use the example value or provide your own. When **set in the target system**, only groups containing the **`IA_`** prefix in their display name will be provisioned to SAP Intelligent Agriculture. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be be provisioned to SAP Intelligent Agriculture.
`ia.include.if.match.wildcard.header`	Makes the connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Intelligent Agriculture system for entity versioning. **Possible values:** • *true* • *false* Default value: *false*
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of group assignments to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of group assignments, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Intelligent Agriculture* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Intelligent Agriculture system. For more information, see Manage Transformations [page 2439].

To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP Intelligent Agriculture entity.

Note

When a user is deleted from the SAP Intelligent Agriculture target system, depending on the off-boarding user handling it can be deleted or set to anonymized.

If you try to read such user, you get only its *id* and the other attributes have value `ANONYMOUS`.

If you try to create again a user with the same unique attributes (`userUuid`, `externalId`, `userName`, and `email`) during the configured retention period, the existed entity is restored to its active state. If the create operation requests changes in the values of the attributes, they will be executed after triggering a resync job.

For more information, see Data Retention Manager.

SAP Intelligent Agriculture SCIM API

Default transformation:

Code Syntax

```
{
  "user": {
    "condition": "isValidEmail($.emails[0].value) && ($.userName
EMPTY false) && ($['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['userUuid'] EMPTY false) && (('%ia.group.prefix%' === 'null') ||
($.groups[?(@.display =~ /%ia.group.prefix%.*/)] empty false))",
    "mappings": [
      {
        "targetPath": "$.id",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.middleName",
        "optional": true,
        "targetPath": "$.name.middleName"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
```

```
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$.userUuid"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.emails[*].value",
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.emails[?(@.value)]"
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
ia.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
ia.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "mappings": [
      {
        "targetPath": "$.id",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schema[0]"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
ia.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%ia.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

5. Add a source system from which to read users and groups. Choose from: Source Systems [page 585]

> ### Note
>
> When Identity Authentication is configured as a source system, the default transformation logic:
>
> - Provisions only groups and user group assignments if they are part of the following predefined group list:
>   - *intagri_Data_Scientist*
>   - *intagri_Farm_Manager*
>   - *intagri_Operator*
>   - *intagri_Operations_Planner*
>   - *intagri_Master_Data_Manager*
>   - *intagri_Administrator*
> - Skips some of the attributes from the identity records.
>
> This way, the transformation logic ensures that the identity data, sent to the Identity Provisioning SCIM REST API, is consistent.

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.33 SAP Jam Collaboration

Follow this procedure to set up SAP Jam Collaboration as a target system.

Prerequisites

You get OAuth credentials for SAP Jam Collaboration. If your SAP Jam tenant is of "SCIM provisioning" type, an OAuth client is automatically created for it, with the name **SCIM API Client**. To find this client:

1. Go to the SAP Jam Collaboration admin panel.
2. Choose *Integrations* > *OAuth Clients*.
3. For *SCIM API Client*, choose *View*.
4. Save the *Key* and *Secret* values – you'll need them later while configuring your SAP Jam Collaboration provisioning system.

To learn more, see: SAP Jam: Add an OAuth Client

Context

After fulfilling the prerequisites, follow the procedure below to create a target SAP Jam Collaboration system to provision users and groups.

These target systems consume SCIM 2.0 API provided by SAP Jam Collaboration.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Jam Collaboration* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL related to your SAP Jam system, in format: **`https://<SAP_Jam_datacenter>.sapjam.com`** For example: *https://jam4.sapjam.com*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth client key, created for your SAP Jam tenant (see **Prerequisites**).
`Password`	(Credential) Enter the OAuth client secret, created for your SAP Jam tenant (see **Prerequisites**).
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Jam instance, in format: **`https://<SAP_Jam_datacenter>/api/v1/auth/token`** For example: *https://jam4.sapjam.com/api/v1/auth/token*

Property Name	Description & Value
(Optional) `jam.group.prefix`	This property distinguishes SAP Jam Collaboration groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **SJC_** You can use the example value or provide your own. When **set in the target system**, only groups containing the **SJC_** prefix in their display name will be provisioned to SAP Jam Collaboration. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be be provisioned to SAP Jam Collaboration.
(Optional) `ips.failed.request.retry.attempts`	Predefined value: *2*
(Optional) `ips.failed.request.retry.attempts.interval`	Predefined value: *30*
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Jam Collaboration* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Jam Collaboration. For more information, see:

   Manage Transformations [page 2439]

   SAP Business Accelerator Hub: SAP Jam Collaboration

- **Mapping logic** – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target entity. If the entity has e-mail addresses, the first entry will be marked as primary.
- **User off-boarding**:
  - Users can be deleted from the SAP Jam Collaboration system via the SCIM REST API. For more information, see SCIM: Deleting Resources .
  - Users can be deactivated by setting the value of their `active` attribute to **false**. For more information, see SCIM: Singular Attributes

Default transformation:

Code Syntax

```
{
    "user": {
        "condition": "('%jam.group.prefix%' === 'null') || ($.groups[?
(@.display =~ /%jam.group.prefix%.*/)] empty false)",
        "mappings": [
            {
                "sourcePath": "$",
                "targetPath": "$"
            },
            {
                "targetPath": "$.id",
                "type": "remove"
            },
            {
                "targetPath": "$.externalId",
                "type": "remove"
            },
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id",
                "scope": "deleteEntity"
            },
            {
                "condition": "$.emails[0].length() > 0",
                "constant": true,
                "targetPath": "$.emails[0].primary"
            },
            {
                "constant": false,
                "targetPath": "$.active",
                "scope": "deleteEntity"
            },
            {
                "targetPath": "$.locale",
                "type": "remove"
            },
            {
                "condition": "($.locale EMPTY false) && ($.addresses[?
(@.type == 'work')].country EMPTY false)",
                "sourcePath": "$.locale",
                "targetPath": "$.locale",
                "functions": [
                    {
                        "function": "toLowerCaseString"
                    },
                    {
```

```
                        "function": "concatString",
                        "suffix": "_"
                    },
                    {
                        "function": "concatString",
                        "suffix": "$.addresses[?(@.type ==
'work')].country"
                    }
                ]
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "optional": true,
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            }
        ]
    },
    "group": {
        "ignore": true,
        "condition": "isAttributeWithOptionalPrefix($.displayName,
jam.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
jam.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
        "mappings": [
            {
                "sourcePath": "$",
                "targetPath": "$"
            },
            {
                "targetPath": "$.id",
                "type": "remove"
            },
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "isAttributeWithMandatoryPrefix(@,
jam.group.prefix)",
                        "function": "replaceFirstString",
                        "regex": "%jam.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "targetPath": "$.members",
                "type": "remove"
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
```

```
                    {
                        "type": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

5. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

1. Before starting a provisioning job, you can first subscribe to the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during your jobs. For more information, see Manage Job Notifications [page 3018].
2. Now, start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

> ⚠ Restriction
>
> Bear in mind the following limitations for the number of sent requests during a provisioning job:
>
> - The **SAP Jam SCIM API** allows up to 13,000 requests per hour and up to 200 requests per minute.
> - The Identity Provisioning service can handle the 200 requests per minute limit. If more requests are sent during the minute, the service will "wait" until it can execute them.

1.6.2.34 SAP Market Communication for Utilities

Follow this procedure to set up SAP Market Communication for Utilities as a target system.

Prerequisites

To establish the connection between Identity Provisioning and SAP Market Communication for Utilities, you need to set up the communication user in SAP BTP ABAP environment. You can do it now (as a prerequisite) or in the process of configuring SAP Market Communication for Utilities as a target system, as described in step 3.

Context

The SAP Market Communication for Utilities application is based on SAP BTP ABAP environment. You can use Identity Provisioning to configure SAP Market Communication for Utilities as a target system to provision entities from a given source system.

This scenario supports writing **users** and **assignments**. In SAP Market Communication for Utilities, groups correspond to roles, thus group members are user assignments of a role.

> ### Note
>
> Identity Provisioning cannot create and delete roles in SAP Market Communication for Utilities target system. It can only create, update and delete user assignments of a role. Therefore, roles must have been created in SAP Market Communication for Utilities target system before you run a provisioning job.
>
> For example, if you try to create or delete a role in SAP Market Communication for Utilities, Identity Provisioning will only add or remove the user assignments of that role, respectively.

The Identity Provisioning service manages the complete set of **business partners** and their relevant **business users** (Employee).

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Market Communication for Utilities* as a target system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP Market Communication for Utilities and configure your authentication method (basic or certificate-based).

   > ### Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP Market Communication for Utilities target system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection [page 2452].

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP BTP ABAP environment backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Create a communication system and assign the created user to the communication system.

For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

d. Create a communication arrangement with the created system.

   For your Identity Provisioning scenario, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

   For more information, see Maintain a Communication Arrangement for Inbound Communication

   > **Note**
   >
   > The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
   >
   > The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the API URL to your SAP Market Communication for Utilities system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **Restriction** Do not use special symbol ',' (comma) as it is not supported.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **Restriction** Do not use special symbol ',' (comma) as it is not supported.

Property Name	Description & Value
`ips.date.variable.format`	*yyyy-MM-dd*
`maco.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP Market Communication for Utilities target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. According to your use case, choose how to set up this property: • Default behavior: This property does not appear in the UI during system creation. Its default value is *personExternalID*. That means, if the service finds an existing user by a *personExternalID* , it updates this user with the data of the conflicting one. If a user with such a *personExternalID* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user matching both unique attributes *email* and *personExternalID*, it updates this user with the data of the conflicting one. If the service finds an existing user matching only the *email*, the update of the existing user fails. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. Possible values: • *personExternalID* • *emails[0].value* Default value: *personExternalID*

Property Name	Description & Value
`maco.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP Market Communication for Utilities target system. • **true** – the current user roles will be deleted in the target system, and the user will be updated only with the roles provisioned by the service. • **false** – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the target system. Default value (if the property is missing during system creation): *true* Default value (if the property appears during system creation): *false* See also: Extended Explanation of the *.user.roles.overwrite Properties
(Optional) `maco.roles.prefix`	This property distinguishes SAP Market Communication for Utilities roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **SMC_** You can use the example value or provide your own. When **set in the target system**, only roles containing the **SMC_** prefix in their role name will be provisioned to SAP Market Communication for Utilities. Roles without this prefix in the role name won't be provisioned. If the property is not set, all roles will be be provisioned to SAP Market Communication for Utilities.
(Optional) `maco.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: APIs for Business User Management If not specified, the default value is *false*.

Property Name	Description & Value
(Optional) `maco.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of users to be provisioned per request. The default value, if not specified, is *20*. The maximum value is **100**. If you enter a number larger than 100, the service will replace it with the default value (20).
(Optional) `ips.delete.threshold.roles`	Use this property to control the number of user assignments of a role to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of user assignments of a role, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*https://12345-aaaaa-3333.abap.hana.ondemand.com*

`User`=*MyMaCoUser*

`Password`=**********

`ips.date.variable.format`=*yyyy-MM-dd*

`maco.user.roles.overwrite`=*false*

`maco.support.bulk.operation` = *true*

`maco.bulk.operations.max.count` = *50*

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default

transformation for the *SAP Market Communication for Utilities* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in SAP Market Communication for Utilities. For more information, see:

Manage Transformations [page 2439]

SAP S/4HANA Cloud API: Business User

Default transformation:

Code Syntax

```
{
  "user": {
    "condition": "isValidEmail($.emails[0].value) &&
(('%maco.roles.prefix%' === 'null') || ($.groups[?(@.display =~ /
%maco.roles.prefix%.*/)] empty false))",
    "mappings": [
      {
        "sourcePath": "$.userName",
        "targetPath": "$.personExternalID"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.personExternalID",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$.personExternalID",
        "optional": true
      },
      {
        "targetPath": "$.personID",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "targetPath": "$.markedForArchivingIndicator",
        "constant": "false"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$.user.globalUserID"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.user.timeZoneCode",
        "functions": [
          {
            "type": "convertTimezoneCode",
            "outputFormat": "sap"
          }
        ],
        "defaultValue": "CET"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
```

```
            "$.userType"
          ],
          "targetPath": "$.businessPartnerRoleCode",
          "defaultValue": "BUP003",
          "valueMappings": [
            {
              "key": [
                "Employee"
              ],
              "mappedValue": "BUP003"
            }
          ]
        },
        {
          "scope": "createEntity",
          "targetPath": "$.validityPeriod.startDate",
          "sourceVariable": "currentDate"
        },
        {
          "scope": "createEntity",
          "targetPath": "$.validityPeriod.endDate",
          "constant": "9999-12-31"
        },
        {
          "scope": "createEntity",
          "sourceVariable": "currentDate",
          "targetPath": "$.user.validityPeriod.startDate"
        },
        {
          "scope": "createEntity",
          "constant": "9999-12-31",
          "targetPath": "$.user.validityPeriod.endDate"
        },
        {
          "sourcePath": "$.name.givenName",
          "targetPath": "$.personalInformation.firstName",
          "optional": true
        },
        {
          "sourcePath": "$.name.familyName",
          "targetPath": "$.personalInformation.lastName"
        },
        {
          "sourcePath": "$.name.middleName",
          "targetPath": "$.personalInformation.middleName",
          "optional": true
        },
        {
          "sourcePath": "$.name.formatted",
          "targetPath": "$.personalInformation.personFullName",
          "optional": true
        },
        {
          "sourcePath": "$.userName",
          "targetPath": "$.user.userName"
        },
        {
          "sourcePath": "$.locale",
          "targetPath": "$.user.logonLanguageCode",
          "optional": true
        },
        {
          "sourcePath": "$.emails[0].value",
          "targetPath": "$.workplaceInformation.emailAddress"
        },
        {
          "targetPath": "$.user.lockedIndicator",
          "constant": "false"
```

```
      },
      {
        "condition": "$.active == false",
        "targetPath": "$.user.lockedIndicator",
        "constant": "true"
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
maco.roles.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
maco.roles.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "mappings": [
      {
        "sourcePath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
maco.roles.prefix)",
            "function": "replaceFirstString",
            "regex": "%maco.roles.prefix%",
            "replacement": ""
          }
        ],
        "targetVariable": "entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
maco.roles.prefix)",
            "function": "replaceFirstString",
            "regex": "%maco.roles.prefix%",
            "replacement": ""
          }
        ],
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

See also: Extended Explanation of the *user.roles.overwrite Properties

6. Now, add a source system from which to read users and roles. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP S/4HANA Cloud Documentation

1.6.2.35 SAP Marketing Cloud

Follow this procedure to set up SAP Marketing Cloud as a target system.

Prerequisites

To establish the connection between Identity Provisioning and SAP Marketing Cloud, you need to set up the communication (user, system and arrangement) on SAP Marketing Cloud. You can do it now (as a prerequisite) or in the process of configuring SAP Marketing Cloud as a target system, as described in step 3.

Context

You can use SAP Marketing Cloud as a target system to provision entities from a certain source system. This scenario supports writing **users** and **assignments**. In SAP Marketing Cloud, groups correspond to roles, thus group members are user assignments of a role.

> ⓘ Note
>
> Identity Provisioning cannot create and delete roles in SAP Marketing Cloud target system. It can only create, update and delete user assignments of a role. Therefore, roles must have been created in SAP Marketing Cloud system before you run a provisioning job.
>
> For example, if you try to create or delete a role in SAP Marketing Cloud, Identity Provisioning will only add or remove the user assignments of that role, respectively.

The Identity Provisioning service manages the complete set of **business partners** and their relevant **business users** (Employee and Contingent Worker).

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Marketing Cloud* as a target system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP Marketing Cloud and configure your authentication method (basic or certificate-based).

   > ### Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP Marketing Cloud target system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP Marketing Cloud backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Create a communication system and assign the created user to the communication system.

      For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

   d. Create a communication arrangement with the created system.

      For your Identity Provisioning scenario, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

      For more information, see: User Provisioning (with Corporate Identity Provider)

      > ### Note
      >
      > The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
      >
      > The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
Type	Enter: *HTTP*

Property Name	Description & Value
`URL`	Specify the URL to your SAP Marketing Cloud system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`ips.date.variable.format`	*yyyy-MM-dd*

Property Name	Description & Value
`marketing.cloud.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP Marketing Cloud target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. According to your use case, choose how to set up this property: • Default behavior: This property does not appear in the UI during system creation. Its default value is *personExternalID*. That means, if the service finds an existing user by a *personExternalID* , it updates this user with the data of the conflicting one. If a user with such a *personExternalID* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user matching both unique attributes *email* and *personExternalID*, it updates this user with the data of the conflicting one. If the service finds an existing user matching only the *email*, the update of the existing user fails. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. Possible values: • *personExternalID* • *emails[0].value* Default value: *personExternalID*
`marketing.cloud.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP Marketing Cloud target system. • **true** – the current user roles will be deleted in the target system, and the user will be updated only with the roles provisioned by the service. • **false** – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the target system. See also: Extended Explanation of the *.user.roles.overwrite Properties

Property Name	Description & Value
(Optional) `marketing.cloud.roles.prefix`	This property distinguishes SAP Marketing Cloud roles by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **SMKC_** You can use the example value or provide your own. When **set in the target system**, only roles containing the **SMKC_** prefix in their role name will be provisioned to SAP Marketing Cloud. Roles without this prefix in the role name won't be provisioned. If the property is not set, all roles will be be provisioned to SAP Marketing Cloud.
(Optional) `marketing.cloud.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: APIs for Business User Management If not specified, the default value is *false*.
(Optional) `marketing.cloud.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of users to be provisioned per request. The default value, if not specified, is *20*. The maximum value is **100**. If you enter a number larger than 100, the service will replace it with the default value (20).
(Optional) `ips.delete.threshold.roles`	Use this property to control the number of user assignments of a role to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of user assignments of a role, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://my1234567-api.s4hana.ondemand.com
User=MyMarketingCloudUser
Password=***********
ips.date.variable.format=yyyy-MM-dd
marketing.cloud.user.roles.overwrite = false
marketing.cloud.support.bulk.operation = true
marketing.cloud.bulk.operations.max.count = 50
```

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Marketing Cloud* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules depending on your setup of entities in your SAP Marketing Cloud. For more information, see:

   Manage Transformations [page 2439]

   SAP Marketing Cloud API: Business User

   Default transformation:

   Code Syntax

```
{
  "user": {
    "condition": "isValidEmail($.emails[0].value) &&
(('%marketing.cloud.roles.prefix%' === 'null') || ($.groups[?(@.display
=~ /%marketing.cloud.roles.prefix%.*/)] empty false))",
    "mappings": [
      {
        "sourcePath": "$.userName",
        "targetPath": "$.personExternalID"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.personExternalID",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$.personExternalID",
        "optional": true
      },
```

```
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$.user.globalUserID"
      },
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.personID"
      },
      {
        "targetPath": "$.markedForArchivingIndicator",
        "constant": "false"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.user.timeZoneCode",
        "functions": [
          {
            "type": "convertTimezoneCode",
            "outputFormat": "sap"
          }
        ],
        "defaultValue": "CET"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.userType"
        ],
        "targetPath": "$.businessPartnerRoleCode",
        "defaultValue": "BUP003",
        "valueMappings": [
          {
            "key": [
              "Employee"
            ],
            "mappedValue": "BUP003"
          },
          {
            "key": [
              "Contingent Worker"
            ],
            "mappedValue": "BBP005"
          }
        ]
      },
      {
        "scope": "createEntity",
        "sourceVariable": "currentDate",
        "targetPath": "$.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "targetPath": "$.validityPeriod.endDate",
        "constant": "9999-12-31"
      },
      {
        "scope": "createEntity",
        "sourceVariable": "currentDate",
        "targetPath": "$.user.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
        "targetPath": "$.user.validityPeriod.endDate"
      },
```

```
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.personalInformation.firstName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.personalInformation.lastName"
      },
      {
        "sourcePath": "$.name.middleName",
        "targetPath": "$.personalInformation.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.name.formatted",
        "targetPath": "$.personalInformation.personFullName",
        "optional": true
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.user.userName"
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.user.logonLanguageCode",
        "optional": true
      },
      {
        "sourcePath": "$.emails[0].value",
        "targetPath": "$.workplaceInformation.emailAddress"
      },
      {
        "targetPath": "$.user.lockedIndicator",
        "constant": "false"
      },
      {
        "condition": "$.active == false",
        "targetPath": "$.user.lockedIndicator",
        "constant": "true"
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
marketing.cloud.roles.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
marketing.cloud.roles.prefix) && (isRegularGroup() ||
isApplicationSpecificGroup())",
    "mappings": [
      {
        "sourcePath": "$.displayName",
        "targetVariable": "entityIdTargetSystem",
        "scope": "createEntity",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
marketing.cloud.roles.prefix)",
            "function": "replaceFirstString",
            "regex": "%marketing.cloud.roles.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
```

```
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
marketing.cloud.roles.prefix)",
            "function": "replaceFirstString",
            "regex": "%marketing.cloud.roles.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

See also: Extended Explanation of the *user.roles.overwrite Properties

6. Now, add a source system from which to read users and roles. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP S/4HANA Cloud Documentation

1.6.2.36 SAP Revenue Growth Management

Follow this procedure to set up SAP Revenue Growth Management as a target system.

Prerequisites

- You have created an SAP Revenue Growth Management consumer application in your SAP Cloud Identity Services tenant and defined a dependency to its API, as described in Configure Integration Between Applications. *UserImport* must be selected for the dependency *API*.
- You have generated API credentials for the consumer application, as described in Generate Credentials to Access the APIs of an Application.

Context

SAP Revenue Growth Management is an SAP BTP SaaS application that provides customers in the Consumer Products industry an easy way to efficiently create, monitor, and maintain account plans.

You can use Identity Provisioning to configure SAP Revenue Growth Management as a target system to provision users and groups. These target systems consume SCIM 2.0 API provided by SAP Revenue Growth Management. For more information, see SAP Revenue Growth Management SCIM API.

> **Note**
>
> Support for groups is intended for future use.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Revenue Growth Management* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
Type	Enter: *HTTP*

Property Name	Description & Value
`URL`	Enter the base URL of your SAP Revenue Growth Management system's SCIM API, excluding any path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the client ID generated when creating API credentials for the consumer application. For more information, see Generate Credentials to Access the APIs of an Application.
`Password`	(Credential) Enter the client secret generated when creating API credentials for the consumer application. For more information, see Generate Credentials to Access the APIs of an Application.
`rgm.userImport.api.dependency.name`	Enter the value of the dependency name used in the configuration of the SAP Revenue Growth Management consumer application, created in the SAP Cloud Identity Services tenant. It is used for access token retrieval from Identity Authentication. For more information, see Integrating Applications.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL.
(Optional) `rgm.user.unique.attribute`	This property defines by which unique attribute(s) an existing user in the target system will be searched and resolved in case Identity Provisioning tries to provision a conflicting user. SAP Revenue Growth Management supports the following unique attributes which are automatically filled in during system creation: **`userName,emails[0].value,externalId`**. If the service finds an existing user by at least one of the unique attributes, it updates this user with the data of the conflicting one. If such a user is not found, the update of the conflicting user fails. If more than one users with these unique attributes are found, the update fails.

Property Name	Description & Value
(Optional) `rgm.group.unique.attribute`	If you try to provision a group that already exists in a target system, the group creation will fail. In this case, the existing group only needs to be updated. This property defines by which unique attribute(s) the existing group will be searched and resolved. The default value is **`['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']`**. Currently, it is the only unique attribute that is supported. If the service finds an existing group by this unique attribute, the group that you try to provision will overwrite the existing one. If such a group is not found, the group that you try to provision will not be created in the target system.
(Optional) `rgm.group.prefix`	This property distinguishes SAP Revenue Growth Management groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **`RGM_`** You can use the example value or provide your own. When **set in the target system**, only groups containing the **`RGM_`** prefix in their display name will be provisioned to SAP Revenue Growth Management. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Revenue Growth Management.
(Optional) `rgm.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. Default value for target systems: *false*
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Property Name	Description & Value
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Revenue Growth Management* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Revenue Growth Management system. For more information, see:

   Manage Transformations [page 2439].

   SAP Revenue Growth Management SCIM API

   To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

   Default transformation:

   Code Syntax

```
{
  "user":{
    "condition": "(('%rgm.group.prefix%' === 'null') || ($.groups[?
(@.display =~ /%rgm.group.prefix%.*/)] empty false))",
    "mappings":[
      {
       "constant": [
         "urn:ietf:params:scim:schemas:core:2.0:User",
         "urn:ietf:params:scim:schemas:extension:sap:2.0:User"
       ],
       "targetPath": "$.schemas"
      },
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['userId']",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
```

```
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.userType",
        "targetPath": "$.userType",
        "optional": true
      },
      {
        "sourcePath": "$.preferredLanguage",
        "targetPath": "$.preferredLanguage",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional":true
      },
      {
        "sourcePath": "$.timezone",
        "targetPath": "$.timezone",
        "optional":true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.addresses",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.addresses"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$.groups[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups[?(@.value)]",
        "functions": [
```

```
          {
            "entityType": "group",
            "type": "resolveEntityIds"
          }
        ]
      }
    ]
  },
  "group": {
   "condition": "isAttributeWithOptionalPrefix($.displayName,
rgm.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
rgm.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "mappings":[
      {
        "targetPath": "$.id",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
       "constant": [
         "urn:ietf:params:scim:schemas:core:2.0:Group",
         "urn:ietf:params:scim:schemas:extension:sap:2.0:Group",
         "urn:sap:cloud:scim:schemas:extension:custom:2.0:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
        "optional": true,
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
rgm.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%rgm.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
rgm.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%rgm.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "entityType": "user",
            "type": "resolveEntityIds"
          }
        ]
      },
      {
```

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

5. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585].

Related Information

SAP Revenue Growth Management

1.6.2.37 SAP Sales Cloud and SAP Service Cloud

Follow this procedure to set up SAP Sales Cloud and SAP Service Cloud, formerly known as *SAP Cloud for Customer* (in short, C4C), as a target system.

Prerequisites

To integrate SAP Sales Cloud and SAP Service Cloud with Identity Provisioning, you need to use SAP Cloud Integration (SAP CI). This service provides a package with integration flows (iFlows) for enabling the creation of users and assignment of users to groups via SCIM API in SAP Sales Cloud and SAP Service Cloud.

- To configure SAP Cloud Integration and SAP Sales Cloud and SAP Service Cloud, see: Identity Provisioning in SAP Cloud for Customer using System for Cross-Domain Identity Management (SCIM)
- To set up and use the *SAP Cloud for Customer Integration with Identity Provisioning via System for Cross-domain Identity Management* package, see: SAP Business Accelerator Hub: SAP Cloud for Customer Integration with Identity Provisioning via System for Cross-domain Identity Management

Context

SAP Sales Cloud and SAP Service Cloud is a cloud-based solution that helps customers manage day-to-day sales and service interactions by sending and receiving signals between front- and back-office solutions and providing a single view of the customer.

You can use Identity Provisioning to configure SAP Sales Cloud and SAP Service Cloud as a target system where you can provision users and group members from source systems. Keep in mind that once you have provisioned the entities to SAP Sales Cloud and SAP Service Cloud, a *business user* and an *employee* are created for every provisioned user. The business user is required for the provisioned user to log into the SAP Sales Cloud and SAP Service Cloud system.

The cloud-based solution provides different APIs for integration with Identity Provisioning, leading to different connector versions for each API. Currently, the following connector versions are available: Version 1 (SOAP-based API), which is deprecated, Version 2 (SOAP-based API), Version 3 (SCIM 2.0 based API) and Version 4 (SCIM 2.0 based API), which is applicable for SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 only.

Each connector version supports specific attribute mappings in transformations and requires particular properties. The latest version is set as the default.

Version 1 (SOAP-based API)

> **Note**
>
> The SOAP-based API version 1 is deprecated.

When created via connector version 1, users are initially transferred to a staging area, and then can be replicated to the C4C system manually or via a job, depending on your tenant setup. This API version is SOAP based. In the SAP Sales Cloud and SAP Service Cloud admin console you can distinguish it by its name – */humancapitalmanagementmasterd6*.

To learn how to replicate users (using API v.1), see: Employee Master Data Replication

Version 2 (SOAP-based API)

When using connector version 2, users are created immediately – there is no need to transfer them to a staging area. This API version is SOAP based. In the C4C admin console, you can distinguish it by its name */employeereplicationin2*.

Version 3 (SCIM 2.0 based API)

When using connector version 3, users are created immediately. There is no need to transfer them to a staging area. For more information, refer to *Identity Provisioning in SAP Cloud for Customer using a System for Cross-domain Identity Management (SCIM)* package in the *Prerequisites* section.

Version 4 (Applicable for SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 only)

When using connector version 4 (SCIM 2.0 based API), only users of type `Employee` are provisioned. This version introduces an enhanced SCIM API that no longer requires SAP Cloud Integration. It supports patch operations and provisioning of application-specific groups. Conflict resolution is handled using not only the username but also based on the user's email and a combination of both username and email.

> ### Note
>
> If you have access to all API versions, you must use **separate** provisioning systems for each version, to avoid data inconsistency errors.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Sales Cloud and SAP Service Cloud* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter SAP Cloud Integration runtime URL. See: How to Get SAP Cloud Integration Runtime URL
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`OAuth2TokenServiceURL`	Enter the URL to the access token provider service. For example: `https://<c4c_instance>/auth/token` endpoint. SAP Sales Cloud and SAP Service Cloud support OAuth token-based authentication, enabling you to obtain an access token using either basic authentication (by setting `User` and `Password`) or certificate authentication. **Relevant for connector version 4**
`User`	Enter SAP Cloud Integration user ID to connect to SAP Cloud Integration. See: • Setting Up Inbound HTTP Connections (with Basic Authentication), Neo Environment • Basic Authentication of IdP User for Integration Flow Processing (Cloud Foundry environment) **Relevant for connector version 3**

Property Name	Description & Value
`Password`	(Credential) Enter SAP Cloud Integration password to connect to SAP Cloud Integration. See: • Setting Up Inbound HTTP Connections (with Basic Authentication), Neo Environment • Basic Authentication of IdP User for Integration Flow Processing (Cloud Foundry environment) **Relevant for connector version 3**
`c4c.api.version`	The version of the SAP Sales Cloud and SAP Service Cloud API you use. • **1** - SOAP-based API (deprecated) • **2** - SOAP-based API • **3** - SCIM 2.0 based API • **4** - SCIM 2.0 based API (enhanced) **Note** After you set up the communication arrangement, you can determine the API version used by your SAP Sales Cloud and SAP Service Cloud system. It represents the ID at the end of your generated URL – the name of API v.1 is *humancapitalmanagementmasterd6*, and for API v.2 is *employeereplicationin2*.
`RemoteSystemID`	Enter the system instance ID, configured for the communication system setting in the C4C system. **Relevant for connector version 1** (deprecated)
`RecipientPartyID`	Enter the recipient system name. Example: **0011SAP** **Relevant for connector version 2**
`SenderPartyID`	Enter the name of the sender system name. It's equal to the value of property RemoteSystemID from version 1. For example: **IPS** **Relevant for connector version 2**

Property Name	Description & Value
`c4c.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. Default value: *false* **Relevant for connector version 4**
`c4c.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. Default value in all versions: *userName* Supported values in version 4: • userName • emails[0].value • userName,emails[0].value
`c4c.group.unique.attribute`	If Identity Provisioning tries to provision a group that already exists in the target system (a conflicting group), this property defines the unique attributes by which the existing group will be searched and resolved. Default value: *displayName*
(Optional) `c4c.group.prefix`	This property distinguishes SAP Sales Cloud and SAP Service Cloud groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **C4C_** You can use the example value or provide your own. When **set in the target system**, only groups containing the **C4C_** prefix in their display name will be provisioned to SAP Sales Cloud and SAP Service Cloud. Groups without this prefix in the display name won't be provisioned. If the property is not set, the SAP Sales Cloud and SAP Service Cloud groups will be read and provisioned to the target system with their actual display names.

Property Name	Description & Value
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Sales Cloud and SAP Service Cloud* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in C4C. For more information, see:

   Manage Transformations [page 2439]

   SAP Business Accelerator Hub: C4C

   SAP Cloud for Customer OData API v2 Reference

   Default transformation for connector version 1: Deprecated

   Code Syntax

```
/* Attribute RemoteObjectID stores the user name from the source system
into C4C. */
{
   "user": {
       "mappings": [
           {
        "sourcePath": "$.userName",
        "targetPath": "$.RemoteObjectID"
       },
/* Statements that start with PersonalDetails are related to the employee
created in C4C. */
       {
          "sourceVariable": "currentDate",
          "targetPath": "$.PersonalDetails.ValidityPeriod.StartDate",
          "functions": [
               {
                  "type": "manipulateDate",
                  "targetDateFormat": "yyyy-MM-dd"
               }
            ]
```

```
        },
      {
          "constant": "9999-12-31",
          "targetPath": "$.PersonalDetails.ValidityPeriod.EndDate"
        },
        {
          "sourcePath": "$.name.givenName",
          "optional": true,
          "targetPath": "$.PersonalDetails.GivenName"
        },
    {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.PersonalDetails.FamilyName"
    },
/* Statements that start with EmployeeType are supported by the C4C system
only for internal employees.
(Service agents are not supported as EmployeeType. The supported employee
types are mandatory and relevant only to lean employees).
The value of the currentDate variable (the date when the provisioning is
executed) is set as validity start date of the employee.
In the default transformation statement, it's converted to the format
required by C4C via a transformation function. */
       {
          "sourceVariable": "currentDate",
          "targetPath": "$.EmployeeType.ValidityPeriod.StartDate",
          "functions": [
             {
              "type": "manipulateDate",
              "targetDateFormat": "yyyy-MM-dd"
             }
             ]
           },
           {
               "constant": "9999-12-31",
               "targetPath": "$.EmployeeType.ValidityPeriod.EndDate"
           },
           {
               "sourcePath": "$.userName",
               "targetPath": "$.Identity.ID"
           },
           {
               "constant": "false",
               "targetPath": "$.Identity.UserAccountsInactiveIndicator"
           },
           {
               "sourcePath": "$.phoneNumbers[?(@.type == 'mobile')].value",
               "optional": true,
               "targetPath":
"$.WorkplaceAddress.MobilePhoneNumberDescription"
           },
           {
               "sourcePath": "$.phoneNumbers[?(@.type == 'work')].value",
               "optional": true,
               "targetPath": "$.WorkplaceAddress.PhoneNumberDescription"
           },
           {
               "sourcePath": "$.emails[0].value",
               "optional": true,
               "targetPath": "$.WorkplaceAddress.EmailURI"
           }
       ]
    }
}
```

If you use the first version of the SAP Sales Cloud and SAP Service Cloud API (SOAP based) – *humancapitalmanagementmasterd6* – your systems will be created with *c4c.api.version=1*. You need to use

the transformation below and specify the mandatory attribute *RemoteSystemID*. The following interface is used for replicating employee master data to SAP Sales Cloud and SAP Service Cloud: Inbound Service HumanCapitalManagementMasterDataReplicationEmployeeMasterDataReplicationIn. Along with replicating employees in SAP Sales Cloud and SAP Service Cloud, a business user is created for every user.

Default transformation for connector version 2

Code Syntax

```
{
    "user": {
        "condition" : "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User'].employeeNumb
er EMPTY false",
        "mappings": [
            {
                "targetPath": "$.ReceiverEmployeeID",
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User'].employeeNumb
er"
            },
            {
                "targetPath": "$.BusinessPartnerID",
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User'].employeeNumb
er"
            },
            {
                "targetPath": "$.EmployeeType.ValidityPeriod.StartDate",
                "sourceVariable": "currentDate",
                "functions": [
                    {
                        "type": "manipulateDate",
                        "targetDateFormat" : "yyyy-MM-dd"
                    }
                ]
            },
            {
                "targetPath": "$.EmployeeType.ValidityPeriod.EndDate",
                "constant": "9999-12-31"
            },
            {
                "targetPath": "$.Common.Name.GivenName",
                "sourcePath": "$.name.givenName",
                "optional": true
            },
            {
                "targetPath": "$.Common.Name.FamilyName",
                "sourcePath": "$.name.familyName"
            },
/* You can set a custom namespace for an attribute. For example, if your
namespace prefix is called a123,
enter the following lines in your transformation:
          {
            "targetPath": "$['a123:PersonalDetails']
['a123:FamilyName']",
            "sourcePath": "$.name.familyName"
          }
When sending the request to C4C, the Identity Provisioning service will
transform this data into XML elements, as follows:
            <a123:PersonalDetails>
                <a123:FamilyName>...</FamilyName>
            </a123:PersonDetails>   */

            {
```

```
                "targetPath": "$.Identity.IdentityID",
                "sourcePath": "$.userName"
            },
            {
                "targetPath": "$.Identity.UserAccountsInactiveIndicator",
                "constant": "false"
            },
            {
                "condition": "$.active == false",
                "targetPath": "$.Identity.UserAccountsInactiveIndicator",
                "constant": "true"
            },
            {
                "targetPath":
"$.WorkplaceAddress.MobilePhoneNumberDescription",
                "sourcePath": "$.phoneNumbers[?(@.type ==
'mobile')].value",
                "optional": true
            },
            {
                "targetPath": "$.WorkplaceAddress.PhoneNumberDescription",
                "sourcePath": "$.phoneNumbers[?(@.type == 'work')].value",
                "optional": true
            },
            {
                "targetPath": "$.WorkplaceAddress.EmailURI",
                "sourcePath": "$.emails[0].value",
                "optional": true
            },
            {
                "constant": "SALES_REP",
                "targetPath": "$.Identity.BusinessRole[0].ID"
            },
            {
                "constant": "SALES_MANAGER",
                "targetPath": "$.Identity.BusinessRole[1].ID"
            }
        ]
    }
}
```

If you want to use the second C4C API (SOAP based) – *employeereplicationin2* – you have to set *c4c.api.version=2*, change the transformation with the one above, and specify the two mandatory attributes – *RecipientPartyID* and *SenderPartyID*.

Default transformation for connector version 3

Code Syntax

```
{
  "user": {
    "condition": "('%c4c.group.prefix%' === 'null') || ($.groups[?
(@.display =~ /%c4c.group.prefix%.*/)] empty false)",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.emails[*].value",
```

```
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.emails[?(@.value)]",
        "optional": true
      },
      {
        "constant": "work",
        "targetPath": "$.emails[0].type"
      },
      {
        "sourcePath": "$.userType",
        "targetPath": "$.userType",
        "optional": true
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.middleName",
        "targetPath": "$.name.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true,
        "defaultValue": true
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
        "targetPath": "$.schemas[0]"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
        "targetPath": "$.schemas[1]"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:extension:sap:2.0:User",
        "targetPath": "$.schemas[2]"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      }
    ]
  },
// You cannot create groups but instead, you can create and update group
members, provided that groups in the source system have the exact same
display name
```

```
// as the groups in C4C. The group deletion operation is skipped.
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
c4c.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
c4c.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "skipOperations": [
      "delete"
    ],
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
c4c.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%c4c.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ],
        "defaultValue": []
      }
    ]
  }
}
```

Default transformation for connector version 4

(Applicable for SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 only)

Code Syntax

```
{
    "user": {
        "condition": "((('%c4c.group.prefix%' === 'null') || ($.groups[?
(@.display =~ /%c4c.group.prefix%.*/)] empty false)) && $.userName empty
false && ($.userType === 'Employee' || $.userType === 'employee'))",
        "mappings": [
            {
                "targetPath": "$.id",
                "sourceVariable": "entityIdTargetSystem"
            },
            {
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:User",
                    "urn:ietf:params:scim:schemas:extension:sap:2.0:User",

"urn:ietf:params:scim:schemas:extension:enterprise:2.0:User"
                ],
```

```
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "preserveArrayWithSingleElement": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true
            },
            {
                "sourcePath": "$.userType",
                "targetPath": "$.userType"
            },
            {
                "sourcePath": "$.displayName",
                "optional": true,
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.externalId",
                "optional": true,
                "targetPath": "$.externalId"
            },
            {
              "sourcePath": "$.name.givenName",
              "optional": true,
              "targetPath": "$.name.givenName"
            },
            {
              "sourcePath": "$.name.middleName",
              "optional": true,
              "targetPath": "$.name.middleName"
            },
            {
              "sourcePath": "$.name.familyName",
              "optional": true,
              "targetPath": "$.name.familyName"
            },
            {
              "sourcePath": "$.name.formatted",
              "optional": true,
              "targetPath": "$.name.formatted"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
                "optional": true,
```

```
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['status']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['status']"
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "optional": true,
                "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']"
            },
            {
                "sourcePath": "$.locale",
                "optional": true,
                "targetPath": "$.locale"
            },
            {
                "sourcePath": "$.timezone",
                "optional": true,
                "targetPath": "$.timezone"
            }
        ]
    },
    "group": {
        "condition": "isAttributeWithOptionalPrefix($.displayName,
c4c.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
c4c.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
        "mappings": [
            {
                "targetPath": "$.id",
                "sourceVariable": "entityIdTargetSystem"
            },
            {
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:Group",
                    "urn:ietf:params:scim:schemas:extension:sap:2.0:Group"
                ],
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "isAttributeWithMandatoryPrefix(@,
c4c.group.prefix)",
                        "function": "replaceFirstString",
                        "regex": "%c4c.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name']",
                "optional": true,
                "targetPath": "$.displayName",
                "functions": [
                    {
```

```
                        "condition": "isAttributeWithMandatoryPrefix(@,
c4c.group.prefix)",
                        "function": "replaceFirstString",
                        "regex": "%c4c.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

5. Add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Blog: Employee Replication FAQ (relevant for version 1)

1.6.2.38 SAP Signavio Process Transformation Suite

Follow this procedure to set up SAP Signavio Process Transformation Suite as a target system.

Prerequisites

> **Note**
>
> The initial setup of SAP Signavio Process Transformation Suite is completed during the onboarding process. This setup includes establishing a connection between Identity Provisioning and SAP Signavio

Process Transformation Suite using certificate-based authentication. Any changes needed after the initial setup should be processed through a support incident.

Context

SAP Signavio Process Transformation Suite helps you understand, improve, and transform your business processes – fast and at scale – with a cloud-based process management platform.

You can use Identity Provisioning to configure SAP Signavio Process Transformation Suite as a target system where you can provision users and groups from source systems. When provisioning users, ensure that the `userName` attribute is formatted as a valid email address.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP Signavio Process Transformation Suite* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	The URL to the SCIM API of your SAP Signavio Process Transformation Suite system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *ClientCertificateAuthentication*
`sig.tenant.id`	The identifier of SAP Signavio Process Transformation Suite tenant

Property Name	Value
`sig.user.unique.attribute`	This property defines the unique attributes used to search and resolve an existing user when Identity Provisioning attempts to provision a user that already exists in the target system (a conflicting user). Default value: *userName* If the service finds an existing user with the same *userName*, it updates that user with the data from the conflicting user. However, if no user with that *userName* exists, the creation of the conflicting user fails.
`sig.group.unique.attribute`	This property defines the unique attributes used to search and resolve an existing group when Identity Provisioning attempts to provision a group that already exists in the target system (a conflicting group). Default value: *displayName*
`sig.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. Default value: *false*
`sig.group.prefix`	This property distinguishes SAP Signavio Process Transformation Suite groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: SIG_ You can use the example value or provide your own. When **set in the target system**, only groups containing the SIG_ prefix in their display name will be provisioned to SAP Signavio Process Transformation Suite. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP Signavio Process Transformation Suite.

Property Name	Value
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Signavio Process Transformation Suite* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Signavio Process Transformation Suite system. For more information, see:

   Manage Transformations [page 2439]

   Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP Signavio Process Transformation Suite entity.

   Default transformation:

   Code Syntax

```
{
  "user": {
    "condition": "($.emails EMPTY false) &&
isValidEmail($.emails[*].value) && (('%sig.group.prefix%' === 'null') ||
($.groups[?(@.display =~ /%sig.group.prefix%.*/)] empty false))",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:User",
          "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
          "urn:ietf:params:scim:schemas:extension:sap:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "condition": "$.emails[?(@.primary == true)].value == []",
        "sourcePath": "$.emails[0].value",
        "targetPath": "$.userName"
      },
      {
```

```
      "condition": "$.emails[?(@.primary == true)].value != []",
      "sourcePath": "$.emails[?(@.primary == true)].value",
      "targetPath": "$.userName"
    },
    {
      "sourcePath": "$.name.givenName",
      "targetPath": "$.name.givenName",
      "optional": true
    },
    {
      "sourcePath": "$.name.middleName",
      "targetPath": "$.name.middleName",
      "optional": true
    },
    {
      "sourcePath": "$.name.familyName",
      "targetPath": "$.name.familyName",
      "optional": true
    },
    {
      "sourcePath": "$.name.honorificPrefix",
      "targetPath": "$.name.honorificPrefix",
      "optional": true
    },
    {
      "sourcePath": "$.name.honorificSuffix",
      "targetPath": "$.name.honorificSuffix",
      "optional": true
    },
    {
      "sourcePath": "$.displayName",
      "targetPath": "$.displayName",
      "optional": true
    },
    {
      "sourcePath": "$.nickName",
      "targetPath": "$.nickName",
      "optional": true
    },
    {
      "sourcePath": "$.profileUrl",
      "targetPath": "$.profileUrl",
      "optional": true
    },
    {
      "sourcePath": "$.title",
      "targetPath": "$.title",
      "optional": true
    },
    {
      "sourcePath": "$.userType",
      "targetPath": "$.userType",
      "optional": true
    },
    {
      "sourcePath": "$.preferredLanguage",
      "targetPath": "$.preferredLanguage",
      "optional": true
    },
    {
      "sourcePath": "$.locale",
      "targetPath": "$.locale",
      "optional": true
    },
    {
      "sourcePath": "$.timezone",
      "targetPath": "$.timezone",
      "optional": true
```

```
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['emails']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['emails']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.phoneNumbers",
        "targetPath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.ims",
        "targetPath": "$.ims",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.photos",
        "targetPath": "$.photos",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.addresses",
        "targetPath": "$.addresses",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.entitlements",
        "targetPath": "$.entitlements",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.roles",
        "targetPath": "$.roles",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.x509Certificates",
        "targetPath": "$.x509Certificates",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
```

```
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['value']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']['manager']
['displayName']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['costCenter']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {
```

```
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['userUuidHistory']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['userUuidHistory']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['status']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['status']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mfaEnabled']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mfaEnabled']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['totpEnabled']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['totpEnabled']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['loginTime']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['loginTime']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['contactPreferences']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['contactPreferences']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
```

```
      {
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['passwordDetails']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['passwordDetails']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sncName']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sncName']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['groupDomains']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['groupDomains']",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",
        "preserveArrayWithSingleElement": true,
        "optional": true
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
sig.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
sig.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [
          "urn:ietf:params:scim:schemas:core:2.0:Group",
          "urn:ietf:params:scim:schemas:extension:sap:2.0:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
sig.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%sig.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.members[*].value",
        "targetPath": "$.members[?(@.value)]",
        "preserveArrayWithSingleElement": true,
        "optional": true,
```

```
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

5. Add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP Signavio Process Transformation Suite

1.6.2.39 SAP S/4HANA Cloud Public Edition

Follow this procedure to set up SAP S/4HANA Cloud Public Edition as a target system.

Prerequisites

To establish the connection between Identity Provisioning and SAP S/4HANA Cloud Public Edition, you need to set up the communication (user, system and arrangement) on SAP S/4HANA Cloud Public Edition. You can do it now (as a prerequisite) or in the process of configuring SAP S/4HANA Cloud Public Edition as a target system, as described in step 3.

Context

SAP S/4HANA Cloud Public Edition is a complete enterprise resource planning (ERP) system with built-in intelligent technologies and advanced analytics.

You can use Identity Provisioning to configure SAP S/4HANA Cloud Public Edition as a target system where you can provision users and group members from source systems of your choice.

SAP S/4HANA Cloud Public Edition provides different APIs for integration with Identity Provisioning, resulting in different connector versions for each API. Each connector version supports specific attribute mappings within the transformations and requires particular property values. The value of the `s4hana.cloud.api.version` property controls which API you use. By default, the Identity Provisioning service uses API version *1*.

- When the value is set to *1* or the property is not present (typical for systems created before versioning was introduced on November 6, 2025), the SOAP API provided by the communication scenario SAP_COM_0193 is used. In SAP S/4HANA Cloud Public Edition connector version 1, groups correspond to business roles, thus group members are user assignments of a business role.

  > ### Note
  >
  > Identity Provisioning cannot create and delete business roles in SAP S/4HANA Cloud Public Edition target system. It can only create, update and delete user assignments of a role. Therefore, business roles must have been created in SAP S/4HANA Cloud Public Edition system before you run a provisioning job.

  When SAP S/4HANA Cloud Public Edition is configured as a target system, the integration with a human resource (HR) system is active by default. In this case, the synchronization of business partners is managed by the HR system and SAP S/4HANA Cloud Public Edition. Identity Provisioning can only be used for managing business users (login users and their login information, such as date/time preferences) and role assignments.
  In SAP S/4HANA Cloud Public Edition, business partners are the central master data objects that hold the complete person profile in the SAP S/4HANA Cloud Public Edition. For example, business partners with role `Employee` and `Contingent Worker`. Business users are persons who can log on to SAP S/4HANA Cloud Public Edition system to complete business tasks.

- When the value is set to *3*, the SCIM interface provided by the communication scenario SAP_COM_0465 is used. It supports provisioning of **business users** with their **assignments** to user groups and roles.
  This version allows you to create, update, and delete user groups, referred to as business user groups in SAP S/4HANA Cloud Public Edition. However, similar to connector version 1, you cannot create or delete groups of type **authorization** (business roles in SAP S/4HANA Cloud Public Edition). You can only update the role assignments. For more information, see Groups.

  > ⓘ Note
  >
  > For groups of the **userGroup** type, a user can have only one assignment at a time. If a user is assigned to a new group of the **userGroup** type, the previous assignment of the same type will be automatically removed, and the user will be exclusively assigned to the newly selected group.

For more information on how to update to version *3*, see Update SAP S/4HANA Cloud Public Edition [page 2435].

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP S/4HANA Cloud Public Edition* as a target system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP S/4HANA Cloud Public Edition and configure your authentication method (basic or certificate-based).

   > ⓘ Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP S/4HANA target system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP S/4HANA backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Create a communication system and assign the created user to the communication system.

      For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

   d. Create a communication arrangement with the created system.

      - When `s4hana.cloud.api.version=1` is set in Identity Provisioning for the SAP S/4HANA Cloud Public Edition target system, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

> **Note**
>
> The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
>
> The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

- When `s4hana.cloud.api.version=3` is set in Identity Provisioning for the SAP S/4HANA Cloud Public Edition target system, choose *Scenario ID* SAP_COM_0465 (System for Cross-domain Identity Management Integration).

> **Note**
>
> The communication scenario *SAP_COM_0465* allows you to maintain data via SCIM (System for Cross-domain Identity Management). For more information, see Integrate Cross-Domain Identity Management.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the SAP S/4HANA Cloud Public Edition API URL. You can find the correct URL in the *API-URL* field of the communication arrangement set up for the communication scenario used. For example: `https://my123456-api.s4hana.ondemand.com`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* **Restriction** Do not use special symbol ',' (comma) as it is not supported.

Property Name	Description & Value
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`s4hana.cloud.api.version`	The version of the system API you use. By default, Identity Provisioning uses version *1*. Version *1* means your SAP S/4HANA Cloud Public Edition system consumes SAP S/4HANA Cloud Public Edition API: Business User. Version *3* means your SAP S/4HANA Cloud Public Edition system consumes System for Cross-domain Identity Management (SCIM) API.
`s4hana.cloud.hr.switch.active`	Defines whether the human resources (HR) integration is active or not. The property is set to `true` by default. This means that the synchronization of business partners is managed by the HR system and SAP S/4HANA Cloud Public Edition. When set to `false`, synchronization is handled by the identity management system instead. **Possible values:** • true (default) • false **Relevant for connector version 1**
`s4hana.cloud.hr.switch.dependent.role.codes`	A default property. As a comma-separated value, add the codes of the roles maintained by the HR integration. Make sure these role codes are part of your *read* and *write* transformations. By default, the following codes are added to your system: **BUP003 and BBP005**. That means, your HR integration will support *employees* and *contingent workers*. **Relevant for connector version 1**
`ips.date.variable.format`	*yyyy-MM-dd*

Property Name	Description & Value

`s4hana.cloud.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP S/4HANA Cloud Public Edition target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved.
	
	Note
	You can configure this property only when the integration between a human resource (HR) system and SAP S/4HANA Cloud Public Edition target system is **OFF**.
	
	The possible values of this property depend on the API version which your SAP S/4HANA Cloud Public Edition system consumes.
	
	Possible values:
	• If your system consumes SAP S/4HANA Cloud Public Edition API: Business User, the service searches for an existing user by *personExternalID* and *emails[0].value*.
	According to your use case, choose how to set up the property:
	  • Default behavior: This property does not appear in the UI during system creation. Its default value is *personExternalID*. That means, if the service finds an existing user by a *personExternalID* , it updates this user with the data of the conflicting one.
	If a user with such a *personExternalID* is not found, the creation of the conflicting user fails.
	  • Value = *emails[0].value*. If the service finds an existing user matching both unique attributes *email* and *personExternalID*, it updates this user with the data of the conflicting one. If the service finds an existing user matching only the *email*, the update of the existing user fails.
	If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails.
	• If your system consumes System for Cross-domain Identity Management (SCIM) API, the service searches for an existing user by *userName* , *emails[0].value*, *externalId*, or another SCIM attribute, or a conjunction of SCIM attributes.

Property Name	Description & Value
	According to your use case, choose how to set up the property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *externalId*. If the service finds an existing user with *externalId* , it updates this user with the data of the conflicting one. If a user with such *externalId* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. For more information, see List of Properties [page 80].
(Optional) `s4hana.cloud.group.prefix`	This property distinguishes SAP S/4HANA Cloud Public Edition groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **S4HANACLOUD_** You can use the example value or provide your own. When **set in the target system**, only groups containing the **S4HANACLOUD_** prefix in their display name will be provisioned to SAP S/4HANA Cloud Public Edition. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to SAP S/4HANA Cloud Public Edition. For more information, see List of Properties [page 80]. **Relevant for connector version 3**

Property Name	Description & Value
`s4hana.cloud.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists in the SAP S/4HANA Cloud Public Edition target system, the creation will fail. In this case, the existing group only needs to be updated. In SAP S/4HANA Cloud Public Edition the uniqueness of a group is defined by the combination of the following two attributes, which are automatically filled in when the target system is created: *externalId, ['urn:ietf:params:scim:schemas:extension:sap:2.0:Group'] ['type']*. To resolve the conflict, an existing group must match both of these attributes. If the group matches only one of the attributes, the conflict is not resolved, and group creation will fail. **Note** Although you can define your own unique combination of attributes (for example, *externalId,displayName*), we recommend that you do not modify the automatically filled value of the `s4hana.cloud.group.unique.attribute` property. If the property is not specified, the search is done by the default attribute *displayName*. **Relevant for connector version 3**

Property Name	Description & Value
`s4hana.cloud.patch.group.members.above.threshold`	Defines the threshold number of group members above which they are provisioned on batches with **PATCH** requests, and below which they are provisioned with **PUT** request. Setting this property allows you to avoid timeouts when updating groups with a large number of group members. The expected value is a positive integer without any separators, such as a space, comma, or period. **Possible values**: integer Default value: *20000* Minimum value: *1* Maximum value: *20000* **Note** If this property is not configured, the default value is applied automatically. The value must be greater than *0* to ensure proper provisioning of group members. For more information, see List of Properties [page 80]. **Note** Regardless of the threshold number you define, when removing group members in SAP S/4HANA Cloud Public Edition, the maximum number of members which can be removed per one **PATCH** request is 50. **Relevant for connector version 3**
`s4hana.cloud.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP S/4HANA Cloud Public Edition target system. - **true** – the current user roles will be deleted in the target system, and the user will be updated only with the roles provisioned by the service. - **false** – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the target system. See also: Extended Explanation of the *.user.roles.overwrite Properties **Relevant for connector version 1**

Property Name	Description & Value
(Optional) `s4hana.cloud.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: APIs for Business User Management If not specified, the default value is *false*. **Relevant for connector version 1**
(Optional) `s4hana.cloud.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to **true**, PATCH operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to **false**, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. Default value: *false* **Relevant for connector version 3**
(Optional) `s4hana.cloud.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of users to be provisioned per request. The default value, if not specified, is *20*. The maximum value is **100**. If you enter a number larger than 100, the service will replace it with the default value (20). **Relevant for connector version 1**
(Optional) `ips.delete.threshold.roles`	Use this property to control the number of user assignments of a role to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of user assignments of a role, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Property Name	Description & Value
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*https://my1234567-api.s4hana.ondemand.com*

`User`=*MyS4HANAuser*

`Password`=**********

`s4hana.cloud.api.version`=*1*

`ips.date.variable.format`=*yyyy-MM-dd*

`s4hana.cloud.hr.switch.active`=*true*

`s4hana.cloud.hr.switch.dependent.role.codes`=*BUP003,BBP005*

`s4hana.cloud.user.roles.overwrite` = *false*

`s4hana.cloud.support.bulk.operation` = *true*

`s4hana.cloud.bulk.operations.max.count` = *50*

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. Identity Provisioning offers a default transformation for the *SAP S/4HANA Cloud Public Edition* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules depending on your setup of entities in SAP S/4HANA Cloud Public Edition. For more information, see:

   Manage Transformations [page 2439]

   SAP S/4HANA Cloud API: Business User

   System for Cross-domain Identity Management (SCIM)

   Integrate Cross-Domain Identity Management

Default transformation for connector version 1:

Code Syntax

```
{
  "user": {
    "condition": "('%s4hana.cloud.roles.prefix%' === 'null') || ($.groups[?
(@.display =~ /%s4hana.cloud.roles.prefix%.*/)] empty false)",
    "mappings": [
      {
        "sourcePath": "$.userName",
        "targetPath": "$.personExternalID"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.personExternalID",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath": "$.personExternalID"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$.user.globalUserID"
      },
      {
        "targetPath": "$.personID",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "targetPath": "$.markedForArchivingIndicator",
        "constant": "false"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.user.timeZoneCode",
        "functions": [
          {
            "type": "convertTimezoneCode",
            "outputFormat": "sap"
          }
        ],
        "defaultValue": "CET"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.userType"
        ],
        "targetPath": "$.businessPartnerRoleCode",
        "defaultValue": "BUP003",
        "valueMappings": [
          {
            "key": [
              "Employee"
            ],
            "mappedValue": "BUP003"
          },
          {
```

```
            "key": [
              "Contingent Worker"
            ],
            "mappedValue": "BBP005"
          }
        ]
      },
      {
        "scope": "createEntity",
        "sourceVariable": "currentDate",
        "targetPath": "$.user.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
        "targetPath": "$.user.validityPeriod.endDate"
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.personalInformation.firstName",
        "optional": true
      },
      {
        "condition": "%s4hana.cloud.hr.switch.active% != null &&
%s4hana.cloud.hr.switch.active% == true",
        "optional": true,
        "sourcePath": "$.name.familyName",
        "targetPath": "$.personalInformation.lastName"
      },
      {
        "condition": "%s4hana.cloud.hr.switch.active% == null ||
%s4hana.cloud.hr.switch.active% == false",
        "sourcePath": "$.name.familyName",
        "targetPath": "$.personalInformation.lastName"
      },
      {
        "sourcePath": "$.name.middleName",
        "targetPath": "$.personalInformation.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.name.formatted",
        "targetPath": "$.personalInformation.personFullName",
        "optional": true
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.user.userName"
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.user.logonLanguageCode",
        "optional": true
      },
      {
        "sourcePath": "$.emails[0].value",
        "targetPath": "$.workplaceInformation.emailAddress",
        "optional": true
      },
      {
        "targetPath": "$.user.lockedIndicator",
        "constant": "false"
      },
      {
        "condition": "$.active == false",
        "targetPath": "$.user.lockedIndicator",
        "constant": "true"
      }
```

```
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
s4hana.cloud.roles.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
s4hana.cloud.roles.prefix) && (isRegularGroup() ||
isApplicationSpecificGroup())",
    "mappings": [
      {
        "sourcePath": "$.displayName",
        "targetVariable": "entityIdTargetSystem",
        "scope": "createEntity",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
s4hana.cloud.roles.prefix)",
            "function": "replaceFirstString",
            "regex": "%s4hana.cloud.roles.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
s4hana.cloud.roles.prefix)",
            "function": "replaceFirstString",
            "regex": "%s4hana.cloud.roles.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

See also: nullExtended Explanation of the *user.roles.overwrite Properties

Default transformation for connector version 3:

Code Syntax

```
{
  "user": {
    "condition": "(('%s4hana.cloud.group.prefix%' === 'null') ||
($.groups[?(@.display =~ /%s4hana.cloud.group.prefix%.*/)] empty false))",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
```

```
        "targetPath": "$.id"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:core:2.0:User","urn:ietf:params:scim:schemas
:extension:sap:2.0:User"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "functions": [
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.externalId",
        "functions": [
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$.externalId",
        "optional": true,
        "functions": [
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true,
        "functions": [
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath": "$.name.formatted",
        "optional": true,
        "targetPath": "$.name.formatted"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
```

```
        "sourcePath": "$.displayName",
        "optional": true,
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails",
        "functions": [
          {
            "function": "putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "sourcePath": "$.userType",
        "optional": true,
        "targetPath": "$.userType"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.preferredLanguage"
      },
      {
        "sourcePath": "$.preferredLanguage",
        "optional": true,
        "targetPath": "$.preferredLanguage"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['language']['correspondence']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['language']['correspondence']",
```

```
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['numericalRepresentation']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['numericalRepresentation']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['date']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['date']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['time']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['localeSettings']['time']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['userUuidHistory']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
['userUuidHistory']",
        "optional": true,
        "preserveArrayWithSingleElement": true
      },
      {
        "sourcePath": "$.groups[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups[?(@.value)]",
        "functions": [
          {
            "entityType": "group",
            "type": "resolveEntityIds"
          }
        ]
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
s4hana.cloud.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
s4hana.cloud.group.prefix) && (isRegularGroup() ||
isApplicationSpecificGroup())",
    "mappings": [
      {
```

```
        "targetPath": "$.id",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:core:2.0:Group","urn:ietf:params:scim:schema
s:extension:sap:2.0:Group"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
s4hana.cloud.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%s4hana.cloud.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.externalId",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
s4hana.cloud.group.prefix)",
            "function": "replaceFirstString",
            "regex": "%s4hana.cloud.group.prefix%",
            "replacement": ""
          },
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true,
        "functions": [
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath": "$.members[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.members[?(@.value)]",
        "functions": [
          {
            "entityType": "user",
            "type": "resolveEntityIds"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']",
        "optional": true,
        "targetPath": "$.externalId",
        "functions": [
```

```
          {
            "function": "toUpperCaseString",
            "locale": "en_EN"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true,
        "defaultValue": "userGroup"
      },
      {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

Group Uniqueness and Provisioning

In the most common scenario, Identity Authentication connector version 2 or Local Identity Directory is used as the source system for groups and SAP S/4HANA Cloud Public Edition connector version 3 is the target system.

In SAP S/4HANA Cloud Public Edition version 3, the uniqueness of a group is defined by the combination of `externalId` and `type` attributes. The system supports two group types: `userGroup` and `authorization` (role). As a result, two different groups can share the same `externalId` while belonging to different types, potentially causing conflicts when provisioned to the target system.

To avoid conflicts, ensure the following:

- Your SAP S/4HANA Cloud Public Edition target system must be updated to version 3, as described in Update SAP S/4HANA Cloud Public Edition [page 2435] and it must have the group uniqueness property: `s4hana.cloud.group.unique.attribute`, set with the following combination of values: `externalId,['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']`. For more information, see List of Properties [page 80].
- The groups in your Identity Authentication connector version 2 or Local Identity Directory source systems must have a value for the `externalName` attribute, which is mapped to the `externalId` attribute.

6. Now, add a source system from which to read users and roles. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].

- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP S/4HANA Cloud Documentation

1.6.2.40 SAP S/4HANA for procurement planning

Follow this procedure to set up SAP S/4HANA for procurement planning as a target system.

Prerequisites

You have technical credentials for SAP S/4HANA for procurement planning. See: Onboadring

Context

SAP S/4HANA for procurement planning is a cloud-based solution designed to help you plan procurement activities with regard to the time schedule, as well as the investment planning of items based on a central bill of material.

You can use Identity Provisioning to configure SAP S/4HANA for procurement planning as a target system where you can provision **users** from source systems.

> **Note**
>
> SAP S/4HANA for procurement planning does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP S/4HANA Procurement Planning* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the SCIM API of your SAP S/4HANA for procurement planning system without path information. For example: `https://procplanning-api.cfapps.eu10.hana.ondemand.com`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, created for your SAP S/4HANA for procurement planning system.
`Password`	Enter the OAuth Client Secret, created for your SAP S/4HANA for procurement planning system.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. For example: **https://procplansecurity.authentication.eu10.hana.ondemand.com/oauth/token**

Property Name	Value
`s4hana.pp.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). **If the service finds such a user on the target system via this filter, then the conflicting user will overwrite the existing one.** If the service does not find such a user, the creation will fail. According to your use case and system type, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value* • *userName,emails[0].value* • *externalId*, or another SCIM attribute, or a conjunction of SCIM attributes Default value: *userName*
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP S/4HANA for procurement planning* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP S/4HANA for procurement planning system. For more information, see:

   Manage Transformations [page 2439]

   SAP Business Accelerator Hub: SAP S/4HANA for Procurement Planning

   Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP S/4HANA for procurement planning entity.

   Default transformation:

   Code Syntax

```
{
    "user": {
        "mappings": [
             {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
             },
             {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
             },
             {
                 "sourcePath": "$.displayName",
                 "targetPath": "$.displayName"
             },
             {
                 "sourcePath": "$.active",
                 "targetPath": "$.active"
             },
             {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails"
            },
            {
                "condition": "$.emails[0].length() > 0",
                "targetPath": "$.emails[0].primary",
                "constant": true
            },
            {
                "targetPath": "$.id",
                "type": "remove"
            },
            {
                "type": "remove",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']"
            }
        ]
    }
}
```

5. Add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP S/4HANA for procurement planning – Product Page

1.6.2.41 SAP S/4HANA On-Premise

Follow the procedure to set up SAP S/4HANA on-premise (also valid for SAP S/4HANA Cloud, private edition) as a target system.

Prerequisites

> ⓘ Note
>
> If you have purchased the Identity Provisioning service between **September 1, 2020** and **October 20, 2020**, and you want to make a connection to this on-premise system, follow the procedure in: Connect to On-Premise Systems in SAP Cloud Identity Infrastructure.

- You have installed the Cloud Connector in your corporate environment and have done the initial configuration. For more information, see: Cloud Connector (Neo)
- You have technical credentials (user and password) for SAP S/4HANA on-premise.
- The SAP S/4HANA on-premise system is version **1809** or higher.
- You have configured your SOA Manager to directly call the following Web services:
  - **ManageBusinessUserIn**
  - **QueryBusinessUserIn**

  For more information, see: Setting Up SOA Manager.

Context

You can use Identity Provisioning to configure SAP S/4HANA on-premise as a target system where you can provision **users** from source systems.

This scenario requires that you first provision the users to SAP AS ABAP. The created ABAP users will have PFCG roles assigned. Once this is done, the users will have ABAP IDs. Then, after the provisioning to SAP S/4HANA on-premise, business users will be created and linked to the relevant ABAP IDs via the *User_Assignment* attribute.

In SAP S/4HANA, a business user is defined as a natural person who is represented by a business partner and a link to a user in the system. Actually, the business user is an AS ABAP (SU01) user who also has a one-to-one relation to a corresponding business partner. For more information on the identity model for business users, see SAP Note 2570961.

SAP S/4HANA on-premise supports provisioning of users with *User UUID* attribute which is generated by Identity Authentication at user creation. The attribute mapping is handled by the default transformation of AS ABAP connector. For more information, see: SAP Application Server ABAP [page 987].

Provisioning of **roles** (considered as *groups*) is also handled by the default transformation of SAP AS ABAP connector.

According to your use case, you can decide whether to use SAP S/4HANA on-premise with HR (human resources) integration active or not.

- System with HR integration – When the HR integration is active, business users in SAP S/4HANA on-premise are automatically created and managed by the HR system (for example, an external data source, such as an identity management system). The Identity Provisioning service can manage only the user-related login information, such as date/time preferences, or role assignments.
  This means that once the users are provisioned to AS ABAP, you need to run the provisioning to SAP S/4HANA on-premise, so that business users (employee, collaboration user, contingent worker, resource) can be linked to the AS ABAP users via the *User_Assignment* attribute.

  > ### Note
  >
  > Identity Provisioning cannot delete business users in SAP S/4HANA Cloud on-premise system with active HR integration. Deletion is possible only in case of erroneous user assignments to business partners. For more information, see 2568251.

- System without HR integration – lean business users will be created after the provisioning job, and the AS ABAP users will be linked to them. The Identity Provisioning service manages the complete set of **business partners** and their relevant **business users**.

To enable HR integration, you need to specify the relevant property in the system configuration. See **step 4** from the procedure below.

Procedure

1. Open the Cloud Connector to add an access control system mapping for **SAP S/4HANA On-Premise**. This is needed to allow the Identity Provisioning service to access SAP S/4HANA On-Premise as a back-end system on the intranet. To learn how, see: Configure Access Control (HTTP)

2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
3. Add *SAP S/4HANA On-Premise* as a target system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to your SAP S/4HANA On-Premise system.
`ProxyType`	Enter: *OnPremise*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the technical user for SAP S/4HANA On-Premise.
`Password`	(Credential) Enter the password for the SAP S/4HANA On-Premise technical user.
`s4hana.onprem.hr.switch.active`	This property is enabled by default. Possible values: • **true** (default value) – HR integration is enabled for your system • **false** – HR integration is disabled for your system
`s4hana.onprem.hr.switch.dependent.role.codes`	A default property. Relevant only for systems with activated HR integration, that is if `s4hana.onprem.hr.switch.active` = *true*. As a comma-separated value, add the codes of the roles maintained by the HR integration. Make sure these role codes are part of your *read* and *write* transformations. By default, the following codes are added to your system: **BUP003, BBP005, BUP012, WFM001**. That means, your HR integration will support *employees*, *contingent worker*, *collaboration user*, and *resource*.
`ips.date.variable.format`	*yyyy-MM-dd*

Property Name	Description & Value
`s4hana.onprem.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the SAP S/4HANA On-Premise target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. According to your use case, choose how to set up this property: • Default behavior: This property does not appear in the UI during system creation. Its default value is *personExternalID*. That means, if the service finds an existing user by a *personExternalID*, it updates this user with the data of the conflicting one. If a user with such a *personExternalID* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user matching both unique attributes *email* and *personExternalID*, it updates this user with the data of the conflicting one. If the service finds an existing user matching only the *email*, the update of the existing user fails. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. Possible values: • *personExternalID* • *emails[0].value* Default value: *personExternalID*
(Optional)`s4hana.onprem.sap-client`	Use this property if you want to specify a particular AS ABAP client to use as the **sap-client** URL parameter. If this property is not specified, the URL will open your default AS ABAP client. To learn more, see: Specifying the Client For more information about **sap-client**, see: SAP URL Parameters
(Optional)`CloudConnectorLocationId`	Relevant when the `ProxyType` property is set to *OnPremise*. Use it in case your SAP Business Technology Platform account uses more than one Cloud Connector or when the Cloud Connector uses `LocationID`.

Property Name	Description & Value
(Optional) `s4hana.onprem.support.bulk.operation`	Set this property to *true* if you want to enable bulk operations for provisioning entities. That means, the Identity Provisioning service can write, update, and delete multiple users or groups in a single request. For more information, see: APIs for Business User Management If not specified, the default value is *false*.
(Optional) `s4hana.onprem.bulk.operations.max.count`	If you have enabled the bulk operations, you can use this property to set the number of users to be provisioned per request. The default value, if not specified, is *20*. The maximum value is **100**. If you enter a number larger than 100, the service will replace it with the default value (20).
(Optional) `ips.delete.threshold.roles`	Use this property to control the number of roles to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of roles, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=OnPremise
URL=http://aaa777.myhost:1234
User=MYS4HANAUSER
Password=************
ips.date.variable.format=yyyy-MM-dd
s4hana.onprem.hr.switch.active=true
s4hana.onprem.hr.switch.dependent.role.codes=BUP003,BBP010,BBP005
s4hana.onprem.sap-client=101
s4hana.onprem.support.bulk.operation = true
s4hana.onprem.bulk.operations.max.count = 50
```

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP S/4HANA On-Premise* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules depending on your setup of entities in SAP S/4HANA On-Premise. For more information, see:

   Manage Transformations [page 2439]

   APIs for Business User Management

   Default transformation:

   Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.userName",
        "targetPath": "$.personExternalID"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.personExternalID",
        "optional": true
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$.personExternalID",
```

```
        "optional": true
      },
      {
        "targetPath": "$.personID",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.userType"
        ],
        "targetPath": "$.businessPartnerRoleCode",
        "defaultValue": "BUP003",
        "valueMappings": [
          {
            "key": [
              "Employee"
            ],
            "mappedValue": "BUP003"
          },
          {
            "key": [
              "Contingent Worker"
            ],
            "mappedValue": "BBP005"
          },
          {
            "key": [
              "Collaboration User"
            ],
            "mappedValue": "BUP012"
          },
          {
            "key": [
              "Resource"
            ],
            "mappedValue": "WFM001"
          }
        ]
      },
      {
        "scope": "createEntity",
        "targetPath": "$.validityPeriod.startDate",
        "sourceVariable": "currentDate"
      },
      {
        "scope": "createEntity",
        "targetPath": "$.validityPeriod.endDate",
        "constant": "9999-12-31"
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.personalInformation.firstName",
        "optional": true
      },
// The following conditions refer to HR integration for your SAP S/4HANA
Cloud system. If HR integration is activated (i.e. property
// s4hana.cloud.hr.switch.active is set to true), then you don't need to
provide family name for the users.
// If it's deactivated (i.e. property s4hana.cloud.hr.switch.active is
missing or set to false,
// then you have to provide family name. You can apply these conditions to
different user attributes, analogically to name.familyName.
      {
        "condition": "%s4hana.onprem.hr.switch.active% != null &&
%s4hana.onprem.hr.switch.active% == true",
        "optional": true,
        "sourcePath": "$.name.familyName",
```

```
        "targetPath": "$.personalInformation.lastName"
      },
      {
        "condition": "%s4hana.onprem.hr.switch.active% == null ||
%s4hana.onprem.hr.switch.active% == false",
        "sourcePath": "$.name.familyName",
        "targetPath": "$.personalInformation.lastName"
      },
      {
        "sourcePath": "$.name.middleName",
        "targetPath": "$.personalInformation.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.name.formatted",
        "targetPath": "$.personalInformation.personFullName",
        "optional": true
      },
      {
        "sourcePath": "$.nickName",
        "targetPath": "$.personalInformation.nickName",
        "optional": true
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userAssignment.userID",
        "optional": true,
        "functions": [
          {
            "function": "toUpperCaseString"
          }
        ]
      },
      {
        "sourcePath": "$.emails[0].value",
        "targetPath": "$.workplaceInformation.emailAddress",
        "optional": true
      }
    ]
  }
}
```

6. Now, add a source system from which to read users and roles. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP S/4HANA On-Premise
APIs for Business User Management

1.6.2.42 SAP SuccessFactors

Follow this procedure to set up SAP SuccessFactors as a target system.

Prerequisites

- You are using **SAP SuccessFactors HCM Suite OData API**
  You have created a technical user with permissions to **call** the SAP SuccessFactors HCM Suite OData API to **import** employee data into an SAP SuccessFactors system. For more information, see Permissions, Setting Up an API User for Sync Jobs in SAP SuccessFactors and URI Conventions (OData Version 2.0) .
- You are using **SAP SuccessFactors Workforce SCIM API**
  Consumers of the SAP SuccessFactors Workforce SCIM API can either create a technical user and assign the necessary permissions, or use the predefined technical user with built-in permissions for communication between SAP SuccessFactors and Identity Provisioning using certificate authentication. For more information, see Upgrade to X.509 Certificate-Based Authentication for Incoming Calls and Permissions.

Context

SAP SuccessFactors provides two APIs for its integration with Identity Provisioning: SAP SuccessFactors HCM Suite OData API and SAP SuccessFactors Workforce SCIM API. The value of `sf.api.version` property controls which API you use.

- When the value is set to **1**, or the property is not defined - SAP SuccessFactors HCM Suite OData API (in short, OData API) is used. This is the default value. SAP SuccessFactors source systems created before the introduction of `sf.api.version` property, use OData API.
  This version allows you to create and update **users**, as well as update **dynamic groups** and **group members**. To update a group from the source system, a group with the same name should already exist in SAP SuccessFactors.

  > ⚠ Restriction
  >
  > Note the following restrictions when using SAP SuccessFactors HCM Suite OData API:
  >
  > - You cannot *create* or *delete* groups as these operations are currently not supported.
  > - Managing SAP SuccessFactors **static groups** is not supported.

- When the value is set to **2** - SAP SuccessFactors Workforce SCIM API (in short, SCIM API) is used.
  This version allows you to provision static permission groups and user's group assignments. Provisioning of user's group assignments from a given source system to SAP SuccessFactors target system is possible if the user is an active employee with a work assignment and has a value of the `perPersonUuid` attribute. Alternatively, `personIdExternal` could be used as a fallback attribute.

This requires that SAP SuccessFactors users are first provisioned with the attribute to the given system, for example Identity Authentication, where the `perPersonUuid` must be mapped to a custom attribute. Afterwards, assigning users to groups in Identity Authentication and running a provisioning job would result in correct user assignments in SAP SuccessFactors target.
To update a group from the source system, a group with the same name should already exist in SAP SuccessFactors.

> **⚠ Restriction**
>
> Note the following restrictions when using SAP SuccessFactors Workforce SCIM API:
>
> - You cannot *create* or *delete* groups using the `Groups` APIs.
> - The `Groups` patch API only supports updating membership of static permission groups.
>
> For more information, see Overview of SAP SuccessFactors Workforce System for Cross-Domain Identity Management API.

For more information about the difference between static and dynamic groups in SAP SuccessFactors, see Permission Groups.

For more information on how to update to version 2, see Update Connector Version [page 2424].

You can configure *SAP SuccessFactors* as a target system to provision entities from a given source system. With regards to user termination, it is a standard HR process that is triggered from the SAP SuccessFactors system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP SuccessFactors* as a target system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP SuccessFactors and configure your authentication method (basic or certificate-based).

   > **ⓘ Note**
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP SuccessFactors target system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip this step if you use basic authentication. The next steps are performed in SAP SuccessFactors Admin Center and are relevant for certificate-based authentication only.

   b. Login to SAP SuccessFactors and go to *Admin Center*. Follow the procedure described in Upgrade to X.509 Certificate-Based Authentication for Incoming Calls.

      Make sure you select *Identity Provisioning Service* in the *Integration Name* field.
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*

URL	Specify the URL to your SAP SuccessFactors API.
	
	For example:
	• For version 1:
	  • When *BasicAuthentication* is configured as authentication method: **`https://apitest.successfactors.com/odata/v2`**
	  • When *ClientCertificateAuthentication* is configured as authentication method: **`https://apitest.cert.successfactors.com/odata/v2`**
	• For version 2:
	  • When *BasicAuthentication* is configured as authentication method: **`https://apitest.successfactors.com`**
	  • When *ClientCertificateAuthentication* is configured as authentication method: **`https://apitest.cert.successfactors.com`**
	
	To see the list of all SAP SuccessFactors data centers, see: SAP SuccessFactors API Reference Guide (OData V2): List of SAP SuccessFactors API Servers and System for Cross-domain Identity Management for Workforce in SuccessFactors

`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*

Property Name	Description & Value
(Optional) `sf.api.version`	Handles the version of the API which is consumed by the SAP SuccessFactors system. **Possible values:** • *1* - Indicates that SAP SuccessFactors HCM Suite OData API (in short, OData API) is used. • *2* - Indicates that SAP SuccessFactors Workforce SCIM API (in short, SCIM API) is used. Default value: *1*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *userID* of your SAP SuccessFactors technical user in the following format: **<user_ID>@<company_ID>**
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the password for your SAP SuccessFactors technical user.
`sf.company.id`	Valid if *ClientCertificateAuthentication* is configured as authentication method. Enter the Company ID of your SAP SuccessFactors system. The Company ID is a short string of characters that identifies each SAP SuccessFactors system. It is like a username for your organization. All users of the same system share the same Company ID.
`sf.user.attributes`	Default property. It's a string representing a comma-separated list of user attributes by which the Identity Provisioning service retrieves the users that have to be updated in SAP SuccessFactors during the writing process. You can leave the default property value (all listed attributes), or leave only some of them. → Remember Always make sure that attribute `lastModifiedDateTime` is in the list. If you don't specify it, the provisioning to SAP SuccessFactors will fail. **Connector version**: SAP SuccessFactors version 1

Property Name	Description & Value
`sf.user.unique.attribute`	When the Identity Provisioning attempts to provision a user for the first time, it may detect that such a user already exists on the target system. Thus, the service needs to retrieve the *entityId* of the existing user via filtering by user unique attribute(s). This property defines by which unique attribute(s) the existing user to be searched (resolved). **If the service finds such a user on the target system via this filter, then the conflicting user will overwrite the existing one.** If the service does not find such a user, the creation will fail. According to your use case and system type, choose how to set up this property: • Default behavior: This property is missing during system creation. Its default value is *userName*. That means, if the service finds an existing user by a *userName*, it updates this user with the data of the conflicting one. If a user with such a *userName* is not found, the creation of the conflicting user fails. • Value = *emails[0].value*. If the service finds an existing user with such *email*, it updates this user with the data of the conflicting one. If a user with such *email* is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. • Value = *userName,emails[0].value*. If the service finds an existing user with both these *userName* and *email*, it updates this user with the data of the conflicting one. If such a user is not found, that means the conflict is due to another reason, so the creation of the conflicting user fails. **Possible values:** • *userName* • *emails[0].value* • *userName,emails[0].value* • *externalId*, or another SCIM attribute, or a conjunction of SCIM attributes Default value: *userName* **Connector version**: SAP SuccessFactors version 2
(Optional) `sf.user.attributes.expand`	This property writes additional user data related to *complex attributes*, which are specified in `sf.user.attributes`. Default value: *personKeyNav* **Connector version**: SAP SuccessFactors version 1

Property Name	Description & Value
`sf.group.unique.attribute`	If the service tries to create a group that already exists in the target system, the creation will fail. In this case, the existing group only needs to be updated. This group can be found via search, based on an attribute (default or specific). To make the search filter by a specific attribute, specify this attribute as a value for the `sf.group.unique.attribute` property. If the property is not specified, the search is done by the default attribute: *displayName* **Connector version**: SAP SuccessFactors version 2
(Optional) `sf.group.prefix`	This property distinguishes SAP SuccessFactors groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **SF_** You can use the example value or provide your own. When **set in the target system**, only groups containing the **SF_** prefix in their display name will be provisioned to SAP SuccessFactors. Groups without this prefix in the display name won't be provisioned. If the property is not set, the SAP SuccessFactors groups will be read and provisioned to the target system with their actual display names. **Connector version**: SAP SuccessFactors version 2

Property Name	Description & Value
(Optional) `sf.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users in the target system. This means, for example, that if a user attribute is modified, only this change will be provisioned and applied in the target system. • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false* **Connector version**: SAP SuccessFactors version 2
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination (configuration):

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*https://apitest.successfactors.com/odata/v2*

`User`=*sfsf_admin@mycompany.com*

`Password`=**********

`sf.user.attributes`=*userId,username,addressLine1,jobTitle,lastName,country,email,location,firstName,lastModifiedDateTime,personKeyNav,manager/username*

`sf.user.attributes.expand`=*personKeyNav,manager*

---

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP SuccessFactors* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   When you configure SAP SuccessFactors as a target system, the Identity Provisioning service will read all user attributes from the selected source system and write them as user records in SAP SuccessFactors. Groups can be only updated if such already exist in SAP SuccessFactors.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP SuccessFactors. For more information, see:

   Manage Transformations [page 2439]

   SAP SuccessFactors HCM Suite OData API

   SAP SuccessFactors Workforce SCIM API

   To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

   Default transformation for SAP SuccessFactors HCM Suite OData API version 1:

   Code Syntax

```
{
    "user": {
        "skipOperations": [
            "delete"
        ],
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.userId"
            },
```

```
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userId",
                "scope": "createEntity"
            },
            {
                "constant": "t",
                "targetPath": "$.status"
            },
            {
                "condition": "$.active == false",
                "constant": "f",
                "targetPath": "$.status"
            },
            {
                "sourcePath": "$.userName",
                "optional": true,
                "targetPath": "$.username"
            },
            {
                "sourcePath": "$.name.familyName",
                "optional": true,
                "targetPath": "$.lastName"
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.firstName"
            },
            {
                "sourcePath": "$.name.honorificPrefix",
                "optional": true,
                "targetPath": "$.salutation"
            },
            {
                "sourcePath": "$.name.honorificSuffix",
                "optional": true,
                "targetPath": "$.suffix"
            },
            {
                "sourcePath": "$.emails[0].value",
                "optional": true,
                "targetPath": "$.email"
            },
            {
                "condition": "$.emails[?(@.primary == true)].value != []",
                "sourcePath": "$.emails[?(@.primary == true)].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.email",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "sourcePath": "$.timezone",
                "optional": true,
                "targetPath": "$.timeZone"
            },
            {
                "sourcePath": "$.nickName",
                "optional": true,
                "targetPath": "$.nickname"
            },
            {
```

```
                "sourcePath": "$.addresses[0].country",
                "optional": true,
                "targetPath": "$.country"
            },
            {
                "condition": "$.addresses[?(@.primary == true)].country !=
[]",
                "sourcePath": "$.addresses[?(@.primary == true)].country",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.country",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "sourcePath": "$.addresses[0].locality",
                "optional": true,
                "targetPath": "$.city"
            },
            {
                "condition": "$.addresses[?(@.primary == true)].locality !
= []",
                "sourcePath": "$.addresses[?(@.primary == true)].locality",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.city",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "sourcePath": "$.addresses[0].formatted",
                "optional": true,
                "targetPath": "$.addressLine1"
            },
            {
                "sourcePath": "$.addresses[1].formatted",
                "optional": true,
                "targetPath": "$.addressLine2"
            },
            {
                "sourcePath": "$.addresses[2].formatted",
                "optional": true,
                "targetPath": "$.addressLine3"
            },
            {
                "condition": "$.phoneNumbers[?(@.type == 'work')].value !=
[]",
                "sourcePath": "$.phoneNumbers[?(@.type == 'work')].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.businessPhone",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
```

```
                "condition": "$.phoneNumbers[?(@.type == 'fax')].value !=
[]",
                "sourcePath": "$.phoneNumbers[?(@.type == 'fax')].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.fax",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "condition": "$.phoneNumbers[?(@.type == 'mobile')].value !
= []",
                "sourcePath": "$.phoneNumbers[?(@.type ==
'mobile')].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.cellPhone",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "condition": "$.phoneNumbers[?(@.type == 'home')].value !=
[]",
                "sourcePath": "$.phoneNumbers[?(@.type == 'home')].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.homePhone",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['employeeNumber']",
                "optional": true,
                "targetPath": "$.empId"
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['division']",
                "optional": true,
                "targetPath": "$.division"
            },
            {
                "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']
['department']",
                "optional": true,
                "targetPath": "$.department"
            }
        ]
    },
// By default, the group mapping is inactive (ignored) but groups are
supported.
```

```
// To start provisioning groups, either delete the statement "ignore":
true, or set its value to false.
    "group": {
        "ignore": true,
        "skipOperations": [
            "delete"
        ],
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.groupID"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.groupName"
            },
            {
                "constant": "DynamicGroup",
                "targetPath": "$.__metadata.uri"
            },
            {
                "constant": "permission",
                "targetPath": "$.groupType"
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "type": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

Default transformation for SCIM API version 2:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [
          "urn:ietf:params:scim:schemas:extension:successfactors:2.0:User",
          "urn:ietf:params:scim:schemas:core:2.0:User",
          "urn:ietf:params:scim:schemas:extension:sap:2.0:User",
          "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
```

```
        "sourcePath": "$.userType",
        "targetPath": "$.userType"
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "condition": "$.emails.length() > 0",
        "targetPath": "$.emails[*].type",
        "constant": "work"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.name.familyName"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.name.givenName"
      },
      {
        "sourcePath": "$.name.middleName",
        "optional": true,
        "targetPath": "$.name.middleName"
      },
      {
        "sourcePath": "$.name.honorificPrefix",
        "optional": true,
        "targetPath": "$.name.honorificPrefix"
      },
      {
        "sourcePath": "$.name.honorificSuffix",
        "optional": true,
        "targetPath": "$.name.honorificSuffix"
      },
      {
        "sourcePath": "$.name.formatted",
        "optional": true,
        "targetPath": "$.name.formatted"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['perPersonUuid']",
        "optional": true,
```

```
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['perPersonUuid']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['loginMethod']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['loginMethod']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['personIdExternal']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['personIdExternal']"
      },
      {
        "sourcePath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['customFields']",
        "optional": true,
        "targetPath":
"$['urn:ietf:params:scim:schemas:extension:successfactors:2.0:User']
['customFields']"
      },
      {
        "sourcePath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.phoneNumbers"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.groups[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups[?(@.value)]",
        "functions": [
          {
            "entityType": "group",
            "function": "resolveEntityIds"
          }
        ]
      }
    ]
  },
  "group": {
    "condition": "('%sf.group.prefix%' === 'null') || ($.displayName =~ /
%sf.group.prefix%.*/)",
    "skipOperations": [
      "delete"
    ],
    "mappings": [
      {
        "targetPath": "$.id",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "sourcePath": "$.displayName",
```

```
        "targetPath": "$.displayName",
        "scope": "createEntity",
        "functions": [
          {
            "condition": "('%sf.group.prefix%' !== 'null') && (@ =~ /
%sf.group.prefix%.*/)",
            "function": "replaceFirstString",
            "regex": "%sf.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "targetPath": "$.members",
        "type": "remove"
      },
      {
        "sourcePath": "$.members[*].value",
        "targetPath": "$.members[?(@.value)]",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "functions": [
          {
            "type": "resolveEntityIds"
          }
        ]
      }
    ]
  }
}
```

6. Now, add a source system to read users from it. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

URI Conventions (OData Version 2.0)
SAP SuccessFactors HCM Suite OData API
SAP SuccessFactors Workforce SCIM API

1.6.2.43 SAP SuccessFactors Employee Central Payroll

Follow this procedure to set up SAP SuccessFactors Employee Central Payroll as a target system.

Context

SAP SuccessFactors Employee Central Payroll is a cloud product for payroll processing. It's a service offering where SAP SuccessFactors Employee Central is used for maintenance of core HR master data and an SAP HCM Payroll system hosted in an SAP or Hyperscaler data center is used for payroll.

You can use Identity Provisioning to configure SAP SuccessFactors Employee Central Payroll as a target system where you provision users and groups from source systems.

> **Note**
>
> When executing a provisioning job, by default users with no `name.familyName` attribute will be skipped, and thus, won't be created in SAP SuccessFactors Employee Central Payroll.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP SuccessFactors Employee Central Payroll* as a target system. For more information, see Add New Systems [page 2415].
3. Set up the communication between Identity Provisioning and SAP SuccessFactors Employee Central Payroll and configure your authentication method (basic or certificate-based).

   > **Note**
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP SuccessFactors Employee Central Payroll target system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next step is performed in SAP SuccessFactors Employee Central Payroll backend system and is relevant for both basic and certificate-based authentication.

   b. For certificate-based authentication, import the certificate generated in the Identity Provisioning UI in your SAP SuccessFactors Employee Central Payroll system, as described in Prerequisite: Finding the Access URL → step 5.

      For basic authentication, provide *User* and *Password*, as described in Prerequisite: Finding the Access URL → step 5.

c. Save your changes.

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP SuccessFactors Employee Central Payroll system. For more information, see Prerequisite: Finding the Access URL → step 8.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication* For more information, see Prerequisite: Finding the Access URL → step 5.
`User`	Valid if *BasicAuthentication* is configured as authentication method. For more information, see Prerequisite: Finding the Access URL → step 5.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. For more information, see Prerequisite: Finding the Access URL → step 5.
`ecp.sap-client`	Enter the SAP client number (a three-digit number) of your SAP SuccessFactors Employee Central Payroll system. For example: **`102`**
(Optional) `ecp.user.unique.attribute`	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. Possible values: *userName* (default value)
(Optional) `ecp.group.unique.attribute`	If the Identity Provisioning tries to create a group that already exists in the target system (a conflicting group), this property defines the unique attributes by which the existing group will be searched and resolved. **Possible values:** *displayName* (default value)

Property Name	Value
(Optional) `ecp.group.prefix`	This property distinguishes SAP SuccessFactors Employee Central Payroll groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **ECP_** You can use the example value or provide your own. When **set in the target system**, only groups containing the **ECP_** prefix in their display name will be provisioned to SAP SuccessFactors Employee Central Payroll. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be be provisioned to SAP SuccessFactors Employee Central Payroll.
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP SuccessFactors Employee Central Payroll* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP SuccessFactors Employee Central Payroll system. For more information, see:

   Manage Transformations [page 2439]

   Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP SuccessFactors Employee Central Payroll entity.

   Default transformation:

Code Syntax

```
{
    "user": {
        "condition": "($.name.familyName EMPTY false) &&
(('%ecp.group.prefix%' === 'null') || ($.groups[?(@.display =~ /
%ecp.group.prefix%.*/)] empty false))",
        "mappings": [
            {
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:User",
                    "urn:ietf:params:scim:schemas:extension:sap:2.0:User"
                ],
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.schemas"
            },
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName",
                "optional": true
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName"
            },
            {
                "sourcePath": "$.name.middleName",
                "targetPath": "$.name.middleName",
                "optional": true
            },
            {
                "sourcePath": "$.nickName",
                "targetPath": "$.nickName",
                "optional": true
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
            },
            {
                "sourcePath": "$.phoneNumbers",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.phoneNumbers",
                "optional": true
            },
            {
                "sourcePath": "$.emails",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.emails",
                "optional": true
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true
            },
            {
                "sourcePath": "$.groups",
```

```
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.groups",
                "optional": true
            }
        ]
    },
    "group": {
        "condition": "isAttributeWithOptionalPrefix($.displayName,
ecp.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
ecp.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:Group",
                    "urn:ietf:params:scim:schemas:extension:sap:2.0:Group"
                ],
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition":
"isAttributeWithOptionalPrefix($.displayName, ecp.group.prefix)",
                        "function": "replaceFirstString",
                        "regex": "%ecp.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            },
            {
                "constant": "User",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[*].type",
                "scope": "createEntity"
            }
        ]
    }
}
```

6. Add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Identity Provisioning for Employee Central Payroll

1.6.2.44 SAP SuccessFactors Incentive Management

Follow this procedure to set up a target connector for SAP SuccessFactors Incentive Management, formerly known as SAP Commissions.

Prerequisites

You have created a technical user with administrator permissions that will be used to call the API of SAP SuccessFactors Incentive Management for creating or updating users and user assignments.

Context

After fulfilling the prerequisites, follow the procedure below to add a target system for SAP SuccessFactors Incentive Management to write users to it. This target system consumes SCIM 2.0 API provided by SAP SuccessFactors Incentive Management.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP SuccessFactors Incentive Management* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the SAP SuccessFactors Incentive Management SCIM API portal.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the user for your SAP SuccessFactors Incentive Management system.
`Password`	(Credential) Enter the password for your SAP SuccessFactors Incentive Management user.
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*https://mycommissions.callidus.run/CallidusPortal*

`User`=*MyCommissionsUser*

`Password`=**********

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP SuccessFactors Incentive Management* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP SuccessFactors Incentive Management. For more information, see:

Manage Transformations [page 2439]

SAP SuccessFactors Incentive Management REST API

Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target entity. If the entity has more than one e-mail addresses, only one of them is used. It is either the e-mail set as primary in the source system, or if no primary e-mail is set, the one that comes first is used.

Default transformation:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.userName",
                "targetPath": "$.id",
                "targetVariable": "entityIdTargetSystem"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$.externalId",
                "targetPath": "$.externalId",
                "optional": true,
                "defaultValue": ""
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName",
                "optional": true,
                "defaultValue": ""
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName",
                "optional": true,
                "defaultValue": ""
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true
            },
// For each user in SAP SuccessFactors Incentive Management, only one
email is written. It can be either the primary email, or the first email
from the "emails[]" list.
            {
                "sourcePath": "$.emails[0].value",
                "targetPath": "$.emails[0].value",
                "optional": true,
                "defaultValue": ""
            },
            {
                "condition": "$.emails[?(@.primary == true)].value != []",
                "sourcePath": "$.emails[?(@.primary == true)].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
```

```
                "targetPath": "$.emails[0].value",
                "functions": [
                    {
                        "function": "elementAt",
                        "index": 0
                    }
                ]
            },
            {
                "constant": "",
                "targetPath": "$.groups[0].value"
            },
            {
                "condition": "$.groups EMPTY false",
                "sourcePath": "$.groups",
                "preserveArrayWithSingleElement": true,
                "targetPath": "$.groups",
                "optional": true
            },
            {
                "sourcePath": "$.locale",
                "targetPath": "$.locale",
                "optional": true,
                "defaultValue": ""
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true,
                "defaultValue": ""
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
                "targetPath": "$.schemas[0]"
            }
        ]
    }
}
```

5. Now, add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

SAP SuccessFactors Incentive Management: Integration with SAP IdP

1.6.2.45 SAP SuccessFactors Learning

Follow this procedure to set up SAP SuccessFactors Learning as a target system.

Prerequisites

You have created a technical user with administrator permissions that will be used to call the SAP SuccessFactors Learning API for provisioning user information.

Context

SAP SuccessFactors Learning is a learning solution which helps organizations to improve employee skills and talent management, align learning outcomes with performance goals, boost compliance, and train external audiences.

You can use the Identity Provisioning user interface (UI) to configure SAP SuccessFactors Learning as a target system where you can provision users from source systems.

> ⚠ Restriction
>
> Provisioning **groups** is not supported.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SAP SuccessFactors Learning* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
Type	Enter: *HTTP*
URL	Specify the URL to the API of your SAP SuccessFactors Learning system. It follows the pattern: `https://<root URL>/learning/public-api/rest/admin/Integration.svc/ias`

Property Name	Value
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the technical user ID for SAP SuccessFactors Learning.
`Password`	(Credential) Enter the password for the SAP SuccessFactors Learning technical user. For more information, see Learning Technical User.
(Optional) `lms.user.unique.attribute`	This property appears by default when the system is created, and its value is set to *userName*. It defines by which unique attribute(s) an existing user to be resolved in the event of conflicting users.
(Optional) `lms.support.patch.operation`	Controls how modified users in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users in the target system. • If set to *false*, PUT operations are used to update users in the target system.
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP SuccessFactors Learning* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP SuccessFactors Learning system. For more information, see:

   Manage Transformations [page 2439]

   SAP Business Accelerator Hub: SAP SuccessFactors Learning

   To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target SAP SuccessFactors Learning entity.

Default transformation:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:core:2.0:User","urn:sap:cloud:scim:schemas:e
xtension:custom:2.0:User","urn:ietf:params:scim:schemas:extension:sap:2.0:U
ser"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName",
        "optional": true
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['siteID']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']['siteID']",
        "optional": true
      },
      {
        "sourcePath": "$.custom-column-path-1",

"targetPath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']
['customColumns']['110']",
        "optional": true
      },
      {
        "sourcePath": "$.custom-column-path-2",

"targetPath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:User']
['customColumns']['120']",
        "optional": true
      },
      {
```

```
        "condition": "$.emails[?(@.primary== true)] empty false",
        "sourcePath": "$.emails[?(@.primary== true)].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails[?(@.value)]"
      },
      {
        "condition": "$.emails[?(@.primary== true)] empty true",
        "sourcePath": "$.emails[0].value",
        "targetPath": "$.emails[0].value",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.name.middleName",
        "targetPath": "$.name.middleName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName"
      },
      {
        "condition": "$.phoneNumbers[?(@.primary== true)] empty false",
        "sourcePath": "$.phoneNumbers[?(@.primary== true)].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.phoneNumbers[?(@.value)]"
      },
      {
        "condition": "$.phoneNumbers[?(@.primary== true)] empty true",
        "sourcePath": "$.phoneNumbers[0].value",
        "targetPath": "$.phoneNumbers[0].value",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      }
    ]
  }
}
```

5. Add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.46 Sales Cloud – Analytics & AI

Follow this procedure to set up a target connector for Sales Cloud – Analytics & AI.

Prerequisites

You have technical user credentials for an Sales Cloud – Analytics & AI (in short, *SCAAI*) system with read and write access permissions.

Context

After fulfilling the prerequisites, follow the procedure to add a target system for Sales Cloud – Analytics & AI to write users and user assignments to groups. This target system consumes SCIM 2.0 API provided by SCAAI.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *Sales Cloud – Analytics & AI* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the SCIM API portal of your SCAAI system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the user for your SCAAI system.
`Password`	(Credential) Enter the password for your SCAAI user.
`OAuth2TokenServiceURL`	Enter the URL to the OAuth2 token service. If not sure about the exact URL, ask your SCAAI administrator.

Property Name	Value
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*http://myscaai:8080/scim_services*

`User`=*MySCAAIUser*

`Password`=**********

`OAuth2TokenServiceURL`=*http://myscaai:8080/gateway_services/api/auth/ips/token*

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Sales Cloud – Analytics & AI* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your Sales Cloud – Analytics & AI. For more information, see: Manage Transformations [page 2439]

   Mapping logic – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target entity. The Identity Provisioning takes a user's original *userName* (from the source system) and assigns it as a user ID in the target SCAAI system. Analogically, the Identity Provisioning takes a group's original *displayName* and assigns it as a group ID in SCAAI.

   Default transformation:

   Code Syntax

   ```
   {
       "user": {
   ```

```
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.externalId"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            }
        ]
    },
    "group": {
        "skipOperations": [
            "delete"
        ],
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.externalId"
            },
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

If you want the users and groups in SCAAI to have the **same** IDs as the respective users and groups in the source system, modify the transformation mappings as follows:

- In the source system:

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetPath": "$.id",
```

```
                "correlationAttribute": true
            },
...
    "group": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetPath": "$.id"
            },
...
}
```

- In the SCAAI target system:

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetPath": "$.externalId"
            },
...
    "group": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetPath": "$.externalId"
            },
...
}
```

5. Now, add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.47 Cloud Foundry UAA Server

Follow this procedure to set up a Cloud Foundry UAA Server as a target system.

Prerequisites

> ⚠ Restriction
>
> This system is available for standalone tenants running on SAP Cloud Identity infrastructure and SAP BTP, Neo environment. Bundle tenants running on SAP Cloud Identity Services infrastructure and Neo environment can use it only through **SAP Identity Access Governance** bundle option.

- You have technical user credentials for a Cloud Foundry system with write access permissions. In case OAuth is used for authentication, client ID and secret are required when creating a destination for access token retrieval. You need Cloud Foundry UAA version **4.2** or higher
- (Optional) You have installed the Cloud Connector in your corporate environment and have done the initial configuration. You need to do this only if the Cloud Foundry UAA server is exposed in a private corporate network. For more information, see Cloud Connector.

Context

User Account and Authentication Service (UAA) is an OAuth2 server that you can use for centralized identity management. It owns the user accounts and authentication sources and supports standard protocols (such as *SAML*, *LDAP*, and *OpenID Connect*) to provide SSO and delegated authorization to Web applications. For more information, see Cloud Foundry: Overview .

> → Tip
>
> This connector is meant for writing users and groups in **general** Cloud Foundry systems (they could be non-SAP ones). If you want to trigger provisioning of entities to SAP Business Technology Platform Cloud Foundry applications, you'd better use SAP BTP XS Advanced UAA (Cloud Foundry) [page 1045] target system.

These target systems consume SCIM 1.1 API provided by Cloud Foundry UAA.

> → Remember
>
> You can write users and groups to Cloud Foundry on **application** level only. You cannot provision or manage them on a *subaccount* level.

Follow the steps below to create Cloud Foundry UAA as a target system to provision users and groups.

Procedure

1. (Optional) If the Cloud Foundry UAA server is exposed in a private corporate network, add an access control system mapping in Cloud Connector. For more information, see Configure Access Control (HTTP).
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
3. Add *Cloud Foundry UAA Server* as a target system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the Cloud Foundry UAA SCIM API. If not sure about the exact URL, ask your Cloud Foundry UAA administrator.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`OAuth2TokenServiceURL`	As you need to make OAuth authentication to the UAA system, enter the URL to the OAuth2 token service. If not sure about the exact URL, ask your Cloud Foundry UAA administrator.
`User`	Enter the OAuth client ID of the Cloud Foundry UAA technical user.
`Password`	(Credential) Enter the OAuth client secret of the technical user.
`uaa.origin`	Enter the location of your Cloud Foundry identity provider. If not sure about the value, ask your Cloud Foundry UAA administrator. The value of this property is a string, which will be used as the *origin* attribute in the system transformation.
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://api.authentication.hana.ondemand.com
OAuth2TokenServiceURL=https://MyCFaccount.authentication.hana.ondemand.com/oauth/token
User=MyCFuser
Password=************
uaa.origin=my_UAA_location
```

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Cloud Foundry UAA Server* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   - **Mapping logic** – The behavior of the default transformation logic is to map all attributes from the internal Cloud Foundry UAA representation to the target entity.
   - **User offboarding** – If a user has been deleted from the source system, this change is recognized, and the user is deleted from the Cloud Foundry UAA target system too.

   You can change the default transformation mapping rules to reflect your current setup of entities in your Cloud Foundry UAA server. For more information, see:

   Manage Transformations [page 2439]

   Cloud Foundry UAA API: Users

   Cloud Foundry UAA API: Groups

   To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

   Default transformation:

   Code Syntax

```
{
  "user": {
    "condition": "($.emails.length() > 0) && (('%uaa.group.prefix%'
=== 'null') || ($.groups[?(@.display =~ /%uaa.group.prefix%.*/)] empty
false))",
    "mappings": [
      {
        "constant": "uaa-dummy-value",
        "targetPath": "$.id",
        "scope": "createEntity"
      },
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
```

```
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.verified",
        "targetPath": "$.verified",
        "optional": true
      },
      {
        "constant": "%uaa.origin%",
        "targetPath": "$.origin"
      },
      {
        "sourcePath": "$.phoneNumbers",
        "targetPath": "$.phoneNumbers",
        "preserveArrayWithSingleElement": true,
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "preserveArrayWithSingleElement": true
      },
      {
        "condition": "$.emails[?(@.primary == true)].value == []",
        "targetPath": "$.emails[0].primary",
        "constant": true
      },
      {
        "constant": "urn:scim:schemas:core:1.0",
        "targetPath": "$.schemas[0]"
      }
    ]
  },
  "group": {
    "condition": "isAttributeWithOptionalPrefix($.displayName,
uaa.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
uaa.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName",
        "functions": [
          {
            "condition": "isAttributeWithMandatoryPrefix(@,
uaa.group.prefix)",
```

```
            "function": "replaceFirstString",
            "regex": "%uaa.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath": "$.description",
        "targetPath": "$.description",
        "optional": true
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "functions": [
          {
            "function": "resolveEntityIds"
          },
          {
            "condition": "@.type EMPTY false",
            "function": "toUpperCaseString",
            "applyOnElements": true,
            "applyOnAttribute": "type",
            "locale": "en_EN"
          }
        ]
      },
      {
        "constant": "urn:scim:schemas:core:1.0",
        "targetPath": "$.schemas[0]"
      }
    ]
  }
}
```

6. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Cloud Foundry UAA: Users
Cloud Foundry UAA: Groups

1.6.2.48 Google G Suite

Follow this procedure to set up Google G Suite as a target system.

Prerequisites

> **Restriction**
>
> This system is available for standalone tenants running on SAP Cloud Identity infrastructure and SAP BTP, Neo environment. Bundle tenants running on SAP Cloud Identity Services infrastructure and Neo environment can use it only through **SAP Identity Access Governance** bundle option.

1. Sign in to the Google API console (https://console.developers.google.com) and create a project.
2. Enable the Admin SDK. To do this, go to *Dashboard* > *ENABLE API* > *Admin SDK* > *ENABLE*.
3. Create a service account for your project. We recommend that you select *Enable G Suite Domain-wide Delegation* during the creation. If you skip this option, you can set it later. For more information, see Creating a service account .
4. Then, in the Google admin console (https://admin.google.com), a user with **Super Admin** role can delegate domain-wide authority to your service account. This way, it will have access to the Google Admin SDK on behalf of your user. For more information, see Delegating domain-wide authority .

   > **Note**
   >
   > When specifying the scopes, the administrator has to enter the following:
   >
   > `https://www.googleapis.com/auth/admin.directory.user, https://www.googleapis.com/auth/admin.directory.group`

Context

A Google service account with delegated domain-wide authority is required for authentication and authorization of the Identity Provisioning service to G Suite domain. The authentication is based on OAuth 2.0 protocol with JSON Web Token (JWT). The private key for the signature is distributed by Google via one-time downloadable JSON data, which is accessible by the domain administrator. The private key is encoded in PKCS8 format and is in the *private_key* field of the JSON data. For more information, see JSON Web Token (JWT) .

- When using it as a source system, you can read both users and groups from Google G Suite and provision them to any target system you have added in the Identity Provisioning user interface.
- When using it as a target system, you can write both users and groups, read from any source system you have added in the Identity Provisioning user interface. Google G Suite can automatically create accounts for your users in the Google Cloud Datastore.

The Identity Provisioning service supports user and group operations based on the following Google Directory API. See the table below.

User Operations	Group Operations
Create a user	Create a group
Retrieve a user	Retrieve a group's properties
Update a user	Update a group's properties
Delete a user	Delete a group

> ⚠ Caution
>
> You can only provision users whose e-mails are from verified domains.

If you have successfully finished with the initial setup (described in the **Prerequisites** section), continue with the procedure below.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *Google G Suite* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
URL	Specify the service URL: **`https://www.googleapis.com/admin/directory`**
`ProxyType`	Enter: **`Internet`**
`Authentication`	Enter: *BasicAuthentication* The authentication type in use is actually **OAuth** with JWT. But for any provisioning system based on OAuth, **BasicAuthentication** is used along with the `OAuth2TokenServiceURL` additional property.
`User`	Enter the service account's ID. You can take it from the *"client_email"* field in the JSON data, downloaded during the setup of Google service account.

Property Name	Description & Value
`Password`	(Credential) Enter the service account's private key, which represents a long string in PKCS8 format. You can take it from the *"private key"* field in the JSON data, downloaded during the setup of Google service account.
`OAuth2TokenServiceURL`	To make OAuth authentication to the Google G Suite system, enter the URL to the access token provider service. For more information, see Using OAuth 2.0 to Access Google APIs .
`jwt.subject`	Enter the Google G Suite user on behalf of which the Google Directory API is called. This user has been assigned the role **User Management Admin**. This property corresponds to "sub" claim in JWT being generated during access token request: JWT: "sub" (Subject) Claim
(Optional) `jwt.scope`	Enter space-separated Google Directory API authorization scopes. For example: **`https://www.googleapis.com/auth/admin.directory.user`**
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary Configuration:

`ProxyType=`*Internet*

`Type=`*HTTP*

`Authentication=`*BasicAuthentication*

`URL=`*https://www.googleapis.com/admin/directory*

`User=`*1234567890-compute@developer.gserviceaccount.com*

`Password=`*-----BEGIN PRIVATE KEY-----\n123ABCDEFG123456789...*

... /123456789ABCDEFG123=\n-----END PRIVATE KEY-----\n

`OAuth2TokenServiceURL=`*https://www.googleapis.com/oauth2/v4/token*

`jwt.subject=`*john.smith@me123.accounts.ondemand.com*

`jwt.scope=`*https://www.googleapis.com/auth/admin.directory.user*

---

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Google G Suite* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   Transformation principles for the target system integration:

   - **Mapping logic** – The provisioning framework reads all attributes from the intermediate JSON data and tries to create consistent records in the Google G Suite target system, using all the available attributes accepted by the Google Directory API. When a required attribute is missing, the default transformation is designed with a condition that will exclude the inconsistent records. Bear in mind the following:
     - Make sure that the JSON data sent by the source system is consistent with the configuration template of the target. For example, if the source system contains mandatory fields and the target one does not support such kind of data, then the target system skips these fields. This may cause crucial data loss.
     - There is a special user status type called **suspended** (temporarily blocks a user without deleting any account data) for the Google directory accounts. When the status of the user account is changed to **suspended**, the Google Directory API will not accept any changes on the user attributes. Once the suspended user is restored by the administrator, all attribute changes pending for the account will be successfully provisioned with the next provisioning job.

     > ⚠ Caution
     >
     > An initial password setup is mandatory for all newly provisioned users. This is required by the Google G Suite API and must be provided when new accounts are created. The constant value that you see as configuration for the password attribute in the default transformation is generated by SAP. You have to change the constant value with another one, known only by the representatives of your company, before starting to use the Identity Provisioning service for creating users in your corporate Google G Suite system automatically.

   - **User off-boarding** – Identity Provisioning service is handling the deletion status of the users. When a user is deleted from the source system, this deletion will be enforced into the Google G Suite system as well.

You can change the default transformation mapping rules to reflect your current setup of entities in your Google G Suite. For more information, see:

Manage Transformations [page 2439]

Google Directory API: Users

Google Directory API: Groups

Default transformation:

Code Syntax

```
{
    "user": {
        "condition": "($.emails.length() > 0) && ($.name.familyName EMPTY
false)",
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "sourcePath": "$.name",
                "targetPath": "$.name"
            },
            {
                "sourcePath": "$.emails[0].value",
                "targetPath": "$.primaryEmail"
            },
            {
                "sourcePath": "$.phoneNumbers",
                "optional": true,
                "targetPath": "$.phones"
            },
            {
                "targetPath": "$.password",
                "scope": "createEntity",
                "functions": [
                    {
                        "type": "randomPassword",
                        "passwordLength": 16,
                        "minimumNumberOfLowercaseLetters": 1,
                        "minimumNumberOfUppercaseLetters": 1,
                        "minimumNumberOfDigits": 1,
                        "minimumNumberOfSpecialSymbols": 0
                    }
                ]
            },
            {
                "constant": "false",
                "targetPath": "$.suspended"
            },
            {
                "condition": "$.active == false",
                "constant": true,
                "targetPath": "$.suspended"
            },
            {
                "constant": "true",
                "targetPath": "$.changePasswordAtNextLogin"
            }
        ]
    },
    "group": {
        "ignore": true,
```

```
        "mappings": [
            {
                "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
                "targetPath": "$.schemas[0]"
            },
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.name"
            },
// Google G Suite requires a group e-mail. By default, the email attribute
is mapped to displayName. If group's Display Name does not contain an e-
mail,
// you can either map email to another attribute, or concatenate
displayName with your domain. To learn how, see the detailed explanation
and example below.

            {
                "sourcePath": "$.displayName",
                "targetPath": "$.email",
                "scope": "createEntity"
            },
            {
                "sourcePath": "$.members[?(@.type == 'User')].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.id)]",
                "functions": [
                    {
                        "entityType": "user",
                        "type": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

If the **displayName** attribute in the source system transformation does not provide group e-mails, you can modify the transformation the following ways:

- Map **email** to another attribute that contains a unique group e-mail.
- Concatenate the **displayName** attribute with your domain. For example:

Sample Code

```
        {
            "sourcePath": "$.displayName",
            "targetPath": "$.email",
            "scope": "createEntity",
            "functions": [
                {
                    "type": "concatString",
                    "suffix": "@test.myaccount.ondemand.com"
                }
            ]
        }
```

5. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.49 LDAP Server

Follow this procedure to set up LDAP Server as a target system.

Prerequisites

> ⚠ Restriction
>
> This system is available for standalone tenants running on SAP Cloud Identity infrastructure and SAP BTP, Neo environment. Bundle tenants running on SAP Cloud Identity Services infrastructure and Neo environment can use it only through **SAP Identity Access Governance** bundle option.

> ⓘ Note
>
> If you have purchased the Identity Provisioning service between **September 1, 2020** and **October 20, 2020**, and you want to make a connection to this on-premise system, follow the procedure in: Connect to On-Premise Systems in SAP Cloud Identity Infrastructure.

- You have installed the Cloud Connector in your corporate environment and have done the initial configuration. For more information, see: Cloud Connector (Neo) or Cloud Connector (Cloud Foundry)
- **For tenants running on the infrastructure of SAP Cloud Identity Services**: You have a multi-environment subaccount in the Cloud Foundry region that maps the region of your Identity Authentication tenant and it is subscribed to the *Cloud Identity Services* application. For more information, see Connect to On-Premise Systems in SAP Cloud Identity Infrastructure.
- You have the credentials of a technical user in the LDAP Server, which is used to call the LDAP Server API to write users and their attributes.

Context

You can use LDAP Server to write entities retrieved from a source system. This scenario supports writing **users** and **group assignments**.

There are two versions of the LDAP Server connector. Both consume the LDAP Server API to read and write users and groups. The versions are handled by the `ldap.api.version` property as follows:

- When the value is set to *1* or the property is not defined (typical for systems created before versioning was introduced on May 25, 2023) LDAP Server API version 1 is used. This is the default value of `ldap.api.version`.
  When using this version of the connector, the entities (users and groups) are read with all attributes.
- When the value is set to *2* – LDAP Server API version 2 is used.
  This version of the connector comes with improved performance of the read operation for user and group attributes. You are now able to define which user and group attributes to be read. This is possible by adding values to the properties `ldap.user.attributes` or `ldap.group.attributes`.
  Via these properties, you are able to add also user and group operational attributes (attributes which the directory organizes for internal use). For more information, refer to the official LDAP server documentation.
  After the additional values of the properties are set, the default read or proxy read transformations should also be adjusted accordingly.
  For more information on how to update to LDAP Server connector version 2, see Update Connector Version [page 2424].

Procedure

1. Add an access control system mapping for the **LDAP Server** in the Cloud Connector. This is needed to allow the Identity Provisioning service to access the LDAP server as a back-end system on the intranet. For more information, see Configure Access Control (LDAP).
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
3. Add *LDAP Server* as a target system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *LDAP*
`ldap.url`	Specify the destination URL. It must be in the following format: `ldap://<external_host>:<external_port>`
`ldap.proxyType`	Enter: *OnPremise*
`ldap.authentication`	Enter: *BasicAuthentication*
`ldap.user`	Enter the *distinguishedName* of the technical LDAP user. This is the user you need to establish the connection and to perform all queries.
`ldap.password`	(Credential) Enter the password for the LDAP technical user.

Property Name	Description & Value
`ldap.group.path`	Enter the complete path to the node containing the groups in the LDAP tree.
`ldap.user.path`	Enter the complete path to the users in the LDAP tree.
(Optional)`ldap.api.version`	Defines the version of LDAP Server API. **Possible values:** • **1** - Indicates that LDAP Server API version 1 is used. • **2** - Indicates that LDAP Server API version 2 is used. If the property is not defined - LDAP Server API version 1 is used.
`CloudConnectorLocationId`	Relevant when the proxy type is *OnPremise*. Use it only if your SAP Business Technology Platform account uses more than one Cloud Connector.
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

> → Remember
>
> We strongly recommend that you enter different paths for LDAP users and groups. That means, the value of `ldap.user.path` should be different than the value of `ldap.group.path`.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

The LDAP Server target system is created by default with the properties listed below:

Default LDAP Properties

`ldap.user.attributes=`

`ldap.user.object.class=` **inetOrgPerson**

`ldap.group.object.class=` **groupOfNames**

`ldap.group.uniquename.attribute=` **cn**

`ldap.attribute.group.id=`**cn**

`ldap.attribute.group.member=` **member**

`ldap.attribute.group.object.class.required=`**cn**

`ldap.attribute.user.object.class.required=`**cn**

`ldap.attribute.user.id=` **uid**

`ldap.attribute.dn=`**distinguishedName**

`ldap.page.size=` **100**

`ldap.attribute.user.mail=` **mail**

`ldap.attribute.user.mobile=`**mobile**

`ldap.attribute.user.givenName=` **givenName**

`ldap.attribute.user.surname=` **sn**

`ldap.attribute.user.groups=` **memberOf**

`ldap.attribute.user.telephoneNumber=` **telephoneNumber**

> ### Note
>
> The **ldap.attribute.*** properties are used as parameterized properties in the default transformation. That is, if a property used in the transformation doesn't have a value, the provisioning job will fail when the transformation is loaded on runtime and the property value is substituted.
>
> Also, you can change a property and use a new one (with a new name). In this case, you must replace the old property with the new one at all corresponding places in the transformation.

> ### Note
>
> The `ldap.attribute.dn` property is auttomatically assigned and cannot be configured. It is used in the target system configuration for conflict resolution during user or group provisioning by the service.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *LDAP Server* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your LDAP server. For more information, see Manage Transformations [page 2439].

Currently, the default write transformations of the two connector versions have no differences, so there is no need to update them.

Before the write transformation (in the intermediate JSON data), the entity attributes are in SCIM format. After the transformation, the attributes in the LDAP Server are represented as arrays (single-element arrays, or multi-value arrays separated by comma (,)). For more information, see the official documentation for LDAP Server schema attributes in the **Related Information** section.

Default transformation:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.userName",
        "targetPath": "$.%ldap.attribute.user.id%[0]",
        "targetVariable": "entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.%ldap.attribute.user.object.class.required%[0]"
      },
      {
        "sourcePath": "$.emails[*].value",
        "optional": true,
        "targetPath": "$.%ldap.attribute.user.mail%"
      },
      {
        "sourcePath": "$.name.givenName",
        "optional": true,
        "targetPath": "$.%ldap.attribute.user.givenName%[0]"
      },
      {
        "sourcePath": "$.name.familyName",
        "optional": true,
        "targetPath": "$.%ldap.attribute.user.surname%[0]"
      },
      {
        "sourcePath": "$.phoneNumbers[*].value",
        "optional": true,
        "targetPath": "$.%ldap.attribute.user.mobile%"
      }
    ]
  },
  "group": {
    "ignore": true,
    "mappings": [
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.%ldap.attribute.group.id%[0]",
        "targetVariable": "entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.%ldap.attribute.group.object.class.required%[0]"
```

```
      },
      {
        "constant": [],
        "targetPath": "$.member"
      },
      {
        "sourcePath": "$.members[*]",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetVariable": "membersVariable",
        "functions": [
          {
            "condition": "@.type != 'Group'",
            "entityType": "user",
            "type": "resolveEntityIds"
          },
          {
            "condition": "@.type == 'Group'",
            "entityType": "group",
            "type": "resolveEntityIds"
          },
          {
            "condition": "@.type != 'Group'",
            "function": "concatString",
            "applyOnElements": true,
            "applyOnAttribute": "value",
            "prefix": "%ldap.attribute.user.id%=",
            "suffix": ",%ldap.user.path%"
          },
          {
            "condition": "@.type == 'Group'",
            "function": "concatString",
            "applyOnElements": true,
            "applyOnAttribute": "value",
            "prefix": "%ldap.attribute.group.id%=",
            "suffix": ",%ldap.group.path%"
          }
        ]
      },
      {
        "sourceVariable": "membersVariable",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.member",
        "variablePath": "$[*].value"
      }
    ]
  }
}
```

Below is illustrated an example of how the data from LDAP Server looks like before and after executing a certain mapping from the write transformation:

Transformation Snippet (from the group mapping)	Intermediate JSON Data (before the transformation)	Target JSON Data (written in the LDAP Server)

Sample Code

```
"group": {
    ...
  {
    "sourcePath":
"$.members[*]",

"preserveArrayWithSi
ngleElement": true,
    "optional":
true,

"targetVariable":
"membersVariable",
    "functions": [
       {

"condition":
"@.type != 'Group'",

"entityType":
"user",
          "type":
"resolveEntityIds"
       },
       {

"condition":
"@.type == 'Group'",

"entityType":
"group",
          "type":
"resolveEntityIds"
       },
       {

"condition":
"@.type != 'Group'",
          "function":
"concatString",

"applyOnElements":
true,

"applyOnAttribute":
"value",
          "prefix":
"%ldap.attribute.use
r.id%=",
          "suffix":
",%ldap.user.path%"
       },
       {
```

Sample Code

```
...
"members":[
  {
    "value":
"SALES_US"
  },
  {
    "value":
"SALES_EU"
  },
  {
    "value":
"SALES_JA"
  }
]
…
```

Sample Code

```
...
"member": [
   "SALES_US",
   "SALES_EU",
   "SALES_JA"
]
…
```

Transformation Snippet (from the group mapping)	Intermediate JSON Data (before the transformation)	Target JSON Data (written in the LDAP Server)

```
"condition":
"@.type == 'Group'",
        "function":
"concatString",

"applyOnElements":
true,

"applyOnAttribute":
"value",
        "prefix":
"%ldap.attribute.gro
up.id%=",
        "suffix":
",%ldap.group.path%"
       }
    ]
   },
   {

"sourceVariable":
"membersVariable",

"preserveArrayWithSi
ngleElement": true,
    "optional":
true,
    "targetPath":
"$.member",
    "variablePath":
"$[*].value"
   }
  ]
 }
}
```

> **Note**
>
> By default, the **cn** attribute is used for writing the groups. An administrator can change this behavior by setting the following properties:
>
> - `ldap.group.uniquename.attribute` – the value can be either the CN or the whole DN (**distinguishedName**) of the group.
> - `ldap.attribute.group.id` – the value can be CN or another attribute to be used as a group ID instead (for example, **displayName** or **description**).
>
> For more information about these properties, see: List of Properties [page 80]

6. Now, add a source system to read users and groups from it. Choose from: Source Systems [page 585]

Related Information

Technical Documents

Setting Timeout for Ldap Operations
Connection Pooling Configuration

1.6.2.50 Microsoft Active Directory

Follow this procedure to set up Microsoft Active Directory as a target system.

Prerequisites

> ⚠ Restriction
>
> This system is available for standalone tenants running on SAP Cloud Identity infrastructure and SAP BTP, Neo environment. Bundle tenants running on SAP Cloud Identity Services infrastructure and Neo environment can use it only through **SAP Identity Access Governance** bundle option.

> ⓘ Note
>
> If you have purchased the Identity Provisioning service between **September 1, 2020** and **October 20, 2020**, and you want to make a connection to this on-premise system, follow the procedure in: Connect to On-Premise Systems in SAP Cloud Identity Infrastructure.

- You have installed the Cloud Connector in your corporate environment and have done the initial configuration. For more information, see: Cloud Connector (Neo)
- You have the credentials of a technical user in the Microsoft Active Directory, which is used to call the Microsoft Active Directory API to write users, groups and their attributes.

Context

You can configure Microsoft Active Directory as a target system to provision identities. Currently, you can provision users, groups and group assignments.

There are two versions of the Microsoft AD connector. Both consume the LDAP Server API to read and write users and groups. The versions are handled by the `ldap.api.version` property as follows:

- When the value is set to *1* or the property is not defined (typical for systems created before versioning was introduced on June 19, 2023) LDAP Server API version 1 is used. This is the default value of `ldap.api.version`.
  When using this version of the connector, the entities (users and groups) are read with all attributes.
  In this version, the `group members` attribute mapping in the proxy read transformation does not include `type` sub-attribute. In this case, all members are considered of type `User`, which is the sub-attribute fallback value. As a consequence, if the external system includes nested groups, they will not be handled properly.
- When the value is set to *2* – LDAP Server API version 2 is used.

This version of the connector is with improved performance of the read operation for user and group attributes. You are now able to define which user and group attributes to be read. This is possible by adding values to the properties `ldap.user.attributes` or `ldap.group.attributes`.
Via these properties, you are able to add also user and group operational attributes (attributes which the directory organizes for internal use). For more information, refer to the official LDAP server documentation.
After the additional values of the properties are set, the default read or proxy read transformations should also be adjusted accordingly.
In this version, the `group members` attribute mapping in the proxy read transformation is enhanced with `type` sub-attribute. The sub-attribute has two possible values – `User` and `Group`. This allows you to read and preserve nested groups.
For more information on how to update to Microsoft Active Directory version 2, see Update Connector Version [page 2424].

Procedure

1. Add an access control system mapping for the **Microsoft Active Directory** in the Cloud Connector. This is needed to allow the Identity Provisioning service to access Microsoft AD as a back-end system on the intranet. For more information, see Configure Access Control (LDAP).
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
3. Add *Microsoft Active Directory* as a target system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *LDAP*
`ldap.url`	Specify a destination URL. It must be in the following format: `ldap://<ext_host>:<ext_port>`
`ldap.proxyType`	Enter: *OnPremise*
`ldap.authentication`	Enter: *BasicAuthentication*
`ldap.user`	Enter the *distinguishedName* or the *userPrincipalName* of the Microsoft AD technical user. This is the user you need to establish the connection and to perform all queries.
`ldap.password`	(Credential) Enter the password for the Microsoft AD technical user.
`ldap.attribute.user.id`	Default property, which denotes the ID of a user. By default, it's set to: *cn*

Property Name	Description & Value
`ldap.attribute.group.id`	Default property, which denotes the ID of a group. By default, it's set to: *cn*
`ldap.attribute.dn`	Default property, which denotes the distinguished name of a user or a group. The distinguished name is auttomatically assigned and cannot be configured. It is used in the target system configuration for conflict resolution during user or group provisioning by the service. Only possible value: *distinguishedName*
`ldap.group.path`	Enter the complete path to the node containing the groups in Microsoft Active Directory.
`ldap.user.path`	Enter the complete path to the users in Microsoft Active Directory.
`CloudConnectorLocationId`	Relevant when the proxy type is *OnPremise*. Use it only if your SAP Business Technology Platform account uses more than one Cloud Connector.
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Example for a destination or a set of properties:

```
Type=LDAP
Name=MyADDestination
ldap.user=john.smith@some.dummy.domain.com
ldap.password=********
ldap.attribute.user.id=cn
ldap.attribute.group.id=cn
ldap.attribute.dn=distinguishedName
ldap.url=ldap://abcd:123
ldap.proxyType=OnPremise
ldap.authentication=BasicAuthentication
ldap.group.path=OU=Groups,OU=IAS,DC=global,DC=corp,DC=mycompany
ldap.user.path=OU=Users,OU=IAS,DC=global,DC=corp,DC=mycompany
```

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Microsoft Active Directory* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   In the intermediate JSON data, the Microsoft Active Directory attributes are in SCIM format. After the write transformation, the attributes are represented as arrays – single-element arrays, or multi-value arrays separated by comma (,). For more information about Microsoft AD schema attributes, see the **Related Information** section.

   Currently, the default write transformations of the two connector versions have no differences, so there is no need to update them.

   You can change the default transformation mapping rules to reflect your current setup of entities in Microsoft AD. For more information, see:

   Manage Transformations [page 2439]

   > ### Note
   >
   > You can provision users with *unicodePwd* attribute to Microsoft Active Directory target systems although it is not included in the default write transformation. This attribute specifies the password of the user in Windows NT operating system one-way format (OWF).

To provision the user password in encrypted format, proceed as follows:

1. Add the unicodePwd attribute mapping in the write transformation. For example:

   Code Syntax

   ```
   {
   "sourcePath": "$.Source_System_Attribute",
   "targetPath": "$.unicodePwd"
   }
   ```

2. Follow the requirements for provisioning users with *unicodePwd*, as described in unicodePwd

Default transformation:

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "condition": "('%ldap.attribute.user.id%' !=
'%ldap.attribute.dn%')",
        "sourcePath": "$.userName",
        "targetPath": "$.%ldap.attribute.user.id%[0]",
        "targetVariable": "entityIdTargetSystem",
        "scope": "createEntity"
      },
// If a user is not a direct member of the configured user base
path, then its distinguishedName is configured to be equal to CN =
<userName>,<nested_path>,<base_path>,
// where <nested_path> is read from "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:ad:2.0:User']['nestedPath']"
      {
        "condition": "('%ldap.attribute.user.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:ad:2.0:User']['nestedPath']",
        "optional": true,
        "targetVariable": "nestedPathVariable",
        "defaultValue": "",
        "scope": "createEntity"
      },
      {
        "condition": "('%ldap.attribute.user.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath": "$.userName",
        "functions": [
          {
            "function": "concatString",
            "prefix": "CN="
          },
          {
            "condition": "('${nestedPathVariable}' != '')",
            "function": "concatString",
            "suffix": ","
          },
          {
            "condition": "('${nestedPathVariable}' != '')",
            "function": "concatString",
            "suffix": "${nestedPathVariable}"
          },
          {
            "function": "concatString",
```

```
            "suffix": ",%ldap.user.path%"
          }
        ],
        "targetPath": "$.%ldap.attribute.user.id%[0]",
        "targetVariable": "entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.cn[0]",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.sAMAccountName[0]"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.displayName[0]",
        "defaultValue": [],
        "optional": true
      },
      {
        "sourcePath": "$.emails[0].value",
        "targetPath": "$.mail[0]",
        "defaultValue": [],
        "optional": true
      },
      {
        "sourcePath": "$.name.givenName",
        "targetPath": "$.givenName[0]",
        "defaultValue": [],
        "optional": true
      },
      {
        "sourcePath": "$.name.familyName",
        "targetPath": "$.sn[0]",
        "defaultValue": [],
        "optional": true
      }
    ]
  },
  "group": {
    "ignore": true,
    "mappings": [
      {
        "condition": "('%ldap.attribute.group.id%' !=
'%ldap.attribute.dn%')",
        "sourcePath": "$.displayName",
        "targetPath": "$.%ldap.attribute.group.id%[0]",
        "targetVariable": "entityIdTargetSystem",
        "scope": "createEntity"
      },
// If a group is not a direct member of the configured group base
path, then its distinguishedName is configured to be equal to CN =
<displayName>,<nested_path>,<base_path>,
// where <nested_path> is read from "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:ad:2.0:Group']['nestedPath']"
      {
        "condition": "('%ldap.attribute.group.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extension:ad:2.0:Group']['nestedPath']",
        "optional": true,
        "targetVariable": "nestedPathVariable",
        "defaultValue": "",
        "scope": "createEntity"
      },
```

```
      {
        "condition": "('%ldap.attribute.group.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath": "$.displayName",
        "functions": [
          {
            "function": "concatString",
            "prefix": "CN="
          },
          {
            "condition": "('${nestedPathVariable}' != '')",
            "function": "concatString",
            "suffix": ","
          },
          {
            "condition": "('${nestedPathVariable}' != '')",
            "function": "concatString",
            "suffix": "${nestedPathVariable}"
          },
          {
            "function": "concatString",
            "suffix": ",%ldap.group.path%"
          }
        ],
        "targetPath": "$.%ldap.attribute.group.id%[0]",
        "targetVariable": "entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.cn[0]",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.displayName",
        "targetPath": "$.sAMAccountName[0]"
      },
      {
        "sourcePath": "$.members[*]",
        "targetVariable": "membersVariable",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "functions": [
          {
            "condition": "@.type != 'Group'",
            "entityType": "user",
            "applyOnElements": true,
            "type": "resolveEntityIds"
          },
          {
            "condition": "@.type == 'Group'",
            "entityType": "group",
            "applyOnElements": true,
            "type": "resolveEntityIds"
          },
          {
            "condition": "(@.type != 'Group') &&
('%ldap.attribute.user.id%' != '%ldap.attribute.dn%')",
            "function": "concatString",
            "applyOnElements": true,
            "applyOnAttribute": "value",
            "prefix": "%ldap.attribute.user.id%=",
            "suffix": ",%ldap.user.path%"
          },
          {
            "condition": "(@.type == 'Group') &&
('%ldap.attribute.group.id%' != '%ldap.attribute.dn%')",
            "function": "concatString",
```

```
            "applyOnElements": true,
            "applyOnAttribute": "value",
            "prefix": "%ldap.attribute.group.id%=",
            "suffix": ",%ldap.group.path%"
          }
        ]
      },
      {
        "sourceVariable": "membersVariable",
        "variablePath": "$[*].value",
        "targetPath": "$.member",
        "defaultValue": [],
        "preserveArrayWithSingleElement": true,
        "optional": true
      }
    ]
  }
}
```

6. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

> **Note**
>
> Identity Provisioning escapes the following special characters: comma (,), plus (+) and semicolon (;) in the CN component of the distinguished name (DN). This means that, if a user attribute in the source system contains a special character (`John, Smith`), and this attribute is mapped to the CN in Microsoft AD target system, the comma will be escaped in the DN.
>
> The following special characters cannot be escaped and result in an error: equal sign (=), less than symbol (<), greater than symbol (>), hash sign (#) and backslash (\).

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Microsoft AD: Technical Documents
Setting Timeout for Ldap Operations
Connection Pooling Configuration
Active Directory Domain Services Overview

1.6.2.51 Microsoft Entra ID

Follow this procedure to set up Microsoft Entra ID (formerly known as Microsoft Azure Active Directory) as a target system.

Prerequisites

> **⚠ Restriction**
>
> This system is available for standalone tenants running on SAP Cloud Identity infrastructure and SAP BTP, Neo environment. Bundle tenants running on SAP Cloud Identity Services infrastructure and Neo environment can use it only through **SAP Identity Access Governance** bundle option.

- You've logged on to Microsoft Azure Portal, with credentials for a user with directory role **Global administrator**. For more information, see Microsoft Entra built-in roles .
- In *Azure Active Directory* > *App registrations*, you've registered an application with a secret key and permissions for Microsoft Graph API. These permissions must be consented by an administrator. For more information, see Microsoft Graph permissions reference .
- (Relevant to target systems) Your registered application is assigned the **User Account Administrator** role. This role allows you to deprovision users. For more information, see Add-MsolRoleMember .

  > **ⓘ Note**
  >
  > If this role isn't assigned, you can only disable users. To do that, set the `accountEnabled` property to **false**. For more information, see MS Graph: user resource type

Permissions

Assign the following permissions to your application, according to your scenario. Also, the permissions have to be of type *Application*.

- Users – *User.ReadWrite.All*, *Directory.AccessAsUser.All*
- Groups – *Group.ReadWrite.All*

For more information, see MS Graph: Users and MS Graph: Groups

Context

When using it as a target system, you can write both users and groups, read from any source system you've added in the Identity Provisioning user interface. The Microsoft Entra ID target systems use Microsoft Graph API. For more information, see Microsoft Graph .

If you've successfully finished with the initial setup (described in the **Prerequisites** section), continue with the procedure.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *Microsoft Entra ID* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter: **https://graph.microsoft.com**
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the application ID registered in your Microsoft Entra ID subscription (see the **Prerequisites** section).
`Password`	(Credential) Enter the secret key associated to your app registration.
`aad.domain.name`	Enter one of the verified domain names from the corresponding Microsoft Entra ID tenant. On this domain, you perform the provisioning operations. For more information, see Managing custom domain names in your Microsoft Entra ID .
`OAuth2TokenServiceURL`	Enter: **https://login.microsoftonline.com/<your_domain>/oauth2/token**, where `<your_domain>` is the domain name you have set in the **`aad.domain.name`** property.
(Optional) `aad.oauth2.resource.name`	Specifies the OAuth2 resource name for Microsoft Graph API. Enter: **https://graph.microsoft.com** You can change the Microsoft Graph endpoint to match your organization's cloud environment. For more information, see Microsoft Graph national cloud deployments.
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

Property Name	Description & Value
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Deprecated) `oauth.resource.name`	Enter: **https://graph.microsoft.com** **Note** The **oauth.resource.name** property has been deprecated, as it is no longer needed by the MS Entra ID connector. Use the **aad.oauth2.resource.name** property instead.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Microsoft Entra ID* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in Microsoft Entra ID. For more information, see:

   Manage Transformations [page 2439]

   MS Graph: Users

   MS Graph: Groups

   Default transformation:

   Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.onPremisesImmutableId",
                "optional": true,
                "targetPath": "$.onPremisesImmutableId"
            },
            {
                "sourcePath": "$.active",
                "optional": true,
                "targetPath": "$.accountEnabled"
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.mailNickname"
            },
```

```
            {
                "sourcePath": "$.displayName",
                "optional": true,
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.name.givenName",
                "optional": true,
                "targetPath": "$.givenName"
            },
            {
                "sourcePath": "$.name.familyName",
                "optional": true,
                "targetPath": "$.surname"
            },
            {
                "sourcePath": "$.addresses[0].locality",
                "optional": true,
                "targetPath": "$.city"
            },
            {
                "sourcePath": "$.addresses[0].country",
                "optional": true,
                "targetPath": "$.country"
            },
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userPrincipalName",
                "scope": "createEntity",
                "functions": [
                    {
                        "type": "concatString",
                        "suffix": "@%aad.domain.name%"
                    }
                ]
            },
            {
                "sourcePath": "$.active",
                "targetPath": "$.accountEnabled",
                "scope": "createEntity"
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.mailNickname",
                "scope": "createEntity"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "scope": "createEntity"
            },
            {
                "targetPath": "$.passwordProfile.password",
                "scope": "createEntity",
                "functions": [
                    {
                        "type": "randomPassword",
                        "passwordLength": 16,
                        "minimumNumberOfLowercaseLetters": 1,
                        "minimumNumberOfUppercaseLetters": 1,
                        "minimumNumberOfDigits": 1,
                        "minimumNumberOfSpecialSymbols": 0
                    }
                ]
            },
            {
                "constant": false,
```

```
                "targetPath":
"$.passwordProfile.forceChangePasswordNextSignIn",
                "scope": "createEntity"
            }
        ]
    },
    "group": {
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "sourcePath": "$.displayName",
                "optional": true,
                "targetPath": "$.displayName"
            },
            {
                "sourcePath": "$.description",
                "optional": true,
                "targetPath": "$.description"
            },
            {
                "sourcePath": "$.allowExternalSenders",
                "optional": true,
                "targetPath": "$.allowExternalSenders"
            },
            {
                "sourcePath": "$.autoSubscribeNewMembers",
                "optional": true,
                "targetPath": "$.autoSubscribeNewMembers"
            },
            {
                "sourcePath": "$.isSubscribedByMail",
                "optional": true,
                "targetPath": "$.isSubscribedByMail"
            },
            {
                "sourcePath": "$.visibility",
                "optional": true,
                "targetPath": "$.visibility"
            },
            {
                "sourcePath": "$.securityEnabled",
                "optional": true,
                "targetPath": "$.securityEnabled"
            },
            {
                "sourcePath": "$.mailEnabled",
                "optional": true,
                "targetPath": "$.mailEnabled"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "scope": "createEntity"
            },
            {
                "sourcePath": "$.externalId",
                "targetPath": "$.mailNickname",
                "scope": "createEntity"
            },
            {
                "constant": true,
                "targetPath": "$.mailEnabled",
                "scope": "createEntity"
            },
            {
```

```
                "constant": false,
                "targetPath": "$.securityEnabled",
                "scope": "createEntity"
            },
            {
                "constant": "Unified",
                "targetPath": "$.groupTypes[0]",
                "scope": "createEntity"
            }
        ]
    }
}
```

5. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.52 Procurement Data Warehouse

Follow this procedure to set up Procurement Data Warehouse as a target system.

Prerequisites

- You obtain credentials to access the Procurement Data Warehouse API.

  > **Note**
  >
  > To obtain credentials to access the Procurement Data Warehouse API, create a case to the component *BNS-ARI-PRI*.
  >
  > In response to the case, SAP provides the certificate needed to access the API as well as the URL for the procurement data warehouse enrollment service, which is landscape-specific and starts with `*.cloud.sap`.

- You established trust between your SAP BTP subaccount and the SAP Cloud Identity Services - Identity Authentication service. For more information, see Establishing Trust Automatically.
- To configure your *Procurement Data Warehouse* provisioning system (see the procedure below), you will need to map your SAP Ariba application parameters to the relevant Identity Provisioning properties. The property mapping between the two systems is as follows:

Procurement Data Warehouse Application	Identity Provisioning	Values
API URL	`URL`	Examples: • US: **https://openapi.ariba.com** • Europe: **https://eu.openapi.ariba.com**
procurement data warehouseOAuth Token URL	`OAuth2TokenServiceURL`	Examples: • US: **https://api.ariba.com/v2/oauth/token** • Europe: **https://api-eu.ariba.com/v2/oauth/token**
OAuth Client ID	`client_id`	Enter the OAuth Client id used for access token retrieval.

- To load transaction data into procurement data warehouse, you must have one or more of the following SAP Ariba applications:
  - SAP Ariba Buying
  - SAP Ariba Buying and Invoicing
  - Guided sourcing capability for SAP Ariba Sourcing
  - SAP Ariba Contracts
  - SAP Ariba Spend Analysis

Context

The Procurement Data Warehouse is a capability that feeds analytics information from SAP Ariba applications, including guided sourcing capability for SAP Ariba Sourcing, SAP Ariba Buying and Invoicing, and SAP Spend Control Tower, to the embedded SAP Analytics Cloud. It provides data transfer, storage, loading, and reporting, and it drives visualization through advanced analytics dashboards.

You can use Identity Provisioning to configure Procurement Data Warehouse as a target system where you can provision users and groups. The target system consumes SCIM 2.0 API provided by Procurement Data Warehouse.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *Procurement Data Warehouse* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the SAP Analytics Cloud URL for your Procurement Data Warehouse.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *ClientCertificateAuthentication*
`client_id`	Enter the OAuth client id, generated for your Procurement Data Warehouse. (see the **Prerequisites** section)
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your Procurement Data Warehouse instance. This token URL is listed in the *OAuth Clients* section of the *App Integration* page.
`pdw.user.unique.attribute`	It defines by which unique attribute(s) an existing user to be resolved in the event of conflicting users. Default value: **userName** Possible values: • **emails[0].value** • **userName,emails[0].value** **Note** If this property is missing, or you've deleted it, and the service does not find such a *userName*, it will try again to resolve the conflicting user – by *email*. If the second attempt for resolution is unsuccessful too, the creation of the conflicting user fails. For more information, see: List of Properties [page 80]
(Optional)`pdw.group.prefix`	This property distinguishes Procurement Data Warehouse groups by specific prefix. It is an optional property which does not appear by default at system creation. Example value: **PDW_** You can use the example value or provide your own. When **set in the target system**, only groups containing the **PDW_** prefix in their display name will be provisioned to Procurement Data Warehouse. Groups without this prefix in the display name won't be provisioned. If the property is not set, all groups will be provisioned to Procurement Data Warehouse.

Property Name	Description & Value
(Optional) `pdw.support.bulk.operation`	Set this property to *true* if you want to enable SCIM bulk operations for provisioning users. That means, the Identity Provisioning service can write, update, and delete a potentially large collection of users in a single request. For more information, see: SCIM Protocol: Bulk Operations If not specified, the default value is *false*. **Note** SCIM bulk operations are not supported for provisioning groups to Procurement Data Warehouse.
(Optional) `pdw.bulk.operations.max.count`	If you have enabled the SCIM bulk operations, you can use this property to set the number of users to be provisioned per request. Default value: **`20`** **Note** The value must not exceed the number of entities defined by the Procurement Data Warehouse system as a SCIM service provider. Otherwise, the provisioning job will fail with HTTP response code 413 (*Request Entity Too Large*).
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the Procurement Data Warehouse target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your Procurement Data Warehouse system. For more information, see: Manage Transformations [page 2439]

> **Note**
>
> To access the Procurement Data Warehouse API documentation, contact the Procurement Data Warehouse technical support.

Default transformation:

Code Syntax

```
{
    "user": {
        "condition": "isValidEmail($.emails[0].value) &&
(('%pdw.group.prefix%' === 'null') || ($.groups[?(@.display =~ /
%pdw.group.prefix%.*/)] empty false))",
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:User",

"urn:ietf:params:scim:schemas:extension:enterprise:2.0:User",
                    "urn:ietf:params:scim:schemas:extension:sap:user-
custom-parameters:1.0"
                ],
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.emails[0].value",
                "targetPath": "$.userName"
            },
            {
                "condition": "$.emails[?(@.primary == true)].value != []",
                "sourcePath": "$.emails[?(@.primary == true)].value",
                "preserveArrayWithSingleElement": false,
                "optional": true,
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "optional": true
            },
            {
                "sourcePath": "$.preferredLanguage",
                "optional": true,
                "targetPath": "$.preferredLanguage"
            },
            {
                "sourcePath": "$.name.givenName",
                "targetPath": "$.name.givenName",
                "optional": true
            },
            {
                "sourcePath": "$.name.familyName",
                "targetPath": "$.name.familyName",
                "optional": true
            },
            {
```

```
                "sourcePath": "$.active",
                "targetPath": "$.active",
                "optional": true
            },
            {
                "sourcePath": "$.emails",
                "targetPath": "$.emails",
                "preserveArrayWithSingleElement": true
            },
            {
                "condition": "$.emails[0].length() > 0",
                "targetPath": "$.emails[0].primary",
                "constant": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:user-custom-parameters:1.0']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:user-custom-parameters:1.0']",
                "optional": true
            },
            {
                "sourcePath": "$.roles",
                "targetPath": "$.roles",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.photos",
                "targetPath": "$.photos",
                "preserveArrayWithSingleElement": true,
                "optional": true
            },
            {
                "sourcePath": "$.groups[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.groups[?(@.value)]",
                "functions": [
                    {
                      "entityType": "group",
                      "function": "resolveEntityIds"
                    }
                ]
      }
        ]
    },
    "group": {
        "condition": "isAttributeWithOptionalPrefix($.displayName,
pdw.group.prefix) && isAttributeWithOptionalPrefix($
['urn:sap:cloud:scim:schemas:extension:custom:2.0:Group']['name'],
pdw.group.prefix) && (isRegularGroup() || isApplicationSpecificGroup())",
        "mappings": [
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "constant": [
                  "urn:ietf:params:scim:schemas:core:2.0:Group",
```

```
                    "urn:ietf:params:scim:schemas:extension:sap:group-custom-
parameters:1.0"
                ],
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.id",
                "scope": "createEntity",
                "functions": [
                    {
                        "condition": "('%pdw.group.prefix%' !== 'null') &&
(@ =~ /%pdw.group.prefix%.*/)",
                        "function": "replaceFirstString",
                        "regex": "%pdw.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "condition": "('%pdw.group.prefix%' !== 'null') &&
(@ =~ /%pdw.group.prefix%.*/)",
                        "function": "replaceFirstString",
                        "regex": "%pdw.group.prefix%",
                        "replacement": ""
                    }
                ]
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "function": "resolveEntityIds"
                    }
                ]
            },
            {

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:group-custom-
parameters:1.0']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:group-custom-
parameters:1.0']",
                "optional": true
            }
        ]
    }
}
```

If you try to provision groups that already exist in Procurement Data Warehouse, your provisioning job may fail with: '*The group already exists on the target system and cannot be provisioned*' error. This happens after a reset of the Procurement Data Warehouse target system or if you create a new target system connected to an existing Procurement Data Warehouse backend.

To avoid this, you have the following options:

- Delete the existing group in Procurement Data Warehouse target system.

- Adapt the Procurement Data Warehouse write transformation. Either ignore provisioning of groups or add temporary the *skipOperations* expression for creating groups.
- Avoid provisioning of already existing groups.

5. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

> ⚠ Caution
>
> The **email** attribute is unique for Procurement Data Warehouse but it might not be unique for other systems, such as *SAP SuccessFactors*. That's why, when you choose a source system, make sure that there are no users with duplicate e-mails. If there are such, then after the provisioning job all users with the same e-mail address will be created as a single user in Procurement Data Warehouse.
>
> To learn more, see *Guided Answers*: Multiple Users from a Source System Are Created as One in the Target
>
> To learn more about excluding users from Procurement Data Warehouse target, see: Exclude Users from Provisioning to Target Systems

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

Related Information

Configure OAuth Certificate Authentication [page 2456]

1.6.2.53 SCIM System

Follow this procedure to set up a SCIM system as a target system.

Prerequisites

> ⚠ Restriction
>
> This system is available for standalone tenants running on SAP Cloud Identity infrastructure and SAP BTP, Neo environment. Bundle tenants running on SAP Cloud Identity Services infrastructure and Neo environment can use it only through **SAP Identity Access Governance** bundle option.

- (Optional) You have installed the Cloud Connector in your corporate environment and have done the initial configuration. You need this only if the SCIM system is exposed in a private corporate network. For more information, see Cloud Connector.
- You have technical user credentials for a SCIM system, with read/write access permissions, depending on the scenario you want to implement. In case OAuth is used for authentication, client ID and secret are required when creating a destination for access token retrieval.

Context

Create a general SCIM 2.0 based target system to write users and groups to it.

Procedure

1. (Optional) If the SCIM system is exposed in a private corporate network, add an access control system mapping in Cloud Connector. For more information, see Configure Access Control (HTTP).
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
3. Add *SCIM System* as a target system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the service URL. For example: **`http://<cloudfoundry_server>.com/api/uaa/`**
`ProxyType`	Depending on your network exposure, enter one of the following: • *Internet* • *OnPremise*
`Authentication`	Enter: *BasicAuthentication*
`User`	You can specify one of the following: • Technical user ID • Client ID for OAuth HTTP destinations. It's used for retrieving of the access token.

Property Name	Value
`Password`	(Credential) You can enter one of the following: • Technical user password • Client secret for OAuth HTTP destinations. It's used for retrieving of the access token.
`OAuth2TokenServiceURL`	If you need to make OAuth authentication to the system, enter the URL to the access token provider service for OAuth HTTP destinations. For example: **`https://<token_provider>.com/api/oauth2/v2.0/token`**
`scim.csrf.token.path`	Specifies the path which is appended to the URL to retrieve the CSRF token when `scim.api.csrf.protection` is enabled.
`scim.api.csrf.protection`	Specifies whether to fetch a CSRF token when sending requests to the system. If enabled, this property requires the configuration of the `scim.csrf.token.path` property to specify the path where the CSRF token can be retrieved. **Possible values:** • *enabled* • *disabled*
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SCIM* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SCIM system. For more information, see Manage Transformations [page 2439].

To make group assignments via the user resource, you need to change the default transformation of the target system as described in Enabling Group Assignment [page 2445].

- **Mapping logic** – The behavior of the default transformation logic is to map all attributes from the internal SCIM representation to the target entity. If the entity has e-mail addresses, the first entry will be marked as primary.
- **User off-boarding** – Users can be deleted from the target system. Depending on the implementation, this could be done through a user interface (if such exists) or the SCIM REST API. Users could be deactivated, depending on the SCIM system implementation. The SCIM core schema defines an attribute "**active**", whose definition depends on the service provider. For more information, see SCIM: Singular Attributes

Default transformation:

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$",
                "targetPath": "$"
            },
            {
                "targetPath": "$.id",
                "type": "remove"
            },
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "condition": "$.emails[0].length() > 0",
                "targetPath": "$.emails[0].primary",
                "constant": true
            },
            {
                "type": "remove",
                "targetPath": "$
['urn:ietf:params:scim:schemas:extension:enterprise:2.0:User']"
            }
        ]
    },
    "group": {
        "ignore": true,
        "mappings": [
            {
                "sourcePath": "$",
                "targetPath": "$"
            },
            {
                "targetPath": "$.id",
                "type": "remove"
            },
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "targetPath": "$.members",
                "type": "remove"
```

```
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
                "functions": [
                    {
                        "type": "resolveEntityIds"
                    }
                ]
            }
        ]
    }
}
```

6. Now, add a source system from which to read users and groups. Choose from: Source Systems [page 585]

Next Steps

- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.2.54 SSH Server (Beta)

Follow this procedure to set up an SSH server (Beta) as a target system.

Prerequisites

- You have credentials for a tenant in SAP Business Technology Platform. For more information, see: Accounts
- (Optional) You have installed the Cloud Connector in your corporate environment and have done the initial configuration. You need this only when your SSH server resides in a remote system, outside your Neo environment. For more information, see Cloud Connector.

> **Note**
>
> This is a beta feature available on SAP Business Technology Platform. For more information, see: *Enable beta features* in Change Subaccount Details

Context

SSH Server is a system (connector) in beta state. It helps you execute bash scripts through SSH connection. The configuration allows you to attach separate scripts per entity lifecycle callback (such as user create, group create/update, and so on). This system helps you connect to remote machines via SSH tunnel, with or without use of the Cloud Connector, depending on whether the SSH port is visible or not.

The bash scripts can take as parameters fields that are coming from the entity JSON data. For example: *sudo su - vcap /home/myscript.sh $.userName $.email*

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Target Systems*.
2. Add *SSH Server (Beta)* as a target system. For more information, see Add New Systems [page 2415].
3. Choose the *Properties* tab to configure the connection settings for your system.

   Below are listed all available SSH Server properties. Some of them can be mandatory and others – optional, depending on your scenario.

   Mandatory Properties

Property Name	Description & Value
`ProxyType`	Possible values: • **Internet** – if the SSH port is visible in your Neo environment • **OnPremise** – if the SSH port is not directly accessible, and you have to use the Cloud Connector. You have to configure TCP protocol connection to the SSH host and port (specify the configuration properties `ssh.host` and `ssh.port`).
`CloudConnectorLocationId`	Relevant when the proxy type is *OnPremise*. Use it only if your SAP Business Technology Platform account uses more than one Cloud Connector.
`ssh.create.user.command`	Path to the bash command you need to execute to create a user.
`ssh.update.user.command`	Path to the bash command you need to execute to update a user.
`ssh.delete.user.command`	Path to the bash command you need to execute to delete a user.

Property Name	Description & Value
`ssh.create.group.command`	Path to the bash command you need to execute to create a group.
`ssh.update.group.command`	Path to the bash command you need to execute to update a group.
`ssh.delete.group.command`	Path to the bash command you need to execute to delete a group.
`ssh.create.user.command.exit.code.already.exists`	Integer number (code). It's generated by the "`create user`" command when the provisioning job tries to create a user that already exists in the target system.
`ssh.update.user.command.exit.code.not.found`	Integer number (code). It's generated by the "`update user`" command when the provisioning job tries to update a user that is missing from the target system.
`ssh.delete.user.command.exit.code.not.found`	Integer number (code). It's generated by the "`delete user`" command when the provisioning job tries to delete a user that is missing from the target system.
`ssh.create.group.command.exit.code.already.exists`	Integer number (code). It's generated by the "`create group`" command when the provisioning job tries to create a group that already exists in the target system.
`ssh.update.group.command.exit.code.not.found`	Integer number (code). It's generated by the "`update group`" command when the provisioning job tries to update a group that is missing from the target system.
`ssh.delete.group.command.exit.code.not.found`	Integer number (code). It's generated by the "`delete group`" command when the provisioning job tries to delete a group that is missing from the target system.
`ssh.auth.type`	Supported SSH authentication types: • **key** • **pwd** • **otp** • **key+otp** • **key+pwd** • **pwd+otp** • **key+pwd+otp**
`ssh.host`	
`ssh.port`	*22*
`ssh.username`	

Property Name	Description & Value
`ssh.password`	(Credential) Taken into account only if the authentication type includes **pwd**. That means any of the following: • `ssh.auth.type` = *pwd* • `ssh.auth.type` = *pwd+otp* • `ssh.auth.type` = *key+pwd* • `ssh.auth.type` = *key+pwd+otp*
`ssh.totp.secret.key`	(Credential) Taken into account only if the authentication type includes **otp**. That means any of the following: • `ssh.auth.type` = *otp* • `ssh.auth.type` = *key+otp* • `ssh.auth.type` = *pwd+otp* • `ssh.auth.type` = *key+pwd+otp*
`ssh.private.key.type`	The type of SSH private key. Possible values: • **ssh-rsa** • **ssh-dsa** Default value: *ssh-rsa* **Note** If you choose *ssh-rsa*, the key should be in format **PKCS #8**, non-encrypted.
`ssh.private.key`	(Credential) Taken into account only if the authentication type includes **key**. That means any of the following: • `ssh.auth.type` = *key* • `ssh.auth.type` = *key+pwd* • `ssh.auth.type` = *key+otp* • `ssh.auth.type` = *key+pwd+otp*
(Optional) `ips.delete.threshold.groups`	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see: List of Properties [page 80]
(Optional) `ips.delete.threshold.users`	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

4. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SSH Server (Beta)* target system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SSH server. For more information, see Manage Transformations [page 2439].

   Default transformation:

   Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.userName",
                "targetPath": "$.userName"
            }
        ]
    }
}
```

5. Now, add a source system from which to read users. Choose from: Source Systems [page 585]

Next Steps

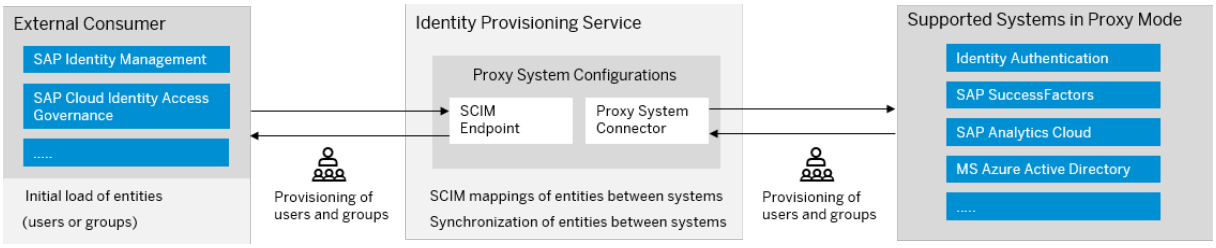
- Before starting a provisioning job, you can first subscribe for e-mail notifications from the source system you use in your scenario. This way, you will be notified by e-mail about eventual failed entities during the jobs. For more information, see Manage Job Notifications [page 3018].
- Start an identity provisioning job. For more information, see Monitor Provisioning Job Logs [page 2982].

1.6.3 Proxy Systems

A proxy system is a special connector type you can use for *hybrid* scenarios.

It exposes any Identity Provisioning supported backend system as a SCIM 2.0 service provider, which can be consumed by any SCIM 2.0 compatible client application, without making a direct connection between them.

After the proxy connector is configured, a consumer application can start sending CRUD requests to the Identity Provisioning proxy connector (which will play the role of a target system). The proxy connector will then read the entities (playing the role of a source system) and provision them to the back-end of the SCIM 2.0 system.



How Proxy Systems Work

Identity Provisioning exposes a SCIM 2.0 based system (connector) as a proxy. Then, an external consumer system regards this proxy connector as its back-end system. To start using a proxy system in your Identity Provisioning tenant, whether it's a bundle or standalone, and depending on the infrastructure or environment it runs on, the following steps need to be performed.

Infrastructure of SAP Cloud Identity Services

Your tenant type meets one of the following requirements:

- Your bundle tenant is created after March 15, 2022.
- Your Identity Provisioning is purchased as a standalone product between September 1, 2020 and October 20, 2020.

1. Sign in to the SAP Cloud Identity Services administration console and navigate to *Users & Authorizations* > *Administrators*.
2. Choose your technical user (administrator user of type **System**). If you don't have a technical user yet, create one. For more information, see Add System as Administrator
3. Configure the technical user authentication.
   - **Certificate**

     Choose *Certificate* > *Configure certificate* and *Upload* your certificate.
     Client certificates are used by HTTP REST clients as a means for SSL certificate authentication.
   - **Secrets**

     Choose *Secrets* > *Add* and provide the required information. After saving it, a Client ID and Client Secret are generated for the technical user. Make sure you copy and save the client secret.
4. Enable the *Access Proxy System API* permission for the technical user.
5. Then navigate to *Identity Provisioning* > *Proxy Systems*, and create a proxy system.
6. (Optional) Export this system as a *.csv* file. This will help an administrator of the external consumer application to import the proxy configuration as a SCIM repository in SAP Identity Management, for example.

   > **Note**
   >
   > The entities exposed by the back-end system will be mapped to SCIM 2.0 entities, if possible. If not possible, the SCIM standard provides a mechanism to define a new resource type with the appropriate schema. You can use the custom resource type to map the back-end entities.
7. Finally, the external application can start sending REST web service requests to the proxy system in order to read identities from the back end of the SCIM 2.0 system. For the authentication, you need to use

the *user ID* and *password* of the Identity Authentication technical user for which you have set permission *Access Proxy System API*. In case you decide to use OAuth authentication, the OAuth token URL should follow the pattern: `https://<ias-tenant-host>/oauth2/token`

How Proxy Transformations Work

The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Example

Conditions in Proxy Scenarios

When conditions are applied to users or groups in the proxy *Read Transformation*, the number of returned resources may be “0” or less than the actual number of read entities. This is because some of the entities are filtered out as they don’t match the applied condition. In this example, the returned resources are "0" because all 5 users (items) returned per page are filtered out.

```
Request: GET /Users?startIndex=6&count=5
Response:
{
  "startIndex": 6,
  "itemsPerPage": 5,
  "totalResults": 10,
  "Resources": [],
  "schemas": ["urn:ietf:params:scim:api:messages:2.0:ListResponse"]
}
```

Example

totalResults in Proxy Scenarios

When reading entities from a proxy system, the `totalResults` value returned by the system depends on whether the system API provides a total count (for example, SCIM-based connectors, SAP BTP Platform Members (Cloud Foundry), SAP BTP Java/HTML5 apps (Neo), SAP Application Server ABAP) and whether a transformation condition is applied.

- When the system API supports `totalResults`, the behavior differs depending on whether conditions are defined in the proxy Read Transformation:
  - **No conditions are applied**
    If no conditions are defined in the proxy Read Transformation, Identity Provisioning returns the backend's `totalResults` value directly. This value represents the full number of entities in the system, regardless of `startIndex` and `count`.

For example: There are 1000 users in the proxy system.

```
Request: GET /Users?startIndex=6&count=5
Response:
{
  "startIndex": 6,
  "itemsPerPage": 5,
  "totalResults": 1000,
  "Resources": [/* Users 6 through 10 */],
  "schemas": ["urn:ietf:params:scim:api:messages:2.0:ListResponse"]
}
```

- **Conditions are applied**
  Identity Provisioning always calculates `totalResults`, based on the number of entities that pass the conditions. The backend's value is not used, because it includes entities that may be filtered out.

- When the system API does not support `totalResults` or an equivalent value, Identity Provisioning always calculates `totalResults`.
  Identity Provisioning derives it as follows:
  - If it is not the last page: `totalResults` = `startIndex` + `count`
  - If it is the last page: `totalResults` = exact number of read entities

  As a result, `totalResults` is only an estimate until the last page is reached. A SCIM client that reads all entities from an IPS SCIM Proxy system should use this logic:

```
startIndex = 1
count = N (where N must be between 1 and 100 inclusive; default: 25)
do {
  page = readPage(startIndex, count)
  startIndex += page.itemsPerPage
} while (startIndex <= page.totalResults)
```

  Example request:

```
GET /Users?startIndex=<startIndex>&count=<count>
```

  Do not use `page.Resources.length` to detect the last page. `page.Resources` contains the entities returned for the requested page, unless a transformation condition filters some of them out.
  If a transformation condition is applied:
  - `page.itemsPerPage` contains number of actually read entities, before the condition.
  - `page.Resources` contains only entities that passed the condition, which could be between 0 and count.
  - Pages may contain an empty `Resources` array even when `page.totalResults` is greater than 0 (when no entity passes the specified condition).

  Examples:
  - The system does not provide `totalResults`, there are 150 users in the proxy system and the page size is 100.

```
GET /Users?startIndex=1&count=100 -> page.totalResults == 101,
page.Resources.size == 100
GET /Users?startIndex=101&count=100 -> page.totalResults == 201,
page.Resources.size == 50
GET /Users?startIndex=201&count=100 -> page.totalResults == 150,
page.Resources.size == 0
```

- The system provides `totalResults`, but there is no transformation condition. There are 150 users in the proxy system and the page size is 100.

```
GET /Users?startIndex=1&count=100 -> page.totalResults == 150,
page.Resources.size == 100
GET /Users?startIndex=101&count=100 -> page.totalResults == 150,
page.Resources.size == 50
GET /Users?startIndex=201&count=100 -> page.totalResults == 150,
page.Resources.size == 0
```

- The system provides `totalResults` with transformation condition that filters out each odd entity. There are 100 users in the proxy system and the page size is 50.

```
GET /Users?startIndex=1&count=50 -> page.totalResults == 51,
page.Resources.size == 25
GET /Users?startIndex=51&count=50 -> page.totalResults == 101,
page.Resources.size == 25
GET /Users?startIndex=101&count=50 -> page.totalResults == 150,
page.Resources.size == 0
```

The following table shows how the `totalResults` value is determined when a SCIM proxy endpoint request is made to read entities from a backend system. The outcome depends on three factors: whether the backend system API returns a `totalResults` value, whether conditions are defined in the proxy Read Transformation, and the value of the `ips.proxy.scim.total.results.strategy` property.

- *Provided* - Identity Provisioning returns the `totalResults` value provided by the backend system API.
- *Calculated* - Identity Provisioning calculates `totalResults`, based on the entities returned after conditions are applied.

`totalResults` Support	Conditions Applied	`ips.proxy.scim.total.results.strategy`	Value
Supported	No	Provided/Calculated	Provided
Supported	Yes	Provided	Provided
Supported	Yes	Calculated	Calculated
Not supported	Yes/No	Provided/Calculated	Calculated

For more information, see List of Properties [page 80] → `ips.proxy.scim.total.results.strategy`

How to call a proxy system

As proxy operations cannot be maintained by the Identity Provisioning UI, you need to manage resources (users and groups) by sending SCIM 2.0 API requests to certain endpoints. Below are listed all endpoints and operations available in the Identity Provisioning service. Each provisioning system, however, supports only a specific set of operations.

Tip: The ID of each proxy system (`system_ID` in the table below) is a dash-separated string. You can see it at the end of the system URL in the Identity Provisioning UI.

> ⚠ Caution
>
> Effective *September 2020*, Shanghai (China) tenants that reside on SAP BTP, Neo should be accessed on the following domain: **dispatcher.cn1.platform.sapcloud.cn**
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*

Depending on the infrastructure/environment your Identity Provisioning tenant (bundle or standalone) runs on, use the relevant URI patterns to call an endpoint:

Resource Operation	HTTP Method	Endpoint
Read users	GET	**Infrastructure of SAP Cloud Identity Services** • **Basic**, **OAuth** and **Certificate** Authentication: `https://<ias-tenant-host>/ipsproxy/service/api/v1/scim/<system_id>/Users`
Read a user	GET	**Infrastructure of SAP Cloud Identity Services** • **Basic**, **OAuth** and **Certificate** Authentication: `https://<ias-tenant-host>/ipsproxy/service/api/v1/scim/<system_id>/Users/<user_id>`
Create a user	POST	**Infrastructure of SAP Cloud Identity Services** • **Basic**, **OAuth** and **Certificate** Authentication: `https://<ias-tenant-host>/ipsproxy/service/api/v1/scim/<system_id>/Users`

Resource Operation	HTTP Method	Endpoint
Update a user (*full*)	PUT	**Infrastructure of SAP Cloud Identity Services** • **Basic**, **OAuth** and **Certificate** Authentication: `https://<ias-tenant-host>/ipsproxy/service/api/v1/scim/<system_id>/Users/<user_id>`
Update a user (*partial*)	PATCH	**Infrastructure of SAP Cloud Identity Services** • **Basic**, **OAuth** and **Certificate** Authentication: `https://<ias-tenant-host>/ipsproxy/service/api/v1/scim/<system_id>/Users/<user_id>`
Delete a user	DELETE	**Infrastructure of SAP Cloud Identity Services** • **Basic**, **OAuth** and **Certificate** Authentication: `https://<ias-tenant-host>/ipsproxy/service/api/v1/scim/<system_id>/Users/<user_id>`
Read groups	GET	**Infrastructure of SAP Cloud Identity Services** • **Basic**, **OAuth** and **Certificate** Authentication: `https://<ias-tenant-host>/ipsproxy/service/api/v1/scim/<system_id>/Groups`
Read a group	GET	**Infrastructure of SAP Cloud Identity Services** • **Basic**, **OAuth** and **Certificate** Authentication: `https://<ias-tenant-host>/ipsproxy/service/api/v1/scim/<system_id>/Groups/<group_id>`

Resource Operation	HTTP Method	Endpoint
Create a group	POST	**Infrastructure of SAP Cloud Identity Services** • **Basic**, **OAuth** and **Certificate** Authentication: `https://<ias-tenant-host>/ipsproxy/service/api/v1/scim/<system_id>/Groups`
Update a group (*full*)	PUT	**Infrastructure of SAP Cloud Identity Services** • **Basic**, **OAuth** and **Certificate** Authentication: `https://<ias-tenant-host>/ipsproxy/service/api/v1/scim/<system_id>/Groups/<group_id>`
Update a group (*partial*)	PATCH	**Infrastructure of SAP Cloud Identity Services** • **Basic**, **OAuth** and **Certificate** Authentication: `https://<ias-tenant-host>/ipsproxy/service/api/v1/scim/<system_id>/Groups/<group_id>`
Delete a group	DELETE	**Infrastructure of SAP Cloud Identity Services** • **Basic**, **OAuth** and **Certificate** Authentication: `https://<ias-tenant-host>/ipsproxy/service/api/v1/scim/<system_id>/Groups/<group_id>`

Query Parameters for Proxy System SCIM API

When using Identity Provisioning Proxy System SCIM API, you can specify the `attributes` and the `excludedAttributes` query parameters to control which user or group resource attributes to be included or excluded from the response. The `membersType` query parameter is applicable for groups and defines the type of members (user or group) to be included in the response.

> **Note**
>
> The query parameters work for first-level attributes only.

> **Code Syntax**
>
> In this example, "name" is the first-level attribute. The "familyName", "givenName" and "middleName" are second-level attributes and cannot be used as the query parameter value.

```
"name": {
        "familyName": "Armstrong",
        "givenName": "Julie",
        "middleName": "Grace"
      },
```

The query parameter value is the resource attribute name. In case you want to specify multiple attribute names, you need to separate them by comma. Using " " (space) or "," (comma) is not a valid value and results in returning all the resource attributes.

In order for an attribute to be included or excluded from the response, the attribute's schema must be defined in the *Schemas* attribute of the user resource. For example, if you want to return all users with the custom attribute *roomNumber* in the response, the custom schema must be defined in the *Schemas* attribute of the user resource.

The `attributes` and the `excludedAttributes` query parameters can be combined with other parameters, such as: filtering, paging of resources, paging of multi-value attributes.

attributes

When specified, the `attributes` query parameter defines which user or group resource attributes to be included in the response.

This request example returns all users with the *active* and *userName* attributes in the response.

> **Code Syntax**

```
GET /Users?attributes=active,userName
```

This request example returns all groups with the *displayName* attribute in the response.

> **Code Syntax**

```
GET /Groups?attributes=displayName
```

This request example combines the *attributes* query parameter with filtering of a user by *userName*. It returns only the TestUser with both attributes: *active* and *displayName* in the response.

> **Code Syntax**

```
GET /Users?filter=userName eq "TestUser"&attributes=active,displayName
```

In case an attribute is defined in two schemas (for example, the *emails* attribute is defined in the core schema and in the custom schema), you need to specify the schema URI and the attribute name as the

query parameter value. This way, the users will be returned with emails attribute from the specified schema. Otherwise, if you only specify *emails* without the schema URI, the *emails* attribute from both schemas will be returned.

Code Syntax

```
GET /Users?
attributes=urn:ietf:params:scim:schemas:extension:sap:2.0:User:emails
```

excludedAttributes

When specified, the `excludedAttributes` query parameter defines which user or group resource attributes to be excluded in the response.

This request example returns all users with all attributes excluding *userType* in the response.

Code Syntax

```
GET /Users?excludedAttributes=userType
```

This request example returns all groups with all attributes excluding *schemas* in the response.

Code Syntax

```
GET /Groups?excludedAttributes=schemas
```

This request example combines the *excludedAttributes* query parameter with paging parameters *startIndex* and *count* of users. It returns 3 users starting with the first one as the first query result with all attributes excluding *schemas* in the response.

Code Syntax

```
GET /Users?startIndex=1&count=3&excludedAttributes=schemas
```

membersType

When specified, the `membersType` query parameter defines the type of group members (user or group) to be included in the response. You can query for members of all groups or within a certain group. This parameter is supported only for SAP Ariba Applications proxy system.

This request example returns only the user members of all groups.

Code Syntax

```
GET /Groups?membersType=user
```

This request example returns only the user members of a specific group.

Code Syntax

```
GET /Groups/{group-id}?membersType=user
```

This request example returns only the group members of a specific group.

Code Syntax

```
GET /Groups/{group-id}?membersType=group
```

This request example returns the group members of all groups matching the specified query.

Code Syntax

```
GET /Groups?<query>&membersType=group
```

Filters

The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query standard) supports the following filtering capabilities for users and groups:

- *Single entity filter* - Refers to a filtering mechanism that is used to identify a specific single resource (user or group) within a system, based on unique identifiers. Typically, these are userName and email attributes for users and displayName attribute for groups. Identity Provisioning supports only the 'eq' (equal) operator. For example:
  `GET .../Users?filter=userName eq "johnsmith03"`
  `GET .../Groups?filter=displayName eq "accountants"`
- *Delta read filter* - Refers to a filter mechanism used to get a list of resources that have been modified after a specified period of time. Identity Provisioning supports only the 'gt' (greater than) operator. For example:
  `GET .../Users?filter=meta.lastModified gt "2024-02-14T04:42:34Z"`

If your system supports native read filtering, the Identity Provisioning proxy application will translate the SCIM filter to the native system filter, and will try to combine it with the relevant read filter property (for example, `<system_prefix>.user.filter`, `<system_prefix>.group.filter` or `<system_prefix>.origin`). For more information about all read filter properties, see List of Properties [page 80].

Example

Combining filters from the properties and the SCIM request

1. You have the following user mapping in the Read Transformation of a proxy system.

Use Case 1

```
{
    "sourcePath": "$.userName",
    "targetPath": "$.userName",
    "correlationAttribute": true
},
```

Use Case 2

```
{
  "sourcePath":
"$.emails[0].value",

"preserveArrayWithSingleElement":
true,
  "targetPath": "$.emails[0].value"
},
```

2. You set a filter on the *Properties* tab of the proxy system. For example: `scim.user.filter` = *timezone eq "US"*. For more information on the system specific filtering properties, see List of Properties.
3. You execute the SCIM Proxy endpoint request to get a user with a given username or email.
   Use Case 1: `GET .../Users?filter=userName eq "johnsmith03"`

> **Use Case 2**: `GET .../Users?filter=emails[0].value eq "john.smith03@example.com"`
> This query request to the proxy system API will result into:
> **Use Case 1**: `/Users?filter=timezone eq "US" and userName eq "johnsmith03"`
> **Use Case 2**: `/Users?filter=timezone eq "US" and emails[0].value eq "john.smith03@example.com"`

If the Identity Provisioning finds:

- **0** users that meet the filtering criteria, the service returns HTTP status code *200* (OK) with '*totalResults*' set to a value of 0.
- **1** user that meets the filtering criteria, the service returns HTTP status code *200* (OK), and includes the result in the body of the response.
- **More than 1** users that meet the filtering criteria, the service returns HTTP status code *400* (Bad Request) with detail error type '*tooMany*'.
- In the *Read Transformation*, there must be a mapping between the attribute names in `"sourcePath"` and `"targetPath"` (see the example mapping above, where targetPath matches the left side of the used '**eq**' filter).
- Fully qualified names (`<schema>:<attribute>`) are not supported.
  For example: `GET .../Users/urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:employee Number eq '<attribute>'`
- If your system supports multivalued e-mails (that is *$.emails[0].value*, *$.emails[1].value*, etc.), the search criteria will always resolve only one user e-mail. This is the first user e-mail (**$.emails[0].value**).

Additional Information

Lean more about system types: System Types [page 70]

Learn more about SCIM 2.0 protocol: SCIM protocol

Proxy scenario example: Hybrid Scenario: SAP Identity Management

SAP Community Blog: Hybrid Scenarios with Identity Provisioning Proxy

1.6.3.1 Identity Authentication

Follow this procedure to set up SAP Cloud Identity Service – Identity Authentication as a proxy system.

Prerequisites

To establish the connection between Identity Provisioning and Identity Authentication, you need to set up the technical user (of type *System*) in Identity Authentication and assign this user the necessary authorizations. You have the option to do it either now, as a prerequisite, or while configuring Identity Authentication as a proxy system, as described in step 5.

Context

Identity Authentication provides authentication and single sign-on for users in the cloud.

You can use the Identity Provisioning user interface (UI) to configure Identity Authentication as a proxy system for hybrid integration, where Identity Authentication is connected to an external identity management system for user and group provisioning. Using the Identity Provisioning you can read corporate users from the external system and provision them to the Identity Authentication user store (and the other way around) without making a direct connection between these systems. This way, you can implement secure authentication, single sign-on (SSO), strong authentication, and mobile SSO so that the provisioned users to Identity Authentication have access to the business applications of your company.

The proxy systems consume SCIM 2.0 API which is provided by Identity Authentication.

There are two versions of the Identity Authentication SCIM API. They are handled by the `ias.api.version` property as follows:

- When the value is set to **1** or the property is not defined (typical for systems created before versioning was introduced on July 9, 2021) - Identity Authentication SCIM API (in short, SCIM API version 1) is used.

  > **Note**
  >
  > The SAP Cloud Identity Services – Identity Authentication SCIM API (version 1) will be decommissioned on November 30, 2028. You must use its successor, the Identity Directory SCIM API. This requires updating your Identity Authentication connector from version 1 to version 2, as described in Update Connector Version [page 2424].

- When the value is set to **2** - Identity Directory SCIM API (in short, SCIM API version 2) is used. This value is set automatically for all manually created systems in the Identity Provisioning UI after versioning was introduced on July 9, 2021.

SCIM API version 2 is enhanced to support patch operations for proxy systems only, paging for group members and user's groups, custom attributes, delta read mode for users. Also, the group resource mapping in the transformation is not ignored by default, as it is in SCIM API version 1.

To create Identity Authentication as a proxy system, proceed as follows:

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.

2. Save your changes.
3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
4. Save your changes.

3. Navigate to *Identity Provisioning* > *Proxy Systems*.
4. Add *Identity Authentication* as a proxy system. For more information, see Add New Systems [page 2415].
5. Set up the communication between Identity Provisioning and Identity Authentication and configure your authentication method (basic or certificate-based).

> ### Note
> We recommend that you use certificate-based authentication.

   a. In your newly added Identity Authentication proxy system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Manage Certificates [page 2450].

      Skip step **a.** if you want to use basic authentication.

      In SAP Cloud Identity Services administration console, perform the next steps. They are relevant for both basic and certificate-based authentication.

   b. Add System as administrator and provide the respective credentials.

      For basic authentication, provide a password. The user ID will be generated automatically when you set the password for the first time.

      For certificate-based authentication, upload the certificate you have generated in SAP Cloud Identity Services administration console on the previous step.

   c. Save your changes.
   d. Make sure *Manage Users* and *Manage Groups* authorization roles are enabled for the technical user. This way, you can create, edit and delete users and groups in the Identity Authentication user store.

6. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
Type	Enter: *HTTP*
URL	Specify the URL of the Identity Authentication tenant of your company. For example: **https://mytenant.accounts.ondemand.com** **Note** If your Identity Authentication Shanghai (China) tenants reside on SAP BTP, Neo, you should use the following URL pattern: *https://<tenant_ID>.accounts.sapcloud.cn/*

Property Name	Description & Value
`ProxyType`	Enter: *Internet* The Identity Authentication is a cloud solution and is outside of your company on-premise infrastructure.
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the Client ID (previously User ID) of the Identity Authentication technical user. It is generated automatically for the administrator of type system, when choosing *Secrets* > *Add* > *Save*. For example: *1ab7c243-5de5-4530-8g14-1234h26373ab* If your technical user was created before *January 2020*, enter the T-user. For example: *T000003*
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the Client Secret (previously Password) of the Identity Authentication technical user. It is generated automatically for the administrator of type system, when choosing *Secrets* > *Add* > *Save*.

Optional Properties

Property Name	Description & Value
• `ias.<property_name>` • `scim.<property_name>`	When using SCIM API version 2, property names start with `ias` prefix, for example: `ias.user.filter`. When using SCIM API version 1, property names start with `scim` prefix, for example: `scim.user.filter`. For more information, see List of Properties [page 80]. Use the main search or filter properties by *Name* or *System Type* columns.

Property Name	Description & Value
`ias.user.filter`	When specified, only those users matching the filter expression will be read. For example: **name.familyName eq "Smith" and addresses.country eq "US"** This filter will read only users whose family name is "*Smith*" and are living in the *United States*. For more information, see Identity Directory SCIM API: User Search.
`ias.group.filter`	When specified, only those groups matching the filter expression will be read. For example: **displayName eq "ProjectTeam1"** This filter will read only groups, whose display name is "*ProjectTeam1*". For more information, see Identity Directory SCIM API: Group Search.

> ### Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

7. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Identity Authentication* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your Identity Authentication system. For more information, see: Manage Transformations [page 2439]

   Identity Authentication: SCIM REST API Identity Authentication: SCIM REST API

   SCIM API version 2: Identity Directory SCIM API

   Default read and write transformations:

   > ### → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Default transformations for SCIM API version 1:

Read Transformation	Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.userUuid",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails"
      },
      {
```

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "condition":
"($.emails.length() > 0) &&
($.name.familyName EMPTY false)",
    "mappings": [
      {
        "sourcePath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.corporateGroups"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
        "targetPath":
"$.schemas[0]"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User",
        "targetPath":
"$.schemas[1]"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "optional": true
      },
      {
        "sourcePath":
"$.emails[*].value",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails[?
(@.value)]"
      },
      {
        "sourcePath":
"$.userType",
        "targetPath":
"$.userType",
```

Read Transformation

```
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value"
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.userType",
        "targetPath":
"$.userType",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath":
"$.addresses",
        "targetPath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.locale",
```

Write Transformation

```
        "defaultValue":
"employee",
        "optional": true
      },
// The userType attribute accepts
different values. The default one
is employee.
// If you set it to public, that
means Identity Authentication is
the default password store. See
the Remember box below.
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "optional": true
      },

// If a user address misses type,
the putIfAbsent function sets the
default address type – work.
// If a user address has type
but it's different than home or
work, the putIfPresent function
sets the default address type –
work.
// If a user address is of type
home or work, the putIfPresent
function keeps this value as is.
      {
        "sourcePath":
"$.addresses",
        "targetPath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "defaultValue": [],
        "optional": true,
        "functions": [
          {
```

Read Transformation

```
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.timeZone",
        "targetPath":
"$.timezone",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath":
"$.sourceSystem",
        "targetPath":
"$.sourceSystem",
        "ignore": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "condition":
"$.displayName EMPTY true",
        "type": "remove",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
      },
      {
```

Write Transformation

```
            "function":
"putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          },
          {
            "condition": "(@.type
NIN ['work', 'home'])",
            "function":
"putIfPresent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional" : true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional" : true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

```
ension:enterprise:2.0:User']
['organization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional" : true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional" : true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional" : true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional" : true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional" : true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "defaultValue": true,
        "optional": true
      },
      {
        "constant": "false",
        "targetPath":
"$.sendMail",
        "scope": "createEntity"
```

Read Transformation

```
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']",
        "optional": true
      },
      {
        "sourcePath": "$.company",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "targetPath":
"$.displayName"
```

Write Transformation

```
      },
      {
        "constant": "true",
        "targetPath":
"$.mailVerified",
        "scope": "createEntity"
      },
// There will be no initial
password provided by default.
That's why passwordStatus is
disabled.
      {
        "constant": "disabled",
        "targetPath":
"$.passwordStatus",
        "scope": "createEntity"
      },
// The sourceSystem attribute
shows the provisioning source of
the users. The supported value is
39.
// That means, a corporate user
is provisioned via the Identity
Authentication REST API. See the
Remember box below.
      {
        "constant": "39",
        "targetPath":
"$.sourceSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.timezone",
        "targetPath":
"$.timeZone",
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "scope": "createEntity",
        "sourcePath":
"$.displayName",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "functions": [
```

Read Transformation	Write Transformation

<pre>{
  "mappings": [
    {
      "type":
"replaceAllString",
      "regex": "[\\s\
\p{Punct}]",
      "replacement": "_"
    }
  ]
},
{
  "scope": "createEntity",
  "optional": true,
  "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group'][
'name']",
  "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group'][
'name']"
},
{
  "optional": true,
  "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']
['description']",
  "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']
['description']"
},
{
  "optional": true,
 	
"preserveArrayWithSingleElement":
true,
  "sourcePath": "$.members",
  "targetPath": "$.members"
}
]
}</pre>
	<pre>{
	  "mappings": [
	    {
	      "type":
	"replaceAllString",
	      "regex": "[\\s\
	\p{Punct}]",
	      "replacement": "_"
	    }
	  ]
	},
	{
	  "scope": "createEntity",
	  "optional": true,
	  "sourcePath": "$
	['urn:sap:cloud:scim:schemas:exten
	sion:custom:2.0:Group'][
	'name']",
	  "targetPath": "$
	['urn:sap:cloud:scim:schemas:exten
	sion:custom:2.0:Group'][
	'name']"
	},
	{
	  "optional": true,
	  "sourcePath": "$
	['urn:sap:cloud:scim:schemas:exten
	sion:custom:2.0:Group']
	['description']",
	  "targetPath": "$
	['urn:sap:cloud:scim:schemas:exten
	sion:custom:2.0:Group']
	['description']"
	},
	{
	  "optional": true,
	
	"preserveArrayWithSingleElement":
	true,
	  "sourcePath": "$.members",
	  "targetPath": "$.members"
	}
	]
	}</pre>

Default transformations for SCIM API version 2:

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails"
      },
      {
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:cor
e:2.0:User","urn:ietf:params:scim:
schemas:extension:enterprise:2.0:U
ser","urn:ietf:params:scim:schemas
:extension:sap:2.0:User"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "optional": true
      },
      {

"sourcePath": "$.emails[*].
['value','display','primary']",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath":
"$.userType",
        "targetPath":
"$.userType",
        "defaultValue":
"employee",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
```

Read Transformation

```
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value"
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.userType",
        "targetPath":
"$.userType",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath":
"$.addresses",
        "targetPath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.locale",
```

Write Transformation

```
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath":
"$.addresses",
        "targetPath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "defaultValue": [],
        "optional": true,
        "functions": [
          {
            "function":
"putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          },
          {
            "condition": "(@.type
NIN ['work', 'home'])",
            "function":
"putIfPresent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation

```
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.timezone",
        "targetPath":
"$.timezone",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath":
"$.sourceSystem",
        "targetPath":
"$.sourceSystem",
        "ignore": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "condition":
"$.displayName EMPTY true",
        "type": "remove",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
      },
      {
```

Write Transformation

```
ension:sap:2.0:User']
['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true
      },
      {
```

Read Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "defaultValue": true,
        "optional": true
      },
      {
        "constant": false,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['sendMail']",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails']",
        "scope": "createEntity",
        "functions": [
          {
            "function":
"putIfAbsent",
            "key": "verified",
            "defaultValue": true
          }
        ]
```

Read Transformation

```
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']",
        "optional": true
      },
      {
        "sourcePath": "$.company",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['workAssignment']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['workAssignment']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['jobDetails']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['jobDetails']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['workforcePersonId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['workforcePersonId']",
        "optional": true
      },
      {
```

Write Transformation

```
      },
      {
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails'][*]
['type']",
        "type": "remove"
      },
      {
        "constant": "disabled",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['passwordDetails']['status']",
        "scope": "createEntity"
      },
      {
        "constant": 39,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['sourceSystem']",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.timezone",
        "targetPath":
"$.timezone",
        "optional": true
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.Operations",
```

Read Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['jobClassification']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['jobClassification']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['organizationalUnit']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['organizationalUnit']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['costCenter']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['plant']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['plant']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['companyCode']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

```
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:cor
e:2.0:Group","urn:sap:cloud:scim:s
chemas:extension:custom:2.0:Group"
,"urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "scope": "createEntity",
        "functions": [
          {
            "type":
"replaceAllString",
            "regex": "[\\s\
\p{Punct}]",
            "replacement": "_"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
```

Read Transformation	Write Transformation

```
ension:sap.odm:2.0:User']
['companyCode']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['generalLedgerAccount']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['generalLedgerAccount']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingOrganization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingOrganization']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingGroup']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingGroup']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
```

```
        "scope": "createEntity"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']
['description']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']
['description']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId'] EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId'] EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId'] EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']"
      },
      {
        "condition": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId'] EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation	Write Transformation

```
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
             "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
         "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
         "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']"
      },
      {
         "sourcePath":
"$.displayName",
         "targetPath":
"$.displayName"
      },
      {
         "sourcePath":
"$.members",
         "targetPath":
"$.members",

"preserveArrayWithSingleElement":
true,
         "optional": true
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
```

```
ension:sap:2.0:Group']
['externalName']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['externalName']"
      }
    ]
  }
}
```

Read Transformation	Write Transformation
`"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']", "optional": true }, { "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']", "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']", "optional": true }, { "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']", "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']", "optional": true }] } }`	

→ Remember

If you set `$.userType` to "**public**", all passwords will be written by default in the Identity Authentication. Thus, all provisioned users will successfully sign in to Identity Authentication target system.

When `$.userType` is set to "**employee**", the sign-in behavior of the provisioned users depends on whether users have been created with or without a password, and where these passwords are stored. Thus, you need to modify the target transformations accordingly in order for the users to successfully sign in to the SAP Cloud Identity Services administration console.

To learn more, see *Guided Answers*: Identity Authentication: Provisioned Users Can't Sign In

8. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

> **Note**
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> **Caution**
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

Identity Authentication: Documentation
Identity Authentication: SCIM REST API
Hybrid Scenario: SAP Identity Management

1.6.3.2 Intelligent Opportunity Analyzer

Follow this procedure to set up intelligent opportunity analyzer as a proxy system.

Prerequisites

- You have added *Intelligent Opportunity Analyzer* with *scim* plan in the *Entitlements* page of your subaccount as described in.
- You have created an instance and generated a service key for the *scim* service plan of *Intelligent Opportunity Analyzer*. For more information, see Assigning the Entitlements for Intelligent Opportunity Analyzer to Your Subaccounts.

Context

Intelligent opportunity analyzer is a service that comes integrated with SAP Ariba Category Management for identifying and managing opportunities that lead to improving the performance of a purchasing category.

You can use Identity Provisioning to configure intelligent opportunity analyzer as a proxy system in hybrid scenarios. For example, when intelligent opportunity analyzer is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users back to the intelligent opportunity analyzer.

> **Note**
>
> Intelligent opportunity analyzer does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.

3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
4. Save your changes.

3. Add *Intelligent Opportunity Analyzer* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL provided by the service key under the *scim-v1* field without adding the path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the value from the *clientid* field of the service key.
`Password`	(Credential) Enter the value from the *clientsecret* field of the service key.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. This is the value from the *url* field of the service key. Ensure to append the value `/oauth/token?grant_type=client_credentials`.
(Optional) `ioa.user.filter`	When specified, only those intelligent opportunity analyzer users matching the filter expression will be read. For example: • *userName eq "SmithJ"* • *emails.value eq "john.doe@example.com"*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users. For more information, see Query Parameters for Proxy System SCIM API.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Intelligent Opportunity Analyzer* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your intelligent opportunity analyzer system. For more information, see:

Manage Transformations [page 2439]

> ⓘ Note
>
> The documentation for the Intelligent Opportunity Analyzer API and its integration with Identity Provisioning is still under development. Refer to the Intelligent Opportunity Analyzer Setup for updates.

Default read and write transformations

> → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta"
      },
      {
        "sourcePath":"$.userName",
        "targetPath":"$.userName",

"correlationAttribute":true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
           "type": "concatString",
           "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {

"sourcePath":"$.loginName",

"targetPath":"$.loginName",
        "optional":true
      },
      {

"sourcePath":"$.name.givenName",

"targetPath":"$.name.givenName",
        "optional":true
      },
      {

"sourcePath":"$.name.familyName",
```

Write Transformation

Code Syntax

```
{
  "user":{
    "mappings":[
     {
       "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
      {

"sourcePath":"$.loginName",

"targetPath":"$.loginName",
        "optional":true
      },
      {

"sourcePath":"$.name.givenName",

"targetPath":"$.name.givenName",
        "optional":true
      },
      {

"sourcePath":"$.name.familyName",

"targetPath":"$.name.familyName",
        "optional":true
      },
      {

"sourcePath":"$.name.formatted",

"targetPath":"$.name.formatted",
        "optional":true
      },
      {

"sourcePath":"$.name.middleName",
```

Read Transformation

```
"targetPath":"$.name.familyName",
        "optional":true
      },
      {

"sourcePath":"$.name.formatted",

"targetPath":"$.name.formatted",
        "optional":true
      },
      {

"sourcePath":"$.name.middleName",

"targetPath":"$.name.middleName",
        "optional":true
      },
      {

"sourcePath":"$.name.honorificPref
ix",

"targetPath":"$.name.honorificPref
ix",
        "optional":true
      },
      {

"sourcePath":"$.name.honorificSuff
ix",

"targetPath":"$.name.honorificSuff
ix",
        "optional":true
      },
      {
        "sourcePath":"$.emails",
        "targetPath":"$.emails",

"preserveArrayWithSingleElement":t
rue
      },
      {
        "sourcePath":"$.emails[?
(@.primary== true)].value",

"correlationAttribute":true
      },
      {
        "sourcePath":"$.active",
        "targetPath":"$.active"
      },
      {

"sourcePath":"$.displayName",

"targetPath":"$.displayName",
        "optional":true
      },
      {
```

Write Transformation

```
"targetPath":"$.name.middleName",
        "optional":true
      },
      {

"sourcePath":"$.name.honorificPref
ix",

"targetPath":"$.name.honorificPref
ix",
        "optional":true
      },
      {

"sourcePath":"$.name.honorificSuff
ix",

"targetPath":"$.name.honorificSuff
ix",
        "optional":true
      },
      {
        "sourcePath":"$.active",
        "targetPath":"$.active"
      },
      {

"sourcePath":"$.displayName",

"targetPath":"$.displayName",
        "optional":true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      }
    ],
    "scimEntityEndpoint": "Users"
  }
}
```

Read Transformation	Write Transformation

<pre>"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
"optional": true,
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']"
}
],
"scimEntityEndpoint": "Users"
}
}</pre>	

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ### Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.

For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*

To learn more, see: Proxy Systems [page 1343]

1.6.3.3 Local Identity Directory

Follow this procedure to set up Local Identity Directory as a proxy system.

Prerequisites

> ⓘ Note
>
> The *Local Identity Directory* connector is available for both bundle and standalone tenants running on SAP Cloud Identity Services infrastructure.

Context

Identity directory is the user store of SAP Cloud Identity Services. It provides a central place for storing and managing users, groups and custom schemas through the System for Cross-domain Identity Management 2.0 REST API, in short Identity Directory SCIM API.

You can use the identity directory for user and group provisioning with an external identity management system without making a direct connection between them. For this, you configure the Local Identity Directory connector as a proxy system in the SAP Cloud Identity Services administration console.

The entities are read and written to and from the identity directory of your current SAP Cloud Identity Services tenant. Therefore, you don't have to set up connectivity and authentication properties for the proxy system. In case you want to use the identity directory in another tenant, add Identity Authentication (version 2) as a proxy system with the respective connectivity set up.

The proxy systems consume Identity Directory SCIM API. It supports patch operations for proxy systems only, paging for group members and user's groups, custom attributes, delta read mode for users. Also, the group resource mapping in the transformation is not ignored by default.

By configuring the directory as a proxy system, you implement secure authentication, single sign-on (SSO), strong authentication, and mobile SSO so that the provisioned users to the directory have access to the business applications of your company.

To create Local Identity Directory as a proxy system, proceed as follows:

Procedure

1. Sign in to SAP Cloud Identity Services administration console and create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to SAP Cloud Identity Services.

   If you already have a technical user, skip this step.

   - For **Certificate-based authentication**, follow the procedure in Manage Certificates for Inbound Connection [page 2456] → *SAP Cloud Identity Infrastructure*
   - For **Basic authentication**, navigate to *Users & Authorizations* > *Administrators* and add an admin user of type **System**. Configure a client ID and secret for this user and enable the *Access Proxy System API* permission.

2. Navigate to *Identity Provisioning* > *Proxy Systems*.
3. Add *Local Identity Directory* as a proxy system.

   For more information, see Add New Systems [page 2415].

4. **Optional:** If needed, choose the *Properties* tab to configure filtering and paging.

   Optional Properties

Property Name	Value
`idds.user.filter`	This property filters users by particular attributes. You can set a single attribute or multiple ones as search criteria. • Value pattern for single attribute: *<user_attribute> eq "<value>"* • Value pattern for multiple attributes: *<user_attribute1> eq "<value1>" and/or <user_attribute2> eq "<value2>"*
`idds.group.filter`	This property filters groups by display name. You can set a single display name or multiple ones as filter criteria. If you enter multiple display names (using **OR** operator), the filter will search for any of them. • Value pattern for single attribute: *displayName eq "<group_name>"* • Value pattern for multiple attributes: *displayName eq "<group_name1>" or displayName eq "<group_name2>"*
`idds.user.groups.paging.enabled`	This property enables paging of user's groups. The maximum number of user's groups returned per request is 1000. To read more than 1000 user's groups, paging must be enabled.

Property Name	Value
`idds.group.members.paging.enabled`	This property enables paging of group members. The maximum number of group members returned per request is 20 000. To read more than 20 000 group members, paging must be enabled.

Local Identity Directory properties are prefixed with `idds.<property_name>`. For more information, see List of Properties [page 80].

> ### Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

5. **Optional:** Configure transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the Local Identity Directory proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your Local Identity Directory. For more information, see:

   - Manage Transformations [page 2439]
   - SCIM API version 2: Identity Directory SCIM API

   Default read and write transformations:

   > ### → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

   Default transformations for Local Identity Directory:

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails"
      },
      {
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:cor
e:2.0:User","urn:ietf:params:scim:
schemas:extension:enterprise:2.0:U
ser","urn:ietf:params:scim:schemas
:extension:sap:2.0:User"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "optional": true
      },
      {
        "sourcePath":
"$.emails[*].value",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails[?
(@.value)]"
      },
      {
        "sourcePath":
"$.userType",
        "targetPath":
"$.userType",
        "defaultValue":
"employee",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
```

Read Transformation

```
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value"
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.userType",
        "targetPath":
"$.userType",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath":
"$.addresses",
        "targetPath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.locale",
```

Write Transformation

```
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath":
"$.addresses",
        "targetPath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "defaultValue": [],
        "optional": true,
        "functions": [
          {
            "function":
"putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          },
          {
            "condition": "(@.type
NIN ['work', 'home'])",
            "function":
"putIfPresent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation	Write Transformation

```
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.timezone",
        "targetPath":
"$.timezone",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath":
"$.sourceSystem",
        "targetPath":
"$.sourceSystem",
        "ignore": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "condition":
"$.displayName EMPTY true",
        "type": "remove",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
      },
      {
```

```
ension:sap:2.0:User']
['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true
      },
      {
```

Read Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "defaultValue": true,
        "optional": true
      },
      {
        "constant": false,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['sendMail']",
        "scope": "createEntity"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails']",
        "scope": "createEntity",
        "functions": [
          {
            "function":
"putIfAbsent",
            "key": "verified",
            "defaultValue": true
          }
        ]
```

Read Transformation

```
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']",
        "optional": true
      },
      {
        "sourcePath": "$.company",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['workAssignment']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['workAssignment']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['jobDetails']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['jobDetails']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['workforcePersonId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['workforcePersonId']",
        "optional": true
      },
      {
```

Write Transformation

```
      },
      {
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails'][*]
['type']",
        "type": "remove"
      },
      {
        "constant": "disabled",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['passwordDetails']['status']",
        "scope": "createEntity"
      },
      {
        "constant": 39,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['sourceSystem']",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.timezone",
        "targetPath":
"$.timezone",
        "optional": true
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.Operations",
```

Read Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['jobClassification']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['jobClassification']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['organizationalUnit']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['organizationalUnit']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['costCenter']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['plant']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['plant']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['companyCode']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

```
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:cor
e:2.0:Group","urn:sap:cloud:scim:s
chemas:extension:custom:2.0:Group"
,"urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "scope": "createEntity",
        "functions": [
          {
            "type":
"replaceAllString",
            "regex": "[\\s\
\p{Punct}]",
            "replacement": "_"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
```

Read Transformation	Write Transformation

ension:sap.odm:2.0:User']
['companyCode']",
 	
"preserveArrayWithSingleElement":
true,
"optional": true
},
{
"sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['generalLedgerAccount']",
"targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['generalLedgerAccount']",
 	
"preserveArrayWithSingleElement":
true,
"optional": true
},
{
"sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingOrganization']",
"targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingOrganization']",
 	
"preserveArrayWithSingleElement":
true,
"optional": true
},
{
"sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingGroup']",
"targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingGroup']",
 	
"preserveArrayWithSingleElement":
true,
"optional": true
}
]
},
"group": {
"scimEntityEndpoint":
"Groups",
"mappings": [
{
"sourcePath": "$.id",
"targetPath": "$.id",
"targetVariable":
"entityIdSourceSystem"
},
{
	"scope": "createEntity"
	},
	{
	"sourcePath": "$
	['urn:sap:cloud:scim:schemas:exten
	sion:custom:2.0:Group']
	['description']",
	"optional": true,
	"targetPath": "$
	['urn:sap:cloud:scim:schemas:exten
	sion:custom:2.0:Group']
	['description']"
	},
	{
	"condition": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group']
	['applicationId'] EMPTY false",
	"sourcePath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group']
	['applicationId']",
	"targetPath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group']
	['applicationId']"
	},
	{
	"condition": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group']
	['applicationId'] EMPTY false",
	"sourcePath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group'][type']",
	"optional": true,
	"targetPath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group'][type']"
	},
	{
	"condition": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group']
	['applicationId'] EMPTY false",
	"sourcePath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group']
	['supportedOperations']",
	"optional": true,
	"targetPath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group']
	['supportedOperations']"
	},
	{
	"condition": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group']
	['applicationId'] EMPTY false",
	"sourcePath": "$
	['urn:ietf:params:scim:schemas:ext

Read Transformation	Write Transformation

```
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
             "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
         "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
         "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']"
      },
      {
         "sourcePath":
"$.displayName",
         "targetPath":
"$.displayName"
      },
      {
         "sourcePath":
"$.members",
         "targetPath":
"$.members",

"preserveArrayWithSingleElement":
true,
         "optional": true
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['applicationId']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
```

```
ension:sap:2.0:Group']
['externalName']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['externalName']"
      }
    ]
  }
}
```

Read Transformation	Write Transformation

<pre>"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
"optional": true
},
{
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
"optional": true
},
{
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName']",
"optional": true
}
]
}
}</pre>	

→ Remember

If you set `$.userType` to "**public**", all passwords will be written by default in the Local Identity Directory. Thus, all provisioned users will successfully sign in to Local Identity Directory target system.

When `$.userType` is set to "**employee**", the sign-in behavior of the provisioned users depends on whether users have been created with or without a password, and where these passwords are stored. Thus, you need to modify the target transformations accordingly in order for the users to successfully sign in to the SAP Cloud Identity Services administration console.

To learn more, see *Guided Answers*: Identity Authentication: Provisioned Users Can't Sign In

6. Connect the external consumer to SAP Cloud Identity Services with the technical user you have created in step 1.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

> **Note**
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management → sections **Next Steps** and **Future Identity Lifecycle**.

1.6.3.4 SAP Advanced Financial Closing

Follow this procedure to set up SAP Advanced Financial Closing as a proxy system.

Prerequisites

- You have created an instance and generated a service key for the standard service plan of SAP Advanced Financial Closing. For more information, see: How to Manage User Access Using the SCIM API Provided. The service key contains the API URL and the OAuth credentials (`clientid` and `clientsecret`) under the `uaa` property.

Context

SAP Advanced Financial Closing allows you to define, automate, process, and monitor the financial closing tasks for the entities of your organization. It is an SAP BTP application that runs in an SAP BTP subaccount.

You can use Identity Provisioning to configure SAP Advanced Financial Closing as a proxy system and configure it in hybrid scenarios. For example, when SAP Advanced Financial Closing is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management and SAP Cloud Identity Access Governance, without making a direct connection between both systems. You can provision users and user groups to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users and user groups back to the SAP Advanced Financial Closing.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Advanced Financial Closing* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL provided by the service key under *endpoints* > *scim2* without adding the path information. For example: `https://afc-production-afc-api.cfapps.eu10.hana.ondemand.com`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the user ID provided by the service key under *uaa* > *clientid*.
`Password`	(Credential) Enter the password provided by the service key under *uaa* > *clientsecret*.

Property Name	Value
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL provided by the service key of your SAP Advanced Financial Closing instance. It follows the pattern: **`<uaa.url>/oauth/token`**, where: • `<uaa.url>` is the URL provided by the service key under *uaa* > *url*. • **`/oauth/token`** is the suffix you need to add.
(Optional) `s4hana.afc.user.filter`	When specified, only those SAP Advanced Financial Closing users matching the filter expression will be read. Supported operators: **`eq`** (equal), **`sw`** (starts with) and **`co`** (contains) For example: • *userName eq "Julie Armstrong"* • *emails eq "julie.armstrong@example.com"* • *name.familyName sw "A"* • *name.givenName co "Ju"*
(Optional) `s4hana.afc.group.filter`	When specified, only those SAP Advanced Financial Closing groups matching the filter expression will be read. Supported operators: **`eq`** (equal), **`sw`** (starts with) and **`co`** (contains) For example: *displayName eq "Administrators"*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> **Note**
>
> The Identity Provisioning implementation of the SCIM Proxy API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy SCIM API.

5. Configure the transformations.

   Transformations are used to map user and group attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Advanced Financial Closing* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Advanced Financial Closing system. For more information, see: Manage Transformations [page 2439].

   How to Manage User Access Using the SCIM API Provided

   SAP Business Accelerator Hub: SAP Advanced Financial Closing

Default read and write transformations:

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.name.formatted",
        "targetPath":
"$.name.formatted",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.name.formatted",
        "targetPath":
"$.name.formatted",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
```

Read Transformation	Write Transformation

```
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.roles",
        "targetPath": "$.roles",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
```

```
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "condition":
"$.emails[0].length() > 0",
        "targetPath":
"$.emails[0].primary",
        "constant": true
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.emails",
        "functions": [
          {
            "function":
"putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "constant":
"urn:ietf:params:scim:api:messages
:2.0:PatchOp",
        "targetPath":
"$.schemas[0]",
        "scope": "patchEntity"
      },
      {
        "sourcePath":
"$.Operations",
```

Read Transformation	Write Transformation

<pre>{
"sourceVariable":
"entityBaseLocation",
"targetPath":
"$.meta.location",
"targetVariable":
"entityLocationSourceSystem",
"functions": [
{
"type":
"concatString",
"suffix": "$
{entityIdSourceSystem}"
}
]
}
]
},
"group": {
"scimEntityEndpoint":
"Groups",
"mappings": [
{
"sourcePath": "$.id",
"targetPath": "$.id",
"targetVariable":
"entityIdSourceSystem"
},
{
"sourcePath":
"$.displayName",
"targetPath":
"$.displayName"
},
{
"sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
"optional": true,
"targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']"
},
{
"sourcePath": "$.members",
"targetPath": "$.members",
 	
"preserveArrayWithSingleElement":
true,
"optional": true
},
{
"sourcePath": "$.schemas",
"targetPath": "$.schemas",
 	
"preserveArrayWithSingleElement":
true
},
{
"sourcePath": "$.meta",
"targetPath": "$.meta",
"optional": true</pre>
	<pre>"preserveArrayWithSingleElement":
	true,
	"targetPath":
	"$.Operations",
	"scope": "patchEntity"
	}
	]
	},
	"group": {
	"scimEntityEndpoint":
	"Groups",
	"mappings": [
	{
	"sourceVariable":
	"entityIdTargetSystem",
	"targetPath": "$.id"
	},
	{
	"constant": [
	
	"urn:ietf:params:scim:schemas:core
	:2.0:Group",
	
	"urn:sap:cloud:scim:schemas:extens
	ion:custom:2.0:Group"
	],
	"targetPath": "$.schemas"
	},
	{
	"sourcePath":
	"$.displayName",
	"targetPath":
	"$.displayName"
	},
	{
	"sourcePath":
	"$.displayName",
	"targetPath": "$
	['urn:sap:cloud:scim:schemas:exten
	sion:custom:2.0:Group']['name']"
	},
	{
	"sourcePath": "$
	['urn:sap:cloud:scim:schemas:exten
	sion:custom:2.0:Group']['name']",
	"optional": true,
	"targetPath": "$
	['urn:sap:cloud:scim:schemas:exten
	sion:custom:2.0:Group']['name']"
	},
	{
	"sourcePath": "$.members",
	"targetPath": "$.members",
	
	"preserveArrayWithSingleElement":
	true,
	"optional": true
	},
	{
	"sourcePath":
	"$.members[*].value",</pre>

Read Transformation	Write Transformation

<pre>},
{
"sourceVariable":
"entityBaseLocation",
"targetPath":
"$.meta.location",
"targetVariable":
"entityLocationSourceSystem",
"functions": [
{
"type":
"concatString",
"suffix": "$
{entityIdSourceSystem}"
}
]
},
{
"sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
"optional": true,
"targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
},
{
"sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
"optional": true,
"targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']"
}
]
}
}</pre>
	<pre>"preserveArrayWithSingleElement":
	true,
	"optional": true,
	"targetPath": "$.members[?
	(@.value)]",
	"functions": [
	{
	"function":
	"resolveEntityIds"
	}
	]
	},
	{
	"sourcePath":
	"$.Operations",
	"targetPath":
	"$.Operations",
	
	"preserveArrayWithSingleElement":
	true,
	"scope": "patchEntity"
	},
	{
	"sourcePath": "$.schemas",
	"targetPath": "$.schemas",
	
	"preserveArrayWithSingleElement":
	true,
	"scope": "patchEntity"
	},
	{
	"sourcePath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group']['type']",
	"targetPath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group']['type']",
	"optional": true
	},
	{
	"sourcePath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group']
	['supportedOperations']",
	"targetPath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:Group']
	['supportedOperations']",
	"optional": true
	}
	]
	}
	}</pre>

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be

automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

- For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
- For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

> ⓘ Note
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.5 SAP Advanced Workflow

Follow this procedure to set up SAP Advanced Workflow as a proxy system.

Prerequisites

- You have technical credentials for SAP Advanced Workflow. Note that SAP Advanced Workflow is available to SAP SuccessFactors Incentive Management customers as an optional add-on. You create an Admin user

in SAP SuccessFactors Incentive Management, which is synchronized with SAP Advanced Workflow. For more information, see: Adding an Admin User and Commissions User Synchronization.

- You have set up SSO between Identity Authentication and SAP Advanced Workflow. For more information, see Integration with SAP IdP.

Context

SAP Advanced Workflow enables you to analyse, organize, and execute business processes to connect people, data, and daily activities. Workflow provides you the tools you need to configure and customize your business processes based on your specific business needs.

Create an SAP Advanced Workflow proxy connector to execute hybrid scenarios. That means, it can provision its users to another (external) back-end system by request, and then can continue executing CRUD operations back to the SAP Advanced Workflow system, whenever the external back-end requests such. This scenario supports provisioning **users**.

> **Note**
>
> SAP Advanced Workflow does not support groups.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Advanced Workflow* as a proxy system. For more information, see Add New Systems [page 2415].

4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the API of your SAP Advanced Workflow system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the user for your SAP Advanced Workflow system.
`Password`	(Credential) Enter the password for your SAP Advanced Workflow user.
`awf.domain`	The domain name is the name of your SAP Advanced Workflow tenant. If you don't know your tenant name, contact your supervisor or administrator, or refer to the email notification you received when your account was created.

   To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Advanced Workflow* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Advanced Workflow. For more information, see:

   Manage Transformations [page 2439]

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourcePath":
"$.schemas",
                "targetPath":
"$.schemas",

"preserveArrayWithSingleElement":
true
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional": true,
                "targetPath":
"$.name.middleName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional": true,
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath":
"$.phoneNumbers[0].value",
                "optional": true,
                "targetPath":
"$.phoneNumbers[0].value"
            },
            {
                "sourcePath":
"$.profileUrl",
```

Write Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap.spm.workflow:2.0:User"
                ],
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional": true,
                "targetPath":
"$.name.middleName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional": true,
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.phoneNumbers"
            },
            {
```

Read Transformation

```
                "optional": true,
                "targetPath":
"$.profileUrl"
            },
            {
                "sourcePath":
"$.active",
                "optional": true,
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.emails[0].value",
                "optional": true,
                "targetPath":
"$.emails[0].value"
            },
            {
                "sourcePath":
"$.timezone",
                "optional": true,
                "targetPath":
"$.timezone"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            }
        ],
        "scimEntityEndpoint":
"Users"
    }
}
```

Write Transformation

```
                "sourcePath":
"$.active",
                "optional": true,
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath":
"$.timezone",
                "optional": true,
                "targetPath":
"$.timezone"
            },
            {
                "constant":
"urn:ietf:params:scim:api:messages
:2.0:PatchOp",
                "targetPath":
"$.schemas[0]",
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.Operations",
                "scope":
"patchEntity"
            }
        ],
        "scimEntityEndpoint":
"Users"
    }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.

- For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PATCH**.

> **Note**
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PATCH** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> **Caution**
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.6 SAP Analytics Cloud

Follow this procedure to set up SAP Analytics Cloud as a proxy system.

Prerequisites

- In SAP Analytics Cloud, you have enabled a custom SAML Identity Provider, for which *User Attribute* is set to **Custom SAML User Mapping**. To learn how, see: Enabling a Custom SAML Identity Provider
- In SAP Analytics Cloud, you have added an OAuth client with authorization grant **Client Credentials**. To learn how, see: Managing OAuth Clients and Trusted Identity Providers

Context

SAP Analytics Cloud is an all-in-one cloud product offered as software as a service for business intelligence, planning, and predictive analytics.

You can use Identity Provisioning to configure SAP Analytics Cloud as a proxy system in hybrid scenarios. For example, when SAP Analytics Cloud is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users and groups to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users and groups back to the SAP Analytics Cloud.

There are two versions of the SAP Analytics Cloud SCIM API. They are handled by the `sac.api.version` property as follows:

- When the value is set to **1** or the property is not defined (typical for systems created before versioning was introduced on April 10, 2023), SAP Analytics Cloud SCIM API version 1 is used. This is the default value.
- When the value is set to **2** - SAP Analytics Cloud SCIM API version 2 is used. This version is released with enhancements, among which the support for patch operations.

For more information on the differences between SAP Analytics Cloud SCIM API version 1 and 2, see Managing Users and Teams.

For more information on how to update to version 2, see Update Connector Version [page 2424].

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Analytics Cloud* as a proxy system. For more information, see Add New Systems [page 2415].

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP Analytics Cloud system without adding the path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the client ID to retrieve the OAuth access token for SAP Analytics Cloud.
`Password`	(Credential) Enter the client secret to retrieve the OAuth access token for SAP Analytics Cloud.
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Analytics Cloud instance. This token URL is listed in the *OAuth Clients* section of the *App Integration* page. For more information, refer to *Authorize API Access for OAuth Clients* in Manage OAuth Clients
`sac.api.csrf.protection`	Specifies whether to fetch a CSRF token when sending requests to the system. This property is automatically added to the system, with default value: **enabled** **Note** If the property is missing, its default value is **disabled**.

Property Name	Description & Value
(Optional) `sac.csrf.token.path`	Path which is appended to the URL to retrieve the CSRF token. The property is automatically added during system creation, when the system in use is SAP Analytics Cloud consuming SCIM API version 2. Based on the version of the SCIM API consumed, the default value for the property is: • SAP Analytics Cloud SCIM API version 1: `/api/v1/scim/Users?count=1` > **Note** > The property is present only for systems created before December 3, 2024. Systems created after this date use the default value for the connector version without adding the property. • SAP Analytics Cloud SCIM API version 2: `/api/v1/scim2/Users?count=1` > **Note** > If you delete the property, the Identity Provisioning will use the default value for the property for SAP Analytics Cloud using SCIM API **version 1**.
(Optional) `sac.group.filter`	When specified, only those SAP Analytics Cloud groups matching the filter expression will be read. > **Note** > You can use this property when SAP Analytics Cloud is based on SCIM API version 2. **Possible values:** For example: *displayName eq "ProjectTeam1"*
(Optional) `sac.api.version`	Handles the version of SAP Analytics Cloud SCIM API. **Possible values:** • *1* - Indicates that SAP Analytics Cloud SCIM API version 1 is used. • *2* - Indicates that SAP Analytics Cloud SCIM API version 2 is used. Default value: *1*

Property Name	Description & Value
(Optional) `ips.http.header.x-ignore-roles-if-missing`	Use this property to control whether users or teams (referred to as groups) should keep their role assignments in SAP Analytics Cloud (SCIM API version 1) when no roles are provisioned. Possible values: • true - Users or teams will keep their role assignments when no roles are provisioned. • false - Users or teams will lose their role assignments when no roles are provisioned. This is the default value. For more information, see SAP note 3092730. In SAP Analytics Cloud (SCIM API version 2), this behavior is managed by `sac.support.patch.operation` property.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Analytics Cloud* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Analytic Cloud. For more information, see:

   Manage Transformations [page 2439]

   SAP Analytics Cloud REST API: Managing Users and Teams

   Managing Users and Teams → api/v1/scim

   Managing Users and Teams → api/v1/scim2

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Default read and write transformations for SAP Analytics Cloud SCIM API version 1:

Read Transformation	Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
```

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "condition":
"($.emails[0].value EMPTY false)",
    "mappings": [
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.schemas"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath": "$.userName"
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value != []",
        "sourcePath": "$.emails[?
(@.primary == true)].value",

"preserveArrayWithSingleElement":
false,
        "optional": true,
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
```

Read Transformation

```
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value"
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.roles",
        "targetPath": "$.roles",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$
['urn:scim:schemas:extension:enter
prise:1.0']['manager']
['managerId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']"
      }
    ]
  },
  "group": {
```

Write Transformation

```
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath": "$.roles",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.roles"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true,
        "targetPath": "$
['urn:scim:schemas:extension:enter
prise:1.0']['manager']
['managerId']"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.schemas"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath": "$.id",
        "scope": "createEntity"
      },
      {
```

Read Transformation

```
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$.roles",
        "targetPath": "$.roles",

"preserveArrayWithSingleElement":
true
     },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
```

Write Transformation

```
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.roles",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.roles"
      },
      {
        "sourcePath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members"
      }
    ]
  }
}
```

Read Transformation	Write Transformation

<pre>"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
"optional": true
}
]
}
}	

Default read and write transformations for SAP Analytics Cloud SCIM API version 2:

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "condition":
"($.emails[0].value EMPTY false)",
    "mappings": [
      {
        "constant": [

"urn:sap:params:scim:schemas:exten
sion:sac:2.0:user-custom-
parameters",

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath": "$.userName"
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value != []",
        "sourcePath": "$.emails[?
(@.primary == true)].value",

"preserveArrayWithSingleElement":
false,
        "optional": true,
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.userName",
        "optional": true,
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
```

Read Transformation

```
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value"
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.roles",
        "targetPath": "$.roles",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']",
```

Write Transformation

```
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.name.middleName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails"
      },
      {
        "condition":
"$.emails[0].length() > 0",
        "constant": true,
        "targetPath":
"$.emails[0].primary"
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']",
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']"
      },
      {
        "sourcePath":
"$.emails[0].value",
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
```

Read Transformation

```
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "optional": true,
```

Write Transformation

```
nsion:sac:2.0:user-custom-
parameters']['idpUserId']"
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value != []",
        "sourcePath": "$.emails[?
(@.primary == true)].value",

"preserveArrayWithSingleElement":
false,
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']['idpUserId']"
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']['idpUserId']",
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']['idpUserId']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "condition":
"('%sac.group.prefix%' ===
'null') || ($.displayName =~ /
%sac.group.prefix%.*/)",
    "mappings": [
      {
        "constant": [
```

Read Transformation

```
"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:group-roles']",
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:group-roles']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:group-custom-
parameters']",
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:group-custom-
parameters']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

Write Transformation

```
"urn:ietf:params:scim:schemas:core
:2.0:Group",

"urn:sap:params:scim:schemas:exten
sion:sac:2.0:group-roles",

"urn:sap:params:scim:schemas:exten
sion:sac:2.0:group-custom-
parameters"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath": "$.id",
        "scope": "createEntity",
        "functions": [
          {
            "condition":
"('%sac.group.prefix%' !==
'null') && (@ =~ /
%sac.group.prefix%.*/)",
            "function":
"replaceFirstString",
            "regex":
"%sac.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "functions": [
          {
```

Read Transformation	Write Transformation

```
            "condition":
"('%sac.group.prefix%' !==
'null') && (@ =~ /
%sac.group.prefix%.*/)",
            "function":
"replaceFirstString",
            "regex":
"%sac.group.prefix%",
            "replacement": ""
          }
        ]
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath": "$.roles",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.roles"
      },
      {
        "sourcePath": "$.members",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.members",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:group-roles']",
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:group-roles']"
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:group-custom-
parameters']",
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:group-custom-
parameters']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']
['description']",
        "optional": true,
```

Read Transformation	Write Transformation

	<pre>"targetPath": "$['urn:sap:params:scim:schemas:extension:sac:2.0:group-custom-parameters']['description']"
	}
	]
	}
	}</pre>

> ⚠ Caution
>
> When running a provisioning job to update users or teams (referred to as groups) with role assignments in SAP Analytics Cloud, these role assignments will be removed as a result of the update. This behavior, which can lead to permission issues, is expected because role assignments in SAP Analytics Cloud are not available as group parameters in some source systems, such as Identity Authentication. To preserve existing role assignments when updating users or teams, proceed as follows based on the version of the SAP Analytics Cloud SCIM API:
>
> - For SCIM API version 1, set the `ips.http.header.x-ignore-roles-if-missing` property to *true* in the SAP Analytics Cloud target system. For more information, see the properties table above and the SAP note 3092730.
> - For SCIM API version 2, set the `sac.support.patch.operation` property to *true* in the SAP Analytics Cloud target system.

> ⓘ Note
>
> Updating a user in SAP Analytics Cloud using SCIM API version 2 depends on whether user attributes in SAP Analytics Cloud are mapped to SAML attributes in your identity provider. If this is the case, the values of those attributes are populated by the identity provider and cannot be changed by the SAP Analytics SCIM API or the UI. For more information, see Mapping SAML Attributes to Users.

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ⓘ Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

7. Run an initial load job.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.7 SAP Application Server ABAP

Follow this procedure to set up SAP Application Server ABAP (AS ABAP) as a proxy system.

Prerequisites

- You have installed the Cloud Connector in your corporate environment and have done the initial configuration. For more information, see: Cloud Connector.
  For more information on how to make a connection to this on-premise system, follow the procedure in Connect to On-Premise Systems [page 2468].
- You have credentials of a technical user with read and write permissions for AS ABAP, which plays both the role of a user data source and a target system. Via this user, the Identity Provisioning service will call the ABAP public API in order to execute a number of function modules. These function modules are listed in **step 1** from the procedure below.
- The technical user must have the required RFC and security authorizations, as described in SAP Note 3666721.

  > ⓘ Note
  >
  > If the technical user has insufficient security authorization, no users or roles can be read.

Context

SAP Application Server ABAP (AS ABAP) offers a user store and user administration capabilities for maintaining users and their authorizations for AS ABAP applications. You can configure AS ABAP as a proxy system so as to provision entities between AS ABAP and another on-premise system.

> **Note**
>
> During the *Initial Load*, only active ABAP users are read. That means, users that have been created before the initial load job is started, and whose expiration date is after the end of the job. To learn about initial load, see **step 8** below – Transformations.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Open Cloud Connector to add an access control system mapping for **AS ABAP**. This is needed to allow the Identity Provisioning service to access AS ABAP as a back-end system on the intranet. To learn how, see: Configure Access Control (RFC)

   Go to *Cloud To On-Premise* > *Access Control* tab and select protocol *RFC SNC*. Then, expose the following *exact names* as accessible resources:

   - `PRGN_ROLE_GETLIST`
   - `BAPI_USER_GETLIST`
   - `BAPI_USER_GET_DETAIL`
   - `BAPI_USER_CREATE1`
   - `BAPI_USER_ACTGROUPS_ASSIGN`
   - `IDENTITY_MODIFY`
   - `BAPI_USER_DELETE`
   - `PRGN_ACTIVITY_GROUPS_LOAD_RFC`
   - `SUSR_USER_GET_DETAIL_MASS` - available as of `SAP_BASIS 7.58`. For more information, see SAP Note 3631587.

     > **Note**
     >
     > The function module `SUSR_USER_GET_DETAIL_MASS`, which is used for mass reading of user details, does not support the `SAPUSER_UUID.SAP_UID` attribute. To read this attribute, use the function module `BAPI_USER_GET_DETAIL` instead.

   These are Business Application Programming Interface (BAPI) functional modules designed to perform certain tasks in the SAP AS ABAP system, such as retrieving, creating, updating or deleting user data.

For more information about each function module and its parameters, refer to the documentation provided in the SAP system. Logon to your SAP system and run transaction code **SE37**. In the *Function Builder: Initial Screen*, enter the function name, choose *Display* and then *Functional Module Documentation*.

2. Open SAP BTP cockpit, and in your Identity Provisioning subaccount create a destination for the AS ABAP system. To learn how, see: Create RFC Destinations

   The destination configuration is required by the Identity Provisioning service to find the back-end system to be used for reading data. It also provides the credentials of the technical user, needed for the connection to the ABAP public API.

   Below are the fields you have to fill in the cockpit destination before using an AS ABAP client as a proxy system.

   > ### Note
   >
   > In the RFC destination, you can set only JCo properties. Properties starting with `abap` are connector specific and must be set in the Identity Provisioning UI.

Field/Property Name	Value
Name	Enter a destination name.
Type	Select *RFC*.
User	Enter the user for AS ABAP. The *User* field corresponds to property `jco.client.user` in the exported RFC destination.
Password	(Credential) Enter the password for the AS ABAP user. The *Password* field corresponds to property `jco.client.passwd` in the exported RFC destination.
jco.client.client	The property is configured during destination creation in SAP BTP cockpit. Provide the client to be used in the ABAP system. Valid format is a three-digit number.
Proxy Type	Defines the proxy type of the connection you need to provide for your ABAP system. The value is set during destination creation in SAP BTP cockpit. The proxy type *OnPremise* requires the Cloud Connector to access resources within your on-premise network. Enter: *OnPremise*
Direct Connection	

Field/Property Name	Value
jco.client.ashost	Provide the virtual host entry that you have configured in the Cloud Connector → *Access Control* configuration. At least one property specifying the target host must be set during destination creation.
jco.client.sysnr	Provide the "system number" of the ABAP system.
Load Balancing Connection	
jco.client.mshost	Represents the message server host to be used. At least one property specifying the target host must be set during destination creation.
jco.client.r3name	Provide the three-character system ID of the ABAP system to be addressed.
jco.client.msservt	Provide the port on which the message server is listening for incoming requests. You can use this property as an alternative to `jco.client.r3name`.
Optional Properties	
`jco.destination.peak_limit`	The value represents the maximum number of active connections that can simultaneously be created for a destination. For example: *10*
`jco.destination.pool_capacity`	The value represents the maximum number of idle connections kept open by the destination. For example: *5*
`abap.user.name.filter`	Filters user names by a regular expression. The regex can define any kind of search pattern. For example, `abap.user.name.filter` = **^MAR.*** reads all user names that start with *MAR*, such as **MARK**, **MARTINA**, and so on. **Note** This property has a higher priority over `abap.user.filter`. That means, if you set both properties in a system, the value of **abap.user.name.filter** will be used. However, if the value of `abap.user.name.filter` is empty, then **abap.user.filter**'s value will be used instead.

Field/Property Name	Value
`abap.role.name.filter`	Filters user roles by a regular expression. The regex can define any kind of search pattern. For example, `abap.role.name.filter` = **^inter.*** reads all roles which start with *inter*, such as **internal**, **internship**, and so on. **Note** This property has a higher priority over `abap.role.filter`. That means, if you set both properties in a system, the value of **abap.role.name.filter** will be used. However, if the value of `abap.role.name.filter` is empty, then **abap.role.filter**'s value will be used instead.
`abap.user.membership.filter`	Filters users by a regular expression, based on their *Role* memberships in AS ABAP. The regex can define any kind of search pattern. For example, `abap.user.membership.filter` = **(?i)^new.*** reads all users who have an assigned role which starts with *new*. This regex is case insensitive, which means the result can be roles starting with **new**, or **New**, or **NEW**, and so on.
`abap.user.page.size`	Use this property to configure the number of users to be read from SAP Application Server ABAP at once. If the property is not set, the function module `SUSR_USER_GET_DETAIL_MASS` **cannot** be used, even if it is available on the ABAP server. **Note** The property can be used only when the function module `SUSR_USER_GET_DETAIL_MASS` is exposed as an accessible resource and is present on the ABAP server. Even when these conditions are fulfilled, if the property is not set, the function module **cannot** be used. For more information, see SAP Note 3631587. **Possible values**: Default value: *100* The maximum allowed number is 100.

> ### Note
>
> If connection properties, like `User` and `Password`, are configured both in the destination (SAP BTP cockpit) and on the *Properties* tab (Identity Provisioning User Interface), the values set in the destination are considered with higher priority.

3. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
4. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
5. Add *SAP Application Server ABAP* as a proxy system. To learn how, see: Add New Systems [page 2415]
6. From the *Destination Name* dropdown, choose the RFC destination you have created in *step 2*.
7. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Application Server ABAP* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your AS ABAP. For more information, see Manage Transformations [page 2439].

   > ### Note
   >
   > The values of the **entityIdSourceSystem** and **entityIdTargetSystem** attributes store the unique ID of the entity. Do not delete or change this statement!

   Default read and write transformations:

   > ### → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

- Use the default transformations if the version of your AS ABAP system is *7.40* or higher. If your system version is *7.31*, you can still use these transformations – just apply SAP Note 1695883.
  NOTE: If a user is *inactive* in the identity management system, the default Write Transformation creates it as *inactive* (locked) in AS ABAP too.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint":
"Users",
    "mappings": [
      {
        "sourcePath":
"$.USERNAME",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id",
        "functions": [
          {
            "type": "encode",
            "algorithm":
"base32",
            "skipPadding": true
          }
        ]
      },
      {
        "sourcePath":
"$.USERNAME",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.ALIAS.USERALIAS",
        "optional": true,
        "targetPath":
"$.externalId",
        "correlationAttribute":
true
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:co
re:2.0:User",
        "targetPath":
"$.schemas[0]"
      },
      {
        "constant": "User",
        "targetPath":
"$.meta.resourceType"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint":
"Users",
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.USERNAME",
        "targetVariable":
"entityIdTargetSystem"
      },
      {
        "scope": "deleteEntity",
        "sourceVariable":
"entityIdTargetSystem",
        "targetVariable":
"entityIdTargetSystem",
        "functions": [
          {
            "type": "decode",
            "algorithm":
"base32",
            "skipPadding": true
          },
          {
            "type": "toString"
          }
        ]
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.ALIAS.USERALIAS"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:e
xtension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath":
"$.SAPUSER_UUID.SAP_UID"
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value !=
[]",
        "sourcePath":
"$.emails[?(@.primary ==
true)].value",

"preserveArrayWithSingleElement"
: false,
        "optional": true,
        "targetPath":
"$.ADDRESS.E_MAIL"
```

Read Transformation

```
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.ADDRESS.E_MAIL",
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "condition":
"$.ADDRESS.E_MAIL EMPTY false",
        "constant": true,
        "targetPath":
"$.emails[0].primary"
      },
      {
        "condition":
"$.ADDRESS.E_MAIL EMPTY false",
        "constant": "work",
        "targetPath":
"$.emails[0].type"
      },
      {
        "sourcePath":
"$.ADDRESS.FIRSTNAME",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.ADDRESS.LASTNAME",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.ADDRESS.MIDDLENAME",
        "optional": true,
        "targetPath":
"$.name.middleName"
      },
      {
        "sourcePath":
"$.ADDRESS.NICKNAME",
        "optional": true,
        "targetPath":
"$.nickName"
      },
      {
        "sourcePath":
"$.ADDRESS.TITLE_P",
        "optional": true,
```

Write Transformation

```
      },
      {
        "condition": "$.emails[?
(@.type == 'work')].value !=
[]",
        "sourcePath":
"$.emails[?(@.type ==
'work')].value",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ADDRESS.E_MAIL",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "sourcePath":
"$.emails[*].value",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ADDSMTP[?(@.E_MAIL)]"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.ADDRESS.FIRSTNAME"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.ADDRESS.LASTNAME"
      },
      {
        "scope": "createEntity",
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.ADDRESS.LASTNAME"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.ADDRESS.MIDDLENAME"
      },
      {
        "sourcePath":
"$.nickName",
```

Read Transformation	Write Transformation

```
        "targetPath":
"$.name.honorificPrefix"
      },
      {
        "sourcePath":
"$.ADDRESS.COUNTRY",
        "optional": true,
        "targetPath":
"$.addresses[0].country"
      },
      {
        "condition":
"$.ADDRESS.COUNTRY EMPTY false",
        "constant": true,
        "targetPath":
"$.addresses[0].primary"
      },
      {
        "condition":
"$.ADDRESS.COUNTRY EMPTY false",
        "constant": "work",
        "targetPath":
"$.addresses[0].type"
      },
      {
        "sourcePath":
"$.ADDRESS.TEL1_NUMBR",
        "optional": true,
        "targetPath":
"$.phoneNumbers[0].value"
      },
      {
        "condition":
"$.ADDRESS.TEL1_NUMBR EMPTY
false",
        "constant": true,
        "targetPath":
"$.phoneNumbers[0].primary"
      },
      {
        "condition":
"$.ADDRESS.TEL1_NUMBR EMPTY
false",
        "constant": "work",
        "targetPath":
"$.phoneNumbers[0].type"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.DEFAULTS.LANGU"
        ],
        "optional": true,
        "targetPath":
"$.locale",
        "defaultValue": "en",
        "valueMappings": [
          {
            "key": [
              "W"
            ],
            "mappedValue": "bg"
```

```
        "optional": true,
        "targetPath":
"$.ADDRESS.NICKNAME"
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "optional": true,
        "targetPath":
"$.ADDRESS.TITLE_P"
      },
      {
        "condition":
"$.phoneNumbers[?(@.primary ==
true)].value != []",
        "sourcePath":
"$.phoneNumbers[?(@.primary ==
true)].value",

"preserveArrayWithSingleElement"
: false,
        "optional": true,
        "targetPath":
"$.ADDRESS.TEL1_NUMBR"
      },
      {
        "condition":
"$.phoneNumbers[?(@.type ==
'work')].value != []",
        "sourcePath":
"$.phoneNumbers[?(@.type ==
'work')].value",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ADDRESS.TEL1_NUMBR",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "sourcePath":
"$.phoneNumbers[*].value",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ADDTEL[?(@.TELEPHONE)]"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.locale"
        ],
        "optional": true,
```

Read Transformation	Write Transformation

```
          }
        ]
      },
      {
        "sourcePath":
"$.ADDRESS.LANGUP_ISO",
        "optional": true,
        "targetPath":
"$.preferredLanguage",
        "functions": [
          {
            "function":
"toLowerCaseString"
          }
        ]
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.LOGONDATA.TZONE"
        ],
        "optional": true,
        "targetPath":
"$.timezone",
        "defaultValue": "Europe/
Berlin",
        "valueMappings": [
          {
            "key": [
              "EET"
            ],
            "mappedValue":
"Europe/Sofia"
          }
        ]
      },
      {
        "constant": false,
        "targetPath": "$.active"
      },
      {
        "condition":
"($.ISLOCKED.LOCAL_LOCK != 'L')
&& ($.ISLOCKED.NO_USER_PW !=
'L') && ($.ISLOCKED.GLOB_LOCK !
= 'L') &&
($.ISLOCKED.WRNG_LOGON != 'L')
&& ($.LOCK != 'L') &&
($.LOCK_LOCALLY != 'X')",
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.ACTIVITYGROUPS[*].AGR_NAME",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.groups[?(@.value)]",
        "functions": [
```

```
        "targetPath":
"$.DEFAULTS.LANGU",
        "defaultValue": "E",
        "valueMappings": [
          {
            "key": [
              "bg"
            ],
            "mappedValue": "W"
          }
        ]
      },
      {
        "sourcePath":
"$.preferredLanguage",
        "optional": true,
        "targetPath":
"$.ADDRESS.LANGUP_ISO",
        "functions": [
          {
            "function":
"toUpperCaseString"
          }
        ]
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.timezone"
        ],
        "optional": true,
        "targetPath":
"$.LOGONDATA.TZONE",
        "defaultValue": "CET",
        "valueMappings": [
          {
            "key": [
              "Europe/Sofia"
            ],
            "mappedValue": "EET"
          }
        ]
      },
      {
        "scope": "createEntity",
        "targetPath":
"$.PASSWORD.BAPIPWD",
        "functions": [
          {
            "type":
"randomPassword",
            "passwordLength":
24,

"minimumNumberOfLowercaseLetters
": 1,

"minimumNumberOfUppercaseLetters
": 1,

"minimumNumberOfDigits": 1,
```

Read Transformation

```
          {
            "type": "encode",
            "algorithm":
"base32",
            "skipPadding": true
          }
        ]
      },
      {
        "constant": "direct",

"preserveArrayWithSingleElement"
: true,
        "targetPath":
"$.groups[*].type",
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath":
"$.ROLE_NAME",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id",
        "functions": [
          {
            "type": "encode",
            "algorithm":
"base32",
            "skipPadding": true
          }
        ]
      },
      {
        "constant": "Group",
        "targetPath":
"$.meta.resourceType"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.ROLE_NAME",
```

Write Transformation

```
"minimumNumberOfSpecialSymbols":
 0
          }
        ]
      },
      {
        "scope": "createEntity",
        "optional": true,
        "ignore": true,
        "sourcePath":
"$.password",
        "targetPath":
"$.PASSWORD.BAPIPWD"
      },
      {
        "constant":
"updateEntity",
        "targetVariable":
"operationTypeVariable"
      },
      {
        "constant":
"createEntity",
        "targetVariable":
"operationTypeVariable",
        "scope": "createEntity"
      },
      {
        "condition":
"$.active == false && '$
{operationTypeVariable}' ==
'createEntity'",
        "constant": "X",
        "targetPath":
"$.LOCK_LOCALLY"
      },
      {
        "condition": "'$
{operationTypeVariable}' ==
'updateEntity'",
        "constant": "U",
        "targetPath": "$.LOCK"
      },
      {
        "condition":
"$.active == false && '$
{operationTypeVariable}' ==
'updateEntity'",
        "constant": "L",
        "targetPath": "$.LOCK"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.ROLE_NAME"
```

Read Transformation	Write Transformation

<pre>"targetPath":
"$.displayName"
},
{
"constant":
"urn:ietf:params:scim:schemas:core:2.0:Group",
"targetPath":
"$.schemas[0]"
},
{
"sourcePath":
"$.USERLIST[*].USERNAME",
 	
"preserveArrayWithSingleElement"
: true,
"targetPath":
"$.members[?(@.value)]",
"optional": true,
"functions": [
{
"type": "encode",
"algorithm":
"base32",
"skipPadding": true
}
]
},
{
"constant": "User",
 	
"preserveArrayWithSingleElement"
: true,
"targetPath":
"$.members[*].type",
"optional": true
}
]
}
}</pre>
	<pre>},
	{
	"sourceVariable":
	"entityIdTargetSystem",
	"targetVariable":
	"entityIdTargetSystem",
	"functions": [
	{
	"type": "decode",
	"algorithm":
	"base32",
	"skipPadding": true
	},
	{
	"type": "toString"
	}
	]
	},
	{
	"sourcePath":
	"$.members[*].value",
	
	"preserveArrayWithSingleElement"
	: true,
	"targetPath":
	"$.USERLIST[?(@.USERNAME)]",
	"optional": true,
	"functions": [
	{
	"type": "decode",
	"algorithm":
	"base32",
	"skipPadding": true
	},
	{
	"type": "toString",
	"applyOnElements":
	true
	}
	]
	}
	]
	}
	}</pre>

- If your AS ABAP version is *7.30* or lower, use the transformations below.
  NOTE: The Write Transformation creates all users as *active* (unlocked) in AS ABAP, regardless if they are *active* or *inactive* in the identity management system.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint":
"Users",
    "mappings": [
      {
        "sourcePath":
"$.USERNAME",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id",
        "functions": [
          {
            "type": "encode",
            "algorithm":
"base32",
            "skipPadding": true
          }
        ]
      },
      {
        "sourcePath":
"$.USERNAME",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.ALIAS.USERALIAS",
        "optional": true,
        "targetPath":
"$.externalId",
        "correlationAttribute":
true
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:co
re:2.0:User",
        "targetPath":
"$.schemas[0]"
      },
      {
        "constant": "User",
        "targetPath":
"$.meta.resourceType"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint":
"Users",
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.USERNAME",
        "targetVariable":
"entityIdTargetSystem"
      },
      {
        "scope": "deleteEntity",
        "sourceVariable":
"entityIdTargetSystem",
        "targetVariable":
"entityIdTargetSystem",
        "functions": [
          {
            "type": "decode",
            "algorithm":
"base32",
            "skipPadding": true
          },
          {
            "type": "toString"
          }
        ]
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.ALIAS.USERALIAS"
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value !=
[]",
        "sourcePath":
"$.emails[?(@.primary ==
true)].value",

"preserveArrayWithSingleElement"
: false,
        "optional": true,
        "targetPath":
"$.ADDRESS.E_MAIL"
      },
      {
        "condition": "$.emails[?
(@.type == 'work')].value !=
[]",
        "sourcePath":
"$.emails[?(@.type ==
'work')].value",
```

Read Transformation

```
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.ADDRESS.E_MAIL",
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "condition":
"$.ADDRESS.E_MAIL EMPTY false",
        "constant": true,
        "targetPath":
"$.emails[0].primary"
      },
      {
        "condition":
"$.ADDRESS.E_MAIL EMPTY false",
        "constant": "work",
        "targetPath":
"$.emails[0].type"
      },
      {
        "sourcePath":
"$.ADDRESS.FIRSTNAME",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.ADDRESS.LASTNAME",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.ADDRESS.MIDDLENAME",
        "optional": true,
        "targetPath":
"$.name.middleName"
      },
      {
        "sourcePath":
"$.ADDRESS.NICKNAME",
        "optional": true,
        "targetPath":
"$.nickName"
      },
      {
        "sourcePath":
"$.ADDRESS.TITLE_P",
        "optional": true,
```

Write Transformation

```
"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ADDRESS.E_MAIL",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "sourcePath":
"$.emails[*].value",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ADDSMTP[?(@.E_MAIL)]"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.ADDRESS.FIRSTNAME"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.ADDRESS.LASTNAME"
      },
      {
        "scope": "createEntity",
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.ADDRESS.LASTNAME"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.ADDRESS.MIDDLENAME"
      },
      {
        "sourcePath":
"$.nickName",
        "optional": true,
        "targetPath":
"$.ADDRESS.NICKNAME"
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "optional": true,
```

Read Transformation	Write Transformation

```
        "targetPath":
"$.name.honorificPrefix"
      },
      {
        "sourcePath":
"$.ADDRESS.COUNTRY",
        "optional": true,
        "targetPath":
"$.addresses[0].country"
      },
      {
        "condition":
"$.ADDRESS.COUNTRY EMPTY false",
        "constant": true,
        "targetPath":
"$.addresses[0].primary"
      },
      {
        "condition":
"$.ADDRESS.COUNTRY EMPTY false",
        "constant": "work",
        "targetPath":
"$.addresses[0].type"
      },
      {
        "sourcePath":
"$.ADDRESS.TEL1_NUMBR",
        "optional": true,
        "targetPath":
"$.phoneNumbers[0].value"
      },
      {
        "condition":
"$.ADDRESS.TEL1_NUMBR EMPTY
false",
        "constant": true,
        "targetPath":
"$.phoneNumbers[0].primary"
      },
      {
        "condition":
"$.ADDRESS.TEL1_NUMBR EMPTY
false",
        "constant": "work",
        "targetPath":
"$.phoneNumbers[0].type"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.DEFAULTS.LANGU"
        ],
        "optional": true,
        "targetPath":
"$.locale",
        "defaultValue": "en",
        "valueMappings": [
          {
            "key": [
              "W"
            ],
            "mappedValue": "bg"
```

```
        "targetPath":
"$.ADDRESS.TITLE_P"
      },
      {
        "condition":
"$.phoneNumbers[?(@.primary ==
true)].value != []",
        "sourcePath":
"$.phoneNumbers[?(@.primary ==
true)].value",

"preserveArrayWithSingleElement"
: false,
        "optional": true,
        "targetPath":
"$.ADDRESS.TEL1_NUMBR"
      },
      {
        "condition":
"$.phoneNumbers[?(@.type ==
'work')].value != []",
        "sourcePath":
"$.phoneNumbers[?(@.type ==
'work')].value",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ADDRESS.TEL1_NUMBR",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "sourcePath":
"$.phoneNumbers[*].value",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ADDTEL[?(@.TELEPHONE)]"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.locale"
        ],
        "optional": true,
        "targetPath":
"$.DEFAULTS.LANGU",
        "defaultValue": "E",
        "valueMappings": [
          {
            "key": [
              "bg"
            ],
            "mappedValue": "W"
```

Read Transformation	Write Transformation

```
            }
          ]
        },
        {
          "sourcePath":
"$.ADDRESS.LANGUP_ISO",
          "optional": true,
          "targetPath":
"$.preferredLanguage",
          "functions": [
            {
              "function":
"toLowerCaseString"
            }
          ]
        },
        {
          "type": "valueMapping",
          "sourcePaths": [
            "$.LOGONDATA.TZONE"
          ],
          "optional": true,
          "targetPath":
"$.timezone",
          "defaultValue": "Europe/
Berlin",
          "valueMappings": [
            {
              "key": [
                "EET"
              ],
              "mappedValue":
"Europe/Sofia"
            }
          ]
        },
        {
          "constant": false,
          "targetPath": "$.active"
        },
        {
          "condition":
"($.ISLOCKED.LOCAL_LOCK != 'L')
&& ($.ISLOCKED.NO_USER_PW !=
'L') && ($.ISLOCKED.GLOB_LOCK !
= 'L') &&
($.ISLOCKED.WRNG_LOGON != 'L')
&& ($.LOCK != 'L') &&
($.LOCK_LOCALLY != 'X')",
          "constant": true,
          "targetPath": "$.active"
        },
        {
          "sourcePath":
"$.ACTIVITYGROUPS[*].AGR_NAME",

"preserveArrayWithSingleElement"
: true,
          "optional": true,
          "targetPath":
"$.groups[?(@.value)]",
          "functions": [
```

```
            }
          ]
        },
        {
          "sourcePath":
"$.preferredLanguage",
          "optional": true,
          "targetPath":
"$.ADDRESS.LANGUP_ISO",
          "functions": [
            {
              "function":
"toUpperCaseString"
            }
          ]
        },
        {
          "type": "valueMapping",
          "sourcePaths": [
            "$.timezone"
          ],
          "optional": true,
          "targetPath":
"$.LOGONDATA.TZONE",
          "defaultValue": "CET",
          "valueMappings": [
            {
              "key": [
                "Europe/Sofia"
              ],
              "mappedValue": "EET"
            }
          ]
        },
        {
          "scope": "createEntity",
          "targetPath":
"$.PASSWORD.BAPIPWD",
          "functions": [
            {
              "type":
"randomPassword",
              "passwordLength":
24,

"minimumNumberOfLowercaseLetters
": 1,

"minimumNumberOfUppercaseLetters
": 1,

"minimumNumberOfDigits": 1,

"minimumNumberOfSpecialSymbols":
 0
            }
          ]
        },
        {
          "scope": "createEntity",
          "optional": true,
          "ignore": true,
```

Read Transformation

```
          {
            "type": "encode",
            "algorithm":
"base32",
            "skipPadding": true
          }
        ]
      },
      {
        "constant": "direct",

"preserveArrayWithSingleElement"
: true,
        "targetPath":
"$.groups[*].type",
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath":
"$.ROLE_NAME",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id",
        "functions": [
          {
            "type": "encode",
            "algorithm":
"base32",
            "skipPadding": true
          }
        ]
      },
      {
        "constant": "Group",
        "targetPath":
"$.meta.resourceType"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.ROLE_NAME",
```

Write Transformation

```
        "sourcePath":
"$.password",
        "targetPath":
"$.PASSWORD.BAPIPWD"
      },
      {
        "scope": "createEntity",
        "sourcePath":
"$.groups[*].value",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ACTIVITYGROUPS[?
(@.AGR_NAME)]",
        "functions": [
          {
            "type": "decode",
            "algorithm":
"base32",
            "skipPadding": true
          },
          {
            "type": "toString",
            "applyOnElements":
true
          }
        ]
      },
      {
        "constant":
"updateEntity",
        "targetVariable":
"operationTypeVariable"
      },
      {
        "constant":
"createEntity",
        "targetVariable":
"operationTypeVariable",
        "scope": "createEntity"
      },
      {
        "condition": "'$
{operationTypeVariable}' ==
'updateEntity'",
        "constant": "U",
        "targetPath": "$.LOCK"
      },
      {
        "condition":
"$.active == false && '$
{operationTypeVariable}' ==
'updateEntity'",
        "constant": "L",
        "targetPath": "$.LOCK"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
```

Read Transformation	Write Transformation

<pre>"targetPath":
"$.displayName"
},
{
"constant":
"urn:ietf:params:scim:schemas:core:2.0:Group",
"targetPath":
"$.schemas[0]"
},
{
"sourcePath":
"$.USERLIST[*].USERNAME",
 	
"preserveArrayWithSingleElement"
: true,
"targetPath":
"$.members[?(@.value)]",
"optional": true,
"functions": [
{
"type": "encode",
"algorithm":
"base32",
"skipPadding": true
}
]
},
{
"constant": "User",
 	
"preserveArrayWithSingleElement"
: true,
"targetPath":
"$.members[*].type",
"optional": true
}
]
}
}</pre>
	<pre>"mappings": [
	{
	"sourcePath":
	"$.displayName",
	"targetPath":
	"$.ROLE_NAME"
	},
	{
	"sourceVariable":
	"entityIdTargetSystem",
	"targetVariable":
	"entityIdTargetSystem",
	"functions": [
	{
	"type": "decode",
	"algorithm":
	"base32",
	"skipPadding": true
	},
	{
	"type": "toString"
	}
	]
	},
	{
	"sourcePath":
	"$.members[*].value",
	
	"preserveArrayWithSingleElement"
	: true,
	"targetPath":
	"$.USERLIST[?(@.USERNAME)]",
	"optional": true,
	"functions": [
	{
	"type": "decode",
	"algorithm":
	"base32",
	"skipPadding": true
	},
	{
	"type": "toString",
	"applyOnElements":
	true
	}
	]
	}
	]
	}
	}</pre>

- **Default transformations supporting User UUID attribute:**

Read Transformation	Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint":
"Users",
    "mappings": [
      {
        "sourcePath":
"$.USERNAME",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id",
        "functions": [
          {
            "type": "encode",
            "algorithm":
"base32",
            "skipPadding": true
          }
        ]
      },
      {
        "sourcePath":
"$.USERNAME",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.ALIAS.USERALIAS",
        "optional": true,
        "targetPath":
"$.externalId",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.SAPUSER_UUID.SAP_UID",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:e
xtension:sap:2.0:User']
['userUuid']"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:co
re:2.0:User",
        "targetPath":
"$.schemas[0]"
      },
      {
        "constant": "User",
```

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint":
"Users",
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.USERNAME",
        "targetVariable":
"entityIdTargetSystem"
      },
      {
        "scope": "deleteEntity",
        "sourceVariable":
"entityIdTargetSystem",
        "targetVariable":
"entityIdTargetSystem",
        "functions": [
          {
            "type": "decode",
            "algorithm":
"base32",
            "skipPadding": true
          },
          {
            "type": "toString"
          }
        ]
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.ALIAS.USERALIAS"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:e
xtension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath":
"$.SAPUSER_UUID.SAP_UID"
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value !=
[]",
        "sourcePath":
"$.emails[?(@.primary ==
true)].value",

"preserveArrayWithSingleElement"
: false,
```

Read Transformation

```
        "targetPath":
"$.meta.resourceType"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.ADDRESS.E_MAIL",
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "condition":
"$.ADDRESS.E_MAIL EMPTY false",
        "constant": true,
        "targetPath":
"$.emails[0].primary"
      },
      {
        "condition":
"$.ADDRESS.E_MAIL EMPTY false",
        "constant": "work",
        "targetPath":
"$.emails[0].type"
      },
      {
        "sourcePath":
"$.ADDRESS.FIRSTNAME",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.ADDRESS.LASTNAME",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.ADDRESS.MIDDLENAME",
        "optional": true,
        "targetPath":
"$.name.middleName"
```

Write Transformation

```
        "optional": true,
        "targetPath":
"$.ADDRESS.E_MAIL"
      },
      {
        "condition": "$.emails[?
(@.type == 'work')].value !=
[]",
        "sourcePath":
"$.emails[?(@.type ==
'work')].value",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ADDRESS.E_MAIL",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "sourcePath":
"$.emails[*].value",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ADDSMTP[?(@.E_MAIL)]"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.ADDRESS.FIRSTNAME"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.ADDRESS.LASTNAME"
      },
      {
        "scope": "createEntity",
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.ADDRESS.LASTNAME"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.ADDRESS.MIDDLENAME"
      },
```

Read Transformation	Write Transformation

```
      },
      {
        "sourcePath":
"$.ADDRESS.NICKNAME",
        "optional": true,
        "targetPath":
"$.nickName"
      },
      {
        "sourcePath":
"$.ADDRESS.TITLE_P",
        "optional": true,
        "targetPath":
"$.name.honorificPrefix"
      },
      {
        "sourcePath":
"$.ADDRESS.COUNTRY",
        "optional": true,
        "targetPath":
"$.addresses[0].country"
      },
      {
        "condition":
"$.ADDRESS.COUNTRY EMPTY false",
        "constant": true,
        "targetPath":
"$.addresses[0].primary"
      },
      {
        "condition":
"$.ADDRESS.COUNTRY EMPTY false",
        "constant": "work",
        "targetPath":
"$.addresses[0].type"
      },
      {
        "sourcePath":
"$.ADDRESS.TEL1_NUMBR",
        "optional": true,
        "targetPath":
"$.phoneNumbers[0].value"
      },
      {
        "condition":
"$.ADDRESS.TEL1_NUMBR EMPTY
false",
        "constant": true,
        "targetPath":
"$.phoneNumbers[0].primary"
      },
      {
        "condition":
"$.ADDRESS.TEL1_NUMBR EMPTY
false",
        "constant": "work",
        "targetPath":
"$.phoneNumbers[0].type"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
```

```
      {
        "sourcePath":
"$.nickName",
        "optional": true,
        "targetPath":
"$.ADDRESS.NICKNAME"
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "optional": true,
        "targetPath":
"$.ADDRESS.TITLE_P"
      },
      {
        "condition":
"$.phoneNumbers[?(@.primary ==
true)].value != []",
        "sourcePath":
"$.phoneNumbers[?(@.primary ==
true)].value",

"preserveArrayWithSingleElement"
: false,
        "optional": true,
        "targetPath":
"$.ADDRESS.TEL1_NUMBR"
      },
      {
        "condition":
"$.phoneNumbers[?(@.type ==
'work')].value != []",
        "sourcePath":
"$.phoneNumbers[?(@.type ==
'work')].value",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ADDRESS.TEL1_NUMBR",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "sourcePath":
"$.phoneNumbers[*].value",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.ADDTEL[?(@.TELEPHONE)]"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.locale"
```

Read Transformation	Write Transformation

```
            "$.DEFAULTS.LANGU"
          ],
          "optional": true,
          "targetPath":
"$.locale",
          "defaultValue": "en",
          "valueMappings": [
            {
              "key": [
                "W"
              ],
              "mappedValue": "bg"
            }
          ]
        },
        {
          "sourcePath":
"$.ADDRESS.LANGUP_ISO",
          "optional": true,
          "targetPath":
"$.preferredLanguage",
          "functions": [
            {
              "function":
"toLowerCaseString"
            }
          ]
        },
        {
          "type": "valueMapping",
          "sourcePaths": [
            "$.LOGONDATA.TZONE"
          ],
          "optional": true,
          "targetPath":
"$.timezone",
          "defaultValue": "Europe/
Berlin",
          "valueMappings": [
            {
              "key": [
                "EET"
              ],
              "mappedValue":
"Europe/Sofia"
            }
          ]
        },
        {
          "constant": false,
          "targetPath": "$.active"
        },
        {
          "condition":
"($.ISLOCKED.LOCAL_LOCK != 'L')
&& ($.ISLOCKED.GLOB_LOCK !=
'L') && ($.ISLOCKED.WRNG_LOGON !
= 'L') && ($.LOCK != 'L') &&
($.LOCK_LOCALLY != 'X')",
          "constant": true,
          "targetPath": "$.active"
        },
```

```
        ],
        "optional": true,
        "targetPath":
"$.DEFAULTS.LANGU",
        "defaultValue": "E",
        "valueMappings": [
          {
            "key": [
              "bg"
            ],
            "mappedValue": "W"
          }
        ]
      },
      {
        "sourcePath":
"$.preferredLanguage",
        "optional": true,
        "targetPath":
"$.ADDRESS.LANGUP_ISO",
        "functions": [
          {
            "function":
"toUpperCaseString"
          }
        ]
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.timezone"
        ],
        "optional": true,
        "targetPath":
"$.LOGONDATA.TZONE",
        "defaultValue": "CET",
        "valueMappings": [
          {
            "key": [
              "Europe/Sofia"
            ],
            "mappedValue": "EET"
          }
        ]
      },
      {
        "scope": "createEntity",
        "targetPath":
"$.PASSWORD.BAPIPWD",
        "functions": [
          {
            "type":
"randomPassword",
            "passwordLength":
24,

"minimumNumberOfLowercaseLetters
": 1,

"minimumNumberOfUppercaseLetters
": 1,

"minimumNumberOfDigits": 1,
```

Read Transformation

```
      {
        "sourcePath":
"$.ACTIVITYGROUPS[*].AGR_NAME",

"preserveArrayWithSingleElement"
: true,
        "optional": true,
        "targetPath":
"$.groups[?(@.value)]",
        "functions": [
          {
            "type": "encode",
            "algorithm":
"base32",
            "skipPadding": true
          }
        ]
      },
      {
        "constant": "direct",

"preserveArrayWithSingleElement"
: true,
        "targetPath":
"$.groups[*].type",
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath":
"$.ROLE_NAME",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id",
        "functions": [
          {
            "type": "encode",
            "algorithm":
"base32",
            "skipPadding": true
          }
        ]
      },
      {
        "constant": "Group",
        "targetPath":
"$.meta.resourceType"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
```

Write Transformation

```
"minimumNumberOfSpecialSymbols":
 0
          }
        ]
      },
      {
        "scope": "createEntity",
        "optional": true,
        "ignore": true,
        "sourcePath":
"$.password",
        "targetPath":
"$.PASSWORD.BAPIPWD"
      },
      {
        "constant":
"updateEntity",
        "targetVariable":
"operationTypeVariable"
      },
      {
        "constant":
"createEntity",
        "targetVariable":
"operationTypeVariable",
        "scope": "createEntity"
      },
      {
        "condition":
"$.active == false && '$
{operationTypeVariable}' ==
'createEntity'",
        "constant": "X",
        "targetPath":
"$.LOCK_LOCALLY"
      },
      {
        "condition": "'$
{operationTypeVariable}' ==
'updateEntity'",
        "constant": "U",
        "targetPath": "$.LOCK"
      },
      {
        "condition":
"$.active == false && '$
{operationTypeVariable}' ==
'updateEntity'",
        "constant": "L",
        "targetPath": "$.LOCK"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.ROLE_NAME"
```

Read Transformation

```
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.ROLE_NAME",
        "targetPath":
"$.displayName"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:co
re:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath":
"$.USERLIST[*].USERNAME",

"preserveArrayWithSingleElement"
: true,
        "targetPath":
"$.members[?(@.value)]",
        "optional": true,
        "functions": [
          {
            "type": "encode",
            "algorithm":
"base32",
            "skipPadding": true
          }
        ]
      },
      {
        "constant": "User",

"preserveArrayWithSingleElement"
: true,
        "targetPath":
"$.members[*].type",
        "optional": true
      },
      {
        "constant":
"authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:e
xtension:sap:2.0:Group']
['type']"
      },
      {
        "constant": "readWrite",
        "targetPath": "$
['urn:ietf:params:scim:schemas:e
xtension:sap:2.0:Group']
['supportedOperations']"
      }
```

Write Transformation

```
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetVariable":
"entityIdTargetSystem",
        "functions": [
          {
            "type": "decode",
            "algorithm":
"base32",
            "skipPadding": true
          },
          {
            "type": "toString"
          }
        ]
      },
      {
        "scope": "deleteEntity",
        "sourceVariable":
"entityIdTargetSystem",
        "targetVariable":
"entityIdTargetSystem",
        "functions": [
          {
            "type": "decode",
            "algorithm":
"base32",
            "skipPadding": true
          },
          {
            "type": "toString"
          }
        ]
      },
      {
        "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement"
: true,
        "targetPath":
"$.USERLIST[?(@.USERNAME)]",
        "optional": true,
        "functions": [
          {
            "type": "decode",
            "algorithm":
"base32",
            "skipPadding": true
          },
          {
            "type": "toString",
            "applyOnElements":
true
          }
        ]
      }
    ]
  }
}
```

Read Transformation	Write Transformation

<pre>]
}
}</pre>	

8. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ⓘ Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.8 SAP Ariba Applications

Follow this procedure to set up SAP Ariba Applications as a proxy system.

Prerequisites

> **Note**
>
> This topic describes customer-managed configurations for the SAP Ariba Applications proxy system. For SAP-managed configurations, refer to the SAP-Managed SAP Cloud Identity [page 520] documentation.

Next-Generation SCIM API

If you are using the next-generation SCIM API provided by SAP Ariba Applications:

- The Identity Provisioning service must be configured to use the OAuth 2.0 client-credentials flow to obtain access tokens from the Identity Authentication service for calling the SAP Ariba Applications next generation SCIM API. The required dependency must be defined in the configuration of the consumer application. For more information, see Configure Form-Based OAuth Authentication [page 2457].

Current-Generation SCIM API

If you are using the current-generation SCIM API provided by SAP Ariba Applications:

- You have created a client application on SAP Ariba APIs Portal for your integration with Identity Provisioning. This is needed to enable SCIM API for user and group for Ariba Applications. For more information, see Register an Application in SAP Ariba Developer Portal .

  > **Note**
  >
  > If you don't have an SAP Ariba APIs account, go to the SAP Ariba Developer Portal and follow the instructions for sign up. For more information, see Get an SAP Ariba APIs Account.

  To find your *realm* name, log in to your SAP Ariba system. It is embedded in your login URL, as illustrated by the following examples:

  - SAP Ariba Buyer example: `https://s1.ariba.com/Buyer/Main/ad/loginPage/...&realm=`*`mycompany-t`*
  - SAP Ariba Sourcing example: `http://`*`mycompany`*`.sourcing.ariba.com/`
- You have enabled the client application for Identity Provisioning on the SAP Ariba Developer Portal.
- You have generated an OAuth secret for your application. For more information, see Generating the OAuth Secret and Base64 Encoded Client ID and Secret.
  To set up your SAP Ariba Applications provisioning system, you'll need to map your SAP Ariba application parameters to the corresponding Identity Provisioning properties. The mapping is as follows:

SAP Ariba	Identity Provisioning	Values
SCIM API URL	URL	Examples: • US: **https://openapi.ariba.com**

SAP Ariba	Identity Provisioning	Values
		• Europe: **https://eu.openapi.ariba.com** • UAE: **https://mn1.openapi.ariba.com** For more information, see SAP Ariba APIs Portal Discover Strategic Sourcing SCIM API for User and Group Master Data Environment Details.
SAP Ariba OAuth 2.0 Token URL	`OAuth2TokenServiceURL`	Examples: • US: **https://api.ariba.com/v2/oauth/token** • Europe: **https://api-eu.ariba.com/v2/oauth/token** • UAE: **https://api.mn1.ariba.com/v2/oauth/token** For more information, see SAP Ariba APIs Portal Discover Strategic Sourcing SCIM API for User and Group Master Data Environment Details.
OAuth Client ID	`User`	Alphanumeric string Example: **aaaa12345-1111-3333-cccc-1234567890**
OAuth Secret	`Password`	Alphanumeric string Example: **aaaGGG1eee12abcdefGHIJK123lmnopTTT**
Application key	`ariba.applications.api.key`	Alphanumeric string Example:**123abc123XYZ000abc123ABC012345**
AN-ID	`ariba.applications.realm.id`	AN<numeric_string> Example: **AN000111222333**

Context

After fulfilling the prerequisites, you can create an SAP Ariba Applications proxy system to load its users into an on-premise system and provision groups and new users back to SAP Ariba Applications.

These proxy systems can consume either the current-generation SCIM API or the next-generation SCIM API provided by SAP Ariba Applications. Identity Provisioning distinguishes between the two by the API URL pattern (the current SCIM API contains **openapi** in the URL, while the next-generation API does not) and by the `AuthType=Form` property, which must be configured only for the next-generation API.

For more information about the SAP Ariba SCIM API scope of support, see SAP Ariba SCIM API.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Ariba Applications* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
URL	Enter the SCIM API URL for your SAP Ariba application (see the **Prerequisites** section). The current-generation SCIM API uses the **openapi** pattern in the URL.
`ProxyType`	Enter: *Internet*

Property Name	Description & Value
`Authentication`	Enter: *BasicAuthentication*
`AuthType`	Enter: *Form* **Relevant for next-generation SCIM API**
`User`	Enter the *Client ID* for the current-generation or the next-generation SCIM API, as described in the corresponding Prerequisites section.
`Password`	(Credential) Enter the *Client Secret* for the current-generation or the next-generation SCIM API, as described in the corresponding Prerequisites section.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL for the current-generation or the next-generation SCIM API, as described in the corresponding Prerequisites section.
`ariba.applications.api.dependency.name`	This property holds the value of the dependency name used in the configuration of the SAP Ariba Applications consumer application, created in the SAP Cloud Identity Services tenant. It is used for access token retrieval from Identity Authentication. **Relevant for next-generation SCIM API**
`ariba.applications.api.key`	(Credential) Enter your application key (see the **Prerequisites** section). **Relevant for current-generation SCIM API**
`ariba.applications.realm.id`	Enter your AN-ID (see the **Prerequisites** section). **Relevant for current-generation SCIM API**

Property Name	Description & Value
(Optional) `ariba.applications.group.flatten`	This property allows or forbids reading "nested groups" (group structures) from SAP Ariba Applications. If enabled (**true**), group members of type *group* will be ignored during read in order not to be provisioned to target systems that do not support nested groups. Thus, leave the default/predefined **false** value only if you are sure that the consuming external application (identity management system) supports nested groups. Default value: *false* Predefined value (during system creation): *false* Leave the default/predefined value only if you are sure that the consuming external application (identity management system) supports nested groups. **Note** If you set this property to *true* in your proxy system, the execution of requests with parameter `membersType=group` will not be supported.
(Optional) `ariba.applications.group.filter`	When specified, only those SAP Ariba Applications groups matching the filter expression will be read. For example: *displayName eq "ProjectTeam1"*
(Optional) `ariba.applications.support.patch.operation`	Default value: *true*

Property Name	Description & Value
`ariba.applications.group.members.page.size`	SAP Ariba Applications has a hardcoded limit of the number of group members returned per request when reading the group SCIM resources. When the Identity Provisioning reads a group, it compares the number of group members with the value configured for `ariba.applications.group.members.page.size`. Depending on the value you defined for `ariba.applications.group.members.paging.enabled`, you can expect the following results: • In case the number of group members equals or exceeds the configured limit and the property `ariba.applications.group.members.paging.enabled` is enabled, the Identity Provisioning will perform full read of all group members via subsequent requests. • In case the number of group members is less than the configured limit or the property `ariba.applications.group.members.paging.enabled` is disabled, the Identity Provisioning will get the returned group members from the target system. Setting the property `ariba.applications.group.members.paging.enabled` allows you to read groups with a large number of group members. For more information, see ariba.applications.group.members.paging.enabled. Default value: *50*
`ariba.applications.group.members.paging.enabled`	This property enables the full read of group members. When it is set to *true*, all groups with members count exceeding the value defined in `ariba.applications.group.members.page.size` will be fully read by the Identity Provisioning via subsequent requests. **Possible values:** • *true* - Full read of group members is enabled. • *false* - Full read of group members is disabled. Default value: *true*

Property Name	Description & Value
`ariba.applications.user.groups.paging.enabled`	SAP Ariba Applications has a hardcoded limitation over the number of user's groups returned per request. This limit is specified by the property `ariba.applications.group.members.page.size`. To trigger a full read of a user with *'groups'* attribute over the configured value, set `ariba.applications.user.groups.paging.enabled` to *true*. **Possible values:** • *true* - the full read of user's groups is enabled. • *false* - the full read of user's groups is disabled. Default value: *true*

Property Name	Description & Value
`ariba.applications.user.groups.page.size`	The number of user groups that SAP Ariba Application returns per request when reading a user. When the Identity Provisioning reads a user, it compares the number groups assigned with the value configured for `ariba.applications.user.groups.page.size`. Depending on the number of groups assigned to the user, you can expect the following results: • If the number of assigned groups is equal or exceeds the configured value, and the `ariba.applications.user.groups.paging.enabled` property is set to *true*, Identity Provisioning will read all user groups in separate requests. For example, if a user has 6 groups assigned and the value of the page size is set to *2*, Identity Provisioning will send 3 requests to read the user groups. • If the number of assigned groups is less than the configured value for `ariba.applications.user.groups.page.size` or the property `ariba.applications.user.groups.paging.enabled` is disabled, the Identity Provisioning will read all user groups via one request. For example, if a user has 2 groups assigned and the value of the page size is set to *6*, Identity Provisioning will read all user groups via one request. Setting the property `ariba.applications.user.groups.paging.enabled` allows you to read users with a large number of groups assigned. For more information, see `ariba.applications.user.groups.paging.enabled`. Default value: *50*

Exemplary destination (property configuration):

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://openapi.ariba.com
User=aaaa12345-1111-3333-cccc-1234567890
Password=************
OAuth2TokenServiceURL=https://api.ariba.com/v2/oauth/token
ariba.applications.group.flatten=false
ariba.applications.support.patch.operation=true
ariba.applications.api.key=123abc123XYZ000abc123ABC012345
ariba.applications.realm.id=AN000111222333
```

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Ariba Applications* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Ariba Applications. For more information, see:

   Manage Transformations [page 2439]

   SCIM API for User and Group Master Data

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value",
        "optional": true
      },
      {
        "sourcePath": "$.emails[?
(@.primary == true)].value",
        "optional": true,
        "correlationAttribute":
true
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "targetPath": "$.id",
        "type": "remove",
        "scope": "patchEntity"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:ariba:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap.odm:2.0:User",

"urn:sap:cloud:scim:schemas:extens
ion:custom:2.0:profile:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
```

Read Transformation

```
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['companyCode']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['companyCode']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['costCenter']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingGroup']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingGroup']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['generalLedgerAccount']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['companyCode']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['companyCode']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['costCenter']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingGroup']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingGroup']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['generalLedgerAccount']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['generalLedgerAccount']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
```

Read Transformation

```
ension:sap.odm:2.0:User']
['generalLedgerAccount']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingOrganization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingOrganization']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['plant']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['plant']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['currency']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['currency']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['deliverTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['deliverTo']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['purchasingUnit']",
```

Write Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingOrganization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.odm:2.0:User']
['purchasingOrganization']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['currency']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['currency']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['deliverTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['deliverTo']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['purchasingUnit']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['purchasingUnit']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['network']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['network']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['purchasingUnit']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['network']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['network']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['addresses']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['addresses']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['passwordAdapter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['passwordAdapter']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['approvalLimit']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['approvalLimit']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['expenseApprovalLimit']",
```

Write Transformation

```
ension:sap:ariba:2.0:User']
['addresses']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['addresses']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['passwordAdapter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['passwordAdapter']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['approvalLimit']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['approvalLimit']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['expenseApprovalLimit']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['expenseApprovalLimit']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:profile:User']
['alternativeDisplayNames']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:profile:User']
['alternativeDisplayNames']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
```

Read Transformation	Write Transformation

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:ariba:2.0:User']
['expenseApprovalLimit']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:profile:User']
['alternativeDisplayNames']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:profile:User']
['alternativeDisplayNames']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.title",
        "optional": true,
        "targetPath": "$.title"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale",
        "functions": [
          {
            "type": "substring",
            "beginIndex": 0,
            "endIndex": 2
          }
        ]
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true,
```

```
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "condition":
"$.emails[0].length() > 0",
        "constant": true,
        "targetPath":
"$.emails[0].primary"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.phoneNumbers",
        "functions": [
          {
            "function":
"putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation

```
        "targetPath":
"$.phoneNumbers"
      },
      {
        "sourcePath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
```

Write Transformation

```
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "targetPath": "$.id",
        "type": "remove",
        "scope": "patchEntity"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:Group",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:Group",

"urn:sap:cloud:scim:schemas:extens
ion:custom:2.0:profile:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
```

Read Transformation	Write Transformation

```
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:profile:Group']
['alternativeDisplayNames']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:profile:Group']
['alternativeDisplayNames']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members"
      }
    ]
  }
}
```

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:profile:Group']
['alternativeDisplayNames']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:profile:Group']
['alternativeDisplayNames']",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.members",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.members",
        "optional": true
      }
    ]
  }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

- For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
- For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PATCH**.

> ⓘ Note
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PATCH** method for modifying entities.

Related Information

The SAP Ariba developer portal

Video: Create application and API approval process

1.6.3.9 SAP Ariba Category Management

Follow this procedure to set up SAP Ariba Category Management as a proxy system.

Prerequisites

You have created an instance and generated a service key for the scim service plan of SAP Ariba Category Management. For more information, see Creating a SAP Ariba Category Management SCIM Service Instance in Your Subaccount.

> ⓘ Note
>
> This connector has been deprecated as of August 25, 2025, and is no longer needed. Customers with existing configurations should stop running jobs to or from SAP Ariba Category Management provisioning systems.

Context

SAP Ariba Category Management is a cloud-based solution that enables organizations to define and manage purchasing categories and provides actionable market intelligence, category insights, and various tools and templates for developing and operationalizing category strategies that drive business value and performance. This business solution is built on SAP Business Technology Platform (BTP).

You can use Identity Provisioning to configure SAP Ariba Category Management as a proxy system in hybrid scenarios. For example, when SAP Ariba Category Management is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct

connection between both systems. You can provision users to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users back to the SAP Ariba Category Management.

> **Note**
>
> SAP Ariba Category Management does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Ariba Category Management* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL provided by the service key under the *scim-v1* field without adding the path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the value from the *clientid* field of the service key.
`Password`	(Credential) Enter the value from the *clientsecret* field of the service key.

Property Name	Value
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. This is the value from the *url* field of the service key plus `/oauth/token`
(Optional) `cm.user.filter`	When specified, only those SAP Ariba Category Management users matching the filter expression will be read. For example: • *userName eq "SmithJ"* • *emails.value eq "john.doe@example.com"* • *emails.value eq "john.doe@example.com" or emails.value eq "john.d@example.com" or emails.value eq "j.doe@example.com"*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users. For more information, see Query Parameters for Proxy System SCIM API.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Ariba Category Management* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Ariba Category Management system. For more information, see:

   Manage Transformations [page 2439]

   SCIM APIs for SAP Ariba Category Management

   User Update and Uniqueness

   The write transformation of SAP Ariba Category Management supports two user unique attributes for managing conflict resolution: *username* and *email*. The username is a mandatory attribute, while the email is an optional, multivalue attribute, indicating that a user may have multiple unique emails.

   The uniqueness of a user in the target system is determined not only by the individual attributes or the combination of both, but also by considering whether one or multiple existing users in the target system share the same values for these unique attributes. Have in mind the following specifics when updating an existing user in the target system, in accordance with the configuration of the `cm.user.unique.attribute` property:

   - When `cm.user.unique.attribute`= *userName*
     If a user in the source system, with a specific username and an email, matches an existing user in the target system by the same email, but a different userName, the user provisioning will fail. In this case,

Identity Provisioning will try to create a new user but will fail, since a user with the same email already exists.

- When `cm.user.unique.attribute`= *emails[*].value*
  If a user in the source system, with a specific username and an email, matches an existing user in the target system by the same username, but a different email, the user provisioning will fail. In this case, Identity Provisioning will try to create a new user but will fail, since a user with the same username already exists.
- When `cm.user.unique.attribute`= *userName, emails[*].value*
  - If a user in the source system, with a specific username and multiple emails, matches an existing user in the target system by the same username and at least one identical email, the user in the target system will be updated.
  - If a user in the source system, with a specific username and multiple emails, matches multiple existing users in the target system (each sharing one of the multiple emails but having different usernames), the user provisioning will fail. In this case, the user will neither be created nor updated.

Default read and write transformations

> → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
```

Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User"
                ],
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "optional": true
            },
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "targetPath":
"$.name.familyName",
                "optional": true
            },
            {
                "sourcePath":
"$.name.givenName",
                "targetPath":
"$.name.givenName",
                "optional": true
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName",
                "optional": true
            },
            {
```

Read Transformation

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      }
    ]
  }
}
```

Write Transformation

```
                "sourcePath":
"$.active",
                "targetPath":
"$.active",
                "optional": true
            },
            {
                "condition":
"$.emails[0].length() > 0",
                "targetPath":
"$.emails[0].primary",
                "constant": true
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.emails"
            }
        ]
    }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.

- For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

> ⓘ Note
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

User Replication Setup

1.6.3.10 SAP Ariba Central Invoice Management

Follow this procedure to set up SAP Ariba Central Invoice Management as a proxy system.

Prerequisites

You have created an instance and generated a service key for the scim service plan of SAP Ariba Central Invoice Management. This step is automated by the booster *Set Up SAP Ariba Central Invoice Management*. For more information, see Setting Up Invoice Processing with SAP Ariba Central Invoice Management (4N6) –> *Set up Configuration Using Booster in SAP BTP Cockpit*.

Context

SAP Ariba Central Invoice Management is an SAP BTP SaaS application running on SAP BTP, Cloud Foundry environment. This application enables central management of supplier invoices from multiple connected systems, such as SAP S/4HANA Cloud Public Edition systems.

You can use Identity Provisioning to configure SAP Ariba Central Invoice Management as a proxy system in hybrid scenarios. For example, when SAP Ariba Central Invoice Management is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users back to the SAP Ariba Central Invoice Management.

This scenario supports provisioning users and groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.

4. Save your changes.

3. Add *SAP Ariba Central Invoice Management* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL provided by the service key under the *ips-connector* field without adding the path information. For example: `https://eu10.cim.cloud.sap`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the value from the *clientid* field of the service key.
`Password`	(Credential) Enter the value from the *clientsecret* field of the service key.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. This is the value from the *url* field of the service key plus `/oauth/token` For example: `https://<my tenant>.authentication.eu10.hana.ondemand.com/oauth/token`
(Optional) `cim.user.filter`	When specified, only those SAP Ariba Central Invoice Management users matching the filter expression will be read. Example: *name.familyName eq "Smith"* and *addresses.country eq "US"*
(Optional) `cim.group.filter`	When specified, only those SAP Ariba Central Invoice Management groups matching the filter expression will be read. Example: *displayName eq "ProjectTeam1"*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> ### Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Ariba Central Invoice Management* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Ariba Central Invoice Management system. For more information, see:

   Manage Transformations [page 2439]

   User Management (SCIM) for SAP Ariba Central Invoice Management

   Default read and write transformations:

   > ### → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
```

Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User"
                ],
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.externalId",
                "targetPath":
"$.externalId",
                "optional": true
            },
            {
                "sourcePath":
"$.name.givenName",
                "targetPath":
"$.name.givenName",
                "optional": true
            },
            {
                "sourcePath":
"$.name.middleName",
                "targetPath":
"$.name.middleName",
                "optional": true
            },
            {
                "sourcePath":
"$.name.familyName",
                "targetPath":
"$.name.familyName",
                "optional": true
            },
            {
                "sourcePath":
"$.name.formatted",
                "targetPath":
"$.name.formatted",
                "optional": true
```

Read Transformation

```
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.formatted",
        "targetPath":
"$.name.formatted",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificSuffix",
        "targetPath":
"$.name.honorificSuffix",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "optional": true,

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value",
        "optional": true
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
```

Write Transformation

```
            },
            {
                "sourcePath":
"$.name.honorificPrefix",
                "targetPath":
"$.name.honorificPrefix",
                "optional": true
            },
            {
                "sourcePath":
"$.name.honorificSuffix",
                "targetPath":
"$.name.honorificSuffix",
                "optional": true
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath":
"$.active",
                "targetPath":
"$.active",
                "optional": true
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName",
                "optional": true
            },
            {
                "sourcePath":
"$.nickName",
                "targetPath":
"$.nickName",
                "optional": true
            },
            {
                "sourcePath":
"$.profileUrl",
                "targetPath":
"$.profileUrl",
                "optional": true
            },
            {
                "sourcePath":
"$.title",
                "targetPath":
"$.title",
                "optional": true
            },
            {
                "sourcePath":
"$.userType",
```

Read Transformation

```
      },
      {
        "sourcePath":
"$.userType",
        "targetPath":
"$.userType",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath":
"$.timezone",
        "targetPath":
"$.timezone",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath":
"$.nickName",
        "targetPath":
"$.nickName",
        "optional": true
      },
      {
        "sourcePath": "$.title",
        "targetPath": "$.title",
        "optional": true
      },
      {
        "sourcePath":
"$.preferredLanguage",
        "targetPath":
"$.preferredLanguage",
        "optional": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",
        "optional": true,

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

```
                "targetPath":
"$.userType",
                "optional": true
            },
            {
                "sourcePath":
"$.preferedLanguage",
                "targetPath":
"$.preferredLanguage",
                "optional": true
            },
            {
                "sourcePath":
"$.locale",
                "targetPath":
"$.locale",
                "optional": true
            },
            {
                "sourcePath":
"$.timezone",
                "targetPath":
"$.timezone",
                "optional": true
            },
            {
                "constant":
"urn:ietf:params:scim:api:messages
:2.0:PatchOp",
                "targetPath":
"$.schemas[0]",
                "scope":
"patchEntity"
            },
            {
        "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.Operations",
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      }
    ]
  },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
```

Read Transformation

```
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
```

Write Transformation

```
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:Group"
                ],
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.externalId",
                "targetPath":
"$.externalId",
                "optional": true
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members[?(@.value)]"
            },
            {
                "sourcePath":
"$.Operations",
                "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.schemas",
                "targetPath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            }
        ]
    }
}
```

Read Transformation	Write Transformation

<pre>"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
"optional": true
},
{
"sourcePath": "$.members",
"targetPath": "$.members",
"optional": true,
"preserveArrayWithSingleElement": true
}
]
}
}</pre>	

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

SAP Ariba Central Invoice Management

1.6.3.11 SAP BTP ABAP environment

Follow this procedure to set up SAP BTP ABAP environment as a proxy system.

Prerequisites

- You have user credentials for an external back-end system with read and write permissions.
- To establish the connection between Identity Provisioning and SAP BTP ABAP environment, you need to set up the communication (user, system and arrangement) on SAP BTP ABAP environment. You can do it now (as a prerequisite) or in the process of configuring SAP BTP ABAP environment as a proxy system, as described in step 5.

Context

You can use SAP BTP ABAP environment as a proxy connector to execute *hybrid* scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to SAP BTP ABAP environment, whenever the external back-end requests such. This scenario supports:

- Reading of **business users** (Employee) and **business roles** (which are considered as *groups*)
- Writing of **users** and **assignments**

> ### Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP BTP ABAP environment* as a proxy system. For more information, see: Add New Systems [page 2415].
4. Set up the communication between Identity Provisioning and SAP BTP ABAP environment and configure your authentication method (basic or certificate-based).

   > ### Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP BTP ABAP environment proxy system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP BTP ABAP environment backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Create a communication system and assign the created user to the communication system.

      For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

d. Create a communication arrangement with the created system.

   For your Identity Provisioning scenario, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

   For more information, see Maintain a Communication Arrangement for Inbound Communication

   > **Note**
   >
   > The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
   >
   > The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the API URL to your SAP BTP ABAP environment system. You can take the URL from the communication scenario SAP_COM_0193.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **Restriction** Do not use special symbol ',' (comma) as it is not supported.

Property Name	Description & Value
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`a4c.skip.read.archived`	In the event of archived (disabled) entities in your SAP BTP ABAP environment system, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to **false**, or delete it.
`ips.date.variable.format`	*yyyy-MM-dd* (needed for the *Read Transformation*)
`a4c.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP BTP ABAP environment proxy system in a hybrid scenario. • **true** – the current user roles will be deleted in the proxy system, and the user will be updated only with the roles provisioned by the service. • **false** – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the proxy system. See also: Extended Explanation of the *user.roles.overwrite Properties
(Optional) `a4c.roles.filter`	Enter OData filtering for reading roles in the SAP BTP ABAP environment system. To learn what criteria you can use, see: OData URI Conventions → **4.5 Filter System Query Option**
(Optional) `a4c.roles.page.size`	Indicate how many business roles (considered as *groups*) per page to be read from your SAP BTP ABAP environment system. The value must be an integer number.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://12345-aaaaa-3333.abap.hana.ondemand.com
User=MyABAPEnvUser
Password=************
ips.date.variable.format=yyyy-MM-dd
a4c.skip.read.archived=true
a4c.user.roles.overwrite = false
a4c.roles.filter=startswith(ID, 'EMPLOYEE_LEVEL_3') eq true
a4c.roles.page.size=30
```

6. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP ABAP environment* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP BTP ABAP environment system. For more information, see:

   Manage Transformations [page 2439]

   Inbound Service: Business User

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.personID",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.personalInformation.firstName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.lastName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.middleName"
,
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.personFullN
ame",
        "targetPath":
"$.name.formatted",
        "optional": true
      },
      {
```

Write Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.personExternalID"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath":
"$.personExternalID"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath":
"$.user.globalUserID"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.personID"
      },
      {
        "targetPath":
"$.businessPartnerRoleCode",
        "type": "valueMapping",
        "sourcePaths": [
          "$.userType"
        ],
        "defaultValue": "BUP003",
        "valueMappings": [
          {
            "key": [
              "Employee"
            ],
            "mappedValue":
"BUP003"
          }
        ]
      },
      {
        "scope": "createEntity",
        "sourceVariable":
"currentDate",
        "targetPath":
"$.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
```

Read Transformation

```
        "sourcePath":
"$.user.userName",
        "targetPath":
"$.userName",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "condition":
"$.user.lockedIndicator ==
'true'",
        "constant": false,
        "targetPath": "$.active",
        "optional": true
      },
      {
        "condition":
"($.user.validityPeriod.startDate
> '${currentDate}') ||
('${currentDate}' >
$.user.validityPeriod.endDate)",
        "constant": false,
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.workplaceInformation.emailAddre
ss",
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.user.logonLanguageCode",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath":
"$.PersonExternalID",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.user.role[*].roleName",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups[?
(@.value)]"
```

Write Transformation

```
        "targetPath":
"$.validityPeriod.endDate"
      },
      {
        "scope": "createEntity",
        "sourceVariable":
"currentDate",
        "targetPath":
"$.user.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
        "targetPath":
"$.user.validityPeriod.endDate"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.personalInformation.firstName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.personalInformation.lastName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.personalInformation.middleName"
      },
      {
        "sourcePath":
"$.name.formatted",
        "optional": true,
        "targetPath":
"$.personalInformation.personFullN
ame"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.user.userName"
      },
      {
        "sourcePath":
"$.nickName",
        "optional": true,
        "targetPath":
"$.user.nickName"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath":
"$.user.logonLanguageCode"
      },
```

Read Transformation

```
      },
      {
        "sourcePath":
"$.user.globalUserID",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.user.timeZoneCode"
        ],
        "targetPath":
"$.timezone",
        "defaultValue": "Europe/
Berlin",
        "valueMappings": [
          {
            "key": [
              "WDFT"
            ],
            "mappedValue":
"Europe/Berlin"
          },
          {
            "key": [
              "ISRAEL"
            ],
            "mappedValue": "Asia/
Jerusalem"
          },
          {
            "key": [
              "RUS03"
            ],
            "mappedValue":
"Europe/Moscow"
          },
          {
            "key": [
              "AUSNSW"
            ],
            "mappedValue":
"Australia/Sydney"
          },
          {
            "key": [
              "UTC+4"
            ],
            "mappedValue": "Asia/
Dubai"
          },
          {
            "key": [
              "BRAZIL"
            ],
            "mappedValue":
"America/Sao_Paulo"
          },
          {
```

Write Transformation

```
      {
        "sourcePath":
"$.emails[0].value",
        "optional": true,
        "targetPath":
"$.workplaceInformation.emailAddre
ss"
      },
      {
        "condition": "$.active ==
false",
        "constant": "true",
        "targetPath":
"$.user.lockedIndicator"
      }
    ],
    "scimEntityEndpoint": "Users"
  },
  "group": {
    "mappings": [
      {
        "sourcePath":
"$.displayName",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members[?
(@.value)]"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ],
    "scimEntityEndpoint": "Groups"
```

Read Transformation	Write Transformation

```
            "key": [
              "BRZLEA"
            ],
            "mappedValue":
"America/Sao_Paulo"
          },
          {
            "key": [
              "MSTNO"
            ],
            "mappedValue":
"America/Phoenix"
          },
          {
            "key": [
              "EST"
            ],
            "mappedValue":
"America/New_York"
          },
          {
            "key": [
              "UTC"
            ],
            "mappedValue": "Etc/
UTC"
          },
          {
            "key": [
              "UTC+3"
            ],
            "mappedValue": "Asia/
Riyadh"
          },
          {
            "key": [
              "EST_"
            ],
            "mappedValue":
"America/Toronto"
          },
          {
            "key": [
              "UTC+8"
            ],
            "mappedValue": "Asia/
Shanghai"
          },
          {
            "key": [
              "JAPAN"
            ],
            "mappedValue": "Asia/
Tokyo"
          }
        ]
      },
      {
        "type": "valueMapping",
        "sourcePaths": [

"$.businessPartnerRoleCode"
```

```
    }
  }
```

Read Transformation	Write Transformation

```
        ],
        "targetPath":
"$.userType",
        "defaultValue":
"Employee",
        "valueMappings": [
          {
            "key": [
              "BUP003"
            ],
            "mappedValue":
"Employee"
          }
        ]
      }
    ]
  },
   "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.ID",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.ID",
        "targetPath":
"$.displayName"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath":
"$.to_BusinessUserAssignment.resul
ts",
        "optional": true,
```

Read Transformation	Write Transformation

```
"preserveArrayWithSingleElement":
true,
        "targetPath": "$.members"
      },
      {
        "type": "remove",
        "targetPath":
"$.members[*].__metadata"
      },
      {
        "type": "remove",
        "targetPath":
"$.members[*].UserName"
      },
      {
        "type": "rename",
        "constant": "value",
        "targetPath":
"$.members[*].PersonID"
      },
      {
        "constant": "User",
        "targetPath":
"$.members[*].type"
      },
      {
        "constant":
"authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
      },
      {
        "constant": "readWrite",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']"
      }
    ]
  }
}
```

By default, Identity Provisioning reads group IDs and members. If you want the service to also read group descriptions, you can add an extra mapping to the *"group"* resource in the *Read Transformation*. To learn how, see Guided Answers: Business Role Description.

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.

- For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PATCH**.

> ⓘ Note
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PATCH** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.12 SAP BTP Account Members (Neo)

Follow this procedure to set up your SAP Business Technology Platform as a proxy system, from and to which you can provision and write members in your Neo account.

Prerequisites

- You have created a new Platform API OAuth client, with API *Account Member Management* and scopes *Manage Account Members* and *Read Account Members*.
  Save the *Client ID* and *Client Secret* as you'll need them when you configure your proxy system. Make sure you save the client secret as you cannot retrieve it later. For more information, see Create a Platform API Client.

Context

The Identity Provisioning service helps companies to automatically manage the user-to-platform roles assignments for SAP Business Technology Platform subaccounts.

You can use SAP Business Technology Platform as a proxy connector to execute *hybrid* scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to SAP Business Technology Platform, whenever the external back-end requests such.

The Identity Provisioning service system SAP Business Technology Platform Account Members represents the various authorization mappings managed by the Platform Authorization Management API of SAP BTP as one generic SCIM 2.0 system, with some limitations. For more information, see: Platform Authorization Management API.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP BTP Account Members (Neo)* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
Type	Enter: *HTTP*

Property Name	Description & Value
URL	Enter: **`https://api.<SAP_BTP_host>/authorization/v1/platform/accounts/<SAP_BTP_account>`** Examples: • (Europe – Rot) *https://api.hana.ondemand.com/authorization/v1/platform/accounts/abc123xyz* • (Japan – Tokyo) *https://api.jp1.hana.ondemand.com/authorization/v1/platform/accounts/abc123xyz*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the Client ID of the new Platform API OAuth client created for the **Account Member Management** API (see the prerequisites).
`Password`	(Credential) Enter the Client Secret of the new Platform API OAuth client created for the **Account Member Management** API (see the prerequisites).
`OAuth2TokenServiceURL`	Enter: **`https://api.<SAP_BTP_host>/oauth2/apitoken/v1`** Examples: • (Europe – Rot) *https://api.hana.ondemand.com/oauth2/apitoken/v1* • (US East – Sterling) *https://api.us3.hana.ondemand.com/oauth2/apitoken/v1*
`scp.user.userbase`	This property specifies the host to the identity provider to be used with this proxy system. All provisioned users can be authenticated only by this identity provider. Default value: *accounts.sap.com* If you use another IdP, enter its value as configured in the SAP BTP cockpit. For example: **`<account_ID>.accounts.ondemand.com`**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP Account Members* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your proxy system. For more information, see:

Manage Transformations [page 2439]

SAP BTP: Authorization Management API

The *SAP BTP Account Members (Neo)* proxy connector supports SCIM PATCH operation.

The read transformation takes the group name from the source system. However, the names of the SAP BTP platform roles sometimes contain characters that do not fit the SCIM URI specification. As Identity Provisioning service cannot directly use these characters as a SCIM resource ID, it needs to first encode them into base32 ASCII format. The intermediate JSON logic of the proxy system will then use the new encoded ID and write it to the target system. That means, the ID will be decoded in the proxy write transformation during the **patchEntity** operation. For more information, see: Transformation Expressions [page 391] → *patchEntity*

Default read and write transformations:

> → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.emails[0].value",
        "optional": true,
```

Write Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.groups[*].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.roles[?
(@.value)]"
      },
      {
        "constant":
"%scp.user.userbase%",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:UserExt']
['userbase']"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
        "targetPath":
"$.schemas[0]"
      },
      {
        "constant":
"urn:sap:cloud:scim:schemas:extens
ion:custom:2.0:UserExt",
        "targetPath":
"$.schemas[1]"
      }
    ],
    "scimEntityEndpoint": "Users"
  },
  "group": {
    "mappings": [
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
```

Read Transformation

```
        "targetPath":
"$.emails[0].value"
      },
      {
        "sourcePath":
"$.description",
        "optional": true,
        "targetPath":
"$.description"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.roles[*].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups[?
(@.value)]"
      }
    ]
  },
 "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
```

Write Transformation

```
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ],
    "scimEntityEndpoint": "Groups"
  }
}
```

Read Transformation	Write Transformation

```
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members[?
(@.value)]"
      },
      {
        "constant": "User",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.members[*].type"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PATCH**.

> ⓘ Note
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PATCH** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.13 SAP BTP Platform Members (Cloud Foundry)

Follow this procedure to set up the SAP BTP Platform Members (Cloud Foundry) as a proxy system.

Prerequisites

- You have a global account in SAP BTP with at least one multi-environment subaccount with enabled Cloud Foundry environment.
- You have established trust between your SAP Cloud Identity Services tenant as custom identity provider for platform users and your global account(s) containing these subaccounts. For more information, see Establish Trust and Federation of Custom Identity Providers for Platform Users.
- You have created a CF provisioning user (which is a regular user of type *Employee* in the local identity directory of your SAP Cloud Identity Services tenant) that will be used for provisioning. Give this user

an email address with the following pattern `cf-user-provisioning-<origin_key>@sap.invalid`, where `<origin_key>` is the origin key of the trust configuration for BTP platform users which points to your SAP Cloud Identity Services tenant. You have set an initial password for the user and marked its email adress as *verified*. For more information, see Create a New User and List and Edit User Details.

> **Note**
>
> The password lifetime for the CF provisioning user depends on the password policy set for the *SAP Business Technology Platform* application in the administration console of SAP Cloud Identity Services. For more information, see Set a Password Policy for an Application.

- You have activated the account of the CF provisioning user by opening the *Profile Page* of SAP Cloud Identity Services, in a separate browser session, and changing its initial password. The URL has the following pattern: `https://<tenant ID>.accounts.ondemand.com` or `https://<tenant ID>.accounts.cloud.sap`
- You have created the group *cf-user-provisioning* in your SAP Cloud Identity Services tenant and added the CF provisioning user to it. For more information, see Create a New Group and Add Users to a Group.
- You have added the CF provisioning user as org member with role **Org Manager** to each Cloud Foundry organization where you want to provision users, in the SAP BTP cockpit For more information, see Add Org Members and About Roles in the Cloud Foundry Environment.

Context

The Cloud Foundry environment enables you to create polyglot cloud applications in Cloud Foundry. It contains the SAP BTP, Cloud Foundry runtime service, which is based on the open-source application platform managed by the Cloud Foundry Foundation. For more information, see Cloud Foundry Environment.

When you enable the Cloud Foundry environment in your subaccount, the system automatically creates a Cloud Foundry organization for you. You are able to add platform users as org members and space members and assign roles to grant these users platform access. For more information, see Valid role types .

SAP BTP Platform Members (Cloud Foundry) connector manages org and space members, as well as their role assignments, in the Cloud Foundry environment of a multi-environment subaccount, where a **single** SAP Cloud Identity Services tenant acts as custom identity provider. We recommend that you use the Identity Provisioning service enabled in this SAP Cloud Identity Services tenant.

In SAP BTP Platform Members (Cloud Foundry), groups correspond to roles in particular Cloud Foundry orgs or spaces, thus group members are user assignments of a role in a specific Cloud Foundry org or space. Group names must follow a defined pattern, explained below, to ensure the correct mapping and provisioning of users and their role assignments to the relevant Cloud Foundry organization or space.

> **Note**
>
> The attribute *name*, defined within the *urn:sap:cloud:scim:schemas:extension:custom:2.0:Group* schema, is used to map the users and user assignments of a role to the relevant Cloud Foundry organization or space. The group names follow the respective patterns:
>
> - organization group name: `<org_ID> <org_role>`
> - space group name: `<org_ID> <space_ID> <space_role>`, where `space_ID` can be retrieved using the Cloud Foundry Command Line Interface by executing the command **`cf space --guid`**.

Whereas the group display names adhere to the following patterns:

- organization group display name:
  `<btp.cf.pm.group.prefix><btp.cf.pm.landscape>:<org_name>:<org_role>`
- space group display name:
  `<btp.cf.pm.group.prefix><btp.cf.pm.landscape>:<org_name>:<space_name>:<space_role>`

You can use SAP BTP Platform Members (Cloud Foundry) as a proxy connector to execute hybrid scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to SAP BTP Platform Members (Cloud Foundry), whenever the external back-end requests such.

→ Remember

This connector enables you to write users and user role assignments to Cloud Foundry on *subaccount* level. For provisioning of users and groups to Cloud Foundry on *application* level, refer to SAP BTP XS Advanced UAA (Cloud Foundry) [page 1045]. For more information, see SAP BTP Integration Scenario.

The proxy system consumes `User Account and Authentication API` and `Cloud Foundry V3 API` provided by Cloud Foundry.

Note

In case you have Cloud Foundry environment enabled in multiple Cloud Foundry landscapes or more than one SAP Cloud Identity Services tenants, you must configure a separate SAP BTP Platform Members (Cloud Foundry) connector for each of them.

Note

The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. To enable provisioning of platform users and user role assignments to and from Cloud Foundry environment, create a Support Ticket on component `BC-CP-CF-SEC-IAM`.

   Specify your SAP Cloud Identity Services tenant ID and the origin key of the trust configuration for BTP platform users which points to your SAP Cloud Identity Services tenant. You can find the origin key value in theSAP BTP cockpit. Go to your SAP BTP subaccount, choose *Trust Configuration* and see the value under *Origin Key* for the relevant *Custom Identity Provider for Platform Users*.
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
3. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.

- For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
- For **Basic authentication**, proceed as follows:
    1. Add an administrator user of type **System** and configure the basic authentication method for this user.
       If you already have a technical user, skip this step.
    2. Save your changes.
    3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
    4. Save your changes.

4. Add *SAP BTP Platform Members (Cloud Foundry)* as a proxy system. For more information, see Add New Systems [page 2415].
5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the `email` of the CF provisioning user (see **Prerequisites**).
`Password`	(Credential) Enter the password for the CF provisioning user (see **Prerequisites**).
`btp.cf.pm.origin`	Enter the origin key of your SAP Cloud Identity Services tenant (see **Step 1**). The value of this property is a string, which always ends with the suffix *-platform*. It will be used as the *origin* attribute in the system transformation.

Property Name	Description & Value
`btp.cf.pm.landscape`	Enter the *technical key* or the *host name* of the landscape in which your multi-environment subaccount with enabled Cloud Foundry environment is located. The techical key is available only for the SAP BTP Cloud Foundry regions. For more information, see Regions and API Endpoints Available for the Cloud Foundry Environment. Please note that when the technical key contains an extension index, the correct value is displayed in the *API Endpoint* column instead of the *Technical Key* column of the table. This is an example of a *technical key*: **`cf-eu10-002`**, corresponding to the API endpoint `api.cf.eu10-002.hana.ondemand.com` and host name `cf.eu10-002.hana.ondemand.com` The host name of the landscape can be found by looking up the *API Endpoint* field in the *Overview* tab of your *Cloud Foundry Environment* on subaccount level of the SAP BTP cockpit. The API endpoint has the following format: `https://api.<landscape-hostname>` **Note** If your provisioning job finishes with error *'Name or service not known'*, the value used for technical key is not correct. In that case you can use the host name of the landscape instead.
(Optional) `btp.cf.pm.user.filter`	When specified, only those SAP BTP Platform Members (Cloud Foundry) users matching the filter expression will be read. For example: *userName eq "SmithJ"*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

6. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP Platform Members (Cloud Foundry)* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

> ### Note
>
> Update operation is skipped for users in the default proxy write transformation.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP BTP Platform Members (Cloud Foundry) system. For more information, see:

Manage Transformations [page 2439]

Cloud Foundry V3 API

> ### Caution
>
> Keep in mind that when you add or delete users and user role assignments in SAP BTP Platform Members (Cloud Foundry), the operations order must adhere to the Cloud Foundry Environment hierarchy. In this case, you can expect the following behavior:
>
> - When adding a user as space member, it must already be present as member of the relevant organization with assigned role **Org User**. When deleting an org member, it should be first removed from the spaces of the relevant organization. In case you end up with erroneous entity structure, you can manually add or delete users via the administration console for SAP Cloud Identity Services.
> - When adding user role assignments, you should first add the org roles and afterwards proceed with the space roles. Also, when deleting user role assignments, you should first delete the space roles and then proceed with the org roles. If this condition is not fullfiled the job will finish with error. In that case, you can rerun the job until it finishes successfully.

Default read and write transformations:

> ### → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.formatted",
        "targetPath":
"$.name.formatted",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
```

Write Transformation

Code Syntax

```
{
  "user": {
    "skipOperations": [
      "update"
    ],
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.formatted",
        "targetPath":
"$.name.formatted",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value == []",
        "targetPath":
"$.emails[0].primary",
        "constant": true
      },
      {
        "constant":
"urn:scim:schemas:core:1.0",
        "targetPath":
"$.schemas[0]"
      }
    ],
    "scimEntityEndpoint": "Users"
  },
```

Read Transformation

```
        "sourcePath":
"$.verified",
        "targetPath":
"$.verified",
        "optional": true
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourcePath": "$.origin",
        "targetPath": "$.origin",
        "optional": true
      },
      {
        "sourcePath": "$.zoneId",
        "targetPath": "$.zoneId",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.id",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:Group"
        ],
        "targetPath": "$.schemas"
```

Write Transformation

```
  "group": {
    "mappings": [
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ],
    "scimEntityEndpoint": "Groups"
  }
}
```

Read Transformation	Write Transformation

```
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "functions": [
          {
            "condition":
"'%btp.cf.pm.group.prefix%' !==
'null'",
            "function":
"concatString",
            "prefix":
"%btp.cf.pm.group.prefix%"
          }
        ]
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "constant":
"authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
      },
      {
        "constant":
"userOnlyMembership",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']"
      }
    ]
  }
}
```

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ⓘ Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

8. Run an initial load job.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

SAP BTP Integration Scenario

1.6.3.14 SAP BTP Java/HTML5 apps (Neo)

Follow this procedure to set up SAP Business Technology Platform as a proxy system, from and to which you can provision user-to-groups assignments for Java/HTML5 applications that run on SAP BTP, Neo environment.

Prerequisites

- You have created a new Platform API OAuth client, with *Authorization Management* scopes. Save the *Client ID* and *Client Secret* as you'll need them when you configure your proxy system. Make sure you save the client secret as you cannot retrieve it later.
  To learn how to create a Platform API client and what are the required platform service roles, see: Create a Platform API Client

Context

The Identity Provisioning service helps companies to automatically manage the user-to-groups assignments for Java/HTML5 applications running on the SAP Business Technology Platform. For this aim, the service reuses data from an active company user store.

For this scenario, SAP Business Technology Platform is a proxy system. That is, the Identity Provisioning service represents the various authorization mappings managed by Authorization Management REST API of SAP BTP as one generic SCIM 2.0 system, with some limitations. For more information, see: Using the Authorization Management REST API

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators* .
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.

- For **Basic authentication**, proceed as follows:
  1. Add an administrator user of type **System** and configure the basic authentication method for this user.
     If you already have a technical user, skip this step.
  2. Save your changes.
  3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
  4. Save your changes.

3. Add *SAP BTP Java/HTML5 apps (Neo)* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter: **`https://api.<SAP_BTP_host>/authorization/v1/accounts/<SAP_BTP_account>`** Examples: • (Europe – Rot) *https://api.hana.ondemand.com/authorization/v1/accounts/abc123xyz* • (Japan – Tokyo) *https://api.jp1.hana.ondemand.com/authorization/v1/accounts/abc123xyz*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the Client ID of the new Platform API OAuth client created for the Authorization Management API (see the prerequisites).
`Password`	(Credential) Enter the Client Secret of the new Platform API OAuth client created for the Authorization Management API (see the prerequisites).
`OAuth2TokenServiceURL`	Enter: **`https://api.<SAP_BTP_host>/oauth2/apitoken/v1`** Examples: • (Europe – Rot) *https://api.hana.ondemand.com/oauth2/apitoken/v1* • (US East – Sterling) *https://api.us3.hana.ondemand.com/oauth2/apitoken/v1*

Property Name	Description & Value
(Optional) `hcp.read.group.roles`	If you set this property to *true*, the Identity Provisioning will read the following additional attributes for a SAP BTP group: • Application roles • Group mappings, defined by your identity provider **⚠ Restriction** This property and the relevant functionality is applicable only if SAP BTP and the external SCIM-based system belong to one and the same region.
(Optional) `hcp.patch.response.with.resource`	Use this property when you execute hybrid scenarios with SAP BTP as a SCIM proxy system, and you update an entity (mostly relevant to groups, like when you change the members of a group) via a PATCH request. Set this property to *true*. This way, the successful PATCH request will return a response code 200 (*OK*) back to the consumer client application with a payload body containing the updated attributes of the relevant group.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP Java/HTML5 apps (Neo)* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your proxy system. For more information, see:

   Manage Transformations [page 2439]

   SAP BTP: Authorization Management API

   The *SAP BTP Java/HTML5 apps (Neo)* proxy connector supports SCIM PATCH operation.

   The read transformation takes the group name from the source system. However, SAP BTP groups names sometimes contain characters that do not fit the SCIM URI specification. As Identity Provisioning service cannot directly use these characters as a SCIM resource ID, it needs to first encode them into base32 ASCII format. The intermediate JSON logic of the proxy system will then use the new encoded ID and write it to the target system. That means, the ID will be decoded in the proxy write transformation during the **patchEntity** operation. For more information, see: Transformation Expressions [page 391] → *patchEntity*

   Default read and write transformations:

→ Tip

The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.name",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem",
        "functions": [
          {
            "type": "encode",
            "algorithm": "base32",
            "skipPadding": true
          }
        ]
      },
      {
        "targetPath":
"$.displayName",
        "sourcePath": "$.name"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "targetPath":
"$.schemas[0]",
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group"
      },
      {
        "sourcePath": "$.members",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.members[?
(@.value)]",
        "optional": true
      },
      {
        "sourcePath": "$.roles",
        "targetPath": "$.roles",
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.name",
        "targetVariable":
"entityIdTargetSystem"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath":
"$.displayName",
        "targetPath": "$.name",
        "targetVariable":
"entityIdTargetSystem"
      },
      {
        "scope": "deleteEntity",
        "sourceVariable":
"entityIdTargetSystem",
        "targetVariable":
"entityIdTargetSystem",
        "functions": [
          {
            "type": "decode",
            "algorithm": "base32",
            "skipPadding": true
          },
          {
            "type": "toString"
          }
        ]
      },
      {
        "scope": "patchEntity",
        "sourceVariable":
"entityIdTargetSystem",
        "targetVariable":
"entityIdTargetSystem",
        "functions": [
          {
            "type": "decode",
            "algorithm": "base32",
            "skipPadding": true
          },
          {
            "type": "toString"
          }
        ]
      },
      {
```

Read Transformation

```
"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.idpGroupMappings",
        "targetPath":
"$.idpGroupMappings",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "constant": "User",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.members[*].type",
        "optional": true
      }
    ]
  },
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.name",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "targetPath":
"$.displayName",
        "sourcePath": "$.name"
      },
      {
        "targetPath":
"$.userName",
        "sourcePath": "$.name",
        "correlationAttribute":
true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
```

Write Transformation

```
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath":
"$.members[*].value",
        "optional": true,
        "targetPath": "$.users"
      }
    ]
  }
}
```

Read Transformation	Write Transformation

```
      },
      {
        "targetPath":
"$.schemas[0]",
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User"
      },
      {
        "sourcePath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.groups[?
(@.value)]",
        "optional": true,
        "functions": [
          {
            "type": "encode",
            "algorithm": "base32",
            "skipPadding": true
          }
        ]
      },
      {
        "constant": "direct",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.groups[*].type",
        "optional": true
      }
    ]
  }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.15 SAP BTP XS Advanced UAA (Cloud Foundry)

Follow this procedure to set up the SAP BTP XS Advanced UAA (running on SAP BTP, Cloud Foundry environment) as a proxy system.

Prerequisites

You have created credentials to call the REST APIs of the SAP Authorization and Trust Management service on global account, subaccount, or directory level. For more information, see: Get Access to the APIs and btp create security/api-credential.

Context

You can use Identity Provisioning to configure SAP BTP XS Advanced UAA as a proxy system in hybrid scenarios. For example, when SAP BTP XS Advanced UAA is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. Upon provisioning platform users, business users and groups (considered role collections in SAP BTP) to the external backend system, it will trigger CRUD (create, read, update, delete) operations on users back to the SAP BTP XS Advanced UAA.

Create a separate proxy system for each global account, multi-environment subaccount and directory on SAP BTP, and configure the respective credentials and connectivity properties. Depending on where you want to provision your entities, you need to:

- Create a proxy system for the global account to provision **platform users** and groups.
- Create a proxy system for the directory to provision **platform users** and groups.
- Create a proxy system for the subaccount to provision **platform users**, **business users** and groups.

> **Note**
>
> Provisioning of platform users as **org members** and **space members** requires the usage of a different connector: SAP BTP Platform Members (Cloud Foundry). For more information, see SAP BTP Platform Members (Cloud Foundry) [page 1499].

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP BTP XS Advanced UAA (Cloud Foundry)* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Properties

Property Name	Description & Value
Type	Enter: *HTTP*

Property Name	Description & Value
`URL`	Enter the API URL in the following format: `https://api.authentication.<region>.hana.ondemand.com` or For more information, see Call an API.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`OAuth2TokenServiceURL`	Enter the URL to the OAuth2 token service in the following format: • `https://<global_account_subdomain>.authentication.<region>.hana.ondemand.com/oauth/token` or • `https://<subaccount_subdomain>.authentication.<region>.hana.ondemand.com/oauth/token` or • `https://<directory_subdomain>.authentication.<region>.hana.ondemand.com/oauth/token` or For more information, see Call an API.
`User`	Enter the respective OAuth client ID that was generated when you created credentials for each global account, subaccount and directory. For more information, see Get Access to the APIs.
`Password`	(Credential) Enter the respective OAuth client secret that was generated when you created credentials for each global account, subaccount and directory. For more information, see Get Access to the APIs.

Property Name	Description & Value
`xsuaa.origin`	Enter the Origin Key of your identity provider. • Identity provider for platform users: In the SAP BTP cockpit, go to your global account (see Navigate in the Cockpit), choose *Security* > *Trust Configuration* and copy the *Origin Key* value used for platform users. This origin key always ends with `-platform`. • Identity provider for business users: In the SAP BTP cockpit, go to your subaccount (see Navigate in the Cockpit), choose *Security* > *Trust Configuration* and copy the *Origin Key* value used for business (application) users. For example: sap.custom For more information about users with multiple origins, see Configure Single and Multiple Origins [page 1526].
`xsuaa.origin.filter.enabled`	This flag property depends on `xsuaa.origin`. Possible values: **true** or **false** • If set to *true*, the Identity Provisioning service will read only users whose identity provider is set as a value of `xsuaa.origin`. • If set to *false*, the Identity Provisioning service will read all users, regardless of their origin. • If set to *true* but the `xsuaa.origin` property is missing, the provisioning will fail.
`scim.support.patch.operation`	Use this property if you want to modify the members of a group. Possible values: • **true** – the Identity Provisioning service will modify the group membership via the *PATCH /Groups* endpoint of UAA. To learn how, see Patch • **false** – the Identity Provisioning service will modify the group membership via the *POST /Groups* or *DELETE /Groups* endpoints of UAA. To learn how, see Add Member and Remove Member .

Property Name	Description & Value
`xsuaa.patch.response.with.resource`	Use this property if you want to retrieve a group whose membership was modified. **Note** This property is usable only when you have configured membership modifications via *Add/Remove Member* UAA endpoints. That is, when the `scim.support.patch.operation` property is set to **false**. Possible values: • **true** – the Identity Provisioning service will return the modified group via the *GET /Groups* endpoint of UAA. To learn how, see Retrieve . • **false** – no modified groups will be returned by the service.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*https://api.authentication.eu10.hana.ondemand.com*

`OAuth2TokenServiceURL`=*https://myaccount.authentication.eu10.hana.ondemand.com/oauth/token*

`User`=*sb-full-access!a76125*

`Password`=**********

`xsuaa.origin`=*sap.custom*

`xsuaa.origin.filter.enabled`=*true*

`scim.support.patch.operation`=*true*

`xsuaa.patch.response.with.resource`=*false*

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP BTP XS Advanced UAA* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP BTP XS Advanced UAA system. For more information, see Manage Transformations [page 2439] and the API documentation.

- User Management (System for Cross-domain Identity Management (SCIM))- SCIM groups - role collections
- User Management (System for Cross-domain Identity Management (SCIM)) - SCIM users - shadow users

Default read and write transformations:

> → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
```

Write Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "constant": "xsuaa-dummy-
value",
        "targetPath": "$.id",
        "scope": "createEntity"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {

"sourcePath":"$.name.givenName",

"targetPath":"$.name.givenName",
        "optional":true
      },
      {

"sourcePath":"$.name.familyName",

"targetPath":"$.name.familyName",
        "optional":true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value == []",
        "targetPath":
"$.emails[0].primary",
        "constant": true
      },
      {
        "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.phoneNumbers",
        "optional": true
      },
      {
        "sourcePath":
"$.externalId",
```

Read Transformation

```
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.origin",
        "targetPath": "$.origin",
        "optional": true
      },
      {
        "sourcePath": "$.zoneId",
        "targetPath": "$.zoneId",
        "optional": true
      },
      {
        "sourcePath":
"$.verified",
        "targetPath":
"$.verified",
        "optional": true
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
        "targetPath":
"$.schemas[0]"
      }
    ],
    "scimEntityEndpoint": "Users"
```

Write Transformation

```
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath":
"$.verified",
        "targetPath":
"$.verified",
        "optional": true
      },
      {
        "constant":
"%xsuaa.origin%",
        "targetPath": "$.origin"
      },
      {
        "constant":
"urn:scim:schemas:core:1.0",
        "targetPath":
"$.schemas[0]"
      }
    ],
    "scimEntityEndpoint": "Users"
  },
  "group": {
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.description",
        "targetPath":
"$.description",
        "optional": true
      },
      {
        "constant":
"urn:scim:schemas:core:1.0",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
```

Read Transformation

```
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "function":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.description",
        "targetPath":
"$.description",
        "optional": true
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.zoneId",
        "targetPath": "$.zoneId"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:Group",
```

Write Transformation

```
      },
      {
        "constant": "USER",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.members[*].type",
        "optional": true
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ],
    "scimEntityEndpoint": "Groups"
  }
}
```

Read Transformation	Write Transformation

<pre>"urn:ietf:params:scim:schemas:extension:sap:2.0:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "constant": "authorization",
        "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {
        "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
        "optional": true,
        "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']"
      }
    ],
    "scimEntityEndpoint": "Groups"
  }
}</pre>	

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation

is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

SAP Authorization and Trust Management Service
btp CLI Command Reference
SAP BTP Integration Scenario

1.6.3.15.1 Configure Single and Multiple Origins

Configure and provision users with single or multiple origins in SAP BTP XS Advanced UAA (Cloud Foundry) proxy system.

Overview

An origin tells you which is the identity provider of a user in SAP BTP XS Advanced UAA (Cloud Foundry). It is defined in the trust configuration in the SAP BTP cockpit under *Security* > *Trust Configuration* > *Origin Key*.

The origin itself is not a concept of Identity Provisioning. The role of the service is to ensure that you can read and provision users with their identity providers. Once you find the identity provider in the Origin Key, you need to set it in the `xsuaa.origin` property. You can configure it in source, target and proxy SAP BTP XS Advanced UAA (Cloud Foundry) systems. Both single and multiple values are supported. The value is a string that usually specifies the name of the identity provider or its location.

For example: `xsuaa.origin=`*ldap* and `xsuaa.origin=` *ldap;myaccount-xsuaa.accounts.ondemand.com*, where the ";" (semicolon) is the only supported delimiter.

Provisioning Users with Single Origin

You want to configure SAP BTP XS Advanced UAA (Cloud Foundry) as a proxy system for provisioning users with a single origin to and from an external identity management system.

1. On the *Properties* tab of the proxy system, set the `xsuaa.origin.filter.enabled` property to *true*.
   This is a prerequisite for enabling the `xsuaa.origin` property in proxy systems.
2. Enter the value for the `xsuaa.origin` property, for example: *idp1*.
   The value of this property acts like a filter.

Expect the following results:

- When executing a GET request, only users with the origin specified in the `xsuaa.origin` property are returned.
- When executing a POST request to create a user, you make one request to the proxy system. The payload contains the *idp1* origin.
- When executing a PUT request to update a user, you make one request to the proxy system. The payload contains the *idp1* origin.

Provisioning Users with Multiple Origins

You want to configure SAP BTP XS Advanced UAA (Cloud Foundry) as a proxy system for provisioning users with multiple origins to and from an external identity management system.

1. On the *Properties* tab of the proxy system, set the `xsuaa.origin.filter.enabled` property to *true*.
   This is a prerequisite for enabling the `xsuaa.origin` property in proxy systems.
2. Enter multiple values for the `xsuaa.origin` property, for example: *idp1;idp2*.
   The value of this property acts like a filter.
3. On the *Transformations* tab, update the SAP BTP XS Advanced UAA (Cloud Foundry) proxy write transformation. Replace the constant and its value with the sourcePath expression pointing to a multivalue origin attribute:

JSON Text Editor	Graphical Editor
Default mapping: Code Syntax `{ "constant":"%xsuaa.origin%", "targetPath":"$.origin" },`	Default mapping:
Expected mapping: Code Syntax `{ "sourcePath":"$.origin", "targetPath":"$.origin" },`	Expected mapping:

4.
   - When executing a GET request, only users with the origins specified in the `xsuaa.origin` property are returned.
   - When executing a POST request to create a user with two origins (*idp1;idp2*), you should make two separate requests to the proxy system with two separate payloads - the first one containing the *idp1* origin and the second one containing *idp2* origin.
     As a result, two entries for one and the same user are created. Both entries contain the same user attributes (for example, username and email address) and differ only in their origins.
   - When executing a PUT request to update a user with two origins (*idp1;idp2*), you should make two separate requests to the proxy system with two separate payloads - the first one containing the *idp1* origin and the second one containing *idp2* origin.
     As a result, two entries for one and the same user are updated.

> ### Note
>
> Multiple origins are not supported in provisioning scenarios between SAP BTP XS Advanced UAA (Cloud Foundry) source system and SAP BTP XS Advanced UAA (Cloud Foundry) target system.

1.6.3.16 SAP Build Work Zone, advanced edition

Follow this procedure to set up SAP Build Work Zone, advanced edition as a proxy system.

Prerequisites

- You have OAuth credentials for SAP Build Work Zone, advanced edition. To learn how, see SAP Build Work Zone, advanced edition: Add an OAuth Client

Context

After fulfilling the prerequisites, follow the procedure below to create a proxy SAP Build Work Zone, advanced edition system to load its users into an on-premise system and provision groups and new users back to SAP Build Work Zone, advanced edition.

> ### Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Build Work Zone, advanced edition* as a proxy system. For more information, see Add New Systems [page 2415].
4. Set up the communication between Identity Provisioning and SAP Build Work Zone, advanced edition and configure the authentication method (basic or OAuth certificate-based).

   > ### Note
   >
   > We recommend that you use OAuth certificate-based authentication in the regions where it is supported by SAP Build Work Zone, advanced edition. This authentication method is supported for subaccounts created on the recommended data centers for new customers according to the SAP Build Work Zone, advanced edition documentation.

   a. In your newly added SAP Build Work Zone, advanced edition proxy system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection [page 2452].

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in the SAP Build Work Zone, advanced edition Admin Console and are relevant for certificate-based authentication only.

   b. Login to the SAP Build Work Zone, advanced edition Admin Console and execute substep *2. f. iii. Use X.509 Certificate-Based Authentication* from *Step 3: Configure the Identity Authentication service and the Identity Provisioning service* in Option 1: Create a new tenant .
5. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
Type	Enter: *HTTP*

Property Name	Description & Value
`URL`	Enter the URL related to your SAP Build Work Zone, advanced edition system, in format: **`https://<account><sap_wz_domain>.workzone.ondemand.com`** For example: *https://mytenant.mydomain123.workzone.ondemand.com*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the OAuth client key, created for your SAP Build Work Zone, advanced edition tenant (see **Prerequisites**).
`Password`	Valid if *BasicAuthentication* is configured as authentication method. (Credential) Enter the OAuth client secret, created for your SAP Build Work Zone, advanced edition tenant (see **Prerequisites**).
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Build Work Zone, advanced edition instance, based on the configured authentication method: • *BasicAuthentication* - **`https://<account><sap_wz_domain>.workzone.ondemand.com/api/v1/auth/token`** For example: *https://myaccountmydomain123.workzone.ondemand.com/api/v1/auth/token* • *ClientCertificateAuthentication* - **`https://<company_uuid>.api-<region_host>.dws.workzone.ondemand.com/api/v2/auth/token`** **Note** The `api-` prefix before `<region_host>` is necessary for using *ClientCertificateAuthentication* authentication. For example: *https://mycompanyuuid123.api-eu-20.dws.workzone.ondemand.com/api/v2/auth/token*

Property Name	Description & Value
`client_id`	Valid if *ClientCertificateAuthentication* is configured as authentication method. Enter the OAuth Client id used for access token retrieval.
Optional Properties	
(Optional) `workzone.content.type`	This property makes the SAP Build Work Zone, advanced edition proxy system to send the specified value for the *Content-Type* HTTP header. Example: **application/json** Default value (when not specified): *application/scim+json*
(Optional) `workzone.support.patch.operation`	The default value of this property is **false**. But for SAP Build Work Zone, advanced edition proxy systems, this property appears during creation and its predefined value is *true*. That means, when the Identity Provisioning identifies a changed entity in the back-end system, it will execute the updates as PATCH requests instead of PUT. That means, only the changes will be written in SAP Build Work Zone, advanced edition, instead of provisioning the whole entity data.
(Optional) `workzone.include.if.match.wildcard.header`	This property makes the SAP Build Work Zone, advanced edition target system to send the *If-Match* HTTP header with a value of "*" for every request to SAP Build Work Zone, advanced edition. This header could be used for entity versioning. Default value (when not specified): *false*
(Optional) `workzone.group.filter`	When specified, only those SAP Build Work Zone, advanced edition groups matching the filter expression will be read. For example: *displayName eq "ProjectTeam1"*
(Optional) `ips.failed.request.retry.attempts`	Predefined value: *2*
(Optional) `ips.failed.request.retry.attempts.interval`	Predefined value: *30*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

6. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default

transformation for the SAP Build Work Zone, advanced edition proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Build Work Zone, advanced edition. For more information, see:

Manage Transformations [page 2439]

SAP Build Work Zone SCIM API

Default read and write transformations:

> → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.emails[0].value",
                "targetPath":
"$.emails[0].value",
                "optional": true
            },
            {
                "sourcePath":
"$.emails[?(@.primary==
true)].value",
                "optional": true,

"correlationAttribute": true
            },
            {

"sourcePath":"$.name.givenName",

"targetPath":"$.name.givenName",
              "optional":true
            },
```

Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id",
                "scope":
"deleteEntity"
            },
// When a user is supposed to be
deleted from SAP Build Work Zone,
advanced edition, it actually has
its status set to inactive instead
of being deleted.
            {
                "constant": false,
                "targetPath":
"$.active",
                "scope":
"deleteEntity"
            },
            {
                "sourcePath":
"$.Operations",
                "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.schemas",
                "targetPath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
                "targetPath":
"$.schemas[0]"
            },
```

Read Transformation

```
            {

"sourcePath":"$.name.familyName",

"targetPath":"$.name.familyName",
              "optional":true
            },
            {
                "sourcePath":
"$.userType",
                "optional": true,
                "targetPath":
"$.userType"
            },
            {
                "sourcePath":
"$.displayName",
                "optional": true,
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.active",
                "optional": true,
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.title",
                "optional": true,
                "targetPath":
"$.title"
            },
            {
                "sourcePath":
"$.locale",
                "optional": true,
                "targetPath":
"$.locale",
                "functions": [
                    {
                        "type":
"substring",

"beginIndex": 0,

"endIndex": 2
                    }
                ]
            },
            {
                "sourcePath":
"$.timezone",
                "optional": true,
                "targetPath":
"$.timezone"
            },
            {
                "sourcePath":
"$.addresses",
```

Write Transformation

```
            {
                "constant":
"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User",
                "targetPath":
"$.schemas[1]"
            },
            {
                 "sourcePath":
"$.userType",
                 "optional":true,
                 "targetPath":
"$.userType"
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas",
                "optional": true
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "targetPath":
"$.userName",
                "optional": true
            },
            {

"sourcePath":"$.name.givenName",

"targetPath":"$.name.givenName",
              "optional":true
            },
            {

"sourcePath":"$.name.familyName",

"targetPath":"$.name.familyName",
              "optional":true
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName",
                "optional": true
            },
            {
                "sourcePath":
"$.emails",
```

Read Transformation	Write Transformation

```
"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.addresses"
            },
            {
                "sourcePath":
"$.groups",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups"
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

```
                "targetPath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "optional": true
            },
            {
                "condition":
"$.emails[0].length() > 0",
                "constant": true,
                "targetPath":
"$.emails[0].primary"
            },
            {
                "targetPath":
"$.locale",
                "type": "remove"
            },
            {
                "condition":
"($.locale EMPTY false)
&& ($.addresses[?(@.type ==
'work')].country EMPTY false)",
                "sourcePath":
"$.locale",
                "targetPath":
"$.locale",
                "functions": [
                    {

"function": "toLowerCaseString"
                    },
                    {

"function": "concatString",
                        "suffix":
"_"
                    },
                    {

"function": "concatString",
                        "suffix":
"$.addresses[?(@.type ==
'work')].country"
                    }
                ]
            },
            {
                "sourcePath":
"$.timezone",
                "optional": true,
                "targetPath":
"$.timezone"
            },
            {
                "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
                "optional": true,
```

Read Transformation

```
ension:enterprise:2.0:User']
['division']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "optional": true
```

Write Transformation

```
                "targetPath":
"$.addresses"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
```

Read Transformation

```
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.members",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members"
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                "optional": true,
```

Write Transformation

```
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
                "optional": true
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.Operations",
                "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.schemas",
                "targetPath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
                "targetPath":
"$.schemas[0]"
            },
            {
                "sourcePath":
"$.schemas",
```

Read Transformation	Write Transformation

<pre>"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
},
{
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
"optional": true,
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']"
}
]
}
}</pre>
	<pre>"preserveArrayWithSingleElement": true,
	"targetPath": "$.schemas",
	"optional": true
	},
	{
	"sourcePath": "$.displayName",
	"targetPath": "$.displayName"
	},
	{
	"sourcePath": "$.members",
	"preserveArrayWithSingleElement": true,
	"targetPath": "$.members",
	"optional": true
	}
	]
	}
	}</pre>

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

1.6.3.17 SAP Build Work Zone, standard edition

Follow this procedure to set up SAP Build Work Zone, standard edition as a proxy system.

Prerequisites

- You have created an instance and generated a service key for the standard service plan of SAP Build Work Zone, standard edition. For more information, see: Initial Setup.
  The service key contains the API URL and the OAuth credentials (`clientid` and `clientsecret`) under the `uaa` property.

Context

The SAP Build Work Zone, standard edition

You can use the Identity Provisioning UI to configure SAP Build Work Zone, standard edition simplifies access to SAP, custom-built, and third party applications and extensions (both on the cloud and on premise) by establishing a central launchpad. as a proxy system and configure it in hybrid scenarios. For example, when SAP Build Work Zone, standard edition is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management and SAP Cloud Identity Access Governance, without making a direct connection between both systems. You can provision users, groups and users' group assignments to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users, groups and users' group assignments back to the SAP Build Work Zone, standard edition.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.

3. Add *SAP Build Work Zone, standard edition* simplifies access to SAP, as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the SAP Build Work Zone, standard edition API URL from the service key of your SAP Build Work Zone, standard edition instance under `endpoints [portal-service]`. It follows the pattern: `https://portal-service.cfapps.sap.hana.ondemand.com`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, from the service key of your SAP Build Work Zone, standard edition instance under `uaa.clientid`.
`Password`	(Credential) Enter the OAuth Client Secret, from the service key of your SAP Build Work Zone, standard edition instance under `uaa.clientsecret`.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL from the service key of your SAP Build Work Zone, standard edition instance. It follows the pattern: `<uaa.url>/oauth/token`. For example: `https://ips-cflp-woaealle.authentication.sap.hana.ondemand.com/oauth/token`

Property Name	Value
(Optional) `cflp.providerId`	Enter a valid providerID value. The provider ID is specified in the Channel Manager of the SAP Build Work Zone, standard edition when defining a new content provider. For more information about configuring the content provider to use the Identity Provisioning service, see: Manage Content Providers (Cloud), Manage Content Providers (On Premise). **Note** All users and groups are provisioned to the proxy SAP Build Work Zone, standard edition system with the providerID defined for this proxy system. If you want to use different providerIDs, you need to create separate proxy systems for every providerID.
(Optional) `cflp.user.filter`	When specified, only those SAP Build Work Zone, standard edition users matching the filter expression will be read. By default, users are always filtered by the providerId. If another filtering attribute is defined, for example the email of the user, both filters are combined. Possible values: • **`emails.value eq "john.smith@example.com"`** **Note** Although, the email is supported as a filtering attribute, it is not returned when searching for the user. • **`urn:ietf:params:scim:schemas:extension:2.0:mapping.providerId eq "ABC123"`** Combining filter attributes is allowed using the and operator, for example: **`emails.value eq "john.smith@example.com" and urn:ietf:params:scim:schemas:extension:2.0:mapping.providerId eq "ABC123"`**

Property Name	Value
(Optional) `cflp.group.filter`	When specified, only those SAP Build Work Zone, standard edition groups matching the filter expression will be read. By default, groups are always filtered by the providerId. Possible values: • **`externalId eq 12345678`** • **`urn:ietf:params:scim:schemas:extension:2.0:mapping.providerId eq 'ABC123'`** • **`meta.isIAG eq true`** This filtering attribute indicates whether the group will be used in a hybrid scenario with SAP Cloud Identity Access Governance.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> **Note**
>
> The Identity Provisioning implementation of the SCIM Proxy API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy SCIM API.

5. Configure the transformations.

   Transformations are used to map user and group attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Build Work Zone, standard edition* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Build Work Zone, standard edition system. For more information, see: Manage Transformations [page 2439].

   ### User Update and Uniqueness

   The proxy write transformation of SAP Build Work Zone, standard edition supports three user unique attributes: *externalId*, *email* and *providerId*. Only providerId is mandatory in combination with at least one of the other attributes. Based on the combinations of these attributes, there can be distinguished three types of users:

   - User 1 - with *email* and *providerId*, but no *externalId*
   - User 2 - with *externalId* and *providerId*, but no *email*
   - User 3 - with all three attributes: *externalId*, *email* and *providerId*

   The uniqueness of the user is ensured by the combination of two unique attributes: the *providerId* and *externalId* or the *providerId* and the *email*.

Updating an existing user in the proxy system depends on how the user is identified there (its combination of unique attributes) and whether Identity Provisioning is aware of this user:

- The existing user was created by a provisioning job (that is, the service is aware of this user).
  If the following attributes are changed in the source system: the *email* of user 1, the *externalId* of user 2, or the *email* or *externalId* of user 3, after running a new job, Identity Provisioning will update the existing user.
- The existing user was not created by a provisioning job or was created by one but afterwards the source and the proxy systems have been reset (that is, the service is not aware of this user).
  If the following attributes are changed in the source system, expect the subsequent results after running a new job:
  - Changing the *email* of user 1 will result in creating a new user in the proxy system, because the user won't match the combination of two unique attributes: *email* and *providerId*.
  - Changing the *externalId* of user 2 will result in creating a new user in the proxy system, because the user won't match the combination of two unique attributes: *externalId* and *providerId*.
  - Changing the *email* or the *externalId* of user 3 will result in updating the user in the proxy system, because the user will still match a combination of two unique attributes. However, changing both, the *email* and the *externalId*, of user 3 will result in creating a new user in the proxy system, because the user will match only one unique attribute - the *providerId*.

Default read and write transformations:

Read Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem",

"correlationAttribute": true
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.emails.value",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.emails.value"
            },
            {
                "sourcePath":
"$.externalId",
                "optional": true,
                "targetPath":
"$.externalId"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.Operations",
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas",
                "scope":
"patchEntity"
            },
            {
                "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:2.0:mapping"
                ],
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:2.0:mapping']
['providerId']",
                "optional": true,
```

Read Transformation

```
ension:2.0:mapping']
['providerId']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:2.0:mapping']
['providerId']"
            },
            {
                "sourcePath":
"$.groups",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups"
            }
        ],
        "scimEntityEndpoint":
"Users"
    },
    "group": {
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:2.0:mapping']
['providerId']",
```

Write Transformation

```
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:2.0:mapping']
['providerId']"
            },
            {
                "condition": "($
['urn:ietf:params:scim:schemas:ext
ension:2.0:mapping']
['providerId'] EMPTY true)",
                "constant":
"%cflp.providerId%",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:2.0:mapping']
['providerId']"
            },
            {
                "sourcePath":
"$.externalId",
                "optional": true,
                "targetPath":
"$.externalId"
            },
            {
                "sourcePath":
"$.groups[*].value",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups[?(@.value)]"
            }
        ],
        "scimEntityEndpoint":
"Users"
    },
    "group": {
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.Operations",
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.schemas",
```

Read Transformation

```
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:2.0:mapping']
['providerId']"
            },
            {
                "sourcePath":
"$.meta.isIAG",
                "targetPath":
"$.meta.isIAG"
            },
            {
                "condition": "
'%cflp.providerId%' != 'null' ",
                "sourcePath":
"$.externalId",
                "targetPath":
"$.externalId"
            },
            {
                "condition": "
'%cflp.providerId%' == 'null' ",
                "sourcePath":
"$.externalId",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.members",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members"
            }
        ],
        "scimEntityEndpoint":
"Groups"
    }
}
```

Write Transformation

```
"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas",
                "scope":
"patchEntity"
            },
            {
                "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:Group",

"urn:ietf:params:scim:schemas:exte
nsion:2.0:mapping"
                ],
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.meta.isIAG",
                "optional": true,
                "targetPath":
"$.meta.isIAG"
            },
            {
                "constant":
"%cflp.providerId%",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:2.0:mapping']
['providerId']"
            },
            {
                "sourcePath":
"$.externalId",
                "optional": true,
                "targetPath":
"$.externalId"
            },
            {
                "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members[?(@.value)]"
                }
        ],
        "scimEntityEndpoint":
"Groups"
    }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

- For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
- For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

> ⓘ Note
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

Configure Integration with the Identity Provisioning Service

1.6.3.18 SAP Business Network

Follow this procedure to set up SAP Business Network as a proxy system.

Context

> **Note**
>
> Currently, the SAP Business Network connector cannot be used by customers due to API access constraints. SAP is working on resolving these constraints to make the connector fully operational and accessible.

SAP Business Network, formerly known as Ariba Network, is a cloud-based offering that makes it possible for buyers and suppliers to collaborate on transactions, strengthen their relationships, and discover new business opportunities.

You can use Identity Provisioning to configure SAP Business Network as a proxy system in hybrid scenarios. For example, when SAP Business Network is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users and groups to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users and group members back to the SAP Business Network.

> **Note**
>
> Identity Provisioning cannot create and delete groups in SAP Business Network target system. It can only update existing groups by adding or removing group members. Therefore, groups must have been created in SAP Business Network.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.

4. Save your changes.

3. Add *SAP Business Network* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the SAP Business Network API URL.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, created for your SAP Business Network system.
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP Business Network system.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL.
`bn.api.key`	An API Key represents the unique key that identifies a particular application as a legitimate consumer of an API.
`bn.realm.id`	The realm name is part of the URL you use to access SAP Business Network.
(Optional) `bn.user.filter`	When specified, only those SAP Business Network users matching the filter expression will be read. For example: • *userName eq "Julie Armstrong"* • *userName sw "J"* • *emails eq "julie.armstrong@example.com"* For more information, see List of Properties [page 80]
(Optional) `bn.group.filter`	When specified, only those SAP Business Network groups matching the filter expression will be read. For example: *displayName eq "Employees"* For more information, see List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> ### Note
>
> The Identity Provisioning implementation of the SCIM Proxy API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy SCIM API.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Business Network* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Business Network system. For more information, see:

   Manage Transformations [page 2439].

   Default read and write transformations:

   > ### → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['sendMail']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['sendMail']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.business.network:2.0:Us
er']['userStatus']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.business.network:2.0:Us
er']['userStatus']",
        "optional": true
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "correlationAttribute":
true,
        "optional": true
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "constant": [

"urn:com.sap.ariba.framework.scim2.co
mmon.types.UserResourceV2",

"urn:ietf:params:scim:schemas:extensi
on:sap.business.network:2.0:User",

"urn:ietf:params:scim:schemas:core:2.
0:User",

"urn:ietf:params:scim:schemas:extensi
on:sap:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extens
ion:sap:2.0:User']['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extens
ion:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extens
ion:sap.business.network:2.0:User']
['userStatus']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extens
ion:sap.business.network:2.0:User']
['userStatus']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extens
ion:sap:2.0:User']['sendMail']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:extens
ion:sap:2.0:User']['sendMail']",
        "optional": true
      },
      {
        "targetPath": "$.id",
        "sourceVariable":
"entityIdTargetSystem"
      },
      {
        "sourcePath": "$.userName",
        "targetPath": "$.userName"
      },
      {
```

Read Transformation

```
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.name.formatted",
        "targetPath":
"$.name.formattedName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificSuffix",
        "targetPath":
"$.name.honorificSuffix",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
```

Write Transformation

```
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificSuffix",
        "targetPath":
"$.name.honorificSuffix",
        "optional": true
      },
      {
        "sourcePath": "$.externalId",
        "targetPath": "$.externalId",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true,
        "defaultValue": true
      },
      {
        "sourcePath": "$.timezone",
        "targetPath": "$.timezone",
        "optional": true
      },
      {
        "sourcePath":
"$.preferredLanguage",
```

Read Transformation

```
      {
        "sourcePath":
"$.emails[0].value",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.preferredLanguage",
        "targetPath":
"$.preferredLanguage",
        "optional": true
      },
      {
        "sourcePath":
"$.addresses",
        "targetPath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.timezone",
        "targetPath":
"$.timezone",
        "optional": true
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
```

Write Transformation

```
        "targetPath":
"$.preferredLanguage",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "condition": "$.addresses[?
(@.type == 'work')].country EMPTY
false",
        "sourcePath": "$.addresses[?
(@.type == 'work')]",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.addresses[0]"
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.Operations",
        "targetPath": "$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint": "Groups",
    "mappings": [
      {
        "targetPath": "$.id",
        "sourceVariable":
"entityIdTargetSystem"
      },
      {
        "sourcePath": "$.Operations",
        "targetPath": "$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
```

Read Transformation

```
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      }
```

Write Transformation

```
        "scope": "patchEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath": "$.displayName"
      },
      {
        "sourcePath": "$.members",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.members",
        "optional": true
      }
    ]
  }
}
```

Read Transformation	Write Transformation

<pre>]
}
}</pre>	

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ⓘ Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

SAP Business Network

1.6.3.19 SAP Central Business Configuration

Follow this procedure to set up SAP Central Business Configuration (in short, CBC) as a proxy system.

Prerequisites

- You have created a technical user with administrator permissions that will be used to call the API of SAP Central Business Configuration for reading, creating and updating user and group member information.

Context

Create an SAP Central Business Configuration proxy connector to execute hybrid scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to the CBC system, whenever the external back-end requests such. This scenario supports provisioning **users** and **group members**.

> ⚠ Caution
>
> You can't create or delete groups on CBC. That means:
>
> - On the attempt to create a group on CBC, Identity Provisioning will only add new members or update existing ones. Also, if you read a group from the external back-end system, there must be a group with the exact same display name (case sensitive) in the CBC system. Otherwise, an error will be thrown and the group members will not be updated.
> - On the attempt to delete a group on CBC, Identity Provisioning will only remove its members (group assignments). And this can happen only if the relevant group assignments have been provisioned/are present in the target system.

> ⓘ Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Central Business Configuration* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the API of your CBC system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Specify the technical user for your CBC system.
`Password`	(Credential) Specify the password for your technical user.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. For example: **https://mycbcaccount.authentication.us2.hana.ondemand.com/oauth/token**
(Optional) `cbc.user.filter`	When specified, only those CBC users matching the filter expression will be read. Example: **name.familyName eq "Smith" and addresses.country eq "US"**

Property Name	Value
(Optional) `cbc.group.filter`	When specified, only those CBC groups matching the filter expression will be read. Example: **displayName eq "ProjectTeam1" or "Employees2020"**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Central Business Configuration* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your CBC system. For more information, see Manage Transformations [page 2439].

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourcePath":
"$.schema",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schema"
            },
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
```

Write Transformation

Code Syntax

```
{
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
                "targetPath":
"$.schema[0]"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members[?(@.value)]"
            }
        ]
    },
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
                "targetPath":
"$.schemas[0]"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User",
                "targetPath":
"$.schemas[1]"
            },
```

Read Transformation

```
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            },
            {
                "sourcePath":
"$.members",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.members"
            }
        ]
    },
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
```

Write Transformation

```
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.name.givenName",
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath":
"$.active",
                "targetPath":
"$.active"
            }
        ]
    }
}
```

Read Transformation	Write Transformation

<pre>{
"sourcePath":
"$.name.givenName",
"optional": true,
"targetPath":
"$.name.givenName"
},
{
"sourcePath":
"$.name.familyName",
"optional": true,
"targetPath":
"$.name.familyName"
},
{
"sourcePath":
"$.active",
"targetPath":
"$.active"
}
]
}
}</pre>	

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

SAP Central Business Configuration – Collection

1.6.3.20 SAP Commerce Cloud

Follow this procedure to set up SAP Commerce Cloud as a proxy system.

Prerequisites

- In SAP Commerce Cloud, you have added an OAuth client with authorization grant **Client Credentials**. To learn how, see: Configuring OAuth Client.

Context

SAP Commerce Cloud is a highly flexible and modular platform for delivering modern customer experiences. It provides a standardized, automated, end-to-end solution that allows your projects to release code from repository to cloud.

You can use Identity Provisioning to configure SAP Commerce Cloud as a proxy system in hybrid scenarios. For example, when SAP Commerce Cloud is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users and groups to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users and group back to the SAP Commerce Cloud.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Commerce Cloud* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP Commerce Cloud system. The URL follows the pattern: `https://backoffice.<tenant>. model-t.cc.commerce.ondemand.com` You can find the correct URL in the *Environments* section of SAP Cloud Portal.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the client ID to retrieve the OAuth access token for SAP Commerce Cloud.

Property Name	Description & Value
`Password`	(Credential) Enter the client secret to retrieve the OAuth access token for SAP Commerce Cloud.
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Commerce Cloud instance. This token URL is listed in the *OAuth Clients* section of the *App Integration* page.
(Optional) `cc.user.filter`	When specified, only those users matching the filter expression will be read. You can filter users by **`userName`**, **`emails.value`**, and **`externalId`**, according to the API syntax of SAP Commerce Cloud. For example: • *userName eq "johnbrown" and externalId eq "P000252"* • *userName eq "johnbrown" and emails.value eq "johnbrown@email.com"* • *userName eq "johnbrown" and emails.value eq "johnbrown@email.com" and externalId eq "P000252"* **Note** These combinations are valid for both 'or' and 'and' operators.
(Optional) `cc.group.filter`	When specified, only those groups matching the filter expression will be read. For example: *displayName eq "ProjectTeam1" or "Students2018"*
(Optional) `cc.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. Default value for proxy systems: *true*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Commerce Cloud* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Commerce Cloud. For more information, see:

Manage Transformations [page 2439]

Commerce Cloud SCIM Web Services API Documentation

Default read and write transformations:

> → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",
                "optional": true,

"correlationAttribute": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath":
"$.externalId",
                "optional": true,
```

Write Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath":
"$.externalId",
                "optional": true,
                "targetPath":
"$.externalId"
            },
            {
                "sourcePath":
"$.displayName",
                "optional": true,
                "targetPath":
"$.displayName"
            },
                    {
                "sourcePath":
"$.nickName",
                "optional": true,
                "targetPath":
"$.nickName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional": true,
```

Read Transformation

```
                "targetPath":
"$.externalId"
            },
            {
                "sourcePath":
"$.displayName",
                "optional": true,
                "targetPath":
"$.displayName"
             },
             {
                "sourcePath":
"$.nickName",
                "optional": true,
                "targetPath":
"$.nickName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional": true,
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional": true,
                "targetPath":
"$.name.middleName"
            },
            {
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.honorificSuffix",
                "optional": true,
                "targetPath":
"$.name.honorificSuffix"
            },
            {
                "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.addresses"
            },
            {
                "sourcePath":
"$.userType",
                "optional": true,
                "targetPath":
"$.userType"
            },
```

Write Transformation

```
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional": true,
                "targetPath":
"$.name.middleName"
            },
            {
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.honorificSuffix",
                "optional": true,
                "targetPath":
"$.name.honorificSuffix"
            },
            {
                "condition":
"($.addresses[*].region EMPTY
false) && ($.addresses[*].country
EMPTY false)",
                "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.addresses",
                "functions": [
                    {
                        "type":
"convertCountryRegion",

"outputFormat": "alpha2",

"inputAttributes": [

"region",

"country"
                        ],

"outputAttribute": "region"
                    }
                ]
            },
            {
                "sourcePath":
"$.userType",
                "optional": true,
                "targetPath":
"$.userType"
            },
            {
```

Read Transformation

```
            {
                "sourcePath":
"$.active",
                "optional": true,
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath":
"$.groups",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups"
            }
        ],
        "scimEntityEndpoint":
"Users"
    },
    "group": {
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.schemas",
```

Write Transformation

```
                "sourcePath":
"$.active",
                "optional": true,
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath":
"$.groups[*].value",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups[?(@.value)]"
            },
            {

"constant":"urn:ietf:params:scim:a
pi:messages:2.0:PatchOp",

"targetPath":"$.schemas[0]",
            "scope":"patchEntity"
            },
            {
                "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.Operations",
                "scope":
"patchEntity"
            }
        ],
        "scimEntityEndpoint":
"Users"
    },
    "group": {
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.displayName",
                "optional": true,
                "targetPath":
"$.displayName"
```

Read Transformation	Write Transformation

<pre>"preserveArrayWithSingleElement": true,
"targetPath": "$.schemas"
},
{
"constant": "authorization",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
},
{
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
"optional": true,
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']"
},
{
"sourcePath": "$.displayName",
"targetPath": "$.displayName"
},
{
"sourcePath": "$.members",
"preserveArrayWithSingleElement": true,
"optional": true,
"targetPath": "$.members"
}
],
"scimEntityEndpoint": "Groups"
}
}</pre>
	<pre>},
	{
	"constant": "urn:ietf:params:scim:schemas:core:2.0:User",
	"targetPath": "$.schemas[0]"
	},
	{
	"sourcePath": "$.Operations",
	"preserveArrayWithSingleElement": true,
	"targetPath": "$.Operations",
	"scope": "patchEntity"
	},
	{
	"sourcePath": "$.schemas",
	"preserveArrayWithSingleElement": true,
	"targetPath": "$.schemas",
	"scope": "patchEntity"
	},
	{
	"sourcePath": "$.members[*].value",
	"preserveArrayWithSingleElement": true,
	"optional": true,
	"targetPath": "$.members[?(@.value)]"
	}
	],
	"scimEntityEndpoint": "Groups"
	}
	}</pre>

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

> ⓘ Note
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

7. Run an initial load job.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

SAP Cloud Identity Services Integration Architecture → cloudscimwebservices extension

1.6.3.21 SAP Concur

Follow this procedure to set up SAP Concur as a proxy system.

Prerequisites

- You are using User v1 API, Identity v4 API or SCIM v4 API.

  > **Note**
  >
  > The **User v1 API** has been decommissioned. For more information, see Concur User API Decommission.

- Your SAP Concur admin user has obtained a *Company Request Token* and a *Company UUID* from the SAP Concur Company Request Token self-service tool.
  For more information, see SAP Concur Company Request Token.

Context

Companies that use SAP Concur for managing and controlling travel expenses, invoices and other can use Identity Provisioning service to automate the identity and access management for the SAP Concur solution. This service allows you to use SAP Concur as a proxy connector to execute *hybrid* scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to SAP Concur, whenever the external back-end requests such. This scenario supports reading and writing of **users**.

The SAP Concur integration with Identity Provisioning is supported with Standard and Professional editions.

SAP Concur provides different APIs for integration with Identity Provisioning, resulting in different connector versions. Each connector version supports specific attribute mappings within the transformations and requires particular property values. The value of the `concur.api.version` property controls which API you use.

- When the value is set to **3**, the SCIM v4 API is used. It supports reading and provisioning of users and groups.
  By default, the Identity Provisioning service uses version 3 for all newly created systems starting December 2, 2025, when version 3 was introduced. The default version is the latest released connector version at the time the system is created.
- When the value is set to **2**, the Identity v4 API is used. This applies to systems created between December 8, 2021, and December 1, 2025. Identity v4 API supports provisioning of users with the `userUUID` attribute, which is generated by Identity Authentication at user creation.
- When the value is set to **1**, or the property is not defined, the User v1 API is used. This applies to systems created before versioning was introduced on December 8, 2021. Version 1 is deprecated.

For more information on how to update to the connector version you need, see: Update Connector Version [page 2424].

To create SAP Concur as a proxy system, proceed as follows:

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Concur* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Version 1 Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	For EU2 data center, enter: **`eu2.concursolutions.com`** For US2 data center, enter: **`www.concursolutions.com`**
`ProxyType`	Enter: **`Internet`**
`concur.api.version`	Defines the version of SAP Concur API. Set the value to *1* to use User v4 API.
`Authentication`	Enter: *BasicAuthentication*
`User`	Deprecated property. Do not use.
`Password`	Deprecated property. Do not use.
`X-ConsumerKey`	Deprecated property. Do not use.

Property Name	Description & Value
`concur.datacenter`	Specify the SAP Concur data center your Identity Provisioning tenant belongs to. The following SAP Concur data centers are available: • EU2 • US2 Based on the provided data center, Identity Provisioning configures the URL of the User v1 API or Identity v4 API as follows: • If you provide **`us2`**, the URL is: `us.api.concursolutions.com`
`concur.authorization.code`	(Credential) Enter the *Company Request Token* and run a provisioning job within 24 hours from generating the token in the SAP Concur Company Request Token self-service tool. Otherwise, the token will expire, and you'll need a new one. After the first run of the job, Identity Provisioning fills in automatically a refresh token as the value of the concur.refresh.token property. If a provisioning job has not been run for six months, you'll again need to generate a new token. → Remember The company request token has a 24 hour validity. If this token expires, you must request a new token. The refresh token has a six month validity. Every time you run a provisioning job, the validity of the refresh token is extended with six months starting from the date of the last run. If you haven't run a provisioning job for six months, your refresh token will expire and you must request a new company request token.
`concur.company.id`	Enter the *Company UUID* as described in the *Prerequisites* section.
`concur.company.domain`	Enter your company domain. The username and the company domain are concatenated in the default transformation in the following format: user@domain Your company domain is the part of your username behind the @ symbol. For example: johnsmith@example.com

Version 2 Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`ProxyType`	Enter: **`Internet`**
`Authentication`	Enter: *BasicAuthentication*
`concur.datacenter`	Specify the SAP Concur data center your Identity Provisioning tenant belongs to. The following SAP Concur data centers are available: • APJ1 • EU2 • US2 Based on the provided data center, Identity Provisioning configures the URL of the User v1 API or Identity v4 API as follows: • If you provide **`us2`**, the URL is: `us.api.concursolutions.com`
`concur.api.version`	Defines the version of SAP Concur API. Set the value to *2* to use the Identity v4 API. This was the default value for systems created between December 8, 2021, and December 1, 2025.
`concur.authorization.code`	(Credential) Enter the *Company Request Token* and run a provisioning job within 24 hours from generating the token in the SAP Concur Company Request Token self-service tool. Otherwise, the token will expire, and you'll need a new one. After the first run of the job, Identity Provisioning fills in automatically a refresh token as the value of the concur.refresh.token property. If a provisioning job has not been run for six months, you'll again need to generate a new token. → Remember The company request token has a 24 hour validity. If this token expires, you must request a new token. The refresh token has a six month validity. Every time you run a provisioning job, the validity of the refresh token is extended with six months starting from the date of the last run. If you haven't run a provisioning job for six months, your refresh token will expire and you must request a new company request token.

Property Name	Description & Value
`concur.company.id`	Enter the *Company UUID* as described in the *Prerequisites* section.
`concur.company.domain`	Enter your company domain. The username and the company domain are concatenated in the default transformation in the following format: user@domain Your company domain is the part of your username behind the @ symbol. For example: johnsmith@example.com
(Optional) `concur.user.filter`	When specified, only those users matching the filter expression will be read. For example: • *userName* eq "johnsmith@example.com" As the userName must be unique in SAP Concur, this filter returns only the user matching this userName. • *companyId* eq "aa067ada-71a9-4f57-8e98-9300b1c3171d" This filter returns all users in the company with this companyId.

Connector Version 3 Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`ProxyType`	Enter: **`Internet`**
`Authentication`	Enter: *BasicAuthentication*
`concur.datacenter`	Specify the SAP Concur data center your Identity Provisioning tenant belongs to. The following SAP Concur data centers are available: • APJ1 • EU2 • US2 Based on the provided data center, Identity Provisioning configures the URL of the User v1 API or Identity v4 API as follows: • If you provide **`us2`**, the URL is: `us.api.concursolutions.com`

Property Name	Description & Value
`concur.api.version`	Defines the version of SAP Concur API. Set the value to *3* to use the SCIM v4 API. By default, the Identity Provisioning service uses version 3 for all newly created systems starting December 2, 2025, when version 3 was introduced.
`concur.authorization.code`	(Credential) Enter the *Company Request Token* and run a provisioning job within 24 hours from generating the token in the SAP Concur Company Request Token self-service tool. Otherwise, the token will expire, and you'll need a new one. After the first run of the job, Identity Provisioning fills in automatically a refresh token as the value of the concur.refresh.token property. If a provisioning job has not been run for six months, you'll again need to generate a new token. → Remember The company request token has a 24 hour validity. If this token expires, you must request a new token. The refresh token has a six month validity. Every time you run a provisioning job, the validity of the refresh token is extended with six months starting from the date of the last run. If you haven't run a provisioning job for six months, your refresh token will expire and you must request a new company request token.
`concur.company.id`	Enter the *Company UUID* as described in the *Prerequisites* section.
`concur.company.domain`	Enter your company domain. The username and the company domain are concatenated in the default transformation in the following format: user@domain Your company domain is the part of your username behind the @ symbol. For example: johnsmith@example.com

Property Name	Description & Value
(Optional) `concur.user.filter`	When specified, only those users matching the filter expression will be read. For example: • *userName* eq "JohnSmith" • *userName* sw "John" • *urn:ietf:params:scim:schemas:extension:enterprise:2.0:employeeNumber* eq "00102345"
(Optional) `concur.group.filter`	When specified, only those groups matching the filter expression will be read. For example: *displayName* eq "ProjectTeam1"
(Optional)`concur.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. Default value for target systems: *false*
(Optional) `concur.group.members.page.size`	The number of group members that SAP Concur returns per request when reading a group. Default value: 1000
(Optional) `concur.group.members.paging.enabled`	This property enables paging of group members. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *true*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users. For more information, see Query Parameters for Proxy System SCIM API.

5. Configure the transformations.

Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Concur* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Concur. For more information, see:

Manage Transformations [page 2439]

SAP Business Accelerator Hub: Concur (Users)

User v1 API: User v1

Identity v4 API: Identity v4

SCIM v4 API: SCIM v4 API

→ Tip

The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Default read and write transformation when using User v1 API (connector version 1):

Read Transformation	Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourcePath":
"$.EmployeeID",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem",
                "functions": [
                    {
                        "type":
"compositeId",
                        "subId":
"$.LoginID"
                    }
                ]
            },
            {
                "sourcePath":
"$.EmployeeID",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
                "targetPath":
"$.schemas[0]"
            },
            {
                "sourcePath":
"$.PrimaryEmail",
                "targetPath":
"$.emails[0].value",
```

Code Syntax

```
{
    "user": {

"scimEntityEndpoint":"Users",
        "mappings": [
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.EmpId"
            },
            {
                "sourcePath":
"$.emails[0].value",
                "targetPath":
"$.EmailAddress"
            },
            {
                "sourcePath":
"$.emails[0].value",
                "targetPath":
"$.LoginId"
            },
            {
                "sourcePath":
"$.name.givenName",
                "targetPath":
"$.FirstName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "targetPath":
"$.LastName"
            },
            {
                "constant": "N",
                "targetPath":
"$.Active"
            },
            {
                "condition":
"$.active == true",
                "constant": "Y",
                "targetPath":
"$.Active"
            },
            {
                "constant": "N",
                "targetPath":
"$.ExpenseApprover"
            },
            {
                "constant": "N",
                "targetPath":
"$.ExpenseUser"
            },
```

Read Transformation

```
"correlationAttribute": true
            },
            {
                "targetPath":
"$.emails[0].primary",
                "constant": true
            },
            {
                "sourcePath":
"$.FirstName",
                "optional": true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.LastName",
                "optional": true,
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath":
"$.CellPhoneNumber",
                "optional": true,
                "targetPath":
"$.phoneNumbers[0].value"
            },
            {
                "sourcePath":
"$.LoginID",

"correlationAttribute": true
            }
        ]
    }
}
```

Write Transformation

```
            {
                "constant": "N",
                "targetPath":
"$.InvoiceApprover"
            },
            {
                "constant": "N",
                "targetPath":
"$.InvoiceUser"
            },
            {
                "constant": "N",
                "targetPath":
"$.IsTestEmp"
            },
            {
                "constant": "N",
                "targetPath":
"$.TripUser"
            },
            {
                "ignore": true,
                "constant": "",
                "targetPath":
"$.Password"
            },
            {
                "constant": "USD",
                "targetPath":
"$.CrnKey"
            },
            {
                "constant": "US",
                "targetPath":
"$.CtryCode"
            },
            {
                "sourcePath":
"$.locale",
                "optional": true,
                "targetPath":
"$.CtryCode",
                "functions": [
                    {
                        "type":
"substring",

"beginIndex": 3
                    }
                ]
            },
            {
                "constant":
"en_US",
                "targetPath":
"$.LocaleName"
            },
            {
                "constant": "US",
                "targetPath":
"$.Custom21"
            },
```

Read Transformation	Write Transformation

	<pre>{
	"constant": "DEFAULT",
	"targetPath": "$.LedgerName"
	},
	{
	"constant": "DEFAULT",
	"targetPath": "$.LedgerCode"
	}
	]
	}
	}</pre>

⚠ Caution

The User v1 API requires an initial password setup for all newly provisioned user accounts. The default transformation offers a statement with an empty string as a value for the password configuration. However, it is ignored in order to prevent from a default setup of a wrong initial password for your systems. While the password statement is ignored, the provisioning will not be working. To enable the provisioning to SAP Concur, you need to perform the following operations:

1. Enable the password statement. To do this, either delete **"ignore": true,** or set it as **"ignore": false**.
2. Set a proper statement for the password attribute value (**"targetPath": "$.Password"**).

(Optional) You can leave the default empty string, or you can use the **randomPassword** function to calculate a random value for the initial password of the newly created SAP Concur accounts. If you choose one of these two options and if you are not using single sign-on solution for SAP Concur, you have to also arrange a password reset support process in your company. This will securely offer an initial password to your corporate users for their newly created SAP Concur accounts. For more information, see Transformation Expressions [page 391] → **Transformation Functions**.

Default read and write transformation when using Identity v4 API (connector version 2):

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails"
      },
      {
        "type": "remove",
        "targetPath":
"$.emails[*].notifications"
      },
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant":
["urn:ietf:params:scim:schemas:cor
e:2.0:User","urn:sap:cloud:scim:sc
hemas:extension:custom:2.0:User","
urn:ietf:params:scim:schemas:exten
sion:sap:2.0:User",
"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User"],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "functions": [
          {
            "type":
"concatString",
            "suffix":
"@%concur.company.domain%"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath":
"$.externalId"
      },
```

Read Transformation

```
      {
        "sourcePath": "$.name",
        "targetPath": "$.name"
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.addresses"
      },
      {
        "sourcePath": "$.title",
        "optional": true,
        "targetPath": "$.title"
      },
      {
        "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.phoneNumbers"
      },
      {
        "sourcePath":
"$.emergencyContacts",
```

Write Transformation

```
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails",
        "functions": [
          {
            "function":
"putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "targetPath":
"$.emails[*].primary",
        "type": "remove"
      },
      {
        "sourcePath": "$.name",
        "targetPath": "$.name"
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.addresses"
      },
      {
        "targetPath":
"$.addresses[*].primary",
        "type": "remove"
```

Read Transformation

```
        "targetPath":
"$.emergencyContacts",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['companyId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['companyId']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']"
      },
      {
```

Write Transformation

```
      },
      {
        "sourcePath": "$.title",
        "optional": true,
        "targetPath": "$.title"
      },
      {
        "sourcePath":
"$.phoneNumbers[?(@.type !=
'mobile' || (@.type == 'mobile'
&& @.primary == false))]",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.phoneNumbers[0]"
      },
      {
        "targetPath":
"$.phoneNumbers[*].primary",
        "type": "remove"
      },
      {
        "sourcePath":
"$.phoneNumbers[?(@.type ==
'mobile' && @.primary == true)]",
        "optional": true,
        "targetPath":
"$.phoneNumbers[0]"
      },
      {
        "sourcePath":
"$.emergencyContacts",
        "targetPath":
"$.emergencyContacts",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "constant":
"%concur.company.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['companyId']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation	Write Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']"
      }
    ]
  }
}
```

```
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "functions": [
          {
            "function":
"resolveEntityIds"
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation	Write Transformation

	<pre>ension:enterprise:2.0:User']
	['manager']['displayName']",
	"optional": true,
	"targetPath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:enterprise:2.0:User']
	['manager']['displayName']"
	}
	]
	}
	}</pre>

Default read and write transformation when using SCIM v4 API (connector version 3):

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.name",
        "targetPath": "$.name"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "functions": [
          {
            "type":
"concatString",
            "suffix":
"@%concur.company.domain%"
          }
        ]
      },
      {
        "sourcePath": "$.name",
        "targetPath": "$.name"
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.addresses",
```

Read Transformation	Write Transformation

```
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "type": "remove",
        "targetPath":
"$.emails[*].notifications"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.addresses"
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.phoneNumbers"
      },
      {
        "sourcePath":
"$.emergencyContacts",
        "targetPath":
"$.emergencyContacts",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.preferredLanguage",
        "optional": true,
        "targetPath":
"$.preferredLanguage"
      },
      {
        "sourcePath": "$.title",
        "optional": true,
        "targetPath": "$.title"
```

```
"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.addresses",
        "functions": [
          {
            "type":
"convertCountryCode",
            "outputFormat":
"alpha2",
            "inputAttributes": [
              "country"
            ],
            "outputAttribute":
"country"
          }
        ]
      },
      {
        "targetPath":
"$.addresses[*].primary",
        "type": "remove"
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath":
"$.timezone",
        "functions": [
          {
            "type":
"convertTimezoneCode",
            "outputFormat": "iana"
          }
        ]
      },
      {
        "sourcePath":
"$.emergencyContacts",
        "targetPath":
"$.emergencyContacts",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "functions": [
          {
            "type":
"convertCountryCode",
            "outputFormat":
"alpha2",
            "inputAttributes": [
              "country"
            ],
            "outputAttribute":
"country"
          }
        ]
      },
      {
```

Read Transformation

```
      },
      {
        "sourcePath":
"$.nickName",
        "optional": true,
        "targetPath": "$.nickName"
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['contactPreferences']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['contactPreferences']"
      },
      {
```

Write Transformation

```
        "sourcePath":
"$.preferredLanguage",
        "optional": true,
        "targetPath":
"$.preferredLanguage"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$.title",
        "optional": true,
        "targetPath": "$.title"
      },
      {
        "sourcePath":
"$.nickName",
        "optional": true,
        "targetPath": "$.nickName"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.emails",
        "functions": [
          {
            "function":
"putIfAbsent",
            "key": "type",
            "defaultValue": "work"
          }
        ]
      },
      {
        "targetPath":
"$.emails[*].primary",
        "type": "remove"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails']",
        "optional": true,

"preserveArrayWithSingleElement":
true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails']"
      },
      {
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails'][*]
['primary']",
        "type": "remove"
      },
      {
```

Read Transformation	Write Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['companyId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['companyId']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails'][*]
['type']",
        "type": "remove"
      },
      {
        "sourcePath":
"$.phoneNumbers[?((@.type !=
'mobile') && (@.type != 'work')
&& (@.type empty false))]",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.phoneNumbers"
      },
      {
        "targetPath":
"$.phoneNumbers[*].primary",
        "type": "remove"
      },
      {
        "targetPath":
"$.phoneNumbers[*].issuingCountry"
,
        "type": "remove"
      },
      {
        "targetPath":
"$.phoneNumbers[*].notifications",
        "type": "remove"
      },
      {
        "sourcePath":
"$.phoneNumbers[?((@.type ==
'mobile') || (@.type empty
true))]",
        "optional": true,

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.phoneNumbers[0]",
        "functions": [
          {
            "function":
"putIfAbsent",
            "key": "type",
            "defaultValue":
"mobile"
          }
        ]
      },
      {
        "targetPath":
"$.phoneNumbers[*].workExtension",
        "type": "remove"
      },
      {
```

Read Transformation

```
ension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']"
      },
      {
        "sourcePath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
```

Write Transformation

```
        "condition":
"$.phoneNumbers[?(@.type ==
'work')] empty false",
        "sourcePath":
"$.phoneNumbers[?(@.type ==
'work')]",
        "optional": true,

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.phoneNumbers[0]",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "condition":
"$.phoneNumbers[?(@.type ==
'work')] empty false",
        "targetPath":
"$.phoneNumbers[0].primary",
        "type": "remove"
      },
      {
        "condition":
"$.phoneNumbers[?(@.type ==
'work')] empty false",
        "targetPath":
"$.phoneNumbers[0].issuingCountry"
,
        "type": "remove"
      },
      {
        "condition":
"$.phoneNumbers[?(@.type ==
'work')] empty false",
        "targetPath":
"$.phoneNumbers[0].notifications",
        "type": "remove"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['phoneNumbers']",
        "optional": true,

"preserveArrayWithSingleElement":
true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['phoneNumbers']"
      },
      {
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation

```
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['externalName']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['externalName']"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "optional": true,

"preserveArrayWithSingleElement":
true
      }
    ]
  }
}
```

Write Transformation

```
ension:sap:2.0:User']
['phoneNumbers'][*]['primary']",
        "type": "remove"
      },
      {
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['phoneNumbers'][*]['type']",
        "type": "remove"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['contactPreferences']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['contactPreferences']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {
```

Read Transformation	Write Transformation

```
        "constant":
"%concur.company.id%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['companyId']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']"
      },
```

Read Transformation	Write Transformation

```
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ]
  },
  "group":  {
    "scimEntityEndpoint":
"Groups",
    "mappings":  [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:Group",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members[?
(@.value)]"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",
```

Read Transformation	Write Transformation

	<pre>"preserveArrayWithSingleElement": true,
	"scope": "patchEntity"
	},
	{
	"sourcePath": "$.schemas",
	"targetPath": "$.schemas",
	
	"preserveArrayWithSingleElement": true,
	"scope": "patchEntity"
	}
	]
	}
	}</pre>

The Identity Provisioning does not provide group resource mapping in the SAP Concur write transformation. Although you cannot provision groups to SAP Concur, you can group the users into organizational units by enhancing your target transformation. For more information, see: SAP Concur (Target System) [page 1085]

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ⓘ Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.22 SAP CPQ

Follow this procedure to set up SAP CPQ (Configure, Price, and Quote) as a proxy system.

Prerequisites

- You have created a technical user with administrator permissions that will be used to call the API of SAP CPQ for reading, creating and updating user and group information.
- Make sure all users from the external consumer system that have been not provisioned via Identity Provisioning and you would like later to be written in SAP CPQ, have an organization unit name. This unit name must correspond to an existing company system ID in SAP CPQ. The *organization unit* and the *company system ID* must be exactly the same. Users with empty (missing) organization units will not be provisioned, as well as users whose organization units don't match any of the SAP CPQ company system IDs for the relevant tenant.
- In order for a created user to be active in SAP CPQ, it should be assigned to an SAP CPQ target group, whose ID ends with suffix *–USERTYPE*. To learn more, see SAP CPQ: SCIM API → section **Mappings between SCIM API and SAP CPQ** → `groups`.

Context

SAP CPQ is a highly configurable system designated to help sales representatives to configure products, apply pricing and generate quotes. SAP CPQ is part of the SAP Sales Cloud portfolio.

Create an SAP CPQ proxy connector to execute hybrid scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to the SAP CPQ system, whenever the external back-end requests such. This scenario supports provisioning **users** and **groups**.

> ⚠ Caution
>
> You can't create or delete groups on SAP CPQ. That means:

- On the attempt to create a group on SAP CPQ, Identity Provisioning will only add new members or update existing ones. Also, if you read a group from the external back-end system, there must be a group with the exact same display name (case sensitive) in the SAP CPQ system. Otherwise, an error will be thrown and the group members will not be updated.
- On the attempt to delete a group on SAP CPQ, Identity Provisioning will only remove its members (group assignments). And this can happen only if the relevant group assignments have been provisioned/are present in the target system.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP CPQ* as a proxy system. For more information, see Add New Systems [page 2415].
4. Set up the communication between Identity Provisioning and SAP CPQ and configure the authentication method (basic or OAuth certificate-based).

   > **Note**
   >
   > We recommend that you use OAuth certificate-based authentication.

   a. In your newly added SAP CPQ proxy system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection [page 2452].

      Skip this step if you use basic authentication. The next steps are relevant for OAuth certificate-based authentication only. For more information, see Configure OAuth Certificate Authentication [page 2456].

   b. Convert your certificate file from `.crt` to `.cer`.

c. Login to SAP CPQ and navigate to *Security* > *Certificate Management* > *User Certificates* > *New Certificate* and upload the certificate to the technical user of SAP CPQ.

   Make sure the *Enable User Certificates* option is turned on. Additionally, the technical user must be set to *Active*.

5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`cpq.user.filter`	(Optional) When specified, only those SAP CPQ users matching the filter expression will be read. Example: **name.familyName eq "Smith" and addresses.country eq "US"**
`OAuth2TokenServiceURL`	Valid if *ClientCertificateAuthentication* is configured as authentication method. Enter the OAuth 2.0 Token Service URL. For example: `https://<cpq_url>/oauth2/token`
`Password`	Valid if *BasicAuthentication* is configured as authentication method. (Credential) Specify the password for your technical user.
`ProxyType`	Enter: *Internet*
`Type`	Enter: *HTTP*
`URL`	Enter the URL to the API of your SAP CPQ system. It is the same as your SAP CPQ tenant URL. It must contain the domain name. For example: **https://sample1234.mycpqdomain.com**
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the Username of your SAP CPQ technical user.

Property Name	Value
`cpq.group.members.page.size`	When the `cpq.group.members.paging.enabled` property is set to true, this setting defines the size of the paged members request. **Note** If you configure `cpq.group.members.page.size` to a number greater than the default value, the complete member retrieval is disabled. Default Value: **100** The number of group members that SAP CPQ returns per request when reading a group.
`cpq.group.members.paging.enabled`	This property enables the complete retrieval of group members through additional sequential requests. The maximum number of group members returned per request is 100 as defined in `cpq.group.members.page.size`. For more information, see `cpq.group.members.page.size`. Possible Values: • **true** - Paging is enabled. • **false** - Paging is disabled. Default Value: **true**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

6. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP CPQ* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP CPQ system. For more information, see:

   Manage Transformations [page 2439].

   SAP CPQ: SCIM API

   Default read and write transformations:

→ Tip

The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.id",
                "targetVariable":
"entityIdSourceSystem",
                "targetPath":
"$.id"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.name.givenName",
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "targetPath":
"$.name.familyName"
            },
            {
```

Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "targetPath":
"$.id",
                "sourceVariable":
"entityIdTargetSystem"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
                "targetPath":
"$.schemas[0]"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User",
                "targetPath":
"$.schemas[1]"
            },
            {
                "sourcePath":
"$.externalId",
                "optional": true,
                "targetPath":
"$.externalId"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.name.givenName",
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.emails"
            },
            {
```

Read Transformation

```
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']"
            },
            {
                "sourcePath":
"$.active",
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath":
"$.emails[?(@.primary ==
true)].value",
                "optional": true,

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.addresses"
            },
            {
                "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.phoneNumbers"
            },
            {
                "sourcePath":
"$.groups",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.groups"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
```

Write Transformation

```
                "sourcePath":
"$.active",
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.addresses"
            },
            {
                "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.phoneNumbers"
            },
            {
                "constant": [],
                "targetPath":
"$.groups",
                "scope":
"createEntity"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']"
            },
            {
                "constant": false,
                "targetPath": "$
['urn:sap:cpq:scim:schemas:extensi
on:custom:2.0:User']['IsSsoUser']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
```

Read Transformation

```
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.id",
                "targetVariable":
"entityIdSourceSystem",
                "targetPath":
"$.id"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.members",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.members"
            },
            {
                "constant":
"authorization",
```

Write Transformation

```
        "mappings": [
            {
                "targetPath":
"$.id",
                "sourceVariable":
"entityIdTargetSystem"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
                "targetPath":
"$.schemas[0]"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.Operations",
                "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.schemas",
                "targetPath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members[?(@.value)]"
            }
        ]
    }
}
```

Read Transformation	Write Transformation

<pre>"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
},
{
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
"optional": true
}
]
}
}</pre>	

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PATCH**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PATCH** method for modifying entities.

Next Steps

After the proxy connector is configured, a consumer application can start sending CRUD requests to it. That means, initially reading all (or a filtered set of) entities, and creating them in the SCIM 2.0 back-end of the consumer application. The proxy connector can continue read newly created or updated entities and provision them to the proxy system's back-end.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management → sections **Next Steps** and **Future Identity Lifecycle**.

REMEMBER! When creating a new user in the consumer application, you have to do the following mandatory settings:

- Enter *organization unit* for the user, because this is a mandatory attribute for SAP CPQ. An organization with that exact name must already exist in SAP CPQ, otherwise the new user will not be provisioned.

- In order the new user to be active, add a group privilege for it of type *–USERTYPE*. To learn more, see SAP CPQ: SCIM API → section **Mappings between SCIM API and SAP CPQ** → `groups`.

> ⚠ Caution
>
> Effective September 2020, Neo tenants from Shanghai (China) can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.23 SAP Data Custodian

Follow this procedure to set up SAP Data Custodian as a proxy system.

Prerequisites

- You have created an SAP Data Custodian tenant.
- You have created a user within your tenant with the required roles for your scenario.
- You have added your user to SAP Identity Service Management (SAP ISM).

Context

SAP Data Custodian is a robust Software as a Service (SaaS) solution that protects sensitive data stored in public, private, hybrid, and multicloud environments. This solution integrates with partnered public hyperscalers, SAP applications, and SAP managed clouds.

You can use Identity Provisioning to configure SAP Data Custodian as a proxy system to execute hybrid scenarios. For example, when SAP Data Custodian is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision entities to the external backend system, which can trigger CRUD (create, read, update, delete) operations back to the SAP Data Custodian. This scenario supports provisioning **users** and **user assignments to groups**.

SCIM API 2.0 does not support managing of group assignments via the SCIM user resource. The "groups" attribute of the user is read-only. This means that the user group assignments should be managed via the SCIM group resources using the "members" attribute (as it is defined by the SCIM standard).

> ### Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add SAP Data Custodian as a proxy system. For more information, see Add New Systems [page 2415].
4. Set up the communication between Identity Provisioning and SAP Data Custodian and configure the authentication method (basic or certificate-based).

   > ### Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP Data Custodian source system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip this step if you use basic authentication. The next steps are performed in SAP Data Custodian and are relevant for certificate-based authentication only.

   b. Login to SAP Data Custodian tenant and create a tenant operations technical user (TOTU), as described in Create a Tenant Operations Technical User in SAP Data Custodian.

   c. Upload the outbound certificate that you downloaded in **substep a.** to the TOTU, as described in Register the Outbound Certificate to Your Tenant Operations Technical User.
5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the SAP Data Custodian SCIM API excluding the path information. `https://<SCIM_API_URL>` For more information, see Identity Provisioning.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* Enter the OAuth client key, created for your SAP Data Custodian tenant. For more information, see Basic Authentication Prerequisites.
`Password`	Valid if *BasicAuthentication* (Credential) Enter the OAuth client secret, created for your SAP Data Custodian tenant. For more information, see Certificate Authentication Prerequisites.
`OAuth2TokenServiceURL`	Specify the URL of the access token provider service for your SAP Data Custodian instance: `https://<SCIM_API_URL>/kms/scim/v1/Oauth2/Token`
`dc.user.filter`	When specified, only those SAP Data Custodian users matching the filter expression will be read. You can filter users by *userName*, *displayName* or *externalId*. **Possible values:** For example: *userName eq "Smith.J"*

Property Name	Value
`dc.group.filter`	This property filters groups by display name or externalId. When specified, only those groups matching the filter expression will be read. **Possible values:** • *SAP_Data_Custodian_Auditor* • *SAP_Data_Custodian_Service_Admin* • *SAP_Data_Custodian_Key_Admin* • *SAP_Data_Custodian_Key_User* For example: *displayName eq "SAP_Data_Custodian_Auditor"*
`dc.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users in the target system. This means, for example, that if a user attribute is modified, only this change will be provisioned and applied in the target system. • If set to *false*, PUT operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. Users can be updated in the target system in various cases, such as: • In the source system, some user attributes are modified, or new attributes are added. • In the source system, a condition or a filter is set for users not to be read anymore. • A user is deleted from the source system. In the last two cases, it's possible to keep the entity in the target system – it will not be deleted but only disabled. To do this, use the `deleteEntity` scope in the transformation of your target or proxy system. See: Transformation Expressions [page 391] → **deleteEntity**. **Possible values:** • *true* • *false* Default value: *false*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

6. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the SAP Data Custodian proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   > ### Note
   >
   > When Identity Authentication is configured as a source system, the default transformation logic:
   >
   > - Provisions only groups and user group assignments if they are part of the following predefined group list:
   >   - *SAP_Data_Custodian_Auditor*
   >   - *SAP_Data_Custodian_Service_Admin*
   >   - *SAP_Data_Custodian_Key_Admin*
   >   - *SAP_Data_Custodian_Key_User*
   > - Skips some of the attributes from the identity records.
   > - Sets primary email for `userName` of the user.
   >
   > This way, the transformation logic ensures that the identity data, sent to the Identity Provisioning SCIM REST API, is consistent.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Data Custodian. For more information, see:

   Manage Transformations [page 2439]

   SAP Data Custodian SCIM 2.0 API

   Default read and write transformations:

   > ### → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation	Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourcePath":
"$.schemas",
                "targetPath":
"$.schemas",

"preserveArrayWithSingleElement":
true
            },
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName",
                "optional": true
            },
            {
              "sourcePath":
"$.displayName",
              "optional":true,
              "targetPath":
"$.displayName"
            },
            {
              "sourcePath":
"$.name.givenName",
              "optional":true,
              "targetPath":
"$.name.givenName"
            },
            {
              "sourcePath":
"$.name.familyName",
              "optional":true,
              "targetPath":
"$.name.familyName"
            },
            {
```

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant":
["urn:ietf:params:scim:schemas:cor
e:2.0:User","urn:ietf:params:scim:
schemas:extension:enterprise:2.0:U
ser","urn:ietf:params:scim:schemas
:extension:sap:2.0:User","urn:sap:
cloud:scim:schemas:extension:custo
m:2.0:User"],
                "targetPath":
"$.schemas"
            },
            {
                "condition":
"$.externalId EMPTY false",
                "sourcePath":
"$.externalId",
                "targetPath":
"$.externalId"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.displayName",
                "optional":true,
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.name.givenName",
                "optional":true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional":true,
                "targetPath":
"$.name.familyName"
            },
            {
```

Read Transformation

```
                "sourcePath":
"$.name.formatted",
                "optional":true,
                "targetPath":
"$.name.formatted"
            },
            {
                "sourcePath":
"$.name.honorificPrefix",
                "optional":true,
                "targetPath":
"$.name.honorificPrefix"
            },
            {
                "sourcePath":
"$.name.honorificSuffix",
                "optional":true,
                "targetPath":
"$.name.honorificSuffix"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional":true,
                "targetPath":
"$.name.middleName"
            },
            {
                    "sourcePath":
"$.externalId",
                    "optional": true,
                    "targetPath":
"$.externalId"
            },
            {
                    "sourcePath":
"$.active",
                    "targetPath":
"$.active"
            },
            {
                    "sourcePath":
"$.emails",
                    "targetPath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                    "optional": true
            },
            {
                    "sourcePath":
"$.groups",
                    "targetPath":
"$.groups",

"preserveArrayWithSingleElement":
true,
                    "optional": true
            },
            {
```

Write Transformation

```
                    "sourcePath":
"$.name.formatted",
                    "optional":true,
                    "targetPath":
"$.name.formatted"
            },
            {
                    "sourcePath":
"$.name.honorificPrefix",
                    "optional":true,
                    "targetPath":
"$.name.honorificPrefix"
            },
            {
                    "sourcePath":
"$.name.honorificSuffix",
                    "optional":true,
                    "targetPath":
"$.name.honorificSuffix"
            },
            {
                    "sourcePath":
"$.name.middleName",
                    "optional":true,
                    "targetPath":
"$.name.middleName"
            },
            {
                    "sourcePath":
"$.active",
                    "optional":true,
                    "targetPath":
"$.active"
            },
            {
                    "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                    "targetPath":
"$.emails"
            },
            {

"constant":"urn:ietf:params:scim:a
pi:messages:2.0:PatchOp",

"targetPath":"$.schemas[0]",

"scope":"patchEntity"
            },
            {
                    "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                    "targetPath":
"$.Operations",
                    "scope":
"patchEntity"
            }
```

Read Transformation

```
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                   {
                      "type":
"concatString",
                      "suffix": "$
{entityIdSourceSystem}"
                   }
                ]
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourcePath":
"$.schemas",
                "targetPath":
"$.schemas",

"preserveArrayWithSingleElement":
true
            },
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.members",
                "targetPath":
"$.members",
                "optional": true,

"preserveArrayWithSingleElement":
true
            },
            {
                "sourcePath":
"$.externalId",
                "targetPath":
"$.externalId",
                "optional": true
            },
            {
```

Write Transformation

```
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
                "targetPath":
"$.schemas[0]"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.members",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.members",
                "optional": true
            },
            {
                "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.Operations",
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.schemas",
                "targetPath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            }
        ]
    }
}
```

Read Transformation	Write Transformation

```
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                   {
                      "type":
"concatString",
                      "suffix": "$
{entityIdSourceSystem}"
                   }
                ]
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
                "optional": true
            }
        ]
    }
}
```

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PATCH**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PATCH** method for modifying entities.

Related Information

SAP Data Custodian

1.6.3.24 SAP Datasphere

Follow this procedure to set up SAP Datasphere as a proxy system.

Prerequisites

- In SAP Datasphere, you have enabled a custom SAML Identity Provider, for which *User Attribute* is set to **Custom SAML User Mapping**. To learn how, see: Enabling a Custom SAML Identity Provider
- In SAP Datasphere, you have added an OAuth client with authorization grant **Client Credentials**. To learn how, see: Create OAuth2.0 Clients to Authenticate Against SAP Datasphere

Context

SAP Datasphere is a solution that harmonizes data management, analytics, and intelligent planning, including capabilities of SAP Data Intelligence Cloud and SAP Analytics Cloud. The solution enables SAP customers to consolidate their SAP data and analytics needs into a single SAP solution.

You can use Identity Provisioning to configure SAP Datasphere as a proxy system in hybrid scenarios. For example, when SAP Datasphere is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users back to the SAP Datasphere.

> **Note**
>
> SAP Datasphere does not support groups.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Datasphere* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP Datasphere system without adding the path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the client ID to retrieve the OAuth access token for SAP Datasphere.
`Password`	(Credential) Enter the client secret to retrieve the OAuth access token for SAP Datasphere.
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Datasphere instance. This token URL is listed in the *Trusted Identity Providers* section of the *App Integration* page. For more information, see Create OAuth2.0 Clients to Authenticate Against SAP Datasphere
`ds.api.csrf.protection`	Specifies whether to fetch a CSRF token when sending requests to the system. This property is automatically added to the system, with default value: **enabled**

Property Name	Description & Value
(Optional) `ds.user.filter`	When specified, only those SAP Datasphere users matching the filter expression will be read For example: *userName eq "SmithJ"*
`ds.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, Identity Provisioning sends a **PATCH** request to the user or group resource in the target system. Only attributes without `"scope": "createEntity"` in the attribute mappings in the write transformation will be updated. For example, if the last name of a user is changed in the source system, the patch operation will update it in the target system and will leave unchanged other attributes with explicitly set "scope": "createEntity". • If set to *false*, **PUT** operations are used to update users in the target system. This means, for example, that if a user attribute is modified, all user attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** • *true* • *false* Default value: *false*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Datasphere* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Datasphere. For more information, see:

   Manage Transformations [page 2439]

   SAP Datasphere SCIM 2.0 API

Default read and write transformations:

> → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Default read and write transformations:

Read Transformation

Code Syntax

```
{
   "user": {
      "scimEntityEndpoint":
"Users",
      "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
```

Write Transformation

Code Syntax

```
{
  "user": {
      "scimEntityEndpoint":
"Users",
      "mappings": [
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:sap:params:scim:schemas:exten
sion:sac:2.0:user-custom-
parameters",

"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User"
      ],
        "targetPath": "$.schemas"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.externalId"
```

Read Transformation

```
        "sourcePath":
"$.preferredLanguage",
        "optional": true,
        "targetPath":
"$.preferredLanguage"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value"
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.roles",
        "targetPath": "$.roles",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']",
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']"
      },
      {
        "sourcePath": "$.photos",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.photos",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']",
        "optional": true
      },
      {
```

Write Transformation

```
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.preferredLanguage",
        "optional": true,
        "targetPath":
"$.preferredLanguage"
      },
      {
        "sourcePath": "$.photos",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.photos",
        "optional": true
      },
      {
        "sourcePath": "$.roles",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.roles"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails"
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value == []",
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value"
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value != []",
        "sourcePath": "$.emails[?
(@.primary == true)].value",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.emails[0].value",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
```

Read Transformation

```
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      }
    ]
  }
}
```

Write Transformation

```
        "condition":
"$.emails[0].length() > 0",
        "constant": true,
        "targetPath":
"$.emails[0].primary"
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']",
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']"
      },
      {
        "sourcePath":
"$.emails[0].value",
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']['idpUserId']"
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value != []",
        "sourcePath": "$.emails[?
(@.primary == true)].value",

"preserveArrayWithSingleElement":
false,
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']['idpUserId']"
      },
      {
        "sourcePath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']['idpUserId']",
        "optional": true,
        "targetPath": "$
['urn:sap:params:scim:schemas:exte
nsion:sac:2.0:user-custom-
parameters']['idpUserId']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']"
      },
```

Read Transformation	Write Transformation

	<pre>{
	"constant":
	"urn:ietf:params:scim:api:messages
	:2.0:PatchOp",
	"targetPath":
	"$.schemas[0]",
	"scope": "patchEntity"
	},
	{
	"sourcePath":
	"$.Operations",
	"preserveArrayWithSingleElement":
	true,
	"targetPath":
	"$.Operations",
	"scope": "patchEntity"
	}
	]
	}
	}</pre>

> ### Note
>
> Updating a user in SAP Datasphere depends on whether user attributes in SAP Datasphere are mapped to SAML attributes in your identity provider. If this is the case, the values of those attributes are populated by the identity provider and cannot be changed by the SAP Datasphere SCIM API or the UI. For more information, see Map SAML Attributes to Users and Create a User.

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

7. Run an initial load job.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is

configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.25 SAP Enable Now

Follow this procedure to set up SAP Enable Now as a proxy system.

Prerequisites

- You have created a technical user in SAP Enable Now with the necessary permissions for establishing a connection with Identity Provisioning.

Context

The SAP Enable Now solution is SAP's digital adoption platform. It can help you improve the adoption, productivity, and efficiency of your software by providing in-application help and e-learning content to users.

You can use Identity Provisioning to configure SAP Enable Now as a proxy system in hybrid scenarios. For example, when SAP Enable Now is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users back to the SAP Enable Now.

> ⓘ Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Enable Now* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP Enable Now system. It follows the pattern: • `https://<company-name>.enable-now.cloud.sap` • `https://<company-name>.sap-enable-now.cn` (relevant for China)
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the username of the SAP Enable Now technical user.
`Password`	(Credential) Enter the password of the SAP Enable Now technical user.
(Optional) `en.user.filter`	When specified, only those SAP Enable Now users matching the filter expression will be read For example: *userName eq "SmithJ"*
(Optional) `en.group.filter`	When specified, only those SAP Enable Now groups matching the filter expression will be read. For example: *userName eq "SmithJ"*

Property Name	Description & Value
`en.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, **PATCH** operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to *false*, **PUT** operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. Default value: *false*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Enable Now* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Enable Now. For more information, see:

   Manage Transformations [page 2439]

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external

application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Default read and write transformations:

Read Transformation	Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourcePath":
"$.meta",
                "optional": true,
                "targetPath":
"$.meta"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.name.givenName",
                "optional": true,
```

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User"
                ],
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.schemas",
                "targetPath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.Operations",
                "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "targetPath":
"$.id",
                "type": "remove",
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
```

Read Transformation

```
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional": true,
                "targetPath":
"$.name.middleName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional": true,
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath":
"$.active",
                "optional": true,
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.groups",
                "targetPath":
"$.groups",
                "optional": true,

"preserveArrayWithSingleElement":
true
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
```

Write Transformation

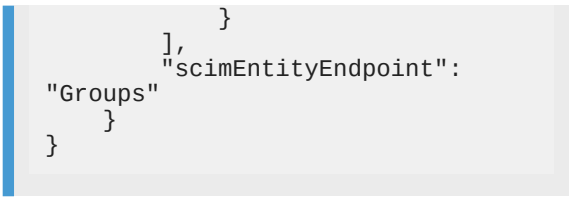
```
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional": true,
                "targetPath":
"$.name.middleName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional": true,
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath":
"$.active",
                "optional": true,
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
            }
        ],
        "scimEntityEndpoint":
"Users"
    },
    "group": {
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant": [
```

Read Transformation

```
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourcePath":
"$.meta",
                "optional": true,
                "targetPath":
"$.meta"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

```
"urn:ietf:params:scim:schemas:core
:2.0:Group",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:Group"
                ],
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.schemas",
                "targetPath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.Operations",
                "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "targetPath":
"$.id",
                "type": "remove",
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                "optional": true
            },
            {
                "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members[?(@.value)]"
```

Read Transformation	Write Transformation
	

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ⓘ Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

7. Run an initial load job.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

SAP Enable Now

1.6.3.26 SAP Fieldglass

Follow this procedure to set up a proxy connector for SAP Fieldglass.

Prerequisites

- You have created an API application key and a web service. To do that, follow the steps on page: Create API Application Key or Web Service and Web Services Setup
  You will need the values of *Virtual Person Name (Username)* and *License Key* for the configuration of your proxy system (**step 6** below).

Context

You can use SAP Fieldglass as a proxy connector to execute *hybrid* scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to SAP Fieldglass, whenever the external back-end requests such. This scenario supports:

- Reading of **users** and **groups**
- Writing of **users** and **assignments**

> ⓘ Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Fieldglass* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify your SAP Fieldglass environment URL. For example: *https://abc123.fgvms.com*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter your *Virtual Person Name (Username)* – see the **Prerequisites** section above.
`Password`	(Credential) Enter your *License Key* – see the **Prerequisites** section above.
`OAuth2TokenServiceURL`	Enter your OAuth token URL in the following format: `https://<Environment_URL>/api/oauth2/v2.0/token` For example: *https://abc123.fgvms.com/api/oauth2/v2.0/token*

Property Name	Value
`fg.support.patch.operation`	This property controls how modified users in the source system are updated in the target system. • If set to *true*, **PATCH** operations are used to update users in the target system. • If set to *false*, **PUT** operations are used to update users in the target system. Default value: *false* For more information, see: List of Properties [page 80]

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Fieldglass* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Fieldglass. For more information, see:

   Manage Transformations [page 2439]

   SAP Business Accelerator Hub: SAP Fieldglass

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.name.formatted",
        "optional": true,
        "targetPath":
"$.name.formatted"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "scope": "createEntity",
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.name.formatted",
        "optional": true,
        "targetPath":
"$.name.formatted"
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "condition":
"$.name.honorificPrefix IN
['Mr.', 'Mrs.', 'Ms.', 'Dr.']",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.title",
```

Read Transformation

```
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.title",
        "optional": true,
        "targetPath": "$.title"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale",
        "functions": [
          {
            "type": "substring",
            "beginIndex": 0,
            "endIndex": 2
          }
        ]
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value"
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.addresses"
      },
      {
        "sourcePath": "$.groups",
```

Write Transformation

```
        "targetPath": "$.title",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value"
      },
      {
        "optional": true,
        "defaultValue": "work",
        "sourcePath":
"$.emails[0].type",
        "targetPath":
"$.emails[0].type"
      },
      {
        "defaultValue": true,
        "optional": true,
        "sourcePath":
"$.emails[0].primary",
        "targetPath":
"$.emails[0].primary"
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.addresses"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
```

Read Transformation	Write Transformation

```
"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$
['resourceExtensions']
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true
      },
      {
        "sourcePath": "$
['resourceExtensions']
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true
      },
      {
        "sourcePath": "$
['resourceExtensions']
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath": "$
['resourceExtensions']
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true
      },
      {
```

```
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation

```
        "sourcePath": "$
['resourceExtensions']
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true
      },
      {
        "sourcePath": "$
['resourceExtensions']
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true
      },
      {
        "sourcePath": "$
['resourceExtensions']
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true
      },
      {
        "sourcePath": "$
['resourceExtensions']
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
```

Write Transformation

```
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "skipOperations": [
      "create",
      "delete"
    ],
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "optional": true,

"preserveArrayWithSingleElement":
true,
        "sourcePath": "$.members",
        "targetPath": "$.members"
      }
    ]
```

Read Transformation	Write Transformation

```
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members"
      },
      {
        "constant": "User",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.members[*].type",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
```

```
    }
  }
```

Read Transformation	Write Transformation

<pre>}
]
}
}</pre>	

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ⓘ Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.27 SAP Field Service Management

Follow this procedure to set up SAP Field Service Management as a proxy system.

Prerequisites

You have OAuth credentials for SAP Field Service Management. For more information, see: Generating Client ID & Secret

Context

SAP Field Service Management is a cloud-based solution that is used to resolve customers issues with end-to-end field service management. For example, it helps customers overcome resource limitations, such as having enough skilled technicians in all locations.

You can use the Identity Provisioning user interface (UI) to configure SAP Field Service Management as a proxy system in hybrid scenarios. For example, when SAP Field Service Management is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users and groups to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users and group members back to the SAP Field Service Management.

This scenario supports provisioning **users** and **group members**. When integrated with Identity Provisioning, SAP Field Service Management supports the following group concept:

- A user is created with a default group assigned. A group is mapped to a company (one of the key organizational units in SAP Field Service Management). When using the Identity Provisioning service API, the group is returned as follows:

  ```
  <GroupName>_<CompanyName>
  ```

- A user can have one group assignment for each company. As in SAP Field Service Management a user can be assigned to multiple companies, this requires a group assignment for each of the companies the user can access. A user cannot be assigned to different groups mapped to one and the same company.

> ⚠ Caution
>
> You cannot create or delete groups in SAP Field Service Management. In this case, you can expect the following behavior:
>
> - If a new group is created in the source system (for example, Identity Authentication) and you run the provisioning job to SAP Field Service Management, the job will fail and no group will be created in the target system.
> - If a group exists both in the source system (for example, Identity Authentication) and the target system - SAP Field Service Management, running a provisioning job will only add new members or update existing ones. In this case, groups in the source and target systems must have the same display name (case sensitive). Otherwise, the job will fail, and no group members will be updated.

- If a group exists in the target SAP Field Service Management system and you try to delete it, Identity Provisioning will only remove its group members. In this case, the relevant group members must exist in the target system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Field Service Management* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the API of your SAP Field Service Management system. It follows the pattern: `https://<cluster>.fsm.cloud.sap`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, created for your SAP Field Service Management system.
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP Field Service Management system.

Property Name	Value
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. For example: `https://<cluster>.fsm.cloud.sap/api/oauth2/v2/token`
(Optional) `fsm.user.filter`	When specified, only those SAP Field Service Management users matching the filter expression will be read. Example: **name.familyName eq "Smith" and addresses.country eq "US"**
(Optional) `fsm.group.filter`	When specified, only those SAP Field Service Management groups matching the filter expression will be read. Example: **displayName eq "ProjectTeam1" or "Employees2020"**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> ⓘ Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Field Service Management* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Field Service Management system. For more information, see:

   Manage Transformations [page 2439].

   SAP Field Service Management - SCIM API

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional": true,
                "targetPath":
"$.name.familyName"
            },
            {
```

Write Transformation

Code Syntax

```
{
    "user":  {
        "mappings":  [
            {

"sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User"
                ],
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.name.givenName",
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath":
"$.active",
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
 true,
                "optional":  true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation

```
                "sourcePath":
"$.active",
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath":
"$.groups",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
            }
        ],
        "scimEntityEndpoint":
"Users"
    },
    "group": {
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
```

Write Transformation

```
ension:sap:2.0:User']
['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
            }
        ],
        "scimEntityEndpoint":
"Users"
    },
    "group":  {
        "mappings":  [
            {

"sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.Operations",
                "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.schemas",
                "targetPath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "scope":
"patchEntity"
            },
            {
                "targetPath":
"$.id",
                "type":  "remove",
                "scope":
"patchEntity"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
                "targetPath":
"$.schemas[0]"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
```

Read Transformation	Write Transformation

<pre>}
]
},
{
"sourcePath":
"$.schemas",
 	
"preserveArrayWithSingleElement":
true,
"targetPath":
"$.schemas"
},
{
"sourcePath":
"$.displayName",
"targetPath":
"$.displayName"
},
{
"sourcePath":
"$.members",
 	
"preserveArrayWithSingleElement":
true,
"optional": true,
"targetPath":
"$.members"
},
{
"sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
"targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
"optional": true
},
{
"sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
"targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
"optional": true
}
],
"scimEntityEndpoint":
"Groups"
}
}</pre>
	<pre>"sourcePath":
	"$.members[*].value",
	
	"preserveArrayWithSingleElement":
	true,
	"optional": true,
	"targetPath":
	"$.members[?(@.value)]"
	}
	],
	"scimEntityEndpoint":
	"Groups"
	}
	}</pre>

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be

automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

- For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
- For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

> ⓘ Note
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.28 SAP HANA Cloud Database Services

Follow this procedure to set up SAP HANA Cloud Database Services as a proxy system.

Prerequisites

You have created a technical user for Identity Provisioning in the SAP HANA database with the OPERATOR and ROLE ADMIN system privileges. For more information, see SAP Identity Provisioning Service → *Prerequisites*.

Context

SAP HANA Cloud allows you to consume the SAP HANA database from cloud applications running on SAP Business Technology Platform, as well as from applications running elsewhere using the standard SAP HANA clients. Every instance of SAP HANA Cloud has its own single SAP HANA Database.

You can use the Identity Provisioning user interface (UI) to configure SAP HANA Cloud Database Services as a proxy system in hybrid scenarios. For example, when SAP HANA Cloud Database Services is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users and roles to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users and user role assignments back to the SAP HANA Cloud Database Services.

The SAP HANA Cloud Database Services connector supports provisioning of entities to the following two database types:

- SAP HANA Database (HDB)
- SAP HANA Data Lake Relational Engine (HDLRE / IQ Database)

You can define the target database type by setting the `hana.cloud.db.instance.type` property to either `hdb` or `hdlre`.

For more information on the supported database types, see SAP HANA Cloud, SAP HANA Database and SAP HANA Cloud, Data Lake.

> **Note**
>
> In SAP HANA Cloud Database Services, groups correspond to roles.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP HANA Cloud Database Services* as a proxy system. For more information, see Add New Systems [page 2415].

4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Authentication`	Enter: *BasicAuthentication*
`hana.cloud.db.instance.type`	Refers to the database type configured or used within SAP HANA Cloud. The value can be set to: • **hdb** - SAP HANA Database • **hdlre** - SAP HANA Data Lake Relational Engine Default value: **hdb**
`hana.cloud.db.instance.id`	Refers to a unique identifier associated with a specific database instance within the SAP HANA Cloud.
`hana.cloud.db.providerName`	Specifies the identity provider (IdP) that users managed through Identity Provisioning will use for authentication. The provider must be configured to support the authentication method specified in `hana.cloud.db.authenticationType`. For more information on how to obtain the value, refer to the create a JWT provider step in the User Provisioning with Identity Provisioning Service.
`hana.cloud.db.authenticationType`	Specifies the method by which users managed through Identity Provisioning authenticate when connecting to the database. The authentication method is set to *JWT* (users authenticate via JWT tokens issued by an identity provider) during system creation.
`hana.cloud.db.user.filter`	(Optional) When specified, only those SAP HANA Cloud Database Services users matching the filter expression will be read. Example: **name.familyName eq "Smith"**
`hana.cloud.db.group.filter`	(Optional) When specified, only those SAP HANA Cloud Database Services roles matching the filter expression will be read. Example: **displayName eq "ProjectTeam1" or "Employees2020"**

Property Name	Description & Value
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. For example: `https://demo-hc-3-test.authentication.sap.hana.ondemand.com/oauth/token` For more information on how to obtain the value, refer to the create a service binding step in the User Provisioning with Identity Provisioning Service.
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP HANA Database system. For more information on how to obtain the value for the technical user, refer to the procedure in the User Provisioning with Identity Provisioning Service.
`ProxyType`	Enter: *Internet*
`Type`	Enter: *HTTP*
`URL`	Enter the SAP HANA Cloud Database Services API URL. For example: `https://api.gateway.orchestration.<cluster>-<datacenter>.hanacloud.ondemand.com`
`User`	Enter the OAuth Client Id, created for your SAP HANA Database system. For more information on how to obtain the value for the technical user, refer to the procedure in the User Provisioning with Identity Provisioning Service.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP HANA Cloud Database Services* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP HANA Cloud Database Services system. For more information, see:

Manage Transformations [page 2439].

> ### Note
>
> Given that the SAP HANA Database username attribute is immutable and the email is not supported, refer to Handling Specific Attributes for your provisioning scenarios, particularly with Identity Authentication.

Default read and write transformations:

> ### → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "targetPath": "$.id",
        "sourceVariable":
"entityIdTargetSystem"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap.hana.db:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userId']",
        "optional": true
      },
      {
        "condition":
"'%hana.cloud.db.providerName%' !
== 'null'",
        "constant":
"%hana.cloud.db.providerName%",
```

Read Transformation

```
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userId']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userId']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.hana.db:2.0:User']
['authenticationDetails']
['providerName']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.hana.db:2.0:User']
['authenticationDetails']
['providerName']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.hana.db:2.0:User']
['userCategory']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.hana.db:2.0:User']
['userCategory']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.hana.db:2.0:User']
['authenticationDetails']
['authenticationType']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.hana.db:2.0:User']
['authenticationDetails']
['authenticationType']",
        "optional": true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",
        "optional": true,

"preserveArrayWithSingleElement":
true
      }
    ]
  },
  "group": {
```

Write Transformation

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.hana.db:2.0:User']
['authenticationDetails']
['providerName']"
      },
      {
        "condition":
"'%hana.cloud.db.authenticationTyp
e%' !== 'null'",
        "constant":
"%hana.cloud.db.authenticationType
%",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap.hana.db:2.0:User']
['authenticationDetails']
['authenticationType']"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "targetPath": "$.id",
        "sourceVariable":
"entityIdTargetSystem"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members[?
(@.value)]"
      }
    ]
  }
}
```

Read Transformation	Write Transformation

```
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
```

Read Transformation	Write Transformation

<pre>},
{
"sourcePath": "$.members",
"targetPath": "$.members",
"optional": true,
 	
"preserveArrayWithSingleElement":
true
}
]
}
}</pre>	

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ### Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.

For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*

To learn more, see: Proxy Systems [page 1343]

Related Information

User Provisioning

1.6.3.29 SAP Integrated Business Planning

Follow this procedure to set up SAP Integrated Business Planning (in short, SAP IBP) as a proxy system.

Prerequisites

- You have user credentials for an external back-end system with read and write permissions.
- To establish the connection between Identity Provisioning and SAP Integrated Business Planning, you need to set up the communication (user, system and arrangement) on SAP Integrated Business Planning. You can do it now (as a prerequisite) or in the process of configuring SAP Integrated Business Planning as a proxy system, as described in step 5.

Context

SAP Integrated Business Planning is a cloud-based solution that combines sales and operations planning (S&OP), forecasting and demand, response and supply, demand-driven replenishment, and inventory planning.

You can use Identity Provisioning to configure SAP IBP as a proxy system to execute *hybrid* scenarios. For example, when SAP IBP is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision entities to the external backend system, which can trigger CRUD (create, read, update, delete) operations back to SAP IBP.

SAP Integrated Business Planning provides different APIs for integration with Identity Provisioning, resulting in different connector versions for each API. Each connector version supports specific attribute mappings within the transformations and requires particular property values. By default, the Identity Provisioning service uses API version *1*. The value of the `ibp.api.version` property controls which API you use.

- When the value is set to *1* or the property is not present (typical for systems created before versioning was introduced on November 6, 2025), the SOAP API provided by the communication scenario SAP_COM_0193 is used. It supports reading of **business users** (employee) and **business roles** and

writing of **business users** and **role assignments**. In SAP Integrated Business Planning version 1, groups correspond to roles, thus group members are user assignments to a role.
For more information on how to update to version *2*, see Update SAP Integrated Business Planning.

- When the value is set to *2*, the SCIM interface provided by the communication scenario SAP_COM_0465 is used. It supports provisioning of **business users** with their **assignments** to user groups and roles.

> ### Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Integrated Business Planning* as a proxy system. For more information, see Add New Systems [page 2415].
4. Set up the communication between Identity Provisioning and SAP Integrated Business Planning and configure your authentication method (basic or certificate-based).

   > ### Note
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP Integrated Business Planning proxy system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP Integrated Business Planning backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

For basic authentication, provide *User Name* and *Password*.

For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

c. Create a communication system and assign the created user to the communication system.

For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

d. Create a communication arrangement with the created system.

- When `ibp.api.version=1` is set in Identity Provisioning for the SAP IBP proxy system, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

> **Note**
>
> The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
>
> The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

- When `ibp.api.version=2` is set in Identity Provisioning for the SAP IBP proxy system, choose *Scenario ID* SAP_COM_0465 (System for Cross-domain Identity Management Integration).

> **Note**
>
> The communication scenario *SAP_COM_0465* allows you to maintain data via SCIM (System for Cross-domain Identity Management). For more information, see Integrate Cross-Domain Identity Management.

5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to your SAP IBP system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*

Property Name	Description & Value
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **⚠ Restriction** Do not use special symbol '**,**' (comma) as it is not supported.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **⚠ Restriction** Do not use special symbol '**,**' (comma) as it is not supported.
`ibp.api.version`	The API version which is consumed by the SAP Integrated Business Planning system. By default, Identity Provisioning uses version *1*.
`ibp.skip.read.archived`	In the event of archived (disabled) entities in your SAP IBP system, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to **false**, or delete it. **Relevant for connector version 1**
`ips.date.variable.format`	*yyyy-MM-dd*

Property Name	Description & Value
`ibp.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP IBP proxy system in a hybrid scenario. • **true** – the current user roles will be deleted in the proxy system, and the user will be updated only with the roles provisioned by the service. • **false** – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the proxy system. See also: Extended Explanation of the *user.roles.overwrite Properties **Relevant for connector version 1**
(Optional)`ibp.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to **true**, PATCH operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to **false**, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. Default value: *false* **Relevant for connector version 2**
(Optional) `ibp.roles.filter`	Enter OData filtering for reading roles in the SAP IBP system. To learn what criteria you can use, see: OData URI Conventions → **4.5 Filter System Query Option** **Relevant for connector version 1**

Property Name	Description & Value
(Optional) `ibp.group.filter`	When specified, only those SAP Integrated Business Planning groups matching the filter expression will be read. Supported operators: eq (equal), co (contains), and sw (starts with). **Possible values::** • `displayName` • `externalId` • `urn:ietf:params:scim:schemas:extension:sap:2.0:Group:type` For example: • *displayName co "ProjectTeam1" or "Students2023"* • *externalId eq "12345678" or externalId sw "123"* • *urn:ietf:params:scim:schemas:extension:sap:2.0:Group:type eq "authorization" and externalId eq "12345678"* **Note** These combinations are valid for both 'or' and 'and' operators. **Relevant for connector version 2**
(Optional) `ibp.roles.page.size`	Indicate how many business roles (considered as *groups*) per page to be read from your SAP IBP system. The value must be an integer number. **Relevant for connector version 1**

Property Name	Description & Value
(Optional) `ibp.user.filter`	When specified, only those SAP Integrated Business Planning users matching the filter expression will be read. You can filter users by **list of attributes** according to the API syntax of SAP Integrated Business Planning. You can set a single attribute or multiple ones as filter criteria. If you enter multiple attributes (using OR operator), the filter will search for any of them. Supported operators: eq (equal), ne (not equal), sw (starts with), ew (ends with) and co (contains) **Possible values:** • `userName` • `externalId` • `emails[0].value` For example: • *userName eq "johnbrown" and externalId eq "P000252"* • *userName sw "J" or externalId eq "P000252"* • *userName eq "johnbrown" and emails[0].value eq "johnbrown@email.com"* • *userName eq "johnbrown" and emails[0].value ne "johnbrown@email.com" or externalId eq "P000252"* **Note** These combinations are valid for both 'or' and 'and' operators. **Relevant for connector version 2**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://my1234567-api.scmibp.ondemand.com
User=MyIBPuser
Password=************
ips.date.variable.format=yyyy-MM-dd
ibp.skip.read.archived=true
ibp.user.roles.overwrite = false
ibp.roles.filter=startswith(ID, 'EMPLOYEE_LEVEL_3') eq true
ibp.roles.page.size=30
```

6. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP IBP* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP IBP. For more information, see:

   Manage Transformations [page 2439]

   SAP Integrated Business Planning API: Business User

   SAP Business Accelerator Hub: SAP IBP

   Integrate Cross-Domain Identity Management

   SCIM Interface for IAM

   Default read and write transformations for connector version 1:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.personID",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.personalInformation.firstName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.lastName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.middleName"
,
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.personFullN
ame",
        "targetPath":
"$.name.formatted",
        "optional": true
      },
      {
```

Write Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.personExternalID"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath":
"$.personExternalID"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath":
"$.user.globalUserID"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.personID"
      },
      {
        "targetPath":
"$.businessPartnerRoleCode",
        "type": "valueMapping",
        "sourcePaths": [
          "$.userType"
        ],
        "defaultValue": "BUP003",
        "valueMappings": [
          {
            "key": [
              "Employee"
            ],
            "mappedValue":
"BUP003"
          }
        ]
      },
      {
        "scope": "createEntity",
        "sourceVariable":
"currentDate",
        "targetPath":
"$.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
```

Read Transformation

```
        "sourcePath":
"$.user.userName",
        "targetPath":
"$.userName",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "condition":
"$.user.lockedIndicator ==
'true'",
        "constant": false,
        "targetPath": "$.active",
        "optional": true
      },
      {
        "targetPath":
"$.schemas[0]",
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User"
      },
      {
        "targetPath":
"$.schemas[1]",
        "constant":
"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User"
      },
      {
        "condition":
"($.user.validityPeriod.startDate
> '${currentDate}') ||
('${currentDate}' >
$.user.validityPeriod.endDate)",
        "constant": false,
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.workplaceInformation.emailAddre
ss",
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.user.logonLanguageCode",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
```

Write Transformation

```
        "targetPath":
"$.validityPeriod.endDate"
      },
      {
        "scope": "createEntity",
        "sourceVariable":
"currentDate",
        "targetPath":
"$.user.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
        "targetPath":
"$.user.validityPeriod.endDate"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.personalInformation.firstName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.personalInformation.lastName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.personalInformation.middleName"
      },
      {
        "sourcePath":
"$.name.formatted",
        "optional": true,
        "targetPath":
"$.personalInformation.personFullN
ame"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.user.userName"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath":
"$.user.logonLanguageCode"
      },
      {
        "sourcePath":
"$.emails[0].value",
        "optional": true,
        "targetPath":
"$.workplaceInformation.emailAddre
ss"
```

Read Transformation

```
        "sourcePath":
"$.personExternalID",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.user.globalUserID",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath":
"$.user.role[*].roleName",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups[?
(@.value)]"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.user.timeZoneCode"
        ],
        "targetPath":
"$.timezone",
        "defaultValue": "Europe/
Berlin",
        "valueMappings": [
          {
            "key": [
              "WDFT"
            ],
            "mappedValue":
"Europe/Berlin"
          },
          {
            "key": [
              "ISRAEL"
            ],
            "mappedValue": "Asia/
Jerusalem"
          },
          {
            "key": [
              "RUS03"
            ],
            "mappedValue":
"Europe/Moscow"
          },
          {
            "key": [
              "AUSNSW"
            ],
            "mappedValue":
"Australia/Sydney"
          },
```

Write Transformation

```
      },
      {
        "condition": "$.active ==
false",
        "constant": "true",
        "targetPath":
"$.user.lockedIndicator"
      }
    ],
    "scimEntityEndpoint": "Users"
  },
  "group": {
    "mappings": [
      {
        "sourcePath":
"$.displayName",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members[?
(@.value)]"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ],
    "scimEntityEndpoint": "Groups"
  }
}
```

Read Transformation	Write Transformation

```
          {
            "key": [
              "UTC+4"
            ],
            "mappedValue": "Asia/
Dubai"
          },
          {
            "key": [
              "BRAZIL"
            ],
            "mappedValue":
"America/Sao_Paulo"
          },
          {
            "key": [
              "BRZLEA"
            ],
            "mappedValue":
"America/Sao_Paulo"
          },
          {
            "key": [
              "MSTNO"
            ],
            "mappedValue":
"America/Phoenix"
          },
          {
            "key": [
              "EST"
            ],
            "mappedValue":
"America/New_York"
          },
          {
            "key": [
              "UTC"
            ],
            "mappedValue": "Etc/
UTC"
          },
          {
            "key": [
              "UTC+3"
            ],
            "mappedValue": "Asia/
Riyadh"
          },
          {
            "key": [
              "EST_"
            ],
            "mappedValue":
"America/Toronto"
          },
          {
            "key": [
              "UTC+8"
            ],
            "mappedValue": "Asia/
Shanghai"
```

Read Transformation	Write Transformation

```
          },
          {
            "key": [
              "JAPAN"
            ],
            "mappedValue": "Asia/
Tokyo"
          }
        ]
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
"$.businessPartnerRoleCode"
        ],
        "targetPath":
"$.userType",
        "defaultValue":
"Employee",
        "valueMappings": [
          {
            "key": [
              "BUP003"
            ],
            "mappedValue":
"Employee"
          }
        ]
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.ID",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.ID",
        "targetPath":
"$.displayName"
```

Read Transformation	Write Transformation

```
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "constant":
"authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "sourcePath":
"$.to_BusinessUserAssignment.resul
ts",
        "optional": true,

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.members"
      },
      {
        "type": "remove",
        "targetPath":
"$.members[*].__metadata"
      },
      {
        "type": "remove",
        "targetPath":
"$.members[*].UserName"
      },
      {
        "type": "rename",
        "constant": "value",
        "targetPath":
"$.members[*].PersonID"
      },
      {
        "constant": "User",
        "targetPath":
"$.members[*].type"
      }
    ]
  }
}
```

By default, Identity Provisioning reads group IDs and members. If you want the service to also read group descriptions, you can add an extra mapping to the *"group"* resource in the *Read Transformation*. To learn how, see Guided Answers: Business Role Description.

Default read and write transformations for connector version 2:

Read Transformation	Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath":
"$.name.formatted",
        "targetPath":
"$.name.formatted",
        "optional": true
```

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath":
"$.externalId",
        "optional": true
```

Read Transformation

```
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails",
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.phoneNumbers"
      },
      {
        "sourcePath":
"$.userType",
        "targetPath":
"$.userType",
        "optional": true
      },
      {
        "condition": "$.locale ==
'not_unique'",
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

```
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath":
"$.name.formatted",
        "optional": true,
        "targetPath":
"$.name.formatted"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails",
        "optional": true
      },
      {
        "sourcePath":
"$.userType",
        "optional": true,
        "targetPath": "$.userType"
      },
      {
        "sourcePath":
"$.preferredLanguage",
        "optional": true,
        "targetPath":
"$.preferredLanguage"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
```

Read Transformation

```
ension:sap:2.0:User']
['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['language']
['correspondence']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['language']
['correspondence']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']
['numericalRepresentation']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']
['numericalRepresentation']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['date']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['date']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['time']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['time']",
        "optional": true
```

Write Transformation

```
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
       {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['language']
['correspondence']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['language']
['correspondence']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']
['numericalRepresentation']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']
['numericalRepresentation']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['date']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['date']",
```

Read Transformation

```
      },
      {
        "sourcePath":
"$.preferredLanguage",
        "targetPath":
"$.preferredLanguage",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuidHistory']",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuidHistory']"
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
```

Write Transformation

```
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['time']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['time']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuidHistory']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuidHistory']",
        "optional": true,

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:Group",
```

Read Transformation	Write Transformation

```
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

```
"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true,
        "defaultValue":
"userGroup"
      },
```

Read Transformation	Write Transformation

	<pre>{
	"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
	"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
	"optional": true
	}
	]
	}
	}</pre>

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PATCH**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PATCH** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ### Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`

So make sure you use the correct domain when you construct your REST API requests.

For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*

To learn more, see: Proxy Systems [page 1343]

1.6.3.30 SAP Intelligent Agriculture

Follow this procedure to set up SAP Intelligent Agriculture as a proxy system.

Prerequisites

- You have onboarded your SAP Intelligent Agriculture tenant as described in the Onboarding with the SAP BTP Cockpit section of the *Administration Guide for SAP Intelligent Agriculture*.
- You have OAuth credentials for SAP Intelligent Agriculture. For more information, see: Getting the Credentials from the SAP Intelligent Agriculture Instance.

Context

SAP Intelligent Agriculture enables agribusinesses to sustainably increase farming efficiency by digitizing their farming processes and services from plan-to-harvest, taking advantage of data science and machine learning capabilities.

You can use Identity Provisioning to configure SAP Intelligent Agriculture as a proxy system in hybrid scenarios. For example, when SAP Intelligent Agriculture is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users and groups to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users and group members back to the SAP Intelligent Agriculture.

⚠ Caution

You can't create or delete groups on SAP Intelligent Agriculture. That means:

- On the attempt to create a group on SAP Intelligent Agriculture, Identity Provisioning will only add new members or update existing ones. Also, if you read a group from the external back-end system, there must be a group with the exact same display name (case sensitive) in the SAP Intelligent Agriculture system. Otherwise, an error will be thrown and the group members will not be updated.
- On the attempt to delete a group on SAP Intelligent Agriculture, Identity Provisioning will only remove its members (group assignments). And this can happen only if the relevant group assignments have been provisioned/are present in the target system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Intelligent Agriculture* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL related to your SAP Intelligent Agriculture system, in format: **`https://*.prod-<region>.ia.agribusiness.cloud.sap`** , where *: is the name of your subaccount. For example: **`https://custom-domain.prod-eu10.ia.agribusiness.cloud.sap`**
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, created for your SAP Intelligent Agriculture instance (see **Prerequisites**).
`Password`	(Credential) Enter the OAuth Client Secret, created for your SAP Intelligent Agriculture instance (see **Prerequisites**).

Property Name	Value
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Intelligent Agriculture instance, in format: **`https://<TENANT>.authentication.<REGION>.hana.ondemand.com/oauth/token`**
(Optional) `ia.user.filter`	When specified, only those SAP Intelligent Agriculture users matching the filter expression will be read. For example: *userName eq "Smith.J"* For more information, see List of Properties [page 80]
(Optional) `ia.group.filter`	When specified, only those SAP Intelligent Agriculture groups matching the filter expression will be read. **Possible values:** • *intagri_Data_Scientist* • *intagri_Farm_Manager* • *intagri_Operator* • *intagri_Operations_Planner* • *intagri_Master_Data_Manager* • *intagri_Administrator* For example: *displayName eq "intagri_Master_Data_Manager"* For more information, see List of Properties [page 80]

Property Name	Value
`ia.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, **PATCH** operations are used to update users and groups in the target system. This means, for example, that if a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to *false*, **PUT** operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. **Possible values:** • *true* • *false* Default value: *false*
`ia.include.if.match.wildcard.header`	Makes the connector send the *If-Match* HTTP header with a value of "*" for every request to the target system. This header could be used by an SAP Intelligent Agriculture system for entity versioning. **Possible values:** • *true* • *false* Default value: *false*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> ⓘ Note
>
> The Identity Provisioning implementation of the SCIM Proxy API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy SCIM API.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Intelligent Agriculture* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

> ### Note
>
> When Identity Authentication is configured as a source system, the default transformation logic:
>
> - Provisions only groups and user group assignments if they are part of the following predefined group list:
>   - *intagri_Data_Scientist*
>   - *intagri_Farm_Manager*
>   - *intagri_Operator*
>   - *intagri_Operations_Planner*
>   - *intagri_Master_Data_Manager*
>   - *intagri_Administrator*
> - Skips some of the attributes from the identity records.
>
> This way, the transformation logic ensures that the identity data, sent to the Identity Provisioning SCIM REST API, is consistent.

> ### Note
>
> When a user is deleted from the SAP Intelligent Agriculture target system, depending on the off-boarding user handling it can be deleted or set to anonymized.
>
> If you try to read such user, you get only its *id* and the other attributes have value `ANONYMOUS`.
>
> If you try to create again a user with the same unique attributes (`userUuid`, `externalId`, `userName`, and `emaill`) during the configured retention period, the existed entity is restored to its active state. If the create operation requests changes in the values of the attributes, they will be executed after triggering a resync job.
>
> For more information, see Data Retention Manager.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Intelligent Agriculture system. For more information, see:

Manage Transformations [page 2439].

SAP Intelligent Agriculture SCIM API

Default read and write transformations:

> ### → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourcePath":
"$.meta",
                "targetPath":
"$.meta"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.userUuid",
```

Write Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
                "targetPath":
"$.schemas[0]"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional": true,
                "targetPath":
"$.name.middleName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional": true,
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath":
"$.active",
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.userUuid",
                "targetPath":
"$.userUuid"
            },
            {
                "sourcePath":
"$.externalId",
                "targetPath":
"$.externalId"
```

Read Transformation

```
                "targetPath":
"$.userUuid"
            },
            {
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional": true,
                "targetPath":
"$.name.middleName"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional": true,
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath":
"$.externalId",
                "targetPath":
"$.externalId"
            },
            {
                "sourcePath":
"$.active",
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath":
"$.emails[0].value",
                "optional": true,

"correlationAttribute": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']"
            },
            {
```

Write Transformation

```
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.Operations",
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas",
                "scope":
"patchEntity"
            }
        ],
        "scimEntityEndpoint":
"Users"
    },
    "group": {
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
                "targetPath":
"$.schema[0]"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
```

Read Transformation

```
                "sourcePath":
"$.groups",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups"
            }
        ],
        "scimEntityEndpoint":
"Users"
    },
    "group": {
    "mappings": [
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
```

Write Transformation

```
                "optional": true,
                "targetPath":
"$.members[?(@.value)]"
            },
            {
                "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.Operations",
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas",
                "scope":
"patchEntity"
            }
        ],
        "scimEntityEndpoint":
"Groups"
    }
}
```

Read Transformation	Write Transformation

<pre>},
{
"sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
"targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
"optional": true
},
{
"sourcePath": "$.members",
 	
"preserveArrayWithSingleElement":
true,
"optional": true,
"targetPath": "$.members"
}
],
"scimEntityEndpoint": "Groups"
}
}</pre>	

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.31 SAP Jam Collaboration

Follow this procedure to set up SAP Jam Collaboration as a proxy system.

Prerequisites

- You get OAuth credentials for SAP Jam Collaboration. If your SAP Jam tenant is of "SCIM provisioning" type, an OAuth client is automatically created for it, with the name **SCIM API Client**. How to find this client:
  1. Go to the SAP Jam Collaboration admin panel.
  2. Choose *Integrations* > *OAuth Clients*.
  3. For *SCIM API Client*, choose *View*.
  4. Save the *Key* and *Secret* values – you'll need them later while configuring your SAP Jam Collaboration provisioning system.

  To learn more, see: SAP Jam: Add an OAuth Client

Context

After fulfilling the prerequisites, follow the procedure below to create a proxy SAP Jam Collaboration system to load its users into an on-premise system and provision groups and new users back to SAP Jam.

> ⚠ Restriction
>
> Bear in mind the following limitations for the number of sent requests during a provisioning job:
>
> - The **SAP Jam SCIM API** allows up to 13,000 requests per hour and up to 200 requests per minute.
> - The Identity Provisioning service can handle the 200 requests per minute limit. If more requests are sent during the minute, the service will "wait" until it can execute them.

> ### Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Jam Collaboration* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL related to your SAP Jam system, in format: **`https://<SAP_Jam_datacenter>.sapjam.com`** For example: *https://jam4.sapjam.com*
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth client key, created for your SAP Jam tenant (see **Prerequisites**).
`Password`	(Credential) Enter the OAuth client secret, created for your SAP Jam tenant (see **Prerequisites**).

Property Name	Description & Value
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your SAP Jam instance, in format: **`https://<SAP_Jam_datacenter>/api/v1/auth/token`** For example: *https://jam4.sapjam.com/api/v1/auth/token*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Jam Collaboration* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Jam Collaboration. For more information, see:

   Manage Transformations [page 2439]

   SAP Business Accelerator Hub: SAP Jam Collaboration

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation	Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourcePath": "$",
                "targetPath": "$"
            },
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.emails[0].value",
                "targetPath":
"$.emails[0].value",
                "optional": true
            },
            {
                "sourcePath":
"$.emails[?(@.primary==
true)].value",
                "optional": true,

"correlationAttribute": true
            }
        ]
    },
    "group": {
```

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourcePath": "$",
                "targetPath": "$"
            },
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id",
                "scope":
"deleteEntity"
            },
            {
                "condition":
"$.emails[0].length() > 0",
                "constant": true,
                "targetPath":
"$.emails[0].primary"
            },
            {
                "constant": false,
                "targetPath":
"$.active",
                "scope":
"deleteEntity"
            },
            {
                "targetPath":
"$.locale",
                "type": "remove"
            },
            {
                "condition":
"($.locale EMPTY false)
&& ($.addresses[?(@.type ==
'work')].country EMPTY false)",
                "sourcePath":
"$.locale",
                "targetPath":
"$.locale",
                "functions": [
                    {

"function": "toLowerCaseString"
                    },
                    {

"function": "concatString",
```

Read Transformation

```
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "targetPath": "$",
                "sourcePath": "$"
            },
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']"
            }
        ]
    }
}
```

Write Transformation

```
                        "suffix":
"_"
                    },
                    {

"function": "concatString",
                        "suffix":
"$.addresses[?(@.type ==
'work')].country"
                    }
                ]
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourcePath": "$",
                "targetPath": "$"
            },
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            }
        ]
    }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be

automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

- For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
- For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

> ⓘ Note
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

1.6.3.32 SAP Market Communication for Utilities

Follow this procedure to set up SAP Market Communication for Utilities as a proxy system.

Prerequisites

- You have user credentials for an external back-end system with read and write permissions.
- To establish the connection between Identity Provisioning and SAP Market Communication for Utilities, you need to set up the communication user in SAP Market Communication for Utilities. You can do it now (as a prerequisite) or in the process of configuring SAP Market Communication for Utilities as a proxy system, as described in step 5.

Context

The SAP Market Communication for Utilities application is based on SAP BTP ABAP environment. You can use Identity Provisioning to configure SAP Market Communication for Utilities as a proxy system to execute *hybrid* scenarios. That means, it can provision its entities to another (external) back-end system by request, and then continue executing CRUD operations back to SAP Market Communication for Utilities, whenever the external back-end requests such. This scenario supports:

- Reading of **business users** (Employee) and **business roles** (which are considered as *groups*)
- Writing of **users** and **assignments**

> ⓘ Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Market Communication for Utilities* as a proxy system. For more information, see: Add New Systems [page 2415].
4. Set up the communication between Identity Provisioning and SAP Market Communication for Utilities and configure your authentication method (basic or certificate-based).

   > **Note**
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP Market Communication for Utilities proxy system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection [page 2452].

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP Market Communication for Utilities backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Create a communication system and assign the created user to the communication system.

      For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

   d. Create a communication arrangement with the created system.

      For your Identity Provisioning scenario, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

      For more information, see Maintain a Communication Arrangement for Inbound Communication

> ### Note
>
> The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
>
> The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the API URL to your SAP Market Communication for Utilities system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.

Property Name	Description & Value
`maco.skip.read.archived`	In the event of archived (disabled) entities in your SAP Market Communication for Utilities system, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to **false**, or delete it.
`ips.date.variable.format`	*yyyy-MM-dd* (needed for the *Read Transformation*)
`maco.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP Market Communication for Utilities proxy system in a hybrid scenario. • **true** – the current user roles will be deleted in the proxy system, and the user will be updated only with the roles provisioned by the service. • **false** – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the proxy system. See also: Extended Explanation of the *user.roles.overwrite Properties
(Optional) `maco.roles.filter`	Enter OData filtering for reading roles in the SAP Market Communication for Utilities system. To learn what criteria you can use, see: OData URI Conventions → **4.5 Filter System Query Option**
(Optional) `maco.roles.page.size`	Indicate how many business roles (considered as *groups*) per page to be read from your SAP Market Communication for Utilities system. The value must be an integer number.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://12345-aaaaa-3333.abap.hana.ondemand.com
User=MyMaCoUser
Password=************
ips.date.variable.format=yyyy-MM-dd
maco.skip.read.archived=true
maco.user.roles.overwrite = false
maco.roles.filter=startswith(ID, 'EMPLOYEE_LEVEL_3') eq true
maco.roles.page.size=30
```

6. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Market Communication for Utilities* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Market Communication for Utilities system. For more information, see:

   Manage Transformations [page 2439]

   SAP S/4HANA Cloud API: Business User

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.personID",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.personalInformation.firstName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.lastName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.middleName"
,
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.personFullN
ame",
        "targetPath":
"$.name.formatted",
        "optional": true
      },
      {
```

Write Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.personExternalID"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath":
"$.personExternalID"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath":
"$.user.globalUserID"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.personID"
      },
      {
        "targetPath":
"$.businessPartnerRoleCode",
        "type": "valueMapping",
        "sourcePaths": [
          "$.userType"
        ],
        "defaultValue": "BUP003",
        "valueMappings": [
          {
            "key": [
              "Employee"
            ],
            "mappedValue":
"BUP003"
          }
        ]
      },
      {
        "scope": "createEntity",
        "sourceVariable":
"currentDate",
        "targetPath":
"$.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
```

Read Transformation

```
        "sourcePath":
"$.user.userName",
        "targetPath":
"$.userName",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "targetPath":
"$.schemas[0]",
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User"
      },
      {
        "targetPath":
"$.schemas[1]",
        "constant":
"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User"
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "condition":
"$.user.lockedIndicator ==
'true'",
        "constant": false,
        "targetPath": "$.active",
        "optional": true
      },
      {
        "condition":
"($.user.validityPeriod.startDate
> '${currentDate}') ||
('${currentDate}' >
$.user.validityPeriod.endDate)",
        "constant": false,
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.workplaceInformation.emailAddre
ss",
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.user.logonLanguageCode",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
```

Write Transformation

```
        "targetPath":
"$.validityPeriod.endDate"
      },
      {
        "scope": "createEntity",
        "sourceVariable":
"currentDate",
        "targetPath":
"$.user.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
        "targetPath":
"$.user.validityPeriod.endDate"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.personalInformation.firstName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.personalInformation.lastName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.personalInformation.middleName"
      },
      {
        "sourcePath":
"$.name.formatted",
        "optional": true,
        "targetPath":
"$.personalInformation.personFullN
ame"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.user.userName"
      },
      {
        "sourcePath":
"$.nickName",
        "optional": true,
        "targetPath":
"$.user.nickName"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath":
"$.user.logonLanguageCode"
      },
```

Read Transformation

```
        "sourcePath":
"$.PersonExternalID",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.user.role[*].roleName",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups[?
(@.value)]"
      },
      {
        "sourcePath":
"$.user.globalUserID",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.user.timeZoneCode"
        ],
        "targetPath":
"$.timezone",
        "defaultValue": "Europe/
Berlin",
        "valueMappings": [
          {
            "key": [
              "WDFT"
            ],
            "mappedValue":
"Europe/Berlin"
          },
          {
            "key": [
              "ISRAEL"
            ],
            "mappedValue": "Asia/
Jerusalem"
          },
          {
            "key": [
              "RUS03"
            ],
            "mappedValue":
"Europe/Moscow"
          },
          {
            "key": [
              "AUSNSW"
            ],
            "mappedValue":
"Australia/Sydney"
          },
```

Write Transformation

```
      {
        "sourcePath":
"$.emails[0].value",
        "optional": true,
        "targetPath":
"$.workplaceInformation.emailAddre
ss"
      },
      {
        "condition": "$.active ==
false",
        "constant": "true",
        "targetPath":
"$.user.lockedIndicator"
      }
    ],
    "scimEntityEndpoint": "Users"
  },
  "group": {
    "mappings": [
      {
        "sourcePath":
"$.displayName",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members[?
(@.value)]"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ],
    "scimEntityEndpoint": "Groups"
```

Read Transformation	Write Transformation

```
          {
            "key": [
              "UTC+4"
            ],
            "mappedValue": "Asia/
Dubai"
          },
          {
            "key": [
              "BRAZIL"
            ],
            "mappedValue":
"America/Sao_Paulo"
          },
          {
            "key": [
              "BRZLEA"
            ],
            "mappedValue":
"America/Sao_Paulo"
          },
          {
            "key": [
              "MSTNO"
            ],
            "mappedValue":
"America/Phoenix"
          },
          {
            "key": [
              "EST"
            ],
            "mappedValue":
"America/New_York"
          },
          {
            "key": [
              "UTC"
            ],
            "mappedValue": "Etc/
UTC"
          },
          {
            "key": [
              "UTC+3"
            ],
            "mappedValue": "Asia/
Riyadh"
          },
          {
            "key": [
              "EST_"
            ],
            "mappedValue":
"America/Toronto"
          },
          {
            "key": [
              "UTC+8"
            ],
            "mappedValue": "Asia/
Shanghai"
```

```
    }
  }
```

Read Transformation	Write Transformation

```
          },
          {
            "key": [
              "JAPAN"
            ],
            "mappedValue": "Asia/
Tokyo"
          }
        ]
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
"$.businessPartnerRoleCode"
        ],
        "targetPath":
"$.userType",
        "defaultValue":
"Employee",
        "valueMappings": [
          {
            "key": [
              "BUP003"
            ],
            "mappedValue":
"Employee"
          }
        ]
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.ID",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.ID",
        "targetPath":
"$.displayName"
```

Read Transformation	Write Transformation

```
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath":
"$.to_BusinessUserAssignment.resul
ts",
        "optional": true,

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.members"
      },
      {
        "type": "remove",
        "targetPath":
"$.members[*].__metadata"
      },
      {
        "type": "remove",
        "targetPath":
"$.members[*].UserName"
      },
      {
        "type": "rename",
        "constant": "value",
        "targetPath":
"$.members[*].PersonID"
      },
      {
        "constant": "User",
        "targetPath":
"$.members[*].type"
      },
      {
        "constant":
"authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

By default, Identity Provisioning reads group IDs and members. If you want the service to also read group descriptions, you can add an extra mapping to the *"group"* resource in the *Read Transformation*. To learn how, see Guided Answers: Business Role Description.

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PATCH**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PATCH** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ### Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.33 SAP Marketing Cloud

Follow this procedure to set up SAP Marketing Cloud as a proxy system.

Prerequisites

- You have user credentials for an external back-end system with read and write permissions.
- To establish the connection between Identity Provisioning and SAP Marketing Cloud, you need to set up the communication (user, system and arrangement) on SAP Marketing Cloud. You can do it now (as a prerequisite) or in the process of configuring SAP Marketing Cloud as a proxy system, as described in step 5.

Context

You can use SAP Marketing Cloud as a proxy connector to execute *hybrid* scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to SAP Marketing Cloud, whenever the external back-end requests such. This scenario supports:

- Reading of **users** and **Business roles** (which are considered as *groups*)
- Writing of **users** and **assignments**

> ⓘ Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators* .
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.

2. Save your changes.
3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
4. Save your changes.

3. Add *SAP Marketing Cloud* as a proxy system. For more information, see Add New Systems [page 2415].
4. Set up the communication between Identity Provisioning and SAP Marketing Cloud and configure your authentication method (basic or certificate-based).

> ### Note
>
> We recommend that you use certificate-based authentication.

   a. In your newly added SAP Marketing Cloud proxy system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP Marketing Cloud backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

      For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

   c. Create a communication system and assign the created user to the communication system.

      For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

   d. Create a communication arrangement with the created system.

      For your Identity Provisioning scenario, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

      > ### Note
      >
      > The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
      >
      > The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to your SAP Marketing Cloud system.
`ProxyType`	Enter: *Internet*

Property Name	Description & Value
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **⚠ Restriction** Do not use special symbol ',' (comma) as it is not supported.
`marketing.cloud.skip.read.archived`	In the event of archived (disabled) entities in your SAP Marketing Cloud system, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to **false**, or delete it.
`ips.date.variable.format`	*yyyy-MM-dd* (needed for the *Read Transformation*)
`marketing.cloud.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP Marketing Cloud proxy system in a hybrid scenario. • **true** – the current user roles will be deleted in the proxy system, and the user will be updated only with the roles provisioned by the service. • **false** – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the proxy system. See also: Extended Explanation of the *user.roles.overwrite Properties

Property Name	Description & Value
(Optional) `marketing.cloud.roles.filter`	Enter OData filtering for reading roles in the SAP Marketing Cloud system. To learn what criteria you can use, see: OData URI Conventions → **4.5 Filter System Query Option**
(Optional) `marketing.cloud.roles.page.size`	Indicate how many business roles (considered as *groups*) per page to be read from your SAP Marketing Cloud system. The value must be an integer number.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*https://my1234567-api.s4hana.ondemand.com*

`User`=*MyMarketingCloudUser*

`Password`=**********

`ips.date.variable.format`=*yyyy-MM-dd*

`marketing.skip.read.archived`=*true*

`marketing.cloud.user.roles.overwrite` = *false*

`marketing.cloud.roles.filter`=*startswith(ID, 'EMPLOYEE_LEVEL_3') eq true*

`marketing.cloud.roles.page.size`=*30*

6. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Marketing Cloud* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Marketing Cloud. For more information, see:

   Manage Transformations [page 2439]

   SAP Marketing Cloud API: Business User

   Default read and write transformations:

→ Tip

The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.personID",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.personalInformation.firstName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.lastName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.middleName"
,
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.personFullN
ame",
        "targetPath":
"$.name.formatted",
        "optional": true
      },
      {
```

Write Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.personExternalID"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath":
"$.personExternalID"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath":
"$.user.globalUserID"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.personID"
      },
      {
        "targetPath":
"$.businessPartnerRoleCode",
        "type": "valueMapping",
        "sourcePaths": [
          "$.userType"
        ],
        "defaultValue": "BUP003",
        "valueMappings": [
          {
            "key": [
              "Employee"
            ],
            "mappedValue":
"BUP003"
          },
          {
            "key": [
              "Contingent Worker"
            ],
            "mappedValue":
"BBP005"
          }
        ]
      },
      {
        "scope": "createEntity",
```

Read Transformation

```
        "sourcePath":
"$.user.userName",
        "targetPath":
"$.userName",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "targetPath":
"$.schemas[0]",
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User"
      },
      {
        "targetPath":
"$.schemas[1]",
        "constant":
"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User"
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "condition":
"$.user.lockedIndicator ==
'true'",
        "constant": false,
        "targetPath": "$.active",
        "optional": true
      },
      {
        "condition":
"($.user.validityPeriod.startDate
> '${currentDate}') ||
('${currentDate}' >
$.user.validityPeriod.endDate)",
        "constant": false,
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.workplaceInformation.emailAddre
ss",
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.user.logonLanguageCode",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
```

Write Transformation

```
        "sourceVariable":
"currentDate",
        "targetPath":
"$.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
        "targetPath":
"$.validityPeriod.endDate"
      },
      {
        "scope": "createEntity",
        "sourceVariable":
"currentDate",
        "targetPath":
"$.user.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
        "targetPath":
"$.user.validityPeriod.endDate"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.personalInformation.firstName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.personalInformation.lastName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.personalInformation.middleName"
      },
      {
        "sourcePath":
"$.name.formatted",
        "optional": true,
        "targetPath":
"$.personalInformation.personFullN
ame"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.user.userName"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath":
"$.user.logonLanguageCode"
```

Read Transformation

```
        "sourcePath":
"$.PersonExternalID",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.user.globalUserID",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.user.timeZoneCode"
        ],
        "targetPath":
"$.timezone",
        "defaultValue": "Europe/
Berlin",
        "valueMappings": [
          {
            "key": [
              "WDFT"
            ],
            "mappedValue":
"Europe/Berlin"
          },
          {
            "key": [
              "ISRAEL"
            ],
            "mappedValue": "Asia/
Jerusalem"
          },
          {
            "key": [
              "RUS03"
            ],
            "mappedValue":
"Europe/Moscow"
          },
          {
            "key": [
              "AUSNSW"
            ],
            "mappedValue":
"Australia/Sydney"
          },
          {
            "key": [
              "UTC+4"
            ],
            "mappedValue": "Asia/
Dubai"
          },
          {
            "key": [
              "BRAZIL"
```

Write Transformation

```
      },
      {
        "sourcePath":
"$.emails[0].value",
        "optional": true,
        "targetPath":
"$.workplaceInformation.emailAddre
ss"
      },
      {
        "condition": "$.active ==
false",
        "constant": "true",
        "targetPath":
"$.user.lockedIndicator"
      }
    ],
    "scimEntityEndpoint": "Users"
  },
  "group": {
    "mappings": [
      {
        "sourcePath":
"$.displayName",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members[?
(@.value)]"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ],
```

Read Transformation	Write Transformation

```
            ],
            "mappedValue":
"America/Sao_Paulo"
          },
          {
            "key": [
              "BRZLEA"
            ],
            "mappedValue":
"America/Sao_Paulo"
          },
          {
            "key": [
              "MSTNO"
            ],
            "mappedValue":
"America/Phoenix"
          },
          {
            "key": [
              "EST"
            ],
            "mappedValue":
"America/New_York"
          },
          {
            "key": [
              "UTC"
            ],
            "mappedValue": "Etc/
UTC"
          },
          {
            "key": [
              "UTC+3"
            ],
            "mappedValue": "Asia/
Riyadh"
          },
          {
            "key": [
              "EST_"
            ],
            "mappedValue":
"America/Toronto"
          },
          {
            "key": [
              "UTC+8"
            ],
            "mappedValue": "Asia/
Shanghai"
          },
          {
            "key": [
              "JAPAN"
            ],
            "mappedValue": "Asia/
Tokyo"
          }
        ]
      },
```

```
    "scimEntityEndpoint": "Groups"
  }
}
```

Read Transformation	Write Transformation

```
      {
        "type": "valueMapping",
        "sourcePaths": [
"$.businessPartnerRoleCode"
        ],
        "targetPath":
"$.userType",
        "defaultValue":
"Employee",
        "valueMappings": [
          {
            "key": [
              "BUP003"
            ],
            "mappedValue":
"Employee"
          },
          {
            "key": [
              "BBP005"
            ],
            "mappedValue":
"Contingent Worker"
          }
        ]
      },
      {
        "sourcePath":
"$.user.role[*].roleName",
"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups[?
(@.value)]"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.ID",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
```

Read Transformation	Write Transformation

```
          }
        ]
      },
      {
        "sourcePath": "$.ID",
        "targetPath":
"$.displayName"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath":
"$.to_BusinessUserAssignment.resul
ts",
        "optional": true,

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.members"
      },
      {
        "type": "remove",
        "targetPath":
"$.members[*].__metadata"
      },
      {
        "type": "remove",
        "targetPath":
"$.members[*].UserName"
      },
      {
        "type": "rename",
        "constant": "value",
        "targetPath":
"$.members[*].PersonID"
      },
      {
        "constant": "User",
        "targetPath":
"$.members[*].type"
      },
      {
        "constant":
"authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
```

Read Transformation	Write Transformation

<pre>"optional": true
}
]
}
}</pre>	

By default, Identity Provisioning reads group IDs and members. If you want the service to also read group descriptions, you can add an extra mapping to the *"group"* resource in the *Read Transformation*. To learn how, see Guided Answers: Business Role Description.

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PATCH**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PATCH** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ### Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*

To learn more, see: Proxy Systems [page 1343]

1.6.3.34 SAP Master Data Integration

Follow this procedure to set up SAP Master Data Integration (in short, MDI) as a proxy system.

Prerequisites

- You have created a tenant in a subaccount on SAP BTP, Cloud Foundry environment. You can create your own tenant (free of charge), or integrate with an existing one.
- You have created a service instance for MDI in the subaccount in order to connect a new system to your tenant and read user account information from it. To learn how, see: Creating Service Instances
- You have created a service key in this instance, which contains the necessary credentials to connect to the MDI service. Creating multiple service keys in the same service instance is not supported. To learn how, see: Creating service Instances

  → Tip

  The *serviceKey* payload provides you with the following properties that you will later need for your system configuration:

  - **uri** =`URL`
  - **uaa.url** = `OAuth2TokenServiceURL`
  - **uaa.clientid** = `User`
  - **clientsecret** = `Password`

Context

As part of SAP's data model and integration unification strategy, SAP BTP Integration Suite has *Master Data Integration* to enable a harmonized integration and distribution of different master data objects and data between SAP solutions. This includes master data for business partners, cost centers, and workforce data. Workforce data is provided for integration scenarios that need data from SAP SuccessFactors Employee Central or other core HR systems.

To learn more, see: Integrating SAP SuccessFactors Employee Central with SAP Master Data Integration

You can use MDI as a proxy connector to execute *hybrid* scenarios. That means, it can provision its entities via API requests sent by an external back-end system (such as SAP Identity Management or SAP IAG).

⚠ Restriction

This scenario supports only executing *read* (GET) operations from MDI to the external back-end system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Master Data Integration* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to the relevant integration application running in the relevant region of SAP BTP. See the **Prerequisites** section → *serviceKey* tip.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the technical user that has access to the API of your MDI service. See the **Prerequisites** section → *serviceKey* tip.
`Password`	(Credential) Enter the password for this technical user. See the **Prerequisites** section → *serviceKey* tip.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. See the **Prerequisites** section → *serviceKey* tip.

   To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> ### Note
>
> The Identity Provisioning implementation of the SCIM Proxy API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy SCIM API.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Master Data Integration* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your MDI proxy system. For more information, see:

   Manage Transformations [page 2439]

   Field Mapping Between Employee Central and SAP Master Data Integration

   Default read and write transformations:

   > ### → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation	Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem",
        "correlationAttribute":
true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
            {
                "type":
"concatString",
                "suffix": "$
{entityIdSourceSystem}"
            }
        ]
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath":
"$.profileDetail[0].content.script
edProfileDetails[0].firstName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.profileDetail[0].content.script
edProfileDetails[0].lastName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.profileDetail[0].content.script
edProfileDetails[0].middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
```

Code Syntax

```
\\ Proxy Write transformation is
currently not supported!
```

Read Transformation	Write Transformation

```
      {
        "sourcePath":
"$.userAccount.userName",
        "targetPath":
"$.userName",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.emails[?
(@.isDefault == true)].address",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "targetPath":
"$.emails[*].usage",
        "type": "remove"
      },
      {
        "targetPath":
"$.emails[*].address",
        "type": "rename",
        "constant": "value"
      },
      {
        "targetPath":
"$.emails[*].isDefault",
        "type": "rename",
        "constant": "primary"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
        "targetPath":
"$.schemas[0]"
      }
    ]
  }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be

automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

- For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
- For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

> ⓘ Note
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

SAP Community: The New Master Data Integration Service for SAP SuccessFactors

1.6.3.35 SAP Revenue Growth Management

Follow this procedure to set up SAP Revenue Growth Management as a proxy system.

Prerequisites

- You have created an SAP Revenue Growth Management consumer application in your SAP Cloud Identity Services tenant and defined a dependency to its API, as described in Configure Integration Between Applications. *UserImport* must be selected for the dependency *API*.
- You have generated API credentials for the consumer application, as described in Generate Credentials to Access the APIs of an Application.

Context

SAP Revenue Growth Management is an SAP BTP SaaS application that provides customers in the Consumer Products industry an easy way to efficiently create, monitor, and maintain account plans.

You can use Identity Provisioning to configure SAP Revenue Growth Management as a proxy system in hybrid scenarios. For example, when SAP Revenue Growth Management is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users and groups to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users and group back to the SAP Revenue Growth Management.

> **Note**
>
> Support for groups is intended for future use.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators* .
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.

- For **Basic authentication**, proceed as follows:
    1. Add an administrator user of type **System** and configure the basic authentication method for this user.
       If you already have a technical user, skip this step.
    2. Save your changes.
    3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
    4. Save your changes.

3. Add *SAP Revenue Growth Management* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the base URL of your SAP Revenue Growth Management system's SCIM API, excluding any path information.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the client ID generated when creating API credentials for the consumer application. For more information, see Generate Credentials to Access the APIs of an Application.
`Password`	(Credential) Enter the client secret generated when creating API credentials for the consumer application. For more information, see Generate Credentials to Access the APIs of an Application.
`rgm.userImport.api.dependency.name`	Enter the value of the dependency name used in the configuration of the SAP Revenue Growth Management consumer application, created in the SAP Cloud Identity Services tenant. It is used for access token retrieval from Identity Authentication. For more information, see Integrating Applications.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL.

Property Name	Description & Value
(Optional) `rgm.user.filter`	When specified, only those users matching the filter expression will be read. You can filter users by **`userName`**, **`emails[0].value`**, and **`externalId`**, according to the API syntax of SAP Revenue Growth Management. For example: • *userName eq "johnbrown" and externalId eq "c167254d-25fd-5fac-af32-0b8c35e0de27"* • *userName sw "J" and externalId eq "c167254d-25fd-5fac-af32-0b8c35e0de27"* • *userName eq "johnbrown" and emails.value eq "johnbrown@email.com"* • *userName eq "johnbrown" and emails.value eq "johnbrown@email.com" and externalId eq "c167254d-25fd-5fac-af32-0b8c35e0de27"* **Note** These combinations are valid for both 'or' and 'and' operators.
(Optional) `rgm.group.filter`	When specified, only those groups matching the filter expression will be read. For example: *urn:sap:cloud:scim:schemas:extension:custom:2.0:Group:name eq "ProjectTeam1" or "Students2018"*
(Optional) `rgm.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. Default value for proxy systems: *true*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Revenue Growth Management* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Revenue Growth Management. For more information, see:

Manage Transformations [page 2439]

SAP Revenue Growth Management SCIM API

Default read and write transformations:

> → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath":"$.userName",
        "targetPath":"$.userName",

"correlationAttribute":true
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
           "type": "concatString",
           "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath": "$.emails[?
(@.primary == true)].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
```

Write Transformation

Code Syntax

```
{
  "user":{
    "scimEntityEndpoint": "Users",
    "mappings":[
     {
       "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "constant":
"urn:ietf:params:scim:api:messages
:2.0:PatchOp",
        "targetPath":
"$.schemas[0]",
        "scope": "patchEntity"
      },
      {
        "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.Operations",
        "scope": "patchEntity"
      },
      {

"sourceVariable":"entityIdTargetSy
stem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
```

Read Transformation

```
"sourcePath":"$.externalId",

"targetPath":"$.externalId",
        "optional":true
      },
      {

"sourcePath":"$.name.givenName",

"targetPath":"$.name.givenName",
        "optional":true
      },
      {

"sourcePath":"$.name.familyName",

"targetPath":"$.name.familyName",
        "optional":true
      },
      {

"sourcePath":"$.displayName",

"targetPath":"$.displayName",
        "optional":true
      },
      {
        "sourcePath":"$.userType",
        "targetPath":"$.userType",
        "optional":true
      },
      {

"sourcePath":"$.preferredLanguage"
,

"targetPath":"$.preferredLanguage"
,
        "optional":true
      },
      {
        "sourcePath":"$.locale",
        "targetPath":"$.locale",
        "optional":true
      },
      {
        "sourcePath":"$.timezone",
        "targetPath":"$.timezone",
        "optional":true
      },
      {
        "sourcePath":"$.active",
        "targetPath":"$.active",
        "optional":true
      },
      {

"sourcePath":"$.addresses",

"preserveArrayWithSingleElement":
true,
```

Write Transformation

```
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {

"sourcePath":"$.name.familyName",

"targetPath":"$.name.familyName",
        "optional":true
      },
      {

"sourcePath":"$.displayName",

"targetPath":"$.displayName",
        "optional":true
      },
      {
        "sourcePath":"$.userType",
        "targetPath":"$.userType",
        "optional":true
      },
      {

"sourcePath":"$.preferredLanguage"
,

"targetPath":"$.preferredLanguage"
,
        "optional":true
      },
      {
        "sourcePath":"$.locale",
        "targetPath":"$.locale",
        "optional":true
      },
      {
        "sourcePath":"$.timezone",
        "targetPath":"$.timezone",
        "optional":true
      },
      {
        "sourcePath":"$.active",
        "targetPath":"$.active",
        "optional":true
      },
      {

"sourcePath":"$.addresses",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":"$.addresses"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
```

Read Transformation

```
        "optional": true,
        "targetPath":"$.addresses"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      }
    ]
  },
   "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
           {
              "type":
"concatString",
              "suffix": "$
{entityIdSourceSystem}"
           }
        ]
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
       {
```

Write Transformation

```
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      }
    ]
  },
   "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "targetPath": "$.id",
        "sourceVariable":
"entityIdTargetSystem"
      },
      {
       "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:Group",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:Group",

"urn:sap:cloud:scim:schemas:extens
ion:custom:2.0:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
```

Read Transformation

```
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",
        "optional": true,

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

Write Transformation

```
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

> ⓘ Note
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

7. Run an initial load job.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

SAP Revenue Growth Management

1.6.3.36 SAP Sales Cloud and SAP Service Cloud

Follow this procedure to set up SAP Sales Cloud and SAP Service Cloud, formerly known as *SAP Cloud for Customer* (in short, C4C), as a proxy system.

Prerequisites

To integrate SAP Sales Cloud and SAP Service Cloud with Identity Provisioning, you need to use SAP Cloud Integration (SAP CI). This service provides a package with integration flows (iFlows) for enabling the creation of users and assignment of users to groups via SCIM API in SAP Sales Cloud and SAP Service Cloud.

- To configure SAP Cloud Integration and SAP Sales Cloud and SAP Service Cloud, see: Identity Provisioning in SAP Cloud for Customer using System for Cross-Domain Identity Management (SCIM)
- To set up and use the *SAP Cloud for Customer Integration with Identity Provisioning via System for Cross-domain Identity Management* package, see: SAP Business Accelerator Hub: SAP Cloud for Customer Integration with Identity Provisioning via System for Cross-domain Identity Management

Context

SAP Sales Cloud and SAP Service Cloud is a cloud-based solution that helps customers manage day-to-day sales and service interactions by sending and receiving signals between front- and back-office solutions and providing a single view of the customer.

You can use Identity Provisioning to configure SAP Sales Cloud and SAP Service Cloud as a proxy system for integration with on-premise or cloud identity management systems that support SCIM 2.0 standard (for example, SAP Identity Management and SAP Cloud Identity Access Governance).

In this scenario Identity Provisioning acts as a proxy between SAP Sales Cloud and SAP Service Cloud and the on-premise or cloud system. Identity Provisioning exposes SAP Sales Cloud and SAP Service Cloud as a proxy system and connects it to the identity management system without making a direct connection between both systems. You can then provision users to the identity management system, which can trigger CRUD (create, read, update, delete) operations on users back to the SAP Sales Cloud and SAP Service Cloud.

The cloud-based solution provides different APIs for integration with Identity Provisioning, leading to different connector versions for each API. Currently, the following connector versions are available: Version 1 (SOAP-based API), which is deprecated, Version 2 (SOAP-based API), Version 3 (SCIM 2.0 based API) and Version 4 (SCIM 2.0 based API), which is applicable for SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 only.

Each connector version supports specific attribute mappings in transformations and requires particular properties. The latest version is set as the default.

Version 4 (Applicable for SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 only)

When using connector version 4 (SCIM 2.0 based API), only users of type Employee are provisioned. This version introduces an enhanced SCIM API that no longer requires SAP Cloud Integration. It supports patch operations and provisioning of application-specific groups. In addition to pagination using `startIndex` and `count`, cursor-based pagination is also supported.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Sales Cloud and SAP Service Cloud* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter SAP Cloud Integration runtime URL. See: How to Get SAP Cloud Integration Runtime URL
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
`OAuth2TokenServiceURL`	Enter the URL to the access token provider service. For example: `https://<c4c_instance>/auth/token` SAP Sales Cloud and SAP Service Cloud support OAuth token-based authentication, enabling you to obtain an access token using either basic authentication (by setting User and Password) or certificate authentication. **Relevant for connector version 4**

Property Name	Description & Value
`User`	Enter SAP Cloud Integration user ID to connect to SAP Cloud Integration. See: • Setting Up Inbound HTTP Connections (with Basic Authentication), Neo Environment • Basic Authentication of IdP User for Integration Flow Processing (Cloud Foundry environment) **Relevant for connector version 3**
`Password`	(Credential) Enter SAP Cloud Integration password to connect to SAP Cloud Integration. See: • Setting Up Inbound HTTP Connections (with Basic Authentication), Neo Environment • Basic Authentication of IdP User for Integration Flow Processing (Cloud Foundry environment) **Relevant for connector version 3**
`c4c.api.version`	The version of the SAP Sales Cloud and SAP Service Cloud API you use. By default, the Identity Provisioning service uses version **4** - the SCIM 2.0 based API.
(Optional) `c4c.user.filter`	When specified, only those users matching the filter expression will be read. Supported user filters in connector version 4: • userName eq "Smith" • userName sw "Sm" • userName eq "Smith" and emails.value eq "smith@example.com" Supported user filter in connector version 3: • addresses.country eq "USA"
(Optional) `c4c.group.filter`	When specified, only those C4C groups matching the filter expression will be read. Example: **displayName eq "ProjectTeam1"**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> ### Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

5. Configure the transformations.

   The Identity Provisioning offers a default transformation for the *SAP Sales Cloud and SAP Service Cloud* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in C4C. For more information, see:

   Manage Transformations [page 2439]

   > ### → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

   Default transformation for connector version 3

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.name.middleName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.name.middleName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.title",
        "optional": true,
        "targetPath": "$.title"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath":
"$.nickName",
        "optional": true,
        "targetPath": "$.nickName"
      },
      {
        "sourcePath":
"$.addresses",
        "optional": true,
```

Read Transformation

```
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.addresses",
        "targetPath":
"$.addresses",
        "optional": true,

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",
        "optional": true,

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.userType",
        "targetPath":
"$.userType",
        "optional": true
      },
      {
        "sourcePath":
"$.nickName",
        "optional": true,
        "targetPath": "$.nickName"
      },
      {
        "sourcePath":
"$.preferredLanguage",
        "optional": true,
        "targetPath":
"$.preferredLanguage"
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.emails[?
(@.primary == true)].value",
        "optional": true,
        "correlationAttribute":
true
```

Write Transformation

```
        "targetPath":
"$.addresses",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "optional": true,
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.userType",
        "optional": true,
        "targetPath": "$.userType"
      },
      {
        "sourcePath":
"$.preferredLanguage",
        "optional": true,
        "targetPath":
"$.preferredLanguage"
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      }
    ]
  },
  "group": {
    "skipOperations": [
      "create",
      "delete"
    ],
    "mappings": [
```

Read Transformation

```
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
```

Write Transformation

```
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      }
    ],
    "scimEntityEndpoint": "Groups"
  }
}
```

Read Transformation	Write Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "sourcePath": "$.members",
        "optional": true,
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      }
    ]
  }
}
```

Default transformation for connector version 4

(Applicable for SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 only)

Read Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.emails[?
(@.primary == true)].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath":
"$.userType",
        "targetPath": "$.userType"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
```

Write Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.userType",
        "targetPath": "$.userType"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.externalId"
      },
      {
```

Read Transformation	Write Transformation

<pre>"targetPath":
"$.displayName"
},
{
"sourcePath":
"$.externalId",
"optional": true,
"targetPath":
"$.externalId"
},
{
"sourcePath":
"$.name.givenName",
"optional": true,
"targetPath":
"$.name.givenName"
},
{
"sourcePath":
"$.name.middleName",
"optional": true,
"targetPath":
"$.name.middleName"
},
{
"sourcePath":
"$.name.familyName",
"optional": true,
"targetPath":
"$.name.familyName"
},
{
"sourcePath":
"$.name.formatted",
"optional": true,
"targetPath":
"$.name.formatted"
},
{
"sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
"optional": true,
"targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
},
{
"sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
"optional": true,
"targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']"
},
{</pre>
	<pre>"sourcePath":
	"$.name.givenName",
	"optional": true,
	"targetPath":
	"$.name.givenName"
	},
	{
	"sourcePath":
	"$.name.middleName",
	"optional": true,
	"targetPath":
	"$.name.middleName"
	},
	{
	"sourcePath":
	"$.name.familyName",
	"optional": true,
	"targetPath":
	"$.name.familyName"
	},
	{
	"sourcePath":
	"$.name.formatted",
	"optional": true,
	"targetPath":
	"$.name.formatted"
	},
	{
	"sourcePath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:User']
	['userUuid']",
	"optional": true,
	"targetPath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:User']['userUuid']"
	},
	{
	"sourcePath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:User']
	['validFrom']",
	"optional": true,
	"targetPath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:User']
	['validFrom']"
	},
	{
	"sourcePath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:User']['validTo']",
	"optional": true,
	"targetPath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:User']['validTo']"
	},
	{
	"sourcePath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:User']['status']",
	"optional": true,</pre>

Read Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['status']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['status']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
```

Write Transformation

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['status']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "constant":
"urn:ietf:params:scim:api:messages
:2.0:PatchOp",
        "targetPath":
"$.schemas[0]",
        "scope": "patchEntity"
      },
      {
        "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.Operations",
        "scope": "patchEntity"
      }
    ],
    "scimEntityEndpoint": "Users"
  },
  "group": {
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:Group",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:Group"
        ],
```

Read Transformation

```
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      }
    ],
    "scimEntityEndpoint": "Users"
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']"
      },
      {
        "sourcePath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members"
```

Write Transformation

```
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:Group']['name']",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members[?
(@.value)]",
        "functions": [
          {
            "function":
"resolveEntityIds"
          }
        ]
      },
      {
        "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.Operations",
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas",
        "scope": "patchEntity"
      }
    ],
    "scimEntityEndpoint": "Groups"
  }
}
```

Read Transformation	Write Transformation

```
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      }
    ],
    "scimEntityEndpoint": "Groups"
  }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.37 SAP Signavio Process Transformation Suite

Follow this procedure to set up SAP Signavio Process Transformation Suite as a proxy system.

Prerequisites

> ⓘ Note
>
> The initial setup of SAP Signavio Process Transformation Suite is completed during the onboarding process. This setup includes establishing a connection between Identity Provisioning and SAP Signavio Process Transformation Suite using certificate-based authentication. Any changes needed after the initial setup should be processed through a support incident.

Context

SAP Signavio Process Transformation Suite helps you understand, improve, and transform your business processes – fast and at scale – with a cloud-based process management platform.

You can use Identity Provisioning to configure SAP Signavio Process Transformation Suite as a proxy system in hybrid scenarios. For example, when SAP Signavio Process Transformation Suite is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users back to the SAP Signavio Process Transformation Suite.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP Signavio Process Transformation Suite* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	The URL to the SCIM API of your SAP Signavio Process Transformation Suite system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *ClientCertificateAuthentication*
`sig.tenant.id`	The identifier of SAP Signavio Process Transformation Suite tenant

Property Name	Value
(Optional) `sig.user.filter`	When specified, only those SAP Signavio Process Transformation Suite users matching the filter expression will be read. For example: • Value = *userName eq "JohnSmith"* • Value = *userName sw "J"*
(Optional) `sig.group.filter`	When specified, only those SAP Signavio Process Transformation Suite groups matching the filter expression will be read. For example: • Value = *displayName eq "ProjectTeam1"* • Value = *displayName sw "P"*
`sig.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to *true*, PATCH operations are used to update users and groups in the target system. • If set to *false*, PUT operations are used to update users and groups in the target system. **Possible values:** • *true* • *false* Default value: *false*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP Signavio Process Transformation Suite* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP Signavio Process Transformation Suite. For more information, see:

   Manage Transformations [page 2439]

Default read and write transformations:

> → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Default read and write transformations:

Read Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "optional": true,
        "targetPath": "$.meta"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
```

Write Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.externalId"
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value == []",
        "sourcePath":
"$.emails[0].value",
        "targetPath": "$.userName"
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value != []",
        "sourcePath": "$.emails[?
(@.primary == true)].value",
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.name.middleName"
      },
```

Read Transformation

```
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.name.middleName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "optional": true,
        "targetPath":
"$.name.honorificPrefix"
      },
      {
        "sourcePath":
"$.name.honorificSuffix",
        "optional": true,
        "targetPath":
"$.name.honorificSuffix"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.nickName",
        "optional": true,
        "targetPath": "$.nickName"
      },
      {
        "sourcePath":
"$.profileUrl",
        "optional": true,
        "targetPath":
"$.profileUrl"
      },
      {
        "sourcePath": "$.title",
        "optional": true,
        "targetPath": "$.title"
      },
      {
        "sourcePath":
"$.userType",
        "optional": true,
        "targetPath": "$.userType"
      },
```

Write Transformation

```
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "optional": true,
        "targetPath":
"$.name.honorificPrefix"
      },
      {
        "sourcePath":
"$.name.honorificSuffix",
        "optional": true,
        "targetPath":
"$.name.honorificSuffix"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.nickName",
        "optional": true,
        "targetPath": "$.nickName"
      },
      {
        "sourcePath":
"$.profileUrl",
        "optional": true,
        "targetPath":
"$.profileUrl"
      },
      {
        "sourcePath": "$.title",
        "optional": true,
        "targetPath": "$.title"
      },
      {
        "sourcePath":
"$.userType",
        "optional": true,
        "targetPath": "$.userType"
      },
      {
        "sourcePath":
"$.preferredLanguage",
        "optional": true,
        "targetPath":
"$.preferredLanguage"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
```

Read Transformation	Write Transformation

```
      {
        "sourcePath":
"$.preferredLanguage",
        "optional": true,
        "targetPath":
"$.preferredLanguage"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath":
"$.emails[0].value",
        "optional": true,
        "targetPath":
"$.emails[0].value"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails']",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails']"
      },
      {
        "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.phoneNumbers"
      },
      {
        "sourcePath": "$.ims",
```

```
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails']",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['emails']"
      },
      {
        "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.phoneNumbers"
      },
      {
        "sourcePath": "$.ims",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.ims"
      },
      {
        "sourcePath": "$.photos",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.photos"
      },
      {
        "sourcePath":
"$.addresses",
```

Read Transformation

```
"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.ims"
      },
      {
        "sourcePath": "$.photos",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.photos"
      },
      {
        "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.addresses"
      },
      {
        "sourcePath":
"$.entitlements",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.entitlements"
      },
      {
        "sourcePath": "$.roles",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.roles"
      },
      {
        "sourcePath":
"$.x509Certificates",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.x509Certificates"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']"
```

Write Transformation

```
"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.addresses"
      },
      {
        "sourcePath":
"$.entitlements",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.entitlements"
      },
      {
        "sourcePath": "$.roles",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.roles"
      },
      {
        "sourcePath":
"$.x509Certificates",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.x509Certificates"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true,
```

Read Transformation

```
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
```

Write Transformation

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['costCenter']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['organization']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation

```
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuidHistory']",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuidHistory']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['status']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['status']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

```
ension:sap:2.0:User']
['userUuidHistory']",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuidHistory']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['status']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['status']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['mfaEnabled']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['mfaEnabled']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['sourceSystem']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['sourceSystem']"
      },
      {
```

Read Transformation

```
ension:sap:2.0:User']
['mfaEnabled']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['mfaEnabled']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['sourceSystem']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['sourceSystem']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['totpEnabled']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['totpEnabled']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['loginTime']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['loginTime']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['contactPreferences']",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['contactPreferences']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['passwordDetails']",
```

Write Transformation

```
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['totpEnabled']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['totpEnabled']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['loginTime']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['loginTime']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['contactPreferences']",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['contactPreferences']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['passwordDetails']",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['passwordDetails']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['sncName']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['sncName']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['groupDomains']",
```

Read Transformation

```
"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['passwordDetails']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['sncName']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['sncName']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['groupDomains']",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['groupDomains']"
      },
      {
        "sourcePath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups"
      }
    ],
    "scimEntityEndpoint": "Users"
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "optional": true,
        "targetPath": "$.meta"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
```

Write Transformation

```
"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['groupDomains']"
      },
      {
        "sourcePath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups"
      },
      {
        "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.Operations",
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas",
        "scope": "patchEntity"
      }
    ],
    "scimEntityEndpoint": "Users"
  },
  "group": {
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",

"preserveArrayWithSingleElement":
true,
```

Read Transformation

```
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']"
      },
      {
        "sourcePath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members"
      }
    ],
    "scimEntityEndpoint": "Groups"
  }
```

Write Transformation

```
        "optional": true,
        "targetPath": "$.members"
      },
      {
        "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.Operations",
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas",
        "scope": "patchEntity"
      }
    ],
    "scimEntityEndpoint": "Groups"
  }
}
```

Read Transformation	Write Transformation
`}`	

> **Note**
>
> Updating a user in SAP Signavio Process Transformation Suite depends on whether user attributes in SAP Signavio Process Transformation Suite are mapped to SAML attributes in your identity provider. If this is the case, the values of those attributes are populated by the identity provider and cannot be changed by the SAP Signavio Process Transformation Suite SCIM API or the UI. For more information, see Map SAML Attributes to Users and Create a User.

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > **Note**
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

7. Run an initial load job.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> **Caution**
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`

So make sure you use the correct domain when you construct your REST API requests.

For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*

To learn more, see: Proxy Systems [page 1343]

1.6.3.38 SAP S/4HANA Cloud Public Edition

Follow this procedure to set up SAP S/4HANA Cloud Public Edition as a proxy system.

Prerequisites

- You have user credentials for an external back-end system with read and write permissions.
- To establish the connection between Identity Provisioning and SAP S/4HANA Cloud Public Edition, you need to set up the communication (user, system and arrangement) on SAP S/4HANA Cloud Public Edition. You can do it now (as a prerequisite) or in the process of configuring SAP S/4HANA Cloud Public Edition as a target system, as described in step 5.

Context

You can use SAP S/4HANA Cloud Public Edition as a proxy connector to execute *hybrid* scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to SAP S/4HANA Cloud Public Edition, whenever the external back-end requests such.

SAP S/4HANA Cloud Public Edition provides different APIs for integration with Identity Provisioning, resulting in different connector versions for each API. Each connector version supports specific attribute mappings within the transformations and requires particular property values. The value of the `s4hana.cloud.api.version` property controls which API you use. By default, the Identity Provisioning service uses version *1*.

- When the value is set to *1* or the property is not present (typical for systems created before versioning was introduced on November 6, 2025), the SOAP API provided by the communication scenario SAP_COM_0193 is used. In SAP S/4HANA Cloud Public Edition version 1, groups correspond to business roles, thus group members are user assignments of a business role.
  In your SAP S/4HANA Cloud Public Edition system version 1, the HCM (HR) integration is active by default. You have **business users** (Employee, Contingent Worker) and **login users** assigned to them. The corresponding HR integration manages business users – it enables you to update these users from your external data source, such as an identity management system. The Identity Provisioning service manages only the user-related login information, such as date/time preferences or role assignments.
  This scenario supports:
  - Reading of **users** and **Business roles** (which are considered as *groups*)
  - Writing of **users** and **role assignments**

- When the value is set to *3*, the SCIM interface provided by the communication scenario SAP_COM_0465 is used. It supports provisioning of **business users** with their **assignments** to groups of type **userGroup** and **authorization**. For more information, see Groups.

For more information on how to update to version *3*, see Update Connector Version [page 2424].

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP S/4HANA Cloud Public Edition* as a proxy system. For more information, see Add New Systems [page 2415].
4. Set up the communication between Identity Provisioning and SAP S/4HANA Cloud Public Edition and configure your authentication method (basic or certificate-based).

   > **Note**
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP S/4HANA Cloud Public Edition proxy system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip step **a.** if you want to use basic authentication.

      The next steps are performed in SAP S/4HANA backend system and are relevant for both basic and certificate-based authentication.

   b. Create a communication user and provide the respective credentials.

      For basic authentication, provide *User Name* and *Password*.

For certificate-based authentication, upload the certificate you have generated in the Identity Provisioning UI on the previous step.

c. Create a communication system and assign the created user to the communication system.

For your Identity Provisioning scenario, provide *System ID*, *System Name* and *Host Name*.

d. Create a communication arrangement with the created system.

- When `s4hana.cloud.api.version=1` is set in Identity Provisioning for the SAP S/4HANA Cloud Public Edition proxy system, choose *Scenario ID* SAP_COM_0193 (SAP Cloud Identity Provisioning Integration).

  > **Note**
  >
  > The communication scenario *SAP_COM_0193* is enhanced to support the User UUID attribute which is generated by Identity Authentication at user creation.
  >
  > The User UUID is universally unique identifier. This attribute is immutable and unique across technology layers, such as user interface, APIs, and security tokens, as well as across products and lines of business contributing to a business process in the Intelligent Enterprise.

- When `s4hana.cloud.api.version=3` is set in Identity Provisioning for the SAP S/4HANA Cloud Public Edition proxy system, choose *Scenario ID* SAP_COM_0465 (System for Cross-domain Identity Management Integration).

  > **Note**
  >
  > The communication scenario *SAP_COM_0465* allows you to maintain data via SCIM (System for Cross-domain Identity Management). For more information, see Integrate Cross-Domain Identity Management.

5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
URL	Enter the SAP S/4HANA Cloud Public Edition API URL. You can find the correct URL in the *API-URL* field of the communication arrangement set up for the communication scenario used. For example: `https://my123456-api.s4hana.ondemand.com`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*

Property Name	Description & Value
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *User Name* from the communication arrangement. **⚠ Restriction** Do not use special symbol '**,**' (comma) as it is not supported.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the *Password* for the user name from the communication arrangement. **⚠ Restriction** Do not use special symbol '**,**' (comma) as it is not supported.
`s4hana.cloud.api.version`	The version of the system API you use. By default, Identity Provisioning uses version *1*. Version *1* means your SAP S/4HANA Cloud Public Edition system consumes SAP S/4HANA Cloud Public Edition API: Business User. Version *3* means your SAP S/4HANA Cloud Public Edition system consumes System for Cross-domain Identity Management (SCIM) API.
`s4hana.cloud.skip.read.archived`	In the event of archived (disabled) entities in your SAP S/4HANA Cloud Public Edition system, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to **false**, or delete it. **Relevant for connector version 1**

Property Name	Description & Value
`s4hana.cloud.hr.switch.active`	Defines whether the human resources (HR) integration is active or not. The property is set to `true` by default. This means that the synchronization of business partners is managed by the HR system and SAP S/4HANA Cloud Public Edition. When set to `false`, synchronization is handled by the identity management system instead. *Possible values*: • true (default) • false **Relevant for connector version 1**
`s4hana.cloud.hr.switch.dependent.role.codes`	A default property. As a comma-separated value, add the codes of the roles maintained by the HR integration. Make sure these role codes are part of your *read* and *write* transformations. By default, the following codes are added to your system: **BUP003, BBP005**. That means, your HR integration will support *employees* and *contingent worker*. **Relevant for connector version 1**
`ips.date.variable.format`	*yyyy-MM-dd* (needed for the read transformation)
`s4hana.cloud.user.roles.overwrite`	This property defines whether the current roles of a user to be preserved or overwritten by the Identity Provisioning service within the SAP S/4HANA Cloud Public Edition proxy system in a hybrid scenario. • **true** – the current user roles will be deleted in the proxy system, and the user will be updated only with the roles provisioned by the service. • **false** – the current user roles will be preserved, and the new roles (if any) will be added for the relevant user in the proxy system. See also: Extended Explanation of the *user.roles.overwrite Properties **Relevant for connector version 1**

Property Name	Description & Value
(Optional) `s4hana.cloud.support.patch.operation`	This property controls how modified entities (users and groups) in the source system are updated in the target system. • If set to **true**, PATCH operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, only these changes will be provisioned and applied in the target system. • If set to **false**, PUT operations are used to update users and groups in the target system. This means, for example, that if a user attribute is modified or a group member is removed from a group, all user attributes and all group attributes are replaced in the target system, instead of updating only the modified ones. Default value: *false* **Relevant for connector version 3**
(Optional) `s4hana.cloud.roles.filter`	Enter OData filtering for reading roles in the S/4HANA system. To learn what criteria you can use, see: OData URI Conventions → **4.5 Filter System Query Option** **Relevant for connector version 1**
(Optional) `s4hana.cloud.group.filter`	When specified, only those SAP S/4HANA Cloud Public Edition groups matching the filter expression will be read. Supported operators: eq (equal), co (contains), and sw (starts with). **Possible values:**: • `displayName` • `externalId` • `urn:ietf:params:scim:schemas:extension:sap:2.0:Group:type` For more information, see List of Properties [page 80]. **Relevant for connector version 3**

Property Name	Description & Value
(Optional) `s4hana.cloud.user.filter`	When specified, only those SAP S/4HANA Cloud Public Edition users matching the filter expression will be read. You can filter users by **list of attributes** according to the API syntax of SAP S/4HANA Cloud Public Edition. You can set a single attribute or multiple ones as filter criteria. If you enter multiple attributes (using OR operator), the filter will search for any of them. Supported operators: eq (equal), ne (not equal), sw (starts with), ew (ends with) and co (contains) **Possible values:** • `userName` • `externalId` • `emails[0].value` For example: • *userName eq "johnbrown" and externalId eq "P000252"* • *userName sw "J" or externalId eq "P000252"* • *userName eq "johnbrown" and emails[0].value eq "johnbrown@email.com"* • *userName eq "johnbrown" and emails[0].value ne "johnbrown@email.com" or externalId eq "P000252"* **Note** These combinations are valid for both 'or' and 'and' operators. **Relevant for connector version 3**
(Optional) `s4hana.cloud.roles.page.size`	Indicate how many business roles (considered as *groups*) per page to be read from your SAP S/4HANA Cloud Public Edition system. The value must be an integer number. **Relevant for connector version 1**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://my1234567-api.s4hana.ondemand.com
User=MyS4HANAuser
Password=************
s4hana.cloud.api.version=1
ips.date.variable.format=yyyy-MM-dd
s4hana.skip.read.archived=true
s4hana.onprem.hr.switch.active=true
s4hana.cloud.roles.filter=startswith(ID, 'EMPLOYEE_LEVEL_3') eq true
s4hana.cloud.user.roles.overwrite = false
s4hana.onprem.hr.switch.dependent.role.codes=BUP003,BBP005
s4hana.cloud.roles.page.size=30
```

6. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP S/4HANA Cloud Public Edition* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP S/4HANA Cloud Public Edition. For more information, see:

   Manage Transformations [page 2439]

   SAP S/4HANA Cloud API: Business User

   System for Cross-domain Identity Management (SCIM)

   Integrate Cross-Domain Identity Management

   Default read and write transformations for connector version 1:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.personID",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.personalInformation.firstName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.lastName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.middleName"
,
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.personFullN
ame",
        "targetPath":
"$.name.formatted",
        "optional": true
      },
      {
```

Write Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.personExternalID"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath":
"$.personExternalID"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath":
"$.user.globalUserID"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.personID"
      },
      {
        "targetPath":
"$.businessPartnerRoleCode",
        "type": "valueMapping",
        "sourcePaths": [
          "$.userType"
        ],
        "defaultValue": "BUP003",
        "valueMappings": [
          {
            "key": [
              "Employee"
            ],
            "mappedValue":
"BUP003"
          },
          {
            "key": [
              "Contingent Worker"
            ],
            "mappedValue":
"BBP005"
          }
        ]
      },
      {
        "scope": "createEntity",
```

Read Transformation

```
        "sourcePath":
"$.user.userName",
        "targetPath":
"$.userName",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "condition":
"$.user.lockedIndicator ==
'true'",
        "constant": false,
        "targetPath": "$.active",
        "optional": true
      },
      {
        "condition":
"($.user.validityPeriod.startDate
> '${currentDate}') ||
('${currentDate}' >
$.user.validityPeriod.endDate)",
        "constant": false,
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.workplaceInformation.emailAddre
ss",
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.user.logonLanguageCode",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath":
"$.PersonExternalID",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.user.globalUserID",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
```

Write Transformation

```
        "sourceVariable":
"currentDate",
        "targetPath":
"$.user.validityPeriod.startDate"
      },
      {
        "scope": "createEntity",
        "constant": "9999-12-31",
        "targetPath":
"$.user.validityPeriod.endDate"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.personalInformation.firstName"
      },
// The following conditions refer
to HR integration for your SAP
S/4HANA Cloud Public Edition
system. If HR integration is
activated
// (i.e. property
s4hana.cloud.hr.switch.active is
set to true), then you don't need
to provide family name for the
users.
// If it's
deactivated (i.e. property
s4hana.cloud.hr.switch.active is
missing or set to false, then you
have to provide family name.
// You can apply these conditions
to different user attributes,
analogically to name.familyName.
{
        "condition":
"%s4hana.cloud.hr.switch.active% !
= null &&
%s4hana.cloud.hr.switch.active%
== true",
        "optional": true,
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.personalInformation.lastName"
      },
      {
        "condition":
"%s4hana.cloud.hr.switch.active%
== null ||
%s4hana.cloud.hr.switch.active%
== false",
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.personalInformation.lastName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
```

Read Transformation

```
        "type": "valueMapping",
        "sourcePaths": [
          "$.user.timeZoneCode"
        ],
        "targetPath":
"$.timezone",
        "defaultValue": "Europe/
Berlin",
        "valueMappings": [
          {
            "key": [
              "WDFT"
            ],
            "mappedValue":
"Europe/Berlin"
          },
          {
            "key": [
              "ISRAEL"
            ],
            "mappedValue": "Asia/
Jerusalem"
          },
          {
            "key": [
              "RUS03"
            ],
            "mappedValue":
"Europe/Moscow"
          },
          {
            "key": [
              "AUSNSW"
            ],
            "mappedValue":
"Australia/Sydney"
          },
          {
            "key": [
              "UTC+4"
            ],
            "mappedValue": "Asia/
Dubai"
          },
          {
            "key": [
              "BRAZIL"
            ],
            "mappedValue":
"America/Sao_Paulo"
          },
          {
            "key": [
              "BRZLEA"
            ],
            "mappedValue":
"America/Sao_Paulo"
          },
          {
            "key": [
              "MSTNO"
            ],
```

Write Transformation

```
        "targetPath":
"$.personalInformation.middleName"
      },
      {
        "sourcePath":
"$.name.formatted",
        "optional": true,
        "targetPath":
"$.personalInformation.personFullN
ame"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.user.userName"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath":
"$.user.logonLanguageCode"
      },
      {
        "sourcePath":
"$.emails[0].value",
        "optional": true,
        "targetPath":
"$.workplaceInformation.emailAddre
ss"
      },
      {
        "condition": "$.active ==
false",
        "constant": "true",
        "targetPath":
"$.user.lockedIndicator"
      }
    ],
    "scimEntityEndpoint": "Users"
  },
  "group": {
    "mappings": [
      {
        "sourcePath":
"$.displayName",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
```

Read Transformation

```
            "mappedValue":
"America/Phoenix"
          },
          {
            "key": [
              "EST"
            ],
            "mappedValue":
"America/New_York"
          },
          {
            "key": [
              "UTC"
            ],
            "mappedValue": "Etc/
UTC"
          },
          {
            "key": [
              "UTC+3"
            ],
            "mappedValue": "Asia/
Riyadh"
          },
          {
            "key": [
              "EST_"
            ],
            "mappedValue":
"America/Toronto"
          },
          {
            "key": [
              "UTC+8"
            ],
            "mappedValue": "Asia/
Shanghai"
          },
          {
            "key": [
              "JAPAN"
            ],
            "mappedValue": "Asia/
Tokyo"
          }
        ]
      },
      {
        "type": "valueMapping",
        "sourcePaths": [

"$.businessPartnerRoleCode"
        ],
        "targetPath":
"$.userType",
        "defaultValue":
"Employee",
        "valueMappings": [
          {
            "key": [
              "BUP003"
            ],
```

Write Transformation

```
        "targetPath": "$.members[?
(@.value)]"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ],
    "scimEntityEndpoint": "Groups"
  }
}
```

Read Transformation	Write Transformation

```
            "mappedValue":
"Employee"
          },
          {
            "key": [
              "BBP005"
            ],
            "mappedValue":
"Contingent Worker"
          }
        ]
      },
      {
        "sourcePath":
"$.user.role[*].roleName",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups[?
(@.value)]"
      }
    ]
  },
   "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.ID",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.ID",
        "targetPath":
"$.displayName"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
```

Read Transformation	Write Transformation

```
      {
        "sourcePath":
"$.to_BusinessUserAssignment.resul
ts",
        "optional": true,

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.members"
      },
      {
        "constant":
"authorization",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']"
      },
      {
        "constant": "readWrite",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']"
      },
      {
        "type": "remove",
        "targetPath":
"$.members[*].__metadata"
      },
      {
        "type": "remove",
        "targetPath":
"$.members[*].UserName"
      },
      {
        "type": "rename",
        "constant": "value",
        "targetPath":
"$.members[*].PersonID"
      },
      {
        "constant": "User",
        "targetPath":
"$.members[*].type"
      }
    ]
  }
}
```

By default, Identity Provisioning reads group IDs and members. If you want the service to also read group descriptions, you can add an extra mapping to the *"group"* resource in the *Read Transformation*. To learn how, see Guided Answers: Business Role Description.

Default read and write transformations for connector version 3:

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
            {
                "type":
"concatString",
                "suffix": "$
{entityIdSourceSystem}"
            }
         ]
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath":
"$.name.formatted",
        "optional": true,
        "targetPath":
"$.name.formatted"
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath":
"$.name.formatted",
        "optional": true,
        "targetPath":
"$.name.formatted"
      },
      {
        "sourcePath":
"$.name.familyName",
```

Read Transformation	Write Transformation

<pre>},
{
"sourcePath":
"$.name.familyName",
"optional": true,
"targetPath":
"$.name.familyName"
},
{
"sourcePath":
"$.name.givenName",
"optional": true,
"targetPath":
"$.name.givenName"
},
{
"sourcePath":
"$.displayName",
"optional": true,
"targetPath":
"$.displayName"
},
{
"sourcePath": "$.emails",
 	
"preserveArrayWithSingleElement":
true,
"optional": true,
"targetPath": "$.emails"
},
{
"sourcePath":
"$.phoneNumbers",
 	
"preserveArrayWithSingleElement":
true,
"targetPath":
"$.phoneNumbers",
"optional": true
},
{
"sourcePath":
"$.userType",
"optional": true,
"targetPath": "$.userType"
},
{
"sourcePath":
"$.preferredLanguage",
"optional": true,
"targetPath":
"$.preferredLanguage"
},
{
"sourcePath": "$.locale",
"optional": true,
"targetPath": "$.locale"
},
{
"sourcePath":
"$.timezone",
"optional": true,</pre>
	<pre>"optional": true,
	"targetPath":
	"$.name.familyName"
	},
	{
	"sourcePath":
	"$.name.givenName",
	"optional": true,
	"targetPath":
	"$.name.givenName"
	},
	{
	"sourcePath":
	"$.displayName",
	"optional": true,
	"targetPath":
	"$.displayName"
	},
	{
	"sourcePath": "$.emails",
	
	"preserveArrayWithSingleElement":
	true,
	"targetPath": "$.emails",
	"optional": true
	},
	{
	"sourcePath":
	"$.userType",
	"optional": true,
	"targetPath": "$.userType"
	},
	{
	"sourcePath":
	"$.preferredLanguage",
	"optional": true,
	"targetPath":
	"$.preferredLanguage"
	},
	{
	"sourcePath": "$.locale",
	"optional": true,
	"targetPath": "$.locale"
	},
	{
	"sourcePath":
	"$.timezone",
	"optional": true,
	"targetPath": "$.timezone"
	},
	{
	"sourcePath": "$.active",
	"optional": true,
	"targetPath": "$.active"
	},
	{
	"sourcePath": "$
	['urn:ietf:params:scim:schemas:ext
	ension:sap:2.0:User']
	['validFrom']",
	"targetPath": "$
	['urn:ietf:params:scim:schemas:ext</pre>

Read Transformation	Write Transformation

```
        "targetPath": "$.timezone"
      },
      {
        "sourcePath": "$.active",
        "optional": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['language']
['correspondence']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['language']
['correspondence']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']
['numericalRepresentation']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']
['numericalRepresentation']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['date']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['date']",
        "optional": true
      },
```

```
ension:sap:2.0:User']
['validFrom']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['validTo']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['language']
['correspondence']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['language']
['correspondence']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']
['numericalRepresentation']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']
['numericalRepresentation']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['date']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['date']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['time']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['time']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation

```
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['time']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['localeSettings']['time']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuidHistory']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuidHistory']",
        "optional": true,

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
```

Write Transformation

```
ension:sap:2.0:User']
['userUuid']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuidHistory']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuidHistory']",
        "optional": true,

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:Group",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:Group"
        ],
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
```

Read Transformation

```
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
           {
               "type":
"concatString",
               "suffix": "$
{entityIdSourceSystem}"
           }
        ]
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
       {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
```

Write Transformation

```
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      }
    ]
  }
}
```

Read Transformation	Write Transformation

<pre>]
}
}</pre>	

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PATCH**.

   > ⓘ Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PATCH** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.39 SAP S/4HANA for procurement planning

Follow this procedure to set up SAP S/4HANA for procurement planning as a proxy system.

Prerequisites

You have technical credentials for SAP S/4HANA for procurement planning. See: Onboadring

Context

SAP S/4HANA for procurement planning is a cloud-based solution designed to help you plan procurement activities with regard to the time schedule, as well as the investment planning of items based on a central bill of material.

You can use Identity Provisioning to configure SAP S/4HANA for procurement planning as a proxy system in hybrid scenarios. For example, when SAP S/4HANA for procurement planning is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users back to the SAP S/4HANA for procurement planning.

This scenario supports provisioning **users**.

> ⓘ Note
>
> SAP S/4HANA for procurement planning does not support groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.

4. Save your changes.

3. Add *SAP S/4HANA Procurement Planning* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the SCIM API of your SAP S/4HANA for procurement planning system without path information. For example: `https://procplanning-api.cfapps.eu10.hana.ondemand.com`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the OAuth Client Id, created for your SAP S/4HANA for procurement planning system.
`Password`	Enter the OAuth Client Secret, created for your SAP S/4HANA for procurement planning system.
`OAuth2TokenServiceURL`	Enter the OAuth 2.0 Token Service URL. For example: **https://procplansecurity.authentication.eu10.hana.ondemand.com/oauth/token**
(Optional) `s4hana.pp.user.filter`	When specified, only those SAP S/4HANA for procurement planning users matching the filter expression will be read. Example: **name.familyName eq "Smith" and addresses.country eq "US"**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users. For more information, see Query Parameters for Proxy System SCIM API.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default

transformation for the *SAP S/4HANA for procurement planningTransformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your SAP S/4HANA for procurement planning system. For more information, see:

Manage Transformations [page 2439].

SAP Business Accelerator Hub: SAP S/4HANA for Procurement Planning

Default read and write transformations:

> → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.userName",

"correlationAttribute": true,
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.emails[0].value",
                "targetPath":
"$.emails[0].value",
                "optional": true
            },
            {
                "sourcePath":
"$.emails[?(@.primary==
true)].value",

"correlationAttribute": true
            },
            {
                 "sourcePath":
"$.displayName",
                 "targetPath":
"$.displayName"
            },
            {
```

Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                 "sourcePath":
"$.displayName",
                 "targetPath":
"$.displayName"
            },
            {
                 "sourcePath":
"$.active",
                 "optional": true,
                 "targetPath":
"$.active"
            },
             {
                 "sourcePath":
"$.externalId",
                 "optional": true,
                 "targetPath":
"$.externalId"
            },
            {
                 "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                 "targetPath":
"$.emails"
            }
        ]
    }
}
```

Read Transformation	Write Transformation
<pre>"sourcePath": "$.active", "targetPath": "$.active" }] } }</pre>	

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ⓘ Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*

To learn more, see: Proxy Systems [page 1343]

Related Information

SAP S/4HANA for procurement planning – Product Page

1.6.3.40 SAP S/4HANA On-Premise

Follow this procedure to set up SAP S/4HANA on-premise (also valid for SAP S/4HANA Cloud, private edition) as a proxy system.

Prerequisites

> **Note**
>
> If you have purchased the Identity Provisioning service between **September 1, 2020** and **October 20, 2020**, and you want to make a connection to this on-premise system, follow the procedure in: Connect to On-Premise Systems in SAP Cloud Identity Infrastructure.

- You have installed the Cloud Connector in your corporate environment and have done the initial configuration. For more information, see: Cloud Connector (Neo)
- You have technical credentials (user and password) for an external back-end system with read and write permissions.
- You have technical credentials (user and password) for SAP S/4HANA on-premise.
- The SAP S/4HANA on-premise system is version **1809** or higher.
- You have configured your SOA Manager to directly call the following Web services:
  - **ManageBusinessUserIn**
  - **QueryBusinessUserIn**

  For more information, see: Setting Up SOA Manager.

Context

You can use SAP S/4HANA on-premise as a proxy connector to execute *hybrid* scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to as a proxy connector to execute SAP S/4HANA on-premise, whenever the external back-end requests such. This scenario supports:

- Reading of **users**

- Writing of **users**

SAP S/4HANA on-premise supports provisioning of users with *User UUID* attribute which is generated by Identity Authentication at user creation. The attribute mapping is handled by the default transformation of AS ABAP. Therefore, to provision a user with *User UUID* to or from S/4HANA on-premise, the user should first be provisioned to AS ABAP and then linked to its corresponding business user in S/4HANA on-premise. For more information, see: SAP Application Server ABAP [page 1424].

According to your use case, you can decide whether to use SAP S/4HANA on-premise with HR (human resources) integration active or not.

- System with HR integration – When the HR integration is active, business users in SAP S/4HANA on-premise are created and managed by the HR system. The Identity Provisioning service can manage only the user-related login information, such as date/time preferences, or role assignments.
  This means that once the business users are created in SAP S/4HANA on-premise, you need to provision the users to AS ABAP so that they are linked to the business users (employee, collaboration user, contingent worker, resource) via the *User_Assignment* attribute.
- System without HR integration – lean business users will be created after the provisioning job, and the AS ABAP users will be linked to them. The Identity Provisioning service manages the complete set of **business partners** and their relevant **business users**.

To enable HR integration, you need to specify the relevant property in the system configuration. See **step 7** from the procedure below.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Open the Cloud Connector to add an access control system mapping for **SAP S/4HANA On-Premise**. This is needed to allow the Identity Provisioning service to access SAP S/4HANA On-Premise as a back-end system on the intranet. To learn how, see: Configure Access Control (HTTP)
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
3. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.

4. Save your changes.

4. Add *SAP S/4HANA On-Premise* as a proxy system. For more information, see Add New Systems [page 2415].
5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to your SAP S/4HANA On-Premise system.
`ProxyType`	Enter: *OnPremise*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the technical user for SAP S/4HANA On-Premise.
`Password`	(Credential) Enter the password for the SAP S/4HANA On-Premise technical user.
`s4hana.onprem.skip.read.archived`	In the event of archived (disabled) entities in a source SAP S/4HANA On-Premise system, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to **false**, or delete it.
`s4hana.onprem.hr.switch.active`	This property is disabled by default. Possible values: • **true** – HR integration is enabled for your system • **false** (default value) – HR integration is disabled for your system
`s4hana.onprem.hr.switch.dependent.role.codes`	A default property. Relevant only for systems with activated HR integration, that is if `s4hana.onprem.hr.switch.active` = *true*. As a comma-separated value, add the codes of the roles maintained by the HR integration. Make sure these role codes are part of your *read* and *write* transformations. By default, the following codes are added to your system: **BUP003, BBP005, BUP012, WFM001**. That means, your HR integration will support *employees*, *contingent worker*, *collaboration user*, and *resource*.
`ips.date.variable.format`	*yyyy-MM-dd* (needed for the read transformation)

Property Name	Description & Value
(Optional) `s4hana.onprem.sap-client`	Use this property if you want to specify a particular AS ABAP client to use as the **sap-client** URL parameter. If this property is not specified, the URL will open your default AS ABAP client. To learn more, see: Specifying the Client For more information about **sap-client**, see: SAP URL Parameters
(Optional)`CloudConnectorLocationId`	Relevant when the `ProxyType` property is set to *OnPremise*. Use it in case your SAP Business Technology Platform account uses more than one Cloud Connector or when the Cloud Connector uses `LocationID`.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*OnPremise*

`URL`=*http://aaa777.myhost:1234*

`User`=*MYS4HANAUSER*

`Password`=**********

`ips.date.variable.format`=*yyyy-MM-dd*

`s4hana.onprem.skip.read.archived`=*true*

`s4hana.onprem.hr.switch.active`=*true*

`s4hana.onprem.hr.switch.dependent.role.codes` = *BUP003,BBP010,BBP005*

`s4hana.onprem.sap-client`=*101*

6. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP S/4HANA On-Premise* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP S/4HANA On-Premise. For more information, see:

   Manage Transformations [page 2439]

Default read and write transformations:

> → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.personID",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.personalInformation.firstName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.lastName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.middleName"
,
        "targetPath":
"$.name.middleName",
        "optional": true
      },
      {
        "sourcePath":
"$.personalInformation.personFullN
ame",
        "targetPath":
"$.name.formatted",
        "optional": true
      },
      {
```

Write Transformation

Code Syntax

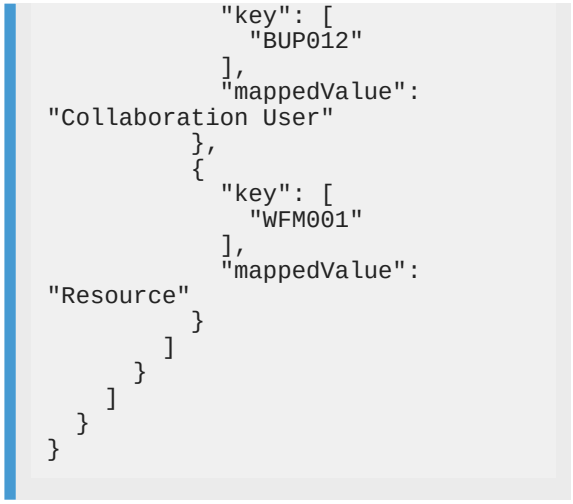
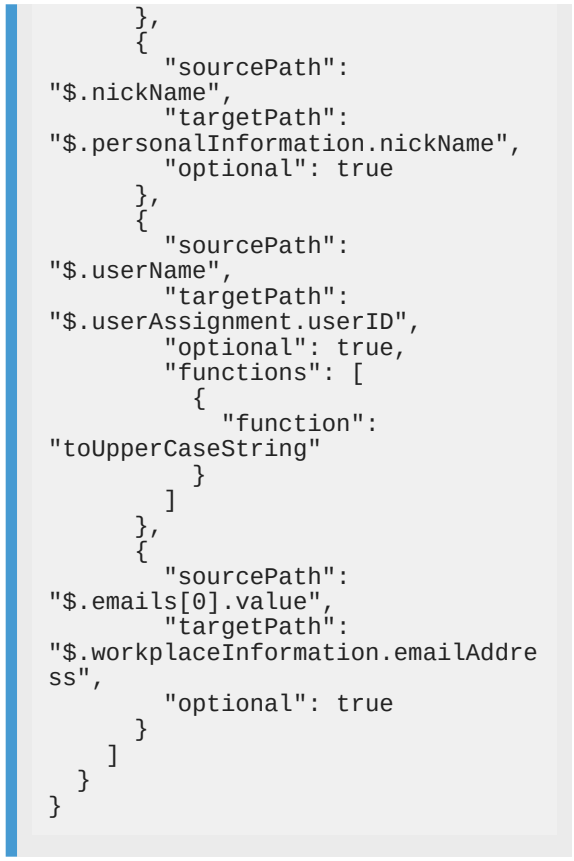
```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.personExternalID"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath":
"$.personExternalID"
      },
      {
        "targetPath":
"$.personID",
        "sourceVariable":
"entityIdTargetSystem"
      },
      {
        "type": "valueMapping",
        "sourcePaths": [
          "$.userType"
        ],
        "targetPath":
"$.businessPartnerRoleCode",
        "defaultValue": "BUP003",
        "valueMappings": [
          {
            "key": [
              "Employee"
            ],
            "mappedValue":
"BUP003"
          },
          {
            "key": [
              "Contingent Worker"
            ],
            "mappedValue":
"BBP005"
          },
          {
            "key": [
              "Collaboration User"
            ],
            "mappedValue":
"BUP012"
          },
          {
            "key": [
              "Resource"
            ],
```

Read Transformation

```
        "sourcePath":
"$.personalInformation.nickName",
        "targetPath":
"$.nickName",
        "optional": true
      },
      {
        "sourcePath":
"$.userAssignment.userID",
        "targetPath":
"$.userName",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "constant": true,
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.workplaceInformation.emailAddre
ss",
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.PersonExternalID",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "type": "valueMapping",
        "sourcePaths": [

"$.businessPartnerRoleCode"
        ],
        "targetPath":
"$.userType",
        "defaultValue":
"Employee",
        "valueMappings": [
          {
            "key": [
              "BUP003"
            ],
            "mappedValue":
"Employee"
          },
          {
            "key": [
              "BBP005"
            ],
            "mappedValue":
"Contingent Worker"
          },
          {
```

Write Transformation

```
            "mappedValue":
"WFM001"
          }
        ]
      },
      {
        "scope": "createEntity",
        "targetPath":
"$.validityPeriod.startDate",
        "sourceVariable":
"currentDate"
      },
      {
        "scope": "createEntity",
        "targetPath":
"$.validityPeriod.endDate",
        "constant": "9999-12-31"
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.personalInformation.firstName",
        "optional": true
      },
      {
        "condition":
"%s4hana.onprem.hr.switch.active%
!= null &&
%s4hana.onprem.hr.switch.active%
== true",
        "optional": true,
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.personalInformation.lastName"
      },
      {
        "condition":
"%s4hana.onprem.hr.switch.active%
== null ||
%s4hana.onprem.hr.switch.active%
== false",
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.personalInformation.lastName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.personalInformation.middleName"
,
        "optional": true
      },
      {
        "sourcePath":
"$.name.formatted",
        "targetPath":
"$.personalInformation.personFullN
ame",
        "optional": true
```

Read Transformation	Write Transformation
	

7. (Optional) If the external consumer system is **SAP Identity Management** (IDM), you can export the newly created proxy system as a *.csv* file. This will reduce the effort of manually entering all the properties again when an IDM administrator import the proxy configuration as a repository.

   To export the proxy system from the UI, choose *Export* > *CSV format*.

> → Tip
>
> When you import the *.csv* file as a SCIM repository in IDM, all the property fields will be automatically filled-in. However:
>
> - You have to manually enter your client ID and secret (AUTH_USER and AUTH_PASSWORD).
> - As currently the scenario does not support groups, you cannot make group assignments for users. Thus, for the SCIM_ASSIGNMENT_METHOD constant, remove the value (leave the field empty).

> ⓘ Note
>
> If you have purchased the Identity Provisioning service after **September 1, 2020**, for AUTH_USER and AUTH_PASSWORD, enter the technical user and password for which you have granted access to the Identity Provisioning Proxy API.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

SAP S/4HANA On-Premise
APIs for Business User Management
Maintain Collaboration Users

1.6.3.41 SAP SuccessFactors

Follow this procedure to set up SAP SuccessFactors as a proxy system.

Prerequisites

- You are using **SAP SuccessFactors HCM Suite OData API**
  You have created a technical user with permissions to **call** the SAP SuccessFactors HCM Suite OData API to **import** employee data into an SAP SuccessFactors system. For more information, see Permissions, Setting Up an API User for Sync Jobs in SAP SuccessFactors and URI Conventions (OData Version 2.0) .
- You are using **SAP SuccessFactors Workforce SCIM API**
  Consumers of the SAP SuccessFactors Workforce SCIM API can either create a technical user and assign the necessary permissions, or use the predefined technical user with built-in permissions for

communication between SAP SuccessFactors and Identity Provisioning using certificate authentication. For more information, see Upgrade to X.509 Certificate-Based Authentication for Incoming Calls and Permissions.

Context

You can use SAP SuccessFactors as a proxy connector to execute *hybrid* scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to SAP SuccessFactors, whenever the external back-end requests such.

SAP SuccessFactors provides two APIs for its integration with Identity Provisioning: SAP SuccessFactors HCM Suite OData API and SAP SuccessFactors Workforce SCIM API. The value of `sf.api.version` property controls which API you use.

- When the value is set to **`1`**, or the property is not defined - SAP SuccessFactors HCM Suite OData API (in short, OData API) is used. This is the default value. SAP SuccessFactors source systems created before the introduction of `sf.api.version` property, use OData API. This version allows you to read users and groups, both static and dynamic. It also supports writing users, updating dynamic groups and group members.

  > ⚠ Restriction
  >
  > Note the following restrictions when using SAP SuccessFactors HCM Suite OData API:
  >
  > - You cannot *create* or *delete* groups as these operations are currently not supported.
  > - Managing SAP SuccessFactors **static groups** is not supported.

- When the value is set to **`2`** - SAP SuccessFactors Workforce SCIM API (in short, SCIM API) is used.
  This version allows you to provision static permission groups and user's group assignments. Provisioning of user's group assignments from a given external system to SAP SuccessFactors proxy system is possible if the user is an active employee with a work assignment.
  To update a group from the external system, a group with the same name should already exist in SAP SuccessFactors. For more information about the difference between static and dynamic groups in SAP SuccessFactors, see Permission Groups.

  > ⚠ Restriction
  >
  > Note the following restrictions when using SAP SuccessFactors Workforce SCIM API:
  >
  > - You cannot *create* or *delete* groups using the `Groups` APIs.
  > - The `Groups` patch API only supports updating membership of static permission groups.
  >
  > For more information, see Overview of SAP SuccessFactors Workforce System for Cross-Domain Identity Management API.

For more information on how to update to version 2, see Update Connector Version [page 2424].

> ⓘ Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP SuccessFactors* as a proxy system. For more information, see Add New Systems [page 2415].
4. Set up the communication between Identity Provisioning and SAP SuccessFactors and configure your authentication method (basic or certificate-based).

   > **Note**
   >
   > We recommend that you use certificate-based authentication.

   a. In your newly added SAP SuccessFactors proxy system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

      Skip this step if you use basic authentication. The next steps are performed in SAP SuccessFactors Admin Center and are relevant for certificate-based authentication only.

   b. Login to SAP SuccessFactors and go to *Admin Center*. Follow the procedure described in Upgrade to X.509 Certificate-Based Authentication for Incoming Calls.

      Make sure you select *Identity Provisioning Service* in the *Integration Name* field.
5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
Type	Enter: *HTTP*

Property Name	Description & Value

URL	Specify the URL to your SAP SuccessFactors API.
	
	For example:
	• For version 1:
	  • When *BasicAuthentication* is configured as authentication method: **`https://apitest.successfactors.com/odata/v2`**
	  • When *ClientCertificateAuthentication* is configured as authentication method: **`https://apitest.cert.successfactors.com/odata/v2`**
	• For version 2:
	  • When *BasicAuthentication* is configured as authentication method: **`https://apitest.successfactors.com`**
	  • When *ClientCertificateAuthentication* is configured as authentication method: **`https://apitest.cert.successfactors.com`**
	
	To see the list of all SAP SuccessFactors data centers, see: SAP SuccessFactors API Reference Guide (OData V2): List of SAP SuccessFactors API Servers and System for Cross-domain Identity Management for Workforce in SuccessFactors

`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication*
(Optional) `sf.api.version`	Handles the version of the API which is consumed by the SAP SuccessFactors system. **Possible values:** • *1* - Indicates that SAP SuccessFactors HCM Suite OData API (in short, OData API) is used. • *2* - Indicates that SAP SuccessFactors Workforce SCIM API (in short, SCIM API) is used. Default value: *1*

Property Name	Description & Value
`User`	Valid if *BasicAuthentication* is configured as authentication method. Enter the *userID* of your SAP SuccessFactors technical user in the following format: **<user_ID>@<company_ID>**
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. Enter the password for your SAP SuccessFactors technical user.
`sf.company.id`	Valid if *ClientCertificateAuthentication* is configured as authentication method. Enter the Company ID of your SAP SuccessFactors system. The Company ID is a short string of characters that identifies each SAP SuccessFactors system. It is like a username for your organization. All users of the same system share the same Company ID.
`sf.user.attributes`	Default property. It's a string representing a comma-separated list of user attributes that have to be loaded (read) from SAP SuccessFactors. You can leave the default property value (all listed attributes), or leave only some of them. → Remember • Always make sure that attribute `lastModifiedDateTime` is in the list of values. If you don't specify it, the provisioning from SAP SuccessFactors will fail. • If a user in SAP SuccessFactors is missing this attribute, it will break the provisioning. You can exclude them from the provisioning (either by using the `sf.user.filter` property, or by setting a condition in the transformation logic). **Connector version**: SAP SuccessFactors version 1
(Optional) `sf.user.attributes.expand`	This property reads additional user data related to *complex attributes*, which are specified in `sf.user.attributes`. Default value: *personKeyNav* **Connector version**: SAP SuccessFactors version 1

Property Name	Description & Value
(Optional) `sf.user.filter`	The possible values of this property depend on the API version which your SAP SuccessFactors system consumes. Use this property to filter users from SAP SuccessFactors. The filter obtains values as described in the OData 2.0 syntax, except any statements with attribute `lastModifiedDateTime`. To learn more, see: • OData version 2 → **4.5. Filter System Query Option ($filter)**. • SAP SuccessFactors API Reference Guide (OData V2) • SAP SuccessFactors Workforce SCIM API and System for Cross-domain Identity Management for Workforce in SuccessFactors

Property Name	Description & Value
(Optional) `sf.user.read.deactivatedafter`	This property filters SAP SuccessFactors inactive users from a particular date on. It is an optional property which does not appear by default at system creation. It accepts a value in the **`yyyy-MM-dd`** format. For example: **`2023-07-17`** The `sf.user.read.deactivatedafter` property works together with the `sf.user.filter` property which is added at system creation with the default value: *active eq true*. Using it can further narrow down the filtering results. To filter active users along with inactive ones from a particular date on, the following configuration must be in place: • Set the `sf.user.read.deactivatedafter` value to a date in the expected format. For example: *2023-07-17* • `sf.user.filter` = *active eq true* As a result, Identity Provisioning reads SAP SuccessFactors active users and the users set to inactive from that date on using the 2023-07-17T00:00:00Z date-time format. Depending on the value you define for `sf.user.filter`, expect the following results: • `sf.user.filter` = *active eq false* All inactive users will be returned. • `sf.user.filter` = *active eq false and userName sw "Test_"* All inactive users with username starting with Test_ will be returned. **Note** When you filter by `sf.user.filter` = *active eq false* along with the property `sf.user.read.deactivatedafter`, the users that match the two critera will be read twice. • `sf.user.filter` = *active eq true and userName sw "Test_"* Inactive users from the provided date on and all active users with username starting with Test_ will be returned.

Property Name	Description & Value
	Connector version: SAP SuccessFactors version 2
(Optional) `sf.group.filter`	The possible values of this property depend on the API version which your SAP SuccessFactors system consumes. Use this property to filter dynamic groups in SAP SuccessFactors. The filter obtains values as described in the OData 2.0 syntax, except any statements with attribute `lastModifiedDateTime`. To learn more, see: • OData version 2 → **4.5. Filter System Query Option ($filter)** • SAP SuccessFactors API Reference Guide (OData V2) → **DynamicGroup** • SAP SuccessFactors Workforce SCIM API and System for Cross-domain Identity Management for Workforce in SuccessFactors
(Optional) `sf.page.size`	Defines the paging size. • Default value: **100** • Maximum value: **1000** **Connector version**: SAP SuccessFactors version 1
(Optional)`sf.group.members.paging.enabled`	This property enables paging of group members. The maximum number of group members returned per request is 100. To read more than 100 group members, paging must be enabled. **Possible values:** • *true* - Paging is enabled. • *false* - Paging is disabled. Default value: *true* **Connector version**: SAP SuccessFactors version 2

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination (configuration):

```
Type=HTTP
Authentication=BasicAuthentication
ProxyType=Internet
URL=https://apitest.successfactors.com/odata/v2
User=sfsf_admin@mycompany.com
Password=************
sf.user.attributes=userId,username,addressLine1,lastName,country,email,location,firstName,lastModifiedDateTime,personKeyNav,manager/username
sf.user.attributes.expand=personKeyNav,manager
sf.user.filter=department ne 'Manufacturing'
sf.group.filter=groupType eq 'permission'
sf.page.size=70
```

6. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP SuccessFactors* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP SuccessFactors. For more information, see:

   Manage Transformations [page 2439]

   SAP SuccessFactors HCM Suite OData API

   SAP SuccessFactors Workforce SCIM API

   Default read and write transformations for SAP SuccessFactors HCM Suite OData API version 1:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external

application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation	Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.userId",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.username",
        "targetPath":
"$.userName",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.personKeyNav.userAccountNav.use
rname",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.firstName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.lastName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.email",
```

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "skipOperations": [
      "delete"
    ],
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.userId"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userId",
        "scope": "createEntity"
      },
      {
        "constant": "t",
        "targetPath": "$.status"
      },
      {
        "condition": "$.active ==
false",
        "constant": "f",
        "targetPath": "$.status"
      },
      {
        "sourcePath":
"$.userName",
        "optional": true,
        "targetPath": "$.username"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.lastName",
        "defaultValue": null
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.firstName",
        "defaultValue": null
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "optional": true,
        "targetPath":
"$.salutation",
        "defaultValue": null
      },
      {
```

Read Transformation

```
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath":
"$.personKeyNav.personIdExternal",
        "targetPath":
"$.externalId",
        "optional": true,
        "correlationAttribute":
true
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.groupID",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
            {
                "type":
"concatString",
                "suffix": "$
{entityIdSourceSystem}"
            }
        ]
      },
      {
        "sourcePath":
"$.groupName",
        "targetPath":
"$.displayName"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
```

Write Transformation

```
        "sourcePath":
"$.name.honorificSuffix",
        "optional": true,
        "targetPath": "$.suffix",
        "defaultValue": null
      },
      {
        "sourcePath":
"$.emails[0].value",
        "optional": true,
        "targetPath": "$.email",
        "defaultValue": null
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value != []",
        "sourcePath": "$.emails[?
(@.primary == true)].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "defaultValue": null,
        "targetPath": "$.email",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath":
"$.timeZone",
        "defaultValue": null
      },
      {
        "sourcePath":
"$.nickName",
        "optional": true,
        "targetPath":
"$.nickname",
        "defaultValue": null
      },
      {
        "sourcePath":
"$.addresses[0].country",
        "optional": true,
        "targetPath": "$.country",
        "defaultValue": null
      },
      {
        "condition":
"$.addresses[?(@.primary ==
true)].country != []",
        "sourcePath":
"$.addresses[?(@.primary ==
true)].country",
```

Read Transformation

```
      },
      {
        "sourcePath":
"$.users[*].userId",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.members[?
(@.value)]",
        "optional": true
      },
      {
        "constant": "User",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.members[*].type"
      }
    ]
  }
}
```

Write Transformation

```
"preserveArrayWithSingleElement":
true,
        "optional": true,
        "defaultValue": null,
        "targetPath": "$.country",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "sourcePath":
"$.addresses[0].locality",
        "optional": true,
        "targetPath": "$.city",
        "defaultValue": null
      },
      {
        "condition":
"$.addresses[?(@.primary ==
true)].locality != []",
        "sourcePath":
"$.addresses[?(@.primary ==
true)].locality",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "defaultValue": null,
        "targetPath": "$.city",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "sourcePath":
"$.addresses[0].formatted",
        "optional": true,
        "defaultValue": null,
        "targetPath":
"$.addressLine1"
      },
      {
        "sourcePath":
"$.addresses[1].formatted",
        "optional": true,
        "targetPath":
"$.addressLine2",
        "defaultValue": null
      },
      {
        "sourcePath":
"$.addresses[2].formatted",
        "optional": true,
```

Read Transformation	Write Transformation

```
        "targetPath":
"$.addressLine3",
        "defaultValue": null
      },
      {
        "condition":
"$.phoneNumbers[?(@.type ==
'work')].value != []",
        "sourcePath":
"$.phoneNumbers[?(@.type ==
'work')].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "defaultValue": null,
        "targetPath":
"$.businessPhone",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "condition":
"$.phoneNumbers[?(@.type ==
'fax')].value != []",
        "sourcePath":
"$.phoneNumbers[?(@.type ==
'fax')].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "defaultValue": null,
        "targetPath": "$.fax",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "condition":
"$.phoneNumbers[?(@.type ==
'mobile')].value != []",
        "sourcePath":
"$.phoneNumbers[?(@.type ==
'mobile')].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "defaultValue": null,
        "targetPath":
"$.cellPhone",
        "functions": [
```

Read Transformation	Write Transformation

```
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "condition":
"$.phoneNumbers[?(@.type ==
'home')].value != []",
        "sourcePath":
"$.phoneNumbers[?(@.type ==
'home')].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "defaultValue": null,
        "targetPath":
"$.homePhone",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['employeeNumber']",
        "optional": true,
        "targetPath": "$.empId",
        "defaultValue": null
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
        "optional": true,
        "targetPath":
"$.division",
        "defaultValue": null
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
        "optional": true,
        "targetPath":
"$.department",
        "defaultValue": null
      }
    ]
  },
  "group": {
```

Read Transformation	Write Transformation
	"scimEntityEndpoint": "Groups", "skipOperations": ["create", "delete"], "mappings": [{ "sourceVariable": "entityIdTargetSystem", "targetPath": "$.groupID" }, { "sourcePath": "$.displayName", "targetPath": "$.groupName" }, { "constant": "DynamicGroup", "targetPath": "$.__metadata.uri" }, { "constant": "permission", "targetPath": "$.groupType" }, { "optional": true, "preserveArrayWithSingleElement": true, "sourcePath": "$.members[*].value", "targetPath": "$.members[?(@.value)]" }] } }

Default read and write transformations for SCIM API version 2:

Read Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.userType",
                "targetPath":
"$.userType"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional": true,
                "targetPath":
"$.name.familyName"
            },
```

Write Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "constant": [

"urn:ietf:params:scim:schemas:exte
nsion:successfactors:2.0:User",

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:enterprise:2.0:User"
                ],
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.nickName",
                "optional": true,
                "targetPath":
"$.nickName"
            },
            {
                "sourcePath":
"$.preferredLanguage",
                "optional": true,
                "targetPath":
"$.preferredLanguage"
            },
            {
                "sourcePath":
"$.userType",
                "targetPath":
"$.userType"
            },
            {
                "sourcePath":
"$.externalId",
                "optional": true,
```

Read Transformation

```
            {
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional": true,
                "targetPath":
"$.name.middleName"
            },
            {
                "sourcePath":
"$.name.honorificPrefix",
                "optional": true,
                "targetPath":
"$.name.honorificPrefix"
            },
            {
                "sourcePath":
"$.name.honorificSuffix",
                "optional": true,
                "targetPath":
"$.name.honorificSuffix"
            },
            {
                "sourcePath":
"$.name.formatted",
                "optional": true,
                "targetPath":
"$.name.formatted"
            },
            {
                "sourcePath":
"$.nickName",
                "optional": true,
                "targetPath":
"$.nickName"
            },
            {
                "sourcePath":
"$.preferredLanguage",
                "optional": true,
                "targetPath":
"$.preferredLanguage"
            },
            {
                "sourcePath":
"$.displayName",
                "optional": true,
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.title",
                "optional": true,
                "targetPath":
"$.title"
```

Write Transformation

```
                "targetPath":
"$.externalId"
            },
            {
                "sourcePath":
"$.displayName",
                "optional": true,
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.locale",
                "optional": true,
                "targetPath":
"$.locale"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.emails"
            },
            {
                "condition":
"$.emails.length() > 0",
                "targetPath":
"$.emails[*].type",
                "constant": "work"
            },
            {
                "sourcePath":
"$.timezone",
                "optional": true,
                "targetPath":
"$.timezone"
            },
            {
                "sourcePath":
"$.name.formatted",
                "optional": true,
                "targetPath":
"$.name.formatted"
            },
            {
                "sourcePath":
"$.name.familyName",
                "optional": true,
```

Read Transformation

```
            },
            {
                "sourcePath":
"$.externalId",
                "optional": true,
                "targetPath":
"$.externalId"
            },
            {
                "sourcePath":
"$.locale",
                "optional": true,
                "targetPath":
"$.locale"
            },
            {
                "sourcePath":
"$.timezone",
                "optional": true,
                "targetPath":
"$.timezone"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['perPersonUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['perPersonUuid']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['loginMethod']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['loginMethod']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['personIdExternal']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

```
                "targetPath":
"$.name.familyName"
            },
            {
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.name.givenName"
            },
            {
                "sourcePath":
"$.name.middleName",
                "optional": true,
                "targetPath":
"$.name.middleName"
            },
            {
                "sourcePath":
"$.name.honorificPrefix",
                "optional": true,
                "targetPath":
"$.name.honorificPrefix"
            },
            {
                "sourcePath":
"$.name.honorificSuffix",
                "optional": true,
                "targetPath":
"$.name.honorificSuffix"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['perPersonUuid']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['perPersonUuid']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['loginMethod']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['loginMethod']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['personIdExternal']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['personIdExternal']"
```

Read Transformation

```
ension:successfactors:2.0:User']
['personIdExternal']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['customFields']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['customFields']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Write Transformation

```
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['customFields']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:successfactors:2.0:User']
['customFields']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['department']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['division']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['value']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['$ref']",
                "optional": true,
```

Read Transformation

```
ension:enterprise:2.0:User']
['manager']['$ref']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['$ref']"
            },
            {
                "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.phoneNumbers"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath":
"$.emails[?(@.primary ==
true)].value",
                "optional": true,

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.active",
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.groups",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups"
            }
        ],
        "scimEntityEndpoint":
"Users"
    },
    "group": {
        "mappings": [
            {
                "sourcePath":
"$.id",
```

Write Transformation

```
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['$ref']"
            },
            {
                "sourcePath":
"$.title",
                "optional": true,
                "targetPath":
"$.title"
            },
            {
                "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.phoneNumbers"
            },
            {
                "sourcePath":
"$.active",
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.groups[*].value",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups[?(@.value)]",
                "functions": [
                    {

"entityType": "group",

"function": "resolveEntityIds"
                    }
                ]
            },
            {
                "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.Operations",
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
```

Read Transformation

```
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.members",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members"
            }
        ],
        "scimEntityEndpoint":
"Groups"
    }
}
```

Write Transformation

```
                "targetPath":
"$.schemas",
                "scope":
"patchEntity"
            }
        ],
        "scimEntityEndpoint":
"Users"
    },
    "group": {
        "skipOperations": [
            "create",
            "delete"
        ],
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.Operations",
                "scope":
"patchEntity"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas",
                "scope":
"patchEntity"
            },
            {
                "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:Group",

"urn:sap:cloud:scim:schemas:extens
ion:custom:2.0:Group"
                ],
                "targetPath":
"$.schemas"
            },
            {
```

Read Transformation	Write Transformation
	<pre>"targetPath": "$.members", "type": "remove" }, { "sourcePath": "$.members[*].value", "preserveArrayWithSingleElement": true, "optional": true, "targetPath": "$.members[?(@.value)]" }], "scimEntityEndpoint": "Groups" } }</pre>

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Related Information

URI Conventions (OData Version 2.0)
SAP SuccessFactors HCM Suite OData API
SAP SuccessFactors Workforce SCIM API

1.6.3.42 SAP SuccessFactors Employee Central Payroll

Follow this procedure to set up SAP SuccessFactors Employee Central Payroll as a proxy system.

Context

SAP SuccessFactors Employee Central Payroll is a cloud product for payroll processing. It's a service offering where SAP SuccessFactors Employee Central is used for maintenance of core HR master data and an SAP HCM Payroll system hosted in an SAP or Hyperscaler data center is used for payroll.

You can use Identity Provisioning to configure SAP SuccessFactors Employee Central Payroll as a proxy system in hybrid scenarios. For example, when SAP SuccessFactors Employee Central Payroll is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users back to the SAP SuccessFactors Employee Central Payroll.

This scenario supports provisioning users and groups.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP SuccessFactors Employee Central Payroll* as a proxy system. For more information, see Add New Systems [page 2415].
4. Set up the communication between Identity Provisioning and SAP SuccessFactors Employee Central Payroll and configure your authentication method (basic or certificate-based).

> **Note**
>
> We recommend that you use certificate-based authentication.

a. In your newly added SAP SuccessFactors Employee Central Payroll proxy system, select the *Certificate* tab and choose *Generate* > *Download*, as described in Generate and Manage Certificates for Outbound Connection.

   Skip step **a.** if you want to use basic authentication.

   The next step is performed in SAP SuccessFactors Employee Central Payroll backend system and is relevant for both basic and certificate-based authentication.

b. For certificate-based authentication, import the certificate generated in the Identity Provisioning UI in your SAP SuccessFactors Employee Central Payroll system, as described in Prerequisite: Finding the Access URL → step 5.

   For basic authentication, provide *User* and *Password*, as described in Prerequisite: Finding the Access URL → step 5.

c. Save your changes.

5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Enter the URL to your SAP SuccessFactors Employee Central Payroll system. For more information, see Prerequisite: Finding the Access URL → step 8.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter your authentication method: • *BasicAuthentication* • *ClientCertificateAuthentication* For more information, see Prerequisite: Finding the Access URL → step 5.
`User`	Valid if *BasicAuthentication* is configured as authentication method. For more information, see Prerequisite: Finding the Access URL → step 5.
`Password`	(Credential) Valid if *BasicAuthentication* is configured as authentication method. For more information, see Prerequisite: Finding the Access URL → step 5.
`ecp.sap-client`	Enter the SAP client number (a three-digit number) of your SAP SuccessFactors Employee Central Payroll system. For example: **`102`**

Property Name	Value
(Optional) `ecp.user.filter`	When specified, only those SAP SuccessFactors Employee Central Payroll users matching the filter expression will be read. Supported operators: eq (equal) and sw (starts with). For example: • Value = *userName eq "JohnSmith"* • Value = *userName sw "J"* • Value = *emails.value eq "john.smith@example.com"*
(Optional) `ecp.group.filter`	When specified, only those SAP SuccessFactors Employee Central Payroll groups matching the filter expression will be read. Supported operators: eq (equal) and sw (starts with). For example: • Value = *displayName eq "ProjectTeam1"* • Value = *displayName sw "P"*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

6. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP SuccessFactors Employee Central PayrollTransformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP SuccessFactors Employee Central Payroll system. For more information, see:

   Manage Transformations [page 2439].

   Default read and write transformations:

   > **→ Tip**
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourcePath":
"$.meta",
                "targetPath":
"$.meta",
                "optional": true
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.schemas",
                "targetPath":
"$.schemas",

"preserveArrayWithSingleElement":
true
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
              "sourcePath":
"$.name.givenName",
               "targetPath":
"$.name.givenName",
               "optional": true
            },
            {
```

Write Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:User",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:User"
                ],

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.name.givenName",
                "targetPath":
"$.name.givenName",
                "optional": true
            },
            {
                "sourcePath":
"$.name.familyName",
                "targetPath":
"$.name.familyName",
                "optional": true
            },
            {
                "sourcePath":
"$.name.middleName",
                "targetPath":
"$.name.middleName",
                "optional": true
            },
            {
                "sourcePath":
"$.nickName",
                "targetPath":
"$.nickName",
                "optional": true
            },
            {
```

Read Transformation

```
                "sourcePath":
"$.name.familyName",
                "targetPath":
"$.name.familyName",
                "optional": true
            },
            {
                "sourcePath":
"$.name.middleName",
                "targetPath":
"$.name.middleName",
                "optional": true
            },
            {
                "sourcePath":
"$.nickName",
                "targetPath":
"$.nickName",
                "optional": true
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName",
                "optional": true
            },
            {
                "sourcePath":
"$.phoneNumbers",
                "targetPath":
"$.phoneNumbers",
                "optional": true,

"preserveArrayWithSingleElement":
true
            },
            {
                "sourcePath":
"$.emails",
                "targetPath":
"$.emails",
                "optional": true,

"preserveArrayWithSingleElement":
true
            },
            {
                "sourcePath":
"$.active",
                "targetPath":
"$.active",
                "optional": true
            },
            {
                "sourcePath":
"$.groups",
                "targetPath":
"$.groups",
                "optional": true,
```

Write Transformation

```
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName",
                "optional": true
            },
            {
                "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.phoneNumbers",
                "optional": true
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.emails",
                "optional": true
            },
            {
                "sourcePath":
"$.active",
                "targetPath":
"$.active",
                "optional": true
            },
            {
                "sourcePath":
"$.groups[*].value",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups[?(@.value)]",
                "functions": [
                    {

"entityType": "group",

"function": "resolveEntityIds"
                    }
                ]
            },
            {
                "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.Operations",
                "scope":
"patchEntity"
            },
            {
```

Read Transformation

```
"preserveArrayWithSingleElement":
true
                }
            ],
            "scimEntityEndpoint":
"Users"
        },
         "group": {
            "condition":
"$.displayName EMPTY false",
            "mappings": [
                {
                    "sourcePath":
"$.id",
                    "targetPath":
"$.id",
                    "targetVariable":
"entityIdSourceSystem"
                },
                {
                    "sourcePath":
"$.meta",
                    "targetPath":
"$.meta",
                    "optional": true
                },
                {
                    "sourceVariable":
"entityBaseLocation",
                    "targetPath":
"$.meta.location",
                    "targetVariable":
"entityLocationSourceSystem",
                    "functions": [
                        {
                            "type":
"concatString",
                            "suffix":
"${entityIdSourceSystem}"
                        }
                    ]
                },
                {
                    "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                    "targetPath":
"$.schemas"
                },
                {
                    "sourcePath":
"$.displayName",
                    "targetPath":
"$.displayName"
                },
                {
                    "sourcePath":
"$.members",
```

Write Transformation

```
                    "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                    "targetPath":
"$.schemas",
                    "scope":
"patchEntity"
                }
            ],
            "scimEntityEndpoint":
"Users"
        },
        "group": {
            "mappings": [
                {
                    "sourceVariable":
"entityIdTargetSystem",
                    "targetPath":
"$.id"
                },
                {
                    "sourcePath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
                    "targetPath":
"$.Operations",
                    "scope":
"patchEntity"
                },
                {
                    "constant": [

"urn:ietf:params:scim:schemas:core
:2.0:Group",

"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:Group"
                    ],
                    "targetPath":
"$.schemas"
                },
                {
                    "sourcePath":
"$.displayName",
                    "targetPath":
"$.displayName"
                },
                {
                    "constant":
"userGroup",
                    "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                    "optional": true
                },
                {
                    "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation	Write Transformation

<pre>"targetPath": "$.members",
"optional": true,
"preserveArrayWithSingleElement": true
},
{
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
"optional": true
},
{
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
"optional": true
}
],
"scimEntityEndpoint": "Groups"
}
}</pre>
	<pre>ension:sap:2.0:Group']['supportedOperations']",
	"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['supportedOperations']",
	"optional": true
	},
	{
	"sourcePath": "$.schemas",
	"preserveArrayWithSingleElement": true,
	"targetPath": "$.schemas",
	"scope": "patchEntity"
	},
	{
	"sourcePath": "$.members[*].value",
	"preserveArrayWithSingleElement": true,
	"targetPath": "$.members[?(@.value)]",
	"optional": true
	},
	{
	"constant": "User",
	"preserveArrayWithSingleElement": true,
	"optional": true,
	"targetPath": "$.members[*].type",
	"scope": "createEntity"
	}
	],
	"scimEntityEndpoint": "Groups"
	}
	}</pre>

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

> **Note**
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> **Caution**
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

Identity Provisioning for Employee Central Payroll

1.6.3.43 SAP SuccessFactors Incentive Management

Follow this procedure to set up SAP SuccessFactors Incentive Management as a proxy system, formerly known as SAP Commissions.

Prerequisites

- You have technical user credentials for an SAP SuccessFactors Incentive Management system with read and write access permissions.

Context

Create a SCIM 2.0 proxy connector for SAP SuccessFactors Incentive Management to execute hybrid scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to the SAP SuccessFactors Incentive Management system, whenever the external back-end requests such. This scenario supports provisioning **users** and **user assignments to groups**.

> ⓘ Note
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.

3. Add *SAP SuccessFactors Incentive Management* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the SAP SuccessFactors Incentive Management SCIM API portal.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the user for your SAP SuccessFactors Incentive Management system.
`Password`	(Credential) Enter the password for your SAP SuccessFactors Incentive Management user.

   To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

   Exemplary destination:

   `Type`=*HTTP*

   `Authentication`=*BasicAuthentication*

   `ProxyType`=*Internet*

   `URL`=*https://mycommissions.callidus.run/CallidusPortal*

   `User`=*MyCommissionsUser*

   `Password`=**********

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP SuccessFactors Incentive Management* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP SuccessFactors Incentive Management. For more information, see:

   Manage Transformations [page 2439]

   SAP SuccessFactors Incentive Management REST API

   Default read and write transformations:

→ Tip

The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "targetVariable":
"entityIdSourceSystem",
        "correlationAttribute":
true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "optional": true
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.middleName",
        "targetPath":
"$.name.middleName",
        "optional": true
      },
       {
        "sourcePath":
"$.name.familyName",
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true,
        "defaultValue": ""
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true,
        "defaultValue": ""
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true,
        "defaultValue": ""
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "defaultValue": ""
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value != []",
        "sourcePath": "$.emails[?
(@.primary == true)].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
```

Read Transformation

```
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.honorificPrefix",
        "targetPath":
"$.name.honorificPrefix",
        "optional": true
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",
        "optional": true,

"preserveArrayWithSingleElement":
true
      },
      {
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value",
        "optional": true
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.timezone",
        "optional": true,
        "targetPath": "$.timezone"
      },
      {
        "sourcePath":
"$.addresses",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.addresses"
      },
      {
```

Write Transformation

```
        "targetPath":
"$.emails[0].value",
        "functions": [
          {
            "function":
"elementAt",
            "index": 0
          }
        ]
      },
      {
        "constant": "",
        "targetPath":
"$.groups[0].value"
      },
      {
        "condition": "$.groups
EMPTY false",
        "sourcePath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.groups",
        "optional": true
      },
      {
        "sourcePath": "$.locale",
        "targetPath": "$.locale",
        "optional": true,
        "defaultValue": ""
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName",
        "optional": true,
        "defaultValue": ""
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
        "targetPath":
"$.schemas[0]"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "skipOperations": [
      "create",
      "delete"
    ],
    "mappings": [
      {
        "sourcePath":
"$.displayName",
        "targetPath": "$.id"
      },
      {
```

Read Transformation

```
        "sourcePath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale",
        "functions": [
          {
            "type": "substring",
            "beginIndex": 0,
            "endIndex": 2
          }
        ]
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath":
"$.displayName",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
```

Write Transformation

```
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "optional": true,

"preserveArrayWithSingleElement":
true,
        "sourcePath": "$.members",
        "targetPath": "$.members"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      }
    ]
  }
}
```

Read Transformation	Write Transformation

```
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "constant": "User",

"preserveArrayWithSingleElement":
true,
        "targetPath":
"$.members[*].type",
        "optional": true
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true
      }
    ]
  }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PATCH**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PATCH** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the

external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

SAP SuccessFactors Incentive Management: Integration with SAP IdP

1.6.3.44 SAP SuccessFactors Learning

Follow this procedure to set up SAP SuccessFactors Learning as a proxy system.

Prerequisites

You have created a technical user with administrator permissions that will be used to call the SAP SuccessFactors Learning API for provisioning user information.

Context

SAP SuccessFactors Learning is a learning solution which helps organizations to improve employee skills and talent management, align learning outcomes with performance goals, boost compliance, and train external audiences.

You can use the Identity Provisioning user interface (UI) to connect to SAP SuccessFactors Learning as a proxy system and configure it in hybrid scenarios. For example, when SAP SuccessFactors Learning is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users and groups

to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users and group members back to the SAP SuccessFactors Learning.

> ⚠ Restriction
>
> Provisioning **groups** is not supported.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *SAP SuccessFactors Learning* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the API of your SAP SuccessFactors Learning system. It follows the pattern: `https://<root URL>/learning/public-api/rest/admin/Integration.svc/ias`
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the technical user ID for SAP SuccessFactors Learning.

Property Name	Value
`Password`	(Credential) Enter the password for the SAP SuccessFactors Learning technical user. For more information, see Learning Technical User.
(Optional) `lms.user.filter`	When specified, only those users matching the filter expression will be read. **Possible values:** • *userName eq "testName"* • *externalID eq "testID"* • *active eq "true"* • *sourceSystem eq "Learning"* - indicates that the user is created directly in SAP SuccessFactors Learning with no involvement of Identity Provisioning. • *sourceSystem eq "Identity Provisioning"* - indicates that the user is created in SAP SuccessFactors Learning by Identity Provisioning.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users. For more information, see Query Parameters for Proxy System SCIM API.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SAP SuccessFactors Learning* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SAP SuccessFactors Learning system. For more information, see:

   Manage Transformations [page 2439].

   SAP Business Accelerator Hub: SAP SuccessFactors Learning

   Default read and write transformations:

   > **→ Tip**
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type": "concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath": "$.userName",
        "optional": true,
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.externalId",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extens
ion:sap:2.0:User']['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extens
ion:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extensio
n:custom:2.0:User']['siteID']",
        "optional": true,
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "optional": true,
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.externalId",
        "optional": true,
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath": "$.locale",
        "optional": true,
        "targetPath": "$.locale"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']
['userUuid']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userUuid']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']['siteID']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']['siteID']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']
['customColumns']['110']",
        "optional": true,
```

Read Transformation

```
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extensio
n:custom:2.0:User']['siteID']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extensio
n:custom:2.0:User']['sourceSystem']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extensio
n:custom:2.0:User']['sourceSystem']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extensio
n:custom:2.0:User']
['applicationID']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extensio
n:custom:2.0:User']['applicationID']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extensio
n:custom:2.0:User']['customColumns']
['110']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extensio
n:custom:2.0:User']['customColumns']
['110']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:extensio
n:custom:2.0:User']['customColumns']
['120']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:extensio
n:custom:2.0:User']['customColumns']
['120']"
      },
      {
        "sourcePath": "$.emails",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.emails"
      },
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "correlationAttribute": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
```

Write Transformation

```
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']
['customColumns']['110']"
      },
      {
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']
['customColumns']['120']",
        "optional": true,
        "targetPath": "$
['urn:sap:cloud:scim:schemas:exten
sion:custom:2.0:User']
['customColumns']['120']"
      },
      {
        "condition": "$.emails[?
(@.primary== true)] empty false",
        "sourcePath": "$.emails[?
(@.primary== true)].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.emails[?
(@.value)]"
      },
      {
        "condition": "$.emails[?
(@.primary== true)] empty true",
        "sourcePath":
"$.emails[0].value",
        "targetPath":
"$.emails[0].value",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.name.middleName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName"
      },
      {
```

Read Transformation

```
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath":
"$.name.middleName",
        "optional": true,
        "targetPath":
"$.name.middleName"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.phoneNumbers"
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      }
    ]
  }
}
```

Write Transformation

```
        "condition":
"$.phoneNumbers[?(@.primary==
true)] empty false",
        "sourcePath":
"$.phoneNumbers[?(@.primary==
true)].value",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.phoneNumbers[?(@.value)]"
      },
      {
        "condition":
"$.phoneNumbers[?(@.primary==
true)] empty true",
        "sourcePath":
"$.phoneNumbers[0].value",
        "targetPath":
"$.phoneNumbers[0].value",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active"
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ]
  }
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be

automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

- For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
- For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

> ⓘ Note
>
> For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.45 Sales Cloud – Analytics & AI

Follow this procedure to set up Sales Cloud – Analytics & AI as a proxy system.

Prerequisites

- You have technical user credentials for an Sales Cloud Analytics & AI (in short, *SCAAI*) system with read and write access permissions.

Context

Create a SCIM 2.0 proxy connector for Sales Cloud – Analytics & AI to execute hybrid scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to the SCAAI system, whenever the external back end requests such. This scenario supports provisioning **users** and **user assignments to groups**.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators* .
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *Sales Cloud – Analytics & AI* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
URL	Specify the URL to the SCIM API portal of your SCAAI system.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*

Property Name	Value
`User`	Enter the user for your SCAAI system.
`Password`	(Credential) Enter the password for your SCAAI user.
`OAuth2TokenServiceURL`	Enter the URL to the OAuth2 token service. If not sure about the exact URL, ask your SCAAI administrator.
(Optional)`sales.cloud.analytics_ai.group.filter`	Enter a group filter criteria, according to the API syntax of SCAAI. For example: **displayName eq "first_group"**
(Optional)`sales.cloud.analytics_ai.user.filter`	Enter a user filter criteria, according to the API syntax of SCAAI. For example: **externalId eq "John123"**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*http://myscaai:8080/scim_services*

`User`=*MySCAAIUser*

`Password`=**********

`OAuth2TokenServiceURL`=*http://myscaai:8080/gateway_services/api/auth/ips/token*

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Sales Cloud – Analytics & AI* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your Sales Cloud – Analytics & AI. For more information, see Manage Transformations [page 2439].

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users**

or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.schemas"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional" : true
      },
      {
        "type": "remove",
        "targetPath":
"$.groups[*].display"
      },
      {
        "type": "remove",
        "targetPath":
"$.groups[*].ref"
      }
```

Write Transformation

Code Syntax

```
{
    "user": {
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.externalId"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
                "targetPath":
"$.schemas[0]"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName"
            }
        ],
        "scimEntityEndpoint":
"Users"
    },
    "group": {
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.externalId"
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
                "targetPath":
"$.schemas[0]"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
```

Read Transformation

```
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "optional": true,

"preserveArrayWithSingleElement":
true,
        "sourcePath": "$.members",
        "targetPath": "$.members"
      },
      {
        "type": "remove",
        "targetPath":
"$.members[*].$ref"
      },
      {
        "type": "remove",
        "targetPath":
"$.members[*].display"
      }
    ]
  }
```

Write Transformation

```
                "sourcePath":
"$.members[*].value",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members[?(@.value)]"
              }
        ],
        "scimEntityEndpoint":
"Groups"
    }
}
```

Read Transformation	Write Transformation
} 	

If you want the users and groups in SCAAI to have the **same** IDs as the respective users and groups in the external back-end system, modify the *Write Transformation* mappings as follows:

```
{
    "user": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetPath": "$.externalId"
            },
...
    "group": {
        "mappings": [
            {
                "sourcePath": "$.id",
                "targetPath": "$.externalId"
            },
...
}
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PATCH**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PATCH** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.46 Cloud Foundry UAA Server

Follow this procedure to set up the Cloud Foundry UAA server as a proxy system.

Prerequisites

> ⚠ Restriction
>
> This system is available for standalone tenants running on SAP Cloud Identity infrastructure and SAP BTP, Neo environment. Bundle tenants running on SAP Cloud Identity Services infrastructure and Neo environment can use it only through **SAP Jam Collaboration** and **SAP Identity Access Governance** bundle options.

- You have a technical user with administrator permissions for Cloud Foundry UAA to read, create, and update user account information. You need Cloud Foundry UAA version **4.2** or higher.
- (Optional) You have installed the Cloud Connector in your corporate environment and have done the initial configuration. You need to do this only if the Cloud Foundry UAA server is exposed in a private corporate network. For more information, see Cloud Connector.

Context

User Account and Authentication Service (UAA) is an OAuth2 server that you can use for centralized identity management. It owns the user accounts and authentication sources, and supports standard protocols (such as *SAML*, *LDAP*, and *OpenID Connect*) to provide SSO and delegated authorization to Web applications. For more information, see Cloud Foundry UAA: Overview .

Cloud Foundry UAA is responsible for the SAP ID service to create and manage platform users (platform administrators and platform developers) in Cloud Foundry.

→ Tip

This connector is meant for provisioning users and groups from/to **general** Cloud Foundry systems (they could be non-SAP ones). If you want to trigger provisioning of entities from/to SAP Business Technology Platform Cloud Foundry applications, you'd better use SAP BTP XS Advanced UAA (Cloud Foundry) [page 1516] proxy system.

These proxy systems consume SCIM 1.1 API provided by Cloud Foundry UAA.

→ Remember

You can manage users and groups to Cloud Foundry on **application** level only. You cannot manage them on a *subaccount* level.

ⓘ Note

The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Follow the steps below to create a SCIM 2.0 representation of your proxy Cloud Foundry UAA system. You can then provision on demand new users and groups back to Cloud Foundry UAA.

Procedure

1. (Optional) Open Cloud Connector to add an access control system mapping for the **Cloud Foundry UAA Server**. This is needed to allow the Identity Provisioning service to access the Cloud Foundry UAA system as a back-end system on the intranet. To learn how, see: Configure Access Control (HTTP)
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators* .
3. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
4. Add *Cloud Foundry UAA Server* as a proxy system. For more information, see Add New Systems [page 2415].
5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the URL to the Cloud Foundry UAA SCIM API. If not sure about the exact URL, ask your Cloud Foundry UAA administrator.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`OAuth2TokenServiceURL`	As you need to make OAuth authentication to the UAA system, enter the URL to the OAuth2 token service. If not sure about the exact URL, ask your Cloud Foundry UAA administrator.
`User`	Enter the OAuth client ID of the Cloud Foundry UAA technical user.
`Password`	(Credential) Enter the OAuth client secret of the technical user.
`uaa.origin`	Enter the location of your Cloud Foundry identity provider. If not sure about the value, ask your Cloud Foundry UAA administrator. The value of this property is a string, which will be used as the *origin* attribute in the system transformations.
`uaa.origin.filter.enabled`	This flag property depends on `uaa.origin`. Possible values: **true** or **false** • If set to *true*, the Identity Provisioning service will read only users whose identity provider is set as a value of `uaa.origin`. • If set to *false*, the Identity Provisioning service will read all users, regardless of their origin. • If set to *true* but the `uaa.origin` property is missing, the provisioning will fail.
`scim.support.patch.operation`	Use this property if you want to modify the members of a group. Possible values: • **true** – the Identity Provisioning service will modify the group membership via the *PATCH /Groups* endpoint of UAA. To learn how, see Patch • **false** – the Identity Provisioning service will modify the group membership via the *POST /Groups* or *DELETE /Groups* endpoints of UAA. To learn how, see Add Member and Remove Member .

Property Name	Description & Value
`uaa.patch.response.with.resource`	Use this property if you want to retrieve a group whose membership was modified. **Note** This property is usable only when you have configured membership modifications via *Add/Remove Member* UAA endpoints. That is, when the `scim.support.patch.operation` property is set to **false**. Possible values: • **true** – the Identity Provisioning service will return the modified group via the *GET /Groups* endpoint of UAA. To learn how, see Retrieve . • **false** – no modified groups will be returned by the service.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary destination:

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`ProxyType`=*Internet*

`URL`=*https://api.authentication.hana.ondemand.com*

`OAuth2TokenServiceURL`=*https://MyCFaccount.authentication.hana.ondemand.com/oauth/token*

`User`=*MyCFuser*

`Password`=**********

`uaa.origin`=*my_UAA_location*

`uaa.origin.filter.enabled`=*true*

`scim.support.patch.operation`=*true*

`uaa.patch.response.with.resource`=*false*

6. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Cloud Foundry UAA Server* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

You can change the default transformation mapping rules to reflect your current setup of entities in your Cloud Foundry UAA server. For more information, see:

Manage Transformations [page 2439]

Cloud Foundry UAA API: Users

Cloud Foundry UAA API: Groups

Default read and write transformations:

> → Tip
>
> The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
>
> The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
>
> However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
```

Write Transformation

Code Syntax

```
{
  "user": {
    "mappings": [
      {
        "constant": "uaa-dummy-
value",
        "targetPath": "$.id",
        "scope": "createEntity"
      },
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.userName"
      },
      {
        "sourcePath":
"$.name.givenName",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.name.familyName",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath": "$.emails",
        "targetPath": "$.emails",

"preserveArrayWithSingleElement":
true
      },
      {
        "condition": "$.emails[?
(@.primary == true)].value == []",
        "targetPath":
"$.emails[0].primary",
        "constant": true
      },
      {
        "sourcePath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath":
"$.phoneNumbers"
      },
      {
        "sourcePath":
"$.externalId",
```

Read Transformation

```
      {
        "sourcePath": "$.emails[?
(@.primary== true)].value",
        "correlationAttribute":
true
      },
      {
        "sourcePath": "$.groups",
        "targetPath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.phoneNumbers",
        "targetPath":
"$.phoneNumbers",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath": "$.active",
        "targetPath": "$.active",
        "optional": true
      },
      {
        "sourcePath":
"$.externalId",
        "targetPath":
"$.externalId",
        "optional": true
      },
      {
        "sourcePath": "$.origin",
        "targetPath": "$.origin",
        "optional": true
      },
      {
        "sourcePath": "$.zoneId",
        "targetPath": "$.zoneId",
        "optional": true
      },
      {
        "sourcePath":
"$.verified",
        "targetPath":
"$.verified",
        "optional": true
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
        "targetPath":
"$.schemas[0]"
      }
    ],
    "scimEntityEndpoint": "Users"
```

Write Transformation

```
        "optional": true,
        "targetPath":
"$.externalId"
      },
      {
        "sourcePath":
"$.verified",
        "targetPath":
"$.verified",
        "optional": true
      },
      {
        "constant":
"%uaa.origin%",
        "targetPath": "$.origin"
      },
      {
        "constant":
"urn:scim:schemas:core:1.0",
        "targetPath":
"$.schemas[0]"
      }
    ],
    "scimEntityEndpoint": "Users"
  },
  "group": {
    "mappings": [
      {
        "sourceVariable":
"entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath":
"$.description",
        "targetPath":
"$.description",
        "optional": true
      },
      {
        "constant":
"urn:scim:schemas:core:1.0",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath": "$.members",
        "targetPath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "functions": [
          {
            "condition": "@.type
EMPTY false",
```

Read Transformation

```
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetPath": "$.id",
        "targetVariable":
"entityIdSourceSystem"
      },
      {
        "sourcePath": "$.meta",
        "targetPath": "$.meta",
        "optional": true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetPath":
"$.meta.location",
        "targetVariable":
"entityLocationSourceSystem",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
        "optional": true
      },
      {
        "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
        "optional": true
      },
      {
        "sourcePath":
"$.description",
        "targetPath":
"$.description",
        "optional": true
```

Write Transformation

```
            "function":
"toUpperCaseString",
            "applyOnElements":
true,
            "applyOnAttribute":
"type",
            "locale": "en_EN"
          }
        ]
      },
      {
        "constant":
"%uaa.origin%",
        "targetPath":
"$.members[*].origin",

"preserveArrayWithSingleElement":
true,
        "optional": true
      },
      {
        "sourcePath":
"$.Operations",
        "targetPath":
"$.Operations",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      },
      {
        "sourcePath": "$.schemas",
        "targetPath": "$.schemas",

"preserveArrayWithSingleElement":
true,
        "scope": "patchEntity"
      }
    ],
    "scimEntityEndpoint": "Groups"
  }
}
```

Read Transformation	Write Transformation

<pre>},
{
"sourcePath": "$.members",
"targetPath": "$.members",
 	
"preserveArrayWithSingleElement":
true,
"optional": true
},
{
"sourcePath": "$.zoneId",
"targetPath": "$.zoneId"
},
{
"constant": [
 	
"urn:ietf:params:scim:schemas:core
:2.0:Group",
 	
"urn:ietf:params:scim:schemas:exte
nsion:sap:2.0:Group"
],
"targetPath": "$.schemas"
}
],
"scimEntityEndpoint": "Groups"
}
}</pre>	

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation

is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

Cloud Foundry UAA: Users
Cloud Foundry UAA: Groups

1.6.3.47 Google G Suite

Follow this procedure to set up Google G Suite as a proxy system.

Prerequisites

1. Sign in to the Google API console (https://console.developers.google.com) and create a project.
2. Enable the Admin SDK. To do this, go to *Dashboard* > *ENABLE API* > *Admin SDK* > *ENABLE* .
3. Create a service account for your project. We recommend that you select *Enable G Suite Domain-wide Delegation* during the creation. If you skip this option, you can set it later. For more information, see Creating a service account .
4. Then, in the Google admin console (https://admin.google.com), a user with **Super Admin** role can delegate domain-wide authority to your service account. This way, it will have access to the Google Admin SDK on behalf of your user. For more information, see Delegating domain-wide authority .
   NOTE: When specifying the scopes, the administrator has to enter the following:
   `https://www.googleapis.com/auth/admin.directory.user, https://www.googleapis.com/auth/admin.directory.group`

Context

A Google service account with delegated domain-wide authority is required for authentication and authorization of the Identity Provisioning service to G Suite domain. The authentication is based on OAuth 2.0 protocol with JSON Web Token (JWT). The private key for the signature is distributed by Google via one-time downloadable JSON data, which is accessible by the domain administrator. The private key is encoded in PKCS8 format and is in the *private_key* field of the JSON data. For more information, see JSON Web Token (JWT) .

- When using it as a source system, you can read both users and groups from Google G Suite and provision them to any target system you have added in the Identity Provisioning user interface.
- When using it as a target system, you can write both users and groups, read from any source system you have added in the Identity Provisioning user interface. Google G Suite can automatically create accounts for your users in the Google Cloud Datastore.

The Identity Provisioning service supports user and group operations based on the following Google Directory API. See the table below.

User Operations	Group Operations
Create a user	Create a group
Retrieve a user	Retrieve a group's properties
Update a user	Update a group's properties
Delete a user	Delete a group

> ⚠ Caution
>
> You can only provision users whose e-mails are from verified domains.

If you have successfully finished with the initial setup (described in the **Prerequisites** section), continue with the procedure below.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.

If you already have a technical user, skip this step.

2. Save your changes.
3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
4. Save your changes.

3. Add *Google G Suite* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Specify the service URL: **`https://www.googleapis.com/admin/directory`**
`ProxyType`	Enter: **`Internet`**
`Authentication`	Enter: *BasicAuthentication* The authentication type in use is actually **OAuth** with JWT. But for any provisioning system based on OAuth, **BasicAuthentication** is used along with the `OAuth2TokenServiceURL` additional property.
`User`	Enter the service account's ID. You can take it from the *"client_email"* field in the JSON data, downloaded during the setup of Google service account.
`Password`	(Credential) Enter the service account's private key, which represents a long string in PKCS8 format. You can take it from the *"private key"* field in the JSON data, downloaded during the setup of Google service account.
`OAuth2TokenServiceURL`	To make OAuth authentication to the Google G Suite system, enter the URL to the access token provider service. For more information, see Using OAuth 2.0 to Access Google APIs .
`jwt.subject`	Enter the Google G Suite user on behalf of which the Google Directory API is called. This user has been assigned the role **User Management Admin**. This property corresponds to "sub" claim in JWT being generated during access token request: JWT: "sub" (Subject) Claim
(Optional) `jwt.scope`	Enter space-separated Google Directory API authorization scopes. For example: **`https://www.googleapis.com/auth/admin.directory.user`**

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

Exemplary Configuration:

`ProxyType`=*Internet*

`Type`=*HTTP*

`Authentication`=*BasicAuthentication*

`URL`=*https://www.googleapis.com/admin/directory*

`User`=*1234567890-compute@developer.gserviceaccount.com*

`Password`=*-----BEGIN PRIVATE KEY-----\n123ABCDEFG123456789...*

... /123456789ABCDEFG123=\n-----END PRIVATE KEY-----\n

`OAuth2TokenServiceURL`=*https://www.googleapis.com/oauth2/v4/token*

`jwt.subject`=*john.smith@me123.accounts.ondemand.com*

`jwt.scope`=*https://www.googleapis.com/auth/admin.directory.user*

---

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Google G Suite* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   > ⚠ Caution
   >
   > An initial password setup is mandatory for all newly provisioned users. This is required by the Google G Suite API and must be provided when new accounts are created. The constant value that you see as configuration for the password attribute in the default transformation is generated by SAP. You have to change the constant value with another one, known only by the representatives of your company, before starting to use the Identity Provisioning service for creating users in your corporate Google G Suite system automatically.

   You can change the default transformation mapping rules to reflect your current setup of entities in your Google G Suite. For more information, see:

   Manage Transformations [page 2439]

   Google Directory API: Users

   Google Directory API: Groups

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application

can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetVariable":
"entityIdSourceSystem",
                "targetPath":
"$.id"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
                "targetPath":
"$.schemas[0]"
            },
            {
                "sourcePath":
"$.primaryEmail",
                "targetPath":
"$.emails[0].value",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.primaryEmail",
                "targetPath":
"$.userName"
            },
            {
                "targetPath":
"$.emails[0].primary",
                "constant": true
            },
            {
                "sourcePath":
"$.name",
```

Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "condition":
"($.emails.length() > 0) &&
($.name.familyName EMPTY false)",
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.name",
                "targetPath":
"$.name"
            },
            {
                "sourcePath":
"$.emails[0].value",
                "targetPath":
"$.primaryEmail"
            },
            {
                "sourcePath":
"$.phoneNumbers",
                "targetPath":
"$.phones",
                "optional": true
            },
            {
                "scope":
"createEntity",
                "targetPath":
"$.password",
                "functions": [
                    {
                        "type":
"randomPassword",

"passwordLength": 16,

"minimumNumberOfLowercaseLetters":
 1,

"minimumNumberOfUppercaseLetters":
 1,

"minimumNumberOfDigits": 1,

"minimumNumberOfSpecialSymbols": 0
                    }
                ]
            },
            {
                "constant":
"false",
```

Read Transformation

```
                "targetPath":
"$.name"
            },
            {
                "constant": true,
                "targetPath":
"$.active"
            },
            {
                "condition":
"$.suspended == true",
                "constant": false,
                "targetPath":
"$.active"
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
                "targetPath":
"$.schemas[0]"
            },
            {
                "sourcePath":
"$.id",
                "targetVariable":
"entityIdSourceSystem",
                "targetPath":
"$.id"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.name",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.members[?((@.type == 'USER')
&& (@.status == 'ACTIVE'))]",
```

Write Transformation

```
                "targetPath":
"$.suspended"
            },
            {
                "condition":
"$.active == false",
                "constant": true,
                "targetPath":
"$.suspended"
            },
            {
                "constant":
"true",
                "targetPath":
"$.changePasswordAtNextLogin"
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
                "targetPath":
"$.schemas[0]"
            },
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.email",
                "scope":
"createEntity"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.name"
            },
            {
                "sourcePath":
"$.members",
                "targetPath":
"$.members",

"preserveArrayWithSingleElement":
true,
                "optional": true
            },
            {
                "targetPath":
"$.members[*].value",
                "type": "rename",
```

Read Transformation	Write Transformation
<pre>"targetPath": "$.members", "optional": true, "preserveArrayWithSingleElement": true }, { "targetPath": "$.members[*].status", "type": "remove" }, { "constant": "value", "targetPath": "$.members[*].id", "type": "rename" }, { "targetPath": "$.members[*].kind", "type": "remove" }, { "targetPath": "$.members[*].etag", "type": "remove" }, { "targetPath": "$.members[*].role", "type": "remove" }, { "constant": "display", "targetPath": "$.members[*].email", "type": "rename" }] } }</pre>	<pre>"constant": "id" }, { "targetPath": "$.members[*].display", "type": "remove" }] } }</pre>

If the **displayName** attribute in the source system transformation does not provide group e-mails, you can modify the transformation the following ways:

- Map **email** to another attribute that contains a unique group e-mail.
- Concatenate the **displayName** attribute with your domain. For example:

Sample Code

```
        {
            "sourcePath": "$.displayName",
            "targetPath": "$.email",
            "scope": "createEntity",
            "functions": [
                {
```

```
                    "type": "concatString",
                    "suffix": "@test.myaccount.ondemand.com"
                }
            ]
        }
```

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ⓘ Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.48 LDAP Server

Follow this procedure to set up LDAP Server as a proxy system.

Prerequisites

> **Note**
>
> This system is available for all standalone tenants. Bundle tenants running on SAP Cloud Identity Services infrastructure and Neo environment can use it as a proxy system for reading entities only.

> **Note**
>
> If you have purchased the Identity Provisioning service between **September 1, 2020** and **October 20, 2020**, and you want to make a connection to this on-premise system, follow the procedure in: Connect to On-Premise Systems in SAP Cloud Identity Infrastructure.

- You have installed the Cloud Connector in your corporate environment and have done the initial configuration. For more information, see: Cloud Connector (Neo) or Cloud Connector (Cloud Foundry)
- **For tenants running on the infrastructure of SAP Cloud Identity Services**: You have a multi-environment subaccount in the Cloud Foundry region that maps the region of your Identity Authentication tenant and it is subscribed to the *Cloud Identity Services* application. For more information, see Connect to On-Premise Systems in SAP Cloud Identity Infrastructure.
- You have the credentials of a technical user in the LDAP Server, which is used to call the LDAP Server API to read and write users and their attributes.

Context

You can use LDAP Server as a proxy connector to execute *hybrid* scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to the LDAP Server, whenever the external back-end requests such.

This scenario supports provisioning **users**, **groups** and **group assignments**.

There are two versions of the LDAP Server connector. Both consume the LDAP Server API to read and write users and groups. The versions are handled by the `ldap.api.version` property as follows:

- When the value is set to *1* or the property is not defined (typical for systems created before versioning was introduced on May 25, 2023) LDAP Server API version 1 is used. This is the default value of `ldap.api.version`.
  When using this version of the connector, the entities (users and groups) are read with all attributes.
  In this version, the `group members` attribute mapping in the proxy read transformation does not include `type` sub-attribute. In this case, all members are considered of type `User`, which is the sub-attribute fallback value. As a consequence, if the external system includes nested groups, they will not be handled properly.

- When the value is set to *2* – LDAP Server API version 2 is used.
  This version of the connector comes with improved performance of the read operation for user and group attributes. You are now able to define which user and group attributes to be read. This is possible by adding values to the properties `ldap.user.attributes` or `ldap.group.attributes`.
  Via these properties, you are able to add also user and group operational attributes (attributes which the directory organizes for internal use). For more information, refer to the official LDAP server documentation.
  After the additional values of the properties are set, the default read or proxy read transformations should also be adjusted accordingly.
  In this version, the `group members` attribute mapping in the proxy read transformation is enhanced with `type` sub-attribute. The sub-attribute has two possible values – `User` and `Group`. This allows you to read and preserve nested groups.
  For more information on how to update to LDAP Server connector version 2, see Update Connector Version [page 2424].

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Open Cloud Connector to add an access control system mapping for **LDAP Server**. This is needed to allow the Identity Provisioning service to access LDAP Server as a back-end system on the intranet. To learn how, see: Configure Access Control (LDAP)
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
3. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
4. Add *LDAP Server* as a proxy system. For more information, see Add New Systems [page 2415].
5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *LDAP*
`ldap.url`	Specify the destination URL. It must be in the following format: `ldap://<external_host>:<external_port>`
`ldap.proxyType`	Enter: *OnPremise*
`ldap.authentication`	Enter: *BasicAuthentication*
`ldap.user`	Enter the *distinguishedName* of the technical LDAP user. This is the user you need to establish the connection and to perform all queries.
`ldap.password`	(Credential) Enter the password for the LDAP technical user.
`ldap.group.path`	Enter the complete path to the node containing the groups in the LDAP tree.
`ldap.user.path`	Enter the complete path to the users in the LDAP tree.
(Optional)`ldap.api.version`	Defines the version of LDAP Server API. **Possible values:** • **1** - Indicates that LDAP Server API version 1 is used. • **2** - Indicates that LDAP Server API version 2 is used. If the property is not defined - LDAP Server API version 1 is used.
(Optional) `ldap.group.filter`	You can optimize the search to return only particular groups. To enter correct group filters, stick to the standard LDAP specification. See: LDAP Representation of Filters – Examples . For example: Value *(cn=mar*)* will return only groups whose CN starts with "mar" (such as *marked*, *March*, or *Marketing*). By default, this filter is empty. That is, if the property is not specified, the filter will search for every group.

> → Remember
>
> We strongly recommend that you enter different paths for LDAP users and groups. That means, the value of `ldap.user.path` should be different than the value of `ldap.group.path`.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

The LDAP Server proxy system is created by default with the properties listed below:

Default LDAP Properties

`ldap.user.object.class=` **inetOrgPerson**

`ldap.group.object.class=` **groupOfNames**

`ldap.group.uniquename.attribute=` **cn**

`ldap.attribute.group.object.class.required=`**cn**

`ldap.attribute.user.object.class.required=`**cn**

`ldap.attribute.group.id=`**cn**

`ldap.attribute.group.member=` **member**

`ldap.attribute.user.id=` **uid**

`ldap.attribute.dn=`**distinguishedName**

`ldap.attribute.user.mail=` **mail**

`ldap.attribute.user.mobile=`**mobile**

`ldap.attribute.user.givenName=` **givenName**

`ldap.attribute.user.surname=` **sn**

`ldap.attribute.user.groups=` **memberOf**

`ldap.attribute.user.telephoneNumber=` **telephoneNumber**

`ldap.respond.with.resource.after.create=`**true**

`ldap.respond.with.resource.after.update=`**true**

> ### Note
>
> The **ldap.attribute.*** properties are used as parameterized properties in the default transformation. That is, if a property used in the transformation doesn't have a value, the provisioning job will fail when the transformation is loaded on runtime and the property value is substituted.
>
> Also, you can change a property and use a new one (with a new name). In this case, you must replace the old property with the new one at all corresponding places in the transformation.

6. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *LDAP Server* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your LDAP Server. For more information, see Manage Transformations [page 2439].

   Default read and write transformations:

→ Tip

The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Default read and write transformations for LDAP Server connector version 1:

Read Transformation	Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.%ldap.attribute.user.id%[0]",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id",
        "correlationAttribute":
true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.id%[0]",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.mail%[0]",
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.givenName%
[0]",
```

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.%ldap.attribute.user.id%[0]",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.%ldap.attribute.user.object.cla
ss.required%[0]"
      },
      {
        "sourcePath":
"$.emails[*].value",
        "optional": true,
        "targetPath":
"$.%ldap.attribute.user.mail%"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.%ldap.attribute.user.givenName%
[0]"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.%ldap.attribute.user.surname%[0
]"
      },
      {
        "sourcePath":
"$.phoneNumbers[*].value",
        "optional": true,
        "targetPath":
"$.%ldap.attribute.user.mobile%"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
```

Read Transformation

```
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.surname%[0
]",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.groups%",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups[?
(@.value)]"
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.mobile%[0]
",
        "optional": true,
        "targetPath":
"$.phoneNumbers[0].value"
      },
      {
        "condition":
"$.%ldap.attribute.user.mobile%.le
ngth() > 0",
        "constant": "mobile",
        "targetPath":
"$.phoneNumbers[0].type"
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.telephoneN
umber%[0]",
        "optional": true,
        "targetPath":
"$.phoneNumbers[1].value"
      },
      {
        "condition":
"$.%ldap.attribute.user.telephoneN
umber%.length() > 0",
        "constant": "work",
        "targetPath":
"$.phoneNumbers[1].type"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
```

Write Transformation

```
        "sourcePath":
"$.displayName",
        "targetPath":
"$.%ldap.attribute.group.id%[0]",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.%ldap.attribute.group.object.cl
ass.required%[0]"
      },
      {
        "constant": [],
        "targetPath": "$.member"
      },
      {
        "sourcePath":
"$.members[*]",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetVariable":
"membersVariable",
        "functions": [
          {
            "condition":
"(@.type != 'Group') &&
('%ldap.attribute.user.id%' !=
'%ldap.attribute.dn%')",
            "function":
"concatString",
            "applyOnElements":
true,
            "applyOnAttribute":
"value",
            "prefix":
"%ldap.attribute.user.id%=",
            "suffix":
",%ldap.user.path%"
          },
          {
            "condition":
"(@.type == 'Group') &&
('%ldap.attribute.group.id%' !=
'%ldap.attribute.dn%')",
            "function":
"concatString",
            "applyOnElements":
true,
            "applyOnAttribute":
"value",
            "prefix":
"%ldap.attribute.group.id%=",
            "suffix":
",%ldap.group.path%"
          }
        ]
      },
```

Read Transformation	Write Transformation
<pre>"sourcePath": "$.%ldap.attribute.group.id%[0]", "targetVariable": "entityIdSourceSystem", "targetPath": "$.id" }, { "sourceVariable": "entityBaseLocation", "targetVariable": "entityLocationSourceSystem", "targetPath": "$.meta.location", "functions": [{ "type": "concatString", "suffix": "${entityIdSourceSystem}" }] }, { "sourcePath": "$.%ldap.attribute.group.id%[0]", "targetPath": "$.displayName" }, { "constant": "urn:ietf:params:scim:schemas:core:2.0:Group", "targetPath": "$.schemas[0]" }, { "sourcePath": "$.%ldap.attribute.group.member%", "preserveArrayWithSingleElement": true, "targetPath": "$.members[?(@.value)]", "optional": true }] } }</pre>	<pre>{ "sourceVariable": "membersVariable", "variablePath": "$[*].value", "preserveArrayWithSingleElement": true, "optional": true, "targetPath": "$.member" }] } }</pre>

Default read transformation for LDAP Server connector version 2:

Read Transformation	Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.%ldap.attribute.user.id%[0]",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id",
        "correlationAttribute":
true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.id%[0]",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.mail%[0]",
        "targetPath":
"$.emails[0].value",
        "optional": true,
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.givenName%
[0]",
```

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.%ldap.attribute.user.id%[0]",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.%ldap.attribute.user.object.cla
ss.required%[0]"
      },
      {
        "sourcePath":
"$.emails[*].value",
        "optional": true,
        "targetPath":
"$.%ldap.attribute.user.mail%"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.%ldap.attribute.user.givenName%
[0]"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath":
"$.%ldap.attribute.user.surname%[0
]"
      },
      {
        "sourcePath":
"$.phoneNumbers[*].value",
        "optional": true,
        "targetPath":
"$.%ldap.attribute.user.mobile%"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
```

Read Transformation

```
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.surname%[0
]",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.groups%",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups[?
(@.value)]"
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.mobile%[0]
",
        "optional": true,
        "targetPath":
"$.phoneNumbers[0].value"
      },
      {
        "condition":
"$.%ldap.attribute.user.mobile%.le
ngth() > 0",
        "constant": "mobile",
        "targetPath":
"$.phoneNumbers[0].type"
      },
      {
        "sourcePath":
"$.%ldap.attribute.user.telephoneN
umber%[0]",
        "optional": true,
        "targetPath":
"$.phoneNumbers[1].value"
      },
      {
        "condition":
"$.%ldap.attribute.user.telephoneN
umber%.length() > 0",
        "constant": "work",
        "targetPath":
"$.phoneNumbers[1].type"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
```

Write Transformation

```
        "sourcePath":
"$.displayName",
        "targetPath":
"$.%ldap.attribute.group.id%[0]",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.%ldap.attribute.group.object.cl
ass.required%[0]"
      },
      {
        "constant": [],
        "targetPath": "$.member"
      },
      {
        "sourcePath":
"$.members[*]",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetVariable":
"membersVariable",
        "functions": [
          {
            "condition":
"(@.type != 'Group') &&
('%ldap.attribute.user.id%' !=
'%ldap.attribute.dn%')",
            "function":
"concatString",
            "applyOnElements":
true,
            "applyOnAttribute":
"value",
            "prefix":
"%ldap.attribute.user.id%=",
            "suffix":
",%ldap.user.path%"
          },
          {
            "condition":
"(@.type == 'Group') &&
('%ldap.attribute.group.id%' !=
'%ldap.attribute.dn%')",
            "function":
"concatString",
            "applyOnElements":
true,
            "applyOnAttribute":
"value",
            "prefix":
"%ldap.attribute.group.id%=",
            "suffix":
",%ldap.group.path%"
          }
        ]
      },
```

Read Transformation	Write Transformation
<pre>"sourcePath": "$.%ldap.attribute.group.id%[0]", "targetVariable": "entityIdSourceSystem", "targetPath": "$.id" }, { "sourceVariable": "entityBaseLocation", "targetVariable": "entityLocationSourceSystem", "targetPath": "$.meta.location", "functions": [{ "type": "concatString", "suffix": "${entityIdSourceSystem}" }] }, { "sourcePath": "$.%ldap.attribute.group.id%[0]", "targetPath": "$.displayName" }, { "constant": "urn:ietf:params:scim:schemas:core:2.0:Group", "targetPath": "$.schemas[0]" }, { "sourcePath": "$.%ldap.attribute.group.member%", "preserveArrayWithSingleElement": true, "targetPath": "$.members", "optional": true }] } }</pre>	<pre>{ "sourceVariable": "membersVariable", "variablePath": "$[*].value", "preserveArrayWithSingleElement": true, "optional": true, "targetPath": "$.member" }] } }</pre>

> **Note**
>
> By default, the **cn** attribute is used for every read or written group. An administrator can change this behavior by setting the following properties:
>
> - `ldap.group.uniquename.attribute` – the value can be either the CN or the whole DN (**distinguishedName**) of the group.
> - `ldap.attribute.group.id` – the value can be CN or another attribute to be used as a group ID instead (for example, **displayName** or **description**).

> For more information about these properties, see: List of Properties [page 80]

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Related Information

Technical Documents
Setting Timeout for Ldap Operations
Connection Pooling Configuration

1.6.3.49 Microsoft Active Directory

Follow this procedure to set up Microsoft Active Directory as a proxy system.

Prerequisites

> ### Note
>
> If you have purchased the Identity Provisioning service between **September 1, 2020** and **October 20, 2020**, and you want to make a connection to this on-premise system, follow the procedure in: Connect to On-Premise Systems in SAP Cloud Identity Infrastructure.

- You have installed the Cloud Connector in your corporate environment and have done the initial configuration. For more information, see: Cloud Connector (Neo)
- You have the credentials of a technical user in the Microsoft Active Directory, which is used to call the Microsoft Active Directory API to read and write users, groups and their attributes.

Context

You can use Microsoft Active Directory as a proxy connector to execute *hybrid* scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to Microsoft Active Directory, whenever the external back-end requests such.

This scenario supports provisioning **users**, **groups** and **group assignments**.

There are two versions of the Microsoft AD connector. Both consume the LDAP Server API to read and write users and groups. The versions are handled by the `ldap.api.version` property as follows:

- When the value is set to *1* or the property is not defined (typical for systems created before versioning was introduced on June 19, 2023) LDAP Server API version 1 is used. This is the default value of `ldap.api.version`.
  When using this version of the connector, the entities (users and groups) are read with all attributes.
  In this version, the `group members` attribute mapping in the proxy read transformation does not include `type` sub-attribute. In this case, all members are considered of type `User`, which is the sub-attribute fallback value. As a consequence, if the external system includes nested groups, they will not be handled properly.
- When the value is set to *2* – LDAP Server API version 2 is used.
  This version of the connector is with improved performance of the read operation for user and group attributes. You are now able to define which user and group attributes to be read. This is possible by adding values to the properties `ldap.user.attributes` or `ldap.group.attributes`.
  Via these properties, you are able to add also user and group operational attributes (attributes which the directory organizes for internal use). For more information, refer to the official LDAP server documentation.
  After the additional values of the properties are set, the default read or proxy read transformations should also be adjusted accordingly.
  In this version, the `group members` attribute mapping in the proxy read transformation is enhanced with `type` sub-attribute. The sub-attribute has two possible values – `User` and `Group`. This allows you to read and preserve nested groups.
  For more information on how to update to Microsoft Active Directory version 2, see Update Connector Version [page 2424].

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Open Cloud Connector to add an access control system mapping for **Microsoft Active Directory**. This is needed to allow the Identity Provisioning service to access Microsoft Active Directory as a back-end system on the intranet. To learn how, see: Configure Access Control (LDAP)
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators* .
3. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.

- For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
- For **Basic authentication**, proceed as follows:
    1. Add an administrator user of type **System** and configure the basic authentication method for this user.
       If you already have a technical user, skip this step.
    2. Save your changes.
    3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
    4. Save your changes.

4. Add *Microsoft Active Directory* as a proxy system. For more information, see Add New Systems [page 2415].
5. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *LDAP*
`ldap.url`	Specify a destination URL. It must be in the following format: `ldap://<ext_host>:<ext_port>`
`ldap.proxyType`	Enter: *OnPremise*
`ldap.authentication`	Enter: *BasicAuthentication*
`ldap.user`	Enter the *distinguishedName* or the *userPrincipalName* of the Microsoft AD technical user. This is the user you need to establish the connection and to perform all queries.
`ldap.password`	(Credential) Enter the password for the Microsoft AD technical user.
`ldap.attribute.user.id`	Default property, which denotes the ID of a user. By default, it's set to: *cn*
`ldap.attribute.group.id`	Default property, which denotes the ID of a group. By default, it's set to: *cn*
`ldap.attribute.dn`	Default property, which denotes the distinguished name of a user or a group. Only possible value: *distinguishedName*
`ldap.respond.with.resource.after.create`	Default property, whose value is *true*.
`ldap.respond.with.resource.after.update`	Default property, whose value is *true*.
`ldap.group.path`	Enter the complete path to the node containing the groups in Microsoft Active Directory.

Property Name	Description & Value
`ldap.user.path`	Enter the complete path to the users in Microsoft Active Directory.
(Optional)`ldap.api.version`	Handles the version of the LDAP Server connector. **Possible values:** • *1* - Indicates that Microsoft AD API version 1 is used. • *2* - Indicates that Microsoft AD API version 2 is used. Default value: *1*
(Optional) `ldap.group.filter`	You can optimize the search to return only particular groups. To enter correct group filters, stick to the standard LDAP specification. See: LDAP Representation of Filters – Examples . For example: Value *(cn=mar*)* will return only groups whose CN starts with "mar" (such as *marked*, *March*, or *Marketing*). By default, this filter is empty. That is, if the property is not specified, the filter will search for every group.

Example for a destination or a set of properties:

```
Type=LDAP
Name=MyADDestination
ldap.user=john.smith@some.dummy.domain.com
ldap.password=********
ldap.attribute.user.id=cn
ldap.attribute.group.id=cn
ldap.attribute.dn=distinguishedName
ldap.url=ldap://abcd:123
ldap.proxyType=OnPremise
ldap.authentication=BasicAuthentication
ldap.group.path=OU=Groups,OU=IAS,DC=global,DC=corp,DC=mycompany
ldap.user.path=OU=Users,OU=IAS,DC=global,DC=corp,DC=mycompany
```

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

6. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Microsoft Active Directory* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your Microsoft AD. For more information, see:

   Manage Transformations [page 2439]

   Default read and write transformations:

   > → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Default read and write transformations for Microsoft AD connector version 1:

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.%ldap.attribute.user.id%[0]",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id",
        "correlationAttribute":
true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.sAMAccountName[0]",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.displayName[0]",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath": "$.mail[0]",
        "targetPath":
"$.emails[0].value",
        "optional": true,
```

Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "condition":
"('%ldap.attribute.user.id%' !=
'%ldap.attribute.dn%')",
        "sourcePath":
"$.userName",
        "targetPath":
"$.%ldap.attribute.user.id%[0]",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },

// If a user is not a direct
member of the configured user
base path, then its
distinguishedName is configured
to be equal to CN =
<userName>,<nested_path>,<base_pat
h>,
// where <nested_path> is read
from "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:ad:2.0:User']['nestedPath']"

      {
        "condition":
"('%ldap.attribute.user.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:ad:2.0:User']['nestedPath']",
        "optional": true,
        "targetVariable":
"nestedPathVariable",
        "defaultValue": "",
        "scope": "createEntity"
      },
      {
        "condition":
"('%ldap.attribute.user.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath":
"$.userName",
        "functions": [
          {
            "function":
"concatString",
            "prefix": "CN="
          },
          {
            "condition": "('$
{nestedPathVariable}' != '')",
```

Read Transformation

```
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.givenName[0]",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.sn[0]",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.memberOf",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups[?
(@.value)]"
      },
      {
        "sourcePath":
"$.mobile[0]",
        "optional": true,
        "targetPath":
"$.phoneNumbers[0].value"
      },
      {
        "condition":
"$.mobile.length() > 0",
        "constant": "mobile",
        "targetPath":
"$.phoneNumbers[0].type"
      },
      {
        "sourcePath":
"$.telephoneNumber[0]",
        "optional": true,
        "targetPath":
"$.phoneNumbers[1].value"
      },
      {
        "condition":
"$.telephoneNumber.length() > 0",
        "constant": "work",
        "targetPath":
"$.phoneNumbers[1].type"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
```

Write Transformation

```
            "function":
"concatString",
            "suffix": ","
          },
          {
            "condition": "('$
{nestedPathVariable}' != '')",
            "function":
"concatString",
            "suffix": "$
{nestedPathVariable}"
          },
          {
            "function":
"concatString",
            "suffix":
",%ldap.user.path%"
          }
        ],
        "targetPath":
"$.%ldap.attribute.user.id%[0]",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.cn[0]",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.sAMAccountName[0]"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName[0]"
      },
      {
        "sourcePath":
"$.emails[0].value",
        "optional": true,
        "targetPath": "$.mail[0]"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.givenName[0]"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath": "$.sn[0]"
      }
```

Read Transformation

```
        "sourcePath":
"$.%ldap.attribute.group.id%[0]",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.sAMAccountName[0]",
        "targetPath":
"$.displayName"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath": "$.member",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.members[?
(@.value)]",
        "optional": true
      }
    ]
  }
}
```

Write Transformation

```
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "condition":
"('%ldap.attribute.group.id%' !=
'%ldap.attribute.dn%')",
        "sourcePath":
"$.displayName",
        "targetPath":
"$.%ldap.attribute.group.id%[0]",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },

// If a group is not a direct
member of the configured group
base path, then its
distinguishedName is configured
to be equal to CN =
<displayName>,<nested_path>,<base_
path>,
// where <nested_path> is read
from "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:ad:2.0:Group']['nestedPath']"

      {
        "condition":
"('%ldap.attribute.group.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:ad:2.0:Group']
['nestedPath']",
        "optional": true,
        "targetVariable":
"nestedPathVariable",
        "defaultValue": "",
        "scope": "createEntity"
      },
      {
        "condition":
"('%ldap.attribute.group.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath":
"$.displayName",
        "functions": [
          {
            "function":
"concatString",
            "prefix": "CN="
          },
          {
            "condition": "('$
{nestedPathVariable}' != '')",
            "function":
"concatString",
```

Read Transformation	Write Transformation

```
            "suffix": ","
          },
          {
            "condition": "('$
{nestedPathVariable}' != '')",
            "function":
"concatString",
            "suffix": "$
{nestedPathVariable}"
          },
          {
            "function":
"concatString",
            "suffix":
",%ldap.group.path%"
          }
        ],
        "targetPath":
"$.%ldap.attribute.group.id%[0]",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath": "$.cn[0]",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.sAMAccountName[0]"
      },
      {
        "constant": [],
        "targetPath": "$.member"
      },
      {
        "sourcePath":
"$.members[*]",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetVariable":
"membersVariable",
        "functions": [
          {
            "condition":
"(@.type != 'Group') &&
('%ldap.attribute.user.id%' !=
'%ldap.attribute.dn%')",
            "function":
"concatString",
            "applyOnElements":
true,
            "applyOnAttribute":
"value",
            "prefix":
"%ldap.attribute.user.id%=",
```

Read Transformation	Write Transformation

	<pre>"suffix":
	",%ldap.user.path%"
	},
	{
	"condition":
	"(@.type == 'Group') &&
	('%ldap.attribute.group.id%' !=
	'%ldap.attribute.dn%')",
	"function":
	"concatString",
	"applyOnElements":
	true,
	"applyOnAttribute":
	"value",
	"prefix":
	"%ldap.attribute.group.id%=",
	"suffix":
	",%ldap.group.path%"
	}
	]
	},
	{
	"sourceVariable":
	"membersVariable",
	
	"preserveArrayWithSingleElement":
	true,
	"optional": true,
	"targetPath": "$.member",
	"variablePath": "$
	[*].value"
	}
	]
	}
	}</pre>

Default read transformation for Microsoft AD connector version 2:

Read Transformation	Write Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "sourcePath":
"$.%ldap.attribute.user.id%[0]",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id",
        "correlationAttribute":
true
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.sAMAccountName[0]",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.displayName[0]",
        "targetPath":
"$.displayName",
        "optional": true
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath": "$.mail[0]",
        "targetPath":
"$.emails[0].value",
        "optional": true,
```

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "mappings": [
      {
        "condition":
"('%ldap.attribute.user.id%' !=
'%ldap.attribute.dn%')",
        "sourcePath":
"$.userName",
        "targetPath":
"$.%ldap.attribute.user.id%[0]",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },

// If a user is not a direct
member of the configured user
base path, then its
distinguishedName is configured
to be equal to CN =
<userName>,<nested_path>,<base_pat
h>,
// where <nested_path> is read
from "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:ad:2.0:User']['nestedPath']"

      {
        "condition":
"('%ldap.attribute.user.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:ad:2.0:User']['nestedPath']",
        "optional": true,
        "targetVariable":
"nestedPathVariable",
        "defaultValue": "",
        "scope": "createEntity"
      },
      {
        "condition":
"('%ldap.attribute.user.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath":
"$.userName",
        "functions": [
          {
            "function":
"concatString",
            "prefix": "CN="
          },
          {
            "condition": "('$
{nestedPathVariable}' != '')",
```

Read Transformation

```
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.givenName[0]",
        "targetPath":
"$.name.givenName",
        "optional": true
      },
      {
        "sourcePath": "$.sn[0]",
        "targetPath":
"$.name.familyName",
        "optional": true
      },
      {
        "sourcePath":
"$.memberOf",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups[?
(@.value)]"
      },
      {
        "sourcePath":
"$.mobile[0]",
        "optional": true,
        "targetPath":
"$.phoneNumbers[0].value"
      },
      {
        "condition":
"$.mobile.length() > 0",
        "constant": "mobile",
        "targetPath":
"$.phoneNumbers[0].type"
      },
      {
        "sourcePath":
"$.telephoneNumber[0]",
        "optional": true,
        "targetPath":
"$.phoneNumbers[1].value"
      },
      {
        "condition":
"$.telephoneNumber.length() > 0",
        "constant": "work",
        "targetPath":
"$.phoneNumbers[1].type"
      }
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
```

Write Transformation

```
            "function":
"concatString",
            "suffix": ","
          },
          {
            "condition": "('$
{nestedPathVariable}' != '')",
            "function":
"concatString",
            "suffix": "$
{nestedPathVariable}"
          },
          {
            "function":
"concatString",
            "suffix":
",%ldap.user.path%"
          }
        ],
        "targetPath":
"$.%ldap.attribute.user.id%[0]",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath": "$.cn[0]",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.userName",
        "targetPath":
"$.sAMAccountName[0]"
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName[0]"
      },
      {
        "sourcePath":
"$.emails[0].value",
        "optional": true,
        "targetPath": "$.mail[0]"
      },
      {
        "sourcePath":
"$.name.givenName",
        "optional": true,
        "targetPath":
"$.givenName[0]"
      },
      {
        "sourcePath":
"$.name.familyName",
        "optional": true,
        "targetPath": "$.sn[0]"
      }
```

Read Transformation

```
        "sourcePath":
"$.%ldap.attribute.group.id%[0]",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.sAMAccountName[0]",
        "targetPath":
"$.displayName"
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath": "$.member",

"preserveArrayWithSingleElement":
true,
        "targetPath": "$.members",
        "optional": true
      }
    ]
  }
}
```

Write Transformation

```
    ]
  },
  "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "condition":
"('%ldap.attribute.group.id%' !=
'%ldap.attribute.dn%')",
        "sourcePath":
"$.displayName",
        "targetPath":
"$.%ldap.attribute.group.id%[0]",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },

// If a group is not a direct
member of the configured group
base path, then its
distinguishedName is configured
to be equal to CN =
<displayName>,<nested_path>,<base_
path>,
// where <nested_path> is read
from "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:ad:2.0:Group']['nestedPath']"

      {
        "condition":
"('%ldap.attribute.group.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath": "$
['urn:sap:cloud:scim:schemas:exten
sion:ad:2.0:Group']
['nestedPath']",
        "optional": true,
        "targetVariable":
"nestedPathVariable",
        "defaultValue": "",
        "scope": "createEntity"
      },
      {
        "condition":
"('%ldap.attribute.group.id%' ==
'%ldap.attribute.dn%')",
        "sourcePath":
"$.displayName",
        "functions": [
          {
            "function":
"concatString",
            "prefix": "CN="
          },
          {
            "condition": "('$
{nestedPathVariable}' != '')",
            "function":
"concatString",
```

Read Transformation	Write Transformation

```
            "suffix": ","
          },
          {
            "condition": "('$
{nestedPathVariable}' != '')",
            "function":
"concatString",
            "suffix": "$
{nestedPathVariable}"
          },
          {
            "function":
"concatString",
            "suffix":
",%ldap.group.path%"
          }
        ],
        "targetPath":
"$.%ldap.attribute.group.id%[0]",
        "targetVariable":
"entityIdTargetSystem",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath": "$.cn[0]",
        "scope": "createEntity"
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.sAMAccountName[0]"
      },
      {
        "constant": [],
        "targetPath": "$.member"
      },
      {
        "sourcePath":
"$.members[*]",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetVariable":
"membersVariable",
        "functions": [
          {
            "condition":
"(@.type != 'Group') &&
('%ldap.attribute.user.id%' !=
'%ldap.attribute.dn%')",
            "function":
"concatString",
            "applyOnElements":
true,
            "applyOnAttribute":
"value",
            "prefix":
"%ldap.attribute.user.id%=",
```

Read Transformation	Write Transformation

	<pre>"suffix":
	",%ldap.user.path%"
	},
	{
	"condition":
	"(@.type == 'Group') &&
	('%ldap.attribute.group.id%' !=
	'%ldap.attribute.dn%')",
	"function":
	"concatString",
	"applyOnElements":
	true,
	"applyOnAttribute":
	"value",
	"prefix":
	"%ldap.attribute.group.id%=",
	"suffix":
	",%ldap.group.path%"
	}
	]
	},
	{
	"sourceVariable":
	"membersVariable",
	
	"preserveArrayWithSingleElement":
	true,
	"optional": true,
	"targetPath": "$.member",
	"variablePath": "$
	[*].value"
	}
	]
	}
	}</pre>

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

Microsoft AD: Technical Documents
Setting Timeout for Ldap Operations
Connection Pooling Configuration
Active Directory Domain Services Overview

1.6.3.50 Microsoft Entra ID

Follow this procedure to set up Microsoft Entra ID (formerly known as Microsoft Azure Active Directory) as a proxy system.

Prerequisites

- You've logged on to Microsoft Azure Portal, with credentials for a user with directory role **Global administrator**. For more information, see Microsoft Entra built-in roles .

- In *Azure Active Directory* > *App registrations*, you've registered an application with a secret key and permissions for Microsoft Graph API. These permissions must be consented by an administrator. For more information, see Microsoft Graph permissions reference .
- (Relevant to target systems) Your registered application is assigned the **User Account Administrator** role. This role allows you to deprovision users. For more information, see Add-MsolRoleMember .

> **Note**
>
> If this role isn't assigned, you can only disable users. To do that, set the `accountEnabled` property to **false**. For more information, see MS Graph: user resource type

Permissions

Assign the following permissions to your application, according to your scenario. Also, the permissions have to be of type *Application*.

- Users – *User.ReadWrite.All*, *Directory.AccessAsUser.All*
- Groups – *Group.ReadWrite.All*

For more information, see MS Graph: Users and MS Graph: Groups

Context

When using it as a proxy system, you can write both users and groups, read from any source system you've added in the Identity Provisioning user interface. The Microsoft Entra ID proxy systems use Microsoft Graph API. For more information, see Microsoft Graph .

If you've successfully finished with the initial setup (described in the **Prerequisites** section), continue with the procedure.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.

- For **Basic authentication**, proceed as follows:
  1. Add an administrator user of type **System** and configure the basic authentication method for this user.
     If you already have a technical user, skip this step.
  2. Save your changes.
  3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
  4. Save your changes.

3. Add *Microsoft Entra ID* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter: **https://graph.microsoft.com**
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *BasicAuthentication*
`User`	Enter the application ID registered in your Microsoft Entra ID subscription (see the **Prerequisites** section).
`Password`	(Credential) Enter the secret key associated to your app registration.
`aad.domain.name`	Enter one of the verified domain names from the corresponding Microsoft Entra ID tenant. On this domain, you perform the provisioning operations. For more information, see Managing custom domain names in your Microsoft Entra ID .
`OAuth2TokenServiceURL`	Enter: **https://login.microsoftonline.com/<your_domain>/oauth2/token**, where `<your_domain>` is the domain name you have set in the **`aad.domain.name`** property.
(Optional) `aad.oauth2.resource.name`	Specifies the OAuth2 resource name for Microsoft Graph API. Enter: **https://graph.microsoft.com** You can change the Microsoft Graph endpoint to match your organization's cloud environment. For more information, see Microsoft Graph national cloud deployments.

Property Name	Description & Value
(Optional) `aad.group.member.attributes`	This property defines the attributes of a group member to be read by the Identity Provisioning. By default, it always reads the **type** and the **id** of a member. If you want the Identity Provisioning to read additional attributes, enter them as a single or a comma-separated value. For example: **Example** • If you want to read the e-mails too, enter: `aad.group.member.attributes=`*mail* This will read a member's type, ID, and e-mail. • If you want to read multiple additional attributes, enter: `aad.group.member.attributes=`*mail,mobilePhone,displayName* This will read a member's type, ID, e-mail, phone, and display name.
(Optional) `aad.user.attributes.membership.active`	Use this property if you want to get information about all the groups to which the users are assigned (if any). • If the property is missing, or is set to *false* – group membership details for the users will not be extracted. • If the property is set to *true* – group membership details for the users will be extracted. To learn more, see: List of Properties [page 80]
(Optional) `aad.user.filter`	Use this property to filter users by specific criteria, according to the Microsoft Graph REST API . **Note** This property replaces the deprecated `msgraph-filter` property. To learn more, see: List of Properties [page 80]
(Optional) `aad.group.filter`	Use this property to filter groups by specific criteria, according to the Microsoft Graph REST API .

Property Name	Description & Value
(Optional) `aad.user.filter.group.filter.combine`	Use this property to filter users based on their group assignments. When set to **true**, this property combines user and group filters defined on the `aad.user.filter` and `aad.group.filter` properties to further narrow the search results. This way, only users that meet the following filtering criteria are returned: • Users that match the user filter and at the same time are members of groups that match the group filter. • Members of the filtered groups that match the user filter. When set to **false**, user and group filters are not combined. To learn more, see: List of Properties [page 80]
(Optional) `aad.user.attributes`	Defines which user attributes are read from Microsoft Entra ID system. The property is set during system creation with the following default value: *id,mail,userPrincipalName,displayName,mailNickname,givenName,surname,mobilePhone,businessPhones* This means that by default, Identity Provisioning will read from Microsoft Entra ID the user attributes defined in the property value. Those attributes are used in the default read transformation. To check the complete set of user attributes (properties) supported by Microsoft Entra ID, see: Microsoft Graph: User Properties To learn more, see: List of Properties [page 80]

Property Name	Description & Value
(Optional) `aad.group.attributes`	Defines which group attributes are read from Microsoft Entra ID system. The property is set during system creation with the following default value: *id,displayName,mailNickname* This means that by default, Identity Provisioning will read from Microsoft Entra ID the group attributes defined in the property value and will also return the *members* attribute. Those attributes are used in the default read transformation. To check the complete set of group attributes (properties) supported by Microsoft Entra ID, see: Microsoft Graph: Group Properties To learn more, see: List of Properties [page 80]
(Optional) `aad.entities.top`	This property defines the number of entities to be read per page. Default value: *100*
(Deprecated) `oauth.resource.name`	Enter: **https://graph.microsoft.com** **Note** The **oauth.resource.name** property has been deprecated, as it is no longer needed by the MS Entra ID connector. Use the **aad.oauth2.resource.name** property instead.

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. (Optional) Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Microsoft Entra ID* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your Microsoft Entra ID. For more information, see:

   Manage Transformations [page 2439]

   MS Graph: Users

   MS Graph: Groups

   Default read and write transformations:

→ Tip

The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
  "user": {
    "scimEntityEndpoint": "Users",
    "condition":
"$.userPrincipalName EMPTY false",
    "mappings": [
      {
        "sourcePath": "$.id",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:User",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath": "$.mail",
        "targetPath":
"$.emails[0].value",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.userPrincipalName",
        "targetPath":
"$.userName",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.displayName",
        "optional": true,
        "targetPath":
"$.displayName"
      },
      {
```

Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "optional": true,
                "sourcePath":
"$.onPremisesImmutableId",
                "targetPath":
"$.onPremisesImmutableId"
            },
            {
                "sourcePath":
"$.active",
                "optional": true,
                "targetPath":
"$.accountEnabled"
            },
            {
                "sourcePath":
"$.name.givenName",
                "optional": true,
                "targetPath":
"$.mailNickname"
            },
            {
                "sourcePath":
"$.displayName",
                "optional": true,
                "targetPath":
"$.displayName"
            },
            {
                 "sourcePath":
"$.name.givenName",
                 "optional": true,
                 "targetPath":
"$.givenName"
            },
            {
                 "sourcePath":
"$.name.familyName",
                 "optional": true,
                 "targetPath":
"$.surname"
            },
            {
                "sourcePath":
"$.addresses[0].locality",
                "optional": true,
                "targetPath":
"$.city"
            },
            {
                "sourcePath":
"$.addresses[0].country",
                "optional": true,
                "targetPath":
"$.country"
```

Read Transformation

```
        "sourcePath":
"$.mailNickname",
        "optional": true,
        "targetPath":
"$.externalId",
        "correlationAttribute":
true
      },
      {
        "sourcePath":
"$.givenName",
        "optional": true,
        "targetPath":
"$.name.givenName"
      },
      {
        "sourcePath": "$.surname",
        "optional": true,
        "targetPath":
"$.name.familyName"
      },
      {
        "sourcePath":
"$.mobilePhone",
        "optional": true,
        "targetPath":
"$.phoneNumbers[0].value"
      },
      {
        "condition":
"$.mobilePhone EMPTY false",
        "constant": "mobile",
        "targetPath":
"$.phoneNumbers[0].type"
      },
      {
        "sourcePath":
"$.businessPhones[0]",
        "optional": true,
        "targetPath":
"$.phoneNumbers[1].value"
      },
      {
        "condition":
"$.businessPhones.length() > 0",
        "constant": "work",
        "targetPath":
"$.phoneNumbers[1].type"
      },
      {
        "sourcePath": "$.groups",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.groups"
      },
      {
        "sourcePath":
"$.manager.id",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

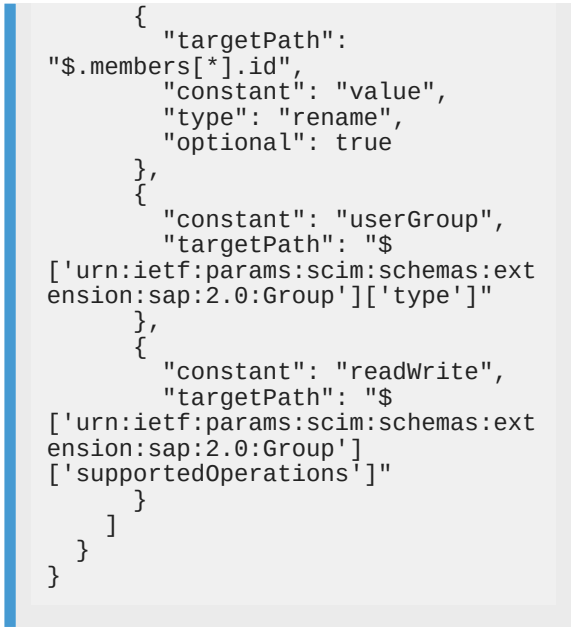
Write Transformation

```
            },
            {
                "scope":
"createEntity",
                "sourcePath":
"$.userName",
                "targetPath":
"$.userPrincipalName",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"@%aad.domain.name%"
                    }
                ]
            },
            {
                "scope":
"createEntity",
                "constant":
"true",
                "targetPath":
"$.accountEnabled"
            },
            {
                "scope":
"createEntity",
                "sourcePath":
"$.active",
                "optional": true,
                "targetPath":
"$.accountEnabled"
            },
            {
                 "scope":
"createEntity",
                 "sourcePath":
"name.givenName",
                 "targetPath":
"$.mailNickname"
            },
            {
                "scope":
"createEntity",
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "scope":
"createEntity",
                "targetPath":
"$.passwordProfile.password",
                "functions": [
                    {
                        "type":
"randomPassword",

"passwordLength": 16,
```

Read Transformation	Write Transformation

```
ension:enterprise:2.0:User']
['manager']['value']",
        "optional": true
      },
      {
        "sourcePath":
"$.manager.displayName",
        "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']
['manager']['displayName']",
        "optional": true
      }
    ]
  },
   "group": {
    "scimEntityEndpoint":
"Groups",
    "mappings": [
      {
        "constant":
"urn:ietf:params:scim:schemas:core
:2.0:Group",
        "targetPath":
"$.schemas[0]"
      },
      {
        "sourcePath": "$.id",
        "targetVariable":
"entityIdSourceSystem",
        "targetPath": "$.id"
      },
      {
        "sourceVariable":
"entityBaseLocation",
        "targetVariable":
"entityLocationSourceSystem",
        "targetPath":
"$.meta.location",
        "functions": [
          {
            "type":
"concatString",
            "suffix": "$
{entityIdSourceSystem}"
          }
        ]
      },
      {
        "sourcePath":
"$.displayName",
        "targetPath":
"$.displayName"
      },
      {
        "sourcePath": "$.members",

"preserveArrayWithSingleElement":
true,
        "optional": true,
        "targetPath": "$.members"
      },
```

```
"minimumNumberOfLowercaseLetters":
 1,

"minimumNumberOfUppercaseLetters":
 1,

"minimumNumberOfDigits": 1,

"minimumNumberOfSpecialSymbols": 0
                    }
                ]
            },
            {
                "scope":
"createEntity",
                "constant": false,
                "targetPath":
"$.passwordProfile.forceChangePass
wordNextSignIn"
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "optional": true,
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "scope":
"createEntity",
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "scope":
"createEntity",
                "sourcePath":
"$.externalId",
                "targetPath":
"$.mailNickname"
            },
            {
                "scope":
"createEntity",
                "constant": true,
                "targetPath":
"$.mailEnabled"
            },
```

Read Transformation	Write Transformation
	

Custom Configurations

Goal	Action	Result
You want Identity Provisioning to read the additional user attributes specified in property `aad.user.attributes` and write them successfully in the external back-end system.	In the *Read Transformation*, extend the "*user*" mapping as follows: `{ "user": { "condition": "$.userPrincipalName EMPTY false", "mappings": [ { "sourcePath": "$", "targetPath": "$" }, { "sourcePath": "$.id", "targetVariable": "entityIdSourceSystem" }, ...`	For example, you specify the `aad.user.attributes` property and set its value to: *id,mail,userPrincipalName,city,department,companyName* As a result, every user in the external back-end system will have the following attributes populated – ID, e-mail, user principle name, city, department, and company name. Returned information of an exemplary user: `... { "Resources":[ { "id":"555-aaaa-333-abcd-111222333", "mail":"john.smith@domain.com", "userPrincipalName":"abc@something.onmicrosoft.com", "city": "Sofia", "department":"029", "companyName":"SAP" } ... ] }`

Goal	Action	Result
You want Identity Provisioning to read the additional group attributes specified in property `aad.group.attributes` and write them successfully in the target system.	Extend the "*group*" mapping as follows: `{ "group": { "ignore": false, "mappings": [ { "sourcePath": "$", "targetPath": "$" }, { "constant": "urn:ietf:params:scim:schemas:core:2.0:Group", "targetPath": "$.schemas[0]" }, ...`	Example: Specify the `aad.group.attributes` property and set its value to: *id,displayName,recommendation,isSubscribedByMail* As a result, every group in the target system will have the following attributes populated – ID, display name, date and time of the last renewal, and information if it's subscribed by e-mail or not. Returned information of an exemplary group: `... { "Resources":[ { "id":"12345-ccc-000-xyz-777888999", "displayName":"ImportantGroup3", "renewedDateTime":"2018-01-01T00:00:00Z", "isSubscribedByMail":"true" } ... ] }`

Goal	Action	Result

You want the returned value of a group member to be not the ID but a different attribute.	Replace **id** with the new attribute. For example:
	If you replace *id* with *mail*, the transformation will look like this:
	`...`
	`{`
	`"sourcePath": "$.members",`
	`"preserveArrayWithSingleElement": true,`
	`"optional": true,`
	`"targetPath": "$.members"`
	`},`
	`{`
	`"targetPath": "$.members[*].mail",`
	`"constant": "value",`
	`"type": "rename",`
	`"optional": true`
	`}`
	`...`
	⚠ Caution
	Make sure that you've added this attribute as a value of property *aad.group.member.attributes*.
		Returned information of an exemplary group member:
		`...`
		`{`
		`"members":[`
		`{`
		`"id":"5555555-aaaa-333-abcd-1111122223333",`
		`"type":"user",`
		`"value":"johnsmith@mail.acme.com"`
		`}`
		`]`
		`}`
		`...`

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   ⓘ Note

   For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.6.3.51 Procurement Data Warehouse

Follow this procedure to set up Procurement Data Warehouse as a proxy system.

Prerequisites

- You obtain credentials to access the Procurement Data Warehouse API.

  > ⓘ Note
  >
  > To obtain credentials to access the Procurement Data Warehouse API, create a case to the component *BNS-ARI-PRI*.
  >
  > In response to the case, SAP provides the certificate needed to access the API as well as the URL for the procurement data warehouse enrollment service, which is landscape-specific and starts with `*.cloud.sap`.

- You established trust between your SAP BTP subaccount and the SAP Cloud Identity Services - Identity Authentication service. For more information, see Establishing Trust Automatically.
- To configure your *Procurement Data Warehouse* provisioning system (see the procedure below), you will need to map your SAP Ariba application parameters to the relevant Identity Provisioning properties. The property mapping between the two systems is as follows:

Procurement Data Warehouse Application	Identity Provisioning	Values
API URL	`URL`	Examples: • US: **https://openapi.ariba.com** • Europe: **https://eu.openapi.ariba.com**
procurement data warehouseOAuth Token URL	`OAuth2TokenServiceURL`	Examples: • US: **https://api.ariba.com/v2/oauth/token** • Europe: **https://api-eu.ariba.com/v2/oauth/token**
OAuth Client ID	`client_id`	Enter the OAuth Client id used for access token retrieval.

- To load transaction data into procurement data warehouse, you must have one or more of the following SAP Ariba applications:
  - SAP Ariba Buying
  - SAP Ariba Buying and Invoicing
  - Guided sourcing capability for SAP Ariba Sourcing
  - SAP Ariba Contracts
  - SAP Ariba Spend Analysis

Context

The Procurement Data Warehouse is a capability that feeds analytics information from SAP Ariba applications, including guided sourcing capability for SAP Ariba Sourcing, SAP Ariba Buying and Invoicing, and SAP Spend Control Tower, to the embedded SAP Analytics Cloud. It provides data transfer, storage, loading, and reporting, and it drives visualization through advanced analytics dashboards.

You can use Identity Provisioning to configure Procurement Data Warehouse as a proxy system in hybrid scenarios. For example, when Procurement Data Warehouse is exposed as a proxy system, you can connect it to an external identity management system, such as SAP Identity Management, without making a direct connection between both systems. You can provision users and groups to the external backend system, which can trigger CRUD (create, read, update, delete) operations on users and groups back to the Procurement Data Warehouse.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
2. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
3. Add *Procurement Data Warehouse* as a proxy system. For more information, see Add New Systems [page 2415].
4. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Description & Value
`Type`	Enter: *HTTP*
`URL`	Enter the SAP Analytics Cloud URL for your Procurement Data Warehouse.
`ProxyType`	Enter: *Internet*
`Authentication`	Enter: *ClientCertificateAuthentication*
`client_id`	Enter the OAuth client id, generated for your Procurement Data Warehouse. (see the **Prerequisites** section)
`OAuth2TokenServiceURL`	Enter the URL of the access token provider service for your Procurement Data Warehouse instance. (see the **Prerequisites** section)
(Optional) `pdw.user.filter`	When specified, only those Procurement Data Warehouse users matching the filter expression will be read. For example: • *userName eq "Julie Armstrong"* • *userName sw "J"* • *emails eq "julie.armstrong@example.com"*

   To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

5. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *Procurement Data Warehouse* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your Procurement Data Warehouse. For more information, see:

   Manage Transformations [page 2439]

   > ### ⓘ Note
   >
   > To access the Procurement Data Warehouse API documentation, contact the Procurement Data Warehouse technical support.

   Default read and write transformations:

   > ### → Tip
   >
   > The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.
   >
   > The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.
   >
   > However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetPath":
"$.meta.location",
                "targetVariable":
"entityLocationSourceSystem",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.preferredLanguage",
                "optional": true,
                "targetPath":
"$.preferredLanguage"
            },
```

Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "condition":
"($.emails[0].value EMPTY false)",
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.emails[0].value",
                "targetPath":
"$.userName"
            },
            {
                "sourcePath":
"$.displayName",
                "optional": true,
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.preferredLanguage",
                "optional": true,
                "targetPath":
"$.preferredLanguage"
            },
            {
                "sourcePath":
"$.name.givenName",
                "targetPath":
"$.name.givenName",
                "optional": true
            },
            {
                "sourcePath":
"$.name.familyName",
                "targetPath":
"$.name.familyName",
                "optional": true
            },
            {
                "sourcePath":
"$.active",
                "optional": true,
```

Read Transformation

```
            {
              "sourcePath":
"$.name.givenName",
              "targetPath":
"$.name.givenName",
              "optional": true
            },
            {
              "sourcePath":
"$.name.familyName",
              "targetPath":
"$.name.familyName",
              "optional": true
            },
            {
                "sourcePath":
"$.active",
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath":
"$.emails[0].value",
                "targetPath":
"$.emails[0].value"
            },
            {
                "sourcePath":
"$.emails[?(@.primary ==
true)].value",
                "optional": true,

"correlationAttribute": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:user-custom-
parameters:1.0']",
                "optional":true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:user-custom-
parameters:1.0']"
```

Write Transformation

```
                "targetPath":
"$.active"
            },
            {
                "sourcePath":
"$.emails",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.emails"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:enterprise:2.0:User']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:user-custom-
parameters:1.0']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:user-custom-
parameters:1.0']"
            },
            {
                "sourcePath": "$
['urn:scim:schemas:extension:enter
prise:1.0']",
                "optional": true,
                "targetPath": "$
['urn:scim:schemas:extension:enter
prise:1.0']"
            },
            {
                "sourcePath":
"$.roles",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.roles"
            },
            {
                "sourcePath":
"$.photos",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.photos"
            },
            {
```

Read Transformation

```
            },
            {
                "sourcePath": "$
['urn:scim:schemas:extension:enter
prise:1.0']",
                "optional":true,
                "targetPath": "$
['urn:scim:schemas:extension:enter
prise:1.0']"
            },
            {
                "sourcePath":
"$.roles",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.roles"
            },
            {
                "sourcePath":
"$.photos",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.photos"
            },
            {
                "sourcePath":
"$.groups",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups"
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
```

Write Transformation

```
                "sourcePath":
"$.groups",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.groups"
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.id",
                "scope":
"createEntity"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.members",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:group-custom-
parameters:1.0']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
```

Read Transformation	Write Transformation

```
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.schemas",

"preserveArrayWithSingleElement":
true,
                "targetPath":
"$.schemas"
            },
            {
                "sourcePath":
"$.displayName",
                "targetPath":
"$.displayName"
            },
            {
                "sourcePath":
"$.members",

"preserveArrayWithSingleElement":
true,
                "optional": true,
                "targetPath":
"$.members"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:group-custom-
parameters:1.0']",
                "optional": true,
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:group-custom-
parameters:1.0']"
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']['type']",
                "optional": true
            },
            {
                "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
                "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:Group']
['supportedOperations']",
```

```
ension:sap:group-custom-
parameters:1.0']"
            }
        ]
    }
}
```

Read Transformation	Write Transformation

<pre>"optional": true
}
]
}
}</pre>	

6. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ### Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

7. Run an initial load job.

   If you try to provision groups that already exist in Procurement Data Warehouse proxy system, your provisioning job may fail with: '*The group already exists on the target system and cannot be provisioned*' error. This happens when you create a new proxy system and connect it to an existing Procurement Data Warehouse backend.

   To avoid this, you have the following options:

   - Delete the existing group in Procurement Data Warehouse proxy system.
   - Adapt the Procurement Data Warehouse write transformation. Either ignore provisioning of groups or add temporary the *skipOperations* expression for creating groups.
   - Avoid provisioning of already existing groups.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

Related Information

Configure OAuth Certificate Authentication [page 2456]

1.6.3.52 SCIM System

Follow this procedure to set up SCIM as a proxy system.

Prerequisites

> ⓘ Note
>
> This system is available for all standalone tenants. Bundle tenants running on SAP Cloud Identity Services infrastructure and Neo environment can use it as a proxy system for reading entities only.

- You have technical user credentials for a SCIM system with read and write access permissions.
- (Optional) You have installed the Cloud Connector in your corporate environment and have done the initial configuration. You need this only if the SCIM system is exposed in a private corporate network. For more information, see Cloud Connector.

Context

Create a general SCIM 2.0 based proxy connector to execute hybrid scenarios. That means, it can provision its entities to another (external) back-end system by request, and then can continue executing CRUD operations back to the SCIM system, whenever the external back-end requests such. This scenario supports provisioning **users** and **groups**.

> **Note**
>
> The Identity Provisioning implementation of the Proxy System SCIM API (based on the SCIM Query) supports single entity and delta read filtering for users and groups. For more information, see Query Parameters for Proxy System SCIM API.

Procedure

1. (Optional) Open Cloud Connector to add an access control system mapping for the **SCIM system**. This is needed to allow the Identity Provisioning service to access the SCIM system as a back-end system on the intranet. To learn how, see: Configure Access Control (HTTP)
2. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Users & Authorizations* > *Administrators*.
3. Create a technical user with the necessary authorizations. It will later be used by the external consumer to connect to Identity Provisioning.
   - For **Certificate-based authentication**, upload the certificate for the technical user of type *System*, as described in Add System as Administrator and enable the *Access Proxy System API* permission.
   - For **Basic authentication**, proceed as follows:
     1. Add an administrator user of type **System** and configure the basic authentication method for this user.
        If you already have a technical user, skip this step.
     2. Save your changes.
     3. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
     4. Save your changes.
4. Add *SCIM System* as a proxy system. For more information, see Add New Systems [page 2415].
5. Choose the *Properties* tab to configure the connection settings for your system.

   Mandatory Properties

Property Name	Value
`Type`	Enter: *HTTP*
URL	Specify the service URL. For example: **`http://<cloudfoundry_server>.com/api/uaa/`**
`ProxyType`	Depending on your network exposure, enter one of the following: • *Internet* • *OnPremise*
`Authentication`	Enter: *BasicAuthentication*

Property Name	Value
`User`	You can specify one of the following: • Technical user ID • Client ID for OAuth HTTP destinations. It's used for retrieving of the access token.
`Password`	(Credential) You can enter one of the following: • Technical user password • Client secret for OAuth HTTP destinations. It's used for retrieving of the access token.
`OAuth2TokenServiceURL`	If you need to make OAuth authentication to the system, enter the URL to the access token provider service for OAuth HTTP destinations. For example: **`https://<token_provider>.com/api/oauth2/v2.0/token`**
`scim.csrf.token.path`	Specifies the path which is appended to the URL to retrieve the CSRF token when `scim.api.csrf.protection` is enabled.
`scim.api.csrf.protection`	Specifies whether to fetch a CSRF token when sending requests to the system. If enabled, this property requires the configuration of the `scim.csrf.token.path` property to specify the path where the CSRF token can be retrieved. **Possible values:** • *enabled* • *disabled*

To learn what additional properties are relevant to this system, see List of Properties [page 80]. You can use the main search, or filter properties by the *Name* or *System Type* columns.

6. Configure the transformations.

   Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the *SCIM* proxy system, whose settings are displayed under the *Transformations* tab after saving its initial configuration.

   You can change the default transformation mapping rules to reflect your current setup of entities in your SCIM system. For more information, see Manage Transformations [page 2439].

   Default read and write transformation:

→ Tip

The proxy *Read Transformation* is used when the external client application (for example, SAP Identity Management) makes initial load. That is, executing GET requests to the resource endpoints (**/Users** or **/Groups**) to retrieve the corresponding entities of the particular type. The external client application can also execute GET requests to a single resource endpoint (querying a single resource is supported). In this case, the proxy system acts as a *source* one.

The proxy *Write Transformation* is used when the external application manages the entities in the proxy system – creates new entities, updates existing ones, or deletes existing ones. In this case, the proxy system acts as a *target* one.

However, after a *Create* or *Update* operation is performed on the proxy system, the *Read Transformation* is applied to the result, so that the created or updated entity is sent back to the external application. This behavior demonstrates that the proxy *Read Transformation* is used for *write* cases, as well.

Read Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "targetPath": "$",
                "sourcePath": "$"
            },
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            },
            {
                "sourcePath":
"$.userName",
                "targetPath":
"$.userName",

"correlationAttribute": true
            },
            {
                "sourcePath":
"$.emails[0].value",
                "targetPath":
"$.emails[0].value",
                "optional": true
            },
            {
                "sourcePath":
"$.emails[?(@.primary==
true)].value",
                "optional": true,

"correlationAttribute": true
            }
        ]
    },
    "group": {
```

Write Transformation

Code Syntax

```
{
    "user": {
        "scimEntityEndpoint":
"Users",
        "mappings": [
            {
                "sourcePath": "$",
                "targetPath": "$"
            },
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            },
            {
                "condition":
"$.emails[0].length() > 0",
                "targetPath":
"$.emails[0].primary",
                "constant": true
            }
        ]
    },
    "group": {
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourcePath": "$",
                "targetPath": "$"
            },
            {
                "sourceVariable":
"entityIdTargetSystem",
                "targetPath":
"$.id"
            }
        ]
    }
}
```

Read Transformation	Write Transformation

```
        "scimEntityEndpoint":
"Groups",
        "mappings": [
            {
                "sourcePath": "$",
                "targetPath": "$"
            },
            {
                "sourcePath":
"$.id",
                "targetPath":
"$.id",
                "targetVariable":
"entityIdSourceSystem"
            },
            {
                "sourceVariable":
"entityBaseLocation",
                "targetVariable":
"entityLocationSourceSystem",
                "targetPath":
"$.meta.location",
                "functions": [
                    {
                        "type":
"concatString",
                        "suffix":
"${entityIdSourceSystem}"
                    }
                ]
            }
        ]
    }
}
```

7. Connect the external consumer to Identity Provisioning with the technical user you have created in step 2.

   If the external consumer system is **SAP Identity Management**, you can export the newly created proxy system as a SCIM repository from Identity Provisioning and import it in SAP Identity Management. This will create a SCIM repository in SAP Identity Management where most of the repository constants will be automatically filled in. You need to provide the technical user credentials that you have set up in step 2 and the SCIM assignment method as described below:

   - For AUTH_USER and AUTH_PASSWORD, enter the user ID and password of the Identity Authentication technical user for which you have set permission *Access Proxy System API*.
   - For the SCIM_ASSIGNMENT_METHOD constant, make sure the value is **PUT**.

   > ⓘ Note
   >
   > For external consumer systems, other than SAP Identity Management, you should also use the **PUT** method for modifying entities.

Next Steps

When a proxy system is connected to an external backend system (in the case of SAP Identity Management this means the exported CSV file is imported into the Identity Management Admin UI and a repository is configured), you can start managing the users and groups into this external system. Usually, the first operation is the initial load of the existing entities into your external system. When this load has finished, changes in the external system, such as creating new users or updating existing ones, can trigger CRUD requests back to the proxy system.

To see an example with SAP Identity Management, see Hybrid Scenario: SAP Identity Management [page 2961] → sections **Next Steps** and **Future Identity Lifecycle**.

> ⚠ Caution
>
> Effective **September 2020**, Shanghai (China) tenants that reside on SAP BTP, Neo can be only accessed on the following domain: `dispatcher.cn1.platform.sapcloud.cn`
>
> So make sure you use the correct domain when you construct your REST API requests.
>
> For example: **GET** *https://ipsproxyabcd12345-xyz789.dispatcher.cn1.platform.sapcloud.cn/ipsproxy/api/v1/scim/bbb111aa-1234-aaaa-7777-1234567abcde/Users/s123456789*
>
> To learn more, see: Proxy Systems [page 1343]

1.7 Administration

The SAP Cloud Identity Services administration and configuration tasks are intended for administrators. They include configuring tenant settings, applications, authorization policies and provisioning, as well as managing users and groups, to ensure proper operations.For these configurations, administrators mainly use the administration console for SAP Cloud Identity Services, a Fiori-based user interface adaptive to most browsers.

Configuring Applications [page 1920]
: This section describes how you can configure the user authentication, access to an application, and use a branding style in accordance with your company requirements. It also explains the trust configuration between Identity Authentication and a service provider or client (relying party).

Configuring Tenant Settings [page 2100]
: Initially, the tenants are configured to use default settings. This section describes how you as a tenant administrator can make custom tenant configurations.

Configuring Password Policies [page 2223]
: Passwords for the authentication of users are subject to certain rules. These rules are defined in the password policy. Identity Authentication provides you with two predefined password policies, in addition to which you can create and configure up to three custom password policies.

Configuring Privacy Policies [page 2235]
: You can configure a custom privacy policy document by creating a new document, adding and editing its language versions, and defining the document for an application.

Configuring Authorization Policies [page 2244]
Authorization Management enables SAP Cloud Identity Services administrators to use authorization policies, customize them, and assign them to users.

Configuring Terms of Use [page 2273]
You can configure a custom terms of use document by creating a new document, adding and editing its language versions, and defining the document for an application.

Configuring Email Templates [page 2282]
Tenant administrators can use the default or a custom email template set for the application processes.

Managing Administrators [page 2301]
This section describes how, as a tenant administrator, you can list all administrators in the administration console for SAP Cloud Identity Services, add new administrators, and edit the administrator authorizations. You can also remove administrators.

Managing Technical Users [page 2329]
As a user with assigned special policy authorizations, you can manage the technical users in the administration console for SAP Cloud Identity Services.

Managing Agents [page 2333]
Agent identities automatically appear in SAP Cloud Identity Services after you deploy agents. You can use SAP Cloud Identity Services to review and manage access to these agents.

Managing Users [page 2342]
Tenant administrators can manage user accounts via the administration console for SAP Cloud Identity Services, and via APIs.

Managing Groups [page 2393]
Tenant administrators can create groups, and assign and unassign these groups to users via the administration console for SAP Cloud Identity Services.

Configuring Provisioning Systems [page 2413]
Configure provisioning systems for synchronizing users and groups between business applications.

Configuring Real-Time Provisioning [page 2487]
As a tenant administrator, you can configure real-time provisioning to immediately provision entities from source to target systems.

Configuring Social Identity Providers [page 2505]
By configuring a social provider, users can log on to applications with their social media credentials by linking their accounts in Identity Authentication to the social media account.

Integrating with Existing Customer Landscape [page 2518]
Identity Authentication can be integrated with already existing customer landscape and supports different types of delegated authentication.

Configuring External Authentication Providers [page 2620]
Configure authentication providers in the administration console for SAP Cloud Identity Services to manage users from external providers.

Configuring OpenID Connect (OIDC) [page 2636]
You can use Identity Authentication for authentication in OpenID Connect (OIDC) protected applications.

1.7.1 Configuring Applications

This section describes how you can configure the user authentication, access to an application, and use a branding style in accordance with your company requirements. It also explains the trust configuration between Identity Authentication and a service provider or client (relying party).

Parent topic: Administration [page 1918]

Related Information

Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

Configuring Applications

Configuring Applications

To learn about	See
How to create an application and configure its display name	• Create a New Application [page 1922] • Configure an Application's Type [page 1935] • Change an Application's Display Name [page 1934] • Configure an Application's Home URL [page 1937] • Visit an Application's Web Page [page 1939]
How to delete an application	• Delete Applications [page 1940]

To learn about	See
How to configure trust	• Configure Trust [page 1941] • Configure the Subject Name Identifier Sent to the Application [page 1988] • Configuring User Attributes from the Identity Directory [page 1973] • Configuring Attributes Based on Flexible Expressions [page 1979]
How to manage authentications	• Authentication [page 1996]
How to configure API Authentication	• Configure Certificates for API Authentication [page 2001] • Configure Secrets for API Authentication [page 1997]
How to manage Terms of Use and Privacy Policies	On a tenant level • Configuring Terms of Use [page 2273] • Configuring Privacy Policies [page 2235] Specific for the application • Define a Terms of Use Document for an Application [page 2278] • Define a Privacy Policy Document for an Application [page 2240]
How to configure branding styles and layout	On a tenant level • Configure Tenant Images [page 2142] Specific for the application • Display Application Name on Logon Page [page 2033] • Configure Logo [page 2030] • Choose and Configure a Branding Style for an Application [page 2035] • Define an Email Template Set for an Application [page 2045] • Configure Registration and Upgrade Forms [page 2046]

To learn about	See
How to configure access	• Configure an Application's Home URL [page 1937] • Visit an Application's Web Page [page 1939] • Configure User Access to the Application [page 2025] • Protecting Application Forms with Google reCAPTCHA [page 2056] **Note** The content in this section is not relevant for China (Shanghai) region. • Set a Password Policy for an Application [page 2228] • Import or Update Users for a Specific Application [page 2348]
How to choose an identity provider for an application	• Authenticating Identity Provider for an Application [page 2065]

1.7.1.1 Create a New Application

You can create a new application and customize it to comply with your company requirements.

Context

You can create two types of applications in Identity Authentication:

> **Note**
>
> There are applications that show the `SAML 2.0 & OpenID Connect` protocol type. You can't create such applications.

Create SAML 2.0 Application [page 1923]
: You can create a new SAML 2.0 application and customize it to comply with your company requirements.

Create OpenID Connect (OIDC) Application [page 1925]
: You can create a new OpenID Connect (OIDC) application.

Related Information

Configure Trust [page 1941]
Configuring Applications [page 1920]

1.7.1.1.1 Create SAML 2.0 Application

You can create a new SAML 2.0 application and customize it to comply with your company requirements.

Context

To create a new SAML 2.0 application, you have to add a new application to the list of applications in Identity Authentication, and make sure that the protocol of the application is set to SAML 2.0.

To create a new SAML 2.0 application, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the *Create* button on the left-hand panel and fill in the information in the dialog box.
4. Provide the following information and save your configuration:

Fields	Notes
Display Name	Required. The display name of the application appears on the logon and registration pages.
Home URL	Optional. Users are redirected to the *Home URL* after activating their accounts, when they're created via a CSV file import or the user registration service of Identity Authentication.
Type	Optional. **⚠ Caution** If the type of the application is not the correct one, the system may change it automatically without a warning.
Parent Application	Optional. **→ Remember** Newly created applications with an assigned parent application will inherit all the configurations from the parent except for the `Client ID` and `Secrets`. The inherited configurations will be marked as such. • If you are an administrator with the *Manage Applications* role, the newly created applications can have as parent any of the parent applications in the

Fields	Notes
	administration console. Choose the parent application from the list that pops up. • If you are an administrator with authorizations based on policies, you can assign as parent only the application to which you have read or manage applications rights. The applications to which you have read or manage rights are restricted on the basis of the `application.organization` attribute in the policy. Choose the parent application from the list that pops up. For more information about how to assign authorizations based on policies, see Configure Application Authorizations [page 2325].
Organization ID	Optional. This field allows you to group the applications under a specific organization and on the basis of the *Organization ID* to restrict the access to these applications. The *Organization ID* can contain lowercase Latin letters (a-z), numbers (0-9), and the underscore (_) character. It can be up to 255 characters. • If you are an administrator with *Manage Applications* role, the newly created applications by default are in the `global` *Organization ID*. You can change the *Organization ID* during the creation of the new application by entering a new name or later you can edit it. • If your authorizations are based on policies, the newly created applications can take the *Organization ID* of the applications to which you have manage rights on the basis of the `application.organization` attribute in the policy. Choose the allowed *Organization ID* from the drop-down. **ⓘ Note** Applications created either as subscriptions within or with the Identity service of an SAP BTP subaccount receive the *Organization ID* of the application that represents the trust configuration of that subaccount.
Protocol Type	Select the *SAML 2.0* radio-button. **→ Remember** Newly created applications with an assigned parent application will inherit the protocol from the parent. To change the protocol, first create the new application and then edit it.

Results

Once the application has been created, the system displays the message `Application <name of application> created`.

The newly created application appears on the list with the applications on the left. It is selected and you can proceed with its configuration.

Next Steps

1. Configure the SAML 2.0 trust with the service provider. For more information, see Configure SAML 2.0 Service Provider [page 1941].
2. (Optional) If necessary, configure the application. For more information about application configuration options, see Configuring Applications [page 1920].

Task overview: Create a New Application [page 1922]

Related Information

Create OpenID Connect (OIDC) Application [page 1925]

1.7.1.1.2 Create OpenID Connect (OIDC) Application

You can create a new OpenID Connect (OIDC) application.

Context

To create a new OpenID Connect (OIDC) application you have to add a new application to the list of applications in Identity Authentication, and set the protocol of the application to OpenID Connect (OIDC).

To create a new OpenID Connect (OIDC) application follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.

3. Choose the *Create* button on the left-hand panel and fill in the information in the dialog box.
4. Provide the following information:

Fields	Notes
Display Name	Required. The display name of the application appears on the logon and registration pages.
Home URL	Optional. Users are redirected to the *Home URL* after activating their accounts, when they are created via a CSV file import or the user registration service of Identity Authentication.
Type	Optional. **⚠ Caution** If the type of the application is not the correct one, the system may change it automatically without a warning.
Parent Application	Optional. **→ Remember** Newly created applications with an assigned parent application will inherit all the configurations from the parent except for the `Client ID` and `Secrets`. The inherited configurations will be marked as such. • If you are an administrator with the *Manage Applications* role, the newly created applications can have as parent any of the parent applications in the administration console. Choose the parent application from the list that pops up. • If you are an administrator with authorizations based on policies, you can assign as parent only the application to which you have read or manage applications rights. The applications to which you have read or manage rights are restricted on the basis of the `application.organization` attribute in the policy. Choose the parent application from the list that pops up. For more information about how to assign authorizations based on policies, see Configure Application Authorizations [page 2325].
Organization ID	Optional. This field allows you to group the applications under a specific organization and on the basis of the *Organization ID* to restrict the access to these applications. The *Organization ID* can contain lowercase Latin letters (a-z), numbers (0-9), and the underscore (_) character. It can be up to 255 characters. • If you are an administrator with *Manage Applications* role, the newly created applications by default are

Fields	Notes
	in the `global` *Organization ID*. You can change the *Organization ID* during the creation of the new application by entering a new name or later you can edit it. • If your authorizations are based on policies, the newly created applications can take the *Organization ID* of the applications to which you have manage rights on the basis of the `application.organization` attribute in the policy. Choose the allowed *Organization ID* from the drop-down. **Note** Applications created either as subscriptions within or with the Identity service of an SAP BTP subaccount receive the *Organization ID* of the application that represents the trust configuration of that subaccount.
Protocol Type	Select the *OpenID Connect(OIDC)* radio-button. **→ Remember** Newly created applications with an assigned parent application will inherit the protocol from the parent. To change the protocol, first create the new application and then edit it.

5. Choose *+Create*.

   Once the application has been created, the system displays the message `Application <name of application> created`.

   The newly created application appears on the list with the applications on the left. It is selected and you can set its protocol to OpenID Connect (OIDC).

Next Steps

(Optional) If necessary, configure additional settings for the application. For more information about the supported configurations for the OpenID Connect (OIDC) applications, see the following links:

- Change an Application's Display Name [page 1934]
- Configure an Application's Home URL [page 1937]
- Visit an Application's Web Page [page 1939]
- Configuring User Attributes from the Identity Directory [page 1973]
- Configuring Attributes Based on Flexible Expressions [page 1979]
- Configure Secrets for API Authentication [page 1997]
- Configure Certificates for API Authentication [page 2001]
- Enable or Disable Kerberos Authentication for an Application [page 2556]

- Configure Risk-Based Authentication for an Application [page 2007]
- Configure User Access to the Application [page 2025]
- Authenticating Identity Provider for an Application [page 2065]
- Enable Email Verification [page 2026]
- Configuring Password Policies [page 2223]
- Configuring Privacy Policies [page 2235]
- Configuring Terms of Use [page 2273]
- Display Application Name on Logon Page [page 2033]
- Configure Logo [page 2030]
- Choose and Configure a Branding Style for an Application [page 2035]
- Configure Registration and Upgrade Forms [page 2046]
- Protecting Self-Registration, Upgrade with Phone Verification [page 2050]
- Protecting Application Forms with Google reCAPTCHA [page 2056]
- Configure Identity Federation [page 2614]
- Configure Authentication Request to Corporate IdPs [page 1966]

Task overview: Create a New Application [page 1922]

Related Information

Create SAML 2.0 Application [page 1923]

1.7.1.2 View Application Configurations

After creating the application, you can view its configuration settings.

Prerequisites

You are assigned the *Manage Applications* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to view.

> ### Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.

Results

The application configuration is displayed. Navigate through the tabs and list items in to check the different configuration settings of the application.

1.7.1.3 Edit Applications

After creating the application, you can change its display name, home URL, type and parent application.

Prerequisites

You are assigned the *Manage Applications* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

The *Display Name* is required. You can't leave that field empty.

The *Home URL* field is optional. Users are redirected to the *Home URL* after activating their accounts, when they are created via a CSV file import or the user registration service of Identity Authentication. Initially, the *Home URL* for an application isn't configured in the administration console for SAP Cloud Identity Services. Once the *Home URL* has been set, you can change it.

> ### → Recommendation
>
> From a usability perspective we recommend you to use URL of a protected page.

> ### → Remember
>
> *Home URL* is necessary when you import new users in Identity Authentication. Identity Authentication needs to send activation emails to the new users and the home URL has to be mentioned in the emails. To access the application, the users have to activate their accounts. For more information see Import or Update Users for a Specific Application [page 2348].

The *Application Type* is optional. You can update the *Application Type* with one of the following:

Name Type	Application Type
SAP Analytics Cloud solution	`bundled`
SAP Ariba solution	`bundled`
SAP Business ByDesign solution	`bundled`
SAP BTP solution	`bundled`
SAP Central Business Configuration solution	`bundled`
SAP Concur solution	`bundled`
SAP Customer Data solution	`bundled`
SAP Data Custodian solution	`bundled`
SAP Datasphere solution	`bundled`
SAP Emarsys Customer Engagement solution	`bundled`
SAP Fieldglass solution	`bundled`
SAP Enable Now solution	`bundled`
SAP Industry Cloud Solution	`bundled`
SAP Integrated Business Planning solution	`bundled`
SAP Jam solution	`bundled`
SAP Marketing Cloud solution	`bundled`
SAP Sales Cloud and SAP Service Cloud solution	`bundled`
SAP Sales Cloud solution	`bundled`
SAP Signavio solution	`bundled`
SAP SuccessFactors solution	`bundled`
SAP S/4HANA Cloud solution	`bundled`
Other SAP cloud solution	`bundled`
SAP on-premise solution	`bundled`
Qualtrics solution	`bundled`
SAP LeanIX solution	`bundled`
SAP Business Data Cloud solution	`bundled`
Non-SAP solution	`charged`
Unknown	`charged`

→ Remember

You can't change the application type of the system applications.

Bundled SAP S/4 HANA Cloud and SAP Integrated Business Planning solutions are managed and configured by SAP and can't be deleted. Part of their SAML 2.0 configurations are read-only. Be careful when you make configuration changes.

The *Parent Application* field is optional.

Changing child application's configuration doesn't affect the configuration of the parent.

Applications that have a parent application configured can't be selected as parent application.

An application that is chosen as a parent application to another can't have a parent application assigned for it.

→ Remember

An application that is chosen as a parent application to another cannot be deleted.

To edit the application configurations, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   ⓘ Note

   Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.

   If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].
4. Choose the *Edit* button on the top right corner.
5. Edit the fields:

Field	Notes
Display Name	Required
Home URL	Optional
Type	Optional ⚠ Caution If the type of the application is not the correct one, the system may change it automatically without a warning.
Parent Application	Optional. • If you are an administrator with the *Manage Applications* role, the newly created applications can

Field	Notes
	have as parent any of the parent applications in the administration console. Choose the parent application from the list that pops up. • If you are an administrator with authorizations based on policies, you can assign as parent only the application to which you have read or manage applications rights. The applications to which you have read or manage rights are restricted on the basis of the `application.organization` attribute in the policy. Choose the parent application from the list that pops up. For more information about how to assign authorizations based on policies, see Configure Application Authorizations [page 2325]. Existing applications that have a parent application assigned to them will inherit only the configurations that have not been changed, including the configurations that are made at the creation of the application. The configurations that are changed, and the configurations made at the creation of the application are not inherited. See the list below for the configurations made at the creation of the application: • Protocol • Subject Name Identifier **Note** When a subscribed application becomes a child application, the configuration of the `Subject Name Identifier` of that child application can be different from the parent application or from the application it is subscribed to. You can make the configuration of the `Subject Name Identifier` the same as the parent application by choosing the *Inherit from Parent* button on the top right-hand corner of the screen. You can also make the configuration of the `Subject Name Identifier` the same as the application it is subscribed to by choosing the *Reset* button on the top right-hand corner of the screen. Depending on your configuration, either the *Inherit from Parent* or the *Reset* button is active, or both buttons are active. • Default Name ID Format • Attributes

Field	Notes
	Note Attributes from different sources are inherited separately. For more information about the different user attribute sources, see User Attributes [page 1972]. • Dependencies • Risk-Based Authentication • Conditional Authentication • Email Verification • Password Policy • Terms Of Use • Privacy Policy • User Application Access • User Attributes for Access • Token Url Separator • Reload Parent Page **→ Tip** A child application can override all the configurations inherited from the parent application. If you change a configuration in the child that is inherited from the parent, and after that you decide to return to the inherited one, go to the respective configuration and choose the *Inherit from Parent* button on the top right-hand corner of the screen. **Caution** If you change a configuration in the child that is inherited from the parent, also change the name of the child, so that it differs from the name of the parent. Saving the configuration without changing the child application's name may result in failing authentication requests.
Organization ID	The *Organization ID* can contain lowercase Latin letters (a-z), numbers (0-9), and the underscore (_) character. It can be up to 255 characters. **→ Remember** The field is visible to users assigned to application authorization policies, or with Manage Applications role.

Field	Notes
	For more information about the application authorization policies, see Configure Application Authorizations [page 2325]. This field allows you to group the applications under a specific organization and on the basis of the *Organization ID* to restrict the access to these applications. By default all applications are in the `global` *Organization ID*. Enter a new name to change the value of *Organization ID*. **Note** Applications created either as subscriptions within or with the Identity service of an SAP BTP subaccount receive the *Organization ID* of the application that represents the trust configuration of that subaccount.
Protocol Type	Choose from: • SAML 2.0 (default selection) • OpenID Connect **Note** There are applications that show the `SAML 2.0 & OpenID Connect` protocol type. You can't create such applications. **→ Remember** If you change the parent application, the child application will inherit the protocol from the new parent. To change the protocol, first finish the editing and then repeat the procedure again, this time changing the protocol.

6. Save your changes.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

1.7.1.4 Change an Application's Display Name

Prerequisites

You are assigned the *Manage Applications* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Edit* button on the top right corner.
5. Change the name in the *Application Display Name* field in the dialog box that appears.
6. Save your changes.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

1.7.1.5 Configure an Application's Type

You can choose and set the *Application Type* in the administration console for SAP Cloud Identity Services.

Context

In some cases Identity Authentication can't determine the correct application type because, for example of unknown URL or custom domain. In such cases you can update the application type with one of the following:

Name Type	Application Type
SAP Analytics Cloud solution	`bundled`
SAP Ariba solution	`bundled`
SAP Business ByDesign solution	`bundled`
SAP BTP solution	`bundled`
SAP Central Business Configuration solution	`bundled`
SAP Concur solution	`bundled`
SAP Customer Data solution	`bundled`
SAP Data Custodian solution	`bundled`
SAP Datasphere solution	`bundled`

Name Type	Application Type
SAP Emarsys Customer Engagement solution	`bundled`
SAP Fieldglass solution	`bundled`
SAP Enable Now solution	`bundled`
SAP Industry Cloud Solution	`bundled`
SAP Integrated Business Planning solution	`bundled`
SAP Jam solution	`bundled`
SAP Marketing Cloud solution	`bundled`
SAP Sales Cloud and SAP Service Cloud solution	`bundled`
SAP Sales Cloud solution	`bundled`
SAP Signavio solution	`bundled`
SAP SuccessFactors solution	`bundled`
SAP S/4HANA Cloud solution	`bundled`
Other SAP cloud solution	`bundled`
SAP on-premise solution	`bundled`
Qualtrics solution	`bundled`
SAP LeanIX solution	`bundled`
SAP Business Data Cloud solution	`bundled`
Non-SAP solution	`charged`
Unknown	`charged`

> → Remember
>
> You can't change the application type of the system applications.
>
> Bundled SAP S/4 HANA Cloud and SAP Integrated Business Planning solutions are managed and configured by SAP and can't be deleted. Part of their SAML 2.0 configurations are read-only. Be careful when you make configuration changes.

To configure the *Application Type*, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> ⓘ Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Edit* button on the top-right corner.
5. Choose the correct type from the drop-down in the *Application Type* field in the dialog box that appears.

> ⚠ Caution
>
> If the type of the application is not the correct one, the system may change it automatically without a warning.

6. Save your changes.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

Related Information

Invitation REST API [page 2802]
User Registration [page 2807]

1.7.1.6 Configure an Application's Home URL

You can configure the *Home URL* of an application in the administration console for SAP Cloud Identity Services.

Context

Users are redirected to the *Home URL* after activating their accounts, when they are created via a CSV file import or the user registration service of Identity Authentication. Initially, the *Home URL* for an application is not configured in the administration console for SAP Cloud Identity Services. Once the *Home URL* has been set, you can change it.

> → Recommendation
>
> From a usability perspective we recommend you to use URL of a protected page.

> → Remember
>
> *Home URL* is necessary when you import new users in Identity Authentication. Identity Authentication needs to send activation emails to the new users and the home URL has to be mentioned in the e-mails.

To access the application, the users have to activate their accounts. For more information see Import or Update Users for a Specific Application [page 2348].

To configure the *Home URL*, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Edit* button on the top right corner.
5. Type the address in the *Application Home URL* field in the dialog box that appears.

   > → Tip
   >
   > From a usability perspective we recommend you to use URL for a protected page.

6. Save your changes.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

Related Information

Invitation REST API [page 2802]
User Registration [page 2807]

1.7.1.7 Visit an Application's Web Page

You can navigate to an application's Web site directly from the administration console for SAP Cloud Identity Services.

Prerequisites

You have configured *Home URL* for the application in question. For more information, see Related Information.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Select the *Home URL* anchor text.

   The Web page for the application opens in a new window.

Related Information

Configure an Application's Home URL [page 1937]
Create a New Application [page 1922]

1.7.1.8 Delete Applications

As a tenant administrator, you can delete one or more charged or bundled applications in a tenant of Identity Authentication.

Prerequisites

- You are assigned the *Manage Applications* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have at least one bundled or charged application that you want to delete.

Context

A *Delete Applications* operation removes the application and all of its configurations from the tenant of Identity Authentication.

> ⓘ Note
>
> You can only delete charged or non-SAP managed bundled applications. The system applications and the bundled applications that are managed and configured by SAP cannot be deleted. For more information, see Application Types [page 63] and Configure an Application's Type [page 1935].

Follow the procedure below for each application that you want to delete:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Select the application that you want to delete.

   > → Tip
   >
   > Type the name of the application in the search field to filter the list items.

4. Choose the *Delete* button in the right-hand panel to delete the selected application.
5. Confirm the operation in the pop-up dialog.

   Once the application has been deleted, the system displays the message `Application <application name> deleted`.

1.7.1.9 Configure Trust

This document is intended to help you configure a trusted service provider (SP) or client (relying party) in the administration console for SAP Cloud Identity Services.

Context

The trust is configured by uploading the service provider metadata (for the SAML 2.0 applications), or by entering the information manually (for the SAML 2.0 or OpenID Connect applications). Depending on the SSO type, you can configure a SAML 2.0 trusted service provider or Open ID Connect trusted client in the administration console for SAP Cloud Identity Services:

Configure SAML 2.0 Service Provider [page 1941]
: This document is intended to help you configure a SAML 2.0 service provider (SP) in the administration console for SAP Cloud Identity Services.

Configure OpenID Connect (OIDC) Application [page 1949]
: This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services.

1.7.1.9.1 Configure SAML 2.0 Service Provider

This document is intended to help you configure a SAML 2.0 service provider (SP) in the administration console for SAP Cloud Identity Services.

Context

The trust is configured from the service provider metadata by uploading it or by accessing it from a URL, or by entering the information manually.

You can enter manually the name of the service provider, its endpoints, and certificates.

To configure the SAML 2.0 settings in the administration console for SAP Cloud Identity Services, choose the following options:

Make a New Trust Configuration [page 1942]
: Configure trust from the service provider metadata by uploading it or by accessing it from a URL, or by entering the information manually.

Configure Signing Certificates and Certificate Options [page 1946]
: You can add up to two signing certificates per application in the administration console for Cloud Identity Services.

Task overview: Configure Trust [page 1941]

Related Information

Configure OpenID Connect (OIDC) Application [page 1949]

1.7.1.9.1.1 Make a New Trust Configuration

Configure trust from the service provider metadata by uploading it or by accessing it from a URL, or by entering the information manually.

Prerequisites

You have the service provider metadata. See the service provider documentation for more information or contact the administrator of the service provider.

→ Tip

For more information how to download the metadata for SAP BTP when it acts as a service provider (SP), see Application Identity Provider. The content in this section is only relevant for SAP BTP Neo environment. The content in this section isn't relevant for China (Shanghai) region.

→ Remember

If your scenario includes the enabling of the *Trust All Corporate Identity Providers* option in the administration console, the service provider metadata must contain the assertion consumer (ACS) endpoint that can process unsolicited SAML responses.

With SAP BTP, the endpoint is the URL of the application's protected page. This endpoint must be either set as a default ACS endpoint of the service provider in Identity Authentication, or chosen by its index when performing IdP-initiated SSO. For more information, see Configure IdP-Initiated SSO [page 2201].

✤ Example

```
<ns3:AssertionConsumerService index="1" isDefault="false"
Binding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-POST" Location="https://
<application URL>/protected.jsp"
/>
```

Context

The trust is configured from the service provider metadata by uploading it or by accessing it from a URL, or by entering the information manually. When the identity provider metadata is uploaded, or the metadata URL is used, the fields are populated with the parsed data from the XML file.

You can enter manually the name of the service provider and its endpoints.

> → Remember
>
> If your scenario includes the enabling of the *Trust All Corporate Identity Providers* option, the assertion consumer (ACS) endpoint with the URL of the application's protected page, and the index must be included in the service provider metadata.

To configure a SAML 2.0 trusted service provider in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *SAML 2.0 Configuration*.
6. Upload the service provider metadata XML file, use the metadata URL, or manually enter the communication settings negotiated between Identity Authentication and the service provider.
   - *Load from URL* - The URL with service provider metadata.

     > → Remember
     >
     > The SAML 2.0 metadata URL is required for the automatic renewal of the automatic renewal of the signing and encryption certificates of the application. When the metadata URL is provided, Identity Authentication will update automatically the expired encryption certificate, and the SAML 2.0 certificate during the first sign in attempt that fails due to the expired certificate.
     >
     > If you update the already provided metadata URL, the SAML configuration will be updated, too.

   - *Load from File* - The metadata file of the service provider.

     > ⓘ Note
     >
     > Use a file with an extension `.xml`.

   - *Configure Manually* - Add the entity ID of the service provider and configure the information in the *Endpoints* tab.

Tab	Description
Endpoints	The SP's endpoint URL that receives the response with the SAML assertion from Identity Authentication. The following options are possible: • *Single Sign-On* Choose *Create* and enter the allowed domain for browser flows. • *Single Logout* Choose *Add* and enter the SP's endpoint URL that receives the logout response or request (for a multiple SPs scenario) from Identity Authentication for the termination of all current sessions. This field has the following attributes: • *Binding* - specifies the SAML binding supported by the logout endpoint. • HTTP-POST • HTTP-REDIRECT • SOAP - The SOAP Endpoint is called only when the user password is changed. **Note** If you have configured a *Warning* concurrent user access option, you must also have a Single Logout Endpoints (SLO) URL with a SOAP binding added for the SAML 2.0 application. Otherwise if a user chooses to sign out and continue to new session, an SLO request for the old session won't be sent to the application. For more information, see Configure Concurrent User Access to the Application [page 2018]. • *URL* - specifies the location of the logout endpoint. • *Response URL* - (optional) specifies a different location to which logout response messages should be sent. • *URL for Principal Propagation* - URL is required for principal propagation scenarios to ABAP applications according to RFC 7522. For the proper URL consult the documentation of the providing application.

Tab	Description
	Note During authentication the ACS endpoint is provided with the request. Through the flow, Identity Authentication is removing the query attributes, and during the response it compares the ACS endpoint with what is configured in the SAML 2.0 configuration of the application. Identity Authentication is looking for an exact match and If there is no such match the authentication will fail.

Restriction

The *Metadata URL*, *Entity ID*, and *Single Logout Endpoint* fields are not editable for the system applications.

Next Steps

Configure trust on the service provider side.

1. Download the SAML 2.0 metadata of Identity Authentication.

   Note

   For more information about how to download the SAML 2.0 metadata describing Identity Authentication as identity provider see Tenant SAML 2.0 Configurations [page 2103].

2. Configure the service provider to trust Identity Authentication.

   Note

   See the service provider documentation for more information about how to configure the trust.

   Tip

   If you use SAP BTP as a service provider, see Integrating the Service with SAP Business Technology Platform, Neo Environment [page 2910].

   The content in this section is only relevant for SAP BTP Neo environment.

   The content in this section is not relevant for China (Shanghai) region.

3. Configure the certificates in the administration console for Cloud Identity Services. For more information, see Configure Signing Certificates and Certificate Options [page 1946].

Task overview: Configure SAML 2.0 Service Provider [page 1941]

Related Information

Configure Signing Certificates and Certificate Options [page 1946]

1.7.1.9.1.2 Configure Signing Certificates and Certificate Options

You can add up to two signing certificates per application in the administration console for Cloud Identity Services.

Context

You can add up to two signing certificates. Both signing certificates are accepted according to the certificate validity.

You can choose the identity provider certificate to be used for signing for each application. For more information about the identity provider certificates, see Tenant SAML 2.0 Configurations [page 2103].

The idea behind the ability to choose the IdP certificate is that when you want to change the default IdP certificate all applications will have downtime since the applications have trust with the current default application on the application side. So, when adding new IdP certificate you can change the applications one by one to trust the new certificate.

> → Tip
>
> You can download the tenant metadata with the specific identity provider (IdP) certificate for the application by choosing the *Download Metadata File* button.

To configure signing certificates and certificate options in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *SAML 2.0 Configuration* > *Certificates*.
6. Under *Signing Certificate*, choose *Add* to add a base64-encoded certificate used by the identity provider to verify the signatures of the SAML protocol messages created by the service provider.

> **Note**
>
> If you have two certificates, you can choose a default one, to mark your primary certificate.
>
> The primary certificate is always validated first. If its validation isn't successful, then the secondary certificate is used. The certificate marked as primary is your main certificate. When it's time to rotate the certificates we recommend you to add a second certificate. When the first certificate expires, and you have a second one, you can safely remove the first one.

7. **Optional:** Under *Signing Options*, choose *Edit* and after that *Save* to configure the signing options for the application. You have the following possibilities:

Option	Default Configuration
Signing Algorithm	Select the digest algorithm for signing outgoing messages from the dropdown list. • *SHA-1* • *SHA-256* - the default option (for applications created after Jun 28, 2021) • *SHA-512*
Sign assertions	*On*
Sign authentication responses	*Off*
Sign single logout messages	*On*
Require signed authentication requests	*Off*
Require signed single logout messages	*On*

8. **Optional:** Under *Encryption Certificate*, add a certificate, if there is no encryption certificates added or you want to add a new certificate.
    a. Choose *Add*.
    b. Provide a file with extension `.crt` or `cer`, or insert the certificate as text.
    c. Choose *Add*.
9. **Optional:** (Visible only if there is and encryption certificate) Under *Encryption Option*, choose *Edit* to configure the encryption of the SAML 2.0 response
    a. Choose the elements to encrypt from the drop-down:
        - *None* - the default option
        - *Whole Assertion*
        - *Subject Name ID*
        - *Subject Name ID and Attributes*
        - *Attributes*

        The method for encryption is `aes-128-cbc`.
    b. Choose *Save*.

10. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *SAML 2.0 Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. **→ Tip** To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *Identity Provider Certificates* > *+Add*. **⚠ Caution** The application will stop working if the configuration of the application is not updated with the new certificate. **ⓘ Note** Your choice of *Active* certificate in the list isn't related to the choice of the *Default* certificate in *Tenant Settings* > *SAML 2.0 Configuration* > *SAML 2.0* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for SAML 2.0 Applications [page 2111].

→ Recommendation

When the default identity provider certificate is changed with a new one, and the old one isn't used anymore, we recommend you to delete the old certificate.

11. Save your selection.

    Once the application has been changed, the system displays the message `Application <name of application> updated`.

Task overview: Configure SAML 2.0 Service Provider [page 1941]

Related Information

Make a New Trust Configuration [page 1942]

1.7.1.9.2 Configure OpenID Connect (OIDC) Application

This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services.

Context

For more information about what Identity Authentication supports for OpenID Connect (OIDC), and how to configure the different flows, see Configuring OpenID Connect (OIDC) [page 2636].

Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 1950]
: This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the authorization code flow.

Configure OpenID Connect (OIDC) Application for Client Credentials Flow [page 1953]
: This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the client credentials flow.

Configure OpenID Connect (OIDC) Application for Resource Owner Password Credentials Flow [page 1955]
: This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the resource owner password credentials flow.

Configure OpenID Connect Application for Implicit Flow [page 1958]
: This document is intended to help you configure an OpenID Connect application in the administration console for SAP Cloud Identity Services for the implicit flow.

Configure OpenID Connect (OIDC) Application for JWT Bearer Flow [page 1960]
: This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the JWT bearer flow.

Configure OpenID Connect Application for Token Exchange [page 1963]
: This document is intended to help you configure an OpenID Connect application in the administration console for SAP Cloud Identity Services for the Token Exchange flow.

Task overview: Configure Trust [page 1941]

Related Information

Configure SAML 2.0 Service Provider [page 1941]

1.7.1.9.2.1 Configure OpenID Connect (OIDC) Application for Authorization Code Flow

This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the authorization code flow.

Prerequisites

You have an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party), and its redirect URIs.

To configure an OpenID Connect (OIDC) trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect (OIDC) Configuration*.
6. Manually enter the communication settings negotiated between Identity Authentication and the client.

Setting	Description
Name (mandatory)	Provide a name of your choice.

Setting	Description
Redirect URIs (mandatory)	The redirection URIs to which the response can be sent. You can add up to 20 redirect URIs.
Front-Channel Logout URIs	(Optional) URIs which will be requested for logout. You can add up to 20 URIs.
Back-Channel Logout URI	(Optional) URIs which will be requested for logout. You can add up to 20 URIs.

> → Tip
>
> For more information about the format of the redirect URIs and post logout redirect URIs, see OpenID Connect (OIDC) Application Configurations [page 2747].

7. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. → Tip To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. ⚠ Caution The application will stop working if the configuration of the application is not updated with the new certificate. ⓘ Note Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

> → Tip
>
> When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

8. Select the following grant types:

Grant Type	Notes
Authorization Code	For the authorization code flow.
Authorization Code > *Authorization Code / Enforce PKCE (S256)*	For the authorization code flow with PKCE.

> ⓘ Note
>
> Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

9. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

> → Remember
>
> Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

- Configure HTTP basic authentication for the application. For more information about the configuration, see Configure Secrets for API Authentication [page 1997].
- Enable the public client flows option for this application. In the administration console, choose *the OpenID Connect (OIDC) application* > *Client Authentication under the Trust tab* > *Enable Public Client Flows under Public Client*. Optionally, you can configure the API permission groups. For more information, see Consuming APIs from Other Applications [page 2889] .

> ⓘ Note
>
> The *Public* client type is used for environments where it is difficult to protect the client credential, such as mobile and desktop applications, and client-side parts of web applications.

Task overview: Configure OpenID Connect (OIDC) Application [page 1949]

Related Information

Configure OpenID Connect (OIDC) Application for Client Credentials Flow [page 1953]
Configure OpenID Connect (OIDC) Application for Resource Owner Password Credentials Flow [page 1955]
Configure OpenID Connect Application for Implicit Flow [page 1958]
Configure OpenID Connect (OIDC) Application for JWT Bearer Flow [page 1960]
Configure OpenID Connect Application for Token Exchange [page 1963]
Configuring OpenID Connect (OIDC) [page 2636]

Handle Single Logout Request from Identity Authentication [page 2743]

1.7.1.9.2.2 Configure OpenID Connect (OIDC) Application for Client Credentials Flow

This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the client credentials flow.

Prerequisites

You have an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party).

To configure an OpenID Connect (OIDC) trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect (OIDC) Configuration*.
6. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. → Tip To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. ⚠ Caution The application will stop working if the configuration of the application is not updated with the new certificate. ⓘ Note Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

→ Tip

When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

7. Select the *Client Credentials* grant type.

ⓘ Note

Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

8. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

→ Remember

Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

1. Configure authentication for the application. For more information about the configuration, see API Authentication [page 1996].

2. Invalidate the tokens that are no longer need. For more information, see Call Identity Authentication Revoke Token Endpoint [page 2736].

Task overview: Configure OpenID Connect (OIDC) Application [page 1949]

Related Information

Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 1950]
Configure OpenID Connect (OIDC) Application for Resource Owner Password Credentials Flow [page 1955]
Configure OpenID Connect Application for Implicit Flow [page 1958]
Configure OpenID Connect (OIDC) Application for JWT Bearer Flow [page 1960]
Configure OpenID Connect Application for Token Exchange [page 1963]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.1.9.2.3 Configure OpenID Connect (OIDC) Application for Resource Owner Password Credentials Flow

This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the resource owner password credentials flow.

Prerequisites

You have an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party).

To configure an OpenID Connect (OIDC) trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.

2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> ### Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect (OIDC) Configuration*.
6. Under the *Configure Manually* section provide a name of your choice.
7. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. **→ Tip** To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. **⚠ Caution** The application will stop working if the configuration of the application is not updated with the new certificate. **Note** Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

> ### → Tip
>
> When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

8. Select the *Password*

> ⓘ Note
>
> Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

9. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

> → Remember
>
> Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

- Configure HTTP basic authentication for the application. For more information about the configuration, see Configure Secrets for API Authentication [page 1997].
- Enable the public client flows option for this application. In the administration console, choose *the OpenID Connect (OIDC) application* > *Client Authentication under the Trust tab* > *Enable Public Client Flows under Public Client*. Optionally, you can configure the API permission groups. For more information, see Consuming APIs from Other Applications [page 2889] .

> ⓘ Note
>
> The *Public* client type is used for environments where it is difficult to protect the client credential, such as mobile and desktop applications, and client-side parts of web applications.

Task overview: Configure OpenID Connect (OIDC) Application [page 1949]

Related Information

Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 1950]
Configure OpenID Connect (OIDC) Application for Client Credentials Flow [page 1953]
Configure OpenID Connect Application for Implicit Flow [page 1958]
Configure OpenID Connect (OIDC) Application for JWT Bearer Flow [page 1960]
Configure OpenID Connect Application for Token Exchange [page 1963]

1.7.1.9.2.4 Configure OpenID Connect Application for Implicit Flow

This document is intended to help you configure an OpenID Connect application in the administration console for SAP Cloud Identity Services for the implicit flow.

Prerequisites

You have an OpenID Connect application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party), and its redirect URIs.

To configure an OpenID Connect trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect Configuration*.
6. Manually enter the communication settings negotiated between Identity Authentication and the client.

Setting	Description
Name (mandatory)	Provide a name of your choice.

Setting	Description
Redirect URIs (optional)	The redirection URIs to which the response can be sent. You can add up to 20 redirect URIs.
Post Logout Redirect URIs (optional)	The redirection URIs where the user can be forwarded after logout. You can add up to 20 redirect URIs.
Front-Channel Logout URIs (optional)	URIs which will be requested for logout. You can add up to 20 URIs.

> → Tip
>
> For more information about the format of the redirect URIs and post logout redirect URIs, see OpenID Connect (OIDC) Application Configurations [page 2747].

7. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. → Tip To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. ⚠ Caution The application will stop working if the configuration of the application is not updated with the new certificate. ⓘ Note Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

> → Tip
>
> When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

8. Selct the *Implicit* grant type.

> ⓘ Note
>
> Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

9. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

> → Remember
>
> Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

Configure HTTP basic authentication for the application. For more information about the configuration, see Configure Secrets for API Authentication [page 1997].

Task overview: Configure OpenID Connect (OIDC) Application [page 1949]

Related Information

Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 1950]
Configure OpenID Connect (OIDC) Application for Client Credentials Flow [page 1953]
Configure OpenID Connect (OIDC) Application for Resource Owner Password Credentials Flow [page 1955]
Configure OpenID Connect (OIDC) Application for JWT Bearer Flow [page 1960]
Configure OpenID Connect Application for Token Exchange [page 1963]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.1.9.2.5 Configure OpenID Connect (OIDC) Application for JWT Bearer Flow

This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the JWT bearer flow.

Prerequisites

You have an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party), and its redirect URIs.

To configure an OpenID Connect (OIDC) trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect (OIDC) Configuration*.
6. Manually enter the communication settings negotiated between Identity Authentication and the client.

Setting	Description
Name (mandatory)	Provide a name of your choice.
Redirect URIs (optional)	The redirection URIs to which the response can be sent. You can add up to 20 redirect URIs.
Post Logout Redirect URIs (optional)	The redirection URIs where the user can be forwarded after logout. You can add up to 20 redirect URIs.
Front-Channel Logout URIs (optional)	URIs which will be requested for logout. You can add up to 20 URIs.

   > → Tip
   >
   > For more information about the format of the redirect URIs and post logout redirect URIs, see OpenID Connect (OIDC) Application Configurations [page 2747].

7. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. → Tip To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. ⚠ Caution The application will stop working if the configuration of the application is not updated with the new certificate. ⓘ Note Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

> → Tip
>
> When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

8. Select the *JWT Bearer* grant type.

> ⓘ Note
>
> Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

9. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

> → Remember
>
> Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

- Configure HTTP basic authentication for the application. For more information about the configuration, see Configure Secrets for API Authentication [page 1997].

- Enable the public client flows option for this application. In the administration console, choose *the OpenID Connect (OIDC) application* > *Client Authentication under the Trust tab* > *Enable Public Client Flows under Public Client*. Optionally, you can configure the API permission groups. For more information, see Consuming APIs from Other Applications [page 2889] .

  > **Note**
  >
  > The *Public* client type is used for environments where it is difficult to protect the client credential, such as mobile and desktop applications, and client-side parts of web applications.

Task overview: Configure OpenID Connect (OIDC) Application [page 1949]

Related Information

Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 1950]
Configure OpenID Connect (OIDC) Application for Client Credentials Flow [page 1953]
Configure OpenID Connect (OIDC) Application for Resource Owner Password Credentials Flow [page 1955]
Configure OpenID Connect Application for Implicit Flow [page 1958]
Configure OpenID Connect Application for Token Exchange [page 1963]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.1.9.2.6 Configure OpenID Connect Application for Token Exchange

This document is intended to help you configure an OpenID Connect application in the administration console for SAP Cloud Identity Services for the Token Exchange flow.

Prerequisites

You have an OpenID Connect application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party), and its redirect URIs.

To configure an OpenID Connect trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> ### Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect Configuration*.
6. Manually enter the communication settings negotiated between Identity Authentication and the client.

Setting	Description
Name (mandatory)	Provide a name of your choice.
Redirect URIs (optional)	The redirection URIs to which the response can be sent. You can add up to 20 redirect URIs.
Post Logout Redirect URIs (optional)	The redirection URIs where the user can be forwarded after logout. You can add up to 20 redirect URIs.
Front-Channel Logout URIs (optional)	URIs which will be requested for logout. You can add up to 20 URIs.

> ### → Tip
>
> For more information about the format of the redirect URIs and post logout redirect URIs, see OpenID Connect (OIDC) Application Configurations [page 2747].

7. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list.

Option	Description
	→ Tip To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. ⚠ Caution The application will stop working if the configuration of the application is not updated with the new certificate. ⓘ Note Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

> → Tip
>
> When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

8. Select the *Token Exchange (RFC 8693)* grant type.

> ⓘ Note
>
> Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

9. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

> → Remember
>
> Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

- Configure HTTP basic authentication for the application. For more information about the configuration, see Configure Secrets for API Authentication [page 1997].
- Enable the public client flows option for this application. In the administration console, choose *the OpenID Connect (OIDC) application* > *Client Authentication under the Trust tab* > *Enable Public Client Flows under Public Client*. Optionally, you can configure the API permission groups. For more information, see Consuming APIs from Other Applications [page 2889] .

> **Note**
>
> The *Public* client type is used for environments where it is difficult to protect the client credential, such as mobile and desktop applications, and client-side parts of web applications.

Task overview: Configure OpenID Connect (OIDC) Application [page 1949]

Related Information

Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 1950]
Configure OpenID Connect (OIDC) Application for Client Credentials Flow [page 1953]
Configure OpenID Connect (OIDC) Application for Resource Owner Password Credentials Flow [page 1955]
Configure OpenID Connect Application for Implicit Flow [page 1958]
Configure OpenID Connect (OIDC) Application for JWT Bearer Flow [page 1960]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.1.10 Configure Authentication Request to Corporate IdPs

Configure Authentication Context (SAML 2.0) [page 1967]
: Tenant administrator can configure the authentication context in the request sent to the corporate identity providers when Identity Authentication acts as a proxy identity provider.

Configure Authentication Context (OpenID Connect (OIDC)) [page 1968]
: Tenant administrator can configure the authentication context in the request sent to the corporate identity providers when Identity Authentication acts as a proxy identity provider.

Configure Different Trust Configurations for the Same Identity Authentication Tenant [page 1969]
: Tenant administrator can configure the issuer name in the request sent to the corporate identity providers when Identity Authentication acts as a proxy identity provider.

Related Information

Corporate Identity Providers [page 2556]

1.7.1.10.1 Configure Authentication Context (SAML 2.0)

Tenant administrator can configure the authentication context in the request sent to the corporate identity providers when Identity Authentication acts as a proxy identity provider.

Context

> **Note**
>
> The following configuration is valid for SAML 2.0 corporate identity providers.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].
4. Choose the *Trust* tab.
5. Under *Conditional Authentication*, choose *Configure Requests to Corporate Identity Providers* > *SAML 2.0*.
6. Choose one of the following options:
   - *None* - Authentication context isn't sent. The requested authentication context from the service provider is ignored.
   - *Service Provider Authentication Context* (Available only for SAML 2.0 applications) - The received authentication context from the service provider is sent.
   - *Password Protected Transport* - Authentication context class `urn:oasis:names:tc:SAML:2.0:ac:classes:PasswordProtectedTransport` is sent. The requested authentication context from the service provider is ignored.
7. Save your configuration.

Task overview: Configure Authentication Request to Corporate IdPs [page 1966]

Related Information

Configure Authentication Context (OpenID Connect (OIDC)) [page 1968]
Configure Different Trust Configurations for the Same Identity Authentication Tenant [page 1969]

1.7.1.10.2 Configure Authentication Context (OpenID Connect (OIDC))

Tenant administrator can configure the authentication context in the request sent to the corporate identity providers when Identity Authentication acts as a proxy identity provider.

Context

> **Note**
>
> The following configuration is valid for the OpenID Connect (OIDC) corporate identity providers.

The authentication context enables administrators to ensure an authentication context. For example, you can require the corporate identity provider to authenticate users for a particular application with multifactor authentication or certificate-based authentication. The corporate identity providers define what authentication contexts that they support as well as the values they require to trigger a particular authentication context. The OpenID Connect (OIDC) standard doesn't define the authentication context values.

For more information, see OpenID Connect Core 1.0 incorporating errata set 2 .

> **Tip**
>
> If the authentication context exists in the metadata of the identity provider it, can be seen in the administration console at *Identity Providers* > *Corporate Identity Providers* > *from the list - the corporate IdP that provides user authentication to the application* > *OpenID Connect (OIDC) Configuration* > *Supported Authentication Contexts section*.

> **Restriction**
>
> You can add up to 20 authentication context references with a length of up to 99 characters each.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.

3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *Conditional Authentication*, choose *Configure Requests to Corporate Identity Providers* > *OpenID Connect (OIDC)*.
6. Choose *Add* > *enter the authentication context class references* > *save your changes*.

Results

The application sends the configured ACRs (authentication context references) to the corporate identity provider as part of the authorization request. These values are included in the `acr_values` query parameter and are separated by spaces ( ). You can see this parameter in the network trace of your browser.

Task overview: Configure Authentication Request to Corporate IdPs [page 1966]

Related Information

Configure Authentication Context (SAML 2.0) [page 1967]
Configure Different Trust Configurations for the Same Identity Authentication Tenant [page 1969]

1.7.1.10.3 Configure Different Trust Configurations for the Same Identity Authentication Tenant

Tenant administrator can configure the issuer name in the request sent to the corporate identity providers when Identity Authentication acts as a proxy identity provider.

Context

> **Note**
>
> To configure Identity Authentication to use a different issuer name when communicating with Microsoft Entra ID (formerly known as Microsoft Azure Active Directory or Azure AD), see (Optional) Configure

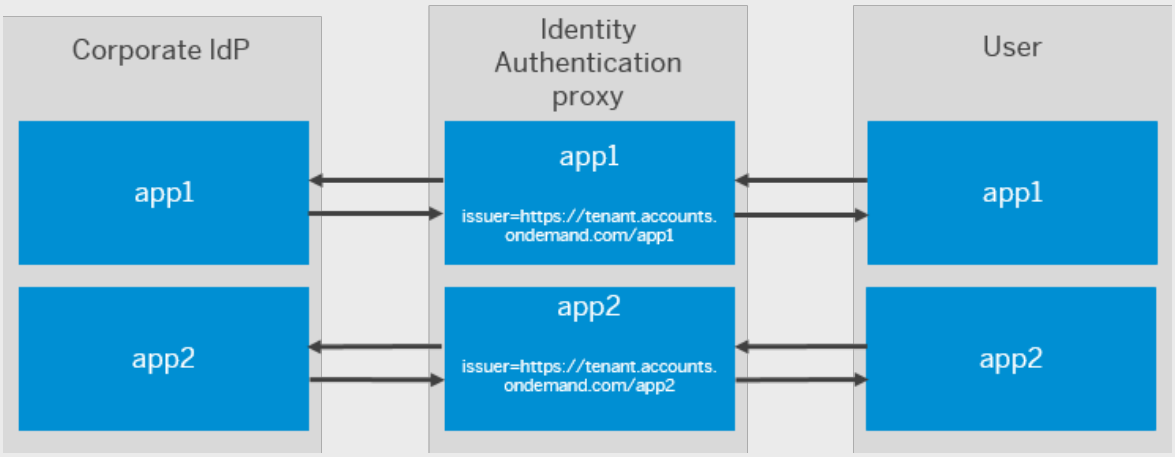
Different Trust Configurations for the Same Identity Authentication Tenant (Microsoft Entra ID Apps) [page 2956].

When Identity Authentication acts as a proxy identity provider you can have different trust configurations for the different applications at the corporate identity provider side. To do this you should configure Identity Authentication to use a different issuer name when communicating with the corporate identity provider, depending on the application for which the authentication is done.

When Identity Authentication issues a SAML2.0 authentication request, logout request or response to the corporate identity provider, then it uses the configured issuer name for the corresponding application.

When Identity Authentication verifies the SAML2.0 message coming from the corporate identity provider, then the verification is done against the configured issuer for the corresponding application.

Example



Identity Authentication acts as identity provider proxy and there are two configured applications app1 and app2. App1 is configured with `issuer=https://tenant.accounts.ondemand.com/app1`. App2 is configured with `issuer=https://tenant.accounts.ondemand.com/app2`. Both applications use one and the same corporate identity provider for authentication.

When App1 issues an authentication request to Identity Authentication, then Identity Authentication sends authentication request to the corporate identity provider, using `issuer=https://tenant.accounts.ondemand.com/app1`. The corporate identity provider authenticates the user and sends authentication response with `audience=https://tenant.accounts.ondemand.com/app1`. Then Identity Authentication sends authentication response to app1.

After that, the same user tries to authenticate to app2 from the same browser session. App2 issues an authentication request to Identity Authentication. Then Identity Authentication again sends authentication request to the corporate identity provider, but this time with `issuer=https://tenant.accounts.ondemand.com/app2`. Thus, the corporate identity provider can apply the specific authentication mechanisms, configured for `issuer=https://tenant.accounts.ondemand.com/app2`. After the authentication, the corporate identity provider issues an authentication response with `audience=https://tenant.accounts.ondemand.com/app2`. Then Identity Authentication sends authentication response to app2.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Choose *Conditional Authentication* > *Configure Requests to Corporate Identity Providers* > *SAML 2.0*.
6. Under *Configure Issuer Name*, type the issuer suffix, which you want to add to the Identity Authentication default issuer name.

   > **Note**
   >
   > The issuer name can contain Latin letters (A-Z and a-z), numbers, and the dash (-) character. It can be up to 36 characters.

   The issuer name, which will be used when sending SAML2.0 requests to the corporate identity provider appears under the input field. It is in the format `<Identity_Authentication_default_issuer_name>/<issuer_name_suffix_typed_in_the_input_field>`. For example, `https://my-tenant.accounts.ondemand.com/suffix/`.
7. Save your changes.

Task overview: Configure Authentication Request to Corporate IdPs [page 1966]

Related Information

Configure Authentication Context (SAML 2.0) [page 1967]

Configure Authentication Context (OpenID Connect (OIDC)) [page 1968]

(Optional) Configure Different Trust Configurations for the Same Identity Authentication Tenant (Microsoft Entra ID Apps) [page 2956]

1.7.1.11 User Attributes

Tenant administrator has an overview of all the attributes provided to the application, regardless of the source of the values, and can provide the attributes needed by the application, specifying the attribute names expected by the application.

Context

> ⚠ Restriction
>
> The attributes configurations in the administration console for SAP Cloud Identity Services are relevant only when the application uses for authentication Identity Authentication, or when it uses a corporate identity provider (IdP), and the *Identity Federation* option is enabled.
>
> When the application uses a corporate IdP for authentication, and *Identity Federation* is disabled, Identity Authentication sends to the application the attributes that come from the corporate identity provider without changing them, and if configured, some of the same values with additional attribute names, namely configured on the trust to the corporate IdP, enriched assertion attributes or enriched token claims.

> → Tip
>
> For OpenID Connect, the supported scopes are `email`, `profile`, `openid`, and `groups`. The usage of these claims ensure uniqueness, especially in proxy mode, they are recommended over the configuration of attributes in the administration console for SAP Cloud Identity Services.

The application can get different values for a certain attribute name. The following options for sources are possible:

- *Identity Directory* - The local user attribute. You choose the value from a drop-down. See Configuring User Attributes from the Identity Directory [page 1973].
- *Corporate IdP* - The user attribute provided by the corporate IdP. You just enter the attribute name provided by the corporate IdP. See Configuring User Attributes from a Corporate Identity Provider [page 1978].
- *Expression* - A static or dynamic value. It can be a user attribute coming from *Identity Directory* or *Corporate Identity Provider*, or even a combination of all sources. See Configuring Attributes Based on Flexible Expressions [page 1979].

> → Tip
>
> The *Identity Directory* source maps to the *Assertion Attributes* term used before in this documentation.
>
> Depending on the scenario, the *Corporate Identity Provider* and *Expression* map to the *Default Attributes* term used before in this documentation.

> ⓘ Note
>
> You can specify multiple user attribute values for each user attribute. Up to 300 attribute values are allowed for self-created customer applications and automatically created single-tenant applications, and up to 50 attribute values for automatically created single-tenant applications.

- For OpenID Connect, the attributes are included in the token as string if there is one value, and array if multiple.
- For SAML 2.0, the attributes are included in the assertion as one attribute statement with multiple values in it.

SAP Cloud Identity Services supports nested attributes. You can use up to five layers of nesting. For example, `"kubernetes.io".serviceaccount.name`.

> ⚠ Restriction
>
> Attribute names that use the prefix `sap_` are reserved.

Self-Defined Attributes

If you have self-created applications or automatically created single-tenant applications in your SAP Cloud Identity Services tenant, you configure the attribute mappings under the *Self-Defined Attributes* section in the administration console. You must know the attributes that the application support, and configure the mappings accordingly.

Application Attributes

If you have subscribed multi-tenant applications in your SAP Cloud Identity Services tenant, the attributes supported by these applications are predefined and listed under the *Primary Attribute* section, with default mappings for source *Identity Directory*. You can see which attributes are supported, and add or remove mappings for the other sources - *Corporate Identity Provider* and *Expression*.

> → Tip
>
> For some subscribed applications, you might need to add custom attribute mappings (under the *Self-Defined Attributes* section). For example, an SAP BTP application connects to another application using a destination that is customer-managed. As *User ID* for principal propagation, the customer decides to use an attribute that is not in the list of predefined application attributes. So, this customer need to add the attribute configured in the destination as a self-defined attribute in the application in the SAP Cloud Identity Services tenant.

1.7.1.11.1 Configuring User Attributes from the Identity Directory

Specify how the local user attributes, configured to be collected by the registration and upgrade forms, are sent to the application.

Context

Identity Authentication defines default names for the user attributes, but you can change them in accordance with your requirements.

You configure the attributes by defining which user attribute corresponds to the user attribute that you set for the registration and upgrade forms. You can also specify multiple user attributes for each user attribute. You perform this mapping to help the application use the same user attribute for different scenarios that require several user attributes.

> **Note**
>
> The user attribute name must match the name that the application is expecting.

The attributes are also put in the `id_token` if the application is OpenID connect. For more information, see Configuring OpenID Connect (OIDC) [page 2636].

By default, Identity Authentication sets only the following user attribute names which are added at the creation of the application:

User Attribute	User Attribute Name
First Name	first_name
Last Name	last_name
Language	locale **Note** `locale` takes as value the language of the user. You can view the configured user language in the administration console for SAP Cloud Identity Services. For more information, see Configuring User Attributes from the Identity Directory [page 1973].
Email	mail
Global User ID	user_uuid

> **Note**
>
> The **Value** column lists the attributes that can be shown on the registration and upgrade forms. The **Name** lists the attributes that are sent in the assertion.

Identity Authentication supports also the following user attributes:

- `Application Activation Time`
- `Application Custom Attribute 1`
- `Application Custom Attribute 2`
- `Application Custom Attribute 3`
- `Application Custom Attribute 4`
- `Application Custom Attribute 5`

> ### Note
>
> The application custom attributes are configured by the application (service provider). They can't be defined for the user.
>
> Custom attributes must not be used to store sensitive personal data.

- `City`
- `Company`
- `Company City`
- `Company Country/Region`
- `Company State/Province`
- `Company Street Address`
- `Company Street Address 2`
- `Company ZIP/Postal Code`
- `Contact by Email`
- `Contact by Telephone`
- `Corporate Groups`

  > ### Note
  >
  > This attribute is applicable for the corporate user store scenarios and contains the groups the user in the corporate user store is assigned to.

- `Cost Center`
- `Country/Region`
- `Department`
- `Display Name`
- `Email Verified`
- `Employee Number`
- `All Groups`

  > ### Note
  >
  > Lists all user groups, all application-specific groups of type `userGroup`, independent of the application name, and all application-specific groups of type `authorization`, belonging to the current application only, that a user is a member of.
  >
  > Use `Groups` as user attribute name for application on the SAP BTP, Cloud Foundry Environment.

- `Application Groups`

  > ### Note
  >
  > Lists only the application-specific groups of type `userGroup`, belonging to the current application only, to which the user is assigned to.

- `Application Groups - external name`
  Unique within the application.

> ⓘ Note
>
> Lists only the application-specific groups of type `userGroup`, belonging to the current application only, to which the user is assigned to.

- `License Groups`

> ⓘ Note
>
> The license group is a special package license group in `Cloud Identity Services` of type `License` for AI features across the SAP portfolio. Only the users that are assigned to these license groups can access the premium Joule scenarios.

> → Remember
>
> The name must be `sap_licenses`.

- `Company Industry`
- `Job Function`
- `Login Name`
- `Middle Name`
- `Company Relationship`
- `State/Province`
- `Street Address`
- `Street Address 2`
- `Telephone Number`
- `Salutation`
- `User ID`
- `ZIP/Postal Code`.

> → Remember
>
> The application custom attributes are configured by the application (service provider). They can't be defined for the user.
>
> Custom attributes must not be used to store sensitive personal data.

> ⓘ Note
>
> The **Value** column lists the attributes that can be shown on the registration and upgrade forms. The **Name** lists the attributes that are sent in the assertion.
>
> The configured custom attributes are also put in the `id_token` if the application is OpenID connect. For more information, see Configuring OpenID Connect (OIDC) [page 2636].

> → Remember
>
> When the application uses a corporate IdP for authentication, and *Identity Federation* is disabled, the user attributes configurations in the administration console for SAP Cloud Identity Services aren't relevant. In such scenarios Identity Authentication sends to the application the user attributes that come from the corporate identity provider without changing them. For more information about the corporate identity

provider scenario, see Corporate Identity Providers [page 2556] and Configure Identity Federation [page 2614].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *Attributes*.
6. Under the *Self-defined Attributes* section, choose *Expand All* to view all the information about the user attributes.
7. **Optional:** Choose the *Add* button:
   a. Provide a name for the attribute.
   b. Choose *Identity Directory* source.
   c. Choose a value from the drop-down list.
8. **Optional:** Choose the plus button next to the attribute to set a new value for the attribute.
9. Save your configuration.

Related Information

Configure Registration and Upgrade Forms [page 2046]
Create a New Application [page 1922]

1.7.1.11.2 Configuring User Attributes from a Corporate Identity Provider

You can configure the value of existing application-defined attributes, like email, first name, last name, for example to values coming from the corporate identity provider (IdP), and add new attributes.

Context

Configure the user attributes as they come from the corporate identity provider (IdP).

> ⓘ Note
>
> For corporate IdPs configured for OpenID Connect (OIDC) , only attributes coming from ID tokens are relevant.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].
4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *Attributes*.
6. Depending on the type of your application go to:
   - *Application Attributes* - for subscribed multitenant applications.
     1. Choose the plus button next to an attribute to set multiple values for the attribute.
     2. Choose *Corporate Identity Provider* source.
     3. Provide the new value.

     > → Tip
     >
     > You can enable or disable the attribute mappings inherited from the subscribed multitenant applications:
     >
     > - To disable the attribute mappings, choose the ⊗ *disable* button on the right of the inherited attribute.

- To enable the attribute, choose ✓ *enable* button on the right of the inherited attribute.

- *Self-defined Attributes* - for self-created applications or automatically created single-tenant applications.
  1. Choose *Add button to add new attribute for the application* > *provide the name* > *Corporate Identity Provider* > *provide the value* .
  2. Choose *plus button next to an attribute to set multiple values for the attribute* > *Corporate Identity Provider* > *provide the new value* > .

7. Save your configuration.

1.7.1.11.3 Configuring Attributes Based on Flexible Expressions

You can configure attributes based on flexible expressions for the application.

Context

The system sends attributes from Identity Authentication to the application in the assertion. You can set static attributes like `location` and `company` with values *Europe* and *Company A*. The application then displays Europe and Company A on its main page.

The attributes are also put in the ID and access tokens if the application is OpenID Connect. For more information, see Configuring OpenID Connect (OIDC) [page 2636].

For both the SAML 2.0 and OpenID Connect applications, you can configure attributes with dynamic values to be added into the assertions in the following pattern: `<prefix> ${attribute_technical_name>} <suffix>`

> ⚠ Restriction
>
> For OpenID Connect applications, you can't set standard claims or SAP-specific claims (prefixed with `sap_`) via the configuration of attributes with default values. For a list of claims supported by SAP Cloud Identity Services, see Claims of SAP Cloud Identity Services OAuth Tokens [page 2744].

Supported Attributes

Supported Attributes

Attribute Display Name	Attribute Technical Name
Salutation	title
First Name	firstName
Middle Name	middleName
Last Name	lastName

Attribute Display Name	Attribute Technical Name
Email	mail
Telephone Number	telephone
Language	language
Logon Name	loginName
Display Name	displayName
User ID	uid
User UUID	userUuid
User Type	type
Street Address	street
Street Address 2	street2
City	city
ZIP/Postal Code	zip
Country	country
State/Province	state
Cost Center	costCenter
Department	department
Division	division
Employee Number	personnelNumber
Company	company
Company Street Address	companyStreet
Company Street Address 2	companyStreet2
Company City	companyCity
Company ZIP/Postal Code	companyZip
Company Country	companyCountry
Company State/Province	companyRegion
Company Industry	industry
Job Function	jobFunction
All Groups	companyGroups → Tip The attributes `companyGroups` and `corporateGroups` support regular expressions, so that they can be filtered.
Application Groups	applicationGroups

Attribute Display Name	Attribute Technical Name
Corporate Groups **Note** This attribute is applicable for the corporate user store scenarios and contains the groups the user in the corporate user store is assigned to. **→ Tip** The `companyGroups` and `corporateGroups` attributes support regular expressions. You can use an expression to filter these attributes. Both the regex filter and the value sent to the application are based on the group's `Name` field. Other group fields, such as `Display Name`, are not considered.	corporateGroups
Contact by Email	contactPreferenceEmail
Contact by Telephone	contactPreferenceTelephone
Application Activation Time	activationTime
Custom Attribute 1	customAttribute1
Custom Attribute 2	customAttribute2
Custom Attribute 3	customAttribute3
Custom Attribute 4	customAttribute4
Custom Attribute 5	customAttribute5
Custom Attribute 6	customAttribute6
Custom Attribute 7	customAttribute7
Custom Attribute 8	customAttribute8
Custom Attribute 9	customAttribute9
Custom Attribute 10	customAttribute10

Examples for attributes with dynamic values:

If you set `${uid}` as a value, the response returns the ID of the user to the application:

Example

```
SAML 2.0
<Attribute Name="User ID">
       <AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                               xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
                               xsi:type="xs:string"
                               >P123456</AttributeValue>
</Attribute>
OpenID Connect
 "User ID": "P123456"
```

If you set `${customAttribute1}` as a value, the response returns the first custom attribute of the user to the application, if there is one. If the user doesn't have a custom attribute, the response contains an empty attribute:

Example

```
SAML 2.0
<Attribute Name="Custom Attribute">
       <AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
                       xsi:type="xs:string"
                       >{customAttribute1}
</AttributeValue>
OpenID Connect
 "Custom Attribute": "{customAttribute1}"
```

If you set `${companyGroups:regex[Admin]}` as a value, the response returns the groups, whose `Name` field contains "Admin". The regex filter applies to the group's `Name` field only. Other fields such as `Display Name` aren't considered. The `Name` field value is also the value sent to the application for each matching group:

Example

```
SAML 2.0
<Attribute Name="Groups">
       <AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
                       xsi:type="xs:string"
                       >Admins</AttributeValue>
       <AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
                       xsi:type="xs:string"
                       >Administrators</AttributeValue>
OpenID Connect
 "Groups": [
    "Admins"
    "Administrators",
 ]
```

Merge User Attributes

For both the SAML 2.0 and OpenID Connect applications, you can define attributes with the same name, but with different values. You can also define a user attribute and an attribute with a default value with the same name. The response then merges these attributes into a multivalue attribute. As a result, several values may appear for a single-value attribute based on the configuration.

The order of the attribute's values in the assertion is arbitrary.

For example, you've defined the `mail` user attribute and at the same time the `mail` attribute with default value `example@example.com`.

Example

When the user Dona Moore logs on, the response returns `mail` as a multivalue attribute with the two values.

```
SAML 2.0
 <Attribute Name="mail">
               <AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                               xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
                               xsi:type="xs:string"
                               >example@example.com</AttributeValue>
               <AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
```

```
                                    xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
                                    xsi:type="xs:string"
                                    >dona.moore@example.com</AttributeValue>
</Attribute>
OpenID Connect
  "mail": [
    "dona.moore@example.com",
    "example@example.com"
  ]
```

Concatenate User Attributes

For both the SAML 2.0 and OpenID Connect applications, you can define attributes by concatenating up to five single-value attributes to create a combined attribute for your application. You can concatenate the attributes in the following pattern: `<prefix> ${<attribute_technical_name1>} <infix>${<attribute_technical_name5>}<suffix>`

⚠ Restriction

You can't concatenate:

- single-value with multi-value attributes
- multi-value with other multi-value attributes
- the attributes: `companyGroups`, `applicationGroups`, `corporateGroups`
- more than five attributes

Example

You want to define a custom work email for the employees at Company A from their first name, corporate number, and the company's domain. You have the following attributes for the user Dona Moore:

Attribute	Value
firstName	Dona
urn:sap:Corporate.numbervalue	123456
customAttribute1	example.com

You set `${firstName:function[lowercase]}_ ${urn:sap:Corporate.numbervalue}@${customAttribute1}` as value, and when Dona Moore logs on, the response returns `custom-mail` as a concatenated attribute.

```
SAML 2.0
<Attribute Name="custom-mail">
                                    <AttributeValue xmlns:xs="http://www.w3.org/
2001/XMLSchema"
                                    xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"xsi:type="xs:string"
                                    >dona_123456@example.com</AttributeValue>
</Attribute>
OpenID Connect
  "custom-mail": "dona_123456@example.com"
```

Identity Federation

- *Identity Federation* not configured
  When the application uses corporate IdP for authentication, and the *Use Identity Authentication user store* option under *Identity Federation* is disabled, the attributes configurations with default values in the administration console for SAP Cloud Identity Services aren't relevant. For more information about the corporate identity provider scenario, see Corporate Identity Providers [page 2556] and Configure Identity Federation [page 2614].
  The configuration of the attributes with default values for the system applications is disabled.
- *Identity Federation* configured
  When the application uses a corporate IdP for authentication, and the *Use Identity Authentication user store* under *Identity Federation* option is enabled, the attributes with default values in the administration console for SAP Cloud Identity Services can be configured to reference attributes coming from the assertion of the corporate IdP for the user and merge them with the attributes coming from Identity Authentication, and thus be sent to the application.
  To configure Identity Authentication to reference attributes coming from the assertion of the corporate IdP, you must use the following format for the attribute:
  `<attribute_name> = <prefix> ${corporateIdP.<corporateIdP_attribute_name><:regex[filter]><:function[<method>]>} <suffix>`

Parameter	Required	Notes
`attribute_name`	Yes	The name of the attribute as defined in the administration console. Free text.
`prefix`	No	Free text.
`corporateIdP`	Yes	Fixed string, indicating that the value is taken from the assertion coming from the corporate IdP.
`corporateIdP_attribute_name`	Yes	The specific attribute from the corporate IdP, whose value is taken.
`:regex[<filter>]`	No	Filter the attributes from the corporate IdP.
`:function[<method>]`	No	Methods to convert the attributes. Currently the supported methods are: • uppercase - the attribute is converted to uppercase letters. • lowercase - the attribute is converted to lowercase letters.
`suffix`	No	Free text.

> **Example**
>
> For example, you have set up a scenario where Identity Authentication acts as a proxy. The default authenticating identity provider is the corporate IdP, and the *Identity Federation* option is configured for that corporate IdP.

The corporate IdP is configured to send the groups with the `group` user attribute.

You want to send the groups coming from the corporate IdP to the application so you have defined the following attribute in the administration console for SAP Cloud Identity Services:

Configuration of Attributes with Default Values in Administration Console

Attribute	Value
groups	Group ${corporateIdP.groups} Member

Dona Moore is assigned to the groups *Management* and *Development* in the corporate IdP. When she logs on to the corporate portal of the company, Identity Authentication sends the `groups` coming from the corporate IdP in the following way:

```
SAML 2.0
<Attribute Name="groups">
                <AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                                xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
                                xsi:type="xs:string"
                                >Group Development Member</AttributeValue>
                <AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                                xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
                                xsi:type="xs:string"
                                >Group Management Member</AttributeValue>
</Attribute>
OpenID Connect
  "groups": [
    "Group Development Member",
    "Group Management Member"
  ]
```

The example can be expanded with the filtering option.

Example

Again, you have a scenario where Identity Authentication is a proxy. The default authenticating identity provider is the corporate IdP, and the *Identity Federation* option is configured for that corporate IdP.

The corporate IdP is configured to send the groups with the `group` user attribute.

You don't want to send all the groups coming from the corporate IdP to the application. One reason for not sending all groups is the 16k size limit on OIDC access tokens. So, you have defined the following attribute in the administration console for SAP Cloud Identity Services:

Configuration of Attributes with Default Values in Administration Console

Attribute	Value
groups	${corporateIdP.groups:regex[ABC-]}

Michael Adams is assigned to the groups *ABC-Management*, *Development* , and *ABC-Everyone* in the corporate IdP. When he logs on to the corporate portal of the company, Identity Authentication sends just those `groups` coming from the corporate IdP that matches *ABC-*, in the following way:

```
SAML 2.0
```

```
</Attribute>
                <Attribute Name="groups"> <AttributeValue xmlns:xs="http://
www.w3.org/2001/XMLSchema"
                                  xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
                                  xsi:type="xs:string"
                                  >ABC-Everyone</AttributeValue>
<AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                                  xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
                                  xsi:type="xs:string"
                                  >ABC-Management</AttributeValue> </
Attribute>
OpenID Connect
  "groups": [
    "ABC-Everyone",
    "ABC-Management"
  ]
```

Example

Another scenario is where Identity Authentication is again a proxy. The default authenticating identity provider is the corporate IdP, and the *Identity Federation* option is configured for that corporate IdP.

The corporate IdP is configured to send the email with the `mail` user attribute. The incoming mail attribute from corporate IdP is in mixed case and it is converted to lower case. For example, Michael Adams' email in the corporate IdP is in mixed case Michael.Adams@example.com, and it is converted into lower case michael.adams@example.com.

Configuration of Attributes with Default Values in Administration Console

Attribute	Value
mail	${corporateIdP.mail:function[lowercase]}

→ Remember

When Identity Authentication is configured to reference an attribute from the corporate IdP, but this attribute isn't sent in the corporate IdP assertion, the attribute isn't sent to the application either.

If the definition of the attribute includes prefix and/or suffix, only the prefix and/or suffix is sent.

However, if the corporate IdP is not configured to send the **phone** attribute, the response includes only the prefix and suffix, defined in the administration console for SAP Cloud Identity Services:

Example

For example, you want to send the **phone** attribute coming from the corporate IdP to the application. You have defined the following attribute in the administration console for SAP Cloud Identity Services:

Configuration of Attributes with Default Values in Administration Console

Attribute	Value
Phone	+49 ${corporateIdP.phone} Corporate Phone

```
SAML 2.0
</Attribute>
                  <Attribute Name="Phone"> <AttributeValue
xmlns:xs="http://www.w3.org/2001/XMLSchema"
                                  xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
                                  xsi:type="xs:string"
                                  >+49 123 456789 Corporate Phone</
AttributeValue>
</Attribute>
OpenID Connect
"Phone": "+49 123 456789 Corporate Phone"
```

Send Identity Directory Custom Schema Attributes

You can configure Identity Authentication to send the Identity Directory custom schema attributes as attributes into authentication tokens to applications.

You can define complex custom schema attributes with single-value child attributes.

Example

For example, you have set up a scenario where Identity Authentication acts as a proxy. The default authenticating identity provider is the corporate IdP, and the *Identity Federation* option is configured for that corporate IdP.

Michael Adams is a user from the corporate identity provider. He is created on the proxy IdP with the custom schema `urn:sap:cloud:scim:schemas:extension:custom:2.0:Profile` with two single-valued attributes, `birthday` and `hobby`, and the complex attribute `name` with its child attribute `firstName`. He logs on to the company's portal with the corporate identity provider.

The default attributes for the application are configured in the administration console as follows:

Configuration of Attributes with Default Values in Administration Console

Attribute	Value
name	${urn:sap:cloud:scim:schemas:extension:custom:2.0:Profile.name.firstName}
hobby	${urn:sap:cloud:scim:schemas:extension:custom:2.0:Profile.hobby}
birthday	${urn:sap:cloud:scim:schemas:extension:custom:2.0:Profile.birthday}

When Michael logs on the application, the values "Michael", "cycling", and "2000-01-02T0:00:00Z" are added to application token during authentication, and received by the application.

The attributes are added both to SAML 2.0 and OIDC tokens.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.

2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *Single Sign-On*, choose *Attributes*.
6. Depending on the type of your application go to:
   - *Application Attributes* - for subscribed multitenant applications.
     1. Choose the plus button next to an attribute to set multiple values for the attribute.
     2. Choose *Corporate Identity Provider* source.
     3. Provide the new value.

     > **→ Tip**
     >
     > To restrict the value of an application attribute:
     >
     > 1. Deactivate the existing attribute.
     > 2. Add a new value with a flexible expression.

   - *Self-defined Attributes* - self-created applications or automatically created single-tenant applications.
     1. Choose *Add button to add new attribute for the application* > *provide the name* > *Expression* > *provide the value*.
     2. Choose *plus button next to an attribute to set multiple values for the attribute* > *Expression* > *provide the new value*.
7. Save your configuration.

1.7.1.12 Configure the Subject Name Identifier Sent to the Application

The `Subject Name Identifier` is a profile attribute that Identity Authentication sends to the application as `name ID` in the SAML 2.0 assertions, and as `subject` in the OpenID Connect tokens. The application then uses this attribute to identify the user.

Context

You can choose primary and fallback attributes to uniquely identify the user during logon.

The *Primary Attribute* value can be chosen from three sources:

- *Identity Directory* - This is the local user attribute. You choose the value from a drop-down.
  The user is identified in one of the following ways:
  - Global User ID
  - User ID
  - Email
  - Display Name
  - Login Name
  - Employee Number.

  > ### Note
  >
  > `Global User ID`, `User ID`, `Login Name` and `Email` can be unique for the tenant. This depends on the tenant configurations.
  >
  > The configuration of the `Subject Name Identifier` attribute for the system applications is disabled. The default setting for these applications is `User ID`.

  > ### Caution
  >
  > Identity Authentication does not check the `Employee Number` attribute for uniqueness. Be sure that the users receive unique employee numbers.

- *Corporate Identity Provider* - The user attribute provided by the corporate identity provider (IdP). You just enter the attribute name provided by the corporate IdP.
- *Expression* - A static or dynamic value. It can be a user attribute coming from:
  - Identity Directory
  - Corporate Identity Provider
  - a combination of the Identity Directory and Corporate Identity Provider
  - a custom single-value attribute defined in a custom schema.

  > ### Caution
  >
  > Attributes of type complex and multi-value attributes are not supported when the source is a custom schema.

  The user is identified by setting an attribute with dynamic value in the following pattern: `<prefix> ${attribute_technical_name} <suffix>`.

  > ### Note
  >
  > If the attribute comes from a custom schema, the pattern is: `<prefix> ${SchemaID.attribute_technical_name} <suffix>` . Attributes of type complex and multi-value attributes are not supported when the source is a custom schema.

  Expand the **Supported Attributes** table below to see the technical names of the attributes that can take dynamic values:

  Supported Attributes

Attribute	Attribute Technical Name
Email	mail

Attribute	Attribute Technical Name
Telephone Number	telephone
Logon Name	loginName
Display Name	displayName
User ID	uid
Global User ID	userUuid
Employee Number	personnelNumber
Custom Attribute 1	customAttribute1
Custom Attribute 2	customAttribute2
Custom Attribute 3	customAttribute3
Custom Attribute 4	customAttribute4
Custom Attribute 5	customAttribute5
Custom Attribute 6	customAttribute6
Custom Attribute 7	customAttribute7
Custom Attribute 8	customAttribute8
Custom Attribute 9	customAttribute9
Custom Attribute 10	customAttribute10
SP (service provider) user nameID	name_id
SP (service provider) user Custom Attribute 1	spCustomAttribute1
SP (service provider) user Custom Attribute 2	spCustomAttribute2
SP (service provider) user Custom Attribute 3	spCustomAttribute3
SP (service provider) user Custom Attribute 4	spCustomAttribute4
SP (service provider) user Custom Attribute 5	spCustomAttribute5
NameID coming from the assertion of the corporate IdP	corporateIdP.NameID
Attribute coming from the assertion of the corporate IdP	corporateIdP.<corporateIDP attribute>

→ Remember

If the *Corporate Identity Provider* or *Expression* configuration options are used, *Use Identity Authentication user store* must be enabled. When the application uses corporate IdP for authentication, and *Use Identity Authentication user store* is disabled, the `Subject Name Identifier` attribute configurations in the administration console for SAP Cloud Identity Services are not relevant. In such scenarios Identity Authentication sends to the application the `Subject Name Identifier` attribute that comes from the corporate identity provider without changing it. For more information about the corporate identity provider scenario, see Corporate Identity Providers [page 2556] and Configure Identity Federation [page 2614].

Apart from the primary attribute configuration you can also configure a fallback subject name identifier. The fallback subject name identifier is used only when the primary identifier is not available. If the primary identifier is not available, and you haven't set a fallback attribute, the authentication of the user fails.

Optionally, you can configure the default name identifier format of the name ID attribute for SAML 2.0 applications.

> ⚠ Caution
>
> When configuring the `Subject Name Identifier`, take into consideration that some of the attributes can be modified by the users.

> ⓘ Note
>
> When a subscribed application becomes a child application, the configuration of the `Subject Name Identifier` of that child application can be different from the parent application or from the application it is subscribed to. You can make the configuration of the `Subject Name Identifier` the same as the parent application by choosing the *Inherit from Parent* button on the top right-hand corner of the screen. You can also make the configuration of the `Subject Name Identifier` the same as the application it is subscribed to by choosing the *Inherit from Source* button on the top right-hand corner of the screen. Depending on your configuration, either the *Inherit from Parent* or the *Inherit from Source* button is active, or both buttons are active.

To set the `Subject Name Identifier` attribute, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].
4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *Subject Name Identifier*.
6. Select one of the following:

   - Identity Directory:
     - Select an attribute from the drop down: `Global User ID`, `User ID` (this is the default setting), `Email`, `Display Name`, `Login Name`, `Employee Number`.

       > ⓘ Note
       >
       > If you select *Login Name*, or *Employee Number*, and the selected attribute has no value set for the user, the user is not able to log on the application. The message "**HTTP Status 401 – Unauthorized**" is displayed when the user provides credentials and logs on. In this case, configure also the fallback attribute.

- (optional) Select a fallback attribute from the drop down: `None` (this is the default setting), `Global User ID`, `User ID`, `Email`, `Display Name`, `Login Name`, `Employee Number`.

- Corporate Identity Provider
  - Enter the attribute name provided by the corporate IdP.

    > → Tip
    >
    > When Identity Authentication acts as proxy IdP, and the users have no profiles in Identity Authentication, you can set an attribute with the value coming from the corporate IdP, or use the *Expression* configuration option.

  - (optional) Select a fallback attribute from the drop down: `None` (this is the default setting), `Global User ID`, `User ID`, `Email`, `Display Name`, `Login Name`, `Employee Number`.

- Expression:
  - Enter a static or dynamic value. It can be a user attribute coming from Identity Directory, or Corporate Identity Provider, or even a combination of all sources. To set an attribute with dynamic value use the following pattern `<prefix> ${attribute_technical_name} <suffix>`. For example: `Phone ${telephone}`

    > ⓘ Note
    >
    > If Identity Authentication acts as proxy IdP, and the users have no profiles in Identity Authentication, set an attribute with the following dynamic value: <prefix> ${corporateIdP.<corporateIDP attribute>} <suffix>

  - (optional) Select a fallback attribute from the drop down: `None` (this is the default setting), `Global User ID`, `User ID`, `Email`, `Display Name`, `Login Name`, `Employee Number`.

7. Save your selection.

   Once the application has been changed, the system displays the message `Application <name of application> updated`.

(Optional) Configure the Default Name ID Format Sent to the Application

The tenant administrator can set a default name ID format via the administration console for SAP Cloud Identity Services.

Context

The name ID format is used in the `Format` attribute of the `NameID` element that Identity Authentication sends to the application with the SAML assertion. This is the default name ID format. It is used only in case the application or the service provider does not request another name ID format with the SAML authentication request.

Identity Authentication supports the following name ID formats:

Name ID Format	URI	Description
Unspecified	urn:oasis:names:tc:SAML:1.1:nameid-format:unspecified	Identity Authentication does not use a specific format to express the identity of the user. The application receives the value of the user attribute that is selected as name ID attribute. By default this is the default Name ID format.
Email address	urn:oasis:names:tc:SAML:1.1:nameid-format:emailAddress	The format of the name ID is an email address. You can configure this setting when you have chosen as name ID attribute a user attribute that contains email. **Note** Identity Authentication does not check if the content of the name ID attribute is email address. You can decide which name ID attribute to configure for the application. Make sure that it contains email address in case you want to conform to the SAML2 specification.

Note

The configuration of the default name ID format for the system applications is disabled. The default setting for these applications is `Unspecified`.

→ Remember

When the application uses corporate IdP for authentication, and *Use Identity Authentication user store* is disabled, the default name ID attribute configurations in the administration console for SAP Cloud Identity Services are not relevant. In such scenarios Identity Authentication sends to the application name ID format that comes from the SAML assertion sent from the corporate IdP without changing it. For more information about the corporate identity provider scenario, see Corporate Identity Providers [page 2556] and Configure Identity Federation [page 2614].

To set the default name ID format, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *Default Name ID Format*.
6. Select the default name ID format from the following:
   - Unspecified
   - Email
7. Save your selection.

   Once the application has been changed, the system displays the message `Application <name of application> updated`.

Related Information

Create a New Application [page 1922]
Configure Trust [page 1941]

1.7.1.13 Convert Subject Name Identifier to Uppercase or Lowercase

Tenant administrator can apply function to change the subject name identifier to uppercase or lowercase letters depending on the need of the application.

Context

The setting is applied in all cases when Identity Authentication issues an assertion or id token - logon in Identity Authentication or delegated authentication to corporate identity provider with or without enabled identity federation.

> **Note**
>
> Conversion for subject name identifier in locale "tr", "az" or "lt" is not supported. If subject name identifier is in "tr", "az" or "lt" locale the conversion may not work properly.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].
4. Under the *Trust* tab, choose *Apply Function to Subject Name Identifier*.
5. Choose one of the options:
   - None - default configuration
   - Uppercase - the subject name identifier is converted to uppercase letters.
   - Lowercase - the subject name identifier is converted to lowercase letters.
6. Save your configuration.

1.7.1.14 Authentication

To learn about	See
How to define rules for authentication in accordance with the risk	Configure Risk-Based Authentication for an Application [page 2007]
Enable social sign-on authentication for an application	Enable or Disable Social Sign-On for an Application [page 2020] **Note** The content in this section is not relevant for China (Shanghai) region.
How to allow users to log on to an application from the corporate network without entering their username and password	Enable or Disable Kerberos Authentication for an Application [page 2556]
How to configure the type of authentication when API methods of Identity Authentication are used	API Authentication [page 1996]

1.7.1.14.1 API Authentication

Developers can choose the type of authentication when API methods of Identity Authentication are used.

The certificate to be used for authentication by the REST APIs of Identity Authentication must be requested from the SAP Support Portal.

Configure Secrets for API Authentication [page 1997]
: This document describes how developers configure secrets with scopes and validity for client authentication.

Unlock Client ID [page 1999]
: Unlock the client ID after five failed logon attempts before the automatic unlock time of 60 minutes has passed.

Disable Client ID Locking [page 2000]
: You can disable the automatic lock of the client ID after five failed logon attempts.

Configure Certificates for API Authentication [page 2001]
: This document describes how developers configure the certificates used for authentication when the API methods and OpenID Connect scenarios of Identity Authentication are used.

Configure JWT for OAuth Client Authentication [page 2003]
: Configure the JSON Web Token (JWT) - the issuer and subject of tokens for JWT client authentication in token requests, or the URI for JSON web key retrieval for client authentication.

SCIM REST API (Deprecated) Authentication Mechanisms [page 2006]
: See how to configure the authentication mechanisms for the Identity Authentication API (Deprecated) [page 2796] methods of Identity Authentication.

1.7.1.14.1.1 Configure Secrets for API Authentication

This document describes how developers configure secrets with scopes and validity for client authentication.

Context

You can use a client ID and a secret to authenticate when REST API calls (Invitation REST API, User management REST API, Password Service Rest API, and Forgot Password REST API) to the tenant of Identity Authentication are used. The client ID and secret can also be used in the OpenID Connect scenarios of Identity Authentication.

The client ID is in the universally unique identifier (UUID) format. For example, `1ab7c243-5de5-4530-8d14-1234a26373ab`.

> → Remember
>
> The User ID and a password (secret) to authenticate the client (relying party) in the OpenID Connect scenario must be encoded using the "application/x-www-form-urlencoded" encoding algorithm.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].
4. Choose the *Trust* tab.
5. Under *Application APIs*, choose *Client Authentication*.

   You can see the *Client ID* generated for the chosen application.
6. Choose the *Add* button in the *Secrets* section.

   > ⚠ Restriction
   >
   > You can generate up to 20 client secrets per application.
7. Provide the required info in the pop up.

Description	This field is optional.
Expire in	You can choose from three options: • 1 year • 2 years • Never
Scope	• Application - Secrets with *Application* scope are used to update the Application configurations. • Application Users - select this option to generate a secret to authenticate when REST API calls (Invitation REST API, User management REST API, Password Service Rest API, and Forgot Service REST API) are used. • OpenID - select this option to generate a secret to authenticate in the OpenID Connect scenario. **Note** By default all options are selected. Your secret can be used for all scopes.

8. Save your configuration.

   A pop up with the generated credentials appears. Make sure that you save the client secret for the client ID. You will need it for the API authentication, but you will not be able to retrieve it from the system later.

Results

Once your secret is generated you can see a table with your secrets and information about them.

Task overview: API Authentication [page 1996]

Related Information

Unlock Client ID [page 1999]
Disable Client ID Locking [page 2000]
Configure Certificates for API Authentication [page 2001]
Configure JWT for OAuth Client Authentication [page 2003]
SCIM REST API (Deprecated) Authentication Mechanisms [page 2006]

1.7.1.14.1.2 Unlock Client ID

Unlock the client ID after five failed logon attempts before the automatic unlock time of 60 minutes has passed.

Context

The client ID locks for 60 minutes after five failed logon attempts. After 60 minutes the client ID is automatically unlocked.

To unlock the client ID before the automatic unlock time of 60 minutes has passed follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application with a locked client ID.
4. Under *API AUTHENTICATION*, choose *Client ID and Secrets*.
5. Choose *Unlock* next to *Client ID Status*.

> → Remember
>
> You can only unlock locked client IDs. If the client ID is not locked, the *Unlock* button is disabled.

Task overview: API Authentication [page 1996]

Related Information

Configure Secrets for API Authentication [page 1997]
Disable Client ID Locking [page 2000]
Configure Certificates for API Authentication [page 2001]
Configure JWT for OAuth Client Authentication [page 2003]
SCIM REST API (Deprecated) Authentication Mechanisms [page 2006]
Disable Client ID Locking [page 2000]

1.7.1.14.1.3 Disable Client ID Locking

You can disable the automatic lock of the client ID after five failed logon attempts.

Context

By default the client ID locks for 60 minutes after five failed logon attempts.

You can disable the automatic lock of the client ID via the administration console for SAP Cloud Identity Services. Use this option to disable the automatic lock in cases when the client ID has a limited scope and the client ID secret(s) are automatically generated.

> ### Note
>
> When you disable the *Client ID Lock* option for a client ID that is already locked, this resets the failed logon attempts counter but the client ID remains locked. If you want to unlock the client ID before the automatic unlock time of 60 minutes has passed you must unlock it manually. For more information, see Unlock Client ID [page 1999].

"

To disable the *Client ID Lock* option, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.
4. Under *API AUTHENTICATION*, choose *Client ID and Secrets*.
5. Disable the *Client ID Lock*.

   If the operation is successful, you receive the following message: `Client ID locking updated.`. The slider next to *Client ID Lock* is switched to *OFF*.

   > ### Note
   >
   > If you want to enable client ID password locking for the application, drag the slider to *ON*.

Task overview: API Authentication [page 1996]

Related Information

Configure Secrets for API Authentication [page 1997]

Unlock Client ID [page 1999]
Configure Certificates for API Authentication [page 2001]
Configure JWT for OAuth Client Authentication [page 2003]
SCIM REST API (Deprecated) Authentication Mechanisms [page 2006]

1.7.1.14.1.4 Configure Certificates for API Authentication

This document describes how developers configure the certificates used for authentication when the API methods and OpenID Connect scenarios of Identity Authentication are used.

Prerequisites

SAP Support Portal Home (For trusted CA Scenario) - If you want to configure a certificate, using your own trusted CA report an incident on SAP Support Portal Home with a component `BC-IAM-IDS`. Attach to the incident the root and intermediate certificates and provide the Identity Authentication tenant ID.

> **Note**
>
> `Tenant ID` is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the `tenant ID`. For more information about your tenants, see View Assigned Tenants and Admins.

Context

You can use a certificate to authenticate when REST API calls (Invitation REST API, User management REST API, Password Service Rest API, and Forgot Service REST API) to the tenant of Identity Authentication are used. The certificate can also be used in the OpenID Connect scenarios of Identity Authentication.

For the configuration, you have to provide the base64-encoded certificate as a file or plain text, or you can generate a certificate via the administration console. You can enter a maximum of 80 certificates.

By default all scope options are selected and the *Scope* field is disabled. Your certificates are used for all scopes.

Identity Authentication, by default, supports the following as trusted certificate authorities (CA): SAP Passport CA G2, SAP SSO CA G2, SAP Cloud Root CA, SAP Global Root CA, DigiCert TLS RSA SHA256 2020 CA1, DigiCert Global Root CA, SAP Cloud Platform Client CA, ConnectivityCA.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.

2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *Application APIs*, choose *Client Authentication*.
6. Choose the *Add* button in the *Certificates for API Authentication* section.
7. **Optional:** Provide description for the certificate.
8. Choose one of the following options:

Certificate Options	Notes
Generate Certificate	Fill in the fields under *Generate Certificate* and choose the *Generate* button. **Note** You must provide a Common Name (CN) and password to generate the certificate. Once the certificate is generated, it is saved as a *.p12* file. The system populates the *Insert as Text* field with it, and provides the certificate attributes in the Subject DN. The common name (CN) in the generated certificate is in the format `<common name>` (`<admin user ID>`), where common name is the CN provided by the administrator, and admin user ID is the administrator's user id. The maximum length of the combination of `<common name>` (`<admin user ID>`) is 64 characters. **→ Remember** You can generate up to 500 certificates per year. If you reach the limit, you can still upload your certificate, or insert it as a text.
Upload Certificate	You must use `.cer` or `.crt` files. **Note** Once the certificate is uploaded, the system populates the Insert as Text field with it
Insert Certificate	Insert the certificate in the text field.

> ⚠ Restriction
>
> You can't use one and the same certificate for application and system as administrator authentication.

9. Choose the *Add* button at the bottom of the page.

   Once the certificate has been configured, the system displays the message `Certificate for API authentication updated`.

> → Tip
>
> When the certificate expires, follow the procedure again and configure a new one.

Results

Once your certificate is added you can see a table with all your certificates and information about them.

Task overview: API Authentication [page 1996]

Related Information

Configure Secrets for API Authentication [page 1997]
Unlock Client ID [page 1999]
Disable Client ID Locking [page 2000]
Configure JWT for OAuth Client Authentication [page 2003]
SCIM REST API (Deprecated) Authentication Mechanisms [page 2006]
Create a New Application [page 1922]
Invitation REST API [page 2802]
User Management REST API [page 2805]

1.7.1.14.1.5 Configure JWT for OAuth Client Authentication

Configure the JSON Web Token (JWT) - the issuer and subject of tokens for JWT client authentication in token requests, or the URI for JSON web key retrieval for client authentication.

Prerequisites

You have an OpenID Connect application.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *Application APIs*, choose *Client Authentication*.
6. Under *JSON Web Tokens*, configure one of the following options and save your changes:

   You can enter a maximum of 20 of each type of trust.

   - (For RFC 7523-based JWT client tokens) Choose the *Add* button for *Configure Trust by Issuer* and provide the required info in the popup:

     Configure Trust by Issuer

Field	Notes
Description	(Optional) - Provide information about the token in the provided field.
Choose *Issuer Configuration* > *Manual Configuration* > radio-button. **Note** When you choose *Manual Configuration* radio-button the *Corporate Identity Provider* section is not configurable.	Under *Manual Configuration* configure: 1. *Issuer* (Required) - The issuer must be OpenID Connect compliant. 2. *JSON Web Key Set URI* (Required) - Choose *Check for Metadata* button. If the entered issuer is verified as OpenID Connect compliant, the JSON Web Key Set URI will be retrieved automatically. Otherwise, you need to explicitly specify the *JSON Web Key Set URI* in the field. 3. *Refresh Interval* (Optional) - Refreshes the URI automatically if it is older than the selected interval. Choose from: • 24 hours (default choice) • 12 hours
Choose *Issuer Configuration* > *Corporate Identity Provider* > radio-button. **Note** When you choose *Corporate Identity Provider* radio-button, the *Manual Configuration* section is not configurable.	*Name* - (Required) Choose from OpenID Connect compliant corporate identity providers.

Field	Notes
General	1. *Subject* (Required) - The sub (subject) that is expected in the token. → Tip If you want to use an OAuth 2.0 token from a corporate identity provider as client credentials for an OpenID Connect application in Identity Authentication, and your OAuth client in the corporate identity provider belongs to a pre-configured application, the subject in the token is the Object ID of that pre-configured application. 2. *Audience* (Optional) - The `audience` claim of the identity provider. 3. `API Permission Groups` (Optional) - The field is enabled only when the *Provided APIs* option is configured. For more information, see Provide APIs for Consumption by Other Applications [page 2891].

- (For OpenID Connect-based JWT client tokens) Choose the *Add* button for *Configure Trust by URI* and provide the required info in the popup:

Configure Trust by URI

Field	Notes
Description	(Optional) You can provide information about the URI here.
URI	(Required) The JSON Web Key Set (JWKS) URI of the trusted party.
Refresh Interval	(Optional) Refreshes the URI automatically if it is older than the selected interval. Choose from: • 24 hours (default choice) • 12 hours
API Access	ⓘ Note This section is read-only. The predefined choice is OpenID.
API Permission Groups	(Optional) `API Permission Groups` field is enabled only when the *Provided APIs* option is configured. For more information, see Provide APIs for Consumption by Other Applications [page 2891].

Task overview: API Authentication [page 1996]

Related Information

Configure Secrets for API Authentication [page 1997]
Unlock Client ID [page 1999]
Disable Client ID Locking [page 2000]
Configure Certificates for API Authentication [page 2001]
SCIM REST API (Deprecated) Authentication Mechanisms [page 2006]
JSON Web Token (JWT) Profile for OAuth 2.0 Client Authentication and Authorization Grants
Proof-of-Possession Key Semantics for JSON Web Tokens (JWTs)
JSON Web Token Client Authentication

1.7.1.14.1.6 SCIM REST API (Deprecated) Authentication Mechanisms

See how to configure the authentication mechanisms for the Identity Authentication API (Deprecated) [page 2796] methods of Identity Authentication.

To call the methods of this SCIM REST API you must have a system as administrator with an assigned *Manage Users* role. For more details about how to add a system as administrator and assign administrator roles, see Add System as Administrator [page 2305], and Edit Administrator Authorizations [page 2309].

Parent topic: API Authentication [page 1996]

Related Information

Configure Secrets for API Authentication [page 1997]
Unlock Client ID [page 1999]
Disable Client ID Locking [page 2000]
Configure Certificates for API Authentication [page 2001]
Configure JWT for OAuth Client Authentication [page 2003]

1.7.1.14.2 Risk-Based Authentication for an Application

Use risk-based authentication to control user access to applications. Define authentication actions based on conditions like IP ranges, user groups, authentication methods, user types, email domains, and corporate attributes.

> → Tip
>
> Risk-based authentication is often confused with conditional authentication.

- Risk-based authentication configures the security level or authentication method you require users to use.
- Conditional authentication configures which identity provider authenticates your users.

Both use similar conditional criteria, such as IP ranges, groups, and user types. However, they serve different purposes in the authentication flow. Conditional authentication happens first to route users to the correct identity provider. Risk-based authentication then determines the security requirements after the identity provider is selected.

For more information about conditional authentication, see Configure Conditional Authentication for an Application [page 2076].

Configure Risk-Based Authentication for an Application [page 2007]
You can define rules for authentication according to different risk factors and apply actions like *Allow*, *Deny*, and *Two-Factor Authentication*.

Create a New Rule [page 2010]
You can create rules for authentication according to different risk factors.

Examples for Risk-Based Authentication Scenarios [page 2014]
Example scenarios for configuring risk-based authentication for an application.

1.7.1.14.2.1 Configure Risk-Based Authentication for an Application

You can define rules for authentication according to different risk factors and apply actions like *Allow*, *Deny*, and *Two-Factor Authentication*.

⚠ Restriction

Not supported for technical communication in OpenID Connect (OIDC) flows.

Authentication Rules

The created rules are displayed sorted by priority. When a user tries to access the application, the rules evaluate if the user meets the criteria of the rule. The evaluation starts with the rule with the highest priority, until the criteria of a rule are met. If the criteria of a rule are met, the rest of the rules aren't evaluated.

Default Authentication Rule

If none of the authentication rules meets the criteria, the default authentication rule is applied. For the default authentication rule, you can configure a *Default Action*, which can be *Allow*, *Deny*, or *Two-Factor Authentication*.

ⓘ Note

If *Two-Factor Authentication* is selected, additionally, you must specify the two-factor method or methods for the user.

The rule is valid for any *IP range*, *Forwarded IP Range*, *Group*, *Authentication Method*, or *User Type*.

Prerequisites

- (For *RADIUS Server Two-Factor Authentication*) You have requested this feature. For more information how to request and configure RADIUS server in Identity Authentication, see Configure RADIUS Server Settings (Beta) [page 2192].
- (For *SMS Two-Factor Authentication*) You have an account in Sinch Service. You have configured Sinch Service in the administration console for SAP Cloud Identity Services. For more information, see Configure Sinch Service in Administration Console [page 2051].

  > ⓘ Note
  >
  > Sinch Verification account is purchased separately. It isn't part of the Identity Authentication contract.

- (For *Email OTP Code*) An Email OTP Code template for the respective languages must exist in the tenant to apply the email OTP code method. If the template does not exist, the user will see the option but when choosing it, the following message will appear: "Sorry, but you are currently not authorized for access". For more information how to add email templates, see Edit or Add an Email Template Set [page 2289].

Context

> → Tip
>
> We recommend you to enable the back-up channels. Thus, the users can use the option as an alternative when they don't have access to the TOTP device or application.. For more information, see Enable Back-Up Channels to Send Passcode for Deactivation of TOTP Two-Factor Authentication Devices [page 2156].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the list item of the application that you want to edit.

   > ⓘ Note
   >
   > If you do not have a created application in your list, you can create one. For more details, see Related Information.

   > ⚠ Caution
   >
   > The list also includes the `Administration Console` application. If you enable risk-based authentication for that application, make sure that you, as a tenant administrator, meet the authentication rules and the default authentication rule. Otherwise when you log out of the administration console you will not be able to log in it again if you don't meet the rules.
   >
   > If `Administration Console` is not in the list of the applications you may request it. To do this, report an incident with a subject on SAP Support Portal Home under the component `BC-IAM-IDS`.

4. Choose the *Authentication and Access* tab.
5. Under *AUTHENTICATION*, choose *Risk-Based Authentication*.
6. **Optional:** Configure the authentication rules. To configure the authentication rules, choose one of the following:

Option	Description
Create a new rule	See Create a New Rule [page 2010]
Edit an existing rule	Choose the icon next to the rule you want to edit.
Delete an existing rule	Choose the delete icon next to the rule you want to delete.
Reprioritize rules	Use the arrows to reprioritize the rules.

> ### Note
>
> By default any user can log on from any IP.

7. **Optional:** Configure the *Default Action*:
   - *Allow* - Any user can log on from any IP. This is the default choice.
   - *Deny* - Nobody can log on.
   - *Two-Factor Authentication* - A drop-down appears when this choice is selected. You must specify the two-factor authentication method or methods for the end user.
8. Save your changes.

   Once the application has been updated, the system displays the message `Authentication rules updated`.

Task overview: Risk-Based Authentication for an Application [page 2006]

Related Information

Create a New Rule [page 2010]
Examples for Risk-Based Authentication Scenarios [page 2014]
SAP Cloud Identity Services Application Directory

1.7.1.14.2.2 Create a New Rule

You can create rules for authentication according to different risk factors.

Context

Each rule contains the following information:

- **Authentication Action**
  This action is performed if the rule conditions meet the defined criteria.
  You can choose one of the following authentication actions:
  - *Allow*
    Identity Authentication allows the authentication of the user in accordance with the rule conditions.
  - *Deny*
    Identity Authentication denies the authentication of the user in accordance with the rule conditions.
    You can set this action for a test application for example, or before an application goes live.
    As long as this rule is valid, when users try to log on to the application, they get the following message: *Sorry, but you are currently not authorized for access*.
  - *Two-Factor Authentication*

    > ⓘ Note
    >
    > If *Two-Factor Authentication* is selected, additionally, you must specify the two-factor method or methods for the user:
    > - *TOTP Two-Factor Authentication*
    >   Identity Authentication asks two factors to authenticate the user.
    >   If you set TOTP two-factor authentication, users are required to provide a time-based one-time password (TOTP) called a passcode in addition to their primary credentials. Users also have to install an authenticator application on their mobile devices to generate TOTP passcodes.
    >   TOTP passcodes are time-based and valid for one logon attempt only.
    > - *SMS Two-Factor Authentication*
    >   Identity Authentication asks two factors to authenticate the user.
    >   If you set SMS two-factor authentication, users are required to provide an SMS code sent to their mobile devices in addition to their primary credentials.
    >
    >   > → Remember
    >   >
    >   > To use *SMS Two-Factor Authentication*, you must have configured Sinch Verification in the administration console for SAP Cloud Identity Services. For more information, see Configure Sinch Service in Administration Console [page 2051].
    >   >
    >   > Users must have their mobile phone numbers verified. The tenant administrator can verify phone numbers manually in the administration console or via the SCIM API. For more information, see List and Edit User Details [page 2361] and Identity Directory API.
    >   >
    >   > If the user does not have a verified phone number, the number is verified during the first log on when SMS code is required. After the user provides user name and credentials, he or she should provide the phone number in the field and request a code. Then provide the

received code in the respective field and choose *Continue*. If the submitted code is correct, the user is allowed access, and the telephone number is verified.

- *Web Two-Factor Authentication*
  Identity Authentication asks two factors to authenticate the user.
  If you set web two-factor authentication, users are required to authenticate with a device such as the built in biometric scanners or USB, Bluetooth or Near-Field Communication (NFC) devices in addition to their primary credentials.
- *Email OTP Code*
  Identity Authentication asks two factors to authenticate the user.
  If you set *Email OTP Code*, users are required to provide the code sent to their email in addition to their primary credentials.

  > ⚠ Caution
  >
  > For security reasons, the Email OTP code is not a recommended two-factor authentication method. You may consider using some of the other methods instead.

  > ⓘ Note
  >
  > In the scenario, when self-registration is enabled and email verification is disabled for the application, if you set only *Email OTP Code* in the rule, users that have registered themselves in the application are required to provide an email OTP code when they activate their account. They provide the received code in the respective field and choose *Continue*. If the submitted code is correct, the user is allowed access, and the email is verified.

  > → Remember
  >
  > An Email OTP Code template for the respective languages must exist in the tenant to apply the email OTP code method. If the template does not exist, the user will see the option but when choosing it, the following message will appear: "Sorry, but you are currently not authorized for access".
  >
  > For more information how to add email templates, see Edit or Add an Email Template Set [page 2289].
- *RADIUS Server Two Factor Authentication*
  If you set *RADIUS Server Two Factor Authentication*, users are required to provide a RADIUS passcode in addition to their primary credentials. Users must have a RADIUS token (hard or soft) configured for them to generate passcodes. For more information about how to configure RADIUS server in Identity Authentication, see Configure RADIUS Server Settings (Beta) [page 2192].

> → Remember
>
> The *Authentication Action* filed is mandatory.

- *IP Range*
  Define a range of IP addresses that authentication requests to Identity Authentication can be sent from. The value has to be specified in Classless Inter-Domain Routing (CIDR) notation.

> ### Note
>
> By default the field is empty, meaning that any IP is allowed.

> ### Example
>
> Enter 123.45.67.1/24 to allow users to log on from any IP starting with 123.45.67.

If no IP range is defined, the rule is valid for all IP ranges.

- *Forwarded IP Range*
  Define a range of IP addresses for the original IP addresses that authentication requests to Identity Authentication can be sent from. The value has to be specified in Classless Inter-Domain Routing (CIDR) notation.

  > ### Caution
  >
  > This range configuration is used in scenarios where authentication requests to Identity Authentication are made by a proxy server on-behalf of the user/client.
  >
  > Configuring the *Forwarded IP Range* for the client/user IPs requires configuring the IP Range for the proxy server IP address(es), that directly communicates with Identity Authentication, first.
  >
  > Identity Authentication expects the client IP to be sent by the proxy server in the **X-Forwarded-For** header of the authentication request.

  > ### Remember
  >
  > Identity Authentication does not have control over the involved proxies and their number, participating in the request between the client/user and Identity Authentication and whether these proxies behave according to the scenario. Therefore it is not guaranteed that the client IP address is sent in the above mentioned header alongside the IP addresses of the proxies (if any) to Identity Authentication.

- *Authentication Method*
  Specify the authenticating method, which the authenticating user has to use. If no method is selected, the rule is valid for any of the methods.
  You can choose from the following:
  - *Client Certificate*
  - *SPNEGO*
  - *User Name and Password*
  - *Token*
  - *Social Identity Provider*
  - *Trusted IdP SAML Assertion*

  > ### Note
  >
  > If the user has an active session with any of the methods, and that method is included in the rule, they can access the application without the need for additional authentication.

- *User Type*
  Specify the type, which the authenticating user must have. If no user type is selected, the rule is valid for any of the types.

- *Group Type*
  - Cloud Group - A group that is created in the Identity Directory of Cloud Identity services. This is the default choice. For more information about the Identity Directory groups, see Groups [page 514].
  - On-Premise Group - A corporate user store group.
- *Group*

  > **Restriction**
  >
  > The administrator must be assigned the *Manage Groups* role or be part of a policy that allows reading groups to be able to view the groups in the list. For more information, see Edit Administrator Authorizations [page 2309] or Configure Group Authorizations [page 2320].

  Specify a cloud or on-premise group, which the authenticating user has to be a member of. If no group is selected, the rule is valid for all users.
  If the rule is valid for an on-premise group, type in the name of the corporate user store group, for which this rule should be valid.
  The cloud groups have to be configured in the administration console for SAP Cloud Identity Services. For more information, see Managing Groups [page 2393].
- *Corporate Attribute*
  Specify an attribute from the corporate identity provider (IdP) assertion, based on which the rule action will be applied.
  The rule must include the attribute name and value. It is valid only when the specified name and value are found in the assertion from the corporate IdP.

  > **Note**
  >
  > For this rule, the *Apply Application Configurations* option of *Identity Federation* must be enabled. For more information, see Configure Identity Federation [page 2614].
- *Email Domain*
  Specify a domain for the email of the authenticating user.

The fields *IP Range*, *Forwarded IP Range*, *Group*, *Authentication Method*, and *User Type* are not mandatory, but at least one of them has to be specified.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the list item of the application that you want to edit.

   > **Note**
   >
   > If you do not have a created application in your list, you can create one. For more details, see Related Information.

   > **Caution**
   >
   > The list also includes the `Administration Console` application. If you enable risk-based authentication for that application, make sure that you, as a tenant administrator, meet the

> authentication rules and the default authentication rule. Otherwise when you log out of the administration console you will not be able to log in it again if you don't meet the rules.
>
> If `Administration Console` is not in the list of the applications you may request it. To do this, you need to report an incident with a subject on SAP Support Portal Home under the component `BC-IAM-IDS`.

4. Choose the *Authentication and Access* tab.
5. Under *AUTHENTICATION*, choose *Risk-Based Authentication*.
6. Choose *Create Rule*.
7. Fill in the fields on the *New Risk-Based Authentication Rule* window.
8. Choose *Create*.
9. Save your changes.

Task overview: Risk-Based Authentication for an Application [page 2006]

Related Information

Configure Risk-Based Authentication for an Application [page 2007]
Examples for Risk-Based Authentication Scenarios [page 2014]
Create a New Application [page 1922]

1.7.1.14.2.3 Examples for Risk-Based Authentication Scenarios

Example scenarios for configuring risk-based authentication for an application.

Example 1 (Setting TOTP Two-Factor Authentication)

Donna Moore is an administrator of company A. She wants to configure Identity Authentication to always ask the company employees for a password and a TOTP passcode (two-factor authentication) to log on to a *Leave Request* application. For this purpose, Donna sets only a *Default Action*:

Default Authentication Rule

Default Action: `Two-Factor Authentication`

Two-Factor Methods: `TOTP`

Michael Adams is an employee of company A and as such he wants to create a leave request. To log on to the Leave Request application, he provides his password. After that he is prompted to activate a mobile device and to provide a second factor for authentication (a passcode generated by an authenticator app on his mobile device). Two factors are required regardless of whether Michael is in the corporate network or on a business

trip. Michael's manager, Julie Armstrong, receives a notification that Michael has created a leave request. She approves it by logging on to the application with two factors (password and passcode generated by her mobile device).

Example 2 (Setting SMS Two-Factor Authentication)

Donna Moore is an administrator of company A. She wants to configure Identity Authentication to always ask the company employees for a password and an SMS code (two-factor authentication) to log on to the *Corporate Page*. For this purpose, Donna first configures Sinch Service in the administration console for SAP Cloud Identity Services. Then in the *Risk-Based Authentication* section in the administration console, he sets only a *Default Action*:

Default Authentication Rule

Default Action: `Two-Factor Authentication`

Two-Factor Methods: `SMS`

John Miller is an employee of company A and as such he wants to access the corporate page of the company. He is prompted to provide two factors (a password and the SMS code sent to his mobile device) to log on to the corporate page. John Miller has his mobile phone verified, so he can receive SMS codes. Two factors are required regardless of whether Miller is in the corporate network or at home.

Example 3 (SPNEGO)

Donna Moore is an administrator of company A. She wants to configure Identity Authentication to allow employees to access the *Leave Request* application from the corporate network with SPNEGO, and from any other network with passcode. All IPs in the company start with 189.101. She would also like to create a rule for the managers to access the application with two authentication factors. In addition, she wants to restrict the access to all the users with type *Customer*. For this purpose, Donna creates the following rules:

Authentication Rules

Authentication Action	IP Range	Group	Authentication Method	User Type
Deny	Any	Any	Any	Customer
Allow	189.101.112.1/16	Employees	SPNEGO	Any
TOTP Two-Factor Authentication	Any	Employees	Any	Any
TOTP Two-Factor Authentication	Any	Managers	Any	Any

Default Authentication Rule

Default Action: `Deny`

Michael Adams, as an employee of company A, accesses the application in his office and logs on with SPNEGO. When he is on a business trip, he can create leave requests by providing two factors. The two factors are SPNEGO and a passcode generated by an authenticator app on his iPhone. Michael's manager, Julie Armstrong, receives a notification that Michael has created a leave request. She approves it by logging on to the application with TOTP Two-Factor Authentication (a password and a passcode generated by her Android phone). Donna Moore, a customer of company A, tries to access the Corporate Portal, and receives a message that she is not authorized for access.

Example 4 (Setting Web Two-Factor Authentication)

Donna Moore is an administrator of company A. She wants to configure Identity Authentication to always, ask the company managers for a password and a web two-factor authentication to log on to a *Leave Request Approval* application. For this purpose, Donna sets only a *Default Action*:

Default Authentication Rule

Default Action: `Two-Factor Authentication`

Two-Factor Methods: `Web Authentication`

Michael Adams is an employee of company A and he creates a leave request. Michael's manager, Julie Armstrong, receives a notification that Michael has created a leave request. To log on to the Leave Request Approval application, she provides her password. After that Julie is prompted to activate a security key as a second factor for authentication. Two factors are required regardless of whether Julie is in the corporate network or on a business trip.

Julie he approves the leave request by logging on to the application with two factors (password and fingerprint).

Example 5 (Setting more than one Two-Factor Authentication method)

Donna Moore is an administrator of company A. She wants to configure Identity Authentication to always ask the company managers for a second factor in addition to their password. She wants to allow the managers to choose between a TOTP and a web two-factor authentication to log on to a *Leave Request Approval* application. For this purpose, Donna sets the *Default Action* to *Two-Factor Authentication* and configures the *Two-Factor Methods*:

Default Authentication Rule

Default Action: `Two-Factor Authentication`

Two-Factor Methods: `TOTP`; `Web Authentication`

Michael Adams is an employee of company A and he creates a leave request. Michael's manager, Julie Armstrong, receives a notification that Michael has created a leave request. To log on to the application *Leave Request Approval* she provides her password. After that she is prompted to choose the two-factor authentication method. She chooses a security key as a second factor for authentication.

Julie approves the leave request by logging on to the application with two factors (password and fingerprint).

In addition to Julie's approval, the leave request must also be approved by the HR Manager of the company, John Miller. To log on to the *Leave Request Approval* application John provides his password. After that he is also prompted to choose the two-factor authentication method. John chooses TOTP. Now, he is prompted to provide a TOTP code from his device. After providing it, he is granted access to the app and approves the leave request.

Two factors are required regardless of whether Julie and John are in the corporate network or on a business trip.

Example 6 (Define IP Addresses)

Donna Moore is the security administrator of company A. She wants to configure Identity Authentication to allow employees to access the company's Critical Infrastructure Depot only from its two office locations. All other access attempts should be denied. The two office locations are with the following IP ranges: the headquarters uses IPs starting with 203.0.113, and the branch office uses IPs starting with 198.51.100. For this purpose, Donna creates the following rules:

Authentication Rules

Authentication Action	IP Range	Group	Authentication Method	User Type
Allow	203.0.113.0/24	Employees	Default	Any
Allow	198.51.100.0/24	Employees	Default	Any

Default Authentication Rule

Default Action: `Deny`

Michael Adams is a security specialist of company A. When he tries to access the Critical Infrastructure Depot from the headquarters office, he successfully logs on with his credentials because his IP address falls within the allowed range. However, when Michael travels and attempts to access the depot from a hotel, he receives a message that access is denied.

Julie Armstrong, a regional security manager of company A, works from the branch office. She can access the Critical Infrastructure Depot without issues because her IP address is also within the allowed range.

John Miller, an employee at Company A, works remotely from home two days per week. He tries to access the Critical Infrastructure Depot from his home office location and receives a message that access is denied. He can access it without problems while he is in the office.

Parent topic: Risk-Based Authentication for an Application [page 2006]

Related Information

Configure Risk-Based Authentication for an Application [page 2007]
Create a New Rule [page 2010]

1.7.1.14.3 Configure Concurrent User Access to the Application

Tenant administrator can allow or restrict user access to an application from one or more browsers or devices.

Context

With the concurrent user access option, you as a tenant administrator can configure the system so that based on the requirements the application can allow or restrict multiple sign-ins by one and the same user.

The concurrent access offers three options:

- **Allow**
  No restrictions are applied. The user can sign in from multiple devices and browsers. This is the default setting.
- **Warning**
- The user can sign in from only one device or browser at a time during the lifetime of the identity provider (IdP) session. If other sessions exist, the user receives a message and will have the option to terminate the other session and sign out.
  To guarantee a consistent behavior for the application, the application session should have the same timeout as the IdP session. For more information about the session management, see Configure Session Timeout [page 2170].

  > → Remember
  >
  > If the user chooses the option *Sign Out and Continue*, they can still choose to continue with the new session without terminating the previous one. The system will not terminate the session automatically. If you want to restrict concurrent access, then you should use the *Error* option to terminate the other session and sign out.

  > ⚠ Caution
  >
  > If you have configured a *Warning* concurrent user access option, you must also have a Single Logout Endpoints (SLO) URL with a SOAP binding added for the SAML 2.0 application. Otherwise if a user chooses to sign out and continue to new session, an SLO request for the old session won't be sent to the application.. For more information, see Configure SAML 2.0 Service Provider [page 1941].

- **Error**
  The user can sign in from only one device or browser at a time during the lifetime of the identity provider (IdP) session. When the user tries to sign in from another device or browser, they see an error message.
  The user can sign out from the unwanted sessions via the profile page. The option is available at *Profile Page* > *Sessions and Tokens* > *Active Sessions*.
  To guarantee a consistent behavior for the application, the application session should have the same timeout as the IdP session. For more information about the session management, see Configure Session Timeout [page 2170].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Authentication and Access* tab.
5. Under *AUTHENTICATION*, choose *Concurrent Access*.
6. Set the radio button for the users you want to allow to log on:
   - Allow
   - Warning
   - Error

   If the application is updated, the system displays the message `Application <name of application> updated`.

1.7.1.14.4 Always Require Password from Users

Always require users to provide password when accessing an application.

Context

By enabling the *Force Authentication* option users must always provide a password when accessing an application. The configuration is per application. The option is switched off by default.

Force authentication can be enabled for both OpenID Connect and SAML 2.0 applications.

> **Note**
>
> In the context of a corporate identity provider scenario, if an application requires force authentication, users have to authenticate themselves against the corporate identity provider each time they access the application even if single sign-on (SSO) is enabled.

To enable force authentication for an application, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Authentication and Access* tab.
5. Enable the *Force Authentication* option under *Authentication*.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

1.7.1.14.5 Enable or Disable Social Sign-On for an Application

Social sign-on allows users to link their Identity Authentication accounts with social network accounts.

> ⓘ Note
>
> The content in this section is not relevant for China (Shanghai) region.

To link the accounts, users have to choose the social provider button on the logon page of a cloud application. When authenticated via their social identity provider, users are prompted to allow their accounts to be linked with the social network accounts. After this initial setup, users can log on to the application without additional authentication.

Prerequisites

You have set the social providers on tenant level. For more information, see Configuring Social Identity Providers [page 2505].

Context

Identity Authentication allows account linking with the following social providers:

- Apple
- Facebook
- Google
- LinkedIn
- X

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Authentication and Access* tab.
5. Under *AUTHENTICATION*, enable or disable social sign-on.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

Results

With *Social Sign-On* users can log on to the application via one of the social providers. They can see this option on the logon page. Which social identity providers logos appear on the logon page of the application depends on the configurations you have made. For more information, see Configuring Social Identity Providers [page 2505].

Related Information

Create a New Application [page 1922]

1.7.1.14.6 Enable or Disable FIDO2 Biometric Authentication for an Application

Allow users to log on to applications with biometric information without the need to provide username and password.

Context

By configuring FIDO2 biometric authentication, users can log on to applications with their biometric information such as fingerprint, face scan, etc., without the need to provide username and password.

Once the biometric authentication is enabled, the user can choose the *Biometric Authentication* button on the logon page. If the user has a configured biometric authentication device, the logon process will require authentication with the registered device. If biometric authentication isn't configured, the user is required to log on with username and password first. After that he or she should visit their profile page and register a device in the *WEB Two-Factor Authentication* section. This is a onetime configuration.

> **Note**
>
> The user can add more than one device, but can't add a device that has already been added, either for biometric authentication or for two-factor authentication. He or she can remove a device from the list on the profile page. The removal of a device is password protected.

Identity Authentication supports the following:

Device	OS	Browser
Face ID	Latest iOS	• Latest Chrome • Latest Safari
Touch ID	Latest macOS	• Latest Chrome • Latest Safari
Windows Hello	Windows 10	• Latest Edge • Latest Chrome

The user can manage the biometric authentication settings at the profile page.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Authentication and Access* tab.
5. Under *AUTHENTICATION*, enable or disable biometric authentication.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

Next Steps

Enable the *Credential change* security alert emailing to inform the user when a biometric authentication device is activated or deactivated. For more information, see Send Security Alert Emails [page 2209].

Related Information

Create a New Application [page 1922]

1.7.1.14.7 Enable Users to Generate and Authenticate with Certificates

Allow users to generate and authenticate with certificates.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

By enabling certificate generation and authentication for the users, they can log on to applications with the generated certificate without the need to provide username and password.

The tenant administrator can configure the system to allow the following types of users to generate their own certificates via the profile page and to authenticate with them:

- *Public*
- *Customer*
- *Partner*
- *Employee*

The certificate generation and authentication is disabled by default for all user types. When you allow it for a user type, the user of this type should go to the profile page and generate a certificate for authentication. Once the certificate is generated, it's downloaded to the user's system. When the user imports it to the certificate store or browser, the newly generated certificate is ready for authentication. When the user navigates to the logon page of the application, he or she is automatically authenticated with the certificate. If the authentication fails, the user is prompted to provide username and password.

> ⚠ Caution
>
> If you have an already configured X.509 client certificate used for user authentication, the users won't be able to authenticate with their own certificates generated via the profile page, and vice versa.

To allow certificate generation and authentication, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Choose the *Certificate Authentication* list item.
4. Use the slider next to a user type to enable certificate authentication.

   If the operation is successful, you receive a confirmation message.

Next Steps

Enable the *Credential change* security alert emailing to inform the user when a certificate generated or removed. For more information, see Send Security Alert Emails [page 2209].

1.7.1.14.8 Configure User Access to the Application

You can configure public access to the application allowing self-registration, or you can restrict the access to existing users or users registered by an application.

Context

You can configure the application to restrict access to specific users only. The following access configurations are possible:

- **Public**
  All users are allowed to log on. Unregistered users start the self-registration process by choosing the *Register Now* link on the logon page.

  > ⓘ Note
  >
  > You have to add as trusted the domains for those applications that allow self-registration to the users. For more information, see Configure Trusted Domains [page 2173].

  > → Tip
  >
  > You can protect the self-registration form against machine registrations via reCAPTCHA. For more information, see Protecting Application Forms with Google reCAPTCHA [page 2056]. The content in this section is not relevant for China (Shanghai) region.

- **Internal** - this is the default setting.
  Only existing users are allowed to log on. Users that are not in the user store of Identity Authentication cannot log on.
  When this option is chosen, self-registration is not possible.

  > → Tip
  >
  > For more information how to add new users in the user store of Identity Authentication, see Create a New User [page 2357].
  >
  > You can also import new users in Identity Authentication via a CSV file. For more information, see Import or Update Users for a Specific Application [page 2348].

- **Private**
  Only users registered by an application can log on. To register users for a specific application, you have to import these users via a CSV file. For more information, about the user import, see Import or Update Users for a Specific Application [page 2348].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.

2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> ### Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Authentication and Access* tab.
5. Under *AUTHENTICATION*, choose *User Application Access*.
6. Set the radio button for the users you want to allow to log on:
   - Public
   - Internal
   - Private
7. Save your selection.

   If the application is updated, the system displays the message `Application <name of application> updated`.

Related Information

Create a New Application [page 1922]
Risk-Based Authentication as an Alternative to Restrict User Access Option in Identity Authentication Service

1.7.1.14.9 Enable Email Verification

Tenant administrators can configure applications to require verification of the user's email address.

Context

Applications can be configured to require users to have their email addresses verified before they can log on. If you want to make sure that a user has a valid email address, even if the email address has not been verified, you should enable the email verification option in the administration console for SAP Cloud Identity Services.

> ### Note
>
> You can check whether the user's email address has been verified by choosing *User Management* > *Details* > *Personal Information* in the administration console. The *Email Verified* box is checked if the

user's email address has been verified. For more information about the user details, see List and Edit User Details [page 2361].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Authentication and Access* tab.
5. Under *AUTHENTICATION*, enable *Email Verification*.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

Results

When the user tries to log on, he or she is asked to verify the email address first. To verify the email, the user must click the link sent to him or her. Once the email address has been verified, the user can log on to the application.

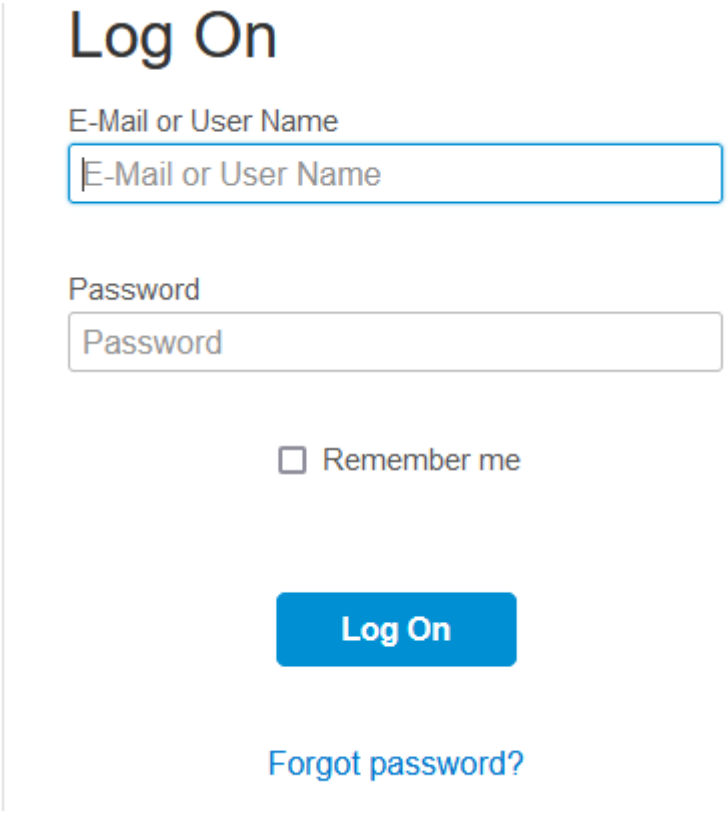
> → Remember
>
> Identity Authentication can send to the user up to three emails (forgot password, reset password, locked password, email verification) per 24 hours.
>
> If the user must receive more than three emails, the administrator must reset the counter for email sending first. For more information, see Reset Counter for Email Sending [page 2383].

1.7.1.14.10 Configure the Remember Me Option

Tenant administrators can configure the *Remember me* option as visible or hidden, and checked or unchecked.

Context

With the remember me option, a cookie is sent to the browser so the users can log on to an application without the need to provide their credentials every time they access it. The default configuration for the remember me option is: visible and unchecked. The default validity for remember me is 3 months. If the default configuration is preferred, users see the remember me option on the logon screen:



You can change the default configuration for the remember me option via the administration console for SAP Cloud Identity Services. The following options are available:

Visible	Checked	Validity	Notes
Yes	No	• 1 month • 3 months - the default choice • 6 months • 12 months • 24 months • 36 months	The default configuration. Users see the option on the logon screen as unchecked. The user is asked every time for credentials by the application. The validity of the cookie is as configured by the administrator. The default validity is 3 months.
Yes	Yes	• 1 month • 3 months - the default choice • 6 months • 12 months • 24 months • 36 months	Users see the option on the logon screen as checked. A cookie is sent to the user's browser at first logon. It automatically logs the user on next time the application is accessed from the same browser on the user's local computer. The validity of the cookie is as configured by the administrator. The default validity is 3 months.
No	No		Users don't see the option on the logon screen.
No	Yes	• 1 month • 3 months - the default choice • 6 months • 12 months • 24 months • 36 months	Users don't see the option on the logon screen, but the option is checked for them. A cookie is sent to the user's browser at first logon. It automatically logs the user on next time the application is accessed from the same browser on the user's local computer. The validity of the cookie is as configured by the administrator. The default validity is 3 months.

Note

Remember me tokens generated for an application are valid for all applications in the tenant. If the option is enabled, the cookie validity of the first application that is accessed by the user is taken into consideration and all other applications have the same cookie validity despite their configuration in the administration console.

Restriction

The remember me option is supported only when Identity Authentication is set as the default local identity provider for the application. When the authentication is done via a corporate identity provider no cookie is sent to the user's browser.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

3. Choose the *Authentication and Access* tab.
4. Under *Authentication*, choose *Remember Me Settings* option is and configure the options according to your needs.
5. Save your changes.

1.7.1.15 Configure Logo

You can upload and configure a custom logo for your applications. Furthermore, you can remove a configured logo and leave the display name only as a title for the application.

The logo is displayed on the application's logon page and can be included into the emails sent to users.

1. Upload Logo in Administration Console [page 2030]
   You can upload a custom logo in the administration console by uploading an image and use it for your applications.
2. Add Logo for an Application [page 2032]
   Configure a custom logo for a specific application by choosing it from the list of your uploaded images in the administration console.

Related Information

Create a New Application [page 1922]
Configuring Email Templates [page 2282]

1.7.1.15.1 Upload Logo in Administration Console

You can upload a custom logo in the administration console by uploading an image and use it for your applications.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Logos* tile.
3. Choose *Add*.
4. Provide a name for the logo in the dialog and choose *+Add*.
5. To upload a logo, choose *Browse* and select an image with the required format.

   You can use one of the following formats for the image: <name>.png, <name>.gif, <name>.jpeg, or <svg>.

   > ⓘ Note
   >
   > The image must be smaller than 100 KB and with a maximum size of 300x100.

   > ⚠ Caution
   >
   > If the image is of format <svg>, please make sure that the attribute `viewBox = "min-x min-y width height"` is present. For proper rendering the width and height values must be 100. For example, `viewBox="0 0 100 100"`
6. Save your configuration.

   > → Tip
   >
   > To remove an uploaded logo from the list, choose the *Delete* button and confirm your choice. Only logos that are not assigned to applications can be removed.

Task overview: Configure Logo [page 2030]

Next task: Add Logo for an Application [page 2032]

Related Information

Create a New Application [page 1922]

1.7.1.15.2 Add Logo for an Application

Configure a custom logo for a specific application by choosing it from the list of your uploaded images in the administration console.

Prerequisites

You have uploaded an image in the administration console. See Upload Logo in Administration Console [page 2030].

Context

The logo is displayed on the application's logon page and can be included into thee mails sent to users. You can also remove a configured logo and leave the display name only as a title for the application.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Branding and Layout* tab.
5. Under *BRANDING*, choose *Logo*.
6. To set a custom logo for the application, choose *Edit*.
7. Choose the *Image* field and select an image from the list that appears.
8. Save your configuration.

   > → Tip
   >
   > To remove a configured logo, choose the *Remove* button and confirm your choice.
   >
   > When you are in *Edit* mode, the *Remove* button is hidden.

Task overview: Configure Logo [page 2030]

Previous task: Upload Logo in Administration Console [page 2030]

Related Information

Create a New Application [page 1922]

1.7.1.16 Display Application Name on Logon Page

This section shows you how to display or hide the name of the application from the logon page.

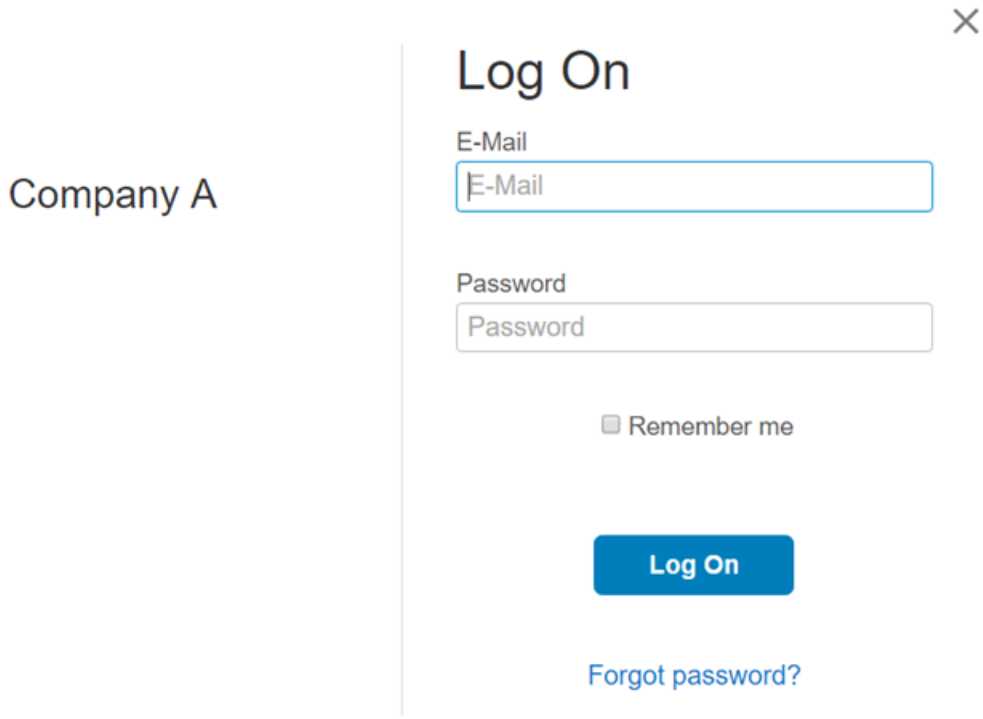
Prerequisites

You are assigned the *Manage Applications* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

The application's display name appears on the left side of the logon and registration pages of the applications that use Identity Authentication for authentication. You set this name when you first create your application in the administration console for SAP Cloud Identity Services, and you can change it later. For more information, see Create a New Application [page 1922] and Change an Application's Display Name [page 1934].

By default, the display name of the application is set to appear on the logon page.



> ⚠ Caution
>
> Be careful when you switch off the display of the application name. The users might not be sure which application they are providing their credentials for.

> → Tip
>
> Instead of leaving the left side of the logon page blank you can add an application's logo. For more information, see Configure Logo [page 2030].

To configure the appearance of the application's name on the logon page, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> ⓘ Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Branding and Layout* tab.
5. Under *BRANDING*, enable or disable the appearance of the application name on the logon page.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

Results

Depending on your choice, the display name of the application is visible or hidden on the logon page.

1.7.1.17 Choose and Configure a Branding Style for an Application

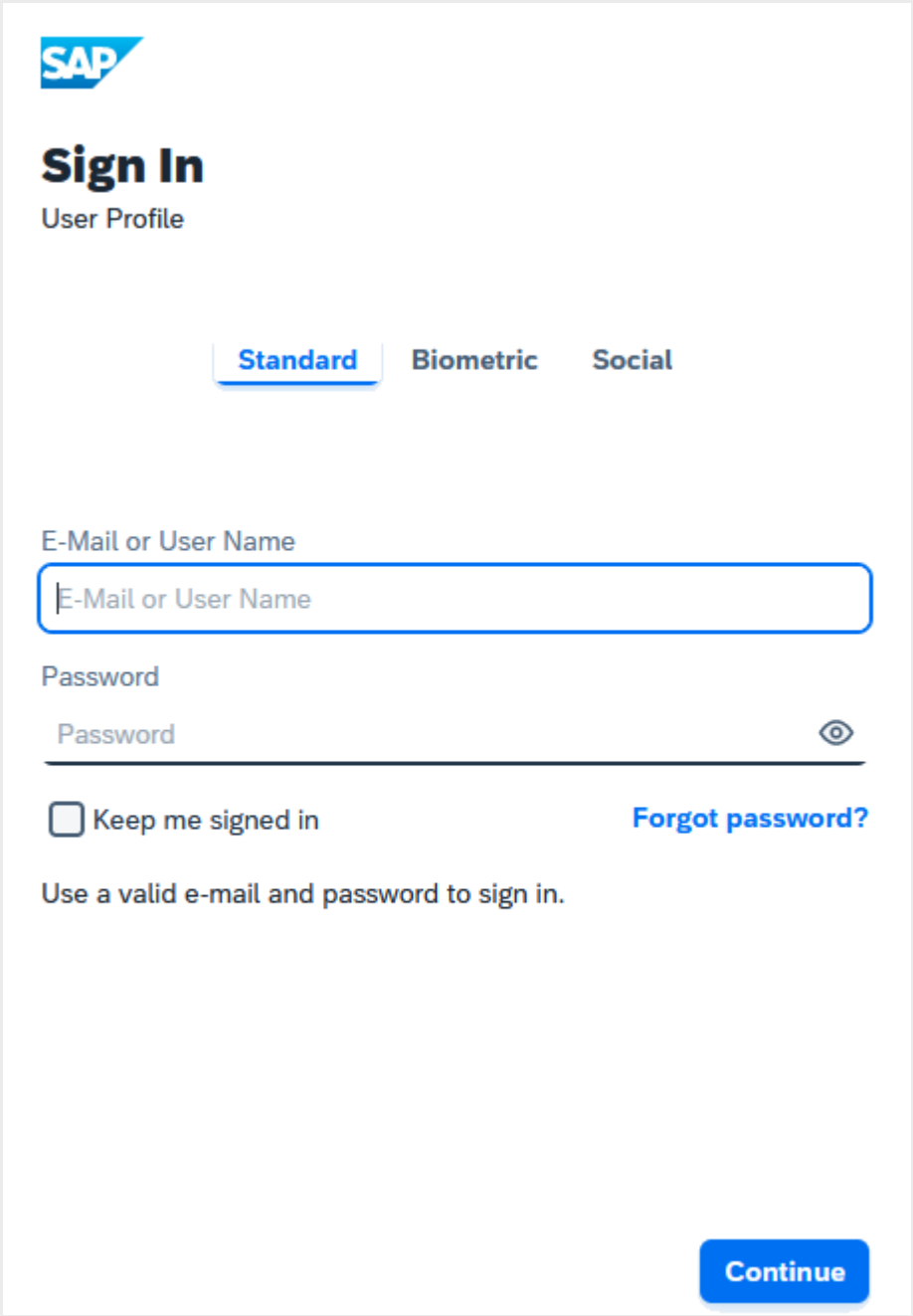
For the configuration of the branding style, you can choose a style for the end-user screens, such as sign-in, registration, upgrade. You can also customize the buttons on these screens.

Context

You can use one of the following themes and customize them:

- *Horizon*:
  - Standard - This predefined theme is an evolution of the SAP Fiori design system. It uses the Horizon theme for the end-user screens. To learn more about the theme, see SAP's UI Technologies supporting the new Horizon visual theme of SAP Fiori.
    You can customize the *Horizon* theme with instructions that appear on the sign in screen.

Example



Note

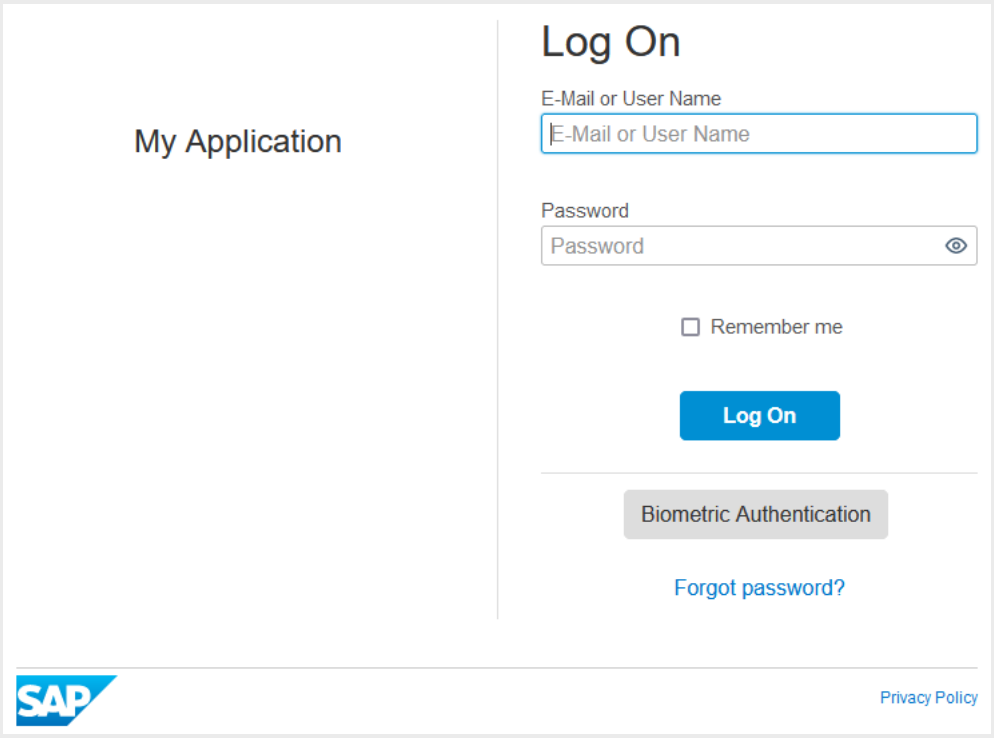
This is the default setting for all new tenants, and for all new applications (service providers) created after Oct 26, 2022 via the administration console, SCIM REST API, or by the Identity Authentication operators.

- Custom Advanced - You can do advanced customization of the end-user screens, by uploading a CSS file and assign that file to an application. The CSS file overrides the default CSS file of the Horizon theme.

- *Quartz*
  - Standard - This predefined theme includes white and brand blue coloring based on SAP Fiori's color palette.

    > **Note**
    >
    > This is the default setting for all tenants and applications created before Oct 26, 2022.

    Example

  - Custom Basic
    The custom basic theme allows you to configure a custom branding style for the buttons and information messages. It uses the Quartz theme for all other elements on the screens and error pages. For this configuration, you can customize:
    - the top and bottom background color of the button
    - the background color of the information message in the forms
    - the button's and information message's border line color
    - and the color of the button's text
  - Custom Advanced
    You can do advanced customization of the end-user screens, by uploading a CSS file and assign that file to an application. The CSS file overrides the default CSS file for the Quartz theme.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.

2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Branding and Layout* tab.
5. Under *Branding*, choose the *Branding Style* list item.
6. Choose your preferred theme and select one of the available options for it:

Theme	Customization
Horizon Theme	• *Standard* For more information about how to customize the theme with instructions that appear on the sign-in screen, see Add Instructions Section on Sign-In Screen [page 2133]. • *Custom Advanced*. For more information, see Use a Custom CSS [page 2040].
Quartz Theme	• *Standard* • *Custom Basic* - For more information, see Use a Custom Basic Quartz Theme [page 2039]. • *Custom Advanced*. For more information, see Use a Custom CSS [page 2040].

7. Save your configuration.

> **→ Tip**
>
> Skip this step if you already saved your configuration in **Step 6**.

Once the application has been changed, the system displays the message `Theme changed to <name of theme>`.

Use a Custom Basic Quartz Theme [page 2039]
Configure a branding style for the buttons and information messages for the Quartz theme.

Use a Custom CSS [page 2040]
Upload a CSS file and assign that file to an application for advanced customization of the end-user screens and the profile page for the Quartz or Horizon Theme.

1.7.1.17.1 Use a Custom Basic Quartz Theme

Configure a branding style for the buttons and information messages for the Quartz theme.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Branding and Layout* tab.
5. Under *Branding*, choose the *Branding Style* list item.
6. Select the *Quartz Theme*
7. Choose the *Custom Basic* radio button.
8. Configure the colors.

   To configure the colors, you can use the color picker or enter the color's hexadecimal value.

   See the table below for more information about the buttons and screens that can be customized.

Color Picker	Customized Area
Background Top Color	• button's background top color in all screens • information's message background color in the *Forget Password* screen
Background Bottom Color	• button's background bottom color in all screens • the button's background color when the mouse is hovered on it in all screens • information's message border color in the *Forget Password* screen
Border Color	the button's border color in all screens
Text Color	the text in the buttons in all screens

9. Save your selection

   Once the application has been changed, the system displays the message `Theme changed to <name of theme>`.

1.7.1.17.2 Use a Custom CSS

Upload a CSS file and assign that file to an application for advanced customization of the end-user screens and the profile page for the Quartz or Horizon Theme.

Context

To use a custom CSS for the application end-user screens and profile page, you must upload your custom CSS file in the administration console for SAP Cloud Identity Services and then assign that file to the application.

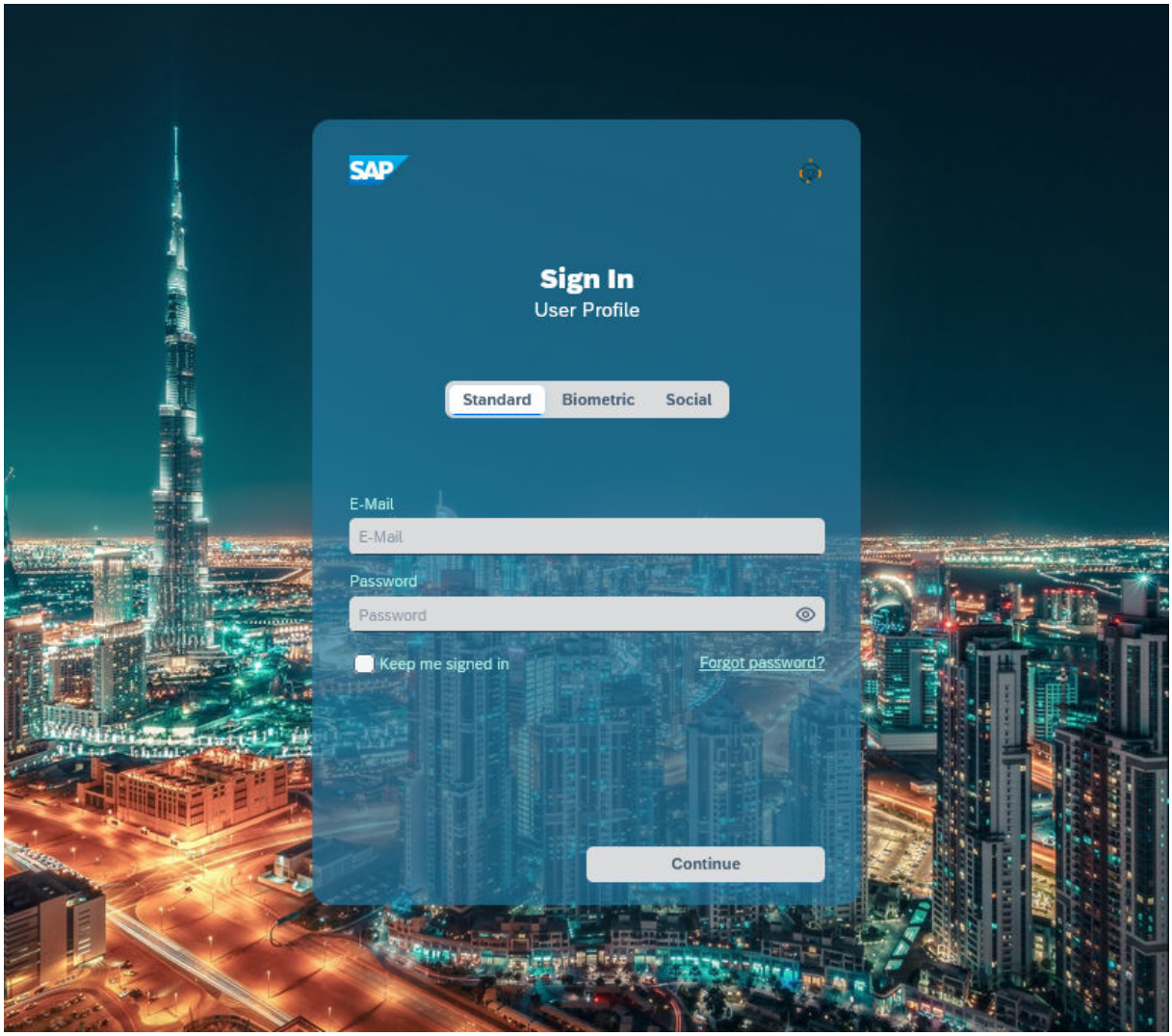
> ⚠ Caution
>
> When working with custom CSS documents or importing CSS documents from external sources, always consider the following safety guidelines:
>
> - *Validate Sources* - Before importing a CSS file from an external source, be sure that you trust the provider.
> - *Review Content* - Inspect the content of the external CSS file to ensure it doesn't contain any security vulnerabilities.
> - *Monitor Regularly* - Periodically review and update the imported CSS documents to prevent any potential security vulnerabilities.

- 1. Upload a Custom CSS File [page 2042]
- 2. Assign the Custom CSS File to an Application [page 2043]

Example

Horizon

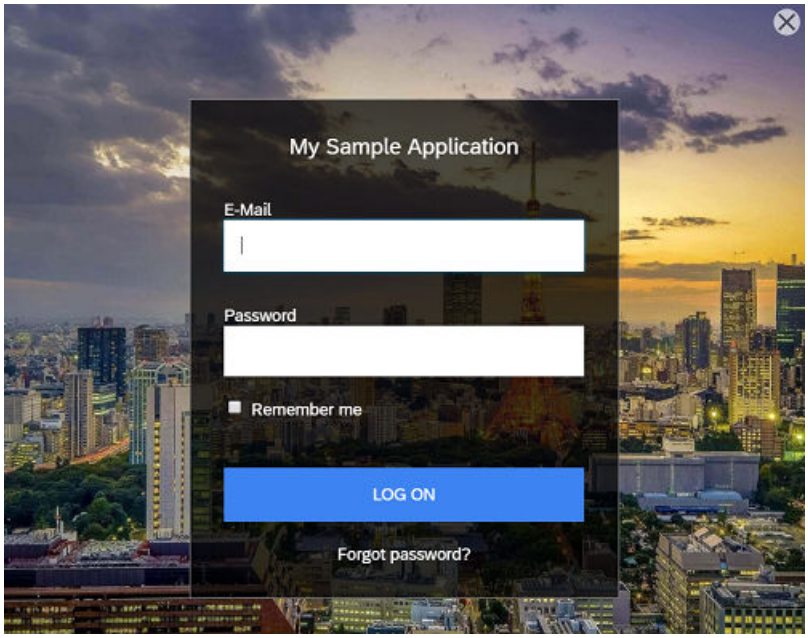
The Horizon-Example.zip contains a CSS file, which you can use to test the Custom CSS option for the Horizon theme. After you follow the procedures for uploading and assigning the example file, the resulting logon page will be:



> **Note**
>
> The background image, and the tenant and application logo are configured separately, and aren't part of the CSS file.

Quartz

The Example.zip contains a CSS file which you can use to test the Custom CSS option for the Quartz theme. After you follow the procedures for uploading and assigning the example file, the resulting logon page will be:



You can download the default CSS file for the Quartz theme at Default.zip.

1.7.1.17.2.1 1. Upload a Custom CSS File

Upload a Quartz or Horizon CSS file for custom end-user screens or the profile page.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Custom CSS* tile.
3. Choose the CSS style that you want to edit.

   > **Note**
   >
   > Type the name of the CSS style in the search field to filter the list items, or choose the CSS style from the list on the left.
   >
   > If you do not have a CSS style list, you can create one by choosing the *Add* button on the screen.

4. Browse for a CSS file and upload it for the respective theme.
5. Save your changes.

   Once the style has been saved, the system displays the message `<name of theme> updated`.

   You can update your custom CSS style as many times as you need. Only the last updated version is used.

   You can delete a CSS style. Select the CSS style and choose the *Delete* button to delete it.

> ⚠ Caution
>
> You can delete a CSS style, only if it is not assigned to an application. If it is assigned, you must assign a new style, and then delete it.

You can download the CSS file for further reference by choosing the *Download* button.

6. **Optional:** To preview the implementation of the CSS on the Sign-In page, copy the preview link and open it in a new private or incognito tab.

1.7.1.17.2.2 2. Assign the Custom CSS File to an Application

Assign the your CSS file to the application or profile page.

Context

Once you have a CSS file uploaded for your theme you can assign it to an application.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> ⓘ Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Branding and Layout* tab.
5. Under *BRANDING*, choose the *Branding Style* list item
6. Select the theme:
   - *Horizon Theme*
   - *Quartz*
7. Choose the *Custom Advanced* radio button.
8. Select the style sheet that you want to apply.
9. Save your selection.

   Once the application has been changed, the system displays the message `Theme changed to <name of theme>`.

> → Remember
>
> The custom CSS file is combined with the default CSS file of Identity Authentication.

1.7.1.18 Enable or Disable Reload Parent Page Option

You can enable or disable the reload of the application's parent page after a successful logon.

Prerequisites

You have an SAML 2.0 application. The *Reload Parent Page Option* is not visible for OpenID Connect applications.

Context

The *Reload Parent Page* option specifies whether the application's parent page reloads or not after a successful logon via an overlay page.

> ⓘ Note
>
> By default the *Reload Parent Page* option is enabled. Disable the option if you would like to embed the logon page in your custom iFrame.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].
4. Choose the *Branding and Layout* tab.
5. Under *LOGIN PAGE BEHAVIOR*, configure the *Reload Parent Page* option.

Once the application has been updated, the system displays the message `Application <name of application> updated`.

1.7.1.19 Define an Email Template Set for an Application

Tenant administrators can define the email template set that the application uses.

Context

Initially, the application uses a default template set with an English language version for all the templates in the set. This template set is named *Default*. If you want to use another email template set, assign it to the respective application.

> **Note**
>
> Both the HTML and TXT formats are included in the emails sent to a user. What the user sees, depends on the settings of his or her email client.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].
4. Choose the *Branding and Layout* tab.
5. Under *EMAIL CONFIGURATIONS*, choose *Email Template Set*.
6. Select the radio button of the email template set the application will use.
7. Save your selection.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

Next Steps

(Optional) Configure custom mail server. To use own mail server for the emails sent for the different application processes, you should configure your mail server in the administration console for SAP Cloud Identity Services.

For more information, see Configure Mail Server for Application Processes [page 2195].

1.7.1.20 Configure Registration and Upgrade Forms

In the administration console, you can configure which user attributes Identity Authentication sends to the service provider or client (relying party) to be displayed on application's registration and upgrade forms. After the user has filled in the form, the information from these attributes is recorded in the user's profile. This is why these attributes are also called profile attributes and are handled by the profile services.

> ⓘ Note
>
> The user ID attribute cannot be controlled by the user. Because of that it is not included on the registration and upgrade forms, but is available in the user's profile.

Context

To configure the profile attributes, you need to specify which personal, company, and contact information the application prompts the user to provide when registering or upgrading. The information that the user has to provide depends on the status of the attribute (required or optional). You configure which attributes are displayed as required or optional in the administration console for SAP Cloud Identity Services.

The list of the attributes includes:

PERSONAL INFORMATION

- `Salutation`
- `First Name`
- `Middle Name`
- `Last Name`

  > ⓘ Note
  >
  > The `Last Name` attribute is always required for user registration or upgrade, so it is not configurable.

- `Email`

  > → Remember
  >
  > The `Email` attribute is required for user registration or upgrade, and as so not configurable, if marked as required on tenant level. If `Email` is marked as not-required on tenant level, it becomes configurable, and must also be changed here.

- `Phone`

> **Note**
>
> The `Phone` attribute becomes required and not configurable for user registration or upgrade, when *Phone Verification via SMS* is enabled. For more information, see Configure Registration and Upgrade Forms [page 2046].
>
> If you disable *Phone Verification via SMS*, the `Phone` attribute remains required for user registration or upgrade. To make it optional, or remove it, follow the procedure in this document.

- `Street Address`
- `Street Address 2`
- `City`
- `Zip/Postal Code`
- `Country`

> **Note**
>
> The `Country` parameter is required when the `Zip/Postal Code` is filled in by the user in the registration form. For example, this is the situation when both `Zip/Postal Code` and `Country` are optional, and the user fills in the `Zip/Postal Code` field. If the user deletes the information in the `Zip/Postal Code` the `Country` parameter becomes optional.

- `State/Province`

> **Note**
>
> The `State/Province` attribute is configurable only if the `Country` attribute is enabled.

COMPANY INFORMATION

- `Company`
- `Street Address`
- `Street Address 2`
- `City`
- `ZIP/Postal Code`
- `Country`

> **Note**
>
> The `Country` parameter is required when the `Zip/Postal Code` is filled in by the user in the registration form. For example, this is the situation when both `Zip/Postal Code` and `Country` are optional, and the user fills in the `Zip/Postal Code` field. If the user deletes the information in the `Zip/Postal Code` the `Country` parameter becomes optional.

- `State/Province`

> **Note**
>
> The `State/Province` attribute is configurable only if the `Country` attribute is enabled.

- `Industry`

- `Relationship`
- `Job Function`
- `Department`
- `Job Title`

CONTACT PREFERENCES

- `By Email`
- `By Telephone`

> **Note**
>
> The *CONTACT PREFERENCES* attributes define if the self-registration form contains a section *Contact Preferences*. This section asks the user if he or she would like to be contacted by email or phone, or both.
>
> The presence of this section depends also on the `Country` attribute. The legislation in some countries requires the user to explicitly agree that he or she would like to be contacted by email or phone, or both.
>
> Based on the specific configuration, the following options are possible:
>
> - If one or both *CONTACT PREFERENCES* parameters are enabled, and both `Country` parameters are disabled the *Contact Preferences* section will appear in the registration form.
> - If one or both `Country` parameters and one or both *CONTACT PREFERENCES* parameters are enabled, the *Contact Preferences* section will appear in the registration form if the user types at least one country which requires the user to explicitly agree that he or she would like to be contacted by email or phone, or both.
> - If both *CONTACT PREFERENCES* parameters are disabled, the "Contact Preferences" section will not appear in the registration form.
>
> For the full set of the countries that do not require the user to explicitly agree that he or she would like to be contacted by email or phone copy the respective URL listed below, replace `<tenant ID>` with your *Tenant ID*, and open the edited URL in a Web browser.

Contact Preference	Countries

By Email	`https://<tenant ID>.accounts.ondemand.com/md/implicitOptInEmailCountryKeys`
	or
	`https://<tenant ID>.accounts.cloud.sap/md/implicitOptInEmailCountryKeys`

Contact Preference	Countries

By Telephone	`https://<tenant ID>.accounts.ondemand.com/md/implicitOptInTelefonCountryKeys`
	or
	`https://<tenant ID>.accounts.cloud.sap/md/implicitOptInTelefonCountryKeys`

In addition to these profile attributes, the registration and upgrade forms include terms of use and privacy policy documents that are configured separately.

To configure profile attributes, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Authentication and Access* tab.
5. Under *Authentication*, choose *User Attributes for Access*.
6. Configure which attributes the form displays as required or optional, and save your selection.

   If the operation is successful, you receive the message `Registration form updated`.
7. **Optional:** To change the phone country selection, choose *Tenant Settings**Tenant Default Values*, select the desired country from the drop-down and save your changes.

   > ⓘ Note
   >
   > The phone country selection appears on the registration and upgrade forms, the verify phone screen, two-factor authentication with SMS screen, and also on the profile page.
   >
   > The default value is United States.

Next Steps

Configure terms of use and privacy policy documents. For more information, see Configuring Privacy Policies [page 2235] and Configuring Terms of Use [page 2273].

Related Information

Configuring User Attributes from the Identity Directory [page 1973]
Configuring Privacy Policies [page 2235]
Configuring Terms of Use [page 2273]
Create a New Application [page 1922]

1.7.1.21 Protecting Self-Registration, Upgrade with Phone Verification

Tenant administrators can configure applications to require new self-registered users and existing users to verify themselves via SMS codes sent to their mobile phones.

Context

Applications with self-registration process can verify newly created users via a code sent as an SMS to their mobile phones. The applications can therefore protect against users creating numerous dummy accounts.

Newly created users are required to provide their phone numbers during the self-registration process. For more information about how to allow self-registration, see Configure User Access to the Application [page 2025].

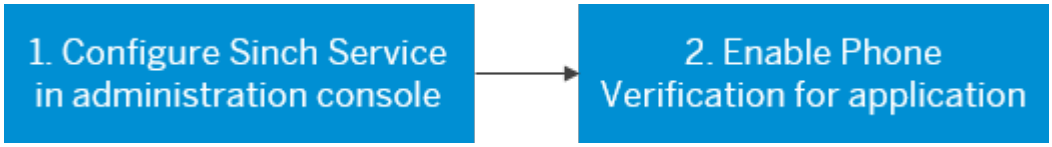
Existing users are required to provide their phone numbers in the upgrade form.

> ✤ Example
>
> Michael Adams is an administrator at retail company A. He has configured his system in such a way that users can register on their own and then purchase from the company's site. He allows users to access his *Company A Purchasing* application by self-registration.
>
> Michael also wants to restrict users to create dummy accounts. To do this, he configures Sinch Service and enables the *Phone Verification via SMS* option in the administration console for SAP Cloud Identity Services.
>
> Donna Moore is a customer who wants to purchase goods from company A for the first time. When she accesses company A's application, she is redirected to the company's logon page. Because she is not registered yet, she has to choose the *Register* button to start the registration process. A registration form then appears, prompting Donna to enter her name, email address, and telephone number, and to accept the organization's terms of use and privacy policy. The telephone number is required when the *Phone Verification via SMS* option is enabled. When she submits the form, she receives an email with instructions

on how to activate her registration. Once she has activated her registration, she receives a code via SMS. At the same time she is redirected to a page where she is prompted to verify her phone number by entering the code she received. Once she has verified the phone number, she is able to log on to the retailing application with her user credentials.

Julie Armstrong is a customer who has an account for company A, but she has never provided her phone number. When she accesses company A's application, she is redirected to the company's logon page. After she provides her credentials, Julie is required to enter her phone number. She receives a code via SMS, and at the same time is redirected to a page where she is prompted to verify the phone by entering the code she received. Once she has verified the phone number, she is able to log on to the retailing application with her user credentials.

Configuration of phone verification in the administration console for SAP Cloud Identity Services comprises two consecutive steps. As a prerequisite, you must have an account in Sinch Service.



- Configure Sinch Service in Administration Console [page 2051]
- Enable Phone Verification for an Application [page 2053]

1. Configure Sinch Service in Administration Console [page 2051]
   Configure Sinch Service to enable *Phone Verification via SMS* or *SMS Two-Factor Authentication* in the administration console.
2. Enable Phone Verification for an Application [page 2053]
   Tenant administrators can enable phone verification via SMS code for an application.

1.7.1.21.1 Configure Sinch Service in Administration Console

Configure Sinch Service to enable *Phone Verification via SMS* or *SMS Two-Factor Authentication* in the administration console.

Prerequisites

- You have an account in Sinch Verification Service. Once you create a Sinch service account, you receive an email with your account information.
- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

Sinch service enables multichannel two-factor authentication (2FA), adding another layer of security to customers' online accounts, beyond their login and password. The service enables you to configure tokens – such as one-time passwords (OTPs), personal identification numbers (PINs), and verification codes – that are tailored to businesses and particular use cases.

For the integration between Identity Authentication and Sinch service, you must provide information in the administration console for SAP Cloud Identity Services. You can have a Sinch Verification configuration. You can clear the fields by choosing the *Remove Configuration* button at the top of the screen.

Provide the following:

Sinch Verification

Configuration	Notes
Application Key	The *Application Key* obtained via Sinch customer dashboard.
Application Secret	The *Application Secret* obtained via Sinch customer dashboard.
Code Timeout	Optional. Request the Sinch Support team to configure this parameter for the client's Sinch account. → Remember Before configuring this property in the administration console for SAP Cloud Identity Services, first request this property to the Sinch Support team.
SMS Text Template	Optional. Request the Sinch Support team to enable this configuration. The SMS text in the configuration must include the `{{CODE}}` placeholder, where Sinch will include the actual code. For example, `"This is your OTP code {{CODE}}!"` → Remember Before configuring this property in the administration console for SAP Cloud Identity Services, first request this property to the Sinch Support team.

→ Remember

It takes 2 minutes for the configuration changes to take place.

To configure the administration console, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Choose the *Sinch Service Configuration* list item.
4. Choose the *Edit* button.
5. Select *Sinch Verification* and enter the information that is required.
6. Save your entries.

Task overview: Protecting Self-Registration, Upgrade with Phone Verification [page 2050]

Next task: Enable Phone Verification for an Application [page 2053]

Related Information

Sinch Verification API
Sinch Verification Developer Documentation

1.7.1.21.2 Enable Phone Verification for an Application

Tenant administrators can enable phone verification via SMS code for an application.

Prerequisites

- You have an account in Sinch Service.
- You have configured Sinch Service. For more information, see Configure Sinch Service in Administration Console [page 2051].

Context

This option requires the user to provide his or her phone number in the registration or upgrade forms. After that the user receives a verification code which must be provided so that the registration or upgrade is completed. If the code fails to arrive, the user has up to five attempts to request a new code via SMS.

After that the user can either contact the system adminstrator or request a code via a phone call. Again the user has up to five attempts to request a new code via phone call. The user can trigger one phone call per session.

When this feature is enabled, the **Phone** attribute becomes required and is not configurable for user registration or upgrade. For more information, see Configure Registration and Upgrade Forms [page 2046].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ### Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Branding and Layout* tab.
5. Under *APPLICATION FORMS CUSTOMIZATION*, enable *Phone Verification via SMS*.

   If the operation is successful, you receive the following message: `Application <Application Name> updated`. The slider is switched to *ON*.

   > ### Note
   >
   > If you do not want to use phone verification via SMS for an application, you can drag the slider next to it to *OFF*.

6. **Optional:** Configure rules to skip phone verification for user types and save your changes. To configure the authentication rules, choose one of the following:

Option	Description
Add a new rule	See Add a New Phone Verification Rule [page 2055].
Delete an existing rule	Choose the delete icon next to the rule you want to delete.

   If the operation is successful, the system displays the message `Verification rules updated`.

Results

Newly created users are required to provide their phone numbers during the self-registration process.

Existing users are required to provide their phone numbers in the upgrade form.

Users should enter the code sent by SMS to their phones in order to finish the validation process.

After the phone number is verified, the user can change it on the profile page, after entering a code sent by SMS to their phones.

Task overview: Protecting Self-Registration, Upgrade with Phone Verification [page 2050]

Previous task: Configure Sinch Service in Administration Console [page 2051]

1.7.1.21.2.1 Add a New Phone Verification Rule

You can add rules to skip phone verification for user types.

Context

Each rule contains the following information:

- **Action**
  You can choose the *Skip* action. Users of the selected user type in the rule will not be prompted to verify their phones when the *Phone Verification Via SMS* option is enabled.
- *User Type*
  Specify the type, which the user must have.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the list item of the application that you want to edit.

> ### Note
>
> If you do not have a created application in your list, you can create one. For more details, see Related Information.

> ### Caution
>
> The list also includes the `Administration Console` application. If you enable risk-based authentication for that application, make sure that you, as a tenant administrator, meet the

authentication rules and the default authentication rule. Otherwise when you log out of the administration console you will not be able to log in it again if you don't meet the rules.

If `Administration Console` is not in the list of the applications you may request it. To do this, you need to report an incident with a subject on SAP Support Portal Home under the component `BC-IAM-IDS`.

4. Choose the *Branding and Layout* tab.
5. Under *Application Forms Customization*, choose *Phone Verification Via SMS*.
6. Choose *Add Rule*.
7. Fill in the fields on the *New Verification Rule* window.
8. Confirm your changes.
9. Choose *Save*.

   If the operation is successful, the system displays the message `Verification rules updated`.

Related Information

Add a New Phone Verification Rule [page 2055]
Protecting Self-Registration, Upgrade with Phone Verification [page 2050]

1.7.1.22 Protecting Application Forms with Google reCAPTCHA

Tenant administrators can use Google reCAPTCHA to protect the application forms of Identity Authentication against spam and abuse.

> ⓘ Note
>
> The content in this section is not relevant for China (Shanghai) region.

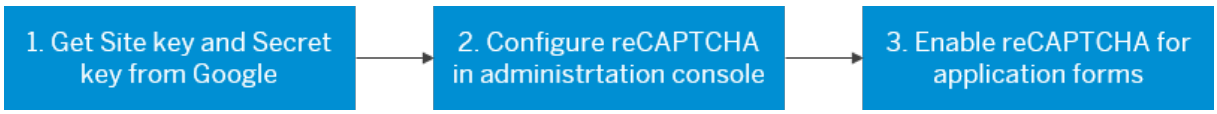
Identity Authentication has integrated Google's Invisible reCAPTCHA service. The aim is to prevent software applications that run automated tasks to flood Identity Authentication.

At present, you can protect the registration form, the login page, and the phone verification pages of Identity Authentication.

> → Remember
>
> To use reCAPTCA:
>
> - for the registration form - you must have allowed self-registration for the application. For more information, see Configure User Access to the Application [page 2025]. Choose *Public* access.
> - for the phone verification pages - you must have enabled phone verification. For more information, see Enable Phone Verification for an Application [page 2053].

reCAPTCHA is provided in the form of a widget that is visible on the protected application form. If the reCAPTCHA service suspects that the user is not a human, a popup with a challenge appears prompting the user to solve a CAPTCHA. By default, only the most suspicious traffic is affected. For more details about Google reCAPTCHA, see **Related Information**.

Configuration of reCAPTCHA for the different application forms of Identity Authentication comprises three consecutive steps:



- Get Site Key and Secret Key from Google for reCAPTCHA [page 2057]
- Configure Google reCAPTCHA in Administration Console [page 2058]
- Enable Google reCAPTCHA for Application Forms [page 2059]

1. Get Site Key and Secret Key from Google for reCAPTCHA [page 2057]
   Register the domain of the application and get a Site key and Secret key from Google.
2. Configure Google reCAPTCHA in Administration Console [page 2058]
   Configure Google reCAPTCHA for your Identity Authentication tenant.
3. Enable Google reCAPTCHA for Application Forms [page 2059]
   Enable Google reCAPTCHA for the registration form, login page, and the phone verification pages of the application.

Related Information

Google reCAPTCHA
reCAPTCHA
Making the Internet safer and faster: Introducing reCAPTCHA Android API

1.7.1.22.1 Get Site Key and Secret Key from Google for reCAPTCHA

Register the domain of the application and get a Site key and Secret key from Google.

Prerequisites

You have a Google account.

Context

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

Procedure

1. Log on to your Google account.
2. Access `https://www.google.com/recaptcha/admin` from your browser.
3. Select the *Invisible reCAPTCHA* radio button.
4. Register your domain.

   > **→ Remember**
   >
   > Your domain is the URL of your Identity Authentication tenant. It has the `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap` pattern.
   >
   > *Tenant ID* is automatically generated by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*.

5. Save your Site key and your Secret key. You need them for the configuration steps in the administration console for SAP Cloud Identity Services.

Task overview: Protecting Application Forms with Google reCAPTCHA [page 2056]

Next task: Configure Google reCAPTCHA in Administration Console [page 2058]

1.7.1.22.2 Configure Google reCAPTCHA in Administration Console

Configure Google reCAPTCHA for your Identity Authentication tenant.

Prerequisites

You have registered the domain of the application and received a Site key and Secret key from Google. For more information, see Get Site Key and Secret Key from Google for reCAPTCHA [page 2057].

Context

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Choose the *CAPTCHA Service* list item.
4. Choose the *Google reCAPTCHA* service.
5. Enter the **`Site Key`** and **`Secret Key`** you received from Google for reCAPTCHA.
6. Save your changes.

Task overview: Protecting Application Forms with Google reCAPTCHA [page 2056]

Previous task: Get Site Key and Secret Key from Google for reCAPTCHA [page 2057]

Next task: Enable Google reCAPTCHA for Application Forms [page 2059]

1.7.1.22.3 Enable Google reCAPTCHA for Application Forms

Enable Google reCAPTCHA for the registration form, login page, and the phone verification pages of the application.

Prerequisites

- You have registered the domain of the application and received a Site key and Secret key from Google. For more information, see Get Site Key and Secret Key from Google for reCAPTCHA [page 2057].
- You have configured the *Google reCAPTCHA* option in the administration console for SAP Cloud Identity Services. For more information, see Configure Google reCAPTCHA in Administration Console [page 2058].

Context

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].
4. Choose the *Branding and Layout* tab.
5. Under *APPLICATION FORMS CUSTOMIZATION*, choose *CAPTCHA Protection*.
6. Enable the CAPTCHA protection for the relevant application form:
   - *Registration Form*

     > **→ Remember**
     >
     > You must have allowed self-registration for the application. For more information, see Configure User Access to the Application. Choose Public access.
   - *Login Page*
   - *Phone Verification*

     > **→ Remember**
     >
     > You must have enabled phone verification for the application. For more information, see Enable Phone Verification for an Application [page 2053].
7. Save your configuration.

   If the operation is successful, you receive the following message: `Application <Application Name> updated`. The slider next to enabled application form is switched to *ON*.

   > **Note**
   >
   > If you do not want to use CAPTCHA protection for an application form, you can drag the slider next to it to *OFF*.

Task overview: Protecting Application Forms with Google reCAPTCHA [page 2056]

Previous task: Configure Google reCAPTCHA in Administration Console [page 2058]

1.7.1.23 Protecting Application Forms with MTCaptcha

Tenant administrators can use MTCaptcha to protect the application forms of Identity Authentication from spam and abuse.

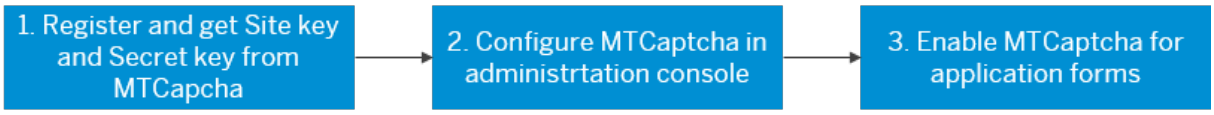
Identity Authentication has integrated the MTCaptcha service. The aim is to prevent software applications that run automated tasks to flood Identity Authentication.

At present, you can protect the registration form, the login page, and the phone verification pages of Identity Authentication.

> → Remember
>
> To use MTCaptcha:
>
> - for the registration form - you must have allowed self-registration for the application. For more information, see Configure User Access to the Application [page 2025]. Choose *Public* access.
> - for the phone verification pages - you must have enabled phone verification. For more information, see Enable Phone Verification for an Application [page 2053].

MTCaptcha is provided in the form of a widget that is visible on the protected application form.

Configuration of MTCaptcha for the different application forms of Identity Authentication comprises three consecutive steps:



- Get Site Key and Private Key for MTCaptcha [page 2062]
- Configure MTCaptcha in Administration Console [page 2062]
- Enable MTCaptcha for Application Forms [page 2063]

1. Get Site Key and Private Key for MTCaptcha [page 2062]
   Register the domain of the application and get a Site key and Private key from MTCaptcha.
2. Configure MTCaptcha in Administration Console [page 2062]
   Configure MTCaptcha for your Identity Authentication tenant.
3. Enable MTCaptcha for Application Forms [page 2063]
   Enable MTCaptcha for the registration form, login page, and the phone verification pages of the application.

1.7.1.23.1 Get Site Key and Private Key for MTCaptcha

Register the domain of the application and get a Site key and Private key from MTCaptcha.

Context

If you have an MTCaptcha registered account and Site Key, skip the procedure below and proceed with Get Site Key and Private Key for MTCaptcha [page 2062].

Procedure

1. Access MTCaptcha website from your browser and sign up..
2. Provide the required information and continue to the next step.

   > → Remember
   >
   > Your domain is the URL of your Identity Authentication tenant. It has the `<tenant ID>.accounts.ondemand.com` pattern.
   >
   > *Tenant ID* is automatically generated by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*.
3. Configure your account.
4. Save your Site key and your Secret key. You need them for the configuration steps in the administration console for SAP Cloud Identity Services.
5. Save your configuration.

Task overview: Protecting Application Forms with MTCaptcha [page 2061]

Next task: Configure MTCaptcha in Administration Console [page 2062]

1.7.1.23.2 Configure MTCaptcha in Administration Console

Configure MTCaptcha for your Identity Authentication tenant.

Prerequisites

You have registered the domain of the application and received a Site key and Private key from MTCaptcha.

Context

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Choose the *CAPTCHA Service* list item.
4. Choose the *MTCaptcha* service.
5. Enter the information in the provided fields:
   - **`Site Key`** - enter the Site Key provided from MTCaptcha
   - **`Secret Key`** - enter the Private Key provided from MTCaptcha
6. Save your changes.

Task overview: Protecting Application Forms with MTCaptcha [page 2061]

Previous task: Get Site Key and Private Key for MTCaptcha [page 2062]

Next task: Enable MTCaptcha for Application Forms [page 2063]

1.7.1.23.3 Enable MTCaptcha for Application Forms

Enable MTCaptcha for the registration form, login page, and the phone verification pages of the application.

Prerequisites

- You have registered the domain of the application and received a Site key and Private key from MTCaptcha. For more information, see Get Site Key and Private Key for MTCaptcha [page 2062].
- You have configured the *MTCaptcha* option in the administration console. For more information, see Configure MTCaptcha in Administration Console [page 2062].

Context

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Branding and Layout* tab.
5. Under *APPLICATION FORMS CUSTOMIZATION*, choose *CAPTCHA Protection*.
6. Enable the CAPTCHA protection for the relevant application form:

   - *Registration Form*

     > → Remember
     >
     > You must have allowed self-registration for the application. For more information, see Configure User Access to the Application. Choose Public access.

   - *Login Page*
   - *Phone Verification*

     > → Remember
     >
     > You must have enabled phone verification for the application. For more information, see Enable Phone Verification for an Application [page 2053].

7. Save your configuration.

   If the operation is successful, you receive the following message: `Application <Application Name> updated`. The slider next to enabled application form is switched to *ON*.

   > ⓘ Note
   >
   > If you do not want to use CAPTCHA protection for an application form, you can drag the slider next to it to *OFF*.

Task overview: Protecting Application Forms with MTCaptcha [page 2061]

Previous task: Configure MTCaptcha in Administration Console [page 2062]

1.7.1.24 Authenticating Identity Provider for an Application

You can choose an authenticating identity provider for the application.

Context

You have the following options for an authenticating identity provider in the administration console for SAP Cloud Identity Services for both bundled and charged applications, and for system applications (`Administration Console`, and `User Profile`):

- Local Identity Provider
  Identity Authentication is set as the default local identity provider for bundled, charged, and system applications.
  The local identity provider gives you access to all application settings in the administration console for SAP Cloud Identity Services.

  > ⓘ Note
  >
  > When Identity Authentication is set as the default authenticating provider, and the service provider (SP) is configured to send a `login_hint` parameter, the user identifier is prefilled on the login screen.
  >
  > The `login_hint` parameter is sent according to the configuration done on tenant level. For more information, see Configure Login Hint Parameter [page 2604].
  >
  > The `login_hint` parameter helps the user when he or she is known to the service provider (SP). Thus it prevents the user from re-typing the user identifier on the logon or conditional screen.

- Corporate Identity Provider
  - Identity Authentication can act as a proxy to delegate authentication to the external corporate identity provider for both bundled and charged applications, and for system applications.

    > ⓘ Note
    >
    > When Identity Authentication is set as proxy, and the service provider (SP) is configured to send a `login_hint` parameter, the `login_hint` coming from the SP will be sent to the corporate IdP in the authentication request.
    >
    > The `login_hint` parameter helps the user when he or she is known to the service provider (SP). Thus it prevents the user from re-typing the user identifier on the logon or conditional screen.

    > → Remember
    >
    > If you want to use a corporate identity provider for a system application, the `name ID` attribute sent in the assertion by that corporate identity provider must match the *Email*, *Login Name* or *User ID* attribute of the user in the user store of Identity Authentication.
    >
    > You should also configure the following options for that corporate provider in the administration console for SAP Cloud Identity Services:
    >
    > 1. Enable *Use Identity Authentication user store*.
    > 2. (Optional) Enable *Apply Application Configurations*. This is needed if you want to apply the custom application configurations for authentication and access policies.

> For more information about how to enable these options, see Configure Identity Federation [page 2614]

- Identity Authentication can act also as a proxy to delegate authentication to multiple external corporate identity providers for again both bundled and charged applications, and for system applications. The scenario includes more than one corporate IdP so the tenant administrator must define URLs that lead users to the application and predefine the IdP that should be used. Thus the users don't need to know about the different IdPs and don't need to choose any. For example one corporate IdP is for employees login, and another is for customers. By default, the application redirects to the IdP for clients, while the employees have a crafted link in their corporate portal page, starting the application with login via the IdP for employees. Thus, neither the employees, nor the customers need to do anything to reach the right IdP for them.
  In this case, the administrator has the following options:
  - Configure rules to decide which identity provider to use in which case. For more information, see Configure Conditional Authentication for an Application [page 2076].

    > **Note**
    >
    > This case requires input from the user at sign in.

  - Provide users with URLs that choose the identity provider. For more information, see Create URLs to Access Applications with Specific Identity Providers [page 2605].

Based on your scenario you can:

- Choose Default Identity Provider for an Application [page 2067]
- Configure Logon via Identity Authentication when a Corporate IdP is Chosen as Default [page 2069]
- Configure Conditional Authentication for an Application [page 2076]
- Enable SSO with Corporate Identity Providers [page 2087]

Choose Default Identity Provider for an Application [page 2067]
: You choose between a local identity provider and a corporate identity provider to be the default identity provider for your application.

Configure Logon via Identity Authentication when a Corporate IdP is Chosen as Default [page 2069]
: You can enable users to log on via Identity Authentication when a corporate identity provider (IdP) is chosen as default.

Configure Conditional Authentication for an Application [page 2076]
: Tenant administrator can define rules for authenticating identity provider according to email domain, user type, user group, and IP range (specified in CIDR notation).

Configure Identity Federation for Applications [page 2084]
: Tenant administrator can enable identity federation for an application to override the identity federation settings on the configured corporate identity provider for the application.

Enable SSO with Corporate Identity Providers [page 2087]
: Tenant administrators can enable IdP-initiated Single Sign-On (SSO) from one, more than one or all configured corporate identity providers (IdPs).

Related Information

Corporate Identity Providers [page 2556]
Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560]
Edit Administrator Authorizations [page 2309]

1.7.1.24.1 Choose Default Identity Provider for an Application

You choose between a local identity provider and a corporate identity provider to be the default identity provider for your application.

Prerequisites

- You are assigned the *Manage Corporate Identity Providers* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have a configured corporate identity provider. For more information how to configure a corporate identity provider, see Related Information.
- You haven't added any rules for authentication. For more information, see Configure Conditional Authentication for an Application [page 2076].

Context

In this scenario you choose which is the default identity provider. It can be either the local identity provider (Identity Authentication) or a corporate identity provider.

Initially Identity Authentication is set as the default local identity provider.

This choice gives you access to all application settings in the administration console for SAP Cloud Identity Services.

If the choice is a corporate identity provider, Identity Authentication acts as a proxy to delegate authentication to the external corporate identity provider. For more information, see Corporate Identity Providers [page 2556]

When you select a corporate identity provider, and you want to apply the custom application configurations for authentication and access policies, you should enable *Apply Application Configurations* for that corporate identity provider. For more information, see Configure Identity Federation [page 2614].

To choose a default identity provider for an application, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under the *Conditional Authentication* section, choose the *Conditional Authentication* list item.
6. Select from the drop down the identity provider that the application will use as the default identity provider.
7. Save your changes.

   Once the application has been updated, the system displays the message `Conditional Authentication updated`.

   The application will use only the chosen identity provider for authentication.

   If you select the local identity provider, you will able to access the custom configurations for the applications.

   If you select a corporate identity provider, you will access only some of the custom configurations for the applications. The configurations under the *Authentication and Access* and *Branding and Layout* tabs will be partially visible. The user will be prompted to provide credentials in a single logon page.
8. **Optional:** (When a corporate identity provider is chosen as default identity provider) Enable the *Allow Identity Authentication Users Log On* option. For more information see, (Optional) Use the Allow Identity Authentication Users Log On Option [page 2074].

Task overview: Authenticating Identity Provider for an Application [page 2065]

Related Information

Configure Logon via Identity Authentication when a Corporate IdP is Chosen as Default [page 2069]
Configure Conditional Authentication for an Application [page 2076]
Configure Identity Federation for Applications [page 2084]
Enable SSO with Corporate Identity Providers [page 2087]
Configure Logon via Identity Authentication when a Corporate IdP is Chosen as Default [page 2069]

1.7.1.24.2 Configure Logon via Identity Authentication when a Corporate IdP is Chosen as Default

You can enable users to log on via Identity Authentication when a corporate identity provider (IdP) is chosen as default.

Sometimes you have multiple identity providers in your landscape, whether they are corporate identity providers or your SAP Cloud Identity Services - Identity Authentication tenant. By providing your users with preconfigured URLs, you can direct them to authenticate with specific identity providers for specific applications.

How you prepare these URLs depends on whether the user starts at the identity provider or at the application. It's further modified by your use case. Some typical use cases are described as follows.

Opting for the Local Identity Provider

You can enable users that are stored in SAP Cloud Identity Services to log on with their Identity Authentication credentials. Thus, the employees log on to the application with their corporate credentials, while the external users, such as clients, partners, or SAP support users are authenticated by the local Identity Authentication tenant. Another example is if you're developing an application and you want to use a test user in the local identity provider instead of creating a test user in the corporate identity provider. The following figure illustrates this architecture.

Sign On With the Local Identity Provider and not the Corporate Identity Provider

Example

Example approuter link:

```
https://approuter-myapplication.cfapps.eu20-001.hana.ondemand.com?
sap_idp=custom.sap
```

Note that the approuter needs the origin key for the custom identity provider for the SAP BTP account.

Example link:

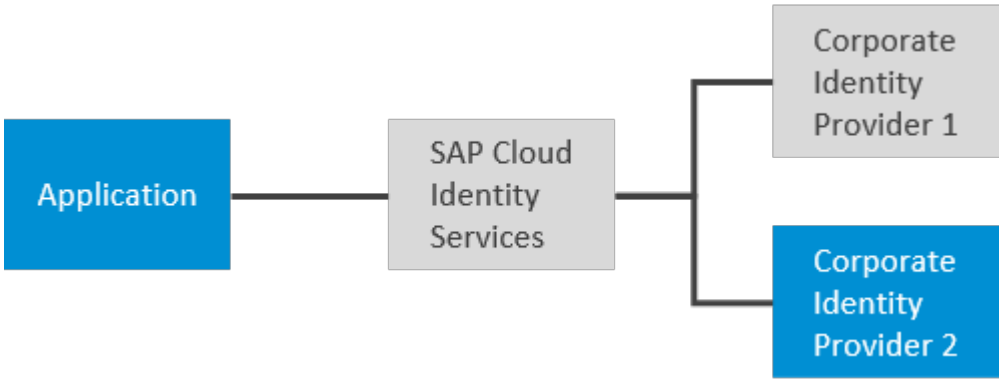
```
https://myapplication.cfapps.eu20-001.hana.ondemand.com?sap_idp=https://
my_sci_tenant.accounts.ondemand.com
```

Example with local value:

```
https://myapplication.cfapps.eu20-001.hana.ondemand.com?sap_idp=local
```

Multiple Corporate Identity Providers

If your application doesn't define rules to determine which identity provider a user authenticates with, SAP Cloud Identity Services offers the choice to your users. To avoid users choosing the wrong identity provider, you can limit their choices. The following figure illustrates this architecture.

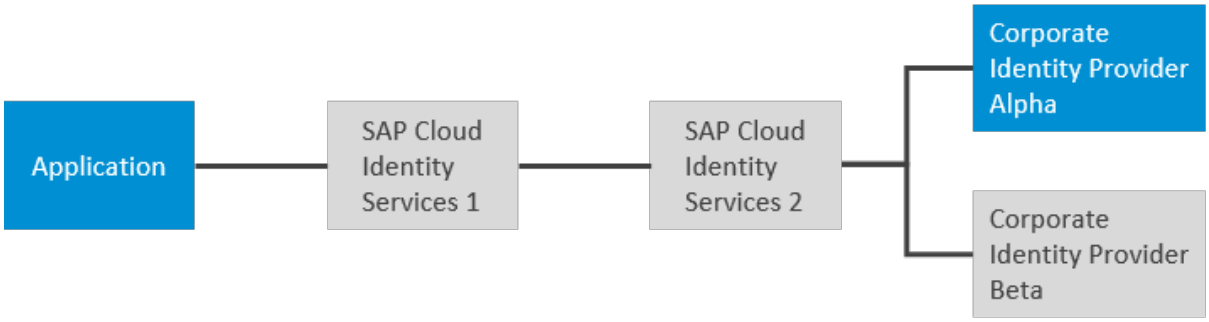


Help Direct Users to the Correct Identity Provider

> ### Example
>
> Example approuter link:
>
> ```
> https://approuter-myapplication.cfapps.eu20-001.hana.ondemand.com?
> sap_idp=https://corp_idp_2.mydomain.com
> ```
>
> Example link:
>
> ```
> https://myapplication.cfapps.eu20-001.hana.ondemand.com?sap_idp=https://
> corp_idp_2.mydomain.com
> ```

Deep Hierarchy of Identity Providers

Your landscape has multiple levels of identity providers, perhaps through subsidiaries and conglomerate organizations. You want to direct your users to a specific identity provider, but this identity provider isn't directly known by your SAP Cloud Identity Services tenant. In such cases, provide the chain of identity providers to which the user must be redirected to reach the authenticating identity provider. The following figure illustrates this architecture.



Deep Structure of Identity Providers

> ### Example
>
> Example approuter link:
>
> ```
> https://approuter-myapplication.cfapps.eu20-001.hana.ondemand.com?
> sap_idp=https://scis_2.mydomain.com,corp_idp_alpha.mydomain.com
> ```
>
> Example link:
>
> ```
> https://myapplication.cfapps.eu20-001.hana.ondemand.com?sap_idp=https://
> scis_2.mydomain.com,corp_idp_alpha.mydomain.com
> ```

Identity-Provider-Initiated Single Sign-On

> ### Restriction
>
> This option is only supported by SAML.

Identity-provider-initiated single sign-on (IdP-initiated SSO) is used for central portals in an enterprise from which users access applications. Users log on to the central portal first. You need to identify the identity provider the user must log on with. You also need to identify the service provider that the user is forwarded to after authenticating. The identity provider can be SAP Cloud Identity Services or a corporate identity provider or a hierarchy of corporate identity providers. Another unique aspect of IdP-initiated SSO is that the URL pattern you use is determined by SAP Cloud Identity Services. Otherwise, the URL is defined by the application. The following figure illustrates this architecture.

Identity-Provider-Initiated Single Sign-On

> **Example**
>
> Example SAML URL:
>
> ```
> https://my_sci_tenant.accounts.ondemand.com/saml2/idp/sso?sp=https://
> myapplication.cfapps.eu20-001.hana.ondemand.com&idp=https://
> corp_idp_1.mydomain.com
> ```

1. Choose a Corporate Identity Provider as Default [page 2072]
   You choose a corporate identity provider to be the default identity provider for your application.
2. (Optional) Use the Allow Identity Authentication Users Log On Option [page 2074]
   Users can log on with their Identity Authentication credentials, when a corporate identity provider is selected as default.
3. Creating URLs for Applications to Authenticate with Specific Identity Providers [page 2075]
   The URL to access an application is generally determined by the application. When you have multiple identity providers available, you can steer users to the correct identity provider with the `idp` parameter.

Parent topic: Authenticating Identity Provider for an Application [page 2065]

Related Information

Choose Default Identity Provider for an Application [page 2067]
Configure Conditional Authentication for an Application [page 2076]
Configure Identity Federation for Applications [page 2084]
Enable SSO with Corporate Identity Providers [page 2087]
Corporate User Store (Neo Environment) [page 2531]

1.7.1.24.2.1 Choose a Corporate Identity Provider as Default

You choose a corporate identity provider to be the default identity provider for your application.

Prerequisites

- You are assigned the *Manage Corporate Identity Providers* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have a configured corporate identity provider. For more information how to configure a corporate identity provider, see Corporate Identity Providers [page 2556].

Context

To choose a corporate identity provider as default, follow the procedures:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under the *Conditional Authentication* section, choose the *Conditional Authentication* list item.
6. Select from the drop down the corporate identity provider that the application will use as the default identity provider.
7. Save your changes.

   Once the application has been updated, the system displays the message `Conditional Authentication updated`.

Results

When the choice is a corporate identity provider, you are not able to access the custom configurations for the applications. The *Authentication and Access* and *Branding and Layout* tabs are not visible.

Task overview: Configure Logon via Identity Authentication when a Corporate IdP is Chosen as Default [page 2069]

Next task: (Optional) Use the Allow Identity Authentication Users Log On Option [page 2074]

1.7.1.24.2.2 (Optional) Use the Allow Identity Authentication Users Log On Option

Users can log on with their Identity Authentication credentials, when a corporate identity provider is selected as default.

Prerequisites

You have chosen a corporate identity provider as default. For more information, see Choose a Corporate Identity Provider as Default [page 2072].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.
4. Choose the *Trust* tab.
5. Under *Conditional Authentication*, select the *Allow users stored in Identity Authentication service to log on* check box.

   > **Note**
   >
   > By default this option is disabled.
   >
   > The check box is visible only if you have chosen a corporate identity provider as default.

6. Save your selection.

Task overview: Configure Logon via Identity Authentication when a Corporate IdP is Chosen as Default [page 2069]

Previous task: Choose a Corporate Identity Provider as Default [page 2072]

Next task: Creating URLs for Applications to Authenticate with Specific Identity Providers [page 2075]

1.7.1.24.2.3 Creating URLs for Applications to Authenticate with Specific Identity Providers

The URL to access an application is generally determined by the application. When you have multiple identity providers available, you can steer users to the correct identity provider with the `idp` parameter.

Prerequisites

- You have set a corporate IdP as the default identity provider in the administration console.
  For more information, see Choose a Corporate Identity Provider as Default [page 2072].
- The application supports selection of the corporate IdP with the idp parameter in the application URL. See the documentation of the application.

  > **Example**
  >
  > Example: Many BTP applications use the application router, which requires the `sap_idp` query parameter.
  >
  > For more information, see @sap/approuter at *npm*.

- If you have enabled the *Allow Identity Authentication Users Log On* option, you can use the reserved name **`local`** to authenticate with your Identity Authentication tenant.
  For more information, see (Optional) Use the Allow Identity Authentication Users Log On Option [page 2074].

Procedure

Use the `idp` parameter in the URL for OIDC or in the URL or body for SAML.

> **Caution**
>
> It depends on the service provider whether the parameter can be passed in body or URL. Identity Authentication doesn't control how the SAML 2.0 request is sent.

Parameters

Parameter	Description
`idp`	Name of the identity provider as defined under the *Name* in the *Trust* configuration of the identity provider. • For SAML, see Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560]. • For OIDC, see Configure Trust with OpenID Connect Corporate Identity Provider [page 2579]. If you need a chain of identity providers, use a comma without spaces (,) to separate them. Use the value **`local`** to override the conditional authentication configuration and authenticate with Identity Authentication instead.

Task overview: Configure Logon via Identity Authentication when a Corporate IdP is Chosen as Default [page 2069]

Previous task: (Optional) Use the Allow Identity Authentication Users Log On Option [page 2074]

1.7.1.24.3 Configure Conditional Authentication for an Application

Tenant administrator can define rules for authenticating identity provider according to email domain, user type, user group, and IP range (specified in CIDR notation).

Prerequisites

You are assigned the *Manage Applications* and *Manage Groups* roles. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

Tenant administrator can control the access to an application by defining different rules for the authenticating identity provider. Based on these rules users are authenticated either via a corporate identity provider or via Identity Authentication.

Rules

The added rules are displayed sorted by priority.

If a rule contains only *IP Range*, and it is first in list or only one, Identity Authentication does not request the user to provide user identifier in the logon screen. The system evaluates whether that user meets all the criteria of the rule. The evaluation starts with the rule with the highest priority, until the criteria of a rule are met. If the criteria of a rule are met, the rest of the conditions are not evaluated

If apart from *IP Range* the rule contains other conditions, or it is not the first in the list, Identity Authentication requests the user to provide user identifier in the logon screen. When a user types the user identifier in the log on page, the system evaluates whether that user meets all the criteria of the rule. The evaluation starts with the rule with the highest priority, until the criteria of a rule are met. If the criteria of a rule are met, the rest of the conditions are not evaluated.

> → Remember
>
> Rules can be defined only if the default identity provider is Identity Authentication.
>
> If you have already defined rules, they won't be considered if you change the default identity provider from Identity Authentication to a corporate identity provider.
>
> To use two or more corporate identity providers and not the default Identity Authentication, your rules must cover all cases so that no user ends up at Identity Authentication. Once you make your other rules, create a catch-all rule with the IP range `0.0.0.0/0`. Such a rule set at the lowest priority ensures that no users end up at Identity Authentication as the default identity provider.

> ⓘ Note
>
> If the user is already existing in Identity Authentication, authentication requests via SPNEGO, existing session, remember me, and token will be processed by Identity Authentication in case of successful authentication no matter of the conditional authentication rules. However, when the tenant has trusted certificate configuration, the application is SAML 2.0 type, and a conditional rule is configured, authentication is done by the local identity provider (IdP) with the certificate, except for the scenario when the conditional rule has an IP Range defined. In this case, if the rule matches, the user will be sent for authentication to the IdP from the rule.

Default Identity Provider

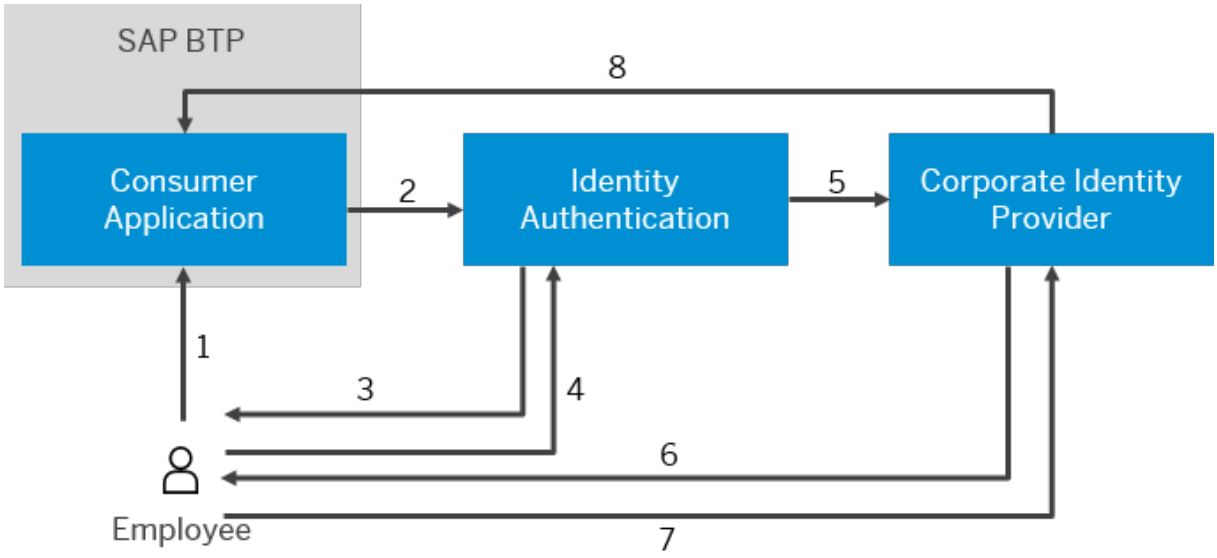
If non-prompt based authentication (SPNEGO, existing session, remember me, and token) is successful, the user is authenticated without checking the rules via Identity Authentication. Otherwise, the user will be asked to provide identifier to evaluate the conditional authentication rules.

If none of the rules meets the criteria, Identity Authentication is chosen for authentication.

Conditional Authentication Flow

The tenant administrator has configured Conditional Authentication in the administration console for SAP Cloud Identity Services. In this setup, the employee should be authenticated via the corporate identity provider, while the customers via Identity Authentication.

Authentication via Corporate Identity Provider



1. The user tries to access the application.
2. The application redirects the request to Identity Authentication to check the rules.
3. Identity Authentication requests the user to provide user identifier in the logon screen.

   > → Remember
   >
   > If the rule contains only *IP Range* Identity Authentication does not request the user to provide user identifier in the logon screen. This step is skipped for this scenario.
   >
   > If apart from *IP Range* the rule contains other conditions, this step is executed.

4. The user provides user identifier. Identity Authentication stores the user identifier in a persistent cookie in the browser. The persistent cookie expires in 36 months.

   > → Remember
   >
   > If the rule contains only *IP Range* Identity Authentication does not request the user to provide user identifier in the logon screen. This step is skipped for this scenario.
   >
   > If apart from *IP Range* the rule contains other conditions, this step is executed.

5. According to the rules, the user should be authenticated via the corporate identity provider. Identity Authentication sends the user identifier as `login_hint` parameter to the corporate identity provider in the authentication request, according to the *Login Hint Configuration* configuration in the administration console.

   > ⓘ Note
   >
   > The `login_hint` parameter helps the user when he or she is known to the service provider (SP). Thus it prevents the user from re-typing the user identifier on the logon or conditional screen.

   > → Remember
   >
   > The user identifier from step 3 is required at the first logon. After that, based on the persistent cookie, the user is redirected directly to the logon screen of the corporate identity provider.
   >
   > If the user wants to log on with another identifier, he or she must delete the browser cookies.

> When the persistent cookie expires, the user is again requested to provide user identifier in the logon screen.

6. If the corporate identity provider supports the `login_hint` parameter, then it requests only the user password. If the corporate identity provider does not support the `login_hint` parameter, it requires both the user identifier and password.
7. Depending on the support of the `login_hint`, user provides only password, or both user identifier and password.

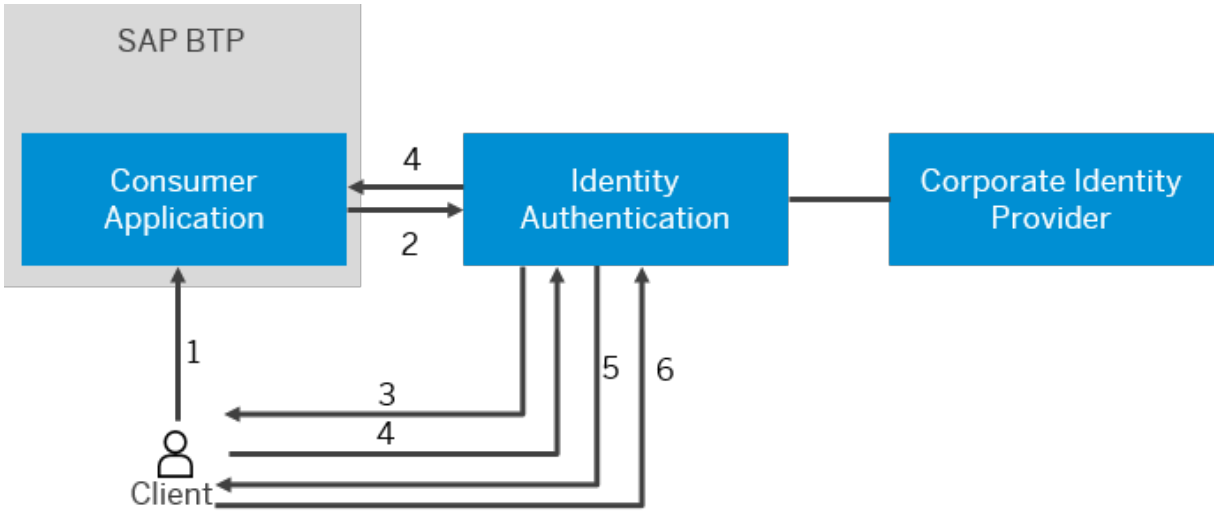
> ### Note
>
> The user can change the prefilled value. The new value is sent to the corporate IdP. If there is an existing conditional authentication cookie, the user identifier stored in that cookie is prefilled on the conditional screen and the `login_hint` is not taken into account.

8. The corporate identity provider grants access to the user.

Result

The user logs on to the application.

Authentication via Identity Authentication



1. The user tries to access the application.
2. The application redirects the request to Identity Authentication to check the rules.
3. Identity Authentication request the user to provide user identifier in the log on screen.

> ### → Remember
>
> If the rule contains only *IP Range* Identity Authentication does not request the user to provide user identifier in the logon screen. This step is skipped for this scenario.
>
> If apart from *IP Range* the rule contains other conditions, this step is executed.

4. The user provides user identifier.

> → Remember
>
> If the rule contains only *IP Range* Identity Authentication does not request the user to provide user identifier in the logon screen. This step is skipped for this scenario.
>
> If apart from *IP Range* the rule contains other conditions, this step is executed.

5. According to the rules, the user should be authenticated via Identity Authentication. The user is prompted to provide password. The user identifier is prefilled in the logon screen.
6. User provides password.
7. Identity Authentication grants access to the user.

Result

The user logs on to the application

> ⓘ Note
>
> Social sign-on and account linking is supported only when the user does not meet the configured rules for conditional authentication.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> ⓘ Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *Conditional Authentication*, choose *Conditional Authentication*.
6. Configure the rules. To configure the rules, choose one of the following options:

Option	Description
Add a new rule	See Add a New Authentication Rule [page 2082].
Edit an existing rule	Choose the icon next to the rule you want to edit.
Delete an existing rule	Choose the delete icon next to the rule you want to delete.

Option	Description
Reprioritize rules	Use the arrows to reprioritize the rules.

7. Choose the default identity provider.

> ⚠ Restriction
>
> Rules can be defined only if the default identity provider is Identity Authentication.

8. Save your changes.

   Once the application has been updated, the system displays the message `Conditional Authentication updated`.

Example: Setting Conditional Authentication

Donna Moore is an administrator of Company A. She wants to configure Identity Authentication so that the employees who log on with their corporate emails to be authenticated via the company's corporate identity provider. The clients and partners of Company A should be authenticated via Identity Authentication. For this purpose, Donna creates the following authentication rule:

Authentication Rule

Identity Provider	Email Domain	User Type	User Groups
Company A Corporate IdP	companya.com		

Default Identity Provider: Identity Authentication

Michael Adams is an employee of Company A and as such he wants to create a leave request. He chooses the logon link on the application and is prompted to provide his email. After that he chooses *Continue* and is redirected to the logon screen of Company A. There he has to provide his user identifier and password. After providing the correct credentials he logs on to the application.

Julie Armstrong is a customer of Company A. She accesses the company's application and chooses the logon link on the application and is prompted to provide her email. After that she chooses *Continue* and is redirected to the log on screen of Identity Authentication where she is prompted to provide her password. Her email is prefilled by the system. After providing the correct password she logs on to the application.

Task overview: Authenticating Identity Provider for an Application [page 2065]

Related Information

Choose Default Identity Provider for an Application [page 2067]
Configure Logon via Identity Authentication when a Corporate IdP is Chosen as Default [page 2069]
Configure Identity Federation for Applications [page 2084]
Enable SSO with Corporate Identity Providers [page 2087]

Configure Login Hint Parameter [page 2604]

1.7.1.24.3.1 Add a New Authentication Rule

You can add rules for authentication against different identity providers.

Prerequisites

You are assigned the *Manage Applications* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

Each rule contains the following information:

- *Identity Provider*
  Define the identity provider that will be used for authentication.
  The dropdown list includes the local identity provider and all corporate identity providers that are created in the administration console for SAP Cloud Identity Services. The corporate identity providers must be configured in the administration. For more information, see Corporate Identity Providers [page 2556].

  > → Remember
  >
  > This field is required.

- *Email Domain*
  Specify the domain of the email address that comes after the @ symbol. You can use asterisk (`*`) in the domain part, for example: `*.example.com`.
  If no domain is entered, the rule is valid for any domain.
- *User Group*

  > ⚠ Restriction
  >
  > The administrator must be assigned the *Manage Groups* role or be part of a policy that allows reading groups to be able to view the groups in the list. For more information, see Edit Administrator Authorizations [page 2309] or Configure Group Authorizations [page 2320].

  Specify the group, which the authenticating user must be part of. If no group is selected, the rule is valid for any of the groups.
  The user groups must be configured in the administration console for SAP Cloud Identity Services. For more information, see Managing Groups [page 2393].

  > ⓘ Note
  >
  > The authenticating user must exist in the user store of Identity Authentication for the *User Group* criteria to be checked properly.

- *User Type*
  Specify the type, which the authenticating user must have. If no user type is selected, the rule is valid for any of the types.

  > **Note**
  >
  > The authenticating user must exist in the user store of Identity Authentication for the *Type* criteria to be checked properly.

- *IP Range*
  Define the range of allowed IP addresses or proxies that the user logs on from. The value must be specified in Classless Inter-Domain Routing (CIDR) notation.

  > **Note**
  >
  > By default the field is empty, meaning that any IP is allowed.

  > **Example**
  >
  > Enter 123.45.67.1/24 to allow users to log on from any IP starting with 123.45.67.

  If no IP range is defined, the rule is valid for all IP ranges.

> **→ Remember**
>
> The fields *Domain*, *Group*, *User Type*, and *IP Range* are not mandatory, but at least one of them has to be specified.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *Conditional Authentication*, choose *Conditional Authenticating Identity Providers*.
6. Choose *Add New Rule*.
7. Fill in the fields on the *New Rule* window.
8. Confirm your changes.

Related Information

Create a New Application [page 1922]

1.7.1.24.4 Configure Identity Federation for Applications

Tenant administrator can enable identity federation for an application to override the identity federation settings on the configured corporate identity provider for the application.

Prerequisites

- You are assigned the *Manage Applications* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have configured Identity Authentication to use a corporate identity provider as an external authenticating authority. For more information, see Create Corporate IdP in Administration Console [page 2558].
- You have selected a configured identity provider as the authenticating identity provider for your application. For more information, see Choose Default Identity Provider for an Application [page 2067].

Context

The *Identity Federation* option allows you to choose whether the user attributes are taken from the corporate IdP assertion or from Identity Authentication user store. Additionally, you can also apply the custom application configurations for the authentication and access policies, as well as allow access based on user groups and email domains.

In scenarios where the application is using a corporate identity provider with *Identity Federation* settings for authentication, the administrator can configure identity federation for the application and override this configuration. This provides a more fine-grained identity federation in situations where multiple applications use the same corporate IdP as an external authenticating authority.

User Store

> → Remember
>
> By default, the *Identity Federation* for an application option is disabled. Enabling the configuration will override the *Identity Federation* settings of the authenticating identity provider.

When *Use Identity Authentication user store* option is enabled, the application checks if the users authenticated by the corporate identity provider exist in the Identity Authentication user store. The existence check is done with the name identifier sent by the corporate identity provider for the identifying attributes `uid`, `loginName`, `emails` and `phoneNumber`.

For users that exist in Identity Authentication, data from Identity Authentication user store is taken and the subject name identifier, assertion and default attributes according to the application configuration are sent. For users with no profile in Identity Authentication, the application receives the nameID attribute from the corporate IdP assertion, and the attributes according to the application configuration.

User Access

By default, *Allow Identity Authentication users only* is disabled, and all users successfully authenticated to the corporate IdP are allowed.

When *Allow Identity Authentication users only* is enabled, only the users that exist in Identity Authentication can access the application.

Restrict Logon to Members of Certain Groups from the Corporate Identity Provider

As a next step, you can assign a group or groups to the corporate IdP. Only users from the Identity Authentication user store that are members of the assigned group can access the application.

> → Remember
>
> You can assign groups, only when *Allow Identity Authentication users only* option is enabled. If disabled, the assigned groups option is not taken into consideration. As a next step, you can assign a group or groups to the corporate IdP or allow certain email domains.

Restrict Logon to Members of Certain Groups

You can assign a group or groups to the corporate identity provider. Only users from the Identity Authentication user store that are members of the assigned groups can access the application.

> → Remember
>
> You can assign groups, only when *Allow Identity Authentication users only* option is enabled. If disabled, the assigned groups option is not taken into consideration.

Restrict Logon to Users with Certain Email Domains

You can add email domains to restrict the access to those specific email domains. Only users from the Identity Authentication user store whose email domains are allowed can access the application.

> → Remember
>
> You can allow email domains from the allow list, only when the *Allow Identity Authentication users only* option is enabled. If disabled, the allow list option is not taken into consideration.

Application Configuration

Additionally, the *Apply Application Configurations* option can be enabled. By default, this option is disabled.

When enabled, the user is authenticated by the corporate IdP, then *Identity Authentication* checks for the configurations for authentication and access policies on application level, and applies them. They include *Risk-Based-Authentication*, *Terms of Use*, *Privacy Policy*, *Upgrade and Registration Forms*.

> → Remember
>
> The concurrent user access configuration is always applied for the application regardless of the *Apply Application Configurations* state.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Select the application that you want to configure for *Identity Federation*.

> **Note**
>
> Type the name of the application in the search field to filter the list items, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose *Identity Federation* to configure the options.
5. Choose *Edit* > *Custom*.
6. Select the desired check boxes:

Option	Description
Use Identity Authentication user store	When you select it, the *Allow Identity Authentication users only* becomes available for selection.
Allow Identity Authentication users only	When you select it, the *Apply Application Configurations* becomes available for selection. Additionally, the *User Groups* and *Allowed Email Domains* sections become visible. There you can see a list of the groups assigned to this corporate IdP and the allow list for email domains. If no groups are assigned or no email domain added, the lists are empty. • To restrict logon to members of certain groups, choose the *Add* button. Select the groups that you want to assign to this corporate identity provider. The list does not include the groups that are already assigned to the corporate identity provider. **Note** To remove groups, choose *Edit*, select the groups you want to remove, choose the *Remove* button, and save your changes. Users that belong only to the unassigned groups will not be able to access the application any more. • To restrict logon to users from certain email domains, under *Email Domains*, choose the *Add* button. Type the email domains that you want to add to the allow list. **Note** To remove email domains, choose *Edit*, select the email domains you want to remove, choose the *Remove* button, and save your changes. Users whose emails contain only the removed email domains will not be able to access the application any more.
Apply Application Configuration	

7. Save your changes.

Task overview: Authenticating Identity Provider for an Application [page 2065]

Related Information

Choose Default Identity Provider for an Application [page 2067]
Configure Logon via Identity Authentication when a Corporate IdP is Chosen as Default [page 2069]
Configure Conditional Authentication for an Application [page 2076]
Enable SSO with Corporate Identity Providers [page 2087]

1.7.1.24.5 Enable SSO with Corporate Identity Providers

Tenant administrators can enable IdP-initiated Single Sign-On (SSO) from one, more than one or all configured corporate identity providers (IdPs).

Prerequisites

- You are assigned the *Manage Applications* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have more than one corporate identity provider, which is configured in the administration console. For more information on how to configure a corporate identity provider, see Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560] or Configure Trust with OpenID Connect Corporate Identity Provider [page 2579].
- (For SAML 2.0 applications) You have added the assertion consumer (ACS) endpoint with the URL of the application's protected page, and the index, in the metadata of that application. The ACS endpoint that you added should look like the following example:

  Sample Code

  ```
  <ns3:AssertionConsumerService index="1" isDefault="false"
  Binding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-POST" Location="https://
  <application URL>/protected.jsp"
  />
  ```

  For more information how to configure your ACS endpoint in the administration for Identity Authentication, see Configure SAML 2.0 Service Provider [page 1941].

Context

Applications can be configured to trust one, more than one, or all corporate identity providers configured in the administration console when identity provider (IdP) initiated Single Sign-On (SSO) is used. The user accesses the application via a URL provided by the corporate identity provider.

To enable IdP-initiated SSO with corporate identity providers configured in the administration console for SAP Cloud Identity Services follow the following procedure:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ### Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under the *Conditional Authentication* section, choose the *Trust Corporate Identity Providers* list item.
6. | Option | Notes |
   | --- | --- |
   | **Enable the slider to allow SSO with all configured corporate identity providers.** | All corporate IdPs are allowed for sign-in.

By default this option is disabled. If enabled, all configured corporate IdPs are hidden, and you can't select a specific IdP from the list. |
   | **Select a specific corporate IdP from the list.** | → Remember

The slider on the top-right corner of the screen must be disabled. If enabled, all configured corporate IdPs are hidden |

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

Results

The application trusts the selected corporate identity providers that are configured in the administration console for SAP Cloud Identity Services.

Task overview: Authenticating Identity Provider for an Application [page 2065]

Related Information

Choose Default Identity Provider for an Application [page 2067]
Configure Logon via Identity Authentication when a Corporate IdP is Chosen as Default [page 2069]
Configure Conditional Authentication for an Application [page 2076]
Configure Identity Federation for Applications [page 2084]
Configure IdP-Initiated SSO with Corporate Identity Providers [page 2564]

1.7.1.25 Use IdP-Initiated Single Logout (SLO)

Use this procedure to start the single logout (SLO) process at Identity Authentication as an identity provider.

Context

Single Logout (SLO) enables users to cleanly close all their sessions in an SSO landscape, even across domains. In this identity provider (IdP) initiated SLO scenario, the user initiates a logout request at Identity Authentication being an identity provider.

SLO can also be initiated at the corporate identity provider (IdP). Identity Authentication accepts SLO requests from corporate identity providers via both SAML and OpenID Connect.

Procedure

1. Direct the user browser to your SLO endpoint.

   Identity Authentication single logout endpoint follows the template `https://<tenant ID>.accounts.ondemand.com/saml2/idp/slo` or `https://<tenant ID>.accounts.cloud.sap/saml2/idp/slo`
2. **Optional:** Add the `RelayState` parameter to provide a relative URL to which the identity provider redirects after the SLO process is complete.

   > → Remember
   >
   > The `RelayState` must be a trusted domain and valid URL. For example, `https://myapplication.ondemand.com/saml2/idp/slo?RelayState=https://example.com`

Results

After successful logout, the identity provider either redirects the user to the URL provided in the RelayState, or if RelayState isn't provided, the user is redirected to the profile page.

Related Information

Configure Trusted Domains [page 2173]

1.7.1.26 Integrating Applications

Enable applications to call each other's APIs, consume reuse services, share authorization context, or hand off authenticated sessions by configuring dependencies in SAP Cloud Identity Services.

SAP Cloud Identity Services supports four types of dependencies. Each type serves a distinct integration scenario as determined by the developer of the application:

Dependency Type	Use When
Provided APIs [page 2090]	One application calls the APIs of another application. The consumer uses technical communication or principal propagation.
Configure Service Dependencies [page 2095]	An application consumes a reuse service, such as the Destination service or Job Scheduling service, using its own token.
Configure Shared Authorization Dependencies [page 2096]	A consumer application, such as a portal or shell, needs to know what a user is authorized to do in a provider application without calling that application directly. The provider's authorization groups are projected as claims into the consumer's token.
Configure Native-Web Trust Dependencies [page 2099]	A native or mobile application hands off an authenticated session to a web application without requiring the user to log in again.

1.7.1.26.1 Provided APIs

Applications sometimes need to propagate principals or have technical communication arrangements between them. To enable one application to consume the APIs of another application, the developer of the application providing APIs defines one or more API permission groups, which the consuming application can consume.

Under this model, applications that can be called by other applications define API permission groups. Administrators grant access to these groups to consuming applications. These permission groups cover a set of related APIs. An application can expose all its APIs without an API permission group when there is no need to restrict consuming applications.

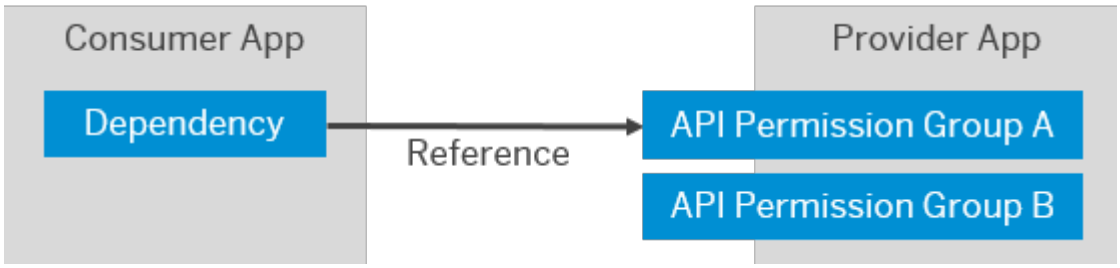
You can set up technical communication or principal propagation.

- With technical communication, you can map authorizations to the communication.
- With principal propagation, user authorizations apply in the decision whether an API call is allowed or not.

In the configuration of the consumer application, the tenant administrator defines the dependencies. Consumer applications use the dependency name to identify to SAP Cloud Identity Services which application the consumer wants to access, including the API permission group to grant.

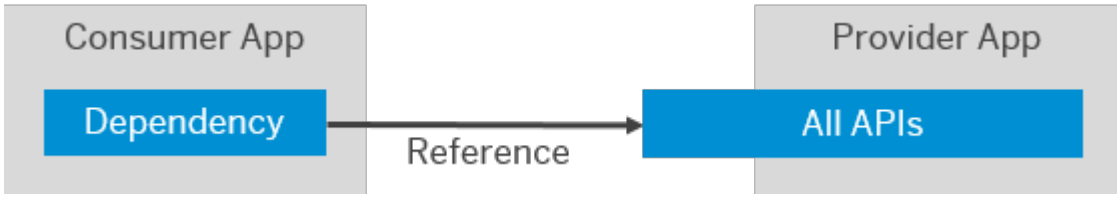
- Typically, an application needs a specific name for integration with a specific provider application. Find the supported dependency names in the provider application or its documentation.
- When an application generically integrates with many arbitrary applications, the application rather enables you to define custom dependency names and maintain them both in SAP Cloud Identity Services as well as in the consuming application itself. The configuration informs the application which dependency to use for which integration scenario.

The following figure shows the relationship between two applications in SAP Cloud Identity Services. The provider application offers two API permission groups. The consumer app is allowed to consume one of these groups. The applications can use technical communication or principal propagation.



App-to-App Integration of API Permission Groups

In the following figure, the provider application doesn't offer any API permission groups. The consumer application references all the available APIs. The provider application depends on the authorizations of the current user when these APIs are accessed.



App-to-App Integration of All APIs

- To configure dependencies, see Configure Integration Between Applications [page 2092].
- To consume APIs from an application that is not registered in SAP Cloud Identity Services, see Generate Credentials to Access the APIs of an Application [page 2093].
- For more information about developing such applications, see Consuming APIs from Other Applications [page 2889].

Configuration Precedence

When you configure both the consumer and provider applications, each application contributes specific OIDC and related configurations. The following table shows which application provides each configuration.

Configuration Source by Application

Configuration	Application	More Information
Subject name identifier	Provider	Configure the Subject Name Identifier Sent to the Application [page 1988]
Token attributes	Provider	User Attributes [page 1972]
Token format	Provider	Token Policy Configuration for Applications [page 2752]
Token grant type	Consumer	Token Policy Configuration for Applications [page 2752]
Token policy	Consumer	Token Policy Configuration for Applications [page 2752]

1.7.1.26.1.1 Configure Integration Between Applications

Define a dependency to the API of another application. With this configuration, you authorize one application to consume the API of another application.

Prerequisites

- You know the name of the provider application and API permission group needed by the consumer application.
  The provider application defines the name of the API permission group during deployment, in the documentation of the application, or from the developers directly for custom developments.
  If the provider application offers all APIs, then you don't need to know the name of a specific API permission group.
  For more information, see Provide APIs for Consumption by Other Applications [page 2891].
- You know the name of the dependency to enter for the consumer application.
  The consumer application must define this name automatically during deployment, in the documentation of the application, or from the developers directly for custom developments.
  For more information, see Integrating Applications [page 2090].

Procedure

1. In the administration console for SAP Cloud Identity Services, choose *Applications and Resources* > *Applications*.
2. Choose the consumer application.
3. Choose the *Trust* tab.

4. Under *Application APIs*, choose *Dependencies*.
5. Enter the required data.

   You can create a maximum of 400 dependencies.

   a. Enter a unique name for the dependency.
   b. Choose the provider application and the API that the consumer application consumes.
6. Save your entries.

Results

The consumer application can consume the specified API of the provider application.

Next Steps

If the applications were developed with a generic implementation, you must configure the consumer application with the dependency name you provided. See the documentation of the consumer application.

For more information about how application developers use this dependency, see Consume an API from a Provider Application [page 2893].

1.7.1.26.1.2 Generate Credentials to Access the APIs of an Application

When integrating applications, create API credentials for the consumer application. From your *Provided APIs*, create a dependent application or create a dependency with an existing application.

Prerequisites

You have created at least one API permission group.

For more information, see Provide APIs for Consumption by Other Applications [page 2891].

Context

The main use cases for this method are as follows:

- You want to integrate an application, which otherwise isn't aware of SAP Cloud Identity Services, such as a non-SAP application. The application requires OAuth credentials, which you provide with this procedure. These credentials enable you to integrate this non-SAP application with the provider application.

- As a developer of a provider application, you want to test your APIs, but don't have a consumer application to integrate with.

Procedure

1. In the administration console for SAP Cloud Identity Services, choose *Applications and Resources* > *Applications*
2. Choose the provider application.
3. Choose the *Trust* tab.
4. Under *Application APIs*, choose *Provided APIs*.
5. Under *Dependent Applications*, choose *Create*.

> ⓘ Note
>
> If you haven't configured an API permission group, the *Dependent Applications* section doesn't appear.

6. Enter the required data.
   a. Enter a unique name for the dependency.
   b. Select the API that the consumer application consumes.
   c. Decide whether you want to create a new application or use an existing one.
      - Create a new application, to create a consumer you can use to test your provided API with. Enable *Generate Secret* to generate client credentials for the new application.
      - Use an existing application to integrate an existing consumer with your provider.
7. Choose *Create*.

> → Remember
>
> If you chose to create a new application and a secret, a dialog summarizes the new dependency. Copy the *Client ID* and *Client Secret* to a safe location. Also copy the *Resource*. You provide this information to the consumer application so it can get a token for the provider application.
>
> For example, using the client credentials flow for a technical communication arrangement, the consumer application can call the `/token` endpoint of Identity Authentication. The service returns a token that the application uses to call the APIs of the provider application. The following is a cURL example of such a call.
>
> ```
> curl  $iashost/oauth2/token -X POST \
>    -H 'Content-Type: application/x-www-form-urlencoded' \
>    -H 'Accept: application/json' \
>    -d "client_id=3ab1ba0e-c573-4ad5-20a4-28aa4e587a19" \
>    -d "client_secret=$secret" \
>    --data-urlencode 'grant_type=client_credentials' \
>    --data-urlencode
> 'resource=urn:sap:identity:application:provider:name:techDependency'
> ```
>
> For more information, see Consume an API from a Provider Application [page 2893].
>
> You can navigate directly to the new application configuration from the dialog.

1.7.1.26.2 Configure Service Dependencies

Enable an application to consume a reuse service using its own token by adding the service to the `consumed-services` property in the Identity service instance parameters.

Prerequisites

- For Cloud Foundry, the reuse service instance exists in the same space as your application.
  For more information, see Creating Service Instances in the SAP BTP documentation.
- You know the name of the service instance of the reuse service.

Context

Use service dependencies when your application needs to call a platform reuse service, such as the Destination service or the Job Scheduling service, using its own application token. The developer of your application must design the application to use mutual TLS when calling the reuse service. The reuse service validates the audience claim of that token to confirm the caller is authorized.

You can reference only reuse services that are designed and documented to support the `consumed-services` parameter. SAP BTP platform services such as the Destination service and Job Scheduling service support this pattern.

> **Note**
>
> You configure service dependencies in the Identity service instance parameters at deployment time. The resulting dependency is visible in the administration console under the application's *Dependencies* > *Services* tab, but you can't modify it there.

Procedure

1. In the Identity service instance parameters of your application, configure the `consumed-services` property.

   In the `consumed-services` array, list each reuse service by its service instance name:

```
{
…
  "consumed-services": [
    {
      "service-instance-name": "<name-of-reuse-service-instance>"
    }
  ],
…
}
```

You can list up to 400 services.

> **Note**
>
> If the token is retrieved based on a public client flow (`public-client` is `true`) without client authentication, SAP Cloud Identity Services doesn't add the client IDs of the dependent services to the audience claim.

2. Create or update the Identity service instance, referencing this parameter file.

   The Identity service instance creation validates that each entry in `consumed-services`refers to an existing, SAP Cloud Identity Services-based service instance. If validation fails, the instance creation fails with an error identifying the unresolvable entry.

Results

SAP Cloud Identity Services adds the client IDs of the named reuse services to the audience (`aud`) claim of tokens issued for this application. Your application can then obtain a token and present it when calling the reuse service with mTLS. The reuse service validates the `aud` claim to confirm the caller is a trusted consumer.

Related Information

Reference Information for the Identity Service of SAP BTP [page 2921]

1.7.1.26.3 Configure Shared Authorization Dependencies

Enable a consumer application to receive authorization group information from a provider application as claims in its access token. With this configuration, the access token of the consumer application includes the authorization groups that the authenticated user holds in the provider application.

Prerequisites

- The provider application has application-specific groups of type authorization.
- You know the display name or application ID of the provider application in your SAP Cloud Identity Services tenant.
- The token claim has been configured by mapping the `dependencyGroups` user attribute to a claim name required by the consumer application.
  To check look for `<consumer-claim-name>` mapped to a self-defined attribute in the application configuration of the consumer application. For more information about self-defined attributes, see User Attributes [page 1972].

If there is no mapping or you haven't deployed the application yet, use one of the following methods:

- When the provider application was deployed, an entry was added to the `assertion-attributes` property in the Identity service instance parameters for the provider application.

  ```
  {
    "assertion-attributes": {
      "<consumer-claim-name>": "dependencyGroups"
    }
  }
  ```

  For more information, refer to the documentation of the consumer and provider applications.

- Send a PATCH request to the Applications REST API of your SAP Cloud Identity Services tenant:

  ```
  PATCH ${iasUrl}/Applications/v1/${consumerAppId}
  Content-Type: application/json
  {
    "operations": [
      {
        "op": "add",
        "path":
  "/urn:sap:identity:application:schemas:extension:sci:1.0:Authentication/
  assertionAttributes",
        "value": [
          {
            "assertionAttributeName": "<consumer-claim-name>",
            "userAttributeName": "dependencyGroups"
          }
        ]
      }
    ]
  }
  ```

  > ⚠ Caution
  >
  > If you set assertion-attributes through the Identity service instance parameters or use the `replace` operation when using the REST API, your map fully replaces the default set of five user profile claims (`groups`, `given_name`, `family_name`, `email`, `user_uuid`). To keep these claims, restate them alongside your new entry.

> → Tip
>
> Replace `<consumer-claim-name>` with the claim name that the consumer application reads from the token, for example, **`dependency_groups`**.
>
> For more information, refer to the documentation of the consumer and provider applications.

Context

Use shared authorization dependencies when a consumer application (such as a portal, shell, or assistant) needs to know what a user is authorized to do in a provider application, without calling that application directly. The provider application's application-specific groups for that user are included as a claim in the consumer's access token. The consumer application reads this claim and uses it to tailor its behavior, for example, to show or hide tiles or features.

The dependency is one-directional: only the consumer's tokens carry the claim. No configuration is required on the provider application. You can add up to 20 shared authorization dependencies per consumer application.

The claim appears only in access tokens, not in ID tokens.

Procedure

1. Add the shared authorization dependency in the administration console.
   a. In the administration console for SAP Cloud Identity Services, choose *Applications and Resources* > *Applications*.
   b. Choose the consumer application.
   c. Choose the *Trust* tab.
   d. Under *Application APIs*, choose *Dependencies*.
   e. Choose *Shared Authorizations*.
   f. Choose *Add* and select the provider application.
   g. Save your entries.
2. Assign the application-specific groups of the provider application to the users who require authorizations.

   To determine which groups to assign, refer to the documentation of the consumer and provider applications.

   For more information about application-specific groups, see Groups [page 514].

Results

When an authenticated user obtains an access token for the consumer application, the token includes a claim named `<consumer-claim-name>` that contains a map of the authorization groups that the user holds in the configured provider applications.

Related Information

Groups [page 514]
Claims of SAP Cloud Identity Services OAuth Tokens [page 2744]

1.7.1.26.4 Configure Native-Web Trust Dependencies

Enable a native or mobile application to hand off an authenticated session to a web application without requiring the user to log in again.

Prerequisites

- The native application has the token exchange grant type enabled (`urn:ietf:params:oauth:grant-type:token-exchange`).
- The native application has an active refresh token for the authenticated user.
- You know the name of the target web application in your SAP Cloud Identity Services tenant.

Context

Use native-web trust dependencies when a user is already authenticated in a native or mobile application and needs to open a web application in a browser. Without this configuration, the browser has no active session and prompts the user to log in again. With this configuration, the native application exchanges its refresh token for a short-lived, one-time SSO token. This SSO token is then passed to the `/saml2/idp/sso` endpoint of the SAP Cloud Identity Services tenant, which establishes a web session for the user and redirects the browser to the target application without a login prompt.

An application can have up to 20 native-web trust dependencies.

Procedure

1. In the administration console for SAP Cloud Identity Services, choose *Applications and Resources* > *Applications*.
2. Choose the native application (the originating application running on the device).
3. Choose the *Trust* tab.
4. Under *Application APIs*, choose *Dependencies*.
5. Choose *Native-Web Trust*.
6. Choose *Add* and select the target web application.
7. Save your entries.

Results

The native-web trust dependency is established. When the native application performs a token exchange with `resource=urn:sap:identity:sso`, the issued SSO token includes the application ID of the configured

target web application in its `sso_application_ids` claim and the trusted redirect URIs of that application in its `sso_redirect_uris` claim. The user can move from the native application to the web application in a browser without being prompted to log in.

Next Steps

After the SSO flow completes, the user is redirected to a trusted URI. For OIDC applications, the configured redirect URIs are used. For SAML applications, the SSO endpoints are used. For both application types, the home URL is also included. To add a specific post-SSO redirect destination, configure the home URL of the target application.

> ⓘ Note
>
> The web session is always established with the `/saml2/idp/sso` endpoint, regardless of whether the target application uses SAML or OIDC.

1.7.2 Configuring Tenant Settings

Initially, the tenants are configured to use default settings. This section describes how you as a tenant administrator can make custom tenant configurations.

Prerequisites

To see and configure the tenant the options in the *Tenant Settings* tile, you must be assigned the Manage Tenant Configuration role. For more information, see Edit Administrator Authorizations [page 2309].

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

Tenant SAML 2.0 Configurations [page 2103]
: You as a tenant administrator can view and download the tenant SAML 2.0 metadata. You can also change the name format and update your certificate used by the identity provider to digitally sign the messages for the applications.

Get SAML 2.0 IdP Metadata via Parameter [page 2107]
: Tenant administrator can get the SAML 2.0 metadata via specific parameters.

Rotate Signing Certificates for Applications [page 2109]
: Tenant administrators must replace existing signing certificates with new ones before they expire.

Tenant OpenID Connect (OIDC) Configurations [page 2117]
: You as a tenant administrator can view and configure the tenant OpenID Connect (OIDC) configurations.

Change Tenant Texts Via Administration Console [page 2121]

The change tenant texts option can be used to change the predefined texts and messages for end-user screens available per tenant in Identity Authentication via the administration console.

Configure Master Data Texts Via Administration Console [page 2125]
The master data texts option can be used to configure the predefined master data for each resource in Identity Authentication via the administration console.

Configure Links Section on Sign-In Screen [page 2129]
You can configure links to appear on the sign-in screen of your applications.

Add Instructions Section on Sign-In Screen [page 2133]
You can customize the sign-in screen of the *Horizon* theme with instructions for the user.

Configure X.509 Client Certificates for User Authentication [page 2136]
Tenant administrators can configure X.509 client certificates for user authentication as an alternative to authenticating with a user name and a password.

Enable Users to Generate and Authenticate with Certificates [page 2140]
Allow users to generate and authenticate with certificates.

Configure Tenant Images [page 2142]
You can configure a custom global logo and, or a background image on the forms for sign-in in, registration, upgrade, password update, and account activation for all applications in a tenant. You can also set a favicon for tenant.

Configure Allowed Logon Identifiers [page 2146]
Tenant administrators can choose the allowed logon identifiers for the users.

Configure User Identifier Attributes [page 2149]
Tenant administrators can configure user identifier attributes as required and unique for the tenant.

Configure Trust this browser Option [page 2154]
Tenant administrator can set the number of days for which the users won't get prompted for second-factor authentication, if they sign in from the same browser.

Enable Back-Up Channels to Send Passcode for Deactivation of TOTP Two-Factor Authentication Devices [page 2156]
Tenant administrator can configure back-up channels to send TOTP deactivation passcodes to the user.

Password Recovery Options [page 2158]
Enable users to reset their password via security questions, PIN code, or email link.

Configure Initial Password and Email Link Validity [page 2168]
As a tenant administrator, you can configure the validity of the initial password and link sent to a user in the various application processes.

Configure Session Timeout [page 2170]
As a tenant administrator, you can configure when the session, created at the Identity Authentication tenant, expires.

Configure Trusted Domains [page 2173]
Service providers that delegate authentication to Identity Authentication can protect their applications when using embedded frames, also called overlays, or when allowing user self-registration.

Use Custom Domain in Identity Authentication [page 2175]
Identity Authentication allows you to use a custom domain that is different from the default ones (`<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`) - for example `www.mytenant.com`.

Change a Tenant's Display Name [page 2181]
You can configure the tenant's name from the administration console for SAP Cloud Identity Services.

Configure Default Risk-Based Authentication for All Applications in the Tenant [page 2183]
You can define rules for authentication according to different risk factors and apply actions like *Allow*, *Deny*, and *Two-Factor Authentication* for all applications in a tenant.

Configure Sinch Service in Administration Console [page 2189]
Configure Sinch Service to enable *Phone Verification via SMS* or *SMS Two-Factor Authentication* in the administration console.

Configure RADIUS Server Settings (Beta) [page 2192]
Configure Remote Authentication Dial-In User Service (RADIUS) server settings in the administration console for SAP Cloud Identity Services.

Configure Mail Server for Application Processes [page 2195]
Configure mail server for the emails sent to the end users in the different application processes.

Configure IdP-Initiated SSO [page 2201]
Enable or disable IdP-Initiated SSO via the administration console for SAP Cloud Identity Services.

Sync SSO Cookie Between Domains [page 2207]
Enable the *Sync SSO Cookie Between Domains Settings* option to allow seamless single sign-on (SSO) and single logout (SLO) across the Cloud Identity Services domains - `ondemand.com` and `cloud.sap`.

Send Security Alert Emails [page 2209]
Send security alert emails to end-users or administrators when changes in their accounts are made.

Enable Password Expiration Reminder [page 2211]
Enable password expiration reminder for SAP Cloud Identity Services to ensure the users are aware that a password change is due.

Send System Notifications via Emails [page 2213]
You can configure the administration console to send emails with information about expiring certificates, system notifications, new administrators, and new applications to specific email addresses or to the emails of all administrators.

Configure Default Language for End User Screens [page 2216]
Select the language that the end user screen uses if the language of the browser isn't in the list of supported languages.

Prompt Users to Explicitly Accept Privacy Policy Document [page 2218]
Enable the privacy policy configuration in SAP Cloud Identity Services to prompt users to accept the privacy policy document. This ensures that users explicitly confirm their understanding of document.

Configure P-User Next Index [page 2220]
Set the value for the P-user next index.

Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]
You as a tenant administrator can reuse an existing tenant for configurations and automated subscriptions.

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.2.1 Tenant SAML 2.0 Configurations

You as a tenant administrator can view and download the tenant SAML 2.0 metadata. You can also change the name format and update your certificate used by the identity provider to digitally sign the messages for the applications.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

> **Note**
>
> For tenants created before 10 September 2024, the signing certificate is one and the same for SAML 2.0 and OpenId Connect (OIDC). Changing one of the configurations causes the service to separate the configuration into two signing certificates, one for SAML and one for OIDC.

→ Remember

The signature and digest methods in the XML of the metadata file depend on the signing certificate, which is configured for the identity provider.

→ Remember

It takes 2 minutes for the configuration changes to take place.

To view and download the tenant SAML 2.0 metadata, or to change the name format, or the default certificate, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Single Sign-On*, choose the *SAML 2.0 Configuration* list item.
   The *SAML 2.0 Configuration* page that opens displays the name of the identity provider, its endpoints, and its signing certificate.
4. **Optional:** To download the identity provider's metadata, press the *Download Metadata File* button and choose one of the options.
   - *Default-certificate*
   - *Non-default certificate*
   - *All certificates*
5. **Optional:** To change the name of the identity provider, choose under the *Identity Provider Settings* tab the *Edit* button next to the *Name* field, select the name from the dropdown list, and save your changes.

   The drop-down list offers the following options:

Issuer	Notes
Default Issuer format	https://<tenant ID>.accounts.ondemand.com
Legacy Issuer format	<tenant ID>.accounts.ondemand.com
Common domain	https://<tenant ID>.accounts.cloud.sap
Custom Domain (if configured)	<custom domain host>
Tenants in China region	https:// <tenant ID>.accounts.sapcloud.cn

> **Note**
>
> Tenant ID is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the tenant ID.

> **→ Remember**
>
> Change the name of the identity provider on the service provider side, or the name of the identity provider on the corporate identity provider side, every time you change the name format of the identity provider in the administration console. If you have set trusts with more than one service provider, or corporate identity provider, change the name in every provider. For more information about how to edit the name, see the documentation of the respective service or corporate identity providers.

If the change of the name is successful, the system displays the message `Tenant <name of tenant> updated`.

6. **Optional:** To update your signing certificate, choose under the *Signing Certificates* tab the *+Add* button on the right. You can choose from the following options:

   - *Regenerate the existing certificate with new validity, reusing the same private key* > *Next Step* > *Choose validity from the drop down* > *Next Step* > *Finish*.

     > **Note**
     >
     > The validity period can be 1, 3, 6 months, 1, 3, 5 and 10 years.

   - *Create new a self-signed certificate with a new private key and the same Subject DN* > *Next Step* > *Select key size* > *Choose validity from the drop down* > *Next Step* > *Finish*.

     > **Note**
     >
     > The key size can be 2048, 3072 and 4096. The validity period can be 1, 3, 6 months, 1, 3, 5, or 10 years.

   - *Download your Certificate Signing Request* > *Next Step* > *add Subject DN and select key size and validity from the options* > *Next Step* > *Download CSR*. Use the downloaded .csr file to generate a certificate from the trusted CA. Copy the newly generated certificate, choose *Edit Certificate*, and paste the certificate as text in the *Certificate Information* field.

     > **Note**
     >
     > The signing algorithm can be SHA-256 and SHA-384. The key size can be 2048, 3072 and 4096.

Next Steps

To change the default certificate for the tenant, choose *Edit* > *the new certificate from the list* > *Save*.

> ⚠ Caution
>
> When you change the default certificate for the tenant, you must also update the trust with the service provider. For more information, see Configure SAML 2.0 Service Provider [page 1941].

Task overview: Configuring Tenant Settings [page 2100]

Related Information

Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
Change Tenant Texts Via Administration Console [page 2121]
Configure Master Data Texts Via Administration Console [page 2125]
Configure Links Section on Sign-In Screen [page 2129]
Add Instructions Section on Sign-In Screen [page 2133]
Configure X.509 Client Certificates for User Authentication [page 2136]
Enable Users to Generate and Authenticate with Certificates [page 2140]
Configure Tenant Images [page 2142]
Configure Allowed Logon Identifiers [page 2146]
Configure User Identifier Attributes [page 2149]
Configure Trust this browser Option [page 2154]
Enable Back-Up Channels to Send Passcode for Deactivation of TOTP Two-Factor Authentication Devices [page 2156]
Password Recovery Options [page 2158]
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1.7.2.2 Get SAML 2.0 IdP Metadata via Parameter

Tenant administrator can get the SAML 2.0 metadata via specific parameters.

Prerequisites

The information in this document is valid only for SAML 2.0 applications.

Context

You can get the SAML 2.0 tenant metadata containing the identity provider (IdP) certificate by calling the `https://<tenant ID>.<tenant domain>/saml2/metadata` end point and providing one of the following parameters:

Parameters

Parameter	Value
`sp_name`	The value is the name of the application as it appears in the *Name* field in the *SAML 2.0 Configuration* screen in the administration console. **Note** If you had added a second signing certificate in the administration console tenant settings for a specific service provider (SP), and you chose one of the certificates to be used for that SP, you can retrieve the SAML 2.0 of the Identity Provider (IdP) metadata for that SP, containing the chosen IDP certificate by providing the SP name in the `sp-name` parameter.

Parameter	Value
`certificates`	• `all` - gets both signing certificates configured for the tenant → Remember If you get the metadata with both certificates to update the trust on the identity provider side, information for the second certificate will be used only. • `default` - gets only the default signing certificate configured for the tenant • `non-default` - gets only the non-default signing certificate configured for the tenant

> ⓘ Note
>
> *Tenant ID* is an automatically generated ID by the system.
>
> The domain can be `accounts.ondemand.com` or `accounts.cloud.sap`

> Example
>
> - `https://mytenant.accounts.ondemand.com/saml2/metadata?sp_name=myapplication` or `https://mytenant.accounts.sap.cloud/saml2/metadata?sp_name=myapplication`
> - `https://mytenant.accounts.ondemand.com/saml2/metadata?certificates=all` or `https://mytenant.accounts.sap.cloud/saml2/metadata?certificates=all`

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Related Information

Tenant SAML 2.0 Configurations [page 2103]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
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1.7.2.3 Rotate Signing Certificates for Applications

Tenant administrators must replace existing signing certificates with new ones before they expire.

Rotate Signing Certificates for SAML 2.0 Applications [page 2111]
: Tenant administrators must replace existing signing certificates with new ones before they expire. This ensures uninterrupted and secure communication between SAML 2.0 applications (referred to as service providers) and Identity Authentication as the identity provider.

Rotate Signing Certificates for OpenID Connect Applications [page 2113]
: Components of SAP Cloud Identity Services use the digital signature of access tokens to verify their validity. Rotate the signing certificates of access tokens periodically to ensure uninterrupted and secure communication between OpenID Connect (OIDC) applications (referred to as relying parties) and Identity Authentication as the identity provider.

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Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]

Tenant OpenID Connect (OIDC) Configurations [page 2117]
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1.7.2.3.1 Rotate Signing Certificates for SAML 2.0 Applications

Tenant administrators must replace existing signing certificates with new ones before they expire. This ensures uninterrupted and secure communication between SAML 2.0 applications (referred to as service providers) and Identity Authentication as the identity provider.

Context

You have received an email notification that your signing certificate is about to expire. The email is sent 30 days, 14 days, and 3 days before the certificate expires. You need to create a new one and configure the service provider to use it. Proceed as follows:

Procedure

1. Create a new signing certificate.
    1. Sign in to the administration console for SAP Cloud Identity Services.
    2. Navigate to *Applications and Resources* > *Tenant Settings* > *Single Sign-On* > *SAML 2.0 Configuration* > *Signing Certificates*.
2. Choose the *+Add* button on the right. You can choose from the following options:
    - *Regenerate the existing certificate with new validity, reusing the same private key* > *Next Step* > *Choose validity from the drop down* > *Next Step* > *Finish*.

      > ⓘ Note
      >
      > The validity period can be 1, 3, 6 months, 1, 3, 5, or 10 years.

    - *Create new a self-signed certificate with a new private key and the same Subject DN* > *Next Step* > *Select key size* > *Choose validity from the drop down* > *Next Step* > *Finish*.

      > ⓘ Note
      >
      > The key size can be 2048, 3072 and 4096. The validity period can be 3, 5 and 10 years.

    - *Download your Certificate Signing Request* > *Next Step* > *add Subject DN and select key size and validity from the options* > *Next Step* > *Download CSR*. Use the downloaded .csr file to generate a certificate from the trusted CA. Copy the newly generated certificate, choose *Edit Certificate*, and paste the certificate as text in the *Certificate Information* field.

      > ⓘ Note
      >
      > The signing algorithm can be SHA-256 and SHA-384. The key size can be 2048, 3072 and 4096.

3. Configure the service provider to use the new certificate.

> **Note**
>
> In case there are many applications, the recommended approach is to renew the certificate for one application at a time following this procedure, as this will minimize or reduce the potential for customer downtime.

1. Navigate to *Applications and Resources* > *Applications* and select the application for which you want to update the trust configuration.
2. Under *Trust* > *Single Sign-On* > *SAML 2.0 Configuration* > *Identity Provider Certificate* > *disable the slider* > *choose Activate next to the new certificate*.

4. Upload the new certificate to the backend of the trusted application.
   1. Navigate to *Applications and Resources* > *Tenant Settings* > *Single Sign-On* > *SAML 2.0 Configuration*.
   2. Proceed according to your use case below.

Applications	Procedure
Your applications support metadata	1. Download the metadata with this certificate in one of the following ways: • Get the SAML 2.0 tenant metadata containing the identity provider certificate by calling the `https://<tenant ID>.<tenant domain>/saml2/metadata` endpoint and provide the `sp_name` parameter, as described in Get SAML 2.0 IdP Metadata via Parameter [page 2107]. • Download the metadata with the new certificate (which is still a non-default one), as described in Tenant SAML 2.0 Configurations [page 2103] → *Step 4* 2. Upload this metadata to the backend of the application for which you have updated the trust configuration in step 2.
Your applications support certificate file	1. Under *Signing Certificates*, choose the *Download* icon next to the new certificate. 2. Upload this certificate file to the backend of the application for which you have updated the trust configuration in step 2.
Your applications support certificate as text	1. Under *Signing Certificates*, choose the *Display* icon next to the new certificate. 2. Copy and paste the certificate to the backend of the application for which you have updated the trust configuration in step 2.

5. After updating the trust configuration for each application, set the new certificate as the default.

Under *Tenant Settings* > *Single Sign-On* > *SAML 2.0 Configuration* > *Signing Certificates*, select the new certificate as default and save your configuration.

6. Navigate to *Applications and Resources* > *Applications* > *Trust* > *Single Sign-On* > *SAML 2.0 Configuration* > *Identity Provider Certificate* > *enable the slider*.

> **Note**
>
> It takes approximately 2 minutes for the new certificate to appear under the trust configuration of the given application.

1.7.2.3.2 Rotate Signing Certificates for OpenID Connect Applications

Components of SAP Cloud Identity Services use the digital signature of access tokens to verify their validity. Rotate the signing certificates of access tokens periodically to ensure uninterrupted and secure communication between OpenID Connect (OIDC) applications (referred to as relying parties) and Identity Authentication as the identity provider.

Prerequisites

Plan for waiting time in hours between the steps of this procedure.

After you activate a new certificate, wait for access tokens signed by the old certificate to expire before deleting the old certificate. The default lifetime of access tokens is 60 minutes.

For more information, see Token Policy Configuration for Applications [page 2752].

Context

Signing certificates exist at the tenant level and are shared by all applications by default. You can also configure application-specific signing certificates. Shared certificates are easier to consume. However, if you need to rotate certificates, you can't delete the old certificate until **all** applications sharing that certificate are configured to use the new certificate.

For more information, see Configure OpenID Connect (OIDC) Application [page 1949].

SAP Cloud Identity Services sends an email notification when your signing certificate is about to expire. The system sends the email 30 days, 14 days, and 3 days before the certificate expires. Create a new certificate and configure the relying party to use it.

> **Note**
>
> It takes approximately two minutes for the new certificate to appear under the trust configuration of the given application.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Navigate to *Applications and Resources* > *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.
3. Check that there's space for a new signing certificate.

   You can store five signing certificates.

   Delete an inactive certificate if you already have five.

   > ⚠ Caution
   >
   > Wait for any existing tokens signed by the inactive signing certificate to expire.
   >
   > By default, access tokens expire after 60 minutes and refresh tokens expire after 12 hours.
   >
   > As long as the old signing certificate exists, the system accepts digital signatures signed by that certificate. After all access tokens and refresh tokens signed by the old certificate expire, you can delete the old certificate. Access tokens have a maximum lifetime of 12 hours. Refresh tokens have a maximum lifetime of 6 months.

4. Add a new signing certificate.

   Choose the *Add* button on the right. You can choose from the following options:

Option	Description
Regenerate	1. Regenerate the existing certificate with new validity, reusing the same private key. 2. Use the wizard to set the validity period and choose *Finish*. → Recommendation Choose a validity period of 3 years.
Create new	1. Create a new self-signed certificate with a new private key and the same Subject Distinguished Name (Subject DN). 2. Use the wizard to set the validity period and key size. Choose *Finish*. → Recommendation Choose a validity period of 3 years. Choose a key size of at least 3072.
Download CSR	1. Download your Certificate Signing Request (CSR). 2. Use the wizard to set the Subject Distinguished Name (Subject DN), signing algorithm, and key size.

Option	Description
	→ Recommendation Choose a key size of at least 3072. 3. Choose *Download CSR*. 4. Use the downloaded `.csr` file to generate a certificate from the trusted CA. 5. Copy the newly generated certificate, choose *Edit Certificate*, and paste the certificate as text in the *Certificate Information* field.

5. Configure your applications that use tenant-level certificates to use the new certificate.

> ⓘ Note
>
> Configuring applications one at a time reduces the potential for downtime.

1. Under *Applications and Resources*, choose the *Applications* tile.
2. Choose the application that you want to edit.

> ⓘ Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

3. Under *Trust* > *Single Sign-On* > *OpenID Connect Configuration* > *Identity Provider Certificates*, select the new certificate in the *Active* column.

6. Upload the new certificate to the backend of the trusted application.

Proceed according to your use case:

Application Backend Supports	Procedure
The discovery endpoint	Refresh or point your application backend to the discovery endpoint: `https://<Cloud Identity Services domain>/.well-known/openid-configuration`
Certificate files	1. Navigate to *Applications and Resources* > *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration*. 2. Under *Signing Certificates*, choose the *Download* icon next to the new certificate.

Application Backend Supports	Procedure
	3. Upload this certificate file to the backend of the application for which you updated the trust configuration in step 5.
Certificates as text	1. Navigate to *Applications and Resources* > *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration*. 2. Under *Signing Certificates*, choose the *Display* icon next to the new certificate. 3. Copy and paste the certificate to the backend of the application for which you updated the trust configuration in step 5.

7. After you update the trust configuration for each application that uses tenant-level certificates, set the new certificate as the default.

   Under *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*, select the new certificate as the default and save your configuration.

   Any new applications use the new signing certificate by default.
8. Wait for any existing tokens signed by the old signing certificate to expire.

   By default, access tokens expire after 60 minutes and refresh tokens expire after 12 hours.

   As long as the old signing certificate exists, the system accepts digital signatures signed by that certificate. After all access tokens and refresh tokens signed by the old certificate expire, you can delete the old certificate. Access tokens have a maximum lifetime of 12 hours. Refresh tokens have a maximum lifetime of 6 months.
9. Deactivate the old certificate.

   Under *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*, choose ⊗ *Deactivate Certificate*.

   The system no longer accepts tokens signed by this certificate. However, the certificate is not deleted. If some applications stop working because of the deactivated certificate, you can reactivate it.
10. Delete the old signing certificate.

    Under *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*, choose 🗑 *Delete*.

1.7.2.4 Tenant OpenID Connect (OIDC) Configurations

You as a tenant administrator can view and configure the tenant OpenID Connect (OIDC) configurations.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

You can change the name format, the certificate used by the identity provider to digitally sign the messages for the applications, set the token policy by configuring the validity of the refresh token, access and id_token, and the maximum sessions per user, and extend the standard metadata with custom values.

> ⓘ Note
>
> For tenants created before 10 September 2024, the signing certificate is one and the same for SAML 2.0 and OpenId Connect (OIDC). Changing one of the configurations causes the service to separate the configuration into two signing certificates, one for SAML and one for OpenID Connect (OIDC).

The *OpenID Connect (OIDC) Configuration* view in the administration console also shows information about the URLs of the *Domain for Browser Flows*, *Authorization Endpoint*, *Token Endpoint*, *UserInfo Endpoint*, and *Logout Endpoint*.

The token policy for the tenant is defined by configuring the validity of the refresh token, access and id_token, and the maximum sessions per user. It's valid for all the applications in the tenant.

> → Remember
>
> If you want a specific token policy per application, you must set a custom token policy for each application. If you modify the settings on the tenant level, all applications without their own custom token policy are affected. For more information, see Token Policy Configuration for Applications [page 2752]

The following table lists the token policy options for OpenID Connect (OIDC) applications.

Token Policy Configuration Parameters

Parameter	Description

Refresh Token	Sets the refresh token lifetime issued by Identity Authentication. The value can range from 1 to 4320 hours, in other words, from 1 hour to 180 days.
	
	The default value is 12 hours.
	
	Note
	• If the validity of refresh tokens is less than the validity of access/ID tokens, access/ID tokens can't be refreshed after the access/ID tokens expire.
	• When using the authorization code flow, if you set the token policy for refresh tokens longer than the session timeout, add the *offline_access* scope to your authorization code request. Without this scope, the service deletes the refresh token from the database when the resource owner ends the session (logs out). Without the refresh token, the OAuth client can't request new tokens anymore. For more information, see:
	  • Configure Session Timeout [page 2170]
	  • Using Authorization Code Flow [page 2638]

Access / ID Token	Sets the access and id_token lifetime issued by Identity Authentication. The value can range from 1 to 720 minutes, in other words, from 1 minute to 12 hours. The default value is 60 minutes.
Max sessions per user	Determines the maximum number of tokens that the service issues for the same session in parallel. Imagine you're logged on to the application through a web interface and a command-line interface in parallel. Then you'd set this parameter to 2. The value can range from 1 to 10. The default value is 1.

The maximum wait time for front-channel logout can be set between 1 and 10 seconds. The default value is 3 seconds.

→ Remember

It takes 2 minutes for the configuration changes to take place.

To view or change the tenant OpenID Connect (OIDC) configurations, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Single Sign-On*, choose the *OpenID Connect (OIDC) Configuration* list item.
   The *OpenID Connect (OIDC) Configuration* page that opens displays the issuer of the identity provider, its endpoints derived from issuer and domain configurations, signing certificate and token policy.

   > ### Note
   >
   > By default, applications use the domain from their trust configuration to access all the endpoints. For browser-based logons and logouts, you can choose another domain. Applications use the chosen domain after reloading the OpenID Connect (OIDC) metadata.
4. **Optional:** To define the token policy, use the slider or provide a number in the input field above the slider. If needed, use the reset button to set to the default value.
5. **Optional:** To change the issuer of the identity provider, choose the *URL* field, select from the dropdown list, and save your changes.

   > ### Caution
   >
   > OpenID Connect (OIDC) applications use the issuer to identify this tenant. Changing the issuer might cause logon issues for existing applications.

   > ### → Remember
   >
   > Change the issuer of the identity provider on the relying party, or the issuer of the identity provider on the corporate identity provider side, every time you change the *URL* format of the issuer in the administration console. If you have set trusts with more than one relying party, or corporate identity provider, change the issuer in every provider. For more information about how to edit the issuer, see the documentation of the respective relying party or corporate identity provider.
6. **Optional:** Update your signing certificate. You can choose from the following options:

   - To regenerate the existing certificate with new validity, reusing the same private key, choose *Add* > *Regenerate* > *Next Step* > *choose validity from the drop down* > *Next Step* > *Finish* > *Save*.
   - To create a new certificate with a new private key and the same Subject DN, choose *Add* > *Create new* > *Next Step* > *choose key size, signing algorithm, and validity from the drop downs* > *Next Step* > *Finish* > *Save*.
   - To create a new certificate, using your own trusted CA, choose *Add* > *Download CSR* > *Next Step* > *add Subject DN, signing algorithm, and key size from the drop downs* > *Next Step* > *Download CSR*. Use the .csr file to generate a certificate from the trusted CA. Copy the newly generated certificate, choose *Edit Certificate*, and paste the certificate as text in the *Certificate Information* field.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113] and Generate Elliptic Curve Signing Certificates [page 2593].

7. **Optional:** Set the maximum wait time for front-channel logout.

Next Steps

To change the default certificate for the tenant, choose the new one from the list, and save your configuration.

> ⚠ Caution
>
> When you change the default certificate for the tenant, you must also update the trust with the service provider. For more information, see Configure OpenID Connect (OIDC) Application [page 1949] or Configure OpenID Connect (OIDC) Application [page 1949].

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Related Information

Tenant SAML 2.0 Configurations [page 2103]
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1.7.2.5 Change Tenant Texts Via Administration Console

The change tenant texts option can be used to change the predefined texts and messages for end-user screens available per tenant in Identity Authentication via the administration console.

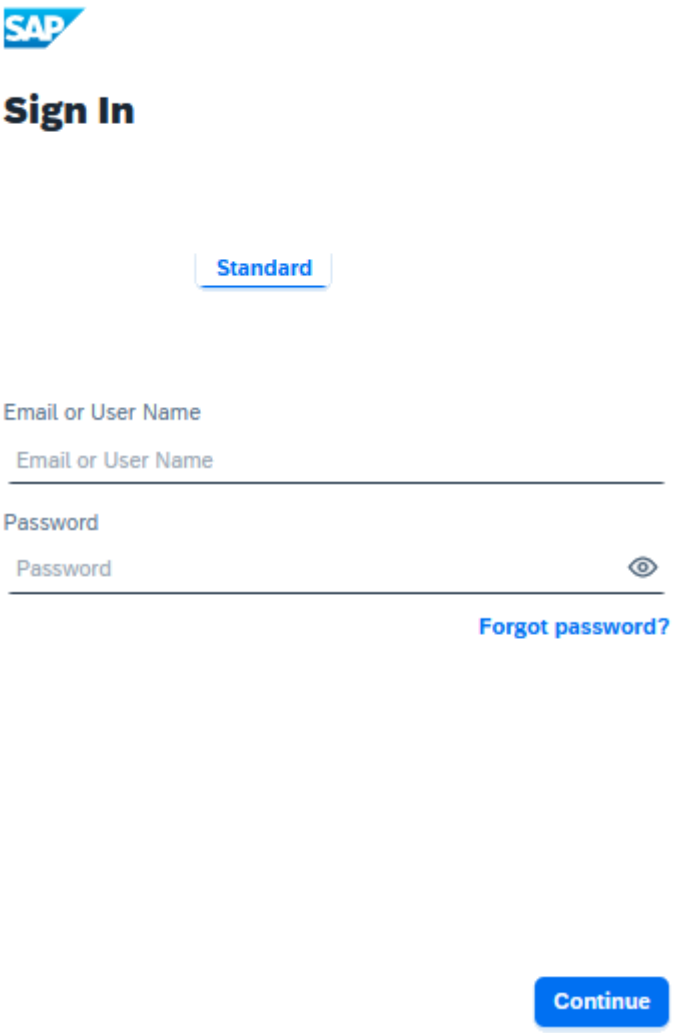
Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

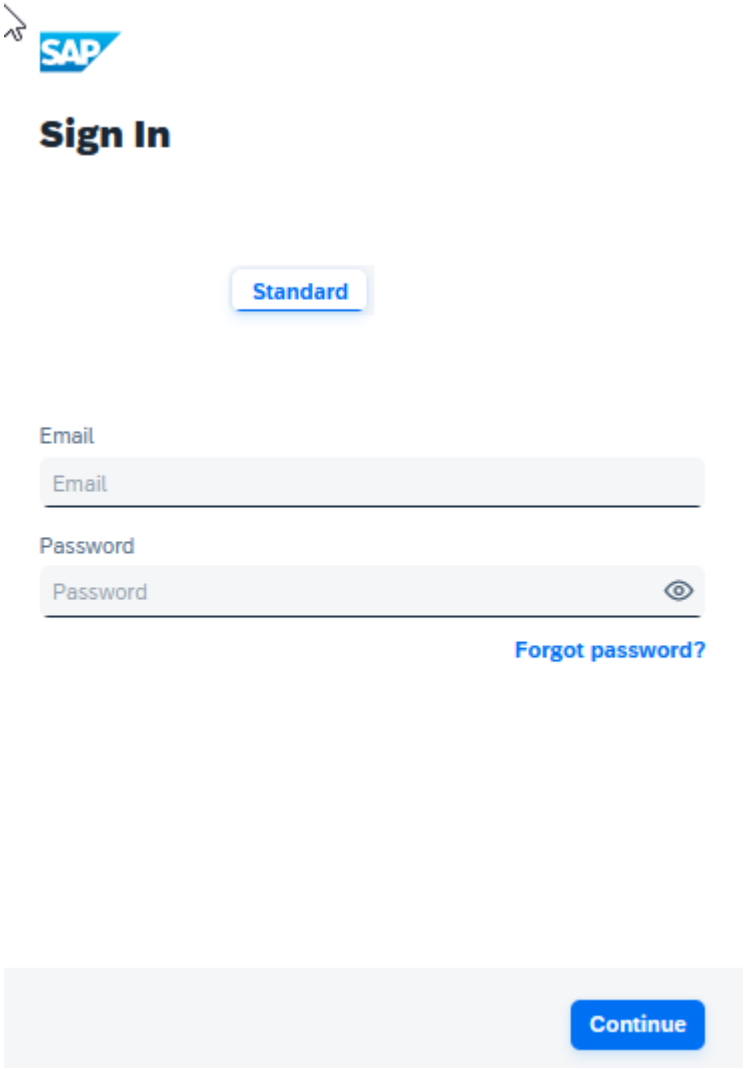
Context

The predefined tenant texts are listed in the administration console for SAP Cloud Identity Services. The texts are classified by category, key, and default value. Each key stores the name of a parameter, and the corresponding value is the text that can be changed and updated as custom value. The keys are self-explanatory and indicate the category and where the texts are used. For example, the

`logon.ui.label.user=Email or User Name` key value pair is for the type of information that the user must provide to sign in to the application. On the *Sign In* screen you see:



If you want to change the user identifier text, you have to update the key value pair `logon.ui.label.user=Email or User Name` to `logon.ui.label.user=Email` to receive:



→ Tip

For more information about the allowed HTML elements in the tenant texts, see Change Tenant Texts REST API [page 2826].

You can browse for specific tenant texts and update only them.

You can also update the tenant texts that you want via a CSV file import for a specific language. The CSV file must contain the following columns: *category*,*key*,*defaultValue*,*customValue*.

You can use again the default tenant texts by simply resetting the custom texts. The *Reset to Default* operation resets all customized tenant texts to the default values.

→ Remember

It takes 2 minutes for the configuration changes to take place.

To change the tenant texts via the administration console, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Texts* tile.
3. Choose the language from the drop-down list.

   The default setting is English (United States).
4. Find the text that you want to change.

   You can use the search field, or the filtering and sorting option, or you can expand the page for more results.
5. Choose the *Edit* button.
6. Add the new text in the *Custom Value* field next to the text that you want to change. Repeat the step for all the texts that you want to change.

   > → Tip
   >
   > You can check the allowed elements at Tenant Texts Allowed HTML Elements [page 2857].
7. Save your changes.

   If the operation is successful, you receive the `Tenant Texts updated` message.

Results

The default tenant texts are changed with the custom ones. It may take up to 2 minutes for the changes to be applied.

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1.7.2.6 Configure Master Data Texts Via Administration Console

The master data texts option can be used to configure the predefined master data for each resource in Identity Authentication via the administration console.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

The predefined master data represents records in Identity Authentication that contain all relevant system data about a resource (Salutations, Functions, Departments, Company Relationships, Industries, Languages, Countries). That data can be used by the system for different classifications in the organization, for example, job titles, departments, or countries. The predefined master data texts are stored in `properties` documents, which can be accessed from the links in the table below.

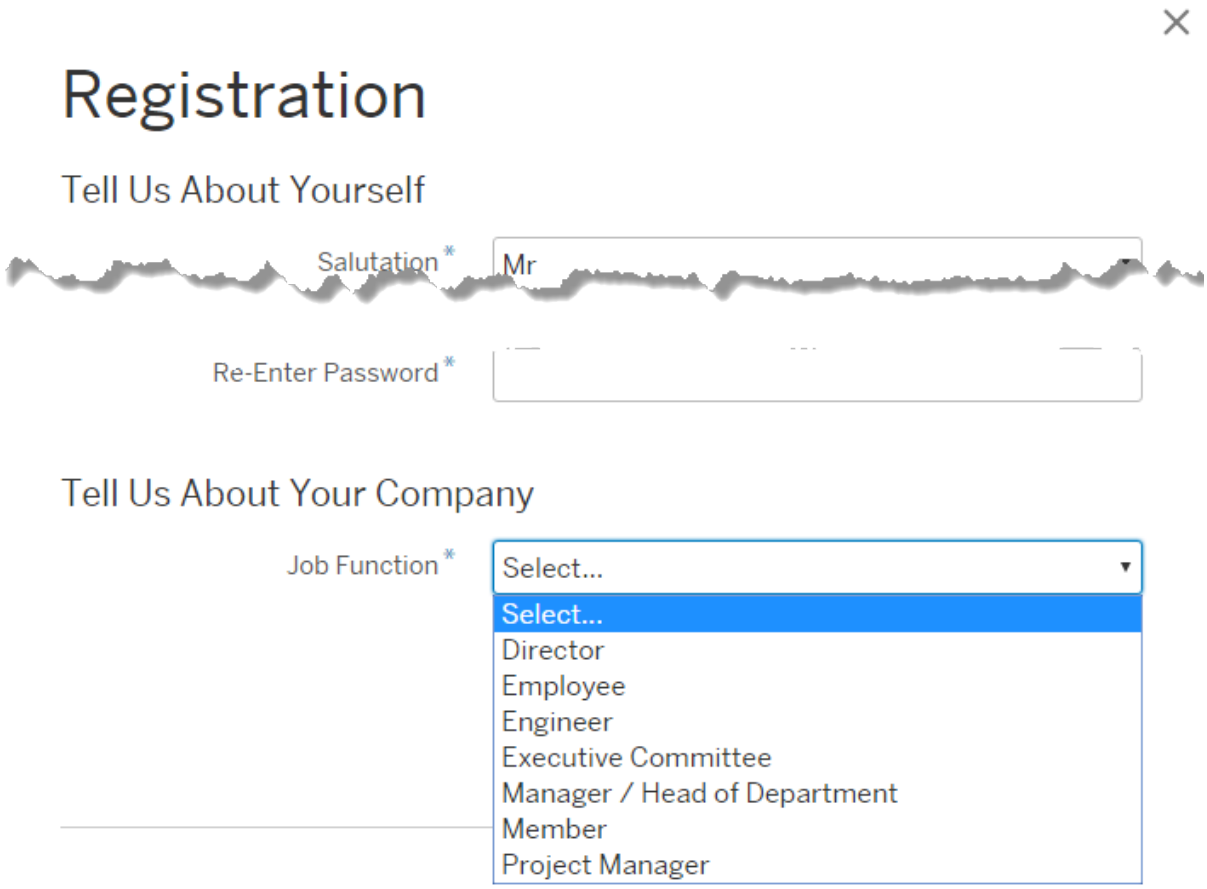
Predefined Master Data Texts

Resource	Link	Notes
Salutations	Salutations.properties [page 2868]	Key value pairs with a predefined set of honorifics.
Functions	Functions.properties [page 2868]	Key value pairs with a predefined set of job titles.
Departments	Departments.properties [page 2868]	Key value pairs with a predefined set of departments.
Company Relationships	Relationships.properties [page 2869]	Key value pairs with a predefined set of business entities, such as customer, partner, employee.
Industries	Industries.properties [page 2870]	Key value pairs with a predefined set of industries.
Languages	Languages.properties [page 2870]	Key value pairs with a predefined set of languages.
Countries	Countries.properties [page 2873]	Key value pairs with a predefined set of countries.

The files contain configurable parameters stored as key value pairs of strings. Each key stores the unified key of the resource, and the corresponding value is the unified value of the resource that can be changed and updated.

The key value pairs are also visible in the administration console under the corresponding tabs.

The example below shows the customized values of the *Functions* file. The dropdown list in the *Job Function* field on the *Registration* form shows the new values that have overwritten the predefined texts in the file.



Depending on your requirements, you can review the texts that you have already overwritten in the predefined master data texts, change or delete existing key value pairs, add new key value pairs, or reset the texts to the default key value pairs.

> ⚠ Caution
>
> Be careful if you change the keys when overwriting data texts, because this can result in invalid values. If a user has a certain value assigned, and the key value pair is overwritten with a different key, then the value for that property of the user will be no longer valid and visualized. For example, if you have set "Mr." to a user, and the key-value pair is "01=Mr.". If you overwrite the key value pair with "03=Mr.", "Mr." will not be visualized for that user.
>
> When overwriting data texts, the keys for the different languages must be one and the same. For example, the master data texts for the German locale are overwritten, and the tenant administrator wants to overwrite the texts for the French locale. The keys for the German locale must be used as keys for the French locale.

The master data is sent in the SAML 2.0 assertion or OpenID Connect (OIDC) token with its key. For example:

```
<Attribute Name="salutation">
            <AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:type="xs:string">0002</
AttributeValue>
</Attribute>
```

→ Remember

It takes 2 minutes for the configuration changes to take place.

To change the master data texts via the administration console, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Master Data* tile.
3. Choose the language from the drop-down list.

   The default setting is English (United States).
4. Choose the tab with the resource that you want.
5. Find the text that you want to change.

   You can use the search field, or the sorting option, or you can expand the page for more results.
6. You have the following options:
   - Change existing texts - Choose the *Edit* button, find the text that you want to change, and replace it in the *Value* field with the new text. You can use the search field, or the sorting option, or you can expand the page for more results.
   - Reset to default values- Choose the *Reset to Default* button. This action resets all master data texts from the chosen resource to their key value pairs.
   - Add a new key value pair - Choose the *Add* button and provide key and value.

     ⓘ Note

     You can't enter duplicate keys.
   - Delete key value pairs - Find the key value pair that you want to delete, select the checkbox next to it and choose the *Delete* button.
7. Save your changes.

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Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
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1.7.2.7 Configure Links Section on Sign-In Screen

You can configure links to appear on the sign-in screen of your applications.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

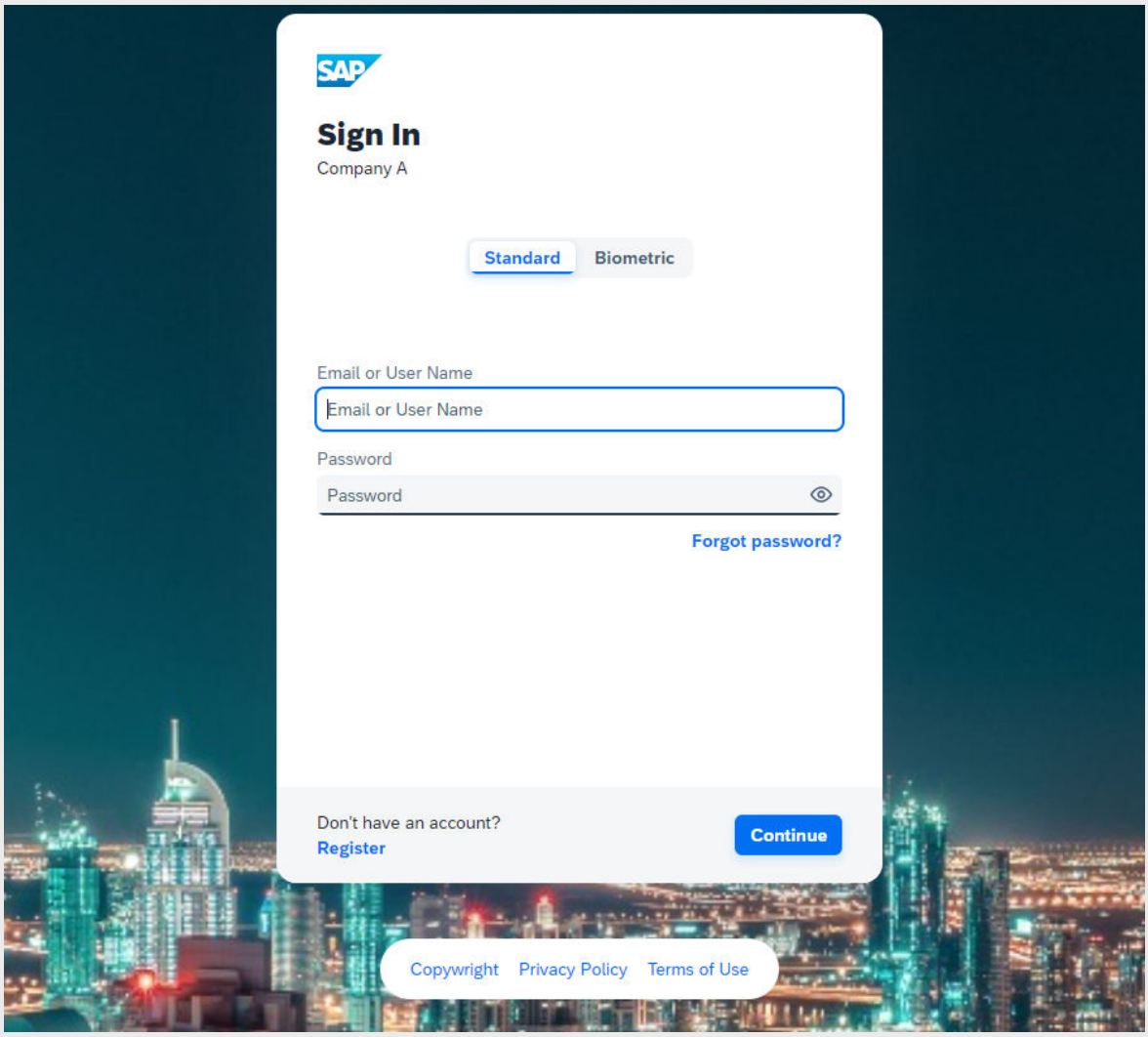
Context

Logon links can be configured via the change tenant texts option in the administration console. Tenant administrator can add three types of links on the logon screen:

- logon.ui.copyright.link - Default Copyright link
- logon.ui.privacy.link - Default Privacy link
- logon.ui.legal.disclosure.link - Default Legal Disclosure link

The default values of the logon link keys are empty and no links are displayed. New values must be in the format `<a href="link_URL">Link Text</a>`, for example `<a href="https://www.example.com" >Privacy Policy</a>`.

Example



The configured links appear on the logon screens of all applications in the tenant.

You can return to the default configuration by resetting the custom texts. The *Reset to Default* operation resets all customized tenant texts to the default values.

> **Note**
>
> If the texts are configured for English (United States) and not configured for another language, and user uses browser with locale that language, the user sees texts customized for English (United States).
>
> > **→ Remember**
> >
> > It takes 2 minutes for the configuration changes to take place.

To configure your logon links, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Texts* tile.
3. Choose the language from the drop-down list.

   The default setting is English (United States).
4. Find the text that you want to change.

   You can use the search field, or the filtering and sorting option, or you can expand the page for more results. The keys for the links are:

Category	Key
Logon	logon.ui.copyright.link
Logon	logon.ui.privacy.link
Logon	logon.ui.legal.disclosure.link

5. Choose *Edit*.
6. Add the new text in the *Custom Value* field next to the text that you want to change.

   > **Note**
   >
   > The value must be in the format `<a href="https:link_URL">Link Text</a>`.
7. Save your changes.

   If the operation is successful, you receive the `Tenant Texts updated` message.

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Get SAML 2.0 IdP Metadata via Parameter [page 2107]

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1.7.2.8 Add Instructions Section on Sign-In Screen

You can customize the sign-in screen of the *Horizon* theme with instructions for the user.

Prerequisites

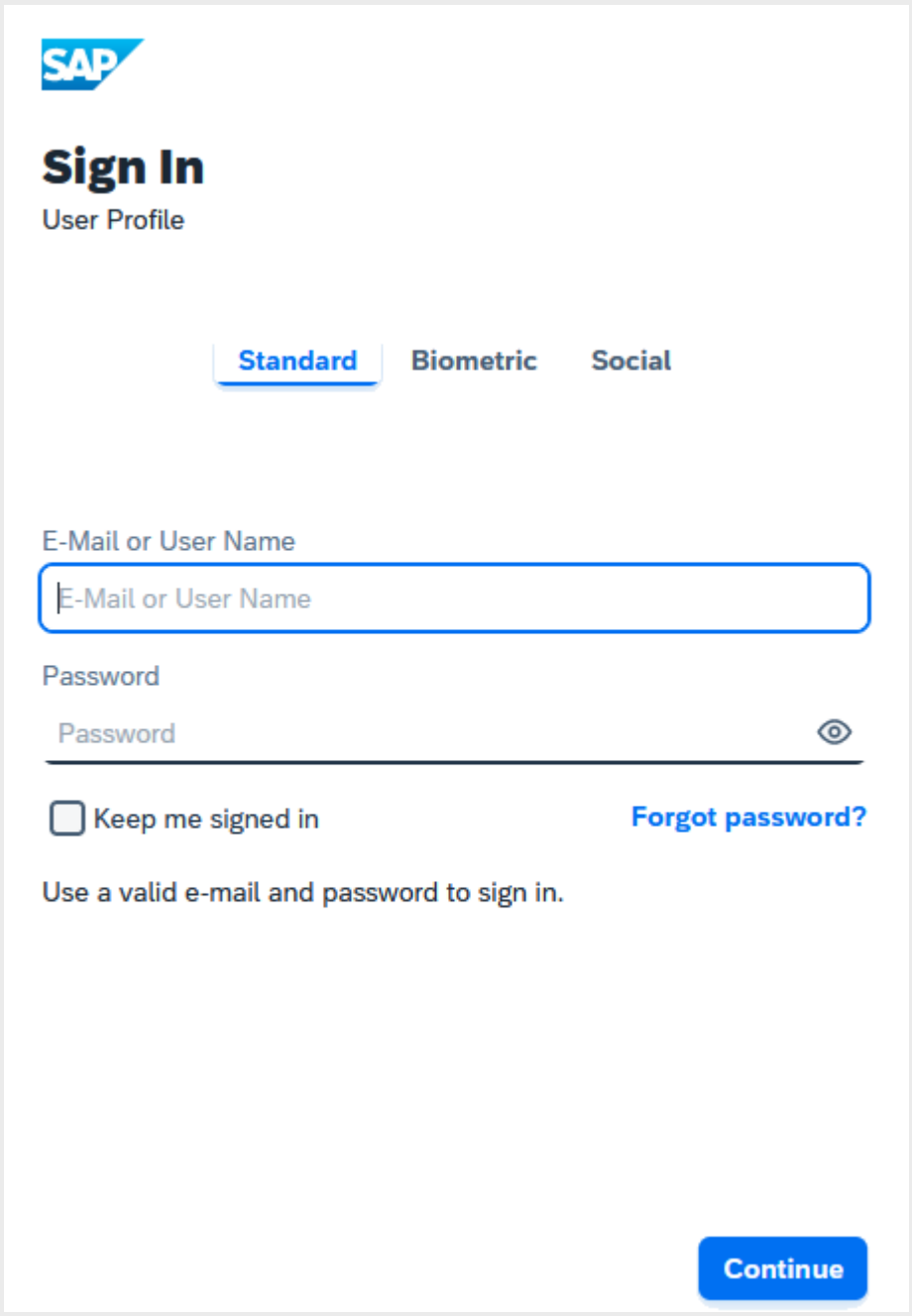
You have chosen the *Horizon* theme for the sign-in screen of your application. For more information, see Choose and Configure a Branding Style for an Application [page 2035].

Context

The section with instructions on the sign-in screen can be configured via the change tenant texts option in the administration console. Tenant administrator can add the section by configuring the `logon.ui.login.instructions` key.

The default value of the `logon.ui.login.instructions` key is empty and nothing is displayed. New values can include text and links.

Example



Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Texts* tile.
3. Choose the language from the drop-down list.

The default setting is English (United States).

4. Find the text that you want to change.

   You can use the search field, or the filtering and sorting option, or you can expand the page for more results. The key for the instruction is: `logon.ui.login.instructions`.

5. Add the new text in the *Custom Value* field.
6. Save your changes.

   If the operation is successful, you receive the `Tenant Texts updated` message.

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Get SAML 2.0 IdP Metadata via Parameter [page 2107]
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1.7.2.9 Configure X.509 Client Certificates for User Authentication

Tenant administrators can configure X.509 client certificates for user authentication as an alternative to authenticating with a user name and a password.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

User authentication with a trusted X.509 certificate takes place using the underlying Secure Sockets Layer (SSL) protocol and users don't need to enter a password for their logon.

Certificates for API Authentication cannot be used for user authentication.

Remember that it may take between two and four weeks to enable the certificate.

> ⓘ Note
>
> If you want to configure a certificate, using your own trusted CA, for example for scenarios like authentication of technical users or OAuth clients, skip the procedure in this document and report an incident on SAP Support Portal Home with a component `BC-IAM-IDS`. Attach to the incident the root and intermediate certificates and provide the Identity Authentication tenant host.
>
> If you already have a trusted CA added to your tenant, but after that a custom domain has been configured for that tenant, report a new incident on SAP Support Portal Home with a component `BC-IAM-IDS` to update the certificate on the custom domain as well.

> ⚠ Caution
>
> If the users have generated their own certificates via the profile page, they won't be able to authenticate with the configured X.509 client certificate, and vice versa.

Edit and Delete Certificates

If you want to edit an already configured certificate, choose certificate, choose the *Edit* button, make your changes and save them.

> → Remember
>
> If you have made changes to the root certificate, report an incident on SAP Support Portal Home with a component `BC-IAM-IDS`, attach to the incident the root and intermediate certificates, and provide the Identity Authentication tenant host.

If you want to delete an already configured certificate, choose the certificate, choose the *Delete* button, and report an incident on SAP Support Portal Home with a component `BC-IAM-IDS` providing the Identity Authentication tenant host.

To configure a trusted X.509 certificate, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Authentication*, choose the *Trusted Certificate Configuration* list item.
4. Choose the *+Add* button.
5. Enter the name of the certificate.

   > ⓘ Note
   >
   > The name and the *Subject DN* must be unique.
6. Choose one of the following options:

Certificate Options	Notes
Upload Certificate	The uploaded certificates must be in **PEM** format. Use `.cer` or `.crt` files.
Root Certificate	Insert the public key in the text field.

7. Choose one of the following source options:
   - *Distinguished Name* - If selected *Distinguished Name* as source, the pattern must match the *Subject DN* of the user certificate. The *CN* attribute from the *DN Pattern* can take the following formats:
     - `CN=${<logonIdentifier>}`. In this case the CN must completely map to one of the supported logon identifiers, `loginName`, `uid`, and `mail`.

       > ✤ Example
       >
       > CN=${loginName},O=Management,C=US.
     - `CN=${mail:<user_email>}`. This case allows you to specify a unique DN pattern for single user authentication.

> **Example**
>
> CN=${mail:dona.moore@example.com},O=Management,C=US.

- *Subject Alternative Name - Other Name* - If selected *Subject Alternative Name - Other Name*, the pattern must match the `subjectAltName` extension entry of type `otherName` (Microsoft User Principal Name form) of the user certificate. The pattern for SAN value must be in format `${<logonIdentifier>}` and must completely map to one of the supported logon identifiers, `loginName`, `uid`, and `mail`.

  > **Example**
  >
  > `${loginName}`

- *Subject Alternative Name - Email (RFC822 Name)* - If selected *Subject Alternative Name - Email (RFC822 Name)*, the pattern must match the `subjectAltName` extension entry of type `rfc822Name` of the user certificate. The pattern for SAN value must be in format `${mail}`.

Two configurations with different source options in one Identity Authentication tenant are not supported.

8. Enter the *Pattern* of the certificate.

> **Note**
>
> If you want to log on with a certificate where the common name contains the user ID (for example: `Subject DN: CN=P000000,O=MyOrg,C=US`) then the pattern value must be: `CN=${uid},O=MyOrg,C=US`.
>
> If you want to log on with a certificate where the common name contains the email of the user (for example: `Subject DN: CN=my_email@example.com,O=MyORG,C=US`), then the pattern value must be: `CN=${mail},O=M,C=US`.

9. Choose the *+Add* button.
10. To add the certificate to your tenant, report an incident on SAP Support Portal Home with a component `BC-IAM-IDS`. The SAP Cloud Root CA certificates are trusted by default.
    a. Attach to the incident the root and intermediate certificates.
    b. Provide the Identity Authentication tenant host.

> **→ Remember**
>
> The following certificates are trusted by default:

CA Display Name	Subject	Serial number	Expiry Date MM-DD-YYYY
SAP Cloud Root CA	CN=SAP Cloud Root CA, O=SAP SE, L=Walldorf, C=DE	18 77 0f be 65 06 6b bf 4c ea 93 38 d9 b1 85 62	02-13-2039
DigiCert Global Root CA	CN=DigiCert Global Root CA, OU=www.digicert.com, O=DigiCert Inc, C=US	08 3b e0 56 90 42 46 b1 a1 75 6a c9 59 91 c7 4a	11-10-2031

CA Display Name	Subject	Serial number	Expiry Date MM-DD-YYYY
DigiCert Global Root G2	CN=DigiCert Global Root G2, OU=www.digicert.com, O=DigiCert Inc, C=US	03 3a f1 e6 a7 11 a9 a0 bb 28 64 b1 1d 09 fa e5	01-15-2038
DigiCert TLS RSA SHA256 2020 CA1	CN=DigiCert Global G2 TLS RSA SHA256 2020 CA1, O=DigiCert Inc, C=US	0c f5 bd 06 2b 56 02 f4 7a b8 50 2c 23 cc f0 66	03-30-2031

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1.7.2.10 Enable Users to Generate and Authenticate with Certificates

Allow users to generate and authenticate with certificates.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

By enabling certificate generation and authentication for the users, they can log on to applications with the generated certificate without the need to provide username and password.

The tenant administrator can configure the system to allow the following types of users to generate their own certificates via the profile page and to authenticate with them:

- *Public*
- *Customer*
- *Partner*
- *Employee*

The certificate generation and authentication is disabled by default for all user types. When you allow it for a user type, the user of this type should go to the profile page and generate a certificate for authentication. Once the certificate is generated, it's downloaded to the user's system. When the user imports it to the certificate store or browser, the newly generated certificate is ready for authentication. When the user navigates to the logon page of the application, he or she is automatically authenticated with the certificate. If the authentication fails, the user is prompted to provide username and password.

> ⚠ Caution
>
> If you have an already configured X.509 client certificate used for user authentication, the users won't be able to authenticate with their own certificates generated via the profile page, and vice versa.

To allow certificate generation and authentication, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Choose the *Certificate Authentication* list item.
4. Use the slider next to a user type to enable certificate authentication.

   If the operation is successful, you receive a confirmation message.

Next Steps

Enable the *Credential change* security alert emailing to inform the user when a certificate generated or removed. For more information, see Send Security Alert Emails [page 2209].

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1.7.2.11 Configure Tenant Images

You can configure a custom global logo and, or a background image on the forms for sign-in in, registration, upgrade, password update, and account activation for all applications in a tenant. You can also set a favicon for tenant.

Prerequisites

- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- (for the background image only) - You use the *Horizon* theme for your application. For more information, see Choose and Configure a Branding Style for an Application [page 2035].

Context

Logo

The logo is displayed in the footer or header of the form that your users access to sign in or to register to an application. If you don't specify a company-specific tenant logo, the forms display the default SAP logo. SAP has configured the following default tenant logo.

You can use an image with a maximum height of 36 pixels.

> ⓘ Note
>
> If the height of the image is larger than 36 pixels, the user forms show the logo proportionally resized to a height of 36 pixels. Thus, the quality will be preserved.
>
> If the height of the image is lower than 36 pixels, the user forms resize the image. Otherwise, the quality is deteriorated.

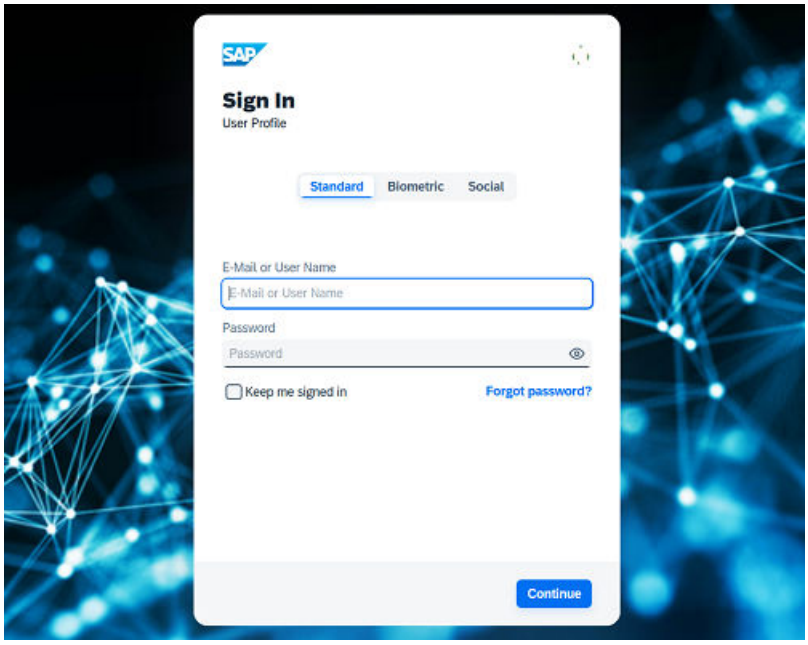
The image file must have one of the following extensions: `.png`, `.gif`, `.jpeg`, or `svg`.

> ⚠ Caution
>
> If the image is of format <svg>, please make sure that the attribute `viewBox = "min-x min-y width height"` is present. For proper rendering the width and height values must not exceed 100. For example, `viewBox="0 0 100 100"`

Background Image

> ⓘ Note
>
> The background image is applied only to applications that use the *Horizon* theme. For more information about the *Horizon* theme, see Choose and Configure a Branding Style for an Application [page 2035].

The background image is displayed behind the form that your users access to sign in or to register to an application.



If you do not specify a background image, the forms won't display anything.

The image file must have one of the following extensions: `.png`, or `.jpg`.

You can use an image with a maximum size of 600 KB (1920x1080 pixels)

> → Remember
>
> The background image option is limited to mobile devices with minimum height of 599 pixels.

Identity Authentication doesn't store old versions the image. Once you upload a new image, the previous image is deleted.

Favicon

Favicon is a shortcut icon in a browser's address bar, for browsers that support them. Browsers that provide favicon support typically display the favicon for a page in the browser's address bar and next to the page title on the tab. Browsers that support a tabbed document interface typically show a favicon for a page next to the page title on the tab. You can upload a file for a custom favicon of Identity Authentication.

The image file must have one of the following extensions: `.png`, or `.jpg` with a maximum size of 100 KB (max 100x100 pixels).

To configure a custom tenant logo a background image, and or a favicon, follow procedure below. It takes up to 2 minutes for the configuration changes to take place.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.

3. Under *Custonization*, choose the *Logo, Background Image and Favicon* list item.
4. **Optional:** Choose the *Logo* tab, upload an image, and save your configuration.

   If the operation is successful, you receive the following message: `Tenant logo updated`.

   > → Tip
   >
   > To restore the default logo, choose *Restore Default*.
   >
   > If this operation is successful, you receive the following message: `Default logo restored`.

5. **Optional:** Choose the *Background Image* tab, upload an image, and save your configuration.

   If the operation is successful, you can see the image applied together with the following message: `Tenant Background Image updated`.

   > → Tip
   >
   > To delete the background image, choose the *Delete* button.
   >
   > If this operation is successful, you receive the following message: `Background image deleted`.

6. **Optional:** Choose the *Favicon* tab, upload an image, and save your configuration.

   If the operation is successful, you can see the image applied together with the following message: Tenant Favicon updated.

Task overview: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
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Configure P-User Next Index [page 2220]
Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]

1.7.2.12 Configure Allowed Logon Identifiers

Tenant administrators can choose the allowed logon identifiers for the users.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

Choose the allowed logon identifier for your users. Once set, the configuration is valid for all applications in the tenant. The options are *User ID*, *Login Name*, *Email*, and *Phone*. *Display Name* is not configurable.

Default Configuration

The default configuration for newly created tenants is the following:

Default Configuration

Logon Identifier	Configuration
User ID	*Off/Configurable*
Login Name	*On/Configurable*

Logon Identifier	Configuration
Email	*On/Configurable* **Note** The email user identifier must be selected unique if you use it for: • logon • sending security alert emails to end-users or administrators when a registration with nonunique email is carried out For more information, see Next Steps at the end.
Display Name	*Off/Not Configurable* **Note** Not allowed for logon.
Phone	*Off/Configurable* **Note** The phone user identifier must be selected unique if you use it for logon. For more information, see Next Steps.

Users can log on only with the enabled logon identifiers.

→ Remember

The logon identifier choice in the administration console does not affect the following scenarios:

- Corporate User Store
- Conditional Authentication
- Forgot Password
- X.509 Client Certificates for User Authentication
- Kerberos Authentication

→ Tip

The texts on the logon screen are predefined. If you change the logon identifiers preference in the tenant, this will not automatically change the text in the logon page. For example if you enable only *Email* as logon identifier, but your previous configuration included *User ID*, *Login Name*, and *Email* the text on the logon screen will remain unchanged.

To change the text, you must update the predefined texts and messages for end-user screens available per tenant in the Identity Authentication. For more information, see Change Tenant Texts REST API [page 2826].

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

If you want to change the allowed logon identifiers for your tenant, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Authentication*, choose the *Logon Alias* list item.
4. Enable or disable the options according to your needs.

   > → Remember
   >
   > You must choose at least one logon identifier.

   If the operation is successful, the system displays the message `Logon alias updated`.

Results

Users can logon to the applications in the tenant only with the selected logon identifiers.

Next Steps

- Change the predefined texts to match the changed logon identifiers. For more information, see Change Tenant Texts REST API [page 2826].
- Configure the user identifier attributes as required and unique. For more information, see Configure User Identifier Attributes [page 2149].

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Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]

Tenant OpenID Connect (OIDC) Configurations [page 2117]
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Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]

1.7.2.13 Configure User Identifier Attributes

Tenant administrators can configure user identifier attributes as required and unique for the tenant.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

Identity Authentication ensures the uniqueness only of the newly set values of the attributes. The system doesn't guarantee the uniqueness of the existing attributes.

The default configuration for the user identifiers are:

User Identifier Default Configuration

User Identifier	Required	Unique
User ID	Yes/Not Configurable	Yes/Not Configurable
Email	Yes/Configurable	Yes/Configurable
Login Name	No/Configurable	Yes/Not Configurable
Display Name	No/Configurable	No/Configurable
Phone	No/Not Configurable	No/Configurable

⚠ Caution

The `User ID` and `Login Name` identifiers of a user can't have values that are equal to the `User ID`, `Email`, `Login Name`, `Display Name`, and `Phone` identifiers of another user.

When `Email` or `Phone` identifiers of a user are set as unique they can't have values that are equal to the `User ID`, `Email`, `Login Name`, `Display Name`, and `Phone` identifiers of another user.

→ Remember

If the `DisplayName` is edited via the profile page, consider the following:

- When the `Display Name` identifier of a user is set as unique it can't have values that are equal to the `User ID`, `Global User ID`, `Email`, `Display Name`, `Login Name`, and `Employee Number` identifiers of another user.
- When the `Display Name` identifier of a user is set as non-unique it can't have values equal to the `User ID`, `Global User ID`, `Email`, `Login Name`, and `Employee Number` identifiers of another user.

ⓘ Note

The `Display Name` user identifier for the tenants created before the system upgrade on May 13, 2020 is configured as required and unique.

The `Phone` user attribute is configured as non-unique by default. If you configure it as unique, all users that are created or updated after this configuration won't be able to have phone numbers taken by someone else.

→ Remember

If `Email` is marked as not-required on tenant level, it becomes configurable on application level, and must also be changed there, too. For more information, see Configure Registration and Upgrade Forms [page 2046].

Email Required/Unique Configurations

Choice	Yes	No
Required	• admin must provide email when creating or editing user in the admin console • `emails.value` attribute is mandatory when creating a user via the SCIM REST API • email appears as required in the registration form of an application • user must provide email, if missing, when an update of the account is triggered • admin can't delete user's email in admin console • user can't delete his/her email in profile page	• admin must provide email when creating user in the admin console only when account activation is *Send activation email* • `emails.value` attribute must be provided when creating a user via the SCIM REST API only when `sendMail` attribute is *true* • email is required in the registration form of the application; the configuration is taken into account for the upgrade process • reset password process will be replaced by change password process for users with no email • user is not prompted to provide email, if missing, when an update of the account is triggered • admin can delete user's email in admin console • user can delete his/her email in profile page • end-user screen texts differ from the actual tenant configuration; admin can change the tenant texts to match the configuration • email verification setting for an application could be skipped for users with no email.

Choice	Yes	No
Unique	• email can be used for logon • email must be unique for security alert emails when a registration with nonunique email is carried out • email must be unique, if provided, when a user is created or edited via the admin console • `emails.value` attribute must be unique, if provided, when a user is created via SCIM REST API • user must provide unique email, if required, when an update of the account is triggered • user import is supported; email must be provided • `email` attribute must be unique, if provided, when a user is registered via User Management REST API **Note** The email user identifier must be selected unique if you use it for logon. For more information about how to configure the allowed logon identifiers, see Next Steps.	• email can't be used for logon • admin can create more than one user with one and the same email in the admin console • `emails.value` attribute may not be unique if provided when a user is created via SCIM REST API • email, if required, may not be unique when an update of the account is triggered • users with non-unique emails can't change their password via the *Forgot Password* process using the email as identifier • end-user screen texts differ from the actual tenant configuration; admin can change the tenant texts to match the configuration • user import is not supported

The texts on the end screen are predefined. If you change the required/unique preference in the tenant, this won't automatically change the texts in the end-user page. To change the text, you must update the predefined texts and messages for end-user screens available per tenant in the Identity Authentication. For more information, see Change Tenant Texts REST API [page 2826].

Although the choice for the `required` attribute is applied for all applications in the tenant, you can still make the *Email* required on the registration and upgrade form for specific applications via a custom configuration. For more information, see Configure Registration and Upgrade Forms [page 2046]

→ Remember

It takes 2 minutes for the configuration changes to take place.

If you want to change the configuration for the user identifier for your tenant, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.

3. Under *Authentication*, choose the *Logon Alias* list item.
4. Select the options for the user identifier according to your needs.
    - Required
    - Unique

    If the operation is successful, the system displays the message `Logon alias updated`. It takes two minutes for the change to be applied.

Next Steps

Choose the allowed logon identifiers for the users. For more information, see Configure Allowed Logon Identifiers [page 2146].

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Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
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Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]

1.7.2.14 Configure *Trust this browser* Option

Tenant administrator can set the number of days for which the users won't get prompted for second-factor authentication, if they sign in from the same browser.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

The validity of *Trust this browser* can be set between 1 and 14 days.

If the option for trust this browser is set at *Disabled*, and a user chooses to always provide second factor when authenticating, the validity of *Trust this browser* is 3 months. But if the administrator sets a validity from 1 to 14 days, the 3-month validity is overridden with the validity set by the tenant administrator.

The *Trust this browser* checkbox appears at sign-in when a second factor is required for the users. When users chose the *Trust this browser* checkbox, the second factor is required only with the first sign-in. The validity depends on the configuration in the administration console.

> **Note**
>
> If the device for time-based one-time password (TOTP) or Web Two-Factor Authentication is deactivated, all the related tokens for the user will be removed.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Authentication*, choose the *Multi-Factor Authentication* list item.
4. Under *Additional Settings for Multi-Factor Authentication*, choose *Edit*.
5. Under *Trusted Browser Setting for Users*, choose *Enabled*.
6. Manually configure the number of days for which users will not get prompted for second-factor authentication, if they sign in from the same browser.
7. Save your changes.

   If the operation is successful, the system displays the message `Additional Multi-Factor Authentication Settings updated`.

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Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
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Configure Risk-Based Authentication for an Application [page 2007]
Configure Default Risk-Based Authentication for All Applications in the Tenant [page 2183]

1.7.2.15 Enable Back-Up Channels to Send Passcode for Deactivation of TOTP Two-Factor Authentication Devices

Tenant administrator can configure back-up channels to send TOTP deactivation passcodes to the user.

Prerequisites

- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- (for the SMS channel) The user must have a verified phone.
- (for the SMS channel) You have configured the Sinch Service option in the administration console for SAP Cloud Identity Services. For more information, see Configure Sinch Service in Administration Console [page 2051].
- (for the email channel) The user must have a verified email address.
- (for the email channel) You have configured an email template for the *Deactivate TOTP Device* process. For more information, see Configuring Email Templates [page 2282].

> → Remember
>
> The *${other.totpResetPasscode}* parameter is mandatory for the *Deactivate TOTP Device* template.

Context

The primary channel to deactivate a device is the passcode generated by the mobile device. If no other back-up channel is enabled, this option is the only one for the user in the *Two-Factor Authentication* section of the administration console for SAP Cloud Identity Services.

When the back-up channels, *SMS* and/or *Email* are enabled in the administration console, the user can choose from the options on the profile page: *Existing Multi-Factor Authentication*, *Passcode by SMS*, and *Passcode by Email*.

> → Tip
>
> We recommend you to enable the back-up channels. Thus the users can use the option as an alternative when they don't have access to the TOTP device or application.

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

To enable or disable back-up channels for deactivation passcode, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Authentication*, choose the *Multi-Factor Authentication* list item.
4. Under *TOTP Deactivation Passcode Channels Configuration*, enable or disable the options according to your needs.
   - *SMS*
   - *Email*

   If the operation is successful, the system displays the message `TOTP Deactivation Options updated`.

Results

Users can choose from the back-up channel options on their profile page if they are enabled via the administration console.

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Related Information

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Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
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1.7.2.16 Password Recovery Options

Enable users to reset their password via security questions, PIN code, or email link.

Enable Users to Recover Password with Security Questions [page 2160]
Users can choose to answer security questions to reset their password.

Enable Users to Recover Password with PIN Code [page 2162]

Users can choose to provide PIN code to reset their password.

Enable Users to Recover Password via Email [page 2164]
Users can choose to receive a link via email to reset their password.

Enable Users to Reset Password via Profile Page [page 2166]
Enable the users to trigger reset password process via the profile page.

Parent topic: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]
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1.7.2.16.1 Enable Users to Recover Password with Security Questions

Users can choose to answer security questions to reset their password.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

You can configure the applications in the tenant to allow users to answer security questions to reset their password instead of receiving an email with a reset password link.

For this setup, you as a tenant administrator, configure the security questions option in the *Tenant Settings* section in the administration console, and the users configure their answers in the profile page. Once the security questions and answers are configured, the user can use this option to reset the password. The user must have configured their answers in the profile page to be able to reset password via security questions.

> ⚠ Caution
>
> The answers to the security questions are case sensitive.

If the security questions option is configured in the admin console, end users that haven't set up their security questions are triggered to do it as a post logon step. Optionally, they can choose to set up the security questions later. If end users choose the *Don't ask me again* check box, the prompt won't be shown anymore. End users still can set up the security questions in their profile page.

> ⓘ Note
>
> You can choose whether the *Setup later* and *Don't ask me again* options are visible for the end users.

Tenant administrator can offer up to 10 predefined questions, and can choose which target users to be able to reset their password via security questions. There are three options:

- *None* - the security questions option isn't offered to the users and all security questions configurations in the administration console are disabled
- *Users without email* - only users that don't have emails can reset password via security questions.
- *All users* - everyone can choose this option to reset the password

- *Specific groups* - only users that belong to the selected groups can reset password via security questions.

If *Home URL* is configured, users are redirected to the URL defined in the *Home URL* after resetting their passwords via Security Questions. If *Home URL* is not configured users are redirected to the profile page.

> ⚠ Caution
>
> The security questions option locks after five wrong answers to the security questions. To unlock it, set an initial password for the user. For more information, see Set Initial Password [page 2384].

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

To configure security questions in the administration console, follow the procedure:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Authentication*, choose the *Password Recovery* list item.
4. Choose the *Security Questions* tab.
5. Choose *Edit* > *Enabled* .
6. Under *Target Users*, choose users that can reset passwords with security questions:
   - *None* - default choice

     > ⓘ Note
     >
     > If selected, the security questions configurations in the administration console are disabled
   - *Users without email*
   - *All users*
   - *Specific groups*

     > ⚠ Restriction
     >
     > The administrator must be assigned the *Manage Groups* role or be part of a policy that allows reading groups to be able to view the groups in the list. For more information, see Edit Administrator Authorizations [page 2309] or Configure Group Authorizations [page 2320].

     > ⓘ Note
     >
     > When you select this option, you must specify the specific group or groups for which you enable password recovery via security questions.
7. Under *Number of Security Questions*, choose how many questions the user must answer.

8. **Optional:** Under *User Options*, select the *Show "Setup later"* check box.

   When selected, the end users see the option to setup the security questions later.
9. **Optional:** Under *User Options*, select the *Show "Don't ask again"* check box.

   When selected, the end users see the option to hide the prompt to setup their security questions later.
10. Save your configuration.

    You can see information about your settings and the 10 predefined questions the user can choose from..

Task overview: Password Recovery Options [page 2158]

Related Information

Enable Users to Recover Password with PIN Code [page 2162]
Enable Users to Recover Password via Email [page 2164]
Enable Users to Reset Password via Profile Page [page 2166]

1.7.2.16.2 Enable Users to Recover Password with PIN Code

Users can choose to provide PIN code to reset their password.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

> ⚠ Caution
>
> For security reasons, the PIN Code is not a recommended password recovery method. The users can follow the *Forgot Password* link on the sign in page to reset their password via email.

You can configure the applications in the tenant to allow users to provide PIN code to reset their password instead of receiving an email with a reset password link.

For this setup, you as a tenant administrator, configure the PIN code option in the *Tenant Settings* section in the administration console, and the users set their PIN code in the profile page.

Once the PIN code is configured, the user can use this option to reset the password. The user must have set their PIN code in the profile page to be able to reset password via this option.

If the PIN code option is configured in the admin console, end users that haven't set up their PIN code are triggered to do it as a post logon step. Optionally, they can choose to set up the PIN code later.

If end users choose the *Don't ask me again* check box, the prompt won't be shown anymore. End users still can set up the PIN code in their profile page.

PIN Code Characteristics

Characteristics	Notes
Allowed Characters	Base 10 digits (0-9)
Min Length	Between 4 and 32 characters; configurable on tenant level
Max Length	32 characters
Required	*Setup later* and *Don't ask me again* are configurable
Target Users	• *Users without email* - only users that don't have emails are able to reset password via PIN code • *All users* - everyone can choose this option to reset the password • *Specific groups* - only users that belong to the selected groups can reset password via PIN code.
Lock	The reset password action with PIN code locks after 5 failed attempts. The logon with correct username and password is not interrupted.
Unlock	To unlock the PIN code, set an initial password for the user.

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

To configure PIN code option in the administration console, follow the procedure:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Authentication*, choose the *Password Recovery* list item.
4. Select the *PIN Code* tab.
5. Choose *Edit* > *Enabled* .
6. Under *Target Users*, choose users that can reset passwords with PIN code:
   - *Users without email*
   - *All users* - default choice
   - *Specific groups*

> ⚠ Restriction
>
> The administrator must be assigned the *Manage Groups* role or be part of a policy that allows reading groups to be able to view the groups in the list. For more information, see Edit Administrator Authorizations [page 2309] or Configure Group Authorizations [page 2320].

> ⓘ Note
>
> When you select this option, you must specify the specific group or groups for which you enable password recovery via PIN code.

7. Under *PIN Length*, choose the minimum required length of the PIN code.
8. **Optional:** Under *User Options*, select the *Show "Setup later"* check box.

   When selected, the end users see the option to set up the PIN code later.
9. **Optional:** Under *User Options*, select the *Show "Don't ask again"* check box.

   When selected, the end users see the option to hide the prompt to set up their PIN code.
10. Save your configuration.

    You can see information about your settings.

Task overview: Password Recovery Options [page 2158]

Related Information

Enable Users to Recover Password with Security Questions [page 2160]
Enable Users to Recover Password via Email [page 2164]
Enable Users to Reset Password via Profile Page [page 2166]

1.7.2.16.3 Enable Users to Recover Password via Email

Users can choose to receive a link via email to reset their password.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

The *Email* option in the administration console is enabled by default. If the *Email* option is enabled and *Profile Page* option is disabled, users can choose to receive a link via email to reset their password when they click the *Forgot password* link when they sign-in.

When the *Email* option is enabled and *Profile Page* option is disabled, users can also change their password directly in the profile page. When accessing the profile page, they see the *Change* button. By choosing *Change*, the user needs to provide the current and the new passwords to the system.

> → Remember
>
> If you disable the *Email* option, at least one of the other two password recovery options must be enabled for a user. Before you disable the email, view your other password recovery settings and select **`All users`** for *Target users*. For more information, see Enable Users to Recover Password with Security Questions [page 2160] or Enable Users to Recover Password with PIN Code [page 2162].

> ⚠ Caution
>
> Be careful also that if you disable the *Email* option, when the *Profile Page* option is enabled, both options will be disabled. For more information, see Enable Users to Reset Password via Profile Page [page 2166].

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

To enable the *Email* option in the administration console, follow the procedure:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Authentication*, choose the *Password Recovery* list item.
4. Select the *Email* tab.
5. Choose *Edit* and select *Enabled*.
6. Save your configuration.

   You can see information about your settings.

Results

Users can choose to receive a link via email to reset their password when click the *Forgot password* link on the sign-in page.

> ⚠ Caution
>
> Be careful if you disable the *Email* option. Disabling the *Email* option when the *Profile Page* option is enabled will disable both options. For more information, see Enable Users to Reset Password via Profile Page [page 2166].

Task overview: Password Recovery Options [page 2158]

Related Information

Enable Users to Recover Password with Security Questions [page 2160]
Enable Users to Recover Password with PIN Code [page 2162]
Enable Users to Reset Password via Profile Page [page 2166]

1.7.2.16.4 Enable Users to Reset Password via Profile Page

Enable the users to trigger reset password process via the profile page.

Prerequisites

- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have enabled the *Email* option for password recovery in the administration console. For more information, see Enable Users to Recover Password via Email [page 2164]

Context

The *Profile Page* option is disabled by default. If the *Email* option is enabled and *Profile Page* option is disabled, users can change their password directly in the profile page. When accessing the profile page, they see the *Change* button. By choosing *Change*, the user needs to provide the current and the new passwords to the system.

Your scenario may require additional confirmation of the password change via email, before the new password is applied. You can do this by enabling the *Profile Page* option, which is visible only when the *Email* option is enabled. When accessing the profile page, users see the *Reset* button. By choosing *Reset*, the user triggers a manual password reset from the profile page. To reset the password the user must have an access to the email.

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

To enable the *Profile Page* option in the administration console, follow the procedure:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Authentication*, choose the *Password Recovery* list item.
4. Select the *Email* tab.

   > → Remember
   >
   > Make sure that the *Email* option is enabled. If not, choose *Edit*, enable it, and save your changes.
5. Next to *Profile Page*, choose *Edit* and select *Enabled*.
6. Save your configuration.

   You can see information about your settings.

Results

When accessing the profile page, users see the *Change* button in the *Password* section.

Task overview: Password Recovery Options [page 2158]

Related Information

Enable Users to Recover Password with Security Questions [page 2160]
Enable Users to Recover Password with PIN Code [page 2162]
Enable Users to Recover Password via Email [page 2164]

1.7.2.17 Configure Initial Password and Email Link Validity

As a tenant administrator, you can configure the validity of the initial password and link sent to a user in the various application processes.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

The tenant administrator can specify how long the link sent to a user in the various application processes will be valid for. The link in the email can be set to expire after between 1 and 23 hours, or 1 and 30 days.

Expired links can't be used. The system automatically sends a new link in an email when a user uses an expired link.

Identity Authentication has predefined the following validity periods:

Default Email Link Validity Periods

Application Process	Default Validity Period
Self-Registration	14 Days
On-Behalf Registration	14 Days
Invitation	14 Days
Forgot Password	2 Hours
Locked Password	2 Hours
Reset Password	2 Hours

The tenant administrator can also set a validity for the initial password. The initial password can be valid between 1 and 365 days depending on the configuration. The default value is 14 days. After the validity of the initial password expires, the user can't log on to the application and must contact the administrator.

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

To change the validity period of the initial password and the links, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Authentication*, choose the *Initial Password and Email Link Validity* list item.
4. Choose *Edit*.
5. **Optional:** (For email link validity) Under *Configure Email Link Validity*, choose an application process and set the validity period of the email link for it.
   a. From the radio buttons on the right, select either *Days* or *Hours*.
   b. From the dropdown list on the left, select a number for this.

   > ⓘ Note
   >
   > You can choose a value between 1 and 23 for *Hours*, and 1 and 30 for *Days*.

   You can repeat the step for all processes.
6. **Optional:** (For initial password validity) Under *Configure Initial Password Validity*, set a value for the validity of the initial password.
7. Save your changes.

Task overview: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
Change Tenant Texts Via Administration Console [page 2121]
Configure Master Data Texts Via Administration Console [page 2125]
Configure Links Section on Sign-In Screen [page 2129]
Add Instructions Section on Sign-In Screen [page 2133]
Configure X.509 Client Certificates for User Authentication [page 2136]
Enable Users to Generate and Authenticate with Certificates [page 2140]
Configure Tenant Images [page 2142]
Configure Allowed Logon Identifiers [page 2146]
Configure User Identifier Attributes [page 2149]
Configure Trust this browser Option [page 2154]

Enable Back-Up Channels to Send Passcode for Deactivation of TOTP Two-Factor Authentication Devices [page 2156]
Password Recovery Options [page 2158]
Configure Session Timeout [page 2170]
Configure Trusted Domains [page 2173]
Use Custom Domain in Identity Authentication [page 2175]
Change a Tenant's Display Name [page 2181]
Configure Default Risk-Based Authentication for All Applications in the Tenant [page 2183]
Configure Sinch Service in Administration Console [page 2189]
Configure RADIUS Server Settings (Beta) [page 2192]
Configure Mail Server for Application Processes [page 2195]
Configure IdP-Initiated SSO [page 2201]
Sync SSO Cookie Between Domains [page 2207]
Send Security Alert Emails [page 2209]
Enable Password Expiration Reminder [page 2211]
Send System Notifications via Emails [page 2213]
Configure Default Language for End User Screens [page 2216]
Prompt Users to Explicitly Accept Privacy Policy Document [page 2218]
Configure P-User Next Index [page 2220]
Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]

1.7.2.18 Configure Session Timeout

As a tenant administrator, you can configure when the session, created at the Identity Authentication tenant, expires.

Prerequisites

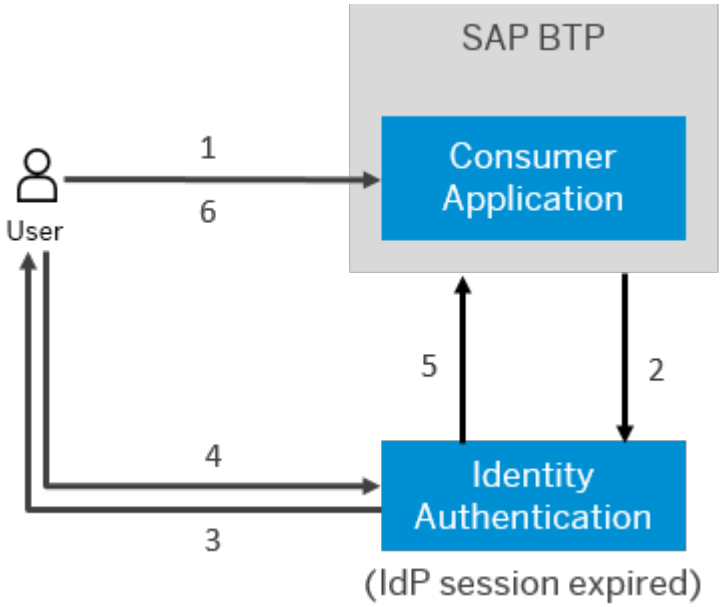
You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

The session timeout feature enables closing of the session at the IdP when the configured period has passed. The configuration determines how long the IdP keeps the session alive.

By default, the timeout session for Identity Authentication is 12 hours.

When the IdP session has expired, and the application session has also expired, or the user tries to access a second application, Identity Authentication forces the user to provide his or her credentials.



1. Tries to access protected resource.
2. Sends request to check if the user has rights to access.
3. Opens Log On screen; forces user to provide credentials.
4. Provides credentials.
5. Grants access.
6. User accesses content.

> ⚠ Caution
>
> For OIDC applications, the Global Session Timeout acts as a hard upper bound on token validity. Access token lifetimes are capped at the remaining session duration, and refresh tokens cannot be issued or renewed beyond the active session, regardless of the token policy configured for the application or tenant. Exchanging a refresh token does not extend the session. If a refresh token is sent with one minute of session lifetime remaining, the resulting token is valid for one minute only.
>
> Configuring short session timeouts (for example, 15–30 minutes) can cause users to reauthenticate more frequently than expected in token-based integrations, even when the application's token policy is set to longer validity periods.

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

To configure the session timeout period via the administration console for SAP Cloud Identity Services follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.

2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Single Sign-On*, choose the *Session Timeout* list item.
4. Set the session timeout period.
   a. From the dropdown list on the right, select either *Minutes* or *Hours*.
   b. From the dropdown list on the left, select a number for your choice.

   > ⓘ Note
   >
   > You can choose a value between *5* and *59* for *Minutes*, and *1* and *12* for *Hours*.
   >
   > The default value is *12* hours.
5. Save your changes.

   If the operation is successful, the system displays the message `Session timeout updated`.

Task overview: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
Change Tenant Texts Via Administration Console [page 2121]
Configure Master Data Texts Via Administration Console [page 2125]
Configure Links Section on Sign-In Screen [page 2129]
Add Instructions Section on Sign-In Screen [page 2133]
Configure X.509 Client Certificates for User Authentication [page 2136]
Enable Users to Generate and Authenticate with Certificates [page 2140]
Configure Tenant Images [page 2142]
Configure Allowed Logon Identifiers [page 2146]
Configure User Identifier Attributes [page 2149]
Configure Trust this browser Option [page 2154]
Enable Back-Up Channels to Send Passcode for Deactivation of TOTP Two-Factor Authentication Devices [page 2156]
Password Recovery Options [page 2158]
Configure Initial Password and Email Link Validity [page 2168]
Configure Trusted Domains [page 2173]
Use Custom Domain in Identity Authentication [page 2175]
Change a Tenant's Display Name [page 2181]
Configure Default Risk-Based Authentication for All Applications in the Tenant [page 2183]
Configure Sinch Service in Administration Console [page 2189]

Configure RADIUS Server Settings (Beta) [page 2192]
Configure Mail Server for Application Processes [page 2195]
Configure IdP-Initiated SSO [page 2201]
Sync SSO Cookie Between Domains [page 2207]
Send Security Alert Emails [page 2209]
Enable Password Expiration Reminder [page 2211]
Send System Notifications via Emails [page 2213]
Configure Default Language for End User Screens [page 2216]
Prompt Users to Explicitly Accept Privacy Policy Document [page 2218]
Configure P-User Next Index [page 2220]
Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]

1.7.2.19 Configure Trusted Domains

Service providers that delegate authentication to Identity Authentication can protect their applications when using embedded frames, also called overlays, or when allowing user self-registration.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

If you want to use overlays in your applications, you must add the domains of these applications as trusted in the administration console for SAP Cloud Identity Services. Otherwise the user will receive an error message when trying to access the overlays of these applications.

You also have to add as trusted the domains for those applications that allow self-registration to the users. For more information about the various access configurations in the administration console for SAP Cloud Identity Services, see Configure User Access to the Application [page 2025].

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.

2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Customization*, choose the *Trusted Domains* list item.
4. Press the *+ Add* button.
5. Type the URL of the trusted host in the field.

   You can type either the full name of the host or you can use the wild card * function.

   > ⚠ Caution
   >
   > Be careful when using the wild card * function. Make sure that you trust the domain when you use it.

   See the examples:

   `mycompany.ondemand.com`

   `*.example.com`
6. Save your changes.

   If the operation is successful the system displays the message `Trusted Domains updated`.

Task overview: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
Change Tenant Texts Via Administration Console [page 2121]
Configure Master Data Texts Via Administration Console [page 2125]
Configure Links Section on Sign-In Screen [page 2129]
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Configure X.509 Client Certificates for User Authentication [page 2136]
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Configure Tenant Images [page 2142]
Configure Allowed Logon Identifiers [page 2146]
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Enable Back-Up Channels to Send Passcode for Deactivation of TOTP Two-Factor Authentication Devices [page 2156]
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Use Custom Domain in Identity Authentication [page 2175]
Change a Tenant's Display Name [page 2181]

Configure Default Risk-Based Authentication for All Applications in the Tenant [page 2183]
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Send System Notifications via Emails [page 2213]
Configure Default Language for End User Screens [page 2216]
Prompt Users to Explicitly Accept Privacy Policy Document [page 2218]
Configure P-User Next Index [page 2220]
Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]
Add Logon Overlays in Customer Applications [page 2877]

1.7.2.20 Use Custom Domain in Identity Authentication

Identity Authentication allows you to use a custom domain that is different from the default ones (`<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`) - for example `www.mytenant.com`.

Prerequisites

- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You must have a custom domain.

  > **Note**
  >
  > Internationalized domain names (IDNs) are not supported.

- You must have configured the CNAME DNS record on your domain to point to the host name used to access Identity Authentication. The host name can be one of the following, depending on where client's tenant is located:

  Host Names

Country/Region	Data Center	Infrastructure	Host Name
Australia	Australia (Sydney)	SAP / Azure	`ap.accounts.ondemand.com.cloud.sap.akadns.net`

Country/Region	Data Center	Infrastructure	Host Name
Brazil	Brazil (São Paulo)	AWS	`br.accounts.ondemand.com.cloud.sap.akadns.net`
China	China (Shanghai)	SAP	`accounts.sapcloud.cn.cloud.sap.akadns.net`
Europe	Germany (Frankfurt)	AWS	`aws-eu-de.accounts.ondemand.com.cloud.sap.akadns.net`
Europe	Europe (Frankfurt / Amsterdam))	SAP	`accounts.ondemand.com.cloud.sap.akadns.net`
India	India (Mumbai)	AWS	`aws-ap-in.accounts.ondemand.com.cloud.sap.akadns.net`
Israel	Israel (Tel Aviv)	Google Cloud	`il.accounts.ondemand.com.cloud.sap.akadns.net`
Japan	Japan (Tokyo / Osaka)	SAP	`jp.accounts.ondemand.com.cloud.sap.akadns.net`
North America (Canada Central)	Canada (Toronto)	Azure	`azr-na-ca.accounts.ondemand.com.cloud.sap.akadns.net`
Saudi Arabia (SA1)	Saudi Arabia (Dammam)	Google Cloud	`sa.accounts.ondemand.com.cloud.sap.akadns.net`
Saudi Arabia (SA2)	Saudi Arabia (Dammam) - Regulated	Google Cloud	`sa2.accounts.ondemand.com.cloud.sap.akadns.net`
Singapore	Singapore	AWS	`aws-ap-se-1.accounts.ondemand.com.cloud.sap.akadns.net`
South Korea	South Korea (Seoul)	AWS	`aws-ap-kr.accounts.ondemand.com.cloud.sap.akadns.net`

Country/Region	Data Center	Infrastructure	Host Name
Switzerland	Switzerland (Zürich)	Azure	`azr-eu-ch.accounts.ondemand.com.cloud.sap.akadns.net`
United Arab Emirates	United Arab Emirates (Dubai)	SAP	`azr-ap-ae.accounts.ondemand.com.cloud.sap.akadns.net`
United Kingdom	United Kingdom (London)	Azure	`uk.accounts.ondemand.com.cloud.sap.akadns.net`
US (North America East)	USA (Sterling, VA / N. Virginia)	SAP	`us-east.accounts.ondemand.com.cloud.sap.akadns.net`
US East **Note** Trial environment.	Trial (USA)	Azure	`trial-accounts.ondemand.com.cloud.sap.akadns.net`
US West	USA (Quincy, WA)	Azure	`azr-us-we.accounts.ondemand.com.cloud.sap.akadns.net`

Context

Note

If you have configured a custom domain and you want to add a deprecation trial token for third party cookies deprecation go to **Step 10**.

→ Remember

It takes 2 minutes for the configuration changes to take place.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.

3. Under *Customization*, choose the *Custom Domain* list item.
4. Provide the required information in the provided fields:

Field	Information
Domain	Host of your custom domain in lowercase
DN **Note** **The DN used for the domain certificate.**	• **CN** (CommonName) - mandatory **Note** The **CN** attribute must match the custom domain used for the domain certificate. • **OU** (OrganizationalUnit) - optional • **O** (Organization) - optional • **L** (Locality) - optional • **S** (StateOrProvinceName) - optional • **C** (CountryName) - optional

5. Save your configuration.
6. Choose the *Download CSR file* used for the domain certificate.
   a. Select the size of the certificate key. The supported key sizes are 2048, 3072 and 4096. The default value is 3072.
   b. **Optional:** Add an additional subject alternative name to the CSR.
   c. Choose the *Download* button and save the generated CSR file.

> **Note**
>
> Use this CSR for the custom domain certificate. Each download generates a new key pair for the CSR. Always use the last downloaded CSR file.

7. Send the CSR to a trusted Certificate Authority to sign the certificate.

> **Note**
>
> The supported algorithms are SHA-256, SHA-384 and SHA-512.

8. Access the tenant's administration console for SAP Cloud Identity Services > *Applications and Resources* > *Tenant Settings* > *Custom Domain* > *Certificate* and upload or insert as text the SSL certificate signed by the trusted CA.

> **Note**
>
> Make sure that the subject DN in the domain certificate and the configured subject DN match exactly.
>
> You can upload the domain certificate or the complete certificate chain. The certificate chain must contain the domain certificate, the intermediate certificate or certificates, and the trusted CA root certificate in the same order.

9. Save your configuration.
10. **Optional:** Add the third-party cookies deprecation trial token in the input field and save the configuration.

> → Tip
>
> For more information about how to get the third-party cookies deprecation trial token, see Deprecation trials .

Results

The custom domain configuration is enabled with the upgrade of Identity Authentication. We recommend you to renew your certificate as early as possible, preferably 30 days before expiration, and no later than the Sunday before productive system upgrade. Identity Authentication has production releases (bi-weekly updates) and immediate updates:

- **Bi-weekly updates** (standard)
  Standard updates are delivered on a bi-weekly basis. You can find the planned schedule for upcoming releases in the SAP Note 3409744 (Release Schedule for SAP Cloud Identity Services (SAP BTP)).
- **Immediate updates**
  Immediate updates are provided when fixes are required for issues that affect productive application operations or when urgent security fixes are necessary.

Next Steps

1. (Optional) Configure tenant's name to be the custom host. Select custom host for the name from the dropdown list in the SAML 2.0 or Open ID Connect Configuration settings. For more information, see Tenant SAML 2.0 Configurations [page 2103] Tenant OpenID Connect (OIDC) Configurations [page 2117].

   > ⚠ Caution
   >
   > After you select the custom host in the administration console, make sure that you also change the name of the identity provider on the service provider side, or the name of the identity provider on the corporate identity provider side. If you have set trusts with more than one service provider, or corporate identity provider, change the name in every provider, otherwise the trusts will not work. For more information about how to edit the name, see the documentation of the respective service or corporate identity providers.

2. Download the new SAML metadata of the identity provider (IdP). Configure the new metadata of the IdP in every application (service provider) you have set trusts with. For more information about how to configure the metadata, see the documentation of the respective service providers.
3. If have you configured social identity providers, please check configuration on the social provider side, and configure correctly the redirect URI, using the new custom host.
4. If have you configured a corporate identity provider, please update the configuration on the corporate identity provider side.
5. If you already have a trusted CA added to your tenant, but after that a custom domain has been configured for that tenant, report a new incident on SAP Support Portal Home with a component `BC-IAM-IDS` to update the certificate on the custom domain as well.

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Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
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Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]

1.7.2.21 Change a Tenant's Display Name

You can configure the tenant's name from the administration console for SAP Cloud Identity Services.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

Information about the tenant is visible in the header of the administration console. The information includes:

- The region, representing the location of data center in which the tenant is. For example, `Cloud Identity Services [Europe]`. This information is provided by the system, and cannot be edited by the tenant administrator. For more information about the regions, see Regional Availability [page 11].
- The tenant display name. This information can be edited. If you have not specified a specific tenant name, you will see the tenant ID instead. You can change the name to make it more understandable for you.
- → Remember

  It takes 2 minutes for the configuration changes to take place.

To edit the tenant's display name, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Choose the edit icon next to the name of the tenant and change it.

   ⓘ Note

   The tenant display name must be between 1 and 50 characters long. You can use upper case (A-Z) and lower case (a-z) Latin characters, numbers 0-9, spaces, and the following symbols: _ - . '
4. Save your changes.
   Once the tenant has been updated, the system displays the message `Tenant <name of tenant> updated`.

Task overview: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
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Configuring Tenant Settings [page 2100]

1.7.2.22 Configure Default Risk-Based Authentication for All Applications in the Tenant

You can define rules for authentication according to different risk factors and apply actions like *Allow*, *Deny*, and *Two-Factor Authentication* for all applications in a tenant.

Prerequisites

- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- (For *RADIUS Server Two-Factor Authentication*) You have requested this feature. For more information how to request and configure RADIUS server in Identity Authentication, see Configure RADIUS Server Settings (Beta) [page 2192].
- (For *SMS Two-Factor Authentication*) You have an account in Sinch. You have configured Sinch Verification in the administration console for Identity Authentication. For more information, see Configure Sinch Service in Administration Console [page 2051].

  > ⓘ Note
  >
  > Sinch Verification account is purchased separately. It is not part of the Identity Authentication contract.

- (For *Email OTP Code*) An Email OTP Code template for the respective languages must exist in the tenant to apply the email OTP code method. If the template does not exist, the user will see the option but when choosing it, the following message will appear: "Sorry, but you are currently not authorized for access". For more information how to add email templates, see Edit or Add an Email Template Set [page 2289].

Context

> ⚠ Caution
>
> Be careful when you set rules for authentication for the tenant. The rules apply to all applications in the tenant, including the `Administration Console`, if those applications have default risk-based authentication (no rules created; default action - Allow). The changes apply immediately when saved, and you may not be able to log in the `Administration Console` again if you don't meet the rules.

On the other hand, if a specific application has risk-based authentication different from the default one (no rules created; default action - Allow), and you apply default risk-based authentication for the tenant, the rules for the tenant won't apply to that specific application.

For more information, how to apply risk-based authentication for a specific application, see Configure Risk-Based Authentication for an Application [page 2007].

The configured rules manage authentication according to IP range (specified in CIDR notation), group membership, authentication method, and type of the authenticating user.

Authentication Rules

The created rules are displayed sorted by priority. When a user tries to access the application, the rules evaluate if the user meets the criteria of the rule. The evaluation starts with the rule with the highest priority, until the criteria of a rule are met. If the criteria of a rule are met, the rest of the rules are not evaluated.

Default Authentication Rule

If none of the authentication rules meets the criteria, the default authentication rule is applied, For the default authentication rule, you can only configure *Default Action*, which can be *Allow*, *Deny*, or *Two-Factor Authentication*.

> ⓘ Note
>
> If *Two-Factor Authentication* is selected, additionally, you must specify the two-factor method or methods for the user.

The rule is valid for any *IP range*, *Forwarded IP Range*, *Group*, *Authentication Method*, or *User Type*.

> ⓘ Note
>
> We recommend you to enable the back-up channels. Thus, the users can use the option as an alternative when they don't have access to the TOTP device or application. For more information, see Enable Back-Up Channels to Send Passcode for Deactivation of TOTP Two-Factor Authentication Devices [page 2156].

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Authentication*, choose *Risk-Based Authentication*.
4. **Optional:** Under *Authentication Rules*, configure the authentication rules by choosing one of the following options:

Option	Description
Create a new rule	See Create a New Rule for Risk-Based Authentication on Tenant Level [page 2186].
Edit an existing rule	Choose the icon next to the rule you want to edit.
Delete an existing rule	*Select the rule you want to delete* > *Delete*
Reprioritize rules	Choose *Edit* > *use the arrows to reprioritize the rules* > *Save*

> **Note**
>
> By default any user can log on from any IP.

5. **Optional:** Under *Default Authentication Rule*, configure the *Default Action*:
   a. Choose *Edit*.
   b. Choose from:
   - *Deny* - Nobody can log on.
   - *Two-Factor Authentication* - A drop-down appears when this choice is selected. You must specify the two-factor authentication method or methods for the end user.
6. Save your changes.

   Once the tenant has been updated, the system displays the message `Authentication rules updated`.

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Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
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Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]

1.7.2.22.1 Create a New Rule for Risk-Based Authentication on Tenant Level

You can create rules for authentication according to different risk factors.

Context

> ⚠ Caution
>
> The rules apply on all applications in the tenant, including the `Administration Console` application. If you enable risk-based authentication for the tenant, make sure that you, as a tenant administrator, meet the authentication rules and the default authentication rule. The changes apply immediately when saved, and you may not be able to log in it again if you don't meet the rules.

Each rule contains the following information:

- **Authentication Action**
  This action is performed if the rule conditions meet the defined criteria.
  You can choose one of the following authenticating actions:
  - *Allow*
    Identity Authentication allows the authentication of the user in accordance with the rule conditions.
  - *Deny*
    Identity Authentication denies the authentication of the user in accordance with the rule conditions.
    You can set this action for a test application for example, or before an application goes live.
    As long as this rule is valid, when users try to log on to the application, they get the following message: *Sorry, but you are currently not authorized for access*.
  - *Two-Factor Authentication*

    > ⓘ Note
    >
    > If *Two-Factor Authentication* is selected, additionally, you must specify the two-factor method or methods for the user:
    > - *TOTP Two-Factor Authentication*
    >   Identity Authentication asks two factors to authenticate the user.

If you set TOTP two-factor authentication, users are required to provide a time-based one-time password (TOTP) called a passcode in addition to their primary credentials. Users also has to install an authenticator application on their mobile devices to generate TOTP passcodes. TOTP passcodes are time-based and valid for one logon attempt only.

- *SMS Two-Factor Authentication*
  Identity Authentication asks two factors to authenticate the user.
  If you set SMS two-factor authentication, users are required to provide an SMS code sent to their mobile devices in addition to their primary credentials.

  > → Remember
  >
  > To use *SMS Two-Factor Authentication*, you must have configured Sinch Verification in the administration console for SAP Cloud Identity Services. For more information, see Configure Sinch Service in Administration Console [page 2051].
  >
  > Users must have their mobile phone numbers verified. The tenant administrator can verify phone numbers manually in the administration console or via the SCIM API. For more information, see List and Edit User Details [page 2361] and Identity Directory API.
  >
  > If the user does not have a verified phone number, the number is verified during the first log on when SMS code is required. After the user provides user name and credentials, he or she should provide the phone number in the field and request a code. Then provide the received code in the respective field and choose *Continue*. If the submitted code is correct, the user is allowed access, and the telephone number is verified.

- *Web Two-Factor Authentication*
  Identity Authentication asks two factors to authenticate the user.
  If you set web two-factor authentication, users are required to authenticate with a device such as the built in biometric scanners or USB, Bluetooth or Near-Field Communication (NFC) devices in addition to their primary credentials.

- *Email OTP Code*
  For security reasons, the Email OTP code is not a recommended two-factor authentication method. You may consider using some of the other methods instead.
  Identity Authentication asks two factors to authenticate the user.
  If you set *Email OTP Code*, users are required to provide the code sent to their email in addition to their primary credentials.

  > → Remember
  >
  > An Email OTP Code template for the respective languages must exist in the tenant to apply the email OTP code method. If the template does not exist, the user will see the option but when choosing it, the following message will appear: "Sorry, but you are currently not authorized for access".
  >
  > For more information how to add email templates, see Edit or Add an Email Template Set [page 2289].

- *RADIUS Server Two Factor Authentication*
  If you set *RADIUS Server Two Factor Authentication*, users are required to provide a RADIUS passcode in addition to their primary credentials. Users must have a RADIUS token (hard or soft) configured for them to generate passcodes. For more information about how to configure

RADIUS server in Identity Authentication, see Configure RADIUS Server Settings (Beta) [page 2192].

The *Action* filed is mandatory.

- *IP Range*
  Define a range of IP addresses that authentication requests to Identity Authentication can be sent from. The value has to be specified in Classless Inter-Domain Routing (CIDR) notation.

  > ### Note
  > By default the field is empty, meaning that any IP is allowed.

  > ### Example
  > Enter 123.45.67.1/24 to allow users to log on from any IP starting with 123.45.67.

  If no IP range is defined, the rule is valid for all IP ranges.
- *Forwarded IP Range*
  Define a range of IP addresses for the original IP addresses that authentication requests to Identity Authentication can be sent from. This range is used in conjunction with IP Range in scenarios where authentication requests to Identity Authentication are made by a proxy on-behalf of the user/client. The value has to be specified in Classless Inter-Domain Routing (CIDR) notation.

  > ### → Remember
  > To specify the *Forwarded IP Range*, the *IP Range* must be defined first.
- *Authentication Method*
  Specify the authenticating method, which the authenticating user has to use. If no method is selected, the rule is valid for any of the methods.
- *User Type*
  Specify the type, which the authenticating user must have. If no user type is selected, the rule is valid for any of the types.
- *Group Type*
  - Cloud Group - A group that is created in the Identity Directory of Cloud Identity services. This is the default choice. For more information about the Identity Directory groups, see Groups [page 514].
  - On-Premise Group - A corporate user store group.
- *Group*

  > ### Restriction
  > The administrator must be assigned the *Manage Groups* role or be part of a policy that allows reading groups to be able to view the groups in the list. For more information, see Edit Administrator Authorizations [page 2309] or Configure Group Authorizations [page 2320].

  Specify a cloud or on-premise group, which the authenticating user has to be a member of. If no group is selected, the rule is valid for all users.
  If the rule is valid for an on-premise group, type in the name of the corporate user store group, for which this rule should be valid.
  The cloud groups have to be configured in the administration console for Identity Authentication. For more information, see Managing Groups [page 2393].

- *Corporate Attribute*
  Specify an attribute from the corporate identity provider (IdP) assertion, based on which the rule action will be applied.
  The rule must include the attribute name and value. It is valid only when the specified name and value are found in the assertion from the corporate IdP.

  > **Note**
  >
  > For this rule, the *Apply Application Configurations* option of *Identity Federation* must be enabled. For more information, see Configure Identity Federation [page 2614].

- *Email Domain*
  Specify a domain for the email of the authenticating user.

The fields *IP Range*, *Forwarded IP Range*, *Group*, *Authentication Method*, and *User Type* are not mandatory, but at least one of them has to be specified.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Authentication*, choose *Risk-Based Authentication*.
4. Choose *+ Create*.
5. Fill in the fields on the *Create Rule* window.
6. Choose *+ Create*.

1.7.2.23 Configure Sinch Service in Administration Console

Configure Sinch Service to enable *Phone Verification via SMS* or *SMS Two-Factor Authentication* in the administration console.

Prerequisites

- You have an account in Sinch Verification Service. Once you create a Sinch service account, you receive an email with your account information.
- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

Sinch service enables multichannel two-factor authentication (2FA), adding another layer of security to customers' online accounts, beyond their login and password. The service enables you to configure tokens – such as one-time passwords (OTPs), personal identification numbers (PINs), and verification codes – that are tailored to businesses and particular use cases.

For the integration between Identity Authentication and Sinch service, you must provide information in the administration console for SAP Cloud Identity Services. You can have a Sinch Verification configuration. You can clear the fields by choosing the *Remove Configuration* button at the top of the screen.

Provide the following:

Sinch Verification

Configuration	Notes
Application Key	The *Application Key* obtained via Sinch customer dashboard.
Application Secret	The *Application Secret* obtained via Sinch customer dashboard.
Code Timeout	Optional. Request the Sinch Support team to configure this parameter for the client's Sinch account. → Remember Before configuring this property in the administration console for SAP Cloud Identity Services, first request this property to the Sinch Support team.
SMS Text Template	Optional. Request the Sinch Support team to enable this configuration. The SMS text in the configuration must include the `{{CODE}}` placeholder, where Sinch will include the actual code. For example, `"This is your OTP code {{CODE}}!"` → Remember Before configuring this property in the administration console for SAP Cloud Identity Services, first request this property to the Sinch Support team.

→ Remember

It takes 2 minutes for the configuration changes to take place.

To configure the administration console, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Choose the *Sinch Service Configuration* list item.
4. Choose the *Edit* button.
5. Select *Sinch Verification* and enter the information that is required.
6. Save your entries.

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Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
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1.7.2.24 Configure RADIUS Server Settings (Beta)

Configure Remote Authentication Dial-In User Service (RADIUS) server settings in the administration console for SAP Cloud Identity Services.

Prerequisites

- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have the `host`, `port`, and `secret` parameters, configured for the RADIUS server which will be used to authenticate users with RADIUS two-factor authentication. You can receive the parameters from the RADIUS server administrator.

Context

> **Note**
>
> This is a beta feature available on Identity Authentication. You as a tenant administrator can request this feature by reporting an incident on SAP Support Portal Home with component `BC-IAM-IDS`.

You can configure more than one RADIUS server in the administration console for SAP Cloud Identity Services to provide backup. Thus, if the first server becomes unavailable, the system sends a request to the next in the list.

> **→ Remember**
>
> It takes 2 minutes for the configuration changes to take place.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *External Services*, choose the *RADIUS Server Configuration* list item.
4. Add a server.

   All fields are required.

Parameter	Details
*Host**	Host address of the RADIUS server (used for user authentication).
*Port**	The port number used by the RADIUS server for authentication requests.
*Secret**	A shared secret is used to encrypt the user password. This shared secret also must be defined in the RADIUS server. Paste the secret defined in the RADIUS server into this field.

5. Choose *User Attribute* and *PAP/CHAP* settings.

   These settings are shared between all configured servers.

Parameter	Details
*User Attribute**	Unique attribute for the user. Configure the supported attribute which the RADIUS server is configured to use for authentication. Possible values are: • **`User ID`** • **`Email`** • **`Login Name`**
*PAP / CHAP**	Authentication method for the RADIUS server. Possible values are: • **`PAP`** • **`CHAP`** The default value is **`PAP`**.

6. Save your entries.

   If the operation is successful, you receive the message `Configuration saved`.

   To delete a server, choose the delete icon 🗑 next to the server, and save your changes.

> ⚠ Caution
>
> Make sure to remove the *RADIUS Server Two-Factor Authentication* rule, when you delete the last RADIUS server configuration. If the configuration is missing, the users cannot log on.

Next Steps

Configure an application to require RADIUS PIN code as a second factor apart from the password for user authentication. For more information, Configure Risk-Based Authentication for an Application [page 2007].

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Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
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Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]

1.7.2.25 Configure Mail Server for Application Processes

Configure mail server for the emails sent to the end users in the different application processes.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

Identity Authentication uses own mail server for the emails sent to users for the different application processes. For example, when a user makes a registration for an application, he or she receives an activation email from *ias@notifications.sap.com*.

Identity Authentication provides also the possibility to configure custom mail server for the application processes emails.

Apart from the Identity Authentication own mail server, which is the default configuration for the mail server, and the custom mail server, you can also use the Amazon Simple Email Service for the emails sent to users for the different application processes. Thus, you can modify the sender and receiver the email without having custom mail server.

To use the Amazon Simple Email Service with custom domain, you must own the domain. Otherwise, you can use the `accounts.ondemand.com` or `accounts.cloud.sap` domain and just modify the *From* and *Reply to* fields.

For more information, about the emails sent for the different application processes, see Configuring Email Templates [page 2282].

→ Remember

Once the tenant administrator configures the mail server, all emails go through this configuration.

If you have created a custom mail configuration, and after that you create an Amazon Simple Email Service configuration, the custom mail configuration will be deleted. And also, if you have created an Amazon Simple Email Service, and after that you create a custom mail configuration, the Amazon Simple Email Service configuration will be deleted.

To return to the default settings, remove the configuration.

→ Remember

It takes 2 minutes for the configuration changes to take place.

> ⚠ Caution
>
> When you are using the default mail server configuration, and you start sending multiple emails to invalid addresses, the mail server will be disabled by SAP. To start using the default mail server again you must stop the actions that caused the disabling of the mail server and report an incident on SAP Support Portal Home with the `BC-IAM-IDS` component.

To configure the mail server, choose one of the procedures below:

Configure Custom Mail Server [page 2197]
: Configure custom mail server for the emails sent to the end-users in the different application processes.

Configure Amazon Simple Email Service [page 2199]
: Configure mail server for the emails sent to the end users in the different application processes.

Task overview: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
Change Tenant Texts Via Administration Console [page 2121]
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Enable Back-Up Channels to Send Passcode for Deactivation of TOTP Two-Factor Authentication Devices [page 2156]
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Configure Default Risk-Based Authentication for All Applications in the Tenant [page 2183]
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Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]
Configuring Email Templates [page 2282]

1.7.2.25.1 Configure Custom Mail Server

Configure custom mail server for the emails sent to the end-users in the different application processes.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

The custom mail server must support SSL (Secure Sockets Layer) and Transport Layer Security 1.2 (TLS). Identity Authentication trusts all certificates from Java SE Runtime Environment 17, therefore the mail server should use only them as a certificate authority when communicating with Identity Authentication. All certificate authorities from the certificate chain must be trusted by Identity Authentication to be able to communicate with the mail server.

> → Tip
>
> You can configure rate limits for your emails between 30 and 600 per minute. Beware that this affects all types of emails, including parallel bulk upload of users.
>
> If a rate limit is not specified, Cloud Identity Services tries to send all emails. You should consider setting the rate limit based on the capabilities of your custom mail server.

> → Remember
>
> You can have only one mail server configuration. Once you configure the custom mail server, all emails will go through this configuration.
>
> To return to the default settings, remove the configuration.

To configure the custom mail server, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *External Services*, choose the *Mail Server Configuration* list item.
4. Choose the *Custom Mail Server* tab.
5. Choose the authentication mechanism, and enter the required information in all the fields.

   > ⓘ Note
   >
   > SAP is not responsible for the information in the fields, and it can't provide technical assistance on the correct values. Please contact your IT Department for the correct custom mail server configurations in the administration console.

   > → Remember
   >
   > The custom mails server host must be reachable over the Internet.
6. Save your entries.

   If the operation is successful, you receive the message `Configuration saved`.

   > → Tip
   >
   > To return to the default settings, remove the configuration.

Example

Testing Mail Server Configuration

Access the profile page at `https://<tenant ID>.accounts.ondemand.com/` from a new browser where you do not have an active session with Identity Authentication. Choose the *Forgot password?* link from the logon page to start the reset password process.

> → Remember
>
> Only three emails are allowed to be sent via Identity Authentication for 24 hour. If you have already reached the limit, the system will not send an email. To check the number of emails sent to the user, access *administration console for SAP Cloud Identity Services* > *Users & Authorizations* > *User Management* > *your user* > *Authentication* > *Password Details*. Next to *Password Reset* you can see the number of emails sent to the user. If necessary, choose *Reset Counter* to be able to send more emails.

If the reset password email is sent successfully, and the configuration is correct, the email that you receive should have as a sender the email specified in the configuration.

If the configuration is not correct, no email is sent.

Task overview: Configure Mail Server for Application Processes [page 2195]

Related Information

Configure Amazon Simple Email Service [page 2199]
Configuring Email Templates [page 2282]

1.7.2.25.2 Configure Amazon Simple Email Service

Configure mail server for the emails sent to the end users in the different application processes.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

By using *Amazon Simple Email Service* you can modify the sender and receiver of the email without the need to have a custom mail server.

To use the Amazon Simple Email Service with custom domain, you must own the domain.

If you don't have an own domain you can use the domain, which is the default one, and just modify the `<username>` part in the *From* and *Reply to* fields. You can add up to two domains in addition to the default one.

Once the tenant administrator configures the mail server, all emails will go through this configuration.

> → Remember
>
> To return to the default settings, remove the configuration.
>
> Note that it takes two minutes for each configuration change to be applied.

To configure the mail server, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *External Services*, choose the *Mail Server Configuration* list item.
4. Choose the *Amazon Simple Email Service* radio-button
5. **Optional:** To use the Amazon Simple Email Service with custom domain, add a valid domain name in the dialog box that appears.
   a. Copy the TXT and CNAME records that are created in the administration console for SAP Cloud Identity Services and enter them in the DNS server.
   b. Choose *Verify* to check the status of the domain in the administration console.

      Domain verification may take up to 72 hours to complete (usually it takes around 5 minutes).

      Status of the domain will change from *Pending* to *Verified* if Amazon has verified the domain, otherwise it will change to *Failed* after 72 hours.

      If you want to start the verification process again, you need to delete the domain and add it again.
6. Configure the <username> parts in the *From* and *Reply to* fields under *Communication Emails*.

   > ⓘ Note
   >
   > The <username> part of the email address can have the following:
   > - uppercase and lowercase Latin letters
   > - digits from 0 to 9
   > - special characters !#$%&'*+-/=?^_`{|}~
   >
   > The <domain> part of the email address can have the following:
   > - uppercase and lowercase Latin letters
   > - digits from 0 to 9
   > - hyphen
   >
   > You can't specify a hyphen at the beginning or end.

   > ⚠ Caution
   >
   > If you have more than one verified domain, make sure that you have selected the correct ones from the drop down in the domain part under *Communication Emails*.
7. Save your entries.

   If the operation is successful, you receive the message `Configuration saved`.

   > → Tip
   >
   > To return to the default settings, remove the configuration.

Example

Testing Mail Server Configuration

Access the profile page at `https://<tenant ID>.accounts.ondemand.com/` from a new browser where you do not have an active session with Identity Authentication. Choose the *Forgot password?* link from the logon page to start the reset password process.

> → Remember
>
> Only three emails are allowed to be sent via Identity Authentication for 24 hour. If you have already reached the limit, the system will not send an email. To check the number of emails sent to the user, access *administration console for SAP Clou Identity Services* > *Users & Authorizations* > *User Management* > *your user* > *Authentication* > *Password Details* . Next to *Password Reset* you can see the number of emails sent to the user. If necessary, choose *Reset Counter* to be able to send more emails.

If the reset password email is sent successfully, and the configuration is correct, the email that you receive should have as a sender the email specified in the configuration.

If the configuration is not correct, no email is sent.

Task overview: Configure Mail Server for Application Processes [page 2195]

Related Information

Configure Custom Mail Server [page 2197]
Configuring Email Templates [page 2282]

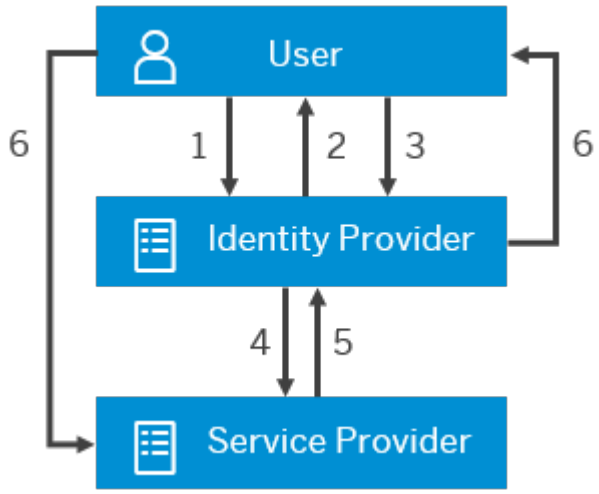
1.7.2.26 Configure IdP-Initiated SSO

Enable or disable IdP-Initiated SSO via the administration console for SAP Cloud Identity Services.

Overview

In the IdP-Initiated Single Sign-On (SSO), the authentication starts at the identity provider (IdP). The user is first authenticated at the identity provider, and after that he or she is allowed to access the protected resource at the application ((service provider (SP)).

> → Remember
>
> IdP-initiated SSO is supported only for SAML 2.0 applications.



1. The user access the identity provider via a link.
2. The identity provider requires credentials.
3. The user provides credentials, and he or she is authenticated.
4. The identity provider sends assertion about the user to the service provider.
5. The service provider validates the assertions and gives access rights to the user.
6. The identity provider redirects the user to the protected resource.

> → Remember
>
> The service provider (SP) metadata that is used to configure the trust must contain the default assertion consumer service (ACS) endpoint that can process unsolicited SAML responses.

> ⓘ Note
>
> When SAP BTP acts as a service provider, the ACS endpoint should be the URL of the protected application resource. To point to the application protected URL, change the ACS endpoint on the identity provider side . For more information about how to change the ACS endpoint in Identity Authentication, see Configure Trust [page 1941].

The link for IdP-Initiated SSO follows the pattern: `https://<tenant_ID>.accounts.ondemand.com/saml2/idp/sso?sp=<sp_name>[&RelayState=<sp_specific_value>&index=<index_number>]`

> ⓘ Note
>
> The following table lists the URL parameters you can use for IdP-initated SSO.

URL Parameters for IdP-Initiated SSO

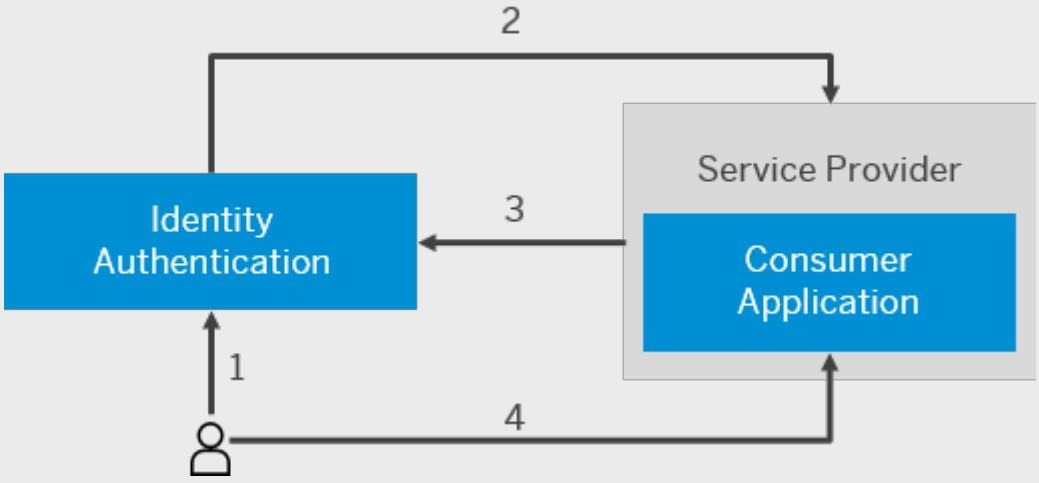
Parameter	Mandatory	Description
`sp`	Yes	Name of the SAML 2 service provider for which SSO is performed. The `sp_name` value of the parameter equals to the `Entity ID` of the service provider. This parameter is needed for Identity Authentication to know which service provider to redirect the user to after successful authentication.
`idp`	No	The technical name of the corporate identity provider as configured in the administration console for SAP Cloud Identity Services. When a chain of identity providers are allowed for an application via conditional authentication, these parameters enable the client to specify which corporate identity providers are to be used. Identity Authentication uses the `idp` parameter to detect the correct corporate identity providers and redirect the request to them. When there is more than one corporate IdP in the IdP-initiated link, they are separated by comma "`,`" without space between them.
`index`	No **Note** You can choose by the index the correct ACS endpoint for unsolicited SAML response processing. Provide the `index` parameter when the default ACS endpoint, which has been configured via the administration console can't process unsolicited SAML responses.	Enter the index number of the endpoint of the assertion consumer service of the service provider as the target of the SAML response. Otherwise, the identity provider uses the default endpoint configured for the trusted service provider. A non-digit value or a value for an index entry that is not configured returns an error message.

Parameter	Mandatory	Description
`login_hint`	No	The `login_hint` parameter helps the user when he or she is known to the service provider (SP). Thus it prevents the user from re-typing the user identifier on the logon or conditional screen. Supported values are the allowed logon identifiers for the users. The options are *User ID*, *Login Name*, and *Email* (For SAML 2.0). For more information, see Configure Allowed Logon Identifiers [page 2146] .
`redirect_uri`	No	URI to redirect the user agent to after a web session is created based on the SSO token. Required only if no `sp` parameter is provided. **⚠ Restriction** The URI must be registered in the applications that participate in the SSO flow.
`RelayState`	No	Relay state forwarded to the service provider with the SAML response. **⚠ Restriction** Not supported for the Cloud Foundry environment.

Parameter	Mandatory	Description
`sso_token`		Value of the SSO token, created in the previous OIDC token exchange call, allows reentry into the web context of Identity Authentication, based on the OIDC runtime. **Note** Create the SSO token with a token exchange call using **`resource=urn:sap:identity:sso`** at the token endpoint. For more information, see Configure the Client to Call Identity Authentication Token Exchange [page 2707].

Example

Richard Wilson, tenant administrator at Company A, would like to set up an IdP-initiated SSO process and has configured the default assertion consumer service (ACS) endpoint correctly at the cloud identity provider. Dona Moore, who is an employee at Company A, tries to access the identity provider, but because she does not have a valid session she is prompted to provide credentials. Once Dona has logged in at the IdP, a session is created for her. She is automatically redirected to her application (the default ACS URL as specified in the service provider (SP) metadata)).



1. User provides credentials; logs on.
2. Identity Authentication sends assertions.
3. Service provider validates assertions; gives access rights.
4. User accesses content.

Enable or Disable IdP-Initiated SSO

Prerequisites

- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have specified the default assertion consumer service (ACS) endpoint in the configuration of a trusted service provider (SP) in the administration console for SAP Cloud Identity Services. For more information, see Configure Trust [page 1941].

Context

By default, IdP-Initiated SSO is enabled in Identity Authentication. The tenant administrator can disable the IdP-Initiated SSO process via the administration console for SAP Cloud Identity Services.

> ⚠ Caution
>
> When IdP-Initiated SSO is disabled, users cannot access their profile page.

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

Use this procedure to disable or enable the IdP-Initiated SSO process.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose *Application Resources* > *Tenant Settings*.
3. Under *Single-Sign-On*, choose *IdP-Initiated SSO*.
4. Use the slider under the *Settings* section to disable or enable it.
   If the operation is successful, you receive a confirmation message.

Parent topic: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
Change Tenant Texts Via Administration Console [page 2121]
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Enable Back-Up Channels to Send Passcode for Deactivation of TOTP Two-Factor Authentication Devices [page 2156]
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Configure P-User Next Index [page 2220]
Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]

1.7.2.27 Sync SSO Cookie Between Domains

Enable the *Sync SSO Cookie Between Domains Settings* option to allow seamless single sign-on (SSO) and single logout (SLO) across the Cloud Identity Services domains - `ondemand.com` and `cloud.sap`.

Prerequisites

- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- The users have allowed third-party cookies in their browsers.
- The `*.cloud.sap` domain is permitted through your corporate firewall. The `cloud.sap` top-level domain (TLD) is non-standard and may be blocked by default in corporate network configurations. Confirm with your network administrator that traffic to `*.cloud.sap` is allowed before enabling this option.

Context

> ⚠ Restriction
>
> This configuration isn't relevant if you're using a custom domain.

Some applications have trust configurations with the `ondemand.com` domain, while others with the new `cloud.sap` domain. This can cause problems with the SSO and SLO between two application using different domains, and trust configuration in the corporate identity provider (IdP).

> → Remember
>
> Use this option when migrating from `ondemand.com` to `cloud.sap`. The long-term usage of the option isn't recommended.

The *Sync SSO Cookie Between Domains Settings* is disabled by default. To enable it, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Single-Sign On*, choose the *Sync SSO Cookie Between Domains Settings* list item.
4. Use the slider next to the option to enable it.

   > → Remember
   >
   > It takes 2 minutes for the configuration changes to take place.

   If the operation is successful, you see a confirmation message.

Task overview: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
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Prompt Users to Explicitly Accept Privacy Policy Document [page 2218]
Configure P-User Next Index [page 2220]
Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]

1.7.2.28 Send Security Alert Emails

Send security alert emails to end-users or administrators when changes in their accounts are made.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

The tenant administrator can configure the system to send security alert emails when some of the following actions are carried out:

- *Email change*
- *Login Name change*
- *Phone change*
- *Credential change*
- *Administrator authorization removal*
- *TFA device activation or deactivation; postpone enabling of Two-Factor Authentication*
- *Registration with nonunique email*

> ⓘ Note
>
> The email user identifier must be selected unique. For more information, see Configure User Identifier Attributes [page 2149].

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

Only the affected user (end-user or administrator) receives the security alert email.

The security alert e-mails are disabled by default. To start sending security alert e-mails, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *General*, choose the *Security Alerting* list item.
4. Use the slider next to an action to enable the sending of security alert emails for it.

   If the operation is successful, you receive a confirmation message.

Task overview: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
Rotate Signing Certificates for Applications [page 2109]
Tenant OpenID Connect (OIDC) Configurations [page 2117]
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Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]

1.7.2.29 Enable Password Expiration Reminder

Enable password expiration reminder for SAP Cloud Identity Services to ensure the users are aware that a password change is due.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

The tenant administrator can configure the system to show a password expiration reminder at the sign-in page of the user. When enabled, the reminder message starts to appear on each sign-in 10 days before the password is due to be changed. The user has the option to change the password immediately or skip the reminder.

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

The password expiration reminder is disabled by default. To enable it, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *General*, choose the *Password Expiration Reminder* list item.
4. Use the slider next to the option to enable it.

   If the operation is successful, you receive a confirmation message.

Task overview: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
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1.7.2.30 Send System Notifications via Emails

You can configure the administration console to send emails with information about expiring certificates, system notifications, new administrators, and new applications to specific email addresses or to the emails of all administrators.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

The tenant administrator can configure the system to send emails to administrators for the following events:

- *Expiring Certificates* - An email is sent 30 days, 14 days, and 3 days before the certificate expires

  > ⚠ Caution
  >
  > The system does not send email notifications for expiring certificates for API authentication. For more information, see Configure Certificates for API Authentication [page 2001].

- *System Notifications* - An email is sent the day after the notification is created. The notifications are created by the SAP Cloud Identity Services team about important updates relevant for everyone.
- *New Administrators* - An email is sent right after the administrator is created
- *New Applications* - An email is sent right after the application is created

> → Remember
>
> It takes 2 minutes for the configuration changes to take place.

> ⓘ Note
>
> With the Jul 5, 2023 upgrade, the first administrator in every new tenant, created after that date, and all newly created administrators are automatically subscribed for system notifications.

To start sending system notifications via emails, proceed as follows:

> ⚠ Caution
>
> When you delete administrators, their emails remain part of the system notifications configuration and they'll continue to receive notifications. To stop sending notifications, remove the emails from the list.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *General*, choose the *System Notifications and Alerts* list item.
4. Choose the *Add* button and select one of the options in the popup that appear:
   - *New email* - enter the email that you want to receive notifications.
   - *All administrators* all tenant administrators will receive the notifications.
5. Confirm your choice by choosing the *Add* button in the pop up.
6. Configure the notifications by selecting the desired check-boxes for each email.

   > ⓘ Note
   >
   > By default all check-boxes are selected.
7. Save your configuration.

   If the operation is successful, you receive a confirmation message.

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Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
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1.7.2.31 Configure Default Language for End User Screens

Select the language that the end user screen uses if the language of the browser isn't in the list of supported languages.

Context

The end user screens of the applications that use Cloud Identity Services for authentication, support the following languages: Arabic (ar), Bulgarian (bg), Catalan (ca), Chinese PRC (zh_CN), Chinese Taiwan (zh_TW), Croatian (hr), Czech (cs), Danish (da), Dutch (nl), English United Kingdom (en_GB), English United States (en_US), Estonian (et), Finnish (fi), French Standard (fr), French Canada (fr_CA), German Standard (de), Greek (el), Hebrew (iw), Hungarian (hu), Italian (it), Japanese (ja), Korean (ko), Latvian (lv), Lithuanian (lt), Malay (s), Norwegian (no), Polish (pl), Portuguese Portugal (pt), Romanian (ro), Russian (ru), Serbian (sr), Slovak (sk), Slovenian (sl), Spanish Standard (es), Spanish Mexico (es_MX), Swedish (sv), Thai (th), Turkish (tr), Ukrainian (uk), Vietnamese (vi), Welsh (cy).

The language for the end user screens is set according to the following order of priorities:

1. If the locale is set, the end user screen uses the language set there.
   Setting the locale, sets an Identity Authentication cookie. This cookie is used for all the applications in this session that are configured to use Identity Authentication as identity provider.

   > **Note**
   >
   > The locale can be set in either of the following ways:
   >
   > - The locale is communicated to Identity Authentication by adding a locale parameter to *SAP_IDS.js*.
   >
   >   **Source Code**
   >
   >   ```
   >   <script src="https://<tenant ID>.accounts.ondemand.com/ui/resources/
   >   javascripts/SAP_IDS.js?locale=en_GB" />
   >   ```
   > - The locale is communicated to Identity Authentication by a direct `GET` request.
   >
   >   **Source Code**
   >
   >   ```
   >   https://<tenant ID>.accounts.ondemand.com/ui/public/setLocale?locale=DE
   >   ```

2. If the locale is not set, the end user screen uses the language that the user's browser is set to.
   - If the locale is in the list of supported locales, the end user screen uses this locale's configurations.
   - If the locale is not among the supported locales, Identity Authentication looks for configuration from the language part of the locale. For example if there is a request with `en_CA`, as it is not supported, Identity Authentication looks for configurations with `en`.
   - If the language is not in the list of supported languages, the end user screen uses the language set by the tenant administrator. The choice is from the list of the supported languages. If no configuration is done, the end user screen uses *English United States* language, which is the default value.

→ Remember

It takes 2 minutes for the configuration changes to take place.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Customization*, choose the *Tenant Default Values* list item.
4. Select from the drop-down the language that the end user screen to use when the language of the browser is not in the list of supported languages.

   The default language is *English (United States)*.
5. Save your changes.

Task overview: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]
Get SAML 2.0 IdP Metadata via Parameter [page 2107]
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1.7.2.32 Prompt Users to Explicitly Accept Privacy Policy Document

Enable the privacy policy configuration in SAP Cloud Identity Services to prompt users to accept the privacy policy document. This ensures that users explicitly confirm their understanding of document.

Context

By default, users are not asked explicitly to confirm that they have read and understood the privacy policy of the application when they are activating their account or when the document is updated. Tenant administrator can enable the option to require the users to confirm the acceptance of the document. A checkbox appears at the registration or update, and users confirm that they have read and understood the privacy policy document before they can continue.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Customization*, choose the *Privacy Policy* list item.
4. Enable the slider for *Privacy Policy Configuration*

Results

Users are explicitly prompted to accept the Privacy Policy document during update or registration. A checkbox appears at the update or registration form, and users must confirm that they have read and understood the privacy policy document before they can continue.

Task overview: Configuring Tenant Settings [page 2100]

Related Information

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Get SAML 2.0 IdP Metadata via Parameter [page 2107]
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1.7.2.33 Configure P-User Next Index

Set the value for the P-user next index.

Context

Every user in Identity Authentication has a `User ID` which is an automatically generated number preceded by the letter P. The first administrator, which is also the first user, receives the number P 000000. The`User ID` of every other user that is created in Identity Authentication is incremented by 1. Thus the second user has a `User ID` P 000001, and so on. When the number reaches P 999999 it is automatically extended and the next number is P 1000000. You can change the P-number index by setting a new number from which the P-number to start. The new value must always be higher than the current one. For example, the User ID of the latest created user is P 000100, which means that the current P-user index is 100. If you want the of the next user to be P 000200,`User ID` then you should make the P-user index 200.

> ⚠ Restriction
>
> Beware, that after the change is saved you won't be able to set the P-number index value to a number lower than the current one.
>
> The P-number index can't be higher than 2147483647.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Customization*, choose the *Tenant Default Values* list item.
4. Enter a number for the *P-user next index*.
5. Save your changes.

Task overview: Configuring Tenant Settings [page 2100]

Related Information

Tenant SAML 2.0 Configurations [page 2103]

Get SAML 2.0 IdP Metadata via Parameter [page 2107]
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Reuse SAP Cloud Identity Services Tenants for Different Customer IDs [page 2222]

1.7.2.34 Reuse SAP Cloud Identity Services Tenants for Different Customer IDs

You as a tenant administrator can reuse an existing tenant for configurations and automated subscriptions.

Prerequisites

- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have a customer ID for the tenant.

Context

Customers that have subsidiaries can reuse existing Identity Authentication tenants for configurations and automated subscriptions in SAP BTP.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *General*, choose the *Tenant Settings and Information* list item.
4. From the drop-down under the *Additional Settings*, choose the customer or customers that will reuse the tenant you are in now for configurations and automated subscriptions.

   The *Customer records* drop-down contains only the customer ID numbers that are assigned to the global ultimate (the most important company within the corporate family).

Task overview: Configuring Tenant Settings [page 2100]

Related Information

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Get SAML 2.0 IdP Metadata via Parameter [page 2107]
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1.7.3 Configuring Password Policies

Passwords for the authentication of users are subject to certain rules. These rules are defined in the password policy. Identity Authentication provides you with two predefined password policies, in addition to which you can create and configure up to three custom password policies.

You have the following options for a password policy:

- Standard
  (Predefined) Use this option to set special rules for changing, resetting, and locking a password.

  > ### Note
  >
  > This is the default setting. It meets the minimum strength requirements.

- Enterprise
  (Predefined) Use this option to set enhanced password management features. It's stronger than the standard policy, but weaker than the custom one.
- Custom
  (Configurable) Use this option to set the strongest password management features for the password policy. It's the responsibility of the tenant administrator to configure the custom password policy stronger than the standard and enterprise ones.

> → Remember
>
> This option is only possible if you've configured a custom password policy in the administration console for SAP Cloud Identity Services. For more information, see Configure Custom Password Policy [page 2230].

Password Policy Requirements

Requirement	Standard	Enterprise	Custom

Content of password	• Minimum length of 8 characters;
	• Maximum length of 255 characters;
	• Characters from at least three of the following groups:
	  • Lower-case Latin characters (a-z)
	  • Upper-case Latin characters (A-Z)
	  • Base 10 digits (0-9)
	  • Non-alphabetic characters
		• Minimum length of 8 characters;
		• Maximum length of 255 characters;
		• Characters from at least three of the following groups:
		  • Lower-case Latin characters (a-z)
		  • Upper-case Latin characters (A-Z);
		  • Base 10 digits (0-9)
		  • Non-alphabetic characters
			• Minimum length of 8 characters;
			• Maximum length of 255 characters;
			• Characters from between 1 and 4 of the following groups:
			  • Lower-case Latin characters (a-z);
			  • Upper-case Latin characters (A-Z);
			  • Base 10 digits (0-9);
			  • Non-alphabetic characters
			Note
			The default is at least three of the groups.

Session time limit Indicates when the current session expires.	Yes, 12 hours	Yes, 12 hours	Yes, 12 hours
"Remember me" option Indicates whether the browser can store a cookie with the credentials.	Yes	Yes	Yes

Requirement	Standard	Enterprise	Custom
Forgot password deactivation period Indicates the period during which users can initiate the number of forgot password emails specified by the forgot password counter.	Yes, 24 hours	Yes, 24 hours	Yes, 24 hours
Forgot password counter Indicates how many times a user can initiate forgot password emails during the deactivation period. For example, a user can initiate up to 3 forgot password emails within 24 hours.	Yes, 3	Yes, 3	Yes, 3
Minimum password age Shows the minimum lifetime of a password before it can be changed.	Unlimited	Yes, 24 hours	Yes, minimum 1 hour, maximum 48 hours
Maximum failed logon attempts Indicates how many logon attempts are allowed before the user password is locked.	Yes, 5	Yes, 5	Yes, minimum 1, maximum 6, default choice 5

Requirement	Standard	Enterprise	Custom
Password locked period Indicates how long a password is locked for.	Yes, 1 hour	Yes, 1 hour	Yes, minimum 1 hour, maximum *unlimited* **Note** If *unlimited* is set, the password can be unlocked only by the tenant administrator. For more information, see Unlock User Password [page 2380]. **Caution** If *unlimited* is selected, email template set must also be changed. Otherwise, the user will receive an email stating that password logon to the account was disabled for - 1 hour. For more information, see Configuring Email Templates [page 2282].
Maximum password age Shows the maximum lifetime of a password before it has to be changed.	Unlimited	Yes, 6 months	Yes, minimum 1 month, maximum *unlimited* Possible values: 1 month, 2 months, ... 6 months; 1 year, 2 years, 3 years; unlimited
Password history Indicates whether a password history is retained, and how many passwords from the history are retained.	Unlimited	Yes, the last 5 passwords are retained.	Yes, minimum the last 5 passwords, and maximum the last 20 passwords are retained.
Maximum unused period Indicates how long the system retains unused passwords for.	Unlimited	Yes, 6 months	Yes, minimum 1 month, maximum 6 months

Requirement	Standard	Enterprise	Custom
Password behavior Indicates possibility to force the user to reset or change password if the applied password policy requires stronger password than the current one.	Not Applicable	Scenario Based • *Reset password* - If a user hasn't used the password during the set user inactivity period, the system will force a password reset at the first logon after the inactivity period expires. • *Change password* - If a user hasn't changed the password within the password lifetime, the system will force a password change at the first logon after the maximum password age period expires.	Yes, administrator can choose from: • *Reset password* • *Change password*

> **Note**
>
> The non-alphabetic characters are all characters that are not lower-case Latin characters (a-z), upper-case Latin characters (A-Z), or base 10 digits (0-9).

As a tenant administrator, you can do the following:

Set a Password Policy for an Application [page 2228]
: As a tenant administrator, you can set a password policy that matches your application logon requirements.

Configure Custom Password Policy [page 2230]
: Tenant administrators can create and configure a custom password policy for scenarios where Identity Authentication is the authenticating authority.

Delete Custom Password Policy [page 2233]
: As a tenant administrator, you can delete the custom password policy that you have created.

Configure Password Exclude List [page 2234]
: As a tenant administrator, you can create a password exclude list to restrict their usage.

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Privacy Policies [page 2235]
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Configuring Terms of Use [page 2273]

Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.3.1 Set a Password Policy for an Application

As a tenant administrator, you can set a password policy that matches your application logon requirements.

Context

You can choose from standard, enterprise, and custom password policies. The standard and enterprise password policies are predefined, and you cannot configure them. You can configure only the custom password policy.

The strength of the policies grows from standard to custom.

When a user is authenticated for specific application, or there is already an active session with that application, Identity Authentication checks the application's requirement for password policy and the current user's password policy and always applies the stronger policy. Thus even if the application doesn't require an enterprise password policy if the user has already authenticated in application which requires enterprise password policy then the enterprise password policy rules will be applied even if the user accesses an application that doesn't require it.

→ Tip

To see the configuration of the password policies in the tenant, go to *Password Polices* tile and choose the icon next to the policy you want to view.

To set a password policy for an application, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Authentication and Access* tab.
5. Under *POLICIES*, choose *Password Policy*.
6. Select the radio button for the password policy.
7. Save your selection.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

   > **Note**
   >
   > Changes may take up to 2 minutes. Users will then sign in according to the new policy. When the user tries to sign in to the application whose password policy has been updated, he or she is prompted to change the password if the current one does not meet the requirements in the updated password policy.

Task overview: Configuring Password Policies [page 2223]

Related Information

Configure Custom Password Policy [page 2230]
Delete Custom Password Policy [page 2233]
Configure Password Exclude List [page 2234]
Create a New Application [page 1922]
Configure Custom Password Policy [page 2230]

1.7.3.2 Configure Custom Password Policy

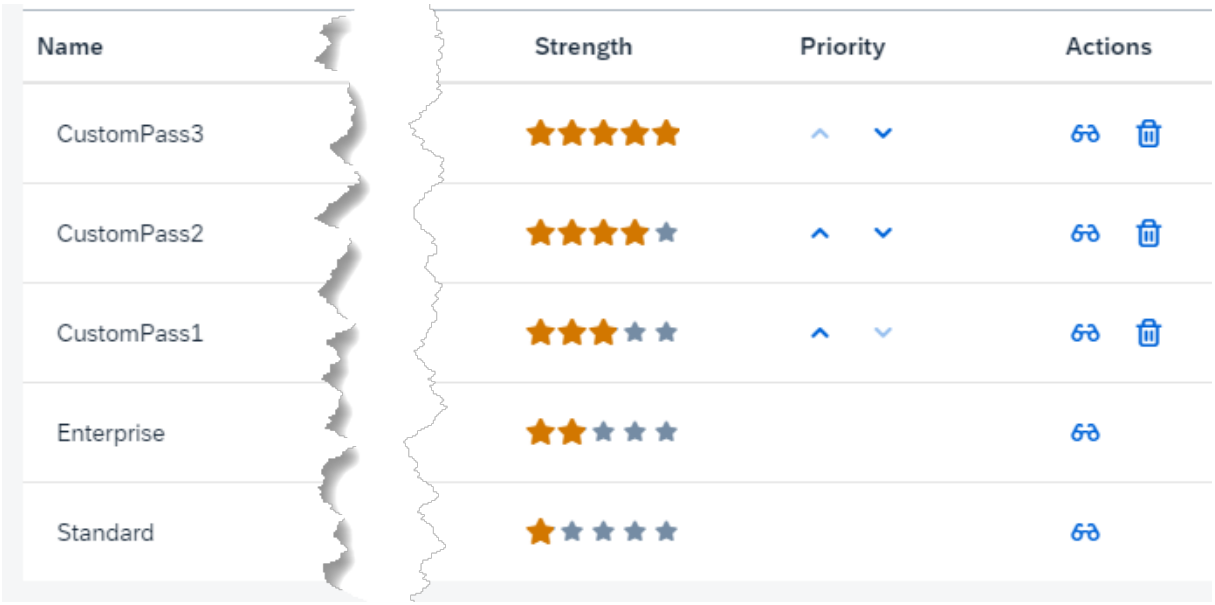
Tenant administrators can create and configure a custom password policy for scenarios where Identity Authentication is the authenticating authority.

Context

Identity Authentication provides you with two predefined password policies, in addition to which you can create and configure up to three custom ones.

The custom password policy by default must be stronger than the enterprise policy, which in turn is stronger than the standard policy. It's the responsibility of the tenant administrator to configure the custom password policy stronger than the standard and enterprise ones. Each password policy has a strength assigned to it, which is visualized by stars in the administration console for SAP Cloud Identity Services. The strength specifies the priority of the password policy. It will define which policy will be enforced for password checks.

It is not possible to have password policies with one and the same strength. Once created and saved, the tenant administrator can reorder the custom password policies in the administration console, thus changing the strength assigned to the custom password policies.



→ Remember

To change the configuration of the custom password policies or to create a new password policy when you have already reached the limit of three custom password policy in the tenant, delete an existing custom policy first, then create the new one. For more information, see Delete Custom Password Policy [page 2233].

Example

Donna Moore is an administrator of company A. She has set a *Standard* password policy for the *Leave Request* application.

Dona Moore wants a stronger password policy for the administration console, and she configured a custom password policy with a minimum password length of 15 characters, and maximum password lifetime of three months. The number of allowed logon attempts is 3. Dona has set a strength of three stars for this password policy.

Dona has set the new custom password policy for the administration console. At his first logon after the setting of the new password policy, Michael Adams, who is also an administrator at Company A, is required to update his password according to the new policy. Thus when he accesses the *Leave Request* application he uses the stronger password. John Miller who is an employee at company A, but not an admin, logs on to the *Leave Request* with a password that is compliant to the *Standard* password policy.

To create and configure a new custom password policy, follow the procedure:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Password Policies* tile.
3. Under *Password Policies*, choose the *+ Create* button.

   > ⚠ Restriction
   >
   > You can create only one custom password policy. If you already have a custom password policy in your tenant, the *+ Create* button is grayed out.

4. Fill in the required information in the fields.

Configuration Options	Information
Policy Name	The name of the password policy that appears in the administration console.
Policy Strength	This strength specifies the priority of the password policy. It will define which policy will be enforced for password checks. It's the responsibility of the tenant administrator to configure the custom password policy stronger than the standard and enterprise ones.
Password Length	The length can be between 8 and 255 characters. The default value is 8 characters.
Password Lifetime	The minimum password lifetime can be between 1 hour and 48 hours. The default value is 24 hours. Possible values are 1 hour, 2 hours, ..., 48 hours. The maximum password lifetime can be between 1 month and unlimited. The default value is 6 months. Possible values are: 1 month, 2 months, ... 6 months; 1 year, 2 years, 3 years; unlimited.

Configuration Options	Information
	Note The *Password Lifetime* configuration for a user starts after the following is done: 1. Create a custom password policy with *Password Lifetime* lifetime configured. 2. Assign the password policy to an application. 3. Authenticate to the application so the policy is assigned.
User Inactivity	The maximum duration of user inactivity can be between 1 and 6 months. The default value is 6 months. **Note** If a user didn't use the password during the set user inactivity period, the system will force either a password reset or change at the first logon. The reset or change depends on the setting for *Password Behavior*.
Password History	The minimum requirement is the last 5 passwords to be retained. The value can't be more than 20.
Failed Sign In Attempts	The number of allowed failed sign in attempts can be between 1 and 6. The default value is 5.
Password Locked Period	The period can be between 1 hour and unlimited. **Tip** When *Unlimited* is selected, we recommend you to mark this password policy the strongest.
Required Character Groups	Specifies the number of required character groups for the password. The value can be between 1 and 4. Based on the value, the users are required to include 1, 2, 3 or 4 of the following in their passwords: • Uppercase letters • Lowercase letters • Numbers • Symbols
Password Behavior	• *Reset password* - At logon, the user is forced to reset password, if the current password is not compliant with the new password policy. This is the default choice. • *Change password* - At logon, the user is redirected to the change password page if the current password is not compliant with the new password policy.

5. Choose + *Create* to save your changes.

   Once the password policy has been created and configured, the system displays the message `Password policies configuration updated`.

The new custom password policy appears in the list of the password policies that you can use for the applications.

Next Steps

To use the custom password policy for your application or applications, you should set it as a password policy for that application or applications. For more information, see Set a Password Policy for an Application [page 2228].

Task overview: Configuring Password Policies [page 2223]

Related Information

Set a Password Policy for an Application [page 2228]
Delete Custom Password Policy [page 2233]
Configure Password Exclude List [page 2234]

1.7.3.3 Delete Custom Password Policy

As a tenant administrator, you can delete the custom password policy that you have created.

Prerequisites

- You have created a custom password policy in the administration console for SAP Cloud Identity Services. For more information, see Configure Custom Password Policy [page 2230].
- The custom password policy should not be set as a password policy for any of the applications in the tenant. For more information about how to set a standard or enterprise policy for an application, see Set a Password Policy for an Application [page 2228].

Context

You can only have one custom password policy for your tenant. To change the configuration of the custom password policy, or to create a new one, delete the existing custom policy first.

To delete the custom password policy, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Password Policies* tile.
3. Press the next to the custom password policy.

> ⚠ Caution
>
> You can only delete the password policy if it is not set as a password policy for any applications in the tenant.

Results

Once the password policy has been deleted, the system displays the message `Password policy <name of policy> deleted`.

Task overview: Configuring Password Policies [page 2223]

Related Information

Set a Password Policy for an Application [page 2228]
Configure Custom Password Policy [page 2230]
Configure Password Exclude List [page 2234]

1.7.3.4 Configure Password Exclude List

As a tenant administrator, you can create a password exclude list to restrict their usage.

Context

The exclude list can include *first name*, *last name* and *login name*, and passwords entered as free text.

You can enter up to 1000 passwords in the list. Each password can be up to 256 characters long.

→ Remember

All new passwords cannot contain configured strings. The restriction is case insensitive. For example, the *qwer* string will exclude passwords like "Qwer&Dfdstg", "TestPasswordqwer", etc.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Users & Authorizations*, choose the *Exclude Lists* tile.
3. Choose the *Password* tab.
4. **Optional:** Under *Dynamic Passwords*, enable the switch next to the options that you want to restrict:

Option	Description
First Name	The password must not include the user's first name.
Last Name	The password must not include the user's last name.
Login Name	The password must not include the user's login name.

5. **Optional:** Under *Password Exclude List* , choose *Create* and enter the password exclude terms as free text in the field.

   For multiple values, use comma between them.
6. Confirm your changes by choosing *Create*.

Task overview: Configuring Password Policies [page 2223]

Related Information

Set a Password Policy for an Application [page 2228]
Configure Custom Password Policy [page 2230]
Delete Custom Password Policy [page 2233]

1.7.4 Configuring Privacy Policies

You can configure a custom privacy policy document by creating a new document, adding and editing its language versions, and defining the document for an application.

Every time you want to update the privacy policy document you add a new language versions for the document.

Each privacy policy document can have one draft and one active version. You can view all language version in the administration console. Optionally you can edit the draft version of a document before making it an active version.

Once a privacy policy document is created it can't be renamed.

For each language version, you have to upload a text file. You can define a privacy policy document in the following languages:

Arabic, Azerbaijani, Bulgarian, Catalan, Chinese (PRC), Chinese (Taiwan), Croatian, Czech, Danish, Dutch, English (United Kingdom), English (United States), Estonian, Finnish, French (Standard), French (Canada), German (Standard), Greek, Hebrew, Hungarian, Italian, Japanese, Korean, Latvian, Malay, Norwegian, Polish, Portuguese (Portugal), Romanian, Russian, Serbian, Slovak, Slovenian, Spanish (Standard), Spanish (Mexico), Swedish, Thai, Turkish, Ukrainian, Vietnamese, Welsh.

The language is defined in the following order of importance:

- By the `locale` parameter in the URL used for accessing the application.

  > **Note**
  >
  > If the SP is configured to support a specific language, only this language is used by the application.

- By the language of the application's browser.

  > **Note**
  >
  > The application takes the browser language only if the SP's language is not selected, and the `locale` parameter isn't set in the URL. The default browser setting is *English*.

> **Note**
>
> If there is only one language for the document, it is recommended to be *English (United States)*.
>
> If the languages are more than one, one of them must be *English (United States)*.

To configure privacy policy in the administration console, do the following:

1. Create a New Privacy Policy Document [page 2237]
   Create a new privacy policy document in the administration console.
2. Add New Language of a Privacy Policy Document [page 2238]
   To add a language version of a privacy policy document, you must upload a UTF-8 encoded plain text file containing the privacy policy text.
3. Add New Version of a Privacy Policy Document [page 2239]
   Add a new version of a privacy policy document and make it active.
4. Define a Privacy Policy Document for an Application [page 2240]
5. (Optional) View and Edit a Privacy Policy Document [page 2241]
   Tenant administrators can view privacy policy documents uploaded in the administration console for SAP Cloud Identity Services and optionally edit the draft versions of those documents.
6. (Optional) Delete a Privacy Policy Document [page 2243]
   Delete a privacy policy document set or a version of a privacy policy document.

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.4.1 Create a New Privacy Policy Document

Create a new privacy policy document in the administration console.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Application & Resources*, choose the *Privacy Policy Documents* tile.

   This operation opens a list of the privacy policy documents.
3. Choose the *+Add* button on the left-hand panel in order to add a new privacy policy document to the list.
4. Specify a name for the document.

   > ⓘ Note
   >
   > The name can contain only Latin letters, numbers or the underscore character, or a combination of them. The lowercase letters are converted to uppercase letters.
5. Save your selection

   Once the document has been created, the system displays the message `Privacy policy document <name of document> created`.

Task overview: Configuring Privacy Policies [page 2235]

Next task: Add New Language of a Privacy Policy Document [page 2238]

1.7.4.2 Add New Language of a Privacy Policy Document

To add a language version of a privacy policy document, you must upload a UTF-8 encoded plain text file containing the privacy policy text.

Prerequisites

You have created a privacy policy document in the administration console. For more information, see Create a New Privacy Policy Document [page 2237] .

Context

You can add one or more languages of the document.

> ⓘ Note
>
> If there is only one language for the document, it is recommended to be *English (United States)*.
>
> If the languages are more than one, one of them must be *English (United States)*.

The plain text document can include the following tags: `<a>`, `<strong>`, and `<em>`.

> ✤ Example
>
> - `See <a href="https://www.example.com">Subscription conditions</a>`
> - `<strong>Additional requirements</strong>`
> - `<em>Important:</em>`

To open the link from the document in the preview in the admin console, right-click the link and open it in a new tab.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Privacy Policy Documents* tile.

   This operation opens a list of the privacy policy documents.

3. To add a language to a privacy policy document, choose that document from the list on the left.

> → Tip
>
> Type the name of the document in the search field to filter the list items.

4. Choose the *Add* button in the details view on the top right-hand corner.
   a. Choose the *Active Language Version* radio button.
   b. Select the language from the drop-down.
   c. Specify the plain text file for the language.

   > ⓘ Note
   >
   > Use a file with an extension `.txt`.

   d. Choose *Add* to save your selection

      Once the file has been uploaded, the system displays the message `Privacy policy file uploaded`.

Task overview: Configuring Privacy Policies [page 2235]

Previous task: Create a New Privacy Policy Document [page 2237]

Next task: Add New Version of a Privacy Policy Document [page 2239]

1.7.4.3 Add New Version of a Privacy Policy Document

Add a new version of a privacy policy document and make it active.

Context

To update an existing privacy policy document you must add a new version for each language of this document by uploading a UTF-8 encoded plain text file containing the updated privacy policy statement. Once the new version is uploaded it is in a status draft. You must activate it to replace the current active version. To make a draft version active, you must have draft versions for all the languages that the current active version has.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Privacy Policy Documents* tile.

   This operation opens a list of the privacy policy documents.

3. To add a new language version to a privacy policy document, choose that document from the list on the left.

   > → Tip
   >
   > Type the name of the document in the search field to filter the list items.

4. Choose the *Add* button in the details view on the top right-hand corner.
   a. Choose the *Draft Language Version* radio button.
   b. Select the language from the drop-down for that version.
   c. Specify the plain text file for the language.

      > ⓘ Note
      >
      > Use a file with an extension `.txt`.

   d. Choose *Add* to save your selection

      Once the file has been uploaded, the system displays the message `Privacy policy file uploaded`.
5. Repeat the procedure for all languages that have active versions.
6. **Optional:** To make the draft version active, choose *Activate* in the details view on the top right-hand corner.

Task overview: Configuring Privacy Policies [page 2235]

Previous task: Add New Language of a Privacy Policy Document [page 2238]

Next task: Define a Privacy Policy Document for an Application [page 2240]

1.7.4.4 Define a Privacy Policy Document for an Application

Context

The privacy policy is displayed on the registration form, which form appears when the user chooses the *Register Now* link on the logon page. It is also shown to the user when the registration information has been upgraded. Initially, the application is set to use a default privacy policy on the registration and upgrade forms. To change this configuration, you have to select a custom privacy policy.

If you have the authorization, you can create a custom privacy policy document. This document is used once you have set it for the respective application.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.

2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Authentication and Access* tab.
5. Under *POLICIES*, choose *Privacy Policy*.
6. Select the radio button for the Privacy Policy.
7. Save your selection.

Task overview: Configuring Privacy Policies [page 2235]

Previous task: Add New Version of a Privacy Policy Document [page 2239]

Next task: (Optional) View and Edit a Privacy Policy Document [page 2241]

1.7.4.5 (Optional) View and Edit a Privacy Policy Document

Tenant administrators can view privacy policy documents uploaded in the administration console for SAP Cloud Identity Services and optionally edit the draft versions of those documents.

Context

The content of the privacy policy documents are visualised in the administration console.

Task overview: Configuring Privacy Policies [page 2235]

Previous task: Define a Privacy Policy Document for an Application [page 2240]

Next task: (Optional) Delete a Privacy Policy Document [page 2243]

View a Privacy Policy Document

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Privacy Policy Documents* tile.

   This operation opens a list of the privacy policy documents.
3. Choose the document from the list on the left.

   > → Tip
   >
   > Type the name of the document in the search field to filter the list items.
4. Choose the version and language from the drop down.

   The content is displayed on the screen.

   To view another privacy policy document, repeat the procedure.

Edit a Draft Version

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Privacy Policy Documents* tile.

   This operation opens a list of the privacy policy documents.
3. Choose the document from the list on the left.

   > → Tip
   >
   > Type the name of the document in the search field to filter the list items.
4. Choose the draft version from the drop down.
5. Choose a language for the draft version from the drop down.
6. Choose *Edit* in the details view on the top right-hand corner.
7. Upload the edited document.
8. Choose *Edit* in the popup to confirm your changes.

1.7.4.6 (Optional) Delete a Privacy Policy Document

Delete a privacy policy document set or a version of a privacy policy document.

Context

You can delete a specific language version or the entire set of privacy policy documents. Deleting the entire set of privacy policy documents, you delete all language versions for this set. The privacy policy document set is removed from the list of the privacy policy documents in the administration console for SAP Cloud Identity Services, when you delete it.

The documents will be deleted no matter if they are accepted or not by any users.

> ⓘ Note
>
> You can delete only documents that are not defined for applications. To delete a document version or entire document set that is defined for an application, you first you must set another privacy policy for the applications. For more information, see Define a Privacy Policy Document for an Application [page 2240].

To delete a privacy policy document set or a specific language version, follow the procedure below:

Task overview: Configuring Privacy Policies [page 2235]

Previous task: (Optional) View and Edit a Privacy Policy Document [page 2241]

Delete Entire Documents Set

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Privacy Policy Documents* tile.

   This operation opens a list of the privacy policy documents.
3. To delete the entire privacy policy document set, choose that set from the list on the left.

   > → Tip
   >
   > Type the name of the document in the search field to filter the list items.
4. Choose *Delete Set* button and confirm the operation in the pop-up that appears.

   If a document is defined for an application, define a new privacy policy document for that application and repeat the step.

   If the document set is deleted, the system displays the message `Privacy Policy document set deleted`.

Delete a Language Version

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Privacy Policy Documents* tile.

   This operation opens a list of the privacy policy documents.
3. To delete a language version to a privacy policy document, choose that document from the list on the left.

   > → Tip
   >
   > Type the name of the document in the search field to filter the list items.

   a. Select the version from the drop-down that you want to delete.
   b. Select the language from the drop-down for that version that you want to delete.
   c. Choose *Delete* button and confirm the operation in the pop-up that appears.

      If the document is defined for an application, define a new privacy policy document for that application and repeat the step.

   If the document version is deleted, the system displays the message `Privacy Policy document deleted`.

1.7.5 Configuring Authorization Policies

Authorization Management enables SAP Cloud Identity Services administrators to use authorization policies, customize them, and assign them to users.

When you subscribe to an application that supports authorization policies, you get a corresponding application in *Applications*, and from there you can assign authorizations.

For more information, see Subscribe to Multitenant Applications Using the Cockpit, Assign Authorization Policies [page 2363], and Add Users to a Group [page 2406].

Prerequisites

- Your application supports authorization policies. Refer to the documentation of your application.
- Your user has administrative permissions in SAP Cloud Identity Services with the following authorizations:
  - *Manage Applications*
  - *Manage Groups*
  - *Read Users*

  For more information, see Edit Administrator Authorizations [page 2309].
- You are using SAP Cloud Identity Services - Identity Authentication. For more information, see Tenants [page 27].

> **Note**
>
> If your application doesn't have an *Authorization Policies* tab, it doesn't support Authorization Management.

Authorization Management

Authorization Management allows SAP Cloud Identity Services administrators to refine authorization policies based on application policy templates with powerful instance restrictions for data access. Developers define and deploy authorization policies with functional checks, instance-based authorizations, and user attributes. They're available in the SAP Cloud Identity Services administration console. If necessary, developers can update existing authorization policies.

For more information, see Developing Authorizations [page 2876].

Authorization Policy Types

An authorization policy is basically a collection of rules, which are applied to resources and restricted by conditions. Authorization policies are defined by developers and come with the respective application.

Setting Up Authorization Policies

Task	User Role	Tool
Define authorization policies	Developer	Development environment
Deploy authorization policies with the application	Developer	Development environment
Modify authorization policies	Administrator	Administration console
Assign authorization policies to users	Administrator	Administration console
Delete authorization policies	Administrator	Administration console

We distinguish between different types of authorization policies. You recognize the different types in the *Package* column.

- The package name of the base policies is defined by the application.
- Customers can deploy their own authorization policies in customer-developed packages, which have the package name *Customer Package* in the list of authorization policies.

Authorization Policy Types

Type	Description
Base authorization policy	Authorization policy delivered by the application. Administrators can copy the authorization policy and modify the copy. The copy, modified or not, is a custom authorization policy. You can't change or delete it in the administration console.
Custom authorization policy	Authorization policy created by administrators. You can change and delete this authorization policy in the administration console.

> ⚠ Restriction
>
> To make sure that Authorization Management is resilient, we've introduced an upper limit of 200 custom authorization policies and 200 base authorization policies that can be defined per application.

Configuration Options

Authorization policy administrators can configure the following in custom authorization policies:

- Combine authorization policies. See Combine Authorization Policies [page 2249].
- Add or delete rules and restrictions and their attribute values. See Edit an Authorization Policy [page 2248].

Refining Authorization Policies

Administrators can edit an existing custom authorization policy. They can add restrictions and manage the values of the restrictions. The restrictions are refined by attributes with fixed values or user attributes. They can also add USE rules to a custom authorization policy and restrict them.

For more information, see Data Control Language (DCL) [page 2251].

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Terms of Use [page 2273]

Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.5.1 Create an Authorization Policy

Identity Authentication administrators can use existing authorization policies to create an authorization policy, configure rules, and assign the authorization policies to users.

Context

You have the following options:

- Create an authorization policy based on the rules of other authorization policies, add restrictions, or change the attribute.
- Create an authorization policy by combining authorization policies.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose *Applications*.
3. Choose an application that supports Authorization Management. For information, see the documentation of the application.

   The details page of your application has an *Authorization Policies* tab.
4. Choose the *Authorization Policies* tab.
5. Choose *Create*.
6. Choose one of the following options:

   - *Create Restriction* Create an authorization policy based on base policies, edit rules, add restrictions, statements, and attributes or only change attributes. See Edit an Authorization Policy [page 2248].

- *Add Combination* Create one authorization policy with an OR combination of the rules of two authorization policies. See Combine Authorization Policies [page 2249].

You can now continue by giving the authorization policy a name, choosing the base policies, assigning it to a user or changing its rule.

1.7.5.2 Edit an Authorization Policy

If SAP Cloud Identity Services administrators want to change restrictions and their attributes, they can edit an existing custom authorization policy.

Context

When you edit an existing custom authorization policy, you can add or delete restrictions and their attribute values. You can also assign users to the authorization policy.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose *Applications*.
3. Choose the application that supports Authorization Management. For more information, see the documentation of the application.

   The details page of your application has an *Authorization Policies* tab.
4. Choose the *Authorization Policies* tab and select a custom authorization policy. Custom authorization policies appear as *Editable* in the *Editable* column of the list of authorization policies. If the policy contains editable restrictions, it is also marked as *Restricted Editing*.
5. Choose the *Edit* button.

   - (Optional) To assign users, choose the *Assignments* tab. See Assign Authorization Policies [page 2363].
   - The rule that came with the custom authorization policy appears in the *Rules* tab.
6. (Optional) You can enter a label and a description of the authorization policy or change them. The description is an optional comment.
7. To edit the rules of the authorization policy, choose the *Rules* tab.
8. Continue to edit the rules. Multiple options are available.

   - Choose + to see the possible `RESTRICT` options. This button is either directly below `RESTRICT` or in an indented row below `RESTRICT`.
   - Choose *Add USE* to add a `USE` statement to the selected authorization policy. Select a `USE` statement from the available authorization policies.

9. For a `RESTRICT` condition, choose one of the available attributes, an operation, and enter a value. You can choose a value from the value help or type it in. To add another `RESTRICT` condition, choose + or use the `Return` key.

   > **Note**
   >
   > All indented rows that appear in a list directly below `RESTRICT` or `USE` have an `AND` conjunction.
   >
   > All rows starting with their own `RESTRICT` condition have an `OR` conjunction with the other `RESTRICT` rows.

   For more information, see Data Control Language (DCL) [page 2251].
10. Choose the conditions that come with the application in the dropdown list. Next, choose an operator and choose or enter a value or choose a user attribute variable such as *$user.country*. In this case, the value is taken from the user attribute.
11. Save your changes.
12. To assign users, choose the *Assignments* tab. See Assign Authorization Policies [page 2363].

1.7.5.3 Combine Authorization Policies

You can combine authorization policies to create a new one. This new authorization policy gets all the rules of the authorization policies that you combined.

Context

> **Note**
>
> When you combine authorization policies, the new authorization policy has no user assignments.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose *Applications*.
3. Choose the *Authorization Policies* tab.
4. Choose *Create*.
5. Choose *Add Combination*.
6. Search for the authorization policies you want to combine, choose them, and enter a name for the new authorization policy.
7. Choose *Confirm*.
8. Select the newly created authorization policy.

9. Choose the *Assignments* tab and assign users. See Assign Authorization Policies [page 2363].
10. Choose the *Rules* tab and edit the label and description of the authorization policy. The description is an optional comment.
11. To change the rules, choose *Edit*. For more information, see Edit an Authorization Policy [page 2248].
12. Save your changes.

1.7.5.4 Delete an Authorization Policy

Identity Authentication administrators can delete an existing custom authorization policy.

Context

> ⚠ Caution
>
> When you delete an authorization policy, remember that you are also removing the user assignments. This means that the users assigned to a deleted authorization policy can't access the application anymore.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose *Applications*.
3. Choose an application that supports Authorization Management. For information, see the documentation of the application.

   The details page of your application has an *Authorization Policies* tab.
4. Choose the *Authorization Policies* tab.
5. Choose the custom authorization policy you want to delete.
6. To delete the authorization policy, choose *Delete* and confirm that you want to delete the authorization policy.

1.7.5.5 Data Control Language (DCL)

Developers define authorization policies in SAP Cloud Identity Services. They use an SQL-like language - the data control language (DCL). Administrators can restrict base policies and combine authorization policies into a new authorization policy.

Prerequisites

- Authorization Management is part of the Identity service. You can activate Authorization Management for all applications that are created in SAP Cloud Identity Services.

Policy Definition in DCL

Using this language, developers define authorizations. Administrators can refine them by combining rules in authorization policies created by them. In the administration console, they assign the authorization policies to users.

This is what a policy definition looks like.

> ⓘ Note
>
> Choose the links in the sample code. They take you to a description of the DCL elements.

> {·} Sample Code

```
{
SCHEMA {
    SalesOrderType: number
}
POLICY readSalesOrders_Type {
    GRANT read ON salesOrders WHERE SalesOrderType BETWEEN 100 AND 500;
}
}
```

Structure of the Data Control Language (DCL) [page 2252]
Use DCL syntax to define authorization policies, including rules, actions, resources, conditions, attributes, and a schema. They are essential for controlling access to data and resources.

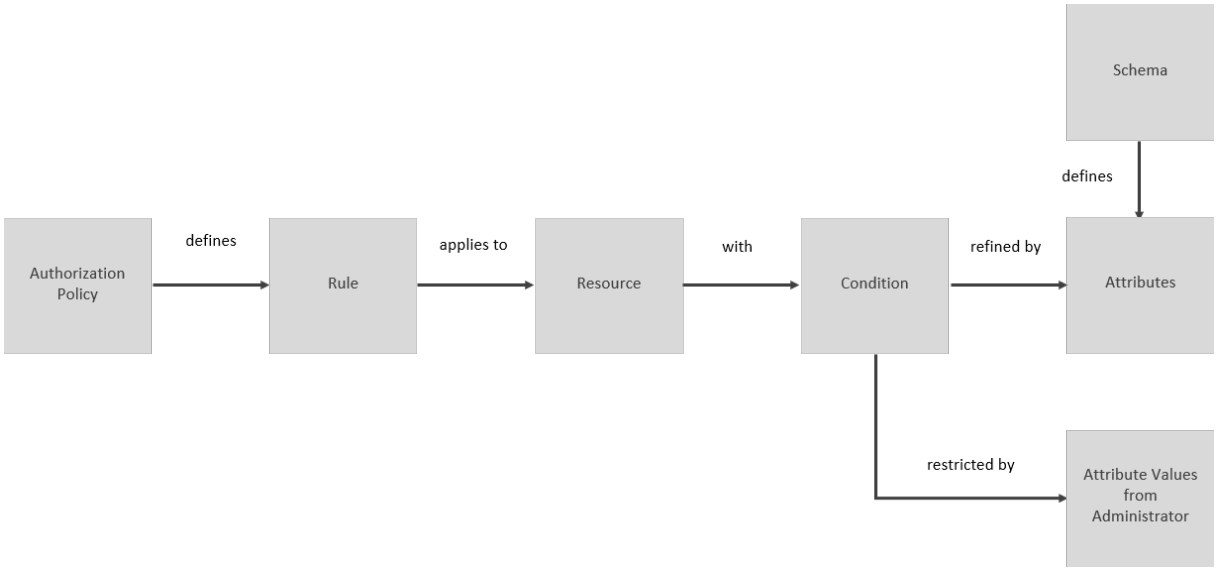
Grammar and Syntax Rules [page 2261]
DCL has a set of grammar and syntax rules you must observe when you define authorization policies.

Advanced Features [page 2269]
You find advanced configuration techniques for authorization policies, focusing on annotations, formatting conventions, and internal functions for enterprise-grade authorization policies.

1.7.5.5.1 Structure of the Data Control Language (DCL)

Use DCL syntax to define authorization policies, including rules, actions, resources, conditions, attributes, and a schema. They are essential for controlling access to data and resources.

The data control language uses the following syntax for the definition of simple and instance-based authorizations.



Authorization Policy

Collection of rules. Administrators can use base authorization policies to create custom authorization policies, change the rules, and assign the policies to users.

Rule

Grants actions such as `read` or `write`. Rules handle actions on resources on resources such as `resource1` or `resource2`. A rule can also be restricted in multiple ways.

An action is granted on a resource. Use the following syntax:

```
GRANT <action> ON <resource> ;
```

- {·} Sample Code

  ```
  GRANT read ON * ;
  ```

Resource

Entities protected by the rule or authorization policy. The application that provides the authorization policy defines the resource. Examples for resources are a data object or an endpoint. A resource has arbitrary attributes that can be used to refine rules.

Condition

A logical expression that is mainly used for filtering. It can be based on attributes with a name and a data type. Before allowing a granted action on a specific entity of a resource, the condition is evaluated with the attribute values of that entity. A condition is a Boolean function over attributes.

```
WHERE <attribute_name> = '<attribute_value>' ;
```

- Sample Code

  ```
  WHERE Country IN ('USA', 'Germany');
  ```

For more information, see Condition Operators [page 2265].

Attribute

You can restrict authorization policies by using a set of conditions with attributes and attribute values. Attributes are defined in a schema or they're provided by an external source, for example an identity provider. When using attributes, you must take care that you declare their data type in a schema. All schema attributes usages are validated to refer to existing names and their usage types.

For more information, see Attribute Constraints [page 2268].

Schema

A schema defines names and types of attributes you want to use for the application. Only one schema can be defined. Define the schema using a file named `schema.dcl` and locate it in the root folder. All schema attributes usages are validated to refer to existing names and their types.

For more information, see Schema Definition [page 2255].

Package

The name of the root folder where the DCL files are located. The package content comprises all of the DCL files in the folder.

For more information, see Package Definition [page 2254].

Parent topic: Data Control Language (DCL) [page 2251]

Related Information

Grammar and Syntax Rules [page 2261]
Advanced Features [page 2269]

1.7.5.5.1.1 Package Definition

A package is the folder where the DCL document is located.

A package can contain multiple DCL documents.

> ⚠ Restriction
>
> - Only `[a-zA-Z_][a-zA-Z_0-9]*` can be used for a package name, which is the name of the path relative to the DCL root.
> - The total length (including package separator) of a package hierarchy is 250 characters.

Cross Package Reuse

Policies from other packages can be reused based on qualified references.

References

Qualified references are used to specify elements across package boundaries. The parts of a qualified reference are connected by a "`.`" (period) in DCL.

Examples

- `sales.SalesOrder:` Package `sales`, Policy: `salesOrder`
- `sales.basic.salesOrder`: Package `sales/basic`, Policy: `salesOrder`

You can introduce qualified references in a compatible way with an additional qualification prefix.

> {·} Sample Code

```
POLICY <policy_name1> {
    USE <package>.<policy_name2>; // <policy_name2> is qualified by <package>
    USE <policy_name3>;           // local policy (same package)
```

```
}
```

- Sample Code

```
POLICY allAdmins {
    USE sales.salesOrders; // fully qualified name to 'salesOrders' in
package 'sales'
    USE readAll;           // local policy
}
```

1.7.5.5.1.2 Schema Definition

A schema defines attribute names and data types that can be used when defining an authorization policy.

To describe the names and types of attributes, you define a schema. You must define a schema in a file called `schema.dcl`. It must be located in the DCL root folder. If attributes are defined in the schema, you can use the grammar element `WHERE` with these attributes. All usages of schema attributes are validated to refer to existing names and that their usage context complies with the respective data types.

Output Code

```
├─ <application>
└─ ams-policies-deployer
    ├─ package.json
    └─ dcl
        ├─ sales
        │  └─ policy_a.dcl
        │  └─ policy_b.dcl
        └─ schema.dcl
```

Attributes in the schema can be used in the `WHERE` grammar element. The usage of all attributes are validated to make sure that they refer to existing names and that their usage context complies with their type.

```
SCHEMA {
    <attribute>: <data_type>
}
```

Data types for attribute values include String, Number, Boolean, and Structure, with options for arrays.

Use the following data types:

- String, String[]
- Number, Number[]
- Boolean, Boolean[]
- Structured type

You can also assign a special null value to an attribute.

A structured type allows specifying hierarchical types.

- The following schema contains an attribute with the data type "number" and an attribute name with the data type "string".

Sample Code

```
SCHEMA {
    salesID: Number,
    Country: String
}
```

- The schema must always define all data types of the attribute values. In the following example code, `Sales` is a structure that groups different attributes related to `Sales`.

Sample Code

```
SCHEMA {
     Sales: {
       Country: String,
       SalesID: Number}
 }
```

There is a predefined default `$user` structure that is populated by the client libraries with OIDC token content in a standard setup.

Sample Code

```
SCHEMA {
    ...
    $user: { // Default $user definition. This is injected if no custom $user
definition is present
        user_uuid: String,
        scim_id: String,
        groups: String[],
        email: String
    }
```

The `$user` structure can be extended with custom attributes.

Related Information

Formats [page 2270]

1.7.5.5.1.3 Authorization Policy Definition

You can grant basic actions like `read` or `write` on resources with an SQL-like syntax. The authorizations are grouped under an authorization policy. You can also combine authorization policies.

You define basic actions in applications. Application developers must make sure that the applications query the actions and resources that are defined in the DCL files. Applications can use actions and resources in the code for authorization checks.

The syntax of an authorization for granting actions looks like this:

```
POLICY <policy_name> {
    GRANT <action> ON <resource> ;
}
```

Examples

- A simple policy that enables the user to perform a `read` action on all resources.

  Sample Code

  ```
  POLICY readAll {
      GRANT read ON * ;
  }
  ```

- A simple policy that enables the user to perform `read` and `write` actions on all resources.

  Sample Code

  ```
  POLICY readAndWriteAll {
      GRANT read, write ON * ;
  }
  ```

- A simple policy, which restricts all resources only on defined attribute values without any action. Use the `WHERE` condition to refer to attributes.

  Sample Code

  ```
  POLICY CountryCode {
      GRANT * ON * WHERE Country = 'US' OR Country = 'DE';
  }
  ```

- A rule that allows the user to do action `read` on all resources with an special attribute value. Use `WHERE` to implement a condition that restricts a `read` action on resources by attributes. You can also use multiple conditions on resources, for example according to a list of attribute values, a range of numbers, or a part of a name.

  Sample Code

  ```
  POLICY salesOrders {
      GRANT read ON * WHERE Country = 'US';
  }
  ```

  Sample Code

  ```
  POLICY salesOrders {
      GRANT read ON * WHERE Country IN ('US','DE','FR');
  }
  ```

  Sample Code

  ```
  POLICY salesOrders {
      GRANT read ON * WHERE Country = 'DE' AND SalesID BETWEEN 100 AND 200;
  ```

```
}
```

Sample Code

```
POLICY salesOrders {
    GRANT read ON SalesOrders WHERE Name LIKE 'Winter%'
}
```

- This is a policy that allows all actions with restriction on special resources. Asterisk (*) means that all actions are granted. No attribute conditions restrict the policy.

Sample Code

```
POLICY salesOrders {
    GRANT * ON SalesOrders;
}
```

Sample Code

```
POLICY salesOrders {
    GRANT * ON SalesOrders, SalesOrderItems, SalesOrderLists;
}
```

- A policy with actions on special resources where additionally attribute values restrict the user authorization.

Sample Code

```
POLICY salesOrders {
    GRANT read, activate ON SalesOrders WHERE SalesOrders.Country = 'US'
                                          AND SalesOrders.SalesID BETWEEN
100 AND 300;
}
```

Sample Code

```
POLICY salesOrders {
    GRANT read, activate ON SalesOrders, SalesOrderItems WHERE
SalesOrders.Country = 'US'
                                                          AND
SalesOrders.SalesID BETWEEN 100 AND 300
                                                          AND
SalesOrderItems.Country = 'US'
                                                          AND
SalesOrderItems.SalesID BETWEEN 100 AND 200;
}
```

- Maybe if all attributes are related to all resources, you can keep the code short by entering only attributes with values which then relates to all resources.

Sample Code

```
POLICY salesOrders {
    GRANT read, activate ON SalesOrders, SalesOrderItems WHERE Country =
'US'
```

```
                                                                AND SalesID
BETWEEN 100 AND 300;
}
```

- A policy with multiple rules.

  Sample Code

  ```
  POLICY salesOrders {
      GRANT read ON * WHERE Country = 'US' AND SalesID BETWEEN 100 AND 200;
      GRANT write ON * WHERE Country = 'DE' AND SalesID BETWEEN 100 AND 200;
  }
  ```

You can create a policy with actions and restrict it to defined resources. Here you use `WHERE` with multiple conditions.

```
POLICY <policy_name> {
    GRANT <action1>, <action2> ON <resource> WHERE <attribute_name> =
'<attribute_value>'
                                        AND <attribute_name> BETWEEN <number>
AND <number>;
}
```

- Sample Code

  ```
  POLICY salesOrders {
      GRANT read, activate ON salesGroup WHERE salesGroup.Country = 'US'
                                           AND salesGroup.SalesID BETWEEN
  100 AND 300;
                                           }
  ```

- Sample Code

  ```
  POLICY salesOrders {
      GRANT read, activate ON salesGroup, salesGroupItems WHERE
  salesGroup.Country = 'US'
                                                          AND
  salesGroup.SalesID BETWEEN 100 AND 300
                                                          AND
  salesGroupItems.Country = 'US'
                                                          AND
  salesGroupItems.SalesID BETWEEN 100 AND 200;
                                                          }
  ```

- Sample Code

  ```
  POLICY salesOrders {
      GRANT read, activate ON salesGroup, salesGroupItems WHERE Country =
  'US'
                                                          AND SalesID
  BETWEEN 100 AND 300;
  }
  ```

You can also create a policy with multiple actions on the same resources depending on different conditions with attribute values.

```
POLICY <policy_name> {
    GRANT <action1> ON <resource> WHERE <attribute1> = '<attribute_value1>' AND
<attribute2> BETWEEN <number1> AND <number2>;
    GRANT <action2> ON <resource> WHERE <attribute1> = '<attribute_value2>' AND
<attribute2> BETWEEN <number1> AND <number2>;
}
```

- Sample Code

```
POLICY salesOrders {
    GRANT read ON * WHERE Country = 'US' AND SalesID BETWEEN 100 AND 200;
    GRANT write ON * WHERE Country = 'DE' AND SalesID BETWEEN 100 AND 200;
    }
```

Combining Authorization Policies

To combine multiple authorization policies, define a new policy, which includes already existing policies by `USE`.

```
POLICY <policy_name> {
    USE <policy_name1>;
    USE <policy_name2>;
}
```

- Sample Code

```
POLICY allAdmins {
    USE salesOrders;
    USE readAll;
    }
```

- You can give this new compound policy its own rules.

  Sample Code

```
POLICY allAdmins {
    USE salesOrders;
    USE readAll;
    GRANT write ON salesOrders WHERE Country IN ('US', 'DE');
    }
```

The `RESTRICT` clause allows you to narrow down the scope of an authorization policy. It enforces that an operation is permitted only when certain attribute-based conditions are true.

Sample Code

```
POLICY "SalesICADelUpdAdmins" {
   USE users.DELETE_USERS
      RESTRICT
         user.costCenter = 'SalesICA'
```

```
   USE users.UPDATE_USERS
      RESTRICT
         user.costCenter = 'SalesICA'
;
}
```

Combination Algorithms

The combination algorithms for policies, rules, and the list entries for actions and resources are fixed.

Combination Algorithms

Syntax Element	Combination Algorithm
Policy	Always **OR**
Rule	**OR**
Action	**OR**
Resource	**OR**

1.7.5.5.1.4 Base Authorization Policies

Base authorization policies are delivered with the respective application.

In the administration console, administrators can use them to create their own refined custom authorization policies, which they assign to users. We recommend that application developers define the base policies with the basic authorizations that are required for the specific application. For more information, see Configuring Authorization Policies [page 2244].

> **Note**
>
> Currently, the administration console doesn't support the creation of tenant-specific base policies.

1.7.5.5.2 Grammar and Syntax Rules

DCL has a set of grammar and syntax rules you must observe when you define authorization policies.

Parent topic: Data Control Language (DCL) [page 2251]

Related Information

Structure of the Data Control Language (DCL) [page 2252]
Advanced Features [page 2269]

1.7.5.5.2.1 Naming Conventions

Naming conventions in Authorization Management involve using regular or delimited identifiers for objects like policies, actions, resources, and attributes, ensuring consistency and clarity in application development. This content is useful for developers to understand how to properly name and manage authorization elements within their applications, promoting uniformity and preventing naming conflicts.

The objects of Authorization Management include authorization policies, actions, resources, and attributes. The names of actions, resources, and attributes are defined in the application and are sent to Authorization Management using the authorization content. A set of valid characters makes it possible for developers to define the names in the same style as all other elements used for development.

The user names and user groups names are defined in the identity provider. There is no naming convention from DCL side. Within DCL, we must handle all kinds of users and group names. It's possible to use unquoted and double-quoted names for users and user groups.

- The name for an object (authorization policy, action, resource, or attribute) can be specified using regular (unquoted) identifiers or delimited (double-quoted) identifiers:
  - Policy name specified using a regular identifier in a `POLICY` statement

    Sample Code

    ```
    POLICY readAll { ... };
    ```

  - Policy name specified using a delimited identifier in a `POLICY` statement:

    Sample Code

    ```
    POLICY "read All"  { ... };
    ```

  - Action name specified using a regular identifier:

    Sample Code

    ```
    GRANT read ON * ;
    ```

  - Action name specified using a delimited identifier:

    Sample Code

    ```
    GRANT "read" ON * ;
    ```

- The maximum length of an authorization policy is 127 characters.
- A name can't be identical to another object of Authorization Management with the same type. For example: there is only one authorization policy with name `Admin`.

String values of attributes are always quoted with `'` (single quotes). This is also valid for identifiers.

Sample Code

```
POLICY salesOrders {
    GRANT read ON SalesOrders, SalesOrderItems WHERE Country = 'DE'
}
```

1.7.5.5.2.2 Input Parameters for Authorization Policies

In authorization policies, you define rules with conditions using attribute constraints like `IS RESTRICTED` and `IS NOT RESTRICTED`, allowing for flexible user permissions based on attributes while restricting actions and resources.

In an authorization policy, you can define rules with conditions that contain the attribute constraint `IS NOT RESTRICTED`. If you assign such a policy to a user, it means that this user is fully authorized based on the attributes. Only the actions and resources are restricted.

In contrast, a condition containing the attribute constraint `IS RESTRICTED` means that this user's attribute-based authorization is restricted. This condition can be refined in a `USE` clause.

See the difference between `IS RESTRICTED` and `IS NOT RESTRICTED` in the example below. It shows how user authorizations only differ if no condition is given. In that case, `IS RESTRICTED` defaults to false, and `IS NOT RESTRICTED` defaults to true. However, if a condition is given (`CountryCode = 'IT'` in the example below), a specified restriction replaces either the general restriction or `IS NOT RESTRICTED`.

Example

Sample Code

```
POLICY readAll {
    GRANT read ON * WHERE CountryCode IS RESTRICTED;
    GRANT write ON * WHERE CountryCode IS NOT RESTRICTED;
}
POLICY readAllNoCondition {
    USE readAll;
}
POLICY readAllItaly {
    USE readAll RESTRICT CountryCode = 'IT';
}
TEST readAll {
    // Policy assigned directly
    EXPECT DENY FOR read ON * POLICY readAll; // If the policy is assigned
directly, IS RESTRICTED evaluates to false
    EXPECT GRANT FOR write ON * POLICY readAll; // If the policy is
assigned directly, IS NOT RESTRICTED evaluates to true
    // Policy USEd, but no condition for CountryCode is given
    EXPECT DENY FOR read ON * POLICY readAllNoCondition; // If the policy
is USEd, but no condition for CountryCode is given, IS RESTRICTED evaluates
to false
    EXPECT GRANT FOR write ON * POLICY readAllNoCondition; // If the policy
is USEd, but no condition for CountryCode is given, IS NOT RESTRICTED
evaluates to true

    // Policy USEd, and a condition for CountryCode is given
    EXPECT GRANT FOR read, write ON * POLICY readAllItaly INPUT
{CountryCode: 'IT'}; // If the policy is USEd and a condition for
CountryCode is given, both IS RESTRICTED and IS NOT RESTRICTED behave the
same way
```

```
    EXPECT DENY FOR read, write ON * POLICY readAllItaly INPUT
{CountryCode: 'DE'} // The same applies if the condition is not met
}
```

In the example below, the user is assigned to the `salesOrders` policy and is thus allowed to read all `SalesOrders`.

Example

Sample Code

```
POLICY salesOrders {
    GRANT read ON SalesOrders WHERE ( Country IS NOT RESTRICTED AND SalesID
IS NOT RESTRICTED )
                                  OR ( Country IS NOT RESTRICTED AND Name IS
NOT RESTRICTED);
}
```

With the assignment to the `salesOrdersWinter` policy, users can read all `SalesOrders`, which include the string '`Winter`' in their name.

Example

Sample Code

```
POLICY salesOrdersWinter {
    GRANT read ON SalesOrders WHERE ( Country IS NOT RESTRICTED
                                      AND SalesID IS NOT RESTRICTED AND
Name IS NOT RESTRICTED )
                                  OR ( Country IS NOT RESTRICTED AND Name
LIKE '%Winter%' );
}
```

These policy parameters can be filled while another policy is used. Here, the assignment of `use_salesOrders` allows users to read `salesOrders` in the countries DE, FR, or IT and with `SalesID 200`, or read `salesOrders` in the countries DE, FR, or IT and with a name that contains the string '`Winter`'.

Note

Include all conditions for a combination of resource and action in a single policy. This way, it's easy for administrators to decide which policy to use. It also ensures that the authorizations remain understandable.

Example

Sample Code

```
POLICY use_salesOrders {
    USE salesOrders RESTRICT Country IN ('DE','FR','IT'), SalesID = 200,
Name LIKE '%WINTER%';
}
```

You can repeat the `RESTRICT` section of a policy multiple times and thus define a list of tuples.

Example

Sample Code

```
POLICY salesOrders{
    GRANT read ON SalesOrders WHERE Country IS NOT RESTRICTED AND SalesID
IS NOT RESTRICTED ;
}
POLICY use_salesOrders {
    USE salesOrders RESTRICT Country = 'DE', SalesID = 300
                    RESTRICT Country = 'FR', SalesID = 200
                    RESTRICT Country = 'IT', SalesID = 100;
}
```

The previous definition is not equivalent to the following since it allows any combination of `Country` and `SalesID` (9 combinations in contrast to 3 combinations).

Example

Sample Code

```
POLICY use_salesOrders {
    USE salesOrders RESTRICT Country IN ('DE','FR','IT'), SalesID IN
(300,200,100);
}
```

The `WHERE` clause is deprecated for `USE` statements.

Caution

The correct syntax is `USE <basePolicy> RESTRICT <condition_for_attribute(s)>`. Don't use `WHERE` in a `USE` statement..

1.7.5.5.2.3 Condition Operators

Condition operators provide a set of tools for evaluating data, including equality, range, and pattern matching.

Condition Operators

Operators	Description
=	Equal
<	Less than
>	Greater than

Operators	Description
<=	Less than or equal
>=	Greater than or equal
<>	Not equal
IN	To specify multiple possible values
BETWEEN lower AND upper	Between a certain range, including upper and lower
LIKE pattern	Search for a string pattern (% any number of characters, _ any single character)
LIKE pattern ESCAPE escape-char	Allows searching for % and _ if they are prefixed and `escape-char.escape-char` is a string constant with exactly one character.
NOT IN	To specify multiple values which are not possible
NOT BETWEEN	Values, which aren't within a certain range
NOT LIKE	Values, which don't match the pattern
IS NULL	Attribute value is empty

LIKE Operators

The LIKE operator matches a given value with a pattern. This pattern can contain % for a sequence of 0 to n arbitrary characters and _ for exactly one character.

The escape clause allows specifying a single character, that escapes the pattern characters and hereby allows searching for this character in the input.

LIKE Operators

Pattern	Escape	Input	Matches	Comment
'AB'		'AB'	true	
		'ABC'	false	
'AB%'		'AB'	true	Implements "Starts with"
		'ABC'	true	
		'XABC'	false	

Pattern	Escape	Input	Matches	Comment
'%AB'		'AB'	true	Implements "Ends with"
		'ABC'	false	
		'XAB'	true	
'%AB%'		'AB'	true	
		'ABCD'	true	
		'XABC'	true	
		'A-B'	false	
'_'		'A'	true	
		'AB'	false	
		'ABC'	false	
'AB'	'!'	'AB'	true	Same as if no pattern were specified
		'ABCD'	false	
'AB!%'	'!'	'AB'	false	
		'ABC'	false	
		'AB%'	true	
'A!B'	'!'	'AB'	false	
		'A!B'	true	
'%%AB'	'%'	'AB'	false	
		'XAB'	false	
		'%AB'	true	

Compatibility of Operators and Data Types

Some combinations of operators and data types aren't allowed in DCL as shown in the table below. The compiler enforces these rules.

Compatibility of Operators and Data Types

Operator	String	Number	Boolean	Array (String[], Number[], Boolean[])	Structure
`a = b`	✓	✓	✓	×	×
`a < b`	✓	✓	✓	×	×
`a > b`	✓	✓	✓	×	×
`a <= b`	✓	✓	✓	×	×
`a >= b`	✓	✓	✓	×	×
`a <> b`	✓	✓	✓	×	×
`a [NOT] IN b`	✓ (for `a`)	✓ (for `a`)	✓ (for `a`)	✓ (for `b`)	×
`a [NOT] BETWEEN b AND a`	✓	✓	×	×	×
`a [NOT] LIKE` pattern	✓	×	×	×	×

1.7.5.5.2.4 Attribute Constraints

Use attribute constraints in authorization policy definitions to define conditions for accessing data, such as `IS NULL` and `IS RESTRICTED`.

Attribute Constraints

Constraint	Description
`IS RESTRICTED`	Denies access until it's relaxed in a `RESTRICT` clause of the `USE` statement
`IS NOT RESTRICTED`	Authorizes access until it's constrained in a `RESTRICT` clause of the `USE` statement

Sample Code

```
POLICY salesOrders {
    GRANT read, write ON SalesOrderItems WHERE Status IS NULL;
}
```

Sample Code

```
POLICY CreateOrders {
    GRANT create ON orders WHERE order.total IS RESTRICTED AND
product.category IS NOT RESTRICTED;
}
POLICY CreateSmallOrders {
     USE CreateOrders RESTRICT order.total < 50, product.category IS NOT
RESTRICTED;
}
POLICY CreateSmallAccessoryOrders {
     USE CreateOrders RESTRICT order.total < 50, product.category =
'accessory';
}
```

Only use the `IS (NOT) RESTRICTED` attribute constraint for attributes of the data type string, number, and Boolean. It's not allowed for arrays and structures.

1.7.5.5.3 Advanced Features

You find advanced configuration techniques for authorization policies, focusing on annotations, formatting conventions, and internal functions for enterprise-grade authorization policies.

Parent topic: Data Control Language (DCL) [page 2251]

Related Information

Structure of the Data Control Language (DCL) [page 2252]
Grammar and Syntax Rules [page 2261]

1.7.5.5.3.1 Annotations

Use annotations, for example, to insert a description of the authorization policy.

@description

Use a `@description` annotation to define a description of the policy.

```
@description: '<policy_description'
POLICY <policy_name> {
    GRANT <action> ON <resource1>, <resource2> WHERE <attribute_name> =
'<attribute_value>'
```

```
}
```

- Sample Code

```
@description: 'Policy for reading sales orders for USA'
POLICY salesOrders {
    GRANT read ON salesGroup, salesGroupItems WHERE Country = 'US'
}
```

@label

Use this annotation to define a policy display name that can be changed by the customer.

Sample Code

```
@label: 'Sales Order Policy'
POLICY salesOrders {
    GRANT read ON salesOrders, salesOrderItems WHERE Country = 'US'
}
```

1.7.5.5.3.2 Formats

Here you find rules for creating valid identifiers in DCL, such as how to quote and how to separate packages. Identifiers are basically similar to Java identifiers.

Identifier Format

1. An unquoted identifier must begin with a Unicode character
   - Uppercase letter
   - Lowercase letter
   - Title case letter
   - Modifier letter
   - Other letter
   - Letter number
   - Currency symbol
   - Connector punctuation
2. Next, you can use the following characters.
   - A start character from above
   - Digit

> ### Note
>
> Don't use the following characters.

- Surrogate
- Formatting character
- Modifier symbol
- Modifier letter

Identifiers are basically similar to Java identifiers.

> ### Example
>
> `µSeconds`, `PublicTransport`

Enclose an identifier by the double quotes (`" ... "`) if you use dots or blank spaces. Remember that this also applies for keywords like `POLICY`.

A quoted identifier can contain any character that isn't a surrogate.

> ### Example
>
> `/value$to`

In a quoted identifier, the character `"` (double quote) must be escaped. See escaping rules.

Identifiers are case-sensitive. For example, `CUSTOMER` and `Customer` are not the same identifiers.

An unquoted identifier is equal to the same text in double quotes.

String Constant Format

A string constant must begin and end with a single quote (`'`).

> ### Sample Code
>
> ```
> POLICY salesOrdersProduct {
>     GRANT read ON product WHERE Country = 'US';
> }
> ```

Keywords

> ### Note
>
> Keywords are case insensitive. This means that `KEYWORD`, `keyword`, `Keyword`, or `KeyWoRd` are all valid notations for the same.

Escaping Rules

The following escape sequences are defined within string constants and quoted identifiers.

Escaping Rules

Sequence	Value
`\"`	`"`
`\\`	`\`
`\t`	Tabulator
`\uHHHH`	Unicode `HHHH`
`\n`	New line
`\r`	Carriage return
`\f`	Form feed
`\b`	Backspace
`\/`	`/` (compliance for JSON)

Hints for Tools to Generate DCL

- Avoid generating an unquoted identifier format. Always use double-quoted strings and escape at least the characters `"` (double quote) and `\` (backspace).
- When generating string constants, escape at least the characters `'` (single quote) and `\` (backspace).
- It's not necessary to quote package qualifiers like `a.b.c.d` because a package name mustn't contain special characters. However, package segments can be quoted, for example `"a"."b"."c"."d"."Policy Name"` is equal to `a.b.c.d."Policy Name"` in a DCL source file.

1.7.5.5.3.3 Default Authorization Policies

Default authorization policies are special base policies applied at runtime without an explicit assignment to a user. They needn't be passed to the DCN runtime explicitly, similar to internal polices.

Both default and internal policies are special cases of base authorization policies. Base policies are all policies that are defined by the developers of the application and deployed together with the application. They exist in

all tenants. However, custom policies are created specifically for a customer by administrators. Custom policies only exist in the customer's SAP Cloud Identity Services tenants.

You can define an authorization policy as a default authorization policy by using the `DEFAULT` element.

Sample Code

```
DEFAULT POLICY defaultSalesOrders {
    GRANT read ON SalesOrders, SalesOrderItems WHERE salesGroup IN
$user.groups
}
```

The `defaultSalesOrders` default policy can't be assigned to a user because it's automatically evaluated by the runtime.

1.7.5.5.3.4 Internal Policies

Developers can create authorization policies for internal purposes. The policies aren't deployed and don't appear in the administration console.

You can use internal authorization policies and map them to a technical client, for example for app-to-app communication. In this case, the policies are set at runtime and are passed to the decision engine in the client libraries. For more information, see SAP Cloud Identity Services Developer Guide .

You can create internal policies by using the keyword `INTERNAL`.

Effect of Using the INTERNAL Keyword

Element	Description	Effect
Base policy	A USE of such a policy isn't allowed.	Compile error
Tenant-specific policy	Annotation isn't allowed at tenant level.	Compile error

In a downstream tool like Authorization Management, an `INTERNAL` element doesn't show in the user interface. This prevents reuse (not allowed as described above) and policy assignment.

You can't combine an `INTERNAL POLICY` with `DEFAULT`. It should be explicitly set with the DCL runtime client API. You can also mark it with `INTERNAL`.

1.7.6 Configuring Terms of Use

You can configure a custom terms of use document by creating a new document, adding and editing its language versions, and defining the document for an application.

Every time you want to update the terms of use document you add a new language versions for the document.

Each terms of use document can have one draft and one active version. You can view all language version in the administration console. Optionally you can edit the draft version of a document before making it an active version.

Once a terms of use document is created it can't be renamed.

For each language version, you have to upload a text file. You can define a terms of use document in the following languages:

Arabic, Azerbaijani, Bulgarian, Catalan, Chinese (PRC), Chinese (Taiwan), Croatian, Czech, Danish, Dutch, English (United Kingdom), English (United States), Estonian, Finnish, French (Standard), French (Canada), German (Standard), Greek, Hebrew, Hungarian, Italian, Japanese, Korean, Latvian, Malay, Norwegian, Polish, Portuguese (Portugal), Romanian, Russian, Serbian, Slovak, Slovenian, Spanish (Standard), Spanish (Mexico), Swedish, Thai, Turkish, Ukrainian, Vietnamese, Welsh.

The language is defined in the following order of importance:

- By the `locale` parameter in the URL used for accessing the application.

  > **Note**
  >
  > If the SP is configured to support a specific language, only this language is used by the application.

- By the language of the application's browser.

  > **Note**
  >
  > The application takes the browser language only if the SP's language is not selected, and the `locale` parameter isn't set in the URL. The default browser setting is *English*.

> **Note**
>
> If there is only one language for the document, it is recommended to be *English (United States)*.
>
> If the languages are more than one, one of them must be *English (United States)*.

To configure terms of use in the administration console, do the following:

1. Create a New Terms of Use Document [page 2275]
   Create a new terms of use document in the administration console.
2. Add New Language of a Terms of Use Document [page 2276]
   To add a new language of a terms of use document, you must upload a UTF-8 encoded plain text file containing the terms of use statement.
3. Add New Version of a Terms of Use Document [page 2277]
   Add a new version of a terms of use document and make it active.
4. Define a Terms of Use Document for an Application [page 2278]
5. (Optional) View and Edit a Terms of Use Document [page 2279]
   Tenant administrators can view terms of use documents uploaded in the administration console for SAP Cloud Identity Services and optionally edit the draft versions of those documents.
6. (Optional) Delete a Terms of Use Document [page 2281]
   Delete a terms of use document set or a version of a terms of use document.

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.6.1 Create a New Terms of Use Document

Create a new terms of use document in the administration console.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Application & Resources*, choose the *Terms of Use Documents* tile.

   This operation opens a list of the terms of use documents.
3. Choose the *+Add* button on the left-hand panel in order to add a new terms of use document to the list.
4. Specify a name for the document.

   > ⓘ Note
   >
   > The name can contain only Latin letters, numbers or the underscore character, or a combination of them. The lowercase letters are converted to uppercase letters.
5. Save your selection

   Once the document has been created, the system displays the message `Terms of use document <name of document> created`.

Task overview: Configuring Terms of Use [page 2273]

Next task: Add New Language of a Terms of Use Document [page 2276]

1.7.6.2 Add New Language of a Terms of Use Document

To add a new language of a terms of use document, you must upload a UTF-8 encoded plain text file containing the terms of use statement.

Prerequisites

You have created a terms of use document in the administration console. For more information, seeCreate a New Terms of Use Document [page 2275] .

Context

You can add one or more languages of the document.

> **Note**
>
> If there is only one language for the document, it is recommended to be *English (United States)*.
>
> If the languages are more than one, one of them must be *English (United States)*.

The plain text document can include the following tags: `<a>`, `<strong>`, and `<em>`.

> **Example**
>
> - `See <a href="https://www.example.com">Subscription conditions</a>`
> - `<strong>Additional requirements</strong>`
> - `<em>Important:</em>`

To open the link from the document in the preview in the admin console, right-click the link and open it in a new tab.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Terms of Use Documents* tile.

   This operation opens a list of the terms of use documents.

3. To add a language to a terms of use document, choose that document from the list on the left.

> → Tip
>
> Type the name of the document in the search field to filter the list items.

4. Choose the *Add* button in the details view on the top right-hand corner.
   a. Choose the *Active Language Version* radio button.
   b. Select the language from the drop-down.
   c. Specify the plain text file for the language.

   > ⓘ Note
   >
   > Use a file with an extension `.txt`.

   d. Choose *Add* to save your selection

      Once the file has been uploaded, the system displays the message `Terms of use file uploaded`.

Task overview: Configuring Terms of Use [page 2273]

Previous task: Create a New Terms of Use Document [page 2275]

Next task: Add New Version of a Terms of Use Document [page 2277]

1.7.6.3 Add New Version of a Terms of Use Document

Add a new version of a terms of use document and make it active.

Context

To update an existing terms of use document you must add a new version for each language of this document by uploading a UTF-8 encoded plain text file containing the updated terms of use statement. Once the new version is uploaded it is in a status draft. You must activate it to replace the current active version. To make a draft version active, you must have draft versions for all the languages that the current active version has.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Terms of Use Documents* tile.

   This operation opens a list of the terms of use documents.
3. To add a new language version to a terms of use document, choose that document from the list on the left.

> → Tip
>
> Type the name of the document in the search field to filter the list items.

4. Choose the *Add* button in the details view on the top right-hand corner.
   a. Choose the *Draft Language Version* radio button.
   b. Select the language from the drop-down for that version.
   c. Specify the plain text file for the language.

      > ⓘ Note
      >
      > Use a file with an extension `.txt`.

   d. Choose *Add* to save your selection

      Once the file has been uploaded, the system displays the message `Terms of use file uploaded`.
5. Repeat the procedure for all languages that have active versions.
6. **Optional:** To make the draft version active, choose *Activate* in the details view on the top right-hand corner.

Task overview: Configuring Terms of Use [page 2273]

Previous task: Add New Language of a Terms of Use Document [page 2276]

Next task: Define a Terms of Use Document for an Application [page 2278]

1.7.6.4 Define a Terms of Use Document for an Application

Context

The terms of use document is displayed on the registration form, which form appears when the user chooses the *Register Now* link on the logon page. It is also shown to the user when the registration information has been upgraded. By default, the application is set not to use a terms of use document on the registration and upgrade forms. To change this configuration, you have to update the *None* setting in the administration console.

If you have the authorization, you can create a custom terms of use document. This document is used once you have set it for the respective application.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Authentication and Access* tab.
5. Under *POLICIES*, choose *Terms of Use*.
6. Select the radio button for the Terms of Use.
7. Save your selection.

Task overview: Configuring Terms of Use [page 2273]

Previous task: Add New Version of a Terms of Use Document [page 2277]

Next task: (Optional) View and Edit a Terms of Use Document [page 2279]

Related Information

Create a New Application [page 1922]

1.7.6.5 (Optional) View and Edit a Terms of Use Document

Tenant administrators can view terms of use documents uploaded in the administration console for SAP Cloud Identity Services and optionally edit the draft versions of those documents.

Context

The content of the terms of use documents are visualised in the administration console

Task overview: Configuring Terms of Use [page 2273]

Previous task: Define a Terms of Use Document for an Application [page 2278]

Next task: (Optional) Delete a Terms of Use Document [page 2281]

View a Terms of Use Document

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Terms of Use Documents* tile.

   This operation opens a list of the terms of use documents.
3. Choose the document from the list on the left.

   > → Tip
   >
   > Type the name of the document in the search field to filter the list items.
4. Choose the version and language from the drop down.

   The content is displayed on the screen.

   To view another terms of use document, repeat the procedure.

Edit a Draft Version

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Terms of Use Documents* tile.

   This operation opens a list of the terms of use documents.
3. Choose the document from the list on the left.

   > → Tip
   >
   > Type the name of the document in the search field to filter the list items.
4. Choose the draft version from the drop down.
5. Choose a language for the draft version from the drop down.
6. Choose *Edit* in the details view on the top right-hand corner.
7. Upload the edited document.
8. Choose *Edit* in the popup to confirm your changes.

1.7.6.6 (Optional) Delete a Terms of Use Document

Delete a terms of use document set or a version of a terms of use document.

Context

You can delete a specific language version or the entire set of terms of use documents. Deleting the entire set of terms of use documents, you delete all language versions for this set. The terms of use document set is removed from the list of the document sets in the administration console for SAP Cloud Identity Services, when you delete it.

The documents will be deleted no matter if they are accepted or not by any users.

> ⓘ Note
>
> You can delete only documents that are not defined for applications. To delete a document version or entire document sett that is defined for an application, you first you must set another terms of use for the applications. For more information, see Define a Terms of Use Document for an Application [page 2278].

Task overview: Configuring Terms of Use [page 2273]

Previous task: (Optional) View and Edit a Terms of Use Document [page 2279]

Delete Entire Documents Set

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Terms of Use Documents* tile.

   This operation opens a list of the terms of use documents.
3. To delete the entire terms of use document set, choose that set from the list on the left.

   > → Tip
   >
   > Type the name of the document in the search field to filter the list items.
4. Choose *Delete Set* button and confirm the operation in the pop-up that appears.

   If a document is defined for an application, define a new terms of use document for that application and repeat the step.

   If the document set is deleted, the system displays the message `Terms of Use document set deleted`.

Delete a Language Version

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Terms of Use Documents* tile.

   This operation opens a list of the terms of use documents.
3. To delete a language version to a terms of use document, choose that document from the list on the left.

   > → Tip
   >
   > Type the name of the document in the search field to filter the list items.

   a. Select the version from the drop-down that you want to delete.
   b. Select the language from the drop-down for that version that you want to delete.
   c. Choose *Delete* button and confirm the operation in the pop-up that appears.

      If the document is defined for an application, define a new terms of use document for that application and repeat the step.

   If the document version is deleted, the system displays the message `Terms of Use document deleted`.

1.7.7 Configuring Email Templates

Tenant administrators can use the default or a custom email template set for the application processes.

Overview

The email template set is a semantic grouping of different email templates. Each email template from the set is used for the respective application process, such as self-registration, for example. When a user makes a registration for an application, Identity Authentication uses the email template for self-registration from the template set to communicate with that user.

The administration console for SAP Cloud Identity Services provides the possibility to use a default email template set with English templates only. The default set is named *Default*.

You can also configure your own templates in a custom template set. There you can customize the texts and branding according to your needs.

You can define a set of email templates with different language versions for the following processes:

Email Template	Request Flow	Description
Self-Registration	User	This email template is used when a user registers via the Registration page. The user then receives an email with instructions about how to activate his or her account. The name of the email template used for this process is *Self-Registration*.
On-Behalf Registration	System	This email template is used when somebody else registers the user on his or her behalf via the import user option for Identity Authentication. In this case, the registered user receives an email with instructions about how to activate his or her account. The name of the email template used for this process is *On-Behalf Registration*.
Invitation	System	This email template is used when a user invites another user for registration via the Invitation REST API. In this case, the invitee receives an email with instructions about how to register. The name of the email template used for this process is *Invitation*.
Forgot Password	• User • System	This email template is used when: • a user wants to change his or her password by going through the Forgot Password page. In this case, the user receives an email with instructions about how to change his or her password. The name of the email template used for this process is *Forgot Password*. • an administrator sends a reset password email via the administration console. For more information, see Send Reset Password Email [page 2381].

Email Template	Request Flow	Description
Locked Password	System	This email template is used when a user locks his or her password by exceeding the allowed number of logon attempts. In this case, the user receives an email with instructions about how to log on. The name of the email template used for this process is *Locked Password*.
Reset Password	System	This email template is used when: a user has to reset his or her password. In this case, the user receives an email with instructions about how to reset his or her password. The name of the email template used for this process is *Reset Password*. The email template for *Reset Password* process are used when the process is triggered from the system, based on the password policy (for example, the user's password length is not enough according to the password policy requirements) and the user receive reset password notification email.
Deactivate TOTP Device	User	This email-template is used when a user has requested to receive a passcode via email. The passcode is used to deactivate the devices used to generate passcodes for TOTP Two-Factor Authentication. The name of the email template used for this process is *Deactivate TOTP Device*.
Email OTP Code	User	This email-template is used when a user has requested to receive an 8-digit code via email. The user needs the code for two-factor authentication.

Email Template	Request Flow	Description
Send Security Alert	System	This email-template is used when the user's password is set, changed, or reset, the email, or login name is changed, or when TOTP or Web TFA devices are activated or deactivated. The name of the email template used for this process is *Send Security Alert*. **Note** The sending of security alert emails is switched off by default. For more information, see Send Security Alert Emails [page 2209].

To activate a user registration or to reset a password, users choose links sent to them in the emails. For these cases, you can use placeholders. For more information about which placeholders can be used, see Edit or Add an Email Template Set [page 2289].

Restriction

If you select a corporate identity provider, the option to configure email templates is not possible. In this case you can access only some of the custom configurations for the applications. The configurations under the *Authentication and Access* and *Branding and Layout* tabs are partially visible. For more information, see Choose Default Identity Provider for an Application [page 2067].

You can also define which languages each email template uses, and you can set custom versions for each language. You can set the following languages:

Arabic, Azerbaijani, Bulgarian, Catalan, Chinese (PRC), Chinese (Taiwan), Croatian, Czech, Danish, Dutch, English (United Kingdom), English (United States), Estonian, Finnish, French (Standard), French (Canada), German (Standard), Greek, Hebrew, Hungarian, Italian, Japanese, Korean, Latvian, Malay, Norwegian, Polish, Portuguese (Portugal), Romanian, Russian, Serbian, Slovak, Slovenian, Spanish (Standard), Spanish (Mexico), Swedish, Thai, Turkish, Ukrainian, Vietnamese, Welsh.

The language for the email template sets is set according to the following order of priorities:

User request flow - when the user requests a process, for example Forgot Password, the emails use the language that the user's browser is set to.

- If the language isn't in the list of supported languages, the emails use *English*.
- If the language is in the list of supported languages, but there isn't a template for that language, the emails use English.
- If the language is in the list of supported languages, and there is a template for that language, the emails use this language.

System request flow - for example when the administrator chooses the Reset Password option in the administration console, the emails use the language set in the profile of the user. If there isn't a template for that language, the emails use *English* instead.

If you want to use a custom email template you should create one if it doesn't exist. Add or edit the email template set, if necessary, and then define that email template set for the application. To add or edit the email template, first you must open the uploaded email templates, and then save a copy. Optionally you can delete an email template set or a language version for a specific application process.

You can perform the following steps:

Create a New Email Template Set [page 2287]
: Tenant administrators can create a new set of email templates so that each template in the set can have a custom language version.

View Email Template Document [page 2288]
: Tenant administrators can view language email templates in the template set uploaded in the administration console for SAP Cloud Identity Services.

Edit or Add an Email Template Set [page 2289]
: Tenant administrators can configure language versions of each template in the template set. They can also set a custom template for each language, and change the name of each template set.

Define an Email Template Set for an Application [page 2295]
: Tenant administrators can define the email template set that the application uses.

Delete an Email Template Set [page 2297]
: Tenant administrators can delete an email template set or a language version for a specific application process.

Allowed Placeholders per Email Template [page 2298]
: This document describes which placeholders can be used in each email template.

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]

Configuring OpenID Connect (OIDC) [page 2636]

1.7.7.1 Create a New Email Template Set

Tenant administrators can create a new set of email templates so that each template in the set can have a custom language version.

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

To create an email template, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Application & Resources*, choose the *Email Template Sets* tile.

   This operation opens a list of the template sets.
3. Press the *Add* button on the left-hand panel to add a new template set to the list.
4. Specify a name for the set.
5. Save your selection

   Once template has been created, the system displays the message `Template set <name of set> created`.

Task overview: Configuring Email Templates [page 2282]

Related Information

View Email Template Document [page 2288]
Edit or Add an Email Template Set [page 2289]
Define an Email Template Set for an Application [page 2295]
Delete an Email Template Set [page 2297]

Allowed Placeholders per Email Template [page 2298]

1.7.7.2 View Email Template Document

Tenant administrators can view language email templates in the template set uploaded in the administration console for SAP Cloud Identity Services.

Prerequisites

- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have added the plain text and HTML template files in the email template set. For more information, see Edit or Add an Email Template Set [page 2289].

> **Note**
>
> Both the TXT and HTML formats are included in the emails sent to a user. What the user sees, depends on the settings of his or her email client.

Context

You can view both the TXT and HTML formats of the emails sent to a user.

To view an email template, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Email Template Sets* tile.

   This operation opens a list of the template sets.
3. Choose the list item of the template set you want to view.
4. Select the tab signifying the process, whose template you want to view.
5. View the file:
   - To view the plain text file, choose the plain text file icon next to the respective language version.
   - To view the HTML file, choose the HTML file icon next to the respective language version.

   The file opens in a new tab.

   To view another template, repeat the procedure.

Task overview: Configuring Email Templates [page 2282]

Related Information

Create a New Email Template Set [page 2287]
Edit or Add an Email Template Set [page 2289]
Define an Email Template Set for an Application [page 2295]
Delete an Email Template Set [page 2297]
Allowed Placeholders per Email Template [page 2298]

1.7.7.3 Edit or Add an Email Template Set

Tenant administrators can configure language versions of each template in the template set. They can also set a custom template for each language, and change the name of each template set.

Prerequisites

- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have prepared the plain text (TXT) and HTML template files that will be added in the email template set.
  For that purpose, you need to open the default template set for the respective application process in the administration console. After that, copy and paste the texts in a new file, edit the texts according to your needs, and save a copy of your versions of the documents. For more information how to open the default templates, see View Email Template Document [page 2288].

  > → Remember
  >
  > .Both, the TXT and HTML template files, must be saved with UTF-8 encoding.

  If you want to use a custom template set in another language, first you need to open the template set for the respective application process in the administration console. After that, copy and paste the texts in a new file, edit the texts in the respective language, and again save a copy of your versions of the documents. For more information how to open the default templates, see View Email Template Document [page 2288]

  > ⚠ Caution
  >
  > You should use only the placeholders used in the template documents. If you use other placeholders, even if they are in comments, emails for the respective process will not be sent. For more information, see Allowed Placeholders per Email Template [page 2298].
  >
  > When you edit texts in languages written in Right-To-Left (RTL) direction check that the placeholders are situated in the right place.

→ Remember

If you do not specify a version for a custom template of a specific process, users will receive the email from the default template set for this process. If you do not set an email template for the self-registration process for example, users will receive the default activation email when they complete the registration.

If you do not specify a language version for a custom template of a specific process, users will receive the email from the default template set for this process.

ⓘ Note

Both the HTML and TXT formats are included in the emails sent to a user. What the user sees, depends on the settings of his or her email client.

Context

To edit or add an email template, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Email Template Sets* tile.

   This operation opens a list of the template sets.
3. Choose the list item of the template set you want to edit.
4. **Optional:** To change the name of the template set, choose it.
5. Select the tab signifying the process, which you want to add or change the template for.
6. Add another language version by choosing the *+Add Language* button or edit an existing language version by selecting the respective list item.
   a. Choose language from the dropdown.

      ⓘ Note

      The dropdown is grayed out when you are editing an existing language version.

   b. Specify the subject.

      This is the subject of the email that the user receives for the respective process. You are allowed to include placeholders in the subject's text.

      ⓘ Note

      The subject must contain at least one non-space character.

   c. Upload the plain text template file that you have prepared.

The text file can contain placeholders and HTML tags. For more information about the allowed placeholders, see Allowed Placeholders per Email Template [page 2298].

> **→ Remember**
>
> The *${other.securityAlertText}* is mandatory for the *Send Security Alert* template. The *${other.securityAlertText}* is customizable. For more information, see Change Tenant Texts REST API [page 2826]. The key-value pairs that should be changed are `security.alert.<name of the flow>`.
>
> The *${other.totpResetPasscode}* is mandatory for the *Deactivate TOTP Device* template.

See examples for the *Deactivate TOTP Device* and *Email OTP Code* templates below:

Deactivate TOTP Device

> **Example**
>
> Dear ${user.firstName} ${user.lastName},
>
> The passcode to deactivate your Two-Factor Authentication device is ${other.totpResetPasscode}.
>
> Best regards,
>
> Your System Administrator
>
> This email may contain trade secrets or privileged, undisclosed, or otherwise confidential information. If you have received this email in error, you are hereby notified that any review, copying, or distribution of it is strictly prohibited. Please inform us immediately and destroy the original transmittal. Thank you for your cooperation.

Email OTP Code Template

> **Example**
>
> Dear ${user.firstName} ${user.lastName},
>
> The code for Email Two-Factor Authentication is ${other.emailOtpCode}.
>
> Best regards,
>
> Your System Administrator
>
> This email may contain trade secrets or privileged, undisclosed, or otherwise confidential information. If you have received this email in error, you are hereby notified that any review, copying, or distribution of it is strictly prohibited. Please inform us immediately and destroy the original transmittal. Thank you for your cooperation.

d. Upload the HTML template file that you have prepared.

The content of the HTML file must comply with the HTML markup requirements, and the file can contain placeholders. For more information about the allowed placeholders, see Allowed Placeholders per Email Template [page 2298].

> **→ Remember**
>
> The *${other.securityAlertText}* is mandatory for the *Send Security Alert* template. The *${other.securityAlertText}* is customizable. For more information, see Change Tenant Texts REST

API [page 2826]. The key-value pairs that should be changed are `security.alert.<name of the flow>`.

The *${other.totpResetPasscode}* is mandatory for the *Deactivate TOTP Device* template.

See examples for the *Deactivate TOTP Device* and *Email OTP Code* templates below:

Deactivate TOTP Device Template

Example

```
html>
<head>
    <meta http-equiv="Content-Type" content="text/html;
charset=UTF-8" />
<script type="text/javascript"
src="/ruxitagentjs_ICA27SVfqrux_10179191120132458.js"
data-dtconfig="uam=1|app=ea7c4b59f27d43eb|
featureHash=ICA27SVfqrux|rdnt=1|uxrgce=1|bp=2|srms=1,1,,,|
uxrgcm=100,25,300,3;100,25,300,3|dpvc=1|lastModification=1574413554517|
dtVersion=10179191120132458|tp=500,50,0,1|uxdcw=1500|agentUri=/
ruxitagentjs_ICA27SVfqrux_10179191120132458.js|reportUrl=/
rb_c52c6336-5dbb-45f1-aa35-08a6f855165d|rid=RID_2026897975|
rpid=-1322452711|domain=sap.corp"></script></head>
<body style="width: 600px;margin: auto;padding: 0;font-family: Arial,
Helvetica, sans-serif;font-size: 14px;color: #333">
    <!-- gold bar -->
    <table cellspacing="0" cellpadding="0" width="600px" class="header"
style="border-bottom-style: solid;border-bottom-color: #F0AB00;border-
bottom-width: 8px;padding-top: 13px;padding-bottom: 12px">
        <tr style="border-collapse: collapse;width: 600px;padding:
0;margin: 0">
            <td style="border-collapse: collapse;width: 600px;padding:
0;margin: 0">
                <table cellspacing="0" cellpadding="0">
                    <tr style="border-collapse: collapse;width:
600px;padding: 0;margin: 0">
                        <td class="logo" style="border-collapse:
collapse;width: 61px;padding: 0;margin: 0">
                            <img src="${user.sap_mailing_logo}"
style="border: 0;height: auto;line-height: 100%;outline: none;text-
decoration: none" /></td>
                        <td class="tagline" style="border-collapse:
collapse;width: 600px;padding: 0;margin: 0;color: #888;font-weight:
bold;font-size: 12px;padding-top: 2px">The Best-Run Businesses Run
SAP</td>
                    </tr>
                </table>
            </td>
            <#if user.company_logo??>
                <td align="right" style="border-collapse:
collapse;width: 600px;padding: 0;margin: 0">
                    <div class="companylogo" style="height:
32px;overflow: hidden"><img src="${user.company_logo}" style="border:
0;height: 32px;line-height: 100%;outline: none;text-decoration:
none;width: auto" /></div>
                </td>
            </#if>
        </tr>
    </table>
    <table cellspacing="0" cellpadding="0" width="600px">
        <tr style="border-collapse: collapse;width: 600px;padding:
0;margin: 0">
            <td class="main_first" style="border-collapse:
collapse;width: 600px;padding: 0;margin: 0;padding-top: 36px;padding-
bottom: 12px">
```

```
                <h1 style="font-family: Arial, Helvetica, sans-
serif;font-weight: bold;font-size: 20px;color: #555;margin-bottom:
24px">Dear ${user.firstName?html} ${user.lastName?html},</h1>
                <p style="font-family: Arial, Helvetica, sans-
serif;font-size: 14px;line-height: 18px">The passcode to deactivate
your Two-Factor Authentication device is ${other.totpResetPasscode}.</p>
            </td>
        </tr>
    </table>
    <table cellspacing="0" cellpadding="0" width="600px">
        <tr style="border-collapse: collapse;width: 600px;padding:
0;margin: 0">
            <td class="main_last" style="border-collapse:
collapse;width: 600px;padding: 0;margin: 0;padding-top: 12px;padding-
bottom: 36px">
                <p style="font-family: Arial, Helvetica, sans-
serif;font-size: 14px;line-height: 18px">Best regards,
                    <br/> Your System Administrator</p>
            </td>
        </tr>
    </table>
    <table cellpadding="0" cellspacing="0" width="600px">
        <tr style="border-collapse: collapse;width: 600px;padding:
0;margin: 0">
            <td class="colophon" style="border-collapse:
collapse;width: 600px;padding: 0;margin: 0;font-size: 11px;color:
#888;line-height: 13px">
                <p style="font-family: Arial, Helvetica, sans-
serif;font-size: 11px;line-height: 13px;color: #888">This email may
contain trade secrets or privileged, undisclosed, or otherwise
confidential information. If you have received this email in error, you
are hereby notified that any review, copying, or distribution of it
is strictly prohibited. Please inform us immediately and destroy the
original transmittal. Thank you for your cooperation.</p>
            </td>
        </tr>
    </table>
</body>
</html>
```

Email OTP Code Template

Example

```
<html>
<head>
    <meta http-equiv="Content-Type" content="text/html;
charset=UTF-8" />
</head>
<body style="width: 600px;margin: auto;padding: 0;font-family: Arial,
Helvetica, sans-serif;font-size: 14px;color: #333">
    <!-- gold bar -->
    <table cellspacing="0" cellpadding="0" width="600px" class="header"
style="border-bottom-style: solid;border-bottom-color: #F0AB00;border-
bottom-width: 8px;padding-top: 13px;padding-bottom: 12px">
        <tr style="border-collapse: collapse;width: 600px;padding:
0;margin: 0">
            <td style="border-collapse: collapse;width: 600px;padding:
0;margin: 0">
                <table cellspacing="0" cellpadding="0">
                    <tr style="border-collapse: collapse;width:
600px;padding: 0;margin: 0">
                        <td class="logo" style="border-collapse:
collapse;width: 61px;padding: 0;margin: 0">
```

```
                            <img src="${user.sap_mailing_logo}"
style="border: 0;height: auto;line-height: 100%;outline: none;text-
decoration: none" /></td>
                        <td class="tagline" style="border-collapse:
collapse;width: 600px;padding: 0;margin: 0;color: #888;font-weight:
bold;font-size: 12px;padding-top: 2px">The Best-Run Businesses Run
SAP</td>
                    </tr>
                </table>
            </td>
            <#if user.company_logo??>
                <td align="right" style="border-collapse:
collapse;width: 600px;padding: 0;margin: 0">
                    <div class="companylogo" style="height:
32px;overflow: hidden"><img src="${user.company_logo}" style="border:
0;height: 32px;line-height: 100%;outline: none;text-decoration:
none;width: auto" /></div>
                </td>
            </#if>
        </tr>
    </table>
    <table cellspacing="0" cellpadding="0" width="600px">
        <tr style="border-collapse: collapse;width: 600px;padding:
0;margin: 0">
            <td class="main_first" style="border-collapse:
collapse;width: 600px;padding: 0;margin: 0;padding-top: 36px;padding-
bottom: 12px">
                <h1 style="font-family: Arial, Helvetica, sans-
serif;font-weight: bold;font-size: 20px;color: #555;margin-bottom:
24px">Dear ${user.firstName?html} ${user.lastName?html},</h1>
                <p style="font-family: Arial, Helvetica, sans-
serif;font-size: 14px;line-height: 18px">The code for Email Two-Factor
Authentication is ${other.emailOtpCode}.</p>
            </td>
        </tr>
    </table>
    <table cellspacing="0" cellpadding="0" width="600px">
        <tr style="border-collapse: collapse;width: 600px;padding:
0;margin: 0">
            <td class="main_last" style="border-collapse:
collapse;width: 600px;padding: 0;margin: 0;padding-top: 12px;padding-
bottom: 36px">
                <p style="font-family: Arial, Helvetica, sans-
serif;font-size: 14px;line-height: 18px">Best regards,
                    <br/> Your System Administrator</p>
            </td>
        </tr>
    </table>
    <table cellpadding="0" cellspacing="0" width="600px">
        <tr style="border-collapse: collapse;width: 600px;padding:
0;margin: 0">
            <td class="colophon" style="border-collapse:
collapse;width: 600px;padding: 0;margin: 0;font-size: 11px;color:
#888;line-height: 13px">
                <p style="font-family: Arial, Helvetica, sans-
serif;font-size: 11px;line-height: 13px;color: #888">This email may
contain trade secrets or privileged, undisclosed, or otherwise
confidential information. If you have received this email in error, you
are hereby notified that any review, copying, or distribution of it
is strictly prohibited. Please inform us immediately and destroy the
original transmittal. Thank you for your cooperation.</p>
            </td>
        </tr>
    </table>
</body>
</html>
```

7. Save your selection.

   Once the email template has been updated, the system displays the message `Email template <name of template> updated`.

Results

When you edit a previous version of a template in the template set, the changes are applied immediately. The email that is sent contains the latest changes.

When you create a new template set, you must assign the new template set to the application. If you do not assign it, the application will use the currently assigned template.

Task overview: Configuring Email Templates [page 2282]

Related Information

Create a New Email Template Set [page 2287]
View Email Template Document [page 2288]
Define an Email Template Set for an Application [page 2295]
Delete an Email Template Set [page 2297]
Allowed Placeholders per Email Template [page 2298]

1.7.7.4 Define an Email Template Set for an Application

Tenant administrators can define the email template set that the application uses.

Context

Initially, the application uses a default template set with an English language version for all the templates in the set. This template set is named *Default*. If you want to use another email template set, assign it to the respective application.

> **Note**
>
> Both the HTML and TXT formats are included in the emails sent to a user. What the user sees, depends on the settings of his or her email client.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Branding and Layout* tab.
5. Under *EMAIL CONFIGURATIONS*, choose *Email Template Set*.
6. Select the radio button of the email template set the application will use.
7. Save your selection.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

Next Steps

(Optional) Configure custom mail server. To use own mail server for the emails sent for the different application processes, you should configure your mail server in the administration console for SAP Cloud Identity Services.

For more information, see Configure Mail Server for Application Processes [page 2195].

Task overview: Configuring Email Templates [page 2282]

Related Information

Create a New Email Template Set [page 2287]
View Email Template Document [page 2288]
Edit or Add an Email Template Set [page 2289]
Delete an Email Template Set [page 2297]
Allowed Placeholders per Email Template [page 2298]

1.7.7.5 Delete an Email Template Set

Tenant administrators can delete an email template set or a language version for a specific application process.

Context

You can delete only email template sets that aren't assigned to applications. If you want to delete an email template that is assigned to an application, first you must choose another email template set for the applications.

You can't delete the default template set.

If you delete a language version for specific application process, users receive the email from the default template set for this process. If you delete the email template for the self-registration process, for example, users receive the default activation email when they complete the registration.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Email Template Sets* tile.

   This operation opens a list of the template sets.
3. Choose the list item of the template set you want to edit.

Option	Result
Select the tab signifying the process, and delete the respective language version of the template. Repeat the operation as many times as necessary.	Deletes a language version for specific application process.
Choose the *Delete* button at the lower right corner of the screen.	Deletes the entire email template set

Task overview: Configuring Email Templates [page 2282]

Related Information

Create a New Email Template Set [page 2287]
View Email Template Document [page 2288]
Edit or Add an Email Template Set [page 2289]
Define an Email Template Set for an Application [page 2295]
Allowed Placeholders per Email Template [page 2298]

1.7.7.6 Allowed Placeholders per Email Template

This document describes which placeholders can be used in each email template.

The allowed placeholders are marked with *Yes* in the table:

Placeholders per Email Template

Placeholder/ Email Template	Self-Registration	On-Behalf Registration	Invitation	Forgot Password	Locked Password	Reset Password	Send Security Alert	Deactivate TOTP Device	Email OTP Code
`${user.sp_name}`	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
`${user.activate_account_link}`	Yes	Yes	Yes	No	No	No	No	No	No
`${user.firstName}`	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
`${user.lastName}`	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
`${user.mail}`	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
`${user.uid}`	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
`${user.ids_home_link}`	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
`${user.loginName}`	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
`${user.company_logo}`	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Placeholder/ Email Template	Self-Registration	On-Behalf Registration	Invitation	Forgot Password	Locked Password	Reset Password	Send Security Alert	Deactivate TOTP Device	Email OTP Code
`${user.sap_mailing_logo}`	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
`${user.target_url}`	No	Yes	Yes	No	No	No	No	No	No
`${user.reset_password_link}`	No	No	No	Yes	No	Yes	No	No	No
`${user.set_locked_password_link}`	No	No	No	No	Yes	No	No	No	No
`${other.securityAlertText}`	No	No	No	No	No	No	Yes/ Mandatory	No	No
`${other.totpResetPasscode}`	No	No	No	No	No	No	No	Yes	No
${other.emailOtpCode}	No	No	No	No	No	No	No	No	Yes
`${user.inviter_name}`	No	No	Yes	No	No	No	No	No	No
`${user.footer}`	No	No	Yes	No	No	No	No	No	No

Placeholder/ Email Template	Self-Registration	On-Behalf Registration	Invitation	Forgot Password	Locked Password	Reset Password	Send Security Alert	Deactivate TOTP Device	Email OTP Code
`${user.header}`	No	No	Yes	No	No	No	No	No	No

Placeholder	Notes
`${user.sp_name}`	The name of the application for which the email is sent.
`${user.activate_account_link}`	The account activation link sent to the user in the email.
`${user.firstName}`	The first name of the user.
`${user.lastName}`	The last name of the user.
`${user.mail}`	The email of the user.
`${user.uid}`	The user universal unique identifier (Global User ID) number.
`${user.ids_home_link}`	
`${user.loginName}`	The login name of the user.
`${user.company_logo}`	The application logo shown on the logon page and in the emails sent to users.
`${user.sap_mailing_logo}`	
`${user.target_url}`	The URL that the user is redirected to after registration.
`${user.reset_password_link}`	The link provided in the email to reset the password.
`${user.set_locked_password_link}`	The link provided in the email to set a new password,
`${other.securityAlertText}`	The text for the specific security alert.
`${other.totpResetPasscode}`	
${other.emailOtpCode}	
`${user.inviter_name}`	
`${user.footer}`	
`${user.header}`	

→ Remember

The *${other.securityAlertText}* is mandatory for the *Send Security Alert* template. The *${other.securityAlertText}* is customizable. For more information, see Change Tenant Texts REST API [page 2826]. The key-value pairs that should be changed are `security.alert.<name of the flow>`.

The *${other.totpResetPasscode}* is mandatory for the *Deactivate TOTP Device* template.

See example for the *Deactivate TOTP Device* template below:

Parent topic: Configuring Email Templates [page 2282]

Related Information

Create a New Email Template Set [page 2287]
View Email Template Document [page 2288]
Edit or Add an Email Template Set [page 2289]
Define an Email Template Set for an Application [page 2295]
Delete an Email Template Set [page 2297]

1.7.8 Managing Administrators

This section describes how, as a tenant administrator, you can list all administrators in the administration console for SAP Cloud Identity Services, add new administrators, and edit the administrator authorizations. You can also remove administrators.

List Administrators [page 2302]
As a tenant administrator, you can list the administrators and their authorizations in the administration console for SAP Cloud Identity Services.

Add Administrators [page 2303]
As a tenant administrator, you can add new administrators in the administration console for SAP Cloud Identity Services.

Edit Administrator Authorizations [page 2309]
As a tenant administrator, you can edit both your own authorizations and other administrators' authorizations in the administration console for SAP Cloud Identity Services. By editing the administrator authorizations you can also delete an administrator.

Configure Authorizations Based on Policies [page 2311]
Enable admin authorizations based on policies to configure a granular access control for the administrators of SAP Cloud Identity Services.

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]

Configuring Email Templates [page 2282]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.8.1 List Administrators

As a tenant administrator, you can list the administrators and their authorizations in the administration console for SAP Cloud Identity Services.

Context

To list all administrators, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Administrators* tile.

   This operation opens a list of all administrators in alphabetical order.

   > ⓘ Note
   >
   > The list also includes the SAP BTP system, which by default has authorizations to set up the trust with Identity Authentication.

3. Choose the name from the list item to view the profile details, such as user ID and email, and the authorizations assigned to the administrator.

   > → Tip
   >
   > Type the name of the administrator in the search field to filter the list items.

4. **Optional:** To change the name of the system as administrator choose the `Rename` button and enter the new display name.

> → Remember
>
> You can change only the name of the system.

Task overview: Managing Administrators [page 2301]

Related Information

Add Administrators [page 2303]
Edit Administrator Authorizations [page 2309]
Configure Authorizations Based on Policies [page 2311]
Identity Authentication Tenant as an Application Identity Provider
Edit Administrator Authorizations [page 2309]

1.7.8.2 Add Administrators

As a tenant administrator, you can add new administrators in the administration console for SAP Cloud Identity Services.

Prerequisites

To add new tenant administrators, you must be assigned the Manage Tenant Configuration role.

Context

You can add both a person and a system in the administration console to act as administrators. The system can receive the same roles and can perform the same actions as the human administrator.

Task overview: Managing Administrators [page 2301]

Related Information

List Administrators [page 2302]
Edit Administrator Authorizations [page 2309]
Configure Authorizations Based on Policies [page 2311]

Add Administrators via SAP for Me

1.7.8.2.1 Add User as Administrator

Context

> → Tip
>
> If you are not a tenant administrator, or you don't have the Manage Tenant Configuration role, as an alternative, you can view the tenant administrators of the tenants that are assigned to you and add new administrators in SAP Cloud Identity Services via the SAP for Me portal. For more information see Cloud Identity Services Administrators Card.

To add a person as a new tenant administrator, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Administrators* tile.

   This operation opens a list of all administrators in alphabetical order.

   > ⓘ Note
   >
   > The list also includes the SAP BTP system, which by default has authorizations to set up the trust with Identity Authentication.

3. On the left-hand panel press *Add* > *User*.
4. Make the appropriate entries in the *Identifier*, *First Name*, and *Last Name* fields for the user you want to add as an administrator.

   The *Identifier* field can take user `email` or `login name`:

   - If `email` is used, it must be unique for the tenant.
   - If `login name` is used, an existing user with that login name must exist.

   > → Remember
   >
   > Once the administrator is created, the *First Name*, *Last Name*, and *Identifier* fields are not editable from the administrator section. If you want to change the information you must go to the *User Management* section. For more information about how to edit the user information, see List and Edit User Details [page 2361].

5. Assign the required administrator roles for the user.

   To be a tenant administrator, a user must be assigned at least one of the following roles.

Administrator Roles

Authorization	Description
Manage Applications	This role gives the tenant administrator permission to configure applications and their authorization policies via the administration console.
Manage Corporate Identity Providers	This role gives the tenant administrator permission to configure the identity providers via the administration console.
Manage Users	This role gives the tenant administrator permission to manage, import, and export users via the administration console.
Read Users	This role gives the tenant administrator permission to retrieve user data and export users via the administration console and the SCIM REST API of Identity Authentication.
Manage Groups	This role gives the tenant administrator permission to create, edit, and delete groups via the administration console. The tenant administrator can also assign authorization policies to users.
Manage Tenant Configuration	This role gives the tenant administrator permission to manage tenant configuration and authorization assignment to users. Tenant administrators with that role can add additional roles to themselves or to other administrators.
Manage Identity Provisioning	This role gives the tenant administrator permission to configure identity provisioning. The tenant administrator is granted the main IPS_ADMIN role.

6. Save your changes.

1.7.8.2.2 Add System as Administrator

Context

To add a system as a new tenant administrator, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Administrators* tile.

   This operation opens a list of all administrators in alphabetical order.

> ### Note
>
> The list also includes the SAP BTP system, which by default has authorizations to set up the trust with Identity Authentication.

3. On the left-hand panel press *Add* > *System* .
4. Enter a name for the system in the provided field.

> ### Caution
>
> Be careful when you choose a name for your system as administrator. Once created, the name cannot be changed.

5. Assign the required administrator roles for the system and save your changes.

   To be a tenant administrator, a user must be assigned at least one of the following roles.

   Administrator Roles

Authorization	Description
Manage Applications	This role gives the tenant administrator permission to configure the applications via the administration console.
Manage Corporate Identity Providers	This role gives the tenant administrator permission to configure the identity providers via the administration console.This role gives the tenant administrator permission to configure the applications via the administration console.
Manage Users	This role gives the tenant administrator permission to manage, import, and export users via the administration console.
Read Users	This role gives the tenant administrator permission to retrieve user data and import users via the administration console and the SCIM REST API of Identity Authentication.
Manage Groups	This role gives the tenant administrator permission to create, edit, and delete user groups via the administration console. The tenant administrator can also assign authorization policies.
Manage Tenant Configuration	This role gives the tenant administrator permission to manage tenant configuration and authorization assignment to users. Tenant administrators with that role can add additional roles to themselves or to other administrators.

Authorization	Description
Access Proxy System API	This role gives the tenant administrator permission to access API for provisioning identities via proxy systems. This role is needed for provisioning scenarios where proxy systems in the Identity Provisioning admin console are configured for synchronizing user data to and from central identity management solutions (such as, the on-premise SAP Identity Management). In this case, you use the credentials of the admin user with *Access Proxy System API* role assigned for setting up the technical user in the identity management solution for communicating with Identity Provisioning.
Access Real-Time Provisioning API	This role gives the tenant administrator permission to access API for real-time provisioning of identities. This role is needed for provisioning scenarios where user data is provisioned real-time without running jobs (manual or scheduled ones) in Identity Provisioning. In this case, you use the credentials of the admin user with *Access Real-Time Provisioning API* role assigned for setting up the authentication mechanism of the provisioning system defined on the *Real-Time Provisioning* tile of the administration console for SAP Cloud Identity Services.
Access Identity Provisioning Tenant Admin API	This role gives the tenant administrator permission to access tenant API for running provisioning jobs. This role is needed for running provisioning jobs from an API client. The API is available on the SAP Business Accelerator Hub: SAP Cloud Identity Services *Identity Provisioning Service* *API Reference* *Jobs*. The URL for accessing the Tenant Admin API follows the pattern: `https://<IPS tenant host>/ips/publicapi/v1/startJob/{SourceSystemId}/jobs/{JobType}`.

6. Configure the method for authentication when the system is used. You can choose from the following two options:

   - *Certificate*
     You can upload a certificate, generate a new one, or insert it as a text.
     You must provide a Common Name (CN) and password to generate the certificate. Once the certificate is generated, it is saved as a *.p12* file. The system populates the *Insert as Text* field with it, and provides the certificate attributes in the Subject DN. The common name (CN) in the generated certificate is in the format `<common name>` (`<admin user ID>`), where common name is the CN provided

by the administrator, and admin user ID is the administrator's user id. The maximum length of the combination of `<common name>` (`<admin user ID>`) is 64 characters.

> **→ Tip**
>
> Identity Authentication, by default, supports the following as trusted certificate authorities (CA): SAP Passport CA G2, SAP SSO CA G2, SAP Cloud Root CA, SAP Global Root CA, DigiCert TLS RSA SHA256 2020 CA1, DigiCert Global Root CA, SAP Cloud Platform Client CA, ConnectivityCA.

> **⚠ Restriction**
>
> You can't use one and the same certificate for application and system as administrator authentication.

> **ⓘ Note**
>
> For real-time provisioning scenarios, you need to upload the certificate generated from the source application that initiated the immediate sync. If Identity Authentication acts as the source application, simply upload the certificate generated in the *Authentication Mechanism* screen under *Users & Authorizations* > *Real-Time Provisioning* > *Add Target System*.

- *Secrets*

> **ⓘ Note**
>
> You need a *Client ID* and *Secret* for system authentication. It is generated automatically with the creation of the system. The client ID is in the universally unique identifier (UUID) format. For example, `1ab7c243-5de5-4530-8d14-1234a26373ab`.

Choose the *+Add* button in the *Secrets* section, provide the required info in the pop-up, and save your entries.

Field	Info
Description	This field is optional.
Expire in	You can choose from three options: • 1 year • 2 years • Never

Make sure you save your client secret. You won't be able to retrieve it from the system later.
Once your secret is generated you can see a table with your secrets and information about them. You can generate up to 2 secrets per system.

1.7.8.3 Edit Administrator Authorizations

As a tenant administrator, you can edit both your own authorizations and other administrators' authorizations in the administration console for SAP Cloud Identity Services. By editing the administrator authorizations you can also delete an administrator.

Prerequisites

To edit tenant administrators' authorizations, you must be assigned the Manage Tenant Configuration role.

Context

To edit an administrator's authorizations, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Administrators* tile.

   This operation opens a list of all administrators in alphabetical order.

   > ⓘ Note
   >
   > The list also includes the SAP BTP system, which by default has authorizations to set up the trust with Identity Authentication.

3. Choose the administrator name whose authorizations you want to edit.

   > → Tip
   >
   > Type the name of the administrator in the search field to filter the list items.

4. Edit the administrator authorizations as required.

   > ⓘ Note
   >
   > To be a tenant administrator, a user must be assigned one or more of the following roles:
   >
   > Administrator Roles

Authorization	Description
Manage Applications	This role gives the tenant administrator permission to configure applications and their authorization policies via the administration console.

Authorization	Description
Manage Corporate Identity Providers	This role gives the tenant administrator permission to configure the identity providers via the administration console.
Manage Users	This role gives the tenant administrator permission to manage, import, and export users via the administration console.
Read Users	This role gives the tenant administrator permission to retrieve user data and export users via the administration console and the SCIM REST API of Identity Authentication.
Manage Groups	This role gives the tenant administrator permission to create, edit, and delete groups via the administration console. The tenant administrator can also assign authorization policies to users.
Manage Tenant Configuration	This role gives the tenant administrator permission to manage tenant configuration and authorization assignment to users. Tenant administrators with that role can add additional roles to themselves or to other administrators.
Manage Identity Provisioning	This role gives the tenant administrator permission to configure identity provisioning. The tenant administrator is granted the main IPS_ADMIN role.

> → Tip
>
> **Remove Administrator**
>
> If you remove all authorizations, the user or system will no longer be an administrator, and the name will be removed from the list on the left. However, removing all authorizations does not delete the user from the user data base of Identity Authentication.

You cannot remove the *Manage Tenant Configuration* role from your own user.

5. Save your changes.

   If the operation is successful, the system displays the message `Tenant administrator <name of tenant administrator> updated`.

Task overview: Managing Administrators [page 2301]

Related Information

List Administrators [page 2302]
Add Administrators [page 2303]
Configure Authorizations Based on Policies [page 2311]
Identity Authentication Tenant as an Application Identity Provider

1.7.8.4 Configure Authorizations Based on Policies

Enable admin authorizations based on policies to configure a granular access control for the administrators of SAP Cloud Identity Services.

Context

> ⚠ Restriction
>
> This feature is relevant only for the Administration Console application.

To be able to configure authorizations based on policies, you must enable the feature via the admin console for SAP Cloud Identity Services. Once it's enabled, it may take up to 60 seconds before the administrator can see the *Authorization Policies* tab when accessing the administration console application.

> → Remember
>
> If your tenant has been migrated to a new region, you need to configure your authorizations based on policies again.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose *Tenant Settings*.
3. Under *General*, choose *Policy-Based Authorizations*.
4. Enable the policy-based authorizations.

   Configure User Authorizations [page 2312]
   : Configure a granular access control based on policies for the administrators of SAP Cloud Identity Services.

   Configure Technical User Authorizations [page 2318]
   : Configure granular access control for the technical users in the SAP Cloud Identity Services administration console.

   Configure Group Authorizations [page 2320]
   : Configure granular access and control over the groups in the administration console of SAP Cloud Identity Services.

   Configure Policy Authorizations [page 2323]
   : Configure granular access and control over the authorization policies in the administration console of SAP Cloud Identity Services.

   Configure Application Authorizations [page 2325]
   : Configure granular access and control over the applications in the administration console of SAP Cloud Identity Services.

   Configure Provisioning Authorizations [page 2327]

Configure granular access control for the Identity Provisioning systems and logs in the SAP Cloud Identity Services administration console.

Task overview: Managing Administrators [page 2301]

Related Information

List Administrators [page 2302]
Add Administrators [page 2303]
Edit Administrator Authorizations [page 2309]

1.7.8.4.1 Configure User Authorizations

Configure a granular access control based on policies for the administrators of SAP Cloud Identity Services.

Prerequisites

You have enabled the authorizations based on policies option in the admin console for SAP Cloud Identity Services. See Configure Authorizations Based on Policies [page 2311].

Context

> ⚠ Restriction
>
> This feature is relevant only for the Administration Console application.

> → Remember
>
> If your tenant has been migrated to a new region, you need to configure your authorizations based on policies again.

Sometimes the administrator authorizations that are predefined in the tenant of SAP Cloud Identity Services aren't enough. The predefined administrator authorizations give unlimited data access. However, you may need to define authorization models with more complex instance restrictions for data access, as is the so-called attribute-based access control (ABAC). Administrators define authorization policies with user attributes and assign these policies to other administrators. Thus, one administrator can have access to a subset of the users in the tenant or to a subset of the attributes of the user.

Once it's enabled, it may take up to 60 seconds before the administrator can see the *Authorization Policies* tab when accessing the administration console application. Initially, only the base

policies are visible: `CREATE_USERS` , `DELETE_USERS`, `MANAGE_USERS`, `READ_USERS`, `UPDATE_USERS`, `CREATE_SCIM_SCHEMAS`, `DELETE_SCIM_SCHEMAS`, `MANAGE_SCIM_SCHEMAS`, `READ_SCIM_SCHEMAS`, `CREATE_GROUPS`, `DELETE_GROUPS`, `MANAGE_GROUPS`, `READ_GROUPS`, and `UPDATE_GROUPS`. You can create new authorization policies on the base of these policies and assign them to administrators.

> ### → Remember
>
> As of February 11, 2025, the `MANAGE_USERS` policy does not contain the read applications permission. You must add the `READ_APPLICATIONS` policy to the user or users if they need to see the *Import Users* tile in the administration console.

> ### ⓘ Note
>
> The Read Users authorization overrides the READ_USERS authorization policy, while the Manage Users authorization overrides all user authorization policies. For example, if administrator users are assigned the Read Users authorizations, they can read all users even if the same administrator users are configured in the authorization policy with a READ_USERS restriction.

When you create a new policy, you can restrict the users on the basis of the following attributes: `user.name`, `user.addresses.country`, `user.costCenter`, `user.division`, `user.department`, `user.organization`, and `user.type`. The subsets of the user attributes are configured via the `user.attributes`.

User Attributes

Attributes	Value
`user.userName`	The *Login Name* of the user as defined in the administration console.
`user.addresses.country`	The value must match the predefined master data one. See Countries.properties [page 2873]. The addresses must be marked as primary via the Identity Directory SCIM REST API. Users who don't have a primary address are excluded even if the `user.addresses.country` attribute matches the address of the user. **→ Tip** Use the key from the key-value pair for the value of the `user.country` attribute. For example, you must use `DE` from the key-value pair `DE=Germany`.
`user.costCenter`	The *Cost Center* of the user as defined in the administration console.
`user.division`	The *Division* of the user as defined in the administration console.
`user.department`	The value must match the predefined master data one. See Departments.properties [page 2868]

Attributes	Value
`user.organization`	The *Company* of the user as defined in the administration console.
`user.type`	The *User Type* as defined in the administration console. The allowed values are `public`, `partner`, `customer`, `external`, `onboardee`, `employee` or `alumni`.
`user.attributes`	The policy allows you to see the attributes that are defined in the value field. The attributes' value format must be according to the SCIM notation. The supported attributes that can be defined in the policy are listed in the **Supported Attributes** section below this table. **Note** If the `user.attributes` is used with the "=" operator, it supports only one attribute. For more attributes, use the "IN" operator adding each attribute separately. **Note** If you use the `password` attribute, you must also add the following two attributes: • `active` • `urn:ietf:params:scim:schemas:extension:sap:2.0:User:status` Enter the attributes separated by comma without using a space. **Example** `active,urn:ietf:params:scim:schemas:extension:sap:2.0:User:status`

Attributes	Value
Deprecated `user.excludedAttributes`	**Note** The `user.excludedAttributes` attribute is deprecated. If you have a policy configured with the `user.excludedAttributes` attribute exchange the `user.excludedAttributes` with the `user.attributes` attribute in combination with the "NOT IN" operator. If the policy is configured with the `user.attributes` attribute used with the "=" operator, it supports only one attribute. For more attributes, use the "IN" operator adding each attribute separately.

Restriction

The "NOT IN" operator can be used only with the `user.attributes` attribute. Do not use "NOT IN" in combination with other attributes or groups.

Expand the **Supported Attributes** section below to see the user attributes that can be configured in the authorization policy:

Supported Attributes

Core Schema

- `name.givenName`
- `name.familyName`
- `userName`
- `displayName`
- `addresses.country`
- `addresses.streetAddress`
- `addresses.streetAddress2`
- `phoneNumbers.value`
- `phoneNumbers.verified`
- `emails.value`
- `emails.verified`
- `password`

Note

For the attributes defined in the core schema, the Schema URI notation `[urn:ietf:params:scim:schemas:core:2.0:User]` is not needed, for all the other attributes, schema URI and the attribute name is required. For example: `user.attributes IN displayName,addresses.country,emails.value;`

EnterpriseUuser Resource Schema

- `costCenter`
- `organization`
- `division`
- `department`

> ⓘ Note
>
> All Enterprise user resource schema attributes require the schema URI urn:ietf:params:scim:schemas:extension:enterprise:2.0:User and the attribute name.
>
> For example: `user.attributes IN urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:costCenter, urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:organization;`

SAP extension schema

- `userUuid`
- `validFrom`
- `validTo`

> ⓘ Note
>
> All SAP extension schema attributes require the schema URI urn:ietf:params:scim:schemas:extension:sap:2.0:User and the attribute name. For example: user.excludedAttributes=urn:ietf:params:`scim:schemas:extension:sap:2.0:User:userUuid, urn:ietf:params:scim:schemas:extension:sap:2.0:User:validFrom;`

Custom Defined Schema

All custom schema defined attributes require a fully qualified attribute name. For example: `user.attributes=urn:sap:cloud:scim:schemas:extension:custom:2.0:MyCustomSchema:CustomString`

Groups of type `Authorization Policy` with names containing the names of the authorization policies are also created in the administration console. You can't delete these groups via the *Groups* section. The groups are related to the authorization policies, and when you delete a policy, the respective group is also removed.

> ⚠ Restriction
>
> You need both read and update access rights to be able to update a field in the administration console. If you can't see a field because of a policy restriction, this field remains also disabled for editing even if update rights are granted to you.

> → Remember
>
> To edit a custom schema custom attribute via the administration console, `users.MANAGE_USERS` without restriction is needed. If the policy has a restriction on the `users.MANAGE_USERS` base policy, you won't be able to edit the custom schema custom attribute.

> ### Example
>
> Michael Adams is an administrator at retail company A. He is located at the company's head office in Germany and as chief administrator of the company he has all the authorizations in the administration console for SAP Cloud Identity Services. Dona Moore is also an administrator at company A. She is responsible for the branch office in the USA. As such she needs to have access only to the users in the USA. Michael Adams creates an authorization policy for read-users access and assigns Dona Moore to that policy. He also removes the *Read Users* and *Manage Users* authorizations that Dona has as an administrator. As a result, now, when Dona accesses the *User Management* section of the administration console, she sees only the users that are located in the USA. All the other users are hidden.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose *Applications*.
3. Under *System Applications*, choose the list item for the administration console.
4. Under the tab *Authorization Policies*, select a custom authorization policy.

   > ### Note
   >
   > Type the name or package in the search field, filter the names or packages, or choose the policy from the list.
   >
   > If you don't have a created authorization policy in your list, you can create one. For more information, see Create an Authorization Policy [page 2247].

5. Choose the *Edit* button.
   a. To customize the rules of the authorization policy, choose the *Rules*.
   b. To assign administrator or administrators to this policy, choose the *Assignments* tab.
6. Save your changes.

Task overview: Configure Authorizations Based on Policies [page 2311]

Related Information

Configure Technical User Authorizations [page 2318]
Configure Group Authorizations [page 2320]
Configure Policy Authorizations [page 2323]
Configure Application Authorizations [page 2325]
Configure Provisioning Authorizations [page 2327]

1.7.8.4.2 Configure Technical User Authorizations

Configure granular access control for the technical users in the SAP Cloud Identity Services administration console.

Prerequisites

You have enabled the authorizations based on policies option in the admin console for SAP Cloud Identity Services. See Configure Authorizations Based on Policies [page 2311].

Context

> ⚠ Restriction
>
> This feature is relevant only for the Administration Console application.

> → Remember
>
> If your tenant has been migrated to a new region, you need to configure your authorizations based on policies again.

Tenant administrators can assign specific authorizations to end users or administrators, allowing them to create, read, update, delete technical users, or a combination of these operations in the administration console of SAP Cloud Identity Services. The distribution of responsibilities enables the authorized users (not necessarily administrators) within your organization to concentrate on their specific areas of access and expertise.

The following authorization policies related to technical users are available in the administration console application under the package name *technical_users*. These policies are non-editable, meaning you can't add or delete rules (i.e., restrictions). You can only assign users to the authorization policy.

Authorization Policy	Description
`CREATE_TECHNICAL_USERS`	Allows users only to create technical users.
`READ_TECHNICAL_USERS`	Allows users only to view the technical users and the information about them. The user with this policy can see the tile *Technical Users* in the administration console for SAP Cloud Identity Services.
`UPDATE_TECHNICAL_USERS`	Allows users only to update the technical users.
`DELETE_TECHNICAL_USERS`	Allows users only to delete the technical users.
`MANAGE_TECHNICAL_USERS`	Allows users to manage the technical users. The `MANAGE_TECHNICAL_USERS` policy

Authorization Policy	Description
	combines the authorizations from the other policies: `CREATE_TECHNICAL_USERS`, `READ_TECHNICAL_USERS`, `UPDATE_TECHNICAL_USERS`, and `DELETE_TECHNICAL_USERS`

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose *Applications*.
3. Under *System Applications*, choose the *Administration Console*.
4. Under the *Authorization Policies* tab, search for the authorization policy and select it.

   You can type the policy name or package in the search field, filter the policy by package type, or choose it directly from the list.

   The policy names follow this convention: `<action>_<functionality>_<resource>`, for example: *MANAGE_TECHNICAL_USERS*
5. Under the *Assignments* tab, choose *Add* and add a user.

   To assign a user to multiple authorization policies, repeat steps 4 and 5 as necessary. Alternatively, you can combine multiple policies by creating a new policy of type *Customer Package* and assign the user to all included policies at once. For more information, see Combine Authorization Policies [page 2249].

Results

When an authorization policy is assigned to a user, a user group named after the policy is also assigned to that user. For example, if a user is assigned the `MANAGE_TECHNICAL_USERS` policy, they become members of the following user group: `technical_users - MANAGE_TECHNICAL_USERS`

Next Steps

Administrators with assigned technical user authorizations can create and manage the technical users. See Managing Technical Users [page 2329].

Task overview: Configure Authorizations Based on Policies [page 2311]

Related Information

Configure User Authorizations [page 2312]
Configure Group Authorizations [page 2320]
Configure Policy Authorizations [page 2323]
Configure Application Authorizations [page 2325]
Configure Provisioning Authorizations [page 2327]

1.7.8.4.3 Configure Group Authorizations

Configure granular access and control over the groups in the administration console of SAP Cloud Identity Services.

Prerequisites

You have enabled the authorizations based on policies option in the admin console for SAP Cloud Identity Services. See Configure Authorizations Based on Policies [page 2311].

Context

> ⚠ Restriction
>
> This feature is relevant only for the Administration Console application.

> → Remember
>
> If your tenant has been migrated to a new region, you need to configure your authorizations based on policies again.

Once it's enabled, it may take up to 60 seconds before the administrator can see the *Authorization Policies* tab when accessing the administration console application. Under the "groups" package the following base policies are visible: `CREATE_GROUPS` , `DELETE_GROUPS`, `MANAGE_GROUPS`, `READ_GROUPS`, and `UPDATE_GROUPS`. You can add users to these policies so that they can have the rights these policies give.

> ⓘ Note
>
> The Manage Groups authorization overrides all "groups" package policies. If you want to configure access to the groups based on policies, you must remove the Manage Groups authorization. For more information, see Edit Administrator Authorizations [page 2309].

You can restrict access on the basis of the base policies or you can use a custom policy, restricting the groups on the basis of the `group.name`, `group.displayName`, `group.type`, or `group.applicationId` attributes.

Group Attributes

Attributes	Value
`group.name`	The name of the group.
`group.displayName`	The display name of the group.
`group.type`	The type of the group. The supported values are: • `userGroup` • `authorization` • `deepLinkActivationPermission`
`group.applicationId`	The ID of the application which is the unique identifier of that application configured in Cloud Identity Services.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose *Applications*.
3. Under *System Applications*, choose the list item for the administration console.
4. Under the tab *Authorization Policies*, you can choose the following options:

Use base policy	1. Filter the policies by the "groups" package. **Note** This limits the policies to the following: *CREATE_GROUPS* , *DELETE_GROUPS*, *MANAGE_GROUPS*, *READ_GROUPS*, and *UPDATE_GROUPS*. 2. Select a policy from the list. 3. Choose the *Add* button. 4. Select the user or users and choose Add.
Use custom policy	1. Select a custom authorization policy. **Note** If you don't have an authorization policy in your list, you can create a restriction policy using the `groups.DELETE_GROUPS`, `groups.READ_GROUPS`, `groups.UPDATE_GROUPS` parameters. For more information about how to create a custom authorization policy, see Create an Authorization Policy [page 2247]. 2. Choose the *Edit* button.

3. To customize the rules of the authorization policy, choose *Rules*.
    1. For each **USE** statement of the selected authorization policy, add a **RESTRICT** condition with `Value`, choose one of the available attributes, an operation, and enter a value.
    2. Save your changes.
4. To assign user to this policy, choose the *Assignments* tab and add users from the list.

    > **Restriction**
    >
    > If the user is a tenant administrator you must remove the *Manage Groups* role from that user, otherwise they won't be able to see and have control (read, update or delete) over the groups for this policy.

5. Save your changes.

    > **Note**
    >
    > Now the assigned users can see and have control over the groups with that are specified in the custom policy.

Task overview: Configure Authorizations Based on Policies [page 2311]

Related Information

Configure User Authorizations [page 2312]
Configure Technical User Authorizations [page 2318]
Configure Policy Authorizations [page 2323]
Configure Application Authorizations [page 2325]
Configure Provisioning Authorizations [page 2327]

1.7.8.4.4 Configure Policy Authorizations

Configure granular access and control over the authorization policies in the administration console of SAP Cloud Identity Services.

Prerequisites

- You have enabled the authorizations based on policies option in the admin console for SAP Cloud Identity Services. See Configure Authorizations Based on Policies [page 2311].
- You need the following authorization policies to fully use Authorization Management in the *Authorization Policies* tab. See Assign Authorization Policies [page 2363].

Authorization Policies	Description
`READ_POLICIES`	Read the authorization policies
`READ_GROUPS`	Read the policy assignments
`READ_USERS`	Display users on the *Assignments* tab
`READ_APPLICATION`	Read the application

Context

> ⚠ Restriction
>
> This feature is relevant only for the Administration Console application.

> → Remember
>
> If your tenant has been migrated to a new region, you need to configure your authorizations based on policies again.

Once it's enabled, it may take up to 60 seconds before the administrator can see the *Authorization Policies* tab when accessing the administration console application. Under the *policies* package the following base policies are visible: `CREATE_POLICIES` , `DELETE_POLICIES`, `MANAGE_POLICIES`, `READ_POLICIES`, and `UPDATE_POLICIES`. You can add users to these policies so that they can have the rights these policies give.

You can restrict access on the basis of the base policies or you can use a custom policy, restricting the authorization policies on the basis of the `policy.name` and `policy.applicationId` attributes.

Policy Attributes

Attributes	Value
`policy.name`	The name of the authorization policy.
`policy.applicationId`	The ID of the application which is the unique identifier of that application configured in SAP Cloud Identity Services.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose *Applications*.
3. Under *System Applications*, choose the list item for the administration console.
4. Under the tab *Authorization Policies*, select an authorization policy with the *policy* package.

   > ⓘ Note
   >
   > Type the name or package in the search field, filter the names or packages, or choose the authorization policy from the list.
   >
   > If you don't have a created authorization policy in your list, you can create one. For more information, see Create an Authorization Policy [page 2247].

5. Expand the authorization policy you want to assign to an administrator.
6. Choose *Add* on the *Assignments* tab.
7. Select the user you want to assign and choose *Add*.

Task overview: Configure Authorizations Based on Policies [page 2311]

Related Information

Configure User Authorizations [page 2312]
Configure Technical User Authorizations [page 2318]
Configure Group Authorizations [page 2320]
Configure Application Authorizations [page 2325]
Configure Provisioning Authorizations [page 2327]

1.7.8.4.5 Configure Application Authorizations

Configure granular access and control over the applications in the administration console of SAP Cloud Identity Services.

Prerequisites

You have enabled the authorizations based on policies option in the admin console for SAP Cloud Identity Services. See Configure Authorizations Based on Policies [page 2311].

Context

> ⚠ Restriction
>
> This feature is relevant only for the Administration Console application.

> → Remember
>
> If your tenant has been migrated to a new region, you need to configure your authorizations based on policies again.

Once it's enabled, it may take up to 60 seconds before the administrator can see the *Authorization Policies* tab when accessing the administration console application. Under the "applications" package the following base policies are visible: `CREATE_APPLICATIONS` , `DELETE_APPLICATIONS`, `MANAGE_APPLICATIONS`, `READ_APPLICATIONS`, and `UPDATE_APPLICATIONS`. You can add users to these policies so that they can have the rights these policies give.

> ⓘ Note
>
> The Manage Applications authorization overrides all "applications" package policies, while the Manage Users authorization overrides the `READ_APPLICATIONS` policy only. If you want to configure access to the applications based on policies, you must remove the Manage Applications and Manage Users authorizations. For more information, see Edit Administrator Authorizations [page 2309].

You can restrict access on the basis of the base policies or you can use a custom policy, restricting the applications on the basis of the `application.organization` attribute.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose *Applications*.
3. Under *System Applications*, choose the list item for the administration console.

4. Under the tab *Authorization Policies*, you can choose the following options:

Use base policy	1. Filter the policies by the "applications" package. **Note** This limits the policies to the following: *CREATE_APPLICATIONS* , *DELETE_APPLICATIONS*, *MANAGE_APPLICATIONS*, *READ_APPLICATIONS*, and *UPDATE_APPLICATIONS*. 2. Select a policy from the list. 3. Choose the *Add* button. 4. Select the user or users and choose Add.
Use custom policy	1. Select a custom authorization policy. **Note** If you don't have a custom authorization policy in your list, you can create a restriction policy using the `applications.CREATE_APPLICATIONS`, `applications.MANAGE_APPLICATIONS`, `applications.DELETE_APPLICATIONS`, `applications.READ_APPLICATIONS`, `applications.UPDATE_APPLICATIONS` parameters. For more information about how to create a custom authorization policy, see Create an Authorization Policy [page 2247]. 2. Choose the *Edit* button. 3. To customize the rules of the authorization policy, choose *Rules*. 4. To assign user to this policy, choose the *Assignments* tab and add users from the list. **Restriction** If the user is a tenant administrator you must remove the *Manage Applications* and *Manage Users* roles from that user, otherwise they won't be able to see and have control (read, update or delete) over the applications for this policy. 5. Save your changes. **Note** Now the assigned users can see and have control over applications with *Organization ID* specified in the custom policy.

Next Steps

View or edit the *Organization ID* of the application. By default, all applications are in the *global* organization. For more information, see Edit Applications [page 1929]

Task overview: Configure Authorizations Based on Policies [page 2311]

Related Information

Configure User Authorizations [page 2312]
Configure Technical User Authorizations [page 2318]
Configure Group Authorizations [page 2320]
Configure Policy Authorizations [page 2323]
Configure Provisioning Authorizations [page 2327]

1.7.8.4.6 Configure Provisioning Authorizations

Configure granular access control for the Identity Provisioning systems and logs in the SAP Cloud Identity Services administration console.

Prerequisites

- You have enabled the *Policy-Based Authorizations* option in the SAP Cloud Identity Services admin console. See Configure Authorizations Based on Policies [page 2311].
- Your tenant administrator has the following authorizations: *Manage Application*, *Manage Groups* and *Read Users*.

Context

> ⚠ Restriction
>
> This feature is relevant only for the Administration Console application.

> → Remember
>
> If your tenant has been migrated to a new region, you need to configure your authorizations based on policies again.

Tenant administrators can assign specific authorizations to users, allowing them to access only source systems, target systems, proxy systems, provisioning logs, or any combination of these. The distribution of responsibilities enables authorized users (not necessarily administrators) within your organization to concentrate on their specific areas of access and expertise.

The following authorization policies related to provisioning are available in the Administration Console application under the package name *provisioning*. These policies are non-editable, meaning you cannot add or delete rules (i.e., restrictions). You can only assign users to the authorization policy.

Authorization Policy	Description
MANAGE_PROVISIONING_SOURCES	Allows users to manage source systems only. This includes creating systems, managing properties, transformations and certificates, as well as running and scheduling provisioning jobs and viewing notifications.
MANAGE_PROVISIONING_TARGETS	Allows users to manage target systems only. This includes creating systems, managing properties, transformations and certificates, as well as viewing notifications.
MANAGE_PROVISIONING_PROXIES	Allows users to manage proxy systems only. This includes creating systems, managing properties, transformations and certificates, as well as viewing notifications.
MANAGE_PROVISIONING_LOGS	Allows users to manage provisioning job logs and real-time provisioning logs, as well as viewing notifications.

> **Note**
>
> The *Manage Identity Provisioning* role, provides full access to all configurations and functionalities of the service. After the implementation of the granular access control on December 3, 2024, users assigned to this role will retain their full access, regardless of any policies assigned.

To assign users to one or more authorization policies, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose *Applications*.
3. Under *System Applications*, choose the *Administration Console*.
4. Under the *Authorization Policies* tab, search for the authorization policy and select it.

   You can type the policy name or package in the search field, filter the policy by package type, or choose it directly from the list.

   The policy names follow this convention: `<action>_<functionality>_<resource>`, for example: *MANAGE_PROVISIONING_LOGS*
5. Under the *Assignments* tab, choose *Add* and add a user.

To assign a user to multiple authorization policies, repeat steps 4 and 5 as necessary. Alternatively, you can combine multiple policies by creating a new policy of type *Customer Package* and assign the user to all included policies at once. For more information, see Combine Authorization Policies [page 2249].

Results

When an authorization policy is assigned to a user, a user group named after the policy is also assigned to that user. For example, if a user is assigned the MANAGE_PROVISIONING_LOGS policy, they will become members of the following user group: MANAGE_PROVISIONING_LOGS - provisioning - Administration Console.

Upon logging in at either `https://<tenant ID>.accounts.ondemand.com/admin` or `https://<tenant ID>.accounts.cloud.sap/admin`, the user will have access specifically to the provisioning logs in the SAP Cloud Identity Services administration console and no other areas.

Task overview: Configure Authorizations Based on Policies [page 2311]

Related Information

Configure User Authorizations [page 2312]
Configure Technical User Authorizations [page 2318]
Configure Group Authorizations [page 2320]
Configure Policy Authorizations [page 2323]
Configure Application Authorizations [page 2325]

1.7.9 Managing Technical Users

As a user with assigned special policy authorizations, you can manage the technical users in the administration console for SAP Cloud Identity Services.

Prerequisites

- The authorizations based on policies option in the admin console for SAP Cloud Identity Services is enabled. See Configure Authorizations Based on Policies [page 2311].
- You are assigned with the respective technical user authorizations. For more information, see Configure Technical User Authorizations [page 2318].

Context

Technical users in SAP Cloud Identity Services are non-human identities that enable automated system-to-system communication and integration. These users are created and managed in the administration console via the *Technical Users* tile that appears under *Users & Authorizations*.

Create a Technical User [page 2331]
: As a user with assigned special policy authorizations, you can create a technical user in the administration console for SAP Cloud Identity Services.

Manage Technical User Details [page 2332]
: As a user with assigned special policy authorizations, you can manage the technical user details and in the administration console for SAP Cloud Identity Services.

Task overview: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]
Users [page 509]

1.7.9.1 Create a Technical User

As a user with assigned special policy authorizations, you can create a technical user in the administration console for SAP Cloud Identity Services.

Prerequisites

- The authorizations based on policies option in the admin console for SAP Cloud Identity Services is enabled. See Configure Authorizations Based on Policies [page 2311].
- You are assigned with a `MANAGE_TECHNICAL_USERS` or `CREATE_TECHNICAL_USERS` authorization policy. For more information, see Configure Technical User Authorizations [page 2318].

Context

You create the new technical user with a *Display Name* as a minimal requirement. The owner's first name, last name, and email are optional and can be added or edited later. The technical user automatically receives an email address, and a SCIM ID that can't be edited.

Later, if you are assigned with the required authorization policy you can edit the technical user attributes, or delete it. For more information, see Configure Technical User Authorizations [page 2318].

To create a technical user, use the following procedure:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose *Users & Authorizations* > *Technical User*.
3. Press *Create*.
4. Fill in the required fields in the dialog box.

Field	Required
Display Name	Yes
Owner First Name	No
Owner Last Name	No
Owner Email	No

5. Press *Create*

Results

If the operation is successful, the system displays the message: `Technical user created`.

Task overview: Managing Technical Users [page 2329]

Related Information

Manage Technical User Details [page 2332]
Users [page 509]

1.7.9.2 Manage Technical User Details

As a user with assigned special policy authorizations, you can manage the technical user details and in the administration console for SAP Cloud Identity Services.

Prerequisites

- The authorizations based on policies option in the admin console for SAP Cloud Identity Services is enabled. See Configure Authorizations Based on Policies [page 2311].
- You are assigned with the respective technical user authorizations, depending on the tasks you need to perform. For more information, see Configure Technical User Authorizations [page 2318].
- There is an existing technical user in the tenant of SAP Cloud Identity Services.

Context

You can view or edit the technical user details, configure authentication, or delete the technical user.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose *Users & Authorizations* > *Technical User*.
3. Select the technical user from the list on the left whose details you want to edit.
4. **Optional:** Under *Details tab* > *General Information* > *Edit* > *choose , change or add new information in the fields* > *Save*.

5. **Optional:** Under *Details tab* > *Client ID*, view and manage the *Client ID* information and status. The *Client ID Lock* option is disabled by default. When enabled, the client ID locks after 5 failed logon attempts using secrets. We recommend you to enable this feature in cases when the client ID secrets are automatically generated.
6. **Optional:** Under *Details tab* > *Configure Authentication*, manage the *Certificates* or *Secrets* that are used for authenticating the client and for API authentication.
7. **Optional:** Under the *Groups tab* > *Assign* > *assign a group from the list* > *Assign*.

Task overview: Managing Technical Users [page 2329]

Related Information

Create a Technical User [page 2331]

1.7.10 Managing Agents

Agent identities automatically appear in SAP Cloud Identity Services after you deploy agents. You can use SAP Cloud Identity Services to review and manage access to these agents.

> ⚠ Caution
>
> This page is still work in progress. This page contains information about product features that are part of an SAP Early Adopter Care (EAC) program. Products or features that are part of an EAC program are not available to users outside of this program.

When you deploy an agent in your SAP landscape, the system automatically registers it in SAP Cloud Identity Services. The agent appears in the *Users & Authorizations* area of the administration console. Assistants, higher-level AI entities that orchestrate one or more agents, also appear in this area and are managed in the same way. From there, you can control which users or groups can invoke the agent. You can't create or delete agent or assistant registrations manually. Nor can you change the agent's authorized API scope. The agent's authorizations are defined by its developer and are read-only for administrators.

Assistants and the agents they invoke are separate registered identities. If an assistant invokes one or more agents, assigning a user to the assistant does not automatically grant them access to those agents. You must assign the user to the assistant and to each agent it invokes separately. If any assignment is missing, the user receives an authorization error when the assistant attempts to invoke the unassigned agent, even if the user has access to all the other entities. When planning access for a user, identify the full chain of entities they need to reach and assign them to each one.

Controlling access to an agent has direct downstream consequences. When you assign a user or group to an agent, you grant those users the right to invoke it. After invocation, the agent acts within the bounds of both its own scoped authorization policies and the invoking user's authorizations in the connected back-end systems. The agent can only call the APIs it was designed to use, and those calls are further limited by what the user is permitted to do in the target application. A user can't use an agent to perform an action that their authorizations wouldn't otherwise allow, and the agent can't access APIs beyond its defined scope regardless

of the user's permissions. Assigning access to an agent is a security decision. It determines not only who can use the feature, but also who can trigger actions in downstream systems through it.

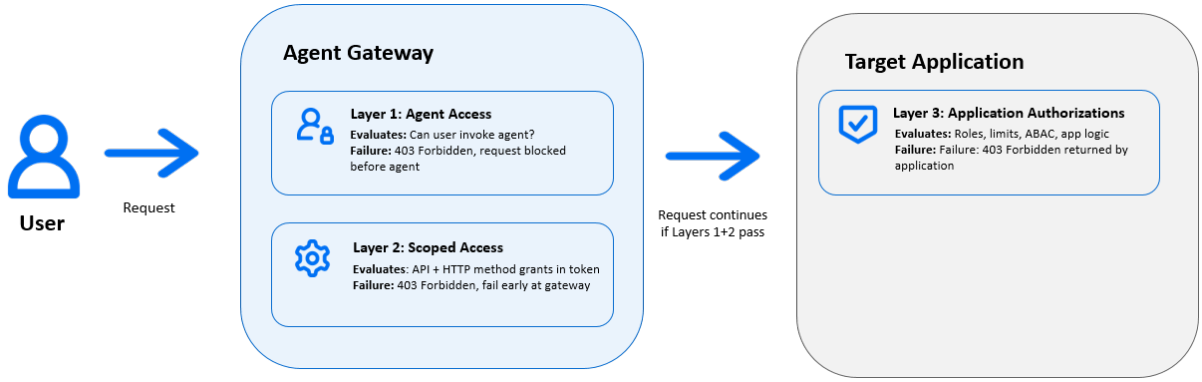
Agent Authorization Policies

Three layers of authorization govern access to and through an agent. The system enforces each layer at a different point in the request path.

The layers are:

1. **Agent access**: whether the requesting user has permission to invoke the agent at all. The system performs this check at the agent gateway on every request. The system automatically generates the policy when you deploy the agent, based on its metadata. The policy appears under the *Authorization Policies* tab of the agent's application entry in the administration console. You can restrict or extend this policy by deciding which users and groups can use the agent.
2. **Scoped access**: whether the agent has permission to call a specific API with a specific HTTP method. The system also enforces this check at the agent gateway, based on grant permissions defined in the agent's authorization token. This check ensures the agent is limited to the API surface it was designed to use, regardless of the invoking user's broader permissions.
3. **Application authorizations**: whether the specific action is permitted by the target application's own authorization logic, for example, ABAP role checks, instance limits, or attribute-based controls in SAP S/4HANA. The target application itself performs this check and returns a 403 response if the action is not permitted. This layer is not configurable in SAP Cloud Identity Services.

The system automatically generates baseline policies for layers one and two from the agent's metadata at deployment time. You can review and adjust these policies in the *Authorization Policies* tab of the agent's application entry. The target application always controls layer three.

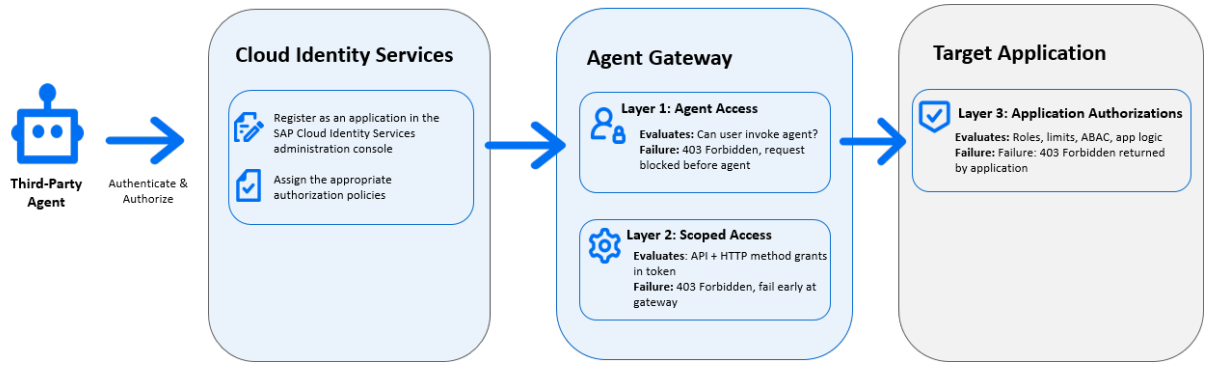


Three Layer Authorization

Agent Identity for Third-Party Tools

Third-party AI tools (external agents not built or deployed by SAP) can also access SAP systems through the agent gateway. To connect a third-party agent, register it as an application in the SAP Cloud Identity Services administration console and assign the appropriate authorization policies. The same three-layer access model applies. The agent must be permitted at the gateway, must have scoped access to the APIs it requires, and its actions remain subject to the target application's own authorization checks.

After you register a third-party agent, it authenticates to the agent gateway using the credentials configured in its application entry in SAP Cloud Identity Services. You can then assign access policies to control which users can invoke the third-party agent and which API operations it is authorized to perform.



Third-Party Agent Authorization Architecture

Review Agent Authorizations [page 2336]
: After agents are deployed, they arrive in your system with the authorizations the developer created for them. Before you enable access to the agent, review its authorizations and decide whether the agent conforms to your authorization policies.

Enable Users to Access Agents [page 2338]
: Assign users to agents to grant users the right to invoke them. You can assign individual users or multiple users with user groups.

Enable Access to Agents for the SAP Early Adopter Care Program [page 2341]
: For the SAP Early Adopter Care (EAC) program, the SAP-managed provisioning jobs create application groups for the Joule agents. Assign these groups to users to enable them to use these agents.

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
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Configuring Social Identity Providers [page 2505]

Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]
Identity and Access Management for Joule [page 580]
Agents [page 511]
Understanding AI Agents Across SAP

1.7.10.1 Review Agent Authorizations

After agents are deployed, they arrive in your system with the authorizations the developer created for them. Before you enable access to the agent, review its authorizations and decide whether the agent conforms to your authorization policies.

Context

> ⚠ Caution
>
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Each agent arrives in your system with a fixed authorization scope, that is, the APIs it is permitted to call at the agent gateway, and whether the permission for each is read or write. This scope is defined by the agent's developer and is read-only for administrators. You can't modify it in SAP Cloud Identity Services. Your role is to evaluate whether the scope is acceptable before you assign any users to the agent. Even if the scope appears broad, actions are also constrained at runtime by the invoking user's own permissions in the target application. That constraint isn't visible here and should not be relied on as a substitute for a careful scope review.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Users & Authorizations*, choose the *Agents* tile.

   The system displays a list of the agents in the tenant.
3. Choose an agent from the list on the left to view its details.
4. Choose the *Authorizations* tab.
5. Review the authorizations of the agent.

   The *Authorizations* tab shows the agent's scoped access as a list of policies, grouped by target application. Each row names an API by its technical identifier and shows whether the permission is Read or Write.

Because the API names are technical identifiers, you may need to look up what each API does before you can evaluate the scope meaningfully. Use the following questions to guide your evaluation.

Review Guidance

Question	Guidance
Does the scope match the agent's purpose?	An agent designed for read-only reporting should not require write access. A scope broader than the agent's stated function warrants further investigation before you proceed.
Does the scope include write access?	Write access can create or modify data in the target system. Confirm that these operations are necessary for the agent's function and that the risk is acceptable in your environment.
Do the APIs cover sensitive data or critical processes?	Even read-only access to APIs that expose personal data, financial records, or privileged configuration may require additional justification before you grant access.
Is the API surface proportionate?	A scope covering many unrelated APIs may indicate the agent was provisioned with more access than it needs. Apply the principle of least privilege when assessing whether to proceed.
Does the scope cover multiple target systems?	The authorization list may include entries from more than one target application. Review each application group independently. Write access in one application doesn't justify write access in another.

Next Steps

Use the authorization information to determine which users should have access to this agent. The agent can only call the APIs defined in its authorization scope, and the user's own authorizations in the target application further limit what the agent can do on their behalf. The Agent Gateway also monitors agents and their requests. If the agent's authorization scope is broader than you want to allow in your environment, consider these options:

- Don't assign any users to the agent. As long as no user is assigned, no one can invoke the agent, regardless of how the landscape is configured. This gives you as the IAM administrator full control.
- For line-of-business agents, consider reducing the authorizations assigned to the users in the target system. Discuss the situation with the landscape administrator in your organization. The landscape administrator enables the agent for your tenant in SAP for Me. They can determine to which systems the agent has access.
- For custom agents, discuss the situation with the developer of the agent within your organization. Then assign users to the agent only after reviewing and accepting its authorization scope.

If this agent is invoked by an assistant, repeat this review for the assistant and every other agent in the chain before assigning any users. Access assignments are not inherited across the chain. For more information about how the chain works and how to plan assignments, see Managing Agents [page 2333].

Task overview: Managing Agents [page 2333]

Related Information

Enable Users to Access Agents [page 2338]
Enable Access to Agents for the SAP Early Adopter Care Program [page 2341]
Identity and Access Management for Joule [page 580]
Agents [page 511]

1.7.10.2 Enable Users to Access Agents

Assign users to agents to grant users the right to invoke them. You can assign individual users or multiple users with user groups.

Prerequisites

> ⚠ Caution
>
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>
> See also Enable Access to Agents for the SAP Early Adopter Care Program [page 2341].

Before you begin, ensure the following are in place:

- The agent is deployed and registered in SAP Cloud Identity Services. Agents appear automatically in the *Users & Authorizations* area after deployment. You can't register them manually.
- The users or user groups you want to assign already exist in SAP Cloud Identity Services. You can't create users or user groups from within this procedure.
  For more information, see Managing Users [page 2342] or Managing Groups [page 2393].
- You're assigned an administrator role with the *Manage Users* or *Manage Groups* authorizations in the SAP Cloud Identity Services administration console.
  For more information, see Add Administrators [page 2303].

Context

This task configures the first of three authorization layers that govern agent access: whether a user can invoke the agent. The agent gateway enforces this check on each incoming request before the agent runs. For an overview of all three layers and how they interact, see Managing Agents [page 2333].

Assigning a user or user group to an agent does not expand that user's permissions in the connected back-end systems. After invocation, the agent acts strictly within the bounds of the invoking user's authorizations. A user can't use an agent to perform an action that their authorizations wouldn't otherwise allow.

Access assignments are not inherited. If an assistant invokes one or more agents, assigning a user to the assistant does not automatically grant them access to those agents. You must assign the user to the assistant and to each agent it invokes separately. If any assignment is missing, the user receives an authorization error when the assistant attempts to call the agent.

You can assign access to individual users or to user groups. Assigning user groups is the recommended approach for most scenarios. Group-based access remains consistent as your user base changes: adding or removing a user from the user group automatically updates their agent access without any change to the agent's configuration. Assign access to individual users only when access is specific to one person and isn't expected to extend to others.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Users & Authorizations*, choose the *Agents* tile.
3. Choose an agent from the list on the left.

   The system displays a list of the agents in the tenant.
4. Choose the *Access* tab.
5. Choose the *Users* or *Groups* tab and then choose *Manage Access*.
6. Add users or user groups to the agent. You have the following options:
   - Select users or groups from the list.
   - Search for users or groups.
     - *Starts With*: (default) Returns only entities that start with the entered value.
     - *Contains*: Returns all entities that contain the entered value. This option may take longer to process.
7. Choose *Review Changes*.
8. Save your changes.

Results

The selected users and members of the selected user groups can now invoke the agent. The change takes effect the next time the user gets a token.

Example: Granting a Team Access to a Joule Assistant and Its Agents

Donna Moore's organization has deployed a Joule assistant for financial analysts in SAP S/4HANA. The assistant invokes two agents: a report generation agent and a data retrieval agent. She wants to grant the entire finance team access to all three.

Before she starts, she confirms that a user group for the finance team already exists in SAP Cloud Identity Services. For example, `Joule_Finance_Analysts`. If the user group doesn't yet exist, she creates it first.

Because access is not inherited, Donna must perform this procedure once for each entity: the assistant and each of its agents. She performs the following steps three times, selecting a different entity each time.

To assign the user group to the agent:

1. In the administration console, Donna goes to *Users & Authorizations* and chooses the *Agents* tile.
2. She selects the financial reporting agent from the list.
3. She chooses the *Access* tab, then the *Groups* tab, and then chooses *Manage Access*.
4. She searches for `Joule_Finance_Analysts`, selects it, chooses *Review Changes*, and then saves.

After Donna completes all three assignments, members of `Joule_Finance_Analysts` can invoke the assistant and the agents it calls. If Donna assigns the group to the assistant but omits one of the agents, users will receive an authorization error when the assistant attempts to call that agent.

When a new analyst joins the team, Donna adds them to the user group `Joule_Finance_Analysts` in SAP Cloud Identity Services. Donna doesn't need to modify the access configuration of the assistant or either agent. When an analyst transfers to another department, removing them from the group revokes their access to all three entities at once.

Each analyst can use the assistant and agents only to perform actions that their own SAP S/4HANA authorizations permit. Assigning the group doesn't grant any additional permissions in the back-end system.

Task overview: Managing Agents [page 2333]

Related Information

Review Agent Authorizations [page 2336]
Enable Access to Agents for the SAP Early Adopter Care Program [page 2341]
Identity and Access Management for Joule [page 580]
Agents [page 511]

1.7.10.3 Enable Access to Agents for the SAP Early Adopter Care Program

For the SAP Early Adopter Care (EAC) program, the SAP-managed provisioning jobs create application groups for the Joule agents. Assign these groups to users to enable them to use these agents.

Prerequisites

> ⚠ Caution
>
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Before you begin, ensure the following are in place:

- SAP-managed provisioning jobs have completed successfully and agent authorization groups are available in SAP Cloud Identity Services.
  See also Set Up Identity and Access Management.
- The users you want to assign already exist in SAP Cloud Identity Services.
- You are assigned an administrator role with the *Manage Users* or *Manage Groups* authorizations in the SAP Cloud Identity Services administration console.

Context

After SAP-managed provisioning jobs create authorization groups for Joule agents in SAP Cloud Identity Services, you must assign users to these groups to grant them access to specific agents. This task is part of Stage 2 deployment, where users with real identities in SAP Cloud Identity Services can be authorized to use agents alongside the conversational Joule assistant.

Each agent has its own authorization group. Users must be members of an agent's group to see and interact with that agent. Without group membership, users can still access conversational Joule functions but can't use agents.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Users & Authorizations*, choose the *Groups* tile.
3. Choose a group from the list on the left.

   Agent authorization groups are created by SAP-managed provisioning jobs. You can sort the list by group type to find the correct authorization group.

4. Choose + *Add*.
5. Add users to the group.
6. Save your changes.

Results

Users assigned to the agent authorization group can now access and interact with the agent. The changes take effect immediately. Users may need to refresh their browser session to see newly available agents.

Next Steps

If users can't access the agent after being assigned to its authorization group:

- Verify that the user has a real identity in SAP Cloud Identity Services with a global user ID assigned.
- Confirm that the user has logged in at least once to the connected line-of-business application using SAP Cloud Identity Services as the identity provider.
- Ask the user to sign out and sign back in to refresh their session.

Task overview: Managing Agents [page 2333]

Related Information

Review Agent Authorizations [page 2336]
Enable Users to Access Agents [page 2338]
Enable Users to Access Agents [page 2338]
Identity and Access Management for Joule [page 580]

1.7.11 Managing Users

Tenant administrators can manage user accounts via the administration console for SAP Cloud Identity Services, and via APIs.

The user management enables you to create, modify, and delete users and their attributes, and manage the user accounts in the user store of Identity Authentication.

> **Note**
>
> For more information about the users that are authenticated with their corporate credentials from the corporate user store, see Corporate User Store (Neo Environment) [page 2531].

→ Remember

To perform user management operations, you must be assigned an administrator role or roles that include the relevant authorizations for the operation. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

With user management, you can perform the following activities:

Activity	Description	Procedure
Create users	Provision users	Centralized Provisioning with Identity Directory [page 554]
	Integrate your identity and access management solution with Cloud Identity Services API	Identity Directory API
	Create users via a CSV file import in the administration console	Import or Update Users for a Specific Application [page 2348]
	Manually create users via the *Add* option in the administration console (not recommended for mass user creation)	Create a New User [page 2357]
Search users	Search users in the administration console	Search Users [page 2359]
	Search users via API	Identity Directory API
List and edit user details	List a specific user and edit the information about that user via the administration console	List and Edit User Details [page 2361]
	List and update user details via API	Identity Directory API
	Update user details via a CSV file import	Import or Update Users for a Specific Application [page 2348]
	Manage user password via the administration console	• Unlock User Password [page 2380] • Send Reset Password Email [page 2381] • Reset Counter for Email Sending [page 2383] • Set Initial Password [page 2384]
Delete users	Delete users via the administration console	Delete Users [page 2367]
	Delete users programmatically via API	Identity Directory API

Activity	Description	Procedure
Manage the user group assignment	Assign and unassign groups via the administration console	• Assign Groups to a User [page 2409] • Unassign Groups from a User [page 2410]

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
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Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.11.1 Import CSV File with Full User Profile

As a tenant administrator, you can create new users or update existing ones with all user data, including attributes from a custom schema, via a CSV file upload.

Prerequisites

- You are assigned the *Manage Users* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

With the CSV file, you can import up to 25000 users to create new users or to update existing users. This is the recommended approach for importing user.

> → Remember
>
> The `SCIM ID` serves as the primary identifier if it is found in the CSV file. The users will only be updated, but not created if there are users with a new `SCIM ID`. This process works within the same tenant because the SCIM IDs differ across tenants.
>
> If the `SCIM ID` doesn't exist in the CSV file, the create and update functionality will function as expected, but the administrator won't be able to modify the `username` or `email`.

The attributes can take one of the following types:

Attribute Type	Column Name (Example)	Value (Example)
`multivalued complex`	emails[0].value	michael.adams@example.com
`complex`	name.familyName	Adams
`single value`	displayName	Michael Adams
`multivalue`	urn:sap:cloud:scim:schemas:extension:custom:2.0:CustomSchema:arrayOfStringsAttr	test1;test2

The following attributes must be a valid master data text:

Attribute	Where to find it
`locale`	• administration console - *Master Data Texts tile* > *Language tab* • API - GET `https://<tenant ID>.accounts.ondemand.com/service/resource?resourceType=RESOURCE_MD_LANGUAGES&locale=en`
`addresses[0].country`	• administration console - *Master Data Texts tile* > *Countries tab* • API - GET `https://<tenant ID>.accounts.ondemand.com/service/resource?resourceType=RESOURCE_MD_COUNTRIES&locale=en`

Attribute	Where to find it
`urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:department`	• administration console - *Master Data Texts tile* > *Departments tab* • API - GET `https://<tenant ID>.accounts.ondemand.com/service/resource?resourceType=RESOURCE_MD_DEPARTMENTS&locale=en`
`name.honorificPrefix`	• administration console - *Master Data Texts tile* > *Salutations tab* • API - GET `https://<tenant ID>.accounts.ondemand.com/service/resource?resourceType=RESOURCE_MD_SALUTATIONS&locale=en`

→ Remember

The CSV file must contain at least the following three columns:

- `userName` or `loginName`
- `emails[0].value` or `mail`
- `name.familyName` or `lastName`

→ Remember

If there is a column with the name of a certain attribute, and you provide a value for that column, the attribute will be updated.

If there is a column with the name of a certain attribute, and you don't provide a value for that column, the attribute will be deleted.

If there is no column with the name of a certain attribute, the value of the attribute remains unchanged.

If the names of the columns are not from the core schema, they must be written with the `urn:attribute name`.

→ Tip

To update the `emails` or `userName` attribute you must provide the `SCIM ID` for the users that you want to update.

ⓘ Note

The user import doesn't assign any special rights or roles to the created or updated users for the specific application.

The CSV file can contain only columns with no spaces between them, with all the attributes whose names are compliant with SCIM specification, plus the attributes from a custom schema that exist in the tenant of Identity Authentication. If you include columns with other attributes, their values in the table are ignored.

If you enter the data in the CSV file as text, you must separate the entries with commas. Beware not to put any spaces before or after the comas. If you enter more than one value in a single entry, separate the values within the entry with semicolons.

Values shouldn't contain semicolons. If there is a semicolon, in the import the value will be split by that semicolon and more than one attribute value will be added.

Depending on your *Logon Alias* configuration in the administration console, you must consider the following:

- If *Email* is required and unique, in the csv file at least one `emails[0].value` must be present and unique.
- If *Login Name* is required and unique, in the csv file at least one `userName` must be present and unique.
- If *Display Name* is required and unique, in the csv file at least one `displayName` must be present unique.

⚠ Restriction

The user import doesn't support the `groups` and `spCustomAttribute1`

You can view all of the supported schemas attributes and their definitions in the administration console of Identity Authentication under schemas. For more information, see Manage Custom Schemas via Administration Console [page 2388].

Example

Sample CSV file:

Sample Code

```
emails[0].value,emails[0].primary,emails[0].type,name.familyName,name.given
Name,name.middleName,name.formatted,name.honorificPrefix,name.honorificSuff
ix,userName,displayName,nickName,profileUrl,title,userType,active,addresse
s[0].type,addresses[0].streetAddress,addresses[0].locality,addresses[0].reg
ion,addresses[0].postalCode,addresses[0].country,addresses[0].formatted,add
resses[0].primary,addresses[1].type,addresses[1].streetAddress,addresses[1]
.locality,addresses[1].region,addresses[1].postalCode,addresses[1].country,
addresses[1].formatted,phoneNumbers[0].value,phoneNumbers[0].type,phoneNumb
ers[1].value,phoneNumbers[1].type,ims[0].value,ims[0].type,ims[1].value,im
s[1].type,photos[0].value,photos[0].type,photos[1].value,photos[1].type,urn
:ietf:params:scim:schemas:extension:enterprise:2.0:User:division,urn:ietf:p
arams:scim:schemas:extension:enterprise:2.0:User:department,urn:ietf:params
:scim:schemas:extension:sap:2.0:User:mailVerified,urn:ietf:params:scim:sche
mas:extension:sap:2.0:User:validFrom,urn:ietf:params:scim:schemas:extension
:sap:2.0:User:validTo,urn:sap:cloud:scim:schemas:extension:custom:2.0:User:
attributes[0].name,urn:sap:cloud:scim:schemas:extension:custom:2.0:User:att
ributes[0].value
dona.moore@example.com,TRUE,work,Moore,Dona,Moore,Dona Moore
Moore,Ms.,III,donaM,Dona Moore Moore,random,https://
test.com,Dr,public,TRUE,work,100 Universal City
Plaza,Hollywood,CA,91608,US,100 Universal City Plaza Hollywood CA 91608
US,true,home,456 Hollywood Blvd,Hollywood,CA,91608,US,456 Hollywood Blvd
Hollywood CA 91608
US,555-555-5555,work,555-555-4444,mobile,someaimhandle,aim,someaimhandle2,a
im,https://photos.example.com/profilephoto/72930000000Ccne/F,photo,https://
photos.example.com/profilephoto/72930000000Ccne/T,thumbnail,Theme
Park,Real Estate
Management,true,2021-10-04T06:48:20Z,2023-10-04T06:48:20Z,customAttribute1,
test
```

To import users into the user store of Identity Authentication, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Users & Authorizations*, choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.
3. Press the *Import* button.
4. Choose the *Browse...* button and specify the location of the CSV file.

   > ⓘ Note
   >
   > Use a UTF-8 encoded file with up to 25000 users and an extension `.csv`. If your file contains more than 25000 users, you have to import the user information in iterations within the user number limit.
5. Choose the *Save* button.

   If the operation is successful, the system displays a message with the number of imported and updated users.

Related Information

SCIM Common Attributes

1.7.11.2 Import or Update Users for a Specific Application

As a tenant administrator, you can import new users or update existing ones for a specific application with a CSV file. You can also send activation emails to the users that have not received activation emails for that application so far.

Prerequisites

- You are assigned the *Manage Users* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have uploaded the service provider's metadata or entered the information manually. For more information see, Configure Trust [page 1941].

  > ⓘ Note
  >
  > You need the metadata to configure the trust between the service provider and Identity Authentication, which is in the role of identity provider.

Context

> **Note**
>
> The recommended approach to import users is via a CSV file upload. For more information, see Import CSV File with Full User Profile [page 2344].

By importing new users with a CSV file, you create user profiles without passwords in Identity Authentication. As a result, the users receive emails with instructions how to activate their accounts. After the users set their passwords, they can log on to the application for which they were imported. Based on the user access configuration of the application, the users can log on to other applications connected with the tenant in Identity Authentication.

In addition to the new user import, you can specify existing users in the imported CSV file. You thus define the users to be updated in Identity Authentication.

By specifying existing users in the imported CSV file you can also restrict the access to a specific application via the *Private* options. For more information, see Configure User Access to the Application [page 2025].

> **Note**
>
> The user import does not assign any special rights or roles to the created or updated users for the specific application.

The CSV file can contain only columns with no spaces between them, with the following attributes. If you include columns with other attributes, their values in the table are ignored.

> **Note**
>
> If you enter the data in the CSV file as text, you must separate the entries with commas. Beware not to put any spaces before or after the commas. If you enter more than one value in a single entry, separate the values within the entry with commas and enclose the entry in quotation marks.

CSV Files Attributes

Attributes	Mandatory	Unique	Valid Values	Notes
`status`	Yes	No	• `active` • `inactive`	The *status* column defines whether the user is still active in the system and is able to work with any tenant applications. When a user is deleted, it's rendered inactive. **Note** When a user's password is not set, if you pass a `status` with value `active`, the status of the user will not change to `active`. It will be set to `new`.
`loginName`	No	Yes		Must be a string value of up to 64 characters. You cannot change the email of an existing user.

Attributes	Mandatory	Unique	Valid Values	Notes
`mail`	Yes	Yes		→ Tip For users without valid email addresses or for testing purposes use the `sap-test.de` domain, for example `<username>@sap-test.de`. Do not use any other existing or non-existing domains. Must be a string value of up to 64 characters. Values that are part of the respective exclude list can't be used. For more information, see Restrict User Attributes Values via Exclude Lists [page 2385].
`firstName`	No	No		Must be a string value of up to 64 characters. Values that are part of the respective exclude list can't be used. For more information, see Restrict User Attributes Values via Exclude Lists [page 2385].
`lastName`	Yes	No		Must be a string value of up to 64 characters. Values that are part of the respective exclude list can't be used. For more information, see Restrict User Attributes Values via Exclude Lists [page 2385].

Attributes	Mandatory	Unique	Valid Values	Notes
`language`	No	No		Must be a string value specified by a two or four-letter code in one of the following formats: XX. Otherwise, the activation email is in English.
`validTo`	No	No		Must be a string value in the Zulu format yyyyMMddHHmmss'Z' . The information in the *validFrom* and *validTo* columns can be processed by the service provider to limit user access. It would affect the authentication of the user, though.
`validFrom`	No	No		Must be a string value in the Zulu format yyyyMMddHHmmss'Z' . The information in the *validFrom* and *validTo* columns can be processed by the service provider to limit user access. It would affect the authentication of the user, though.
`userType`	No	No	• public • employee • customer • partner • external • onboardee • alumni	

Attributes	Mandatory	Unique	Valid Values	Notes
`urn:ietf:params:scim:schemas:extension:sap:2.0:User:mailVerified`	No	No	• true • false - the default value	
`spCustomAttribute1`	No	No		
`spCustomAttribute2`	No	No		
`spCustomAttribute3`	No	No		
`spCustomAttribute4`	No	No		
`spCustomAttribute5`	No	No		
`groups`	No	No	The `name` attribute defined in the Group schema (`urn:sap:cloud:scim:schemas:extension:custom:2.0:Group`)	The groups in the *groups* column must be existing. You can't add a user to a user group that is not existing.

Examples

Example

A tenant administrator decides to import three new users (Michael, Julie, Donna) and to update two others (John and Denise) that will use the company's applications. Michael is a member of three groups, namely *Employees*, *Managers*, and *HR*. John and Denise were inactive users that now use tenant's applications. The administrator would also like to update another user (Richard) who currently does not work for the company. To do this, the administrator uploads a CSV file with the following information:

Sample Code

```
status,loginName,mail,firstName,lastName,language,validFrom,validTo,spCustomAttribute1,groups
active,EID00001,michael.adams@example.com,Michael,Adams,en,20110901120000Z,20150901120000Z,Industry,"Managers, Employees, HR"
active,EID00002,julie.armstrong@example.com,Julie,Armstrong,en,20110901120000Z,20150901120000Z,Department,Employees
active,EID00003,donna.moore@example.com,Donna,Moore,de,20110901120000Z,20160901120000Z,Shift,HR
active,EID00004,john.miller@example.com,John,Miller,en,20110901120000Z,20180901120000Z,Unit,IT
active,EID00005,denise.smith@example.com,Denise,Smith,en,20110901120000Z,20140901120000Z,,Administrators
inactive,EID00006,richard.wilson@example.com,Richard,Wilson,en,20110901120000Z,20160901120000Z,,All
```

The administrator receives the following message: `3 users will be created. 3 users will be updated because they already exist. Do you want to continue?`

The users that have not received activation emails will receive such emails, and then can activate their accounts and log on.

Example

A sales company has employed two new people (Michael Adams and Julie Armstrong) in the HR department. The tenant administrator has to import these two new employees that will use the company's applications. Michael and Julie are members of the groups **Employees** and **HR**. To do this, the administrator uploads a CSV file with the following information:

Sample Code

```
status,mail,firstName,lastName,groups
active,michael.adams@example.com,Michael,Adams,"Employees, HR"
active,julie.armstrong@example.com,Julie,Armstrong,"Employees, HR"
```

Six months later, Julie is promoted as HR Manager, while Michael moves to the Sales department. The tenant administrator has to update the profiles of Michael and Julie to represent the changes. Julie remains in the current groups and has to be included in the **Managers** group. Michael has to be removed from the **HR** group, and to be included in the **Sales** group. To do this, the administrator uploads a CSV file with the following information:

Sample Code

```
status,mail,firstName,lastName,groups
active,michael.adams@example.com,Michael,Adams,"Employees, Sales"
active,julie.armstrong@example.com,Julie,Armstrong,"Employees, HR,
Management"
```

→ Remember

When tenant administrators must update a user via a CSV file, they must include the new information, and repeat the information that remains unchanged.

If you, as a tenant administrator, include only the new information, the information that is not repeated will be deleted.

In this example, the administrator imports Michael with the **Employees** group, and Julie with the **Employees** and **HR** groups in the new table.

To import users for an application into Identity Authentication, and to send activation emails, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.

2. Choose the *Import Users* tile.

   This operation opens the *Import Users* page.
3. Choose the application that you want to edit.

> ### Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Under *Upload CSV File*, choose the *Browse...* button and specify the location of the CSV file.

> ### Note
>
> Use a file smaller than 100 KB and with an extension `.csv`. If your file is 100 KB or larger, you have to import the user information in iterations with smaller size files.

5. Choose the *Import* button.

   If the operation is successful, the system displays the message `Users imported or updated`.
6. Choose the one of the following options:

Option	Description
Do nothing	The users are imported or updated for the selected application, but they will not receive activation emails. The activation emails will be sent when you choose *Send Emails* > *Send*.
Repeat steps 1 to 5	The users are imported or updated for the selected application, but they will not receive activation emails. The activation emails will be sent when you choose the *Send* button.
Under *Send Emails* choose the *Send* button	This will send activation emails to all users that are imported for the selected application, but have not received activation emails so far. **Note** The *Send* button is inactive if *Home URL* or SAML 2.0 configuration of the application is missing. You can only import users, but you cannot send activation emails. You need the *Home URL* configured for the specific application to be able to send the activation emails to the imported new users. For more information, see Configure an Application's Home URL [page 1937]. To access the application, the users have to activate their accounts by following the link they receive in the emails.

Related Information

Configure an Application's Home URL [page 1937]
Configure User Access to the Application [page 2025]
List and Edit Groups [page 2400]

Create a Group [page 2398]
Edit Administrator Authorizations [page 2309]
Export Existing Users of a Tenant of Identity Authentication [page 2356]

1.7.11.3 Export Existing Users of a Tenant of Identity Authentication

Prerequisites

- You are assigned the *Manage Users* or *Read Users* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- (For administrators assigned to a custom authorization policy) You are assigned to `READ_USERS` and `READ_SCIM_SCHEMAS` authorization policies. Otherwise, you won't be able to see and access the *Export Users* tile.

Context

You can export a CSV file containing information of all tenant users in Identity Authentication including the tenant administrators. The CSV file can contain columns with all user supported SCIM attributes. You can filter the user attributes that you want to include in the exported file. The default columns are: *active*, *userName*, *userType*, *emails[0].display*, *emails[0].primary*, *emails[0].type*, and *emails[0].value*.

If the values for the *validTo* and *validFrom* attributes for a specific user aren't in the correct Zulu format yyyyMMddHHmmss'Z', Identity Authentication returns empty values for these attributes in the CSV file.

> → Remember
>
> The `SCIM ID` serves as the primary identifier if it is found in the CSV file. The users will only be updated, but not created if there are users with a new `SCIM ID`. This process works within the same tenant because the `SCIM ID`s differ across tenants.
>
> If the `SCIM ID` doesn't exist in the CSV file, the create and update functionality will function as expected, but the administrator won't be able to modify the username or email.

> ⚠ Restriction
>
> Each multivalued attribute is limited to 3 fixed values, and each custom attribute is limited to 10 values.
>
> For example, if a user in the administration console has more than 3 emails, the export functionality will export only 3 emails. On the other hand, if a user has only one email, then the csv file will show 3 emails, but two of them will be empty.
>
> If you need to export multivalued attributes with more than 3 values, or custom attributes with more than 10 values, you can use the Identity Provisioning service. For more information, see Real-Time Provisioning from Identity Authentication or Provisioning from Source to Target Systems.

> **Note**
>
> If the *active* attribute of a user is FALSE, he or she can't perform any operations on the tenant.

> **Example**
>
> A tenant administrator exports a CSV file with the current users in the system and only the following attributes: *active*, *userName*, *userType*, and *emails[0].value*. As a result, the administrator receives the following information:

active	userName	userType	emails[0].value
TRUE	EID00001	public	michael.adams@example.com
TRUE	EID00002	employee	julie.armstrong@example.com
TRUE	EID00003	partner	donna.moore@example.com
FALSE	EID00004	customer	john.miller@example.com

To export tenant users from Identity Authentication, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Users & Authorizations*, choose the *Export Users* tile.

   This operation opens the *Export Users* page.
3. Choose the attributes that you want to include in the CSV file.

   The default choice is: *active*, *userName*, *userType*, *emails[0].display*, *emails[0].primary*, *emails[0].type*, and *emails[0].value*.
4. Choose the *Export* button.

1.7.11.4 Create a New User

As a tenant administrator, you can create a new user in the administration console for SAP Cloud Identity Services.

Prerequisites

You are assigned the *Manage Users* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

The tenant administrator creates the new user with a minimum set of attributes and can set an initial password.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.
3. Press *Add*.
4. Fill in the required fields in the dialog box.

   > ⓘ Note
   >
   > By default the *User Type* field is *Employee*. To change the default setting, choose user type from the drop-down list. The available user types are: *Customer*, *Employee*, *Partner*, *Public*, *External*, *Onboardee*, and *Alumni*.

   Email field can be skipped when *Set initial password* or *Set status active* is selected, and *Email* is configured as not required in the *Logon Alias* view.

   Values for *Email*, *First Name* and *Last Name* that are part of the respective exclude list can't be used. For more information, see Restrict User Attributes Values via Exclude Lists [page 2385].

   > → Tip
   >
   > If email is mandatory, for users without valid email addresses or for testing purposes use the `sap-test.de` domain, for example `<username>@sap-test.de`. Do not use any other existing or nonexisting domains.
5. Select one of the following options:
   - Send activation email - The tenant administrator creates a user with status `New`. The user receives an email with instructions how to activate the user account. After activating the account, the user status changes to `Active`.
   - Set initial password - The tenant administrator creates a user with status `Active` and sets the password for the user.

     > ⚠ Restriction
     >
     > The initial password can be valid between 1 and 365 days depending on the configuration made by the administrator. The default value is 14 days. The user is prompted to reset the password during the first authentication. After the validity of the initial password expires, the user can't log on to the application and must contact an administrator. For more information for more information about how to configure the initial password validity, see Configure Initial Password and Email Link Validity [page 2168].

- Set status active - The tenant administrator creates a user with status `Active`. The tenant administrator does not set an initial password for the user, and the user does not receive an email with instructions how to activate the user account.
  This option can be used in the following scenarios:
  - Identity Authentication acting as identity provider (IdP) proxy
  - certificate authentication

6. Save your entries.

Results

If the operation is successful, the system displays the message: `User added`. Identity Authentication creates the new user and assigns `User ID` (P user) and `Global User ID` (universally unique identifier (UUID) format). The `User ID` field is unique and not editable. The `Global User ID`, on the other hand, is unique, but editable. You can change it via the user management field in the administration console.

Related Information

Configure User Identifier Attributes [page 2149]
Restrict User Attributes Values via Exclude Lists [page 2385]
List and Edit User Details [page 2361]

1.7.11.5 Search Users

As a tenant administrator, you can search for a specific user or users in the administration console for SAP Cloud Identity Services.

Prerequisites

You are assigned the *Manage Users* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

You can list all users in the tenant for Identity Authentication or filter your search by *User ID*, *Global User ID*, *Scim ID*, *First Name*, *Last Name*, *Email*, or *Login Name*.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.
3. **Optional:** You can choose one of the following:

   Search Options

List All Users Press *More*	This will expand the list by 20 users. **Note** This option is available only if there are more than 20 users in the tenant.
Unfiltered Search • (default mode) Type your search criteria string in the search field and press the *Enter* key • (when in *Filtered Search* mode) Press *Hide Filters*, type your search criteria string in the search field, and press the *Enter* key	Once the search is completed, the system will list the users whose *User ID*, *Global User ID*, *SCIM ID*, *Email*, or *Login Name* match your search criteria string. In this case the system doesn't include the *First Name* and *Last Name* fields in the search. If you aren't satisfied with the search result, edit your search criteria and repeat the step again.
Filtered Search Press *Show Filters*, type your search criteria strings in at least one of the search fields, and press the *Enter* key	The system checks the search fields simultaneously. Once the search is completed, it will list the users that match the search criteria from all the fields.

4. **Optional:** To configure the user details columns, choose the *Settings* icon and select the columns that you want to see.

Related Information

Edit Administrator Authorizations [page 2309]

1.7.11.6 List and Edit User Details

As a tenant administrator, you can view detailed information about the users in the administration console for SAP Cloud Identity Services. Optionally you can edit this information.

Context

User ID (P user) and *Global User ID* (universally unique identifier (UUID) format) are automatically generated by Identity Authentication at user creation. The `User ID` field is unique and not editable. The *Global User ID*, on the other hand, is unique, and editable. Thus you can use the generated one, or change it with your own one. You can also reuse previous versions of the *Global User ID* for one and the same user.

By default the *User Type* field is *Employee*. The available user types are: *Customer*, *Employee*, *Partner*, *Public*, *External*, *Onboardee*, and *Alumni*.

Last Name and *Email* fields are mandatory.

Vaues for *Email*, *First Name* and *Last Name* that are part of the respective exclude list can't be used. For more information, see Restrict User Attributes Values via Exclude Lists [page 2385].

Display Name can be configured to be mandatory or not. For more information see Configure User Identifier Attributes [page 2149].

Email and *Login Name* can be used as unique identifiers. Be sure to enter unique values if you edit these two fields.

Only administrators with *Manage Tenant Configuration* role can edit the *Email* attribute.

Display Name can be configured to be unique or not. For more information see Configure User Identifier Attributes [page 2149].

If the check box below the *Email* is selected, the `email` of the user is verified. For more information about how to require email verification of the user, see, Enable Email Verification [page 2026].

If the check box below the *Telephone Verified* is selected, the phone of the user is verified. The tenant administrator can select it manually, to mark the phone as verified.

> ⚠ Caution
>
> Be careful when there is no Sinch configuration for the tenant and you manually verify the phone via the administration console. In this case users can't change their phones via the profile page. Make sure to inform the users how to change the phone on the profile page.

User ID, and *Manager Display Name* fields cannot be edited. They are filled automatically by the system.

The *Valid From* and *Valid To* fields are empty if no validity dates are set for the user.

> → Remember
>
> When the *Display Name* is set as unique:
>
> - If the user is created via the User Registration service of the User Management Rest API, and after successful activation of the user, the display name is generated in the following way: *Display Name* is `first_name last_name`.

- If there is already another user with the same display name, the *Display Name* is `first_name last_name user_profile_id`.

The display name can be changed also by the user via the user profile page.

Attributes from Custom Schema

The attributes assigned from custom schema are visible at the bottom of the *Details* tab under a section with the name of the custom schema.

You can also update an attribute by calling the PATCH method of Resource Management API of SAP Cloud Identity Services.

Procedure

1. Find the user whose details you want to view or edit.

   For more information about how to find a user in Identity Authentication, see Search Users [page 2359].
2. Click the user to view his or her details.

   This operation opens the *Details* view.
3. **Optional:** Choose the *Details* tab.
   a. Press the *Edit* button at the top of the page.
   b. Edit the information in the relevant fields and save your changes.

      If the operation is successful, the system displays the message `User <user ID> updated`.

   > ### Note
   >
   > To exit edit mode, press the *Cancel* button.
4. **Optional:** Choose the *Applications* tab, to view details specific for the applications that the user has logged on, and the applications that the user was imported to via a CSV file import.

   > ### Note
   >
   > The tab does not appear if the user hasn't logged on, or wasn't imported to an application via a CSV file import. The tab *Applications* is replaced by *Administration Console* if the user is admin and the only application that they have logged on is the administration console.
5. **Optional:** Choose the *Legal* tab to view audit information about the user, such as the policies accepted, the last log on, and password related information.

   > ### → Remember
   >
   > The *Last Log On* and *Password Login Time* fields are updated once per 24 hours.

   > ### Note
   >
   > The tenant administrators can view a history of the signed **Terms of Use** and **Privacy Policy** documents of the users in the administration console. The information is in the format *<document name><document version><time stamp>*.

> The information about when the user has accepted a particular document is available for documents accepted after May 16, 2018.

6. **Optional:** Choose the *Authentication* tab to manage the two-factor authentication for the user. You have the following options:

Option	Additional Information
User Status	Unblock a User [page 2369]
Authentication > *Password Details*	• Unlock user password. For more information, see Unlock User Password [page 2380]. • Send reset password email. For more information, see Send Reset Password Email [page 2381]. • Reset counter for email sending. For more information, see Reset Counter for Email Sending [page 2383]. • Set initial password to the user. For more information, see Set Initial Password [page 2384]. **Note** Only administrators with *Manage Tenant Configuration* role can set initial password of other administrators.
Authentication > *Multi-Factor Authentication*	If the user has enabled two-factor authentication, you can do the following: • Deactivate the user devices. For more information, see Deactivate User Devices for TOTP Two-Factor Authentication [page 2371]. • Unlock User TOTP Passcode [page 2372].

7. **Optional:** Choose the *User Groups* tab to manage the group assignments of the user.

   You can see the user groups assigned to the user. You have two options:

   - Assign groups. For more information, see Assign Groups to a User [page 2409].
   - Unassign one or more groups that are assigned to the user Unassign Groups from a User [page 2410].

1.7.11.7 Assign Authorization Policies

As an administrator, you can assign authorizations to users.

Prerequisites

- Your application supports authorization policies. Refer to the documentation of your application.
- Your user has administrative permissions in Identity Authentication with the the following authorizations:
  - *Manage Applications*

- *Manage Groups*

  For more information, see Edit Administrator Authorizations [page 2309].
- You have a subscription to SAP Cloud Identity Services. For more information, see SAP Cloud Identity Services.

> **Note**
>
> If your application doesn't have an *Authorization Policies* tab, it doesn't support authorization management.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose the *Applications* tile.
3. Choose your application that supports authorization management. For more information, see the documentation of the application.

   The details page of your application has an *Authorization Policies* tab.
4. Choose the *Authorization Policies* tab.
5. Choose an authorization policy that you want to assign. The administration console opens the *Assignments* pane.
6. Choose *Add*.
7. Select the users that you want to add to the authorization policy and choose *Add*.

   You have added the selected users to an authorization policy. These users are authorized to access and use the resources with the rules and restrictions defined in the authorization policy.

Related Information

Configuring Authorization Policies [page 2244]

1.7.11.7.1 Manage Authorization Policies for a Reuse Service Across Multiple Applications

The authorization data for a reuse service is shared among multiple business applications in the same tenant, ensuring consistent authorization policies across different applications for users logged on to these applications.

Context

A reuse service is consumed by multiple business applications, for example application 1 and application 2. All applications and reuse services can have their own set of authorization policies that are always tied to the application or to the reuse service. For easy maintenance, it is possible to manage the authorization policies of the reuse service in the *Authorization Policies* tab of the application that consumes the reuse services.

When an administrator assigns a user to an authorization policy of the reuse service, the reuse grants this permission regardless of the authorization context of the business application. That means, if the user logs on to application 1 or application 2, the permissions for the reuse service are the same, meaning the authorization policies of the reuse service.

The same is reflected on the administration side. Users assigned to authorization policies of the reuse service can always consume the same content. It doesn't matter whether they have logged on to application 1 or application 2.

If you assign the authorization policies of the reuse service to users of application 1 or 2, you enable these users to access the reuse service. They can simply log on to application 1 or 2 and immediately use the reuse service.

To assign authorization policies of the reuse service to users, take the following steps.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications & Resources*, choose the *Applications* tile.
3. Choose the application that supports authorization management and authorization policies for the reuse service.
4. Choose the *Authorization Policies* tab. You see the list of the authorization policies of your application. Next to the table title, there's a dropdown list where you can select the reuse service. Currently, it displays the name of your application.

   > ⓘ Note
   >
   > If the consuming application doesn't have its own policies, the administration console directly displays the policies of the reuse service instead.

5. Open the drowdown list.
6. Select the reuse service. The table displays the list of the policies that belong to the reuse service.

7. Choose an authorization policy you want to assign. The *Assignments* pane opens.
8. Choose *Add*.
9. Select the users that you want to add to the authorization policy and choose *Add*.

   You have added the selected users to an authorization policy. These users are now authorized to access and use the reuse service with the rules and restrictions defined in the authorization policy.

1.7.11.8 Block or Delete Users Due to Inactivity

Tenant administrator can configure specific user types to be blocked and deleted, or only blocked, or only deleted after a certain period of inactivity.

Context

The users for which blocking can be enabled are of the type:

- *Public*
- *Customers*
- *Partners*
- *Employees*

You can enable the chosen user type or types to be only blocked, or only deleted, or both blocked and deleted after a certain period of inactivity. The period spans from 14 up to 2000 days.

To enable blocking and or deletion, you must choose at least one user type and set at least one period.

After a user has exceeded the set period of inactivity, they will be blocked and their status will be updated on the next sign-in attempt.

You can also configure when the deletion script starts.

> ⓘ Note
>
> If you configure both *Block after* and *Delete after* periods, the *Delete after* period must be longer than the *Block after* period.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Choose the *Blocking and Deletion* list item.
4. Choose *Edit*.

5. Select a user type.
6. Enable the options under *Inactivity Periods*. By default the options are disabled. At least one of the status fields *Block after* or *Delete after* must be enabled.

Option	Description
Blocking status	When enabled, configure the *Block after* period. The *Block after* period can be between 14 and 2000 days.
Deletion status	When enabled, configure the *Delete after* period. The *Delete after* period can be between 14 and 2000 days. → Remember If you configure both *Block after* and *Delete after* periods, the *Delete after* period must be longer than the *Block after* period.
Deletion script start date	This status is optional and it appears only after the *Deletion status* is enabled. Configure the date after which the deletion script starts. The automatic configuration is based on the current date plus the days specified in the *Delete after* period. The deletion process starts from the chosen date and is runs nightly at 04:00 UTC.
Creation grace period status	This status is optional and it can be configured only after the *Deletion status* is enabled. When enabled, configure when users who have been created but never logged in will be deleted. The grace period can be between 2 and 30 days. → Remember If the status is disabled, users that are created before the start of the deletion script, but have never logged in, they will be deleted even if the period of their inactivity is less than the delete after period that is configured in the administration console.

7. Save your changes.

Next Steps

To unblock users, follow the procedure in Unblock a User [page 2369]

1.7.11.9 Delete Users

As a tenant administrator, you can delete users in the administration console.

Context

→ Remember

You are not allowed to delete your own user profile.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.
3. Select the checkbox next to the user or users that you want to delete.

   > → Tip
   >
   > If you don't see the user or users that you want to delete, you can expand the list or use the search field. For more information, see Search Users [page 2359].
4. Press *Delete*.
5. Confirm the operation.

   > ⚠ Caution
   >
   > If the user you delete is also an administrator, he or she will be removed from the administrator list too. For more information about how to remove and administrator, without deleting it from the user data base, see Edit Administrator Authorizations [page 2309].

1.7.11.10 Deactivate Users

As a tenant administrator, you can deactivate users in the administration console for SAP Cloud Identity Services.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.
3. Find the user that you want to deactivate.

   For more information about how to find a user in Identity Authentication, see Search Users [page 2359].
4. Select the user that you want to deactivate.

   This operation opens the *Details* view.
5. **Optional:** Choose the *Details* tab.
   a. Expand the *Personal Information*
   b. Press the *Edit* button at the top of the screen.
   c. Select *Inactive* from the dropdown next to the *Status* field.
6. Save your changes.

   If the operation is successful, the system displays the message `User <user ID> updated`.

Results

When a user with status `Inactive` tries to log on to an application the following message appears: `Sorry, we could not authenticate you. Try again.`

When a user is with a status `Inactive`, all actions in the *Password Details* page under the *Authentication* tab of *User Management* are disabled.

> ⓘ Note
>
> To active a user again, set the *Status* to `Active`.

1.7.11.11 Unblock a User

Unblock a user that has been blocked due to inactivity.

Context

The inactivity period is configurable and can be between 14 days and 2000 days. For more information, see Block or Delete Users Due to Inactivity [page 2366].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.
3. Find and select the user that you want to unblock.

   For more information about how to find a user in Identity Authentication, see Search Users [page 2359].

   This operation opens the *Details* view.
4. Choose *Authentication* > *User Status* > *Unblock*.

1.7.11.12 Multi-Factor Authentication

This section provides information about how to manage the multi-factor authentication user settings in the administration console for SAP Cloud Identity Services.

Deactivate Two-Factor Authentication [page 2370]

You can deactivate the second factor (passcode or security key) if the user has activated it via the profile page.

Deactivate User Devices for TOTP Two-Factor Authentication [page 2371]
This document shows you how to deactivate the mobile devices used by a user to generate passcodes for access to applications requiring time-based one-time (TOTP) as two-factor authentication. You deactivate the user mobile devices from the administration console for SAP Cloud Identity Services.

Unlock User TOTP Passcode [page 2372]
You can unlock a user passcode when the user must log on to the application before the automatic unlock time of 60 minutes has passed.

Remove User Device for Web Two-Factor Authentication [page 2373]
This document shows you how to remove the registered devices used by a user for access to applications requiring web two-factor authentication (FIDO2 standard).

Unlock User SMS Code [page 2374]
You can unlock a user SMS code when the user must log on to the application before the automatic unlock time of 60 minutes has passed.

Allow Users to Protect Accounts with Second Factor for Authentication [page 2376]
Tenant administrator can allow users to decide whether to protect their own accounts with second factor for authentication or not.

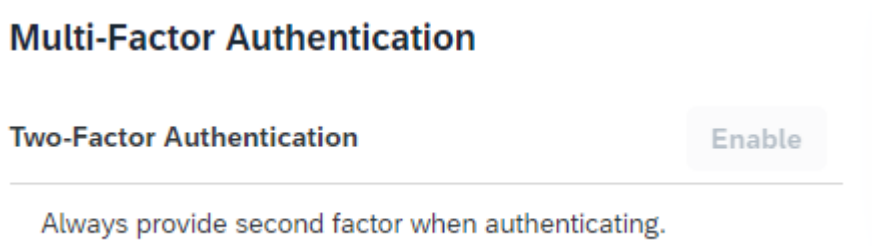
Allow Users To Skip Two-Factor Authentication Setup [page 2378]
You can set the number of days for which the users can postpone the enabling of second factor for authentication.

1.7.11.12.1 Deactivate Two-Factor Authentication

You can deactivate the second factor (passcode or security key) if the user has activated it via the profile page.

Context

When you deactivate the two-factor authentication via the administration console for SAP Cloud Identity Services, the user sees the option disabled on the profile page under the *Multi-Factor Authentication* section:



Once the two factor authentication option is deactivated, the user won't be required to provide a second factor for stronger authentication but the registered devices for access to applications requiring TOTP two-factor or web two-factor authentication (FIDO2 standard) will be preserved in the user's account. Thus, if the two-factor authentication is enabled again, the user will be able to use the already registered devices.

To deactivate the *Two-Factor Authentication* option, proceed as follows:

Procedure

1. Find and select the user whose device you want to remove.

   For more information about how to find a user in Identity Authentication, see Search Users [page 2359].
2. Choose the *Authentication* tab.
3. Choose *Multi-Factor Authentication*.
4. Under *Two-Factor Authentication*, use the slider next to *Status* to deactivate the option.

Task overview: Multi-Factor Authentication [page 2369]

Related Information

Deactivate User Devices for TOTP Two-Factor Authentication [page 2371]
Unlock User TOTP Passcode [page 2372]
Remove User Device for Web Two-Factor Authentication [page 2373]
Unlock User SMS Code [page 2374]
Allow Users to Protect Accounts with Second Factor for Authentication [page 2376]
Allow Users To Skip Two-Factor Authentication Setup [page 2378]

1.7.11.12.2 Deactivate User Devices for TOTP Two-Factor Authentication

This document shows you how to deactivate the mobile devices used by a user to generate passcodes for access to applications requiring time-based one-time (TOTP) as two-factor authentication. You deactivate the user mobile devices from the administration console for SAP Cloud Identity Services.

Context

You deactivate all user's mobile devices that generate passcodes if a single device has been lost or stolen. You cannot deactivate a single device.

> **Note**
>
> If you deactivate the mobile devices, the user will no longer be able to log on to applications that require passcodes. To be able to access them again the user has to activate a new mobile device on the user profile page. For more information, see the Related Information.

To deactivate the mobile devices that generate passcodes, proceed as follows:

Procedure

1. Find the user whose device you want to deactivate.

   For more information about how to find a user in Identity Authentication, see Search Users [page 2359].
2. Select the user whose device you want to deactivate.
3. Choose the *Authentication* tab.
4. Choose *Multi-Factor Authentication*.
5. Under *TOTP*, use the slider next to *Status* to deactivate two-factor authentication.

Task overview: Multi-Factor Authentication [page 2369]

Related Information

Deactivate Two-Factor Authentication [page 2370]
Unlock User TOTP Passcode [page 2372]
Remove User Device for Web Two-Factor Authentication [page 2373]
Unlock User SMS Code [page 2374]
Allow Users to Protect Accounts with Second Factor for Authentication [page 2376]
Allow Users To Skip Two-Factor Authentication Setup [page 2378]

1.7.11.12.3 Unlock User TOTP Passcode

You can unlock a user passcode when the user must log on to the application before the automatic unlock time of 60 minutes has passed.

Context

The user locks his or her passcode after submitting five incorrect passcodes when trying to log on to an application that requires time-based one-time (TOTO) two-factor authentication. The passcode is unlocked automatically after 60 minutes.

> **Note**
>
> When the user locks his or her TOTP passcode, he or she will not be able to log on only to applications that require TOTP passcodes. The applications that do not require two-factor authentication will be accessible only with the user password.

To unlock the user TOTP passcode manually, proceed as follows:

Procedure

1. Find the user whose TOTP passcode you want to unlock.

   For more information about how to find a user in Identity Authentication, see Search Users [page 2359].
2. Select the user that you want to unlock.

   > **Note**
   >
   > You can only unlock locked user TOTP passcodes.
3. Choose the *Authentication* tab.
4. Choose *Multi-Factor Authentication*.
5. Press *Unlock* under *Passcode Status*.

   The system displays the message `Account <name of user> unlocked`.

Task overview: Multi-Factor Authentication [page 2369]

Related Information

Deactivate Two-Factor Authentication [page 2370]
Deactivate User Devices for TOTP Two-Factor Authentication [page 2371]
Remove User Device for Web Two-Factor Authentication [page 2373]
Unlock User SMS Code [page 2374]
Allow Users to Protect Accounts with Second Factor for Authentication [page 2376]
Allow Users To Skip Two-Factor Authentication Setup [page 2378]

1.7.11.12.4 Remove User Device for Web Two-Factor Authentication

This document shows you how to remove the registered devices used by a user for access to applications requiring web two-factor authentication (FIDO2 standard).

Context

You remove the user's device by deleting it from the administration console for SAP Cloud Identity Services.

> **Note**
>
> When you remove the device, the user will no longer be able to log on to applications that require web authentication. To be able to access them again the user has to activate a new device on the user profile page or when prompted when logging on to an application. For more information, see the Related Information.

To remove the devices registered for web authentication, proceed as follows:

Procedure

1. Find and select the user whose device you want to remove.

   For more information about how to find a user in Identity Authentication, see Search Users [page 2359].
2. Choose the *Authentication* tab.
3. Choose *Multi-Factor Authentication*.
4. Under *Web* delete the device that you want to remove.

Task overview: Multi-Factor Authentication [page 2369]

Related Information

Deactivate Two-Factor Authentication [page 2370]
Deactivate User Devices for TOTP Two-Factor Authentication [page 2371]
Unlock User TOTP Passcode [page 2372]
Unlock User SMS Code [page 2374]
Allow Users to Protect Accounts with Second Factor for Authentication [page 2376]
Allow Users To Skip Two-Factor Authentication Setup [page 2378]

1.7.11.12.5 Unlock User SMS Code

You can unlock a user SMS code when the user must log on to the application before the automatic unlock time of 60 minutes has passed.

Context

The user locks his or her SMS code after requesting five codes when trying to log on to an application that requires SMS two-factor authentication. The code is unlocked automatically after 60 minutes.

> ⚠ Caution
>
> If the user phone is not verified, the code will not be unlocked after 60 minutes. In this case you should unlock the code via the procedure below.

> ⓘ Note
>
> When the user locks his or her SMS code, he or she will not be able to log on only to applications that require SMS codes. The applications that do not require SMS two-factor authentication will be accessible only with the user password.

To unlock the user SMS code manually, proceed as follows:

Procedure

1. Find the user whose SMS code you want to unlock.

   For more information about how to find a user in Identity Authentication, see Search Users [page 2359].
2. Select the user that you want to unlock.

   > ⓘ Note
   >
   > You can only unlock locked user SMS codes.
3. Choose the *Authentication* tab.
4. Choose the *Authentication* tab.
5. Choose *Multi-Factor Authentication*.
6. Press *Unlock* under *SMS Status*.

   The system displays the message `Account <name of user> unlocked`.

Task overview: Multi-Factor Authentication [page 2369]

Related Information

Deactivate Two-Factor Authentication [page 2370]
Deactivate User Devices for TOTP Two-Factor Authentication [page 2371]
Unlock User TOTP Passcode [page 2372]
Remove User Device for Web Two-Factor Authentication [page 2373]
Allow Users to Protect Accounts with Second Factor for Authentication [page 2376]
Allow Users To Skip Two-Factor Authentication Setup [page 2378]

1.7.11.12.6 Allow Users to Protect Accounts with Second Factor for Authentication

Tenant administrator can allow users to decide whether to protect their own accounts with second factor for authentication or not.

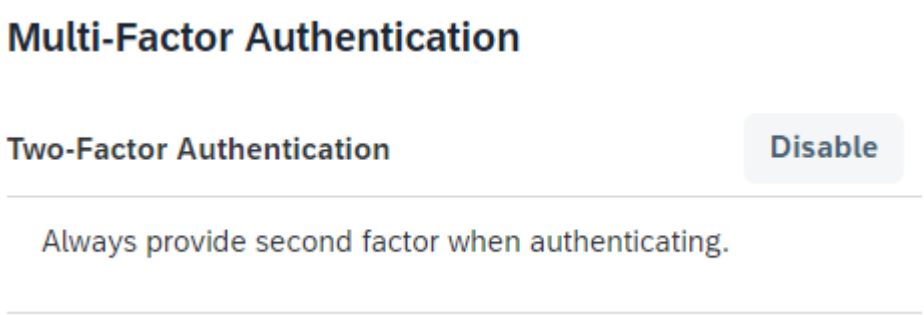
Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

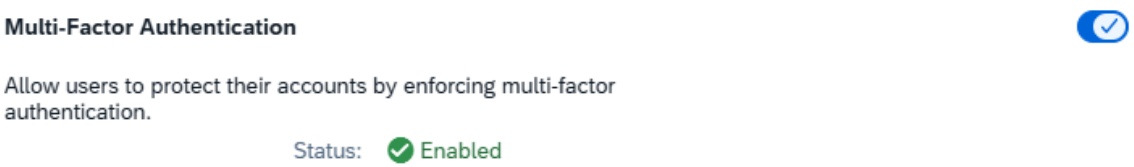
Context

If applications don't require second factor for authentication, or the applications have configured risk-based authentication rules requiring second factor, but the users don't fall into these rules that require second factor for authentication, the users can enhance their security by switching on the *Multi-Factor Authentication* option on the profile page.

> **Note**
>
> The option is switched on by default for all tenants that are created after the February 26, 2026 upgrade of SAP Cloud Identity Services.



This option is visible on the profile page, when it is enabled on the administration console.



When the *Multi-Factor Authentication* option in the administration console is switched on, the users can choose to always provide second factor when authenticating.

If users chose the *Trust this browser* checkbox, the second factor is required only with the first logon. The validity of *Trust this browser* depends on the configuration in the administration console. It can be set between 1 and 14 days. If the additional option for trust this browser is set at *Disabled* the validity of *Trust this browser* is 3 months. But if the administrator sets a validity from 1 to 14 days, the 3-month validity is overridden with the validity set by the tenant administrator.

To enable or disable the *Multi-Factor Authentication* option in the administration console, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Choose the *Multi-Factor Authentication* list item.
4. Enable or disable the *Multi-Factor Authentication* option.

   If the operation is successful, the system displays the message `Multi-Factor authentication updated`.

Results

Users can choose from the *Multi-Factor Authentication* section on their profile page to protect their accounts with additional forms of authentication.

Task overview: Multi-Factor Authentication [page 2369]

Related Information

Deactivate Two-Factor Authentication [page 2370]
Deactivate User Devices for TOTP Two-Factor Authentication [page 2371]
Unlock User TOTP Passcode [page 2372]
Remove User Device for Web Two-Factor Authentication [page 2373]
Unlock User SMS Code [page 2374]
Allow Users To Skip Two-Factor Authentication Setup [page 2378]
Configure Risk-Based Authentication for an Application [page 2007]
Configure Default Risk-Based Authentication for All Applications in the Tenant [page 2183]

1.7.11.12.7 Allow Users To Skip Two-Factor Authentication Setup

You can set the number of days for which the users can postpone the enabling of second factor for authentication.

Prerequisites

- You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have configured the application or the tenant to require two-factor authentication. For more information, see Configure Risk-Based Authentication for an Application [page 2007] or Configure Default Risk-Based Authentication for All Applications in the Tenant [page 2183].

Context

The number of days for which users can postpone the enabling of second factor can be set between 1 and 14 days. The configuration is valid for all applications in the tenant. The default setting is 14 days.

> **Note**
>
> The option is switched on by default for all tenants that are created after the February 26, 2026 upgrade of SAP Cloud Identity Services.

If the tenant or application requires two-factor authentication, and the users don't have an enabled two-factor authentication method, or the method that is enabled is not among the ones required by the application, at sign-in users can choose to skip the enabling of two-factor authentication for the period defined by the tenant administrator. The users can sign in without providing a second factor until the grace period expires.

Users can be notified for their choice, if the security alert notification is switched on. They receive a security alert email after they first skip the enabling of two-factor. For more information about how to configure the e-mail alerts, see Send Security Alert Emails [page 2209].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Choose the *Multi-Factor Authentication* list item.
4. Under *Additional Settings for Multi-Factor Authentication*, choose *Edit*.

5. Under *Second Factor Activation Postponement*, choose *Enabled*.
6. Manually configure the number of days for which users can postpone the enabling of second factor.
7. Save your changes.

   If the operation is successful, the system displays the message `Additional Multi-Factor Authentication Settings updated`.

Task overview: Multi-Factor Authentication [page 2369]

Related Information

Deactivate Two-Factor Authentication [page 2370]
Deactivate User Devices for TOTP Two-Factor Authentication [page 2371]
Unlock User TOTP Passcode [page 2372]
Remove User Device for Web Two-Factor Authentication [page 2373]
Unlock User SMS Code [page 2374]
Allow Users to Protect Accounts with Second Factor for Authentication [page 2376]

1.7.11.13 Manage User Passwords

This section provides information about how to unlock user passwords, trigger email reset, manage email sending limits, and set initial password.

Unlock User Password [page 2380]
: You can unlock a user password when the user must log on to the application before the automatic unlock time of 60 minutes has passed, or if the *Password Locked Period* in the password policy is set at *unlimited*.

Send Reset Password Email [page 2381]
: You can trigger the sending of an email to the user with reset password information.

Reset Counter for Email Sending [page 2383]
: You can reset the counter for the number of emails that can be sent to the user.

Set Initial Password [page 2384]
: You can set initial password for a user that has already activated his or her account.

1.7.11.13.1 Unlock User Password

You can unlock a user password when the user must log on to the application before the automatic unlock time of 60 minutes has passed, or if the *Password Locked Period* in the password policy is set at *unlimited*.

Context

The user locks the password after submitting five incorrect passwords when trying to log on to the account. The user receives a notification email that the password log on to the account has been disabled for 60 minutes. The user can either choose the link provided in the email to set a new password, or wait for 60 minutes and then log on with his or her current password.

If the *Password Locked Period* in the password policy is set at *unlimited*, the password can be unlocked only by the tenant administrator. When a user locks the password and then tries to reset it, they get the following message : "Your password is locked. Please contact your system administrator to unlock it." In this case the email template set must also be changed. Otherwise, the user will receive an email stating that password logon to the account was disabled for - 1 hour. For more information, see Configuring Email Templates [page 2282].

> → Remember
>
> Identity Authentication can send to the user up to three emails (forgot password, reset password, locked password, email verification) per 24 hours.
>
> If the user must receive more than three emails, the administrator must reset the counter for email sending first. For more information, see Reset Counter for Email Sending [page 2383].

The tenant administrator can unlock the logon to the user account via the administration console.

> ⚠ Restriction
>
> When a user is with a status `Inactive`, unlocking the user password is disabled. For more information, see Deactivate Users [page 2368].

To unlock the user password manually, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.
3. Choose the user whose password you want to unlock.

   For more information about how to find a user in Identity Authentication, see Search Users [page 2359].
4. Choose *Password Details* under the *Authentication* tab.
5. Choose *Unlock*.

> → Remember
>
> You can only unlock locked user passwords.

Results

The password is unlocked. The user can now access his or her account after providing the correct credentials.

Task overview: Manage User Passwords [page 2379]

Related Information

Send Reset Password Email [page 2381]
Reset Counter for Email Sending [page 2383]
Set Initial Password [page 2384]
Send Reset Password Email [page 2381]

1.7.11.13.2 Send Reset Password Email

You can trigger the sending of an email to the user with reset password information.

Context

Tenant administrator can trigger the sending of an email to the user with reset password information. When the user follows the link provided in the email, the reset password screen appears and the user is prompted to set a new password.

> ⓘ Note
>
> In this scenario, the email that is sent uses the Forgot Password template set that is defined for the User Prfile application. For more information, see View Email Template Document [page 2288].

Identity Authentication can send to the user up to three emails (forgot password, reset password, email verification) per 24 hours. This counter includes also the emails sent with passcode for TOTP deactivation. If you send three emails within 24 hours, the user will not be able to request a passcode for TOTP deactivation via the profile page during these 24 hours.

If the user must receive more than three emails, the administrator must reset the counter for email sending first. For more information, see Reset Counter for Email Sending [page 2383].

→ Tip

Configure SAML 2.0 Service Provider [page 1941].

⚠ Restriction

When a user is with a status `Inactive`, sending reset password email to that user is disabled. For more information, see Deactivate Users [page 2368].

To send a reset password email, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.
3. Choose the user whose password needs to be reset.

   For more information about how to find a user in Identity Authentication, see Search Users [page 2359].

   → Remember

   The maximum number of emails that can be sent to a user within 24 hours is three. This counter includes also the emails sent with passcode for TOTP deactivation. If you send three emails within 24 hours, the user will not be able to request a passcode for TOTP deactivation via the profile page during these 24 hours.

   If you need to send more emails, first reset the password counter. For more information, see Reset Counter for Email Sending [page 2383].
4. Choose *Password Details* under the *Authentication* tab.
5. Press *Send Email*.

Results

The user receives an email with information how to reset his or her password. After resetting the password, he or she is able to log on to the account.

Task overview: Manage User Passwords [page 2379]

Related Information

Unlock User Password [page 2380]

Reset Counter for Email Sending [page 2383]
Set Initial Password [page 2384]

1.7.11.13.3 Reset Counter for Email Sending

You can reset the counter for the number of emails that can be sent to the user.

Context

By default, Identity Authentication sends up to three emails per 24 hours. If three emails are sent, but it is necessary to send a new email, the tenant administrator must first reset the counter for email sending.

> ⚠ Restriction
>
> When a user is with a status `Inactive`, resetting the counter for that user is disabled. For more information, see Deactivate Users [page 2368].

To reset the counter for email sending, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.
3. Choose the user whose email counter needs to be reset.

   For more information about how to find a user in Identity Authentication, see Search Users [page 2359].
4. Choose *Password Details* under the *Authentication* tab.
5. Press *Reset Counter*.

Results

The counter for the number of emails that can be sent to the user is reset to zero.

Task overview: Manage User Passwords [page 2379]

Related Information

Unlock User Password [page 2380]
Send Reset Password Email [page 2381]
Set Initial Password [page 2384]
Send Reset Password Email [page 2381]
Enable Email Verification [page 2026]

1.7.11.13.4 Set Initial Password

You can set initial password for a user that has already activated his or her account.

Prerequisites

You are assigned the *Manage Tenant Configuration* and *Manage Users* roles. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

Tenant administrator can set an initial password for a user that has already activated the account. When the user logs on with the new password, he or she is prompted to change the password with a new one.

> ⚠ Restriction
>
> The initial password can be valid between 1 and 365 days depending on the configuration made by the administrator. The default value is 14 days. The user is prompted to reset the password during the first authentication. After the validity of the initial password expires, the user can't log on to the application and must contact an administrator. For more information for more information about how to configure the initial password validity, see Configure Initial Password and Email Link Validity [page 2168].

> → Tip
>
> Configure SAML 2.0 Service Provider [page 1941].

> ⚠ Restriction
>
> When a user is with a status `Inactive`, setting initial password to that user is disabled. For more information, see Deactivate Users [page 2368].

To set an initial password, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.
3. Choose the user who needs an initial password.

   For more information about how to find a user in Identity Authentication, see Search Users [page 2359].
4. Choose *Password Details* under the *Authentication* tab.
5. Press *Set Initial*.

Results

The user logs on to the application with the new password set by the tenant administrator. He or she is prompted to change the password with a new one. After changing the password, the user is able to log on to the application.

Task overview: Manage User Passwords [page 2379]

Related Information

Unlock User Password [page 2380]
Send Reset Password Email [page 2381]
Reset Counter for Email Sending [page 2383]

1.7.11.14 Restrict User Attributes Values via Exclude Lists

You can restrict the values that can be set on the email, first and last name attributes of the user.

Context

The configured exclude lists for email, first and last name attributes are considered in the user creation and user update via the Identity Authentication SCIM API, Identity Directory SCIM API, the administration console for SAP Cloud Identity Services, and user import.

If the exclude lists are empty, you can create and update a user with any value for email, first and last name attributes.

To create exclude lists for email, first and last name attributes, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Users & Authorizations*, choose the *Exclude Lists* tile.
3. Choose the tab for the attribute that you want to restrict:
   - First Name
   - Last Name
   - Email
4. Choose *Create* and enter the exclude terms as free text in the field.

   For multiple values, use comma between them.
5. Confirm your changes by choosing *Create*.

Related Information

Import or Update Users for a Specific Application [page 2348]
Create a New User [page 2357]
List and Edit User Details [page 2361]
Identity Authentication API (Deprecated) [page 2796]
Identity Directory Service SCIM API

1.7.11.15 Restrict User Security Keys for Web Two-Factor Authentication

This document shows you how to restrict the security keys used by users of type employee for access to applications requiring web two-factor authentication (FIDO2 standard).

Context

You restrict the employee user's security key by creating a list of security keys that are allowed for authentication. All other security keys that aren't in the allow list are forbidden for usage by the users.

→ Remember

The allow list is valid only for users of the type employee. The other user types aren't affected by the creation of a security key allow list.

When you include a security key in the list, the users will be able to log on to applications that require web authentication only with the security keys are in the list. To be able to access the applications with other security keys, the employee user has to activate a new one on the user profile page or when prompted when logging on to an application. For more information, see the Related Information.

When the allow list is empty, all security keys are permitted for usage by the user.

When you add security keys in the administration console, you can choose from a predefined list, or you can add custom security keys. The AaGuid of the security keys must be in the format `XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX`.

✤ Example

Michael Adams is an administrator at retail company A. He would like to restrict the access to the company's internal portal page only to employees using security keys controlled by the company.

✤ Example

In the administration console, Michael creates the following rule:

Default Authentication Rule

Action	IP Range	Group	Authentication Method	User Type
Web Two-Factor Authentication	Any	Any	Any	Employee

Default Authentication Rule

Default Action: `Deny`

Michael Adams restricts also the Web Two-Factor Authentication to only the company's custom security key by providing `ab1c23de-4f56-7890-1g23-4h56a7869012` for the AaGuid.

Dona Moore is working at Company A and as such her user type is `Employee`. As an employee she has a company's security key. She has registered it at her profile page. So when she logs on to the portal page she provides her user name and password, and then she proves her identity with the company's security key.

Julie Armstrong is also an employee at company A, but she does't have a custom security key like Dona. When she is prompted to prove her identity, she tries another security key, but the authentication fails.

Richard Wilson is a supplier to Company A. His user type is `Partner`. When he tries to log on to the portal page with his user name and password, his authentication fails.

To create an allow list of the security keys, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under *Authentication*, choose *Multi-Factor Authentication* > *WebAuthn Security Keys*.
4. Choose the *Add* button.
5. Provide the required information in the input fields.
6. Confirm the information with the *Add* button.
7. Repeat the procedure for all security keys that you want to put into the allow list.

1.7.11.16 Manage Custom Schemas via Administration Console

You can view and define your custom schema via the administration console for SAP Cloud Identity Services.

Context

The administration console shows information about all existing schemas, predefined, and custom, in the tenant.

If you need your own custom attributes for users you can define your own custom schema, and once the schema is defined, the custom attributes that it defines can be used.

To be used, the custom attributes must be assigned to the user first. For more information, see Configuring Attributes Based on Flexible Expressions [page 1979].

> **Note**
>
> When the attributes are assigned to a user, they can be sent to the application. For more information, see Configuring User Attributes from the Identity Directory [page 1973].

You can define a schema either by importing a JSON file, which must comply with the SCIM standard, or by creating it manually via the administration console.

You can check the available schemas in the tenant, and use one of the predefined schemas. You can also import, or manually create your custom schema. The maximum number of all custom schemas, imported and manually created, per tenant is 20. Each schema can have up to 20 attributes. Each multivalued attribute of type `complex` can have up to 20 subattributes.

You can also download a schema template or any of the already existing schemas. You can use the template to define your own schema extensions.

> ⚠ Caution
>
> The schemas in the administration console are read-only. To edit a custom schema, you must delete it and import a new one with the necessary changes. If you delete a schema this will also delete the attributes and subattributes in this schema and their values, and remove all already assigned attributes to users from this schema.
>
> You can delete only custom schemas.

> ⓘ Note
>
> For more information about the SCIM Core Schema, see 7. Schema Definition .

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Schemas* tile.

   This operation opens the *Schemas* page with all the schemas in the tenant, predefined, and custom.
3. Choose *Actions* button, then choose one of the options:
   - *Template* - Download a schema template. You can use the template to define your own schema extensions.

     > ⓘ Note
     >
     > To download an existing scema, select the schema in the list on the left and choose the *Export* button in the right top corner.

   - *Import* - Upload the JSON file with your SCIM-compliant schema.
     The following singular and multivalued attributes are defined:

     Singular Attributes

Attributes	Notes
`id`	Required Example: `"id": "urn:sap:cloud:scim:schemas:extension:custom:2.0:MyCustomSchema",`

Attributes	Notes
`name`	Optional **Note** If present, it's validated against the `id`, and must match it. Example: `"name": "MyCustomSchema",`
`description`	Optional Example: `"description": "Custom schema description",`

Multi-Valued Attributes

Attribute	Subattributes	Notes
`attributes`		Required. A set of schema attribute objects which contains the attribute qualifications.
	`name`	Required
	`type`	Required Values: "string", "boolean", "decimal", "integer", "dateTime", "reference", and "complex". **→ Remember** "type=complex" fits only to attribute. If an attribute is marked as "type=complex" and "multiValued = true" this attribute must have at least one subattribute.
	`subAttributes`	Optional. **→ Remember** Required, when main attribute is of type "complex". "subAttributes" has the same schema qualification settings as "attributes".

Attribute	Subattributes	Notes
	`multiValued`	Required Boolean → Remember When attribute is marked as "multiValued=true", the type can only be "complex". If an attribute is marked as "type=complex" and "multiValued = true" this attribute must have at least one subattribute.
	`description`	Optional
	`required`	Required Boolean
	`canonicalValues`	Optional A set of supported values for the attribute.
	`caseExact`	Required Boolean
	`mutability`	Required Values: "readOnly", "readWrite", "immutable", and "writeOnly".
	`returned`	Required Valid key words: "always", "default", "never", and "request".
	`uniqueness`	Required Valid key words: "none", "server", and "global".

Attribute	Subattributes	Notes
	`referenceTypes`	Optional Valid values: • A SCIM resource type (for example "User" or "Group") • "external" - indicating that the resource is an external resource • "uri" - indicating that the reference is to a service endpoint or an identifier

- *Create* - Manually create a custom schema. Fill in the required information and add the attributes. The following singular and multivalued attributes are defined:

Attribute Name	Provide a name for the attribute. Each attribute must be unique within the schema.
Parent Attribute	Choose a parent for the attribute from the existing attributes of type complex. **Note** A parent attribute is a previously configured "multiValued" attribute of type "complex".
Type	Select a value from the drop-down: • "string" (default value) • "boolean" • "decimal" • "integer" • "dateTime" • "reference" • "complex" **→ Remember** If an attribute is marked as "Type=complex" and "Multi-Valued = true" this attribute must have at least one subattribute.
Mutable	Select a value from the drop-down: • "readOnly" (default value) • "readWrite" • "immutable" • "writeOnly"

Returned	Select a value from the drop-down: • "always" (default value) • "default" • "never" • "request"
Unique	Select a value from the drop-down: • "none" (default value) • "server" • "global"
Description	Optional.
Canonical	Optional.
Required	Boolean. When checkbox is selected, the value is "true". Default value is "false".
Case Exact	Boolean. When checkbox is selected, the value is "true". Default value is "false".
Multi-Valued	Boolean. When checkbox is selected, the value is "true". Default value is "false".

Results

The newly created schemas appear in the list on the left. The schemas are listed in alphabetical order.

Related Information

System for Cross-domain Identity Management: Core Schema
Attribute Data Types
Identity Directory API - Create Users

1.7.12 Managing Groups

Tenant administrators can create groups, and assign and unassign these groups to users via the administration console for SAP Cloud Identity Services.

A group is a collection of users. Groups serve to create sets of users who have something in common. For example, users who work in the same department or users who have similar tasks in a company.

Import Groups via CSV File [page 2395]

As a tenant administrator, you can create new groups or update existing ones with the assigned users, via a CSV file upload.

Create a Group [page 2398]
: As a tenant administrator you can create groups in the tenant to organize users based on common characteristics, authorization, or application via the administration console for SAP Cloud Identity Services.

List and Edit Groups [page 2400]
: As a tenant administrator, you can list and edit information about the groups in a tenant in the administration console for SAP Cloud Identity Services.

Enable or Disable Central Store-Based Provisioning [page 2401]
: You can enable or disable the *Central Store-Based Provisioning* option in the administration console for SAP Cloud Identity Services.

Manage Application-Specific Groups by Identity Provisioning [page 2402]
: By running provisioning jobs, you can create application-specific groups in the Identity Directory of your SAP Cloud Identity Services tenant or Identity Authentication (SCIM API version 2) target system and provision them afterward to target systems of your choice.

List and Search Users in Groups [page 2405]
: As a tenant administrator, you can list and view information about the users in a group in a tenant in the administration console for SAP Cloud Identity Services.

Add Users to a Group [page 2406]
: As a tenant administrator, you can add one or more users created for a specific tenant to a group via the administration console for SAP Cloud Identity Services.

Remove Users from a Group [page 2408]
: As a tenant administrator, you can remove one, more than one, or all users added to a group via the administration console for SAP Cloud Identity Services.

Assign Groups to a User [page 2409]
: As a tenant administrator, you can assign one or more groups created for a specific tenant to a user via the administration console for SAP Cloud Identity Services.

Unassign Groups from a User [page 2410]
: As a tenant administrator, you can unassign one or more groups that are assigned to a user via the administration console for SAP Cloud Identity Services.

Delete Groups [page 2411]
: As a tenant administrator, you can delete one or more groups in administration console for SAP Cloud Identity Services.

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]

Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.12.1 Import Groups via CSV File

As a tenant administrator, you can create new groups or update existing ones with the assigned users, via a CSV file upload.

Prerequisites

- You are assigned the *Manage Groups* and *Manage Users* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

With the CSV file, you can import up to 25000 groups to create new groups or to update existing ones with their user assignments.

If the group does not exist in the tenant, and you provide that group in the CSV file, the group will be created. The CSV file must contain the `groups` column. The value of the `groups` column is recorded as `name` and `displayName` in the newly created group. You cant set `description`, or separately provide `name` or `displayName` in the CSV file. No user assignment is required when creating a new group.

You can only add users to the groups via the import groups option, you don't remove users from the groups.

> ⚠ Caution
>
> The supported number of users in a group is 344,000 members. If you exceed this number, the operations with the group might not work properly.

→ Remember

When you create new groups with user assignments, or update existing ones with their user assignments, make sure that the CSV file contains unique users with unique identifiers.

The CSV file must contain the `groups` column and a column with at least one of the following identifiers:

- *Global User ID*
- *Email* - `emails[0].value`

  #### ⓘ Note

  - If the `emails[0].value` is not unique, and there are more than one users with one and the same email, all the users with that email in the CSV will be assigned to the group.
  - If the `emails[0].value` is not unique, and there are more than one users with one and the same email, you can distinguish the users by the `userName` or `userUuid` attribute. If you use the `userUuid` attribute, make sure that the user that you want to assign the group has it.
- *User Name*

If the file contains more than one identifier then the priority will be:

1. Global User ID
2. Email
3. User Name

If the file contains email and user name, a combination of both will be used as identifier. Thus the user will be matched with the email and user name at the same time.

If the CSV file contains one and the same email identifier, then the group will be assigned to all users that has this email.

ⓘ Note

The user import doesn't assign any special rights or roles to the users for the specific application.

The CSV file can contain only columns with no spaces between the values.

If you enter the data in the CSV file as text, you must separate the entries with commas. Beware not to put any spaces before or after the commas. If you enter more than one value in a single entry, separate the values within the entry with semicolons.

✤ Example

{·} Sample Code

```
groups,urn:ietf:params:scim:schemas:extension:sap:2.0:User:userUuid,emails[
0].value
HR,1ab2e307-1c7c-4c37-9bcf-ed0123456789,dona.moore@example.com
Management,1ab2e307-1c7c-4c37-9bcf-ed0123456700,management@example.com
IT,1ab2e307-1c7c-4c37-9bcf-ed0123456712,management@example.com
```

To import groups via CSV file into Identity Authentication, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Groups* tile.
3. Press the *Import* button.
4. Choose the *Browse...* button and specify the location of the CSV file.

   > ⓘ Note
   >
   > Use a UTF-8 encoded file with up to 25000 groups and an extension `.csv`. If your file contains more than 25000 groups, you have to import the groups information in iterations within the group number limit.

   > → Tip
   >
   > If you import groups from one tenant to another, you can first export the users from the first tenant via the export users option, and then use the exported CSV file for the import to the other tenant or tenants. For more information, see Export Existing Users of a Tenant of Identity Authentication [page 2356].

5. Choose the *Save* button.

   If the operation is successful, the system displays a message with the number of imported and updated groups.

Task overview: Managing Groups [page 2393]

Related Information

Create a Group [page 2398]
List and Edit Groups [page 2400]
Enable or Disable Central Store-Based Provisioning [page 2401]
Manage Application-Specific Groups by Identity Provisioning [page 2402]
List and Search Users in Groups [page 2405]
Add Users to a Group [page 2406]
Remove Users from a Group [page 2408]
Assign Groups to a User [page 2409]
Unassign Groups from a User [page 2410]
Delete Groups [page 2411]

1.7.12.2 Create a Group

As a tenant administrator you can create groups in the tenant to organize users based on common characteristics, authorization, or application via the administration console for SAP Cloud Identity Services.

Prerequisites

You are assigned the *Manage Groups* and *Manage Users* roles. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

You can choose whether to create just a user group or a user group that is application-specific. For more information about the different groups in the Identity Directory of Cloud Identity Services, see Groups [page 514].

To create a new user group, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Groups* tile.

   This operation opens a list of the groups in the tenant.
3. Press the *+ Create* button at the top of the page.
4. In the *General* screen, fill in the required information, and choose *Next Step*.

   You have the following options:

Field	Notes
Name	Required. The name of the group. → Tip To ensure successful group provisioning from SAP Cloud Identity Access Governance, create the groups in Identity Authentication with length of their names within 40 characters.
Display Name	Required. The display name of the group.
Description	Optional. Additional information for the group.

5. **Optional:** In the *Application (Optional)* screen, fill in the information if you want to create an application-specific group.

a. Select an application from the list to create an application-specific group.
b. Choose a group type:
   - *userGroup* - A group of users who have something in common.
   - *authorization* - A group related to authorization of users, groups or applications.
   - *deepLinkActivationPermission* - A group of applications that support deep linking.

   For more information about the group, see Groups [page 514].

   > → Remember
   >
   > At the creation of the group, the *Supported Operations* field is set to `readWrite` and is read-only. For more information about supported operations, refer to the *Application-Specific Groups Supported Operations* table in Groups [page 514].

6. Choose *Next Step* > *Finish*.

Results

If the operation is successful, the system displays the message: `Group "<name of the group>" updated`. Identity Authentication creates the new group and assigns the *Group ID* to it. *Group ID* is in the universally unique identifier (UUID) format. If the group is application-specific the *Application Name* is also assigned to the group.

Task overview: Managing Groups [page 2393]

Related Information

Import Groups via CSV File [page 2395]
List and Edit Groups [page 2400]
Enable or Disable Central Store-Based Provisioning [page 2401]
Manage Application-Specific Groups by Identity Provisioning [page 2402]
List and Search Users in Groups [page 2405]
Add Users to a Group [page 2406]
Remove Users from a Group [page 2408]
Assign Groups to a User [page 2409]
Unassign Groups from a User [page 2410]
Delete Groups [page 2411]
Groups [page 514]

1.7.12.3 List and Edit Groups

As a tenant administrator, you can list and edit information about the groups in a tenant in the administration console for SAP Cloud Identity Services.

Prerequisites

- You are assigned the *Manage Groups* and *Manage Users* roles. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have created groups in your tenant. For more details on how to create groups, see Create a Group [page 2398].

Context

The groups are sorted in alphabetical order of the display name.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Groups* tile.

   This operation opens a list of the groups in the tenant.
3. Choose a group from the list on the left to view its details.

   > → Tip
   >
   > Type a `Group ID`, `Display Name`, or `Name` of a group in the search field to filter the list items. Choose *Show Filter Bars* for advanced search. You must be assigned to a policy that allows reading applications to view the information for *Application Name*.
4. **Optional:** Press the *Edit* button at the top-right corner of the screen.
   a. Edit the *Display name* or *Description* fields, or both.
   b. Save your changes.

Task overview: Managing Groups [page 2393]

Related Information

Import Groups via CSV File [page 2395]

Create a Group [page 2398]
Enable or Disable Central Store-Based Provisioning [page 2401]
Manage Application-Specific Groups by Identity Provisioning [page 2402]
List and Search Users in Groups [page 2405]
Add Users to a Group [page 2406]
Remove Users from a Group [page 2408]
Assign Groups to a User [page 2409]
Unassign Groups from a User [page 2410]
Delete Groups [page 2411]
Create a Group [page 2398]
Edit Administrator Authorizations [page 2309]

1.7.12.4 Enable or Disable Central Store-Based Provisioning

You can enable or disable the *Central Store-Based Provisioning* option in the administration console for SAP Cloud Identity Services.

Prerequisites

- You have an application specific group in your tenant. For more details on how to create groups, see Create a Group [page 2398].
- You have configured the **Central Store-Based Provisioning** in the administration console for SAP Cloud Identity Services. For more information, see Central Store-Based Provisioning [page 579].

Context

When you enable the *Central Store-Based Provisioning* for a specific application, whenever you update an application-specific group associated with this application, a provisioning of the updates is triggered from the Local Identity Directory to a target system of your choice. There is no need to run manual or scheduled jobs in the Identity Provisioning.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Under the *Provisioning* tab, enable or disable the *Central Store-Based Provisioning* option.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

Task overview: Managing Groups [page 2393]

Related Information

Import Groups via CSV File [page 2395]
Create a Group [page 2398]
List and Edit Groups [page 2400]
Manage Application-Specific Groups by Identity Provisioning [page 2402]
List and Search Users in Groups [page 2405]
Add Users to a Group [page 2406]
Remove Users from a Group [page 2408]
Assign Groups to a User [page 2409]
Unassign Groups from a User [page 2410]
Delete Groups [page 2411]

1.7.12.5 Manage Application-Specific Groups by Identity Provisioning

By running provisioning jobs, you can create application-specific groups in the Identity Directory of your SAP Cloud Identity Services tenant or Identity Authentication (SCIM API version 2) target system and provision them afterward to target systems of your choice.

Prerequisites

You have an existing application in your SAP Cloud Identity Services tenant. For more information, see Create a New Application [page 1922].

Context

The application-specific groups are a special kind of groups which can be created in the Identity Directory of SAP Cloud Identity Services or Identity Authentication (SCIM API version 2) target system by running provisioning jobs, or via the administration console for SAP Cloud Identity Services. For more information, see Create a Group [page 2398].

Operating with application-specific groups by the Identity Provisioning service requires having a source system with set property `ips.application.id`. By running provisioning jobs from such source system you can create, update, and delete application-specific groups in the Identity Directory of your SAP Cloud Identity Services tenant or Identity Authentication (SCIM API version 2) target system, depending on the values of their attributes.

Each application-specific group contains information about the application to which it is associated through the `application ID`. That is how you can distinguish which group comes from which application (or provisioning system in the context of Identity Provisioning) in the Identity Directory or Identity Authentication (SCIM API version 2) target system.

Another specific attribute for these groups is `supportedOperations`. The purpose of this attribute is to specify what are the supported operations by the Identity Provisioning for each application-specific group. For more information, see Groups [page 514].

> **Note**
>
> To find out which connectors support application-specific groups, check the list of systems under *System Types* for the `ips.application.id` property in List of Properties [page 80].

The result from each operation with an application-specific group executed by the Identity Provisioning, depending on the value of its attribute `supportedOperations`, is described below.

Create Application-Specific Groups

By running provisioning jobs from your source system, associated to an application, you can create application-specific groups in the Identity Directory of your SAP Cloud Identity Services tenant or Identity Authentication (SCIM API version 2) target system.

When you configure Local Identity Directory or Identity Authentication (SCIM API version 2) as source system and you try to create a group that already exists in the target system with matching attributes and `supportedOperations` attribute value `userOnlyMembership` or `membership`, its group members are updated. If you try to provision a group with value `readOnly` for the attribute `supportedOperations`, it is displayed as *Skipped* in the job log statistics for the source system. In case the group has value `readWrite` for the attribute `supportedOperations`, it is created in the target system.

Update Application-Specific Groups

If you update a user group assignment or a group itself in the application to which is associated your source system and you run a read or resync job, the Identity Provisioning tries to search and resolve this group in the

Identity Directory of your SAP Cloud Identity Services tenant or the Identity Authentication (SCIM API version 2) target system.

In case such application-specific group is found, depending on the value of the attribute `supportedOperations` this will result in:

- `readOnly` - The group is displayed as *Skipped* in the job log statistics for the source system.
- `readWrite` - The attributes of the group and its members are updated.
- `userOnlyMembership` - The group members that are users are updated.
- `membership` - The group members, both users and groups, are updated.

If such group is not found, the job fails with an exception.

Delete Application-Specific Groups

The deletion of application-specific groups from the local directory after they have been deleted from your source system requires setting the property `ips.delete.existedbefore.entities` = *true* in the target system. The setting of the property must be done **before** the job to delete those groups from the target system is executed. For more information, see List of Properties [page 80].

In case you have set the `ips.delete.existedbefore.entities` property before running the provisioning job, depending on the value of the attribute `supportedOperations`, expect the following results:

- `readOnly` - The group is displayed as *Skipped* in the job log statistics.
- `readWrite` - The group is deleted.
- `userOnlyMembership` - The group members that are users are deleted. The group is displayed as *Deleted* in the job log statistics.
- `membership` - The group members, both users and groups, are deleted. The group is displayed as *Deleted* in the job log statistics.

If the property `ips.delete.existedbefore.entities` is missing or set to *false*, the Identity Provisioning will not attempt to delete the group and no record will be displayed in the job log statistics.

Parent topic: Managing Groups [page 2393]

Related Information

Import Groups via CSV File [page 2395]
Create a Group [page 2398]
List and Edit Groups [page 2400]
Enable or Disable Central Store-Based Provisioning [page 2401]
List and Search Users in Groups [page 2405]
Add Users to a Group [page 2406]
Remove Users from a Group [page 2408]
Assign Groups to a User [page 2409]

Unassign Groups from a User [page 2410]
Delete Groups [page 2411]

1.7.12.6 List and Search Users in Groups

As a tenant administrator, you can list and view information about the users in a group in a tenant in the administration console for SAP Cloud Identity Services.

Prerequisites

- You are assigned the *Manage Groups* and *Manage Users* roles. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have created at least one user group in the tenant. For more details about how to create groups, see Related Information.
- You have assigned at least one user to the selected user group. For more details about how to assign groups to a user, see Related Information.

Context

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Groups* tile.

   This operation opens a list of the groups in the tenant.
3. Type a `Group ID`, `Display Name`, or `Name` of a group in the search field in order to filter the list items.

   > → Tip
   >
   > Choose *Show Filter Bars* for advanced search. You must be assigned to a policy that allows reading applications to view the information for *Application Name*.
4. Choose a group from the list.

   This will open the groups details (*Group ID*, *Name*, *Display Name*, *Description*, *Type*, and *Application Name*) on the right. The users that are part of the group are listed below the user group details.
5. To search for a specific user, type the *SCIM ID*, *Email*, *Login Name* of the user in the search field and press the *Enter* key.

Task overview: Managing Groups [page 2393]

Related Information

Import Groups via CSV File [page 2395]
Create a Group [page 2398]
List and Edit Groups [page 2400]
Enable or Disable Central Store-Based Provisioning [page 2401]
Manage Application-Specific Groups by Identity Provisioning [page 2402]
Add Users to a Group [page 2406]
Remove Users from a Group [page 2408]
Assign Groups to a User [page 2409]
Unassign Groups from a User [page 2410]
Delete Groups [page 2411]
Create a Group [page 2398]
Assign Groups to a User [page 2409]
Edit Administrator Authorizations [page 2309]

1.7.12.7 Add Users to a Group

As a tenant administrator, you can add one or more users created for a specific tenant to a group via the administration console for SAP Cloud Identity Services.

Prerequisites

- You are assigned the *Manage Groups* and *Manage Users* roles. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have groups created in your tenant.

Context

> ⚠ Caution
>
> The supported number of users in a group is 344,000 members. If you exceed this number, the operations with the group might not work properly.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Groups* tile.

   This operation opens a list of the groups in the tenant.
3. Choose a group from the list on the left.

   This will show the group details and the users that are added to that group.
4. Press the *Add* button.

   > **Restriction**
   >
   > The *Add* button is hidden for application-specific groups with `readOnly` supported operations.
5. Add users to the group. You have the following options:
   - Select users from the list.
   - Search for the users.
     - *starts with* - (default/recommended operator) returns only users that start with the entered value.
     - *contains* - returns all the users that contain the entered value. Using the *contains* operator may result in longer processing time.

   > **Note**
   >
   > You can filter your search by *First Name*, *Last Name*, *Scim ID*, *Email*, or *Login Name*.
6. Save your changes.

Next Steps

Configure the attributes that are sent to the application in the assertion. For more information, see Configuring User Attributes from the Identity Directory [page 1973]

Task overview: Managing Groups [page 2393]

Related Information

Import Groups via CSV File [page 2395]
Create a Group [page 2398]
List and Edit Groups [page 2400]
Enable or Disable Central Store-Based Provisioning [page 2401]
Manage Application-Specific Groups by Identity Provisioning [page 2402]
List and Search Users in Groups [page 2405]
Remove Users from a Group [page 2408]

Assign Groups to a User [page 2409]
Unassign Groups from a User [page 2410]
Delete Groups [page 2411]
Create a Group [page 2398]

1.7.12.8 Remove Users from a Group

As a tenant administrator, you can remove one, more than one, or all users added to a group via the administration console for SAP Cloud Identity Services.

Prerequisites

- You are assigned the *Manage Groups* and *Manage Users* roles. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

> ⚠ Restriction
>
> The removal of users is not possible for application-specific groups with `readOnly` supported operations.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Groups* tile.

   This operation opens a list of the groups in the tenant.
3. Choose a group from the list on the left.

   This will show the group details and the users that are added to that group.
4. Select the users that you want to remove from the group.

   The selected users are highlighted.
5. Press the *Remove* button and confirm the operation.

Task overview: Managing Groups [page 2393]

Related Information

Import Groups via CSV File [page 2395]
Create a Group [page 2398]
List and Edit Groups [page 2400]
Enable or Disable Central Store-Based Provisioning [page 2401]
Manage Application-Specific Groups by Identity Provisioning [page 2402]
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Add Users to a Group [page 2406]
Assign Groups to a User [page 2409]
Unassign Groups from a User [page 2410]
Delete Groups [page 2411]

1.7.12.9 Assign Groups to a User

As a tenant administrator, you can assign one or more groups created for a specific tenant to a user via the administration console for SAP Cloud Identity Services.

Prerequisites

- You have created user groups in your tenant. For more details how to create user groups, see Related Information.
- You are assigned the *Manage Users* and *Manage Groups* roles. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

> ⚠ Caution
>
> The supported number of users in a group is 344,000 members. If you exceed this number, the operations with the group might not work properly.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.

3. Choose the user that you want to assign to a group or groups.
4. Choose the *Groups* tab.
5. Press the *Assign* button at the top of the page.
6. In the dialog box, select the group or groups that you want to assign to the user.
7. Press *Assign*.

Task overview: Managing Groups [page 2393]

Related Information

Import Groups via CSV File [page 2395]
Create a Group [page 2398]
List and Edit Groups [page 2400]
Enable or Disable Central Store-Based Provisioning [page 2401]
Manage Application-Specific Groups by Identity Provisioning [page 2402]
List and Search Users in Groups [page 2405]
Add Users to a Group [page 2406]
Remove Users from a Group [page 2408]
Unassign Groups from a User [page 2410]
Delete Groups [page 2411]

1.7.12.10 Unassign Groups from a User

As a tenant administrator, you can unassign one or more groups that are assigned to a user via the administration console for SAP Cloud Identity Services.

Prerequisites

- You are assigned the *Manage Groups* and *Manage Users* roles. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have created groups in your tenant. For more details how to create groups, see Related Information.
- You have assigned groups to the user. For more details how to assign a group or groups to a user, see Related Information.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.
3. Choose the user whose group assignments you want to edit.
4. Choose *Groups* tab.
5. Select the group or groups that you want to unassign.
6. Press the *Unassign Groups* button at the bottom of the page.
7. Confirm your changes.

Task overview: Managing Groups [page 2393]

Related Information

Import Groups via CSV File [page 2395]
Create a Group [page 2398]
List and Edit Groups [page 2400]
Enable or Disable Central Store-Based Provisioning [page 2401]
Manage Application-Specific Groups by Identity Provisioning [page 2402]
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Remove Users from a Group [page 2408]
Assign Groups to a User [page 2409]
Delete Groups [page 2411]
Create a Group [page 2398]
Assign Groups to a User [page 2409]

1.7.12.11 Delete Groups

As a tenant administrator, you can delete one or more groups in administration console for SAP Cloud Identity Services.

Prerequisites

You are assigned the *Manage Groups* and *Manage Users* roles. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

> ⚠ Restriction
>
> This document does not apply to:
>
> - groups of type *Authorization* and groups associated with an authorization policy. These groups are related to the authorization policies, and when you delete a policy, the respective group is also removed. For more information, see Delete an Authorization Policy [page 2250].
> - application-specific groups with supported operations: `readOnly`, `userOnlyMembership` and `membership`.

The delete groups operation removes groups and unassigns all users from them.

Follow the procedure below for each group that you want to delete:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Groups* tile.

   This operation opens a list of the groups in the tenant.
3. Select the user group that you want to delete.
4. Press the *Delete* button in the right-hand panel.
5. Confirm the operation in the pop-up dialog.

   If the operation is successful, the system displays the message `Group <group name> deleted.`.

Task overview: Managing Groups [page 2393]

Related Information

Import Groups via CSV File [page 2395]
Create a Group [page 2398]
List and Edit Groups [page 2400]
Enable or Disable Central Store-Based Provisioning [page 2401]
Manage Application-Specific Groups by Identity Provisioning [page 2402]
List and Search Users in Groups [page 2405]
Add Users to a Group [page 2406]
Remove Users from a Group [page 2408]
Assign Groups to a User [page 2409]
Unassign Groups from a User [page 2410]

1.7.13 Configuring Provisioning Systems

Configure provisioning systems for synchronizing users and groups between business applications.

Administrators create source, target, and proxy systems, configure properties and transformations when necessary. They manage certificates, update connector versions, schedule and run jobs. Other administrator tasks include enabling or disabling systems as well as exporting and importing them.

For more information about the provisioning systems (connectors) available for use as source, target, and proxy systems, refer to Source Systems [page 585], Target Systems [page 936] and Proxy Systems [page 1343], respectively.

> **Note**
>
> This documentation refers to Identity Provisioning in the SAP Cloud Identity Services infrastructure. To access the service documentation in the Neo environment, go to Identity Provisioning Service in the Neo Environment.

Add New Systems [page 2415]
: You can add source, target, and proxy systems for your provisioning scenarios.

Search and Edit Systems [page 2418]
: You can search and edit source, target, and proxy systems in the Identity Provisioning user interface.

Enable and Disable Systems [page 2419]
: You can enable and disable source and target systems in Identity Provisioning.

Export and Import Systems [page 2421]
: You can export and import source, target and proxy systems in Identity Provisioning.

Delete Systems [page 2423]
: You can delete a source, target, or proxy system from Identity Provisioning.

Update Connector Version [page 2424]
: Update a connector version to allow your provisioning system to use a new API.

Manage Properties [page 2438]
: You can add, delete and modify properties for a system in Identity Provisioning.

Manage Transformations [page 2439]
: You can manage transformations with graphical and JSON text editor. Regardless of which one you choose, the following initial steps are the same.

Manage Certificates [page 2450]
: Identity Provisioning supports certificate-based authentication for secure communication with the provisioning systems (connectors) provided by the service.

Configure Form-Based OAuth Authentication [page 2457]
: Configure Identity Provisioning to use form-based OAuth 2.0 authentication to call SAP Ariba Applications APIs.

Manage Full and Delta Read [page 2459]
: When you set up your systems and start a scheduled provisioning task, the standard behavior of the process reads all the entities from the source system. This mode prevents data loss and always keeps your target system synchronized with the source. However, it may take a long time for every job to be executed.

Manage Deleted Entities [page 2463]

Manage deletion of entities (users or groups) in the target system after they have been deleted from the source system.

Connect to On-Premise Systems [page 2468]
: Set up the connection to on-premise systems when your Identity Provisioning bundle or standalone tenant is running on the infrastructure of SAP Cloud Identity Services.

Start and Stop Provisioning Jobs [page 2472]
: You can start and stop a provisioning job from the Identity Provisioning user interface (UI) or from an API client by using the Identity Provisioning tenant admin API.

Handle Rate Limits [page 2481]
: Identity Provisioning APIs implement rate limits to control the number of incoming requests for a given time.

Handle Failed Operations [page 2482]
: In certain cases, you can set a retry for a failed operation due to an occurred exception.

Reset Identity Provisioning Settings [page 2484]
: Resetting your provisioning settings will delete all provisioning systems configured for your tenant, along with the related job execution logs.

Reset Identity Provisioning System [page 2485]
: Resetting an Identity Provisioning system (source or target) deletes all Identity Provisioning operational data.

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.13.1 Add New Systems

You can add source, target, and proxy systems for your provisioning scenarios.

To provision entities (users, groups, roles) from one system to another across your enterprise, you first need to add and configure these systems as source and target connectors in Identity Provisioning.

> **⚠ Restriction**
>
> The maximum number of systems you are allowed to add is:
>
> - **20** sources systems
> - **50** target systems
>
> This restriction is valid for *Customer Managed* systems only. If your business requires using more systems, create an incident for component *BC-IAM-IPS* to request them. Describe your scenarios and provide a reason why you need the additional systems.

When you add a system, it is created with its default properties and transformations. If the system has different versions (based on the APIs it provides), you can specify which one you want to use, so that the system is created with the version specific properties and transformations.

Versioning is supported for Identity Authentication, SAP SuccessFactors, SAP Concur, SAP Analytics Cloud and SAP Sales Cloud and SAP Service Cloud. It is controlled by the `<system prefix>.api.version` property.

To add a system, proceed as follows:

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*, *Target Systems* or *Proxy Systems*.
2. Choose the *Add* button at the top left corner of your screen.
3. On the *Details* tab, provide the following information:

Field	Description
Type	Select the system type that you want to configure.
System Name	Add a name for your system. Make sure it does not duplicate another system's name in the UI. **ⓘ Note** System names can have a length of up to 100 characters. Only the following characters are allowed: (a-z), (A-Z), (0-9), (-), (_), (.) and spaces.

Field	Description
Destination Name	(Optional) Select a destination. If you have previously created a connectivity destination in SAP BTP cockpit on subaccount level, you can access it from the Identity Provisioning UI. **→ Remember** • When you select a connectivity destination, it must be compliant to the relevant system type. • The destination should specify all the connection settings required for your identity provisioning scenario. • For *SAP Application Server ABAP* systems, creating a destination is **mandatory**. If you skip the *Destination Name* field, you can enter the connection and configuration properties, needed for your scenario, on the *Properties* tab. **ⓘ Note** If you use both a connectivity destination and the *Properties* tab, and one and the same property exists in both places, the value set in the *Properties* tab will be considered with higher priority. If you leave both the *Destination Name* field and the *Properties* tab empty and then run a job, no identity provisioning will be performed.
Description	(Optional) Enter a description. It will help you easily distinguish your systems in the list later on.
Source Systems	This field is displayed only for target systems. Select a source system whose identities you want to read and provision to the target one. You can select multiple source systems. **→ Tip** If you had previously added one or more source systems but some of them were later deleted in your Identity Provisioning UI, an error message will appear. To correct this inconsistency, edit the target system configuration (select active source systems), and save the changes.

4. Decide on the version of the system you want to add.

- Choose *Save*, if the system you add has no version, or it has two or more versions and you want the default one. The new system appears in the left-side panel. The default transformations and properties are displayed under the respective tabs.
- Do not choose *Save*, if the system you add has two or more versions and you want to specify a particular one. In this case, proceed as follows:
  1. From the *Details* tab (without saving your configurations), move to the *Properties* tab and select it.
  2. Add the API version property for your system and a value. For example, if you add SAP SuccessFactors on the *Details* tab, add `sf.api.version` and the desired version **2**.
  3. Now, choose *Save*.

  This creates an SAP SuccessFactors system with specific properties and transformation for version 2, which is based on SAP SuccessFactors Workforce SCIM API. Providing value 1, would result in creating a system with specific properties and transformation for version 1, which is based on SAP SuccessFactors HCM Suite OData API.

  > **Note**
  >
  > Once you save your configuration, switching between versions is possible but requires manual work, mostly adding the version specific properties and transformations. For more information, see Update Connector Version [page 2424].

5. To add connection and configuration properties, choose *Properties* > *Edit*. See List of Properties [page 80]
6. To modify your default system transformation (if needed), choose *Transformations* > *Edit*.
7. Save your changes.

At the end of the Identity Provisioning URL, a dash-separated string appears. This is the automatically generated unique ID of the newly created system.

Parent topic: Configuring Provisioning Systems [page 2413]

Related Information

Search and Edit Systems [page 2418]
Enable and Disable Systems [page 2419]
Export and Import Systems [page 2421]
Delete Systems [page 2423]
Update Connector Version [page 2424]
Manage Properties [page 2438]
Manage Transformations [page 2439]
Manage Certificates [page 2450]
Configure Form-Based OAuth Authentication [page 2457]
Manage Full and Delta Read [page 2459]
Manage Deleted Entities [page 2463]
Connect to On-Premise Systems [page 2468]
Start and Stop Provisioning Jobs [page 2472]
Handle Rate Limits [page 2481]

Handle Failed Operations [page 2482]
Reset Identity Provisioning Settings [page 2484]
Reset Identity Provisioning System [page 2485]
Manage Properties [page 2438]
Manage Transformations [page 2439]

1.7.13.2 Search and Edit Systems

You can search and edit source, target, and proxy systems in the Identity Provisioning user interface.

Prerequisites

Searching and paging of provisioning systems is supported for tenants running on SAP Cloud Identity infrastructure.

Context

Paging is enabled by default. The number of systems loaded per page is 20. They are sorted by type (*Customer Managed* and *SAP Initiated*) and state (*Enabled* and *Disabled*) in the following order:

1. Customer Managed, enabled systems are displayed alphabetically
2. Customer Managed, disabled systems are displayed alphabetically
3. SAP Initiated, enabled systems are displayed alphabetically
4. SAP Initiated, disabled systems are displayed alphabetically

> **Note**
>
> If you can't find the system you are searching for, choose *More* to expand the list of systems as many times as needed until it is displayed.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*, *Target Systems* or *Proxy Systems*.
2. From the list on the left, either directly select a system, or search for it and select it.
3. Select a tab and edit the configurations.
   - *Details*
   - *Transformations*

- *Properties*
- *Outbound Certificates*

4. Choose the *Edit* button and make your changes.
5. Save your changes.

Task overview: Configuring Provisioning Systems [page 2413]

Related Information

Add New Systems [page 2415]
Enable and Disable Systems [page 2419]
Export and Import Systems [page 2421]
Delete Systems [page 2423]
Update Connector Version [page 2424]
Manage Properties [page 2438]
Manage Transformations [page 2439]
Manage Certificates [page 2450]
Configure Form-Based OAuth Authentication [page 2457]
Manage Full and Delta Read [page 2459]
Manage Deleted Entities [page 2463]
Connect to On-Premise Systems [page 2468]
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Handle Rate Limits [page 2481]
Handle Failed Operations [page 2482]
Reset Identity Provisioning Settings [page 2484]
Reset Identity Provisioning System [page 2485]
Manage Properties [page 2438]
Manage Transformations [page 2439]

1.7.13.3 Enable and Disable Systems

You can enable and disable source and target systems in Identity Provisioning.

Context

To use a system for provisioning purposes, its status has to be **Enabled**. When you add a new system, it is enabled by default. If one of your added systems is configured and you currently do not need it, but would like to use it later, you can disable it.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*, *Target Systems* or *Proxy Systems*.
2. From the list on the left, select a system.
3. At the top right corner of your screen, choose one of the following options:
   - If the system is currently disabled, choose the *Enable* button and confirm the action.
   - If the system is currently enabled, choose the *Disable* button and confirm the action.
4. Save your changes.

Task overview: Configuring Provisioning Systems [page 2413]

Related Information

Add New Systems [page 2415]
Search and Edit Systems [page 2418]
Export and Import Systems [page 2421]
Delete Systems [page 2423]
Update Connector Version [page 2424]
Manage Properties [page 2438]
Manage Transformations [page 2439]
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Connect to On-Premise Systems [page 2468]
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Handle Rate Limits [page 2481]
Handle Failed Operations [page 2482]
Reset Identity Provisioning Settings [page 2484]
Reset Identity Provisioning System [page 2485]

1.7.13.4 Export and Import Systems

You can export and import source, target and proxy systems in Identity Provisioning.

Context

If you have added and configured a system, you can export it for further use. The export-import function comes in handy in the following use cases:

- You need to back up your system before updating to a new connector version. See: Update Connector Version [page 2424]
- You need another system of the same type but with slightly different setup, and you don't want to manually enter all data and configuration properties all over again.
- You need to reuse an existing system in the Identity Provisioning UI but for another subaccount.
- You have reached the maximum number of systems you are allowed to add. You need to add one more system, which means you must delete some of the previous ones. However, you don't want to lose their configurations, thus you export these systems.

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Related Information

Add New Systems [page 2415]
Search and Edit Systems [page 2418]
Enable and Disable Systems [page 2419]
Delete Systems [page 2423]
Update Connector Version [page 2424]
Manage Properties [page 2438]
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Reset Identity Provisioning System [page 2485]

Procedure

Export a System

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*, *Target Systems* or *Proxy Systems*.
2. From the list on the left, select the system you want to export.
3. Choose the *Export* button.
4. The exported system configuration depends on your scenario. If your system is a *source* or a *target* one, it will be exported as a JSON file. If it's a *proxy* one, you have two options:
   - Select *JSON format* – the system configuration will be exported as a *.json* file, which you can later import back in the Identity Provisioning UI.
   - Select *CSV format* – the system configuration will be exported as a *.csv* file, which you can later import in the SAP Identity Management UI as a SCIM repository.
5. Save the file on your local file system.

Import a System

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*, *Target Systems* or *Proxy Systems*.
2. Choose the *Add* button.
3. In section *Define from File*, choose the *Browse* button.
4. Browse and select the file with system configuration you need on your local file system. You can import files with extension **.json** as well as files with no extension.
5. The system configuration is displayed in the *Details* editor. You can also see the imported transformations and properties of this system in the respective UI tabs.
6. Change the *System Name*, otherwise an error message will appear warning you that a system with this name already exists.
7. The *Properties* tab will prompt you to enter the credentials (like passwords or client secrets). When you export a system, credentials are skipped (not displayed as plain text in the *.json* file). Therefore, when you import it, you have to manually enter the passwords/secrets.
8. Save your changes.
   The imported system appears in the left-side panel. Its ID is different than the one of the "original" system (you can see it in the URL).

> ⚠ Caution
>
> You cannot export a target system and import it back as a source, nor the other way around.

1.7.13.5 Delete Systems

You can delete a source, target, or proxy system from Identity Provisioning.

Context

> **Note**
>
> Before you delete a system, make sure you don't need it anymore. If you think you might need it in future, export it first as a JSON or a CSV file. To learn how, see: Export and Import Systems [page 2421]

To delete a system, proceed as follows:

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*, *Target Systems* or *Proxy Systems*.
2. From the list on the left, select a system.
3. At the top right corner of your screen, choose the *Delete* button and confirm the action.
4. Save your changes.

Results

The system disappears from the panel.

Task overview: Configuring Provisioning Systems [page 2413]

Related Information

Add New Systems [page 2415]
Search and Edit Systems [page 2418]
Enable and Disable Systems [page 2419]
Export and Import Systems [page 2421]
Update Connector Version [page 2424]
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1.7.13.6 Update Connector Version

Update a connector version to allow your provisioning system to use a new API.

When an SAP cloud solution or service provides a new API for integrating with Identity Provisioning, you can update your respective connector (provisioning system) to use this API by configuring a version property and replacing its transformations.

For example, SAP Sales Cloud and SAP Service Cloud (formerly known as SAP Cloud for Customer) initially provided two SOAP-based APIs for integrating with Identity Provisioning and later introduced a SCIM-based API. Likewise, Identity Authentication service initially provided a SCIM-based API and later introduced an Identity Directory SCIM API.

Version property set to **`1`** means that your connector is using the initial API. You can continue using it as-is or update your connector to a new version.

> **Note**
>
> Before updating your connector to a new version, it is always a good practice to export the system for backup purposes. See: Export and Import Systems [page 2421]

We recommend proceeding with the update of the connector version after executing successfully a provisioning job in **full read** mode for the existing version of the provisioning system.

To update your connector to use a new API, proceed as follows:

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*, *Target Systems* or *Proxy Systems*.
2. From the list on the left, select a system.
3. On the *Properties* tab, configure the following:
    1. **Version Property** - Add or update the **`<system_prefix>.api.version`** property and set its value accordingly.

Connector	Property	Value
Identity Authentication	`ias.api.version`	• **1** - Identity Authentication SCIM API (in short, SCIM API version 1) **Note** When the property is not defined - Identity Authentication SCIM API is used. • **2** - Identity Directory SCIM API (in short, SCIM API version 2) For more information, see Update Identity Authentication [page 2429].
SAP Concur	`concur.api.version`	• **1** - User v1 API This is the default value for systems created before versioning was introduced on December 8, 2021. • **2** - Identity v4 API This is the default value for systems created between December 8, 2021, and December 1, 2025. • **3** - SCIM v4 API By default, the Identity Provisioning service uses version 3 for all newly created systems starting December 2, 2025, when version 3 was introduced. Systems created earlier continue to follow the previous defaults.
SAP Integrated Business Planning	`ibp.api.version`	• *1* - SAP Integrated Business Planning API: Business User (Version 1) is used. This is the default value. • *2* - Identity v4 API (Version 2) For more information, see Update SAP Integrated Business Planning [page 2432].

Connector	Property	Value
SAP Sales Cloud and SAP Service Cloud	`c4c.api.version`	• **1** - Version 1 (SOAP-based API) > **Note** > The SOAP-based API version 1 is depricated. • **2** - Version 2 (SOAP-based API) • **3** - Version 3 (SCIM 2.0 based API) • **4** - Version 4 (SCIM 2.0 based API) applicable for SAP Sales Cloud Version 2 and SAP Service Cloud Version 2 only
SAP SuccessFactors	`sf.api.version`	• **1** - SAP SuccessFactors HCM Suite OData API (Version 1) > **Note** > When the property is not defined - SAP SuccessFactors HCM Suite OData API is used. • **2** - SAP SuccessFactors Workforce SCIM API (Version 2)
SAP Analytics Cloud	`sac.api.version`	• **1** - SAP Analytics Cloud SCIM API version 1. This is the default value. • **2** - SAP Analytics Cloud SCIM API version 2
SAP S/4HANA Cloud Public Edition	`s4hana.cloud.api.version`	• **1** - SAP S/4HANA Cloud Public Edition API: Business User is used. This is the default value. • **3** - System for Cross-domain Identity Management (SCIM) API is used. For more information, see Update SAP S/4HANA Cloud Public Edition [page 2435].
LDAP Server	`ldap.api.version`	• *1* - LDAP Server API version 1 is used. This is the default value. • *2* - LDAP Server API version 2 is used.

Connector	Property	Value
Microsoft Active Directory	`ldap.api.version`	• *1* - LDAP API version 1 is used. This is the default value. • *2* - LDAP API version 2 is used.

2. **Properties with Version-Specific Values** - Update connector properties which have version-specific values and configure any new properties introduced in the updated version.

Connector	Property	Value
SAP SuccessFactors	`URL`	• Version 1: `https://apitest.successfactors.com/odata/v2` • Version 2: `https://apitest.successfactors.com`
	`sf.user.filter`	• Version 1: **`username eq 'cbraun' status eq 'active'`** • Version 2: **`userName eq "cbraun" active eq true`**
SAP Analytics Cloud	`csrf.token.path`	• Version 1: `/api/v1/scim/Users?count=1` • Version 2: `/api/v1/scim2/Users?count=1`

SAP Integrated Business Planning	`ibp.user.unique.attribute`	• Version 1:
		  • `personExternalID`
		  • `emails[0].value`
		• Version 2:
		  • `userName`
		  • `emails[0].value`
		  • `externalId`

SAP S/4HANA Cloud Public Edition	`s4hana.cloud.user.unique.attribute`	• Version 1:
		  • `personExternalID`
		  • `emails[0].value`
		• Version 3:
		  • `userName`
		  • `emails[0].value`
		  • `externalId`

Connector	Property	Value
SAP Concur	`concur.user.unique.attribute`	Version 1 and 2: • `userName` • `externalId` Version 3: • `userName` • `externalId` • `urn:ietf:params:scim:schemas:extension:enterprise:2.0:User:employeeNumber`
	`concur.user.filter`	Version 1 and 2: • **`userName eq "johnsmith@example.com"`** • **`companyId eq "aa067ada-71a9-4f57-8e98-9300b1c3171d"`** Version 3: • **`userName eq "JohnSmith"`** • **`userName sw "John"`** • **`urn:ietf:params:scim:schemas:extension:enterprise:2.0:employeeNumber eq "00102345"`**

4. On the *Transformations* tab, if you've customized your transformation logic, copy and save it first, and then replace it with the default transformation provided for the respective API version. Use the Identity Provisioning connector documentation as a source of information for the transformation you need.
5. Reset the system [page 2485] to clear the operational data. It is assumed that you have already run provisioning jobs to target systems.
   - If you reset a target system, set the `ips.delete.existedbefore.entities` to **true** in the target system itself. This ensures that, if from now on you delete entities in the source system that is connected to your target system, those entities will be recognized as previously existed entities in the target system and will be deleted there.
   - If you reset a source system, set the `ips.delete.existedbefore.entities` to **true** in every target system connected to the given source system. This ensures that, if from now on you delete entities in the source system that is connected to your target system, those entities will be recognized as previously existed entities in the target system and will be deleted there.
6. Adapt your new transformation, that is, apply the customizations from your previous transformation.
7. Save your changes and run a provisioning job.

Parent topic: Configuring Provisioning Systems [page 2413]

Related Information

Add New Systems [page 2415]
Search and Edit Systems [page 2418]
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Update SAP S/4HANA Cloud Public Edition [page 2435]

1.7.13.6.1 Update Identity Authentication

Update the Identity Authentication connector to a new version.

Following the deprecation of the Identity Authentication SCIM API (SCIM API v1), the API will be decommissioned on **November 30, 2028**. You must migrate to its successor, the Identity Directory SCIM API (SCIM API v2). This migration requires updating all Identity Authentication provisioning systems (source, target, and proxy) from connector version 1 to connector version 2.

> **Note**
>
> Migrate each provisioning system separately and complete one migration before proceeding to the next.

Prerequisites

- When Identity Authentication is the source system (connector version 1) and you want to update its version: Verify that the last provisioning job for the Identity Authentication source system using SCIM API v1 completes successfully before starting the update.
- When Identity Authentication is the target system (connector version 1) and you want to update its version: Verify that the last provisioning job for the source systems completes successfully before starting the update.

> **Note**
>
> The update requires a system reset. If the last provisioning job has failed entities in the source system, resolve them before proceeding. Otherwise, Identity Provisioning will not be able to remove them from the target system after the update, and obsolete users may remain indefinitely.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.

   > **Note**
   >
   > Repeat the steps below for each Identity Authentication source, target, and proxy system.

2. From the list on the left, select your system of type *Identity Authentication*.
3. Select the *Properties* tab and add the following property `ias.api.version`=2.

   > **Note**
   >
   > The property prefix depends on the SCIM API version being used.
   >
   > - For SCIM API v2, property names use the `ias` prefix. For example: `ias.user.unique.attribute`
   > - For SCIM API v1, property names use the `scim` prefix. For example: `scim.user.unique.attribute`

   When migrating from SCIM API v1 to SCIM API v2, ensure that all configuration properties using the `scim` prefix are updated to their corresponding `ias`-prefixed properties.
   In addition, configure the following Identity Provisioning properties:

   - `ips.application.id=<application_ID>`
     The ID of the application created under *Applications & Resources* for the corresponding Identity Authentication provisioning system.
   - `ips.delete.existedbefore.entities=true`
     This property must be set for all Identity Authentication target systems.
   - `ias.support.patch.operation=true`
     This property must be set for all Identity Authentication target systems.
   - `ias.user.automatic.conflict.resolution=true`

This property must be set for all Identity Authentication target systems.

For more information, see List of Properties [page 80].

4. Select the *Transformations* tab and update the user and group attribute mappings that have changed in SCIM API v2 as explained in Attributes Mapping.

Example

Replace the following targetPath:

`$.mailVerified` with `$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mailVerified']`

In addition to updating existing mappings, review the default transformations delivered for SCIM API v2 and add newly supported attributes that are not present in your current configuration. The Identity Authentication connector documentation provides default transformation mappings for both SCIM API v1 and SCIM API v2.
Compare the mappings and include the additional SCIM API v2 attributes required for your scenario.

Example

Examples of attributes that may need to be added when migrating to SCIM API v2 include:

Code Syntax

```
{

"condition": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId'] EMPTY false",

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId']",

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId']"
      },
      {

"condition": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId'] EMPTY false",
        "sourcePath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']",
        "optional": true,
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']"
      },
      {

"condition": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId'] EMPTY false",

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['supportedOperations']"
      },
      {

"condition": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId'] EMPTY false",
```

```
"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalName']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalName']"
      },
      {

"condition": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId'] EMPTY false",

"sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalLocation']",
        "optional": true,

"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalLocation']",
        "scope": "createEntity"
      }
```

If your provisioning system uses customized transformations, manually compare them with the SCIM API v2 default transformations to ensure that both updated and newly supported attributes are included after the migration.

For more information on the default transformations, see:

- Identity Authentication source system
- Identity Authentication target system
- Identity Authentication proxy system

5. Reset the system. This is necessary because it is assumed that you have already run provisioning jobs to the target systems.
6. Save your changes and run a provisioning job.

1.7.13.6.2 Update SAP Integrated Business Planning

Update the SAP Integrated Business Planning connector to a new version.

The SAP Integrated Business Planning connector supports two versions, corresponding to the APIs used for integration with Identity Provisioning:

- **Version 1**: Uses the SOAP-based API provided through the communication scenario SAP_COM_0193.
- **Version 2**: Uses the SCIM-based API provided through the communication scenario SAP_COM_0465.

Updating the connector to version 2 depends on your provisioning flow, specifically whether the SAP Integrated Business Planning system is configured as a source or a target system in that flow.

Update Source System

> **Note**
>
> If you have only an SAP Integrated Business Planning source system, or if it is connected in a flow with a target system such as Identity Authentication version 2 or Local Identity Directory, always start the update from the source system.

Proceed as follows:

1. Open your SAP Integrated Business Planning source system and verify that it is configured for version 1. The system is on version 1 if the `ibp.api.version` property is either not set or set to `1`.
2. Select the *Properties* tab and set the following:
   - *ibp.api.version*=2
   - *ips.application.id*=`<The ID of the application that is created under Applications & Resources for the corresponding SAP Integrated Business Planning provisioning system.>`
3. Select the *Transformations* tab and replace the read transformation with the default transformation provided for version 2. Use the Identity Provisioning connector documentation as a source of information for the transformation you need.
4. Reset the source system. This is necessary because it is assumed that you have already run provisioning jobs to the target systems.
   The reset will clear the operational data. After the reset, running subsequent provisioning jobs will update the existing users and groups instead of creating new ones.
5. Open the target system that is connected to your source system (Identity Authentication version 2 or Local Identity Directory). Select the *Transformations* tab and add the mapping for `externalName` if it is missing:

```
{
   "condition":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId'] EMPTY false",
   "sourcePath":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalName']",
   "optional":true,
   "targetPath":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalName']"
}
```

6. Open the source system and run a provisioning job.
7. Open the target system, select the *Properties* tab and add the properties for handling conflict resolution for the respective target system and the property for deleting entities.
   - *ias.group.unique.attribute*=`['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type'],['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName'],['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']`
   - *idds.group.unique.attribute*=`['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type'],['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName'],['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']`
   - *ips.delete.existedbefore.entities*=`true`
     When you reset a source system, this property must be configured in every connected target system. This ensures that if you later delete entities in the source system, those entities are recognized as previously existing in the target system and are deleted there as well.

Result: Following a successful run, it is expected that business roles are provisioned as groups of type `Authorization` for the respective application in SAP Cloud Identity Services, as well as new user groups are successfully created as groups of type `User Group`.

Update Target System

You have an SAP Integrated Business Planning version 1 target system, that is connected with source systems such as Identity Authentication version 2 or Local Identity Directory.

Proceed as follows:

1. Open your source system, respectively Identity Authentication version 2 or Local Identity Directory and ensure that it has the following mapping for `externalName` attribute.

   ```
   {
      "sourcePath":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
   ['externalName']",
      "targetPath":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
   ['externalName']",
      "optional":true
   }
   ```

   > ### Note
   >
   > Groups in Identity Authentication version 2 or Local Identity Directory source systems must have the `externalName` attribute to be provisioned successfully to an SAP Integrated Business Planning version 2 target system.

2. Open the SAP Integrated Business Planning version 1 target system and set the following on the *Properties* tab:
   - *ibp.api.version*=`2`
   - *ibp.group.unique.attribute*=`externalId, ['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']`
   - *ips.application.id*=`<The ID of the application that is created under Applications & Resources for the corresponding SAP Integrated Business Planning provisioning system.>`
   - *ips.delete.existedbefore.entities*=`true`
     When you reset the target system (in step 4), this property will ensure that if you later delete entities in the source system, those entities are recognized as previously existing in the target system and are deleted there as well.
   - Remove unnecessary properties that are valid for SAP Integrated Business Planning version 1.
3. Select the *Transformations* tab and replace the write transformation with the default transformation provided for version 2. Use the Identity Provisioning connector documentation as a source of information for the transformation you need.
4. Reset the target system. This is necessary because it is assumed that you have already run provisioning jobs to the target systems.
   The reset will clear the operational data. After the reset, running subsequent provisioning jobs will update the existing users and groups instead of creating new ones.
5. Open the source system and run a provisioning job.

Update Proxy System

Proceed as follows:

1. Open your SAP Integrated Business Planning proxy system and verify that it is configured for version 1. The system is on version 1 if the `ibp.api.version` property is either not set or set to `1`.
2. Select the *Properties* tab and set the following:
   - *ibp.api.version*=2
   - Remove unnecessary properties that are valid for SAP Integrated Business Planning version 1.
3. Select the *Transformations* tab and replace the proxy read and proxy write transformations with the default transformations provided for version 2. Use the Identity Provisioning connector documentation as a source of information for the transformation you need.

1.7.13.6.3 Update SAP S/4HANA Cloud Public Edition

Update the SAP S/4HANA Cloud Public Edition connector to a new version.

The SAP S/4HANA Cloud Public Edition connector supports two versions, corresponding to the APIs used for integration with Identity Provisioning:

- **Version 1**: Uses the SOAP-based API provided through the communication scenario SAP_COM_0193.
- **Version 3**: Uses the SCIM-based API provided through the communication scenario SAP_COM_0465.

Updating the connector to version 3 depends on your provisioning flow, specifically whether the SAP S/4HANA Cloud Public Edition system is configured as a source or a target system in that flow.

Update Source System

> **Note**
>
> If you have only an SAP S/4HANA Cloud Public Edition source system, or if it is connected in a flow with a target system such as Identity Authentication version 2 or Local Identity Directory, always start the update from the source system.

Proceed as follows:

1. Open your SAP S/4HANA Cloud Public Edition source system and verify that it is configured for version 1. The system is on version 1 if the `ibp.api.version` property is either not set or set to `1`.
2. Select the *Properties* tab and set the following:
   - *s4hana.cloud.api.version*=3
   - *ips.application.id*=`<The ID of the application that is created under Applications & Resources for the corresponding SAP S/4HANA Cloud Public Edition provisioning system>`.
3. Select the *Transformations* tab and replace the read transformation with the default transformation provided for version 3. Use the Identity Provisioning connector documentation as a source of information for the transformation you need.

4. Reset the source system. This is necessary because it is assumed that you have already run provisioning jobs to the target systems.
   The reset will clear the operational data. After the reset, running subsequent provisioning jobs will update the existing users and groups instead of creating new ones.
5. Open the target system that is connected to your source system (Identity Authentication version 2 or Local Identity Directory). Select the *Transformations* tab and add the mapping for `externalName` if it is missing:

```
{
   "condition":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['applicationId'] EMPTY false",
   "sourcePath":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalName']",
   "optional":true,
   "targetPath":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalName']"
}
```

6. Open the source system and run a provisioning job.
7. Open the target system, select the *Properties* tab and add the properties for handling conflict resolution for the respective target system and the property for deleting entities.
   - *ias.group.unique.attribute*=`['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type'],['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName'],['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']`
   - *idds.group.unique.attribute*=`['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type'],['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['externalName'],['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['applicationId']`
   - *ips.delete.existedbefore.entities*=`true`
     When you reset a source system, this property must be configured in every connected target system. This ensures that if you later delete entities in the source system, those entities are recognized as previously existing in the target system and are deleted there as well.

Result: Following a successful run, it is expected that business roles are provisioned as groups of type `Authorization` for the respective application in SAP Cloud Identity Services, as well as new user groups are successfully created as groups of type `User Group`.

Update Target System

You have an SAP S/4HANA Cloud Public Edition version 1 target system, that is connected with source systems such as Identity Authentication version 2 or Local Identity Directory.

Proceed as follows:

1. Open your source system, respectively Identity Authentication version 2 or Local Identity Directory and ensure that it has the following mapping for `externalName` attribute.

```
{
   "sourcePath":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalName']",
   "targetPath":"$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']
['externalName']",
   "optional":true
```

```
}
```

> **Note**
>
> Groups in Identity Authentication version 2 or Local Identity Directory source systems must have the `externalName` attribute to be provisioned successfully to an SAP S/4HANA Cloud Public Edition version 2 target system.

2. Open the SAP S/4HANA Cloud Public Edition version 1 target system and set the following on the *Properties* tab:
   - *s4hana.cloud.api.version*=3
   - *s4hana.cloud.group.unique.attribute*=`externalId, ['urn:ietf:params:scim:schemas:extension:sap:2.0:Group']['type']`
   - *ips.application.id*=`<The ID of the application that is created under Applications & Resources for the corresponding SAP S/4HANA Cloud Public Edition provisioning system>`
   - *ips.delete.existedbefore.entities*=`true`
     When you reset the target system (in step 4), this property will ensure that if you later delete entities in the source system, those entities are recognized as previously existing in the target system and are deleted there as well.
   - Remove unnecessary properties that are valid for SAP S/4HANA Cloud Public Edition version 1.
3. Select the *Transformations* tab and replace the write transformation with the default transformation provided for version 3. Use the Identity Provisioning connector documentation as a source of information for the transformation you need.
4. Reset the target system. This is necessary because it is assumed that you have already run provisioning jobs to the target systems.
   The reset will clear the operational data. After the reset, running subsequent provisioning jobs will update the existing users and groups instead of creating new ones.
5. Open the source system and run a provisioning job.

Update Proxy System

Proceed as follows:

1. Open your SAP S/4HANA Cloud Public Edition proxy system and verify that it is configured for version 1. The system is on version 1 if the `s4hana.cloud.api.version` property is either not set or set to `1`.
2. Select the *Properties* tab and set the following:
   - *s4hana.cloud.api.version*=3
   - Remove unnecessary properties that are valid for SAP S/4HANA Cloud Public Edition version 1.
3. Select the *Transformations* tab and replace the proxy read and proxy write transformations with the default transformations provided for version 3. Use the Identity Provisioning connector documentation as a source of information for the transformation you need.

1.7.13.7 Manage Properties

You can add, delete and modify properties for a system in Identity Provisioning.

Prerequisites

You have added a system (source, target, or proxy) in the Identity Provisioning user interface. To learn how, see Add New Systems [page 2415].

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*, *Target Systems* or *Proxy Systems*.
2. Select a system from the left panel and go to the *Properties* tab.
3. To modify the current properties, choose *Edit* in the bottom right corner.

   > **Note**
   >
   > When you update the URL or the host name of an existing provisioning system, you must re-enter the values of the configured credential properties. The only exception to this are the credential properties of systems that are created with a connectivity destination.

4. Add the properties required by your scenario to make successful connection to the selected system. You can use two types of properties:
   - *Standard*: These are properties, whose values are displayed as numbers or plain text strings. For example: **Type**, **ProxyType**, **URL**, **Authentication**.
   - *Credential*: These are properties whose values contain sensitive information that must not be displayed as plain text. For example: **Password** (standard passwords, private keys, or OAuth client secrets), **ssh.private.key** (relevant to SSH Server).

     > **Note**
     >
     > Properties whose values contain sensitive information can be added only as *Credential* in the Identity Provisioning UI. The values of these properties are stored as encrypted data. They are excluded from the file during system configuration export.

5. Save your changes.

Task overview: Configuring Provisioning Systems [page 2413]

Related Information

Add New Systems [page 2415]
Search and Edit Systems [page 2418]
Enable and Disable Systems [page 2419]
Export and Import Systems [page 2421]
Delete Systems [page 2423]
Update Connector Version [page 2424]
Manage Transformations [page 2439]
Manage Certificates [page 2450]
Configure Form-Based OAuth Authentication [page 2457]
Manage Full and Delta Read [page 2459]
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Connect to On-Premise Systems [page 2468]
Start and Stop Provisioning Jobs [page 2472]
Handle Rate Limits [page 2481]
Handle Failed Operations [page 2482]
Reset Identity Provisioning Settings [page 2484]
Reset Identity Provisioning System [page 2485]
List of Properties [page 80]

1.7.13.8 Manage Transformations

You can manage transformations with graphical and JSON text editor. Regardless of which one you choose, the following initial steps are the same.

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*, *Target Systems* or *Proxy Systems*.
2. Select a system from the left panel and go to the *Transformations* tab.
   The graphical editor is displayed by default. You can switch to the JSON editor by choosing the code-bracket icon.
3. Choose *Edit*. You need to work in edit mode to add, modify and delete entities and their configurations.

   > **Note**
   >
   > When editing the transformations, if you add an entity - such as a user, group or role - that is not supported for the given system type, the related configurations may be accepted but will not be applied.

   - Working with the JSON editor allows you to type changes and perform operations like select, cut, copy and paste the transformation code.
   - Working with the graphical editor allows you to graphically model your changes.
4. Save your changes.

Parent topic: Configuring Provisioning Systems [page 2413]

Related Information

Add New Systems [page 2415]
Search and Edit Systems [page 2418]
Enable and Disable Systems [page 2419]
Export and Import Systems [page 2421]
Delete Systems [page 2423]
Update Connector Version [page 2424]
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Reset Identity Provisioning System [page 2485]
Transformation Editors [page 505]
Working with Graphical Editor [page 2440]

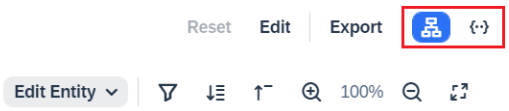
1.7.13.8.1 Working with Graphical Editor

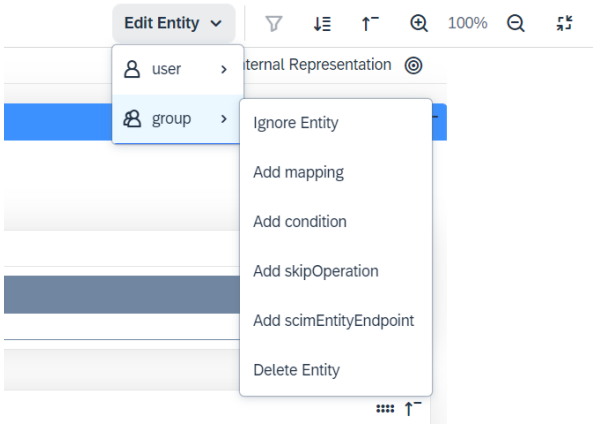
You can create, update and delete entities and their attribute mappings with a handy and easy to use graphical editor. It provides typical operations for an editor, like adding new data, editing and deleting existing data and saving changes. And what's more, it brings improved user experience, requires less typing and more choosing from a list of prefilled values.

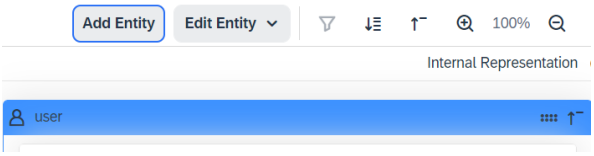
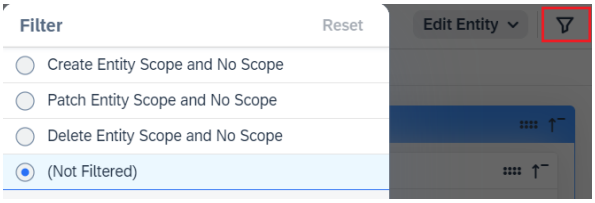
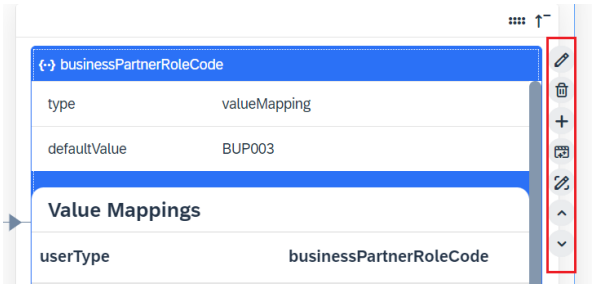
Basic Operations

> ⓘ Note
>
> The graphical editor is available only for Identity Provisioning tenants running on SAP Cloud Identity infrastructure. It is the default editor.

> **Note**
>
> By default, the transformation opens in the graphical editor. However, the availability of the graphical editor depends on the total number of mappings across all entities. The JSON editor is always available.
>
> - 500 mappings or fewer: Both the graphical and JSON editors are available, and you can switch between them.
> - More than 500 mappings: The transformation can be viewed and edited only in the JSON editor. Switching to the graphical editor is not supported.

Description	Operation
Switch Editor When selecting the *Transformations* tab, the graphical editor is displayed by default. It is represented by a tree icon (in the red rectangle). You can switch to the JSON text editor by choosing the code-bracket icon. Switching between editors is allowed in view and edit mode.	

Description	Operation
Edit Entity > *user/group* > *Ignore Entity* Use this option to ignore the provisioning of an entity. When selected, a check mark appears in front of the option and the entity is greyed out. To enable it again, deselect the option. *Edit Entity* > *user/group* > *Add mapping* Use this option to add attribute mapping for an entity. When selected, the *Add Mapping* dialog appears. For more information, see Add or Edit Attribute Mappings. New mappings are added at the end of the list of mappings for the selected entity. As the order matters, you can move them up and down to define the proper place for processing them. *Edit Entity* > *user/group* > *Add condition* Use this option to add a condition for an entity. When selected, the *Add Condition* dialog appears. You need to provide a value. Conditions defined on an entity level are displayed right under the entity name in orange box. *Edit Entity* > *user/group* > *Add skipOperation* Use this option to add skipOperation for an entity. It is always defined on an entity level and in target system transformations only. When selected, the *Add skipOperation* dialog appears with a dropdown list of prefilled values. *Edit Entity* > *user/group* > *Add scimEntityEndpoint* Use this option to add the entity endpoint which is needed in proxy systems only. *Edit Entity* > *user/group* > *Delete Entity* Use this option to delete an entire entity. Confirmation is required. When an entity is deleted, you cannot restore it. You can only add a new one.	

Description	Operation
Add Entity **Note** The *Add Entity* button is displayed only if an entity has been deleted. When selected, an *Add Entity* dialog appears. You are allowed to select only an entity of the type that has been deleted. The new entity appears in its dedicated colored box and has no content.	
Filter The *Filter* icon allows you to search for entity attributes by their scope: create, patch and delete. Filtering is enabled only in view mode, meaning you must not choose *Edit* on the *Transformations* tab. When selected, the *Filter* dialog appears with the default value *(Not Filtered)*. Selecting a filter results in returning all entity attributes with the given scope plus all attributes without any scope. **→ Remember** Attribute mappings with no scope are always returned regardless of the selected filter. Choosing *Reset* returns you to the default value *(Not Filtered)*.	
Operations on an attribute mapping level: - *Pencil* icon - Edit an attribute mapping - *Trash* icon - Delete an attribute mapping - *Plus* sign icon - Adds an attribute mapping - *Function* icon - Add or edit a function Functions appear only for attributes on the right side of the editor. - *Pencil* icon - Edit value mappings This icon appears only for attributes with `type` set to `valueMapping`, for example: `businessPartnerRoleCode`, `timezone` and others. - *Up arrow* icon - Move an attribute mapping one level up. - *Down arrow* icon - Move an attribute mapping one level down.	

Add or Edit Attribute Mappings

When you select *Add mapping* for an entity or choose the *Plus* icon next to an attribute, the *Add Mapping* or *Edit Mapping* dialog appears, respectively. Use it to configure the following:

- Add *Mapping Paths* - A mapping path is a name-value pair (that is, source and target path expressions and their values) based on a selected type of value.
  In the *Name* field, you can choose one of the predefined `sourcePath` and `targetPath` expressions, select your *Type* of value: *attribute*, *property* or *value* and finally, provide the value itself.
  Attribute is the default type value.
  If you want to configure a complex attribute, for example **`name.familyName`**, add the **`name`** attribute first and then choose the *Plus* sign icon to add the **`familyName`** subattribute second. You can add as many subattributes as you need.
- Add *Mapping Configurations* - A mapping configuration is a name-value pair (that is, a transformation expression and its value).
  In the *Name* field, you can choose one of the predefined expressions: *condition*, *scope*, *constant*, *ignore*, *correlationAttribute* and others, define the data type by selecting *Type* and provide a value.
  Some expressions have predefined types and values. For example, ignore and optional are always set with Boolean data type and have true or false as predefined values. The skipOperations are always set with String data type and have: create, update and delete as predefined values.
- View your configuration in the *Mapping Representation in JSON* view.

Add or Edit Functions

When you choose the *Function* icon on an attribute mapping level, the *Add Functions* dialog appears. If a function is already configured for a mapping, the *Edit Functions* dialog appears. Use it to configure the following:

- Choose *Add* > *Function* and select one from the list.
  Most functions have predefined parameters, listed in the *Name* field. For example, if you add the `replaceLastString` function, the `regex` and `replacement` parameters are prefilled. You only need to provide a value.
  If no parameters are listed in the *Name* field, choose *Add* next to the selected function to add a name-value pair.
  You can add as many functions as you need. The number of functions you've added is displayed under the title of the dialog.
- Order the functions if needed. As the order of the functions matter, you can choose the arrows to move them up and down to define the proper place for processing them.

Related Information

Blog Post: Identity Provisioning Graphical Editor – the Game Changer for Transformation Configurations

1.7.13.8.2 Enabling Group Assignment

When provisioning users from source to target systems, in addition to replicating the user data, you can also assign them to groups in the target system. This works for standard provisioning with or without configured bulk operation on the target, and real-time provisioning.

Prerequisites

- You have one of the following target systems configured:
  - Identity Authentication [page 937]
  - Local Identity Directory [page 957]
  - Microsoft Entra ID [page 1323]
  - SAP Advanced Financial Closing [page 966]
  - SAP Analytics Cloud [page 975]
  - SAP Ariba Applications [page 1003]
  - SAP Ariba Central Invoice Management [page 1016]
  - SAP BTP ABAP environment [page 1021]
  - SAP Build Work Zone, standard edition [page 1062]
  - SAP BTP XS Advanced UAA (Cloud Foundry) [page 1045]
  - SAP Commerce Cloud [page 1079]
  - SAP Concur [page 1085]
  - SAP Data Custodian [page 1113]
  - SAP Enable Now [page 1129]
  - SAP Field Service Management [page 1140]
  - SAP HANA Cloud Database Services [page 1145]
  - SAP Integrated Business Planning [page 1157]
  - SAP Intelligent Agriculture [page 1175]
  - SAP Marketing Cloud [page 1196]
  - SAP Revenue Growth Management [page 1205]
  - SAP Sales Cloud and SAP Service Cloud [page 1211]
  - SAP S/4HANA Cloud Public Edition [page 1234]
  - SAP SuccessFactors Employee Central Payroll [page 1279]
  - SAP SuccessFactors Incentive Management [page 1284]
  - Cloud Foundry UAA Server [page 1295]
  - SCIM System [page 1335]
- The groups that you want to assign exist on the target system.

Context

To enable the group assignment, you need to modify the target system transformation by adding a mapping under the user resource containing the following configuration parts:

- **Condition** - Defines for which users the condition will apply. For example, all users with emails.
- **Constant** - Holds the IDs of the groups in the target system. Currently, you can only identify groups by ID.
- **targetVariable** - Specifies whether you want to assign users to groups or unassign users from groups.

Assign Users to Groups

This mapping will result in assigning all users with emails to two groups in a target system.

Code Syntax

```
{
   "user":{
      "mappings":[

<Existing transformation>
{
            "condition":"($.emails
EMPTY false)",
            "constant":[
               {
                  "id":"00f8ab94-
a732-48fa-9169-e51f87b8dcd5"
               },
               {

"id":"01231139-4711-4a28-8f9d-67458
43ef716"
               }
            ],

"targetVariable":"assignGroup"
         }
      ]
   },
   "group":{
      "mappings":[
```

Unassign Users from Groups

This mapping will result in unassigning all users with emails from two groups in a target system.

Code Syntax

```
{
   "user":{
      "mappings":[
<Existing transformation>
{
            "condition":"($.emails
EMPTY false)",
            "constant":[
               {

"id":"006aeaea-3486-479e-8c65-
eabd2868653e"
               },
               {

"id":"00bbae67-6b9d-4f46-9306-6aad2
8c40861"
               }
            ],

"targetVariable":"unassignGroup"
         }
      ]
   },
   "group":{
      "mappings":[
```

Groups which are assigned to users through the user resource are referred to as user managed groups. Following a successful provisioning, if the user managed groups are managed through the group resource, the assignments made through the user managed groups will be overwritten.

Note

If the group doesn't exist in the target system during user creation or update, the user provisioning fails.

If the group doesn't exist in the target system during user deletion, the user deletion succeeds.

> **Note**
>
> If you use different conditions within the group assignment mappings and the same user matches more than one condition, only the last matching condition is applied. As a result, only the groups defined in the last condition are assigned to the user.

If you want to remove the group assignments of a deleted user from the source system, you need to modify the target system transformation by adding a mapping under the user resource containing the following configuration parts:

- **Constant** - Holds the IDs of the groups in the target system which will be updated. Currently, you can only identify groups by ID.
- **targetVariable** - Use **`"unassignGroup"`** to specify that you want to unassign users from the defined groups.
- **scope** - Use **`"deleteEntity"`** to define that the deleted user from the source system should be persisted in the target system. For detailed information depending on the target system in use, see Transformation Expressions [page 391] → `deleteEntity`.

```
{
    "constant": [
        {
            "id": "groupID"
        }
    ],
    "targetVariable": "unassignGroup",
    "scope": "deleteEntity"
}
```

This mapping will result in updating the assignments to the defined group in the target system.

> **Note**
>
> If you try to assign or unassign a user managed group to a user for a provisioning system that does not support it and trigger a provisioning job, the entity will fail and the job will finish with error.

When using the group unassignment for users that are deleted from the source system, these are the possible scenarios based on the system where the user was created and the value of the property `ips.delete.existedbefore.entities`:

- When the user was provisioned from another system by the Identity Provisioning service or it was initially created in the target system and the property `ips.delete.existedbefore.entities` is set to **`true`** In this case only transformation mappings for the affected user containing scope `deleteEntity` are executed. This mapping could be used, for example, for disabling a user account in the target system. For more information, see Transformation Expressions [page 391] → `deleteEntity`.

  > **Note**
  >
  > The update is executed via PUT or PATCH operation, based on the configuration of the property `*.support.patch.operation` in the target system.

- When the user was initially created in the target system and the property `ips.delete.existedbefore.entities` is set to **`false`**, the group assignments of the user in the target system will be updated.

Related Information

Manage Transformations [page 2439]
Transformation Expressions [page 391]
Blog Post: Group Assignments Based on User Attributes – a Flexible Solution for Managing Conditional & Risk-Based Authentication and Much More

1.7.13.8.3 Manage Transformations History

Manage the history of transformations, review and restore them to a previous version of your choice.

> **Note**
>
> This functionality is available for Identity Provisioning tenants running on SAP Cloud Identity infrastructure.

Context

Every time you modify the transformation of a provisioning system and save your changes, a new version is created and displayed in the *Transformation History* screen. It lists all versions - from the initial to the current one, and gives you additional information about the last modification date, the person who modified it and a description. Within this screen, you can perform a number of actions: reset the transformation to its initial version, apply the current default transformation or apply a version of your choice, as well as download previous versions of the transformations for the current system. In case you need to compare two or more versions, you can do it in a text editor, once you download the respective versions.

Transformation History

Column	Description
Version	Transformation version **Note** Version 1 is the initial version. The version with the highest number is the current version. The **initial version** is the one that you get when the system is created (manually or automatically). If you don't make any changes to the transformation while creating the system, the initial transformation matches the default one. If you or a provisioning tool make changes to the transformation while creating the system, the initial transformation doesn't match the default one. There is no restriction on the number of versions you can store and manage. By default, 20 versions are displayed on the screen. You need to expand the list until all your versions are displayed.
Last Modified	Date and time the transformation was last modified
Modified By	Who modified the transformation: • The userID of the person who modified the transformation • *SAP* - Indicates that this transformation has been migrated by SAP. This could be your initial transformation or your current one.
Description	(Mandatory) Description of the changes When modifying the transformation, saving your changes is only allowed if you provide a description. **→ Recommendation** Keep the transformation change description as short as possible. Note that some languages and special characters may consume more space than others.
Actions	Actions you can perform for a selected version: • Apply the selected version. This restores the transformation to the selected version. • Download the selected version.
Default Version	Apply the current default version.
Reset to Initial	Reset the transformation to its initial version.

Following the introduction of transformations history on November 27, 2023, all your systems will have initial version listed in the *Transformation History*. What this initial version means for you, however, depends on the release of the reset transformation functionality on November 1, 2021. And that is because with implementing the reset, Identity Provisioning began storing the initial version of modified transformations.

- Systems created before November 1, 2021, will get the current transformation as initial version.
- Systems created after November 1, 2021, will get their actual initial transformation (the one they got at system creation) as initial version. If this transformation has been changed, the *Transformation History* will also list the current transformation as version 2.
- As of November 27, 2023, every newly created system is created with initial version. Disabled systems will get the initial version upon enabling them.

> **Note**
>
> Deleting a system or resetting the tenant results in deleting the transformation history. Resetting the system itself does not delete the history.

Procedure

To manage your transformations history and apply the version you need, proceed as follows:

1. Sign in to SAP Cloud Identity Services administration console and navigate to *Identity Provisioning*.
2. Select a system under *Source Systems*, *Target Systems* or *Proxy Systems* and choose the *Transformations* tab.
   If the transformations have been modified, an info message is displayed in the message view.
3. Select the *Edit* button and choose the clock icon.
   This opens the *Transformation History* screen.
   Without entering the edit mode, you are only allowed to view the transformation versions and download them.
4. Choose the action that you want to execute.
5. Choose *Close*.

1.7.13.9 Manage Certificates

Identity Provisioning supports certificate-based authentication for secure communication with the provisioning systems (connectors) provided by the service.

Context

Certificates can be used in outbound and inbound connections to Identity Provisioning.

In **outbound connections**, Identity Provisioning acts as an SSL client. The service generates an X.509 client certificate for mutual Transport Layer Security (mTLS) authentication against a given provisioning system acting as a server. The Identity Provisioning client certificate must be uploaded to the given provisioning

system for configuring the certificate-based authentication there. For example, in SAP BTP ABAP Environment, the Identity Provisioning certificate must be uploaded to the communication user used in the communication arrangement.

> **Note**
>
> Outbound certificates generated by Identity Provisioning can also be used for OAuth authentication with provisioning systems. In OAuth certificate authentication, Identity Provisioning acts as a client application that sends requests to the URL of the access token provider service (the authorization server). The server issues access tokens following a successful authentication with certificate (instead of using client ID and client secret).
>
> Currently, OAuth certificate authentication is supported for SAP Business Network and Procurement Data Warehouse.

In **inbound connections**, Identity Provisioning acts as a server whereas the given provisioning system acts as a client and must present a client certificate for establishing the communication to the service. Customers of bundle and standalone tenants running on SAP BTP, Neo environment import client certificates in the Identity Provisioning admin console. Customers of bundle and standalone tenants running on SAP Cloud Identity infrastructure upload the client certificates in the Identity Authentication admin console on the technical user of type *System*.

Inbound certificates are supported for source and proxy systems in the following scenarios: configuring proxy systems and real-time provisioning.

> **Note**
>
> Client certificate authentication is not supported for systems where the *ProxyType* and *ldap.proxyType* properties, required for the HTTP and LDAP connection respectively, are set to `OnPremise`.

> **Note**
>
> If you are using Identity Provisioning proxy and real-time provisioning scenarios, ensure that you trust the new root CA: DigiCert Global Root G2 as the old one DigiCert Global Root CA will be deprecated. For more information, see Root Certificate Replacement.

Task overview: Configuring Provisioning Systems [page 2413]

Related Information

Add New Systems [page 2415]
Search and Edit Systems [page 2418]
Enable and Disable Systems [page 2419]
Export and Import Systems [page 2421]
Delete Systems [page 2423]
Update Connector Version [page 2424]
Manage Properties [page 2438]
Manage Transformations [page 2439]

Configure Form-Based OAuth Authentication [page 2457]
Manage Full and Delta Read [page 2459]
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Reset Identity Provisioning System [page 2485]
Generate and Manage Certificates for Outbound Connection [page 2452]
Manage Certificates for Inbound Connection [page 2456]

1.7.13.9.1 Generate and Manage Certificates for Outbound Connection

Identity Provisioning handles the following tasks related to X.509 client certificates for outbound connection: generating, automatic regenerating, activating, deactivating, downloading and deleting.

Administrators can manage up to two certificates for secure outbound connection with a given provisioning system. Only one certificate can be active at a time.

Generating a second certificate might be needed in case your active certificate is about to expire. Currently, renewing or extending the validity of a certificate is handled by generating a second one. This will ensure that the communication between Identity Provisioning and the given provisioning system won't be disrupted.

Both certificates - the first and the second one, have the same distinguished name (DN), issued by the same certificate authority (CA).

Thirty days before the expiration of your active certificate you'll start receiving a warning message in the *Outbound Certificate* tab of your Identity Provisioning user interface (UI). If you allow the certificate to expire, it becomes invalid. It is not possible to extend its validity.

> → Recommendation
>
> Enable *Automatic Regeneration* to ensure your certificate does not expire. This option is supported for tenants running on SAP Cloud Identity Services infrastructure.

You can subscribe to receive notification e-mails for expiration of your outbound certificate. Based on the enablement of the automatic regeneration option, the e-mail provides the following information:

- When the automatic regeneration option is enabled, 14 days prior to the certificate expiration you will receive an e-mail with subject *Outbound Certificate Renewed. <system_type> System: <system_name>* containing details about the regenerated outbound certificate.
- When the automatic regeneration option is not enabled, on the 30th, 14th, and 3rd day before the outbound certificate expiration, you will receive an e-mails with subject *Outbound Certificate Expiration. <system_type> System: <system_name>* containing information about the certificate.

The following table explains the key tasks available:

Task	Description
Automatic Regeneration	Turn automatic regeneration option on or off: • *ON* - enables the automatic regeneration and activation of outbound certificates. The certificate will be automatically regenerated and activated within 14 days prior to its expiration. → Recommendation This option is recommended for systems validating certificates based on their subject and issuer. In such cases, no further manual steps are required. This option is not recommended for systems validating certificates based on fingerprint. When you regenerate the certificate, the fingerprint is changed. This will cause the authentication to fail until the certificate is downloaded and uploaded to the connected system. If your active certificate has expired and you have manually generated a second one without activating it, enabling the automatic regeneration option will result in the following behavior: 1. A new certificate will be generated and set to active. 2. The expired certificate will be set to inactive. 3. The initially inactive certificate will be deleted. • *OFF* - disables the automatic regeneration. Certificate regeneration will not be done automatically, therefore checking certificate expiration and regeneration should be a manual process.
Generate	You can generate a maximum of two certificates. After reaching the limit, the *Generate* button is greyed out. The status of the first certificate you generate is set to *Active*. The status of the second certificate is set to *Inactive*. Certificates are distinguished by their **fingerprints** (the unique identifier of the certificate).
Subscribe	You can subscribe yourself or another user or a group to receive notification e-mails for expiring or automatically regenerated outbound certificates.

Task	Description
Activate	Activate an inactive certificate. → Recommendation We recommend that you activate your certificate after you have uploaded it in the backend provisioning system. This will ensure that the communication between Identity Provisioning and the backend system won't be disrupted. Only one certificate can be active at a time.
Download	You can download both active and inactive certificates.
Expand	You can expand active and inactive certificates to view their details. If a certificate is about to expire, you see a warning message in the details.
Delete	You can delete only inactive certificates.

Generate Certificate

To generate a client certificate, proceed as follows:

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*, *Target Systems* or *Proxy Systems*.
2. Select your provisioning system and choose *Outbound Certificate* > *Generate*.
   If the certificate is generated successfully, the toast message `Certificate generated successfully` is displayed on the screen.
3. View the certificate information.
   Each certificate contains fields specifying the subject, the name of the CA issuing the certificate, the algorithm used by the issuer to sign the certificate, validity period, key size and the certificate unique identifier.
4. Download the certificate.
5. Log on to the provisioning system you want to authenticate to (in this case, SAP BTP ABAP Environment) and upload the certificate to configure the certificate-based authentication there. For more information, see: How to Create Communication Users
6. Return to the Identity Provisioning UI and select the *Properties* tab of the provisioning system.
7. Set the *Authentication* property to **`ClientCertificateAuthentication`**.
8. Save your configuration.

Manage Certificate Validity

To manage the certificate validity, choose your applicable approach: automatic or manual.

Manage Validity Automatically

Enable the *Automatic Regeneration* option.

The expiring active certificate is set to inactive. A new certificate is automatically regenerated and set to active.

For systems validating certificates based on their subject and issuer, no further manual steps are required.

Manage Validity Manually

1. Periodically check the validity of your active certificate.
2. When the certificate approaches its expiration date (or it has already expired), generate a new certificate, as described above. The status of this certificate will be *Inactive*.
3. Continue with downloading the inactive certificate and uploading it to the backend provisioning system.
4. Once ready, activate the second (inactive) certificate. As a result, your current active certificate will be set to inactive.
5. You can safely delete the inactive certificate.

Subscribe for Expiring Outbound Certificate Notification

To avoid the need of checking manually the validity of your active outbound certificate, you are able to subscribe yourself or another user or a group to receive notification e-mails for its expiration. This way, you will be notified by e-mail on the 30th, 14th, and 3rd day before it expires.

To get subscribed, proceed as follows:

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*, *Target Systems* or *Proxy Systems*.
2. Select the system you need to watch and choose *Outbound Certificates*.
3. Choose *Subscribe*.
   - To subscribe yourself, choose *Subsribe* > *Subscribe me*.
   - To subscribe another user or a group (distribution list), choose *Subscribe others*. Fill in the required fields *Display name* and *E-Mail* and choose *Add*.

     > **Note**
     >
     > From the *Recipients* list, you can remove existing subscribers. To do that, go to the *Action* column and choose the icon.

   - If you no longer need to be subscribed to a source system, choose *Subscribe* > *Unsubscribe me*.

1.7.13.9.2 Manage Certificates for Inbound Connection

Identity Provisioning handles the following tasks related to X.509 client certificates for inbound connection: importing and deleting.

Importing client certificates for inbound connection to Identity Provisioning is used in two scenarios: configuring real-time provisioning and configuring proxy systems for provisioning user data to and from a central identity management solution. Therefore, inbound client certificates can be configured for source and proxy systems only.

Certificate files with the following filename extensions are supported: `.crt`, `.pem` and `.der`.

Depending on the infrastructure/environment your Identity Provisioning tenant is running, you manage the inbound certificates in the administration console of Identity Provisioning or SAP Cloud Identity Services.

SAP Cloud Identity Infrastructure

Bundle or standalone tenants running on SAP Cloud Identity infrastructure manage certificates for inbound connections in the SAP Cloud Identity Services admin console, where the certificate must be uploaded for the technical user of type *System*. For more information, see Add System as Administrator.

If this technical user will be used to connect to Identity Provisioning proxy system, enable the *Access Proxy System API* permission.

1.7.13.9.3 Configure OAuth Certificate Authentication

You can use outbound certificates generated by Identity Provisioning for OAuth authentication with provisioning systems.

In OAuth certificate authentication, Identity Provisioning acts as a client application that sends requests to the URL of the access token provider service (the authorization server). The server issues access tokens following a successful authentication with certificate (instead of using client ID and client secret).

> **Note**
>
> Currently, OAuth certificate authentication is supported for SAP Build Work Zone, advanced edition, SAP CPQ, SAP Business Network and Procurement Data Warehous.

To configure OAuth certificate authentication for communication between Identity Provisioning and a given provisioning system, proceed as follows:

1. Select the *Properties* tab of the provisioning system and make sure the *Authentication* property is set to **`ClientCertificateAuthentication`**.

   > **Note**
   >
   > This authentication method can only be configured using properties. It doesn't work with destinations.

2. Add the *OAuth2TokenServiceURL* property and provide the URL of the access token provider service for your provisioning system.
3. Add the *client_id* property and provide the OAuth Client id used for access token retrieval.
4. Select the *Certificates* tab and generate the certificate for outbound connection, as described in Generate and Manage Certificates for Outbound Connection [page 2452].
5. Log in to the provisioning system and upload the certificate. The specific location may vary depending on the system, but it is typically found in the security or certificate management section.
6. Run a provisioning job.

Results

During the job execution, Identity Provisioning authenticates to the authorization server using the client certificate and gets the access token issued by the server.

1.7.13.10 Configure Form-Based OAuth Authentication

Configure Identity Provisioning to use form-based OAuth 2.0 authentication to call SAP Ariba Applications APIs.

In this authentication scenario, the Identity Provisioning Service uses the OAuth 2.0 client credentials flow to obtain an access token from the Identity Authentication Service. The access token is retrieved using a form-encoded request with a client ID and client secret that are generated for the Administration Console system application of SAP Cloud Identity Services. This token is then used to call the next-generation SCIM API of SAP Ariba Applications.

Prerequisites

- SAP Ariba Applications configuration exists in SAP Cloud Identity Services.
- The required Provided APIs are configured for the SAP Ariba Applications application. For more information, see Provide APIs for Consumption by Other Applications
- A source, target, or proxy system for SAP Ariba Applications is created in Identity Provisioning. For more information, see SAP Ariba Applications (Source [page 641], Target [page 1003], Proxy [page 1450]).

Application Configuration

To enable token retrieval for the next-generation SCIM API of SAP Ariba Applications, a client ID and client secret must be generated in the Administration Console system application under *Trust* > *Application APIs* > *Client Authentication*. These credentials are later used in the SAP Ariba Applications provisioning system properties for the technical user (`User` and `Password`).

For more information, see Configure Secrets for API Authentication.

Additional configuration in the Administration Console system application requires establishing a dependency to the SAP Ariba Applications API. This is maintained under *Trust* > *Application APIs* > *Dependencies*, where the dependency name, the application, and the corresponding API are defined.

For more information, see Integrating Applications Configure Integration Between Applications.

The value provided for the *Dependency Name* is used to identify the API that the consumer application consumes.

> **Note**
>
> The dependency name must match the value of the `<system_prefix>.api.dependency.name` property in the provisioning system configuration.

For more information, see List of Properties [page 80].

Provisioning System Configuration

To configure your SAP Ariba Applications provisioning system, the following properties must be in place:

- `User` - the Client ID generated in the Administration Console system application.
- `Password` - the Client Secret generated in the Administration Console system application.
- `ariba.applications.api.dependency.name` - the dependency name provided in the Administration Console system application.
- `AuthType` - set to Form
- `Authentication` - set to BasicAuthentication
- `OAuth2TokenServiceURL` - the `/oauth2/token` Service URL of Identity Authentication: `https://<Identity Authentication tenant host>/oauth2/token`

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1.7.13.11 Manage Full and Delta Read

When you set up your systems and start a scheduled provisioning task, the standard behavior of the process reads all the entities from the source system. This mode prevents data loss and always keeps your target system synchronized with the source. However, it may take a long time for every job to be executed.

Context

Delta read is a concept for optimizing the amount of data retrieved from the source system. Delta read is much faster, but sometimes might have limitations. In order for a source system to support delta read mode, its API should allow the implementation of this feature.

For example, the *Microsoft Active Directory* source system uses the **uSNChanged** attribute. For more information, see Microsoft: Polling for Changes Using USNChanged .

The main difference between delta and full read is:

- *Delta read* – only modified data is read from the source system and triggered to the target one. Modified data means: new entities and updates on existing entities. Entities deleted from the source system will not be deleted from the target. They can be deleted only during a *full read* job.
- *Full read* – all entities (new, updated, deleted, and existing unchanged ones) are read and checked every time a provisioning job is triggered to the target system.

To keep source and target systems completely synchronized, you can use the *Resync* type of provisioning job.

> → Tip
>
> We recommend that you enforce full reads from time to time if the connector is in delta read mode. To achieve this, you need to set up the following source system property: `ips.full.read.force.count`. For example, `ips.full.read.force.count` = **10** will result in alternating full reads after every 10 delta reads are performed.
>
> This property only impacts scheduled runs; manually triggered runs are ignored. In case it is not set, only delta read jobs will be executed.

> → Remember
>
> When the Identity Provisioning reads entities from a source system for the first time, it always triggers a **full read** job. If the job is successful, the service can then continue with delta read jobs (if such are activated).

During a delta read job, the service reads only the entities that are new or have been modified after the last successful job.

> **Note**
>
> Note that Identity Provisioning will not delete entities in the target system if the job runs in **delta read** mode, if entities fail to be read from the source system or if both conditions apply. Deletion is only possible when the job runs in **full read** mode and the entities are successfully read from the source system.
>
> In cases where deletion does not occur automatically, you must decide on the next steps. You can either resolve the failed entities in the source system and run a full read job, or proceed with deletion as described in Manage Deleted Entities [page 2463].

Below are listed all source systems that currently support *delta read* mode.

Supported Systems

System Type	Details
Microsoft Active Directory	Default mode: *Full read* You can switch to delta read, if you set up the relevant property: `ips.delta.read` = **enabled** Bear in mind the following specifics and limitations: • Make sure that the service user, which is used in the AD destination, has a **Domain Admin** role, otherwise the connector won't be able to extract any data from the recycle bin. • Due to the *linked attributes* concept of AD, there is a limitation in the Microsoft Active Directory read connector, when performing in delta read mode. We recommend that you enforce full reads periodically in order to avoid data loss. See: Microsoft: Linked Attributes • You need to set limitations about which particular attributes to be read. For this purpose, set the properties `ldap.user.attributes` and `ldap.group.attributes` and add **uSNChanged** to the attributes list. Otherwise, the provisioning job will run in *full read* mode. • If an entity is moved outside the base path (another directory context), the connector won't recognize this change during delta read.
SAP SuccessFactors	Default mode: *Delta read* You can switch to full read, if you set up the relevant property: `ips.delta.read` = **disabled**
SAP SuccessFactors Learning	Default mode: *Delta read* You can switch to full read, if you set up the relevant property: `ips.delta.read` = **disabled**
Identity Authentication	Default mode: *Full read*

System Type	Details
Intelligent Opportunity Analyzer	You can switch to delta read, if you set up the relevant property: `ips.delta.read` = **enabled** **Note** When using SAP Central Business Configuration and Identity Directory SCIM API (in short, SCIM API version 2), delta read mode is only supported for user resources. For delta read of resources (users and groups), bear in mind the following API requirements: • The system API should return **lastModified**, which is a subattribute of the **meta** attribute. The **lastModified** subattribute denotes the most recent date and time when the resource details were updated at the service provider. See: SCIM: Common Attributes • The system API has to also support filtering by the **lastModified** attribute, and the system should support the **gt** operator in filter expressions. See: SCIM: Filtering
Local Identity Directory	
SAP Central Business Configuration	
SAP Data Custodian	
SAP Enterprise Portal	
SAP SuccessFactors Employee Central Payroll	
SAP Sales Cloud and SAP Service Cloud	
SAP Field Service Management	
SAP CPQ	
SAP Concur	
SAP Advanced Financial Closing	
SAP Advanced Workflow	
SAP Intelligent Agriculture	
SAP HANA Cloud, SAP HANA Database	
SCIM System (General SCIM system, if fulfills the API requirements)	

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1.7.13.12 Manage Deleted Entities

Manage deletion of entities (users or groups) in the target system after they have been deleted from the source system.

Scenario 1: Entity exists in the source and the target system

An entity exists both in the source and the target system.

1. You run a provisioning job for the first time.
   As a result, Identity Provisioning reads this entity from the source and updates it on the target system.
2. You delete the relevant entity from the source system.
3. You run another provisioning job, which finishes successfully.
   However, the service recognizes the relevant entity as a "previously existed one" and **does not delete** it from the target.

Solution

To delete entities from a target system after they have been deleted from the source system, you need to set the following property: **`ips.delete.existedbefore.entities`** = *true* in the target system. This must be done **before** the job to delete those entities from the target system is executed.

> → Recommendation
>
> The following sequence of steps is recommended for synchronizing deletion of entities between source and target systems, as in **Scenarios 1, 2** and **3**:
>
> You have run successful provisioning jobs (*Read* or *Resync*) between the systems.
>
> 1. Delete an entity from the source system.
> 2. On the *Properties* tab of the target system, add the **`ips.delete.existedbefore.entities`** property and set its value to *true*.
> 3. Run a provisioning job.
> 4. Verify that the relevant entity has been deleted from the target system.

If the property is set **afterwards**, entities recognized as "previously existed ones" cannot be deleted from the target system anymore. In this case, you need to delete them from the target system (for example, manually or via script).

The **`ips.delete.existedbefore.entities`** is an optional property which can be set on every target system. You can use it to control whether recognized entities as "previously existed ones" should be deleted from the target system.

This is important for security and legal reasons in cases when users (for example, employees) are no longer active in the source system, and their availability and permissions must be removed from the relevant target system(s).

For more information about this property, see: List of Properties [page 80], where you can search it by *Name* or use the general table search.

Scenario 2: Entity does not exist in the source and the target system

An entity does not exist in either system (neither source, nor target).

1. You run provisioning jobs (*Read* or *Resync*) between the systems.
2. You add this entity to the source system.
3. The same entity is added (manually or via script) to the target system.
4. You run a new provisioning job.
   As a result, Identity Provisioning reads this entity from the source and updates it in the target system.
5. You delete the relevant entity from the source system.
6. You run another provisioning job, which finishes successfully.
   However, the service recognizes the relevant entity as a "previously existed one" and **does not delete** it from the target.

Solution

To delete entities from a target system after they have been deleted from the source system, you need to set the following property: **`ips.delete.existedbefore.entities`** = *true* in the target system. This must be done **before** the job to delete those entities from the target system is executed.

> → Recommendation
>
> The following sequence of steps is recommended for synchronizing deletion of entities between source and target systems, as in **Scenarios 1, 2** and **3**:
>
> You have run successful provisioning jobs (*Read* or *Resync*) between the systems.
>
> 1. Delete an entity from the source system.
> 2. On the *Properties* tab of the target system, add the **`ips.delete.existedbefore.entities`** property and set its value to *true*.
> 3. Run a provisioning job.
> 4. Verify that the relevant entity has been deleted from the target system.

If the property is set **afterwards**, entities recognized as "previously existed ones" cannot be deleted from the target system anymore. In this case, you need to delete them from the target system (for example, manually or via script).

The **`ips.delete.existedbefore.entities`** is an optional property which can be set on every target system. You can use it to control whether recognized entities as "previously existed ones" should be deleted from the target system.

This is important for security and legal reasons in cases when users (for example, employees) are no longer active in the source system, and their availability and permissions must be removed from the relevant target system(s).

For more information about this property, see: List of Properties [page 80], where you can search it by *Name* or use the general table search.

Scenario 3: Entity exists in the source system only

An entity exists in the source system only.

1. You run at least one provisioning job.
   As a result, Identity Provisioning reads this entity and creates it in the target system.
2. You reset one of these systems. See: Reset Identity Provisioning System [page 2485]
3. You run a new provisioning job.
   As a result, Identity Provisioning reads this entity from the source system (but is not "aware" of it, that is, it behaves like reading it for the first time) and makes a full update of it in the target system.
4. You delete the relevant entity from the source system.
5. You run another provisioning job, which finishes successfully.
   However, the service recognizes the relevant entity as a "previously existed one" and **does not delete** it from the target.

Solution

To delete entities from a target system after they have been deleted from the source system, you need to set the following property: **`ips.delete.existedbefore.entities`** = *true* in the target system. This must be done **before** the job to delete those entities from the target system is executed.

> → Recommendation
>
> The following sequence of steps is recommended for synchronizing deletion of entities between source and target systems, as in **Scenarios 1, 2** and **3**:
>
> You have run successful provisioning jobs (*Read* or *Resync*) between the systems.
>
> 1. Delete an entity from the source system.
> 2. On the *Properties* tab of the target system, add the **`ips.delete.existedbefore.entities`** property and set its value to *true*.
> 3. Run a provisioning job.
> 4. Verify that the relevant entity has been deleted from the target system.

If the property is set **afterwards**, entities recognized as "previously existed ones" cannot be deleted from the target system anymore. In this case, you need to delete them from the target system (for example, manually or via script).

The **`ips.delete.existedbefore.entities`** is an optional property which can be set on every target system. You can use it to control whether recognized entities as "previously existed ones" should be deleted from the target system.

This is important for security and legal reasons in cases when users (for example, employees) are no longer active in the source system, and their availability and permissions must be removed from the relevant target system(s).

For more information about this property, see: List of Properties [page 80], where you can search it by *Name* or use the general table search.

Scenario 4: Entity not created in the target system by Identity Provisioning

An entity exists both in the source and the target system. (It has **not** been created on the target by the Identity Provisioning service.)

Conditions or expressions, such as (*ignore* or *skipOperations*), are not set in the target transformation.

1. You run a successful *Read* job. As a result, Identity Provisioning updates the existing entity on the target system.
2. You delete this entity from the source system.
3. You run a provisioning job, which finishes with error.
   As a result, the relevant entity has not been deleted from the target system.
4. In the job log, you see that there are failed entities (users or groups) on the source system. That means, the job has failed trying to read them from the source.

Solution

1. Resolve the failed entities in the source system.
2. On the *Properties* tab of the target system, add the **`ips.delete.existedbefore.entities`** property and set its value to *true*.
3. Run a successful *Read* job between the systems.
4. Verify that the relevant entity has been deleted from the target system.

> → Tip
>
> Even if the job fails due to errors on the target system, if the read from the source is successful, the service will still delete the entity from the target.

Scenario 5: Entity created in the target system by Identity Provisioning

An entity exists in the source system and has been provisioned to the target by the Identity Provisioning service.

Conditions or expressions, such as (*ignore* or *skipOperations*), are not set in the target transformation.

1. You delete this entity from the source system.
2. You run a provisioning job, which finishes with error.
   As a result, the relevant entity has not been deleted from the target system.
3. In the job log, you see that there are failed entities (users or groups) on the source system. That means, the job has failed trying to read them from the source.

Solution

1. Resolve the failed entities in the source system.

2. Run a successful *Read* job between the systems.
3. Verify that the relevant entity has been deleted from the target system.

> → Tip
>
> Even if the job fails due to errors on the target system, if the read from the source is successful, the service will still delete the entity from the target.

Scenario 6: Delta read job has been run

An entity exists in the source system and has been provisioned to the target by the Identity Provisioning service.

Conditions or expressions, such as (*ignore* or *skipOperations*), are not set in the target transformation.

1. You delete an entity from the source system.
2. You run a **delta read** job, which finishes successfully.
   However, the relevant entry has not been deleted from the target system. That's because *delta read* jobs do not take deleted users into consideration. To learn more, see: Manage Full and Delta Read [page 2459]

Solution

1. On the *Properties* tab of the source system, set the `ips.delta.read` property to *false*.
   Alternatively, you can wait for the next scheduled *full job* to start (if it's coming soon), according to the number you have set for property `ips.full.read.force.count`.
2. Run a new provisioning job (or wait for it to run automatically). It will be a *full read* job.
3. Verify that the relevant entity has been deleted from the target system.
4. (Optional) If you want to continue running *delta read* jobs, go to the `ips.delta.read` property and set it back to *true*.

Blog Post:To Delete or Not to Delete: No Room for Drama in Identity Provisioning

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1.7.13.13 Connect to On-Premise Systems

Set up the connection to on-premise systems when your Identity Provisioning bundle or standalone tenant is running on the infrastructure of SAP Cloud Identity Services.

Prerequisites

- You have installed the Cloud Connector (for SAP BTP multi-environment subaccount), and have done the initial configuration. For more information, see Cloud Connector.

Context

If your provisioning scenarios involve on-premise systems, this requires a separate configuration in three places:

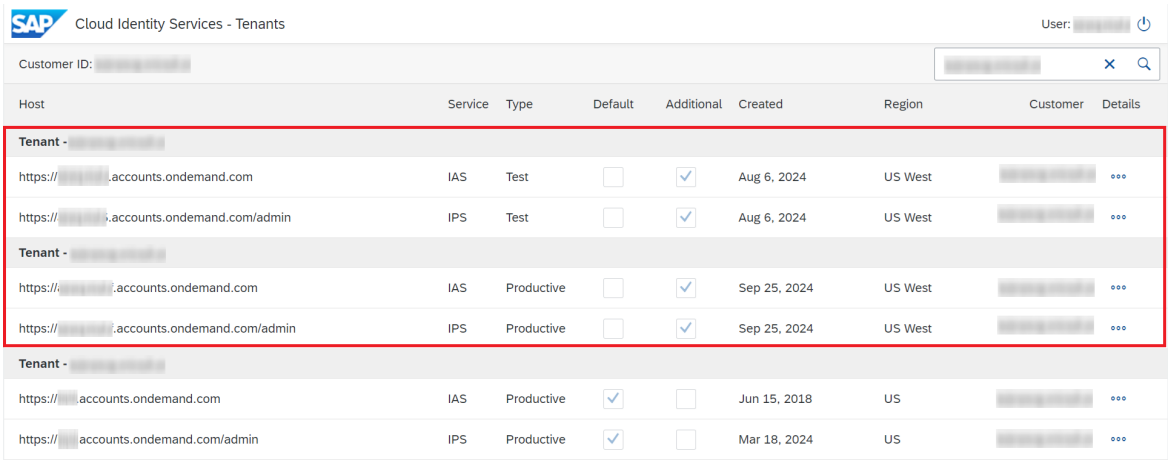
- SAP BTP cockpit, where you subscribe to the Cloud Identity Services connectivity plan in your multi-environment subaccount.
- SAP Cloud Connector, where the connection to your multi-environment subaccount is established, and the backend (on-premise) system is defined.
- The Identity Provisioning section of SAP Cloud Identity Services administration console, where you configure the on-premise provisioning systems.

Procedure

1. Before you start actual configuration, access the *SAP Cloud Identity Services - Tenants* application at the following URL: https://iamtenants.accounts.cloud.sap/ to view the region and the type of Identity Authentication and Identity Provisioning tenants assigned to your customer ID. You will need this information later when creating multi-environment subaccount.

   For more information, see Viewing Assigned Tenants and Administrators.

The screenshot below illustrates a customer landscape with Identity Authentication and Identity Provisioning tenants running on their common SAP Cloud Identity Services infrastructure in one region US West (in the red frame). The first pair is used for testing purposes and the second one for productive purposes.



2. Log on to SAP BTP cockpit and choose your global account.

   If you have only one global account, you are automatically taken there. If you have multiple ones, select the global account you want to set up connection to on-premise system.

   For more information on how to navigate in the SAP BTP cockpit, refer to Navigate in the Cockpit.

3. Create a multi-environment subaccount in the Cloud Foundry region that maps the region of the Identity Authentication tenant (where the Identity Provisioning is also running). See the mapping table below.

Identity Authentication Region	Cloud Foundry Region
Netherlands (Amsterdam) / Germany (Frankfurt)	Europe (Frankfurt) AWS
Germany (Frankfurt)	Europe (Frankfurt) AWS
UAE (Dubai)	Europe (Frankfurt) AWS
Australia (Sydney)	Australia (Sydney) AWS
China (Shanghai)	Singapore AWS
Singapore	Singapore AWS
South Korea (Seoul)	South Korea (Seoul) AWS
Japan (Tokyo) / Japan (Osaka)	Japan (Tokyo) AWS
Brazil (São Paulo)	Brazil (São Paulo) AWS
United States (Sterling) / United States (Colorado)	US East (VA) AWS
Canada (Toronto)	Canada (Montreal) AWS

Identity Authentication Region	Cloud Foundry Region
US West	US West (WA) Azure
Switzerland (Zürich)	Switzerland Azure
India (Mumbai)	India (Mumbai) GCP
Saudi Arabia (Dammam) - Regulated	KSA (Dammam) GCP
Saudi Arabia (Dammam)	Europe (Frankfurt) AWS

For example, customers with Identity Authentication and Identity Provisioning tenants in US West region must create a Cloud Foundry subaccount in US West (WA) Azure region.

Once you create the subaccount, you must enable Cloud Foundry for this subaccount.

> **Note**
>
> If you already have a Cloud Foundry subaccount, you can use it.

4. Open your subaccount, and from the left-side panel choose *Service Marketplace* > *Foundation / Cross Services* > *Cloud Identity Services*.
5. Choose *Create*, select the *connectivity* plan and then *Next*.

   If the connectivity plan is not present, navigate to *Entitlements* > *Configure Entitlements* > *Add Service Plans*. Search for *Cloud Identity Services* > *connectivity plan*, add it as service plan and save your changes.

   > **Note**
   >
   > You can create only one subscription plan per subaccount. This means, you cannot have connectivity along with additional-tenant plan in the same subaccount.

   > **Note**
   >
   > The Cloud Foundry subaccount with the connectivity plan is involved only in setting up the connection, not in transferring data. It is used to establish the tunnel between the SAP Cloud Identity Services tenants (Identity Authentication and Identity Provisioning) and the SAP Cloud Connector. No business data passes through the Cloud Foundry subaccount or its region. It flows directly between the on-premise system and SAP Cloud Identity Services.

6. From the *Cloud Service Type* dropdown, choose for what type of tenant (test or productive) you want to use the subaccount.
   - *Test*
   - *Productive* - default value
7. Choose *Next* and then *Create*.

   This creates a subscription to Cloud Identity Services connectivity plan and creates a binding to your Identity Authentication tenant and your Identity Provisioning service.

> ### Note
>
> You can have two separate SAP BTP multi-environment subaccounts per region: one of them configured for *Productive* connections and the other for *Test* connections. While it is allowed to create multiple Cloud Foundry subaccounts per one region and per one type (test or productive), be aware that in this case only the first one where the connectivity plan was enabled will be used by Identity Provisioning.
>
> The service instance types *Test* and *Productive* map to the tenant types displayed in the *SAP Cloud Identity Services - Tenants* application which is accessible at https://iamtenants.accounts.cloud.sap/.

8. Now, you need to add and connect your subaccounts to the Cloud Connector. Log on to the Cloud Connector administration UI and choose *Add Subaccount*.

   For more information, see Managing Subaccounts.

9. Verify that the Cloud Connector is connected to your Cloud Foundry subaccount. In SAP BTP cockpit, from the left-side panel, choose *Connectivity* > *Cloud Connectors* to see your up and running cloud connector tunnels.
10. Return to the Cloud Connector and define the backend (on-premise) system.
11. Return to your subaccount in SAP BTP cockpit and navigate to *Connectivity* > *Destinations* to configure the destination for your on-premise system.

    > ### Note
    >
    > This step is mandatory only for SAP Application Server ABAP. For the other on-premise systems, you can use the *Properties* tab.

    For more information, see: Create RFC Destinations and SAP Application Server ABAP → step 2.

12. Sign in to SAP Cloud Identity Services administration console.

    The URL follows the pattern: `https://<tenant_id>.accounts.ondemand.com/admin`

13. In the Identity Provisioning section, add an on-premise system (source, target or proxy).
14. If you want to create a connection to SAP AS ABAP, from the *Destination Name* combo box, select the destination you have created in the cockpit and save your configurations.

    If you want to create a connection to other on-premise systems, configure the connection details on the *Properties* tab of the given system.

15. (Optional) If your Cloud Connector is configured with *Location ID*, this location identifier must also be set for the respective on-premise system. You have the following options:

    - Connectivity destination - Create it in your subaccount in the SAP BTP cockpit and provide the Location ID there.

      > ### Note
      >
      > Using connectivity destination is mandatory only for SAP Application Server ABAP.

    - `CloudConnectorLocationId` property - Configure the property in the Identity Provisioning section of SAP Cloud Identity Services administration console for all HTTP and LDAP-based systems, SSH Server (Beta) and SAP HANA Database with ProxyType set to **`OnPremise`**.
      Alternatively, for HTTP-based systems only, you can add the Identity Provisioning `ips.http.header.<header_name>` property, where the header name is: **`SAP-Connectivity-`**

`SCC-Location_ID` and the value is the Location ID. For example: `ips.http.header.SAP-Connectivity-SCC-Location_ID=<LocationID>`

16. Add another provisioning system, connect it to your on-premise one, and run a provisioning job.

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1.7.13.14 Start and Stop Provisioning Jobs

You can start and stop a provisioning job from the Identity Provisioning user interface (UI) or from an API client by using the Identity Provisioning tenant admin API.

Prerequisites

- Your source and target systems are configured and enabled.
- (Optional) You have run a *Simulate* and/or a *Validate* job before you run the actual provisioning job to verify that Identity Provisioning configurations produce the desired result in the target systems.

Job Types

The Identity Provisioning service provides the following types of provisioning jobs:

Run From	Job Type	Real Provisioning
Admin console	*Read Job* - Reads all entities from the source system and provisions only new or updated entities to the target system. If the job is run in **delta read** mode, it reads and provisions only new or updated entities in the source system. See Read Provisioning Job [page 2474]	Yes
	Resync Job - Reads all entities from the source system and provisions all entities to the target system. See Resync Provisioning Job [page 2476]	
	Simulate Job - Estimates the number of entities that will be created, updated, deleted or skipped in the target system. Provides the expected results of a resync job without modifying the target system. See Simulate Provisioning Jobs [page 2477]	No
	Validate Job - Verifies how entities (users and groups) would be mapped from source to target systems without modifying them. See Validate Provisioning Jobs [page 2478]	
API client	Use the Identity Provisioning tenant admin API to run a provisioning job from an API client. The API is available on the SAP Business Accelerator Hub. See Run Provisioning Jobs via API [page 2480]	Yes

Start a Job

To run a job, select a source system and choose *Jobs* > `<Job_Type>` > *Run Now*.

Schedule a Job

To schedule a job run, select a source system and choose *Jobs* > *Read Job* > *Schedule*.

Stop a Job

To stop a running job, select a source system and choose the *Stop Job* button in the *Action* column.

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Monitor Provisioning Job Logs [page 2982]
Manage Provisioning Job Logs [page 3010]

1.7.13.14.1 Read Provisioning Job

The *Read* job reads all entities from the source system and provisions only new or updated entities to the target one. If the job is run in **delta read** mode - meaning the `ips.delta.read` property is enabled - it reads and provisions only new or updated entities in the source system.

> **Note**
>
> A Read job checks only for changes in the source system. If there have been changes in the target system, they are not affected by the job.

Although the name of the job implies reading users, a Read job could also lead to deleting users from the target system. This could happen if any of the following changes occur after you have run a provisioning job (Read or Resync):

- A user has been deleted in the source system. As a result, on the next run of the Read job, the deleted user won't be read from the source system and will be deleted in the target system.
- A user filter is applied in the source system. As a result, on the next run of the Read job, the users not matching the filtering criteria will be deleted in the target system.
- A condition is applied either in the source or in the target system. As a result, on the next run of the Read job, the users not matching the condition criteria will be skipped or deleted in the target system.
- The mapping of the targetVariable `entityIdSourceSystem` or `entityIdTargetSystem` has been changed.

> **Example**
>
> You have a source system with users A, B, C and D.

1. Run a Read Job.
   Users A, B, C and D are created in the target system.
2. In the source system, update user C, delete user D and add a new user E.
3. Run a second job.
   - If the second job is a Read Job: It reads users A, B, C and E from the source system and then creates user E, updates user C, and deletes user D in the target system.
   - If the second job is a Delta Read job: It reads users C and E from the source system and then creates user E and updates user C.
   - If the second job is a Resync job: It reads users A, B, C and E from the source system and then creates user E, updates users A, B and C, and deletes user D.

Run Now

To run a read job, select a source system and choose *Jobs* > *Read Job* > *Run Now*. This starts a read job immediately.

Schedule

To schedule a read job, select a source system and choose *Jobs* > *Read Job* > *Schedule*. This runs the job at the scheduled time period.

Option	Description
ON/OFF	Turn the job scheduler on or off: • *ON* - enables the job scheduler. After that, jobs run at the scheduled time period. This option does not run the job immediately. • *OFF* - stops the job scheduler. After that, jobs are stopped from running at the scheduled time. This option does not stop/pause a running job.

Option	Description
Run job every (minutes)	Schedule how often a read job to be run. The number must be larger than **30** (minutes). This option sets the time period but does not run the job immediately. Thus, after you set a scheduled period, turn *ON* the job scheduler. Use cases: • If the scheduled job is finished **before** the next time point, it will start again on the upcoming time point as scheduled, at the precise second. • If the scheduled job is finished **after** the next time point, it will skip the relevant upcoming time point(s) and will start again on the next available one, at the precise second. • If **during** a running scheduled job, you click *Run Now*, this will trigger a manual job, which will start immediately after the current scheduled job has finished. The next scheduled job will start at the upcoming regular time point, at the precise second. → Remember Before you schedule a job, make sure it's not been paused (that is, the job scheduler has not been turned off). Otherwise, the job will not be executed.
Run job on a specific day of the week and time:	Specify the day and time to run the job. You must select at least one day of the week and the exact time. Day and time fields cannot be empty. The job is run in customer's time zone. ⓘ Note This functionality is available only for Identity Provisioning bundle and standalone tenants running on SAP Cloud Identity infrastructure.

1.7.13.14.2 Resync Provisioning Job

The *Resync* job reads all entities from the source system and provisions all entities to the target system.

This job performs a full replace of entities in the target system with entities from the source system. You can use it to fix inconsistent data between both systems. For example, when an entity has been changed or deleted in the target system only.

A Resync job overwrites changes in the target system. As a result, the data between the source and the target system becomes consistent again.

To run a resync job, select a source system and choose *Jobs* > *Resync Job* > *Run Now*. This starts a resync job immediately.

ⓘ Note

Normally, you don't schedule a *Resync Job* as it is expected to be run on demand.

1.7.13.14.3 Simulate Provisioning Jobs

You can simulate a provisioning job before you actually run it.

Prerequisites

The source system where you run the job must be enabled.

Context

Simulating a provisioning job allows you to test your Identity Provisioning configurations and see whether they produce the desired result in the target system. In case the result is not what you have expected, you can identify the wrong configurations and correct them before you run the actual provisioning job.

The simulate job reads the data from the source system, applies the source and target system transformation and provides the expected results of a resync job without actually modifying the target system. The simulation is based on the Identity Provisioning operational data from previously provisioned entities using the same source and target system configurations.

Run this job whenever you change filtering properties, apply conditions or modify attribute mappings in the transformation code of a source and/or target system because these changes affect the data you want to provision. As a result, in the *Job Execution Logs* screen, you will see an estimation of the numbers of users and groups that will be created, updated, deleted or skipped in the target system.

Note that, if some of the entities already exists on the target system, the simulation won't detect them. In such cases during the real provisioning Identity Provisioning will detect them and try to update them, so they may appear as updated entities. The simulation can detect some wrong attribute mappings as missing attributes that are defined as required in the transformation. However, it cannot predict errors that may occur on the target system as duplicate identifiers or not supported values.

Proceed as follows:

Procedure

1. Open the source system and choose the *Jobs* tab.
2. Choose *Run Now* for the *Simulate Job*.
3. Open the *Job Execution Logs* and verify the provisioned entities.

Results

If the number of created, updated, deleted or skipped entities is what you have expected, run the read or resync job.

If the number of created, updated, deleted or skipped entities is not what you have expected, correct your configurations and run the simulate job again.

Related Information

Blog Post: Simulate it until you make it! Try out the Identity Provisioning job that tests your configuration.

1.7.13.14.4 Validate Provisioning Jobs

You can validate a provisioning job before you actually run it.

Prerequisites

- You have enabled the source system where you want to run the validate job from.
- You have created separate CSV input files for validating users and groups.

> **Note**
>
> This functionality is available only for Identity Provisioning tenants running on SAP Cloud Identity infrastructure.

Context

The validate job allows you to test how entities (users and groups) would be mapped from source to target systems before you run the actual provisioning job. You can use it to validate both - default and modified transformations, and see what would be the expected result in the target system.

Like the simulate job, the validate job predicts results in the target system without modifying it. However, unlike the simulate job (which estimates the number of entities that will be created, updated, deleted or skipped), the validate job verifies the content of the entities. For example, you can use it in the following cases:

- You want to run your first provisioning job from system A to system B and see what will be the result when the default transformations are applied.

- You have modified the **`sourcePath`** or the **`targetPath`** attributes in the transformations and you want to see how they will be mapped in the target system. For example, instead of populating the username in SAP Analytics Cloud target system with the email, you want to provision the username itself:

Default Mapping	Changed Mapping

Code Syntax
`{`
`"sourcePath": "$.emails[0].value",`
`"targetPath": "$.userName"`
`},`
	Code Syntax
	`{`
	`"sourcePath": "$.userName",`
	`"targetPath": "$.userName"`
	`},`

- You want to test if conditions are met or if all required user and group attributes are available

To run a validate job, you need to import one or two CSV files - one for users and/or one for groups. Each file must contain a maximum of 10 users or groups with attributes as defined in the source system. The attribute names must use the JSONPath dot-notation. For example: **`emails[0].value`** attribute indicating the value of the first email within an email array, or **`name.familyName`** - a complex **`name`** attribute containing a **`familyName`** sub-attribute. If the file has more than 10 users or groups, the job will validate the first 10 and will ignore the rest.

You can create a CSV file using a text editor, for example Notepad. Make sure the number of fields match the number of headers in the file. Otherwise, you will get an error when importing the file and won't be able to proceed until you correct it.

Example

The following example illustrates that the number of fields does not match the number of headers (columns 1 to 5) at row 3.

```
col1,col2,col3,col4,col5
A,B,,D,E
L,M,N,,
P,Q,
```

Example

The following example illustrates how to validate two users with attributes defined in the MS Entra ID source system.

```
userPrincipalName,id,mail,givenName,surname,groups[0].value,groups[1].value
ChrisEvans,ec2c28da-
a9fe-46c9-876e-564317f9a3af,chris@example.com,Christopher,Evans,Employee,Suppo
rt
MaryWatson,b016ecad-2c30-4fdf-8fb3-117db4911256,mary@example.com,Mary,Watson,E
mployee,Developer
```

For more information on how to create the CSV files and end-to-end examples, see the blog post After simulation, try out validation. Identity Provisioning closes the loop with a fresh new test job.

The validate job is triggered manually. There is no option to schedule it. Proceed as follows:

Procedure

1. Open the source system and choose the *Jobs* tab.
2. For *Validate Job*, choose *Run Now*.
3. In the *Import Entities* dialog box, browse for and select the CSV files for testing users and groups.

   You can import and validate one of the files or both.
4. Choose *Validate*.

Results

The result of a validate job is not displayed in the Job Logs, as it is with the read, resync and simulate jobs. It is provided in a downloadable ZIP file. Each administrator who runs the validate job, can see only his or her test results.

The ZIP file contains one or more CSV files, and a trace.log file in case the validation job identified failed or skipped entities. You can expect the following zipped files: one for each of the entities (users and groups) you have tested, and one for each of the target systems connected to the source system where you run the job from.

For example, if you have tested users and groups from source system A, which is connected to three target systems B, C, and D, expect a ZIP file with six CSV files.

If the result of a validate job is not what you have expected or finished with errors and skipped entities, you can identify where the wrong configurations or missing values come from and correct them.

1.7.13.14.5 Run Provisioning Jobs via API

You can run a provisioning job via API requests.

Prerequisites

You need a technical user with *Access Identity Provisioning Tenant Admin API* permission assigned. For more information, see Managing Administrators.

Context

Use the Identity Provisioning tenant admin API to run a provisioning job from an API client. The API is available on the SAP Business Accelerator Hub: SAP Cloud Identity Services *Identity Provisioning Service* *API Reference* *Jobs*. The URL for accessing the Tenant Admin API follows the pattern: `https://<ias-tenant-host>/ips/service/publicapi/v1/startJob/{SourceSystemId}/jobs/{JobType}`, where:

- `<SourceSystemId>` is the ID of the source system, displayed at the end of the system URL in the SAP Cloud Identity Services administration console.
- `<JobType>` is **READ** or **RESYNC**.

> **Note**
>
> This functionality is available only for Identity Provisioning tenants running on SAP Cloud Identity infrastructure.

1.7.13.15 Handle Rate Limits

Identity Provisioning APIs implement rate limits to control the number of incoming requests for a given time.

If an external system sends requests to Identity Provisioning proxy or real-time APIs and gets the *429* status code *Too Many Requests*, this means that the specified rate limits of the service are exceeded. As a result, Identity Provisioning rejects the requests for a period of time. Rate limits help avoid overloading, misuse of APIs and performance issues.

Identity Provisioning Rate Limits

Value	Description
150	Allowed requests per minute. When the limit is reached, the requests are slowed down.
200	Maximum requests per minute. When the limit is reached, further requests are immediately rejected.

The Identity Provisioning rate limits are enforced at tenant level to the total number of incoming requests (that is, incoming calls for real-time provisioning and proxy systems).

For more information on how Identity Provisioning handles failed operations when the service sends requests to external systems, see Handle Failed Operations [page 2482].

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1.7.13.16 Handle Failed Operations

In certain cases, you can set a retry for a failed operation due to an occurred exception.

If an entity operation (*get*, *create*, *update*, or *delete*) fails due to an occurred exception (rate limit, bad gateway, missing authorization, or timeout), you can set it for retry by the Identity Provisioning. You can specify a number of retries by setting the property `ips.failed.request.retry.attempts`. You can also specify a time interval (in seconds) between the retries via the property `ips.failed.request.retry.attempts.interval`. For more information, see List of Properties [page 80].

Based on the system you use, the type of failed operation, and the occurred exception, the option to retry is possible in the following cases:

Identity Provisioning Retry Support

Exception	Write Operation via job or Proxy	Read Operation via job	Read Operation via Proxy
Timeout exception	All connectors	All connectors **Note** The retry is supported only for full read.	The retry is not supported.

Exception	Write Operation via job or Proxy	Read Operation via job	Read Operation via Proxy
Too many requests (429)	• Identity Authentication • Microsoft Entra ID • SAP Analytics Cloud • SAP BTP XS Advanced UAA (Cloud Foundry) • SAP Sales Cloud and SAP Service Cloud • SAP SuccessFactors version 2 • SAP Jam Collaboration • SCIM System • SAP Build Work Zone, advanced edition	• SCIM-based systems (listed in System Types [page 70]) • Microsoft Entra ID **Note** The retry is supported only for full read.	• SCIM-based systems (listed in System Types [page 70]) • Microsoft Entra ID **Note** The retry is supported for single entity read by `ID` and for full read.
Bad gateway (502)	• SAP Build Work Zone, standard edition • SAP BTP XS Advanced UAA (Cloud Foundry) **Note** For SAP Build Work Zone, standard edition and SAP BTP XS Advanced UAA (Cloud Foundry) is supported retry only for patch operation. • SAP Sales Cloud and SAP Service Cloud	SAP Sales Cloud and SAP Service Cloud	SAP Sales Cloud and SAP Service Cloud
Forbidden (403)	SAP Analytics Cloud	The retry is not supported.	The retry is not supported.

For more information, see:

Rate and Burst Limits for SAP Build Work Zone, advanced edition

Understanding Rate Limits of SAP Ariba SCIM API

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1.7.13.17 Reset Identity Provisioning Settings

Resetting your provisioning settings will delete all provisioning systems configured for your tenant, along with the related job execution logs.

Context

Be careful with this option. If you reset your provisioning settings, you will lose all systems, configurations, scheduled jobs, source system subscriptions, and job execution logs. To use the service again afterward, you will need to set up new systems.

If want to reset your provisioning settings, proceed as follows:

Procedure

Sign in to the administration console of SAP Cloud Identity Services, navigate to *Identity Provisioning* and choose *Reset Provisioning Settings*.

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1.7.13.18 Reset Identity Provisioning System

Resetting an Identity Provisioning system (source or target) deletes all Identity Provisioning operational data.

Context

There might be times when you would like to delete the current Identity Provisioning operational data for a particular system. For example, clearing entities that were read from the source system and were then mapped to SCIM specific attributes via the intermediate transformation logic.

This operation is called *system reset*. If you choose it, you only clear the Identity Provisioning operational data. The system configurations and all existing read and provisioned entities, along with their authorizations, will be preserved in the back-end source and target systems. To learn more, see: Transformations [page 388]

Cases that require reset of the provisioning system are:

- Updating the version of the provisioning source or target system; For more information, see Update Connector Version [page 2424].
- Changing the attribute mapped to the variable *entityIdSourceSystem* (for a source system) or *entityIdTargetSystem* (for a target system); For more information, see Transformation Variables [page 490].
- Resetting Identity Authentication target system might be needed after an SAP SuccessFactors instance refresh. For more information, see SAP Note 2954491.

- Database restore of the back-end target system (for example, Identity Authentication). After restoring a previously saved state of the database, some data or configurations might be out of sync. It's recommended that you reset the system before running a provisioning job.

In all other cases, reset is not required, nor recommended. If you have any concerns, report an incident on SAP Support Portal Home for the *BC-IAM-IPS* component.

If you want to reset your system, proceed as follows:

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems* or *Target Systems*
2. Select the relevant source or target system.

   > **Restriction**
   >
   > This *reset* operation is not applicable to proxy systems.

3. At the top right corner of your screen, choose the *Reset* button and confirm the action.

Next Steps

Regardless of the type of system you have reset - source or target ones, continue with the following steps:

1. Start a provisioning job. See: Start and Stop Provisioning Jobs [page 2472]
2. Set the `ips.delete.existedbefore.entities` to **true** on all affected target systems. This ensures that, if from now on you delete entities in the source system, those entities will be recognized as previously existed entities in the target systems and will be deleted there.
3. Start a provisioning job again.

> **Note**
>
> Following a reset, scheduled jobs preserve their defined time period.

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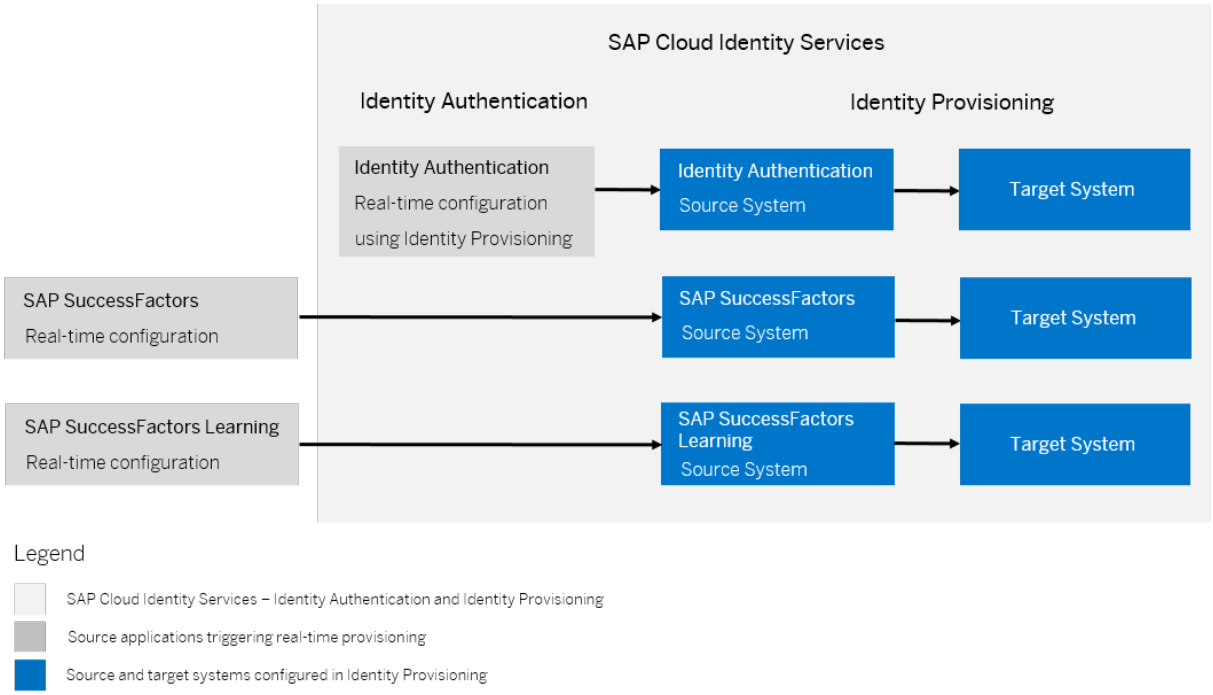
1.7.14 Configuring Real-Time Provisioning

As a tenant administrator, you can configure real-time provisioning to immediately provision entities from source to target systems.

This feature allows you to synchronize newly created, updated or deleted users and groups without the need to run manual or scheduled jobs in Identity Provisioning. It is useful in scenarios where synchronous provisioning is required, such as: user self-registration that needs immediate system access.

The real-time provisioning process involves the following set-ups. First, you need a source application which triggers the real-time sync when changes occur. Second, you must configure the source application as a source system in the Identity Provisioning. Finally, you need to add one or more target systems in the Identity Provisioning, where the real-time changes will be synchronized and reflected immediately.

The following diagram illustrates the process.



For more information on how to enable real-time sync in SAP SuccessFactors, see Managing Identity Authentication/Identity Provisioning Real Time Sync.

For more information on the real-time sync configuration in SAP SuccessFactors Learning, see Learning Configuration Procedure.

Changes related to users in Identity Authentication can also be provisioned real-time directly to SAP Jam without using Identity Provisioning. For more information, see Configure SAP Jam for Real-Time Provisioning [page 2492].

Standard vs Real-Time

Provisioning Mode	Use Case	Key Differences	Source Systems
Standard	Use it for initial, regular and scheduled *Read* and *Resync* provisioning jobs of users and groups from any supported source system to any target system.	• Starting a provisioning job in Identity Provisioning is required. • Users and groups (roles) are provisioned. • Filtering properties are considered.	**All source systems** (supported by Identity Provisioning)

Provisioning Mode	Use Case	Key Differences	Source Systems
Real-Time	Use it for instant provisioning of a single or a number of users or groups that were newly created or updated in the supported source system to any target system.	• You don't start a provisioning job. Changes in the user account (create, update, delete) trigger real-time sync. • Users and groups are provisioned real time. **Note** Real-time provisioning of groups can be configured for each source system that supports the execution of requests to the `/Groups` endpoint of the Real-time provisioning API. • Filtering properties are not considered.	• Identity Authentication • SAP SuccessFactors • SAP SuccessFactors Learning

Real-Time Workflow

Configuring real-time provisioning involves the following steps:

1. Create a technical user for accessing the real-time provisioning API and assign it the *Access Real-Time Provisioning API* permission. This user must be an administrator of type *System*.
   For more information, see Add System as Administrator [page 2305].
2. Configure the application which triggers real-time provisioning as a source system in Identity Provisioning, for example SAP SuccessFactors.
   For more information, see Source Systems [page 585].
3. Connect this source system to one or more target systems, for example Identity Authentication.
   For more information, see Target Systems [page 936] and Provision Users to Target Systems [page 2498].
4. Enable the real-time provisioning configuration in the source application that triggers it, for example SAP SuccessFactors.
   For more information, see Enable Real-Time Provisioning in Source Applications [page 2491].
5. Create or update a user in the source application. For example, create a user with *Onboardee* user account type in SAP SuccessFactors.
   As a result, the onboardee user will be immediately provisioned to Identity Authentication.
   When the source and target systems are configured for real-time provisioning scenarios, the following behavior can be expected:
   - **If a condition is applied in the source transformation:** Entities that do not meet the filtering criteria will either be deprovisioned or not updated in the target system.

- **If a condition is applied in the target transformation:** Entities that do not meet the filtering criteria will either be deprovisioned or skipped in the target system.

Enable Real-Time Provisioning in Source Applications [page 2491]
: Enable the real-time provisioning configuration in the source applications that trigger immediate sync when changes occur.

Configure SAP Jam for Real-Time Provisioning [page 2492]
: Tenant administrators can configure SAP Jam target systems for real-time provisioning via the administration console for SAP Cloud Identity Services.

Configure Identity Authentication for Real-Time Provisioning [page 2493]
: Enable real-time provisioning in Identity Authentication to trigger immediate synchronization of user changes to target systems configured in Identity Provisioning.

Provision Users to Target Systems [page 2498]
: Tenant administrators can provision users of Identity Authentication to SAP Jam and Identity Provisioning target systems.

Delete Target System [page 2503]
: As a tenant administrator, you can delete one or more target systems in a tenant of Identity Authentication.

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.14.1 Enable Real-Time Provisioning in Source Applications

Enable the real-time provisioning configuration in the source applications that trigger immediate sync when changes occur.

> ⓘ Note
>
> In the context of real-time provisioning, Identity Authentication, SAP SuccessFactors and SAP SuccessFactors Learning are considered source applications.

The way you enable real-time provisioning differs among applications. SAP SuccessFactors requires that you turn on the Enable Real Time Sync option in the Admin Center. Identity Authentication requires that you configure Identity Provisioning as a target system in the SAP Cloud Identity Services admin console.

Regardless of application-specific configurations, you are always required to provide the real-time provisioning endpoint URL, the authentication type and credentials.

Identity Authentication

- Configure Identity Authentication for Real-Time Provisioning [page 2493]

SAP SuccessFactors

- Manage Real-Time Sync of New Hires from SAP SuccessFactors to Identity Authentication with Identity Provisioning
- Managing Identity Authentication/Identity Provisioning Real Time Sync

SAP SuccessFactors Learning

- Learning Configuration Procedure

  > ⚠ Restriction
  >
  > Real-time provisioning of **groups** is not supported for this system.

Parent topic: Configuring Real-Time Provisioning [page 2487]

Related Information

Configure SAP Jam for Real-Time Provisioning [page 2492]
Configure Identity Authentication for Real-Time Provisioning [page 2493]
Provision Users to Target Systems [page 2498]
Delete Target System [page 2503]

1.7.14.2 Configure SAP Jam for Real-Time Provisioning

Tenant administrators can configure SAP Jam target systems for real-time provisioning via the administration console for SAP Cloud Identity Services.

Prerequisites

You are assigned the *Manage Users* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

> ⓘ Note
>
> Currently, Identity Authentication supports real-time provisioning to SAP Jam and SAP Cloud Identity Services - Identity Provisioning instances.

> → Remember
>
> Use the Identity Authentication real-time provisioning feature for up to 500 users.
>
> If you want to provision more than 500 user, use Identity Provisioning. For more information, see Configure Identity Authentication for Real-Time Provisioning [page 2493]. The content in this section is not relevant for China (Shanghai) region.

Context

To configure an SAP Jam target system, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Users & Authorizations*, choose the *Real-Time Provisioning* tile.

   This operation opens a list of the target systems.
3. Press the *+Add* button on the left-hand panel to add a new target system to the list.
4. Make the corresponding entries in the *Target Configurations* and *Authentication Configurations* fields for the target system you want to add.

   All fields are obligatory.

   > → Remember
   >
   > Choose *SAP Jam* from the drop-down.

5. Save your changes.

   If the operation is successful, the system displays the message `System <name of system> configuration updated.`
6. (Optional) To check the SAP Jam Target System configuration press the *Test Connection* button.

   If the operation is successful, the system displays the message `Connection to the selected target system was established successfully.`

   > **Note**
   >
   > To change the configuration, select the target system, press *Edit*, fill in the fields with the new entries, and save your changes.

Next Steps

Provision Users to Target Systems [page 2498]

Task overview: Configuring Real-Time Provisioning [page 2487]

Related Information

Enable Real-Time Provisioning in Source Applications [page 2491]
Configure Identity Authentication for Real-Time Provisioning [page 2493]
Provision Users to Target Systems [page 2498]
Delete Target System [page 2503]
Configuring Real-Time Provisioning [page 2487]

1.7.14.3 Configure Identity Authentication for Real-Time Provisioning

Enable real-time provisioning in Identity Authentication to trigger immediate synchronization of user changes to target systems configured in Identity Provisioning.

Prerequisites

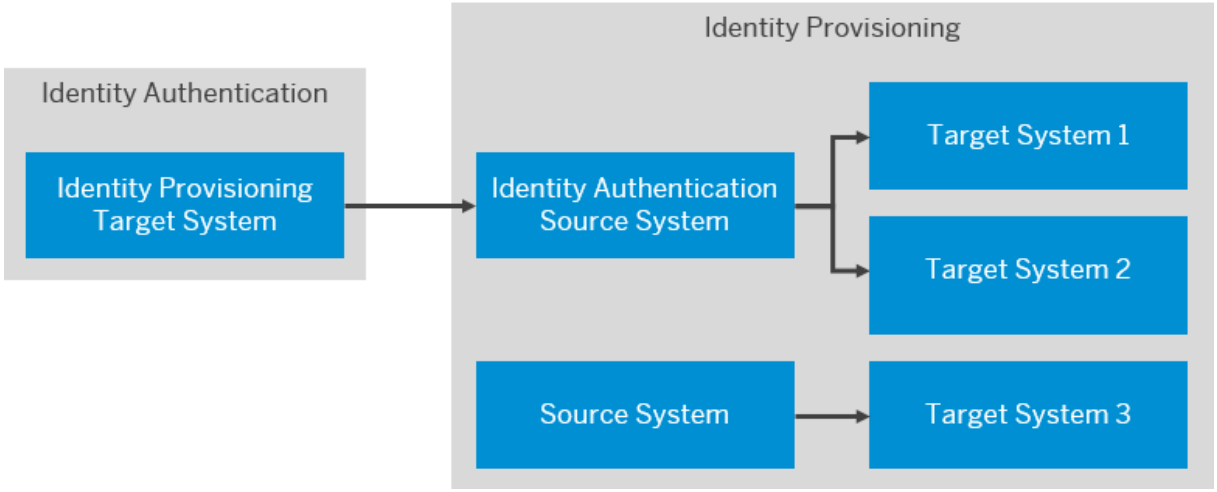
- You have created a technical user for accessing the real-time provisioning API and assigned it the *Access Real-Time Provisioning API* permission. This user must be an administrator of type *System*. You have also configured *Secret* for the technical user.
  For more information, see Add System as Administrator [page 2305].

For OAuth authentication scenario for tenants running on Neo environment, you've made the required configurations in the SAP BTP cockpit. For more information, see Create OAuth Client Credentials in SAP BTP Cockpit.

- You have created Identity Authentication source system in Identity Provisioning and connected it to target systems of your choice.
  For more information, see Identity Authentication [page 586] and Target Systems [page 936].

Context

In this scenario you configure Identity Provisioning as a target system in Identity Authentication real-time configuration, and Identity Authentication as a source system in Identity Provisioning. As a result, when users are created, updated or deleted in Identity Authentication, the changes will be immediately provisioned to the target systems configured in Identity Provisioning.



Real-time provisioning is applicable for users that have been manually created or updated in Identity Authentication, using SCIM API (version 1 or 2), or via CSV file upload. It is not applicable for users updated by a provisioning job.

> ⚠ Caution
>
> Users coming from a source (system A) which are created or updated in Identity Authentication target (system B) using a provisioning job, **cannot** be later synchronized from Identity Authentication to another target (system C) using real-time provisioning. This is a precaution behavior, preventing any data collisions if you later decide to run a provisioning job for the same systems: Identity Authentication source (system B) and target (system C). See the example below.

> ✤ Example
>
> 1. You enable real-time provisioning for Identity Authentication and configure it as a source system that points to SAP Marketing Cloud target system.
>   As a result, Identity Authentication users are immediately provisioned to SAP Marketing Cloud.
> 2. You configure Microsoft Active Directory as a source system that points to Identity Authentication target system and run a provisioning job.

As a result, Microsoft Active Directory users are provisioned to Identity Authentication. Even though new users are created in Identity Authentication, they are not provisioned further to SAP Marketing Cloud by real-time provisioning.

→ Tip

If you want to provision groups in real time, you need to use a REST client for initiating POST and DELETE requests to the Real-time provisioning API: `https://<tenantId>.<host>/ipsproxy/service/api/v1/systems/<system-id>/entities/group`.

To configure real-time provisioning in Identity Authentication to trigger immediate synchronization of user changes to target systems configured in Identity Provisioning follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Users & Authorizations*, choose the *Real-Time Provisioning* tile.

   This operation opens a list of the target systems.
3. Choose the *+Add* button to add a new target system and provide the following information:

New System	Field	Value
Target Configurations	*Display Name*	Provide a name for the target system. It can be identical with the source system you have created in Identity Provisioning.
	Type	From the dropdown, select *Identity Provisioning*.
	SCIM URL	Provide the SCIM URL in the following pattern: `https://<ias-tenant-host>/ipsproxy/service/api/v1/systems/<system-id>/entities/user` The `<system-id>` is the ID of the *Identity Authentication* source system you have added in Identity Provisioning. It is displayed at the end of the system URL.

New System	Field	Value
	Version • *1* • *2*	Defines the version of Identity Authentication SCIM API. *1* - the Identity Authentication SCIM API (in short, SCIM API version 1) is used. *2* - the Identity Directory SCIM API (in short, SCIM API version 2) is used. **Note** The version that you specify here must be the same as the version of the `ias.api.version` property in Identity Authentication source system.
Authentication Mechanism	*OAuth*	For tenants running on SAP Cloud Identity infrastructure, use the **user ID** and **password** of the Identity Authentication technical user. Provide an OAuth token URL following the pattern: `https://<ias-tenant-host>/oauth2/token` **Note** OAuth authentication is only supported when calling the Real-Time Provisioning API. It cannot be configured in the administration console of SAP Cloud Identity Services. For more information for tenants running on Neo environment, see Create OAuth Client Credentials in SAP BTP Cockpit.

New System	Field	Value
	Basic	If you choose this option, provide the following information: In the *Username* field, enter the client ID of the technical user. In the *Password* field, enter the client secret of the technical user. For more information, see *step 6* in Add System as Administrator.
	Certificate	If you choose this option, proceed as follows: 1. Provide a common name and a password. 2. Generate and download the certificate. 3. Import it in the *Configure System Authentication* screen of the technical user for real-time provisioning, as described in *step 6* in Add System as Administrator. The maximum length of the CN is 64 characters. Once the certificate is generated, it is saved as a `.crt` file. The common name is in the format `<common name>(<admin user ID>)`, where common name is the CN provided by the administrator, and admin user ID is the administrator's user id. Identity Authentication supports SAP Passport CA as trusted certificate authority (CA).

4. Save your changes.

   If the operation is successful, the system displays the message: `Target system <name of system> created`.

5. **Optional:** Choose *Test Connection* before executing the provisioning.

   If the setup is correct, the following message is displayed: `Connection to the selected target system was established successfully.`

6. Choose *Provision*.

Next Steps

Verify that the changes in Identity Authentication are provisioned to the target systems configured in Identity Provisioning. Each subsequently created or updated users will be provisioned instantly and automatically.

Task overview: Configuring Real-Time Provisioning [page 2487]

Related Information

Enable Real-Time Provisioning in Source Applications [page 2491]
Configure SAP Jam for Real-Time Provisioning [page 2492]
Provision Users to Target Systems [page 2498]
Delete Target System [page 2503]

1.7.14.4 Provision Users to Target Systems

Tenant administrators can provision users of Identity Authentication to SAP Jam and Identity Provisioning target systems.

Prerequisites

- You are assigned the *Manage Users* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have configured a target system in the administration console for SAP Cloud Identity Services. For more details about how to configure target systems, see Configure SAP Jam for Real-Time Provisioning [page 2492] or Configure Identity Authentication for Real-Time Provisioning [page 2493].

> ⓘ Note
>
> Currently, Identity Authentication supports real-time provisioning to instances of:
>
> - SAP Jam
>
>   > → Remember
>   >
>   > Use the Identity Authentication real-time provisioning feature for provisioning of up to 500 users. If you want to provision more than 500 user, use Identity Provisioning. For more information, see SAP Cloud Identity Services - Identity Provisioning
>
> - SAP Cloud Identity Services - Identity Provisioning

Context

Identity Authentication supports the following scenarios for real-time provisioning to target systems:

- Provision all users to selected target systems - In this scenario, all users in the user store of Identity Authentication are provisioned to a specific target system. The tenant administrator chooses the target system that all users will be provisioned to.
- Provision selected users to all target systems - In this scenario, the tenant administrator provisions particular users to all target systems that are configured. The administrator chooses which users to provision.
- Provision users from corporate user store - In this scenario, the users who are stored in a corporate user store are provisioned during authentication.
- Provision users at user creation and update - In this scenario, all newly created or updated the users are automatically provisioned to all target systems.

→ Remember

The users that are in the user store of Identity Authentication are provisioned to the SAP Jam target system with a certain set of attributes. The table below shows the attributes taken from Identity Authentication and their mapping to the SAP Jam target system.

Attribute Mapping Between Identity Authentication and SAP Jam

Identity Authentication Attribute	SAP Jam Attribute
Core Schema	
firstName	name.givenName
lastName	name.familyName
uid	userName
title	jobFunction
userType	type
mail	emails.value
status	active **Note** `active` is *true* only when `status` in Identity Authentication is equal to `active`. In the other case `active` is false.
telephone	phoneNumbers[work].value
street	addresses[home].streetAddress
city	addresses[home].locality

Identity Authentication Attribute	SAP Jam Attribute
zip	addresses[home].postalCode
country	addresses[home].country
companyStreet	addresses[work].streetAddress
companyCity	addresses[work].locality
companyZip	addresses[work].postalCode
companyCountry	addresses[work].country
locale	locale **Note** The locale must be of the format ll_CC where: • `ll` is the language two letter code in small letters • `CC` is the country (region) two letter code in capital letters **Caution** Do not send `locale` if `language` or `country` user attribute is missing.
Enterprise User Schema Extension	
employeeNumber	personnelNumber
costCenter	costCenter
organization	company
division	division
department	department

Note

If Identity Authentication is used as proxy to delegate the authentication to a corporate identity provider, the users that are authenticated by the corporate identity provider will not be provisioned during authentication.

When you delete a user, he or she is automatically deprovisioned from the configured target systems.

To provision users, choose one of the options below and follow the corresponding procedure.

Task overview: Configuring Real-Time Provisioning [page 2487]

Related Information

Enable Real-Time Provisioning in Source Applications [page 2491]
Configure SAP Jam for Real-Time Provisioning [page 2492]
Configure Identity Authentication for Real-Time Provisioning [page 2493]
Delete Target System [page 2503]

Provision All Users to a Selected Target System

Context

The tenant administrator can select the target systems that all users will be provisioned to.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Users & Authorizations*, choose the *Real-Time Provisioning* tile.

   This operation opens a list of the target systems.
3. Choose the list item of the target system that you want to provision the users to.

   > ⓘ Note
   >
   > If you do not have a configured target system in your list, you can add one. For more details, see Configure SAP Jam for Real-Time Provisioning [page 2492].

   > ⓘ Note
   >
   > The configuration of OAuth authentication mechanism via the administration console of SAP Cloud Identity Services is not supported.
4. Press *Provision*.

   If the operation is successful, the system displays the message `<number of users> provisioned`.

Provision Selected Users to All Target Systems

Context

The tenant administrator can choose which of the users to be provisioned to the configured target systems.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *User Management* tile.

   The system displays the first 20 users in the tenant sorted by their user ID number.
3. **Optional:** You can choose one of the following:

   Search Options

List All Users Press *More*	This will expand the list by 20 users. **Note** This option is available only if there are more than 20 users in the tenant.
Unfiltered Search • (default mode) Type your search criteria string in the search field and press the *Enter* key • (when in *Filtered Search* mode) Press *Hide Filters*, type your search criteria string in the search field, and press the *Enter* key	Once the search is completed, the system will list the users whose *User ID*, *Global User ID*, *SCIM ID*, *Email*, or *Login Name* match your search criteria string. In this case the system doesn't include the *First Name* and *Last Name* fields in the search. If you aren't satisfied with the search result, edit your search criteria and repeat the step again.
Filtered Search Press *Show Filters*, type your search criteria strings in at least one of the search fields, and press the *Enter* key	The system checks the search fields simultaneously. Once the search is completed, it will list the users that match the search criteria from all the fields.

4. Select the checkbox next to the user or users that you want to provision.
5. Press *Provision Users*.
6. Confirm the operation.

   If the operation is successful, the system displays the message `<number of users> provisioned`.

Provision Users from Corporate User Store

> **Note**
>
> The content in this section is only relevant for SAP BTP, Neo environment.

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

A user is automatically provisioned after he or she has been authenticated.

For more information about how to configure Identity Authentication to use a corporate user store in addition to its own user store, see Configure Connection to a Corporate User Store [page 2535].

Provision Users at User Create and User Update

All newly created and updated users are automatically provisioned to the target systems configured in the administration console for SAP Cloud Identity Services. The users that use the self-registration service will be automatically provisioned to the target systems too.

For more information about user creation or user update, see **Related Information**

Related Information

Create a New User [page 2357]
Add Administrators [page 2303]
Import or Update Users for a Specific Application [page 2348]
Scenarios [page 544]
Identity Directory API

1.7.14.5 Delete Target System

As a tenant administrator, you can delete one or more target systems in a tenant of Identity Authentication.

Prerequisites

You are assigned the *Manage Users* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Task overview: Configuring Real-Time Provisioning [page 2487]

Related Information

Enable Real-Time Provisioning in Source Applications [page 2491]
Configure SAP Jam for Real-Time Provisioning [page 2492]
Configure Identity Authentication for Real-Time Provisioning [page 2493]
Provision Users to Target Systems [page 2498]

Delete Multiple Target Systems

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Users & Authorizations*, choose the *Real-Time Provisioning* tile.

   This operation opens a list of the target systems.
3. Choose the icon in the left-hand panel.

   This operation activates the *Delete Target Systems* mode.
4. Select the target system or systems that you want to delete.
5. Choose the *Delete* button.
6. Confirm the operation in the pop-up dialog.

   Once the target system or systems have been deleted, the system displays the message `<number> target systems deleted`.

Delete Single Target System

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Users & Authorizations*, choose the *Real-Time Provisioning* tile.

   This operation opens a list of the target systems.
3. Select the target system that you want to delete.
4. Choose the *Delete* button in the right-hand panel to delete the selected target system.
5. Confirm the operation in the pop-up dialog.

   Once the application has been deleted, the system displays the message `1 target system deleted`.

Related Information

Configuring Real-Time Provisioning [page 2487]

1.7.15 Configuring Social Identity Providers

By configuring a social provider, users can log on to applications with their social media credentials by linking their accounts in Identity Authentication to the social media account.

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

Identity Authentication uses the OAuth protocol for social sign-on via one of the following predefined social providers:

- Apple ID
- Google
- Facebook
- LinkedIn
- X

Once a user has allowed Identity Authentication to link his or her account with the social provider accounts, the user can log on to applications via these social providers.

Configure Google as Identity Provider [page 2506]
: By configuring Google as a social identity provider, users can log on to applications with their Google credentials by liking their accounts in Identity Authentication to the Google account.

Configure Facebook as Identity Provider [page 2508]
: By configuring Facebook as a social identity provider, users can log on to applications with their social media credentials by liking their accounts in Identity Authentication to the social media account.

Configure LinkedIn as Identity Provider [page 2510]
: By configuring LinkedIn as social identity provider, users can log on to applications with their LinkedIn credentials by linking their accounts in Identity Authentication to the LinkedIn account.

Configure X as Identity Provider [page 2512]
: By configuring X as social provider, users can log on to applications with their X credentials by liking their accounts in Identity Authentication to the X account.

Configure Apple as Identity Provider [page 2514]
: Users can log on to applications with their Apple ID credentials by linking their accounts in Identity Authentication to their Apple account.

Remove Social Identity Providers Configuration [page 2517]
: You can remove the configurations of the social providers in the administration console for SAP Cloud Identity Services.

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.15.1 Configure Google as Identity Provider

By configuring Google as a social identity provider, users can log on to applications with their Google credentials by liking their accounts in Identity Authentication to the Google account.

Context

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

Identity Authentication uses the OAuth protocol for social sign-on via Google as a social identity provider.

Once a user has allowed Identity Authentication to link his or her account with the Google account, the user can log on to applications via the social provider.

To configure Google as social identity provider for the tenant, you have to register new applications on the Google site. For more details, see Verify your site ownership .

> **Note**
>
> You need to type **`https://<tenant ID>.accounts.ondemand.com/ui/oauth/googleCallback`** in the *Authorized redirect URIs* field when you create your client ID in Google Developers Console. For more information about the redirect URIs for your OAuth 2.0 credentials, see Set a redirect URI .

To perform the social identity provider configuration in the administration console for SAP Cloud Identity Services, you have to provide the following data:

Required Google Settings

Authentication Attribute	Description
Google's client ID	The Google OAuth 2.0 credential after you set a project in the Google Developers Console.
Google's client secret	The Google OAuth 2.0 credential after you set a project in the Google Developers Console.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Social Identity Providers* tile.

   This operation opens a list of the social providers.
3. Choose the Google list item.
4. Under *Configuration*, choose *Edit*.
5. Enter the generated authentication attributes from the social provider.

Option	Description
Client ID	The Google OAuth 2.0 credential after you set a project in the Google Developers Console.
Client Secret	The Google OAuth 2.0 credential after you set a project in the Google Developers Console.

   > ### Note
   >
   > Check for leading or trailing spaces in the authentication attributes fields, and delete them. Sign-on through the social identity provider will not work if there are blank spaces before or after the strings in the fields.

6. Save your configuration.

   If the operation is successful, you will receive the following message: `Google authentication configured`. The slider next to the social provider is switched to *ON*.

   > ### Note
   >
   > With the slider *ON* the social identity provider is shown as authentication option on the logon screen. If you do not want to show any of the social providers on the logon screen, you can drag the slider next to the social provider to *OFF*. The configuration for this social provider will be preserved, but the social provider will not appear on the logon pages of the applications in the tenant.
   >
   > To disable the social sign option for a specific social provider, remove the configuration for that provider. See Related Information for more details.

Next Steps

The above configurations are valid for the whole tenant. They will take effect for a specific application if you enable the *Social Sign-On* option via the administration console. For more information about how to enable social sign on for a specific application, see Enable or Disable Social Sign-On for an Application [page 2020].

Task overview: Configuring Social Identity Providers [page 2505]

Related Information

Configure Facebook as Identity Provider [page 2508]
Configure LinkedIn as Identity Provider [page 2510]
Configure X as Identity Provider [page 2512]
Configure Apple as Identity Provider [page 2514]
Remove Social Identity Providers Configuration [page 2517]
Obtain OAuth 2.0 credentials
Social Authentication [page 2771]

1.7.15.2 Configure Facebook as Identity Provider

By configuring Facebook as a social identity provider, users can log on to applications with their social media credentials by liking their accounts in Identity Authentication to the social media account.

Context

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

Identity Authentication uses the OAuth protocol for social sign-on via Facebook as a social identity provider.

Once a user has allowed Identity Authentication to link his or her account with the Facebook account, the user can log on to applications via the social provider.

To configure Facebook as social identity provider for the tenant, you have to register new applications on the Facebook site. For more details, see Related Information.

> **Note**
>
> You need to type **`https://<tenant ID>.accounts.ondemand.com/ui/oauth/oauthCallback`** in the *Valid OAuth redirect URIs* field in the Settings section of Facebook Login in the App Dashboard. For more information about Facebook Valid OAuth redirect URIs, see Enable Strict Mode .

To perform the social identity provider configuration in the administration console for SAP Cloud Identity Services, you have to provide the following data:

Required Facebook Settings

Authentication Attribute	Description
Facebook's application ID	The ID generated after you add an application on Facebook.
Facebook's application secret	The secret generated after you add an application on Facebook.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Social Identity Providers* tile.

   This operation opens a list of the social providers.
3. Choose the Facebook list item.
4. Under *Configuration*, choose *Edit*.
5. Enter the generated authentication attributes from the social provider.

   > ### Note
   >
   > Check for leading or trailing spaces in the authentication attributes fields, and delete them. Sign-on through the social identity provider will not work if there are blank spaces before or after the strings in the fields.
6. Save your configuration.

   If the operation is successful, you will receive the following message: `Facebook authentication configured`. The slider next to the social provider is switched to *ON*.

   > ### Note
   >
   > With the slider *ON* the social identity provider is shown as authentication option on the logon screen. If you do not want to show any of the social providers on the logon screen, you can drag the slider next to the social provider to *OFF*. The configuration for this social provider will be preserved, but the social provider will not appear on the logon pages of the applications in the tenant.
   >
   > To disable the social sign option for a specific social provider, remove the configuration for that provider. See Related Information for more details.

Next Steps

The above configurations are valid for the whole tenant. They will take effect for a specific application if you enable the *Social Sign-On* option via the administration console. For more information about how to enable social sign on for a specific application, see Enable or Disable Social Sign-On for an Application [page 2020].

Task overview: Configuring Social Identity Providers [page 2505]

Related Information

Configure Google as Identity Provider [page 2506]
Configure LinkedIn as Identity Provider [page 2510]
Configure X as Identity Provider [page 2512]
Configure Apple as Identity Provider [page 2514]
Remove Social Identity Providers Configuration [page 2517]
Facebook Login
Enable or Disable Social Sign-On for an Application [page 2020]
Remove Social Identity Providers Configuration [page 2517]
Social Authentication [page 2771]

1.7.15.3 Configure LinkedIn as Identity Provider

By configuring LinkedIn as social identity provider, users can log on to applications with their LinkedIn credentials by linking their accounts in Identity Authentication to the LinkedIn account.

Context

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

Identity Authentication uses the OAuth protocol for social sign-on via LinkedIn as a social identity provider.

Once a user has allowed Identity Authentication to link his or her account with the LinkedIn account, the user can log on to applications via the social provider.

To configure LinkedIn as social identity provider for the tenant, you have to register new applications on the LinkedIn site.

> **Note**
>
> You need to type **`https://<tenant ID>.accounts.ondemand.com/ui/oauth/oauthCallback`** in the *Redirect URLs* field in the *Auth* tab in the LinkedIn Developer Platform. For more information about the redirect URLs for your OAuth 2.0 credentials, see Step 1: Configure Your Application .

To perform the social identity provider configuration in the administration console for SAP Cloud Identity Services, you have to provide the following data:

Required LinkedIn Settings

Authentication Attribute	Description
LinkedIn's Client ID (API key)	The key generated by LinkedIn to identify a user application and for API calls.
LinkedIn's Client Secret (API secret)	The secret generated by LinkedIn to identify a user application and for API calls.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Social Identity Providers* tile.

   This operation opens a list of the social providers.
3. Choose the LinkedIn list item.
4. Under *Configuration*, choose *Edit*.
5. Enter the generated authentication attributes from the social provider.

   > ### Note
   >
   > Check for leading or trailing spaces in the authentication attributes fields, and delete them. Sign-on through the social identity provider will not work if there are blank spaces before or after the strings in the fields.
6. Save your configuration.

   If the operation is successful, you will receive the following message: `LinkedIn authentication configured`. The slider next to the social provider is switched to *ON*.

   > ### Note
   >
   > With the slider *ON* the social identity provider is shown as authentication option on the logon screen. If you do not want to show any of the social providers on the logon screen, you can drag the slider next to the social provider to *OFF*. The configuration for this social provider will be preserved, but the social provider will not appear on the logon pages of the applications in the tenant.
   >
   > To disable the social sign option for a specific social provider, remove the configuration for that provider. See Related Information for more details.

Next Steps

The above configurations are valid for the whole tenant. They will take effect for a specific application if you enable the *Social Sign-On* option via the administration console. For more information about how to enable social sign on for a specific application, see Enable or Disable Social Sign-On for an Application [page 2020].

Task overview: Configuring Social Identity Providers [page 2505]

Related Information

Configure Google as Identity Provider [page 2506]
Configure Facebook as Identity Provider [page 2508]
Configure X as Identity Provider [page 2512]
Configure Apple as Identity Provider [page 2514]
Remove Social Identity Providers Configuration [page 2517]
Authentication (Linkedin)
Enable or Disable Social Sign-On for an Application [page 2020]
Remove Social Identity Providers Configuration [page 2517]
Social Authentication [page 2771]

1.7.15.4 Configure X as Identity Provider

By configuring X as social provider, users can log on to applications with their X credentials by liking their accounts in Identity Authentication to the X account.

Context

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

Identity Authentication uses the OAuth protocol for social sign-on via X as a social identity provider.

Once a user has allowed Identity Authentication to link his or her account with the X account, the user can log on to applications via the social provider.

To configure X as social identity provider for the tenant, you have to register new applications on the X site. For more details, see Create a Project and connect an App .

> **Note**
>
> You need to type **`https://<tenant ID>.accounts.ondemand.com/ui/oauth/oauthCallback`** in the *Callback URLs* field in the Create an application section of X Application Management page. For more information about X Callback URL, see Callback URL .

To perform the social identity provider configuration in the administration console for SAP Cloud Identity Services, you have to provide the following data:

Required X Settings

Authentication Attribute	Description
X consumer key	The key generated by X to identify which user application is making the request.
X consumer secret	The secret generated by X to identify which user application is making the request.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Social Identity Providers* tile.

   This operation opens a list of the social providers.
3. Choose the X list item.
4. Under *Configuration*, choose *Edit*.
5. Enter the generated authentication attributes from the social provider.

   > **Note**
   >
   > Check for leading or trailing spaces in the authentication attributes fields, and delete them. Sign-on through the social identity provider will not work if there are blank spaces before or after the strings in the fields.
6. Save your configuration.

   If the operation is successful, you will receive the following message: `X authentication configured`. The slider next to the social provider is switched to *ON*.

   > **Note**
   >
   > With the slider *ON* the social identity provider is shown as authentication option on the logon screen. If you do not want to show any of the social providers on the logon screen, you can drag the slider next to the social provider to *OFF*. The configuration for this social provider will be preserved, but the social provider will not appear on the logon pages of the applications in the tenant.
   >
   > To disable the social sign option for a specific social provider, remove the configuration for that provider. See Related Information for more details.

Next Steps

The above configurations are valid for the whole tenant. They will take effect for a specific application if you enable the *Social Sign-On* option via the administration console. For more information about how to enable social sign on for a specific application, see Enable or Disable Social Sign-On for an Application [page 2020].

Task overview: Configuring Social Identity Providers [page 2505]

Related Information

Configure Google as Identity Provider [page 2506]
Configure Facebook as Identity Provider [page 2508]
Configure LinkedIn as Identity Provider [page 2510]
Configure Apple as Identity Provider [page 2514]
Remove Social Identity Providers Configuration [page 2517]
Authorizing a request
Enable or Disable Social Sign-On for an Application [page 2020]
Remove Social Identity Providers Configuration [page 2517]
Social Authentication [page 2771]

1.7.15.5 Configure Apple as Identity Provider

Users can log on to applications with their Apple ID credentials by linking their accounts in Identity Authentication to their Apple account.

Prerequisites

You have a developer account at Apple.

Context

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

Identity Authentication uses the OAuth protocol for social sign-on via Apple as a social identity provider.

Once a user has allowed Identity Authentication to link his or her account with the Apple ID account, the user can log on to applications via the social provider.

To configure sign in with Apple for the tenant, first you have to register an Application ID (App ID) and Services ID. For more details, see Configure Sign in with Apple for the web .

> **Note**
>
> You must type **`https://<tenant ID>.accounts.ondemand.com`** in the *Domains and Subdomains* box, and **`https://<tenant ID>.accounts.ondemand.com/ui/oauth/oauthCallback`** in the *Return URLs* box when you enable and configure the *Sign in with Apple* option at the Apple Developer site.

To perform the social identity provider configuration in the administration console for SAP Cloud Identity Services, you have to provide the following data:

Required Apple Settings

Authentication Attribute	Description
Client ID	The *Services ID* that you registered at Apple site.
Team ID	A 10-digit key generated by Apple. You can see your Team ID under the membership information at Apple site.
Key ID	For the Key ID, you must register a key at the apple site. → Remember You must provide a name for your key and select the *Sign in with Apple* check box. Once you configure your key, it needs to be bound to your primary App ID.
Client Certificate	The private key provided by Apple in a `.p8` file. To use the key for the configuration in the administration console for SAP Cloud Identity Services, copy and paste the text from the `.p8` file.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Social Identity Providers* tile.

   This operation opens a list of social providers.
3. Choose the *Apple* list item.
4. Under *Configuration*, choose *Edit*.
5. Enter the generated authentication attributes from Apple.

   > **Note**
   >
   > Check for leading or trailing spaces in the authentication attributes fields, and delete them. Sign-on through the social identity provider will work if there are blank spaces before or after the strings in the fields.

6. Save your configuration.

   If the operation is successful, you receive the following message: `Apple authentication configured`. The slider next to the social provider is switched to *ON*.

   > **Note**
   >
   > With the slider *ON* the social identity provider is shown as authentication option on the logon screen. If you do not want to show any of the social providers on the logon screen, you can drag the slider next to the social provider to *OFF*. The configuration for this social provider will be preserved, but the social provider will not appear on the logon pages of the applications in the tenant.
   >
   > To disable the social sign option for a specific social provider, remove the configuration for that provider. See Related Information for more details.

Next Steps

The above configurations are valid for the whole tenant. They will take effect for a specific application if you enable the *Social Sign-On* option via the administration console. For more information about how to enable social sign-on for a specific application, see Enable or Disable Social Sign-On for an Application [page 2020].

Task overview: Configuring Social Identity Providers [page 2505]

Related Information

Configure Google as Identity Provider [page 2506]
Configure Facebook as Identity Provider [page 2508]
Configure LinkedIn as Identity Provider [page 2510]
Configure X as Identity Provider [page 2512]
Remove Social Identity Providers Configuration [page 2517]
Enable or Disable Social Sign-On for an Application [page 2020]
Remove Social Identity Providers Configuration [page 2517]
Social Authentication [page 2771]

1.7.15.6 Remove Social Identity Providers Configuration

You can remove the configurations of the social providers in the administration console for SAP Cloud Identity Services.

Prerequisites

You have a configured social provider in the administration console.

Context

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Social Identity Providers* tile.

   This operation opens a list of the social providers.
3. Choose the list item of the social provider whose configuration you want to remove.
4. Choose *<Social Provider Name> Sign-On*.
5. Choose the *Remove Configuration* button.
6. Confirm your choice.

Results

The configuration for the selected social provider will be removed. The social provider will not appear on the logon pages of the applications in the tenant.

Task overview: Configuring Social Identity Providers [page 2505]

Related Information

Configure Google as Identity Provider [page 2506]

Configure Facebook as Identity Provider [page 2508]
Configure LinkedIn as Identity Provider [page 2510]
Configure X as Identity Provider [page 2512]
Configure Apple as Identity Provider [page 2514]

1.7.16 Integrating with Existing Customer Landscape

Identity Authentication can be integrated with already existing customer landscape and supports different types of delegated authentication.

The table shows the available options and what are the suitable use-cases:

Use Case	Configuration
Corporate User Store	• Corporate User Store (Cloud Foundry Environment) [page 2519] • Corporate User Store (Neo Environment) [page 2531]
Kerberos Authentication	• Configure Kerberos Authentication [page 2548] • Enable or Disable Kerberos Authentication for an Application [page 2556]
Identity Provider Proxy	Corporate Identity Providers [page 2556]

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]

Configuring External Authentication Providers [page 2620]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.16.1 Corporate User Store

If you have an existing on-premise user store, you can configure Identity Authentication to use the corporate user store in addition to its own cloud user store.

The integration between Identity Authentication and the corporate user store allows users to authenticate with their corporate credentials from the corporate user store, without the need to use another set of credentials for their cloud access.

You can have two configurations, depending on the SAP BTP environment, you are using:

- Corporate User Store (Cloud Foundry Environment) [page 2519]
  Recommended.
- Corporate User Store (Neo Environment) [page 2531]
  Recommended if you are using SAP NetWeaver AS JAVA, with the following variants:
  - SAP NetWeaver AS JAVA - UME
  - SAP NetWeaver AS JAVA - UME
  - Multiple Active Directories connected to SAP NetWeaver AS JAVA - UME
  - SAP NetWeaver AS ABAP connected to SAP NetWeaver AS JAVA - UME

1.7.16.1.1 Corporate User Store (Cloud Foundry Environment)

Configure corporate user store for applications in the Cloud Foundry environment to allow users to authenticate with their corporate credentials, without the need to use another set of credentials for their cloud access.

> ⓘ Note
>
> The content in this section is only relevant for SAP BTP, Cloud Foundry environment.

> → Tip
>
> The Cloud Foundry setup can be used also for applications that are in the Neo environment.

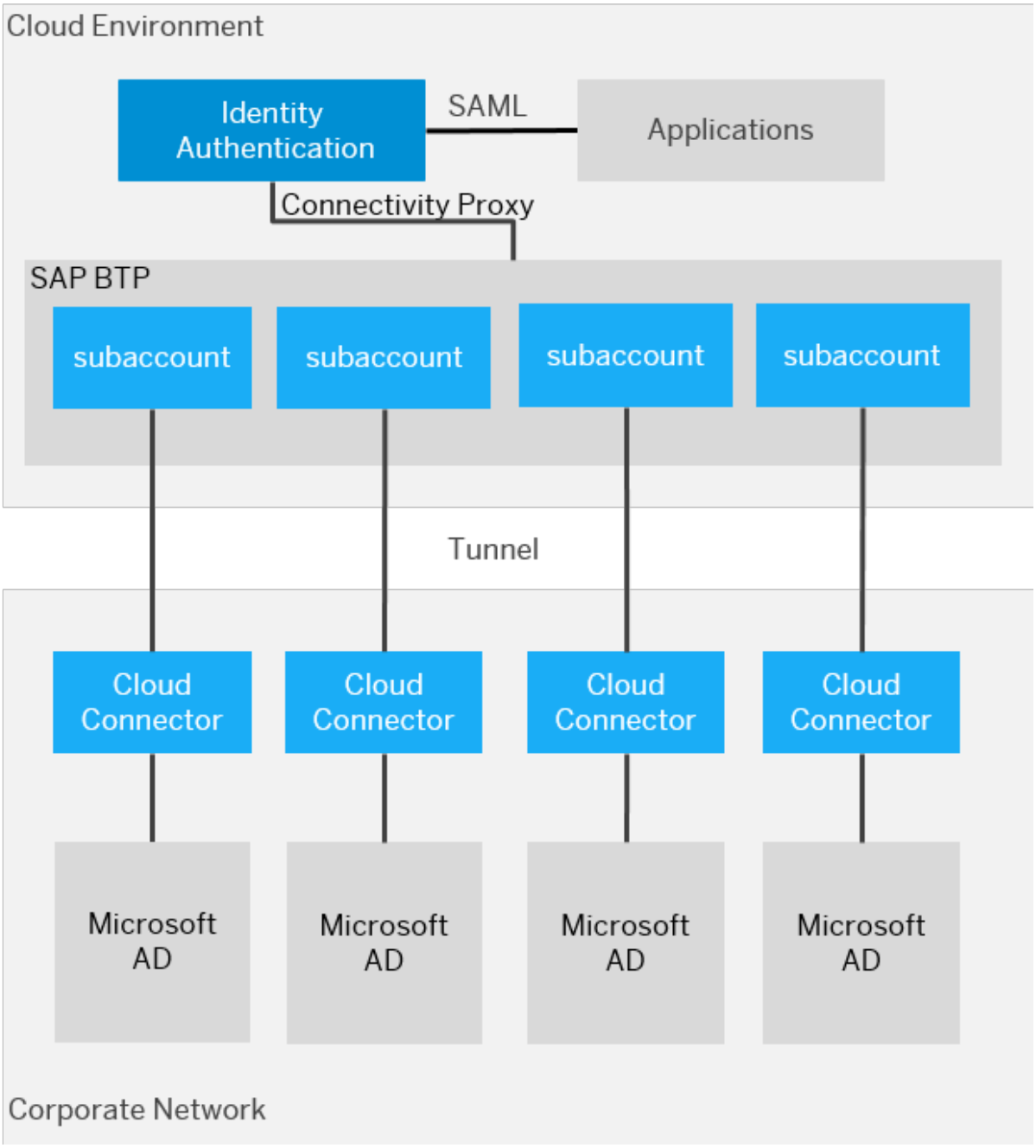
Overview

Identity Authentication can connect with the following corporate user stores:

- Microsoft Active Directory

This scenario works with the connectivity proxy component (logically part of the Connectivity service). For more information about the connectivity proxy, see How the Connectivity Proxy Works

The connection between SAP BTP and the corporate user store is carried out with a Cloud Connector.



Authentication Flow

The *Corporate User Store* option is configured properly. A user tries to access a trusted application for the first time with the on-premise credentials, *Login Name* and *Password* entered correctly. This user is authenticated successfully against the corporate user store. With this initial successful authentication of the user, a partial user record is created in the user store for Identity Authentication. It is created with user details taken from the corporate user store. The cloud user store doesn't copy the user's credentials. For more details about what

data is copied from the corporate user store, see User Records [page 2535]. With subsequent logins, the user is always authenticated against the corporate user store, and the user record is updated.

For the first logon with on-premise credentials, the user enters his or her *Login Name* and a *Password*. For subsequent logins the user can use either his or her *Login Name*, *Email*, or *User ID*, and the *Password*.

> → Remember
>
> If the user has enabled the *Remember me* option, the following limitations are observed:
>
> - when a user is authenticating with remember me cookie, his or her user record in Identity Authentication is not updated.
> - if the user changes his or her password in the corporate network, the remember me cookie is not invalidated.
>
> For more information, see Use the Remember Me Option [page 2774].

> ⓘ Note
>
> The user in the corporate user store must have the `mail` attribute.
>
> The tenant administrator needs to monitor and prevent the coexistence of a cloud and on-premise user with one and the same email address. The tenant administrator has to instruct the users to logon for the first time with their *Login Name*, not with the *Email*.

If a user with a user record in the cloud user store is deleted in the corporate user store, the user can't authenticate using Identity Authentication. The user record for this user remains in the cloud user store, and the tenant administrator can delete it via the administration console for SAP Cloud Identity Services. For more information, see Delete Users [page 2367].

For all users from the corporate user store, a second factor for authentication can be enabled for some applications, or cloud user groups can be assigned. For more details, see Configure Risk-Based Authentication for an Application [page 2007] and Assign Groups to a User [page 2409].

Control Access to Applications and Resources

In the scope of the *Corporate User Store* scenario, you can manage access to applications and their resources based on the groups available in the corporate user store.

The corporate user groups are sent to an application in the SAML 2.0 assertion. `corporate_groups` is the attribute that contains the groups that the user in the corporate user store is assigned to. For more details about how the groups are sent to the application in the SAML 2.0 assertion, see Configuring User Attributes from the Identity Directory [page 1973].

> ⓘ Note
>
> If your application is deployed on the SAP BTP, the corporate user store groups, relevant for the application, and contained in the `corporate_groups` attribute in the SAML 2.0 assertion, can be mapped to assertion-based groups created in SAP BTP cockpit. For more information, see Map Role Collections to User Attributes.

You can also restrict access to applications based on membership in a corporate user group by setting different rules via risk-based authentication. For more information, see Configure Risk-Based Authentication for an Application [page 2007].

1.7.16.1.1.1 User Records

When a user has been successfully authenticated for the first time with the credentials from the corporate user store, a record for that user is created in Identity Authentication. That user record is created with details from the corporate user store. In this record, the user is created with a *User Type*: `employee`, `Password Authentication Provider Type (39)`, and `Password Authentication Provider ID`.

This *User Type* cannot be changed.

For more information about the attributes taken from the Active Directory and their mapping to the user store of Identity Authentication, see **Configure SAP BTP When Connecting to an LDAP User Store** in Configure SAP BTP [page 2522].

Related Information

Security Information [page 2897]

1.7.16.1.1.2 Configure Connection to a Corporate User Store

To configure connection to a corporate user store, you have to make the following configurations in SAP BTP and in Identity Authentication.

For more details about how to configure these systems, see:

- Configure SAP BTP [page 2522]
- Configure Identity Authentication [page 2530]

1.7.16.1.1.3 Configure SAP BTP

Context

The configuration of SAP BTP depends on the type of the user store. You have the option to connect to a Microsoft Active Directory user store.

Data Center Mapping

Identity Authentication Data Center	Cloud Foundry Data Center
eu-nl - Amsterdam	cf-eu10
ap-cn - Shanghai	cf-ap11 - Singapore
na-us - Sterling/Toronto	cf-us10 - US East (VA)
ap-jp - Tokyo/Osaka	cf-jp10 - Japan (Tokyo)
ap-au Sydney	cf-ap10 - Australia (Sydney)
la-br - São Paulo	cf-br10 - Brazil (São Paulo)
aws-ap-se-1 Singapore	cf-ap11 - Singapore
azr-us-we - US West	cf-us20 - US West (WA)
azr-na-ca - Toronto	cf-ca10 - Canada (Montreal)
aws-fr - Frankfurt	cf-eu10 - Europe (Frankfurt)
aws-ap-northeast-2 - Seoul	cf-ap12 - South Korea (Seoul)
ap-sa - Riyadh	cf-eu10 - Europe (Frankfurt)
ap-ae - Dubai	cf-eu10 - Europe (Frankfurt)
ap-in - India (Mumbai)	cf-in30 - India (Mumbai)

Configure SAP BTP When Connecting to a Microsoft Active Directory User Store

Procedure

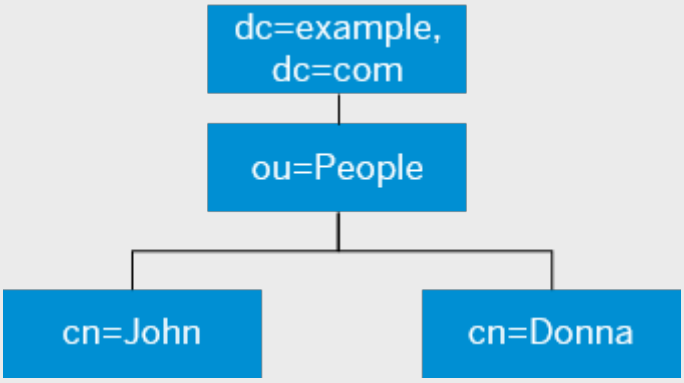
1. Sign in SAP BTP cockpit with the cockpit administrator role. For more information, see Subaccounts.
2. In the SAP BTP cockpit, choose in the navigation area *Services* > *Instances and Subscriptions*.
3. Choose *Create* and select the following info from the dropdowns:
   - *Service* - **`Cloud Identity Services`**
   - *Plan* - **`Connectivity`**

   > **Note**
   >
   > If the *Connectivity* plan is not available check the entitlements configurations of your subaccount. For more information, see Managing Entitlements and Quotas Using the Cockpit.

   The Cloud Identity Services appears under the *Subscriptions* tab.
4. From the *Cloud Service Type* dropdown, choose for what type of tenant (test or productive) you want to use the subaccount.
   - *Test*
   - *Productive* - default value
5. Install an Cloud Connector in your corporate network.

For more information, see Installation.

6. Connect the Cloud Connector between your corporate user store and SAP BTP account.
   - If you haven't used your Cloud Connector before, see Initial Configuration.
   - If you have used your Cloud Connector before, you can start the configuration from Set up Connection Parameters and HTTPS proxy.
7. Connect SAP BTP with your corporate user store.

> ### Note
>
> You have to specify the SAP BTP settings. For more information, see Configure the On-Premise User Store, without the *Prerequisites* section.
>
> The *User Name* field must be in the **`<service_user_name>@<domain>`** format.
>
> For the *User Path* and *Group Path* fields, specify the Microsoft Active Directory tree that contains the users and groups, respectively. For example, if the tree has the following structure:
>
> 
>
>
> The user and group paths should appear as in the table below:
>
> | | |
> |---|---|
> | User Path | ou=People,dc=example,dc=com |
> | Group Path | ou=People,dc=example,dc=com |

8. **Optional:** Include additional attributes.

> ### Note
>
> You can add the `employeeNumber`, `division`, `department`, and `organization` attributes that are defined in the SCIM Enterprise User Schema Extension.
>
> Cloud Connector uses the SCIM protocol to transfer the data, so the Active Directory attributes are mapped first to the SCIM attributes. When the data is provisioned, the SCIM attributes are mapped to the user store attributes of Identity Authentication.

   a. In your system go to `/sapcc-<version>/config_master/com.sap.core.connectivity.protocol.scim/`
   b. On that level, create a new *idstorage.cfg* file based on the *idstorage_extended_schema.cfg* file which is given as an example in the folder.
   c. Edit the newly created file. For more details, see the information below:

To add new user attributes you have to edit the whole file.

> ⚠ Caution
>
> This file overwrites the configurations you made in Configure the User Store Be careful not to change the user attributes taken from Microsoft Active Directory.

In this section, provide the same information as when you specified the SAP BTP settings in the previous step.

```
 {
 "LDAPServers": [
    {
      "Host": "<The host name of the LDAP server to be contacted>",
      "Port": "<The port where the LDAP service is running. If omitted
then the default LDAP port will be used - 389 for plain connections and
636 for SSL connections>"
    }
 ],
 "UserPath": "<LDAP subtree containing the users. Example
"DC=users,DC=organisation,DC=location">",
 "GroupPath": "<LDAP subtree containing the groups. Example
"DC=groups,DC=organisation,DC=location">",
 "ServiceUser": {
    "Name": "<The name of the user that will be used to establish
communication with the LDAP. In case of Active Directory the user name
should contains Domain suffix, e.g. "john@ACME.COM">",
    "Password": "<Password of this user>"
 },
```

If you want to use SSL, we recommend that you configure this section.

```
 "UseSSL": "<Possible values are "true" or "false". If true then the
communication to LDAP will go over SSL>",
 "IdentityKeystorePath": "<File system path to the client identity
keystore - must be set if the used LDAP server requires client certificate
authentication>",
 "IdentityKeystorePassword": "<The password of the client identity
keystore>",
 "TrustKeystorePath": "<File system path to the trusted CAs keystore -
must be set if UseSSL is true>",
 "TrustKeystorePassword": "<The password of the trusted CAs keystore>",
 "IsActiveDirectory": "<Possible values are "true" (default value if
missing) or "false". "true" indicates that the LDAP server is Active
Directory>",
 "ExcludeUsersAttribute": {
    "AttributeName": "<Name of user attribute that will be used to exclude
some users from the result depending on their type. Attribute is treated
as bitwise. Such attribute for Active Directory is "UserAccountControl">",
    "AttributeMask": "<Bitwise mask represented as decimal value. In case
any of the high bits of this mask match with the corresponding bit of
the value of the above attribute, the user will be excluded from the
result. Example mask for Active Directory is "67121154" - it is the sum
of the following flags ACCOUNTDISABLE(2), WORKSTATION_TRUST_ACCOUNT(4096),
SERVER_TRUST_ACCOUNT(8192) and PARTIAL_SECRETS_ACCOUNT(67108864)>"
 },
```

In this section add the additional attributes `employeeNumber`, `division`, `department`, and `organization`, defined in the SCIM Enterprise User Schema Extension.

```
 {
  "SingularAttributes": [
      {
```

```
            "SCIMAttribute": "userName",
            "mappings": [
                {
                    "LDAPAttribute": {
                        "name": "sAMAccountname"
                    }
                }
            ]
        },
            "SCIMAttribute": "name",
            "mappings": [
                {
                    "SCIMSubAttribute": "givenName",
                    "LDAPAttribute": {
                        "name": "givenName"
                    }
                },
                {
                    "SCIMSubAttribute": "familyName",
                    "LDAPAttribute": {
                        "name": "sn"
                    }
                },
                {
                    "SCIMSubAttribute": "honorificPrefix",
                    "LDAPAttribute": {
                        "name": "personalTitle"
                    }
                }
            ]
        },
        {
            "SCIMAttribute": "displayName",
            "mappings": [
                {
                    "LDAPAttribute": {
                        "name": "displayName"
                    }
                }
            ]
        },
        {
            "SCIMAttribute": "locale",
            "mappings": [
                {
                    "LDAPAttribute": {
                        "name": "locale"
                    }
                }
            ]
        },
        {
            "SCIMAttribute": "timeZone",
            "mappings": [
                {
                    "LDAPAttribute": {
                        "name": "timezone"
                    }
                }
            ]
        },
        {
            "SCIMAttribute": "employeeNumber",
            "mappings": [
                {
                    "LDAPAttribute": {
                        "name": "<LDAP property containing the employee
number"
```

```
                }
            }
        ]
    },
    {
        "SCIMAttribute": "division",
        "mappings": [
            {
                "LDAPAttribute": {
                    "name": "<LDAP property containing the division>"
                }
            }
        ]
    },
    {
        "SCIMAttribute": "department",
        "mappings": [
            {
                "LDAPAttribute": {
                    "name": "<LDAP property containing the department>"
                }
            }
        ]
    },
    {
        "SCIMAttribute": "organization",
        "mappings": [
            {
                "LDAPAttribute": {
                    "name": "company"
                }
            }
        ]
    },
   {
        "SCIMAttribute": "costCenter",
        "mappings": [
            {
                "LDAPAttribute": {
                    "name": "<LDAP property containing the cost center>"
                }
            }
        ]
    }
],
"MultiValuedAttributes": [
    {
        "SCIMAttribute": "emails",
        "values": [
            {
                "primary": "true",
                "mappings": [
                    {
                        "SCIMSubAttribute": "value",
                        "LDAPAttribute": {
                            "name": "mail"
                        }
                    }
                ]
            }
        ]
    },
    {
        "SCIMAttribute": "phoneNumbers",
        "values": [
            {
                "type": "work",
                "primary": "true",
```

```
                "mappings": [
                    {
                        "SCIMSubAttribute": "value",
                        "LDAPAttribute": {
                            "name": "telephoneNumber"
                        }
                    }
                ]
            },
            {
                "type": "fax",
                "mappings": [
                    {
                        "SCIMSubAttribute": "value",
                        "LDAPAttribute": {
                            "name": "facsimileTelephoneNumber"
                        }
                    }
                ]
            },
            {
                "type": "cell",
                "mappings": [
                    {
                        "SCIMSubAttribute": "value",
                        "LDAPAttribute": {
                            "name": "mobile"
                        }
                    }
                ]
            }
        ]
    },
    {
        "SCIMAttribute": "addresses",
        "values": [
            {
                "primary": "true",
                "mappings": [
                    {
                        "SCIMSubAttribute": "streetAddress",
                        "LDAPAttribute": {
                            "name": "streetAddress"
                        }
                    },
                    {
                        "SCIMSubAttribute": "locality",
                        "LDAPAttribute": {
                            "name": "l"
                        }
                    },
                    {
                        "SCIMSubAttribute": "region",
                        "LDAPAttribute": {
                            "name": "st"
                        }
                    },
                    {
                        "SCIMSubAttribute": "postalCode",
                        "LDAPAttribute": {
                            "name": "postalCode"
                        }
                    },
                    {
                        "SCIMSubAttribute": "country",
                        "LDAPAttribute": {
                            "name": "co"
                        }
```

```
                    }
                ]
            }
        ]
    }
]
}
```

d. Save your changes.

e. Restart Cloud Connector to apply the new attributes mapping, which comes from *idstorage.cfg*

This file overwrites the configurations you made in Configuring the User Store.

Next Steps

The following table shows the default mapping between the Active Directory user attributes and the SCIM attributes. It also shows the existing mapping between the SCIM attributes and the attributes in the user store of Identity Authentication.

Detailed Attribute Mapping Between Active Directory and SCIM, and between SCIM and the User Store of Identity Authentication

Microsoft Active Directory Attributes	SCIM Attributes	Identity Authentication User Store Attribute
sAMAccountname	userName	loginName
givenName	givenName	firstName
sn	familyName	lastName
personalTitle	honorificPrefix	title
displayName	displayName	displayName
locale	locale	language
timezone	timeZone	timeZone
employeeNumber	employeeNumber	personnelNumber
division	division	division
department	department	department
costCenter	costCenter	costCenter
company	organization	company
mail	emails.value	mail
telephoneNumber	phoneNumbers[work].value	telephone
facsimileTelephoneNumber	phoneNumbers[fax].value	fax
mobile	phoneNumbers[cell].value	mobile
streetAddress	addresses. streetAddress	street
l	Addresses.locality	city
st	Addresses.region	state

Microsoft Active Directory Attributes	SCIM Attributes	Identity Authentication User Store Attribute
postalCode	Addresses. postalCode	zip
co	Addresses.country	country

> **Note**
>
> The attributes `employeeNumber`, `division`, `department`, `costCenter` in the *Microsoft Active Directory Attributes* column are given as examples. They can differ according to the specific LDAP properties containing these attributes.
>
> If the attribute `language` or `country` comes from the corporate user store in:
>
> - small letters, then the respective keys in the master data service must be updated to small letters to be sent in this way to the service provider.
> - capital letters, then the respective keys in the master data service must be updated to capital letters to be sent in this way to the service provider. By default, `language` and `country` are in capital letters in the master date service.
>
> For more information how to update the master data service, see Change Master Data Texts REST API [page 2860].

Configure Identity Authentication [page 2547]

1.7.16.1.1.4 Configure Identity Authentication

Context

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Authentication Providers* tile.
3. Press the *Create* button on the left-hand panel to add a new source system to the list.
4. Make the corresponding entries in the configuration for the target system you want to add:
   - *Source System*

Configuration	Description
Display Name	(optional) The name of the configuration.

Configuration	Description
Type	Select the *Microsoft Active Directory* type.

- *Configuration*

Configurations	Description
Environment	Cloud Foundry
System ID	This is the name of the system. Provide a name of your choice as *System ID*.
Subaccount domains	Select your SAP BTP subaccount's domain from the drop-down.

5. Save your configuration.

   If the operation is successful, you receive the message `Source system <System ID name> created`.

Results

When the configuration is complete, the user can log in to the application with the on-premise credentials. The first logon requires *Login Name* and password. After successful authentication, a new user record is created in Identity Authentication with type `employee`.

1.7.16.1.2 Corporate User Store (Neo Environment)

Configure corporate user store for applications in the Neo environment to allow users to users to authenticate with their corporate credentials, without the need to use another set of credentials for their cloud access.

> ⓘ Note
>
> The content in this section is only relevant for SAP BTP, Neo environment.

> ⓘ Note
>
> The content in this section is not relevant for China (Shanghai) region.

> → Tip
>
> The recommended configuration of the corporate user store is in the Cloud Foundry environment. The Cloud Foundry setup can be used also for applications that are in the Neo environment. For more information, see Corporate User Store (Cloud Foundry Environment) [page 2519].

Overview

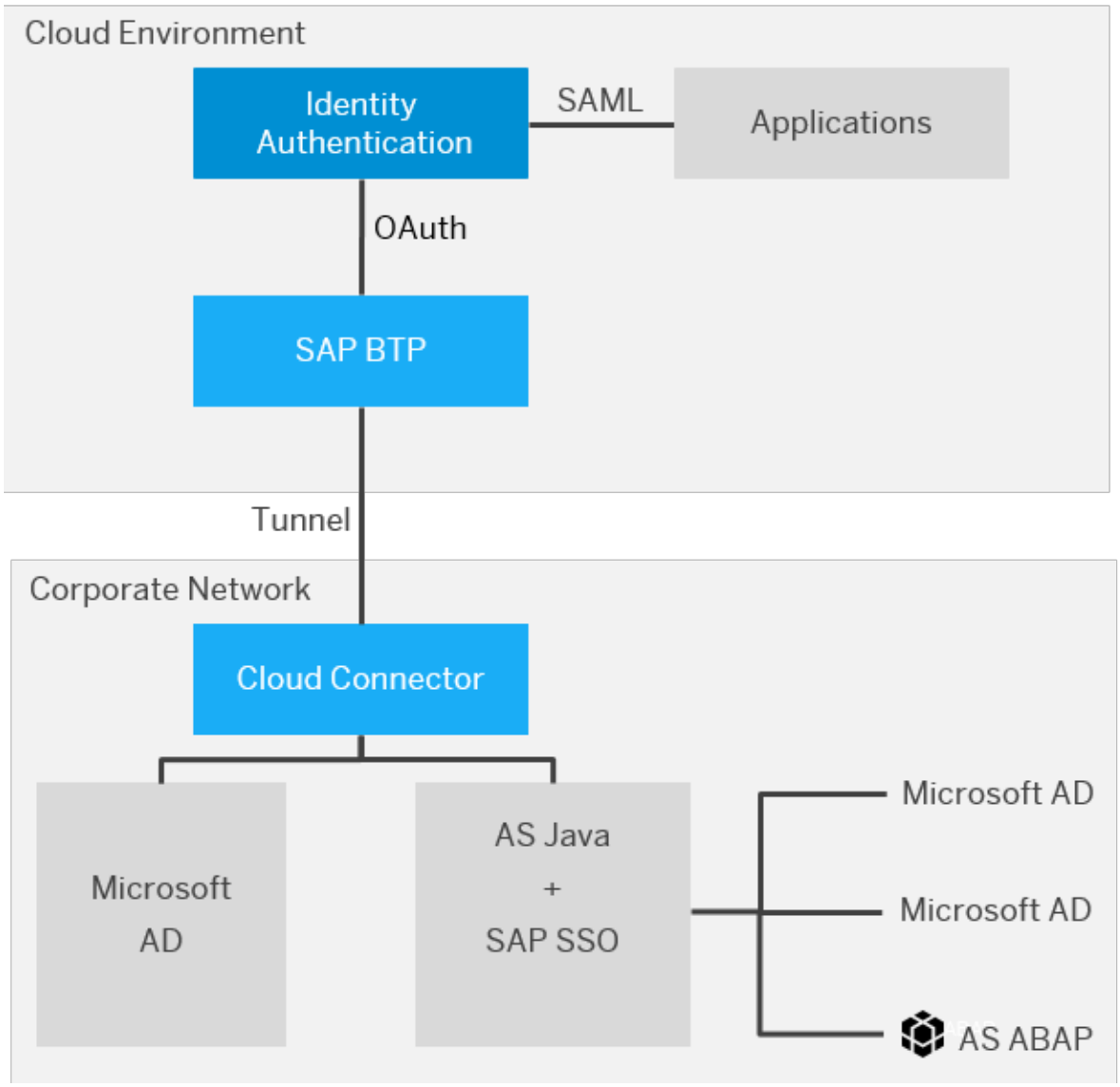
Identity Authentication can connect with the following corporate user stores:

- Microsoft Active Directory
- SAP NetWeaver AS JAVA, with the following variants:
  - SAP NetWeaver AS JAVA - UME
  - Multiple Active Directories connected to SAP NetWeaver AS JAVA - UME
  - SAP NetWeaver AS ABAP connected to SAP NetWeaver AS JAVA - UME

This scenario works with an SAP BTP application named *proxy* and provided by an SAP BTP subaccount named *sci*. The proxy application on SAP BTP uses the OAuth authentication mechanism when communicating with Identity Authentication.

> ⚠ Restriction
>
> The *sci proxy* application is not supported for China.

The connection between SAP BTP and the corporate user store is carried out with an Cloud Connector.



Authentication Flow

The *Corporate User Store* option is configured properly. A user tries to access a trusted application for the first time with the on-premise credentials, *Login Name* and *Password* entered correctly. This user is authenticated successfully against the corporate user store. With this initial successful authentication of the user, a partial user record is created in the user store for Identity Authentication. It is created with user details taken from the corporate user store. The cloud user store does not copy the user's credentials. For more details about what data is copied from the corporate user store, see User Records [page 2535]. With subsequent logins, the user is always authenticated against the corporate user store, and the user record is updated.

For the first logon with on-premise credentials, the user enters his or her *Login Name* and a *Password*. For subsequent logins the user can use either his or her *Login Name*, *Email*, or *User ID*, and the *Password* .

→ Remember

If the user has enabled the *Remember me* option, the following limitations are observed:

- when a user is authenticating with remember me cookie, his or her user record in Identity Authentication is not updated.
- if the user changes his or her password in the corporate network, the remember me cookie is not invalidated.

For more information, see Use the Remember Me Option [page 2774].

ⓘ Note

The user in the corporate user store must have the `mail` attribute.

The tenant administrator needs to monitor and prevent the coexistence of a cloud and on-premise user with one and the same email address. The tenant administrator has to instruct the users to logon for the first time with their *Login Name*, not with the *Email*.

If a user with a user record in the cloud user store is deleted in the corporate user store, the user cannot authenticate using Identity Authentication. The user record for this user remains in the cloud user store, and the tenant administrator can delete it via the administration console for SAP Cloud Identity Services. For more information, see Delete Users [page 2367].

For all users from the corporate user store, a second factor for authentication can be enabled for some applications, or cloud user groups can be assigned. For more details, see Configure Risk-Based Authentication for an Application [page 2007] and Assign Groups to a User [page 2409].

Control Access to Applications and Resources

In the scope of the *Corporate User Store* scenario, you can manage access to applications and their resources based on the groups available in the corporate user store.

The corporate user groups are sent to an application in the SAML 2.0 assertion. `corporate_groups` is the attribute that contains the groups that the user in the corporate user store is assigned to. For more details about how the groups are sent to the application in the SAML 2.0 assertion, see Configuring User Attributes from the Identity Directory [page 1973].

ⓘ Note

If your application is deployed on the SAP BTP, the corporate user store groups, relevant for the application, and contained in the `corporate_groups` attribute in the SAML 2.0 assertion, can be mapped to assertion-based groups created in SAP BTP cockpit. For more information, see the *4. (If Using an Identity Provider) Define the Group-to-Role Mapping* section in Managing Roles.

You can also restrict access to applications based on membership in a corporate user group by setting different rules via risk-based authentication. For more information, see Configure Risk-Based Authentication for an Application [page 2007].

1.7.16.1.2.1 User Records

When a user has been successfully authenticated for the first time with the credentials from the corporate user store, a record for that user is created in Identity Authentication. That user record is created with details from the corporate user store. In this record, the user is created with a *User Type* `employee`. This *User Type* cannot be changed.

For more information about the attributes taken from the Active Directory and their mapping to the user store of Identity Authentication, see **Configure SAP BTP When Connecting to an LDAP User Store** in Configure SAP BTP [page 2535].

Related Information

Security Information [page 2897]

1.7.16.1.2.2 Configure Connection to a Corporate User Store

To configure connection to a corporate user store, you have to make the following configurations in SAP BTP and in Identity Authentication.

For more details about how to configure these systems, see:

- Configure SAP BTP [page 2535]
- Configure Identity Authentication [page 2547]

1.7.16.1.2.3 Configure SAP BTP

Context

The configuration of SAP BTP depends on the type of the user store. You have two options: Microsoft Active Directory user store and SAP NetWeaver AS Java user store.

Configure SAP BTP When Connecting to a Microsoft Active Directory User Store

Procedure

1. Sign in SAP BTP cockpit with the cockpit administrator role. For more information, see Subaccounts.

2. In the SAP BTP cockpit, choose *Services in the navigation area* > *Identity Authentication Add-On* > *Enable in the detailed view of the service*.

   This enables the extension service of Identity Authentication named *proxy* and provided by an SAP BTP subaccount named *sci*.

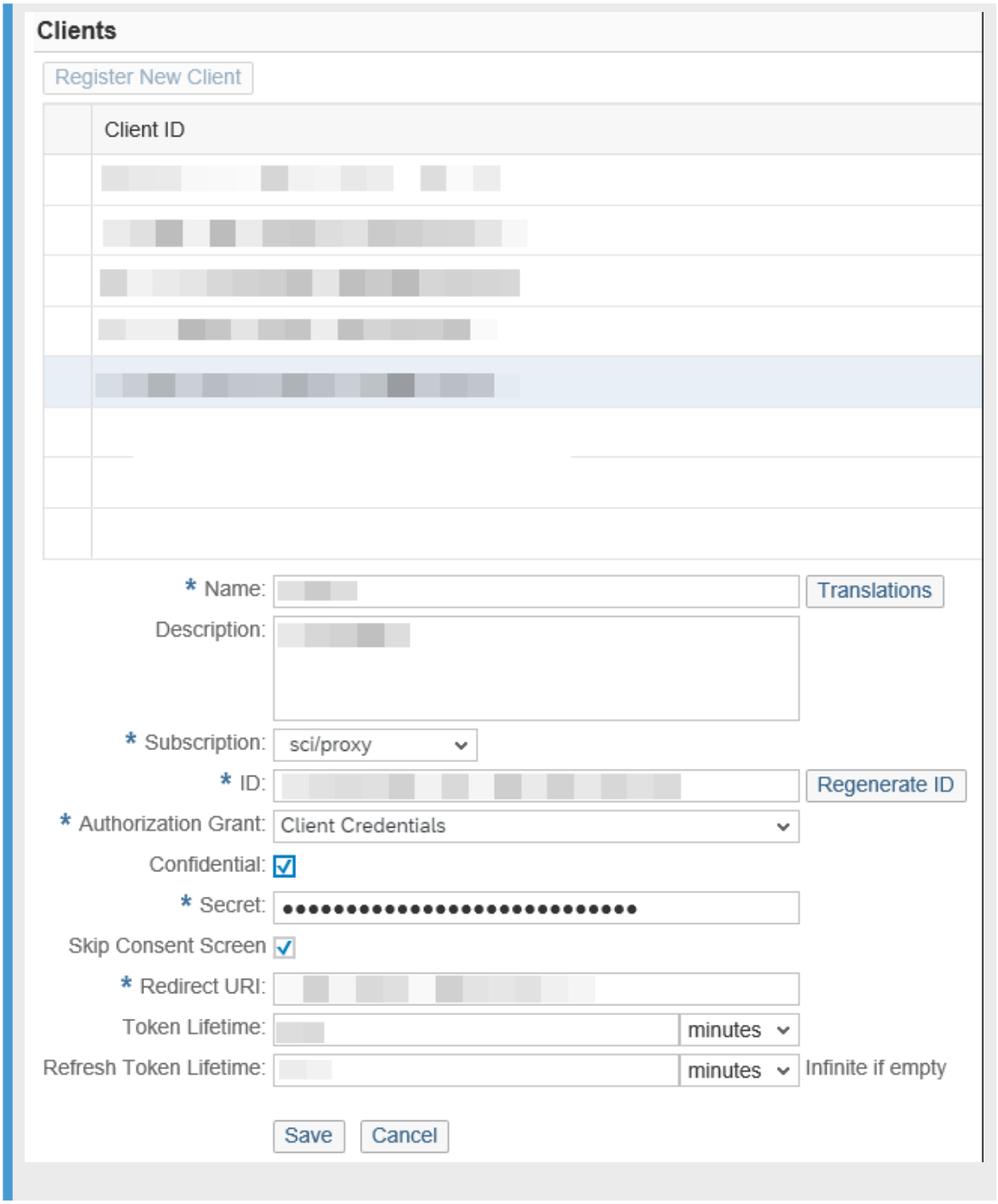
   > ⚠ Caution
   >
   > If you don't see the *Identity Authentication Add-On* tile in the cockpit, you need to report an incident with a subject "Enable Corporate User Store Feature" on SAP Support Portal Home under the component `BC-IAM-IDS`. You have to provide information about your SAP BTP subaccount name and region.

3. In your subaccount on SAP BTP, register an OAuth client for the subscribed *proxy* application provided by the *sci* subaccount.

   The procedure is described in the documentation of SAP BTP in the link below.

   > ⓘ Note
   >
   > Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

   > ⓘ Note
   >
   > Since Identity Authentication will create the subscription to the *proxy* application, the *Prerequisites* section in the respective document isn't relevant for the current scenario.
   >
   > For the *Authorization Grant* field in the SAP BTP cockpit, choose *Client Credentials* from the dropdown, and for the *Subscription* field, choose *sci/proxy*.



For more information about how to register an OAuth client, see Register an OAuth Client.

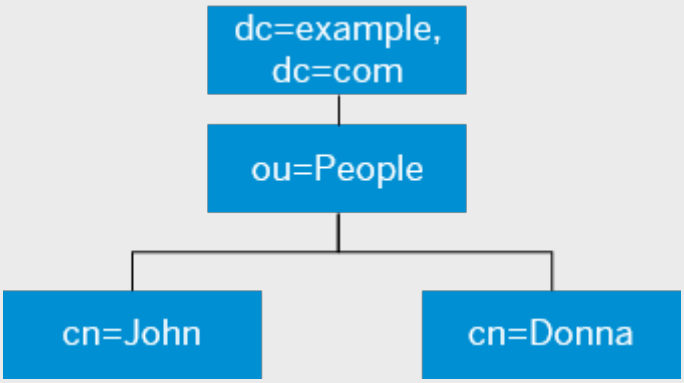
4. Install a Cloud Connector in your corporate network.

   For more information, see Installation.

5. Connect the Cloud Connector with your SAP BTP account.
   - If you haven't used your Cloud Connector before, see Initial Configuration.
   - If you have used your Cloud Connector before, you can start the configuration from Set up Connection Parameters and HTTPS proxy.

6. Connect SAP BTP with your corporate user store.

> **Note**
>
> You have to specify the SAP BTP settings. The *Prerequisites* section in the document describing the configuration is already configured for the *proxy* application, and you should proceed with the configuration steps. For more information, see Configure the User Store.
>
> The *User Name* field must be in the **`<service_user_name>@<domain>`** format.
>
> For the *User Path* and *Group Path* fields, specify the Microsoft Active Directory tree that contains the users and groups, respectively. For example, if the tree has the following structure:



> The user and group paths should appear as in the table below:

User Path	ou=People,dc=example,dc=com
Group Path	ou=People,dc=example,dc=com

7. **Optional:** Include additional attributes.

> **Note**
>
> You can add the `employeeNumber`, `division`, `department`, and `organization` attributes that are defined in the SCIM Enterprise User Schema Extension.
>
> Cloud Connector uses the SCIM protocol to transfer the data, so the Active Directory attributes are mapped first to the SCIM attributes. When the data is provisioned, the SCIM attributes are mapped to the user store attributes of Identity Authentication.

a. In your system go to `/sapcc-<version>/config_master/com.sap.core.connectivity.protocol.scim/`
b. On that level, create a new *idstorage.cfg* file based on the *idstorage_extended_schema.cfg* file which is given as an example in the folder.
c. Edit the newly created file. For more details, see the information below:

   To add new user attributes you have to edit the whole file.

> ⚠ Caution
>
> This file overwrites the configurations you made in Configure the User Store Be careful not to change the user attributes taken from Microsoft Active Directory.

In this section, provide the same information as when you specified the SAP BTP settings in the previous step.

```
 {
 "LDAPServers": [
    {
      "Host": "<The host name of the LDAP server to be contacted>",
      "Port": "<The port where the LDAP service is running. If omitted
then the default LDAP port will be used - 389 for plain connections and
636 for SSL connections>"
    }
 ],
 "UserPath": "<LDAP subtree containing the users. Example
"DC=users,DC=organisation,DC=location">",
 "GroupPath": "<LDAP subtree containing the groups. Example
"DC=groups,DC=organisation,DC=location">",
 "ServiceUser": {
    "Name": "<The name of the user that will be used to establish
communication with the LDAP. In case of Active Directory the user name
should contains Domain suffix, e.g. "john@ACME.COM">",
    "Password": "<Password of this user>"
 },
```

If you want to use SSL, we recommend that you configure this section.

```
 "UseSSL": "<Possible values are "true" or "false". If true then the
communication to LDAP will go over SSL>",
 "IdentityKeystorePath": "<File system path to the client identity
keystore - must be set if the used LDAP server requires client certificate
authentication>",
 "IdentityKeystorePassword": "<The password of the client identity
keystore>",
 "TrustKeystorePath": "<File system path to the trusted CAs keystore -
must be set if UseSSL is true>",
 "TrustKeystorePassword": "<The password of the trusted CAs keystore>",
```

In this section add the additional attributes `employeeNumber`, `division`, `department`, and `organization`, defined in the SCIM Enterprise User Schema Extension.

```
 {
  "SingularAttributes": [
      {
          "SCIMAttribute": "userName",
          "mappings": [
              {
                 "LDAPAttribute": {
                     "name": "sAMAccountname"
                 }
              }
          ]
       },
          "SCIMAttribute": "name",
          "mappings": [
              {
                 "SCIMSubAttribute": "givenName",
                 "LDAPAttribute": {
                     "name": "givenName"
                 }
```

```
                },
                {
                    "SCIMSubAttribute": "familyName",
                    "LDAPAttribute": {
                        "name": "sn"
                    }
                },
                {
                    "SCIMSubAttribute": "honorificPrefix",
                    "LDAPAttribute": {
                        "name": "personalTitle"
                    }
                }
            ]
        },
        {
            "SCIMAttribute": "displayName",
            "mappings": [
                {
                    "LDAPAttribute": {
                        "name": "displayName"
                    }
                }
            ]
        },
        {
            "SCIMAttribute": "locale",
            "mappings": [
                {
                    "LDAPAttribute": {
                        "name": "locale"
                    }
                }
            ]
        },
        {
            "SCIMAttribute": "timeZone",
            "mappings": [
                {
                    "LDAPAttribute": {
                        "name": "timezone"
                    }
                }
            ]
        },
        {
            "SCIMAttribute": "employeeNumber",
            "mappings": [
                {
                    "LDAPAttribute": {
                        "name": "<LDAP property containing the employee
number"
                    }
                }
            ]
        },
        {
            "SCIMAttribute": "division",
            "mappings": [
                {
                    "LDAPAttribute": {
                        "name": "<LDAP property containing the division>"
                    }
                }
            ]
        },
        {
            "SCIMAttribute": "department",
```

```
          "mappings": [
              {
                 "LDAPAttribute": {
                     "name": "<LDAP property containing the department>"
                 }
              }
          ]
       },
       {
          "SCIMAttribute": "organization",
          "mappings": [
              {
                 "LDAPAttribute": {
                     "name": "company"
                 }
              }
          ]
       },
      {
          "SCIMAttribute": "costCenter",
          "mappings": [
              {
                 "LDAPAttribute": {
                     "name": "<LDAP property containing the cost center>"
                 }
              }
          ]
       }
  ],
  "MultiValuedAttributes": [
      {
         "SCIMAttribute": "emails",
         "values": [
             {
                "primary": "true",
                "mappings": [
                    {
                        "SCIMSubAttribute": "value",
                        "LDAPAttribute": {
                            "name": "mail"
                        }
                    }
                ]
             }
         ]
      },
      {
         "SCIMAttribute": "phoneNumbers",
         "values": [
             {
                 "type": "work",
                 "primary": "true",
                 "mappings": [
                     {
                         "SCIMSubAttribute": "value",
                         "LDAPAttribute": {
                             "name": "telephoneNumber"
                         }
                     }
                 ]
             },
             {
                 "type": "fax",
                 "mappings": [
                     {
                         "SCIMSubAttribute": "value",
                         "LDAPAttribute": {
                             "name": "facsimileTelephoneNumber"
```

```
                        }
                    }
                ]
            },
            {
                "type": "cell",
                "mappings": [
                    {
                        "SCIMSubAttribute": "value",
                        "LDAPAttribute": {
                            "name": "mobile"
                        }
                    }
                ]
            }
        ]
    },
    {
        "SCIMAttribute": "addresses",
        "values": [
            {
                "primary": "true",
                "mappings": [
                    {
                        "SCIMSubAttribute": "streetAddress",
                        "LDAPAttribute": {
                            "name": "streetAddress"
                        }
                    },
                    {
                        "SCIMSubAttribute": "locality",
                        "LDAPAttribute": {
                            "name": "l"
                        }
                    },
                    {
                        "SCIMSubAttribute": "region",
                        "LDAPAttribute": {
                            "name": "st"
                        }
                    },
                    {
                        "SCIMSubAttribute": "postalCode",
                        "LDAPAttribute": {
                            "name": "postalCode"
                        }
                    },
                    {
                        "SCIMSubAttribute": "country",
                        "LDAPAttribute": {
                            "name": "co"
                        }
                    }
                ]
            }
        ]
    }
]
}
```

d. Save your changes.

e. Restart Cloud Connector to apply the new attributes mapping, which comes from *idstorage.cfg*

This file overwrites the configurations you made in Configuring the User Store.

Next Steps

The following table shows the default mapping between the Active Directory user attributes and the SCIM attributes. It also shows the existing mapping between the SCIM attributes and the attributes in the user store of Identity Authentication.

Detailed Attribute Mapping Between Active Directory and SCIM, and between SCIM and the User Store of Identity Authentication

Microsoft Active Directory Attributes	SCIM Attributes	Identity Authentication User Store Attribute
sAMAccountname	userName	loginName
givenName	givenName	firstName
sn	familyName	lastName
personalTitle	honorificPrefix	title
displayName	displayName	displayName
locale	locale	language
timezone	timeZone	timeZone
employeeNumber	employeeNumber	personnelNumber
division	division	division
department	department	department
costCenter	costCenter	costCenter
company	organization	company
mail	emails.value	mail
telephoneNumber	phoneNumbers[work].value	telephone
facsimileTelephoneNumber	phoneNumbers[fax].value	fax
mobile	phoneNumbers[cell].value	mobile
streetAddress	addresses. streetAddress	street
l	Addresses.locality	city
st	Addresses.region	state
postalCode	Addresses. postalCode	zip
co	Addresses.country	country

> ### Note
>
> The attributes `employeeNumber`, `division`, `department`, `costCenter` in the *Microsoft Active Directory Attributes* column are given as examples. They can differ according to the specific LDAP properties containing these attributes.
>
> If the attribute `language` or `country` comes from the corporate user store in:
>
> - small letters, then the respective keys in the master data service must be updated to small letters to be sent in this way to the service provider.

- capital letters, then the respective keys in the master data service must be updated to capital letters to be sent in this way to the service provider. By default, `language` and `country` are in capital letters in the master date service.

For more information how to update the master data service, see Change Master Data Texts REST API [page 2860].

Configure Identity Authentication [page 2547]

Configure SAP BTP When Connecting to an SAP NW AS Java User Store

Prerequisites

- You have an SAP NetWeaver 7.2 or higher Application Server for the Java system.
- You have SAP Single Sign-On (SAP SSO) 2.0 or higher installed in your system landscape.
- You have installed and deployed federation software component archive (SCA) from SAP Single Sign-On (SSO) 2.0. For more information, see Downloading and Installing the Federation Software.

Procedure

1. Sign in SAP BTP cockpit with the cockpit administrator role. For more information, see Subaccounts.
2. In the SAP BTP cockpit, choose *Services in the navigation area* > *Identity Authentication Add-On* > *Enable in the detailed view of the service*.

   This will enable the extension service of Identity Authentication named *proxy* and provided by an SAP BTP subaccount named *sci*.

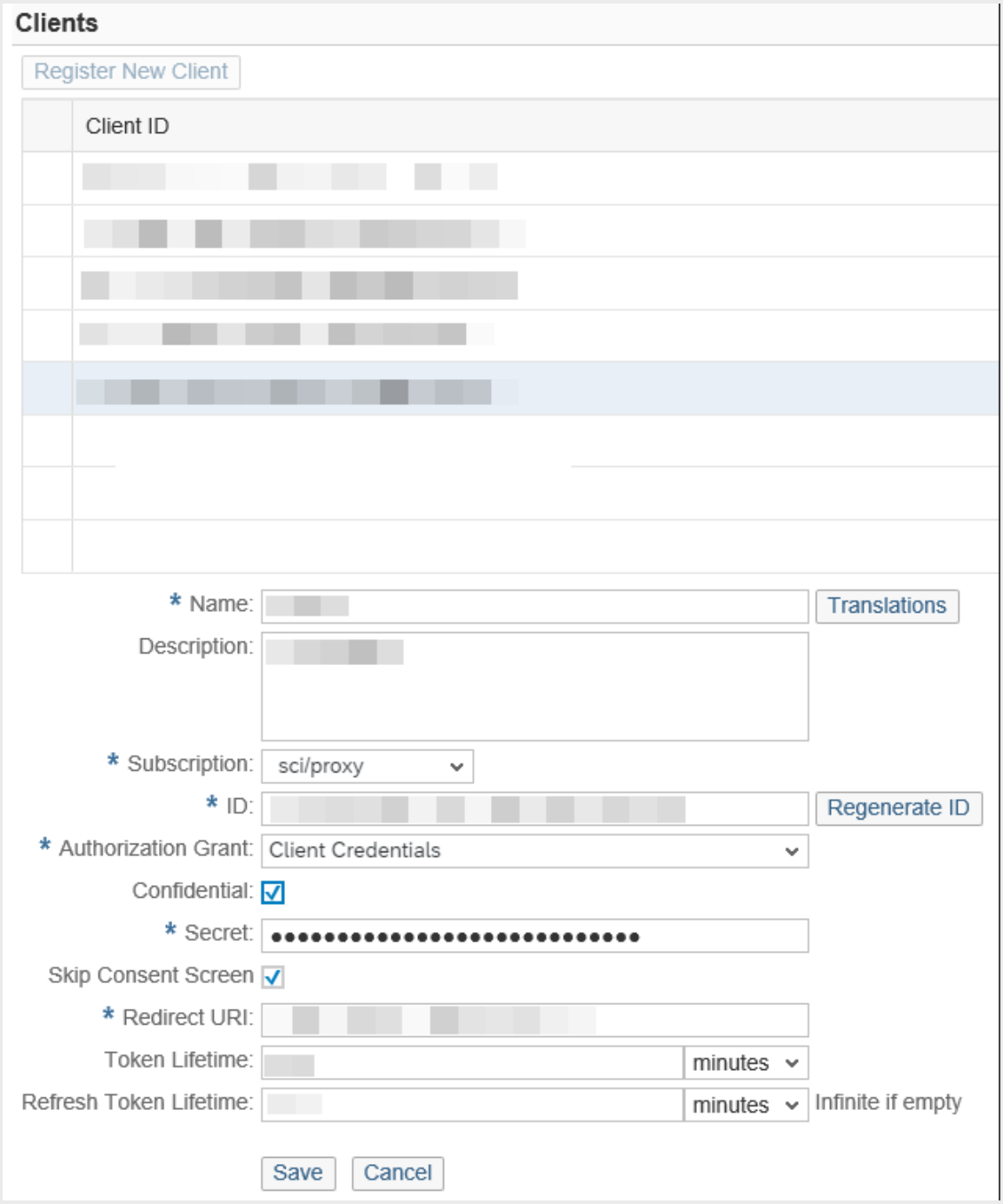
   > ⚠ Caution
   >
   > If you don't see the *Identity Authentication Add-On* tile in the cockpit, you need to report an incident with a subject "Enable Corporate User Store Feature" on SAP Support Portal Home under the component **BC-IAM-IDS**. You have to provide information about your SAP BTP subaccount name and region.

3. In your subaccount on SAP BTP, register an OAuth client for the subscribed *proxy* application provided by the *sci* subaccount.

   The procedure is described in the documentation of SAP BTP in the link below.

   > ⓘ Note
   >
   > Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

> ### Note
>
> Since Identity Authentication will create the subscription to the *proxy* application, the *Prerequisites* section in the respective document isn't relevant for the current scenario.
>
> For the *Authorization Grant* field in the SAP BTP cockpit, choose *Client Credentials* from the dropdown, and for the *Subscription* field, choose *sci/proxy*.



For more information about how to register an OAuth client, see Register an OAuth Client.

4. Install a Cloud Cloud Connector in your corporate network.

For more information, see Installation.

5. Connect the Cloud Connector with your SAP BTP account.
   - If you haven't used your Cloud Connector before, see Initial Configuration.
   - If you have used your Cloud Connector before, you can start the configuration from Set up Connection Parameters and HTTPS proxy.
6. Connect SAP BTP with your corporate user store.
   a. In the configuration of Cloud Connector, configure the host mapping to the on-premise system. For more information, see Configure Access Control (HTTP). For the *Limiting the Accessible Services for HTTP(S)* section, be sure that the `URL Path` is `/scim/v1`, and `Path and all Subpaths` radio button is chosen for `Access Policy`.
   b. Create a destination to the on-premise system. In the SAP BTP cockpit, choose *Services in the navigation area* > *Identity Authentication Add-On* > *Configure Identity Authentication Add-On* > *New Destination*.

> **Note**
>
> When configuring the destination to the on-premise system, make sure of the following:
>
> - The *Name* is **`SAPCloudIdentityUserStore`**.
> - The *Type* is **`HTTP`**.
> - The *URL* of the destination, the host name, and the port should coincide with the virtual host name and virtual port from the setup of the access control in Cloud Connector.
>
>   > → Remember
>   >
>   > The protocol of the *URL* must be **`HTTP`**.
>
>   The *URL* of the destination should be in the following pattern: *http:// <Virtual host configured in Cloud Connector>:<virtual Port>/scim/v1/*
> - The *Proxy Type* is **`OnPremise`**.
> - The *Authentication* is **`BasicAuthentication`**.
>
> For more information, see Using an SAP System as an On-Premise User Store. Since Identity Authentication has already deployed the *proxy* application, you should start from the **2. Configure the On-Premise System** section in the documentation.

Next Steps

Configure Identity Authentication [page 2547]

1.7.16.1.2.4 Configure Identity Authentication

Context

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Authentication Providers* tile.
3. Press the *Create* button on the left-hand panel to add a new source system to the list.
4. Make the corresponding entries in the configuration for the target system you want to add:
   - *Source System*

Configuration	Description
Display Name	(optional) The name of the configuration.
Type	Select the *Microsoft Active Directory* type.

   - *Configuration*

Configurations	Description
Environment	Neo
Data Center	Select your SAP BTP subaccount's region.
Technical Name	The *Technical Name* must match your SAP BTP subaccount *Technical Name*.
Client ID	The *Client ID* must match the *ID* registered on SAP BTP under the *OAuth Settings* tab for your subaccount.
Client Secret	The *Client Secret* must match the *Secret* registered on SAP BTP under the *OAuth Settings* tab for your subaccount.

5. Save your configuration.

   If the operation is successful, you receive the message `Source system <System ID name> created`.

Results

When the configuration is complete, the user can log in to the application with the on-premise credentials. The first logon requires *Login Name* and password. After successful authentication, a new user record is created in Identity Authentication with type `employee`.

1.7.16.2 Kerberos Authentication

Configure Kerberos authentication for Identity Authentication to allow users to log on without a username and password when they are in the corporate network, and enable this authentication to allow users to log on to an application from the corporate network without entering their username and password.

- Configure Kerberos Authentication [page 2548]
- Enable or Disable Kerberos Authentication for an Application [page 2556]

1.7.16.2.1 Configure Kerberos Authentication

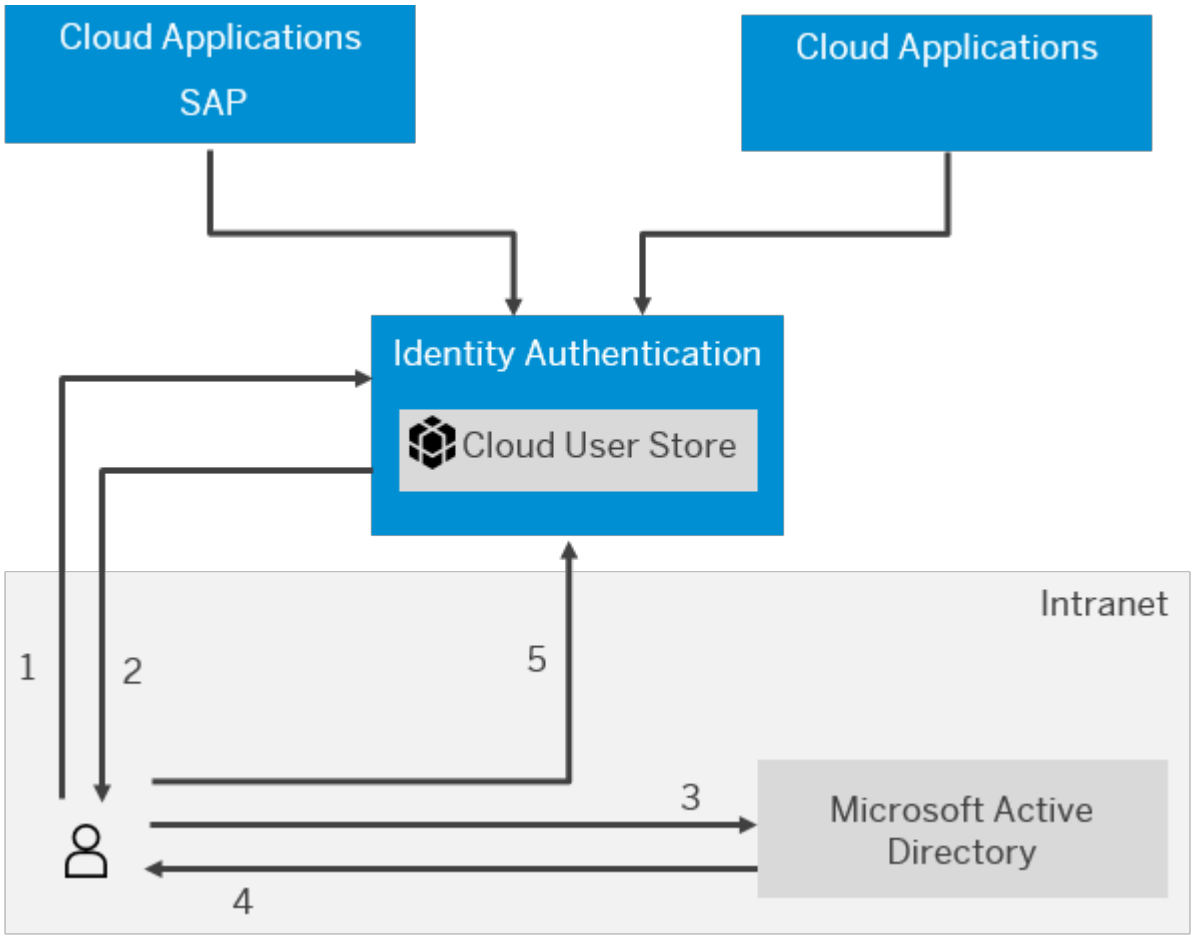
Overview

You configure Kerberos authentication for Identity Authentication to allow users to log on without a username and password when they are in the corporate network. Identity Authentication supports Kerberos with Simple and Protected GSS-API Negotiation Mechanism (SPNEGO).

Kerberos authentication with Identity Authentication requires the following systems:

- Web client
  The web client requests a protected resource of an application configured to use Identity Authentication as an identity provider. It authenticates against the Key Distribution Center (KDC). For example, users can use the web browser to access cloud applications using Identity Authentication.
- Key Distribution Center (KDC)
  It authenticates the user and grants a ticket that is used for the communication between the web client and Identity Authentication.
- Identity Authentication
  Identity Authentication accepts the ticket issued by the KDC and checks the authenticating user in its cloud user store.

The communication flow is as follows:



1. A user accesses a cloud application on his or her web browser (web client) in the corporate network, and the web client sends a request to Identity Authentication.
2. Identity Authentication returns a response with status code *401 Unauthorized* .
   The response contains a *WWW-Authenticate: Negotiate* header.
3. The Web client requests a Kerberos ticket from the Microsoft Active Directory (KDC).
4. The Microsoft Active Directory (KDC) responds with the ticket.
5. The Web client sends the ticket to Identity Authentication.
6. Identity Authentication validates the ticket, checks that the user exists in its cloud user store by the user's logon name, and authenticates the user.

Related Information

Kerberos: The Network Authentication Protocol

1.7.16.2.1.1 Prerequisites

- You have configured the web browser (web client) to use Kerberos authentication.
  For more information about this procedure, see the corresponding browser (client) documentation.

  > → Tip
  >
  > This setting is usually found under the *Local Intranet* tab or the *Trusted Sites* list in the browser. Search on the Internet or in your browser documentation for information about how Kerberos authentication is enabled.

- You have a tenant for Identity Authentication.
- The trust with the service provider of Identity Authentication is configured.
  For more information, see Integrating the Service [page 2910].
- The users logging on with Kerberos authentication exist in the cloud user store with the required details. Each user has to have a logon name as a user attribute. This is specified under the *loginName* column in the imported CSV file. For details, see Related Information.

Related Information

Import or Update Users for a Specific Application [page 2348]

1.7.16.2.1.2 Configure Key Distribution Center (KDC)

Context

This procedure is performed by the domain administrator. If you are not a domain administrator, skip to the next section.

Procedure

1. Create a service user in KDC (in Microsoft Active Directory for example).

   > ⓘ Note
   >
   > A service user is associated with one tenant only.

2. Generate a keytab file and provide it to the tenant administrator.

   When you create the keytab file, the password you specify for the service user is used to generate a key. A setting on the service user also allows you to configure a key type to be derived and used for the

encryption of the Kerberos ticket. You have to provide this keytab file as well as the key type to the tenant administrator so that he or she can configure Identity Authentication.

The key type has to match the encryption type of the corresponding service user that contains the tenant as a service principal name.

Key Type	Supported by Identity Authentication
RC4_HMAC	Yes
AES128_CTS_HMAC_SHA1_96	Yes
AES256_CTS_HMAC_SHA1_96	Yes
NTLMSSP	No
NEGOEXT	No

For more information about these key types, see RFC 4757 and RFC 3962.

> ⚠ Caution
>
> The realm you specify to generate the keytab file has to be in capital letters.

> ✤ Example
>
> You can derive the key by using your *Java* installation. To derive the key, proceed as follows:
>
> 1. In the command prompt, run the `ktab -help` command to see the list of available commands.
>
>   > → Tip
>   >
>   > Go to your Java `bin` folder or set the Java path as an environment variable.
>
> 2. Enter `ktab -a <service user>@<realm> -k <path>\<keytab filename>.ktab` with `<realm>` in capital letters to create a new keytab file .
>   You are also prompted to enter the service user password that is used to derive the key.

3. Register a service principal name (SPN) associated with the service user for the host name used to access Identity Authentication. The SPN has to be unique.

> ✤ Example
>
> The command line `setspn -A HTTP/<tenantID>.accounts.ondemand.com <service user>` registers an SPN for the *<tenantID>.accounts.ondemand.com* host associated with the service user.

> ⚠ Caution
>
> If you have one of the following combination of an operating system and web browser:
>
> | Operating System | Browser |
> |---|---|
> | Microsoft Windows 10 | Microsoft Internet Explorer 11 |
> | Microsoft Windows | Google Chrome |

Operating System	Browser
Microsoft Windows	Mozilla Firefox

you must ensure a registration of an SPN for the root host name of Identity Authentication.

Root host names per tenant locations:

Host Names

Country/Region	Data Center	Infrastructure	Host Name
Australia	Australia (Sydney)	SAP / Azure	`ap.accounts.ondemand.com.cloud.sap.akadns.net`
Brazil	Brazil (São Paulo)	AWS	`br.accounts.ondemand.com.cloud.sap.akadns.net`
China	China (Shanghai)	SAP	`accounts.sapcloud.cn.cloud.sap.akadns.net`
Europe	Germany (Frankfurt)	AWS	`aws-eu-de.accounts.ondemand.com.cloud.sap.akadns.net`
Europe	Europe (Frankfurt / Amsterdam))	SAP	`accounts.ondemand.com.cloud.sap.akadns.net`
India	India (Mumbai)	AWS	`aws-ap-in.accounts.ondemand.com.cloud.sap.akadns.net`
Israel	Israel (Tel Aviv)	Google Cloud	`il.accounts.ondemand.com.cloud.sap.akadns.net`
Japan	Japan (Tokyo / Osaka)	SAP	`jp.accounts.ondemand.com.cloud.sap.akadns.net`
North America (Canada Central)	Canada (Toronto)	Azure	`azr-na-ca.accounts.ondemand.com.cloud.sap.akadns.net`
Saudi Arabia (SA1)	Saudi Arabia (Dammam)	Google Cloud	`sa.accounts.ondemand.com.cloud.sap.akadns.net`

Country/Region	Data Center	Infrastructure	Host Name
Saudi Arabia (SA2)	Saudi Arabia (Dammam) - Regulated	Google Cloud	`sa2.accounts.ondemand.com.cloud.sap.akadns.net`
Singapore	Singapore	AWS	`aws-ap-se-1.accounts.ondemand.com.cloud.sap.akadns.net`
South Korea	South Korea (Seoul)	AWS	`aws-ap-kr.accounts.ondemand.com.cloud.sap.akadns.net`
Switzerland	Switzerland (Zürich)	Azure	`azr-eu-ch.accounts.ondemand.com.cloud.sap.akadns.net`
United Arab Emirates	United Arab Emirates (Dubai)	SAP	`azr-ap-ae.accounts.ondemand.com.cloud.sap.akadns.net`
United Kingdom	United Kingdom (London)	Azure	`uk.accounts.ondemand.com.cloud.sap.akadns.net`
US (North America East)	USA (Sterling, VA / N. Virginia)	SAP	`us-east.accounts.ondemand.com.cloud.sap.akadns.net`
US East **Note** Trial environment.	Trial (USA)	Azure	`trial-accounts.ondemand.com.cloud.sap.akadns.net`
US West	USA (Quincy, WA)	Azure	`azr-us-we.accounts.ondemand.com.cloud.sap.akadns.net`

Example

The command line `setspn -A HTTP/accounts.ondemand.com.cloud.sap.akadns.net <service user>` registers an SPN for the root host *accounts.ondemand.com.cloud.sap.akadns.net* associated with the service user.

1.7.16.2.1.3 Configure Identity Authentication

Prerequisites

You are assigned the *Manage Tenant Configuration* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

This procedure is performed by the tenant administrator.

Procedure

1. Extract the key from the keytab file provided by the domain administrator.

   > ### Example
   >
   > You can extract the key by using your *Java* installation.
   >
   > 1. In the command prompt, run the `ktab -help` command to see the list of available commands.
   >
   >   > ### → Tip
   >   >
   >   > Go to your Java `bin` folder or set the Java path as an environment variable.
   >
   > 2. Enter `klist -e -f -k -K <path>\<keytab filename>.ktab` to list the derived keys corresponding to the key types.
   > 3. Copy the key according to the number of the provided key type: *23* corresponds to *RC4*, *17* corresponds to *AES128*, and *18* corresponds to *AES256*.

2. Access the tenant's administration console for SAP Cloud Identity Services by using the console's URL.

   > ### Note
   >
   > The URL has the `https://<tenant ID>.accounts.ondemand.com/admin` pattern.

3. Under *Applications and Resources*, choose the *Tenant Settings* tile.

   At the top of the page, you can view the administrative and license relevant information of the tenant.

4. Choose the *SPNEGO* list item.
5. Choose *+Add* and enter the following information:
   - Realm
     The Kerberos realm (domain) which the SPN and service user reside in.

> ⓘ Note
>
> The realm (domain) has to be in capital letters.

- Key type
  The key type has to match the encryption type of the corresponding service user that contains the tenant as a service principal name.

Key Type	Supported by Identity Authentication
RC4_HMAC	Yes
AES128_CTS_HMAC_SHA1_96	Yes
AES256_CTS_HMAC_SHA1_96	Yes
NTLMSSP	No
NEGOEXT	No

  For more information about these key types, see RFC 4757 and RFC 3962.

- Key
  The key derived with the password of the service user in the KDC.
- IP white list
  A comma-separated list of ranges of the web client's IPs or proxies allowed for the Kerberos authentication. The IP white list has to contain ranges in Classless Inter-Domain Routing (CIDR) notation.

> ⓘ Note
>
> By default the field is empty, which means any client IP is allowed.

> ✤ Example
>
> Enter *123.45.67.1/24, 189.101.112.1/16* to allow the web client to use any client IP starting with 123.45.67 or with 189.101.

6. Choose *+Add*.

   If the operation is successful, you receive the message `SPNEGO settings saved`.

> → Tip
>
> To update the saved settings, choose *Edit* and save your changes.

Next Steps

Enable Kerberos authentication for specific applications. For more information, see Enable or Disable Kerberos Authentication for an Application [page 2556].

1.7.16.3 Enable or Disable Kerberos Authentication for an Application

You enable Kerberos authentication to allow users to log on to an application from the corporate network without entering their username and password.

Prerequisites

You have configured Kerberos authentication for the tenant. For more information, see Configure Kerberos Authentication [page 2548].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Authentication and Access* tab.
5. Under *AUTHENTICATION*, enable or disable SPNEGO authentication.

   Once the application has been updated, the system displays the message `Application <name of application> updated`.

Related Information

Create a New Application [page 1922]

1.7.16.4 Corporate Identity Providers

Initially, Identity Authentication is set as the default identity provider for the applications. This section describes the scenarios in which Identity Authentication acts as a proxy to delegate the authentication to a corporate identity provider.

A corporate identity provider is an enterprise-controlled service that verifies user identities and issues standards-based security tokens (such as SAML or OpenID Connect tokens) for single sign-on to corporate

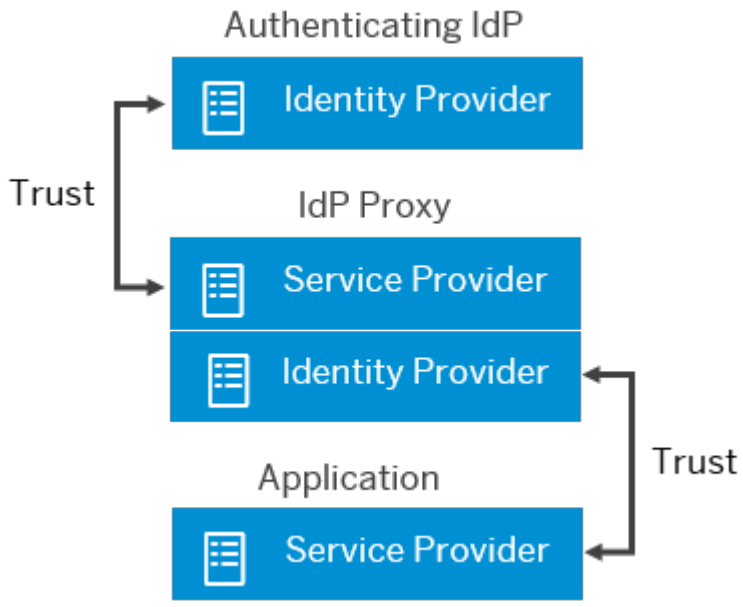
applications. It centralizes authentication and access policies (such as MFA and conditional access), manages trust with service providers, and can federate with other identity providers to provide a consistent, secure sign-in experience across the enterprise.

Identity Provider Proxy Overview

An identity provider can function as a proxy for another identity provider. An identity provider proxy enables you to create structures of trust relationships that ultimately simplify the management of your applications.

A proxy relationship involves the following participants:

- Corporate Identity Provider
  The identity provider proxy trusts the authenticating identity provider.
- Identity Provider Proxy
  The identity provider proxy is both an identity provider and a service provider. The service provider of the identity provider proxy trusts the authenticating identity provider.
- Application
  A service provider hosts a service that users want to access. This service provider trusts the identity provider of the identity provider proxy.



There's no direct trust relationship between the authenticating identity provider and the service provider that the user is trying to access.

Identity Authentication acting as a proxy can use a SAML 2.0 or an OpenID Connect (OIDC) identity provider as an external authenticating authority for the application. The choice of one or the other protocol depends on your scenario and needs.

> → Tip
>
> If you use Microsoft Entra ID as your corporate identity provider with SAML, you can additionally enable OIDC for your applications. Instead of looking under *Enterprise Applications* for the SAML configuration,

find the OIDC configuration for Entra ID under *App Registrations* and search for the application ID. Configure the required redirect URIs.

For more information, see the documentation for Microsoft Entra ID.

1.7.16.4.1 Create Corporate IdP in Administration Console

Create a new corporate identity provider (IdP) and customize it or copy the configuration from an existing one.

Prerequisites

- You are assigned the *Manage Corporate Identity Providers* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Context

When you create a new corporate IdP in the administration console for SAP Cloud Identity Services you can also change the default SAML 2.0 Compliant type choice offered to you to Single Sign-On (SAML 2.0), Microsoft ADFS / Entra ID (SAML 2.0) or OpenID Connect Compliant types.

You can also copy the settings from an IdP that is already existing in your tenant to the new corporate IdP. For example, you can have separate SAML 2.0 and OpenID Connect IdP trust configurations for one and the same corporate IdP. Thus, you can migrate your applications from one protocol to another by switching application by application to the new trust.

→ Remember

If you are copying the settings from one type of corporate IdP to another, all settings are copied apart from the ones that are protocol-specific (SAML 2.0 or OpenID Connect).

After copying the setting, configure the trust manually for the newly created IdP, and adjust the key value pairs in the *Enriched Token Claims* section.

To create a new corporate IdP, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Choose the *Create* button on the left-hand panel.

4. Provide the following information in the dialog box:

Fields	Notes
Display Name	Required. The display name of the identity provider.
Identity Provider Type	Optional. You can choose one of the following options: • SAML 2.0 Compliant • SAP Single Sign-On (SAML 2.0) • Microsoft ADFS / Entra ID (SAML 2.0) **⚠ Caution** For this type, the digest algorithm of the corporate identity provider must be *SHA-256*. For more information, see 2. Configure Trust on Identity Authentication Side in Configure Trust with Corporate Identity Provider [page 2560]. • OpenID Connect Compliant
Copy Settings from Identity Provider	Optional. You can create a brand new corporate IdP or copy the configurations from an already existing one. Possible options: • Don't copy (default) - create a corporate IdP and after that configure it. • Choose from a list with all corporate IdPs that are created in the admin console - the new corporate IdP gets all configuration settings apart from the protocol-specific ones (SAML or OpenID Connect) of the selected corporate IdP.

5. Save your changes.

Results

Once the identity provider has been created, the system displays the message `Identity provider <name of identity provider> created`.

The newly created identity provider appears on the list on the left. It is selected and you can proceed with its configuration.

1.7.16.4.2 Configure Trust with SAML 2.0 Corporate Identity Provider

This document is intended to help you configure trust with a SAML 2.0 corporate identity provider. In this scenario Identity Authentication acts as a proxy to delegate the authentication to the SAML 2.0 corporate identity provider.

Context

Identity Authentication can use a SAML 2.0 identity provider as an external authenticating authority. Identity Authentication thus acts as a proxy to delegate authentication to the external corporate identity provider. The requests for authentication sent by a service provider will be forwarded to the corporate identity provider.

As an identity provider proxy, Identity Authentication will act as an SAML 2.0 identity provider to the service provider, and as an SAML 2.0 service provider to the corporate identity provider. Once a user is authenticated at the corporate identity provider, successive authentication requests from service providers, which use the same corporate identity provider will not be forwarded to it as long as the session at Identity Authentication is active. Identity Authentication will issue assertions based on the user data received during the first authentication.

> ⓘ Note
>
> If an application requires force authentication (ForceAuthn="true"), users have to authenticate themselves against the corporate identity provider each time they access the application even if single sign-on (SSO) is enabled.

To use Identity Authentication as a proxy to delegate authentication to an external corporate identity provider you have to configure trust with that corporate identity provider.

> → Tip
>
> If you want to change one corporate identity provider with another, for example move from a SAML identity provider to an OpenID Connect (OIDC) one, it's helpful to know the applications that the corporate identity provider uses. To see the applications that have established trust with a specific corporate identity provider, sign in to the administration console and go to *Identity Providers* > *Corporate Identity Providers* > *identity provider from the list* > *Trusting Applications*.

To configure trust with the corporate identity provider, follow the procedures below:

1. Configure Trust on the Corporate Identity Provider Side

Set up trust with Identity Authentication as a service provider.

Prerequisites

You have the SAML 2.0 metadata of Identity Authentication. For more information how to download the metadata, see Tenant SAML 2.0 Configurations [page 2103].

Context

Procedure

1. Register Identity Authentication as a service provider at the corporate identity provider.

   → Remember

   If you want to use IdP-initiated single sign-on (SSO) from your corporate identity provider, you have to add the parameter `sp=<sp_name>` to the assertion consumer service (ACS) endpoint configured on your corporate identity provider side for Identity Authentication.

   ✤ Example

   https://<the current ACS endpoint URL>?sp=<sp_name>>

   `sp` is the name of the SAML 2 service provider for which SSO is performed.

   To see how to download the SAML 2.0 metadata of Identity Authentication read Tenant SAML 2.0 Configurations [page 2103].

2. **Optional:** Download the corporate identity provider SAML 2.0 metadata.

   You need the corporate SAML 2.0 metadata for the setup of the trust on Identity Authentication. Optionally, you can make the configurations manually.

2. Configure Trust on Identity Authentication Side

Set up trust with a corporate identity provider in the administration console for SAP Cloud Identity Services.

Prerequisites

- You are assigned the *Manage Corporate Identity Providers* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have registered Identity Authentication as service provider at the corporate identity provider.
- You have the corporate identity provider SAML 2.0 metadata.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.

   > → Tip
   >
   > If you need to change the protocol, see Choose Identity Provider Type [page 2602].

4. Under *SAML 2.0*, choose *SAML 2.0 Configuration*.
5. Upload the corporate identity provider metadata XML file, use metadata URL, or manually enter the communication settings negotiated between Identity Authentication and the identity provider.

   > ⓘ Note
   >
   > Use a file with an extension `.xml`.

   When the identity provider metadata is uploaded, or the metadata URL is used, the fields are populated automatically with the parsed data from the XML file. The minimum configuration is to complete the *Name* field, add at least one single sign-on endpoint, and provide a signing certificate.

   You can add up to two signing certificates. Both signing certificates are accepted according to the certificate validity.

Field	Description
Choose: • Metadata File • Metadata URL	• The metadata XML file of the identity provider. • The URL with identity provider metadata. > ⚠ Restriction > The metadata URL must not contain query parameter.

Field	Description
Name	The entity ID of the identity provider.
Single Sign-On Endpoint URL	The URL of the identity provider single sign-on endpoint that receives authentication requests. → Remember Use *HTTP-Redirect* as default binding if you have a link which takes the users to a specific application with a specific corporate identity provider when `Identity Authentication` acts as proxy to multiple corporate identity providers (IdP), and the identity providers are more than one. For more information, see Create URLs to Access Applications with Specific Identity Providers [page 2605].
Single Logout Endpoint URL	The URL of the identity provider's single logout endpoint that receives logout messages.
Binding	The SAML-specified HTTP binding used by the identity provider showing how the various SAML protocol messages can be carried over underlying transport protocols.
Signing Certificate	A base64-encoded certificate used by the service provider to sign digitally SAML protocol messages sent to Identity Authentication. Use the *Add* button to add a second signing certificate. ⓘ Note If you have two certificates, you can choose a default one, to mark your primary certificate.

6. **Optional:** Choose the digest algorithm for signing outgoing messages from the dropdown list in the *Algorithm* section. You have the following options:

   - *SHA-1*
   - *SHA-256* - the default option

     → Remember

     The algorithm must be *SHA-256* if the Identity provider type is set at *Microsoft ADFS / Entra ID*.
   - *SHA-512*

7. **Optional:** Sign SAML 2.0 authentication requests. You have the following options:

Option	Notes
Sign authentication requests	Enabled - default configuration
Sign single logout messages	Enabled - default configuration

8. **Optional:** Enable or disable the *Include scoping* attribute to include or exclude the Scoping element in the SAML 2.0 request.

   > → Remember
   >
   > The default setting for the *Include scoping* is enabled. The Scoping element sent in the SAML 2.0 request is 1.
   >
   > If the identity provider type is set at *Microsoft ADFS / Entra ID* the default setting for the *Include scoping* is disabled and the Scoping element is not sent in the SAML 2.0 request.

9. Save your selection.

   Once the identity provider has been updated, the system displays the message `Identity provider <name of identity provider> updated`.

Next Steps

- Select the configured identity provider as the authenticating identity provider for the application. For more information, see Choose Default Identity Provider for an Application [page 2067].
- (Optional) Configure the Name ID Format Attribute Sent to the SAML 2.0 Corporate IdP [page 2577]

Related Information

Edit Administrator Authorizations [page 2309]
Configure IdP-Initiated SSO [page 2201]
Choose Identity Provider Type [page 2602]

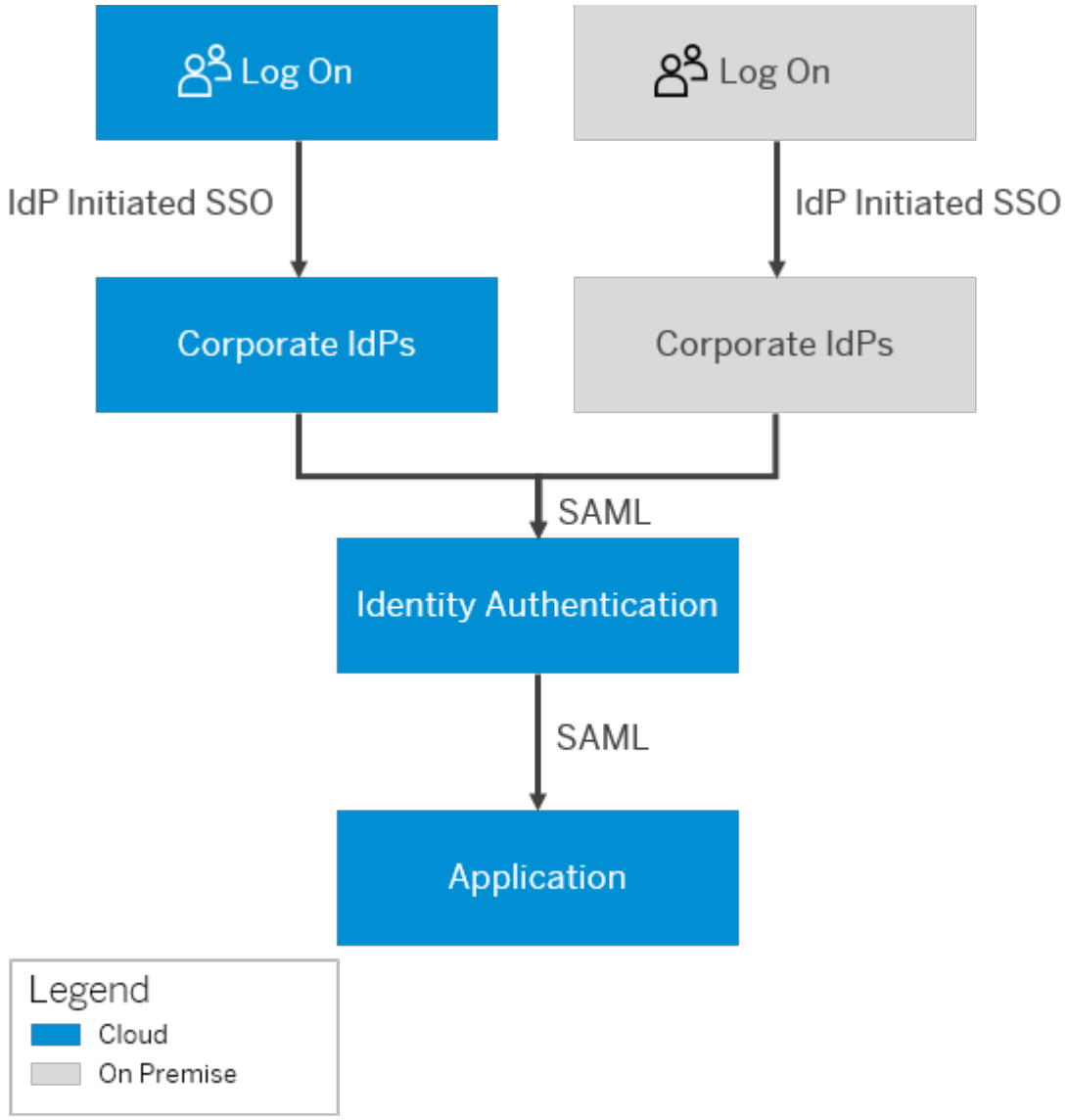
1.7.16.4.2.1 Configure IdP-Initiated SSO with Corporate Identity Providers

This document shows you how to configure identity provider (IdP) initiated single sign-on (SSO) with corporate identity providers.

This use case is suitable for customers and partners who need to provide access to a cloud application for their employees via their corporate identity providers. When Identity Authentication acts as proxy to multiple identity providers you can allow the partner users to login via their corporate identity providers.

In this scenario the authentication starts at the corporate identity provider (IdP), with Identity Authentication being in the role of an identity provider proxy. As such, Identity Authentication will act as an SAML 2.0 identity provider to the service provider, and as an SAML 2.0 service provider to the corporate identity provider or providers. Once a user is authenticated at the corporate identity provider, successive authentication requests from the service provider, which use the same corporate identity provider will not be forwarded to it while the session at Identity Authentication is active. Identity Authentication will issue assertions based on the user data received during the first authentication.

This scenario also supports authentication with more than one corporate identity provider.



Configuration Steps

For this scenario, the configurations are made by users with different roles in different systems.

Configuration and Roles

Role	System to be configured	What has to be configured	More Information
Application administrator	Consumer application	Trust	Configure Identity Authentication as a Trusted Identity Provider for the Application [page 2566]

Role	System to be configured	What has to be configured	More Information
Tenant administrator	Tenant of Identity Authentication	• Authenticating identity providers • (Optional) Configure the *Trust Corporate Identity Providers* feature in the administration console.	• Configure Identity Authentication To Trust the Corporate Identity Provider [page 2567] • Enable SSO with Corporate Identity Providers [page 2087]
Tenant administrator	Tenant of Identity Authentication	(Optional) User attributes and access for users. Enable the *Identity Federation* options in the administration console.	(Optional) Configure Additional Settings in Identity Authentication [page 2571]
Corporate IdP administrator	Corporate IdP	• Trust • Assertion consumer service endpoint	Configure the Corporate Identity Provider [page 2574]

For more information about the configuration steps, follow the links in the table or see the sections below.

Prerequisites

The IDP-Initiated SSO option in the tenant of Identity Authentication must be enabled.

> ⓘ Note
>
> IDP-Initiated SSO is enabled by default in Identity Authentication. For more information about how the tenant administrator can enable or disable this option, see Configure IdP-Initiated SSO [page 2201].

1.7.16.4.2.1.1 Configure Identity Authentication as a Trusted Identity Provider for the Application

Context

The following configuration is made by the administrator of the application.

As an administrator of the application, you have to configure Identity Authentication as a trusted identity provider for the application.

> ⓘ Note
>
> To do this, you will need the SAML 2.0 metadata of Identity Authentication. To receive the metadata, contact the tenant administrator of Identity Authentication.

Next Steps

Send the metadata of the service provider to the administrator of Identity Authentication. This is required for setting up the trust on the Identity Authentication provider side. For more information about how to set up th etrust, see Configure SAML 2.0 Service Provider [page 1941].

1.7.16.4.2.1.2 Configure Identity Authentication To Trust the Corporate Identity Provider

Context

In the scenario where Identity Authentication acts as an identity provider proxy, it is in the role of an identity provider for the application, and a service provider for the corporate identity provider. You should configure trusts with the service provider and the corporate identity provider.

The following configuration is made by the tenant administrator of Identity Authentication.

Procedure

1. Configure trust with the service provider via the administration console. For more information, see Configure SAML 2.0 Service Provider [page 1941].

   For this procedure, you will need the metadata from the service provider. If you do not have this, contact the administrator of the application.

   The service provider metadata that is used to configure the trust must contain the default assertion consumer service (ACS) endpoint that can process unsolicited SAML responses.

   > → Remember
   >
   > With SAP BTP, the endpoint is the URL of the application's protected page.
   >
   > If your scenario includes the configuration of the *Trust Corporate Identity Providers* option in the administration console, the service provider metadata that is used to configure the trust must contain also the assertion consumer (ACS) endpoint with the URL of the application's protected page and the index.

> ### Sample Code
>
> ```
> <ns3:AssertionConsumerService index="1" isDefault="false"
> Binding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-POST"
> Location="https://<application URL>/protected.jsp"
> />
> ```

2. Configure a trust with the corporate identity provider. For more information, see Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560].

   For this procedure, you will need the metadata from the corporate identity provider. If you do not have this, contact the administrator of the corporate identity provider.

3. Select identity provider. You have the following options:

   - If your scenario includes more than one corporate identity provider, configure the *Trust Corporate Identity Providers* feature in the administration console. For more information, see Enable SSO with Corporate Identity Providers [page 2087].
   - If your scenario includes only one corporate identity provider, set the configured identity provider as the authenticating identity provider for the application. For more information, see Choose Default Identity Provider for an Application [page 2067].

Next Steps

1. Send the `Entity ID` of the service provider to the administrator of the corporate identity provider. The administrator needs this information for the consumer assertion endpoint configuration.

   > ### → Tip
   >
   > The `Entity ID` of the service provider is in the administration console in *Applications* > *<application_name>* > *Trust* > *SAML 2.0 Configuration* > *Name*.

   > ### → Remember
   >
   > If your scenario includes the configuration of the *Trust Corporate Identity Providers* option in the administration console, send also the index of the ACS endpoint of the application's protected page. The administrator needs this information for the consumer assertion endpoint configuration.

2. Send the metadata of the tenant of Identity Authentication to the administrator of the service provider and the administrator of the corporate identity provider. They need the metadata for the trust configurations of the systems. For more information about how to download the tenant metadata, see Tenant SAML 2.0 Configurations [page 2103].

1.7.16.4.2.1.3 Enable SSO with Corporate Identity Providers

Tenant administrators can enable IdP-initiated Single Sign-On (SSO) from one, more than one or all configured corporate identity providers (IdPs).

Prerequisites

- You are assigned the *Manage Applications* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have more than one corporate identity provider, which is configured in the administration console. For more information on how to configure a corporate identity provider, see Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560] or Configure Trust with OpenID Connect Corporate Identity Provider [page 2579].
- (For SAML 2.0 applications) You have added the assertion consumer (ACS) endpoint with the URL of the application's protected page, and the index, in the metadata of that application. The ACS endpoint that you added should look like the following example:

  Sample Code

  ```
  <ns3:AssertionConsumerService index="1" isDefault="false"
  Binding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-POST" Location="https://
  <application URL>/protected.jsp"
  />
  ```

  For more information how to configure your ACS endpoint in the administration for Identity Authentication, see Configure SAML 2.0 Service Provider [page 1941].

Context

Applications can be configured to trust one, more than one, or all corporate identity providers configured in the administration console when identity provider (IdP) initiated Single Sign-On (SSO) is used. The user accesses the application via a URL provided by the corporate identity provider.

To enable IdP-initiated SSO with corporate identity providers configured in the administration console for SAP Cloud Identity Services follow the following procedure:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> ### Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under the *Conditional Authentication* section, choose the *Trust Corporate Identity Providers* list item.
6.

Option	Notes
Enable the slider to allow SSO with all configured corporate identity providers.	All corporate IdPs are allowed for sign-in. By default this option is disabled. If enabled, all configured corporate IdPs are hidden, and you can't select a specific IdP from the list.
Select a specific corporate IdP from the list.	→ Remember The slider on the top-right corner of the screen must be disabled. If enabled, all configured corporate IdPs are hidden

Once the application has been updated, the system displays the message `Application <name of application> updated`.

Results

The application trusts the selected corporate identity providers that are configured in the administration console for SAP Cloud Identity Services.

Related Information

Configure IdP-Initiated SSO with Corporate Identity Providers [page 2564]

1.7.16.4.2.1.4 (Optional) Configure Additional Settings in Identity Authentication

Context

The following configurations are made by the tenant administrator of Identity Authentication.

Follow the procedures in this document if you want to:

- **Restrict Access to Users in Identity Authentication User Store**
- **Restrict Access to Users in Certain Groups**
- **Send Specific Assertion and Name ID Attributes to the Application**
- **Apply Application Configurations**

Send Specific Assertion and Name ID Attributes to the Application

Context

When the *Use Identity Authentication user store* option is disabled, Identity Authentication sends to the application the same attributes it has received from the corporate identity provider.

When the *Use Identity Authentication user store* is enabled, Identity Authentication checks if a user with the respective unique identifier, written in the NameID attribute in the assertion coming from the corporate identity provider, exists in the user store of Identity Authentication. The following options exist:

- If the user exists, Identity Authentication issues a new nameID, assertion and default attributes, configured for the application.
- If the user does not exist in the user store of Identity Authentication, the application receives the nameID from the corporate IdP assertion and the attributes according to the application configuration.

This configuration can be used if you want the application to receive assertions and name ID attributes that are different from those sent by the corporate identity provider.

If you want the application to receive assertions and name ID attributes that are different from those sent by the corporate identity provider, do the following: Configuring User Attributes from the Identity Directory [page 1973] and Configure the Subject Name Identifier Sent to the Application [page 1988].

Procedure

1. Switch on the *Use Identity Authentication user store* option in the administration console. For more information, see Configure Identity Federation [page 2614].

2. Configure the user attributes and Name ID attribute sent to the application. For more information, see:
    - Configuring User Attributes from the Identity Directory [page 1973]
    - Configure the Subject Name Identifier Sent to the Application [page 1988]

Results

The application will receive in the assertion the attributes and name ID attribute that you have configured in Identity Authentication. They will be different from those that the corporate identity provider sent to Identity Authentication.

Restrict Access to Users in Identity Authentication User Store

Context

This configuration allows you to restrict access to the application to users who are in the Identity Authentication user store. Users who are not in the user store of Identity Authentication will not be able to access the application.

Procedure

1. Provision, or import the users that you will grant access to the application to via a CSV file import.

    For more information about how to import users in Identity Authentication, see Import or Update Users for a Specific Application [page 2348].
2. Switch on the *Use Identity Authentication user store* and *Allow Identity Authentication users only* options under *Identity Federation* in the administration console. For more information, see Configure Identity Federation [page 2614].

Results

Only users that are in the user store of Identity Authentication will be able to access the application. If a user is not part of the user store of Identity Authentication, this user receives the following message: `Sorry, but you are currently not authorized for access.`

The settings in the application configuration for assertion attributes and name ID attribute will be used for issuing the assertion. For more information, see Configuring User Attributes from the Identity Directory [page 1973]and Configure the Subject Name Identifier Sent to the Application [page 1988].

Restrict Access to Users in Certain Groups

Context

This configuration allows you to restrict the access to the application only to users who belong to certain groups. This option is suitable for scenarios with more than one corporate identity providers. The tenant administrator assigns different groups to different identity providers and users, and can thus specify that certain users come from specific identity providers by assigning the same groups to them. Identity Authentication can thus check if the users are authenticated by the identity provider that they belong to.

Procedure

1. Make sure that the required groups are created in the administration console of Identity Authentication. For more information, see Create a Group [page 2398].
2. Provision, or import the users that you will grant access to the application via a CSV file import.

   > ⚠ Caution
   >
   > The groups in the *Groups* column in your CSV file must match the groups that you created via the administration console.

   For more information about how to import users in Identity Authentication, see Import or Update Users for a Specific Application [page 2348].
3. Switch on the *Use Identity Authentication user store* and *Allow Identity Authentication users only* options under *Identity Federation* in the administration console. For more information, see Configure Identity Federation [page 2614].
4. Assign groups to the corporate identity provider. For more information, see Configure Identity Federation [page 2614].

Results

Only the members of these groups will be authorized to access applications after successful authentication. If a user is not part of the groups, this user receives the following message: `Sorry, but you are currently not authorized for access.`

Apply Application Configurations

Prerequisites

Context

This configuration allows you to apply the following application configurations, when Identity Authentication is used as a proxy.

Procedure

1. Provision, or import the users that you will grant access to the application to via a CSV file import.

   For more information about how to import users in Identity Authentication, see Import or Update Users for a Specific Application [page 2348].
2. Switch on the *Use Identity Authentication user store* and *Allow Identity Authentication users only* options under *Identity Federation* in the administration console. For more information, see Configure Identity Federation [page 2614].
3. Switch on the *Apply Application Configurations* option. For more information, see Configure Identity Federation [page 2614].

Results

The application configuration will be taken into consideration, when the users are authenticated by the corporate IdP.

1.7.16.4.2.1.5 Configure the Corporate Identity Provider

Context

The following configuration is made by the administrator of the corporate identity provider.

Procedure

1. Register Identity Authentication as a service provider.

   > ⓘ Note
   >
   > To do this, you will need the SAML 2.0 metadata of Identity Authentication. If you do not have this, contact the tenant administrator of Identity Authentication.

> → Tip
>
> For more information about how to register Identity Authentication as a service provider, consult the corporate identity provider documentation. If the corporate identity provider is also a tenant of Identity Authentication, see Configure Trust [page 1941].

2. Add the `sp=<sp_name>` parameter, and the index of the application's assertion consumer service (ACS) endpoint URL, if provided. Replace the `sp_name` with the `Entity ID` of the service provider. This parameter is needed for Identity Authentication to know which service provider to redirect the user to after successful authentication.

   > → Tip
   >
   > The ACS endpoint URL should have the following format: `https://<the current ACS endpoint URL>?sp=<sp_name>&index=<index_number>`. Request the `Entity ID` of the service provider, and the index of the application's protected page from the tenant administrator of Identity Authentication.
   >
   > The index is required when the application has multiple ACS endpoint URLs and the default one is not the one that should be used.

Results

Once the trust is configured, the user can access the application via the link sent by the corporate identity provider administrator. For more information about how to configure the link for the IdP-initiated SSO scenario, consult the corporate identity provider documentation.

> → Tip
>
> If your corporate identity provider is Identity Authentication, the link for IdP-Initiated SSO follows the pattern: https://<tenant_ID>.accounts.ondemand.com/saml2/idp/sso?sp=<sp_name>[&RelayState=<sp_specific_value>&index=<index_number>]. In this use case, replace the `sp_name` with the `Entity ID` of the tenant of Identity Authentication acting as the service provider. The `RelayState` is not mandatory and can be skipped. If your scenario includes only one corporate identity provider, the `index` parameter is not mandatory and also can be skipped. If your scenario includes more than one corporate identity provider, and the *Trust Corporate Identity Providers* feature in the administration console is enabled, the `index` is used to determine the ACS endpoint which could process unsolicited SAML responses. For more information about the configuration, see Configure IdP-Initiated SSO [page 2201].

Next Steps

Send the metadata of the corporate identity provider to the administrator of Identity Authentication. This will be needed for setting up the trust.

1.7.16.4.2.2 Enrich Assertion Attributes Coming from Corporate IdP

Tenant administrator can modify the assertion attributes received from the corporate identity provider (IdP) before they are sent to the application (service provider) that uses the corporate IdP for authentication.

Context

The attributes received from the corporate IdP by Identity Authentication are modified and thus sent to the application in the assertion.

The attributes are also put in the `id_token` if the application is OpenID connect. For more information, see Configuring OpenID Connect (OIDC) [page 2636].

For both the SAML 2.0 and OpenID Connect applications, you can configure attributes with:

- dynamic values - The assertion attributes can take up to two dynamic values. They are added into the assertions in the following pattern: `<prefix> ${<received_attribute>} <suffix>`. You can also use multivalue attributes.

  > ⚠ Restriction
  >
  > The combination of two multivalue attributes is not allowed. The assertion attributes can take two single value attributes or a single value and a multivalue attribute.

- static values

> ⚠ Restriction
>
> The `sap_licenses` attribute is not supported for the enrich assertion attributes scenario.

You can add up to 30 attributes per corporate IdP.

Identity Federation

When the application uses corporate IdP for authentication, the assertion attributes enriched in the administration console for SAP Cloud Identity Services are taken into consideration and sent to the application in the modified form, if the *Use Identity Authentication user store* option under *Identity Federation* is disabled.

If *Identity Federation* is configured, use the modified attributes in the *Default Attributes* section for the applications that use the corporate IdP for authentication. For more information, see Configuring Attributes Based on Flexible Expressions [page 1979].

> ⓘ Note
>
> You can also overwrite the `Subject Name Identifier` via the *Enrich Assertion Attributes* option. Identity Authentication sends the modified attribute to the application as `NameID` in the SAML 2.0 assertions, and as `sub` in the OpenID Connect tokens.
>
> When overwriting the `Subject Name Identifier` you can use the `${NameID}` variable to include the `NameID` received from the Identity Provider. You can't use the `${NameID}` variable for any other attributes.
>
> The `Subject Name Identifier` must be added in the following pattern:

Attribute	Value
NameID	`<prefix>${NameID}<suffix>`

To modify the assertion attributes coming from the corporate IdP, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.

   > → Tip
   >
   > If you need to change the protocol, see Choose Identity Provider Type [page 2602].

4. Under *Trust*, choose the *Enriched Assertion Attributes* list item.
5. Add the attributes as received from the corporate IdP with the new values to be sent to the application.
6. Save your configuration.

   If the operation is successful, you receive the message `Identity provider "<name of identity provider>" updated`.

1.7.16.4.2.3 (Optional) Configure the Name ID Format Attribute Sent to the SAML 2.0 Corporate IdP

Context

When Identity Authentication uses a SAML 2.0 identity provider (IdP) as an external authenticating authority it acts as a proxy to delegate authentication to that external corporate identity provider. As an identity provider proxy, Identity Authentication acts as a SAML 2.0 identity provider to the service provider, and as a SAML 2.0 service provider to the corporate identity provider. The requests for authentication sent by the service provider are forwarded by Identity Authentication to the corporate identity provider.

In this scenario, the tenant administrator can configure the name ID format that is sent to the corporate identity provider. The name ID format is used in the `Format` attribute of the `NameID` element that Identity Authentication sends to the corporate identity provider with the SAML 2.0 authentication request.

The tenant administrator can also configure if and how the `AllowCreate` attribute is sent to the corporate identity provider.

> → Remember
>
> To use Identity Authentication as a proxy to delegate authentication to an external corporate identity provider you have to configure trust with that corporate identity provider. For more information, see Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560].

To set the name ID format to be sent to the corporate identity provider, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.

   > → Tip
   >
   > If you need to change the protocol, see Choose Identity Provider Type [page 2602].
4. Under *SAML 2.0*, choose *Name ID Policy*.
5. Select the name ID format from the following:
   - *None* - Name ID format is not sent. Corporate identity provider decides which attribute to use.
   - *Default* - The name ID format received from the application is sent.
   - *Unspecified* - The name ID format `urn:oasis:names:tc:SAML:1.1:nameid-format:unspecified` is always sent.
   - *Email* - The name ID format `urn:oasis:names:tc:SAML:1.1:nameid-format:emailAddress` is always sent.
6. Select the AllowCreate configuration from the following:
   - *None* - `AllowCreate` attribute sent by the Service Provider is ignored.
   - *Default* - `AllowCreate` attribute sent by the Service Provider is forwarded to the Corporate IdP.
   - *True* - `AllowCreate` attribute with value "true" is always forwarded to the Corporate IdP.
   - *False* - `AllowCreate` attribute with value "false" is always forwarded to the Corporate IdP.
7. Save your selection.

   Once the identity provider has been updated, the system displays the message `Identity provider <name of identity provider> updated`.

Related Information

Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560]
Choose Identity Provider Type [page 2602]

1.7.16.4.3 Configure Trust with OpenID Connect Corporate Identity Provider

This document is intended to help you configure trust with an OpenID Connect (OIDC) corporate identity provider. In this scenario, Identity Authentication acts as a proxy to delegate the authentication to an OIDC-compliant corporate identity provider.

Context

Identity Authentication can use OIDC-compliant identity providers as external authenticating authorities. The service acts as a proxy to delegate authentication to the external corporate identity provider. The requests for authentication sent by the relying party are forwarded to the corporate identity provider.

As an identity provider proxy, Identity Authentication acts as an OpenID identity provider to the relying party, and as a relying party to the corporate identity provider. Once a user is authenticated at the corporate identity provider, successive authentication requests from relying parties, which use the same corporate identity provider aren't forwarded to it as long as the session at Identity Authentication is active. Identity Authentication issues JSON Web Tokens (JWTs) based on the user data received during the first authentication.

To use Identity Authentication as a proxy to delegate authentication to an external OpenID Connect corporate identity provider, configure trust with that corporate identity provider.

> → Tip
>
> If you want to change one corporate identity provider with another, for example move from a SAML identity provider to an OpenID Connect (OIDC) one, it's helpful to know the applications that the corporate identity provider uses. To see the applications that have established trust with a specific corporate identity provider, sign in to the administration console and go to *Identity Providers* > *Corporate Identity Providers* > *identity provider from the list* > *Trusting Applications*.

To configure trust with the corporate identity provider, use the following procedures:

1. Configure the Corporate Identity Provider Side

Configuration of Identity Authentication as a client in the corporate identity provider establishes the necessary trust and connectivity for delegating authentication through OpenID Connect. This step is required to enable secure user authentication with the corporate identity provider.

Prerequisites

- You have registered Identity Authentication as an application of the corporate identity provider.

> ### Note
>
> The corporate identity provider complies with OpenID Connect Discovery .

- You have created a client secret.

  > ### Note
  >
  > Use the client credentials of the application's registration for Client ID and Client Secret. Use corporate identity provider tenant as Issuer. You can retrieve the information by calling the discovery endpoint of the corporate identity provider:
  >
  > ```
  > https://<OpenID-Connect-IdP>/.well-known/openid-configuration
  > ```

- (For the authorization code flow) - You have configured the callback endpoint of the Identity Authentication tenant as `Redirect URI`

  ```
  https://<Cloud Identity Services domain>/oauth2/callback
  ```

  > ### Note
  >
  > The domain part has the following pattern:
  >
  > `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
  >
  > *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

> ### Note
>
> Use the *OIDC Callback URL* button to copy the callback URL of the Identity Authentication tenant.

Context

The information retrieved from `.well-known/openid-configuration` endpoint can contain:

Attribute		Notes
`issuer`		Required. Must be a valid URI. **Note** If you already have a corporate identity provider configuration with this issuer, configure your corporate identity provider to include an audience claim using the client ID, so the correct configuration can be identified. See also 3718653.
`jwks_uri`		Required. Must be a valid HTTPS URI.
	`id_token_signing_alg_values_supported`	Required. The JSON array must list the JWS signing algorithms (alg values) the OpenID provider supports for encoding ID token claims in a JWT.
	`response_types_supported`	Required. The JSON array lists the supported OAuth 2.0 `response_type` values.
	`subject_types_supported`	Required. The JSON array lists the subject identifier types this OpenID provider supports.
`authorization_endpoint`		Optional. If present it must be a valid HTTPS URI.
`end_session_endpoint`		Optional. If present it must be a valid HTTPS URI. **→ Remember** If the `end_session_endpoint` isn't supported by the OpenID Connect corporate identity provider, the corporate identity provider can't participate in single logout flows.
`token_endpoint`		Optional. If present it must be a valid HTTPS URI. **Note** If the `token_endpoint` isn't supported, it can't participate in browser-based SSO flows. JWT Bearer flow is still possible.

Attribute	Notes
`userinfo_endpoint`	Optional. If present it must be a valid HTTPS URI.
`grant_types_supported`	Optional. → Remember If `grant_types_supported` is provided in the metadata, it must contain `authorization_code`, which means that the OpenID Connect provider must support the authorization code flow.

2. Configure the Identity Authentication Side

Configure the corporate identity provider in the administration console for SAP Cloud Identity Services.

Prerequisites

- You are assigned the *Manage Corporate Identity Providers* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have registered Identity Authentication as an application at the corporate identity provider.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.
4. Under *Provider Configuration* enter the following information for the corporate identity provider:

Provider Configuration	Notes
Discovery URL	Required. Issuer or metadata URL of the corporate identity provider.

Provider Configuration	Notes
Name	Required. Enter a text string. Use a value that can serve as a URL parameter, such as a URI. These values don't need to be escaped. There is no default value. If you load the discovery URL before you enter a name, the system uses the name you chose when you created this identity provider configuration. For more information, see Create Corporate IdP in Administration Console [page 2558].
Metadata Refresh Interval	Refreshes the OpenID Connect metadata of the corporate identity provider automatically if it's older than the selected interval and there are logons which forward the request to the corporate identity provider. Optional. Choose from: • 24 hours (default choice) • 12 hours

5. Under *Client Authentication* enter the following information for the corporate identity provider:

Client Authentication Method	Optional. Choose from: • Client secret in body (default choice) • Client secret in authorization header • Private key JWT • Client assertion in body → Tip If possible, choose **`Private key JWT`** or **`Client assertion in body`**. → Recommendation If you connect Microsoft EntraID, use **`Client assertion in body`** and configure the trust in EntraID. This configuration enables an automatic trust update, because EntraID fetches the keys for signature validation automatically. Use the read-only *JWT Issuer* and *JWT Audience* values displayed below to complete the certificate credential configuration in Entra ID.
Client ID	Required. The Client ID of the application on the corporate identity provider side.

Client Secret	Required when *Client Authentication Method* is **`Client secret in body`** or **`Client secret in authorization header`**. The Client Secret of the application on the corporate identity provider side.
Audience	Required when *Client Authentication Method* is **`Private key JWT`**. The audience value that SAP Cloud Identity Services includes in the `aud` claim of the JWT client assertion sent to the corporate identity provider. Choose the value that the corporate identity provider expects: • *Issuer (default)*: the audience is set to the issuer identifier of the corporate identity provider, as retrieved from its OpenID Connect discovery document. • *Token endpoint*: the audience is set to the token endpoint URL of the corporate identity provider. Consult the documentation of your corporate identity provider to determine the correct value. For example, Microsoft Entra ID requires *Token endpoint*.
JWT Subject	Required when *Client Authentication Method* is **`Client assertion in body`**. The subject of the JWT client assertion sent to the corporate identity provider. By default, the system pre-fills this field with the name of the corporate identity provider. If the name is not available, the display name is used.
JWT Issuer	Read-only. Applies when *Client Authentication Method* is **`Client assertion in body`**. The issuer value that Identity Authentication includes in the JWT client assertion. Enter this value in the configuration of the corporate identity provider, for example when configuring certificate credentials in Microsoft Entra ID.
JWT Audience	Read-only. Applies when *Client Authentication Method* is **`Client assertion in body`**. The audience value that Identity Authentication includes in the JWT client assertion. Enter this value in the configuration of the corporate identity provider. **Note** The JWT Issuer and JWT Audience values are always identical.

6. **Optional:** If needed for your scenario, choose from the *Advanced Settings*.

   For more information, see:

   - Configure PKCE for Corporate Identity Providers [page 2588]
   - Enforce Nonces for Corporate Identity Providers [page 2589]

- Disable ID Token Hints for Corporate Identity Providers [page 2590]

7. **Optional:** View the OpenID Connect issuer and endpoints under the *Endpoints* section.

   The *Endpoints* section is read-only. The *Issuer* shown here is the issuer of the corporate identity provider retrieved from its metadata, not the Identity Authentication Issuer used in client assertions.

8. **Optional:** Add additional scopes if needed.

   You can have up to 20 scopes. The `openid` scope is added by default. Each scope can have a length of up to 99 characters.

   > → Tip
   >
   > If you're using Microsoft Entra ID as corporate identity provider, the recommended claims are:
   > - openid
   > - email
   > - profile

9. **Optional:** Choose the *Validate* button to check the configuration.

   An authorization code flow is run against the corporate identity provider. The configuration is validated in a new tab where additional information about the authorization code, the token with all the claims and scopes, and the token verification is provided.

   > → Tip
   >
   > When SAP Cloud Identity Services works with an OIDC corporate identity provider, we expect a response time within 10 seconds. The response time can be a problem for on-premise identity providers, which don't have defined service level agreements like cloud-based identity providers. To use this configuration, you must ensure that the corporate identity provider can respond within this timeframe. When the corporate identity provider doesn't respond quickly enough, SAP Cloud Identity Services presents the following error message:
   >
   > `OpenID provider cannot process the request because the configuration is incorrect. Please contact your system administrator.`

10. Save your configuration.

    Once the identity provider has been updated, the system displays the message `Identity provider <name of identity provider> updated`.

Next Steps

- Select the configured identity provider as the authenticating identity provider for the application. For more information, see Choose Default Identity Provider for an Application [page 2067].
- (Optional) Configure the Subject Name Identifier Sent to the OpenID Connect Corporate IdP [page 2587]

Related Information

Edit Administrator Authorizations [page 2309]

Logging OpenID Connect (OIDC) Tokens [page 2980]

Microsoft identity platform application authentication certificate credentials

1.7.16.4.3.1 Enrich Token Claims Coming from Corporate IdP

Tenant administrator can modify the token claims received from the corporate identity provider (IdP) before they are sent to the application that uses the corporate IdP for authentication.

Context

The claims are put in the JSON Web Tokens (JWTs) for the OpenID Connect application. For more information, see Configuring OpenID Connect (OIDC) [page 2636].

For the OpenID Connect applications, you can configure claims with:

- dynamic values - The claims can take up to two dynamic values. They are added into the tokens in the following pattern: `<prefix> ${<received_attribute>} <suffix>`. You can use also multivalue claims.

  > ⚠ Restriction
  >
  > The combination of two multivalue claims is not allowed.

- static values

You can add up to 30 claims per corporate IdP.

SAP Cloud Identity Services supports nested attributes. You can use up to five layers of nesting. For example, `${"kubernetes.io".serviceaccount.name}`.

Identity Federation

If *Use Identity Authentication user store* under *Identity Federation* is disabled, modify the token claims received from the corporate identity provider (IdP). The claims enriched in the administration console for SAP Cloud Identity Services are thus taken into consideration and sent to the application in the modified form. The application specific settings, the claims in the *Default Attributes* section, are ignored.

If *Use Identity Authentication user store* under *Identity Federation* is enabled, use the modified claims in the *Default Attributes* section for the applications that use the corporate IdP for authentication. For more information, see Configuring Attributes Based on Flexible Expressions [page 1979].

> ⓘ Note
>
> You can also overwrite the `Subject Name Identifier` via the Enriched Token Claims option. Identity Authentication sends the modified claim to the application as `subject` in the OpenID Connect tokens.
>
> The `Subject Name Identifier` must be added in the following pattern:

Claim	Value
NameID	`<prefix> ${<NameID>} <suffix>`

To modify the claims coming from the corporate IdP, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.
4. Choose the *Enriched Token Claims* list item.
5. Add the claims as received from the corporate IdP with the new values to be sent to the application.
6. Save your configuration.

   If the operation is successful, you receive the message `Identity provider "<name of identity provider>" updated`.

1.7.16.4.3.2 (Optional) Configure the Subject Name Identifier Sent to the OpenID Connect Corporate IdP

Define the claim which is used as subject name identifier.

Prerequisites

You have enabled the *Use Identity Authentication user store* option. For more information, see Configure Identity Federation [page 2614].

Context

In this scenario, the tenant administrator can define the claim from the token of the corporate identity provider which Identity Authentication should use as subject name identifier. The Subject Name Identifier configuration defines with which value the identity provider user will be searched in the Identity Authentication user store. For more information, see the Context of Configure Identity Federation.

To define the claim which is used as subject name identifier, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.
4. Under *Trust*, choose *Subject Name Identifier*.
5. Select the Subject Name Identifier from the following:
   - *None* - Subject claim (`sub`) is used as subject name identifier. Corporate IdP decides which attribute to use.
   - *Email* - Email claim (`email`) is used as subject name identifier.
6. Save your selection.

   Once the identity provider has been updated, the system displays the message `Identity provider <name of identity provider> updated`.

Related Information

Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560]
Choose Identity Provider Type [page 2602]

1.7.16.4.3.3 Configure PKCE for Corporate Identity Providers

If your corporate identity provider supports Proof Key for Code Exchange (PKCE) and you have public applications that aren't capable of keeping client secrets, such as mobile applications, native applications, and single page applications, we recommend enabling this feature.

Context

Proof Key for Code Exchange (PKCE) is an enhancement of the authorization code flow to mitigate the threat of malicious actors from intercepting the authorization code.

> **Note**
>
> We only support the code challenge method S256.

For more information, see RFC7636 .

Procedure

1. In the administration console for SAP Cloud Identity Services, navigate to the OpenID Connect configuration of your corporate identity provider.

   For more information, see Configure Trust with OpenID Connect Corporate Identity Provider [page 2579].
2. Enable the *Enable PKCE (S256)* check box.

   Use the *Validate* button to confirm that the configuration is still working.
3. Save your entries.

Related Information

Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow with PKCE [page 2650]

1.7.16.4.3.4 Enforce Nonces for Corporate Identity Providers

Some corporate identity providers require that applications send nonces. If your applications don't comply, enable this feature to ensure that one is sent.

Context

A nonce is a string associated with a client session and token and is used to mitigate replay attacks. If an application supplies a nonce, Identity Authentication forwards the nonce with requests to the corporate identity provider. By default, if there's no nonce, then Identity Authentication just forwards the request as is. When this feature is enabled and an application doesn't supply a nonce, Identity Authentication generates nonces before forwarding requests to the corporate identity provider.

Procedure

1. In the administration console for SAP Cloud Identity Services, navigate to the OpenID Connect configuration of your corporate identity provider.

   For more information, see Configure Trust with OpenID Connect Corporate Identity Provider [page 2579].
2. Enable the *Enforce Nonce* check box.

   Use the *Validate* button to confirm that the configuration is still working.
3. Save your entries.

1.7.16.4.3.5 Disable ID Token Hints for Corporate Identity Providers

Corporate identity providers sometimes have trouble when the ID token is included as a URL parameter in logout requests. Use this option to disable the inclusion of the ID token.

Context

Identity Authentication includes the ID token in the `id_token_hint` URL parameter. This parameter can help the corporate identity provider identify the applications relevant to this call. For example, the user triggered a logout and the identity provider needs to identify all the applications related to the session. Including the ID token can exceed the length of URL that the corporate identity provider can consume. Your best option is then to disable the `id_token_hint`.

Procedure

1. In the administration console for SAP Cloud Identity Services, navigate to the OpenID Connect configuration of your corporate identity provider.

   For more information, see Configure Trust with OpenID Connect Corporate Identity Provider [page 2579].
2. Enable the *Disable Logout ID Token Hint* check box.
3. Save your entries.

1.7.16.4.3.6 Enforce Issuer Validation for Corporate Identity Providers

To mitigate mix-up attacks, require corporate identity providers acting as authorization servers to identify themselves in their responses. The OAuth 2.0 Authorization Framework, as defined in RFC6749 , doesn't require the authorization server to identify itself in its responses for all grant types. Enable this option to require the corporate identity provider to identify itself in the `iss` parameter of the response.

Prerequisites

> ⚠ Caution
>
> Make sure that your corporate identity provider supports and is configured to issue the `iss` parameter. If the corporate identity provider doesn't include this parameter, SAP Cloud Identity Services rejects the response. You can't use this corporate identity provider for authentication as long as issuer validation is enforced.

To check if you corporate identity provider supports this feature, visit its discovery endpoint and look for `"authorization_response_iss_parameter_supported":true` . For an example of the discovery endpoint, see `https://accounts.sap.com/.well-known/openid-configuration`.

The service also rejects responses that include values in `iss` that aren't trusted identity providers. Verify that the corporate identity provider sends the issuer in your trust configuration.

For more information, Configure Trust with OpenID Connect Corporate Identity Provider [page 2579]

Context

With this implementation, SAP Cloud Identity Services supports RFC9207 .

Procedure

1. In the administration console for SAP Cloud Identity Services, navigate to the OpenID Connect configuration of your corporate identity provider.

   For more information, see Configure Trust with OpenID Connect Corporate Identity Provider [page 2579].
2. Enable the *Enforce Issuer Validation* check box.
3. Use the *Validate* button to confirm that the configuration is still working.
4. Save your entries.

1.7.16.4.3.7 Elliptic Curve Signing Certificates

SAP Cloud Identity Services supports elliptic curve (EC) signing certificates for use with OpenID Connect corporate identity providers that require EC-based signatures.

What Are Elliptic Curve Signing Certificates

Signing certificates allow SAP Cloud Identity Services to sign tokens and requests using a private key so that the recipient can verify the signature using the corresponding public key. By default, SAP Cloud Identity Services uses RSA-based signing certificates (RS256).

Elliptic curve (EC) signing certificates use the ECDSA algorithm family. The following EC algorithms are supported:

Algorithm	Standard Name	Curve
ES256	ECDSA using P-256 and SHA-256	P-256 (secp256r1)

Algorithm	Standard Name	Curve
ES384	ECDSA using P-384 and SHA-384	P-384 (secp384r1)
ES512	ECDSA using P-521 and SHA-512	P-521 (secp521r1)

EC algorithms produce shorter signatures and keys than RSA at equivalent security levels. Some corporate identity providers, particularly those that follow national digital identity standards, accept only EC-signed tokens.

Once you create an EC certificate, you assign it as the *Active* signing certificate in the OpenID Connect configuration of a specific corporate identity provider. This design allows you to share a single EC certificate across multiple corporate identity providers, while each corporate identity provider can independently switch to a different certificate without affecting others.

You can also generate EC certificates through the SAP Cloud Identity Services REST API. Specify `ES256`, `ES384`, or `ES512` as the value of the algorithm parameter in the certificate generation request.

Scope and Constraints

EC signing certificates in SAP Cloud Identity Services have the following scope:

Scope of EC Signing Certificates

Scope	Description
Supported use	Signing of tokens and requests sent toward an OpenID Connect corporate identity provider when you select the EC certificate as the active certificate for that corporate identity provider.
Not supported	You cannot use EC certificates for application token signing (tokens issued to end-user applications). They don't appear in application-level signing certificate lists. **Note** You cannot set EC certificates as the tenant default signing certificate. The tenant default must remain an RSA certificate (RS256 or RS512). If you don't explicitly select a certificate for a corporate identity provider, the system uses the tenant default RSA certificate.

Configuration Overview

To use an EC signing certificate with a corporate identity provider:

1. Generate an EC signing certificate at tenant level. See Generate Elliptic Curve Signing Certificates [page 2593].

2. Assign the certificate as the active signing certificate for the corporate identity provider. See Configure Identity Provider Certificates for Corporate Identity Providers [page 2594].
3. Provide the public key of the certificate to the corporate identity provider for signature verification. The public key is available in the JWKS endpoint of your SAP Cloud Identity Services tenant.

1.7.16.4.3.7.1 Generate Elliptic Curve Signing Certificates

Generate a signing certificate that uses an elliptic curve (EC) algorithm so that it can be assigned as the active signing certificate for a specific corporate identity provider.

Prerequisites

You understand the following constraints before generating an EC certificate:

- EC signing certificates can only be used as the active signing certificate for corporate identity providers. They are not available for application token signing and cannot be set as the tenant default.
- To assign the generated certificate to a corporate identity provider, see Configure Identity Provider Certificates for Corporate Identity Providers [page 2594].

Context

SAP Cloud Identity Services supports elliptic curve (EC) signing certificates at the tenant level. EC certificates use the ECDSA algorithm family (ES256, ES384, ES512) and can be assigned per corporate identity provider, allowing each corporate identity provider to use a different signing algorithm and key if required.

> ⚠ Restriction
>
> EC certificates cannot be configured as the *Default* signing certificate on tenant level. The tenant default must use an RSA algorithm (RS256 or RS512). EC certificates are only applied when explicitly assigned as the active certificate in the configuration of a specific corporate identity provider.

Procedure

1. In the SAP Cloud Identity Services admin console, go to *Tenant Settings* > *Single Sign-On*.
2. Under *OpenID Connect Configuration*, find the *Signing Certificates* section and choose *Generate New Certificate*.
3. From the *Algorithm* dropdown list, select the elliptic curve algorithm to use.

Algorithm	Curve	Hash
ES256	P-256	SHA-256
ES384	P-384	SHA-384
ES512	P-521	SHA-512

> **Note**
>
> For elliptic curve algorithms, the *Key Size* field is not applicable. The key size is determined by the selected curve and is not configurable.

4. Select the certificate *Validity* (1, 3, 6 months, 1, 3, 5, or 10 years).
5. Choose *Generate* to create the certificate.

Results

The new certificate appears in the *Signing Certificates* list with its algorithm (for example, *ES256*) shown in the *Algorithm* column. The certificate is not set as *Default* and is not shown in application-level signing certificate lists.

You can now assign this certificate as the active signing certificate for a corporate identity provider.

Next Steps

Assign the certificate to one or more corporate identity providers. See Configure Identity Provider Certificates for Corporate Identity Providers [page 2594].

1.7.16.4.3.7.2 Configure Identity Provider Certificates for Corporate Identity Providers

Select the signing certificate that SAP Cloud Identity Services uses when communicating with a specific OpenID Connect corporate identity provider.

Prerequisites

Generate the signing certificate you want to use. The certificate must exist in the tenant-level signing certificate list before you can assign it here. For more information, see the following:

- For RSA certificates (RS256, RS512): Tenant OpenID Connect (OIDC) Configurations [page 2117].
- For elliptic curve certificates (ES256, ES384, ES512): Generate Elliptic Curve Signing Certificates [page 2593].

Context

The *Identity Provider Certificates* section in the OpenID Connect configuration of a corporate identity provider controls which signing certificate SAP Cloud Identity Services uses when sending signed requests or tokens to that corporate identity provider. All signing certificates created at tenant level are available for selection here, regardless of algorithm.

Select an elliptic curve certificate (ES256, ES384, or ES512) to meet the signing algorithm requirements of corporate identity providers that only accept EC-based signatures.

> **Note**
>
> You cannot generate new signing certificates from this section. Certificate generation is only available at tenant level under *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration*.

Procedure

1. In the SAP Cloud Identity Services admin console, go to *Applications and Resources* > *Identity Providers*.
2. Select the corporate identity provider whose signing certificate you want to configure.
3. Choose the *OpenID Connect Configuration* tab.
4. In the *Identity Provider Certificates* section, review the list of available signing certificates.

   The table displays all signing certificates configured at tenant level. Each row includes the following information:

Column	Description
Certificate	The subject name or thumbprint of the certificate.
Algorithm	The signing algorithm, such as RS256, RS512, ES256, ES384, or ES512.
Valid Until	The expiry date of the certificate.
Active	Indicates the certificate currently selected as active for this corporate identity provider.

5. Select *Active* for the certificate you want to assign to this corporate identity provider.

   > **Note**
   >
   > Only one certificate can be active at a time for a given corporate identity provider. When you select a new certificate, the system deactivates the previously active one for this identity provider.

6. Save your changes.

Results

SAP Cloud Identity Services uses the selected certificate when signing requests sent to this corporate identity provider. If you selected an elliptic curve certificate, the system uses the EC algorithm defined by that certificate (ES256, ES384, or ES512) for signing.

The tenant *Default* certificate continues to be used for all other signing operations not covered by a corporate identity provider-specific selection, such as application token signing.

Next Steps

Verify that the corporate identity provider accepts tokens signed with the selected algorithm by testing the authentication flow.

Related Information

Generate an Elliptic Curve Signing Certificate [page 2593]
Elliptic Curve Signing Certificates [page 2594]
API Reference: Certificate Generation — Algorithm Parameter

1.7.16.4.4 Service Provider-Initiated Logout with Corporate Identity Providers

In this scenario, Identity Authentication is configured as an identity provider proxy and the corporate identity provider acts as an authenticating identity provider to the application. When a user triggers logout at an application, this request is propagated to Identity Authentication and any corporate identity providers, while the response returns by the same route.

The logout procedure is triggered by the user at the service provider and results in a logout request sent to the identity provider proxy. Consequently, the identity provider proxy processes the request and destroys any local session information about the user. The identity provider proxy then checks whether there are other service providers in the single sign-on (SSO) session and sends logout requests to all of them. In return, the service providers send logout responses to the identity provider proxy informing it that the logout process is successful. Finally, the identity provider proxy sends a logout response to the original requesting service provider or the service provider of the application.

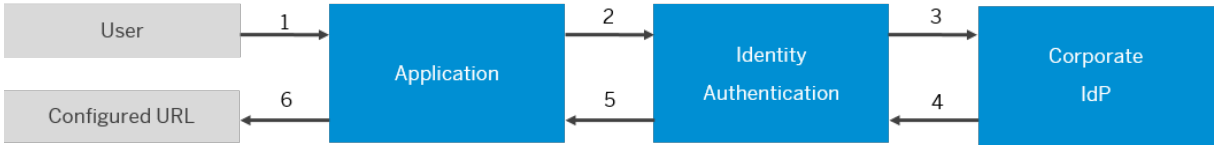
Depending on the protocol used to trust the corporate identity provider, you have different options for configuring logout redirect URLs. See the following table for details.

Protocol Differences for Logout Redirect URLs

Protocol of Service Provider	Options for Logout Redirect URL
SAML	You can configure a URL which is sent in the SAML 2.0 Logout Response as an extension. The URL can be used to redirect the users after logging out of the application. The URL is specific for each corporate identity provider with which Identity Authentication has established trust.
OIDC	If a client OpenID Connect (OIDC) application doesn't specify a post logout redirect URI in its logout request, you can specify a fallback URL. The service redirects the client to this URL after a successful logout.

Corporate Identity Provider Logout Flow

When Identity Authentication acts as a proxy to delegate authentication to a corporate identity provider, and the user who is logged on to one or more applications chooses the *Log Out* link in one of the applications, the following flow is in applies:



Logout Flow with Corporate Identity Providers

1. The user triggers a logout from the application.
2. Identity Authentication checks if the user is logged on to other applications. If yes, Identity Authentication sends logout requests to the applications, and closes the sessions.
3. Identity Authentication sends logout requests to the corporate identity provider or corporate identity providers, if there are more than one.
4. The user is logged out of the corporate identity providers.
5. The user is logged out of Identity Authentication.
    - For SAML: If the Redirect URL option is configured in the administration console for SAP Cloud Identity Services, the URL is sent as an extension in the SAML 2.0 Logout Response. This option can be used by the application to redirect the user to the destination from the link configured for the specific corporate identity provider.

      > ### Note
      >
      > This step is valid only if the application supports the redirect of the user, such as SAP SuccessFactors HCM Suite.

    - For OIDC: Applications should provide the `post_logout_redirect_uri` parameter with their logout request. If applications don't provide this parameter, you can configure a fallback URL. Otherwise, this configuration is ignored.
      For more information, see Call Identity Authentication End Session Endpoint [page 2740].

6. A logout response is sent to the initiating application.
   The response can differ depending on the protocol.
   - For SAML: The SAML service provider redirects to the Logout Redirect URL which was sent as an extension.
   - For OIDC: The application can redirect the user further.

1.7.16.4.4.1 Configure the Logout Redirect URL for Corporate Identity Providers

As a tenant administrator, you can configure an optional logout redirect URL to be used when the conditions apply.

Context

For more information about the conditions, see Service Provider-Initiated Logout with Corporate Identity Providers [page 2596].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.

   > → Tip
   >
   > If you need to change the protocol, see Choose Identity Provider Type [page 2602].
4. Choose *Logout Redirect URL*.
5. Enter the required data.
6. Save your entries.

Results

Once the user has logged out of the application, Identity Authentication sends:

- For SAML: The logout redirect URL in the SAML 2.0 Response as an extension to the application. The logout redirect URL can be used by the application to redirect the user to the URL configured for the corporate identity provider.
- For OIDC: The logout response to the provided `post_logout_redirect_uri` or fallbacks to logout redirect URL configured for the corporate identity provider.

Related Information

Single Logout Flows [page 2887]

1.7.16.4.5 Switch Protocols for Corporate Identity Providers

To take advantage of features of other protocols, switch the protocol of your corporate identity provider, for example from SAML to OpenID Connect (OIDC).

SAML isn't obsolete, but the trend is clear that many corporate identity providers are offering more new features with OIDC and not SAML. OIDC also enables you to take advantage of token exchange scenarios to support principal propagation. However, switching from OIDC to SAML is also possible.

To support your move to a new protocol for your corporate identity provider, you can switch all your applications at once or move one application at a time.

1.7.16.4.5.1 Impact on Applications

While this configuration doesn't necessarily affect your applications directly, what is application-specific are the attribute mappings between the attributes the corporate identity provider sends and the attributes the application needs. The identity provider might send different attribute names with the new protocol or even different values, for example, a different subject.

When identity federation is enabled, these mappings are application-specific in Identity Authentication.

For more information, see Configure Identity Federation [page 2614].

For each application, check the configurations, such as:

- Attribute mappings
- Configuration of SAML 2.0 request to corporate identity providers that includes authentication context and issuer.
- Identity provider-initiated single sign-on
  OIDC identity providers don't support such options. Switching to OIDC can break a scenario.
- You can't configure the signing and encryption methods for JWT tokens in OIDC identity providers.
- Single logout isn't supported by all OIDC providers.

1.7.16.4.5.2 Switch All Applications at Once

If you're confident that the new integration with the corporate identity provider will work immediately or any issues can be solved in your maintenance window, you can switch to the new corporate identity provider all at once.

Prerequisites

Plan for some downtime during the switch over.

Context

- Advantage: Efficiency. Move many applications to the new protocol at once.
- Disadvantage: Risk. Unless you've configured a test tenant with your corporate identity provider and your applications, you risk downtime if the applications encounter problems with the new setup.

Procedure

1. Go to the *Identity Provider Type* of your identity provider configuration and change the protocol.

   For more information, see Choose Identity Provider Type [page 2602].
2. Make any protocol-specific configurations.

   See also:

   - Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560]
   - Configure Trust with OpenID Connect Corporate Identity Provider [page 2579]
3. Make any application-specific configurations.

   For more information, see Impact on Applications [page 2599].

1.7.16.4.5.3 Switch One Application at a Time

If you have many applications, this approach requires more time. However, you can observe any changes in the behavior of your applications and fix them.

Context

- Advantage: Lower risk. If you have many applications, the potential for downtime is lower. You only migrate one application at a time.
- Disadvantage: Time. It takes longer to migrate all applications to the new protocol, because you migrate one application at a time.

Procedure

1. Create a new identity provider configuration under the new protocol by copying the configuration of your existing corporate identity provider under the old protocol.

   Use the *Copy Settings from Identity Provider* option.

   For more information, see Create Corporate IdP in Administration Console [page 2558].
2. Make any protocol-specific configurations.

   See also:

   - Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560]
   - Configure Trust with OpenID Connect Corporate Identity Provider [page 2579]
3. In each application configuration, change the identity provider to the new copy under the new protocol.

   For more information, see Authenticating Identity Provider for an Application [page 2065].
4. Make any application-specific configurations.

   For more information, see Impact on Applications [page 2599].
5. Check that you've moved all your applications from the old identity provider configuration.

   To see a list of the applications that use your old identity provider configuration, go to the identity provider configuration and choose *Trusting Applications*.
6. When the list of *Trusting Applications* is empty, delete the old identity provider configuration.

   For more information, see Delete Corporate Identity Providers [page 2619].

1.7.16.4.6 Choose Identity Provider Type

This topic shows you how to choose a type for the corporate identity provider.

Prerequisites

You are assigned the *Manage Corporate Identity Providers* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.

   > → Tip
   >
   > If you need to change the protocol, see Choose Identity Provider Type [page 2602].

4. Choose *Identity Provider Type*.
5. Select the radio button for the type of the identity provider.

   You can choose one of the following options:

   - SAML 2.0 Compliant
   - SAP Single Sign-On (SAML 2.0)
   - Microsoft ADFS / Entra ID (SAML 2.0)

     > ⚠ Caution
     >
     > For this type the digest algorithm of the corporate identity provider must be *SHA-256*. For more information, see 2. Configure Trust on Identity Authentication Side in Configure Trust with Corporate Identity Provider [page 2560].

   - OpenID Connect Compliant
6. Save your selection.

   Once the identity provider has been updated, the system displays the message `Identity provider <name of identity provider> updated`.

Related Information

Corporate Identity Providers [page 2556]
Edit Administrator Authorizations [page 2309]

1.7.16.4.7 Find Which Applications Trust a Corporate Identity Provider

The administration console for SAP Cloud Identity Services enables you to quickly find applications which trust a given corporate identity provider. This information enables you to find the applications you need to consider when you change or remove a corporate identity provider.

Context

Applications trust in corporate identity providers is founded in one or more of the following ways:

- Client authentication
  This source is configured by trust by issuer. For more information, see Configure JWT for OAuth Client Authentication [page 2003].
- Conditional authentication
  For more information, see Configure Conditional Authentication for an Application [page 2076].
- Trust identity providers
  For more information, see Enable SSO with Corporate Identity Providers [page 2087].

> ⓘ Note
>
> Only conditional authentication and trust in identity providers can't be combined.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.

   > → Tip
   >
   > If you need to change the protocol, see Choose Identity Provider Type [page 2602].

4. Choose *Trust* > *Trusting Applications*.
   The system displays a list of applications which trust this corporate identity provider. Search for a specific application or sort by the source of trust.

Next Steps

To navigate to the source of the trust configuration in the relevant application, choose the ⊘ *(check mark)* icon.

1.7.16.4.8 Configure Login Hint Parameter

Tenant administrator can configure the login hint parameter via the administration console for SAP Cloud Identity Services.

Context

The configuration of the login hint parameter affects the following scenarios:

- Identity Authentication acting as proxy to delegate authentication to an external corporate identity provider (IdP). The user is authenticated via that corporate identity provider.
- Identity Authentication acting as a conditional proxy where an external corporate identity provider is configured to be used as an authenticating IdP under certain conditions.

The login hint parameter helps the user when he or she is known to the service provider (SP) or relying party. Thus it prevents the user from re-typing the user identifier on the logon. If the corporate IdP supports the login hint parameter, then it requests only the user credentials.

If the corporate IdP does not support the login hint parameter, it requires both the user identifier and password. You can configure the login hint parameter for corporate IdP via the administration console for SAP Cloud Identity Services.

There are two aspects in the configuration of the login hint parameter: the value of the parameter, and how it is sent to the corporate IdP. The *How it is sent to the corporate IdP* configuration is only relevant for SAML 2.0 corporate identity providers and depends on the *Value* configuration:

Configuration Options

Value	How it is sent to the corporate IdP	Additional Info
User Input	• Login Hint URL parameter - the login hint is sent in the URL as `login_hint parameter` • Authentication Request - the login hint is sent in the SAML 2.0 authentication request as <`Subject`> attribute (for SAML 2.0 only)	As `login_hint` will be sent the user identifier provided at the conditional authentication sign in screen of Identity Authentication.
None	no login hint parameter is sent	The user will need to provide user identifier and password at the corporate identity provider sign in screen.
User Attributes • Login Name • Email	• Login Hint URL parameter - the login hint is sent in the URL as `login_hint parameter` • Authentication Request - the login hint is sent in the SAML 2.0 authentication request as <`Subject`> attribute (for SAML 2.0 only)	As `login_hint` will be sent one of the following: • Login Name • Email .

> **Note**
>
> The *User Input* is the default value option.
>
> The *User Attributes* option is valid only for users that exist in the Identity Authentication local user store. If the user is missing the specified attribute in the configuration, then login hint parameter will not be sent to the corporate IdP.
>
> If the user does not exist in Identity Authentication local user store, then the User Input will be sent as login hint to the corporate IdP.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.

   > → Tip
   >
   > If you need to change the protocol, see Choose Identity Provider Type [page 2602].

4. Set a value for the login hint parameter:
   - User Input
   - None
   - User Attribute (local users only)
5. (For SAML 2.0 providers only) Set how the login hint parameter should be sent to the corporate IdP:
   - *Login Hint URL parameter* -
   - *Authentication Request*

   > → Remember
   >
   > The options are valid only if *User Input* or *User Attribute (local users only)* values are set.

6. Save your configuration.

1.7.16.4.9 Create URLs to Access Applications with Specific Identity Providers

Enhance the user experience by eliminating the need for users to determine which identity provider to use. Users often don't have the context to choose the correct identity providers.

Sometimes you have multiple identity providers in your landscape, whether they are corporate identity providers or your SAP Cloud Identity Services - Identity Authentication tenant. By providing your users with preconfigured URLs, you can direct them to authenticate with specific identity providers for specific applications.

How you prepare these URLs depends on whether the user starts at the identity provider or at the application. It's further modified by your use case. Some typical use cases are described as follows.

Opting for the Local Identity Provider

You can enable users that are stored in SAP Cloud Identity Services to log on with their Identity Authentication credentials. Thus, the employees log on to the application with their corporate credentials, while the external users, such as clients, partners, or SAP support users are authenticated by the local Identity Authentication tenant. Another example is if you're developing an application and you want to use a test user in the local identity provider instead of creating a test user in the corporate identity provider. The following figure illustrates this architecture.

Sign On With the Local Identity Provider and not the Corporate Identity Provider

Example

Example approuter link:

```
https://approuter-myapplication.cfapps.eu20-001.hana.ondemand.com?
sap_idp=custom.sap
```

Note that the approuter needs the origin key for the custom identity provider for the SAP BTP account.

Example link:

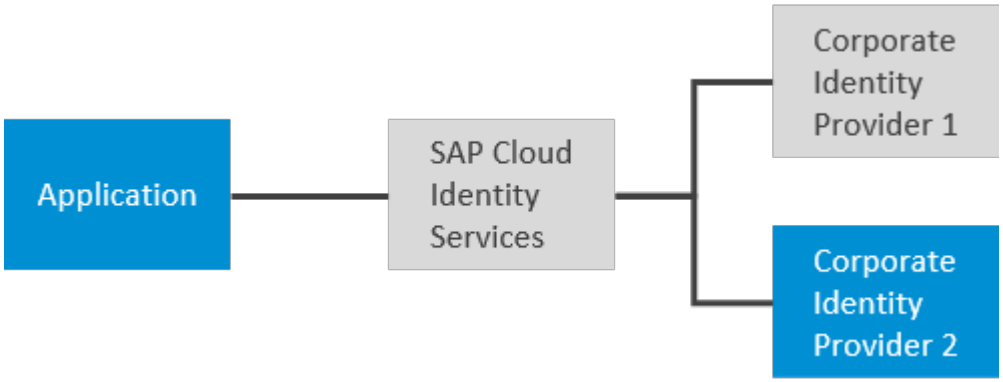
```
https://myapplication.cfapps.eu20-001.hana.ondemand.com?sap_idp=https://
my_sci_tenant.accounts.ondemand.com
```

Example with local value:

```
https://myapplication.cfapps.eu20-001.hana.ondemand.com?sap_idp=local
```

Multiple Corporate Identity Providers

If your application doesn't define rules to determine which identity provider a user authenticates with, SAP Cloud Identity Services offers the choice to your users. To avoid users choosing the wrong identity provider, you can limit their choices. The following figure illustrates this architecture.

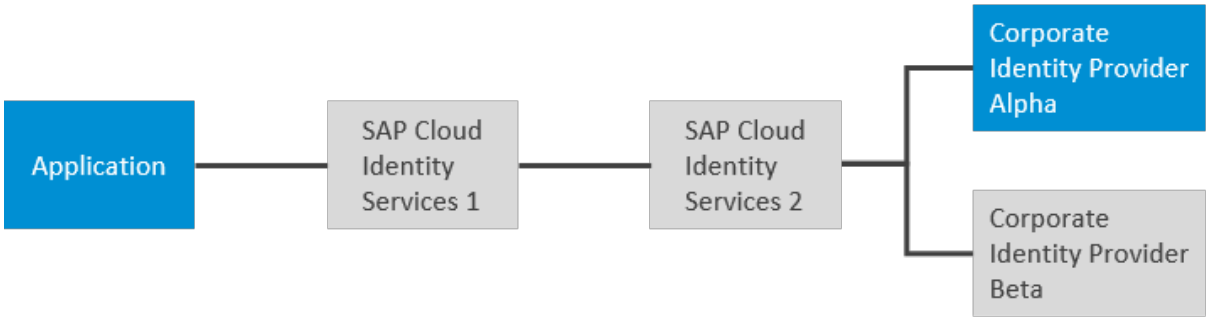


Help Direct Users to the Correct Identity Provider

> ### Example
>
> Example approuter link:
>
> ```
> https://approuter-myapplication.cfapps.eu20-001.hana.ondemand.com?
> sap_idp=https://corp_idp_2.mydomain.com
> ```
>
> Example link:
>
> ```
> https://myapplication.cfapps.eu20-001.hana.ondemand.com?sap_idp=https://
> corp_idp_2.mydomain.com
> ```

Deep Hierarchy of Identity Providers

Your landscape has multiple levels of identity providers, perhaps through subsidiaries and conglomerate organizations. You want to direct your users to a specific identity provider, but this identity provider isn't directly known by your SAP Cloud Identity Services tenant. In such cases, provide the chain of identity providers to which the user must be redirected to reach the authenticating identity provider. The following figure illustrates this architecture.



Deep Structure of Identity Providers

Example

Example approuter link:

```
https://approuter-myapplication.cfapps.eu20-001.hana.ondemand.com?
sap_idp=https://scis_2.mydomain.com,corp_idp_alpha.mydomain.com
```

Example link:

```
https://myapplication.cfapps.eu20-001.hana.ondemand.com?sap_idp=https://
scis_2.mydomain.com,corp_idp_alpha.mydomain.com
```

Identity-Provider-Initiated Single Sign-On

Restriction

This option is only supported by SAML.

Identity-provider-initiated single sign-on (IdP-initiated SSO) is used for central portals in an enterprise from which users access applications. Users log on to the central portal first. You need to identify the identity provider the user must log on with. You also need to identify the service provider that the user is forwarded to after authenticating. The identity provider can be SAP Cloud Identity Services or a corporate identity provider or a hierarchy of corporate identity providers. Another unique aspect of IdP-initiated SSO is that the URL pattern you use is determined by SAP Cloud Identity Services. Otherwise, the URL is defined by the application. The following figure illustrates this architecture.

Identity-Provider-Initiated Single Sign-On

Example

Example SAML URL:

```
https://my_sci_tenant.accounts.ondemand.com/saml2/idp/sso?sp=https://
myapplication.cfapps.eu20-001.hana.ondemand.com&idp=https://
corp_idp_1.mydomain.com
```

1.7.16.4.9.1 Creating URLs for Applications to Authenticate with Specific Identity Providers

The URL to access an application is generally determined by the application. When you have multiple identity providers available, you can steer users to the correct identity provider with the `idp` parameter.

Prerequisites

- You have set a corporate IdP as the default identity provider in the administration console.
  For more information, see Choose a Corporate Identity Provider as Default [page 2072].
- The application supports selection of the corporate IdP with the idp parameter in the application URL. See the documentation of the application.

  > **Example**
  >
  > Example: Many BTP applications use the application router, which requires the `sap_idp` query parameter.
  >
  > For more information, see @sap/approuter at *npm*.

- If you have enabled the *Allow Identity Authentication Users Log On* option, you can use the reserved name **`local`** to authenticate with your Identity Authentication tenant.
  For more information, see (Optional) Use the Allow Identity Authentication Users Log On Option [page 2074].

Procedure

Use the `idp` parameter in the URL for OIDC or in the URL or body for SAML.

> **Caution**
>
> It depends on the service provider whether the parameter can be passed in body or URL. Identity Authentication doesn't control how the SAML 2.0 request is sent.

Parameters

Parameter	Description
`idp`	Name of the identity provider as defined under the *Name* in the *Trust* configuration of the identity provider. • For SAML, see Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560]. • For OIDC, see Configure Trust with OpenID Connect Corporate Identity Provider [page 2579]. If you need a chain of identity providers, use a comma without spaces (,) to separate them. Use the value **`local`** to override the conditional authentication configuration and authenticate with Identity Authentication instead.

1.7.16.4.9.2 Creating URLs for IdP-Initiated Single Sign-On

Direct users to a authenticate at an IdP before accessing an application. For IdP-Initiated SSO, the URL syntax is determined by SAP Cloud Identity services.

Prerequisites

- You have enabled IdP-initiated Single Sign-On (SSO) from all configured corporate IdPs. For more information, see Enable SSO with Corporate Identity Providers [page 2087] .
- You have configured one of the following options:
- The application supports IdP-initiated Single Sign-On (SSO). For more information, see Security Assertion Markup Language (SAML) V2.0 Technical Overview .

Context

This option is available to SAML applications.

Procedure

Contruct the URL from the following pattern.

```
https://<tenant_ID>.accounts.ondemand.com/saml2/idp/sso?
sp=<sp_name>&idp=<corporateIdP1_name>,<corporateIdP2_name>
```

Parameters

Parameter	Description
`sp`	Redirect target after authentication. The `<sp_name>` value equals to the Entity ID of the service provider.
`idp`	Name of the identity provider as defined under the *Issuer* in the *Tenant Settings*. For more information, see Tenant SAML 2.0 Configurations [page 2103]. If you need a chain of identity providers, use a comma without spaces (,) to separate them.

> → Remember
>
> If there is more than one corporate IdP in the IdP-initiated link, use *HTTP-Redirect* as default binding in the trust configuration with SAML 2.0 Corporate Identity Provider. For more information, see Configure Trust on Identity Authentication Side [page 2562].

1.7.16.4.10 Forward All SSO Requests to Corporate IdP

Identity Authentication can be configured to forward all single sign-on(SSO) requests to the corporate identity provider.

Prerequisites

- You are assigned the *Manage Corporate Identity Providers* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have configured Identity Authentication to use a corporate identity provider as an external authenticating authority. For more information, see Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560] or Forward All SSO Requests to Corporate IdP [page 2611].
- You have selected the configured identity provider as the authenticating identity provider for your applications. For more information, see Choose Default Identity Provider for an Application [page 2067].

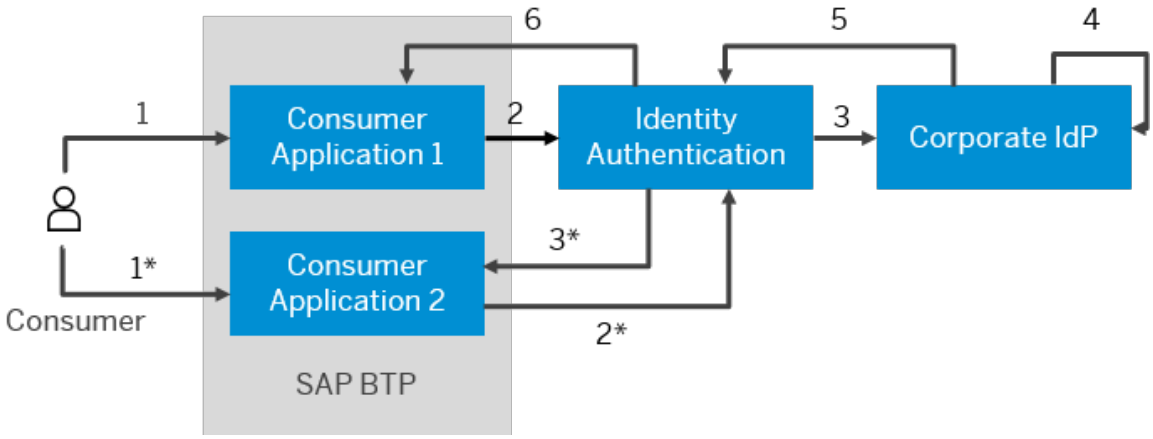
Context

The *Forward All SSO Requests to Corporate IdP* option allows the corporate identity provider (IdP) to process all single sign-on (SSO) requests when Identity Authentication acts as a proxy.

By default, the *Forward All SSO Requests to Corporate IdP* option is disabled. In this case Identity Authentication reuses the active session of the user.

When *Forward All SSO Requests to Corporate IdP* option is enabled, Identity Authentication sends new authentication request for each application.

Forward All SSO Requests to Corporate IdP Disabled



1. User tries to access *Application 1*.

2. *Application 1* sends authentication request to Identity Authentication.

3. Identity Authentication acts as proxy; forwards the request to corporate IdP.

4. Corporate IdP logs on the user.

5. Corporate IdP returns authentication response to Identity Authentication.

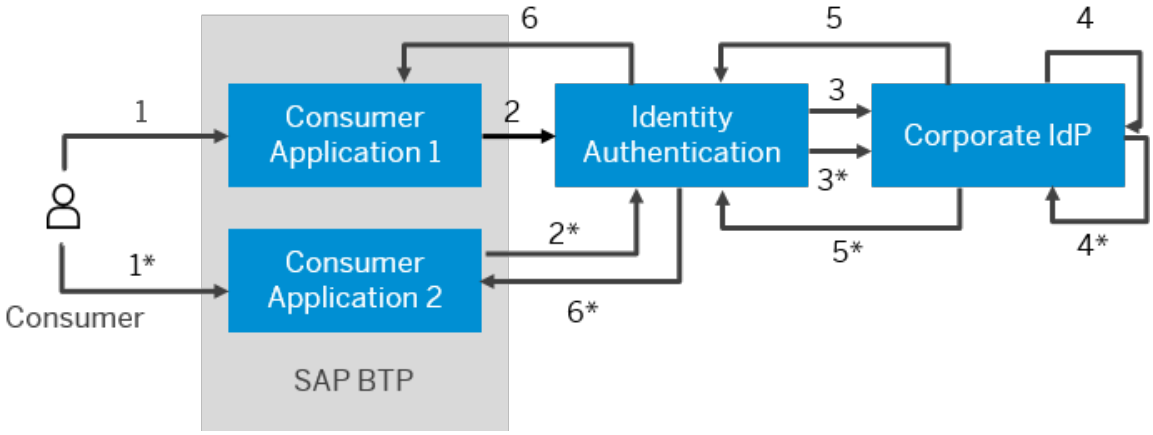
6. Identity Authentication returns authentication response to *Application 1*.

1*. User tries to access *Application 2*.

2*. *Application 2* sends authentication request to Identity Authentication.

3*. The user has an active session. Identity Authentication reuses the session and returns the authentication response to *Application 2*.

Forward All SSO Requests to Corporate IdP Enabled



1. User tries to access *Application 1*.

2. *Application 1* sends authentication request to Identity Authentication.

3. Identity Authentication acts as proxy; forwards the request to corporate IdP.

4. Corporate IdP logs on the user.

5. Corporate IdP returns authentication response to Identity Authentication.

6. Identity Authentication returns authentication response to *Application 1*.

1*. User tries to access *Application 2*.

2*. *Application 2* sends authentication request to Identity Authentication.

3*. Identity Authentication acts as proxy; it forwards the request to corporate IdP, though the user has an active session..

4*. Corporate IdP logs on the user.

5*. Corporate IdP returns new authentication response to Identity Authentication.

6*. Identity Authentication returns new authentication response to *Application 2*.

To enable the *Forward All SSO Requests to Corporate IdP*, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure for *Forward All SSO Requests to Corporate IdP*.
    - If you do not have an identity provider in your list, click the *Add* button to create one, and proceed with the configuration.
    - If you have an identity provider in your list, choose the one that you want to configure.
4. Use the slider next to *Forward All SSO Requests to Corporate IdP* to enable or disable it.

    If the operation is successful, the system displays the message `Identity provider <name of identity provider> updated`.

1.7.16.4.11 Configure Identity Federation

Tenant administrators can configure whether the attributes are taken from the assertion of the corporate identity provider or from the user store of Identity Authentication, and can restrict access based on the user profile.

Prerequisites

- You are assigned the *Manage Corporate Identity Providers* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have configured Identity Authentication to use a corporate identity provider as an external authenticating authority. For more information, see Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560] or Configure Identity Federation [page 2614].
- You have selected the configured identity provider as the authenticating identity provider for your application. For more information, see Choose Default Identity Provider for an Application [page 2067].
- You have imported the users, authenticated by the corporate identity provider, in Identity Authentication. For more information about how to import users, see Import or Update Users for a Specific Application [page 2348].

Context

The *Identity Federation* option allows you to choose whether the user attributes are taken from the corporate IdP assertion or from Identity Authentication user store. You can also restrict access based on the user profile, and additionally apply the custom application configurations for the authentication and access policies.

User Store

By default, *Use Identity Authentication user store* is disabled.

In scenarios when the application is using for authentication a corporate identity provider, and the *Use Identity Authentication user store* option is disabled, the user attributes, the name ID attribute, and the default attributes configurations in the administration console for SAP Cloud Identity Services are not relevant. In such scenarios, Identity Authentication sends to the application the same attributes it has received from the corporate identity provider. For more information about the corporate identity provider scenario, see Corporate Identity Providers [page 2556].

When *Use Identity Authentication user store* option is enabled, the application checks if the users authenticated by the corporate identity provider exist in the Identity Directory, the user store of Identity Authentication. The existence check is done with the name identifier sent by the corporate identity provider for the identifying attributes `uid`, `loginName`, `emails` and `phoneNumber`.

For users that exist in the Identity Directory, data from the user store is taken and the subject name identifier, assertion and default attributes according to the application configuration are sent. For users with no profile in Identity Authentication, the application receives the nameID attribute from the corporate IdP assertion, and the attributes according to the application configuration.

User Access

When *Allow Identity Authentication users only* is enabled, only the users that exist in Identity Authentication can access the application.

By default, *Allow Identity Authentication users only* is disabled, and all users successfully authenticated to the corporate IdP are allowed.

Restrict Logon to Members of Certain Groups

You can assign a group or groups to the corporate identity provider. Only users from the Identity Authentication that are members of the assigned group can access the application.

> → Remember
>
> You can assign groups, only when *Allow Identity Authentication users only* option is enabled. If disabled, the assigned groups option is not taken into consideration.

Restrict Logon to Users with Certain Email Domains

Only users from the Identity Authentication user store whose email domains are allowed can access the application.

> → Remember
>
> You can allow email domains from the allow list, only when the *Allow Identity Authentication users only* option is enabled. If disabled, the allow list option is not taken into consideration.

Application Configuration

By default, *Apply Application Configurations* is disabled.

When enabled, the custom application configurations for authentication and access policies are applied.

When this option is enabled, the user is authenticated by the corporate IdP, then Identity Authentication checks for the configurations for authentication and access policies on application level, and applies them. They include Risk-Based-Authentication, Terms of Use, Privacy Policy, Upgrade and Registration Forms.

> → Remember
>
> The concurrent user access configuration is always applied for the application regardless of the *Apply Application Configurations* state.

Use Identity Authentication user store

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure for *Identity Federation*.

- If you do not have an identity provider in your list, click the *Add* button to create one, and proceed with the configuration.
- If you have an identity provider in your list, choose the one that you want to configure.

4. Choose *Identity Federation* to configure the options.
5. Use the slider next to *Use Identity Authentication user store* to enable it.

   If the operation is successful, the system displays the message `Identity provider <name of identity provider> updated`.

Allow Identity Authentication users only

Prerequisites

You have enabled *Use Identity Authentication user store* option. For more information about how to enable the option, see *Use Identity Authentication user store* in this topic.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure for *Identity Federation*.
   - If you do not have an identity provider in your list, click the *Add* button to create one, and proceed with the configuration.
   - If you have an identity provider in your list, choose the one that you want to configure.
4. Choose *Identity Federation* to configure the options.
5. Use the slider next to *Allow Identity Authentication users only* to enable it.

   If the operation is successful, the system displays the message `Identity provider <name of identity provider> updated`.

Apply Application Configurations

Prerequisites

You have enabled *Allow Identity Authentication users only* option. For more information about how to enable the option, see *Allow Identity Authentication users only* in this topic.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure for *Identity Federation*.
   - If you do not have an identity provider in your list, click the *Add* button to create one, and proceed with the configuration.
   - If you have an identity provider in your list, choose the one that you want to configure.
4. Choose *Identity Federation* to configure the options.
5. Use the slider next to *Apply Application Configurations* to enable it.

   If the operation is successful, the system displays the message `Identity provider <name of identity provider> updated`.

(Optional) Restrict Logon Only to Members of Certain Groups

Prerequisites

You have enabled *Allow Identity Authentication users only* option. For more information about how to enable the option, see *Allow Identity Authentication users only* in this topic.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.

   > → Tip
   >
   > If you need to change the protocol, see Choose Identity Provider Type [page 2602].

4. Choose *Identity Federation*.

   You will see a list of the groups assigned to this corporate identity provider. If no groups are assigned, the list will be empty.
5. Choose the *Add* button.
6. Select the groups that you want to assign to this corporate identity provider.

   The list does not include the groups that are already assigned to the corporate identity provider.

   If you do not have any groups in your list, you can create one. For more details about how to create groups and assign the group to a user, see Related Information.
7. Save your changes.

If the operation is successful, the system displays the message `Identity provider <name of identity provider> updated`.

> **Note**
>
> To unassign groups, select the groups you want to unassign, choose the *Remove* button, and confirm the operation in the dialog.
>
> Users that belong only to the unassigned groups will not be able to access the application any more.

(Optional) Restrict Logon Only to Users with Specific Email Domains

Prerequisites

You have enabled *Allow Identity Authentication users only* option. For more information about how to enable the option, see *Allow Identity Authentication users only* in this topic.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.

   > **→ Tip**
   >
   > If you need to change the protocol, see Choose Identity Provider Type [page 2602].

4. Choose *Identity Federation*.

   You will see a list of the allowed email domains for this corporate identity provider. If no email domains are allowed, the list will be empty.
5. Choose the *Add* button.
6. Type the email domains that you want to allow for this corporate identity provider.
7. Save your changes.

   If the operation is successful, the system displays the message `Identity provider <name of identity provider> updated`.

   > **Note**
   >
   > To remove the email domains, select them from the list, choose the *Remove* button, and confirm the operation in the dialog.

1.7.16.4.12 Delete Corporate Identity Providers

As a tenant administrator, you can delete one or more corporate identity providers in a tenant of Identity Authentication.

Prerequisites

- You are assigned the *Manage Corporate Identity Providers* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have at least one corporate identity provider that you want to delete.
- The identity provider you want to delete must not be used by an application. For more information about how to choose an identity provider for an application, see Authenticating Identity Provider for an Application [page 2065].

Context

A *Delete Identity Providers* operation removes the identity providers and all of their configurations from the tenant of Identity Authentication.

> **Note**
>
> If you want to delete an identity provider that is in use for conditional authentication, client authentication, or is set as a trusted identity provider or a default identity provider for an application, edit the rules or the client authentication, or change the trusted or default identity provider for the application. For more information see Authenticating Identity Provider for an Application [page 2065].

To delete one or more identity providers, not used not used by applications in the tenant, choose one of the following options:

Delete Multiple Corporate Identity Providers

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Choose the icon in the left-hand panel.

   This operation activates the *Delete Identity Providers* mode.
4. Select the identity provider or identity providers that you want to delete.

5. Choose the *Delete* button.
6. Confirm the operation in the pop-up dialog.

   The system deletes only the identity providers that are not used by applications.

   Once the identity provider or identity providers have been deleted, the system displays the message `<number> identity providers deleted`.

Delete Single Corporate Identity Provider

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the identity provider that you want to delete.

   > → Tip
   >
   > Type the name of the identity provider in the search field to filter the list items.

4. Choose the *Delete* button in the right-hand panel to delete the selected corporate identity provider.
5. Confirm the operation in the pop-up dialog.

   The system deletes only the identity provider that is not used by applications.

   Once the identity provider has been deleted, the system displays the message `Identity provider deleted`.

Related Information

Edit Administrator Authorizations [page 2309]
Choose Default Identity Provider for an Application [page 2067]

1.7.17 Configuring External Authentication Providers

Configure authentication providers in the administration console for SAP Cloud Identity Services to manage users from external providers.

Configure Authentication Provider To Migrate User Passwords from SAP SuccessFactors Systems to Identity Authentication [page 2621]

Configure Authentication Provider To Migrate User Passwords from SAP Learning Management System to Identity Authentication [page 2626]

Configure Authentication Provider To Migrate User Passwords from SAP Fieldglass to Identity Authentication [page 2629]

Configure Authentication Provider To Migrate User Passwords from SAP Concur to Identity Authentication [page 2632]

SAP SuccessFactors Data Centers Mapping to Authentication URL [page 2634]

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring OpenID Connect (OIDC) [page 2636]

1.7.17.1 Configure Authentication Provider To Migrate User Passwords from SAP SuccessFactors Systems to Identity Authentication

Prerequisites

- You have provisioned the users from `SAP SuccessFactors` to Identity Authentication.
- Users provisioned to Identity Authentication have the `sourceSystem` attribute with value `100`, and the `sourceSystemId` with value equal to the SAP SuccessFactors company ID.
  You can find information about the user's authentication provider type and system ID under the legal tab in the user details section. For more information, see List and Edit User Details [page 2361].

Context

In this scenario, you have an SAP SuccessFactors instance integrated with Identity Authentication. In the SAP SuccessFactors instance, there are users that log on with username and password (also known as password or non-sso users). The authentication provider opportunity gives the possibility these users to be migrated and to use Identity Authentication without the need to change the passwords that they already have. The password of each SAP SuccessFactors user is migrated once only during his or her first successful logon after the configuration of the authentication provider scenario in Identity Authentication. After that the user passwords are managed by Identity Authentication.

The first logon after the migration must be with a username and password. After this first successful logon, the user can use any other allowed logon identifier.

> → Remember
>
> If a user has enabled the *Remember me* option, when that user changes his or her password in the external authentication provider, the remember me cookie isn't invalidated. For more information, see Use the Remember Me Option [page 2774].

To configure a authentication provider, follow the steps below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Authentication Providers* tile.
3. Press the *Create* button on the left-hand panel to add a new authentication provider to the list.
4. Under *Configuration*, make the corresponding entries in the configuration for the target system you want to create:

Configurations	Description
Display Name	The name of the configuration.
Type	Select the *SuccessFactors* type.
Company ID	The company of the authentication provider.

Configurations	Description
Password Validation URL	The URL endpoint for validation of the users name and password. It can be provided by the authentication provider administrator. **Note** The URl must have the following pattern: `https://<SuccessFactors_API_Host>/odata/v2/restricted/validateUser` Replace the host `<SuccessFactors_API_host>` with the Authentication URL host from the SAP SuccessFactors Data Centers Mapping to Authentication URL [page 2634] document. Plain HTTP is supported for testing purposes only. Make sure that you use the encrypted HTTPS protocol for productive systems. For more information about the mapping between the SAP SuccessFactors data centers and the password validation URL, see SAP SuccessFactors Data Centers Mapping to Authentication URL [page 2634]. **→ Remember** If you choose the *X.509* certificate as your *Authentication Type*, you'll need to add ".cert" after the subdomain part of the URL. For example: `https://exampleapi.cert.sapsf.com/odata/v2/restricted/validateUser`

Configurations	Description
• *Authentication Type*	Choose: • *Basic* - provide *Technical User* and *Technical User Secret*: *Technical User* Technical user added in the authentication provider that has administrator permissions to access the OData API. It can be provided by the external authentication provider administrator. For more information of the permission settings of the user, see Setting Up an API User for Sync Jobs in SAP SuccessFactors (Basic Authentication Method). *Technical User Secret* Password set on the technical user. It can be provided by the authentication provider administrator. → Tip For productive systems, we recommend that you use passwords that are difficult to be guessed. • *X.509* Enter CN for the certificate in the provided field. Once the certificate is generated, you can view its details. The validity of the certificate is 1 year. ⓘ Note You can choose the option for automatic regeneration of the certificate by selecting the *Automatic Renewal* checkbox. Two weeks before the expiry of the certificate, it is regenerated. The renewed certificate will have the same DN. → Remember If you choose the *X.509* certificate as your *Authentication Type*, you'll need to upload that certificate into SAP SuccessFactors to register Identity Authentication for incoming calls using *X.509* certificate-based authentication. Refer to Upgrade to X.509 Certificate-Based Authentication for Incoming Calls for the steps to complete the upload. → Tip To create a *X.509* certificate to use it as *Authentication Type* for the authentication provider:

Configurations	Description

	1. Open the authentication provider created in the administration console.
	2. Choose the *display button* under the *Actions* column.
	3. Copy content of the *X.509* certificate from the popup window.
	4. Create a `cer.` file in the following way:
	Sample Code
	`-----BEGIN CERTIFICATE-----`
	`<Copied content of the X.509 certificate>`
	`-----END CERTIFICATE-----`

First Logon Behavior	Choose if a user whose password doesn't meet the password policy requirements of the application must reset or change it after the first successful logon. • *Change password* - The user will be presented a dialog to change their password. The new password must meet the current password policy. This is the default choice. • *Reset password* - The user receives the reset password link by email.

5. Save your configuration.

   The system displays the message `Authentication provider <id of the system> created`.

   If you have chosen *Authentication Type - X.509*, the certificate is generated after saving the configuration.
6. **Optional:** Choose *Test Connection* to test the authentication provider configuration.

Next Steps

> **Note**
>
> If an application requires force authentication (ForceAuthn="true"), users have to authenticate themselves against the corporate identity provider each time they access the application even if single sign-on (SSO) is enabled.

- (Optional) To edit an existing authentication provider configuration, select *provider you want to edit* > *Edit button* > *make the necessary changes* > *Save*.
- (Optional) To change the display name of an existing authentication provider, select the authentication provider whose name you want to change, choose the *Edit icon*, provide the new name, and save your changes.

Task overview: Configuring External Authentication Providers [page 2620]

Related Information

Configure Authentication Provider To Migrate User Passwords from SAP Learning Management System to Identity Authentication [page 2626]
Configure Authentication Provider To Migrate User Passwords from SAP Fieldglass to Identity Authentication [page 2629]
Configure Authentication Provider To Migrate User Passwords from SAP Concur to Identity Authentication [page 2632]
SAP SuccessFactors Data Centers Mapping to Authentication URL [page 2634]
Migrating Passwords from SAP SuccessFactors to Identity Authentication Service

1.7.17.2 Configure Authentication Provider To Migrate User Passwords from SAP Learning Management System to Identity Authentication

Prerequisites

- You have provisioned the users from `SAP Learning Management System` to Identity Authentication.
- Users provisioned to Identity Authentication have the `sourceSystem` attribute with value `101`, and the `sourceSystemId` with value equal to the SAP Learning Management System tenant ID.
  You can find information about the user's authentication provider type and system ID under the legal tab in the user details section. For more information, see List and Edit User Details [page 2361].

Context

In this scenario, you have an SAP Learning Management System instance integrated with Identity Authentication. In the SAP Learning Management System instance there are users that log on with username and password (also known as External Learners). The authentication provider opportunity gives the possibility these users to be migrated and to use Identity Authentication without the need to change the passwords that they already have. The password of each SAP Learning Management System user is migrated once only during his or her first successful logon after the configuration of the authentication provider scenario in Identity Authentication. After that the user passwords are managed by Identity Authentication.

> → Remember
>
> If a user has enabled the *Remember me* option, when that user changes his or her password in the external authentication provider, the remember me cookie is not invalidated. For more information, see Use the Remember Me Option [page 2774].

To configure a authentication provider, follow the steps below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Authentication Providers* tile.
3. Press the *Create* button on the left-hand panel to add a new authentication provider to the list.
4. Under *Configuration*, make the corresponding entries in the configuration for the target system you want to add:

Configurations	Description
Display Name	The name of the configuration.
Type	Select the *Learning Management System* type.
System ID	The ID of the authentication provider.
Password Validation URL	The URL endpoint for validation of the users name and password. It is automatically generated and has the following pattern: `https://<lms-system-id>.plateau.com/learning/rest/integrated/admin/Learner.svc/ias/v1/~validateCredentials` The *Password Validation URL* can be provided also by the authentication provider administrator. It must have the following pattern: `https://<lms-system-id>.plateau.com/learning/rest/integrated/admin/Learner.svc/ias/v1/~validateCredentials` Plain HTTP is supported for testing purposes only. Make sure that you use the encrypted HTTPS protocol for productive systems.

Configurations	Description
Authentication Type	Choose: • *Basic* - provide *Technical User* and *Technical User Secret*: *Technical User* Technical user added in the authentication provider that has administrator permissions to access the OData API. It can be provided by the external authentication provider administrator. *Technical User Secret* Secret set on the technical user. It can be provided by the authentication provider administrator. → Tip For productive systems, we recommend that you use passwords that are difficult to be guessed. • *X.509* Enter CN for the certificate in the provided field. Once the certificate is generated, you can view its details. The validity of the certificate is one year. ⓘ Note You can choose the option for automatic regeneration of the certificate by selecting the *Automatic Renewal* checkbox. Two weeks before the expiry of the certificate, it will be regenerated. The renewed certificate will have the same DN.
First Logon Behavior	Choose: • if a user whose password does not meet the password policy requirements of the application must reset or change it after the first successful logon. • if a user whose LMS password is expired must reset or change it after the first successful logon. The default choice is *Change password*.

5. Save your configuration.

   The system displays the message `Authentication provider <id of the system> created`.

   If you have chosen *Authentication Type - X.509*, the certificate is generated after saving the configuration.
6. **Optional:** Choose *Test Connection* to test the authentication provider configuration.

Next Steps

> **Note**
>
> If an application requires force authentication (ForceAuthn="true"), users have to authenticate themselves against the corporate identity provider each time they access the application even if single sign-on (SSO) is enabled.

- (Optional) To edit an existing authentication provider configuration, select *provider you want to edit* > *Edit button* > *make the necessary changes* > *Save*.
- (Optional) To change the display name of an existing authentication provider, select the authentication provider whose name you want to change, choose the *Edit icon*, provide the new name, and save your changes.

Task overview: Configuring External Authentication Providers [page 2620]

Related Information

Configure Authentication Provider To Migrate User Passwords from SAP SuccessFactors Systems to Identity Authentication [page 2621]
Configure Authentication Provider To Migrate User Passwords from SAP Fieldglass to Identity Authentication [page 2629]
Configure Authentication Provider To Migrate User Passwords from SAP Concur to Identity Authentication [page 2632]
SAP SuccessFactors Data Centers Mapping to Authentication URL [page 2634]

1.7.17.3 Configure Authentication Provider To Migrate User Passwords from SAP Fieldglass to Identity Authentication

Prerequisites

- You have provisioned the users from SAP Fieldglass to Identity Authentication.
- Users provisioned to Identity Authentication have the `sourceSystem` attribute with value `102`, and the `sourceSystemId` with value equal to the SAP Fieldglass tenant ID.
  You can find information about the user's authentication provider type and system ID under the legal tab in the user details section. For more information, see List and Edit User Details [page 2361].

Context

In this scenario, you have an SAP Fieldglass instance integrated with Identity Authentication. In the SAP Fieldglass instance there are users that log on with username and password. The authentication provider opportunity gives the possibility these users to be migrated and to use Identity Authentication without the need to change the passwords that they already have. The password of each SAP Fieldglass user is migrated once only during his or her first successful logon after the configuration of the authentication provider scenario in Identity Authentication. After that the user passwords are managed by Identity Authentication.

> → Remember
>
> If a user has enabled the *Remember me* option, when that user changes his or her password in the external authentication provider, the remember me cookie is not invalidated. For more information, see Use the Remember Me Option [page 2774].

To configure a authentication provider, follow the steps below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Authentication Providers* tile.
3. Press the *Create* button on the left-hand panel to add a new authentication provider to the list.
4. Under *Configuration*, make the corresponding entries in the configuration for the target system you want to add:

Configurations	Description
Display Name	The name of the configuration.
Type	Select the **`Fieldglass`** type.
Company Code	The company code of the authentication provider.
Password Validation URL	The URL endpoint for validation of the users name and password. It must have the following pattern: `https://<fieldglass_host>/api/v1/ias/passwordsync` Plain HTTP is supported for testing purposes only. Make sure that you use the encrypted HTTPS protocol for productive systems.
Technical User	Technical user added in the authentication provider that has administrator permissions to access the OData API. It can be provided by the external authentication provider administrator.

Configurations	Description
Technical User Secret	Secret set on the technical user. It can be provided by the authentication provider administrator. → Tip For productive systems, we recommend that you use passwords that are difficult to be guessed.
Token URL	Token endpoint to which the technical user is authenticated.
First Logon Behavior	Choose if a user whose password does not meet the password policy requirements of the application must reset or change it after the first successful logon. The default choice is *Change password*.

5. Save your configuration.

   The system displays the message `Authentication provider <id of the system> created`.
6. **Optional:** Choose *Test Connection* to test the authentication provider configuration.

Next Steps

> ⓘ Note
>
> If an application requires force authentication (ForceAuthn="true"), users have to authenticate themselves against the corporate identity provider each time they access the application even if single sign-on (SSO) is enabled.

- (Optional) To edit an existing authentication provider configuration, select *provider you want to edit* > *Edit button* > *make the necessary changes* > *Save*.
- (Optional) To change the display name of an existing authentication provider, select the authentication provider whose name you want to change, choose the *Edit icon*, provide the new name, and save your changes.

Task overview: Configuring External Authentication Providers [page 2620]

Related Information

Configure Authentication Provider To Migrate User Passwords from SAP SuccessFactors Systems to Identity Authentication [page 2621]

Configure Authentication Provider To Migrate User Passwords from SAP Learning Management System to Identity Authentication [page 2626]

Configure Authentication Provider To Migrate User Passwords from SAP Concur to Identity Authentication [page 2632]
SAP SuccessFactors Data Centers Mapping to Authentication URL [page 2634]

1.7.17.4 Configure Authentication Provider To Migrate User Passwords from SAP Concur to Identity Authentication

Prerequisites

- You have provisioned the users from SAP Concur to Identity Authentication.
- Users provisioned to Identity Authentication have the `sourceSystem` attribute with value `103`, and the `sourceSystemId` with value equal to the SAP Concur tenant ID.
  You can find information about the user's authentication provider type and system ID under the legal tab in the user details section. For more information, see List and Edit User Details [page 2361].

Context

> → Remember
>
> If you have completed automated identity provisioning with SAP Cloud Identity Services for your organization, skip the procedure below. For more information, see Process.

In this scenario, you have an SAP Concur instance integrated with Identity Authentication. In the SAP Concur instance, there are users that log on with username and password. The authentication provider opportunity gives the possibility these users to be migrated and to use Identity Authentication without the need to change the passwords that they already have. The password of each SAP Concur user is migrated once only during his or her first successful logon after the configuration of the authentication provider scenario in Identity Authentication. After that the user passwords are managed by Identity Authentication.

> → Remember
>
> - If a user has enabled the *Remember me* option, when that user changes his or her password in the external authentication provider, the remember me cookie is not invalidated. For more information, see Use the Remember Me Option [page 2774].
> - If a user has a second-factor authentication configured for his profile, they will be asked to provide it together with the password.

To configure an authentication provider, follow the steps below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Authentication Providers* tile.
3. Press the *Create* button on the left-hand panel to add a new authentication provider to the list.
4. Under *Configuration*, make the corresponding entries in the configuration for the target system you want to add:

Configurations	Description
Display Name	The name of the configuration.
Type	Select the **`Concur`** type.
Company ID	The company ID of the authentication provider.
Password validation URL	The URL endpoint for validation of the users name and password. It must have the following pattern: `https://<concur_host>/password/v0/ias/passwordcheck` Plain HTTP is supported for testing purposes only. Make sure that you use the encrypted HTTPS protocol for productive systems.
Common Name (CN)	
First Logon Behavior	Choose if a user whose password does not meet the password policy requirements of the application must reset or change it after the first successful logon. The default choice is *Change password*.

5. Save your configuration.

   The system displays the message `Authentication provider <id of the system> created`.
6. **Optional:** Choose *Test Connection* to test the authentication provider configuration.

Next Steps

> ⓘ Note
>
> If an application requires force authentication (ForceAuthn="true"), users have to authenticate themselves against the corporate identity provider each time they access the application even if single sign-on (SSO) is enabled.

- (Optional) To edit an existing authentication provider configuration, select *provider you want to edit* > *Edit button* > *make the necessary changes* > *Save*.

- (Optional) To change the display name of an existing authentication provider, select the authentication provider whose name you want to change, choose the *Edit icon*, provide the new name, and save your changes.

Task overview: Configuring External Authentication Providers [page 2620]

Related Information

Configure Authentication Provider To Migrate User Passwords from SAP SuccessFactors Systems to Identity Authentication [page 2621]

Configure Authentication Provider To Migrate User Passwords from SAP Learning Management System to Identity Authentication [page 2626]

Configure Authentication Provider To Migrate User Passwords from SAP Fieldglass to Identity Authentication [page 2629]

SAP SuccessFactors Data Centers Mapping to Authentication URL [page 2634]

1.7.17.5 SAP SuccessFactors Data Centers Mapping to Authentication URL

SAP SuccessFactors Data Centers	Authentication URL
DC2 - Production	https://api2.successfactors.eu/odata/v2/restricted/validateUser
DC2 - SalesDemo	https://apisalesdemo2.successfactors.eu/
DC2 - Preview	https://api2preview.sapsf.eu/odata/v2/restricted/validateUser
DC4 - Production	https://api4.successfactors.com/odata/v2/restricted/validateUser
DC4 - SalesDemo	https://apisalesdemo4.successfactors.com/
DC4 - Preview	https://api4preview.sapsf.com/odata/v2/restricted/validateUser
DC5 - Production	https://api5.successfactors.eu/odata/v2/restricted/validateUser
DC8 - Production	https://api8.successfactors.com/odata/v2/restricted/validateUser
DC8 - SalesDemo	https://apisalesdemo8.successfactors.com/
DC8 - Preview	https://api8preview.sapsf.com/odata/v2/restricted/validateUser
DC10 - Production	https://api10.successfactors.com/odata/v2/restricted/validateUser

SAP SuccessFactors Data Centers	Authentication URL
DC10 - Preview	https://api10preview.sapsf.com/odata/v2/restricted/validateUser
DC12 - Production	https://api012.successfactors.eu/odata/v2/restricted/validateUser
DC12 - Rot	https://apirot.successfactors.eu/odata/v2/restricted/validateUser
DC12 - Preview	https://api12preview.sapsf.eu/odata/v2/restricted/validateUser
DC15 - Production	https://api15.sapsf.cn/odata/v2/restricted/validateUser
DC16 - Production	https://api16.sapsf.eu/odata/v2/restricted/validateUser
DC17 - Preview	https://api17preview.sapsf.com/odata/v2/restricted/validateUser
DC17 - Production	https://api17.sapsf.com/odata/v2/restricted/validateUser
DC18 - Preview	https://api18preview.sapsf.com/odata/v2/restricted/validateUser
DC18 - Production	https://api18.sapsf.com/odata/v2/restricted/validateUser
DC19 - Preview	https://api19preview.sapsf.com/
DC19 - Production	https://api19.sapsf.com/
DC22 Preview	https://api22preview.sapsf.com/
DC22 - Production	https://api22.sapsf.com/
DC23 - Preview	https://api23preview.sapsf.com/
DC23 - Production	https://api23.sapsf.com/
DC41 - Preview	https://api41preview.sapsf.com
DC41 - Production	https://api41.sapsf.com
DC42 - Preview	https://api42preview.sapsf.com/
DC42 - Production	https://api42.sapsf.com/
DC44 - Preview	https://api44preview.sapsf.com/
DC44 - Production	https://api44.sapsf.com/
DC47 - Preview	https://api47preview.sapsf.com/
DC47 - Production	https://api47.sapsf.com/
DC55 - Preview	https://api55preview.sapsf.eu/
DC55 - Production	https://api55.sapsf.eu/

Parent topic: Configuring External Authentication Providers [page 2620]

Related Information

Configure Authentication Provider To Migrate User Passwords from SAP SuccessFactors Systems to Identity Authentication [page 2621]

Configure Authentication Provider To Migrate User Passwords from SAP Learning Management System to Identity Authentication [page 2626]

Configure Authentication Provider To Migrate User Passwords from SAP Fieldglass to Identity Authentication [page 2629]

Configure Authentication Provider To Migrate User Passwords from SAP Concur to Identity Authentication [page 2632]

1.7.18 Configuring OpenID Connect (OIDC)

You can use Identity Authentication for authentication in OpenID Connect (OIDC) protected applications.

OpenID Connect (OIDC) is a simple identity layer on top of the OAuth 2.0 protocol. Clients can verify the identity of the end-user based on the authentication performed by an authorization server, as well as to obtain basic profile information about the end-user in an interoperable and REST-like manner. For more information about the OpenID Connect (OIDC) specification, see OpenID Connect Core 1.0 .

The OpenID Connect (OIDC) implementation of Identity Authentication supports all flow from OpenID Connect (OIDC) Core 1.0.

In addition to the core, the service also supports the following:

- Discovery
- Front-Channel Logout
- OAuth 2.0 Dynamic Client Registration Protocol

Based on your scenario, choose one of the flows to see how to authorize access to your application:

- Using Authorization Code Flow [page 2638]
- Call Identity Authentication Refresh Token [page 2725]
- Using Client Credentials Flow [page 2660]
- Using Resource Owner Password Credentials Flow [page 2669]
- Using Implicit Flow [page 2679]
- Call Identity Authentication Revoke Token Endpoint [page 2736]
- Call Identity Authentication Discovery Endpoint [page 2729]
- Call Identity Authentication UserInfo Endpoint [page 2735]
- Call Identity Authentication Certificate Endpoint [page 2731]
- Call Identity Authentication List Sessions Endpoint [page 2733]
- Call Identity Authentication Introspect Token Endpoint [page 2738]
- Using JWT Bearer Flow [page 2689]
- Call Identity Authentication End Session Endpoint [page 2740]
- Call Identity Authentication Introspect Token Endpoint [page 2738]
- Using Token Exchange [page 2701]
- Call Identity Authentication Revoke Token Endpoint [page 2736]
- Call Identity Authentication List Sessions Endpoint [page 2733]
- Using Token Exchange [page 2701]
- Logging OpenID Connect (OIDC) Tokens [page 2980]

Parent topic: Administration [page 1918]

Related Information

Configuring Applications [page 1920]
Configuring Tenant Settings [page 2100]
Configuring Password Policies [page 2223]
Configuring Privacy Policies [page 2235]
Configuring Authorization Policies [page 2244]
Configuring Terms of Use [page 2273]
Configuring Email Templates [page 2282]
Managing Administrators [page 2301]
Managing Technical Users [page 2329]
Managing Agents [page 2333]
Managing Users [page 2342]
Managing Groups [page 2393]
Configuring Provisioning Systems [page 2413]
Configuring Real-Time Provisioning [page 2487]
Configuring Social Identity Providers [page 2505]
Integrating with Existing Customer Landscape [page 2518]
Configuring External Authentication Providers [page 2620]

1.7.18.1 Using Authorization Code Flow

To authenticate using the authorization code flow, follow the procedures below. Tasks 1 and 2 are configurations on Identity Authentication side. Tasks 3 and 4 are configurations, done on the client (relying party) side.

> → Recommendation
>
> Follow the best practices of RFC 9700 and use PKCE for confidential clients. You must use PKCE for public clients.

1. Create OpenID Connect (OIDC) Application [page 2639]
   You can create a new OpenID Connect (OIDC) application.
2. Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 2642]
   This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the authorization code flow.
3. Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow [page 2645]
   The authorize endpoint is used for authentication and returns authorization code when using the authorization code flow. It's recommended for confidential clients that can keep the client secret.
4. Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow with PKCE [page 2650]
   The authorization code flow with PKCE is recommended for public clients that aren't capable of keeping the client secrets.

5. Configure the Client to Call Identity Authentication Token Endpoint for Authorization Code Flow [page 2656]
   The token endpoint is used to get the user's access token, id token and refresh token.

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.1.1 Create OpenID Connect (OIDC) Application

You can create a new OpenID Connect (OIDC) application.

Context

To create a new OpenID Connect (OIDC) application you have to add a new application to the list of applications in Identity Authentication, and set the protocol of the application to OpenID Connect (OIDC).

To create a new OpenID Connect (OIDC) application follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the *Create* button on the left-hand panel and fill in the information in the dialog box.
4. Provide the following information:

Fields	Notes
Display Name	Required. The display name of the application appears on the logon and registration pages.
Home URL	Optional. Users are redirected to the *Home URL* after activating their accounts, when they are created via a CSV file import or the user registration service of Identity Authentication.
Type	Optional.

Fields	Notes
	⚠ Caution If the type of the application is not the correct one, the system may change it automatically without a warning.
Parent Application	Optional. **→ Remember** Newly created applications with an assigned parent application will inherit all the configurations from the parent except for the `Client ID` and `Secrets`. The inherited configurations will be marked as such. • If you are an administrator with the *Manage Applications* role, the newly created applications can have as parent any of the parent applications in the administration console. Choose the parent application from the list that pops up. • If you are an administrator with authorizations based on policies, you can assign as parent only the application to which you have read or manage applications rights. The applications to which you have read or manage rights are restricted on the basis of the `application.organization` attribute in the policy. Choose the parent application from the list that pops up. For more information about how to assign authorizations based on policies, see Configure Application Authorizations [page 2325].
Organization ID	Optional. This field allows you to group the applications under a specific organization and on the basis of the *Organization ID* to restrict the access to these applications. The *Organization ID* can contain lowercase Latin letters (a-z), numbers (0-9), and the underscore (_) character. It can be up to 255 characters. • If you are an administrator with *Manage Applications* role, the newly created applications by default are in the `global` *Organization ID*. You can change the *Organization ID* during the creation of the new application by entering a new name or later you can edit it. • If your authorizations are based on policies, the newly created applications can take the *Organization ID* of the applications to which you have manage rights on the basis of the `application.organization` attribute in the policy. Choose the allowed *Organization ID* from the drop-down.

Fields	Notes
	Note Applications created either as subscriptions within or with the Identity service of an SAP BTP subaccount receive the *Organization ID* of the application that represents the trust configuration of that subaccount.
Protocol Type	Select the *OpenID Connect(OIDC)* radio-button. **→ Remember** Newly created applications with an assigned parent application will inherit the protocol from the parent. To change the protocol, first create the new application and then edit it.

5. Choose *+Create*.

   Once the application has been created, the system displays the message `Application <name of application> created`.

   The newly created application appears on the list with the applications on the left. It is selected and you can set its protocol to OpenID Connect (OIDC).

Next Steps

(Optional) If necessary, configure additional settings for the application. For more information about the supported configurations for the OpenID Connect (OIDC) applications, see the following links:

- Change an Application's Display Name [page 1934]
- Configure an Application's Home URL [page 1937]
- Visit an Application's Web Page [page 1939]
- Configuring User Attributes from the Identity Directory [page 1973]
- Configuring Attributes Based on Flexible Expressions [page 1979]
- Configure Secrets for API Authentication [page 1997]
- Configure Certificates for API Authentication [page 2001]
- Enable or Disable Kerberos Authentication for an Application [page 2556]
- Configure Risk-Based Authentication for an Application [page 2007]
- Configure User Access to the Application [page 2025]
- Authenticating Identity Provider for an Application [page 2065]
- Enable Email Verification [page 2026]
- Configuring Password Policies [page 2223]
- Configuring Privacy Policies [page 2235]
- Configuring Terms of Use [page 2273]
- Display Application Name on Logon Page [page 2033]

- Configure Logo [page 2030]
- Choose and Configure a Branding Style for an Application [page 2035]
- Configure Registration and Upgrade Forms [page 2046]
- Protecting Self-Registration, Upgrade with Phone Verification [page 2050]
- Protecting Application Forms with Google reCAPTCHA [page 2056]
- Configure Identity Federation [page 2614]
- Configure Authentication Request to Corporate IdPs [page 1966]

Task overview: Using Authorization Code Flow [page 2638]

Next task: Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 2642]

1.7.18.1.2 Configure OpenID Connect (OIDC) Application for Authorization Code Flow

This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the authorization code flow.

Prerequisites

You have an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party), and its redirect URIs.

To configure an OpenID Connect (OIDC) trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> ### Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect (OIDC) Configuration*.
6. Manually enter the communication settings negotiated between Identity Authentication and the client.

Setting	Description
Name (mandatory)	Provide a name of your choice.
Redirect URIs (mandatory)	The redirection URIs to which the response can be sent. You can add up to 20 redirect URIs.
Front-Channel Logout URIs	(Optional) URIs which will be requested for logout. You can add up to 20 URIs.
Back-Channel Logout URI	(Optional) URIs which will be requested for logout. You can add up to 20 URIs.

> ### → Tip
>
> For more information about the format of the redirect URIs and post logout redirect URIs, see OpenID Connect (OIDC) Application Configurations [page 2747].

7. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. **→ Tip** To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. **⚠ Caution** The application will stop working if the configuration of the application is not updated with the new certificate.

Option	Description
	Note Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

> **Tip**
>
> When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

8. Select the following grant types:

Grant Type	Notes
Authorization Code	For the authorization code flow.
Authorization Code > *Authorization Code / Enforce PKCE (S256)*	For the authorization code flow with PKCE.

> **Note**
>
> Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

9. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

> **Remember**
>
> Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

- Configure HTTP basic authentication for the application. For more information about the configuration, see Configure Secrets for API Authentication [page 1997].
- Enable the public client flows option for this application. In the administration console, choose *the OpenID Connect (OIDC) application* > *Client Authentication under the Trust tab* > *Enable Public Client Flows under Public Client*. Optionally, you can configure the API permission groups. For more information, see Consuming APIs from Other Applications [page 2889] .

> **Note**
>
> The *Public* client type is used for environments where it is difficult to protect the client credential, such as mobile and desktop applications, and client-side parts of web applications.

Task overview: Using Authorization Code Flow [page 2638]

Previous task: Create OpenID Connect (OIDC) Application [page 2639]

Next: Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow [page 2645]

Related Information

Configuring OpenID Connect (OIDC) [page 2636]
Handle Single Logout Request from Identity Authentication [page 2743]

1.7.18.1.3 Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow

The authorize endpoint is used for authentication and returns authorization code when using the authorization code flow. It's recommended for confidential clients that can keep the client secret.

This document explains how to call the `authorize` endpoint and what are the authorize request parameters supported by Identity Authentication for the authorization code flow.

> ⓘ Note
>
> Confidential clients are applications that can keep the client secret. These clients are the so-called web applications.

Prerequisites

For your business application, there's an OpenID Connect (OIDC) application in Identity Authentication.

For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Request

URI: `https://<Cloud Identity Services domain>/oauth2/authorize`

> ⓘ Note
>
> The domain part has the following pattern:

`<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.

Tenant ID is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *GET*

Request Parameters

Parameter	Required	Data Type	Additional Information	Parameter Type
`client_id`	Yes	string	Used to identify the corresponding Identity Authentication application. **Note** The `client_id` parameter is optional if the request is provided with authentication.	Query
`redirect_uri`	Yes	string	• Configure OpenID Connect (OIDC) Application [page 1949] • Redirect URIs, Post Logout Redirect URI Rules [page 2747]	Query
`response_type`	Yes	string	The supported value is `code`.	Query

Parameter	Required	Data Type	Additional Information	Parameter Type
`scope`	Yes	string	The supported values are: • `openid` • `email` • `profile` • `groups` • `offline_access` **Note** The new tokens are independently created from the Identity Authentication Web session. This means that even if a user logs out from Identity Authentication the `refresh_token` will exist in the database until it expires, and can be used to perform the refresh token flow if the user is not present in Identity Authentication with a Web session. Values must be space-delimited parameter, for example: `scope=openid email`.	Query
`state`	Yes	string	Free text.	Query
`app_tid`	No	string	Reserved.	Query

Parameter	Required	Data Type	Additional Information	Parameter Type
`idp`	No	string	The technical name of the corporate identity provider as configured in the administration console for SAP Cloud Identity Services. When a chain of identity providers are allowed for an application via conditional authentication, these parameters enable the client to specify which corporate identity providers are to be used. Identity Authentication uses the `idp` parameter to detect the correct corporate identity providers and redirect the request to them. When there is more than one corporate IdP in the IdP-initiated link, they are separated by comma "`,`" without space between them. For applications that allow authentication with OIDC, use the value **`local`** to override the conditional authentication configuration and authenticate with Identity Authentication instead.	Query
`login_hint`	No	string	The `login_hint` parameter helps the user when he or she is known to the service provider (SP). Thus it prevents the user from re-typing the user identifier on the logon or conditional screen. Supported values are the allowed logon identifiers for the users. The options are *User ID*, *Login Name*, and *Email* (For SAML 2.0). For more information, see Configure Allowed Logon Identifiers [page 2146] .	Query
`logout_uri`	No	string	• Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 2642] • Logout URI Rules [page 2750]	Query
`max_age`	No	integer	Maximum time in seconds since the user was last authenticated. When `max_age` has been reached, the user must re-authenticate.	Query

Parameter	Required	Data Type	Additional Information	Parameter Type
`nonce`	No	string	Free text.	Query
`prompt`	No	string	This parameter enables the client to determine if the user is still present in the current session. Supported values are: • *login* Force reauthentication of the user. • *none* Without user interaction, check if there is an exisiting user session. If there is a session, continue as normal. If there is no session or if the user is forced to authenticate for other reasons, return the error code `login_required`.	Query

Request Example

```
https://my-tenant.ondemand.com/oauth2/authorize?
response_type=code&scope=openid+email&client_id=94ff0b4b0baa45a893c7cd24254b72b7&
state=state&nonce=m-0G6_FaS3Kg&redirect_uri=https://
example.com&login_hint=dona.moore@example.com
```

Response

Response Headers

Header	Description
`code`	The `code` is generated by Identity Authentication and is returned in the URL as a parameter. It must be used when calling the token endpoint. **Note** The parameter can be used within two minutes after it is generated and returned in the URL. It can be used only once.

Response Status and Error Codes

Code	Reason	Description
302 Found	Successful operation.	Additionally provides a URL in the header field Location. **Note** The URL contains `code` necessary for the token endpoint.
400 Bad Request	Missing or wrong parameter	

Response Example

```
Location:
 https://www.example.com/?code=4454554df477w01s34540672dc462e6f0&state=state
```

Parent topic: Using Authorization Code Flow [page 2638]

Previous task: Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 2642]

Next: Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow with PKCE [page 2650]

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.1.4 Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow with PKCE

The authorization code flow with PKCE is recommended for public clients that aren't capable of keeping the client secrets.

Prerequisites

- For your business application, there's an OpenID Connect (OIDC) application in Identity Authentication. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

- For your OpenID Connect (OIDC) application in Identity Authentication, you've prepared an authentication credential for the API call. The API supports the following authentication schemes:
  - Basic
    For more information, see `Authorization` in the *Request Header* table below.
  - X.509
    For more information, see `Authorization` in the *Request Header* table below.
  - JWT (client_assertion)
    - Trust by issuer: You must include the `client_id` in the request.
    - Trust by URI: Including `client_id` in the request is optional.

    For more information, see the *Request Parameters* table below.

  For more information, see API Authentication [page 1996].

This document explains how to call the `authorize` endpoint and what are the authorize request parameters supported by Identity Authentication for the authorization code flow with PKCE.

> ### Note
>
> Public clients are applications that aren't capable of keeping the client secret, such as mobile applications, native applications, and single page applications.

Request

URI: `https://<Cloud Identity Services domain>/oauth2/authorize`

> ### Note
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *GET*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Request Parameters

Parameter	Required	Data Type	Additional Information	Parameter Type
`client_id`	Yes	string	Used to identify the corresponding Identity Authentication application. **Note** The `client_id` parameter is optional if the request is provided with authentication.	Query
`code_challenge`	Yes	string	Client created code. **→ Tip** You can use online PKCE generator tools to generate the `code_challenge` values.	Query
`code_challenge_method`	Yes	string	Code verifier transformation method is "S256" or "plain".	Query
`redirect_uri`	Yes	string	• Configure OpenID Connect (OIDC) Application [page 1949] • Redirect URIs, Post Logout Redirect URI Rules [page 2747]	Query
`response_type`	Yes	string	The supported value is `code`.	Query

Parameter	Required	Data Type	Additional Information	Parameter Type
`scope`	Yes	string	Value must be space delimited parameter, for example: `scope=openid email`. The supported values are: • `openid` • `email` • `profile` • `groups` • `offline_access` **Note** The new tokens are independently created from the Identity Authentication Web session. This means that even if a user logs out from Identity Authentication the `refresh_token` will exist in the database until it expires, and can be used to perform the refresh token flow if the user is not present in Identity Authentication with a Web session.	Query
`state`	Yes	string	Free text.	Query
`app_tid`	No	string	Reserved.	Query
`login_hint`	No	string	The `login_hint` parameter helps the user when he or she is known to the service provider (SP). Thus it prevents the user from re-typing the user identifier on the logon or conditional screen. Supported values are the allowed logon identifiers for the users. The options are *User ID*, *Login Name*, and *Email* (For SAML 2.0). For more information, see Configure Allowed Logon Identifiers [page 2146] .	Query

Parameter	Required	Data Type	Additional Information	Parameter Type
`logout_uri`	No	string	• Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 2642] • Logout URI Rules [page 2750]	Query
`max_age`	No	integer	Maximum time in seconds since the user was last authenticated. When `max_age` has been reached, the user must re-authenticate.	Query
`nonce`	No	string	Free text.	Query
`prompt`	No	string	This parameter enables the client to determine if the user is still present in the current session. Supported values are: • *login* Force reauthentication of the user. • *none* Without user interaction, check if there is an exisiting user session. If there is a session, continue as normal. If there is no session or if the user is forced to authenticate for other reasons, return the error code `login_required`.	Query
`refresh_expiry`	No	string	Reduces the expiration of a refresh token. It's useful if your application is called from mobile and web applications, and both have different session requirements. If you set the token lifetime to 0, you won't receive a `refresh_token` in response.	Query

Request Example

Request Example (`code_challenge` and `code_challenge_method` are required)

```
https://my-tenant.ondemand.com/oauth2/authorize?
response_type=code&scope=openid&client_id=94ff0b4b0baa45a893c7cd24254b72b7&state=
state&redirect_uri=https://example.com&code_challenge=MCVg2_O79Qpr6E7HDA0ik5R-
ylwjT-_H3oUgdDQ9W50&code_challenge_method=S256
```

Response

Response Headers

Header	Description
`code`	The `code` is generated by Identity Authentication and is returned in the URL as a parameter. It must be used when calling the token endpoint. **Note** The parameter can be used within two minutes after it is generated and returned in the URL. It can be used only once.

Response Status and Error Codes

Code	Reason	Description
302 Found	Successful operation.	Additionally provides a URL in the header field Location. **Note** The URL contains `code` necessary for the token endpoint.
400 Bad Request	Missing or wrong parameter	

Response Example

```
Location:
 https://www.example.com/?code=4454554df477w01s34540672dc462e6f0&state=state
```

Parent topic: Using Authorization Code Flow [page 2638]

Previous: Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow [page 2645]

Next: Configure the Client to Call Identity Authentication Token Endpoint for Authorization Code Flow [page 2656]

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

Proof Key for Code Exchange by OAuth Public Clients (IETF)

1.7.18.1.5 Configure the Client to Call Identity Authentication Token Endpoint for Authorization Code Flow

The token endpoint is used to get the user's access token, id token and refresh token.

This document explains how to call the token endpoint and what are the parameters supported by Identity Authentication.

Prerequisites

- For your business application, there's an OpenID Connect (OIDC) application in Identity Authentication. For more information, see Create OpenID Connect (OIDC) Application [page 1925].
- For your OpenID Connect (OIDC) application in Identity Authentication, you've prepared an authentication credential for the API call. The API supports the following authentication schemes:
  - Basic
    For more information, see `Authorization` in the *Request Header* table below.
  - X.509
    For more information, see `Authorization` in the *Request Header* table below.
  - JWT (client_assertion)
    - Trust by issuer: You must include the `client_id` in the request.
    - Trust by URI: Including `client_id` in the request is optional.

    For more information, see the *Request Parameters* table below.

  For more information, see API Authentication [page 1996].

Request

URI: `https://<Cloud Identity Services domain>/oauth2/token`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`grant_type`	Yes	string	authorization_code	Request body
`code`	Yes	string	The `code` generated by Identity Authentication in the `authorize` endpoint. It's returned with the response from the `authorize` endpoint. **Note** The `code` parameter can be used within 2 minutes after it generated and returned in the URL. It can be used only once.	Request body
`redirect_uri`	Yes	string	The same URL that has been sent with the authorization request. For more information, see Configure OpenID Connect (OIDC) Application [page 1949].	Request body
`client_id`	Yes	string	Used to identify the corresponding Identity Authentication application. **Note** The `client_id` parameter is optional if the request is provided with authentication.	Request body
`code_verifier`	No	string	Validation value for the `code_challenge`. Required for PKCE.	Request body
`refresh_expiry`	No	string	Reduces the expiry of a refresh token. It is useful if your application is called from mobile and web applications, and both have different session requirements. If you set the token lifetime to 0 or less, you won't receive a `refresh_token` in response.	Request body

Parameter	Required	Data Type	Description	Parameter Type
`max_exchange_period`	No	string	You can set a limit on how long the application can exchange user tokens without re-authenticating after having acquired the initial token. The `max_exchange_period` parameter has a numeric value which can be set at between 1 hour and 4320 hours (six months). → Remember If the `max_exchange_period` value is smaller than the configuration defined in token policy, this value overrides the configuration defined in token policy configuration and propagates it once the token is exchanged or refreshed. If the value is bigger, it is ignored and the configuration defined in token policy is taken into consideration. For more information, see Token Policy Configuration for Applications [page 2752]	Request body
`token_format`	No	string	The `token_format` can be set to `opaque` to retrieve an opaque access token or to `jwt` to retrieve a JWT-based access token. If not set, the current defaults per grant type are used.	Request body
`app_tid`	No	string	Reserved.	Path

Request Example

```
grant_type=authorization_code&client_id=12b52d2c-1q34-5r5t-
a576-75e85asdf523&code=768fsd1206we951236729134fa4a247ee&redirect_uri=https://
www.example.com
```

Request Example with PKCE

```
grant_type=authorization_code&client_id=23q45d2c-1e98-4c7f-
a576-69e85dddb456&code=b1d87f103c9f4c8a936a2f164c8728a0&redirect_uri=https://
www.example.com&code_verifier=e4yAeaa_OLJ-8ZC1sj6JWFqgRK8QR-
egfqG4lQlsy0~NOrjOcVdIpPxPvUEsG1Il1LP76F0MGr4a0miVD2Z84Gn_Lw67BhZ9GQDTF6wN_fUwQtJ
JeN3bnvk7J.Ty6jDi
```

Response

Response Status and Error Codes

Code	Reason	Description
200 OK	Successful operation.	Returns `access_token`, `refresh_token`, and `id_token`. **Note** The `refresh_token` is used to obtain a new `id_token` and `access_token` when the current token becomes invalid or expires. For more information, see Call Identity Authentication Refresh Token [page 2725]. The `id_token` is in the form of a JWT (JSON Web Token) and contains information about the user. The `id_token` is valid for 60 minutes.
400 Bad Request	Missing or wrong parameter.	Returns information about the error.

Response Payload Example

```
{
 "access_token":
"eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdWQiOiIxMmEzNGI1Yy02ZDc4LTl
lMWYtZzM0NS02N2g4OWlqa2wxMjMiLCJzdWIiOiJQMTIzNDU2IiwibWFpbCI6ImRvbmEu
bW9vcmVAZXhhbXBsZS5jb20iLCJpc3MiOiJodHRwczovL215LXRlbmFudC5hY2NvdW50c
y5vbmRlbWFuZC5jb20iLCJsYXN0X25hbWUiOiJNb29yZSIsInNhcF91aWQiOiIxMjM0NT
ZhYmM3ZGU4LWZnaGktOTEyMy1qNDU2LTc4OTEya2wzNG01NiIsImV4cCI6MTU4ODAxOTA
0NCwiaWF0IjoxNTg4MDEzNzYwLCJub25jZSI6Im0tMEc2X0ZhUzNLZyIsImZpcnN0X25h
bWUiOiJEb25hIiwianRpIjoiMzhlNDIzMzAtZGU3YS00MTMwLWEzYTEtYjU4MmI1MjhkY
Tk4In0.J-aygbQ8IQlw4KD2t1hTPB2gyu3NdTbINDJ9eRs997g",
 "refresh_token": "d12a12abcd198765dd54r456e98321",
 "id_token":
"eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdWQiOiIxMmEzNGI1Yy02ZDc4LTl
lMWYtZzM0NS02N2g4OWlqa2wxMjMiLCJzdWIiOiJQMTIzNDU2IiwibWFpbCI6ImRvbmEu
bW9vcmVAZXhhbXBsZS5jb20iLCJpc3MiOiJodHRwczovL215LXRlbmFudC5hY2NvdW50c
y5vbmRlbWFuZC5jb20iLCJsYXN0X25hbWUiOiJNb29yZSIsInNhcF91aWQiOiIxMjM0NT
ZhYmM3ZGU4LWZnaGktOTEyMy1qNDU2LTc4OTEya2wzNG01NiIsImV4cCI6MTU4ODAxOTA
0NCwiaWF0IjoxNTg4MDEzNzYwLCJub25jZSI6Im0tMEc2X0ZhUzNLZyIsImZpcnN0X25h
bWUiOiJEb25hIiwianRpIjoiMzhlNDIzMzAtZGU3YS00MTMwLWEzYTEtYjU4MmI1MjhkY
Tk4In0.J-aygbQ8IQlw4KD2t1hTPB2gyu3NdTbINDJ9eRs997g",
 "token_type": "Bearer",
 "expires_in": 3600
}
```

Note

The format of the `issuer` depends on the configuration in the administration console for SAP Cloud Identity Services. For more information, see Tenant OpenID Connect (OIDC) Configurations [page 2117].

Parent topic: Using Authorization Code Flow [page 2638]

Previous: Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow with PKCE [page 2650]

Related Information

Configuring OpenID Connect (OIDC) [page 2636]
Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow with PKCE [page 2650]

1.7.18.2 Using Client Credentials Flow

To authenticate using the client credentials flow, follow the procedures below. Tasks 1 and 2 are configurations on Identity Authentication side. Task 3 is a configuration, done on the client (relying party) side.

1. Create OpenID Connect (OIDC) Application [page 2660]
   You can create a new OpenID Connect (OIDC) application.
2. Configure OpenID Connect (OIDC) Application for Client Credentials Flow [page 2663]
   This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the client credentials flow.
3. Configure the Client to Call Identity Authentication Token Endpoint for Client Credentials Flow [page 2666]
   The token endpoint is used to get the user's access token.

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.2.1 Create OpenID Connect (OIDC) Application

You can create a new OpenID Connect (OIDC) application.

Context

To create a new OpenID Connect (OIDC) application you have to add a new application to the list of applications in Identity Authentication, and set the protocol of the application to OpenID Connect (OIDC).

To create a new OpenID Connect (OIDC) application follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the *Create* button on the left-hand panel and fill in the information in the dialog box.
4. Provide the following information:

Fields	Notes
Display Name	Required. The display name of the application appears on the logon and registration pages.
Home URL	Optional. Users are redirected to the *Home URL* after activating their accounts, when they are created via a CSV file import or the user registration service of Identity Authentication.
Type	Optional. **⚠ Caution** If the type of the application is not the correct one, the system may change it automatically without a warning.
Parent Application	Optional. **→ Remember** Newly created applications with an assigned parent application will inherit all the configurations from the parent except for the `Client ID` and `Secrets`. The inherited configurations will be marked as such. • If you are an administrator with the *Manage Applications* role, the newly created applications can have as parent any of the parent applications in the administration console. Choose the parent application from the list that pops up. • If you are an administrator with authorizations based on policies, you can assign as parent only the application to which you have read or manage applications rights. The applications to which you have read or manage rights are restricted on the basis of the `application.organization` attribute in the policy. Choose the parent application from the list that pops up. For more information about how to assign authorizations based on policies, see Configure Application Authorizations [page 2325].
Organization ID	Optional.

Fields	Notes
	This field allows you to group the applications under a specific organization and on the basis of the *Organization ID* to restrict the access to these applications. The *Organization ID* can contain lowercase Latin letters (a-z), numbers (0-9), and the underscore (_) character. It can be up to 255 characters. • If you are an administrator with *Manage Applications* role, the newly created applications by default are in the `global` *Organization ID*. You can change the *Organization ID* during the creation of the new application by entering a new name or later you can edit it. • If your authorizations are based on policies, the newly created applications can take the *Organization ID* of the applications to which you have manage rights on the basis of the `application.organization` attribute in the policy. Choose the allowed *Organization ID* from the drop-down. **ⓘ Note** Applications created either as subscriptions within or with the Identity service of an SAP BTP subaccount receive the *Organization ID* of the application that represents the trust configuration of that subaccount.
Protocol Type	Select the *OpenID Connect(OIDC)* radio-button. **→ Remember** Newly created applications with an assigned parent application will inherit the protocol from the parent. To change the protocol, first create the new application and then edit it.

5. Choose *+Create*.

   Once the application has been created, the system displays the message `Application <name of application> created`.

   The newly created application appears on the list with the applications on the left. It is selected and you can set its protocol to OpenID Connect (OIDC).

Next Steps

(Optional) If necessary, configure additional settings for the application. For more information about the supported configurations for the OpenID Connect (OIDC) applications, see the following links:

- Change an Application's Display Name [page 1934]

- Configure an Application's Home URL [page 1937]
- Visit an Application's Web Page [page 1939]
- Configuring User Attributes from the Identity Directory [page 1973]
- Configuring Attributes Based on Flexible Expressions [page 1979]
- Configure Secrets for API Authentication [page 1997]
- Configure Certificates for API Authentication [page 2001]
- Enable or Disable Kerberos Authentication for an Application [page 2556]
- Configure Risk-Based Authentication for an Application [page 2007]
- Configure User Access to the Application [page 2025]
- Authenticating Identity Provider for an Application [page 2065]
- Enable Email Verification [page 2026]
- Configuring Password Policies [page 2223]
- Configuring Privacy Policies [page 2235]
- Configuring Terms of Use [page 2273]
- Display Application Name on Logon Page [page 2033]
- Configure Logo [page 2030]
- Choose and Configure a Branding Style for an Application [page 2035]
- Configure Registration and Upgrade Forms [page 2046]
- Protecting Self-Registration, Upgrade with Phone Verification [page 2050]
- Protecting Application Forms with Google reCAPTCHA [page 2056]
- Configure Identity Federation [page 2614]
- Configure Authentication Request to Corporate IdPs [page 1966]

Task overview: Using Client Credentials Flow [page 2660]

Next task: Configure OpenID Connect (OIDC) Application for Client Credentials Flow [page 2663]

1.7.18.2.2 Configure OpenID Connect (OIDC) Application for Client Credentials Flow

This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the client credentials flow.

Prerequisites

You have an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party).

To configure an OpenID Connect (OIDC) trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect (OIDC) Configuration*.
6. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. → Tip To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. ⚠ Caution The application will stop working if the configuration of the application is not updated with the new certificate.

Option	Description
	Note Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

> **Tip**
>
> When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

7. Select the *Client Credentials* grant type.

> **Note**
>
> Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

8. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

> **Remember**
>
> Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

1. Configure authentication for the application. For more information about the configuration, see API Authentication [page 1996].
2. Invalidate the tokens that are no longer need. For more information, see Call Identity Authentication Revoke Token Endpoint [page 2736].

Task overview: Using Client Credentials Flow [page 2660]

Previous task: Create OpenID Connect (OIDC) Application [page 2660]

Next: Configure the Client to Call Identity Authentication Token Endpoint for Client Credentials Flow [page 2666]

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.2.3 Configure the Client to Call Identity Authentication Token Endpoint for Client Credentials Flow

The token endpoint is used to get the user's access token.

This document explains how to call the token endpoint and what are the parameters supported by Identity Authentication - for a relying party client and system as administrator.

Prerequisites

- For your business application, there's an OpenID Connect (OIDC) application in Identity Authentication. For more information, see Create OpenID Connect (OIDC) Application [page 1925].
- For your OpenID Connect (OIDC) application in Identity Authentication, you've prepared an authentication credential for the API call. The API supports the following authentication schemes:
  - Basic
    For more information, see `Authorization` in the *Request Header* table below.
  - X.509
    For more information, see `Authorization` in the *Request Header* table below.
  - JWT (client_assertion)
    - Trust by issuer: You must include the `client_id` in the request.
    - Trust by URI: Including `client_id` in the request is optional.

    For more information, see the *Request Parameters* table below.

  For more information, see API Authentication [page 1996].

Request - Relying Party

URI: `https://<Cloud Identity Services domain>/oauth2/token`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`grant_type`	Yes	string	client_credentials	Request body
`client_id`	Yes	string	Used to identify the corresponding Identity Authentication application. **Note** The `client_id` parameter is optional if the request is provided with authentication.	Request body
`token_format`	No	string	The `token_format` can be set to `opaque` to retrieve an opaque access token or to `jwt` to retrieve a JWT-based access token. If not set, the current defaults per grant type are used.	Request body
`app_tid`	No	string	Reserved	Request body

Request Example

```
grant_type=client_credentials&client_id=12b52d2c-1q34-5r5t-a576-75e85asdf523
```

Request - System as Administrator

> **Note**
>
> Use the received `access_token` to authenticate when accessing the Application Configurations API and Identity Directory API with a system as administrator.

URI: `https://<Cloud Identity Services domain>/oauth2/token`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.

Tenant ID is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`grant_type`	Yes	string	client_credentials	Request body
`client_id`	Yes	string	Used to identify the corresponding Identity Authentication application. **Note** The `client_id` parameter is optional if the request is provided with authentication.	Request body

Request Example

```
Client - Relying Party
Request:
grant_type=client_credentials&client_id=12b52d2c-1q34-5r5t-a576-75e85asdf523
```

```
Client - System as Administrator
Request:
grant_type=client_credentials
```

Response

Response Status and Error Codes

Code	Reason	Description
200 OK	Successful operation.	Returns `access_token`.
400 Bad Request	Missing or wrong parameter.	Returns information about the error.

Response Payload Example

```
{
  "access_token":
"eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdWQiOiIxMnM1NHc2YS0xdzU
2LTVlOGItZjI1Ni0wN3YwNXZmZ2IzMTIiLCJzdWIiOiIxMnM1NHc2YS0xdzU2LTVl
```

```
OGItZjI1Ni0wN3YwNXZmZ2IzMTIiLCJpc3MiOiJodHRwczovLzx0ZW5hbnQgSUQ-L
mFjY291bnRzLm9uZGVtYW5kLmNvbSIsImV4cCI6MTU4ODAyNzk3MCwiaWF0IjoxNT
g4MDI0MTEyLCJqdGkiOiIzMDhiYTg5NC0zMGE0LTQ4MDUtYWUzOS1vc2EyYXNkMjU
2ZTIifQ.9IJLkgsKQtYzMRrHUY004X7OJYWI-L8QV1oG42EZ0Kc",
  "token_type": "Bearer",
  "expires_in": 3600
}
}
```

> **Note**
>
> The format of the `issuer` depends on the configuration in the administration console for SAP Cloud Identity Services. For more information, see Tenant OpenID Connect (OIDC) Configurations [page 2117].

Parent topic: Using Client Credentials Flow [page 2660]

Previous task: Configure OpenID Connect (OIDC) Application for Client Credentials Flow [page 2663]

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.3 Using Resource Owner Password Credentials Flow

To get access using the resource owner password credentials flow, follow the procedures below.

Tasks 1 and 2 are configurations on Identity Authentication. Task 3 is a configuration on the client (relying party).

> **Restriction**
>
> Only corporate identity providers configured with OpenID Connect (OIDC) support this flow.
>
> If you've configured a corporate identity provider for your application, multifactor authentication isn't supported with this flow. For human users we recommend that you use a web-based flow like the authorization code flow.

1. Create OpenID Connect (OIDC) Application [page 2670]
   You can create a new OpenID Connect (OIDC) application.
2. Configure OpenID Connect (OIDC) Application for Resource Owner Password Credentials Flow [page 2673]
   This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the resource owner password credentials flow.
3. Configure the Client to Call Identity Authentication Token Endpoint for Resource Owner Password Credentials Flow [page 2675]

The token endpoint is used to get the user's access token, id token and refresh token.

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.3.1 Create OpenID Connect (OIDC) Application

You can create a new OpenID Connect (OIDC) application.

Context

To create a new OpenID Connect (OIDC) application you have to add a new application to the list of applications in Identity Authentication, and set the protocol of the application to OpenID Connect (OIDC).

To create a new OpenID Connect (OIDC) application follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the *Create* button on the left-hand panel and fill in the information in the dialog box.
4. Provide the following information:

Fields	Notes
Display Name	Required. The display name of the application appears on the logon and registration pages.
Home URL	Optional. Users are redirected to the *Home URL* after activating their accounts, when they are created via a CSV file import or the user registration service of Identity Authentication.
Type	Optional. **⚠ Caution** If the type of the application is not the correct one, the system may change it automatically without a warning.
Parent Application	Optional.

Fields	Notes
	→ Remember Newly created applications with an assigned parent application will inherit all the configurations from the parent except for the `Client ID` and `Secrets`. The inherited configurations will be marked as such. • If you are an administrator with the *Manage Applications* role, the newly created applications can have as parent any of the parent applications in the administration console. Choose the parent application from the list that pops up. • If you are an administrator with authorizations based on policies, you can assign as parent only the application to which you have read or manage applications rights. The applications to which you have read or manage rights are restricted on the basis of the `application.organization` attribute in the policy. Choose the parent application from the list that pops up. For more information about how to assign authorizations based on policies, see Configure Application Authorizations [page 2325].
Organization ID	Optional. This field allows you to group the applications under a specific organization and on the basis of the *Organization ID* to restrict the access to these applications. The *Organization ID* can contain lowercase Latin letters (a-z), numbers (0-9), and the underscore (_) character. It can be up to 255 characters. • If you are an administrator with *Manage Applications* role, the newly created applications by default are in the `global` *Organization ID*. You can change the *Organization ID* during the creation of the new application by entering a new name or later you can edit it. • If your authorizations are based on policies, the newly created applications can take the *Organization ID* of the applications to which you have manage rights on the basis of the `application.organization` attribute in the policy. Choose the allowed *Organization ID* from the drop-down. **ⓘ Note** Applications created either as subscriptions within or with the Identity service of an SAP BTP subaccount receive the *Organization ID* of the application that represents the trust configuration of that subaccount.

Fields	Notes
Protocol Type	Select the *OpenID Connect(OIDC)* radio-button. → Remember Newly created applications with an assigned parent application will inherit the protocol from the parent. To change the protocol, first create the new application and then edit it.

5. Choose *+Create*.

   Once the application has been created, the system displays the message `Application <name of application> created`.

   The newly created application appears on the list with the applications on the left. It is selected and you can set its protocol to OpenID Connect (OIDC).

Next Steps

(Optional) If necessary, configure additional settings for the application. For more information about the supported configurations for the OpenID Connect (OIDC) applications, see the following links:

- Change an Application's Display Name [page 1934]
- Configure an Application's Home URL [page 1937]
- Visit an Application's Web Page [page 1939]
- Configuring User Attributes from the Identity Directory [page 1973]
- Configuring Attributes Based on Flexible Expressions [page 1979]
- Configure Secrets for API Authentication [page 1997]
- Configure Certificates for API Authentication [page 2001]
- Enable or Disable Kerberos Authentication for an Application [page 2556]
- Configure Risk-Based Authentication for an Application [page 2007]
- Configure User Access to the Application [page 2025]
- Authenticating Identity Provider for an Application [page 2065]
- Enable Email Verification [page 2026]
- Configuring Password Policies [page 2223]
- Configuring Privacy Policies [page 2235]
- Configuring Terms of Use [page 2273]
- Display Application Name on Logon Page [page 2033]
- Configure Logo [page 2030]
- Choose and Configure a Branding Style for an Application [page 2035]
- Configure Registration and Upgrade Forms [page 2046]
- Protecting Self-Registration, Upgrade with Phone Verification [page 2050]
- Protecting Application Forms with Google reCAPTCHA [page 2056]
- Configure Identity Federation [page 2614]

- Configure Authentication Request to Corporate IdPs [page 1966]

Task overview: Using Resource Owner Password Credentials Flow [page 2669]

Next task: Configure OpenID Connect (OIDC) Application for Resource Owner Password Credentials Flow [page 2673]

1.7.18.3.2 Configure OpenID Connect (OIDC) Application for Resource Owner Password Credentials Flow

This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the resource owner password credentials flow.

Prerequisites

You have an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party).

To configure an OpenID Connect (OIDC) trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect (OIDC) Configuration*.
6. Under the *Configure Manually* section provide a name of your choice.
7. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. → Tip To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. ⚠ Caution The application will stop working if the configuration of the application is not updated with the new certificate. ⓘ Note Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

→ Tip

When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

8. Select the *Password*

ⓘ Note

Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

9. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

→ Remember

Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

- Configure HTTP basic authentication for the application. For more information about the configuration, see Configure Secrets for API Authentication [page 1997].
- Enable the public client flows option for this application. In the administration console, choose *the OpenID Connect (OIDC) application* > *Client Authentication under the Trust tab* > *Enable Public Client Flows under Public Client*. Optionally, you can configure the API permission groups. For more information, see Consuming APIs from Other Applications [page 2889] .

  > ### Note
  >
  > The *Public* client type is used for environments where it is difficult to protect the client credential, such as mobile and desktop applications, and client-side parts of web applications.

Task overview: Using Resource Owner Password Credentials Flow [page 2669]

Previous task: Create OpenID Connect (OIDC) Application [page 2670]

Next: Configure the Client to Call Identity Authentication Token Endpoint for Resource Owner Password Credentials Flow [page 2675]

1.7.18.3.3 Configure the Client to Call Identity Authentication Token Endpoint for Resource Owner Password Credentials Flow

The token endpoint is used to get the user's access token, id token and refresh token.

Prerequisites

- For your business application, there's an OpenID Connect (OIDC) application in Identity Authentication. For more information, see Create OpenID Connect (OIDC) Application [page 1925].
- For your OpenID Connect (OIDC) application in Identity Authentication, you've prepared an authentication credential for the API call. The API supports the following authentication schemes:
  - Basic
    For more information, see `Authorization` in the *Request Header* table below.
  - X.509
    For more information, see `Authorization` in the *Request Header* table below.
  - JWT (client_assertion)
    - Trust by issuer: You must include the `client_id` in the request.
    - Trust by URI: Including `client_id` in the request is optional.

    For more information, see the *Request Parameters* table below.

For more information, see API Authentication [page 1996].

This document explains how to call the token endpoint and what are the parameters supported by Identity Authentication.

Request

URI: `https://<Cloud Identity Services domain>/oauth2/token`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Request Parameters

Parameter	Required	Data Type	Additional Information	Parameter Type
`grant_type`	Yes	string	The supported value is `password`	Request body
`username`	Yes	string	The user identifier.	Request body
`password`	Yes	string	The user password, and the one-time password (OTP) generated by the user's registered device if the application is configured to require two-factor authentication. **Note** If the application requires two-factor authentication, the OTP code must be linked to the password. For example, `password=mypassword123456`.	Request body

Parameter	Required	Data Type	Additional Information	Parameter Type
`client_id`	Yes	string	Used to identify the corresponding Identity Authentication application. **Note** The `client_id` parameter is optional if the request is provided with authentication.	Request body
`refresh_expiry`	No	string	Reduces the expiration of a refresh token. It's useful if your application is called from mobile and web applications, and both have different session requirements. If you set the token lifetime to 0, you won't receive a `refresh_token` in response.	Request body
`max_exchange_period`	No	string	You can set a limit on how long the application can exchange user tokens without re-authenticating after having acquired the initial token. The `max_exchange_period` parameter has a numeric value which can be set at between 1 hour and 4320 hours (six months). **→ Remember** If the `max_exchange_period` value is smaller than the configuration defined in token policy, this value overrides the configuration defined in token policy configuration and propagates it once the token is exchanged or refreshed. If the value is bigger, it is ignored and the configuration defined in token policy is taken into consideration. For more information, see Token Policy Configuration for Applications [page 2752]	Request body

Parameter	Required	Data Type	Additional Information	Parameter Type
`token_format`	No	string	The `token_format` can be set to `opaque` to retrieve an opaque access token or to `jwt` to retrieve a JWT based access token. If not set, the current defaults per grant type are used.	Request body
`app_tid`	No	string	Reserved.	Request body
`scope`	No	string	Value must be space delimited parameter, for example: `scope=openid email`. The supported values are: • `openid` • `email` • `profile` • `groups`	Request body

Request Example

```
grant_type=password&username=<user identifier>&password=<user
password>[<otpcode>]
```

Response

Response Status and Error Codes

Code	Result or X-Message Code	Description
200 OK	Successful operation.	Returns `access_token`, `refresh_token`, and `id_token`. **Note** The `refresh_token` is used to obtain a new `id_token` and `access_token` when the current token becomes invalid or expires. For more information, see Call Identity Authentication Refresh Token [page 2725]. The `id_token` is in the form of a JWT (JSON Web Token) and contains information about the user. The `id_token` is valid for 60 minutes.
400 Bad Request	Missing or wrong parameter.	Returns an information about the error.

Code	Result or X-Message Code	Description
401 Unauthorized	Wrong user ID or password parameters passed for the basic authentication. For more information, see Configure Secrets for API Authentication [page 1997].	The authentication of the client (relying party) failed.
	PASSWORD_CHANGE_REQUIRED	When the user must change his or her password before logon.
	INITIAL_PASSWORD_EXPIRED	When the initial password of the user has expired. After the validity of the initial password expires, the user can't log on to the application and must contact the administrator.

Response Payload Example


```
{
 "access_token": "387qb8bc-7t78-4eb8-8a8c-cfbe31860811",
 "refresh_token": "d12a12abcd198765dd54r456e98321"
 "id_token":
"eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdWQiOiIxMmEzNGI1Yy02ZDc4LTll
MWYtZzM0NS02N2g4OWlqa2wxMjMiLCJzdWIiOiJQMTIzNDU2IiwibWFpbCI6ImRvbmEubW
9vcmVAZXhhbXBsZS5jb20iLCJpc3MiOiJodHRwczovL215LXRlbmFudC5hY2NvdW50cy5v
bmRlbWFuZC5jb20iLCJsYXN0X25hbWUiOiJNb29yZSIsInNhcF91aWQiOiIxMjM0NTZhYmM
3ZGU4LWZnaGktOTEyMy1qNDU2LTc4OTEya2wzNG01NiIsImV4cCI6MTU4ODAxODkyNSwiaW
F0IjoxNTg4MDEzNzYwLCJmaXJzdF9uYW1lIjoiRG9uYSIsImp0aSI6IjM4ZTQyMzMwLWRlN
2EtNDEzMC1hM2ExLWI1ODJiNTI4ZGE5OCJ9.-LSwBN2WSqnnqSkzSbg9iRmtAMR4moU5TpE
40mX0Umwg",
 "token_type": "Bearer",
 "expires_in": 300
}
```


> ### Note
>
> The format of the `issuer` depends on the configuration in the administration console for SAP Cloud Identity Services. For more information, see Tenant OpenID Connect (OIDC) Configurations [page 2117].

Parent topic: Using Resource Owner Password Credentials Flow [page 2669]

Previous task: Configure OpenID Connect (OIDC) Application for Resource Owner Password Credentials Flow [page 2673]

1.7.18.4 Using Implicit Flow

To authenticate using the implicit flow, follow the procedures below. Tasks 1 and 2 are configurations on Identity Authentication side. Task 3 is a configuration, done on the client (relying party) side.

> ### Note
>
> For applications such as mobile applications, native applications and single page applications, the authorization code flow with PKCE is recommended. For more information, see Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow with PKCE [page 2650].

1. Create OpenID Connect (OIDC) Application [page 2680]
   You can create a new OpenID Connect (OIDC) application.
2. Configure OpenID Connect Application for Implicit Flow [page 2683]
   This document is intended to help you configure an OpenID Connect application in the administration console for SAP Cloud Identity Services for the implicit flow.
3. Configure the Client to Call Identity Authentication Authorize Endpoint for Implicit Flow [page 2686]
   The `authorize` endpoint is used for authentication and returns the `id_token` when using the implicit flow.

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.4.1 Create OpenID Connect (OIDC) Application

You can create a new OpenID Connect (OIDC) application.

Context

To create a new OpenID Connect (OIDC) application you have to add a new application to the list of applications in Identity Authentication, and set the protocol of the application to OpenID Connect (OIDC).

To create a new OpenID Connect (OIDC) application follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the *Create* button on the left-hand panel and fill in the information in the dialog box.
4. Provide the following information:

Fields	Notes
Display Name	Required. The display name of the application appears on the logon and registration pages.
Home URL	Optional. Users are redirected to the *Home URL* after activating their accounts, when they are created via a

Fields	Notes
	CSV file import or the user registration service of Identity Authentication.
Type	Optional. **⚠ Caution** If the type of the application is not the correct one, the system may change it automatically without a warning.
Parent Application	Optional. **→ Remember** Newly created applications with an assigned parent application will inherit all the configurations from the parent except for the `Client ID` and `Secrets`. The inherited configurations will be marked as such. • If you are an administrator with the *Manage Applications* role, the newly created applications can have as parent any of the parent applications in the administration console. Choose the parent application from the list that pops up. • If you are an administrator with authorizations based on policies, you can assign as parent only the application to which you have read or manage applications rights. The applications to which you have read or manage rights are restricted on the basis of the `application.organization` attribute in the policy. Choose the parent application from the list that pops up. For more information about how to assign authorizations based on policies, see Configure Application Authorizations [page 2325].
Organization ID	Optional. This field allows you to group the applications under a specific organization and on the basis of the *Organization ID* to restrict the access to these applications. The *Organization ID* can contain lowercase Latin letters (a-z), numbers (0-9), and the underscore (_) character. It can be up to 255 characters. • If you are an administrator with *Manage Applications* role, the newly created applications by default are in the `global` *Organization ID*. You can change the *Organization ID* during the creation of the new application by entering a new name or later you can edit it. • If your authorizations are based on policies, the newly created applications can take the *Organization ID* of the applications to which you have manage rights on

Fields	Notes
	the basis of the `application.organization` attribute in the policy. Choose the allowed *Organization ID* from the drop-down. **Note** Applications created either as subscriptions within or with the Identity service of an SAP BTP subaccount receive the *Organization ID* of the application that represents the trust configuration of that subaccount.
Protocol Type	Select the *OpenID Connect(OIDC)* radio-button. **→ Remember** Newly created applications with an assigned parent application will inherit the protocol from the parent. To change the protocol, first create the new application and then edit it.

5. Choose *+Create*.

   Once the application has been created, the system displays the message `Application <name of application> created`.

   The newly created application appears on the list with the applications on the left. It is selected and you can set its protocol to OpenID Connect (OIDC).

Next Steps

(Optional) If necessary, configure additional settings for the application. For more information about the supported configurations for the OpenID Connect (OIDC) applications, see the following links:

- Change an Application's Display Name [page 1934]
- Configure an Application's Home URL [page 1937]
- Visit an Application's Web Page [page 1939]
- Configuring User Attributes from the Identity Directory [page 1973]
- Configuring Attributes Based on Flexible Expressions [page 1979]
- Configure Secrets for API Authentication [page 1997]
- Configure Certificates for API Authentication [page 2001]
- Enable or Disable Kerberos Authentication for an Application [page 2556]
- Configure Risk-Based Authentication for an Application [page 2007]
- Configure User Access to the Application [page 2025]
- Authenticating Identity Provider for an Application [page 2065]
- Enable Email Verification [page 2026]
- Configuring Password Policies [page 2223]

- Configuring Privacy Policies [page 2235]
- Configuring Terms of Use [page 2273]
- Display Application Name on Logon Page [page 2033]
- Configure Logo [page 2030]
- Choose and Configure a Branding Style for an Application [page 2035]
- Configure Registration and Upgrade Forms [page 2046]
- Protecting Self-Registration, Upgrade with Phone Verification [page 2050]
- Protecting Application Forms with Google reCAPTCHA [page 2056]
- Configure Identity Federation [page 2614]
- Configure Authentication Request to Corporate IdPs [page 1966]

Task overview: Using Implicit Flow [page 2679]

Next task: Configure OpenID Connect Application for Implicit Flow [page 2683]

1.7.18.4.2 Configure OpenID Connect Application for Implicit Flow

This document is intended to help you configure an OpenID Connect application in the administration console for SAP Cloud Identity Services for the implicit flow.

Prerequisites

You have an OpenID Connect application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party), and its redirect URIs.

To configure an OpenID Connect trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.

2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> ⓘ Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect Configuration*.
6. Manually enter the communication settings negotiated between Identity Authentication and the client.

Setting	Description
Name (mandatory)	Provide a name of your choice.
Redirect URIs (optional)	The redirection URIs to which the response can be sent. You can add up to 20 redirect URIs.
Post Logout Redirect URIs (optional)	The redirection URIs where the user can be forwarded after logout. You can add up to 20 redirect URIs.
Front-Channel Logout URIs (optional)	URIs which will be requested for logout. You can add up to 20 URIs.

> → Tip
>
> For more information about the format of the redirect URIs and post logout redirect URIs, see OpenID Connect (OIDC) Application Configurations [page 2747].

7. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. → Tip To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. ⚠ Caution The application will stop working if the configuration of the application is not updated with the new certificate.

Option	Description
	Note Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

> **Tip**
>
> When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

8. Selct the *Implicit* grant type.

> **Note**
>
> Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

9. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

> **Remember**
>
> Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

Configure HTTP basic authentication for the application. For more information about the configuration, see Configure Secrets for API Authentication [page 1997].

Task overview: Using Implicit Flow [page 2679]

Previous task: Create OpenID Connect (OIDC) Application [page 2680]

Next: Configure the Client to Call Identity Authentication Authorize Endpoint for Implicit Flow [page 2686]

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.4.3 Configure the Client to Call Identity Authentication Authorize Endpoint for Implicit Flow

The `authorize` endpoint is used for authentication and returns the `id_token` when using the implicit flow.

Prerequisites

- For your business application, there's an OpenID Connect (OIDC) application in Identity Authentication. For more information, see Create OpenID Connect (OIDC) Application [page 1925].
- For your OpenID Connect (OIDC) application in Identity Authentication, you've prepared an authentication credential for the API call. The API supports the following authentication schemes:
  - Basic
    For more information, see `Authorization` in the *Request Header* table below.
  - X.509
    For more information, see `Authorization` in the *Request Header* table below.
  - JWT (client_assertion)
    - Trust by issuer: You must include the `client_id` in the request.
    - Trust by URI: Including `client_id` in the request is optional.

    For more information, see the *Request Parameters* table below.

  For more information, see API Authentication [page 1996].

This document explains how to call the `authorize` endpoint and what are the request parameters supported by Identity Authentication for the implicit flow.

Request

URI: `https://<Cloud Identity Services domain>/oauth2/authorize`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *GET*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Request Parameters

Parameter	Required	Data Type	Additional Information	Parameter Type
`client_id`	Yes	string	Used to identify the corresponding Identity Authentication application. **Note** The `client_id` parameter is optional if the request is provided with authentication.	Request
`nonce`	Yes	string	Free text.	Request
`redirect_uri`	Yes	string	One of the valid URIs configured for the application. You can configure up to five URIs. For more information, see Configure OpenID Connect (OIDC) Application [page 1949].	Request
`response_type`	Yes	string	The supported value is `id_token`	Request
`scope`	Yes	string	The supported value is `openid`	Request
`login_hint`	No	string	The `login_hint` parameter helps the user when he or she is known to the service provider (SP). Thus it prevents the user from re-typing the user identifier on the logon or conditional screen. Supported values are the allowed logon identifiers for the users. The options are *User ID*, *Login Name*, and *Email* (For SAML 2.0). For more information, see Configure Allowed Logon Identifiers [page 2146] .	Request
`logout_uri`	No	string	• Create OpenID Connect (OIDC) Application [page 1925] • Logout URI Rules [page 2750]	Request
`max_age`	No	integer	Maximum time in seconds since the user was last authenticated. When `max_age` has been reached, the user must re-authenticate.	Request

Parameter	Required	Data Type	Additional Information	Parameter Type
`prompt`	No	string	This parameter enables the client to determine if the user is still present in the current session. Supported values are: • *login* Force reauthentication of the user. • *none* Without user interaction, check if there is an exisiting user session. If there is a session, continue as normal. If there is no session or if the user is forced to authenticate for other reasons, return the error code `login_required`.	Request
state	No	string	Free text.	Request

Request Example

```
https://my-tenant.ondemand.com/oauth2/authorize?
response_type=id_token&scope=openid&client_id=T000000&state=state&nonce=m-0G6_FaS
3Kg&redirect_uri=https://example.com&login_hint=dona.moore@example.com
```

Response

Response Status and Error Codes

Code	Reason	Description
200 OK	Successful operation.	Additionally provides a URL in the header field Location. **Note** The URL contains `id_token`. The `id_token` is in the form of a JWT (JSON Web Token) and contains information about the user. The `id_token` is valid for 60 minutes.
400	Missing or wrong parameter	

Response Example

```
https://www.example.com/#id_token=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.ey
```

```
JhdWQiOiIxMmEzNGI1Yy02ZDc4LTllMWYtZzM0NS02N2g4OWlqa2wxMjMiLCJzdWIiOiJQMTIzNDU2Iiw
ibWFpbCI6ImRvbmEubW9vcmVAZXhhbXBsZS5jb20iLCJpc3MiOiJodHRwczovL215LXRlbmFudC5hY2Nv
dW50cy5vbmRlbWFuZC5jb20iLCJsYXN0X25hbWUiOiJNb29yZSIsInNhcF91aWQiOiIxMjM0NTZhYmM3Z
GU4LWZnaGktOTEyMy1qNDU2LTc4OTEya2wzNG01NiIsImV4cCI6MTU4ODAxODkyNSwiaWF0IjoxNTg4MD
EzNzYwLCJmaXJzdF9uYW1lIjoiRG9uYSIsImp0aSI6IjM4ZTQyMzMwLWRlN2EtNDEzMC1hM2ExLWI1ODJ
iNTI4ZGE5OCJ9.-
LSwBN2WSqnnqSkzSbg9iRmtAMR4moU5TpE40mX0Umw&state=state&token_type=Bearer
```

Parent topic: Using Implicit Flow [page 2679]

Previous task: Configure OpenID Connect Application for Implicit Flow [page 2683]

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.5 Using JWT Bearer Flow

To authenticate using the JWT Bearer flow, follow the procedures below. Tasks 1 and 2 are configurations on Identity Authentication side. Task 3 is a configuration, done on the client (relying party) side.

The JWT Bearer flow is used for token exchanges from a server application, where the server already has a token. For example:

- To switch for a public client with restricted audience to a confidential client with full audience list
- To authenticate with a token coming from a different identity provider (IdP) to an Identity Authentication token

> → Remember
>
> If the `id_token` is issued by another OpenID Connect provider, like Microsoft Entra ID for example, you must configure that provider as a corporate identity provider in the administration console of Identity Authentication. For more information, see Configure Trust with OpenID Connect Corporate Identity Provider [page 2579].

1. Create OpenID Connect (OIDC) Application [page 2690]
   You can create a new OpenID Connect (OIDC) application.
2. Configure OpenID Connect (OIDC) Application for JWT Bearer Flow [page 2693]
   This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the JWT bearer flow.
3. Configure the Client to Call Identity Authentication JWT Bearer Token [page 2696]
   With the JWT bearer flow you can use an `id_token` ID from an application which is in the same Identity Authentication tenant, or an external `id_token` whose issuer is trusted by Identity Authentication.

1.7.18.5.1 Create OpenID Connect (OIDC) Application

You can create a new OpenID Connect (OIDC) application.

Context

To create a new OpenID Connect (OIDC) application you have to add a new application to the list of applications in Identity Authentication, and set the protocol of the application to OpenID Connect (OIDC).

To create a new OpenID Connect (OIDC) application follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the *Create* button on the left-hand panel and fill in the information in the dialog box.
4. Provide the following information:

Fields	Notes
Display Name	Required. The display name of the application appears on the logon and registration pages.
Home URL	Optional. Users are redirected to the *Home URL* after activating their accounts, when they are created via a CSV file import or the user registration service of Identity Authentication.
Type	Optional. **⚠ Caution** If the type of the application is not the correct one, the system may change it automatically without a warning.
Parent Application	Optional. **→ Remember** Newly created applications with an assigned parent application will inherit all the configurations from the parent except for the `Client ID` and `Secrets`. The inherited configurations will be marked as such. • If you are an administrator with the *Manage Applications* role, the newly created applications can have as parent any of the parent applications in the

Fields	Notes
	administration console. Choose the parent application from the list that pops up. • If you are an administrator with authorizations based on policies, you can assign as parent only the application to which you have read or manage applications rights. The applications to which you have read or manage rights are restricted on the basis of the `application.organization` attribute in the policy. Choose the parent application from the list that pops up. For more information about how to assign authorizations based on policies, see Configure Application Authorizations [page 2325].
Organization ID	Optional. This field allows you to group the applications under a specific organization and on the basis of the *Organization ID* to restrict the access to these applications. The *Organization ID* can contain lowercase Latin letters (a-z), numbers (0-9), and the underscore (_) character. It can be up to 255 characters. • If you are an administrator with *Manage Applications* role, the newly created applications by default are in the `global` *Organization ID*. You can change the *Organization ID* during the creation of the new application by entering a new name or later you can edit it. • If your authorizations are based on policies, the newly created applications can take the *Organization ID* of the applications to which you have manage rights on the basis of the `application.organization` attribute in the policy. Choose the allowed *Organization ID* from the drop-down. **ⓘ Note** Applications created either as subscriptions within or with the Identity service of an SAP BTP subaccount receive the *Organization ID* of the application that represents the trust configuration of that subaccount.
Protocol Type	Select the *OpenID Connect(OIDC)* radio-button. **→ Remember** Newly created applications with an assigned parent application will inherit the protocol from the parent. To change the protocol, first create the new application and then edit it.

5. Choose *+Create*.

Once the application has been created, the system displays the message `Application <name of application> created`.

The newly created application appears on the list with the applications on the left. It is selected and you can set its protocol to OpenID Connect (OIDC).

Next Steps

(Optional) If necessary, configure additional settings for the application. For more information about the supported configurations for the OpenID Connect (OIDC) applications, see the following links:

- Change an Application's Display Name [page 1934]
- Configure an Application's Home URL [page 1937]
- Visit an Application's Web Page [page 1939]
- Configuring User Attributes from the Identity Directory [page 1973]
- Configuring Attributes Based on Flexible Expressions [page 1979]
- Configure Secrets for API Authentication [page 1997]
- Configure Certificates for API Authentication [page 2001]
- Enable or Disable Kerberos Authentication for an Application [page 2556]
- Configure Risk-Based Authentication for an Application [page 2007]
- Configure User Access to the Application [page 2025]
- Authenticating Identity Provider for an Application [page 2065]
- Enable Email Verification [page 2026]
- Configuring Password Policies [page 2223]
- Configuring Privacy Policies [page 2235]
- Configuring Terms of Use [page 2273]
- Display Application Name on Logon Page [page 2033]
- Configure Logo [page 2030]
- Choose and Configure a Branding Style for an Application [page 2035]
- Configure Registration and Upgrade Forms [page 2046]
- Protecting Self-Registration, Upgrade with Phone Verification [page 2050]
- Protecting Application Forms with Google reCAPTCHA [page 2056]
- Configure Identity Federation [page 2614]
- Configure Authentication Request to Corporate IdPs [page 1966]

Task overview: Using JWT Bearer Flow [page 2689]

Next task: Configure OpenID Connect (OIDC) Application for JWT Bearer Flow [page 2693]

1.7.18.5.2 Configure OpenID Connect (OIDC) Application for JWT Bearer Flow

This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the JWT bearer flow.

Prerequisites

You have an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party), and its redirect URIs.

To configure an OpenID Connect (OIDC) trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect (OIDC) Configuration*.
6. Manually enter the communication settings negotiated between Identity Authentication and the client.

Setting	Description
Name (mandatory)	Provide a name of your choice.

Setting	Description
Redirect URIs (optional)	The redirection URIs to which the response can be sent. You can add up to 20 redirect URIs.
Post Logout Redirect URIs (optional)	The redirection URIs where the user can be forwarded after logout. You can add up to 20 redirect URIs.
Front-Channel Logout URIs (optional)	URIs which will be requested for logout. You can add up to 20 URIs.

> → Tip
>
> For more information about the format of the redirect URIs and post logout redirect URIs, see OpenID Connect (OIDC) Application Configurations [page 2747].

7. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. → Tip To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. ⚠ Caution The application will stop working if the configuration of the application is not updated with the new certificate. ⓘ Note Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

> → Tip
>
> When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

8. Select the *JWT Bearer* grant type.

> ⓘ Note
>
> Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

9. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

> → Remember
>
> Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

- Configure HTTP basic authentication for the application. For more information about the configuration, see Configure Secrets for API Authentication [page 1997].
- Enable the public client flows option for this application. In the administration console, choose *the OpenID Connect (OIDC) application* > *Client Authentication under the Trust tab* > *Enable Public Client Flows under Public Client*. Optionally, you can configure the API permission groups. For more information, see Consuming APIs from Other Applications [page 2889] .

> ⓘ Note
>
> The *Public* client type is used for environments where it is difficult to protect the client credential, such as mobile and desktop applications, and client-side parts of web applications.

Task overview: Using JWT Bearer Flow [page 2689]

Previous task: Create OpenID Connect (OIDC) Application [page 2690]

Next: Configure the Client to Call Identity Authentication JWT Bearer Token [page 2696]

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.5.3 Configure the Client to Call Identity Authentication JWT Bearer Token

With the JWT bearer flow you can use an `id_token` ID from an application which is in the same Identity Authentication tenant, or an external `id_token` whose issuer is trusted by Identity Authentication.

> ⚠ Restriction
>
> Token exchange to external tokens is not supported for public clients and is intentionally restricted for security reasons.
>
> If your use case requires external token exchange, use a **confidential client** instead.

Prerequisites

- For your business application, there's an OpenID Connect (OIDC) application in Identity Authentication. For more information, see Create OpenID Connect (OIDC) Application [page 1925].
- For your OpenID Connect (OIDC) application in Identity Authentication, you've prepared an authentication credential for the API call. The API supports the following authentication schemes:
  - Basic
    For more information, see `Authorization` in the *Request Header* table below.
  - X.509
    For more information, see `Authorization` in the *Request Header* table below.
  - JWT (client_assertion)
    - Trust by issuer: You must include the `client_id` in the request.
    - Trust by URI: Including `client_id` in the request is optional.

    For more information, see the *Request Parameters* table below.

  For more information, see API Authentication [page 1996].

> ⓘ Note
>
> You can't exchange external tokens in a public client.

Request

URI: `https://<Cloud Identity Services domain>/oauth2/token`

> ⓘ Note
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.

Tenant ID is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

Content-Type `application/x-www-form-urlencoded`

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Request Parameters

Parameter	Required	Data Type	Additional Information	Parameter Type

`assertion`	Yes	string	**Same tenant ID token:**
			• user identifier:
			  • if "`user_uuid`" claim is provided, it will be used to search the user by user_uuid
			  • otherwise "`sub`" claim will be used, therefore the subject name identifier from the other application will be checked to accordingly search the user in the tenant
			• audience: must contain the client ID of the requested application in "`aud`" claim
			External ID token:
			• user identifier: according to the Subject Name Identifier configuration of the used corporate identity provider, the value of "email" or "sub" claim will be used to search the user by email
			• audience: must contain either the requested Identity Authentication issuer or the client ID of the corporate identity provider in "`aud`" claim
				Request body

Parameter	Required	Data Type	Additional Information	Parameter Type
`client_id`	Yes	string	Used to identify the corresponding Identity Authentication application. **Note** The `client_id` parameter is optional if the request is provided with authentication.	Request body
`grant_type`	Yes	string	`urn:ietf:params:oauth:grant-type:jwt-bearer`	Request body
`app_tid`	No	string	Reserved.	Request body
`client_secret`	No	string	The client secret configured for basic authentication for the application. For more information, see Configure Secrets for API Authentication [page 1997].	Request body
`max_exchange_period`	No	string	You can set a limit on how long the application can exchange user tokens without re-authenticating after having acquired the initial token. The `max_exchange_period` parameter has a numeric value which can be set at between 1 hour and 4320 hours (six months). **→ Remember** If the `max_exchange_period` value is smaller than the configuration defined in token policy, this value overrides the configuration defined in token policy configuration and propagates it once the token is exchanged or refreshed. If the value is bigger, it is ignored and the configuration defined in token policy is taken into consideration. For more information, see Token Policy Configuration for Applications [page 2752]	Request body

Parameter	Required	Data Type	Additional Information	Parameter Type
`refresh_expiry`	No	string	Reduces the expiration of a refresh token. It's useful if your application is called from mobile and web applications, and both have different session requirements. If you set the token lifetime to 0, you won't receive a `refresh_token` in response.	Request body
`resource`	No	string	Identifies the dependency consumed by the application. The resource parameter is a Uniform Resource Name (URN). For more information, see Consuming APIs from Other Applications [page 2889].	Request body
`scope`	No	string	Value must be space delimited parameter, for example: `scope=openid email`. The supported values are: • `openid` • `email` • `profile` • `groups` • `offline_access` **Note** The new tokens are independently created from the Identity Authentication Web session. This means that even if a user logs out from Identity Authentication the `refresh_token` will exist in the database until it expires, and can be used to perform the refresh token flow if the user is not present in Identity Authentication with a Web session.	Request body

Parameter	Required	Data Type	Additional Information	Parameter Type
`token_format`	No	string	The `token_format` can be set to `opaque` to retrieve an opaque access token or to `jwt` to retrieve a JWT based access token. If not set, the current defaults per grant type are used.	Request body

Request Example

```
POST /oauth2/token
Content-Type: application/x-www-form-urlencoded
grant_type=urn:ietf:params:oauth:grant-type:jwt-bearer&
client_id=c95ad226-fg34-abc6-abc6-6e8a6b9f2442&
client_secret=PASu9/0sTUeUCG1LAYmSQ18Ut0zrfMz&
assertion=eyJraWQiOiJva3JFVVNsbHRIbGlL...hoXvzy2TgmdPS0LlAXA
```

Response

Response Example

```
Content-Type: application/json

{
    "access_token": "dzEyYTM0YmM2OWZhZnU3NmE3OThzNjZmN3NmNjg3MQ",
    "refresh_token": "s95s27af4fefae76c798a66f7ef9034",
    "id_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJQMT
    IzNDU2IiwiYXVkIjoiMjAzZjYyMDktMjRmZi00MmY1LWE0MTMtNmRlNWE5M2ZjM
    jYxIiwidXNlcl91dWlkIjoiMDEyM2E0NS1mMWFmLTFmMWMtYTMxMy03NDZkMTIw
    YWJjZDkiLCJtYWlsIjoiZG9uYS5tb29yZUBleGFtcGxlLmNvbSIsImlzcyI6Imh
    0dHBzOi8vdGVzdHRlbmFudC50ZXN0LmNvbSIsImxhc3RfbmFtZSI6Ik1vb3JlIi
    wiZXhwIjoxNjM5MTQ0MDEwLCJpYXQiOjE2MzkxNDA0MTAsImZpcnN0X25hbWUiO
    iJEb25hIiwianRpIjoiMzFlMzJjNWYtNjljC00NmE2LWE5MDctOTExZDExMjA1
    czBzIn0.QYftNy7WgISWoKpRXh4_RX5UDNXsXDTndYuIA85L2II",
    "token_type": "Bearer",
    "expires_in": 3600
}
```

Parent topic: Using JWT Bearer Flow [page 2689]

Previous task: Configure OpenID Connect (OIDC) Application for JWT Bearer Flow [page 2693]

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

JSON Web Token (JWT) Profile for OAuth 2.0 Client Authentication and Authorization Grants

1.7.18.6 Using Token Exchange

To authenticate using the Token Exchange flow, follow the procedures below. Tasks 1 and 2 are configurations on Identity Authentication side. Task 3 is a configuration, done on the client (relying party) side.

The Token Exchange is used, as the name states, for token exchanges. For more information about this flow, see the OAuth 2.0 Token Exchange specification.

1.7.18.6.1 Create OpenID Connect (OIDC) Application

You can create a new OpenID Connect (OIDC) application.

Context

To create a new OpenID Connect (OIDC) application you have to add a new application to the list of applications in Identity Authentication, and set the protocol of the application to OpenID Connect (OIDC).

To create a new OpenID Connect (OIDC) application follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the *Create* button on the left-hand panel and fill in the information in the dialog box.
4. Provide the following information:

Fields	Notes
Display Name	Required. The display name of the application appears on the logon and registration pages.
Home URL	Optional. Users are redirected to the *Home URL* after activating their accounts, when they are created via a CSV file import or the user registration service of Identity Authentication.
Type	Optional. **⚠ Caution** If the type of the application is not the correct one, the system may change it automatically without a warning.
Parent Application	Optional.

Fields	Notes
	→ Remember Newly created applications with an assigned parent application will inherit all the configurations from the parent except for the `Client ID` and `Secrets`. The inherited configurations will be marked as such. • If you are an administrator with the *Manage Applications* role, the newly created applications can have as parent any of the parent applications in the administration console. Choose the parent application from the list that pops up. • If you are an administrator with authorizations based on policies, you can assign as parent only the application to which you have read or manage applications rights. The applications to which you have read or manage rights are restricted on the basis of the `application.organization` attribute in the policy. Choose the parent application from the list that pops up. For more information about how to assign authorizations based on policies, see Configure Application Authorizations [page 2325].
Organization ID	Optional. This field allows you to group the applications under a specific organization and on the basis of the *Organization ID* to restrict the access to these applications. The *Organization ID* can contain lowercase Latin letters (a-z), numbers (0-9), and the underscore (_) character. It can be up to 255 characters. • If you are an administrator with *Manage Applications* role, the newly created applications by default are in the `global` *Organization ID*. You can change the *Organization ID* during the creation of the new application by entering a new name or later you can edit it. • If your authorizations are based on policies, the newly created applications can take the *Organization ID* of the applications to which you have manage rights on the basis of the `application.organization` attribute in the policy. Choose the allowed *Organization ID* from the drop-down. **ⓘ Note** Applications created either as subscriptions within or with the Identity service of an SAP BTP subaccount receive the *Organization ID* of the application that represents the trust configuration of that subaccount.

Fields	Notes
Protocol Type	Select the *OpenID Connect(OIDC)* radio-button. → Remember Newly created applications with an assigned parent application will inherit the protocol from the parent. To change the protocol, first create the new application and then edit it.

5. Choose *+Create*.

   Once the application has been created, the system displays the message `Application <name of application> created`.

   The newly created application appears on the list with the applications on the left. It is selected and you can set its protocol to OpenID Connect (OIDC).

Next Steps

(Optional) If necessary, configure additional settings for the application. For more information about the supported configurations for the OpenID Connect (OIDC) applications, see the following links:

- Change an Application's Display Name [page 1934]
- Configure an Application's Home URL [page 1937]
- Visit an Application's Web Page [page 1939]
- Configuring User Attributes from the Identity Directory [page 1973]
- Configuring Attributes Based on Flexible Expressions [page 1979]
- Configure Secrets for API Authentication [page 1997]
- Configure Certificates for API Authentication [page 2001]
- Enable or Disable Kerberos Authentication for an Application [page 2556]
- Configure Risk-Based Authentication for an Application [page 2007]
- Configure User Access to the Application [page 2025]
- Authenticating Identity Provider for an Application [page 2065]
- Enable Email Verification [page 2026]
- Configuring Password Policies [page 2223]
- Configuring Privacy Policies [page 2235]
- Configuring Terms of Use [page 2273]
- Display Application Name on Logon Page [page 2033]
- Configure Logo [page 2030]
- Choose and Configure a Branding Style for an Application [page 2035]
- Configure Registration and Upgrade Forms [page 2046]
- Protecting Self-Registration, Upgrade with Phone Verification [page 2050]
- Protecting Application Forms with Google reCAPTCHA [page 2056]
- Configure Identity Federation [page 2614]

- Configure Authentication Request to Corporate IdPs [page 1966]

1.7.18.6.2 Configure OpenID Connect Application for Token Exchange

This document is intended to help you configure an OpenID Connect application in the administration console for SAP Cloud Identity Services for the Token Exchange flow.

Prerequisites

You have an OpenID Connect application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party), and its redirect URIs.

To configure an OpenID Connect trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > **Note**
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect Configuration*.
6. Manually enter the communication settings negotiated between Identity Authentication and the client.

Setting	Description
Name (mandatory)	Provide a name of your choice.
Redirect URIs (optional)	The redirection URIs to which the response can be sent. You can add up to 20 redirect URIs.
Post Logout Redirect URIs (optional)	The redirection URIs where the user can be forwarded after logout. You can add up to 20 redirect URIs.
Front-Channel Logout URIs (optional)	URIs which will be requested for logout. You can add up to 20 URIs.

> → Tip
>
> For more information about the format of the redirect URIs and post logout redirect URIs, see OpenID Connect (OIDC) Application Configurations [page 2747].

7. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. → Tip To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. ⚠ Caution The application will stop working if the configuration of the application is not updated with the new certificate. ⓘ Note Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

> → Tip
>
> When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

8. Select the *Token Exchange (RFC 8693)* grant type.

> ⓘ Note
>
> Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

9. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

> → Remember
>
> Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

- Configure HTTP basic authentication for the application. For more information about the configuration, see Configure Secrets for API Authentication [page 1997].
- Enable the public client flows option for this application. In the administration console, choose *the OpenID Connect (OIDC) application* > *Client Authentication under the Trust tab* > *Enable Public Client Flows under Public Client*. Optionally, you can configure the API permission groups. For more information, see Consuming APIs from Other Applications [page 2889] .

> ⓘ Note
>
> The *Public* client type is used for environments where it is difficult to protect the client credential, such as mobile and desktop applications, and client-side parts of web applications.

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.6.3 Configure the Client to Call Identity Authentication Token Exchange

With the Token Exchange flow you can exchange one token for another. For more information, see OAuth 2.0 Token Exchange .

Prerequisites

- For your business application, there's an OpenID Connect (OIDC) application in Identity Authentication. For more information, see Create OpenID Connect (OIDC) Application [page 1925].
- For your OpenID Connect (OIDC) application in Identity Authentication, you've prepared an authentication credential for the API call. The API supports the following authentication schemes:
  - Basic
    For more information, see `Authorization` in the *Request Header* table below.
  - X.509
    For more information, see `Authorization` in the *Request Header* table below.
  - JWT (client_assertion)
    - Trust by issuer: You must include the `client_id` in the request.
    - Trust by URI: Including `client_id` in the request is optional.

    For more information, see the *Request Parameters* table below.

  For more information, see API Authentication [page 1996].

> **Restriction**
>
> Token exchange to external tokens is not supported for public clients and is intentionally restricted for security reasons.
>
> If your use case requires external token exchange, use a **confidential client** instead.

Request

URI: `https://<Cloud Identity Services domain>/oauth2/token`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

Content-Type `application/x-www-form-urlencoded`

HTTP Method: *POST*

Request Headers

Request Parameters

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Parameter	Required	Data Type	Additional Information	Parameter Type
`grant_type`	Yes	string	`urn:ietf:params:oauth:grant-type:token-exchange`	Request body
`client_id`	Yes	string	Used to identify the corresponding Identity Authentication application. **Note** The `client_id` parameter is optional if the request is provided with authentication.	Request body
`client_secret`	No	string	The client secret configured for basic authentication for the application. For more information, see Configure Secrets for API Authentication [page 1997].	Request body
`subject_token`	Yes	string	A security token that represents the identity of the party on behalf of whom the request is being made, as stated in the OAuth 2.0 Token Exchange specification.	Request body

Parameter	Required	Data Type	Additional Information	Parameter Type
`subject_token_type`	Yes	string	The type of the security token in the `subject_token`. Allowed values: • `urn:sap:identity:oauth:token-type:saml2-session` • `urn:ietf:params:oauth:token-type:access_token` • `urn:ietf:params:oauth:token-type:refresh_token` • `urn:ietf:params:oauth:token-type:id_token` • `urn:ietf:params:oauth:token-type:jwt` → Tip You can use either a JWT-based access token or an id_token. If you want to exchange a JWT token of a corporate identity provider (IdP), you should do it before an exchange with a JWT bearer token.	Request body

Parameter	Required	Data Type	Additional Information	Parameter Type
`requested_token_type`	No	string	The type of the requested security token. Allowed values for `requested_token_type` parameter: • `urn:ietf:params:oauth:token-type:access_token` • `urn:ietf:params:oauth:token-type:id_token` • `urn:ietf:params:oauth:token-type:saml2` • `urn:sap:identity:oauth:token-type:saml2-header` **Note** The exchange of an OpenID Connect token with SAML 2.0 is possible in two scenarios, depending on the requested token type string: • `urn:sap:identity:oauth:token-type:saml2-header` - 2043039 • `urn:ietf:params:oauth:token-type:saml2` - SAML 2.0 Bearer Assertion Flow for OAuth 2.0 Client	Request body

Parameter	Required	Data Type	Additional Information	Parameter Type
`resource`	No	string	Identifies the single sign-on scenario you want to consume. To consume the APIs of other applications, enter the provided API names. For more information, see Consuming APIs from Other Applications [page 2889]. **Note** If `resource` parameter is used to consume APIs from other applications, then the returned token is always of type `urn:ietf:params:oauth:token-type:access_token` even if `requested_token_type` is requested for type `id_token`. To exchange a refresh token for an opaque access token in mobile to web application scenarios, enter `urn:sap:identity:sso`. The access token is only valid for five minutes.	Request body
`refresh_expiry`	No	string	Reduces the expiry of a refresh token. It is useful if your application is called from mobile and web applications, and both have different session requirements. If you set the token lifetime to 0 or less, you won't receive a `refresh_token` in response.	Request body

Parameter	Required	Data Type	Additional Information	Parameter Type
`max_exchange_period`	No	string	You can set a limit on how long the application can exchange user tokens without re-authenticating after having acquired the initial token. The `max_exchange_period` parameter has a numeric value which can be set at between 1 hour and 4320 hours (six months). → Remember If the `max_exchange_period` value is smaller than the configuration defined in token policy, this value overrides the configuration defined in token policy configuration and propagates it once the token is exchanged or refreshed. If the value is bigger, it is ignored and the configuration defined in token policy is taken into consideration. For more information, see Token Policy Configuration for Applications [page 2752]	Request body
`token_format`	No	string	The `token_format` can be set to `opaque` to retrieve an opaque access token or to `jwt` to retrieve a JWT-based access token. If not set, the current defaults per grant type are used.	Request body
`app_tid`	No	string	Reserved.	Request body

Parameter	Required	Data Type	Additional Information	Parameter Type
`scope`	Yes	string	Value must be space delimi-nated parameter, for example: `scope=openid email`. The supported values are: • `openid` • `email` • `profile` • `groups` • `offline_access` **Note** The new tokens are inde-pendently created from the Identity Authentication Web session. This means that even if a user logs out from Identity Authentication the `refresh_token` will exist in the database un-til it expires, and can be used to perform the re-fresh token flow if the user is not present in Identity Authentication with a Web session.	Request body

Examples

Example

Request to exchange an opaque access token for a new token set.

Request

```
POST /oauth2/token
Content-Type: application/x-www-form-urlencoded
grant_type=urn:ietf:params:oauth:grant-type:token-exchange&
client_id=a90ca226sbc34-soc5-dcf6-6k8a6b9f2469&
client_secret=OWSu0/0sSUeUCG1LAYmSQ10Ut0yrfPz&
subject_token=Zjk1YTI3YERzNGZlZmTlNzZjNzk4YTY2ZjdlZjYwMacw&
subject_token_type=urn:ietf:params:oauth:token-type:access_token&
scope=openid
```

Response

```
Content-Type: application/json
```

```
{
    "access_token": "dzEyYTM0YmM2OWZhZnU3NmE3OThzNjZmN3NmNjg3MQ",
    "refresh_token": "s95s27af4fefae76c798a66f7ef9034",
    "issued_token_type": "urn:ietf:params:oauth:token-type:access_token",
    "id_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.ey
JzdWIiOiJQMTIzNDU2IiwiYXVkIjoiMjAzZjYyMDktMjRmZi00MmY1LW
E0MTMtNmRlNWE5M2ZjMjYxIiwidXNlcl91dWlkIjoiMDEyM2E0NS1mMW
FmLTFmMWMtYTMxMy03NDZkMTIwYWJjZDkiLCJtYWlsIjoiZG9uYS5tb2
9yZUBleGFtcGxlLmNvbSIsImlzcyI6Imh0dHBzOi8vdGVzdHRlbmFudC
50ZXN0LmNvbSIsImxhc3RfbmFtZSI6Ik1vb3JlIiwiZXhwIjoxNjM5MT
Q0MDEwLCJpYXQiOjE2MzkxNDA0MTAsImZpcnN0X25hbWUiOiJEb25hIi
wianRpIjoiMzFlMzJjNWYtNjljC00NmE2LWE5MDctOTExZDExMjA1cz
BzIn0.QYftNy7WgISWoKpRXh4_RX5UDNXsXDTndYuIA85L2II",
    "token_type": "Bearer",
    "expires_in": 3600
}
```

Example

Request to exchange an `id_token` for an SAML2 bearer token.

Request

```
https://my-tenant.ondemand.com/oauth2/token?
grant_type=urn:ietf:params:oauth:grant-type:token-exchange&
client_id=a90ca226sbc34-soc5-dcf6-6k8a6b9f2469&
client_secret=OWSu0/0sSUeUCG1LAYmSQ10Ut0yrfPz&
subject_token=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJQMTIzNDU2IiwiYXV
kIjoiMjAzZjYyMDktMjRmZi00MmY1LWE0MTMtNmRlNWE5M2ZjMjYxIiwidXNlcl91dWlkIjoiMDEyM
2E0NS1mMWFmLTFmMWMtYTMxMy03NDZkMTIwYWJjZDkiLCJtYWlsIjoiZG9uYS5tb29yZUBleGFtcGx
lLmNvbSIsImlzcyI6Imh0dHBzOi8vdGVzdHRlbmFudC50ZXN0LmNvbSIsImxhc3RfbmFtZSI6Ik1vb
3JlIiwiZXhwIjoxNjM5MTQ0MDEwLCJpYXQiOjE2MzkxNDA0MTAsImZpcnN0X25hbWUiOiJEb25hIiw
ianRpIjoiMzFlMzJjNWYtNjljC00NmE2LWE5MDctOTExZDExMjA1czBzIn0.QYftNy7WgISWoKpRX
h4_RX5UDNXsXDTndYuIA85L2II&
subject_token_type=urn:ietf:params:oauth:token-type:id_token&
requested_token_type=urn:ietf:params:oauth:token-type:saml2&
resource=urn:sap:identity:application:provider:name:SAML2SSO
```

Response

```
Content-Type: application/json
{
"access_token": "<Base64Url-Encoded-SAML2-Bearer-Assertion>",
"issued_token_type": "urn:ietf:params:oauth:token-type:saml2",
"token_type": "Bearer",
"expires_in": 600
}
```

Example

Request to exchange an opaque `access_token` for an opaque SSO token using the SSO resource flow.

Request

```
https://my-tenant.ondemand.com/oauth2/token?
grant_type=urn:ietf:params:oauth:grant-type:token-exchange&
client_id=a30ca226sbc21-soc5-dcf6-7k8a6b9f24
client_secret=OWSu0/0sSUeUCG1LAYmSQ10Ut0yrfz&
subject_token=Zjk1YTI3YERzNGZlZmTlNzZjNzk4YTY2ZjdlZjYwacw&
subject_token_type=urn:ietf:params:oauth:token-type:access_token&
requested_token_type=urn:ietf:params:oauth:token-type:acces_token&
resource=urn:sap:identity:sso
```

Response

```
Content-Type: application/json
{
"access_token": "MDFmMTNmY2MtMmViMC00ZWI5LWI0YWQtYjlkZjc5NmE1Mz",
"issued_token_type": "urn:ietf:params:oauth:token-type:access_token",
"scope": "openid",
"token_type": "Bearer",
"expires_in": 300
}
```

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

JSON Web Token (JWT) Profile for OAuth 2.0 Client Authentication and Authorization Grants

1.7.18.7 Using SAML 2.0 Bearer Flow

To authenticate using the SAML 2.0 Bearer flow, follow these procedures. Tasks 1 and 2 are configurations on the Identity Authentication side. Task 3 is a configuration on the client (relying party) side.

Use this flow to exchange a SAML assertion from your corporate identity provider for an OAuth token. This flow implements RFC 7522 .

Your implementation must include the following requirements:

- The issuer of the SAML assertion must match the name of a trusted SAML corporate identity provider.
- The SAML audience must specify the Identity Authentication issuer.
  `<saml:Audience>https://<IAS_Issuer>.<domain></saml:Audience>`
- In the `subject` element, the `subjectConfirmation` uses the method `urn:oasis:names:tc:SAML:2.0:cm:bearer`. In this element, `SubjectConfirmationData` specifies the recipient as the Identity Authentication issuer combined with the token path.

  Example

  ```
  <saml:Subject>
      <saml:NameID Format="urn:oasis:names:tc:SAML:1.1:nameid-
  format:emailAddress">my.name@example.com</saml:NameID>
      <saml:SubjectConfirmation
  Method="urn:oasis:names:tc:SAML:2.0:cm:bearer">
          <saml:SubjectConfirmationData NotOnOrAfter="2026-03-24T11:42:49Z"
  Recipient="https://my_issuer.cloud.sap/oauth2/token"/>
      </saml:SubjectConfirmation>
  </saml:Subject>
  ```

1. Create OpenID Connect (OIDC) Application [page 2716]
   You can create a new OpenID Connect (OIDC) application.

2. Configure OpenID Connect (OIDC) Application for SAML 2.0 Bearer Flow [page 2719]
   This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the SAML 2.0 bearer flow.
3. Configure the Client to Call Identity Authentication SAML 2.0 Bearer Token [page 2722]
   In the SAML 2.0 Bearer Token flow, you exchange one token for another. For more information, see RFC 7522 .

1.7.18.7.1 Create OpenID Connect (OIDC) Application

You can create a new OpenID Connect (OIDC) application.

Context

To create a new OpenID Connect (OIDC) application you have to add a new application to the list of applications in Identity Authentication, and set the protocol of the application to OpenID Connect (OIDC).

To create a new OpenID Connect (OIDC) application follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the *Create* button on the left-hand panel and fill in the information in the dialog box.
4. Provide the following information:

Fields	Notes
Display Name	Required. The display name of the application appears on the logon and registration pages.
Home URL	Optional. Users are redirected to the *Home URL* after activating their accounts, when they are created via a CSV file import or the user registration service of Identity Authentication.
Type	Optional. **⚠ Caution** If the type of the application is not the correct one, the system may change it automatically without a warning.
Parent Application	Optional.

Fields	Notes
	→ Remember Newly created applications with an assigned parent application will inherit all the configurations from the parent except for the `Client ID` and `Secrets`. The inherited configurations will be marked as such. • If you are an administrator with the *Manage Applications* role, the newly created applications can have as parent any of the parent applications in the administration console. Choose the parent application from the list that pops up. • If you are an administrator with authorizations based on policies, you can assign as parent only the application to which you have read or manage applications rights. The applications to which you have read or manage rights are restricted on the basis of the `application.organization` attribute in the policy. Choose the parent application from the list that pops up. For more information about how to assign authorizations based on policies, see Configure Application Authorizations [page 2325].
Organization ID	Optional. This field allows you to group the applications under a specific organization and on the basis of the *Organization ID* to restrict the access to these applications. The *Organization ID* can contain lowercase Latin letters (a-z), numbers (0-9), and the underscore (_) character. It can be up to 255 characters. • If you are an administrator with *Manage Applications* role, the newly created applications by default are in the `global` *Organization ID*. You can change the *Organization ID* during the creation of the new application by entering a new name or later you can edit it. • If your authorizations are based on policies, the newly created applications can take the *Organization ID* of the applications to which you have manage rights on the basis of the `application.organization` attribute in the policy. Choose the allowed *Organization ID* from the drop-down. **ⓘ Note** Applications created either as subscriptions within or with the Identity service of an SAP BTP subaccount receive the *Organization ID* of the application that represents the trust configuration of that subaccount.

Fields	Notes
Protocol Type	Select the *OpenID Connect(OIDC)* radio-button. → Remember Newly created applications with an assigned parent application will inherit the protocol from the parent. To change the protocol, first create the new application and then edit it.

5. Choose *+Create*.

   Once the application has been created, the system displays the message `Application <name of application> created`.

   The newly created application appears on the list with the applications on the left. It is selected and you can set its protocol to OpenID Connect (OIDC).

Next Steps

(Optional) If necessary, configure additional settings for the application. For more information about the supported configurations for the OpenID Connect (OIDC) applications, see the following links:

- Change an Application's Display Name [page 1934]
- Configure an Application's Home URL [page 1937]
- Visit an Application's Web Page [page 1939]
- Configuring User Attributes from the Identity Directory [page 1973]
- Configuring Attributes Based on Flexible Expressions [page 1979]
- Configure Secrets for API Authentication [page 1997]
- Configure Certificates for API Authentication [page 2001]
- Enable or Disable Kerberos Authentication for an Application [page 2556]
- Configure Risk-Based Authentication for an Application [page 2007]
- Configure User Access to the Application [page 2025]
- Authenticating Identity Provider for an Application [page 2065]
- Enable Email Verification [page 2026]
- Configuring Password Policies [page 2223]
- Configuring Privacy Policies [page 2235]
- Configuring Terms of Use [page 2273]
- Display Application Name on Logon Page [page 2033]
- Configure Logo [page 2030]
- Choose and Configure a Branding Style for an Application [page 2035]
- Configure Registration and Upgrade Forms [page 2046]
- Protecting Self-Registration, Upgrade with Phone Verification [page 2050]
- Protecting Application Forms with Google reCAPTCHA [page 2056]
- Configure Identity Federation [page 2614]

- Configure Authentication Request to Corporate IdPs [page 1966]

Task overview: Using SAML 2.0 Bearer Flow [page 2715]

Next task: Configure OpenID Connect (OIDC) Application for SAML 2.0 Bearer Flow [page 2719]

1.7.18.7.2 Configure OpenID Connect (OIDC) Application for SAML 2.0 Bearer Flow

This document is intended to help you configure an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services for the SAML 2.0 bearer flow.

Prerequisites

You have an OpenID Connect (OIDC) application in the administration console for SAP Cloud Identity Services. For more information, see .

Context

The trust is configured by entering the information manually. You can enter manually the name of the client (relying party), and its redirect URIs.

To configure an OpenID Connect (OIDC) trusted application in the administration console for SAP Cloud Identity Services, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.

5. Under *SINGLE SIGN-ON*, choose *OpenID Connect (OIDC) Configuration*.
6. Manually enter the communication settings negotiated between Identity Authentication and the client.

Setting	Description
Name (mandatory)	Provide a name of your choice.
Redirect URIs (optional)	The redirection URIs to which the response can be sent. You can add up to 20 redirect URIs.
Post Logout Redirect URIs (optional)	The redirection URIs where the user can be forwarded after logout. You can add up to 20 redirect URIs.
Front-Channel Logout URIs (optional)	URIs which will be requested for logout. You can add up to 20 URIs.

> → Tip
>
> For more information about the format of the redirect URIs and post logout redirect URIs, see OpenID Connect (OIDC) Application Configurations [page 2747].

7. Select the *SAML 2.0 Bearer (RFC 7522)* grant type.

> ⓘ Note
>
> Beware that for each flow the respective grant type must be selected. All other grant types can be deselected if they aren't required by the application.

8. **Optional:** (If you have added signing certificates in *Tenant Settings* or an application certificate in *Applications*.) Under *Identity Provider Certificates*, choose the certificate to be used when a request to the application is signed.

Option	Description
ON	This is the default setting. When the option is enabled, the certificate that is set as *Default* in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates* is used when a request to the application is signed.
OFF	When this option is disabled, the certificate that is set as *Active* in the list is used when a request to the application is signed. You can choose the active certificate from the list. → Tip To add an application certificate for the specific application, go to *Applications* > *choose the application you want to edit* > *Single Sign-On* > *OpenID Connect Configuration* > *Certificates* > *Add*. ⚠ Caution The application will stop working if the configuration of the application is not updated with the new certificate. ⓘ Note Your choice of *Active* certificate in the list is not related with the choice of *Default* certificate in *Tenant Settings* > *Single Sign-On* > *OpenID Connect Configuration* > *Signing Certificates*.

For more information, see Rotate Signing Certificates for OpenID Connect Applications [page 2113].

> → Tip
>
> When the default signing certificate is changed to a new one, and the old one isn't used anymore, we recommend you delete the old certificate.

9. Save your selection. Once the application has been changed, the system displays the message `Application <name of application> updated`.

> → Remember
>
> Configure trust on the client side. See the client documentation for more information about how to configure the trust.

Next Steps

- Configure HTTP basic authentication for the application. For more information about the configuration, see Configure Secrets for API Authentication [page 1997].
- Enable the public client flows option for this application. In the administration console, choose *the OpenID Connect (OIDC) application* > *Client Authentication under the Trust tab* > *Enable Public Client Flows under Public Client*. Optionally, you can configure the API permission groups. For more information, see Consuming APIs from Other Applications [page 2889] .

> ⓘ Note
>
> The *Public* client type is used for environments where it is difficult to protect the client credential, such as mobile and desktop applications, and client-side parts of web applications.

Task overview: Using SAML 2.0 Bearer Flow [page 2715]

Previous task: Create OpenID Connect (OIDC) Application [page 2716]

Next: Configure the Client to Call Identity Authentication SAML 2.0 Bearer Token [page 2722]

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.7.3 Configure the Client to Call Identity Authentication SAML 2.0 Bearer Token

In the SAML 2.0 Bearer Token flow, you exchange one token for another. For more information, see RFC 7522 .

Prerequisites

- For your business application, there's an OpenID Connect (OIDC) application in Identity Authentication. For more information, see Create OpenID Connect (OIDC) Application [page 1925].
- For your OpenID Connect (OIDC) application in Identity Authentication, you've prepared an authentication credential for the API call. The API supports the following authentication schemes:
  - Basic
    For more information, see `Authorization` in the *Request Header* table below.
  - X.509
    For more information, see `Authorization` in the *Request Header* table below.
  - JWT (client_assertion)
    - Trust by issuer: You must include the `client_id` in the request.
    - Trust by URI: Including `client_id` in the request is optional.

    For more information, see the *Request Parameters* table below.

  For more information, see API Authentication [page 1996].

Request

URI: `https://<Cloud Identity Services domain>/oauth2/token`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

Content-Type `application/x-www-form-urlencoded`

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Request Parameters

Parameter	Required	Data Type	Additional Information	Parameter Type
`assertion`	Yes	string	Enter a Base64URL-encoded SAML assertion.	Request body
`client_id`	Yes	string	Enter the identifier for the Identity Authentication application. **Note** This parameter is optional if the request uses authentication.	Request body
`grant_type`	Yes	string	`urn:ietf:params:oauth:grant-type:saml2-bearer`	Request body
`app_tid`	No	string	The parameter is reserved.	Request body
`client_secret`	No	string	Enter the client secret for basic authentication. For more information, see Configure Secrets for API Authentication [page 1997].	Request body
`max_exchange_period`	No	string	Set a limit for how long the application can exchange user tokens without re-authenticating after the initial token is acquired. You select a numeric value between one hour and 4320 hours (six months). **→ Remember** If the `max_exchange_period` value is smaller than the configuration in token policy, the value overrides the token policy configuration and applies when the token is exchanged or refreshed. If the value is larger, the system ignores it and uses the existing token policy configuration. For more information, see Token Policy Configuration for Applications [page 2752].	Request body

Parameter	Required	Data Type	Additional Information	Parameter Type
`refresh_expiry`	No	string	Reduces the expiration of a refresh token. It's useful if your application is called from mobile and web applications, and both have different session requirements. If you set the token lifetime to 0, you won't receive a `refresh_token` in response.	Request body
`resource`	No	string	Enter the Uniform Resource Name (URN) for the dependency used by the application. For more information, see Consuming APIs from Other Applications [page 2889].	Request body
`scope`	No	string	Parameter values must be space-delimited. For example: `scope=openid email`. Supported values: • `openid` • `email` • `profile` • `groups` • `offline_access` **Note** Tokens are created independently from the Identity Authentication web session. Even if a user logs out from Identity Authentication, the `refresh_token` remains in the database until it expires. You can use it for the refresh token flow if the user doesn't have an active session in Identity Authentication.	Request body
`token_format`	No	string	You can set `token_format` to `opaque` to retrieve an opaque access token, or to `jwt` to retrieve a JWT-based access token. If you don't set the parameter, the system uses the current defaults for each grant type.	Request body

Request Example

```
POST /oauth2/token
Content-Type: application/x-www-form-urlencoded
grant_type=urn:ietf:params:oauth:grant-type:saml2-bearer&
client_id=c95ad226-fg34-abc6-abc6-6e8a6b9f2442&
client_secret=PASu9/0sTUeUCG1LAYmSQ18Ut0zrfMz&
assertion=PHNhbWw6QXNzZXJ0aW9uIElEPSJB...1sOkFzc2VydGlvbj4=
```

Response

Response Example

```
Content-Type: application/json

{
    "access_token": "dzEyYTM0YmM2OWZhZnU3NmE3OThzNjZmN3NmNjg3MQ",
    "refresh_token": "s95s27af4fefae76c798a66f7ef9034",
    "id_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJQMT
    IzNDU2IiwiYXVkIjoiMjAzZjYyMDktMjRmZi00MmY1LWE0MTMtNmRlNWE5M2ZjM
    jYxIiwidXNlcl91dWlkIjoiMDEyM2E0NS1mMWFmLTFmMWMtYTMxMy03NDZkMTIw
    YWJjZDkiLCJtYWlsIjoiZG9uYS5tb29yZUBleGFtcGxlLmNvbSIsImlzcyI6Imh
    0dHBzOi8vdGVzdHRlbmFudC50ZXN0LmNvbSIsImxhc3RfbmFtZSI6Ik1vb3JlIi
    wiZXhwIjoxNjM5MTQ0MDEwLCJpYXQiOjE2MzkxNDA0MTAsImZpcnN0X25hbWUiO
    iJEb25hIiwianRpIjoiMzFlMzJjNWYtNjljC00NmE2LWE5MDctOTExZDExMjA1
    czBzIn0.QYftNy7WgISWoKpRXh4_RX5UDNXsXDTndYuIA85L2II",
    "token_type": "Bearer",
    "expires_in": 3600
}
```

Parent topic: Using SAML 2.0 Bearer Flow [page 2715]

Previous task: Configure OpenID Connect (OIDC) Application for SAML 2.0 Bearer Flow [page 2719]

1.7.18.8 Call Identity Authentication Refresh Token

The refresh token is used to receive a new id token and access token, to be used when the old token is not valid or expires.

To get an id token or access token via the refresh token for the first time the client must have a refresh token received via either the authorization code flow or the resource owner password credentials flow. For more information, see Call Identity Authentication Refresh Token [page 2725], or Configure the Client to Call Identity Authentication Token Endpoint for Resource Owner Password Credentials Flow [page 2675]. Once the client has a valid refresh token it can be used to call the token endpoint to receive a new id token and access token.

Prerequisites

- For your business application, there's an OpenID Connect (OIDC) application in Identity Authentication. For more information, see Create OpenID Connect (OIDC) Application [page 1925].
- For your OpenID Connect (OIDC) application in Identity Authentication, you've prepared an authentication credential for the API call. The API supports the following authentication schemes:
  - Basic
    For more information, see `Authorization` in the *Request Header* table below.
  - X.509
    For more information, see `Authorization` in the *Request Header* table below.
  - JWT (client_assertion)
    - Trust by issuer: You must include the `client_id` in the request.
    - Trust by URI: Including `client_id` in the request is optional.

    For more information, see the *Request Parameters* table below.

  For more information, see API Authentication [page 1996].

Request

URI: `https://<Cloud Identity Services domain>/oauth2/token`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`grant_type`	Yes	string	refresh_token	Request body

Parameter	Required	Data Type	Description	Parameter Type
`refresh_token`	Yes	string	The `refresh_token` generated by Identity Authentication in the token endpoint of the supported flows. It is returned with the request to the token endpoint. If `token_format` was specified in `grant_type` before, for example `authorization_code`, this format is re-used in the refresh token endpoint. **Note** The validity of the `refresh_token` after it is generated and returned in the URL is the same as timeout session for Identity Authentication. For more information about the session timeout feature, see Configure Session Timeout [page 2170].	Request body
`refresh_expiry`	No	string	Reduces the expiration of a refresh token. It's useful if your application is called from mobile and web applications, and both have different session requirements. If you set the token lifetime to 0, you won't receive a `refresh_token` in response.	Request body
`token_format`	No	string	The `token_format` can be set to `opaque` to retrieve an opaque access token or to `jwt` to retrieve a JWT based access token. If not set, the current defaults per grant type are used.	Request body
`app_tid`	No	string	Reserved.	Request body

Request Example

```
grant_type=refresh_token&refresh_token=d12a12abcd198765dd54r456e98321
```

Response

Response Status and Error Codes

Code	Reason	Description
200 OK	Successful operation.	Returns `access_token`, `refresh_token`, and `id_token`. The `id_token` is valid for 60 minutes.
400 Bad Request	Missing or wrong parameter.	Returns an information about the error.
401 Unauthorized	Wrong user ID or password parameters passed for the basic authentication. For more information, see Configure Secrets for API Authentication [page 1997].	The authentication of the client (relying party) failed.

Response Payload Example

```
{
 "access_token": "123ab87c-8p54-7ab9-7a9s-sfde12345678",
 "refresh_token": "s34s32as23r12345dt85r178e30701"
 "id_token":
"eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdWQiOiIxMmEzNGI1Yy02ZDc4LTllMW
YtZzM0NS02N2g4OWlqa2wxMjMiLCJzdWIiOiJQMTIzNDU2IiwibWFpbCI6ImRvbmEubW9vcm
VAZXhhbXBsZS5jb20iLCJpc3MiOiJodHRwczovL215LXRlbmFudC5hY2NvdW50cy5vbmRlbW
FuZC5jb20iLCJsYXN0X25hbWUiOiJNb29yZSIsInNhcF91aWQiOiIxMjM0NTZhYmM3ZGU4LW
ZnaGktOTEyMy1qNDU2LTc4OTEya2wzNG01NiIsImV4cCI6MTU4ODAxOTc4NywiaWF0IjoxNT
M5MDUyMDA2LCJmaXJzdF9uYW1lIjoiRG9uYSIsImp0aSI6IjI1MmViZjQ1LTEwYTgtNGU5My
04ZTZhLTBmZGE2YTNlNzFlYyJ9.D--uBbATOVX259bIJU6TgRTP26fD-jy-plwWzns5lyI",
 "token_type": "Bearer",
 "expires_in": 3600
}
or
{
 "access_token":
"eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdWQiOiIxMmEzNGI1Yy02ZDc4LTl
lMWYtZzM0NS02N2g4OWlqa2wxMjMiLCJzdWIiOiJQMTIzNDU2IiwibWFpbCI6ImRvbmEu
bW9vcmVAZXhhbXBsZS5jb20iLCJpc3MiOiJodHRwczovL215LXRlbmFudC5hY2NvdW50c
y5vbmRlbWFuZC5jb20iLCJsYXN0X25hbWUiOiJNb29yZSIsInNhcF91aWQiOiIxMjM0NT
ZhYmM3ZGU4LWZnaGktOTEyMy1qNDU2LTc4OTEya2wzNG01NiIsImV4cCI6MTU4ODAxOTA
0NCwiaWF0IjoxNTg4MDEzNzYwLCJub25jZSI6Im0tMEc2X0ZhUzNLZyIsImZpcnN0X25h
bWUiOiJEb25hIiwianRpIjoiMzhlNDIzMzAtZGU3YS00MTMwLWEzYTEtYjU4MmI1MjhkY
Tk4In0.J-aygbQ8IQlw4KD2t1hTPB2gyu3NdTbINDJ9eRs997g",
 "refresh_token": "d12a12abcd198765dd54r456e98321",
 "id_token":
"eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdWQiOiIxMmEzNGI1Yy02ZDc4LTl
lMWYtZzM0NS02N2g4OWlqa2wxMjMiLCJzdWIiOiJQMTIzNDU2IiwibWFpbCI6ImRvbmEu
bW9vcmVAZXhhbXBsZS5jb20iLCJpc3MiOiJodHRwczovL215LXRlbmFudC5hY2NvdW50c
y5vbmRlbWFuZC5jb20iLCJsYXN0X25hbWUiOiJNb29yZSIsInNhcF91aWQiOiIxMjM0NT
ZhYmM3ZGU4LWZnaGktOTEyMy1qNDU2LTc4OTEya2wzNG01NiIsImV4cCI6MTU4ODAxOTA
0NCwiaWF0IjoxNTg4MDEzNzYwLCJub25jZSI6Im0tMEc2X0ZhUzNLZyIsImZpcnN0X25h
bWUiOiJEb25hIiwianRpIjoiMzhlNDIzMzAtZGU3YS00MTMwLWEzYTEtYjU4MmI1MjhkY
Tk4In0.J-aygbQ8IQlw4KD2t1hTPB2gyu3NdTbINDJ9eRs997g",
 "token_type": "Bearer",
 "expires_in": 3600
}
```

> ### Note
>
> The format of the `issuer` depends on the configuration in the administration console for SAP Cloud Identity Services. For more information, see Tenant OpenID Connect (OIDC) Configurations [page 2117].

Related Information

Call Identity Authentication Refresh Token [page 2725]
Call Identity Authentication Refresh Token [page 2725]

1.7.18.9 Call Identity Authentication Discovery Endpoint

This document explains how to call the discovery endpoint.

OpenID Connect Discovery enables clients to verify the identity of the end user based on the authentication performed by Identity Authentication.

Call Discovery Endpoint

URI: `https://<Cloud Identity Services domain>/.well-known/openid-configuration`

> ⓘ Note
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *GET*

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`appid`	No	string	Used to identify the corresponding Identity Authentication application.	Request body
`app_tid`	No	string	Reserved.	Request body
`client_id`	No	string	Used to identify the corresponding client of the Identity Authentication application.	Request body

Request Examples

```
https://my-tenant.ondemand.com/.well-known/openid-configuration
```

Result

Result


```
{
  "issuer" : "https://my-tenant.ondemand.com",
  "authorization_endpoint" : "https://my-tenant.ondemand.com/oauth2/authorize",
  "token_endpoint" : "https://my-tenant.ondemand.com/oauth2/token",
  "userinfo_endpoint" : "https://my-tenant.ondemand.com/oauth2/userinfo",
  "end_session_endpoint" : "https://my-tenant.ondemand.com/oauth2/logout",
  "jwks_uri" : "https://my-tenant.ondemand.com/oauth2/certs",
  "introspection_endpoint":"https://my-tenant.ondemand.com/oauth2/introspect",
  "revocation_endpoint":"https://my-tenant.ondemand.com/oauth2/revoke",
  "response_types_supported" : [ "code", "id_token", "token" ],
  "grant_types_supported" : [ "password", "authorization_code",
"refresh_token", "client_credentials", "urn:ietf:params:oauth:grant-type:jwt-
bearer", "urn:ietf:params:oauth:grant-type:token-exchange"],
  "subject_types_supported" : [ "public" ],
  "id_token_signing_alg_values_supported" : [ "RS256" ],
  "scopes_supported" : [ "openid", "email", "profile", "offline_access" ],
  "token_endpoint_auth_methods_supported" : [ "tls_client_auth",
"client_secret_basic", "client_secret_post", "private_key_jwt" ],
  "claims_supported" : [ "iss", "ias_iss", "iat", "exp", "auth_time", "sid",
"jti", "nonce", "sub", "email", "email_verified", "given_name",
        "middle_name", "family_name", "preferred_username", "name", "scim_id",
"groups", "aud", "azp", "app_tid", "ias_apis", "at_hash",
        "sap_id_type", "sap_gtid"],
  "code_challenge_methods_supported" : [ "plain", "S256" ],
  "tls_client_certificate_bound_access_tokens" : true,
  "frontchannel_logout_supported" : true,
  "frontchannel_logout_session_supported" : true,
  "backchannel_logout_supported": true,
  "backchannel_logout_session_supported": true,
  "authorization_response_iss_parameter_supported": true
}
```


> **Note**
>
> The format of the `issuer` depends on the configuration in the administration console for SAP Cloud Identity Services. For more information, see Tenant OpenID Connect (OIDC) Configurations [page 2117].

> **Restriction**
>
> - If the `issuer` doesn't use HTTPS, then this tenant isn't OIDC-compliant. Not using HTTPS can cause integration failures. Changing the protocol at the tenant level can also cause integration failures. You can change the protocol at the application level to make the issuer OIDC compliant for that application. In the *OpenID Connect Configuration* of the application, set *Enforce Compliant OIDC Issuer*.
> - If *Enforce Compliant OIDC Issuer* is enabled, you must provide one of the following parameter combinations in your discovery endpoint request:
>   - `client_id` and optional `app_tid`
>   - `appid`
>
>   Without one of these parameter combinations, the discovery endpoint returns an issuer URL without the protocol, which causes OIDC validation to fail.

> **→ Remember**
>
> Identity Authentication supports only the assertion created by the security token service (STS) model. For more information, see OAuth Assertion Framework .

Related Information

Configuring OpenID Connect (OIDC) [page 2636]
Configure OpenID Connect (OIDC) Application [page 1949]
OpenID Connect Discovery 1.0 incorporating errata set 1

1.7.18.10 Call Identity Authentication Certificate Endpoint

The certificate endpoint contains information about the OpenID provider public keys.

The public keys are encoded as JSON Web Key (JWK).

Request

URI: `https://<Cloud Identity Services domain>/oauth2/certs`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *GET*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/json
`x-app_tid`	No	Reserved.
`x-client_id`	No	Reserved.

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`appid`	No	string	Used to identify the corresponding Identity Authentication application.	Request body
`app_tid`	No	string	Reserved.	Request body

Parameter	Required	Data Type	Description	Parameter Type
client_id	No	string	Used to identify the corresponding the client of the Identity Authentication application.	Request body

Request Example

```
https://my-tenant.ondemand.com/oauth2/certs
```

Response

Response Status and Error Codes

Code	Description
200	Success

Response Payload Example


```
{
  "keys": [{
      "kty": "RSA",
      "x5t#S256" : "kAQDSY3vl3BuRFLOl-6CaB_VlIWWzcupzYjnWV4BHow",
      "e": "AQAB",
      "use" : "sig",
      "kid" : "VEwv92eXR9jDN9o94GXL8pkWa9Y",
      "x5c" : [
"MIIDFjCCAf6gAwIBAQIGAWm+n1AfMA0GCSqGSIb3DQEBBQUAMEoxCzAJBgNVBAYTAkRFMQ8wDQY
DVQQKEwZTQVAtU0UxKjAoBgNVBAMTIXNjaS1kb2N1LmFjY291bnRzNDAwLm9uZGVtYW5kLmNvbTA
eFw0xOTAzMjcxMDA4MjVaFw0yOTAzMjcxMDA4MjVaMEoxCzAJBgNVBAYTAkRFMQ8wDQYDVQQKEwZ
TQVAtU0UxKjAoBgNVBAMTIXNjaS1kb2N1LmFjY291bnRzNDAwLm9uZGVtYW5kLmNvbTCCASIwDQY
JKoZIhvcNAQEBBQADggEPADCCAQoCggEBAKqMOaHOUai+p9wUPO5kARMRkExJfyUwrwHYLCfoGtv
dos74MseRWUdMcMTxlSaVyOSX4eGv9cRW1yQNarounGkTrVhmAkLdhFVcIytdOd1DOKAEvLFGjHX
P5tR0HsFvmGaRQzSS1nTfmW+qvEU6H8GPXnIypx/8wNXBFO52CS3Bx4IioaMSLWY3qJFRHt2paw1
VY/9L8rHGyMAjE458kHjhGXCf1S4iw5Yf79ruIno6PlNx1+QlR90BSy5BheNrDaTqNalnRNmaPt5
55Z4ptSooFumi5Ixw2ngqwtrdCRM6FIWYYLm8/SBzeRWm7N40f3EWbQsLaNyojnBWjIgXGEcCAwE
AAYICAAAwDQYJKoZIhvcNAQEFBQADggEBADNDQvM7ly5LSeYC+a/RfcumB2UnEaw9YZ3o2ymFywW
NstqNsfZ3HTRikAxmtB12wHlKwZLhUaEf3gMAGarPbh+kFbnRHoWOYLQEl+x+Aj7krdul58LnACX
wwTnGeRoZ5M/LlWqTZPDc+lNCIuubwU3F/LIUvXRZOLu/5T3ltjVV3yZ0bM/yxCvXxmX9pv2BmeH
vUYmH9bP7mJcGhbp/puUe9f5YmAur4GbveVDZaL4N2+4qC++dRad1kGzOqIhwXIHzNKt81Hwgll3
DuQcuKvM+I+I9nGXzwpTHB06HmVBNgSofS26bUzi20rD6StOdOIAPFmiimy/r1D6Nmw/Mk50=" ],
      "n":
"jjNiLjC91AWhAhsTWxCZ5DmljUioVNzrvtYwpm5nZL3rYVUXfIFGrE5mmAM6zzkcEP9ioPSYsFoPll8_
R3akYYi2iKbwEaEoOcSly-
ddL0LGkDIN8A5DDAhpnR49anElaJV87JzuxqpeazY30kw91Wrcohc22rs4vFUXhm4ZCrzK6nmfwTmHSty
Zb5QiQUGgRMjzbGJ_nOIBdsiMFAZf5BUMkFHjNb45YAlnZrstS2Is-
M6YA_79YntqbPlY2fRZOfEtj5evy6gaj8aCQANzI51Dh6aW3ZpRXphLbYZCLdod05ROTgw-
adR_7jKdQCoxNg71qIR_Fka4O8_lugpQ"
  }]
}
```

Related Information

JSON Web Key (JWK)

1.7.18.11 Call Identity Authentication List Sessions Endpoint

The service endpoint returns the list of sessions associated with the OpenID Connect (OIDC) token submitted with the request. Use this endpoint to enable your application to see how many tokens are still active after the application has performed a token exchange.

> **Note**
>
> The result is similar to when the authenticated user looks at their sessions and tokens in their user profile. For an example, view the self-service application.
>
> 1. Navigate to `https://<my_tenant>.sap.com/ui/protected/profilemanagement`.
> 2. Under *Sessions and Tokens*, choose *Show*.

Prerequisites

- For your business application, there's an OIDC application in Identity Authentication.
  For more information, see Create OpenID Connect (OIDC) Application [page 1925].
- For your OIDC application in Identity Authentication, you've prepared an authentication credential for the API call. The API supports the following authentication schemes:
  - Basic
  - X.509
  - JWT Bearer

  For more information, see API Authentication [page 1996].
- Your application has an access, ID, or refresh token.

Call Session List Endpoint

URI: `https://<Cloud Identity Services domain>/oauth2/token/list`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`token`	Yes	string	Provide the Identity Authentication token. The endpoint supports ID tokens, access tokens, and refresh tokens.	Request body
`client_id`	No	string	Used to identify the corresponding Identity Authentication application. **Note** The `client_id` parameter is mandatory if the request is provided without authentication.	Request body

Request Example

```
POST /oauth2/token/list
Content-Type: application/x-www-form-urlencoded
Authorization: Basic eyJbiOi…

token=eyJ0eab…
```

Result

The result is an array of sessions for the user associated with token your application submitted.

```
Content-Type: application/json

[
    {
        "sub": "dona.moore@example.com",
        "iss": "https://my_tenant.accounts.ondemand.com",
        "client_id": "a1b23cd4-f2d5-45d8-9937-3a4bebd6598c",
        "uid": "dona.moore@example.com",
        "sp_name": "XSUAA_1a2b3c45-ae93-4450-ba12-4becb1c345ab",
        "client_agent": "PostmanRuntime/7.29.2",
        "grant_type": "AUTHORIZATION_CODE",
        "scope": "openid",
        "client_ip": "10.10.10.201",
        "exp": 1669151188,
        "iat": 1669107988,
        "value": "eyJraW…_3421YaZjwFNVoZucjs-PU32SZ7Yd33BihtGczV8bkig",
        "jti": "ab123c45-e3d5-441c-ba07-b404e3d36465",
```

```
        "client_auth_method": "client_secret_basic"
    }
]
```

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.7.18.12 Call Identity Authentication UserInfo Endpoint

OpenID Connect UserInfo endpoint enables the clients to retrieve information about the authenticated user.

The client makes a request to the UserInfo endpoint using the access token obtained from the OpenID Connect authentication from the user-based flows.

Request

URI: `https://<Cloud Identity Services domain>/oauth2/userinfo`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *GET* or *POST*

Request Headers

Header	Required	Values
`Authorization`	Yes	Bearer <access_token>

Request Example

```
GET /oauth2/userinfo
  Host: https://<tenant ID>.accounts.ondemand.com
  Authorization: Bearer eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9
```

Response

Response Status and Error Codes

Code	Reason	Description
200	Successful operation.	Returns information about the user.
400	Bad request	Returns an error with error description. For example, expired token.
401	Unauthorized	The client is not authenticated.
403	Forbidden	Returns an error with error description.

Response Example

```
{
 "aud": "a123456e-314a-4923-899c-05c535c3d7f9",
 "sub": "P12345",
 "user_uuid": "158a1b2c7-981az-69ab9-8079-d824da69c681",
 "mail": "dona.moore@example.com",
 "iss": "https://my-_tenant.accounts.ondemand.com",
 "last_name": "Moore",
 "exp": 1588019044,
 "iat": 158801376,
 "first_name": "Dona",
 "jti": "38e42330-de7a-4130-a3a1-b582b528da98"
}
```

1.7.18.13 Call Identity Authentication Revoke Token Endpoint

The revoke token endpoint invalidates any access and refresh tokens issued to the client for the same end-user and session.

To revoke tokens from other sessions belonging to the same end-user, find the other session and revoke tokens for those sessions separately. For more information about finding user sessions, see Call Identity Authentication List Sessions Endpoint [page 2733].

The token revocation endpoint is implemented according to RFC 7009 OAuth 2.0 Token Revocation .

Request

URI: `https://<Cloud Identity Services domain>/oauth2/revoke`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.

Tenant ID is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded
`Authentication`	Yes	• Basic Authentication - Client ID and a Secret to authenticate the client (relying party). For more information, see Configure Secrets for API Authentication [page 1997]. **Note** The client ID and secret must be encoded using the "application/x-www-form-urlencoded" encoding algorithm. • X.509 Certificate • Bearer `<client_credential_token>`)

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`token`	Yes	string	Must contain the JWT or opaque token from the issuer.	Path
`token_type_hint`	No	string		Path
`client_id`	Yes	string	Used to identify the corresponding Identity Authentication application. **Note** The `client_id` parameter is optional if the request is provided with authentication.	Request body

Request Example

```
token=0ab12345978c51ffe7c4e9a2b1158bb4
```

Response

Token is successfully revoked.

Response Status and Error Codes

Code	Reason
200 OK	Successful operation.
401 Unauthorized	The client is not authenticated.

1.7.18.14 Call Identity Authentication Introspect Token Endpoint

The introspect token endpoint is an option for clients to get long expiring refresh tokens, opaque access and id tokens in JWT format validated.

The introspect token endpoint is implemented according to RFC 7662 OAuth 2.0 Token Introspection .

Request

URI: `https://<Cloud Identity Services domain>/oauth2/introspect`

> ⓘ Note
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Header	Required	Values
`Authorization`	Yes	• Basic - Client ID and a Secret to authenticate the client (relying party). For more information, see Configure Secrets for API Authentication [page 1997]. **Note** The client ID and secret must be encoded using the "application/x-www-form-urlencoded" encoding algorithm. • Bearer <client_credential_token>) • X.509 Certificate

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`token`	Yes	string	Must contain the JWT or opaque token from the issuer.	Request body
`token_type_hint`	No	string		Request body
`client_id`	Yes	string	Used to identify the corresponding Identity Authentication application. **Note** The `client_id` parameter is optional if the request is provided without authentication.	Request body

Request Example

```
token=0ab82505123c12ffe7c4e9a2b0158cn7
```

Response

Response Example

If token is valid:

```
Content-Type: application/json
{
 "active": true,
 "aud": "a123456e-314a-4923-899c-05c535c3d7f9",
 "username": "P12345",
 "sub": "P12345",
 "user_uuid": "158a1b2c7-981az-69ab9-8079-d824da69c681",
 "mail": "dona.moore@example.com",
 "iss": "https://my-_tenant.accounts.ondemand.com",
```

```
 "last_name": "Moore",
 "exp": 1588019044,
 "iat": 158801376,
 "first_name": "Dona",
 "jti": "38e42330-de7a-4130-a3a1-b582b528da98"
}
```

If token is not valid/expired/not known or any other token related issue:

```
Content-Type: application/json
{
 "active": false
}
```

Response Status and Error Codes

Code	Reason
200 OK	Successful operation.
401 Unauthorized	The client is not authenticated.

1.7.18.15 Call Identity Authentication End Session Endpoint

To propagate a user's logout to other applications and any corporate identity providers used, an OIDC application can call the end session endpoint of Identity Authentication.

After the entire single logout (SLO) operation has completed, by default, Identity Authentication redirects the user agent to the Identity Authentication profile page of the user. To redirect the user after logout to a different page of the application that triggered the logout, include a post logout URI in your request. The URI must be among the allowed post logout redirect URIs configured for the application.

For more information, see OpenID Connect (OIDC) Application Configurations [page 2747].

If you include a post logout redirect URI in your request, include the client ID. Instead of including the client ID, you can include the ID token and the service reads the client ID from the token.

Call End Session Endpoint

URI: `https://<Cloud Identity Services domain>/oauth2/logout`

> ⓘ Note
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *GET*

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`id_token_hint`	No → Remember The parameter is optional, but either `id_token` or `client_id` must be present.	string	Previously issued `id_token`. **Note** The token must be from the current session.	Query
`post_logout_redirect_uri`	No	string	The redirection URI where the user will be forwarded after logout.	Query
`state`	No	string	Used by the application to maintain state between the logout request and the callback to the endpoint specified by the `post_logout_redirect_uri` query parameter. If included in the logout request, the identity provider passes this value back to the application using the `state` query parameter when redirecting the browser back to the application.	Query

Parameter	Required	Data Type	Description	Parameter Type
`client_id`	No → Remember The parameter is optional, but either `id_token` or `client_id` must be present.	string	Used to identify the corresponding Identity Authentication application.	Query
`app_tid`	No	string	Required for multitenant scenarios to identify corresponding Identity Authentication application.	Query

> ### Note
>
> All request parameters can be combined in any order.

Request Examples

```
https://my-tenant.ondemand.com/oauth2/logout?
id_token_hint=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJhdWQiOiJUMTIzNDU2Iiwic3ViIj
oiUDEyMzQ1NiIsIm1

haWwiOiJkb25hLm1vb3JlQGV4YW1wbGUuY29tIiwiaXNzIjoiaHR0cHM6Ly9teS10ZW5hbnQuYWNjb3Vu
dHMu

b25kZW1hbmQuY29tIiwibGFzdF9uYW1lIjoiTW9vcmUiLCJleHAiOjE1MzkwNzQyMDYsImlhdCI6MTUzO
TAyO

TU0NCwiZmlyc3RfbmFtZSI6IkRvbmEiLCJqdGkiOiJjODM1NjZlZi0yZDc4LTQyZTgtOTk1M2M1NGZiZD
kyIn
                              0.n2KkMD9MUPekF_chd65wMdAu1Dv-m4yPE6i6ZeMOtAg
```

```
https://my-tenant.ondemand.com/oauth2/logout?
post_logout_redirect_uri=https%3A%2F%2Fexample.com&client_id=a0b22cd0-
f2d5-45d8-9937-3a4bebd6598c
```

```
https://my-tenant.ondemand.com/oauth2/logout?state=state&client_id=a0b22cd0-
f2d5-45d8-9937-3a4bebd6598c
```

Result

Response Status and Error Codes

Identity Authentication terminates all user sessions and finally redirects the user to the post logout page.

Related Information

Configuring OpenID Connect (OIDC) [page 2636]
Configure OpenID Connect (OIDC) Application [page 1949]
RP-Initiated Logout (Specification from OpenID Foundation)
Single Logout Flows [page 2887]

1.7.18.16 Handle Single Logout Request from Identity Authentication

When a user logs out from an application, the other applications that participate in the same user session must also be notified, so that it can log out the user as well. The application that doesn't trigger the logout, but must be notified, should do the following:

- Provide your own logout endpoint. For more information, see OpenID Connect Front-Channel Logout 1.0 .
- Configure the endpoint (*Front-Channel Logout URIs* or *Back-Channel Logout URIs*) in the respective Identity Authentication application. For more information, see Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 2642] or Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow with PKCE [page 2650].

> **Note**
>
> If the application is a multitenant one, it must include the tenant-specific logout endpoint during the login flow. This endpoint must follow the rules, specified in Logout URI Rules [page 2750]. The logout endpoint is sent during login (see Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow [page 2645] or Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow with PKCE [page 2650]).

Related Information

Configuring OpenID Connect (OIDC) [page 2636]
Configure OpenID Connect (OIDC) Application [page 1949]
Configure Trusted Domains [page 2173]
RP-Initiated Logout (Specification from OpenID Foundation)
Single Logout Flows [page 2887]

1.7.18.17 Claims of SAP Cloud Identity Services OAuth Tokens

Understanding OAuth token claims helps you to achieve secure and efficient authentication and authorization. This knowledge helps you use tokens correctly, reduce security risks, and maintain trust in your system.

Viewing Claims Supported by SAP Cloud Identity Services

View the claims issued by SAP Cloud Identity Services by calling the discovery endpoint and checking the `claims_supported` property.

URI: `https://<Cloud Identity Services domain>/.well-known/openid-configuration`

For more information, see Call Identity Authentication Discovery Endpoint [page 2729].

Supported Claims

The following table describes the claims supported by SAP Cloud Identity Services.

Claims Supported by SAP Cloud Identity Services

Claim	Description
`app_tid`	An SAP claim that identifies the tenant of the application. This claim is used in multitenant scenarios, for example, if you subscribed to a SAP BTP application.
`at_hash`	This claim contains a hash of the access token. Clients can hash the access token that it received and compare it to the `at_hash` value in the ID token to ensure they belong together and prevent token substitution attacks.
`aud`	Identifies the recipient of the token. It can be a string or array of audience identifiers, such as the client ID of the application requesting access.
`auth_time`	This claim shows the time when the client authenticated, in seconds since the Unix epoch in UTC. Applications can use this claim to determine how long ago a user authenticated and to enforce reauthentication.
`azp`	The authorized party identifies the client ID to which the ID token was issued.

Claim	Description
`azpacr`	An SAP claim that identifies the authentication context class reference of the authorized party. It indicates the strength of authentication used by the client application when it obtained the token. A value of `1` indicates the client used a client secret, and a value of `2` indicates the client used a certificate. This claim is present in tokens issued through the client credentials grant type.
`azp_app_tid`	An SAP claim that identifies the tenant of the authorized party in app-to-app token exchange scenarios. In multitenant SAP BTP environments, this claim represents the application tenant identifier of the consumer application that initiated the token exchange. The `app_tid` claim represents the provider application's tenant.
`cnf`	Binds a token to a specific cryptographic key held by the client. In proof-of-possession token scenarios, only the client holding the corresponding private key can use the token.
`email`	Email address of the user for whom the token was issued. This claim is included when the `email` scope is requested.
`email_verified`	Indicates whether or not the email has been verified. It's a boolean. This claim is included when the `email` scope is requested.
`exp`	The expiration time after which the token must not be accepted.
`family_name`	The surname (last name) of the user for whom the token was issued. This claim is included when the `profile` scope is requested.
`given_name`	The given name (first name) of the user for whom the token was issued. This claim is included when the `profile` scope is requested.
`groups`	The groups the user belongs to. These groups are associated with authorizations. This claim is included when the `groups` scope is requested.
`ias_apis`	An SAP claim that lists the API permission groups or a fixed value for when all APIs of an application are consumed. They're used to make authorization decisions. See also Consuming APIs from Other Applications [page 2889].
`ias_iss`	An SAP claim that identifies an SAP tenant even if the token was issued from a custom domain in a non SAP domain.

Claim	Description
`iat`	The time when the token was issued in seconds of the Unix epoch. Used to determine the age of the token.
`iss`	Identifies the issuer of the token. Typically a URL, for example, `https://<tenant>.accounts.ondemand.com`.
`jti`	A unique identifier for the JWT. Used to avoid replay attacks.
`middle_name`	Middle name of the user for whom the token was issued.
`name`	Full display name of the user for whom the token was issued.
`nonce`	A string associated with the client session to help mitigate against replay attacks. Clients can verify that it matches was the client sent in its authentication request.
`preferred_username`	A human-readable display name for the user for whom the token was issued. The claim shows what the user recognizes as their username.
`sap_azp_gtid`	An SAP claim that provides a globally unique tenant identifier of the authorized party in app-to-app token exchange scenarios. In multitenant SAP BTP environments, this claim represents the globally unique tenant identifier of the consumer application that initiated the token exchange.
`sap_gtid`	An SAP claim that provides a globally unique tenant identifier.
`sap_intospect`	An SAP claim that holds a list of claim names that were omitted from the token due to size restrictions. You can retrieve the full values of these claims from the `/oauth2/userinfo` or `/oauth2/introspect` endpoint.
`sap_id_type`	An SAP claim that identifies the type of token. Values: • `app`: The token is a credential for an application. • `user`: The token is a credential for a user.
`sap_scope`	An SAP claim that lists the scopes granted to the token. Currently, only the `offline_access` scope is stored in this claim.
`sap_user_type`	An SAP claim that identifies the type of user for whom the token was issued.
`scim_id`	An SAP claim that identifies the user with their SCIM ID in SAP Cloud Identity Services.

Claim	Description
`sid`	The session ID is used to track the session of a user with that specific application.
`sub`	The subject is a unique identifier for the user for whom the token is issued. Identity Authentication rejects a token request when the subject claim of the token matches the client ID of the requesting Identity Authentication application.

1.7.18.18 OpenID Connect (OIDC) Application Configurations

Redirect URIs, Post Logout Redirect URI Rules [page 2747]
Rules for the redirect URIs or post logout redirect URIs.

Logout URI Rules [page 2750]
Rules for the front and back-channel URIs.

Token Policy Configuration for Applications [page 2752]
Set the token policy for a specific OpenID Connect application. Configure the validity of the refresh token, access and id_token, and the maximum sessions per user.

Configure Grant Types [page 2755]
Configure the allowed grant type for your OpenID Connect application.

1.7.18.18.1 Redirect URIs, Post Logout Redirect URI Rules

Rules for the redirect URIs or post logout redirect URIs.

The redirect URI defines the redirection URIs to which the response can be sent. The post logout redirect URI defines the redirection URIs where the user can be forwarded after logout.

The redirect URI or post logout redirect URI must be in the following format:

`protocol://domain<:port>/<path>`

For example: `https://example.com:70/path`.

> ⓘ Note
>
> The URL parameters are not taken into consideration during runtime.

When you construct the redirect URIs or post logout redirect URIs have the following in mind:

Number

You may have a maximum of 20 redirect URIs and 20 post logout redirect URIs.

Length

The length is limited to 499 characters.

Domains

Top level domains are allowed. The domain part can't end with a hyphen "-".

- Protocols - Use the HTTPS protocol. The HTTP protocol is only allowed for localhost.

  > ✤ Example
  >
  > https://example.com
  >
  > https://10.11.12.13

- Localhost is allowed in the domain part.

  > ✤ Example
  >
  > http://localhost
  >
  > https://localhost
  >
  > http://127.0.0.1
  >
  > https://127.0.0.1

- Non-ascii characters are not allowed in the domain part.
- Custom schemas are supported, except `ftp`, `sftp`, `tftp`, `file`, `telnet` and `http`.
  Custom schemas must have at least 2 characters. The following are allowed: lower case characters, numbers, and the special characters "`.`", "`-`", and "`+`" .

  > ✤ Example
  >
  > custom-schema://example

Wildcards

- You can use asterisk (*) as first subdomain for localhost addresses. No other subdomains allowed.

  > ✤ Example
  >
  > http://*.localhost
  >
  > https://*.localhost

- You can use asterisk (`*`) in the domain part, or you can have a subdomain and add an asterisk at the beginning or at the end of it. There must be at least 3 subdomains left after the subdomain with the asterisk, and there can only be one subdomain with an asterisk.

> ### Example
>
> https://*.test.example.com
>
> https://app.*foo.test.example.com
>
> https://app.foo*.test.example.com

- You can use asterisk (`*`) at the end of the path, or between paths.

  > ### Example
  >
  > https://example.com/path/*
  >
  > https://example.com/path/*/path2

- You can use double wildcard, two asterisks (`**`), at the end of a path.

  > ### Example
  >
  > https://example.com/**
  >
  > http://localhost/path/*/path2/**

Ports (optional)

After the domain you can specify a port number, always preceded by a leading colon (`:`).

> ### Example
>
> https://example.com:8080

We support port wildcards for `localhost` and IP address `127.0.0.1` in redirect URIs of OIDC applications. Port wildcards enable tools, such as ABAP Development Tools, which use variable ports.

> ### Restriction
>
> Port wildcards aren't supported for other hosts.

> ### Example
>
> ```
> http://localhost:*
> https://localhost:*
> https://localhost:*/some/path
> http://*.localhost:*
> http://127.0.0.1:*/some/path
> https://127.0.0.1:*/some/path
> ```

Parent topic: OpenID Connect (OIDC) Application Configurations [page 2747]

Related Information

Logout URI Rules [page 2750]
Token Policy Configuration for Applications [page 2752]
Configure Grant Types [page 2755]

1.7.18.18.2 Logout URI Rules

Rules for the front and back-channel URIs.

Identity Authentication triggers a logout request to the front or back-channel URI.

The front and back-channel logout URI must be in the following format:

`protocol://domain<:port>/<path><?query parameters>`

For example: `https://example.com:70/logout?abc=123`.

When you construct the front and back-channel URIs, have the following in mind:

Number

You may have a maximum of 20 front and 20 back-channel URIs.

Length

The length is limited to 499 characters.

Domains

- Protocols - Use the HTTPS protocol. The HTTP protocol is only allowed for localhost.

  > ✤ Example
  >
  > https://example.com/logout

- Localhost - It's allowed in the domain part.

> **Example**
>
> http://localhost/logout
>
> https://localhost/logout

- Wildcard - It's allowed in the domain part. Wildcards are only supported in front-channel flows. Although there's no restriction on using wildcards with single tenant apps, the use of wildcards was developed with multitenant applications in mind. When using wildcards, the `logout_uri` parameter is required.

  > **Example**
  >
  > https://*.example.com/logout
  >
  > Allow during authorize call to register a URI with parameter `logout_uri`, for example: `https://app1.example.com/logout`.

- IP Addresses - Usage of IP addresses isn't allowed.

Ports (optional)

After the domain part, you can put the port numbers. Always use a leading colon (`:`).

> **Example**
>
> https://example.com:8080/logout

Fragments

> **Restriction**
>
> Usage of fragment identifier (#) isn't allowed. For example, you can't use `https://example.com/path#index.html`.

Parent topic: OpenID Connect (OIDC) Application Configurations [page 2747]

Related Information

Redirect URIs, Post Logout Redirect URI Rules [page 2747]
Token Policy Configuration for Applications [page 2752]
Configure Grant Types [page 2755]
OpenID Connect Front-Channel Logout 1.0 Specification
OpenID Connect Back-Channel Logout 1.0 Specification

1.7.18.18.3 Token Policy Configuration for Applications

Set the token policy for a specific OpenID Connect application. Configure the validity of the refresh token, access and id_token, and the maximum sessions per user.

Context

Token Policy

The following table lists the token policy options for OIDC applications.

Token Policy Configuration Parameters

Parameter	Description

Refresh Token	Sets the refresh token lifetime issued by Identity Authentication. The value can range from 1 to 4320 hours, in other words, from 1 hour to 180 days.
	
	The default value is 12 hours.
	
	Note
	• If the validity of refresh tokens is less than the validity of access/ID tokens, access/ID tokens can't be refreshed after the access/ID tokens expire.
	• When using the authorization code flow, if you set the token policy for refresh tokens longer than the session timeout, add the *offline_access* scope to your authorization code request. Without this scope, the service deletes the refresh token from the database when the resource owner ends the session (logs out). Without the refresh token, the OAuth client can't request new tokens anymore. For more information, see:
	  • Configure Session Timeout [page 2170]
	  • Using Authorization Code Flow [page 2638]

Access / ID Token	Sets the access and id_token lifetime issued by Identity Authentication. The value can range from 1 to 720 minutes, in other words, from 1 minute to 12 hours. The default value is 60 minutes.

Parameter	Description
Max sessions per user	Determines the maximum number of tokens that the service issues for the same session in parallel. Imagine you're logged on to the application through a web interface and a command-line interface in parallel. Then you'd set this parameter to 2. The value can range from 1 to 10. The default value is 1.

> → Remember
>
> The default configurations for the token policy per application are the tenant token policy. If you configure new values for an application, the service ignores the tenant token policy for that application. For more information about the tenant token policy, see Tenant OpenID Connect (OIDC) Configurations [page 2117]

> ⓘ Note
>
> The effective token validity for OIDC applications cannot exceed the Global Session Timeout configured at the tenant level. If the session timeout is shorter than the token policy values configured here, access tokens are capped at the remaining session duration, and refresh tokens cannot be renewed beyond session expiry. Exchanging a refresh token does not extend the session. For more information, see Configure Session Timeout [page 2170].

Offline Access

Disable this option to stop users from using this application beyond the regular session validity. You can configure this only for source applications. It is turned off by default. This control isn't visible for single-tenant applications and is enabled by default. Use this option to enable users to access a mobile or native application, such as a command-line interface, without having to authenticate each time.

Enforce Reauthentication

Use this option to limit how long the application can refresh user tokens without reauthenticating. This function helps you meet compliance requirements that restrict tokens from being valid beyond a specific period. By default this option is enabled and set to 3 months. When enabled, the maximum possible value is 2,147,483,647 seconds.

Advanced Settings

Refresh Token Usage After Renewal

When refresh token flows fail, you can enable the server to accept a refresh token that was already submitted. Normally, a refresh token can only be used once. The configuration aims to solve issues, for example network problems in refresh token calls, and allow an application to retry this call. Define the behavior of clients depending on your scenario and the risk. If you extend the rotation life time, we recommend revoking existing tokens with a separate call.

Refresh Token Usage After Renewal

Settings	Description
Off (Default)	The new refresh token immediately invalidates the old one.

Settings	Description
Online scenarios	The new refresh token is created and the old one is still active for 5 minutes.
Mobile scenarios	The new and old refresh token are valid during the configured refresh token life time.

> **Note**
>
> For online and mobile scenarios, calls to refresh the old refresh token end in the new token. Calls with the new refresh token, invalidates the old refresh token.

Access Token Format

Settings	Description
Grant-Type Dependent (Default)	The format of the token depends on the grant type. It is as folows: • Authorization Code Flow - `JWT` • Client Credentials Flow - `JWT` • Resource Owner Password Credentials Flow - `opaque` • Token Exchange - `opaque` • JWT Bearer Flow - `opaque` • Refresh Token Endpoint - `opaque`
JSON Web Token	All tokens are generated in the JWT format.
Opaque	All tokens are generated in the opaque format.

Enforce Compliant OIDC Issuer

If you notice this configuration option in your application, the default configuration of the tenant issuer is not OIDC compliant. This configuration can cause your application to

- Fail OIDC discovery.
- Fail token validation in standard libraries.
- Be unable to interoperate with other OIDC-compliant systems.

When enabled, the issuer of tokens for this application always starts with "`https://`".

To configure an application-specific token policy, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *Single Sign-On*, choose *OpenID Connect Configuration*.
6. Under the *Authentication* tab, choose *Authentication Policy (`<Default or Custom>`)*.
7. Choose *Edit*.
8. Configure the token policy for the application.

   If needed, use the reset button to set to the default value.
9. **Optional:** Choose values for the *Advanced Settings* configuration.
10. Save your changes.

Task overview: OpenID Connect (OIDC) Application Configurations [page 2747]

Related Information

Redirect URIs, Post Logout Redirect URI Rules [page 2747]
Logout URI Rules [page 2750]
Configure Grant Types [page 2755]
Tenant OpenID Connect (OIDC) Configurations [page 2117]

1.7.18.18.4 Configure Grant Types

Configure the allowed grant type for your OpenID Connect application.

Prerequisites

You have an OpenID Connect application in the administration console for SAP Cloud Identity Services. For more information, see Create OpenID Connect (OIDC) Application [page 1925].

Context

To configure the grant type, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *OpenID Connect Configuration*.
6. Configure the allowed grant types for the application.

Grant Type	Default Behavior
Authorization Code	selected
Authorization Code / Enforce PKCE (S256)	not-selected
Password	selected
JWT Bearer	selected
Implicit	not-selected
Refresh	selected
Client Credentials	selected
Token Exchange (RFC 8693)	not-selected

> **Note**
>
> Beware that for each flow the respective grant type must be selected.

7. Save your selection.

   Once the application has been changed, the system displays the message `Application <name of application> updated`.

Task overview: OpenID Connect (OIDC) Application Configurations [page 2747]

Related Information

Redirect URIs, Post Logout Redirect URI Rules [page 2747]
Logout URI Rules [page 2750]
Token Policy Configuration for Applications [page 2752]

1.8 User Guide

This user guide describes the scenarios in the Identity Authentication from a user's perspective. It's aimed at consumers, partners, and employees.

For the scenarios, users can use services to maintain or update their user profiles and to log on to applications. User services are divided into profile and sign-on services. For configuration of certain profile services, users access the profile page.

> **Note**
>
> You have received the link to your profile page in an email with information how to activate your account for user profile. If you can't find this email, and you don't know how to access your profile page, please contact your system administrator for assistance.

Users can use the following sign-on services:

- Basic authentication
  Users can log on to applications by providing basic credentials, such as user name and password.
- Single Sign-On authentication
  Users can access cloud applications via SAML 2.0 or OpenID Connect based single sign-on (SSO).
- Multi-factor authentication
  If the application requires second factor for authentication, users should provide an SMS code, or a passcode generated by a mobile device as an addition to the basic authentication. For more information, see Multi-Factor Authentication [page 2761].

Users can use the following self-services:

On the Logon Page

- Self-registration
  If this option is activated by an administrator, a user can register when he or she accesses the application's logon page. When the user clicks on the *Register* button, a registration form appears, where he or she needs to provide some personal data. The user also has to accept the organization's terms of use and privacy policy.
  By clicking the *Register* button, the users acknowledge that he or she has read the organization's privacy policy document.
  Once the user has submitted the form, he or she receives an email explaining how to activate the account. Successful activation of the user account completes the registration, thus allowing the user to log on to the application.

  > **Note**
  >
  > If the user hasn't received the activation email, he or she can do the following:

- Check the *Deleted*, *Junk*, or other folders in his or her mail box.
- Follow the *Forgot Password* link in the logon page of the application. If the registration is successful, this will trigger the *Forgot Password* process, and the user will receive an email with instructions how to reset the password.
- Choose the *Register Now* link, and fill in the registration form again.
- Contact the system administrator of the application.

- Forgot password
  A user can change his or her password when he or she accesses the application's logon page. When the user clicks on the *Forgot password?* link, the user triggers the forgot password process.
- Social sign-on
  If this feature is enabled by an administrator, users can link their Identity Authentication accounts with social network accounts. That way they can authenticate through a social provider by choosing the social network button on the logon page. They can also unlink their accounts on the profile page.

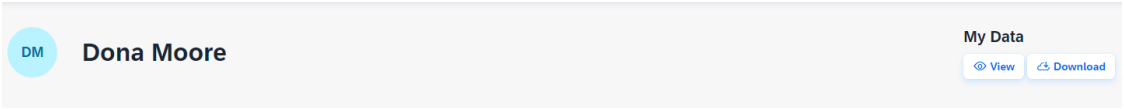
> ### Note
>
> The content in this section is not relevant for China (Shanghai) region.

On the Profile Page

- Changing profile data
  Users can change their profile information on the profile page. This includes Personal Information, Company Information, and Contact Preferences.

> ### Note
>
> The `Login Name` field can't be edited on the profile page. This field can be edited by the tenant administrator via the administration console for SAP Cloud Identity Services.
>
> The `Valid From` and `Valid To` fields are empty if no validity dates are set for the user. The dates are not editable via the profile page.

- Viewing profile data
  - Users can open a printable overview of all the data on file for them by visiting their account information in the upper right corner of the screen and choosing the *User Data* section from the menu.



> ### Note
>
> The data includes the **Terms of Use** and **Privacy Policy** documents singed by the user. The documents can be seen by clicking on their names. They open in a new tab.

  - Users can download an overview of all the data for them on a JSON file by clicking the download button in the *User Data* section.
- Self verifying user phone
  The phone verification possibility on the Profile Page is optional and must be enabled by a system administrator.
  If phone verification is enabled by a system administrator, the users provide their phone number in the respective field, after that they receive an SMS code and enters it in the provided field to complete the

verification. Once the phone number is verified, users can change it on the profile by choosing *Edit* in the *Personal Information* section. They will receive a verification code on the phone which must be provided so that the change is completed. If the code fails to arrive, users have up to five attempts to request a new code via SMS. They can also trigger one phone call per session.
When CAPTCHA (Completely Automated Public Turing test to tell Computers and Humans Apart) protection is enabled, users must solve the CAPTCHA to verify their phone.
If the phone number is not verified, users can change it without a verification code.

> ⚠ Restriction
>
> A phone number can be verified by only one user. If another user has already verified the phone number that is provided, the user is prompted to choose a different one or contact an administrator.

- Changing user passwords
  Users can change their passwords on the profile page.
  By clicking the change password button in the *Password* section, users trigger the change password process. That way, they are asked to provide the current and the new passwords. Users must also comply with the password requirements.

  > ⓘ Note
  >
  > Users must familiarize themselves with the password policy for logging on to the respective application. If no password policy is set for the application, users are able to modify their passwords with any characters of their choosing.

- Viewing sessions and tokens information
  Users can view their sessions and tokens on the profile page and delete those that they don't want or they don't trust. There is information about the:
  - browser
  - operating system
  - IP address
  - credentials or token type
  - expiration period
- Unlinking social sign-on
  Users can unlink their accounts on the profile page.

  > ⓘ Note
  >
  > The content in this section is not relevant for China (Shanghai) region.

Related Information

Scenarios [page 544]
Multi-Factor Authentication [page 2761]
Social Authentication [page 2771]
Use the Remember Me Option [page 2774]

1.8.1 Passwordless Authentication

This document provides information about the passwordless authentication option.

Based on the configurations for the application, you may authenticate with a certificate or biometric information (something that you are) without the need to provide your username and password.

FIDO2 Biometric Authentication

If biometric authentication is allowed for the application, you can choose the *Biometric Authentication* button on the logon page. If you have a configured FIDO2 biometric authentication device, the logon process will require authentication with the registered device. If biometric authentication isn't configured, you are required to register a device by providing your username and password first. This is a one time configuration.

> **Note**
>
> You can add more than one device via your profile page, but you can't add a device that has already been added, either for biometric authentication or for two-factor authentication.
>
> You can also remove a device from the list on the profile page. The removal of a device is password protected.

Identity Authentication supports the following:

Device	OS	Browser
Face ID	Latest iOS	• Latest Chrome • Latest Safari
Touch ID	Latest macOS	• Latest Chrome • Latest Safari
Windows Hello	Windows 10	• Latest Edge • Latest Chrome

Certificate Authentication

If certificate authentication is allowed for the application, you can generate your own certificate and authenticate with it without the need to provide your username and password.

When certificate authentication is allowed, you can see the *Certificates* section on your profile page. You should go to the profile page and under the *Certificates* section generate your certificate for authentication. You must provide your user password, then the certificate password, and the certificate validity. You can choose from 12 months, three months, and one month.

Once the certificate is generated, it's downloaded to your device. Import the certificate to your certificate store, and it is ready for authentication. Now, when you navigate to the logon page of the application, you

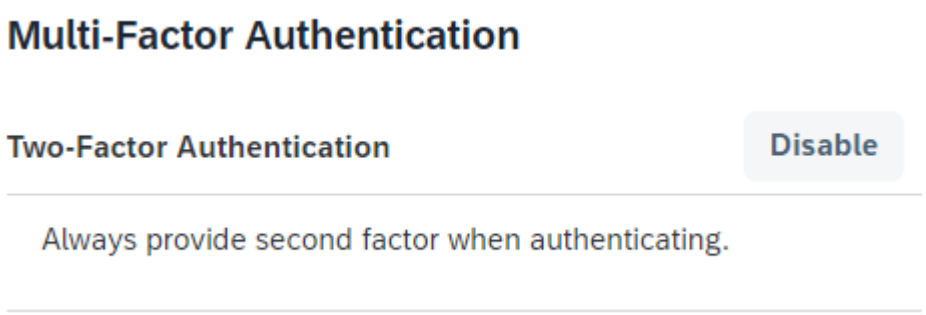
are automatically authenticated with the certificate. If the authentication fails, you're prompted to provide username and password.

1.8.2 Multi-Factor Authentication

This document provides information about the second factor for authentication or how to sign in if you are asked to provide a second factor to your primary credentials.

Based on the configurations for the application, you can be asked to provide a time-based, one time (TOTP) passcode, generated on a mobile device, a one-time (OTP) passcode sent to your email, a passcode sent via SMS, or a PIN security key. You can choose one of the enabled second factor authentication methods or press the button to enable a new method.

You can also choose to enhance your security with a second factor, although the application may not explicitly require it. This choice is available only if the system administrator has allowed it for your profile. If enabled, you see the following option on the profile page:



For more information about this option, see Always Authenticate with Second Factor [page 2763].

Authenticate with TOTP Two-Factor Authentication

TOTP passcodes are time-based and are valid for one logon attempt only, thus providing additional security to the common static passwords. Passcodes are generated by an authenticator application. The authenticator is a mobile application that you install on your mobile devices. For more information about how to install and configure authenticators, see their documentation.

To sign in when the application requires TOTP two-factor authentication, first you provide your primary credentials, choose the method or enable it, then provide the passcode generated on your mobile device.

If you don't have a device, activated for TOTP two-factor authentication, see Activate TOTP Two-Factor Authentication [page 2765].

Authenticate with SMS Two-Factor Authentication

SMS passcodes are sent to a mobile device.

To sign in when the application requires SMS two-factor authentication, first you provide your primary credentials, choose the method, or enable it, then you provide the passcode you received via SMS.

Authenticate with Web Two-Factor Authentication

To sign in when the application requires web authentication (FIDO2 standard), first you provide your primary credentials, choose the method, or enable it, then you provide the PIN code on your security key hardware USB authenticator. If you don't have a registered security key, you must register it first.

If you don't have a device activated for web authentication, see Add a Device for Web Two-Factor Authentication [page 2766].

Authenticate with Email Two-Factor Authentication

To sign in when the application requires email two-factor authentication, first you provide your primary credentials, choose the method, or enable it, then enter the 8-digit code sent to your email. You can request a new code after 3 minutes from your last request.

> → Remember
>
> Your session is valid for 10 minutes. If you don't use the code within the ten-minute period, the session expires and you must provide your primary credentials, and enter the new 8-digit code sent to your email again.

If you enter five wrong codes in row, the email two-factor authentication will be temporarily locked because of too many failed sign in attempts. It will be automatically unlocked in 60 minutes. If the application allows other two-factor authentication methods, you can use them.

If your email is not verified, the system won't send you a code, and you must contact the application's administrator for assistance.

Always Authenticate with Second Factor [page 2763]
: This document provides information about how to enhance the security of your account by always providing second factor in addition to your primary credentials.

Activate TOTP Two-Factor Authentication [page 2765]
: To log on to applications that require time-based one-time password (TOTP) as two-factor authentication, first you have to activate a mobile device that will generate TOTP passcodes.

Add a Device for Web Two-Factor Authentication [page 2766]
: To log on to applications that require web two-factor authentication (FIDO2 standard), first you have to activate an authenticator device.

Deactivate TOTP Two-Factor Authentication [page 2768]
: This document shows you how to deactivate the TOTP two-factor authentication that you use to access applications requiring passcodes for stronger authentication.

Remove Web Two-Factor Authentication [page 2770]
: This document shows you how to remove the web two-factor authentication (FIDO2 standard) that you use to access applications requiring it for stronger authentication.

1.8.2.1 Always Authenticate with Second Factor

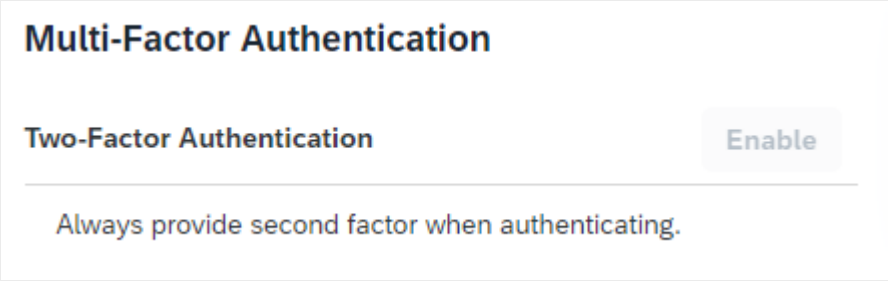
This document provides information about how to enhance the security of your account by always providing second factor in addition to your primary credentials.

Prerequisites

You have a device activated for time-based one-time password (TOTP) or Web Two-Factor Authentication. For more information, seeActivate TOTP Two-Factor Authentication [page 2765] and Add a Device for Web Two-Factor Authentication [page 2766].

Context

You can choose to enhance your security with additional authentication, second factor (passcode or security key), although the application may not explicitly require it. This option is available only if the system administrator has allowed it for your profile.

> **Note**
>
> If the option is allowed by the administrator, but you don't have an activated device, you won't be able to enable it. When the option is disabled, but you dont't have on the profile page, under *Multi-Factor Authentication*, you see:
>
> 
>
>
> To use this option you must first activate a device for time-based one-time password (TOTP) or Web Two-Factor Authentication.

To enable the *Multi-Factor Authentication* option, proceed as follows:

Procedure

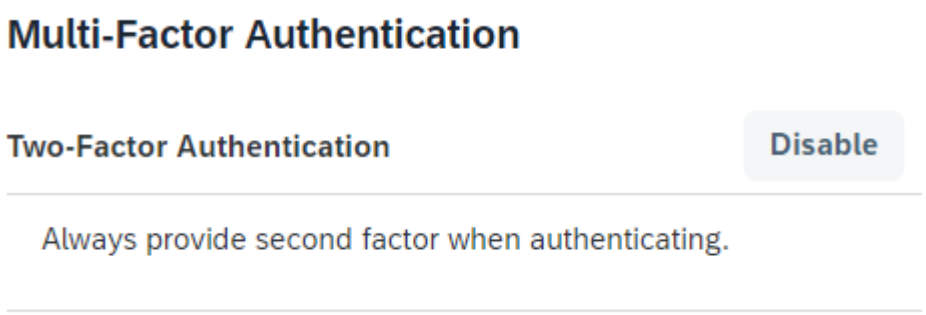
1. Access your profile page.

> **Note**
>
> You have received the link to your profile page in an email with information how to activate your account for user profile. If you can't find this email, and you don't know how to access your profile page, please contact your system administrator for assistance.

2. Under *Multi-Factor Authentication*, enable *Two-Factor Authentication* to always provide second factor when authenticating.
3. Verify your identity if you have already activated multi-factor authentication.

Results

When enabled, you see the following option on your profile page:



Next Steps

Activate multifactor authentication methods. For more information, see Activate TOTP Two-Factor Authentication [page 2765] or Add a Device for Web Two-Factor Authentication [page 2766].

Task overview: Multi-Factor Authentication [page 2761]

Related Information

Activate TOTP Two-Factor Authentication [page 2765]
Add a Device for Web Two-Factor Authentication [page 2766]
Deactivate TOTP Two-Factor Authentication [page 2768]
Remove Web Two-Factor Authentication [page 2770]

1.8.2.2 Activate TOTP Two-Factor Authentication

To log on to applications that require time-based one-time password (TOTP) as two-factor authentication, first you have to activate a mobile device that will generate TOTP passcodes.

Prerequisites

- You have installed a QR code scanner and an authenticator application on your mobile device.

  > **Note**
  >
  > You can use third-party authenticators such as Google Authenticator or Microsoft Authenticator. For more information about how to install and configure the authenticators see their documentation.

Context

Some applications require two-factor authentication as an additional security to the common static passwords. They will ask you to provide your password and a passcode, generated by a mobile device.

> **Note**
>
> Passcodes are only necessary for applications that require two-factor authentication. You do not need to activate a device for applications that only require passwords for authentication.

You activate a mobile device that generates passcodes.

Procedure

1. Access your profile page.

   > **Note**
   >
   > You have received the link to your profile page in an email with information how to activate your account for user profile. If you can't find this email, and you don't know how to access your profile page, please contact your system administrator for assistance.

2. Under the *Multi-Factor Authentication* section, choose *Activate* for *TOTP Two-Factor Authentication*.
3. Open the authenticator application.
4. Provide your security key and follow the onscreen instructions on the profile page.

   > **Note**
   >
   > Skip this step if you activate two-factor authentication for the first time.

5. Do one of the following on your mobile device:
    - Scan the QR code from the profile page screen.
    - Enter the secret key from the profile page screen manually.
6. Complete the setup on your mobile device.
7. Enter the passcode generated by the authenticator application in the space provided on the screen on your profile page.
8. Press *Activate*.

Results

Now, you can log on to applications that require passcode as an additional security for authentication.

> **Note**
>
> Optionally, you can activate TOTP two-factor authentication when you log on to an application that requires passcode as a second factor. Access the protected page of the application and follow the onscreen instructions.

Task overview: Multi-Factor Authentication [page 2761]

Related Information

Always Authenticate with Second Factor [page 2763]
Add a Device for Web Two-Factor Authentication [page 2766]
Deactivate TOTP Two-Factor Authentication [page 2768]
Remove Web Two-Factor Authentication [page 2770]

1.8.2.3 Add a Device for Web Two-Factor Authentication

To log on to applications that require web two-factor authentication (FIDO2 standard), first you have to activate an authenticator device.

Prerequisites

- Use a browser and platform that supports web authentication.

Context

Some applications require web two-factor authentication as an additional security to the common static passwords. They ask you to authenticate to the application and use your device to continue the logon.

> **Note**
>
> You can add more than one device for web two-factor authentication.

Procedure

1. Access your profile page.

   > **Note**
   >
   > You have received the link to your profile page in an email with information how to activate your account for user profile. If you can't find this email, and you don't know how to access your profile page, please contact your system administrator for assistance.

2. Press plus button next to *Web Two-Factor Authentication*.
3. Verify your identity by choosing a verification method and follow the onscreen instructions.

   > **Note**
   >
   > Skip this step if you activate two-factor authentication for the first time.

TOTP Two-Factor Authentication	(if configured) Provide the passcode and choose *Send* after selecting that option.
WEB Two-Factor Authentication	(if configured) Your security key. Follow the onscreen instructions after selecting that option.

4. Choose *Activate* and follow the instructions.

   Connect your authenticator device if not connected.
5. In the popup, allow the site to see the make and model of your security key.
6. Provide a name for your device.

Results

Now, you can log on to applications that require web authentication as an additional security for authentication.

> **Note**
>
> Optionally, you can activate a web two-factor authentication when you log on to an application that require it as a second factor. Access the protected page of the application and follow the onscreen instructions.

Task overview: Multi-Factor Authentication [page 2761]

Related Information

Always Authenticate with Second Factor [page 2763]
Activate TOTP Two-Factor Authentication [page 2765]
Deactivate TOTP Two-Factor Authentication [page 2768]
Remove Web Two-Factor Authentication [page 2770]

1.8.2.4 Deactivate TOTP Two-Factor Authentication

This document shows you how to deactivate the TOTP two-factor authentication that you use to access applications requiring passcodes for stronger authentication.

Context

You can deactivate TOTP two-factor authentication if you don't want to use it anymore to generate passcodes.

If you deactivate the TOTP two-factor authentication, you won't be able to log on to applications that require only passcodes. To be able to access these applications again, you have to activate again TOTP. For more information, see Activate TOTP Two-Factor Authentication [page 2765].

> → Tip
>
> If your device that generates passcodes has been lost or stolen, or you can't provide a valid passcode, you can still deactivate the TOTP via the channels offered on the profile page, or contact a system administrator of the application.

Based on the configuration of the account made by the administrator, you can choose among three deactivation channels:

- *Existing Multi-Factor Authentication* - to verify your identity, choose a verification method:
  - *TOTP Two-Factor Authentication* - use a passcode generated on your registered device
  - (if configured) *WEB Two-Factor Authentication* - provide security key
- *Passcode via SMS* - (if configured) use a passcode sent to you on your verified phone
- *Passcode via Email* - (if configured) use a passcode sent to you on your verified email

> ⚠ Caution
>
> The system can send you up to three emails within 24 hours. After that you won't be able to request a passcode for TOTP deactivation via email during these 24 hours.

To deactivate your mobile device that generates passcodes, proceed as follows:

Procedure

1. Access your profile page.

   > **Note**
   >
   > You have received the link to your profile page in an email with information how to activate your account for user profile. If you can't find this email, and you don't know how to access your profile page, please contact your system administrator for assistance.

2. Press the deactivate button next to *TOTP Two-Factor Authentication*.
3. Select a deactivation channel:

Existing Multi-Factor Authentication	Choose one of the verification methods: • *TOTP Two-Factor Authentication* - This is the default choice. Provide the passcode and choose *Send* after selecting that option. • *WEB Two-Factor Authentication* - Your security key (if configured.). Follow the onscreen instructions after selecting that option.
Passcode via SMS	Provide the passcode and choose *Send* after selecting that option.
Passcode via Email	Provide a passcode and choose *Send* after selecting that option.

   > **Note**
   >
   > The deactivation channels depend on the configuration of the account made by the administrator. You may not see all deactivation channels.

4. Press *Deactivate*.

Task overview: Multi-Factor Authentication [page 2761]

Related Information

Always Authenticate with Second Factor [page 2763]
Activate TOTP Two-Factor Authentication [page 2765]
Add a Device for Web Two-Factor Authentication [page 2766]
Remove Web Two-Factor Authentication [page 2770]

1.8.2.5 Remove Web Two-Factor Authentication

This document shows you how to remove the web two-factor authentication (FIDO2 standard) that you use to access applications requiring it for stronger authentication.

Context

You can remove web two-factor authentication if you don't want to use it anymore to verify your identity.

> **Note**
>
> If you remove the web two-factor authentication, you won't be able to log on to applications that require it as additional authentication. To be able to access these applications again, you have to activate again web authentication. For more information, see Add a Device for Web Two-Factor Authentication [page 2766].

Based on the configuration of the account made by the administrator, you can choose among three channels:

- *Existing Multi-Factor Authentication* - choose a verification method:
  - (if configured) *TOTP Two-Factor Authentication* - use a passcode generated on your registered device
  - *WEB Two-Factor Authentication* - provide security key
- *Passcode via SMS* - (if configured) use a passcode sent to you on your verified phone
- *Passcode via Email* - (if configured) use a passcode sent to you on your verified email

> **Caution**
>
> The system can send you up to three emails within 24 hours. After that you won't be able to request a passcode for TOTP deactivation via email during these 24 hours.

To remove your mobile device that generates passcodes, proceed as follows:

Procedure

1. Access your profile page.

   > **Note**
   >
   > You have received the link to your profile page in an email with information how to activate your account for user profile. If you can't find this email, and you don't know how to access your profile page, please contact your system administrator for assistance.

2. Under *Web Two-Factor Authentication*, press the delete button next to the device that you want to remove.
3. Select a channel:

Existing Multi-Factor Authentication	Choose one of the verification methods:

	• *TOTP Two-Factor Authentication* - Provide the pass-code and choose *Send* after selecting that option. • *WEB Two-Factor Authentication* - This is the default choice. Your security key (if configured.). Follow the onscreen instructions after selecting that option.
Passcode via SMS	Provide the passcode and choose *Send* after selecting that option.
Passcode via Email	Provide a passcode and choose *Send* after selecting that option.

> ⓘ Note
>
> The deactivation channels depend on the configuration of the account made by the administrator. You may not see all deactivation channels.

4. Press *Remove* if required.

Task overview: Multi-Factor Authentication [page 2761]

Related Information

Always Authenticate with Second Factor [page 2763]
Activate TOTP Two-Factor Authentication [page 2765]
Add a Device for Web Two-Factor Authentication [page 2766]
Deactivate TOTP Two-Factor Authentication [page 2768]

1.8.3 Social Authentication

Context

> ⓘ Note
>
> The content in this section is not relevant for China (Shanghai) region.

Log on with a Social Provider Account

You can log on to applications that use Identity Authentication via your accounts in X, Facebook, LinkedIn, or Google.

Prerequisites

The applications must be configured to allow logon via social networks.

Context

Using the social network authentication, you link your Identity Authentication account with your social network account or accounts. After the initial setup, when you link the accounts, you can log on to the applications with your social network credentials.

Identity Authentication has access to the following data from the social providers:

Social Identity Provider	Data Used by Identity Authentication
Apple	ID of your user - used to link your Apple account with your Identity Authentication account
Facebook	• ID of your user - used to link your Facebook account with your Identity Authentication account • your display name - used to prefill the registration form in case of non-existing user
Google	• ID of your user - used to link your Google account with your Identity Authentication account • your display name, first name, and last name - used to prefill the registration form in case of non-existing user
LinkedIn	• ID of your user - used to link your LinkedIn account with your Identity Authentication account • your first name and last name - used to prefill the registration form in case of non-existing user
X	• ID of your user - used to link your X account with your Identity Authentication account • your display name - used to prefill the registration form in case of non-existing user • your logon name

This data is used for the initial linking of your Identity Authentication account with the social network account. Later, if you update the personal information in your social account, the updated information will not be copied by Identity Authentication.

To link your Identity Authentication account with a social network account, proceed as follows:

Procedure

1. Access the application's logon page.
2. Choose the icon of the social network provider you want to log on with.

   > ### Note
   >
   > Which social networks are displayed on the page depends on the application.

3. Sign in to your social network account.
4. Choose one of the following options:
   - Link your Identity Authentication account with your social network account.
   - Create a new Identity Authentication account that will be linked with your social network account.

     > ### Note
     >
     > This option appears only for applications that allow user registration.

Results

Once you allow Identity Authentication to link your account with the social providers' accounts, you can log on to the applications via the social providers.

Unlink a Social Provider Account

You can unlink your social provider account via the profile page.

Context

To remove your social network logon information from your Identity Authentication account, proceed as follows:

Procedure

1. Access the profile page.

   > ### Note
   >
   > You have received the link to your profile page in an email with information how to activate your account for user profile. If you can't find this email, and you don't know how to access your profile page, please contact your system administrator for assistance.

2. Press *Edit* under *Social Sign-On*.
3. Press *Unlink* to remove your social network logon information from your account.

   If the operation is successful, the system displays the message Profile updated .

1.8.4 Use the Remember Me Option

With the *Remember me* functionality enabled, you can log on to an application without the need to provide your credentials every time you access it.

Context

If you enable the *Remember me* functionality, the application saves a cookie in the browser on your local computer. It automatically logs you on next time you access the application from the same browser on your local computer. Once enabled, the *Remember me* functionality is valid for 3 months unless you log out from the application.

> ### Note
>
> If the application requires two-factor authentication, you must provide a valid one-time password (passcode) generated by a mobile device every time you access the application. For more details about how to use two-factor authentication on your mobile device, see **Related Information**.

Procedure

1. Access the application that you want to log on to with *Remember me*.
2. Provide your credentials and select the *Remember me* checkbox.
3. **Optional:** Provide a passcode if required.

Results

Next time you access the application, you will be logged on automatically. If the application requires two-factor authentication, you need to provide a valid passcode, generated by a mobile device.

> ⓘ Note
>
> If you want to disable *Remember me* for an application log out from the application, and do not select the checkbox next time you log on.

> → Remember
>
> If you are authenticating with credentials from an external system (authentication provider or corporate user store), the following limitations are observed:
>
> - (in the case of corporate user store) when you are authenticating with remember me cookie, your cloud user record in Identity Authentication is not updated.
> - (in the case of authentication provider or corporate user store) if you change your password in the external system, the remember me cookie is not invalidated.
>
> For more information about the type of your system, contact your system administrator.

> → Remember
>
> If you are authenticating with credentials from an external authentication provider, if you change your password in the external provider, the remember me cookie is not invalidated.
>
> For more information about the type of your provider, contact your system administrator.

Related Information

Multi-Factor Authentication [page 2761]

1.8.5 Change or Reset Password via Profile Page

You, as an end-user, can change your current password or request a reset password email via your profile page.

Context

Your password grants you access to any platform connected to Identity Authentication. Depending on the configuration of your profile page, you can either change your password directly on the profile page or request a reset password email via your profile page. The configuration is done by an administrator, and you can't choose which option to use.

To change your current password, you need to provide the current and the new passwords to the system. To reset your password, you must have access to your email. You can find information about the email of your account in the *Personal Information* section of your profile page. For both cases you must comply with the password requirements.

> → Tip
>
> If you don't know your current password, follow the *Forgot Password* link at the sign-in page of the application. This triggers the **Forgot Password** process, and receive an email with instructions how to reset your password. For more information, see Reset Your Forgotten Password [page 2777].
>
> If you don't see the *Forgot Password* link at the sign-in page, contact the system administrator of your application.

To change or reset your current password, follow the procedure below:

Procedure

1. Access your profile page.

   > ⓘ Note
   >
   > You have received the link to your profile page in an email with information how to activate your account for user profile. If you can't find this email, and you don't know how to access your profile page, please contact your system administrator for assistance.

2. Under *Authentication*, you see only one of the following options in the *Password* section:
   - *Change* - When you press the *Change* button, a pop-up requiring input from you appears. Provide your current and new password and save your changes. If the operation is successful, you sign in with your new password.
   - *Reset* - When you press the button *Reset* an email is sent to your inbox. To reset your password, click on the link that is in the email, provide the new password on screen that appears, and save your changes. If the operation is successful, you can sign in with your new password.

Results

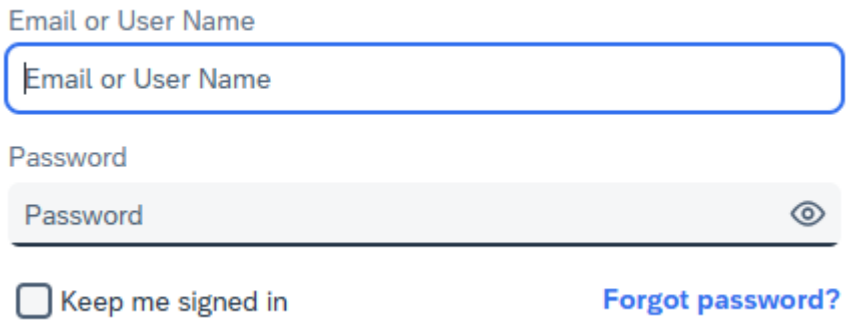
> → Remember
>
> Your password can only be changed once every 24 hours.

1.8.6 Reset Your Forgotten Password

You can reset your password if you have forgotten it.

Prerequisites

- You have forgotten your current password.
- You see the *Forgot Password* link on the sign-in page of your application:



> ⓘ Note
>
> The image is just for an illustration purpose and may differ from your sign-in page.

Some applications don't offer the reset password option. If you can't see the *Forgot Password* link, contact the system administrator of your application with a request to reset your password.

Context

> → Tip
>
> If you know your password, but for security or any other reasons you want to change it, follow the procedure in Change or Reset Password via Profile Page [page 2776].

If you don't remember your current password, choose the *Forgot Password* link in the sign-in page of the application, and follow the onscreen instructions. Depending on the configurations made by your organization, you may have one, two, or all three options:

- Receive an email with instructions how to reset your password.
- Answer security questions. You must have already configured your answers in the profile page to be able to reset password via this option.
- Provide a PIN code. You must have already set your PIN code in the profile page to be able to reset the password via this option.

Passwords can be reset and changed once in 24 hours.

To reset your password, follow the steps below:

Procedure

1. Access the sign-in page of the application.
2. Choose the *Forgot Password* link.
3. Choose one of the options available on your screen. Be aware that the options are visible only if the application supports them:
   - **Optional:** *Email* - Provide the email address from your account and press *Continue*. An email with a link to a page where you can reset your password will be sent to you. The email might take a few minutes to reach your inbox.

     > → Tip
     >
     > If you don't receive the email with the reset password link, check your spam folder. If the email isn't there, choose *Return to Sign In* and start the process again. If the problem persists, contact your administrator for further assistance.

   - **Optional:** *Security Questions* - Provide the required information and answers, and press *Continue*.

     > → Remember
     >
     > To use this option, you must have configured the security questions section in your profile page. Note that the answers to the security questions are case sensitive. You must provide exactly the same answers that you have configured.

   - **Optional:** *PIN Code* - Provide the required information and PIN code, and press *Continue*.

     > → Remember
     >
     > To use this option, you must have configured the PIN code section in your profile page.

1.8.7 Error Messages on End User Screens

This section aims to help end users to deal with error messages when using applications for logon, registration, invitation, password update, and account activation.

Error Codes

Solutions to Error Messages

Error Message	Description	Solution
Authentication failed; try again	• You have entered the wrong credentials; • Your account is not activated.	• Try again; • Open your email and activate your account; • Register on the logon page; • Contact your administrator.
An error occurred; try again	An unexpected error occurred.	• Try again; • Contact your administrator.
Insufficient password complexity; check password requirements	Your password does not comply with the application's password policy.	Familiarize yourself with the password policy for the application.
Cannot verify your password; try again	The old password for the password update is invalid.	Enter your correct current password.
The password has already been changed in the last 24 hours	According to the application's policy, you are not allowed to change the password right now.	• Familiarize yourself with the application's policy for changing the password; • Try again later.
Your email activation link is invalid.	The email activation link that you have received is expired or already used by you.	• *Visit the application > choose Log On > follow Forgot Password link* • Contact your administrator.

Related Information

Getting Support [page 3141]

1.9 Development

The developer guide is aimed mainly at organization developers who can implement configurations in addition to the ones in the administration console of Identity Authentication. Developers can use REST API services

to configure various authentication and registration mechanisms for their applications. The applications that administrators configure use different application services for all user-related processes.

Application Services

The application services are used by the cloud services and cloud applications to interact with Identity Authentication with regard to user records in the tenant.

The following APIs are offered to cloud applications:

- An API to register users
  For more information, see User Registration [page 2807]
- An API to invite users
  For more information, see Invitation REST API [page 2802]
- Identity Authentication implementation of the System for Cross-domain Identity Management (SCIM) REST API protocol.
  For more information, see Identity Authentication API (Deprecated) [page 2796].
- An API to change the predefined texts and messages for end-user screens available per tenant in the Identity Authentication.
  For more information, see Change Tenant Texts REST API [page 2826].
- An API to change the predefined master data texts.
  For more information, see Change Master Data Texts REST API [page 2860].

Related Information

Error and Success Codes [page 2883]
Configure Certificates for API Authentication [page 2001]

1.9.1 API Authentication

Developers can choose the type of authentication when API methods of Identity Authentication are used.

The certificate to be used for authentication by the REST APIs of Identity Authentication must be requested from the SAP Support Portal.

Configure Secrets for API Authentication [page 2781]
This document describes how developers configure secrets with scopes and validity for client authentication.

Unlock Client ID [page 2783]
Unlock the client ID after five failed logon attempts before the automatic unlock time of 60 minutes has passed.

Disable Client ID Locking [page 2784]
You can disable the automatic lock of the client ID after five failed logon attempts.

Configure Certificates for API Authentication [page 2785]
: This document describes how developers configure the certificates used for authentication when the API methods and OpenID Connect scenarios of Identity Authentication are used.

Configure JWT for OAuth Client Authentication [page 2787]
: Configure the JSON Web Token (JWT) - the issuer and subject of tokens for JWT client authentication in token requests, or the URI for JSON web key retrieval for client authentication.

1.9.1.1 Configure Secrets for API Authentication

This document describes how developers configure secrets with scopes and validity for client authentication.

Context

You can use a client ID and a secret to authenticate when REST API calls (Invitation REST API, User management REST API, Password Service Rest API, and Forgot Password REST API) to the tenant of Identity Authentication are used. The client ID and secret can also be used in the OpenID Connect scenarios of Identity Authentication.

The client ID is in the universally unique identifier (UUID) format. For example, `1ab7c243-5de5-4530-8d14-1234a26373ab`.

> → Remember
>
> The User ID and a password (secret) to authenticate the client (relying party) in the OpenID Connect scenario must be encoded using the "application/x-www-form-urlencoded" encoding algorithm.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   > ⓘ Note
   >
   > Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
   >
   > If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *Application APIs*, choose *Client Authentication*.

   You can see the *Client ID* generated for the chosen application.

6. Choose the *Add* button in the *Secrets* section.

> **Restriction**
>
> You can generate up to 20 client secrets per application.

7. Provide the required info in the pop up.

Description	This field is optional.
Expire in	You can choose from three options: • 1 year • 2 years • Never
Scope	• Application - Secrets with *Application* scope are used to update the Application configurations. • Application Users - select this option to generate a secret to authenticate when REST API calls (Invitation REST API, User management REST API, Password Service Rest API, and Forgot Service REST API) are used. • OpenID - select this option to generate a secret to authenticate in the OpenID Connect scenario. **Note** By default all options are selected. Your secret can be used for all scopes.

8. Save your configuration.

   A pop up with the generated credentials appears. Make sure that you save the client secret for the client ID. You will need it for the API authentication, but you will not be able to retrieve it from the system later.

Results

Once your secret is generated you can see a table with your secrets and information about them.

Task overview: API Authentication [page 2780]

Related Information

Unlock Client ID [page 2783]
Disable Client ID Locking [page 2784]
Configure Certificates for API Authentication [page 2785]
Configure JWT for OAuth Client Authentication [page 2787]

1.9.1.2 Unlock Client ID

Unlock the client ID after five failed logon attempts before the automatic unlock time of 60 minutes has passed.

Context

The client ID locks for 60 minutes after five failed logon attempts. After 60 minutes the client ID is automatically unlocked.

To unlock the client ID before the automatic unlock time of 60 minutes has passed follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application with a locked client ID.
4. Under *API AUTHENTICATION*, choose *Client ID and Secrets*.
5. Choose *Unlock* next to *Client ID Status*.

> → Remember
>
> You can only unlock locked client IDs. If the client ID is not locked, the *Unlock* button is disabled.

Task overview: API Authentication [page 2780]

Related Information

Configure Secrets for API Authentication [page 2781]
Disable Client ID Locking [page 2784]
Configure Certificates for API Authentication [page 2785]
Configure JWT for OAuth Client Authentication [page 2787]
Disable Client ID Locking [page 2000]

1.9.1.3 Disable Client ID Locking

You can disable the automatic lock of the client ID after five failed logon attempts.

Context

By default the client ID locks for 60 minutes after five failed logon attempts.

You can disable the automatic lock of the client ID via the administration console for SAP Cloud Identity Services. Use this option to disable the automatic lock in cases when the client ID has a limited scope and the client ID secret(s) are automatically generated.

> ### Note
>
> When you disable the *Client ID Lock* option for a client ID that is already locked, this resets the failed logon attempts counter but the client ID remains locked. If you want to unlock the client ID before the automatic unlock time of 60 minutes has passed you must unlock it manually. For more information, see Unlock Client ID [page 1999].

"

To disable the *Client ID Lock* option, follow the procedure below:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.
4. Under *API AUTHENTICATION*, choose *Client ID and Secrets*.
5. Disable the *Client ID Lock*.

   If the operation is successful, you receive the following message: `Client ID locking updated.`. The slider next to *Client ID Lock* is switched to *OFF*.

   > ### Note
   >
   > If you want to enable client ID password locking for the application, drag the slider to *ON*.

Task overview: API Authentication [page 2780]

Related Information

Configure Secrets for API Authentication [page 2781]

Unlock Client ID [page 2783]
Configure Certificates for API Authentication [page 2785]
Configure JWT for OAuth Client Authentication [page 2787]

1.9.1.4 Configure Certificates for API Authentication

This document describes how developers configure the certificates used for authentication when the API methods and OpenID Connect scenarios of Identity Authentication are used.

Prerequisites

SAP Support Portal Home (For trusted CA Scenario) - If you want to configure a certificate, using your own trusted CA report an incident on SAP Support Portal Home with a component `BC-IAM-IDS`. Attach to the incident the root and intermediate certificates and provide the Identity Authentication tenant ID.

> **Note**
>
> `Tenant ID` is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the `tenant ID`. For more information about your tenants, see View Assigned Tenants and Admins.

Context

You can use a certificate to authenticate when REST API calls (Invitation REST API, User management REST API, Password Service Rest API, and Forgot Service REST API) to the tenant of Identity Authentication are used. The certificate can also be used in the OpenID Connect scenarios of Identity Authentication.

For the configuration, you have to provide the base64-encoded certificate as a file or plain text, or you can generate a certificate via the administration console. You can enter a maximum of 80 certificates.

By default all scope options are selected and the *Scope* field is disabled. Your certificates are used for all scopes.

Identity Authentication, by default, supports the following as trusted certificate authorities (CA): SAP Passport CA G2, SAP SSO CA G2, SAP Cloud Root CA, SAP Global Root CA, DigiCert TLS RSA SHA256 2020 CA1, DigiCert Global Root CA, SAP Cloud Platform Client CA, ConnectivityCA.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.

3. Choose the application that you want to edit.

> ### Note
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *Application APIs*, choose *Client Authentication*.
6. Choose the *Add* button in the *Certificates for API Authentication* section.
7. **Optional:** Provide description for the certificate.
8. Choose one of the following options:

Certificate Options	Notes
Generate Certificate	Fill in the fields under *Generate Certificate* and choose the *Generate* button. **Note** You must provide a Common Name (CN) and password to generate the certificate. Once the certificate is generated, it is saved as a *.p12* file. The system populates the *Insert as Text* field with it, and provides the certificate attributes in the Subject DN. The common name (CN) in the generated certificate is in the format `<common name>` (`<admin user ID>`), where common name is the CN provided by the administrator, and admin user ID is the administrator's user id. The maximum length of the combination of `<common name>` (`<admin user ID>`) is 64 characters. **→ Remember** You can generate up to 500 certificates per year. If you reach the limit, you can still upload your certificate, or insert it as a text.
Upload Certificate	You must use `.cer` or `.crt` files. **Note** Once the certificate is uploaded, the system populates the Insert as Text field with it
Insert Certificate	Insert the certificate in the text field.

> ⚠ Restriction
>
> You can't use one and the same certificate for application and system as administrator authentication.

9. Choose the *Add* button at the bottom of the page.

   Once the certificate has been configured, the system displays the message `Certificate for API authentication updated`.

> → Tip
>
> When the certificate expires, follow the procedure again and configure a new one.

Results

Once your certificate is added you can see a table with all your certificates and information about them.

Task overview: API Authentication [page 2780]

Related Information

Configure Secrets for API Authentication [page 2781]
Unlock Client ID [page 2783]
Disable Client ID Locking [page 2784]
Configure JWT for OAuth Client Authentication [page 2787]
Create a New Application [page 1922]
Invitation REST API [page 2802]
User Management REST API [page 2805]

1.9.1.5 Configure JWT for OAuth Client Authentication

Configure the JSON Web Token (JWT) - the issuer and subject of tokens for JWT client authentication in token requests, or the URI for JSON web key retrieval for client authentication.

Prerequisites

You have an OpenID Connect application.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

> **Note**
>
> Filter the list items in the search field by typing the name, display name, application ID, organization ID, or client ID, or choose the application from the list on the left.
>
> If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].

4. Choose the *Trust* tab.
5. Under *Application APIs*, choose *Client Authentication*.
6. Under *JSON Web Tokens*, configure one of the following options and save your changes:

   You can enter a maximum of 20 of each type of trust.

   - (For RFC 7523-based JWT client tokens) Choose the *Add* button for *Configure Trust by Issuer* and provide the required info in the popup:

     Configure Trust by Issuer

Field	Notes
Description	(Optional) - Provide information about the token in the provided field.
Choose *Issuer Configuration* > *Manual Configuration* > radio-button. **Note** When you choose *Manual Configuration* radio-button the *Corporate Identity Provider* section is not configurable.	Under *Manual Configuration* configure: 1. *Issuer* (Required) - The issuer must be OpenID Connect compliant. 2. *JSON Web Key Set URI* (Required) - Choose *Check for Metadata* button. If the entered issuer is verified as OpenID Connect compliant, the JSON Web Key Set URI will be retrieved automatically. Otherwise, you need to explicitly specify the *JSON Web Key Set URI* in the field. 3. *Refresh Interval* (Optional) - Refreshes the URI automatically if it is older than the selected interval. Choose from: • 24 hours (default choice) • 12 hours
Choose *Issuer Configuration* > *Corporate Identity Provider* > radio-button. **Note** When you choose *Corporate Identity Provider* radio-button, the *Manual Configuration* section is not configurable.	*Name* - (Required) Choose from OpenID Connect compliant corporate identity providers.

Field	Notes
General	1. *Subject* (Required) - The sub (subject) that is expected in the token. → Tip If you want to use an OAuth 2.0 token from a corporate identity provider as client credentials for an OpenID Connect application in Identity Authentication, and your OAuth client in the corporate identity provider belongs to a pre-configured application, the subject in the token is the Object ID of that pre-configured application. 2. *Audience* (Optional) - The `audience` claim of the identity provider. 3. `API Permission Groups` (Optional) - The field is enabled only when the *Provided APIs* option is configured. For more information, see Provide APIs for Consumption by Other Applications [page 2891].

- (For OpenID Connect-based JWT client tokens) Choose the *Add* button for *Configure Trust by URI* and provide the required info in the popup:

Configure Trust by URI

Field	Notes
Description	(Optional) You can provide information about the URI here.
URI	(Required) The JSON Web Key Set (JWKS) URI of the trusted party.
Refresh Interval	(Optional) Refreshes the URI automatically if it is older than the selected interval. Choose from: • 24 hours (default choice) • 12 hours
API Access	ⓘ Note This section is read-only. The predefined choice is OpenID.
API Permission Groups	(Optional) `API Permission Groups` field is enabled only when the *Provided APIs* option is configured. For more information, see Provide APIs for Consumption by Other Applications [page 2891].

Task overview: API Authentication [page 2780]

Related Information

Configure Secrets for API Authentication [page 2781]
Unlock Client ID [page 2783]
Disable Client ID Locking [page 2784]
Configure Certificates for API Authentication [page 2785]
JSON Web Token (JWT) Profile for OAuth 2.0 Client Authentication and Authorization Grants
Proof-of-Possession Key Semantics for JSON Web Tokens (JWTs)
JSON Web Token Client Authentication

1.9.2 API Documentation

This document contains references to the API Documentation of Identity Authentication.

> **Note**
>
> To view the error codes that may be returned from any method on any resource URI, see Error and Success Codes [page 2883].

Application Configurations API [page 2791]
: Manage application configurations.

Identity Directory API [page 2792]
: Manage users, groups and custom schemas in the cloud.

Identity Provisioning API [page 2795]
: Manage identity lifecycle processes for cloud and on-premise systems.

Identity Authentication API (Deprecated) [page 2796]
: Deprecated.

Migrating Deprecated Identity Authentication API to Identity Directory API [page 2797]
: Migrating from the Identity Authentication API to Identity Directory API.

Invitation REST API [page 2802]
: The invitation service allows you to implement a request for user invitations.

User Management REST API [page 2805]
: This REST API allows you to implement a request for user management, such as user registration, as well as SP user retrieval, deactivation and deletion.

Password Service REST API [page 2819]
: The password service is used for operations related to user passwords, such as verification of the user name and the password combination.

Forgot Password REST API [page 2821]
: The forgot password REST API sends a reset password email.

TOTP Validation Service [page 2824]
: Validation of time-based one-time password (TOTP).

Change Tenant Texts REST API [page 2826]

The Change Tenant Texts REST API of Identity Authentication can be used to change the predefined texts and messages for end-user screens available per tenant in the Identity Authentication.

Change Master Data Texts REST API [page 2860]
The Change Master Data Texts REST API can be used to change the predefined master data for each resource in Identity Authentication.

1.9.2.1 Application Configurations API

Manage application configurations.

> **Note**
>
> The API is available on the SAP Business Accelerator Hub: Application Configurations API.
>
> Note that the Application Configurations API may return a higher number of applications than those displayed in the administration console. Some applications are used for internal purposes and are not shown in the admin console.

Prerequisites

To call the methods of this SCIM REST API you must have a system as administrator (technical user) with an assigned *Manage Users* role.

For more information, see Add System as Administrator [page 2305], and Edit Administrator Authorizations [page 2309].

Parent topic: API Documentation [page 2790]

Related Information

Identity Directory API [page 2792]
Identity Provisioning API [page 2795]
Identity Authentication API (Deprecated) [page 2796]
Migrating Deprecated Identity Authentication API to Identity Directory API [page 2797]
Invitation REST API [page 2802]
User Management REST API [page 2805]
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TOTP Validation Service [page 2824]
Change Tenant Texts REST API [page 2826]
Change Master Data Texts REST API [page 2860]

1.9.2.2 Identity Directory API

Manage users, groups and custom schemas in the cloud.

> → Remember
>
> The API is available on the SAP Business Accelerator Hub: Identity Directory Service

Prerequisites

To call the methods of this SCIM REST API you must have a system as administrator (technical user) with an assigned *Manage Users* role.

For more information, see Add System as Administrator [page 2305], and Edit Administrator Authorizations [page 2309].

Parent topic: API Documentation [page 2790]

Related Information

Application Configurations API [page 2791]
Identity Provisioning API [page 2795]
Identity Authentication API (Deprecated) [page 2796]
Migrating Deprecated Identity Authentication API to Identity Directory API [page 2797]
Invitation REST API [page 2802]
User Management REST API [page 2805]
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TOTP Validation Service [page 2824]
Change Tenant Texts REST API [page 2826]
Change Master Data Texts REST API [page 2860]

1.9.2.2.1 Extensions

The Identity Directory SCIM REST API can be manually extended by adding user attribute from a custom schema. The steps include:

1. Create custom schema in the administration console for SAP Cloud Identity Services. See Manage Custom Schemas via Administration Console [page 2388].

Example

You have set up a scenario where Identity Authentication acts as a proxy. The default authenticating identity provider is the corporate IdP, and the *Identity Federation* option is configured for that corporate IdP.

Michael Adams is a user from the corporate identity provider. He is created on the proxy IdP with the custom schema `urn:sap:cloud:scim:schemas:extension:custom:2.0:Profile` with two single-valued attributes, `birthday` and `hobby`, and the complex attribute `name` with its child attribute `firstName`.

Sample Code

```
{
  "id": "urn:sap:cloud:scim:schemas:extension:custom:2.0:Profile",
  "schemas": [
    "urn:ietf:params:scim:schemas:core:2.0:Schema"
  ],
  "name": "Profile",
  "description": "Custom schema description",
  "attributes": [
    {
      "name": "hobby",
      "description": "Custom attribute description",
      "canonicalValues": [

      ],
      "multiValued": false,
      "required": false,
      "caseExact": false,
      "type": "string",
      "mutability": "readWrite",
      "returned": "default",
      "uniqueness": "none"
    },
    {
      "name": "birthday",
      "description": "Custom attribute description",
      "canonicalValues": [

      ],
      "multiValued": false,
      "required": false,
      "caseExact": false,
      "type": "string",
      "mutability": "readWrite",
      "returned": "default",
      "uniqueness": "none"
    },
    {
      "name": "name",
      "description": "Custom complex attribute description",
      "multiValued": false,
      "required": false,
      "caseExact": false,
      "type": "complex",
      "mutability": "readWrite",
      "returned": "default",
      "uniqueness": "none",
      "subAttributes": [
        {
          "name": "firstName",
          "description": "Custom subAttribute description",
          "canonicalValues": [
```

```
            ],
            "multiValued": false,
            "required": false,
            "caseExact": false,
            "type": "string",
            "mutability": "readWrite",
            "returned": "default",
            "uniqueness": "none"
          }
        ]
      }
    ]
}
```

2. Assign a custom schema to the user via one of the following options:
   - Assign an attribute from the custom schema via the administration console:
     1. *Access the administration console for Cloud Identity services* > *Users & Authorizations* > *User Management*.
     2. Select a user and choose the *Extensions* tab.

        > **Note**
        >
        > The attributes assigned from the custom schemas are visible there.

     3. Press the *Assign* button at the top of the page and choose which custom schema or schemas to add.
     4. Add values for for the attributes in the custom schema.

        > **→ Remember**
        >
        > To save your configuration you must add at least one value.

     5. Save your configuration.

        > **→ Tip**
        >
        > To unassign the already assigned schema, choose *Unassign* > *select the schema or schemas you want to unassign* > *Unassign*.

   - Assign an attribute form custom schema to the user by calling the Identity Directory SCIM REST API. See Identity Directory Service.

     > **Example**
     >
     > You have updated Michael's `hobby` attribute, by calling the *PATCH* method of Identity Directory Service.
     >
     > **Sample Code**
     >
     > ```
     > content-type: application/scim+json
     > Body:
     > {
     >     "schemas": [
     >         "urn:ietf:params:scim:api:messages:2.0:PatchOp",
     >         "urn:sap:cloud:scim:schemas:extension:custom:2.0:Profile"
     >     ],
     >     "Operations": [
     > ```

```
        {
            "op": "add",
            "path":
"urn:sap:cloud:scim:schemas:extension:custom:2.0:Profile:hobby",
            "value": "Chess playing."
        }
    ]
}
```

3. Send the already assigned custom attribute by configuring it in the default attributes sent to the application. See Configuring Attributes Based on Flexible Expressions [page 1979] under the *Send Identity Directory Custom Schema Attributes* section.

> ### Example
>
> The default attributes for the application are configured in the administration console as follows:
>
> Default Attributes Configuration in Administration Console
>
> | Attribute | Value |
> | --- | --- |
> | name | ${urn:sap:cloud:scim:schemas:extension:custom:2.0:Profile.name.firstName} |
> | hobby | ${urn:sap:cloud:scim:schemas:extension:custom:2.0:Profile.hobby} |
> | birthday | ${urn:sap:cloud:scim:schemas:extension:custom:2.0:Profile.birthday} |

1.9.2.3 Identity Provisioning API

Manage identity lifecycle processes for cloud and on-premise systems.

> ### Note
>
> The API is available on the SAP Business Accelerator Hub: Identity Provisioning API.

Prerequisites

To call the methods of this SCIM REST API you must have a system as administrator (technical user) with the following roles assigned:

- *Access Proxy System API*
- *Access Real-Time Provisioning API*
- *Access Identity Provisioning Tenant Admin API*

For more information, see Add System as Administrator [page 2305], and Edit Administrator Authorizations [page 2309].

Parent topic: API Documentation [page 2790]

Related Information

Application Configurations API [page 2791]
Identity Directory API [page 2792]
Identity Authentication API (Deprecated) [page 2796]
Migrating Deprecated Identity Authentication API to Identity Directory API [page 2797]
Invitation REST API [page 2802]
User Management REST API [page 2805]
Password Service REST API [page 2819]
Forgot Password REST API [page 2821]
TOTP Validation Service [page 2824]
Change Tenant Texts REST API [page 2826]
Change Master Data Texts REST API [page 2860]

1.9.2.4 Identity Authentication API (Deprecated)

Deprecated.

> ⓘ Note
>
> This API is deprecated. Please use Identity Directory API instead. For more information, see Migrating Deprecated Identity Authentication API to Identity Directory API [page 2797].

> → Tip
>
> Information about the deprecated API is available on the SAP Business Accelerator Hub: Identity Authentication API (Deprecated).

Parent topic: API Documentation [page 2790]

Related Information

Application Configurations API [page 2791]
Identity Directory API [page 2792]
Identity Provisioning API [page 2795]
Migrating Deprecated Identity Authentication API to Identity Directory API [page 2797]
Invitation REST API [page 2802]

User Management REST API [page 2805]
Password Service REST API [page 2819]
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Change Tenant Texts REST API [page 2826]
Change Master Data Texts REST API [page 2860]
System for Cross-Domain Identity Management
SCIM Data Types

1.9.2.5 Migrating Deprecated Identity Authentication API to Identity Directory API

Migrating from the Identity Authentication API to Identity Directory API.

Documentation

Documentation Links

Identity Authentication Service API (Deprecated)	Identity Directory Service API (Successor)
Identity Authentication API	Identity Directory API

Endpoints Mapping

User Management

HTTP Method	Identity Authentication Service API (Deprecated)	Identity Directory Service API (Successor)
GET	`/service/scim/Users/`	`/scim/Users/`
GET	`/service/scim/Users/<id*>`	`/scim/Users/<id**>`
POST	`/service/scim/Users`	`/scim/Users`
PUT	`/service/scim/Users/<id*>`	`/scim/Users/<id**>`
DELETE	`/service/scim/Users/<id*>`	`/scim/Users/<id**>`

> **Note**
>
> *The <id*> is legacy `User ID`. It is a number starting with a letter. For example, `P123456`.

The <id> is in the universally unique identifier (UUID) format. For example, `1234fa2b-16or-4no7-87t7-26a02bf0bk69`.

Group Management Endpoints

HTTP Method	Identity Authentication Service API (Deprecated)	Identity Directory Service API (Successor)
GET	`/service/scim/Groups/`	`/scim/Groups/`
GET	`/service/scim/Groups/<id of the group*>`	`/scim/Groups/<id of the group**>`
POST	`/service/scim/Groups`	`/scim/Groups`
PUT	`/service/scim/Groups/<id of the group*>`	`/scim/Groups/<id of the group**>`
DELETE	`/service/scim/Groups/<id of the group*>`	`/scim/Groups/<id of the group**>`

ⓘ Note

The <id of the group> is in a legacy ID format. For example, `63c468ode3c07406970900kc`.

The <id of the group> is in the universally unique identifier (UUID) format. For example, `7369fa2b-16as-4sd7-87c7-26a01of0bk78`.

→ Remember

To assign group to a user, use the groups endpoint.

Attributes Mapping

Manage Users Attributes Mapping

Identity Authentication Service API (Deprecated)	Identity Directory Service API (Successor)
`id`	`urn:ietf:params:scim:schemas:extension:sap:2.0:User.userId`
`emails.value`	`emails.value`
`name.honorificPrefix`	`name.honorificPrefix`
`name.givenName`	`name.givenName`
`name.familyName`	`name.familyName`
`name.middleName`	`name.middleName`
`userName`	`userName`

Identity Authentication Service API (Deprecated)	Identity Directory Service API (Successor)
addresses.streetAddressaddresses.locality addresses.region addresses.postalCode addresses.country	addresses.streetAddressaddresses.locality addresses.region addresses.postalCode addresses.country urn:ietf:params:scim:schemas:extension:sap:2.0:User.addresses.streetAddress2
addresses.streetAddressaddresses.locality addresses.region addresses.postalCode addresses.country	addresses.streetAddressaddresses.locality addresses.region addresses.postalCode addresses.country urn:ietf:params:scim:schemas:extension:sap:2.0:User.addresses.streetAddress2
locale	locale
phoneNumbers[work].value	phoneNumbers[work].value
phoneNumbers[mobile].value	phoneNumbers[mobile].value
phoneNumbers[fax].value	phoneNumbers[fax].value
timeZone	timezone
password	password
passwordStatus	urn:ietf:params:scim:schemas:extension:sap:2.0:User.passwordDetails.status
active	active
displayName	displayName
contactPreferenceEmail	urn:ietf:params:scim:schemas:extension:sap:2.0:User.contactPreferences.email
contactPreferenceTelephone	urn:ietf:params:scim:schemas:extension:sap:2.0:User.contactPreferences.telephone
industryCrm	urn:ietf:params:scim:schemas:extension:sap:2.0:User.industry
company	urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.organization
companyRelationship	urn:ietf:params:scim:schemas:extension:sap:2.0:User:companyRelationship
department	urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.department
title	
groups	groups
passwordLoginTime	urn:ietf:params:scim:schemas:extension:sap:2.0:User.passwordDetails.loginTime
loginTime	urn:ietf:params:scim:schemas:extension:sap:2.0:User.loginTime

Identity Authentication Service API (Deprecated)	Identity Directory Service API (Successor)
passwordSetTime	urn:ietf:params:scim:schemas:extension:sap:2.0:User.passwordDetails.setTime
	urn:ietf:params:scim:schemas:extension:sap:2.0:User.validFrom
	urn:ietf:params:scim:schemas:extension:sap:2.0:User.validTo
scenarios	
termsOfUse	termsOfUse
privacyPolicy	privacyPolicy
newsletters	
userType	userType
sourceSystem	urn:ietf:params:scim:schemas:extension:sap:2.0:User.sourceSystem
	urn:ietf:params:scim:schemas:extension:sap:2.0:User.sourceSystemId
socialIdentities	
sendMail	urn:ietf:params:scim:schemas:extension:sap:2.0:User.sendMail
mailVerified	urn:ietf:params:scim:schemas:extension:sap:2.0:User.emails.verified
telephoneVerified	urn:ietf:params:scim:schemas:extension:sap:2.0:User.phoneNumbers.verified
telephoneVerificationAttempts	
passwordPolicy	urn:ietf:params:scim:schemas:extension:sap:2.0:User.passwordDetails.policy
passwordStatus	urn:ietf:params:scim:schemas:extension:sap:2.0:User.passwordDetails.status
passwordFailedLoginAttempts	urn:ietf:params:scim:schemas:extension:sap:2.0:User.passwordDetails.failedLoginAttempts
applicationId	urn:ietf:params:scim:schemas:extension:sap:2.0:User.applicationId
	urn:ietf:params:scim:schemas:extension:sap:2.0:User.activationUrl
targetUrl	urn:ietf:params:scim:schemas:extension:sap:2.0:User.targetUrl
emailTemplateSetId	urn:ietf:params:scim:schemas:extension:sap:2.0:User.emailTemplateSetId

Identity Authentication Service API (Deprecated)	Identity Directory Service API (Successor)
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.employeeNumber	urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.employeeNumber
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.organization	urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.organization
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.costCenter	urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.costCenter
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.division	urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.division
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.manager.value	urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.manager.value
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.manager.$ref	urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.manager.$ref
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.manager.displayName	urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.manager.displayName
urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.department	urn:ietf:params:scim:schemas:extension:enterprise:2.0:User.department
urn:sap:cloud:scim:schemas:extension:custom:2.0:User.attributes.	urn:sap:cloud:scim:schemas:extension:custom:2.0:User.attributes.
customAttribute1	customAttribute1
customAttribute2	customAttribute2
customAttribute3	customAttribute3
customAttribute4	customAttribute4
customAttribute5	customAttribute5
customAttribute6	customAttribute6
customAttribute7	customAttribute7
customAttribute8	customAttribute8
customAttribute9	customAttribute9
customAttribute10	customAttribute10

Manage Groups Attributes Mapping

Identity Authentication Service API (Deprecated)	Identity Directory Service API (Successor)
id	urn:sap:cloud:scim:schemas:extension:custom:2.0:Group.additionalId
displayName	displayName
members.value	members.value
members.$ref	members.$ref
members.display	

Identity Authentication Service API (Deprecated)	Identity Directory Service API (Successor)
	`members.type`
`urn:sap:cloud:scim:schemas:extension:custom:2.0:Group.name`	`urn:sap:cloud:scim:schemas:extension:custom:2.0:Group.name`
`urn:sap:cloud:scim:schemas:extension:custom:2.0:Group.description`	`urn:sap:cloud:scim:schemas:extension:custom:2.0:Group.description`
`urn:sap:cloud:scim:schemas:extension:custom:2.0:Group.groupId`	`id`
	`externalId`

Parent topic: API Documentation [page 2790]

Related Information

Application Configurations API [page 2791]
Identity Directory API [page 2792]
Identity Provisioning API [page 2795]
Identity Authentication API (Deprecated) [page 2796]
Invitation REST API [page 2802]
User Management REST API [page 2805]
Password Service REST API [page 2819]
Forgot Password REST API [page 2821]
TOTP Validation Service [page 2824]
Change Tenant Texts REST API [page 2826]
Change Master Data Texts REST API [page 2860]

1.9.2.6 Invitation REST API

The invitation service allows you to implement a request for user invitations. The invitees then receive an email containing information about how to register.

Prerequisites

You need to set up the authentication type to access this API. You can use a *client ID* and a *secret*, or a certificate. For more information, see Configure Secrets for API Authentication [page 1997] or Configure Certificates for API Authentication [page 2001].

Resources

To configure the invitation service, you use a `POST` request with the following URI: `https://<Cloud Identity Services domain>/cps/invite/`.

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

Representation

You have to use a JSON representation of the invitation request by specifying *application/json* content type. All declared parameters in the request must also be JSON encoded.

Parameters

Required Parameters for the POST Method

Required Parameter	Description
`inviteeEmail`	The email of the invitee **Note** Only `inviteeEmail` or `inviteeUserId` should be used, not both.
`inviteeUserId`	The user ID of the invitee **Note** Only `inviteeEmail` or `inviteeUserId` should be used, not both.
`inviterName`	The display name of the user who sends the invitation.

Required Parameter	Description
`targetUrl`	The URL that the user is redirected to after registration. **→ Recommendation** From a usability perspective we recommend you to use URL of a protected page. **Note** The `targetUrl` parameter is optional if a *Home URL* is set for the application, and the application does not use overlay. If `targetUrl` is not specified, or the application uses overlay, the user is redirected to the application's *Home URL*, which must be set. For more information how to configure *Home URL*, see Configure an Application's Home URL [page 1937].
`sourceUrl`	The URL for the invitation link in the email sent to the invitee. The URL must be a public page. **Note** The `sourceUrl` parameter is optional if a `Home URL` is set for the application, and the application does not use overlay. If `sourceUrl` is not specified, or the application uses overlay, the user is redirected to the application's *Home URL*, which must be set. For more information how to configure *Home URL*, see Configure an Application's Home URL [page 1937].

Optional Parameters for the POST Method

Optional Parameter	Description
`inviteeFirstName`	The first name of the invitee
`inviteeLastName`	The last name of the invitee
`footerText`	The footer text of the invitation email
`headerText`	The header text of the invitation email
`language`	The preferred language for the invitation email. The usage of this parameter does not affect the end-user screens. Must be a string value specified by a two or four-letter code in one of the following formats: XX, xx, xx-XX or xx_XX. Otherwise the invitation email is in English.

Example

```
POST /cps/invite/
Content-Type: application/json
 {
  "inviteeEmail": "john.miller@company.com",
  "inviteeFirstName": "John",
  "inviteeLastName": "Miller",
  "inviterName": "Donna Moore",
  "footerText": "Invitation footer sample text",
  "headerText": "Invitation header sample text",
  "targetUrl": "http://www.myserviceprovider.com/protected_home_page/",
  "sourceUrl": "http://www.myserviceprovider.com/public_home_page/"
 }
```

Parent topic: API Documentation [page 2790]

Related Information

Application Configurations API [page 2791]
Identity Directory API [page 2792]
Identity Provisioning API [page 2795]
Identity Authentication API (Deprecated) [page 2796]
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Change Master Data Texts REST API [page 2860]
Add Logon Overlays in Customer Applications [page 2877]
Configure an Application's Home URL [page 1937]
Configure Certificates for API Authentication [page 2001]
Error and Success Codes [page 2883]
Configure an Application's Home URL [page 1937]

1.9.2.7 User Management REST API

This REST API allows you to implement a request for user management, such as user registration, as well as SP user retrieval, deactivation and deletion.

Prerequisites

You need to set up the authentication type to access this API. You can use a *client ID* and a *secret*, or a certificate. For more information, see Configure Secrets for API Authentication [page 1997] or Configure Certificates for API Authentication [page 2001].

Methods

HTTP Method	Action	URI
POST	User Registration [page 2807]	`https://<tenant ID>.accounts.ondemand.com/service/users`
GET	SP User ID Retrieval [page 2811]	`https://<tenant ID>.accounts.ondemand.com/service/users`
GET	SP User Information [page 2813]	`https://<tenant ID>.accounts.ondemand.com/service/users/<SP user ID>`
PUT	SP User Deactivation [page 2815]	`https://<tenant ID>.accounts.ondemand.com/service/users/<SP user ID>`
DELETE	SP User Deletion [page 2817]	`https://<tenant ID>.accounts.ondemand.com/service/users/<SP user ID>`

Parent topic: API Documentation [page 2790]

Related Information

Application Configurations API [page 2791]
Identity Directory API [page 2792]
Identity Provisioning API [page 2795]
Identity Authentication API (Deprecated) [page 2796]
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1.9.2.7.1 User Registration

The user registration is used for registration of new users or for on-behalf registration of partners.

Prerequisites

You need to set up the authentication type to access this API. You can use a *client ID* and a *secret*, or a certificate. For more information, see Configure Secrets for API Authentication [page 1997] or Configure Certificates for API Authentication [page 2001].

Request

URI: `https://<Cloud Identity Services domain>/service/users`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Authorization`	Yes	`Basic` • via application ((service provider (SP)) Authentication certificate • via SP REST API username and password For more information, see API Authentication [page 1996].
`Content-Type`	Yes	application/x-www-form-urlencoded

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`email`	Yes	String	The email of the user you register. The email must be unique. → Tip If email is mandatory, for users without valid email addresses or for testing purposes use the `sap-test.de` domain, for example `<username>@sap-test.de`. Do not use any other existing or nonexisting domains.	Request body
`user_profile_id`	No	String	The user ID	Request body
`login_name`	No	String	The logon name of the user	Request body
`applicationId`	No	String	The ID of the subscribed application. The user is created for the application with the provided ID.	Request body
`first_name`	No	String	The first name of the user you register. The allowed maximum length for the first name is 32 characters.	Request body
`last_name`	Yes	String	The last name of the user you register. The allowed maximum length for the last name is 64 characters.	Request body
`language`	No	String	The preferred language for the activation email of the user you register. The usage of this parameter does not affect the end-user screens. Must be a string value specified by a two or four-letter code in one of the following formats: XX, xx, xx-XX or xx_XX. Otherwise the activation email is in English.	Request body
`valid_from`	No	String	The *valid from* date	Request body
`valid_to`	No	String	The *valid until* date	Request body
`name_id`	No	String	The name ID of the user you register.	Request body

Parameter	Required	Data Type	Description	Parameter Type
`source_url`	No	String	The URL to the public page of the application where the Identity Authentication overlays are integrated. If not provided, the activation screen is shown without overlays. This parameter value must be URL-encoded. **⚠ Caution** Put `/` at the end of the source URL.	Request body
`target_url`	No	String	The URL to the application page that the user should be redirected to after he or she has completed account activation. If `target_url` is not provided, the user is redirected to the home URL configured for the service provider. For more information how to configure *Home URL*, see Configure an Application's Home URL [page 1937].	Request body
`send_email`	No	Boolean	Values: true (default value), false • If **true** - activation email is sent to the user. • If **false** - activation email is not sent to the user. **ⓘ Note** If the user is new, the link for account activation is returned with the **201** response.	Request body
`spCustomAttribute1`	No	String	Custom attributes are used to store additional information for the SP users. It is allowed to pass five customer attributes for a user	Request body
`spCustomAttribute2`	No	String	Custom attributes are used to store additional information for the SP users. It is allowed to pass five customer attributes for a user	Request body

Parameter	Required	Data Type	Description	Parameter Type
`spCustomAttribute3`	No	String	Custom attributes are used to store additional information for the SP users. It is allowed to pass five customer attributes for a user	Request body
`spCustomAttribute4`	No	String	Custom attributes are used to store additional information for the SP users. It is allowed to pass five customer attributes for a user	Request body
spCustomAttribute5	No	String	Custom attributes are used to store additional information for the SP users. It is allowed to pass five customer attributes for a user	Request body

Request Example

Example

> ⚠ Caution
>
> All parameters for the POST method must be written on one line.

```
POST /service/users
Content-Type: application/x-www-form-urlencoded
name_id=johns&user_profile_id=p987654&email=john.smith@sap.com&first_name=John&last_name=Smith&language=en&valid_from=20110901120000Z&valid_to=20120901110000Z&source_url=http%3A%2F%2Fwww.myapp.com%2Fpublic.jsp&target_url=http%3A%2F%2Fwww.myapp.com%2Fprotected.jsp&spCustomAttribute1=Industry
```

Response

Response Status and Error Codes

Code	Reason	Description
201	Created	The user is successfully created.
400	Bad Request	The user for whom you are trying to create an SP user is `inactive`.
409	Conflict	The SP user already exists.

For more information about the general error codes that may be returned, see Error and Success Codes [page 2883].

Response Example

Example

The URI of the created user is returned in the location header of the HTTP Response.

```
Location: https://<tenant ID>.accounts.ondemand.com/service/users/0800200c9a66
```

Example

In case of conflict, the URI of the conflicting user is returned in the location header of the HTTP Response.

```
Location: https://<tenant ID>.accounts.ondemand.com/service/users/467345637aa
```

Example

In case of creating a new user with "send_email=false", the activation link is returned in the HTTP Response body.

```
Content-Type: application/json
    {
         "activationLink" : "https://<tenant ID>.accounts.ondemand.com/ids/
activation?
token=I1830C497BF9B857B7D6298E5F117AF397I1F59A28838A3276E8B68FFFF54414C8843ACF395
A05401ABE6568FF3659D6EE96"
    }
```

Related Information

User Management REST API [page 2805]

1.9.2.7.2 SP User ID Retrieval

To retrieve an SP (service provider) user's ID, you use the `GET` method.

Prerequisites

You need to set up the authentication type to access this API. You can use a *client ID* and a *secret*, or a certificate. For more information, see Configure Secrets for API Authentication [page 1997] or Configure Certificates for API Authentication [page 2001].

Request

URI: `https://<Cloud Identity Services domain>/service/users`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *GET*

Request Headers

Header	Required	Values
`Authorization`	Yes	`Basic` • via application ((service provider (SP)) Authentication certificate • via SP REST API username and password For more information, see API Authentication [page 1996].
`Content-Type`	Yes	application/vnd.sap-id-service.sp-user-id+xml

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`name_id`	Yes	String	The name ID of the user you register. The `name_id` can be retrieved via the SP User Information [page 2813] API. It can be a User ID, Login name, etc.	Query string
`applicationId`	No	String	The ID of the subscribed application. The user is retrieved for the application with the provided ID.	Query string

Request Example

```
GET /service/users?name_id=johns
```

Response

Response Headers

Header	Description
`Location`	The ID of the created SP user is in the location header of the HTTP Response.

Response Example

```
Location: https://<tenant ID>.accounts.ondemand.com/service/users/<SP user ID>
```

Related Information

User Management REST API [page 2805]

1.9.2.7.3 SP User Information

To retrieve information for an SP (service provider) user, you use a `GET` method.

Prerequisites

You need to set up the authentication type to access this API. You can use a *client ID* and a *secret*, or a certificate. For more information, see Configure Secrets for API Authentication [page 1997] or Configure Certificates for API Authentication [page 2001].

Request

URI: `https:/<Cloud Identity Services domain>.`

> ⓘ Note
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *GET*

Request Headers

Header	Required	Values
`Authorization`	Yes	`Basic` • via application ((service provider (SP)) Authentication certificate • via SP REST API username and password For more information, see API Authentication [page 1996].
`Content-Type`	Yes	application/vnd.sap-id-service.sp-user-id+xml

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`<SP user ID>`	Yes	String	The SP user id.	Path
`applicationId`	No	String	The ID of the subscribed application. The information for the SP useruser is retrieved for the application with the provided ID.	Query string

Request Example

```
GET /service/users/<the ID of the SP user>
```

Response

Response Attributes

The response returns the following attributes. Only the attributes that exist for the user are returned.

- user_profile_id
- name_id
- status
- profile_status
- valid_from
- valid_to
- activation_time
- email
- language
- first_name
- last_name
- sp_name
- login_name

- country
- city
- company_city
- spCustomAttributeX

Response Status and Error Codes

Code	Reason	Description
200	OK	The information for the user is returned in the response.
404	Not Found	The SP user with the provided ID does not exist.

Related Information

User Management REST API [page 2805]

1.9.2.7.4 SP User Deactivation

To deactivate an SP (service provider) user, you use a `PUT` method.

Prerequisites

You need to set up the authentication type to access this API. You can use a *client ID* and a *secret*, or a certificate. For more information, see Configure Secrets for API Authentication [page 1997] or Configure Certificates for API Authentication [page 2001].

Request

URI: `https://<Cloud Identity Services domain>/service/users/<SP user ID>`

> ⓘ Note
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *PUT*

Request Headers

Header	Required	Values
`Authorization`	Yes	`Basic` • via application ((service provider (SP)) Authentication certificate • via SP REST API username and password For more information, see API Authentication [page 1996].
`Content-Type`	Yes	application/vnd.sap-id-service.sp-user-id+xml

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`status`	Yes	String	When the user activates his or her email, the `status` is set to **active**. To deactivate the SP user, update the `status` parameter by setting it to **inactive**. To activate the SP user, set it to **active**.	Request body
`applicationId`	No	String	The ID of the subscribed application. The user is deactivated for the application with the provided ID.	Query string

Request Example

```
PUT /service/users/<the ID of the SP user>
Content-type: application/vnd.sap-id-service.sp-user-id+xml; version=1.0
<user>
  <status>inactive</status>
</user>
```

Response

Response Status and Error Codes

Code	Reason	Description
200	Deactivated	SP user is successfully deactivated

For more information about the general error codes that may be returned, see Error and Success Codes [page 2883].

> **Note**
>
> If a user logs into an application he or she is deactivated for, the identity provider will block the logon and notify the application.

Response Example

```
<Response Consent="urn:oasis:names:tc:SAML:2.0:consent:unspecified"
          Destination="https://example.com/saml2/acs"
          Version="2.0"
          xmlns:ns2="urn:oasis:names:tc:SAML:2.0:assertion"
          xmlns="urn:oasis:names:tc:SAML:2.0:protocol"
          xmlns:ns4="http://www.w3.org/2001/04/xmlenc#"
          xmlns:ns3="http://www.w3.org/2000/09/xmldsig#"
          >
  <ns2:Issuer>accounts.sap.com</ns2:Issuer>
  <Status>
    <StatusCode Value="urn:oasis:names:tc:SAML:2.0:status:Responder">
      <StatusCode Value="urn:oasis:names:tc:SAML:2.0:status:AuthnFailed" />
    </StatusCode>
    <StatusMessage>The SP user [URI of the sp user] is with status inactive for
Service Provider [example.com]</StatusMessage>
  </Status>
</Response>
```

Related Information

User Management REST API [page 2805]

1.9.2.7.5 SP User Deletion

Delete an SP (service provider) user.

Prerequisites

You need to set up the authentication type to access this API. You can use a *client ID* and a *secret*, or a certificate. For more information, see Configure Secrets for API Authentication [page 1997] or Configure Certificates for API Authentication [page 2001].

> ⚠ Caution
>
> If the last SP user is deleted, and the user is of type `type public`, then the whole user profile is deleted.

Request

URI: `https://<Cloud Identity Services domain>/service/users/<SP user ID>`

> ⓘ Note
>
> The domain part has the following pattern:

`<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.

Tenant ID is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *DELETE*

Request Headers

Header	Required	Values
`Authorization`	Yes	`Basic` • via application ((service provider (SP)) Authentication certificate • via SP REST API username and password For more information, see API Authentication [page 1996].
`Content-Type`	Yes	application/vnd.sap-id-service.sp-user-id+xml

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`applicationId`	No	String	The ID of the subscribed application. The user is deleted for the application with the provided ID.	Query string

Request Example

```
DELETE /service/users/0800200c9a66
```

Response

Response Status and Error Codes

Code	Reason	Description
200	The operation is successful.	The SP user is deleted.

For more information about the general error codes that may be returned, see Error and Success Codes [page 2883].

Related Information

User Management REST API [page 2805]

1.9.2.8 Password Service REST API

The password service is used for operations related to user passwords, such as verification of the user name and the password combination.

Verify Username and Password Combination

Verify the username and password combination, or verify the thing ID and password combination.

Request

URI: `https://<Cloud Identity Services domain>/service/users/password`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Authorization`	Yes	`Basic` - Username and password are provided by the user. **Note** Depending on the allowed logon identifiers for the user, the username can be the `User ID`, `Login Name`, or `Email`. For more information, see Configure Allowed Logon Identifiers [page 2146]. **Caution** If the user provides wrong password, then each verification counts as a failed logon attempt. The password locks when the number of the allowed failed logon attempts is reached. The number depends on the password policy applied for the application. For more information, see Configuring Password Policies [page 2223].

Header	Required	Values
`Content-Type`	Yes	application/json

Request Example

```
POST /service/users/password
Authorization: Basic <Username/Password>
```

Response

Response Status and Error Codes

Code	Result or X-Message Code	Description
200 OK	Success	When the username and password combination or thing ID and password combination is verified.
401 Unauthorized	PASSWORD_LOCKED	When the password is locked for 60 minutes.
	PASSWORD_DISABLED	When the password is disabled.
	USER_INACTIVE	When the user is not in status active.
	PASSWORD_RESET_REQUIRED	When the user must reset his or her password before logon.
	PASSWORD_CHANGE_REQUIRED	When the user must change his or her password before logon.
	INITIAL_PASSWORD_EXPIRED	When the initial password of the user has expired. After the validity of the initial password expires, the user can't log on to the application and must contact the administrator.
	INVALID_AUTHORIZATION_HEADER_LENGTH	The time-based one-time password (TOTP) code is not provided, but Two-Factor Authentication (TFA) with TOTP is enabled.
	INVALID_OTP_CODE	Wrong TOTP code was provided.
	RBA_RULE_ACTION_DENY	Denied because of a risk-based authentication rule

Response Example

On success, the HTTP Response Body contains:

```
{
      "uid": "P000000",
      "first_name": "Dona",
      "last_name": "Moore",
      "mail": "dona.moore@example.com"
      "type": "employee"
   }
```

Parent topic: API Documentation [page 2790]

Related Information

Application Configurations API [page 2791]
Identity Directory API [page 2792]
Identity Provisioning API [page 2795]
Identity Authentication API (Deprecated) [page 2796]
Migrating Deprecated Identity Authentication API to Identity Directory API [page 2797]
Invitation REST API [page 2802]
User Management REST API [page 2805]
Forgot Password REST API [page 2821]
TOTP Validation Service [page 2824]
Change Tenant Texts REST API [page 2826]
Change Master Data Texts REST API [page 2860]
Configure Risk-Based Authentication for an Application [page 2007]
Configure User Access to the Application [page 2025]
Error and Success Codes [page 2883]

1.9.2.9 Forgot Password REST API

The forgot password REST API sends a reset password email.

The language of the reset password email is defined in the following order of importance:

Custom Template Sets Configured for Application

1. If the user is set with a specific language:
    1. if the language exists in the custom template, the system sends the reset password email in that language
    2. if the language does not exist in the custom template, the system sends the reset password email in English, if English exists in the custom template set
    3. if the language of the user, and the English language do not exist in the custom template, the system sends the reset password email in English from the *Default* template set
2. If the user is not set with a specific language:
    1. if English exists in the custom template set, the system sends the reset password email in English from the custom template set
    2. if English does not exist in the custom template set, the system sends the reset password email in English from the *Default* template set

Default Template Set Configured for Application

If the user is set with a specific language, the system sends the reset password email in that language, if it exists in the *Default* set. Otherwise, it sends the reset password email in English from the *Default* template set.

For more information about the email template sets, see Configuring Email Templates [page 2282].

→ Tip

Configure SAML 2.0 Service Provider [page 1941].

Request

URI: `https://<Cloud Identity Services domain>/service/users/forgotPassword`

ⓘ Note

The domain part has the following pattern:

`<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.

Tenant ID is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Authorization`	Yes	`Basic` • via application ((service provider (SP)) Authentication certificate • via SP REST API username and password For more information, see API Authentication [page 1996].
`Content-Type`	Yes	application/json

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`identifier`	Yes	string	supported attributes: • email → Tip If email is mandatory, for users without valid email addresses or for testing purposes use the `sap-test.de` domain, for example `<username>@sap-test.de`. Do not use any other existing or nonexisting domains. • loginName • uid	Request body

Request Example

```
{
    "identifier": "dona.moore@example.com"
}
```

Response

Response Status and Error Codes

Code	Reason	Description
200 OK	The user exists and password reset is allowed for the user profile.	Forgot password email is sent to the provided user. ⓘ Note Forgot password email is not sent when the response is **200 OK** in following cases: • the user does not exist • the limit of three forgot password e-mails sent for the last 24 hours has been reached
400 Bad Request	PASSWORD_TOO_NEW	When the password has already been changed for the last 24 hours..
	USER_INACTIVE	When the user status is `inactive`.

Code	Reason	Description
	PASSWORD_DISABLED	When the password is disabled.

Parent topic: API Documentation [page 2790]

Related Information

Application Configurations API [page 2791]
Identity Directory API [page 2792]
Identity Provisioning API [page 2795]
Identity Authentication API (Deprecated) [page 2796]
Migrating Deprecated Identity Authentication API to Identity Directory API [page 2797]
Invitation REST API [page 2802]
User Management REST API [page 2805]
Password Service REST API [page 2819]
TOTP Validation Service [page 2824]
Change Tenant Texts REST API [page 2826]
Change Master Data Texts REST API [page 2860]
List and Edit User Details [page 2361]
Error and Success Codes [page 2883]

1.9.2.10 TOTP Validation Service

Validation of time-based one-time password (TOTP).

The API is called on behalf of an application ((service provider (SP)).

Request

URI: `https://<Cloud Identity Services domain>/service/users/otp`

> ⓘ Note
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Authentication mechanisms:

- Client certificate
- Basic authentication

> **Note**
>
> For more information about the authentication mechanisms, see API Authentication [page 1996].

Request Headers

Header	Required	Values
`Content Type`	Yes	application/json

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`userName`	Yes	String	Provided by the user. The `userName` can be either the user email, the user login name, or the user profile ID.	Request body
`otpCode`	Yes	String	Provided by the user. The `otpCode` is the time-based one-time password (TOTP) to be validated.	Request body

Request Example

```
POST /service/users/otp
Content-Type: application/json
{
 "userName": "michael.adams@example.com",
 "otpCode": "123456"
}
```

Response

Response Status and Error Codes

Code	Result or X-Message Code	Description
200 OK	Success	The one-time password is valid.
401 Unauthorized	BAD_CREDENTIALS	The user isn't authenticated or the authentication failed due to wrong credentials.
	INVALID_OTP_CODE	The TOTP isn't valid.
	LOCKED_OTP_CODE	The user provided wrong TOTP five times. The user's passcode authentication is locked.

Code	Result or X-Message Code	Description
	USED_OTP_CODE	The TOTP is already used.

Parent topic: API Documentation [page 2790]

Related Information

Application Configurations API [page 2791]
Identity Directory API [page 2792]
Identity Provisioning API [page 2795]
Identity Authentication API (Deprecated) [page 2796]
Migrating Deprecated Identity Authentication API to Identity Directory API [page 2797]
Invitation REST API [page 2802]
User Management REST API [page 2805]
Password Service REST API [page 2819]
Forgot Password REST API [page 2821]
Change Tenant Texts REST API [page 2826]
Change Master Data Texts REST API [page 2860]
Error and Success Codes [page 2883]

1.9.2.11 Change Tenant Texts REST API

The Change Tenant Texts REST API of Identity Authentication can be used to change the predefined texts and messages for end-user screens available per tenant in the Identity Authentication.

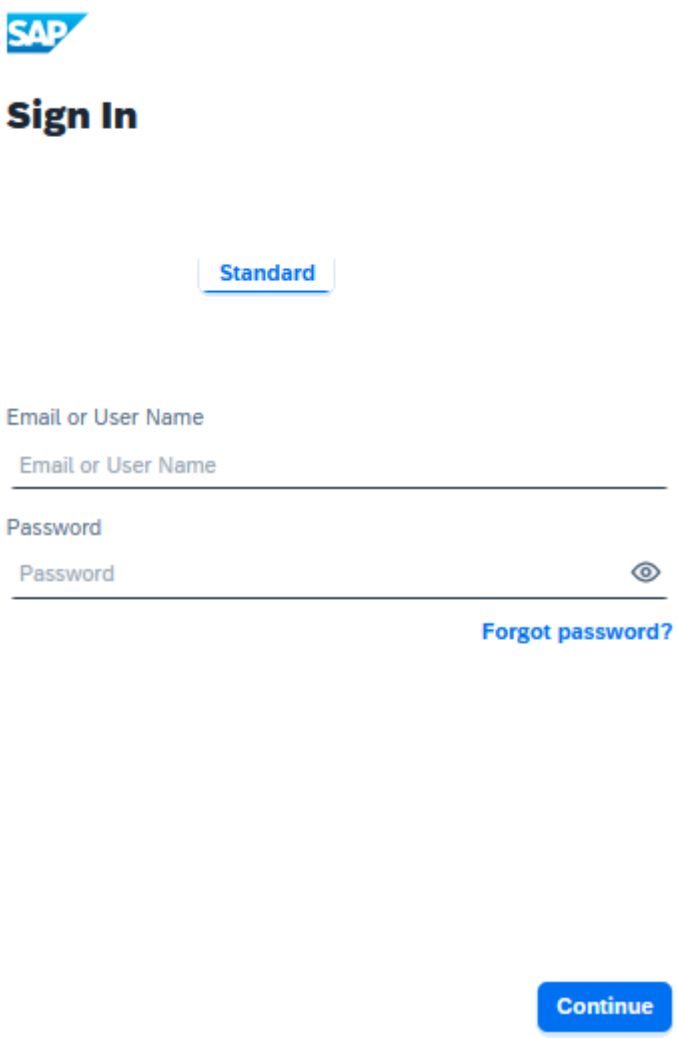
> **Note**
>
> We recommend you to change the tenant texts via the administration console of Identity Authentication. For more information, see Change Tenant Texts Via Administration Console [page 2121].

Prerequisites

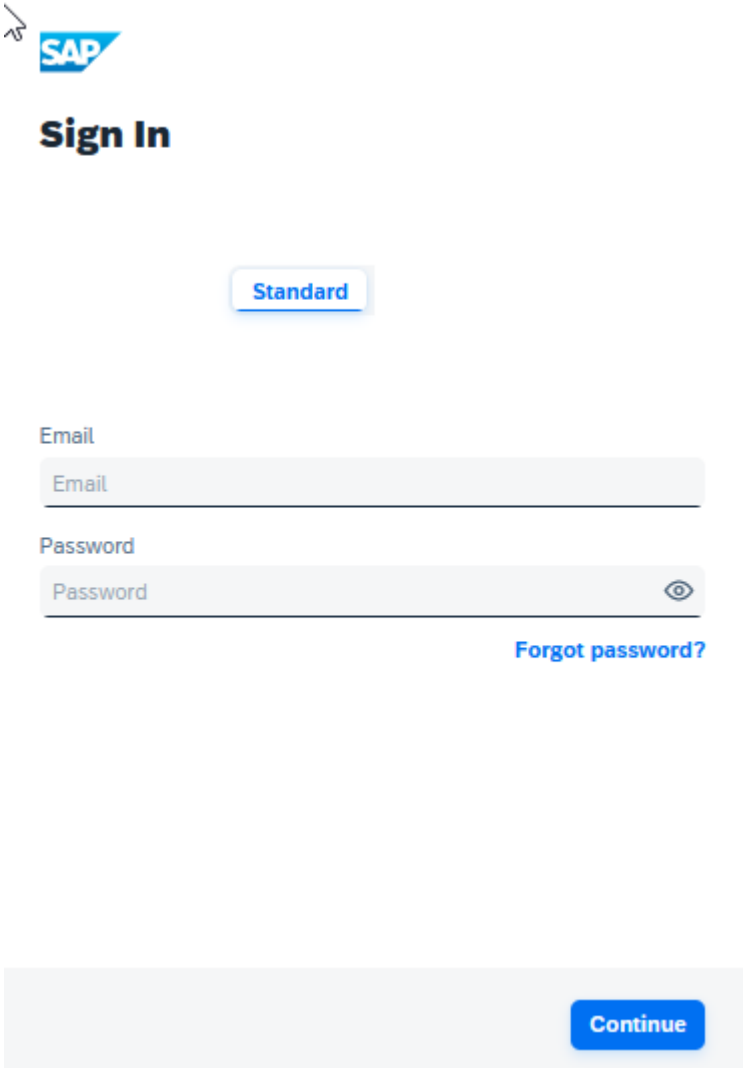
To call the methods of this Change Tenant Texts REST API you must have a system as administrator (technical user) with an assigned *Manage Tenant Configuration* role. For more information, see Add System as Administrator [page 2305], and Edit Administrator Authorizations [page 2309].

Usage

The predefined tenant texts are stored in the Tenant Texts [page 2835] document. The document contains configurable parameters stored as key value pairs of strings. Each key stores the name of a parameter, and the corresponding value is the text that can be changed and updated. The keys are self-explanatory and show where the texts are used. For example, the `logon.ui.label.user=Email or User Name` key value pair is for the type of information that the user must provide to sign in to the application. On the *Sign In* screen you see:



If you want to change the user identifier text you have to update the key value pair `logon.ui.label.user=Email or User Name` to `logon.ui.label.user=Email` to receive:



> **Note**
>
> For some tenants some changed predefined changed texts may exist. If your tenant contains predefined changed texts, you can:
>
> - Use the **GET** method to obtain the texts that overwrote part of the predefined tenant texts when your custom tenant was created, change the texts that you want, add them to the **POST** request and upload them.
> - Use the **GET** method to obtain the texts that overwrote part of the predefined tenant texts when your custom tenant was created, delete a key value pair, add the texts without this line to the **POST** request and upload them. This will replace the deleted key value pair with the predefined one.
> - Open the Tenant Texts [page 2835] document, use the **GET** method to obtain the texts that overwrote part of the predefined tenant texts when your custom tenant was created, from the Tenant Texts [page 2835] document copy the key value pairs that were not included in the response, change the texts in the copied key value pairs, add these new key value pairs to the **POST** request, and execute it.
> - Open the Tenant Texts [page 2835] document, copy the texts to the **POST** request, and execute it. This will replace all texts in the tenant with the predefined ones.

→ Remember

If there are no predefined changed texts for your tenant, the GET Tenant Texts [page 2830] returns the *404 Not Found* code. In this case, proceed with the POST Tenant Texts [page 2833] method.

Methods

HTTP Method	See	URI
GET	GET Tenant Texts [page 2830]	`https://<tenant ID>.accounts.ondemand.com/service/resource?resourceType=RESOURCE_I18N_BUNDLE&locale=<value>`
POST	POST Tenant Texts [page 2833]	`https://<tenant ID>.accounts.ondemand.com/service/resource/SAP_DEFAULT`

ⓘ Note

Tenant ID is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with an URL in it. This URL contains the *tenant ID*.

Parent topic: API Documentation [page 2790]

Related Information

Application Configurations API [page 2791]
Identity Directory API [page 2792]
Identity Provisioning API [page 2795]
Identity Authentication API (Deprecated) [page 2796]
Migrating Deprecated Identity Authentication API to Identity Directory API [page 2797]
Invitation REST API [page 2802]
User Management REST API [page 2805]
Password Service REST API [page 2819]
Forgot Password REST API [page 2821]
TOTP Validation Service [page 2824]
Change Master Data Texts REST API [page 2860]

1.9.2.11.1 GET Tenant Texts

For tenants that contain predefined changed texts, download the texts that overwrote part of the predefined tenant texts when your custom tenant was created.

Request

URI: `https://<Cloud Identity Services domain>/service/resource?resourceType=RESOURCE_I18N_BUNDLE&locale=<value>`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *GET*

Content-Type: application/json

Permissions: You must have a system as administrator with an assigned *Manage Tenant Configuration* role. For more details about how to add a system as administrator and assign administrator roles, see Add System as Administrator [page 2305], and Edit Administrator Authorizations [page 2309].

URL Parameters

Parameter	Required	Description	Notes
`setId`	No	The identifier of the scenario that the resource is related to.	The default value is `SAP_DEFAULT`
`resourceType`	Yes	The type of the resource.	Use `RESOURCE_I18N_BUNDLE`

Parameter	Required	Description	Notes
`locale`	Yes	The locale of the resource.	The end user screens of the applications that use Cloud Identity Services for authentication, support the following languages: Arabic (ar), Bulgarian (bg), Catalan (ca), Chinese PRC (zh_CN), Chinese Taiwan (zh_TW), Croatian (hr), Czech (cs), Danish (da), Dutch (nl), English United Kingdom (en_GB), English United States (en_US), Estonian (et), Finnish (fi), French Standard (fr), French Canada (fr_CA), German Standard (de), Greek (el), Hebrew (iw), Hungarian (hu), Italian (it), Japanese (ja), Korean (ko), Latvian (lv), Lithuanian (lt), Malay (s), Norwegian (no), Polish (pl), Portuguese Portugal (pt), Romanian (ro), Russian (ru), Serbian (sr), Slovak (sk), Slovenian (sl), Spanish Standard (es), Spanish Mexico (es_MX), Swedish (sv), Thai (th), Turkish (tr), Ukrainian (uk), Vietnamese (vi), Welsh (cy).

Request Example

```
GET /service/resource?resourceType=RESOURCE_I18N_BUNDLE&locale=en
```

Response

Response Status and Error Codes

Success Codes

Code	Meaning	Description
200 OK	The request was successful.	OK

Error Codes

Response Code	Meaning	Description
400	Bad Request	Validate JSON data. Validate that the special characters are escaped properly and new lines are added.

Response Code	Meaning	Description
401	Unauthorized	The client is not authenticated.
403	Forbidden	Access to the resource is denied.
404	Not Found	The requested resource cannot be found.
405	Method Not Allowed	The requested method is not supported for the given resource.
415	Unsupported Media Type	The REST service does not support the API version requested by the REST client.
500	Internal Server Error	The operation cannot be completed due to a service error.
503	Service Unavailable	The service is currently unavailable.

Response Example

```
registerSuccess.thankyou=Thank you for registering with {0}
registerForm.validation.accept.privacy.statement=To accept the privacy
statement, click the box above
registerSuccess.activateAccount=To activate your account, click the link
contained in the e-mail. Note that it might take a few minutes for the e-mail
to reach your inbox.
accountActivation.ui.label.activationsuccesstext=Thank you for registering and
activating your account
accountActivation.ui.label.activationtext=This account can also be used to
access other applications
createForgottenPasswordMail.user=E-Mail
logon.ui.label.user=E-Mail
profileManagement.socialSignOn.unlinkWarning=Choose "Unlink" to remove your
social logon information from your account. You can re-link your account the
next time you log on with a social sign-on-enabled site.
accountActivation.ui.label.notlinkedtext=Your account does not currently use
social sign-on. To link your account with a social sign-on account, click one
of the social network buttons next time you log on.
profileManagement.socialSignOn.nolinktext=Your account does not currently use
social sign-on. To link your account with a social sign-on provider, click one
of the social network buttons next time you log on.
linkSocialSignOnAccount.headline=Link your account with {0}
linkSocialSignOnAccount.message.profile=Profile Page
linkSocialSignOnAccount.message=As a final step, you can either link an existing
account to your {0} account, or create a new account that will be linked to your
{0} account.
logon.ui.tooltip.SOCIAL_NOT_ALLOWED=The site you are attempting to access
requires your password.  Enter your password and click Continue.
oauth.network=The selected social sign-on service is currently unavailable. Use
a different social sign-on or your credentials to log on.
logon.ui.tooltip.MISSING_GOOGLE_PLUS_PROFILE=This Google account has not joined
Google yet. Join Google and link your account, or log on with an existing
account.
general.auth.request.error=Identity Provider could not process the
authentication request received. Delete your browser cache and stored cookies,
and restart your browser. If you still experience issues after doing this,
please contact your administrator.
```

1.9.2.11.2 POST Tenant Texts

Update the tenant texts.

> ⚠ Caution
>
> Make sure that you have applied the proper escaping in your **POST** request.
>
> You should add *\n* at the end of each line in the data property content.
>
> Escape Sequences

Value	Escape Sequences
\b	Backspace (ascii code 08)
\f	Form feed (ascii code 0C)
\n	New line
\r	Carriage return
\t	Tab
\"	Double quote
\\	Backslash character

> → Tip
>
> Instead of editing the request manually, you can use an on-line tool for converting a normal string into a quoted one.

Request

URI: `https://<Cloud Identity Services domain>/service/resource/SAP_DEFAULT`

> ⓘ Note
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: POST

Content-Type: application/json

Permissions: You must have a system as administrator with an assigned *Manage Tenant Configuration* role. For more details about how to add a system as administrator and assign administrator roles, see Add System as Administrator [page 2305], and Edit Administrator Authorizations [page 2309].

Request Example

```
[{
"resourceType": "RESOURCE_I18N_BUNDLE",
"locale": "en",
"contentType": "text/html;charset=UTF-8",
"data": "registerSuccess.thankyou=Thank you for registering with {0}\n
registerForm.validation.accept.privacy.statement=To accept the privacy
statement, click the box above\n
registerSuccess.activateAccount=To activate your account, click the link
contained in the e-mail. Note that it might take a few minutes for the e-mail
to reach your inbox.\n
accountActivation.ui.label.activationsuccesstext=Thank you for registering and
activating your account\n
accountActivation.ui.label.activationtext=\n
createForgottenPasswordMail.user=E-Mail\n
logon.ui.label.user=E-Mail\n
logon.ui.label.forgotpassword=Forgot/Reset Password\n
profileManagement.socialSignOn.unlinkWarning=Choose \"Unlink\" to remove your
social logon information from your account. You can re-link your account the
next time you log on with a social sign-on-enabled site.\n
accountActivation.ui.label.notlinkedtext=Your account does not currently use
social sign-on. To link your account with a social sign-on account, click one
of the social network buttons next time you log on.\n
profileManagement.socialSignOn.nolinktext=Your account does not currently use
social sign-on. To link your account with a social sign-on provider, click one
of the social network buttons next time you log on.\n
linkSocialSignOnAccount.headline=Link your account with {0}\n
linkSocialSignOnAccount.message.profile=Profile Page\n
linkSocialSignOnAccount.message=As a final step, you can either link an existing
account to your {0} account, or create a new account that will be linked to your
{0} account.\n
logon.ui.tooltip.SOCIAL_NOT_ALLOWED=The site you are attempting to access
requires your password.  Enter your password and click Continue.\n
oauth.network=The selected social sign-on service is currently unavailable. Use
a different social sign-on or your credentials to log on.\n
logon.ui.tooltip.MISSING_GOOGLE_PLUS_PROFILE=This Google account has not joined
Google yet. Join Google and link your account, or log on with an existing
account.\n
general.auth.request.error=Identity Provider could not process the
authentication request received. Delete your browser cache and stored cookies,
and restart your browser. If you still experience issues after doing this,
please contact your administrator.\n"
}]
```

Response

Response Status and Error Codes

Success Codes

Code	Meaning	Description
200 OK	The request was successful.	OK

Error Codes

Response Code	Meaning	Description
400	Bad Request	Validate JSON data. Validate that the special characters are escaped properly and new lines are added.
401	Unauthorized	The client is not authenticated.
403	Forbidden	Access to the resource is denied.
404	Not Found	The requested resource cannot be found.
405	Method Not Allowed	The requested method is not supported for the given resource.
415	Unsupported Media Type	The REST service does not support the API version requested by the REST client.
500	Internal Server Error	The operation cannot be completed due to a service error.
503	Service Unavailable	The service is currently unavailable.

1.9.2.11.3 Tenant Texts

Predefined tenant texts for Identity Authentication

Tenant Texts Key Value Pairs

```
sap.id.service=Identity Authentication
logon.ui.label.register=Register
logon.ui.label.login=Log On
logon.ui.label.signin=Sign In
logon.ui.label.logout=Log Out
logon.ui.label.linkAccounts=Link Accounts
logon.ui.label.continue=Continue
logon.ui.label.user=Email or User Name
logon.ui.label.userPhone=Email or User Name or Phone
logon.ui.label.emailPhone=Email or Phone
logon.ui.label.fioritheme.user=Email or User Name
logon.ui.label.email=Email
logon.ui.label.change.user=Change
logon.ui.label.password=Password
logon.ui.label.tooltip.hide=Hide Password
logon.ui.label.tooltip.show=Show Password
logon.ui.label.otpCode=Passcode
logon.ui.label.emailCode=Email Code
logon.ui.label.returnToLogonTitle=Get access another way?
logon.ui.label.forgotpassword=Forgot password?
logon.ui.label.returnToLogonLink=Return to Sign In
logon.ui.label.forgotpasswordLink=Forgot Password
logon.ui.label.rememberme=Remember me
logon.ui.label.keepMeSignedIn=Keep me signed in
logon.ui.label.remembermeConsent=Your credentials will be stored on this device
for a long period of time
```

```
logon.ui.label.trustbrowser=Trust this browser
logon.ui.label.notRegisteredYet=Not registered yet?
logon.ui.label.registerNow=Register Now
logon.ui.label.twoFactorAuthenticationTitle=Two-Factor Authentication
logon.ui.label.twoFactorChoiceTitle=Choose Two-Factor Authentication Method
logon.ui.label.twoFactorAuthenticationDescription=To proceed, please enter the
time-based passcode generated by your mobile device.
logon.ui.label.twoFactorAuthenticationEnabledDescription=The {0} application
requires a time-based one-time passcode as a second factor for authentication.
You need to activate a mobile device to generate passcodes. No devices are
currently activated.
logon.ui.label.rsaTwoFactorAuthenticationDescription=To proceed, please enter
the code generated by your RADIUS Server token. Enter the code according to the
token type requirements.
logon.ui.label.emailTwoFactorAuthenticationDescription=We have sent you a code
by email. Please enter this code and choose Continue.
logon.ui.label.smsTwoFactorAuthenticationDescription=The {0} application
requires telephone verification. We will send a code to your telephone number.
Please enter the code you have received and choose Continue.
logon.ui.label.smsTwoFactorAuthenticationVerifyPhoneDescription=The {0}
application requires telephone verification. Please enter your telephone number
and request a code.
logon.ui.label.twoFactorChoiceDescription=Please choose one of the enabled two-
factor authentication methods or press the button to enable a new method.
logon.ui.label.twoFactorChoiceAllMethodsEnabledDescription=Please choose a
method and follow the instructions.
logon.ui.label.twoFactorChoiceSelectMethodForVerification=To continue, one of
your active two-factor authentication methods must be validated. Please choose
a method and follow the instructions.
logon.ui.label.twoFactorChoiceSelectMethodForActivation=Please choose a two-
factor authentication method to enable it.
logon.ui.label.twoFactorChoiceTOTP=TOTP Two-Factor Authentication
logon.ui.label.twoFactorChoiceRSA=RADIUS Server Two-Factor Authentication
logon.ui.label.twoFactorChoiceWEB=WEB Two-Factor Authentication
logon.ui.label.twoFactorChoiceSMS=SMS Two-Factor Authentication
logon.ui.label.twoFactorChoiceEMAIL=Email Two-Factor Authentication
logon.ui.tfa.choice.button.activate.device=Enable Method
logon.ui.tfa.choice.skip.button=Skip
logon.ui.tfa.choice.skip.warningMessage=You can skip enabling the second factor
for %{param0} days.
logon.ui.tfa.choice.skip.warningMessageSingular=You can skip enabling the second
factor for 1 day.
logon.ui.tfa.back.button=Back
logon.ui.label.otpTwoFactorAuthenticationActivateDeviceRequired=An authenticator
app is required in order to enable two-factor authentication.
logon.ui.label.otpTwoFactorAuthenticationActivateDeviceStepOne=Scan QR Code
logon.ui.label.otpTwoFactorAuthenticationActivateDeviceStepTwo=Enter passcode
logon.ui.label.otpTwoFactorAuthenticationActivateDeviceUseSecurityKey=No QR
scanner? Use this security key instead.
logon.ui.label.otpTwoFactorAuthenticationActivateDeviceEnterCodeMessage=The
Authenticator app will generate a passcode which you need to enter in the field
below. Note that the passcode is valid for several seconds.
logon.ui.label.web.activateTwoFactorAuthenticationDescription=There are no web
authentication devices registered for "{0}". Choose "Activate" to register a new
one.
logon.ui.label.web.activateTwoFactorAuthenticationButton=Activate
logon.ui.label.web.verifyTwoFactorAuthenticationDescription=Use your device to
continue logon as "{0}".
logon.ui.label.web.verifyTwoFactorAuthenticationButton=Verify
logon.ui.label.web.unsupported=Your browser does not support web authentication.
logon.ui.label.passcodeAccepted=Passcode accepted. Enter your logon password.
logon.ui.label.newUserRegisterWithSocialSignOn=Register Now
logon.ui.label.verifyTelephoneTitle=Verify Your Telephone Number
logon.ui.label.verifyTelephoneDescription=The {0} application requires telephone
verification. We will send a code to your telephone number. Please enter the
code you have received and choose Continue.
```

```
logon.ui.label.verifyTelephoneCaptchaDescription=The {0} application requires
telephone verification. If CAPTCHA is required, please verify it and choose {1}.
Enter the code you receive in the field below and choose {2}.
logon.ui.label.changeTelephoneMessage=You can change your telephone number.
Please type: +(country code) followed by the area code without the leading zero,
followed by the subscriber number. Example: +(XXX) XXX XXXXXXX
logon.ui.label.telephone=Telephone
logon.ui.label.code=Code
logon.ui.label.sendNewCode=Request New Code
logon.ui.label.sendCode=Request Code
logon.ui.label.callPhone=Request Phone Call
logon.ui.label.logonToLinkWithSocialSignOn=Enter your logon details and choose
Link Accounts.  Alternatively, to create a new account that is linked to your
{0} account, click the Register Now link below.
logon.ui.label.linkAccountWithTwitter=Log On with X
logon.ui.label.linkAccountWithFacebook=Log On with Facebook
logon.ui.label.linkAccountWithGoogle=Log On with Google
logon.ui.label.linkAccountWithLinkedIn=Log On with LinkedIn
logon.ui.label.linkAccountWithApple=Log On with Apple
logon.ui.label.or=or
logon.ui.label.orSocialSignIn=or Sign In with
logon.ui.label.dontHaveAccount=Don't have an account?
logon.ui.label.notReceivingText=Not receiving texts?
logon.ui.loginSection.button.standard=Standard
logon.ui.loginSection.button.biometric=Biometric
logon.ui.loginSection.button.social=Social
logon.ui.loginSection.passwordlessAuth.title=To sign in with biometrics, choose
Continue.
logon.ui.messageStrip.closeButton.error=Close error message
logon.ui.messageStrip.closeButton.info=Close information message
logon.ui.messageStrip.closeButton.warning=Close warning message
logon.ui.messageStrip.closeButton.success=Close success message
logon.ui.messageStrip.closeButton.default=Close message
logon.ui.tooltip.SOCIAL_NOT_ALLOWED=The site you are attempting to access
requires your password.  Enter your password and click Continue.
logon.ui.tooltip.MISSING_GOOGLE_PLUS_PROFILE=This Google account has not joined
Google yet. Join Google and link your account, or log on with an existing
account.
logon.ui.tooltip.otpTwoFactorAuthenticationActivateDeviceStepTwo=Once you have
scanned or entered the key, enter the passcode generated by the authenticator
app on your device below and click "Continue".
logon.ui.errormessage.EXPIRED_INITIAL_PASSWORD=Your initial password has
expired. Please contact your system administrator.
logon.ui.errormessage.BAD_CREDENTIALS=Sorry, we could not authenticate you. Try
again.
logon.ui.errormessage.ACCOUNT_LOCKED=Sorry, your account is currently locked and
cannot be used to log on.
logon.ui.errormessage.NOT_AUTHORIZED=Sorry, but you are currently not authorized
for access
logon.ui.errormessage.AUTHORIZATION_CONFLICT=You are trying to log on with an
account other than the one you are currently logged on with. You can either stay
logged on with your current account and re-authenticate to your application,
or log off and authenticate with the other account. If you choose the second
option, you have to log off manually and then log on to the application with the
new account.
logon.ui.errormessage.GBB451_GEO_IP_POLICY_SUSPEND_USER_ERROR=Your login has
been denied due to attempted access from a country or region where this product
or service is not available. Any attempt to circumvent this restriction could
result in your account being permanently disabled.
logon.ui.errormessage.GBC451_GEO_IP_POLICY_RESUSPEND_USER_ERROR=Your login has
been denied due to attempted access from a country or region where this product
or service is not available. Any attempt to circumvent this restriction could
result in your account being permanently disabled.
logon.ui.errormessage.GBA503_GEO_IP_POLICY_UNAVAILABLE_ERROR=Service is
unavailable. We apologize for the inconvenience. We are actively working to
resolve this issue. Please try again later.
logon.ui.errormessage.TECHNICAL_ERROR=An error occurred, try again.
```

```
logon.ui.errormessage.INVALID_CERTIFICATE=Sorry, the certificate you have used
to log on is not valid.
logon.ui.errormessage.INVALID_TICKET=Sorry, the link you have clicked is no
longer valid.
logon.ui.errormessage.PASSWORD_LOCKED=Sorry, your account was temporarily locked
because of too many failed attempts to log on. Your account will be
automatically unlocked in {0} minutes.
logon.ui.errormessage.GENERIC=An error occurred. Try again.
logon.ui.errormessage.LOCKED_OTP_CODE=Sorry, two-factor authentication is
temporarily locked because of too many failed logon attempts. It will be
automatically unlocked in 60 minutes.
logon.ui.errormessage.INVALID_OTP_CODE=Wrong passcode; enter the passcode again
logon.ui.errormessage.INVALID_PASSWORD=Wrong password; enter the password again
logon.ui.errormessage.ALREADY_SEND_EMAIL_OTP_CODE=Please wait for three minutes
from the last request before you request a new code.
logon.ui.errormessage.LOCKED_EMAIL_OTP_CODE=Sorry, email two-factor
authentication is temporarily locked because of too many failed logon attempts.
It will be automatically unlocked in 60 minutes.
logon.ui.errormessage.INVALID_EMAIL_OTP_CODE=You have entered the wrong code or
a code which is not valid any more. Please try again or request a new code.
logon.ui.errormessage.NO_EMAIL_OTP_CODE_TEMPLATE=Sorry, but sending a code via
email is not possible right now. Please contact your administrator before
continuing.
logon.ui.errormessage.UNVERIFIED_EMAIL_FOR_OTP_CODE=We did not send you a code
because your email has not been verified. Please contact your administrator for
assistance.
logon.ui.errormessage.PASSCODE_LOCKED=Your account has been temporarily locked
because of too many failed attempts. Your account will be automatically unlocked
in 10 minutes.
logon.ui.errormessage.INVALID_CAPTCHA_PROTECTION=We are unable to validate your
Captcha protection at the moment. Please try again later.
logon.ui.errormessage.SESSION_EXPIRED=Your session has expired. Please sign in
again.
logon.ui.errormessage.RSA_CODE_NOT_PROVIDED=RADIUS Server code not provided.
Please enter your code.
logon.ui.errormessage.RSA_CODE_REJECTED=Wrong RADIUS Server code. Please enter
new code and try again.
logon.ui.errormessage.SECURE_ID_SERVER_TIMEOUT=A problem has occurred. Please
try again later or contact your system administrator.
logon.ui.errormessage.SECURE_ID_SERVER_COMMUNICATION_ERROR=A problem has
occurred. Please try again later or contact your system administrator.
logon.ui.errormessage.SECURE_ID_SERVER_NOT_CONFIGURED=The SecurID server is not
configured. Please contact your system administrator.
logon.ui.errormessage.WRONG_USERNAME_PASSCODE=Wrong user name or passcode
logon.ui.errormessage.MAX_SENT_REACHED_SMS_CODE=We did not send you a code
because you have reached the limit of 5 SMS codes. Please request a phone call
or contact your system administrator.
logon.ui.errormessage.SMS_CODE_NOT_SENT_DUPLICATE=Your telephone number has
already been verified by another user. Please change your telephone number or
contact your system administrator.
logon.ui.errormessage.SMS_CODE_NOT_SENT_NOT_UNIQUE_PHONE=We could
no\u0080\u0099t send you a code because this telephone number is already used
as a user identifier by someone else. Please use another telephone number, or
contact your administrator.
logon.ui.errormessage.SMS_SERVER_NOT_CONFIGURED=SMS service configuration is
missing. Please contact your system administrator.
logon.ui.errormessage.INVALID_PHONE_NUMBER=Your telephone number is invalid.
Please change and try again.
logon.ui.errormessage.INVALID_PHONE_CAPTCHA=The captcha you entered is
incorrect. Please try again.
logon.ui.errormessage.FAILED_SEND_SMS_CODE=We couldn't send you a code. Please
check your telephone number and try again.
logon.ui.errormessage.PHONE_CALL_NOT_ALLOWED=We did not send you a code because
you are not allowed to request phone calls. Please contact your system
administrator.
logon.ui.errormessage.MAX_SENT_PHONE_CALL_CODE=We did not send you a code
because you have reached the limit of 5 SMS codes and 5 phone calls. Please
contact your system administrator.
```

```
logon.ui.errormessage.FAILED_VALIDATE_SMS_CODE=We couldn't validate your code.
Please try again.
logon.ui.errormessage.INVALID_SMS_CODE=You have entered the wrong code or a code
which is not valid any more. Please try again or request a new code.
logon.ui.errormessage.INVALID_SMS_CODE_MAX_ATTEMPTS=You have entered a wrong
code. You have no more attempts for this code. Please choose New Code to receive
a new one.
logon.ui.infomessage.NEW_SMS_CODE_SENT=We have sent you a new code. The code
will be valid for {0} minutes.
logon.ui.infomessage.NEW_EMAIL_OTP_CODE_SENT=We have sent you a new code. The
code will be valid for 10 minutes.
logon.ui.infomessage.NEW_PHONE_CALL_CODE_SENT=We have triggered a call to your
phone to provide you with a code. The code will be valid for {0} minutes.
logon.ui.errormessage.SMS_TFA_SEND_CODE_LIMIT_REACHED=The {0} application
requires telephone verification. We did not send you a code because you have
reached the limit of 5 codes. The SMS code request service has been locked for
one hour. Please wait until the service is unlocked and try again.
logon.ui.errormessage.SMS_TFA_INVALID_PHONE_NUMBER=The {0} application requires
telephone verification. Your telephone number is not valid. Please contact your
administrator.
logon.ui.errormessage.SMS_TFA_NO_PHONE_AVAILABLE=The {0} application requires
telephone verification. Your telephone number is not in our system. Please
contact your administrator.
logon.ui.errormessage.SMS_TFA_PHONE_ASSIGNED_TO_MULTIPLE_USERS=The {0}
application requires telephone verification. Your telephone number has already
been verified by another user. Please contact your administrator.
logon.ui.infomessage.NEW_MAIL_CODE_SENT=We have sent you a new code by email.
logon.ui.errormessage.FAILED_SEND_MAIL_CODE=We couldn't send you a code. Please
check your email and try again.
logon.ui.errormessage.USER_TIMESTAMP_INACTIVE_UNSTARTED=Your user will be valid
from {0}. For assistance, please contact your administrator.
logon.ui.errormessage.USER_TIMESTAMP_INACTIVE_EXPIRED=Your user was valid until
{0}. For assistance, please contact your administrator.
logon.ui.errormessage.CONFLICTING_USER_MAIL=You cannot log on. Your email
address is associated with another account. Please use the other account or
contact your system administrator.
logon.ui.securityQuestion.title=Security Questions
logon.ui.securityQuestion.hint=Follow the link to reset your password using
logon.ui.securityQuestion.link=Security Questions
logon.ui.securityQuestion.description=To reset your password provide the
required information and answers below.
logon.ui.securityQuestion.select=Choose a question
logon.ui.securityQuestion.question=Question %{param0}
logon.ui.securityQuestion.error=You have provided a wrong combination of user,
questions, and answers. Please try again.
logon.ui.passwordResetPin.title=PIN Code for Password Reset
logon.ui.passwordResetPin.link=PIN Code
login.ui.forgotPassword.create.or=or
logon.ui.passwordResetPin.error=You have provided a wrong combination of user
and PIN. Please try again.
logon.ui.passwordResetPin.description=To reset your password provide the
required information.
logon.ui.copyright.link=Default Copyright Link
logon.ui.privacy.link=Default Privacy Link
logon.ui.legal.disclosure.link=Default Legal Disclosure Link
logon.ui.legal.cookie.link=Default Cookie Notice Link
logon.ui.login.instructions=Default Login Instructions
salutation.ui.welcome=Welcome
accountActivation.accountSuccessfullyActivated=Account Successfully Activated
accountActivation.accountSuccessfullyLinked=Account Successfully Linked
accountActivation.accountSuccessfullyLinkedWith=Account Linked to {0}
accountActivation.ui.label.activationsuccesstext=Thank you for registering and
activating your account {0}
accountActivation.ui.label.activation.error.text.user.inactive=Your user account
is inactive. For more information, please contact your administrator.
accountActivation.ui.label.linkingtext=Your accounts are now linked and {0} log
on is enabled for supported sites.
```

```
accountActivation.ui.label.fioritheme.linkingtext=Your accounts are now linked
and {0} sign-in is enabled for supported sites.
accountActivation.ui.label.continue=Continue
accountActivation.mail.subject=Identity Authentication Account Activation
invitation.mail.subject=You are invited to {0}
tou.termsAndConditions=Terms and Conditions
tou.termsOfUse=Terms of Use
tou.privacyPolicy=Privacy Policy
tou.ui.label.version=Version {0}
tou.ui.label.confirmationtext=I have read and understood the <a target\="_blank"
id\="touLink" href\="{1}">Terms and Conditions</a> of {0}.
tou.ui.label.accept=Accept
tou.ui.message.scrollToBottom=Scroll to the bottom of the terms of use and click
the box to accept.
securityQuestions.ui.label.description=Set up security questions for your
profile. Choose a security question and provide an answer to it.
passwordResetPin.ui.label.description=Set up PIN for password reset for your
profile.
password.recovery.ui.label.setUpLater=Set Up Later
password.recovery.ui.label.dontAskAgain=Don't ask me again
onbehalfupdate.ui.label.headline=Activate Your Account
onbehalfupdate.ui.label.message1=An account has been created for you with
Identity Authentication for use with {0}. The account information we already
have for you is below.
onbehalfupdate.ui.label.message2=To begin using your account for {0}, set a
password below.
onbehalfupdate.ui.label.firstname=First Name
onbehalfupdate.ui.label.lastname=Last Name
onbehalfupdate.ui.label.email=Email
onbehalfupdate.ui.label.copyright=Copyright
onbehalfupdate.ui.label.privacy=Privacy
onbehalfupdate.ui.label.legaldisclosure=Legal Disclosure
onbehalfupdate.ui.errormessage.retypeDoesNotMatch=Entered passwords do not
match; enter passwords again
onbehalfupdate.ui.errormessage.insufficientPasswordComplexity=Insufficient
password complexity; check password requirements
onbehalfupdate.ui.errormessage.passwordCouldNotBeValidated=Could not update user
forgottenpassword.resetPassword=Reset Your Password
forgottenpassword.resetPassword.text=Reset the password for your account {0}:
forgottenpassword.typeRetypePassword=Enter and re-enter your new password.
changepassword.changePassword=Change Your Password
changepassword.typeRetypePassword=Enter and re-enter your new password.
changepassword.ui.infomessage.insufficientPasswordComplexity=Your password is
not strong enough.
changepassword.ui.infomessage.passwordLocked=Your password has been locked.
changepassword.ui.infomessage.passwordTooNew=Your password can only be changed
once every {0} hours and has already been changed during this period.
changepassword.ui.infomessage.passwordTooOld=Your password is more than {0}
months old.
changepassword.ui.infomessage.passwordNotUsed=You have not logged on with your
password during the last {0} months.
forgottenpassword.validation.passwordNotComplexEnough=Password must comply with
the requirements
forgottenpassword.validation.passwordsDoNotMatch=Must match the password
forgottenpassword.validation.invalidInput=Invalid input
forgottenpassword.ui.errormessage.emailNotFound=No account found for this email
address.
forgottenpassword.ui.errormessage.accountNotFound=No account found for the data
you entered.
forgottenpassword.ui.errormessage.forbidden=Passwords can only be changed once
every {0} hours.
forgottenpassword.ui.errormessage.couldNotBeValidated=Could not validate
password.
forgottenpassword.ui.errormessage.passwordInHistory=This password is already in
your password history. Choose a different password.
forgottenPasswordMailConfirm.forgotPasswordConfirm=Confirm Forgotten Password
forgottenPasswordMailConfirm.forgotYourPassword=Forgot My Password
```

```
forgottenPasswordMailConfirm.emailSent=If an account exists, an email with a
link to reset your password has been sent.
forgottenPasswordMailConfirm.clickTheLink.days=Click the link in the email. You
will be forwarded to a page where you can reset your password. The link in the
email will expire in {0} days.
forgottenPasswordMailConfirm.clickTheLink.hours=Click the link in the email. You
will be forwarded to a page where you can reset your password. The link in the
email will expire in {0} hours.
forgottenPasswordMailConfirm.closeButton=Close
informAboutPasswordStatus.infoPasswordStatusPage=Password Status Page
informAboutPasswordStatus.passwordNeedsReset=Your Password Needs to Be Reset
informAboutPasswordStatus.emailSent=An email with a link to reset your password
has been sent to {0}.
informAboutPasswordStatus.text=Your password has not been used for more than {0}
months. To help maintain your security, your password needs to be reset. We have
sent you an email with a link to a page where you can reset your password. The
link in the email expires on {0}.
informAboutPasswordStatus.reason.passwordNotUsed=Your password has not been used
for more than {0} months.
informAboutPasswordStatus.reason.insufficientComplexity=Your password does not
meet the password complexity requirements.
informAboutPasswordStatus.maintainSecurity.days=To help maintain your security,
your password needs to be reset. We have sent an email to your email address
containing a link to a page where you can reset your password. The link will
expire in {0} days.
informAboutPasswordStatus.maintainSecurity.hours=To help maintain your security,
your password needs to be reset. We have sent an email to your email address
containing a link to a page where you can reset your password. The link will
expire in {0} hours.
passwordForm.invalidInput=Invalid input
passwordForm.required=Required
passwordForm.save=Save
policyaccept.acceptPolicy=Accept Policy
policyaccept.ui.label.headline=Privacy policy for {0}
policyaccept.ui.label.accessMessage=To access and use {0}, read and accept the
privacy policy below.
policyaccept.ui.label.version=Version {0}
policyaccept.ui.label.required=Required
policyaccept.ui.label.accept=Accept
policyaccept.ui.label.decline=Decline
policyaccept.ui.errormessage.accepttoproceed=To proceed, accept the privacy
policy.
policyaccept.ui.errormessage.markcheckbox=To accept the privacy policy, click
the checkbox.
document.ui.label.headline=Legal Disclaimers
document.ui.label.accessUse=We have updated our legal documents. To access and
use {0}, please read and accept the documents below:
privacy.link.text=Familiarize yourself with our <a target\="_blank"
href\="{0}">Privacy Statement</a>.
privacy.link.updatedText=Note: There is a new version of our privacy statement
available <a target\="_blank" href\="{0}">here</a>.
privacy.link.ppCheckEnabled=By checking this box, I confirm that I have read
and understood the above <a target\="_blank" id\="pdLink" href\="{0}">Privacy
Statement</a>, which explains how my personal data will be processed in
accordance with applicable data protection laws.
upgraderegistration.upgradeAccount=Upgrade Your Account
upgraderegistration.ui.label.register=Save and Continue
upgraderegistration.ui.label.upgradeAccountFor=Upgrade Your Account for {0}
upgraderegistration.ui.label.upgraderegistrationtext=To use your account with
{0}, provide the required information below.
upgradeRegistrationConfirmation.accountSuccessfullyActivated=Account
Successfully Activated
upgradeRegistrationConfirmation.accountUpgraded=Thank you for registering and
activating your account with SAP Store.
upgradeRegistrationConfirmation.accountUpgraded.text=This account can also be
used to access other SAP platforms such as SAP.com, SAP Community Network, and
SAP Cloud applications.
upgradeRegistrationConfirmation.continue=Continue
```

```
upgradeRegistrationInformMailStatus.mailNotVerifiedStatus=Verify Your Email
upgradeRegistrationInformMailStatus.mailSentTo=A link to verify your email has
been sent to {0}.
upgradeRegistrationInformMailStatus.mailSentTo.text=To verify your email, click
the link sent to you. Note that it might take a few minutes for the email with
the link to reach your inbox.
salutation.mr=Mr.
salutation.mrs=Mrs.
salutation.ms=Ms.
relationshipToSAP.Customer=Customer
relationshipToSAP.prospectiveCustomer=Prospective Customer
relationshipToSAP.partner=Partner
relationshipToSAP.prospectivePartner=Prospective Partner
relationshipToSAP.consultant=Consultant
relationshipToSAP.pressAnalyst=Press/Analyst
relationshipToSAP.investorStakeholder=Investor/Stakeholder
relationshipToSAP.student=Student
relationshipToSAP.employee=SAP Employee
select.number_employees.UP_TO_50=1 - 49
select.number_employees.UP_TO_100=50 - 99
select.number_employees.UP_TO_250=100 - 249
select.number_employees.UP_TO_500=250 - 500
select.number_employees.UP_TO_1000=501 - 999
select.number_employees.MORE_THAN_1000=1000+
registerForm.registration=Registration
registerForm.createAccount=Create Account
registerForm.validation.max.size=The value must not have more than {max}
characters.
registerForm.validation.required=Required field
registerForm.validation.zip.country.required=The zip code can't be validated
because the country is not provided. Please contact your administrator.
registerForm.validation.region.country.required=The region can't be validated
because the country is not provided. Please contact your administrator.
registerForm.validation.invalidMail=Enter a valid email address.
registerForm.validation.invalidTelephone=Enter a valid telephone number, using
only brackets and hyphens to separate digits.
registerForm.validation.notUniqueTelephone=This telephone number is already used
as a user identifier by someone else. Please use another one.
registerForm.validation.duplicateDisplayName=The display name you have chosen is
already in use. Choose a different display name.
registerForm.validation.duplicateUserName=The user name you have chosen is
already in use. Choose a different user name.
registerForm.please_select=Select
registerForm.please_select_special=Select...
registerForm.validation.accept.terms.of.use=To accept the terms of use, check
the box
registerForm.validation.accept.privacy.policy=To accept the privacy policy,
check the box
registerForm.validation.passwordsDoNotMatch=Must match the password
registerForm.validation.passwordNotComplexEnough=The password must comply with
the requirements
registerForm.validation.passwordPolicyLoadError=The required password policy
cannot be loaded. Please contact your system administrator.
registerForm.validation.contactPrefEmail=Do you want to be contacted by email?
registerForm.validation.contactPrefPhone=Do you want to be contacted by phone?
registerForm.validation.invalidPostalCode=ZIP/postal code must match country
registerForm.validation.select_state=Select a state
registerForm.validation.userNameCharRestriction=Please do not use a user ID in
the user name field. The user name must start with lower case characters a-z or
numbers 0-9, and can also contain the following symbols: .-_
registerForm.validation.userNameUniqueRestriction=An account already exists with
this user name
registerForm.validation.userNameConflictWithUsersUserId=Please do not use your
user ID in the user name field
registerForm.ui.label.required=Required
registerForm.ui.label.submit=Register
registerForm.registerForSPinfo=To register for {0}, provide the required
information below.
```

```
registerForm.register=Register
registerForm.tellUsAboutYourself=Tell Us About Yourself
registerForm.salutation=Salutation
registerForm.firstName=First Name
registerForm.middleName=Middle Name
registerForm.lastName=Last Name
registerForm.userName=User Name
registerForm.email=Email
registerForm.phone=Phone
registerForm.chooseUsernamePassword=Set Password
registerForm.loginname=Login Name
registerForm.displayname=Display Name
registerForm.password=Password
registerForm.repeatpassword=Re-Enter Password
registerForm.confirmPassword=Confirm Password
registerForm.passwordHint.musstText=Your password must:
registerForm.passwordHint.length=Your password must be between {0} and {1}
characters long
registerForm.passwordHint.includeText=and include {0} of the following:
registerForm.passwordHint.includeAllText=and include all of the following:
registerForm.passwordHint.uppercaseLetters=Uppercase letters
registerForm.passwordHint.lowercaseLetters=Lowercase letters
registerForm.passwordHint.numbers=Numbers
registerForm.passwordHint.symbols=Symbols
registerForm.tellUsAboutYourCompany=Tell Us About Your Company
registerForm.company=Company
registerForm.streetAddress=Street Address
registerForm.companyStreetAddress=Street Address
registerForm.streetAddress2=Street Address 2
registerForm.companyStreetAddress2=Street Address 2
registerForm.city=City
registerForm.companyCity=City
registerForm.postalcode=ZIP/Postal Code
registerForm.companyPostalcode=ZIP/Postal Code
registerForm.country=Country/Region
registerForm.companyCountry=Country/Region
registerForm.state=State/Province
registerForm.industry=Industry
registerForm.relationshipToSAP=Relationship to SAP
registerForm.jobFunction=Job Function
registerForm.department=Department
registerForm.futureContactPrefs=Contact Preferences
registerForm.futureContactPrefs.text=In addition to communications that will
result from this registration, would you also like to receive news and event
notifications from SAP that are specific to your interests?
registerForm.viaEmail=By email
registerForm.viaTelephone=By telephone
registerForm.yes=Yes
registerForm.no=No
registerForm.privacy.text=I acknowledge that I have read <a
id\="pdLink" href\="http://www.sap.com/about-sap/company/legal/privacy.epx"
target\="_blank">SAP's Privacy Statement</a> and consent to the processing of
my personal data in accordance with the terms of the Privacy Statement. This
includes, without limitation, that SAP may collect, store and process any
personal data voluntarily provided by me on this Web site and aggregate it
with other personal data that I provided to SAP on earlier occasions by
the <a href\="http://www.sap.com/about-sap/company/legal/privacy.epx\#purpose"
target\="_blank">methods and for the purposes</a> described in the Privacy
Statement and on this Web site. In accordance with the terms of the Privacy
Statement and on this Web site, SAP may further track my use of SAP's Web sites
and aggregate it with my personal data. I acknowledge that I can, at any time,
request information on my personal data held by SAP and that I can have SAP
update and correct such data and withdraw my consent given hereby by contacting
SAP at <span class\="safermailto"></span>.
registerForm.privacy.text.enterprise=Familiarize yourself with our <a
id\="pdLink" href\="{0}" target\="_blank"">Privacy Statement</a>.
registerSuccess.thankyou=Thank you for registering with {0}
registerSuccess.thankyou.enterprise=Thank you for registering with {0}
```

```
registerSuccess.emailSent=An email with a link to activate your account has been
sent to {0}
registerSuccess.activateAccount=To activate your account, click the link
contained in the email. Note that it might take a few minutes for the email to
reach your inbox.
registerSuccess.activateAccount.enterprise=To activate your account, click the
link contained in the email. Note that it might take a few minutes for the email
to reach your inbox.
validation.passwordInExcludeList=You have chosen a password that is not allowed.
Please choose a different password and try again.
validation.firstNameInExcludeList=You have chosen a first name that is not
allowed. Please choose a different first name and try again.
validation.lastNameInExcludeList=You have chosen a last name that is not
allowed. Please choose a different last name and try again.
validation.emailInExcludeList=You have chosen an email address that is not
allowed. Please choose a different one and try again.
general.ui.help.title=Help for
ResourceType.RESOURCE_TERMS_OF_USE=Terms Of Use
ResourceType.RESOURCE_TERMS_OF_USE.for=Terms of Use for {0}
ResourceType.RESOURCE_PRIVACYPOLICY=Privacy Policy
ResourceType.RESOURCE_PRIVACYPOLICY.for=Privacy Policy for {0}
createForgottenPasswordMail.forgotPassword=Forgot My Password
createForgottenPasswordMail.forgotPassword.text=Enter your credentials below and
click Send. An email will be sent containing a link to a page where you can
reset your password. Note that the email might take a few minutes to reach your
inbox.
createForgottenPasswordMail.forgotPassword.fioritheme.text=To reset your
password, enter your email address and choose Continue.
createForgottenPasswordMail.user=Email or User Name
createForgottenPasswordMail.email=Email
createForgottenPasswordMail.send=Send
createForgottenPasswordMail.continue=Continue
errorPage.errorPage=Error Page
errorPage.errorTitle=Error
errorPage.warningTitle=Warning
errorPage.error={0}
errorPage.errorId=Error ID: {0}
errorPage.errorMessage=An error occurred. If you need support, send an email
to <span class\="safermailto" data-safermailto-recipient\="sso" data-safermailto-
subject\="Identity Authentication Error"></span>
errorPage.troubleshootingTitle=Troubleshooting
errorPage.troubleshootingMessage=You can provide the following correlation ID to
your administrator: %{param0}
oauth.network=The selected social sign-on service is currently unavailable. Use
a different social sign-on or your credentials to log on.
error.oauth.social.sign-on.failed.error=Social sign-on failed. Please try again.
error.oauth.signing.process.failed.unapproved.callback.url.error=Social sign-on
failed due to unapproved callback URL in the social provider application. Please
contact your system administrator.
error.oauth.signing.process.failed.wrong.client.secret.error=Social sign-on
failed due to wrong client secret. Please contact your system administrator.
error.oauth.access.denied.errorMessage=Access denied from social provider.
token.invalid=Your email activation link is invalid or has already been used.
token.expired=Your email activation link has expired. A new email has been sent
automatically. To activate your account, click the link in the new email.
token.expired.resetPassword=Your reset password link has expired. A new email
has been sent automatically. To reset your password, click the link in the new
email.
general.auth.request.error=Identity Provider could not process the
authentication request received. Delete your browser cache and stored cookies,
and restart your browser. If you still experience issues after doing this,
please contact your administrator.
no.saml2.message.in.auth.request.warn=You have followed an invalid link. If you
have bookmarked this link, please replace it with your application URL.
idp.proxy.response.error=Identity provider cannot process the response due to
wrong configuration. Please contact your system administrator.
idp.initiated.sso.disabled.error=IdP-initiated SSO has been disabled by your
system administrator
```

```
provisioningInfo.accountProvisioning=Account Activation
provisioningInfo.enableAccount=Activating Your Account
provisioningInfo.enablingAccount=Activating your account for {0}...
provisioningInfo.enablingAccount.text=Your account is currently being enabled
for {0}. This may take a few seconds, please wait.
provisioningTimeout.errorPage=Error Page
provisioningTimeout.errorOccurred=Account Activation Failure
provisioningTimeout.accountNotActivated=Your account could not be activated.
provisioningTimeout.loginIn15minutes=To activate your account, contact {0}.
profileManagement.title=Profile Management
profileManagement.authentication=Authentication
profileManagement.main.headline=Profile
profileManagement.main.description=Your global Identity Authentication profile
information is displayed below
profileManagement.multipleaaccounts.message=There are multiple accounts using
the email address {0}:
profileManagement.multipleaaccounts.headline=Your Accounts
profileManagement.multipleaaccounts.table.heading.userId=User ID
profileManagement.multipleaaccounts.table.heading.activeFor=Active for
profileManagement.multipleaaccounts.table.heading.status=Status
profileManagement.multipleaaccounts.table.lineMarker.loggedOn=Logged On
profileManagement.multipleaaccounts.selectionMessage=When logging on with email
and password, log on to:
profileManagement.multipleaaccounts.active={0} Active
profileManagement.multipleaaccounts.notActive=Not yet active for {0}
profileManagement.global.label.yes=Yes
profileManagement.global.label.no=No
profileManagement.global.label.required=Required
profileManagement.global.label.verified=Verified
profileManagement.global.label.notVerified=Not Verified
profileManagement.global.label.verify=Verify
profileManagement.global.label.submit=Submit
profileManagement.global.label.change=Change
profileManagement.global.label.reset=Reset
profileManagement.global.button.save=Save
profileManagement.global.button.close=Close
profileManagement.global.button.cancel=Cancel
profileManagement.global.button.unlink=Unlink
profileManagement.global.button.edit=Edit
profileManagement.global.button.downloadUserData=Download
profileManagement.global.button.view=View
profileManagement.global.button.delete=Delete
profileManagement.global.button.remove=Remove
profileManagement.global.button.apply=Apply
profileManagement.global.button.add=Add
profileManagement.global.button.enable=Enable
profileManagement.global.button.disable=Disable
profileManagement.global.button.signOut=Sign Out
profileManagement.mydata.headline=My Data
profileManagement.mydata.description=Click "View" to open a printable overview
of all the data on file for you or click "Download" to save it in a machine-
readable JSON format.
profileManagement.editPersonalInfo.headline=Personal Information
profileManagement.editPersonalInfo.label.firstName=First Name
profileManagement.editPersonalInfo.label.lastName=Last Name
profileManagement.editPersonalInfo.label.userId=User ID
profileManagement.editPersonalInfo.label.language=Language
profileManagement.editPersonalInfo.phone=Phone
profileManagement.editPersonalInfo.timezone=Time Zone
profileManagement.editPersonalInfo.label.validFrom=Valid From
profileManagement.editPersonalInfo.label.validTo=Valid to
profileManagement.editPersonalInfo.button.edit=Edit
profileManagement.editPersonalInfo.changeNameFailed=Could not change your
personal information.
profileManagement.editPersonalInfo.phoneVerify=We have sent a code to your
telephone number. Please enter the code here.
profileManagement.editPersonalInfo.phoneVerifyCaptchaRequestCode=Please verify
the CAPTCHA if required and request a verification code.
```

```
profileManagement.editPersonalInfo.phoneVerifyCaptcha=Please verify the captcha
if required and request a verification code.
profileManagement.editPersonalInfo.phoneVerifyRequestCode=Please request a
verification code and enter it in the field below.
profileManagement.editPersonalInfo.phoneVerifyTitle=Verify your phone number
profileManagement.editPersonalInfo.phoneChangeTitle=Change your phone number
profileManagement.emailInfo.label.email=Email
profileManagement.companyInformation.headline=Company Information
profileManagement.companyInformation.companyCountry=Company Country/Region
profileManagement.companyInformation.companyCity=Company City
profileManagement.contactPreferences.headline=Contact Preferences
profileManagement.contactPreferences.message.explanatoryText=In addition to
communications that will result from this registration, would you also like
to receive news and event notifications from SAP that are specific to your
interests?
profileManagement.socialSignOn.headline=Social Sign-On
profileManagement.socialSignOn.linktext.withHandle=Your account is currently
linked to {1} user <strong>{0}</strong>.
profileManagement.socialSignOn.linktext=Your account is currently linked to {1}.
profileManagement.socialSignOn.unlinkWarning=Choose "Unlink" to remove your
social logon information from your account. You can re-link your account the
next time you log on with a social sign-on-enabled site.
profileManagement.socialSignOn.nolinktext=Your account does not currently use
social sign-on. To link your account with a social sign-on provider, click one
of the social network buttons next time you log on.
profileManagement.passwordChange.headline=Password
profileManagement.passwordChange.description.allowed=Your password grants you
access to any platform connected to Identity Authentication. Note that your
password can only be changed once every {0} hours.
profileManagement.passwordChange.description.standard=Your password grants you
access to any platform connected to Identity Authentication.
profileManagement.passwordChange.button.changePassword=Change Password
profileManagement.passwordChange.label.currentPassword=Current Password
profileManagement.passwordChange.button.disablePassword=Disable Password
profileManagement.passwordChange.message.disablePassword=Your password is
enabled. You can disable it if you are not using it for authentication. You can
always enable it again from this page or using the Forgot Password link on the
login page.
profileManagement.passwordChange.label.passwordStatus=Password Status
profileManagement.passwordChange.label.enabled=Enabled
profileManagement.passwordChange.label.disabled=Disabled
profileManagement.passwordChange.button.enablePassword=Enable Password
profileManagement.passwordChange.label.newPassword=New Password
profileManagement.passwordChange.label.retypeNewPassword=Re-enter New Password
profileManagement.passwordChange.label.confirmNewPassword=Confirm New Password
profileManagement.passwordChange.userDisabledPasswordInfo=You can't change your
password because it is managed by an external system. For more information,
please contact your administrator.
profileManagement.passwordChange.passwordRequirements.mustContain=Your password
must contain at least eight characters including three of:
profileManagement.passwordChange.passwordRequirements.upperCase=Uppercase letters
profileManagement.passwordChange.passwordRequirements.lowerCase=Lowercase letters
profileManagement.passwordChange.passwordRequirements.numbers=Numbers
profileManagement.passwordChange.passwordRequirements.symbols=Symbols
profileManagement.errormessage.changePasswordFailed=Could not change your
password.
profileManagement.errormessage.changePasswordFailed.notAuthenticated=Could not
be verified.
profileManagement.errormessage.changePasswordFailed.retypeDoesNotMatch=Does not
match the new password.
profileManagement.errormessage.changePasswordFailed.passwordUnlimitedlyLocked=You
r password is locked. Please contact your system administrator to unlock it.
profileManagement.errormessage.changePasswordFailed.insufficientPasswordComplexit
y=Insufficient password complexity.
profileManagement.errormessage.changePasswordFailed.passwordPolicyLoadError=The
required password policy cannot be loaded. Please contact your system
administrator.
```

```
profileManagement.errormessage.changePasswordFailed.passwordInHistory=The
password must be different from the last {0} passwords.
profileManagement.errormessage.changePasswordFailed.passwordLocked=The account
is locked. The password must be reset.
profileManagement.errormessage.changePasswordFailed.passwordDisabled=Password
has been disabled. You need to reset it.
profileManagement.errormessage.changePasswordFailed.passwordNotUsed=The password
has expired. It must be reset.
profileManagement.errormessage.changePasswordFailed.passwordTooNew=The password
has already been changed in the last {0} hours.
profileManagement.errormessage.changePasswordFailed.passwordTooOld=The password
has expired. It must be reset.
profileManagement.errormessage.changeCompanyInformationFailed=Could not change
your company information.
profileManagement.errormessage.changeContactPreferenceFailed=Could not change
your contact preferences
profileManagement.errormessage.userUpdateError=Could not update the user. Please
try again.
profileManagement.errormessage.socialUnlinkError=Could not unlink the social
account. Please try again.
profileManagement.errormessage.duplicateDisplayName=The display name you have
chosen is already in use. Choose a different display name.
profileManagement.errormessage.displayNameRequired=A display name is required
for the account. Please enter one.
profileManagement.errormessage.duplicateMail=The email address you have chosen
is already in use. Please choose a different one.
profileManagement.errormessage.duplicate.username=An account already exists with
that user name; register a different user name or sign on with that user name
profileManagement.errormessage.duplicate.telephone=An account has already
verified this telephone number. Choose a different one or contact an
administrator.
profileManagement.errormessage.securityQuestionRequired=Please select a question.
profileManagement.errormessage.securityQuestionAnswerRequired=Please answer the
question.
profileManagement.errormessage.securityQuestionAnswerLengthConstraint=Your
answer should contain between {0} and {1} characters.
profileManagement.errormessage.duplicatedSecurityQuestions=You have one or more
repeated questions. Please answer each question only once.
profileManagement.errormessage.authTokenExpired=The generated secret has
expired. It was valid for five minutes. Please start the process again.
profileManagement.errormessage.webDeviceWithNameExists=A device already exists
with this name.
profileManagement.errormessage.mailLimitExceeded=We did not send you an email
because you have reached the limit of three emails in any 24 hour period.
profileManagement.errormessage.passwordResetPinLengthInsufficient=PIN too short.
It should contain between {0} and {1} digits.
profileManagement.errormessage.wrong.format=Password Reset PIN must contain
digits only.
profileManagement.successmessage.changePassword=Your password was changed.
profileManagement.successmessage.disablePassword=Your password has been disabled.
profileManagement.passwordChange.infoMessage.disablePassword.terminateSessions=On
ce you have disabled the password, you will be logged off from all sessions that
you opened with the old password
profileManagement.passwordChange.infoMessage.changePassword.terminateSessions=Onc
e you have changed the password, you will be logged off from all sessions that
you opened with the old password
profileManagement.platforms.headline=Your Platforms
profileManagement.platforms.name.scn=SAP Community Network
profileManagement.platforms.name.bc=SAP Business Center
profileManagement.platforms.name.store=SAP Store
profileManagement.platforms.name.sapcom=SAP.com
profileManagement.platforms.name.ecohub=SAP EcoHub
profileManagement.platforms.name.search=SAP Search
profileManagement.platforms.name.streamwork=SAP Streamwork
profileManagement.multiFactor.title.section=Multi-Factor Authentication
profileManagement.multiFactor.switch.enableMfaSwitch=To enable this option,
please register a TOTP or web two-factor device.
```

```
profileManagement.multiFactor.description=Different types of multi-factor
authentication require different capabilities and services to operate.
profileManagement.multiFactor.switch.description=Always provide second factor
when authenticating.
profileManagement.multiFactor.deactivationChannels.title=Deactivation Channels
profileManagement.multiFactor.deactivationChannels.existingMfa=Existing Multi-
Factor Authentication
profileManagement.multiFactor.editMfa.verifyIdentity=Verify your identity
profileManagement.multiFactor.editMfa.chooseMfaMethod=Choose verification method
profileManagement.multiFactor.editMfa.chooseMfa.web.totpActivate=To proceed with
the activation of the TOTP device, please enter your security key.
profileManagement.multiFactor.editMfa.chooseMfa.web.totpDeactivate=To proceed
with the deactivation of the TOTP device, please enter your security key.
profileManagement.multiFactor.editMfa.chooseMfa.web.addWebDevice=To add a new
device, please enter your security key.
profileManagement.multiFactor.editMfa.chooseMfa.web.deleteWebDevice=To remove
the selected device, please enter your security key.
profileManagement.multiFactor.editMfa.chooseMfa.web.toggleMfaSwitch=To enable
multi-factor authentication across all applications, please enter your security
key.
profileManagement.multiFactor.editMfa.chooseMfa.web.toggleMfaSwitchDisable=To
disable multi-factor authentication across all applications, please enter your
security key. You might still need to enter the second factor if the application
requires it.
profileManagement.multiFactor.editMfa.chooseMfa.totp.totpDeactivate=To proceed
with the deactivation of the TOTP device, please enter your TOTP passcode.
profileManagement.multiFactor.editMfa.chooseMfa.totp.addWebDevice=To add a new
device, please enter your TOTP passcode.
profileManagement.multiFactor.editMfa.chooseMfa.totp.deleteWebDevice=To remove
the selected device, please enter your TOTP passcode.
profileManagement.multiFactor.editMfa.chooseMfa.totp.toggleMfaSwitch=To enable
multi-factor authentication across all applications, please enter your TOTP
passcode.
profileManagement.multiFactor.editMfa.chooseMfa.totp.toggleMfaSwitchDisable=To
disable multi-factor authentication across all applications, please enter your
TOTP passcode. You might still need to enter the second factor if the
application requires it.
profileManagement.multiFactor.editMfa.chooseMfa.rsa.totpActivate=To proceed with
the activation of the TOTP device, please enter the code generated by your
RADIUS Server token. Enter the code according to the token type requirements.
profileManagement.multiFactor.editMfa.chooseMfa.rsa.totpDeactivate=To proceed
with the deactivation of the TOTP device, please enter the code generated
by your RADIUS Server token. Enter the code according to the token type
requirements.
profileManagement.multiFactor.editMfa.chooseMfa.rsa.addWebDevice=To add a new
device, please enter the code generated by your RADIUS Server token. Enter the
code according to the token type requirements.
profileManagement.multiFactor.editMfa.chooseMfa.rsa.deleteWebDevice=To remove
the selected device, please enter the code generated by your RADIUS Server
token. Enter the code according to the token type requirements.
profileManagement.multiFactor.editMfa.chooseMfa.rsa.toggleMfaSwitch=To enable
multi-factor authentication across all applications, please enter the code
generated by your RADIUS Server token. Enter the code according to the token
type requirements.
profileManagement.multiFactor.editMfa.chooseMfa.rsa.toggleMfaSwitchDisable=To
disable multi-factor authentication across all applications, please enter the
code generated by your RADIUS Server token. Enter the code according to the
token type requirements. You may still need to enter the second factor if the
application requires it.
profileManagement.twoFactor.title.section=TOTP Two-Factor Authentication
profileManagement.twoFactor.noActiveDevices=To access applications that require
TOTP two-factor authentication, you need to have a device activated. No devices
are currently activated for TOTP two-factor authentication.
profileManagement.twoFactor.deactivateDevice=If you want to deactivate your
device, choose a deactivation channel and follow the instructions. If you cannot
enter your passcode or security key, contact your system administrator.
profileManagement.twoFactor.title.howto=How to activate Two-Factor
Authentication for your device
```

```
profileManagement.twoFactor.active=You have a device configured for TOTP two-
factor authentication
profileManagement.twoFactor.deactivateDevices=Deactivate
profileManagement.twoFactor.activateDevice=Activate
profileManagement.twoFactor.resetOtpSms=Passcode by SMS
profileManagement.twoFactor.resetOtpMail=Passcode by Email
profileManagement.twoFactor.resetOtpSend=Send
profileManagement.twoFactor.resetOtpTelephone=Telephone {0}
profileManagement.twoFactor.resetOtpUserMail=Email {0}
profileManagement.twoFactor.passcode.totp=TOTP Passcode
profileManagement.twoFactor.message.sapAuthenticator=This requires a device with
a time-based authentication application installed (such as Google Authenticator
or Microsoft Authenticator).
profileManagement.twoFactor.message.appStoreDownload=Download on the App Store
profileManagement.twoFactor.message.googlePlayDownload=Download on Google Play
profileManagement.twoFactor.message.trademarks=Apple and iTunes are trademarks
of Apple Inc. App Store is a service mark of Apple Inc. Android and Google Play
are trademarks of Google Inc.
profileManagement.twoFactor.step.scanCode=Scan the QR code using the
authenticator app on your device, or enter the key manually.
profileManagement.twoFactor.step.generateCode=Once you have scanned or entered
the key, enter the passcode generated by the authenticator app on your device
below and click "Activate".
profileManagement.twoFactor.message.deactivateDevices=Are you sure you want to
deactivate all mobile devices configured for two-factor authentication? If you
cannot provide a passcode for deactivation, contact your system administrator.
Enter a passcode to proceed.
profileManagement.twoFactor.yourSecretKey=Your Secret Key
profileManagement.twoFactor.passcode=Passcode
profileManagement.twoFactor.web.title=Web Two-Factor Authentication
profileManagement.twoFactor.web.addDevice=Add Device
profileManagement.twoFactor.web.register.title=Register New Device
profileManagement.twoFactor.web.register.description=Choose "Activate" to
register a new device.
profileManagement.twoFactor.web.noDevicesMessage=To access applications that
require Web two-factor authentication, you need to have a device activated. No
devices are currently activated for Web two-factor authentication.
profileManagement.twoFactor.web.activate.text=Your account does not use web
authentication. Choose "Activate" to register a new device.
profileManagement.twoFactor.web.activate.popup=Name your device:
profileManagement.twoFactor.web.activate.noName=No device name provided.
profileManagement.twoFactor.web.devicesMessage=You have the following devices
registered for web authentication:
profileManagement.twoFactor.web.deviceRegistered=Registered
profileManagement.twoFactor.web.deviceLastUsed=Last used
profileManagement.twoFactor.web.registrationDeviceNotSupported=The device you
are trying to register is not supported. Please choose a new device or contact
your system administrator for more information.
profileManagement.twoFactor.web.verificationDeviceNotSupported=Your device is
not supported anymore. Please register a new device and try again.
profileManagement.twoFactor.web.removeDeviceNotSupported=Unfortunately, this
device is not supported anymore. Please contact your system administrator to
remove it.
profileManagement.twoFactor.web.verificationDeviceNotSupportedDuringAdditionalReg
istration=The device you are trying to verify is not supported anymore. Please
contact your system administrator.
profileManagement.sessionManager.headline=Sessions and Tokens
profileManagement.sessionManager.description=You are able to see and control
your active sessions and tokens.
profileManagement.sessionManager.sessions=Active Sessions
profileManagement.sessionManager.authTokens=Authentication Tokens
profileManagement.sessionManager.refreshTokens=Application Access Tokens
profileManagement.sessionManager.expirationDate=Expires in {0}
profileManagement.sessionManager.expired=Expired
profileManagement.sessionManager.signWith=Sign with: {0}
profileManagement.sessionManager.type=Type: {0}
profileManagement.sessionManager.sessionDetails=IP: {0} | Sign with: {1}
profileManagement.sessionManager.authTokenDetails=IP: {0} | Type: {1}
```

```
profileManagement.sessionManager.current=Current
profileManagement.sessionManager.applicationName=Application: {0}
profileManagement.sessionManager.credentialType.OTP=One-time password
profileManagement.sessionManager.credentialType.RSA=RSA
profileManagement.sessionManager.credentialType.WEB=Web authentication
profileManagement.sessionManager.credentialType.SMS=SMS
profileManagement.sessionManager.credentialType.TFA=Two-factor authentication
profileManagement.sessionManager.credentialType.MOBILE_SSO_PASSCODE=Mobile
single sign-on
profileManagement.sessionManager.credentialType.CERT=Certificate
profileManagement.sessionManager.credentialType.SPNEGO=SPNEGO
profileManagement.sessionManager.credentialType.UID_PW=User name and password
profileManagement.sessionManager.credentialType.EMAIL_PW=Email and password
profileManagement.sessionManager.credentialType.TOKEN=Token
profileManagement.sessionManager.credentialType.PREVIOUS_SESSION=Previous session
profileManagement.sessionManager.credentialType.REMEMBERME_COOKIE=Remember me
cookie
profileManagement.sessionManager.credentialType.TRUSTED_IDP_SAML_ASSERTION=Truste
d identity provider session assertion
profileManagement.sessionManager.credentialType.TRUSTED_IDP_SESSION=Trusted
identity provider session
profileManagement.sessionManager.credentialType.SOCIAL_IDENTITY=Social identity
provider
profileManagement.sessionManager.credentialType.PASSWORD_NO_XSRF=Password
profileManagement.sessionManager.mailingAction.REMEMBERME=Remember me
profileManagement.sessionManager.mailingAction.MFATrust=MFA Trust browser
profileManagement.sessionManager.mailingAction.TOTP=OTP code
profileManagement.sessionManager.mailingAction.SMS=SMS code
profileManagement.sessionManager.mailingAction.Mobile=Mobile SSO
profileManagement.sessionManager.mailingAction.RSA=RSA code
profileManagement.sessionManager.mailingAction.WEB=Web authentication
profileManagement.sessionManager.mailingAction.EMAIL=Email authentication
profileManagement.sessionManager.mailingAction.ONBEHALF_REGISTRATION=On-behalf
registration
profileManagement.sessionManager.mailingAction.RESET_PASSWORD=Reset password
profileManagement.sessionManager.mailingAction.SELF_REGISTRATION=Self-
registration
profileManagement.sessionManager.mailingAction.INVITATION=Invitation
profileManagement.sessionManager.mailingAction.FORGOTTEN_PASSWORD=Forgot password
profileManagement.sessionManager.mailingAction.LOCKED_PASSWORD=Locked password
profileManagement.generateCertificates.headline=Certificates
profileManagement.generateCertificates.description=Generate X.509 certificate.
profileManagement.generateCertificates.notBefore=Not Before
profileManagement.generateCertificates.notAfter=Not After
profileManagement.generateCertificates.addCertificate=Generate
profileManagement.generateCertificates.label=Generating
profileManagement.generateCertificates.message=You can have up to two
certificates.
profileManagement.generateCertificates.deleteMessage=To remove the selected
certificate with serial number: %{param0}, please enter your password.
profileManagement.generateCertificates.yourPassword=Your Password
profileManagement.generateCertificates.certificatePassword=Certificate Password
profileManagement.generateCertificates.validity=Validity
profileManagement.generateCertificates.defaultValidity=12 Months (Default)
profileManagement.generateCertificates.threeMonthsValidity=3 Months
profileManagement.generateCertificates.oneMonthValidity=1 Month
profileManagement.generateCertificates.insufficientPasswordComplexity=Insufficien
t password complexity
profileManagement.securityQuestions.title=Security Questions
profileManagement.passwordResetPin.title=PIN Code
profileManagement.securityQuestions.noQuestionsConfigured=You can use security
questions to reset your password. Choose Edit to configure your questions.
profileManagement.securityQuestions.questionsSuccessfullyConfigured=You have
configured your security questions. You can use them to reset your password.
Choose Edit to change your security questions configuration.
profileManagement.securityQuestions.question=Question {0}
profileManagement.securityQuestions.answer=Answer {0}
profileManagement.passwordResetPin.pin=PIN
```

```
profileManagement.passwordResetPin.noPinConfigured=You can use your PIN to reset
your password. Choose Edit to configure your PIN.
profileManagement.passwordResetPin.pinSuccessfullyConfigured=You have configured
your PIN. You can use it to reset your password. Choose Edit to change your PIN
configuration.
ui.unique.checks.MESSAGE_TO_SHORT=Query too short
ui.unique.checks.MESSAGE_UNKNOWN_FIELD=Unknown field
ui.unique.checks.MESSAGE_NOT_IMPLEMENTED_QUERY_FOR_GIVEN_FIELD=Query not
implemented for given field
ui.unique.checks.duplicate.mail=This email address has already been registered.
{0} or use {1} to access your account.
ui.unique.checks.duplicate.username=An account already exists with that user
name; register a different user name or sign on with that user name
ui.unique.checks.duplicate.displayname=An account already exists with that
display name; enter a different display name
mmc.titleTextKey=MMC Form
mmc.headlineTextKey=We Are Updating Our Records
mmc.mainMessageTextKey=So that we can continue to provide you with the most
relevant content for your business needs, please update your information and
accept our Privacy and Web Tracking Policy.
mmc.buttonMessageKey=Save
resendMailConfirm.emailSent=Your activation token has expired. A new email has
been sent to you.
resendMailConfirm.headline=Activate Your Account
multiaccount.alert.title=Multiple Accounts
multiaccount.alert.message=To log onto the SAP Store, click Continue. If you
encounter a problem, it could be because there are multiple accounts registered
to your email address. Set or change your default SAP Store account from your <a
href\="/" target\="_blank">Identity Authentication Profile</a>. <strong>If you
change your default SAP Store account, you will need to log off and log on
again.</strong>
multiaccount.alert.checkbox=Do not show this message again.
samlRedirect.authnRequest.title=SAML Authentication Request from Identity
Authentication
samlRedirect.authnRequest.message.default=Your authentication request has been
forwarded to the target system for processing.
samlRedirect.response.title=SAML Response from Identity Authentication
samlRedirect.response.message.default=You have now been authenticated. Your
request has been forwarded to the target system for processing.
samlRedirect.logoutRequest.title=SAML Logout Request from Identity Authentication
samlRedirect.logoutRequest.message.default=Your logout request has been
forwarded to the target system for processing.
samlRedirect.logoutResponse.title=SAML Logout Response from Identity
Authentication
samlRedirect.logoutResponse.message.default=You have now been logged out. Your
request has been forwarded to the target system for processing.
oidcLogout.frontchannel.title=OIDC Frontchannel Logout Request from Identity
Authentication
oidcLogout.frontchannel.heading=Logout from Identity Authentication
oidcLogout.frontchannel.errorMessage=We haven't received confirmation that all
applications associated with your user session have logged you out. To ensure
that you're completely logged out, please delete any session cookies and close
all browser windows.
oidcLogout.frontchannel.continue=OK
userDataViewer.title=User Data Viewer
userDataViewer.searchbox.label.userid=User ID
userDataViewer.searchbox.label.email=Email
userDataViewer.searchbox.button=Search
userDataViewer.resultlist.label.userid=User ID
userDataViewer.resultlist.label.email=Email
userDataViewer.resultlist.label.firstName=First Name
userDataViewer.resultlist.label.lastName=Last Name
userDataViewer.resultlist.label.scenarios=Scenarios
userDataViewer.resultlist.label.status=Status
userDataViewer.resultlist.label.company=Company
userDataViewer.resultlist.label.SAPStoreRole=SAP Store Role
profileManagement.mobileapps.headline=Authorized Mobile Applications
profileManagement.mobileapps.button.revoke=Revoke
```

```
linkSocialSignOnAccount.headline=Link your account with {0}
linkSocialSignOnAccount.button.register=Create New Linked Account
linkSocialSignOnAccount.headline2=You were authenticated by {0}, {1}.
linkSocialSignOnAccount.headline3=You were authenticated by {0}.
linkSocialSignOnAccount.message=As a final step, you can either link an existing
account to your {0} account, or create a new account that will be linked to your
{0} account.
linkSocialSignOnAccount.button.cancel=Cancel
linkSocialSignOnAccount.label.or=or
linkSocialSignOnAccount.message.unlink=You can unlink your accounts again at any
time on your
linkSocialSignOnAccount.message.profile=Profile Page
invitationConflict.page=Account Conflict
invitationConflict.headline=You are already logged in
invitationConflict.fioritheme.headline=You are already signed in
invitationConflict.question1=You are trying to activate an account other than
the one you are currently logged in to. You can either stay logged in to your
current account and abort the activation, or log out and continue activating
the other account. If you choose the second option, you will have to log out
manually and then log in to your application with the new account.
invitationConflict.fioritheme.question1=You are trying to activate an account
other than the one you are currently signed in to. You can either stay signed
in to your current account and abort the activation, or sign out and continue
activating the other account. If you choose the second option, you will have to
sign out manually and then sign in to your application with the new account.
invitationConflict.question2=You are trying to reset a password for an account
other than the one you are currently logged in to. You can either stay logged
in to your current account and abort the activation, or log out and continue
resetting the password of the other account.
invitationConflict.continueInvitationProcess=Log Out & Continue
invitationConflict.fioritheme.continueInvitationProcess=Sign Out & Continue
invitationConflict.cancelInvitationProcess=Stay Logged In
invitationConflict.fioritheme.cancelInvitationProcess=Stay Signed In
invitationConflictCancel.page=Account Conflict
invitationConflictCancel.headline1=Activation Canceled
invitationConflictCancel.headline2=Password Reset Canceled
invitationConflictCancel.message1=You have chosen to stay logged in and to
cancel activating an account. You can use the link provided in the email later.
invitationConflictCancel.message2=You chose to stay logged in and to cancel
resetting a password. You can use the link to reset your password provided in
the email later.
accessConflict.error=You have an active session on another device or browser.
Please sign out from your current session before continuing with the new sign in.
accessConflict.page=Session Conflict
accessConflict.headline=You are already signed in
accessConflict.message=You have an active session. You can either stay signed
in, or sign out and continue with the new sign-in.
accessConflict.continueLogin=Sign Out & Continue
accessConflict.cancelLogin=Stay Signed In
accessConflictCancel.page=Session Conflict
accessConflictCancel.headline=Sign-in canceled
accessConflictCancel.message=You have chosen to stay signed in to your other
session.
userData.title=User Data
userData.main.headline=Your Profile Information
userData.label.uid=User ID
userData.label.userUuid=Global User ID
userData.label.mid=MID
userData.label.mail=Email
userData.label.mailVerified=Mail Verified
userData.label.mailVerifiedTime=Email Verified Time
userData.label.valid_from=User Profile Valid from
userData.label.valid_to=User Profile Valid to
userData.label.title=Salutation
userData.label.status=Status
userData.label.type=Type
userData.label.relationshipToSAP=Relationship
userData.label.firstName=First Name
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userData.label.middleName=Middle Name
userData.label.lastName=Last Name
userData.label.loginName=Login Name
userData.label.displayName=Display Name
userData.label.userName=User Name
userData.label.hasPassword=Password Set
userData.label.hasOtpSecretKey=OTP Secret Set
userData.label.contactId=Contact ID
userData.label.city=City
userData.label.zip=ZIP/Postal Code
userData.label.country=Country/Region
userData.label.state=State/Province
userData.label.street=Street Address
userData.label.street2=Street Address 2
userData.label.telephone=Telephone
userData.label.telephoneVerified=Telephone Verified
userData.label.telephoneVerificationAttempts=Telephone Verification Attempts
userData.label.mobile=Mobile
userData.label.fax=Fax
userData.label.language=Preferred Language
userData.label.personnelNumber=Personal Number
userData.label.sapName=SAP Name
userData.label.timeZone=Time Zone
userData.label.industry=Industry
userData.label.industryCrm=Industry CRM
userData.label.company=Company
userData.label.department=Department
userData.label.companyEmployees=Company Number of Employees
userData.label.companyCity=Company City
userData.label.companyCountry=Company Country/Region
userData.label.companyRegion=Company State/Province
userData.label.companyStreet=Company Street Address
userData.label.companyStreet2=Company Street Address 2
userData.label.companyZip=Company ZIP/Postal Code
userData.label.companyurl=Company URL
userData.label.companyCode=Company Code
userData.label.costCenter=Cost Center
userData.label.managerUid=Manager ID
userData.label.division=Division
userData.label.addinfo=Add Information
userData.label.areasofexpertise=Areas of Expertise
userData.label.certifications=Certifications
userData.label.instantMessagingID=Instant Messaging ID
userData.label.instantmessagingtype=Instant Messaging Type
userData.label.otherinstantmessagingid=Other Instant Messaging ID
userData.label.otherinstantmessagingtype=Other Instant Messaging Type
userData.label.weblogurl=Weblog URL
userData.label.personalurl=Personal URL
userData.label.othermail=Other Mail
userData.label.discipline=Discipline
userData.label.jobFunction=Job Function
userData.label.photoUrl=Photo URL
userData.label.contactPreferenceEmail=Contact preference by email
userData.label.contactPreferenceTelephone=Contact preference by telephone
userData.label.modifyTime=Modification Time
userData.label.createTime=Creation Time
userData.label.loginTime=Login Time
userData.label.updateTime=Update Time
userData.label.passwordPolicy=Password Policy
userData.label.passwordSetTime=Password Set At
userData.label.passwordLoginTime=Password Login at
userData.label.passwordLockedTime=Password Locked at
userData.label.passwordStatus=Password Status
userData.label.passwordFailedLoginAttempts=Password Failed Login Attempts
userData.label.otpFailedLoginAttempts=OTP Failed Login Attempts
userData.label.otpUsedPasscodeTime=OTP Used at
userData.label.otpLockedTime=OTP Locked at
userData.label.spUser.ServiceProviderName=SP User Service Provider Name
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userData.label.spUser.NameID=SP User NameID
userData.label.spUser.Status=SP User Status
userData.label.spUser.Role=SP User Role
userData.label.spUser.valid_to=SP User Valid to
userData.label.spUser.valid_from=SP User Valid from
userData.label.spUser.ActivationTime=SP User Activation Time
userData.label.spUser.spCustomAttribute1=Application Custom Attribute 1
userData.label.spUser.spCustomAttribute2=Application Custom Attribute 2
userData.label.spUser.spCustomAttribute3=Application Custom Attribute 3
userData.label.spUser.spCustomAttribute4=Application Custom Attribute 4
userData.label.spUser.spCustomAttribute5=Application Custom Attribute 5
userData.label.companyGroups=Company Groups
userData.label.corporateGroups=Corporate Groups
userData.label.customAttribute1=User Custom Attribute 1
userData.label.customAttribute2=User Custom Attribute 2
userData.label.customAttribute3=User Custom Attribute 3
userData.label.customAttribute4=User Custom Attribute 4
userData.label.customAttribute5=User Custom Attribute 5
userData.label.customAttribute6=User Custom Attribute 6
userData.label.customAttribute7=User Custom Attribute 7
userData.label.customAttribute8=User Custom Attribute 8
userData.label.customAttribute9=User Custom Attribute 9
userData.label.customAttribute10=User Custom Attribute 10
userData.label.newsletters=Newsletter
userData.label.privacyPolicy=Privacy Policy
userData.label.termsOfUse=Terms of Use
userData.label.account=Account
userData.label.alternativeLoginId=Alternative Login ID
userData.label.publicAttributes=Public Attributes
userData.label.socialIdentities=Social Identities
userData.label.socialIdentitiesMap=Social Identities Map
userData.label.validFrom=Valid From
userData.label.validTo=Valid to
error.ui.errormessage.idpSsesionExpired=Your session has expired. Please log on
again.
error.ui.errormessage.idpSessionExpiredUserActivation=Your session has expired.
Open the email for your account activation and follow the link it contains to
reactivate your account.
error.ui.errormessage.idpSessionExpiredPasswordReset=Your session has expired.
Open the email for your password reset and follow the link in it to reset your
password.
error.ui.link.not.found.error=The requested URL could not be found on this
server.
error.ui.clickjacking.error=Your domain is not trusted. Please contact your
system administrator.
error.ui.unknown.application.error=An error has occurred. Please visit the
application and log on again.
error.ui.unexistent.text.resource.error=The requested document does not exist.
Please contact your system administrator.
sci_eu_data_protection.headline=European Union (EU) Access Mode
sci_eu_data_protection.message=<p>This instance is operated and supported
in the European Union EU Access Service.<br>By selecting <strong>Confirm</
strong>, you confirm that you are either:<br>(i) physically located
in the EU or<br>(ii) you have verified that EU Access restrictions
do not apply in your situation or<br>(iii) you are a Technical
Support engineer located outside the EEA or Switzerland, and you have
read and accepted the<a href\="https://wiki.one.int.sap/wiki/display/idservice/
SCI+EU+DP+Working+Instructions" target\="_blank">SCI EUDP Working Instructions</
a>.<br>Your confirmation will be logged.</p>
sci_eu_data_protection.confirm=Confirm
sci_eu_data_protection.leave=Quit
error.ui.errormessage.untrustedTargetUrl=Your request failed because the page
you have to be redirected to is not trusted. Please contact your system
administrator.
error.ui.errormessage.invalidTargetUrl=Your registration failed because the URL
of the redirect page is invalid. Please contact your system administrator.
error.ui.errormessage.disabledUserRegistration=The application does not allow
user self-registration. Please contact your system administrator.
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error.ui.errormessage.invalidServiceProvider=Identity provider cannot process
the request because the configuration is incorrect. Please contact your system
administrator.
error.ui.errormessage.missingMail=An email address is required in order to
continue. Please contact your system administrator.
error.ui.errormessage.invalidClientId=SAP Cloud Identity Services can't process
this OpenID Connect request. The client_id sent by the client application must
match the configuration for the application in SAP Cloud Identity Services.
error.ui.errormessage.invalidRedirectUri=SAP Cloud Identity Services can't
process this OpenID Connect request. The redirect_uri sent by the client
application must match the configuration for the application in SAP Cloud
Identity Services.
error.ui.errormessage.invalidOIDCParameters=SAP Cloud Identity Services can't
process this OpenID Connect request. The parameters sent by the client
application must match the configuration for the application in SAP Cloud
Identity Services.
error.ui.errormessage.missingOIDCParameters=SAP Cloud Identity Services can't
process this OpenID Connect request. Some of the parameters required by SAP
Cloud Identity Services are missing.
error.ui.errormessage.invalidOpenIDProvider=OpenID provider cannot process the
request because the configuration is incorrect. Please contact your system
administrator.
error.ui.errormessage.invalidOpenIDPostLogoutRedirectUri=OpenID provider cannot
process the logout request because the post_logout_redirect_uri is invalid.
Please contact your system administrator.
error.ui.errormessage.unknownOpenIDPostLogoutRedirectUri=OpenID provider cannot
process the logout request because the post_logout_redirect_uri is unknown.
Please contact your system administrator.
error.ui.errormessage.invalidOpenIDTokenHint=OpenID provider cannot process the
logout request because the id_token_hint is invalid. Please contact your system
administrator.
error.ui.errormessage.unknownClientId=OpenID provider cannot process the logout
request because the client_id is unknown. Please contact your system
administrator.
error.ui.errormessage.clientIdNotUnique=OpenID provider cannot process the
logout request because the client_id of the application is not unique. Please
also provide [app_tid].
error.ui.errormessage.missingOpenIDTokenHintAndClientID=OpenID provider cannot
process the logout request because neither client_id nor id_token_hint parameter
has been provided with the post_logout_redirect_uri. Please contact your system
administrator.
flashes.error_validating_captcha=We are unable to validate your Captcha
protection at the moment. Please try again later.
security.alert.email.changed=Your email address in Identity Authentication
tenant {0} has been changed. The new email address is {1}.
security.alert.telephone.number.changed=Your telephone number in Identity
Authentication tenant {0} has been changed. The new number is {1}.
security.alert.mobile.number.changed=Your mobile number in Identity
Authentication tenant {0} has been changed. The new number is {1}.
security.alert.fax.number.changed=Your fax number in Identity Authentication
tenant {0} has been changed. The new number is {1}.
security.alert.telephone.number.deleted=Your telephone number in Identity
Authentication tenant {0} has been deleted.
security.alert.mobile.number.deleted=Your mobile number in Identity
Authentication tenant {0} has been deleted.
security.alert.fax.number.deleted=Your fax number in Identity Authentication
tenant {0} has been deleted.
security.alert.login.name.changed=Your login name attribute in Identity
Authentication tenant {0} has been changed. The new login name attribute is {1}.
security.alert.email.deleted=Your email address in Identity Authentication
tenant {0} has been deleted.
security.alert.login.name.deleted=Your login name attribute in Identity
Authentication tenant {0} has been deleted.
security.alert.initial.password.set=The administrator has set an initial
password for your account in Identity Authentication tenant {0}.
security.alert.new.password.set=A new password has been set for your account in
Identity Authentication tenant {0}.
```

```
security.alert.password.change=Your password has been changed in Identity
Authentication tenant {0}.
security.alert.password.reset=Your password has been reset in Identity
Authentication tenant {0}.
security.alert.register.same.email=Someone has tried to register a new user with
your email address in Identity Authentication tenant {0}.
security.alert.remove.authorizations=Your authorizations {1} in Identity
Authentication tenant {0} have been removed.
security.alert.secondFactor.skipped=The setup of Two-Factor Authentication for
your user in Identity Authentication tenant {0} has been skipped. You can
postpone the setup of Two-Factor Authentication for {1} more days.
security.alert.totpDevice.activated=A TOTP Two-Factor Authentication device has
been activated for your user in Identity Authentication tenant {0}.
security.alert.totpDevice.deactivated=A TOTP Two-Factor Authentication device
has been deactivated for your user in Identity Authentication tenant {0}.
security.alert.webAuthnDevice.activated=A Web Two-Factor Authentication device
has been activated for your user in Identity Authentication tenant {0}.
secutity.alert.webAuthnDevice.deactivated=A Web Two-Factor Authentication device
has been deactivated for your user in Identity Authentication tenant {0}.
security.alert.passwordlessAuthnDevice.activated=A Biometric Authentication
device has been activated for your user in Identity Authentication tenant {0}.
security.alert.passwordlessAuthnDevice.deactivated=A Biometric Authentication
device has been deactivated for your user in Identity Authentication tenant {0}.
security.alert.certificate.generation=A X.509 certificate has been generated for
your user in Identity Authentication tenant {0}.
security.alert.certificate.deletion=A X.509 certificate has been deleted from
your user in Identity Authentication tenant {0}.
logon.ui.label.passwordlessAuthn=Biometric Authentication
logon.ui.errormessage.NO_PASSWORDLESS_AUTHN_DEVICE=No device. There is no
biometric authentication device configured for this user. To configure one,
please sign in with your password.
#XTOL
logon.error.emptyField=Please enter the required information in the field.
profileManagement.passwordlessAuthn.title.section=Biometric Authentication
profileManagement.passwordlessAuthn.devicesMessage=You have the following
devices registered for biometric authentication:
profileManagement.passwordlessAuthn.noDevicesMessage=To access applications that
require biometric authentication, you need to have a device activated. No
devices are currently activated for biometric authentication.
profileManagement.passwordlessAuthn.deviceRegistered=Registered
profileManagement.passwordlessAuthn.deviceLastUsed=Last used
profileManagement.passwordlessAuthn.addWebDevice=To add a device, please enter
your password.
profileManagement.passwordlessAuthn.deleteWebDevice=To remove the selected
device, please enter your password.
profileManagement.passwordlessAuthn.addDevice=Add Device
profileManagement.passwordlessAuthn.configure.title=Configure your biometric
authentication device
profileManagement.passwordlessAuthn.configure.description=There are no biometric
authentication devices registered for "%{param0}". Choose "Activate" to register
a new one.
profileManagement.passwordlessAuthn.configure.skip=Skip
profileManagement.passwordlessAuthn.configure.activate=Activate
profileManagement.passwordlessAuthn.activate.popup=Name your device:
profileManagement.passwordlessAuthn.activate.noName=No device name provided.
security.question.1=What was the name of the city your parent was born in?
security.question.2=What was the name of your first pet?
security.question.3=What was the make and model of your first car?
security.question.4=What was the name of your favorite teacher?
security.question.5=What was the last name of your first friend?
security.question.6=What was the destination of your first airplane trip?
security.question.7=What is the middle name of your sibling?
security.question.8=What is the middle name of your oldest grandchild?
security.question.9=What was your first phone number?
security.question.10=What is your vehicle registration number?
requestStorageAccess.grantAccess=Grant Access
requestStorageAccess.confirmation=Confirmation
```

```
requestStorageAccess.message=To sign in to the application you have to grant the
application access to your browser storage. By pressing Continue, a new popup
will appear where you can grant the access. If you block this you will not be
able to sign in.
requestStorageAccess.errorMessage=You have either blocked access to your browser
storage or you have never opened the current page before. To continue, please go
to your browser settings and reset the embedded content permissions, or open the
requestStorageAccess.confirmationPage=confirmation page.
requestStorageAccess.confirmationMessage=You have been redirected to this page
because your browser has never interacted with the authentication service
application. Since you have already opened the current page, the browser can
further process the sign-in request. Please choose Confirm to go back to the
previous page and continue with the sign-in.
requestStorageAccess.continue=Continue
requestStorageAccess.comfirm=Confirm
```

Related Information

Change Tenant Texts REST API [page 2826]

1.9.2.11.4 Tenant Texts Allowed HTML Elements

Allowed Elements

Allowed Elements

Category	Description	Tags
Inline text & emphasis	Phrasing-level formatting.	`span`, `strong`, `em`, `b`, `i`, `u`, `s`, `mark`, `small`, `sub`, `sup`
Inline semantic	Phrasing with semantic meaning.	`abbr`, `cite`, `q`, `time`, `data`, `bdi`, `bdo`, `ins`, `del`
Code & technical phrasing	Inline and preformatted technical text.	`code`, `kbd`, `samp`, `var`, `pre`
Block-level text	Paragraphs and block quotations.	`p`, `blockquote`
Headings	Section headings, plus heading group.	`h1`, `h2`, `h3`, `h4`, `h5`, `h6`, `hgroup`
Sectioning & generic containers	Document outline and grouping.	`div`, `header`, `footer`, `main`, `article`, `section`, `aside`, `nav`, `address`
Lists	Ordered, unordered, description, and command lists.	`ul`, `ol`, `li`, `dl`, `dt`, `dd`, `menu`

Category	Description	Tags
Tables	Tabular data and column groups.	`table`, `thead`, `tbody`, `tfoot`, `tr`, `th`, `td`, `caption`, `colgroup`, , `col`
Figures	Figure containers. **Note** `img` is not permitted	`figure`, `figcaption`, `picture`
Line breaks & separators	Soft and hard breaks; thematic break.	`hr`, `br`, `wbr`
Ruby annotations	East Asian pronunciation guides.	`ruby`, `rt`, `rp`, `rb`, `rtc`
Interactive disclosure & gauges	Native disclosure widget and progress/meter indicators.	`details`, `summary`, `progress`, `meter`
Anchors	Hyperlinks (may appear with or without attributes).	`a`
Legacy representative	Deprecated HTML4 styling tag.	`font`

Element Specific Attributes

Element Specific Attributes

Element(s)	Attribute(s)	Constraint
`abbr`	`title`	No
`time`, `ins`, `del`	`datetime`	No
`blockquote`, `q`	`cite`	No
`data`	`value`	No
`bdo`, `bdi`	`dir`	`ltr` or `rtl` (case-insensitive)
`ol`	`start`, `type`, `reversed`	No
`li`	`value`	No
`th`, `td`	`colsoan`, `rowspan`, `scope`, `headers`	No
`th`	`abbr`, `scope`	No
`col`, `colgroup`	`span`	No
`table`	`summary`	No
`details`	`open`	No
`progress`	`value`, `max`	No
`meter`	`value`, `min`, `max`, `low`, `high`, `optimum`	No

Element(s)	Attribute(s)	Constraint
`font`	`size`, `color`, `face`	No
`a`	`href`	Permitted URL schemes only (see table below)
`a`	`target`	`_blank`, `_self`, `_parent`, `_top` (case-insensitive)
`a`	`rel`	Must start with a letter; letters, white-space, and hyphens only

Global Attributes (allowed on every element)

Global Attributes

Attribute	Notes
`id`	Element identifier.
`class`	CSS class names.
`data-safermailto-subject`	Custom data attribute used by the email-link protection feature.
`data-safermailto-recipient`	Custom data attribute used by the email-link protection feature.
`data-safermailto-domain`	Custom data attribute used by the email-link protection feature.

Permitted URL Schemes

Permitted URL Schemes

Scheme	Status
`http`	Permitted
`https`	Permitted
`mailto`	Permitted
`ftp`	Permitted
`javascript`	Rejected
`data`	Rejected

> **Note**
>
> The `style` attribute is not in the allow list. Visual styling must be applied via `class`.
>
> If you want to add style to an HTML element, you should use a custom CSS. For more information, see Use a Custom CSS [page 2040].

1.9.2.12 Change Master Data Texts REST API

The Change Master Data Texts REST API can be used to change the predefined master data for each resource in Identity Authentication.

Prerequisites

To call the methods of this Change Master Data Texts REST API you must have a system as administrator (technical user) with an assigned *Manage Tenant Configuration* role. For more information, see Add System as Administrator [page 2305], and Edit Administrator Authorizations [page 2309].

Usage

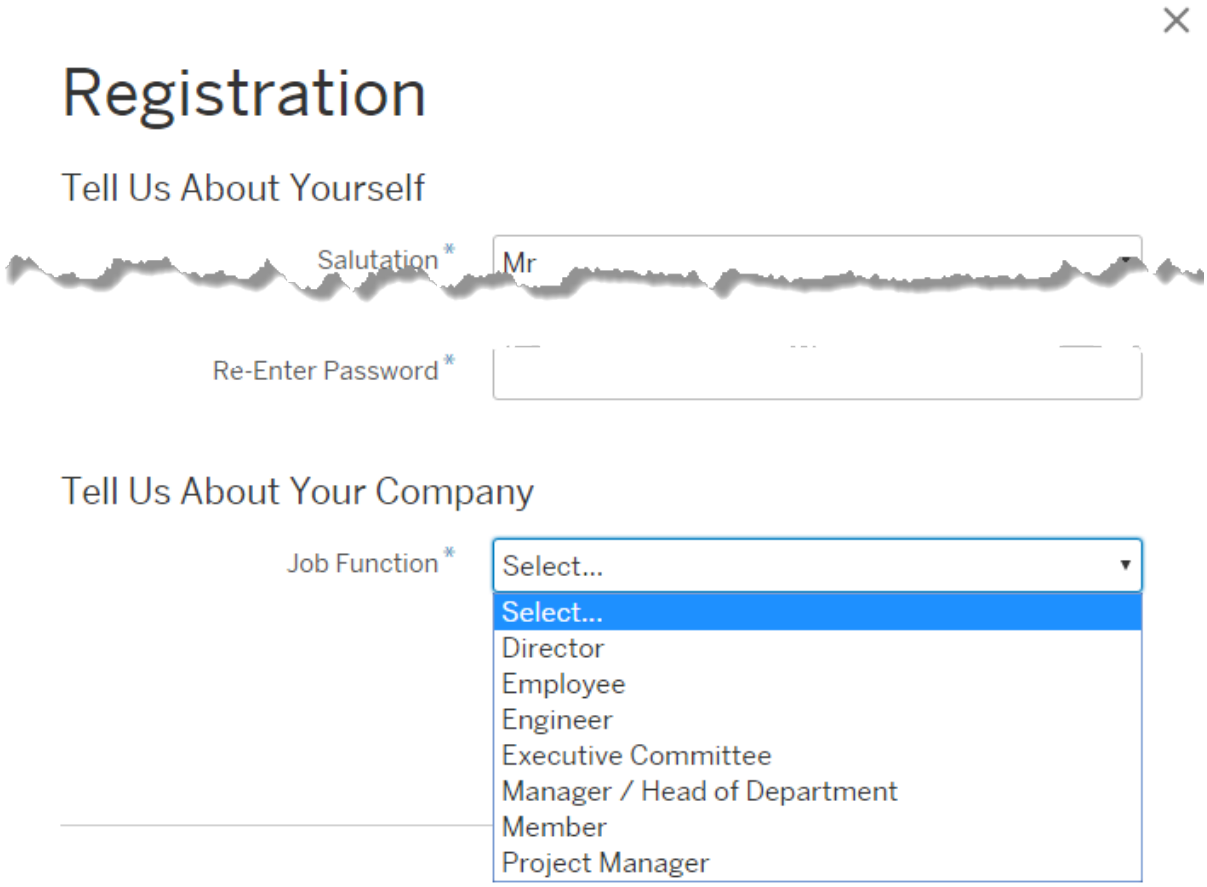
The predefined master data represents records in Identity Authentication that contain all relevant system data about a resource (Salutations, Functions, Departments, Company Relationships, Industries, Languages, Countries). That data can be used by the system for different classifications in the organization, for example, job titles, departments, or countries. The predefined master data texts are stored in `properties` documents which can be accessed from the links in the table below.

Predefined Master Data Texts

Resource	Link	Notes
Salutations	Salutations.properties [page 2868]	Key value pairs with a predefined set of honorifics.
Functions	Functions.properties [page 2868]	Key value pairs with a predefined set of job titles.
Departments	Departments.properties [page 2868]	Key value pairs with a predefined set of departments.
Company Relationships	Relationships.properties [page 2869]	Key value pairs with a predefined set of business entities, such as customer, partner, employee.
Industries	Industries.properties [page 2870]	Key value pairs with a predefined set of industries.
Languages	Languages.properties [page 2870]	Key value pairs with a predefined set of languages.
Countries	Countries.properties [page 2873]	Key value pairs with a predefined set of countries.

The files contain configurable parameters stored as key value pairs of strings. Each key stores the unified key of the resource, and the corresponding value is the unified value of the resource that can be changed and updated.

The example below shows the customized values of the *Functions* file. The dropdown list in the *Job Function* field on the *Registration* form shows the new values that have overwritten the predefined texts in the file.



Depending on your requirements, you can:

- Use the **GET** method to obtain the texts that you have already overwritten in the predefined master data texts, change the texts that you want, add them to the **POST** request and upload them.
- Use the **GET** method to obtain the texts that you have already overwritten in the predefined master data texts, delete a key value pair, add the texts without this line to the **POST** request and upload them. This will replace the deleted key value pair with the predefined one.
- Open the respective `properties` document from the link in the table, use the **GET** method to obtain the texts that overwrote part of the predefined master data texts when your custom tenant was created, from the downloaded file copy the key value pairs that were not included in the response, change the texts in the copied key value pairs, add these new key value pairs to the **POST** request, and execute it.
- Open the respective `properties` document from the link in the table, copy it to the **POST** request, and execute it. This will replace all texts with the predefined ones.

> ⚠ Caution
>
> Be careful if you change the keys when overwriting data texts, because this can result in invalid values. If a user has a certain value assigned, and the key value pair is overwritten with a different key, then the value for that property of the user will be no longer valid and visualized. For example, if you have set "Mr." to a user, and the key-value pair is "01=Mr.". If you overwrite the key value pair with "03=Mr.", "Mr." will not be visualized for that user.
>
> When overwriting data texts, the keys for the different languages must be one and the same. For example, the master data texts for the German locale are overwritten, and the tenant administrator wants to overwrite the texts for the French locale. The keys for the German locale should be obtained first and used as keys for the French locale. After that the values can be translated in French.

Methods

HTTP Method	See	URI
GET	GET Master Data Texts [page 2863]	`https://<tenant ID>.accounts.ondemand.com/service/resource?resourceType=RESOURCE_MD_<VALUE>&locale=<value>`
POST	POST Master Data Texts [page 2866]	`https://<tenant ID>.accounts.ondemand.com/service/resource/SAP_DEFAULT`

> ⓘ Note
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*.

Parent topic: API Documentation [page 2790]

Related Information

Application Configurations API [page 2791]
Identity Directory API [page 2792]
Identity Provisioning API [page 2795]
Identity Authentication API (Deprecated) [page 2796]
Migrating Deprecated Identity Authentication API to Identity Directory API [page 2797]
Invitation REST API [page 2802]
User Management REST API [page 2805]
Password Service REST API [page 2819]

Forgot Password REST API [page 2821]
TOTP Validation Service [page 2824]
Change Tenant Texts REST API [page 2826]

1.9.2.12.1 GET Master Data Texts

Download the texts that you have already overwritten in the predefined master data texts.

> → Remember
>
> If you haven't overwritten the predefined master data texts the GET request will return the *404 Not Found Resource not found for scenario* response.

Request

URI: `https://<Cloud Identity Services domain>/service/resource?resourceType=<value>&locale=<value>`

> ⓘ Note
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *GET*

Permissions: You must have a system as administrator with an assigned *Manage Tenant Configuration* role. For more details about how to add a system as administrator and assign administrator roles, see Add System as Administrator [page 2305], and Edit Administrator Authorizations [page 2309].

URL Parameters

Parameter	Required	Description	Notes
`setId`	No	The identifier of the scenario that the resource is related to.	The default value is `SAP_DEFAULT`

Parameter	Required	Description	Notes
`resourceType`	Yes	The type of the resource.	Use: • `RESOURCE_MD_SALUTATIONS` • `RESOURCE_MD_FUNCTIONS` • `RESOURCE_MD_DEPARTMENTS` • `RESOURCE_MD_RELATIONSHIPS` • `RESOURCE_MD_INDUSTRIES` • `RESOURCE_MD_LANGUAGES` • `RESOURCE_MD_COUNTRIES`
`locale`	Yes	The locale of the resource.	The default languages are: • Chinese (zh_CN) • Dutch (nl) • English (en) • French (fr) • German (de) • Hebrew (iw) • Italian (it) • Japanese (ja) • Korean (ko) • Polish (pl) • Portuguese (pt) • Russian (ru) • Spanish (es) • Welsh (cy) • Swedish (sv)

Request Example

```
GET /service/resource?resourceType=RESOURCE_MD_SALUTATIONS&locale=en
Content-Type: application/json
```

Response

Response Status and Error Codes

Success Codes

Code	Meaning	Description
200 OK	The request was successful.	OK

Error Codes

Response Code	Meaning	Description
400	Bad Request	Validate JSON data. Validate that the special characters are escaped properly and new lines are added.
401	Unauthorized	The client is not authenticated.
403	Forbidden	Access to the resource is denied.
404	Not Found	The requested resource cannot be found.
405	Method Not Allowed	The requested method is not supported for the given resource.
415	Unsupported Media Type	The REST service does not support the API version requested by the REST client.
500	Internal Server Error	The operation cannot be completed due to a server error.
503	Service Unavailable	The service is currently unavailable.

Response Example

Example of response that returns overwritten values of the *Salutations* file.

```
{
  0001=Dr.,
  0002=Prof.
}
```

Example of response that returns overwritten keys of the *Salutations* file. The keys can differ from the original in the master data file. It is not necessary to be in sequence.

```
{
  01=Ms.,
  05=Mr.
}
```

1.9.2.12.2 POST Master Data Texts

Update the master data texts.

> ⚠ Caution
>
> Make sure that you have applied the proper escaping in your **POST** request.
>
> You should add *\n* at the end of each line in the data property content.
>
> Escape Sequences

Value	Escape Sequences
\b	Backspace (ascii code 08)
\f	Form feed (ascii code 0C)
\n	New line
\r	Carriage return
\t	Tab
\"	Double quote
\\	Backslash character

> → Tip
>
> Instead of editing the request manually, you can use an on-line tool for converting a normal string into a quoted one.

Request

URI: `https://<Cloud Identity Services domain>/service/resource/SAP_DEFAULT`

> ⓘ Note
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: POST

Permissions: You must have a system as administrator with an assigned *Manage Tenant Configuration* role. For more details about how to add a system as administrator and assign administrator roles, see Add System as Administrator [page 2305], and Edit Administrator Authorizations [page 2309].

Request Example

```
https://<tenantId>.accounts.ondemand.com/service/resource/SAP_DEFAULT
Content-Type: application/json
Body:
[{
"resourceType": "RESOURCE_MD_DEPARTMENTS",
"locale": "en",
"contentType": "text/html;charset=UTF-8",
"data": "01=HR\n02=IT"
}]
```

Response

Response Status and Error Codes

Success Codes

Code	Meaning	Description
200 OK	The request was successful.	OK

Error Codes

Response Code	Meaning	Description
400	Bad Request	Validate JSON data. Validate that the special characters are escaped properly and new lines are added.
401	Unauthorized	The client is not authenticated.
403	Forbidden	Access to the resource is denied.
404	Not Found	The requested resource cannot be found.
405	Method Not Allowed	The requested method is not supported for the given resource.
415	Unsupported Media Type	The REST service does not support the API version requested by the REST client.
500	Internal Server Error	The operation cannot be completed due to a server error.
503	Service Unavailable	The service is currently unavailable.

1.9.2.12.3 Salutations.properties

Salutations Default Master Data

```
0001=Ms.
0002=Mr.
```

1.9.2.12.4 Functions.properties

Functions Default Master Data

```
1=Executive Board
2=Managing Director
4=Director
9=Project Manager
11=Manager/Head of Department
12=Team Lead/Supervisor
21=Chief Digital Officer
22=Chief Technology Officer
23=Chief Procurement Officer
24=Chief Transformation Officer
25=Chief Sales Officer
40=Employee
49=Chief HR Officer
50=Chief Information Officer
51=Chief Financial Officer
52=Chief Executive Officer
53=Chief Operating Officer
54=President
55=Vice President
56=Chief Marketing Officer
57=Consultant
59=Analyst
60=System Administrator
81=Delegate
82=Member
83=Architect
84=Developer
87=Engineer
CO=Chief Compliance Officer
```

1.9.2.12.5 Departments.properties

Departments Default Master Data

```
0002=Management
0004=Administration
0006=Training
0008=Corporate Services
```

```
0010=Information Technology
0012=SAP System Support
0020=Marketing
0030=Sales
0034=Legal
0038=Treasury
0040=Finance/Accounting
0041=Risk Management
0043=Sustainability
0048=Controlling
0049=Research/Development
0050=Human Resources
0051=Pension Fund Administration
0060=Materials Management
0062=Procurement
0065=Business Intelligence
0067=Business Planning
0070=Service
0071=Customer Services
0072=Consulting
0080=Logistics
0100=Technical Area
0101=Environmental
0110=Product Planning
0120=Engineering/Design
0160=Quality Assurance
0165=Customer Compliance Center
0170=Plant Maintenance
0200=Healthcare
0290=Operations/Manufacturing
0300=Operations/Business
0310=Administrative Board
0311=Insurance/Claims
0390=Editorial
3010=E-Commerce
3070=Real Estate Management
3110=Product Management
3150=Private Clients
3190=Health and Safety
4010=Audit
```

1.9.2.12.6 Relationships.properties

Relationships Default Master Data

```
01=Customer
02=Prospective Customer
03=Partner
04=Prospective Partner
05=Consultant
06=Press/Analyst
07=Investor/Shareholder
08=Student
09=Employee
```

1.9.2.12.7 Industries.properties

Industries Default Master Data

```
01=Aerospace and Defense
02=Automotive
03=Banking
04=Chemicals
05=Consumer Products
06=Industrial Machinery and Components
07=Healthcare
08=High Tech
09=Insurance
10=Oil and Gas
11=Life Sciences
12=Public Sector
13=Retail
14=Telecommunications
15=Utilities
17=Building Materials
19=Forest Products, Furniture, and Textiles
21=Fabricated Metal Products
22=Primary Metals
23=Mining
25=Sports and Entertainment
27=Media
28=Passenger and Cargo Services
30=Higher Education and Research
31=Professional Services
32=Transportation (Travel and Logistics)
35=Wholesale Distribution
36=Defense
37=Engineering, Construction, and Operations
38=Postal
```

1.9.2.12.8 Languages.properties

Languages Default Master Data

```
AB=Abkhazian
AA=Afar
AF=Afrikaans
SQ=Albanian
AM=Amharic
AR=Arabic
HY=Armenian
AS=Assamese
AY=Aymara
AZ=Azerbaijani
BA=Bashkir
EU=Basque
BE=Belarusian
BN=Bengali
```

```
BH=Bihari
BI=Bislama
BR=Breton
BG=Bulgarian
MY=Burmese
CA=Catalan
ZH=Chinese
zh_TW=Chinese (Taiwan)
CO=Corsican
HR=Croatian
CS=Czech
DA=Danish
NL=Dutch
DZ=Dzongkha
EN=English
English (United Kingdom)
EO=Esperanto
ET=Estonian
FO=Faroese
FJ=Fijian
FI=Finnish
FR=French
fr_CA=French (Canada)
FY=Frisian
GL=Galician
KA=Georgian
DE=German
EL=Greek
KL=Greenlandic
GN=Guarani
GU=Gujarati
HA=Hausa
HE=Hebrew
HI=Hindi
HU=Hungarian
IS=Icelandic
ID=Indonesian
IA=Interlingua
IE=Interlingue
IU=Inuktitut
IK=Inupiak
GA=Irish
IT=Italian
JA=Japanese
JW=Javanese
KN=Kannada
KS=Kashmiri
KK=Kazakh
KM=Khmer
RW=Kinyarwanda
KY=Kirghiz
KO=Korean
KU=Kurdish
LO=Laotian
LA=Latin
LV=Latvian
LN=Lingala
LT=Lithuanian
MK=Macedonian
MG=Malagasy
MS=Malay
ML=Malayalam
MT=Maltese
MI=Maori
MR=Marathi
MO=Moldavian
MA=Mongolian
NA=Nauruan
```

```
NE=Nepali
NO=Norwegian
OC=Occitan
OR=Oriya
OM=Oromo
PS=Pashto
FA=Persian
PL=Polish
PT=Portuguese
pt_BR=Portuguese (Brazil)
PA=Punjabi
QU=Quechua
RM=Rhaeto-Romance
RO=Romanian
RN=Rundi
RU=Russian
SM=Samoan
SG=Sango
SA=Sanskrit
GD=Scottish Gaelic
SR=Serbian
SH=Serbo-Croatian
ST=Sesotho
TN=Setswana
SN=Shona
SD=Sindhi
SI=Sinhalese
SS=Siswati
SK=Slovak
SL=Slovenian
SO=Somali
ES=Spanish
es_MX=Spanish (Mexico)
SU=Sudanese Arabic
SW=Swahili
SV=Swedish
TL=Tagalog
TG=Tajik
TA=Tamil
TT=Tatar
TE=Telugu
TH=Thai
BO=Tibetan
TI=Tigrinya
TO=Tonga
TS=Tsonga
TR=Turkish
TK=Turkmen
TW=Twi
UK=Ukrainian
UR=Urdu
UG=Uyghur
UZ=Uzbek
VI=Vietnamese
VO=Volapuk
CY=Welsh
WO=Wolof
XH=Xhosa
YI=Yiddish
YO=Yoruba
ZA=Zhuang
ZU=Zulu
```

1.9.2.12.9 Countries.properties

Countries Default Master Data

```
AF=Afghanistan
AX=Aland Islands
AL=Albania
DZ=Algeria
AS=American Samoa
AD=Andorra
AO=Angola
AI=Anguilla
AQ=Antarctica
AG=Antigua and Barbuda
AR=Argentina
AM=Armenia
AW=Aruba
AU=Australia
AT=Austria
AZ=Azerbaijan
BS=Bahamas
BH=Bahrain
BD=Bangladesh
BB=Barbados
BY=Belarus
BE=Belgium
BZ=Belize
BJ=Benin
BM=Bermuda
BT=Bhutan
BO=Bolivia
BQ=Bonaire, Sint Eustatius and Saba
BA=Bosnia and Herzegovina
BW=Botswana
BV=Bouvet Island
BR=Brazil
IO=British Indian Ocean Territory
BN=Brunei Darussalam
BG=Bulgaria
BF=Burkina Faso
BI=Burundi
KH=Cambodia
CM=Cameroon
CA=Canada
CV=Cape Verde
KY=Cayman Islands
CF=Central African Republic
TD=Chad
CL=Chile
CN=China
CX=Christmas Island
CC=Cocos (Keeling) Islands
CO=Colombia
KM=Comoros
CG=Congo
CD=Congo DRC
CK=Cook Islands
CR=Costa Rica
CI=Côte d'Ivoire
HR=Croatia
CY=Cyprus
CZ=Czech Republic
DK=Denmark
DJ=Djibouti
DM=Dominica
```

```
DO=Dominican Republic
EC=Ecuador
EG=Egypt
SV=El Salvador
GQ=Equatorial Guinea
ER=Eritrea
EE=Estonia
ET=Ethiopia
FK=Falkland Islands (Malvinas)
FO=Faroe Islands
FJ=Fiji
FI=Finland
FR=France
GF=French Guiana
PF=French Polynesia
TF=French Southern Territories
GA=Gabon
GM=Gambia
GE=Georgia
DE=Germany
GH=Ghana
GI=Gibraltar
GR=Greece
GL=Greenland
GD=Grenada
GP=Guadeloupe
GU=Guam
GT=Guatemala
GN=Guinea
GW=Guinea-Bissau
GY=Guyana
HT=Haiti
HN=Honduras
HK=Hong Kong, China
HU=Hungary
IS=Iceland
IN=India
ID=Indonesia
IQ=Iraq
IE=Ireland
IL=Israel
IT=Italy
JM=Jamaica
JP=Japan
JO=Jordan
KZ=Kazakhstan
KE=Kenya
KI=Kiribati
KR=Korea, South
KW=Kuwait
KG=Kyrgyzstan
LA=Laos
LV=Latvia
LB=Lebanon
LS=Lesotho
LR=Liberia
LY=Libyan Arab Jamahiriya
LI=Liechtenstein
LT=Lithuania
LU=Luxembourg
MO=Macao, China
MK=Macedonia
MG=Madagascar
MW=Malawi
MY=Malaysia
MV=Maldives
ML=Mali
MT=Malta
```

```
MH=Marshall Islands
MQ=Martinique
MR=Mauritania
MU=Mauritius
YT=Mayotte
MX=Mexico
FM=Micronesia
MD=Moldova, Republic Of
MC=Monaco
MN=Mongolia
ME=Montenegro
MS=Montserrat
MA=Morocco
MZ=Mozambique
MM=Myanmar
NA=Namibia
NR=Nauru
NP=Nepal
NL=Netherlands
NC=New Caledonia
NZ=New Zealand
NI=Nicaragua
NE=Niger
NG=Nigeria
NU=Niue
NF=Norfolk Island
MP=Northern Mariana Islands
NO=Norway
OM=Oman
PK=Pakistan
PW=Palau
PS=Palestinian Territory
PA=Panama
PG=Papua New Guinea
PY=Paraguay
PE=Peru
PH=Philippines
PN=Pitcairn
PL=Poland
PT=Portugal
PR=Puerto Rico
QA=Qatar
RE=Reunion
RO=Romania
RU=Russian Federation
RW=Rwanda
SH=Saint Helena
KN=Saint Kitts and Nevis
PM=Saint Pierre and Miquelon
WS=Samoa
SM=San Marino
ST=São Tomé and Príncipe
SA=Saudi Arabia
SN=Senegal
RS=Serbia
SC=Seychelles
SL=Sierra Leone
SG=Singapore
SK=Sint Maarten
SK=Slovakia
SI=Slovenia
SB=Solomon Islands
SO=Somalia
ZA=South Africa
ES=Spain
LK=Sri Lanka
SR=Suriname
SJ=Svalbard and Jan Mayen
```

```
SZ=Swaziland
SE=Sweden
CH=Switzerland
TW=Taiwan, China
TJ=Tajikistan
TZ=Tanzania
TH=Thailand
TL=Timor-Leste
TG=Togo
TK=Tokelau
TO=Tonga
TT=Trinidad and Tobago
TN=Tunisia
TR=Turkey
TM=Turkmenistan
TC=Turks and Caicos Islands
TV=Tuvalu
UG=Uganda
UA=Ukraine
AE=United Arab Emirates
GB=United Kingdom
US=United States
UY=Uruguay
UZ=Uzbekistan
VU=Vanuatu
VA=Vatican City State
VE=Venezuela
VN=Vietnam
VG=Virgin Islands, British
VI=Virgin Islands, U.S.
WF=Wallis and Futuna
EH=Western Sahara
YE=Yemen
ZM=Zambia
ZW=Zimbabwe
```

1.9.3 Developing Authorizations

Developers define, update, and create authorizations. These are the basis for Authorization Management, which administrators can use to grant users access to resources.

Authorization Management enables SAP Cloud Identity Services administrators to use instance-based authorizations in the form of authorization policies in multiple environments. They assign authorization policies to users and thus manage user access to resources.

Developers define and deploy an application that supports authorization policies in SAP Cloud Identity Services. They include functional checks, instance-base authorizations, and user attributes. Authorization Management is automatically enabled for a new tenant and can be used as soon as an application requests it. Service instances and bindings are added automatically. The authorization policies are available in the SAP Cloud Identity Services administration console.

Developing Authorization Policies

Developers can use the Cloud Application Programming model for authorization policies. We've provided a sample application, which you can use for the creation of authorization policies.

- Authorization and Access Control with the Cloud Application Programming model
- Sample apps in GitHub

> **Note**
>
> Changes in the assignment of authorization policies usually take up to 15 seconds to propagate due to caching delays. The delay can also be longer. So, respect this waiting time.

Defining Authorization Policies

The Data Control Language (DCL) enables developers to define authorization policies and rules using an SQL-like syntax in SAP Cloud Identity Services. This declarative language allows them to specify access controls for data and resources through structured authorization policy definitions.

- Define authorization policies using SQL-like syntax.
- Create reusable authorization policies with attributes and conditions.
- Enable administrators to combine and refine rules into custom authorization policies.
- Support user assignment in the administration console.

DCL provides essential elements including rules, resources, conditions, attributes, schemas, and packages for enterprise-grade access control implementation. For more information, see SAP Cloud Identity Services Developer Guide .

1.9.4 Add Logon Overlays in Customer Applications

This document describes how service providers that delegate authentication to Identity Authentication can use embedded frames, also called overlays, for the logon pages of their applications.

Prerequisites

- You have a customer application. For more information, see Application Types [page 63].
- You have added the domains of the applications for which you want to use overlays as trusted in the administration console for SAP Cloud Identity Services. For more information, see Configure Trusted Domains [page 2173].

Context

The use of overlays maintains the application context, by keeping the application page as dimmed background, to provide for minimum disturbance to the workflow. By default, after a successful logon via an overlay page,

the application's parent page reloads. For more information how to configure that option, see Enable or Disable Reload Parent Page Option [page 2044].

> **Note**
>
> When the application uses overlay for the logon page, but the client's browser doesn't accept third party cookies, the logon page opens in fullscreen.
>
> To open the logon page of the application in an overlay instead of in fullscreen when the browser is set not to accept third-party cookies, the user has to add an exception for the domain of this application. The users can consult the documentations of the different browsers for more information about how to enable third-party cookies for specific Web sites and domains.

To add a logon overlay into your application proceed as follows:

Procedure

1. Include the following libraries to the landing page of your application:
   - *jQuery*
     If you have already included *jQuery* you don't need to do it again.
     You can download the *jQuery* from a Content Delivery Network (CDN), or you can use the following pattern:

     > **Sample Code**
     >
     > ```
     > <script src="https://<tenant ID>.accounts.ondemand.com/ui/resources/
     > javascripts/jquery.min.js"></script>
     > ```

   - SAP_IDS.js
     Use the following pattern:

     > **Sample Code**
     >
     > ```
     > <script src="https://<tenant ID>.accounts.ondemand.com/ui/resources/
     > javascripts/SAP_IDS.js"></script>
     > ```

   > **Caution**
   >
   > Make sure that the reference to the javascript file is pointing to the same Identity Authentication tenant that is used for authentication of this application. Have this in mind when you migrate your applications from quality to productive environment, if different Identity Authentication tenants are used.

2. Add a logon link.

> ### Note
>
> The logon link must be an HTML anchor with the following attributes:

Attribute	Value
rel	IDS_login
href	• points to an actual resource in your application that generates SAML 2.0 authentication request to Identity Authentication • the name of the resource isn't important

> ### Sample Code
>
> ```
> <a href="/login.jspa" rel="IDS_Login">Log in</a>
> ```

Results

When the user chooses the logon link, the following happens:

- If the user isn't logged on, the log on overlay is displayed.
- If the user is logged on to an application with the same Identity Authentication tenant, in other words he or she has an active application session, the user is automatically logged on to the second application.

> ### Caution
>
> The logon link should be visible only if the user doesn't have an active application session.
>
> If you still want to show the logon link when the user has an active application session, you must change the logic for the logonon link for this case. The logon link should not contain `rel="IDS_Login"` in this case. For example, the logic could be that, when the user chooses the logon link, he or she is directly redirected to the protected resource.

Next Steps

Protect applications against clickjacking when using overlays. For more information, see Configure Clickjacking Protection [page 2882].

Further Options

Locale

If the locale is known, this can be communicated to Identity Authentication by adding a locale parameter to *SAP_IDS.js*.

Sample Code

```
<script src="https://<tenant ID>.accounts.ondemand.com/ui/resources/
javascripts/SAP_IDS.js?locale=en_GB"></script>
```

Note

The locale parameter follows the Java specifications for a locale and must be of the format `ll_CC` where:

- `ll` is the language two letter code in small letters
- `CC` is the country (region) two letter code in capital letters

Token URL Separator

If your application is using overlays, you can configure the delimiter used in the token URL sent for the different application processes, such as forgot password, for example.

You can choose from:

- `;` - parameter, the default configuration
- # - fragment
- `?/&` - query string parameter

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the application that you want to edit.

   ### Note

   Type the name of the application in the search field to filter the list items, or choose the application from the list on the left.

   If you don't have a created application in your list, you can create one. For more information, see Create a New Application [page 1922].
4. Choose the *Branding and Layout* tab.
5. Under *Email Configurations*, choose the *Token Url Separator* list item.
6. Choose a deliminator from the list:
   - `;` - parameter, the default configuration

- # - fragment
- ?/& - query string parameter

7. Save your selection.

1.9.5 Rate Limiting and Throttling

This section provides information on the rate limiting and throttling in Identity Authentication.

Requests

To ensure a safe and stable environment, all requests have a limit of 50 concurrent requests per second. The requests are associated with the originating IP address, and not with the user making the requests.

When the limit is exceeded, the client receives the `HTTP 429 Too Many Requests` response status code.

> ⚠ Restriction
>
> Executing performance tests on Cloud Identity Services tenants requires approval from SAP. For more information, see KBA 3667040 - Performance and Stress tests for SAP Cloud Identity Services.

SCIM API

To prevent Identity Authentication from being overloaded by too many requests, we have introduced SCIM API throttling to regulate the API usage per tenant.

SCIM API Throttling

Requests per minute	Notes
up to 200	The requests are carried out.
201–280	The requests are delayed by 1 second.
281 and more	The client receives the `HTTP 429 Too Many Requests` response status code for 60 seconds.

OpenID Connect (OIDC) Endpoints

To ensure a safe and stable environment, all requests to the `/oauth2` endpoints have a limit of 1000 requests per minute per tenant. When the limit is exceeded, the client receives the `HTTP 429 Too Many Requests` response status code.

For a list of the relevant endpoints, use the following URL:

`https://<tenant ID>.accounts.ondemand.com/.well-known/openid-configuration`

Tenant ID is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

If you've a custom domain configured, the URL has the following pattern:

`https://<your custom domain>/.well-known/openid-configuration`

SAML 2.0 Endpoints

To ensure a safe and stable environment, all requests to the `/saml2` endpoints have a limit of between 12000 and 15000 requests per minute per tenant. When the limit is exceeded, the client receives the `HTTP 429 Too Many Requests` response status code.

1.9.6 Configure Clickjacking Protection

Clickjacking is an attempt to trick users into clicking hidden user interface elements without the user realizing it. The user thinks he or she is clicking on the underlying frame, but is actually clicking on an action chosen by the attacker.

You have two options to protect your applications against clickjacking when using embedded frames, also called overlays, for the logon pages of the applications:

- If the applications are SAP UI5 or Web Dynpro, or they use the overlays of Identity Authentication, add the domains of these applications as trusted in the administration console for SAP Cloud Identity Services. For more information, see Configure Trusted Domains [page 2173].
- If the applications are not SAP UI5 or Web Dynpro, or they do not use the overlays of Identity Authentication, add the following code to your message handler:

Sample Code

```
function messageHandler(oEvent){
        if(oEvent.data=='SAPFrameProtection*require-origin'){
                oEvent.source.postMessage('SAPFrameProtection*parent-
origin','*');
        }
}
```

Related Information

Add Logon Overlays in Customer Applications [page 2877]

1.9.7 Error and Success Codes

This section is to help developers with solutions to the REST API response codes.

Error Codes

General Error Codes

Response Code	Meaning	Description
301	Permanent Location	The requested resource resides on a URI other than the requested one.You need to set up the authentication type to access the API.
400	Bad Request	The requested operation cannot be executed because the service cannot understand the data sent in the entity body of the request.
401	Unauthorized	The client is not authenticated.
403	Forbidden	Access to the resource is denied.
404	Not Found	The requested resource cannot be found.
405	Method Not Allowed	The requested method is not supported for the given resource.
406	Not Acceptable	You need to set up the authentication type to access the API.The requested method does not produce any of the media types requested in the HTTP request.
409	Conflict	The operation cannot be completed because it conflicts with an existing resource.
415	Unsupported Media Type	The REST service does not support the API version requested by the REST client.
500	Internal Server Error	The operation cannot be completed due to a service error.
503		The service is currently unavailable.

In addition to the general error codes, the REST APIs return one of the following detailed error codes as an *X-message-code* HTTP response header:

Detailed Error Codes

REST API	X-message code	Description
Invitation REST API	INVITATION_API_PARAMETER_INCORRECT	You have used an invalid parameter.
	INVITATION_API_EMAIL_INCORRECT	You have specified an incorrect address or the email address does not exist.
	PASSWORD_LOCKED	The password is locked for 60 minutes after 5 failed logon attempts with wrong value.
User Management REST API	PASSWORD_LOCKED	The password is locked for 60 minutes after 5 failed logon attempts with wrong value.
Update Group Resource SCIM REST API	Mismatched group id	The **id of the group** in the URI and the `id` of the group in the request body are different. Make sure that the `id` of the group is one and the same in both places.

Success Codes

Response Code	Meaning	Description
200	OK	Operation successful.
201	Created	Entity created successfully.
204	No Content	The service received and understood the request, but there is no need to send any content back.

Check also the troubleshooting scenarios available in the Guided Answers tool.

See Guided Answers.

Related Information

Getting Support [page 3141]

1.9.8 Exchanging Identity Authentication Tokens for Tokens from Corporate Identity Providers

This service endpoint returns the tokens issued by the corporate identity provider during the OpenID Connect (OIDC) authentication process. Use this endpoint when you integrate with external applications that don't accept tokens from Identity Authentication and require tokens issued from the corporate identity provider instead.

Prerequisites

- For your business application, there's an OIDC application in Identity Authentication.
  For more information, see Create OpenID Connect (OIDC) Application [page 1925].
- You prepared an authentication credential for the API call for your OIDC application in Identity Authentication. The API supports the following authentication schemes:
  - Basic
  - X.509
  - JWT Bearer

  For more information, see API Authentication [page 1996].
- Your business application has a corporate identity provider as the default identity provider using OIDC.
  For more information, see Configure Trust with OpenID Connect Corporate Identity Provider [page 2579] and Choose Default Identity Provider for an Application [page 2067].
- The user has authenticated to this application using Identity Authentication and the corporate OIDC identity provider.

> **Note**
>
> This endpoint works only with a corporate identity provider configured for OIDC. It isn't supported for SAML 2.0 corporate identity providers.

Call Corporate Identity Provider Endpoint

URI: `https://<Cloud Identity Services domain>/oauth2/exchange/corporateidp`

> **Note**
>
> The domain part has the following pattern:
>
> `<tenant ID>.accounts.ondemand.com` or `<tenant ID>.accounts.cloud.sap`. If you have a configured custom domain, the domain has the following pattern: <your custom domain>.
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].

HTTP Method: *POST*

Request Headers

Header	Required	Values
`Content-Type`	Yes	application/x-www-form-urlencoded

Request Parameters

Parameter	Required	Data Type	Description	Parameter Type
`assertion`	Yes	string	Provide the Identity Authentication user token that identifies Identity Authentication session or refresh token.	Request body
`response_type`	No	string	Specifies the token requested from the corporate identity provider. By default, only the `access_token` is requested. Otherwise, you can request the ID token or both. Possible values: • `token` • `id_token` • `token id_token` Separate multiple values with a space ( ).	Request body
`scope`	No	string	Overwrites the default scopes defined in the Corporate IdP configuration. This allows to receive only specific scopes from the IdP.	Request body

Request Example

```
POST /oauth2/exchange/corporateidp
Content-Type: application/x-www-form-urlencoded
Authorization: Bearer eyJbiOi…

response_type=token id_token&
assertion=eyJ0eab…
```

Result

```
Content-Type: application/json

{
    "access_token": "YTU1OGU0NzdjYzRkNjA4YzI3ODk4MWEzM2UyYmFlYTI
w",
    "id_token": "eyJraWQiOiJUdDV0Wkt5RHlxSzV4em93VUhrdkNBS3c3bkE
iLCJhbGciOiJSUzI1NiJ9.eyJhdWQiOiJUMTIzNDU2Iiwic3ViIjoibWFuYWdlci
IsIm1haWwiOiJkb25hLm1vb3JlQGV4YW1wbGUuY29tIiwiaXNzIjoiZG9tYS1tYW
```

```
4ubWFuLXVuaS51cy5ldS1wdC0xLmNsb3VkLmNvbSIsImxhc3RfbmFtZSI6Ik1vb3
JlIiwiZXhwIjoxNTQ4NjgzNzE4LCJpYXQiOjE1NDg2ODM0MTgsImZpcnN0X25hbW
UiOiJEb25hIiwianRpIjoiZmIxMjM0YTUtMTIzYS0xYmQzLWExYWItMGExMjNiYz
Q1ZDZlIn0.3ormYv_nxnimm6xB972qSGOk_YTc_gpE4wVhgaA7q0I",
    "token_type": "Bearer",
    "expires_in": 3569
}
```

Related Information

Configuring OpenID Connect (OIDC) [page 2636]

1.9.9 Single Logout Flows

Encourage users of your applications to log out at the end of their session. Malicious users can impersonate session owners if they gain access to session data (such as cookies) or find unattended clients.

To counter this threat, ensure that your application participates in single logout (SLO) flows. When users log out from your application, your application notifies Identity Authentication. The service cleans up the user session and triggers logout on any identity providers and applications associated with the session. The service forwards users of your application to a URL.

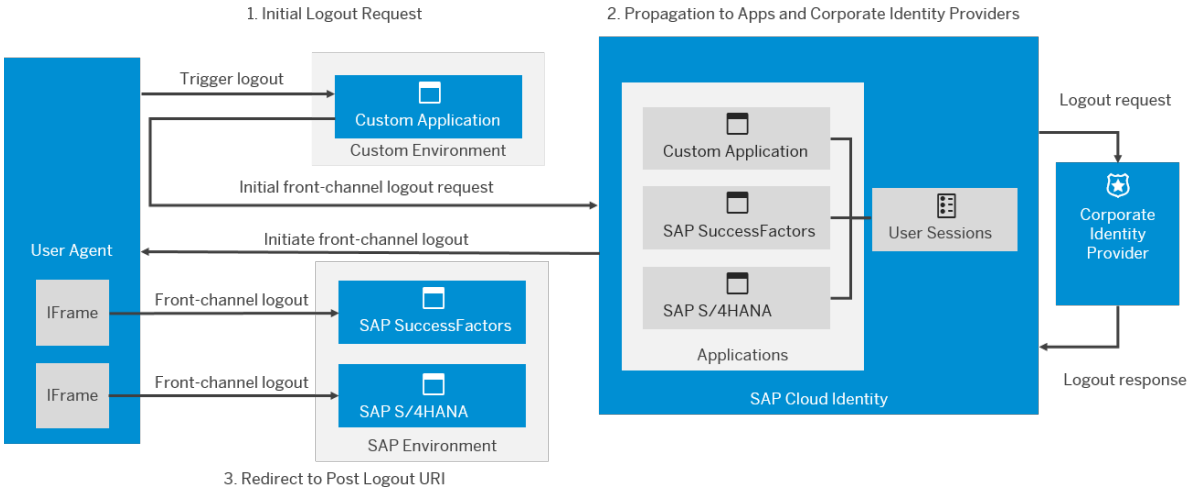
The service supports SAML 2.0 and OIDC logout flows. Regardless of which protocol your applications and identity providers use, the service forwards the logout requests to any corporate identity providers and applications associated with the user session.

> ⓘ Note
>
> SLO ends the sessions of all applications under the same domain as Identity Authentication (`ondemand.com` and `cloud.sap`) and under custom domains configured for the service. To have applications participate in SLO, ensure that the applications operate under these domains (recommended) or allow third-party cookies.

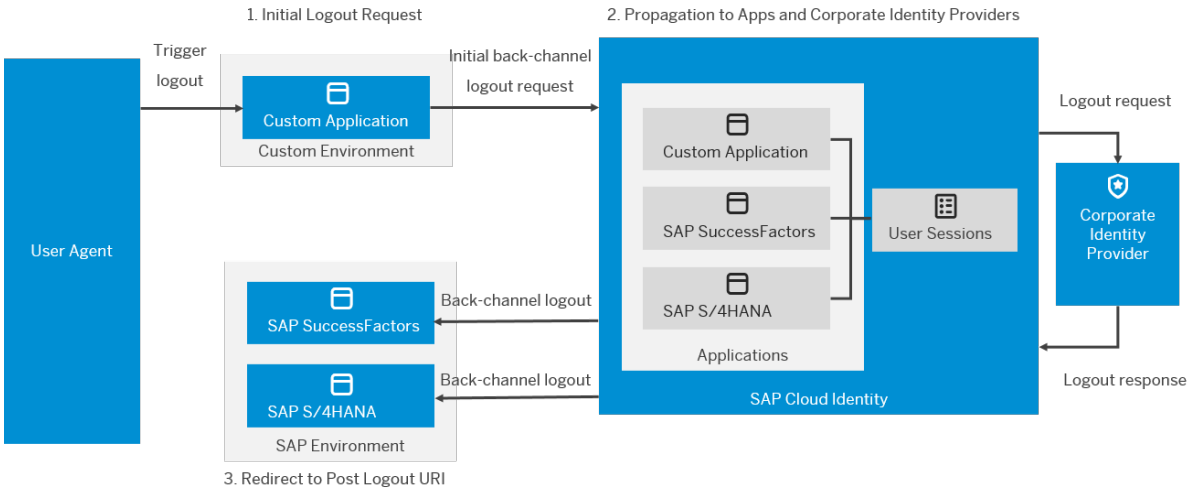
If you define both channels, the system executes both channels.

> → Recommendation
>
> Enabling both channels sends unnecessary messages over the network, since applications only need one logout message. Applications could return unexpected results when they receive a second logout request unless they were designed to handle such requests.
>
> We recommend back-channel logout flows where possible, but applications must be designed to support them.
>
> If you have to manually configure logout flows, check the documentation of your application to determine which logout flows it supports.

The following figures illustrate a logout scenario with multiple applications and a corporate identity provider:



Front-Channel Logout Flow



Back-Channel Logout Flow

- SAML applications: Service Provider-Initiated Logout with Corporate Identity Providers [page 2596]
- OIDC applications: Call Identity Authentication End Session Endpoint [page 2740], Handle Single Logout Request from Identity Authentication [page 2743]
- Cloud Identity Services-initiated SLO: Use IdP-Initiated Single Logout (SLO) [page 2089]

Related Information

Use Custom Domain in Identity Authentication [page 2175]

1.9.10 Consuming APIs from Other Applications

Applications sometimes need to propagate principals or have technical communication arrangements between them. To enable OpenID Connect (OIDC) applications to consume the APIs of other applications, the developer defines API permission groups for a provider application, which a consumer application can consume through a defined dependency.

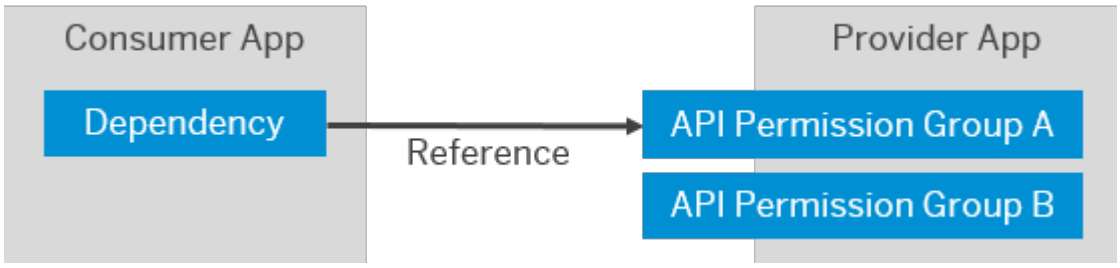
Design Options

In this scenario, you have a provider application that provides a group of APIs and a consumer application that consumes them. Tenant administrators grant access to the consumer applications for this group. You can take one of the following approaches:

- Technical communication that retrieves a new token with a client credential flow
- Principal propagation with a choice between the JWT bearer and token exchange flows

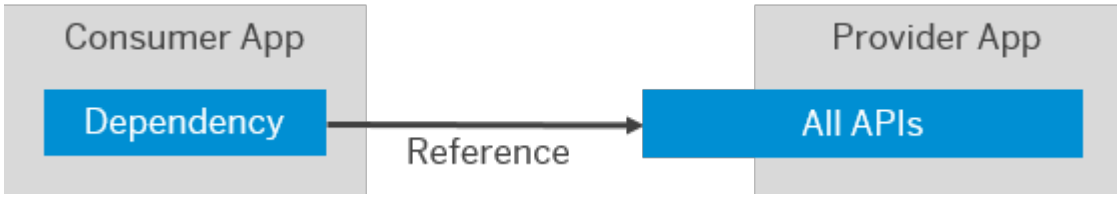
In either case, the audience of the token matches the provider application. In addition, the `ias_apis` claim lists the granted API permission groups or a fixed value for when all APIs are consumed.

The following figure shows the relationship between two applications in SAP Cloud Identity Services. The provider application offers two API permission groups. The dependency of the consumer application refers to one of these groups.



App-to-App Integration of API Permission Groups

Alternatively, if your provider application doesn't need to distinguish different API permission groups and is consumed with principal propagation, you can make all APIs available. The provider application must check the principal propagated by the consumer application for the authorizations required by the APIs. In the following figure, the provider application doesn't offer any API permission groups. The consumer application refers to all the available APIs. The provider application depends on the authorizations of the current user when these APIs are accessed.



App-to-App Integration of All APIs

Configuration Precedence

When you configure both the consumer and provider applications, each application contributes specific OIDC and related configurations. The following table shows which application provides each configuration.

Configuration Source by Application

Configuration	Application	More Information
Subject name identifier	Provider	Configure the Subject Name Identifier Sent to the Application [page 1988]
Token attributes	Provider	User Attributes [page 1972]
Token format	Provider	Token Policy Configuration for Applications [page 2752]
Token grant type	Consumer	Token Policy Configuration for Applications [page 2752]
Token policy	Consumer	Token Policy Configuration for Applications [page 2752]

At Runtime

At runtime, the consumer application fetches a token according to the appropriate flow:

- Use the client credential flow for technical communication.
  For more information, see Using Client Credentials Flow [page 2660].
- Choose the appropriate flow for principal propagation.
  - If the consumer application already issues a JWT token, use the JWT bearer flow.
    For more information, see Using JWT Bearer Flow [page 2689].
  - If the consumer application issues another token type, such as SAML2 assertion to bearer token, use the token exchange flow.
    For more information, see Using Token Exchange [page 2701].

In either case, use the `resource` parameter to identify the dependency consumed by the consumer application. The `resource` parameter is a Uniform Resource Name (URN).

For example: `urn:sap:identity:application:provider:name:<Dependency_name>`

The dependency name is the name that the tenant administrator configures for the dependency between the consumer application and the provider application.

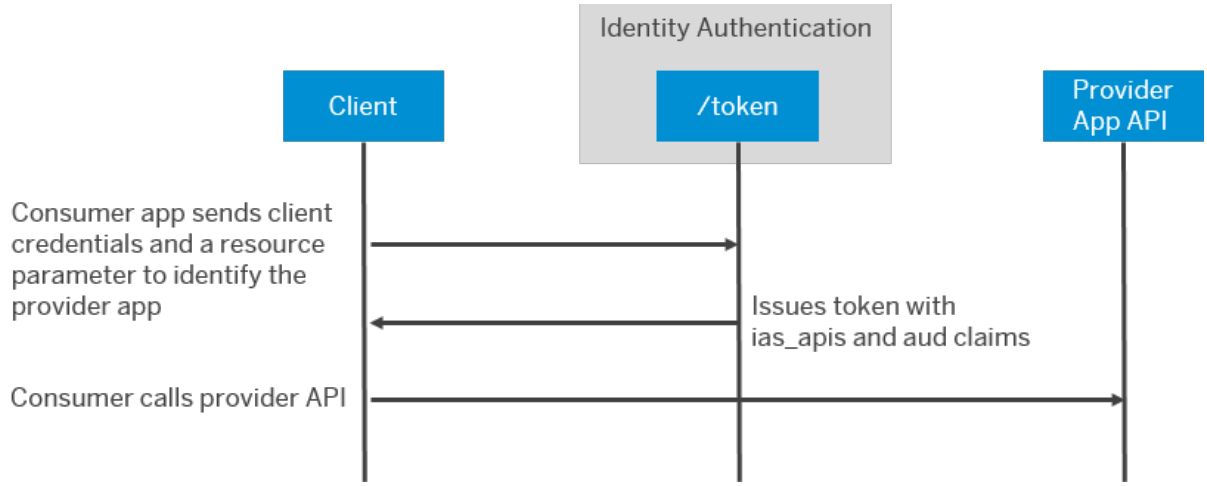
> → Tip
>
> Document the name in a scenario guide so that administrators know which name to use when creating dependencies. If you have experience working with destinations, you're already used to this approach.

In the unusual case that you need a token that reflects multiple dependencies to the same provider application, you can replace `name:<Dependency_name>` with `clientid:<client_id>`. Use the client ID of the provider application in Identity Authentication.

The service returns a token, which includes the following relevant claims:

- The audience (`aud`) claim of the provider application.
- The `ias_api` claim, which includes an array of API permission groups of the provider application.

The consumer application uses this token to call the API endpoint of the provider application. The provider application validates the token and its claims and processes the call. The following figure illustrates this scenario for technical communication between systems.



Technical Communication Sequence Between Applications

Related Information

Provide APIs for Consumption by Other Applications [page 2891]
Consume an API from a Provider Application [page 2893]

1.9.10.1 Provide APIs for Consumption by Other Applications

SAP Cloud Identity Services can help you expose APIs of your application to other applications. You can expose APIs with an API permission group or tie the access to the authorizations of the current user (principal propagation).

Context

You can protect APIs by adding a name for an API permission group. Other applications that are allowed to consume this API permission group pass a token issued by SAP Cloud Identity Services with this name as a claim. Your application must evaluate the `ias_apis` claim for this name to allow access.

If the consuming application is using a principal propagation scenario and not a technical user scenario, your application can check the authorizations assigned to the current user. You can also offer all APIs of the provider application without permission groups. In this case, `principal-propagation` is written in the `ias_apis` claim. Your application must rely on the authorizations available to the current user alone.

As the providing application, you have the following options for configuring identity authentication for API permission groups:

- Use the administration console for SAP Cloud Identity Services as described in the following procedure.
- The provider application can set the API permission groups with the parameters of the Identity service of SAP BTP.
  The following example shows an API permission group named `sales-orders`.

  ```
  "provided-apis": [
    {
      "name": "sales-orders",
      "description": "Access sales orders"
    }
  ]
  ```

  For more information, see Reference Information for the Identity Service of SAP BTP [page 2921].

> → Remember
>
> The provided APIs for the `Administration Console` for Cloud Identity Services is read-only.

Procedure

1. In the administration console for SAP Cloud Identity Services, choose *Applications and Resources* > *Applications*
2. Choose the provider application.
3. Choose the *Trust* tab.
4. Under *Application APIs*, choose *Provided APIs*.
5. Verify that the name of the API permission group that you check in your application is listed.

   Otherwise enter the required data and save your entries.

   > ⚠ Caution
   >
   > The API name must match exactly what is expected by consumer applications. The name must be unique within all APIs provided by the same provider application. Consumer applications use this name to determine if their application has the rights to access the provider application.
   >
   > The name can be any URN-compliant string of up to 50 characters. You can define a maximum of 50 APIs.
   >
   > For more information about URNs, see RFC 8141 .

   To rely solely on the permissions of the current user, select the *Allow all APIs for principal propagation* option. Identity Authentication adds `principal-propagation` to the `ias_apis` claim.

1.9.10.2 Consume an API from a Provider Application

Your consumer application can request an access token from Identity Authentication to consume the API of a provider application.

Prerequisites

You know the name of the dependency defined in the application configuration of SAP Cloud Identity Services. The tenant administrator enters the dependency name in the application configuration. As the owner of the consumer application, you must provide documentation to enable the administrator to do the following:

- Enter the name that you expect.
- Configure the same dependency name in your application.

For more information, see Configure Integration Between Applications [page 2092].

Procedure

1. Develop your consumer application to call the `/oauth2/token` endpoint of Identity Authentication with the dependency name.

   Example: Technical Communication

   A consumer application (client ID 3ab1ba0e-c573-4ad5-20a4-28aa4e587a19) has a dependency to the provider application (client ID 77f42d85-7da7-1aa1-84a2-e65a76308cd3) named **`techDependency`**. The application calls the `/oauth2/token` endpoint with its client ID and the `resource` parameter with the dependency name used to identify the API that the consumer application consumes.

   ```
   curl  $iashost/oauth2/token -X POST \
      -H 'Content-Type: application/x-www-form-urlencoded' \
      -H 'Accept: application/json' \
      -d "client_id=3ab1ba0e-c573-4ad5-20a4-28aa4e587a19" \
      -d "client_secret=$secret" \
      --data-urlencode 'grant_type=client_credentials' \
      --data-urlencode
   'resource=urn:sap:identity:application:provider:name:techDependency'
   ```

   Identity Authentication returns a token with the access control for the APIs and audience claim. The audience claim contains the client ID of the provider application. The following is an example of a token with the relevant claims.

   ```
   {
       "ias_apis": [
           "apiPermissionGroup"
       ],
       "sub": "3ab1ba0e-c573-4ad5-20a4-28aa4e587a19",
       "aud": "77f42d85-7da7-1aa1-84a2-e65a76308cd3",
       "azp": "3ab1ba0e-c573-4ad5-20a4-28aa4e587a19",
   }
   ```

Example: Principal Propagation:

A consumer application (client ID e7c7e327-86c0-48a6-af57-a1234b567869) has a dependency to the provider application (client ID 7ab2cd3e-c573-4ad5-9007-28aa4e587a19) named **`myDependency`**. The application calls the `/oauth2/token` endpoint with its client ID and the `resource` parameter with the dependency name used to identify the API that the consumer application consumes.

```
curl $iashost/oauth2/token  -X POST \
    -H 'Content-Type: application/x-www-form-urlencoded' \
    -H 'Accept: application/json' \
    -d "client_id=e7c7e327-86c0-48a6-af57-a1234b567869" \
    -d "client_secret=$secret" \
    -d "grant_type=urn%3Aietf%3Aparams%3Aoauth%3Agrant-type%3Ajwt-bearer" \
    -d "assertion=$BEARERTOKEN" \
    -d "resource=urn:sap:identity:application:provider:name:myDependency"
```

Identity Authentication returns a token with the access control for the APIs and audience claim. The audience claim contains the client ID of the provider application. The following is an example of a token with the relevant claims.

```
{
    "ias_apis": [
        "apiPermissionGroup"
    ],
    "sub": "P123456",
    "mail": "donna.moore@example.com",
    "last_name": "Moore",
    "aud": "7ab2cd3e-c573-4ad5-9007-28aa4e587a19",
    "scim_id": "1a2345bc-4f6b-456b-9bfb-5d58beb97fb7",
    "user_uuid": "1a2345bc-4f6b-456b-9bfb-5d58beb97fb7",
    "azp": "e7c7e327-86c0-48a6-af57-a1234b567869",
    "first_name": "Donna",
    …
}
```

2. Send the token to the API endpoint of the provider application.

Related Information

Consuming APIs from Other Applications [page 2889]

1.9.10.3 Consume APIs from the Administration Console

Your consumer application can request an access token from Identity Authentication to consume the APIs of the Administration Console for Cloud Identity Services acting as a provider application.

Prerequisites

You have configured the integration between the `Administration Console` application in the tenant and your consumer application that will use the APIs of the `Administration Console`. For more information, see Configure Integration Between Applications [page 2092].

Context

Cloud Identity Services have exposed the Identity Directory API, Corporate Identity Providers API, and SAP Cloud Identity Services Application Directory as provided APIs for the Administration Console application. These APIs can be used for principal propagation. For more information, see Consuming APIs from Other Applications [page 2889].

> **Note**
>
> Tenant administrators with **Manage Users** and **Manage Groups** roles have full access to the Identity Directory API. For all other users, the Identity Directory API supports *GET* and *PATCH* operations about the user for whom the token is issued, and the following attributes of the user can be returned or updated:
>
> - `urn:ietf:params:scim:schemas:core:2.0:User:locale`
> - `urn:ietf:params:scim:schemas:extension:sap:2.0:User:localeSettings.currency`
> - `urn:ietf:params:scim:schemas:extension:sap:2.0:User:termsOfUse`
> - `urn:ietf:params:scim:schemas:extension:sap:2.0:User:privacyPolicy`

Procedure

1. Develop your consumer application to call the `/oauth2/token` endpoint of Identity Authentication with the respective dependency name.

   > **Note**
   >
   > The available names are: `scim`, `identityproviders`, `applications`.

   Example: Principal Propagation:

   A consumer application (client ID 8bbc6f92-0222-46f1-b9c5-b2b48825e78b) has a dependency to the Administration Console application acting as a provider application (client ID 09b90b70-3e4c-4fbc-b4ac-

ef66848b7fbc) named **`applications`**. The application calls the `/oauth2/token` endpoint with its client ID and the `resource` parameter with the dependency name used to identify the API that the consumer application consumes.

```
curl $iashost/oauth2/token  -X POST \
                        -H 'Content-Type: application/x-www-form-urlencoded' \
                        -H 'Accept: application/json' \
                        -d "client_id=8bbc6f92-0222-46f1-b9c5-b2b48825e78b" \
                        -d "client_secret=$secret" \
                        -d "grant_type=urn%3Aietf%3Aparams%3Aoauth%3Agrant-
type%3Ajwt-bearer" \
                        -d "assertion=$BEARERTOKEN" \
                        -d
"resource=urn:sap:identity:application:provider:name:applications"
```

Identity Authentication returns a token with the access control for the API and audience claim. The audience claim contains the client ID of the provider application. The following is an example of a token with the relevant claims.

```
{
                            "ias_apis": [
                            "applications"
                            ],
                        "sub": "P123456",
                        "mail": "donna.moore@example.com",
                        "last_name": "Moore",
                        "aud": "09b90b70-3e4c-4fbc-b4ac-ef66848b7fbc",
                        "scim_id": "1a2345bc-4f6b-456b-9bfb-5d58beb97fb7",
                        "user_uuid": "1a2345bc-4f6b-456b-9bfb-5d58beb97fb7",
                        "azp": "8bbc6f92-0222-46f1-b9c5-b2b48825e78b",
                        "first_name": "Donna",
                        …
                        }
```

2. Send the token to the API endpoint of the provider application.

Related Information

Consuming APIs from Other Applications [page 2889]

1.10 Security

This section describes the security features of SAP Cloud Identity Services.

In this section:

- Security Information [page 2897]
- Data Protection and Privacy [page 2901]

1.10.1 Security Information

This document is an overview of security-relevant information that applies to Identity Authentication, and contains recommendations about how administrators should secure it.

While this document is intended to support administrators and security responsible specialists to use Identity Authentication in a secure manner, there may be further elements to consider depending on usage, integrations as well as applicable industry or local legal regulations.

Updates and Notifications

For more information, see Updates and Notifications [page 3142].

Before You Start

Before you secure Identity Authentication, protect the cloud application that trusts Identity Authentication. For more information about protecting SAP BTP applications, see Security, Neo Environment.

Security Recommendations

SAP BTP Security Recommendations collects information which helps you to secure the configuration and operation of SAP BTP services in your landscape.

For more information, see Security Recommendations - Identity Authentication and Security Recommendations - Identity Provisioning and Monitoring with SAP Cloud ALM [page 2991].

Communication Protocol

Identity Authentication is fully web browser based application, with all access over HTTPS. Every page of the Identity Authentication application is currently delivered via Transport Layer Security (TLS). Access to Identity Authentication is encrypted-in-transit over HTTPS using 256-bit TLS encryption.

By default, the Identity Provisioning uses secure communication channels. Still, when connecting to customer systems, you decide (define) what the communication channel to be.

Recommendations

- Always use secure protocols when specifying your connection details (in the cockpit → *Destinations* section, in the Identity Provisioning UI → *Properties* tab).
- Always re-enter the values of the configured credential properties when you update the **URL** or host name of an existing provisoning system in the *Properties* tab of the Identity Provisioning UI. The only exception to this are the credential properties of systems that are created with a connectivity destination.

- Avoid using property `TrustAll` in productive scenarios. When it's set to **true**, the SSL server certificate is not verified, and thus the server is not authenticated.

User Administration

Set user permissions in accordance with the scenario you are configuring. For more information, see Scenarios [page 544].

You have the following options:

- For a business-to-consumer scenario, allow *Public* user application access.
- For a business-to-business scenario, allow *Private* user application access.
- For a business-to-employee scenario, allow *Internal* user application access.

For more information about the settings for user application access, see Configure User Access to the Application [page 2025].

You can also define rules for authentication according to different risk factors. For more information, see Configure Risk-Based Authentication for an Application [page 2007].

User Authentication

Identity Authentication protects your users during authentication in the following ways:

- With SAML 2.0 or OpenID Connect
  Identity Authentication supports the single sign-on (SSO) mechanism. Every user with an account is able to use SSO for the cloud applications that use Identity Authentication.
- With an application certificate
  The Identity Authentication authentication interfaces that receive REST API calls are protected, since they require an application certificate. Use the certificate when implementing REST APIs for your application. For more information, see the REST API configurations in Development [page 2779].

Authorizations

Applications configured in Identity Authentication can manage business user authorizations for applications. For such applications, the *Authorization* tab shows a list of authorization policies created by the application. You can also create custom authorization policies. These authorization policies are stored as user groups. You can assign authorization policies to business users with the admin console, but you can also use a SCIM API.

Password Security

Identity Authentication does not store plain text passwords in the database, but only their iterated random-salted secure hash values. The random salt is at least 512 bits, and it is different for each password. Only

generic hash functions are used with minimum 512 bits key length. No default passwords are delivered, used, or accepted anywhere.

Identity Authentication can use also passwords from on-premise systems for user authentication. These passwords are not stored by Identity Authentication. It sends the user ID and the password for authentication to the on-premise system via the Transport Layer Security (TLS) connection. The management of these passwords depends on the integrated on-premise system that supports them, for example Microsoft Active Directory.

Identity Authentication supports three levels of password security. You should use the highest level of security that matches the requirements of your application. The passwords are managed based on password policy rules. For more information, see Configuring Password Policies [page 2223].

Encryption

When using Identity Provisioning to configure a provisioning system, always set credentials (such as passwords, private keys and OAuth secrets) as *Credential* properties. When you add a credential property, its value is displayed as an encrypted string. For better security, the encrypted string is always displayed as 40 characters, no matter how long your real password is.

Session Security

Session cookies in Identity Authentication are protected with a Transport Layer Security (TLS) and with the *Secure* and *HttpOnly* attributes. You don't need to make any additional configurations for Identity Authentication.

Network and Communication Security

Identity Authentication is setup in a fenced network, separated from the SAP internal network.

Customer applications run in a shared environment where the business data is isolated from each other, the SAP BTP services uses a shared SAP BTP infrastructure. The internal traffic is controlled by firewalls. SAP administrative access is done via terminal service that requires strong authentication.

All communication channels are protected with TLS, and you should configure the cloud application to use TLS and to check the SAML 2.0 signature.

Data Storage Security

Data storage security is about how Identity Authentication protects its own database. Data storage security is ensured by the isolated tenant that each customer receives. Only tenant-specific requests can access the tenant's database. These requests are performed by a tenant service, which works with a dependency injection

framework and makes sure that all the services, for example the persistence service and the mail service, are injected with the instances dedicated to the given tenant.

In Identity Provisioning, no personal or sensitive information about the provisioned entities is saved. To check whether any changes have been made to an entity after the initial provisioning, Identity Provisioning uses strong hashed algorithm for the provisioned entities.

If a provisioning job repeatedly fails and you need problem investigation, you can enable detailed entity tracing. That means, the Identity Provisioning service will log the complete information (general, personal and sensitive data) of your provisioned entities. For example, if *"groups": XXX* are not an array but have a string value instead; or an attribute that must have a string value, has an integer value instead. Tracing such data could help you identify potential incorrect attribute values for a certain entity, which you can correct in the source system or via the transformation functions.

To activate entity tracing, proceed as follows:

1. In your source system, set property **ips.trace.failed.entity.content** to *true*.
2. Run again the provisioning job.
3. Open the *Job Logs* section, select your job, and under *Failed Entities*, choose an entity and find the log information about it.
4. If you cannot resolve the problem, contact the Identity Provisioning support.

> **Note**
>
> The operators may need the full trace content, so they can ask you to set the property in your target system as well, and once again run the provisioning job.

Security-Relevant Logging and Tracing

You can download a CSV file with a history of operations performed by administrators. For more information, see Export Change Logs with a History of Administration Operations [page 2976].

You can retrieve the statistics on the number of user logon request per month. This number is counted on each single authentication managed via Identity Authentication. For more information, see View Usage Statistics [page 2966].

Related Information

Security on SAP Community Network
SAP Data Center
SAP Security Notes
SAP Security Certificates

1.10.2 Data Protection and Privacy

Governments place legal requirements on industry to protect data and privacy. We provide features and functions to help you meet these requirements.

> ⓘ Note
>
> SAP does not provide legal advice in any form. SAP software supports data protection compliance by providing security features and data protection-relevant functions, such as blocking and deletion of personal data. In many cases, compliance with applicable data protection and privacy laws is not covered by a product feature. Furthermore, this information should not be taken as advice or a recommendation regarding additional features that would be required in specific IT environments. Decisions related to data protection must be made on a case-by-case basis, taking into consideration the given system landscape and the applicable legal requirements. Definitions and other terms used in this documentation are not taken from a specific legal source.

Handle personal data with care. You as the data controller are legally responsible when processing personal data. It is not permitted to deal with sensitive personal data in Identity Authentication.

> → Remember
>
> Custom attributes must not be used to store sensitive personal data. For more information, see Configuring User Attributes from the Identity Directory [page 1973].

Glossary for Data Protection and Privacy [page 2902]
The following terms are general to SAP products. Certain terms might not be relevant for Identity Authentication.

Change Logging and Audit Logging [page 2904]
Change logging records changes to personal data, while audit logging provides access to personal data, successful, and failed authentications. You may be required to gather this information for auditing purposes or legal requirements.

Information Report [page 2904]
An information report is a collection of data relating to a data subject. A data privacy specialist may be required to provide such a report or an application may offer a self-service.

Erasure [page 2905]
When handling personal data, consider the legislation in the different countries where your organization operates. After the data has passed the end of purpose, regulations may require you to delete the data. However, additional regulations may require you to keep the data longer. During this period, you must block access to the data by unauthorized persons until the end of the retention period, when the data is finally deleted.

Consent [page 2906]
We assume that software operators, such as SAP customers, collect and store the consent of data subjects, before collecting personal data from data subjects. A data privacy specialist can later determine whether data subjects have granted, withdrawn, or denied consent.

Customer Data [page 2906]

1.10.2.1 Glossary for Data Protection and Privacy

The following terms are general to SAP products. Certain terms might not be relevant for Identity Authentication.

Term	Definition
Blocking	A method of restricting access to data for which the primary business purpose has ended.
Business purpose	A legal, contractual, or in other form justified reason for the processing of personal data. The assumption is that any purpose has an end that is usually already defined when the purpose starts.
Consent	The action of the data subject confirming that the usage of his or her personal data shall be allowed for a given purpose. A consent functionality allows the storage of a consent record in relation to a specific purpose and shows if a data subject has granted, withdrawn, or denied consent.
Deletion	Deletion of **personal data** so that the data is no longer available.
End of business	Date where the business with a data subject ends, for example the order is completed, the subscription is canceled, or the last bill is settled.
End of purpose (EoP)	End of purpose and start of blocking period. The point in time, when the primary processing purpose ends (for example contract is fulfilled).
End of purpose (EoP) check	A method of identifying the point in time for a data set when the processing of **personal data** is no longer required for the primary **business purpose**. After the **EoP** has been reached, the data is **blocked** and can only be accessed by users with special authorization (for example, tax auditors).
Personal data	Any information relating to an identified or identifiable natural person ("data subject"). An identifiable natural person is one who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural, or social identity of that natural person
Purpose	The information that specifies the reason and the goal for the processing of a specific set of personal data. As a rule, the purpose references the relevant legal basis for the processing of personal data.

Term	Definition
Residence period	The period of time between the end of business and the end of purpose (EoP) for a data set during which the data remains in the database and can be used in case of subsequent processes related to the original purpose. At the end of the longest configured residence period, the data is blocked or deleted. The residence period is part of the overall retention period.
Retention period	The period of time between the end of the last business activity involving a specific object (for example, a business partner) and the deletion of the corresponding data, subject to applicable laws. The retention period is a combination of the residence period and the blocking period.
Sensitive personal data	A category of personal data that usually includes the following type of information: • Special categories of personal data, such as data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic data, biometric data, data concerning health or sex life or sexual orientation. • Personal data subject to professional secrecy • Personal data relating to criminal or administrative offenses • Personal data concerning insurances and bank or credit card accounts
Where-used check (WUC)	A process designed to ensure data integrity in the case of potential blocking of business partner data. An application's where-used check (WUC) determines if there is any dependent data for a certain business partner in the database. If dependent data exists, this means the data is still required for business activities. Therefore, the blocking of business partners referenced in the data is prevented.

1.10.2.2 Change Logging and Audit Logging

Change logging records changes to personal data, while audit logging provides access to personal data, successful, and failed authentications. You may be required to gather this information for auditing purposes or legal requirements.

Audit Logs

Tenant administrators can access the audit logs for changes in the personal data, successful, and failed authentications in Identity Authentication. If you want to view the audit logs, you should generate Client ID and Client Secret for audit logs in the administration console for SAP Cloud Identity Services, obtain an access token, and call the audit log retrieval API to access the data. For more information, see Access Audit Logs (Audit Log Service in SAP BTP, Neo) [page 2972].

The audit log entries are retained for:

- SAP BTP, Cloud Foundry environment (for tenants on the AWS and Azure infrastructure.) - 90 days
- SAP BTP, Neo environment (for tenants on the SAP infrastructure)- 201 days

Change Logs

Tenant administrators can access information about configuration changes made by administrators in Identity Authentication. You can download a CSV file with a history of the operations performed by administrators. The change logs are stored in the data base for 24 months. For more information, see Export Change Logs with a History of Administration Operations [page 2976].

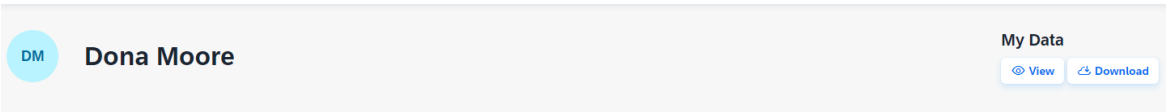
1.10.2.3 Information Report

An information report is a collection of data relating to a data subject. A data privacy specialist may be required to provide such a report or an application may offer a self-service.

Information About Own Personal Data

Data subjects can obtain information about their personal data in Identity Authentication, via the profile page at https://<tenant ID>.accounts.ondemand.com/.

- Users can open an overview of all the data for them by visiting their account and chosing the *View* button under the *My Data* section on the screen.



> ### Note
>
> The data includes the **Terms of Use** and **Privacy Policy** documents singed by the user. The documents can be seen by clicking on their names. They open in a new tab.

- Users can download an overview of all the data for them on a JSON file by clicking the *Download* button in the *My Data* section.

Information About Users' Data

Tenant administrators can view detailed information about the users in the administration console for SAP Cloud Identity Services. Optionally they can edit this information. For more information, see List and Edit User Details [page 2361].

> ### Note
>
> Currently, the Information Report is not applicable for Identity Provisioning.

The Identity Provisioning only transfers entities from a source system to a target one. The applications representing these source and target systems may provide data about the provisioned entities but this data is only stored in the systems, not in the Identity Provisioning itself.

1.10.2.4 Erasure

When handling personal data, consider the legislation in the different countries where your organization operates. After the data has passed the end of purpose, regulations may require you to delete the data. However, additional regulations may require you to keep the data longer. During this period, you must block access to the data by unauthorized persons until the end of the retention period, when the data is finally deleted.

Only the service provider is aware of the retention policies of the user, and thus whether a user record has to be blocked first, and deleted at a certain time in the future, or whether a user record can be deleted immediately.

The end user who wants to cancel his or her registration from the service should start the process via the service provider, not via Identity Authentication. Identity Authentication keeps user information for every service provider. Each service provider deletes its specific data stored about the user in Identity Authentication via an API call. If the user has accounts in more than one service provider, his or her whole user profile in Identity Authentication is deleted when the information for the last service provider is deleted. For more information, see SP User Deletion [page 2817].

Apart from deleting, users can be blocked. When users are blocked, they exist in Identity Authentication, but cannot be authenticated when they try to log on. For more information, see SP User Deactivation [page 2815].

Tenant administrators can delete and block users via the administration console for SAP Cloud Identity Services. For more information, see Delete Users [page 2367] and Deactivate Users [page 2368].

Tenant administrators can also configure blocking and deletion for specific user types and after certain period of time, For more information, see Block or Delete Users Due to Inactivity [page 2366].

1.10.2.5 Consent

We assume that software operators, such as SAP customers, collect and store the consent of data subjects, before collecting personal data from data subjects. A data privacy specialist can later determine whether data subjects have granted, withdrawn, or denied consent.

Every (service provider) application that trusts Identity Authentication can configure its own privacy policy and terms of use documents. For more information, see Configuring Privacy Policies [page 2235], and Configuring Terms of Use [page 2273].

Identity Authentication takes care the documents to be accepted by the users. Whenever a new version of the document becomes available, an upgrade process is started.

The end users can view the signed **Terms of Use** and **Privacy Policy** documents by them at the profile page. For more information, see Information Report [page 2904].

The tenant administrators can view a history of the signed **Terms of Use** and **Privacy Policy** documents of the users in the administration console for SAP Cloud Identity Services. The information is in the format *<document name><document version><time stamp>*.

> **Note**
>
> The information about when the user has accepted a particular document is available for documents accepted after May 16, 2018.

To view that information, access *Administration Console* > *Users & Authorizations* > *User Management* > *User Name* > *Legal tab*. For more information about how to view the user information in the administration console, see List and Edit User Details [page 2361].

1.10.2.6 Customer Data

Customer can request export of its customer data in a JSON format by reporting an incident on SAP Support Portal Home with a component `BC-IAM-IDS`. The request can be for full data export or selected data export according to the customer requirements.

When using Identity Provisioning, you can clear the customer data (systems, jobs, execution logs). You need to sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Tenant Reset*. For more information, see Reset Identity Provisioning Settings [page 2484].

1.10.3 Auditing and Logging Information

Here you can find a list of the security events that are logged by Identity Authentication.

Security events written in audit logs

Event grouping	What events are logged	How to identify related log events	Additional information
Authentication	All authentication and authorization checks.	`audit.authentication`	Formatting each log in key-value pairs simplifies a lot displaying logs in systems, searching by predefined key or value, downloading logging files, etc. In the Identity Authentication audit logging interface, the following log content is available: Caller `<caller>` did action `<action>` related to objectType `<objectType>` with unique identifier `<objectId>`. The result of this action is state `<state>` which is caused by cause `<cause>` and the message is `<message>`. Each audit log entry consists of the following key-value pairs: • state* = [successful | aborted | failed | ignored | process] • action* = [create | update | search | delete | upsert | add | validate | validate_sms_code | reset | export | import | verifyPhone | testConnection | read | restore | upload | list | provision | login | authenticate | logout | sendMail(s) | passwordCheck | load | rbaRuleCheck | otpCodeCheck | authenticityCheck | grantPermissions | count | setOtpAttributes | getStatistics | getUsage | setAttributes | provisionAllUsers | provisionUser(s) | forgotPassword | setIni-
Configuration	All property/properties of an entity that presents kind of configuration changes that affect the system behavior.	`audit.configuration`	
Data Change	All attribute/attributes of an entity changes.	`audit.data-change`	

Event grouping	What events are logged	How to identify related log events	Additional information
Change Logs	All attribute/attributes of change logs.	audit.config-change	tialPassword | process | setConsolidationStatus | mark | sendSmsCode | issueAssertion | issueAuthorizationCode | issueJwtToken | linkUser | returningConsolidationStatus | returnungResetPasswordStatus] • objectType* = [company | tenant(s) | tenantAdmin | serviceProvider(s) | identityProvider(s) | socialProvider(s) | service | scimGroup(s) | passwordPolicy(ies) | provisioningSystem(s) | allProvisioningSystems | user(s) | spUser |group(s) | thing(s) | contact | emailTemplates | resource | tenantLogo | scimConfiguration | smsConfiguration | spnegoConfiguration | termsOfUse | privacyPolicy | tenantSettings | trustedDomain(s) | tenantSamlConfiguration | auditLogClients | scimUser(s) | otpCode | forgotPasswordMailCounter | cockpit | usageStatistics | activityAggregator | corporateUserStore | channel | token | trustedIdp | newSpSession | activeSpAndIdpSessions | rbaAction | tenantRbaConfiguration I socialIdentity | openIdClient | googleRecaptcha | companyGroup | psrMailService | entity.Policy | entity.UserPolicy]

Event grouping	What events are logged	How to identify related log events	Additional information
			• objectId* = [unique attribute of the objectType – either id, or name, etc...] • caller = [The identifier of the subject] • cause = [callerUnauthorized | userNotFound | validationFailure, interrupted, otpCheckFailure, passwordCheckFailure | mailNotVerified | notSupported | alreadyUsed | maximumNumberReached | invalidCode | thirdInvalidCode | alreadyInUse | missingUserPassword | missingTargetSystemConfig] • message = [Exception message or additional message which specifies the result of the action] • category* = [audit.authentication | audit.configuration | audit.dataChange] • additionalAttributes – key-value formatted attributes which are specific for each objectType • all change log resource types **Note** * Available in each log entry.

Related Information

Audit Logging in the Cloud Foundry Environment
Audit Logging in the Neo Environment

1.11 Integrating the Service

You can integrate Identity Authentication with SAP and non-SAP systems as service providers.

Integrating the Service with SAP Business Technology Platform, Neo Environment [page 2910]
: SAP BTP acts as a service provider, and Identity Authentication acts as an identity provider in this setup.

Integrating the Service with the Identity Service of SAP BTP [page 2918]
: The Identity service of SAP BTP enables you to delegate authentication and authorization to Cloud Identity Services. When you register an application, the Identity service automatically creates an OpenID Connect (OIDC) application in Cloud Identity Services.

Integrating the Service with SAP Web IDE Full-Stack [page 2932]
: You can use Identity Authentication as identity provider for SAP Web IDE Full-Stack.

Integrating the Service with SAP Document Center [page 2936]
: You can use Identity Authentication as identity provider for SAP Document Center.

Integrating the Service with SAP Identity Management 8.0 [page 2941]

Integrating the Service with SAP S/4HANA Cloud Public Edition, SAP Integrated Business Planning and SAP Analytics Cloud [page 2941]
: This integration document aims to provide information about single sign-on (SSO) options for SAP S/4HANA Cloud Public Edition or SAP Integrated Business Planning and SAP Analytics Cloud, that use Identity Authentication as an authenticating or proxy identity provider.

Integrating the Service with Microsoft Entra ID [page 2952]

Integrating the Service with SAP Task Center [page 2959]

Hybrid Scenario: SAP Identity Management [page 2961]
: You can execute hybrid scenarios between provisioning systems from the Identity Provisioning UI and external systems that support SCIM 2.0 protocol.

Blogs [page 2964]
: Links to blogs and documents about integration scenarios with Identity Authentication.

1.11.1 Integrating the Service with SAP Business Technology Platform, Neo Environment

SAP BTP acts as a service provider, and Identity Authentication acts as an identity provider in this setup.

Context

> **Note**
>
> The content in this section is only relevant for SAP BTP, Neo environment.

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

For the integration, you must set the trust on both SAP BTP and Identity Authentication. As a result of the trust setting, when you have deployed an application to SAP BTP that has protected resources and requires SAML authentication, the user is redirected to the logon page of Identity Authentication to provide credentials.

> **Note**
>
> Once setting Identity Authentication as a trusted identity provider for SAP BTP all the services in the SAP BTP would be authenticated via Identity Authentication. For more information about the services provided by SAP BTP, see Services.

For the integration you need to make configurations in the cockpit of SAP BTP and in the administration console for Identity Authentication. The configurations made in the administration console do not affect the authentication for the cockpit, which is carried out via the SAP-defined tenant, SAP ID service.

> **→ Tip**
>
> If you want to use a custom Identity Authentication tenant as an identity provider for the cockpit, see Platform Identity Provider (for the cockpit or console client).

For more information about the access to SAP BTP and Identity Authentication, see the following sections:

- Access to SAP BTP Cockpit
- Access to Identity Authentication

Task overview: Integrating the Service [page 2910]

Related Information

Integrating the Service with the Identity Service of SAP BTP [page 2918]
Integrating the Service with SAP Web IDE Full-Stack [page 2932]
Integrating the Service with SAP Document Center [page 2936]
Integrating the Service with SAP Identity Management 8.0 [page 2941]
Integrating the Service with SAP S/4HANA Cloud Public Edition, SAP Integrated Business Planning and SAP Analytics Cloud [page 2941]
Integrating the Service with Microsoft Entra ID [page 2952]
Integrating the Service with SAP Task Center [page 2959]
Hybrid Scenario: SAP Identity Management [page 2961]
Blogs [page 2964]

Access to SAP BTP Cockpit

Once you purchase a customer or partner account of SAP BTP, an email is sent to the contact person from your company with a link to your SAP BTP cockpit. The contact person is specified in the Order Form for SAP Cloud Services. He or she is the first subaccount member of the SAP BTP cockpit.

> ⓘ Note
>
> For more information how to add other users for the subaccount, see Managing Members in the Neo Environment

The cockpit is the central point for managing all activities associated with your cloud-based business applications. For more information about the cockpit, see Cloud Cockpit.

To deploy applications on SAP BTP and to make configurations in the cockpit, you need a subaccount that corresponds to your role. For more information, see Getting a Global Account.

Access to Identity Authentication

Identity Authentication does not use for authentication the users registered in the SAP Service Marketplace, but maintains an own user store for administrators and users.

Once you purchase a customer or partner account of SAP BTP, a user account for Identity Authentication is created for the same contact person, specified in the Order Form. The contact person is the first administrator in the administration console for Identity Authentication. He or she receives an activation email for the administration console account. The subject of the email is: *Activate Your Account for Administration Console*. Following the required steps, the administrator activates the account and can continue to the administration console for SAP Cloud Identity Services via the console's URL.

> ⓘ Note
>
> The URL has the `https://<tenant ID>.accounts.ondemand.com/admin` pattern.
>
> *Tenant ID* is an automatically generated ID by the system. The URL is in the activation email received by the first administration contains the *tenant ID*.

> ⚠ Caution
>
> If new SAP BTP members are added into the Members page of the SAP BTP cockpit these members are not added as administrators of Identity Authentication. This is done only for the first account member. For more information about how to add new administrators in Identity Authentication, see Add User as Administrator [page 2304].

1.11.1.1 Configure SAP BTP

Prerequisites

- You have a subaccount for SAP BTP.
  For more information, see Getting a Paid Enterprise Global Account.
- You have a tenant of Identity Authentication.
- You have protected your SAP BTP application.
  For more information about how to protect your resource, see Enabling Authentication.

> **Note**
>
> If you want to use SAP BTP in a productive environment, you should purchase a customer account or join a partner account. The free tier model for SAP BTP uses the default SAP tenant.
>
> By default, SAP BTP uses `SAP ID service` as a trusted identity provider. `SAP ID service` is an SAP-defined tenant that cannot be configured by external administrators. If your environment contains SAP applications such as SAP Jam, SAP Community Network or Success Map that use authentication through `SAP ID service`, you can use the default tenant.

Context

You have two configuration options based on the following cases:

You want to	Procedure
Add a tenant of Identity Authentication registered for your company or organization as an identity provider.	Follow the procedure in: Identity Authentication Tenant as an Application Identity Provider. **Note** In this case, the trust is established automatically upon registration in both the SAP BTP and the Identity Authentication tenant. Automatically the SAP BTP Account is registered as an application in the tenant of Identity Authentication. You can find it in the administration console under the *CUSTOM APPLICATIONS* list, representing your SAP BTP account.

You want to	Procedure
Configure manual trust	1. Download the metadata file of Identity Authentication. For more details, see: Tenant SAML 2.0 Configurations [page 2103]. 2. Use this metadata to configure the trust on SAP BTP. For more details about this configuration, see Application Identity Provider.

→ Tip

Once setting Identity Authentication as a trusted identity provider for SAP BTP all SAP BTP applications and services use the trust and configuration settings. If you need different settings for the different SAP BTP applications or services, open a new subaccount. For more information, see Create Subaccounts. Once you have created the new subaccount, add the tenant of Identity Authentication in the new subaccount. Repeat the procedure from the table above to set the trust for each subaccount.

1.11.1.2 Configure Identity Authentication

Prerequisites

- You have a tenant of Identity Authentication.
- You have added the tenant of Identity Authentication as an identity provider in SAP BTP cockpit. For more information, see: Identity Authentication Tenant as an Application Identity Provider.
- You have downloaded the metadata for SAP BTP as a service provider (SP). For more information, see Application Identity Provider.

Context

You need this configuration if you have added a tenant of Identity Authentication which is not registered for the organization or company for which the SAP BTP account is created.

If you have added a tenant of Identity Authentication registered for your company or organization as an identity provider, see Configure SAP BTP [page 2913].

Procedure

1. Set the trust with SAP BTP. For more details, see Configure Trust [page 1941]
2. **Optional:** Customize the settings for the application. For more information, see Configuring Applications [page 1920].

1.11.1.3 Configure IdP-Initiated SSO with SAP BTP

Context

You can perform IdP-initiated single sign-on (SSO) in SAP BTP by sending the SAML Response directly to the application URL that is protected with SAML. This requires you to change the assertion consumer service (ACS) endpoint on the identity provider side to point to this URL.

To configure IdP-initiated SSO with SAP BTP, follow the steps below:

Procedure

1. Change the ACS endpoint on the identity provider side to point to the application protected URL. For more information, see Step 6 in Configure Trust [page 1941].
2. Configure IdP-initiated SSO. For more information, see Configure IdP-Initiated SSO [page 2201].

1.11.1.4 Configure User Attributes Mapping

You have to specify how the user attributes are sent to SAP BTP in the assertion, and define their mapping.

1. Configure User Attributes in Identity Authentication

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose your SAP BTP application from the list.
4. Choose the *Trust* tab.

5. Under *SINGLE SIGN-ON*, choose *Attributes*.
6. Configure the attributes.
7. Save your configuration.

   If the operation is successful, you receive the message `Attributes updated`.

Related Information

Configuring User Attributes from the Identity Directory [page 1973]

2. Configure Mapping in SAP BTP

Procedure

1. Sign in SAP BTP cockpit with the cockpit administrator role. For more information, see Subaccounts.
2. Select the subaccount and choose *Trust* in the navigation bar.
3. Choose the *Application Identity Provider* > *the identity provider that the platform uses for authentication* .
4. Choose *Attributes* > *Add Assertion-Based Attribute* .
5. Enter the fields as follows:

Assertion Attribute	Principal Attribute
*	*

> **Note**
>
> This specifies that all assertion attributes will be mapped to the corresponding principal attributes without a change
>
> The *Assertion Attribute* field is for the attribute that comes from the assertion.
>
> The *Principal Attribute* field is the user attribute that the users will have at SAP BTP.

6. Save your changes.

Related Information

Application Identity Provider

1.11.1.5 Configure SAP BTP Custom Domains

You can make your SAP BTP applications accessible on your own domain different from hana.ondemand.com - for example www.myshop.com.

Prerequisites

You have configured your application's custom domain using the SAP BTP console client. For more information, see Configuring Custom Domains.

Context

When a custom domain is used, both the domain name and the server certificate for this domain are owned by the customer.

To use a custom domain for the application that uses your Identity Authentication tenant for authentication, follow the procedure as described in the current document.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the list item of the application that will use a custom domain.
4. Choose the *Trust* tab.
5. Under *SAML 2.0*, choose *SAML 2.0 Configuration*.
6. Edit the *Assertion Consumer Service Endpoint* URL field and the *Single Logout Endpoint* URLs fields by replacing `authn.hana.ondemand.com` with your custom domain name. In the example above,this would be `www.myshop.com`.
7. Save your changes.

Results

Once the configuration has been changed, the system displays the message `Application <name of application> updated`. The application can only be accessed via the custom domain.

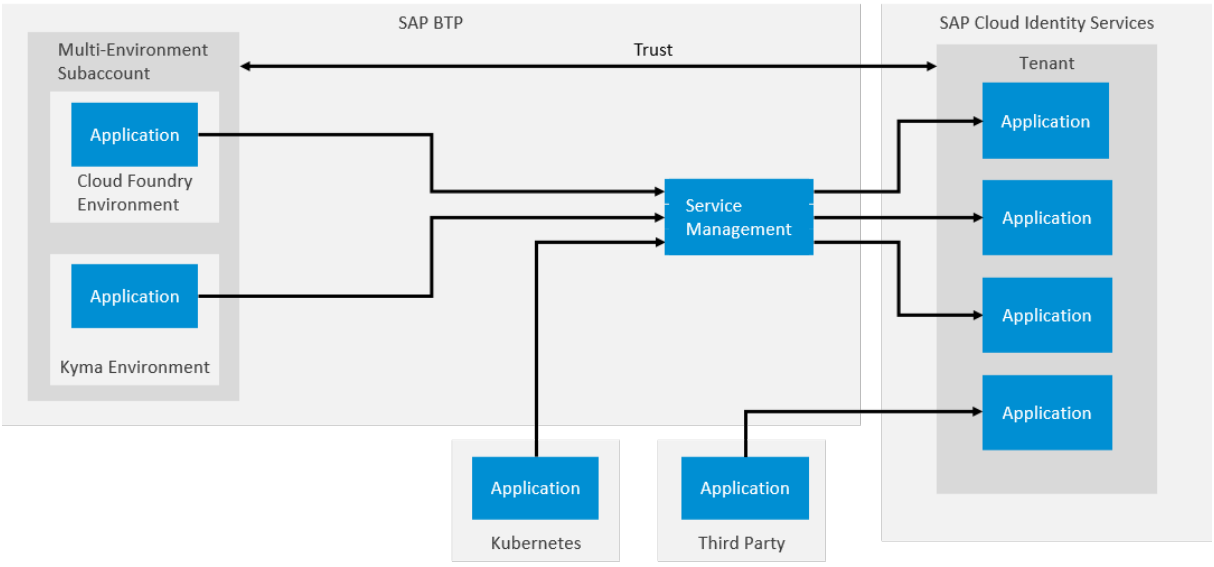
1.11.2 Integrating the Service with the Identity Service of SAP BTP

The Identity service of SAP BTP enables you to delegate authentication and authorization to Cloud Identity Services. When you register an application, the Identity service automatically creates an OpenID Connect (OIDC) application in Cloud Identity Services.

The Identity service of SAP BTP enables you to realize authentication for applications built with SAP BTP regardless of whether your application resides in an environment of SAP BTP or not. While you can manually configure an application of Cloud Identity Services to serve the function of an OIDC provider, the Identity service automates this process. You can preconfigure aspects of the Cloud Identity Services application, such as the display name of the application and redirect URIs. The Identity service enables an application to exchange information such as URL and credentials with the application of the Cloud Identity Services tenant.

This integration supports principal propagation to services consumed by your application.

The following figure shows each application integrating with the Cloud Identity Services tenant linked to your subaccount. The Identity service instance is represented by an application in the Cloud Identity Services tenant.



Applications Using the Identity Service

Each Identity instance has its own OAuth client for the Cloud Identity Services application serving as the OAuth and OIDC server. The OAuth client has its own credentials; either client ID and client secret or X.509 certificate. To use an existing X.509 client certificate from a trusted Certification Authority, provide the certificate when binding the instance to your application. Alternatively, you can request a new certificate as part of the binding. Access these credentials from the service binding. No matter which type of credentials you use, the Identity service supports code flows for users and technical users.

> ### Note
>
> The credentials only enable you to request a token and not use any other APIs of the Cloud Identity Services.

Parent topic: Integrating the Service [page 2910]

Related Information

Integrating the Service with SAP Business Technology Platform, Neo Environment [page 2910]
Integrating the Service with SAP Web IDE Full-Stack [page 2932]
Integrating the Service with SAP Document Center [page 2936]
Integrating the Service with SAP Identity Management 8.0 [page 2941]
Integrating the Service with SAP S/4HANA Cloud Public Edition, SAP Integrated Business Planning and SAP Analytics Cloud [page 2941]
Integrating the Service with Microsoft Entra ID [page 2952]
Integrating the Service with SAP Task Center [page 2959]
Hybrid Scenario: SAP Identity Management [page 2961]
Blogs [page 2964]
Getting Started with the Identity Service of SAP BTP [page 2919]
Reference Information for the Identity Service of SAP BTP [page 2921]

1.11.2.1 Getting Started with the Identity Service of SAP BTP

To create OpenID Connect (OIDC) applications in the Cloud Identity Services service using SAP Cloud Service Management service, instantiate the Identity service and bind your service instance to an application. The Identity service automates the manual creation of Cloud Identity Services OIDC applications.

Prerequisites

- You've established trust between your subaccount and your tenant of the Cloud Identity Services service using OpenID Connect.
  For more information, see Establish Trust and Federation Between UAA and Identity Authentication Service.
- You've prepared your application to consume the services of SAP BTP
  For more information, see Consuming Services in Other Environments Using the Service Management Instances.

Context

You can use Service Manager capabilities to create and manage instances of the Identity service from any runtime environment.

In the following procedure, we provide the syntax using the Cloud Foundry command-line interface (cf cli) and the Kubernetes command-line tool (kubectl).

Procedure

1. Create the identity instance.

   The following examples

   - cf cli: **`cf create-service identity application <instance_name>`**
   - kubectl:

   ```
   cat <<EOF | kubectl apply -f -
   apiVersion: servicecatalog.k8s.io/v1beta1
   kind: ServiceInstance
   metadata:
     name: {<instance_name>}
     namespace: {<namespace>}
   spec:
     clusterServiceClassExternalName: identity
     clusterServicePlanExternalName: application
   EOF
   ```

2. Bind your service instance to your application.

   - cf cli: **`cf bind-service <my_app_name> <instance_name>`**
   - kubectl:

   ```
   cat <<EOF | kubectl apply -f -
   apiVersion: servicecatalog.kyma-project.io/v1alpha1
   kind: ServiceBindingUsage
   metadata:
    name: {<instance_name>}
    namespace: {NAMESPACE_NAME}
   spec:
    serviceBindingRef:
      name: {<binding_name>}
    usedBy:
      kind: deployment
      name: {<my_app_name>}
    parameters:
      envPrefix:
        name: "PREFIX_"
   EOF
   ```

3. Consume the binding data.

Binding data consumed by a ...	Then ...
Web application	Use the approuter to authenticate the user. Otherwise, authenticate the user in the application code, according to the OIDC specification using information from service binding (URL, client credentials). For more information, see: • @sap/approuter at *npm*.

Binding data consumed by a ...	Then ...
	• Basic Client Implementer's Guide at *OpenID*.
Service consumed by application (principal propagation)	Validate the token validation and evaluate the binding data using client libraries. For more information, see Reference Information for the Identity Service of SAP BTP [page 2921].

Related Information

Using Services in the Kyma Environment

1.11.2.2 Reference Information for the Identity Service of SAP BTP

Properties enable you to customize the configuration of the Identity service.

Service Properties

> → Tip
>
> If you're creating a new tenant via subscription, `Cloud Identity Services` supports only the `cloud_service` configuration parameter for the subscription creation. The supported values are: **`PRODUCTIVE`** and **`TEST`**. If you provide a different parameter or values, the tenant isn't created. For more information about the tenant creation, see Get Your Tenant [page 528].
>
> If you want to customize the configuration of the Identity service, follow the rest of the information in this document.

Provide properties in JSON format.

The syntax of the properties is as follows:

```
{
  "name": "opportunity-management",
  "authorization": {
    "enabled": true,
    "value_help_url": "https://
myapp_namespace.cert.cfapps.eu10.hana.ondemand.com/odata/v4/ValueHelpService/"
  },
  "oauth2-configuration": {
    "redirect-uris": [
      "https://*.myapp_namespace.cfapps.eu10.hana.ondemand.com/**"
    ],
    "post-logout-redirect-uris": [
```

```
      "https://*.myapp_namespace.cfapps.eu10.hana.ondemand.com/logout/**"
    ],
    "front-channel-logout-uris": ["https://myapp.mydomain.com/fc-logout"],
    "back-channel-logout-uris": ["https://myapp.mydomain.com/fc-logout"],
    "public-client": false,
    "grant-types": ["client_credentials", "authorization_code"],
    "token-policy": {
      "token-validity": 1800,
      "refresh-validity": 7776000,
      "refresh-parallel" : 3,
      "refresh-usage-after-renewal" : "off"
    }
  },
  "consumed-services": [
    {
      "service-instance-name": "app_log"
    }
  ],
  "display-name": "Opportunity Management",
  "subject-name-identifier": {
    "attribute": "userUuid",
    "fallback-attribute": "uid"
  },
  "provided-apis": [
    {
      "name": "write-access",
      "description": "grants access to write APIs"
    }
  ]
}
```

Properties for the Creation of an Instance of the Identity Service

Property	Description
`name`	Sets the technical name of the application. The admin console shows this name below the display name in the list of applications and as the name of the *OpenID Connect Configuration*. **⚠ Restriction** The name must be unique in your tenant. By default, the Cloud Identity Services service uses the instance ID generated by the `create-instance` command.

Property	Description
`authorization`	Integrates the SAP Cloud Identity Services - Authorization Management service, enabling instance-based authorizations for your application. For more information about developing applications with Authorization Management, see Developing Authorizations [page 2876]. The `enabled` property is false by default. Set this value to true to create an instance of the service for your SAP Cloud Identity Services tenant. The `value_help_url` property is an absolute URL that points to an OData service of your application. The OData service serves values for attributes defined in the DCL schema of your application. Limits: • The format must be `https://host/path` No fragments (#) allowed HTTPS only
`redirect-uris`	An array of up to 20 redirect URIs that are explicitly allowed for the authorization code flow. **⚠ Restriction** Redirect URIs are required for the authorization code flow to redirect back to your application. This property is empty by default. For more information, see OpenID Connect (OIDC) Application Configurations [page 2747].
`post-logout-redirect-uris`	An array of up to 20 redirect URIs, where users can be forwarded after logout. This property is empty by default. For more information, see Call Identity Authentication End Session Endpoint [page 2740] and OpenID Connect (OIDC) Application Configurations [page 2747].

Property	Description
`front-channel-logout-uris`	An array of up to 20 logout URIs, which are called in a logout flow. This property is empty by default. > **Note** > You can't combine front-channel and back-channel logout URIs in the same configuration. For more information, see OpenID Connect (OIDC) Application Configurations [page 2747].
`back-channel-logout-uris`	An array of up to 20 logout URIs, which are called in a logout flow. This property is empty by default. > **Note** > You can't combine front-channel and back-channel logout URIs in the same configuration. For more information, see OpenID Connect (OIDC) Application Configurations [page 2747].
`public-client`	Set to `true` to enable OAuth flows with public clients. Use public clients in environments where it's difficult to protect client credentials, such as mobile and desktop applications or client-side parts of web applications. This property is `false` by default. For more information, see: - Configure OpenID Connect (OIDC) Application for Authorization Code Flow [page 2642] - Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow with PKCE [page 2650]
`grant-types`	An array of allowed grant types. By default, the following grant types are allowed: `client_credentials`, `password`, `authorization_code`, `refresh_token`, and `urn:ietf:params:oauth:grant-type:token-exchange`. The following grant types can be added: `implicit`, `urn:ietf:params:oauth:grant-type:jwt-bearer`, and `authorization_code_pkce_s256`.

Property	Description
`token-policy`	Defines the token policy for the application as determined by the optional properties `token-validity`, `refresh-validity`, and `refresh-parallel`. The admin console also supports the configuration of token policies. For more information, see Token Policy Configuration for Applications [page 2752].
`token-validity`	Sets the lifetime in seconds of tokens issued by the service. The value can range from 60 to 43,200, in other words, from 1 minute to 12 hours. The default value is 3600 seconds, which translates to 1 hour. **Note** The tenant-level Global Session Timeout is an upper bound on the effective token lifetime. If the configured `token-validity` exceeds the remaining session duration, the issued access token is valid only for the time left in the session.
`refresh-validity`	Sets the refresh lifetime of tokens issued by the service. During the refresh lifetime of a token, the service can reissue the token for a user. The value can range from 3600 to 15552000, in other words, from 1 hour to 180 days. Setting the value to 0 disables the refreshing of tokens. The default value is 43200 seconds, which translates to 12 hours. **Note** The tenant-level Global Session Timeout is an upper bound on refresh token validity. Refresh tokens cannot be issued or renewed beyond the remaining session lifetime. Exchanging a refresh token does not extend the session. The resulting token's validity is limited to the time remaining when the exchange occurs.
`refresh-parallel`	Determines the number of tokens that can be issued for the same session in parallel. Imagine you're logged on to the application through a web interface and a command-line interface in parallel. Then you want this property to be set to 2. The value can range from 1 to 10. The default value is 1.

Property	Description
`refresh-usage-after-renewal`	Defines the validity of the old refresh token after requesting a new one through the refresh token grant type. The attribute supports the following values: • `off` - The new refresh token immediately invalidates the old one. • `online` - The new refresh token is created and the old one is still active for 5 minutes. • `mobile` - The new and old refresh tokens are valid during the configured refresh token life time.
`access-token-format`	Defines the format of access tokens. • `default` - The access token format is dependent on the grant type. For more information, see Token Policy Configuration for Applications [page 2752]. • `jwt` - The access token format is JWT. • `opaque` - The access token format is opaque.
`consumed-services`	Use this array to enable the application to consume the APIs of reuse services. Name the reuse services consumed by the application you're protecting. The system adds the client IDs of the reuse services to the audience claim of the identity token. You can name up to 400 services. **Note** If the token is retrieved based on a public flow (`public-client` is true) without client authentication, Cloud Identity Services doesn't add the client IDs of the dependent services to the audience claim. This property is empty by default.

Property	Description
`xsuaa-cross-consumption`	Set to `true` to add the client ID of the Cloud Identity Services application created by the SAP Authorization and Trust Management service (XSUAA) to the audience claim. Setting `xsuaa-cross-consumption` enables cross-consumption of SAP Authorization and Trust Management service. **Note** If the token is retrieved based on a public flow (`public-client` is true) without client authentication, Cloud Identity Services doesn't add the client IDs of the dependent services to the audience claim. The default value is `false`.
`display-name`	Sets the name of the application that you create with the Cloud Identity Services service. Enter a maximum of 99 characters. **→ Recommendation** Provide a display name, which helps the person who administrates the Cloud Identity Services service to understand the purpose of the application. By default, the Cloud Identity Services service uses the name of the instance that you provide in the `create-instance` command.
`subject-name-identifier`	Sets the value to use to fill the subject claim in tokens issued by the service. The `attribute` attribute identifies the user attribute used to fill the subject claim. The attribute supports the following values: • `userUuid` • `uid` • `mail` • `displayName` • `loginName` • `personnelNumber` For more information about attributes for the subject name identifier, see Configure the Subject Name Identifier Sent to the Application [page 1988].

Property	Description
`provided-apis`	An array of API names (`name`) and descriptions (`description`) which this application makes available for other applications to consume. **⚠ Caution** The API name must match exactly what is expected by consumer applications. The name must be unique within all APIs provided by the same provider application. Consumer applications use this name to determine if their application has the rights to access the provider application. The name can be any URN-compliant string of up to 50 characters. You can define a maximum of 50 APIs. For more information about URNs, see RFC 8141 . For more information, see Consuming APIs from Other Applications [page 2889].

Binding Properties

Provide binding properties in JSON format. If you provide no properties, the binding uses the `SECRET` credential type. The syntax of the properties is as follows:

```
{"credential-type": "SECRET"}
```

Or:

```
{
    "credential-type": "X509_GENERATED",
    "key-length": 2048,
    "validity": 30,
    "validity-type": "DAYS",
    "app-identifier":"myApplication"
}
```

Or:

```
{
    "credential-type": "X509_PROVIDED",
    "certificate": "-----BEGIN CERTIFICATE-----\nMIID6zC…\n-----END
CERTIFICATE-----"
}
```

Or:

```
{"credential-type": "NONE"}
```

> ⚠ Restriction
>
> You can bind a maximum of 1000 applications to a single service instance.
>
> You can have a maximum of 20 different certificate subject DNs.

Credential Types for the Binding of an Instance of the Identity Service

Credential Type	Credential Properties	Description
SECRET		Generates a client secret. If no properties are provided, SECRET is the default type.
X509_GENERATED		Generates an X.509 certificate and private key.
	`key-length`	Specifies the byte length of the generated private key. The default length is `2048` bytes. You can also choose `4096` bytes.
	`validity`	Specifies the number of time units for `validity-type`. The default value is `30`. Together with the `validity-type` the range of validity runs from `1 DAYS` to `30 DAYS`.
	`validity-type`	Specifies the time unit for `validity`. Supported values are DAYS. The default value is DAYS.
	`app-identifier`	Enables you to generate a certificate with stable subject. The identifier eases the rotation of the certificates for your microservice while keeping the subject stable. The maximum length is 20 characters.

Credential Type	Credential Properties	Description
`X509_PROVIDED`		Creates a binding using a certificate that you provide from a trusted certificate authority (Certification Authority). This certificate can be used to request tokens. **⚠ Restriction** Don't forget the new lines (`\n`) after `-----BEGIN CERTIFICATE-----` and before `-----END CERTIFICATE-----`. See the previous example. The certificate you provide must be unique in the Cloud Identity Services tenant.
`NONE`		No credentials are added to the binding. Used for public clients. For more information, see Configure the Client to Call Identity Authentication Authorize Endpoint for Authorization Code Flow with PKCE [page 2650].

Binding Information

The service key or binding provides the following information.

- Client ID
- Secrets
  Secrets provided depend on how the service was configured.
  - By default the service generates a client secret.
  - If you had the service generate an X.509 certificate, the binding includes the key and the certificate.
    If you provided the service with your own X.509 certificate, the binding includes the certificate.
  - For public clients, you can suppress the creation of a secret.
- Cloud Identity Services domains
- Cloud Identity Services URL

The following examples show the information stored in the binding:

Binding with client ID and client secret:

```
{
  "clientid": "1aa234ca-faed-4fb2-966e-7c86262ba93c",
```

```
  "clientsecret": "<secret>",
  …
  "domains": ["accounts.ondemand.com", "accounts.cloud.sap"],
  "url": "https://caev6ngwk.accounts.ondemand.com"
}
```

Binding with generated X.509 certificate:

```
{
  "clientid": "1aa234ca-faed-4fb2-966e-7c86262ba93c",
  "key": "<content>",
  "certificate": "<content>",
  …
  "domains": ["accounts.ondemand.com", "accounts.cloud.sap"],
  "url": "https://caev6ngwk.accounts.ondemand.com"
}
```

Binding with provided X.509 certificate:

```
{
  "clientid": "1aa234ca-faed-4fb2-966e-7c86262ba93c",
  "certificate": "<content>",
  …
  "domains": ["accounts.ondemand.com", "accounts.cloud.sap"],
  "url": "https://caev6ngwk.accounts.ondemand.com"
}
```

Binding with no credentials:

```
{
  "clientid": "1aa234ca-faed-4fb2-966e-7c86262ba93c",
  "credential-type": "NONE",
  …
  "domains": ["accounts.ondemand.com", "accounts.cloud.sap"],
  "url": "https://caev6ngwk.accounts.ondemand.com"
}
```

Code Flows

For information about how the Cloud Identity Services service supports code flows, see Configuring OpenID Connect (OIDC) [page 2636].

Token Validation

To validate tokens, we provide client libraries to support the authentication of tokens.

- For Java, see SAP BTP Java Security Client Library on *GitHub*. The Java library supports authentication of tokens from version 2.8.0.
- For node.js, see @sap/xssec: XS Advanced Container Security API for node.js on *npm*.

Token Attributes

The administrator of the Cloud Identity Services service determines what attributes are available in tokens. For more information, see Configuring User Attributes from the Identity Directory [page 1973].

1.11.3 Integrating the Service with SAP Web IDE Full-Stack

You can use Identity Authentication as identity provider for SAP Web IDE Full-Stack.

Prerequisites

- You have a subaccount for SAP BTP.
  For more information, see Getting a Paid Enterprise Global Account.
- You have an Identity Authentication tenant. For more information about how to get Identity Authentication, see SAP BTP Pricing and Packaging Options, or contact your SAP sales representative.
- You have registered the Identity Authentication as a trusted identity provider for your SAP BTP account. See Identity Authentication Tenant as an Application Identity Provider.
- You have set the trust with SAP BTP. For more details, see Configure Trust [page 1941].
- You have switched off the *Reload Parent Page* option for SAP Web IDE Full-Stack in the administration console for SAP Cloud Identity Services. For more information, see Enable or Disable Reload Parent Page Option [page 2044].
- You have added your SAP Web IDE Full-Stack host to the list of the trusted domains in the administration console for SAP Cloud Identity Services. For more information, see Configure Trusted Domains [page 2173].
- You have checked the **User ID** of the users that will be able to access SAP Web IDE Full-Stack. The **User ID** is a six-digit number preceded by the letter **P**. For more information about how to check the **User ID**, see List and Edit User Details [page 2361].

Context

> **Note**
>
> The content in this section is only relevant for SAP BTP, Neo environment.

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

The integration between SAP Web IDE Full-Stack and Identity Authentication enables users to access SAP Web IDE Full-Stack with their Identity Authentication credentials.

SAP Web IDE Full-Stack access is protected by permissions. To grant a user the permission to access a protected resource, you can assign one of the predefined roles to such a permission. The following predefined roles are available:

- `DiDeveloper` - To develop with SAP Web IDE Full-Stack, the DiDeveloper role needs to be assigned.
- `DiAdministrator` - To be able to manage user data, a user has to be assigned to the DiAdministrator role.
- `DiScpGitAdministrator`- To be able to manage Git repositories, a user has to be assigned to the DiScpGitAdministrator role.

> → Remember
>
> You should use the `DiDeveloper` or `DiAdministrator` role to enable users to access SAP Web IDE Full-Stack with their Identity Authentication credentials. For more information, see Assign DiDeveloper or DiAdministrator Roles [page 2933].

Task overview: Integrating the Service [page 2910]

Related Information

Integrating the Service with SAP Business Technology Platform, Neo Environment [page 2910]
Integrating the Service with the Identity Service of SAP BTP [page 2918]
Integrating the Service with SAP Document Center [page 2936]
Integrating the Service with SAP Identity Management 8.0 [page 2941]
Integrating the Service with SAP S/4HANA Cloud Public Edition, SAP Integrated Business Planning and SAP Analytics Cloud [page 2941]
Integrating the Service with Microsoft Entra ID [page 2952]
Integrating the Service with SAP Task Center [page 2959]
Hybrid Scenario: SAP Identity Management [page 2961]
Blogs [page 2964]

1.11.3.1 Assign `DiDeveloper` or `DiAdministrator` Roles

Context

If you want to use the `DiDeveloper` or `DiAdministrator` role together with Identity Authentication as an identity provider, complete the following steps:

Procedure

1. Sign in SAP BTP cockpit with the cockpit administrator role. For more information, see Subaccounts.
2. In the navigation area, choose *Services* > *SAP Web IDE Full-Stack tile*.

   > ⓘ Note
   >
   > If not enabled, choose *Enable*.

3. Choose *Configure Service* under *Take Action*.
4. On the *Configure Service - Roles* tab, under *New Role*, select either `DiDeveloper` or `DiAdministrator` from the list of predefined roles.
5. Choose *Assign* from the *Individual User* section.
6. Enter the **User ID** (the **P number)** of the user in question and choose *Assign*.

   > ⓘ Note
   >
   > For more information how to check the *User ID*, see List and Edit User Details [page 2361]. The *User ID* is a six-digit number preceded by the letter **P**.

Results

The assigned users can log on to SAP Web IDE Full-Stack with their credentials for Identity Authentication (email and password).

1.11.3.2 Show User's Name on SAP Web IDE Full-Stack Home Page

Context

When you use Identity Authentication as a trusted identity provider for SAP Web IDE Full-Stack you can configure the application to display the first name of the user that is logged on in the **menu bar** and the welcome screen.

For this scenario, you have to configure the user attribute `First Name` in Identity Authentication to be sent to SAP Web IDE Full-Stack in the assertion attribute. You also have to configure the First Name user attribute mapping for Identity Authentication in the SAP BTP cockpit.

Configure `First Name` User Attribute in Identity Authentication

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose your SAP Web IDE Full-Stack application from the list.
4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *Attributes*.
6. Configure the `First Name` attribute.
7. Save your configuration.

   If the operation is successful, you receive the message `Attributes updated`.

Configure Mapping in SAP BTP

Procedure

1. In your Web browser, open the SAP BTP cockpit using the following URLs:
   - Europe: https://account.hana.ondemand.com/cockpit
   - United States: https://account.us1.hana.ondemand.com/cockpit
   - Australia: https://account.ap1.hana.ondemand.com/cockpit
2. Select the customer account and choose *Trust* in the navigation bar.
3. Choose the *Application Identity Provider* subtab, and then choose the identity provider that SAP Web IDE Full-Stack uses for authentication.
4. Choose *Attributes* > *Add Assertion-Based Attribute* > .
5. Enter the fields as follows:

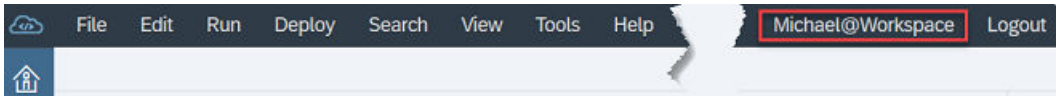
Assertion Attribute	Principal Attribute
first_name	firstname

> **Note**
>
> The *Assertion Attribute* field is for the attribute that comes from the SAML Assertion.
>
> The *Principal Attribute* field is the user attribute that the users will have at SAP BTP.

6. Save your changes.

Results

Once the user has been successfully authenticated to SAP Web IDE Full-Stack, his or her name will appear in *Menu bar* on the right.



1.11.4 Integrating the Service with SAP Document Center

You can use Identity Authentication as identity provider for SAP Document Center.

Prerequisites

- You have a subaccount for SAP BTP.
  For more information, see Getting a Paid Enterprise Global Account.
- You have an Identity Authentication tenant. For more information about how to get Identity Authentication, see SAP BTP Pricing and Packaging Options, or contact your SAP sales representative.
- You have registered the Identity Authentication as a trusted identity provider for your SAP BTP account. See Identity Authentication Tenant as an Application Identity Provider.
- You have set the trust with SAP BTP. For more details, see Configure Trust [page 1941].
- You have added your SAP Document Center host to the list of the trusted domains in the administration console for SAP Cloud Identity Services. For more information, see Configure Trusted Domains [page 2173].
  You have checked the **User ID** of the users that can access SAP Document Center. The **User ID** is a six-digit number preceded by the letter **P**. For more information about how to check the **User ID**, see List and Edit User Details [page 2361].
- *Optional (for the group assignment):* You have created a user group in Identity Authentication and assigned that group to the users you want to give authorizations to access SAP Document Center.

Context

> **Note**
>
> The content in this section is only relevant for SAP BTP, Neo environment.

> **Note**
>
> The content in this section is not relevant for China (Shanghai) region.

The integration between SAP Document Center and Identity Authentication enables users to access SAP Document Center with their Identity Authentication credentials. Identity Authentication users must be assigned to the specific *Web* roles for SAP Document Center. The specific roles contain the access authorizations for the user interfaces (UIs). For more information about the specific *Web* roles of SAP Document Center see, Assigning Users to Roles

The configuration steps are done in the administration console for SAP Cloud Identity Services and in the cockpit of SAP BTP.

Task overview: Integrating the Service [page 2910]

Related Information

Integrating the Service with SAP Business Technology Platform, Neo Environment [page 2910]
Integrating the Service with the Identity Service of SAP BTP [page 2918]
Integrating the Service with SAP Web IDE Full-Stack [page 2932]
Integrating the Service with SAP Identity Management 8.0 [page 2941]
Integrating the Service with SAP S/4HANA Cloud Public Edition, SAP Integrated Business Planning and SAP Analytics Cloud [page 2941]
Integrating the Service with Microsoft Entra ID [page 2952]
Integrating the Service with SAP Task Center [page 2959]
Hybrid Scenario: SAP Identity Management [page 2961]
Blogs [page 2964]

1.11.4.1 Assign Identity Authentication Users to Roles

Context

You can use Java EE roles to define access to SAP Document Center. You can assign the respective roles for SAP Document Center to users or to groups of users of Identity Authentication. For more information about the specific web roles for SAP Document Center, see Assigning Users to Roles.

You have three options to define access:

- Assign Users to Roles
- Assign Groups to Roles
- Assign Default Groups

Assign Users to Roles

Context

Assign the respective roles for SAP Document Center to individual users of Identity Authentication.

Procedure

1. Sign in SAP BTP cockpit with the cockpit administrator role. For more information, see Subaccounts.
2. In the navigation area go to *Services* > *SAP Document Center*.

   > → Tip
   >
   > Choose *Enable* if the service is not enabled for this account.
3. Under *Service Configuration* choose *Assign Roles & Set Destinations*.
4. Under *Roles* select the role you want to manage assignments for. For more information about the specific web roles for SAP Document Center, see Assigning Users to Roles.
5. Choose *Assign* for the *Individual Users* section.
6. Enter the **User ID** (the **P number)** of the user in question and choose *Assign*.

   > ⓘ Note
   >
   > For more information how to check the *User ID*, see List and Edit User Details [page 2361]. The *User ID* is a six-digit number preceded by the letter **P**.

Results

The assigned user can log on to SAP Document Center with their credentials for Identity Authentication.

Assign Groups to Roles

Context

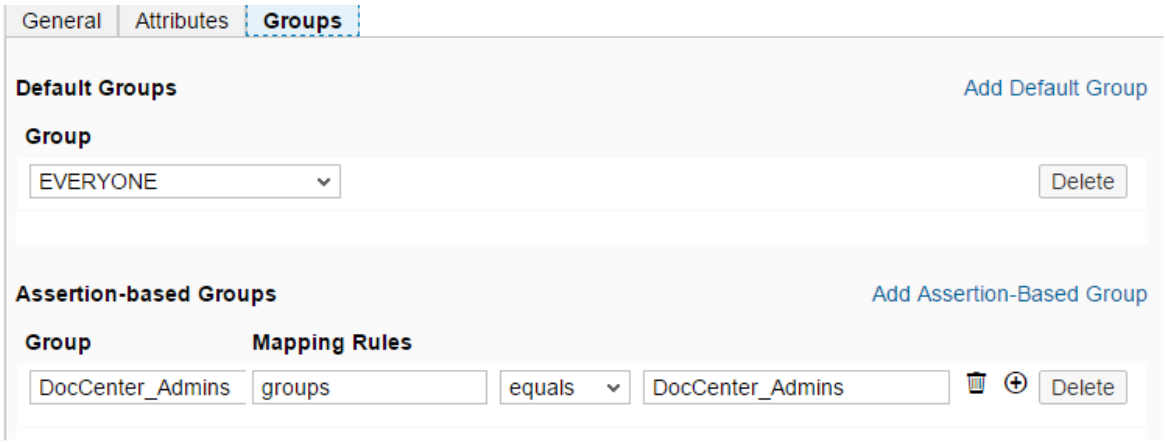
Assign the respective roles for SAP Document Center to collections of users of Identity Authentication instead of individual users. Groups allow you to easily manage the role assignments.

Assertion-based groups are groups determined by values of attributes in the SAML 2.0 assertion.

Procedure

1. Assign a group to users of Identity Authentication. For more information, see Assign Groups to a User [page 2409].

   For example, you can have a group *EVERYONE* for all the users of Identity Authentication, and a group *DocCenter_Admins* just for the administrators. In this case, you should assign the group EVERYONE to all the users of Identity Authentication, and just the administrators to the *DocCenter_Admins* group.
2. Configure the `groups` attribute that is sent to SAP Document Center in the SAML 2.0 assertion. For more information, see Configuring User Attributes from the Identity Directory [page 1973]
3. In the cockpit of SAP BTP define the assertion based groups for the group-to-role mapping in the cockpit. For more information, see 4. (If Using an Identity Provider) Define the Group-to-Role Mapping.



Results

All users that are members of the group can access SAP Document Center.

1.11.4.2 Configure User Attributes Mapping

You have to specify how the user attributes are sent to SAP BTP in the assertion, and define their mapping.

1. Configure User Attributes in Identity Authentication

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.

2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose your SAP BTP application from the list.
4. Choose the *Trust* tab.
5. Under *SINGLE SIGN-ON*, choose *Attributes*.
6. Configure the attributes.
7. Save your configuration.

   If the operation is successful, you receive the message `Attributes updated`.

Related Information

Configuring User Attributes from the Identity Directory [page 1973]

2. Configure Mapping in SAP BTP

Procedure

1. Sign in SAP BTP cockpit with the cockpit administrator role. For more information, see Subaccounts.
2. Select the subaccount and choose *Trust* in the navigation bar.
3. Choose the *Application Identity Provider* > *the identity provider that the platform uses for authentication*.
4. Choose *Attributes* > *Add Assertion-Based Attribute*.
5. Enter the fields as follows:

Assertion Attribute	Principal Attribute
*	*

> **Note**
>
> This specifies that all assertion attributes will be mapped to the corresponding principal attributes without a change
>
> The *Assertion Attribute* field is for the attribute that comes from the assertion.
>
> The *Principal Attribute* field is the user attribute that the users will have at SAP BTP.

6. Save your changes.

Related Information

Application Identity Provider

1.11.5 Integrating the Service with SAP Identity Management 8.0

SAP Identity Management can provision users to and from Identity Authentication via the Identity Authentication connector. For more information about the setting up of an Identity Authentication system, see Setting up an Identity Authentication System.

Parent topic: Integrating the Service [page 2910]

Related Information

Integrating the Service with SAP Business Technology Platform, Neo Environment [page 2910]
Integrating the Service with the Identity Service of SAP BTP [page 2918]
Integrating the Service with SAP Web IDE Full-Stack [page 2932]
Integrating the Service with SAP Document Center [page 2936]
Integrating the Service with SAP S/4HANA Cloud Public Edition, SAP Integrated Business Planning and SAP Analytics Cloud [page 2941]
Integrating the Service with Microsoft Entra ID [page 2952]
Integrating the Service with SAP Task Center [page 2959]
Hybrid Scenario: SAP Identity Management [page 2961]
Blogs [page 2964]

1.11.6 Integrating the Service with SAP S/4HANA Cloud Public Edition, SAP Integrated Business Planning and SAP Analytics Cloud

This integration document aims to provide information about single sign-on (SSO) options for SAP S/4HANA Cloud Public Edition or SAP Integrated Business Planning and SAP Analytics Cloud, that use Identity Authentication as an authenticating or proxy identity provider.

SAP Analytics Cloud, is integreted in SAP S/4HANA Cloud Public Edition or in SAP Integrated Business Planning, or it can be part of SAP Datasphere, that is integrated with SAP S/4HANA Cloud Public Edition as of SAP S/4HANA Cloud Public Edition 2208.

- Use Identity Authentication as Authenticating IdP [page 2942]

- Use Your Own Corporate IdP for Authentication [page 2944]

Parent topic: Integrating the Service [page 2910]

Related Information

Integrating the Service with SAP Business Technology Platform, Neo Environment [page 2910]
Integrating the Service with the Identity Service of SAP BTP [page 2918]
Integrating the Service with SAP Web IDE Full-Stack [page 2932]
Integrating the Service with SAP Document Center [page 2936]
Integrating the Service with SAP Identity Management 8.0 [page 2941]
Integrating the Service with Microsoft Entra ID [page 2952]
Integrating the Service with SAP Task Center [page 2959]
Hybrid Scenario: SAP Identity Management [page 2961]
Blogs [page 2964]

1.11.6.1 Use Identity Authentication as Authenticating IdP

In this scenario, to log on to an application, users provide their credentials to Identity Authentication, and Identity Authentication asserts their identity to the application.

Every SAP S/4HANA system or SAP Analytics Cloud embedded come with an Identity Authentication as a trusted identity provider. You've received a provisioning email with initial access information for the Identity Authentication tenant.

> → Tip
>
> If you need your users in the tenant of Identity Authentication, you can import them, or create them via the administration console or the Identity Directory API.

The tables below show the pre-configured settings for your system that you have in the Identity Authentication administration console.

> ⓘ Note
>
> The tenant URL has the following pattern:
>
> `https://<tenant ID>.accounts.ondemand.com/admin`
>
> *Tenant ID* is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*. For more information about your tenants, see View Assigned Tenants and Administrators [page 525].
>
> If you have a configured custom domain, the URL has the following pattern: `<your custom domain>/admin`.

ABAP tenants provisioned before SAP S/4HANA Cloud Public Edition 2208

System	Trust-Related Setting	Value
S/4HANA Cloud Public Edition or Integrated Business Planning	`Subject Name Identifier`	**`Login Name`** **Note** For successful authentication, the user name of the business user in S/4HANA Cloud Public Edition or Integrated Business Planning must be identical to the login name.
	`Default Name ID Format`	**`Unspecified`**
	`Apply Function to Subject Name Identifier`	**`None`** (ABAP does implicit uppercase conversion)
	`Conditional Authentication`	Identity Authentication
SAP Analytics Cloud, embedded edition	`Subject Name Identifier`	**`Login Name`**
	`Default Name ID Format`	**`Unspecified`**
	`Apply Function to Subject Name Identifier`	**`Uppercase`**
	`Conditional Authentication`	Identity Authentication

ABAP tenants provisioned from SAP S/4HANA Cloud Public Edition 2208 onwards

Application	Trust-related setting	Value
S/4HANA Cloud Public Edition or SAP Integrated Business Planning	`Subject Name Identifier`	**`Email`** **Note** For successful authentication, the email address of the business user in S/4HANA Cloud Public Edition or Integrated Business Planning must be identical to the email.
	`Default Name ID Format`	**`Email`**
	`Apply Function to Subject Name Identifier`	**`None`** (email addresses are compared case-insensitively)
	`Conditional Authentication`	**`Identity Authentication`**
SAP Analytics Cloud, embedded edition	`Subject Name Identifier`	**`Email`**
	`Default Name ID Format`	**`Unspecified`**
	`Apply Function to Subject Name Identifier`	**`Lowercase`**
	`Conditional Authentication`	**`Identity Authentication`**

Application	Trust-related setting	Value
SAP Datasphere	`Subject Name Identifier`	**`Email`**
	`Default Name ID Format`	**`Email`**
	`Apply Function to Subject Name Identifier`	**`None`** (email addresses are compared case-insensitively)
	`Conditional Authentication`	**`Identity Authentication`**

> **Note**
>
> The same identifying attribute (login name or email address) is used for both ABAP and SAP Analytics Cloud. User records must be created in ABAP and are then automatically replicated to SAP Analytics Cloud:

- Replication from SAP S/4HANA Cloud Public Edition to SAP Analytics Cloud happens via a pre-integrated Identity Provisioning tenant (communication scenario *SAP_COM_0193*).
- Replication from SAP Integrated Business Planning to SAP Analytics Cloud happens via the SAP-managed communication scenario *SAP_COM_1187*.

Related Information

Tenants [page 27]
Identity Directory REST API
Create a New User [page 2357]
Import CSV File with Full User Profile [page 2344]
Import or Update Users for a Specific Application [page 2348]
Configure SAML 2.0 Service Provider [page 1941]
Communication Scenarios Managed by SAP
Initial System Access for SAP S/4HANA Cloud Public Edition in Your 3-system Landscape

1.11.6.2 Use Your Own Corporate IdP for Authentication

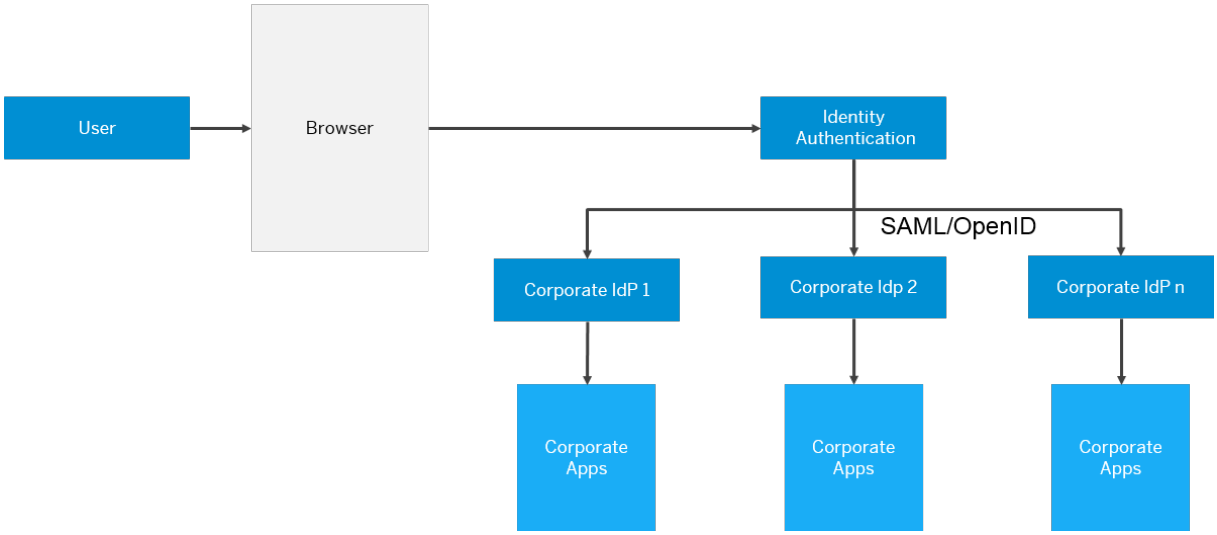
This document is intended to help you configure trust with a corporate identity provider. In this scenario Identity Authentication acts as a proxy to delegate the authentication to the corporate identity provider.

Context

In this scenario Identity Authentication acts as a proxy to delegate the authentication to the corporate identity provider. The requests for authentication sent by a service provider/relying party will be forwarded to the corporate identity provider.

Identity Authentication can use a SAML 2.0 or OpenID Connect identity provider as an external authenticating authority. The choice of one or the other protocol depends on your scenario and needs.

As an identity provider proxy, Identity Authentication will act as an identity provider to the service provider/ relying party, and as a service provider/relying party to the corporate identity provider. Once a user is authenticated at the corporate identity provider, successive authentication requests from service providers/ relying parties, which use the same corporate identity provider will not be forwarded to it as long as the session at Identity Authentication is active. Identity Authentication will issue assertions or JSON Web Tokens (JWTs) based on the user data received during the first authentication.



→ Remember

For the embeded version of SAP Analyics Cloud, the *Forward All SSO Requests to Corporate IdP* option must be disabled. Identity Authentication forwards the SSO request to the corporate identity provider only during the first sign in flow to SAP S/4HANA Cloud Public Edition, but not during the subsequent sign in flow to SAP Analytics Cloud.

To use Identity Authentication as a proxy to delegate authentication to an external corporate identity provider choose your scenario and follow the procedures below:

- Configure Trust with SAML 2.0 Corporate Identity Provider [page 2946]
- Configure Trust with OpenID Connect Corporate Identity Provider [page 2949]
- Configure Identity Federation [page 2949]

1.11.6.2.1 Configure Trust with SAML 2.0 Corporate Identity Provider

This document is intended to help you configure trust with a SAML 2.0 corporate identity provider.

Context

> ⓘ Note
>
> If an application requires force authentication (ForceAuthn="true"), users have to authenticate themselves against the corporate identity provider each time they access the application even if single sign-on (SSO) is enabled.

To configure trust with the SAML 2.0 corporate identity provider, follow the procedures below:

1. Configure Trust on the Corporate Identity Provider Side

Set up trust with Identity Authentication as a service provider.

Prerequisites

You have the SAML 2.0 metadata of Identity Authentication. For more information how to download the metadata, see Tenant SAML 2.0 Configurations [page 2103].

Procedure

1. Register Identity Authentication as a service provider at the corporate identity provider.

   > → Remember
   >
   > If you want to use IdP-initiated single sign-on (SSO) from your corporate identity provider, you have to add the parameter `sp=<sp_name>` to the assertion consumer service (ACS) endpoint configured on your corporate identity provider side for Identity Authentication.
   >
   > > ✤ Example
   > >
   > > https://<the current ACS endpoint URL>?sp=<sp_name>>
   > >
   > > `sp` is the name of the SAML 2 service provider for which SSO is performed.

> To see how to download the SAML 2.0 metadata of Identity Authentication read Tenant SAML 2.0 Configurations [page 2103].

2. **Optional:** Download the corporate identity provider SAML 2.0 metadata.

   You need the corporate SAML 2.0 metadata for the setup of the trust on Identity Authentication. Optionally, you can make the configurations manually.

2. Configure Trust on Identity Authentication Side

Set up trust with your corporate identity provider in the administration console for SAP Cloud Identity Services.

Prerequisites

- You are assigned the *Manage Corporate Identity Providers* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have registered Identity Authentication as service provider at the corporate identity provider.
- You have the corporate identity provider SAML 2.0 metadata.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure.

   > → Tip
   >
   > If you need to change the protocol, see Choose Identity Provider Type [page 2602].

4. Under *SAML 2.0*, choose *SAML 2.0 Configuration*.
5. Upload the corporate identity provider metadata XML file or manually enter the communication settings negotiated between Identity Authentication and the identity provider.

   > ⓘ Note
   >
   > Use a file with an extension `.xml`.

   When the identity provider metadata is uploaded, the fields are populated automatically with the parsed data from the XML file. The minimum configuration is to complete the *Name* field, add at least one single sign-on endpoint, and provide a signing certificate.

   You can add up to two signing certificates. Both signing certificates are accepted according to the certificate validity.

Field	Description
Metadata File	The metadata XML file of the identity provider.
Name	The entity ID of the identity provider.
Single Sign-On Endpoint URL	The URL of the identity provider single sign-on endpoint that receives authentication requests.
Single Logout Endpoint URL	The URL of the identity provider's single logout endpoint that receives logout messages.
Binding	The SAML-specified HTTP binding used by the identity provider showing how the various SAML protocol messages can be carried over underlying transport protocols.
Signing Certificate	A base64-encoded certificate used by the service provider to sign digitally SAML protocol messages sent to Identity Authentication. Use the *Add* button to add a second signing certificate. **Note** If you have two certificates, you can choose a default one, to mark your primary certificate.

6. **Optional:** Choose the digest algorithm for signing outgoing messages from the dropdown list in the *Algorithm* section. You have the following options:
   - *SHA-1*
   - *SHA-256* - the default option

     → Remember

     The algorithm must be *SHA-256* if the Identity provider type is set at *Microsoft ADFS / Entra ID*.
   - *SHA-512*
7. **Optional:** Enable or disable the *Include scoping* attribute to include or exclude the Scoping element in the SAML 2.0 request.

   → Remember

   The default setting for the *Include scoping* is enabled. The Scoping element sent in the SAML 2.0 request is 1.

   If the identity provider type is set at *Microsoft ADFS / Entra ID* the default setting for the *Include scoping* is disabled and the Scoping element is not sent in the SAML 2.0 request.
8. Save your selection.

   Once the identity provider has been updated, the system displays the message `Identity provider <name of identity provider> updated`.

Next Steps

- Select the configured identity provider as the authenticating identity provider for the application. For more information, see Choose Default Identity Provider for an Application [page 2067].
- (Optional) Configure the Name ID Format Attribute Sent to the SAML 2.0 Corporate IdP [page 2577]
- Configure Identity Federation [page 2949]

1.11.6.2.2 Configure Trust with OpenID Connect Corporate Identity Provider

This document is intended to help you configure trust with a corporate identity provider.

Context

In this scenario Identity Authentication acts as a proxy to delegate the authentication to the corporate identity provider. To use Identity Authentication as a proxy to delegate authentication to an OpenID Connect corporate identity provider you have to configure trust with that corporate identity provider. For more information, see Configure Trust with OpenID Connect Corporate Identity Provider [page 2579].

Next Steps

Configure Identity Federation [page 2949]

1.11.6.2.3 Configure Identity Federation

Prerequisites

- You are assigned the *Manage Corporate Identity Providers* role. For more information about how to assign administrator roles, see Edit Administrator Authorizations [page 2309].
- You have configured Identity Authentication to use the OpenID Connect corporate identity provider as an external authenticating authority. For more information, see Configure Trust with OpenID Connect Corporate Identity Provider [page 2579].
- You have selected the configured identity provider as the authenticating identity provider for your application. For more information, see Choose Default Identity Provider for an Application [page 2067].
- You have imported the users, authenticated by the corporate identity provider, in Identity Authentication. For more information about how to import users, see Import or Update Users for a Specific Application [page 2348].

Context

By default, *Use Identity Authentication user store* is disabled.

In scenarios when the application is using for authentication a corporate identity provider, and the *Use Identity Authentication user store* option is disabled, the user attributes, the name ID attribute, and the default attributes configurations in the administration console for SAP Cloud Identity Services are not relevant. In such scenarios, Identity Authentication sends to the application the same attributes it has received from the corporate identity provider (no change in the Subject Name Identifier or its format - Unspecified or Email). As a consequence, the same format is used not only for the ABAP and SAP Analytics Cloud tenants but also for all other SAP applications connected to the same Identity Authentication tenant, which may lead to contradictions if some applications require a different format.

When *Use Identity Authentication user store* option is enabled, the application checks if the users authenticated by the corporate identity provider exist in the Identity Authentication user store. For users that exist in Identity Authentication, data from Identity Authentication user store is taken and the subject name identifier, assertion and default attributes according to the application configuration are sent. For users with no profile in Identity Authentication, the application receives the nameID attribute from the corporate IdP assertion, and the attributes according to the application configuration.

When *Use Identity Authentication user store* is enabled, it is possible to configure different name ID Formats per application. The values for all necessary formats must be contained in the assertion issued by the corporate identity provider:

- as Subject Name Identifier in the SAML 2.0 assertion
- or as a SAML Assertion Attribute.
- or as a claim in an OpenID Connect token

Example

SAML 2.0 Scenario

The ABAP and SAP Analytics Cloud tenants require email address (in our example *dona.moore@test.com*) for logon, and another application requires the login name. If the corporate identity provider issues an assertion containing the login name as Subject Name Identifier and an Attribute Statement containing an email address, the SAML assertion XML document would contain:

Sample Code

```
<Subject>
  <NameID Format="urn:oasis:names:tc:SAML:1.1:nameid-
format:unspecified">P012345</NameID>
  ...
</Subject>
...
<AttributeStatement>
  <Attribute Name="mail">
    <AttributeValue xsi:type="xs:string">dona.moore@test.com</
AttributeValue>
  </Attribute>
  ...
</AttributeStatement>
```

Note

In this case, the setting Subject Name Identifier = Login Name is relevant only for Identity Authentication administrators, for whom a user record exists in Identity Authentication. These will be logged on to the other application with the login name from their Identity Authentication user record, which must therefore be the same as in their corporate identity provider user record. For all other users, the Subject Name Identifier asserted by the corporate identity provider (in the example, `P012345`) will be used for logon to the other application.

Example

OpenID Connect Scenario

If the corporate identity provider issues an OpenID Connect token containing the login name as Subject Name Identifier and the email address as an Assertion Attribute. The JSON web token would contain claims like:

Sample Code

```
{"sub": "P012345",
 "mail": "dona.moore@test.com",
 ...}
```

If the ABAP and SAP Analytics Cloud tenants require the email address (in our example *dona.moore@test.com*) for logon, but another application requires the login name, this can be achieved with the following configuration in the administration console for SAP Cloud Identity Services:

Scenario Settings

Application	Trust-Related Setting	Value (SAML 2.0 Scenario)	Value (OpenID Connect Scenario)
ABAP	Subject Name Identifier	*Source*: **Corporate Identity** *Value*: **mail**	*Source*: **Corporate Identity** *Value*: **mail**
	Default Name ID Format	**Email**	**Email**
	Apply Function to Subject Name Identifier	**None**	**None**
	Conditional Authentication	**<your corporate identity provider>**	**<your corporate identity provider>**
SAP Analytics Cloud	Subject Name Identifier	*Source*: **Corporate Identity** *Value*: **mail**	*Source*: **Corporate Identity** *Value*: **mail**
	Default Name ID Format	**Email**	**Email**
	Apply Function to Subject Name Identifier	**None**	**None**

Application	Trust-Related Setting	Value (SAML 2.0 Scenario)	Value (OpenID Connect Scenario)
	`Conditional Authentication`	**`<your corporate identity provider>`**	**`<your corporate identity provider>`**
Other Application	`Subject Name Identifier`	**`Login Name`**	*Source*: **`Corporate Identity`** *Value*: **`sub`**
	`Default Name ID Format`	**`Unspecified`**	**`Unspecified`**
	`Apply Function to Subject Name Identifier`	**`None`**	**`None`**
	`Conditional Authentication`	**`<your corporate identity provider>`**	**`<your corporate identity provider>`**

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Select the corporate identity provider that you want to configure for *Identity Federation*.
    - If you do not have an identity provider in your list, click the *Add* button to create one, and proceed with the configuration.
    - If you have an identity provider in your list, choose the one that you want to configure.
4. Choose *Identity Federation* to configure the options.
5. Use the slider next to *Use Identity Authentication user store* to enable or disable it.

1.11.7 Integrating the Service with Microsoft Entra ID

Identity Authentication is part of the application gallery of Microsoft Entra ID (formerly known as Microsoft Azure Active Directory or Azure AD) under the name Identity Authentication.

This integration aims to provide single sign-on (SSO) between applications using Microsoft Entra ID as authenticating identity provider and applications using Identity Authentication as proxy identity provider.

Prerequisites

- You have a valid Microsoft Entra ID subscription.
- You have a subscription for Identity Authentication. For more information how to get Identity Authentication, see Initial Setup [page 523].

Overview

In this scenario Identity Authentication acts as a proxy identity provider and Microsoft Entra ID as the main authentication authority for the applications. The authentication requests sent to Identity Authentication are redirected to Microsoft Entra ID. User management and authentication is done on Microsoft Entra ID side.

> ⓘ Note
>
> Users who are in the Microsoft Entra ID user store can use the single sign-on (SSO) functionality.
>
> Users who are provisioned to Identity Authentication, but not to Microsoft Entra ID, are not able to log on.

> → Tip
>
> Identity Authentication supports the *Identity Federation* option. This option allows the application to check if the users authenticated by the corporate identity provider exist in the user store of Identity Authentication. In the default setting, the *Identity Federation* option is disabled. If *Identity Federation* is enabled, only the users that are imported in Identity Authentication are able to access the application. For more information about how to enable or disable *Identity Federation* with Identity Authentication, see **Enable Identity Federation with Identity Authentication** in Configure Identity Federation [page 2614].

For this scenario, the configurations are made in Microsoft Entra ID classic portal and in the administration console for SAP Cloud Identity Services.

1. Configure Microsoft Entra ID [page 2954]
   Configure Microsoft Entra ID to integrate with Identity Authentication.
2. Configure Microsoft Entra ID as Corporate Identity Provider at Identity Authentication [page 2955]
   Create and configure Microsoft Entra ID as a corporate identity provider in the administration console for SAP Cloud Identity Services.
3. (Optional) Configure Different Trust Configurations for the Same Identity Authentication Tenant (Microsoft Entra ID Apps) [page 2956]
   Configure issuer name on Identity Authentication and update the name and certificate on Microsoft Entra ID side.

Parent topic: Integrating the Service [page 2910]

Related Information

Integrating the Service with SAP Business Technology Platform, Neo Environment [page 2910]
Integrating the Service with the Identity Service of SAP BTP [page 2918]
Integrating the Service with SAP Web IDE Full-Stack [page 2932]
Integrating the Service with SAP Document Center [page 2936]
Integrating the Service with SAP Identity Management 8.0 [page 2941]
Integrating the Service with SAP S/4HANA Cloud Public Edition, SAP Integrated Business Planning and SAP Analytics Cloud [page 2941]
Integrating the Service with SAP Task Center [page 2959]

Hybrid Scenario: SAP Identity Management [page 2961]
Blogs [page 2964]

1.11.7.1 Configure Microsoft Entra ID

Configure Microsoft Entra ID to integrate with Identity Authentication.

Prerequisites

You have a valid Microsoft Entra ID subscription.

Context

For the configuration of Microsoft Entra ID see Tutorial: Microsoft Entra single sign-on (SSO) integration with SAP Cloud Identity Services .

> ⓘ Note
>
> For the configuration you need the metadata of your Identity Authentication tenant. For more information about how to get it, see Tenant SAML 2.0 Configurations [page 2103].

> → Tip
>
> Download and save the Microsoft Entra ID Metadata XML file. You will need it for the configuration of Microsoft Entra ID as corporate identity provider at Identity Authentication.

Task overview: Integrating the Service with Microsoft Entra ID [page 2952]

Next task: Configure Microsoft Entra ID as Corporate Identity Provider at Identity Authentication [page 2955]

1.11.7.2 Configure Microsoft Entra ID as Corporate Identity Provider at Identity Authentication

Create and configure Microsoft Entra ID as a corporate identity provider in the administration console for SAP Cloud Identity Services.

Prerequisites

- You have a subscription for Identity Authentication. For more information how to get Identity Authentication, see Initial Setup [page 523].
- You have configured Microsoft Entra ID.

Context

To use Identity Authentication as a proxy, create, and configure Microsoft Entra ID as a corporate identity provider in the administration console for SAP Cloud Identity Services. This corporate identity provider is used as an authenticating authority for the applications.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Identity Providers*, choose the *Corporate Identity Providers* tile.
3. Choose the *Add* button to create an Microsoft Entra ID corporate identity provider.

   > **Note**
   >
   > If you have an Microsoft Entra ID corporate identity provider in your list, choose it, and proceed with its configuration. Type the name of the identity provider in the search field to filter the list items.

4. Under *SAML 2.0*, choose *SAML 2.0 Configuration*.
5. Upload Microsoft Entra ID metadata XML file or configure manually the following fields:

Field	Information
Name	Provide the `entity ID` of the corporate identity provider.
Single Sign-On Endpoint URL	Provide the URL of the identity provider single sign-on endpoint that receives authentication requests. For *Binding*, choose the one that corresponds to respective single sign-on endpoint.

Field	Information
Single Logout Endpoint URL	Provide the URL of the identity provider's single logout endpoint that receives logout messages. For *Binding*, choose the one that corresponds to respective single logout endpoint.
Signing Certificate	Provide the base64-encoded certificate used by the identity provider to digitally sign SAML protocol messages sent to Identity Authentication.

→ Tip

The information for the manual configuration is in the metadata XML file of Microsoft Entra ID.

Next Steps

1. Choose **Microsoft ADFS / Entra ID** as the type for the configured corporate identity provider. For more information, see Choose Identity Provider Type [page 2602].
2. Select the configured identity provider as the authenticating identity provider for the desired application. For more information, see Choose Default Identity Provider for an Application [page 2067].

Task overview: Integrating the Service with Microsoft Entra ID [page 2952]

Previous task: Configure Microsoft Entra ID [page 2954]

Next task: (Optional) Configure Different Trust Configurations for the Same Identity Authentication Tenant (Microsoft Entra ID Apps) [page 2956]

1.11.7.3 (Optional) Configure Different Trust Configurations for the Same Identity Authentication Tenant (Microsoft Entra ID Apps)

Configure issuer name on Identity Authentication and update the name and certificate on Microsoft Entra ID side.

Context

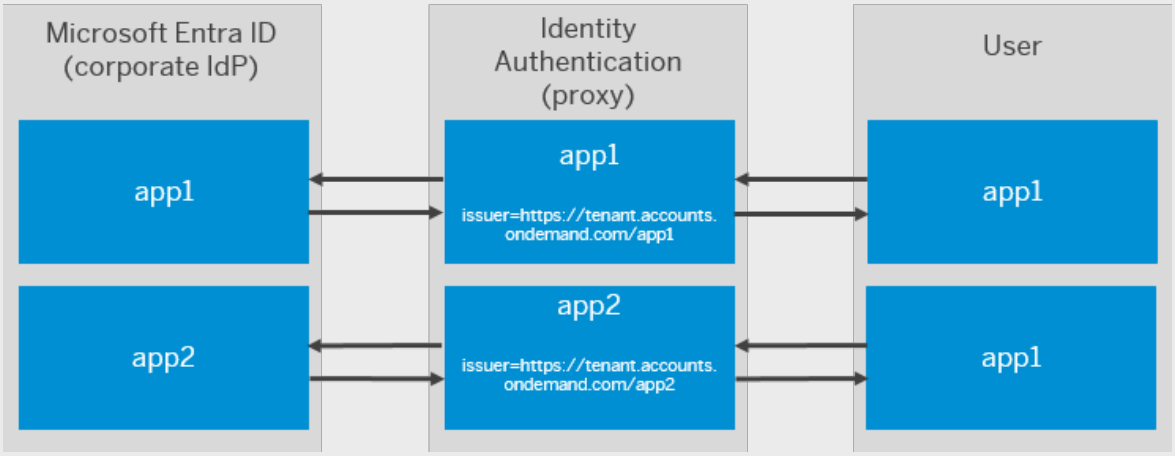
When Identity Authentication acts as a proxy identity provider you can have different trust configurations for the different applications at Microsoft Entra ID side, acting as a corporate identity provider. To do this you should configure Identity Authentication to use a different issuer name when communicating with Microsoft Entra ID, depending on the application for which the authentication is done.

When Identity Authentication issues a SAML2.0 authentication request, logout request or response to Microsoft Entra ID, then it uses the configured issuer name for the corresponding application.

When Identity Authentication verifies the SAML2.0 message coming from Microsoft Entra ID, then the verification is done against the configured issuer for the corresponding application.

The configuration is done both in Identity Authentication and Microsoft Entra ID.

Example



Identity Authentication acts as identity provider proxy and there are two configured applications app1 and app2. App1 is configured with `issuer=https://tenant.accounts.ondemand.com/app1`. App2 is configured with `issuer=https://tenant.accounts.ondemand.com/app2`. Both applications use one and the same corporate identity provider for authentication.

When App1 issues an authentication request to Identity Authentication, then Identity Authentication sends authentication request to Microsoft Entra ID, using `issuer=https://tenant.accounts.ondemand.com/app1`. Microsoft Entra ID authenticates the user and sends authentication response with `audience=https://tenant.accounts.ondemand.com/app1`. Then Identity Authentication sends authentication response to app1.

After that, the same user tries to authenticate to app2 from the same browser session. App2 issues an authentication request to Identity Authentication. Then Identity Authentication again sends authentication request to Microsoft Entra ID, but this time with `issuer=https://tenant.accounts.ondemand.com/app2`. Thus, Microsoft Entra ID can apply the specific authentication mechanisms, configured for `issuer=https://tenant.accounts.ondemand.com/app2` . After the authentication,Microsoft Entra ID issues an authentication response with `audience=https://tenant.accounts.ondemand.com/app2`. Then Identity Authentication sends authentication response to app2.

Task overview: Integrating the Service with Microsoft Entra ID [page 2952]

Previous task: Configure Microsoft Entra ID as Corporate Identity Provider at Identity Authentication [page 2955]

1. Configure Issuer Name at Identity Authentication

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under *Applications and Resources*, choose the *Applications* tile.
3. Choose the *Trust* tab.
4. Choose the application that you want to edit.
5. Under *Conditional Authentication*, choose *Configure Requests to Corporate Identity Providers*.
6. Under *Configure Issuer Name*, type the issuer suffix, which you want to add to the Identity Authentication default issuer name.

   > ⓘ Note
   >
   > The issuer name can contain Latin letters (A-Z and a-z), numbers, and the dash (-) character. It can be up to 36 characters.

   The issuer name, which will be used when sending SAML2.0 requests to the corporate identity provider appears under the input field. It is in the format `<Identity_Authentication_default_issuer_name>/<issuer_name_suffix_typed_in_the_input_field>` . For example, `https://my-tenant.accounts.ondemand.com/issuername/>`.
7. Save your changes.

2. Configure Microsoft Entra ID

Context

> ⓘ Note
>
> If you want to register Identity Authentication tenant more than once in Microsoft Entra ID, you must configure issuer name per Identity Authentication application and change the name and certificate on Microsoft Entra ID side.
>
> You have to change the certificates, because by default Microsoft Entra ID generates application specific certificates. On Identity Authentication side there is one corporate identity provider with one certificate which means that all applications on Microsoft Entra ID side for the same Identity Authentication tenant should use the same certificate for SAML 2.0.

Procedure

1. Create Microsoft Entra ID applications.

2. Update the issuer names.
3. Generate a certificate with a private key for the Microsoft Entra ID applications.

   You can generate a certificate with a private key using OpenSSL with the following command:

   ```
   openssl req -x509 -days 365 -newkey rsa:2048 -keyout cert.pem -out cert.pem
   ```

   The commands are executed in the MS Windows or Mac OS Terminal.
4. Convert the `.pem` file to `.pfx` format.

   You can convert the `.pem` file to `.pfx` format, using the following command:

   ```
   openssl pkcs12 -export -in cert.pem -inkey cert.pem -out cert.pfx
   ```

   The commands are executed in the MS Windows or Mac OS Terminal.
5. Import the generated certificate to your Microsoft Entra ID applications.

Next Steps

Add the generated certificate for the configured corporate identity provider in your Identity Authentication tenant. For more information, see Configure Trust with SAML 2.0 Corporate Identity Provider [page 2560].

1.11.8 Integrating the Service with SAP Task Center

Overview

For the integration scenario of SAP Task Center, you must use the Global User ID of SAP Cloud Identity Services - Identity Authentication as the federation identifier.

The Global User ID specifies an identifier for a user that is unique across technology layers such as user interface, APIs, and security tokens, as well as across products and lines of businesses contributing to a business process in the Intelligent Enterprise.

This attribute is automatically generated by Identity Authentication at user creation, and it can be changed by the admin after that. It can be provisioned to various SAP cloud solutions by Identity Provisioning.

> → Remember
>
> When Identity Authentication uses a corporate identity provider (IdP) to authenticate the users, the *Enable the Use Identity Authentication user store* option under the *Identity Federation* configuration of the corporate IdP must be enabled, and the users must exist in the Identity Directory, the local user store of Identity Authentication.

The Global User ID addresses the challenge of integrating user-related data across system boundaries. SAP Task Center is an example of a service, which requires the use of the Global User ID as a common user identifier. SAP Task Center provides a single entry point for business users to access their tasks. This scenario needs an enterprise-wide mapping of users to relate tasks to each other in different systems.

> ### Note
>
> Use the Global User ID as a common user identifier for SAP if you expect to use applications, which require one identifier in all solutions like SAP Task Center. For more information, see System Integration Guide for SAP Cloud Identity Services and SAP Task Center.

Usage

Identity Authentication

For every newly created user (self-registered, imported, or manually created - via the administration console or API), Identity Authentication generates a Global User ID. It is unique and can be changed later by the tenant administrator.

The system generated attribute value is 36 characters long (32 hexadecimal characters and 4 hyphens).

> ### Example
>
> f81d4fae-7dec-11d0-a765-00a0c91e6bf6

This attribute can be sent from Identity Authentication to applications as a user attribute, `Subject Name Identifier`, and default attribute in the SAML assertion. The Global User ID is also put in the `id_token` if the application uses OpenID connect.

Identity Provisioning

Identity Provisioning supports provisioning users with Global User ID for the following SAP cloud solutions:

- SAP Ariba
- SAP Fieldglass
- SAP SuccessFactors
- SAP S/4HANA Cloud Public Edition
- SAP S/4HANA through SAP Application Server ABAP
- SAP Marketing Cloud
- SAP Integrated Business Planning
- SAP BTP ABAP Environment

> ### Note
>
> In the context of SAP Task Center, those solutions are known as SAP task providers. For more information about the availability of the SAP task provider solutions integrated with SAP Task Center, refer to SAP Note: 2977611.

The Global User ID is provisioned by the Identity Provisioning using the SCIM attribute `userUuid` as part of the `urn:ietf:params:scim:schemas:extension:sap:2.0:User` namespace.

SAP Identity Management

SAP Identity Management (on-premise solution) supports reading users with Global User ID from Identity Authentication using the Identity Management *SCI* connector. SCI stands for SAP Cloud Identity, the former

name of Identity Authentication. It is not possible to use the Identity Management SCIM connector in a hybrid scenario with Identity Provisioning for that purpose.

Once loaded in SAP Identity Management, users with Global User ID can be provisioned to SAP S/4HANA on-premise systems only. In this case, the Identity Management *ABAP* or *BusinessSuite* connectors are used.

When SAP Identity Management provisions a new user to Identity Authentication, the Global User ID is generated by the service and returned back as a response. It is stored in the Identity Management MX_USER_UUID attribute.

For more information on how to provision the Global User ID from your SAP Identity Management system to your SAP S/4HANA on-premise system, refer to SAP Note 3047993.

Parent topic: Integrating the Service [page 2910]

Related Information

Integrating the Service with SAP Business Technology Platform, Neo Environment [page 2910]
Integrating the Service with the Identity Service of SAP BTP [page 2918]
Integrating the Service with SAP Web IDE Full-Stack [page 2932]
Integrating the Service with SAP Document Center [page 2936]
Integrating the Service with SAP Identity Management 8.0 [page 2941]
Integrating the Service with SAP S/4HANA Cloud Public Edition, SAP Integrated Business Planning and SAP Analytics Cloud [page 2941]
Integrating the Service with Microsoft Entra ID [page 2952]
Hybrid Scenario: SAP Identity Management [page 2961]
Blogs [page 2964]
System Integration Guide for SAP Cloud Identity Services and SAP Task Center
SAP Business Accelerator Hub
Global User ID
Blog Post: Why SAP Cloud Identity Matters for SAP Task Center: It's About Trust and Global User ID

1.11.9 Hybrid Scenario: SAP Identity Management

You can execute hybrid scenarios between provisioning systems from the Identity Provisioning UI and external systems that support SCIM 2.0 protocol.

Prerequisites

- You have an Identity Provisioning service standalone or bundle tenant.
- You have the user credentials to connect SAP Identity Management to SAP Cloud Identity Services - Identity Provisioning. Your user has read and write permissions.

Context

A proxy system is a special connector type you can use for *hybrid* scenarios. It exposes any Identity Provisioning supported backend system as a SCIM 2.0 service provider, which can be consumed by any SCIM 2.0 compatible client application, without making a direct connection between them.

The Identity Provisioning proxy system will initially provision entities to an external consumer system and then will start executing CRUD operations backwards, whenever the external system requests such. For more information, see Proxy Systems [page 1343].

The scenario below is exemplary. It involves a proxy connector you have added in the Identity Provisioning UI, and **SAP Identity Management** as an external leading identity management system.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services
2. Create a technical user with the necessary authorizations. It will later be used by the external leading identity management system to connect to Identity Provisioning.
    1. In SAP Cloud Identity Services admin console, navigate to *Users & Authorizations* > *Administrators*.
    2. Add an administrator user of type **System** and configure the basic authentication method for this user. If you already have a technical user, skip this step.
    3. Save your changes.
    4. Select your administrator user of type **System** and enable the *Access Proxy System API* permission.
    5. Save your changes.
3. Access Identity Provisioning. The access URL depends on the productive type of your Identity Provisioning. See: Access Admin Console [page 542]
4. Add a proxy system. You can choose among a list of systems available for hybrid scenarios with SAP Identity Management. To find the system you need, see Proxy Systems [page 1343] → **More Information**.
5. Open the *Properties* tab of the selected proxy system to configure its connection settings.
6. If necessary, modify the default *Read* and *Write* transformation mapping rules to reflect the current setup of entities in your proxy system.
7. Save your changes.

Next Steps

1. Now, you can export the newly created proxy system. To do that, choose *Export* → *CSV format*.
2. Then, go to SAP Identity Management to register or import a SCIM repository.

> **Note**
>
> If you import the *.csv* file, you will have all the fields automatically filled-in. However:
>
> - You need to manually enter your client ID and secret (AUTH_USER and AUTH_PASSWORD).

- For the SCIM_ASSIGNMENT_METHOD constant, leave its default value: **PATCH**.
  However, check explicitly in the documentation of the relevant proxy system type whether you should leave the default value **PATCH**, or change it to **PUT**. Find your system under section Proxy Systems [page 1343].

3. Then start an *Initial load* job. After the initial load is done, you can create new users or update existing ones in SAP Identity Management.

Future Identity Lifecycle

What happens when you make new changes (create/update/delete an entity)?

- If a new change is made in SAP Identity Management, a new job is automatically triggered. It applies the changes in the external back-end system so that both systems become up-to-date and synchronized.
- If a new change is made in the back-end system, you have to start a new *Initial load* job so that the changes can apply in SAP Identity Management.

Related Information

SAP Identity Management: Setting Up Hybrid Integration with a Cloud System

SAP Community Blog: Hybrid Scenarios with Identity Provisioning Proxy

Parent topic: Integrating the Service [page 2910]

Related Information

Integrating the Service with SAP Business Technology Platform, Neo Environment [page 2910]
Integrating the Service with the Identity Service of SAP BTP [page 2918]
Integrating the Service with SAP Web IDE Full-Stack [page 2932]
Integrating the Service with SAP Document Center [page 2936]
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Integrating the Service with SAP S/4HANA Cloud Public Edition, SAP Integrated Business Planning and SAP Analytics Cloud [page 2941]
Integrating the Service with Microsoft Entra ID [page 2952]
Integrating the Service with SAP Task Center [page 2959]
Blogs [page 2964]

1.11.10 Blogs

Links to blogs and documents about integration scenarios with Identity Authentication.

> **Note**
>
> The following links are not part of the official documentation of Identity Authentication and some of the information may be outdated.

Blogs

- Configure SAP SuccessFactors Business Execution (SAP SuccessFactors BizX) SSO to use Identity Authentication
- Identity and Access Management across SAP and Google
- Connecting SAP IAS as a proxy to Azure AD using OpenID Connect
- Why SAP Cloud Identity Matters for SAP Task Center: It's About Trust and Global User ID

Parent topic: Integrating the Service [page 2910]

Related Information

Integrating the Service with SAP Business Technology Platform, Neo Environment [page 2910]
Integrating the Service with the Identity Service of SAP BTP [page 2918]
Integrating the Service with SAP Web IDE Full-Stack [page 2932]
Integrating the Service with SAP Document Center [page 2936]
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Integrating the Service with SAP S/4HANA Cloud Public Edition, SAP Integrated Business Planning and SAP Analytics Cloud [page 2941]
Integrating the Service with Microsoft Entra ID [page 2952]
Integrating the Service with SAP Task Center [page 2959]
Hybrid Scenario: SAP Identity Management [page 2961]

1.12 Monitoring and Troubleshooting

This section describes how you can get usage statistics, access audit logs, export change logs or get support if things start to go wrong.

View Usage Statistics [page 2966]

You can view statistical information for a tenant in the administration console for SAP Cloud Identity Services.

Audit Logs [page 2966]
: Tenant administrators can access the audit logs for changes in the personal data, and successful, and failed authentications in Identity Authentication.

Export Change Logs with a History of Administration Operations [page 2976]
: You can download a CSV file with a history of the operations performed by administrators in the administration console for SAP Cloud Identity Services.

Logging OpenID Connect (OIDC) Tokens [page 2980]
: Tenant administrator can view the JWT payload of the OpenID Connect (OIDC) tokens issued or received by Identity Authentication.

Monitor Provisioning Job Logs [page 2982]
: Job logs display information about the execution of provisioning jobs. Each row in the list of job logs shows information about one execution of a job.

Monitor Real-Time Logs [page 2986]
: Real-time provisioning logs display information about the execution of a real-time sync of a user or a group entity.

Monitor Central Store Logs [page 2988]
: Central Store Logs provide information about the application-specific groups that have been provisioned from the Identity Directory to your target systems.

Monitoring with SAP Cloud ALM [page 2991]
: SAP Cloud Identity Services are integrated with SAP Cloud ALM, which is a cloud-based application lifecycle management offering that enables you to implement, operate, and monitor your cloud solutions.

Manage Provisioning Job Logs [page 3010]
: After you view and analyze the provisioning job logs, you can download or delete them.

Manage Job Notifications [page 3018]
: You can subscribe to a source system to receive notification e-mails. They provide information about the job execution and links to download job logs, in case of created, updated, failed, skipped, or deleted entities.

View Troubleshooting Logs [page 3020]
: Tenant administrator can filter and view troubleshooting logs directly in the administration console for SAP Cloud Identity Services to diagnose problems with authentication.

Download Troubleshooting Logs [page 3021]
: Tenant administrator can view Identity Authentication troubleshooting logs to monitor the system, and diagnose and solve problems.

Analyze Troubleshooting Logs with Support Log Assistant 2.0 [page 3022]

Troubleshooting Issues with Authentication and User and Group Management [page 3023]
: While using the Identity Authentication service, you encounter issues or have questions regarding specific settings, limitations, or other aspects.

Troubleshooting Provisioning Issues [page 3070]
: While using the Identity Provisioning service, you might encounter issues or have questions regarding specific settings, limitations, or other aspects.

1.12.1 View Usage Statistics

You can view statistical information for a tenant in the administration console for SAP Cloud Identity Services.

Context

The *Reporting* view displays a chart of statistical information with the number of the user logon requests. A logon request is a single authentication request managed via Identity Authentication. Identity Authentication counts only one logon request per user per day. Logon requests are independent of the authentication mechanism and user type.

The default statistical information shows the usage of the bundled, charged and system applications for the last six months. You can customize what you see by requesting information about specific applications and modifying the period.

Identity Authentication keeps the usage statistics for three years.

> ⓘ Note
>
> If you need statistics for the period before August 2015, you can report an incident on SAP Support Portal Home with a component `BC-IAM-IDS`.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Reporting* tile.
3. **Optional:** Specify the applications that you want to view from the *Logon counters* drop-down.
4. **Optional:** Modify the statistics period.

1.12.2 Audit Logs

Tenant administrators can access the audit logs for changes in the personal data, and successful, and failed authentications in Identity Authentication.

> → Tip
>
> You can find a list of the security events that are logged by Identity Authentication on Auditing and Logging Information [page 2906].

There are two different procedures for accessing the audit logs:

- Access Audit Logs (Audit Log Service in SAP BTP, Cloud Foundry) [page 2967]
- Access Audit Logs (Audit Log Service in SAP BTP, Neo) [page 2972]

1.12.2.1 Access Audit Logs (Audit Log Service in SAP BTP, Cloud Foundry)

Access the audit logs for changes in the personal data, successful, and failed authentications for Identity Authentication tenants on both the SAP, and the AWS and Azure infrastructures in the Audit Log Service in SAP BTP, Cloud Foundry.

Prerequisites

- You have a subaccount in your global account in SAP BTP, Cloud Foundry. For more information, see Create a Subaccount.
- The subaccount region must map your Identity Authentication region.

Context

> ⓘ Note
>
> The content in this document is both for tenants on the SAP, and AWS and Azure infrastructures.

> → Tip
>
> If your tenant is on the SAP infrastructure, when you access the administration console for SAP Cloud Identity Services, the *Audit and Change Logs* tile, you see the *Cloud Foundry*, *NEO*, and *Change Logs* options for configurations. If your tenant is on the AWS, Azure infrastructure, you see the *Audit Logs* and *Change Logs* options for configurations.
>
> Information about the region, representing the data center in which the tenant is located is visible in the header of the administration console. For example, *Cloud Identity Services* `Europe (Frankfurt / Amsterdam)`. For more information about the regions, see Regional Availability [page 11].

The audit log entries in the Audit Log Service in SAP BTP, Cloud Foundry are retained for 90 days. For more information, see Audit Log Retention for the Cloud Foundry Environment.

To view the audit logs, follow the procedures below:

1. Configure the Service Plans

Procedure

1. In the SAP BTP cockpit, navigate to the corresponding subaccount. For more information, see Navigate in the Cockpit.

2. In the navigation area on the left, choose *Entitlements*, then choose *Edit* > *Add Service Plans*.
3. From the dialog, search for and select *Audit Log Viewer Service*, select the *free (Application)* checkbox, and then choose *Add 1 service plan*.
4. Save your changes.

2. Configure Audit Log Viewer Instance

Procedure

1. In the SAP BTP cockpit, navigate to the corresponding subaccount. For more information, see Navigate in the Cockpit.

   > ⚠ Caution
   >
   > The subaccount region must map your Identity Authentication region. Otherwise you'll get an error when you try to access the audit logs directly from the administration console.

2. Navigate to *Services* > *Service Marketplace*, and select *Audit Log Viewer Service*.
3. Choose *Create*.

   In the *New Instance or Subscription* dialog box, **`Audit Log Viewer Service`** for *Service*, and **`free`** for *Plan* are already preselected.
4. Choose *Create* and close the information message.

3. Create and Assign Role Collection and Roles

Procedure

1. In the SAP BTP cockpit, navigate to the corresponding subaccount. For more information, see Navigate in the Cockpit.
2. Choose *Security* > *Roles Collections* in the navigation area.
3. Choose *Create*.
4. Enter name for the new role in the *Create Role Collection* popup and choose *Create*.

   The new role appears in the *Role Collections* list.
5. Select the newly created role collection from the list.
6. Choose the *Edit* button.
7. Under the *Roles* tab, expand the *Role Name* list and select the auditlog viewer and auditlog-management roles.

   If the roles don't exist, create them. For more information, see Configure Application Roles and Assign Roles to Users.

8. Choose *Add*.
9. Navigate back to the subaccount and choose *Security* > *Users* and assign the role collection to the user.

4. Add Configuration in Administration Console and View Audit Logs

Context

> → Remember
>
> You can have only one configuration in the administration for Cloud Identity services. If you want to add a new configuration, you must remove the already configured one, and then follow the steps below to add the new one.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Audit and Change Logs* tile.
3. Choose the *Audit Logs* tab.
4. You have the following options:
   - If your tenant is on the SAP infrastructure, choose the *Cloud Foundry* tab.
   - If your tenant is on the AWS, Azure infrastructure, choose the *Audit Logs* tab.
5. Choose *+Add*.
6. Fill in the required information in the pop up and save your changes.

Configuration	Notes
Tenant ID	Required. The tenant ID of your Cloud Foundry account.
Region	SAP BTP, Cloud Foundry region. You can choose a region from the options in the dropdown. For more information, see the mapping table.
Subdomain	Optional. If you provide it, a link to the *Audit Log Viewer* is added in the *Audit Service Configuration*, and you can access the audit logs directly from the administration console. ⚠ Caution The subaccount region must map your Identity Authentication region. Otherwise you'll get an error when you try to access the audit logs directly from the administration console.

Identity Authentication - Cloud Foundry Regions Mapping

Identity Authentication		Cloud Foundry Regions		
				Default **Note** **The default Cloud Foundry region where logs are sent.**
Region	**Infrastructure**	**Technical Name**	**Name**	
North America (Canada Central) / Canada (Toronto)	azure-canadacentral	cf-ca10	Canada (Montreal)	Yes
US West / West US 2	azure-westus2	cf-us20	US West (WA)	Yes
		cf-us21	US East (VA)	No
		cf-us10	US East (VA)	No
		cf-us30	US Central (IA)	No
US West / East US	azure-eastus	cf-us20	US West (WA)	Yes
Singapore	aws-ap-southeast-1	cf-ap11	Singapore	Yes
		cf-ap21	Singapore	No
South Korea / South Korea (Seoul)	aws-ap-northeast-2	cf-ap12	South Korea (Seoul)	Yes
Europe / Germany (Frankfurt)	aws-eu-central-1	cf-eu11	Europe (Frankfurt) EU Access	Yes
		cf-eu10	Europe (Frankfurt)	No
		cf-eu20	Europe (Netherlands)	No
		cf-eu30	Europe (Frankfurt) GCP	No
Europe / Switzerland	azure-switzerland-north	cf-ch20	Switzerland (Zurich) Azure EU Access	Yes
India (Mumbai)	aws-ap-south-1	cf-in30	India (Mumbai)	Yes
Brazil	aws-sa-east-1	cf-br10	Brazil (São Paulo)	Yes
Europe / Germany (Frankfurt)	eu-de-2	cf-eu10	Europe (Frankfurt)	Yes
		cf-eu11	Europe (Frankfurt) EU Access	No
		cf-eu20	Europe (Netherlands)	No
		cf-eu30	Europe (Frankfurt) GCP	No

Identity Authentication		Cloud Foundry Regions		
				Default **Note** **The default Cloud Foundry region where logs are sent.**
Region	**Infrastructure**	**Technical Name**	**Name**	
Australia (Sydney)	ap-au-1	cf-ap10	Australia (Sydney)	Yes
		cf-ap20	Australia (Sydney)	No
Japan (Tokyo)	ap-jp-1	cf-jp10	Japan (Tokyo)	Yes
		cf-jp20	Japan (Tokyo)	No
US East / East US	na-us-2	cf-us10	US East (VA)	Yes
		cf-us21	US East (VA)	No
Europe / Netherlands (Amsterdam)	eu-nl-1	cf-eu10	Europe (Frankfurt)	Yes
		cf-eu11	Europe (Frankfurt) EU Access	No
		cf-eu20	Europe (Netherlands)	No
		cf-eu30	Europe (Frankfurt) GCP	No
US East / East US	na-us-1	cf-us10	US East (VA)	Yes
		cf-us21	US East (VA)	No

7. Save your changes.

> **Note**
>
> The configuration will be enabled with the next 15 minutes.

> **Caution**
>
> If your SAP Cloud Identity Services tenant is migrated to a new region, you must remove the current configuration and repeat procedure with the new region.

8. View the audit logs. You have two options:
   - (if subdomain is configured) choose the link to the *Audit Log Viewer* in the *Audit Service Configuration* in the administration console.
   - in the cockpit, navigate to *Services* > *Instances and Subscriptions* > *Audit Log Viewer*.

Results

Upon accessing the Audit Log Viewer, you have the option to filter the logs based on date and keyword filters.

The audit logs provide information about the event category and timestamp, the event and object type, who performed the action and others.

> Example
>
> - *Category*: audit.security-events (logged as security event message), audit.configuration (logged as configuration modification message), audit.data-access (logged as data access message).
> - *Event Type*: JOB_TRIGGERED, SYSTEM_UPDATED, SYSTEM_CREATED, SYSTEM_DELETED
> - *Object Type*: Job, System
> - *ObjectAttribute.performed-by-user*: P123456

For more information about the security events that are logged by Identity Authentication, see Auditing and Logging Information [page 2906].

Next Steps

(Optional) Retrieve the audit logs via the Audit Log Retrieval API. See Audit Log Retrieval API Usage for Subaccounts in the Cloud Foundry Environment.

1.12.2.2 Access Audit Logs (Audit Log Service in SAP BTP, Neo)

View the audit logs for changes in the personal data, successful, and failed authentications for Identity Authentication tenants on the SAP infrastructure in the Audit Log Service in SAP BTP, Neo.

Context

> Note
>
> The content in this section is only relevant for tenants on the SAP infrastructure.

> Caution
>
> The deprecation of the access of the audit logs for Identity Authentication tenants on the SAP infrastructure in the Audit Log Service in SAP BTP, Neo has been postponed and will not be carried out according to the previously announced timeline. A new timeline will be announced accordingly.
>
> You can continue using the Audit Log Service in SAP BTP, Neo to access the Identity Authentication tenants on the SAP infrastructures.

→ Tip

If your tenant is on the SAP infrastructure, when you access the administration console for SAP Cloud Identity Services, the *Audit and Change Logs* tile, you see the *Cloud Foundry*, *NEO*, and *Change Logs* options for configurations. If your tenant is on the AWS, Azure infrastructure, you see the *Audit Logs* and *Change Logs* options for configurations.

To view the audit logs, you should generate Client ID and Client Secret for audit logs in the administration console for SAP Cloud Identity Services first. After that you should obtain an access token, and then call the audit log retrieval API to access the data.

The audit log entries in the Audit Log Service in SAP BTP, Neo are retained for 201 days. For more information, see Audit Log Retention API Usage for the Neo Environment.

To access the audit logs, follow the procedures below:

1. Generate Client Credentials for Audit Logs

Context

The Client ID and Client Secret for the current tenant are generated in the administration console for SAP Cloud Identity Services.

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Audit and Change Logs* tile.
3. Choose the *NEO* tab.
4. Under *Generate Client Credentials for Audit Logs* choose the *Generate* button.

   A dialog box with the generated Client ID and Client Secret appears.

   → Remember

   Make sure that you copy the Client Secret from the dialog box. When you choose *OK*, the dialog box closes, and you can't retrieve the Client Secret from the system anymore.

   The generated Client ID can be seen in the *Generate Client Credentials for Audit Logs* section under the *NEO* tab.

   → Tip

   To delete the client credentials, choose the icon next to the generated Client ID. This deletes the Client ID and Client Secret from the system.

2. Obtain an Access Token

Context

Use the Client ID and Client Secret generated for the current tenant in the administration console for SAP Cloud Identity Services to obtain an access token. For more information about how to obtain the access token see **2. Get an OAuth Access Token** in Using Platform APIs.

→ Tip

The URL for the POST request looks like this: `https://api.<SAP BTP Host>/oauth2/apitoken/v1?grant_type=client_credentials`

Cluster	Tenant Region	SAP BTP Host in Request
Europe	Amsterdam	`eu1.hana.ondemand.com` (primary)
		`eu3.hana.ondemand.com`
Australia	Sydney	`ap1.hana.ondemand.com` (primary)
		`jp1.hana.ondemand.com`
Japan	Tokyo/Osaka	`jp1.hana.ondemand.com`
US-East	Sterling/Toronto	`us3.hana.ondemand.com` (primary)
		`us1.hana.ondemand.com`
Saudi Arabia	Riyadh/Dammam	`sa1.hana.ondemand.com`

Use the primary host for your search. If the audit log is not found there, use the other hosts.

For more information, see Regional Availability [page 11] and Regions and Hosts Available for the Neo Environment.

3. Call the Audit Log Retrieval API

Context

To access the audit logs, you should call the audit log retrieval API. You need the Client ID and Client Secret for the current tenant, generated in the administration console for SAP Cloud Identity Services, and the access token. For more information, see Audit Log Retrieval API Usage.

You can filter the audit logs by time and/or categories. The categories that you can filter are:

- audit.authentication - audit logs related to authentication (failed, successful, type)
- audit.data-change - audit logs related to changes in personal data

→ Remember

The URI for the GET request looks like this:

```
https://api.<SAP BTP Host>/auditlog/v1/accounts/<Tenant ID>/AuditLogRecordsIds?
$filter=(Time ge '2018-05-17T13.00.00' and Time le '2018-05-18T05.00.00') and
Category eq '<category>'
```

The timestamp is in Coordinated Universal Time (UTC).

Replace the URI from the examples in Audit Log Retrieval API Usage with the provided one for your Identity Authentication tenant.

Tenant ID is an automatically generated ID by the system. The first administrator created for the tenant receives an activation email with a URL in it. This URL contains the *tenant ID*.

Cluster	Tenant Region	SAP BTP Host in Request	Deprecated
Europe	Amsterdam	`eu1.hana.ondemand.com` (primary)	Oct 23, 2024
		`eu3.hana.ondemand.com`	
Asia Pacific	Sydney/Tokyo	`ap1.hana.ondemand.com` (primary)	Oct 2, 2024
		`jp1.hana.ondemand.com`	Sep 11, 2024
Japan	Tokyo/Osaka	`jp1.hana.ondemand.com`	Sep 11, 2024
US-East	Sterling/Toronto	`us3.hana.ondemand.com` (primary)	Oct 9, 2024
		`us1.hana.ondemand.com`	
Saudi Arabia	Riyadh/Dammam	`sa1.hana.ondemand.com`	Jul 31, 2024

Use the primary host for your search. If the audit log is not found there, use the other hosts.

For more information, see Regional Availability [page 11] and Regions and Hosts Available for the Neo Environment.

Results

For more information about the security events that are logged by Identity Authentication, see Auditing and Logging Information [page 2906].

The procedure outlined above does not display logging information related to Identity Provisioning.

1.12.3 Export Change Logs with a History of Administration Operations

You can download a CSV file with a history of the operations performed by administrators in the administration console for SAP Cloud Identity Services.

Context

The exported change logs are saved in a CSV file and contain information about *CREATE*, *UPDATE*, or *DELETE* operations performed by administrators. The change logs are stored in the data base for 24 months.

Each record should contain the following data:

- *Date*
  The date when the resource was created, updated, or deleted.
- *Administrator's ID*
  The ID of the user or the system that made the change.
- *Administrator's Name*
  The name of the user or the system that made the change.
- *Resource Type*
  This indicates which type of resource was created, updated, or deleted, for example service provider.
- *Resource ID*
  An identifier for the resource.
- *Resource Name*
  The name of the resource that was created, updated, or deleted.
- *Action*
  - *CREATE*
    The log shows the value of the attribute when the resource was created.
  - *UPDATE*
    The log shows the changed value of the attribute when the resource was updated.
  - *DELETE*
    The log shows which resource was deleted.
- *Attribute*
  The attribute of the resource that was created or updated with the relevant operation.
- *Value*
  The new value of the attribute.

> **Note**
>
> There are attributes whose values cannot be displayed in the logs. In such cases the value fields are left blank.

> **Example**
>
> The table shows change logs for the following operations:
>
> 1. A tenant administrator, Donna Moore, creates a Company A application.

2. The tenant administrator uploads service provider metadata for the SAML 2.0 trust.
3. The tenant administrator creates custom Company A terms of use by uploading a plain text file.
4. The tenant administrator creates a custom Company A privacy policy by uploading a plain text file.
5. The tenant administrator allows user self-registration for an application of the configured service provider as she uses the default email template for the self-registration process.
6. The tenant administrator configures a branding style for the application.
7. The tenant administrator sets *SAP Community Password Policy v.1.1* for the application .
8. The tenant administrator sets *Company A Terms of Use* for the application.
9. The tenant administrator sets *Company A Privacy Policy* for the application.

Sample Change Logs Data

Date	Administrator's ID	Administrator's Name	Resource Type	Resource ID	Resource Name	Action	Attribute	Value
16-05-2014 10:21:21	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	sp.example.com	UPDATE	Privacy Policy	Company A Privacy Policy
16-05-2014 10:21:15	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	sp.example.com	UPDATE	Terms of Use	Company A Terms of Use
16-05-2014 10:21:08	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	sp.example.com	UPDATE	Password Policy	https://accounts.sap.com/policy/passwords/sap/web/1.1
16-05-2014 10:20:18	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	sp.example.com	UPDATE	Branding Style	Text Color=#ffffff, Background Top Color=#E3E3E3, Border Color=#7B9EB3, show_header=false, Background Bottom Color=#E3E3E3
16-05-2014 10:15:10	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	sp.example.com	UPDATE	Branding Type	Custom Theme
16-05-2014 10:14:29	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	sp.example.com	UPDATE	Self-Registration	Allowed

Date	Administrator's ID	Administrator's Name	Resource Type	Resource ID	Resource Name	Action	Attribute	Value
16-05-2014 10:06:24	P123456	Donna Moore	Privacy Policy Document	53da05f9e4b0732235f24b8e	English Version of Company A Privacy Policy	CREATE	Plain Text Privacy Policy File	
16-05-2014 10:05:11	P123456	Donna Moore	Privacy Policy Document	53da05f9e4b0732235f24b8a	Company A Privacy Policy	CREATE	Display Name	Company A Privacy Policy
16-05-2014 10:03:18	P123456	Donna Moore	Terms of Use Document	53da0595e4b0732235f24b04	English Version of Company A Terms of Use	CREATE	Plain Text Terms of Use File	
16-05-2014 10:02:21	P123456	Donna Moore	Terms of Use Document	53da0595e4b0732235f24b00	Company A Terms of Use	CREATE	Display Name	Company A Terms of Use
15-05-2014 11:32:18	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	UPDATE	SAML 2.0 Provider Name	company_a_service_provider
15-05-2014 11:32:18	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	UPDATE	Authenticating Authority	company_a_tenant_id
15-05-2014 11:32:18	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	UPDATE	Assertion Consumer Service Endpoint	service_provider_acs_endpoint, HTTP-POST
15-05-2014 11:32:18	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	UPDATE	Single Logout Endpoint	service_provider_slo_endpoint, HTTP-POST
15-05-2014 11:32:18	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	UPDATE	Single Logout Endpoint	service_provider_slo_endpoint, HTTP-Redirect

Date	Administrator's ID	Administrator's Name	Resource Type	Resource ID	Resource Name	Action	Attribute	Value
15-05-2014 11:31:11	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	CREATE	Name ID Attribute	User ID
15-05-2014 11:31:11	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	CREATE	Default Name ID Format	urn:oasis:names:tc:SAML:1.1:nameid-format:unspecified
15-05-2014 11:31:11	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	CREATE	Self-Registration	Inherit
15-05-2014 11:31:11	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	CREATE	Certificate	CN=sp.example.com
15-05-2014 11:31:11	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	CREATE	Assertion Consumer Service Endpoint	
15-05-2014 11:31:11	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	CREATE	Single Logout Endpoint	
15-05-2014 11:31:11	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	CREATE	User Attribute	
15-05-2014 11:31:11	P123456	Donna Moore	Service Provider	500964f6e4b0f3cba101f700	Company A (company_a_service_provider)	CREATE	Constant Attribute	

To download a CSV file with change logs information, proceed as follows:

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose the *Audit and Change Logs* tile.
3. Under *Export Change Logs* choose the *Export* button.

Results

You have downloaded a CSV file containing history of operations for the tenant administration, and for the configurations of identity providers and service providers.

1.12.4 Logging OpenID Connect (OIDC) Tokens

Tenant administrator can view the JWT payload of the OpenID Connect (OIDC) tokens issued or received by Identity Authentication.

Prerequisites

You have the Identity Authentication troubleshooting logs with severity **`INFO`**. For more information, see View Troubleshooting Logs [page 3020] or Download Troubleshooting Logs [page 3021].

Context

Identity Authentication can issue OpenID Connect (OIDC) tokens to the applications, and receive tokens. The received tokens can be either tokens from the corporate identity provider, or from Identity Authentication itself. When Identity Authentication issues or receives tokens, it logs the payload of that token.

Message Information

Key	Value	Additional Info
`state`	successful	The token is issued or received.
`action`	issueJwtToken | receiveJwtToken	For issued or received tokens.
`objectType`	openIdClient	

Key	Value	Additional Info
`objectId`	<the `client ID` of the application>	Unique attribute of the objectType – either id, or name, etc.
`category`	audit.authentication	
`jwtPayload`	<JWT claims>	A JSON string, which contains the token claims of issued or received OpenID Connect (OIDC) tokens. **Note** Claims that aren't relevant for troubleshooting are removed from the log for the sake of brevity: `iat`, `exp` and `jti`.
`workflow`	openIdConnect	

`userIdentifier`	<the user identifier>	• The local user ID (P number generated by Identity Authentication).
		• For proxy scenarios:
		  • the local user ID (if *Use Identity Authentication user store* option is enabled and the user is found locally)
		  • the user identifier received from the corporate identity provider (if *Use Identity Authentication user store* is disabled or the user isn't found locally)

`grant_type`	<the respective grant type>	
`serviceProvider`	<the name of the application>	

Example

Message

```
state=successful, action=issueJwtToken, objectType=openIdClient,
objectId="e8b77d3e-e6ba-4fb6-a577-e92d9ecab567", category=audit.authentication,
jwtPayload="{"sub":"P123456","aud":["<the audience claim>"],"scim_id":"b12a0234-
b1b0-41f8-9a2b-dc7395aaaa77","user_uuid":" b12a0234-b1b0-41f8-9a2b-
dc7395aaaa77","user_name":"P123456","azp":"e2b34d3e-e6ab-4fb6-s37-
e92d9ecab567","last_name":"Moore","first_name":"Dona","email":"dona.moore@example
.com","sid":"S-SP-fbf690ce-ce77-464f-b4b3-06adf3312c86"}",
workflow=openIdConnect, userIdentifier=P123456, grant_type=authorization_code,
serviceProvider=myapplication
```

1.12.5 Monitor Provisioning Job Logs

Job logs display information about the execution of provisioning jobs. Each row in the list of job logs shows information about one execution of a job.

Prerequisites

- You have enabled and set up a source and a target system in the Identity Provisioning UI. Make sure all mandatory properties are configured correctly. See: Enable and Disable Systems [page 2419]
- You have run a provisioning job. See: Start and Stop Provisioning Jobs [page 2472]
- Your Identity Provisioning tenant must run on SAP Cloud Identity Services infrastructure to be able to search for job logs.

Procedure

To search and view the job logs, proceed as follows:

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Provisioning Logs* > *Job Logs*.
2. In the *Job Execution Logs* screen, search for a job log and select it. You can search by system name, job type and status.
   Each row in the list of job logs displays the following information:

Column	Details
Source System	The source system the job was triggered for.
Job Type	The job type can be **READ**, **RESYNC** and **SIMULATE**.
Trigger Type	The triggering type for the job. It can be: • **IMMEDIATE** – for manually triggered jobs after choosing *Run Now* or for restarted jobs which were in previous status **Pending Restart**. • **REPEAT** – for scheduled jobs after choosing *Schedule*.

Column	Details
Status	The status of the job. • **Success** – provisioning job has finished successfully. • **Finished with Error** – provisioning job has finished with error. Some entities failed to be read or written, connection to the source system failed or reading from the source system encountered an issue. • **Running** – provisioning job is still running and has not encountered any issues so far. • **Running with Error** – provisioning job is still running, but some entities already failed. • **Manually Terminated** – provisioning job is manually stopped by Identity Provisioning administrator. • **Pending Restart** – provisioning job is temporary paused due to external reasons, not related to your direct interaction (for example, when Identity Provisioning is currently down). When the state gets back to normal, the job will automatically resume with a new job ID, continuing from the last processed entity. Its status will switch over to **Running** or **Running with Error**. Note that, if a provisioning job from a source system to multiple target systems stops due to an error in one of the targets, it will not resume for that specific target upon restart.
Start Time	Date, time, and timezone in UTC format when the job is started.
End Time	Date, time, and timezone in UTC format when the job is finished.
Action	Allows you to stop a running provisioning job by choosing the *Stop Job* button .

3. In the *Job Execution Details* screen, view the following details about a given job:
   - **<System_Name>** – Shows the source system name.
   - **Job Type** - Shows the type of the job, as described in the table above.
   - **Read Mode** - Shows the mode of the job. It can be:
     - **Full Read**
     - **Delta Read**
       The read mode considers how job-related properties, such as: `ips.delta.read` and `ips.full.read.force.count`, are configured on the source system.
       For example, if your system works in delta read mode and you set up ips.full.read.force.count=10, the *Job Execution Details* screen will display 10 consecutive jobs with *Read Mode:Delta Read* followed by one with *Read Mode:Full Read*.

       > ### Note
       >
       > **Resync** and **Simulate** jobs (the latter is available for Identity Provisioning tenants on SAP Cloud Identity Services infrastructure) will always have Read Mode: Full Read. Migration jobs does not display read mode.
   - **Trigger Type** - Shows the trigger type, as described in the table above.

- **Triggered By** - Shows who triggered the job. This field is displayed for jobs of type `Read`, `Resync`, and `Simulate` when the trigger type is IMMEDIATE.
  - User ID (for example, *P000000*) – The ID of the user who manually triggered a `Read`, `Resync`, and `Simulate` job.
  - *Identity Provisioning API* – `<uuid/username>` – The UUID or username of the technical user who triggered a `Read` or `Resync` job via an API client. For more information, see Run Provisioning Jobs via API [page 2480]
  - *Identity Provisioning Internal API* – SAP internal API.

> **Note**
>
> Scheduled jobs don't display the *Triggered By* information.

- **Start Time** and **End Time** - Shows the job start and end time, as described in the table above.
- **Error Message** – Shows the error message.
- **Job Execution Outcome** - Shows the job execution specifics that have influenced the deletion of entities from the target system(s) at the end of the provisioning job.
- **Statistics** – Shows the job statistics, that is, details about how entities are handled.

> **Note**
>
> Note that if an entity type is set to be ignored in the transformations, you won't get any data in the job log statistics for this entity type.

Column	Source System	Target System
Entity	Type of the entity (user, group)	Type of the entity (user, group)
System	Name of the source system	Name of the target system
Action	Action executed on the system: *Read*	Action executed on the system: *Write*
Read	Number of read entities	Not applicable
Created	Not applicable	Number of created entities
Updated	Not applicable	Number of updated entities
Deleted	Not applicable	Number of deleted entities

Column	Source System	Target System
Skipped	Number of skipped entities from the source system Entities can be skipped if they don't fulfill a condition in the read transformation. **Example** Setting this group condition in the read transformation results in skipping all read groups that do not match the specified display name. **Code Syntax** `"condition": "$.displayName == 'Employee'",`	Number of skipped entities in the target system Entities can be skipped if they don't fulfill a condition in the write transformation. An application-specific group is skipped if you try to perform an operation which is not supported for it. For more information about the *supportedOperations* attribute, see Groups [page 514]. **Example** Setting this group condition in the write transformation results in skipping all groups that do not match the specified display name. **Code Syntax** `"condition": "$.displayName == 'Employee'",`
Failed	Number of entities failed during the read operation from the source system	Number of entities failed to be created, updated or deleted in the target system

- **Failed Entities** – Shows additional information about the failed entities. A maximum number of ten entities can be displayed per job log. In case you need to view the full job log, download it from the *Download All Error Logs for This Job* button. For more information see: Manage Provisioning Job Logs [page 3010]

Related Information

Start and Stop Provisioning Jobs [page 2472]
Manage Job Notifications [page 3018]

1.12.6 Monitor Real-Time Logs

Real-time provisioning logs display information about the execution of a real-time sync of a user or a group entity.

Prerequisites

- Your Identity Provisioning tenant is running on SAP Cloud Identity infrastructure.
- You have configured the real-time provisioning scenario. For more information, see Real-Time Provisioning in SAP Cloud Identity Infrastructure
- Your source and target systems are enabled. For more information, see Enable and Disable Systems [page 2419]

Procedure

You can search, view, refresh and configure a retention period of real-time provisioning logs. Proceed as follows:

1. Sign in to the SAP Cloud Identity Services administration console and select *Identity Provisioning* > *Provisioning Logs* > *Real-Time Logs* .
2. In the *Real-Time Provisioning Execution Logs* screen, search for a log and select it. You can search by source system name, entity ID, entity type and status.
   Each row in the list of logs displays the following information about one execution of a real-time sync.

Column	Details
Source System	The name of the source system the real-time provisioning is triggered from. For example: *IAS_Source*
Entity ID	The entity ID of the user or the group in the target system. For example: *baba3ba7-35f0-4567-b507-ce5555c11bbb* If the entity ID is missing, this means that the real-time provisioning has failed. It happens when something goes wrong with the source system. For example, the source system is disabled or establishing the connection to the source system has failed due to incorrect URL. In this case, you get the log without the entity ID.
Entity Type	The entity type: user or group.

Column	Details
Status	The status of the real-time provisioning. For example: • *Finished Successfully* - Real-time provisioning finished successfully. A user or a group is successfully created, updated or deleted real time. • *Finished with Error* - Real-time provisioning finished with error. A user or a group failed to be created, updated or deleted real time.
Start Time	Date, time, and time zone in UTC format when the real-time sync is triggered. For example: *28/Jun/2023 09:19:00 +03*

3. In the *Real-Time Log Details* screen, view the following details under the `<Source System>` name: log type (*Real Time*), entity ID (if it is available), start time, status, error message and *Result on Source and Target System* tables.

Column	Details
Source System	The name of the source system from which the real-time provisioning was triggered. For example: *IAS_Source*
Target System	The name of the target system for which the real-time provisioning was triggered. For example: *Concur_Target*
Operation	The type of the operation. • *CREATE* • *UPDATE* • *PATCH* • *DELETE*
State	The state of the entity when real-time provisioning is triggered. • *Entity Created* • *Entity Updated* • *Entity Skipped* • *Entity Already Provisioned* • *Entity Failed* • *Entity Read* • *Entity Deleted* • *Entity Not Provisioned*

Column	Details
Additional Information	Additional information about the failed entity. If the target system is SAP BTP XS Advanced UAA (Cloud Foundry), the *Additional Information* column shows the origin of the users that have been created, updated or deleted.
Reason	• **Reason (Source System)**: Specifies that the entity was skipped and will to be deprovisioned from the target system. • **Reason (Target System)**: Specifies that the entity was skipped and has been deprovisioned from the target system.

4. (Optional) Choose *Refresh* to get the latest updates of the real-time provisioning logs.
5. Choose *Configure* to define the retention period of real-time provisioning logs.
   Set a period (7, 14 or 30 days). Logs which are older than this period will be automatically deleted. By default, job logs are kept for 7 days.

> ⚠ Restriction
>
> Note the following restrictions about the real-time provisioning logs:
>
> - You cannot delete them manually.
> - You cannot download them.
> - You cannot subscribe to a source system to receive notifications about their status.

1.12.7 Monitor Central Store Logs

Central Store Logs provide information about the application-specific groups that have been provisioned from the Identity Directory to your target systems.

Prerequisites

- You have configured the Local Identity Directory as a source system, linked it to a target system, and run at least one provisioning job. For more information, see Start and Stop Provisioning Jobs [page 2472].
- You have created an application under *Applications & Resources*, and set its application ID as the value of the `ips.application.id` property in the target system.
- You have enabled the *Central Store-Based Provisioning* option under *Applications & Resources* > *Application* > *Provisioning*.

Procedure

You can search, view and refresh central store logs.

1. Sign in to the SAP Cloud Identity Services administration console and select *Identity Provisioning* > *Provisioning Logs* > *Central Store Logs*.

   > ⓘ Note
   >
   > *Central Store Logs* are enabled and will appear under *Provisioning Logs* whenever changes to application-specific groups occur in the Identity Directory, such as assigning users or modifying group attributes.

2. In the *Central Store Provisioning Logs* screen, search for a log and select it. You can search by target system name, application ID, target entity ID, source entity ID and status.
   Each row in the log list provides information about a single execution of central store-based provisioning.

Column	Details
Target System	The name of the target system to which the central store-based provisioning is triggered. For example: *Fieldglass_Target*
Application ID	The ID of the application the application-specific group belongs to. It is configured as a value of the `ips.application.id` property in the target system.
Source Entity ID	The entity ID of the group in the source system, that is Identity Directory. For example: *4200a79f-228b-4fcd-ab15-46d2d78360f6*
Target Entity ID	The entity ID of the group in the target system. For example: *baba3ba7-35f0-4567-b507-ce5555c11bbb* If the target entity ID is missing, it means that the central store-based provisioning has failed. This can occur, for example, when the group is not found or when multiple groups with the same display name are found in the target system.
Entity Type	The entity type: group.

Column	Details
Status	The status of the central store-based provisioning For example: • *Finished Successfully* - Central store-based provisioning finished successfully. A group is successfully updated. • *Finished with Error* - Central store-based provisioning finished with error. A group failed to be updated.
Start Time	Date, time, and time zone in UTC format when the central store-based provisioning is triggered For example: *28/Jan/2025 09:34:21 +02*

3. In the *Central Store Provisioning Log Details* screen, view the following details under the `<Target System>` name:
   - *Log Type*: Central Store Provisioning Log Details
   - *Source System*: The name of the source system
   - *Entity Type*: group
   - *Source Entity ID*: The group ID in the source system
   - *Target Entity ID*: The group ID in the target system
   - *Application ID*: The ID of the application the application-specific group belongs to.
   - *Start Time*: Date, time, and time zone in UTC format when the central store-based provisioning is triggered
   - *Status*: The status of the central store-based provisioning

Column	Details
Target System	The name of the target system to which the central store-based provisioning is triggered. For example: *Fieldglass_Target*
Operation	The type of the operation. • *CREATE* - This operation occurs when a group is created in the target system. • *UPDATE* - This operation occurs when a group is updated in the target system. • *DELETE*- This operation occurs when a group is deleted in the target system.

Column	Details
State	The state of the entity when central store-based provisioning is triggered. • *Entity Created* • *Entity Updated* • *Entity Failed* • *Entity Deleted*
Additional Information	Additional information about the failed entity.

4. (Optional) Choose *Refresh* to get the latest updates of the central store-based provisioning logs.

1.12.8 Monitoring with SAP Cloud ALM

SAP Cloud Identity Services are integrated with SAP Cloud ALM, which is a cloud-based application lifecycle management offering that enables you to implement, operate, and monitor your cloud solutions.

Initial Configuration

You can request SAP Cloud ALM on SAP for Me for yourself or for all customers for whom you have permissions.

After you've requested SAP Cloud ALM, follow the steps to set it up. For more information, see Required Setup for SAP Cloud ALM.

To use the SAP Cloud ALM apps you must activate the data collection for the *Identity Authentication* and/or *Identity Provisioning* component. For more information, see Setup for SAP Cloud Identity Services – Identity Authentication and Setup for SAP Cloud Identity Services – Identity Provisioning.

Configuration & Security Analysis

With the Configuration & Security Analysis app of SAP Cloud ALM, you can check if your Cloud Identity tenants and provisioning systems are configured in alignment with the Security Recommendations - Identity Authentication and Security Recommendations - Identity Provisioning and Monitoring with SAP Cloud ALM [page 2991].

> **Note**
>
> Identity Authentication and Identity Provisioning report data to SAP Cloud ALM daily. For more information, see Monitor Security Recommendations with SAP Cloud ALM.

Cloud Identity services report security configurations that are directly linked to the SAP BTP Security Recommendations. You can find details about the security configurations in the Security Recommendation

Compliance [page 2992] table below. The Configuration & Security Analysis Details [page 3002] table, in turn, shows the security configurations on your tenant that are not linked to the security recommendations.

Security Recommendation Compliance

Service	Name	Value	Security Recommendation Status	Security Recommendation Index
Identity Provisioning	`audit_logs`	The *Data Record* displays the following information as displayed in the *Audit Logs* tab in the administration console for SAP Cloud Identity Services: • Tenant ID • Region • Subdomain If there is no audit logs configuration, the value is empty.	*COMPLIANT* - in case there are audit logs configurations *NONCOMPLIANT* in the following cases: • there is no audit log configuration • you have a stand-alone tenant	BTP-IPS-0001
Identity Provisioning	`trace_setting`	The *Data Record* displays the values set for the properties that enable logging and tracing for personal and sensitive data of the provisioned entities: • `ips.trace.failed.entity.content` = *<value>* • `ips.trace.skipped.entity.content` = *<value>* • `ips.trace.created.entity.content` = *<value>* • `ips.trace.updated.entity.content` = *<value>*	*COMPLIANT* - when all listed properties are set to *false* *NONCOMPLIANT* – if one or more of these properties are set to *true*	BTP-IPS-0002

Service	Name	Value	Security Recommendation Status	Security Recommendation Index
Identity Provisioning	`outbound_authentication_method`	The *Data Record* displays the values set for the properties `Authentication` or `ldap.authentication`. If there is no authentication property configured, the value is empty.	*COMPLIANT* in the following cases: • The system supports certificate authentication and the authentication property is set to *ClientCertificateAuthentication*. • The system supports **only** basic authentication and the `Authentication` property is set to *BasicAuthentication*. *NONCOMPLIANT* in the following cases: • There is no authentication property configured. **Note** This requirement is not applicable to the Local Identity Directory, as it does not require configuration of an authentication property. Therefore, it should not be considered noncompliant.	BTP-IPS-0003

Service	Name	Value	Security Recommendation Status	Security Recommendation Index
			• The system supports certificate authentication, but the value of the `Authentication` property is different from *ClientCertificateAuthentication* or the certificate is not configured.	
Identity Provisioning	`inbound_authentication_method`	• *Basic Authentication* • *Certificate Authentication* **Note** The compliance check is relevant on the tenant level.	*COMPLIANT* in the following cases: There is a system user with configured certificate that possesses at least one of the roles: *Access Real-Time Provisioning API* or *Access Proxy System API*. *NONCOMPLIANT* when: There is no system user with configured certificate that possesses at least one of the roles: *Access Real-Time Provisioning API* or *Access Proxy System API*.	BTP-IPS-0004

Service	Name	Value	Security Recommendation Status	Security Recommendation Index
Identity Provisioning	`secure_protocols`	The *Data Record* displays the values of all configured **URL** properties. If no URL properties are configured, the value is empty.	*COMPLIANT* in the following cases: • The system uses `ProxyType` *Internet* and the secure protocols used are *HTTPS* or *SSH*. • The system uses `ProxyType` *OnPremise*, hence insecure protocols are accepted. • If there are no **URL** properties configured. *NONCOMPLIANT* when the system uses `ProxyType` *Internet* and the protocol is insecure (for example *HTTP*).	BTP-IPS-0005

Service	Name	Value	Security Recommendation Status	Security Recommendation Index
Identity Provisioning	`credential_properties`	The check validates whether for the configuration of sensitive information are used credential properties. The information is displayed in the following format: `sensitive property = <value>` The possible values are: • `true` - when the property is created as `Credential` For example, if you have configured `Password` as `Credential`, the result is `Password = true` • `false` - if the property is not created as `Credential` For example, if you have configured `ssh.password` as `Standard`, the result is `ssh.password = false` empty - if there are no sensitive properties configured If there are no sensitive properties configured, the value is empty.	*COMPLIANT* in the following cases: • All sensitive properties are configured as type *Credential*. • There are no sensitive properties configured. *NONCOMPLIANT* if one or more sensitive properties are configured as type *Standard*.	BTP-IPS-0006

Service	Name	Value	Security Recommendation Status	Security Recommendation Index
Identity Provisioning	`trust_all`	The check evaluates whether the `TrustAll` property is used in productive scenarios. The information is displayed in the following format: `Value` = *<property value>*	*COMPLIANT* when the property `TrustAll` is set to *false*. *NONCOMPLIANT* - when the property `TrustAll` is set to *true*. **Note** The security compliance check is not supported for connectivity destinations in the SAP BTP cockpit.	BTP-IPS-0007
Identity Provisioning	`ips_admin_user_count`	The *Data Record* displays the number of assigned tenant administrators.	*COMPLIANT* when the number of tenant administrators assigned is greater than one. *NONCOMPLIANT*: • In case there is only one tenant administrator assigned.	BTP-IPS-0008
Identity Authentication	`certificate_authentication`	Checks whether the certificate generation and authentication is enabled for each of the user types. If it is, the user types are displayed in a comma separated string: `usertypes`: `employee`,`partner`. If there are no user types for which the setting is enabled, the value is empty.	*COMPLIANT* - if there is at least 1 user type for which the setting is enabled. *NONCOMPLIANT* - if there are no user types for which the setting is enabled.	BTP-IAS-0006

Service	Name	Value	Security Recommendation Status	Security Recommendation Index
Identity Authentication	`audit_logs`	Checks whether there is an audit log configuration. If there is, the value is *CONFIGURED*. If there is not, the value is `empty`.	*COMPLIANT* - in case there are audit logs configurations. *NONCOMPLIANT* - in case there are *NO* audit logs configurations.	BTP-IAS-0011
Identity Authentication	`custom_domain`	Checks whether the custom domain is configured. If it is, the value is *CONFIGURED*. If it is not, the value is empty.	*COMPLIANT* - if the custom domain is configured. *NONCOMPLIANT* - if the custom domain is *NOT* configured.	BTP-IAS-0019
Identity Authentication	`mail_server_configuration`	Checks whether there is a custom mail server configured. If the custom mail server is of type AWS, then the value is `aws`, otherwise the value is `custom`. If the there is no custom mail server configuration, the value is empty.	*COMPLIANT* - if the custom mail server is configured. *NONCOMPLIANT* - if the custom mail server is NOT configured.	BTP-IAS-0020
Identity Authentication	`system_notification`	Checks whether the admin console is configured to send emails about expiring certificates, system notifications, and new administrators to specific e-mail addresses, to the email addresses of all administrators or to none. If there are no email recipients, the value is empty. If the number of configured recipients is equal to the number of user admins or the setting to send the notifications to all admins is enabled, the value is `configured`. Otherwise it is `misconfigured`.	*COMPLIANT* - If the number of configured recipients is equal to the number of user admins or the setting to send the notifications to all admins is `enabled`. *NONCOMPLIANT* - If there are no email recipients or if it is `misconfigured`.	BTP-IAS-0022

Service	Name	Value	Security Recommendation Status	Security Recommendation Index
Identity Authentication	`admin_console_mfa`	Checks whether multi-factor authentication (MFA) is configured for the administration console. The possible values are: • `delegated_to_corp_idp` - if the administration console is set in full proxy mode and the "Apply Application Configurations" setting of the corporate IdP is disabled. • `configured` - if there is a TFA action configured on either service provider (SP) or on a tenant level. • `misconfigured` - if there is no TFA action configured.	*COMPLIANT* - if there is a TFA action configured. *NONCOMPLIANT* - if there is no TFA action configured. *UNRATED* - if it is delegated to the corporate IdP.	BTP-IAS-0001

Service	Name	Value	Security Recommendation Status	Security Recommendation Index
Identity Authentication	`password_policy`	Checks whether a password policy is configured. The possible values are: • `delegated_to_corp_idp` - if the SP is set in full proxy mode. • `standard` - if the password policy is set to the default standard one. • `enterprise` - if the password policy is set to the default enterprise one. • `custom` - if a custom password policy is configured. • `empty` - if there is no password policy set.	*COMPLIANT* - if the password policy is either custom or enterprise. *NONCOMPLIANT* - if the password policy is either standard or none is set. *UNRATED* - if it is delegated to the corporate IdP.	BTP-IAS-0002
Identity Authentication	`user_application_access`	Checks the access of end users to the application. The possible values are: • `public` - Everyone can access the application; new users can register using the self-registration process. • `internal` - Only existing users can access the application. • `private` - Only users registered by the application can log on. • empty - If the property is inherited from the parent application.	*COMPLIANT* - if the access is restricted only to existing users. `value=internal`. *NONCOMPLIANT* - `value=public` or `value=private` or the value is inherited from the parent application.	BTP-IAS-0003

Service	Name	Value	Security Recommendation Status	Security Recommendation Index
Identity Authentication	`social_sign_on`	Checks whether social sign-on (SSO) is enabled. If it is enabled, the value is *ON*. If it is disabled, the value is *OFF*. The value is empty if the property is inherited from the parent application.	*COMPLIANT* - if SSO is disabled. *NONCOMPLIANT* - if SSO is enabled. *UNRATED* - if it is inherited from the parent application.	BTP-IAS-0005
Identity Authentication	`disable_password_authentication`	Checks whether the default authentication method for end users to authenticate with the service is *user name* and *password*. The possible values are: • empty - if the configured IdP is not the default local IdP, or there is no RBA configuration, or there are no RBA rules. • `configured` - in case the configuration matches the described in the *COMPLIANT* status . • `partially configured` - the default action of the RBA configuration is either *Allow* or *TFA* and there is a RBA rule with action *Deny* which only has authentication method set to *User Name and Password* and there is a RBA rule with action different from *Deny*.	*COMPLIANT* • if the default action of the RBA configuration and the actions of all the RBA rules is *Deny*. • the default action of the RBA configuration is either *Allow* or *TFA* and there is a RBA rule with action *Deny* which only has authentication method set to *User Name and Password*. NONCOMPLIANT • if the default action is *Deny* and the action of any of the RBA rules is *Allow* or *TFA* and the authentication method is either not configured or *User Name and Password*. • the default action of the RBA configuration is either *Allow* or *TFA* and there are no RBA rules.	BTP-IAS-0006

Service	Name	Value	Security Recommendation Status	Security Recommendation Index
Identity Authentication	`branding`	Checks whether there is branding of the organization added to the logon screen. The possible values are: • empty - if there is no logo resource configured and the custom CSS is not enabled. • `configured` - if there is logo resource configured or the custom CSS is enabled.	*COMPLIANT* - if there is logo resource configured or the custom CSS is enabled. *NONCOMPLIANT* - if there is no logo resource configured and the custom CSS is not enabled.	BTP-IAS-0021
Identity Authentication	`reCAPTCHA`	Checks whether self-registration and CAPTCHA are enabled. The possible values are: 1. empty if CAPTCHA is not enabled. 2. configured: if CAPTCHA is enabled for at least the registration form or the login page.	COMPLIANT - if the self-registration is disabled or CAPTCHA is enabled. NONCOMPLIANT - if self-registration is enabled or CAPTCHA is disabled.	BTP-IAS-0025
Identity Authentication	`use_identity_auth_user_store`	Reports whether the *Use Identity Authentication user store* setting is enabled. The possible values are: • *ON*- if enabled • *OFF* - if disabled	*COMPLIANT* - if enabled. *NONCOMPLIANT* - if disabled.	BTP-IAS-0015

Configuration & Security Analysis Details

Service	Name	Value
Identity Authentication	`ID`	Reports the Id of the service provider or corporate identity provider (IdP).
Identity Authentication	`TYPE`	Reports the type of the service provider (`SYSTEM_APP`, `BUNDLED_APP` or `CHARGED_APP`) or `CORPORATE_IDP` for corporate IdPs.
Identity Authentication	`NAME`	Reports the name of the service provider or corporate IdP.

Service	Name	Value
Identity Authentication	`days_without_mfa_prompt`	Reports the number of days for which the users will not get prompted for second-factor authentication, if they sign in from the same browser. If the property is not configured, the value is empty.
Identity Authentication	`self_registration_link_lifetime_seconds`	Reports the validity of the link sent to user for self registration in seconds.
Identity Authentication	`on_behalf_link_lifetime_seconds`	Reports the validity of the link sent to user for on-behalf registration in seconds.
Identity Authentication	`invitation_link_lifetime_seconds`	Reports the validity of the link sent to user for invitation in seconds.
Identity Authentication	`forgot_password_lifetime_seconds`	Reports the validity of the link sent to user for forgotten password in seconds.
Identity Authentication	`reset_password_lifetime_seconds`	Reports the validity of the link sent to user for password reset in seconds.
Identity Authentication	`locked_password_lifetime_seconds`	Reports the validity of the link sent to user for locked password in seconds.
Identity Authentication	`initial_password_token_lifetime_seconds`	Reports the validity of the initial password set to user in seconds.
Identity Authentication	`idp_initiated_sso`	Reports whether the IdP-initiated Single Sign On (SSO) is enabled. If it is enabled, the value is *ON*. Otherwise the value is *OFF*.
Identity Authentication	`manage_identity_provisioning_admin_count`	Reports the count of users and systems authorized to access the Application Configuration API for running Identity Provisioning jobs or downloading job logs. **Example** `systems=0;users=1`
Identity Authentication	`read_users_admin_count`	Reports the count of users and systems authorized to read and export users. **Example** `systems=0;users=1`

Service	Name	Value
Identity Authentication	`manage_applications_admin_count`	Reports the count of users and systems authorized to configure applications. **Example** `systems=0;users=1`
Identity Authentication	`manage_users_admin_count`	Reports the count of users and systems authorized to manage, export, and import users. **Example** `systems=0;users=1`
Identity Authentication	`manage_corp_idp_admin_count`	Reports the count of users and systems authorized to configure corporate identity providers. **Example** `systems=0;users=1`
Identity Authentication	`manage_groups_admin_count`	Reports the count of users and systems authorized to manage user groups. **Example** `systems=0;users=1`
Identity Authentication	`manage_tenant_configuration_admin_count`	Reports the count of users and systems authorized to manage tenant configuration and authorization assignment to users. **Example** `systems=0;users=1`
Identity Authentication	`default_rba_auth_action`	Reports the default action set in the RBA configuration of the application. The possible values are: • *Allow* • *Deny* • *TFA*
Identity Authentication	`default_authenticating_idp_id`	Reports the `id` of the configured corporate IdP.
Identity Authentication	`default_authenticating_idp_name`	Reports the name of the configured corporate IdP.

Service	Name	Value
Identity Authentication	`default_authenticating_idp_display_name`	Reports the display name of the configured corporate IdP.
Identity Authentication	`default_authenticating_idp_type`	Reports the type of the configured corporate IdP.
Identity Authentication	`trust_all_corporate_idp`	Reports whether the *Trust All Corporate Identity Providers* setting is enabled. The possible values are: • *ON*- if enabled • *OFF* - if disabled
Identity Authentication	`allow_ias_users_log_on`	Reports whether the *Allow Identity Authentication Users Log On* setting is enabled. The possible values are: • *ON*- if enabled • *OFF* - if disabled
Identity Authentication	`spnego_enabled`	Reports whether the *SPNEGO Authentication* setting is enabled. The possible values are: • *ON*- if enabled • *OFF* - if disabled
Identity Authentication	`biometric_authentication_enabled`	Reports whether the *Biometric Authentication* setting is enabled. The possible values are: • *ON*- if enabled • *OFF* - if disabled
Identity Authentication	`force_authentication`	Reports whether the *Force Authentication* setting is enabled. The possible values are: • *ON*- if enabled • *OFF* - if disabled
Identity Authentication	`default_signing_algorithm`	Reports the *Signing Algorithm* configuration. The possible values are: • *SHA-1* • *SHA-256* • *SHA-512*
Identity Authentication	`concurrent_access`	Reports the Concurrent Access configuration. The possible values are: • *Allow* • *Warning* • *Error*
Identity Authentication	`identity_provider_id`	Reports the `id` of the corporate IdP.

Service	Name	Value
Identity Authentication	`identity_provider_name`	Reports the name of the corporate IdP. The value is empty if not configured.
Identity Authentication	`identity_provider_display_name`	Reports the display name of the corporate idP.
Identity Authentication	`identity_provider_type`	Reports the type of the corporate IdP. Possible values are: • *MS_ADFS_2.0 (Microsoft ADFS / Entra ID (SAML 2.0))* • *SAML_2.0 (SAML 2.0 Compliant)* • *OPENID_CONNECT (OpenID Connect Compliant)NW_SSO (SAP Single Sign-On (SAML 2.0))*
Identity Authentication	`forward_all_sso_requests`	Reports whether the *Forward All SSO Requests to Corporate IdP* setting is enabled. The possible values are: • *ON*- if enabled • *OFF* - if disabled
Identity Authentication	`allow_identity_auth_users_only`	Reports whether the *Allow Identity Authentication users only* setting is enabled. The possible values are: • *ON*- if enabled • *OFF* - if disabled
Identity Authentication	`apply_app_auth_configs`	Reports whether the "Apply Application Configurations" setting is enabled. The possible values are: • *ON*- if enabled • *OFF* - if disabled
Identity Authentication	`logout_redirect_url`	Reports *Logout Redirect URL*. The value is empty if not configured.
Identity Authentication	`default_signing_algorithm`	Reports the *Signing Algorithm* configuration only if the type is `NOT OPENID_CONNECT`. The possible values are: • *SHA-1* • *SHA-256* • *SHA-512*
Identity Authentication	`sign_authentication_requests`	Reports whether the *Sign authentication requests* setting is enabled only if the type is `NOT OPENID_CONNECT`. • *ON*- if enabled • *OFF* - if disabled

Service	Name	Value
Identity Authentication	`sign_logout_messages`	Reports whether the *Sign single logout messages* setting is enabled only if the type is `NOT OPENID_CONNECT`. The possible values are: • *ON*- if enabled • *OFF* - if disabled
Identity Authentication	`PRIORITY`	Reports the index of the rule based on priority.
Identity Authentication	`IDENTITY_PROVIDER_ID`	Reports the id of the corporate IdP.
Identity Authentication	`IDENTITY_PROVIDER_NAME`	Reports the name of the corporate IdP. Empty if not configured.
Identity Authentication	`IDENTITY_PROVIDER_DISPLAY_NAME`	
Identity Authentication	`USER_TYPE`	Reports the user type of the rule. Possible values are: `employee`, `public`, `partner`, `customer`, `external`, `onboardee`, `alumni`. The value is empty if not configured.
Identity Authentication	`USER_GROUP`	Reports the name of the group. The value is empty if not configured.
Identity Authentication	`USER_EMAIL_DOMAIN`	Reports the email domain of the rule. The value is empty if not configured.
Identity Authentication	`IP_NETWORK_RANGE`	Reports the IP network range of the rule in CIDR format. **Example** `10.10.10.10/11` The value is empty if not configured.
Identity Authentication	`PRIORITY`	Reports the index of the rule based on priority.
Identity Authentication	`IP_NETWORK_RANGE`	Reports the IP network range of the rule in CIDR format. **Example** `10.10.10.10/11` Empty if not configured.

Service	Name	Value
Identity Authentication	IP_FORWARD_RANGE	Reports the IP forward range of the rule in CIDR format. **Example** `10.10.10.10/11` The value is empty if not configured.
Identity Authentication	ACTION	Reports the action of the rule. The possible values are: - *Allow* - *Deny* - *TFA*
Identity Authentication	GROUP_TYPE	Reports the name of the group. Possible values are: - *Cloud (Cloud Group)* - *Corporate(On-Premise Group)* - The value is empty if not configured.
Identity Authentication	AUTH_METHOD	Reports the authentication method for the rule. Possible values are: - *CERT(Client Certificate)SPNEGO* - *UID_PW (User Name and Password)* - *TOKEN (Token)* - *SOCIAL_IDENTITY (Social identity Provider)* - *TRUSTED_IDP_SAML_ASSERTION (Trusted IdP SAML Assertion)* - The value is empty if not configured.
Identity Authentication	USER_ATTRIBUTES	Reports the user attributes for the rule in format of `key=value`. The value is empty if not configured.
Identity Authentication	CORPORATE_IDP_ATTRIBUTE	Reports the corporate IdP attribute for the rule in format of `key=value`. The value is empty if not configured.
Identity Authentication	USER_EMAIL_DOMAIN	Reports the email domain of the rule. The value is empty if not configured.
Identity Authentication	PASSWORD_POLICY_NAME	Reports the name of the password policy.
Identity Authentication	PASSWORD_POLICY_STRENGTH	Reports the strength of the password policy.
Identity Authentication	MINIMUM_PASSWORD_LENGTH	Reports the minimum length of characters of the password policy.

Service	Name	Value
Identity Authentication	`MINIMUM_PASSWORD_AGE_HOURS`	Reports the minimum password age in hours.
Identity Authentication	`USER_INACTIVITY_MONTHS`	Reports the maximum user inactivity in months.
Identity Authentication	`PASSWORD_HISTORY_ENTRY_COUNT`	Reports the password history entry count.
Identity Authentication	`FAILED_SIGN_IN_ATTEMPTS`	Reports the failed sign in attempts.
Identity Authentication	`PASSWORD_LOCKED_PERIOD_HOURS`	Reports the password locked period in hours.
Identity Authentication	`REQUIRED_CHARACTER_GROUPS`	Reports the number of the required character groups.
Identity Authentication	`PASSWORD_BEHAVIOUR`	Reports the password behavior of the policy. Possible values are: • *reset* • *change*

Job & Automation Monitoring

In the Job & Automation Monitoring app, you can monitor the execution of your provisioning jobs using specific metrics. The supported provisioning job types are *Read* (immediate and scheduled), *Resync* and *Simulate*. In the Job & Automation Monitoring app they are grouped by system name and job type. You have the option to drill down from a specific row to the list of executed jobs.

For more information, see Job & Automation Monitoring and the relevant section on the SAP Support Portal.

Getting Support

In case you encounter problems with setting up or using the apps, create a case under the relevant component:

App Name	Component
Configuration & Security Analysis	`SV-CLM-OP-CSA`
Job and Automation Monitoring	`SV-CLM-OP-JM`

Related Information

SAP Cloud ALM

1.12.9 Manage Provisioning Job Logs

After you view and analyze the provisioning job logs, you can download or delete them.

Prerequisites

- You have enabled and set up a source and a target system in the Identity Provisioning UI. Make sure all mandatory properties are configured correctly. See: Enable and Disable Systems [page 2419]
- You have run a provisioning job. See: Start and Stop Provisioning Jobs [page 2472]

Download Execution Logs for All Jobs

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Provisioning Logs* > *Job Logs*.
2. From the upper right corner, choose the *Download Execution Logs for All Jobs* button.
3. Choose *Download*. If the number of logs is too large, the execution logs will be downloaded in parts. Each part (a ZIP archive) contains 3000 logs, by default.
4. Save all ZIP files in your local file system.

Download All Error Logs for a Single Job

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Provisioning Logs* > *Job Logs*.
2. Select a job that has finished with error.
3. Above the **Failed Entities** table, choose the button and select *Download All Error Logs for This Job*.
4. Choose *Download*. The job log is downloaded as a ZIP archive. Its name pattern is: **ips_jobErrorLogs_<job ID>_<date>_<time>**
   Hint: The job ID is at the end of the URL of the selected job in the Identity Provisioning admin console.
5. Save the ZIP file in your local file system, and then open it to view the log records of all failed entities from this job.

> **Note**
>
> If your provisioning job is in *Pending Restart* status (that is, temporary paused due to external reasons), the logs for this job will be generated after it is resumed and completed. Be aware that logs for created, skipped, deleted, and failed entities, on the page where the job was paused, will be duplicated, assuming the page size property is configured for the given provisioning system. For more information, see List of Properties [page 80].
>
> For example, if you have set `concur.page.size`=30 and your job paused while processing the 20th entity on the second page, the logs for those 20 entities will be duplicated. The number of duplicated

logs can be equal to or less than the configured page size. If page size is not configured, the default value of 100 will be used.

Download All Skipped Entity Logs for a Single Job

> **Note**
>
> This functionality is available only for Identity Provisioning bundle and standalone tenants running on SAP Cloud Identity infrastructure.

Prerequisites

- You have set the `ips.trace.skipped.entity` property to **`true`** in the source system.
- You have set the `ips.trace.skipped.entity.content` property to **`true`** in the source system.

For more information, see List of Properties [page 80]

Download the skipped entities for a single job to identify the entities themselves, the systems they are skipped from, the reason behind this, as well as to view the content of the entities.

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Provisioning Logs* > *Job Logs*.
2. Select a job that has at least one skipped entity.
3. On the right-hand side of the *Statistics* section, choose the button and select *Download All Skipped Entity Logs for This Job*.
   The log is downloaded as a `zip` archive. The name of the file follows the pattern: `ips_jobSkippedEntitiesLogs_<job ID>_<date>_<time>`.
4. Save the `zip` file in your local file system, and then open it to view the log records of all skipped entities for this job.
   The log displays the following information:

Entity Details	Value
User	The ID of the skipped entity For example: **`12345678-1a2b-1bc2-3cd4-1234567890ef`**
Group	
System	The system the entity is skipped from. This could be either a source system or a target system. For example: **`IAS.target`**

Entity Details	Value
Skip Reason	The reason the entity is skipped Normally, a user or a group is skipped because it does not fulfill a condition in the source read transformation or the target write transformation. If this is the case, you will see the exact condition the entity does not fulfill. For example: **Code Syntax** `The condition: ($.emails EMPTY false) && ($.userName EMPTY false), is not fulfilled.` A user or a group can also be skipped if you use the `skipOperations` expression in the transformations to avoid creating, deleting, or updating entities in target systems. If this is the case, you will see the following message: `Operation [<create><update><delete>] is skipped in target transformation`
Content	The content (attributes of the skipped entity For example: **Code Syntax** `{"active":true,"displayName":"John Smith","emails":[{"value":"john.smith@example.com"}],"name":{"familyName":"Smith","givenName":"John"},"schemas":["urn:ietf:params:scim:schemas:core:2.0:User","urn:ietf:params:scim:schemas:extension:sap:2.0:User"],"urn:ietf:params:scim:schemas:extension:sap:2.0:User":{"userUuid":"0036024b-0ede-4fc3-9ed7-c55632de8246"},"urn:sap:cloud:scim:schemas:extension:custom:2.0:User":{"userId":"12345678-1a2b-1bc2-3cd4-1234567890ef"},"userName":"","userType":"public"}`

The log is organized in sections which start with the ID of the skipped entity. If a user or a group is skipped in more than one systems, the log will display the ID of the skipped entity as many times as there are systems where the entity is skipped from, the skip reason and the entity's content.

Download All Created Entity Logs for a Single Job

> **Note**
>
> This functionality is available only for Identity Provisioning bundle and standalone tenants running on SAP Cloud Identity infrastructure.

Prerequisites

- You have set the `ips.trace.created.entity` property to **`true`** in the source system.
- You have set the `ips.trace.created.entity.content` property to **`true`** in the source system.

For more information, see List of Properties [page 80]

Download the created entities for a single job to identify the system in which they are created and to view the content of the entities.

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Provisioning Logs* > *Job Logs*.
2. In the *Job Execution Logs* screen, search for a job log and select it.
3. If the job has created at least one entity, you are able to choose the button on the right-hand side of the *Statistics* section and select *Download All Created Entity Logs for This Job*.
   The log is downloaded as a `zip` archive. The name of the file follows the pattern: `ips_jobCreatedEntitiesLogs_<job ID>_<date>_<time>`.
4. Save the `zip` file in your local file system, and then open it to view the log records of all created entities for this job.
   The log displays the following information:

Entity Details	Value
User	The ID of the created entity in the target or proxy system For example: **`12345678-1a2b-1bc2-3cd4-1234567890ef`**
Group	
System	The system in which the entity is created. This could be either a target system or a proxy system. For example: **`IAS.target`**

Entity Details	Value
Content	The content attributes of the created entity For example: Code Syntax `{"active":true,"displayName":"John Smith","emails":[{"value":"john.smith@example.com"}],"name":{"familyName":"Smith","givenName":"John"},"schemas":["urn:ietf:params:scim:schemas:core:2.0:User","urn:ietf:params:scim:schemas:extension:sap:2.0:User"],"urn:ietf:params:scim:schemas:extension:sap:2.0:User":{"userUuid":"0036024b-0ede-4fc3-9ed7-c55632de8246"},"urn:sap:cloud:scim:schemas:extension:custom:2.0:User":{"userId":"12345678-1a2b-1bc2-3cd4-1234567890ef"},"userName":"","userType":"public"}`

The log is organized in sections which start with the ID of the created entity. If a user or a group is created in more than one systems, the log will display the ID of the created entity as many times as the number of systems in which it is created and the entity's content.

Download All Updated Entity Logs for a Single Job

> **Note**
>
> This functionality is available only for Identity Provisioning bundle and standalone tenants running on SAP Cloud Identity infrastructure.

Prerequisites

- You have set the `ips.trace.updated.entity` property to **`true`** in the source system.
- You have set the `ips.trace.updated.entity.content` property to **`true`** in the source system.

For more information, see List of Properties [page 80]

Download the updated entities for a single job to identify the system in which they are updated and to view the content of the entities.

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Provisioning Logs* > *Job Logs*.

2. In the *Job Execution Logs* screen, search for a job log and select it.
3. If the job has updated at least one entity, you are able to choose the button on the right-hand side of the *Statistics* section and select *Download All Updated Entity Logs for This Job*.
   The log is downloaded as a `zip` archive. The name of the file follows the pattern: `ips_jobUpdatedEntitiesLogs_<job ID>_<date>_<time>`.
4. Save the `zip` file in your local file system, and then open it to view the log records of all updated entities for this job.
   The log displays the following information:

Entity Details	Value
User *Group*	The ID of the updated entity in the target or proxy system For example: **`12345678-1a2b-1bc2-3cd4-1234567890ef`**
System	The system in which the entity is updated. This could be either a target system or a proxy system. For example: **`IAS.target`**

Content	The content attributes of the updated entity
	For example:
	Code Syntax
	`{"active":true,"displayName":"John Smith","emails":[{"value":"john.smith@example.com"}],`
	`"name":{"familyName":"Smith","givenName":"John"},"schemas":["urn:ietf:params:scim:schemas:core:2.0:User","urn:ietf:params:scim:schemas:extension:sap:2.0:User"],"urn:ietf:params:scim:schemas:extension:sap:2.0:User":{"userUuid":"0036024b-0ede-4fc3-9ed7-c55632de8246"},"urn:sap:cloud:scim:schemas:extension:custom:2.0:User":{"userId":"12345678-1a2b-1bc2-3cd4-1234567890ef"},"userName":"","userType":"public"}`

The log is organized in sections which start with the ID of the updated entity. If a user or a group is updated in more than one systems, the log will display the ID of the updated entity as many times as the number of systems in which it is updated and the entity's content.

Download All Deleted Entity Logs for a Single Job

> **Note**
>
> This functionality is available only for Identity Provisioning bundle and standalone tenants running on SAP Cloud Identity infrastructure.

Prerequisites

You have set the `ips.trace.deleted.entity` property to **`true`** in the source system.

For more information, see List of Properties [page 80]

Download the deleted entities for a single job to identify the system from which they are deleted.

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Provisioning Logs* > *Job Logs*.
2. In the *Job Execution Logs* screen, search for a job log and select it.
3. If the job has deleted at least one entity, you are able to choose the button on the right-hand side of the *Statistics* section and select *Download All Deleted Entity Logs for This Job*.
   The log is downloaded as a `zip` archive. The name of the file follows the pattern: `ips_jobDeletedEntitiesLogs_<job ID>_<date>_<time>`.
4. Save the `zip` file in your local file system, and then open it to view the log records of all deleted entities for this job.
   The log displays the following information:

Entity Details	Value
User *Group*	The ID of the deleted entity in the target or proxy system For example: **`12345678-1a2b-1bc2-3cd4-1234567890ef`**
System	The system from which the entity is deleted. This could be either a target system or a proxy system. For example: **`IAS.target`**

   The log is organized in sections which start with the ID of the deleted entity. If a user or a group is deleted from more than one systems, the log will display the ID of the deleted entity as many times as the number of systems from which it is deleted.

Download Job Logs via API

> **Note**
>
> This functionality is available only for Identity Provisioning bundle and standalone tenants running on SAP Cloud Identity infrastructure.

- You need a technical user (a user of type System in SAP Cloud Identity Services admin console) with *Access Identity Provisioning Tenant Admin API* permission assigned. For more information, see Managing Administrators.

Use the Identity Provisioning tenant admin API to download job logs via API calls for a single provisioning job. The API is available on the SAP Business Accelerator Hub: SAP Cloud Identity Services *Identity Provisioning Service* > *API Reference* > *JobLogs*. The URL for accessing the Tenant Admin API follows the pattern: `https://<ias-tenant-host>/ips/service/publicapi/v1/jobLogs/<JobId>?action=export&logType=<logType>`, where:

Parameter	Value
`<JobId>`	The job identifier
Action	The action to be executed Only possible value: **`export`**
`<logType>`	The type of the job log to be downloaded Possible values: • **`commonLog`** - Download logs containing the following details for this specific job execution (start time, end time, statistics, type, mode). These details are currently included in the common logs for all job executions. • **`failedEntitiesLog`** - Download logs containing details about the failed entities for this specific job execution. • **`skippedEntitiesLog`** - Download logs containing details about the skipped entities for this specific job execution. • **`createdEntitiesLog`** - Download logs containing details about the created entities for this specific job execution. • **`updatedEntitiesLog`** - Download logs containing details about the updated entities for this specific job execution. • **`deletedEntitiesLog`** - Download logs containing details about the deleted entities for this specific job execution.

For example: `https: //<ias-tenant-host>/ips/service/publicapi/v1/jobLogs/cbc1eb01686664856111?action=export&logType=allErrorsLog`

Save the `zip` file in your local file system, and then open it to view the log records for the log type you have defined.

Delete Job Logs

If you don't need your job logs anymore, you can delete them. You can do this manually or automatically (by setting a retention period).

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Provisioning Logs* > *Job Logs*.
2. Choose the *Delete Logs* icon.

   > ⚠ Caution
   >
   > This deletes the logs for all finished jobs. If a job is still running though, it will stay along with its logs.

3. You can set a time period for keeping the job logs available for monitoring.
   1. From the upper right corner, choose *Configure Job Logs Settings*.
   2. Set a period (**7**, **14** or **30** days). Logs which are older than this period will be automatically deleted. By default, job logs are kept for 7 days.
   3. If you want to keep the logs longer, you can download and save them in your local file system.

1.12.10 Manage Job Notifications

You can subscribe to a source system to receive notification e-mails. They provide information about the job execution and links to download job logs, in case of created, updated, failed, skipped, or deleted entities.

Context

When you subscribe to a source system, you can receive notification e-mails in the following cases:

- You start or schedule a provisioning job and it fails.
  You'll receive an e-mail with subject *Provisioning Running with Error. Source System: <name>*. You receive one e-mail per job, after the first failed entity. If more entities fail during this job, no additional e-mails will be sent. This behavior is related to property `ips.job.notification.skip.intermediate.notifications`.
- The failed job has finished.
  You'll receive an e-mail with subject *Provisioning Finished with Error. Source System: <name>*. By default, if the same job runs again and keeps failing, no further notifications will be sent to your e-mail. However, you can control the notifications via properties `ips.job.notification.ignored.consecutive.failures` and `ips.job.notification.repeat.on.failure`. For more information, see: List of Properties [page 80]
- The job is back to normal (the problem with the failed entities has been resolved).
  After a new run, the job has successfully finished. You'll receive only one e-mail with subject *Provisioning Success. Source System: <name>*.

The notification e-mail tells you which are the source and the target systems, what are the job type and the read mode, its start time and status. In some cases the e-mail contains job execution specifics that have

influenced the deletion of entities from the target system(s). It includes also the `Navigate to details` link that opens the *Job Execution Details* screen in the Identity Provisioning UI.

In case your job finished with created, updated, failed, skipped, or deleted entities, links to `Download created entities logs`, `Download updated entities logs`, `Download error logs`, `Download skipped entities logs`, and `Download deleted entities logs` are provided, respectively. The content of the downloaded log files depends on your configuration of the `ips.trace.created.entity`, `ips.trace.created.entity.content`, `ips.trace.updated.entity`, `ips.trace.updated.entity.content`, `ips.trace.skipped.entity`, `ips.trace.skipped.entity.content`, `ips.trace.failed.entity.content`, and `ips.trace.deleted.entity` properties. For more information, see: List of Properties [page 80]

> **Note**
>
> Sending navigation links to the download page of created, updated, failed, skipped, and deleted entities is supported for tenants running on SAP Cloud Identity Services infrastructure.

> **Note**
>
> If you subscribe to a source system, and then run a successful provisioning job, no notification e-mails will be sent.

Procedure

1. Sign in to the administration console of SAP Cloud Identity Services and navigate to *Identity Provisioning* > *Source Systems*.
2. Select the system you need to watch and choose *Jobs*.

   > **Note**
   >
   > Notifications are supported only for read and resync jobs.

3. Choose *Subscribe*.
   - To subscribe yourself, choose *Subscribe me*.
   - To subscribe another user or a group (distribution list), choose *Subscribe others*. Fill in the required fields and choose *Add*.

     > **Note**
     >
     > From the *Recipients* list, you can remove existing subscribers. To do that, go to the *Action* column and choose the icon.

4. You can now run or schedule a provisioning job.
5. If you no longer need to be subscribed to a source system, choose *Subscribe* > *Unsubscribe me*.

Related Information

Monitor Provisioning Job Logs [page 2982]

1.12.11 View Troubleshooting Logs

Tenant administrator can filter and view troubleshooting logs directly in the administration console for SAP Cloud Identity Services to diagnose problems with authentication.

Context

> → Tip
>
> Use the audit logs option to track changes in the personal data.

The maximum number of log entries is 1000. The search criteria you can choose include:

Criteria	Notes
Time period	The time period is in Coordinated Universal Time (UTC). • You have one hour time period for searching logs. • The date is limited back to 14 days from the current date **Note** The default selection is the current date, with a start period - one hour back from your current time in UTC. **→ Remember** The start and end periods can't be later than the current UTC time. The end period can't also be earlier than the start period. The search period can't exceed 60 minutes.
Severity	• All - default • Error • Info • Warning

Once you have the results you can filter them by *Time*, *Severity*, *IP Address*, *CorrelationID*, or *Message.*

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose *Monitoring & Reporting* > *Troubleshooting Logs*.
3. Define the criteria and start your search.
4. **Optional:** Export the troubleshooting logs in an Excel file.

Next Steps

- View information about the JWT payload of the OpenID Connect tokens issued or received by Identity Authentication. For more information, see Logging OpenID Connect (OIDC) Tokens [page 2980]

Related Information

Audit Logs [page 2966]

1.12.12 Download Troubleshooting Logs

Tenant administrator can view Identity Authentication troubleshooting logs to monitor the system, and diagnose and solve problems.

Context

The maximum number of log entries is 10000. The criteria you can choose include:

Criteria	Notes
Time period	• 1 hour - default • 4 hours • 24 hours
Severity	• All • Error - default • Info • Warning

Criteria	Notes
File format	• JSON • XML • CSV - default

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Choose *Monitoring & Reporting* > *Troubleshooting Logs*.
3. Choose the *Download* button.
4. Define your criteria and download the log entries.

Next Steps

- Use the Support Log Assistant 2.0 tool to analyze your troubleshooting logs. For more information, see Analyze Troubleshooting Logs with Support Log Assistant 2.0 [page 3022].
- View information about the JWT payload of the OpenID Connect tokens issued or received by Identity Authentication. For more information, see Logging OpenID Connect (OIDC) Tokens [page 2980]

1.12.13 Analyze Troubleshooting Logs with Support Log Assistant 2.0

Context

The Support Log Assistant 2.0 is a tool that allows you to automatically scan and analyze text files such as logs, configuration files, or traces. The tool then suggests solutions to known issues found in the files and highlights important details that it finds. The tool is integrated into the incident logging procedure and is also available as a standalone tool.

Procedure

1. Access the tool via the following link: Support Log Assistant 2.0.
2. Choose *Add Files*, and select the files that you want to analyze.

No files are uploaded to SAP during this entire process. Your files are analyzed locally within your browser.

Once the log files have been selected, they are scanned to identify what type of log file they are. The files that are identified are listed on the top. Under them are the files that can't be identified and scanned.

3. Choose *Scan Files*, and select the *ALERTS*, *FILES*, or *ADDITIONAL DETAILS* tab.
4. Choose *Summary* to see a short summary of the issues found and their solutions.

Next Steps

If you haven't found a solution, you can choose the *Contact SAP Support* link to access other ways to receive support with your issue.

1.12.14 Troubleshooting Issues with Authentication and User and Group Management

While using the Identity Authentication service, you encounter issues or have questions regarding specific settings, limitations, or other aspects.

If you experience issues when using Identity Authentication service, check the following topics for help.

General: What information to include in a support ticket

You can report an incident on SAP Support Portal Home with the `BC-IAM-IDS` component for issues related to authentication and user and group management.

Please include the following information:

- - Your tenant ID from `https://tenant ID>.accounts.ondemand.com/admin` or `https://<tenant ID>.accounts.cloud.sap/admin`
  - Your email address
  - The error message you see
  - Attachments of screenshot that show the reproduced error, including the exact timestamp
  - If you solved your issue, a description of the steps you took
  - List of Knowledge Base Articles (KBAs) and SAP Notes, you have used to solve the issue
  - Attachments of SAML traces as per KBA: 2461862 - Collecting SAML traces with Chrome or Firefox

Accessing the Administration Console [page 3024]
Problems with the signing into the administration console for SAP Cloud Identity Services.

User Import [page 3031]
Problems with the user import in the administration console for SAP Cloud Identity Services.

Emails [page 3038]
Problems with emails sent for the different application processes.

Authentication [page 3042]
Problems with the authentication of the user and administrator.

Application Integration [page 3045]
Problems that different applications integrated with Cloud Identity Services may face.

Request, Create, and Delete Identity Authentication Tenant [page 3050]
Problems related to requesting, creating, or deleting a tenant.

End user screens [page 3051]
Problems that you may face when working with the end user screens.

REST APIs [page 3052]
Problems that you may face when using the REST APIs of Cloud Identity Services.

Corporate Identity Providers [page 3055]
Problems that you may face with corporate identity providers (IdPs) when using Cloud Identity services as a proxy.

Corporate User Store [page 3063]
Problems with corporate user store scenarios.

Kerberos Authentication [page 3066]
Problems with Kerberos authentication scenarios.

Risk-Based Authentication [page 3067]
Problems that you may face when configuring or using with risk-based authentication.

Custom Domains [page 3069]
Problems that you may face when using custom domains in Identity Authentication.

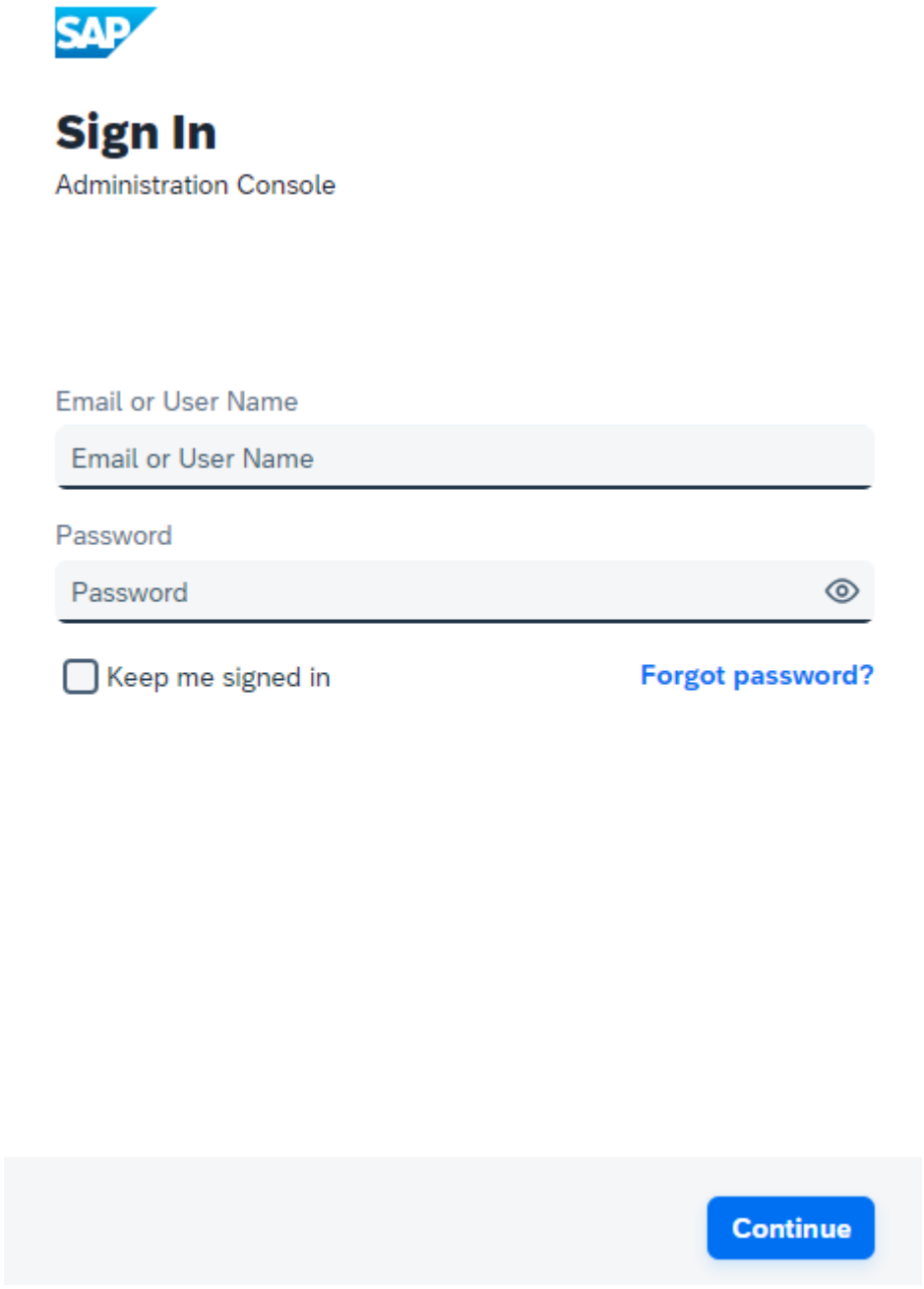
1.12.14.1 Accessing the Administration Console

Problems with the signing into the administration console for SAP Cloud Identity Services.

You can reach the tenant administration console via the URL `https://<tenant ID>.accounts.ondemand.com/admin` or `https://<tenant ID>.accounts.cloud.sap/admin` pattern. For more information, see Application Types [page 63].

You must be an administrator user.

The Sign In screen appears correctly



but you face one of the following problems:

I don't know who the tenant administrator is

Symptom:

You don't know who the tenant administrator is

Solution:

Contact an existing administrator for the tenant.

Cloud Identity Services doesn't use for authentication the users registered in the SAP Service Marketplace, but maintains an own user store for administrators and users.

Once you purchase a customer or partner account in SAP BTP, a user account for Cloud Identity Services is created for the contact person specified in the *Order Form*. The contact person is the first tenant administrator in the administration console for Cloud Identity Services. That person receives an activation email for the administration console account. The subject of the email is: **Activate Your Account for Identity Authentication Service**. The first administrator activates the account and continues to the administration console for Identity Authentication via the console's URL.

The first administrator can add new tenant administrators.

See also:

- KBA 3035908 - How to Find the Identity Authentication / Identity Provisioning Tenant Administrator Information or Tenant URL?
- Identity Authentication Service tenant specific request only possible for customer owning the tenant
- Access Admin Console [page 542]
- Reset Your Forgotten Password [page 2777]
- Activate Your Account [page 540]
- Access Admin Console [page 542]
- Manage Administrators [page 543]

I can't sign in to admin console of custom tenant with S-user

Symptom:

Logging into the administration console of a Cloud Identity services tenant with an S-user is unsuccessful.

Solution:

See KBA **2424064** - Cannot login to Administration Console of custom Identity Authentication tenant with S-user.

Administration Console isn't accessible due to network problem

Symptom:

The administration console can't be accessed, the browser returns `ERR_EMPTY_RESPONSE`, or similar error message `ERR_TIMED_OUT`, `took too long to respond`, and so on.

Timeout-related errors can also be seen on the browser.

Solution:

See KBA 2918278 - Identity Authentication tenant is not accessible due to network problem.

Email activation problem

As a tenant administrator you can reach the administration console via the URL https://<tenant ID>.accounts.ondemand.com/admin or https://<tenant ID>.accounts.cloud.sap/admin pattern.

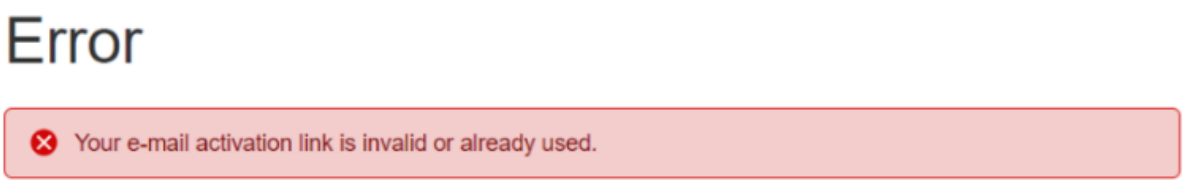
`Tenant ID` is an automatically generated ID by the system. The URL is in the activation e-mail received by you.

Check your inbox for an e-mail from **ias@notifications.sap.com** or **notification@sapnetworkmail.com**.

See also: Activate Your Account [page 540].

I have activated my account but I still face problems

You see the following error after you've chosen the activation email link:



This message means that the user or someone else clicked the activation link after it had already expired or the user account at the Identity Authentication tenant had already been activated.

Use the "Forgot Password" functionality to gain access to your tenant. For more information, see Reset Your Forgotten Password [page 2777].

I haven't activated my account or didn't receive any email

Open your inbox and search for an email from **ias@notifications.sap.com** or *notification@sapnetworkmail.com*. The email contains the URL to activate your account.

- **If the URL is expired**
  - The system sends you a new activation e-mail. Follow the procedure to activate your account.
- **If the link is invalid or already used, or you haven't received an email**
  - Check your spam folder
  - Contact an existing administrator for the tenant to ask the following
    - Whether an activation email was sent (or if only an initial password was set, since in this case no email is sent).
    - To resend the activation email (*Admin Console* > *User Management* > *select the user* > *Authentication* > *Password Details* > *Password Details* > *Send Email*. See also Send Reset Password Email [page 2381].

See also: **KBA 2517844** - How to get the activation e-mail of an Identity Authentication tenant.

Error "Sorry, we could not authenticate you. Try again."

This is a general error, and it can be a symptom of many different situations. You try to sign in...:

... after an LDAP user store is configured.

Symptom:

The user is locked when configuring the LDAP user store on SAP BTP.

Solution:

You have configured an LDAP scenario based on the Configure SAP BTP When Connecting to an LDAP User Store documentation. However, when you attempt to sign in, you see **"Sorry, we could not authenticate you. Try again."**

There is an issue in the connection between the SAP Cloud Connector and the corporate user store due to an issue with the system user which is used to access the corporate user store.

See KBA: 2680867 - Check if the system user is locked when configuring and LDAP user store on SAP BTP

... with S-user credentials

Symptom:

You're trying, without success, to sign in to the administration console as an S-user.

Solution:

There are no S-users used in custom tenants. S-users can only log on to `accounts.sap.com`, which belongs to SAP. In a custom tenant, you can sign in using your email address and password.

See KBA 2424064 - Cannot login to admin console of custom Identity Authentication tenant with S-user

... but sign-in fails after successful activation with error: Sorry, we could not authenticate you. Try again.

Symptom:

The user sign-in fails after the activation link is called and the password update was successful.

The user receives the following message:

Sorry, we could not authenticate you. Try again.

In the Troubleshooting log, the following error is displayed:

Identity Provider could not process the authentication request received due to error on its own side. The SP user [<user_uuid>] is with status inactive for Service Provider [<service_provider_url>] Caused by: javax.security.auth.login.AccountException: The SP user [<user_uuid>] is with status inactive for Service Provider [<service_provider_url>] Caused by: The SP user [<user_uuid>] is with status inactive for Service Provider [<service_provider_url>]

Solution:

See KBA 2770797 - The IAS tenant user login fails after successful activation: Sorry, we could not authenticate you. Try again.

but ... you forgot your password

Use the "Forgot Password" functionality to gain access to your tenant. For more information, see Reset Your Forgotten Password [page 2777].

If you have an account as administrator in Cloud Identity Services, an email with a link to a page where you can reset your password is sent. Note that the email might take a few minutes to reach your inbox. If you don't have an account as administrator, you won't receive an e-mail. In this case, contact an existing administrator for the tenant.

See also KBA 2517844 - How to get the activation e-mail of an Identity Authentication tenant.

... but you aren't an administrator

Cloud Identity Services does not use for authentication the users registered in the SAP Service Marketplace, but maintains an own user store for administrators and users. Only administrators can access the administration console.

Ask an existing administrator to add new tenant administrators. See Manage Administrators [page 543] and 2570572 - How to add Administrators to Identity Authentication tenants.

... but you receive error Used logon identifier is not allowed

Symptom

Sign in to the administration console for Cloud Identity services is failing with error: **Sorry, we could not authenticate you. Try again.**. Using the forgot password link allows you to reset the password and log in. However, signing out and trying to sign in again the same error is shown as previously. Only a password reset allows login again for just one user session.

The Troubleshooting log shows the following error:

User authentication failed. Reason: Used logon identifier isn't allowed; Identifier: [<logon_identifier>]

Solution:

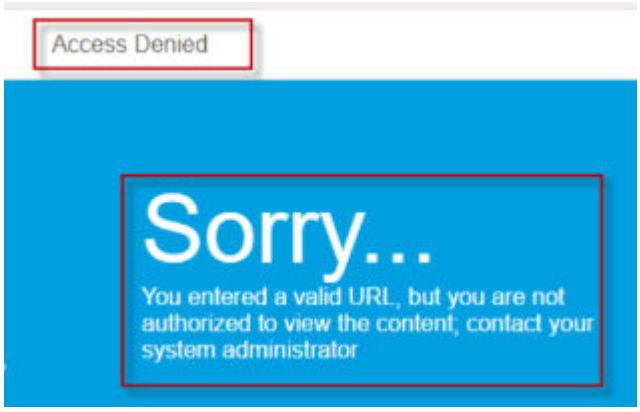
See KBA 2895349 - Cannot logon to Identity Authentication Administration Console with valid logon identifier and password.

Error "Access denied. Sorry... You entered a valid URL, but you are not authorized to view the content."

Symptom

Immediately after you've signed in to the administration console for Cloud Identity Services, you see the message:

Access Denied. Sorry... You entered a valid URL, but you are not authorized to view the content, contact your system administrator

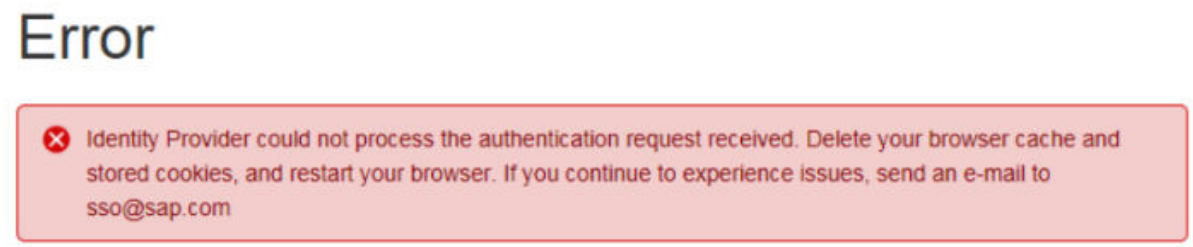


Ask an administrator to configure the required authorization, as explained in the following KBA:

2579343 - Accessing /admin of custom Identity Authentication tenant ends with "you are not authorized to view the content"

Error "The identity provider could not process the authentication request received"

Cloud Identity Services using the Security Assertion Markup Language (SAML) 2.0 as identity provider (IdP) fails to process the authentication request.



Make sure the service provider's name is the same as configured in Cloud Identity Services, acting as IdP:

2260000 - Identity provider could not process the authentication request received.

If this doesn't solve your problem please open a ticket on BC-IAM-IDS component, and provide the following information:

- Your tenant ID from `https://tenant ID>.accounts.ondemand.com/admin`
- Your email address
- The error message that you see
- Attachments of screenshot that show the reproduced error, including the exact timestamp
- If you solved your issue, a description of the steps you took
- List of KBAs and SAP Notes, you have used to solve the issue
- Attachments of SAML traces as per KBA: 2461862 - Collecting SAML traces with Chrome or Firefox

Error "Access to this Identity Authentication tenant is blocked, please contact your administrator"

Symptom:

Accessing the SAP Cloud Platform Identity Authentication Service (IAS) tenant fails with the following error:

"HTTP Status 403 – Access to this Identity Authentication tenant is blocked, please contact your administrator."

Solution:

See **KBA** - https://launchpad.support.sap.com/#/notes/2909142

Parent topic: Troubleshooting Issues with Authentication and User and Group Management [page 3023]

Related Information

User Import [page 3031]
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1.12.14.2 User Import

Problems with the user import in the administration console for SAP Cloud Identity Services.

As a tenant administrator, you can import new users or update existing ones for a specific application with a CSV file via the administration console. You can also send activation emails to the users that haven't received activation emails for that application so far.

The scenario, requirements, and configurations are described in Import or Update Users for a Specific Application.

> → Tip
>
> You as a tenant administrator can view Identity Authentication troubleshooting logs to monitor the system, and diagnose and solve problems. For more information, see View Troubleshooting Logs and Analyze

> Troubleshooting Logs with Support Log Assistant 2.0. See also **KBA 2942816** - How to export and self-analyze Troubleshooting logs from Identity Authentication Service.

You may face one of the following problems:

Users don't receive emails from `notification@sapnetworkmail.com` or `ias@notifications.sap.com` after a CSV user import

Symptom:

You import users by uploading a CSV file successfully, but users don't receive email for that.

Solution:

See KBA 2940801 - E-mail is not received after "Import Users"

Email sending is interrupted

Symptom:

There is a need to import users to an application and to send the initial emails, but the second step fails with the following message:

"First 0 mails are sent. Mailing has been interrupted because sending e-mail to user with e-mail address <email> is not possible. Please update the user e-mail and trigger e-mail sending again."

Solution:

See KBA 2833007 - First 0 mails are sent.

Users have to be redirected to a specific URL after successful registration

Symptom:

There is a need to redirect users to a specific URL after a successful registration.

Solution:

See **KBA 2942954** - Redirect users after successful registration in Identity Authentication

Import users to Identity Authentication with additional attributes

Symptom:

You have to import user data into Identity Authentication using the *Import Users* option of the administration console, but it only supports some certain attributes of a user, not all.

All supported attributes of a user for this function are mentioned in the SAP Help Document: Import or Update Users for a Specific Application.

Solution:

SAP provides SCIM REST API which allows to set all other user attributes in a programmatic way.

See SCIM REST API and **KBA 2533357** -Import users to Identity Authentication with additional attributes

Alternatively, SAP provides SCIM REST API which allows to set all other user attributes in a programmatic way.

You receive error: "Provided file has the wrong type"

Symptom:

When importing users to Identity Authentication, the import fails with the following error:

"Provided file has the wrong type. Please specify a different file and try again."

Solution:

See **KBA 2461975** - Import users fails with error: Provided file has the wrong type

You receive error: "User with email '' contains empty values - error during user import"

Symptom:

You see the following error at user import in the Identity Authentication service:

"User with email '' contains empty values."

Solution:

See **KBA 2859442** - User with email '' contains empty values - error during user import

You receive error: "User import has been interrupted for users with email address "<email_address>".

Symptom:

During user import (update), the following error is shown when the user already exists in the userbase:

User import has been interrupted for users with e-mail address "<email_address>".

The reason for this error is as follows: **"The import reported error".**

Solution:

See KBA 2958785 - Identity Authentication Import Users throws Attribute '<attributeName>' has wrong format error

You receive error: "Home URL or SAML 2.0 configuration of the application missing. You can only import users, but you cannot send activation emails."

Symptom:

You have imported users into the Identity Authentication tenant, but the activation e-mails to them can't be sent. The following message is shown:

"Home URL or SAML 2.0 configuration of the application missing. You can only import users, but you cannot send activation emails."

Solution:

See **KBA 2591826 -** Activation e-mails cannot be sent - Home URL or SAML 2.0 configuration of the application missing

You receive error: "The CSV file failed to upload..."

Symptom:

You, as an administrator, imported a CSV user file in Identity Authentication and it returns the following error:

"The CSV file failed to upload. Contact your system administrator for assistance."

Solution:

See **KBA 2975595** - Import users CSV file in IAS fails with error - The CSV file failed to upload.

You receive error: "0 emails sent when trying to send activation e-mails on Identity Authentication"

Symptom:

Users are imported to Identity Authentication, but when choosing "Send Emails", a message is received, stating that **"0 emails sent when trying to send activation e-mails on Identity Authentication".**

Solution:

See **KBA 2471540** - 0 emails sent when trying to send activation e-mails on Identity Authentication

You receive error: "First 0 users are imported. User import has been interrupted for users with email address . The reason for this error is as follows: "The import reported error".

Symptom

Importing users from a CSV file in Identity Authentication Service is failing with the following error message:

First 0 users are imported. User import has been interrupted for users with e-mail address <user e-mail>. The reason for this error is as follows: "The import reported error".

Solution:

See **KBA 2717186** - "User import has been interrupted" in Identity Authentication Service

You receive error: "First <number> users are imported. User import has been interrupted due to an invalid entity."

Symptom:

The first users in the CSV file are already imported for the tenant. The next user was not imported because the CSV file contains an incorrect data for that user.

Solution:

Delete the first <number> users from the CSV file, correct the invalid row, and re-import the updated file.

You receive error: "First <number> users are imported. User import has been interrupted for user with email address ."

Symptom:

The first <number> users in the CSV file are already imported for the tenant. The next user was not imported because its data in the CSV file conflicts with the data in the database.

Solution:

- Correct the email for the logon name under the mail column, delete the first <number> users from the CSV file, and re-import the updated file.

  > ⚠ Caution
  >
  > During CSV import, you can't change the email of an existing user.

- Change the user logon name because the same logon name is used for another user in the database, delete the first <number> users from the CSV file, and re-import the updated file.

  > ⓘ Note
  >
  > You can't have two users with the same logon name, but with different emails.

- Contact a tenant administrator of Identity Authentication.

You receive error: "First <number> users are imported. User import has been interrupted due to multiple users in the database with the same email address <email address>. The email address must be unique for each user."

Symptom:

The first <number> users in the CSV file are already imported for the tenant. The next user was not imported because its email matches with the email of other users.

Solution:

- Correct the user data, delete the first number> users from the CSV file, and re-import the updated file.
- Contact an operator of Identity Authentication to check if your emails exist in the database.

You receive error: "The imported CSV file contains entries with a duplicate email <email address>. The values under mail column must be unique."

Symptom:

You have provided a CSV file with a `mail` column that has the same email entries.

Solution:

Correct the file so that the mail column contains unique values.

You receive error: "The imported CSV file contains entries with a duplicate login name <login name>. The values under loginName column must be unique."

Reason:

You have provided a CSV file with a `loginName` column that has the same entries.

Solution:

Correct the file so that the `loginName` column contains unique values.

You receive error: "The imported CSV file does not contain one of the mandatory columns: status, mail and lastName. Make sure that the columns status, mail, lastName are in the CSV file."

Symptom

An administrator imports the CSV user file in Identity Authentication and it returns the following error:

The imported CSV file does not contain one of the mandatory columns: status, mail and lastName.

Make sure that the columns `status`, `mail`, `lastName` are in the CSV file.

Solution:

Follow the "Import or Update Users for a Specific Application" documentation document which explains the pre-requisites, possible columns with the attributes, mandatory columns, possible values for columns.

Follow the example in the documentation which shows you how your CSV file should look like.

See also KBA 2607696 - Import users CSV file in Cloud Identity Authentication Service and it fails with error

Parent topic: Troubleshooting Issues with Authentication and User and Group Management [page 3023]

Related Information

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1.12.14.3 Emails

Problems with emails sent for the different application processes.

Check the following entries for issues with emails sent for the different application processes:

Email Activation

You have received an email from **notification@sapnetworkmail.com** or:

- If you have already activated your account - You see the following error "E-Mail Already Activated" after you've clicked the activation email link." E-Mail Already Activated. This message means that the Cloud Identity Services has already been activated. Use the "Forgot Password" option to gain access to your tenant.
- If you haven't activated your account or didn't receive an email - Open your mail box and search for an e-mail from **ias@notifications.sap.com** or **notification@sapnetworkmail.com**. The email contains a link to activate your account. The following is possible:
  - The link has expired - The system sends you a new activation email. Follow the procedure to activate your account.
  - The link is invalid or already used, or you haven't received an email:
    - Check your spam folder
    - Contact an existing tenant administrator for the tenant to ask the following.
      - Whether an activation email was sent (or if only an initial password was set, since this case no email is sent).
      - To resend the activation email.

      > **Note**
      >
      > Who is my tenant administrator?

See also: **KBA 2517844** - How to get the activation e-mail of an Identity Authentication tenant

Check if emails are sent

Symptom:

End-users state that they don't receive emails from Identity Authentication for various application processes such as account activation, password reset, and so on.

Solution:

There is a need to troubleshoot if the e-mails are sent from Identity Authentication to determine if the issue is with Identity Authentication or the email server.

See **KBA 2978346** - How to check if e-mails are sent and the sender from Identity Authentication Service.

Error: "Internal error; contact system administrator" in Identity Authentication

Symptom:

An email template in the *Default* template of an application is uploaded. During the attempt to manually send it, the following pop-up warning is shown:

"Internal error; contact system administrator"

In the browser logs, the following error is visible:

"500 Internal Server Error"

Solution:

See **KBA 2573314** - E-mail template error: Internal error; contact system administrator in Identity Authentication (IAS)

Emails are in English language, which isn't the desired language

Symptom:

Activate Your Account or *Password Reset* email is in English, which isn't the desired language.

Solution:

See **KBA 3023020** - "Password Reset" e-mail is in English

User receives error: "Your e-mail activation link is invalid or already used"

Symptom:

Account activation of the Identity Authentication service isn't possible. The following error is displayed:

"Your e-mail activation link is invalid or already used."

Solution:

See **KBA 2637748** - Identity Authentication - Your email activation link is invalid or already used

Identity Authentication didn't send Forgot Password email

Symptom:

After requesting a new password with the *Forgot Password* option, the email isn't received.

The troubleshooting log (see **KBA 2942816** - How to export and self-analyze Troubleshooting logs from Identity Authentication Service) is showing the following error message:

"No Forgot password message sent to <e-mail address>. Limit exceeded!"

Solution:

Solution KBA 2981630 - No Forgot password message sent from Identity Authentication

Emails sent still contain the SAP logo

Symptom:

After you have configured the email templates in the administration console for Identity Authentication, the emails sent still contain the SAP logo.

Solution:

See **KBA 2854429** - How to Set Custom Company Logo for IAS Template E-mails

Email template set in Identity Authentication tenant isn't working

Symptom:

An email template set within the Identity Authentication tenant has been configured but the templates aren't applied when testing them.

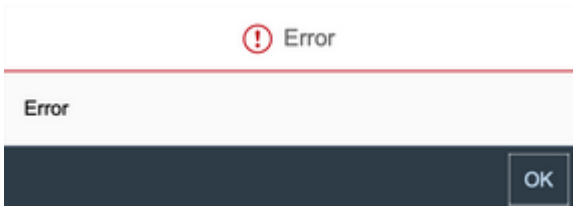
Solution:

See **KBA 2675079** - E-mail template set in Identity Authentication tenant not working

Error when uploading custom email template sets in Identity Authentication

Symptom:

The following error message is received when uploading custom email template sets in the administration console for Identity Authentication:



Solution:

See **KBA 2716042** - Error when uploading custom E-Mail Template Sets in Identity Authentication.

Changing the email ias@notifications.sap.com or notification@sapnetworkmail.com in Identity Authentication

Symptom:

When a user requests to reset their password in an Identity Authentication tenant, they receive an email from **notification@sapnetworkmail.com** (as of Sep 29, 2021 - **ias@notifications.sap.com**). There is a need to change this e-mail to another sender/server. Identity Authentication also provides the possibility to configure a custom mail server for the application processes e-mails.

Solution:

See **KBA** - Changing the e-mail ias@notifications.sap.com (notification@sapnetworkmail.com) in Identity Authentication.

Parent topic: Troubleshooting Issues with Authentication and User and Group Management [page 3023]

Related Information

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1.12.14.4 Authentication

Problems with the authentication of the user and administrator.

The following authentication issues may occur:

Error: "Sorry, we could not authenticate you. Try again"

Symptom

When you try to sign in to an SAP Cloud Identity Services tenant, the following error message can be seen: **"Sorry, we could not authenticate you. Try again"**

Solution

See KBA **3291844** - Sorry, we could not authenticate you. Try again" error message seen in Identity Authentication login

Error: "Your domain is not trusted. Please contact your system administrator"

Symptom::

When using embedded frames or overlays in an application that authenticates against the Identity Authentication, the following error is displayed when trying to log in via the domain of the overlay:

"Your domain is not trusted. Please contact your system administrator."

The other error can be related to trusted domains:

"Your request failed because the page you have to be redirected to is not trusted. Please contact your system administrator."

Solution:

See KBA **2739473** - "Your domain is not trusted. Please contact your system administrator" in Identity Authentication Service

Error: "The digital signature of the received SAML2 message is invalid"

Symptom:

Identity Authentication tenant returns the message in subject when an authentication request is done from a specific service provider.

This can be checked via a SAML tracer. For more information, see **KBA** 2461862 - Collecting SAML traces with Chrome or Firefox.

If the service provider is an SAP BTP subaccount, an Error 500 can be seen on the screen.

In the Troubleshooting log, the following error is displayed:

"message=Identity Provider could not process the authentication request received due to error on its own side. An unexpected exception occurred. See call stack for details. Caused by: Signature of the SAML2 protocol token cannot be validated because neither primary nor secondary certificates are available in the configuration"

Solution:

See KBA **2645425** - The digital signature of the received SAML2 message is invalid

Reset Password isn't working

Symptom:

The Reset Password option isn't working for Identity Authentication and in the Troubleshooting Logs you find the following error:

"Identifier xxx@xx.xx not unique"

Solution:

See KBA 3030889 - Reset password for IAS fails: "Identifier xxx@xx.xx not unique"

Attribute with name loginName isn't supported

Symptom:

Authentication through Identity Authentication is failing with the following error message:

"Identity provider cannot process the request because the configuration is incorrect. Please contact your system administrator."

The Troubleshooting Log shows the following error:

"Attribute with name loginName is not supported. Reason: loginName"

The error message can show other attributes as well, depending on the scenario.

Solution::

See KBA 2954081 - Attribute with name loginName is not supported.

Error: "Not supported attribute"

Symptom::

User is trying to sign in to an application through Identity Authentication however the authentication is failing. The application sign-in page is coming up after authentication in Identity Authentication.

In the Troubleshooting log, the following error is displayed:

"message=Identity Provider could not process the authentication request received due to error on its own side.com.sap.security.saml2.lib.common.exceptions.SAML2ErrorResponseException: Not supported attribute.loginName"

Meanwhile SAML trace has the following `nameid` format:

```
<NameID Format="urn:oasis:names:tc:SAML:1.1:nameid-
format:emailAddress"><email_address></NameID>
```

Solution:

See KBA 2954035 - IAS error: Not supported attribute.

Error: "Cookie name is not valid"

Symptom:

Authentication through Identify Authentication is failing. The troubleshooting logs of Identity Authentication, shows the following error:

"... message=""User authentication failed.Cookie name is not valid: ..."

Solution:

See KBA 2698571 - User authentication failed in IAS due to error "Cookie name is not valid".

Error: "Your user was valid until. For assistance, contact your administrator."

Symptom:

When trying to sign in an application protected by an Identity Authentication tenant, the following error message is displayed:

"Your user was valid until <date>. For assistance, contact your administrator."

Solution:

See KBA **2950819 -** "Your user was valid until <date>. For assistance, contact your administrator." error when logging on via Identity Authentication Service.

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1.12.14.5 Application Integration

Problems that different applications integrated with Cloud Identity Services may face.

You receive error: "Your registration failed because the URL of the redirect page is invalid. Please contact your system administrator"

Symptom:

User registration on the SAP Cloud Platform Identity Authentication Service tenant fails and the following error is shown:

"Your registration failed because the URL of the redirect page is invalid. Please contact your system administrator"

Solution:

See KBA **2743394** - Your registration failed because the URL of the redirect page is invalid.

You receive error: Error: "Your domain is not trusted. Please contact your system administrator"

Symptom::

When using embedded frames or overlays in an application that authenticates against the Identity Authentication, the following error is displayed when trying to log in via the domain of the overlay:

"Your domain is not trusted. Please contact your system administrator."

The other error can be related to trusted domains:

"Your request failed because the page you have to be redirected to is not trusted. Please contact your system administrator."

Solution:

See KBA **2739473** - "Your domain is not trusted. Please contact your system administrator" in Identity Authentication Service

You receive error: "The digital signature of the received SAML2 message is invalid"

Symptom:

An Identity Authentication tenant returns the message: "The digital signature of the received SAML2 message is invalid" in subject when an authentication request is done from a specific service provider. This can be checked via a SAML tracer (see SAP KBA 2461862 Collecting SAML traces with Chrome or Firefox).

In case the service provider is an SAP BTP subaccount, the **Error 500** can be seen on the screen.

In the Troubleshooting log, the following error is displayed:

"message=Identity Provider could not process the authentication request received due to error on its own side. An unexpected exception occurred. See call stack for details. Caused by: Signature of the SAML2 protocol token cannot be validated because neither primary nor secondary certificates are available in the configuration"

Solution:

See KBA 2645425 - The digital signature of the received SAML2 message is invalid

Integration with SAP IBP. IBP logon screen is received after authentication (SSO doesn't work)

Symptom:

- You need to Integrate SAP Integrated Business Planning (SAP IBP) with either Identity Authentication or a corporate identity provider (IdP).
- The authentication happens to IBP via **Identity Authentication** or corporate IdP, but the IBP sign-in screen is received after authentication (SSO doesn't work).

Solution:

See KBA 2659221 - User authentication prerequisites for IBP with Identity Authentication Service or Corporate IdP.

Proxy scenario with S4/HANA

Symptom:

There is a need to access the S/4HANA application with Identity Authentication that is used as a proxy.

At sign in attempt, an error message is shown depending on the application is used to log in.

Solution:

See KBA 2762530 - https://launchpad.support.sap.com/#/notes/2762530.

SFSF password migration isn't taking effect

Symptom:

After provisioning users to Identity Authentication from SuccessFactors (through Identity Provisioning), users can't sign in to SuccessFactors with a `username` and `password`. They use the forgot password link and get an error message on the Identity Authentication sign-in page:

Passwords can only be changed once every %{param0} hours.

or

Passwords can only be changed once every 24 hours.

In the troubleshooting logs of Identity Authentication with "WARN" Severity, you see the error:

"Cannot send Forgot Password Mail to user with uid <P_UserID>. Reason: PASSWORD_DISABLED. "

other factors are:

- SuccessFactors password migration to Identity Authentication has been configured Migrating Passwords from SAP SuccessFactors to the Identity Authentication Service .
- Users with the issue are existing already in Identity Authentication prior to running the Identity Provisioning sync job.
- Users with the issue initially try to sign in but their password doesn't work. They reset the password and try to sign in again. This is when the error is received.

Solution:

See KBA 2957801 - SFSF password migration is not taking effect - "Passwords can only be changed once every %{param0} hours" upon forgot password.

Authentication is returning error in incognito mode, through Identity Authentication overlay sign-in page

Symptom:

The authentication through Identity Authentication service is working fine in Google Chrome (or any other browser) through embedded frame, also called overlay, for the sign-in page of their application.

However, the incognito mode shows "HTTP Status 400 – Bad Request error." The error may occur only at the second login attempt.

Solution:

See KBA

Partial SSO after Identity Authentication implementation on SuccessFactors

Symptom:

- On SuccessFactors instances integrated with Identity Authentication, how no-SSO users can access the instance?
- How to implement a similar feature to the Partial SSO from SuccessFactors on Identity Authentication.

Solution:

See KBA **2954556** - How to implement Partial SSO after Identity Authentication implementation on SuccessFactors

Browser Back button isn't supported when using SAML

Symptom:

Using the browser's *Back* function while applying the Security Assertion Markup Language (SAML) authentication leads to inconsistent results.

Solution:

See KBA 2677653 - Browser Back button is not supported when using SAML

You receive error: "Authentication Token's `targetUrl` is not part of the trusted domains"

Symptom:

There is an application using overlay in Identity Authentication for authentication, and the authentication fails with the error: **Your domain is not trusted, please contact your system administrator**.

Meanwhile, in the Troubleshooting log, the following error can be seen: **Authentication Token's targetUrl is not part of the trusted domains. Redirecting to error page.**

Solution:

See KBA **2944969** - Authentication Token's targetUrl is not part of the trusted domains

S-user provided in the Upgrade Center isn't assigned Identity Authentication admin role

Symptom:

The integration between SuccessFactors (SFSF) and Identity Authentication has been initiated.

After the integration, the administration console for Identity Authentication (`https://<tenantID>.accounts.ondemand.com/admin/`) with the user provided in the `Upgrade Center` can't be accessed.

Error: **"Sorry, we could not authenticate you. Try again."** is shown in the popup.

Solution:

See KBA **2983379** - S-user provided in the Upgrade Center is not assigned IAS admin role

User presence prerequisites for SAP Hybris Cloud with Identity Authentication Service

Symptom:

There is a need to Integrate SAP Hybris Cloud with Identity Authentication and would like to know where users will be maintained.

Solution:

See KBA **2665797** - User presence prerequisites for Hybris Cloud with Cloud Identity Authentication Service

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1.12.14.6 Request, Create, and Delete Identity Authentication Tenant

Problems related to requesting, creating, or deleting a tenant.

Identity Authentication provides one productive tenant per customer, regardless of the number of contracts signed in which Identity Authentication is included or bundled.

For more information about the tenant model and tenant licensing of Identity Authentication, see Tenant Model and Licensing. For the obtaining a tenant, see Get Your Tenant [page 528].

If you don't find your solution in the documentation, see the following suggestions:

I purchased a tenant, but it hasn't been provisioned yet

Symptom:

An Identity Authentication tenant has been purchased but hasn't been provisioned yet.

Solution:

See KBA 2751968 - Details required when requesting an Identity Authentication tenant

How to request additional Identity Authentication Service tenants

Symptom:

There is a need to request additional Identity Authentication tenants besides already existing ones.

Solution:

See KBA - How to request additional Identity Authentication Service tenants.

I can't find Identity Authentication tile in SAP BTP cockpit

Symptom:

You can't find the Identity Authentication tile in *SAP BTP cockpit* > *Services*.

Solution:

See **KBA 2711456** - Identity Authentication tile in SAP BTP Cockpit

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1.12.14.7 End user screens

Problems that you may face when working with the end user screens.

You need to remove the "Forgot password" link from the sign in screen

Symptom:

You need to remove the *Forgot Password* link from the Identity Authentication sign-in screen. This can be a goal when the Corporate User Store is used, and the password is managed on the Corporate User Store itself, so the forgot password functionality would not be needed.

Solution:

See **KBA 2607588** - How to remove Forgot Password link from Identity Authentication logon page.

Missing X-Frame-Options from accounts.ondemand.com HTTP header

Symptom:

X-Frame-Options HTTP header is missing from the IAS tenant sign in page, `https://<tenantID>.accounts.ondemand.com`

Solution:

See KBA **2989320** - Missing X-Frame-Options from accounts.ondemand.com HTTP header.

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1.12.14.8 REST APIs

Problems that you may face when using the REST APIs of Cloud Identity Services.

SCIM REST API request fails with Status code 415 Unsupported Media Type

Symptom:

Your SCIM requests are failing with **Status code 415 Unsupported Media Type** error when you use one of the Manage Users SCIM REST API.

Solution:

See **KBA 2942542** - SCIM REST API request fails with Status code 415 Unsupported Media Type

You receive error: "500 Internal Server Error"

Symptom:

While creating a user with Cloud Identity Services SCIM REST API, the REST client returns the following: **"500 Internal Server Error"**

Solution:

500 Internal Server Error usually can be caused by an incorrectly created script. See: KBA **2745866** - SAP Cloud Platform Identity Authentication Service - "500 Internal Server Error" while creating user with SCIM REST API

Change Tenant Texts REST API fails with "403 Forbidden"

Symptom:

Logon labels can't be customized since the POST API isn't reachable due to **"403 Forbidden"** error:

This is a legal request, but the server is refusing to respond to it. Unlike the **401 Unauthorized** response, where authenticating will make no difference.

Solution:

See KBA **2650614** - Identity Authentication Service - Cannot change logon screen labels - 403 Forbidden

User update fails with "400 Bad Request"

Symptom:

User update via the Identity Authentication API or the administration console fails with 400 Bad Request or Bad Request. The user can't be updated.

Solution:

See KBA 2895736 - 400 Bad Request when updating user in Identity Authentication Service

403 Forbidden error

Symptom:

Calling the SCIM REST API of Identity Authentication via a REST client returns a 403 Forbidden error.https://launchpad.support.sap.com/Create%20User%20Resource

Solution:

See KBA 403 Forbidden when requesting URI: https://<tenant ID>.accounts.ondemand.com/service/scim/Users

"Value not supported for attribute department" when updating a user via the SCIM API

Symptom:

When updating a user in Identity Authentication using the SCIM API, the following error is displayed:

"StatusCode: BadRequest Message: Processing of the HTTP request resulted in an exception. Please see the HTTP response returned by the 'Response' property of this exception for details. Web Response: Value not supported for attribute department. For more information about supported values, see Identity Authentication Documentation. This operation was retried 0 times. It will be retried again after this date: 2020-05-26T18:22:53.1296153Z UTC"

Solution:

See KBA 2931057 - "Value not supported for attribute department" when updating an user via the SCIM API.

A new Identity Authentication tenant user activation link redirects to a password change screen

Symptom

A new user activation link redirects to a password change screen, however, the activation link is correct.

Solution:

See KBA **2770762** - The Identity Authentication activation link redirects to password change screen.

How to retrieve users who enabled TFA (Two-Factor Authentication) in Identity Authentication using SCIM REST API

Symptom:

I don't know how to retrieve/search users who enabled TFA (Two-Factor Authentication) in Identity Authentication by using SCIM REST API.

Solution:

See KBA **2942076** - How to retrieve users who enabled TFA (Two-Factor Authentication) in IAS using SCIM REST API.

How to mass update user password in Identity Authentication

Symptom:

A large number of passwords need to be updated. One by one isn't a solution.

Solution:

See KBA 3001615 - How to mass update user password in Identity Authentication.

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1.12.14.9 Corporate Identity Providers

Problems that you may face with corporate identity providers (IdPs) when using Cloud Identity services as a proxy.

SSO with Corporate Identity Provider fails in some browsers/devices

Symptom:

One or both of the following issues are occurring:

- Single Sign-On (SSO) with corporate IdPs fails in the Cloud Identity services with the following error **"Identity provider cannot process the response due to wrong configuration. Please contact your system administrator".**
- SAML2.0 Authentication on SAP Business Technology Platform (SAP BTP) fails with the following error **"HTTP 400 Bad Request."**

In addition, one or both of the following statements are true:

- The problem occurs in some browsers/devices, but not in others
- The problem occurs only in a normal browser window, but not in private/incognito, or the other way around.

Solution:

See KBA **2946518** - SSO with Corporate Identity Provider fails in some browsers/devices.

Given URL does not contain SAML2 authentication request for validation

Symptom:

SAML 2.0 authentication when using Cloud Identity services fails and errors similar to the following ones are recorded in the Troubleshooting log:

"ERROR#com.sap.security.saml2.idp.endpoints.sso.SSOEndpoint##"

"#Identity Provider could not process SAML2 authentication request sent over Redirect due to SAML2 core error that prevents sending error response to the caller."

"Reason: [Given url: <URL DETAILS> does not contain SAML2 authentication request for validation.]"

Solution:

See KBA **2698094 -** Given url does not contain SAML2 authentication request for validation.

Cloud Identity services as proxy to Google fails with 403 app_not_configured_for_user

Symptom:

Cloud Identity services as proxy to Google Workspace fails with the following errors:

"Error: app_not_configured_for_user"

"Service is not configured for this user."

Solution:

See KBA **3030557** - Identity Authentication as proxy to Google fails with 403 app_not_configured_for_user.

Signature validation of SAML2Assertion failed

Symptom:

Cloud Identity services is acting as a proxy, the authentication happens through a corporate identity provider.

When logging in, users are redirected back to the login page and get in a loop.

In the Troubleshooting log, the following error is displayed:

"Signature validation of SAML2Assertion failed. Signature validation of SAML2Assertion failed. Caused by: Signature not valid!"

Solution:

See **KBA** - 3026204 Signature validation of SAML2Assertion failed

Name id attribute must match at least one valid and supported user attribute value

Symptom:

Cloud Identity services is acting as a proxy, the authentication happens through a corporate identity provider.

Subject Name Identifier is set to `${corporateIdP.<corporateIdP attribute>}`. Identity Federation is enabled - user does exist in the Identity Directory user store.

Authentication is failing with such user, and the Troubleshooting Log is showing the following error message:

SAML configuration error. Name id attribute `${corporateIdP.<corporateIdP attribute>}` must match at least one valid and supported user attribute value

Solution:

See KBA **2976923** - **Name id attribute ${corporateIdP.<corporateIDP attribute>} must match at least one valid and supported user attribute value**.

HTTP Status 400 - Identity Provider could not process the logout message received - Identity Authentication

Symptom:

After logging out of an application via SAML configured for the Cloud Identity services, the logout fails with the following error message:

"HTTP Status 400 - Identity Provider could not process the logout message received"

Solution:

See **KBA** - 2682874 HTTP Status 400 - Identity Provider could not process the logout message received - Identity Authentication.

SAML2Assertion doesn't specify `Subject NameID`

Symptom:

SAML 2.0 authentication when using Cloud Identity services or SAP NetWeaver Application Server for Java (SAP AS Java) as the service provider (SP) fails and errors similar to those below are recorded in the Troubleshooting log:

Authentication error.SAML2Assertion does not specify Subject NameID.

authenticationMethod=saml2Assertion

Solution:

See KBA **2908718** - SAML2Assertion does not specify Subject NameID.

Service Provider does not match specified audience in the SAML2Assertion

Symptom:

SAML 2.0 authentication when using Cloud Identity services fails and errors similar to below are recorded in the Troubleshooting log:

ERROR#com.sap.security.saml2.idp.endpoints.sso.ACSEndpoint#

Authentication error. Reason: [Service Provider does not match specified audience in the SAML2Assertion.]

Solution:

See KBA **2693814** - Service Provider does not match specified audience in the SAML2Assertion.

Logout Redirect URL is not working when Cloud Identity services is acting as a proxy

Symptom:

Logout Redirect URL is not working when Cloud Identity services is acting as a proxy. Logout Redirect URL is configured, but the logout is ending at the Single Logout Endpoint HTTP-Redirect URL.

Solution:

See KBA **2976906** - Logout Redirect URL is not working when IAS is acting as a proxy.

IdP not returning attribute `InResponseTo` in the response with the ID

Symptom:

A corporate identity provider is configured and Cloud Identity services is used as a proxy. Cloud Identity services sends an authentication request to the corporate IdP with an ID, but the IdP is not returning the attribute `InResponseTo` in the response with the ID.

Meanwhile, Cloud Identity services is showing the following error:

"Identity provider cannot process the response due to wrong configuration. Please contact your system administrator."

Solution:

See KBA **2926867** - InResponseTo attribute - What does IAS expect when it is used as a proxy - How To.

User attribute configured for `name-id format unspecified` is not supported

Symptom:

Authentication to an application fails when using Cloud Identity services. In the application logs and, or in the SAML trace, the error **"User attribute configured for name-id format unspecified is not supported"** can be seen.

Solution:

See KBA **2594230** - User attribute configured for name-id format unspecified is not supported.

Error "Identity Provider could not process SAML2 logout message. RedirectPayload is not signed."

Symptom:

SAML 2.0 authentication when using Cloud Identity services as the IdP fails and errors similar to those below are recorded in the Troubleshooting log:

ERROR#com.sap.security.saml2.idp.endpoints.slo.SLOEndpoint##

Identity Provider could not process SAML2 logout message. RedirectPayload is not signed.

Solution:

See KBA **2811941** - **Identity Provider could not process SAML2 logout message.RedirectPayload is not signed.**

Cloud Identity services as proxy to ADFS or Azure fails because of the Scoping tag

Symptom:

Microsoft ADFS or Azure is set as a Corporate Identity Provider to delegate authentication from a Cloud Identity services tenant.

Authentication is not working on the ADFS/Azure side and either of the following messages is seen:

The SAML authentication request element 'Scoping' is not supported.

The SAML authentication request property 'Scoping/ProxyCount' is not supported.

Solution:

See KBA **2615705** - Identity Authentication Service as proxy to ADFS or Azure fails because of the Scoping tag.

How to connect Ping Identity to Cloud Identity services

Symptom:

There is a need to configure Ping Identity as a corporate identity provider for Cloud Identity services.

Solution:

See KBA **2968405 -** How to connect Ping Identity to Identity Authentication Service.

How to connect Azure Active Directory to Cloud Identity services

Symptom:

There is a need to configure Azure as a corporate IdP to SAP Cloud Identity Services.

Solution:

See KBA **2945035** - How to connect Azure Active Directory to Identity Authentication Service.

How to configure Okta as corporate identity provider with Cloud Identity services

Symptom:

There is a need to configure Okta as corporate IdP for an application with SAP Cloud Identity Services as proxy.

Solution:

See KBA **2943651** - How to configure Okta as corporate identity provider with Identity Authentication.

Incorrect destination when using Okta corporate IdP with Cloud Identity services

Symptom:

When using theSAP Cloud Identity Services tenant as a proxy in a corporate IdP-initiated login scenario with the corporate IdP Okta, the following error appears:

Error:**Identity Provider could not process the authentication request received. Delete your browser cache and stored cookies, and restart your browser. If you still experience issues after doing this, please contact your administrator.**

Solution:

See KBA **2753454** - Incorrect destination when using Okta corporate IDP with Identity Authentication Service.

Error with ADFS: "SAML2Assertion does not specify Subject NameID"

Symptom:

SAP Cloud Identity Services is acting as a proxy with Active Directory Federation Services (AD FS). After authentication to an application, it fails with the error HTTP Status 500.

Meanwhile, in the Troubleshooting log, the following errors can be seen:

- **Authentication error. The authentication process did not set an authenticated principal in the current thread.**
- **state=failed, action=login, objectType=user, cause=authenticationStepFailure, category=audit.authentication, credentialType="{TRUSTED_IDP_SAML_ASSERTION=rejected}**
- **SAML2Assertion does not specify Subject NameID.com.sap.security.saml2.sp.sso.exception.BadCredentialsException: SAML2Assertion does not specify Subject NameID.**

Solution:

See KBA **2945414** - 'SAML2Assertion does not specify Subject NameID' error with AD FS.

Note: This topic fails to consulting category. Microsoft is responsible to do this configuration. However, this KBA provides some hints to troubleshoot and solve this issue.

Error: "Identity provider cannot process the response due to wrong configuration. Please contact your system administrator."

Symptom:

A corporate identity provider is configured and IAS is used as a proxy. The authentication is failing and IAS is showing the following error:

"Identity provider cannot process the response due to wrong configuration. Please contact your system administrator."

Solution:

See KBA **2926891 -** RelayState attribute - What does IAS expect when it is used as a proxy - How To.

Error: "Could not find user xxx in mongo DB."

Symptom:

- Corporate Identity Provider is configured for authentication in Identity Authentication, but authentication is failing.
- The Troubleshooting log is showing the following error:
  "Authentication error.User not found: xxx Caused by: Could not find user <test@mail.com> in mongo DB." ensure that the Identity Authentication Troubleshooting log to contains "Could not find user <test@mail.com> in mongo DB" error, before proceeding with the resolution of this KBA.

- You may see the response from the service provider :
  "StatusCode in ResponseMessage != OK; please refer to the database trace for more information"

Solution:

See KBA **2908064** - Identity Authentication: Could not find user xxx in mongo DB.

HTTP 500 error from corporate identity provider - Certificate used to validate the signature cannot be null

Symptom

Sign-in to Corporate Identity Provider (IdP) does not work with the Identity Authentication Service (IAS) functioning as a proxy. Corporate IdP login screen shows an "HTTP 500" error.

In Troubleshooting Logs, the following entries can be seen:

"POST /saml2/idp/acs/<TenantID>.accounts.ondemand.com HTTP/1.1" 200

severity=INFO, location=umtrace, crtAccount=<TenantID>, authenticatedSubject="anonymous", state=failed, action=authenticate, objectType=user, authenticationMethod=saml2Assertion, category=audit.configuration, correlationId

#ERROR#com.sap.security.saml2.idp.endpoints.sso.ACSEndpoint##<TenantID>#anonymous#http-bio-127.0.0.1-8080-exec-5#na#N/A#N/A#N/A#Authentication error.SAML2Response signature verification failed. Caused by: The certificate used to validate the signature cannot be null

However, SAML response is successful:

<...>

<samlp:Response xmlns:samlp="urn:oasis:names:tc:SAML:2.0:protocol" xmlns:saml="urn:oasis:names:tc:SAML:2.0:assertion"

<...>

*<dsig:X509Certificate><...></dsig:X509Certificate><...><samlp:Status><samlp:StatusCode Value="urn:oasis:names:tc:SAML:2.0:***status:Success***"*

<...>

Solution:

See KBA **2758293** - IAS proxy scenario: HTTP 500 error from corporate identity provider - Certificate used to validate the signature cannot be null.

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1.12.14.10 Corporate User Store

Problems with corporate user store scenarios.

Corporate user store in Cloud Foundry environment

The corporate user store feature is supported in the Cloud Foundry subaccount.

For more information about the configuration steps, see Corporate User Store (Cloud Foundry Environment).

See also KBA **3058176** - Corporate User Store Feature in Cloud Foundry environment.

Corporate user store in Neo subaccount

The corporate user store feature is supported in the Neo subaccount.

For more information about the configuration steps, see Corporate User Store (Neo Environment).

See also KBA **2768285** - OAuth Client subscription type "sci/proxy" not visible.

How to connect Identity Authentication to on-premise user store

Symptom:

There is a need to integrate Identity Authentication to an on-premise user store so that users can use their corporate credentials to access cloud applications.

Solution:

See KBA **2461375** - How to connect Identity Authentication to on-premise user store.

You need to remove the "Forgot password" link from the sign in screen

Symptom:

You need to remove the *Forgot Password* link from the Identity Authentication sign-in screen. This can be a goal when the Corporate User Store is used, and the password is managed on the Corporate User Store itself, so the forgot password functionality would not be needed.

Solution:

See **KBA 2607588** - How to remove Forgot Password link from Identity Authentication logon page.

How to reset password for users from Corporate User Store

Symptom:

There is a need to reset the Identity Authentication service password for a user coming from the Corporate User Store. When the e-mail received to reset the password is followed, the sign-in is successful. However, at the next sign-in attempt, the newly set password does not work anymore.

Solution:

See KBA **2757021** - Identity Authentication - Password reset for users from Corporate User Store.

ABAP as Corporate User Store for Identity Authentication Service

Symptom:

There is a need to use SAP NetWeaver AS ABAP on-premise user store in addition to Identity Authentication own cloud user store.

Solution:

See KBA **2942094** - SAP AS ABAP as Corporate User Store for SAP Identity Authentication Service (IAS).

SAML 2.0 Response issue instant is not valid

Symptom:

A corporate identity provider is configured for Identity Authentication. The authentication fails with an error, meanwhile, in the Troubleshooting log, the following error is shown:

SAML2Response issue instant is not valid.com.sap.security.saml2.sp.sso.exception.BadCredentialsException: SAML2Response issue instant is not valid.Issue Instant is not valid anymore. IssueInstant: <Date>Curent time: <Date>

The caused by can be the following as well, depending on the time setting on the AD FS: **Caused by: Issue Instant is not valid yet.**

Solution:

See KBA **2945159** - SAML2Response issue instant is not valid.

OAuth Client subscription type "sci/proxy" not visible

Symptom:

There is a need to set a subscription type to "sci/proxy" during the OAuth 2.0 configuration.

The "sci/proxy" option in the drop-down list is not present.

Solution:

See KBA **2768285** - OAuth Client subscription type "sci/proxy" not visible.

Error: "End-user gets a "Sorry, we could not authenticate you. Try again" error when attempting to log in with LDAP user store"

Symptom:

An LDAP user store on Identity Authentication is configured, but the end-user is getting a "**Sorry, we could not authenticate you. Try again**" error when attempting to log in.

Solution:

See KBA **2680867** - Check if the system user is locked when configuring and LDAP user store.

Parent topic: Troubleshooting Issues with Authentication and User and Group Management [page 3023]

Related Information

Accessing the Administration Console [page 3024]
User Import [page 3031]
Emails [page 3038]
Authentication [page 3042]
Application Integration [page 3045]
Request, Create, and Delete Identity Authentication Tenant [page 3050]
End user screens [page 3051]
REST APIs [page 3052]
Corporate Identity Providers [page 3055]
Kerberos Authentication [page 3066]
Risk-Based Authentication [page 3067]
Custom Domains [page 3069]

1.12.14.11 Kerberos Authentication

Problems with Kerberos authentication scenarios.

Error "Kerberos Authentication fails with Could not decrypt the encrypted data error"

Symptom:

Kerberos authentication is configured for Identity Authentication to allow users to log on without a username and password when they are in the corporate network based on Configure Kerberos Authentication' SAP Help documentation.

However, users are still prompted to authenticate with a username and password.

The Troubleshooting Log shows the following error:

Could not validate SPNEGO token: Could not decrypt the encrypted data

Solution:

See KBA **2958786** - Kerberos Authentication fails with Could not decrypt the encrypted data error.

Kerberos Authentication fails with Unsupported encryption SPNEGO token type error

Symptom:

Kerberos authentication is configured for Identity Authentication to allow users to log on without a username and password when they are in the corporate network based on Configure Kerberos Authentication SAP Help documentation.

However, users are still prompted to authenticate with a username and password.

The Troubleshooting Log shows the following error:

Unsupported encryption SPNEGO token type

Solution:

See KBA **2958806** - Kerberos Authentication fails with Unsupported encryption SPNEGO token type error.

Supported number of realms or domains for Kerberos/SPNEGO in Identity Authentication

Symptom:

There is a need to know how many realms or domains are supported for Kerberos/SPNEGO Authentication in Identity Authentication.

Solution:

See KBA **2726591** - Supported number of realm or domain for Kerberos/SPNEGO in IAS.

Parent topic: Troubleshooting Issues with Authentication and User and Group Management [page 3023]

Related Information

Accessing the Administration Console [page 3024]
User Import [page 3031]
Emails [page 3038]
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1.12.14.12 Risk-Based Authentication

Problems that you may face when configuring or using with risk-based authentication.

Error: "Sorry, but you are currently not authorized for access"

Symptom:

SAML 2.0 authentication when using Identity Authentication fails and a similar error to below is shown:

Sorry, but you are currently not authorized for access

The Troubleshooting log is showing one the following errors:

cause=rbaRulesCheckFailure, message="Denied by RBA rules" or **"Authentication error. User not found: xxx Caused by: Could not find user <e-mail> in mongo DB."**

Solution:

See KBA **2891275** - Sorry, but you are currently not authorized for access: IAS issue.

Two-Factor Authentication is not taking effect in Identity Authentication

Symptom:

Enable two-factor authentication for an application via risk-based authentication rules.

There is no change in the behavior during sign-in.

Solution:

See KBA **2941888** - Two-Factor Authentication is not taking effect in IAS.

Passcode not valid upon Two-Factor Authentication

Symptom:

The passcode generated for two-factor authentication is invalid for a mobile device.

Solution:

See KBA **2968616** - Passcode not valid upon Two-Factor Authentication.

Parent topic: Troubleshooting Issues with Authentication and User and Group Management [page 3023]

Related Information

Accessing the Administration Console [page 3024]
User Import [page 3031]
Emails [page 3038]
Authentication [page 3042]
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Custom Domains [page 3069]

1.12.14.13 Custom Domains

Problems that you may face when using custom domains in Identity Authentication.

Error: "Invalid certificate Subject DN. Please make sure that the Subject DN of the uploaded certificate is the same as the configured one."

Symptom:

The following error is shown after the certificate, received from the CA, is uploaded:

Invalid certificate Subject DN. Please make sure that the Subject DN of the uploaded certificate is the same as the configured one.

Solution:

See KBA **2922845** - Invalid certificate Subject DN. Please make sure that the Subject DN of the uploaded certificate is the same as the configured one.

It is not possible to rename the Identity Authentication tenant

Symptom:

There is a need to change the `tenant ID` of your Identity Authentication tenant that is part of the URL:

`tenant ID>.accounts.ondemand.com/admin`

Solution:

See KBA 2774169 - Rename Identity Authentication tenant is not possible.

Custom domain in Identity Authentication using a wild-card certificate

Symptom:

There is a need to configure a custom domain in Identity Authentication, using an existing wild card certificate.

Solution:

See KBA 2922427 - Custom domain on IAS using wildcard certificate.

Parent topic: Troubleshooting Issues with Authentication and User and Group Management [page 3023]

Related Information

Accessing the Administration Console [page 3024]
User Import [page 3031]
Emails [page 3038]
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Risk-Based Authentication [page 3067]

1.12.15 Troubleshooting Provisioning Issues

While using the Identity Provisioning service, you might encounter issues or have questions regarding specific settings, limitations, or other aspects.

For a step-by-step walkthrough of the administration console for SAP Cloud Identity Services, refer to the following video on using guided tours while navigating within the provisioning systems and logs.

The following topics provide detailed information and steps to help you identify and resolve common issues that may arise during your experience with the Identity Provisioning service.

- Administration Issues [page 3071]
- Job and Transformation Issues [page 3077]
- Error Messages [page 3111]
- Entity Deletion Issues [page 3121]
  Learn how to troubleshoot the most frequent reasons for deletion of users or groups from a target system by the Identity Provisioning.
- Additional Information [page 3130]
- Handling Specific Attributes [page 3137]

1.12.15.1 Administration Issues

Can't Enable a Disabled System

Problem:

- A previously enabled system (source, target or proxy) is now disabled.
- You choose the icon to edit the system status.
- The *Enable* button is inactive (grayed out).

Find the root cause for your problem in the cases below:

A Feature Was Explicitly but Only Temporarily Enabled

Problem:

(Bundle, Standalone) A particular system is disabled because it uses a feature that was only temporarily enabled.

Solution:

If you still need this feature, you can request it again by contacting the Support team. Create an incident for component *BC-IAM-IPS*. In the incident, specify also:

- The IDs of your global account and subaccount, or your bundle tenant.
- To which region your account/tenant belongs. You can find it in this list: (Discovery Center) Identity Provisioning -> *Service Plan*

One of Your Bundles Has Expired

Problem:

(Bundle) You use the Identity Provisioning service as part of an SAP cloud solution. The systems belonging to the relevant bundle are disabled, indicating that the license for the relevant product has expired.

Solution:

To enable your inactive systems and continue using them, extend your bundle license.

Beta Features Are Disabled

Problem:

(Standalone) A particular system is disabled because it's available only when Beta features are enabled.

Solution:

Beta features might be disabled for your subaccount. To verify this, follow the steps below:

1. In the SAP BTP cockpit, navigate to your global account.
2. On your subaccount tile and choose *Edit*.

3. For *Beta Features*, select the *Enable* checkbox.
4. Refresh your administration console for SAP Cloud Identity Services to see if the disabled system is now available.

System Is Still Disabled

Your problem requires deeper analysis by SAP. Please create an incident for component **BC-IAM-IPS**.

In the incident, specify also:

- The IDs of your global account and subaccount, or your bundle tenant.
- To which region your account/tenant belongs. You can find it in this list: (Discovery Center) Identity Provisioning-> Service Plan.

Before you open an incident, read:

2654164 - What to consider before opening a Support Ticket for SAP Cloud Platform

For contract-related questions, read:

1660069 - How to contact SAP Contracts Department for questions or issues

Can't Create Another System

Problem:

1. You have added a certain number of source or target systems in the administration console for SAP Cloud Identity Services.
2. You choose *Add* to create a new one.
3. A warning message appears, saying that you have reached the maximum number of source or target systems allowed for your subaccount.

Reason

By default, you are allowed to create up to **20** source and **50** target systems for your subaccount.

Solution:

1. Check if there is a system that is no longer relevant to your business scenario.
2. If you think you might still need it in future, export it first as a JSON or a CSV file.
3. Delete that system from the UI.
4. Add a new source/target system.

If your business case requires more productive source systems, create an incident for component **BC-IAM-IPS** to request the number of additional systems. In the incident, specify also:

- The IDs of your global account and subaccount, or your bundle tenant.
- To which region your account/tenant belongs. You can find it in this list: (Discovery Center) Identity Provisioning -> *Service Plan*.

Can't Get or Grant Admin Permissions for a Bundle

You use the Identity Provisioning service as integrated in an SAP cloud product. To learn more, see: Get Your Tenant [page 528]

You don't know who is the administrator of your bundle tenant, or how to grant other users *Administrator* permissions.

Depending on your case, you can find more information below:

You Don't Know Who Is Administrator

Problem:

You need admin rights for your Identity Provisioning bundle tenant. You have to ask the initial (or a secondary) administrator for your bundle tenant but you don't know who they are.

Solution:

You can view all your SAP Cloud Identity Services tenants by using the *SAP Cloud Identity Services - Tenants* application. You can access it at the following URL: https://iamtenants.accounts.cloud.sap/Information. This tool displays the tenant administrators, the type of your tenant (test or productive), the date it was created, and the region where it is available.

If you still struggle and can't find any administrator yourself, create an incident for component **BC-IAM-IPS** and request the contacts of the initial tenant administrator.

You Are Administrator

Problem:

You are either the initial administrator of your Identity Provisioning bundle tenant, or another admin user has granted you such permissions. Now, other users ask you to grant them admin permissions too, but you don't know how to do it, or what tool to use.

Solution:

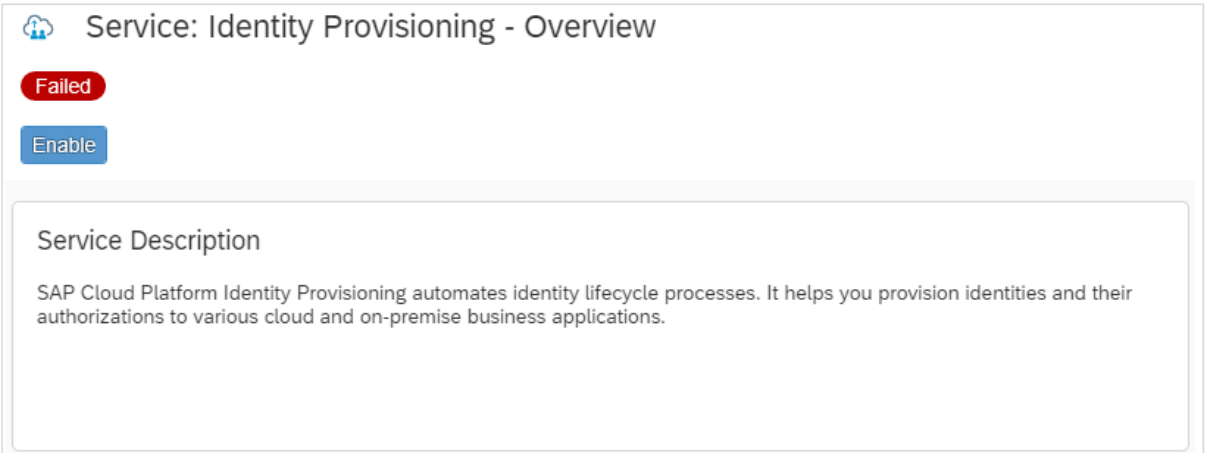
Open the administration console for SAP Cloud Identity Services and choose the *Authorizations* tile. It allows you to grant admin rights to as many additional users as you want.

To learn how, see: Manage Administrators [page 543].

(Standalone) Enabling the Identity Provisioning Service Fails

Problem:

1. You have a global account in the SAP BTP cockpit.
2. In one of your subaccounts, you open the *Identity Provisioning* tile.
3. You choose *Enable* but the operation fails.
4. Even if you try several times, you cannot enable the service.



Reason and Solution

There might be different reasons for failed enabling. The most common ones are:

Reason	Solution
The Identity Provisioning service is currently down.	Create an incident for component *BC-IAM-IPS* and we'll investigate the issue.
In your global account, you have enabled the Identity Provisioning for two subaccounts, and you're trying to enable it for a third one. You are not allowed to do this because, by default, you can enable the service only for *two* subaccounts per global account.	If you have already enabled the service for two subaccounts, and want to raise the quota for your global account to three or more subaccounts, create an incident for component *BC-IAM-IPS*. Request the number of additional subaccounts.

In the created incident, specify also:

- The IDs of your global account and subaccount.
- To which region your account belongs. You can find it in this list: (Discovery Center) Identity Provisioning -> *Service Plan*

Identity Provisioning Tile Is Missing In the SAP BTP Cockpit

Your access to the *Identity Provisioning* tile in SAP BTP cockpit depends mostly on the type of product you have purchased. You have one of the following:

- *Identity Provisioning service integrated (bundled) in an SAP cloud product*
  Problem:
  1. You already have a production global account and use at least one subaccount for non-identity provisioning purposes (that is, other services and cloud tasks).
  2. After purchasing an SAP cloud product (see the list of available bundle options), you request and receive access to the Identity Provisioning service. You can open the administration console for SAP Cloud Identity Services by using two URLs related to two subaccounts (one for testing purposes and one for productive use). These subaccounts are new, and not to be confused with any you already have.

3. You try to access the *Identity Provisioning* tile via the SAP BTP cockpit, using your "previous" subaccount:
    - In the cockpit, you choose the previous subaccount.
    - From the left-side navigation, you choose *Services*.
    - You don't see a *Provisioning Service* tile.

Reason & Solution:

Even though you have obtained the Identity Provisioning service as part of a bundle product, it's not bound to your previous subaccount. The service is bound only to the subaccounts you've received from SAP, which have been specifically created for your Identity Provisioning scenarios.

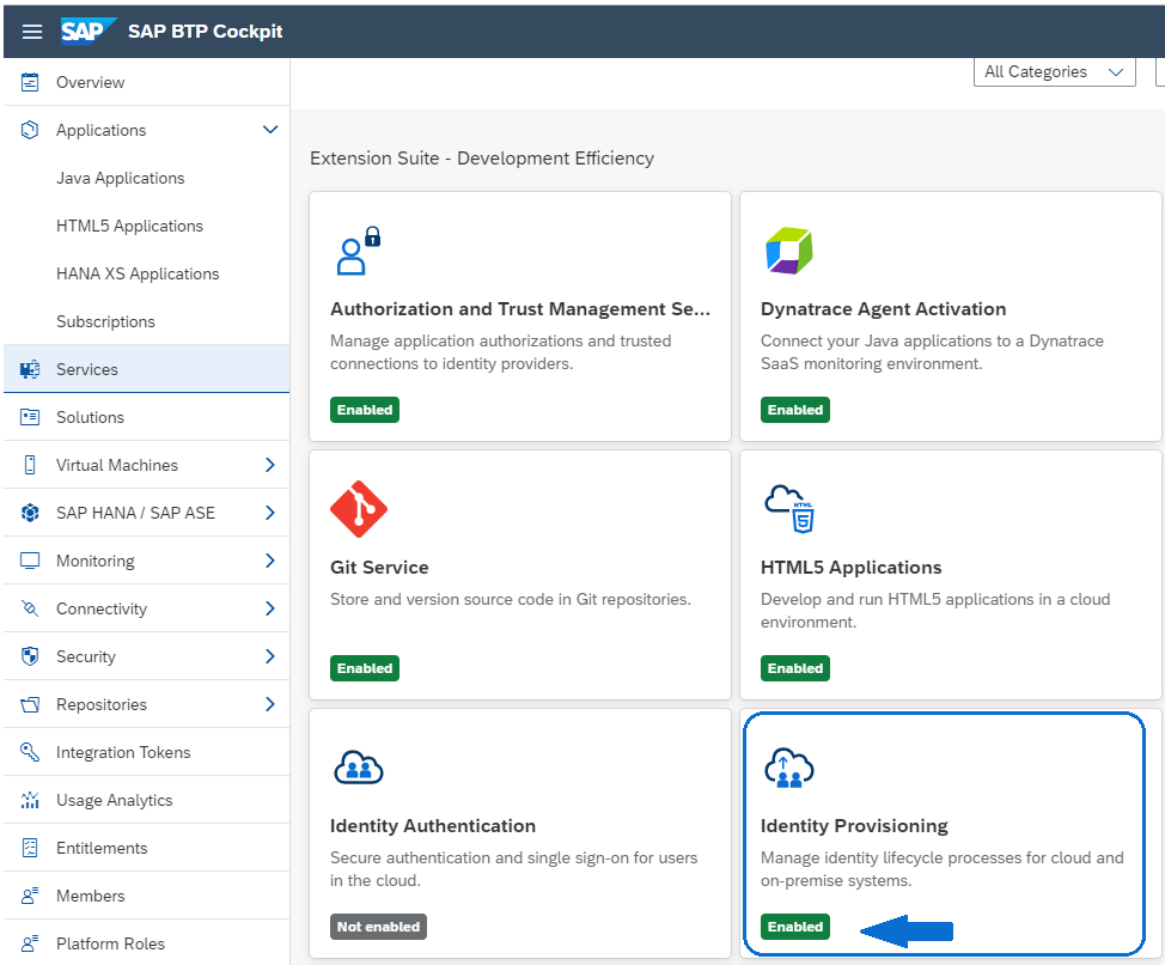
Also, you won't be able to select the *Provisioning Service* tile from either of these two subaccounts. That's because you're not intended to control the service from the cockpit.

- *Identity Provisioning service as a standalone product*

  Make sure you use the correct (sub)account that is bound to your Identity Provisioning license. Then:

  1. Log on to the SAP BTP cockpit: https://<region>.hana.ondemand.com.
  2. Go to your global account and choose your subaccount.
  3. From the navigation area, choose Your problem requires deeper analysis by SAP. Please create an incident for component *Services*.
  4. Browse through this page and find the *Extension Suite – Development Efficiency* section.
  5. You should see an *Identity Provisioning* tile.
  6. The default status of the service is *Not enabled*. Choose *Enable*.

Your *Services* screen should look like this:



Still Cannot See the Identity Provisioning Tile

If your problem requires deeper analysis by SAP, please create an incident for component *BC-IAM-IPS*.

In the incident, specify also:

- The IDs of your global account and subaccount, or your bundle tenant.
- To which region your account/tenant belongs. You can find it in this list: (Discovery Center) Identity Provisioning -> *Service Plan*.

Before you open an incident, read:

- 2654164 - What to consider before opening a Support Ticket for SAP BTP

For contract-related questions, read:

- 1660069 - How to contact SAP Contracts Department for questions or issues

Parent topic: Troubleshooting Provisioning Issues [page 3070]

Related Information

Job and Transformation Issues [page 3077]
Error Messages [page 3111]
Entity Deletion Issues [page 3121]
Additional Information [page 3130]
Handling Specific Attributes [page 3137]

1.12.15.2 Job and Transformation Issues

You have a struggle with a transformation, a property or a provisioning job?

Find your particular problem below:

Entities Are Not Updated or Deleted in the Target System

When you set up your systems and start a manual or scheduled provisioning job, the standard behavior of the process reads all the entities from the source system. This mode prevents data loss and always keeps your target system synchronized with the source. You can also choose to optimize the amount of retrieved data and read only the updated entities (new, updated, deleted). For more information about these modes, see Manage Full and Delta Read [page 2459].

To keep source and target systems completely synchronized, you can use the **Resync** type of provisioning job.

In some cases though, specific entities should be updated or deleted, but during the provisioning job they are not read at all. They are therefore not updated in the target system.

Entities Are Not Updated

Problem:

1. Run a provisioning job (manual or scheduled).
2. The job finishes successfully but some entities have not been updated.

Reason:

You might have set special conditions or expressions in your transformations.

Solution:

1. Make sure that no special conditions or expressions (ignore or skipOperations) are set in the JSON transformation.
2. Run again a **Read** or **Resync** provisioning job.
3. Check to see if the relevant entities have been updated in the target system.

Entities are Still Not Updated in the Target System

If your problem requires deeper analysis by SAP, please create an incident for component BC-IAM-IPS.

In the incident, specify also:

- The IDs of your global account and subaccount, or your bundle tenant.
- To which region your account/tenant belongs. You can find it in this list: (Discovery Center) Identity Provisioning -> *Service Plan*

Before you create a support case, read:

2654164 - What to consider before opening a Support Ticket for SAP Business Technology Platform

For contract-related questions, read:

1660069 - How to contact SAP Contracts Department for questions or issues

Entities Are Not Deleted

Problem

Your business case requires you to delete some entities (users or groups) from both the source and the target system. You proceed as follows:

1. Delete these entities from the source system.
2. Run a provisioning job (manual or scheduled).

As a result, the job finishes either **successfully** or **with error**, and in both cases - the relevant entities are not deleted from the target system.

Job is successful but entities are not deleted

Problem

You have deleted some entities from the source system, and then run a provisioning job.

The job finishes successfully but the relevant entities have not been deleted from the target system.

Reason

You either have set special conditions in your transformations, or have missed to add the `ips.delete.existedbefore.entities` property in your target system configuration.

> **Note**
>
> If the property `ips.delete.existedbefore.entities` has not been added before the job attempting the deletion is run, adding it later will not resolve the problem.

Solution

Resolving this issue depends on whether you have added the `ips.delete.existedbefore.entities` property in your target system before or after running the job attempting the deletion.

The property is added before running the provisioning job:

1. Make sure that no special conditions or expressions are set in the target transformation. That includes: *ignore expression*, *skipOperations* (skipping delete), or *deleteEntity* scope.
2. Run a **Read** or **Resync** provisioning job.
3. Check if the relevant entities have been deleted from the target system.

The property is added after running the provisioning job:

If the property is added after the provisioning job finished successfully, entities recognized as "previously existed ones" cannot be deleted from the target system anymore. In this case, you need to delete them from the target system (for example, manually or via script).

For more information on how to properly delete entities in the target system, see Manage Deleted Entities [page 2463].

EntitiesAre Still Available in the Target System

If your problem requires deeper analysis by SAP, please create an incident for component **BC-IAM-IPS**.

In the incident, specify also:

- The IDs of your global account and subaccount, or your bundle tenant.
- To which region your account/tenant belongs. You can find it in this list: (Discovery Center) Identity Provisioning -> *Service Plan*

Before you create a support case, read:

2654164 - What to consider before opening a Support Ticket for SAP Business Technology Platform

For contract-related questions, read:

1660069 - How to contact SAP Contracts Department for questions or issues

Job finishes with error and entities are not deleted

Problem

You have deleted some entities from the source system. You have not set any special conditions or expressions (*ignore*, *skipOperations*, or *deleteEntity* scope) in the target transformation. You have set property `ips.delete.existedbefore.entities` to *true* in the target system, and then run a provisioning job.

The job finishes with error, and the relevant entities have not been deleted from the target system.

Reason

In the job log, you see that there are failed entities when the job has tried to read them from the source system.

Solution

1. Resolve the failed entities in the source system.
2. Run again a **Read** or a **Resync** provisioning job.
3. Check if the relevant entities have been deleted from the target system.

Entities Are Still Available in the Target System

If your problem requires deeper analysis by SAP, please create an incident for component **BC-IAM-IPS**.

In the incident, specify also:

- The IDs of your global account and subaccount, or your bundle tenant.
- To which region your account/tenant belongs. You can find it in this list: (Discovery Center) Identity Provisioning -> *Service Plan*

Before you create a support case, read:

2654164 - What to consider before opening a Support Ticket for SAP Business Technology Platform

For contract-related questions, read:

1660069 - How to contact SAP Contracts Department for questions or issues

Entities Should Not Be Deleted in the Target System

Problem & Reason

The default behavior of the Identity Provisioning service is:

1. It reads entities from a source system and provisions (writes) them in a target system.
2. If, on a next reading job, the service detects that some previously existed entities are now missing, it considers them as **deleted** (or not active anymore).
3. Thus, the next provisioning job (Read or Resync) deletes such entities from the target system.

However, these entities might still be existing and active but are not read because you have set a filter or condition for some business reason.

You **do not want** the "unread" entities (the ones that do not fulfill the filter/condition) to be deleted from the target.

Solution

To prevent deletion of existing active users in your target system, proceed as follows:

1. Select your target system in the administration console for SAP Cloud Identity Services .
2. Choose the *Transformations* tab and then *Edit*.
3. If you want to keep existing/active users, in the user section, add the following mapping:

Sample Code

```
{
 "constant": false,
 "targetPath": "$.active",
 "scope": "deleteEntity"
 }
```

This way, every time you use a filter or a condition, only the relevant filtered entities (users) will be created or updated.

The rest users will remain as is - not updated but not deleted as well.

To learn more about this scope and see specific examples for SCIM systems, see: Transformation Expressions [page 391] ▶ *scope* ▶ *deleteEntity* ◀.

Exclude Users from Provisioning to Target Systems

You want to know how to set up your provisioning scenario so that some users are not replicated to the target system based on given criteria.

Let's assume the employees of your company (with email domain `parentCompany.com`) are stored in SAP SuccessFactors along with other users (for example, contractors). You want to replicate some of them to Identity Authentication or SAP Analytics Cloud and exclude the rest (with email domain `contractorsCompany.com`) because they don't need access to those systems to do their job.

For this, you need to specify a condition in the target system that is based on something common the users share, like their email domain.

Proceed as follows:

1. In SAP SuccessFactors, verify that the users you want to exclude have email address.
   The user email is optional and may not be unique in SAP SuccessFactors. However, it's mandatory and unique in Identity Authentication and SAP Analytics Cloud.
2. In the administration console for SAP Cloud Identity Services, select your target system and open its transformation in *Edit* mode.
   Identity Provisioning is embedded in SAP Cloud Identity Services admin console (formerly known as Identity Authentication admin console). If your Identity Provisioning tenant is running on SAP Cloud Identity infrastructure, you can configure and work with provisioning systems there.
3. Depending on your scenario, add one of the following conditions within the user entity:

   Sample Code

   ```
   {
       "user": {
           "condition": "!($.emails[0].value =~ /.*contractorsCompany.com/)",
           "mappings": [
   ```

   Based on this condition, Identity Provisioning will provision only users that do not have email domain `contractorsCompany.com`. Users with this domain will be read from the source but will be skipped in the target system.

   Sample Code

   ```
   {
       "user": {
           "condition": "$.emails[0].value!= 'user123@contractorsCompany.com'
   ",
           "mappings": [
   ```

   Based on this condition, Identity Provisioning will provision only users that do not have the specified email. Users with this email will be read from the source but will be skipped in the target system. Use this condition in case the users you want to exclude have the same dummy email.
4. (Optional) Run the **Simulate** job from the connected SAP SuccessFactors source system to verify that excluded users are skipped, therefore not provisioned. This test job is available for tenants running on SAP Cloud Identity infrastructure.
5. If the **Simulate** job displays the expected result, run a provisioning job (**Read** or **Resync**).
   Adding a similar condition for excluding users is also applicable to other target systems.

Ensure SAP S/4HANA Cloud Public Edition Users are Replicated During Their Validity Period

You want to ensure that Identity Provisioning service replicates SAP S/4HANA Cloud Public Edition users to the embedded SAP Analytics Cloud during their validity period.

This is handled by the following changed condition in the default read transformation of both - the standard (manually created) and the preconfigured (automatically created) SAP S/4HANA Cloud Public Edition source systems in Identity Provisioning bundle tenants.

Changed condition:

Sample Code

```
{
"user": {
"condition": "($.user.validityPeriod.startDate <= '${currentDate}') &&
($.user.validityPeriod.endDate > '${currentDate}')",
"mappings": [
```

For comparison, the initial condition ensured that SAP S/4HANA Cloud Public Edition users are replicated to the SAP Analytics Cloud during the validity of the *Employee* role of their business partner.

Initial condition:

Sample Code

```
{
"user": {
"condition": "($.validityPeriod.startDate <= '${currentDate}') &&
($.validityPeriod.endDate > '${currentDate}')",
"mappings": [
```

How to proceed:

1. In the administration console for SAP Cloud Identity Services, open the transformation of your SAP S/4HANA Cloud Public Edition source system and check the condition under the user entity.
   - If the read transformation contains the changed condition, no action is required.
   - If the read transformation contains the initial condition, replace it with the changed one. Alternatively, you can create a new SAP S/4HANA Cloud Public Edition source system and configure it manually. The newly created system will contain the changed condition by default.
2. Save your configurations.

Ensure deleted SAP S/4HANA Cloud Public Edition Users are Also Deleted in SAP Analytics Cloud

You want to ensure that users who were deleted in SAP S/4HANA Cloud Public Edition source system will also be deleted in the embedded SAP Analytics Cloud target system.

This is handled by the `ips.delete.existedbefore.entities` property which is now set to *true* in the preconfigured SAP Analytics Cloud target system of the Identity Provisioning bundle tenants for SAP S/4HANA Cloud Public Edition release 1911 or higher.

Initially, this property was not preconfigured in the SAP Analytics Cloud target system and had to be added manually.

How to proceed:

1. In the administration console for SAP Cloud Identity Services, open the *Properties* tab of the embedded SAP Analytics Cloud target system and check if `ips.delete.existedbefore.entities` is set to *true*.
   - If the property is set, no action is required.

- If the property is not set, add it and set it to *true*.

2. Save your configurations.

> **Note**
>
> After successful user provisioning from SAP S/4HANA Cloud Public Edition to SAP Analytics Cloud, if you delete users in SAP S/4HANA Cloud Public Edition and want to provision the change (that is, to delete them) in the target system, you need to set the **`ips.delete.existedbefore.entities=true`** before running the job to delete the users in the target system. If you set it afterwards, the users won't be deleted in SAP Analytics Cloud as they will be recognized as "previously existed ones". In this case, to have them deleted, you need to create an incident on component **LOD-ANA-OEM**.

For more information on the required sequence of steps for deleting users, see Manage Deleted Entities [page 2463].

Groups Are Not Provisioned to the Target System

Problem:

1. You have added a source and a target system in the administration console for SAP Cloud Identity Services. Both systems support provisioning of groups.
2. You have correctly set up these systems. For the source one, you use the default group transformation.
3. Start a provisioning job.
4. View the job results.

In the target system, users are provisioned but groups are missing.

Reason:

In the source system transformations, group provisioning is disabled (ignored) by default.

Solution:

To enable group provisioning, manually change the *ignore* condition to:

> **Sample Code**
>
> ```
> "group": {
>        "ignore": false,
>        "mappings": [
>           {
> ...
> ```

> **Sample Code**
>
> Related Information:
>
> Transformation Expressions [page 391]

Identity Authentication: Provisioned Users Can't Log On

In Identity Authentication target systems, the behavior of the provisioning jobs depends on the constant value set for `$.userType` (in the JSON transformation).

- `"constant": "public"`
  This means that all passwords will be written by default in the Identity Authentication. All provisioned users will therefore be able to successfully log into the administration console for SAP Cloud Identity Services.
- `"constant": "employee"`
  In this case, the login behavior of the provisioned users depends on whether users have been created with or without a password, and where these passwords are stored. You therefore need to modify the target transformations accordingly in order for the users to successfully log in to the administration console for SAP Cloud Identity Services.
  Find your use case from the options below:

Users Are Created Without a Password

This occurs in scenarios in which Identity Authentication acts as a proxy to delegate the authentication to a corporate identity provider.

Solution

Open the Identity Authentication target system transformation and remove the `"constant": "39"` mapping below.

As a result, you should have:

Identity Authentication (SCIM API version 1)	Identity Authentication (SCIM API version 2)

<pre>{
    "constant": "false",
    "targetPath": "$.sendMail",
    "scope": "createEntity"
},
{
    "constant": "true",
    "targetPath": "$.mailVerified",
    "scope": "createEntity"
},
{
    "constant": "disabled",
    "targetPath": "$.passwordStatus",
    "scope": "createEntity"
},
// Remove:
{
    "constant": "39",
    "targetPath": "$.sourceSystem",
    "scope": "createEntity"
},
// Make sure you have this
target path with "constant" set to
"employee".
{
    "constant": "employee",
    "targetPath": "$.userType"
},</pre>
	<pre>{
	    "constant": false,
	    "targetPath": "$
	['urn:ietf:params:scim:schemas:extensi
	on:sap:2.0:User']['sendMail']",
	    "scope": "createEntity"
	},
	{
	    "constant": true,
	    "targetPath": "$
	['urn:ietf:params:scim:schemas:extensi
	on:sap:2.0:User']['mailVerified']",
	    "scope": "createEntity"
	},
	{
	    "constant": "disabled",
	    "targetPath": "$
	['urn:ietf:params:scim:schemas:extensi
	on:sap:2.0:User']['passwordDetails']
	['status']",
	    "scope": "createEntity"
	},
	// Remove the mapping starting with
	**"constant": "<your-source-system-type-
	code>"**, if its value is set to **39**.
	// Alternatively, set "ignore" to
	true.
	{
	    "constant": **39**,
	    "targetPath": "$
	['urn:ietf:params:scim:schemas:extensi
	on:sap:2.0:User']['sourceSystem']",
	    "scope": "createEntity",
	    "ignore": **true**
	},
	// Remove this mapping.
	Alternatively, set "ignore" to **true**.
	{
	     "constant": "<your-source-system-
	id>",
	     "targetPath": "$
	['urn:ietf:params:scim:schemas:extensi
	on:sap:2.0:User']['sourceSystemId']",
	     "scope": "createEntity",
	     "ignore": **true**
	}
	// Make sure you have this
	target path with "constant" set to
	"employee".
	{
	    "constant": "employee",
	    "targetPath": "$.userType"
	},</pre>

→ Tip

If you don't want to delete the `"constant": "39"` section from the transformation, you can configure the *Corporate User Store* in the administration console for SAP Cloud Identity Services so as to reuse the corporate passwords. To learn how, see: Corporate User Store.

Related information: Identity Authentication

Users Still Cannot Log on

Your problem requires deeper analysis by SAP. Please create an incident for component BC-IAM-IPS.

In the incident, specify also:

- The IDs of your global account and subaccount, or your bundle tenant.
- To which region your account/tenant belongs. You can find it in this list: (Discovery Center) Identity Provisioning -> *Service Plan*

Before you create a support case, read:

2654164 - What to consider before opening a Support Ticket for SAP Business Technology Platform

For contract-related questions, read:

1660069 - How to contact SAP Contracts Department for questions or issues

Passwords Are Stored in Corporate User Store

In this case, users are authenticated by Corporate User Store. This is relevant to scenarios that require integration with Cloud Connector. The source system can be AS ABAP, Microsoft Active Directory, or LDAP.

Solution

Open the Identity Authentication target system transformation. For mapping `"targetPath": "$.passwordStatus"`, change `"constant": "disabled"` to `"enabled"`.

Also, if using Identity Authentication (SCIM API version 2), set the value of the `"constant": "<your-source-system-type-code>",` mapping to `39` and set your source system identifier in `"constant": "<your-source-system-id>"`.

As a result, you should have:

Identity Authentication (SCIM API version 1)	Identity Authentication (SCIM API version 2)

<pre>{
"constant": "false",
"targetPath": "$.sendMail",
"scope": "createEntity"
},
{
"constant": "true",
"targetPath": "$.mailVerified",
"scope": "createEntity"
},
{
"constant": "enabled",
"targetPath": "$.passwordStatus",
"scope": "createEntity"
},
{
"constant": "39",
"targetPath": "$.sourceSystem",
"scope": "createEntity"
},
// Make sure you have this target path with "constant" set to "employee".
{
"constant": "employee",
"targetPath": "$.userType"
},</pre>
	<pre>{
	"constant": false,
	"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sendMail']",
	"scope": "createEntity"
	},
	{
	"constant": true,
	"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mailVerified']",
	"scope": "createEntity"
	},
	{
	"constant": "enabled",
	"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['passwordDetails']['status']",
	"scope": "createEntity"
	},
	// Set the value of the **"constant": "<your-source-system-type-code>"**, mapping to **39**.
	// Make sure "ignore" is set to **false**.
	{
	"constant": **39**,
	"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
	"scope": "createEntity",
	"ignore": **false**
	},
	// Set the value of the **"constant": "<your-source-system-id>"**, mapping.
	// Make sure "ignore" is set to **false**.
	{
	"constant": "<your-source-system-id>",
	"targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystemId']",
	"scope": "createEntity",
	"ignore": **false**
	}
	// Make sure you have this target path with "constant" set to "employee".
	{
	"constant": "employee",
	"targetPath": "$.userType"
	},</pre>

Related information: Identity Authentication

Passwords Are Stored in Identity Authentication (initial password)

This use case requires creating an initial password. Users will enter it on their logon to the Identity Authentication service. They can later change this password.

Solution

In the Identity Authentication target system transformation, modify the following JSON mappings:

- For `"$.passwordStatus"`, set "constant" to `"initial"`
- Enter an additional mapping, starting with `"constant": "<initial password defined by customer>"`. Replace `"<initial password defined by customer>"` with an actual initial password. In Identity Authentication (SCIM API version 2), the mapping, starting with `"constant": "<your-initial-password>"`, is part of the default transformation. You only need to replace `<your-initial-password>` with an actual initial password and set `"ignore": "true"` to `"false"`.
- Remove the mapping in bold below.

As a result, you should have:

Identity Authentication (SCIM API version 1)	Identity Authentication (SCIM API version 2)

<pre>{
    "constant": "initial",
    "targetPath": "$.passwordStatus",
    "scope": "createEntity"
},
{
    "constant": "<initial password defined by customer>",
    "targetPath": "$.password",
    "scope": "createEntity"
},
{
    "constant": "false",
    "targetPath": "$.sendMail",
    "scope": "createEntity"
},
{
    "constant": "true",
    "targetPath": "$.mailVerified",
    "scope": "createEntity"
},
// Make sure you have this target path with "constant" set to "employee".
{   "constant": "employee",
    "targetPath": "$.userType"
},
// Remove this mapping:
{
**   "constant": "39",**
**   "targetPath": "$.sourceSystem",**
**   "scope": "createEntity"**
},</pre>
	<pre>{
	    "constant": "initial",
	    "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['passwordDetails']['status']",
	    "scope": "createEntity"
	},
	{
	    "constant": "<your-initial-password>",
	    "targetPath": "$.password",
	    "scope": "createEntity",
	    "ignore": "false"
	},
	{
	    "constant": false,
	    "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sendMail']",
	    "scope": "createEntity"
	},
	{
	    "constant": true,
	    "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['mailVerified']",
	    "scope": "createEntity"
	},
	// Remove the mapping starting with "constant": "<your-source-system-type-code>", if its value is set to 39.
	// Alternatively, set "ignore" to true.
	{
	    "constant": 39,
	    "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystem']",
	    "scope": "createEntity",
	    "ignore": true
	},
	// Remove this mapping. Alternatively, set "ignore" to true.
	{
	     "constant": "<your-source-system-id>",
	     "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['sourceSystemId']",
	     "scope": "createEntity",
	     "ignore": true
	}
	// Make sure you have this target path with "constant" set to "employee".
	{
	 "constant": "employee",
	 "targetPath": "$.userType"
	},</pre>

→ Tip

If you don't want the initial password to be visible in the transformation, you can create a credential property called *initial.password* in the *Properties* section of the target system. Then, use the following substitution of the property as a mapping in the transformation:

Identity Authentication (SCIM API version 1)	Identity Authentication (SCIM API version 2)
`{ "constant": "%initial.password%", "targetPath": "$.password", "scope": "createEntity" }`	`{ "constant": "%initial.password%", "targetPath": "$.password", "scope": "createEntity", "ignore": "false" }`

→ Tip

If you don't want to delete the `"constant": "39"` section from the transformation, you can configure the Corporate User Store in the administration console for SAP Cloud Identity Services so as to reuse the corporate passwords. To learn how, see: Corporate User Store

Related information: Identity Authentication

Passwords Are Stored in Identity Authentication (activation e-mail)

This use case requires sending an activation e-mail (without verifying it). Via this e-mail, users can logon to the Identity Authentication service. They can later change their password.

Solution

Open the Identity Authentication target system transformation and modify the following JSON mappings:

- For `"$.sendMail"`, change `"constant": "false"` to **true**.
- For `"$.mailVerified"`, change `"constant": "true"` to **false**.
- Remove the mappings in bold below.

As a result, you should have:

Identity Authentication (SCIM API version 1)	Identity Authentication (SCIM API version 2)

<pre>{
"constant": "true",
"targetPath": "$.sendMail",
"scope": "createEntity"
},
{
"constant": "false",
"targetPath": "$.mailVerified",
"scope": "createEntity"
},
// **Remove these mappings:**
{
"constant": true,
"targetPath": "$.active"
},
{
"constant": "enabled",
"targetPath": "$.passwordStatus",
"scope": "createEntity"
},
{
"constant": "39",
"targetPath": "$.sourceSystem",
"scope": "createEntity"
}, */</pre>
	<pre>{
	"constant": true,
	"targetPath": "$
	['urn:ietf:params:scim:schemas:extensi
	on:sap:2.0:User']['sendMail']",
	"scope": "createEntity"
	},
	{
	"constant": false,
	"targetPath": "$
	['urn:ietf:params:scim:schemas:extensi
	on:sap:2.0:User']['mailVerified']",
	"scope": "createEntity"
	},
	// **Remove these mappings:**
	{
	"constant": true,
	"targetPath": "$.active"
	},
	{
	"constant": "enabled",
	"targetPath": "$
	['urn:ietf:params:scim:schemas:extensi
	on:sap:2.0:User']['passwordDetails']
	['status']",
	"scope": "createEntity"
	},
	// Remove the mapping starting with
	"constant": "<your-source-system-type-code>", if its value is set to *39*.
	// Alternatively, set "ignore" to *true*.
	{
	"constant": **39**,
	"targetPath": "$
	['urn:ietf:params:scim:schemas:extensi
	on:sap:2.0:User']['sourceSystem']",
	"scope": "createEntity",
	"ignore": **true**
	},
	// Remove this mapping.
	Alternatively, set "ignore" to *true*.
	{
	"constant": "<your-source-system-
	id>",
	"targetPath": "$
	['urn:ietf:params:scim:schemas:extensi
	on:sap:2.0:User']['sourceSystemId']",
	"scope": "createEntity",
	"ignore": **true**
	}</pre>

→ Tip

If you don't want to delete the `"constant": "39"` section from the transformation, you can configure the Corporate User Store in the administration console for SAP Cloud Identity Services so as to reuse the corporate passwords. To learn how, see: Corporate User Store

Related information: Identity Authentication

Passwords are Migrated from SAP SuccessFactors

Problem

1. You have SAP SuccessFactors (SFSF) as a source system and Identity Authentication as a target system.
2. You have configured your *SFSF* application so as to migrate user passwords to the administration console for SAP Cloud Identity Services, as described on Configure Authentication Provider To Migrate User Passwords from SAP SuccessFactors Systems to Identity Authentication.
3. You might have **missed** to make the additional transformation modifications, and then you have run your first provisioning job between SFSF and Identity Authentication.
4. The provisioning job is successful but the migrated passwords of SFSF users are disabled in Identity Authentication, thus users cannot login to the administration console for SAP Cloud Identity Services.

The reason: In the Identity Authentication target transformation, the `“passwordStatus”` mapping has been left with its default value - `"disabled"`.

Solution

1. You have properly configured your SFSF source system so as to migrate user passwords to Identity Authentication. To learn how, see: Configure Source System To Migrate User Passwords from SAP SuccessFactors Systems to Identity Authentication
   Migrating Passwords from SAP SuccessFactors to the SAP Cloud Identity Services
2. Make sure you have done the following modifications in the Identity Authentication transformation:
   - You have added two extra mappings with constants `<SF_Company_ID>` and *100* (see the code block below).
   - You have removed the `"constant": "39"` mapping.
   - For the `passwordStatus` mapping, you have changed `"disabled"` to `"enabled"`
3. To fix your issue, you also need to remove the `"createEntity"` scope from the `passwordStatus` mapping!
   So, after you have done the modifications listed above, your Identity Authentication transformation should look like this:

   Explanation:

   This way, on the next provisioning jobs, the Identity Provisioning will always update the password status of the users, even if they get locked in Identity Authentication. Also, the migrated passwords will be set first in Identity Authentication when a user logs in for the first time to one of their assigned applications (like SAP SuccessFactors or *User Profile*).

Identity Authentication (SCIM API version 1)	Identity Authentication (SCIM API version 2)

<pre>{
"constant": "<SF_Company_ID>",
"targetPath": "$.sourceSystemId"
},
{
"constant": "100",
"targetPath": "$.sourceSystem"
},
{
"constant": "enabled",
"targetPath": "$.passwordStatus"
 	
},
{
"constant": "false",
"targetPath": "$.sendMail",
"scope": "createEntity"
},
{
"constant": "true",
"targetPath": "$.mailVerified",
"scope": "createEntity"
},</pre>
	<pre>{
	"constant": "<SF_Company_ID>",
	"targetPath": "$
	['urn:ietf:params:scim:schemas:extens
	ion:sap:2.0:User']
	['sourceSystemId']",
	"scope": "createEntity",
	"ignore": false
	},
	{
	"constant": 100,
	"targetPath": "$
	['urn:ietf:params:scim:schemas:extens
	ion:sap:2.0:User']['sourceSystem']",
	"scope": "createEntity",
	"ignore": false
	},
	{
	"constant": "enabled",
	"targetPath": "$
	['urn:ietf:params:scim:schemas:extens
	ion:sap:2.0:User']['passwordDetails']
	['status']"
	
	},
	{
	"constant": false,
	"targetPath": "$
	['urn:ietf:params:scim:schemas:extens
	ion:sap:2.0:User']['sendMail']",
	"scope": "createEntity"
	},
	{
	"constant": true,
	"targetPath": "$
	['urn:ietf:params:scim:schemas:extens
	ion:sap:2.0:User']['mailVerified']",
	"scope": "createEntity"
	},</pre>

4. Run a successful **Read** job.
5. Make sure that SFSF users can now successfully login to the administration console for SAP Cloud Identity Services.
6. Then open the Identity Authentication transformation again, and add `"scope": "createEntity"` back to the `passwordStatus` mapping. This way, newly created users will get their password status set by the Identity Provisioning. And since this status stays *"enabled"*, their passwords won't be locked.

Additional Information

Another (more simple) way to have all users provisioned and their passwords - successfully migrated to Identity Authentication, is to delete all SFSF users from Identity Authentication, and recreate them again. Also, you might need to change the value of the `sf.user.filter` property so as to read only SFSF users who have already existed in Identity Authentication and have *Administrator* role there. This way, the admin users will be kept and all the other SFSF users will be deleted from Identity Authentication.

> ⚠ Caution
>
> You must perform the user deletion via the Identity Provisioning service! If you delete them manually from Identity Authentication, the Identity Provisioning will still "remember" that these users were previously

created by it, thus - will have marked them with a flag that would not allow their passwords to be migrated. **Password migration can happen only during user creation via an Identity Provisioning job!**

To learn more, see:Identity Authentication: Provisioned Users Are Duplicated in SAP Jam or SAP Work Zone [page 3095]

Recreate users that have been provisioned with disabled passwords

Problem

You have provisioned users from a source system to Identity Authentication. When users try to login to the administration console for SAP Cloud Identity Services though, they cannot do that because their passwords are locked.

The reason for that is improper transformation mappings, which have caused users to be provisioned with password status *"disabled"*.

Solution

A simple solution would be to delete all provisioned users from Identity Authentication and re-create them again, this time - with active passwords. You can do that via an Identity Provisioning job, and changing the value of the user filter property, so as to read only users who have already existed in Identity Authentication and have *Administrator* role there. To do that, proceed as follows:

1. Open your Identity Authentication target transformation.
2. Search for the `"user"` mapping below and change `"disabled"` to **`"enabled"`**:

Identity Authentication (SCIM API version 1)	Identity Authentication (SCIM API version 2)
`{ "constant": "disabled", "targetPath": "$.passwordStatus", "scope": "createEntity" },`	`{ "constant": "disabled", "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['passwordDetails']['status']", "scope": "createEntity" },`

3. Go to the source system and change the user filter so as to read only users who have already existed in Identity Authentication and have *Administrator* role there.
   For example, if you have an SAP SuccessFactors (SFSF) source system, go to the `sf.user.filter` property and extend its value to: `status eq 'active' and username in '<username1>','<username2>',...`
   This way, the Identity Authentication admin users will be kept, and all the other SFSF users will be deleted from Identity Authentication.
4. Run a successful **Read** job. It will delete all users that **Identity Provisioning** has previously created and that are not in the filter list.
5. Open the user filter property again and change it back to its default value. For example, change the `sf.user.filter` value back to: `status eq 'active'`
6. Run a successful **Read** job again to have all the SFSF users re-created.

Check if users can now perform their first login using their source system passwords.

Identity Authentication: Provisioned Users Are Duplicated in SAP Jam or SAP Work Zone

Problem

You have executed direct user provisioning from the administration console for SAP Cloud Identity Services to SAP Jam.

Then you go to the administration console for SAP Cloud Identity Services, set up Identity Authentication as a source system, and SAP Jam Collaboration as a target system, and run a job.

As a result, multiple users are created twice (duplicated) in SAP Jam.

The same issue may happen when the target system is SAP Work Zone. In this case, you can use the same solution here.

Reason

- When provisioned from Identity Authentication to SAP Jam via Identity Authentication console, a user `id` in Identity Authentication is mapped to a `userName` in SAP Jam.
- When provisioned from Identity Authentication to SAP Jam via IPS console, an `userName` in Identity Authentication is mapped to a `userName` in SAP Jam.

As a result, one and the same user can be created twice in SAP Jam - by its `id`, and by its `userName`.

1. Stop provisioning users to SAP Jam from the administration console for SAP Cloud Identity Services. See: Configure SAP Jam Target Systems for User Provisioning
2. Then navigate to the *Identity Provisioning* > *Target Systems*.
3. Open your SAP Jam Collaboration target system and modify its transformation removing the following two mappings:

```
{
             "sourceVariable": "entityIdTargetSystem",
             "targetPath": "$.id",
             "scope": "deleteEntity"
         },
        {
             "constant": false,
             "targetPath": "$.active",
             "scope": "deleteEntity"
        },
```

4. Now open your Identity Authentication source system and reset it. To do that, select the system and choose *Edit* > *Reset*.
5. Then modify the Identity Authentication transformation the following way:
   Change this mapping:

```
{
     "sourcePath": "$.userName",
     "optional": true,
     "targetPath": "$.userName",
     "correlationAttribute": true
  }
```

   To:

```
{
     "sourcePath": "$.id",
```

```
    "targetPath": "$.userName",
    "correlationAttribute": true
  }
```

6. Run a **Read** job for the Identity Authentication system. All entities created before directly from the Identity Authentication console to SAP JAM will be updated in SAP JAM. This way, the Identity Provisioning already knows about these entities and can manage them. Entities created from provisioning through the administration console for SAP Cloud Identity Services will remain.
7. Go to the SAP Jam target system, and add the following (Standard) system property: `ips.delete.existedbefore.entities` = **`true`**
8. Now go back to the Identity Authentication source system and add a special condition in the `"user"` section. Your mapping should look like this:

```
"user":
 {
  "condition": "($.name.familyName EMPTY false) && ($.name.familyName ==
'DUMMY_NAME')",
"mappings": [
```

9. Run a new **Read** job for the Identity Authentication system. At this step, all users provisioned through Identity Authentication console should be deleted from SAP JAM.
10. Now go to the SAP Jam target system and return the mappings you have removed on step 3.
11. Then go back to the Identity Authentication source system and delete the condition from step 8.

Next steps

From now on, use only the administration console for SAP Cloud Identity Services for provisioning users (and groups) from Identity Authentication to SAP Jam. All future *Read/Resync* jobs will provision your users correctly, mapping `userName` to `userName`, without duplicating entities anymore.

Identity Authentication: Some Additional Users Should Not Be Provisioned to SAP Jam

Problem

1. You have an SAP Jam product, which contains Identity Authentication and Identity Provisioning.
2. When your product is activated, all users from Identity Authentication are initially provisioned to SAP Jam, which is a default and desired behavior.
3. Later on, additional users were added to Identity Authentication (either created manually, or come from another product).
4. All these new users are automatically provisioned to SAP Jam too, but some of them **should not be**.

Reason

The default SAP Jam transformation is set so as to be synchronized with all changes that take place in the source system (Identity Authentication). That means - both updating existing users, or creating newly added ones.

However, you want some of these users to stay only in Identity Authentication and not to be provisioned to SAP Jam. For example, specific users coming from another products.

Solution 1

You can set a filter that will read, create and update only **"pure Identity Authentication"** users (originally existing in Identity Authentication, or manually created later).

1. You have to assign all these Identity Authentication users to a specific group. Let's say, the group name is called **`IAS_Specific`**.
2. All additional users (coming from other products) must belong to other groups or to no groups at all.
3. To set a filter, open your SAP Jam target system.
4. Go to the *Properties* tab and choose *Edit*.
5. Add property `scim.user.filter` and set its value to **`groups eq 'IAS_Specific'`**

This way, if a new user appears in Identity Authentication, and it doesn't belong to *IAS_Specific*, it will not be provisioned to SAP Jam.

Solution 2

If it's too complicated or cumbersome to assign all **"pure Identity Authentication"** users to a single group, you can instead assign the newly come **foreign** users to a specific group. Then, you can set a condition in the transformation so as to skip only this specific group.

1. Let's say, the **foreign** users come from SAP Marketing Cloud, so you assign the relevant users to a group named **`Marketing_Cloud`**.
2. To set a condition, open your Identity Authentication source system.
3. Go to the *Transformations* tab and choose *Edit*.
4. Under `"user":{` enter the following condition:

```
{
   "user": {
       "condition": "$.groups EMPTY true || $.groups[?(@.value ==
'Marketing_Cloud' )] == []",
       "mappings": [
       {
           "sourcePath": "$.id",
           "targetVariable": "entityIdSourceSystem"
       },
. . .
```

This way, the Identity Provisioning service:

- Will provision all users that don't belong to any groups.
- Will **not** provision anything from group *Marketing_Cloud*.
- Will provision all users that belong to any group different than *Marketing_Cloud*.

See also: Identity Authentication: Provisioned Users Are Duplicated in SAP Jam or SAP Work Zone [page 3095]

Identity Authentication: Send activation email in users preferred language

When provisioning users from SAP SuccessFactors to Identity Authentication, you want the newly created users in Identity Authentication to get the activation emails in their preferred language. For this, you need to configure the transformations of both SAP SuccessFactors (source) and Identity Authentication (target) systems.

Proceed as follows:

1. In your SAP SuccessFactors system, select the *Transformations* tab and add one of the following transformation codes in the user mapping section:

Use `preferredLanguage` attribute	Use `defaultLocale` attribute
`{ "sourcePath": "$.preferredLanguage", "optional": true, "targetPath": "$.locale" },`	`{ "sourcePath": "$.defaultLocale", "optional": true, "targetPath": "$.locale" },`

   Normally, `defaultLocale` is used for onboardees (new hires). If a user in SAP SuccessFactors does not have the `defaultLocale` attribute filled in as desired, you can set it by following the procedure described in KBA: 2395445 - Default Locale | How to control the user language in SuccessFactors
   For more information, see: Configuring Email Templates

2. Select the *Properties* tab and append the attribute of your choice (`preferredLanguage` or `defaultLocale`) to the already listed attributes in the value field of the `sf.user.attributes` property.
3. Save your changes.
4. In your Identity Authentication target system, select the *Transformations* tab and add the following transformation code in the user mapping section for `"locale"`:

```
{
   "sourcePath":"$.locale",
   "optional":true,
   "targetPath":"$.locale",
   "functions":[
   {
      "function":"toUpperCaseString"
   },
   {
      "function":"substring",
      "beginIndex":0,
      "endIndex":"2"
   }
  ]
},
```

   This is needed because language formats differ in SAP SuccessFactors and Identity Authentication.
   For more information, see: 2269945 - List of Supported Languages with Locales for SuccessFactors
5. Save your changes.
6. Run a provisioning job in SAP SuccessFactors source system.
7. Verify that the newly created users in Identity Authentication receive the activation emails in their preferred language.

For other solutions on how to send welcome e-mails in a predefined language, see SAP KBA 3035482

Job Is Successful but Statistics Are Missing

Problem

1. You've added a source and a target system in your administration console for SAP Cloud Identity Services.
2. You've started a provisioning job (*Read* or *Sync*).
3. In the *Job Logs* section, you see that the job has finished successfully. However, the *Statistics* table is empty.

Reason	Solution
You have set a filter in the *Properties* tab that does not match any of the entities in the system.	Change the filter so that it matches at least one entity.
No changes have been made in the source system since the last provisioning job.	You'll see the statistics if you change at least one entity in the source system, and then run a new job.
Your source system is SAP SuccessFactors, and the technical user is missing *Export* permissions to read employee data from the SAP SuccessFactors system.	Obtain *Export* permissions for SAP SuccessFactors. To learn how, see: Permissions.

Microsoft AD: Nested Groups Are Not Resolved in the Target System

Problem

Your source system is Microsoft Active Directory and you want to provision all users and groups, along with all group members. However, some of the groups contain "nested" subgroups. All these groups and subgroups contain users.

When the provisioning job is done, in the target system only the direct members of the parent group are provisioned. The user assignments from all subgroups are missing.

Reason

Some target systems don't support nested groups (group structures). Such target systems are:

- SAP Jam Collaboration
- Identity Authentication

Therefore, if you provision group structures to one of these target system types, the user assignments from the subgroups will be missing.

Solution

Luckily, you can enable recursive reading of such group structures. All you need is to set the following two properties:

- `ad.group.flatten` = **`true`**
- `ldap.group.filter` = **`(cn=<parent_group>)`**

As a result, in the target system all users will be written in one parent group, which has the same name as the source parent group.

Example

- Let's say, in your Microsoft AD system there is a parent group *"Canteen"* that contains nested subgroups. If you set `ldap.group.filter` = **`(cn=Canteen)`**, the Identity Provisioning will resolve all the direct users and groups of *"Canteen"*, along with all the users of its subgroups (and their subgroups). In the target system, all users will be written in one parent group named also *"Canteen"*.
- If you have multiple parent groups (for example, *Canteen*, *Finances* and *Support_Team*) that contain nested subgroups, you can set the filter property like this:

```
ldap.group.filter = (|(cn=Canteen)(cn=Finance)(cn=Support_Team))
```

Microsoft Azure: Emails Are Missing

Problem

When you add Microsoft Active Director as a source system and you run a provisioning job, some of the populated users in the target system have missing e-mail addresses. This issue can be a provisioning blocker for target systems that require a mail attribute (for example, Identity Authentication).

Reason

> **Note**
>
> The image/data used is from SAP internal systems, sample data, or demo systems. Any resemblance to real data is purely coincidental.

There can be two reasons for such a behavior:

- Some users don't have a Microsoft Office 365 license.
- They have this license but their `mail` attribute is not tied to Microsoft Exchange Online, which leads to lack of synchronization between user data on the on-premise Microsoft AD and Azure AD.

In both cases, after the provisioning job, the value of the user's emails attribute in both the source and the target system is *null*. For example:

User details in the source system	User details in the target system

`"displayName": "Johnny123",`
`"givenName": "John",`
`"surname": "Smith",`
`"mail": "null",`
`"userPrincipalName":`
`"JohnSmith",`
`...`
	`"displayName": "Johnny123",`
	`"givenName": "John",`
	`"familyName": "Smith",`
	`"emails": null,`
	`"userName": "JohnSmith",`
	`...`

Solution

You need to map the user e-mail to another attribute, for example `userPrincipalName`. In the source transformation, modify the following JSON lines:

```
/* Remove this part:
    {
        "sourcePath": "$.mail",
```

```
        "targetPath": "$.emails[0].value"
    }, */
// Change the following part to:
    }
        "sourcePath": "$.userPrincipalName",
        "targetPath": "$.emails[0].value"
    },
    {
        "sourcePath": "$.displayName",
        "optional": true,
        "targetPath": "$.displayName"
    },
...
```

Result

User details in the target system (after modifying the transformation):

```
"displayName": "Johnny123",
"givenName": "John",
"familyName": "Smith",
"emails": null,
"userName": "john.smith@acme.com",
...
```

Related information: Microsoft AD (Source System)

Multiple Users from a Source System Are Created as One in the Target

The problem described on this page illustrates a scenario involving SAP SuccessFactors (SFSF) as a source system and Identity Authentication as a target one. However, this issue may occur to any source system while provisioning users to the following targets:

- Identity Authentication
- SAP Analytics Cloud
- SAP Datasphere

> **Note**
>
> The problem will happen only if you use *email* as a unique attribute in the target.
>
> If you have modified the *email* to **NOT** be unique (for example - in Identity Authentication), **ignore this section!**

Problem

1. You've added SAP SuccessFactors as a source system and Identity Authentication as a target one.
2. You haven't set any specific filters. Thus, the provisioned users are expected to be the same number as the ones in SFSF..
3. You've started a **Read Job**, which finishes successfully.
4. When you check the populated entities, though, you notice that the users created in Identity Authentication are fewer (or significantly fewer) than the original number of users in SFSF.

Reason

The user e-mail is optional for SFSF (it might not be unique). However, it's mandatory and unique for Identity Authentication.

Therefore, it's very likely that some users have **duplicate** e-mail addresses in SFSF, which causes the improper provisioning. The Identity Provisioning service stores a **relation** for every source entity and its corresponding target entity. When multiple users have duplicate e-mails and the provisioning is triggered, the stored relation is "**many** source entities that point to **a single** target entity".

> → Remember
>
> Before you start provisioning SFSF users to Identity Authentication for the first time, make sure that all users **have** e-mails, which are **unique**, even if these entities are only meant for test purposes. This way, you won't encounter the problem described in this guided answer.

Prerequisites

To solve the occurred problem, you first need to edit the e-mails of all affected SFSF users. You can do this in two ways - manually or programmatically.

- Manually
  Edit the e-mail of each affected user in SFSF.
- Programmatically
  You have to modify the SF source system transformation so as to create a unique e-mail address for each conflicting user during the next **Read** Job.
  - If you use SFSF source system with SAP SuccessFactors Workforce SCIM API (SF API version 2), follow the procedure in KBA: 3308246
  - If you use SFSF source system with SAP SuccessFactors HCM Suite OData API (SF API version 1), follow the procedure below:
    To do this, enhance the default email mapping:

```
{
  "sourcePath": "$.email",
    "targetPath": "$.emails[0].value"
 },
```

    to:

```
{
   "sourcePath": "$.email",
   "optional": true,
   "targetPath": "$.emails[0].value"
},
{
   "condition": "!($.email EMPTY false) || $.email == '' || $.email == ' '
|| $.email == 'user123@companyname.com' ",
   "sourcePath": "$.userId",
   "targetPath": "$.emails[0].value",
   "functions": [
      {
        "type": "concatString",
        "suffix": "@sap-test.de"
      }]
}
```

    What does the condition above mean?

- `$.email == !($.email EMPTY false)` means that the user lacks an email attribute.
- `$.email == ''` means the SFSF user has no business e-mail set.
- `$.email == ' '` means the SFSF user has a space character set as an e-mail address.
- `$.email == 'user123@companyname.com'` means that multiple users in SFSF have a business e-mail *user123@companyname.com*, which results in errors in the Identity Authentication target system. This email address is just an example! In your condition, the listed emails can refer to your actual company domain, or to another non-existing one.

> ### Note
>
> **@sap-test.de** is the recommended test domain approved by SAP for customer dummy (testing) requests. Thus, please use only this one for the *concatString* function! If you use another non-existing domain, it might get blocked for security reasons.

> ### Note
>
> Bare in mind that the condition is case sensitive! Thus, make sure that all SFSF e-mails contain low-case letters only. If some of them contain capital letters, you can either go and manually change them to low case, or list all emails like that in the condition.

Using the code above, each provisioned user will be created in Identity Authentication with a unique e-mail in format **`<userId>@sap-test.de`**. Users that have valid unique business e-mails set in SFSF will be provisioned with the same valid e-mails in the administration console for SAP Cloud Identity Services.

Solution

After you have executed the prerequisite steps, follow the final steps below:

1. Reset the Identity Authentication target system. To do this, select the system and choose *Edit* > *Reset*. This way, all previous entity relations will be deleted. System configurations and all existing provisioned entities, along with their authorizations, will be preserved.
2. In the *Properties* tab, add a standard property `ips.delete.existedbefore.entities` and set it to *true*. This property provides correct behavior if a user is later deleted from SFSF. If this happens, during the next **Read/Resync** Job, the Identity Provisioning will recognize this user as **"existed before but now deleted from the source system"** and will delete it from the target system as well. To learn more, see List of Properties.
3. Add a standard property `scim.user.unique.attribute` and set it to **`userName`**. This way, if the Identity Provisioning reads two different users with the same e-mail, it will try to resolve the second one by *user name*. Since these two users have different user names, this second entity will fail with an error for duplicate e-mail. You will see it in the *Job Logs*. This approach is safe because it won't affect the successfully created users in the target system, and will only prompt you to edit the SFSF e-mails of newly added conflicting users.
4. Now open the SAP SuccessFactors source system.
5. Start a new **Read** Job.
6. As a result, all source entities will be read and then created/updated in the Identity Authentication target system. The Identity Provisioning will store a new unique relation between each source entity and its corresponding target entity.

Provisioning Jobs Take Extremely Long Time

Problem

1. You have added a source and a target system in your administration console for SAP Cloud Identity Services.
2. You start a **Resync** job.
3. The job is taking an extremely long time to execute, even if it finishes successfully.

Reasons

There can be a few reasons for such a behavior, depending on your type of system, or the location of your business applications and the data centers where your tenants reside.

Choose the option that applies to your scenario:

Your source system is SAP SuccessFactors, SuccessFactors LMS, Microsoft AD, or Identity Authentication

- **Overview Information**
  Your source system type is one of the following:
  - SAP SuccessFactors
  - Microsoft Active Directory
  - SAP Cloud Identity Authentication

  These three systems support **delta read** provisioning.
  Delta read is a concept for optimizing the amount of data retrieved from the source system. Delta read is much faster than **full read** as only modified data is read from the source system and triggered to the target one. Modified data means: new entities, deleted entities, and updates on existing entities. This is useful when you schedule the read jobs. For example, every 10th job can be a full read, and the rest will be delta read jobs, which can significantly improve the execution time of the provisioning job.

SAP SuccessFactors **SAP SuccessFactors LMS**	By default, delta read is **enabled** for these systems. That means, the first ever **Read** job will be a full one, meaning - reading all source entities. If it's successful and you have set schedule jobs to be executed on a certain time interval, the next executed job will be a **delta** one (reading only modified data).
Microsoft Active Directory **SAP Cloud Identity Authentication**	By default, delta read is **disabled** for these systems. That means, the first ever **Read** job will be a full one, meaning - reading all source entities. If it's successful and you have set schedule jobs to be executed on a certain time interval, the next executed job will be again a **full** one (reading again everything from the source system). To activate delta read, go to the *Properties* tab, and set the following property: `ips.delta.read=enabled`

- **Possible Reason**
  Instead of **Read**, you might have run a **Resync** job. What's the difference?
  - A **Read** job will read all users, but provision only the **delta** to the target systems. That means, only newly created, updated, or deleted users in the source system be respectively created, updated, or

deleted in the target systems by the Identity Provisioning service. When the first full provisioning is successful, the next read jobs will work in **delta read** mode.

- A **Resync** job would read all users from the source system (based on the configured filter), and provision all of them to the target system, even if previous provisioning job already did this and most of the users have not changed since the previous job. It is like executing a provisioning job for the first time. If some users were already provisioned before, the Identity Provisioning service will perform update for them.
  In addition, if the users are already present in the target systems by other means, and not from an Identity Provisioning job, the service will try to create them - which operation would fail - and then the service will fetch every user from the target and update it. This additional operation could further slow down the job execution.

Recommendation:
We recommend that you use a **Resync** job when manual changes have been made in the target systems, which the Identity Provisioning is unaware of. In this case, the service can sync all users data from the source to the target with a **Resync** job. In all other cases, it is sufficient to execute **Read** jobs only.
Solution
Next time, manually execute a **Read** job. Make sure the delta read is set for your system!
If the result is sufficient to you, you can schedule your read jobs in order to be automatically executed on a certain period of time.
Recommendation:
We recommend that you enforce full reads once in a while if your provisioning system is set in **delta read** mode. To achieve this, you need to set up the following source system property like, for example:
`ips.full.read.force.count=10`
This will result in alternating full reads after every **10 delta reads** are performed. This property only impacts scheduled runs. Manually triggered runs are ignored.

Your Source System is of Another Type

Reason

Your source and target business applications are placed in one location, and the data center where your SAP Cloud Identity Services tenant resides is in a very distant region.

For example, your SAP Cloud Identity Services tenant is located in Europe, while your business application instances are located in the US. This makes the connection between the system significantly slower.

Solution

- Migrate your tenants, or request new ones.

  > ### Note
  >
  > The provided solution is only applicable to **bundle tenants**!

- You can migrate your SAP Cloud Identity Services tenant to another data center - where your source system is. To request that change, create an incident to component **BC-IAM-IDS**. You will receive details about the procedure, or you can request a new SAP Cloud Identity Services tenant to be created in the region where your source system is.
- Currently you cannot migrate your SAP Cloud Identity Services tenants. But you can request a new one for the region where your source system is. The old SAP Cloud Identity Services tenant can be deleted or preserved - depending on whether you will need it in future for business applications whose instances are located in the initial region. To request that change, create an incident to component BC-IAM-IPS.

Provisioning Job Fails for One of Multiple Target Systems

Problem

1. You added one source and multiple target systems in your administration console for SAP Cloud Identity Services.
2. You started a provisioning job.
3. The job fails for one of the multiple target systems.

Reason

Provisioning to one of the target systems may fail due to exception.

Provisioning Job Fails with Exception

Problem

1. You added one source and multiple target systems in your administration console for SAP Cloud Identity Services.
2. You started a provisioning job.
3. The job fails with one of the following exceptions:

- *Caused by: Timeout exception has occurred while processing entity for <URL of the source system>*
- *Caused by: Timeout exception occurred*

Reason

Those exceptions occur when provisioning to one of the multiple target systems fails, for example, due to connectivity issues.

Solution

Start the provisioning job again.

Not All Users are Provisioned to the Target Systems

Problem

1. You added one source and multiple target systems in your administration console for SAP Cloud Identity Services.
2. You started a provisioning job.
3. You get one of the following results and you want to know why:
    - Not all users are read from the source system, and yet, their number is greater than the number of users provisioned to the target system.
    - All users are read from the source system but less are provisioned to the target system.

Reasons

Possible reasons when **not all users** are read from the source system, and yet, their number is greater than the number of users provisioned to the target system.

- There are inactive users in the source system. Identity Provisioning only provisions active users.

- Provisioning job is run in delta read mode.
- Connection to the source system fails.

Possible reasons when **all users** are read from the source system but less are provisioned to the target system:

- Provisioning to one of the multiple target systems has failed. See: Provisioning Job Fails for One of Multiple Target Systems [page 3106]
- A full Read job has been run. After that, there were no changes on users in the source system. Next run of a full Read job reads all users from the source system but provisions "0" users to the target due to lack of changes.

Provisioning Job Fails for One of Multiple Target Systems

Problem

1. You added one source and multiple target systems in your administration console for SAP Cloud Identity Services.
2. You started a provisioning job.
3. The job fails for one of the multiple target systems.

Reason

Provisioning to one of the target systems may fail due to exception. See: Provisioning Job Fails with Exception [page 3116]

Scheduled Jobs Are Not Running

Problem

1. In the administration console for SAP Cloud Identity Services, add a source and a target system.
2. Open the source system, then the *Jobs* tab.
3. Choose *Schedule* and enter a number (of minutes).
4. Choose *OK* and wait for a while.
5. Open *Job Logs* to check the job's status. However, you don't even see the job, which means it's not started yet.
6. No matter how long you wait, the job does not appear on the screen.

Solution

After you've set up a schedule job, you must always choose the **On** button to actually start the job. This is a one-time manual step.

Once you've selected *On*, the job starts automatically, according to the predefined period of time.

I Still Have Problems with Job Scheduling

Your problem requires deeper analysis by SAP. Please create an incident for component BC-IAM-IPS.

In the incident, specify also:

- The IDs of your global account and subaccount, or your bundle tenant.
- To which region your account/tenant belongs. You can find it in this list: (Discovery Center) Identity Provisioning -> *Service Plan*

Before you create a support case, read:

2654164 - What to consider before opening a Support Ticket for SAP Business Technology Platform

For contract-related questions, read:

1660069 - How to contact SAP Contracts Department for questions or issues

SAP IBP: Avoid Restoring Decimal Format, Time Zone and Language to Their Defaults

When updating a user in SAP Integrated Business Planning (or any other SAP S/4HANA Cloud Public Edition-based system), you want to make sure that only specified attribute values are updated, while others remain unchanged.

> **Note**
>
> In addition to SAP IBP, this troubleshooting information is also applicable to SAP S/4HANA Cloud Public Edition, SAP BTP ABAP environment , SAP Marketing Cloud and SAP Market Communication for Utilities.

Problem & Reason

Currently, if a user is created in SAP IBP target system and later on you update the user without providing values for attributes in the User node (such as, decimal format, time zone and language), these values are restored to their defaults.

To reproduce this behavior, proceed as follows:

1. Run a provisioning job from a source system (for example, Identity Authentication) to SAP IBP. Users are created with the default values:
   - **Decimal Format**: 1.234.567,89
   - **Time Zone**: No value
   - **Language**: English United States
2. In SAP IBP, modify the attributes of a given user:
   - Change the **Decimal Format**, for example to **`1,234,567.89`**
   - Select a **Time Zone**, for example **`Eastern Europe`**.
   - Leave the English United States **Language** as-is or change it.
3. In Identity Authentication, update the user, for example change its last name only.
4. In Identity Provisioning, run a provisioning job.
   As a result, the user's last name in SAP IBP is updated. Since no values are provided for decimal format, time zone and language, in SAP IBP expect the following:
   - **Time Zone and Decimal Format** are restored to their default values.
   - **Language** is removed.

Solution

To ensure that only specified attribute values are updated, while others remain unchanged, you need to modify the SAP IBP target transformation.

1. Add the bolded attribute mapping at the end of the user entity as follows:

```
{
                "condition": "$.active == false",
                "constant": "true",
                "targetPath": "$.user.lockedIndicator"
            },
        {
                "constant": "02",
                "targetPath": "$.user.actionCode"
            },
            {
                "constant": "01",
                "targetPath": "$.user.actionCode",
                "scope": "createEntity"
            }
        ]
    },
```

   - `"constant": "02"`
     This means that Action code 02 (representing update operation) is applied for attributes in the User node.
   - `"constant": "01"`
     This means that Action code 01 (representing create operation) is applied for attributes in the User node. The scope of this mapping is createEntity, which means that the entity attribute is only processed during creation.
2. Modify some user's attributes and run a provisioning job.

As a result, only the specified attribute values are updated. Others for which no value is provided remain unchanged.

SAP SuccessFactors: Some Entities are Unexpectedly Deleted from Target System

Problem

1. You set SAP SuccessFactors as a source system and have enabled **delta read** for it.
2. You set a target system (any kind).
3. You have been running manual or scheduled jobs for some time. When a job finishes with an error, you have noticed that no entities have been deleted.
4. If you run a new job after **October 1, 2020**, you see that some or many entities are unexpectedly deleted.

Reason

Expected behavior:

If a provisioning job has successfully read all entities from a source system (with **delta read** enabled) but fails due to problems with provisioning of some entities, the entities that have to be deleted **should be deleted**.

Actual behavior before October 1, 2020

Entities that should have been deleted were **not deleted** during delta read. This behavior was incorrect but is now fixed.

> **Note**
>
> If a provisioning job was successful, this problem would not have been observed. Read entities were created/updated, and unread entities were deleted from the target.

Solution

Effective **October 1, 2020** this bug has been fixed in the Identity Provisioning service, and now the default behavior happens. That's why you observe "unusual deletion" of entities in your target system, which is how it's supposed to be.

Related Information

If you want the service to read only part of the entities in your source system but to **save** the unread ones from deletion in the target, see: Entities Should Not Be Deleted in the Target System [page 3080]

Users Are Not Being Assigned to Groups in the Target System

Problem:

After running a provisioning job, some users are not assigned to groups in the target system.

Reason:

By default, Identity Provisioning provisions users and groups separately in one job. User assignments are provisioned as part of the group membership.

Identity Provisioning cannot assign a user to a group if:

- The user failed or was skipped during provisioning.
- The user was not read from the source system.

This applies even if the user already exists in the target system (for example, created manually).

Solution:

To ensure users are assigned to groups:

- Verify that all users are successfully provisioned by Identity Provisioning.
- Check the job logs for failed or skipped users. If issues exist, follow Manage Provisioning Job Logs [page 3010] to identify the reason for failures and download error logs and skipped entity logs for the job.

Parent topic: Troubleshooting Provisioning Issues [page 3070]

Related Information

Administration Issues [page 3071]
Error Messages [page 3111]
Entity Deletion Issues [page 3121]

Additional Information [page 3130]
Handling Specific Attributes [page 3137]

1.12.15.3 Error Messages

[Job Logs] Error statusCode: 401

Problem

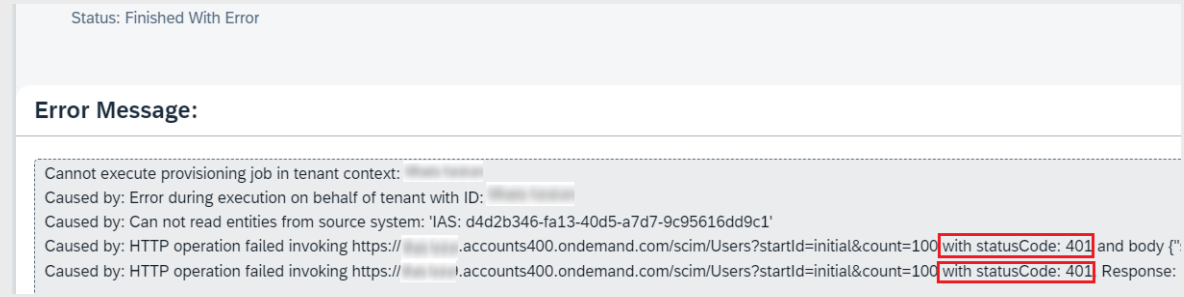
1. You added a source and a target system in the administration console for SAP Cloud Identity Services.
2. You started a provisioning job (**Read** or **Resync**).
3. In the *Job Logs* section, you see that the job has finished with error.
4. You find an error with status code **401** - either in the main *Error Message* filed, or in the details of a particular failed entity.

The error looks similar to the following ones:

- In the main *Error Message* field, you find: *"HTTP operation failed invoking <system_URL> with statusCode: 401"*
- In the logs of a failed entity, you find: *"OAuth request failed with status: 401 and body: {"error":"unauthorized","error":"description":"Bad credentials"}"*

Example

1



Example

2

Reason

This error mostly appears when the credentials of the relevant system are wrong. However, the reason can also be incorrect names or values of the following properties:

- `Authentication` (*BasicAuthentication*, *NoAuthentication*, *ClientCertificateAuthentication*)
- `ProxyType` (*Internet*, *OnPremise*)
- `Type` (*HTTP*, *LDAP*, *RFC*)
- `URL`

> **Note**
>
> These properties and values are case-sensitive.

Bear in mind the place you've found the error! That means:

- If the **401** error appears in the main *Error Message* field - the wrong credentials or incorrect property values are in the source system.
- If the **401** error appears in the details of a failed entity - the wrong credentials or incorrect property values are in the target system.

Solution

Go to the *Properties* tab of the relevant source or target system and correct the credentials (`User` and `Password`), or the other affected properties.

Error statusCode: 432

Problem

1. You use SAP SuccessFactors as a source system, and Identity Authentication as a target one.
2. You run a provisioning job between the two systems.
3. In the administration console for SAP Cloud Identity Services you see that all users have been successfully created or updated.
4. Nevertheless, in the *Job Logs* statistics, the job has finished with error due to some failed users. You can find the following error message for the relevant users:

statusCode: 432, Response: User is provisioned to Identity Authentication but its UUID is not successfully updated back on SAP SuccessFactors.

Reason

Effective February 16, 2021, Identity Provisioning supports the SAP SuccessFactors attribute `sapGlobalUserId`. That means, your provisioning logic will be as follows:

1. You start a new Read or Resync job.
2. The Identity Provisioning service reads the users from SAP SuccessFactors and writes/updates them in Identity Authentication.
3. The Identity Authentication generates a `UUID` for each newly created user.
4. During the runtime of the same job, the service will also send a request back to SAP SuccessFactors, which aims to update all users whose `sapGlobalUserId` attribute is either *null* or already exists.

- When the value is *null* - Identity Provisioning reads the `sapGlobalUserId` attribute of every user in SAP SuccessFactors, and if for some user this attribute is missing, the service will update it with the value of the `userUUID` attribute in Identity Authentication.
- When the value already exists - Identity Provisioning updates the existing `sapGlobalUserId` attribute value of a given user with the new value. Typically, this could happen if a user is provisioned from SAP SuccessFactors to Identity Authentication and later on deleted from the target system. After running a Resync Job, the deleted user is created again in Identity Authentication. A new `userUUID` is generated for this user and Identity Provisioning updates the old value in SAP SuccessFactors.

So, if for some reason the update request to the source system fails, you will get this general *432* error code.

Reason

For your SAP SuccessFactors system, send the following REST API request:

```
https://<system_host>.successfactors.eu/odata/v2/Users/('user_D')?
$expand=personKeyNav/userAccountNav&$select=personKeyNav/userAccountNav/
sapGlobalUserId,personKeyNav/userAccountNav/accountUuid, personKeyNav/
userAccountNav/username, &$format=json
```

In addition to username, this request returns also the following specific attributes: `accountUuid` and `sapGlobalUserId`

- `accountUuid` – this attribute is used by Identity Provisioning to update the `userUuid` attribute of a user in the SAP SuccessFactors system. The value of `accountUuid` must not be *null* or invalid! Thus, make sure all users have a valid `accountUuid` value.
- `sapGlobalUserId` – this attribute stores the value of `userUuid`. Identity Provisioning will try to provision the `userUuid` if the value of `sapGlobalUserId` is *null* (missing) or if it already exists. Thus, the update of `userUuid` will be completed successfully when the `sapGlobalUserId` attribute has/gets a value, or when the old value is updated with a new one.

User 'v4admin' Causes Job Error

Problem

1. You've added SAP SuccessFactors as a source system and Identity Authentication as a target one.
2. You've started a **Read** Job, which finishes with error.
3. In the *Job Logs* section, you see that the error mentions user "*v4admin*" and contains the following error message:

Property lastModifiedDateTime is not available for {"lastName":"System","empId":null,"lastModifiedDateTime":null,...,"userId:"v4admin"...}

Reason

1. Open the SAP SuccessFactors source system and choose *Properties*.
2. Modify the `sf.user.filter` property, extending its current value by adding the following string at the end: `and userId ne 'v4admin'`
3. Save your changes.
4. Run a new Read Job.
5. The provisioning job should now finish successfully.

> ### Example
>
> If the previous value of your `sf.user.filter` property is, for example: *location eq 'US'*, then you should change it to: **`location eq 'US' and userId ne 'v4admin'`**.

Invalid Property Names: User/usersSysIdOfSecondaryAssignmentsItemNav

Problem

A provisioning job that reads users from SAP SuccessFactors fails with the following error:

*Response: {"error" : {"code" : "COE_PROPERTY_NOT_FOUND", "message" : {"lang" : "en-US", "value" : "***[COE0021]Invalid property names: User/usersSysIdOfSecondaryAssignmentsItemNav***. Please check the property name in Admin Center > OData API Data Dictionary or entity metadata. Ensure there were no data model changes that removed this field, and please execute a refresh metadata to ensure the cache is not corrupted."}}}*

Reason

The attribute `usersSysIdOfSecondaryAssignmentsItemNav` is present when options *Concurrent Employment* and *Global Assignment* are enabled in the SAP SuccessFactors system. The default Identity Provisioning configuration considers that those options are enabled.

If they are not enabled, the above mentioned error appears. To solve this problem, follow the steps below.

Solution

1. Open the administration console for SAP Cloud Identity Services and navigate to the SAP SuccessFactors source system.
2. Go to the *Properties* tab and set the following attributes:
   - `sf.user.attributes` = *userId,username,status,email,lastName,firstName,lastModifiedDateTime,personKeyNav*
   - `sf.user.attributes.expand` = *personKeyNav,personKeyNav/userAccountNav*
   - `sf.user.filter` = *status in 'active','active_external','active_external_suite'*
3. Go to the *Transformations* tab and remove the following condition from the transformation (everywhere you find it):
   `"condition": "!($.usersSysIdOfSecondaryAssignmentsItemNav.results.length() > 0)",`
4. Run a new provisioning job.

Provisioning to S4 failed: [Employee ONPREMXXXXX: Insertion or modification with errors

Problem

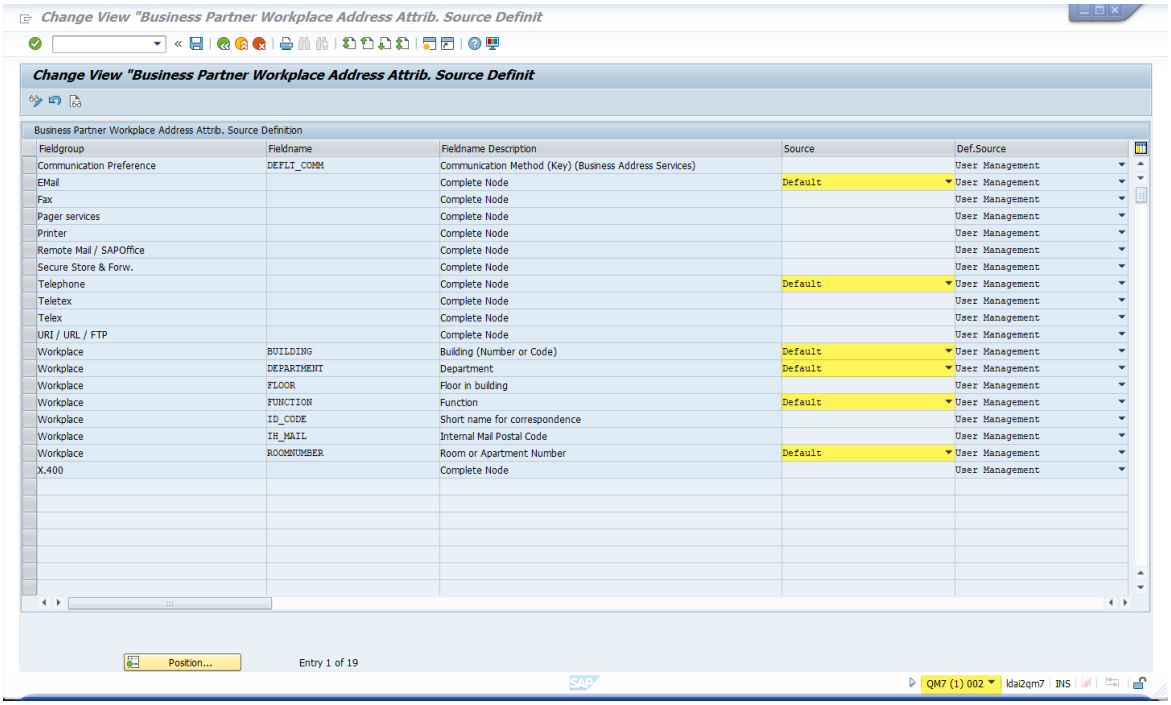
1. You configure an SAP S/4HANA on-premise system (in SAP ABAP).
2. You add a target system for it in the administration console for SAP Cloud Identity Services.

3. You add a source system and connect it to the target S/4HANA On-Premise target system.
4. Run a provisioning (Read) job.
5. The job fails with an error of type: *Provisioning to S4 failed for the following reason(s): [Employee ONPREMXXXXX: Insertion or modification with errors.]*

The error message contains one of the following reasons (or similar):
[BP 0000019334: Change of field EMAIL not permitted using data source BP], [ONPREM16772: Error while inserting / updating workplace address.],
[BP 0000019334: Change of field WORKPLACE - BUILDING not permitted using data source BP], [ONPREM16772: Error while inserting / updating workplace address.],
[BP 0000019334: Change of field WORKPLACE - FUNCTION not permitted using data source BP], [ONPREM16772: Error while inserting / updating workplace address.],
[BP 0000019334: Change of field WORKPLACE - DEPARTMENT not permitted using data source BP], [ONPREM16772: Error while inserting / updating workplace address.],
[BP 0000019334: Change of field WORKPLACE - ROOMNUMBER not permitted using data source BP], [ONPREM16772: Error while inserting / updating workplace address.],

Problem

1. Now go to your S/4HANA on-premise system and open transaction */nSPRO*.
2. Then navigate to *Cross-Application Components* > *SAP Business Partner* > *Addition Customizing for SAP Business Partners* > *Workplace Address Attribute Source Definition*.
3. (Optional) You can reach to the same place via transaction */nSM30*, and then for Table/View enter: **TBZ_V_EEWA_SRC**
4. Your table looks similar to this:



Reason

Your user attributes are of type *Default*. But they should be of type *Business Partner*.

Solution

1. You have opened the table view in *"Workplace Address Attribute Source Definition"* of your S/4HANA on-premise system.
2. Now go to column *Source*, and by the drop-down menu, change *Default* to *Business Partner*.

Provisioning Job Fails with Exception

Problem

1. You added one source and multiple target systems in the administration console for SAP Cloud Identity Services.
2. You started a provisioning job.
3. The job fails with one of the following exceptions:
   - *Caused by: Timeout exception has occurred while processing entity for <URL of the source system>*
   - *Caused by: Timeout exception occurred*

Reason

Those exceptions occur when provisioning to one of the multiple target systems fails, for example, due to connectivity issues.

Solution

Start the provisioning job again.

Provisioning Job Fails with Error Unsuccessful Provisioning of the Entity

Problem

1. You added a source and a target system in your administration console for SAP Cloud Identity Services.
2. You started a provisioning job (**Read** or **Resync**).
3. In the *Job Logs* section, you see that the job has finished with error.
   In the details of the failed entity, you find the error *"Unsuccessful provisioning of the entity"*.

Solution

Your problem requires deeper analysis by SAP. Please create an incident for component BC-IAM-IPS.

In the incident, specify also:

- The IDs of your global account and subaccount, or your bundle tenant.
- To which region your account/tenant belongs. You can find it in this list: (Discovery Center) Identity Provisioning -> *Service Plan*
- The Job log with information about the failed entity

Before you create a support case, read:

2654164 - What to consider before opening a Support Ticket for SAP Business Technology Platform

For contract-related questions, read:

1660069 - How to contact SAP Contracts Department for questions or issues

Display All Failed or Skipped Entities

Your provisioning job fails or results in skipped entities. Users and groups are not provisioned to the target system. In the *Job Execution Details* screen, you see a maximum number of ten failed entities (although they might be more) and the content of each failed or skipped entity is hidden, that is: *content=<hidden content>*.

To further investigate the issue, you must enable tracing and logging.

1. Select the *Properties* tab of your source system and set the `ips.trace.failed.entity.content`, `ips.trace.skipped.entity` and `ips.trace.skipped.entity.content` properties to *true*.
2. Run again the provisioning job.
3. Navigate to *Provisioning Logs* > *Job Logs* and select the job that finished with error.
4. Choose the *Download* icon and select the respective log: *Download All Error Logs for This Job* or *Download All Skipped Entity Logs for This Job*.

Now, when you open the log file, you see all failed and skipped entities and their content (attributes). For example, if the failed entity is a group, you see its *displayName* and *members*.

For more information, see:

Monitor Provisioning Job Logs [page 2982]

Manage Provisioning Job Logs [page 3010]

List of Properties [page 80]

How to Resolve 409 Error in SAP Analytics Cloud Target System

You want to know how to resolve a 409 error when using Identity Provisioning for provisioning users to SAP Analytics Cloud target system.

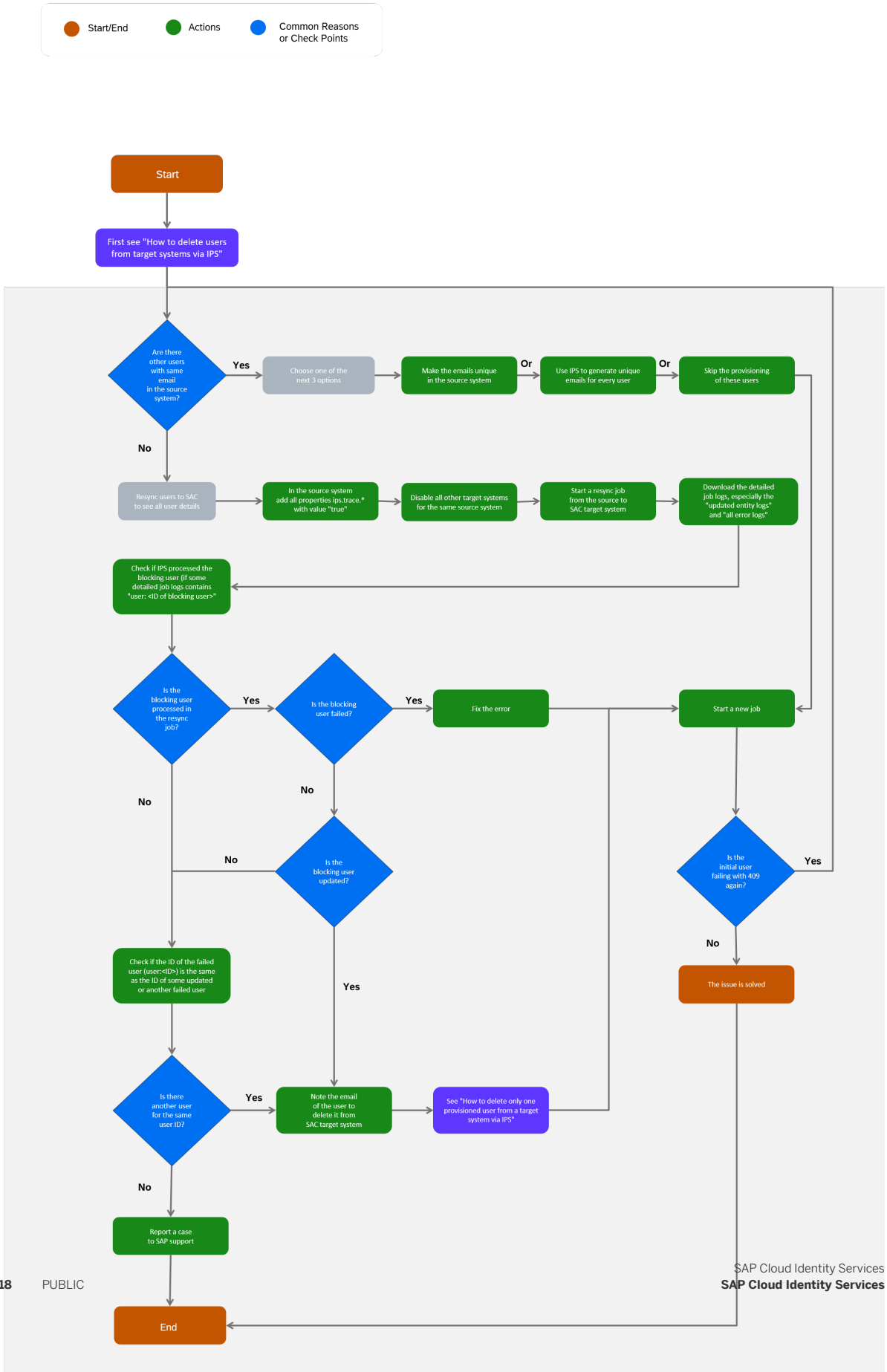
First, determine whether user creation or user update fails with HTTP 409. A creation attempt fails, if the job error log for the user contains the error: `Entity create failed with "Entity already exists"`, but the entity was not found by the unique attribute configured. An update attempt fails, if the job error log for the user contains the following path: `/api/v1/scim/Users/<SOME USER ID>`.

In both cases - the create and update failure scenarios, the underlying issue is that another user in SAP Analytics Cloud already uses the same email address. When this is the case, the error log includes a message like: `Email address is already used for <ID of the blocking user>`.

Before proceeding with the troubleshooting step, make sure to note the following:

- Whether the create or update action failed
- The ID of the blocking user (the user who owns the conflicting email)

Solving error 409 in Identity Provisioning for SAC target system

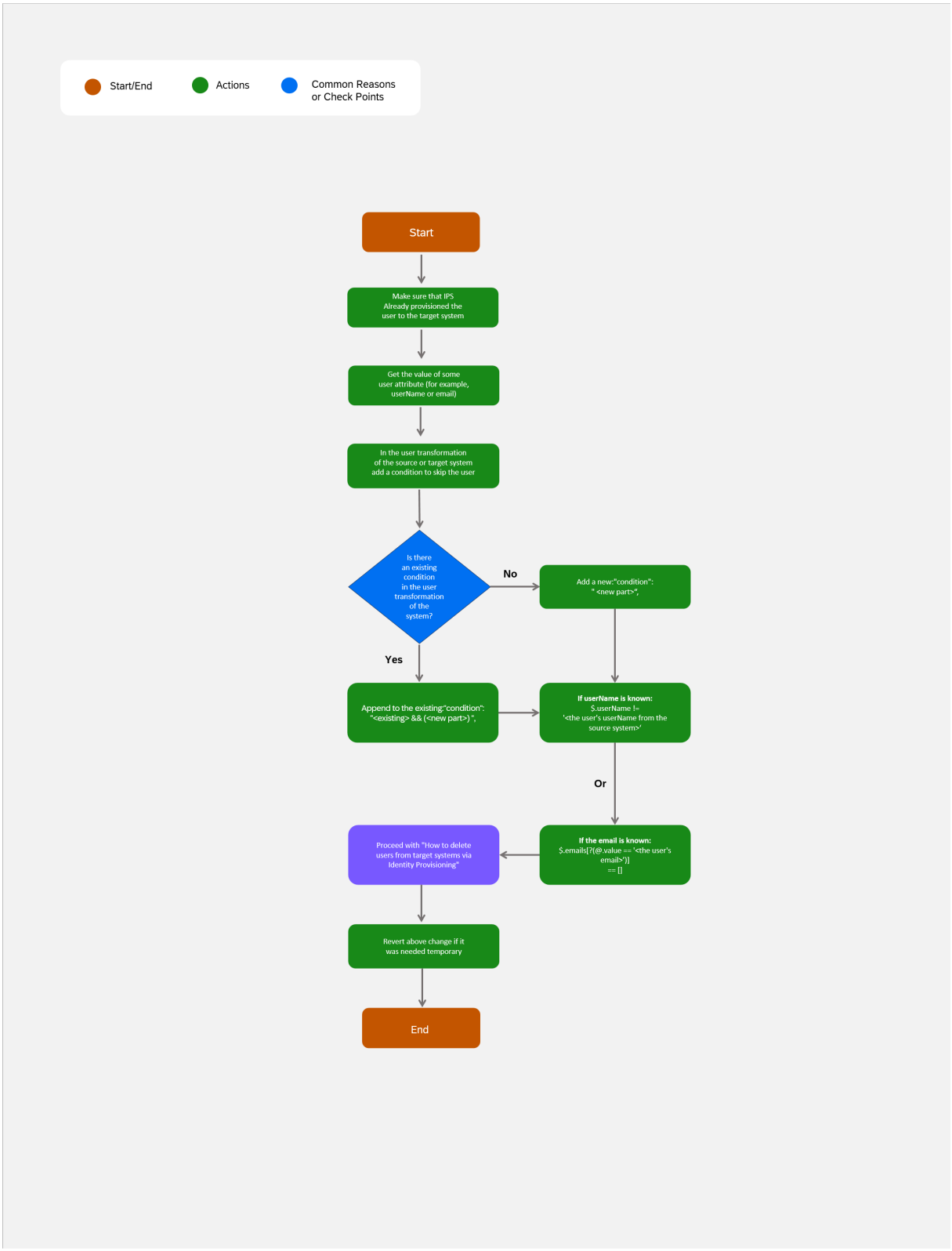


- Manage Deleted Entities [page 2463]
- List of Properties [page 80]
- Enable and Disable Systems [page 2419]
- Resync Provisioning Job [page 2476]
- Manage Provisioning Job Logs [page 3010]
- Start and Stop Provisioning Jobs [page 2472]
- Getting Support [https://help.sap.com/docs/cloud-identity-services/cloud-identity-services/getting-support?version=Cloud]

How to Delete One Provisioned User From a Target System

You want to know how to delete only one provisioned user by Identity Provisioning from a given target system.

Deleting one provisioned user from a target system via Identity Provisioning



- Transformation Expressions [page 391]

- Transformation Conditions [https://help.sap.com/docs/cloud-identity-services/cloud-identity-services/condition-examples-in-transformations?version=Cloud]
- Working with Graphical Editor [page 2440]

Parent topic: Troubleshooting Provisioning Issues [page 3070]

Related Information

Administration Issues [page 3071]
Job and Transformation Issues [page 3077]
Entity Deletion Issues [page 3121]
Additional Information [page 3130]
Handling Specific Attributes [page 3137]

1.12.15.4 Entity Deletion Issues

Learn how to troubleshoot the most frequent reasons for deletion of users or groups from a target system by the Identity Provisioning.

In the scenarios described below, the entity is already provisioned by the Identity Provisioning via a provisioning job from a source to a target system.

Since the topic is focused on occurrences of entity deletion, it only covers cases where deletion is expected. Deletion is not expected, when the provisioning job:

- has been run in a delta read mode;
- has failed entities for the source system;
- has an irreparable error for the source system;
- has irreparable errors for all target systems, which the job has to provision to.

It is important to clarify that even if an entity existed in the target system before its provisioning for the first time, it could still be deleted by the Identity Provisioning if the property `ips.delete.existedbefore.entities` is set to *true* in the target system. However, if the entity is created in the target system by provisioning in the first place, then the service is allowed to delete it , unless there is a threshold set for deletion of entities from the target or `“skipOperations”:[“delete”]` is present in the transformation of the target system.

Search among the most frequent causes for entity deletion by the Identity Provisioning and learn what should be checked to make sure that this is the situation you run into.

> → Recommendation
>
> For troubleshooting purposes, set the `ips.trace.deleted.entity` property to `true` on the source system. This enables you to download a ZIP archive containing details of all entities deleted during a given job. For more information, see List of Properties [page 80].

The Entity is Deleted From the Source System or Re-Created with a New ID

Access the audit logs for the source system and check for the deletion of the entity and eventually its re-creation with a new ID. If an entity previously provisioned to the target system was deleted in the source system, the deletion timestamp in the source system should be before the start time of the provisioning job that deleted the entity from the target.

Bellow you can find an exemplary audit log for deleted user from an Identity Authentication source system:

```
category=audit.data-change
crtAccount=<IAS tenant ID>
action=delete
objectType=user
objectId=<the SCIM ID of the deleted user in IAS>
```

For more information, see Audit Logs.

Entity is not Returned from the API of the Source System When a Read or a Resync Job has Performed the Deletion

In the *Job Execution Details* screen, compare the *Read* statistics for the job, which deleted the entity, and the previous job for the same source-target systems pair. If you download all job logs, these statistics are displayed as value of the `numberRead` attribute. Check also if the next provisioning job for the same source-target systems re-created the entity in the target system.

In case of further investigation needed, please contact the corresponding API experts.

Entity Does not Meet the Filtering Criteria Defined in the Source System Anymore

Check if there is a filtering property set in the provisioning source system (`*.user.filter` or `*.group.filter`). Then look through the entity details in the source system itself and evaluate if it meets the defined filter.

You can also check the audit logs from the source system for entity changes that lead to not meeting the filtering criteria. The audit logs should span the period before the provisioning job that deleted the entity was run.

The entity may have temporarily failed to meet the defined filter while the deletion job was running in the target system. For exmaple, a user has became inactive in the source system and then active again after the job which deleted it from the target system has finished.

In the *Job Execution Details* screen, compare the *Read* statistics for the job, which deleted the user or the group, and the previous job for the same source-target systems pair. If you download all job logs, these statistics are displayed as value of the `numberRead` attribute. Check also if the next provisioning job for the same source-target systems has re-created the entity in the target system.

Bellow you can find an exemplary audit log for updated user that was deleted from Identity Authentication source system:

```
category=audit.data-change
crtAccount=<IAS tenant ID>
action=update
objectType=user
objectId=<the SCIM ID of the deleted user in IAS>
payload: […]
//each changed attribute has its “oldValue” and "newValue" displayed
```

The Filter Set in the Source System was Changed and the Entity is no Longer Read by the Identity Provisioning

First, you can check if the filter property set in the source system has been changed. To do so, check the audit logs for any manual changes of the provisioning source system. The audit logs should span the period before the provisioning job that deleted the user.

After that, in the *Job Execution Details* screen, compare the *Read* statistics for the job, which deleted the entity, and the previous job for the same source-target systems pair. If you download all job logs, these statistics are displayed as the value of `numberRead` attribute. Check also if the next provisioning job for the same source-target systems re-created the entity in the target system.

For example, in case your SAP Cloud Identity Services tenant is running on SAP Cloud Identity Services Infrastructure, refer to Access Audit Logs (Audit Log Service in SAP BTP, Cloud Foundry) [page 2967] for more information. The procedure should have been applied before the Identity Provisioning sends audit logs to the Audit Log service.

Bellow you can find an exemplary audit log for updated filter of an Identity Authentication source system:

```
category=audit.configuration
crtAccount=<IPS tenant ID>
action=update
objectType=System
ObjectAttribute.system-old-details
ProvisioningSystem [id=<the ID of the provisioning system>, …, name=<the name of
the provisioning system>, …]
ObjectAttribute.performed-by-user
ObjectAttribute.system-new-details
```

You can see the difference between `system-old-details` and `system-new-details` for exact changes in the source system. The changes in the filter are part of the `“properties={…}”` sections of the old and the new details for the system.

The Entity Does not Meet an Existing Condition in the Read Transformation Anymore

Start by checking what is the condition in the read transformation of the source system. Then check the entity details in the source system itself and if it meets the condition.

You can also check on the *Job Execution Details* screen if the next provisioning job displays the entity as skipped in the statistics of the source system. You can download the job log for skipped entities and search for the entity in the downloaded file. For more information, see Download All Skipped Entity Logs for a Single Job [page 3011].

The entity may have temporarily failed to meet the defined condition while the deletion job was running in the target system. You can see audit logs from the source system for entity changes that lead to not applying to the condition. The audit logs should span the period before the provisioning job that deleted the entity was run.

For example, if a user is skipped, in the downloaded skipped entities job log you will see a similar content:

```
user:<user ID>,
system=<system name>,
skip reason=Entity condition [($.emails EMPTY false) && ($.userName EMPTY false)
&& isValidEmail($.emails[0].value)] is not fulfilled
//the skip reason could contain a different condition, but the sentence is
“Entity condition… is not fulfilled
```

In case you need to check audit logs, because the entity did not meet the condition only while one provisioning job was running, then for Identity Authentication source system, see Audit Logs. The procedure should have been applied, before Identity Provisioning sends audit logs to the Audit Log service.

Find bellow an exemplary content of an audit log for update of a user in an Identity Authentication source system that caused not meeting the defined condition:

```
category=audit.data-change
crtAccount=<IAS tenant ID>
action=update
objectType=user
objectId=<the SCIM ID of the updated user in IAS>
payload: […]
//each changed attribute has its old and new values displayed (“oldValue” and
“newValue”)
```

The Entity Does not Meet an Existing Condition in the Target System Anymore

Start by checking what is the condition in the write transformation of the target system. Then check the entity details in the source system itself. Consider the transformation of the source system and how the values of the attributes are mapped to each `targetPath` there. Then, evaluate if the entity meets the condition set in the target system.

For example, if the user attribute for email in the source system is `email`, and there is a similar mapping in the transformation of the source system:

```
{
“sourcePath”: “$.email”,
“targetPath”: “$.emails[0].value”
}
```

And the condition in the target system using the email is the following:

```
"condition":"$.emails[0].value =~ /.*@domain.com/”,
//this condition checks if the email is on a particular domain
```

Then you should validate that the value of the user's attribute `email` in the source system ends with the specified domain in the condition of the target system.

You can also check if the next provisioning job for the same source-target systems show the user as skipped in the statistics of the target system. You can download the job log for skipped entities and search for the entity in the downloaded file. For more information, see Download All Skipped Entity Logs for a Single Job [page 3011].

The entity may have temporarily failed to meet the defined condition while the deletion job was running in the target system. You can see audit logs from the source system for entity changes that lead to not applying to the condition. The audit logs should span the period before the provisioning job that deleted the entity was run.

For example, if a user is skipped due to this reason, the downloaded skipped entities job log will include similar content:

```
user:<user ID>,
system=<system name>,
skip reason=Entity condition [($.emails EMPTY false) && ($.userName EMPTY false)
&& isValidEmail($.emails[0].value)] is not fulfilled
//the skip reason may contain a different condition, but the sentence is 'Entity
condition… is not fulfilled'
```

In case you need to check audit logs, because the entity did not meet the condition only while one provisioning job was running, then for Identity Authentication source system, see Audit Logs. The procedure should have been applied, before Identity Provisioning sends audit logs to the Audit Log service.

Find bellow an exemplary content of an audit log for update of a user in Identity Authentication source system that caused not meeting the set condition:

```
category=audit.data-change
crtAccount=<IAS tenant ID>
action=update
objectType=user
objectId=<the SCIM ID of the updated user in IAS>
payload: […]
//each changed attribute has its old and new values displayed (“oldValue” and
“newValue”)
```

The Entity does not Meet a new Condition in the Source System

In that case, the entity is skipped after the Identity Provisioning reads it from the source system and it is deleted from the target.

You can check the transformation history and whether the condition in the read transformation was changed.

You can also check if the next provisioning job for the same source-target systems show the entity as skipped in the statistics of the target system. You can download the job log for skipped entities and search for the entity in the downloaded file. For more information, see Download All Skipped Entity Logs for a Single Job [page 3011].

The entity may have temporarily failed to meet the defined condition while the deletion job was running in the target system. You can see audit logs from the source system for entity changes that lead to not applying to the condition. The audit logs should span the period before the provisioning job that deleted the entity was run.

For example, if a user is skipped due to this reason, the downloaded skipped entities job log will include similar content:

```
user:<user ID>,
system=<system name>,
skip reason=Entity condition [($.emails EMPTY false) && ($.userName EMPTY false)
&& isValidEmail($.emails[0].value)] is not fulfilled
//the skip reason may contain a different condition, but the sentence is 'Entity
condition… is not fulfilled'
```

In case you need to check audit logs, because the entity did not meet the condition only while one provisioning job was running, then for Identity Authentication source system, see Audit Logs. The procedure should have been applied, before Identity Provisioning sends audit logs to the Audit Log service.

Find bellow an exemplary content of an audit log for a user update in Identity Authentication source system that caused not meeting the set condition:

```
category=audit.data-change
crtAccount=<IAS tenant ID>
action=update
objectType=user
objectId=<the SCIM ID of the updated user in IAS>
payload: […]
//each changed attribute has its old and new values displayed (“oldValue” and
“newValue”)
```

The Entity does not Meet a new Condition set in the Target System

In that case, the entity is skipped during the provisioning to the target system and is deleted from it.

You can check the transformation history and whether the condition in the transformation of the target system was changed. After that, check the entity details in the source system itself. Consider the transformation of the source system and how the values of the entity attributes are mapped to each `“targetPath”` in the target system. Then, validate if the entity does not apply to the condition in the target system based on the mappings in the transformation of the source system and the values of the entity attributes in the source system.

For example, if the user attribute for email in the source system is `email`, and there is a mapping in the transformation of the source system like the following:

```
{
“sourcePath”: “$.email”,
“targetPath”: “$.emails[0].value”
}
```

If the condition in the target system using the `email` is the following:

```
“condition”: “$.emails[0].value =~ /.*@domain.com/”,
//checks if the email ends with the specified domain
```

Then validate that the value of the user attribute `email` in the source system ends with the specified domain in the condition of the target system.

You can also check if the next provisioning job for the same source-target systems show the entity as skipped in the statistics of the target system. You can download the job log for skipped entities and search for the entity in the downloaded file. For more information, see Download All Skipped Entity Logs for a Single Job [page 3011].

The entity may have temporarily failed to meet the defined condition while the deletion job was running in the target system. You can see audit logs from the source system for entity changes that lead to not applying to the condition. The audit logs should span the period before the provisioning job that deleted the entity was run.

If a user is skipped for not meeting the set condition in the target system, it will be seen in the downloaded skipped entities job log like:

```
user:<user ID>,
system=<system name>,
skip reason=Entity condition [($.emails EMPTY false) && ($.userName EMPTY false)
&& isValidEmail($.emails[0].value)] is not fulfilled
//the skip reason could contain a different condition, but the sentence is
'Entity condition… is not fulfilled'
```

In case you need to check audit logs, because the entity did not meet the condition only while one provisioning job was running, then for Identity Authentication source system, see Audit Logs. The procedure should have been applied, before the Identity Provisioning sends audit logs to the Audit Log service.

Find bellow an exemplary content of an audit log for user update in Identity Authentication source system that caused not meeting the set condition in the target system:

```
category=audit.data-change
crtAccount=<IAS tenant ID>
action=update
objectType=user
objectId=<the SCIM ID of the updated user in IAS>
payload: […]
// each changed attribute has its old and new values displayed (“oldValue” and
“newValue”)
```

Changed Mapping for `"entityIdTargetSystem"` or `"entityIdTargetSystem"` in the Transformation of the Source or Target System Without System Reset

This issue occurs most commonly, though not exclusively, while changing the version of the provisioning system, as this can lead to differences in the versions of the default transformations used. The symptom is mass entity update and mass entity deletion of the same entities in the target system.

If you think that this is probably your case, check the transformation history for changes in the mapping for one of the variables `"entityIdSourceSystem"` (part of the transformation of the source system) or `"entityIdTargetSystem"` (part of the transformation of the target system). Then, check in the audit logs whether a reset was performed for the provisioning system, in which one of the above mentioned variables was changed. The audit logs should span the period before the provisioning job that deleted the entity was run.

For example, in case your SAP Cloud Identity Services tenant is running on SAP Cloud Identity Services Infrastructure, refer to Access Audit Logs (Audit Log Service in SAP BTP, Cloud Foundry) [page 2967] for more information. The procedure should have been applied before the Identity Provisioning sends audit logs to the Audit Log service.

Bellow you can find an exemplary audit log for system reset of Identity Authentication source system:

```
category=audit.configuration
crtAccount=<IPS tenant ID>
action=update
objectType=System
ObjectAttribute.event-type=SYSTEM_RESET
```

```
ObjectAttribute.performed-by-user
ObjectAttribute.system-id=<the ID of the provisioning system>
```

If the provisioning system was reset, this has not caused the entity deletion.

If the provisioning system was not reset but one of the above mentioned variables was changed, that caused the entities deletion.

Removed `"delete"` Operation from skipOperations Expression or the skipOperations Expression Itself from the Transformation of the Target System

All entities, which were due to be deleted from the target system while skip of `"delete"` operation was present, were reported in previous jobs as skipped. As a result from removing the `"delete"` operation or the entire skipOperations expression, they can now be deleted from the target system by the Identity Provisioning.

In that case, you can compare the skipped statistics of the job, which deleted the entities, and the previous job for the same source-target systems. If you download all job logs, these statistics are shown in the value of `“numberSkipped”` attribute. The skipped statistics should be checked for the target system, where the entity was deleted from. You can also download the job log for skipped entities of both jobs (the one that deleted the entity from the target system and the previous one) and find the entity in the downloaded file of the older job. For more information, see Download All Skipped Entity Logs for a Single Job [page 3011].

You can also check the transformation history and whether the skipOperations expression contained `“delete”` before the provisioning job that deleted the entity from the target system was triggered.

If a user was skipped due to this reason, it will be seen in the downloaded skipped entities job log like:

```
user:<user ID>,
system=<target system name>,
skip reason=Operation [delete] is skipped in target transformation,
```

Single or Mass Entity Deletion Because the Entities Were not Deleted Earlier

Single or, more likely, mass entity deletion may occur if any of the cases described above happened during an earlier provisioning job, where an error prevented that job from deleting the entities from the target system.

Some of these errors could be:

- failed entities for the source system
- irreparable issue during provisioning, like a connection issue, which stopped the job before it reached the point of deletion of the entities
- delete operation failure
- other

As a result, the entities are queued for deletion by the next provisioning job, which will try again to delete them once the above mentioned issues are resolved.

You can start the investigation by checking the job logs, statistics, and job error logs of the provisioning jobs prior the one that deleted the entities. For more information, see Manage Provisioning Job Logs [page 3010].

Proof that you have run into some of the listed situations could include:

- If there were failed entities for the source system in the previous jobs, and then there were no failed entities for the source system in the job, which deleted the users from the target system.
- If there were some irreparable errors for the source system or for the target system (where the entities were deleted from) in the previous jobs, and then there were no such irreparable errors in the job, which deleted the entities from the target system. Irreparable errors are connection issues, like connection reset, some timeout exception, rate limits (response code 429 from the API of a provisioning system), and others.
- If there were failed entities for the target system in the previous jobs, the failures were for the entities that got deleted later from the target system, and the error job logs show failed delete operation.
  For example, if a user deletion failed, it will be displayed in the downloaded error job log as follows:

  ```
  user:<user ID>,
  system=<system name>,
  time=<time error happened>
  error=<the error details>
  content=null
  ```

As a last step, you can analyze the job logs of the previous jobs for some issues or patterns, which are not observed in the job, which deleted the entities from the target system.

How to Prevent Incorrect Entity Deletion in These Cases

By following some of the hints bellow, you can avoid unintended deletion of entities from your target systems:

- Add the standard property `ips.delete.threshold.users` (or `ips.delete.threshold.groups` for groups) to your target system configuration before running a provisioning. For more information, see the blog Thresholds Prevent Unintended Deletion of Users when Provisioning with SAP Cloud Identity Services.
- Migrate your Identity Provisioning tenant from SAP BTP Neo environment to SAP Cloud Identity Infrastructure, where you will be able to use more features (like the validate job, seeing skipped entities for a job execution, and many others). For more information, see the blog Go for your quick win! Migrate Identity Provisioning tenants to SAP Cloud Identity infrastructure.
- Always execute a validate job first, after doing a change in the transformation or properties of a provisioning system. For more information, see the blog Simulate it until you make it! Try out the Identity Provisioning job that tests your configuration.
- Follow all steps of the process for updating the connector version, decribed in Update Connector Version [page 2424].

Parent topic: Troubleshooting Provisioning Issues [page 3070]

Related Information

Administration Issues [page 3071]
Job and Transformation Issues [page 3077]
Error Messages [page 3111]
Additional Information [page 3130]
Handling Specific Attributes [page 3137]

1.12.15.5 Additional Information

Extended Explanation of the *.user.roles.overwrite Properties

Job Logs

You have a target or a proxy system of one of the following types:

- SAP S/4HANA Cloud Public Edition
- SAP Marketing Cloud
- SAP Market Communication for Utilities
- SAP BTP ABAP Environment
- SAP SAP Integrated Business Planning Cloud

You want to know the correct way to preserve or overwrite user roles on your target or proxy system. Thus, you need to know how to use the relevant **.user.roles.overwrite* properties, and whether or not to make additional changes in your transformations.

Scenarios

1. You want to keep existing user assignments (roles) on the target system
   Meaning:
   During the provisioning, you want to keep existing roles assigned to users on the target system, and add new assignments, if any . That means:
   - Keeping user assignments (roles) that exist only on the target system
   - Keeping user assignments (roles) that exist on both the source and the target system
   - Assigning new roles, which are read from the source system (if any), to users on the target system

   > **Example**
   >
   > Role A and B exist on the source system, and roles A, B, and C exist on the target system. Let's say a user is assigned to role (group) A in the source system, and the same user is assigned to roles B and C in the target system. You want role A to be added, and roles B and C to be preserved as assignments for this user in the target system. That is, after the provisioning, this user has to have roles A, B and C assigned to it on the target/proxy system.

   What do you need to do?

1. Set the relevant **.user.roles.overwrite* property to false.
2. Make sure you have the following mapping in the `"user"` object of your target/proxy write transformation:

   Sample Code

   ```
   {
       "sourcePath": "$.groups[*].value",
       "preserveArrayWithSingleElement": true,
       "targetPath": "$.user.role[?(@.roleName)]",
       "optional": true
   },
   ```

3. Remove the whole `"group"` object from your target transformation. That is:

   Sample Code

   ```
   "group": {
     "mappings": [
       {
           "sourcePath": "$.displayName",
           "targetVariable": "entityIdTargetSystem",
           "scope": "createEntity"
       },
       {
           "sourcePath": "$.displayName",
           "targetPath": "$.displayName"
       },
       {
           "sourcePath": "$.displayName",
           "targetPath": "$.ID"
       },
       {
           "sourcePath": "$.members[*].value",
           "preserveArrayWithSingleElement": true,
           "optional": true,
           "targetPath": "$.members[?(@.value)]",
           "functions": [
              {
                  "function": "resolveEntityIds"
              }
           ]
       }
     ]
   }
   ```

   Note

   For proxy systems, the `"group"` object looks a bit different.

2. You want to overwrite existing user assignments (roles) on the target system

   Meaning:

   During the provisioning, you want to unassign roles assigned to users on the target system, and assign only roles read from the source system. That means:

   - Unassigning all user assignments (roles) that are assigned to the user on the target system (unassigning roles which exist on both source and target system, as well as roles existing only on the target system)
   - Assigning user assignments (roles) read from the source system to the user on the target system

Example

Role A and B exist on the source system, and roles A, B, and C exist on the target system. Let's say a user is assigned to role (group) A in the source system, and the same user is assigned to roles B and C in the target system. You want role A to be added, and roles B and C to be unassigned from this user in the target system. That is, after the provisioning, this user has to have only role A assigned to it on the target/proxy system.

What do you need to do?

1. Set the relevant **.user.roles.overwrite* property to true.
2. Make sure you have the following mapping in the `"user"` object of your target/proxy write transformation:

   ### Sample Code

   ```
   {
       "sourcePath": "$.groups[*].value",
       "preserveArrayWithSingleElement": true,
       "targetPath": "$.user.role[?(@.roleName)]",
       "optional": true
   },
   ```

3. Remove the whole `"group"` object from your target transformation. That is:

   ### Sample Code

   ```
   "group": {
     "mappings": [
       {
           "sourcePath": "$.displayName",
           "targetVariable": "entityIdTargetSystem",
           "scope": "createEntity"
     },
     {
           "sourcePath": "$.displayName",
           "targetPath": "$.displayName"
     },
     {
           "sourcePath": "$.displayName",
           "targetPath": "$.ID"
     },
     {
           "sourcePath": "$.members[*].value",
           "preserveArrayWithSingleElement": true,
           "optional": true,
           "targetPath": "$.members[?(@.value)]",
           "functions": [
              {
                  "function": "resolveEntityIds"
              }
           ]
       }
     ]
   ```

   ### Note

   For proxy systems, the `"group"` object looks a bit different.

3. You want to update groups, read from the source system, and their assignments (users) on the target system

Meaning:
During the provisioning, you want to update groups, read from the source system, and their assignments on the target system. That means:

- Updating groups, read from the source system, and their assignments (users) on the target system
- Groups that are not read from the source system (existing only on the target system) and their assignments will not be updated on the target system

> ### Example
>
> Role A exists on the source system, and roles A and B exist on the target system. Let's say, on the source system role A has assigned member user X, and on the target system it has no members assigned. Role B has assigned member user X on the target system. You want user X to be added as a member to role A on the target, and don't want to update role B. That is, after the provisioning, role A will have member X assigned, and role B will not be updated, thus it will still have member user X assigned.

What do you need to do?

1. Set the relevant **.user.roles.overwrite* property to false.
2. Delete the following mapping in the `"user"` object from your target/proxy write transformation:

   > ### Sample Code
   >
   ```
   {
       "sourcePath": "$.groups[*].value",
       "preserveArrayWithSingleElement": true,
       "targetPath": "$.user.role[?(@.roleName)]",
       "optional": true
   },
   ```

3. Add the following `"group"` object to your target transformation.

   > ### Sample Code
   >
   ```
   "group": {
     "mappings": [
       {
          "sourcePath": "$.displayName",
          "targetVariable": "entityIdTargetSystem",
          "scope": "createEntity"
       },
       {
          "sourcePath": "$.displayName",
          "targetPath": "$.displayName"
       },
       {
          "sourcePath": "$.displayName",
          "targetPath": "$.ID"
       },
       {
          "sourcePath": "$.members[*].value",
          "preserveArrayWithSingleElement": true,
          "optional": true,
          "targetPath": "$.members[?(@.value)]",
          "functions": [
             {
                 "function": "resolveEntityIds"
             }
          ]
       }
   ```

```
]
```

> **Note**
>
> For proxy systems, look at the default write transformation and copy the `"group"` object from there.

> **Caution**
>
> When provisioning user assignments through the `"user"` object, make sure that the existing target roles have the exact same names as the relevant source groups! Otherwise, the default mapping won't work.
>
> Still, if you need to map source groups whose names are different than the relevant target roles,configure the relevant mapping the following way for all affected group-role pairs:
>
> **Sample Code**

```
...
   {
        "condition": "$.groups[*].value contains
'<sourceGroupName1>'",
        "constant": "<targetGroupName1>",
        "targetPath": "$.user.role[0].roleName"
   },
   {
        "condition": "$.groups[*].value contains
'<sourceGroupName2>'",
        "constant": "<targetGroupName2>",
        "targetPath": "$.user.role[1].roleName"
   }
...
```

Related Information

List of Properties [page 80]

SAP S/4HANA Cloud Public Edition [page 1234]

SAP Marketing Cloud [page 1196]

SAP Market Communication for Utilities [page 1187]

SAP BTP ABAP environment [page 1021]

SAP Integrated Business Planning [page 1157]

Negative Condition Statements in Transformations (Examples)

Problem

You want to use negations (**"not"** conditions) in the transformations of your provisioning systems but you struggle with finding the right way to do it.

Take a look at the examples below - they represent the most frequently used negations, according or regardless of the system type.

Examples

Negative Condition	Meaning
`"condition": "$.userName != 'John Smith'"`	"If the user name is not John Smith, then..."
`"condition": "%s4hana.cloud.hr.switch.active% != null && %s4hana.cloud.hr.switch.active% == true",`	"If the `s4hana.cloud.hr.switch.active` property is set, and its value is *TRUE* (which means HR integration is enabled), then ..."
`"condition": "('${nestedPathVariable}' != '')",`	"If the transformation variable `<nestedPathVariable>` does not have an empty value, then ..."
`"condition": "('%ldap.attribute.user.id%' != '%ldap.attribute.dn%')",`	"If the system property `ldap.attribute.user.id` does not have the same value as the system property `ldap.attribute.dn`, then ..."
`"condition": "$.emails[?(@.type != 'work')].value != []",`	"If the values of emails of type different than `'work'` are different than empty array, then ..."

SAP S/4HANA Based Systems: Business Role Description

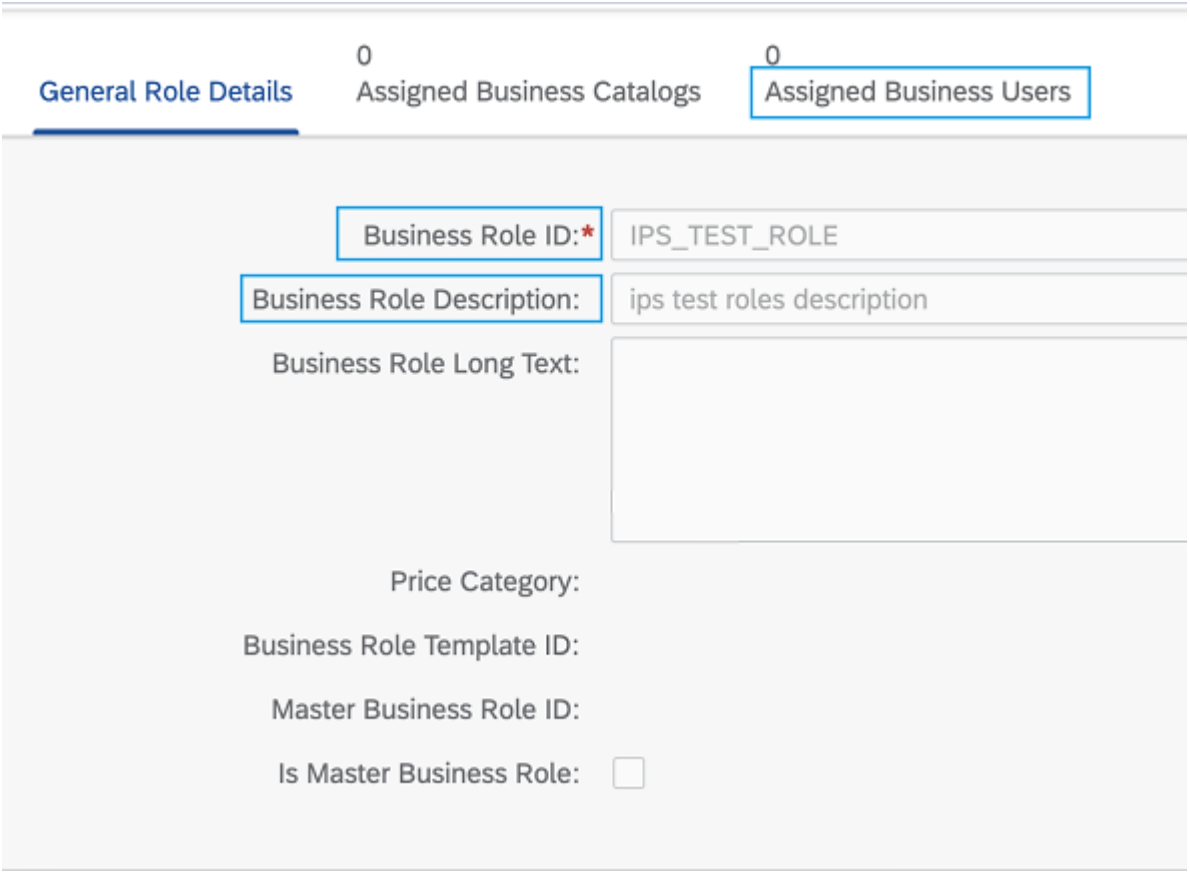
Scenario

You have a source or a proxy system of the following types:

- SAP S/4HANA Cloud Public Edition
- SAP Marketing Cloud
- SAP BTP, ABAP Environment
- SAP Integrated Business Planning Cloud

By default, Identity Provisioning reads *group ID* (= Business Role ID) and *members* (= Assigned Business Users).

Now, the service can also read *group descriptions* (= Business Role Description). To learn how to do this, see the *Solution* section below.



All this group information can be provisioned to a target system, or read by a consuming application (like SAP Identity Management) in proxy scenarios.

Solution

To make Identity Provisioning read the descriptions of groups (roles), you need to add the following mapping in the `"group"` resource of your source/proxy system:

Sample Code

```
...
   {
      "sourcePath": "$.Description",
      "optional": true,
      "targetPath": "$.description"
  },
...
```

Therefore, if you place it for example after the *ID* mapping, your `"group"` resource should look like this:

Sample Code

```
"group": {
      "mappings": [
         {
             "sourcePath": "$.ID",
             "targetPath": "$.displayName",
             "targetVariable": "entityIdSourceSystem"
         },
```

```
          {
              "sourcePath": "$.Description",
              "optional": true,
              "targetPath": "$.description"
          },
          {
              "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
              "targetPath": "$.schemas[0]"
        },
...
```

Parent topic: Troubleshooting Provisioning Issues [page 3070]

Related Information

Administration Issues [page 3071]
Job and Transformation Issues [page 3077]
Error Messages [page 3111]
Entity Deletion Issues [page 3121]
Handling Specific Attributes [page 3137]

1.12.15.6 Handling Specific Attributes

Manage specific user and group attributes during the provisioning of entities to different systems.

SAP HANA Database Username and Email

In SAP HANA Cloud, SAP HANA Database, users have an immutable HANA database username and are not expected to have an associated email. This affects provisioning scenarios, particularly with Identity Authentication, and requires changes in the transformations.

When users are provisioned from Identity Authentication to SAP HANA Cloud, SAP HANA Database, the **userID** attribute (P-user in Identity Authentication) is mapped to the **userId** attribute (HANA database username in SAP HANA Cloud, SAP HANA Database) and provisioned accordingly. The HANA database username should not to be confused with UserName. Given that the User ID in Identity Authentication is unique only within a specific tenant, consider the following cases when your organization uses multiple tenants:

Use Case 1 - Change tenants

- You have two SAP Cloud Identity Services tenants (tenant 1 and tenant 2).
- You have provisioned the users from tenant 1 to SAP HANA Cloud, SAP HANA Database.
- Now you want to change tenant 1 with tenant 2 and start provisioning the users from tenant 2 to SAP HANA Cloud, SAP HANA Database.

In this case, one and the same user in tenant 1 and tenant 2 might have different P user values. For example: John Smith (P000001) exists in tenant 1 and John Smith (P000002) exists in tenant 2. The Identity Provisioning won't be able to transfer the value from tenant 2, because the HANA database username is immutable.

We suggest handling this as follows:

1. Delete the affected users in the target SAP HANA Cloud, SAP HANA Database system. That is, John Smith (P000001).
2. Run a provisioning job from the Identity Authentication source system in your tenant 2.
   As a result, the user John Smith will be provisioned with his P000002 P user.

Use Case 2 - Use more than one tenant

- You have two SAP Cloud Identity Services tenants (tenant 1 and tenant 2) in different regions, for example, and you want to use both.

In this case, different users in tenant 1 and tenant 2 might have the same P user value. For example, John Smith (P000001) in tenant 1 (Europe) and Julie Armstrong (P000001) in tenant 2 (Asia).

We suggest handling this as follows:

1. Modify the read transformation in the Identity Authentication source system of your tenant 1 by adding, for example, a prefix or a suffix to the P user value.

Code Syntax

```
{
  "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userId']",
  "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userId']",
  "functions": [
    {
      "function": "concatString",
      "prefix": "Europe_"
    }
  ]
}
```

Code Syntax

```
{
  "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userId']",
  "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userId']",
  "functions": [
    {
      "function": "concatString",
      "suffix": "_Europe"
    }
  ]
}
```

2. Modify the read transformation in the Identity Authentication source system of your tenant 2 by adding, for example, a prefix or a suffix to the P user value.

Code Syntax

```
{
  "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userId']",
  "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userId']",
  "functions": [
```

Code Syntax

```
{
  "sourcePath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userId']",
  "targetPath": "$
['urn:ietf:params:scim:schemas:ext
ension:sap:2.0:User']['userId']",
  "functions": [
```

```
    {
      "function": "concatString",
      "prefix": "Asia_",
    }
  ]
}
```

```
    {
      "function": "concatString",
      "suffix": "_Asia"
    }
  ]
}
```

3. Run provisioning jobs from both Identity Authentication source systems in tenant 1 and tenant 2.
   As a result, you will distinguish the different users with the same P user value by adding a prefix.

Use Case 3 - Email of SAP HANA Database User

- You want to provision SAP HANA Database users to Identity Authentication. The users don't have email addresses, as this is neither expected nor needed. However, email is a required user attribute in Identity Authentication.

We suggest handling this as follows:

1. Modify the read transformation in the SAP HANA Cloud, SAP HANA Database source system to construct a fictitious email address by appending the username with a suffix that includes the "@" sign and a domain.

   Code Syntax

   ```
   {
     "sourcePath": "$.userName",
     "targetPath": "$.emails[0].value",
     "functions": [
       {
         "function": "concatString",
         "suffix": "@test.test"
       }
     ]
   }
   ```

2. Run a provisioning job from the source system.
   As a result, you will meet the following condition of the target Identity Authentication: `"condition": "($.emails EMPTY false) && ($.userName EMPTY false) && isValidEmail($.emails[0].value)"`, and the job will not fail due to a missing required attribute.

Parent topic: Troubleshooting Provisioning Issues [page 3070]

Related Information

Administration Issues [page 3071]
Job and Transformation Issues [page 3077]
Error Messages [page 3111]
Entity Deletion Issues [page 3121]
Additional Information [page 3130]

1.13 Submitting Improvement Requests

You can submit your improvement request for Identity Authentication. It's a single central place for all product improvement request related activities. You can browse improvement requests, submit improvement requests, comment, vote, receive updates, and browse who has voted.

Context

You can also browse improvement requests, comment, vote, receive updates, and browse who has voted.

> **Note**
>
> Before you submit your improvement, check if a similar improvement hasn't already been submitted requested. If yes, vote for it, instead of submitting a new one.

Procedure

1. Access the SAP Continuous Influence site from the following link: https://cis.influence.sap.com/
2. Sign in with your email if required.
3. Choose *SAP Business Technology Platform - Foundation* > *Security Services>Add an improvement request* .
4. Enter the details for the improvement request that you want to submit.

Field	Notes
Choose a session for this Improvement Request	SAP Business Technology Platform. It's already prefilled from the drop-down list.
Your Improvement Request	Enter a one-sentence summary here clearly stating your request and area of improvement.
Choose a category for this Improvement Request	*Security Services*. It's already prefilled from the drop-down list.
Problem description	Provide details about your idea.
Solution proposal	Propose a solution.
Expected benefit	Explain what is the benefit from this improvement.

   You can also provide links and attachments.

5. Choose the *ADD IMPROVEMENT REQUEST* button when you're ready.

1.14 Getting Support

This document is to help users, administrators, and developers deal with issues from SAP Cloud Identity Services.

SAP Support Portal

You can report an incident on SAP Support Portal Home with the following components:

- `BC-IAM-IDS` - for issues related to authentication and user and group management
- `BC-IAM-IPS` - for issues related to user and group provisioning
- `BC-IAM-OID` - for issues related to OpenID Connect tokens

> **Note**
>
> Please include in the incident the following information:
>
> - Your tenant ID from `https://tenant ID>.accounts.ondemand.com/admin` or `https://<tenant ID>.accounts.cloud.sap/admin`
> - Your email address
> - The error message you see
> - Attachments of screenshot that show the reproduced error, including the exact timestamp
> - If you solved your issue, a description of the steps you took
> - List of Knowledge Base Articles (KBAs) and SAP Notes, you have used to solve the issue
> - (if relevant) Attachments of SAML traces as per KBA: 2461862 - Collecting SAML traces with Chrome or Firefox
>
> See also: 2654164 (What to consider before opening a Support Incident for SAP BTP issue).

SAP Community Network

You can start a discussion in the Security Community.

Related Information

Error and Success Codes [page 2883]

1.15 Updates and Notifications

SAP Cloud Identity Services production updates follow a regular biweekly release schedule and are generally delivered mid-week. For more information about the newly introduced features, see the What's New for Cloud Identity Services [page 26] page.

Subscription for Notifications

To get notifications about new releases and related updates, subscribe to the *What's New* section of Cloud Identity Services.

Reasons for Updates

- **Bi-weekly updates** (standard)
  Standard updates are delivered on a bi-weekly basis. You can find the planned schedule for upcoming releases in the SAP Note 3409744 (Release Schedule for SAP Cloud Identity Services (SAP BTP)).
- **Immediate updates**
  Immediate updates are provided when fixes are required for issues that affect productive application operations or when urgent security fixes are necessary.

> **Note**
>
> No service interruption is expected as a result of the updates.

Service Availability

You can monitor the availability of Identity Authentication in Cloud Availability Center CAC/SAP4Me. For that you can see a T-installation for Identity Services in the support tools. The Identity Authentication tenants of a customer are assigned to this T-installation. Based on these T-installations you can request outage notifications, and see potential Root Cause Analysis (RCA) information.

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